Seminar on Algebraic Geometry 3

Group Schemes

English Translation and TEX Edition
Exposés I–XXVI,
Tome-I subject index, Tome-III mathematical guide, and terminal index

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Editorial Notice

We present here a slightly revised edition of the original Seminar, whose purpose and contents are described in the Introduction. The revision consisted mainly of correcting typographical errors, adding a few supplementary remarks or references in footnotes, dividing the work in its present form into three volumes, each with a detailed table of contents and an index of notation, and adding a terminological index at the end of Volume 3. In addition, Exposé $\mathrm{VI_B}$, by J.-E. Bertin, was partially rewritten by the author, especially Sections 5 and 10, so some references to that Exposé differ from the references to the original version. A list of the Exposés in the Seminar appears at the beginning of this volume.

Since the first edition of the present Seminar appeared, all of Chapter IV of the *Éléments de Géométrie Algébrique* has been published, making certain passages of the Seminar unnecessary. We have occasionally indicated in footnotes the relevant references to EGA IV that allow those passages to be bypassed.

For another treatment of algebraic groups that systematically uses the language of schemes, we refer to M. Demazure and P. Gabriel, *Groupes Algébriques* (North-Holland & Masson et Cie). Unlike the present Seminar, that book assumes no knowledge of algebraic geometry, but includes all the required preliminaries from scheme theory; it can therefore serve as an introduction to the study of this Seminar. (It also treats subjects not covered in the Seminar, such as the Dieudonné-style structure theory of commutative affine algebraic groups, in *loc. cit.*, Chapter V.)

Bures-sur-Yvette, March 1970

Introduction

1. The purpose of this Seminar is twofold.

On the one hand, we aim to provide convenient foundations for the theory of group schemes in general. Exposés I–IV will supply the indispensable preliminary exercises in schematic and categorical syntax. To obtain a language that fits geometric intuition without effort, and to avoid circumlocutions that become intolerable in the long run, we always identify a prescheme X over another prescheme S with the functor

$$(\mathbf{Sch})^{\circ}_{/S} \longrightarrow \mathbf{Set}$$

that it represents*. It is therefore necessary to formulate many definitions so that they apply to arbitrary functors, whether representable or not. Moreover, nearly all the functors we shall use will be "sheaves" (for the "faithfully flat quasi-compact topology"). Exposé IV, which treats groups only incidentally, sketches the language of "localization" and sheaves; this language also proves very convenient in questions concerning the representability of functors. Above all, that Exposé will provide the most convenient framework for the questions of passage to a quotient that arise later.

Exposé V gives some general results on the existence of quotients. They are taken up again in Exposé VI_{A} for the quotient of an algebraic group over a field (or, more generally, over an Artinian ring) by a subgroup**. That Exposé and the following Exposé VI_{B} also contain various elementary results specific to algebraic groups over a field that are used routinely later.

Exposé VII studies certain phenomena related to the characteristic of the base field and, in particular, develops with the appropriate generality the correspondence between radicial group schemes of height 1 and restricted p-Lie algebras.

Finally, Exposé XVIII contains the scheme-theoretic generalization of Weil's theorem on the "birational" definition of algebraic groups.

On the other hand, we intend to generalize the Borel–Chevalley structure theory of affine algebraic groups to groups over an arbitrary base prescheme. In fact, while the Seminar notes were being written, it became apparent that the affine hypothesis was unnecessary for many results of the theory. The most complete results are of course obtained for "semisimple

*Such a viewpoint seems to have been considered for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to state it precisely and systematize it as it deserved.

**For a more thorough study of passage to the quotient, especially by reductive groups, see D. Mumford's important work *Geometric Invariant Theory*, Ergebnisse der Mathematik, vol. 34, Springer, 1965. We note that on one important point the terminology of that book does not agree with ours: over a field of characteristic $p > 0$, the groups Mumford calls "reductive"(1) are the smooth groups of multiplicative type in the sense of the Seminar (cf. Exposé IX). One may reasonably suppose that Mumford adopted this meaning of "reductive," which loses its significance over a base that is not a field, provisionally and as a kind of stopgap (and this is approximately what Mumford explains, for other reasons, as early as the second paragraph of his preface). (1) *Editor's note:* See the remarks added at the end of this Introduction.

group schemes," or more generally "reductive group schemes," with which we shall be exclusively concerned beginning with Exposé XIX. Chevalley himself had already constructed the "Tôhoku groups" over the ring of integers, and that construction will be taken up in the present Seminar. The principal uniqueness theorem[(2)] gives a simple characterization of the "twisted" variants of these Tôhoku groups over a base prescheme S: they are the affine, smooth group schemes over S whose geometric fibers are connected semisimple groups in the usual sense[∗∗∗(2)].

2.

As in the familiar theory over an algebraically closed field, the subtori of the group schemes under consideration play a crucial role. Thus the preliminary study of tori and, more generally, of "group schemes of multiplicative type"—both intrinsically and as multiplicative-type subgroups of a given group—occupies a fairly large part of this Seminar (Exposés VIII–XII). Their remarkable rigidity (in some respects even greater than that of abelian schemes or semisimple group schemes) makes them highly effective tools for studying certain more general groups.

3.

Beginning with Exposé XII (apart from Exposé XVIII, already mentioned), we shall routinely use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the 1956 Chevalley Seminar, especially in Exposés IV–IX of that Seminar. We shall also use, though to a lesser extent, the later Exposés of the Chevalley Seminar devoted to the structure of semisimple algebraic groups. Indeed, we shall redevelop Chevalley's theory directly within the framework of schemes; it will be seen that this approach makes the exposition simpler and more precise, even over a base field.

4.

The principal purpose of this Seminar is, of course, to develop techniques that apply to the study of group schemes over an arbitrary base, that is, essentially, to the study of families of algebraic groups. The infinitesimal properties of such families, and in particular the case of an Artinian base scheme, therefore play an important role. These properties arise even in the study of algebraic groups over a field k when the field is not perfect, especially when applying descent in the non-Galois case. Among the new results obtained in this setting, let us mention the existence of maximal tori and Cartan subgroups in an arbitrary smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exposé XIV[(3)]); another example is the correspondence between the "forms" of a semisimple group and homogeneous principal bundles under a suitable semisimple algebraic group (generally not connected) (Exposé XXIV, 1.16–1.20). In general, one may say that the methods required for work over a nonperfect base field are essentially those used over arbitrary base preschemes and thus lie outside the framework of classical algebraic geometry.

[(2)] *Editor's note:* This refers to Corollary XXIII.5.6, which follows easily from the uniqueness theorem for split reductive groups (cf. XXIII, Theorem 4.1 and Corollary 5.2), since every semisimple S-group is a "twisted form" (for the *étale* topology) of a Tôhoku group (cf. XXII, Corollary 2.3).

[∗∗∗] This is the essential result of M. Demazure's thesis, *Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413.

[(3)] *Editor's note:* In particular, Theorems 1.1 and 6.1.

5.

It did not seem useful to indicate, at the beginning of the written Exposés, the date or dates of the corresponding oral presentations in the Seminar. We shall merely say that the order of the mimeographed Exposés (I–XXVI) does indeed agree with the order of the oral presentations. Moreover, the final written text was sometimes prepared considerably later than the oral presentation and often differs from it quite substantially: the written text is generally more detailed and more complete (as in Exposés IV and $\mathrm{VII_B}$), or even appreciably more general (as in Exposés XII and $\mathrm{VII_B}$), than the oral presentation. Other written Exposés correspond to no oral presentation ($\mathrm{VI_B}$, $\mathrm{VII_A}$, XV, XVI, XVII, and most of XXVI); they were written and inserted into the mimeographed Seminar either to provide convenient references for various other Exposés (notably in the case of $\mathrm{VI_B}$) or because they form a natural continuation of the concepts and techniques already developed. Consequently, Exposés $\mathrm{VII_A}$, $\mathrm{VII_B}$, XV, XVI, and XVII need not be read in order to study the rest of the Seminar, particularly the part devoted to reductive group schemes.

6.

From scheme theory we shall use primarily the general language of schemes set out in EGA I; the notions of flat, *é*tale, and smooth morphisms set out in SGA 1, I–V; and the theory of faithfully flat descent in SGA 1, VIII. As far as possible, we have avoided imposing unnecessary Noetherian hypotheses. In return, this has required us to replace the usual hypothesis "of finite type" with the hypothesis "of finite presentation." For the notion of a morphism of finite presentation, the reader should consult EGA IV, 1.4 and 1.6. The results of SGA 1, I–IV, which are most often stated in the Noetherian setting, will be developed in general in EGA IV[* * **]; standard methods for reducing certain kinds of statements involving finite-presentation hypotheses to the Noetherian case will also be developed there in detail (EGA IV, Sections 8, 9, and 11). A reader who does not wish to assume these results from EGA IV may simplify certain statements or their proofs by assuming the base prescheme to be locally Noetherian. This does, however, lead to difficulties when a proof proceeds by descent from $\widehat{A}$ to A, where $\widehat{A}$ is the completion of a Noetherian local ring A, because that method introduces the generally non-Noetherian ring

$$\widehat{A} \otimes_A \widehat{A}.$$

7.

References follow the usual decimal system: the reference 5.7.11 denotes the proposition (or lemma, definition, and so forth) with that number in the same Exposé; in a reference such as XVII 7.8, the Roman numeral denotes the Exposé. We use the following abbreviations for our standard references:

Bible Chevalley Seminar, *Groupes de Lie algébriques*, 1956/58.

EGA X**,** $x.y.z$ J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chapter X, statement $x.y.z$ (or subsection $x.y$, and so forth).

SGA n**,** X $y.z$ *Séminaire de Géométrie Algébrique du Bois-Marie*, year n, statement $y.z$ of Exposé X.

[* * **]Since this Introduction was written, all four parts (Sections 1–21) of EGA IV have appeared.

TDTE A. Grothendieck, *Techniques de descente et Théorèmes d'existence en Géométrie Algébrique*, Exposés delivered in the Bourbaki Seminar between 1959 and 1962.

(1) *Editor's note.* Concerning quotients by a reductive group, the situation has evolved considerably since A. Grothendieck wrote Note ($**$). For an affine algebraic group over an arbitrary field k, the appropriate notion, introduced by Mumford, is that of a "geometrically reductive" group. (On the other hand, G is said to be "linearly reductive" if every rational representation of G is completely reducible; as noted in Note ($**$), however, that condition is too restrictive when $\mathrm{char}(k) > 0$.) By results of M. Nagata and W. J. Haboush ([Na64], [NM64], [Ha75]), the geometrically reductive k-groups are precisely the reductive k-groups in the sense of the present Seminar. For all of this, see the later editions of Mumford's book ([MF82]). The notion of geometric reductivity over an arbitrary base, together with its consequences for passage to a quotient, was extended to this setting by C. S. Seshadri ([Se77]); additions due to M. Raynaud appear in the article [CTS79] by J.-L. Colliot-Thélène and J.-J. Sansuc. Finally, for more recent developments on passage to the quotient by an algebraic group or, more generally, by a groupoid (cf. Exposé V of the present Seminar), we refer to the articles [Ko97] and [KM97] by J. Kollár and by S. Keel and S. Mori.

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Exposé I
Algebraic Structures. Group Cohomology

by M. Demazure

This exposé consists of two parts: the first collects a number of general definitions and establishes notation that will often be used in the sequel; the second treats group cohomology and culminates in Theorem 5.3.3, on the vanishing of the cohomology of diagonalizable groups. [1]

We choose once and for all a Universe.[2] All definitions and constructions will be relative to this Universe. We shall systematically allow ourselves the following abuse of language. To define a functor $f : \mathscr{C} \to \mathscr{C}'$, we shall be content to define the object $f(S)$ of $\mathscr{C}'$ for every object S of $\mathscr{C}$, whenever there is no ambiguity about how to define $f(h)$ for an arrow h of $\mathscr{C}$. In practice, we shall say: let $f : \mathscr{C} \to \mathscr{C}'$ be the functor defined by $f(S) = \cdots$.

1. Generalities

1.1.

Let $\mathscr{C}$ be a category. We denote by $\widehat{\mathscr{C}}$ the category

$$\mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

of contravariant functors from $\mathscr{C}$ to the category **Set** of sets.[3] There is a canonical functor

$$\mathbf{h} : \mathscr{C} \longrightarrow \widehat{\mathscr{C}}$$

which associates with every $X \in \mathrm{Ob}(\mathscr{C})$ the functor $\mathbf{h}_X$ such that

$$\mathbf{h}_X(S) = \mathrm{Hom}(S, X).$$

For every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, one has (see, for example, EGA 0_{III}, 8.1.4) a bijection

$$\mathrm{Hom}(\mathbf{h}_X, \mathbf{F}) \xrightarrow{\sim} \mathbf{F}(X).$$

[4]

[1] Editor's note: version of October 13, 2024.

[2] Editor's note: see SGA 4, Exposé I, §0 and the Appendix; see also the discussion in [DG70], p. xxvi.

[3] Editor's note: this is called the category of presheaves on $\mathscr{C}$; see IV, 4.3.1.

[4] Editor's note: this result is often called the Yoneda lemma; we use that terminology in other editor's notes.

In particular, for every pair X, Y of objects of $\mathscr{C}$, the canonical map

$$\operatorname{Hom}_{\mathscr{C}}(X,Y) \xrightarrow{\sim} \operatorname{Hom}_{\widehat{\mathscr{C}}}(\mathbf{h}_X, \mathbf{h}_Y)$$

is bijective; that is, the functor $\mathbf{h}$ is fully faithful. It therefore defines an isomorphism of $\mathscr{C}$ onto a full subcategory of $\widehat{\mathscr{C}}$, and an equivalence of $\mathscr{C}$ with the full subcategory of $\widehat{\mathscr{C}}$ formed by the representable functors (that is, the functors isomorphic to one of the form $\mathbf{h}_X$). In the sequel we shall often identify X with $\mathbf{h}_X$. The following paragraphs are intended to show that this identification can be made safely.

Remark 1.1.1.[5] We sometimes need the following variant. Let $\mathscr{D}$ be a full subcategory of $\mathscr{C}$, and let $X, Y \in \operatorname{Ob}(\mathscr{D})$. Denote by $\mathbf{h}'_X$ and $\mathbf{h}'_Y$ the restrictions of $\mathbf{h}_X$ and $\mathbf{h}_Y$ to $\mathscr{D}$. Then

$$\operatorname{Hom}_{\mathscr{C}}(X,Y) = \operatorname{Hom}_{\mathscr{D}}(X,Y) = \operatorname{Hom}_{\widehat{\mathscr{D}}}(\mathbf{h}'_X, \mathbf{h}'_Y).$$

Thus, to give a morphism $X \to Y$ "is the same thing" as to give, for every $T \in \operatorname{Ob}(\mathscr{D})$, a map

$$\phi(T) : \operatorname{Hom}(T,X) \longrightarrow \operatorname{Hom}(T,Y)$$

functorially in T.

1.2.

We shall say that $\mathbf{F}$ is a *subobject* (or a subfunctor) of $\mathbf{G}$ if $\mathbf{F}(S)$ is a subset of $\mathbf{G}(S)$ for every S.[6]

In $\widehat{\mathscr{C}}$, arbitrary inverse limits exist and are computed pointwise:

$$\left(\varprojlim_i \mathbf{F}_i\right)(S) = \varprojlim_i \mathbf{F}_i(S).$$

[7] In particular, fiber products are defined by

$$(\mathbf{F} \times_{\mathbf{G}} \mathbf{F}')(S) = \mathbf{F}(S) \times_{\mathbf{G}(S)} \mathbf{F}'(S).$$

[8]

As a final object of $\widehat{\mathscr{C}}$, we choose the functor $\underline{\mathbf{e}}$ such that $\underline{\mathbf{e}}(S) = \{\varnothing\}$.[9] Every $\mathbf{F} \in \operatorname{Ob}(\widehat{\mathscr{C}})$ has a unique morphism to $\underline{\mathbf{e}}$, and we set

$$\mathbf{F} \times \mathbf{F}' = \mathbf{F} \times_{\underline{\mathbf{e}}} \mathbf{F}'.$$

The functor $\mathbf{h}$ commutes with inverse limits. In particular, for $X \times X'$ to exist, where $X, X' \in \operatorname{Ob}(\mathscr{C})$, respectively for $\mathscr{C}$ to admit a final object e, it is necessary and sufficient that $\mathbf{h}_X \times \mathbf{h}_{X'}$, respectively $\underline{\mathbf{e}}$, be representable; in that case

$$\mathbf{h}_X \times \mathbf{h}_{X'} \simeq \mathbf{h}_{X \times X'}, \qquad \mathbf{h}_e \simeq \underline{\mathbf{e}}.$$

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[5] Editor's note: this remark has been added.

[6] Editor's note: the order has been changed so that fiber products are introduced before monomorphisms; see editor's note (9).

[7] Editor's note: similarly, arbitrary direct limits exist and are computed pointwise: $\left(\varinjlim_i \mathbf{F}_i\right)(S) = \varinjlim_i \mathbf{F}_i(S)$. In general, however, the functor $\mathbf{h}$ does not commute with direct limits.

[8] Editor's note: in particular, the equalizer of a pair of morphisms $u, v : \mathbf{F} \rightrightarrows \mathbf{G}$ is the subfunctor $\operatorname{Ker}(u,v)$ of $\mathbf{F}$ defined by $\operatorname{Ker}(u,v)(S) = \{x \in \mathbf{F}(S) \mid u(x) = v(x)\}$.

[9] Editor's note: $\{\varnothing\}$, the power set of the empty set, is a one-element set.

A monomorphism in $\widehat{\mathscr{C}}$ is precisely a morphism $\mathbf{F} \to \mathbf{G}$ such that, for every $S \in \mathrm{Ob}(\mathscr{C})$, the corresponding map of sets $\mathbf{F}(S) \to \mathbf{G}(S)$ is injective.[10]

The functor Γ. For every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, set

$$\Gamma(\mathbf{F}) = \mathrm{Hom}(\underline{\mathbf{e}}, \mathbf{F}).$$

An element of $\Gamma(\mathbf{F})$ is therefore a family $(\gamma_S)_{S \in \mathrm{Ob}(\mathscr{C})}$, with $\gamma_S \in \mathbf{F}(S)$, such that for every arrow $f : S' \to S''$ of $\mathscr{C}$,

$$\mathbf{F}(f)(\gamma_{S''}) = \gamma_{S'}.$$

For $X \in \mathrm{Ob}(\mathscr{C})$, set $\Gamma(X) = \Gamma(\mathbf{h}_X)$. If $\mathscr{C}$ has a final object e, then

$$\Gamma(X) \simeq \mathrm{Hom}(e, X).$$

1.3.

Let $S \in \mathrm{Ob}(\mathscr{C})$. We denote by $\mathscr{C}_{/S}$ the category of objects of $\mathscr{C}$ over S: its objects are the arrows $f : T \to S$ of $\mathscr{C}$, and $\mathrm{Hom}(f, f')$ is the subset of $\mathrm{Hom}(T, T')$ formed by the arrows u such that $f = f' \circ u$. If $\mathscr{C}$ has a final object e, then $\mathscr{C}_{/e}$ is isomorphic to $\mathscr{C}$. The category $\mathscr{C}_{/S}$ has a final object, namely the identity arrow $S \to S$.

If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, one may form the category $(\mathscr{C}_{/S})_{/f}$, which by abuse of language we denote by $(\mathscr{C}_{/S})_{/T}$, and there is a canonical isomorphism

$$\mathscr{C}_{/T} \simeq (\mathscr{C}_{/S})_{/T}.$$

This construction also applies to $\widehat{\mathscr{C}}$; in particular, one obtains the category $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. On the other hand, one may form the category $\widehat{\mathscr{C}_{/S}}$.

If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S, that is, arrows $S \to T$ that are right inverses to f. Moreover, $\mathbf{h}_f : \mathbf{h}_T \to \mathbf{h}_S$ is then an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and

$$\Gamma(\mathbf{h}_f) \simeq \Gamma(\mathbf{h}_T/\mathbf{h}_S) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

1.4.

We now define an equivalence between the categories $\widehat{\mathscr{C}_{/S}}$ and $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. In other words, we shall prove that to give a functor on the category of objects of $\mathscr{C}$ over S is "the same thing" as to give a functor on $\mathscr{C}$ equipped with a morphism to $\mathbf{h}_S$.

(i) Construction of $\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \to \widehat{\mathscr{C}_{/S}}$.

First let $H : \mathbf{F} \to \mathbf{h}_S$ be an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. We must define a functor $\alpha_S(H)$ on $\mathscr{C}_{/S}$. For an object $f : T \to S$ of $\mathscr{C}_{/S}$, define $\alpha_S(H)(f)$ to be the inverse image of $f \in \mathbf{h}_S(T)$ under the map

$$H(T) : \mathbf{F}(T) \longrightarrow \mathbf{h}_S(T).$$

[11]

[10]Editor's note: if $\mathbf{F}(S) \to \mathbf{G}(S)$ is injective for every S, it is clear that $\mathbf{F} \to \mathbf{G}$ is a monomorphism. The converse follows by considering $\mathbf{F} \times_{\mathbf{G}} \mathbf{F} \rightrightarrows \mathbf{F} \to \mathbf{G}$. Thus $\mathbf{F} \to \mathbf{G}$ is a monomorphism if and only if the diagonal morphism $\mathbf{F} \to \mathbf{F} \times_{\mathbf{G}} \mathbf{F}$ is an isomorphism (see EGA I, 5.3.8). Similarly, a pointwise surjective morphism is an epimorphism, and the converse follows by considering the pushout $\mathbf{G} \amalg_{\mathbf{F}} \mathbf{G}$; see the proof of Lemma 4.4.4 in Exposé IV.

[11]Editor's note: for example, if $\mathbf{F} = \mathbf{h}_X$, then H corresponds to a morphism $h : X \to S$, and $H(T) : \mathrm{Hom}_{\mathscr{C}}(T, X) \to \mathrm{Hom}_{\mathscr{C}}(T, S)$ is the map $g \mapsto h \circ g$. Thus $\alpha_S(\mathbf{h}_X) = \mathrm{Hom}_{\mathscr{C}_{/S}}(-, X)$.

Next, let $u : f \to f'$ be an arrow of $\mathscr{C}_{/S}$. Then $\mathbf{F}(u) : \mathbf{F}(T') \to \mathbf{F}(T)$ induces a map from $\alpha_S(H)(f')$ to $\alpha_S(H)(f)$, denoted $\alpha_S(H)(u)$. It is immediate that the assignments

$$f \longmapsto \alpha_S(H)(f), \qquad u \longmapsto \alpha_S(H)(u)$$

define a functor on $\mathscr{C}_{/S}$, hence an object $\alpha_S(H)$ of $\widehat{\mathscr{C}_{/S}}$.

Finally, let $H : \mathbf{F} \to \mathbf{h}_S$ and $H' : \mathbf{F}' \to \mathbf{h}_S$ be two objects of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and let $U : H \to H'$ be a morphism in that category:

$$\begin{tikzcd}
\mathbf{F} \arrow[rr, "U"] \arrow[dr, "H"'] & & \mathbf{F}' \arrow[dl, "H'"] \\
& \mathbf{h}_S &
\end{tikzcd}.$$

For every $f : T \to S$, the map $U(T) : \mathbf{F}(T) \to \mathbf{F}'(T)$ induces a map

$$\alpha_S(U)(f) : \alpha_S(H)(f) \longrightarrow \alpha_S(H')(f),$$

which defines a morphism of functors

$$\alpha_S(U) : \alpha_S(H) \longrightarrow \alpha_S(H').$$

It is easy to verify that the assignments

$$H \longmapsto \alpha_S(H), \qquad U \longmapsto \alpha_S(U)$$

define a functor

$$\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \longrightarrow \widehat{\mathscr{C}_{/S}}.$$

(ii) Proposition 1.4.1. *The functor $\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \to \widehat{\mathscr{C}_{/S}}$ is an equivalence of categories.*

We indicate only the principle of the construction of a quasi-inverse functor

$$\beta_S : \widehat{\mathscr{C}_{/S}} \longrightarrow \widehat{\mathscr{C}}_{/\mathbf{h}_S}.$$

Let $\mathbf{G}$ be a functor on $\mathscr{C}_{/S}$. For every object T of $\mathscr{C}$, set

$$\beta_S(\mathbf{G})(T) = \coprod_{f \in \mathrm{Hom}(T,S)} \mathbf{G}(f), \qquad \mathrm{Hom}(T,S) = \mathbf{h}_S(T).$$

This defines a functor $\beta_S(\mathbf{G})$ on $\mathscr{C}$, equipped with the evident projection to $\mathbf{h}_S$.

1.5.

The equivalence α_S commutes with the functors Γ. In other words, if $H : \mathbf{F} \to \mathbf{h}_S$ is an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$ and $\alpha_S(H)$ is the corresponding object of $\widehat{\mathscr{C}_{/S}}$, then

$$\Gamma(\alpha_S(H)) \simeq \Gamma(H) \simeq \Gamma(\mathbf{F}/\mathbf{h}_S).$$

The equivalence α_S also commutes with the functors $\mathbf{h}$. That is, if $f : T \to S$ is an object of $\mathscr{C}_{/S}$, then $\mathbf{h}_f : \mathbf{h}_T \to \mathbf{h}_S$ is an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and its image under α_S is precisely $\mathbf{h}_{\mathscr{C}_{/S}}(f)$, where

$$\mathbf{h}_{\mathscr{C}_{/S}} : \mathscr{C}_{/S} \longrightarrow \widehat{\mathscr{C}_{/S}}$$

is the canonical functor.[12] Consequently:

Proposition 1.5.1. *Let $H : \mathbf{F} \to \mathbf{h}_S$ be an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. The functor*

$$\alpha_S(H) : (\mathscr{C}_{/S})^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if $\mathbf{F} : \mathscr{C}^\circ \to \mathbf{Set}$ is representable. If $\mathbf{F} \simeq \mathbf{h}_T$, then $\alpha_S(H)$ is represented by the object $T \to S$ of $\mathscr{C}_{/S}$.

The equivalence α_S is transitive in S. If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, there is a commutative diagram of equivalences:

$$\begin{tikzcd}
(\widehat{\mathscr{C}}_{/\mathbf{h}_S})_{/\mathbf{h}_T} \arrow[r, "\alpha_{S/\mathbf{h}_T}"] \arrow[dr, "\simeq"'] & (\widehat{\mathscr{C}_{/S}})_{/\mathbf{h}_T} \arrow[r, "\alpha_f"] & \widehat{(\mathscr{C}_{/S})_{/T}} \arrow[dl, "\simeq"] \\
& \widehat{\mathscr{C}}_{/\mathbf{h}_T} \arrow[r, "\alpha_T"] & \widehat{\mathscr{C}_{/T}}
\end{tikzcd},$$

Here $\alpha_{S/\mathbf{h}_T}$ denotes, provisionally, the restriction (see 1.6) of the functor α_S to the objects over $\mathbf{h}_T$.

1.6. Base change in a functor

For every $S \in \mathrm{Ob}(\mathscr{C})$, there is a canonical functor

$$i_S : \mathscr{C}_{/S} \longrightarrow \mathscr{C}$$

defined by $i_S(f) = T$ when f is the arrow $T \to S$. If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, we denote by $i_{T/S} = i_f$ the functor

$$i_{T/S} : (\mathscr{C}_{/S})_{/T} \longrightarrow \mathscr{C}_{/S},$$

and there is a commutative diagram:

$$\begin{tikzcd}
(\mathscr{C}_{/S})_{/T} \arrow[r, "i_{T/S}"] \arrow[dr, "\simeq"'] & \mathscr{C}_{/S} \arrow[r, "i_S"] & \mathscr{C} \\
& \mathscr{C}_{/T} \arrow[ur, "i_T"'] &
\end{tikzcd},$$

Thus, identifying $(\mathscr{C}_{/S})_{/T}$ with $\mathscr{C}_{/T}$, as we shall henceforth do,

$$i_S \circ i_{T/S} = i_T.$$

Likewise, if $\mathscr{C}$ is identified with $\mathscr{C}_{/e}$ when $\mathscr{C}$ has a final object e, then $i_{S/e} : \mathscr{C}_{/S} \to \mathscr{C}_{/e}$ is identified with i_S.

For $X \in \mathrm{Ob}(\mathscr{C})$ (respectively, $Y \in \mathrm{Ob}(\mathscr{C}_{/S})$), let $p_S(X)$ (respectively, $p_{T/S}(Y)$) be the object of $\mathscr{C}_{/S}$ (respectively, of $\mathscr{C}_{/T}$), when it exists, defined by $X \times S$ (respectively, $Y \times_S T$) equipped with its second projection:

[12] Editor's note: see editor's note (10).

$$\begin{array}{ccc} X \times S & \text{resp.} & Y \times_S T \\ \Big\downarrow{\scriptstyle p_S(X)} & & \Big\downarrow{\scriptstyle p_{T/S}(Y)} \\ S & & T \quad . \end{array}$$

The partially defined functor p_S (respectively, $p_{T/S}$) is called the *base-change functor*. By the definition of a product (respectively, a fiber product), it is right adjoint to i_S (respectively, to $i_{T/S}$). [13] We also write

$$p_S(X) = X_S, \qquad p_{T/S}(Y) = Y_T.$$

The functor i_S defines a restriction functor

$$i_S^* : \widehat{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}_{/S}}.$$

We write

$$\mathbf{F}_S = i_S^*(\mathbf{F}) = \mathbf{F} \circ i_S.$$

Plainly,

$$i_{T/S}^* \circ i_S^* = i_T^*,$$

that is, for every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$,

$$(\mathbf{F}_S)_T = \mathbf{F}_T.$$

The notation is justified by the following result.

Proposition 1.6.1. *The functor*

$$(\mathbf{h}_X)_S : (\mathscr{C}_{/S})^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if the product $X \times S$ exists. In that case

$$(\mathbf{h}_X)_S \simeq \mathbf{h}_{X_S}.$$

This shows that $\mathbf{F}_S$ has two interpretations: it is the restriction of $\mathbf{F}$ to $\mathscr{C}_{/S}$, and it is the functor obtained by the base change $e \leftarrow S$. This leads to the following notation:

$$\begin{array}{ccccc} \mathbf{F} & \text{———} & \mathbf{F}_S & \text{———} & \mathbf{F}_T \\ \Big| & & \Big| & & \Big| \\ e & \longleftarrow & S & \longleftarrow & T \end{array}$$

This diagram reflects both of the preceding interpretations. Notice that

$$\Gamma(\mathbf{F}_S) \simeq \mathrm{Hom}(\mathbf{h}_S, \mathbf{F}) \simeq \mathbf{F}(S),$$

and, in particular,

$$\Gamma(X_S) \simeq \mathrm{Hom}(S, X).$$

[13] Editor's note: that is, if $g : U \to S$ (respectively, $h : V \to T$) is an object of $\mathscr{C}_{/S}$ (respectively, of $\mathscr{C}_{/T}$), then $i_S(g) = U$, while $i_{T/S}(h)$ is the object $f \circ h : V \to S$ of $\mathscr{C}_{/S}$, and

$$\mathrm{Hom}_{\mathscr{C}_{/S}}(U, X \times S) \simeq \mathrm{Hom}_{\mathscr{C}}(U, X),$$

respectively,

$$\mathrm{Hom}_{\mathscr{C}_{/T}}(V, X \times_S T) \simeq \mathrm{Hom}_{\mathscr{C}_{/S}}(V, X).$$

1.7.0.

Let $\mathbf{E}$ be an object of $\widehat{\mathscr{C}}$. [14] Consider the category $\mathscr{C}_{/\mathbf{E}}$ of *objects of* $\mathscr{C}$ over $\mathbf{E}$. Its objects are pairs (V,ρ) formed by an object V of $\mathscr{C}$ and a morphism $\rho : \mathbf{h}_V \to \mathbf{E}$ in $\widehat{\mathscr{C}}$, that is, $\rho \in \mathbf{E}(V)$. A morphism from (V,ρ) to (V',ρ') is a morphism $f : V \to V'$ in $\mathscr{C}$ such that

$$\rho' \circ f = \rho, \qquad \text{that is,} \qquad \mathbf{E}(f)(\rho') = \rho.$$

Let $\mathbf{L}$ be the functor

$$\mathbf{L} = \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V.$$

Thus, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\mathbf{L}(S) = \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V(S)$$

is the set of equivalence classes of triples (V,ρ,v), where $v : S \to V$ is a morphism in $\mathscr{C}$, and where (V,ρ,v) is identified with $(V',\rho',f\circ v)$ for every morphism $f : V \to V'$ in $\mathscr{C}$ such that $\rho' \circ f = \rho$.

The map $\phi_{\mathbf{E}}(S)$ which associates with the class of (V,ρ,v) the element $\rho\circ v$ of $\mathbf{E}(S)$ is well defined and gives a morphism of functors

$$\phi_{\mathbf{E}} : \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V \longrightarrow \mathbf{E}.$$

Lemma. *The morphism $\phi_{\mathbf{E}}$ is an isomorphism.*

Indeed, let $S \in \mathrm{Ob}(\mathscr{C})$. Every $x \in \mathbf{E}(S)$ is the image under $\phi_{\mathbf{E}}(S)$ of the triple (S,x,id_S); hence $\phi_{\mathbf{E}}(S)$ is surjective. Conversely, let $\ell_1 = (V_1,\rho_1,v_1)$ and $\ell_2 = (V_2,\rho_2,v_2)$ be two elements of $\mathbf{L}(S)$ having the same image in $\mathbf{E}(S)$, and set

$$\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2.$$

Then ℓ_1 and ℓ_2 are both equal in $\mathbf{L}(S)$ to the class of (S,γ,id_S). Thus $\phi_{\mathbf{E}}(S)$ is injective.

Corollary. *For every object $\mathbf{F}$ of $\widehat{\mathscr{C}}$,*

$$\mathrm{Hom}(\mathbf{E},\mathbf{F}) = \varprojlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{F}(V).$$

1.7. The objects $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Isom}}$, etc.

Let $\mathbf{F}$ and $\mathbf{G}$ be two objects of $\widehat{\mathscr{C}}$. We define another object of $\widehat{\mathscr{C}}$ by

$$\begin{aligned}\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})(S) &= \mathrm{Hom}_{\widehat{\mathscr{C}_{/S}}}(\mathbf{F}_S,\mathbf{G}_S)\\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}_{/\mathbf{h}_S}}(\mathbf{F}\times\mathbf{h}_S,\mathbf{G}\times\mathbf{h}_S)\\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}\times\mathbf{h}_S,\mathbf{G}).\end{aligned}$$

The object $\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})$ has the following properties:

[14] Editor's note: the lemma below has been added (see SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful repeatedly in the sequel.

(i) $\underline{\mathrm{Hom}}(\underline{\mathbf{e}}, \mathbf{G}) \simeq \mathbf{G}$. [15]

(ii) The formation of $\underline{\mathrm{Hom}}$ commutes with base extension:

$$\underline{\mathrm{Hom}}(\mathbf{F}_S, \mathbf{G}_S) \simeq \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})_S.$$

(iii) $(\mathbf{F}, \mathbf{G}) \mapsto \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$ is a bifunctor, contravariant in $\mathbf{F}$ and covariant in $\mathbf{G}$.

These three properties follow immediately from the definitions.

We shall prove that, for every object $\mathbf{E}$ of $\widehat{\mathscr{C}}$,

$$\mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

Let $\phi : \mathbf{E} \times \mathbf{F} \to \mathbf{G}$. We must associate with it a morphism from $\mathbf{E}$ to $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$. For an arrow $S' \to S$ of $\mathscr{C}$, there are maps

$$\mathbf{E}(S) \times \mathbf{F}(S') \longrightarrow \mathbf{E}(S') \times \mathbf{F}(S') \xrightarrow{\phi(S')} \mathbf{G}(S').$$

Every element e of $\mathbf{E}(S)$ therefore defines, for each $S' \to S$, a map $\mathbf{F}(S') \to \mathbf{G}(S')$ functorial in S'; this is an element $\theta_\phi(e)$ of $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})(S)$. We have thus obtained a map

$$\phi \longmapsto \theta_\phi, \qquad \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \longrightarrow \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big),$$

which is functorial in $\mathbf{E}$.

Proposition 1.7.1. [16] *Let* $\mathbf{E}, \mathbf{F}, \mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$.

(a) The map $\phi \mapsto \theta_\phi$ *is a bijection:*

$$\mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \xrightarrow{\sim} \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big).$$

(b) Moreover, there is an isomorphism of functors

$$\underline{\mathrm{Hom}}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \underline{\mathrm{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

For (a), regard both sides as functors of $\mathbf{E}$. The asserted result is true when $\mathbf{E} = \mathbf{h}_X$, since it is then precisely the definition of the functor $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$. Moreover, both sides, as functors of $\mathbf{E}$, take direct limits to inverse limits. Finally, by Lemma 1.7.0, every object $\mathbf{E}$ of $\widehat{\mathscr{C}}$ is isomorphic to the direct limit of the $\mathbf{h}_X$, as X ranges over $\mathscr{C}_{/\mathbf{E}}$. This proves (a).

Let us also sketch a direct proof of (a). To every

$$\theta \in \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big)$$

associate $\phi_\theta \in \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G})$ as follows. For every $S \in \mathrm{Ob}(\mathscr{C})$, there is a map

$$\theta(S) : \mathbf{E}(S) \longrightarrow \mathrm{Hom}(\mathbf{F} \times S, \mathbf{G}),$$

[15] Editor's note: if $\mathbf{E}$ is a third object of $\widehat{\mathscr{C}}$, then

$$\underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{F} \times \mathbf{G}) \simeq \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{F}) \times \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G}).$$

[16] Editor's note: part (b), which will be useful in II.1 and II.3.11, has been added. A second, more direct proof of part (a) has also been sketched.

functorial in S. If $(e, f) \in \mathbf{E}(S) \times \mathbf{F}(S)$, then f is a morphism $S \to \mathbf{F}$, so $f \times \mathrm{id}_S$ is a morphism $S \to \mathbf{F} \times S$. On the other hand, $\theta(S)(e)$ is a morphism $\mathbf{F} \times S \to \mathbf{G}$. Composition therefore gives a morphism

$$\theta(S)(e) \circ (f \times \mathrm{id}_S) : S \longrightarrow \mathbf{G},$$

that is, an element $\phi_\theta(S)(e, f)$ of $\mathbf{G}(S)$. It is easy to check that $S \mapsto \phi_\theta(S)$ is functorial in S, and therefore defines a morphism ϕ_θ from $\mathbf{E} \times \mathbf{F}$ to $\mathbf{G}$. We leave to the reader the verification that $\theta \mapsto \phi_\theta$ and $\phi \mapsto \theta_\phi$ are mutually inverse bijections.

For (b), if $S \in \mathrm{Ob}(\mathscr{C})$, then by 1.7(ii) and part (a), applied to $\mathscr{C}_{/S}$,

$$\begin{aligned}
&\underline{\mathrm{Hom}}(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}))(S) \\
&\quad \simeq \mathrm{Hom}_S(\mathbf{E}_S, \underline{\mathrm{Hom}}_S(\mathbf{F}_S, \mathbf{G}_S)) \\
&\quad \simeq \mathrm{Hom}_S(\mathbf{E}_S \times_S \mathbf{F}_S, \mathbf{G}_S) \\
&\quad \simeq \mathrm{Hom}(\mathbf{E} \times \mathbf{F} \times S, \mathbf{G}) \\
&\quad \simeq \underline{\mathrm{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G})(S),
\end{aligned}$$

and these isomorphisms are functorial in S.

Corollary 1.7.2. *There are isomorphisms*

$$\mathrm{Hom}(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \mathrm{Hom}(\mathbf{F}, \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G})),$$

and

$$\underline{\mathrm{Hom}}(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\mathrm{Hom}}(\mathbf{F}, \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G})).$$

In particular, taking $\mathbf{E} = \underline{\mathbf{e}}$ and using $\underline{\mathrm{Hom}}(\underline{\mathbf{e}}, \mathbf{G}) \simeq \mathbf{G}$, we obtain

$$\Gamma(\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \mathrm{Hom}(\mathbf{F}, \mathbf{G}).$$

Composition supplies functorial morphisms

$$\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{H}) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{H}).$$

If $\mathbf{F}$ and $\mathbf{G}$ are two objects of $\widehat{\mathscr{C}}$, let $\mathrm{Isom}(\mathbf{F}, \mathbf{G})$ denote the subset of $\mathrm{Hom}(\mathbf{F}, \mathbf{G})$ formed by the isomorphisms from $\mathbf{F}$ onto $\mathbf{G}$. Define a subobject $\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})$ of $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$ by

$$\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})(S) = \mathrm{Isom}(\mathbf{F}_S, \mathbf{G}_S).$$

Then

$$\Gamma(\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})) \simeq \mathrm{Isom}(\mathbf{F}, \mathbf{G}), \qquad \underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G}) \simeq \underline{\mathrm{Isom}}(\mathbf{G}, \mathbf{F}).$$

In the special case $\mathbf{F} = \mathbf{G}$, set

$$\begin{aligned}
\underline{\mathrm{End}}(\mathbf{F}) &= \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{F}), & \mathrm{End}(\mathbf{F}) &= \mathrm{Hom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\mathrm{End}}(\mathbf{F})), \\
\underline{\mathrm{Aut}}(\mathbf{F}) &= \underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{F}), & \mathrm{Aut}(\mathbf{F}) &= \mathrm{Isom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\mathrm{Aut}}(\mathbf{F})).
\end{aligned}$$

Formation of the objects $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Isom}}$, $\underline{\mathrm{Aut}}$, and $\underline{\mathrm{End}}$ commutes with base change.

Remark 1.7.3. [17] An object isomorphic to $\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})$ may also be constructed as follows. There is a morphism

$$\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\mathrm{End}}(\mathbf{F}).$$

[17] Editor's note: the numbering 1.7.3 has been added for later references.

Interchanging $\mathbf{F}$ and $\mathbf{G}$ gives a morphism

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F}) \longrightarrow \underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G}).$$

On the other hand, the identity morphism of $\mathbf{F}$ is an element of $\mathrm{End}(\mathbf{F})$, and hence defines a morphism $\underline{\mathbf{e}} \to \underline{\mathrm{End}}(\mathbf{F})$. Doing the same for $\mathbf{G}$ and taking the product gives a morphism

$$\underline{\mathbf{e}} \longrightarrow \underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G}).$$

It is then immediate that the fiber product of $\underline{\mathbf{e}}$ and

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F})$$

over $\underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G})$ is isomorphic to $\underline{\mathrm{Isom}}(\mathbf{F},\mathbf{G})$.

All these definitions apply, in particular, when $\mathbf{F} = \mathbf{h}_X$ and $\mathbf{G} = \mathbf{h}_Y$. If $\underline{\mathrm{Hom}}(\mathbf{h}_X, \mathbf{h}_Y)$ is representable by an object of $\mathscr{C}$, we denote that object by $\underline{\mathrm{Hom}}(X,Y)$. It has the following property: if $Z \times X$ exists, then

$$\mathrm{Hom}(Z, \underline{\mathrm{Hom}}(X,Y)) \simeq \mathrm{Hom}(Z \times X, Y).$$

This property characterizes it when products exist in $\mathscr{C}$.

Similarly, one defines, when they exist, the objects

$$\underline{\mathrm{Isom}}(X,Y), \qquad \underline{\mathrm{End}}(X), \qquad \underline{\mathrm{Aut}}(X).$$

We merely note that the construction above shows that $\underline{\mathrm{Isom}}(X,Y)$ exists whenever fiber products exist in $\mathscr{C}$ and the objects

$$\underline{\mathrm{Hom}}(X,Y), \quad \underline{\mathrm{Hom}}(Y,X), \quad \underline{\mathrm{End}}(X), \quad \text{and} \quad \underline{\mathrm{End}}(Y)$$

exist.

Everything above also applies to categories of the form $\mathscr{C}_{/S}$. The corresponding objects will be denoted as explicitly as possible by appropriate symbols. For example, if T and T' are two objects of $\mathscr{C}$ over S, we denote by $\underline{\mathrm{Hom}}_S(T,T')$ the object

$$\underline{\mathrm{Hom}}_{\mathscr{C}_{/S}}(T/S, T'/S).$$

1.8. Constant objects

Let $\mathscr{C}$ be a category in which direct sums and fiber products exist and in which direct sums commute with base change (for example, the category of schemes).[18] For every set E and every object S of $\mathscr{C}$, set

$$E_S = \Big\{ \text{the direct sum of a family } (S_i)_{i\in E} \text{ of objects} \quad \text{of } \mathscr{C}, \text{ all isomorphic to } S. \tag{1.8.1}$$

This object is characterized by the formula

$$\mathrm{Hom}_{\mathscr{C}}(E_S, T) = \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}_{\mathscr{C}}(S,T)), \tag{1.8.2}$$

for every $T \in \mathrm{Ob}(\mathscr{C})$.

[18] Editor's note: the former terminology "preschemes/schemes" has been replaced by the current terminology "schemes/separated schemes."

The object E_S has a canonical projection to S, so that $E \mapsto E_S$ is in fact a functor from **Set** to $\mathscr{C}_{/S}$.[19] If $S' \to S$ is an arrow of $\mathscr{C}$, then, because direct sums commute with base change,

$$E_{S'} = (E_S)_{S'}.$$

In particular, if $\mathscr{C}$ has a final object e, then

$$E_S = (E_e)_S.$$

The functor $E \mapsto E_S/S$, from **Set** to $\mathscr{C}_{/S}$, commutes with finite products. It is enough to prove that

$$E_S \times_S F_S = (E \times F)_S. \tag{$\times$}$$

Indeed, by the results of 1.7 applied to $\mathscr{C}_{/S}$, for every $T \in \mathrm{Ob}(\mathscr{C}_{/S})$ there are natural isomorphisms (all unspecified Hom's being taken in $\mathscr{C}_{/S}$)

$$\begin{aligned} \mathrm{Hom}((E \times F)_S, T) &\simeq \mathrm{Hom}_{\mathbf{Set}}(E \times F, \mathrm{Hom}(S, T)) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}_{\mathbf{Set}}(F, \mathrm{Hom}(S, T))) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}(F_S, T)), \end{aligned}$$

and

$$\begin{aligned} \mathrm{Hom}(E_S \times_S F_S, T) &\simeq \mathrm{Hom}(E_S, \underline{\mathrm{Hom}}(F_S, T)) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T))). \end{aligned}$$

But

$$\mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T)) \simeq \mathrm{Hom}(F_S, T),$$

which proves $(\times)$.

Suppose that, with $\varnothing$ denoting an initial object of $\mathscr{C}$, the following diagram is cartesian:

$$(*) \qquad \begin{array}{ccc} \varnothing & \longrightarrow & S \\ \big\downarrow & & \big\downarrow \\ S & \longrightarrow & S \amalg S \end{array}$$

This is the case for the category of schemes.[20] Then the functor $E \mapsto E_S$ commutes with finite inverse limits.

Indeed, in view of $(\times)$, it is enough to show that $E \mapsto E_S$ commutes with fiber products. Let $u : E \to G$ and $v : F \to G$ be two maps of sets. Since direct sums in $\mathscr{C}$ commute with base change, one has

$$F_S \times_{G_S} E_S \simeq \coprod_{f \in F} (S_f \times_{G_S} E_S) \simeq \coprod_{\substack{f \in F \\ x \in E}} (S_f \times_{G_S} S_x). \tag{1}$$

If $v(f) \neq u(x)$, there is a morphism in $\mathscr{C}_{/S}$

$$S_f \times_{G_S} S_x \longrightarrow S_f \times_{S_{v(f)} \coprod S_{u(x)}} S_x.$$

[19]Editor's note: in the next three paragraphs, the order of the sentences has been changed and a few details have been added concerning the role of condition $(*)$ below.

[20]Editor's note: Condition $(*)$ is not satisfied if $\mathscr{C}$ is the category whose arrows are $A \to B$, id_A, and id_B. In that case $B \coprod B = B$ and $B \times_B B = B \not\simeq A$.

By condition $(*)$, the term on the right is $\varnothing$. Hence

$$S_f \times_{G_S} S_x \simeq \varnothing \qquad \text{if } v(f) \neq u(x). \tag{2}$$

On the other hand, if $v(f) = u(x)$, there is a morphism in $\mathscr{C}_{/S}$

$$S \longrightarrow S_f \times_{G_S} S_x.$$

Since $S \xrightarrow{\mathrm{id}} S$ is the final object of $\mathscr{C}_{/S}$, it follows that

$$S_f \times_{G_S} S_x \simeq S \qquad \text{if } v(f) = u(x). \tag{3}$$

Combining (1), (2), and (3), we obtain an isomorphism functorial in S:

$$F_S \times_{G_S} E_S \simeq \coprod_{F \times_G E} S = (F \times_G E)_S.$$

An object of the form E_S is called a *constant object*. There is a morphism, functorial in E,

$$E \longrightarrow \Gamma(E_S/S),$$

which associates with each $i \in E$ the section of E_S over S defined by the isomorphism from S onto S_i. Suppose condition $(*)$ is satisfied for every object S of $\mathscr{C}$. Then $E \to \Gamma(E_S/S)$ is a monomorphism whenever $S \not\simeq \varnothing$.

If $\mathscr{C}$ is the category of schemes, then $\Gamma(E_S/S)$ is identified with the locally constant maps from the topological space S to the set E, the preceding map associating with each element of E the corresponding constant map. It follows that E_S may equivalently be defined as representing the functor that assigns to every S' over S the set of locally constant functions from the topological space S' to E.[21]

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we shall extend it to the category $\mathscr{C}$. We first treat the example of groups.

2.1. Group structures

We retain the notation of the preceding paragraph.

Definition 2.1.1. Let $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$. A *$\widehat{\mathscr{C}}$-group structure* on $\mathbf{G}$ is the datum, for every $S \in \mathrm{Ob}(\mathscr{C})$, of a group structure on the set $\mathbf{G}(S)$, such that for every arrow $f : S' \to S''$ of $\mathscr{C}$, the map

$$\mathbf{G}(f) : \mathbf{G}(S'') \longrightarrow \mathbf{G}(S')$$

is a group homomorphism. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathscr{C}}$-groups, a *morphism of $\widehat{\mathscr{C}}$-groups* from $\mathbf{G}$ to $\mathbf{H}$ is a morphism $u \in \mathrm{Hom}(\mathbf{G}, \mathbf{H})$ such that $u(S) : \mathbf{G}(S) \to \mathbf{H}(S)$ is a group homomorphism for every $S \in \mathrm{Ob}(\mathscr{C})$.

We denote by $\mathrm{Hom}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ the set of these morphisms and by $(\widehat{\mathscr{C}}\text{-Gr})$ their category.

Examples. For $\mathbf{E} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, the object $\underline{\mathrm{Aut}}(\mathbf{E})$ has an evident $\widehat{\mathscr{C}}$-group structure. The final object $\underline{\mathbf{e}}$ has a unique $\widehat{\mathscr{C}}$-group structure, making it a final object of $(\widehat{\mathscr{C}}\text{-Gr})$.

[21] Editor's note: the diagonal morphism $E_S \to E_S \times_S E_S = (E \times E)_S$ is also a closed immersion; that is, E_S is separated over S.

For every $S \in \mathrm{Ob}(\mathscr{C})$, let $e_{\mathbf{G}}(S)$ be the identity element of $\mathbf{G}(S)$. The family of elements $e_{\mathbf{G}}(S)$ defines

$$e_{\mathbf{G}} \in \Gamma(\mathbf{G}) = \mathrm{Hom}(\underline{\mathbf{e}}, \mathbf{G}).$$

It is a morphism of $\widehat{\mathscr{C}}$-groups $\underline{\mathbf{e}} \to \mathbf{G}$, called the *unit section* of $\mathbf{G}$.

To give a $\widehat{\mathscr{C}}$-group structure on $\mathbf{G}$ is equivalently to give a composition law on $\mathbf{G}$, that is, a $\widehat{\mathscr{C}}$-morphism

$$\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}$$

such that $\pi_{\mathbf{G}}(S)$ equips $\mathbf{G}(S)$ with a group structure for every $S \in \mathrm{Ob}(\mathscr{C})$.

Similarly, $f : \mathbf{G} \to \mathbf{H}$ is a morphism of $\widehat{\mathscr{C}}$-groups if and only if the following diagram is commutative:

$$\begin{tikzcd}
\mathbf{G} \times \mathbf{G} \arrow[r, "\pi_{\mathbf{G}}"] \arrow[d, "{(f,f)}"'] & \mathbf{G} \arrow[d, "f"] \\
\mathbf{H} \times \mathbf{H} \arrow[r, "\pi_{\mathbf{H}}"] & \mathbf{H}
\end{tikzcd} .$$

A subobject $\mathbf{H}$ of $\mathbf{G}$ such that $\mathbf{H}(S)$ is a subgroup of $\mathbf{G}(S)$ for every S has the evident $\widehat{\mathscr{C}}$-group structure induced by that of $\mathbf{G}$. It is the unique structure for which the monomorphism $\mathbf{H} \to \mathbf{G}$ is a morphism of $\widehat{\mathscr{C}}$-groups. With this structure, $\mathbf{H}$ is called a *sub-$\widehat{\mathscr{C}}$-group* of $\mathbf{G}$.

If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathscr{C}}$-groups, then $\mathbf{G} \times \mathbf{H}$ has the evident product structure: for every S, the set $\mathbf{G}(S) \times \mathbf{H}(S)$ is given the product of the prescribed group structures. The resulting object is called the *product $\widehat{\mathscr{C}}$-group* of $\mathbf{G}$ and $\mathbf{H}$; it is their product in the category of $\widehat{\mathscr{C}}$-groups.

If $\mathbf{G}$ is a $\widehat{\mathscr{C}}$-group, then $\mathbf{G}_S$ is a $\widehat{\mathscr{C}_{/S}}$-group for every $S \in \mathrm{Ob}(\mathscr{C})$. For two $\widehat{\mathscr{C}}$-groups $\mathbf{G}$ and $\mathbf{H}$, define the object $\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ of $\widehat{\mathscr{C}}$ by

$$\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})(S) = \mathrm{Hom}_{\widehat{\mathscr{C}_{/S}}\text{-Gr}}(\mathbf{G}_S, \mathbf{H}_S).$$

Notice that $\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}$ is not in general a $\widehat{\mathscr{C}}$-group, still less the internal $\underline{\mathrm{Hom}}$ object in the category ($\widehat{\mathscr{C}}$-Gr).

One similarly defines

$$\underline{\mathrm{Isom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H}), \qquad \underline{\mathrm{End}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}), \qquad \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}).$$

Definition 2.1.2. Let $G \in \mathrm{Ob}(\mathscr{C})$. A *$\mathscr{C}$-group structure* on G is a $\widehat{\mathscr{C}}$-group structure on $\mathbf{h}_G \in \mathrm{Ob}(\widehat{\mathscr{C}})$. A morphism from the $\mathscr{C}$-group G to the $\mathscr{C}$-group H is an element

$$u \in \mathrm{Hom}(G, H) \simeq \mathrm{Hom}(\mathbf{h}_G, \mathbf{h}_H)$$

that defines a morphism of $\widehat{\mathscr{C}}$-groups from $\mathbf{h}_G$ to $\mathbf{h}_H$.

We denote the category of $\mathscr{C}$-groups by ($\mathscr{C}$-Gr). There is a cartesian square in **Cat**:

$$\begin{tikzcd}
(\mathscr{C}\text{-Gr.}) \arrow[r] \arrow[d] & (\widehat{\mathscr{C}}\text{-Gr.}) \arrow[d] \\
\mathscr{C} \arrow[r, "\mathbf{h}"] & \widehat{\mathscr{C}}
\end{tikzcd} .$$

All the preceding definitions and constructions therefore carry over at once to ($\mathscr{C}$-Gr) whenever the functors they use (products, internal $\underline{\mathrm{Hom}}$ objects, and so on) are representable. They also apply to the categories $\mathscr{C}_{/S}$. In that case we write $\underline{\mathrm{Hom}}_{S\text{-Gr}}$ for $\underline{\mathrm{Hom}}_{\mathscr{C}_{/S}\text{-Gr}}$, and similarly for the other objects.

2.2.

More generally, suppose (T) is a species of structure on n base sets defined by finite inverse limits—for example, by commutative diagrams constructed from cartesian products, as in the structures of a monoid, a group, a set with operators, a module over a ring, or a Lie algebra over a ring. The preceding construction defines the notion of a "structure of species (T)" on n objects $\mathbf{F}_1, \ldots, \mathbf{F}_n$ of $\widehat{\mathscr{C}}$. Such a structure consists, for every $S \in \mathrm{Ob}(\mathscr{C})$, of a structure of species (T) on the sets $\mathbf{F}_1(S), \ldots, \mathbf{F}_n(S)$, in such a way that for every arrow $S' \to S''$ of $\mathscr{C}$, the family of maps

$$\left(\mathbf{F}_i(S'')\right)_i \longrightarrow \left(\mathbf{F}_i(S')\right)_i$$

is a polyhomomorphism for the species (T).

Morphisms of structures of species (T) are defined similarly, giving a category

$$(\widehat{\mathscr{C}} \times \cdots \times \widehat{\mathscr{C}})^{(T)}.$$

The fully faithful functor $\mathbf{h} \times \cdots \times \mathbf{h}$ then defines, by inverse image, the category

$$(\mathscr{C} \times \cdots \times \mathscr{C})^{(T)}.$$

Since it commutes with inverse limits, all the functorial properties, notions, and notation introduced in $\widehat{\mathscr{C}}$ carry over to this category.

Now suppose fiber products exist in $\mathscr{C}$, and let (T) be a species of algebraic structure specified by certain morphisms between cartesian products, subject to axioms consisting of commutative diagrams formed with those morphisms. A structure of species (T) on a family of objects of $\mathscr{C}$ is then defined by corresponding morphisms between cartesian products satisfying the corresponding commutativity conditions. Consequently, if $\mathscr{C}$ and $\mathscr{C}'$ are categories with products and $f : \mathscr{C} \to \mathscr{C}'$ is a functor commuting with products, then a structure of species (T) on a family (F_i) of objects of $\mathscr{C}$ induces one on the family $(f(F_i))$ in $\mathscr{C}'$. Thus every $\mathscr{C}$-group becomes a $\mathscr{C}'$-group, every pair consisting of a $\mathscr{C}$-ring and a module over it becomes the analogous pair in $\mathscr{C}'$, and so on.

In particular, let $\mathscr{C}$ satisfy the conditions of 1.8.[22] The functor $E \mapsto E_S$ defined there commutes with finite inverse limits. It therefore carries groups to S-groups (that is, $\mathscr{C}_{/S}$-groups), rings to S-rings, and so forth.

Remark. It is worth observing that the preceding construction, when applied to $\widehat{\mathscr{C}}$, gives back precisely the notions already defined there. In other words, to give an object of $\widehat{\mathscr{C}}$ a structure of species (T) when it is viewed as a functor on $\mathscr{C}$ is the same as to give the representable functor on $\widehat{\mathscr{C}}$ defined by that object a structure of species (T).[23]

We shall treat two further special cases of this construction: structures with operator groups and modules.

[22] Editor's note: including condition $(*)$.

[23] Editor's note: for group structures, for example, let $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$. If the functor

$$\widehat{\mathscr{C}} \longrightarrow \mathbf{Set}, \qquad \mathbf{F} \longmapsto \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}, \mathbf{G})$$

has a group structure, then so does its restriction to $\mathscr{C}$, namely $X \mapsto \mathbf{G}(X)$. Conversely, if $\mathbf{G}$ is a $\widehat{\mathscr{C}}$-group, its multiplication morphism $\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ induces, for every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, a group structure on $\mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}, \mathbf{G})$, functorially in $\mathbf{F}$.

2.3. Structures with operator groups

Definition 2.3.1. Let $\mathbf{E} \in \mathrm{Ob}(\widehat{\mathscr{C}})$ and $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}}\text{-}\mathbf{Gr})$. A structure on $\mathbf{E}$ of an object with $\widehat{\mathscr{C}}$-operator group $\mathbf{G}$ (or of a $\mathbf{G}$-object) consists in giving $\mathbf{E}(S)$, for every $S \in \mathrm{Ob}(\mathscr{C})$, the structure of a set with operator group $\mathbf{G}(S)$, in such a way that, for every arrow $S' \to S''$ in $\mathscr{C}$, the map of sets $\mathbf{E}(S'') \to \mathbf{E}(S')$ is compatible with the homomorphism of operator groups $\mathbf{G}(S'') \to \mathbf{G}(S')$.

As usual, this is equivalent to giving a morphism

$$\mu : \mathbf{G} \times \mathbf{E} \longrightarrow \mathbf{E}$$

which, for every S, equips $\mathbf{E}(S)$ with the structure of a set on which $\mathbf{G}(S)$ acts. But

$$\mathrm{Hom}(\mathbf{G} \times \mathbf{E}, \mathbf{E}) \simeq \mathrm{Hom}(\mathbf{G}, \underline{\mathrm{End}}(\mathbf{E})),$$

so μ defines a morphism $\mathbf{G} \to \underline{\mathrm{End}}(\mathbf{E})$. It is immediate that this morphism maps $\mathbf{G}$ into $\underline{\mathrm{Aut}}(\mathbf{E})$ and is a morphism of $\widehat{\mathscr{C}}$-groups. Consequently, giving $\mathbf{E}$ the structure of an object with $\widehat{\mathscr{C}}$-operator group $\mathbf{G}$ is equivalent to giving a morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}(\mathbf{E}).$$

In particular, every element $g \in \mathbf{G}(S)$ defines an automorphism $\rho(g)$ of the functor $\mathbf{E}_S$, that is, an automorphism of $\mathbf{E} \times \mathbf{h}_S$ commuting with the projection $\mathbf{E} \times \mathbf{h}_S \to \mathbf{h}_S$, and hence, in particular, an automorphism of the set $\mathbf{E}(S')$ for every $S' \to S$.

Definition 2.3.2. We denote by $\mathbf{E}^{\mathbf{G}}$ the subobject of $\mathbf{E}$ defined as follows:

$$\mathbf{E}^{\mathbf{G}}(S) = \{x \in \mathbf{E}(S) \mid x_{S'} \text{ is invariant under } \mathbf{G}(S') \text{ for every } S' \to S\},$$

where $x_{S'}$ denotes the image of x under $\mathbf{E}(S) \to \mathbf{E}(S')$.

Thus $\mathbf{E}^{\mathbf{G}}$, the "subobject of invariants of $\mathbf{G}$," is the largest subobject of $\mathbf{E}$ on which $\mathbf{G}$ acts trivially.

Definition 2.3.3. Let $\mathbf{F}$ be a subobject of $\mathbf{E}$. We denote by $\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F}$ and $\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F}$ the sub-$\widehat{\mathscr{C}}$-groups of $\mathbf{G}$ defined by

$$\begin{aligned}
(\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}_S = \mathbf{F}_S\} \\
&= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}(S') = \mathbf{F}(S') \text{ for every } S' \to S\}, \\
(\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}_S} = \mathrm{id}\} \\
&= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}(S')} = \mathrm{id} \ \text{ for every } S' \to S\}.
\end{aligned}$$

Here the vertical bar after $\rho(g)$ denotes restriction. [24]

Scholium 2.3.3.1. [25] In particular, let $x \in \Gamma(\mathbf{E})$, that is (cf. 1.2), a collection of elements $x_S \in \mathbf{E}(S)$, $S \in \mathrm{Ob}(\mathscr{C})$, such that for every arrow $f : S' \to S$ one has $\mathbf{E}(f)(x_S) = x_{S'}$. If $\mathscr{C}$ has a final object S_0, then $\Gamma(\mathbf{E}) = \mathbf{E}(S_0)$. The element x then defines a subfunctor of $\mathbf{E}$, denoted by $\mathbf{x}$, and

$$\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{x} = \underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{x}.$$

[24] Editor's note: the original has been corrected by replacing the inclusion $\rho(g)\mathbf{F}_S \subset \mathbf{F}_S$ with an equality, so as to ensure that $\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{F})$ is indeed a group. See $\mathrm{VI_B}$, 6.4 for conditions under which the "transporter" coincides with the "strict transporter."

[25] Editor's note: Scholium 2.3.3.1 and Remark 2.3.3.2 have been added.

We denote this functor by $\underline{\mathrm{Stab}}_{\mathbf{G}}(x)$ and call it the *stabilizer* of x; thus, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\underline{\mathrm{Stab}}_{\mathbf{G}}(x)(S) = \{g \in \mathbf{G}(S) \mid \rho(g)x_S = x_S\}.$$

Suppose that fiber products exist in $\mathscr{C}$. If $\mathbf{G} = \mathbf{h}_G$ (respectively, $\mathbf{E} = \mathbf{h}_E$), where G is a $\mathscr{C}$-group (respectively, $E \in \mathrm{Ob}(\mathscr{C})$), and if $\mathscr{C}$ has a final object S_0, so that x is a morphism $S_0 \to E$, then $\underline{\mathrm{Stab}}_{\mathbf{G}}(x)$ is represented by the fiber product $G \times_E S_0$, where the morphism $G \to E$ is the composite of

$$\mathrm{id}_G \times x : G = G \times S_0 \longrightarrow G \times E$$

and

$$\mu : G \times E \longrightarrow E.$$

Remark 2.3.3.2.[25] The formation of $\mathbf{E}^{\mathbf{G}}$, $\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F}$, and $\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F}$ commutes with base change; that is, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$(\mathbf{E}^{\mathbf{G}})_S = (\mathbf{E}_S)^{\mathbf{G}_S}, \qquad (\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\mathrm{Norm}}_{\mathbf{G}_S}\mathbf{F}_S, \qquad (\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\mathrm{Centr}}_{\mathbf{G}_S}\mathbf{F}_S.$$

Definition 2.3.4. If G is a $\mathscr{C}$-group and $\mathbf{E}$ is an object of $\widehat{\mathscr{C}}$ (respectively, E is an object of $\mathscr{C}$), a G-object structure on $\mathbf{E}$ (respectively, on E) is an $\mathbf{h}_G$-object structure on $\mathbf{E}$ (respectively, on $\mathbf{h}_E$).

In view of this definition, all the notions and notation defined above carry over to $\mathscr{C}$ whenever they involve only representable functors. For example, if $\underline{\mathrm{Norm}}_{\mathbf{h}_G}(\mathbf{h}_F)$ is representable, then there is a unique subobject of G representing it; it is then a sub-$\mathscr{C}$-group of G, and is denoted by $\mathrm{Norm}_G(F)$, and so forth.

Definition 2.3.5.

(a) We say that the $\widehat{\mathscr{C}}$-group $\mathbf{G}$ acts on the $\widehat{\mathscr{C}}$-group $\mathbf{H}$ if $\mathbf{H}$ is equipped with a $\mathbf{G}$-object structure such that, for every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{H}(S)$ defined by g is a group automorphism.

Equivalently, for every $g \in \mathbf{G}(S)$, the automorphism $\rho(g)$ of $\mathbf{H}_S$ is an automorphism of $\widehat{\mathscr{C}}_{/S}$-groups; or equivalently, the morphism of $\widehat{\mathscr{C}}$-groups $\mathbf{G} \to \underline{\mathrm{Aut}}(\mathbf{H})$ maps $\mathbf{G}$ into

$$\underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-}\mathbf{Gr}}(\mathbf{H}).$$

(b) In the situation above, there is a unique $\widehat{\mathscr{C}}$-group structure on $\mathbf{H} \times \mathbf{G}$ such that, for every S, $(\mathbf{H} \times \mathbf{G})(S)$ is the semidirect product of the groups $\mathbf{H}(S)$ and $\mathbf{G}(S)$ for the given action of $\mathbf{G}(S)$ on $\mathbf{H}(S)$. We denote this $\widehat{\mathscr{C}}$-group by $\mathbf{H} \cdot \mathbf{G}$ and call it the *semidirect product of* $\mathbf{H}$ *by* $\mathbf{G}$. Thus, by definition,

$$(\mathbf{H} \cdot \mathbf{G})(S) = \mathbf{H}(S) \cdot \mathbf{G}(S).$$

Let $\mathbf{G}$ be a $\widehat{\mathscr{C}}$-group. For every arrow $S' \to S$ in $\mathscr{C}$ and every $g \in \mathbf{G}(S)$, let $\mathrm{Int}(g)$ be the automorphism of $\mathbf{G}(S')$ defined by

$$\mathrm{Int}(g)h = ghg^{-1}.$$

This definition extends to a morphism of $\widehat{\mathscr{C}}$-groups

$$\mathrm{Int} : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-}\mathbf{Gr}}(\mathbf{G}) \subset \underline{\mathrm{Aut}}(\mathbf{G}).$$

Definition 2.3.3 therefore applies, and for every subobject $\mathbf{E}$ of $\mathbf{G}$ one has the sub-$\widehat{\mathscr{C}}$-groups of $\mathbf{G}$

$$\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{E}) \qquad \text{and} \qquad \underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{E}).$$

Definition 2.3.6. The *center* of $\mathbf{G}$, denoted $\underline{\mathrm{Centr}}(\mathbf{G})$, is the sub-$\widehat{\mathscr{C}}$-group $\underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{G})$ of $\mathbf{G}$. We say that $\mathbf{G}$ is commutative if $\underline{\mathrm{Centr}}(\mathbf{G}) = \mathbf{G}$, or, equivalently, if $\mathbf{G}(S)$ is commutative for every S.

We say that the sub-$\widehat{\mathscr{C}}$-group $\mathbf{H}$ of $\mathbf{G}$ is invariant in $\mathbf{G}$ if $\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is invariant in $\mathbf{G}(S)$ for every S. [26]

Definition 2.3.6.1. [27] Let $f : \mathbf{G} \to \mathbf{G}'$ be a morphism of $\widehat{\mathscr{C}}$-groups. The *kernel* of f, denoted $\operatorname{Ker} f$, is the sub-$\widehat{\mathscr{C}}$-group of $\mathbf{G}$ defined by

$$(\operatorname{Ker} f)(S) = \{x \in \mathbf{G}(S) \mid f(S)(x) = 1\} = \operatorname{Ker} f(S)$$

for every $S \in \mathrm{Ob}(\mathscr{C})$; it is an invariant sub-$\widehat{\mathscr{C}}$-group.

Notice that if $\mathbf{G} = \mathbf{h}_G$ and $\mathbf{G}' = \mathbf{h}_{G'}$, if $\mathscr{C}$ has a final object S_0, and if fiber products exist in $\mathscr{C}$, then $\mathrm{Ker}(f)$ is represented by $S_0 \times_{G'} G$.

Definition 2.3.6.2.[27] Let $\mathbf{E} \in \widehat{\mathscr{C}}$, and let $\mathbf{G}$ be a $\widehat{\mathscr{C}}$-group acting on $\mathbf{E}$. We say that the action of $\mathbf{G}$ on $\mathbf{E}$ is *faithful* if the kernel of the morphism $\mathbf{G} \to \underline{\mathrm{Aut}}(\mathbf{E})$ is trivial; that is, if, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $g \in \mathbf{G}(S)$, the condition

$$g_{S'} \cdot x = x \qquad \text{for every } S' \to S \text{ and every } x \in \mathbf{E}(S')$$

implies $g = 1$.

Many definitions and propositions from elementary group theory carry over readily. We record only the following one, which will be useful to us.

Proposition 2.3.7. *Let $f : \mathbf{W} \to \mathbf{G}$ be a morphism of $\widehat{\mathscr{C}}$-groups. Put $\mathbf{H}(S) = \operatorname{Ker} f(S)$. Let $u : \mathbf{G} \to \mathbf{W}$ be a morphism of $\widehat{\mathscr{C}}$-groups that is a section of f (and is therefore necessarily a monomorphism). Then $\mathbf{W}$ is identified with the semidirect product of $\mathbf{H}$ by $\mathbf{G}$ for the action of $\mathbf{G}$ on $\mathbf{H}$ defined by*

$$(g, h) \longmapsto \mathrm{Int}(u(g))h$$

for $g \in \mathbf{G}(S)$, $h \in \mathbf{H}(S)$, and $S \in \mathrm{Ob}(\mathscr{C})$.

All these definitions and propositions carry over, as usual, to $\mathscr{C}$. In particular, the semidirect product of two $\mathscr{C}$-groups H and G, with G acting on H, is defined whenever the Cartesian product $H \times G$ exists, and the analog of Proposition 2.3.7 takes the following form.

Proposition 2.3.8. *Let*

$$H \xrightarrow{i} W \xrightarrow{f} G$$

be a sequence of morphisms of $\mathscr{C}$-groups such that, for every $S \in \mathrm{Ob}(\mathscr{C})$, the pair $(H(S), i(S))$ is a kernel of $f(S) : W(S) \to G(S)$. Let $u : G \to W$ be a morphism of $\mathscr{C}$-groups that is a section of f. Then W is identified with the semidirect product of H by G for the action of G on H such that, if $S \in \mathrm{Ob}(\mathscr{C})$, $g \in G(S)$, and $h \in H(S)$, then

$$\mathrm{Int}(u(g))i(h) = i({}^{g}h).$$

[26] Editor's note: in addition, $\mathbf{H}$ is said to be *central* in $\mathbf{G}$ if $\underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is central in $\mathbf{G}(S)$ for every S.

[27] Editor's note: Definitions 2.3.6.1 and 2.3.6.2 have been added.

3. The category of O-modules, the category of G-O-modules

Definition 3.1. Let $\mathbf{O}$ and $\mathbf{F}$ be two objects of $\widehat{\mathscr{C}}$. We say that $\mathbf{F}$ is a $\widehat{\mathscr{C}}$-module over the $\widehat{\mathscr{C}}$-ring $\mathbf{O}$, or, for short, an $\mathbf{O}$-module, if, for every $S \in \mathrm{Ob}(\mathscr{C})$, the set $\mathbf{O}(S)$ has been endowed with a ring structure and $\mathbf{F}(S)$ with a module structure over that ring, in such a way that for every arrow $S' \to S''$ in $\mathscr{C}$, the map $\mathbf{O}(S'') \to \mathbf{O}(S')$ is a ring homomorphism and $\mathbf{F}(S'') \to \mathbf{F}(S')$ is a homomorphism of abelian groups compatible with that ring homomorphism.

Once $\mathbf{O}$ is fixed, one defines in the usual way a morphism between two $\mathbf{O}$-modules $\mathbf{F}$ and $\mathbf{F}'$, hence the abelian group $\mathrm{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and the category of $\mathbf{O}$-modules, denoted ($\mathbf{O}$-Mod.).

Lemma 3.1.1.[28] *The category* ($\mathbf{O}$*-Mod.*) *has the structure of an abelian category, defined objectwise. Moreover,* ($\mathbf{O}$*-Mod.*) *satisfies axiom* (AB 5) *(see* [Gr57, 1.5]*); that is, arbitrary direct sums exist, and if* $\mathbf{M}$ *is an* $\mathbf{O}$*-module,* $\mathbf{N}$ *a submodule, and* $(\mathbf{F}_i)_{i\in I}$ *an increasing filtered family of submodules of* $\mathbf{M}$*, then*

$$\bigcup_{i\in I}(\mathbf{F}_i \cap \mathbf{N}) = \left(\bigcup_{i\in I}\mathbf{F}_i\right) \cap \mathbf{N}.$$

Indeed, let $f : \mathbf{F} \to \mathbf{F}'$ be a morphism of $\mathbf{O}$-modules. Define the $\mathbf{O}$-modules $\operatorname{Ker} f$ (respectively, $\operatorname{Im} f$ and $\operatorname{Coker} f$) by setting, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$(\operatorname{Ker} f)(S) = \operatorname{Ker} f(S)$$

(respectively, the analogous formulas). Then $\operatorname{Ker} f$ (respectively, $\operatorname{Coker} f$) is a kernel (respectively, a cokernel) of f, and there is an isomorphism of $\mathbf{O}$-modules

$$\mathbf{F}/\operatorname{Ker} f \xrightarrow{\sim} \operatorname{Im} f.$$

This proves that ($\mathbf{O}$-Mod.) is an abelian category.

Arbitrary direct sums exist and are defined objectwise. Finally, if $\mathbf{M}$ is an $\mathbf{O}$-module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i\in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then the inclusion

$$\bigcup_{i\in I}(\mathbf{F}_i \cap \mathbf{N}) \subset \left(\bigcup_{i\in I}\mathbf{F}_i\right) \cap \mathbf{N}$$

is an equality. Indeed, if $S \in \mathrm{Ob}(\mathscr{C})$ and

$$x \in \mathbf{N}(S) \cap \bigcup_i \mathbf{F}_i(S),$$

there is an $i \in I$ such that $x \in \mathbf{N}(S) \cap \mathbf{F}_i(S)$.

Proposition 3.1.2.[28] *Suppose that the category* $\mathscr{C}$ *is small, that is, that* $\mathrm{Ob}(\mathscr{C})$ *is a set. Then* ($\mathbf{O}$*-Mod.*) *has a generator, and consequently has enough injective objects.*

Proof. For every object S of $\mathscr{C}$, define the $\mathbf{O}$-module $\mathbf{O}_S$ as follows. For every object S', $\mathbf{O}_S(S')$ is a direct sum of copies of $\mathbf{O}(S')$ indexed by $\mathrm{Hom}(S', S)$, that is,

$$\mathbf{O}_S(S') = \bigoplus_{g:S'\to S} \mathbf{O}(S')g.$$

[28] Editor's note: Statements 3.1.1 and 3.1.2 have been added to show that the category ($\mathbf{O}$-Mod.) is abelian and satisfies axiom (AB 5), and moreover has enough injective objects when the category $\mathscr{C}$ is small. An error in 3.1.2, pointed out in 2016 by Linyuan Liu, is corrected here.

For every morphism $f : S'' \to S'$, writing f^* for the ring homomorphism $\mathbf{O}(S') \to \mathbf{O}(S'')$, the morphism

$$\mathbf{O}_S(f) : \mathbf{O}_S(S') \longrightarrow \mathbf{O}_S(S'')$$

sends each element ag of $\mathbf{O}_S(S')$, where $a \in \mathbf{O}(S')$ and $g : S' \to S$, to the element $f^*(a)(g \circ f)$ of $\mathbf{O}_S(S'')$. It is readily checked that this defines a functor in $\mathbf{O}$-modules.

Since $\mathrm{Ob}(\mathscr{C})$ is a set by hypothesis, we may form the direct sum

$$\mathbf{G} = \bigoplus_{S \in \mathrm{Ob}(\mathscr{C})} \mathbf{O}_S.$$

Lemma 3.1.2.1. *Let $\mathbf{F}$ be an $\mathbf{O}$-module. For every $S \in \mathrm{Ob}(\mathscr{C})$, every element $x \in \mathbf{F}(S)$ defines a unique morphism of $\mathbf{O}$-modules*

$$\mathbf{x} : \mathbf{O}_S \longrightarrow \mathbf{F}$$

such that $\mathbf{x}(1\,\mathrm{id}_S) = x$.

Proof. Let $\phi : \mathbf{O}_S \to \mathbf{F}$ be a morphism of $\mathbf{O}$-modules such that $\phi(1\,\mathrm{id}_S) = x$. Let $g : S' \to S$ and $a \in \mathbf{O}(S')$. The $\mathbf{O}(S')$-linearity and the commutativity of the diagram

$$\begin{tikzcd}
\mathbf{O}_S(S) \arrow[r, "\phi"] \arrow[d, "\mathbf{O}_S(g)"'] & \mathbf{F}(S) \arrow[d, "\mathbf{F}(g)"] \\
\mathbf{O}_S(S') \arrow[r, "\phi"] & \mathbf{F}(S')
\end{tikzcd}$$

imply that

$$\phi(ag) = a\phi(1g) = a\mathbf{F}(g)\big(\phi(1\,\mathrm{id}_S)\big) = a\mathbf{F}(g)(x).$$

Thus ϕ, if it exists, is determined by the preceding equality. Conversely, if ϕ is defined in this way, then for every $f : S'' \to S'$ one has

$$\begin{aligned}
\phi \circ \mathbf{O}_S(f)(ag) &= \phi\big(f^*(a)(g \circ f)\big) \\
&= f^*(a)\mathbf{F}(g \circ f)(x) \\
&= \mathbf{F}(f)\big(a\mathbf{F}(g)(x)\big) \\
&= \mathbf{F}(f) \circ \phi(ag),
\end{aligned}$$

that is, the diagram

$$\begin{tikzcd}
\mathbf{O}_S(S') \arrow[r, "\phi"] \arrow[d, "\mathbf{O}_S(f)"'] & \mathbf{F}(S') \arrow[d, "\mathbf{F}(f)"] \\
\mathbf{O}_S(S'') \arrow[r, "\phi"] & \mathbf{F}(S'')
\end{tikzcd}$$

is commutative. This proves the lemma.

Now let $\mathbf{F}$ be an $\mathbf{O}$-module. For each $S \in \mathrm{Ob}(\mathscr{C})$, let $F_0(S)$ be a set of generators of the $\mathbf{O}(S)$-module $\mathbf{F}(S)$, and let $\mathbf{L}_S$ be the direct sum, indexed by $F_0(S)$, of copies of $\mathbf{O}_S$. By the lemma, we obtain a morphism of $\mathbf{O}$-modules

$$\mathbf{L}_S \longrightarrow \mathbf{F}$$

such that the morphism $\mathbf{L}_S(S) \to \mathbf{F}(S)$ is surjective, and hence is an epimorphism in the category of $\mathbf{O}(S)$-modules.

Set

$$I = \coprod_{S\in \mathrm{Ob}(\mathscr{C})} F_0(S).$$

We then obtain a morphism of $\mathbf{O}$-modules

$$\mathbf{G}^{\oplus I} \longrightarrow \bigoplus_{S\in \mathrm{Ob}(\mathscr{C})} \mathbf{L}_S \longrightarrow \mathbf{F}$$

which is an epimorphism objectwise, and hence an epimorphism of ($\mathbf{O}$-Mod.). This shows that $\mathbf{G}$ is a generator of ($\mathbf{O}$-Mod.) (see [Gr57, 1.9.1]). Since ($\mathbf{O}$-Mod.) satisfies (AB 5), it follows from [Gr57, 1.10.1] that ($\mathbf{O}$-Mod.) has enough injective objects.

Remark 3.1.3. If $\mathbf{O}_0$ is the $\widehat{\mathscr{C}}$-ring defined by

$$\mathbf{O}_0(S) = \mathbb{Z}$$

(which must not be confused with the functor associated with the constant object $\mathbb{Z}$), then the category of $\mathbf{O}_0$-modules is isomorphic to the category of commutative $\widehat{\mathscr{C}}$-groups.

Definition 3.1.4. Notice that if $\mathbf{F}$ is an $\mathbf{O}$-module, then for every $S \in \mathrm{Ob}(\mathscr{C})$, $\mathbf{F}_S$ is an $\mathbf{O}_S$-module. We may therefore define an abelian $\widehat{\mathscr{C}}$-group $\underline{\mathrm{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$ by

$$\underline{\mathrm{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')(S) = \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{F}_S, \mathbf{F}'_S).$$

In the same way one defines the objects

$$\underline{\mathrm{Isom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}'), \qquad \underline{\mathrm{End}}_{\mathbf{O}}(\mathbf{F}), \qquad \text{and} \qquad \underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F}),$$

the last of which is a $\widehat{\mathscr{C}}$-group under the structure induced by composition of automorphisms.

Definition 3.2. Let $\mathbf{O}$ be a $\widehat{\mathscr{C}}$-ring, $\mathbf{F}$ an $\mathbf{O}$-module, and $\mathbf{G}$ a $\widehat{\mathscr{C}}$-group. A $\mathbf{G}$-$\mathbf{O}$-module structure on $\mathbf{F}$ is a $\mathbf{G}$-object structure such that, for every $S \in \mathrm{Ob}(\mathscr{C})$ and every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{F}(S)$ defined by g is an automorphism of its $\mathbf{O}(S)$-module structure.

Equivalently, the morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}(\mathbf{F})$$

defined in 2.3 maps $\mathbf{G}$ into the $\widehat{\mathscr{C}}$-subgroup $\underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F})$ of $\underline{\mathrm{Aut}}(\mathbf{F})$. Thus, giving a $\mathbf{G}$-$\mathbf{O}$-module structure on the $\mathbf{O}$-module $\mathbf{F}$ is the same as giving a morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F}).$$

One defines naturally the abelian group $\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and hence the additive category of $\mathbf{G}$-$\mathbf{O}$-modules, denoted ($\mathbf{G}$-$\mathbf{O}$-Mod.).

Remark 3.2.1. The reader will notice that ($\mathbf{G}$-$\mathbf{O}$-Mod.) may also be defined as the category of $\mathbf{O}[\mathbf{G}]$-modules, where $\mathbf{O}[\mathbf{G}]$ is the algebra of the $\widehat{\mathscr{C}}$-group $\mathbf{G}$ over the $\widehat{\mathscr{C}}$-ring $\mathbf{O}$, whose definition is clear. [29] Consequently, by 3.1.1 and 3.1.2, ($\mathbf{G}$-$\mathbf{O}$-Mod.) is an abelian category satisfying axiom (AB 5); moreover, if $\mathscr{C}$ is small, ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects.

All the constructions of this section specialize immediately to the case where $\mathbf{G}$ (or $\mathbf{O}$, or both) is representable by an object of $\mathscr{C}$, which is thereby endowed with the corresponding algebraic structure.

[29] Editor's note: the sentence which follows has been added; it will be used in Section 5.

We have treated briefly the principal algebraic structures that occur later in this seminar. For the others (the structure of an **O**-Lie algebra, for example), we believe that the reader has now seen enough examples in this section to make the general mechanism sketched in 2.2 work in each particular case.

We shall now apply what we have just done to the category of schemes, denoted (**Sch**), and more generally to the categories $(\mathbf{Sch})_{/S}$, also denoted $(\mathbf{Sch}_{/S})$.

4. Algebraic structures in the category of schemes

Whenever there is no ambiguity, we shall permit ourselves the following abuses of language. We shall denote by T the object $T \xrightarrow{f} S$ of $\mathscr{C}_{/S}$, with the datum of f (the structural morphism of T) understood, and we shall identify $\mathscr{C}$ with a subcategory of $\widehat{\mathscr{C}}$. In view of the compatibilities stated in the preceding sections, these identifications may be made without danger.

We shall also simplify terminology according to the following pattern: a (**Sch**)-group will also be called a *group scheme*; a $(\mathbf{Sch})_{/S}$-group will be called a *group scheme over S*, an *S-group scheme*, or an *S-group*; and it will be called an *A-group* when S is the spectrum of the ring A.

4.1. Constant schemes

The category of schemes satisfies the conditions required in 1.8. Constant objects may therefore be defined in it. For every set E, there is a scheme $E_{\mathbb{Z}}$, and for every scheme S, an S-scheme

$$E_S = (E_{\mathbb{Z}})_S.$$

Recall (cf. *loc. cit.*) that for every S-scheme T,

$$\mathrm{Hom}_S(T, E_S)$$

is the set of locally constant maps from the topological space underlying T to E.

The functor $E \mapsto E_S$ commutes with finite inverse limits. In particular, if G is a group, then G_S is an S-group scheme; if O is a ring, then O_S is an S-ring scheme; and similarly for other algebraic structures.

4.2. Affine S-groups

Let us recall some facts about affine S-schemes (EGA II, Section 1). An S-scheme T is said to be *affine over S* if the inverse image of every affine open subset of S is affine. The $\mathcal{O}_S$-algebra $f_*\mathcal{O}_T$, denoted by $\mathscr{A}(T)$, is then quasi-coherent, where f denotes the structural morphism of T. Conversely, to every quasi-coherent $\mathcal{O}_S$-algebra $\mathscr{A}$, one can associate an S-scheme affine over S, denoted $\mathrm{Spec}(\mathscr{A})$. The functors

$$T \longmapsto \mathscr{A}(T), \qquad \mathscr{A} \longmapsto \mathrm{Spec}(\mathscr{A})$$

are quasi-inverse equivalences between the category of S-schemes affine over S and the category opposite to that of quasi-coherent $\mathcal{O}_S$-algebras.

It follows that giving an algebraic structure on an S-scheme T affine over S is equivalent to giving the corresponding structure on $\mathscr{A}(T)$ in the category opposite to that of quasi-coherent $\mathcal{O}_S$-algebras. In particular, if G is an S-group affine over S, then $\mathscr{A}(G)$ is endowed with the structure of an augmented $\mathcal{O}_S$-bialgebra: there are two morphisms of $\mathcal{O}_S$-algebras

$$\Delta : \mathscr{A}(G) \longrightarrow \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G), \qquad \varepsilon : \mathscr{A}(G) \longrightarrow \mathcal{O}_S,$$

corresponding to the morphisms of S-schemes

$$\pi : G \times G \longrightarrow G, \qquad e_G : S \longrightarrow G.$$

The maps Δ and ε satisfy the following conditions, which express that G is an S-monoid.[30]

(HA 1) Δ is coassociative: the following diagram is commutative.

$$\begin{tikzcd}
& \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \arrow[dr, "\mathrm{Id} \otimes \Delta"] & \\
\mathscr{A}(G) \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \\
& \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \arrow[ur, "\Delta \otimes \mathrm{Id}"'] &
\end{tikzcd}.$$

(HA 2) Δ is compatible with ε: the following two composites are the identity:

$$\mathscr{A}(G) \xrightarrow{\Delta} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\mathrm{Id} \otimes \varepsilon} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathscr{A}(G),$$

$$\mathscr{A}(G) \xrightarrow{\Delta} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\varepsilon \otimes \mathrm{Id}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\sim} \mathscr{A}(G).$$

Let us take this opportunity to note once again that it follows from the definition of an S-group structure that, in order to give such a structure on an S-scheme G affine over S, it is not necessary to verify anything on $\mathscr{A}(G)$; one need only endow every $G(S')$, for S' over S, with a group structure functorial in S'. This observation applies *mutatis mutandis* to morphisms.

4.3. The groups $\mathbb{G}_a$ and $\mathbb{G}_m$. The ring O

4.3.1.

Let $\mathbb{G}_a$ be the functor from $(\mathbf{Sch})^\circ$ to $\mathbf{Set}$ defined by

$$\mathbb{G}_a(S) = \Gamma(S, \mathcal{O}_S),$$

endowed with the $\widehat{(\mathbf{Sch})}$-group structure defined by the additive-group structure of the ring $\Gamma(S, \mathcal{O}_S)$. It is representable by an affine scheme, which we denote by $\mathbb{G}_a$, and which is therefore a group scheme:

$$\mathbb{G}_a = \operatorname{Spec} \mathbb{Z}[T].$$

Indeed,

$$\mathbb{G}_a(S) = \operatorname{Hom}(S, \mathbb{G}_a) = \operatorname{Hom}_{\mathrm{Alg.}}\big(\mathbb{Z}[T], \Gamma(S, \mathcal{O}_S)\big) \simeq \Gamma(S, \mathcal{O}_S).$$

For every scheme S, one therefore has an S-group affine over S, denoted $\mathbb{G}_{a,S}$, corresponding to the $\mathcal{O}_S$-bialgebra $\mathcal{O}_S[T]$, whose diagonal map and augmentation are defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \qquad \varepsilon(T) = 0.$$

[30] Editor's note: of course, the inversion morphism $G \to G$ induces a morphism of $\mathcal{O}_S$-algebras $\tau : \mathscr{A}(G) \to \mathscr{A}(G)$ which, together with Δ and ε, makes $\mathscr{A}(G)$ into a Hopf $\mathcal{O}_S$-algebra.

4.3.2.

Let $\mathbb{G}_m$ be the functor from $(\mathbf{Sch})^\circ$ to **Set** defined by

$$\mathbb{G}_m(S) = \Gamma(S, \mathcal{O}_S)^\times,$$

where $\Gamma(S, \mathcal{O}_S)^\times$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S, \mathcal{O}_S)$, endowed with its natural $\widehat{(\mathbf{Sch})}$-group structure. It is representable by an affine scheme denoted by $\mathbb{G}_m$:

$$\mathbb{G}_m = \operatorname{Spec} \mathbb{Z}[T, T^{-1}] = \operatorname{Spec} \mathbb{Z}[\mathbb{Z}],$$

where $\mathbb{Z}[\mathbb{Z}]$ denotes the group algebra of $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed,

$$\mathbb{G}_m(S) = \operatorname{Hom}_{\text{Alg.}}(\mathbb{Z}[T, T^{-1}], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)^\times.$$

For every scheme S, one therefore has an S-group affine over S, denoted $\mathbb{G}_{m,S}$, corresponding to the $\mathcal{O}_S$-algebra $\mathcal{O}_S[\mathbb{Z}]$, whose diagonal map and augmentation are defined by

$$\Delta(x) = x \otimes x, \qquad \varepsilon(x) = 1, \qquad x \in \mathbb{Z}.$$

4.3.3.

Let $\mathbf{O}$ be the functor $\mathbb{G}_a$ endowed with its $\widehat{(\mathbf{Sch})}$-ring structure. It is represented by the scheme $\operatorname{Spec} \mathbb{Z}[T]$, which we denote by $\mathbf{O}$ when it is regarded as endowed with its ring-scheme structure.

For every scheme S,

$$\mathbf{O}_S = S \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Z}[T] = \operatorname{Spec}(\mathcal{O}_S[T])$$

is therefore an S-ring scheme affine over S. (Note: in EGA II, 1.7.13, $\mathbf{O}_S$ is denoted by $S[T]$.)

4.3.3.1. [31] For every object X of $\widehat{(\mathbf{Sch})}$,

$$\mathbf{O}(X) = \operatorname{Hom}(X, \mathbf{O})$$

has a ring structure, functorial in X. In particular, if X' is a scheme and morphisms $x : X' \to X$ and $f : X \to \mathbf{O}$ are given (that is, $f \in \mathbf{O}(X)$), then

$$f(x) = f \circ x \in \mathbf{O}(X') = \Gamma(X', \mathcal{O}_{X'}).$$

Definition. Let $\pi : M \to X$ be a morphism in $\widehat{(\mathbf{Sch})}$, and set

$$\mathbf{O}_X = \mathbf{O} \times X.$$

We say that M is an $\mathbf{O}_X$-module if, for every X-scheme X', the set

$$\operatorname{Hom}_X(X', M)$$

has been endowed with an $\mathbf{O}(X')$-module structure, functorial in X'.

This is equivalent to giving an X-abelian-group law

$$\mu : M \times_X M \longrightarrow M$$

[31] Editor's note: this paragraph has been added; it will be useful later (cf. II, 1.3).

and an *external law*

$$\mathbf{O} \times M = \mathbf{O}_X \times_X M \longrightarrow M, \qquad (f, m) \longmapsto f \cdot m,$$

which is an X-morphism, that is, $\pi(f \cdot m) = \pi(m)$, and which, for every $x \in X(S)$, endows

$$M(x) = \{m \in M(S) \mid \pi(m) = x\}$$

with an $\mathbf{O}(S)$-module structure.

In that case, for every $Y \in \mathrm{Ob}(\widehat{(\mathbf{Sch})}_{/X})$, not necessarily representable,

$$\mathrm{Hom}_X(Y, M) = \Gamma(M_Y/Y)$$

is an $\mathbf{O}(Y)$-module, functorially in Y.

4.4. Diagonalizable groups

4.4.1. The construction of $\mathbb{G}_m$ generalizes as follows. Let M be a commutative group, and let $M_{\mathbb{Z}}$ be the group scheme associated with it by the procedure of 4.1. Consider the functor defined by

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{groups}}(M, \mathbb{G}_m(S)) \simeq \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This is a commutative $\widehat{(\mathbf{Sch})}$-group represented by a group scheme denoted $D(M)$; thus, by definition,

$$D(M) \simeq \underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m).$$

Indeed, set

$$D(M) = \mathrm{Spec}\, \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the group algebra of M over $\mathbb{Z}$. Then

$$D(M)(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[M], \Gamma(S, \mathcal{O}_S)) \simeq \mathrm{Hom}_{\mathrm{groups}}(M, \Gamma(S, \mathcal{O}_S)^{\times})$$

by the very definition of the algebra $\mathbb{Z}[M]$.

4.4.2. For every scheme S, one therefore has an S-group scheme affine over S,

$$D_S(M) = D(M)_S = \big(\underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m)\big)_S = \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

It is associated with the $\mathcal{O}_S$-bialgebra $\mathcal{O}_S[M]$, endowed with the diagonal map and augmentation defined by

$$\Delta(x) = x \otimes x, \qquad \varepsilon(x) = 1, \qquad x \in M.$$

4.4.3. If $f : M \to N$ is a homomorphism of commutative groups, one defines in the evident way a morphism of S-groups

$$D_S(f) : D_S(N) \longrightarrow D_S(M),$$

and hence a functor

$$D_S : M \longmapsto D_S(M)$$

from the category of abelian groups to the category of S-groups affine over S. It may also be defined as the composite of the functor $M \mapsto M_S$ and the functor

$$M_S \longmapsto \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This functor commutes with base extension.

An S-group isomorphic to a group of the form $D_S(M)$ is said to be *diagonalizable*. Notice that the elements of M are interpreted as certain characters of $D_S(M)$, that is, as certain elements of

$$\mathrm{Hom}_{S\text{-}\mathbf{Gr}}(D_S(M), \mathbb{G}_{m,S}).$$

(Compare VIII, 1.)

4.4.4. Let us give some examples of diagonalizable groups. First,

$$D(\mathbb{Z}) = \mathbb{G}_m, \qquad D(\mathbb{Z}^r) = (\mathbb{G}_m)^r.$$

Set

$$\boldsymbol{\mu}_n = D(\mathbb{Z}/n\mathbb{Z}),$$

and call it the *group of n-th roots of unity*. Indeed,

$$\boldsymbol{\mu}_n(S) = \mathrm{Hom}_{\mathrm{groups}}(\mathbb{Z}/n\mathbb{Z}, \Gamma(S, \mathcal{O}_S)^\times) = \{f \in \Gamma(S, \mathcal{O}_S) \mid f^n = 1\}.$$

The S-group $\boldsymbol{\mu}_{n,S}$ corresponds to the $\mathcal{O}_S$-algebra $\mathcal{O}_S[T]/(T^n - 1)$. Suppose, in particular, that S is the spectrum of a field k of characteristic $p = n$. On setting $T - 1 = s$, one obtains

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\boldsymbol{\mu}_{p,k}$ consists of one point, whose local ring is the Artinian k-algebra $k[s]/(s^p)$. In the same vein, note that the S-schemes $\mathbb{G}_{a,S}$, $\mathbb{G}_{m,S}$, and $\mathbf{O}_S$ are smooth over S, and that $D_S(M)$ is

flat over S. Moreover, it is formally smooth (respectively, smooth) over S if and only if no residue characteristic of S divides the torsion of M (respectively, and if in addition M is of finite type); see Exposé VIII, 2.1.

4.5. Other examples of groups

The preceding procedure applies to the "classical groups" (linear groups GL_n, symplectic groups Sp_n, orthogonal groups O_n, and so forth). For example, one defines GL_n as representing the functor $\mathbf{GL}_n$ such that

$$\mathbf{GL}_n(S) = \mathrm{GL}(n, \Gamma(S, \mathcal{O}_S)) = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{O}_S^n).$$

It may be constructed, for example, as the open subscheme of $\mathrm{Spec}\,\mathbb{Z}[T_{ij}]$, $1 \le i, j \le n$, defined by the function

$$f = \det\big((T_{ij})_{i,j=1}^n\big),$$

that is, as $\mathrm{Spec}\,\mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Functor-modules in the category of schemes

We shall associate with every $\mathcal{O}_S$-module on a scheme S a $\mathbf{O}_S$-module, where $\mathbf{O}_S$ denotes the functor-ring introduced in 4.3.3. This can be done in two different ways, as follows.

Definition 4.6.1. Let S be a scheme. For every $\mathcal{O}_S$-module $\mathscr{F}$, denote by $\mathbf{V}(\mathscr{F})$ and $\mathbf{W}(\mathscr{F})$ the contravariant functors on $(\mathbf{Sch})_{/S}$ defined by

$$\mathbf{V}(\mathscr{F})(S') = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}, \mathcal{O}_{S'}),$$
$$\mathbf{W}(\mathscr{F})(S') = \Gamma(S', \mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

where $\mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ denotes the inverse image on S' of the $\mathcal{O}_S$-module $\mathscr{F}$.

Then $\mathbf{V}(\mathscr{F})$ and $\mathbf{W}(\mathscr{F})$ are endowed in the evident way with $\mathbf{O}_S$-module structures (recall that $\mathbf{O}_S(S') = \Gamma(S', \mathcal{O}_{S'}) = \mathbf{W}(\mathcal{O}_S)(S')$). Thus one has in fact defined two functors $\mathbf{V}$ and $\mathbf{W}$ from the category of $\mathcal{O}_S$-modules to the category of $\mathbf{O}_S$-modules, with $\mathbf{V}$ contravariant and $\mathbf{W}$ covariant.

For the remainder of this paragraph, we restrict to the case in which the $\mathcal{O}_S$-modules in question are quasi-coherent. Thus we regard $\mathbf{V}$ and $\mathbf{W}$ as functors

$$\mathbf{V} : (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.})^{\circ} \longrightarrow (\mathbf{O}_S\text{-}\mathbf{Mod.}), \qquad \mathbf{W} : (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \longrightarrow (\mathbf{O}_S\text{-}\mathbf{Mod.}).$$

Remark 4.6.1.1. [32] The reader will notice that, in what follows, every proposition involving only the functor $\mathbf{W}$ remains valid for arbitrary $\mathcal{O}_S$-modules, not necessarily quasi-coherent.

Proposition 4.6.2.

(i) The functors $\mathbf{V}$ and $\mathbf{W}$ commute with base extension: if S' lies over S and $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_S$-module, then

$$\mathbf{V}(\mathscr{F} \otimes \mathcal{O}_{S'}) \simeq \mathbf{V}(\mathscr{F})_{S'}, \qquad \mathbf{W}(\mathscr{F} \otimes \mathcal{O}_{S'}) \simeq \mathbf{W}(\mathscr{F})_{S'}.$$

(ii) The functors $\mathbf{V}$ and $\mathbf{W}$ are fully faithful: the canonical maps

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F})),$$

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}'))$$

are bijective.

(iii) The functors $\mathbf{V}$ and $\mathbf{W}$ are additive:

$$\mathbf{V}(\mathscr{F} \oplus \mathscr{F}') \simeq \mathbf{V}(\mathscr{F}) \times_S \mathbf{V}(\mathscr{F}'), \qquad \mathbf{W}(\mathscr{F} \oplus \mathscr{F}') \simeq \mathbf{W}(\mathscr{F}) \times_S \mathbf{W}(\mathscr{F}').$$

Parts (i) and (iii) are immediate from the definitions. For (ii), take the S' to be open subschemes of S. We leave the proof to the reader (for $\mathbf{V}$, use EGA II, 1.7.14).

Recall (cf. 3.1.4) that if $\mathbf{F}$ and $\mathbf{F}'$ are $\mathbf{O}_S$-modules, then $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{F}, \mathbf{F}')$ denotes the S-functor in abelian groups that assigns to every $S' \to S$ the group

$$\mathrm{Hom}_{\mathbf{O}_{S'}}(\mathbf{F}_{S'}, \mathbf{F}'_{S'}).$$

Proposition 4.6.3. *There are canonical morphisms in* $(\mathbf{O}_S\text{-}\mathbf{Mod.})$*:* [33]

$$\begin{tikzcd}
\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}')) \arrow[rr, equal] & & \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F})) \\
& \mathbf{W}(\mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}')) \arrow[ul] \arrow[ur] &
\end{tikzcd}.$$

This follows immediately from 4.6.2(i) and (ii).

[32] Editor's note: the numbering 4.6.1.1 has been added for later references.

[33] Editor's note: the isomorphism $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}')) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F}))$ has been added.

Notation 4.6.3.1. Let $\mathscr{F}$ be a quasi-coherent $\mathcal{O}_S$-module. By EGA II, 1.7.8, the S-functor $\mathbf{V}(\mathscr{F})$ is represented by an S-scheme affine over S, denoted $\mathbb{V}(\mathscr{F})$, and called the *vector fibration* [34] defined by $\mathscr{F}$:

$$\mathbb{V}(\mathscr{F}) = \mathrm{Spec}(\mathcal{S}(\mathscr{F})),$$

where $\mathcal{S}(\mathscr{F})$ denotes the symmetric algebra of the $\mathcal{O}_S$-module $\mathscr{F}$. [35]

Proposition 4.6.4. *Let $\mathscr{F}$ and $\mathscr{F}'$ be two quasi-coherent $\mathcal{O}_S$-modules, and let $\mathscr{A}$ be a quasi-coherent $\mathcal{O}_S$-algebra. There is a functorial isomorphism*

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathscr{A}), \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}'), \mathbf{W}(\mathscr{F}))) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}', \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}).$$

Indeed, put $X = \mathrm{Spec}(\mathscr{A})$. The left-hand side is canonically isomorphic to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}'), \mathbf{W}(\mathscr{F}))(X),$$

that is, by definition, to

$$\begin{aligned} &\mathrm{Hom}_{\mathbf{O}_X}(\mathbf{W}(\mathscr{F}')_X, \mathbf{W}(\mathscr{F})_X) \\ &\qquad \simeq \mathrm{Hom}_{\mathbf{O}_X}(\mathbf{W}(\mathscr{F}' \otimes \mathcal{O}_X), \mathbf{W}(\mathscr{F} \otimes \mathcal{O}_X)) \end{aligned}$$

(cf. 4.6.2(i)). By 4.6.2(ii), this may also be written as

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}' \otimes \mathcal{O}_X, \mathscr{F} \otimes \mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}', \pi_*\pi^*(\mathscr{F})),$$

where $\pi : X \to S$ is the structural morphism. But EGA II, 1.4.7 gives

$$\pi_*\pi^*(\mathscr{F}) \simeq \mathscr{F} \otimes \mathscr{A},$$

which completes the proof.

Corollary 4.6.4.1. *There is a canonical isomorphism*

$$\mathbf{W}(\mathscr{F} \otimes \mathscr{A}) \simeq \underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathscr{A}), \mathbf{W}(\mathscr{F})).$$

Indeed, [36] let $f : S' \to S$ be an S-scheme and put $X' = X \times_S S'$. There is a Cartesian square

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\ S' \arrow[r, "f"] & S \end{tikzcd}$$

By EGA II, 1.5.2, X' is affine over S' and $\pi'_*\mathcal{O}_{X'} = f^*(\mathscr{A})$. Hence

$$\begin{aligned} &\underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathscr{A}), \mathbf{W}(\mathscr{F}))(S') \\ &\qquad = \mathrm{Hom}_{S'}(\mathrm{Spec}(f^*\mathscr{A}), \mathbf{W}(f^*\mathscr{F})). \end{aligned}$$

[34] Editor's note: "vector bundle" has been replaced by "vector fibration." Current usage calls a vector fibration that is locally trivial of rank r a "vector bundle of rank r," that is, one whose sheaf of sections is locally isomorphic to $\mathcal{O}_S^{\oplus r}$.

[35] Editor's note: the articles [Ni04] and [Ni02], respectively, show that if S is Noetherian and $\mathscr{F}$ is a coherent $\mathcal{O}_S$-module, then $\mathbf{W}(\mathscr{F})$, respectively the S-group assigning $\mathrm{Aut}_{\mathcal{O}_T}(\mathscr{F} \otimes \mathcal{O}_T)$ to every $T \to S$, is representable if and only if $\mathscr{F}$ is locally free.

[36] Editor's note: the original has been expanded in what follows.

By 4.6.4, applied to $f^*\mathscr{F}$, $\mathscr{F}' = \mathcal{O}_{S'}$, and $f^*\mathscr{A}$, this is equal to

$$\begin{aligned}\Gamma(S', f^*\mathscr{F} \otimes f^*\mathscr{A}) &= \Gamma(S', f^*(\mathscr{F} \otimes \mathscr{A})) \\ &= \mathbf{W}(\mathscr{F} \otimes \mathscr{A})(S').\end{aligned}$$

Proposition 4.6.5. *If $\mathscr{F}$ and $\mathscr{F}'$ are two $\mathcal{O}_S$-modules locally free of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \to S$,

$$\begin{aligned}&\mathbf{W}(\mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}'))(S') \\ &\quad = \Gamma(S', \mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \otimes \mathcal{O}_{S'}) \\ &\quad = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathscr{F} \otimes \mathcal{O}_{S'}, \mathscr{F}' \otimes \mathcal{O}_{S'}).\end{aligned}$$

The last member is isomorphic both to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}'))(S')$$

and to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F}))(S')$$

by 4.6.2(i) and (ii).

Corollary 4.6.5.1. *Let $\mathscr{F}$ be an $\mathcal{O}_S$-module locally free of finite type. Put $\mathscr{F}^\vee = \mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathcal{O}_S)$. There are canonical isomorphisms*

$$\mathbf{W}(\mathscr{F}^\vee) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{O}_S) \simeq \mathbf{V}(\mathscr{F}),$$

$$\mathbf{V}(\mathscr{F}^\vee) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}), \mathbf{O}_S) \simeq \mathbf{W}(\mathscr{F}).$$

Finally, we have the following proposition.

Proposition 4.6.6. *Let $f : \mathscr{F} \to \mathscr{F}'$ be a morphism of $\mathcal{O}_S$-modules locally free of finite rank. The morphism*

$$\mathbf{W}(f) : \mathbf{W}(\mathscr{F}) \longrightarrow \mathbf{W}(\mathscr{F}')$$

is a monomorphism if and only if f identifies $\mathscr{F}$ locally with a direct summand of $\mathscr{F}'$.

The direct assertion is essentially contained in EGA 0_{I}, 5.5.5. [37] Conversely, if $\mathscr{F}$ is locally a direct summand of $\mathscr{F}'$, then for every $\pi : S' \to S$, the module $\pi^*\mathscr{F}$ is a submodule of $\pi^*\mathscr{F}'$, since this holds locally as a direct summand. Thus

$$\mathbf{W}(\mathscr{F})(S') = \Gamma(S', \pi^*\mathscr{F})$$

is a submodule of

$$\mathbf{W}(\mathscr{F}')(S') = \Gamma(S', \pi^*\mathscr{F}').$$

4.7. The category of G-$\mathcal{O}_S$-modules

Let G be an S-group and $\mathscr{F}$ an $\mathcal{O}_S$-module; $\mathbf{W}(\mathscr{F})$ has a $\mathbf{O}_S$-module structure.

Definition 4.7.1. A *G-$\mathcal{O}_S$-module structure* on $\mathscr{F}$ is an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{W}(\mathscr{F})$ (cf. 3.2). A morphism of G-$\mathcal{O}_S$-modules is, by definition, a morphism of the associated $\mathbf{h}_G$-$\mathbf{O}_S$-modules. This gives the category (G-$\mathcal{O}_S$-**Mod**.); we denote by (G-$\mathcal{O}_S$-**Mod.q.c.**) its full subcategory consisting of those G-$\mathcal{O}_S$-modules that are quasi-coherent as $\mathcal{O}_S$-modules.

[37] Editor's note: the following sentence has been expanded.

By 3.2, giving a G-$\mathcal{O}_S$-module structure on $\mathscr{F}$ is therefore equivalent to giving a morphism of $\widehat{(\mathbf{Sch})}_{/S}$-groups

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F})).$$

Remark 4.7.1.0. [38] Since 4.6.2 gives an anti-isomorphism of S-functors in groups

$$(\dagger) \qquad \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F})) \simeq \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F})),$$

it is equivalent to give an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{W}(\mathscr{F})$ or on $\mathbf{V}(\mathscr{F})$. Indeed, let

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}))$$

be as above. For every $T \to S$ and $g \in G(T)$, denote by $\rho^*(g)$ the image of $\rho(g)$ under the anti-isomorphism $(\dagger)$. Then

$$\rho^*(gh) = \rho^*(h) \circ \rho^*(g),$$

so ρ^* defines a *right* $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{V}(\mathscr{F})$. Setting

$$\rho^\vee(g) = \rho^*(g^{-1}),$$

one obtains an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{V}(\mathscr{F})$, whose datum is equivalent to that of ρ.

Remark 4.7.1.1. One may say that the categories just defined have been constructed by the following Cartesian squares.

$$\begin{tikzcd}
(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \arrow[r, hook] \arrow[d] & (G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.}) \arrow[r] \arrow[d] & (\mathbf{h}_G\text{-}\mathbf{O}_S\text{-}\mathbf{Mod.}) \arrow[d, "\text{forget}"] \\
(\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \arrow[r, hook] & (\mathcal{O}_S\text{-}\mathbf{Mod.}) \arrow[r, "\mathbf{W}"] & (\mathbf{O}_S\text{-}\mathbf{Mod.})
\end{tikzcd} .$$

The categories $(\mathcal{O}_S$-$\mathbf{Mod.})$ and $(\mathbf{O}_S$-$\mathbf{Mod.})$ are abelian, but one should remember that in general the functor $\mathbf{W}$ is neither left nor right exact. [39]

Remark 4.7.1.2. [40] Let $\mathscr{F}$ be a G-$\mathcal{O}_S$-module. The subsheaf of invariants $\mathscr{F}^G$ is defined as follows: for every open subset U of S,

$$\mathscr{F}^G(U) = \mathbf{W}(\mathscr{F})^G(U) = \{x \in \mathscr{F}(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U,\ g \in G(S')\},$$

where $x_{S'}$ denotes the image of x in

$$\Gamma(S', f^*\mathscr{F}) = \Gamma(U, f_*f^*\mathscr{F}).$$

One should note that the natural morphism

$$\mathbf{W}(\mathscr{F}^G) \longrightarrow \mathbf{W}(\mathscr{F})^G$$

is not an isomorphism in general. For example, let $S = \mathrm{Spec}(\mathbb{Z})$, and let G be the constant group $\mathbb{Z}/2\mathbb{Z} = \{1, \tau\}$ acting on $\mathscr{F} = \mathcal{O}_S$ by $\tau \cdot 1 = -1$. Then $\mathscr{F}^G = 0$, whereas, if R is an $\mathbb{F}_2$-algebra,

$$\mathbf{W}(\mathscr{F})^G(\mathrm{Spec}(R)) = R.$$

[38] Editor's note: this remark has been added from [DG], II, Section 2, 1.1.
[39] Editor's note: the original has been corrected by deleting the assertion that the category $(G$-$\mathcal{O}_S$-$\mathbf{Mod.})$ is abelian; see 4.7.2.1 below.
[40] Editor's note: this remark has been added.

4.7.2. From now until the end of 4.7, suppose that G is affine over S. [41] By 4.6.4, giving a morphism of S-functors

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{End}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}))$$

is equivalent to giving a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathscr{F} \longrightarrow \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G).$$

Saying that ρ is a morphism of $\widehat{(\mathbf{Sch})}_{/S}$-groups is then equivalent to saying that μ satisfies the following axioms.

(CM 1) The following diagram is commutative.

$$\begin{tikzcd} \mathscr{F} \arrow[r, "\mu"] \arrow[d, "\mu"'] & \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \arrow[d, "\mathrm{Id} \otimes \Delta"] \\ \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \arrow[r, "\mu \otimes \mathrm{Id}"] & \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \end{tikzcd} .$$

(CM 2) The following composite is the identity.

$$\mathscr{F} \xrightarrow{\ \mu\ } \mathscr{F} \otimes \mathscr{A}(G) \xrightarrow{\mathrm{Id} \otimes \varepsilon} \mathscr{F} \otimes \mathcal{O}_S \xrightarrow{\ \sim\ } \mathscr{F} .$$

These axioms (CM 1) and (CM 2) are those of a right *comodule* structure over the bialgebra $\mathscr{A}(G)$. [42]

Put $\mathscr{A} = \mathscr{A}(G)$. If $\mathscr{F}$ and $\mathscr{F}'$ are $\mathscr{A}$-comodules, a morphism of comodules $f : \mathscr{F} \to \mathscr{F}'$ is a morphism of $\mathcal{O}_S$-modules such that the following diagram is commutative.

$$\begin{tikzcd} \mathscr{F} \arrow[r, "f"] \arrow[d, "\mu_{\mathscr{F}}"'] & \mathscr{F}' \arrow[d, "\mu_{\mathscr{F}'}"] \\ \mathscr{F} \otimes \mathscr{A} \arrow[r, "f \otimes \mathrm{Id}"] & \mathscr{F}' \otimes \mathscr{A} \end{tikzcd} .$$

This gives the category ($\mathscr{A}$-**Comod**.); we denote by ($\mathscr{A}$-**Comod.q.c.**) its full subcategory consisting of those $\mathscr{A}$-comodules that are quasi-coherent as $\mathcal{O}_S$-modules. We have therefore proved the following statement.

Proposition 4.7.2. *Let G be an S-group affine over S. There are equivalences of categories*

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.}) \simeq (\mathscr{A}(G)\text{-}\mathbf{Comod.}),$$

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \simeq (\mathscr{A}(G)\text{-}\mathbf{Comod.q.c.}).$$

If, in addition, $S = \mathrm{Spec}(\Lambda)$ is affine and one sets $\Lambda[G] = \Gamma(S, \mathscr{A}(G))$, there is an equivalence of categories

$$(\mathscr{A}(G)\text{-}\mathbf{Comod.q.c.}) \simeq (\Lambda[G]\text{-}\mathbf{Comod.}).$$

[41] Editor's note: see $\mathrm{VI_B}$, Sections 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine but G and $\mathscr{F}$ are assumed flat over S.

[42] Editor's note: left G-$\mathcal{O}_S$-modules correspond naturally to right $\mathscr{A}(G)$-comodules.

[43]

Suppose, moreover, that $\mathscr{A} = \mathscr{A}(G)$ is a flat $\mathcal{O}_S$-module. [44] Let $\mathscr{E}$ be an $\mathscr{A}$-comodule and $\mathscr{F}$ an $\mathcal{O}_S$-submodule of $\mathscr{E}$. Since $\mathscr{A}$ is flat over $\mathcal{O}_S$, one may identify $\mathscr{F}\otimes\mathscr{A}$ (respectively, $\mathscr{F}\otimes\mathscr{A}\otimes\mathscr{A}$) with an $\mathcal{O}_S$-submodule of $\mathscr{E}\otimes\mathscr{A}$ (respectively, of $\mathscr{E}\otimes\mathscr{A}\otimes\mathscr{A}$). Suppose that $\mu_{\mathscr{E}}$ maps $\mathscr{F}$ into $\mathscr{F}\otimes\mathscr{A}$. Then its restriction

$$\mu_{\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{F}\otimes\mathscr{A}$$

endows $\mathscr{F}$ with an $\mathscr{A}$-comodule structure; one says that $\mathscr{F}$ is a *subcomodule* of $\mathscr{E}$. Passing to the quotient, $\mu_{\mathscr{E}}$ defines a morphism of $\mathcal{O}_S$-modules

$$\mathscr{E}/\mathscr{F} \longrightarrow (\mathscr{E}/\mathscr{F})\otimes\mathscr{A},$$

which endows $\mathscr{E}/\mathscr{F}$ with an $\mathscr{A}$-comodule structure. If $f : \mathscr{E} \to \mathscr{E}'$ is a morphism of $\mathscr{A}$-comodules, then $\operatorname{Ker} f$ (respectively, $\operatorname{Im} f$) is an $\mathscr{A}$-subcomodule of $\mathscr{E}$ (respectively, of $\mathscr{E}'$), and f induces an isomorphism of $\mathscr{A}$-comodules

$$\mathscr{E}/\operatorname{Ker} f \xrightarrow{\sim} \operatorname{Im} f.$$

Moreover, if $\mathscr{E}$ and $\mathscr{E}'$ are quasi-coherent $\mathcal{O}_S$-modules, then so are $\operatorname{Ker} f$ and $\operatorname{Im} f$. Consequently, ($\mathscr{A}$-**Comod**.) and ($\mathscr{A}$-**Comod.q.c**.) are abelian categories.

Corollary 4.7.2.1. *Suppose that G is affine and flat over S. Then the category (G-$\mathcal{O}_S$-**Mod.q.c**.) (respectively, (G-$\mathcal{O}_S$-**Mod**.)), equivalent to the category of $\mathscr{A}(G)$-comodules that are quasi-coherent over $\mathcal{O}_S$ (respectively, of all $\mathscr{A}(G)$-comodules), is abelian.*

4.7.3. Suppose now that G is a *diagonalizable group*, that is, that $\mathscr{A}(G)$ is the algebra of a commutative group M over the sheaf of rings $\mathcal{O}_S$. If $\mathscr{F}$ is an $\mathcal{O}_S$-module, then

$$\mathscr{F}\otimes\mathscr{A}(G) = \coprod_{m\in M} \mathscr{F}\otimes m\mathcal{O}_S.$$

Giving a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathscr{F} \longrightarrow \mathscr{F}\otimes\mathscr{A}(G)$$

is therefore equivalent to giving $\mathcal{O}_S$-endomorphisms $(\mu_m)_{m\in M}$ of $\mathscr{F}$ such that, for every section x of $\mathscr{F}$ on an open subset of S, the family $(\mu_m(x))$ is a section of the direct sum $\coprod_{m\in M}\mathscr{F}$. This means that on every sufficiently small open subset, only finitely many restrictions of the $\mu_m(x)$ are nonzero.

For μ defined by

$$\mu(x) = \sum_{m\in M} \mu_m(x)\otimes m,$$

condition (CM 1), respectively (CM 2), holds if and only if

$$\mu_m\circ\mu_{m'} = \delta_{mm'}\mu_m, \qquad \left(\text{respectively, } \sum_{m\in M}\mu_m = \operatorname{Id}_{\mathscr{F}}\right).$$

Thus the μ_m are pairwise orthogonal projectors whose sum is the identity. We have proved the following result.

[43] Editor's note: the preceding sentence has been added.

[44] Editor's note: the following paragraph has been added from [Se68, Section 1.3].

Proposition 4.7.3. *If $G = D_S(M)$ is a diagonalizable S-group, the category of quasi-coherent G-$\mathscr{O}_S$-modules (respectively, of all G-$\mathscr{O}_S$-modules) is equivalent to the category of quasi-coherent $\mathscr{O}_S$-modules (respectively, of all $\mathscr{O}_S$-modules) graded by M.*

Remark. If $\mathscr{F}$ has an $\mathscr{O}_S$-algebra structure preserved by the operations of G, then the grading of $\mathscr{F}$ is an algebra grading. More precisely:

Corollary 4.7.3.1. *The functor $\mathscr{A} \mapsto \mathrm{Spec}(\mathscr{A})$ induces an equivalence between the category of quasi-coherent $\mathscr{O}_S$-algebras graded by M and the category opposite to that of S-schemes affine over S with S-operator group $G = D_S(M)$.*

Proposition 4.7.4. *Let G be a diagonalizable S-group. If*

$$0 \longrightarrow \mathscr{F}_1 \longrightarrow \mathscr{F}_2 \longrightarrow \mathscr{F}_3 \longrightarrow 0$$

is an exact sequence of quasi-coherent G-$\mathscr{O}_S$-modules that splits as a sequence of $\mathscr{O}_S$-modules, then it also splits as a sequence of G-$\mathscr{O}_S$-modules.

Indeed, if $G = D_S(M)$, each $\mathscr{F}_i$ is graded by the $(\mathscr{F}_i)_m$, and for every $m \in M$ the sequence

$$0 \longrightarrow (\mathscr{F}_1)_m \longrightarrow (\mathscr{F}_2)_m \longrightarrow (\mathscr{F}_3)_m \longrightarrow 0$$

of $\mathscr{O}_S$-modules is split. The preceding proposition then gives the result.

5. Group cohomology

5.1. The standard complex

Let $\mathscr{C}$ be a category, $\mathbf{G}$ a $\widehat{\mathscr{C}}$-group, $\mathbf{O}$ a $\widehat{\mathscr{C}}$-ring, and $\mathbf{F}$ a $\mathbf{G}$-$\mathbf{O}$-module.[45] For $n \geq 0$, set

$$C^n(\mathbf{G}, \mathbf{F}) = \mathrm{Hom}(\mathbf{G}^n, \mathbf{F}), \qquad \underline{C}^n(\mathbf{G}, \mathbf{F}) = \underline{\mathrm{Hom}}(\mathbf{G}^n, \mathbf{F}),$$

where $\mathbf{G}^0$ is the final object $\underline{\mathbf{e}}$. Then $\underline{C}^n(\mathbf{G}, \mathbf{F})$ (respectively, $C^n(\mathbf{G}, \mathbf{F})$) is endowed in the evident way with an $\mathbf{O}$-module structure (respectively, a $\Gamma(\mathbf{O})$-module structure), and

$$C^n(\mathbf{G}, \mathbf{F}) \simeq \Gamma(\underline{C}^n(\mathbf{G}, \mathbf{F})), \qquad \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S).$$

To give an element of $C^n(\mathbf{G}, \mathbf{F})$ is to give, for every $S \in \mathrm{Ob}(\mathscr{C})$, an n-cochain of $\mathbf{G}(S)$ with values in $\mathbf{F}(S)$, functorially in S. Recall that the boundary operator

$$\partial : C^n(\mathbf{G}(S), \mathbf{F}(S)) \longrightarrow C^{n+1}(\mathbf{G}(S), \mathbf{F}(S))$$

is given by

$$\begin{aligned} \partial f(g_1, \dots, g_{n+1}) = {} & g_1 f(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \dots, g_n). \end{aligned}$$

It is functorial in S, and hence defines a homomorphism

$$\partial : C^n(\mathbf{G}, \mathbf{F}) \longrightarrow C^{n+1}(\mathbf{G}, \mathbf{F})$$

[45] Editor's note: This complex is often called the Hochschild complex; see, for example, Section II.3 of [DG70].

such that $\partial \circ \partial = 0$. We have thus defined a complex of abelian groups (indeed, of $\Gamma(\mathbf{O})$-modules), denoted $C^*(\mathbf{G}, \mathbf{F})$. In the same way one defines the complex of $\mathbf{O}$-modules $\underline{C}^*(\mathbf{G}, \mathbf{F})$, and

$$C^*(\mathbf{G}, \mathbf{F}) = \Gamma(\underline{C}^*(\mathbf{G}, \mathbf{F})).$$

We denote by $H^n(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{H}^n(\mathbf{G}, \mathbf{F})$) the groups (respectively, the $\widehat{\mathscr{C}}$-groups) of cohomology of $C^*(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{C}^*(\mathbf{G}, \mathbf{F})$). In particular,

$$\underline{H}^0(\mathbf{G}, \mathbf{F}) = \mathbf{F}^{\mathbf{G}}, \qquad H^0(\mathbf{G}, \mathbf{F}) = \Gamma(\mathbf{F}^{\mathbf{G}}).$$

Remark 5.1.1. The "set-theoretic" description of ∂ given above is convenient for checking that $\partial \circ \partial = 0$. One may also define ∂ in terms of the multiplication $m : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ and the action $\mu : \mathbf{G} \times \mathbf{F} \to \mathbf{F}$ as follows: for every $f \in C^n(\mathbf{G}, \mathbf{F})$,

$$\partial f = \mu \circ (\mathrm{id}_{\mathbf{G}} \times f) + \sum_{i=1}^{n} (-1)^i f \circ \left(\mathrm{id}_{\mathbf{G}^{i-1}} \times m \times \mathrm{id}_{\mathbf{G}^{n-i}}\right) + (-1)^{n+1} f \circ \mathrm{pr}_{[1,n]},$$

where $\mathrm{pr}_{[1,n]}$ denotes the projection of $\mathbf{G}^{n+1} = \mathbf{G}^n \times \mathbf{G}$ onto $\mathbf{G}^n$. Similarly, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $f \in \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S)$,

$$\partial f = \mu_S \circ (\mathrm{id}_{\mathbf{G}_S} \times f) + \sum_{i=1}^{n} (-1)^i f \circ \left(\mathrm{id}_{\mathbf{G}_S^{i-1}} \times m_S \times \mathrm{id}_{\mathbf{G}_S^{n-i}}\right) + (-1)^{n+1} f \circ \mathrm{pr}_{[1,n]},$$

where m_S and μ_S are obtained from m and μ by base change. [46]

5.2.

Recall (see Section 3) that ($\mathbf{G}$-$\mathbf{O}$-Mod.) is an abelian category, with its structure defined objectwise. Thus

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

is an exact sequence of $\mathbf{G}$-$\mathbf{O}$-modules if and only if the sequence of abelian groups

$$0 \longrightarrow \mathbf{F}'(S) \longrightarrow \mathbf{F}(S) \longrightarrow \mathbf{F}''(S) \longrightarrow 0$$

is exact for every $S \in \mathrm{Ob}(\mathscr{C})$. [47]

Suppose that $\mathscr{C}$ is small. Then, by 3.2.1, ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects, so the derived functors of the left exact functors $\underline{H}^0$ and H^0 are defined. We shall now show that the functors $\underline{H}^n$ (respectively, H^n) are indeed the derived functors of $\underline{H}^0$ (respectively, H^0). [48]

Definition 5.2.0. For every $\mathbf{O}$-module $\mathbf{P}$, let $E(\mathbf{P})$ denote the object $\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})$ of $\widehat{\mathscr{C}}$, endowed with the following $\mathbf{G}$-$\mathbf{O}$-module structure. For every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})(S) = \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S),$$

and $g \in \mathbf{G}(S)$ and $a \in \mathbf{O}(S)$ act on $\phi \in \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ by

$$(g \cdot \phi)(h) = \phi(hg), \qquad (a \cdot \phi)(h) = a\phi(h),$$

[46] Editor's note: This remark has been added.
[47] Editor's note: The preceding reminder has been added.
[48] Editor's note: The preceding sentence has been added.

for every $h \in \mathbf{G}(S')$, $S' \to S$. Moreover, for every $\phi \in \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$, set

$$\varepsilon(\phi) = \phi(1) \in \mathbf{P}(S),$$

where 1 denotes the unit element of $\mathbf{G}(S)$. This defines a functor

$$E : (\mathbf{O}\text{-Mod.}) \longrightarrow (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$$

and a natural transformation $\varepsilon : E \to \mathrm{Id}$, where Id denotes the identity functor of ($\mathbf{O}$-Mod.).[49]

Remark 5.2.0.1. In what follows, let $\mathbf{G}_1$ and $\mathbf{G}_2$ denote two copies of $\mathbf{G}$. The morphism

$$\mathbf{G}_1 \times E(\mathbf{P}) \longrightarrow E(\mathbf{P}), \qquad (g_1, \phi) \longmapsto (g_2 \longmapsto \phi(g_2 g_1))$$

corresponds, via the isomorphisms

$$\begin{aligned} \mathrm{Hom}(\mathbf{G}_1 \times E(\mathbf{P}), E(\mathbf{P})) &\simeq \mathrm{Hom}\Big(E(\mathbf{P}), \underline{\mathrm{Hom}}(\mathbf{G}_1, \underline{\mathrm{Hom}}(\mathbf{G}_2, \mathbf{P}))\Big) \\ &\simeq \mathrm{Hom}\Big(E(\mathbf{P}), \underline{\mathrm{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})\Big), \end{aligned}$$

to the morphism

$$\phi \longmapsto ((g_2, g_1) \longmapsto \phi(g_2 g_1)),$$

that is, to the morphism

$$\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})$$

induced by the multiplication

$$\mu_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}, \qquad (g_2, g_1) \longmapsto g_2 g_1.$$

Lemma 5.2.0.2. *(i) The functor E is right adjoint to the forgetful functor*

$$(\mathbf{G}\text{-}\mathbf{O}\text{-}\mathit{Mod.}) \longrightarrow (\mathbf{O}\text{-}\mathit{Mod.}).$$

More precisely, $\varepsilon : E \to \mathrm{Id}$ induces, for every $\mathbf{M} \in (\mathbf{G}\text{-}\mathbf{O}\text{-}\mathit{Mod.})$ and $\mathbf{P} \in \mathrm{Ob}(\mathbf{O}\text{-}\mathit{Mod.})$, a bijection

$$\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{O}\text{-Mod.}}(\mathbf{M}, \mathbf{P})$$

functorial in $\mathbf{M}$ and $\mathbf{P}$.

(ii) Consequently, if $\mathbf{I}$ is an injective object of ($\mathbf{O}$-Mod.), then $E(\mathbf{I})$ is an injective object of ($\mathbf{G}$-$\mathbf{O}$-Mod.).

Proof. To every $\mathbf{O}$-morphism $f : \mathbf{M} \to \mathbf{P}$, associate the element ϕ_f of $\mathrm{Hom}_{\mathbf{O}}(\mathbf{M}, E(\mathbf{P}))$ defined as follows. For every $S \in \mathrm{Ob}(\mathscr{C})$ and $m \in \mathbf{M}(S)$, the element $\phi_f(m)$ of $\mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $g \in \mathbf{G}(S')$, $S' \to S$, by

$$\phi_f(m)(g) = f(gm) \in \mathbf{P}(S').$$

Then, for every $h \in \mathbf{G}(S)$, one has $\phi_f(hm) = h \cdot f(m)$; that is, ϕ_f is an element of

$$\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})).$$

[49]Editor's note: The original has been modified so as to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

If $\phi \in \mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-}\mathrm{Mod.}}(\mathbf{M}, E(\mathbf{P}))$, and if for every $m \in \mathbf{M}(S)$ we write $f(m) = \phi(m)(1)$, then

$$\phi_f(m)(g) = f(gm) = \phi(gm)(1) = (g \cdot \phi(m))(1) = \phi(m)(g);$$

thus $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows at once.

Definition 5.2.0.3. Let $\mathbf{M}$ be a $\mathbf{G}$-$\mathbf{O}$-module. The identity map of $\mathbf{M}$, considered as an $\mathbf{O}$-module, corresponds under the adjunction to the morphism of $\mathbf{G}$-$\mathbf{O}$-modules

$$\mu_{\mathbf{M}} : \mathbf{M} \longrightarrow E(\mathbf{M})$$

such that, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $m \in \mathbf{M}(S)$, $\mu_{\mathbf{M}}(m)$ is the morphism $\mathbf{G}_S \to \mathbf{M}_S$ defined as follows: for every $S' \to S$ and $g \in \mathbf{G}(S')$,

$$\mu_{\mathbf{M}}(m)(g) = g \cdot m_{S'} \in \mathbf{M}(S').$$

Notice that $\mu_{\mathbf{M}}$ is a monomorphism. Indeed, $\varepsilon_{\mathbf{M}} : E(\mathbf{M}) \to \mathbf{M}$ is a morphism of $\mathbf{O}$-modules such that $\varepsilon_{\mathbf{M}} \circ \mu_{\mathbf{M}} = \mathrm{id}_{\mathbf{M}}$. Consequently, as an $\mathbf{O}$-module, $\mathbf{M}$ is a direct summand of $E(\mathbf{M})$.

Proposition 5.2.1. *Suppose that $\mathscr{C}$ is small, that finite products exist in $\mathscr{C}$, and that $\mathbf{G}$ is representable. Then the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) are the derived functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$) on the category of $\mathbf{G}$-$\mathbf{O}$-modules.*

By the well-known general results, [50] it suffices to verify that the $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0. Let

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

be an exact sequence of $\mathbf{G}$-$\mathbf{O}$-modules, and let $S \in \mathrm{Ob}(\mathscr{C})$. By hypothesis, $\mathbf{G}$ is represented by an object $G \in \mathrm{Ob}(\mathscr{C})$, and finite products exist in $\mathscr{C}$; in particular, $\mathscr{C}$ has a final object e. Hence every $\mathbf{G}^n \times \mathbf{h}_S$ is represented by $G^n \times S$ (with $G^0 = e$), and the sequence

$$0 \longrightarrow \mathbf{F}'(G^n \times S) \longrightarrow \mathbf{F}(G^n \times S) \longrightarrow \mathbf{F}''(G^n \times S) \longrightarrow 0$$

is exact. This shows that the sequence of $\mathbf{O}$-modules

$$0 \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}') \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}) \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}'') \longrightarrow 0$$

is exact. It follows that $\underline{C}^*(\mathbf{G}, -)$, regarded as a functor on ($\mathbf{G}$-$\mathbf{O}$-Mod.) with values in the category of complexes of ($\mathbf{O}$-Mod.), is exact. Thus the functors $\underline{H}^n(\mathbf{G}, -)$ are cohomological.

Since Γ is exact, the same is true of the functors $H^n(\mathbf{G}, -)$. It remains only to prove the following.

Lemma 5.2.2. *For every $\mathbf{P} \in \mathrm{Ob}(\mathbf{O}\text{-}Mod.)$, one has*

$$\underline{H}^n(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \qquad H^n(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \qquad n > 0.$$

It is enough to show that $\underline{C}^*(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ and $C^*(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ are homotopically trivial in degrees > 0. It is enough, in fact, to do this for the second complex, since the corresponding result for the first follows by base change. [51] For every $n \geq 0$, define a morphism

$$\sigma : C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})) \longrightarrow C^n(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$$

[50] Editor's note: See [Gr57, 2.2.1] and [Gr57, 2.3]. The original has also been expanded in what follows.

[51] Editor's note: The original has been expanded in what follows.

as follows. Let $f \in C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$. For every $S \in \mathrm{Ob}(\mathscr{C})$ and $g_1, \ldots, g_n \in \mathbf{G}(S)$, the element $\sigma(f)(g_1, \ldots, g_n)$ of $\mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $S' \to S$ and $x \in \mathbf{G}(S')$, by

$$\sigma(f)(g_1, \ldots, g_n)(x) = f(x, g_1, \ldots, g_n)(e) \in \mathbf{P}(S'),$$

where e denotes the unit element of $\mathbf{G}(S')$. Then σ is a homotopy operator in degrees > 0. Indeed, for every $g_1, \ldots, g_{n+1} \in \mathbf{G}(S)$ and $x \in \mathbf{G}(S')$, on the one hand,

$$\begin{aligned}\partial\sigma(f)(g_1, \ldots, g_{n+1})(x) &= f(xg_1, g_2, \ldots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^{n}(-1)^i f(x, g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1})(e) \\ &\quad + (-1)^{n+1} f(x, g_1, \ldots, g_n)(e),\end{aligned}$$

and, on the other hand,

$$\begin{aligned}\sigma(\partial f)(g_1, \ldots, g_{n+1})(x) &= \big(xf(g_1, g_2, \ldots, g_{n+1})\big)(e) - f(xg_1, g_2, \ldots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^{n}(-1)^{i+1} f(x, g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ &\quad + (-1)^{n+2} f(x, g_1, \ldots, g_n)(e).\end{aligned}$$

Consequently,

$$\big(\partial\sigma(f) + \sigma(\partial f)\big)(g_1, \ldots, g_{n+1})(x) = f(g_1, \ldots, g_{n+1})(x);$$

that is, $\partial\sigma + \sigma\partial$ is the identity map of $C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ for every $n \geq 0$.

Remark 5.2.3. The hypothesis that $\mathscr{C}$ be small is used only to ensure the existence of the derived functors $R^n\underline{H}^0$ and R^nH^0. In every case, what precedes shows that the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0, and hence are the (right) satellite functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$), in the sense of [Gr57, 2.2]. If ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects (as it does when $\mathscr{C}$ is small), they coincide with the derived functors (*loc. cit.*, 2.3). [52]

5.3. Cohomology of G-$\mathscr{O}_S$-modules

Let S be a scheme, G an S-group, and $\mathscr{F}$ a quasi-coherent G-$\mathscr{O}_S$-module. The cohomology groups of G with values in $\mathscr{F}$ are defined by

$$H^n(G, \mathscr{F}) = H^n(\mathbf{h}_G, \mathbf{W}(\mathscr{F}))$$

(for the notation, see 4.6).

Suppose that G is affine over S. In view of Proposition 4.6.4, this cohomology is computed as follows: $H^n(G, \mathscr{F})$ is the n-th cohomology group of the complex $C^*(G, \mathscr{F})$, whose n-th term is

$$C^n(G, \mathscr{F}) = \Gamma\Big(S, \mathscr{F} \otimes \underbrace{\mathscr{A}(G) \otimes \cdots \otimes \mathscr{A}(G)}_{n \text{ times}}\Big).$$

If f (respectively, a_i) is a section of $\mathscr{F}$ (respectively, of $\mathscr{A}(G)$) over an open subset of S, then

$$\begin{aligned}\partial(f \otimes a_1 \otimes \cdots \otimes a_n) &= \mu_{\mathscr{F}}(f) \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \\ &\quad + \sum_{i=1}^{n}(-1)^i f \otimes a_1 \otimes \cdots \otimes \Delta a_i \otimes \cdots \otimes a_n \\ &\quad + (-1)^{n+1} f \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \otimes 1,\end{aligned}$$

[52] Editor's note: This remark has been added.

where
$$\Delta : \mathscr{A}(G) \longrightarrow \mathscr{A}(G) \otimes \mathscr{A}(G), \qquad \mu_{\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A}(G)$$
describe the coalgebra structure of $\mathscr{A}(G)$ and the comodule structure of $\mathscr{F}$. Notice in passing that the cohomology of G with values in $\mathscr{F}$ therefore depends only on the comodule structure of $\mathscr{F}$, and in particular only on the S-monoid structure of G.

In particular,
$$H^0(G, \mathscr{F}) = \Gamma(S, \mathscr{F}^G),$$
where $\mathscr{F}^G$, the *sheaf of invariants of* $\mathscr{F}$, is the sheaf whose sections over an open subset U of S are those sections of $\mathscr{F}$ over U whose inverse image in every S' over U is invariant under $G(S')$ (see 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme and G an S-group that is affine and flat over S. The functors $H^n(G, -)$ are the derived functors of $H^0(G, -)$ on the category of quasi-coherent G-$\mathscr{O}_S$-modules.*

Proof. Since G is affine and flat over S, 4.7.2.1 shows that the category (G-$\mathscr{O}_S$-Mod.q.c.) is equivalent to the category ($\mathscr{A}(G)$-Comod.q.c.) of quasi-coherent $\mathscr{A}(G)$-comodules over $\mathscr{O}_S$, and is therefore abelian. Moreover, since $\mathscr{A}(G)$ is a flat $\mathscr{O}_S$-module, each functor
$$\mathscr{F} \longmapsto \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{A}(G)^{\otimes n}$$
is exact; since S is affine, it follows that $C^*(G, -)$ is an exact functor on (G-$\mathscr{O}_S$-Mod.q.c.). [53]

Let Δ (respectively, η) denote the comultiplication (respectively, the augmentation) of $\mathscr{A}(G)$. For every quasi-coherent $\mathscr{O}_S$-module $\mathscr{P}$, let
$$\mathrm{Ind}(\mathscr{P}) = \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G),$$
endowed with the $\mathscr{A}(G)$-comodule structure defined by
$$\mathrm{id}_{\mathscr{P}} \otimes \Delta : \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \otimes_{\mathscr{O}_S} \mathscr{A}(G).$$

This defines a functor
$$\mathrm{Ind} : (\mathscr{O}_S\text{-Mod.q.c.}) \longrightarrow (G\text{-}\mathscr{O}_S\text{-Mod.q.c.}).$$
It follows from 4.6.4.1, 5.2.0, and 5.2.0.1 that there is an isomorphism of $\mathbf{G}$-$\mathbf{O}_S$-modules
$$\mathbf{W}(\mathrm{Ind}(\mathscr{P})) \simeq E(\mathbf{W}(\mathscr{P})) = \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{W}(\mathscr{P})). \tag{*}$$
Under this identification, the morphism $\varepsilon : E(\mathbf{W}(\mathscr{P})) \to \mathbf{W}(\mathscr{P})$ corresponds to the morphism of $\mathscr{O}_S$-modules
$$\mathrm{id}_{\mathscr{P}} \otimes \eta : \mathrm{Ind}(\mathscr{P}) \longrightarrow \mathscr{P}.$$

We have already used the fact that the functor
$$\mathbf{W} : (\mathscr{O}_S\text{-Mod.}) \longrightarrow (\mathbf{O}_S\text{-Mod.})$$
is fully faithful. By Definition 4.7.1, the same is true of its restriction to (G-$\mathscr{O}_S$-Mod.): if $\mathscr{M}$ and $\mathscr{M}'$ are G-$\mathscr{O}_S$-modules, there is a functorial isomorphism
$$\mathrm{Hom}_{G\text{-}\mathscr{O}_S\text{-Mod.}}(\mathscr{M}, \mathscr{M}') \simeq \mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}_S\text{-Mod.}}(\mathbf{W}(\mathscr{M}), \mathbf{W}(\mathscr{M}')).$$

[53] Editor's note: The original has been modified to show that the category (G-$\mathscr{O}_S$-Mod.q.c.) is abelian and has enough injective objects. Compare [Ja03], Part I, 3.3–3.4, 3.9, 4.2, and 4.14–4.16 (bearing in mind that "k-group scheme" there means "affine k-group scheme"; see *loc. cit.*, 2.1).

Consequently, Lemma 5.2.0.2 gives the following result.

Corollary 5.3.1.1. *(i) The functor* Ind *is right adjoint to the forgetful functor*

$$(G\text{-}\mathscr{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathscr{O}_S\text{-Mod.q.c.}).$$

More precisely, the map $\mathrm{id}_{\mathscr{P}} \otimes \eta : \mathrm{Ind}(\mathscr{P}) \to \mathscr{P}$ *induces, for every object* $\mathscr{M}$ *of* $(G\text{-}\mathscr{O}_S\text{-Mod.q.c.})$*, a bijection*

$$\mathrm{Hom}_{G\text{-}\mathscr{O}_S\text{-Mod.}}(\mathscr{M}, \mathrm{Ind}(\mathscr{P})) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_S}(\mathscr{M}, \mathscr{P}).$$

(ii) Consequently, if $\mathscr{I}$ *is an injective object of* $(\mathscr{O}_S\text{-Mod.q.c.})$*, then* $\mathrm{Ind}(\mathscr{I})$ *is an injective object of* $(G\text{-}\mathscr{O}_S\text{-Mod.q.c.})$*.*

Let $\mathscr{F}$ be a G-$\mathscr{O}_S$-module, and let $\mu_{\mathscr{F}} : \mathscr{F} \to \mathrm{Ind}(\mathscr{F})$ be the map defining its $\mathscr{A}(G)$-comodule structure. It follows from 5.2.0.3 (or directly from axioms (CM 1) and (CM 2) of 4.7.2) that $\mu_{\mathscr{F}}$ is a morphism of G-$\mathscr{O}_S$-modules, and that

$$(\mathrm{id}_{\mathscr{F}} \otimes \eta) \circ \mu_{\mathscr{F}} = \mathrm{id}_{\mathscr{F}}\,.$$

Thus $\mathscr{F}$ is a direct summand of $\mathrm{Ind}(\mathscr{F})$ as an $\mathscr{O}_S$-module; in particular, $\mu_{\mathscr{F}}$ is a monomorphism. By (*) and 5.2.2,

$$H^n(\mathbf{G}, \mathbf{W}(\mathrm{Ind}(\mathscr{F}))) \simeq H^n\Big(\mathbf{G}, \underline{\mathrm{Hom}}_S(\mathbf{G}, \mathbf{W}(\mathscr{F}))\Big) = 0 \qquad (n > 0).$$

Therefore $H^n(G, -)$ is effaceable for $n > 0$.

Finally, since S is affine, $(\mathscr{O}_S\text{-Mod.q.c.})$ has enough injective objects. Let $\mathscr{F} \hookrightarrow \mathscr{I}$ be a monomorphism of $\mathscr{O}_S$-modules, where $\mathscr{I}$ is injective in $(\mathscr{O}_S\text{-Mod.q.c.})$. Since $\mathscr{A}(G)$ is flat over $\mathscr{O}_S$, $\mathrm{Ind}(\mathscr{F})$ is a G-$\mathscr{O}_S$-submodule of $\mathrm{Ind}(\mathscr{I})$. Hence:

Corollary 5.3.1.2. *The abelian category* $(G\text{-}\mathscr{O}_S\text{-Mod.q.c.})$ *has enough injective objects.*

Taking account of [Gr57, 2.2.1 and 2.3], already used in the proof of 5.2.1, completes the proof of Theorem 5.3.1.

Remark 5.3.1.3. Corollary 5.3.1.1 may also be proved by the following calculation. To every morphism of G-$\mathscr{O}_S$-modules

$$\phi : \mathscr{M} \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G)$$

associate the $\mathscr{O}_S$-morphism

$$(\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \phi : \mathscr{M} \longrightarrow \mathscr{P}.$$

Conversely, to every $\mathscr{O}_S$-morphism $f : \mathscr{M} \to \mathscr{P}$, associate the morphism of G-$\mathscr{O}_S$-modules

$$(f \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} : \mathscr{M} \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G).$$

One sees at once that

$$(\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ (f \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} = (f \otimes \mathrm{id}_{\mathscr{O}_S}) \circ (\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \mu_{\mathscr{M}} = f.$$

On the other hand, for every ϕ, the following diagram is commutative.

$$\begin{tikzcd}
\mathscr{M} \arrow[r, "\phi"] \arrow[d, "\mu_{\mathscr{M}}"'] & \mathscr{P} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \arrow[d, "\mathrm{Id}_{\mathscr{P}} \otimes \Delta"] \\
\mathscr{M} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \arrow[r, "\phi \otimes \mathrm{Id}_{\mathscr{A}(G)}"] & \mathscr{P} \otimes_{\mathcal{O}_S} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G)
\end{tikzcd} \quad .$$

It follows that

$$\begin{aligned}\Big(((\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \phi) \otimes \mathrm{id}_{\mathscr{A}(G)}\Big) &\circ \mu_{\mathscr{M}} \\ &= (\mathrm{id}_{\mathscr{P}} \otimes \eta \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ (\phi \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} \\ &= (\mathrm{id}_{\mathscr{P}} \otimes \eta \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ (\mathrm{id}_{\mathscr{P}} \otimes \Delta) \circ \phi = \phi.\end{aligned}$$

This proves Corollary 5.3.1.1(i), and (ii) follows.

Let $\mathscr{F}$ be a G-$\mathscr{O}_S$-module. We saw above that axiom (CM 2) of 4.7.2 shows that, as an $\mathscr{O}_S$-module, $\mathscr{F}$ is a direct summand of $E(\mathscr{F})$. This gives:

Proposition 5.3.2. *Let S be an affine scheme and G an S-group that is affine and flat. Suppose that every exact sequence*

$$0 \longrightarrow \mathscr{F}_1 \longrightarrow \mathscr{F}_2 \longrightarrow \mathscr{F}_3 \longrightarrow 0$$

of quasi-coherent G-$\mathscr{O}_S$-modules which splits as a sequence of $\mathscr{O}_S$-modules also splits as a sequence of G-$\mathscr{O}_S$-modules. Then the functors $H^n(G,-)$ vanish for $n > 0$ (or, equivalently, the functor $H^0(G,-)$ is exact).

Indeed, by hypothesis, the sequence of G-$\mathscr{O}_S$-modules

$$0 \longrightarrow \mathscr{F} \longrightarrow E(\mathscr{F}) \longrightarrow E(\mathscr{F})/\mathscr{F} \longrightarrow 0$$

is split. Thus $\mathscr{F}$ is a direct summand of $E(\mathscr{F})$ as a G-$\mathscr{O}_S$-module, while the cohomology of the latter vanishes.

Combining this with Proposition 4.7.4 immediately gives:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S-group. For every quasi-coherent G-$\mathscr{O}_S$-module $\mathscr{F}$, one has*

$$H^n(G,\mathscr{F}) = 0 \qquad (n > 0).$$

Remark. Proposition 5.3.2 remains valid when G is not necessarily flat over S; the proof then uses relative cohomology. [54]

6. G-equivariant objects and modules

Let $\mathscr{C}$ be a category having a final object and in which fiber products exist.[55] Let G be a $\widehat{\mathscr{C}}$-group, $\pi : M \to X$ a morphism in $\widehat{\mathscr{C}}$, and $\lambda = \lambda_X : G \times X \to X$ an action of G on X. In what follows, $Y \times_f M$ denotes the fiber product of $\pi : M \to X$ and an X-functor $f : Y \to X$.

For every $U \in \mathrm{Ob}(\mathscr{C})$ and $x \in X(U)$, write $M_x = U \times_x M$; thus, for every $\phi : U' \to U$,

$$M_x(U') = \{m \in M(U') \mid \pi(m) = x_{U'} = \phi^*(x)\}.$$

Finally, if $g \in G(U)$, we also write gx for the element $\lambda(g,x)$ of $X(U)$.

Definition 6.1. *(a) We say that M is a G-equivariant X-object if an action $\Lambda : G \times M \to M$ of G on M, lifting λ, has been given; that is, if the following square commutes.*

[54] Editor's note: The editors have not attempted to develop this remark.

[55] Editor's note: This section has been added.

$$\begin{array}{ccc} G\times M & \xrightarrow{\Lambda} & M \\ {\scriptstyle \mathrm{id}_G\times\pi}\big\downarrow & & \big\downarrow{\scriptstyle \pi} \\ G\times X & \xrightarrow{\lambda} & X \end{array}$$

Equivalently, for every morphism $(g,x): U \to G\times X$, one is given maps

$$\Lambda_x^U(g) : M_x(U) \longrightarrow M_{gx}(U), \qquad m \longmapsto g\cdot m,$$

which satisfy $1\cdot m = m$ and $g\cdot(h\cdot m) = (gh)\cdot m$, and which are functorial in the $(G\times X)$-object U. Equivalently again, one is given morphisms of U-objects

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}$$

satisfying $\Lambda_x(1) = \mathrm{id}$ and $\Lambda_{hx}(g)\circ\Lambda_x(h) = \Lambda_x(gh)$.

(b) Now let $\mathbf{O}$ be a $\widehat{\mathscr{C}}$-ring and put $\mathbf{O}_X = \mathbf{O}\times X$. Under the hypotheses of (a), we say that M is a G-equivariant $\mathbf{O}_X$-module if it is an $\mathbf{O}_X$-module (cf. Definition 4.3.3.1, which is valid for every ring functor on a category $\mathscr{C}$) and if the action Λ is compatible with the $\mathbf{O}_X$-module structure on M. Thus, for every $(g,x)\in G(U)\times X(U)$, $U\in\mathrm{Ob}(\mathscr{C})$, the map

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \qquad m \longmapsto g\cdot m,$$

is a morphism of $\mathbf{O}_U$-modules.

Remark 6.2. (i) In (a), the conditions $\Lambda_x(1) = \mathrm{id}$ and $\Lambda_{hx}(g)\circ\Lambda_x(h) = \Lambda_x(gh)$ plainly imply that each $\Lambda_x(g)$ is an isomorphism, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if each $\Lambda_x(g)$ is an isomorphism, applying the second condition with $h=1$ gives $\Lambda_x(1) = \mathrm{id}$.

(ii) Let M be an $\mathbf{O}_X$-module. Giving a morphism $\Lambda : G\times M\to M$ which makes the diagram of Definition 6.1 commute and for which every morphism

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \qquad m \longmapsto g\cdot m,$$

is an isomorphism of $\mathbf{O}_U$-modules is equivalent to giving an isomorphism of $\mathbf{O}_{G\times X}$-modules

$$\theta : G\times M = (G\times X)\times_{\mathrm{pr}_X} M \xrightarrow{\sim} (G\times X)\times_\lambda M, \qquad (g,x,m)\longmapsto (g,x,g\cdot m).$$

Since each $\Lambda_x(g)$ is assumed to be an isomorphism, the equality $\Lambda_x(1) = \mathrm{id}$ follows from $\Lambda_{hx}(g)\circ\Lambda_x(h) = \Lambda_x(gh)$ with $h=1$. Consequently, Λ is an action of G on M, that is, $g\cdot(h\cdot m) = (gh)\cdot m$, if and only if the following diagram of $(G\times G\times X)$-isomorphisms commutes. Here μ denotes multiplication in G, and $f^*(\theta)$ is obtained from θ by base change along $f : G\times G\times X\to G\times X$.

$$\begin{array}{ccc} (G\times G\times X)\underset{\mathrm{pr}_X\circ\mathrm{pr}_{2,3}}{\times} M & \xrightarrow[\sim]{\mathrm{pr}_{2,3}^*(\theta)} & (G\times G\times X)\underset{\lambda\circ\mathrm{pr}_{2,3}}{\times} M \\ \Big\| & & \Big\| \\ (G\times G\times X)\underset{\mathrm{pr}_X\circ(\mu\times\mathrm{id}_X)}{\times} M & & (G\times G\times X)\underset{\mathrm{pr}_X\circ(\mathrm{id}_G\times\lambda)}{\times} M \\ {\scriptstyle (\mu\times\mathrm{id}_X)^*(\theta)}\big\downarrow{\scriptstyle\sim} & & {\scriptstyle\sim}\big\downarrow{\scriptstyle (\mathrm{id}_G\times\lambda)^*(\theta)} \\ (G\times G\times X)\underset{\lambda\circ(\mu\times\mathrm{id}_X)}{\times} M & = & (G\times G\times X)\underset{\lambda\circ(\mathrm{id}_G\times\lambda)}{\times} M \end{array}.$$

Thus, giving a G-equivariant $\mathbf{O}_X$-module structure on M is equivalent to giving an isomorphism θ of $\mathbf{O}_{G\times X}$-modules as above for which this diagram commutes.

(iii) Everything above extends to the case in which G is merely a $\widehat{\mathscr{C}}$-monoid. In that case, giving an action $\Lambda : G \times M \to M$ lifting λ, such that every $\Lambda_x(g) : M_x \to M_{gx}$ is a morphism of $\mathbf{O}_U$-modules, is equivalent to giving a morphism θ of $\mathbf{O}_{G\times X}$-modules as in (ii), for which the diagram above commutes without the isomorphism signs under the arrows and which satisfies

$$p_M \circ \theta \circ (\varepsilon_G \times \mathrm{id}_M) = \mathrm{id}_M,$$

where ε_G is the unit section of G and p_M is the projection onto M.

6.3. G-equivariant morphisms

Let Y be a second object of $\widehat{\mathscr{C}}$, endowed with an action $\lambda_Y : G \times Y \to Y$, and let N be a second G-equivariant $\mathbf{O}_X$-module. A morphism $f : Y \to X$ in $\widehat{\mathscr{C}}$ (respectively, a morphism of $\mathbf{O}_X$-modules $\phi : M \to N$) is called *G-equivariant* if it commutes with the action of G. In set-theoretic notation, this says

$$f(g \cdot y) = g \cdot f(y) \qquad \text{(respectively, } \phi(g \cdot m) = g \cdot \phi(m)\text{)},$$

or, equivalently,

$$f \circ \lambda_Y = \lambda_X \circ (\mathrm{id}_G \times f), \qquad \phi \circ \Lambda_M = \Lambda_N \circ (\mathrm{id}_G \times \phi).$$

The following lemma is immediate.

Lemma 6.3.1. *Let $f : Y \to X$ be a G-equivariant morphism and let M be a G-equivariant $\mathbf{O}_X$-module. Then the inverse image $f^*(M) = Y \times_f M$ is a G-equivariant $\mathbf{O}_Y$-module.*

On the other hand, G acts on $\mathrm{Hom}_{\widehat{\mathscr{C}}}(Y, X)$. Indeed, let $T \in \mathrm{Ob}(\mathscr{C})$, $g \in G(T)$, and let ϕ be a T-morphism $Y_T \to X_T$. Then g defines automorphisms $\lambda_Y(g)$ and $\lambda_X(g)$ of Y_T and X_T, and we write $g \cdot \phi$, or $g\phi g^{-1}$, for the morphism

$$\lambda_X(g) \circ \phi \circ \lambda_Y(g^{-1}).$$

This defines an action of $G(T)$ on $\mathrm{Hom}_T(Y_T, X_T)$, functorially in T.

Definition 6.3.2. *If $\phi : Y \to X$ is an arbitrary morphism in $\widehat{\mathscr{C}}$, we may therefore consider its stabilizer* $\mathrm{Stab}_G(\phi)$ *(cf. 2.3.3.1). For every $T \in \mathrm{Ob}(\mathscr{C})$, $\mathrm{Stab}_G(\phi)(T)$ is the subgroup of $G(T)$ formed by those g for which $g \circ \phi_T = \phi_T \circ g$, that is, for which the following diagram commutes.*

$$\begin{array}{ccc} Y_T & \xrightarrow{\phi_T} & X_T \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g} \\ Y_T & \xrightarrow{\phi_T} & X_T \end{array}$$

Then $\phi : Y \to X$ is equivariant under $\mathrm{Stab}_G(\phi)$.

6.4. Global sections

Let M be a G-equivariant $\mathbf{O}_X$-module. Let S_0 denote the final object of $\mathscr{C}$, and denote by $\prod_{X/S_0} M$ the *functor of sections of M over X* (cf. Exposé II, 1.1): to every $T \in \mathrm{Ob}(\mathscr{C})$ it assigns

$$\mathrm{Hom}_X(X_T, M) = \mathrm{Hom}_{X_T}(X_T, M_T) = \Gamma(M_T/X_T).$$

Recall also that every morphism $g : Z \to Y$ of $\widehat{\mathscr{C}}$-objects over X induces a morphism of abelian groups
$$M(g) : M(Y) \longrightarrow M(Z),$$
compatible with the ring homomorphism $g^* : \mathbf{O}(Y) \to \mathbf{O}(Z)$. In particular, when $Z = Y$, so that g is an X-endomorphism of Y, one obtains a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Y)$$

which is not generally a morphism of $\mathbf{O}(Y)$-modules, but which satisfies

$$M(g)(\alpha \cdot m) = g^*(\alpha) \cdot M(g)(m)$$

for every $m \in M(Y)$ and $\alpha \in \mathbf{O}(Y)$. In what follows, we simply write g^* instead of $M(g)$.

Let $T \in \mathrm{Ob}(\mathscr{C})$, put $X_T = X \times T$, and let $\mathrm{pr}_X : X_T \to X$ be the projection. For $\alpha \in \mathbf{O}(X_T)$ and $g \in G(T)$, set
$$g(\alpha) = (g^{-1})^*(\alpha).$$
Set-theoretically, this element of $\mathbf{O}(X_T)$ is defined, for every $S \in \mathrm{Ob}(\mathscr{C})$, $x \in X(S)$, and $t \in T(S)$, by
$$g(\alpha)(x, t) = \alpha(g^{-1}x, t).$$
This gives a left action of $G(T)$ by ring automorphisms on $\mathbf{O}(X_T)$, functorially in T, and $g(\alpha) = \alpha$ when α is the image in $\mathbf{O}(X_T)$ of an element of $\mathbf{O}(T)$.

Let ϕ now be the identity morphism of X_T (cf. 6.3 and the generalization below to a morphism $\phi : Y \to X$). Denote $\mathrm{Hom}_X(X_T, M)$ by $M(\phi g^{-1})$ or $M(\phi)$ according as X_T is regarded as an X-object through $\mathrm{pr}_X \circ \lambda(g^{-1})_T$ or through pr_X, respectively.

Then $\lambda(g^{-1})_T$ is an X-morphism between these two X-objects. Hence there is a morphism of abelian groups

$$(g^{-1})^* : M(\phi) \longrightarrow M(\phi g^{-1}), \qquad m \longmapsto m \circ \lambda(g^{-1})_T,$$

and
$$(g^{-1})^*(\alpha \cdot m) = (g\alpha) \cdot (g^{-1})^*(m)$$
for $m \in M(\phi)$ and $\alpha \in \mathbf{O}(X_T)$. If m is a section of M_T over X_T, then $(g^{-1})^*(m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$. In particular, $(g^{-1})^*$ is a morphism of $\mathbf{O}(T)$-modules.

The functoriality of the $\mathbf{O}(X_T)$-module morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{tikzcd}
M(\phi) \arrow[r, "(g^{-1})^*"] \arrow[d, "\Lambda_\phi(g)"'] & M(\phi g^{-1}) \arrow[d, "\Lambda_{\phi g^{-1}}(g)"] \\
M(g\phi) \arrow[r, "(g^{-1})^*"] & M(g\phi g^{-1})
\end{tikzcd}$$

Moreover, $g\phi g^{-1} = \phi$, because ϕ is the identity. Writing

$$A(g) = (g^{-1})^* \circ \Lambda_\phi(g),$$

we obtain a morphism of abelian groups

$$A(g) : M(\phi) \longrightarrow M(\phi)$$

which is compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$, namely

$$A(g)(\alpha \cdot m) = (g\alpha) \cdot A(g)(m).$$

Finally, if h is a second element of $G(T)$, functoriality of M and of the morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{tikzcd}
M(\phi) \arrow[d, "\Lambda_\phi(g)"'] \arrow[dr, "\Lambda_\phi(hg)"] & & \\
M(g\phi) \arrow[r, "\Lambda_\phi(h)"] \arrow[d, "(g^{-1})^*"'] & M(hg\phi) \arrow[d, "(g^{-1})^*"'] \arrow[dr, "((hg)^{-1})^*"] & \\
M(\phi) \arrow[r, "\Lambda_\phi(h)"] & M(h\phi) \arrow[r, "(h^{-1})^*"] & M(\phi)
\end{tikzcd}$$

It follows that $A(h) \circ A(g) = A(hg)$, and hence:

Proposition 6.4.1. *For every T, the $\mathbf{O}(X_T)$-module*

$$\left(\prod_{X/S_0} M\right)(T) = \Gamma(M_T/X_T)$$

is endowed, functorially in T, with an action of $G(T)$ compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$; that is,

$$A(g)(\alpha \cdot m) = g(\alpha) \cdot A(g)(m).$$

Since $g(\alpha) = \alpha$ for $\alpha \in \mathbf{O}(T)$, it follows in particular that $\prod_{X/S_0} M$ is a G-$\mathbf{O}_{S_0}$-module.

More generally, let Y be a second G-equivariant $\widehat{\mathscr{C}}$-object, let $\phi : Y \to X$ be a morphism in $\widehat{\mathscr{C}}$, and put $H = \mathrm{Stab}_G(\phi)$, as in 6.3. Then

$$M_\phi = Y \times_\phi M$$

is an H-equivariant $\mathbf{O}_Y$-module. We obtain:

Corollary 6.4.2. *The functor $\prod_{Y/S_0} M_\phi$ is a $\mathrm{Stab}_G(\phi)$-$\mathbf{O}_{S_0}$-module.*

6.5. G-equivariant $\mathscr{O}_X$-modules

Apply the preceding discussion in the following situation. Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be an $\mathscr{O}_X$-module, not necessarily quasi-coherent.

Definition 6.5.1. *We say that $\mathcal{F}$ is a G-equivariant $\mathscr{O}_X$-module if the $\mathbf{O}_X$-module $M = W(\mathcal{F})$ is G-equivariant. Thus, for every morphism $(g, x) : U \to G \times_S X$, one is given isomorphisms of $\mathscr{O}_U$-modules*

$$\Lambda_x(g) : W\big(x^*(\mathcal{F})\big) \xrightarrow{\sim} W\big((gx)^*(\mathcal{F})\big),$$

functorial in U and satisfying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. Because W is fully faithful (cf. 4.6.1.1 and 4.6.2), these yield isomorphisms of $\mathscr{O}_U$-modules

$$\Lambda_x(g) : x^*(\mathcal{F}) \xrightarrow{\sim} (gx)^*(\mathcal{F}),$$

where gx denotes $\lambda \circ (g, x) : U \to X$. Applying this to the identity morphism of $G \times_S X$ gives an isomorphism of $\mathcal{O}_{G\times_S X}$-modules

$$(\star) \qquad \theta : \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

such that the following diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ commutes.

$$\begin{tikzcd}
\mathrm{pr}_{2,3}^* \circ \mathrm{pr}_X^*(\mathcal{F}) \arrow[r, "\mathrm{pr}_{2,3}^*(\theta)", "\sim"'] \arrow[d, equal] & \mathrm{pr}_{2,3}^* \circ \lambda^*(\mathcal{F}) \arrow[r, equal] & (\mathrm{id}_G \times \lambda)^* \circ \mathrm{pr}_X^*(\mathcal{F}) \arrow[d, "(\mathrm{id}_G \times\lambda)^*(\theta)", "\sim"'] \\
(\mu \times \mathrm{id}_X)^* \circ \mathrm{pr}_X^*(\mathcal{F}) \arrow[r, "(\mu\times\mathrm{id}_X)^*(\theta)", "\sim"'] & (\mu \times \mathrm{id}_X)^* \circ \lambda^*(\mathcal{F}) \arrow[r, equal] & (\mathrm{id}_G \times\lambda)^* \circ \lambda^*(\mathcal{F})
\end{tikzcd} .$$

If $\mathcal{E}$ is a second G-equivariant $\mathcal{O}_X$-module, a morphism $\phi : \mathcal{E} \to \mathcal{F}$ of $\mathcal{O}_X$-modules is called G-*equivariant* if $W(\phi) : W(\mathcal{E}) \to W(\mathcal{F})$ is G-equivariant. With the notation above, this is equivalent to

$$\theta_{\mathcal{F}} \circ \mathrm{pr}_X^*(\phi) = \lambda^*(\phi) \circ \theta_{\mathcal{E}}.$$

Remark 6.5.2. If $f : Y \to X$ is G-equivariant, Lemma 6.3.1 shows that $f^*(\mathcal{F})$ is a G-equivariant $\mathcal{O}_Y$-module.

Remark 6.5.3. Suppose in addition that $\mathcal{F}$ is quasi-coherent. Giving a G-equivariant $\mathcal{O}_X$-module structure on $M = W(\mathcal{F})$ is then equivalent to giving such a structure on $V(\mathcal{F})$, and this in turn is equivalent to giving an action of G on the vector fibration $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

Indeed, let

$$\varphi : \lambda^*(\mathcal{F}) \xrightarrow{\sim} \mathrm{pr}_X^*(\mathcal{F})$$

be the inverse of θ. For every morphism $(g, x) : U \to G \times_S X$, the isomorphism of $\mathcal{O}_U$-modules

$$\varphi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$$

from $(gx)^*(\mathcal{F})$ to $x^*(\mathcal{F})$ induces an isomorphism of $\mathcal{O}_U$-modules

$$V\big((gx)^*(\mathcal{F})\big) \xrightarrow{\sim} V\big(x^*(\mathcal{F})\big).$$

Denote the latter by ${}^t\varphi_x(g)$. Then

$${}^t\varphi_{gx}(h) \circ {}^t\varphi_x(g) = {}^t\big(\Lambda_{gx}(h) \circ \varphi_x(g)\big)^{-1} = \Lambda_x(hg)^{-1} = {}^t\varphi_x(hg).$$

For every X-scheme $f : Y \to X$, one has

$$V(\mathcal{F}) \times_f Y \simeq V\big(f^*(\mathcal{F})\big) \qquad \text{(cf. 4.6.2)}.$$

Therefore the isomorphism

$$\begin{aligned} {}^t\varphi : \quad G \times_S V(\mathcal{F}) &= (G \times_S X) \times_{\mathrm{pr}_X} V(\mathcal{F}) = V\big(\mathrm{pr}_X^*(\mathcal{F})\big) \\ &\xrightarrow{\sim} V\big(\lambda^*(\mathcal{F})\big) = (G \times_S X) \times_\lambda V(\mathcal{F}) \end{aligned}$$

endows $V(\mathcal{F})$ with a G-equivariant $\mathcal{O}_X$-module structure. Identifying every functor $V(-)$ with the vector fibration which represents it, and composing ${}^t\varphi$ with the projection onto $V(\mathcal{F})$, gives an action of G on $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

This recovers the definition in [GIT, Chapter 1, Section 3], except that Mumford there considers a locally free $\mathcal{O}_X$-module $\mathcal{E}$ of finite rank and defines an action of G on $V(\mathcal{E}) = W(\mathcal{E}^\vee)$. Indeed, the preceding diagram ($\star\star$), with θ replaced by φ and the directions of the arrows reversed, is exactly the diagram in *loc. cit.*, Definition 1.6, and the isomorphism ${}^t\varphi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. Consider in particular the case $X = S$, with the trivial action of G. A G-equivariant $\mathcal{O}_S$-module $\mathcal{F}$ is then the same thing as a G-$\mathcal{O}_S$-module (cf. 4.7.1). If $f : G \to S$ denotes the morphism which here equals both pr_X and λ, the isomorphism

$$(\star) \qquad \theta : f^*(\mathcal{F}) \xrightarrow{\sim} f^*(\mathcal{F})$$

is an element of

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{F}), f^*(\mathcal{F})) &= \mathrm{Hom}_{\mathcal{O}_G}(W(f^*(\mathcal{F})), W(f^*(\mathcal{F})))\\ &= \mathrm{End}_{\mathcal{O}_S}(W(\mathcal{F}))(G).\end{aligned}$$

It is precisely the morphism

$$\rho : G \longrightarrow \mathrm{End}_{\mathcal{O}_S}(W(\mathcal{F}))$$

which defines the action of G on $W(\mathcal{F})$. Moreover, θ corresponds by adjunction to the morphism of $\mathcal{O}_S$-modules

$$f_*(\theta) \circ \tau : \mathcal{F} \longrightarrow f_*f^*(\mathcal{F}),$$

where $\tau : \mathcal{F} \to f_*f^*(\mathcal{F})$ is the unit morphism of the adjunction. This will be used in Exposé $\mathrm{VI_B}$, 11.10 bis.

6.6. The functors $\prod_{X/S} W(\mathcal{F})$ and $W(p_*(\mathcal{F}))$

Let S be a scheme, let G be an S-group scheme acting on S-schemes X and Y through

$$\lambda : G \times_S X \to X, \qquad \mu : G \times_S Y \to Y,$$

and let $p : X \to Y$ be a G-equivariant morphism. Suppose that $p : X \to Y$ and $\nu_Y : Y \to S$ are quasi-compact and quasi-separated and that $\pi : G \to S$ is flat.

The projection $\mathrm{pr}_Y : G \times_S Y \to Y$ is then flat, as is μ, since μ is the composite of pr_Y and the automorphism $(g, y) \mapsto (g, gy)$. For a variable S-scheme $f : T \to S$, denote by $p_T : X_T \to T$ and $f_X : X_T \to X$ the morphisms obtained from p and f by base change.

Let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module, and let

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

be the isomorphism of 6.5.1 ($\star$). Since p is quasi-compact and quasi-separated, $p_*(\mathcal{F})$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$-module $p_*(\mathcal{F})$ is G-equivariant.*

Indeed, one has the following two cartesian squares.

$$\begin{array}{ccccc} G \times_S X & \xrightarrow{\mathrm{pr}_X} & X & \xleftarrow{\lambda} & G \times_S X \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ G \times_S Y & \xrightarrow{\mathrm{pr}_Y} & Y & \xleftarrow{\mu} & G \times_S Y \end{array} \quad .$$

Since p is quasi-compact and quasi-separated and pr_Y and μ are flat, EGA III, 1.4.15, completed by EGA IV_1, 1.7.21, gives

$$q_*\mathrm{pr}_X^*(\mathcal{F}) = \mathrm{pr}_Y^* p_*(\mathcal{F}), \qquad q_*\lambda^*(\mathcal{F}) = \mu^* p_*(\mathcal{F}).$$

Consequently, θ induces an isomorphism

$$\theta_Y : \mathrm{pr}_Y^* p_*(\mathcal{F}) \xrightarrow{\sim} \mu^* p_*(\mathcal{F}).$$

The same argument shows that θ_Y satisfies the cocycle condition 6.5.1 ($\star\star$), because the morphisms $G \times_S G \times_S Y \to G \times_S Y$ involved are flat. This proves the lemma.

In particular, take $Y = S$ with the trivial action of G. By Remark 6.5.4, $p_*(\mathcal{F})$ is then a G-$\mathcal{O}_S$-module. If, in addition, G is affine over S, and if

$$\mathscr{A}(G) = \pi_*(\mathcal{O}_G),$$

then $p_*(\mathcal{F})$ is an $\mathscr{A}(G)$-comodule, by 4.7.2.

On the other hand, by Proposition 6.4.1, the functor $\prod_{X/S} W(\mathcal{F})$, which assigns to every $f : T \to S$

$$W(f_X^*(\mathcal{F}))(X_T) = \Gamma(X_T, f_X^*(\mathcal{F})) = \Gamma(T, p_{T*} f_X^*(\mathcal{F})),$$

is a G-$\mathcal{O}_S$-module. There is moreover a canonical morphism

$$\tau : W(p_*(\mathcal{F})) \longrightarrow \prod_{X/S} W(\mathcal{F}),$$

which, for each $f : T \to S$, is given by the canonical morphism

$$\Gamma(T, f^* p_*(\mathcal{F})) \longrightarrow \Gamma(T, p_{T*} f_X^*(\mathcal{F})).$$

It is an isomorphism when restricted to the full subcategory of schemes T flat over S, and it is immediate that τ is a morphism of G-$\mathcal{O}_S$-modules. We have therefore proved the following proposition; for part (ii), compare [GIT], p. 32.

Proposition 6.6.2. *Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and $p : X \to S$ is quasi-compact and quasi-separated.*

(i) Then $p_*(\mathcal{F})$ is a quasi-coherent G-$\mathcal{O}_S$-module. The canonical morphism

$$W(p_*(\mathcal{F})) \longrightarrow \prod_{X/S} W(\mathcal{F})$$

is a morphism of G-$\mathcal{O}_S$-modules, and the two functors coincide on the category of flat S-schemes.

(ii) If, in addition, G is affine over S, and if $\mathscr{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ has the structure of an $\mathscr{A}(G)$-comodule.[56]

[56] Editor's note: The same holds if G is flat, quasi-compact, and quasi-separated over S, and if $p_*(\mathcal{F})$ is a flat $\mathcal{O}_S$-module; cf. Exposé $\mathrm{VI_B}$, 11.6.1(ii).

6.7. Stabilizers

Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Let Y be a second S-scheme with an action of G, possibly the trivial action, let $\phi : Y \to X$ be an S-morphism, not necessarily G-equivariant, and let $\mathrm{Stab}_G(\phi)$ be the stabilizer of ϕ in G (cf. 6.3.2).

Let us point out at once (see Exposé $\mathrm{VI_B}$, Section 6) that $\mathrm{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S (cf. *loc. cit.*, Definition 6.2.1). Indeed, consider

$$r : G \times_S Y \longrightarrow X \times_S X, \qquad r(g, y) = \big(\phi(y), g\phi(g^{-1}y)\big),$$

put $P = G \times_S Y$, and let P' be the inverse image under r of the diagonal $\Delta_{X/S}$. Then (cf. *loc. cit.*, 6.2.4(a))

$$\mathrm{Stab}_G(\phi) = \prod_{P/G} P'.$$

It follows that $\mathrm{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S. The latter condition is automatic if S is the spectrum of a field or if $Y = S$. Under these hypotheses, $\phi^*(\mathcal{F})$ is a quasi-coherent H-equivariant $\mathcal{O}_Y$-module (cf. 6.5.2).

If, in addition, $\pi : G \to S$ and $p : Y \to S$ are quasi-compact and quasi-separated over S, and π is flat, then $p_*\phi^*(\mathcal{F})$ is an H-$\mathcal{O}_S$-module by Proposition 6.6.2. In particular:

Corollary 6.7.1. *Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and $X \to S$ is quasi-compact and separated. Let $\tau : S \to X$ be a section of X over S.*

(i) The stabilizer $H = \mathrm{Stab}_G(\tau)$ is a closed subgroup scheme of G, and $\tau^*(\mathcal{F})$ is a quasi-coherent H-$\mathcal{O}_S$-module.

(ii) If H is moreover affine over S, for example if G is, and if $\mathscr{A}(H) = \pi_*(\mathcal{O}_H)$, then $\tau^*(\mathcal{F})$ is an $\mathscr{A}(H)$-comodule.

6.8. G-equivariant sheaves on G

To conclude, we give two results, 6.8.1 and 6.8.6, which will be used in Exposés II and III, in particular in III, 4.25.

Proposition 6.8.1. *Let S be a scheme, let $\pi : G \to S$ be an S-group scheme, and let $\varepsilon : S \to G$ be the unit section. Consider the action of G on itself by left translations. The functors*

$$\mathcal{E} \longmapsto \pi^*(\mathcal{E}), \qquad \mathcal{F} \longmapsto \varepsilon^*(\mathcal{F})$$

induce mutually quasi-inverse equivalences between the category of quasi-coherent $\mathcal{O}_S$-modules and the category of quasi-coherent G-equivariant $\mathcal{O}_G$-modules.

Proof. Let μ denote multiplication in G, and let $\mathrm{pr}_2 : G \times_S G \to G$ be the second projection. Since $\pi \circ \mu = \pi \circ \mathrm{pr}_2$, every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$ has a canonical isomorphism

$$\mu^*\pi^*(\mathcal{E}) = \mathrm{pr}_2^*\pi^*(\mathcal{E}).$$

It is immediate that this isomorphism satisfies the cocycle condition 6.5.1 $(\star\star)$, so $\pi^*(\mathcal{E})$ is a G-equivariant $\mathcal{O}_G$-module. Since $\varepsilon^*\pi^*(\mathcal{E}) = \mathcal{E}$, the functor $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ is

fully faithful. It remains to show that, for every G-equivariant $\mathcal{O}_G$-module $\mathcal{F}$,

$$\mathcal{F} \simeq \pi^*\varepsilon^*(\mathcal{F}).$$

By hypothesis, there is an isomorphism

$$\theta : \mathrm{pr}_2^*(\mathcal{F}) \xrightarrow{\sim} \mu^*(\mathcal{F}).$$

Pulling θ back along the morphism

$$\tau : G \longrightarrow G \times_S G, \qquad \tau = (\mathrm{id}_G, \varepsilon \circ \pi),$$

gives an isomorphism $\mathcal{F} \xrightarrow{\sim} \pi^*\varepsilon^*(\mathcal{F})$.

Remarks 6.8.2. (a) Consider the action of $G \times_S G$ on G defined by

$$(g_1, g_2) \cdot g = g_1 g g_2^{-1}.$$

The stabilizer of the unit section $\varepsilon : S \to G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if $\mathcal{F}$ is a quasi-coherent $(G \times_S G)$-equivariant $\mathcal{O}_G$-module, then $\mathcal{E} = \varepsilon^*(\mathcal{F})$ has the structure of an H-$\mathcal{O}_S$-module, by 6.5.2.

(b) One can show that $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ is an equivalence between the category of quasi-coherent $(G \times_S G)$-equivariant $\mathcal{O}_G$-modules and the category of quasi-coherent H-$\mathcal{O}_S$-modules. This is a special case of more general descent results (cf. Exposé IV, Section 2, and SGA 1, VIII); see, for example, [Th87], 1.2–1.3.

Remark 6.8.3. Retain the notation of 6.8.1. For every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, let $\pi_*^G\pi^*(\mathcal{E})$ be the $\mathcal{O}_S$-submodule of $\pi_*\pi^*(\mathcal{E})$ whose sections over an open subset V of S are the elements

$$\gamma \in \Gamma(\pi^{-1}(V), \pi^*(\mathcal{E}))$$

such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \to V$ and $g \in G(S')$. Then the natural morphism

$$\mathcal{E} \longrightarrow \pi_*^G\pi^*(\mathcal{E})$$

is an isomorphism. This is immediate if

$$\pi_*\pi^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_S} \pi_*(\mathcal{O}_G),$$

for example if $G \to S$ is affine, or if $G \to S$ is quasi-compact and quasi-separated and $\mathcal{E}$ is flat; the general case is verified without difficulty.

Remark 6.8.4. Let S be a scheme, let H be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Suppose H is flat over S, and let $W_P(\mathcal{F})$ denote the restriction of $W(\mathcal{F})$ to the full subcategory of flat S-schemes. By 6.5.1, endowing $\mathcal{F}$ with an H-equivariant module structure is equivalent to giving an isomorphism

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \longrightarrow \lambda^*(\mathcal{F})$$

of sheaves on $H \times_S X$ which satisfies the cocycle condition ($\star\star$). Hence, to endow $\mathcal{F}$ with an H-equivariant module structure, it suffices to endow $W_P(\mathcal{F})$ with such a structure.

Recollection 6.8.5. Let X be an S-scheme and Y a closed S-subscheme of X. Let $\mathcal{N}_{Y/X}$ denote the conormal sheaf of the immersion $i : Y \hookrightarrow X$ (cf. EGA IV$_4$, 16.1.2). If $S' \to S$ is flat and $i' : Y' \hookrightarrow X'$ is obtained from i by base change, then *loc. cit.*, 16.2.2(iii), gives

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} = \mathcal{N}_{Y'/X'}.$$

Proposition 6.8.6. *Let S be a scheme, let X be an S-group scheme, and let Y be an S-subgroup scheme of X. Suppose that Y is flat over S. Then the conormal sheaf $\mathcal{N}_{Y/X}$ is a $(Y \times_S Y)$-equivariant $\mathcal{O}_Y$-module.*

Indeed, $H = Y \times_S Y$ is flat over S. By Remark 6.8.4, it is therefore enough to endow $W_P(\mathcal{N}_{Y/X})$ with an H-equivariant module structure. Let S' be a flat S-scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and put

$$\mathcal{N}' = \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}.$$

By 6.8.5, $\mathcal{N}' = \mathcal{N}_{Y'/X'}$. Every $h \in H(S')$ induces an automorphism of Y', and hence, for every $y \in Y'(S')$, isomorphisms

$$\Gamma(S', y^*(\mathcal{N}')) \xrightarrow{\sim} \Gamma(S', y^*h^*(\mathcal{N}'))$$

which endow $W_P(\mathcal{N}_{Y/X})$ with an H-equivariant module structure (cf. 6.1).

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Exposé II
Tangent Bundles – Lie Algebras

by M. Demazure

We propose in this exposé to construct, in the theory of schemes, the analogues of the tangent bundles and Lie algebras of classical theory. It will, however, be useful not to restrict ourselves to schemes proper, but also to consider certain functors on the category of schemes that are not necessarily representable (for example, the functors $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Norm}}$, and so forth). As announced in the preceding exposé (cf. I 1.1), we shall identify a scheme with its associated functor.[0]

On the other hand, the constructions presented below go beyond the setting of scheme theory. They are equally valid, for example, in the theory of analytic spaces with nilpotent elements, apart from a few changes of detail.

Before beginning this construction, we must give a few general definitions that supplement those of I 1.7.

1. The functors $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$

We resume the notation of I 1.1. We identify the category $\mathscr{C}$ with a full subcategory of

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

(in particular, we suppress the underlinings that allowed us to distinguish graphically an object of $\widehat{\mathscr{C}}$ from an object of $\mathscr{C}$).[1]

Consider the following situation: four objects of $\widehat{\mathscr{C}}$, denoted by S, X, Y, Z, the first of which is in fact an object of $\mathscr{C}$, with X and Y over Z, and Z over S.

$$\begin{tikzcd} X \arrow[dr, "p_X"'] & & Y \arrow[dl, "p_Y"] \\ & Z \arrow[d] & \\ & S & \end{tikzcd} .$$

Figure 1: The objects X and Y over Z, and Z over S.

[0]Editor's note: version of October 14, 2024.

[1]Editor's note: in this edition these are rendered by bold letters $\mathbf{F}, \mathbf{G}, \mathbf{V}, \mathbf{W}$; cf. Exposé I. For functors such as $\underline{\mathrm{Norm}}$ and $\underline{\mathrm{Centr}}$ (cf. 5.2), however, the underlinings of the original have been retained, and one has been added to the functor $\underline{\mathrm{Lie}}$, in order to distinguish the functor $\underline{\mathrm{Lie}}(G/S)$ from the $\Gamma(S, \mathcal{O}_S)$-Lie algebra $\mathrm{Lie}(G/S) = \underline{\mathrm{Lie}}(G/S)(S)$, used, for example, in Exposé $\mathrm{VII_A}$.

Definition 1.1.

We define an object of $\widehat{\mathscr{C}}_{/S}$, denoted $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$, by

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}}(X_{S'}, Y_{S'}) = \mathrm{Hom}_Z(X \times_S S', Y), \tag{1}$$

for every object S' of $\mathscr{C}_{/S}$. One sees at once that $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ is simply the subobject of $\underline{\mathrm{Hom}}_S(X, Y)$ formed by the morphisms compatible with p_X and p_Y; that is, the kernel of the pair of morphisms

$$\underline{\mathrm{Hom}}_S(X, Y) \rightrightarrows \underline{\mathrm{Hom}}_S(X, Z),$$

the first defined by composition with p_Y, and the second being the constant morphism whose "image" is p_X.

As in I 1.7, for every object T of $\widehat{\mathscr{C}}$ over S, there is a natural bijection

$$\mathrm{Hom}_S(T, \underline{\mathrm{Hom}}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_Z(X \times_S T, Y). \tag{2}$$

Moreover, by I 1.7.1, if E, F are objects of $\widehat{\mathscr{C}}$ over Z, then

$$\mathrm{Hom}_Z\big(E, \underline{\mathrm{Hom}}_Z(F, Y)\big) \simeq \mathrm{Hom}_Z(E \times_Z F, Y) \simeq \mathrm{Hom}_Z\big(F, \underline{\mathrm{Hom}}_Z(E, Y)\big).$$

Applying this with $E = X$ and $F = Z \times_S T$, one obtains natural bijections, for every object T of $\widehat{\mathscr{C}}_{/S}$,

$$\begin{aligned} \mathrm{Hom}_S\big(T, \underline{\mathrm{Hom}}_{Z/S}(X, Y)\big) &\simeq \mathrm{Hom}_Z(X \times_S T, Y) \\ &\simeq \begin{cases} \mathrm{Hom}_Z\big(Z \times_S T, \underline{\mathrm{Hom}}_Z(X, Y)\big), \\ \mathrm{Hom}_Z\big(X, \underline{\mathrm{Hom}}_Z(Z \times_S T, Y)\big). \end{cases} \end{aligned} \tag{3}$$

These bijections are functorial in T, and hence give isomorphisms of S-functors.[2]

$$\begin{tikzcd} \underline{\mathrm{Hom}}_S(T, \underline{\mathrm{Hom}}_{Z/S}(X, Y)) \arrow[rr, "\sim"] \arrow[dr, "\simeq"'] & & \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(Z \times_S T, Y)) \\ & \underline{\mathrm{Hom}}_{Z/S}(X \times_S T, Y) \arrow[ur, "\simeq"'] & \end{tikzcd} \tag{4}$$

Figure 2: The functorial isomorphisms (4).

Let us point out two special cases of the definition. If $Z = S$, then

$$\underline{\mathrm{Hom}}_{S/S}(X, Y) = \underline{\mathrm{Hom}}_S(X, Y).$$

On the other hand, when $X = Z$, we set

$$\prod_{Z/S} Y = \underline{\mathrm{Hom}}_{Z/S}(Z, Y), \tag{5}$$

so that, by definition,

$$\Big(\prod_{Z/S} Y\Big)(S') = \mathrm{Hom}_Z(Z \times_S S', Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

[2]Editor's note: points (2), (3), and (4) have been expanded; in particular, (4) will be used in 3.11.

The functor $\prod_{Z/S} : \widehat{\mathscr{C}}_{/Z} \to \widehat{\mathscr{C}}_{/S}$ is right adjoint to the base-change functor from S to Z: for every S-functor U,

$$\mathrm{Hom}_S\left(U, \prod_{Z/S} Y\right) = \mathrm{Hom}_Z(U \times_S Z, Y).$$

If $\mathscr{C} = \mathbf{Sch}$ and Z is an S-scheme, the functor $\prod_{Z/S}$ is called the "Weil restriction of scalars." [3] One also has an isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y) \simeq \underline{\mathrm{Hom}}_{X/S}(X, Y \times_Z X) = \prod_{X/S} (Y \times_Z X), \tag{6}$$

which gives in particular, when $Z = S$, an isomorphism

$$\underline{\mathrm{Hom}}_S(X, Y) \simeq \prod_{X/S} Y_X. \tag{7}$$

Remark 1.2.

The functor $Y \mapsto \underline{\mathrm{Hom}}_{Z/S}(X, Y)$ commutes with products in the following sense: there is a functorial isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y \times_Z Y') \simeq \underline{\mathrm{Hom}}_{Z/S}(X, Y) \times_S \underline{\mathrm{Hom}}_{Z/S}(X, Y'). \tag{*}$$

It follows that if Y is a Z-group, respectively a Z-ring, and so on, then $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ is an S-group, respectively an S-ring, and so on.

Remark 1.3.

Let $\pi : M \to Y$ be a Y-functor in $\mathcal{O}_Y$-modules (cf. I 4.3.3.1), and set $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. Then $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ has a natural $\mathcal{O}_H$-module structure; more precisely, for every H' over H, the set

$$\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$$

has a natural $\mathcal{O}(X \times_S H')$-module structure. [4]

Indeed, let $m : M \times_Y M \to M$ and $\lambda : \mathcal{O}_Y \times_Y M \to M$ denote the morphisms defining the structures of a Y-abelian group and an $\mathcal{O}_Y$-module. Let H' be an S-scheme over $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$; equivalently, one has been given a Z-morphism $f : X \times_S H' \to Y$, which makes $X \times_S H'$ a Y-object. Then

$$\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$$

is the set of Z-morphisms $\phi : X \times_S H' \to M$ such that $\pi \circ \phi = f$, that is, the set of Y-morphisms $X \times_S H' \to M$.

If ϕ, ψ are two such morphisms, define $\phi + \psi$ to be the composite Y-morphism

$$X \times_S H' \xrightarrow{\phi \times \psi} M \times_Y M \xrightarrow{m} M.$$

This makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ an abelian group over $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. [5]

Similarly, if $a \in \mathcal{O}(X \times_S H')$, that is, if $a : X \times_S H' \to \mathcal{O}_S$ is an S-morphism, define $a\phi$ to be the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ is the Y-morphism from $X \times_S H'$ to $\mathcal{O}_Y \times_Y M \simeq \mathcal{O}_S \times_S M$ with components a and ϕ. This gives $\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$ an $\mathcal{O}(X \times_S H')$-module structure, functorial in the H-object H'.

[3] Editor's note: the preceding two sentences have been added.

[4] Editor's note: this remark has been added; it is useful for Corollary 3.11.1.

[5] Editor's note: if M is a Y-group, not necessarily abelian, and one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ a group over $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$.

2. The schemes $I_S(\mathcal{M})$

Definition 2.1.

Let S be a scheme and $\mathcal{M}$ a quasi-coherent $\mathcal{O}_S$-module. Denote by $\mathscr{D}_{\mathcal{O}_S}(\mathcal{M})$ the quasi-coherent $\mathcal{O}_S$-algebra $\mathcal{O}_S \oplus \mathcal{M}$, where $\mathcal{M}$ is regarded as an ideal of square zero. Denote by

$$I_S(\mathcal{M}) = \operatorname{Spec} \mathscr{D}_{\mathcal{O}_S}(\mathcal{M})$$

the corresponding S-scheme.[6] In particular, put

$$\mathscr{D}_{\mathcal{O}_S} = \mathscr{D}_{\mathcal{O}_S}(\mathcal{O}_S), \qquad I_S = I_S(\mathcal{O}_S),$$

and call these, respectively, the algebra of dual numbers over S and the scheme of dual numbers over S.

The assignment $\mathcal{M} \mapsto I_S(\mathcal{M})$ is a contravariant functor from quasi-coherent $\mathcal{O}_S$-modules to S-schemes. In particular, the morphisms $0 \to \mathcal{M}$ and $\mathcal{M} \to 0$ define, respectively, the structural morphism

$$\rho : I_S(\mathcal{M}) \longrightarrow I_S(0) = S$$

and a section $\varepsilon_{\mathcal{M}}$ of it, called the *zero section*.[7]

2.1.1.

Since $\mathcal{M} \mapsto I_S(\mathcal{M})$ is contravariant, every $a \in \operatorname{End}_{\mathcal{O}_S}(\mathcal{M})$ defines an S-endomorphism a^* of $I_S(\mathcal{M})$, and

$$1^* = \mathrm{id}, \qquad (ab)^* = b^* \circ a^*, \qquad 0^* = \varepsilon_{\mathcal{M}} \circ \rho, \qquad a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}.$$

Consequently, $I_S(\mathcal{M})$ carries a right action of the multiplicative monoid of $\operatorname{End}_{\mathcal{O}_S}(\mathcal{M})$, commuting with the S-morphisms $I_S(\mathcal{M}) \to I_S(\mathcal{M}')$ arising from morphisms $\mathcal{M}' \to \mathcal{M}$; in particular, the operators a^* preserve the zero section.[8]

For every $a \in \operatorname{End}_{\mathcal{O}_S}(\mathcal{M})$, every $f : S' \to S$, and every $m \in I_S(\mathcal{M})(S')$, write $m \cdot a = a^*(m)$. Then

$$m \cdot 1 = m, \qquad (m \cdot a) \cdot b = m \cdot (ab), \qquad m \cdot 0 = \varepsilon_{\mathcal{M}}(\rho(m)),$$

and if $m = \varepsilon_{\mathcal{M}}(f)$, then $m \cdot a = m$. [9]

Remark 2.1.2.

Formation of $I_S(\mathcal{M})$ commutes with base extension: there are canonical isomorphisms

$$I_S(\mathcal{M})_{S'} \simeq I_{S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

For simplicity write $I_{S'}(\mathcal{M}) = I_S(\mathcal{M})_{S'}$. More generally, if X is an S-functor, not necessarily representable, write $I_X(\mathcal{M}) = I_S(\mathcal{M}) \times_S X$.

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[6] Editor's note: the underlying space of $I_S(\mathcal{M})$ is the same as that of S.

[7] Editor's note: compare 3.1.1.

[8] Editor's note: what follows expands the original; the numbering 2.1.1–2.1.3 has also been added.

[9] Editor's note: later we shall be chiefly interested in $a \in \mathcal{O}(S)$, acting by homotheties on $\mathcal{M}$. For example, if $\mathcal{M}$ is free of rank r, then for every $S' \to S$, $\operatorname{Hom}_S(S', I_S(\mathcal{M}))$ identifies with the set of r-tuples $(\epsilon_1, \dots, \epsilon_r) \in \mathcal{O}(S')^r$ satisfying $\epsilon_i \epsilon_j = 0$ for all i, j, and $(\epsilon_1, \dots, \epsilon_r) \cdot a = (a\epsilon_1, \dots, a\epsilon_r)$ for every $a \in \mathcal{O}(S)$.

2.1.3.

By the preceding discussion, the multiplicative monoid of $\mathcal{O}(S')$ acts on the S'-scheme $I_{S'}(\mathcal{M})$, functorially in $\mathcal{M}$. In other words, the S-scheme $I_S(\mathcal{M})$ is an object with an operator-monoid structure $\mathcal{O}_S$, functorial in $\mathcal{M}$. Thus there is a morphism of S-schemes

$$\lambda : I_S(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_S(\mathcal{M}),$$

satisfying the evident conditions. For every S-functor X, base change gives a morphism of X-functors

$$\lambda_X : I_X(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_X(\mathcal{M}),$$

making $I_X(\mathcal{M})$ an object with operator monoid $\mathcal{O}(X)$. Every $a \in \mathcal{O}(X) = \mathrm{Hom}_S(X, \mathcal{O}_S)$ defines the X-endomorphism

$$a^* = \lambda_X \circ (\mathrm{id}_{I_X(\mathcal{M})} \times (a \circ \mathrm{pr}_X)).$$

Explicitly, if $x \in X(S')$ and $m \in I_S(\mathcal{M})(S') = I_{S'}(\mathcal{M})(S')$, then $a(x) = a \circ x \in \mathcal{O}(S')$ and

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in $\mathcal{M}$ and preserves the zero section $\varepsilon_{\mathcal{M}} : X \to I_X(\mathcal{M})$; thus $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$. [10]

The operation is also "functorial in X" in the following sense. If $\pi : Y \to X$ is a morphism of S-functors and $u : \mathcal{O}(X) \to \mathcal{O}(Y)$ the corresponding ring morphism, so that $u(a) = a \circ \pi$, then the following diagram is commutative.

$$\begin{tikzcd}
I_X(\mathcal{M}) \arrow[r, "a^*"] & I_X(\mathcal{M}) \\
I_Y(\mathcal{M}) \arrow[u, "\pi"] \arrow[r, "u(a)^*"] & I_Y(\mathcal{M}) \arrow[u, "\pi"']
\end{tikzcd} .$$

Figure 3: Functoriality of the operator action under $Y \to X$.

2.2.

Let $\mathcal{M}$ and $\mathcal{N}$ be two quasi-coherent $\mathcal{O}_S$-modules. The commutative diagram below induces the corresponding diagram of S-schemes on the following page.

$$\begin{tikzcd}
& \mathcal{M} \oplus \mathcal{N} \arrow[dl] \arrow[dr] & \\
\mathcal{M} \arrow[dr] & & \mathcal{N} \arrow[dl] \\
& 0 &
\end{tikzcd}$$

Figure 4: The direct-sum diagram of $\mathcal{O}_S$-modules.

[10] Editor's note: this paragraph has been added; it will be useful in 3.4.2 and 4.6.2.

$$\begin{tikzcd}
 & \mathrm{I_S}(\mathscr{M} \oplus \mathscr{N}) & \\
\mathrm{I_S}(\mathscr{M}) \arrow[ur] & & \mathrm{I_S}(\mathscr{N}) \arrow[ul] \\
 & \mathrm{S} \arrow[ul, "\varepsilon_{\mathscr{M}}"] \arrow[uu, "\varepsilon_{\mathscr{M}\oplus\mathscr{N}}"'] \arrow[ur, "\varepsilon_{\mathscr{N}}"'] &
\end{tikzcd} .$$

Figure 5: The induced commutative diagram of S-schemes.

Proposition 2.2.

For every S-scheme X, the diagram of functors over S obtained by applying $\underline{\mathrm{Hom}}_S(-, X)$ to the preceding diagram is cartesian.

$$\begin{tikzcd}
 & \underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I_S}(\mathscr{M} \oplus \mathscr{N}), \mathrm{X}) \arrow[dl] \arrow[dr] & \\
\underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I_S}(\mathscr{M}), \mathrm{X}) \arrow[dr] & & \underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I_S}(\mathscr{N}), \mathrm{X}) \arrow[dl] \\
 & \underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{S}, \mathrm{X}) = \mathrm{X} &
\end{tikzcd} .$$

Figure 6: The cartesian diagram of $\underline{\mathrm{Hom}}_S$-functors.

Indeed, for every $S' \to S$, one must verify that the diagram of sets obtained by evaluating the functors on S' is cartesian. Since formation of $I_S(\mathcal{P})$ commutes with base extension, it is enough to take $S' = S$, and hence to verify that the following diagram of sets is cartesian.

$$\begin{tikzcd}
 & \mathrm{X}(\mathrm{I_S}(\mathscr{M} \oplus \mathscr{N})) \arrow[dl] \arrow[dr] \arrow[dd, "\mathrm{X}(\varepsilon_{\mathscr{M}\oplus\mathscr{N}})"] & \\
\mathrm{X}(\mathrm{I_S}(\mathscr{M})) \arrow[dr, "\mathrm{X}(\varepsilon_{\mathscr{M}})"'] & & \mathrm{X}(\mathrm{I_S}(\mathscr{N})) \arrow[dl, "\mathrm{X}(\varepsilon_{\mathscr{N}})"] \\
 & \mathrm{X}(\mathrm{S}) &
\end{tikzcd} .$$

Figure 7: The cartesian diagram after evaluation on S.

If $x \in X(S)$, it follows from SGA 1, III 5.1 that $X(\varepsilon_{\mathcal{M}})^{-1}(x)$ is isomorphic, functorially in $\mathcal{M}$, to

$$\mathrm{Hom}_{\mathcal{O}_S}(x^*(\Omega^1_{X/S}), \mathcal{M}),$$

where $\Omega^1_{X/S}$ denotes the sheaf of relative differentials of X over S. This latter functor in $\mathcal{M}$ plainly carries a direct sum of $\mathcal{O}_S$-modules to the product of the corresponding sets, which proves the proposition.[11]

[11] Editor's note: see also the addition 0.1.8 in Exposé III.

Corollary 2.2.1.

Let X be an S-scheme and $\mathcal{M}$ a finite free $\mathcal{O}_S$-module. The functor $\underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)$, as a functor over X, is isomorphic to a finite product over X of copies of $\underline{\mathrm{Hom}}_S(I_S, X)$.

Nota 2.2.2.

It follows from the proof of the proposition that $\underline{\mathrm{Hom}}_S(I_S, X)$, as an X-functor, is isomorphic to $\mathbf{V}(\Omega^1_{X/S})$ (I 4.6.1), and is therefore represented by the vector fibration[12]

$$V(\Omega^1_{X/S}).$$

3. The tangent bundle, condition (E)

Throughout this section, unless otherwise stated, the letters $\mathcal{M}, \mathcal{M}', \mathcal{N}, \dots$ will denote finite free $\mathcal{O}_S$-modules (that is, modules isomorphic to finite direct sums of copies of $\mathcal{O}_S$).

We shall systematically use the identifications established in Exposé I. Thus, by a "functor over S" we shall mean indifferently either a functor equipped with a morphism to $S = h_S$, or a functor on the category of objects over S. We shall use "group functor over S," and similar expressions, in the same way.

Definition 3.1.

Let S be a scheme, let $\mathcal{M}$ be a finite free $\mathcal{O}_S$-module, and let X be a functor over S. The *tangent bundle of X over S, relative to the $\mathcal{O}_S$-module* $\mathcal{M}$, is the S-functor

$$T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X).$$

In particular, the *tangent bundle of X over S* is [13]

$$T_{X/S} = T_{X/S}(\mathcal{O}_S) = \underline{\mathrm{Hom}}_S(I_S, X).$$

The assignment $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$-modules to S-functors. In particular, the morphisms $\mathcal{M} \to 0$ and $0 \to \mathcal{M}$ define, respectively, an S-morphism

$$\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \longrightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of it, called the *zero section.* [14]

Remark 3.1.1.

[15] The projection $\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \to X$ is induced by the zero section $\varepsilon_{\mathcal{M}} : S \to I_S(\mathcal{M})$, whereas the zero section $\tau_0 : X \to T_{X/S}(\mathcal{M})$ is induced by the structural morphism $\rho : I_S(\mathcal{M}) \to S$. More explicitly, let $t \in T_{X/S}(\mathcal{M})(S')$ correspond to the S-morphism $f : S' \times_S I_S(\mathcal{M}) \to X$, and let $x \in X(S')$ correspond to $g : S' \to X$. Then

$$\pi(t) = f \circ (\mathrm{id}_{S'} \times \varepsilon_{\mathcal{M}}), \qquad \tau_0(x) = g \circ (\mathrm{id}_{S'} \times \rho).$$

It follows from 3.1 that $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$-modules to functors over X. In particular, $\mathcal{O}(S)$ is an operator monoid of the X-functor $T_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$.

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[12] Editor's note: cf. editor's note (33) of Exposé I.

[13] Editor's note: when $S = \mathrm{Spec}(k)$ and X is a k-scheme, one has $T_{X/S}(S') = \mathrm{Hom}_S(S' \otimes_k k[\epsilon], X) = X(S' \otimes_k k[\epsilon])$; this recovers one of the usual definitions of the tangent bundle.

[14] Editor's note: the original order "$0 \to \mathcal{M}$ and $\mathcal{M} \to 0$" has been corrected to "$\mathcal{M} \to 0$ and $0 \to \mathcal{M}$."

[15] Editor's note: paragraphs 3.1.1 and 3.1.2 have been added.

Scholium 3.1.2.

[15] The preceding discussion means, in particular, the following. For every S-morphism $\varphi : X' \to X$, put

$$\Sigma(X', \mathcal{M}) = \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})).$$

The multiplicative monoid $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ acts on $\Sigma(X', \mathcal{M})$; write the action as $(\lambda, x) \mapsto \lambda * x$. Then

$$\lambda * (\mu * x) = (\lambda\mu) * x, \qquad 1 * x = x, \qquad 0 * x = \tau_0 \circ \varphi.$$

There is likewise an action of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M} \oplus \mathcal{M})$ on $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$. [16]

Let $m : \mathcal{M} \oplus \mathcal{M} \to \mathcal{M}$ be addition, and let $\delta : \mathcal{M} \to \mathcal{M} \oplus \mathcal{M}$ be the diagonal. Write

$$m_{X'} : \Sigma(X', \mathcal{M} \oplus \mathcal{M}) o \Sigma(X', \mathcal{M}), \qquad \delta_{X'} : \Sigma(X', \mathcal{M}) o \Sigma(X', \mathcal{M} \oplus \mathcal{M})$$

for the induced maps. For $\lambda, \mu \in \mathcal{O}(S)$, let ℓ_λ denote multiplication by λ on $\mathcal{M}$, and let $\ell_{\lambda,\mu}$ denote multiplication by (λ, μ) on $\mathcal{M} \oplus \mathcal{M}$. Since

$$m \circ \ell_{\lambda,\lambda} = \ell_\lambda \circ m, \qquad m \circ \ell_{\lambda,\mu} \circ \delta = \ell_{\lambda+\mu},$$

we have, for every $z \in \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ and $x \in \Sigma(X', \mathcal{M})$,

$$\lambda * m(z) = m((\lambda, \lambda) * z), \qquad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x. \tag{$\dagger$}$$

Definition 3.2.

Let $u \in X(S) = \mathrm{Hom}_S(S, X) = \Gamma(X/S)$. The *tangent space of X over S at u, relative to* $\mathcal{M}$, denoted $L^u_{X/S}(\mathcal{M})$, is the S-functor obtained from the X-functor $T_{X/S}(\mathcal{M})$ by pullback along $u : S \to X$.

$$\begin{tikzcd}
\mathrm{L}^u_{\mathrm{X/S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{T}_{\mathrm{X/S}}(\mathscr{M}) \arrow[d, "\pi"] \\
\mathrm{S} \arrow[r, "u"] & \mathrm{X}
\end{tikzcd}$$

Figure 8: The cartesian square defining $L^u_{X/S}(\mathcal{M})$.

In particular, $L^u_{X/S}(\mathcal{O}_S)$ is denoted $L^u_{X/S}$ and is called the *tangent space of X over S at u*.

Remark 3.2.1.

[17] It follows from 3.1.1 that, for every $t : S' \to S$, $L^u_{X/S}(\mathcal{M})(S')$ is the set of S-morphisms $f : I_{S'}(\mathcal{M}) \to X$ such that $f \circ \varepsilon_{\mathcal{M}} = u \circ t$, where $\varepsilon_{\mathcal{M}} : S' \to I_{S'}(\mathcal{M})$ is the zero section.

[16] Editor's note: in fact, only the actions of $\mathcal{O}(S)$ (respectively $\mathcal{O}(S) \times \mathcal{O}(S)$) on $\Sigma(X', \mathcal{M})$ (respectively $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$) will be used; see below and the proof of 3.6.

[17] Editor's note: this remark has been added.

Proposition 3.3.

If X is represented by an S-scheme $\widetilde{X}$, then $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ are representable. In particular, $T_{X/S}$ and $L^u_{X/S}$ are represented by the vector fibrations

$$V(\Omega^1_{\widetilde{X}/S}) \qquad \text{and} \qquad V(u^*\Omega^1_{\widetilde{X}/S})$$

over $\widetilde{X}$ and S, respectively.

It plainly suffices to prove the assertion for $T_{X/S}(\mathcal{M})$, since the corresponding assertion for $L^u_{X/S}(\mathcal{M})$ follows by pullback. By Corollary 2.2.1, it even suffices to treat $T_{X/S}$, in which case the result is exactly Nota 2.2.2.

Remark 3.3.1.

The proposition gives a particularly simple description of the vector fibration representing $L^u_{X/S}$. The image of the section u of X over S is locally closed, [18] and hence is defined by a quasi-coherent ideal $\mathfrak{m}$ in the scheme induced on an open subscheme of X. The quotient $\mathfrak{m}/\mathfrak{m}^2$ may be regarded as a quasi-coherent module on S; it is this module that defines the vector fibration in question.

For example, let X be an algebraic scheme over a field k, and let u be a k-rational point of X. Let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_{X,u}$, and let $\mathfrak{t}$ be the k-vector space dual to $\mathfrak{m}/\mathfrak{m}^2$; this is the Zariski tangent space of $\mathcal{O}_{X,u}$ at u. In the notation of I 4.6.5.1,

$$L^u_{X/k} = V(\mathfrak{m}/\mathfrak{m}^2) = W(\mathfrak{t}).$$

Returning to the general situation, observe first that $L^u_{X/S}(\mathcal{M})$ is covariant from finite free $\mathcal{O}_S$-modules to functors over S. In particular, $\mathcal{O}(S)$ is a set of operators of the S-functor $L^u_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$. [19]

Proposition 3.4.

Formation of $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ commutes with base extension. For every S-scheme S', there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} T_{X/S}(\mathcal{M})_{S'},$$

$$L^{u'}_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} L^u_{X/S}(\mathcal{M})_{S'}, \qquad u' = u_{S'}.$$

This follows immediately from the compatibility of Hom with base extension.

Corollary 3.4.1.

The X-functor $T_{X/S}(\mathcal{M})$, respectively the S-functor $L^u_{X/S}(\mathcal{M})$, is naturally an object with operators $\mathcal{O}_X$, respectively $\mathcal{O}_S$, and these structures are functorial in $\mathcal{M}$.

For $L^u_{X/S}(\mathcal{M})$, this is immediate. For each S' over S, the ring $\mathcal{O}(S')$ acts on $\mathcal{M} \otimes \mathcal{O}_{S'}$, hence on

$$L^{u'}_{X_{S'}/S'}(\mathcal{M} \otimes \mathcal{O}_{S'}) = L^u_{X/S}(\mathcal{M})_{S'};$$

this action is functorial in S'.

[18] Editor's note: cf. EGA I, 5.3.11.
[19] Editor's note: cf. 3.1.2.

The case of $T_{X/S}(\mathcal{M})$ requires a little more detail. For an X-object $f : X' \to X$, put $T_{X/S}(\mathcal{M})_{X'} = T_{X/S}(\mathcal{M}) \times_X X'$. We must equip

$$T_{X/S}(\mathcal{M})_{X'}(X') = \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$$

with an action of $\mathcal{O}(X')$, functorially in X'. Let $X_{X'} = X \times_S X'$, and let f' be the section of $X_{X'}$ over X' defined by f. The following diagram provides the required identification.

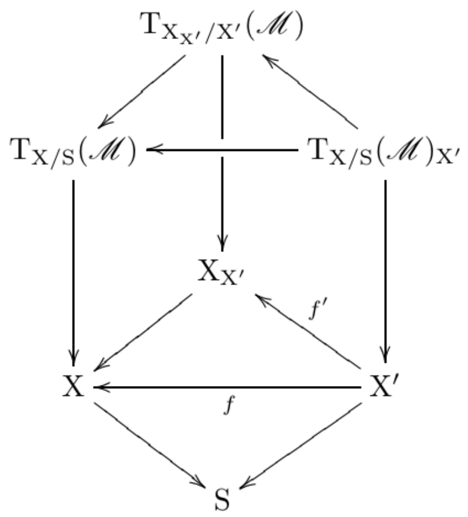

Figure 9: The base-change diagram used to define the $\mathcal{O}_X$-operator structure.

Together with 3.2.1, the diagram identifies $T_{X/S}(\mathcal{M})_{X'}(X')$ with

$$L^{f'}_{X_{X'}/X'}(\mathcal{M})(X') = \Big\{ \ X'\text{-morphisms } \psi : I_{X'}(\mathcal{M}) \to X_{X'} \mid \psi \circ \varepsilon_{\mathcal{M}} = f' \ \Big\}.$$

Every $a \in \mathcal{O}(X')$ acts through its action on $I_{X'}(\mathcal{M})$: in the notation of 2.1.1,

$$a\psi = \psi \circ a^*, \qquad (a\psi)(x) = \psi(x \cdot a)$$

for every $X'' \to X'$ and every $x \in I_{X'}(\mathcal{M})(X'')$. This construction is plainly functorial in X'. [20] The isomorphisms of Proposition 3.4 are therefore, by construction, isomorphisms for the corresponding $\mathcal{O}_{X_{S'}}$- and $\mathcal{O}_{S'}$-operator structures.

Remark 3.4.2.

[21] The action of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$ can be seen more directly as follows. For every $f : X' \to X$,

$$\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})) = \{\varphi \in \mathrm{Hom}_S(I_{X'}(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = f\}.$$

By 2.1.3, the S-functor $I_{X'}(\mathcal{M})$ carries an action of the monoid $\mathcal{O}(X')$ preserving the zero section. Thus, if a^* denotes the X'-endomorphism defined by $a \in \mathcal{O}(X')$, then

$$a\varphi = \varphi \circ a^*, \qquad (a\varphi)(m, x') = \varphi(m \cdot a(x'), x')$$

for every $S' \to S$ and every $(m, x') \in \mathrm{Hom}_S(S', I_S(\mathcal{M}) \times_S X')$. Notice that $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$, so that $(a\varphi) \circ \varepsilon_{\mathcal{M}} = f$.

Likewise, for $t : S' \to S$, the set $L^u_{X/S}(\mathcal{M})(S')$ consists of the S-morphisms $\varphi : I_{S'}(\mathcal{M}) \to X$ satisfying $\varphi \circ \varepsilon_{\mathcal{M}} = u \circ t$, and $a\varphi = \varphi \circ a^*$ for $a \in \mathcal{O}(S')$.

[20] Editor's note: the original has been expanded in what follows.

[21] Editor's note: Remarks 3.4.2 and 3.4.3 have been added.

Remark 3.4.3.

[21] The observations of 3.1.2 apply equally to the action of $\mathcal{O}_S$ on $L^u_{X/S}(\mathcal{M})$ and to that of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$.

Definition 3.5.

Let S be a scheme and X an S-functor. We say that X *satisfies condition (E) relative to S* if, for every S' over S and every pair of finite free $\mathcal{O}_{S'}$-modules $\mathcal{M}, \mathcal{N}$, the diagram of sets obtained by applying X to diagram $(*)$ of 2.1 is cartesian. [22]

$$\begin{tikzcd}
 & \mathrm{X}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M}\oplus\mathscr{N})) \arrow[dl] \arrow[dr] & \\
\mathrm{X}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[dr] & & \mathrm{X}(\mathrm{I}_{\mathrm{S}'}(\mathscr{N})) \arrow[dl] \\
 & \mathrm{X}(\mathrm{S}') & ,
\end{tikzcd}$$

Figure 10: The cartesian diagram in condition (E).

3.5.1.

[23] Equivalently, the functor $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ carries direct sums of finite free $\mathcal{O}_S$-modules to products of X-functors. In that case, the same is true of

$$\mathcal{M} \longmapsto L^u_{X/S}(\mathcal{M}) = S \times_X T_{X/S}(\mathcal{M})$$

for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that every representable functor satisfies condition (E). We shall sometimes abbreviate "X satisfies condition (E) relative to S" to "X/S satisfies condition (E)."

If X/S satisfies (E), then $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ commutes with products and hence carries groups to groups. In particular, $T_{X/S}(\mathcal{M})$ is a commutative X-group; for the same reason, $L^u_{X/S}(\mathcal{M})$ is a commutative S-group.

Proposition 3.6.

If X/S satisfies (E), then the abelian-group structure on $T_{X/S}(\mathcal{M})$, respectively on $L^u_{X/S}(\mathcal{M})$, together with the action of $\mathcal{O}_X$, respectively of $\mathcal{O}_S$, makes it an $\mathcal{O}_X$-module, respectively an $\mathcal{O}_S$-module.

The operator action is functorial in $\mathcal{M}$, and therefore respects the abelian-group structure obtained by functoriality from that of $\mathcal{M}$. [24] Indeed, retaining the notation of 3.1.2, the abelian X-group law on $T_{X/S}(\mathcal{M})$ is the composite

$$T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M}) \simeq T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \xrightarrow{m} T_{X/S}(\mathcal{M}),$$

whereas

$$T_{X/S}(\mathcal{M}) \xrightarrow{\delta} T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \simeq T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M})$$

[22] Editor's note: for examples of S-functors and S-groups that do not satisfy (E), see 6.2 and 6.3.
[23] Editor's note: what follows has been added.
[24] Editor's note: what follows has been expanded.

is the diagonal morphism.

By Remark 3.4.3 and the identities (†) of 3.1.2,

$$\lambda(x+y) = \lambda x + \lambda y, \qquad (\lambda+\mu)x = \lambda x + \mu x$$

for every $f : X' \to X$, every $x, y \in \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$, and every $\lambda, \mu \in \mathcal{O}(X')$.

Remark 3.6.1.

If X is representable, then it satisfies (E), and $T_{X/S}$ and $L^u_{X/S}$ are represented by vector fibrations. The preceding laws are then exactly those induced by the vector-fibration structures (cf. I 4.6). [25]

Proposition 3.4 bis.

If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E), and the isomorphisms of 3.4 respect the $\mathcal{O}_{X_{S'}}$-module and $\mathcal{O}_{S'}$-module structures, respectively.

No further comment is needed.

Proposition 3.7.

The functors $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ are functorial in X. Thus, if $f : X \to X'$ is an S-morphism, the following diagrams commute.

$$\begin{tikzcd} \mathrm{T}_{\mathrm{X/S}}(\mathscr{M}) \arrow[r, "\mathrm{T}(f)"] \arrow[d] & \mathrm{T}_{\mathrm{X'/S}}(\mathscr{M}) \arrow[d] \\ \mathrm{X} \arrow[r] & \mathrm{X'} \end{tikzcd} \quad , \quad \begin{tikzcd} \mathrm{L}^u_{\mathrm{X/S}}(\mathscr{M}) \arrow[rr, "\mathrm{L}(f)"] \arrow[dr] & & \mathrm{L}^{f\circ u}_{\mathrm{X'/S}} \arrow[dl] \\ & \mathrm{S} & \end{tikzcd}$$

Figure 11: Functoriality of tangent bundles and tangent spaces in X.

Moreover, if f is a monomorphism, then so are $T(f)$ and $L(f)$. [26] The existence of $T(f)$ and $L(f)$, and the last assertion, follow immediately from the definitions. Commutativity follows from functoriality in $\mathcal{M}$ and from $T_{X/S}(0) = X$.

Remark 3.7.1.

[27] Suppose that X and X' are representable, and let r be the rank of $\mathcal{M}$. By 2.2.2, $T_{X/S}(\mathcal{M})$ is isomorphic to the product over X of r copies of $V(\Omega^1_{X/S})$, and similarly for $T_{X'/S}(\mathcal{M})$. Hence the square in 3.7 is cartesian when f is an open immersion, or more generally when $f^*\Omega^1_{X'/S} = \Omega^1_{X/S}$, for example when f is étale. Under these conditions there is an isomorphism of S-functors

$$L^u_{X/S}(\mathcal{M}) \xrightarrow{\sim} L^{f\circ u}_{X'/S}(\mathcal{M}).$$

[25] Editor's note: explicitly, for every S-morphism $f : X' \to X$, the action of $\mathcal{O}(X')$ on $\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$ corresponds, under the identification

$$\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})) \simeq \mathrm{Hom}_{\mathcal{O}_{X'}}(f^*\Omega^1_{X/S}, \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{X'}),$$

to the natural action on the right. This follows from the proof of SGA 1, III 5.1; see also the addition 0.1.8 in Exposé III.

[26] Editor's note: the preceding sentence has been added.

[27] Editor's note: the hypothesis that X and X' be representable has been added, and what follows has been expanded.

In general, the square in 3.7 defines the following morphism of X-functors.

$$\begin{tikzcd} \mathrm{T}_{\mathrm{X}/\mathrm{S}}(\mathscr{M}) \arrow[rr] \arrow[dr] & & \mathrm{T}_{\mathrm{X}'/\mathrm{S}}(\mathscr{M}) \times_{\mathrm{X}'} \mathrm{X} \arrow[dl] \\ & \mathrm{X} & \end{tikzcd}.$$

Figure 12: The induced morphism over X in the general case.

Proposition 3.7 bis.

Let $f : X \to X'$ be an S-morphism. If X and X' satisfy (E) relative to S, then

$$T_{X/S}(\mathcal{M}) \xrightarrow{T(f)} T_{X'/S}(\mathcal{M})_X, \qquad \text{respectively} \qquad L^u_{X/S}(\mathcal{M}) \xrightarrow{L(f)} L^{f\circ u}_{X'/S}(\mathcal{M}),$$

is a morphism of $\mathcal{O}_X$-modules, respectively of $\mathcal{O}_S$-modules.

This follows from Proposition 3.7 by functoriality in $\mathcal{M}$.

Proposition 3.8.

For two functors X, Y over S, there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X/S}(\mathcal{M}) \times_S T_{Y/S}(\mathcal{M}) \xrightarrow{\sim} T_{(X\times_S Y)/S}(\mathcal{M}), \tag{3.8.1}$$

$$L^u_{X/S}(\mathcal{M}) \times_S L^v_{Y/S}(\mathcal{M}) \xrightarrow{\sim} L^{(u,v)}_{(X\times_S Y)/S}(\mathcal{M}). \tag{3.8.2}$$

The first isomorphism follows from 1.2 $(*)$; [28] the second follows from it by base change along $(u, v) : S \to X \times_S Y$.

Remark 3.8.0.

The isomorphism (3.8.1) may also be read as an isomorphism of $X \times_S Y$-functors

$$(T_{X/S}(\mathcal{M}) \times_X (X \times_S Y)) \underset{X\times_S Y}{\times} (T_{Y/S}(\mathcal{M}) \times_Y (X \times_S Y)) \xrightarrow{\sim} T_{(X\times_S Y)/S}(\mathcal{M}).$$

Corollary 3.8.1.

If X/S carries an algebraic structure defined by finite cartesian products, then $T_{X/S}(\mathcal{M})$ carries a structure of the same kind, and the projection $T_{X/S}(\mathcal{M}) \to X$ is a morphism for that structure.

Proposition 3.8 bis.

If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E), and (3.8.1), respectively (3.8.2), is an isomorphism of $\mathcal{O}_{X\times_S Y}$-modules, respectively of $\mathcal{O}_S$-modules.

Proof. [29] By 3.5.1 and (3.8.1), the product $(X \times_S Y)/S$ satisfies (E). Let us prove the assertion about (3.8.1). Given an S-morphism $(x, y) : Z \to X \times_S Y$, Remark 3.4.2 reduces the claim to the $\mathcal{O}(Z)$-linearity of the map

$$\{\varphi \in \mathrm{Hom}_S(I_Z(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = x\} \times \{\psi \in \mathrm{Hom}_S(I_Z(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = y\}$$
$$\longrightarrow \{\theta \in \mathrm{Hom}_S(I_Z(\mathcal{M}), X \times_S Y) \mid \theta \circ \varepsilon_{\mathcal{M}} = (x, y)\},$$

[28] Editor's note: what follows has been expanded.
[29] Editor's note: this proof has been added.

which sends (φ, ψ) to $\varphi \times \psi$. This is clear, since for $a \in \mathcal{O}(Z)$,

$$(\varphi \circ a^*) \times (\psi \circ a^*) = (\varphi \times \psi) \circ a^* = a \cdot (\varphi \times \psi).$$

The assertion about (3.8.2) follows similarly from 3.2.1.

3.9.0.

[30] If X is an S-group with unit section $e : S \to X$, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L^e_{X/S}(\mathcal{M}).$$

Thus $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is defined by the following cartesian square.

$$\begin{tikzcd}
\underline{\mathrm{Lie}}(\mathrm{X/S}, \mathscr{M}) \arrow[r, "i"] \arrow[d] & \mathrm{T}_{\mathrm{X/S}}(\mathscr{M}) \arrow[d, "p"] \\
\mathrm{S} \arrow[r, "e"] & \mathrm{X}.
\end{tikzcd}$$

Figure 13: The cartesian square defining $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$.

By Corollary 3.8.1, the projection $p : T_{X/S}(\mathcal{M}) \to X$ is a morphism of S-groups. Hence $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is an S-group, identified by i with the kernel of p.

If X/S also satisfies (E), Proposition 3.9 will show that this group structure, induced by that of X, agrees with the abelian-group structure induced by functoriality in $\mathcal{M}$ (cf. 3.5.1). In fact, the result holds under the weaker assumption that X be an S-functor in monoids, or more generally an S-functor in H-sets, as defined next.

Definitions 3.9.0.1.

(a) An *H-set* is a set X equipped with a composition law having a two-sided identity, denoted e_X or simply e. [31] If $f : X \to Y$ is a morphism of H-sets, its kernel is $\ker f = f^{-1}(e_Y)$, a sub-H-set of X.

(b) An *H-object* in a category $\mathscr{C}$ is an object X equipped with a morphism $X \times X \to X$ and a section of X over the final object that is a two-sided identity. Every $\mathscr{C}$-monoid, and in particular every $\mathscr{C}$-group, is a $\mathscr{C}$-H-object. An H-object in the category of functors over S will be called an *S-H-functor*.

(c) If X is an S-H-functor (for example, an S-group), and $e : S \to X$ is its identity section, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L^e_{X/S}(\mathcal{M}), \qquad \underline{\mathrm{Lie}}(X/S) = \underline{\mathrm{Lie}}(X/S, \mathcal{O}_S).$$

We now spell out the following special case of 3.8.1. [32]

[30] Editor's note: this paragraph has been added to explain the introduction of the notion of an S-H-functor.
[31] Editor's note: this terminology is inspired by the notion of an H-space in topology.
[32] Editor's note: Corollary 3.9.0.2 has been added; it will be used in 4.1.

Corollary 3.9.0.2.

If X is an S-H-functor, respectively an S-group, then $T_{X/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(X/S,\mathcal{M})$ are likewise S-H-functors, respectively S-groups, and there are morphisms of S-H-functors, respectively of S-groups, as follows.

$$\underline{\mathrm{Lie}}(\mathrm{X}/\mathrm{S},\mathscr{M}) \xrightarrow{i} \mathrm{T}_{\mathrm{X}/\mathrm{S}}(\mathscr{M}) \underset{s}{\overset{p}{\rightleftarrows}} \mathrm{X} \quad ,$$

Figure 14: The inclusion, projection, and zero section for the tangent functor.

Here i identifies $\underline{\mathrm{Lie}}(X/S,\mathcal{M})$ with $\ker p$, and s is a section of p.

Proposition 3.9.

Let X be an S-H-functor satisfying (E) relative to S. The S-H-functor structure on $\underline{\mathrm{Lie}}(X/S,\mathcal{M})$ induced by that of X agrees with the S-group structure defined in 3.5.1.

The preceding discussion shows that $\underline{\mathrm{Lie}}(X/S,\mathcal{M})$ is an H-object in the category of $\mathcal{O}_S$-modules. The proposition follows from the next lemma. [33]

Lemma 3.10.

Let $\mathscr{C}$ be a category. Let G be an H-object in the category of $\mathscr{C}$-H-objects. Thus G is a $\mathscr{C}$-H-object, whose composition law we denote $f : G \times G \to G$, together with a morphism of $\mathscr{C}$-H-objects $h : G \times G \to G$. [34] Then $f = h$, and f is commutative.

Evaluating the functors on a variable argument reduces the proof, as usual, to the case in which $\mathscr{C}$ is the category of sets. We then have a set G and two maps $f, h : G \times G \to G$ satisfying

$$h(f(x,y), f(z,t)) = f(h(x,z), h(y,t)).$$

Let $e, u \in G$ be the respective identities for f and h, so that

$$f(e,x) = f(x,e) = x, \qquad h(u,x) = h(x,u) = x.$$

First,

$$h(f(u,y), f(x,u)) = f(x,y) = h(f(x,u), f(u,y)).$$

Putting $y = e$, respectively $x = e$, gives

$$\begin{aligned} x = f(x,e) = h(f(u,e), f(x,u)) = h(u, f(x,u)) = f(x,u), \\ y = f(e,y) = h(f(e,u), f(u,y)) = h(u, f(u,y)) = f(u,y). \end{aligned}$$

Substituting these identities into the original relation yields

$$h(y,x) = f(x,y) = h(x,y).$$

This proves the lemma and Proposition 3.9. The following corollaries now follow from 3.9.

Corollary 3.9.1.

If X is an S-H-functor satisfying (E) relative to S, every element of $X(I_S(\mathcal{M}))$ that maps to the identity element of $X(S)$ is invertible.

[33] Editor's note: this is inspired by the standard proof that the fundamental group of an H-space is abelian.

[34] Editor's note: $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \to G \times G$, $((x,z),(y,t)) \mapsto (f(x,y), f(z,t))$.

Corollary 3.9.2.

If X is an S-monoid satisfying (E) relative to S, an element of $X(I_S(\mathcal{M}))$ is invertible if and only if its image in $X(S)$ is invertible.

Corollary 3.9.3.

If X is an S-group satisfying (E) relative to S, the two S-group laws on $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ coincide.

Corollary 3.9.4.

[35] Let G be an S-group satisfying (E) relative to S. For $n \in \mathbb{Z}$, let $n_G : G \to G$ be the morphism of S-functors defined by $g \mapsto g^n$. Then the derived morphism

$$L(n_G) : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S)$$

is multiplication by n; that is, it sends $x \in \underline{\mathrm{Lie}}(G/S)(S')$ to nx.

The map n_G is not generally a group homomorphism, but it preserves the identity section $e : S \to G$, so its derivative indeed carries $\underline{\mathrm{Lie}}(G/S) = L^e_{G/S}$ to itself. If i denotes the inclusion $\underline{\mathrm{Lie}}(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is characterized by

$$i(L(n_G)(x)) = i(x)^n.$$

By 3.9, the group laws on $\underline{\mathrm{Lie}}(G/S)$ coming from (E) and from the law on G agree. Hence $i(x)^n = i(nx)$, and therefore $L(n_G)(x) = nx$.

Before drawing further consequences from Proposition 3.9, we prove one more functoriality result.

Proposition 3.11.

In the situation of Section 1, there is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Indeed, by definition,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X,Y)).$$

Applying the isomorphism (4) of 1.1 with $T = I_S(\mathcal{M})$ gives

$$\underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X,Y)) \\ \simeq \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(Z \times_S I_S(\mathcal{M}), Y)).$$

Since $Z \times_S I_S(\mathcal{M}) \simeq I_Z(\mathcal{M})$, this is

$$\underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(I_Z(\mathcal{M}), Y)) = \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

[35] Editor's note: this corollary has been added; it amplifies Remark 4.1.1.2 below.

Corollary 3.11.1.

If Y/Z satisfies (E), then $\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S$ satisfies (E), and the isomorphism of 3.11 respects the module structures over $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$. [36]

Proof. Let $\mathcal{M},\mathcal{N}$ be finite free $\mathcal{O}_S$-modules. If Y/Z satisfies (E), then

$$T_{Y/Z}(\mathcal{M}\oplus\mathcal{N}) \simeq T_{Y/Z}(\mathcal{M}) \times_Y T_{Y/Z}(\mathcal{N}).$$

The right-hand side is a subfunctor of $T_{Y/Z}(\mathcal{M}) \times_S T_{Y/Z}(\mathcal{N})$, and 1.2 $(*)$ therefore gives

$$\begin{aligned}&\underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}\oplus\mathcal{N}))\\ &\qquad\simeq \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \underset{\underline{\mathrm{Hom}}_{Z/S}(X,Y)}{\times} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{N})).\end{aligned}$$

Together with 3.11, this says that $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$ satisfies (E), proving the first assertion.

For the second, put $H = \underline{\mathrm{Hom}}_{Z/S}(X,Y)$, and let $\Delta : H' \to H$ be an S-morphism. Equivalently, let $\delta : H' \times_S X \to Y$ be the corresponding Z-morphism; it makes $H' \times_S X$ a Y-object. On the one hand, we have the following commutative diagram.

$$\begin{tikzcd}
\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathscr{M}))) \arrow[r, hook] \arrow[d, equal] & \mathrm{Hom}_S(H', \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathscr{M}))) \arrow[d, equal] \\
\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathscr{M})) \arrow[r, hook] \arrow[d, equal] & \mathrm{Hom}_Z(H' \times_S X, T_{Y/Z}(\mathscr{M})) \arrow[d, equal] \\
\{\psi \in \mathrm{Hom}_Z(I_{H'\times_S X}(\mathscr{M}), Y) \mid \psi\circ\varepsilon_{\mathscr{M}} = \delta\} \arrow[r, hook] & \mathrm{Hom}_Z(I_{H'\times_S X}(\mathscr{M}), Y) .
\end{tikzcd}$$

Figure 15: The first comparison diagram in the proof of Corollary 3.11.1.

By 1.3, the action of $\alpha \in \mathcal{O}(H' \times_S X)$ on $\Psi \in \mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M}))$ is given, for every $U \to S$ and every $(h,x) \in \mathrm{Hom}_S(U, H' \times_S X)$, by

$$(\alpha\Psi)(h,x) = \alpha(h,x)\Psi(h,x),$$

where U is viewed over Y through $\delta\circ(h,x)$, and $\alpha(h,x) \in \mathcal{O}(U)$ acts through the $\mathcal{O}_Y$-module structure on $T_{Y/Z}(\mathcal{M})$. By 3.4.2, under the identification

$$\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M})) = \{\psi \in \mathrm{Hom}_Z(I_{H'\times_S X}(\mathcal{M}), Y) \mid \psi\circ\varepsilon_{\mathcal{M}} = \delta\},$$

this action is

$$(\alpha\psi)(m,h,x) = \psi(m\cdot\alpha(h,x), h, x) \tag{1}$$

for every $(m,h,x) \in \mathrm{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$.

On the other hand, consider $T_{H/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), H)$. We have a second commutative diagram.

[36] Editor's note: writing $H = \underline{\mathrm{Hom}}_{Z/S}(X,Y)$, this implies in particular that $T_{H/S}(\mathcal{M})$ is naturally a module over $\prod_{X\times_S H/H} \mathcal{O}_H$. This result appeared somewhat surprising to the editors, so the proof has been given in detail.

$$\begin{tikzcd}
\mathrm{Hom}_{\mathrm{H}}(\mathrm{H}', \mathrm{T}_{\mathrm{H}/\mathrm{S}}(\mathscr{M})) \arrow[r, hook] \arrow[d, equal] & \mathrm{Hom}_{\mathrm{S}}(\mathrm{H}', \mathrm{T}_{\mathrm{H}/\mathrm{S}}(\mathscr{M})) \arrow[d, equal] \\
\{\Phi \in \mathrm{Hom}_{\mathrm{S}}(\mathrm{I}_{\mathrm{H}'}(\mathscr{M}), \mathrm{H}) \mid \Phi \circ \varepsilon_{\mathscr{M}} = \Delta\} \arrow[r, hook] \arrow[d, equal, "(*)"'] & \mathrm{Hom}_{\mathrm{S}}(\mathrm{I}_{\mathrm{H}'}(\mathscr{M}), \mathrm{H}) \arrow[d, equal] \\
\{\phi \in \mathrm{Hom}_{\mathrm{Z}}(\mathrm{I}_{\mathrm{H}' \times_{\mathrm{S}} \mathrm{X}}(\mathscr{M}), \mathrm{Y}) \mid \phi \circ \varepsilon_{\mathscr{M}} = \delta\} \arrow[r, hook] & \mathrm{Hom}_{\mathrm{Z}}(\mathrm{I}_{\mathrm{H}' \times_{\mathrm{S}} \mathrm{X}}(\mathscr{M}), \mathrm{Y}),
\end{tikzcd}$$

Figure 16: The second comparison diagram in the proof of Corollary 3.11.1.

The bijection $(*)$ in the diagram is described as follows (cf. 1.1(2) and I 1.7.1). For every $U \to S$ and $(m, h, x) \in \mathrm{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$, where U lies over Z through $U \xrightarrow{x} X \to Z$, one has $\Phi(m, h) \in \mathrm{Hom}_Z(X \times_S U, Y)$ and

$$\varphi(m, h, x) = \Phi(m, h) \circ (x \times \mathrm{id}_U) \in \mathrm{Hom}_Z(U, Y). \tag{†}$$

By 3.4.2, with X replaced by H and X' by H', the action of $a \in \mathcal{O}(H')$ on $\Phi \in \mathrm{Hom}_S(I_{H'}(\mathcal{M}), H)$ is

$$(a\Phi)(m, h) = \Phi(m \cdot a(h), h).$$

Consequently, if φ, respectively $a\varphi$, corresponds to Φ, respectively $a\Phi$, then (†) gives

$$(a\varphi)(m, h, x) = \Phi(m \cdot a(h), h) \circ (x \times \mathrm{id}_U) = \varphi(m \cdot a(h), h, x). \tag{2}$$

Together with (1), this proves that the isomorphism

$$T_{H/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))$$

of 3.11.1 is an isomorphism of $\mathcal{O}(H)$-modules. Moreover, for every $H' \to H$, the $\mathcal{O}(H')$-module structure on $\mathrm{Hom}_H(H', T_{H/S}(\mathcal{M}))$ extends, functorially in H', to an $\mathcal{O}(H' \times_S X)$-module structure.

Taking $Z = S$ yields the next corollary.

Corollary 3.11.2.

There is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_S(X, T_{Y/S}(\mathcal{M})).$$

If Y/S satisfies (E), then $\underline{\mathrm{Hom}}_S(X, Y)/S$ satisfies (E), and this isomorphism respects the module structures over $\underline{\mathrm{Hom}}_S(X, Y)$.

Let $u : X \to Y$ be an S-morphism, identified with the constant morphism $u : S \to \underline{\mathrm{Hom}}_S(X, Y)$ such that $u(f) = u$ for every $f : S' \to S$. The fiber product of u with

$$\underline{\mathrm{Hom}}_S(X, T_{Y/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{Hom}}_S(X, Y)$$

identifies with $\underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M}))$, where X lies over Y through u. Hence the definition of the tangent space and Corollary 3.11.2 give the following result. [37]

[37] Editor's note: the preceding sentences have been added.

Corollary 3.11.3.

For an S-morphism $u : X \to Y$, there is an isomorphism, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})),$$

where on the right X lies over Y through u. If Y/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$-modules. [38]

In particular, for $Y = X$, the functor $\underline{\mathrm{End}}_S(X)$ is an S-functor in monoids, hence an S-H-functor. Recall that $\underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M})$ denotes $L^e_{\underline{\mathrm{End}}_S(X)/S}(\mathcal{M})$, with e the identity section.

Corollary 3.11.4.

There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}).$$

If X/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$-modules. [38]

Remark 3.11.5.

[39] Suppose that X/S satisfies (E). Then

$$\prod_{X/S} T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))$$

is a module over $\prod_{X/S} \mathcal{O}_X$. Explicitly, for every $S' \to S$,

$$\begin{aligned}&\underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))(S') \\ &\qquad = \{\psi \in \mathrm{Hom}_X(I_{S'}(\mathcal{M}) \times_S X, X) \mid \psi \circ (\varepsilon_{\mathcal{M}} \times \mathrm{id}_X) = \mathrm{pr}_X\}\end{aligned}$$

is functorially an $\mathcal{O}(X \times_S S')$-module. This follows either from 3.6 and the properties of $\prod_{X/S}$ (cf. 1.2), or from the proof of 3.11.1.

We now give a geometric interpretation of the tangent bundle. Let U be an S-functor. [40] By I 1.7.2, there are isomorphisms, functorial in $\mathcal{M}$,

$$\begin{aligned} T_{X/S}(\mathcal{M})(U) &= \mathrm{Hom}_S(U, \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)) \\ &\simeq \mathrm{Hom}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_S(U, X)) \\ &\simeq \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}). \end{aligned}$$

In particular, the morphism $\mathcal{M} \to 0$ gives the following commutative diagram.

$$\begin{tikzcd} \mathrm{Hom}_S(U, T_{X/S}(\mathscr{M})) \arrow[r, "\sim"] \arrow[d, "\pi_{\mathscr{M}}"'] & \mathrm{Hom}_{I_S(\mathscr{M})}(U_{I_S(\mathscr{M})}, X_{I_S(\mathscr{M})}) \arrow[d] \\ \mathrm{Hom}_S(U, X) \arrow[r, "\text{identité}"] & \mathrm{Hom}_S(U, X) \end{tikzcd} ,$$

Figure 17: The geometric interpretation of the projection $\pi_{\mathcal{M}}$.

[38] Editor's note: the preceding sentence has been added.
[39] Editor's note: this remark has been added.
[40] Editor's note: the original has been simplified in what follows.

The second vertical arrow is obtained by base change along $\varepsilon_{\mathcal{M}} : S \to I_S(\mathcal{M})$. [41]

Proposition 3.12.

Let $h_0 : U \to X$ be an S-morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(\mathcal{M}))$ identifies with the set of $I_S(\mathcal{M})$-morphisms

$$U_{I_S(\mathcal{M})} \longrightarrow X_{I_S(\mathcal{M})}$$

whose restriction to U is h_0, where U is regarded as a subobject of $U \times_S I_S(\mathcal{M})$ through $\mathrm{id}_U \times_S \varepsilon_{\mathcal{M}}$.

Taking $U = X$ and $h_0 = \mathrm{id}_X$ gives the following. [42]

Corollary 3.12.1.

The set $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$-endomorphisms ϕ of $X_{I_S(\mathcal{M})}$ inducing the identity on X; equivalently, the following diagram commutes. [43]

$$\begin{tikzcd}
\mathrm{I}_{\mathrm{X}}(\mathscr{M}) \arrow[r, "\phi"] & \mathrm{I}_{\mathrm{X}}(\mathscr{M}) \\
\mathrm{X} \arrow[u, "\varepsilon_{\mathscr{M}}"] \arrow[r, "\mathrm{id}"] & \mathrm{X} \arrow[u, "\varepsilon_{\mathscr{M}}"']
\end{tikzcd}$$

Figure 18: An infinitesimal endomorphism inducing the identity on X.

On the other hand, by 3.11.2,

$$\Gamma(T_{X/S}(\mathcal{M})/X) \simeq \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M})(S).$$

If X/S satisfies (E), then $\underline{\mathrm{End}}_S(X)/S$ satisfies (E), so the latter Lie functor is an $\mathcal{O}_S$-module by 3.6 (indeed, by 3.11.5, it is a $\prod_{X/S} \mathcal{O}_X$-module). Applying 3.9 gives the next proposition. [44]

Proposition 3.13.

If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$-endomorphisms of $X_{I_S(\mathcal{M})}$ that induce the identity on X. Consequently, every such endomorphism is an automorphism.

Corollary 3.13.1.

Let $u : X \to Y$ be an S-isomorphism, with Y/S satisfying (E). Every $I_S(\mathcal{M})$-morphism $X_{I_S(\mathcal{M})} \to Y_{I_S(\mathcal{M})}$ extending u is an isomorphism.

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[41] Editor's note: via

$$\mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \simeq \mathrm{Hom}_S(U \times_S I_S(\mathcal{M}), X),$$

this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

[42] Editor's note: the original has been expanded in what follows.

[43] Editor's note: when $\mathcal{M} = \mathcal{O}_X$, these are called *infinitesimal endomorphisms* of X; see 6.2 at the end of this exposé.

[44] Editor's note: the original has been expanded in what follows.

Corollary 3.13.2.

If Y/S satisfies (E), the monomorphism

$$\underline{\mathrm{Isom}}_S(X,Y) \longrightarrow \underline{\mathrm{Hom}}_S(X,Y)$$

induces, for every $u \in \underline{\mathrm{Isom}}_S(X,Y)$, an isomorphism

$$L^u_{\underline{\mathrm{Isom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}).$$

Proof. [45] For every $S' \to S$, we must show that this map is a bijection on S'-points. By base change (cf. 3.4), it suffices to take $S' = S$. The right-hand side then parametrizes the $I_S(\mathcal{M})$-morphisms $X_{I_S(\mathcal{M})} \to Y_{I_S(\mathcal{M})}$ extending u, whereas the left-hand side parametrizes those that are isomorphisms. The assertion is therefore Corollary 3.13.1.

Corollary 3.13.3.

If X/S satisfies (E), the monomorphism $\underline{\mathrm{Aut}}_S(X) \to \underline{\mathrm{End}}_S(X)$ induces, for every $u \in \underline{\mathrm{Aut}}_S(X)$, an isomorphism

$$L^u_{\underline{\mathrm{Aut}}_S(X)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{End}}_S(X)/S}(\mathcal{M}).$$

In particular,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S,\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S,\mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}),$$

so that $\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S,\mathcal{M})$ is a $\prod_{X/S}\mathcal{O}_X$-module. [46]

3.14.

Finally, suppose that X is representable. [47] By 2.2.2, the X-functor $T_{X/S}$ is represented by $V(\Omega^1_{X/S})$, whence bijections

$$\Gamma(T_{X/S}/X) \simeq \mathrm{Hom}_X(\Omega^1_{X/S},\mathcal{O}_X) \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X).$$

This also follows from the preceding results. By 3.13, $\Gamma(T_{X/S}/X)$ identifies with the set of infinitesimal endomorphisms of X, that is, the I_S-endomorphisms of X_{I_S} that induce the identity on X. The schemes X and X_{I_S} have the same underlying topological space, and their structure sheaves are $\mathcal{O}_X$ and

$$\mathscr{D}_{\mathcal{O}_X} = \mathcal{O}_X \oplus \mathcal{M}, \qquad \mathcal{M} = \mathcal{O}_X,$$

where $\mathcal{M}$ is a square-zero ideal. Let $\pi : \mathscr{D}_{\mathcal{O}_X} \to \mathcal{O}_X$ be the morphism of $\mathcal{O}_X$-algebras vanishing on $\mathcal{M}$. Giving an infinitesimal endomorphism of X is equivalent to giving a morphism of $\mathcal{O}_S$-algebras $\phi : \mathcal{O}_X \to \mathscr{D}_{\mathcal{O}_X}$ satisfying $\pi \circ \phi = \mathrm{id}_{\mathcal{O}_X}$, hence to giving an $\mathcal{O}_S$-derivation of $\mathcal{O}_X$.

If $D, D' \in \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$, and ϕ_D denotes the infinitesimal endomorphism corresponding to D, then

$$\phi_{D+D'} = \phi_D \circ \phi_{D'}.$$

[45] Editor's note: the following proof has been added.

[46] Editor's note: the preceding assertion has been added.

[47] Editor's note: the reminder of 2.2.2 has been added, and the sequel has been expanded.

Thus the identification

$$\{\text{infinitesimal endomorphisms of } X\} \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$$

is an isomorphism of abelian groups. In view of 3.13 and 3.11.5, we have therefore constructed an isomorphism of abelian groups, indeed of $\mathcal{O}(X)$-modules,

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X),$$

recovering the classical interpretation of tangent vector fields as derivations of the structure sheaf. [48] Finally,

$$\Gamma(T_{X/S}/X) = H^0(X, \mathfrak{g}_{X/S}),$$

where $\mathfrak{g}_{X/S}$ is the dual of $\Omega^1_{X/S}$.

4. Tangent Space to a Group — Lie Algebras

4.1.

Let G be a group functor over S. By 3.9.0.2, $T_{G/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ are endowed with group structures over S, and there are group morphisms

$$\underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M}) \xrightarrow{\ i\ } \mathrm{T}_{\mathrm{G/S}}(\mathscr{M}) \underset{s}{\overset{p}{\rightleftarrows}} \mathrm{G} \quad ;$$

Figure 19: The split tangent sequence of the group functor G.

By definition, i identifies $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ with the kernel of p, and s is a section of p. It follows from I 2.3.7 that this sequence identifies $T_{G/S}(\mathcal{M})$ with the semidirect product of G by $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Definition 4.1.A.

The corresponding action of G on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is denoted

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

and is called the *adjoint representation* of G, relative to $\mathcal{M}$. Thus, by definition, for $x \in G(S')$ and $X \in \underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$,

$$\mathrm{Ad}(x)X = i^{-1}(s(x)i(X)s(x)^{-1}).$$

If G is commutative, then $T_{G/S}(\mathcal{M})$ is commutative as well and $\mathrm{Ad}(x)X = X$. [49]

[48] Editor's note: this isomorphism is also functorial in S. If $S' \to S$ and $X' = X_{S'}$, then

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq \mathrm{Der}_{\mathcal{O}_{S'}}(\mathcal{O}_{X'}).$$

[49] Editor's note: the numbering 4.1.A and 4.1.B has been added in order to bring these definitions into relief.

Definition 4.1.B.

Let G, H be group functors over S, and let $f : G \to H$ be a group morphism. Functoriality gives a morphism of exact sequences compatible with the canonical sections.

$$\begin{tikzcd}
1 \arrow[r] & \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M}) \arrow[r] \arrow[d, "\mathrm{L}(f)"'] & \mathrm{T}_{\mathrm{G/S}}(\mathscr{M}) \arrow[r] \arrow[d, "\mathrm{T}(f)"'] & \mathrm{G} \arrow[r] \arrow[d, "f"'] & 1 \\
1 \arrow[r] & \underline{\mathrm{Lie}}(\mathrm{H/S}, \mathscr{M}) \arrow[r] & \mathrm{T}_{\mathrm{H/S}}(\mathscr{M}) \arrow[r] & \mathrm{H} \arrow[r] & 1
\end{tikzcd} \quad ;$$

Figure 20: Functoriality of the split tangent sequence.

The morphism $L(f)$, also denoted $\underline{\mathrm{Lie}}(f)$, is called the *derived morphism* of f. [50]

Remark 4.1.C.

If G/S and H/S satisfy condition E, then $L(f)$ respects the $\mathcal{O}_S$-module structures obtained from "functoriality in $\mathcal{M}$" (cf. 3.6). [51]

Proposition 4.1.1.

Let $g \in G(S)$. Then
$$\mathrm{Ad}(g) : \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$
is the derived morphism of $\mathrm{Int}(g) : G \to G$.

Indeed, $\mathrm{Ad}(g)X = i^{-1}(\mathrm{Int}(g)i(X))$, which is precisely $L(\mathrm{Int}(g))(X)$ by the definition of the derived morphism.

Suppose that G/S satisfies E. By Proposition 3.9, the group structure just defined on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is exactly the one induced by its $\mathcal{O}_S$-module structure (which is defined by (E)). Proposition 4.1.1 and the functoriality of the $\mathcal{O}_S$-action (cf. 3.6) therefore yield the following corollary.

Corollary 4.1.1.1.

Suppose that G/S satisfies E. Then Ad maps G into the subgroup
$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$
of $\underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$. In other words, for every $g \in G(S')$, the automorphism $\mathrm{Ad}(g)$ respects the $\mathcal{O}_{S'}$-module structure on $\underline{\mathrm{Lie}}(G'/S', \mathcal{M})$, where $G' = G \times_S S'$. Thus Ad is a *linear representation* (cf. I 3.2) of G on the $\mathcal{O}_S$-module $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Remarks 4.1.1.2.

a) The functor G/S satisfies E if and only if, for every pair $(\mathcal{M}, \mathcal{N})$ of finite free $\mathcal{O}_S$-modules, the following diagram, obtained by applying $\underline{\mathrm{Lie}}(G/S, -)$ to diagram $(*)$ of 2.1, is cartesian.

[50] Editor's note: if f is a monomorphism, then so are $L(f)$ and $T(f)$; cf. 3.7.
[51] Editor's note: this remark has been added.

$$\begin{tikzcd} & \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M} \oplus \mathscr{N}) \arrow[dl] \arrow[dr] & \\ \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M}) \arrow[dr] & & \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{N}) \arrow[dl] \\ & \underline{\mathrm{Lie}}(\mathrm{G/S}, 0) = \mathrm{S} & \end{tikzcd},$$

Figure 21: The cartesian criterion for condition E.

b) Suppose that G/S satisfies E. The derived morphism of the group law $\pi : G \times_S G \to G$ is the addition law on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$. Notice that π is not a group morphism, but $\pi(e, e) = e$; hence the derived morphism $L(\pi)$ maps

$$T^{(e,e)}_{(G \times_S G)/S}(\mathcal{M}) = \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

to $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$; cf. 3.7 and 3.8. Likewise, for $n \in \mathbb{Z}$, let $n_G : G \to G$ denote the S-functor morphism $g \mapsto g^n$. Then $L(n_G)$ is multiplication by n on $\underline{\mathrm{Lie}}(G/S)$; cf. 3.9.4.

4.1.2.0.

Consider the S-functor $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$.[52] For every $S' \to S$, one has $T_{G/S}(\mathcal{M})_{S'} \simeq T_{G_{S'}/S'}(\mathcal{M})$ (cf. 3.4), and therefore

$$\begin{aligned} &\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))(S') \\ &\qquad \simeq \mathrm{Hom}_{G_{S'}}(G_{S'}, T_{G_{S'}/S'}(\mathcal{M})) = \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}). \end{aligned}$$

First note the isomorphism, functorial in S',

$$\mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})) \xrightarrow{\sim} \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}), \tag{*}$$

which sends $f : G_{S'} \to \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ to the section $s_f : G_{S'} \to T_{G_{S'}/S'}(\mathcal{M})$ defined, for every $S'' \to S'$ and $g \in G(S'')$, by

$$s_f(g) = i(f(g))s(g).$$

Let h be an automorphism of the functor $G_{S'}$ over S', not necessarily preserving the group structure. To every section τ of $T_{G_{S'}/S'}(\mathcal{M})$ one associates, by transport of structure, a section $h(\tau)$: it is the unique section that makes the following diagram commute.

$$\begin{tikzcd} \mathrm{G}_{\mathrm{S}'} \arrow[r, "\tau"] \arrow[d, "h"'] & \mathrm{T}_{\mathrm{G}_{\mathrm{S}'}/\mathrm{S}'}(\mathscr{M}) \arrow[d, "\mathrm{T}(h)"] \\ \mathrm{G}_{\mathrm{S}'} \arrow[r, "h(\tau)"] & \mathrm{T}_{\mathrm{G}_{\mathrm{S}'}/\mathrm{S}'}(\mathscr{M}) \end{tikzcd}.$$

Figure 22: Transport of a tangent section by an automorphism h.

In particular, take $h = t_x$, right translation by an element $x \in G(S')$; thus $h(g) = t_x(g) = g \cdot x$ for all $g \in G(S'')$, $S'' \to S'$. One then has

$$t_x(s_f) = s_{t_x(f)},$$

[52] Editor's note: the original has been expanded in what follows to display the action of G on the functors $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$ and $\underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$.

where $t_x(f) : G_{S'} \to \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ is given by

$$t_x(f)(g) = f(g \cdot x^{-1}).$$

Let G act by right translations on

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \quad \text{and} \quad \underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

as follows. For $S' \to S$ and $x \in G(S')$, let

$$\begin{aligned} &\sigma \in \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}),\\ &f \in \mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})). \end{aligned}$$

Set

$$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x), \qquad (f \cdot x)(g) = f(g \cdot x^{-1})$$

for every $g \in G(S'')$, $S'' \to S'$. The isomorphism $(*)$ respects these G-actions.

Consequently, under this isomorphism, the elements of

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$$

correspond to the constant morphisms from $G_{S'}$ to $\underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$, that is, to the morphisms factoring through $G_{S'} \to S'$; equivalently, they correspond to the elements of $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$.

Terminology. The elements of $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ are called the sections of $T_{G_{S'}/S'}(\mathcal{M})$ invariant under right translation.

Proposition 4.1.2.

The map

$$\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S) \longrightarrow \Gamma(T_{G/S}(\mathcal{M})/G), \qquad X \longmapsto (x \mapsto X \cdot x),$$

is a bijection onto the set of sections invariant under right translation. [53]

Likewise, let G act on the right on $\underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$ by

$$(u \cdot x)(g) = u(g \cdot x^{-1}) \cdot x,$$

for $S' \to S$, $x \in G(S')$, $u \in \mathrm{End}_{I_{S'}(\mathcal{M})}(G_{I_{S'}(\mathcal{M})})$, and $g \in G(S'')$, $S'' \to I_{S'}(\mathcal{M})$. The morphism of 3.12.1

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$$

respects the G-actions. For each $S' \to S$, it therefore induces a bijection between $\Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and the set of $I_{S'}(\mathcal{M})$-endomorphisms u of $G_{I_{S'}(\mathcal{M})}$ that induce the identity on G and "commute with right translations," meaning that $u_{S''} \cdot x = u_{S''}$ for every $S'' \to S'$ and $x \in G(S'')$.

Proposition 4.1.3.

There is a bijection, functorial in G, between $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S)$ and the set of $I_S(\mathcal{M})$-endomorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G, in the sense just specified.

Taking 3.13 into account gives the following theorem.

[53] Statements 4.1.2, 4.1.3, and 4.1.4 can be obtained more simply by observing that the automorphisms of G invariant under right translations are the left translations. Editor's note: see, for example, the proofs of Propositions 4.6 and 2.5 of [DG70], §II.4.

Theorem 4.1.4.

Let G be an S-functor in groups, and suppose that G/S satisfies E. Then $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S)$ is identified, functorially in G, with the group of $I_S(\mathcal{M})$-automorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G.

For $\mathcal{M} = \mathcal{O}_S$, this recovers one of the classical definitions of the Lie algebra of a group.

4.2.0.

Before proceeding, let us establish further consequences of 3.11.[54] Let X, Y be over Z, with Z over S, as in Section 1. As was seen in 3.11, the isomorphisms 1.1(4)

$$\begin{tikzcd}
\underline{\mathrm{Hom}}_S(I_S(\mathscr{M}), \underline{\mathrm{Hom}}_{Z/S}(X, Y)) \arrow[rr, "\sim"] \arrow[dr, "\simeq"'] & & \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(I_Z(\mathscr{M}), Y)) \\
& \underline{\mathrm{Hom}}_{Z/S}(X \times_S I_S(\mathscr{M}), Y) \arrow[ur, "\simeq"'] &
\end{tikzcd}$$

Figure 23: The adjunction isomorphisms of 1.1(4).

induce the following isomorphism θ.

$$\begin{tikzcd}
T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)}(\mathscr{M}) \arrow[rr, "\theta", "\sim"'] \arrow[dr, "\simeq"'] & & \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathscr{M})) \\
& \underline{\mathrm{Hom}}_{Z/S}(X \times_S I_S(\mathscr{M}), Y) \arrow[ur, "\simeq"'] &
\end{tikzcd} .$$

Figure 24: The induced tangent-space isomorphism θ.

By 1.3, if Y is a Z-group, then $\underline{\mathrm{Hom}}_Z(V, Y)$ is a Z-group for every $V \to Z$, in particular for $V = I_Z(\mathcal{M})$. Explicitly, if $Z'' \to Z' \to Z$ and $\varphi, \psi \in \mathrm{Hom}_Z(V_{Z'}, Y)(Z')$, their product is defined by

$$(\varphi \cdot \psi)(v) = \varphi(v)\psi(v) \qquad (v \in V_{Z'}(Z'')).$$

Suppose now that X and Y are Z-groups.

Definition 4.2.0.1.

We denote by $\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)$ the subfunctor of $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ defined, for every $S' \to S$, by

$$\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}). \tag{3}$$

The same definition applies with Y replaced by the Z-group $T_{Y/Z}(\mathcal{M})$.

Under the isomorphisms (2) above, $T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M})(S')$ corresponds to the $Z_{S'}$-morphisms

$$\varphi : X_{S'} \times_{S'} I_{S'}(\mathcal{M}) \longrightarrow Y_{S'}$$

that are multiplicative in X, that is,

$$\varphi(x_1 x_2, m) = \varphi(x_1, m)\varphi(x_2, m).$$

These in turn correspond to the morphisms of $Z_{S'}$-groups $X_{S'} \to T_{Y/Z}(\mathcal{M})_{S'}$. Hence:

[54] Editor's note: paragraph 4.2.0 has been added; its results are used implicitly in 4.2 and 4.7 of the original.

Proposition 4.2.0.2.

Let X, Y be Z-groups, where Z is over S. There are isomorphisms of S-functors, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, T_{Y/Z}(\mathcal{M})).$$

Taking $Z = S$ gives the following consequence. If Y is a commutative S-group, then so is $T_{Y/S}(\mathcal{M})$, and hence so are $H = \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, Y)$, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M}))$, and $T_{H/S}(\mathcal{M})$.

Corollary 4.2.0.3.

Let X, Y be S-groups. There are isomorphisms of S-functors, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M})).$$

If Y is commutative, these are moreover isomorphisms of abelian S-groups.

Definition 4.2.0.4.

If Y is an $\mathcal{O}_S$-module, the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$, carries an $\mathcal{O}_S$-module structure induced by that of Y. Equipped with this structure, it will be denoted $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}'(Y/S, \mathcal{M})$. [55]

Consequently, if X, Y are $\mathcal{O}_S$-modules, then $T'_{Y/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), Y)$. If $H = \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)$, then $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M}))$ and $T'_{H/S}(\mathcal{M})$ also carry $\mathcal{O}_S$-module structures.

Corollary 4.2.0.5.

If X, Y are $\mathcal{O}_S$-modules, there are isomorphisms of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$T'_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})).$$

Definition 4.2.A.

Let X, L be S-groups, with X acting on L by group automorphisms (cf. I 2.3.5). We define the subfunctor $\underline{Z}^1_S(X, L)$ of $\underline{\mathrm{Hom}}_S(X, L)$ by

$$\begin{aligned}\underline{Z}^1_S(X, L)(S') = \{\, \varphi \in \mathrm{Hom}_{S'}(X_{S'}, L_{S'}) \mid \varphi(x_1x_2) = \varphi(x_1)(x_1 \cdot \varphi(x_2)),& \\ x_1, x_2 \in X(S''),\ S'' \to S' \,\}.&\end{aligned}$$

It is called the *functor of crossed homomorphisms* from X to L. [56]

Remark 4.2.B.

a) If L' is another S-group on which X acts by group automorphisms, then

$$\underline{Z}^1_S(X, L \times_S L') \simeq \underline{Z}^1_S(X, L) \times_S \underline{Z}^1_S(X, L').$$

b) If L is an X-$\mathcal{O}_S$-module, then $\underline{Z}^1_S(X, L)$ is the kernel of the differential

$$\partial : \underline{\mathrm{Hom}}_S(X, L) \longrightarrow \underline{\mathrm{Hom}}_S(X^2, L)$$

defined in I 5.1. In particular, $\underline{Z}^1_S(X, L)$ is then an $\mathcal{O}_S$-module.

[55] Editor's note: the prime is used to distinguish the $\mathcal{O}_S$-module structure induced by that of Y from the one that will be obtained from condition E.

[56] Editor's note: the original has been expanded below; in particular, Definition 4.2.A and Remark 4.2.B have been added.

4.2.

Let $u : X \to Y$ be a morphism of S-groups. We describe $L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M})$. First, 3.11.3 gives an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})). \tag{†}$$

Since Y is an S-group,

$$T_{Y/S}(\mathcal{M}) = \underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y = \underline{\mathrm{Lie}}(Y/S,\mathcal{M})_Y,$$

and one obtains an isomorphism, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_S(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M})).$$

For $S' \to S$, let $u' : X' \to Y'$ be the base change of u. The S-functor[57]

$$\underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y)$$

is defined by

$$\begin{aligned}&\underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y)(S')\\ &\quad = \mathrm{Hom}_{Y'\text{-gr.}}(X', (\underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y)_{S'})\\ &\quad = \mathrm{Hom}_{Y'\text{-gr.}}(X', \underline{\mathrm{Lie}}(Y'/S',\mathcal{M}) \cdot Y').\end{aligned}$$

Thus (†) induces an isomorphism

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y). \tag{†′}$$

On the other hand, the composite

$$\mathrm{X} \xrightarrow{u} \mathrm{Y} \xrightarrow{\mathrm{Ad}} \underline{\mathrm{Aut}}_{\mathrm{S\text{-}gr.}}(\underline{\mathrm{Lie}}(\mathrm{Y/S}, \mathscr{M}))$$

Figure 25: The action of X on $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$.

defines an action of X on $L = \underline{\mathrm{Lie}}(Y/S,\mathcal{M})$ by group automorphisms.

Let $\Phi \in \underline{\mathrm{Hom}}_{Y/S}(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M}) \cdot Y)$. For $S'' \to S' \to S$ and $x \in X(S'')$, there is a unique expression

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \qquad \varphi(S')(x) \in \underline{\mathrm{Lie}}(Y'/S',\mathcal{M})(S'').$$

This determines an element $\varphi \in \underline{\mathrm{Hom}}_S(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M}))$. The map $\Phi(S')$ is a group morphism if and only if, for all $x_1, x_2 \in X(S'')$,

$$\begin{aligned}\varphi(S')(x_1x_2) &= \varphi(S')(x_1)\big(u(x_1)\varphi(S')(x_2)u(x_1)^{-1}\big)\\ &= \varphi(S')(x_1)\big(x_1 \cdot \varphi(S')(x_2)\big),\end{aligned}$$

that is, precisely when $\varphi \in \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S,\mathcal{M}))$.

[57] Editor's note: this is Definition 4.2.0.1 applied with $Z = Y$, to the Y-groups $u : X \to Y$ and $T_{Y/S}(\mathcal{M}) \to Y$.

Proposition 4.2.

Let $u : X \to Y$ be a morphism of S-groups. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

Suppose in addition that Y/S satisfies E. It follows from 4.2.0.3, just as in the proof of 3.11.1, that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S$ satisfies E. Thus (cf. 3.5.1)

$$\begin{aligned} L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M} \oplus \mathcal{N}) \simeq L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \\ \times_S L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{N}). \end{aligned}$$

This also follows directly from Proposition 4.2 and Remark 4.2.Ba. Consequently, both sides of the isomorphism in Proposition 4.2 carry $\mathcal{O}_S$-module structures obtained from functoriality in $\mathcal{M}$, and that isomorphism respects them.

Proposition 4.2 bis.

Let $u : X \to Y$ be a morphism of S-groups, and suppose that Y/S satisfies E. There is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

[58]

When Y/S satisfies E, 3.13.1 also gives, just as in the proof of 3.13.2, an isomorphism functorial in $\mathcal{M}$, for every $u \in \underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)$,

$$L^u_{\underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}). \qquad (*)$$

This yields the following two corollaries.

Corollary 4.2.1.

Let $u : X \to Y$ be a morphism of S-groups. If Y/S satisfies E, there is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

Corollary 4.2.2.

Let X be an S-group. If X/S satisfies E, there is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{S\text{-gr.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(X/S, \mathcal{M})).$$

If Y is commutative, its adjoint action on $L = \underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ is trivial, so $\underline{Z}^1_S(X, L) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, L)$. [59]

Corollary 4.2.3.

Let Y be a commutative S-group. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

[58] Editor's note: the original has been expanded from this point; in particular, Proposition 4.2 bis has been added.

[59] Editor's note: the following sentence has been added.

4.3.

We now consider the case in which X and Y are $\mathcal{O}_S$-modules. Recall (cf. 4.2.0.4) that $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$, denotes the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$, equipped with the $\mathcal{O}_S$-module structure induced by that of Y.

When Y/S satisfies E, we shall always write $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$ for this functor equipped instead with the $\mathcal{O}_S$-module structure defined for every functor satisfying E. By 3.9, the underlying abelian-group structures of $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$ and $\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$ coincide, but their module structures need not coincide (see the counterexample in 6.3). For $S'\to S$, $a\in\mathcal{O}(S')$, and $m\in\underline{\mathrm{Lie}}(Y/S,\mathcal{M})(S')$, we write $a\cdot' m$, respectively $a\cdot m$, for the action coming from Y, respectively the action obtained from condition E. We use the same convention for the two actions on $T_{Y/S}(\mathcal{M})$.

There is an isomorphism of $\mathcal{O}_S$-modules

$$T'_{Y/S}(\mathcal{M})\simeq \underline{\mathrm{Lie}}'(Y/S,\mathcal{M})\oplus Y.$$

Exactly as for Propositions 4.2 and 4.2 bis, one therefore obtains:

Proposition 4.3.

Let $u: X\to Y$ be a morphism of $\mathcal{O}_S$-modules. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M})\xrightarrow{\sim}\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})).\qquad(*)$$

If Y/S satisfies E, then $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S$ satisfies E, and $(*)$ is an isomorphism of $\mathcal{O}_S$-modules when both sides carry the structures obtained from functoriality in $\mathcal{M}$. [60] [61]

Remark 4.3 bis.

Let $u: X\to Y$ be a morphism of $\mathcal{O}_S$-modules.[62] Denote by τ_u the map that sends each $\mathcal{O}_S$-linear morphism $\varphi: X\to\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$ to

$$u\oplus\varphi: X\longrightarrow T'_{Y/S}(\mathcal{M})=Y\oplus\underline{\mathrm{Lie}}'(Y/S,\mathcal{M}).$$

The isomorphism of 4.3 belongs to the following commutative diagram, functorial in $\mathcal{M}$.

$$\begin{tikzcd}
L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \arrow[r, "\sim"] \arrow[d, hook] & \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})) \arrow[d, hook, "\tau_u"] \\
T_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \arrow[r, "\sim"] & \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,T'_{Y/S}(\mathcal{M})).
\end{tikzcd}$$

Figure 26: Compatibility of the isomorphism of 4.3 with τ_u.

Moreover, when Y/S satisfies E, 3.13.1 gives, just as in the proof of 3.13.2, for every $u\in\underline{\mathrm{Isom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)$,

$$L^u_{\underline{\mathrm{Isom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M})=L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}).\qquad(*)$$

[60] Editor's note: the following sentence has been added.

[61] Editor's note: on the right-hand side, the scalar action is $(a\varphi)(x)=a\varphi(x)$, where $a\varphi(x)$ denotes the action of a on $\varphi(x)\in\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$. This differs from the action $(a\cdot'\varphi)(x)=a\cdot'\varphi(x)=\varphi(ax)$, but the two agree when $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})=\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$.

[62] Editor's note: this remark has been added; it will be useful in 4.5.1.

Corollary 4.3.1.

Let X be an $\mathcal{O}_S$-module satisfying E over S. There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\mathrm{Lie}}'(X/S, \mathcal{M})),$$

and this isomorphism respects the $\mathcal{O}_S$-module structures obtained from functoriality in $\mathcal{M}$. In particular, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S$ satisfies E. [63]

Proof. The first assertion follows from $(*)$ and 4.3. For the second, since X/S satisfies E, there are isomorphisms of $\mathcal{O}_S$-modules

$$\underline{\mathrm{Lie}}'(X/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(X/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(X/S, \mathcal{N}).$$

Consequently,

$$\begin{aligned}&\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M} \oplus \mathcal{N})\\ &\quad\simeq \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{N}).\end{aligned}$$

Remark 4.1.1.2a now proves the assertion.

4.3.2.

Before continuing in this direction, let us examine more closely the relations among Y, $\underline{\mathrm{Lie}}(Y/S)$, and $\underline{\mathrm{Lie}}'(Y/S)$. First,

$$\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(\mathcal{O}_S/S, \mathcal{M}) = \mathbf{W}(\mathcal{M}), \tag{1}$$

where $\mathbf{W}(\mathcal{M})$ is defined in I 4.6. There is therefore a canonical isomorphism

$$d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S). \tag{2}$$

Let F be an $\mathcal{O}_S$-module. For every $S_2 \to S_1 \to S$, there is a dihomomorphism

$$\begin{cases} F(S_1) \longrightarrow F(S_2), \\ \mathcal{O}(S_1) \longrightarrow \mathcal{O}(S_2), \end{cases} \tag{3}$$

and hence a morphism of $\mathcal{O}(S_2)$-modules

$$F(S_1) \otimes_{\mathcal{O}(S_1)} \mathcal{O}(S_2) \longrightarrow F(S_2).$$

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In particular, take $S_1 = S'$ and $S_2 = I_{S'}(\mathcal{M})$. One obtains morphisms of $\mathcal{O}(S')$-modules, functorial in $\mathcal{M}$,

$$F(S') \otimes_{\mathcal{O}(S')} T_{\mathcal{O}_S/S}(\mathcal{M})(S') \longrightarrow T'_{F/S}(\mathcal{M})(S').$$

As S' varies, these give morphisms of $\mathcal{O}_S$-modules

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}), \tag{4}$$

functorial in $\mathcal{M}$. Compatibility with the tangent-bundle projections onto their bases then yields morphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}), \tag{5}$$

[63] Editor's note: compare editor's note (61). The final sentence and its proof have also been added.

[64] Editor's note: $S' \to S$ has been replaced by $S_2 \to S_1 \to S$, since both S_2 and S_1 must vary below. The original has also been expanded in what follows.

and the following diagram commutes.

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M}) \arrow[r] & \mathrm{T}'_{\mathrm{F}/\mathrm{S}}(\mathscr{M}) \arrow[r] & \mathrm{F} \arrow[r] & 0.
\end{tikzcd}$$

Figure 27: The morphisms (4) and (5) in the split tangent sequences.

The morphisms (5) may be regarded as morphisms of abelian S-groups

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}). \tag{6}$$

Tensoring F with the isomorphism $d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S)$ gives, when $\mathcal{M} = \mathcal{O}_S$, a morphism of abelian S-groups

$$F \xrightarrow{\sim} F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S) \longrightarrow \underline{\mathrm{Lie}}(F/S), \tag{7}$$

also denoted $d : F \to \underline{\mathrm{Lie}}(F/S)$.

Remark 4.3.3.

When F/S satisfies E, the morphisms (6) and (7) need not be $\mathcal{O}_S$-linear if both sides are equipped with the module structures obtained from $\mathcal{M}$ by condition E.

For example, let k be a field of characteristic $p > 0$, let $S = \mathrm{Spec}(k)$, and let F be the $\mathcal{O}_S$-module that assigns to every S-scheme T the set $F(T) = \Gamma(T, \mathcal{O}_T)$, with scalars acting through the p-th power:

$$r \cdot f = r^p f \qquad (r \in \mathcal{O}(T),\ f \in F(T)).$$

As an S-functor in groups, $F \simeq \mathbf{G}_{a,S}$. Hence F satisfies E and $\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbf{G}_{a,S}/S) \simeq \mathcal{O}_S$. For every $T \to S$, the canonical morphism $d : F(T) \to \mathcal{O}(T)$ is the identity: it respects the abelian-group structure but not the $\mathcal{O}_S$-module structure. [65]

Remark 4.3.4.

The morphisms (4) and (5) can be made explicit as follows. The morphism

$$\Theta : F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), F)$$

is defined, for $S' \to S$, $\alpha \in \mathcal{O}(I_{S'}(\mathcal{M}))$, and $f : S' \to F$, by

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where $\tau_0 : F \to T'_{F/S}(\mathcal{M})$ is the zero section and $\rho : I_{S'}(\mathcal{M}) \to S'$ is the structural morphism.

The morphism Θ induces

$$\theta : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}).$$

This follows from the functoriality in $\mathcal{M}$ noted after (4), and can also be seen directly. Indeed,

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M})(S') = \{\, \varphi \in \mathrm{Hom}_S(I_{S'}(\mathcal{M}), F) \mid \varphi \circ \varepsilon_{\mathcal{M}} = e \,\},$$

[65] Editor's note: the original asserted that, when F/S satisfies E, (6), and hence (7), are morphisms of $\mathcal{O}_S$-modules. The counterexample in 6.3 contradicts this. That assertion has been removed, the preceding example inserted, and Definition 4.4 modified accordingly. This does not affect what follows.

where $e : S' \to F$ is the unit section. On the other hand, $\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})(S')$ equals $\Gamma(S', \mathcal{M})$, the kernel of the augmentation

$$\eta : \mathcal{O}(I_{S'}(\mathcal{M})) \longrightarrow \mathcal{O}(S').$$

It remains to see that if $f \in F(S')$ and $\alpha \in \Gamma(S', \mathcal{M})$, then $\alpha \cdot (f \circ \rho) \circ \varepsilon_{\mathcal{M}} = e$.

Apply the dihomomorphism (3) to the S-morphism $\varepsilon_{\mathcal{M}} : S' \to I_{S'}(\mathcal{M})$. The following diagram commutes.

$$\begin{tikzcd}
\mathrm{F}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[r, "\mathrm{F}(\varepsilon_{\mathscr{M}})"] \arrow[d, "\alpha"'] & \mathrm{F}(\mathrm{S}') \arrow[d, "\eta(\alpha)"] \\
\mathrm{F}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[r, "\mathrm{F}(\varepsilon_{\mathscr{M}})"] & \mathrm{F}(\mathrm{S}')
\end{tikzcd}.$$

Figure 28: Compatibility of scalar action with $F(\varepsilon_{\mathcal{M}})$.

Thus, for every $\varphi : I_{S'}(\mathcal{M}) \to F$,

$$(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = \eta(\alpha) \cdot (\varphi \circ \varepsilon_{\mathcal{M}}),$$

and therefore $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = e$ when $\eta(\alpha) = 0$.

In particular, taking $\mathcal{M} = t\mathcal{O}_S$, one has $\underline{\mathrm{Lie}}(\mathcal{O}_S/S) = t\mathcal{O}_S$, and the morphism $F \to \underline{\mathrm{Lie}}'(F/S)$ is $f \mapsto t \cdot (f \circ \rho)$.
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Remark 4.3.5.

Let F be an $\mathcal{O}_S$-module, set

$$E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

and denote by d_F and d_E the $\mathcal{O}_S$-linear morphisms given by 4.3.2(5):

$$d_F : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}),$$
$$d_E : E \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(E/S, \mathcal{M}).$$

Remark 4.3.4 gives the following commutative diagram of $\mathcal{O}_S$-linear morphisms.

$$\begin{tikzcd}
\mathrm{E} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M}) \arrow[r, "d_\mathrm{E}"] \arrow[d, equal] & \underline{\mathrm{Lie}}'(\mathrm{E}/\mathrm{S}, \mathscr{M}) \\
\underline{\mathrm{Hom}}_{\mathbf{O}_\mathrm{S}}(\mathrm{F}, \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M})) \arrow[r, "d_\mathrm{F}"] & \underline{\mathrm{Hom}}_{\mathbf{O}_\mathrm{S}}(\mathrm{F}, \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M})) \arrow[u, "\simeq"', "(*)"]
\end{tikzcd}$$

Figure 29: Compatibility of d_E and d_F.

The right vertical arrow is the isomorphism $(*)$ of 4.3. Thus, by that result, if F/S satisfies E, then E/S satisfies E, and $(*)$ is also an isomorphism of $\mathcal{O}_S$-modules when the right-hand terms are equipped with the module structures obtained from E. [67]

[66] Editor's note: this remark has been added; it will be useful in 4.6.2.
[67] Editor's note: this remark has been added; it will be useful in 4.5 and 4.7.

Remark 4.4.0.

In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. If these equivalent conditions hold, then F/S satisfies E. Indeed, it is enough to verify that

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(F/S, \mathcal{N}).$$

Under the hypothesis, the horizontal arrows in the following commutative diagram are isomorphisms.

$$\begin{tikzcd}
\mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M} \oplus \mathscr{N}) \arrow[r, "\sim"] \arrow[d, "\wr"] & \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M} \oplus \mathscr{N}) \arrow[d] \\
\mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \big(\underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M}) \underset{\mathrm{S}}{\times} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{N})\big) \arrow[r, "\sim"] & \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M}) \underset{\mathrm{S}}{\times} \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{N});
\end{tikzcd}$$

Figure 30: The direct-sum comparison used to verify condition E.

The second vertical arrow is therefore an isomorphism as well, which is exactly condition E for F/S. [68]

Definition 4.4.

An $\mathcal{O}_S$-module F is called *good* if the morphisms

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T_{F/S}(\mathcal{M}),$$

or, equivalently,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}),$$

are isomorphisms of abelian S-groups (so that F/S satisfies (E)), and if moreover they respect the $\mathcal{O}_S$-module structures obtained from condition E.

Corollary 4.4.1.

Let F be an $\mathcal{O}_S$-module, and consider the following conditions:[69]

(i) F is a good $\mathcal{O}_S$-module;

(ii) F/S satisfies E, and $d : F \to \underline{\mathrm{Lie}}(F/S)$ is an isomorphism of $\mathcal{O}_S$-modules;

(iii) $\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$.

Then

$$\text{(i)} \Longleftrightarrow \text{(ii)} \Longrightarrow \text{(iii)}.$$

[68] Editor's note: in view of the changes made in Definition 4.4 (cf. editor's note (65)), this remark has been inserted here; in the original it occurred in the proof of 4.4.1.

[69] Editor's note: the original proved that (i) implies (ii) and (iii); the implication (ii) implies (i) has been added.

Proof. The implication (i) ⇒ (ii) follows from the definition. To prove (ii) ⇒ (i), it remains to show that the morphisms of abelian S-groups, functorial in $\mathcal{M}$,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S, \mathcal{M})$$

are isomorphisms of $\mathcal{O}_S$-modules. Since F/S satisfies E, both sides carry finite direct sums of copies of $\mathcal{O}_S$ to finite products of abelian S-groups. We are therefore reduced to the case $\mathcal{M} = \mathcal{O}_S$, which is precisely the hypothesis.

Finally, (i) ⇒ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$$

of 4.3.2(5) are $\mathcal{O}_S$-linear.

Examples 4.4.2.

For every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, the $\mathcal{O}_S$-modules $\mathbf{V}(\mathcal{E})$ and $\mathbf{W}(\mathcal{E})$ defined in I 4.6 are good.

Indeed, for every $f : S' \to S$, the morphisms

$$\begin{aligned} \mathbf{V}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) &\longrightarrow T_{\mathbf{V}(\mathcal{E})/S}(\mathcal{M})(S'), \\ \mathbf{W}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) &\longrightarrow T_{\mathbf{W}(\mathcal{E})/S}(\mathcal{M})(S') \end{aligned}$$

correspond respectively to

$$\begin{aligned} &\mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathcal{O}_{S'}) \otimes_{\mathcal{O}(S')} \Gamma(S', D_{\mathcal{O}_{S'}}(\mathcal{M})) \\ &\qquad \longrightarrow \mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), D_{\mathcal{O}_{S'}}(\mathcal{M})) \qquad (\mathrm{V}) \\ &\Gamma(S', f^*(\mathcal{E})) \otimes_{\mathcal{O}(S')} \Gamma(S', D_{\mathcal{O}_{S'}}(\mathcal{M})) \\ &\qquad \longrightarrow \Gamma(S', f^*(\mathcal{E}) \otimes_{\mathcal{O}_{S'}} D_{\mathcal{O}_{S'}}(\mathcal{M})) \qquad (\mathrm{W}). \end{aligned}$$

These are isomorphisms because, as an $\mathcal{O}_{S'}$-module, $D_{\mathcal{O}_{S'}}(\mathcal{M})$ is isomorphic to a finite direct sum of copies of $\mathcal{O}_{S'}$. [70]

Proposition 4.5. *Let F be a good $\mathcal{O}_S$-module. Then:*

(i) The S-group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ satisfies condition (E), and there is a functorial isomorphism

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}\left(F, \underline{\mathrm{Lie}}(F/S, \mathcal{M})\right),$$

which respects the $\mathcal{O}_S$-module structures obtained from condition (E). In particular, there is an isomorphism of $\mathcal{O}_S$-modules

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \xrightarrow{\sim} \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

(ii)[71] *Moreover, $\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$ is a good $\mathcal{O}_S$-module.*

Indeed, by 4.4.1, F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \simeq F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathbf{O}_S/S, \mathcal{M}). \qquad (1)$$

Assertion (i) now follows from 4.3.1. Put $E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$. By (1) and Remark 4.3.5, one has the following commutative diagram of morphisms of abelian S-groups.

[70] Editor's note: the preceding proof has been added.

[71] Editor's note: part (ii), an immediate consequence of what precedes, has been added.

$$
\begin{tikzcd}
\underline{\mathrm{End}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F}) \otimes_{\mathbf{O}_{\mathrm{S}}} \underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{M}) \arrow[r, "d_{\mathrm{E}}"] \arrow[d, equal] & \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F})/\mathrm{S}, \mathscr{M}) \\
\underline{\mathrm{Hom}}_{\mathbf{O}_{\mathrm{S}}}(\mathrm{F}, \mathrm{F} \otimes_{\mathbf{O}_{\mathrm{S}}} \underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{M})) \arrow[r, "d_{\mathrm{F}}"] & \underline{\mathrm{Hom}}_{\mathbf{O}_{\mathrm{S}}}(\mathrm{F}, \underline{\mathrm{Lie}}(\mathrm{F}/\mathrm{S}, \mathscr{M})) \arrow[u, "\simeq"', "(*)"]
\end{tikzcd}
$$

Figure 31: The comparison diagram for the differential d_E.

Here d_F and the vertical isomorphism $(*)$ are isomorphisms of $\mathcal{O}_S$-modules; consequently, so is d_E. This proves (ii).

Scholium 4.5.1. [72] Put (cf. 2.1) $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$, and let F be a good $\mathcal{O}_S$-module. For every $S' \to S$, the morphism

$$F(S') \oplus tF(S') = F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}) \longrightarrow F(I_{S'}) = F(S') \oplus \underline{\mathrm{Lie}}(F/S)(S'),$$

which is the identity on $F(S')$, induces an isomorphism of $\mathcal{O}(S')$-modules

$$tF(S') \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S)(S').$$

As S' varies, this gives an isomorphism that we may write

$$\underline{\mathrm{Lie}}(F/S) \simeq tF.$$

Thus, for every $S' \to S$, Proposition 4.5 gives a commutative diagram.

$$
\begin{tikzcd}
\mathrm{End}_{\mathbf{O}_{\mathrm{S}'}\text{-mod.}}(\mathrm{F}_{\mathrm{S}'}) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathbf{O}_{\mathrm{S}'}\text{-mod.}}(\mathrm{F}_{\mathrm{S}'}, t\,\mathrm{F}_{\mathrm{S}'}) \arrow[r, "\sim"] \arrow[d, hook] & \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F})/\mathrm{S})(\mathrm{S}') \arrow[d, hook] \\
& \mathrm{Aut}_{\mathbf{O}_{\mathrm{I}_{\mathrm{S}'}}\text{-mod.}}(\mathrm{F}_{\mathrm{I}_{\mathrm{S}'}}) \arrow[r, equal] & \mathrm{T}_{\underline{\mathrm{Aut}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F})/\mathrm{S}}(\mathrm{S}')
\end{tikzcd}
$$

Figure 32: Endomorphisms and automorphisms over the dual numbers.

It follows from 4.3 bis that every

$$X \in \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$$

corresponds to $\mathrm{id} + tX \in \underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$.

Definition 4.6. An S-functor in groups G is called *good* if G/S satisfies condition (E) and $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module.

[73] Notice that if F is a good $\mathcal{O}_S$-module, then it is a good S-group: indeed, F/S satisfies (E), and $\underline{\mathrm{Lie}}(F/S) \simeq F$ is a good $\mathcal{O}_S$-module.

Example 4.6.1. *If G is representable, it is good. Indeed, G/S satisfies (E), and $\underline{\mathrm{Lie}}(G/S)$ has the form $V(E)$, hence is good by 4.4.2.*

Lemma 4.6.2. [74] *Let G be an S-functor in groups such that G/S satisfies (E), and put $F = \underline{\mathrm{Lie}}(G/S)$. Then F satisfies (E), and the morphism of abelian groups*

$$d : F \longrightarrow \underline{\mathrm{Lie}}(F/S)$$

[72] Editor's note: this scholium, implicit in the original, has been added for use in 4.7.1. Here and below, t, t', and so forth denote square-zero variables; the letter ε is reserved for the unit section of a group.

[73] Editor's note: the following sentence has been added.

[74] Editor's note: this lemma has been added.

respects the $\mathcal{O}_S$-module structures. Consequently, G is good if and only if $F \to \underline{\mathrm{Lie}}(F/S)$ is bijective.

Proof. Suppose that G/S satisfies (E). Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$-modules, write

$$F(\mathcal{N}) = \underline{\mathrm{Lie}}(G/S, \mathcal{N}),$$

and let e denote the unit section of G. For every $S' \to S$,

$$F(\mathcal{N})(S') = \{g \in \mathrm{Hom}_S(I_{S'}(\mathcal{N}), G) \mid g \circ \varepsilon_{\mathcal{N}} = e\}.$$

Moreover, $\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M})(S')$ identifies with the set of morphisms

$$\left\{ \Phi \in \mathrm{Hom}_S(I_{S'}(\mathcal{N}) \times_{S'} I_{S'}(\mathcal{M}), G) \;\middle|\; \begin{array}{l} \Phi \circ (\varepsilon_{\mathcal{N}} \times \mathrm{id}_{I_{S'}(\mathcal{M})}) = e \in G(I_{S'}(\mathcal{M})), \\ \Phi \circ (\mathrm{id}_{I_{S'}(\mathcal{N})} \times \varepsilon_{\mathcal{M}}) = e \in G(I_{S'}(\mathcal{N})). \end{array} \right\}.$$

This proves

$$\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}) \simeq \underline{\mathrm{Lie}}(F(\mathcal{M})/S, \mathcal{N}). \tag{1}$$

Since G/S satisfies (E), it follows that

$$\begin{aligned} \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) &\simeq \underline{\mathrm{Lie}}(F(\mathcal{M}_1 \oplus \mathcal{M}_2)/S, \mathcal{N}) \\ &\simeq \underline{\mathrm{Lie}}((F(\mathcal{M}_1) \times_S F(\mathcal{M}_2))/S, \mathcal{N}). \end{aligned}$$

By 3.8, the last term is isomorphic to

$$\begin{aligned} &\underline{\mathrm{Lie}}(F(\mathcal{M}_1)/S, \mathcal{N}) \times_S \underline{\mathrm{Lie}}(F(\mathcal{M}_2)/S, \mathcal{N}) \\ &\quad \simeq \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2). \end{aligned}$$

Hence

$$\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) \simeq \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2), \tag{2}$$

so $F(\mathcal{N})/S$ satisfies (E), by Remark 4.1.1.2(a).

It remains to show that the morphism of abelian groups

$$d : F(\mathcal{N}) \longrightarrow \underline{\mathrm{Lie}}(F(\mathcal{N})/S)$$

respects the $\mathcal{O}_S$-module structures. Consider the free $\mathcal{O}_S$-module $\mathcal{M} = t\mathcal{O}_S$. Then

$$\mathcal{O}(I_{S'}(\mathcal{M})) = \mathcal{O}(S')[t]/(t^2) = \mathcal{O}(S') \oplus t\mathcal{O}(S'), \qquad \underline{\mathrm{Lie}}(\mathbf{O}_S/S) \simeq t\mathcal{O}_S.$$

Let $\rho_t : I_S(\mathcal{M}) \to S$ be the structural morphism, corresponding to the injection

$$u_t : \mathcal{O}_S \hookrightarrow D_{\mathcal{O}_S}(\mathcal{M}) = \mathcal{O}_S \oplus t\mathcal{O}_S.$$

Recall (cf. 3.4.2) that for every S-functor X, $F(\mathcal{N})(X)$ is the set of S-morphisms $\phi : I_X(\mathcal{N}) \to G$ such that $\phi \circ \varepsilon_{\mathcal{N}} = e$, and the action of $a \in \mathcal{O}(X)$ is

$$a \cdot \phi = \phi \circ a^*,$$

where a^* is the endomorphism of $I_X(\mathcal{N})$ associated with a; see 2.1.3.

Consequently, by 4.3.4, the composite

$$d : F(\mathcal{N}) \xrightarrow{\sim} F(\mathcal{N}) \otimes_{\mathcal{O}_S} t\mathcal{O}_S \longrightarrow \underline{\mathrm{Lie}}(F(\mathcal{N})/S)$$

is given, for $S' \to S$ and $f \in \mathrm{Hom}_S(S', F(\mathcal{N}))$, by

$$f \longmapsto f \otimes t \longmapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*.$$

Here $f \circ \rho_t \in F(\mathcal{N})(I_{S'}(\mathcal{M}))$ is viewed as an S-morphism

$$\phi : I_{I_{S'}(\mathcal{M})}(\mathcal{N}) \longrightarrow G$$

such that $\phi \circ \varepsilon_{\mathcal{N}} = e$. We must prove that d is $\mathcal{O}_S$-linear, or equivalently that

$$d(a \cdot f) = u_t(a) \cdot d(f) \qquad (a \in \mathcal{O}(S')).$$

Viewing $f \in F(\mathcal{N})(S')$ as a morphism $I_{S'}(\mathcal{N}) \to G$, one likewise has $a \cdot f = f \circ a^*$. Functoriality in X of the action of $\mathcal{O}(X)$ on $I_X(\mathcal{N})$ gives the following commutative diagram.

$$\begin{tikzcd}
\mathrm{I}_{S'}(\mathcal{N}) \arrow[r, "a^*"] & \mathrm{I}_{S'}(\mathcal{N}) \\
\mathrm{I}_{\mathrm{I}_{S'}(\mathcal{M})}(\mathcal{N}) \arrow[u, "\rho_t"] \arrow[r, "u_t(a)^*"] & \mathrm{I}_{\mathrm{I}_{S'}(\mathcal{M})}(\mathcal{N}) \arrow[u, "\rho_t"'] .
\end{tikzcd}$$

Figure 33: Compatibility of scalar action with the structural morphism ρ_t.

Therefore

$$\begin{aligned} d(a \cdot f) &= f \circ a^* \circ \rho_t \circ t^* \\ &= f \circ \rho_t \circ u_t(a)^* \circ t^* \\ &= f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f). \end{aligned}$$

The penultimate equality follows from the commutativity of $\mathcal{O}$. This completes the proof of Lemma 4.6.2.

Theorem 4.7. *If F is a good $\mathcal{O}_S$-module, then the S-group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-}mod.}(F)$ is good.*

[75] Indeed, by 4.5, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S$ satisfies (E), while

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a good $\mathcal{O}_S$-module.

4.7.1.

[76] Let G be an S-group and F a good $\mathcal{O}_S$-module. Suppose given a linear representation of G in F, that is (I 4.7.1), a morphism of S-groups

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F).$$

If G/S satisfies (E), then 4.1.C and 4.5 give a morphism of $\mathcal{O}_S$-modules, denoted ρ' or $d\rho$,

$$\underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

[75] Editor's note: the original has been simplified here by using the additions made in 4.3.1 and 4.5.
[76] Editor's note: the numbering 4.7.1, 4.7.2, and 4.7.3 has been added.

[77] Furthermore, put $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$. It follows from 4.5.1 that, if $S' \to S$ and

$$X \in \underline{\mathrm{Lie}}(G/S)(S') \subset G(I_{S'}),$$

then in $\underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$ one has

$$\rho(X) = \mathrm{id} + t\rho'(X). \tag{*}$$

In other words, for every $S'' \to I_{S'}$ and $f \in F(S'')$,

$$\rho(X)(f) = f + t\rho'(X)(f) \quad \text{in } F(S'').$$

Definition 4.7.2. Let G be a good S-group. Then $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module, and there is a morphism of S-groups

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S)).$$

By 4.7.1, it induces a morphism of $\mathcal{O}_S$-modules

$$\mathrm{ad} : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S)),$$

or, equivalently, an $\mathcal{O}_S$-bilinear morphism

$$\underline{\mathrm{Lie}}(G/S) \times_S \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S), \qquad (x, y) \longmapsto [x, y] = \mathrm{ad}(x) \cdot y.$$

Here x, y are arbitrary elements of

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S').$$

If G is commutative, then $[x, y] = 0$.

4.7.3.

The bracket admits the following equivalent description. It is enough to consider sections over S, that is,

$$x, y \in \underline{\mathrm{Lie}}(G/S)(S).$$

There is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$. To avoid confusion, let I and I' denote two copies of I_S, and put

$$\mathcal{O}_I = \mathcal{O}_S[t], \qquad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \qquad t^2 = 0 = (t')^2.$$

There is then a commutative square

$$\begin{tikzcd} \mathrm{I} \times \mathrm{I}' \arrow[r] \arrow[d] & \mathrm{I}' \arrow[d] \\ \mathrm{I} \arrow[r] & \mathrm{S} \end{tikzcd},$$

Figure 34: The two dual-number structures on $I \times_S I'$.

[77] Editor's note: the following sentence has been added.

in which the two arrows from $I \times I'$ identify it with the scheme of dual numbers over I or over I', respectively. Writing $L = \underline{\mathrm{Lie}}(G/S)$, one obtains the following commutative diagram of groups.

(1)
$$\begin{tikzcd}
 & & 1 \arrow[d] & 1 \arrow[d] & \\
 & & \mathrm{L(I)} \arrow[r] \arrow[d] & \mathrm{L(S)} \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathrm{L(I')} \arrow[r] \arrow[d] & \mathrm{G(I\times I')} \arrow[r] \arrow[d] & \mathrm{G(I')} \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathrm{L(S)} \arrow[r] \arrow[d] & \mathrm{G(I)} \arrow[r] \arrow[d] & \mathrm{G(S)} \arrow[r] \arrow[d] & 1 \\
 & 1 & 1 & 1 & .
\end{tikzcd}$$

Figure 35: The first exact-sequence diagram used to construct the bracket.

The ninth object in this diagram is $\underline{\mathrm{Lie}}(L/S)(S)$. If G is good, this is $L(S)$, and hence one obtains the next commutative diagram. Its rows and columns are exact sequences of groups; the five groups denoted $L(-)$ are commutative.[78] Under the identifications

$$L(I) = L(S) \oplus tL(S), \qquad L(I') = L(S) \oplus t'L(S),$$

the injections $L(S) \hookrightarrow L(I)$ and $L(S) \hookrightarrow L(I')$ are $u \mapsto tu$ and $u \mapsto t'u$, respectively.

(2)
$$\begin{tikzcd}
z \in \mathrm{L(S)} \arrow[r, "t"] \arrow[d, "t'"'] & \mathrm{L(I)} \arrow[r] \arrow[d] & \mathrm{L(S)} \ni x \arrow[d] \\
\mathrm{L(I')} \arrow[r] \arrow[d] & \mathrm{G(I\times I')} \arrow[r] \arrow[d] & \mathrm{G(I')} \arrow[d] \\
y \in \mathrm{L(S)} \arrow[r] & \mathrm{G(I)} \arrow[r] & \mathrm{G(S)} \quad .
\end{tikzcd}$$

Figure 36: The exact-sequence diagram defining $[x, y]$.

In such a diagram, take x, y as indicated and choose arbitrary lifts $\widetilde{x} \in L(I)$ of x and $y' \in L(I')$ of y. The commutator

$$\widetilde{x}\, y'\, \widetilde{x}^{-1} (y')^{-1} \quad \text{in } G(I \times I')$$

is independent of the chosen lifts and is the image of a uniquely determined element z as shown. One has $z = [x, y]$. [79]

Indeed, use the same letters x and y for their images under the canonical sections $L(S) \to L(I)$ and $L(S) \to L(I')$. Then

$$\widetilde{x} = xu, \qquad y' = yv, \qquad u, v \in L(S) = L(I) \cap L(I').$$

[78] Editor's note: the original has been expanded in what follows.

[79] Editor's note: the original said in a footnote that the author, at Gabriel's request, acknowledged that the exercise was not immediate, which was precisely why it had not been placed in the text. The details have been supplied here, taking account of the addition in 4.7.1; see also [DG70], §II.4, 4.2.

Since $L(I)$ and $L(I')$ are commutative,

$$\begin{aligned}\widetilde{x}\, y'\, \widetilde{x}^{-1}(y')^{-1} &= xuyvu^{-1}x^{-1}v^{-1}y^{-1}\\ &= xuyu^{-1}vx^{-1}v^{-1}y^{-1}\\ &= xyx^{-1}y^{-1}.\end{aligned}$$

This element maps to the unit in both $G(I)$ and $G(I')$, and therefore comes from the unique z indicated above. Finally, viewing y as an element of $L_{I'}(I')$, and x as an element of $L(S) \subset G(I')$, formula 4.7.1 $(*)$ gives

$$xyx^{-1} = \mathrm{Ad}(x)(y) = (\mathrm{id} + t' \,\mathrm{ad}(x))(y) = y + t'[x,y].$$

Thus $xyx^{-1}y^{-1} \in L(I')$ is the image of $z = [x,y] \in L(S)$.

Two properties are now apparent from this construction:

(i) the bracket is functorial in G. More precisely,

$$G \longmapsto \underline{\mathrm{Lie}}(G/S)$$

is a functor from good S-groups to good $\mathcal{O}_S$-modules equipped with an $\mathcal{O}_S$-bilinear composition law;

(ii) $[x,y] + [y,x] = 0$, since the diagram is symmetric about its main diagonal.[80]

Proposition 4.8. *Let F be a good $\mathcal{O}_S$-module. Under the identification*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

one has

$$\mathrm{Ad}(g) \cdot Y = g \circ Y \circ g^{-1}, \qquad [X,Y] = X \circ Y - Y \circ X,$$

for every $S' \to S$, $g \in \underline{\mathrm{Aut}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$, and

$$X, Y \in \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S)\,(S') = \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}).$$

Proof.[81] After base change, we may assume $S' = S$. Put $I = I_S$ and $\mathcal{O}_I = \mathcal{O}_S[t]$, with $t^2 = 0$. Recall from 4.5.1 that the inclusion

$$i : \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F) \hookrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_I\text{-mod.}}(F_I)$$

sends Y to $\mathrm{id} + tY$. By the definition of $\mathrm{Ad}(g)$ in 4.1.A,

$$\mathrm{id} + t\,\mathrm{Ad}(g)(Y) = g \circ (\mathrm{id} + tY) \circ g^{-1} = \mathrm{id} + t(g \circ Y \circ g^{-1}),$$

and therefore $\mathrm{Ad}(g)(Y) = g \circ Y \circ g^{-1}$.

Let I' be a second copy of I_S, and put $\mathcal{O}_{I'} = \mathcal{O}_S[t']$, with $(t')^2 = 0$. Apply 4.7.3 to

$$G = \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F), \qquad L = \underline{\mathrm{Lie}}(G/S) = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

Here $L = \mathrm{End}(F)$ is the differential identification of Proposition 4.5. The re-edition prints $\mathrm{Aut}(F)$ at this point, but $\mathrm{End}(F)$ is the type-correct object used by the ensuing endomorphism calculation. Identify X with its image under the canonical section $L(S) \hookrightarrow L(I)$. Its

[80] Editor's note: that is, x and y play symmetric roles. The image of $[y,x]$ in $G(I \times I')$ is the commutator $y'\widetilde{x}(y')^{-1}\widetilde{x}^{-1}$, the inverse of $\widetilde{x}y'\widetilde{x}^{-1}(y')^{-1} = [x,y]$. On the other hand (cf. 4.10 below), it is not known at this point whether $[x,x] = 0$ necessarily holds: if x is represented by $\widetilde{x} \in L(I)$ and by $x' \in L(I')$, it is not a priori clear that $\widetilde{x}$ and x' commute.

[81] Editor's note: the original has been expanded and simplified in what follows.

image in $G(I \times I')$ is $\mathrm{id} + t'X$, whose inverse is $\mathrm{id} - t'X$. Similarly, Y lifts to $\mathrm{id} + tY$, with inverse $\mathrm{id} - tY$. Their commutator is

$$\begin{aligned}(\mathrm{id} + t'X) \circ (\mathrm{id} + tY) \circ (\mathrm{id} - t'X) \circ (\mathrm{id} - tY)\\ = \mathrm{id} + tt'(X \circ Y - Y \circ X).\end{aligned}$$

This is the image in $G(I \times I')$ of $Z = X \circ Y - Y \circ X \in L(S)$: first Z maps to $tZ \in L(I)$, and then to $\mathrm{id} + t'tZ \in G(I \times I')$. By 4.7.3, $[X, Y] = X \circ Y - Y \circ X$.

Corollary 4.8.1. *Let G be a good S-group and let $x, y, z \in \underline{\mathrm{Lie}}(G/S)(S')$. Then*

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

[82] Indeed, since G is good, $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module; hence Theorem 4.7 shows that

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S))$$

is a good S-group. The morphism

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S))$$

and the functoriality in 4.7.3(i) yield

$$\mathrm{ad}[x, y] = [\mathrm{ad}\, x, \mathrm{ad}\, y].$$

Together with Proposition 4.8, this gives

$$\mathrm{ad}[x, y] = \mathrm{ad}\, x \circ \mathrm{ad}\, y - \mathrm{ad}\, y \circ \mathrm{ad}\, x.$$

Applying both sides to z gives the Jacobi identity.

Corollary 4.8.2. *Let G be a good S-group acting linearly on a good $\mathcal{O}_S$-module F; equivalently, let F be a G-$\mathcal{O}_S$-module, with G and F good. Then the linear map*

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a representation; that is,

$$\rho'([x, y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholium 4.9. To every good S-group G (for example, every representable one) we have functorially associated a good $\mathcal{O}_S$-module $\underline{\mathrm{Lie}}(G/S)$ equipped with a bilinear map satisfying

$$[x, y] + [y, x] = 0, \qquad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

We shall call $\underline{\mathrm{Lie}}(G/S)$, with this structure, the *Lie algebra* of G over S. The qualification is that it is not yet known whether it is an $\mathcal{O}_S$-Lie algebra in the strict sense.[83] Every linear representation of G in a good $\mathcal{O}_S$-module F induces a representation of its Lie algebra. In particular, the adjoint representation of G induces the adjoint representation of its Lie algebra.

Definition 4.10. An S-functor in groups G is called *very good* if it is good and $\underline{\mathrm{Lie}}(G/S)$ is an $\mathcal{O}_S$-Lie algebra, that is, if $[x, x] = 0$ identically.

Examples 4.10.1. The following S-groups are very good: $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ for every good $\mathcal{O}_S$-module F (by 4.7 and 4.8); every representable group (as shown below); every good S-group admitting a monomorphism into a very good S-group, in particular every good subgroup functor of a representable group; and every good S-group admitting a faithful linear representation in a good $\mathcal{O}_S$-module, for example every good S-group for which Ad is a monomorphism.

[82] Editor's note: the original has been expanded in what follows.

[83] Editor's note: the unresolved point is whether $[x, x] = 0$; see 4.10.

4.11.

Suppose now that G is a group scheme over S. By 4.1.4, $\underline{\mathrm{Lie}}(G/S)(S)$ identifies with the group of infinitesimal automorphisms of G/S invariant under right translation; by 3.14, this is the group of derivations of $\mathcal{O}_G$ over $\mathcal{O}_S$ invariant under right translation. This identification respects the module structure and is an anti-isomorphism of Lie algebras.[84]

To see this, argue as in 4.7.3. Put

$$\mathcal{O}_I = \mathcal{O}_S[t], \qquad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \qquad t^2 = 0 = (t')^2,$$

and let $x \in L(I)$, $y \in L(I')$, where $L = \underline{\mathrm{Lie}}(G/S)$. Left translation λ_x, respectively λ_y, is an S-automorphism of $G_{I\times I'}$ inducing the identity on $G_{I'}$, respectively on G_I. It corresponds to the $\mathcal{O}_S$-automorphism

$$u = \mathrm{id} + t\, d_x, \qquad \text{respectively} \qquad v = \mathrm{id} + t'\, d_y$$

of

$$\mathcal{O}_{G_{I\times I'}} = \mathcal{O}_G[t,t']/(t^2,(t')^2),$$

where d_x, d_y are $\mathcal{O}_S$-derivations of $\mathcal{O}_G$ invariant under right translation. The correspondence between S-automorphisms of $G_{I\times I'}$ and $\mathcal{O}_S$-automorphisms of $\mathcal{O}_{G_{I\times I'}}$ is contravariant. Therefore

$$\lambda_x\lambda_y\lambda_x^{-1}\lambda_y^{-1}$$

corresponds to

$$v^{-1}u^{-1}vu = \mathrm{id} + tt'(d_y d_x - d_x d_y).$$

It follows from 4.7.3 that

$$x \longmapsto -d_x$$

is an isomorphism of Lie algebras; for further details, see [DG70], §II.4, 4.4 and 4.6. The same argument applies to

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S')$$

for every $S' \to S$. Thus one recovers the classical definition.[85]

Scholium 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $\underline{\mathrm{Lie}}(G/S)$ identifies with the functor that assigns to each $S' \to S$ the $\mathcal{O}(S')$-Lie algebra of derivations of $G_{S'}$ relative to S' invariant under right translation.*

Since every representable group is already known to be good by 4.6.1, the preceding discussion gives:

Corollary 4.11.2. *Every representable group is very good.*

Let $\varepsilon : S \to G$ be the unit section of G, and put

$$\omega^1_{G/S} = \varepsilon^*(\Omega^1_{G/S}).$$

Recall from 3.3 that $\underline{\mathrm{Lie}}(G/S)$ is represented by the vector fibration $V(\omega^1_{G/S})$.

Scholium 4.11.3. *Thus to every S-group scheme G we have functorially associated a vector fibration*

$$\mathrm{Lie}(G/S) = V(\omega^1_{G/S})$$

[84]Editor's note: a sign error in the original has been corrected here; cf. [DG70], §II.4, 4.4 and 4.6.

[85]Editor's note: the original has been expanded in what follows. In particular, the numbering 4.11.1–4.11.8 has been added for later reference.

over S, representing the functor $\underline{\mathrm{Lie}}(G/S)$, and hence endowed with the structure of an S-scheme in $\mathcal{O}_S$-Lie algebras. Moreover, this construction commutes with base extension and finite products (cf. 3.4 and 3.8).

Remarks 4.11.4. [86] Let $\pi : G \to S$ be the structural morphism.

(a) The $\mathcal{O}_G$-module $\Omega^1_{G/S}$ is naturally $(G\times_S G)$-equivariant (cf. I, §6). Hence, by I 6.8.1,

$$\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S}).$$

For example, $\Omega^1_{G/S}$ is locally free, respectively locally free of finite type, whenever $\omega^1_{G/S}$ is. This holds in particular when S is the spectrum of a field, respectively when S is the spectrum of a field and G is locally of finite type over S.

(b) By I 6.8.2, $\omega^1_{G/S}$ has a canonical G-$\mathcal{O}_S$-module structure, which induces the adjoint action on

$$V(\omega^1_{G/S}) = \mathrm{Lie}(G/S).$$

[87]

(c) By EGA I, 5.3.11 and 5.4.6, $\varepsilon : S \to G$ is an immersion; it is a closed immersion if G is separated over S. Thus $\omega^1_{G/S}$ identifies with $\mathcal{I}/\mathcal{I}^2$, where $\mathcal{I}$ is the quasi-coherent ideal defining $\varepsilon(S)$ in an open subscheme $U \subset G$ in which $\varepsilon(S)$ is closed. If $G \to S$ is separated, one may take $U = G$. If $G = \operatorname{Spec} A(G)$ is affine over S, then $\mathcal{I}$ is the augmentation ideal of $A(G)$, namely the kernel of

$$\varepsilon^\natural : A(G) \longrightarrow \mathcal{O}_S$$

(cf. I 4.2).[88]

Remark 4.11.5. The isomorphism $\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S})$ identifies the $\mathcal{O}_S$-module $\omega^1_{G/S}$ with the sheaf

$$\pi_*(\Omega^1_{G/S})^G$$

of differentials of G relative to S invariant under right translation: its sections over an open $U \subset S$ are the sections of $\Omega^1_{G/S}$ over $\pi^{-1}(U)$ invariant under right translation (cf. I 6.8.3; compare also $\mathrm{VII_A}$, 2.4).[89]

Notation 4.11.6. Write $\mathscr{L}ie(G/S)$ for the sheaf of sections of the vector fibration $\mathrm{Lie}(G/S) \to S$. It is the $\mathcal{O}_S$-module

$$\mathscr{L}ie(G/S) = (\omega^1_{G/S})^\vee = \mathcal{H}om_{\mathcal{O}_S}(\omega^1_{G/S}, \mathcal{O}_S),$$

dual to $\omega^1_{G/S}$ (cf. EGA II, 1.7.9), and it carries a structure of $\mathcal{O}_S$-Lie algebra.

In general this construction does not commute with base extension. Consequently, the Lie-algebra structure on this module does not determine the structure of an S-scheme in $\mathcal{O}_S$-Lie algebras on $\mathrm{Lie}(G/S)$.[90]

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[86] Editor's note: what follows has been expanded, taking account of the additions made in Expose I, §6.8.

[87] Editor's note: for this reason, the linear action of G on $\omega^1_{G/S}$ could be called the pre-adjoint action; by an abuse of terminology it will still be called the adjoint action. A slightly more general construction, used in Expose III (especially III 0.8), is as follows. Suppose that to every $Y \to S$ one assigns an $\mathcal{O}_Y$-module $\mathcal{N}(Y)$, and to every S-morphism $\phi : Z \to Y$ a functorial morphism of $\mathcal{O}_Z$-modules $\mathcal{N}(\phi) : \phi^*\mathcal{N}(Y) \to \mathcal{N}(Z)$. For example, if $\mathcal{J}$ is a quasi-coherent ideal of $\mathcal{O}_S$, one may take $\mathcal{N}(Y) = \mathcal{J}\mathcal{O}_Y$. The action of G on $\omega^1_{G/S}$ then induces a generalized adjoint action of G on the S-functor in abelian groups that assigns to $f : Y \to S$ the group $\mathrm{Hom}_{\mathcal{O}_Y}(f^*\omega^1_{G/S}, \mathcal{N}(Y))$.

[88] Editor's note: the preceding description of $\omega^1_{G/S}$, which is useful later (for example in $\mathrm{VII_A}$, 5.5), has been added.

[89] Editor's note: this description of $\omega^1_{G/S}$ in terms of invariant differentials is not used below.

[90] Editor's note: the reason is that $\mathscr{L}ie(G/S) \simeq (\omega^1_{G/S})^\vee$ need not determine $\omega^1_{G/S}$. For example, let $S = \mathbb{A}^1_k$, where k is a field, and let G be the S-group whose fiber at $s = 0$ is $\mathbb{G}_{a,k}$, while all other fibers are the unit group. Then $\mathscr{L}ie(G/S) = 0$, but $\mathrm{Lie}(G_s/k)(k) = k$.

Nevertheless:

Lemma 4.11.7. *Suppose that $\omega^1_{G/S}$ is locally free of finite type. This is the case, in particular, if G is smooth over S (cf. EGA IV$_4$, 17.2.4), or if S is the spectrum of a field and G is locally of finite type over S. Then*

$$\mathscr{L}ie(G/S)^\vee \simeq (\omega^1_{G/S})^{\vee\vee} \simeq \omega^1_{G/S},$$

and therefore

$$\mathrm{Lie}(G/S) = V(\omega^1_{G/S}) = V(\mathscr{L}ie(G/S)^\vee) = W(\mathscr{L}ie(G/S)),$$

where the final equality follows from I 4.6.5.1.

Finally, let $G \to H$ be a monomorphism of functors in groups. Then

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

is also a monomorphism (cf. 3.7). Every vector monomorphism of vector fibrations is a closed immersion.[91] It follows that:

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S-group schemes.*

(i) The morphism

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

is a closed immersion; consequently

$$\omega^1_{H/S} \longrightarrow \omega^1_{G/S}$$

is an epimorphism.

(ii) If $\omega^1_{G/S}$ is locally free of finite type, then the corresponding morphism

$$\mathscr{L}ie(G/S) \longrightarrow \mathscr{L}ie(H/S)$$

identifies $\mathscr{L}ie(G/S)$ with an $\mathcal{O}_S$-submodule of $\mathscr{L}ie(H/S)$ that is locally a direct summand.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: diagonalizable groups

Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). Since the formation of $\underline{\mathrm{Lie}}(G/S)$ commutes with base extension, it is enough to carry out the construction for $G = D(M)$. We then have

$$G(I_S) = \mathrm{Hom}_{\mathrm{gr}}(M, \Gamma(I_S, \mathscr{O}_{I_S})^\times) = \mathrm{Hom}_{\mathrm{gr}}(M, \Gamma(S, \mathscr{D}_{\mathscr{O}_S})^\times).$$

There is a split exact sequence

$$1 \longrightarrow \Gamma(S, \mathscr{O}_S) \longrightarrow \Gamma(S, \mathscr{D}_{\mathscr{O}_S})^\times \longrightarrow \Gamma(S, \mathscr{O}_S)^\times \longrightarrow 1,$$

which shows that $\underline{\mathrm{Lie}}(G)(S)$ identifies with $\mathrm{Hom}_{\mathrm{gr}}(M, \mathscr{O}(S))$, endowed with its evident $\mathscr{O}(S)$-module structure. After base change we therefore obtain:

[91] Editor's note: let $f : M \to N$ be a morphism of $\mathcal{O}_S$-modules and let $P = \mathrm{Coker}(f)$. If $V(N) \to V(M)$ is a monomorphism, the surjective morphism $\mathrm{Sym}(N) \to \mathrm{Sym}(P)$ factors through $\mathcal{O}_S$; hence $P = 0$.

Proposition 5.1. *There are isomorphisms*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, \mathbf{O}_S) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S)$$

and

$$\mathscr{H}om_{\mathrm{gr}}(\widetilde{M}_S, \mathscr{O}_S) \xrightarrow{\sim} \mathscr{L}ie(D_S(M)/S),$$

where, in the second isomorphism, $\widetilde{M}_S$ denotes the constant sheaf of groups on S defined by M, and $\mathscr{H}om_{\mathrm{gr}}$ denotes the sheaf of homomorphisms of sheaves of groups.

Corollary 5.1.1. *If M is free of finite type—or, as we shall say later, if $D_S(M)$ is a split torus—then, with W as in I 4.6.5,*

$$W(\mathscr{L}ie(D_S(M)/S)) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S), \qquad M^\vee \otimes_{\mathbb{Z}} \mathscr{O}_S \xrightarrow{\sim} \mathscr{L}ie(D_S(M)/S).$$

Here $M^\vee$ denotes the dual of the abelian group M. In particular,

$$\mathbf{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathbb{G}_{m,S}/S), \qquad \mathscr{O}_S \xrightarrow{\sim} \mathscr{L}ie(\mathbb{G}_{m,S}/S).$$

5.2. Normalizers and centralizers

We first prove a few lemmas. Recall (I 3.1.1) that a sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

of $\mathbf{O}_S$-modules is called *exact* if, for every $S' \to S$, the sequence

$$0 \longrightarrow F'(S') \longrightarrow F(S') \longrightarrow F''(S') \longrightarrow 0$$

of $\mathbf{O}(S')$-modules is exact. Likewise, a sequence $1 \to G' \to G \to G'' \to 1$ of S-groups is called exact if the corresponding sequence of groups on S'-points is exact for every $S' \to S$.

Lemma 5.2.1. *Let $1 \to G' \to G \to G'' \to 1$ be an exact sequence of S-groups.* [92]

(i) *The sequences*

$$1 \longrightarrow T_{G'/S}(\mathscr{M}) \longrightarrow T_{G/S}(\mathscr{M}) \longrightarrow T_{G''/S}(\mathscr{M}) \longrightarrow 1$$

and

$$1 \longrightarrow \underline{\mathrm{Lie}}(G'/S, \mathscr{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathscr{M}) \longrightarrow \underline{\mathrm{Lie}}(G''/S, \mathscr{M}) \longrightarrow 1$$

are exact.

(ii) *Let $1 \to H' \to H \to H'' \to 1$ be a second sequence of groups. It is exact if and only if*

$$1 \longrightarrow G' \times_S H' \longrightarrow G \times_S H \longrightarrow G'' \times_S H'' \longrightarrow 1$$

is exact.

(iii) *If two of G', G, G'' satisfy condition (E), then so does the third.*

(iv) *If $0 \to F' \to F \to F'' \to 0$ is exact and two of the $\mathbf{O}_S$-modules F', F, F'' are good, then so is the third.*

(v) *If two of the S-groups G', G, G'' are good, then so is the third.*

[92] Editor's note: The statement and proof of this lemma have been expanded; this will be useful in 5.3.1.

The first part of (i) is immediate, and the second follows from it. Likewise, (ii) is immediate. We prove (iii). For brevity, write $L(\mathscr{M}) = \underline{\mathrm{Lie}}(G, \mathscr{M})$, $L'(\mathscr{M}) = \underline{\mathrm{Lie}}(G', \mathscr{M})$, and similarly for L''. We have the following commutative diagram. [93]

$$\begin{tikzcd}
1 \arrow[r] & \mathrm{L}'(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & \mathrm{L}(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & \mathrm{L}''(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathrm{L}'(\mathscr{M}) \underset{\sim}{\times} \mathrm{L}'(\mathscr{N}) \arrow[r] & \mathrm{L}(\mathscr{M}) \underset{\sim}{\times} \mathrm{L}(\mathscr{N}) \arrow[r] & \mathrm{L}''(\mathscr{M}) \underset{\sim}{\times} \mathrm{L}''(\mathscr{N}) \arrow[r] & 1
\end{tikzcd}$$

Figure 37: The exact-sequence diagram used to prove Lemma 5.2.1(iii).

The upper row is exact by (i), and the lower row is exact by (i) and (ii). Assertion (iii) follows.

We prove (iv). There is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{F}' \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F}'' \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{T}_{\mathrm{F}'/\mathrm{S}}(\mathscr{M}) \arrow[r] & \mathrm{T}_{\mathrm{F}/\mathrm{S}}(\mathscr{M}) \arrow[r] & \mathrm{T}_{\mathrm{F}''/\mathrm{S}}(\mathscr{M}) \arrow[r] & 0
\end{tikzcd}$$

Figure 38: The exact-sequence diagram used to prove Lemma 5.2.1(iv).

with exact rows: the first because $T_{\mathbf{O}_S/S}(\mathscr{M})$ is a free, hence flat, $\mathbf{O}_S$-module, and the second by (i). Thus, if two of F', F, F'' are good, so is the third. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let G be an S-group, let E, H be two G-objects, and let F be a G-$\mathbf{O}_S$-module.*

(i) *The canonical homomorphism $E^G \times_S H^G \to (E \times_S H)^G$ is an isomorphism.*

(ii) *If F is a good $\mathbf{O}_S$-module, then so is F^G.*

The first assertion is immediate; we prove the second. For every $S' \to S$, there is a commutative diagram

$$\begin{tikzcd}
\mathrm{F}^\mathrm{G}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[rr, hook] & & \mathrm{F}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \\
\mathrm{F}^\mathrm{G}(\mathrm{S}') \otimes_{\mathbf{O}(\mathrm{S}')} \mathbf{O}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[u, "\phi"] \arrow[r, hook] & \mathrm{F}(\mathrm{S}') \otimes_{\mathbf{O}(\mathrm{S}')} \mathbf{O}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[ur, "\phi_1", "\simeq"'] &
\end{tikzcd}$$

Figure 39: The invariants-and-base-change square in the proof of Lemma 5.2.2.

[93]Editor's note: The proofs of (iii) and (v) have been added.

and we must prove that ϕ is bijective. It is plainly injective. To prove surjectivity, let $(t_0, \dots, t_n)$ be a basis of the free $\mathbf{O}(S')$-module $\mathbf{O}(I_{S'}(\mathscr{M}))$, and let

$$u = \sum_{i=0}^{n} f_i \otimes t_i$$

be an element of $F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathscr{M}))$ such that $\phi_1(u)$ belongs to $F^G(I_{S'}(\mathscr{M}))$. We show that each f_i belongs to $F^G(S')$.

Let $S'' \to S'$. Via the zero section $\varepsilon_{\mathscr{M}}$, we may regard S'' as lying over $I_{S'}(\mathscr{M})$. Then, for every $g \in G(S'')$,

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^{n} g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

Since the $(t_i)_{S''}$ form a basis of

$$\mathbf{O}(I_{S''}(\mathscr{M})) \simeq \mathbf{O}(I_{S'}(\mathscr{M})) \otimes_{\mathbf{O}(S')} \mathbf{O}(S'')$$

over $\mathbf{O}(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, hence $f_i \in F^G(S')$ for all i.

Notation. If E is an S-group and F a sub-S-group, write E/F for the S-functor sending $S' \to S$ to the set $E(S')/F(S')$ of classes $\bar{x} = xF(S')$. If E is commutative, then E/F is a commutative S-group. [94]

Now let G be an S-group and K a sub-S-group, and set $E = \underline{\mathrm{Lie}}(G/S, \mathscr{M})$ and $F = \underline{\mathrm{Lie}}(K/S, \mathscr{M})$. The adjoint action of K on E stabilizes F, and therefore induces an action on E/F. For every $S' \to S$,

$$\begin{aligned}(E/F)^K(S') = \{\bar{x} \in E(S')/F(S') \mid \text{for every } f : S'' \to S' \text{ and } k \in K(S''),\\ f^*(\bar{x}^{-1}) \operatorname{Ad}(k)(f^*(\bar{x})) \in F(S'')\},\end{aligned}$$

where $f^*(\bar{x})$ denotes the image of $\bar{x}$ in $E(S'')$.

Theorem 5.2.3. *Let G be an S-group and K a sub-S-group. Put (I 2.3.3)*

$$N = \underline{\mathrm{Norm}}_G(K), \qquad Z = \underline{\mathrm{Centr}}_G(K),$$

and let K act on $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ through the adjoint representation of G.

(i) *If the group law of $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ is commutative,*[95][96] *then*

$$\frac{\underline{\mathrm{Lie}}(N/S, \mathscr{M})}{\underline{\mathrm{Lie}}(K/S, \mathscr{M})} = \left(\frac{\underline{\mathrm{Lie}}(G/S, \mathscr{M})}{\underline{\mathrm{Lie}}(K/S, \mathscr{M})}\right)^K.$$

(ii) *If the same group law is commutative, then*

$$\underline{\mathrm{Lie}}(Z/S, \mathscr{M}) = \underline{\mathrm{Lie}}(G/S, \mathscr{M})^K.$$

(iii) *If G satisfies (E), then Z satisfies (E); if G and K satisfy (E), then N satisfies (E).*

(iv) *If G is good, then Z is good; if, moreover, G is very good, then Z is very good.*

(v) *If G and K are good, then N is good; if, moreover, G is very good, then N is very good.*

[94] Editor's note: These notations have been expanded.

[95] This condition is automatic when G satisfies (E), for example when G is representable.

[96] Editor's note: For an example in which the S-group $\underline{\mathrm{Lie}}(G/S)$ is noncommutative, see 6.3.

To prove (i) and (ii),[97] we use the following lemma, which follows from the diagram of exact sequences considered in 4.1.B, with G and H interchanged.

Lemma 5.2.3.0. *Let G be an S-group, H a sub-S-group, and $\mathscr{M}$ a free $\mathscr{O}_S$-module of finite type. Then $T_{H/S}(\mathscr{M})$ and $\underline{\mathrm{Lie}}(G/S,\mathscr{M})$ are sub-S-groups of $T_{G/S}(\mathscr{M})$, and*

$$T_{H/S}(\mathscr{M}) \cap \underline{\mathrm{Lie}}(G/S,\mathscr{M}) = \underline{\mathrm{Lie}}(H/S,\mathscr{M}),$$

where

$$T_{H/S}(\mathscr{M}) \cap \underline{\mathrm{Lie}}(G/S,\mathscr{M}) := T_{H/S}(\mathscr{M}) \times_{T_{G/S}(\mathscr{M})} \underline{\mathrm{Lie}}(G/S,\mathscr{M}).$$

Since the functors in (i) and (ii) commute with base extension, it is enough to prove the equalities on S-points. Put $H = N$ (respectively, $H = Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definitions of N and Z (I 2.3.3),

$$\begin{aligned} \underline{\mathrm{Lie}}(H/S,\mathscr{M})(S) = \{X \in \underline{\mathrm{Lie}}(G/S,\mathscr{M})(S) \subset G(I_S(\mathscr{M}))\ \mid\ & \\ &\text{for every } f : S' \to I_S(\mathscr{M}) \text{ and } u \in K(S'),\\ & f^*(X^{\alpha})\, u\, f^*(X^{-\alpha})\, u^{-1} \in K(S'),\\ & \text{respectively } f^*(X)\, u\, f^*(X^{-1})\, u^{-1} = 1\}. \end{aligned} \tag{*}$$

Here $f^*(X)$ denotes the image of X in $G(S')$.

For brevity, write

$$\mathfrak{g} = \underline{\mathrm{Lie}}(G/S,\mathscr{M}), \quad \mathfrak{k} = \underline{\mathrm{Lie}}(K/S,\mathscr{M}), \quad \mathfrak{n} = \underline{\mathrm{Lie}}(N/S,\mathscr{M}), \quad \mathfrak{z} = \underline{\mathrm{Lie}}(Z/S,\mathscr{M}).$$

If $X \in \underline{\mathrm{Lie}}(H/S,\mathscr{M})(S)$, the equalities (*) hold for every $f : S' \to S$, because f factors through $I_S(\mathscr{M})$ as

$$f = \rho \circ \varepsilon_{\mathscr{M}} \circ f,$$

where $\rho : I_S(\mathscr{M}) \to S$ is the structural morphism and $\varepsilon_{\mathscr{M}} : S \to I_S(\mathscr{M})$ is the zero section. Consequently,

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S), \qquad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

For the reverse inclusions, assume henceforth that $\mathfrak{g}$ is commutative. Then $\mathfrak{g}^K$ and $(\mathfrak{g}/\mathfrak{k})^K$ are commutative S-groups. Let $X \in \mathfrak{g}(S)$, and write $\bar{X}$ for its image in $(\mathfrak{g}/\mathfrak{k})(S)$. Suppose that $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$ (respectively, $X \in \mathfrak{g}^K(S)$); we show that $X \in \mathfrak{n}(S)$ (respectively, $X \in \mathfrak{z}(S)$).

Let $f : S' \to I_S(\mathscr{M})$. Consider the cartesian square

$$\begin{tikzcd} I_{S'}(\mathscr{M}) \arrow[r, "p"] \arrow[d, "\rho'"'] & I_S(\mathscr{M}) \arrow[d, "\rho"] \\ S' \arrow[r, "\rho\circ f"] & S \end{tikzcd}$$

Figure 40: The cartesian square used in the reverse-inclusion argument.

[97]Editor's note: The proof has been expanded, and in particular Lemma 5.2.3.0, implicit in the original, has been added.

and let h be the section of ρ' defined by f. It is enough to show that, for every $v \in K(I_{S'}(\mathscr{M}))$,

$$\left.\begin{aligned} p^*(X^\alpha)\, v\, p^*(X^{-\alpha})\, v^{-1} &\in K(I_{S'}(\mathscr{M})), \\ \text{respectively} \qquad p^*(X)\, v\, p^*(X^{-1})\, v^{-1} &= 1. \end{aligned}\right\} \tag{**}$$

Indeed, take $v = \rho'^*(u)$ and apply h^* to (**). This gives (*), because $\rho' \circ h = \mathrm{id}_{S'}$ and $p \circ h = f$.

We now prove (**), writing X for $p^*(X)$. Every $v \in K(I_{S'}(\mathscr{M}))$ is uniquely of the form Yk, with $Y \in \underline{\mathrm{Lie}}(K/S, \mathscr{M})(S')$ and $k \in K(S')$. The expression $X^\alpha v X^{-\alpha} v^{-1}$ becomes

$$X^\alpha Y (k X^{-\alpha} k^{-1}) Y^{-1} = X^\alpha \operatorname{Ad}(k)(X^{-\alpha}),$$

because $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ is commutative. This element lies in $\underline{\mathrm{Lie}}(G/S, \mathscr{M})(S')$. By Lemma 5.2.3.0, condition (**) is therefore equivalent, for every $k \in K(S')$, to

$$\begin{cases} \operatorname{Ad}(k)(X) = X, & H = Z, \\ X^\alpha \operatorname{Ad}(k)(X^{-\alpha}) \in \underline{\mathrm{Lie}}(K/S, \mathscr{M})(S'), & H = N. \end{cases}$$

For $H = Z$, this follows from $X \in \mathfrak{g}^K(S)$. For $H = N$, the condition may equally be written

$$\operatorname{Ad}(k)(\bar{X}) = \bar{X}, \qquad \operatorname{Ad}(k)(\bar{X}^{-1}) = \bar{X}^{-1}. \tag{*'}$$

The second equality follows from the first because the action of K on $\mathfrak{g}/\mathfrak{k}$ respects its group structure. Thus (**) also follows from $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$, proving (i) and (ii).

For (iii)–(v), put

$$\mathfrak{g}(\mathscr{M}) = \underline{\mathrm{Lie}}(G/S, \mathscr{M})$$

and define $\mathfrak{k}(\mathscr{M})$, $\mathfrak{z}(\mathscr{M})$, and $\mathfrak{n}(\mathscr{M})$ similarly. If G/S satisfies (E), then

$$\mathfrak{g}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{g}(\mathscr{M}) \times_S \mathfrak{g}(\mathscr{N}),$$

and hence, by Lemma 5.2.2(i),

$$\mathfrak{g}(\mathscr{M} \oplus \mathscr{N})^K \simeq \mathfrak{g}(\mathscr{M})^K \times_S \mathfrak{g}(\mathscr{N})^K.$$

It follows that $\mathfrak{z}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{z}(\mathscr{M}) \times_S \mathfrak{z}(\mathscr{N})$, so Z satisfies (E). If K/S also satisfies (E), then

$$(\mathfrak{g}/\mathfrak{k})(\mathscr{M} \oplus \mathscr{N}) \simeq (\mathfrak{g}/\mathfrak{k})(\mathscr{M}) \times_S (\mathfrak{g}/\mathfrak{k})(\mathscr{N}),$$
$$(\mathfrak{g}/\mathfrak{k})^K(\mathscr{M} \oplus \mathscr{N}) \simeq (\mathfrak{g}/\mathfrak{k})(\mathscr{M})^K \times_S (\mathfrak{g}/\mathfrak{k})(\mathscr{N})^K,$$

and we have the following commutative diagram.

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{k}(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d, "\wr"] & \mathfrak{n}(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & (\mathfrak{n}/\mathfrak{k})(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d, "\wr"] & 0 \\
0 \arrow[r] & \mathfrak{k}(\mathscr{M}) \underset{S}{\times} \mathfrak{k}(\mathscr{N}) \arrow[r] & \mathfrak{n}(\mathscr{M}) \underset{S}{\times} \mathfrak{n}(\mathscr{N}) \arrow[r] & (\mathfrak{n}/\mathfrak{k})(\mathscr{M}) \underset{S}{\times} (\mathfrak{n}/\mathfrak{k})(\mathscr{N}) \arrow[r] & 0
\end{tikzcd}$$

Figure 41: The exact-sequence comparison proving condition (E) for N.

It follows that

$$\mathfrak{n}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{n}(\mathscr{M}) \times_S \mathfrak{n}(\mathscr{N}),$$

so N satisfies (E).

Now write $\mathfrak{g} = \mathfrak{g}(\mathscr{O}_S) = \underline{\mathrm{Lie}}(G/S)$, and similarly $\mathfrak{k}, \mathfrak{z}, \mathfrak{n}$. If G is good, then $\mathfrak{g}$ is a good $\mathbf{O}_S$-module, and Lemma 5.2.2(ii) shows that $\mathfrak{z} \simeq \mathfrak{g}^K$ is good; since Z satisfies (E), Z is good. If K is also good, then $\mathfrak{k}$ is good, and Lemmas 5.2.1(iv) and 5.2.2(ii) show that $\mathfrak{g}/\mathfrak{k}$ and $(\mathfrak{g}/\mathfrak{k})^K$ are good. From the exact sequence

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{n} \longrightarrow (\mathfrak{g}/\mathfrak{k})^K \longrightarrow 0$$

and Lemma 5.2.1(iv), $\mathfrak{n}$ is good. Finally, if G is very good, so that $[x, x] = 0$ identically for $x \in \mathfrak{g}$, then Z and N are plainly very good. This proves (iii)–(v).

Corollary 5.2.3.1. *If the group law of $\underline{\mathrm{Lie}}(G/S)$ is commutative, then*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Centr}}(G)/S) = \underline{\mathrm{Lie}}(G/S)^G.$$

Corollary 5.2.3.2. *If the group law of $\underline{\mathrm{Lie}}(G/S)$ is commutative and K is a normal subgroup of G, then*

$$(\underline{\mathrm{Lie}}(G/S)/\underline{\mathrm{Lie}}(K/S))^K = \underline{\mathrm{Lie}}(G/S)/\underline{\mathrm{Lie}}(K/S).$$

5.3. Linear representations

Let G be a good S-group acting linearly on a good $\mathbf{O}_S$-module F through

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S\text{-mod}}(F).$$

In 4.7.1 and 4.8.2 we defined the corresponding linear representation

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathbf{O}_S\text{-mod}}(F).$$

The sub-S-groups $\underline{\mathrm{Norm}}_G(E)$ and $\underline{\mathrm{Centr}}_G(E)$ are defined for every subset E of F, and in particular for every $\mathbf{O}_S$-submodule E.

Definition 5.3.0. For every $S' \to S$, put

$$\begin{aligned}\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') &= \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} \subset E_{S'}\},\\ \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') &= \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} = 0\}.\end{aligned}$$

This construction works for every linear representation of an $\mathbf{O}_S$-Lie algebra in the sense of 4.9, and the two subobjects obtained are $\mathbf{O}_S$-submodules stable under the bracket.

Theorem 5.3.1. *Let G be a good S-group acting linearly on a good $\mathbf{O}_S$-module F, and let E be an $\mathbf{O}_S$-submodule of F.*

(i) *One has*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Centr}}_G(E)/S) = \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E),$$

and $\underline{\mathrm{Centr}}_G(E)$ is a good S-group; it is very good if G is very good.

(ii) *Suppose that E is a good $\mathbf{O}_S$-module. Then*[98]

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Norm}}_G(E)/S) = \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E),$$

and $\underline{\mathrm{Norm}}_G(E)$ is a good S-group; it is very good if G is very good.

[98] Editor's note: The original stated the following equality without a hypothesis on E; this has been corrected. See 6.5.1 for counterexamples.

The proof is left to the reader.

5.3.2. Let G be a good S-group. The preceding results apply in particular to the adjoint representation. Let E be a good $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G/S)$.[99] It has a centralizer and a normalizer in G, and by 5.3.1 their Lie algebras are the usual centralizer and normalizer of E, computed using the bracket:

$$\begin{aligned}\underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') &= \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] = 0\},\\ \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') &= \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] \subset E_{S'}\}.\end{aligned}$$

5.3.3. Let K be a sub-S-group of G. Then $\underline{\mathrm{Lie}}(K/S)$ is an $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G/S)$. Suppose that it is a good $\mathbf{O}_S$-module,[100] as happens when K is a good S-group. Plainly,

$$\underline{\mathrm{Centr}}_G(K) \subset \underline{\mathrm{Centr}}_G(\underline{\mathrm{Lie}}(K/S)),$$

$$\underline{\mathrm{Norm}}_G(K) \subset \underline{\mathrm{Norm}}_G(\underline{\mathrm{Lie}}(K/S)).$$

Hence, by 5.3.1,

$$\begin{aligned}\underline{\mathrm{Lie}}(\underline{\mathrm{Centr}}_G(K)/S) &\subset \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(\underline{\mathrm{Lie}}(K/S)),\\ \underline{\mathrm{Lie}}(\underline{\mathrm{Norm}}_G(K)/S) &\subset \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(\underline{\mathrm{Lie}}(K/S)).\end{aligned}$$

None of these four inclusions is an equality in general; many examples will appear later. In particular, if K is a normal subgroup of G, then $\underline{\mathrm{Lie}}(K/S)$ is an ideal of $\underline{\mathrm{Lie}}(G/S)$.

6. Miscellaneous remarks

6.1.

The bracket of two infinitesimal automorphisms can be defined for an S-functor X which is not necessarily a group: apply the results of this exposé to $\underline{\mathrm{Aut}}_S(X)$. To obtain a convenient formalism, one is led to assume that X is good, meaning that the $\mathbf{O}_X$-module $T_{X/S}$ is good. If X is an S-group, this agrees with Definition 4.6.

6.2.

There exist functors with infinitesimal endomorphisms that are not automorphisms and hence, a fortiori, do not satisfy condition (E). [101] For a pointed set (E, x_0), let $M_A(E)$ be the free abelian monoid generated by E, and let $M_{A,P}(E, x_0)$ be the quotient of $M_A(E)$ by the equivalence relation generated by $m \sim x_0 + m$. The assignment

$$(E, x_0) \longmapsto M_{A,P}(E, x_0)$$

is left adjoint to the forgetful functor from abelian monoids to pointed sets; $M_{A,P}(E, x_0)$ is the free abelian monoid on the pointed set (E, x_0).

Now let X be the functor assigning to every scheme S the free abelian monoid on $\mathbf{O}(S)$, pointed by zero. Every morphism

$$f : S \longrightarrow I_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[\varepsilon])$$

[99] Editor's note: This hypothesis has been added; cf. 6.5.1.

[100] Editor's note: This hypothesis has been added; cf. 6.5.1.

[101] Editor's note: The original has been expanded here. In particular, O. Gabber pointed out the correct definition of the free abelian monoid on a pointed set.

corresponds to a square-zero element $u_f \in \mathbf{O}(S)$, and hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$, where the sum is taken in $M_{A,P}(\mathbf{O}(S), 0)$. Thus we obtain an endomorphism ϕ of

$$X_{I_\mathbb{Z}} = X \times_\mathbb{Z} I_\mathbb{Z}$$

defined, for $f \in I_\mathbb{Z}(S)$ and $x \in X(S)$, by

$$\phi((x, f)) = (x + u_f, f).$$

If $f_0 : S \to I_\mathbb{Z}$ is the composite of the structural morphism $S \to \operatorname{Spec}(\mathbb{Z})$ and the zero section, then $u_{f_0} = 0$, so $\phi((x, f_0)) = (x, f_0)$. Thus ϕ induces the identity on X: it is an infinitesimal endomorphism of X which is plainly not an automorphism.

6.3.

There exist modules which are not good. First, the preceding counterexample can be modified slightly.[102] Let $F(S)$ be the free $\mathbf{O}(S)$-module with basis the elements of $\mathbf{O}(S)$. Then F does not satisfy (E) relative to $\operatorname{Spec}(\mathbb{Z})$. Moreover, let G be the $\mathbb{Z}$-group defined by

$$G(S) = \mathrm{GL}_n(R(S)), \qquad n \geq 2,$$

where $R(S)$ is the $\mathbf{O}(S)$-algebra of the abelian group $\mathbf{O}(S)$. Then the $\mathbb{Z}$-group $\underline{\mathrm{Lie}}(G/\mathbb{Z})$ is not commutative.

There are also the following counterexamples.[103] Let $S = \operatorname{Spec}(k)$, where k is a field of characteristic $p > 0$.

(a) Let F be the $\mathbf{O}_S$-module assigning to every S-scheme T the group

$$F(T) = \Gamma(T, \mathscr{O}_T),$$

with its $\mathbf{O}(T)$-module structure defined by $r \cdot f = r^p f$. As an S-functor of groups, F is isomorphic to $\mathbb{G}_{a,S}$. Thus F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbb{G}_{a,S}/S) \simeq \mathbf{O}_S.$$

For each $T \to S$, the canonical morphism $F \to \underline{\mathrm{Lie}}(F/S)$ is the identity map $F(T) \to \mathbf{O}(T)$. It respects the abelian-group structure but not the module structure, so F is not good (4.4.1).

(b) Let $\alpha_{p,k}$ be the k-functor of groups defined by

$$\alpha_{p,k}(T) = \{x \in \mathbf{O}(T) \mid x^p = 0\}.$$

It is represented by $\operatorname{Spec}(k[X]/(X^p))$, hence is a very good S-group. It also carries an $\mathbf{O}_S$-module structure, but is not a good $\mathbf{O}_S$-module, because the canonical morphism

$$\alpha_{p,k} \longrightarrow \underline{\mathrm{Lie}}(\alpha_{p,k}/k) = \mathbb{G}_{a,k}$$

is not bijective.

[102] Editor's note: The following observation has been added.
[103] Editor's note: Example (a) has been expanded and example (b) has been added.

6.4.

For a group-valued functor G over S, the definitions give

$$(G/S \text{ satisfies (E)}) \Longleftarrow (G \text{ is good}) \Longleftarrow (G \text{ is very good}).$$

It would be interesting to prove or disprove the reverse implications. [104]

6.5.

Let Nil be the $\mathbb{Z}$-functor of groups defined by

$$\operatorname{Nil}(S) = \{x \in \mathbf{O}(S) \mid \text{there exists } n \in \mathbb{N} \text{ with } x^n = 0\}.$$

[105] The functor Nil is very good but not representable. Let Nil^2, $\mathbf{O}_{\mathrm{red}}$, and F be the $\mathbb{Z}$-functors of groups assigning to S, respectively, the ideal $\operatorname{Nil}(S)^2$ and the groups

$$\mathbf{O}_{\mathrm{red}}(S) = \mathbf{O}(S)/\operatorname{Nil}(S), \qquad F(S) = \mathbf{O}(S)/\operatorname{Nil}(S)^2.$$

It is easy to see that

$$\underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{red}}/\mathbb{Z}) = 0,$$

so the $\mathbf{O}_{\mathbb{Z}}$-module $\mathbf{O}_{\mathrm{red}}$ is not good, although it is a good $\mathbb{Z}$-group. If $\mathscr{M}, \mathscr{N}$ are free $\mathbb{Z}$-modules of finite rank, then

$$\begin{aligned}\operatorname{Nil}^2(I_S(\mathscr{M} \oplus \mathscr{N})) \simeq\ & \operatorname{Nil}^2(S) \oplus \operatorname{Nil}(S) \otimes_{\mathbb{Z}} \mathscr{M} \\ & \oplus \operatorname{Nil}(S) \otimes_{\mathbb{Z}} \mathscr{N},\end{aligned}$$

and hence

$$\begin{aligned}F(I_S(\mathscr{M} \oplus \mathscr{N})) \simeq\ & F(S) \oplus \mathbf{O}_{\mathrm{red}}(S) \otimes_{\mathbb{Z}} \mathscr{M} \\ & \oplus \mathbf{O}_{\mathrm{red}}(S) \otimes_{\mathbb{Z}} \mathscr{N}.\end{aligned}$$

Thus F satisfies condition (E), while

$$\underline{\mathrm{Lie}}(F/\mathbb{Z}) = \mathbf{O}_{\mathrm{red}}.$$

Since $\mathbf{O}_{\mathrm{red}}$ is not a good $\mathbf{O}_{\mathbb{Z}}$-module, F is a $\mathbb{Z}$-group which satisfies (E) but is not good.

6.5.1. We now give the counterexamples announced in 5.3.1–5.3.3. Let S be a scheme, let $F = \mathbf{O}_S^{\oplus 2}$ with the natural action of the good S-group $G = \mathrm{GL}_{2,S}$, and let E be the $\mathbf{O}_S$-submodule of F formed by the pairs (x_1, x_2) for which x_2 is nilpotent. Put

$$N = \underline{\mathrm{Norm}}_G(E).$$

Then

$$\underline{\mathrm{Lie}}(N/S) = \underline{\mathrm{Lie}}(G/S),$$

whereas, for every $S' \to S$,

$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \left\{ \begin{pmatrix} a & b \\ x & c \end{pmatrix} \,\middle|\, a, b, c, x \in \mathbf{O}(S'),\ x \text{ nilpotent} \right\}.$$

[104] Editor's note: A representable G is very good (4.11). For representability criteria see, for example, [MO67]. For automorphisms of an affine algebraic group over an algebraically closed field of characteristic zero, see [HM69].

[105] Editor's note: This paragraph has been added.

Consequently,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Norm}}_G(E)/S) \neq \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E).$$

Taking the semidirect product $G_1 = F \rtimes G$ gives an analogous counterexample in which E is an $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G_1/S)$. Moreover, with the notation above,

$$E = \underline{\mathrm{Lie}}(K/S),$$

where K is the subgroup $\mathbf{O}_S \oplus \mathrm{Nil}^2$ of F; that is, $K(S')$ consists of the pairs (x_1, x_2) with $x_2 \in \mathrm{Nil}(S')^2$.

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[106]

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[106]Editor's note: Additional references cited in this exposé.

Exposé III
Infinitesimal Extensions

by M. Demazure

In this exposé we work in the following general situation. We have a scheme S and a coherent nilpotent ideal $\mathscr{I}$ on S. We denote by S_n the closed subscheme of S defined by the ideal $\mathscr{I}^{n+1}$, for $n \geq 0$. In particular, S_0 is defined by $\mathscr{I}$. Since $\mathscr{I}$ is nilpotent, $S_n = S$ for all sufficiently large n, and the S_i have the same underlying topological space. A typical example is the following: S is the spectrum of a local Artinian ring A, and $\mathscr{I}$ is the ideal defined by the radical of A, so that S_0 is the spectrum of the residue field of A.[0]

In this situation one is given certain data over S_0 and seeks data over S that lift them, that is, data whose base change from S to S_0 recovers the original data. This is done step by step, through the schemes S_n. At each step we shall define the obstructions that arise and classify the resulting solutions when they exist.

The passage from S_n to S_{n+1} may be generalized as follows. We have a scheme S and two ideals $\mathscr{I} \supset \mathscr{J}$ such that $\mathscr{I}\mathscr{J} = 0$. In the preceding case S, $\mathscr{I}$, and $\mathscr{J}$ are respectively

$$S_{n+1}, \qquad \mathscr{I}/\mathscr{I}^{n+2}, \qquad \mathscr{I}^{n+1}/\mathscr{I}^{n+2}.$$

We denote by S_0, respectively $S_{\mathscr{J}}$, the closed subscheme of S defined by $\mathscr{I}$, respectively $\mathscr{J}$, and pose a problem of extending data from $S_{\mathscr{J}}$ to S.

SGA 1, Exposé III, treated the problems of extending morphisms of schemes and of extending schemes. Here we consider the problems of extending group morphisms, group structures, and subgroups.

In §0 we have collected those results of SGA 1, Exposé III, that will be useful to us, putting them into the form best suited to our purpose and sparing the reader constant reference to that exposé.[1] Section 1 collects computations in group cohomology that will be used later and have nothing to do with scheme theory. Sections 2 and 3 treat, respectively, the extension of group morphisms and the extension of group structures. In §4 we briefly recall the proof of a result stated in TDTE IV concerning the extension of subschemes, and apply it to the problem of extending subgroups. For the remainder of the Seminar, only the result of §2 concerning the extension of group morphisms is indispensable.[2]

The idea of reducing problems of infinitesimal extension to the usual cohomological computations in group extensions was suggested by J. Giraud during the oral exposé, whose computations were appreciably more complicated and less transparent. Unfortunately, this

[0] Editor's note: version of October 14, 2024.

[1] Editor's note: the reader may prefer to begin with §1, which is easier and may serve as motivation and a guide to the results of §0.

[2] Editor's note: nevertheless, 3.10 is used in XXIV 1.13. Moreover, 4.37 is used in the proof of IX 3.2 bis and 3.6 bis, although those results also follow from results of Exposé X that do not use 4.37.

method seems to work well only for the first two problems studied, and we have been unable to avoid rather laborious computations for extensions of subgroups.

To simplify the language, we shall call a Y-functor, respectively a Y-scheme, and so on, a functor, respectively a scheme, and so on, equipped with a morphism to the functor Y. This extends the definitions of Exposé I, which concerned only representable Y.

0. Reminders from SGA 1, Exposé III, and miscellaneous remarks

We begin with a general definition.

Definition 0.1. *Let $\mathscr{C}$ be a category, X an object of $\widehat{\mathscr{C}}$, and G a $\widehat{\mathscr{C}}$-group acting on X. We say that X is* formally principal homogeneous[3] *under G if the following equivalent conditions hold:*

(i) *for every object T of $\mathscr{C}$, the set $X(T)$ is either empty or principal homogeneous under $G(T)$;*

(ii) *the morphism of functors*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is an isomorphism.

We shall now put the results of SGA 1, III §5[4] into the form most useful here. Throughout this section we use the following notation. On a scheme S we have two quasi-coherent ideals $\mathscr{I}$ and $\mathscr{J}$ such that

$$\mathscr{I} \supset \mathscr{J}, \qquad \mathscr{I}\mathscr{J} = 0.$$

In particular, $\mathscr{J}^2 = 0$. We denote by S_0, respectively $S_{\mathscr{J}}$, the closed subscheme of S defined by $\mathscr{I}$, respectively $\mathscr{J}$. For every S-functor X, we systematically denote by X_0 and $X_{\mathscr{J}}$ the functors obtained by base change from S to S_0 and $S_{\mathscr{J}}$. We use the same notation for a morphism.

Definition 0.1.1.[5] *Let X be an S-functor. Define a functor X^+ over S by*

$$\mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}}) = \mathrm{Hom}_S(Y_{\mathscr{J}}, X)$$

for a variable S-scheme Y. In the notation of Exposé II, §1,

$$X^+ \simeq \prod_{S_{\mathscr{J}}/S} X_{\mathscr{J}}.$$

The identity morphism of $X_{\mathscr{J}}$ defines, by construction, an S-morphism

$$p_X : X \longrightarrow X^+.$$

Explicitly,[6] for every S-scheme Y, the map

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_S(Y_{\mathscr{J}}, X)$$

[3]Editor's note: one also calls such an object a *pseudo-torsor*; see EGA IV_4, 16.5.15. The more general notion of a formally homogeneous object, not necessarily principal homogeneous, is defined in Exposé IV, 6.7.1.

[4]Editor's note: see also EGA IV_4, 16.5.14–18.

[5]Editor's note: the numbering 0.1.1 through 0.1.13 has been added to make the definitions and results contained here more visible.

[6]Editor's note: the following sentence has been added, and the next two remarks have been expanded.

is induced by the morphism $Y_{\mathscr{J}} \to Y$.

Remark 0.1.2. If X is an S-group functor, then $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group functor. The defining formula for X^+ therefore equips X^+ with an S-group-functor structure, and the description above shows that

$$p_X : X \longrightarrow X^+$$

is a morphism of S-group functors.

Remark 0.1.3. For every S-group functor Y, one has

$$\mathrm{Hom}_{S\text{-gr}}(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}\text{-gr}}(Y_{\mathscr{J}}, X_{\mathscr{J}}).$$

Indeed, let $f \in \mathrm{Hom}_S(Y, X^+)$ correspond to $f_{\mathscr{J}} \in \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}})$. The condition that f be an S-group morphism is that, for every $T \to S$ and every $y, y' \in Y(T)$,

$$f(y \cdot y') = f(y) \cdot f(y'),$$

which is equivalent to

$$f_{\mathscr{J}}(y_{\mathscr{J}}) \cdot f_{\mathscr{J}}(y'_{\mathscr{J}}) = f_{\mathscr{J}}((y \cdot y')_{\mathscr{J}}).$$

Since $(y \cdot y')_{\mathscr{J}} = y_{\mathscr{J}} \cdot y'_{\mathscr{J}}$, this is precisely the condition that $f_{\mathscr{J}}$ be a group morphism. Applying this to $Y = X$, we again see that p_X, which corresponds to $\mathrm{id}_{X_{\mathscr{J}}}$, is a morphism of S-group functors.

Return now to the general case and suppose that X is an S-scheme. An S-morphism from a variable S-scheme Y to X^+ is, by definition, an $S_{\mathscr{J}}$-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$. We define an X^+-functor in abelian groups L_X as follows.

Scholium 0.1.4.[7] If $\pi : Y \to S$ is a morphism of schemes and M is an $\mathscr{O}_S$-module, we denote the inverse image $\pi^* M$ by

$$M \otimes_{\mathscr{O}_S} \mathscr{O}_Y.$$

If $\mathscr{J}$ is an ideal of $\mathscr{O}_S$, we denote by $\mathscr{J}\mathscr{O}_Y$ the ideal of $\mathscr{O}_Y$ that is the image of

$$\pi^*(\mathscr{J}) = \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Y \longrightarrow \mathscr{O}_Y.$$

For every morphism of S-schemes $f : Z \to Y$, there is an epimorphism of $\mathscr{O}_Z$-modules

$$f^*(\mathscr{J}\mathscr{O}_Y) \longrightarrow \mathscr{J}\mathscr{O}_Z. \tag{0.1.4}$$

It follows from the commutative diagram below.

$$\begin{tikzcd}
\mathcal{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Y \otimes_{\mathscr{O}_Y} \mathscr{O}_Z \arrow[r, equal] \arrow[d] & \mathcal{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Z \arrow[d] \\
f^*(\mathcal{J}\mathscr{O}_Y) = (\mathcal{J}\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{O}_Z \arrow[r] & \mathcal{J}\mathscr{O}_Z.
\end{tikzcd}$$

Figure 42: The tensor-product diagram yielding the epimorphism (0.1.4).

[7]Editor's note: this scholium has been added.

Definition 0.1.5.[8] Let X be an S-scheme. For every X^+-scheme Y, given by a morphism

$$g_{\mathscr{J}} : Y_{\mathscr{J}} \longrightarrow X_{\mathscr{J}},$$

set

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y),$$

where $\Omega^1_{X_0/S_0}$ is the module of relative differentials of X_0 over S_0, and $\mathscr{J}\mathscr{O}_Y$ is regarded as an $\mathscr{O}_{Y_0}$-module because it is annihilated by $\mathscr{I}$.

Then L_X is an X^+-functor in abelian groups.[9] Indeed, for every X^+-morphism $f : Z \to Y$, the functor f_0^* and the morphism

$$f_0^*(\mathscr{J}\mathscr{O}_Y) \longrightarrow \mathscr{J}\mathscr{O}_Z$$

of (0.1.4) induce a natural homomorphism $L_X(f)$:

$$\begin{aligned}
&\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y)\\
&\quad\longrightarrow \mathrm{Hom}_{\mathscr{O}_{Z_0}}(f_0^* g_0^*(\Omega^1_{X_0/S_0}), f_0^*(\mathscr{J}\mathscr{O}_Y))\\
&\quad\longrightarrow \mathrm{Hom}_{\mathscr{O}_{Z_0}}(f_0^* g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Z).
\end{aligned}$$

Finally, $L_X(f)$ has the following simple local description. The schemes Y and $Y_{\mathscr{J}}$ have the same underlying topological space, as do V and $V_{\mathscr{J}}$ for every open subset V of Y. Let

$$U = \mathrm{Spec}(A), \qquad V = \mathrm{Spec}(B), \qquad W = \mathrm{Spec}(C)$$

be affine opens such that U lies over $\mathrm{Spec}(\Lambda) \subset S$, $g_{\mathscr{J}}(V_{\mathscr{J}}) \subset U$, and $W \subset f^{-1}(V)$. Write I, J for the ideals of Λ corresponding to $\mathscr{I}, \mathscr{J}$. Then f, respectively $g_{\mathscr{J}}$, induces a morphism of Λ-algebras

$$\theta : B \longrightarrow C, \qquad \text{respectively} \qquad \varphi : A \longrightarrow B/JB,$$

and clearly $\theta(JB) \subset JC$. On the other hand,

$$m \in \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y)$$

induces an element D of

$$\begin{aligned}
\mathrm{Hom}_{\mathscr{O}_{V_0}}(g_0^*(\Omega^1_{U/S}), \mathscr{J}\mathscr{O}_V) &= \mathrm{Hom}_{B/IB}(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB)\\
&= \mathrm{Der}_\Lambda(A, JB),
\end{aligned}$$

and the image of $L_X(f)(m)$ in

$$\begin{aligned}
\mathrm{Hom}_{\mathscr{O}_{W_0}}(f_0^* g_0^*(\Omega^1_{U/S}), \mathscr{J}\mathscr{O}_Z) &= \mathrm{Hom}_{C/IC}(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC)\\
&= \mathrm{Der}_\Lambda(A, JC)
\end{aligned}$$

is simply $\theta \circ D$.

[8] Editor's note: introducing the functor L_X gives an extension, functorial in Y and in X, of SGA 1, III, Proposition 5.1; see 0.1.8 and 0.1.9 below, as well as 0.1.10 and 0.2. Moreover, when X is an S-group functor, L_X gives rise, under certain hypotheses, to an S-group functor L'_X and to an exact sequence $1 \to L'_X \to X \to X^+$; see 0.4, 0.9, and 0.11. These play an essential role in this exposé; see Theorems 2.1, 3.5, and 4.21.

[9] Editor's note: what follows has been added.

Remark 0.1.6.[10] Let $f : X \to W$ be an S-morphism. It induces an S-morphism

$$f^+ : X^+ \longrightarrow W^+$$

defined as follows. If $g \in \mathrm{Hom}_S(Y, X^+)$ corresponds to an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, then $f^+(g)$ is the element $f \circ g_{\mathscr{J}}$ of $\mathrm{Hom}_S(Y, W^+)$. The following diagram is commutative.

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "p_X"'] & W \arrow[d, "p_W"] \\
X^+ \arrow[r, "f^+"] & W^+
\end{tikzcd}.$$

Figure 43: Naturality of the morphisms p_X and p_W.

Reminders 0.1.7.[11] In this paragraph, given an S-scheme X, we "recall" certain functorial properties of the module of differentials $\Omega^1_{X/S}$ and of the first infinitesimal neighborhood of the diagonal, $\Delta^{(1)}_{X/S}$. These properties follow readily from EGA IV_4, §§16.1–16.4, but do not appear there explicitly.

(a) Let us begin by recalling the following facts (cf. EGA II, §§1.2–1.5). Let $g : Y \to X$ be a morphism of schemes, let $\pi : X' \to X$ be an affine X-scheme, and let $\mathscr{B} = \pi_*(\mathscr{O}_{X'})$ be the corresponding quasi-coherent $\mathscr{O}_X$-algebra. Then the Y-scheme $Y \times_X X'$ is affine and corresponds to the quasi-coherent $\mathscr{O}_Y$-algebra $g^*(\mathscr{B})$, and there is a commutative diagram of bijections:

$$\begin{tikzcd}
\mathrm{Hom}_X(Y, X') \arrow[r, "\sim"] \arrow[d, "\wr"'] & \mathrm{Hom}_Y(Y, Y \times_X X') \arrow[d, "\wr"] \\
\mathrm{Hom}_{\mathscr{O}_X\text{-alg.}}(\mathscr{B}, g_*(\mathscr{O}_Y)) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathscr{O}_Y\text{-alg.}}(g^*(\mathscr{B}), \mathscr{O}_Y).
\end{tikzcd}$$

Figure 44: The affine-scheme/algebra bijections after base change by g.

Moreover, these bijections are functorial in the pair (X, X'). More precisely, let W' be an affine W-scheme corresponding to the quasi-coherent $\mathscr{O}_W$-algebra $\mathscr{A}$, and suppose that one has a commutative diagram of morphisms of schemes as follows.

[10] Editor's note: this remark has been expanded.

[11] Editor's note: This paragraph has been added (see also [DG70], §I.4, nos. 1–2).

$$
\begin{tikzcd}
 & \mathrm{X}' \arrow[r, "f'"] \arrow[d] & \mathrm{W}' \arrow[d] \\
\mathrm{Y} \arrow[r, "g"] \arrow[ur, dashed, "g'"] & \mathrm{X} \arrow[r, "f"] & \mathrm{W}
\end{tikzcd}
$$

Figure 45: The scheme morphisms used to express functoriality in (X, X').

Denote by

$$\varphi : \mathscr{A} \longrightarrow f_*(\mathscr{B}), \qquad \varphi^\sharp : f^*(\mathscr{A}) \longrightarrow \mathscr{B}$$

the algebra morphisms associated with f', and by

$$\theta : \mathscr{B} \longrightarrow g_*(\mathscr{O}_{\mathrm{Y}}), \qquad \theta^\sharp : g^*(\mathscr{B}) \longrightarrow \mathscr{O}_{\mathrm{Y}}$$

those associated with a variable X-morphism $g' : Y \to X'$. Then the following diagram is commutative, where Y is viewed as a W-scheme through $f \circ g$.

$$
\begin{tikzcd}
\mathrm{Hom}_{\mathrm{X}}(\mathrm{Y}, \mathrm{X}') \arrow[r, "g' \mapsto f' \circ g'"] \arrow[dd, "\wr"'] & \mathrm{Hom}_{\mathrm{W}}(\mathrm{Y}, \mathrm{W}') \arrow[dr, "\simeq"] & \\
 & & \mathrm{Hom}_{\mathscr{O}_{\mathrm{W}}\text{-alg.}}(\mathscr{A}, f_* g_*(\mathscr{O}_{\mathrm{Y}})) \arrow[d, "\wr"] \\
\mathrm{Hom}_{\mathscr{O}_{\mathrm{X}}\text{-alg.}}(\mathscr{B}, g_*(\mathscr{O}_{\mathrm{Y}})) \arrow[rr, "\theta \mapsto \theta \circ \phi^\sharp"] \arrow[urr, "\theta \mapsto f_*(\theta) \circ \phi"] \arrow[d, "\wr"'] & & \mathrm{Hom}_{\mathscr{O}_{\mathrm{X}}\text{-alg.}}(f^*(\mathscr{A}), g_*(\mathscr{O}_{\mathrm{Y}})) \arrow[d, "\wr"] \\
\mathrm{Hom}_{\mathscr{O}_{\mathrm{Y}}\text{-alg.}}(g^*(\mathscr{B}), \mathscr{O}_{\mathrm{Y}}) \arrow[rr, "\theta^\sharp \mapsto \theta^\sharp \circ g^*(\phi^\sharp)"] & & \mathrm{Hom}_{\mathscr{O}_{\mathrm{Y}}\text{-alg.}}(g^* f^*(\mathscr{A}), \mathscr{O}_{\mathrm{Y}}).
\end{tikzcd}
$$

Figure 46: Functoriality of the affine-scheme/algebra correspondences.

(b) Let now X be an S-scheme. Let $\Omega^1_{X/S}$ be the module of differentials of X over S, and let $\Delta^{(1)}_{X/S}$ be the first infinitesimal neighborhood of the diagonal immersion $\delta_X : X \to X \times_S X$. It is a subscheme of $X \times_S X$, in which the diagonal $\Delta_{X/S}$ is a closed subscheme. Write pr^i_X, $i = 1, 2$, for the two projections $X \times_S X \to X$, and write π_X for the restriction of pr^1_X to $\Delta^{(1)}_{X/S}$.

On the one hand, every morphism $f : X \to W$ of S-schemes induces an S-morphism

$$\Delta^{(1)} f : \Delta^{(1)}_{X/S} \longrightarrow \Delta^{(1)}_{W/S}$$

for which the following diagram is commutative.

$$\begin{tikzcd}
\mathrm{X} \arrow[r, "\delta_{\mathrm{X}}"] \arrow[d, "f"'] & \Delta^{(1)}_{\mathrm{X/S}} \arrow[r] \arrow[d, "\Delta^{(1)} f"] & \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[r, "\mathrm{pr}^i_{\mathrm{X}}"] \arrow[d, "f \times f"] & \mathrm{X} \arrow[d, "f"] \\
\mathrm{W} \arrow[r, "\delta_{\mathrm{W}}"] & \Delta^{(1)}_{\mathrm{W/S}} \arrow[r] & \mathrm{W} \times_{\mathrm{S}} \mathrm{W} \arrow[r, "\mathrm{pr}^i_{\mathrm{W}}"] & \mathrm{W}.
\end{tikzcd}$$

Figure 47: Functoriality of the first infinitesimal neighborhood of the diagonal.

On the other hand, through the projection π_X, the scheme $\Delta^{(1)}_{X/S}$ is an affine X-scheme: it is the spectrum of the augmented quasi-coherent $\mathscr{O}_X$-algebra

$$\mathscr{P}^1_{X/S} = \mathscr{O}_X \oplus \Omega^1_{X/S},$$

where $\Omega^1_{X/S}$ is a square-zero ideal. The augmentation is the $\mathscr{O}_X$-algebra morphism

$$\varepsilon_X : \mathscr{P}^1_{X/S} \longrightarrow \mathscr{O}_X$$

that vanishes on $\Omega^1_{X/S}$ and corresponds to the closed immersion $\delta_X : X \hookrightarrow \Delta^{(1)}_{X/S}$. Hence every morphism of S-schemes $f : X \to W$ induces a morphism of augmented $\mathscr{O}_X$-algebras

$$f^*(\mathscr{P}^1_{W/S}) = \mathscr{O}_X \oplus f^*(\Omega^1_{W/S}) \longrightarrow \mathscr{P}^1_{X/S} = \mathscr{O}_X \oplus \Omega^1_{X/S},$$

or, equivalently, a morphism of $\mathscr{O}_X$-modules

$$f_{X/W/S} : f^*(\Omega^1_{W/S}) \longrightarrow \Omega^1_{X/S};$$

see EGA IV_4, (16.4.3.6), and (16.4.18.2) for the notation $f_{X/W/S}$.

Since $\pi_X : \Delta^{(1)}_{X/S} \to X$ is affine, part (a) shows that $\Delta^{(1)} f$ is completely determined by $f_{X/W/S}$. For every X-scheme $g : Y \to X$, the set

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \simeq \mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{O}_Y \oplus g^*(\Omega^1_{X/S}), \mathscr{O}_Y)$$

identifies with a subset of $\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y)$, namely

$$\widetilde{\mathrm{Hom}}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y),$$

the subset formed by the $\mathscr{O}_Y$-morphisms $\psi : g^*(\Omega^1_{X/S}) \to \mathscr{O}_Y$ for which $\mathrm{Im}(\psi)$ is a square-zero ideal of $\mathscr{O}_Y$.[12]

Consequently, apply part (a) to the following diagram.

[12]Editor's note: In the proof of SGA 1, III 5.1, the sentence "But the algebra homomorphisms $\mathscr{A} \to \mathscr{O}_Y$ correspond . . . " must be corrected accordingly; the remainder of the cited passage is unchanged.

$$\begin{tikzcd}
 & \Delta^{(1)}_{\mathrm{X/S}} \arrow[r, "\Delta^{(1)} f"] \arrow[d] & \Delta^{(1)}_{\mathrm{W/S}} \arrow[d] \\
\mathrm{Y} \arrow[r, "g"] \arrow[ur, dashed, "g'"] & \mathrm{X} \arrow[r, "f"] & \mathrm{W}
\end{tikzcd}$$

Figure 48: The diagram to which the affine functoriality statement is applied.

Taking into account that $\Delta^{(1)} f$ is the restriction of $f \times f$ to $\Delta^{(1)}_{X/S}$, one obtains the following commutative diagram, functorial in the X-scheme $g : Y \to X$.

$$(0.1.7\,(*))\qquad \begin{tikzcd}
\mathrm{Hom}_{\mathrm{X}}(\mathrm{Y}, \mathrm{X} \times_{\mathrm{S}} \mathrm{X}) \arrow[r, "(g,g') \mapsto (f \circ g, f \circ g')"] & \mathrm{Hom}_{\mathrm{W}}(\mathrm{Y}, \Delta^{(1)}_{\mathrm{W/S}}) \\
\mathrm{Hom}_{\mathrm{X}}(\mathrm{Y}, \Delta^{(1)}_{\mathrm{X/S}}) \arrow[u, hook] \arrow[r, "g' \mapsto (\Delta^{(1)} f) \circ g'"] \arrow[d, "\simeq"'] & \mathrm{Hom}_{\mathrm{W}}(\mathrm{Y}, \Delta^{(1)}_{\mathrm{W/S}}) \arrow[u, hook] \arrow[d, "\simeq"'] \\
\mathrm{Hom}^{\square}_{\mathscr{O}_{\mathrm{Y}}}(g^*(\Omega^1_{\mathrm{X/S}}), \mathscr{O}_{\mathrm{Y}}) \arrow[r, "\psi \mapsto \psi \circ g^*(f_{\mathrm{X/W/S}})"] & \mathrm{Hom}^{\square}_{\mathscr{O}_{\mathrm{Y}}}(g^* f^*(\Omega^1_{\mathrm{W/S}}), \mathscr{O}_{\mathrm{Y}}).
\end{tikzcd}$$

Figure 49: The functorial comparison of first-neighborhood morphisms and differential maps, equation 0.1.7 $(*)$.

Remark 0.1.7.1. Let us conclude this paragraph with the following remark, which will be useful later (cf. 0.1.10). Denote by $\widetilde{L}_X$ the X-functor that associates to every X-scheme $g : Y \to X$ the set

$$\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y),$$

and denote by $\widetilde{L}_f : \widetilde{L}_X \to \widetilde{L}_W$ the morphism of functors defined above: it sends $\psi \in \widetilde{L}_X(Y)$ to $\psi \circ g^*(f_{X/W/S})$. The preceding discussion gives a commutative diagram of functors.

$$\begin{tikzcd}
\mathrm{X} \times_{\mathrm{X}} \mathrm{L}^{\square}_{\mathrm{X}} \arrow[d, "f \times \mathrm{L}^{\square}_f"'] & \Delta^{(1)}_{\mathrm{X/S}} \arrow[l, "\sim"'] \arrow[r, hook] \arrow[d, "\Delta^{(1)} f"] & \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[d, "f \times f"] \\
\mathrm{W} \times_{\mathrm{W}} \mathrm{L}^{\square}_{\mathrm{W}} & \Delta^{(1)}_{\mathrm{W/S}} \arrow[l, "\sim"'] \arrow[r, hook] & \mathrm{W} \times_{\mathrm{S}} \mathrm{W}.
\end{tikzcd}$$

Figure 50: Functoriality of $\widetilde{L}_X$ and the first infinitesimal neighborhood.

Theorem 0.1.8. (SGA 1, III 5.1)[13] *Let Y and X be two S-schemes, let $\mathscr{J}$ be a quasi-*

[13]Editor's note: The statement and proof of SGA 1, III 5.1 are reproduced here, since they are needed

coherent square-zero ideal of $\mathscr{O}_Y$, let $Y_{\mathscr{J}}$ be the closed subscheme of Y defined by $\mathscr{J}$, and let $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$ be an S-morphism.

(a) *The set $P(g_{\mathscr{J}})$ of S-morphisms $g : Y \to X$ extending $g_{\mathscr{J}}$ is either empty or principal homogeneous under the abelian group*

$$\mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g_{\mathscr{J}}^*(\Omega^1_{X/S}), \mathscr{J}).$$

(b) *Suppose that $i : Y_0 \hookrightarrow Y_{\mathscr{J}}$ is the closed immersion defined by a quasi-coherent ideal $\mathscr{I} \supset \mathscr{J}$ such that $\mathscr{I}\mathscr{J} = 0$, and set $g_0 = g_{\mathscr{J}} \circ i$. Then the preceding abelian group is isomorphic to*

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X/S}), \mathscr{J}).$$

Proof. Part (b) follows at once from part (a). Indeed, since $\mathscr{J}$ is annihilated by $\mathscr{I}$, it may be regarded as an $\mathscr{O}_{Y_0}$-module, and adjunction gives

$$\mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g_{\mathscr{J}}^*(\Omega^1_{X/S}), \mathscr{J}) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(i^* g_{\mathscr{J}}^*(\Omega^1_{X/S}), \mathscr{J}).$$

To prove part (a), we may suppose that $P(g_{\mathscr{J}}) \neq \emptyset$, that is, that an S-morphism $g : Y \to X$ extending $g_{\mathscr{J}}$ exists. Write $j : Y_{\mathscr{J}} \hookrightarrow Y$ for the immersion. Then $P(g_{\mathscr{J}})$ is the set of S-morphisms $g' : Y \to X$ such that $g' \circ j = g_{\mathscr{J}}$. Giving such a g' is equivalent to giving an S-morphism

$$h : Y \longrightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ h = g$ and $h_{\mathscr{J}} = \delta \circ g_{\mathscr{J}}$, where $h_{\mathscr{J}} = h \circ j$ and $\delta : X \hookrightarrow X \times_S X$ is the diagonal immersion.

$$\begin{tikzcd}
X \times_S X \arrow[d, "\mathrm{pr}_1"'] & Y_{\mathfrak{J}} \arrow[l, "h_{\mathfrak{J}} = \delta \circ g_{\mathfrak{J}}"'] \arrow[d, "j"] \\
X & Y. \arrow[l, "g"] \arrow[ul, dashed, "h"']
\end{tikzcd}$$

Figure 51: The morphism h determined by two extensions of $g_{\mathscr{J}}$.

Since $h_{\mathscr{J}}$ factors through δ, and since Y lies in the first infinitesimal neighborhood of the immersion $j : Y_{\mathscr{J}} \to Y$ because $\mathscr{J}^2 = 0$, functoriality (EGA IV$_4$, 16.2.2(i)) shows that the desired morphisms h factor uniquely through $\Delta^{(1)}_{X/S}$; see 0.1.7. Put

$$Y' = \Delta^{(1)}_{X/S} \times_X Y, \qquad Y'_{\mathscr{J}} = Y' \times_Y Y_{\mathscr{J}} = \Delta^{(1)}_{X/S} \times_X Y_{\mathscr{J}}.$$

below to prove Proposition 0.2; see already the following Corollary 0.1.9. Moreover, $\mathscr{J}$ (respectively $\mathscr{I}$) is another calligraphic form of the letter J (respectively I). Returning to the earlier notation, with $\mathscr{J} \subset \mathscr{I}$ ideals of $\mathscr{O}_S$ such that $\mathscr{I}\mathscr{J} = 0$, this result will be applied with $\mathscr{J} = \mathscr{J}\mathscr{O}_Y$ (respectively $\mathscr{I} = \mathscr{I}\mathscr{O}_Y$).

The desired morphisms h are then in bijection with the sections u of $Y' \to Y$ extending the section $u_{\mathscr{J}} = (\delta \circ g_{\mathscr{J}}, \mathrm{id})$ of $Y'_{\mathscr{J}} \to Y_{\mathscr{J}}$. On the other hand, Y' (respectively $Y'_{\mathscr{J}}$) is affine over Y (respectively $Y_{\mathscr{J}}$) and corresponds to the quasi-coherent algebra

$$\mathscr{A} = \mathscr{O}_Y \oplus g^*(\Omega^1_{X/S}), \qquad \text{respectively} \qquad \mathscr{A}_{\mathscr{J}} = \mathscr{A} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_{Y_{\mathscr{J}}} \oplus g^*_{\mathscr{J}}(\Omega^1_{X/S}).$$

Let $\varepsilon : \mathscr{A} \to \mathscr{O}_Y$ be the canonical augmentation, that is, the $\mathscr{O}_Y$-algebra morphism that vanishes on $g^*(\Omega^1_{X/S})$, and define $\varepsilon_{\mathscr{J}} : \mathscr{A}_{\mathscr{J}} \to \mathscr{O}_{Y_{\mathscr{J}}}$ similarly. Then

$$\Gamma(Y'/Y) \simeq \mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{A}, \mathscr{O}_Y), \qquad \Gamma(Y'_{\mathscr{J}}/Y_{\mathscr{J}}) \simeq \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}\text{-alg}}(\mathscr{A}_{\mathscr{J}}, \mathscr{O}_{Y_{\mathscr{J}}}).$$

Under these isomorphisms, the section $u = (\delta \circ g, \mathrm{id})$ (respectively $u_{\mathscr{J}}$) corresponds to ε (respectively $\varepsilon_{\mathscr{J}}$). Consequently, $P(g_{\mathscr{J}})$ is in bijection with the algebra morphisms $\mathscr{A} \to \mathscr{O}_Y$ that reduce to $\varepsilon_{\mathscr{J}}$; under this bijection, g corresponds to ε.

Put $M = g^*(\Omega^1_{X/S})$. Then $\mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{A}, \mathscr{O}_Y)$ identifies with the set of $\mathscr{O}_Y$-morphisms $\psi : M \to \mathscr{O}_Y$ whose image is a square-zero ideal. We are interested in those morphisms that induce the zero morphism $M \to \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_Y/\mathscr{J}$, that is, those that carry M into $\mathscr{J}$. Conversely, since $\mathscr{J}^2 = 0$, every $\mathscr{O}_Y$-morphism $\phi : M \to \mathscr{J}$ comes from a unique algebra morphism $\mathscr{A} \to \mathscr{O}_Y$ reducing to $\varepsilon_{\mathscr{J}}$. Finally,

$$\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{J}) = \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J})$$

because $\mathscr{J}^2 = 0$, as in the proof of part (b). Thus one obtains a bijection

$$P(g_{\mathscr{J}}) \simeq \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J})$$

under which g corresponds to the zero morphism.

For every

$$m \in \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}),$$

write $m \cdot g$ for the element of $P(g_{\mathscr{J}})$ associated with g and m by the preceding bijection. We have already seen that $0 \cdot g = g$; it remains to prove

$$m' \cdot (m \cdot g) = (m + m') \cdot g. \tag{0.1.8 (*)}$$

This may be checked locally.[14] Indeed, the two preceding morphisms $Y \to X$ induce the same continuous map as g on the underlying topological spaces. It is therefore enough to check that, for every affine open $U = \mathrm{Spec}(A)$ of X lying over an affine open $\mathrm{Spec}(\Lambda)$ of S, and every affine open $V = \mathrm{Spec}(B)$ of $g^{-1}(U)$, they induce the same morphism of Λ-algebras $A \to B$.

Set $J = \Gamma(V, \mathscr{J})$, and let $\varphi, \psi, \eta : A \to B$ be the morphisms induced respectively by g, $m \cdot g$, and $m' \cdot (m \cdot g)$. They agree modulo J. There are unique expressions

$$\psi = \varphi + D, \qquad \eta = \psi + D',$$

where D belongs to

$$\mathrm{Der}_\varphi(A, J) = \{\delta \in \mathrm{Hom}_\Lambda(A, J) \mid \delta(ab) = \varphi(a)\delta(b) + \varphi(b)\delta(a)\},$$

and D' belongs to $\mathrm{Der}_\psi(A, J)$. But $\mathrm{Der}_\varphi(A, J) = \mathrm{Der}_\psi(A, J)$, since $J^2 = 0$, and both identify with

$$\mathrm{Hom}_{B/J}(\Omega^1_{A/\Lambda} \otimes_A B/J, J).$$

[14] Editor's note: The preceding argument is reproduced almost word for word from SGA 1, III 5.1; what follows has been expanded.

Under this identification, D corresponds to m and D' to m'. Hence $\eta = \varphi + D + D'$, while $D + D'$ corresponds to $m + m'$, proving (0.1.8 (*)).

Corollary 0.1.9.[15] *Let X be an S-scheme, and retain the notation of 0.1.5. Then X carries a left action of the abelian X^+-group L_X, making X formally principal homogeneous under L_X over X^+. In other words, there is an isomorphism of X^+-functors*

$$L_X \times_{X^+} X \xrightarrow{\sim} X \times_{X^+} X, \qquad (m, x) \longmapsto (x, m \cdot x).$$

Proof. Let $i_0 : X_0 \hookrightarrow X$ be the immersion. Since $X_0 = X \times_S S_0$, one first has

$$i_0^*(\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0}$$

(EGA IV, 16.4.5).

Let Y be an X^+-scheme, given by an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, and let $g_0 : Y_0 \to X_0$ be obtained by base change. By 0.1.8, if $\mathrm{Hom}_{X^+}(Y, X)$ is nonempty, it is principal homogeneous under

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^* i_0^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y),$$

which identifies with

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y) = L_X(Y).$$

Thus there is a bijection

$$L_X(Y) \times \mathrm{Hom}_{X^+}(Y, X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(Y, X \times_{X^+} X), \qquad (m, g) \longmapsto (g, m \cdot g).$$

We show that this is functorial in Y. Let $f : Z \to Y$ be a morphism of S-schemes. It is enough to prove that the following diagram is commutative.

$$\begin{tikzcd}
L_X(Y) \times \mathrm{Hom}_{X^+}(Y, X) \arrow[r, "\sim"] \arrow[d, "L_X(f) \times f"'] & \mathrm{Hom}_{X^+}(Y, X \times_{X^+} X) \arrow[d, "f \times f"] \\
L_X(Z) \times \mathrm{Hom}_{X^+}(Z, X) \arrow[r, "\sim"] & \mathrm{Hom}_{X^+}(Z, X \times_{X^+} X).
\end{tikzcd}$$

Figure 52: Functoriality in Y of the principal-homogeneous bijection.

If $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to prove. It is therefore enough to show that, for every S-morphism $g : Y \to X$ extending $g_{\mathscr{J}}$ and every $m \in L_X(Y)$, one has

$$(m \cdot g) \circ f = L_X(f)(m) \cdot (g \circ f). \qquad (0.1.9\,(*))$$

These two S-morphisms $Z \to X$ agree on $Z_{\mathscr{J}}$ with $g_{\mathscr{J}} \circ f_{\mathscr{J}}$; in particular, they induce the same continuous map as $g \circ f$ on the underlying topological spaces. It therefore suffices to check the equality on local rings. Let $z \in Z$, put $y = f(z)$, $x = g(y)$, and let A, B, C denote respectively the local rings $\mathscr{O}_{X,x}, \mathscr{O}_{Y,y}, \mathscr{O}_{Z,z}$. Let s be the image of x in S, put $\Lambda = \mathscr{O}_{S,s}$, let J and I be the ideals of Λ corresponding to $\mathscr{J}$ and $\mathscr{I}$, and let $\varphi, \psi : A \to B$ and $\theta : B \to C$ be the Λ-algebra morphisms induced by g, $m \cdot g$, and f, respectively. Then m induces an element D of

$$\mathrm{Hom}_{B/IB}(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB) = \mathrm{Der}_\Lambda(A, JB),$$

[15] Editor's note: This corollary has been added to SGA 1, III 5.1; it proves part (i) of Proposition 0.2 below.

and $\psi = \varphi + D$. Thus $(m \cdot g) \circ f$ corresponds to

$$\theta \circ \psi = \theta \circ \varphi + \theta \circ D.$$

By 0.1.5, $\theta \circ D$ is the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{C/IC}(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC) = \mathrm{Der}_\Lambda(A, JC).$$

Consequently, $\theta \circ \varphi + \theta \circ D$ is the image of $L_X(f)(m) \cdot (g \circ f)$. This proves (0.1.9 (*)).

Corollary 0.1.10.[16]

(a) *The functor L_X depends functorially on X. For every S-morphism $f : X \to W$, there is an S-morphism $L_f : L_X \to L_W$ that is a morphism of abelian groups over f^+; that is, the following diagram is commutative,*

$$\begin{tikzcd}
\mathrm{L_X} \arrow[r, "\mathrm{L}_f"] \arrow[d] & \mathrm{L_W} \arrow[d] \\
\mathrm{X}^+ \arrow[r, "f^+"] & \mathrm{W}^+
\end{tikzcd}$$

Figure 53: The morphism L_f over f^+.

and, for every $Y \to X^+$, the map

$$L_f(Y) : \mathrm{Hom}_{X^+}(Y, L_X) \longrightarrow \mathrm{Hom}_{W^+}(Y, L_W),$$

where Y lies over W^+ through f^+, is a homomorphism of abelian groups.

(b) *Moreover, the following diagram is commutative.*

$$\begin{tikzcd}
\mathrm{L_X} \times_{\mathrm{X}^+} \mathrm{X} \arrow[r, "\sim"] \arrow[d, "\mathrm{L}_f \times f"'] & \mathrm{X} \times_{\mathrm{X}^+} \mathrm{X} \arrow[d, "f \times f"] \\
\mathrm{L_W} \times_{\mathrm{W}^+} \mathrm{W} \arrow[r, "\sim"] & \mathrm{W} \times_{\mathrm{W}^+} \mathrm{W}
\end{tikzcd} .$$

Figure 54: Compatibility of L_f with the formally principal homogeneous actions.

Proof. For part (a), L_f is induced by the morphism of $\mathscr{O}_{X_0}$-modules

$$f_{X_0/W_0/S_0} : f_0^*(\Omega^1_{W_0/S_0}) \longrightarrow \Omega^1_{X_0/S_0}$$

(cf. 0.1.7(b)). For every X^+-scheme Y, given by an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, one has the following commutative diagram, functorial in Y.

[16] Editor's note: The functoriality of L_X in X has been made explicit, in particular in part (b) below.

$$\begin{tikzcd}
\mathrm{Hom}_{\mathscr{O}_{\mathrm{Y}_0}}(g_0^*(\Omega^1_{\mathrm{X}_0/\mathrm{S}_0}), \mathcal{J}\mathscr{O}_{\mathrm{Y}}) \arrow[r, "\mathrm{L}_f(\mathrm{Y})"] \arrow[d] & \mathrm{Hom}_{\mathscr{O}_{\mathrm{Y}_0}}(g_0^* f_0^*(\Omega^1_{\mathrm{W}_0/\mathrm{S}_0}), \mathcal{J}\mathscr{O}_{\mathrm{Y}}) \arrow[d] \\
\{g_{\mathcal{J}}\} \arrow[r] & \{f_{\mathcal{J}} \circ g_{\mathcal{J}}\}
\end{tikzcd}$$

Figure 55: The homomorphism $L_f(Y)$ on the differential-module descriptions.

Here $L_f(Y)$ is the map
$$\psi \longmapsto \psi \circ g_0^*(f_{X_0/W_0/S_0}),$$
which is indeed a homomorphism of abelian groups.[17]

We now prove part (b). Let Y be an X^+-scheme. If $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to prove. Thus let $g \in \mathrm{Hom}_{X^+}(Y, X)$. We must show that, for every $m \in L_X(Y)$,

$$f \circ (m \cdot g) = L_f(Y)(m) \cdot (f \circ g). \tag{0.1.10 (*)}$$

With g fixed, $\mathrm{Hom}_X(Y, X \times_{X^+} X)$ is a subset of $\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S})$, and

$$\begin{aligned} L_X(Y) &= \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y) \\ &= \mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y), \end{aligned}$$

which is a subset of $\widetilde{L}_X(Y)$; see 0.1.7. Moreover, $L_f(Y)$ is the restriction of $\widetilde{L}_f(Y)$ to $L_X(Y)$. Finally, the bijection

$$L_X(Y) \xrightarrow{\sim} \mathrm{Hom}_X(Y, X \times_{X^+} X), \qquad m \longmapsto (g, m \cdot g),$$

is the inverse of the restriction to $L_X(Y) \subset \widetilde{L}_X(Y)$ of the bijection

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \xrightarrow{\sim} \{g\} \times \widetilde{L}_X(Y)$$

considered in 0.1.7.1. Consequently, equality (0.1.10 (*)) follows from (0.1.7 (*)). Indeed, let g' be the X-morphism $Y \to \Delta^{(1)}_{X/S}$ defined by $(g, m \cdot g)$. The element of $\widetilde{L}_W(Y)$ corresponding to $(f \circ g, f \circ g')$ is $\widetilde{L}_f(Y)(m) = L_f(Y)(m)$. Thus

$$L_f(m) \cdot (f \circ g) = f \circ (m \cdot g).$$

Lemma 0.1.11. *Let X, X' be two S-schemes. There is a commutative diagram*

$$\begin{tikzcd}
\mathrm{L}_{\mathrm{X}} \times_{\mathrm{S}} \mathrm{L}_{\mathrm{X}'} \arrow[r, "\sim"] \arrow[d] & \mathrm{L}_{\mathrm{X} \times_{\mathrm{S}} \mathrm{X}'} \arrow[d] \\
\mathrm{X}^+ \times_{\mathrm{S}} \mathrm{X}'^+ \arrow[r, "\sim"] & (\mathrm{X} \times_{\mathrm{S}} \mathrm{X}')^+
\end{tikzcd} \quad .$$

Figure 56: Compatibility of L_X and X^+ with products.

[17]Editor's note: Observe that L_f depends only on $f_0 : X_0 \to W_0$.

Proof.[18] For every S-scheme Y,

$$\begin{aligned}\mathrm{Hom}_S(Y, X^+ \times_S X'^+) &= \mathrm{Hom}_S(Y, X^+) \times \mathrm{Hom}_S(Y, X'^+)\\ &\simeq \mathrm{Hom}_S(Y_{\mathscr{J}}, X) \times \mathrm{Hom}_S(Y_{\mathscr{J}}, X')\\ &= \mathrm{Hom}_S(Y, (X \times_S X')^+).\end{aligned}$$

Thus

$$X^+ \times_S X'^+ \simeq (X \times_S X')^+.$$

Now let Y be a scheme over $X^+ \times_S X'^+$, represented by a morphism

$$h : Y_{\mathscr{J}} \longrightarrow X \times_S X'.$$

Let p, q be the projections from $X \times_S X'$ to X, X', and put $f = p \circ h$, $g = q \circ h$. Since

$$\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0} \simeq p_0^*(\Omega^1_{X_0/S_0}) \oplus q_0^*(\Omega^1_{X'_0/S_0})$$

(EGA IV_4, 16.4.23), there is a natural isomorphism

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_{Y_0}}(f_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y) \times \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X'_0/S_0}), \mathscr{J}\mathscr{O}_Y)\\ &\quad\simeq \mathrm{Hom}_{\mathscr{O}_{Y_0}}(h_0^*(\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0}), \mathscr{J}\mathscr{O}_Y).\end{aligned}$$

In other words,

$$L_X(Y) \times L_{X'}(Y) \simeq L_{X \times_S X'}(Y).$$

Remark 0.1.12.[19] Let $\mathscr{C}$ be a category stable under fiber products, let S be an object of $\mathscr{C}$, and let T_1, T_2 be two objects over S. For $i = 1, 2$, let L_i and X_i be two objects over T_i, as in the diagram below.

$$\begin{tikzcd}[column sep=small] L_1 \arrow[dr] & & X_1 \arrow[dl] & & L_2 \arrow[dr] & & X_2 \arrow[dl] \\ & T_1 \arrow[drr] & & & & T_2 \arrow[dll] & \\ & & & S & & & \end{tikzcd}.$$

Figure 57: The objects L_i, X_i over T_i, and T_i over S.

There is then a natural isomorphism

$$(L_1 \times_{T_1} X_1) \times_S (L_2 \times_{T_2} X_2) \simeq (L_1 \times_S L_2) \times_{T_1 \times_S T_2} (X_1 \times_S X_2).$$

Consequently, the preceding lemma gives:

Corollary 0.1.13. *Let X_1, X_2 be two S-schemes. There is a commutative diagram of isomorphisms:*

[18] Editor's note: the proof has been expanded.

[19] Editor's note: this remark and the corollary that follows have been added.

$$\begin{tikzcd}
\mathrm{L}_{\mathrm{X}_1\times_{\mathrm{S}}\mathrm{X}_2} \times_{(\mathrm{X}_1\times_{\mathrm{S}}\mathrm{X}_2)^+} (\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[d, "(0.1.11)"', "\simeq"] \arrow[dr, "\simeq"] & \\
(\mathrm{L}_{\mathrm{X}_1} \times_{\mathrm{S}} \mathrm{L}_{\mathrm{X}_2}) \times_{(\mathrm{X}_1^+\times_{\mathrm{S}}\mathrm{X}_2^+)} (\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r, "\sim", "(0.1.12)"'] & (\mathrm{L}_{\mathrm{X}_1} \times_{\mathrm{X}_1^+} \mathrm{X}_1) \times_{\mathrm{S}} (\mathrm{L}_{\mathrm{X}_2} \times_{\mathrm{X}_2^+} \mathrm{X}_2).
\end{tikzcd}$$

Figure 58: The product isomorphisms of Corollary 0.1.13.

We can now state:

Proposition 0.2. *For every S-scheme X, one can define a left action of the X^+-abelian group L_X on the X^+-object X, with the following properties:*

(i) *this action makes X formally principal homogeneous under L_X over X^+; that is, the morphism*

$$L_X \times_{X^+} X \longrightarrow X \times_{X^+} X$$

is an isomorphism of X^+-functors;

(ii) *the action is functorial in the S-scheme X: for every S-morphism $f : X \to W$, the following diagram is commutative;*

$$\begin{tikzcd}
\mathrm{L}_{\mathrm{X}} \times_{\mathrm{X}^+} \mathrm{X} \arrow[r] \arrow[d, "\mathrm{L}_f \times f"'] & \mathrm{X} \arrow[d, "f"] \\
\mathrm{L}_{\mathrm{W}} \times_{\mathrm{W}^+} \mathrm{W} \arrow[r] & \mathrm{W}
\end{tikzcd} \quad ;$$

Figure 59: Functoriality of the action in Proposition 0.2(ii).

(iii) *the action commutes with fiber products: for all S-schemes X_1, X_2, the following diagram is commutative.*

$$\begin{tikzcd}
\mathrm{L}_{\mathrm{X}_1\times_{\mathrm{S}}\mathrm{X}_2} \times_{(\mathrm{X}_1\times_{\mathrm{S}}\mathrm{X}_2)^+} (\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r] \arrow[d, "\simeq"'] \arrow[dr, "\simeq"] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2 \\
(\mathrm{L}_{\mathrm{X}_1} \times_{\mathrm{S}} \mathrm{L}_{\mathrm{X}_2}) \times_{(\mathrm{X}_1\times_{\mathrm{S}}\mathrm{X}_2)^+} (\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r, "\sim"] & (\mathrm{L}_{\mathrm{X}_1} \times_{\mathrm{X}_1^+} \mathrm{X}_1) \times_{\mathrm{S}} (\mathrm{L}_{\mathrm{X}_2} \times_{\mathrm{X}_2^+} \mathrm{X}_2). \arrow[u]
\end{tikzcd}$$

Figure 60: Compatibility of the action with fiber products.

Proof.[20] Parts (i) and (ii) follow respectively from Corollaries 0.1.9 and 0.1.10. To prove (iii), write

$$P(X) = L_X \times_{X^+} X$$

[20]Editor's note: the original has been modified and expanded to account for the preceding additions; in particular, those additions incorporate, with details, the content of original page 89.

for every S-scheme X. Applying (ii) to the projections

$$p_i : X_1 \times_S X_2 \longrightarrow X_i$$

gives the following commutative squares.

$$\begin{tikzcd}
\mathrm{P}(\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r] \arrow[d, "\mathrm{L}_{p_i} \times p_i"'] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2 \arrow[d, "p_i"] \\
\mathrm{P}(\mathrm{X}_i) \arrow[r] & \mathrm{X}_i
\end{tikzcd}$$

Figure 61: The square induced by each projection p_i.

For $i = 1, 2$, these yield the commutative square

$$\begin{tikzcd}
\mathrm{P}(\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r] \arrow[d] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2 \arrow[d, equal] \\
\mathrm{P}(\mathrm{X}_1) \times_{\mathrm{S}} \mathrm{P}(\mathrm{X}_2) \arrow[r] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2.
\end{tikzcd}$$

Figure 62: The product square in the proof of Proposition 0.2(iii).

Combining this square with Corollary 0.1.13 shows that its vertical arrow is an isomorphism and gives precisely the commutative diagram asserted in (iii).

Remark 0.3. Suppose that the X^+-scheme Y is flat over S (cf. SGA 1, IV). We may then write

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}).$$

Remark 0.4. Let $\pi_0 : X_0 \to S_0$ be the structure morphism, and suppose that there is an $\mathscr{O}_{S_0}$-module $\omega^1_{X_0/S_0}$ such that

$$\Omega^1_{X_0/S_0} \simeq \pi_0^*(\omega^1_{X_0/S_0}).$$

(This occurs, in particular, when X_0 is an S_0-group; cf. II 4.11.) Define a functor L'_X over S by

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_Y). \tag{0.4.1}$$

Then

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X)$$

for every X^+-scheme Y; in other words,

$$L_X = L'_X \times_S X^+.$$

Since $L_X \times_{X^+} X = L'_X \times_S X$, the action of L_X on X induces an action of L'_X on X, and this action respects $p_X : X \to X^+$.[21] Indeed, if Y is an S-scheme, $h : Y \to X$ is an

[21] Editor's note: what follows has been expanded.

S-morphism, and $m \in L'_X(Y)$, then h and $m \cdot h$ have the same restriction to $Y_{\mathscr{J}}$; that is,

$$p_X(m \cdot h) = p_X(h).$$

Remark 0.5. Keep the hypotheses and notation of 0.4, and suppose in addition that Y is an S-scheme flat over S. Then

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{S_0}(Y_0, L_{0X}),$$

where the S_0-functor in abelian groups L_{0X} is defined by the following identity (with T the variable S_0-scheme):

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{\mathscr{O}_T}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T). \tag{0.5.1}$$

Thus, in the notation of II 1, we have shown that L'_X and $\prod_{S_0/S} L_{0X}$ have the same restriction to the full subcategory of $(\mathbf{Sch})/S$ whose objects are the S-schemes flat over S.

Remark 0.6. Keep the hypotheses and notation of 0.5,[22] and suppose in addition that $\pi_0 : X_0 \to S_0$ has a section ε_0. Then

$$\omega^1_{X_0/S_0} \simeq \varepsilon_0^*(\Omega^1_{X_0/S_0}).$$

First, independently of this last hypothesis, one has

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma\Big(T, \mathscr{H}om_{\mathscr{O}_T}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T)\Big).$$

Suppose now that $\omega^1_{X_0/S_0}$ admits a finite presentation (cf. EGA 0_{I}, 5.2.5), as is in particular the case when X_0 is locally of finite presentation over S_0 (cf. EGA IV_4, 16.4.22). If T is flat over S_0, it then follows from EGA 0_{I}, 6.7.6 that

$$\mathscr{H}om_{\mathscr{O}_T}(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T) \simeq \mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T,$$

whence

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma\Big(T, \mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\Big).$$

Using the notation $\mathbf{W}(\,\cdot\,)$ of I 4.6.1, we have therefore proved that, for every S_0-scheme T flat over S_0,

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{S_0}\Big(T, \mathbf{W}\Big(\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big)\Big).$$

In summary, if $\omega^1_{X_0/S_0}$ admits a finite presentation and one restricts to the category of S-schemes flat over S, then

$$L'_X = \prod_{S_0/S} \mathbf{W}\Big(\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big), \tag{0.6.1}$$

and $\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})$ is a quasi-coherent $\mathscr{O}_{S_0}$-module, by EGA I, 9.1.1.

Finally, suppose moreover that $\omega^1_{X_0/S_0}$ is locally free of finite rank, for example that X_0 is smooth over S_0 (in which case it is automatically locally of finite presentation over S_0). Then

$$\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \simeq \underline{\mathrm{Lie}}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}, \tag{0.6.2}$$

[22] Editor's note: the following hypothesis, which was implicit in the original, has been added.

where, by an abuse of language (since X_0 need not be an S_0-group), $\underline{\mathrm{Lie}}(X_0/S_0)$ denotes the dual of the $\mathscr{O}_{S_0}$-module $\omega^1_{X_0/S_0}$.[23]

Proposition 0.2 (and its proof) has two important corollaries.[24]

Corollary 0.7. *Let X be an S-scheme.*

(a) *Every S-endomorphism of X inducing the identity on $X_{\mathscr{J}}$ is an automorphism.*

(b) *There is an exact sequence of groups*

$$0 \longrightarrow \mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathscr{J}}}(X_{\mathscr{J}}).$$

(c) *Let $\mathrm{Aut}_S(X)$ act on the first group by transport of structure. Then, for every $u \in \mathrm{Aut}_S(X)$ and $m \in \mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X)$,*

$$i(u\cdot m) = u\, i(m)\, u^{-1}.$$

Proof. By 0.2(i), $\mathrm{Hom}_{X^+}(X,X)$ is a principal homogeneous set under $\mathrm{Hom}_{X^+}(X,L_X)$, since it is certainly nonempty: it contains the marked point given by the identity automorphism of X.[25] Consequently, the map $m \mapsto m\cdot \mathrm{id}_X$ induces a bijection

$$\mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X) = L_X(X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X,X).$$

Let $m \in L_X(X)$, and let $f = m'\cdot \mathrm{id}_X$ be an element of $\mathrm{Hom}_{X^+}(X,X)$. Applying 0.2(ii) to f gives

$$f\circ(m\cdot\mathrm{id}_X) = L_f(X)(m)\cdot f = L_f(X)(m)\cdot(m'\cdot \mathrm{id}_X).$$

On the other hand, because f is an X^+-endomorphism of X, one has $f_{\mathscr{J}} = \mathrm{id}_{X_{\mathscr{J}}}$, hence $f_0 = \mathrm{id}_{X_0}$. Since L_f depends only on f_0 (cf. the editor's note (17) in 0.1.10), it follows that $L_f(X)(m) = m$. Thus the preceding equality becomes

$$(m'\cdot\mathrm{id}_X)\circ(m\cdot\mathrm{id}_X) = m\cdot(m'\cdot\mathrm{id}_X) = (m+m')\cdot\mathrm{id}_X\,.$$

This shows that the bijection $m \mapsto m\cdot\mathrm{id}_X$ carries the group law of $\mathrm{Hom}_{X^+}(X,L_X)$ to composition of X^+-endomorphisms of X.

It follows first that every element of $\mathrm{Hom}_{X^+}(X,X)$ is invertible, proving part (a), and then that there is an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{X^+}(X,L_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathscr{J}}}(X_{\mathscr{J}}),$$

which proves part (b).

The morphism i just defined is functorial in X with respect to isomorphisms, because it is defined structurally from the action of L_X on X over X^+, an action that is itself functorial in X by Proposition 0.2(ii).[26] Hence every automorphism u of X over S induces, by transport of structure, isomorphisms

$$h : \mathrm{Hom}_{X^+}(X,L_X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X,L_X) \quad\text{and}\quad f : \mathrm{Aut}_S(X) \xrightarrow{\sim} \mathrm{Aut}_S(X)$$

[23] Editor's note: this is justified by II 3.3 and 4.11. Indeed, the functor $L^{\varepsilon_0}_{X_0/S_0}$, the "tangent space to X_0 over S_0 at ε_0," is represented over S_0 by the vector bundle $\mathbf{V}(\varepsilon_0^*(\Omega^1_{X_0/S_0})) = \mathbf{V}(\omega^1_{X_0/S_0})$. Its sheaf of sections is the dual module $\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{O}_{S_0})$; this is $\underline{\mathrm{Lie}}(X_0/S_0)$ when X_0 is an S_0-group and ε_0 is the identity section.

[24] Editor's note: Corollary 0.7 bis, which was used implicitly in the proof of 0.8, has been added to 0.7. We point out that 0.8–0.12, as well as 0.17, which play an important technical role later in this exposé, are consequences of 0.7 and 0.7 bis.

[25] Editor's note: what follows has been expanded.

[26] Editor's note: what follows has been expanded.

such that the following diagram commutes.

$$\begin{tikzcd} \mathrm{Hom}_{X^+}(X, L_X) \arrow[r, "i"] \arrow[d, "h"'] & \mathrm{Aut}_S(X) \arrow[d, "f"] \\ \mathrm{Hom}_{X^+}(X, L_X) \arrow[r, "i"] & \mathrm{Aut}_S(X) \end{tikzcd}$$

Figure 63: Functoriality of i under transport of structure.

Thus $f \circ i = i \circ h$. On the other hand, f is described by the following commutative diagram.

$$\begin{tikzcd} X \arrow[r, "a"] \arrow[d, "u"'] & X \arrow[d, "u"] \\ X \arrow[r, "f(a)"] & X \end{tikzcd}$$

Figure 64: The automorphism f acts by conjugation with u.

In other words,

$$f(a) = u \circ a \circ u^{-1} \qquad \text{for every } a \in \mathrm{Aut}_S(X).$$

Writing $i(h(m)) = f(i(m))$ now gives the asserted formula.

Corollary 0.7 bis. *Let X be an S-scheme such that $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-monoid. Then L_X has the structure of an S-monoid, and there is a split exact sequence of S-monoids:*

$$\begin{tikzcd} 1 \arrow[r] & L'_X \arrow[r, "i"] & L_X \arrow[r, shift left, "p"] & X^+ \arrow[l, shift left, "s"] \arrow[r] & 1 \end{tikzcd}$$

Figure 65: The split exact sequence of S-monoids in Corollary 0.7 bis.

The monoid law induced on L'_X coincides with its abelian-group structure. In particular, if $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group, then L_X is an S-group and is the semidirect product of X^+ and L'_X.

Proof. Since $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-monoid,

$$X^+ = \prod_{S_{\mathscr{J}}/S} X_{\mathscr{J}}$$

is an S-monoid; indeed, $X^+(Y) = X_{\mathscr{J}}(Y_{\mathscr{J}})$ for every $Y \to S$. For every S-scheme Y, let $\widetilde{Y}_{\mathscr{J}}$ denote the affine $Y_{\mathscr{J}}$-scheme corresponding to the quasi-coherent $\mathscr{O}_{Y_{\mathscr{J}}}$-algebra

$$\mathscr{O}_{Y_{\mathscr{J}}} \oplus \mathscr{J}\mathscr{O}_Y,$$

that is, the graded algebra associated with the filtration $\mathscr{O}_Y \supset \mathscr{J}\mathscr{O}_Y$. Then $L_X(Y)$ identifies with $X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}})$, and $L'_X(Y)$ identifies with the kernel of the morphism

$$p : X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}}) \longrightarrow X_{\mathscr{J}}(Y_{\mathscr{J}})$$

induced by the *zero section* $Y_{\mathscr{J}} \to \widetilde{Y}_{\mathscr{J}}$. Equivalently, that section is induced by the morphism of $\mathscr{O}_{Y_{\mathscr{J}}}$-algebras

$$\mathscr{O}_{\widetilde{Y}_{\mathscr{J}}} \longrightarrow \mathscr{O}_{Y_{\mathscr{J}}}$$

that vanishes on the ideal $\mathscr{J}\mathscr{O}_Y$. Thus, for every $Y \to S$, there is a split exact sequence of monoids, functorial in Y:

$$\begin{tikzcd} 1 \arrow[r] & \mathrm{L}'_{\mathrm{X}}(\mathrm{Y}) \arrow[r, "i"] & \mathrm{L}_{\mathrm{X}}(\mathrm{Y}) \arrow[r, shift left, "p"] & \mathrm{X}^{+}(\mathrm{Y}) \arrow[l, shift left, "s"] \arrow[r] & 1 . \end{tikzcd}$$

Figure 66: The pointwise split exact sequence for $L_X(Y)$.

It remains to show that the monoid law induced on L'_X coincides with its abelian-group structure. Let μ be the monoid law of L_X, and let e be its unit section. We must prove that, for every $m, m' \in L'_X(Y)$,

$$\mu(m \cdot e, m' \cdot e) = (m + m') \cdot e.$$

This may be seen in either of two ways. First, repeat the proof of (0.1.10 $(*)$), replacing the morphism $f : X \to W$ there by

$$\mu : L_X \times_S L_X \longrightarrow L_X.$$

Identify $X^+(Y) = X_{\mathscr{J}}(Y_{\mathscr{J}})$ with its image under s in $L_X(Y) = X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}})$. For all $g, g' \in X_{\mathscr{J}}(Y_{\mathscr{J}})$ and $m, m' \in L'_X(Y)$, one then obtains

$$\mu(m \cdot g, m' \cdot g') = L_\mu^{(g,g')}(m, m') \cdot \mu(g, g'). \tag{$\star$}$$

Here $L_\mu^{(g,g')}$ denotes the morphism derived from μ at the point (g, g'); in other words, $\widetilde{Y}_{\mathscr{J}}$ lies over $L_X \times_S L_X$ through (g, g'). In particular,

$$\mu(m \cdot e, m' \cdot e) = L_\mu^{(e,e)}(m, m') \cdot e.$$

But

$$L_\mu^{(e,e)}(m, m') = L_{\ell_e}(m') + L_{r_e}(m),$$

where ℓ_e, respectively r_e, denotes left, respectively right, translation by e. Each is the identity map of $X_{\mathscr{J}}$, and therefore $L_\mu^{(e,e)}(m, m') = m + m'$.

Alternatively, one may argue as follows (cf. the proof of [DG70], §II.4, Theorem 3.5). By Lemma 0.1.11, formation of X^+ and of L_X commutes with products, and hence so does formation of L'_X. It follows that the morphism

$$\mu' : L'_X \times L'_X \longrightarrow L'_X$$

induced by μ is a homomorphism for the abelian-group structure. Lemma 3.10 of Exposé II then shows that μ' coincides with the abelian-group law.

0.8.[27] Now let X be an S-scheme such that $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group. Suppose that there is an S-morphism

$$P : X \times_S X \longrightarrow X$$

[27] Editor's note: the number 0.8 has been inserted here to mark the return to the original text.

whose base change

$$P_{\mathscr{I}} : X_{\mathscr{I}} \times_{S_{\mathscr{I}}} X_{\mathscr{I}} \longrightarrow X_{\mathscr{I}}$$

is the group law of $X_{\mathscr{I}}$. An important special case is that in which X is an S-group and P is its group law. We obtain a morphism

$$L_P : L_X \times_S L_X \simeq L_{X \times_S X} \longrightarrow L_X.$$

In fact, this morphism does not depend on P, since it can be computed from the group law $P_{\mathscr{I}}$ of $X_{\mathscr{I}}$, as we now show.[28]

By Proposition 0.2(ii) and (iii), for every $Y \to S$, every $x, x' \in X(Y)$, and every $m, m' \in L'_X(Y)$,

$$\begin{aligned} P(m \cdot x, m' \cdot x') &= P((m, m') \cdot (x, x')) \\ &= L_P^{(x,x')}(m, m') \cdot \mu(g, g'), \end{aligned}$$

where g, respectively g', is the image of x, respectively x', in $X^+(Y)$. Moreover (cf. the proof of 0.10), $L_P^{(x,x')} = L_\mu^{(g,g')}$, and by 0.7 bis ($\star$) this is the element of $L'_X(Y)$ defined by the equality

$$L_\mu^{(g,g')}(m, m') \cdot \mu(g, g') = \mu(m \cdot g, m' \cdot g')$$

in $L_X(Y)$.

Write $\times$, rather than μ, for the group law of L_X, and write Ad for the *adjoint action* of X^+ on L'_X. This action factors through X_0 and is induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$. Then

$$\begin{aligned} L_\mu^{(g,g')}(m, m') \times g \times g' &= m \times g \times m' \times g' \\ &= (m \times \mathrm{Ad}(g)(m')) \times g \times g', \end{aligned}$$

and hence

$$L_P^{(x,x')}(m, m') = m \times \mathrm{Ad}(g)(m').$$

We have therefore proved:

Proposition 0.8. *Let $P : X \times_S X \to X$ be an S-morphism such that $P_{\mathscr{I}}$ equips $X_{\mathscr{I}}$ with an $S_{\mathscr{I}}$-group structure. Let $\times$ denote the group law of L'_X, let $(m, x) \mapsto m \cdot x$ denote the morphism $L'_X \times_S X \to X$ defining the action of L'_X on X, and let*

$$\mathrm{Ad} : X^+ \longrightarrow \mathrm{Aut}_{S\text{-gr}}(L'_X)$$

be the adjoint action of X^+ on L'_X, induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$. Then, for every $S' \to S$, every $x, x' \in X(S')$, and every $m, m' \in L'_X(S')$,

$$P(m \cdot x, m' \cdot x') = (m \times \mathrm{Ad}(p_X(x))(m')) \cdot P(x, x'). \tag{0.8.1}$$

If X is an S-group, we shall write $*$ for its law, e for its unit section, and i for the S-morphism defined by

$$i(m) = m \cdot e$$

for every $S' \to S$ and every $m \in L'_X(S')$.

Corollary 0.9. *Let X be an S-group. Then X^+ is naturally an S-group, and p_X is a morphism of S-groups. Moreover, the S-morphism*

$$i : L'_X \longrightarrow X, \qquad m \longmapsto m \cdot e,$$

[28] Editor's note: the original has been expanded in what follows.

is an isomorphism of S-groups from L'_X onto $\operatorname{Ker}(p_X)$*, and, for every $S' \to S$, every $x' \in X(S')$, and every $m \in L'_X(S')$,*

$$m \cdot x' = (m \cdot e) * x' = i(m) * x'. \tag{0.9.1}$$

The first two assertions were already proved in 0.1.2. Since X is formally principal homogeneous over X^+ under $L_X = L'_X \times_S X^+$, the morphism i is indeed an isomorphism of S-functors from L'_X onto the kernel of p_X. The fact that i is a group morphism and formula (0.9.1) follow from (0.8.1), applied respectively with $x = x' = e$, and with $x = e,\ m' = 1$.

Corollary 0.10. *Let X be an S-group. With the preceding notation, for every $S' \to S$, every $x \in X(S')$, and every $m' \in L'_X(S')$,*

$$x * i(m') * x^{-1} = i\big(\operatorname{Ad}(p_X(x))(m')\big). \tag{0.10.1}$$

This follows from the equality $i(m') * x^{-1} = m' \cdot x^{-1}$ and from (0.8.1), applied with $m = 1$ and $x' = x^{-1}$.

Thus, when X is an S-group, we have explicitly determined the kernel of $X \to X^+$ and the action of the inner automorphisms of X on this kernel. We shall now see that the same can be done for certain, not necessarily representable, S-group functors. One case will be useful to us, namely the functors $\underline{\operatorname{Aut}}$ of Exposé I, 1.7. We state it at once in the next proposition.

Proposition 0.11. *Let E be an S-scheme, and put $X = \underline{\operatorname{Aut}}_S(E)$. The kernel of the morphism of S-group functors*

$$p_X : X \longrightarrow X^+$$

is canonically identified with the commutative S-group functor L'_X defined by

$$\operatorname{Hom}_S(Y, L'_X) = \operatorname{Hom}_{\mathscr{O}_{E_0 \times_{S_0} Y_0}}\left(\Omega^1_{E_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{E \times_S Y}\right),$$

where Y denotes a variable S-scheme.

Indeed, for a variable S-scheme Y, we have

$$\operatorname{Hom}_S(Y, X) = \operatorname{Aut}_Y(E \times_S Y)$$

and

$$\begin{aligned}\operatorname{Hom}_S(Y, X^+) &= \operatorname{Hom}_S(Y_{\mathscr{J}}, X)\\ &= \operatorname{Aut}_{Y_{\mathscr{J}}}(E \times_S Y_{\mathscr{J}})\\ &= \operatorname{Aut}_{Y_{\mathscr{J}}}\big((E \times_S Y) \times_Y Y_{\mathscr{J}}\big).\end{aligned}$$

Applying 0.7 (b) to the Y-scheme $E \times_S Y$, we obtain an isomorphism of groups

$$\operatorname{Hom}_S(Y, L'_X) \simeq \operatorname{Ker}\Big(\operatorname{Hom}_S(Y, X) \longrightarrow \operatorname{Hom}_S(Y, X^+)\Big).$$

This isomorphism is readily checked to be functorial in the S-scheme Y. Hence it yields an isomorphism of S-groups

$$L'_X \simeq \operatorname{Ker}(X \longrightarrow X^+),$$

which completes the proof of Proposition 0.11.

Corollary 0.12.[29] *Retain the notation of 0.11: E is an S-scheme and $X = \underline{\operatorname{Aut}}_S(E)$. There is a natural action f of X on L'_X, defined as follows. For every S-scheme Y,*

$$\operatorname{Hom}_S(Y, X) = \operatorname{Aut}_Y(E \times_S Y)$$

[29] Editor's note: "Remark" has been changed to "Corollary".

and

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{E_0\times_{S_0}Y_0}}\left(\Omega^1_{E_0\times_{S_0}Y_0/Y_0}, \mathscr{J}\,\mathscr{O}_{E\times_S Y}\right).$$

Here

$$\Omega^1_{E_0/S_0}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}\simeq\Omega^1_{E_0\times_{S_0}Y_0/Y_0}$$

(see EGA IV$_4$*, 16.4.5). The first group acts on the second by transport of structure, and this action is functorial in* Y*. One then has*

$$x\,i(m)\,x^{-1} = i(f(x)m), \tag{0.12.1}$$

for every $Y\to S$*,* $x\in\mathrm{Hom}_S(Y,X)$*, and* $m\in\mathrm{Hom}_S(Y,L'_X)$*.*

Indeed, this follows from 0.7 (c), applied to the Y-scheme $E\times_S Y$.

Reminder 0.13. The direct image of a quasi-coherent module under a morphism of finite presentation is quasi-coherent. Under the same hypotheses, formation of the direct image commutes with flat base change. In the situation

$$\begin{tikzcd} \mathrm{T} \arrow[d, "f"'] & \mathrm{T}' = \mathrm{T}\times_{\mathrm{S}}\mathrm{S}' \arrow[l, "g'"'] \arrow[d, "f'"] \\ \mathrm{S} & \mathrm{S}' \arrow[l, "g"'] \end{tikzcd}$$

Figure 67: The flat base-change square in Reminder 0.13.

if f (and hence f') is of finite presentation and g (and hence g') is flat, then for every quasi-coherent $\mathscr{O}_T$-module $\mathscr{F}$,

$$f_*(\mathscr{F})\otimes_{\mathscr{O}_S}\mathscr{O}_{S'} = f'_*(\mathscr{F}\otimes_{\mathscr{O}_S}\mathscr{O}_{S'}),$$

or, more elegantly,

$$g^*(f_*\mathscr{F}) = f'_*(g'^*\mathscr{F}).$$

These two facts hold more generally for a quasi-compact and quasi-separated morphism f; see EGA I, 9.2.1, and EGA III$_1$, 1.4.15 in the quasi-compact separated case (taking EGA III$_2$, $\mathrm{Err}_{\mathrm{III}}$ 25 into account), as well as EGA IV$_1$, 1.7.4 and 1.7.21.

Remark 0.14.[30] Return to the notation of 0.11: let E be an S-scheme, put $X = \underline{\mathrm{Aut}}_S(E)$, and let L'_X be the commutative S-group functor defined by

$$\begin{aligned} L'_X(Y) &= \mathrm{Hom}_{\mathscr{O}_{E_0\times_{S_0}Y_0}}\left(\Omega^1_{E_0/S_0}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}, \mathscr{J}\,\mathscr{O}_{E\times_S Y}\right) \\ &= \mathrm{Hom}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, \mathscr{J}\,\mathscr{O}_{E\times_S Y}\right) \\ &= \Gamma\Big(E\times_S Y, \widetilde{\mathrm{Hom}}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, \mathscr{J}\,\mathscr{O}_{E\times_S Y}\right)\Big). \end{aligned}$$

Suppose that Y is flat over S. There are then isomorphisms

$$\mathscr{J}\,\mathscr{O}_{E\times_S Y}\xleftarrow{\ \sim\ }(\mathscr{J}\,\mathscr{O}_E)\otimes_{\mathscr{O}_S}\mathscr{O}_Y\xrightarrow{\ \sim\ }(\mathscr{J}\,\mathscr{O}_E)\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}.$$

[30] Editor's note: The first part of this remark has been expanded, and a second part has been added.

Suppose in addition that E is of finite presentation over S. Then $\Omega^1_{E/S}$ is an $\mathscr{O}_E$-module of finite presentation (EGA IV_4, 16.4.22), and therefore, by EGA 0_{I}, 6.7.6,

$$\begin{aligned}&\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y,(\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_S}\mathscr{O}_Y\right)\\ &\simeq \widetilde{\mathrm{Hom}}_{\mathscr{O}_E}\left(\Omega^1_{E/S},\mathscr{J}\mathscr{O}_E\right)\otimes_{\mathscr{O}_S}\mathscr{O}_Y.\end{aligned}$$

Let $\pi : E \to S$ and $g : Y \to S$ be the structural morphisms. Apply 0.13 to the diagram

$$\begin{tikzcd} E & E\times_S Y \arrow[l, "g'"'] \arrow[d, "\pi'"] \\ S \arrow[from=u, "\pi"'] & Y \arrow[l, "g"] \end{tikzcd}$$

Figure 68: The structural base-change square used in Remark 0.14.

and to the $\mathscr{O}_E$-module

$$\mathscr{F} = \widetilde{\mathrm{Hom}}_{\mathscr{O}_E}\left(\Omega^1_{E/S},\mathscr{J}\mathscr{O}_E\right).$$

We obtain

$$\begin{aligned}\Gamma(E\times_S Y, g'^*\mathscr{F}) &= \Gamma(Y,\pi'_* g'^*\mathscr{F})\\ &= \Gamma(Y, g^*\pi_*\mathscr{F})\\ &= W(\pi_*\mathscr{F})(Y).\end{aligned}$$

Thus, if E is of finite presentation over S, we have

$$L'_X = W\Big(\pi_*\widetilde{\mathrm{Hom}}_{\mathscr{O}_E}(\Omega^1_{E/S},\mathscr{J}\mathscr{O}_E)\Big) \tag{0.14.1}$$

on the category of S-schemes flat over S. Notice also that the module to which W is applied is quasi-coherent, by EGA I, 9.1.1 and 9.2.1.

Let L_0 be the S_0-functor[31]

$$L_0 = W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0},\mathscr{J}\mathscr{O}_E)\Big).$$

Returning to the definition of $L'_X(Y)$, and using the isomorphism

$$\mathscr{J}\mathscr{O}_{E\times_S Y}\simeq(\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0},$$

the same reasoning gives

$$L'_X(Y) = L_0(Y_0) = L_0(Y\times_S S_0) = \left(\prod_{S_0/S} L_0\right)(Y).$$

Consequently, on the category of S-schemes flat over S,

$$L'_X = \prod_{S_0/S} W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0},\mathscr{J}\mathscr{O}_E)\Big).$$

[31] Editor's note: What follows has been added.

It is not obvious that the action of X on L'_X defined in 0.12 comes from an action of $X_0 = \underline{\mathrm{Aut}}_{S_0}(E_0)$ on L_0. This is nevertheless the case when E is also flat over S.

Indeed, in that case there are canonical isomorphisms

$$\mathscr{J}\mathscr{O}_E \simeq \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_E \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0},$$

and hence

$$L_0 \simeq W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}\left(\Omega^1_{E_0/S_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0}\right)\Big),$$

and

$$L'_X = \prod_{S_0/S} W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}\left(\Omega^1_{E_0/S_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0}\right)\Big). \tag{0.14.2}$$

For every S_0-scheme T, we then have

$$L_0(T) \simeq \mathrm{Hom}_{\mathscr{O}_{E_0 \times_{S_0} T}}\left(\Omega^1_{E_0 \times_{S_0} T/T}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0 \times_{S_0} T}\right).$$

Moreover,

$$\mathrm{Hom}_{S_0}(T, X_0) = \mathrm{Aut}_T(E_0 \times_{S_0} T)$$

acts on $L_0(T)$ by transport of structure, functorially in T. Finally, for every S-scheme Y flat over S, the action by transport of structure of

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$

on $L'_X(Y) = L_0(Y_0)$ factors through $\mathrm{Aut}_{Y_0}(E_0 \times_{S_0} Y_0)$.

Finally, let us extract the following two propositions from SGA 1, Exposé III.

Proposition 0.15. *(SGA 1, Exposé III, 6.8)*[32] *For every affine $S_{\mathscr{J}}$-scheme Y smooth over $S_{\mathscr{J}}$, there exists an S-scheme X, smooth over S, such that $X \times_S S_{\mathscr{J}} \simeq Y$; such an X is unique up to a non-unique isomorphism.*

Proposition 0.16. *(SGA 1, Exposé III, 5.5)*[33] *Let X be an S-scheme smooth over S. For every affine S-scheme Y, the canonical map*

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}})$$

is surjective.

Corollary 0.17. *Let E be an affine S-scheme smooth over S, and put $X = \underline{\mathrm{Aut}}_S(E)$. For every affine S-scheme Y, the canonical map*

$$\begin{aligned} \mathrm{Aut}_Y(E \times_S Y) = \mathrm{Hom}_S(Y, X) &\longrightarrow \mathrm{Hom}_S(Y, X^+) \\ &= \mathrm{Aut}_{Y_{\mathscr{J}}}(E_{\mathscr{J}} \times_{S_{\mathscr{J}}} Y_{\mathscr{J}}) \end{aligned}$$

is surjective.

Indeed, $Y \times_S E$ is affine over the affine scheme Y, hence is affine. Applying 0.16, we see that every $S_{\mathscr{J}}$-morphism

$$Y_{\mathscr{J}} \times_{S_{\mathscr{J}}} E_{\mathscr{J}} \longrightarrow E_{\mathscr{J}}$$

extends to an S-morphism

$$Y \times_S E \longrightarrow E.$$

In other words,[34] every $Y_{\mathscr{J}}$-endomorphism of $Y_{\mathscr{J}} \times_{S_{\mathscr{J}}} E_{\mathscr{J}}$ lifts to a Y-endomorphism of $Y \times_S E$. Then 0.7 (a) shows that every $Y_{\mathscr{J}}$-automorphism of $Y_{\mathscr{J}} \times_{S_{\mathscr{J}}} E_{\mathscr{J}}$ lifts to a Y-automorphism of $Y \times_S E$, which is the asserted property.

[32] Editor's note: This "global" lifting result uses the "local" lifting result of loc. cit., 4.1, which is stated for S locally Noetherian; see EGA IV_4, 18.1.1 for the general case.

[33] Editor's note: This is an immediate consequence of the definition of "(formally) smooth" adopted in EGA IV_4, 17.3.1 (and 17.1.1).

[34] Editor's note: The original has been modified slightly so that the hypotheses of 0.7 apply exactly.

1. Extensions and Cohomology

1.1.

Let $\mathscr{C}$ be a category stable under fiber products.[35] Let S be an object of $\mathscr{C}$, let G be a representable S-group, and let F be an S-functor in commutative groups on which G acts. The cohomology groups $H^n(G, F)$ were defined in I 5.1. Recall that they are the cohomology groups of a complex denoted $C^*(G, F)$. Writing

$$(G/S)^n = G \times_S \cdots \times_S G \qquad (n \text{ factors}),$$

one has

$$C^n(G, F) = \mathrm{Hom}_S((G/S)^n, F).$$

Since G, and hence each $(G/S)^n$, is representable, one also has

$$C^n(G, F) = F((G/S)^n).$$

It follows from this and from the definition of the coboundary operator that the complex $C^*(G, F)$ depends only on the restriction of F to the full subcategory of $\mathscr{C}/S$ whose objects are the Cartesian powers of G over S. Consequently:

Lemma 1.1.1. *Let $\mathscr{C}$ be a category stable under fiber products, let S be an object of $\mathscr{C}$, and let G be a representable S-group. Denote by $\mathscr{C}(G)$ the full subcategory of $\mathscr{C}/S$ whose objects are the Cartesian powers of G over S. Let F and F' be two S-functors in commutative groups on which G acts. If F and F' have the same restriction to $\mathscr{C}(G)$, there is a canonical isomorphism*

$$H^*(G, F) \xrightarrow{\sim} H^*(G, F').$$

1.1.2. Cohomology and restriction of scalars.[36]

We record another comparison result. Let $T \to S$ be a morphism in $\mathscr{C}$. If F is a T-functor in commutative groups, then the functor obtained by restriction of scalars (cf. Exposé II, 1),

$$F_1 = \prod_{T/S} F,$$

is an S-functor in commutative groups, and there is a morphism of S-functors in groups[37]

$$u : \prod_{T/S} \underline{\mathrm{Aut}}_{T\text{-gr.}}(F) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(F_1).$$

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[35] Editor's note: the hypothesis that $\mathscr{C}$ is stable under fiber products has been added. It holds in the applications in which $\mathscr{C}$ is the category of schemes (**Sch**), or the category of functors $(\mathbf{Sch})^\circ \to (\mathbf{Ens})$. If this hypothesis is omitted, one must instead assume below that the fiber products $G \times_S \cdots \times_S G$ are representable.

[36] Editor's note: the heading of this paragraph has been added.

[37] Editor's note: explicitly, let $S' \to S$ and $\alpha \in \underline{\mathrm{Aut}}_{(T\times_S S')\text{-gr.}}(F_{T\times_S S'})$. For every $U \to T \times_S S'$, let $\alpha_U \in \mathrm{Aut}_{\mathrm{gr.}}(F_{T\times_S S'}(U))$ be the element defined by α. Then $u(S')(\alpha)$ is the element β of

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(F_1)(S') = \underline{\mathrm{Aut}}_{S'\text{-gr.}}\left(\prod_{T\times_S S'/S'} F_{T\times_S S'}\right)$$

such that $\beta_{S''} = \alpha_{T\times_S S''}$ for every $S'' \to S'$.

Now let G be an S-functor in groups, and suppose an action of G_T on F is given by

$$G_T \longrightarrow \underline{\mathrm{Aut}}_{T\text{-gr.}}(F).$$

By the definition of $\prod_{T/S}$, this gives a morphism of S-functors in groups

$$G \longrightarrow \prod_{T/S} \underline{\mathrm{Aut}}_{T\text{-gr.}}(F).$$

Composing with u therefore gives an action of G on $F_1 = \prod_{T/S} F$.[38]

Lemma 1.1.2. *Under the preceding assumptions, there is a canonical isomorphism*

$$H^*\Big(G, \prod_{T/S} F\Big) \simeq H^*(G_T, F).$$

Indeed, by the definition of cohomology, the standard complexes are canonically isomorphic.

1.2. Lifting group morphisms.[39]

Following the general principles, we make the following definition.

Definition 1.2.1. *Let*

$$1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be a sequence of morphisms of $\widehat{\mathscr{C}}$-groups. It is called exact *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the following sequence of ordinary groups is exact:*

$$1 \longrightarrow M(S) \xrightarrow{u(S)} E(S) \xrightarrow{v(S)} G(S);$$

(ii) *for every object H of $\widehat{\mathscr{C}}$, the following sequence of ordinary groups is exact:*

$$1 \longrightarrow \mathrm{Hom}(H, M) \xrightarrow{u(H)} \mathrm{Hom}(H, E) \xrightarrow{v(H)} \mathrm{Hom}(H, G).$$

Taking $H = G$ in (ii), one sees that the set of sections of v (which need not preserve the group structures) is either empty or principal homogeneous under $\mathrm{Hom}(G, M)$. Suppose that it is nonempty, and let

$$s : G \longrightarrow E$$

be a section of v. For every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $x \in G(S)$, the element $s(x) \in E(S)$ defines an inner automorphism of E_S that normalizes M_S (more precisely, the image of M_S under u_S), and hence an automorphism of M_S.

[38] Editor's note: explicitly, for every $S' \to S$, the action of $G(S')$ on $F_1(S') = F(T \times_S S')$ is given by the action of $G_T(T \times_S S') = G(T \times_S S')$ on $F(T \times_S S')$, together with the group morphism $G(S') \to G(T \times_S S')$ corresponding to the projection $T \times_S S' \to S'$. Equivalently, the action of G_T on F is a morphism $G_T \times_T F \to F$. Applying $\prod_{T/S}$, and writing $G_1 = \prod_{T/S} G_T$, gives, by II 1.2, a morphism $G_1 \times_S F_1 \to F_1$, hence an action of G_1 on F_1. The action of G is obtained through the natural morphism $G \to G_1$, which corresponds by adjunction to id_G.

[39] Editor's note: this heading has been added.

Scholium 1.2.1.1. [40] *If M is commutative, one sees set-theoretically that this automorphism is independent of the chosen section and depends only on x, and that it depends multiplicatively on x. In summary, every exact sequence*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

in which M is commutative and v admits a section determines a morphism of $\widehat{\mathscr{C}}$-groups

$$G \longrightarrow \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M),$$

called the action of G on M defined by the extension (E).

Definition 1.2.1.2. We saw in I 2.3.7 that v admits a section that is a morphism of $\widehat{\mathscr{C}}$-groups if and only if the extension (E) is isomorphic, as an extension, to the semidirect product of M by G for the preceding action. Such a section of v is called a *section of the extension* (E), or simply a *section of* (E).

If s is a section of (E) and

$$m \in \Gamma(M) \simeq \mathrm{Ker}(\Gamma(E) \to \Gamma(G))$$

(for the definition of Γ, see I 1.2), then the morphism $G \to E$ defined by

$$x \longmapsto u(m)s(x)u(m)^{-1}$$

is also a section of (E).[41] It is called the section deduced from s by the inner automorphism defined by m, or by $u(m)$.

Lemma 1.2.2. *Let*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups in which M is commutative and v admits a section. Let G act on M in the manner defined by (E).

(i) *The extension (E) canonically defines a class $c(E) \in H^2(G, M)$, whose vanishing is necessary and sufficient for the existence of a section of (E).*

(ii) *If $c(E) = 0$, the set of sections of (E) is principal homogeneous under $Z^1(G, M)$, and the set of sections of (E), modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$, is principal homogeneous under $H^1(G, M)$.*

(iii) [42] *Let s be a section of (E). The set of conjugates of s by the inner automorphisms defined by $\Gamma(M)$ is in bijection with $\Gamma(M)/H^0(G, M)$.*

The proof is exactly the same as for ordinary groups; starting from a section of v ensures the functoriality of the set-theoretic computations. We briefly indicate the principal steps.

(a) To every section s of v, associate the morphism

$$D_s : G \times G \longrightarrow M$$

defined set-theoretically by

$$u(D_s(x, y)) = s(xy)s(y)^{-1}s(x)^{-1}.$$

[40] Editor's note: the numbering 1.2.1.1 and 1.2.1.2 has been added to highlight the notions introduced here.

[41] Editor's note: in the right-hand side, x has been corrected to $s(x)$.

[42] Editor's note: this item has been added as the counterpart of 1.2.4(iii). The assertion is valid for every section of v; see the proof.

The following computation shows that D_s is a 2-cocycle.[43] By the definition of the differential in the standard complex (I 5.1),

$$(\partial^2 D_s)(x,y,z) = \big(s(x)D_s(y,z)s(x)^{-1}\big) \cdot D_s(x,y)^{-1} \cdot D_s(xy,z)^{-1} \cdot D_s(x,yz).$$

Substituting the definition of D_s into this formula gives, without using commutativity,

$$D_s \in Z^2(G,M).$$

(b) If s and s' are two sections of v, there is a morphism $f : G \to M$ such that $s(x) = f(x)s'(x)$. Then

$$\begin{aligned} D_{s'}(x,y) &= f^{-1}(xy)D_s(x,y)s(x)f(y)s(x)^{-1}f(x) \\ &= D_s(x,y) \cdot f^{-1}(xy) \cdot \big(s(x)f(y)s(x)^{-1}\big) \cdot f(x), \end{aligned}$$

and therefore

$$D_{s'} = D_s \cdot \partial^1 f.$$

This shows that the class of D_s in $H^2(G,M)$ is independent of the chosen section s of v; it is the class $c(E)$ of the extension (E).[44]

(c) Let s and s' be two sections of v, and let $m \in \Gamma(M)$. The equality

$$s(x) = m^{-1}s'(x)m \qquad (x \in G(S),\ S \in \operatorname{Ob}\mathscr{C})$$

is equivalent to

$$s(x) = m^{-1}s'(x)m\, s'(x)^{-1}s'(x), \qquad \text{that is,} \qquad s = \partial^0 m \cdot s'.$$

In particular, the stabilizer of s in $\Gamma(M)$ is the subgroup of elements m such that $\partial^0 m = e_M$, namely $H^0(G,M)$. This already proves (iii).

(d) The argument is now familiar.[45] Let s_0 be an arbitrary section of v. There exists a section s, necessarily of the form $s = f \cdot s_0$, that is a group morphism, that is, satisfies $D_s = 0$, if and only if

$$(D_{s_0})^{-1} = \partial^1 f,$$

for some $f : G \to M$, or equivalently if and only if $c(E) = 0$. This proves (i).

In that case, the sections of (E) are precisely the sections $s' = h \cdot s$, where $h : G \to M$ satisfies $\partial^1 h = 0$, that is, $h \in Z^1(G,M)$. Moreover, by (c), two such sections $h_1 \cdot s$ and $h_2 \cdot s$ are conjugate under $\Gamma(M)$ if and only if h_1 and h_2 have the same image in $H^1(G,M)$. This proves (ii).

Still let

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups with M commutative, and let

$$f : H \longrightarrow G$$

be a morphism of $\widehat{\mathscr{C}}$-groups.

Consider $E_f = H \times_G E$. It is a $\widehat{\mathscr{C}}$-group, and the projection $v_f : E_f \to H$ is a morphism of $\widehat{\mathscr{C}}$-groups; the same holds for $e_f : E_f \to E$. Sending M into E by u and into H by

[43] Editor's note: the original has been corrected to make it compatible with I 5.1.
[44] Editor's note: the preceding sentence has been added.
[45] Editor's note: what follows has been expanded.

the unit morphism also defines a morphism of $\widehat{\mathscr{C}}$-groups $u_f : M \to E_f$. We have therefore constructed the following commutative diagram of $\widehat{\mathscr{C}}$-groups.

$$\begin{tikzcd}
(\mathrm{E}) & 1 \arrow[r] & \mathrm{M} \arrow[r, "u"] & \mathrm{E} \arrow[r, "v"] & \mathrm{G} \\
(\mathrm{E}_f) & 1 \arrow[r] & \mathrm{M} \arrow[r, "u_f"] \arrow[u, "\mathrm{id}"] & \mathrm{E}_f \arrow[r, "v_f"] \arrow[u, "e_f"] & \mathrm{H} \arrow[u, "f"]
\end{tikzcd}$$

Figure 69: The pullback extension (E_f) and its morphism to (E).

It follows immediately that:

Lemma 1.2.3.

(i) *The sequence (E_f) is exact.*

(ii) *The map $s \mapsto e_f \circ s = f'$ gives a bijective correspondence between the morphisms*

$$s : H \longrightarrow E_f$$

such that $v_f \circ s = \mathrm{id}$, that is, the sections of v_f, and the morphisms

$$f' : H \longrightarrow E$$

such that $v \circ f' = f$, that is, the morphisms f' lifting f.

(iii) *Under this correspondence, the sections of (E_f) correspond to the group morphisms f' lifting f.*

Applying Lemma 1.2.2 to the extension (E_f), and taking 1.2.3 into account, gives the following proposition, which formally contains Lemma 1.2.2.

Proposition 1.2.4. *Let*

$$(E) \qquad 1 \longrightarrow M \longrightarrow E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups with M commutative. Let $f : H \to G$ be a morphism of $\widehat{\mathscr{C}}$-groups, and suppose that it lifts to a morphism, not necessarily a group morphism, $f' : H \to E$. Let H act on M by the composite morphism, which is multiplicative and independent of the choice of f',

$$H \xrightarrow{f'} E \xrightarrow{\text{int}} \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M).$$

(i) *The morphism f canonically defines a class $c(f) \in H^2(H, M)$, whose vanishing is necessary and sufficient for the existence of a morphism of $\widehat{\mathscr{C}}$-groups*

$$f' : H \longrightarrow E$$

lifting f.

(ii) *If $c(f) = 0$, the set of morphisms of $\widehat{\mathscr{C}}$-groups f' lifting f, modulo the action of the inner automorphisms defined by elements of $\Gamma(M)$ (that is, by elements $m \in \Gamma(E)$ such that $v(m) = e$), is principal homogeneous under $H^1(H, M)$.*

(iii) *If $f' : H \to E$ is a group morphism lifting f, the set of transforms of f' by the inner automorphisms defined by the elements of $\Gamma(M)$ is isomorphic to*

$$\Gamma(M)/\Gamma(M^H) = \Gamma(M)/H^0(H, M).$$

1.3. Extensions of group laws.

Consider the following situation: one has a morphism in $\widehat{\mathscr{C}}$,

$$(\dagger) \qquad p : X \longrightarrow Y,$$

and a commutative $\widehat{\mathscr{C}}$-group M acting on X, such that X is formally principal homogeneous over Y under M_Y.

If $g : Y \to Z$ is an arbitrary morphism in $\widehat{\mathscr{C}}$, then $g \circ p : X \to Z$ is invariant under M: for every $S \in \operatorname{Ob}\mathscr{C}$, the map $(g \circ p)(S)$ is invariant under the action of $M(S)$ on $X(S)$. Conversely, we shall assume that the following condition holds for $n = 1, 2, 3, 4$:

$(+)_n$: *Every morphism from X^n to M that is invariant under the action of M^n on X^n factors uniquely through $p^n : X^n \to Y^n$, where the superscript n denotes a Cartesian power.*

Lemma 1.3.1.

(i) *If h is a morphism from Y to M, the automorphism u_h of X, defined set-theoretically by $x \mapsto h(p(x)) \cdot x$, preserves the fibers of p and commutes with the operations of M on X.*[46] *Thus, for every $S \in \operatorname{Ob}\mathscr{C}$, $x \in X(S)$, and $m \in M(S)$, one has*

$$p(h(p(x)) \cdot x) = p(x), \qquad m \cdot h(p(x)) \cdot x = h(p(m \cdot x)) \cdot m \cdot x.$$

(ii) *This construction gives a bijective correspondence between morphisms from Y to M and automorphisms of X that preserve the fibers of p and commute with the operations of M.*

The first assertion is clear, since $p(m \cdot x) = p(x)$ and M is commutative. Conversely, an automorphism u of X preserving the fibers of p can be written set-theoretically as $x \mapsto g(x)x$, where g is a certain morphism from X to M. If u commutes with the operations of M, then g is invariant under M,[47] and the conclusion follows from condition $(+)_1$.

Suppose now, in addition, that a group law on Y and an action of Y on M are given, that is, a morphism of $\widehat{\mathscr{C}}$-groups

$$(\ddagger) \qquad f : Y \longrightarrow \underline{\operatorname{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M).$$

Definition 1.3.2. *A composition law on X,*

$$P : X \times X \longrightarrow X,$$

is called admissible *if it satisfies the following two conditions:*

(i) *P lifts the group law of Y; that is, the following diagram is commutative.*

$$\begin{tikzcd}
\mathrm{X} \times \mathrm{X} \arrow[r, "\mathrm{P}"] \arrow[d, "{(p,p)}"'] & \mathrm{X} \arrow[d, "p"] \\
\mathrm{Y} \times \mathrm{Y} \arrow[r] & \mathrm{Y}
\end{tikzcd}$$

Figure 70: The compatibility of an admissible composition law with the group law on Y.

[46] Editor's note: here and below, "commutes with p and the operations of M" has been replaced by "preserves the fibers of p and commutes with the operations of M." The equalities that follow have also been added.

[47] Editor's note: indeed, $mg(x)x = mu(x) = u(mx) = g(mx)mx$, and hence $g(mx) = g(x)$.

(ii) *For every $S \in \mathrm{Ob}\,\mathscr{C}$ and all $x, y \in X(S)$, $m, n \in M(S)$, one has*

$$(++) \qquad P(m \cdot x, n \cdot y) = m \cdot f(p(x))(n) \cdot P(x, y).$$

Proposition 1.3.3. *For a group law $*$ on X to be admissible, it is necessary and sufficient that the following four conditions hold:*

(i) *$p : X \to Y$ is a group morphism.*

(ii) *The morphism $i : M \to X$, defined by $i(m) = m \cdot e_X$, is a group isomorphism from M onto* $\ker(p)$*; in other words, set-theoretically,*

$$(m \cdot e_X) * (n \cdot e_X) = (mn) \cdot e_X.$$

(iii) *One has $m \cdot x = (m \cdot e_X) * x = i(m) * x$ for every $m \in M(S)$ and $x \in X(S)$.*

(iv) *The inner automorphisms of X act on* $\ker(p)$ *according to the set-theoretic formula*

$$x * i(m) * x^{-1} = i(f(p(x))m).$$

The proof is immediate.

Lemma 1.3.4.[48] *Let $h : Y \to M$ be a morphism and let u_h be the automorphism $x \mapsto h(p(x)) \cdot x$ of X (cf. 1.3.1). Let P be an admissible composition law (respectively, an admissible group law) on X, and let P' be the composition law on X obtained from P by means of u_h, that is,*

$$P'(x, y) = u_h^{-1}(P(u_h(x), u_h(y))).$$

Then:

(i) *P' is an admissible composition law (respectively, an admissible group law).*

(ii) *For every $x, y \in X(S)$, with $S \in \mathrm{Ob}\,\mathscr{C}$, set $v = p(x)$ and $w = p(y)$. Then*

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

Proof. One has $u_h^{-1} = u_{h^\vee}$, where $h^\vee : Y \to M$ is defined by $h^\vee(y) = h(y)^{-1}$. By 1.3.2(i) and (ii),

$$P(h(v) \cdot x, h(w) \cdot y) = h(v) \cdot f(v)(h(w)) \cdot P(x, y), \qquad p(P(x, y)) = vw,$$

and hence

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

It follows immediately that P' satisfies conditions (i) and (ii) of 1.3.2.

Definition 1.3.5. *Two admissible composition laws obtained from one another by the procedure of 1.3.4 are called* equivalent.[49]

Proposition 1.3.6. *Suppose that an admissible composition law on X exists. Then:*

[48] Editor's note: the statement and its proof have been expanded with a view to step (e) in the proof of 1.3.6.

[49] Editor's note: this is indeed an equivalence relation, since $u_h^{-1} = u_{h^\vee}$.

(i) *There exists a canonically determined class $c \in H^3(Y, M)$, whose vanishing is necessary and sufficient for the existence of an admissible associative composition law on X.*

(ii) *If $c = 0$, the set of admissible associative composition laws on X (respectively, the set of their equivalence classes) is principal homogeneous under $Z^2(Y, M)$ (respectively, $H^2(Y, M)$).*

The proof proceeds in several steps.

(a) Let P be an admissible composition law on X. Since P lifts the associative composition law of Y, there is a unique morphism $a : X^3 \to M$ such that

$$(*) \qquad P(x, P(y, z)) = a(x, y, z) \cdot P(P(x, y), z).$$

Applying conditions 1.3.2(i) and (ii), one sees at once that a is invariant under the action of M^3 on X^3.[50] Condition $(+)_3$ therefore gives:

(1) There exists a unique morphism $DP : Y^3 \to M$ such that

$$P(x, P(y, z)) = DP(p(x), p(y), p(z)) \cdot P(P(x, y), z),$$

and P is associative if and only if $DP = 0$.

(b) Let us compute $P(P(P(x, y), z), t)$ step by step using the preceding formula. Put

$$p(x) = u, \qquad p(y) = v, \qquad p(z) = w, \qquad p(t) = h.$$

One obtains[51] the following pentagonal diagram, in which an arrow $a \xrightarrow{m} b$ means that $b = m \cdot a$.

$$\begin{tikzcd}
 & & \mathrm{P}(x, \mathrm{P}(y, \mathrm{P}(z, t))) \arrow[dll, "\mathrm{DP}(u,v,wh)"'] \arrow[drr, "f(u)(\mathrm{DP}(v,w,h))"] & & \\
\mathrm{P}(\mathrm{P}(x, y), \mathrm{P}(z, t)) \arrow[d, "\mathrm{DP}(uv,w,h)"'] & & & & \mathrm{P}(x, \mathrm{P}(\mathrm{P}(y, z), t)) \arrow[d, "\mathrm{DP}(u,vw,h)"] \\
\mathrm{P}(\mathrm{P}(\mathrm{P}(x, y), z, t) & & & & \mathrm{P}(\mathrm{P}(x, \mathrm{P}(y, z)), t) \arrow[llll, "\mathrm{DP}(u,v,w)"']
\end{tikzcd}$$

Figure 71: The pentagonal comparison used to compute the associativity obstruction.

Thus one finds

$$\begin{aligned} DP(u, v, w) \cdot DP(u, vw, h) \cdot f(u)(DP(v, w, h)) \\ \cdot DP(u, v, wh)^{-1} \cdot DP(uv, w, h)^{-1} = e_M, \end{aligned}$$

[50] Editor's note: for $x, y, z \in X(S)$, set $u = p(x)$, $v = p(y)$, and $w = p(z)$. Then

$$\begin{aligned} P(mx, P(m'y, m''z)) &= P(mx, m'f(v)(m'')P(y, z)) \\ &= mf(u)(m')f(uv)(m'')P(x, P(y, z)). \end{aligned}$$

On the other hand, since $p(P(x, y)) = uv$, one also has

$$\begin{aligned} P(P(mx, m'y), m''z) &= P(mf(u)(m')P(x, y), m''z) \\ &= mf(u)(m')f(uv)(m'')P(P(x, y), z). \end{aligned}$$

Comparison of these equalities with $(*)$ gives $a(mx, m'y, m''z) = a(x, y, z)$.

[51] Editor's note: the diagram that follows has been added.

that is, $\partial^3 DP(u,v,w,h) = e_M$. Moreover, the left-hand side of the preceding formula can be expressed, by means of P and a, as the value at (x,y,z,t) of a certain morphism $X^4 \to M$.

It follows from the uniqueness assertion in $(+)_4$ that $\partial^3 DP$ and e_M, which factor the same morphism, are equal. Hence:

(2) DP is a cocycle; that is, $DP \in Z^3(Y,M)$.

(c) If P and P' are two admissible composition laws on X, there is a unique morphism

$$b : X^2 \longrightarrow M$$

such that $P'(x,y) = b(x,y)P(x,y)$. Applying 1.3.2(ii) to P and P', one sees that b is invariant under M^2, and hence, by $(+)_2$:

(3) For every pair (P,P') of admissible composition laws there exists a unique morphism $d(P,P') : Y^2 \to M$ such that

$$P'(x,y) = d(P,P')(p(x),p(y)) \cdot P(x,y),$$

and the set of admissible composition laws is thereby made principal homogeneous under

$$\mathrm{Hom}(Y^2, M) = C^2(Y,M).$$

(d) Under the preceding conditions one has

$$(4) \qquad DP' - DP = \partial^2 d(P,P').$$

(e) The laws P and P' are equivalent if and only if there is a morphism

$$h \in C^1(Y,M) = \mathrm{Hom}(Y,M)$$

such that $d(P,P') = \partial^1 h$. This follows from the definition of equivalence and from 1.3.4(ii).

(f) It remains only to conclude. We seek an associative P', that is, one for which $DP' = e_M$. Now DP is a cocycle whose class in $H^3(Y,M)$ does not depend on the chosen admissible composition law P, by (3) and (4). This class is the required obstruction c. An appropriate P' can be chosen if and only if $c = 0$. Indeed, after choosing an arbitrary P, one must solve, by (1),

$$0 = DP' = DP + \partial^2 d(P,P'),$$

which is possible, by (3) and (4), if and only if $c = 0$. Again by (3) and (4), the set of associative laws P' is principal homogeneous under $Z^2(Y,M)$. By (e), the set of associative laws P' modulo equivalence is principal homogeneous under $H^2(Y,M)$.

2. Infinitesimal extensions of a morphism of group schemes

Retain the notation of §0. Let Y and X be two S-group functors, and let M be the kernel of the group homomorphism $p_X : X \to X^+$. Thus there is an exact sequence of S-group functors

$$1 \longrightarrow M \longrightarrow X \xrightarrow{p_X} X^+.$$

By the definition of X^+, there are isomorphisms

$$\mathrm{Hom}_S(Y, X^+) \xrightarrow{\sim} \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}}),$$
$$\mathrm{Hom}_{S\text{-gr}}(Y, X^+) \xrightarrow{\sim} \mathrm{Hom}_{S_{\mathscr{J}}\text{-gr}}(Y_{\mathscr{J}}, X_{\mathscr{J}}).$$

The map

$$\mathrm{Hom}_S(Y, p_X) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)$$

sends an S-morphism $f : Y \to X$ to the S-morphism $f^+ : Y \to X^+$ corresponding under these isomorphisms to the $S_{\mathscr{J}}$-morphism $f_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$ obtained from f by base change. If M is commutative, Proposition 1.2.4 applies to this situation.

2.0.[52] In what follows we shall consider the case in which Y is flat over S, and X is an S-group functor of one of the following two types:

a X is an S-group scheme;

b $X = \underline{\mathrm{Aut}}_S(E)$, where E is an S-scheme of finite presentation over S.

Write (Flat)/S for the category of S-schemes flat over S. In case *a*, respectively *b*, the S-group functor $M = \mathrm{Ker}(X \to X^+)$, its restriction L to (Flat)/S, and the actions on M of the inner automorphisms of X were calculated in 0.9, 0.5, and 0.10, respectively in 0.11, 0.14, and 0.12.

More explicitly, in case *a*, let L_0 be the S_0-functor in commutative groups defined, for every S_0-scheme T_0, by

$$\mathrm{Hom}_{S_0}(T_0, L_0) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\right).$$

The group X_0 acts on this functor through its adjoint representation on $\omega^1_{X_0/S_0}$, and

$$L = \prod_{S_0/S} L_0; \qquad L(T) = L_0(T \times_S S_0)$$

for every S-scheme T.

In case *b*, let $\pi : E \to S$ be the structural morphism. Then L is the abelian-group functor on (Flat)/S defined by

$$\mathrm{Hom}_S(T, L) = \Gamma\Big(T, \pi_* \mathrm{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E) \otimes_{\mathscr{O}_S} \mathscr{O}_T\Big),$$

on which X, regarded as a functor on (Flat)/S, acts as described in 0.12. In either case there is an exact sequence of group functors on (Flat)/S:

$$1 \longrightarrow L \longrightarrow X \longrightarrow X^+. \tag{E}$$

Since Y is flat over S, Lemma 1.1.1 shows that the groups $H^i(Y, M)$ depend only on the restriction L of M to (Flat)/S. In case *a*, where $L = \prod_{S_0/S} L_0$, Lemma 1.1.2 gives isomorphisms

$$H^i(Y, L) \simeq H^i(Y_0, L_0).$$

Taking all this into account, Proposition 1.2.4 yields the following result.[53]

Theorem 2.1. *Let S be a scheme, and let $\mathscr{I} \supset \mathscr{J}$ be two quasi-coherent ideals satisfying $\mathscr{I}\mathscr{J} = 0$, defining the closed subschemes S_0 and $S_{\mathscr{J}}$. Let*

X be an S-group functor of type a or b, with L_0 and L as above;

Y be an S-group scheme flat over S, and $f_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$ a morphism of $S_{\mathscr{J}}$-groups.

[52] Editor's note: The numbering 2.0 has been added for later references.

[53] Editor's note: The definitions that appeared in the original statement of Theorem 2.1 and occupied most of original page 113 have been moved above.

Then:

i *The morphism $f_{\mathscr{J}}$ lifts to a morphism of S-groups $Y \to X$ if and only if the following two conditions hold:*

  i_1 *$f_{\mathscr{J}}$ lifts to a morphism of S-functors $Y \to X$. By Proposition 1.2.4 this defines an action of Y on L, independent of the chosen lift. In case a, the resulting action of Y_0 on L_0 comes from $f_0 : Y_0 \to X_0$ and the adjoint action of X_0 on L_0.*

  i_2 *The canonically defined obstruction $c(f_{\mathscr{J}}) \in H^2(Y, L)$ vanishes. In case a, this group is isomorphic to $H^2(Y_0, L_0)$.*

(ii) *If the conditions in i hold, the set $\mathcal{E}$ of morphisms of S-group functors $Y \to X$ extending $f_{\mathscr{J}}$ is principal homogeneous under $Z^1(Y, L)$, isomorphic in case a to $Z^1(Y_0, L_0)$. Moreover, the quotient of $\mathcal{E}$ by the action of inner automorphisms of X defined by sections over S that induce the unit section of $X_{\mathscr{J}}$ over $S_{\mathscr{J}}$ is principal homogeneous under $H^1(Y, L)$, isomorphic in case a to $H^1(Y_0, L_0)$.*

(iii) *If $f : Y \to X$ is a morphism of S-group functors extending $f_{\mathscr{J}}$, the set of morphisms $Y \to X$ obtained from f by those inner automorphisms is isomorphic to*

$$\Gamma(L)/H^0(Y, L),$$

and in case a to $\Gamma(L_0)/H^0(Y_0, L_0)$.

Remark 2.1.1.[54] If $f, f' : Y \to X$ are morphisms of S-group functors extending $f_{\mathscr{J}}$, one obtains a cocycle

$$d(f, f') \in Z^1(Y, L) \quad (\simeq Z^1(Y_0, L_0) \text{ in case (a)})$$

such that

$$f' = d(f, f') \cdot f. \tag{*}$$

[55] Write $\bar{d}(f, f')$ for the image of $d(f, f')$ in $H^1(Y, L)$, or equivalently in $H^1(Y_0, L_0)$ in case a.

Remark 2.2. Keep the preceding notation; in particular, Y is flat over S. In case b, formula (0.14.1) identifies L with the restriction to (Flat)$/S$ of

$$\mathbf{W}\Big(\pi_* \operatorname{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big).$$

In case a, suppose in addition that X is locally of finite presentation over S. Formula (0.6.1) then identifies L with the restriction to (Flat)$/S$ of

$$\prod_{S_0/S} \mathbf{W}\Big(\operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big).$$

In both cases the module to which $\mathbf{W}$ is applied is quasi-coherent, by EGA I, 9.1.1. Suppose further that Y is affine over S.[56] By I, 5.3, one obtains

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i\Big(Y_0, \operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big) \qquad \text{in case (a)},$$
$$H^i(Y, L) = H^i\Big(Y, \pi_* \operatorname{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big) \qquad \text{in case (b)}.$$

Remark 2.3.

[54] Editor's note: This remark, analogous to 4.5.1, has been added to introduce the notation $d(f, f')$ and $\bar{d}(f, f')$, used in 4.38. Consequently, the part of 2.1(ii) concerning $\mathcal{E}$ itself was also added.

[55] Editor's note: We follow the sign conventions of the original. One could instead write $f' = d(f', f) \cdot f$, but then, in order to retain the equality $d_1(d(i, i')) = d(Y, i'(Y))$ in 4.27 for two immersions $i, i' : Y \hookrightarrow X$, the sign of the class $d(Y, Y')$ introduced in 4.5.1 would have to be changed. This would in turn alter the signs in 4.8, 4.14, and 4.17. We have preferred to retain all the signs of the original, which are correct.

[56] Editor's note: This hypothesis and the reference to I, 5.3 have been added.

(1) By 0.16 and 0.17, condition i_1 is automatic when Y is affine and

$$\begin{cases} X \text{ is smooth over } S, & \text{in case (a)},\\ E \text{ is smooth and affine over } S, & \text{in case (b)}.\end{cases}$$

(2) Under these hypotheses, with Y still flat over S as in 2.0, formulas 2.2(a) and (0.6.2) give in case *a*

$$H^i(Y,L) = H^i(Y_0,L_0) = H^i\Big(Y_0, \operatorname{Lie}(X_0/S_0)\otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),$$

and in case *b*, by (0.14.2), 1.1.2, and I, 5.3,

$$H^i(Y,L) = H^i\Big(Y_0, \pi_{0*}\operatorname{Hom}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J}\otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0})\Big).$$

[57]

We now state several corollaries for a diagonalizable S-group Y (I, 4.4). If S is affine, then, by I, 5.3.3, $H^n(Y,F)=0$ for every $n>0$ and every quasi-coherent $\mathscr{O}_S$-module F.

Corollary 2.4. *Let $S_0 \hookrightarrow S$ be a closed subscheme defined by a nilpotent ideal, and let Y be a diagonalizable S-group. Let*

a X be an S-group locally of finite presentation over S, or

b $X = \underline{\mathrm{Aut}}_S(E)$, where E is an S-scheme locally of finite presentation.

If $f : Y \to X$ is a morphism of S-groups whose base change $f_0 : Y_0 \to X_0$ is the unit morphism, then f is the unit morphism.

Indeed, the question is local on S, and in case *b* also local on E. We may therefore suppose that S is affine and, in case *b*, that E is of finite presentation over S. Introduce the closed subschemes $S_n \subset S$ defined by the powers of the ideal defining S_0. This reduces the assertion to the case in which S_0 is defined by a square-zero ideal, where it follows from Theorem 2.1 and Remark 2.2.

If f_0 is not required to be the unit morphism, one has the following statement.

Corollary 2.5. *Let S and S_0 be as in 2.4, with S affine. Let Y be a diagonalizable S-group, X an S-group functor, and $f_0 : Y_0 \to X_0$ a morphism of S_0-group functors.*

i [58]

(ii) *Suppose either that X is a smooth S-group, or that $X = \underline{\mathrm{Aut}}_S(E)$ with E smooth and affine over S. Then f_0 extends to a morphism of S-groups $Y \to X$, and any two extensions are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.*

[57] Editor's note: The preceding material has been added; see Editor's Note 31 in 0.14.

[58] Editor's note: The original stated: "Suppose one of the following two properties holds: *a* X is an S-group locally of finite presentation; *b* $X = \underline{\mathrm{Aut}}_S(E)$, where E is of finite presentation over S. Then f_0 extends to a morphism of S-groups $Y \to X$ if and only if it extends to a morphism of S-functors $Y \to X$." It added that this followed step by step from part (ii) of the theorem. That argument appears insufficient. For example, if $\mathscr{I}^3 = 0$ and $\mathscr{J} = \mathscr{I}^2$, and if $f : Y \to X$ is an S-functor morphism lifting f_0, then f_0 lifts to an $S_{\mathscr{J}}$-group morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$; but must $g_{\mathscr{J}}$ then lift to an S-functor morphism $g : Y \to X$? In any event, assertion *i* is not used later, where X is always assumed smooth over S.

Introduce S_n as above.[59] For (ii), a scheme smooth over S is necessarily locally of finite presentation. Thus, in case b, a smooth affine E/S is of finite presentation, so the hypothesis in 2.0(b) is indeed satisfied.

Condition i_1 of Theorem 2.1 is automatic by 0.16 and 0.17. Moreover, every section of X_{S_n} over S_n lifts to a section of $X_{S_{n+1}}$ over S_{n+1}: this follows from smoothness in case a, and from 0.17 in case b. Consequently, if f and f' are two lifts of f_0, one may arrange step by step that $f_n = f'_n$ by lifting the inner automorphism supplied by Theorem 2.1(ii). This proves the assertion.

The same argument, together with Remark 2.3, gives:

Corollary 2.6. *Let $\mathscr{I}$ be a nilpotent ideal of S defining S_0, let Y be an affine S-group flat over S, and let X be a smooth S-group.*

i If, for every $n \geq 0$,

$$H^2\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then every S_0-group morphism $f_0 : Y_0 \to X_0$ lifts to an S-group morphism $f : Y \to X$.

(ii) *If, for every $n \geq 0$,*

$$H^1\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then any two such lifts are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.

Lemma 2.7. *Let S be affine, G an affine S-group, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_S$-module, and $\mathscr{L}$ a locally free $\mathscr{O}_S$-module. Suppose G acts on $\mathscr{F}$ in the sense of Exposé I, hence acts on $\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{L}$. Let Λ be the ring of S, and let L be the projective Λ-module defining $\mathscr{L}$. Then there is a canonical isomorphism*

$$H^*(G, \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{L}) \simeq H^*(G, \mathscr{F}) \otimes_\Lambda L.$$

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Let $\mathscr{A}$ be the $\mathscr{O}_S$-algebra $\mathscr{A}(G)$, and consider the complex $\mathscr{C}$ of quasi-coherent $\mathscr{O}_S$-modules

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A} \longrightarrow \mathscr{F} \otimes \mathscr{A} \otimes \mathscr{A} \longrightarrow \cdots .$$

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By I, 5.3, $H^*(G, \mathscr{F})$, respectively $H^*(G, \mathscr{F} \otimes \mathscr{L})$, is the cohomology of $\Gamma(S, \mathscr{C})$, respectively $\Gamma(S, \mathscr{C} \otimes \mathscr{L})$. Since S is affine, EGA I, 1.3.12 gives

$$\Gamma(S, \mathscr{C} \otimes \mathscr{L}) \simeq \Gamma(S, \mathscr{C}) \otimes_\Lambda L.$$

Because L is projective, hence flat,

$$H^*(\Gamma(S, \mathscr{C}) \otimes_\Lambda L) \simeq H^*(\Gamma(S, \mathscr{C})) \otimes_\Lambda L,$$

which proves the lemma.

Using the lemma, Corollary 2.6 becomes:

Corollary 2.8. *Let S be affine and let $\mathscr{I}$ be a nilpotent ideal defining S_0. Suppose every $\mathscr{I}^{n+1}/\mathscr{I}^{n+2}$ is locally free on S_0. Let Y be an affine S-group flat over S, let X be a smooth S-group, and let $f_0 : Y_0 \to X_0$ be an S_0-group morphism.*

[59] Editor's note: What follows has been slightly modified and expanded from the original.

[60] Editor's note: Here $\mathscr{L}$ is given the trivial action of G.

[61] Editor's note: The original proof has been expanded and simplified. It also invoked the isomorphisms $H^n(\mathscr{C} \otimes \mathscr{L}) \simeq H^n(\mathscr{C}) \otimes \mathscr{L}$ and $H_n(\Gamma(-)) \simeq \Gamma(H_n(-))$, where $H^n(\mathscr{C})$ denotes the cohomology sheaf of the complex $\mathscr{C}$.

i If $H^2(Y_0, \mathrm{Lie}(X_0/S_0)) = 0$, then f_0 lifts to an S-group morphism $Y \to X$.

(ii) *If $H^1(Y_0, \mathrm{Lie}(X_0/S_0)) = 0$, then any two such lifts are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.*

Taking $Y = X$ gives:

Corollary 2.9. *Let S and S_0 be as above, and let X be a smooth affine S-group.*

i If $H^1(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, every endomorphism of X over S that induces the identity on X_0 is the inner automorphism defined by a section over S inducing the unit section of X_0 over S_0.

(ii) *If $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, every S_0-automorphism of X_0 extends to an S-automorphism of X.*[62]

Remark 2.10. The assertions involving H^1 have converses by Theorem 2.1. For example, let $S = I_{S_0}$ be the dual-number scheme over S_0 (II, 2.1), and let X be a flat S-group such that every automorphism over S inducing the identity over S_0 is an inner automorphism defined by a section over S inducing the unit section of X_0. Then

$$H^1(X_0, \mathrm{Lie}(X_0/S_0)) = 0.$$

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Corollary 2.11. *Let $S, \mathscr{I}, \mathscr{J}$ be as in 2.1. Let Y be an S-group scheme flat over S, let X be an S-group scheme, and let $f : Y \to X$ be an S-group morphism. The set of morphisms from Y to X obtained from f by conjugation with elements $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$ is isomorphic to the quotient*

$$\mathcal{E} = \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) / \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})^{\mathrm{ad}(Y_0)}.$$

Here the second group consists of those $\mathscr{O}_{S_0}$-morphisms $\omega^1_{X_0/S_0} \to \mathscr{J}$ which, after every base change $S' \to S_0$, yield morphisms

$$\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{S'} \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{S'}$$

invariant under the action of $Y_0(S')$ on the first factor.

By Theorem 2.1(iii), this set is $\Gamma(L_0)/H^0(Y_0, L_0)$. Now

$$\Gamma(L_0) = \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}),$$

and $H^0(Y_0, L_0)$ is precisely $\Gamma(L_0)^{\mathrm{ad}(Y_0)}$ in the sense just stated.

Corollary 2.12. *Under the hypotheses of 2.11, suppose in addition that $\omega^1_{X_0/S_0}$ is locally free of finite rank. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big) / H^0\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

[62] Editor's note: Let f_0 be an S_0-automorphism of X_0, with inverse g_0. By 2.8(i), lift f_0 and g_0 to S-endomorphisms f and g of X. Then $g \circ f$ and $f \circ g$ induce the identity on X_0, so by 0.7 they are S-automorphisms. Hence f and g are S-automorphisms as well.

[63] Editor's note: This is used in Exposé XXIV, 1.13.

Indeed,[64] local freeness of finite rank gives

$$\mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \simeq \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}.$$

Corollary 2.13. *Suppose in addition that Y_0 is diagonalizable. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big) / \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),$$

where $\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}$ *is the factor of the decomposition in I, 4.7.3 corresponding to the zero character of Y_0.*

Indeed, if $Y_0 \simeq D_{S_0}(M)$, the cited result gives a direct-sum decomposition

$$\mathrm{Lie}(X_0/S_0) = \mathrm{Lie}(X_0/S_0)_0 \oplus \bigoplus_{\substack{m\in M\\ m\neq 0}} \mathrm{Lie}(X_0/S_0)_m.$$

Tensoring with $\mathscr{J}$ gives the analogous decomposition for $\mathrm{Lie}(X_0/S_0) \otimes \mathscr{J}$, whence

$$H^0(Y_0, \mathrm{Lie}(X_0/S_0) \otimes \mathscr{J}) \simeq \Gamma\Big(S_0, \mathrm{Lie}(X_0/S_0)_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

Corollary 2.14. *Suppose in addition that S_0 is affine. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \Big[\mathrm{Lie}(X_0/S_0)/\,\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}\Big] \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

3. Infinitesimal extensions of a group scheme

Still using the notation of §0 ($S, \mathscr{I}, \mathscr{J}$, and so on), let X be an S-scheme, and suppose that $X_{\mathscr{J}}$ is endowed with a group structure. We propose to determine the S-group structures on X that induce the given structure on $X_{\mathscr{J}}$.

From now on, assume that X is flat over S. Let $\mathscr{C}$ be the category of S-schemes flat over S; thus $X \in \mathrm{Ob}\,\mathscr{C}$. Denote by Y, respectively M, the functor on $\mathscr{C}$ defined by X^+, respectively L'_X. The canonical morphism $p_X : X \to X^+$ defines a morphism in $\widehat{\mathscr{C}}$

$$p : X \longrightarrow Y,$$

and the action of L'_X on X in $\widehat{(\mathrm{Sch})/S}$ defines an action of M on X in $\widehat{\mathscr{C}}$. One checks at once that X is thereby formally principal homogeneous under M_Y over Y (cf. 0.2(i) and 0.4).

The action of X^+ on L'_X defined in 0.8 (denoted Ad there) defines an action, denoted $\widetilde{f}$, of Y on M. On the other hand, by 0.5,

$$\mathrm{Hom}_{\widehat{\mathscr{C}}}(Z, M) \simeq \mathrm{Hom}_{S_0}(Z_0, L_0), \qquad Z \in \mathrm{Ob}\,\mathscr{C},$$

where L_0 is the functor defined in 0.5.

Lemma 3.1.

(i) *Condition $(+)_n$ of 1.3 holds for every positive integer n.*

(ii) *Let the S_0-group X_0 act on the S_0-functor L_0 through its adjoint representation. Then there is a canonical isomorphism*

$$H^*(X_0, L_0) \simeq H^*(Y, M),$$

where the first cohomology is computed in $(\mathrm{Sch})/S_0$*, and the second in* $\mathscr{C}$.

[64] Editor's note: The following sentence has been added.

Both parts of the lemma follow from the relation

$$\begin{aligned}\operatorname{Hom}_{\widehat{\mathscr{C}}}(Y, M) &\simeq \operatorname{Hom}_{\widehat{(\mathrm{Sch})/S_0}}(X^+ \times_S S_0, L_0)\\ &\simeq \operatorname{Hom}_{S_0}(X_0, L_0)\\ &\simeq \operatorname{Hom}_{\widehat{\mathscr{C}}}(X, M),\end{aligned}$$

which follows at once from the definition of M as $\prod_{S_0/S} L_0$. More generally, the same relation holds with X, Y replaced by X^n, Y^n. Consequently, every morphism $X^n \to M$ factors uniquely through Y^n, which gives $(+)_n$. It also gives the relation $C^*(Y, M) = C^*(X_0, L_0)$, and hence (ii).

We may therefore apply the constructions of 1.3. In particular:

Lemma 3.2. *Let $P : X \times_S X \to X$ be a morphism. Then P induces the group law of $X_{\mathscr{J}}$ if and only if P is an admissible composition law on X (cf. 1.3.2).*

Indeed, P induces the group law of $X_{\mathscr{J}}$ if and only if it lifts the group law of X^+, or equivalently that of Y. Thus it remains only to show that every morphism P lifting the group law of $X_{\mathscr{J}}$ satisfies identity ((++)) of 1.3.2(ii), which is exactly what was proved in 0.8.

Proposition 3.3. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let X be an S-scheme that is flat and either quasi-compact or locally of finite presentation over S. Let $P : X \times_S X \to X$ be a composition law on X. Then P is a group law if and only if the following two conditions hold:*

(i) *P is associative;*

(ii) *P induces a group law on $X_0 = X \times_S S_0$.*

These conditions are clearly necessary. We prove that they are sufficient. Suppose first that $X \to S$ has a section. Since $X(S')$ is then nonempty for every $S' \to S$, it suffices [65] to show that, for every $x \in X(S')$, left and right translation by x are isomorphisms of $X_{S'}$. [66]

We may plainly assume $S' = S$. The translation in question, t, induces on X_0 a translation t_0, which is an automorphism since X_0 is a group. The conclusion follows by flatness (SGA 1 III, 4.2). [67]

Now drop the assumption that X has a section over S, and suppose that there exists $S' \to S$ such that $X_{S'}$ has a section over S'. By what has just been proved, $X_{S'}$ is then an S'-group; let e' be its unit section. The inverse image of e' under $\mathrm{pr}_i : S' \times_S S' \to S'$, $i = 1, 2$, is the unit section of $X_{S''}$ for the group law obtained by pulling $P_{S'}$ back along

[65] Editor's note: The original has been corrected by deleting the inappropriate reference to an exercise of Bourbaki on semigroups (cf. [BAlg], §I.2, Exercises 9–13) and by indicating the role of the left and right translations; see the following editor's note.

[66] Editor's note: Let E be a nonempty set endowed with an associative composition law such that every left translation ℓ_x is bijective, and fix $x_0 \in E$. There is a unique $e \in E$ such that $x_0 e = x_0$; then $x_0 e x = x_0 x$ implies $ex = x$ for every $x \in E$. Moreover, for every x there is a unique x' such that $xx' = e$. Suppose in addition that there is a $b \in E$ for which right translation r_b is injective. Then $xeb = xb$ gives $xe = x$, so e is a unit, and $xx'x = x = xe$ gives $x'x = e$; hence x' is both a left and a right inverse of x, and E is a group.

The hypothesis that r_b be injective is necessary: on any set E, define $xy = y$ for all $x, y \in E$. Every left translation is then the identity (and the law is associative), whereas $r_y(E) = \{y\}$ for every y; thus E is not a group when $|E| > 1$.

[67] Editor's note: Since X and X_0 have the same underlying topological space and t_0 is an automorphism, t is a homeomorphism, hence an affine morphism; cf. Exposé $\mathrm{VI_B}$, 2.9.1, or EGA IV_4, 18.12.7.1. It therefore suffices to show the following. Let $\mathscr{J}$ be a nilpotent ideal of a ring Λ, and let $\varphi : A \to B$ be a morphism of Λ-algebras, with B flat over Λ, such that $\varphi \otimes_\Lambda (\Lambda/\mathscr{J})$ is bijective. Then φ is bijective. By the "nilpotent Nakayama lemma," φ is surjective; moreover, since B is flat over Λ, one also has $\operatorname{Ker}(\varphi) \otimes_\Lambda (\Lambda/\mathscr{J}) = 0$, whence $\operatorname{Ker}(\varphi) = 0$. Thus φ is bijective.

pr_i. But since P is "defined over S," these two group laws coincide, and so do their unit sections. Hence

$$\mathrm{pr}_1^*(e') = \mathrm{pr}_2^*(e').$$

If $S' \to S$ is a descent morphism (cf. Exposé IV, no. 2), there is a section of X whose base extension is e', and the proof is complete. Since X_X has a section over X, namely the diagonal section, it is now enough to prove that $X \to S$ is a descent morphism. It is flat and surjective, and is either quasi-compact or locally of finite presentation; it is therefore an (fpqc) covering, hence a descent morphism (Exposé IV, no. 6).

Remark. In fact, the hypothesis that $X \to S$ be quasi-compact or locally of finite presentation is superfluous, by the following result, which the reader may prove as an exercise on Exposé IV:

Under the stated hypotheses on S and S_0, if $X \to S$ is flat and $X_0 \to S_0$ is an (fpqc) covering, then $X \to S$ is a descent morphism.

Lemma 3.4. *Two admissible composition laws on X are equivalent (cf. 1.3.5) if and only if one is obtained from the other by an automorphism of X over S that induces the identity on $X_{\mathscr{J}}$.*

Indeed, the morphisms constructed in 1.3.1 are precisely those in the preceding statement, by 0.7. [68]

Taking account of all the preceding results, Proposition 1.3.6 gives:

Theorem 3.5. *Let S be a scheme, and let $\mathscr{I}$ and $\mathscr{J}$ be two ideals on S such that $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let S_0 and $S_{\mathscr{J}}$ be the closed subschemes of S that they define. Let X be an S-scheme flat over S (and either locally of finite presentation or quasi-compact over S), and let X_0 and $X_{\mathscr{J}}$ be the schemes obtained by base change. Suppose $X_{\mathscr{J}}$ is endowed with an $S_{\mathscr{J}}$-group structure, and let L_0 denote the S_0-functor in commutative groups defined by*

$$\mathrm{Hom}_{S_0}(T, L_0) = \mathrm{Hom}_{\mathscr{O}_T}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right),$$

on which X_0 acts through its adjoint representation.

(i) *An S-group structure exists on X that induces the given structure on $X_{\mathscr{J}}$ if and only if the following conditions hold:*

  (i_1) *there is a morphism of S-schemes $X \times_S X \to X$ inducing the group law of $X_{\mathscr{J}}$;*

  (i_2) *a certain obstruction class in $H^3(X_0, L_0)$, canonically defined by X and the group law of $X_{\mathscr{J}}$, is zero.*

(ii) *If the conditions in (i) hold, the set $\mathcal{E}$ of group laws on X inducing the given law on $X_{\mathscr{J}}$ is principal homogeneous under $Z^2(X_0, L_0)$. Moreover, the quotient of $\mathcal{E}$ by the S-automorphisms of X inducing the identity on $X_{\mathscr{J}}$ is principal homogeneous under $H^2(X_0, L_0)$.*

[69] Indeed, by 3.2 every morphism of S-schemes $f : X \times_S X \to X$ inducing the group law of $X_{\mathscr{J}}$ is an admissible composition law on X. By 1.3.6(i), the existence of an associative admissible composition law $P : X \times_S X \to X$ is equivalent to the vanishing of a certain

[68] Editor's note: Indeed, by the proof of 0.7, the S-endomorphisms of X inducing the identity on $X_{\mathscr{J}}$ are the automorphisms $m \cdot \mathrm{id}_X$, where m ranges over $M(X) = \mathrm{Hom}_S(X, M)$. For every $S' \to S$ and every $x \in X(S')$, one has $(m \cdot \mathrm{id}_X)(x) = m(x) \cdot x$. By the proof of 3.1, each $m : X \to M$ factors uniquely through a morphism $h : Y = X^+ \to M$; hence $m \cdot \mathrm{id}_X$ is the automorphism u_h introduced in 1.3.1. The lemma now follows from the definition of equivalence; cf. 1.3.4 and 1.3.5.

[69] Editor's note: The following argument has been added.

class $c(f) \in H^3(X_0, L_0)$; in that case, 3.3 shows that P is a group law. This proves (i), and (ii) then follows from 3.3 and 1.3.6(ii).

Remark 3.5.1. [70] If μ and μ' are group laws on X inducing the given law on $X_{\mathscr{J}}$, one obtains a cocycle $\delta(\mu, \mu') \in Z^2(X_0, L_0)$. The chosen sign convention is $\mu' = \delta(\mu, \mu') \cdot \mu$; that is, for every $S' \to S$ and $x, y \in X(S')$,

$$\mu'(x, y) = \delta(\mu, \mu')(x_0, y_0) \cdot \mu(x, y).$$

[71]

Write $\bar{\delta}(\mu, \mu')$ for the image of $\delta(\mu, \mu')$ in $H^2(X_0, L_0)$. Finally, if X endowed with μ, respectively μ', is denoted simply by X, respectively X', write $\delta(X, X')$ in place of $\delta(\mu, \mu')$, and similarly for $\bar{\delta}(X, X')$.

Remark 3.6. Let $X_{\mathscr{J}}$ be a smooth affine $S_{\mathscr{J}}$-scheme. By 0.15, up to isomorphism there is a unique S-scheme X, smooth over S, whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$. If $X_{\mathscr{J}}$ is endowed with an $S_{\mathscr{J}}$-group structure, condition (i_1) is automatic by 0.16. Moreover, 0.6 simplifies the definition of L_0 and gives:

Corollary 3.7. *Let $S, \mathscr{I}, \mathscr{J}$ be as in 3.1, and let $X_{\mathscr{J}}$ be a smooth affine $S_{\mathscr{J}}$-group.*

(i) *The set of smooth S-groups whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$, up to isomorphism inducing the identity on $X_{\mathscr{J}}$, is either empty or principal homogeneous under*

$$H^2\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

(ii) *A smooth S-group whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$ exists if and only if a certain obstruction in*

$$H^3\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big)$$

is zero.

As usual, one obtains the following corollaries.

Corollary 3.8. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Let X_0 be a smooth affine S_0-group.*

(i) *If, for every $n \geq 0$,*

$$H^2\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then any two smooth S-groups whose reduction is X_0 are isomorphic by an isomorphism inducing the identity on X_0.

(ii) *If, for every $n \geq 0$,*

$$H^3\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then there exists a smooth S-group whose reduction is X_0.

Corollary 3.9. *Let S be affine and S_0 a closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Suppose that every $\mathscr{I}^{n+1}/\mathscr{I}^{n+2}$ is locally free on S_0. Let X_0 be a smooth affine S_0-group.*

[70] Editor's note: This remark, analogous to 4.5.1, has been added to introduce the notation $\delta(\mu, \mu')$, or $\delta(X, X')$, used in 4.38. Consequently, the part of 3.5(ii) concerning $\mathcal{E}$ itself has also been added.

[71] Editor's note: We follow the sign conventions of the original in order to have, in 4.38(5), the equality $\partial^1 \bar{d}(X, X') = \bar{\delta}(X, X')$; see also Editor's Note 54.

(i) *If* $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0$*, then any two smooth* S*-groups whose reduction is* X_0 *are isomorphic.*

(ii) *If* $H^3(X_0, \mathrm{Lie}(X_0/S_0)) = 0$*, then there exists a smooth* S*-group whose reduction is* X_0*.*

Corollary 3.10. *Let* S_0 *be a scheme and let* $S = I_{S_0}$ *be the dual-number scheme over* S_0*. Let* X_0 *be a smooth* S_0*-group. Then every smooth* S*-group* Y *for which* Y_0 *is* S_0*-isomorphic to* X_0 *is* S*-isomorphic to* $X = X_0 \times_{S_0} S$ *if and only if*

$$H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0.$$

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Indeed, by 3.5 the set of classes of such groups Y, up to an S-group isomorphism inducing the identity on X_0, is in bijection with $H^2(X_0, \mathrm{Lie}(X_0/S_0))$. Hence their classes up to an arbitrary S-group isomorphism are in bijection with

$$H^2(X_0, \mathrm{Lie}(X_0/S_0))/\Gamma_0,$$

where

$$\Gamma_0 = \mathrm{Aut}_{S_0\text{-gr.}}(X_0),$$

which acts in the evident way on H^2. The conclusion follows at once. [73]

4. Infinitesimal extensions of closed subgroups

We first state a result valid in an arbitrary abelian category.

Lemma 4.1. *Let*

$$0 \longrightarrow A' \xrightarrow{i} A \xrightarrow{p} A'' \longrightarrow 0$$

be an exact sequence, let $\phi : A' \to Q$ *be a morphism, and let* $\pi : A'' \to P$ *be an epimorphism with kernel* C*. Let* E *be the set, up to isomorphism, of quadruples* (B, f, g, h) *such that the sequence*

$$0 \longrightarrow Q \xrightarrow{f} B \xrightarrow{g} P \longrightarrow 0$$

is exact and the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "\phi"'] & A \arrow[r, "p"] \arrow[d, "h"'] & A'' \arrow[r] \arrow[d, "\pi"] & 0 \\
0 \arrow[r] & Q \arrow[r, "f"] & B \arrow[r, "g"] & P \arrow[r] & 0.
\end{tikzcd}$$

(i) *The set* E *is nonempty if and only if the image in* $\mathrm{Ext}^1(C, Q)$ *of the element* A *of* $\mathrm{Ext}^1(A'', A')$ *is zero.*

(ii) *Under these conditions,* E *is a principal homogeneous set under the abelian group* $\mathrm{Hom}(C, Q)$*.*

[72] Editor's note: This result is used in XXIV, 1.13.

[73] Editor's note: Indeed, $\mathrm{Aut}_{S_0\text{-gr.}}(X_0)$ acts by group automorphisms on the abelian group $H^2(X_0, \mathrm{Lie}(X_0/S_0))$, so the orbit of 0 is the singleton $\{0\}$. Consequently, the quotient set is a singleton if and only if $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = \{0\}$.

Introduce the amalgamated sum

$$B' = A \coprod_{A'} Q.$$

There is then a commutative diagram with exact rows: [74]

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "\phi"'] & A \arrow[r, "p"] \arrow[d] & A'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Q \arrow[r, "j"] & B' \arrow[r] & A'' \arrow[r] & 0,
\end{tikzcd}$$

It is clear that the category of solutions to the problem is canonically isomorphic to the category of solutions to the corresponding problem for the sequence

$$0 \longrightarrow Q \longrightarrow B' \longrightarrow A'' \longrightarrow 0$$

and the morphisms id_Q and $\pi : A'' \to P$. [75] In this case, E is in bijection with the set of subobjects N of B' for which $B' \to A''$ induces an isomorphism of N with the kernel C of $A'' \to P$; equivalently, it is the set of morphisms $e : C \to B'$ lifting the canonical morphism $C \to A''$. The abelian group $G = \mathrm{Hom}(C, Q)$ acts on E by $g \cdot e = g + e$, with addition in $\mathrm{Hom}(C, B')$; if $E \neq \varnothing$, this makes E a principal homogeneous set under G.

We deduce the following.

Proposition 4.2. [76] *Let S be a scheme, let $S_{\mathscr{J}}$ be the closed subscheme defined by a quasi-coherent square-zero ideal $\mathscr{J}$, let X be an S-scheme, and let $\mathscr{F}$ be an $\mathscr{O}_X$-module. Put*

$$X_{\mathscr{J}} = X \times_S S_{\mathscr{J}}, \qquad \mathscr{F}_{\mathscr{J}} = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_{\mathscr{J}}},$$

and let $\mathscr{G}_{\mathscr{J}} = \mathscr{F}_{\mathscr{J}}/\mathscr{H}_{\mathscr{J}}$ be a quotient module of $\mathscr{F}_{\mathscr{J}}$. Suppose given a morphism of $\mathscr{O}_{X_{\mathscr{J}}}$-modules

$$f : \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}} \longrightarrow \mathscr{Q}.$$

Let $\mathcal{E}$ be the sheaf of sets on X defined as follows: for each open subset U of X, $\mathcal{E}(U)$ is the set of quotient modules $\mathscr{G}$ of $\mathscr{F}|_U$ such that $\mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$ and for which there exists an isomorphism

$$h : \mathscr{J}\mathscr{G} \xrightarrow{\sim} \mathscr{Q}|_U$$

making the following diagram commute:

$$\begin{tikzcd}
\mathcal{J} \otimes_{\mathscr{O}_{S_{\mathcal{J}}}} (\mathscr{G}_{\mathcal{J}}|_U) \arrow[r, "f|_U"] \arrow[d, "\text{can.}"'] & \mathscr{Q}|_U \\
\mathcal{J}\mathscr{G} \arrow[ur, "h", "\simeq"'] &
\end{tikzcd} .$$

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[74] Editor's note: One has $\mathrm{Coker}(j) = B' \coprod_Q 0 = A \coprod_{A'} 0 = A''$, and $\mathrm{Ker}(j) \simeq \mathrm{Ker}(i) = 0$, as one sees by reasoning "as if the category were a category of modules." For a proof entirely in terms of arrows, see for example [Fr64], Theorem 2.5.4 (*).

[75] Editor's note: In what follows, A has been replaced by B', and the end of the argument has been expanded.

[76] Editor's note: The statement has been rewritten so as to fit exactly the application made in 4.3; the proof has also been expanded, following indications supplied by M. Demazure.

The isomorphism h is then unique, since $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{G}_{\mathscr{J}}|_U) \to \mathscr{J}\mathscr{G}$ is an epimorphism. Then $\mathcal{E}$ is a sheaf formally principal homogeneous under the sheaf of commutative groups

$$\mathscr{A} = \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q}) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q}).$$

Proof. If $\mathcal{E}(U) = \varnothing$, there is nothing to prove; we may therefore suppose that $\mathcal{E}(U)$ contains an element $\widetilde{\mathscr{G}}$. In the following diagram, h is an isomorphism and all the arrows are epimorphisms:

$$\begin{tikzcd}
\mathcal{J} \otimes_{\mathscr{O}_{S_{\mathcal{J}}}} (\mathscr{F}_{\mathcal{J}}|_U) \arrow[r] \arrow[d, "\text{can.}"'] & \mathcal{J} \otimes_{\mathscr{O}_{S_{\mathcal{J}}}} (\mathscr{G}_{\mathcal{J}}|_U) \arrow[r, "f|_U"] \arrow[d, "\text{can.}"'] & \mathcal{Q}|_U \\
\mathcal{J}\mathscr{F}|_U \arrow[r] & \mathcal{J}\widetilde{\mathscr{G}} \arrow[ur, "h", "\simeq"'] &
\end{tikzcd}.$$

Thus the morphism

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{F}_{\mathscr{J}}|_U) \longrightarrow \mathscr{Q}|_U$$

induces a necessarily unique epimorphism $\phi : \mathscr{J}\mathscr{F}|_U \to \mathscr{Q}|_U$. If $\mathscr{G}$ is an $\mathscr{O}_U$-module such that $\mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$ and there is a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{J}\mathscr{F}|_U \arrow[r] \arrow[d, "\phi"'] & \mathscr{F}|_U \arrow[r] \arrow[d] & \mathscr{F}_{\mathcal{J}}|_U \arrow[r] \arrow[d, "\pi"] & 0 \\
0 \arrow[r] & \mathcal{Q}|_U \arrow[r] & \mathscr{G} \arrow[r, "p_{\mathcal{J}}"] & \mathscr{G}_{\mathcal{J}}|_U \arrow[r] & 0
\end{tikzcd}$$

(where $p_{\mathscr{J}}$ is the projection $\mathscr{G} \to \mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$, so that $\mathscr{Q}|_U = \mathrm{Ker}(p_{\mathscr{J}}) = \mathscr{J}\mathscr{G}$), then $\mathscr{G}$ may be identified with a quotient module of $\mathscr{F}|_U$. Consequently, by 4.1(ii), the set $\mathcal{E}(U)$ is principal homogeneous under the abelian group

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q})(U) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q})(U).$$

Proposition 4.3. *(TDTE IV 5.1) Let S be a scheme, let $S_{\mathscr{J}}$ be the closed subscheme defined by a quasi-coherent square-zero ideal $\mathscr{J}$, let X be an S-scheme, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. Put*

$$X_{\mathscr{J}} = X \times_S S_{\mathscr{J}}, \qquad \mathscr{F}_{\mathscr{J}} = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_{\mathscr{J}}}.$$

Let $\mathscr{G}_{\mathscr{J}} = \mathscr{F}_{\mathscr{J}}/\mathscr{H}_{\mathscr{J}}$ be a quasi-coherent quotient module of $\mathscr{F}_{\mathscr{J}}$, flat over $S_{\mathscr{J}}$. For each open subset U of X, let $\mathcal{E}(U)$ be the set of quasi-coherent quotient modules $\mathscr{G}$[77] of $\mathscr{F}|_U$, flat over S, such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}|_U$. Then the $\mathcal{E}(U)$ form a sheaf of sets $\mathcal{E}$ on X, formally principal homogeneous under the sheaf of commutative groups

$$\mathscr{A} = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{H}_{\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}}\right).$$

[77] Editor's note: To lighten the statement, the hypothesis that $\mathscr{G}$ be quasi-coherent has been added here; the remark that this hypothesis is automatically satisfied has been moved to the proof, which has been expanded accordingly.

Proof. Let $\pi : X \to S$ be the structural morphism. Let U be an open subset of X, and let $\mathscr{G}$ be an $\mathscr{O}_U$-module flat over S, such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}|_U$. For every $x \in U$, the stalk $\mathscr{G}_x$ is a flat module over the local ring $\mathscr{O}_{S,s}$, where $s = \pi(x)$. Hence the morphism

$$\mathscr{J}_s \otimes_{\mathscr{O}_{S,s}} (\mathscr{G}/\mathscr{J}\mathscr{G})_x = \mathscr{J}_s \otimes_{\mathscr{O}_{S,s}} \mathscr{G}_x \longrightarrow (\mathscr{J}\mathscr{G})_x$$

is bijective. We therefore have an exact sequence

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_S} (\mathscr{G}_{\mathscr{J}}|_U) \longrightarrow \mathscr{G} \longrightarrow \mathscr{G}_{\mathscr{J}}|_U \longrightarrow 0.$$

Since $\mathscr{J} \otimes_{\mathscr{O}_S} (\mathscr{G}_{\mathscr{J}}|_U)$ and $\mathscr{G}_{\mathscr{J}}|_U$ are quasi-coherent $\mathscr{O}_U$-modules, $\mathscr{G}$ is quasi-coherent as well (EGA III, 1.4.17).

Conversely, since $\mathscr{G}_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$, if $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_U$-module such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}$ and the morphism

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}} \longrightarrow \mathscr{J}\mathscr{G}$$

is bijective, then $\mathscr{G}$ is flat over S, by the "fundamental criterion of flatness" (SGA 1, IV, 5.5). [78]

Consequently, the set $\mathcal{E}(U)$ considered here is the set considered in 4.2 when f is the identity morphism of $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}}$; the conclusion therefore follows from 4.2. □

Retain the preceding notation. [79] Let $Y_{\mathscr{J}}$ be a closed subscheme of $X_{\mathscr{J}}$, defined by a quasi-coherent ideal $\mathscr{I}_{Y_{\mathscr{J}}}$, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Applying 4.3 to $\mathscr{F} = \mathscr{O}_X$ and

$$\mathscr{G}_{\mathscr{J}} = \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_{X_{\mathscr{J}}}/\mathscr{I}_{Y_{\mathscr{J}}},$$

we obtain the following corollary.

Corollary 4.3.1. *Let $S, S_{\mathscr{J}}, \mathscr{J}, X, X_{\mathscr{J}}, Y_{\mathscr{J}}$, and $\mathscr{I}_{Y_{\mathscr{J}}}$ be as above, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Let $\mathscr{A}_{\mathscr{J}}$ be the sheaf of commutative groups*

$$\mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right)$$

on $X_{\mathscr{J}}$, and put $\mathscr{A} = i_*(\mathscr{A}_{\mathscr{J}})$, where $i : X_{\mathscr{J}} \hookrightarrow X$ is the immersion. For each open subset U of X, let $\mathcal{E}(U)$ be the set of closed subschemes Y of U, flat over S, such that

$$Y \times_S S_{\mathscr{J}} = Y_{\mathscr{J}} \cap U.$$

Then $\mathcal{E}$ is an $\mathscr{A}$-pseudo-torsor.

If, moreover, such a Y exists locally—that is, if every $x \in X$ has an open neighborhood U with $\mathcal{E}(U) \neq \varnothing$—then $\mathcal{E}$ is an $\mathscr{A}$-torsor. The $\mathscr{A}$-torsors on X are parametrized by

$$H^1(X, \mathscr{A}) = H^1(X_{\mathscr{J}}, \mathscr{A}_{\mathscr{J}})$$

(see, for example, EGA IV_4, 16.5.15), and $\mathcal{E}$ has a global section—that is, $\mathcal{E}(X) \neq \varnothing$—if and only if its corresponding cohomology class is zero. Thus we obtain:

Corollary 4.4. *Let $S, S_{\mathscr{J}}, \mathscr{J}, X, X_{\mathscr{J}}, Y_{\mathscr{J}}$, and $\mathscr{I}_{Y_{\mathscr{J}}}$ be as above, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Let E be the set of closed subschemes Y of X, flat over S, such that $Y \times_S S_{\mathscr{J}} = Y_{\mathscr{J}}$.*

[78] Editor's note: See also [BAC], §III.5, Theorem 1.

[79] Editor's note: The discussion that follows has been expanded and Corollary 4.3.1 has been added. Recall also that "pseudo-torsor" is synonymous with "formally principal homogeneous."

(i) *The set E is either empty or principal homogeneous under the abelian group*

$$H^0(X,\mathscr{A}) = H^0(X_{\mathscr{J}},\mathscr{A}_{\mathscr{J}}) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}},\mathscr{J}\otimes_{\mathscr{O}_{S_{\mathscr{J}}}}\mathscr{O}_{Y_{\mathscr{J}}}\right).$$

(ii) *The set E is nonempty if and only if both of the following conditions hold:*

(a) *The problem has a solution locally on X.*

(b) *A certain obstruction vanishes; it lies in*

$$H^1\Big(X_{\mathscr{J}},\mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}},\mathscr{J}\otimes_{\mathscr{O}_{S_{\mathscr{J}}}}\mathscr{O}_{Y_{\mathscr{J}}}\right)\Big).$$

Complement 4.4.1. [80] Retain the notation of 4.4, and suppose that E contains an element Y. Let $\mathscr{I}_Y$ be the ideal of $\mathscr{O}_X$ defining Y, and let $\mathscr{I}_{Y_{\mathscr{J}}}$ be its image in $\mathscr{O}_{X_{\mathscr{J}}}$. As we saw in the proof of 4.2, there is a commutative diagram

$$\begin{tikzcd}
\mathcal{J}\otimes_{\mathscr{O}_{S_{\mathcal{J}}}}\mathscr{O}_{X_{\mathcal{J}}} \arrow[r, two heads] \arrow[d] & \mathcal{J}\mathscr{O}_X \arrow[d] \arrow[dl, dashed] \\
\mathcal{J}\otimes_{\mathscr{O}_{S_{\mathcal{J}}}}\mathscr{O}_{Y_{\mathcal{J}}} \arrow[r, "\sim"] & \mathcal{J}\mathscr{O}_Y
\end{tikzcd}$$

and hence an epimorphism of $\mathscr{O}_X$-modules

$$\phi : \mathscr{J}\mathscr{O}_X \longrightarrow \mathscr{J}\otimes_{\mathscr{O}_{S_{\mathscr{J}}}}\mathscr{O}_{Y_{\mathscr{J}}}.$$

Let $\mathscr{K}$ be its kernel. For every element Y' of E, the morphism $\mathscr{O}_X\to\mathscr{O}_{Y'}$ then factors through $\mathscr{O}_X/\mathscr{K}$, which is the amalgamated sum B' in the proof of Lemma 4.1. If $\mathscr{I}_{Y'}$ denotes the ideal of Y' in $\mathscr{O}_X$, there is a commutative diagram

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
 & & \mathscr{I}_{Y'}/\mathscr{K} \arrow[r, "\sim"] \arrow[d] & \mathscr{I}_{Y_{\mathcal{J}}} \arrow[d] & \\
0 \arrow[r] & (\mathcal{J}\mathscr{O}_X)/\mathscr{K} \arrow[r] \arrow[d, "\wr"] & \mathscr{O}_X/\mathscr{K} \arrow[r] \arrow[d] & \mathscr{O}_{X_{\mathcal{J}}} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{J}\otimes_{\mathscr{O}_{S_{\mathcal{J}}}}\mathscr{O}_{Y_{\mathcal{J}}} \arrow[r] \arrow[d] & \mathscr{O}_{Y'} \arrow[r] \arrow[d] & \mathscr{O}_{Y_{\mathcal{J}}} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 & .
\end{tikzcd}$$

[80] Editor's note: This complement has been added; it is useful in proving Proposition 4.8(ii).

Replacing X by the closed subscheme defined by $\mathscr{K}$, we are thus reduced to the case $\mathscr{K} = 0$. The datum of Y' is then equivalent to that of the $\mathscr{O}_X$-submodule $\mathscr{I}_{Y'}$ of $\mathscr{O}_X$ which maps bijectively onto $\mathscr{I}_{Y_\mathscr{J}}$ under the projection $p : \mathscr{O}_X \to \mathscr{O}_{X_\mathscr{J}}$. Let

$$f' : \mathscr{I}_{Y_\mathscr{J}} \xrightarrow{\sim} \mathscr{I}_{Y'}, \qquad f : \mathscr{I}_{Y_\mathscr{J}} \xrightarrow{\sim} \mathscr{I}_{Y}$$

denote the respective inverse isomorphisms. Then $f' - f$ is an element of

$$\mathrm{Hom}_{\mathscr{O}_{X_\mathscr{J}}}\left(\mathscr{I}_{Y_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right) = \mathrm{Hom}_{\mathscr{O}_{X_\mathscr{J}}}(\mathscr{I}_{Y_\mathscr{J}}, \mathscr{J}\mathscr{O}_Y),$$

which we denote by $d(Y', Y)$. Notice that $d(Y, Y') = -d(Y', Y)$.

For our fixed Y and variable Y', consider the morphism

$$\mathscr{I}_{Y'} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_Y = \mathscr{O}_X/\mathscr{I}_Y.$$

Since its composite with $\mathscr{O}_Y \to \mathscr{O}_{Y_\mathscr{J}}$ is zero, it takes values in

$$\mathscr{J}\mathscr{O}_Y = \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}.$$

More precisely, let V be an open subset of X, let x' be a section of $\mathscr{I}_{Y'}$ on V, and let $x_\mathscr{J}$ be its image in $\Gamma(V, \mathscr{I}_{Y_\mathscr{J}})$. Then

$$x' = f'(x_\mathscr{J}) = f(x_\mathscr{J}) + (f' - f)(x_\mathscr{J}) = f(x_\mathscr{J}) + d(Y', Y)(x_\mathscr{J}).$$

Consequently, the morphism $\mathscr{I}_{Y'} \to \mathscr{J}\mathscr{O}_Y$ is given by $d(Y', Y)$.

4.5.0.[81] Retain the notation of 4.3.1 and 4.4, and make the following transformations. The quasi-coherent $\mathscr{O}_{X_\mathscr{J}}$-module

$$\mathscr{I}_{Y_\mathscr{J}}/\mathscr{I}^2_{Y_\mathscr{J}}$$

is annihilated by $\mathscr{I}_{Y_\mathscr{J}}$, and is therefore the direct image of a quasi-coherent $\mathscr{O}_{Y_\mathscr{J}}$-module, denoted by $\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}$, and called the *conormal sheaf* of $Y_\mathscr{J}$ in $X_\mathscr{J}$.[82] Since $\mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}$ is annihilated by $\mathscr{I}_{Y_\mathscr{J}}$, the sheaf of commutative groups $\mathscr{A}_\mathscr{J}$ of 4.3.1 identifies with

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{I}_{Y_\mathscr{J}}/\mathscr{I}^2_{Y_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\\ &\quad= \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right),\end{aligned}$$

and hence, for every $i \geq 0$,

$$H^i(X_\mathscr{J}, \mathscr{A}_\mathscr{J}) = H^i\Big(Y_\mathscr{J}, \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\Big).$$

[83] One may now drop the hypothesis that Y be closed, as follows. First observe that every open subset $U_\mathscr{J}$ of $X_\mathscr{J}$ is obtained by base change from the open subscheme U of X having the same underlying topological space as $U_\mathscr{J}$. Let $Y_\mathscr{J}$ now be a closed subscheme of $U_\mathscr{J}$, flat over $S_\mathscr{J}$, and let $\mathscr{I}_{Y_\mathscr{J}}$ be the quasi-coherent ideal of $\mathscr{O}_{U_\mathscr{J}}$ defining $Y_\mathscr{J}$. If $Y_\mathscr{J}$ lifts to a subscheme Y of X, then Y, having the same underlying topological space as $Y_\mathscr{J}$, is a closed subscheme of U. Consequently, the obstruction to lifting $Y_\mathscr{J}$ to a subscheme, flat over S, of X or of U is the same; it lies in

$$H^1\Big(Y_\mathscr{J}, \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_\mathscr{J}}} \mathscr{O}_{Y_\mathscr{J}}\right)\Big).$$

[81] Editor's note: The numbering 4.5.0 has been added to mark the return to the original.
[82] Editor's note: The following sentence has been corrected.
[83] Editor's note: The discussion that follows has been expanded.

Finally, return to the notation of §0. Let $\mathscr{I}$ be a quasi-coherent ideal of $\mathscr{O}_S$ such that $\mathscr{J} \subset \mathscr{I}$ and $\mathscr{I}\mathscr{J} = 0$, and let S_0 be the closed subscheme of $S_{\mathscr{J}}$ defined by $\mathscr{I}$. For every S-scheme Z, write

$$Z_{\mathscr{J}} = Z \times_S S_{\mathscr{J}}, \qquad Z_0 = Z \times_S S_0.$$

Since $\mathscr{J}$ is annihilated by $\mathscr{I}$, the notation of 4.4 gives

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0},$$

and

$$\begin{aligned} &\operatorname{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right) \\ &\qquad = \operatorname{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right), \end{aligned}$$

and similarly for the other terms. Thus one obtains:

Proposition 4.5. *Let S be a scheme, and let S_0 and $S_{\mathscr{J}}$ be the closed subschemes defined by quasi-coherent ideals $\mathscr{I}$ and $\mathscr{J}$, where $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let X be an S-scheme and let $Y_{\mathscr{J}}$ be a subscheme of $X_{\mathscr{J}}$, flat over $S_{\mathscr{J}}$. Define the $\mathscr{O}_{Y_0}$-module*

$$\mathscr{A}_0 = \operatorname{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right).$$

Then:

(i) *A subscheme Y of X, reducing to $Y_{\mathscr{J}}$ and flat over S, exists if and only if the following two conditions hold:*

  (a) *Such a Y exists locally on X.*
  (b) *A certain obstruction in $R^1\Gamma(Y_0, \mathscr{A}_0)$ vanishes.*[84]

(ii) *Under these conditions, the set of subschemes Y satisfying the stated requirements is principal homogeneous under the abelian group $\Gamma(Y_0, \mathscr{A}_0)$.*

Remark 4.5.1.[85] It follows from 4.5(ii) that for every pair (Y, Y') of subschemes[86] of X, flat over S and reducing to $Y_{\mathscr{J}}$, there is a "deviation"

$$\begin{aligned} d(Y', Y) &\in \Gamma(Y_0, \mathscr{A}_0) \\ &= \operatorname{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right). \end{aligned}$$

With the sign convention adopted in 4.4.1, $d(Y', Y)$ corresponds to the morphism of $\mathscr{O}_X$-modules

$$\mathscr{I}_{Y'} \hookrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_Y,$$

which takes values in $\mathscr{J}\mathscr{O}_Y \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}$, and factors through $\mathscr{I}_{Y'_{\mathscr{J}}} = \mathscr{I}_{Y_{\mathscr{J}}}$, then through $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$.

[84] Editor's note: Here $R^1\Gamma(Y_0, \mathscr{A}_0)$ denotes the "coherent" cohomology group $H^1(Y_0, \mathscr{A}_0)$ of the $\mathscr{O}_{Y_0}$-module $\mathscr{A}_0$, in order to distinguish it from the "Hochschild" cohomology groups $H^i(Y_0, M_0)$, where Y_0 is an S_0-group and M_0 an $\mathscr{O}_{S_0}$-module, which will be considered beginning in 4.16.

[85] Editor's note: This remark has been placed here; it replaces Remark 4.7 of the original.

[86] Editor's note: "closed subschemes" has been corrected to "subschemes."

Remark 4.6.[87] *If X is flat over S and $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$, then condition (a) is always satisfied, and every Y flat over S lifting $Y_{\mathscr{J}}$ is locally complete intersection in X. If, moreover, Y_0 is affine, condition (b) is also satisfied.*

Definition 4.6.1. *(cf. SGA 6, VII 1.1) Let B be a commutative ring, let $f : E \to B$ be a B-linear morphism, where E is a free B-module of finite rank d, and let $I = f(E)$. If a basis of E is chosen, f is given by a d-tuple $(f_1, \dots, f_d)$ of elements of B, and I is the ideal generated by the f_i. The* Koszul complex *$K_\bullet(f)$ is the graded B-module $\bigwedge_B^\bullet E$, equipped with the differential of degree -1*

$$x_1 \wedge \cdots \wedge x_i \longmapsto \sum_{j=1}^{i} (-1)^{j-1} f(x_j) x_1 \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_i.$$

Thus there is an augmented chain complex, with B/I in degree -1,

$$\cdots \longrightarrow \bigwedge^2 E \longrightarrow E \xrightarrow{f} B \longrightarrow B/I \longrightarrow 0,$$

which is exact in degree 0 by definition, since $I = f(E)$. The morphism f is said to be *regular* if $K_\bullet(f)$ is acyclic in degrees > 0, that is, if the augmented complex above is a resolution of $C = B/I$.

In that case, the proof of SGA 6, VII 1.2(b) shows that the C-modules I^n/I^{n+1}, for $n \in \mathbb{N}$, are free, with I/I^2 of rank d.

Definition 4.6.2. *(cf. SGA 6, VII 1.4) Let X be a scheme, Y a subscheme, and U an open subset of X such that Y is a closed subscheme of U, defined by the quasi-coherent ideal $\mathscr{I}_Y$.*

The subscheme Y is said to be *locally complete intersection* in X if $Y \hookrightarrow X$ is a regular immersion in the sense of SGA 6, VII 1.4; that is, if for every $y \in Y$ there exist an affine open neighborhood V of y in U, a finite free $\mathscr{O}_V$-module $\mathscr{E}$, and a regular morphism $f : \mathscr{E} \to \mathscr{O}_V$ with image $\mathscr{I}_Y|_V$, such that $K_\bullet(f)$ is a resolution of $\mathscr{O}_{Y\cap V}$.

This implies that the immersion $Y \hookrightarrow X$ is locally of finite presentation and, by 4.6.1, that the conormal sheaf

$$\mathscr{N}_{Y/X} = \mathscr{I}_Y/\mathscr{I}_Y^2$$

is a finite locally free $\mathscr{O}_Y$-module.

Lemma 4.6.3.[88] *Let A be a ring, let J be a square-zero ideal of A, put $\overline{A} = A/J$, let B be a flat A-algebra, let E be a free B-module of finite rank, and let $f : E \to B$ be a morphism of B-modules. Suppose that the morphism*

$$g : \overline{E} = E \otimes_A \overline{A} \longrightarrow \overline{B} = B \otimes_A \overline{A}$$

induced by f is regular and that $\overline{B}/g(\overline{E})$ is flat over $\overline{A}$. Then f is regular and $B/f(E)$ is flat over A.

Proof. Put

$$C = B/f(E), \qquad \overline{C} = C \otimes_A \overline{A} = \overline{B}/g(\overline{E}).$$

First, the $\bigwedge_B^i E$ are free B-modules, and hence flat A-modules, since B is flat over A. Since

$$\bigwedge_B^\bullet E \otimes_A \overline{A} \simeq \bigwedge_{\overline{B}}^\bullet \overline{E},$$

[87]Editor's note: We have retained, for reference, Remark 4.6 of the original, which did not include the definition of "locally complete intersection." It is followed by the "correct" definition, taken from SGA 6, VII 1.4 (and replacing that of EGA IV_4, 16.9.2), and by proofs of the three assertions made in the remark.

[88]Editor's note: To prove the results stated in Remark 4.6, Lemmas 4.6.3 and 4.6.4, Proposition 4.6.5, and Remark 4.6.6 have been added.

there is an exact sequence of complexes

$$0 \longrightarrow J \otimes_A \bigwedge_B^{\bullet} E \longrightarrow \bigwedge_B^{\bullet} E \longrightarrow \bigwedge_{\overline{B}}^{\bullet} \overline{E} \longrightarrow 0.$$

Moreover, since $J^2 = 0$, for every A-module M one has

$$J \otimes_A M \simeq J \otimes_{\overline{A}} (\overline{A} \otimes_A M).$$

Writing the augmentation arrows as dashed arrows, and denoting the rank of E by d, one obtains the following bicomplex, whose rows are exact.

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \bigwedge_{\overline{\mathrm{B}}}^{d} \overline{\mathrm{E}} \arrow[r] \arrow[d] & \bigwedge_{\mathrm{B}}^{d} \mathrm{E} \arrow[r] \arrow[d] & \bigwedge_{\overline{\mathrm{B}}}^{d} \overline{\mathrm{E}} \arrow[r] \arrow[d] & 0 \\
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
0 \arrow[r] & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \overline{\mathrm{E}} \arrow[r] \arrow[d, "\mathrm{id} \otimes g"'] & \mathrm{E} \arrow[r] \arrow[d, "f"] & \overline{\mathrm{E}} \arrow[r] \arrow[d, "g"] & 0 \\
0 \arrow[r] & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \overline{\mathrm{B}} \arrow[r] \arrow[d, dashed] & \mathrm{B} \arrow[r] \arrow[d, dashed] & \overline{\mathrm{B}} \arrow[r] \arrow[d, dashed] & 0 \\
 & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \overline{\mathrm{C}} \arrow[r] \arrow[d] & \mathrm{C} \arrow[r] \arrow[d] & \overline{\mathrm{C}} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 & .
\end{tikzcd}$$

The right and left columns are also exact, since $K_\bullet(g)$ is a resolution of $\overline{C}$, and $\overline{C}$ is flat over $\overline{A}$. The long exact homology sequence associated with the exact sequence of unaugmented complexes therefore shows that $K_\bullet(f)$ is acyclic in degrees > 0, and in degree 0 gives an exact sequence

$$0 \longrightarrow J \otimes_{\overline{A}} \overline{C} \longrightarrow C \longrightarrow \overline{C} \longrightarrow 0.$$

It follows from the "fundamental criterion of flatness" that C is flat over A (cf. [BAC], §III.5, Theorem 1). $\square$

Lemma 4.6.4.[88] *Let A be a commutative ring, let J be a nilpotent ideal, and let $N \subset M$ be A-modules such that M/N is flat over A. If $x_1, \dots, x_n$ are elements of N whose images generate the image $\overline{N}$ of N in M/JM, then they generate N.*

Indeed, let N' be the submodule of N generated by the x_i, and put $Q = N/N'$. The morphism $N' \otimes_A (A/J) \to \overline{N}$ is surjective. On the other hand, since M/N is flat over A, the morphism $N \otimes_A (A/J) \to \overline{N}$ is bijective. Thus $Q \otimes_A (A/J) = 0$, whence $Q = 0$ by the "nilpotent Nakayama lemma":

$$Q = JQ = J^2Q = \cdots = 0.$$

We can now prove the following proposition.

Proposition 4.6.5.[88] *Let $S, \mathscr{I}, \mathscr{J}$, and $X, Y_{\mathscr{J}}$ be as in 4.5. Suppose in addition that X is flat over S and that $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$.*

(a) *Condition (a) of 4.5(i) is then satisfied; moreover, every Y flat over S lifting $Y_{\mathscr{J}}$ is locally complete intersection in X.*

(b) *If, in addition, Y_0 is affine, condition (b) of the cited result is also satisfied.*

Proof. The first assertion of (a) follows from Lemma 4.6.3, and the second then follows from Lemma 4.6.4. Moreover, the hypothesis implies (cf. 4.6.2) that $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$ is a finite locally free $\mathscr{O}_{Y_{\mathscr{J}}}$-module. Hence the $\mathscr{O}_{Y_0}$-module

$$\mathscr{A}_0 = \operatorname{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right)$$

is quasi-coherent (cf. EGA I, 1.3.12), and therefore $R^1\Gamma(Y_0, \mathscr{A}_0) = 0$ when Y_0 is affine. □

Remark 4.6.6.[88] Let us conclude this paragraph with an example showing that, under the hypotheses of Lemma 4.6.3, a regular sequence $(\overline{g}_1, \overline{g}_2)$ generating the ideal $\overline{I} = g(\overline{E})$ need not lift to a regular sequence in B.

Let k be a field, put $\overline{A} = k[X, Y]$, let $k\varepsilon$ denote the $\overline{A}$-module $\overline{A}/(X, Y)$, so that

$$P\varepsilon = P(0,0)\varepsilon \qquad (P \in \overline{A}),$$

and let $A = \overline{A} \oplus k\varepsilon$, where $J = k\varepsilon$ is a square-zero ideal. Thus $A/J = \overline{A}$.

The algebra $B = A \otimes_k k[Z, T]$ is free, and hence flat, over A, and $\overline{B} = k[X, Y, Z, T]$. Put

$$\overline{g}_1 = XZ - YT, \qquad \overline{g}_2 = XZ - 1.$$

Since $\overline{g}_1$ is irreducible, $\overline{B}/(\overline{g}_1)$ is an integral domain; hence $(\overline{g}_1, \overline{g}_2)$ is a regular sequence in $\overline{B}$, generating the ideal

$$\overline{I} = (XZ - 1, YT - 1).$$

Therefore

$$\overline{C} = \overline{B}/\overline{I} = k[X, Y, X^{-1}, Y^{-1}] = \overline{A}[X^{-1}, Y^{-1}]$$

is a flat $\overline{A}$-algebra, and also a flat A-algebra. But every lift of $\overline{g}_1$ to B has the form

$$XZ - YT + \lambda\varepsilon, \qquad \lambda \in k[Z, T],$$

and therefore annihilates ε.

4.7. Remark 4.7 has been suppressed here and placed in 4.5.1.

Remark 4.8.0.[89] *Let S be a scheme, S' a closed subscheme of S, X an S-scheme, Y an S-subscheme of X, and put $X' = X \times_S S'$ and $Y' = Y \times_S S'$. Then there is a surjective morphism of $\mathscr{O}_{Y'}$-modules*

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \twoheadrightarrow \mathscr{N}_{Y'/X'}.$$

Indeed, after replacing X by a suitable open subscheme, one may suppose that Y is closed and is defined by an ideal $\mathscr{I}_Y$ of $\mathscr{O}_X$. The image of $\mathscr{I}_Y$ in $\mathscr{O}_{X'}$ is then the ideal $\mathscr{I}_{Y'}$ defining Y', and there is a surjective morphism of $\mathscr{O}_{Y'}$-modules

$$\pi : (\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \twoheadrightarrow \mathscr{I}_{Y'}/\mathscr{I}_{Y'}^2.$$

Suppose in addition that $\mathscr{O}_Y = \mathscr{O}_X/\mathscr{I}_Y$ is flat over $\mathscr{O}_S$. The natural morphism

$$\mathscr{I}_Y \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{I}_{Y'}$$

is then bijective (cf. EGA IV_2, 2.1.8).

There is therefore the following commutative diagram with exact rows:

$$\begin{tikzcd}
 & \mathscr{I}_Y^2 \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] \arrow[d, "\text{surj.}"'] & \mathscr{I}_Y \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] \arrow[d, "\wr"] & (\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d, "\pi"', "\text{surj.}"] & 0 \\
0 \arrow[r] & \mathscr{I}_{Y'}^2 \arrow[r] & \mathscr{I}_{Y'} \arrow[r] & \mathscr{I}_{Y'}/\mathscr{I}_{Y'}^2 \arrow[r] & 0
\end{tikzcd}$$

It follows from the snake lemma that[90]

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \xrightarrow{\sim} \mathscr{N}_{Y'/X'} \qquad \text{if } Y \text{ is flat over } S. \tag{4.8.0}$$

Proposition 4.8. *Let $S, S_0, S_{\mathscr{J}}$, and $\mathscr{I}, \mathscr{J}$ be as in 4.5.*[91] *Let X be an S-scheme, let Y be a subscheme of X, and let $i : Y \hookrightarrow X$ be the immersion.*

(i) *For every S-morphism $f : T \to X$ such that $f_{\mathscr{J}} : T_{\mathscr{J}} \to X_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, one can define an obstruction*

$$c(X, Y, f) \in \mathrm{Hom}_{\mathscr{O}_{T_0}}\left(f_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\right) \tag{*}$$

whose vanishing is equivalent to the existence of a factorization of f through Y.

(ii) *Let Y' be a second subscheme of X. Suppose that $Y'_{\mathscr{J}} = Y_{\mathscr{J}}$ and that Y and Y' are flat over S. Then there are isomorphisms (cf. 4.8.0)*

$$\mathscr{J}\mathscr{O}_Y \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0} \simeq \mathscr{J}\mathscr{O}_{Y'}, \qquad \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_{\mathscr{J}}} \xrightarrow{\sim} \mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}},$$

and hence an isomorphism

$$\begin{aligned} u :\ & \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_Y\right) \\ & \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{Y'}\right). \end{aligned}$$

[89] Editor's note: This remark, used in the proposition that follows, has been inserted here; it appeared in 4.10 of the original.

[90] Editor's note: In the original, this was stated in Remark 4.10 under the additional hypothesis that Y' be locally complete intersection in X'. Consequently, that hypothesis also appeared in statements 4.12–4.14; it seems in fact superfluous, and has been deleted from those statements.

[91] Editor's note: The hypothesis that $\mathscr{I}$ be nilpotent has been deleted, since it appears to be superfluous (cf. the proof).

Writing $i' : Y' \hookrightarrow X$ for the canonical immersion and $d(Y, Y')$ for the deviation of 4.5.1, one has[92]

$$c(X, Y, i') = u\big(d(Y, Y')\big). \tag{**}$$

(iii) *The canonical morphism*

$$\mathscr{N}_{Y/X} \xrightarrow{D} i^*(\Omega^1_{X/S})$$

(cf. SGA 1, II, formula 4.3)[93] *induces a morphism*

$$D_0 : \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0},$$

and hence, for every S-morphism $f : T \to X$ as in (i), a morphism

$$\begin{aligned} v_{f_0} :\ &\mathrm{Hom}_{\mathscr{O}_{T_0}}\Big(f_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_T\Big) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\Big(f_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\Big), \\ &\qquad a \longmapsto a \circ f_0^*(D_0), \end{aligned}$$

where the first group above is $\mathrm{Hom}_{X^+}(T, L_X)$*, cf. 0.1.5. For* $a \in \mathrm{Hom}_{X^+}(T, L_X)$*, one has*

$$c(X, Y, a \cdot f) - c(X, Y, f) = v_{f_0}(a), \tag{***}$$

where $a \cdot f$ denotes the composite morphism

$$T \xrightarrow{a \times f} L_X \times_{X^+} X \longrightarrow X.$$

We shall prove part (i) of the proposition, leaving it to the reader to (not) verify assertions (ii) and (iii); this verification proceeds by reduction to the affine case and then by comparison of the explicit definitions.[94]

Let us therefore prove (i). The morphism $f : T \to X$ defines a morphism of sheaves of rings[95]

$$\phi : \mathscr{O}_X \longrightarrow f_*(\mathscr{O}_T).$$

Let U be an open subscheme of X in which Y is closed. Since T, respectively $Y_{\mathscr{J}}$, has the same underlying space as $T_{\mathscr{J}}$, respectively Y, the continuous map underlying f sends T into U. Since U is open in X, the morphism ϕ induces a morphism of sheaves of rings

$$\mathscr{O}_U = \mathscr{O}_X|_U \longrightarrow f_*(\mathscr{O}_T),$$

that is, f factors through U.

We may therefore restrict to the case in which Y is closed, hence defined by a sheaf of ideals $\mathscr{I}_Y$. The morphism f factors through Y if and only if the composite

$$\mathscr{I}_Y \longrightarrow \mathscr{O}_X \longrightarrow f_*(\mathscr{O}_T)$$

is zero. Since $f_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, the composite

$$\mathscr{I}_{Y_{\mathscr{J}}} \longrightarrow \mathscr{O}_{X_{\mathscr{J}}} \longrightarrow f_*(\mathscr{O}_{T_{\mathscr{J}}})$$

is zero. Consider the following commutative diagram, whose first row is exact:

[92] Editor's note: See also 4.27 below.

[93] Editor's note: See also EGA IV$_4$, 16.4.21. Recall that if U is an affine open subset of X such that $Y \cap U$ is defined by the ideal I of $A = \mathscr{O}_X(U)$, if d denotes the differential $A \to \Gamma(U, \Omega^1_{X/S})$, and if $x \in I$, then $D(x + I^2)$ is the element $d(x) \otimes 1$ of $\Gamma(U, \Omega^1_{X/S}) \otimes_A (A/I)$.

[94] Editor's note: These verifications have been carried out below.

[95] Editor's note: On the one hand, the hypothesis that $\mathscr{J}$ be nilpotent—that is, that X_0 have the same underlying topological space as X—has been deleted; on the other hand, the sentence that follows has been expanded.

$$\begin{tikzcd}
0 \arrow[r] & f_*(\mathcal{J}\mathscr{O}_{\mathrm{T}}) \arrow[r] & f_*(\mathscr{O}_{\mathrm{T}}) \arrow[r] & f_*(\mathscr{O}_{\mathrm{T}_{\mathcal{J}}}) \\
 & & \mathscr{O}_{\mathrm{X}} \arrow[u, "\phi"] \arrow[r] & \mathscr{O}_{\mathrm{X}_{\mathcal{J}}} \arrow[u, "\phi_{\mathcal{J}}"'] \\
 & & \mathscr{I}_{\mathrm{Y}} \arrow[u] \arrow[r] \arrow[uul, dashed, "\phi"] & \mathscr{I}_{\mathrm{Y}_{\mathcal{J}}} \arrow[u]
\end{tikzcd}$$

It follows that ϕ maps $\mathscr{I}_Y$ into $f_*(\mathscr{J}\mathscr{O}_T)$.[96] Since $\mathscr{J}^2 = 0$, the $\mathscr{O}_X$-module $f_*(\mathscr{J}\mathscr{O}_T)$, with its $\mathscr{O}_X$-module structure induced by ϕ, is annihilated by $\mathscr{I}_Y$. Consequently, ϕ induces a morphism of $\mathscr{O}_X$-modules

$$h : i_*(\mathscr{N}_{Y/X}) = \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow f_*(\mathscr{J}\mathscr{O}_T).$$

On the other hand, one has the following cartesian squares:

$$\begin{tikzcd}
\mathrm{T}_0 \arrow[r, "f_0"] \arrow[d, "\tau_{\mathrm{T}_0}"] & \mathrm{X}_0 \arrow[d, "\tau_{\mathrm{X}_0}"] & \mathrm{Y}_0 \arrow[l, "i_0"'] \arrow[d, "\tau_{\mathrm{Y}_0}"] \\
\mathrm{T} \arrow[r, "f"] & \mathrm{X} & \mathrm{Y}. \arrow[l, "i"']
\end{tikzcd}$$

Here $\tau_{T_0}, \tau_{X_0}, \tau_{Y_0}$ are the closed immersions obtained by base change from $S_0 \hookrightarrow S$. Since $\mathscr{J}\mathscr{O}_T$ is a quasi-coherent $\mathscr{O}_T$-module annihilated by $\mathscr{I}$, there is an isomorphism

$$\mathscr{J}\mathscr{O}_T \simeq (\tau_{T_0})_*\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T),$$

and hence

$$f_*(\mathscr{J}\mathscr{O}_T) \simeq (\tau_{X_0})_*(f_0)_*\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T).$$

Thus, by adjunction, h corresponds to a morphism of $\mathscr{O}_{T_0}$-modules

$$h_0 : f_0^*\tau_{X_0}^* i_*(\mathscr{N}_{Y/X}) \longrightarrow \tau_{T_0}^*(\mathscr{J}\mathscr{O}_T).$$

Now

$$\tau_{X_0}^* i_*(\mathscr{N}_{Y/X}) \simeq (i_0)_*\tau_{Y_0}^*(\mathscr{N}_{Y/X}) = \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}.$$

Returning to the customary abuse of notation $\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T) = \mathscr{J}\mathscr{O}_T$, the morphism h_0 identifies with a morphism of $\mathscr{O}_{T_0}$-modules

$$h_0 : f_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \longrightarrow \mathscr{J}\mathscr{O}_T,$$

which is the required obstruction $c(X, Y, f)$. This proves (i).

When f is the immersion $i' : Y' \hookrightarrow X$, one sees that $c(X, Y, i')$ comes from the morphism

$$\mathscr{I}_Y \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_{Y'},$$

[96] Editor's note: The argument that follows has been expanded.

and hence corresponds, by 4.4.1 and 4.5.1, to the class $d(Y, Y')$. This proves (ii).

Let us prove (iii). First, $D : \mathscr{N}_{Y/X} \to i^*(\Omega^1_{X/S})$ induces a morphism

$$\begin{aligned} D_0 : \quad \tau^*_{Y_0}(\mathscr{N}_{Y/X}) &\longrightarrow \tau^*_{Y_0} i^*(\Omega^1_{X/S}) \\ &= i_0^* \tau^*_{X_0}(\Omega^1_{X/S}). \end{aligned}$$

Since $X_0 = X \times_S S_0$, one has

$$\tau^*_{X_0}(\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0} \qquad \text{(cf. EGA IV}_4\text{, 16.4.5).}$$

This gives the asserted morphism

$$D_0 : \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0}.$$

We shall verify equality $(***)$ after the following remark.

Remark 4.9. The obstruction $c(X, Y, f)$ is computed locally on T. Let $U = \operatorname{Spec}(C)$ be an affine open subset of T lying over an affine open subset $\operatorname{Spec}(A)$ of X, itself lying over an affine open subset $\operatorname{Spec}(\Lambda)$ of S. Let $J \subset I \subset \Lambda$, respectively $I_Y \subset A$, be the ideals corresponding to $\mathscr{J} \subset \mathscr{I}$, respectively $\mathscr{I}_Y$, put $B = A/I_Y$, and let $\phi : A \to C$ be the morphism of Λ-algebras corresponding to $f : T \to X$. Since $f(T_{\mathscr{J}}) \subset Y_{\mathscr{J}}$, one has $\phi(I_Y) \subset JC$, and ϕ therefore induces a morphism of Λ-algebras

$$B \longrightarrow C/JC \longrightarrow C_0 = C/IC.$$

The obstruction $c = c(X, Y, f)$ is then computed by the following commutative diagram:

$$\begin{tikzcd}
\mathrm{I_Y} \arrow[rr] \arrow[d] & & \mathrm{A} \arrow[r, "\phi"] & \mathrm{C} \\
\mathrm{I_Y/I_Y^2} \arrow[r] & \mathrm{I_Y/I_Y^2 \otimes_B C_0} \arrow[rr, "c"] & & \mathrm{JC} \arrow[u]
\end{tikzcd} \quad ,$$

That is, over U, it is the unique element of

$$\operatorname{Hom}_{C_0}\left(I_Y/I_Y^2 \otimes_B C_0, JC\right) = \operatorname{Hom}_{B_0}\left(I_Y/I_Y^2 \otimes_B B_0, JC\right)$$

such that, with the evident notation,

$$c(\overline{x} \otimes_B 1) = \phi(x) \qquad \text{for every } x \in I_Y.$$

[97] We can now complete the proof of 4.8(iii). Equality $(***)$ may be checked locally on T, so we are reduced to the affine situation described above. Let $d_{A/\Lambda} : A \to \Omega^1_{A/\Lambda}$ denote the differential. Over U, the element a corresponds to an element a_U of

$$\begin{aligned} &\operatorname{Hom}_{C_0}\left(\Omega^1_{A_0/\Lambda_0} \otimes_{A_0} C_0, JC\right) \\ &\qquad \simeq \operatorname{Hom}_{B_0}\left(\Omega^1_{A/\Lambda} \otimes_A B_0, JC\right) \simeq \operatorname{Hom}_A(\Omega^1_{A/\Lambda}, JC). \end{aligned}$$

[97] Editor's note: What follows has been added.

On the one hand, $v_{f_0}(a)$ corresponds over U to $a_U \circ D_0$, where D_0 is the morphism of B-modules[98]

$$I_Y/I_Y^2 \longrightarrow \Omega^1_{A/\Lambda} \otimes_A B_0,$$
$$x + I_Y^2 \longmapsto d_{A/\Lambda}(x) \otimes 1.$$

On the other hand (cf. the proofs of 0.1.8 and 0.1.9), the morphism of Λ-algebras $\phi' : A \to C$ corresponding to $a \cdot f$ differs from ϕ by the Λ-derivation $A \to JC$ associated with a_U; that is,

$$\phi' = \phi + a_U \circ d_{A/\Lambda} = \phi + a_U \circ (d_{A/\Lambda} \otimes 1).$$

Consequently, writing $c' = c(X, Y, a \cdot f)$, for every $x \in I_Y$, and writing $\overline{x}$ for its image in I_Y/I_Y^2, one has

$$\begin{aligned}(c' - c)(\overline{x} \otimes 1) &= a_U(d_{A/\Lambda}(x) \otimes 1) \\ &= (a_U \circ D_0)(\overline{x}) = v_{f_0}(a)(\overline{x}).\end{aligned}$$

This proves that $c' - c = v_{f_0}(a)$.

4.10. Remark 4.10 of the original has been suppressed, since it was made obsolete by the addition of Remark 4.8.0.

4.11. We now propose to study the following situation. Let S, $S_{\mathscr{J}}$, and S_0 be as in 4.8. We have three S-schemes X, X', T, a subscheme Y of X (respectively a subscheme Y' of X'), and morphisms $f : T \to X'$ and $g : X' \to X$.

$$\begin{tikzcd} & Y' \arrow[d, hook, "i'"'] & Y \arrow[d, hook, "i"'] \\ T \arrow[r, "f"] & X' \arrow[r, "g"] & X \end{tikzcd} \quad .$$

We suppose that, after reduction modulo $\mathscr{J}$, this diagram can be completed to a commutative diagram

$$\begin{tikzcd} & Y'_{\mathcal{J}} \arrow[r, dashed] \arrow[d, hook, "i'_{\mathcal{J}}"'] & Y_{\mathcal{J}} \arrow[d, hook, "i_{\mathcal{J}}"'] \\ T_{\mathcal{J}} \arrow[ur, dashed] \arrow[r, "f_{\mathcal{J}}"] & X'_{\mathcal{J}} \arrow[r, "g_{\mathcal{J}}"] & X_{\mathcal{J}} \end{tikzcd} \quad .$$

By 4.8 we therefore have the obstructions

$$\begin{aligned} c(X, Y, g \circ i') &\in \operatorname{Hom}_{\mathscr{O}_{Y'_0}}(i_0'^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{Y'}), \\ c(X', Y', f) &\in \operatorname{Hom}_{\mathscr{O}_{T_0}}(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_T), \\ c(X, Y, g \circ f) &\in \operatorname{Hom}_{\mathscr{O}_{T_0}}(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T), \end{aligned}$$

[98] Editor's note: Cf. Editor's Note 93.

whose relations we seek to compute. [99]

Lemma 4.12. *Suppose that Y' is flat over S, so that*

$$\mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y'_0} = \mathscr{J}\mathscr{O}_{Y'}.$$

Then:

(i) *There is a natural morphism*

$$\begin{aligned} b_{f_0}:\ &\mathrm{Hom}_{\mathscr{O}_{Y'_0}}(i_0'^{*} g_0^{*}(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{Y'}) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^{*} g_0^{*}(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{T}). \end{aligned}$$

(ii) *There is also a natural morphism, functorial in T,* [100]

$$\begin{aligned} a_{g_0}(f_0):\ &\mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^{*}(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_{T}) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^{*} g_0^{*}(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{T}). \end{aligned}$$

Proof. [101] First note that, with X, X', Y, Y' fixed, giving a T as above is equivalent to giving a morphism $(f, f_{\mathscr{J}}) : T \to X' \times_{X'^{+}} Y'^{+}$. Put $Z = X' \times_{X'^{+}} Y'^{+}$, and let M and M' be the Z-functors defined, for every $f : T \to Z$, by

$$\begin{aligned} M(T) &= \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^{*} g_0^{*}(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{T}), \\ M'(T) &= \mathrm{Hom}_{\mathscr{O}_{T_0}}(f_0^{*}(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_{T}). \end{aligned}$$

[102]

In all cases there is a commutative diagram

$$\begin{tikzcd} f_0^{*}(\mathcal{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y'_0}) \arrow[r, equal] \arrow[d] & \mathcal{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0} \arrow[d] \\ f_0^{*}(\mathcal{J}\mathscr{O}_{Y'}) \arrow[r, dashed] & \mathcal{J}\mathscr{O}_{T} \end{tikzcd}$$

Since Y' is flat over S, the left vertical arrow is an isomorphism, and hence we obtain a morphism of $\mathscr{O}_{T_0}$-modules

$$f_0^{*}(\mathscr{J}\mathscr{O}_{Y'}) \longrightarrow \mathscr{J}\mathscr{O}_{T}.$$

[99]Editor's note: Starting with 4.17, this will be applied when X is an S-group, $g : X \times_S X \to X$ is multiplication, Y is a subscheme of X such that $Y_{\mathscr{J}}$ is a subgroup of $X_{\mathscr{J}}$, $Y' = Y \times_S Y$, and the two morphisms $Y^3 \to X^2$ send (y_1, y_2, y_3) respectively to $(y_1 y_2, y_3)$ and $(y_1, y_2 y_3)$. In that case, comparison of the obstructions above will show that the obstruction to Y being a subgroup of X lies in a certain Hochschild cohomology group $H^2(Y_0, N_0)$.

[100]Editor's note: We have removed the hypothesis that Y'_0 be locally complete intersection in X'_0, which is superfluous by 4.8.0. We have also added that $a_{g_0}(f_0)$ is "functorial in T", a fact that plays a crucial role in the proof of 4.17.

[101]Editor's note: The proof has been expanded to make the "functoriality in T" of a_{g_0} explicit.

[102]Editor's note: The situation simplifies from 4.16 onward: we shall restrict to schemes flat over S, Y will be a flat S-group, and $Y' = Y \times_S Y$; this will give S_0-functors N_0 and N'_0.

It induces a morphism of abelian groups

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), f_0^*(\mathscr{J}\mathscr{O}_{Y'})\big) \longrightarrow M(T),$$

and, on composing with the morphism

$$M(Y') \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), f_0^*(\mathscr{J}\mathscr{O}_{Y'})\big)$$

induced by f_0^*, we obtain the morphism $b_{f_0} : M(Y') \to M(T)$.

Likewise, in all cases there is a diagram

$$\begin{tikzcd}
g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \arrow[r, dashed] \arrow[d] & \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0} \arrow[d] \\
g_0^*(\mathscr{N}_{Y_0/X_0}) \arrow[r] & \mathscr{N}_{Y'_0/X'_0}
\end{tikzcd}$$

Since Y' is flat over S, the second vertical arrow is an isomorphism by 4.8.0. We therefore obtain an $\mathscr{O}_{Y'_0}$-morphism

$$i_0'^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \longrightarrow \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}.$$

This induces a morphism $a_{g_0}(\mathrm{id}_{Y'}) : M'(Y') \to M(Y')$ and, for every $f : T \to Z$, a morphism $a_{g_0}(f) : M'(T) \to M(T)$, such that the following diagram commutes:

$$\begin{tikzcd}
M'(Y') \arrow[r, "a_{g_0}(\mathrm{id}_{Y'_0})"] \arrow[d, "b'_{f_0}"'] & M(Y') \arrow[d, "b_{f_0}"] \\
M'(T) \arrow[r, "a_{g_0}(f_0)"] & M(T)
\end{tikzcd}$$

Here b'_{f_0} is defined in the same way as b_{f_0}. Q.E.D.

Remark 4.12.1. [103] *Let M_0 and M'_0 be the Y'_0-functors defined, for every $f : T_0 \to Y'_0$, by*

$$M_0(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big),$$
$$M'_0(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big).$$

Note at once that $Z_0 = Y'_0$, and that on the category of Z-schemes T flat over S, the functors M and M' coincide respectively with $\prod_{S_0/S} M_0$ and $\prod_{S_0/S} M'_0$. In that case, b_{f_0} is simply the morphism

$$f_0^* : M_0(Y'_0) \longrightarrow M(T_0)$$

induced by f_0. Moreover, for every morphism $u : U \to T$, if $h = f \circ u$, the following diagram commutes:

[103] Editor's note: This remark has been added for use in the proof of 4.17.

$$\begin{tikzcd}
\mathrm{M}'_0(\mathrm{T}_0) \arrow[r, "a_{g_0}(f_0)"] \arrow[d, "u_0^*"'] & \mathrm{M}_0(\mathrm{T}_0) \arrow[d, "u_0^*"] \\
\mathrm{M}'_0(\mathrm{U}_0) \arrow[r, "a_{g_0}(h_0)"] & \mathrm{M}_0(\mathrm{U}_0)
\end{tikzcd}$$

Thus a_{g_0} becomes a morphism of functors

$$\prod_{S_0/S} M'_0 \longrightarrow \prod_{S_0/S} M_0.$$

Proposition 4.13. *Suppose that Y' is flat over S. Then*

$$c(X, Y, g \circ f) = a_{g_0}(c(X', Y', f)) + b_{f_0}(c(X, Y, g \circ i')).$$

Since the definitions of the various obstructions and of the morphisms a_{g_0} and b_{f_0} are local, it plainly suffices to verify the formula when all the schemes in question are affine. Write

$$S = \operatorname{Spec}(\Lambda), \qquad S_{\mathscr{J}} = \operatorname{Spec}(\Lambda/\mathscr{J}), \qquad S_0 = \operatorname{Spec}(\Lambda/\mathscr{I}),$$
$$T = \operatorname{Spec}(C), \qquad X = \operatorname{Spec}(A), \qquad Y = \operatorname{Spec}(A/I_Y) = \operatorname{Spec}(B),$$
$$X' = \operatorname{Spec}(A'), \qquad Y' = \operatorname{Spec}(A'/I_{Y'}) = \operatorname{Spec}(B').$$

There is therefore a diagram of rings and ideals [104]

$$\begin{tikzcd}
 & \mathrm{B}' & \mathrm{B} \\
\mathrm{C} & \mathrm{A}' \arrow[l, "f"'] \arrow[u, "\pi'"] & \mathrm{A} \arrow[l, "g"'] \arrow[u, "\pi"] \\
 & \mathrm{I}_{\mathrm{Y}'} \arrow[u, hook] & \mathrm{I}_{\mathrm{Y}} \arrow[u, hook]
\end{tikzcd}$$

Let us study the terms of the formula to be proved. In what follows, for $x \in I_Y$ (respectively $u \in I_{Y'}$), write $\bar{x}$ (respectively $\bar{u}$) for its image in I_Y/I_Y^2 (respectively $I_{Y'}/I_{Y'}^2$). If m belongs to a Λ-module M, write $\bar{m}_0$ for its image in $M_0 = M/\mathscr{I}M$.

We saw that $c = c(X, Y, g \circ f)$ is the unique C_0-morphism

$$I_Y/I_Y^2 \otimes_B C_0 \longrightarrow \mathscr{J}C$$

[104] Editor's note: We retain the notation of the original, writing $f : A' \to C$ and $g : A \to A'$ for the ring homomorphisms corresponding to $f : T \to X'$ and $g : X' \to X$. This explains the formula $c(X, Y, g \circ f)(\bar{x} \otimes 1) = f(g(x))$, for $x \in I_Y$.

such that $c(\bar{x} \otimes 1) = f(g(x))$ for every $x \in I_Y$.

Fix $x \in I_Y$. Since $g_{\mathscr{J}}(Y'_{\mathscr{J}}) \subset Y_{\mathscr{J}}$, one has $g(x) \in I_{Y'} + \mathscr{J} A'$. Write

$$g(x) = x' + \sum_i \lambda_i a'_i, \qquad x' \in I_{Y'}, \quad \lambda_i \in \mathscr{J}, \quad a'_i \in A'.$$

Then

$$c(X, Y, g \circ f)(\bar{x} \otimes 1) = f(g(x)) = f(x') + \sum_i \lambda_i f(a'_i). \tag{1}$$

Consider $a_{g_0}(c(X', Y', f))$. By the definitions just given, it is defined by the following diagram:

$$\begin{tikzcd}
 & \mathrm{I}_{\mathrm{Y}'} \arrow[r, "f"] \arrow[d] & \mathrm{C} \\
\mathrm{I}_{\mathrm{Y}'_0}/\mathrm{I}^2_{\mathrm{Y}'_0} \otimes_{\mathrm{B}'_0} \mathrm{C}_0 & \mathrm{I}_{\mathrm{Y}'}/\mathrm{I}^2_{\mathrm{Y}'} \otimes_{\mathrm{B}'} \mathrm{C}_0 \arrow[l, "\sim"'] \arrow[r, "c(\mathrm{X}',\mathrm{Y}',f)"] & \mathrm{JC} \arrow[u, hook] \\
\mathrm{I}_{\mathrm{Y}_0}/\mathrm{I}^2_{\mathrm{Y}_0} \otimes_{\mathrm{B}_0} \mathrm{C}_0 \arrow[u, "\overline{g}_0"] & \mathrm{I}_{\mathrm{Y}}/\mathrm{I}^2_{\mathrm{Y}} \otimes_{\mathrm{B}} \mathrm{C}_0 \arrow[l, two heads] \arrow[u] \arrow[ur, "a_{g_0}(c(\mathrm{X}',\mathrm{Y}',f))"'] &
\end{tikzcd}.$$

Thus $a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(u)$, where $u \in I_{Y'}$ has image $\bar{u}$ in $I_{Y'_0}/I^2_{Y'_0}$ satisfying

$$\bar{u}_0 \otimes 1 = \bar{g}_0(\bar{x}_0) \otimes 1 = g_0(\bar{x}_0) \otimes 1.$$

We may therefore take $u = x'$, which gives

$$a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(x'). \tag{2}$$

Finally consider $b_{f_0}(c(X, Y, g \circ i'))$. By hypothesis, the morphism of Λ_0-algebras $f_0 : A'_0 \to C_0$ factors through B'_0. Since Y' is flat, and hence

$$\mathscr{J} \otimes_{\Lambda_0} B'_0 \xrightarrow{\sim} \mathscr{J} B',$$

we obtain a morphism of B'_0-modules $\psi : \mathscr{J} B' \to \mathscr{J} C$ making the following diagram commute:

$$\begin{tikzcd}
\mathrm{J} \otimes_{\Lambda_0} \mathrm{A}'_0 \arrow[r, "\mathrm{id} \otimes \pi'"] \arrow[d] & \mathrm{J} \otimes_{\Lambda_0} \mathrm{B}'_0 \arrow[r, "\mathrm{id} \otimes f_0"] \arrow[d, "\wr"] & \mathrm{J} \otimes_{\Lambda_0} \mathrm{C}_0 \arrow[d] \\
\mathrm{JA}' \arrow[r, "\pi'"] & \mathrm{JB}' \arrow[r, dashed, "\psi"] & \mathrm{JC}.
\end{tikzcd}$$

Let

$$\varphi : \mathscr{J} B' \otimes_{B'_0} C_0 \longrightarrow \mathscr{J} C$$

be the morphism of C_0-modules induced by ψ. Then, for every $a' \in A'$ and $\lambda \in \mathscr{J}$,

$$\varphi(\lambda\pi'(a') \otimes 1) = \lambda f(a'). \tag{†}$$

The element $b_{f_0}(c(X, Y, g \circ i'))$ is then defined by the commutative diagram

$$\begin{tikzcd}
\mathrm{I_Y} \arrow[rr, "\pi' \circ g"] \arrow[d] & & \mathrm{B'} \\
\mathrm{I_Y/I_Y^2 \otimes_B B'_0} \arrow[rr, "c(\mathrm{X},\mathrm{Y},g\circ i')"] \arrow[d] & & \mathrm{JB'} \arrow[u, hook] \arrow[d] \\
\mathrm{I_Y/I_Y^2 \otimes_B C_0} \arrow[rr] \arrow[drr, "b_{f_0}(c(\mathrm{X},\mathrm{Y},g\circ i'))"'] & & \mathrm{JB' \otimes_{B'_0} C_0} \arrow[d, "\phi"] \\
& & \mathrm{JC.}
\end{tikzcd}$$

Consequently,

$$b_{f_0}(c(X, Y, g \circ i'))(\bar{x} \otimes 1) = \varphi\left(\sum_i \lambda_i \pi'(a'_i) \otimes 1\right) = \sum_i \lambda_i f(a'_i), \tag{3}$$

the last equality following from (†). Comparison of the three explicit formulas (1), (2), and (3) gives the required formula.

Corollary 4.14. *Let Y, Y' be two flat subschemes of X, both reducing to $Y_{\mathscr{J}}$, and suppose that Y_0 is locally complete intersection in X_0. If $f : T \to X$ is an S-morphism such that $f_{\mathscr{J}}$ factors through $Y_{\mathscr{J}} \to X_{\mathscr{J}}$, then*

$$c(X, Y, f) - c(X, Y', f) = b_{f_0}(d(Y, Y')).$$

Indeed, applying the preceding formula to the diagram

$$\begin{tikzcd}
& \mathrm{Y'} \arrow[d, hook, "i'"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\
\mathrm{T} \arrow[r, "f"] & \mathrm{X} \arrow[r, "\mathrm{id}"] & \mathrm{X}
\end{tikzcd}$$

we find $c(X, Y, f) - c(X, Y', f) = b_{f_0}(c(X, Y, i'))$. Moreover, by 4.8(ii), $c(X, Y, i') = d(Y, Y')$.

Proposition 4.15. *Let X be an S-group smooth over S, and let Y be a sub-S-group flat and locally of finite presentation over S. Then Y is locally complete intersection in X (cf. 4.6.2).*

Proof. [105] We show that the immersion $Y \to X$ is regular in the sense of EGA IV_4, 16.9.2. By EGA IV_4, 19.5.1, this implies that it is regular also in the sense of 4.6.2 (indeed, by loc. cit., the two definitions are equivalent when S is locally noetherian). Thus "regular immersion" will here mean regular in the sense of EGA IV_4, 16.9.2. Since X and Y are flat and locally of finite presentation over S, EGA IV_4, 19.2.4 reduces the assertion to proving that $Y_s \to X_s$ is a regular immersion for every $s \in S$. By EGA IV_4, 19.1.5(ii), it is enough to check this on the geometric fibers of S, and hence when S is the spectrum of an algebraically closed field k.

By Exposé $\mathrm{VI_A}$, 3.2, the quotient X/Y exists and is smooth, the morphism $\pi : X \to X/Y$ is flat, and the following square is cartesian:

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[d] & X \arrow[d, "\pi"] \\ \bar{e} \arrow[r, "i"] & X/Y \end{tikzcd}$$

Here $\bar{e}$ is the image in X/Y of the unit point of X. By flat base change (cf. EGA IV_4, 19.1.5(iii)), it therefore suffices to show that i is a regular immersion. This is immediate: the noetherian local ring $\mathscr{O}_{X/Y,\bar{e}}$ is smooth (hence regular), so its maximal ideal is generated by a regular sequence.

4.16. [106] Let X be an S-group smooth over S, and denote its group law by $\mu : X \times_S X \to X$. Let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. By 4.15, $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$. Hence, by 4.6.5, every flat [107] S-scheme Y lifting

[105] Editor's note: We have added to the statement the hypothesis that Y be locally of finite presentation over S, and supplied the proof that follows, which is more direct than the one sketched in the original. For completeness, we also spell out the latter. As in the proof above, first reduce to the case $S = \mathrm{Spec}(k)$, where k is algebraically closed. By EGA IV_4, 16.9.10 and 19.3.2, it suffices to see that, for every $y \in Y$, the completion of the local ring $\mathscr{O}_{Y,y}$ is a quotient of a complete noetherian local ring by a regular sequence. By loc. cit., 19.3.3, the set of $y \in Y$ with this property is an open subset $U \subset Y$. Since Y is of finite type over k, it suffices to show that U contains every closed point. Because Y is a k-group, a translation argument reduces this to the completion of $\mathscr{O}_{Y,e}$, that is, to the formal group $\widehat{Y}$ corresponding to Y (cf. Exposé $\mathrm{VII_B}$). Since X is smooth, the affine algebra $\mathscr{A}(\widehat{X})$ is a power series algebra $k[[X_1, \ldots, X_n]]$. The Dieudonné structure theorem then shows that $\mathscr{A}(\widehat{Y})$ is isomorphic to a quotient

$$k[[X_1, \ldots, X_{r+s}]]/(X_1^{p^{n_1}}, \ldots, X_r^{p^{n_r}}),$$

where p is the characteristic exponent of k and $r + s \leq n$; cf. Exposé $\mathrm{VII_B}$, Remark 5.5.2(b).

[106] Editor's note: We have reorganized 4.16 by collecting here, on the one hand, the hypotheses stated at the end of 4.15 and, on the other, the definition of the obstruction DY.

[107] Editor's note: The original has been corrected by adding the word "flat".

$Y_{\mathscr{J}}$ is locally complete intersection in X. For such a Y, 4.8.0 gives

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} = \mathscr{N}_{Y_0/X_0} = \mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}. \tag{4.16.1}$$

On the other hand, let $\varepsilon_0 : S_0 \to Y_0$ be the unit section of Y_0, and let $\mathbf{n}_{Y_0/X_0}$ be the quasi-coherent $\mathscr{O}_{S_0}$-module

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathscr{N}_{Y_0/X_0}).$$

Since Y_0 and X_0 are S_0-groups, it is immediate that $\mathscr{N}_{Y_0/X_0}$ is invariant under, say, left translations by Y_0. Hence [108] it is the inverse image of $\mathbf{n}_{Y_0/X_0}$ under $Y_0 \to S_0$; that is,

$$\mathscr{N}_{Y_0/X_0} = \mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}. \tag{4.16.2}$$

Taking (4.16.1) and (4.16.2) into account, we deduce first from 4.5 that the set of subschemes Y of X, flat over S and lifting $Y_{\mathscr{J}}$, is either empty or principal homogeneous under

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}). \tag{4.16.3}$$

On the other hand, 4.8(i) shows that, for every such Y and every S-morphism $f : T \to X$ such that $f_{\mathscr{J}} : T_{\mathscr{J}} \to X_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, the obstruction $c(X, Y, f)$ to f factoring through Y belongs to

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J}\mathscr{O}_T).$$

If T is also flat over S, this last group is equal to

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}).$$

This leads us to introduce the following group functor N_0.

Definition 4.16.1. *Let N_0 be the S_0-functor in commutative groups defined, for every $Z \in \mathrm{Ob}(\mathbf{Sch})_{/S_0}$, by*

$$\mathrm{Hom}_{S_0}(Z, N_0) = \mathrm{Hom}_{\mathscr{O}_Z}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_Z, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_Z). \tag{*}$$

Then the set of subschemes Y of X, flat over S and lifting $Y_{\mathscr{J}}$, is either empty or principal homogeneous under

$$\mathrm{Hom}_{S_0}(Y_0, N_0) = C^1(Y_0, N_0).$$

For every such Y, consider the diagram

$$\begin{tikzcd}
Y \times_S Y \arrow[d, hook, "{(i,i)}"'] & & Y \arrow[d, hook, "i"'] \\
X \times_S X \arrow[rr, "\mu"] & & X
\end{tikzcd}$$

[108] Editor's note: See 4.25 below.

and let

$$DY = c(X, Y, \mu \circ (i, i))$$

be the obstruction to $\mu \circ (i, i)$ factoring through Y, that is, to Y being stable under the group law of X. By what precedes, DY is an element of

$$N_0(Y_0 \times_{S_0} Y_0) = C^2(Y_0, N_0).$$

Lemma 4.17.[109] *Let X be an S-group smooth over S, and let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. For every subscheme Y of X, flat over S and lifting $Y_{\mathscr{J}}$, consider the obstruction defined in 4.16.1:*

$$DY \in \operatorname{Hom}_{S_0}(Y_0 \times_{S_0} Y_0, N_0) = C^2(Y_0, N_0).$$

(i) *For Y to be a sub-S-group of X, it is necessary and sufficient that $DY = 0$.*

(ii) *If Y_0 is made to act on N_0, by functoriality, through the inner automorphisms of Y_0, then $DY \in Z^2(Y_0, N_0)$.*

(iii) *If Y and Y' are two subschemes of X, flat over S, lifting $Y_{\mathscr{J}}$ (so that the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ is defined; cf. 4.5.1), then*

$$DY' - DY = \partial^1 d(Y, Y').$$

[110]

We prove these assertions in succession.

4.18. — Proof of 4.17(i). By definition, $DY = 0$ if and only if Y is stable under the group law of X. Thus $DY = 0$ if Y is a subgroup of X. Conversely, if $DY = 0$, then Y is endowed with the induced law μ_Y, which is associative and reduces modulo $\mathscr{J}$ to the group law on $X_{\mathscr{J}}$. Since Y is flat and locally of finite presentation over S, it follows from 3.3 that μ_Y is a group law.

4.19. — Proof of 4.17(ii). We compare the two values of $u = c(X, Y, \mu^2 \circ (i, i, i))$ computed in the following two diagrams (D_j), $j = 1, 2$:

$$(\mathrm{D}_j) \qquad \begin{tikzcd} Y \times_S Y \times_S Y \arrow[d, "{(i,i,i)}"'] & Y \times_S Y \arrow[d, hook, "{(i,i)}"'] & Y \arrow[d, hook, "i"'] \\ X \times_S X \times_S X \arrow[r, "f_j"] & X \times_S X \arrow[r, "\mu"] & X \end{tikzcd}$$

Here $f_1 = (1, \pi)$, $f_2 = (\pi, 1)$, and μ^2 denotes the morphism

$$\mu \circ f_1 = \mu \circ f_2 : X \times_S X \times_S X \longrightarrow X.$$

[111] Put

$$\mu_Y = \mu \circ (i, i), \qquad f_j^Y = f_j \circ (i, i, i), \qquad \mu^{2,Y} = \mu^2 \circ (i, i, i).$$

[109] Editor's note: Sections 4.17 and 4.18 have been modified to take account of the additions made in 4.16.

[110] Editor's note: The original has $DY' - DY = -\partial d(Y, Y')$, but the operator ∂ there is the negative of the differential ∂^1 defined in I, 5.1.

[111] Editor's note: The notation has been slightly modified, and the beginning of the argument has been expanded.

For $j = 1, 2$, let a_j and b_j be the morphisms

$$a_j = a_{\mu_0}((f_j^Y)_0), \qquad b_j = b_{(f_j^Y)_0},$$

associated by Lemma 4.12 with the pair of morphisms (f_j^Y, μ). Thus

$$\begin{cases} a_j : \operatorname{Hom}_{\mathscr{O}_{Y_0^3}}\left((f_j^Y)_0^*\mathscr{N}_{Y_0\times Y_0/X_0\times X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\right) \longrightarrow \\ \qquad\qquad \operatorname{Hom}_{\mathscr{O}_{Y_0^3}}\left((\mu^{2,Y})_0^*\mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\right), \\ b_j : \operatorname{Hom}_{\mathscr{O}_{Y_0^2}}\left((\mu_Y)_0^*\mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^2}\right) \longrightarrow \\ \qquad\qquad \operatorname{Hom}_{\mathscr{O}_{Y_0^3}}\left((\mu^{2,Y})_0^*\mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\right). \end{cases} \tag{$\dagger$}$$

Since

$$\mathscr{N}_{Y_0\times Y_0/X_0\times X_0} \simeq \operatorname{pr}_1^*\mathscr{N}_{Y_0/X_0} \oplus \operatorname{pr}_2^*\mathscr{N}_{Y_0/X_0}$$

(because X_0 and Y_0 are flat over S_0), and

$$\mathscr{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0},$$

we have

$$(f_j^Y)_0^*\mathscr{N}_{Y_0\times Y_0/X_0\times X_0} \simeq (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathscr{O}_{Y_0^3},$$

and likewise

$$\begin{aligned} (\mu^{2,Y})_0^*\mathscr{N}_{Y_0/X_0} &\simeq \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^3}, \\ (\mu_Y)_0^*\mathscr{N}_{Y_0/X_0} &\simeq \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^2}. \end{aligned}$$

Moreover, since Y_0^2 and Y_0^3 are flat over S_0, ($\dagger$) may be rewritten as

$$\begin{cases} a_j : \operatorname{Hom}_{S_0}(Y_0^3, N_0 \oplus N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0), \\ b_j : \operatorname{Hom}_{S_0}(Y_0^2, N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0). \end{cases} \tag{$\ddagger$}$$

Applying 4.13 twice to $c(X, Y, \mu^{2,Y}) = u$, we obtain

$$\begin{aligned} a_1\big(c(X^2, Y^2, f_1)\big) + b_1\big(c(X, Y, \mu_Y)\big) &= u \\ &= a_2\big(c(X^2, Y^2, f_2)\big) + b_2\big(c(X, Y, \mu_Y)\big). \end{aligned}$$

Now $c(X, Y, \mu_Y) = DY$, and, since $f_1 = (1, \mu)$ and $f_2 = (\mu, 1)$, we have, with the evident notation,

$$c(X^2, Y^2, f_1) = (0, DY), \qquad c(X^2, Y^2, f_2) = (DY, 0).$$

Consequently,

$$u = a_1((0, DY)) + b_1(DY) = a_2((DY, 0)) + b_2(DY).$$

The first point to note is that b_j is simply $\operatorname{Hom}_{S_0}((f_j^Y)_0, N_0)$, that is, the morphism deduced from $(f_j^Y)_0$ by functoriality. The preceding identity therefore becomes

$$\begin{aligned} a_1((0, DY)) - \operatorname{Hom}((\mu, 1), N_0)(DY) + \operatorname{Hom}((1, \mu), N_0)(DY) \\ - a_2((DY, 0)) = 0. \end{aligned}$$

The two middle terms are the $DY(xy, z)$ and $DY(x, yz)$ parts of the formula for the 2-coboundary. It remains only to identify the other two terms.

We first compute the map a_j. It is obtained, by pullback along $(f_j^Y)_0$, from the morphism of $\mathscr{O}_{Y_0^2}$-modules

$$P : \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^2} \longrightarrow (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathscr{O}_{Y_0^2}$$

induced by multiplication in Y_0. Consider the vector bundle $V = \mathbf{V}(\mathbf{n}_{Y_0/X_0})$. By duality, P gives

$$\mathbf{V}(P) : V \times_{S_0} V \times_{S_0} Y_0 \times_{S_0} Y_0 \longrightarrow V \times_{S_0} Y_0 \times_{S_0} Y_0,$$

which is expressed set-theoretically by

$$\mathbf{V}(P)(u, v, a, b) = (u + \mathrm{Ad}(a)v, ab, b).$$

[112] This is proved exactly as is the corresponding fact for Lie algebras, that is, for the module $\omega^1_{Y_0/S_0}$. First observe that, by functoriality in Y_0, V is endowed with a group structure in the category of vector bundles over S_0. By the lemma already used for Lie algebras (Exposé II, 3.10), this structure coincides with the group structure underlying its $\mathscr{O}_S$-module structure. Next,

$$\mathbf{V}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}) = \mathbf{V}(\mathscr{N}_{Y_0/X_0})$$

is itself endowed with an S_0-group structure, namely the semidirect product of the structure on V by that on Y_0. It remains only to identify the operations of Y_0 on V to obtain the desired formula.

We now compute the two remaining terms. Consider first $a_1((0, DY))$. It is computed by the following diagram, in which $\mathbf{n}$ denotes $\mathbf{n}_{Y_0/X_0}$:

$$\begin{tikzcd}
\mathfrak{n} \otimes \mathscr{O}_{\mathrm{Y}_0^2} \arrow[r, "\mathrm{P}"] \arrow[d, "(f_1^{\mathrm{Y}})_0^*"] & (\mathfrak{n} + \mathfrak{n}) \otimes \mathscr{O}_{\mathrm{Y}_0^2} \arrow[d, "(f_1^{\mathrm{Y}})_0^*"] \\
\mathfrak{n} \otimes \mathscr{O}_{\mathrm{Y}_0^3} \arrow[dr, "a_1((0,\mathrm{DY}))"'] & (\mathfrak{n} + \mathfrak{n}) \otimes \mathscr{O}_{\mathrm{Y}_0^3} \arrow[d, "(0,\mathrm{DY})"] \\
 & \mathcal{J} \otimes \mathscr{O}_{\mathrm{Y}_0^3}
\end{tikzcd}.$$

Regard the vector bundles defined by these modules as schemes over S_0, and take points with values in an arbitrary scheme. If (u, x, y, z) denotes a point of $\mathbf{V}(\mathscr{J}) \times Y_0^3$, we obtain the following diagram:

[112]Editor's note: The entries a, b have been replaced by ab, b to show that $\mathbf{V}(P)$ is obtained, by pullback to Y_0^2, from the multiplication morphism $V_{Y_0} \times_{S_0} V_{Y_0} \to V_{Y_0}$.

$$\begin{tikzcd}
(\mathrm{Ad}(x)\mathrm{DY}_{y,z}(u), x, yz) & (0+\mathrm{DY}_{y,z}(u), x, yz) \arrow[l, mapsto] \\
& (0+\mathrm{DY}_{y,z}(u), x, y, z) \arrow[u, mapsto] \\
(\mathrm{Ad}(x)\mathrm{DY}_{y,z}(u), x, y, z) \arrow[uu, mapsto] & (u, x, y, z) \arrow[l, mapsto] \arrow[u, mapsto]
\end{tikzcd} .$$

Thus
$$a_1((0, DY))(x, y, z) = \mathrm{Ad}(x)DY(y, z),$$
which is indeed the first term of the coboundary. Likewise, $a_2((DY, 0))(x, y, z) = DY(x, y)$, and therefore
$$\begin{aligned} 0 =&\ \mathrm{Ad}(x)DY(y, z) - DY(xy, z) + DY(x, yz) - DY(x, y) \\ =&\ (\partial^2 DY)(x, y, z). \end{aligned}$$

[113]

4.20. — Proof of 4.17(iii). [114] We compare the two values of $v = c(X, Y, \mu \circ (i', i'))$ computed in the following two diagrams:

$$(*) \qquad \begin{tikzcd}
& \mathrm{Y}' \arrow[d, hook, "i'"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\
\mathrm{Y}' \times_{\mathrm{S}} \mathrm{Y}' \arrow[r, "\mu\circ(i',i')"] & \mathrm{X} \arrow[r, equal] & \mathrm{X}
\end{tikzcd}$$

$$(\dagger) \qquad \begin{tikzcd}
& \mathrm{Y} \times_{\mathrm{S}} \mathrm{Y} \arrow[d, hook, "(i,i)"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\
\mathrm{Y}' \times_{\mathrm{S}} \mathrm{Y}' \arrow[r, "(i',i')"] & \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[r, "\mu"] & \mathrm{X}.
\end{tikzcd}$$

Let $f = \mu \circ (i', i')$. Diagram $(*)$ gives
$$v = DY' + f_0^*(c(X, Y, i')). \tag{1}$$
Now $Y_0' = Y_0$, and f_0 is the multiplication $Y_0^2 \to Y_0$; hence
$$f_0^*(c(X, Y, i'))(x_0, y_0) = c(X, Y, i')(x_0 y_0). \tag{2}$$
Put $c = c(X, Y, i')$. Under the identification $N_0' \simeq N_0 \oplus N_0$, the element
$$c(X \times_S X, Y \times_S Y, (i', i'))$$

[113]Editor's note: The signs have been changed to make them compatible with I, 5.1.
[114]Editor's note: The argument that follows has been expanded from the original.

identifies with the pair (c, c). Thus, putting $h = (i', i')$, diagram (†) gives

$$v = h_0^*(DY) + a_{\mu_0}(c, c). \tag{3}$$

Since h_0 is the identity map on Y_0^2, we have $h_0^*(DY) = DY$. Finally, by the preceding calculation of a_{μ_0}, for every $S' \to S$ and every $x_0, y_0 \in Y_0(S_0')$,

$$a_{\mu_0}(c, c)(x_0, y_0) = c(x_0) + \mathrm{Ad}(x_0)(c(y_0)). \tag{4}$$

It follows that

$$\begin{aligned}(DY' - DY)(x_0, y_0) &= \mathrm{Ad}(x_0)(c(X, Y, i')(y_0)) - c(X, Y, i')(x_0 y_0) + c(X, Y, i')(x_0) \\ &= (\partial^1 c(X, Y, i'))(x_0, y_0).\end{aligned} \tag{5}$$

Since $c(X, Y, i') = d(Y, Y')$ (cf. 4.8(iii)), this proves that

$$DY' - DY = \partial^1 d(Y, Y').$$

Theorem 4.21. *Let S be a scheme and let $\mathscr{I}$ and $\mathscr{J}$ be two ideals*[115] *on S, such that $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let X be an S-group smooth over S, and let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. Consider the S_0-functor of commutative groups N_0 defined by*

$$\mathrm{Hom}_{S_0}(T, N_0) = \mathrm{Hom}_{\mathscr{O}_T}\left(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right), \qquad T \in \mathrm{Ob}(\mathbf{Sch}/S_0),$$

on which Y_0 acts through the inner automorphisms of X_0.

(i) *For there to exist a sub-S-group of X, flat over S, which reduces to $Y_{\mathscr{J}}$, it is necessary and sufficient that the following two conditions hold:*

  (i_1) *There exists a subscheme Y of X, flat over S, lifting $Y_{\mathscr{J}}$. This condition is automatically satisfied when Y_0 is affine (cf. 4.6.5).*

  (i_2) *A certain canonical obstruction, an element of $H^2(Y_0, N_0)$, vanishes.*

(ii) *If the conditions in (i) hold, the set of sub-S-groups Y of X, flat over S and reducing to $Y_{\mathscr{J}}$, is a principal homogeneous set under $Z^1(Y_0, N_0)$.*[116]

Indeed, condition (i_1) is necessary. Suppose that it holds, and let Y, flat over S, lift $Y_{\mathscr{J}}$. We must find a Y', also flat over S and lifting $Y_{\mathscr{J}}$, such that $DY' = 0$,[117] cf. 4.17(i). By 4.17(iii), this amounts to finding an element $d(Y', Y) \in C^1(Y_0, N_0)$ such that

$$DY = \partial^1 d(Y', Y).$$

[118] Let $c \in H^2(Y_0, N_0)$ be the class represented by DY, which is a cocycle by 4.17(ii). By 4.17(iii), this class does not depend on the choice of Y, and its vanishing is necessary and sufficient for the existence of $d(Y', Y)$ satisfying the preceding equation. This proves (i).

If Y has now been chosen so that $DY = 0$, the equation to be solved becomes $\partial^1 d(Y', Y) = 0$, which proves (ii).

[115] Editor's note: The hypothesis that $\mathscr{I}$ be nilpotent, which appears superfluous, has been removed.

[116] Editor's note: The question whether this set, modulo conjugation by the elements $x \in X(S)$ inducing the identity element of $X(S_{\mathscr{J}})$, is a principal homogeneous set under $H^1(Y_0, N_0)$, occupies 4.23–4.36.

[117] Editor's note: The expression $\partial DY'$ in the original has been corrected to DY'.

[118] Editor's note: See the editor's note to 4.17(iii).

Remark 4.22. Retain the notation of 4.21. By 4.15, Y_0 is locally a complete intersection in X_0; hence $\mathscr{N}_{Y_0/X_0}$ is a finite locally free $\mathscr{O}_{Y_0}$-module, and therefore

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathscr{N}_{Y_0/X_0})$$

is a finite locally free $\mathscr{O}_{S_0}$-module. Thus, writing

$$\mathbf{n}^\vee_{Y_0/X_0} = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{O}_{Y_0}),$$

we have

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_T}\left(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right)\\ &\simeq \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\end{aligned}$$

for every $T \to S_0$.[119] Consequently, the S_0-functor N_0 is isomorphic to

$$\mathbf{W}\left(\mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\right) \simeq \mathbf{W}\left(\mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\right).$$

It follows that there are isomorphisms[120]

$$\begin{aligned}H^2(Y_0, N_0) &\simeq H^2\Big(Y_0, \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\Big)\\ &\simeq H^2\Big(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),\\ Z^1(Y_0, N_0) &\simeq Z^1\Big(Y_0, \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\Big)\\ &\simeq Z^1\Big(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).\end{aligned}$$

4.23. — Still under the hypotheses of 4.21, we shall now study how the set of subgroups Y lifting $Y_{\mathscr{J}}$ behaves under conjugation by sections of X. If x is a section of X over S inducing the unit section of $X_{\mathscr{J}}$, the inner automorphism $\mathrm{Int}(x)$ defined by x sends flat subgroups of X lifting $Y_{\mathscr{J}}$ to flat subgroups of X lifting $Y_{\mathscr{J}}$. Under the conditions of 4.21(ii), the set of these subgroups is a principal homogeneous space under $Z^1(Y_0, N_0)$. We shall see that there is then a subgroup Γ of $B^1(Y_0, N_0)$[121] such that two subgroups of X, flat over S and lifting $Y_{\mathscr{J}}$, are conjugate by an $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$ if and only if their *difference* in $Z^1(Y_0, N_0)$ belongs to Γ. In the best cases we shall show that $\Gamma = B^1(Y_0, N_0)$. Thus the set of flat subgroups of X lifting $Y_{\mathscr{J}}$, modulo conjugation by the elements $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$, is either empty or a principal homogeneous space under $H^1(Y_0, N_0)$ (cf. 4.29 and 4.36).

4.24. — We retain the notation of 4.21. Let Y be a flat subgroup of X whose reduction is $Y_{\mathscr{J}}$. Recall that in 0.5 we introduced the functor L_0^X (respectively L_0^Y) defined, for a variable S_0-scheme T, by the identity

$$\mathrm{Hom}_{S_0}(T, L_0^X) = \mathrm{Hom}_{\mathscr{O}_T}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right),$$

[119]Editor's note: The preceding argument has been expanded. It shows that the following isomorphism is valid without a flatness hypothesis. On the other hand, since 4.16 the discussion has been restricted to S-schemes $f : T \to S$ flat over S, in order to ensure that the group

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\left(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J}\mathscr{O}_T\right),$$

in which the obstruction $c(X, Y, f)$ lies, coincides with $N_0(T_0)$; see the end of 4.16.

[120]Editor's note: With the notation of I, 5.3, and assuming that Y_0 is affine over S_0.

[121]Editor's note: $Z^1(Y_0, N_0)$ has been replaced by $B^1(Y_0, N_0)$, since the proof shows that Γ is a subgroup of $B^1(Y_0, N_0)$; cf. 4.27–4.29.

and similarly with Y in place of X, as well as the functor

$$L'_X = \prod_{S_0/S} L_0^X.$$

We then have:

Lemma 4.25. *There is a canonical exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules*

$$\mathbf{n}_{Y_0/X_0} \xrightarrow{d} \omega^1_{X_0/S_0} \longrightarrow \omega^1_{Y_0/S_0} \longrightarrow 0 \tag{+}$$

having the following properties:

(i) *After pullback to Y_0, d gives the morphism $\overline{D}_0$ of 4.8(iii).*

(ii) *If X_0 and Y_0 are smooth over S_0, then d is injective. Since the two modules of differentials are then locally free of finite type, the same is true of $\mathbf{n}_{Y_0/X_0}$, and the sequence is locally split.*

Proof.[122] Let $\pi_0 : Y_0 \to S_0$ denote the structure morphism. By SGA 1 II, formula (4.3) (see also EGA IV$_4$, 16.4.21), there is a canonical exact sequence of $\mathscr{O}_{Y_0}$-modules

$$\mathscr{N}_{Y_0/X_0} \xrightarrow{\overline{D}_0} \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{Y_0/S_0} \longrightarrow 0. \tag{†}$$

Since this sequence consists of $(Y_0 \times_S Y_0)$-equivariant modules and morphisms, its pullback (+) by ε_0^* is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules, and (†) is the pullback of (+) by π_0^* (cf. Exposé I, Section 6.8). This proves (i).

Suppose in addition that X_0 and Y_0 are smooth over S_0. Then, by SGA 1 II, 4.10 (see also EGA IV$_4$, 17.2.3(i) and 17.2.5), $\overline{D}_0$ is injective and the sequence (†) consists of $\mathscr{O}_{Y_0}$-modules locally free of finite type (and is therefore locally split). By the equivalence of categories I, 6.8.1, d is also injective, and consequently the sequence (+) has the asserted properties.

4.26. —[123] For every S_0-scheme $f : T \to S_0$, the sequence (+) gives an exact sequence of $Y_0(T)$-$\mathscr{O}(T)$-modules

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\omega^1_{Y_0/S_0}, f^*\mathscr{J}\right) &\longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\omega^1_{X_0/S_0}, f^*\mathscr{J}\right) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\mathbf{n}_{Y_0/X_0}, f^*\mathscr{J}\right). \end{aligned}$$

Thus there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \xrightarrow{d} N_0. \tag{4.26.1}$$

It gives an exact sequence of complexes of abelian groups

$$0 \longrightarrow C^*(Y_0, L_0^Y) \longrightarrow C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0),$$

and, in particular, the following commutative diagram with exact rows:

[122] Editor's note: The proof has been expanded to take account of the additions made in Exposé I, Section 6.8.

[123] Editor's note: The original has been expanded to make clear that this is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules.

$$\begin{array}{ccccccc}
0 \longrightarrow & C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \xrightarrow{d^0} & C^0(Y_0, N_0) \\
& \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\
0 \longrightarrow & C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0) & .
\end{array}$$

Observe that $C^0(Y_0, L_0^Y)$ (respectively $C^0(Y_0, L_0^X)$) is just

$$\mathrm{Hom}_{S_0}(S_0, L_0^Y) = \mathrm{Hom}_S(S, L'_Y)$$

(respectively the analogous group with X in place of Y); by 0.9, it is the group of sections of Y (respectively X) over S inducing the unit section of $X_{\mathscr{J}}$. Also, d^1 is precisely the morphism $v_{i_{Y_0}}$ of 4.8(iii), where $i_{Y_0} : Y_0 \to X_0$ is the canonical immersion.[124]

Lemma 4.27. *Under the conditions of 4.21 for S, $\mathscr{I}$, $\mathscr{J}$, and X, let Y be a subgroup of X, flat over S and lifting $Y_{\mathscr{J}}$. Let $i : Y \hookrightarrow X$ denote the canonical immersion.*[125]

(i) *Let $i' : Y \to X$ be a morphism of S-schemes lifting i_0 (so that i' is also an immersion), let $Y' = i'(Y)$, and let $d(i, i')$ be the element of $C^1(Y_0, L_0^X)$ such that $i' = d(i, i') \cdot i$ (cf. 1.2.4). Then the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by*

$$d(Y, Y') = d^1(d(i, i')).$$

(ii) *Let $x \in C^0(Y_0, L_0^X)$ be a section of X over S inducing the unit section of $X_{\mathscr{J}}$ over $S_{\mathscr{J}}$. Then the deviation $d(Y, \mathrm{Int}(x)Y) \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by*

$$-d(Y, \mathrm{Int}(x)Y) = d^1 \partial x = \partial d^0 x.$$

Indeed, Y' is the image of Y under the composite morphism [126]

$$Y \xrightarrow{(d(i,i'),i)} L'_X \times_S X \longrightarrow X,$$

which is denoted $d(i, i') \cdot i$ in 4.8(iii). By loc. cit. and the equality $v_{i_0} = d^1$, we therefore have

$$\begin{aligned} &c(X, Y', d(i, i') \cdot i) - c(X, Y', i) \\ &\qquad = v_{i_0}(d(i, i')) = d^1(d(i, i')). \end{aligned}$$

But $d(i, i') \cdot i = i'$ factors through Y' by definition, so the first term is zero. Moreover, by 4.8(ii),

$$c(X, Y', i) = d(Y', Y) = -d(Y, Y').$$

Hence $d(Y, Y') = d^1(d(i, i'))$, which proves (i).

Now let x be as in (ii). From the formula

$$xyx^{-1} = xyx^{-1}y^{-1}y = (x - \mathrm{Ad}(y)x)y = (-\partial x)(y) \cdot y,$$

we see that Y' is the image of Y under the immersion $i' = (-\partial x) \cdot i_Y$. Part (i) therefore gives

$$-d(Y, \mathrm{Int}(x)Y) = d^1 \partial x = \partial d^0 x.$$

[124] Editor's note: This follows from the definition of $d : \mathbf{n}_{Y_0/X_0} \to \Omega^1_{X_0/S_0}$ (cf. 4.25) and from the definition of $v_{i_{Y_0}}$ (cf. 4.8).

[125] Editor's note: Part (i) below has been added; it will be useful in 4.35.1 and then in 4.38(4) and (5).

[126] Editor's note: Recall that $L'_X = \prod_{S_0/S} L_0^X$.

Corollary 4.28. *Two subgroups Y and Y' of X, flat over S and lifting $Y_{\mathscr{J}}$, are conjugate by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ if and only if*

$$d(Y,Y') \in \partial d^0 C^0(Y_0, L_0^X) \subset \partial C^0(Y_0, N_0) = B^1(Y_0, N_0).$$

Corollary 4.29. *If d^0 is surjective, then Y and Y' as above are conjugate by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ if and only if $d(Y,Y') \in B^1(Y_0, N_0)$.*

Corollary 4.30. *Let Y be as in 4.27. The set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$d^1\partial C^0(Y_0, L_0^X) = C^0(Y_0, L_0^X)/\ker(d^1\partial).$$

Observe now that $C^0(Y_0, L_0^X)/\ker(d^1\partial)$ is computed using only the left-hand square of the commutative diagram in 4.26. In particular, it can also be computed in any diagram of the same type having the same left-hand square. Consider the functor L_0^X/L_0^Y over S_0, defined by

$$\operatorname{Hom}_{S_0}(T, L_0^X/L_0^Y) = \operatorname{Hom}_{S_0}(T, L_0^X)/\operatorname{Hom}_{S_0}(T, L_0^Y).$$

There is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & C^0(Y_0, L_{0Y}) \arrow[r] \arrow[d, "\partial"] & C^0(Y_0, L_{0X}) \arrow[r] \arrow[d, "\partial"] & C^0(Y_0, L_{0X}/L_{0Y}) \arrow[r] \arrow[d, "\partial"] & 0 \\
0 \arrow[r] & C^1(Y_0, L_{0Y}) \arrow[r] & C^1(Y_0, L_{0X}) \arrow[r] & C^1(Y_0, L_{0X}/L_{0Y}) \arrow[r] & 0
\end{tikzcd},$$

whence, by the preceding observation:

Corollary 4.31. *Let Y be as in 4.27. The set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$E = \partial C^0(Y_0, L_0^X/L_0^Y) = C^0(Y_0, L_0^X/L_0^Y)/H^0(Y_0, L_0^X/L_0^Y).$$

Corollary 4.32. *Suppose in addition that S_0 is affine and that $\omega^1_{Y_0/S_0}$ is finite locally free.*[127] *If*

$$\mathscr{F}_0 = \big(\operatorname{Lie}(X_0/S_0)/\operatorname{Lie}(Y_0/S_0)\big) \otimes_{\mathscr{O}_{S_0}} \mathscr{J},$$

then $E = \Gamma(S_0, \mathscr{F}_0)/H^0(Y_0, \mathscr{F}_0)$.

Indeed,[128] since $\omega^1_{Y_0/S_0}$ is finite locally free, as is $\omega^1_{X_0/S_0}$ (because X is assumed smooth over S), 0.6 gives

$$L_0^Y = \mathbf{W}\big(\operatorname{Lie}(Y_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\big), \qquad L_0^X = \mathbf{W}\big(\operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\big).$$

On the other hand, by 4.25 there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow \mathscr{K} \longrightarrow \omega^1_{X_0/S_0} \xrightarrow{\varphi} \omega^1_{Y_0/S_0} \longrightarrow 0, \qquad \mathscr{K} = \ker(\varphi).$$

Since $\omega^1_{Y_0/S_0}$ and $\omega^1_{X_0/S_0}$ are finite locally free, there is a locally split exact sequence

$$0 \longrightarrow \operatorname{Lie}(Y_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \longrightarrow \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \longrightarrow \mathscr{F}_0 \longrightarrow 0.$$

[127] Editor's note: $\operatorname{Lie}(Y_0/S_0)$ has been corrected to $\omega^1_{Y_0/S_0}$.
[128] Editor's note: The original has been expanded in what follows.

It follows that there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \longrightarrow \mathbf{W}(\mathscr{F}_0).$$

By the argument used to prove 4.31, we may compute E as the image of the composite map

$$C^0(Y_0, L_0^X) \xrightarrow{\pi} C^0(Y_0, \mathbf{W}(\mathscr{F}_0)) \xrightarrow{\partial} C^1(Y_0, \mathbf{W}(\mathscr{F}_0)).$$

Now π is the map

$$\Gamma(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}) \longrightarrow \Gamma(S_0, \mathscr{F}_0).$$

Since S_0 is affine, π is surjective, and this proves the assertion.

Corollary 4.33. *Let $S, \mathscr{I}, \mathscr{J}$, and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose that $\omega^1_{Y_0/S_0}$ is finite locally free and that S_0 is affine.*[129] *The set of subgroups of X conjugate to Y by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$E = \Gamma\Big(S_0, \frac{\mathrm{Lie}(X_0/S_0)}{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}} \otimes_{\Gamma(S_0, \mathscr{O}_{S_0})} \Gamma(S_0, \mathscr{J})\Big),$$

that is, to $L_0^X(Y_0)/H^0(Y_0, L_0^X)$.[130]

Indeed, by I, 4.7.3 (cf. 2.13), write

$$\mathrm{Lie}(X_0/S_0) = \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)} \oplus \mathscr{R}.$$

Since Y_0 is commutative,

$$\mathrm{Lie}(Y_0/S_0) \subset \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)},$$

and hence

$$\mathscr{F}_0 = \left(\frac{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}}{\mathrm{Lie}(Y_0/S_0)}\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \ \oplus\ \mathscr{R} \otimes_{\mathscr{O}_{S_0}} \mathscr{J},$$

$$\mathscr{F}_0^{\mathrm{ad}(Y_0)} = \left(\frac{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}}{\mathrm{Lie}(Y_0/S_0)}\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}.$$

By 4.32, therefore, $E \simeq \Gamma(S_0, \mathscr{R} \otimes_{\mathscr{O}_{S_0}} \mathscr{J})$. Returning to the definition of $\mathscr{R}$ completes the proof.

Corollary 4.34. *Let $S, \mathscr{I}, \mathscr{J}$, and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose that $\omega^1_{Y_0/S_0}$ is finite locally free and that S_0 is affine.*[131] *If $x \in X(S)$ induces the unit section of $X_{\mathscr{J}}$ and normalizes Y, then it centralizes Y.*

This follows immediately by comparing the preceding corollary with 2.14. Indeed, 4.33 shows that the elements of $C^0(Y_0, L_0^X)$ which globally preserve Y are the elements of $H^0(Y_0, L_0^X)$, and 2.14 shows that these are precisely the elements which act trivially on the canonical immersion $Y \longrightarrow X$.

4.35. — Return to the general situation of 4.21 and suppose that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$. By 4.25(ii), there is then an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow \mathrm{Lie}(Y_0/S_0) \longrightarrow \mathrm{Lie}(X_0/S_0) \longrightarrow \mathbf{n}^{\vee}_{Y_0/X_0} \longrightarrow 0, \qquad (*)$$

and these are finite locally free $\mathscr{O}_{S_0}$-modules.

[129] Editor's note: The hypothesis on $\omega^1_{Y_0/S_0}$ has been added, and the hypothesis "S affine" has been replaced by "S_0 affine."
[130] Editor's note: What follows has been added; cf. 4.34.
[131] Editor's note: Cf. Editor's Note 129.

On the other hand, by SGA 1, II, 4.10, every subscheme Y of X lifting $Y_{\mathscr{J}}$ and flat over S is smooth over S.[132] Suppose in addition that S_0 and $Y_{\mathscr{J}}$ are affine. Since $\mathbf{n}^{\vee}_{Y_0/X_0}$ is a locally free $\mathscr{O}_{S_0}$-module, the sequence $(*)$ remains exact after tensoring with $\mathscr{J}$ over $\mathscr{O}_{S_0}$, and then pulling back to Y_0^n. Since the Y_0^n are affine, this gives an exact sequence of complexes of abelian groups

$$0 \longrightarrow C^*(Y_0, L_0^Y) \longrightarrow C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0) \longrightarrow 0,$$

and, in particular, a commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{Y}}) \arrow[r] \arrow[d, "\partial"] & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}) \arrow[r, "d^1"] \arrow[d, "\partial"] & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{N}_0) \arrow[r] \arrow[d, "\partial"] & 0 \\
0 \arrow[r] & \mathrm{C}^2(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{Y}}) \arrow[r] & \mathrm{C}^2(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}) \arrow[r, "d^2"] & \mathrm{C}^2(\mathrm{Y}_0, \mathrm{N}_0) \arrow[r] & 0
\end{tikzcd} \quad .$$

Let now Y and Y' be two subgroups of X lifting $Y_{\mathscr{J}}$ and flat, hence smooth, over S. Since $Y_{\mathscr{J}}$ is affine, 0.15 shows that Y and Y' are isomorphic as schemes lifting $Y_{\mathscr{J}}$; that is, there is an isomorphism of S-schemes

$$f : Y \xrightarrow{\sim} Y'$$

inducing the identity on $Y_{\mathscr{J}}$. On the one hand, by 1.2.4, f defines an element $a \in C^1(Y_0, L_0^X)$ such that

$$f(y) = a(y_0)y$$

for every $y \in Y(S')$, $S' \to S$, and by 4.27(i),

$$d^1(a) = d(Y, Y').$$

Moreover, because Y and Y' are subgroups of X, the last element belongs to $Z^1(Y_0, N_0)$ (cf. 4.21). Thus ∂a is an element of $Z^2(Y_0, L_0^Y)$, whose image $\overline{\partial a}$ in $H^2(Y_0, L_0^Y)$ depends only on the class $\overline{d(Y, Y')} \in H^1(Y_0, N_0)$. This is precisely the definition of the connecting map

$$\partial^1 : H^1(Y_0, N_0) \longrightarrow H^2(Y_0, L_0^Y),$$

so that

$$\partial^1\big(\overline{d(Y, Y')}\big) = \overline{\partial a}.$$

On the other hand, transport the group structure of Y' by f, and let Y_1 be the resulting group, whose underlying scheme is therefore Y. Thus its group law μ_1 is defined, for every $S' \to S$ and $x, y \in Y(S')$, by

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

By 3.5.1, Y_1 defines a cocycle

$$\delta(Y, Y_1) \in Z^2(Y_0, \mathrm{Lie}(Y_0/S_0))$$

[132]Editor's note: What follows, together with Proposition 4.35.1, has been added; it is implicit in the original. Cf. 4.38(5).

such that, for every $S' \to S$ and $x, y \in Y(S')$,

$$\delta(Y, Y_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Put $b = \delta(Y, Y_1)$. For every $S' \to S$ and $x, y \in Y(S')$, we have $(b(x_0, y_0)xy)_0 = x_0 y_0$. Hence $f(b(x_0, y_0)xy)$ is equal, on the one hand, to

$$a(x_0 y_0)b(x_0, y_0)xy,$$

and, on the other hand, to

$$f(x)f(y) = a(x_0)x\, a(y_0)y = a(x_0)\,\mathrm{Ad}(x_0)(a(y_0))xy.$$

Comparing these two expressions, we find that $b(x_0, y_0)$ equals

$$\begin{aligned} &a(x_0 y_0)^{-1} a(x_0)\,\mathrm{Ad}(x_0)(a(y_0)) \\ &\quad = \mathrm{Ad}(x_0)(a(y_0)) - a(x_0 y_0) + a(x_0) = (\partial a)(x_0, y_0). \end{aligned}$$

Thus $\delta(Y, Y_1) = \partial a$, and we have proved:

Proposition 4.35.1.[132] *Under the hypotheses of 4.21, suppose in addition that S_0 is affine and that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$ and affine. Let Y, Y' be two subgroups of X lifting $Y_{\mathscr{J}}$ and flat (hence smooth) over S. Let $f : Y \xrightarrow{\sim} Y'$ be an isomorphism of S-schemes inducing the identity on $Y_{\mathscr{J}}$, and let Y_1 denote the group obtained by transporting the group structure of Y' by f. Then*

$$\partial^1(\overline{d(Y, Y')}) = \overline{\delta(Y, Y_1)}.$$

Proposition 4.36. *Under the hypotheses of 4.21, suppose in addition that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$ and that S_0 is affine. The set of sub-S-groups Y of X, flat (or smooth) over S, reducing to $Y_{\mathscr{J}}$, modulo conjugation by sections of X over S inducing the unit section of $X_{\mathscr{J}}$, is either empty or a principal homogeneous set under the group*

$$H^1\Big(Y_0, [\mathrm{Lie}(X_0/S_0)/\,\mathrm{Lie}(Y_0/S_0)] \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

It is enough to verify that Corollary 4.29 applies, that is, that

$$d^0 : \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \longrightarrow \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})$$

is surjective. This follows from the fact that the sequence (+) of 4.25(ii) is split, since S_0 is affine. [133]

Let us finally state a corollary common to 4.21 and 4.36. In fact, this will be the only form in which we shall subsequently use the general results of this section. [134]

Corollary 4.37. *Let S be a scheme and let S_0 be the closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Let X be an S-group smooth over S, and let Y_0 be a sub-S_0-group of X_0, flat over S_0.*

(i) *If S_0 is affine, Y_0 is smooth over S_0, and if*

$$H^1\Big(Y_0, [\mathrm{Lie}(X_0/S_0)/\,\mathrm{Lie}(Y_0/S_0)] \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0$$

for every $n \geq 0$, then any two sub-S-groups of X, flat (or smooth) over S and reducing to Y_0, are conjugate by a section of X over S inducing the unit section of X_0.

[133] Editor's note: This also follows from the proof of 4.32.

[134] Editor's note: For example, 4.37 is used in Exposé IX to prove statements 3.2 bis and 3.6 bis.

(ii) *If Y_0 is affine and of finite presentation, and if*[135]

$$H^2\Big(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0$$

for every $n \geq 0$, then there exists a sub-S-group of X, flat over S, reducing to Y_0.

4.38. It remains to relate the three constructions made in this Exposé. To avoid inessential complications, we shall place ourselves in the following situation: S_0 is the spectrum of a field k, S is the spectrum of the dual numbers $D(k)$, G is an S-group smooth over S, and K is a sub-S-group, smooth over S and affine. [136]

Put

$$\mathfrak{g}_0 = \operatorname{Lie} G_0 = \Gamma(S_0, \operatorname{Lie} G_0) = \operatorname{Lie}(G_0/S_0)(S_0), \qquad \mathfrak{k}_0 = \operatorname{Lie} K_0.$$

We have an exact sequence of k-vector spaces [137]

$$0 \longrightarrow \mathfrak{k}_0 \xrightarrow{i} \mathfrak{g}_0 \xrightarrow{d} \mathbf{n}^\vee_{K_0/G_0} \longrightarrow 0,$$

which gives rise to an exact cohomology sequence

$$\begin{aligned} 0 \longrightarrow H^0(K_0, \mathfrak{k}_0) \xrightarrow{i^0} H^0(K_0, \mathfrak{g}_0) \xrightarrow{d^0} H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) \\ \xrightarrow{\partial^0} H^1(K_0, \mathfrak{k}_0) \xrightarrow{i^1} H^1(K_0, \mathfrak{g}_0) \xrightarrow{d^1} H^1(K_0, \mathbf{n}^\vee_{K_0/G_0}) \xrightarrow{\partial^1} H^2(K_0, \mathfrak{k}_0). \end{aligned}$$

These various groups all have a geometric meaning.

a) $H^0(K_0, \mathfrak{k}_0) = \operatorname{Lie} \underline{\mathrm{Centr}}(K_0)$ [138] by II, 5.2.3.

b) $H^0(K_0, \mathfrak{g}_0) = \operatorname{Lie} \underline{\mathrm{Centr}}_{G_0}(K_0)$ [138] (idem).

c) $H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) = \operatorname{Lie} \underline{\mathrm{Norm}}_{G_0}(K_0)/\mathfrak{k}_0$[138] (idem).

d)

$$H^1(K_0, \mathfrak{k}_0) = \operatorname{Lie}\Big(\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)\Big) \,/\, \operatorname{Im}(\mathfrak{k}_0),$$

where $\operatorname{Im}(\mathfrak{k}_0)$ denotes the image of $\mathfrak{k}_0$ under the morphism $\operatorname{Lie}(\operatorname{Int}_0)$ deduced from

$$\operatorname{Int}_0 : K_0 \longrightarrow \underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0).$$

Indeed, it follows from 2.1(ii), applied with $Y = X = K$ and $f_0 = \mathrm{id}_{K_0}$, that $Z^1(K_0, \mathfrak{k}_0)$ is the group of infinitesimal automorphisms of the S_0-group K_0, and that $H^1(K_0, \mathfrak{k}_0)$ is obtained by quotienting by inner infinitesimal automorphisms, that is, by the image of $\mathfrak{k}_0$. Moreover, by II, 4.2.2, we also have

$$Z^1(K_0, \mathfrak{k}_0) = \operatorname{Lie}\Big(\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)\Big)\,.$$

[139]

[135] Editor's note: We have replaced $\operatorname{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{I}^{n+1}/\mathscr{I}^{n+2})$ by $\mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}$, in accordance with Remark 4.22.

[136] Editor's note: We have slightly modified the original in what follows. In particular, we have replaced X by G and Y by K, and denoted their Lie algebras by $\mathfrak{g}_0$ and $\mathfrak{k}_0$. We have also written $H^i(K_0, \cdot)$ explicitly in place of the abbreviation $H^i(\cdot)$ used in the original.

[137] Editor's note: Equipped with the adjoint action of K_0.

[138] Editor's note: Since the formation of centralizers and normalizers commutes with base change (cf. I, 2.3.3.1), we have written $\underline{\mathrm{Centr}}(K_0)$ in place of $\underline{\mathrm{Centr}}(K)_0$ in the original, and likewise $\underline{\mathrm{Centr}}_{G_0}(K_0)$ and $\underline{\mathrm{Norm}}_{G_0}(K_0)$ in place of $\underline{\mathrm{Centr}}_G(K)_0$ and $\underline{\mathrm{Norm}}_G(K)_0$.

[139] Editor's note: This is the Lie algebra $\mathrm{Der}_k(\mathfrak{k}_0)$ of derivations of $\mathfrak{k}_0$; hence $H^1(K_0, \mathfrak{k}_0)$ is the quotient of $\mathrm{Der}_k(\mathfrak{k}_0)$ by the inner derivations (that is, by the image of $\operatorname{ad} : \mathfrak{k}_0 \to \mathrm{Der}_k(\mathfrak{k}_0)$).

e) By 2.1(ii), $H^1(K_0, \mathfrak{g}_0)$ is the group of deviations between homomorphisms $K \to G$ extending the canonical immersion $i_0 : K_0 \to G_0$, modulo the deviations obtained through the action of the inner automorphisms of G defined by elements of $G(S)$ which give the unit of $G(S_0)$ (that is, elements of $\mathfrak{g}_0$).

f) By 4.36, $H^1(K_0, \mathbf{n}^{\vee}_{K_0/G_0})$ is the group of deviations between subgroups K' of G extending K_0 and flat over S (hence smooth over S, cf. SGA 1, II, 4.10), modulo the deviations obtained through the action of the inner automorphisms of G constructed as above.

g) By 3.5(ii), $H^2(K_0, \mathfrak{k}_0)$ is the group of deviations between group structures on K extending that of K_0, modulo the S-automorphisms of K inducing the identity on K_0.

We now propose to make explicit the six morphisms of the preceding exact sequence in terms of the geometric interpretation just given.

1) The morphisms i^0 and d^0 are simply those obtained by passing to the Lie algebra (and then, for d^0, to the quotient) in the canonical monomorphisms

$$\underline{\mathrm{Centr}}(K_0) \longrightarrow \underline{\mathrm{Centr}}_{G_0}(K_0) \longrightarrow \underline{\mathrm{Norm}}_{G_0}(K_0).$$

This follows immediately from the definition of the identifications **a)**, **b)**, and **c)**.

2) The morphism ∂^0 is constructed as follows. Let

$$\bar{x} \in \mathrm{Lie}\,\underline{\mathrm{Norm}}_{G_0}(K_0)/\mathfrak{k}_0.$$

Lift it to an element

$$x \in \mathrm{Lie}\,\underline{\mathrm{Norm}}_{G_0}(K_0) \subset \underline{\mathrm{Norm}}_G(K)(S).$$

Then $\mathrm{Int}(x)$ defines an automorphism of K inducing the identity on K_0, hence an element of $\mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)$. Denote by $\overline{\mathrm{Int}(x)}$ the image of this element in

$$\mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)/\,\mathrm{Im}(\mathfrak{k}_0).$$

Then

$$\partial^0(\bar{x}) = -\overline{\mathrm{Int}(x)} = \overline{\mathrm{Int}(x^{-1})}. \tag{*}$$

Indeed, let us compute the element of $\mathrm{Lie}\,\underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)$ defined by $\mathrm{Int}(x)$. By definition it corresponds to an element $a \in Z^1(K_0, \mathfrak{k}_0)$ such that

$$xyx^{-1} = a(y_0)y, \qquad y \in K(S'), \quad S' \longrightarrow S.$$

But this can also be written

$$a(y_0) = xyx^{-1}y^{-1} = x - \mathrm{Ad}(y)x = -\partial(x)(y_0),$$

whence $a = -\partial(x)$.

On the other hand,[140] the image under ∂ of

$$x \in \mathrm{Lie}\,\underline{\mathrm{Norm}}_{G_0}(K_0) \subset \mathfrak{g}_0$$

is an element of $Z^1(K_0, \mathfrak{k}_0)$, whose image $\overline{\partial(x)}$ in $H^1(K_0, \mathfrak{k}_0)$ depends only on $\bar{x}$. By definition of the connecting map ∂^0,

$$\partial^0(\bar{x}) = \overline{\partial(x)}.$$

Combined with $a = -\partial(x)$, this proves $(*)$.

[140] Editor's note: We have added the following sentence.

3) [141] Let $i : K \to G$ denote the canonical immersion. Let $\bar{u} \in H^1(K_0, \mathfrak{k}_0)$ be the image of an element
$$u \in \operatorname{Lie} \underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(K).$$
Then
$$i^1(\bar{u}) = \bar{d}(i, i \circ u), \tag{**}$$
where $\bar{d}(i, i \circ u)$ is the class defined in 2.1.1.

Indeed, $\bar{u}$ is the image of an element $v \in Z^1(K_0, \mathfrak{k}_0)$ such that $u(y) = v(y_0)y$, and $i^1(\bar{u})$ is the image in $H^1(K_0, \mathfrak{g}_0)$ of the cocycle $i \circ v \in Z^1(K_0, \mathfrak{g}_0)$. Since i is a group morphism, the equality $u(y) = v(y_0)y$ implies
$$i(u(y)) = i(v(y_0))i(y).$$
It follows that $i \circ v = d(i, i \circ u)$, proving $(**)$.

4) Let $i' : K \to G$ be a group morphism lifting i_0, let $\bar{d}(i, i')$ be the class defined in 2.1.1, and let
$$d(K, i'(K)) \in C^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$$
be the deviation defined in 4.5.1. By 4.21, $d(K, i'(K))$ belongs to $Z^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$. Denote its image in $H^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$ by $\bar{d}(K, i'(K))$. Then, by 4.27(i),
$$d^1(\bar{d}(i, i')) = \bar{d}(K, i'(K)). \tag{†}$$

5) Finally, let K' be a subgroup of G lifting K_0 and flat, hence smooth, over S. We have assumed that K_0 is affine. We therefore know that K and K' are isomorphic as schemes extending K_0 (cf. 0.15), so there exists an isomorphism of S-schemes
$$f : K \xrightarrow{\sim} K'$$
inducing the identity on K_0. Transport the group structure of K' by f, and let K_1 be the group thereby obtained (whose underlying scheme is thus K); that is, the group law μ_1 of K_1 is defined, for every $S' \to S$ and every $x, y \in K(S')$, by
$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$
By 3.5.1,[142] K_1 defines a cocycle $\delta(K, K_1) \in Z^2(K_0, \mathfrak{k}_0)$ such that, for every $S' \to S$ and every $x, y \in K(S')$,
$$\delta(K, K_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$
Then, by 4.35.1,
$$\partial^1 \bar{d}(K, K') = \bar{\delta}(K, K_1). \tag{‡}$$

[141] Editor's note: We have expanded the original in what follows, and in $(**)$ we have corrected $u \circ i$ to $i \circ u$.
[142] Editor's note: We have modified the original in what follows, taking account of the additions made in 3.5.1 and 4.35.1.

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Exposé IV
Topologies and Sheaves

by M. Demazure[*]

This exposé is intended to acquaint the reader with the essentials of the language of topologies and sheaves (without cohomology), a language particularly convenient for questions concerning the formation of quotients, among others.[0]

The first three sections develop the language of forming quotients. The fourth, which is the central part, presents the theory of sheaves, oriented mainly toward its application to questions about quotients; the fifth applies this theory to the formation of quotients in groups and to principal homogeneous bundles. The final section concerns the category of schemes more specifically and defines several useful topologies on that category.

The reader will profitably consult [AS], [MA], [D], and SGA 4; in particular, [D] for the applications of topologies to descent theory, and SGA 4 for questions concerning universes (which are treated especially poorly in this exposé).

1. Universal Effective Epimorphisms

Throughout the remainder of this exposé, a category $\mathscr{C}$ is fixed.

Definition 1.1. *A morphism $u : T \to S$ is called an* epimorphism *if, for every object X, the corresponding map*

$$X(S) = \mathrm{Hom}(S, X) \longrightarrow X(T) = \mathrm{Hom}(T, X)$$

is injective.[1] *It is called a* universal epimorphism *if, for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists and $u' : T' \to S'$ is an epimorphism.*

Definition 1.2. *A diagram of maps of sets*

$$A \xrightarrow{u} B \underset{v_2}{\overset{v_1}{\rightrightarrows}} C$$

is said to be exact *if u is injective and its image consists of those elements $b \in B$ for which $v_1(b) = v_2(b)$. A diagram of the same form in $\mathscr{C}$ is said to be* exact *if, for every object X of $\mathscr{C}$, the corresponding diagram of sets*

$$A(X) \longrightarrow B(X) \rightrightarrows C(X)$$

[*]This text develops the substance of two lectures given by A. Grothendieck, supplementing them at several important points that had been passed over in silence or only briefly touched upon.

[0]Editor's note: version of October 13, 2024.

[1]Editor's note: that is, if u is right-cancellable.

is exact; one then also says that u makes A a kernel *of the pair of arrows* (v_1, v_2)*. Dually, a diagram in* $\mathscr{C}$

$$C \underset{v_2}{\overset{v_1}{\rightrightarrows}} B \xrightarrow{u} A$$

is said to be exact if it is exact as a diagram in the opposite category $\mathscr{C}^\circ$; that is, if for every object X of $\mathscr{C}$, the corresponding diagram of sets

$$X(A) \longrightarrow X(B) \rightrightarrows X(C)$$

is exact.[2] *One also says that u makes A a* cokernel *of the pair of arrows* (v_1, v_2)*.*

Definition 1.3. *A morphism $u : T \to S$ is called an* effective epimorphism *if the fiber square $T \times_S T$ exists and the diagram*

$$T \times_S T \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} T \xrightarrow{u} S$$

is exact; that is, if u makes S a cokernel of $(\mathrm{pr}_1, \mathrm{pr}_2)$*. It is called a* universal effective epimorphism *if, for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists and the morphism $u' : T' \to S'$ is an effective epimorphism.*

There are evidently implications

$$\begin{array}{ccc} \text{universal effective epimorphism} & \Longrightarrow & \text{effective epimorphism} \\ \Downarrow & & \Downarrow \\ \text{universal epimorphism} & \Longrightarrow & \text{epimorphism,} \end{array}$$

but in general no other implication holds.[3]

Definition 1.4.0.[4] We "recall" that a morphism $u : T \to S$ is said to be *squarable* if, for every morphism $S' \to S$, the fiber product $T \times_S S'$ exists.

Lemma 1.4. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then:*

(a) *if u and v are epimorphisms, then uv is an epimorphism; and if uv is an epimorphism, then u is an epimorphism;*

(b) *if u and v are universal epimorphisms, then uv is a universal epimorphism; and if uv is a universal epimorphism and u is squarable, then u is a universal epimorphism.*

Lemma 1.4 follows immediately from the definitions. We deduce:

Corollary 1.5. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

[2]Editor's note: this implies, in particular, that u is an epimorphism.

[3]Editor's note: for example, if $\mathscr{C} = \mathbf{Sch}$ is the category of schemes, it is easy to see that every universal epimorphism is surjective. Let

$$T = \coprod_{p \text{ prime}} \mathrm{Spec}(\mathbf{F}_p), \qquad S = \mathrm{Spec}(\mathbf{Z}).$$

Then the morphism $u : T \to S$ is an epimorphism that is not universal. On the other hand, $T \times_S T$ identifies with T, so the two projections $T \times_S T \to T$ coincide. Since id_T does not descend to a morphism $S \to T$, this shows that u is not an effective epimorphism.

[4]Editor's note: the numbering 1.4.0 has been added for later reference.

is a universal epimorphism.

We also record the following.

Definition 1.6.0.[5] An object S of $\mathscr{C}$ is called *squarable* if its product with every object of $\mathscr{C}$ exists. (If $\mathscr{C}$ has a final object e, this is equivalent to saying that the morphism $S \to e$ is squarable; cf. 1.4.0.)

Lemma 1.6. *Let $u : X \to Y$ be a morphism in $\mathscr{C}_{/S}$. For it to be an epimorphism (respectively, a universal epimorphism, an effective epimorphism, or a universal effective epimorphism), it is sufficient that the corresponding morphism in $\mathscr{C}$ have the same property; it is also necessary if S is assumed to be a squarable object of $\mathscr{C}$.*

The proof is immediate and is left to the reader. The hypothesis that S is squarable is used to interpret the $\mathscr{C}$-morphisms from an object Y of $\mathscr{C}_{/S}$ to an object Z of $\mathscr{C}$ as the $\mathscr{C}_{/S}$-morphisms from Y to $Z \times S$.

Lemma 1.7. *With the notation of 1.4, if u and v are effective epimorphisms and v is a universal epimorphism, then uv is an effective epimorphism.*

To see this, consider the diagram

$$\begin{array}{ccccc}
S & \overset{u}{\leftarrow} & T & \leftleftarrows & T \times_S T \\
 & & {\scriptstyle v}\uparrow & & {\scriptstyle v \times_S v}\uparrow \\
 & & U & \leftleftarrows & U \times_S U \\
 & & \upuparrows & \nearrow & \\
 & & U \times_T U & &
\end{array} \tag{1.7.1}$$

By hypothesis, the first row and the first column are exact; moreover, by 1.5 and 1.6, $v \times_S v$ is an epimorphism, since v is a universal epimorphism. The conclusion follows by an evident diagram chase. Indeed, if an element of $X(U)$ has the same two images in $X(U \times_S U)$, then it has, *a fortiori*, the same two images in $X(U \times_T U)$, and hence comes from an element of $X(T)$, since the first column is exact.

Since the first row is exact, it remains only to verify that the element in question has the same two images in $X(T \times_S T)$. Because $v \times_S v$ is an epimorphism, it is enough to verify that their images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. If u and v are universal effective epimorphisms, then uv is a universal effective epimorphism. If uv is a universal effective epimorphism and u is squarable, then u is a universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, consider the following diagram (of "bisimplicial" type):

$$\begin{array}{ccccc}
S & \overset{u}{\leftarrow} & T & \leftleftarrows & T \times_S T \\
{\scriptstyle uv}\uparrow & & \uparrow & & \uparrow \\
U & \longleftarrow & U \times_S T & \leftleftarrows & U \times_S T \times_S T \\
\upuparrows & & \upuparrows & & \upuparrows \\
U \times_S U & \longleftarrow & U \times_S U \times_S T & \leftleftarrows & U \times_S U \times_S T \times_S T.
\end{array} \tag{1.8.1}$$

The first, second, and third columns are exact by the hypothesis that uv is a universal effective epimorphism. The second row is exact because $U \times_S T \to U$ is an effective epimorphism (it has a section over U), and the same is true of the third row, for the same reason. An evident diagram chase then shows that the first row is exact, that is, that u is an effective epimorphism. Since the hypotheses are invariant under an arbitrary base change $S' \to S$, it follows that u is in fact a universal effective epimorphism.

[5]Editor's note: the numbering 1.6.0 has been added for later reference.

Corollary 1.9. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal effective epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

is a universal effective epimorphism.

The proof is the same as for 1.5, using the diagram

$$\begin{tikzcd} & Y' & X' \arrow[l, "u'"'] \\ X \arrow[d, "u"'] & X \times Y' \arrow[l] \arrow[u] \arrow[d] & X \times X' \arrow[l] \arrow[u] \arrow[dl, "u \times u'"] \\ Y & Y \times Y' \arrow[l] & . \end{tikzcd} \tag{1.9.1}$$

Corollary 1.10. *Let $u : T \to S$ be a squarable morphism, and let $S' \to S$ be a base-change morphism that is a universal effective epimorphism. Then u is a universal effective epimorphism if and only if*

$$u' : T' = T \times_S S' \longrightarrow S'$$

is one:

$$\begin{tikzcd} T \arrow[d, "u"'] & T' \arrow[l, "v'"'] \arrow[d, "u'"] \\ S & S'. \arrow[l, "v"'] \end{tikzcd} \tag{1.10.1}$$

Only sufficiency requires proof. If u' is a universal effective epimorphism, then so is vu', by 1.8. Since $vu' = uv'$, another application of 1.8 shows that u is a universal effective epimorphism.

Remark 1.11. *The same reasoning shows that in 1.10 one may replace "universal effective epimorphism" by "universal epimorphism," by "universal and effective epimorphism," or simply by "epimorphism" (and in the last case the hypothesis that u be squarable is plainly unnecessary).*

In the proof of 1.8 we used the following result, which deserves to be stated explicitly.

Proposition 1.12. *Let $u : T \to S$ be a morphism admitting a section. Then u is an epimorphism; if $T \times_S T$ exists, it is an effective epimorphism; and if, in addition, u is squarable, it is a universal effective epimorphism.*

The first assertion is contained in 1.4(a), and the third follows at once from the second, so it is enough to establish the second. In fact, there is a stronger conclusion: for every functor $F : \mathscr{C}^\circ \to \mathbf{Set}$, not necessarily representable, the diagram of sets

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be regarded as a special case of the formalism of Čech cohomology (in dimension 0!), which we merely recall here. Assume only that $T \times_S T$ exists, and set

$$\check{H}^0(T/S, F) = \mathrm{Ker}\big(F(T) \rightrightarrows F(T \times_S T)\big).$$

One may plainly regard $\check{H}^0(T/S, F)$ as a contravariant functor of the variable T in $\mathscr{C}_{/S}$: every S-morphism $T' \to T$ defines a map

$$\check{H}^0(T/S, F) \longrightarrow \check{H}^0(T'/S, F). \tag{+}$$

Fix T, T' in $\mathscr{C}_{/S}$. A familiar calculation shows that, if there exists an S-morphism $T' \to T$, the corresponding map (+) is independent of the choice of that morphism.[6]

Thus $\check{H}^0(T/S, F)$ may be regarded as a functor on the category associated with the set $\mathrm{Ob}(\mathscr{C}_{/S})$, preordered by the relation of domination: T' dominates T if there exists an S-morphism $T' \to T$. In particular, if T and T' are isomorphic in this latter category, that is, if each dominates the other, then (+) is an isomorphism of sets. This applies in particular when T' is the final object of $\mathscr{C}_{/S}$, essentially S itself. The object T always dominates $T' = S$, while the converse holds precisely when T/S has a section. This proves 1.12 in the stronger form announced.

Remark 1.13. *For various applications, the notions introduced and the results stated in this exposé must be developed more generally for a family of morphisms $u_i : T_i \to S$ with the same target, rather than for a single morphism $u : T \to S$. Such a family is called* epimorphic *if, for every object X of $\mathscr{C}$, the corresponding map*

$$X(S) \longrightarrow \prod_i X(T_i)$$

is injective. One similarly introduces the notion of an effective epimorphic family and the universal variants of these notions. Whenever needed below, we shall take for granted that the results of this exposé extend to this more general situation.

Remark 1.14. *Every morphism that is both a monomorphism and an effective epimorphism is an isomorphism.*

Indeed, in the notation of 1.3, the fact that $T \to S$ is a monomorphism implies that the two morphisms

$$\mathrm{pr}_1, \mathrm{pr}_2 : T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). But a diagram

$$C \overset{v}{\underset{v}{\rightrightarrows}} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.[7]

2. Descent Morphisms

Recall the following definitions.

Definition 2.1. *Let $f : S' \to S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u' : X' \to S'$ be an object over S'. A* gluing datum *on X'/S', relative to f, is an S''-isomorphism*

$$c : X''_1 \xrightarrow{\sim} X''_2,$$

[6] Editor's note: this is the following argument, communicated by M. Demazure. Let $f, g : T' \to T$, and let $\phi : T' \to T \times_S T$ be the morphism with components f and g, so that $p_1 \circ \phi = f$ and $p_2 \circ \phi = g$. Then $F(\phi) : F(T \times_S T) \to F(T')$ satisfies $F(\phi) \circ F(p_1) = F(f)$ and $F(\phi) \circ F(p_2) = F(g)$. For every $x \in \check{H}^0(T/S, F)$, however, one has $F(p_1)(x) = F(p_2)(x)$. Applying $F(\phi)$ to both sides gives $F(f)(x) = F(g)(x)$, which shows that f and g induce the same map.

[7] Editor's note: a monomorphism that is an epimorphism need not be an isomorphism. For example, in $\mathscr{C} = \mathbf{Sch}$, the morphism

$$\mathrm{Spec}(\mathbf{F}_p) \amalg \mathrm{Spec}(\mathbf{Z}[1/p]) \longrightarrow \mathrm{Spec}(\mathbf{Z})$$

is a monomorphism and a surjective epimorphism, but is not an isomorphism.

where X''_i, for $i = 1, 2$, denotes the inverse image, assumed to exist, of X'/S' under the projection $\mathrm{pr}_i : S'' \to S'$. The gluing datum c is called a descent datum *if it satisfies the "cocycle condition"*

$$\mathrm{pr}^*_{3,1}(c) = \mathrm{pr}^*_{3,2}(c)\ \mathrm{pr}^*_{2,1}(c),$$

where $\mathrm{pr}_{i,j}$, for $1 \leq j < i \leq 3$, are the canonical projections from

$$S''' = S' \times_S S' \times_S S'$$

to S'' (and we now assume that S''' also exists), where $\mathrm{pr}^*_{i,j}(c)$ is the inverse image of c, viewed as an S'''-morphism from X'''_j to X'''_i, and where, for each integer k from 1 to 3, X'''_k denotes the inverse image, assumed to exist, of X'/S' under the k-th projection $q_k : S''' \to S'$.

In the second part of the definition we have therefore used customary identifications and abuses of notation,[8] which experience shows to be harmless, but which must plainly be avoided in a rigorous exposition of descent theory—a theory that must itself justify, to some extent, these customary abuses of language. Such a formal exposition ([D]) was written by Giraud, with a view to justifying and making precise SGA 1, Exposé VII, which was never written. For a detailed account of the faithfully flat descent results used constantly in this Seminar, one may consult SGA 1, Exposé VIII.

Continue to let $f : S' \to S$ be a morphism for which $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist. The inverse images of X' under pr_i, for $i = 1, 2$, then exist and are canonically isomorphic; consequently, X'/S' carries a canonical gluing datum relative to f. When S''' and $X''' = X \times_S S'''$ exist, this is in fact a descent datum. If Y is another object over S satisfying the same conditions as X/S, then for every S-morphism $X \to Y$, the corresponding S'-morphism $X' \to Y'$ is compatible with the canonical gluing data on X' and Y'. In particular, if $S' \to S$ is squarable, then

$$X \longmapsto X' = X \times_S S'$$

is a functor from $\mathscr{C}_{/S}$ to the "category of objects over S' equipped with a descent datum relative to f"—a category whose definition is left to the reader and which is a full subcategory of the "category of objects over S' equipped with a gluing datum relative to f."

With this understood, we make the following definition.

Definition 2.2. *A morphism $f : S' \to S$ is called a* descent morphism *(respectively, an* effective descent morphism*) if f is squarable (that is, $X \times_S S'$ exists for every $X \to S$) and the preceding functor*

$$X \longmapsto X' = X \times_S S'$$

from the category $\mathscr{C}_{/S}$ of objects over S to the category of objects over S' equipped with a descent datum relative to f is fully faithful (respectively, an equivalence of categories).

The first of these notions can be expressed using gluing data alone, and therefore without invoking the triple fiber product S''': f is a descent morphism precisely when it is squarable and $X \mapsto X'$ is a fully faithful functor from $\mathscr{C}_{/S}$ to the category of objects over S' equipped with a gluing datum relative to f. Written out, this means that for any two objects X, Y over S, the following diagram of sets is exact:

$$\mathrm{Hom}_S(X,Y) \longrightarrow \mathrm{Hom}_{S'}(X',Y') \underset{p_2}{\overset{p_1}{\rightrightarrows}} \mathrm{Hom}_{S''}(X'',Y''). \tag{x}$$

[8] Editor's note: for example, we have identified, on the one hand, $\mathrm{pr}^*_{2,1}(X''_1) = \mathrm{pr}^*_{3,1}(X''_1)$, and, on the other hand, $\mathrm{pr}^*_{2,1}(X''_2) = \mathrm{pr}^*_{3,2}(X''_2)$.

Here $p_i(u')$ denotes the inverse image of $u' \in \mathrm{Hom}_{S'}(X', Y')$ under the projection $\mathrm{pr}_i : S'' \to S'$, for $i = 1, 2$. Indeed, by definition, the kernel of the pair (p_1, p_2) is precisely the set of S'-morphisms $X' \to Y'$ compatible with the canonical gluing data.

By the definition of the inverse images Y' and Y'', there are canonical bijections

$$\mathrm{Hom}_{S'}(X', Y') \simeq \mathrm{Hom}_S(X', Y), \qquad \mathrm{Hom}_{S''}(X'', Y'') \simeq \mathrm{Hom}_S(X'', Y).$$

Thus exactness of (x) is equivalent to exactness of

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_S(X', Y) \rightrightarrows \mathrm{Hom}_S(X'', Y), \tag{xx}$$

which is obtained by applying $\mathrm{Hom}_S(-, Y)$ to the diagram in $\mathscr{C}_{/S}$

$$X'' \rightrightarrows X' \longrightarrow X, \tag{xxx}$$

itself obtained from

$$S'' \rightrightarrows S' \longrightarrow S$$

by the base change $X \to S$. In view of 1.6, this proves the first part of the following proposition.

Proposition 2.3. *Let $f : S' \to S$ be a morphism. If f is a universal effective epimorphism, then it is a descent morphism. Conversely, every descent morphism is a universal effective epimorphism if S is a squarable object of $\mathscr{C}$, that is, if its product with every object of $\mathscr{C}$ exists.*

It remains to prove the converse. If f is a descent morphism, we must show that it is a universal effective epimorphism: for every morphism $X \to S$, diagram (xxx) must be exact, or equivalently, its image under $\mathrm{Hom}(-, Z)$ must be an exact diagram of sets for every object Z of $\mathscr{C}$. By hypothesis, $Z \times S$ exists. Let Y be the object of $\mathscr{C}_{/S}$ that it defines. The image of (xxx) under $\mathrm{Hom}(-, Z)$ is then isomorphic to its image under $\mathrm{Hom}_S(-, Y)$, and the latter is exact by the hypothesis on f.

The results concerning descent morphisms may therefore be applied to universal effective epimorphisms. For example:

Proposition 2.4. *Let $f : S' \to S$ be a descent morphism (for example, a universal effective epimorphism). Then:*

(a) *For every S-morphism $u : X \to Y$, the morphism u is an isomorphism (respectively, a monomorphism) if and only if $u' : X' \to Y'$ is one.*

(b) *Let X, Y be two subobjects of S, and let X', Y' be their inverse images in S'. Then X is contained in Y (respectively, is equal to Y) if and only if the same is true of X' and Y'.*

For (a), the definition shows that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism. Every isomorphism of objects over S' that is compatible with gluing data is, however, an isomorphism of objects with gluing datum: its inverse is also compatible with the gluing data. For (b), it remains to show that if X' is contained in Y', that is, if there exists an S'-morphism $X' \to Y'$, then the same is true of X, Y over S. Since $Y' \to S'$, and hence $Y'' \to S''$, is a monomorphism, the morphism $X' \to Y'$ is automatically compatible with the gluing data and therefore comes from an S-morphism $X \to Y$.

The proof applies more generally to two objects X, Y over S, with $Y \to S$ a monomorphism, when one asks whether $X \to S$ factors through Y: it is enough that $X' \to S'$ factor through Y'.

Corollary 2.5. *Let $f : S' \to S$ be a universal effective epimorphism and $g : S \to T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' with itself over T, that is, that the canonical morphism*

$$S' \times_S S' \xrightarrow{\sim} S' \times_T S'$$

is an isomorphism. Then $g : S \to T$ is a monomorphism (and the converse is of course true).

Indeed, consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \xrightarrow{\sim} & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \longrightarrow & S \times_T S, \end{array} \tag{2.5.1}$$

where the second horizontal arrow is the diagonal morphism. By 1.9, the second vertical arrow f is a universal effective epimorphism; by hypothesis, the first horizontal arrow is an isomorphism. It follows from 2.4(a), or equally from 2.4(b),[9] that $S \to S \times_T S$ is also an isomorphism, which says precisely that $g : S \to T$ is a monomorphism.

Remark 2.6. *The notions introduced in 2.1 in terms of morphisms between certain inverse limits have an evident expression in terms of the contravariant functors defined by the objects S, S', X' under consideration. Subject to the existence of the fiber products considered in 2.1, there is a one-to-one correspondence between gluing data (respectively, descent data) on X'/S' relative to $S' \to S$, and gluing data (respectively, descent data) for the corresponding objects of*

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set}).$$

This makes it possible, if desired, to extend these notions to the case in which no existence hypothesis is imposed on fiber products in $\mathscr{C}$.

Remark 2.7. *The notions introduced in this section generalize to families of morphisms. They can also be presented more intrinsically by means of the notion of a sieve (4.1). For these matters the reader is referred to [D].*

3. Universal Effective Equivalence Relations

3.1. Equivalence Relations: Definitions

Definition 3.1.1. *A $\mathscr{C}$-equivalence relation in an object $X \in \mathrm{Ob}\,\mathscr{C}$ is a representable subfunctor R of the functor $X \times X$ such that, for every $S \in \mathrm{Ob}\,\mathscr{C}$, $R(S)$ is the graph of an equivalence relation on $X(S)$.*

This definition applies in particular to the category $\widehat{\mathscr{C}}$. If X is regarded as an object of $\widehat{\mathscr{C}}$, then a $\widehat{\mathscr{C}}$-equivalence relation in X is simply a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation on $X(S)$, for every object S of $\mathscr{C}$.[10]

If R is a $\mathscr{C}$-equivalence relation in X, let p_i denote the morphism from R to X induced by the projection $\mathrm{pr}_i : X \times X \to X$. Thus there is a diagram

$$p_1, p_2 : R \rightrightarrows X.$$

[9] Editor's note: applied to $f : S' \times_T S' \to S \times_T S = Y$.

[10] Editor's note: the condition is plainly necessary. Conversely, if $R(S)$ is the graph of an equivalence relation for every $S \in \mathrm{Ob}\,\mathscr{C}$, this equivalence relation extends to $R(F)$ for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$: two morphisms $\varphi, \psi : F \to R$ are declared equivalent if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $x \in F(S)$, the elements $\varphi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

Definition 3.1.2. *A morphism $u : X \to Z$ is said to be* compatible with R *if $up_1 = up_2$. An object of $\mathscr{C}$ that is a cokernel of the pair (p_1, p_2) is also called a* quotient object *of X by R, and is denoted X/R.*

Thus one has an exact diagram

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} X/R,$$

and X/R represents the covariant functor

$$\mathrm{Hom}_{\mathscr{C}}(X/R, Z) = \{\text{morphisms from } X \text{ to } Z \text{ compatible with } R\}.$$

Since quotient objects are taken in $\mathscr{C}$, the quotient X/R is unique whenever it exists.

These definitions extend at once to a $\widehat{\mathscr{C}}$-equivalence relation in X. One should note, however, that the identification of the objects of $\mathscr{C}$ with their images in $\widehat{\mathscr{C}}$ does not commute with the formation of quotients: the quotient X/R of X by R in $\mathscr{C}$ is not *a priori* a quotient of X by R in $\widehat{\mathscr{C}}$. One must therefore avoid identifying $\mathscr{C}$ uncritically with its image in $\widehat{\mathscr{C}}$ in questions involving passage to a quotient.[11]

In what follows, we shall simply say *equivalence relation* to mean a $\widehat{\mathscr{C}}$-equivalence relation; whenever necessary, we shall specify that a $\mathscr{C}$-equivalence relation is intended.[12]

Definition 3.1.3. *If X is an object of $\mathscr{C}$ over S, an* equivalence relation in X over S *is an equivalence relation R in X for which the structure morphism $X \to S$ is compatible with R.*

The canonical monomorphism $R \to X \times X$ then factors through the monomorphism

$$X \times_S X \longrightarrow X \times X$$

and defines an equivalence relation in the object $X \to S$ of $\mathscr{C}_{/S}$. Whenever the quotient X/R exists, it is equipped with a canonical morphism to S, and the corresponding object of $\mathscr{C}_{/S}$ is a quotient of $X \in \mathrm{Ob}\,\mathscr{C}_{/S}$ by the preceding equivalence relation. Conversely, if S is a squarable object of $\mathscr{C}$ and $Y \to S$ is a quotient of X by this equivalence relation in $\mathscr{C}_{/S}$, then Y is a quotient of X by R in $\mathscr{C}$. In any event, we shall never have to consider quotients in $\mathscr{C}_{/S}$ that are not already quotients in $\mathscr{C}$.

Definition 3.1.4. *Let X (respectively X') be an object of $\mathscr{C}$ equipped with an equivalence relation R (respectively R'). A morphism*

$$u : X \longrightarrow X'$$

is said to be compatible with R and R' *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, two points of $X(S)$ congruent modulo $R(S)$ are carried by u to two points of $X'(S)$ congruent modulo $R'(S)$;*

[11] Editor's note: here is an illustration that also previews the sequel of this exposé. Let G be a $\mathscr{C}$-group, let H be a $\mathscr{C}$-subgroup, and let R be the equivalence relation in G defined by $G \times H \to G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\widehat{\mathscr{C}}$ (by 4.4.9 applied to the coarsest topology; cf. 4.4.2), but it is not in general the quotient one wants. For example, when $\mathscr{C} = \mathbf{Sch}$, there is an exact sequence of affine group schemes

$$1 \longrightarrow \mu_2 \longrightarrow \mathbf{G}_m \xrightarrow{p} \mathbf{G}_m \longrightarrow 1$$

that identifies $\mathbf{G}_m$ with the quotient $\mathbf{G}_m/\mu_2$. Moreover, because p is finite and locally free, $\mathbf{G}_m$ is the sheaf quotient of $\mathbf{G}_m$ by μ_2 in the larger category of sheaves for the fppf topology; cf. 4.6.6(ii) and 6.3.2. By contrast, the quotient Q in $\widehat{\mathscr{C}}$ is not isomorphic to $\mathbf{G}_m$, since, for example, $Q(\mathbf{Z}) = \{1\}$, whereas $\mathbf{G}_m(\mathbf{Z}) = \{\pm 1\}$. Thus Q is not an fppf sheaf and, *a fortiori*, is not representable.

[12] Editor's note: even for a $\widehat{\mathscr{C}}$-equivalence relation in X, the question of interest is the existence of a quotient in $\mathscr{C}$.

(ii) *there exists a necessarily unique morphism $R \to R'$ making the diagram*

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow{u \times u} & X' \times X' \end{array}$$

commute.

By the universal property of X/R, whenever X/R and X'/R' exist there is a unique morphism v making the diagram

$$\begin{array}{ccc} X & \xrightarrow{p} & X/R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ X' & \xrightarrow{p'} & X'/R' \end{array}$$

commute.

Definition 3.1.5. *A subobject (that is, a representable subfunctor) Y of X is said to be* stable under the equivalence relation R *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$;*

(ii) *the two subobjects of R that are the inverse images of Y under p_1 and p_2 are identical.*

An important special case of a stable subobject of X is the following: the quotient X/R exists and Y is the inverse image in X of a subobject of X/R.

Definition 3.1.6. *Let R be an equivalence relation in X, and let $X' \to X$ be a morphism. The equivalence relation R' in X' obtained from the cartesian diagram*

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ X' \times X' & \longrightarrow & X \times X \end{array}$$

is called the equivalence relation in X' that is the inverse image *of the equivalence relation R in X. In particular, if X' is a subobject of X, it is called the equivalence relation* induced *on X' by R, and is denoted $R_{X'}$.*

The morphism $X' \to X$ is compatible with R' and R; hence, when the quotients exist, it induces a morphism $X'/R' \to X/R$ (3.1.4). If X' is a subobject of X, we shall see later that in certain cases one can prove that $X'/R' \to X/R$ is a monomorphism and therefore identifies X'/R' with a subobject of X/R. In that situation, the inverse image of this subobject in X is a subobject containing X' and stable under R: the *saturation* of X' with respect to the equivalence relation R.

Proposition 3.1.7. *If the subobject Y of X is stable under R, then for $i = 1, 2$ the following two squares are cartesian:*

$$\begin{array}{ccc} R_Y & \longrightarrow & R \\ {\scriptstyle p_i}\downarrow & & \downarrow{\scriptstyle p_i} \\ Y & \longrightarrow & X. \end{array}$$

The proof is immediate.

3.2. Equivalence Relation Defined by a Group Acting Freely

Definition 3.2.1. *Let X be an object of $\mathscr{C}$, and let H be a $\mathscr{C}$-group acting on X, that is, equipped with a morphism of $\widehat{\mathscr{C}}$-groups*

$$H \longrightarrow \mathrm{Aut}(X).$$

The group H is said to act freely *on X if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the group $H(S)$ acts freely on $X(S)$;*

(ii) *the morphism of functors*
$$H \times X \longrightarrow X \times X$$
defined on sets by $(h, x) \mapsto (hx, x)$ is a monomorphism.

In the preceding definition the group H was assumed to act on X on the left. There is of course an analogous notion when the group acts on the right, that is, when one is given a group morphism from the opposite group H° to $\mathrm{Aut}(X)$.

If H acts freely on X, the image of $H \times X$ under the morphism in (ii) is an equivalence relation in X, called the *equivalence relation defined by the action of H on X*. When the quotient of X by this equivalence relation exists, it is denoted $H\backslash X$ (or X/H when H acts on the right). It represents the following covariant functor: for every object Z of $\mathscr{C}$,

$$\mathrm{Hom}(H\backslash X, Z) = \{\text{morphisms from } X \text{ to } Z \text{ invariant under } H\},$$

where a morphism $f : X \to Z$ is said to be invariant under H if, for every $S \in \mathrm{Ob}\,\mathscr{C}$, the corresponding morphism $X(S) \to Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the assumptions of 3.2.1, let Y be a subobject of X. The following conditions are equivalent:*

(i) *Y is stable under the equivalence relation defined by H (3.1.5);*

(ii) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;*

(iii) *there exists a necessarily unique morphism f making the diagram*

$$\begin{array}{ccc} H \times Y & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ H \times X & \longrightarrow & X \end{array}$$

commute.

Under these conditions, f defines a morphism of $\widehat{\mathscr{C}}$-groups

$$H \longrightarrow \mathrm{Aut}(Y),$$

and the equivalence relation in Y defined by this action of H is precisely the equivalence relation induced on Y by the equivalence relation in X defined by the action of H.

The proof is immediate. There is of course an analogous statement for a right action. The action of H on Y defined above will be called the action induced on Y by the given action of H on X.

Consider now two $\mathscr{C}$-groups H and G, together with a group morphism

$$u : H \longrightarrow G.$$

Then H acts on G by translations (set-theoretically, $hg = u(h)g$), and it acts freely if and only if u is a monomorphism. When it exists, the quotient of G by this action of H is denoted $H\backslash G$. Similarly, one defines a right action of H on G and a quotient G/H. These quotients are functorial with respect to the groups involved. More precisely, one has the following lemma, stated for right quotients.

Lemma 3.2.3. *Let $u : H \to G$ and $u' : H' \to G'$ be monomorphisms of $\mathscr{C}$-groups, and suppose that a morphism of $\mathscr{C}$-groups*

$$f : G \longrightarrow G'$$

is given. The following conditions are equivalent:

(i) *f is compatible with the equivalence relations in G and G' defined by H and H';*

(ii) *for every $S \in \operatorname{Ob}\mathscr{C}$, one has $fu(H(S)) \subseteq u'(H'(S))$;*

(iii) *there exists a necessarily unique multiplicative morphism $g : H \to H'$ such that the diagram*

$$\begin{array}{ccc} H & \xrightarrow{g} & H' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u'} \\ G & \xrightarrow{f} & G' \end{array}$$

commutes.

Under these conditions, if the quotients G/H and G'/H' exist, there is a unique morphism $\overline{f}$ making the diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p'} \\ G/H & \xrightarrow{\overline{f}} & G'/H' \end{array}$$

commute.

The first assertion is proved by reduction to the set-theoretic case. The second follows immediately from (i).

The notions introduced above for general equivalence relations could all be translated into the present setting. We record only the following lemma, whose proof is immediate by reduction to the set-theoretic case.

Lemma 3.2.4. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups, and let G' be a $\mathscr{C}$-subgroup of G. The subobject G' of G is stable under the equivalence relation defined by H if and only if u factors through the canonical monomorphism $G' \to G$. Under this condition, the action of H on G' induced by the action on G defined by u is precisely the action arising from the monomorphism $H \to G'$ through which u factors.*

3.3. Universally Effective Equivalence Relations

Definition 3.3.1. *Let $f : X \to Y$ be a morphism. The* equivalence relation defined by f *in X, denoted $\mathscr{R}(f)$, is the $\widehat{\mathscr{C}}$-equivalence relation in X that is the image of the canonical monomorphism*

$$X \times_Y X \longrightarrow X \times X.$$

Definition 3.3.2. *An equivalence relation R in X is said to be* effective *if:*

(i) *R is representable, that is, it is a $\mathscr{C}$-equivalence relation;*

(ii) *the quotient* $Y = X/R$ *exists in* $\mathscr{C}$;[13]

(iii) *the diagram*

$$R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X \xrightarrow{p} Y$$

makes R *the fiber square of* X *over* Y, *that is,* R *is the equivalence relation defined by* p.

Scholium 3.3.2.1.[14] *If* R *is an effective equivalence relation in* X, *then* p *is an effective epimorphism (1.3). If* $f : X \to Y$ *is an effective epimorphism, then* $\mathscr{R}(f)$ *is an effective equivalence relation in* X, *one quotient of which is* Y. *Thus there is a one-to-one "Galois" correspondence between effective equivalence relations in* X *and effective quotients of* X, *that is, equivalence classes of effective epimorphisms with source* X.

Definition 3.3.3. *An equivalence relation* R *in* X *is said to be* universally effective *if the quotient* $Y = X/R$ *exists and, for every* $Y' \to Y$, *the fiber products* $X' = X \times_Y Y'$ *and* $R' = R \times_Y Y'$ *exist and* R' *is the fiber square of* X' *over* Y'. *Equivalently,* R *is effective and* $p : X \to X/R$ *is a universal effective epimorphism.*

Scholium 3.3.3.1.[14] *As above, there is a one-to-one correspondence between universally effective equivalence relations in* X *and universal effective quotients of* X.

Remark 3.3.3.2.[14] Suppose that $\mathscr{C}$ is the category of S-schemes, and write $\mathbf{A}^1_S$ for affine one-space over S. Let $R \subset X \times_S X$ be a universally effective equivalence relation, and let $p : X \to Y$ be its quotient. Then, for every open subset U of Y,

$$\mathcal{O}(U) = \mathrm{Hom}_S(U, \mathbf{A}^1_S)$$

is the set of elements φ of

$$\mathcal{O}(p^{-1}(U)) = \mathrm{Hom}_S(p^{-1}(U), \mathbf{A}^1_S)$$

such that $\varphi \circ \mathrm{pr}_1 = \varphi \circ \mathrm{pr}_2$. In particular, if R is defined by the action of a group H acting freely on the right on X (cf. 3.2.1), then $\mathcal{O}(U)$ is the set of all $\varphi \in \mathcal{O}(p^{-1}(U))$ such that

$$\varphi(xh) = \varphi(x)$$

for every $S' \to S$, $x \in X(S')$, and $h \in H(S')$.

Proposition 3.3.4. *Let* R *be a universally effective equivalence relation in* X. *Let* $f : X \to Z$ *be a morphism compatible with* R, *hence factoring through a morphism* $g : X/R \to Z$. *The following conditions are equivalent:*

(i) g *is a monomorphism;*

(ii) R *is the equivalence relation defined by* f.

Indeed, (i) plainly implies (ii), while the converse is precisely 2.5.

Definition 3.3.5. *Let* H *be a* $\mathscr{C}$*-group acting freely on* X. *One says that* H *acts* effectively, *or that the action of* H *on* X *is* effective *(respectively* universally effective*), if the equivalence relation in* X *defined by the action of* H *is effective (respectively universally effective).*

[13] Editor's note: the words "in $\mathscr{C}$" have been added.

[14] Editor's note: the numbering 3.3.2.1 (respectively 3.3.3.1) has been added for later reference; Remark 3.3.3.2 has also been added.

3.4. (M)-Effectivity

In practice, universal effective epimorphisms are most often difficult to characterize. Nevertheless, one frequently has at hand a number of morphisms of this kind–in scheme theory, for example, faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. We first state several conditions on a family (M) of morphisms of $\mathscr{C}$:

(a) (M) is stable under base change: every $u : T \to S$ belonging to (M) is squarable (cf. 1.4.0), and for every $S' \to S$, the base change
$$u' : T \times_S S' \longrightarrow S'$$
belongs to (M);

(b) the composite of two elements of (M) belongs to (M);

(c) every isomorphism belongs to (M);

(d) every element of (M) is an effective epimorphism.

Conditions (a) and (b) imply:

(a′) the cartesian product of two elements of (M) belongs to (M). More precisely, let $u : X \to Y$ and $u' : X' \to Y'$ be two S-morphisms belonging to (M). If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ belongs to (M).

This follows from the diagram

$$\begin{tikzcd}
 & Y' & X' \arrow[l, "u'"'] \\
X \arrow[d, "u"'] & X \times_S Y' \arrow[l] \arrow[u] \arrow[d] & X \times_S X' \arrow[l] \arrow[u] \arrow[dl, "u \times_S u'"] \\
Y & Y \times_S Y' \arrow[l] &
\end{tikzcd}$$

Likewise, (a) and (d) imply:

(d′) every element of (M) is a universal effective epimorphism.

3.4.2. The family (M_0) of universal effective epimorphisms satisfies conditions (a)–(d) of 3.4.1. Indeed, (a), (c), and (d) hold by definition, and (b) follows from 1.8. Henceforth we assume given a family (M) of morphisms of $\mathscr{C}$ satisfying these conditions; our results apply in particular to the family (M_0).

Definition 3.4.3. An equivalence relation R in X is said to be *of type* (M) if it is representable and $p_1 \in (M)$; conditions (b) and (c) then imply that $p_2 \in (M)$.

The relation R is said to be (M)-*effective* if it is effective and the canonical morphism $X \to X/R$ belongs to (M).

A quotient Y of X is said to be (M)-*effective* if the canonical morphism $X \to Y$ belongs to (M).

The following consequences are immediate from the definition.[15]

Proposition 3.4.3.1.

[15] Editor's note: the numbering 3.4.3.1 has been added for later reference, and the proof of part (i) has been expanded.

(i) *An (M)-effective equivalence relation is of type (M) and is universally effective.*

(ii) *An (M)-effective quotient is universally effective (cf. 3.3.3).*

(iii) *The maps $R \mapsto X/R$ and $p \mapsto \mathscr{R}(p)$ establish a one-to-one correspondence between (M)-effective equivalence relations in X and (M)-effective quotients of X.*

(iv) *To be (M_0)-effective is equivalent to being universally effective.*

We prove (i). Since R is (M)-effective, there is a cartesian square

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow{p} & X/R, \end{array}$$

and $p \in (M)$. By 3.4.1(a), p_1 and p_2 belong to (M), so R is of type (M).

Set $Y = X/R$, and let $Y' \to Y$ be any morphism. By 3.4.1(a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist, and the morphisms $X' \to Y'$ and $p_i' : R' \to X'$ belong to (M). Finally, since $R = X \times_Y X$, associativity of the fiber product gives

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

This shows that R' is (M)-effective; in particular, R is universally effective. This proves (i), as well as (iv). Parts (ii) and (iii) follow, taking Definition 3.3.2 into account.

3.4.4. Let H be an S-group whose structure morphism belongs to (M). If H acts freely on the S-object X, then it defines an equivalence relation of type (M) in X. Indeed, by (a), the fiber product $H \times_S X$ exists and $\mathrm{pr}_2 : H \times_S X \to X$ belongs to (M). The action of H is said to be (M)-*effective* if the equivalence relation in X defined by this action is (M)-effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M)-effective equivalence relation in X over S, and set $Y = X/R$. Let $S' \to S$ be a base change for which $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M)-effective equivalence relation in X' over S', and*

$$X'/R' \simeq (X/R)'.$$

Indeed, the canonical morphisms $X \to Y$ and $R \to Y$ belong to (M); hence, by (a′), X' and R' are representable. By associativity of the fiber product, R' is the equivalence relation in X' defined by the canonical morphism $X' \to Y'$, which belongs to (M). This proves the assertion.

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (respectively R') be an (M)-effective equivalence relation in X (respectively X') over S. If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M)-effective equivalence relation in $X \times_S X'$ over S, and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

Set $Y = X/R$ and $Y' = X'/R'$. By (a′), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q : X \times_S X' \longrightarrow Y \times_S Y'$$

belongs to (M). The formula

$$(X \times_S X') \times_{(Y \times_S Y')} (X \times_S X') \simeq (X \times_Y X) \times_S (X' \times_{Y'} X')$$

(all unlabelled fiber products being taken over S) shows that $R \times_S R'$ is the equivalence relation in $X \times_S X'$ defined by q, which completes the proof.

3.4.7.[16] Suppose that $\mathscr{C}$ has a final object e, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups belonging to (M). By 3.4.1(a), the kernel $\operatorname{Ker}(f)$ is represented by $e \times_{G'} G$, and the morphism $\operatorname{Ker}(f) \to e$ belongs to (M).

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $\operatorname{Ker}(f)$, say on the right, on G: it is the image of the morphism

$$G \times \operatorname{Ker}(f) \longrightarrow G \times G$$

defined on sets by $(g, h) \mapsto (g, gh)$. Consequently, Scholium 3.3.2.1 gives the following corollary.

Corollary 3.4.7.1. *Suppose that $\mathscr{C}$ has a final object e, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups belonging to (M). Then the action of* $\operatorname{Ker}(f)$ *on G is (M)-effective, and G' is the quotient $G/\operatorname{Ker}(f)$.*

3.5. Construction of Quotients by Descent

It frequently happens that one cannot construct a quotient directly, but can do so after a suitable base change. This subsection gives a useful criterion for that situation.

In 2.1 we defined a descent datum on an object X' over S', relative to a morphism $S' \to S$.

Definition 3.5.1. *A descent datum on X', relative to $S' \to S$, is said to be* effective *if X', equipped with this descent datum, is isomorphic to the inverse image over S' of an object X over S, equipped with its canonical descent datum.*

If $S' \to S$ is a descent morphism (2.2), then the object X in the definition is unique up to unique isomorphism. To say that $S' \to S$ is an effective descent morphism (2.2) means that it is a descent morphism and that every descent datum relative to it is effective.

Now consider an equivalence relation R in an object X over S. Let X', X'', and X''' be the inverse images of X over

$$S', \qquad S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S',$$

respectively, and let R', R'', and R''' be the equivalence relations obtained from R by inverse image. Suppose that R' in X' is (M)-effective, and consider the quotient $Y' = X'/R'$, an object over S'. By 3.4.5, its two inverse images over S'' are isomorphic to X''/R''. Thus the S'-object Y' has a canonical gluing datum. Applying in the same way the uniqueness of X'''/R''', one sees that this is in fact a descent datum. (In this argument it has been tacitly assumed that all the displayed fiber products exist. This holds, in particular, if $S' \to S$ is squarable, for example if it is a descent morphism.)

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S, and let $S' \to S$ be a universal effective epimorphism. Assume that every S-morphism whose inverse image over S' belongs to (M) itself belongs to (M). The following conditions are equivalent:*

(i) *R is (M)-effective in X;*

(ii) *R' is (M)-effective in X', and the canonical descent datum on X'/R' is effective.*

[16] Editor's note: this paragraph has been added.

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R.

The implication (i)⇒(ii) is simply 3.4.4 translated into the language of descent. Once the converse is proved, the last assertion of the proposition follows from the fact that a universal effective epimorphism is a descent morphism (2.3), and hence the descended object is unique up to unique isomorphism.

We prove (ii)⇒(i). Let $Y' = X'/R'$, and let Y be the descended object. The canonical morphism $p' : X' \to X'/R' = Y'$ is compatible by construction with the descent data: its two inverse images over S'' coincide with the canonical morphism $X'' \to X''/R''$. It is therefore the inverse image over S' of an S-morphism $p : X \to Y$. Since p' belongs to (M), the assumption on $S' \to S$ implies that p also belongs to (M). Moreover, because p' is compatible with the equivalence relation R', the morphism p is compatible with R, again because a universal effective epimorphism is a descent morphism. Hence there is a morphism

$$R \longrightarrow X \times_Y X.$$

To prove that R is (M)-effective and that Y is isomorphic to X/R, it suffices to prove that this morphism is an isomorphism. After base change from S to S', it becomes an isomorphism because R' is effective; it is therefore an isomorphism for the same reason as in 2.4.

Notice that the hypothesis in the proposition is satisfied if (M) is the family (M_0) of universal effective epimorphisms and if $\mathscr{C}$ has fiber products (1.10). We obtain the following.

Corollary 3.5.3. *Suppose that $\mathscr{C}$ has fiber products (fiber products over S would suffice). Let R be an equivalence relation in X over S, and let $S' \to S$ be a universal effective epimorphism. The following conditions are equivalent:*

(i) *R is universally effective in X;*

(ii) *R' is universally effective in X', and the canonical descent datum on X'/R' is effective.*

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R.

4. Topologies and Sheaves

The notion of a sieve, and the presentation of the notion of a topology adopted here (4.2.1), which is more intrinsic and in many respects more convenient than the presentation by covering families in [MA], are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. *A* sieve *of the category $\mathscr{C}$ is a subfunctor C of the final functor*

$$e : \mathscr{C}^\circ \longrightarrow \mathbf{Set}.$$

To every sieve C of $\mathscr{C}$ one associates the set $E(C)$ of objects X of $\mathscr{C}$ such that $C(X) \neq \varnothing$, that is, such that the structural morphism $X \to e$ factors through C. Thus

$$\begin{cases} X \in E(C) & \iff \quad C(X) = e(X) = \{\varnothing\}, \\ X \notin E(C) & \iff \quad C(X) = \varnothing. \end{cases} \tag{+}$$

The set $E = E(C)$ has the following property:

$$X \in E \text{ and } \operatorname{Hom}(Y, X) \neq \varnothing \quad \Longrightarrow \quad Y \in E. \tag{++}$$

(If the set $\operatorname{Ob}\mathscr{C}$ is endowed with its natural preorder, with Y dominating X when there is an arrow from Y to X, then the sets E satisfying (++) are precisely the subsets of $\operatorname{Ob}\mathscr{C}$ that contain every object dominating one of their elements.)[17]

Conversely, if E is a subset of $\operatorname{Ob}\mathscr{C}$ having property (++), then E can be written uniquely in the form $E(C)$, with C defined by the formulas (+). Thus sieves of $\mathscr{C}$ correspond bijectively to subsets of $\operatorname{Ob}\mathscr{C}$ satisfying (++). By abuse of language, we shall sometimes call the set $E(C)$ itself a sieve of $\mathscr{C}$.

Let C and C' be two sieves of $\mathscr{C}$.[18] Since both are subfunctors of the final functor e, it is equivalent to say that C dominates C' (for the domination relation in $\operatorname{Ob}\widehat{\mathscr{C}}$), that C is a subfunctor of C', or that $E(C) \subset E(C')$. In this case we say that C is *finer* than C'. This is an order relation on the set of sieves of $\mathscr{C}$. Moreover,

$$E(C) \cap E(C') = E(C \times C'),$$

so the set of sieves of $\mathscr{C}$ is filtered for the order relation "is finer than."

Every subset E of $\operatorname{Ob}\widehat{\mathscr{C}}$, and in particular every subset of $\operatorname{Ob}\mathscr{C}$, defines a sieve $C(E)$: the set of $X \in \operatorname{Ob}\mathscr{C}$ such that $F(X) \neq \varnothing$ for at least one $F \in E$ satisfies (++), and hence defines the desired sieve.

This sieve may also be defined as the image of the family of morphisms $\{F \to e \mid F \in E\}$, in the sense of the following definition.

Definition 4.1.2. *Let $\{F_i \to F\}$ be a family of morphisms of $\widehat{\mathscr{C}}$ with common target F. The* image *of this family is the subfunctor of F defined by*

$$S \longmapsto \bigcup_i \operatorname{Im} F_i(S) \subset F(S).$$

Proposition 4.1.3. *Formation of the image commutes with base change. In the preceding notation let I be the image of the family $\{F_i \to F\}$. For every morphism $G \to F$ in $\widehat{\mathscr{C}}$, the image of the family*

$$\{F_i \times_F G \longrightarrow G\}$$

is the subfunctor $I \times_F G$ of G.

Definition 4.1.4.0.[19] *Let C be a sieve of $\mathscr{C}$. If E is a subset of $\operatorname{Ob}\mathscr{C}$ such that $C(E) = C$, then E is called a* base *of C. Every sieve C has a base, for example $E(C)$.*

We now describe $\operatorname{Hom}(C, F)$, where C is a sieve of $\mathscr{C}$ and $F \in \operatorname{Ob}\widehat{\mathscr{C}}$, by means of a base $\{S_i\}$ of C. For every pair (i, j) there is a diagram in $\widehat{\mathscr{C}}$:

$$\begin{array}{ccc} S_i \times S_j & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ S_j & \longrightarrow & e. \end{array}$$

[17] Editor's note: Here "dominating" is meant in the sense of the preorder just mentioned: Y dominates X if there is an arrow $Y \to X$. On the other hand, if X, Y are two subobjects of an object Z, one says (cf. 2.4) that Y contains X if $X \subset Y$. To avoid any ambiguity between these usages, the sequel uses "dominating" in the first case and "containing" in the second.

[18] Editor's note: The following sentence has been expanded.

[19] Editor's note: The numbering 4.1.4.0 has been added to highlight this definition.

It gives a diagram of sets

$$\Gamma(F) = \operatorname{Hom}(e, F) \xrightarrow{\sigma} \prod_i \operatorname{Hom}(S_i, F) \underset{\tau_2}{\overset{\tau_1}{\rightrightarrows}} \prod_{i,j} \operatorname{Hom}(S_i \times S_j, F),$$

with $\tau_1\sigma = \tau_2\sigma$. Hence there is a morphism

$$\operatorname{Hom}(e, F) \longrightarrow \operatorname{Ker}\left(\prod_i \operatorname{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \operatorname{Hom}(S_i \times S_j, F)\right).$$

One checks immediately:

Proposition 4.1.4. *There is an isomorphism, functorial in F,*

$$\operatorname{Hom}(C, F) \xrightarrow{\sim} \operatorname{Ker}\left(\prod_i \operatorname{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \operatorname{Hom}(S_i \times S_j, F)\right),$$

for which the diagram

$$\begin{array}{ccc}
\operatorname{Hom}(e, F) & \longrightarrow & \operatorname{Ker}\left(\prod_i \operatorname{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \operatorname{Hom}(S_i \times S_j, F)\right) \\
\| & & \uparrow \wr \\
\operatorname{Hom}(e, F) & \longrightarrow & \operatorname{Hom}(C, F)
\end{array}$$

is commutative; the bottom arrow is induced by the canonical morphism $C \to e$.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example that the S_i are squarable. Then, for every $F \in \operatorname{Ob}\widehat{\mathscr{C}}$, there is an isomorphism*

$$\operatorname{Hom}(C, F) \xrightarrow{\sim} \operatorname{Ker}\left(\prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times S_j)\right).$$

Remark 4.1.6. Let R be a sieve of $\mathscr{C}$, let $\overline{R}$ denote the full subcategory of $\mathscr{C}$ whose object set is $E(R)$, and let

$$i_R : \overline{R} \longrightarrow \mathscr{C}$$

be the inclusion functor. There is an isomorphism, functorial in $F \in \operatorname{Ob}\widehat{\mathscr{C}}$,

$$\operatorname{Hom}(R, F) \xrightarrow{\sim} \Gamma(F \circ i_R),$$

for which the diagram

$$\begin{array}{ccc}
\operatorname{Hom}(e, F) & \longrightarrow & \operatorname{Hom}(R, F) \\
\wr\downarrow & & \wr\downarrow \\
\Gamma(F) & \longrightarrow & \Gamma(F \circ i_R)
\end{array}$$

is commutative; the bottom arrow is induced by i_R.

Definition 4.1.7. *Let $\mathscr{C}$ be a category. A* sieve of the object *S of $\mathscr{C}$ is a sieve of the category $\mathscr{C}_{/S}$.*

A sieve of S is therefore a subobject of S in $\widehat{\mathscr{C}}$. It corresponds canonically to a subset of $\operatorname{Ob}\mathscr{C}_{/S}$ that contains the source of every arrow whose target it contains. By abuse of language, such a subset will also be called a sieve of S.

4.2. Topologies: Definitions

Definition 4.2.1. *A* topology *on a category $\mathscr{C}$ consists, for every object S of $\mathscr{C}$, of a set $J(S)$ of sieves of S, called* covering sieves *or* refinements *of S, satisfying the following axioms:*

(T 1) *For every refinement R of S and every morphism $T \to S$, the sieve $R \times_S T$ of T is covering ("stability under base change").*

(T 2) *If R, C are two sieves of S, if R is covering, and if for every $T \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $T \to R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S.*[20]

(T 3) *If $C \supset R$ are two sieves of S and R is covering, then C is covering.*

(T 4) *For every S, S is a refinement of S.*

These axioms may be reformulated as follows. Suppose that a topology $S \mapsto J(S)$ on $\mathscr{C}$ is given. For every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, let $J(F)$ be the set of subfunctors R of F such that, for every morphism $T \to F$ in $\widehat{\mathscr{C}}$ with T representable, the sieve $R \times_F T$ of T is covering. By (T 1), this notation agrees with the preceding one. An $R \in J(F)$ is again called a *refinement* of F. The preceding axioms immediately imply:

($T'0$) If $F \supset G$ are objects of $\widehat{\mathscr{C}}$, and if $G \times_F S \in J(S)$ for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $S \to F$, then $G \in J(F)$.

($T'1$) If $G \in J(F)$ and $H \to F$ is a morphism in $\widehat{\mathscr{C}}$, then $G \times_F H \in J(H)$.

($T'2$) If $F \supset G \supset H$, if $G \in J(F)$, and if $H \in J(G)$, then $H \in J(F)$.

($T'3$) If $F \supset G \supset H$ and $H \in J(F)$, then $G \in J(F)$.

($T'4$) For every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, one has $F \in J(F)$.

Conversely, if for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$ one is given a set $J(F)$ of subobjects of F satisfying $T'0$–$T'4$, then $S \mapsto J(S)$ defines a topology on $\mathscr{C}$, and these two constructions are inverse to each other.

Properties $T'1$, $T'2$, and $T'3$ imply the further property[21]:

($T'5$) If G and H are two subobjects of F and $G, H \in J(F)$, then $G \cap H \in J(F)$.

Thus $J(F)$, ordered by $\supset$, is filtered. This observation will be useful below.

4.2.2. The topology defined by J is said to be *finer* than the topology defined by J' if $J(S) \supset J'(S)$ for every $S \in \mathrm{Ob}\,\mathscr{C}$. Equivalently, $J(F) \supset J'(F)$ for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$.

Every set of topologies on $\mathscr{C}$ has a greatest lower bound. More precisely, let I be an index set and, for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on $\mathscr{C}$. Then

$$J(S) = \bigcap_{i \in I} J_i(S)$$

defines a topology, and it is the greatest lower bound of the given set.

In particular, suppose that for every $S \in \mathrm{Ob}\,\mathscr{C}$ a set $E(S)$ of sieves of S is given. The *topology generated by these sets* is the least fine topology for which every element of $E(S)$ is a refinement of S.

[20] Editor's note: In other words, if C is covering "locally with respect to the covering sieve R," then C is covering.

[21] Editor's note: $T'1$ and $T'2$ suffice: $G \cap H = G \times_F H$ belongs to $J(H)$ by $T'1$, and hence to $J(F)$ by $T'2$.

Definition 4.2.3. *Let $\{F_i \to F\}$ be a family of morphisms in $\widehat{\mathscr{C}}$, and let $G \subset F$ be its image (4.1.2). The family is* covering *if $G \in J(F)$. A morphism is covering if the one-member family consisting of that morphism is covering.*

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \to S$ is covering.

Axioms $T'0$–$T'5$ give the following properties of covering families:

(C 0) If $\{F_i \to F\}$ is a family in $\widehat{\mathscr{C}}$ such that every representable base change $\{F_i \times_F S \to S\}$ is covering, then the original family is covering.

(C 1) If $\{F_i \to F\}$ is covering and $G \to F$ is any morphism, then $\{F_i \times_F G \to G\}$ is covering ("stability under base change").

(C 2) If $\{F_i \to F\}$ is covering and, for every i, $\{F_{ij} \to F_i\}$ is covering, then the composite family $\{F_{ij} \to F\}$ is covering ("stability under composition").

(C 3) If $\{G_j \to F\}$ is covering and $\{F_i \to F\}$ is a family such that for every j there is an i for which $G_j \to F$ factors through $F_i \to F$, then $\{F_i \to F\}$ is covering ("saturation").

(C 4) Every one-member family consisting of an isomorphism is covering.

Notice that (C 2) and (C 3) also imply:

(C 5) If $\{F_i \to F\}$ is a family for which there exists a covering family $\{G_j \to F\}$ such that $\{F_i \times_F G_j \to G_j\}$ is covering for every j, then $\{F_i \to F\}$ is covering ("a locally covering family is covering").

4.2.4. Conversely, let $\mathscr{C}$ be a category having fiber products, and suppose that for every $S \in \operatorname{Ob}\mathscr{C}$ one is given a set of families of morphisms in $\mathscr{C}$ with target S, called covering families, satisfying (C 1)–(C 4), and hence also (C 5). Let $J(S)$ be the set of sieves of S having a covering base (or, equivalently by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on $\mathscr{C}$, and the two preceding constructions are inverse to one another.

In applications it is inconvenient to consider all covering families, since one often has a fairly simple description of a "sufficient" number of them. This leads to the following definitions.

Definition 4.2.5.0.[22] *Let $\mathscr{C}$ be a category, and suppose given, for every $S \in \operatorname{Ob}\mathscr{C}$, a set $P(S)$ of families of morphisms in $\mathscr{C}$ with target S. The* topology generated by P *is the least fine topology for which the given families are covering.*

Definition 4.2.5. *A* pretopology *on a category $\mathscr{C}$ consists, for every $S \in \operatorname{Ob}\mathscr{C}$, of a set $R(S)$ of families $\{S_i \to S\}$, called* covering for the pretopology*, satisfying:*

(P 1) *For every $\{S_i \to S\} \in R(S)$ and every morphism $T \to S$, the fiber products $S_i \times_S T$ exist and $\{S_i \times_S T \to T\} \in R(T)$.*

(P 2) *If $\{S_i \to S\} \in R(S)$ and, for each i, $\{T_{ij} \to S_i\} \in R(S_i)$, then the composite family $\{T_{ij} \to S\}$ belongs to $R(S)$.*

(P 3) *Every one-member family consisting of an isomorphism is covering.*

[22] Editor's note: This definition, which appeared after 4.2.5 in the original, has been placed here because it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

Proposition 4.2.6. *For each S, let $J(S)$ be the set of sieves of S that are covering for the topology generated by the pretopology R. Let $J_R(S)$ be the subset of $J(S)$ consisting of the sieves defined by the families in $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family in $R(S)$.*

For every S, let $J'(S)$ be the set of sieves of S containing a sieve in $J_R(S)$. Clearly $J'(S) \subset J(S)$. To prove $J(S) = J'(S)$, it suffices to prove that the $J'(S)$ define a topology, that is, satisfy (T 1)–(T 4). Axioms (T 1), (T 3), and (T 4) are immediate; only (T 2) remains.[23]

Let $U \in J'(S)$, let C be a sieve of S, and suppose that $C \times_S T \in J'(T)$ for every $T \to U$. We must prove $C \in J'(S)$. By definition, U contains a refinement U' defined by a family $\{S_i \to S\} \in R(S)$. Since (T 3) has already been verified, it is enough to prove $U' \cap C \in J'(S)$; hence we may suppose $U = U'$. By hypothesis, $C \times_S S_i \in J'(S_i)$ for every i. Thus for each i there is a covering family $\{T_{ij} \to S_i\} \in R(S_i)$ such that $T_{ij} \to S_i$ factors through $C \times_S S_i \to S_i$. Consequently $T_{ij} \to S$ factors through $C \to S$. Hence C contains the sieve defined by the composite family $\{T_{ij} \to S\}$, and (P 2) completes the proof.

Axioms (P 1)–(P 3) are those of [MA]. Because pretopologies are so useful in practice, we shall interpret every important result in terms of a pretopology defining the topology under consideration.

Remark 4.2.7. One may introduce a slightly more general notion: for every S, one specifies a set of covering families satisfying (P 1), (P 3), and the conclusion of Proposition 4.2.6. This occurs in particular when the given families satisfy (P 1), (P 3), and (C 5). See [D].

Definition 4.2.8. *Let $\mathscr{C}$ be equipped with a topology, let S be an object of $\mathscr{C}$, and let $P(S')$ be a condition on an argument $S' \in \operatorname{Ob}\mathscr{C}_{/S}$. Suppose that $\operatorname{Hom}(S'', S') \neq \varnothing$ implies $P(S') \Rightarrow P(S'')$. The condition P is said to hold* locally *on S for the topology under consideration if the following equivalent conditions hold:*

(i) *The set of $S' \to S$ for which $P(S')$ holds is a refinement of S.*

(ii) *There is a refinement of S such that $P(S')$ holds for every S' in that refinement.*

(iii) *If the topology is defined by a pretopology, there is a covering family for that pretopology such that $P(S')$ holds for every S' in the family.*

Example 4.2.9. *Let $f : X \to Y$ be an S-morphism. We say that f is* locally an isomorphism *if there is a covering family $\{S_i \to S\}$ such that $f \times_S S_i$ is an isomorphism for every i. Equivalently, there is a refinement R of S such that $X(T) \to Y(T)$ is an isomorphism for every $T \to R$.*

Many other examples of this “local” language will appear below.

4.3. Presheaves, Sheaves, and the Sheaf Associated to a Presheaf

Definition 4.3.1. *Let $\mathscr{C}$ be a category. A* presheaf of sets *on $\mathscr{C}$ is a contravariant functor from $\mathscr{C}$ to the category of sets. The category*

$$\widehat{\mathscr{C}} = \operatorname{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

is called the category of presheaves *on $\mathscr{C}$. If $\mathscr{C}$ is equipped with a topology, a presheaf P is* separated *(respectively, is a* sheaf*) if, for every $S \in \operatorname{Ob}\mathscr{C}$ and every $R \in J(S)$, the canonical map*

$$P(S) = \operatorname{Hom}(S, P) \longrightarrow \operatorname{Hom}(R, P) \tag{+}$$

[23] Editor's note: Typographical errors in the original have been corrected in what follows.

is injective (respectively, bijective). The full subcategory of $\widehat{\mathscr{C}}$ *whose objects are the sheaves is called the* category of sheaves *and is denoted by* $\widetilde{\mathscr{C}}$.[24]

Proposition 4.3.2. *Let P be a separated presheaf (respectively, a sheaf). For every $H \in \operatorname{Ob}\widehat{\mathscr{C}}$ and every $R \in J(H)$, the canonical map*

$$\operatorname{Hom}(H, P) \longrightarrow \operatorname{Hom}(R, P) \tag{+}$$

is injective (respectively, bijective).

Let P be separated, let H be a presheaf, let $R \in J(H)$, and let $u, v : H \to P$ satisfy $uj = vj$. For every $f : S \to H$, with $S \in \operatorname{Ob}\mathscr{C}$, the sieve $R \times_H S$ is a refinement of S, and $ufj_S = vfj_S$. This is expressed by the diagram

$$\begin{tikzcd} R \times_H S \arrow[r, "j_S"] \arrow[d, "f_R"] & S \arrow[d, "f"] & \\ R \arrow[r, "j"] & H \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & P. \end{tikzcd}$$

Since P is separated, it follows that $uf = vf$. This holds for every representable S, so $u = v$.

Now suppose that P is a sheaf. Let $g : R \to P$; we show that it factors through H. For every $f : S \to H$, with $S \in \operatorname{Ob}\mathscr{C}$, the morphism $g \circ f_R : R \times_H S \to P$ factors uniquely through S, thereby defining a morphism $h : S \to P$. By uniqueness, h is plainly functorial in f:

$$\begin{tikzcd} R \times_H S \arrow[r, "f_R"] \arrow[d, "j_S"] & R \arrow[r, "g"] \arrow[d, "j"] & P \\ S \arrow[r, "f"] \arrow[rru, bend right, "h"'] & H \arrow[ru, dashed] & \end{tikzcd}$$

Thus, for every S, we have defined a map $H(S) \to P(S)$ functorial in S, and hence a morphism $H \to P$ with the required properties.

Corollary 4.3.2.1.[25] *Let R and F be two sheaves. If R is a refinement of F, then $R = F$.*

Indeed, suppose that R is a refinement of F, and write $j : R \hookrightarrow F$ for the inclusion. By 4.3.2, $\operatorname{Hom}(F, R) = \operatorname{End}(R)$; hence there is a $\pi : F \to R$ such that $\pi j = \mathrm{id}_R$. Likewise, $\operatorname{End}(F) = \operatorname{Hom}(R, F)$, and $j\pi j = j$ implies $j\pi = \mathrm{id}_F$. Thus j is an isomorphism.

Proposition 4.3.3 ([AS], 1.3). *Let $\mathscr{C}$ be a category and P a presheaf on $\mathscr{C}$. For every $S \in \operatorname{Ob}\mathscr{C}$, let $J(S)$ be the set of sieves R of S such that, for every $T \to S$, the map*

$$\operatorname{Hom}(T, P) \longrightarrow \operatorname{Hom}(R \times_S T, P) \tag{+}$$

is injective (respectively, bijective). Then the $J(S)$ define a topology on $\mathscr{C}$; that is, they satisfy (T 1)–(T 4).

Corollary 4.3.4. *For every $S \in \operatorname{Ob}\mathscr{C}$, let $K(S)$ be a family of sieves satisfying (T 1), and let P be a presheaf on $\mathscr{C}$. Then P is separated (respectively, a sheaf) for the topology generated by the $K(S)$ if and only if, for every $S \in \operatorname{Ob}\mathscr{C}$ and every $R \in K(S)$, the canonical map*

$$\operatorname{Hom}(S, P) \longrightarrow \operatorname{Hom}(R, P) \tag{+}$$

[24] Editor's note: If Q is a subpresheaf of a separated presheaf P, then Q is separated. Indeed, for every sieve R of S, the composite $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \to Q(R)$.

[25] Editor's note: This corollary has been added.

is injective (respectively, bijective).

Corollary 4.3.5. *For each $S \in \operatorname{Ob}\mathscr{C}$, let $R(S)$ be a set of families of morphisms in $\mathscr{C}$ with target S, satisfying (P 1), for example a pretopology. Let P be a presheaf on $\mathscr{C}$. Then P is separated (respectively, a sheaf) for the topology generated by R if and only if, for every $S \in \operatorname{Ob}\mathscr{C}$ and every family $\{S_i \to S\} \in R(S)$, the map*

$$P(S) \longrightarrow \prod_i P(S_i)$$

is injective (respectively, the diagram

$$P(S) \longrightarrow \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

is exact).

Definition 4.3.6. *The* canonical topology *on a category $\mathscr{C}$ is the finest topology for which every representable functor is a sheaf.*

Corollary 4.3.7. *A sieve R of S is a refinement for the canonical topology if and only if, for every morphism $T \to S$ in $\mathscr{C}$ and every $X \in \operatorname{Ob}\mathscr{C}$, the canonical map*

$$\operatorname{Hom}(T, X) \longrightarrow \operatorname{Hom}(R \times_S T, X)$$

is bijective.

Definition 4.3.8. *A sieve covering for the canonical topology is called a* universal effective epimorphic sieve*.*

Corollary 4.3.9. *A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every squarable family defining a universal effective epimorphic sieve is universal effective epimorphic.*

Return now to a category $\mathscr{C}$ equipped with an arbitrary topology and to the construction of the sheaf associated to a presheaf P. Let S be an object of $\mathscr{C}$. If $R \supset R'$ are two refinements of S, there is a diagram

$$\begin{tikzcd} \operatorname{Hom}(S, P) \arrow[r] \arrow[dr] & \operatorname{Hom}(R, P) \arrow[d] \\ & \operatorname{Hom}(R', P). \end{tikzcd}$$

As already observed, the ordered set $J(S)$ is filtered. Since $S \in J(S)$, there is an evident morphism

$$\operatorname{Hom}(S, P) \longrightarrow \varinjlim_{R \in J(S)} \operatorname{Hom}(R, P).$$

Definition 4.3.10.0.[26] Set

$$\check{H}^0(S, P) = \varinjlim_{R \in J(S)} \operatorname{Hom}(R, P).$$

One checks that $\check{H}^0(S, P)$ depends functorially on S, and hence defines a functor LP by

$$\operatorname{Hom}(S, LP) = \check{H}^0(S, P) = \varinjlim_{R \in J(S)} \operatorname{Hom}(R, P). \tag{++}$$

[26]Editor's note: The numbering 4.3.10.0 has been added for later reference. Moreover, it follows from the definition that if $Q \to P$ is a monomorphism, then the same is true of $LP \to LQ$; thus L "preserves monomorphisms" (see also 4.3.16 for the more general statement that L "commutes with finite inverse limits").

By construction there are morphisms

$$\ell_P : P \longrightarrow LP, \qquad z_R : \mathrm{Hom}(R, P) \longrightarrow \mathrm{Hom}(S, LP).$$

Lemma 4.3.10.

(i) *For every refinement R of S and every $u : R \to P$, the diagram*

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ {\scriptstyle u}\uparrow & & \uparrow{\scriptstyle z_R(u)} \\ R & \xrightarrow{i_R} & S \end{array}$$

is commutative.

(ii) *For every morphism $v : S \to LP$, there exist a refinement R of S and a morphism $u : R \to P$ such that $v = z_R(u)$.*

(iii) *Let Q be a functor and let $u, v : Q \to P$ satisfy $\ell_P u = \ell_P v$. Then the kernel of the pair (u, v) is a refinement of Q.*

(iv) *Let $u : R \to P$ and $u' : R' \to P$. The equality $z_R(u) = z_{R'}(u')$ holds if and only if there is a refinement $R'' \subset R \cap R'$ of S on which u and u' coincide.*

Proof. For (i), we must prove $z_R(u)i_R = \ell_P u$. It is enough to compare the composites of these two morphisms with every morphism $g : T \to R$, where T is representable. Put $f = i_R g$ and $R' = R \times_S T$.[27] We have

$$\begin{array}{ccc} P & \xrightarrow{\ell_P} & LP \\ {\scriptstyle u}\uparrow & & \uparrow{\scriptstyle z_R(u)} \\ R & \xrightarrow{i_R} & S \\ {\scriptstyle p}\uparrow & \nwarrow{\scriptstyle g} & \uparrow{\scriptstyle f} \\ R' & \xrightarrow[\sim]{i_{R'}} & T. \end{array}$$

By the definition of ℓ_P, $\ell_P ug = z_R(u)f$; this is the special case of the assertion in which i_R is an isomorphism. But $z_R(u)f = z_R(u)i_R g$, proving (i).

Assertions (ii) and (iv) merely express the definition of $\mathrm{Hom}(S, LP)$ as a direct limit.

For (iii), let K be the kernel of (u, v). For every morphism $f : S \to Q$, with S representable, $K \times_Q S$ is a subfunctor of the kernel of $uf, vf : S \to P$. By $T'0$, it is therefore enough to prove the assertion when $Q = S$ is representable. In that case, (ii) and (iv) show that K contains a refinement of S, and hence is itself a refinement of S.

One finally checks that $P \mapsto LP$ defines a functor

$$L : \widehat{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}},$$

and that $P \mapsto \ell_P$ defines a morphism of functors

$$\ell : \mathrm{Id}_{\widehat{\mathscr{C}}} \longrightarrow L.$$

[27]Editor's note: Let $p : R' \to R$ be the projection. Since i_R is a monomorphism, $p = gi_{R'}$. Let g' be the section of $i_{R'}$ defined by g. Then $pg' = g$, hence $pg' i_{R'} = gi_{R'} = p$. This implies $g' i_{R'} = \mathrm{id}_{R'}$, so $i_{R'} : R' \to T$ is an isomorphism with inverse g'.

We now state the essential result.

Proposition 4.3.11.

(i) *For every presheaf P, the presheaf LP is separated and $\ell_P : P \to LP$ is covering (4.2.3).*

(ii) *If P is a sheaf, then $P \to LP$ is an isomorphism.*

(iii) *For every presheaf P and every separated presheaf (respectively, sheaf) F, the map*

$$\mathrm{Hom}(\ell_P, F) : \mathrm{Hom}(LP, F) \longrightarrow \mathrm{Hom}(P, F)$$

is injective (respectively, bijective).

(iv) *If P is separated, then $\ell_P : P \to LP$ is a covering monomorphism (so P is a refinement of LP), and LP is a sheaf.*

Proof. (i) First, $P \to LP$ is covering. Indeed, for every $v : S \to LP$, Lemma 4.3.10(i),(ii) supplies a refinement R of S and a commutative diagram (the map from R to the fiber product is induced by $v' : R \to P$ and $i_R : R \to S$)

$$\begin{tikzcd}
 & P \arrow[r, "\ell_P"] & LP \\
 & P \times_{LP} S \arrow[u] \arrow[r] & S \arrow[u, "v"'] \\
R \arrow[ur, bend left, "v'"] \arrow[u, phantom] \arrow[ur] \arrow[urr, bend right, "i_R"'] & &
\end{tikzcd}$$

so $P \times_{LP} S \to S$ is covering. By (C 0), $P \to LP$ is covering.

Now suppose that two morphisms $v_1, v_2 : S \to LP$ induce the same morphism on a refinement R of S. There are refinements R_i, $i = 1, 2$, and morphisms $u_i : R_i \to P$ with $z_{R_i}(u_i) = v_i$. Replacing R by a smaller refinement, we may suppose $R_1 = R_2 = R$. The commutative diagram of Lemma 4.3.10(i) then gives $\ell_P u_1 = \ell_P u_2$. By 4.3.10(iii), u_1 and u_2 coincide on a refinement of R, hence on a refinement of S; by 4.3.10(iv), $z_R(u_1) = z_R(u_2)$. Thus LP is separated.

(ii) This is clear: if P is a sheaf, then $\mathrm{Hom}(S, P) \to \mathrm{Hom}(R, P)$ is already an isomorphism for every refinement R of S.

(iii) Let $u, v : LP \to F$ satisfy $u\ell_P = v\ell_P$. To show $u = v$, it is enough to prove $uf = vf$ for every $f : S \to LP$ with S representable. There are a refinement R of S and a morphism $g : R \to P$ with $f = z_R(g)$. The morphisms uf and vf agree on R, where both equal $u\ell_P g = v\ell_P g$. If F is separated, they therefore agree on S, proving injectivity. If F is a sheaf, the commutative diagram

$$\begin{tikzcd}
P \arrow[r, "\ell_P"] \arrow[d] & LP \arrow[d] \\
F \arrow[r, "\sim"] & LF
\end{tikzcd}$$

shows that $\mathrm{Hom}(\ell_P, F)$ is also surjective.

(iv) Suppose that P is separated. To show that $P \to LP$ is a monomorphism, it is enough to consider $u, v : S \to P$, with S representable, satisfying $\ell_P u = \ell_P v$. By 4.3.10(iii),

u and v agree on a refinement of S, and therefore agree because P is separated. Thus $P \to LP$ is a monomorphism; since it is covering by (i), P is a refinement of LP.

It remains to show that LP is a sheaf. We already know from (i) that it is separated. Let $S \in \operatorname{Ob}\mathscr{C}$, let R be a refinement of S, and let $h : R \to LP$. We must find $u : S \to LP$ with $ui_R = h$. Since P is a refinement of LP, $R' = P \times_{LP} R$ is a refinement of R, and hence of S. Put $u = z_{R'}(h')$; the situation is

$$\begin{tikzcd} P \arrow[r, "\ell_P"] & LP & \\ R' \arrow[u, "h'"] \arrow[r, "j"] & R \arrow[u, "h"] \arrow[r, "i_R"] & S. \arrow[ul, "u"'] \end{tikzcd}$$

Here the diagonal arrow is $u : S \to LP$. We have $ui_{R'} = \ell_P h' = hj$, hence $ui_R j = hj$. Since R' is a refinement of R and LP is separated, Proposition 4.3.2 gives $ui_R = h$.

Corollary 4.3.12.[28] *Let F be a sheaf and let R be a subobject of F in $\widehat{\mathscr{C}}$. Then R is a separated presheaf, $\ell_R : R \to LR$ is a covering monomorphism, and there is a commutative diagram*

$$\begin{tikzcd} R \arrow[r, "i"] \arrow[dr, "\ell_R"'] & F \\ & LR. \arrow[u, "j"'] \end{tikzcd}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

We already observed that R is separated and that j is a monomorphism; see editor notes (24) and (26). By Proposition 4.3.11(iv), ℓ_R is a covering monomorphism. Thus, if j is an isomorphism, R is a refinement of F. Conversely, if i is covering, then so is j, and Proposition 4.3.2.1 shows that j is an isomorphism.

Remark 4.3.13. If $J'(S)$ is a cofinal subset of $J(S)$, then

$$\operatorname{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \operatorname{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L can be described using the covering families belonging to $R(S)$. Writing out the preceding formula recovers the construction of [MA].

Let $i : \widetilde{\mathscr{C}} \to \widehat{\mathscr{C}}$ be the inclusion functor. Proposition 4.3.11 gives the following theorem.

Theorem 4.3.14. *There is a unique functor $a : \widehat{\mathscr{C}} \to \widetilde{\mathscr{C}}$ for which the diagram*

$$\begin{tikzcd} \widehat{\mathscr{C}} \arrow[r, "L"] \arrow[d, "a"] & \widehat{\mathscr{C}} \arrow[d, "L"] \\ \widetilde{\mathscr{C}} \arrow[r, "i"] & \widehat{\mathscr{C}} \end{tikzcd}$$

is commutative; that is, $L(L(P))$ is a sheaf for every presheaf P. The functors i and a are adjoint: for every presheaf P and every sheaf F, there is an isomorphism, functorial in P and F,

$$\operatorname{Hom}_{\widehat{\mathscr{C}}}(P, i(F)) \simeq \operatorname{Hom}_{\widetilde{\mathscr{C}}}(a(P), F),$$

or simply

$$\operatorname{Hom}(P, F) \simeq \operatorname{Hom}(a(P), F).$$

[28] Editor's note: The statement and proof of this corollary have been expanded.

Definition 4.3.15. *The sheaf $a(P)$ is called the sheaf* associated *to the presheaf P.*

Remark 4.3.16. Since L is constructed using inverse limits and filtered direct limits, it commutes with finite inverse limits.[29]

Moreover, under the identification $L(P\times P) = LP\times LP$, the morphism $\ell_{P\times P}$ is identified with $\ell_P \times \ell_P$. Thus, for example, if P is a presheaf of groups, then LP is canonically a presheaf of groups and the canonical morphism $P \to LP$ is a morphism of groups. The same holds for the functor a. Consequently, if P is a presheaf of groups and F a sheaf of groups, there is an isomorphism

$$\mathrm{Hom}_{\widehat{\mathscr{C}}\text{-gr.}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathscr{C}}\text{-gr.}}(a(P), F).$$

See [D] for further details.

4.3.17. Let $\mathscr{V}$ be any category. A *presheaf on $\mathscr{C}$ with values in $\mathscr{V}$* is a contravariant functor from $\mathscr{C}$ to $\mathscr{V}$. Before defining sheaves with values in $\mathscr{V}$, we recall the definition of the inverse limit of a functor. Let $\mathscr{R}$ and $\mathscr{V}$ be categories, and let

$$F : \mathscr{R}^\circ \longrightarrow \mathscr{V}$$

be a contravariant functor. Write $\varprojlim F$ for the object of $\widehat{\mathscr{V}}$ defined by

$$\begin{aligned}(\varprojlim F)(X) &= \mathrm{Hom}_{\widehat{\mathscr{V}}}(X, \varprojlim F)\\ &= \varprojlim_{U\in \mathrm{Ob}\,\mathscr{R}} \mathrm{Hom}_{\mathscr{V}}(F(U), X)\\ &= \mathrm{Hom}(c_X, F),\end{aligned}$$

where X varies over the objects of $\mathscr{V}$, where c_X denotes the contravariant functor from $\mathscr{R}$ to $\mathscr{V}$ that sends every object to X and every arrow to id_X, and where the last Hom is taken in $\mathrm{Hom}(\mathscr{R}^\circ, \mathscr{V})$. If $\mathscr{R}$ has a final object $e_{\mathscr{R}}$, then $\varprojlim F = F(e_{\mathscr{R}})$. If $\mathscr{V} = \mathbf{Set}$, the functor $\varprojlim$ is identified with Γ.

Let S be an object of $\mathscr{C}$ and R a sieve of S. Let $\overline{R}$ be the full subcategory of $\mathscr{C}_{/S}$ whose object set is $E(R)$, and let $i_R : \overline{R} \to \mathscr{C}_{/S}$ be the canonical functor. A presheaf P on $\mathscr{C}$ with values in $\mathscr{V}$ restricts to a functor

$$P_S : (\mathscr{C}_{/S})^\circ \longrightarrow \mathscr{V}.$$

The functor i_R induces a morphism in $\widehat{\mathscr{V}}$:

$$P(S) = P_S(S) = \varprojlim P_S \longrightarrow \varprojlim (P_S \circ i_R).$$

We write

$$\varprojlim_R P = \varprojlim (P_S \circ i_R)$$

for the latter object of $\widehat{\mathscr{V}}$. In view of 4.1.6, Definition 4.3.1 generalizes as follows.

Definition 4.3.18. *A presheaf P on $\mathscr{C}$ with values in $\mathscr{V}$ is* separated *(respectively, a* sheaf*) if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $R \in J(S)$, the canonical morphism in $\widehat{\mathscr{V}}$*

$$P(S) \longrightarrow \varprojlim_R P$$

[29] Editor's note: In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$. This will be used in 4.4.5.

is a monomorphism (respectively, an isomorphism).

When $\mathscr{V}$ is the category **Gr** of groups (or any other category of sets endowed with algebraic structures defined by finite inverse limits), the following notions are equivalent (cf. [D]): a presheaf on $\mathscr{C}$ with values in **Gr** whose underlying presheaf of sets is a sheaf, and a group object in the category of sheaves of sets. With these identifications understood, we shall always regard sheaves with values in a category of sets endowed with algebraic structures defined by finite inverse limits as sheaves of sets equipped, in $\widetilde{\mathscr{C}}$, with the corresponding algebraic structure.

4.4. Exactness Properties of the Category of Sheaves

Theorem 4.4.1.

(i) *Arbitrary inverse limits exist in $\widetilde{\mathscr{C}}$, and "are computed in $\widehat{\mathscr{C}}$"; that is, the inclusion functor*

$$i : \widetilde{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}}$$

commutes with inverse limits. More explicitly, if (X_α) is an inverse system of sheaves, the presheaf

$$\varprojlim_{\alpha} i(X_\alpha) : S \longmapsto \varprojlim_{\alpha} X_\alpha(S)$$

is a sheaf, and

$$i\left(\varprojlim_{\alpha} X_\alpha\right) = \varprojlim_{\alpha} i(X_\alpha).$$

(ii) *Arbitrary direct limits exist in $\widetilde{\mathscr{C}}$. If (X_α) is a direct system of sheaves, then*

$$\varinjlim_{\alpha} X_\alpha = a\left(\varinjlim_{\alpha} i(X_\alpha)\right),$$

where $\varinjlim_\alpha i(X_\alpha)$ is the presheaf direct limit of the $i(X_\alpha)$:

$$\varinjlim_{\alpha} i(X_\alpha) : S \longmapsto \varinjlim_{\alpha} X_\alpha(S).$$

(iii) *The functor*

$$a : \widehat{\mathscr{C}} \longrightarrow \widetilde{\mathscr{C}}$$

commutes with arbitrary direct limits and finite inverse limits.

Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and[30] the first assertion of (iii) follows from (ii). The second assertion of (iii) was already noted in 4.3.16.

Scholium 4.4.2. *This theorem makes it possible to use the following method for proving in $\widetilde{\mathscr{C}}$ a statement involving both arbitrary direct limits and finite inverse limits (for example, "every epimorphism is universally effective"; cf. below). One begins by proving the corresponding statement in the category of sets, then extends it pointwise to the category of presheaves. One then uses the preceding theorem to pass from the category of presheaves to the category of sheaves. Many examples of this method will occur below (4.4.3, 4.4.6, 4.4.9, and so forth).*

[30] Editor's note: the original has been modified here.

Finally, observe that statements concerning the category of presheaves are formally corollaries of the corresponding statements concerning the category of sheaves. It is enough to take the coarsest ("chaotic") topology, namely the topology defined by

$$J(S) = \{S\} \qquad \text{for every } S \in \operatorname{Ob}\mathscr{C};$$

every functor is a sheaf for this topology.

Proposition 4.4.3. *Let $\mathscr{F} = \{F_i \to F\}$ be a family of morphisms of sheaves. The following conditions are equivalent:*

(i) *$\mathscr{F}$ is an epimorphic family;*

(ii) *$\mathscr{F}$ is a universal effective epimorphic family (1.13);*

(iii) *$\mathscr{F}$ is covering (4.2.3);*

(iv) *the sheaf image of $\mathscr{F}$, that is, the sheaf associated with the presheaf image of $\mathscr{F}$ (4.1.2), is F.*

The equivalence of (iii) and (iv) follows from 4.3.12. The remaining equivalences will follow from the next lemmas.

Lemma 4.4.4. *Let $f : X \to Y$ be a monomorphism of sheaves that is also an epimorphism. Then f is an isomorphism.*

The lemma is clear first of all in the category of sets. Let us next prove it in the category of presheaves. Consider the presheaf

$$V : S \longmapsto Y(S) \amalg_{X(S)} Y(S);$$

this is the amalgamated sum of Y and Y over X in the category of presheaves.[31] Let $i_1, i_2 : Y \rightrightarrows V$ be the two coordinate morphisms. If $X \to Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In that case, for every S, the map $X(S) \to Y(S)$ is surjective; since it is also injective, it is bijective, and hence $X \to Y$ is an isomorphism.

Finally, work in the category $\widetilde{\mathscr{C}}$ of sheaves. By 4.4.1(ii), the amalgamated sum in $\widetilde{\mathscr{C}}$ of the sheaves Y and Y over X is the sheaf Z associated with the presheaf V. Consider the diagram

$$X \xrightarrow{f} Y \underset{i_2}{\overset{i_1}{\rightrightarrows}} V \xrightarrow{\tau} Z = a(V).$$

We have $i_1 f = i_2 f$, hence $\tau i_1 f = \tau i_2 f$, and therefore $\tau i_1 = \tau i_2$, since f is an epimorphism in $\widetilde{\mathscr{C}}$. By part (iii) of the following lemma, the presheaf V is separated, so τ is a monomorphism (4.3.11(iv)). Thus $i_1 = i_2$, and we saw above that this implies that $f : X \to Y$ is an isomorphism. This proves Lemma 4.4.4, once it has been verified that V is separated.

Let $R_\varnothing$ denote the presheaf that assigns the empty set to every $S \in \operatorname{Ob}\mathscr{C}$; it is an initial object of $\widehat{\mathscr{C}}$.[32]

Lemma 4.4.5.

[31] Editor's note: the remainder of the proof has been slightly modified.

[32] Editor's note: part (i) of Lemma 4.4.5 has been corrected, and the proofs of all three parts have been given in detail.

(i) *Suppose that $R_\varnothing \notin J(S)$ for every $S \in \operatorname{Ob}\mathscr{C}$. If (X_i) is a family of separated presheaves, then the coproduct presheaf $\coprod_i X_i$ is separated.*[33]

(ii) *Consider an equivalence relation in the category of presheaves*

$$X \underset{v}{\overset{u}{\rightrightarrows}} Y,$$

and let $w : Y \to Z$ be the quotient. If X is a sheaf and Y is separated, then Z is separated.

(iii) *Consider an amalgamated sum in the category of presheaves, in which u and u' are monomorphisms:*

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle u'}\big\downarrow & & \big\downarrow \\ Y' & \longrightarrow & V. \end{array}$$

If Y and Y' are separated and X is a sheaf, then V is separated.

Proof. For (i), set $X = \coprod_i X_i$. Let $S \in \operatorname{Ob}\mathscr{C}$, let $\tau : R \hookrightarrow S$ be a refinement of S, and let $x_1, x_2 \in X(S)$ satisfy $x_1 \circ \tau = x_2 \circ \tau$. There are indices i, j such that $x_1 \in X_i(S)$ and $x_2 \in X_j(S)$. Since $R \neq R_\varnothing$, there is a morphism $\phi : T \to R$ with $T \in \operatorname{Ob}\mathscr{C}$. Then

$$x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi.$$

Since $X(T)$ is the disjoint union of the $X_k(T)$, it follows that $i = j$. Because X_i is separated and is a subobject of X, the map

$$X_i(S) \longrightarrow X_i(R) \longrightarrow X(R)$$

is injective. Hence $x_1 = x_2$, which proves that X is separated.

We next prove (iii). Let $i : Y \to V$ and $j : Y' \to V$ be the canonical morphisms, and let K be the kernel of the pair

$$Y \times Y' \underset{q}{\overset{p}{\rightrightarrows}} V, \qquad p = i \circ \mathrm{pr}_1, \quad q = j \circ \mathrm{pr}_2\,.$$

For $S \in \operatorname{Ob}\mathscr{C}$, the monomorphisms u, u' allow us to identify $X(S)$ with its image in $Y(S)$, respectively in $Y'(S)$. Write $Z(S)$, respectively $Z'(S)$, for the complement. Then $V(S)$ identifies with the disjoint union of $Z(S)$, $Z'(S)$, and $X(S)$. It follows at once that

$$i(S) : Y(S) \longrightarrow V(S), \qquad j(S) : Y'(S) \longrightarrow V(S)$$

are injective and that

$$(u \times u')(S) : X(S) \longrightarrow K(S)$$

is bijective. Consequently i and j are monomorphisms, and $u \times u' : X \to K$ is an isomorphism.

[33]Editor's note: in general, the coproduct of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in \operatorname{Ob}\mathscr{C}$. Suppose that the coproduct $S = S_1 \coprod S_2$ exists in $\mathscr{C}$, and that the fiber product $S_1 \times_S S_2$ is an initial object $\varnothing$ of $\mathscr{C}$ (cf. I, 1.8). Let R be the sieve of S with basis $\{S_1, S_2\}$. Then $(F \coprod G)(R)$ is the disjoint union of $F(S) \coprod G(S)$ and the sets $F(S_i) \times G(S_j)$, for $i \neq j$; thus $F \coprod G$ is not a sheaf in general. On the other hand, let $\mathscr{C}$ be the category with a single object S and id_S as its only morphism, equipped with the topology defined by $J(S) = \{R_\varnothing, S\}$. Then the only separated presheaves are $R_\varnothing$ and $X = \mathbf{h}_S$ (which is a sheaf), and $X \coprod X$ is not separated.

Set $U = Y \coprod Y'$. For a refinement $\tau : R \hookrightarrow S$, there is a commutative diagram

$$\begin{array}{ccc} U(S) & \xrightarrow{U(\tau)} & U(R) \\ \downarrow & & \downarrow \\ V(S) & \xrightarrow{V(\tau)} & V(R). \end{array}$$

Let $v_1, v_2 \in V(S)$ have the same image in $V(R)$. By the definition of V, they lift to elements $y_1, y_2 \in U(S)$; let z_1, z_2 denote their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \to V$ and $j : Y' \to V$ are monomorphisms, the maps $Y(R) \to V(R)$ and $Y'(R) \to V(R)$ are injective. Thus, because Y and Y' are separated, if y_1, y_2 both belong to $Y(S)$, or both belong to $Y'(S)$, then $y_1 = y_2$. Otherwise we may suppose that $y_1 \in Y(S)$ and $y_2 \in Y'(S)$, so that $z_1 \in Y(R)$ and $z_2 \in Y'(R)$. Since $i \circ z_1 = j \circ z_2$, the morphism

$$z_1 \boxtimes z_2 : R \longrightarrow Y \times Y'$$

factors through $K = X$. Moreover, since X is a sheaf, the restriction map identifies $X(S)$ with $X(R)$. Hence there is an $x \in X(S)$ such that

$$u(x) \circ \tau = z_1 = y_1 \circ \tau, \qquad u'(x) \circ \tau = z_2 = y_2 \circ \tau.$$

Since Y and Y' are separated, $u(x) = y_1$ and $u'(x) = y_2$, and therefore $v_1 = v_2$. This proves that V is separated.

Finally, we prove (ii). First observe that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K is the kernel of the pair

$$w \circ \mathrm{pr}_i : Y \times Y \rightrightarrows Z,$$

is an isomorphism. Indeed, for every $T \in \mathrm{Ob}\,\mathscr{C}$, $X(T)$ is an equivalence relation on $Y(T)$, and hence the diagram

$$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \underset{w \circ \mathrm{pr}_2}{\overset{w \circ \mathrm{pr}_1}{\rightrightarrows}} Z(T)$$

is exact in the category of sets.

Now let $S \in \mathrm{Ob}\,\mathscr{C}$, and let $g_1, g_2 : S \to Z$ be two morphisms that agree on a refinement $\tau : R \hookrightarrow S$. By the construction of Z,

$$Z(S) = Y(S)/X(S),$$

so there are morphisms $f_1, f_2 : S \to Y$ such that $w f_i = g_i$. We have $w f_1 \tau = w f_2 \tau$; by the preceding observation, there is a morphism $\phi : R \to X$ such that

$$u\phi = f_1 \tau, \qquad v\phi = f_2 \tau.$$

Since X is a sheaf, there is a morphism $\psi : S \to X$ with $\phi = \psi\tau$. Thus, in $Y(R)$,

$$u\psi\tau = f_1\tau, \qquad v\psi\tau = f_2\tau.$$

Because $Y(S) \to Y(R)$ is injective, Y being separated, it follows that $u\psi = f_1$ and $v\psi = f_2$, hence $g_1 = g_2$. Thus Z is separated.

Lemma 4.4.6.[34]

[34] Editor's note: the statement of the lemma and its proof have been given in detail.

(i) *Let $\{F_i \xrightarrow{f_i} F\}$ be a family of morphisms of presheaves, and let $G \subset F$ be its presheaf image. Then the following diagram in $\widehat{\mathscr{C}}$ is exact:*

$$\left(\coprod_i F_i\right) \times_G \left(\coprod_j F_j\right) \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} \coprod_i F_i \longrightarrow G.$$

Equivalently, for every presheaf H, the following diagram of sets is exact:

$$\mathrm{Hom}(G, H) \longrightarrow \prod_i \mathrm{Hom}(F_i, H) \underset{q}{\overset{p}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(F_i \times_G F_j, H). \qquad (*)$$

(ii) *Every covering family of morphisms of sheaves is a universal effective epimorphic family.*

Proof. For (i), let H be a presheaf. The map

$$f^* : \mathrm{Hom}(G, H) \longrightarrow \prod_i \mathrm{Hom}(F_i, H), \qquad \phi \longmapsto (\phi \circ f_i)_i,$$

is injective. Indeed, for each $S \in \mathrm{Ob}\,\mathscr{C}$, the map $\phi(S)$ is determined by the maps $(\phi \circ f_i)(S)$, since the family

$$f_i(S) : F_i(S) \longrightarrow G(S)$$

is surjective. It is clear that the image of f^* is contained in $\ker(p, q)$.

Conversely, let $\phi_i : F_i \to H$ be a family of morphisms such that, for every i, j, the diagram

$$\begin{array}{ccc} F_i \times_G F_j & \longrightarrow & F_j \\ \downarrow & & \downarrow{\scriptstyle \phi_j} \\ F_i & \xrightarrow{\phi_i} & H \end{array}$$

commutes. For every S, the map

$$\coprod_i F_i(S) \longrightarrow H(S)$$

then factors uniquely through a map $\phi(S) : G(S) \to H(S)$. These maps define a morphism $\phi : G \to H$ satisfying $f^*(\phi) = (\phi_i)$. This proves the exactness of $(*)$, and hence (i).

We prove (ii). Since covering families are stable under base change, it is enough to show that every covering family is effective epimorphic. Let $\{F_i \to F\}$ be a covering family of morphisms of sheaves, and let G be its presheaf image. Because the family is covering, the monomorphism $G \hookrightarrow F$ is covering as well; therefore, by 4.3.12,

$$a(G) = F.$$

Moreover, since $G \hookrightarrow F$ is a monomorphism, the fiber products $F_i \times_G F_j$ and $F_i \times_F F_j$ coincide. Part (i) therefore shows that, for every presheaf H, the diagram

$$\mathrm{Hom}(G, H) \longrightarrow \prod_i \mathrm{Hom}(F_i, H) \underset{q}{\overset{p}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(F_i \times_F F_j, H) \qquad (**)$$

is exact. If, in addition, H is a sheaf, then

$$\mathrm{Hom}(G, H) = \mathrm{Hom}(a(G), H) = \mathrm{Hom}(F, H).$$

Thus $(**)$ shows that $\{F_i \to F\}$ is an effective epimorphic family in the category $\widetilde{\mathscr{C}}$ of sheaves.

Lemma 4.4.7. *Every family of morphisms of sheaves $\{F_i \to F\}$ factors as a covering family $\{F_i \to G\}$ followed by a monomorphism $G \to F$.*

Indeed, it is enough to take G to be the sheaf image of the given family.

Proof of Proposition 4.4.3. We saw in 4.4.6 that (iii) implies (ii), while (ii) evidently implies (i). Finally, let $\{F_i \to F\}$ be an epimorphic family. By Lemma 4.4.7 it factors as a covering family followed by a monomorphism. The latter monomorphism, being dominated[35] by an epimorphic family, is itself an epimorphism; hence it is an isomorphism by 4.4.4.

Remark 4.4.8.[36] Because the presheaf image of the family $\mathscr{F} = \{F_i \to F\}$ is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:

(v) for every $S \in \operatorname{Ob}\mathscr{C}$, each $f \in F(S)$ lies locally in the image of $\mathscr{F}$; that is,

(vi) for every $S \in \operatorname{Ob}\mathscr{C}$ and every $f \in F(S)$, the set of morphisms $S' \to S$ such that the image of f in $F(S')$ belongs to the image of one of the $F_i(S')$ is a refinement of S;

(vii) if the topology is defined by a pretopology $S \mapsto \mathscr{R}(S)$, then, for every $S \in \operatorname{Ob}\mathscr{C}$ and every $f \in F(S)$, there is a family $\{S_j \to S\} \in \mathscr{R}(S)$ such that, for each j, the image f_j of f in $F(S_j)$ belongs to the image of one of the $F_i(S_j)$.

Remark 4.4.8 bis. If the sheaf F is representable, the preceding conditions are also equivalent to:

(viii) the set of morphisms $T \to F$, with $T \in \operatorname{Ob}\mathscr{C}$, for which there exist an i and a commutative diagram

$$\begin{array}{ccc} & F_i & \\ & \downarrow & \\ T & \longrightarrow & F \end{array}$$

is a refinement of F.

Indeed, if (viii) holds, the presheaf image of the F_i is dominated by[37] a refinement of F, which implies that the family is covering. Conversely, apply (vi) to $\mathrm{id}_F \in F(F)$.

In pictorial language, this condition says: *locally on F, there is an i for which $F_i \to F$ has a section.* In particular, a morphism

$$G \longrightarrow F,$$

where G is a sheaf and F is a representable sheaf, is covering if and only if it has a section locally on F.

The following lemmas will be useful in $\mathrm{VI_B}$, 8.1 and 8.2.[38]

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on $\mathscr{C}$, with Q normal in P. Suppose that P is separated and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently,*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

[35] Editor's note: recall (cf. editor's note 17) that a morphism $g : G \to H$ is said to be *dominated* by a family of morphisms $F_i \to H$ if that family factors through g.

[36] Editor's note: the numbering of the original, which had two paragraphs numbered 4.4.7, has been corrected to agree with later references.

[37] Editor's note: "contains" has been replaced by "is dominated by"; cf. editor's note 17.

[38] Editor's note: Lemmas 4.4.8.1 and 4.4.8.2 have been added.

It is enough to prove the first assertion, since the second follows from it by 4.3.11(iv) and (ii). Let $S \in \mathrm{Ob}\,\mathscr{C}$, let R be a sieve of S, and let $y \in P(S)/Q(S)$ have identity image in $(P/Q)(R)$. We must show that $y = 1$.

By hypothesis, the morphism $P(S) \to P(R)$ is injective, while $Q(S) \to Q(R)$ is an isomorphism, and the upper row in the following commutative diagram is exact:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Q(S) & \longrightarrow & P(S) & \longrightarrow & P(S)/Q(S) & \longrightarrow & 1 \\ & & \downarrow\wr & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & Q(R) & \longrightarrow & P(R) & \longrightarrow & (P/Q)(R). & & \end{array}$$

The result follows if the lower row is exact. Let $f : R \to P$ have identity image in $(P/Q)(R)$, and let $\phi : T \to R$ with $T \in \mathrm{Ob}\,\mathscr{C}$. Then $f \circ \phi$ is the identity in

$$(P/Q)(T) = P(T)/Q(T),$$

that is, $f \circ \phi \in Q(T)$. Hence $\phi \mapsto f \circ \phi$ is a functorial map $R(T) \to Q(T)$, and so defines a morphism of functors $\pi : R \to Q$ such that $\pi(\mathrm{id}_R) = f$. Thus $f \in Q(R)$. This proves the exactness of the lower row and establishes the lemma.

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on $\mathscr{C}$.*

(i) *If H is normal in G, then $L(H)$ is normal in $L(G)$.*

(ii) *If H is central in G, then $L(H)$ is central in $L(G)$.*

Let $S \in \mathrm{Ob}\,\mathscr{C}$. We must show (cf. I 2.3.6) that $L(H)(S)$ is normal, respectively central, in $L(G)(S)$. Let

$$h \in L(H)(S), \qquad g \in L(G)(S).$$

There are a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$ such that

$$h = z_R(h'), \qquad g = z_R(g')$$

in the notation of 4.3.10. Since z_R is a group morphism,

$$ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1}).$$

In case (i),

$$g'h'g'^{-1}h'^{-1} \in H(R),$$

and therefore $ghg^{-1}h^{-1} \in L(H)(S)$. In case (ii), $g'h'g'^{-1}h'^{-1} = 1$, and hence $ghg^{-1}h^{-1} = 1$.

Proposition 4.4.9. *Every equivalence relation in $\widetilde{\mathscr{C}}$ is universally effective (3.3.3). More precisely, let R be a $\widetilde{\mathscr{C}}$-equivalence relation on the sheaf X. Then the sheaf associated with the separated presheaf*

$$i(X)/i(R) : S \longmapsto X(S)/R(S)$$

is a universal effective quotient of X by R.

Let X/R denote the sheaf quotient of X by R, which exists by 4.4.1(ii):

$$X/R = a\big(i(X)/i(R)\big).$$

We must show that $X \to X/R$ is a universal effective epimorphism and that the morphism

$$f : R \longrightarrow X \times_{X/R} X$$

is an isomorphism. The first assertion has already been proved in 4.4.3. As for f, it is obtained by applying the functor a to the morphism

$$i(R) \longrightarrow i(X) \times_{i(X/R)} i(X),$$

or, since $i(X)/i(R)$ is separated by 4.4.5(ii), so that

$$i(X)/i(R) \longrightarrow i(X/R)$$

is a monomorphism, to the canonical morphism

$$i(R) \longrightarrow i(X) \times_{i(X)/i(R)} i(X).$$

Thus we are reduced to proving the same statement in the category of presheaves. But $i(X)/i(R)$ is the presheaf

$$S \longmapsto X(S)/R(S),$$

so the assertion reduces to its set-theoretic analogue, where it is immediate.

Proposition 4.4.10. *Under the hypotheses of 4.4.9, let Y be a subsheaf of X, and write R_Y for the equivalence relation induced on Y by R. Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. It identifies Y/R_Y with a subsheaf of X/R, namely the sheaf image of the composite

$$Y \longrightarrow X \longrightarrow X/R.$$

The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it carries monomorphisms to monomorphisms; hence

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The final assertion follows from the commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y/R_Y & \longrightarrow & X/R \end{array}$$

and the fact that $Y \to Y/R_Y$ is covering.

In view of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R.

Proposition 4.4.11. *Let R be a $\widetilde{\mathscr{C}}$-equivalence relation on the sheaf X. For every subsheaf Y of X stable under R, let $Y' = Y/R_Y$, regarded as a subsheaf of $X' = X/R$. Then*

$$Y = Y' \times_{X'} X,$$

and the maps

$$Y \longmapsto Y/R_Y, \qquad Y' \longmapsto Y' \times_{X'} X$$

define a bijective correspondence between the subsheaves Y of X stable under R and the subsheaves Y' of X'.

If Y' is a subsheaf of X', then $Y' \times_{X'} X$ is a subsheaf of X stable under R.[39] If Y' is obtained by passage to the quotient from a subsheaf Y of X, then Y is a subobject of $Y' \times_{X'} X$. It is therefore enough to prove the following: if Y and Y_1 are two subsheaves of X, both stable under R, with Y_1 containing Y, and if their quotients Y/R_Y and Y_1/R_{Y_1} are identical, then $Y = Y_1$.

We may plainly reduce to the case $Y_1 = X$. Write

$$P = i(X)/i(R), \qquad Q = i(Y)/i(R_Y)$$

for the corresponding presheaves. The diagram

$$\begin{array}{ccc} Y & \longrightarrow & Q \\ \downarrow & & \downarrow \\ X & \longrightarrow & P \end{array}$$

is Cartesian. On the other hand, there is a commutative diagram

$$\begin{array}{ccc} Q & \longrightarrow & a(Q) \\ \downarrow & & \downarrow\wr \\ P & \longrightarrow & a(P). \end{array}$$

Since $Q \hookrightarrow a(Q)$ is covering by 4.3.11, the monomorphism $Q \hookrightarrow P$ is covering, so Q is a refinement of P. After base change, Y is a refinement of X. Because X and Y are sheaves, 4.3.12 then gives $Y = X$.

4.4.12. In particular, if Y is a subsheaf of X and $Y' = Y/R_Y$, the preceding correspondence defines a subsheaf $\overline{Y}$ of X, stable under R, containing Y, and minimal with these properties. It is called the *saturation* of Y for the equivalence relation R.

4.5. The Case of a Topology Coarser than the Canonical Topology

By 4.3.6 and 4.3.8, the following conditions on a topology $\mathcal{T}$ on $\mathscr{C}$ are equivalent:

(i) $\mathcal{T}$ is coarser than the canonical topology of $\mathscr{C}$;

(ii) every representable presheaf is a sheaf for $\mathcal{T}$;

(iii) every covering sieve for $\mathcal{T}$ is universal effective epimorphic.

If $\mathcal{T}$ is defined by a pretopology $S \mapsto \mathscr{R}(S)$, these conditions are also equivalent to:

(iv) every family belonging to $\mathscr{R}(S)$ is universal effective epimorphic.

When these conditions hold, the canonical functor $\mathscr{C} \to \widehat{\mathscr{C}}$ factors through a functor

$$j_{\mathscr{C}} = j : \mathscr{C} \longrightarrow \widetilde{\mathscr{C}};$$

we also write $j(S) = \widetilde{S}$.[40]

Proposition 4.5.1. *The functor $j : \mathscr{C} \to \widetilde{\mathscr{C}}$ is fully faithful and commutes with arbitrary inverse limits. In particular, it is left exact and therefore preserves algebraic structures defined by finite inverse limits.*

[39] Editor's note: moreover, $(Y' \times_{X'} X)/R = Y'$.

[40] Editor's note: we shall also write $\mathbf{h}_S = \widehat{S}$; cf. the first commutative diagram in 4.5.4.

This follows immediately from the commutative factorization

$$\begin{tikzcd} & \mathscr{C} \arrow[dl, "j"'] \arrow[dr, "\text{canonical}"] & \\ \widetilde{\mathscr{C}} \arrow[rr, "i"] & & \widehat{\mathscr{C}} \end{tikzcd}$$

and from 4.4.1(i).

Before giving further properties of j, we must define the topology induced on a category $\mathscr{C}/S$. We no longer suppose here that the given topology is necessarily coarser than the canonical topology. A sieve of an object T in $\mathscr{C}$, together with a morphism $T \to S$, naturally defines a sieve of T in $\mathscr{C}/S$, because the definition of a sieve of T depends only on the category

$$\mathscr{C}/T = (\mathscr{C}/S)/T.$$

For example, if the sieve is defined by a family $\{T_i \to T\}$, its image in $\mathscr{C}/S$ is defined by the same family, now regarded as a family of morphisms in $\mathscr{C}/S$.

Declaring these images of covering sieves to be covering defines a topology on $\mathscr{C}/S$, called the *topology induced* by the given topology. In the terminology of [AS], 2.3, it is the coarsest topology on $\mathscr{C}/S$ for which the canonical functor

$$i_S : \mathscr{C}/S \longrightarrow \mathscr{C}$$

is a comorphism.[41] By definition, the identifications

$$(\mathscr{C}/S)/T = \mathscr{C}/T$$

respect these topologies.

Proposition 4.5.2. *Let S be a representable sheaf on $\mathscr{C}$, and let $F \to S$ be a morphism in $\widehat{\mathscr{C}}$. The presheaf*

$$S' \longmapsto \mathrm{Hom}_S(S', F)$$

on $\mathscr{C}/S$ is separated (respectively, is a sheaf) if and only if F is a separated presheaf (respectively, a sheaf) on $\mathscr{C}$.

For every functor P, one has (I, 1.4.1)

$$\mathrm{Hom}(P, F) = \coprod_{f \in \mathrm{Hom}(P,S)} \mathrm{Hom}_f(P, F).$$

[42] Let $S' \in \mathrm{Ob}(\mathscr{C})$, and let $\eta : C' \to S'$ be a covering sieve. Since S is a sheaf, the map $f \mapsto f \circ \eta$ is a bijection

$$\mathrm{Hom}(S', S) \xrightarrow{\sim} \mathrm{Hom}(C', S).$$

Consequently, the map

$$\mathrm{Hom}_{\widehat{\mathscr{C}}}(S', F) \longrightarrow \mathrm{Hom}_{\widehat{\mathscr{C}}}(C', F)$$

decomposes as the "disjoint union" of the maps

$$\mathrm{Hom}_f(S', F) \longrightarrow \mathrm{Hom}_{f\circ\eta}(C', F).$$

The proposition follows.

[41] Editor's note: that is, for every object $T \to S$ of $\mathscr{C}/S$, every covering sieve R' of $i_S(T \to S) = T$, when regarded as a sieve of the object $(T \to S) \in \mathrm{Ob}(\mathscr{C}/S)$, is covering.

[42] Editor's note: the argument implicit in the original has been expanded below.

Corollary 4.5.3. *If a topology on $\mathscr{C}$ is coarser than the canonical topology, then the topology it induces on $\mathscr{C}/S$ is coarser than the canonical topology of $\mathscr{C}/S$.*

Corollary 4.5.4. *Suppose that the given topology on $\mathscr{C}$ is coarser than the canonical topology. For every $S \in \mathrm{Ob}(\mathscr{C})$, there is an equivalence of categories*

$$\widetilde{\mathscr{C}}_{/\widetilde{S}} \xrightarrow{\sim} \widetilde{\mathscr{C}/S}.$$

The following diagrams commute up to isomorphism; every unlabelled arrow in them is an equivalence:

$$\begin{tikzcd}
\mathscr{C}_{/S} \arrow[r, "(j_{\mathscr{C}})_{/S}"] \arrow[d, equal] & \widetilde{\mathscr{C}}_{/\widetilde{S}} \arrow[r, "(i_{\mathscr{C}})_{/\widetilde{S}}"] \arrow[d, "\wr"] & \widehat{\mathscr{C}}_{/\widehat{S}} \arrow[r, "(a_{\mathscr{C}})_{/\widehat{S}}"] \arrow[d, "\wr"] & \widetilde{\mathscr{C}}_{/\widetilde{S}} \arrow[d, "\wr"] \\
\mathscr{C}_{/S} \arrow[r, "j_{\mathscr{C}/S}"] & \widetilde{\mathscr{C}/S} \arrow[r, "i_{\mathscr{C}/S}"] & \widehat{\mathscr{C}/S} \arrow[r, "a_{\mathscr{C}/S}"] & \widetilde{\mathscr{C}/S}.
\end{tikzcd}$$

If $T \to S$ is an object of $\mathscr{C}/S$, one also has the commutative diagram

$$\begin{tikzcd}
(\widetilde{\mathscr{C}}_{/\widetilde{S}})_{/\widetilde{T}} \arrow[r] \arrow[d, "\wr"] & (\widetilde{\mathscr{C}/S})_{/\widetilde{T}} \arrow[dr] & \\
& & (\widetilde{\mathscr{C}/S})_{/T}\,. \\
\widetilde{\mathscr{C}}_{/\widetilde{T}} \arrow[r] & \widetilde{\mathscr{C}/T} \arrow[ur, "\sim"] &
\end{tikzcd}$$

The commutativity of the first two squares follows from the definition of the equivalence $\widetilde{\mathscr{C}}_{/\widetilde{S}} \to \widetilde{\mathscr{C}/S}$. To prove the commutativity of the last square, it is enough to observe that the following square commutes:

$$\begin{tikzcd}
\widehat{\mathscr{C}}_{/\widehat{S}} \arrow[r, "L'"] \arrow[d] & \widehat{\mathscr{C}}_{/\widehat{S}} \arrow[d] \\
\widehat{\mathscr{C}/S} \arrow[r, "L''"] & \widehat{\mathscr{C}/S}.
\end{tikzcd}$$

Here L' is the restriction of the functor $L_{\mathscr{C}}$ to $\widehat{\mathscr{C}}_{/\widehat{S}}$, and L'' is the functor $L_{\mathscr{C}/S}$. This is seen immediately by returning to the definition of the functors L, given after 4.3.9. The second diagram is simply the restriction to the categories of sheaves of the corresponding diagram for the categories of presheaves (Exposé I, no. 1), which is commutative.

Scholium 4.5.5. The preceding assertions show that, when the given topology is coarser than the canonical topology, one may identify $\mathscr{C}$ with a full subcategory of $\widetilde{\mathscr{C}}$, itself a full subcategory of $\widehat{\mathscr{C}}$. Under these identifications one may make the customary abuses of language concerning $\mathscr{C} \to \widehat{\mathscr{C}}$, justified by the preceding commutative diagrams. In particular, the first diagram of 4.5.4 shows that the functor a may be used without any special precaution.

In the next subsection we shall see that the identification of $\mathscr{C}$ with a full subcategory of $\widetilde{\mathscr{C}}$, unlike the corresponding identification with $\widehat{\mathscr{C}}$, commutes with the formation of certain direct limits; we shall then explain how this fact is to be used.

From now on, unless expressly stated otherwise, we assume that the given topology is coarser than the canonical topology, and we systematically make the identifications described above.

Proposition 4.5.6. *Let F and G be two sheaves over S, and let $f : F \to G$ be an S-morphism. The following conditions on f are equivalent:*

(i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\widetilde{\mathscr{C}}$;*

(ii) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\widetilde{\mathscr{C}}_{/S} = \widetilde{\mathscr{C}/S}$.*

For monomorphisms and isomorphisms this is immediate, since it is a question about presheaves. For epimorphisms it follows from the description of epimorphisms as covering morphisms (4.4.3) and from the fact that, by the definition of the induced topology, these are the same in $\widetilde{\mathscr{C}}$ and in $\widetilde{\mathscr{C}/S}$.

Proposition 4.5.7. *Let $f : F \to G$ be a morphism of sheaves. The following conditions are equivalent:*

(i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism);*

(ii) *for every $S \in \mathrm{Ob}(\mathscr{C})$, the morphism $f_S : F_S \to G_S$ is locally a monomorphism (respectively, an epimorphism, respectively, an isomorphism), that is:*

(iii) *for every $S \in \mathrm{Ob}(\mathscr{C})$, the set of morphisms $T \to S$ for which $F_T \to G_T$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) is a refinement of S.*

If the given topology is defined by a pretopology $\mathscr{R}$, these conditions are also equivalent to:

(iv) *for every $S \in \mathrm{Ob}(\mathscr{C})$, there is a covering family $\{S_i \to S\} \in \mathscr{R}(S)$ such that, for every i, $F_{S_i} \to G_{S_i}$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism).*

If $\mathscr{C}$ has a final object e, it is enough in conditions (ii), (iii), and (iv) to take $S = e$.

Clearly (ii), (iii), and (iv) are equivalent. To prove the equivalence of (i) and (ii), as well as the assertion concerning a final object, it is enough to show that (ii) implies (i) and that the properties in question are stable under base change. We prove the latter point first. For monomorphisms and isomorphisms it is immediate, since it is a question about presheaves. For epimorphisms it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

It remains to show that (ii) implies (i). Suppose first that $f_S : F_S \to G_S$ is locally a monomorphism (respectively, an isomorphism). There is then a covering sieve C of S such that f_T is a monomorphism (respectively, an isomorphism) for every $T \to S$ belonging to C. Since an inverse limit of monomorphisms (respectively, isomorphisms) is again one, $\mathrm{Hom}(C, f)$ is a monomorphism (respectively, an isomorphism); see 4.1.4. The commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(C,F) & \xrightarrow{\mathrm{Hom}(C,f)} & \mathrm{Hom}(C,G) \\ \uparrow\wr & & \uparrow\wr \\ F(S) & \xrightarrow{f(S)} & G(S) \end{array}$$

therefore shows that $f(S)$ is injective (respectively, bijective).

Finally, suppose that $f : F \to G$ is locally an epimorphism, and let $H \subset G$ be its image. For every $S \to G$, the sheaf $H \times_G S$ is the image of

$$f \times_G S : F \times_G S \longrightarrow S.$$

To prove that f is an epimorphism, it is enough to prove that H is a refinement of G, or equivalently that $H \times_G S$ is a refinement of S for every S. But this holds after each base change $T \to S$ belonging to a refinement of S, since $f \times_G S$ is locally covering. Hence $H \times_G S$ is itself a refinement of S, by Axiom (T 2).

Corollary 4.5.8. *Let F and G be two sheaves over S, and let $f : F \to G$ be an S-morphism. Then f is a monomorphism, respectively an epimorphism, respectively an isomorphism, if and only if it is so locally on S.*

Remark 4.5.9. The proof of the proposition shows that its assertion about monomorphisms (respectively, isomorphisms) remains valid if one assumes only that F is a separated presheaf (respectively, a sheaf) and G is an arbitrary presheaf (respectively, a separated presheaf).

We return temporarily to the case of an arbitrary topology and make the following definition.

Definition 4.5.10. *Let $G \to F$ be a morphism in $\widehat{\mathscr{C}}$. We say that G is a* relative sheaf over F *if, for every F-functor H and every refinement R of H, the canonical map*

$$\mathrm{Hom}_F(H, G) \xrightarrow{\sim} \mathrm{Hom}_F(R, G) \tag{+}$$

is bijective.

Proposition 4.5.2 immediately gives the following generalization.

Proposition 4.5.11. *If F is a sheaf, then G is a relative sheaf over F if and only if G is a sheaf.*

Lemma 4.5.12. *In a sequence*

$$X \longrightarrow T \longrightarrow S,$$

where X, T, S are three objects of $\widehat{\mathscr{C}}$, suppose that X is a relative sheaf over T. Then

$$U = \prod_{T/S} X$$

is a relative sheaf over S.

Indeed, for every S-functor Y, one has

$$\mathrm{Hom}_S(Y, U) = \mathrm{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y, then $T \times_S C$ is a sieve of $T \times_S Y$, and the conclusion is immediate.

Corollary 4.5.13. *The presheaves $\mathrm{Hom}_{T/S}(X, Y)$, $\mathrm{Isom}_S(X, Y)$, and the analogous presheaves are sheaves whenever all the arguments occurring in them are sheaves.*

Indeed, all these presheaves are constructed using fiber products and presheaves $\prod_{T/S} X$ (I, 1.7 and II, 1). It is therefore enough to verify the assertion for a presheaf $\prod_{T/S} X$; in that case it follows from 4.5.11 and 4.5.12.

4.6. Description of the Quotient of a Sheaf by an Equivalence Relation

Recall that the given topology $\mathcal{T}$ is assumed to be coarser than the canonical topology.

Proposition 4.6.1. *Let $R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X$ be a $\widetilde{\mathscr{C}}$-equivalence relation in the sheaf X. Define $F \in \mathrm{Ob}(\widehat{\mathscr{C}})$ as follows: for each $S \in \mathrm{Ob}(\mathscr{C})$,*[43]

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S \text{, stable under } R \times S \text{, whose quotient} \\ \text{by } R_Z \text{ is } S \text{, that is, for which } R_Z \rightrightarrows Z \to S \text{ is exact} \end{array} \right\}.$$

[43]Editor's note: $R \times S$ denotes the equivalence relation in $X \times S$ defined by $R \times S_{\mathrm{diag}} \subset X \times X \times S \times S$, and R_Z denotes the equivalence relation that it induces on Z (cf. 3.1.6).

For every sheaf Y, $\mathrm{Hom}(Y, F)$ *is naturally identified with the set*

$$\left\{ \begin{array}{l} \text{sub-}Y\text{-sheaves of } X \times Y, \text{ stable under } R \times Y, \\ \text{and having quotient } Y \end{array} \right\}.$$

In particular, the subsheaf $R \subset X \times X$ corresponds to an element $p \in \mathrm{Hom}(X, F)$, and the exact diagram

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} F$$

identifies F with the sheaf quotient X/R.

Indeed, put $Q = X/R$. For every sheaf Y and every morphism $f \in \mathrm{Hom}(Y, Q)$, corresponding to a section $s : Y \hookrightarrow Q \times Y$, consider the diagram

$$\begin{array}{ccccc} R \times Y & \rightrightarrows & X \times Y & \longrightarrow & Q \times Y \\ & & \uparrow & & \uparrow{\scriptstyle s} \\ & & Z & \longrightarrow & Y, \end{array} \tag{*}$$

in which the right-hand square is cartesian. By 4.4.11, Z is immediately seen to be a sub-Y-sheaf of $X \times Y$, stable under $R \times Y$, with quotient Y. Conversely, every Z of this kind arises from a unique section of $Q \times Y$ over Y. Taking Y first to be representable gives an isomorphism $Q \simeq F$; taking arbitrary Y then gives the asserted description of $\mathrm{Hom}(Y, F)$. Finally, the canonical morphism $X \to Q$ corresponds to the sub-X-sheaf $R \subset X \times X$, which completes the proof.

Corollary 4.6.2. *Let G be any subfunctor of F such that* $\mathrm{Hom}(X, G) \subset \mathrm{Hom}(X, F)$ *contains R. Then the canonical morphism $p : X \to F$ factors through G.*

Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F. In particular, every subsheaf G of F satisfying the preceding condition equals F (4.3.12).

4.6.3. We now consider the case in which X and R are representable. We begin with some terminology. Besides the conditions (a)–(d) introduced in 3.4.1, we shall use the following conditions on a family (M) of morphisms of $\mathscr{C}$; for completeness, we repeat conditions (a)–(c):

(a) (M) is stable under base change.

(b) The composite of two elements of (M) belongs to (M).

(c) Every isomorphism belongs to (M).

$(d_{\mathcal{T}})$ Every element of (M) is covering. [44]

$(e_{\mathcal{T}})$ Let $f : X \to Y$ be a morphism of $\mathscr{C}$. If there is a refinement R of Y such that, for every $Y' \to R$, the morphism $X \times_Y Y' \to Y'$ belongs to (M), then f belongs to (M).

Conditions (a) and (b) imply

(a') the fiber product of two elements of (M) belongs to (M).

On the other hand, by 4.3.9, conditions (a) and $(d_{\mathcal{T}})$ imply

(d') every element of (M) is a universal effective epimorphism.

[44] Editor's note: $\mathcal{T}$ denotes the given topology on $\mathscr{C}$, which is coarser than the canonical topology.

4.6.4. When $\mathscr{C}$ has fiber products, the family of covering morphisms, denoted $(M_{\mathcal{T}})$, satisfies the preceding conditions. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), $(d_{\mathcal{T}})$ from the definition, and $(e_{\mathcal{T}})$ from (C 5). Thus the results established below for a family satisfying these conditions apply in particular to $(M_{\mathcal{T}})$. In particular, $\mathcal{T}$ may be taken to be the canonical topology and (M) to be the family of universal effective epimorphisms.

Lemma 4.6.5. *Let (M) be a family of morphisms satisfying conditions (a)–$(e_{\mathcal{T}})$ above. Let R be a $\mathscr{C}$-equivalence relation of type (M) in $X \in \mathrm{Ob}(\mathscr{C})$. Let $\widetilde{X}$ be the sheaf defined by X, let $\widetilde{R}$ be the $\widetilde{\mathscr{C}}$-equivalence relation in $\widetilde{X}$ defined by R, and let $\widetilde{X}/\widetilde{R}$ be the sheaf quotient. Then R is (M)-effective if and only if $\widetilde{X}/\widetilde{R}$ is representable. When these conditions hold, $\widetilde{X}/\widetilde{R}$ is represented by the quotient X/R.*

Suppose first that R is (M)-effective, and put $Y = X/R$. The canonical morphism $p : X \to Y$ belongs to (M), hence is covering by $(d_{\mathcal{T}})$. The corresponding morphism

$$\widetilde{p} : \widetilde{X} \longrightarrow \widetilde{X}/\widetilde{R}$$

is therefore a universal effective epimorphism in $\widetilde{\mathscr{C}}$ (4.4.3). Consequently it identifies $\widetilde{X}/\widetilde{R}$ with the quotient of $\widetilde{X}$ by the equivalence relation R' defined by $\widetilde{p}$. Since the canonical functor $\mathscr{C} \to \widetilde{\mathscr{C}}$ commutes with fiber products, R' is precisely $\widetilde{R}$, because R is the equivalence relation defined by p.

Conversely, suppose that $\widetilde{X}/\widetilde{R}$ is represented by an object Y of $\mathscr{C}$. Let $p : X \to Y$ be the morphism deduced from the canonical morphism $\widetilde{X} \to \widetilde{X}/\widetilde{R}$; by 4.4.3 it is covering. As before, R is plainly the equivalence relation defined by p. It remains only to show that $p \in (M)$. The cartesian diagram

$$\begin{tikzcd} R \arrow[r, "\sim"] \arrow[dr, "p_2"'] & X \times_Y X \arrow[r] \arrow[d, "\mathrm{pr}_2"] & X \arrow[d, "p"] \\ & X \arrow[r, "p"] & Y \end{tikzcd}$$

shows that, after the covering base change p, the morphism p becomes p_2, which belongs to (M). The assertion follows from $(e_{\mathcal{T}})$.

Corollary 4.6.5.1. [45] *Let (M) satisfy conditions (a)–$(e_{\mathcal{T}})$ above, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups such that $f \in (M)$. Suppose that $\mathrm{Ker}(f)$ is representable (as is the case when $\mathscr{C}$ has a final object e). Then the equivalence relation in G defined by $H = \mathrm{Ker}(f)$ is (M)-effective, and G' represents the sheaf quotient $\widetilde{G}/\widetilde{H}$ for the topology $\mathcal{T}$.*

This follows from 4.6.5 and 3.4.7.1.

We can now state the main result of this subsection.

Theorem 4.6.6. *Let (M) be a family of morphisms satisfying Axioms (a)–$(e_{\mathcal{T}})$ of 4.6.3. Let R be a $\mathscr{C}$-equivalence relation of type (M) (cf. 3.4.3) in an object X of $\mathscr{C}$. Consider the functor $F \in \mathrm{Ob}(\widehat{\mathscr{C}})$ defined by*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S, \text{ stable under } R \times S, \\ \text{whose quotient by } R_Z \text{ is } S \end{array} \right\}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the representable $Z \in F(S)$, that is,

$$F_0(S) = \left\{ \begin{array}{l} \text{sub-}\mathscr{C}/S\text{-objects } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ R_Z \text{ is } (M)\text{-effective with quotient } S, \text{ i.e. such that} \\ Z \to S \text{ belongs to } (M) \text{ and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

Then:

[45] Editor's note: This corollary has been added.

(i) *The morphism $p \in \mathrm{Hom}(X, F) = F(X)$ defined by the subobject $R \subset X \times X$ identifies F with the sheaf quotient of X by R.*

(ii) *The following conditions are equivalent:*

(a) *F is representable;*

(b) *F_0 is representable;*

(c) *R is (M)-effective.*

Under these conditions, $F = F_0 = X/R$.

(iii) *Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S_i \to S\}$ and every family $\{T_i \to S_i\}$ of morphisms in (N), every descent datum on the T_i relative to $\{S_i \to S\}$ is effective. Suppose that X is squarable (cf. 1.6.0) and that $R \to X \times X$ belongs to (N). Then $F_0 = F$.*

Proof. Assertion (i) was proved in 4.6.1.

For (ii), we have already proved the equivalence of (a) and (c), as well as the equality $F = X/R$. It remains to prove that either (b) or (c) implies $F_0 = F$. First note, as stated in the theorem, that F_0 is indeed a subfunctor of F: if $S \in \mathrm{Ob}(\mathscr{C})$ and $Z \in F_0(S)$, then $Z \to S$ is squarable, and hence $Z \times_S S' \in F_0(S')$ for every $S' \to S$. Since $R \in F(X)$ belongs to $F_0(X)$, Corollary 4.6.2 shows that (b) implies $F_0 = F$.

Now suppose that (c) holds, and let Q be an object of $\mathscr{C}$ representing X/R. Then $X \to Q$ belongs to (M). For every $S \in \mathrm{Ob}(\mathscr{C})$ and every $Z \in F(S)$, diagram $(*)$ of 4.6.1 gives

$$Z = S \times_{Q \times S} (X \times S).$$

Thus Z is representable and $Z \to S$ belongs to (M), so $Z \in F_0(S)$.

For (iii), let $f \in \mathrm{Hom}(S, F)$ correspond to $Z \in F(S)$. We must show that f factors through F_0, or equivalently that Z is representable. This is immediate when f factors through X, by the following lemma.

Lemma 4.6.7. *Let $x_0 \in X(S)$. The image of x_0 in $F(S)$ corresponds to the subsheaf $Z \subset X_S$ defined by the two cartesian squares*

$$\begin{array}{ccccc} X_S & \xrightarrow{\mathrm{id}_{X_S} \times x_0} & X_S \times_S X_S & \longrightarrow & X \times X \\ \uparrow & & \uparrow & & \uparrow \\ Z & \longrightarrow & R_S & \longrightarrow & R. \end{array}$$

This follows at once from the description of the morphism $X \to F$.

Returning to the proof of the theorem, suppose first that f factors through X. Then Z is representable; moreover, since $R \to X \times X$ belongs to (N), so does $Z \to X_S$.

In general, f need not factor through X. Since $X \to F$ is covering (4.4.3), however, 4.4.8(vii) gives a covering family $\{S_i \to S\}$ and, for each i, a morphism $S_i \to X$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & F \\ \uparrow & & \uparrow f \\ S_i & \longrightarrow & S \end{array}$$

commute. By the preceding argument, the morphism $f_i : S_i \to F$ defined by this diagram belongs to $\mathrm{Hom}(S_i, F_0)$ and corresponds to the subsheaf $Z \times_S S_i \subset X_{S_i}$. The morphism $Z \times_S S_i \to X_{S_i}$ belongs to (N), and the family $\{X_{S_i} \to X_S\}$ is covering. It therefore remains only to prove the following proposition.

Proposition 4.6.8. *Let $\{S_i \to S\}$ be a covering family and let Z be a sheaf over S. Suppose that, for every i, the S_i-functor $Z \times_S S_i$ is represented by an object T_i. Then the family (T_i) carries a canonical descent datum relative to $\{S_i \to S\}$. The sheaf Z is representable if and only if this datum is effective; in that case the descended object represents Z.*

First observe that, by 4.4.3, $\{S_i \to S\}$ is a universal effective epimorphic family in $\widetilde{\mathscr{C}}$, and hence a descent family in $\widetilde{\mathscr{C}}$ (2.3). If Z is represented by an object T, then $T \times_S S_i$, regarded as a sheaf, is isomorphic to $Z \times_S S_i$. Thus the descent datum on the T_i is effective, and its uniquely determined descended object is isomorphic to T, hence represents Z.

Conversely, suppose that the canonical descent datum on the T_i is effective, and let T be the descended object. Because $\{S_i \to S\}$ is a descent family in $\widetilde{\mathscr{C}}$, there is an S-morphism $T \to Z$ whose base change to each S_i is the canonical morphism $T_i \to Z \times_S S_i$. This morphism is locally an isomorphism; since both T and Z are sheaves, 4.5.8 implies that it is an isomorphism.

Corollary 4.6.9. *Let R be an (M)-effective equivalence relation in X. For every sheaf F, the map*

$$\mathrm{Hom}(X/R, F) \longrightarrow \mathrm{Hom}(X, F)$$

identifies the first set with the subset of the second consisting of morphisms compatible with R.

Corollary 4.6.10. *Let $\mathcal{T}'$ be a topology coarser than $\mathcal{T}$, for which the morphisms in (M) are covering. Under the hypotheses of 4.6.6(iii), X/R is also the sheaf quotient of X by R for every topology lying between $\mathcal{T}'$ and the canonical topology.*

Remark 4.6.11. If, in the statement of 4.6.6(iii), one assumes in addition that the descended morphism $T \to S$ belongs to (N) whenever the hypotheses there are satisfied, then the inclusion morphisms $Z \hookrightarrow X_S$ also belong to (N). This follows immediately from the construction of Z by descent.

Remark 4.6.12. The implications

$$\text{(c)} \Longrightarrow \text{(b)} \Longrightarrow \text{(a)} \qquad \text{and} \qquad \text{(c)} \Longrightarrow [F_0 = F = X/R]$$

were proved without using the sufficiency direction of Lemma 4.6.5, the only place where condition $(e_{\mathcal{T}})$ enters. They therefore remain valid when (M) satisfies only conditions (a)–$(d_{\mathcal{T}})$. The family of squarable covering morphisms is an example of such a family (M) (compare 4.6.4). For the canonical topology these are precisely the universal effective epimorphisms. Hence:

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X. Then the object X/R of $\mathscr{C}$ is the sheaf quotient of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub-$\mathscr{C}/S$-objects Z of X_S, stable under $R \times S$, for which the induced equivalence relation is universal effective and has quotient S.*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let (M) be the family of squarable covering morphisms. If R is an (M)-effective equivalence relation in X, then the object X/R of $\mathscr{C}$ is the sheaf quotient of X by R, and it represents the functor F_0 of 4.6.6.*

Scholium 4.6.15. We can now make the following clarifications to 4.5.5. In questions involving inverse limits alone—fiber products, algebraic structures, and so forth—the results of Exposé I and 4.5.5 allow us to identify $\mathscr{C}$, without distinction, with a full subcategory of either $\widetilde{\mathscr{C}}$ or $\widehat{\mathscr{C}}$. The same is not true for questions that mix inverse and direct limits.

Whenever both inverse and direct limits occur, and especially when passing to quotients (for example, when putting a group structure on the quotient of a group by an invariant subgroup), we shall regard the given category as embedded in the category of sheaves. Thus, if R is a $\mathscr{C}$-equivalence relation in an object X of $\mathscr{C}$, the notation X/R will mean the sheaf quotient of X by R, formerly denoted $j(X)/j(R)$, and, when this sheaf is representable, also the object that represents it. The preceding results show that, in the most important cases, a quotient in $\mathscr{C}$ is also a quotient in the category of sheaves. In any event, we shall not use the notation X/R for a quotient in $\mathscr{C}$ that does not coincide with the quotient in $\widetilde{\mathscr{C}}$—for instance, one that is not universal. This modifies the definitions of no. 3.

To study a problem of this kind, one therefore works first in the category of sheaves, where all the usual results hold (cf. no. 4.4), and then specializes the resulting statements to the original category, using the results of the present subsection and, when available, criteria for the effectivity of descent. Examples of this method will appear in the following subsections.

4.7. Use of Effectivity Criteria: Isomorphism Theorem

In this subsection we give an example of the use of effectivity criteria. The initial data are a topology $\mathcal{T}$ on $\mathscr{C}$, still assumed coarser than the canonical topology; a family (M) of morphisms of $\mathscr{C}$ satisfying Axioms (a)–$(e_{\mathcal{T}})$ of 4.6.3; and a family (N) of morphisms of $\mathscr{C}$ which may satisfy the following axioms:

(a) (N) is stable under base change;

$(f_{\mathcal{T}})$ morphisms in (N) descend for the given topology. More precisely, for every $S \in \mathrm{Ob}(\mathscr{C})$, every covering family $\{S_i \to S\}$, and every family $\{T_i \to S_i\}$ of morphisms in (N), every descent datum on the T_i relative to $\{S_i \to S\}$ is effective; if T denotes the descended object, then $T \to S$ belongs to (N).

Since every element of (M) is covering (condition 4.6.3 $(d_{\mathcal{T}})$), Axiom $(f_{\mathcal{T}})$ implies the following axiom:[46]

(f_M) if $Y \to X$ belongs to (N) and $X \to X_1$ belongs to (M), every descent datum on Y relative to $X \to X_1$ is effective; if Y_1 denotes the descended object, then $Y_1 \to X_1$ belongs to (N).

Here is an example, to be treated later: $\mathscr{C}$ is the category of schemes, $\mathcal{T}$ is the faithfully flat quasi-compact topology, (M) is the family of faithfully flat quasi-compact morphisms, and (N) is the family of closed immersions, or the family of quasi-compact immersions.[47]

Taking 4.6.11 into account, the principal result of 4.6.6 may be recalled as follows.

Proposition 4.7.1. *Let X be a squarable object of $\mathscr{C}$, and let R be an equivalence relation of type (M) in X. Suppose that $R \to X \times X$ belongs to (N), where (N) satisfies (a) and $(f_{\mathcal{T}})$. Then the sheaf quotient X/R is given by*

$$\begin{aligned}(X/R)(S) = \{Z \subseteq X_S \mid\ & Z \text{ is an } S\text{-subobject stable under } R \times S,\\ & Z \to X_S \text{ belongs to } (N), \quad Z \to S \text{ is covering or belongs to } (M),\\ & R_Z \simeq Z \times_S Z\,\}.\end{aligned}$$

We also have the following result.

[46] Editor's note: Axiom (f_M), which originally appeared before Proposition 4.7.2, has been moved here.
[47] Editor's note: cf. §6.4; see also VI_A, 5.3.1.

Proposition 4.7.2. *Let $X \in \mathrm{Ob}(\mathscr{C})$, let R be an (M)-effective equivalence relation in X, and let (N) be a family of morphisms satisfying (a) and (f_M). For every subobject Y of X which is stable under R and for which $Y \to X$ belongs to (N), the equivalence relation R_Y induced in Y by R is (M)-effective. Moreover, the quotient*

$$Y/R_Y = Y'$$

is a subobject of $X' = X/R$, and $Y' \to X'$ belongs to (N).

The map $Y \mapsto Y'$ is a bijection from the set of subobjects Y of X which are stable under R and for which $Y \to X$ belongs to (N), onto the set of subobjects Y' of X' for which $Y' \to X'$ belongs to (N). Its inverse is

$$Y' \longmapsto Y' \times_{X'} X.$$

Indeed, since R is (M)-effective, the morphism $X \to X'$ belongs to (M). Let Y' be a subobject of X' such that the canonical morphism $Y' \to X'$ belongs to (N). Then $Y = Y' \times_{X'} X$ is a subobject of X stable under R, and the morphisms $Y \to X$ and $Y \to Y'$ belong respectively to (N) and (M), since both families are stable under base change. Let R_Y denote the equivalence relation induced in Y by R. By 4.4.11, the sheaf quotient Y/R_Y is represented by Y'; hence 4.6.5 shows that R_Y is (M)-effective.

Conversely, let Y be a subobject of X, stable under R, such that $Y \to X$ belongs to (N). Since Y is stable under R, its two inverse images in

$$R = X \times_{X'} X$$

are identical, and Y is equipped with a descent datum relative to $X \to X'$. The required conclusion follows from Axiom (f_M) for the family (N).

Corollary 4.7.3. *Let $X \in \mathrm{Ob}(\mathscr{C})$, and let R be an (M)-effective equivalence relation in X. Suppose in addition that $R \to X \times X$ belongs to (N), where (N) satisfies (a) and $(f_{\mathcal{T}})$. Then, for every Y as in 4.7.2, $R_Y \to Y \times Y$ also belongs to (N), and consequently 4.7.1 gives*

$$\begin{aligned}(Y/R_Y)(S) = \{Z \subseteq Y_S \mid\ & Z \text{ is an } S\text{-subobject stable under } R_Y \times S,\\ & Z \to Y_S \text{ belongs to } (N) \text{ (hence so does } Z \to X_S),\\ & Z \to S \text{ is covering}, \quad R_Z \simeq Z \times_S Z\}.\end{aligned}$$

5. Passage to the Quotient and Algebraic Structures

5.1. Principal Homogeneous Bundles

Definition 5.1.0.[48] We recall (III 0.1) that an object X equipped with a right action of H is said to be *formally principal homogeneous*[49] under H if the canonical morphism of functors

$$X \times H \longrightarrow X \times X, \qquad (x, h) \longmapsto (x, xh),$$

is an isomorphism. Equivalently (cf. *loc. cit.*), for every $S \in \mathrm{Ob}\,\mathscr{C}$, the set $X(S)$ is formally principal homogeneous under $H(S)$: it is either empty or principal homogeneous under

[48] Editor's note: the numbering 5.1.0 has been introduced for later reference.

[49] Editor's note: one also says "pseudo-torsor"; cf. EGA IV_4, 16.5.15. The more general notion of a formally homogeneous object (not necessarily principal homogeneous) is defined in Addendum 6.7.1 at the end of this exposé.

$H(S)$. In particular, when H acts on itself by right translations, it is formally principal homogeneous under itself.

Definition 5.1.1. An object X with an action of H is said to be *trivial* if it is isomorphic, as an H-object, to H with its translation action.

Proposition 5.1.2. *If X is formally principal homogeneous under H, there is an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} \mathrm{Isom}_{H\text{-obj.}}(H, X)$$

of principal homogeneous sets under $\Gamma(H)$.

To a section x of X, associate the morphism $H \to X$ defined on sets by $h \mapsto xh$. The assertion follows immediately by reduction to the set-theoretic case.

Corollary 5.1.3. *There is an isomorphism of H-objects*

$$X \xrightarrow{\sim} \mathrm{Isom}_{H\text{-obj.}}(H, X).$$

Corollary 5.1.4. *An object with a group action is trivial if and only if it is formally principal homogeneous and has a section.*

Definition 5.1.5. Let $\mathscr{C}$ be a category equipped with a topology. An S-object X with an action of the S-group H is called a *principal homogeneous bundle under H* if it is locally trivial, that is, if the following equivalent conditions hold:[50]

(i) the collection of morphisms $T \to S$ for which the functor $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S;

(ii) for a pretopology defining the given topology, there is a covering family $\{S_i \to S\}$ such that, for every i, the S_i-functor $X \times_S S_i$ is trivial under the S_i-group functor $H \times_S S_i$, equivalently, it has a section over S_i.

Proposition 5.1.6. *Let $\mathscr{C}$ be a category equipped with a topology $\mathcal{T}$, and let (M) be a family of morphisms of $\mathscr{C}$ satisfying axioms (a)–($e_{\mathcal{T}}$) of 4.6.3. Let H be an S-group whose structure morphism $H \to S$ belongs to (M), and let P be an S-object with an action of H. The following conditions are equivalent:*

(i) *P is a principal homogeneous bundle under H (Definition 5.1.5);*

(ii) *P is formally principal homogeneous under H, and its structure morphism $P \to S$ belongs to (M);*

(iii) *there exists a morphism $S' \to S$ belonging to (M) such that, after base change from S to S', P becomes trivial; that is, $P \times_S S'$ is trivial under $H \times_S S'$;*

(iv) *H acts freely and (M)-effectively on P, and the quotient P/H is isomorphic to S.*

First, (ii) and (iv) are equivalent. In either case $P \to S$ belongs to (M), hence is squarable, which ensures that the fiber products $H \times_S P$ and $P \times_S P$ are representable. Condition (ii) plainly implies (iii): one may take $S' = P$, since formal principal homogeneity implies that $P \times_S P$ is trivial under $H \times_S P$ by 5.1.4, its diagonal section providing a section. Condition (iii) plainly implies (i), because $\{S' \to S\}$ is a covering family by axiom ($d_{\mathcal{T}}$).

It remains to prove (i)⇒(ii). The morphism of sheaves

$$P \times_S H \longrightarrow P \times_S P$$

[50] Editor's note: in this case one also says that X is an H-torsor.

is locally an isomorphism and hence is an isomorphism (4.5.8); thus P is formally principal homogeneous. The structure morphism $P \to S$ is locally isomorphic to the structure morphism $H \to S$, which belongs to (M). It therefore belongs to (M) by axiom $(e_{\mathcal{T}})$.

The equivalence between (i) and (iv) admits the following generalization.

Proposition 5.1.7. *Under the same assumptions on $\mathscr{C}$ and (M), let H be an S-group and let X be an S-object on which H acts on the right. Suppose that $H \to S$ belongs to (M). The following conditions are equivalent:*

(i) *H acts freely and (M)-effectively on X;*

(ii) *there exists an S-morphism $p : X \to Y$, compatible with the equivalence relation on X defined by the action of H, such that the induced action of $H \times_S Y$ on X over Y makes X a principal homogeneous bundle under H_Y over Y.*

Under these conditions, p identifies Y with the quotient X/H.

If $p : X \to Y$ is compatible with the action of H, then the induced action of $H \times_S Y$ on X over Y defines the same equivalence relation on X as the original action of H, by the isomorphism

$$H_Y \times_Y X \xrightarrow{\sim} H \times_S X.$$

The proposition follows from this observation and the preceding equivalence (iv)$\Leftrightarrow$(i).

Corollary 5.1.7.1.[51] *Let $\mathscr{C}$ be a category having a final object and fiber products, and let it be equipped with a topology $\mathcal{T}$ coarser than the canonical topology. Let $f : G \to H$ be a morphism of $\mathscr{C}$-groups, put $K = \mathrm{Ker}(f)$, and suppose that f is covering for $\mathcal{T}$. Then H represents the sheaf quotient G/K, and f is a K_H-torsor. Equivalently, one may say that G is a K-torsor over H.*

Indeed, because f is covering, it is a universal effective epimorphism (4.4.3). Hence, by 3.3.3.1, H is the quotient of G by the equivalence relation

$$\mathscr{R}(f) = G \times_H G.$$

Moreover, the morphism

$$G \times K \longrightarrow G \times_H G, \qquad (g,k) \longmapsto (g, gk),$$

is an isomorphism of $K_G = G \times_H K_H$-objects; its inverse is $(g,g') \mapsto (g, g^{-1}g')$. Thus $\mathscr{R}(f)$ is the equivalence relation defined by K. Since $f : G \to H$ is covering, it is a K_H-torsor by 5.1.6(ii), or directly by Definition 5.1.5(ii).

We can now sharpen Theorem 4.6.6 in the case of a quotient by a group action.

Proposition 5.1.8. *Under the assumptions of 5.1.7, let F_0 denote the functor over S defined as follows: for each $S' \to S$, $F_0(S')$ is the set of representable S'-subfunctors Z of $X \times_S S'$ that are stable under $H \times_S S'$ and are principal homogeneous bundles under this S'-group for the induced action (3.2.2).*

(i) *The following conditions are equivalent:*

(a) *the action of H on X is (M)-effective and free;*[52]

(b) *F_0 is representable.*

[51] Editor's note: this special case has been added as a corollary; it will be used several times in the following exposés.

[52] Editor's note: the words "and free" have been added.

Under these conditions, $F_0 = X/H$.

(ii) *Let (N) be a family of morphisms stable under base change, such that for every covering family $\{S'_i \to S'\}$ and every family of morphisms $\{T_i \to S'_i\}$ in (N), every descent datum on the T_i relative to $\{S'_i \to S'\}$ is effective. Suppose that $X \times_S H \to X \times_S X$ belongs to (N), and that X is squarable. Then the element $p \in \operatorname{Hom}(X, F_0)$ corresponding to the subobject $X \times_S H$ of $X \times_S X$ identifies F_0 with the sheaf quotient X/H.*

5.2. Group Structures and Passage to the Quotient

In this subsection we are concerned with the algebraic structures that can be placed on the quotient (G/H) of a group by a subgroup. We first work in the category of sheaves on $\mathscr{C}$ for an arbitrary topology. Taking the canonical topology and using 4.5.12 will then give results concerning universal effective passage to the quotient in $\mathscr{C}$.

Proposition 5.2.1. *Let $u : H \to G$ be a monomorphism of sheaves of groups. There is a unique G-object structure on the quotient sheaf G/H for which the canonical morphism*

$$p : G \longrightarrow G/H$$

is a morphism of G-objects. This structure is functorial in the pair (G, H): given a commutative diagram

$$\begin{array}{ccc} H & \longrightarrow & G \\ \downarrow & & \downarrow f \\ H' & \longrightarrow & G', \end{array}$$

the morphism $f : G/H \to G'/H'$ of 3.2.3 is compatible with the morphism f between the acting groups.

Indeed, G/H is the sheaf associated with the presheaf

$$i(G)/i(H) : S \longmapsto G(S)/H(S).$$

Since the functor a is left exact, it carries objects with a group action to objects with a group action. The presheaf $i(G)/i(H)$ is an $i(G)$-object, and therefore

$$G/H = a(i(G)/i(H))$$

is an $a(i(G)) = G$-object. This construction plainly has all the stated properties.

Corollary 5.2.2. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups, and suppose that the action of H on G is universally effective. There is a unique G-object structure on the quotient object $G/H \in \operatorname{Ob}\mathscr{C}$ for which $p : G \to G/H$ is a morphism of objects with an action. In the sense of the preceding proposition, this structure is functorial in the pair (H, G), with H acting universally effectively on G.*

Proposition 5.2.3. *Let $u : H \to G$ be a monomorphism of sheaves of groups that identifies H with an invariant subgroup sheaf of G. There is a unique sheaf-of-groups structure on the quotient sheaf G/H for which the canonical morphism $p : G \to G/H$ is a group morphism. This structure is functorial in the pair (H, G), with H invariant.*

The proof is similar to that of 5.2.1.

Corollary 5.2.4. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups that identifies H with an invariant subgroup of G. Suppose that the action of H on G is universally effective. There* *is a unique group structure on the quotient object $G/H \in \operatorname{Ob}\mathscr{C}$ for which the canonical morphism $G \to G/H$ is a group morphism. This structure is functorial in the pair (H, G), with H invariant and acting universally effectively.*

The group structure of G/H can be characterized more explicitly.

Proposition 5.2.5. *Under the assumptions of 5.2.4, let K be a $\mathscr{C}$-group and let $f : G \to K$ be a morphism. The following conditions are equivalent:*

(i) *f is a group morphism compatible with the equivalence relation defined by H;*

(ii) *f is a group morphism whose restriction $H \to K$ is the trivial morphism;*

(iii) *f factors through a group morphism $G/H \to K$.*

In particular, there is an isomorphism, functorial in the group K,

$$\operatorname{Hom}_{\mathscr{C}\text{-gr.}}(G/H, K) \xrightarrow{\sim} \{f \in \operatorname{Hom}_{\mathscr{C}\text{-gr.}}(G, K) \mid f \circ u = e\}.$$

The equivalence of (i) and (ii) is proved set-theoretically, and (iii)⇒(ii) is immediate. The equivalence of (iii) and (ii) follows from the formula

$$\operatorname{Hom}(G/H, K) \simeq \operatorname{Hom}(i(G)/i(H), K)$$

and the definition of the group structure on G/H.

Remark 5.2.6. In the preceding situation, if the kernel of f is exactly H, then the morphism $G/H \to K$ through which f factors is a monomorphism. This follows at once from 3.3.4.

For sheaves of groups, 4.4.11 can be sharpened as follows.

Proposition 5.2.7. *Let G be a sheaf of groups and H an invariant subgroup sheaf. For every subgroup sheaf K of G containing H, let $K' = K/H$, regarded as a subgroup of $G' = G/H$. Then*

$$K = K' \times_{G'} G,$$

and the maps

$$K \longmapsto K/H, \qquad K' \longmapsto K' \times_{G'} G$$

give a bijection between the subgroup sheaves of G containing H and the subgroup sheaves of G'. Under this correspondence, the invariant subgroup sheaves of G and of G' correspond.

The first assertion follows readily from 4.4.11 and 3.2.4. It remains to show that K is invariant in G if and only if K' is invariant in G'. If K is invariant in G, then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$; by the usual argument, the same is true of the associated sheaves. Conversely, if K' is invariant in G', then the fiber product $K' \times_{G'} G$ is invariant in G, as is immediate.

Now let L be an arbitrary subgroup sheaf of G. Denote by $\overline{L}$ the saturation of L for the equivalence relation defined by H; we also write $\overline{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding assumptions, $L \cdot H$ is a subgroup sheaf of G containing H, and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

Indeed, let L' denote the image sheaf of L in G/H. Under the correspondence of the preceding proposition, it is the subgroup sheaf of G/H corresponding to $L \cdot H$. Since $L \to L'$ is covering, and hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \to L'$, which is plainly $H \cap L$.

Finally, consider a sheaf of groups G, a subgroup sheaf K, and an invariant subgroup sheaf H of K. We first define a right action of the sheaf of groups $H\backslash K$ $(= K/H)$ on G/H. The group K acts on G by right translations. Since H is invariant in K, this action is compatible with the equivalence relation defined by the action of H, and therefore induces an action of K on G/H, that is, a morphism

$$K^\circ \longrightarrow \mathrm{Aut}(G/H)$$

from the group opposite to K. Since the target is a sheaf (4.5.13) and this morphism is trivial on H, it factors through K/H and defines the desired action. The right and left actions of G on itself commute; hence the actions of G and K/H on G/H commute.

Proposition 5.2.9. *Under the preceding assumptions, K/H acts freely on the right on G/H, and there is a canonical isomorphism of G-objects*

$$(G/H)/(K/H) \simeq G/K.$$

If K is invariant in G, in which case K/H is invariant in G/H by 5.2.7, this isomorphism respects the group structures on both sides.

There is an isomorphism of presheaves

$$(i(G)/i(H))/(i(K)/i(H)) \xleftarrow{\sim} i(G)/i(K),$$

which respects the $i(G)$-object structures. Applying the functor a to this relation gives the asserted result.

Corollary 5.2.10. *Let G be a $\mathscr{C}$-group, K a $\mathscr{C}$-subgroup of G, and H an invariant $\mathscr{C}$-subgroup of K. Let (M) be a family of morphisms of $\mathscr{C}$ satisfying axioms (a)–($e_{\mathcal{T}}$). Suppose that the right action of H on G (respectively, on K) is (M)-effective. Then K/H acts naturally and freely on the right on G/H, and this action commutes with that of G. The following conditions are equivalent:*

(i) *the action of K on G is (M)-effective;*

(ii) *the action of K/H on G/H is (M)-effective.*

Under these conditions, there is an isomorphism of objects[53] *with a G-action*

$$(G/H)/(K/H) \simeq G/K.$$

5.3. Use of Effectivity Criteria: Noether's Theorem

[54] Let $\mathscr{C}$, $\mathcal{T}$, and (M) be as usual. Let (N) be a family of morphisms satisfying axioms (a) and (f_M) of 4.7. Combining 5.2.7 and 4.7.2 gives the following result.

Proposition 5.3.1. *Let G be a $\mathscr{C}$-group, and let H be an invariant $\mathscr{C}$-subgroup of G that acts (M)-effectively on G. For every $\mathscr{C}$-subgroup K of G containing H such that $K \to G$ belongs to (N), the action of H on K is (M)-effective, and the quotient $K/H = K'$ is a $\mathscr{C}$-subgroup of $G/H = G'$ such that $K' \to G'$ belongs to (N). The map $K \mapsto K'$ is a bijection*

[53] Editor's note: of $\mathscr{C}$.

[54] Editor's note: in fact, the isomorphisms established in 5.2.8–5.2.10 deserve to be called "Noether isomorphism theorems."

between the set of $\mathscr{C}$-subgroups K of G that contain H and for which $K \to G$ belongs to (N), and the set of $\mathscr{C}$-subgroups K' of G' for which $K' \to G'$ belongs to (N). Its inverse is

$$K' \longmapsto K \times_{G'} G.$$

Under this correspondence, the invariant subgroups of G and of G' correspond.

Corollary 5.3.2. *If $H \to G$ belongs to (N), then $\mathscr{C}$ has a final object e, and the unit section $e \to G/H$ belongs to (N).*[55]

6. Topologies in the Category of Schemes

6.1. The Zariski Topology

This is the topology generated by the following pretopology: a family of morphisms $\{S_i \to S\}$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S. It is denoted by (Zar).

Definition 6.1.1. *A sheaf for the Zariski topology is also called a* functor of local nature*: it is a contravariant functor from* **Sch** *to* **Set** *such that, for every scheme S and every covering of S by open subschemes S_i, the diagram*

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j)$$

is exact.

In particular, a functor of local nature carries direct sums to products. Since every representable functor is a sheaf, the Zariski topology is coarser than the canonical topology.

As a matter of terminology, whenever we say "local" or "locally" without further qualification, the reference is to the Zariski topology and hence to the usual meaning of these words.

6.2. A Procedure for Constructing Topologies

Proposition 6.2.1. *Let $\mathscr{C}$ be a category, let $\mathscr{C}'$ be a full subcategory, and let P be a set of families of morphisms in $\mathscr{C}$ with common target, stable under base change and composition (that is, satisfying axioms (P 1) and (P 2) of 4.2.5). Let P' be a set of families of morphisms in $\mathscr{C}'$ that contains every singleton family consisting of an identity isomorphism. Equip $\mathscr{C}$ with the topology generated by P and P' (cf. 4.2.5.0), and suppose that the following three conditions hold:*

(a) *If $\{S_i \to S\} \in P'$, so that $S_i, S \in \operatorname{Ob}\mathscr{C}'$, and $T \to S$ is a morphism in $\mathscr{C}'$, then the fiber products $S_i \times_S T$, formed in $\mathscr{C}$, exist and the family $\{S_i \times_S T \to T\}$ belongs to P'; in particular, $S_i \times_S T \in \operatorname{Ob}\mathscr{C}'$.*

    This condition implies that P' is stable under base change in $\mathscr{C}'$, but is not equivalent to that assertion: it also requires the inclusion functor $\mathscr{C}' \hookrightarrow \mathscr{C}$ to commute with certain fiber products.

(b) *For every $S \in \operatorname{Ob}\mathscr{C}$, there exists a family $\{S_i \to S\} \in P$ with $S_i \in \operatorname{Ob}\mathscr{C}'$ for every i.*

[55] Editor's note: this follows by applying the proposition with $K = H$.

(c) *In a situation*

$$\begin{tikzcd} & S_{ijk} \arrow[d, "(P')"'] \\ S_i \arrow[d, "(P')"'] & S_{ij} \arrow[l, "(P)"'] \\ S \end{tikzcd} \tag{6.2.1.c}$$

where S, S_i, S_{ij}, S_{ijk} are objects of $\mathscr{C}'$, where $\{S_i \to S\} \in P'$, $\{S_{ij} \to S_i\} \in P$ for each i, and $\{S_{ijk} \to S_{ij}\} \in P'$ for each pair ij, there exists a family $\{T_n \to S\} \in P'$, and for each n a multi-index (ijk), fitting into a commutative diagram

$$\begin{tikzcd} S_{ijk} \arrow[dr] & T_n \arrow[l] \arrow[d] \\ & S. \end{tikzcd} \tag{6.2.1.d}$$

Under these hypotheses, a sieve R of an object $S \in \operatorname{Ob}\mathscr{C}$ is covering if and only if there is a composite family

$$\begin{tikzcd} R \arrow[d] & S_{ij} \arrow[l, dashed] \arrow[d, "(P')"'] \\ S & S_i \arrow[l, "(P)"'] \end{tikzcd} \tag{6.2.1.e}$$

where $S_i, S_{ij} \in \operatorname{Ob}\mathscr{C}'$, $\{S_i \to S\} \in P$, and $\{S_{ij} \to S_i\} \in P'$ for every i, such that the resulting morphisms $S_{ij} \to S$ factor through R. Equivalently, the sieve generated by this composite family is contained in R.

Proof. Families belonging to P and P' are covering, and therefore so is a composite of such families, by (C 2). A sieve of the displayed form is thus covering because it contains a covering sieve.

Conversely, it is enough to show that the sieves of this form constitute a topology, that is, to verify axioms (T 1)–(T 4) of 4.2.1.

Axiom (T 4). Let $S \in \operatorname{Ob}\mathscr{C}$. By (b), there exists a family $\{S_i \to S\} \in P$ with $S_i \in \operatorname{Ob}\mathscr{C}'$. The singleton families $\{S_i \xrightarrow{\mathrm{id}_{S_i}} S_i\}$ belong to P' by hypothesis. Hence the maximal sieve S of S has the required form:

$$\begin{tikzcd} S_i \arrow[r] \arrow[d, "(P')"', "\mathrm{id}"] & S \arrow[d, "\mathrm{id}_S"] \\ S_i \arrow[r, "(P)"] & S. \end{tikzcd} \tag{6.2.1.T4}$$

Axiom (T 3). This is evident.

Axiom (T 2). Let R be a sieve of S of the required form, and let C be a sieve[56] of S such that, for every $T \in \operatorname{Ob}\mathscr{C}$ and every morphism $T \to S$ factoring through R, the sieve $C \times_S T$ of T has the required form. Since $S_{ij} \to S$ factors through R, the sieve

$$C_{ij} = C \times_S S_{ij}$$

[56]Editor's note: the original has been corrected here.

of S_{ij} has the required form; it is represented schematically by

$$\begin{tikzcd}
& S_{ij} \arrow[ddl] \arrow[d, "(P')"'] & C_{ij} \arrow[l] \arrow[dd] \\
& S_i \arrow[d, "(P)"'] & \\
R \arrow[r] & S & C. \arrow[l]
\end{tikzcd} \tag{6.2.1.T2a}$$

Consequently, for every pair ij, there is a diagram of the form

$$\begin{tikzcd}
S_{ijkl} \arrow[d, "(P')"'] \arrow[ddr] & \\
S_{ijk} \arrow[d, "(P)"'] & \\
S_{ij} & C_{ij}. \arrow[l]
\end{tikzcd} \tag{6.2.1.T2b}$$

We have therefore produced a composite family

$$S_{ijkl} \xrightarrow{(P')} S_{ijk} \xrightarrow{(P)} S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S \tag{6.2.1.T2c}$$

belonging to $P \circ P' \circ P \circ P'$, factoring through C, and having every object other than S in $\mathscr{C}'$.

Apply condition (c) to each family $\{S_{ijkl} \to S_i\}$. For every i we obtain a family $\{T_{in} \to S_i\} \in P'$ such that $T_{in} \to S$ factors through one of the S_{ijkl}, and hence through C:

$$\begin{tikzcd}
T_{in} \arrow[r] \arrow[d, "(P')"'] & S_{ijkl} \arrow[d] \\
S_i \arrow[d, "(P)"'] & C \arrow[dl] \\
S &
\end{tikzcd} \tag{6.2.1.T2d}$$

Thus the sieve C of S has the required form, completing the verification of (T 2).

Axiom (T 1). Let R be a sieve of S of the prescribed form and let $T \to S$ be a morphism in $\mathscr{C}$. We show that the sieve $R \times_S T$ of T again has the prescribed form. Consider the commutative configuration

$$\begin{tikzcd}
& S_{ij} \arrow[ddl] \arrow[d, "(P')"'] & & U_{ikj} \arrow[ll] \arrow[d, "(P')"'] \\
& S_i \arrow[d, "(P)"'] & T_i \arrow[l] \arrow[d, "(P)"'] & U_{ik} \arrow[l, "(P)"'] \arrow[dl, "(P)"] \\
R \arrow[r] & S & T \arrow[l] & .
\end{tikzcd} \tag{6.2.1.T1a}$$

Put $T_i = S_i \times_S T$. The family $\{T_i \to T\}$ belongs to P, by (P 1). By (b), choose families $\{U_{ik} \to T_i\} \in P$ with $U_{ik} \in \operatorname{Ob} \mathscr{C}'$. By (P 2), $\{U_{ik} \to T\} \in P$. Condition (a) shows that

$$U_{ik} \times_{S_i} S_{ij} = U_{ikj}$$

is an object of $\mathscr{C}'$ and that, for each ik, $\{U_{ikj} \to U_{ik}\} \in P'$. The commutative diagram

$$\begin{tikzcd}
R \arrow[d] & U_{ikj} \arrow[l] \arrow[d, "(P')"'] & \\
S & T \arrow[l] & U_{ik} \arrow[l, "(P)"']
\end{tikzcd} \tag{6.2.1.T1b}$$

shows that the morphisms $U_{ikj} \to T$ factor through the sieve $T \times_S R$ of T. This sieve therefore has the required form, and the proof is complete.

Corollary 6.2.2. *Let $S \in \mathrm{Ob}\,\mathscr{C}'$, and let R be a sieve of S. Then R is covering if and only if there exists a family $\{T_i \to S\} \in P'$ that factors through R.*

Such a sieve is plainly covering. Conversely, apply condition (c) with the family $\{S_i \to S\}$ reduced to the identity isomorphism of S; the proposition then shows that every covering sieve has the stated form.

Corollary 6.2.3. *A presheaf F on $\mathscr{C}$ is separated (respectively, is a sheaf) if and only if the map*

$$F(S) \longrightarrow \prod_i F(S_i)$$

is injective (respectively, the diagram

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

is exact) in each of the following two cases:

(i) $\{S_i \to S\} \in P$;

(ii) $S, S_i \in \mathrm{Ob}\,\mathscr{C}'$ *and* $\{S_i \to S\} \in P'$.

The conditions are necessary because the families in question are covering. Let C be the sieve of S that is the image of a composite family

$$\{S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S\}.$$

A straightforward diagram chase shows that the conditions of the corollary imply that

$$\mathrm{Hom}(S, F) \xrightarrow{g} \mathrm{Hom}(C, F)$$

is injective, respectively bijective. Every covering sieve R of S contains a sieve C of this form, and there is a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(S,F) \arrow[r, "f"] \arrow[dr, "g"'] & \mathrm{Hom}(R,F) \arrow[d, "h"] \\ & \mathrm{Hom}(C,F). \end{tikzcd} \tag{6.2.3.1}$$

Since g is injective, so is f, and hence F is separated. Moreover, R refines C, so h is also injective. If g is bijective, then f is bijective as well, and F is a sheaf.

Remark 6.2.4. *The preceding corollary does not follow from 4.3.5, since P' is not stable under base extension.*

Remark 6.2.5. *Condition (c) holds in particular when:*

(i) *P' is stable under composition;*

(ii) *whenever $\{S_j \to S\}$ is a family of morphisms in $\mathscr{C}'$ belonging to P, it contains a subfamily belonging to P'.*

6.3. Application to the Category of Schemes

Take $\mathscr{C}$ to be the category of schemes, $\mathscr{C}'$ the full subcategory of affine schemes, and P the set of surjective families of open immersions. We shall consider the following sets P'_i:

> P'_1 : finite surjective families of flat morphisms;
>
> P'_2 : finite surjective families of flat, finitely presented, quasi-finite morphisms;
>
> P'_3 : finite surjective families of étale morphisms;
>
> P'_4 : finite surjective families of finite étale morphisms.

The description of P'_2 is accompanied by the following clarification.[57]

For each P'_i, except P'_4, the conditions of Proposition 6.2.1 hold. Condition (c) follows from 6.2.5: since an affine scheme is quasi-compact, every family of morphisms in $\mathscr{C}'$ belonging to P contains a finite subfamily that still belongs to P, and hence belongs to P'_i for $i = 1, 2, 3$. The topology T_i generated by P and P'_i is denoted and named as follows:[57]

T_1	= (fpqc)	the faithfully flat quasi-compact topology,
T_2	= (fppf)	the faithfully flat topology (locally of finite presentation),
T_3	= (ét)	the étale topology,
T_4	= (étf)	the finite étale topology.

Since $P'_1 \supset P'_2 \supset P'_3 \supset P'_4$, one has

$$\text{(fpqc)} \geq \text{(fppf)} \geq \text{(ét)} \geq \text{(étf)} \geq \text{(Zar)}.$$

Proposition 6.3.1.

(i) *A sieve R of S is covering for T_i, $1 \leq i \leq 3$, if and only if there exist a covering (S_p) of S by affine open subschemes and, for every p, a family $\{S_{pq} \to S_p\} \in P'_i$, with all S_{pq} affine, such that every morphism $S_{pq} \to S$ factors through R.* [58]

(ii) *A presheaf F on* **Sch** *is a sheaf for* (fpqc) *(respectively,* (fppf), (ét), *or* (étf)*) if and only if:*

(a) *F is a sheaf for* (Zar)*, that is, a functor of local nature;*

(b) *for every faithfully flat (respectively, faithfully flat, finitely presented and quasi-finite; surjective étale; or surjective finite étale) morphism $T \to S$, with T and S affine, the diagram*

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact.

(iii) *The topologies T_i, for $i = 1, 2, 3, 4$, are coarser than the canonical topology.*

(iv) *Every surjective family of flat open morphisms (respectively, flat morphisms locally of finite presentation; étale morphisms; or finite étale morphisms) is covering for* (fpqc) *(respectively,* (fppf), (ét), *or* (étf)*).*

[57] Editor's note: let P''_2 be the set of finite surjective families of morphisms in $\mathscr{C}'$ that are flat and of finite presentation. By Proposition 6.3.1 below, the topology T_2 generated by P and P'_2 coincides with the topology generated by P and P''_2. This follows from the results of EGA IV$_4$, §17.16 on quasi-sections; see the proof of 6.3.1. We recall the following special case of EGA IV$_4$, 17.16.2. Let S be affine and let $f : X \to S$ be a surjective flat morphism locally of finite presentation. Then there exist a faithfully flat, finitely presented, quasi-finite morphism $S' \to S$, with S' affine, and an S-morphism $S' \to X$.

[58] Editor's note: every family $\{S_{pq} \to S_p\} \in P'_i$ is finite. Hence $S'_p = \coprod_q S_{pq}$ is affine, and the family may be replaced by the morphism $S'_p \to S_p$, which still belongs to P'_i.

(v) *Every finite surjective family of flat quasi-compact morphisms is covering for* (fpqc). *In particular, every faithfully flat quasi-compact morphism is covering for* (fpqc).

Proof. Part (i) follows from 6.2.1 and part (ii) from 6.2.3, taking into account that a sheaf for the Zariski topology carries direct sums to products. Every representable functor is a sheaf for (Zar) and satisfies condition (ii)(b), by SGA 1, VIII, 5.3. Thus T_1 is coarser than the canonical topology, which proves (iii).

We prove (iv). Let $\{S_i \to S\}$ be a family as in the statement. By covering S with affine open subschemes, we reduce immediately to the case in which S is affine.[59]

First suppose that the morphisms $S_i \to S$ are flat and open (respectively, étale). Cover each S_i by affine open subschemes S_{ij}. Since the morphisms are open, the images T_{ij} of the S_{ij} form an open covering of S. The affine scheme S is quasi-compact, so finitely many T_{ij}, indexed by a finite set E, cover it. Then

$$S' = \coprod_{(i,j)\in E} S_{ij}$$

is affine, and $S' \to S$ belongs to P_1', respectively P_3', and is therefore covering. Since it factors through the given family, that family is covering.

In the (étf) case, every S_i is finite over S and hence affine. In the preceding argument one may take the singleton covering $\{S_i\}$ of S_i, which gives $S' \to S \in P_4'$.

Now suppose that the morphisms $f_i : S_i \to S$ are flat and locally of finite presentation. For every $s \in S$, the proof of EGA IV_4, 17.16.2 supplies an affine subscheme $X(s)$ of some S_i, with $s \in f_i(X(s))$, such that the restriction

$$g_i : X(s) \longrightarrow S$$

is flat, finitely presented, and quasi-finite. Its image $g_i(X(s))$ is an open neighborhood $U(s)$ of s, by EGA IV_2, 2.4.6. Since S is affine, finitely many such opens $U(s_j)$, $j = 1, \ldots, n$, cover S. Consequently,

$$X' = \coprod_{j=1}^{n} X(s_j)$$

is affine and $X' \to S$ is surjective, flat, finitely presented, and quasi-finite, so it belongs to P_2'.[60] This completes the proof of (iv).

We prove (v). Let $\{S_i \to S\}$ be a finite surjective family of flat quasi-compact morphisms, and cover S by affine open subschemes T_j. The schemes

$$S_{ij} = T_j \times_S S_i$$

are quasi-compact, and therefore admit finite affine open coverings T_{ijk}. Each $T_{ijk} \to T_j$ is flat, and the family $\{T_{ijk} \to T_j\}$ is finite and surjective, hence covering for T_1. By composition, $\{T_{ijk} \to S\}$ is covering as well. It factors through the given family, which is therefore covering:

$$\begin{tikzcd} S_i \arrow[d] & S_{ij} \arrow[l] \arrow[d] & T_{ijk} \arrow[l] \arrow[dl] \\ S & T_j \arrow[l] & \end{tikzcd} \tag{6.3.1.v}$$

[59]Editor's note: the argument that follows has been simplified by using the fact that S is now assumed affine.

[60]Editor's note: this shows that if P_2'' denotes the set of finite surjective families of flat, finitely presented morphisms in $\mathscr{C}'$, then the topology generated by P and P_2'' equals T_2. Moreover, in the notation used at the beginning of the proof of (iv), if each S_i is covered by affine opens of finite presentation over S, then the resulting morphism $S' \to S$ belongs to P_2''.

Corollary 6.3.2. *Let (M_i) be the following families of morphisms:*

(M_1) : faithfully flat quasi-compact morphisms;

(M_2) : faithfully flat morphisms locally of finite presentation;

(M_3) : surjective étale morphisms;

(M_4) : surjective finite étale morphisms.

The family (M_i) satisfies axioms (a), (b), (c), (d_{T_i}), and (e_{T_i}) of 4.6.3. [61]

Conditions (a), (b), and (c) are classical (see EGA and SGA 1, passim).[62] By 6.3.1(iv),(v), (M_i) satisfies (d_{T_i}). It remains to verify (e_{T_i}). It is enough to verify (e_{T_1}), which implies all the others; this follows from SGA 1, VIII, §§4–5.

Corollary 6.3.3. *Let X be a scheme and let R be an equivalence relation in X of type (M_i). Then R is (M_i)-effective if and only if the sheaf quotient of X by R for T_i is representable; in that case it is represented by the quotient X/R.*

This is 4.6.5.

6.4. Effectivity Conditions

We now seek families (N) of morphisms satisfying axiom (f_T) of 4.7. Observe first that (f_{T_1}) implies (f_{T_i}), so it is enough to consider the (fpqc) topology.

Lemma 6.4.1. *The following families of morphisms satisfy axiom (f_{T_1}) of 4.7, that is, they "descend by* (fpqc)*":*

(N) : open immersions;

(N') : closed immersions;

(N'') : quasi-compact immersions.

By 6.3.1(ii), it is enough to verify that these families descend for the Zariski topology and along a faithfully flat quasi-compact morphism. The first assertion is clear. For (N), the second is SGA 1, VIII, 4.4; for (N'), it is loc. cit., 1.9; for (N''), argue as in loc. cit., 5.5, using the two preceding results.

Corollary 6.4.2. *The same result holds for quasi-compact open immersions.*

These results make the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, and so on applicable to the present situation. We state the first as an example.

Corollary 6.4.3. (= 4.7.1 + 4.6.10). *Let X be a scheme and R an equivalence relation in X. Suppose that $R \to X$ is faithfully flat and quasi-compact, and that $R \to X \times X$ is a closed immersion (respectively, open, quasi-compact, or quasi-compact open). Then the sheaf quotient X/R is the same for the* (fpqc) *topology as for the canonical topology. For every scheme S,*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{closed (respectively, open, retrocompact, or open retrocompact)} \\ \text{subschemes } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ Z \to S \text{ is faithfully flat and quasi-compact and} \\ R_Z \rightrightarrows Z \to S \text{ is exact} \end{array} \right\}. \tag{6.4.3.1}$$

[63]

[61] Editor's note: the original has been corrected by adding surjectivity to (M_3) and (M_4); it is automatically satisfied in the other two cases.

[62] Editor's note: see EGA I, 6.6.4 for "quasi-compact"; EGA II, 6.1.5 for "finite"; EGA IV_1, 1.6.2 for "locally of finite presentation"; EGA IV_2, 2.2.13 for "faithfully flat"; and EGA IV_4, 17.3.3 for "étale".

[63] Editor's note: recall that a subscheme Z of a scheme T is called *retrocompact* if the immersion $Z \hookrightarrow T$ is quasi-compact; see EGA 0_{III}, 9.1.1.

6.5. Principal Homogeneous Bundles

We merely record the terminology:

topology	principal homogeneous bundles
(fpqc)	principal homogeneous bundles (without qualifier)
(ét)	quasi-isotrivial principal homogeneous bundles
(étf)	locally isotrivial principal homogeneous bundles
(Zar)	locally trivial principal homogeneous bundles.

(6.5.1)

6.6. Other Topologies

Other topologies on the category of schemes are sometimes useful. One is the *global finite étale topology* (étfg), generated by the pretopology whose covering families are the surjective families of finite étale morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called *isotrivial*.

The position of this topology, and of the chaotic topology, in the lattice is as follows:

$$\begin{tikzcd}[column sep=small]
 & \text{(canonical)} \arrow[d] & \\
 & \text{(fpqc)} \arrow[d] & \\
 & \text{(fppf)} \arrow[d] & \\
 & \text{(ét)} \arrow[d] & \\
 & \text{(étf)} \arrow[dl] \arrow[dr] & \\
\text{(Zar)} \arrow[dr] & & \text{(étfg)} \arrow[dl] \\
 & \text{(chaotic)} &
\end{tikzcd} \tag{6.6.1}$$

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6.7. Homogeneous Spaces

Let[65] G be an S-group scheme, let X be an S-scheme equipped with a left action of G, and let

$$\Phi : G \times_S X \longrightarrow X \times_S X$$

be the morphism of S-schemes defined on points by

$$(g, x) \longmapsto (gx, x).$$

Recall (cf. 5.1.0 and III.0.1) that X is called a *formally principal homogeneous space* under G if the following equivalent conditions hold:

(i) for every $T \to S$, the set $X(T)$ is empty or is a principal homogeneous space under $G(T)$;

[64] Editor's note: recall (cf. 4.4.2) that the chaotic topology is the coarsest topology, defined by $J(S) = \{S\}$ for every $S \in \mathrm{Ob}\,\mathscr{C}$.

[65] Editor's note: the numbering that follows has been added.

(ii) Φ is an isomorphism of S-functors;

(iii) Φ is an isomorphism of S-schemes.

The equivalence (i)$\Longleftrightarrow$(ii) is clear, and (ii)$\Longleftrightarrow$(iii) holds because $\mathscr{C} = (\mathbf{Sch}/S)$ is a full subcategory of $\widehat{\mathscr{C}}$.

The notion of a *formally homogeneous space*, not necessarily principal homogeneous, is obtained by requiring Φ to be an epimorphism in the category of sheaves for an appropriate topology $\mathcal{T}$. Indeed, requiring Φ to be an epimorphism of S-functors is equivalent to requiring that, for every $T \to S$, the set $X(T)$ be empty or homogeneous (not necessarily principal homogeneous) under $G(T)$. This is too restrictive, as the following simple example shows.

Let

$$S = \operatorname{Spec} R, \qquad G = \mathbf{G}_{m,R}, \qquad X = \mathbf{G}_{m,R},$$

and let G act on X by $t \cdot x = t^2 x$. Then Φ is surjective, finite, and étale, and hence is an epimorphism in the category of sheaves for the (étf) topology (and, *a fortiori*, an epimorphism of S-schemes). On the other hand, the points 1 and -1 of $X(R)$ are not conjugate under any element of $G(R)$. Thus

$$G(R) \times X(R) \longrightarrow X(R) \times X(R)$$

is not surjective.[66] We are therefore led to the following definition.

Definition 6.7.1. *Let G be an S-group, let X be an S-scheme equipped with an action of G, and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The scheme X is called a* formally homogeneous space under G, *relative to $\mathcal{T}$, if the following equivalent conditions hold:*

(i) *the morphism $\Phi : G \times_S X \to X \times_S X$ is an epimorphism in the category of sheaves for $\mathcal{T}$;*

(ii) *for every $T \to S$ and every $x, y \in X(T)$, there exist a $\mathcal{T}$-covering morphism $T' \to T$ and an element $g \in G(T')$ such that $y_{T'} = g \cdot x_{T'}$.*

Remark 6.7.2. *Condition (i) implies in particular that Φ is a universal effective epimorphism in $(\mathbf{Sch}/S)$, by 4.4.3. As is readily seen, it follows that Φ is surjective; cf. 1.3, Editor's note (3).*

Proposition and Definition 6.7.3. [67] *Let G be an S-group, let X be an S-scheme equipped with an action of G, and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The following conditions are equivalent:*

(i) *X satisfies both of the following conditions:*

 (1) *the morphism $\Phi : G \times_S X \to X \times_S X$ is covering, that is, X is a formally homogeneous G-space;*

[66] Editor's note: this difficulty plainly comes from the following fact. Let $\mathscr{C}'$ be a full subcategory of $\widehat{\mathscr{C}}$ containing $\mathscr{C}$, for example the category $\widetilde{\mathscr{C}}_{\mathcal{T}}$ of sheaves on $\mathscr{C}$ for a topology $\mathcal{T}$ coarser than the canonical topology. If $f : X \to Y$ is a morphism in $\mathscr{C}$, then the implications

$$f \text{ epimorphic in } \widehat{\mathscr{C}} \Longrightarrow f \text{ epimorphic in } \mathscr{C}' \Longrightarrow f \text{ epimorphic in } \mathscr{C}$$

are generally strict.

[67] Editor's note: cf. [Ray70], Definition VI.1.1.

(2) *the structural morphism $X \to S$ is also covering, that is, locally for $\mathcal{T}$ it has a section (cf. 4.4.8 bis, (b)).*

(ii) *Locally on S for the topology $\mathcal{T}$, the scheme X, with its G-action, is isomorphic to the quotient sheaf of G by a subgroup scheme H, for $\mathcal{T}$. In other words, there is a covering family $\{S_i \to S\}$ such that every $X \times_S S_i$ represents the sheaf quotient of $G \times_S S_i$ by some subgroup scheme H_i.*

*Under these conditions, X is called a G-*homogeneous space*, relative to $\mathcal{T}$.*

Proof. Assume (ii). Put

$$G_i = G \times_S S_i, \qquad X_i = X \times_S S_i.$$

The scheme X_i has a section over S_i, namely the composite of the unit section $\varepsilon_i : S_i \to G_i$ with the projection $\pi_i : G_i \to X_i = G_i/H_i$. It follows that $X \to S$ is covering.

Moreover, π_i is covering, so $\pi_i \times \pi_i$ is covering as well, by 4.2.3 (C 1),(C 2). There is a commutative diagram

$$\begin{array}{ccc} G_i \times_{S_i} X_i & \xrightarrow{\Phi_i} & X_i \times_{S_i} X_i \\ {\scriptstyle \mathrm{id} \times \pi_i}\big\uparrow & & {\scriptstyle \pi_i \times \pi_i}\big\uparrow \\ G_i \times_{S_i} G_i & \xrightarrow[\sim]{\Psi_i} & G_i \times_{S_i} G_i, \end{array} \tag{6.7.3.1}$$

where Φ_i is obtained from Φ by the base change $S_i \to S$, and Ψ_i is the isomorphism defined on points by

$$(g, g') \longmapsto (gg', g).$$

The composite $(\pi_i \times \pi_i) \circ \Psi_i$ is covering; hence Φ_i is covering by 4.2.3 (C 3). Thus Φ is locally covering and is therefore covering, by 4.2.3 (C 5). This proves (ii)$\Longrightarrow$(i).

Conversely, assume (i), and first suppose in addition that the structural morphism $X \to S$ has a section σ. By EGA I, 5.3.13, σ is an immersion. Define $H = G \times_X S$ by the following diagram, whose two squares are cartesian:

$$\begin{array}{ccccc} H & \longrightarrow & G & \xrightarrow{\mathrm{id}_G \boxtimes \sigma} & G \times_S X \\ \big\downarrow & & {\scriptstyle \pi}\big\downarrow & & {\scriptstyle \Phi}\big\downarrow \\ S & \xrightarrow{\sigma} & X & \xrightarrow{\mathrm{id}_X \boxtimes \sigma} & X \times_S X. \end{array} \tag{6.7.3.2}$$

Here π, $\mathrm{id}_G \boxtimes \sigma$, and $\mathrm{id}_X \boxtimes \sigma$ are the morphisms defined on points, for $T \to S$, $g \in G(T)$, and $x \in X(T)$, by

$$\pi(g) = g \cdot \sigma_T, \qquad (\mathrm{id}_G \boxtimes \sigma)(g) = (g, \sigma_T), \qquad (\mathrm{id}_X \boxtimes \sigma)(x) = (x, \sigma_T).$$

Then π is covering, and H is a subgroup scheme of G representing the stabilizer $\mathrm{Stab}_G(\sigma)$ of σ (cf. I, 2.3.3). Thus, for every $T \to S$,

$$H(T) = \{g \in G(T) \mid g \cdot \sigma_T = \sigma_T\}.$$

Let G/H denote the presheaf

$$T \longmapsto G(T)/H(T),$$

and let $a(G/H)$ be its associated sheaf for $\mathcal{T}$. The preceding construction gives a commutative diagram of presheaves equipped with a G-action:

$$\begin{array}{ccc} G & \xrightarrow{\pi} & X \\ \big\downarrow & \nearrow_{\bar{\pi}} & \\ G/H & & \end{array} \tag{6.7.3.3}$$

where $\bar{\pi}$ is a monomorphism. Since π is covering, $\bar{\pi}$ is covering as well. By 4.3.12, $\bar{\pi}$ therefore induces an isomorphism

$$a(G/H) \xrightarrow{\sim} X.$$

We have proved: if X is a G-homogeneous space such that $X \to S$ has a section σ, then X represents the sheaf quotient G/H, where $H = G \times_X S$ is the stabilizer of σ.

In general, by hypothesis there is a covering family $\{S_i \to S\}$ such that every morphism

$$X_i = X \times_S S_i \longrightarrow S_i$$

has a section σ_i. Put $G_i = G \times_S S_i$. The morphism

$$\Phi_i : G_i \times_{S_i} X_i \longrightarrow X_i \times_{S_i} X_i$$

obtained from Φ by base change along $S_i \to S$ is still covering.

The preceding argument then gives

$$X_i \simeq G_i/H_i,$$

where H_i is the stabilizer of σ_i in G_i. This completes the proof of (i)$\Longrightarrow$(ii).

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[68]Editor's note: this reference has been added.

Exposé V
Construction of Quotient Schemes

by P. Gabriel

The purpose of this exposé is to prove the theorems stated in TDTE III.[1] If X and T are two objects of a category $\mathscr{C}$, we write $X(T)$ instead of $\mathrm{Hom}_{\mathscr{C}}(T, X)$. Similarly, if $\varphi : Y \to X$ is an arrow (respectively, if T is an object) of $\mathscr{C}$, then $\varphi(T)$ denotes the map $g \mapsto \varphi \circ g$ from $Y(T)$ to $X(T)$:

$$\begin{tikzcd}
T \arrow[d, "g"'] \arrow[dr, dashed, "\varphi\circ g"] & \\
Y \arrow[r, "\varphi"'] & X
\end{tikzcd}$$

Figure 1: The map $\varphi(T) : Y(T) \to X(T)$.

Likewise, $T(\varphi)$ denotes the map $g \mapsto g \circ \varphi$ from $T(X)$ to $T(Y)$:

$$\begin{tikzcd}
Y \arrow[r, "\varphi"] \arrow[dr, dashed, "g\circ\varphi"'] & X \arrow[d, "g"] \\
& T
\end{tikzcd}$$

Figure 2: The map $T(\varphi) : T(X) \to T(Y)$.

Finally, if P is a scheme, we denote by $\underline{P}$ the underlying set of P.

As an exception, in the present exposé we do not follow the convention on quotient notation stated in IV 4.6.15 (*loc. cit.*, top of p. 227 of the original), because we wish to give here a construction of quotients that also applies to "pre-equivalence relations"[2] that are not equivalence relations.

[1] Editor's note: namely, Theorems 5.1, 5.3, 6.1, 6.2, and 7.2 of TDTE III. The first two (respectively, the next two) correspond to Theorem 4.1 (respectively, Theorems 7.1 and 8.1) of this exposé. Theorem 7.2 of TDTE III is proved in Exposé $\mathrm{VI_A}$, 3.2 and 3.3.

[2] Editor's note: that is, groupoids with base X; see the terminology at the end of Section 1. When $\mathscr{C}$ is the category of schemes, the quotient $p : X \to Y$ of a groupoid X_* with base X exists under certain hypotheses (see 4.1, 6.1, and 7.1). If, in addition, X_* is an equivalence relation, then under the same hypotheses p is faithfully flat and quasi-compact, hence a universal epimorphism; see *loc. cit.*.

1. $\mathscr{C}$-groupoids

a) Let $\mathscr{C}$ be a category in which products and fiber products exist. Recall first that a diagram

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0 \xrightarrow{\ p\ } Y$$

Figure 3: A parallel pair followed by its proposed cokernel.

in $\mathscr{C}$ is said to be *exact* if $pd_0 = pd_1$ and if, for every $T \in \mathscr{C}$, the map $T(p)$ is a bijection from $T(Y)$ onto the subset of $T(X_0)$ consisting of the arrows $f : X_0 \to T$ such that $fd_0 = fd_1$. One also says that (Y, p) is the *cokernel* of (d_0, d_1), and writes

$$(Y, p) = \operatorname{Coker}(d_0, d_1).$$

b) For example, let $\mathscr{C}$ be the category $(\mathbf{Esp}.\mathbf{An})$ of ringed spaces. In this case a cokernel (Y, p) always exists, and it may be described as follows. The underlying topological space of Y is obtained from X_0 by identifying the points $d_0(x)$ and $d_1(x)$, and by giving Y the quotient topology. The canonical map $\pi : X_0 \to Y$, together with d_0, d_1, then induces a pair of arrows of sheaves of rings on Y:

$$\pi_*(\mathcal{O}_0) \underset{\delta_1}{\overset{\delta_0}{\rightrightarrows}} \pi_*(d_{0*}\mathcal{O}_1) = \pi_*(d_{1*}\mathcal{O}_1)$$

Figure 4: The pair of induced arrows between direct-image sheaves.

Here $\mathcal{O}_i$ is the structure sheaf of X_i. As the sheaf of rings on Y, one chooses the subsheaf of $\pi_*(\mathcal{O}_0)$ whose sections s satisfy $\delta_0(s) = \delta_1(s)$. The arrow p is defined in the evident way.

Consider[3] the following diagram in $(\mathbf{Esp}.\mathbf{An})$:

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0$$

Figure 5: The pair $(d_0, d_1) : X_1 \rightrightarrows X_0$.

Let (Y, p) be its cokernel. An open subset U of X_0 is said to be *saturated* if

$$d_0^{-1}(U) = d_1^{-1}(U),$$

which is equivalent to saying $U = p^{-1}(p(U))$. In this case, since Y has the quotient topology, $p(U)$ is an open subset of Y.

Lemma 1.1. *Let U be a saturated open subset of X, and set $V = p(U)$. Denote by $U_1 = d_0^{-1}(U) = d_1^{-1}(U)$ the corresponding open subset of X_1, and by $\widetilde{d}_0, \widetilde{d}_1$, and $\widetilde{p}$ the restrictions of d_0, d_1 to U_1, and of p to U, respectively. Then $(V, \widetilde{p})$ is a cokernel in* $(\mathbf{Esp}.\mathbf{An})$ *of:*[4]

[3]Editor's note: Lemmas 1.1 and 1.2 have been added; they are used several times in Sections 5–9.

[4]Editor's note: this is not the case in the category of schemes. For example, let $S = \operatorname{Spec}(\mathbf{C})$, let $X_0 = \mathbf{A}^2_S = \operatorname{Spec}(\mathbf{C}[x_1, x_2])$, let $d_1 : \mathbf{G}_{m,S} \times_S \mathbf{A}^2_S \to \mathbf{A}^2_S$ be the action of $\mathbf{G}_{m,S}$ on $\mathbf{A}^2_S$ by homotheties, let d_0 be the projection onto the second factor, and let $U = \mathbf{A}^2_S \setminus \{m\}$, where $m = (0, 0)$. Then the projective line

$$U_1 \overset{\widetilde{d_1}}{\underset{\widetilde{d_0}}{\rightrightarrows}} U \xrightarrow{\widetilde{p}} V$$

Figure 6: The restricted pair and its cokernel in Lemma 1.1.

The verification is easy.

Lemma 1.2.— *Let $d_0, d_1 : X_1 \rightrightarrows X_0$ be a diagram in $(\mathbf{Sch})$, and let (Y, p) be its cokernel in $(\mathbf{Esp.An})$.*

(i) If Y is a scheme and p is a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch})$.

(ii) Suppose that every point of X_0 has a saturated open neighborhood U with the following property. Let $\widetilde{d_0}$ and $\widetilde{d_1}$ be the restrictions of d_0 and d_1 to $d_0^{-1}(U) = d_1^{-1}(U)$, and let (Q, q) be the cokernel of $(\widetilde{d_0}, \widetilde{d_1})$ in $(\mathbf{Esp.An})$. If Q is a scheme and q is a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch})$.

Part (i) is proved in §4(c); since the proof is short, let us repeat it here. Let $f : X_0 \to Z$ be a morphism of schemes such that $fd_0 = fd_1$. By hypothesis there is a unique morphism of ringed spaces $r : Y \to Z$ such that $f = rp$. It remains to show that, for every $y \in Y$, the homomorphism $\mathscr{O}_{r(y)} \to \mathscr{O}_y$ induced by r is local. This follows because p is surjective, so $y = p(x)$ for some x, and because the homomorphism $\mathscr{O}_{f(x)} \to \mathscr{O}_x$ induced by f is local.

Part (ii) follows from part (i) and the preceding lemma.

c) In this exposé we study the existence of $\mathrm{Coker}(d_0, d_1)$ when the double arrow (d_0, d_1) is placed in a richer context. More precisely, let

$$X_2 = X_1 \times_{d_1, d_0} X_1$$

be the fiber product of the diagram

$$\begin{tikzcd} & X_1 \arrow[d, "d_1"'] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \tag{*}$$

and let $d_0', d_2' : X_2 \to X_1$ be the two canonical projections. Thus, by definition, the square

$$\begin{tikzcd} X_2 \arrow[r, "d_0'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"'] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \tag{0}$$

is Cartesian.

$\mathbf{P}^1_S$ is the cokernel of $(\widetilde{d_0}, \widetilde{d_1})$ in both $(\mathbf{Esp.An})$ and $(\mathbf{Sch})$, whereas the cokernel Y of (d_0, d_1) in $(\mathbf{Esp.An})$ is the union of $\mathbf{P}^1_S$ and the point $y_0 = \{p(m)\}$. The only open subset containing y_0 is Y, and $\Gamma(Y, \mathcal{O}_Y) = \mathbf{C}$. If $f : \mathbf{A}^2_S \to T$ is a morphism of S-schemes such that $fd_0 = fd_1$, and if $V = \mathrm{Spec}(A)$ is an affine open subset of T containing $t_0 = f(y_0)$, then $f^{-1}(V) = \mathbf{A}^2$, and the ring homomorphism $A \to \mathbf{C}[x_1, x_2]$ factors through $\mathbf{C}$. This shows that $S = \mathrm{Spec}(\mathbf{C})$ is the cokernel of (d_0, d_1) in the category $(\mathbf{Sch}/S)$.

In addition, suppose that a third arrow $d_1' : X_2 \to X_1$ is given. We say that

$$(d_0, d_1 : X_1 \rightrightarrows X_0, d_1')$$

is a *$\mathscr{C}$-groupoid* if, for every object T of $\mathscr{C}$, $X_1(T)$ is the set of arrows of a groupoid $X_*(T)$ whose set of objects is $X_0(T)$, whose source map is $d_1(T)$, whose target map is $d_0(T)$, and whose composition map is $d_1'(T)$. As usual, we identify

$$(X_1 \times_{d_1,d_0} X_1)(T) \simeq X_1(T) \times_{d_1(T),d_0(T)} X_1(T),$$

and recall that a groupoid is a category in which every arrow is invertible.[5]

If φ is an arrow of the groupoid $X_*(T)$, the map $f \mapsto \varphi \circ f$ is a bijection from the set of arrows f whose target equals the source of φ onto the set of arrows having the same target as φ. This is equivalently expressed by saying that the square

$$\begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_0'"'] & X_1 \arrow[d, "d_0"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \tag{1}$$

is Cartesian.

Similarly, $g \mapsto g \circ \varphi$ is a bijection from the set of arrows g of $X_*(T)$ whose source equals the target of φ onto the set of arrows having the same source as φ. Equivalently, the square

$$\begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_1"'] & X_0 \end{tikzcd} \tag{2}$$

is Cartesian.

On the other hand, let $s : X_0 \to X_1$ be the unique arrow of $\mathscr{C}$ such that, for every T, the map $s(T) : X_0(T) \to X_1(T)$ sends each object of $X_*(T)$ to its identity arrow.[6] The arrow s satisfies

$$d_1 s = \mathrm{id}_{X_0}, \tag{3}$$
$$d_0 s = \mathrm{id}_{X_0}\,. \tag{3 bis}$$

Finally, associativity of the composition maps $d_1'(T)$ is equivalent to commutativity of the diagram

$$\begin{tikzcd} X_1 \times_{d_1,d_0} X_1 \times_{d_1,d_0} X_1 \arrow[r, "d_1' \times X_1"] \arrow[d, "X_1 \times d_1'"'] & X_1 \times_{d_1,d_0} X_1 \arrow[d, "d_1'"] \\ X_1 \times_{d_1,d_0} X_1 \arrow[r, "d_1'"'] & X_1 \end{tikzcd} \tag{4}$$

[5] Editor's note: in this case $X_2(T)$ is the set of pairs (f_2, f_1) of composable arrows, that is, pairs such that $d_0(f_1) = d_1(f_2)$; the maps d_0', d_1', d_2' send (f_2, f_1), respectively, to $f_2, f_2 \circ f_1, f_1$.

[6] Editor's note: $T \mapsto s(T)$ defines an element of $\mathrm{Hom}(h_{X_0}, h_{X_1})$, and the latter is equal to $\mathrm{Hom}(X_0, X_1)$ by the Yoneda lemma.

Conversely, conditions (1), (2), and (4), together with the existence of an arrow s satisfying (3), imply that $(X_1 \rightrightarrows X_0, d_1')$ is a $\mathscr{C}$-groupoid. Condition (3) is harmless: it merely ensures that $d_1(T) : X_1(T) \to X_0(T)$ is surjective for every $T \in \mathscr{C}$. In the sequel we shall mainly use the Cartesian squares (0), (1), and (2), summarized in the diagram

$$\begin{tikzcd} X_2 \arrow[r, shift left, "d_1'"] \arrow[r, shift right, "d_0'"'] \arrow[d, "d_2'"'] & X_1 \arrow[r, "d_0"] \arrow[d, "d_1"] & X_0 \\ X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 & \end{tikzcd} \tag{0,1,2}$$

The two left-hand squares in this diagram (that is, squares (0) and (2)) are Cartesian; the first row is exact; and X_2 is identified with the fiber product $X_1 \times_{d_0,d_0} X_1$.

We use associativity only indirectly, for example to ensure the existence of an arrow s satisfying (3) and (3 bis), or to ensure the existence of an arrow

$$\sigma : X_1 \longrightarrow X_1, \qquad d_0\sigma = d_1, \quad d_1\sigma = d_0. \tag{†}$$

Here σ is chosen so that $\sigma(T) : X_1(T) \to X_1(T)$ sends every arrow of $X_*(T)$ to its inverse.[7]

By an abuse of language, we shall sometimes call a diagram

$$X_2 \overset{d_0',d_1',d_2'}{\Rrightarrow} X_1 \overset{d_0,d_1}{\rightrightarrows} X_0$$

a $\mathscr{C}$-groupoid when (0), (1), and (2) are Cartesian, (4) is commutative, and there exists an s satisfying (3). Thus X_2 may be "a" fiber product of (*) without being "the" fiber product of (*).[8]

Terminology. In place of the $\mathscr{C}$-groupoid X_*, we shall also speak of the *groupoid X_* with base X_0*, or of the *pre-equivalence relation X_* in X_0*.

2. Examples of $\mathscr{C}$-groupoids

a) Let X be an object of $\mathscr{C}$, and let G be a $\mathscr{C}$-group acting on the left on X. We denote by $d_0 : G \times X \to X$ the arrow defining the action of G on X, by $d_1 : G \times X \to X$ the projection onto the second factor, and by $\mu : G \times G \to G$ the arrow defining

the $\mathscr{C}$-group structure on G, and finally by $\mathrm{pr}_{2,3}$ the projection of $G \times G \times X = G \times (G \times X)$ onto the second factor. Then

$$G \times G \times X \xrightarrow[G\times d_0]{\mathrm{pr}_{2,3},\ \mu\times X} G \times X \overset{d_1}{\underset{d_0}{\rightrightarrows}} X$$

Figure 7: The groupoid attached to the action of the $\mathscr{C}$-group G on X.

is a $\mathscr{C}$-groupoid.

b) Let $d_0, d_1 : X_1 \to X_0$ be an *equivalence pair*. In other words, if $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is the arrow with components d_0 and d_1, we suppose that, for every object T of $\mathscr{C}$, the map $(d_0 \boxtimes d_1)(T)$ is a bijection from $X_1(T)$ onto the graph of an equivalence relation on $X_0(T)$.

[7] Editor's note: the Yoneda lemma shows that σ is an involutive automorphism of X_1; this will be used, for example, in 3(e) and in Theorem 4.1.

[8] Editor's note: see Example 2(a) below.

Consequently, $X_1(T)$ is identified with the set of pairs (x, y) of elements of $X_0(T)$ such that $x \sim y$. Likewise,

$$X_2(T) = (X_1 \times_{d_1, d_0} X_1)(T)$$

is identified with the set of triples (x, y, z) of elements of $X_0(T)$ such that $x \sim y$ and $y \sim z$. There is therefore a unique arrow $d'_1 : X_2 \to X_1$ making squares (1) and (2) commute: $d'_1(T)$ must send $(x, y, z) \in X_2(T)$ to $(x, z) \in X_1(T)$. With this choice of d'_1,

$$(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$$

is a $\mathscr{C}$-groupoid.

Conversely, consider a $\mathscr{C}$-groupoid X_* for which $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is a monomorphism. Then (d_0, d_1) is an equivalence pair, and X_* can be reconstructed from (d_0, d_1) as explained a few lines above.[9]

c) If $p : X \to Y$ is any arrow of $\mathscr{C}$, and if pr_1 and pr_2 are the two projections of $X \times_{p,p} X$ onto X, then

$$(\mathrm{pr}_1, \mathrm{pr}_2) : X \times_{p,p} X \rightrightarrows X$$

is an equivalence pair. The arrow p is called an *effective epimorphism* if the diagram

$$X \times_{p,p} X \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} X \xrightarrow{\ p\ } Y$$

Figure 8: The defining diagram for an effective epimorphism.

is exact, that is, if

$$(Y, p) = \mathrm{Coker}(\mathrm{pr}_1, \mathrm{pr}_2).$$

For example, let S be a Noetherian scheme and let $\mathscr{C}$ be the category of schemes finite over S. We show that an epimorphism in $\mathscr{C}$ need not be effective. Take

$$S = \mathrm{Spec}\, k[T^3, T^5], \qquad Y = S, \qquad X = \mathrm{Spec}\, k[T],$$

where k is a commutative field. If i is the inclusion of $B = k[T^3, T^5]$ into $A = k[T]$, take $p = \mathrm{Spec}\, i$. Then $X \times_{p,p} X$ is identified with $\mathrm{Spec}(A \otimes_B A)$, and $\mathrm{Coker}(\mathrm{pr}_1, \mathrm{pr}_2)$ with $\mathrm{Spec}\, B'$, where B' is the subring of A consisting of the elements a such that $a \otimes_B 1 = 1 \otimes_B a$. But

$$T^7 \otimes_B 1 = (T^2 T^5) \otimes_B 1 = T^2 \otimes_B T^5 = T^2 \otimes_B (T^3 T^2) = T^5 \otimes_B T^2 = 1 \otimes_B T^7.$$

Thus T^7 belongs to B', does not belong to B, and $\mathrm{Spec}\, B'$ is different from $\mathrm{Spec}\, B$, which gives the counterexample.[10]

3. Some sorites concerning $\mathscr{C}$-groupoids

Here are, in no particular order, some remarks used below.

a) Let

[9]Editor's note: in particular, if G is a $\mathscr{C}$-group acting on the left on an object X of $\mathscr{C}$, and if X_* is the $\mathscr{C}$-groupoid defined in 2(a), then (d_0, d_1) is an equivalence pair if and only if G acts freely on X; cf. Exposé III, 3.2.1.

[10]Editor's note: the same argument applies to $B = k[T^3, T^4]$ and $T^5 \otimes_B 1$; more generally, it applies to $B = k[T^n, T^{n+r}]$ and the element $T^{n+2r} \otimes_B 1$, provided that n does not divide $2r$.

$$X_2 \overset{d'_0,d'_1,d'_2}{\rightrightarrows\!\!\!\to} X_1 \overset{d_0,d_1}{\rightrightarrows} X_0$$

Figure 9: The $\mathscr{C}$-groupoid X_*.

be a $\mathscr{C}$-groupoid, and let $f_0 : Y_0 \to X_0$ be an arrow of $\mathscr{C}$. We shall define a $\mathscr{C}$-groupoid with base Y_0,

$$Y_2 \overset{e'_0,e'_1,e'_2}{\rightrightarrows\!\!\!\to} Y_1 \overset{e_0,e_1}{\rightrightarrows} Y_0$$

Figure 10: The induced $\mathscr{C}$-groupoid Y_*.

which will be said to be *induced by X_* and f_0*. We also say that Y_* is the *inverse image of X_* under the base change f_0*.

We choose Y_1 to be the fiber product of the diagram

$$\begin{tikzcd} Y_1 \arrow[r, dashed, "f_1"] \arrow[d, dashed] & X_1 \arrow[d, "d_0 \boxtimes d_1"] \\ Y_0 \times Y_0 \arrow[r, "f_0 \times f_0"'] & X_0 \times X_0 \end{tikzcd}$$

Figure 11: The fiber-product definition of Y_1.

and take e_0 and e_1 to be the composites of the canonical arrow $Y_1 \to Y_0 \times Y_0$ with the first and second projections of $Y_0 \times Y_0$, respectively. The morphism $Y_1 \to Y_0 \times Y_0$ is then $e_0 \boxtimes e_1$, and

$$f_0 \circ e_i = d_i \circ f_1 \qquad (i = 0, 1),$$

where f_1 denotes the projection of Y_1 onto X_1.

Set $Y_2 = Y_1 \times_{e_0,e_1} Y_1$, cf. 1(c). One may say that the pair (e_0, e_1) is defined so that, for every $T \in \mathscr{C}$ and every pair (y, x) of elements of $Y_0(T)$, there is a certain bijective correspondence

$$\psi \longmapsto {}_y\psi_x$$

between the arrows ψ of $X_*(T)$ with source $f_0(x)$ and target $f_0(y)$, and the arrows ${}_y\psi_x$ of $Y_*(T)$ with source x and target y. Thus $e'_1 : Y_2 \to Y_1$ is determined by defining, for every $T \in \mathscr{C}$, the composition of arrows in $Y_*(T)$ by the formula

$$_z\psi_y \circ {}_y\varphi_x = {}_z(\psi \circ \varphi)_x.$$

It is clear that this definition makes each $Y_*(T)$ a groupoid.

b) Given the $\mathscr{C}$-groupoid X_* and the base change $f_0 : Y_0 \to X_0$, one can reconstruct the pair $(e_0, e_1) : Y_1 \rightrightarrows Y_0$ in another way.[11] Construct $Y_0 \times_{X_0} X_1$, pr_1, and pr_2 so that the square

$$\begin{tikzcd} Y_0 \times_{X_0} X_1 \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & X_1 \arrow[d, "d_0"] \\ Y_0 \arrow[r, "f_0"'] & X_0 \end{tikzcd}$$

Figure 12: The Cartesian square defining $Y_0 \times_{X_0} X_1$.

[11] Editor's note: this second viewpoint is used in 3(f) and in the proof of 6.1.

is Cartesian. A straightforward reduction to the set-theoretic case then shows that the following square is Cartesian:

$$\begin{tikzcd} Y_1 \arrow[r, "e_0 \boxtimes f_1"] \arrow[d, "e_1"'] & Y_0 \times_{X_0} X_1 \arrow[d, "d_1 \circ \mathrm{pr}_2"] \\ Y_0 \arrow[r, "f_0"'] & X_0 \end{tikzcd}$$

Figure 13: The second Cartesian square in the reconstruction of (e_0, e_1).

Here f_1 denotes the canonical projection of

$$Y_1 = (Y_0 \times Y_0) \times_{(X_0 \times X_0)} X_1$$

onto X_1.

c) We give two examples of inverse images of a $\mathscr{C}$-groupoid. Take $Y_0 = X_1$ and $f_0 = d_0$. For every object T of $\mathscr{C}$, the set $Y_1(T)$ is then identified with the set of diagrams of the form

$$\begin{tikzcd} b \arrow[r, "\varphi"] & d \\ a \arrow[u, "f"] & c \arrow[u, "g"'] \end{tikzcd}$$

Figure 14: An arrow of the first inverse-image groupoid.

in $X_*(T)$. The source of such a diagram is the arrow f, and its target is the arrow g. These diagrams compose in the evident way.

Now set $Y_0' = X_1$ and $f_0' = d_1$ (we add the primes[12] to avoid any confusion with the preceding example). In this case, for every $T \in \mathscr{C}$, the set $Y_1'(T)$ is identified with the set of diagrams of the form

$$\begin{tikzcd} b & d \\ a \arrow[u, "f"] \arrow[r, "\psi"'] & c \arrow[u, "g"'] \end{tikzcd}$$

Figure 15: An arrow of the second inverse-image groupoid.

in the groupoid $X_*(T)$. The source of such a diagram is f, its target is g, and these diagrams again compose in the evident way.

It is now clear that the identity map of $Y_0(T)$, together with the map

$$\begin{tikzcd} b \arrow[r, "\varphi"] & d \\ a \arrow[u, "f"] & c \arrow[u, "g"'] \end{tikzcd} \quad \mapsto \quad \begin{tikzcd} b & d \\ a \arrow[u, "f"] \arrow[r, "g^{-1} \circ \varphi \circ f"'] & c \arrow[u, "g"'] \end{tikzcd}$$

Figure 16: The map from $Y_1(T)$ to $Y_1'(T)$.

[12] Editor's note: "accents" in the original.

defines an isomorphism from the groupoid $Y_*(T)$ onto $Y'_*(T)$. Moreover, this isomorphism depends functorially on T, so the $\mathscr{C}$-groupoids Y_* and Y'_* are isomorphic.[13]

d)

Proposition 3.1.— *Keep the notation of 3(a), and suppose that f_0 is an effective and universal epimorphism. Then* $\mathrm{Coker}(d_0, d_1)$ *exists if and only if* $\mathrm{Coker}(e_0, e_1)$ *exists.*[14] *Moreover, in this case f_0 induces an isomorphism*

$$\mathrm{Coker}(d_0, d_1) \xrightarrow{\sim} \mathrm{Coker}(e_0, e_1).$$

Recall first that an epimorphism $f_0 : Y_0 \to X_0$ is called *universal* if, in every Cartesian square

$$\begin{tikzcd} Y' \arrow[r] \arrow[d, "f'"'] & Y_0 \arrow[d, "f_0"] \\ X' \arrow[r] & X_0 \end{tikzcd}$$

Figure 17: A base change of the universal epimorphism f_0.

the arrow f' is an epimorphism. Now let $C(d_0, d_1)$ denote the covariant functor from $\mathscr{C}$ to sets which assigns to every $T \in \mathscr{C}$ the kernel of the pair

$$T(d_0), T(d_1) : T(X_0) \rightrightarrows T(X_1),$$

and define $C(e_0, e_1)$ in the same way. For every $T \in \mathscr{C}$, there is therefore a commutative diagram

$$\begin{tikzcd} C(d_0,d_1)(T) \arrow[r] \arrow[d, "T(f)"'] & T(X_0) \arrow[r, shift left, "T(d_1)"] \arrow[r, shift right, "T(d_0)"'] \arrow[d, "T(f_0)"'] & T(X_1) \arrow[d, "T(f_1)"'] \\ C(e_0,e_1)(T) \arrow[r] & T(Y_0) \arrow[r, shift left, "T(e_1)"] \arrow[r, shift right, "T(e_0)"'] & T(Y_1) \end{tikzcd}$$

Figure 18: The comparison of the functors $C(d_0, d_1)$ and $C(e_0, e_1)$.

Here $T(f)$ is the injection induced by the injection $T(f_0)$. If we show that $T(f)$ is surjective for every T, we obtain a functorial isomorphism

$$f : C(d_0, d_1) \xrightarrow{\sim} C(e_0, e_1).$$

Thus one of these functors is representable if and only if the other is, which proves the proposition.

To prove that $T(f)$ is surjective, consider the diagram

$$\begin{tikzcd} & Y_1 \arrow[r, "f_1"] \arrow[d, shift right, "e_0"'] \arrow[d, shift left, "e_1"] & X_1 \arrow[d, shift right, "d_0"'] \arrow[d, shift left, "d_1"] \\ Y_0 \times_{X_0} Y_0 \arrow[ur, "\Delta"] \arrow[r, shift left, "\mathrm{pr}_2"] \arrow[r, shift right, "\mathrm{pr}_1"'] & Y_0 \arrow[r, "f_0"'] & X_0 \end{tikzcd}$$

Figure 19: The diagram used to prove the surjectivity of $T(f)$.

[13]Editor's note: this plays a crucial role in the proof of Lemma 6.1.
[14]Editor's note: the original has been modified to display explicitly the isomorphism below.

Here Δ is the section of $Y_1 \to Y_0 \times Y_0$ defined by the morphism

$$s \circ f_0 \circ \mathrm{pr}_1 : Y_0 \times Y_0 \longrightarrow X_1,$$

where $s : X_0 \to X_1$ satisfies (3) and (3 bis) of Section 1.

If $g : Y_0 \to T$ satisfies $g \circ e_0 = g \circ e_1$, then

$$g \circ e_0 \circ \Delta = g \circ e_1 \circ \Delta,$$

so $g \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$. Since f_0 is an effective epimorphism, g factors as $h \circ f_0$ for an arrow $h : X_0 \to T$; in other words, $g = T(f_0)(h)$. It remains to prove that $h \in C(d_0, d_1)(T)$, or equivalently that $hd_0 = hd_1$. But

$$hd_0f_1 = hf_0e_0 = ge_0 = ge_1 = hf_0e_1 = hd_1f_1,$$

and the desired equality follows because f_1 is an epimorphism (since f_0 is a universal epimorphism).

e) Now let S be a scheme and take $\mathscr{C} = (\mathbf{Sch}/S)$. The data of a $\mathscr{C}$-groupoid

$$\begin{tikzcd} X_2 \arrow[r, shift left=2, "d'_2"] \arrow[r, "d'_1" description] \arrow[r, shift right=2, "d'_0"'] & X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \end{tikzcd}$$

Figure 20: A $(\mathbf{Sch}/S)$-groupoid.

defines an equivalence relation on the set $\underline{X_0}$ underlying the scheme X_0: for $x, y \in \underline{X_0}$, write $x \sim y$ if there exists $z \in \underline{X_1}$ such that $x = d_1z$ and $y = d_0z$. Reflexivity and symmetry are clear.[15] Let us prove transitivity. If $x \sim y$ and $y \sim z$, there exist $u, v \in \underline{X_1}$ such that

$$x = d_1u, \qquad y = d_0u, \qquad y = d_1v, \qquad z = d_0v.$$

It follows that (v, u) belongs to the set-theoretic fiber product $\underline{X_1} \times_{\underline{d_1},\underline{d_0}} \underline{X_1}$. Since the canonical map

$$\underline{X_1 \times_{d_1,d_0} X_1} \longrightarrow \underline{X_1} \times_{\underline{d_1},\underline{d_0}} \underline{X_1}$$

Figure 21: The canonical map from the underlying set of the fiber product to the fiber product of the underlying sets.

is surjective, (v, u) is the image of some $w \in \underline{X_2}$. Hence

$$x = d_1d'_1w, \qquad z = d_0d'_1w,$$

and therefore $x \sim z$.

f) Keep the notation of 3(a) and 3(b), still taking $\mathscr{C} = (\mathbf{Sch}/S)$. If x, y are points of Y_0, then

$$x \sim y \iff f_0(x) \sim f_0(y).$$

Thus the inverse image of the equivalence relation defined by a groupoid is the equivalence relation defined by the inverse image of that groupoid.

[15] Editor's note: reflexivity follows from the existence of $s : X_0 \to X_1$, which is a section of both d_0 and d_1; symmetry follows from the existence of the involution σ of X_1 which "exchanges d_0 and d_1", that is, which satisfies $d_0\sigma = d_1$ and $d_1\sigma = d_0$; cf. Section 1, (3), (3 bis), and (†).

Indeed, suppose that $x \sim y$. There is a $z \in \underline{Y_1}$ such that $x = e_1(z)$ and $y = e_0(z)$. Since $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, it follows that

$$f_0(x) = d_1 f_1(z), \qquad f_0(y) = d_0 f_1(z),$$

and hence $f_0(x) \sim f_0(y)$.

Conversely, suppose that $f_0(x) \sim f_0(y)$, and let $z \in \underline{X_1}$ satisfy

$$f_0(y) = d_1(z), \qquad f_0(x) = d_0(z).$$

Using the construction and notation of 3(b), there is a point $t \in \underline{Y_0 \times_{X_0} X_1}$ such that $\mathrm{pr}_1(t) = x$ and $\mathrm{pr}_2(t) = z$. Since $f_0(y) = d_1 \, \mathrm{pr}_2(t)$, there is likewise an $s \in \underline{Y_1}$ such that

$$y = e_1(s), \qquad (e_0 \boxtimes f_1)(s) = t.$$

It follows that

$$e_0(s) = \mathrm{pr}_1(e_0 \boxtimes f_1)(s) = \mathrm{pr}_1(t) = x,$$

so $x \sim y$.

4. Passage to the quotient by a finite flat groupoid (proof of a special case)

Theorem 4.1.— *Consider a* $(\mathbf{Sch}/S)$*-groupoid*

$$X_2 \overset{d_2'}{\underset{d_0'}{\overset{d_1'}{\rightrightarrows\!\!\to}}} X_1 \overset{d_1}{\underset{d_0}{\rightrightarrows}} X_0$$

and suppose that the following conditions hold:[16]

a) d_1 *is finite locally free;*

b) for every $x \in X_0$*, the set* $d_0 d_1^{-1}(x)$ *is contained in an affine open subset of* X_0*.*[17]

Then:

(i) *there exists a cokernel* (Y, p) *of* (d_0, d_1) *in* $(\mathbf{Sch}/S)$*. Moreover, such a pair* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of all ringed spaces;*

(ii) p *is integral and open, and* Y *is affine when* X_0 *is affine;*[18]

(iii) *the morphism*

$$X_1 \longrightarrow X_0 \times_Y X_0$$

with components d_0 *and* d_1 *is surjective;*

[16] Editor's note: since $d_0 = d_1\sigma$, where σ is an involutive automorphism of X_1, these two conditions are symmetric in d_1 and d_0. Moreover, $d_0 d_1^{-1}(x) = d_1 d_0^{-1}(x)$.

[17] Editor's note: condition b) cannot be omitted. H. Hironaka gave an example of an action of the finite group $G = \mathbb{Z}/2\mathbb{Z}$ on a proper complex variety X_0 for which the quotient X_0/G is an algebraic space that is not a scheme; see [Hi62], and also [Mum65], Chapter 4, §3.

[18] Editor's note: the assertion that p is open has been added by repeating the analogous argument given in 6.1.

(iv) *if* (d_0, d_1) *is an equivalence pair, then* $X_1 \to X_0 \times_Y X_0$ *is an isomorphism*[19] *and* $p : X_0 \to Y$ *is finite locally free.*[20] *Moreover,* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of sheaves for the fppf topology and, for every base change* $Y' \to Y$*, the scheme* Y' *is the cokernel of the groupoid* $X_* \times_Y Y'$ *obtained from* X_* *by the base change* $X_0 \times_Y Y' \to X_0$*.*

In particular, for every base change $S' \to S$*, the scheme* $Y' = Y \times_S S'$ *is the cokernel of the* S'*-groupoid* $X'_* = X_* \times_S S'$*. Thus, in this case, formation of the quotient commutes with base change.*

It follows immediately from (i) that the underlying topological space of Y is the quotient of the underlying topological space of X_0 by the equivalence relation defined by the $(\mathbf{Sch}/S)$-groupoid X_*.

We shall first prove the theorem when X_0 is affine and d_1 is locally free of constant rank n. We shall then see how to reduce the general situation to this special case.

In the case now under consideration, X_0, X_1, X_2 are affine. We may therefore suppose that

$$X_i = \operatorname{Spec} A_i, \qquad d_j = \operatorname{Spec} \delta_j, \qquad d'_k = \operatorname{Spec} \delta'_k,$$

where the A_i are commutative rings and the δ_j, δ'_k are ring homomorphisms. The diagram (0,1,2) may then be replaced by the following dual diagram:

$$\begin{tikzcd}
A_2 & A_1 \arrow[l, shift right, "\delta'_1"'] \arrow[l, shift left, "\delta'_0"] & A_0 \arrow[l, "\delta_0"'] \\
A_1 \arrow[u, "\delta'_2"] & A_0 \arrow[u, "\delta_1"'] \arrow[l, shift right, "\delta_1"'] \arrow[l, shift left, "\delta_0"]
\end{tikzcd} \tag{$(0,1,2)^*$}$$

The two left-hand squares in this diagram are cocartesian.

Let B be the subring of A_0 consisting of the elements $a \in A_0$ for which $\delta_0(a) = \delta_1(a)$.

a) We first show that A_0 is integral over B. For $a \in A_0$, let

$$P_{\delta_1}(T, \delta_0(a)) = T^n - \sigma_1 T^{n-1} + \cdots + (-1)^n \sigma_n \tag{1}$$

be the characteristic polynomial of $\delta_0(a)$, where A_1 is regarded as an A_0-algebra through δ_1; see Bourbaki, *Algèbre*, VIII, §12, and *Algèbre commutative*, II, §5, Exercise 9. Since the two left-hand squares of the diagram $(0,1,2)^*$ are cocartesian, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_0\delta_0(a)), \tag{2}$$

$$\delta_1(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_1\delta_0(a)). \tag{3}$$

Because $\delta'_0\delta_0 = \delta'_1\delta_0$, it follows that

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = \delta_1(P_{\delta_1}(T, \delta_0(a)))\,;$$

equivalently,

$$\delta_0(\sigma_i) = \delta_1(\sigma_i) \qquad \text{for every } i.$$

[19] Editor's note: in this case $X_1 \to X_0 \times_S X_0$ is therefore an immersion (EGA I, 5.3.10); see also Exposé $\mathrm{VI_B}$, 9.2.1. For existence of the quotient—in the category of schemes or in that of algebraic spaces—under the weaker hypothesis that d_0 and d_1 are finite but not necessarily flat, see [An73], §1.1, [Fe03], [Ko08], and the references there.

[20] Editor's note: the consequences that follow have been made explicit; see [Ray67a], Theorem 1(iii), and the proof given below at the end of Section 5.

On the other hand, the Cayley–Hamilton theorem gives

$$\delta_0(a)^n - \delta_1(\sigma_1)\delta_0(a)^{n-1} + \cdots + (-1)^n\delta_1(\sigma_n) = 0.$$

Since $\delta_1(\sigma_i) = \delta_0(\sigma_i)$, one also has

$$\delta_0(a)^n - \delta_0(\sigma_1)\delta_0(a)^{n-1} + \cdots + (-1)^n\delta_0(\sigma_n) = 0.$$

Hence

$$a^n - \sigma_1 a^{n-1} + \cdots + (-1)^n\sigma_n = 0,$$

because there is a homomorphism $\tau : A_1 \to A_0$ such that $\tau\delta_0 = \mathrm{id}_{A_0}$, so δ_0 is injective. It follows that A_0 is integral over B.

b) Consider two prime ideals x and y of A_0. We shall show that the equality

$$x \cap B = y \cap B$$

implies the existence of a prime ideal z of A_1 such that

$$x = d_0(z), \qquad y = d_1(z).$$

Indeed, if this assertion were false, x would be distinct from $\delta_0^{-1}(t)$ for every prime ideal t of A_1 such that $\delta_1^{-1}(t) = y$. For every such t, one would have

$$\delta_0^{-1}(t) \cap B = \delta_1^{-1}(t) \cap B = y \cap B = x \cap B.$$

It would follow from Cohen–Seidenberg (see Bourbaki, *Algèbre commutative*, V, §2, Corollary 1 to Proposition 1) that x is contained in none of the ideals $\delta_0^{-1}(t)$.[21] There are at most n prime ideals t of A_1 satisfying $\delta_1^{-1}(t) = y$ (see *loc. cit.*, Proposition 3). Hence, by the Prime Avoidance Lemma (*loc. cit.*, II, §1, Proposition 3), there would be an element $a \in x$ belonging to none of the ideals $\delta_0^{-1}(t)$.

Consequently, $\delta_0(a)$ would belong to none of these prime ideals t, and the lemma below would imply that the norm $N_{\delta_1}(\delta_0(a))$ does not belong to y. Here the norm is computed by regarding A_1 as an A_0-algebra through δ_1; with the notation of part a),

$$N_{\delta_1}(\delta_0(a)) = \sigma_n.$$

On the other hand,

$$(-1)^{n-1}\sigma_n = a^n + \sum_{i=1}^{n-1}(-1)^i\sigma_i a^{n-i},$$

so this norm belongs to $B \cap x = B \cap y$, a contradiction.

Lemma 4.1.1.— *Let $A \to A'$ be a homomorphism of commutative rings which makes A' a projective A-module of rank n. Let $p \in \mathrm{Spec}(A)$, let $q_1, \ldots, q_r$ be the points of $\mathrm{Spec}(A')$ lying above p, and let $a \in A'$. Then $a \in q_1 \cup \cdots \cup q_r$ if and only if its norm $N(a)$ belongs to p.*

Indeed, after replacing A and A' by the localizations A_p and A'_p, we reduce to the case in which (A, p) is local and A' is semilocal, with

$$\mathrm{Spec}(A') = \{q_1, \ldots, q_r\}.$$

[21] Editor's note: the original has been expanded in what follows; in particular, Lemma 4.1.1, taken from [DG70], III, §2.4, Lemma 4.3, has been added.

In this case A' is a free A-module of rank n (see Bourbaki, *Algèbre commutative*, II, §3.2, Corollary 2 to Proposition 5), and $N(a)$ is the determinant of the endomorphism

$$\ell_a : A' \longrightarrow A', \qquad a' \longmapsto aa'.$$

Thus one has the equivalences

$$N(a) \notin p \quad \Longleftrightarrow \quad N(a) \text{ is invertible} \quad \Longleftrightarrow \quad \ell_a \text{ is invertible} \quad \Longleftrightarrow \quad a \notin q_1 \cup \cdots \cup q_r.$$

c) Proof of (i). Set $Y = \operatorname{Spec} B$ and $p = \operatorname{Spec} i$, where $i : B \hookrightarrow A_0$ is the inclusion. By part a), the morphism $p : X_0 \to Y$ is surjective. We first show that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. Indeed, part b) shows that the set underlying $\operatorname{Spec} B$ is obtained from the set underlying X_0 by identifying the points x and y for which there exists $z \in X_1$ such that

$$d_1 z = y, \qquad d_0 z = x.$$

Moreover, since i is integral, $p = \operatorname{Spec} i$ is closed, and therefore Y carries the quotient topology induced by X_0.

It follows that p is open. Indeed, let U' be any open subset of X_0. Since d_1 is surjective and finite locally free, it is faithfully flat and of finite presentation, hence open. Thus

$$U = d_1\Big(d_0^{-1}(U')\Big)\,,$$

the saturation of U' for the equivalence relation defined by X_*, is open. Since Y has the quotient topology, $p(U') = p(U)$ is open.

Finally, the choice of B, together with the fact that p, d_0, and d_1 are affine, shows that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*\mathcal{O}_{X_0} \underset{p_*(\delta_0)}{\overset{p_*(\delta_1)}{\rightrightarrows}} p_* d_{0*}\mathcal{O}_{X_1} = p_* d_{1*}\mathcal{O}_{X_1}$$

is exact.

It remains to show that (Y, p) is also the cokernel of (d_0, d_1) in the category of schemes—more generally, in the category of locally ringed spaces. Let $q : X_0 \to Z$ be a morphism of schemes such that $qd_0 = qd_1$. By what precedes, there is a unique morphism of ringed spaces $r : Y \to Z$ such that $q = rp$. We must show that, for every $y \in Y$, the homomorphism

$$\mathcal{O}_{Z,r(y)} \longrightarrow \mathcal{O}_{Y,y}$$

induced by r is local. Since p is surjective, $y = p(x)$ for some $x \in X_0$, and the assertion follows from the fact that the homomorphism

$$\mathcal{O}_{Z,q(x)} \longrightarrow \mathcal{O}_{X_0,x}$$

induced by q is local.

d) Proof of (ii). This follows from parts a) and c).

e) Proof of (iii). Recall that $\underline{P}$ denotes the set underlying a scheme P, and that $\underline{d} : \underline{P} \to \underline{Q}$ denotes the map induced by a morphism $d : P \to Q$.

Lemma 4.1.2.—[22] *Let $(A, \mathfrak{m})$ be a local ring, let k be its residue field, and let K/k be a field extension. There exists a local, flat A-algebra B such that $B/\mathfrak{m}B$ is k-isomorphic to K. If K/k is finite, B may moreover be chosen finite and free over A.*

[22] Editor's note: this lemma, which is used repeatedly in this Exposé and in later Exposés ($\mathrm{VI_A}$, $\mathrm{VI_B}$), has been inserted here. It appeared as Lemma $\mathrm{VI_B}$, 4.5.1, in the original 1965 edition of SGA 3.

This is proved in EGA 0_{III}, 10.3.1, where it is also shown that B may be chosen Noetherian when A is Noetherian. For the reader's convenience, we recall the proof.

Put $A' = A[T]$, where T is an indeterminate. If $K = k(T)$, let $\mathfrak{p} = \mathfrak{m}A'$ and $B = A'_{\mathfrak{p}}$. Then

$$B/\mathfrak{m}B \simeq k[T]_{(0)} = k(T).$$

The ring B is flat over A', which is a free A-module, so B is flat over A.

Suppose next that $K = k(t) = k[t]$, where t is algebraic over k. Set $B = A'/(F)$, where $F \in A'$ is a monic polynomial whose image in $k[T]$ is the minimal polynomial f of t over k. Then B is a free A-module of finite rank

$$\deg(F) = \deg(f).$$

In particular, B is integral over A, and hence every maximal ideal of B contains $\mathfrak{m}$. Since

$$B/\mathfrak{m}B \simeq k[T]/(f) \simeq K,$$

B is local, with maximal ideal $\mathfrak{m}B$. This already proves that B can be chosen finite and free over A when $[K : k] < \infty$.

In the general case, choose a family $(t_i)_{i \in I}$ of generators of K/k, and well-order I; that is, give I a total order $\leq$ in which every nonempty subset has a least element. For $i \in I$, let k_i (respectively $k_{<i}$) be the subfield of K generated by the t_j with $j \leq i$ (respectively $j < i$). By adding one element if necessary, we may suppose that I has a greatest element ξ, so that $K = k_\xi$.

Let $J \subseteq I$ be the set of indices i for which there exists a direct system $(A_j)_{j \leq i}$ of local, flat A-algebras such that

$$A_j/\mathfrak{m}A_j \simeq k_j$$

and A_j is flat over A_ℓ whenever $\ell < j$. Suppose that $I \setminus J$ is nonempty, let i be its least element, and put

$$A' = \varinjlim_{j<i} A_j.$$

Since tensor products commute with direct limits, A' is flat over A and over every A_j, $j < i$, and

$$A'/\mathfrak{m}A' \simeq A' \otimes_A (A/\mathfrak{m}) \simeq k_{<i}.$$

Moreover, A' is local, with maximal ideal $\mathfrak{m}A'$. Indeed, if $x = f_j(x_j)$ is not invertible, then x_j is not invertible and hence belongs to the maximal ideal $\mathfrak{m}A_j$ of A_j; therefore $x \in \mathfrak{m}A'$.

The monogenic case treated above now gives a local, flat A'-algebra A_i such that

$$A_i/\mathfrak{m}A_i \simeq k_{<i}(t_i) = k_i.$$

The algebra A_i is flat over every A_j, $j < i$, so $i \in J$, contrary to the choice of i. This contradiction proves that $J = I$, and A_ξ has the required properties. This proves Lemma 4.1.2.

Let us now prove Theorem 4.1(iii). Recall once more that $\underline{P}$ denotes the set underlying a scheme P, and that $\underline{d} : \underline{P} \to \underline{Q}$ is the map induced by a morphism $d : P \to Q$. Part b) can then be restated by saying that the map

$$\underline{d_0} \boxtimes \underline{d_1} : \underline{X_1} \longrightarrow \underline{X_0} \times_{\underline{Y}} \underline{X_0}$$

with components $\underline{d_0}$ and $\underline{d_1}$ is surjective. This map factors as

$$\underline{X_1} \xrightarrow{\underline{d_0} \boxtimes \underline{d_1}} \underline{X_0} \times_{\underline{Y}} \underline{X_0} \xrightarrow{q} \underline{X_0 \times_Y X_0},$$

where q is the canonical map. The image of $\underline{d_0} \boxtimes \underline{d_1}$ therefore contains every point $v \in \underline{X_0 \times_Y X_0}$ such that

$$\{v\} = q^{-1}(q(v)).$$

This condition is satisfied in particular when v is *rational over* Y, that is, when the residue field $\kappa(v)$ is identified with the residue field $\kappa(w)$ of the image w of v in Y.[23]

If $v \in \underline{X_0 \times_Y X_0}$ is not rational over Y, again let w be its image in Y. By Lemma 4.1.2, there exist a local ring C with radical $\mathfrak{m}$ and a local, flat homomorphism

$$f : \mathcal{O}_{Y,w} \longrightarrow C$$

such that $C/\mathfrak{m}$ is isomorphic to $\kappa(v)$ as a $\kappa(w)$-algebra. Put $Y' = \operatorname{Spec} C$, and let $\pi : Y' \to Y$ be the morphism induced by f. The canonical projection

$$(X_0 \times_Y X_0) \times_Y Y' \longrightarrow X_0 \times_Y X_0$$

sends some point v', rational over Y', to v. Since

$$(X_0 \times_Y X_0) \times_Y Y' \simeq (X_0 \times_Y Y') \times_{Y'} (X_0 \times_Y Y'),$$

and since the hypotheses of Theorem 4.1 and the preceding results, in particular part b), remain valid after the base change $\pi : Y' \to Y$, the point v' is the image of an element

$$u' \in \underline{X_1 \times_Y Y'}$$

under the base change of $d_0 \boxtimes d_1$. If u is the image of u' in X_1, then

$$v = (d_0 \boxtimes d_1)(u),$$

as required.

f) Proof of (iv).[24]

Lemma 4.1.3.— *If a monomorphism of schemes $f : T \to Z$ is finite, it is a closed immersion.*

Covering Z by affine open subsets Z_i, and replacing f by the induced morphisms $f^{-1}(Z_i) \to Z_i$, reduces us—since a finite morphism is affine—to the case

$$Z = \operatorname{Spec} B, \qquad T = \operatorname{Spec} A.$$

Because f is a monomorphism, the diagonal morphism

$$T \longrightarrow T \times_Z T$$

is an isomorphism (EGA I, 5.3.8); equivalently, multiplication gives an isomorphism

$$A \otimes_B A \xrightarrow{\sim} A.$$

Consequently, for every maximal ideal $\mathfrak{m}$ of B, there is an isomorphism

$$(A/\mathfrak{m}A) \otimes_k (A/\mathfrak{m}A) \xrightarrow{\sim} A/\mathfrak{m}A, \qquad k = B/\mathfrak{m}.$$

[23] Editor's note: the pages of Lecture Notes in Mathematics 151 are permuted here; their actual order is 265–266–268–269–267–270–271.

[24] Editor's note: the following lemma has been added from [DG70], I, §5, Proposition 1.5; see also the proof of EGA IV$_3$, 8.11.5. It was used implicitly in the original and explicitly in [DG70], III, §2, 4.6. It is also useful in Theorem 7.1 below.

Since A is finite over B, the k-vector space $A/\mathfrak{m}A$ has finite dimension d, and this isomorphism implies $d = 0$ or $d = 1$. Hence the map

$$k = B/\mathfrak{m} \longrightarrow A/\mathfrak{m}A$$

is surjective. Nakayama's lemma, applied after localization at $\mathfrak{m}$, shows that $B_\mathfrak{m} \to A_\mathfrak{m}$ is surjective. It follows that $B \to A$ is surjective: its cokernel C satisfies $C_\mathfrak{m} = 0$ for every maximal ideal $\mathfrak{m}$, and therefore vanishes. This proves the lemma.

Let us now prove (iv). By hypothesis,

$$X_0 = \operatorname{Spec} A_0, \qquad X_1 = \operatorname{Spec} A_1,$$

and, for $i = 0, 1$, the homomorphism $\delta_i : A_0 \to A_1$ makes A_1 a finite A_0-module. Therefore, a fortiori, the homomorphism

$$A_0 \otimes_B A_0 \longrightarrow A_1$$

is finite.

We assume in addition that

$$d = d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_Y X_0$$

is a monomorphism. By Lemma 4.1.3, the corresponding homomorphism $A_0 \otimes_B A_0 \to A_1$ is surjective. We shall prove that it is an isomorphism; in the course of doing so, we shall also prove that $p : X_0 \to Y$ is finite locally free.

It is enough to prove that, for every prime ideal $\mathfrak{p}$ of B, the homomorphism

$$(A_0)_\mathfrak{p} \otimes_{B_\mathfrak{p}} (A_0)_\mathfrak{p} \longrightarrow (A_1)_\mathfrak{p},$$

with components $\delta_{0\mathfrak{p}}$ and $\delta_{1\mathfrak{p}}$, is bijective. Thus we may suppose that B is local. Part b) then shows that $(A_0)_\mathfrak{p}$ is semilocal. Indeed, if $\mathfrak{m}$ is one of its maximal ideals, the others have the form $\delta_0^{-1}(\mathfrak{n})$, where $\mathfrak{n}$ ranges over the prime ideals of A_1 satisfying $\delta_1^{-1}(\mathfrak{n}) = \mathfrak{m}$. There are at most

$$n = [A_1 : A_0]$$

such prime ideals $\mathfrak{n}$.

After a faithfully flat base change, we may also suppose that the residue field of B is infinite.[25] We can then use the following lemma.

Lemma 4.2.— *Let B be a local ring with infinite residue field, let A be a semilocal ring, and let $i : B \to A$ be a homomorphism which sends the maximal ideal $\mathfrak{n}$ of B into the radical $\mathfrak{r}$ of A. Let M be a free A-module of rank n, and let N be a B-submodule of M which generates M as an A-module. Then N contains an A-basis of M.*

Recall that a sequence $m_1, \ldots, m_n$ of elements of M is an A-basis if and only if its canonical images in $M/\mathfrak{r}M$ form a basis over $A/\mathfrak{r}$. We may therefore replace

$$M \text{ by } M/\mathfrak{r}M, \quad N \text{ by } N/(N \cap \mathfrak{r}M), \quad A \text{ by } A/\mathfrak{r}, \quad B \text{ by } B/\mathfrak{n}.$$

In this case the lemma is elementary. Namely, if

$$A = K_1 \times \cdots \times K_r, \qquad M = K_1^n \times \cdots \times K_r^n,$$

choose, for each j, an element $x_j \in N$ whose j-th component is nonzero. Because the field B is infinite, a suitable B-linear combination x of the x_j has all its components nonzero. Replace M by M/Ax, and proceed by induction on n.

[25] Editor's note: see Lemma 4.1.2.

Apply the lemma with $A = A_0$, with $i : B \hookrightarrow A_0$, with $M = A_1$ regarded as an A_0-module through δ_1, and with $N = \delta_0(A_0)$. Indeed, since

$$d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_Y X_0$$

is a closed immersion, the homomorphism

$$A_0 \otimes_B A_0 \longrightarrow A_1$$

with components δ_0, δ_1 is surjective. This says exactly that $\delta_0(A_0)$ generates the A_0-module A_1.

Choose $a_1, \ldots, a_n \in A_0$ so that $\delta_0(a_1), \ldots, \delta_0(a_n)$ form an A_0-basis of A_1. If $a_1, \ldots, a_n$ form a B-basis of A_0, then $A_0 \otimes_B A_0 \to A_1$ sends the basis

$$(1 \otimes a_i)_{1 \leq i \leq n}$$

to the basis

$$(\delta_0(a_i))_{1 \leq i \leq n},$$

and is therefore bijective. Thus, if

$$\varepsilon : \mathbb{Z}^n \longrightarrow A_0$$

is the homomorphism of abelian groups which sends the standard basis of $\mathbb{Z}^n$ to $a_1, \ldots, a_n$, it remains only to prove that the map

$$B \otimes_{\mathbb{Z}} \mathbb{Z}^n \longrightarrow A_0$$

with components i and ε is bijective.

The dual diagram $(0, 1, 2)^*$ considered at the beginning of this proof induces the following commutative diagram:

$$\begin{tikzcd}
A_2 & & A_1 \arrow[ll, shift right, "\delta'_1"'] \arrow[ll, shift left, "\delta'_0"'] & & A_0 \arrow[ll, "\delta_0"] \\
A_1 \otimes_{\mathbb{Z}} \mathbb{Z}^n \arrow[u, "u_2"] & & A_0 \otimes_{\mathbb{Z}} \mathbb{Z}^n \arrow[ll, shift right, "\delta_1 \otimes \mathbb{Z}^n"'] \arrow[ll, shift left, "\delta_0 \otimes \mathbb{Z}^n"'] \arrow[u, "u_1", "\cong"'] & & B \otimes_{\mathbb{Z}} \mathbb{Z}^n \arrow[ll, "i \otimes \mathbb{Z}^n"] \arrow[u, "u_0"]
\end{tikzcd}$$

Figure 22: The commutative ring-and-module diagram concluding the affine proof of Theorem 4.1(iv).

Here u_0, u_1, u_2 have, respectively, the component pairs

$$(i, \varepsilon), \qquad (\delta_1, \delta_0\varepsilon), \qquad (\delta'_2, \delta'_0\delta_0\varepsilon).$$

We know that u_1 is an isomorphism. Since the two left-hand squares of $(0, 1, 2)^*$ are cocartesian, u_2 is bijective. The two horizontal rows of the diagram are exact, and therefore u_0 is bijective.[26] Thus A_0 is a locally free B-module of rank n. By the preceding reductions, it follows that

$$\delta_0 \otimes \delta_1 : A_0 \otimes_B A_0 \xrightarrow{\sim} A_1$$

is an isomorphism. This completes the proof of Theorem 4.1 in the special case under consideration: X_0 is affine and d_1 is locally free of constant rank n.

[26] Editor's note: the argument which follows has been added.

5. Passage to the quotient by a finite flat groupoid (the general case)

a) Let $U^{(n)}$ be the largest open subset of X_0 over which d_1 is finite locally free of rank n. The scheme X_0 is the disjoint union of the $U^{(n)}$. The two Cartesian squares

$$\begin{tikzcd} X_2 \arrow[r, "d_0'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \qquad \textit{and} \qquad \begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_1"'] & X_0 \end{tikzcd}$$

Figure 23: The two Cartesian squares controlling the rank strata $U^{(n)}$.

show that the inverse images of $U^{(n)}$ under d_0 and d_1 both coincide with the largest open subset of X_1 over which d_2' is locally free of rank n.[27] Thus

$$d_0^{-1}(U^{(n)}) = d_1^{-1}(U^{(n)}),$$

and the groupoid X_* is the disjoint union of the groupoids $X_*^{(n)}$ induced by X_* on the open-and-closed subschemes $U^{(n)}$. It is therefore enough to prove Theorem 4.1 for each $X_*^{(n)}$. We are reduced to the case in which d_1 is finite locally free of rank n.

b) *We can now prove the theorem in the general case.*

By part a), we may suppose that d_1 is locally free of rank n. Let (Y, p) be a cokernel of (d_0, d_1) in the category of all ringed spaces. The argument at the end of 4(c) shows that, in order to prove Theorem 4.1(i), it is enough to show that Y is a scheme and that $p : X_0 \to Y$ is a morphism of schemes. By Lemma 1.2, this question is local on Y. Let $y \in Y$ and choose $x \in X_0$ with $p(x) = y$. If x has a saturated affine open neighborhood U, then Section 4 shows that $p(U)$ is an affine open subset of Y, and $p|_U : U \to p(U)$ is a morphism of schemes. Thus it remains to prove that every $x \in X_0$ has a saturated affine open neighborhood. We proceed as follows; the argument is taken from SGA 1, VIII, Corollary 7.6.

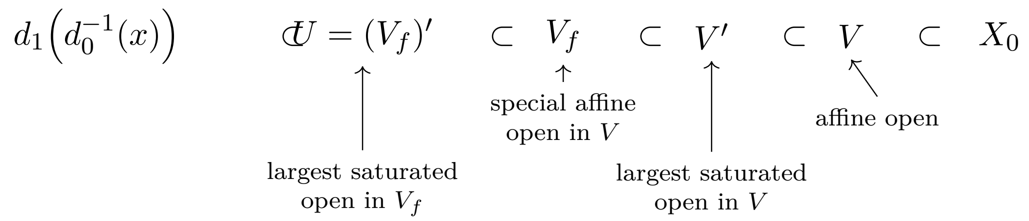

Figure 24: The nested open subsets used to construct a saturated affine neighborhood of x.

By condition 4.1(b), there is an affine open subset $V \subseteq X_0$ which contains $d_1(d_0^{-1}(x))$.[28] Put $F = X_0 \setminus V$. Since d_1 is integral, $d_1(d_0^{-1}(F))$ is closed, and

$$V' = X_0 \setminus d_1(d_0^{-1}(F))$$

[27] Editor's note: since d_0 (respectively d_1) is surjective, flat, and finite, hence faithfully flat and affine, d_2' has rank n over a neighborhood of a point $x \in X_1$ if and only if d_1 has rank n over a neighborhood of $d_0(x)$ (respectively $d_1(x)$).

[28] Editor's note: $d_1(d_0^{-1}(x)) = d_0(d_1^{-1}(x))$; see editor's note 16 in Theorem 4.1.

is the largest saturated open subset contained in V. The set V' is a neighborhood of the finite set $d_1(d_0^{-1}(x))$. Hence there is a section f of the structure sheaf of V, vanishing on $V \setminus V'$, such that $d_1(d_0^{-1}(x))$ is contained in the principal open subset $V_f \subseteq V$. We shall prove that the largest saturated open subset $(V_f)'$ contained in V_f is affine; it will then be the required neighborhood.

Put $Z(f) = V' \setminus V_f$. Then $d_0^{-1}(Z(f))$ is the set of points of

$$d_0^{-1}(V') = d_1^{-1}(V')$$

at which $d_0^*(f)$ vanishes. The morphism induced by d_1 from this common inverse image to V' is locally free of rank n.[29] Lemma 4.1.1 therefore shows that $d_1(d_0^{-1}(Z(f)))$ is precisely the zero locus of the norm N of $d_0^*(f)$ relative to d_1. It follows that

$$(V_f)' = V' \setminus d_1(d_0^{-1}(Z(f)))$$

is the set of points of V_f at which N does not vanish. Consequently, $(V_f)'$ is affine.

This proves Theorem 4.1(i); assertions (ii), (iii), and the first part of (iv) are then clear. It remains to prove the consequences stated at the end of (iv); see [Ray67a], Theorem 1(iii).

By hypothesis, the groupoid X_* comes from an equivalence relation

$$i : R \longrightarrow X_0 \times X_0,$$

where i is an immersion (see editor's note 19). We have shown that R is effective (see Exposé IV, 3.3.2), and that

$$p : X_0 \longrightarrow Y = X_0/R$$

is surjective and finite locally free; in particular, it is faithfully flat and of finite presentation.

Consequently, if (M) denotes the family of faithfully flat morphisms locally of finite presentation, then R is (M)-effective. By Exposé IV, 6.3.3, (Y, p) therefore represents the quotient sheaf of X_0 by R for the fppf topology, and the base-change assertions follow from Exposé IV, 3.4.3.1.

Remark 5.1.—[30] Keep the hypotheses and notation of Theorem 4.1, and suppose in addition that S is locally Noetherian and that $\pi_0 : X_0 \to S$ is quasi-projective. We shall prove that $\pi : Y \to S$ is quasi-projective.

The preceding hypotheses imply that $Y \to S$ is of finite type; see the proof of 6.1(ii). Let $\mathscr{A}$ be an invertible $\mathcal{O}_{X_0}$-module which is ample for π_0. By EGA II, 6.1.12, $p_*\mathscr{A}$ is an invertible $p_*\mathcal{O}_{X_0}$-module. Hence Y has an affine open cover $(V_i)_{i\in I}$ such that $\mathscr{A}$ is trivial over each of the saturated affine open subsets

$$U_i = p^{-1}(V_i).$$

For each i, put

$$A_{i,0} = \mathcal{O}_{X_0}(U_i),$$

and let $A_{i,1}$ be the ring of the affine open subset $d_0^{-1}(U_i) = d_1^{-1}(U_i)$ of X_1. Let $\delta_{i,0}$ (respectively $\delta_{i,1}$) be the homomorphism $A_{i,0} \to A_{i,1}$ induced by d_0 (respectively d_1), and put

$$B_i = \mathcal{O}_Y(V_i) = \{b \in A_{i,0} \mid \delta_{i,0}(b) = \delta_{i,1}(b)\}.$$

Following EGA II, §6.5, consider the invertible $\mathcal{O}_{X_0}$-module

$$N_{d_1}(d_0^*\mathscr{A}),$$

[29] Editor's note: the reference to Lemma 4.1.1 has been added; see [DG70], III, §5.2, p. 313.

[30] Editor's note: this paragraph has been added.

the norm, relative to the finite locally free morphism $d_1 : X_1 \to X_0$, of the invertible $\mathcal{O}_{X_1}$-module $d_0^*\mathscr{A}$. Suppose that $\mathscr{A}$, relative to the cover $(U_i)_{i\in I}$, is given by transition functions

$$c_{ij} \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times.$$

Then $N_{d_1}(d_0^*\mathscr{A})$ is given by the transition functions

$$N_{\delta_1}(\delta_0(c_{ij})) \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times.$$

By 4(a), these elements belong to $\mathcal{O}_Y(V_i \cap V_j)^\times$, and hence define an invertible $\mathcal{O}_Y$-module $\mathscr{L}$ such that

$$p^*\mathscr{L} = N_{d_1}(d_0^*\mathscr{A}).$$

Moreover, for every $n \in \mathbb{N}^*$,

$$p^*(\mathscr{L}^n) = N_{d_1}(d_0^*(\mathscr{A}^n))\,;$$

see *loc. cit.*, (6.5.2.1).

We now show that $\mathscr{L}$ is ample for $\pi : Y \to S$. Replacing S by an affine open subset, we may suppose that S is affine. Let $y \in Y$, choose $x \in X_0$ with $p(x) = y$, let V be an affine open neighborhood of y in Y, and put $U = p^{-1}(V)$. Since $\mathscr{A}$ is π_0-ample, there exist $n \in \mathbb{N}^*$ and a section

$$s \in \Gamma(X_0, \mathscr{A}^n)$$

such that

$$x \in (X_0)_s \subseteq U.$$

With the notation above, s is represented by sections

$$a_i \in A_{i,0} = \mathcal{O}_{X_0}(U_i), \qquad a_i = c_{ij}a_j \ \text{ on } \ U_i \cap U_j,$$

and $(X_0)_s$ is the union of the open subsets

$$U_i' = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid a_i \notin \mathfrak{p}\}.$$

For every i, put

$$N(a_i) = N_{\delta_1}(\delta_0(a_i)) \in B_i.$$

Theorem 4.1(i) and Lemma 4.1.1 give

$$p(U_i') = p\, d_1 d_0^{-1}(U_i') = p\, d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\})\,,$$

while

$$d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\}) = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid N_{\delta_1}(\delta_{i,0}(a_i)) \notin \mathfrak{p}\}.$$

It follows that

$$p(U_i') = \{\mathfrak{p} \in \operatorname{Spec}(B_i) \mid N(a_i) \notin \mathfrak{p}\}.$$

Consequently,

$$p((X_0)_s) = Y_{N(s)},$$

where $N(s)$ is the section of $\mathscr{L}^n$ on Y defined by the sections $N(a_i) \in \mathcal{O}_Y(V_i)$. Thus

$$y \in p((X_0)_s) = Y_{N(s)} \subseteq p(U) = V. \tag{*}$$

This proves that $\mathscr{L}$ is ample for $\pi : Y \to S$, and hence completes the proof that π is quasi-projective.

6. Passage to the quotient when a quasi-section exists

We shall now prove a technical lemma that will be used in the proof of the two theorems we have in view. Let S be a scheme and let

$$X_2 \overset{d_0',d_1',d_2'}{\rightrightarrows} X_1 \overset{d_0,d_1}{\rightrightarrows} X_0$$

Figure 25: The groupoid X_*.

be a $(\mathbf{Sch}/S)$-groupoid. We call a *quasi-section* of the groupoid X_* any subscheme U of X_0 for which the following two conditions hold:

(1) The restriction v of d_1 to $d_0^{-1}(U)$ is a finite, locally free, surjective morphism from $d_0^{-1}(U)$ onto X_0.

(2) Every subset E of U consisting of points that are pairwise equivalent for the equivalence relation defined by X_* (§3.e)) is contained in an affine open subset of U.

If U is a quasi-section of X_*, the $(\mathbf{Sch}/S)$-groupoid

$$U_2 \overset{u_0',u_1',u_2'}{\rightrightarrows} U_1 \overset{u_0,u_1}{\rightrightarrows} U$$

Figure 26: The groupoid U_* induced by X_*.

induced by X_* and the inclusion $U \hookrightarrow X_0$ (see §3.a)) satisfies the hypotheses of Theorem 4.1. Indeed, put

$$V = d_0^{-1}(U),$$

and let u and v be the morphisms with source V induced by d_0 and d_1, respectively:

$$X_0 \xleftarrow[v]{} V \xrightarrow{u} U$$

Figure 27: The morphisms $v : V \to X_0$ and $u : V \to U$.

By §3.b), there is a Cartesian square

$$\begin{tikzcd} U_1 \arrow[r] \arrow[d, "u_1"'] & V \arrow[d, "v"] \\ U \arrow[r, "\text{inclusion}"'] & X_0 \end{tikzcd}$$

Figure 28: The Cartesian square defining U_1.

so u_1 is surjective and finite locally free by condition (1). Together with condition (2), condition (1) therefore ensures that the groupoid U_* satisfies the hypotheses of Theorem 4.1. In particular, $\mathrm{Coker}(u_0, u_1)$ exists in $(\mathbf{Sch}/S)$. Moreover, d_0 has a section, so u is a universal effective epimorphism (see Exposé III, 1.12). Proposition 3.1 then shows that $\operatorname{Coker}(u_0, u_1)$ coincides with the cokernel $\operatorname{Coker}(v_0, v_1)$ of the groupoid V_*:

$$\begin{tikzcd} V_2 \arrow[r, shift left=2, "v_2'"] \arrow[r, "v_1'" description] \arrow[r, shift right=2, "v_0'"'] & V_1 \arrow[r, shift left, "v_1"] \arrow[r, shift right, "v_0"'] & V \end{tikzcd}$$

Figure 29: The groupoid V_*.

This is the inverse image of U_* under the base change $u : V \to U$, and hence also the inverse image of X_* under the base change

$$V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_0} X_0$$

Figure 30: The base change defining V_*.

By §3.c), V_* is isomorphic to the groupoid V'_*, the inverse image of X_* under the base change

$$v\colon \qquad V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_1} X_0$$

Figure 31: The base change defining V'_*.

Thus V'_* admits a cokernel in $(\mathbf{Sch}/S)$. Since $v : V \to X_0$ is flat, surjective, and finite, it is faithfully flat and quasi-compact, and therefore a universal effective epimorphism by Exposé III, 6.3.2. Proposition 3.1 consequently shows that X_* admits a cokernel $\operatorname{Coker}(d_0, d_1)$ in $(\mathbf{Sch}/S)$. We have therefore proved the first assertion of part (i) of the following lemma.[31]

Lemma 6.1.—*Suppose that the $(\mathbf{Sch}/S)$-groupoid X_* possesses a quasi-section. Then:*

(i) *There exists a cokernel (Y, p) of (d_0, d_1) in $(\mathbf{Sch}/S)$. Moreover, every such (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*

(i') *The morphism p is surjective; it is open (respectively, universally closed) if d_0 is.*

(ii) *Suppose that S is locally Noetherian and that X_0 is locally of finite type (respectively, of finite type) over S. Then p and $Y \to S$ are locally of finite presentation (respectively, of finite presentation).*

(iii) *The morphism $X_1 \to X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*

(iv) *If (d_0, d_1) is an equivalence pair, then $X_1 \to X_0 \times_Y X_0$ is an isomorphism. Moreover, if $d_0 : X_1 \to X_0$ is flat, then p is faithfully flat.*

Before proving the second assertion of part (i), we shall prove parts (i'), (ii), and (iii).

a) Proof of (i') and (ii).—We have just seen that (Y, p) identifies both with $\operatorname{Coker}(v_0, v_1)$ and with $\operatorname{Coker}(u_0, u_1)$. Let q and r be the canonical epimorphisms from U and V, respectively, to Y:

[31] Editor's note: The argument that follows has been slightly modified. In particular, the additional hypothesis that d_0 be flat has been moved to part (iv); the former part (ii) has been split into parts (i') and (ii); and the second assertion of part (i') has been added.

$$\begin{tikzcd} X_0 \arrow[dr, "p"'] & V \arrow[l, "v"'] \arrow[r, "u"] \arrow[d, "r"] & U \arrow[dl, "q"] \\ & Y & \end{tikzcd}$$

Figure 32: The canonical epimorphisms to Y.

By hypothesis, v is surjective and finite locally free, hence open.

On the other hand, if $d_0 : X_1 \to X_0$ is open (respectively, universally closed), then u, being obtained from d_0 by base change, has the same property. By Theorem 4.1, q is surjective, integral, and open. It follows that r is surjective and is open (respectively, universally closed) if d_0 is. Since v is surjective, the same is true of p. This proves part (i').

Now suppose that S is locally Noetherian and that X_0 is locally of finite type over S, so that X_0 is locally Noetherian.

We show that Y is locally of finite presentation over S. Let

$$S' = \operatorname{Spec} R$$

be an affine open subset of S, let

$$Y' = \operatorname{Spec} B$$

be an affine open subset of Y mapping into S', and let

$$U' = \operatorname{Spec} A$$

be the inverse image of Y' in U. Since R is Noetherian, it is enough to show that B is a finitely generated R-algebra. By §§4 and 5, B is contained in A, which is a finitely generated R-algebra; the assertion therefore follows because R is Noetherian and A is integral over B.

Finally, since $X_0 \to S$ is locally of finite type, so is p (EGA I, 6.6.6). Hence p is locally of finite presentation because Y is locally Noetherian.

It remains to prove the final assertion of part (ii). Suppose in addition that X_0 is of finite type over S. Since p is surjective, Y is then quasi-compact over S, and hence of finite type over S. Because S is locally Noetherian, the morphisms $X_0 \to S$ and $Y \to S$ are of finite presentation, and therefore so is $p : X_0 \to Y$ (EGA IV$_1$, 1.6.2(v)).

b) Proof of (iii).—The groupoid V_* with base V is isomorphic both to the inverse image of U_* under the base change u and to the inverse image of X_* under the base change v. We therefore have a double Cartesian square:

$$\begin{tikzcd} X_1 \arrow[d, "d_0 \boxtimes d_1"'] & V_1 \arrow[l] \arrow[r] \arrow[d, "v_0 \boxtimes v_1"'] & U_1 \arrow[d, "u_0 \boxtimes u_1"'] \\ X_0 \times_Y X_0 & V \times_Y V \arrow[l, "v \times v"'] \arrow[r, "u \times u"] & U \times_Y U \end{tikzcd}$$

Figure 33: The double Cartesian square used in the proof of Lemma 6.1(iii).

Since $u_0 \boxtimes u_1$ is surjective, so is $v_0 \boxtimes v_1$. Since $v \times v$ is surjective, the composite morphism

$$V_1 \longrightarrow X_0 \times_Y X_0$$

is surjective, and hence so is $d_0 \boxtimes d_1$.

c) Proof of (i).—It remains to prove that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. We first show that Y is obtained from X_0 by identifying those points x and y for which there exists $z \in X_1$ with

$$d_0(z) = x, \qquad d_1(z) = y.$$

Indeed, p is surjective and $pd_0 = pd_1$. Conversely, if $p(x) = p(y)$, there is a point $z' \in X_0 \times_Y X_0$ whose first projection is x and whose second projection is y. If $z \in X_1$ satisfies

$$(d_0 \boxtimes d_1)(z) = z',$$

then $d_0(z) = x$ and $d_1(z) = y$, as required.

On the other hand, if W is a saturated open subset of X_0, then $W \cap U$ is a saturated open subset of U. By Theorem 4.1, $q(W \cap U)$ is open in Y. Since $q(W \cap U) = p(W)$, the topology on Y is the quotient topology induced by X_0.

It remains to show that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*\mathcal{O}_{X_0} \rightrightarrows p_*d_{0*}\mathcal{O}_{X_1} = p_*d_{1*}\mathcal{O}_{X_1} \tag{4}$$

is exact.

Let Y' be an open subset of Y, and put

$$U' = q^{-1}(Y'), \qquad X_0' = p^{-1}(Y'),$$

and so forth.[32] Then U' is an open subset of U saturated for the equivalence relation defined by the groupoid U_*. It follows from Lemmas 1.1 and 1.2 that Y' is the cokernel, both in $(\mathbf{Sch}/S)$ and in the category of ringed spaces, of the groupoid induced by U_* on U'. Likewise, X_0' is an open subset of X_0 saturated for the equivalence relation defined by X_*, and we have the following commutative diagram, in which both squares are Cartesian:

$$\begin{tikzcd}
X_0' & V' = d_0^{-1}(U') \arrow[l, "\widetilde{d_1}"'] \arrow[r, "\widetilde{d_0}"] \arrow[d] & U' \arrow[d] \\
X_0 & V = d_0^{-1}(U) \arrow[l, "d_1"'] \arrow[r, "d_0"'] & U
\end{tikzcd}$$

Figure 34: The restricted groupoids over X_0' and X_0.

Here $\widetilde{d_1}$ is surjective and finite locally free. On the other hand, let $x \in U'$. Since U is a quasi-section, the set

$$E = d_0 d_1^{-1}(x) \cap U$$

is finite and is contained in an affine open subset W of U. Then $E' = E \cap U'$ is a finite set contained in the quasi-affine open subset $W \cap U'$. Consequently there is an affine open subset W' of $W \cap U'$ that contains E'. This proves that U' is a quasi-section of the groupoid X_*' induced by X_* on X_0'. The first assertion of part (i), applied to X_*' and U', now shows that Y' is the cokernel of X_*' in $(\mathbf{Sch}/S)$.

In particular, for every S-scheme T, the sequence

$$T(Y') \xrightarrow{T(p|_{X_0'})} T(X_0') \underset{T(d_0|_{X_1'})}{\overset{T(d_1|_{X_1'})}{\rightrightarrows}} T(X_1') \tag{5}$$

[32]Editor's note: The argument that follows has been expanded, in particular the verification that U' is a quasi-section of the groupoid induced on X_0'.

is exact. If T is the affine line $\mathbf{G}_{a,S}$ (Exposé I, 4.3), this sequence identifies with

$$\begin{aligned}\Gamma(Y', \mathcal{O}_Y) \longrightarrow \Gamma(p^{-1}(Y'), \mathcal{O}_{X_0}) \underset{\delta_0}{\overset{\delta_1}{\rightrightarrows}} &\Gamma(d_0^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}) \\ &= \Gamma(d_1^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}),\end{aligned} \tag{6}$$

which is therefore exact for every open subset Y'. This completes the proof of Lemma 6.1(i).

d) Proof of (iv).—If (d_0, d_1) is an equivalence pair, then so is (u_0, u_1). It follows from Theorem 4.1 that

$$u_0 \boxtimes u_1 : U_1 \longrightarrow U \times_Y U$$

is an isomorphism, and hence so is $v_0 \boxtimes v_1$; see the Cartesian squares in part (b). Since $v \times v$ is faithfully flat and quasi-compact, $d_0 \boxtimes d_1$ is an isomorphism (SGA 1, VIII, 5.4).

Moreover, if d_0 is flat, then u is flat. The morphism q is flat by Theorem 4.1, so r is flat as well. Since v is faithfully flat, p is flat; being also surjective, it is faithfully flat.

7. Quotient by a proper and flat groupoid

Theorem 7.1.—[33] *Let S be a locally Noetherian scheme and let*

$$X_2 \overset{d'_2}{\underset{d'_0}{\overset{d'_1}{\substack{\longrightarrow\\\longrightarrow\\\longrightarrow}}}} X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0$$

Figure 35: The groupoid occurring in Theorem 7.1.

be a $(\mathbf{Sch}/S)$-groupoid such that d_1 is proper and flat, X_0 is quasi-projective over S,[34] *and the morphism*

$$d : X_1 \longrightarrow X_0 \times_S X_0$$

with components d_0 and d_1 is quasi-finite. Then:

(i) *There exists a cokernel (Y, p) of (d_0, d_1) in $(\mathbf{Sch}/S)$. Moreover, every such (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*

(ii) *The morphism p is surjective, open, proper, and of finite presentation, and $Y \to S$ is separated and of finite presentation.*[35]

[33] Editor's note: We mention here the article of S. Keel and S. Mori [KM97], which establishes the following theorem. Let X be an algebraic space of finite type over a locally Noetherian base S, and let $j : R \to X \times_S X$ be a flat groupoid whose stabilizer $j^{-1}(\Delta_X)$ is finite over X. There then exists an algebraic space that is a geometric quotient of X by R and a uniform categorical quotient; moreover, if j is separated, this quotient is separated. In particular, if a flat S-group scheme G acts properly on X, with finite stabilizer (that is, if

$$G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (x, g \cdot x)$$

is proper and the stabilizer of the diagonal is finite over X), then there exists a geometric quotient $X \to X/G$. For a reductive S-group scheme G, this is a result of J. Kollár [Ko97].

[34] Editor's note: This hypothesis on X_0 is necessary; see editor's note (17) to Theorem 4.1.

[35] Editor's note: TDTE III, Theorem 6.1, states that $Y \to S$ is quasi-projective if S is Noetherian. The editors did not see how to deduce this from the local existence of quasi-sections.

(iii) *The morphism*

$$X_1 \longrightarrow X_0 \times_Y X_0$$

with components d_0 *and* d_1 *is surjective.*

(iv) *If* (d_0, d_1) *is an equivalence pair, then* $X_1 \to X_0 \times_Y X_0$ *is an isomorphism and* p *is faithfully flat.*[36] *Moreover,* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of sheaves for the fppf topology, and, for every base change* $Y' \to Y$*, the scheme* Y' *is the cokernel of the groupoid* $X_* \times_Y Y'$ *obtained from* X_* *by the base change* $X_0 \times_Y Y' \to X_0$*.*

In particular, for every base change $S' \to S$*, the scheme*

$$Y' = Y \times_S S'$$

is the cokernel of the S'*-groupoid*

$$X'_* = X_* \times_S S'.$$

Thus, in this case, "formation of the quotient commutes with base change".

Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Lemma 1.2 shows that, to prove part (i), it is enough to show that every point $z \in X_0$ has a saturated open neighborhood U_z such that, if $\widetilde{d_0}$ and $\widetilde{d_1}$ denote the restrictions of d_0 and d_1 to

$$d_0^{-1}(U_z) = d_1^{-1}(U_z),$$

and if (Q, q) denotes the cokernel of $(\widetilde{d_0}, \widetilde{d_1})$ in the category of ringed spaces, then Q is a scheme and q is a morphism of schemes.

By Lemma 6.1(i), it is therefore enough to show that every point $z \in X_0$ has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section. We may even assume that z is closed in its fiber over S (we shall say that z is *closed relative to* S).[37] The existence of U_z follows from the lemmas below.

Lemma 7.2.—*Let* T *be an affine Noetherian scheme, let* X, Y, Z *be* T*-schemes of finite type, with* X *quasi-projective over* T*, and let*

$$\begin{tikzcd} Y \arrow[r, "u"] \arrow[d, "v"'] & X \arrow[d, dashed] \\ Z \arrow[r, dashed] & T \end{tikzcd}$$

Figure 36: The diagram in Lemma 7.2.

[36]Editor's note: The consequences that follow have been made explicit; see [Ray67a], Theorem 1(iii), and the end of the proof of the theorem. We also mention that another proof of Theorem 7.1 in the case of an equivalence relation, based on the existence of Hilbert schemes, is given in [AK80], Theorem 2.9; see also [BLR90], §8.2, Theorem 12. In that case it is furthermore shown that $Y \to S$ is quasi-projective.

[37]Editor's note: Indeed, if such an open neighborhood U_z has been constructed for every point z that is closed relative to S, then the union of these U_z covers X_0: every fiber over S of the closed complement is a Noetherian scheme with no closed points, and is therefore empty.

be a diagram in $(\mathbf{Sch}/T)$*. Let* z *be a point of* $v(Y)$ *that is closed relative to* T*, and suppose that* v *is flat at the points of* $v^{-1}(z)$*. Then there exists a closed subscheme* F *of* X *such that*

$$u(u^{-1}(F) \cap v^{-1}(z))$$

is finite and nonempty, and such that the restriction of v *to* $u^{-1}(F)$ *is flat at the points of* $v^{-1}(z)$*.*

Put $T = \operatorname{Spec} A$. We may suppose that

$$X = \operatorname{Proj} \mathscr{S},$$

where $\mathscr{S}$ is the symmetric algebra of a finitely generated A-module E.

If $u(v^{-1}(z))$ is finite, we may take $F = X$. Otherwise, let

$$y_1, \ldots, y_n$$

be the points of the fiber $v^{-1}(z)$ associated with its structure sheaf $\mathcal{O}_{v^{-1}(z)}$. Thus, if $\mathcal{O}_i$ denotes the local ring of $v^{-1}(z)$ at y_i, the maximal ideal of $\mathcal{O}_i$ consists of zero divisors. Let t be the image of z in T. The set $u(v^{-1}(z))$ is an infinite constructible subset of the fiber of X over t. Consequently, that fiber contains a closed point $x \in u(v^{-1}(z))$ distinct from $u(y_1), \ldots, u(y_n)$. The open subset $X - \{x\}$ is a neighborhood of $u(y_1), \ldots, u(y_n)$, and hence contains an open neighborhood of the form $D_+(f)$, where $f \in \mathscr{S}$ is homogeneous of degree d (we use the notation of EGA II, §2.3).

The closed subscheme

$$X_1 = V_+(f)$$

defined by f therefore contains x and avoids the points $u(y_1), \ldots, u(y_n)$. Its inverse image

$$Y_1 = u^{-1}(V_+(f))$$

is consequently distinct from Y and meets $v^{-1}(z)$. We shall in fact prove that the restriction v_1 of v to Y_1 is flat at the points of $v^{-1}(z)$. If $u(v_1^{-1}(z))$ is finite, we may then take $F = X_1$. Otherwise, we repeat the argument with Y_1, v_1, and the morphism u_1 induced by u on Y_1, in place of Y, v, u. In this way we obtain a descending sequence

$$X \supset X_1 \supset \cdots$$

of closed subschemes of X. Since such a sequence terminates,

$$u(u^{-1}(X_n) \cap v^{-1}(z))$$

is finite and nonempty for some n, and we then take $F = X_n$.

It remains to show that v_1 is flat at the points of $v^{-1}(z)$. Let $y \in Y_1$ lie over z, let $\mathcal{O}_y$ be the local ring of Y at y, let $\overline{\mathcal{O}}_y$ be the local ring of $v^{-1}(z)$ at y, and let $\mathcal{O}_{v(y)}$ be the local ring of Z at $v(y)$. Choose $g \in \mathscr{S}_1$ such that $D_+(g)$ is a neighborhood of $u(y)$ in X, and let φ and $\overline{\varphi}$ be the respective images of f/g^d in $\mathcal{O}_y$ and $\overline{\mathcal{O}}_y$. By the construction of f, the element $\overline{\varphi}$ is not a zero divisor in $\overline{\mathcal{O}}_y$. Since $\mathcal{O}_y$ is flat over $\mathcal{O}_z = \mathcal{O}_{v(y)}$, the element φ is not a zero divisor in $\mathcal{O}_y$, and

$$\mathcal{O}_y/\mathcal{O}_y\varphi$$

is flat over $\mathcal{O}_z$ (SGA 1, IV, 5.7). This quotient is precisely the local ring of Y_1 at y.

Lemma 7.3.—*Retain the notation and hypotheses of Theorem 7.1. Every point* $z \in X_0$ *that is closed relative to* S *has a saturated open neighborhood* U_z *such that the groupoid induced by* X_* *on* U_z *has a quasi-section.*

The assertion is local on S, so we may suppose that S is affine and Noetherian and apply the preceding lemma to the diagram

$$\begin{tikzcd} X_1 \arrow[r, "d_0"] \arrow[d, "d_1"'] & X_0 \arrow[d, dashed] \\ X_0 \arrow[r, dashed] & S \end{tikzcd}$$

Figure 37: The diagram to which Lemma 7.2 is applied.

in $(\mathbf{Sch}/S)$. Thus let F be a closed subscheme of X_0 such that

$$d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$$

is finite and nonempty, and such that the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $d_1^{-1}(z)$.

Let F_1 and F_2 be the inverse images of F under d_0 and $d_0 d_0' = d_0 d_1'$, respectively, and let $\widetilde{d}_0, \widetilde{d}_1, \ldots$ denote the morphisms induced by $d_0, d_1, \ldots$. We then have a commutative diagram

$$\begin{tikzcd} F_2 \arrow[r, shift left, "\widetilde{d_1'}"] \arrow[r, shift right, "\widetilde{d_0'}"'] \arrow[d, "\widetilde{d_2'}"'] & F_1 \arrow[r, "\widetilde{d_0}"] \arrow[d, "\widetilde{d_1}"] & F \arrow[d, dashed, "\widetilde{q}"] \\ X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \arrow[r, dashed, "q"] & S \end{tikzcd}$$

Figure 38: The induced diagram on F_2, F_1, F.

in which the two left-hand squares are Cartesian, the first row is exact (cf. the diagram $(0, 1, 2)$ in §1), and $q, \widetilde{q}$ are the structure morphisms.

We first show that only finitely many points of F_1 lie over z.[38] Let s be the image of z in S. Since F is of finite type over S, the fiber $\widetilde{q}^{-1}(s)$ is a Noetherian scheme. Since $\widetilde{d}_0$ is proper,

$$\widetilde{d}_0(\widetilde{d}_1^{-1}(z))$$

is a closed subscheme of $\widetilde{q}^{-1}(s)$ consisting of finitely many points. Consequently (cf. EGA I, 6.2.2), those points are closed in $\widetilde{q}^{-1}(s)$, and also in the fiber $q^{-1}(s)$ of X_0, because F is closed in X_0.

[38] Editor's note: Details have been added, and the role of the properness hypothesis on d_0 and d_1 in Theorem 7.1 has been made explicit. Compare the statement and proof of Theorem 8.1, where this properness hypothesis is omitted.

Let y be one of these points. Since $q^{-1}(s)$ is of finite type over $\kappa(s)$, it has affine open neighborhoods

$$\operatorname{Spec} B \quad \text{and} \quad \operatorname{Spec} C$$

of y and z, respectively, with B and C finitely generated $\kappa(s)$-algebras. The points y and z correspond to maximal ideals

$$\mathfrak{p} \subset B, \qquad \mathfrak{q} \subset C.$$

The fields $B/\mathfrak{p}$ and $C/\mathfrak{q}$ are finite over $\kappa(s)$, so

$$(B/\mathfrak{p}) \otimes_{\kappa(s)} (C/\mathfrak{q})$$

is a finite-dimensional $\kappa(s)$-algebra. Its maximal ideals correspond precisely to the points of $X_0 \times_S X_0$ whose second projection is z and whose first projection is y. Thus there are only finitely many points $u \in X_0 \times_S X_0$ whose second projection is z and whose first projection belongs to

$$\widetilde{d}_0(\widetilde{d}_1^{-1}(z)).$$

Finally, since $X_1 \to X_0 \times_S X_0$ has finite fibers, each such point u is the image of only finitely many points of X_1. This proves the assertion.

The morphism $\widetilde{d}_1$ is therefore quasi-finite and flat at the points of F_1 lying over z. Since $\widetilde{d}_1$ is of finite type, SGA 1, IV, 6.10 and EGA III, 4.4.10 imply that the set Φ of points of F_1 at which $\widetilde{d}_1$ is not both flat and quasi-finite is closed in F_1, and hence in X_1 because F_1 is closed in X_1.[39] Since d_1 is proper, $\widetilde{d}_1(\Phi)$ is closed and, by what precedes, does not contain z. Put

$$W = \widetilde{d}_1(F_1) - \widetilde{d}_1(\Phi).$$

The restriction of $\widetilde{d}_1$ to $\widetilde{d}_1^{-1}(W)$ is of finite presentation (by the Noetherian hypotheses), flat, proper, and quasi-finite.[40] It is therefore finite, locally free, and open, by EGA III, 4.4.2 and EGA IV$_2$, 2.1.12 and 2.4.6. Consequently, $\widetilde{d}_1(F_1)$ is a neighborhood of z, and W is the largest open subset of X_0 contained in $\widetilde{d}_1(F_1)$ over which $\widetilde{d}_1$ is both flat and quasi-finite.

We shall see in Lemma 7.4 that the inverse images of Φ under $\widetilde{d}'_1$ and $\widetilde{d}'_0$ both identify with the set of points of F_2 at which $\widetilde{d}'_2$ is not both flat and quasi-finite. It follows that

$$d_0^{-1}(W) = \widetilde{d}'_2(F_2) - \widetilde{d}'_2((\widetilde{d}'_0)^{-1}(\Phi)),$$
$$d_1^{-1}(W) = \widetilde{d}'_2(F_2) - \widetilde{d}'_2((\widetilde{d}'_1)^{-1}(\Phi))$$

coincide. Thus W is saturated.

Set

$$W_1 = \widetilde{d}_1^{-1}(W).$$

The equality $d_0^{-1}(W) = d_1^{-1}(W)$ implies

$$(\widetilde{d}'_2)^{-1}(d_0^{-1}(W)) = (\widetilde{d}'_2)^{-1}(d_1^{-1}(W)),$$

that is,

$$(\widetilde{d}'_0)^{-1}(W_1) = (\widetilde{d}'_1)^{-1}(W_1).$$

[39] Editor's note: If $f : X \to Y$ is locally of finite type, the set of points $x \in X$ that are isolated in their fiber $f^{-1}(f(x))$ is open in X. In EGA III, 4.4.10, this is deduced from Zariski's Main Theorem when Y is locally Noetherian. For arbitrary Y, it follows from Chevalley's semicontinuity theorem (EGA IV$_3$, 13.1.3 and 13.1.4). Consequently, f is quasi-finite at x if and only if it is of finite type at x and x is isolated in $f^{-1}(f(x))$. This will be used below; see editor's note (42).

[40] Editor's note: The original has been expanded in what follows.

The morphism $\widetilde{d_0}$ is faithfully flat and quasi-compact, because d_0, like d_1, is surjective, proper, and flat. Moreover, the square

$$\begin{tikzcd} F_2 \arrow[r, "\widetilde{d_1'}"] \arrow[d, "\widetilde{d_0'}"'] & F_1 \arrow[d, "\widetilde{d_0}"] \\ F_1 \arrow[r, "\widetilde{d_0}"] & F \end{tikzcd}$$

Figure 39: The Cartesian square used to descend W_1.

is Cartesian. It follows that

$$W_1 = \widetilde{d_0}^{-1}(U)$$

for an open subset U of F (SGA 1, VIII, 4.4). This open subset U is a quasi-section of the groupoid with base W induced by X_*. We may therefore take $U_z = W$.

It remains to state the following lemma, whose proof is classical.

Lemma 7.4.—*Consider a Cartesian square of schemes*

$$\begin{tikzcd} F_2 \arrow[r, "v"] \arrow[d, "d'"'] & F_1 \arrow[d, "d"] \\ X_1 \arrow[r, "u"] & X_0 \end{tikzcd}$$

Figure 40: The Cartesian square in Lemma 7.4.

and let x be a point of F_2.

(i) If u is flat, d' is flat at x if and only if d is flat at $v(x)$.

(ii) If d is locally of finite type, d' is quasi-finite at x if and only if d is quasi-finite at $v(x)$.[41]

We have thus proved that X_0 has a covering by saturated open subsets W such that the groupoid W_* induced by X_* on W possesses a quasi-section.[42]

By Lemma 6.1 and the reductions stated after Theorem 7.1, this proves assertions (i) and (iii) of that theorem, as well as the facts that p is surjective and open and that p and

[41]Editor's note: The conditions are sufficient by base change (EGA II, 6.2.4(iii), and EGA IV$_2$, 2.1.4). Conversely, put $y = d'(x)$ and $z = u(y) = d(v(x))$, and suppose that d' is flat at x and that u (hence also v) is flat. Then $\mathcal{O}_{v(x)} \to \mathcal{O}_x$ is faithfully flat, as is $\mathcal{O}_z \to \mathcal{O}_y \to \mathcal{O}_x$. Consequently, $\mathcal{O}_z \to \mathcal{O}_{v(x)}$ is faithfully flat (EGA IV$_2$, 2.2.11(iv)). Finally, suppose that d is locally of finite type and that d' is quasi-finite at x. Then $v(x)$ is isolated in its fiber $d^{-1}(z)$, since x is isolated in its fiber $d'^{-1}(y) = d^{-1}(z) \otimes_{\kappa(z)} \kappa(y)$. By Chevalley's semicontinuity theorem there is therefore an open neighborhood of $v(x)$ every point of which is isolated in its fiber (EGA IV$_3$, 13.1.3 and 13.1.4), and hence d is quasi-finite at $v(x)$.

[42]Editor's note: The sequel has been modified to take advantage of the additions made to Lemma 6.1.

$Y \to S$ are locally of finite presentation. Moreover, because $X_0 \to S$ is quasi-projective, and hence separated and of finite type, p is separated; the proof of Lemma 6.1(ii) shows that p and $Y \to S$ are of finite presentation.

To prove that p is proper, it remains to prove that it is universally closed. Since this assertion is local on Y, we may work over a saturated open subset W such that the groupoid W_* induced by X_* on W possesses a quasi-section U (because X_0 is covered by such open subsets). With the notation of 6(a), there is a commutative diagram

$$\begin{tikzcd} W \arrow[dr, "p"'] & V \arrow[l, "v"'] \arrow[r, "u"] \arrow[d, "r"] & U \arrow[dl, "q"] \\ & Z & \end{tikzcd}$$

Figure 41: The properness diagram over the open subset $Z \subseteq Y$.

where Z is an open subset of Y, all the arrows are surjective, and q is integral. By hypothesis $d_0 : X_1 \to X_0$ is proper, so u, which is obtained from d_0 by base change, is proper as well. Consequently r is universally closed, and hence so is p, since v is surjective.

Finally, because p is surjective and universally closed and X_0 is quasi-projective, hence separated, the diagonal $\Delta_{Y/S}(Y)$ is closed in $Y \times_S Y$: it is the image under $p \times p$ of the diagonal $\Delta_{X_0/S}(X_0)$. Thus Y is separated over S. This completes the proof of Theorem 7.1(ii).

The assertions in Theorem 7.1(iv) are local on Y. Since the saturated open subsets U_z cover X_0, it is enough to verify them after replacing X and Y by U_z and $V_z = p(U_z)$, respectively. As observed at the beginning of the proof of Theorem 7.1, Lemmas 1.1, 1.2, and 6.1(i) show that $(V_z, p|_{U_z})$ is the cokernel, both in $(\mathbf{Sch})$ and in the category of ringed spaces, of the groupoid induced by X_* on U_z. The hypothesis that $d = (d_0, d_1)$ is a monomorphism is preserved under the base change $U_z \to X_0$. The first two assertions of Theorem 7.1(iv) therefore follow from Lemma 6.1(iv).

Let us finally prove the consequences stated at the end of part (iv) (see [Ray67a], Theorem 1(iii)). By hypothesis, the groupoid X_* comes from an equivalence relation $R \to X_0 \times X_0$. We have proved that R is effective (Exposé IV, 3.3.2) and that $p : X_0 \to Y = X_0/R$ is faithfully flat and of finite presentation. Consequently, if (M) denotes the family of faithfully flat morphisms locally of finite presentation, then R is (M)-effective. By Exposé IV, 6.3.3, the pair (Y, p) therefore represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from Exposé IV, 3.4.3.1.

8. Passage to the quotient by a flat groupoid that is not necessarily proper

Theorem 8.1.—[43]*Let S be a Noetherian scheme and*

[43]Editor's note: There is a largest open subset W of X_0 satisfying the conclusions of the theorem. Indeed, let W be an open subset as in the theorem and let $W^\sharp$ be a dense saturated open subset of W. Since p is open, $V^\sharp = p(W^\sharp)$ is open in V, and $W^\sharp = p^{-1}(V^\sharp)$, because $W^\sharp$ is saturated. By Lemma 1.1, $V^\sharp$ is a cokernel for the groupoid induced on $W^\sharp$. Thus two open subsets W and W' satisfying the conclusions of the theorem may be glued along their intersection $W^\sharp$, and conditions (i)–(iv), together with the property that p and $V \to S$ are locally of finite presentation, are preserved. Conclusion (ii') follows, as in the proof of Lemma 6.1(ii), from the hypothesis that X_0 is of finite type over the Noetherian scheme S.

$$\begin{tikzcd} X_2 \arrow[r, shift left=2, "d'_2"] \arrow[r, "d'_1"] \arrow[r, shift right=2, "d'_0"'] & X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \end{tikzcd}$$

Figure 42: The $(\mathbf{Sch}/S)$-groupoid of Theorem 8.1.

a $(\mathbf{Sch}/S)$-groupoid such that d_1 is flat and of finite type, X_0 is of finite type over S, and the morphism $X_1 \to X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite.

There exists a dense saturated open subset $W \subseteq X_0$ having the following properties:

(i) *if $W_2 \overset{w'_i}{\rightrightarrows} W_1 \overset{w_j}{\rightrightarrows} W$ is the groupoid induced by X_* on W, then (w_0, w_1) has a cokernel (V, p) in $(\mathbf{Sch}/S)$; moreover, (V, p) is a cokernel of (w_0, w_1) in the category of all ringed spaces;*

(ii) *p is surjective and open;*

(ii') *p and $V \to S$ are of finite presentation;*

(iii) *the morphism $W_1 \to W \times_V W$ with components w_0 and w_1 is surjective;*

(iv) *if (d_0, d_1) is an equivalence pair, then $W_1 \to W \times_V W$ is an isomorphism and p is faithfully flat.*

We shall prove that W may be chosen so that the $(\mathbf{Sch}/S)$-groupoid W_* induced by X_* possesses a quasi-section (see §7). Theorem 8.1 will then follow from Lemma 6.1.

Suppose provisionally that for every point $z \in X_0$ closed relative to S (see §7) there is a saturated open subset W_z that possesses a quasi-section and meets every irreducible component of X_0 passing through z. Then the complement $X_0 - W_z$ is saturated: its saturation $d_1(d_0^{-1}(X_0 - W_z))$ is open and does not meet W_z. If this complement is nonempty, choose in it a point z' closed relative to S, and associate to z' an open subset $W_{z'}$ as above. We may further suppose that $W_{z'} \subseteq X_0 - W_z$. Then W_z and $W_{z'}$ are disjoint, and the groupoid induced by X_* on $W_z \cup W_{z'}$ possesses a quasi-section. The process must stop, because X_0 has only finitely many irreducible components. It therefore remains to construct W_z.

For this purpose we may suppose that S is affine. Let $y \in X_1$ satisfy $d_1(y) = z$, let X be an affine open subset of X_0 containing $d_0(y)$, let Y be the inverse image of X under $d_0 : X_1 \to X_0$, and let $u : Y \to X$ and $v : Y \to X_0$ be the morphisms induced by d_0 and

Lemmas 1.1 and 1.2 also show that the union of all saturated open subsets $U \subseteq X_0$ such that the open subset $p(U) \subseteq Y$ is a scheme and $p|_U : U \to p(U)$ is a morphism of schemes is the largest saturated open subset $\Omega \subseteq X_0$ satisfying condition (i) of Theorem 8.1. The theorem shows that Ω contains a dense open subset W, but it is not immediate that Ω satisfies properties (ii)–(iv). For more precise results, and for the representability of the fppf quotient sheaf $\widetilde{X/R}$ (where R denotes the groupoid X_* with base $X = X_0$), see [Ray67a], [Ray67b], and Appendix I of [An73]. These works use weaker hypotheses: S may be arbitrary, X is locally of finite type over S, and R is an S-groupoid with base X such that d_0 (and hence d_1) is flat and of finite presentation. In particular, if $\widetilde{X/R}$ is represented by an S-scheme Y, then Y is also the cokernel in the category of ringed spaces. The converse is false in general (see [Mum65], Chapter 0, §3, Example 0.4, cited in [Ray67a], Remark 1), but it is true when $d = (d_0, d_1)$ is an immersion. Under this hypothesis, $p : \Omega \to Z := \Omega/R_\Omega$ is faithfully flat and of finite presentation. If, in addition, S is locally Noetherian, then a codimension-one point x of X belongs to Ω if and only if the graph of the groupoid induced on $\mathrm{Spec}(\mathcal{O}_{X,x})$ is closed. See [Ray67a], Proposition 1; [Ray67b], Proposition 1 and Theorems 2, 1, and 4; and [An73], Theorems 5 and 6 on pages 66–67 and Proposition 3.3.1 on page 49. For an action of an algebraic group over an algebraically closed field k, see also [DR81].

d_1. Since X is affine, hence quasi-projective, Lemma 7.2 provides a subscheme $F \subseteq X_0$ such that $d_0^{-1}(F) \cap d_1^{-1}(z)$ is nonempty, $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite, and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $v^{-1}(z)$. We can therefore resume the notation of Lemma 7.3, writing F_1 and F_2 for the inverse images of F in X_1 and X_2, and so on:

$$\begin{tikzcd}
F_2 \arrow[r, shift left, "\widetilde{d}'_1"] \arrow[r, shift right, "\widetilde{d}'_0"'] \arrow[d, "\widetilde{d}_2"'] & F_1 \arrow[r, "\widetilde{d}_0"] \arrow[d, "\widetilde{d}_1"] & F \\
X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 &
\end{tikzcd}$$

Figure 43: The induced diagram on F_2, F_1, F used in the proof of Theorem 8.1.

As in 7.3, one proves that $\widetilde{d}_1$ is quasi-finite at the points of $\widetilde{d}_1^{-1}(z)$. It is therefore natural to consider the open subset $F'_1 \subseteq F_1$ consisting of the points at which $\widetilde{d}_1$ is both flat and quasi-finite. By Lemma 7.4, the inverse images of F'_1 under $\widetilde{d}'_1$ and $\widetilde{d}'_0$ both consist of the points of F_2 at which $\widetilde{d}'_2$ is flat and quasi-finite. Thus these inverse images coincide, and F'_1 is of the form $\widetilde{d}_0^{-1}(F')$ for an open subset $F' \subseteq F$ (SGA 1, VIII, 4.4). After replacing F by F', we may therefore suppose that $\widetilde{d}_1$ is quasi-finite and flat. Let W_z be the largest open subset of $\widetilde{d}_1(F_1)$ over which $\widetilde{d}_1$ is finite and flat.

The open subset W_z need not contain z, but it contains the generic points of the irreducible components of X_0 passing through z.[44] Since d_0 (and likewise d_1) is faithfully flat and of finite presentation, hence open, SGA 1, VIII, 5.7 shows that $d_0^{-1}(W_z)$ and $d_1^{-1}(W_z)$ both coincide with the largest open subset of $\widetilde{d}'_2(F_2)$ over which $\widetilde{d}'_2$ is finite and flat. As in 7.3, the inverse images of $\widetilde{d}_1^{-1}(W_z)$ under $\widetilde{d}'_0$ and $\widetilde{d}'_1$ therefore coincide. Hence $\widetilde{d}_1^{-1}(W_z) = \widetilde{d}_0^{-1}(U)$ for an open subset $U \subseteq F$, and U is a quasi-section for the groupoid induced by X_* on W_z.

9. Elimination of the Noetherian hypotheses in Theorem 7.1

a) Resume the notation and hypotheses of Lemma 6.1, and let $\pi : S' \to S$ be an arbitrary base change. If $f : X \to Y$ is a morphism of S-schemes, write $f' : X' \to Y'$ for the morphism of S'-schemes obtained from f by extension of the base along π. With this convention, $p' : X'_0 \to Y'$ is surjective, as is the morphism

$$X'_1 \longrightarrow X'_0 \times_{Y'} X'_0$$

with components d'_0 and d'_1. Thus the set underlying Y' is the quotient of the set underlying X'_0 by the equivalence relation on X'_0 defined by the S'-groupoid X'_*. Moreover, $q' : U' \to Y'$ is integral and surjective. The topology of Y' is consequently the quotient topology of that of U', and hence also of that of X'_0; see the proof in §6(c).

On the other hand, U' is plainly a quasi-section of the S'-groupoid X'_*, so Lemma 6.1 applies. In particular, X'_* has a cokernel (Y_1, p_1), and the topological space underlying Y_1

[44] Editor's note: Indeed, let η be such a generic point. The hypotheses imply that $\mathcal{O}_{X_0,\eta}$ is an Artinian local ring and that $\mathcal{O}_{F_1,\eta}$ is a finitely generated $\mathcal{O}_{X_0,\eta}$-module. By SGA 1, VIII, 6.5, there is therefore an open neighborhood of η over which $\widetilde{d}_1$ is finite.

is obtained from that underlying X'_0 by identifying the points equivalent under the relation defined by X'_*. It follows that the canonical morphism $Y_1 \to Y'$ is a homeomorphism. In fact, it is a universal homeomorphism. Indeed, if S'' is an S'-scheme, let Y_2 be the cokernel of $(d_0 \times_S S'', d_1 \times_S S'')$. The preceding argument, applied to the base changes $S'' \to S'$ and $S'' \to S$, shows that

$$Y_2 \longrightarrow Y_1 \times_{S'} S'' \qquad \text{and} \qquad Y_2 \longrightarrow Y \times_S S'' \simeq Y' \times_{S'} S''$$

are homeomorphisms. Hence

$$Y_1 \times_{S'} S'' \longrightarrow Y' \times_{S'} S''$$

is a homeomorphism as well.

b) The same remarks plainly apply when the groupoid X_* *locally* possesses quasi-sections (see the proof of Theorem 7.1).[45] For example, we have the following theorem:

Theorem 9.0.—*Let S be an arbitrary scheme and*

$$X_2 \xrightarrow{d'_j} X_1 \xrightarrow{d_i} X_0$$

a $(\mathbf{Sch}/S)$-groupoid such that X_0 is of finite presentation and quasi-projective over S, d_1 is of finite presentation, proper, and flat, and the morphism

$$d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_S X_0$$

is quasi-finite. Then:

(1) *Every point x of X_0 has a saturated open neighborhood W such that the groupoid induced by X_* on W has a quasi-section.*

(2) *Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Then Y is a scheme, p is a morphism of schemes, and (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch}/S)$.*

(3) *The morphism p is surjective, open, and universally closed.*

(4) *The morphism*
$$d : X_1 \longrightarrow X_0 \times_Y X_0$$
with components d_0 and d_1 is surjective.

(5) *If (d_0, d_1) is an equivalence pair, then:*

   (a) *the morphism*
   $$d : X_1 \longrightarrow X_0 \times_Y X_0$$
   is an isomorphism and p is faithfully flat;

   (b) *the morphisms p and $Y \to S$ are of finite presentation, and (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.*

Proof. For (1), the question is local on S, so we may suppose that $S = \operatorname{Spec} B$ is affine. There then exist a ring A of finite type over $\mathbb{Z}$, a morphism

$$S \longrightarrow T = \operatorname{Spec} A,$$

[45] Editor's note: What follows has been expanded in order to bring out Theorem 9.0 below.

and a $(\mathbf{Sch}/T)$-groupoid Z_* such that

$$X_* \simeq Z_* \times_T S.$$

Indeed, apply EGA IV$_3$, 8.8.3 to $S_0 = \operatorname{Spec} \mathbb{Z}$ and $S_i = \operatorname{Spec} A_i$, where the A_i range over the finitely generated $\mathbb{Z}$-subalgebras of B. Moreover, by EGA IV$_3$, 8.10.5, we may suppose that Z_* satisfies the hypotheses of Theorem 7.1. Consequently, Z_* has quasi-sections locally.

By part (a), the same is therefore true of X_*, and assertions (2), (3), (4), and (5)(a) follow from Lemma 6.1, just as in the proof of Theorem 7.1.

c) Proof of (5)(b).—Let us show that $Y \to S$ is of finite presentation.[46] By hypothesis, $(d_0^{X_*}, d_1^{X_*})$ is an equivalence pair; that is,

$$d^{X_*} : X_1 \longrightarrow X_0 \times_S X_0$$

is a monomorphism. By EGA IV$_3$, 8.10.5, after enlarging A if necessary, we may suppose that

$$d^{Z_*} : Z_1 \longrightarrow Z_0 \times_T Z_0$$

is a monomorphism. Since $T = \operatorname{Spec} A$, with A Noetherian, Theorem 7.1 then shows that the groupoid Z_* has a cokernel (Q, q) in $(\mathbf{Sch}/T)$, that q and $Q \to T$ are of finite presentation, and, moreover, that

$$q : Z_0 \longrightarrow Q$$

is faithfully flat and d^{Z_*} induces an isomorphism

$$Z_1 \xrightarrow{\sim} Z_0 \times_Q Z_0.$$

Put $Q_S = Q \times_T S$. Since $X_i \simeq Z_i \times_T S$, we obtain an isomorphism

$$d^{Z_*} \times_T S : X_1 \xrightarrow{\sim} X_0 \times_{Q_S} X_0.$$

Denote its inverse by ϕ, and let π be the canonical morphism

$$X_0 \times_Y X_0 \longrightarrow X_0 \times_{Q_S} X_0.$$

Then $\phi \circ \pi$ is the inverse of

$$d_0 \boxtimes d_1 : X_1 \xrightarrow{\sim} X_0 \times_Y X_0.$$

It follows that the equivalence relation defined by X_*, that is, the monomorphism shown in Figure 44,

$$X_1 \xrightarrow[\sim]{d_0 \boxtimes d_1} X_0 \times_Y X_0 \hookrightarrow X_0 \times_S X_0$$

Figure 44: The equivalence-relation monomorphism defined by X_*.

identifies with the equivalence relation $\mathscr{R}(q_S)$ defined by $q_S : X_0 \to Q_S$. Since q_S is faithfully flat and of finite presentation, hence a universal effective epimorphism, $\mathscr{R}(q_S)$ has quotient Q_S; see Exposé IV, 3.3.2. Consequently,

$$Y \simeq Q \times_T S,$$

[46] Editor's note: The original states that this follows from Proposition 9.1 below. We were unable to reconstruct that argument. The proof that follows was communicated to us by O. Gabber.

so $Y \to S$ and p are of finite presentation. Moreover, by Exposé IV, 6.3.3, (Y, p) is also a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.

Proposition 9.1.—*Consider morphisms of schemes*

$$X_0 \xrightarrow{p} Y \xrightarrow{q} S$$

such that qp is of finite type (respectively, of finite presentation) and p is faithfully flat and of finite presentation. Then q is of finite type (respectively, of finite presentation).[*]

Since p is surjective and qp is quasi-compact, q is quasi-compact. We may therefore suppose that S, Y, and X_0 are affine, with rings A, B, and C, respectively. We have

$$B = \varinjlim_i B_i,$$

where the B_i range over the finitely generated A-subalgebras of B. Since C is of finite presentation over B, there exist an index i_0, a B_{i_0}-algebra C_{i_0} of finite presentation, and an isomorphism

$$C \simeq C_{i_0} \otimes_{B_{i_0}} B.$$

For $i \geq i_0$, put

$$C_i = C_{i_0} \otimes_{B_{i_0}} B_i;$$

then

$$C \simeq C_i \otimes_{B_i} B.$$

$$\begin{tikzcd}
B \arrow[r] & C \\
B_i \arrow[u] \arrow[r] & C_i \arrow[u] \\
A \arrow[u] &
\end{tikzcd}$$

Figure 45: The compatible A-algebras used in the limit argument.

Since C is faithfully flat over B, EGA IV_3, 11.2.6 and 8.10.5(vi) give an index $i_1 \geq i_0$ such that C_{i_1} is faithfully flat over B_{i_1}. Consequently, C_i is faithfully flat over B_i for every $i \geq i_1$. For $i \geq i_1$, the canonical map $C_i \to C$ is then injective, because it is obtained from $B_i \to B$ by faithfully flat base extension.

If C is of finite type over A, it follows that there is an index $j \geq i_1$ such that $C_j = C$, whence $B_j = B$, because C_j is faithfully flat over B_j. Thus B is of finite type over A.

Now suppose that C is of finite presentation over A. By what precedes, B is of finite type over A, and hence has the form

$$B = \overline{B}/I,$$

where $\overline{B}$ is a polynomial algebra over A in finitely many indeterminates and I is an ideal of $\overline{B}$. The ideal I is the union of its finitely generated subideals I_α; consequently,

$$B = \varinjlim_\alpha B_\alpha, \qquad B_\alpha = \overline{B}/I_\alpha.$$

[*]See EGA IV_4, 17.7.5 for a more general result.

Proceeding as above, there exist an index α_0, a B_{α_0}-algebra C_{α_0} of finite presentation, and an isomorphism

$$C \simeq C_{\alpha_0} \otimes_{B_{\alpha_0}} B.$$

For $\alpha \geq \alpha_0$, put

$$C_\alpha = C_{\alpha_0} \otimes_{B_{\alpha_0}} B_\alpha,$$

so that

$$C \simeq C_\alpha \otimes_{B_\alpha} B \qquad (\alpha \geq \alpha_0).$$

Again by EGA IV_3, 11.2.6 and 8.10.5(vi), C_α is faithfully flat over B_α for all sufficiently large α. In that case, the kernels of the maps $C_\alpha \to C$ and $C_\alpha \to C_\beta$, respectively, identify with

$$C_\alpha \otimes_{B_\alpha} (I/I_\alpha) \quad \text{and} \quad C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) \qquad (\beta \geq \alpha).$$

Since C_α and C are of finite presentation over A and $C_\alpha \to C$ is surjective,

$$C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$$

is a finitely generated ideal,[47] and it is the union of the ideals

$$C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha).$$

Thus, for all sufficiently large β,

$$C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) = C_\alpha \otimes_{B_\alpha} (I/I_\alpha).$$

Since C_α is faithfully flat over B_α, this also gives $I_\beta = I$. Therefore B is of finite presentation over A.

10. Supplement: quotients by a group scheme

Sections 10.2–10.4 below, written following suggestions of M. Raynaud, apply the preceding theorems to the case of an action of a group scheme. For the reader's convenience, Section 10.1 first reproduces statements 2.1–2.3 of Exposé XVI.

10.1. Representability theorems for quotients.—We first recall the following result.

Theorem 10.1.1.—*Let S be a scheme, let X and Y be two S-schemes, and let $f : X \to Y$ be an S-morphism. Suppose that one of the following two conditions holds:*

α) *the morphism f is locally of finite presentation;*

β) *the scheme S is locally Noetherian and X is locally of finite type over S.*

Then the following conditions are equivalent:

(i) *there exist an S-scheme X' and a factorization*

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism locally of finite presentation and f'' is a monomorphism;

[47] Editor's note: see EGA IV_1, 1.4.4.

(ii) *the first projection*

$$p_1 : X \times_Y X \longrightarrow X$$

is flat.

Moreover, if these conditions hold, then (X', f') is a quotient of X by the equivalence relation defined by f for the fppf topology. Thus the factorization $f = f'' \circ f'$ in (i) is unique up to isomorphism.

The case in which Y is locally Noetherian and X is of finite type over Y is treated in [Mur65], Corollary 2 to Theorem 2. We shall see that the general case reduces to that one.

We first make a few remarks.

a) The implication (i)⇒(ii) is immediate. Indeed, the first projection

$$p'_1 : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism because f'' is a monomorphism, and p'_1 is flat because f' is flat; hence p_1 is flat.

b) The assertions of Theorem 10.1.1 are local on Y, and hence local on S. They are also local on X, as follows readily from the fact that a flat morphism locally of finite presentation is open (EGA IV_3, 11.3.1).

c) Under hypothesis α of Theorem 10.1.1, the preceding remarks reduce us to the case where X and Y are affine and f is of finite presentation. Replacing S by Y, if necessary, we may suppose that X and Y are of finite presentation over S. EGA IV_3, 11.2.6 then reduces us to the case where S is Noetherian.

d) Under hypothesis β of Theorem 10.1.1, we may suppose that S, X, and Y are affine, that S is Noetherian, and that X is of finite type over S. Write Y as a filtered inverse limit of affine schemes Y_i of finite type over S. The schemes

$$X \times_{Y_i} X$$

form a filtered decreasing family of closed subschemes of $X \times_S X$, whose inverse limit is $X \times_Y X$. Since $X \times_S X$ is Noetherian, for all sufficiently large i we have

$$X \times_{Y_i} X = X \times_Y X.$$

It follows that

$$f_i : X \xrightarrow{f} Y \longrightarrow Y_i$$

satisfies condition (ii) of Theorem 10.1.1 whenever f does. The equivalence relation on X defined by f is the same as the one defined by f_i. Thus it suffices to prove (ii)⇒(i) for f_i, reducing us to the case where Y is of finite type over S.

Application to group schemes. Let S be a scheme, let G be an S-group scheme locally of finite presentation over S, and suppose that G acts on the left on an S-scheme X. If $X \to S$ has a section ξ, recall that the stabilizer $\mathrm{Stab}_G(\xi)$ is representable by a subgroup scheme of G; see Exposé I, 2.3.3.

Theorem 10.1.2.—*Let S be a scheme and let G be an S-group scheme locally of finite presentation over S, acting on an S-scheme X. Suppose that $X \to S$ has a section ξ whose stabilizer H in G is flat over S. If one of the following hypotheses holds:*

a) X is locally of finite type over S;

b) S is locally Noetherian,

then the fppf quotient sheaf G/H is representable by an S-scheme, locally of finite presentation over S, and the S-morphism

$$f : G \longrightarrow X, \qquad g \longmapsto g \cdot \xi$$

factors as

$$\begin{tikzcd} G \arrow[d, "p"'] \arrow[dr, "f"] & \\ G/H \arrow[r, hook, "i"] & X \end{tikzcd}$$

Figure 46: The factorization of the orbit morphism in Theorem 10.1.2.

where p is the canonical projection, which is faithfully flat and locally of finite presentation, and i is a monomorphism.

Proof.— The morphism f makes G into an X-scheme. By definition of the stabilizer of ξ, the morphism

$$G \times_S H \longrightarrow G \times_X G, \qquad (g, h) \longmapsto (g, gh)$$

is an isomorphism. Since H is flat over S, the scheme $G \times_S H$ is flat over G; hence the first projection

$$p_1 : G \times_X G \longrightarrow G$$

is flat.Moreover, if X is locally of finite type over S, then f is locally of finite presentation (EGA IV_1, 1.4.3(v)); otherwise, S is assumed locally Noetherian. It therefore suffices to apply Theorem 10.1.1 to f. It remains only to see that G/H is locally of finite presentation over S, and this follows immediately from Proposition 9.1.

Corollary 10.1.3.— *Let S be a scheme and $u : G \to H$ a morphism of S-group schemes. Suppose that G is locally of finite presentation over S, and that either H is locally of finite type over S or S is locally Noetherian.*

If $K = \ker(u)$ is flat over S, then the quotient group G/K is representable by an S-group scheme locally of finite presentation over S, and u factors as

$$\begin{tikzcd} G \arrow[rr, "u"] \arrow[dr, "p"'] & & H \\ & G/K \arrow[ur, "i"'] & \end{tikzcd}$$

Figure 47: The factorization of u through G/K.

where p is the canonical projection and i is a monomorphism.

Proof.— Apply Theorem 10.1.2 with $X = H$ and with the unit section of H as ξ.

10.2. Stabilizer of the diagonal

Let S be a Noetherian scheme, X an S-scheme of finite type, and G a flat S-group scheme of finite type acting on the left on X. Thus we have an S-action $d_0 : G \times_S X \to X$. Let $d_1 : G \times_S X \to X$ be projection onto the second factor. As in §2(a), we have the groupoid

$$G \times_S G \times_S X \overset{\mathrm{pr}_{2,3}}{\underset{G\times d_0}{\overset{\mu\times X}{\rightrightarrows}}} G \times_S X \underset{d_0}{\overset{d_1}{\rightrightarrows}} X$$

Figure 48: The action groupoid attached to (X, G).

whose cokernel, when it exists, is denoted $G\backslash X$.

Definition 10.2.1.— *We denote by $F \subset G \times_S X$ the* stabilizer of the diagonal section*; that is, F is the X-scheme defined by the Cartesian square*

$$\begin{tikzcd} F \arrow[r] \arrow[d] & X \arrow[d, "\Delta"] \\ G \times_S X \arrow[r, "(d_0, d_1)"] & X \times_S X \end{tikzcd}$$

Figure 49: The stabilizer F of the diagonal section.

Then F is an X-subgroup scheme of $G \times_S X$. Since $G \times_S X$ is of finite type over the Noetherian scheme S, it is Noetherian; consequently F is of finite type both over S and over X (EGA I, 6.3.5 and 6.3.6). If $X \to S$ is separated, then F is a closed X-subgroup scheme of $G \times_S X$.

Recall that G is said to *act freely* on X when

$$(d_0, d_1) : G \times_S X \longrightarrow X \times_S X$$

is a monomorphism (Exposé III, 3.2.1). Equivalently, F is the trivial group scheme over X.

10.3. The case in which F is quasi-finite over X

Since F is of finite type over X, it is quasi-finite over X if and only if the stabilizers of the geometric points of X are finite.

Theorem 10.3.1.—[48] *Under the hypotheses of 10.2, suppose that F is quasi-finite over X. Then there is a dense, G-saturated open subset $U \subset X$ having the following properties:*

(i) *in* $(\mathbf{Sch}/S)$, *the cokernel $V = G\backslash U$ exists; moreover, V is a quotient in the category of ringed spaces;*

(ii) *$p : U \to V$ is surjective, open, and of finite presentation;*

[48] Editor's note: here too there is a largest open subset $U \subset X$ satisfying the conclusions of the theorem; see editor's note 44.

(iii) *V is of finite presentation over S;*

(iv) *the morphism*

$$G \times_S U \longrightarrow U \times_V U, \qquad (g, x) \longmapsto (gx, x),$$

is surjective;

(v) *if, in addition, G acts freely on X, then $U \to V$ is a left G-torsor, locally trivial for the fppf topology. In particular, $U \to V$ is faithfully flat.*[49]

Proof.— By hypothesis, the morphism

$$G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

is quasi-finite. Theorem 8.1 therefore applies to the groupoid defined by (X, G). Hence there is a dense saturated open subset $U \subset X$ for which the quotient $G\backslash U$ exists, and it has properties (i), (ii), and (iii); the same theorem also gives (iv).

To prove (v), recall that G acts freely on X if and only if (d_0, d_1) is an equivalence pair (Exposé III, 3.2.1).In that case Theorem 8.1(iv) shows that

$$G \times_S U \longrightarrow U \times_V U$$

is an isomorphism and that p is faithfully flat and of finite presentation. Thus U is a G-torsor over V, locally trivial for the fppf topology.

10.4. The case in which F is flat over X

Write

$$d = (d_0, d_1) : G \times_S X \longrightarrow X \times_S X, \qquad d(g, x) = (gx, x).$$

Recall that the sheaf-theoretic graph $\widetilde{\Gamma}$ of the equivalence relation associated with (X, G) is the fppf S-subsheaf of $X \times_S X$ given by the image of (d_0, d_1). It is the fppf sheaf associated with the graph functor

$$T \longmapsto \Gamma(T) = \big\{(x_0, x_1) \in X(T) \times X(T) \mid x_0 \in G(T)x_1\big\}.$$

Put $G_X = G \times_S X$. For every S-scheme T there is a surjective map

$$G_X(T) \longrightarrow \Gamma(T), \qquad (g, x) \longmapsto (gx, x),$$

which induces a bijection

$$\phi(T) : G_X(T)/F(T) \xrightarrow{\sim} \Gamma(T).$$

Indeed, if $(g, x), (g', x') \in G_X(T)$ have the same image, then $x' = x$ and $g^{-1}g'x = x$, so that $(g^{-1}g', x) \in F(T)$, and (g, x) and (g', x) have the same image in $G_X(T)/F(T)$.By definition (see Exposé IV, 4.4.1(ii), or the proof of 5.2.1), the quotient sheaf G_X/F is the fppf sheaf associated with the presheaf

$$T \longmapsto G_X(T)/F(T) \simeq \Gamma(T).$$

We therefore obtain an isomorphism of sheaves

$$\phi : G_X/F \xrightarrow{\sim} \widetilde{\Gamma}.$$

Theorem 10.4.1.—[50] *Under the hypotheses of 10.2, the following assertions hold:*

[49] Editor's note: if one assumes in addition that G is a reductive S-group scheme and that its free action on X is linearizable, then $G\backslash X$ is known to be representable and $X \to G\backslash X$ is a left G-torsor. This follows from results of Raynaud and Seshadri and appears in [CTS79], Proposition 6.11.

[50] Editor's note: this is part (2) of Theorem 3 of [Ray67b]. That note also sketches another proof of Theorem 10.1.1.

a) the sheaf $\widetilde{\Gamma}$ *is representable if and only if* F *is flat over* X*;*

b) suppose that F *is flat over* X*. Then the morphisms induced by* d_1 *and* d_0

$$\begin{tikzcd} G_X/F \arrow[r, shift left, "\overline{d}_1"] \arrow[r, shift right, "\overline{d}_0"'] & X \end{tikzcd}$$

Figure 50: The two maps induced by d_1 and d_0.

are faithfully flat and of finite presentation.

Proof of a).— Suppose that the fppf sheaf G_X/F is represented by an X-scheme Y. Then, by Exposé IV, 6.3.3, the morphism $p : G_X \to Y$ is faithfully flat and locally of finite presentation, and the second square in the following diagram is Cartesian:

$$\begin{tikzcd} F \arrow[r] \arrow[d] & F \times_X G_X \arrow[r] \arrow[d] & G_X \arrow[d, "p"] \\ X \arrow[r, "e_X"] & G_X \arrow[r] & Y \end{tikzcd}$$

Figure 51: The Cartesian square used in the proof of Theorem 10.4.1(a).

The first square is obtained by base change along the unit section $e_X : X \to G_X$. Since p is faithfully flat and locally of finite presentation, the same is true of $F \to X$.

Conversely, suppose F is flat over X, and put $X^2 = X \times_S X$. The morphism $d : G_X \to X^2$ gives the fiber square

$$\begin{tikzcd} G_X \times_{X^2} G_X \arrow[r] \arrow[d] & G_X \arrow[d] \\ G_X \arrow[r] & X^2 \end{tikzcd}$$

Figure 52: The fiber square associated with $d : G_X \to X^2$.

The first projection

$$\mathrm{pr}_1 : G_X \times_{X^2} G_X \longrightarrow G_X$$

is then an $F \times_X G_X$-torsor over G_X, and is therefore flat and of finite type, because F is.By Theorem 10.1.1, d factors uniquely as

$$G_X \xrightarrow{\psi} Y \xrightarrow{\tau} X \times_S X$$

Figure 53: The unique factorization of d.

where ψ is faithfully flat and of finite type, and τ is a monomorphism of schemes.

It follows that the morphism of sheaves $\psi : G_X \to Y$ is F-invariant, and hence induces a morphism

$$\bar{\psi} : G_X/F \longrightarrow Y.$$

Moreover, because ψ is faithfully flat and of finite type, the monomorphism of sheaves τ factors through the sheaf image of d, namely $\widetilde{\Gamma}$. Thus the isomorphism $G_X/F \simeq \widetilde{\Gamma}$ factors through the monomorphism $Y \to \widetilde{\Gamma}$. We conclude that Y represents G_X/F.

Proof of b).— Suppose F is flat over X. By part (a) and its proof, G_X/F is representable, and

$$p : G_X \longrightarrow G_X/F$$

is faithfully flat and of finite presentation. On the other hand, the morphisms $d_i : G_X \to X$, for $i = 0, 1$, are faithfully flat and of finite presentation by hypothesis. Since $d_i = \bar{d}_i \circ p$, it follows from EGA IV_2, 2.2.13(iii), and EGA IV_3, 11.3.16, that each $\bar{d}_i$ is faithfully flat and of finite presentation.

Theorem 10.4.2.—[51] *Under the hypotheses of 10.2, suppose that F is flat over X. Then there is a dense saturated open subset $U \subset X$ such that the fppf quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.*

Proof.— Theorem 10.4.1 shows that $G_X/F \simeq \widetilde{\Gamma}$ is representable. The fppf sheaf $G\backslash X$ is therefore the quotient sheaf of

$$G_X/F \overset{\overline{d}_1}{\underset{\overline{d}_0}{\rightrightarrows}} X$$

Figure 54: The equivalence pair whose quotient is $G\backslash X$.

By what precedes,

$$\bar{d}_i : G_X/F \longrightarrow X$$

is faithfully flat and of finite presentation for $i = 0, 1$, and the morphism

$$G_X/F \xrightarrow{\ \sim\ } \widetilde{\Gamma} \hookrightarrow X \times_S X$$

Figure 55: The monomorphism from the quotient G_X/F to $X \times_S X$.

is a monomorphism. In other words, $(\bar{d}_0, \bar{d}_1)$ is an equivalence pair. Consequently, Theorem 8.1 applies. There is therefore a dense saturated open subset $U \subset X$ such that the fppf quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.

Taking the generic-flatness theorem into account (EGA IV_2, 6.9.3), we obtain the following result.

[51] Editor's note: here too there is a largest open subset $U \subset X$ satisfying the conclusions of the theorem. Moreover, a codimension-one point $x \in X$ belongs to U if and only if

$$(G_X/F) \times_X \mathrm{Spec}(\mathcal{O}_{X,x}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{X,x}) \times_S \mathrm{Spec}(\mathcal{O}_{X,x})$$

is a closed immersion; see editor's note 44.

Corollary 10.4.3.— *Under the hypotheses of 10.2, suppose that X is reduced. Then there is a dense saturated open subset $U \subset X$ such that the fppf quotient $G\backslash U$ is an S-scheme of finite type and $U \to G\backslash U$ is faithfully flat and of finite presentation.*

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[52] Editor's note: additional references cited in this exposé.

Exposé VI_{A}
Generalities on Algebraic Groups

by P. Gabriel

[0]Throughout this chapter, A will denote an Artinian local ring with residue field k. A group scheme G over $\operatorname{Spec} A$ will be called simply an A-*group*. This A-group is said to be *locally of finite type* if its underlying scheme is locally of finite type over A; it is said to be *algebraic* if its underlying scheme is of finite type over A.

0. Preliminary Remarks

0.1. Let us first consider a group scheme G over an arbitrary scheme S. We call the structural morphism

$$\mu : G \times_S G \longrightarrow G$$

multiplication, and we call the morphism $c : G \to G$, defined by $c(T)(x) = x^{-1}$ for every scheme T over S and every $x \in G(T)$, *inversion*. If U and V are subsets of the underlying set of G, we denote by $U \cdot V$ the image under multiplication of the subset of $G \times_S G$ consisting of those points whose first projection belongs to U and whose second projection belongs to V. Likewise, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 be the projection of $G \times_S G$ onto the first factor, and let

$$\sigma : G \times_S G \longrightarrow G \times_S G$$

be the morphism with components pr_1 and μ. For every S-scheme T, the map $\sigma(T)$ is

$$(x, y) \longmapsto (x, xy);$$

it follows that σ is an automorphism. The composite of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S, pr_2, and hence μ, are flat morphisms; when G is smooth over S, pr_2, and hence μ, are smooth morphisms, and so on.

0.2. From now on we assume that S is the spectrum of an Artinian local ring A with residue field k. We denote by $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ the category of reduced schemes over k. For every scheme X over A, the reduced scheme X_{red} is an object of $(\mathbf{Sch}_{/k})_{\mathrm{red}}$, and the functor

$$X \longmapsto X_{\mathrm{red}}$$

is right adjoint to the inclusion of $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ into $\mathbf{Sch}_{/A}$. It follows that, for every A-group G, G_{red} is a group in the category $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.[1] In other words, for every reduced k-scheme

[0]Editor's note: version of October 13, 2024.
[1]Editor's note: the sentences that follow have been added.

T, $G_{\mathrm{red}}(T)$ is endowed, functorially in T, with a group structure. One must bear in mind that G_{red} is not necessarily a k-group, because its "multiplication" is only a morphism

$$(G_{\mathrm{red}} \times_k G_{\mathrm{red}})_{\mathrm{red}} \longrightarrow G_{\mathrm{red}}.$$ [2]

[3] However, if k is a perfect field, the inclusion of $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ into $\mathbf{Sch}_{/k}$ commutes with products. Groups in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ may therefore be identified with group schemes over k whose underlying schemes are reduced. In this case, if G is a k-group, G_{red} is a group subscheme of G; in general, this subgroup is not normal in G.

For example, if k is a field of characteristic 3, the constant group $(\mathbb{Z}/2\mathbb{Z})_k$ acts nontrivially on the diagonalizable group $D_k(\mathbb{Z}/3\mathbb{Z})$ (cf. Exposé I, 4.1 and 4.4). If G denotes the semidirect product of $D_k(\mathbb{Z}/3\mathbb{Z})$ by $(\mathbb{Z}/2\mathbb{Z})_k$ defined by this action, then G_{red} is identified with $(\mathbb{Z}/2\mathbb{Z})_k$ and is not normal in G.[4]

Let k be an arbitrary field, let $k^{p^{-\infty}}$ be its perfect closure, and let H be a group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$. Then

$$(H \otimes_k k^{p^{-\infty}})_{\mathrm{red}}$$

is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{\mathrm{red}}$ have the same underlying topological space, groups in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ share with k-groups certain topological properties that are invariant under extension of the base field. For example, it will follow from 0.3 and the preceding remarks that every group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ is separated.

In what follows, we shall encounter groups in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$ whenever we deal with a nonempty locally closed subset U of an A-group G such that

$$U \cdot U = U \qquad \text{and} \qquad U^{-1} = U.$$

Indeed, the reduced subscheme of G defined by U is then a group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.

0.3. *An A-group G is always separated, because the unit section $e : \operatorname{Spec} A \to G$ is a closed immersion.* Indeed, let x be the unique point of $\operatorname{Spec} A$, and let $\eta : G \to \operatorname{Spec} A$ be the structural morphism. Since $\eta \circ e = \mathrm{id}_{\operatorname{Spec} A}$, for every affine open $U = \operatorname{Spec} B$ of G containing $e(x)$, the homomorphism $B \to A$ has a section and is therefore surjective. It follows that e is a closed immersion.[5]

The diagonal in $G \times_A G$ is identified with the functor from $(\mathbf{Sch}_{/A})^{\circ}$ to **Sets** which associates to every A-scheme S the inverse image of the identity element of $G(S)$ under the map

$$\varphi(S) : G(S) \times G(S) \longrightarrow G(S), \qquad (x, y) \longmapsto x \cdot y^{-1}.$$

We therefore have the following cartesian square. Since the diagonal morphism is obtained from a closed immersion by base change, it is itself a closed immersion.

$$\begin{tikzcd}
G \times_S G \arrow[r, "\varphi"] & G \\
G \arrow[u, "\text{diagonal morphism}"] \arrow[r] & \operatorname{Spec} A \arrow[u, "\text{unit section}"']
\end{tikzcd}$$

[2]Editor's note: for examples of group schemes G over a nonperfect field k for which G_{red} is not a group scheme over k, see 1.3.2 below.

[3]Editor's note: the remainder of 0.2, as well as 0.3, has been slightly modified.

[4]Editor's note: for an analogous example with G connected, one may take, when $\mathrm{char}(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $\mathbb{G}_{m,k}$.

[5]Editor's note: this argument is valid for every local ring A of dimension zero, and shows that, if S is a discrete scheme, every S-group is separated; cf. $\mathrm{VI_B}$, 5.2. On the other hand (cf. $\mathrm{VI_B}$, 5.6.4), if S contains a nonisolated closed point s, the S-scheme G obtained by gluing two copies of S along the open subset $S - \{s\}$ is not separated over S, although it carries an S-group structure.

0.4.[6] Let G be an A-scheme. We shall say that a point g of G is *strictly rational over A* if there exists an A-morphism
$$s : \operatorname{Spec} A \longrightarrow G$$
which sends the unique point of $\operatorname{Spec} A$ to g; that is, if the homomorphism $A \to \mathcal{O}_{G,g}$ admits a retraction. One then has $\kappa(g) = k$, so g is a closed point of G. Indeed, if B is the ring of an affine open neighborhood of g, and if $\mathfrak{p}$ is the prime ideal of B corresponding to g, then $B/\mathfrak{p} \subset k$ is a finite A-algebra, hence an integral Artinian ring, and therefore a field.

Suppose henceforth that G is an A-group. Such a morphism $s : \operatorname{Spec} A \to G$ defines an automorphism r_s of the scheme G over A, called *right translation by s*. For every morphism $\pi : S \to \operatorname{Spec} A$, the automorphism $r_s(\pi)$ of $G(S)$ is defined by
$$x \longmapsto x \cdot G(\pi)(s), \qquad x \in G(S).$$
Similarly, ℓ_s denotes *left translation by s*, namely the automorphism of G defined by
$$\ell_s(\pi)(x) = G(\pi)(s) \cdot x, \qquad x \in G(S).$$

The schemes $G \otimes_A k$ and G have the same underlying topological space $\underline{G}$, $G \otimes_A k$ is a k-group, and $s \otimes_A k$ depends only on g, not on s. Consequently, the automorphisms of $\underline{G}$ induced by r_s and ℓ_s (or by $r_{s\otimes k}$ and $\ell_{s\otimes k}$) depend only on g, not on s. Thus, if P is a subset of $\underline{G}$, we may write
$$r_g(P) \text{ or } P \cdot g \qquad \text{(respectively, } \ell_g(P) \text{ or } g \cdot P)$$
instead of $r_s(P)$ (respectively, $\ell_s(P)$), in agreement with 0.1.

Remark 0.4.1. *If g is a strictly rational point of G, and if $A \to A'$ is a morphism of Artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g, and g' is strictly rational over A'. Moreover, if P' denotes the inverse image of P in G', then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).*

Proposition 0.5.[7] *Let G be an A-group, and let U, V be two dense open subsets of G. Then $U \cdot V$, that is, the image of $U \times_A V$ under multiplication, is the entire underlying space of G.*

Proof. Since G and $G \otimes_A k$ have the same underlying space, after replacing A by k and G by $G \otimes_A k$, if necessary, we may assume that $A = k$. Let $g \in G$, and set $K = \kappa(g)$. Then left translation ℓ_g is an automorphism of G_K. Since the projection $G_K \to G$ is open, U_K and V_K are dense open subsets of G_K, as is the image of V_K under ℓ_g. Consequently there exists $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K.

Let L be an extension of K containing $\kappa(v)$, and hence $\kappa(u)$, and let g_L and v_L be the L-points of G_L induced by g and v. Then
$$g_L \cdot v_L = u'$$
is a point of G_L above u, and therefore
$$g_L = u' \cdot v_L^{-1}$$
lies above $U \cdot V$. Thus $g \in U \cdot V$, proving the proposition. □

[6] Editor's note: the original has been expanded in what follows; in particular, Remark 0.4.1 has been added.

[7] Editor's note: the original stated this result under the hypothesis that G be locally of finite type over A. Since the result is useful in the general case, and since the proof is essentially the same, it has been stated and proved here in full generality. It will be used several times below.

Corollary 0.5.1.[8] *If G is an irreducible A-group, then G is quasi-compact.*

Proof. Let U be a nonempty affine open subset of G. Then U is dense in G, so by 0.5 the morphism

$$\mu : U \times_A U \longrightarrow G$$

is surjective. Hence G is quasi-compact, since $U \times_A U$ is. □

Corollary 0.5.2.[9] *Let G be an A-group scheme, and let H be a group subscheme of G over A. Then H is closed.*

Proof. Let k' be the perfect closure of the residue field k of A. Since the underlying topological spaces of G and H are unchanged under the base change $A \to k \to k'$, we may assume that $A = k$ is a perfect field. We may then assume that G and H are reduced, hence geometrically reduced.

Let $\overline{H}$ be the closure of H. Then $\mu^{-1}(\overline{H})$ is a closed subset of $G \times G$ containing $H \times H$. The morphisms

$$H \longrightarrow \operatorname{Spec} k \qquad \text{and} \qquad \overline{H} \longrightarrow \operatorname{Spec} k$$

are universally open. Since H is dense in $\overline{H}$, $H \times H$ is dense in $\overline{H} \times H$, and $\overline{H} \times H$ is dense in $\overline{H} \times \overline{H}$. Thus $H \times H$ is dense in $\overline{H} \times \overline{H}$. Consequently

$$\mu(\overline{H} \times \overline{H}) \subset \overline{H}.$$

Since $\overline{H} \times \overline{H}$ is reduced, μ induces a morphism

$$\mu' : \overline{H} \times \overline{H} \longrightarrow \overline{H}.$$

Now let $g \in \overline{H}$, and put $K = \kappa(g)$. The projection $\overline{H}_K \to \overline{H}$ is open, so H_K and $\ell_g(H_K)$ are two dense open subsets of $\overline{H}_K$. Hence there exist $u, v \in H_K$ such that $\ell_g(v) = u$. As in the proof of 0.5, it follows that g belongs to $H \cdot H = H$. Therefore $\overline{H} = H$. □

1. Local properties of an A-group locally of finite type

We shall first see that, if G is locally of finite type and flat over A, one can "make every closed point of G strictly rational" by means of a finite flat extension of the base.

1.1. Unless expressly stated otherwise, from now on we assume that G is an A-group locally of finite type. When A is a field k, we shall obtain in Exposé VII_B very precise results on the local rings of G.[10] Here we content ourselves with a few elementary results.

Proposition 1.1.1. *Let x be a point of an A-group G locally of finite type and flat over A. Then the local ring $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and there is a system of parameters $a_1, \ldots, a_n$ of $\mathcal{O}_{G,x}$ such that $\mathcal{O}_{G,x}/(a_1, \ldots, a_n)$ is a finite flat A-module (hence finite free).*

We first suppose that A is equal to its residue field k. It then suffices to prove that $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and we may restrict to the case in which x is a closed point (cf. EGA 0_{IV}, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $\mathcal{O}_{G,y}$ is Cohen–Macaulay. By SGA 1, I, § 9, this is equivalent to saying that, for every finite extension K of the base field k and every point $\overline{y}$ of $\overline{G} = G \otimes_k K$ lying above y, the local

[8]Editor's note: this corollary has been inserted here; cf. 2.4 and 2.6.2 below.

[9]Editor's note: this corollary, used implicitly in VIII, 6.7, has been added; see also VI_B, 6.2.5.

[10]Editor's note: for example, they are always complete intersections; see Exposé VII_B, 5.5.1. Moreover, if $\operatorname{char}(k) = 0$, then G is smooth (Exposé VI_B, 1.6.1; see also Exposé VII_B, 3.3.1).

ring $\mathcal{O}_{\overline{G},\overline{y}}$ is Cohen–Macaulay. If K has been chosen sufficiently large, that is, if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$, then $\overline{y}$ is (strictly) rational over K, as is every point $\overline{x}$ of $\overline{G}$ lying above x.[11] Since the automorphism

$$r_{\overline{x}} \circ (r_{\overline{y}})^{-1}$$

sends $\overline{y}$ to $\overline{x}$, it follows that $\mathcal{O}_{\overline{G},\overline{x}}$, and hence $\mathcal{O}_{G,x}$ (SGA 1, I, § 9), are Cohen–Macaulay.

Now let A again be arbitrary. The preceding argument applies to $k \otimes_A G$, so that $k \otimes_A \mathcal{O}_{G,x}$ is Cohen–Macaulay. If $a_1, \dots, a_n$ is a sequence of elements of $\mathcal{O}_{G,x}$ whose image in $k \otimes_A \mathcal{O}_{G,x}$ is a system of parameters, it follows from SGA 1, IV, 5.7, or EGA 0_{IV}, 15.1.16, that $a_1, \dots, a_n$ is an $\mathcal{O}_{G,x}$-regular sequence and that $\mathcal{O}_{G,x}/(a_1, \dots, a_n)$ is finite and flat (hence finite and free) over A.

Lemma 1.1.2. *Every nonempty scheme X, locally of finite type over an Artinian ring A, contains a closed point x whose local ring is Cohen–Macaulay.*

We may plainly suppose that X is affine, with algebra B, and argue by induction on $\dim X$ (the assertion is clear if X is discrete, since all its local rings are then Artinian). Since B is of finite type over A, if $\dim B > 0$, then B contains an element a that is not invertible and is not a zero divisor.[12] The closed subscheme

$$X' = \operatorname{Spec} B/(a)$$

of X then has dimension strictly less than $\dim X$, and by the induction hypothesis it contains a closed point x such that $\mathcal{O}_{X',x}$ is Cohen–Macaulay. Since $\mathcal{O}_{X',x} = \mathcal{O}_{X,x}/(a)$, and a is neither invertible nor a zero divisor in $\mathcal{O}_{X,x}$, the ring $\mathcal{O}_{X,x}$ is Cohen–Macaulay (see also EGA IV_2, 6.11.3).

Proposition 1.2. *Let A be an Artinian local ring, let G be an A-group locally of finite type and flat over A, and let x be a closed point of G. Then there exists a local A-algebra A', finite and free over A, such that every point of $G \otimes_A A'$ lying above x is strictly rational over A'.*

[13] Indeed, let k_1 be a finite normal extension of k containing the residue field $\kappa(x)$ of x. By Lemma V.4.1.2, there exists a local A-algebra A_1, finite and free over A, with residue field k_1. In this case (cf. Editor's note 11), all the points $g_1, \dots, g_n$ of $G \otimes_A A_1$ lying above $x \in G$ have k_1 as residue field; that is, $g_1, \dots, g_n$ are rational over A_1 in the sense of Exposé V, § 4.e).

Let $B_1, \dots, B_n$ be the local rings of $g_1, \dots, g_n$. By 1.1.1, the rings $B_1, \dots, B_n$ have quotients $B'_1, \dots, B'_n$ that are Artinian and finite and free over A_1. Set

$$A' = B'_1 \otimes_{A_1} \cdots \otimes_{A_1} B'_n.$$

Then A' is local, finite, and free over A_1, and, for each $i = 1, \dots, n$, there are surjective homomorphisms

$$B_i \otimes_{A_1} A' \twoheadrightarrow B'_i \otimes_{A_1} A' \twoheadrightarrow A',$$

[11] Editor's note: indeed, the hypothesis on K implies that, for every extension L of K, every k-morphism $\kappa(x) \to L$ (respectively, $\kappa(y) \to L$) factors through K. Consequently, all the points of $G \otimes_k K$ lying above x or y have K as residue field; that is, they are (strictly) rational over K.

[12] Editor's note: indeed, B is a Noetherian Jacobson ring (cf. [BAC], V, § 3.4). If every noninvertible element is a zero divisor, then every prime ideal is an associated prime ideal of B; in particular, B has only finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B. It follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B, and hence that $\dim B = 0$.

[13] Editor's note: the original has been expanded in what follows.

the second induced by the multiplication map

$$B'_i \otimes_{A_1} B'_i \twoheadrightarrow B'_i.$$

Consequently, A' has the required property.

1.3. Let e be the unit element (or origin) of G, that is, the image of the unique point of $\operatorname{Spec} A$ under the unit section $\operatorname{Spec} A \to G$. By definition, e is strictly rational over A.

Proposition 1.3.1.[14] *Let G be a group locally of finite type and flat over an Artinian ring A, and let K (respectively, $\overline{k}$) be the perfect closure (respectively, an algebraic closure) of the residue field k of A.*

(1) *For every closed point x of $\overline{G} = G \otimes_k \overline{k}$, the local rings $\mathcal{O}_{\overline{G},e}$ and $\mathcal{O}_{\overline{G},x}$ are isomorphic. In particular, the tangent spaces $T_e\overline{G}$ and $T_x\overline{G}$ have the same dimension.*

(2) *The following assertions are equivalent:*

   (i) *$G \otimes_A K$ is reduced.*

   (i bis) *$\mathcal{O}_{G,e} \otimes_A K$ is reduced.*

   (ii) *G is smooth over A.*

   (ii bis) *G is smooth over A at the origin.*

Proof. (1) Let x be a closed point of $\overline{G}$. There is exactly one $\overline{k}$-morphism

$$s : \operatorname{Spec} \overline{k} \longrightarrow \overline{G}$$

whose image is x. Right translation r_s then induces an isomorphism from

$$\mathcal{O}_{\overline{G},e} = \mathcal{O}_{G,e} \otimes_k \overline{k}$$

onto $\mathcal{O}_{\overline{G},x}$, which proves (1).

We now prove (2). By SGA 1, II.2.1, we reduce at once to the case in which A is a field ($A = k$). The implications

$$\text{(i)} \Rightarrow \text{(i bis)}, \qquad \text{(ii)} \Rightarrow \text{(ii bis)}, \qquad \text{(ii)} \Rightarrow \text{(i)}, \qquad \text{(ii bis)} \Rightarrow \text{(i bis)}$$

are evident, so it remains only to prove that (i bis) implies (ii).

Now (i bis) implies that $\mathcal{O}_{G,e} \otimes_k \overline{k}$ is reduced. Hence, by (1), $\mathcal{O}_{\overline{G},x}$ is reduced for every closed point x of $\overline{G}$, and therefore $\overline{G}$ is reduced. Since $\overline{G}$ is locally of finite type over $\overline{k}$, there is at least one closed point y such that $\mathcal{O}_{\overline{G},y}$ is regular. By (1), the local rings at all closed points of $\overline{G}$ are isomorphic to $\mathcal{O}_{\overline{G},e}$; thus all these local rings are regular. Consequently, $\overline{G}$ is smooth over $\overline{k}$, and hence G is smooth over k.

[15] We may now give the examples below, pointed out by M. Raynaud, of group schemes G over a nonperfect field k for which G_{red} is not a k-group scheme.

Examples 1.3.2. Let k be a nonperfect field of characteristic $p > 0$, let $t \in k - k^p$, let $\overline{k}$ be an algebraic closure of k, and let $\alpha \in \overline{k}$ satisfy $\alpha^p = t$.

[14] Editor's note: part (1), which is useful in Examples 1.3.2 below, has been added.

[15] Editor's note: the following sentence and Examples 1.3.2 have been added.

(1) Consider the additive group $\mathbb{G}_{a,k} = \operatorname{Spec} k[X]$, and let G be the group subscheme, finite over k, defined by the additive polynomial $X^{p^2} - tX^p$. Then

$$G_{\text{red}} = \operatorname{Spec} k[X]/X\bigl(X^{p(p-1)} - t\bigr)$$

is étale at the origin. If it were a group scheme over k, it would be smooth (by 1.3.1); but G_{red} is not geometrically reduced, so it is not a group scheme over k.

(2) Consider

$$\mathbb{G}^4_{a,k} = \operatorname{Spec} k[X, Y, U, V],$$

and let G be the group subscheme defined by the ideal I generated by the additive polynomials

$$P = X^p - tY^p \qquad \text{and} \qquad Q = U^p - tV^p.$$

Then G is two-dimensional and irreducible, since

$$(G_{\overline{k}})_{\text{red}} \simeq \operatorname{Spec} \overline{k}[Y, V]$$

is irreducible.

Let $A = k[X, Y, U, V]$, and let $\mathfrak{m}$ be its augmentation ideal. Denote by x, y, u, v the images of dX, dY, dU, dV in

$$\Omega^1_{A/k} \otimes_A (A/\mathfrak{m}),$$

viewed as linear forms on the tangent space

$$k^4 = T_0\mathbb{G}^4_{a,k}.$$

We shall show that the subspace $E = T_0G_{\text{red}}$ equals k^4. Otherwise, there would be a linear form

$$f = ax + by + a'u + b'v, \qquad a, b, a', b' \in k,$$

not all of whose coefficients are zero, that vanishes on E. Recall that the formation of $\Omega^1_{A/k}$, and hence that of tangent spaces, commutes with base change (cf. EGA IV_4, 16.4.5), and identify f with its image in $(\overline{k}^4)^*$. Since

$$(G_{\overline{k}})_{\text{red}} \subset (G_{\text{red}})_{\overline{k}},$$

the form f vanishes on the subspace $T_0(G_{\overline{k}})_{\text{red}}$ of $\overline{k}^4$, which is defined by the equations

$$g_1 = x - \alpha y \qquad \text{and} \qquad g_2 = u - \alpha v.$$

Thus

$$f = \lambda g_1 + \mu g_2, \qquad \lambda, \mu \in \overline{k}.$$

But $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$. This contradiction shows that $E = k^4$, and hence

$$T_0(G_{\text{red}})_{\overline{k}} = \overline{k}^4.$$

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, since

$$R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p).$$

Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{\text{red}})_{\overline{k}}$ is contained in the hyperplane H of $\overline{k}^4$ with equation

$$\alpha\, dV + dX - dU - \alpha\, dY = 0,$$

and therefore has dimension at most 3.[16] It follows from part (1) of 1.3.1 that G_{red} is not a group scheme over k.

[16] Editor's note: in fact, it is easy to see that $\sqrt{I}$ is generated by P, Q, R at every point other than 0, so $F = H$.

2. Connected components of an A-group locally of finite type

2.1. Let us first consider an arbitrary A-group G, and let G' be the connected component of the origin e in G. This connected component is plainly closed, so we may identify it with the reduced closed subscheme of G having G' as its underlying space.[17]

Proposition 2.1.1. *For every extension K of the residue field k of A, the underlying space of $G' \otimes_A K$ is the connected component of the origin in the K-group $G \otimes_A K$; in other words, G' is geometrically connected.*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the unit element of G, that image is contained in G'. Thus $(G \otimes_A K)'$ is contained in the inverse image $G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from the connectedness of $G' \otimes_A K$, which is proved in the following lemma.

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k. If X contains a rational point, then $X \times_k Y$ is connected.*

We give a direct proof below of this result from EGA IV$_2$ (4.5.8 and 4.5.14).

Suppose first that Y is nonempty, connected, and affine, with algebra B. Then $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$-algebra

$$\mathcal{B} = \mathcal{O}_X \otimes_k B.$$

We wish to show that every nonempty subset U of $X \times_k Y$ that is both open and closed equals $X \times_k Y$. Now U is affine over X, and its affine $\mathcal{O}_X$-algebra is a direct factor of $\mathcal{B}$. It follows from Lemma 2.1.3 below that the image of U in X is open and closed, and hence is all of X. In particular, this image contains a rational point x of X, so U meets the inverse image of x in $X \times_k Y$. Since this inverse image is isomorphic to Y, and hence connected, U contains the whole inverse image. The same conclusion would hold for the complement of U in $X \times_k Y$ if U were different from $X \times_k Y$, which is absurd.

If Y is now an arbitrary k-scheme, the preceding argument shows that the fibers of the canonical projection

$$X \times_k Y \longrightarrow Y$$

are connected. If x is a rational point of X, all these fibers meet the subscheme $\{x\} \times_k Y$, which is itself connected. This proves the assertion.

Lemma 2.1.3. *Let X be a scheme, and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_X$-algebra that is locally*[18] *a direct factor of a free $\mathcal{O}_X$-module. Then the image of* Spec $\mathcal{A}$ *in X is open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, then $\mathcal{A}_x$ is nonzero and is a free $\mathcal{O}_{X,x}$-module, since, by Kaplansky, every projective module over a local ring is free. Consequently, the fiber of Spec $\mathcal{A}$ at x, which is affine with algebra

$$\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

is nonempty. Thus the image of Spec $\mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s, and hence $\mathcal{A}$, vanish in a neighborhood of x. Therefore the support of $\mathcal{A}$ is closed. On the other hand, we may

[17] Editor's note: we shall see below (2.2) that G' is a group in the category $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.

[18] Editor's note: the word "locally" has been added. The reference to Kaplansky is *Projective modules*, Ann. of Math. **68** (1958), 372–377; see also [BAC], § II.3, Ex. 3.

assume that $\mathcal{A}$ is a direct factor of the direct sum $\mathcal{L}$ of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.[19] For each β, let φ^β denote the restriction to $\mathcal{A}$ of the canonical projection

$$\mathcal{L} = \bigoplus_\alpha \mathcal{O}_X^\alpha \longrightarrow \mathcal{O}_X^\beta.$$

For $x \in X$, let $\mathcal{A}_x^*$ denote the dual module

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}).$$

Since $\mathcal{A}_x$ is a direct factor of $\mathcal{L}_x$, the restriction map

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathcal{O}_{X,x}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$, then, since $\mathcal{A}_x$ is free and nonzero, there exist $a \in \mathcal{A}_x$ and a linear form $\varphi \in \mathcal{A}_x^*$ such that $\varphi(a) = 1$. There is therefore a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$,

$$\varphi(u) = \sum_\alpha \xi^\alpha \varphi^\alpha(u);$$

this sum is finite because $\varphi^\alpha(u) = 0$ except for finitely many α. Applying this to $u = a$, we find $\alpha_1, \dots, \alpha_n$ such that

$$1 = \varphi(a) = \xi^{\alpha_1}\varphi^{\alpha_1}(a) + \cdots + \xi^{\alpha_n}\varphi^{\alpha_n}(a).$$

There is then an open neighborhood U of x on which a and the ξ^{α_i} come from sections

$$\widetilde{a} \in \mathcal{A}(U) \qquad \text{and} \qquad \widetilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U).$$

The equality

$$\sum_i \widetilde{\xi}^{\alpha_i}\varphi^{\alpha_i}(\widetilde{a}) = 1 \quad \text{on } U$$

shows that $\widetilde{a}_y \neq 0$ for every $y \in U$. ("The support of a projective module is open.")

2.2. The notation remains that of 2.1. It is clear that G' is a reduced k-scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so $(G' \times_k G')_{\mathrm{red}}$ is the reduced subscheme of $G \times_A G$ whose underlying space is the connected component of the origin. In particular, the multiplication morphism

$$\mu : G \times_A G \longrightarrow G$$

induces a morphism

$$\mu' : (G' \times_k G')_{\mathrm{red}} \longrightarrow G'$$

that makes G' a group in $(\mathbf{Sch}_{/k})_{\mathrm{red}}$.

2.2.bis.[20] Recall (cf. Exposé V) that, if P is a scheme, $\underline{P}$ denotes the underlying topological space of P. We define an A-subfunctor G^0 of G by setting, for every A-scheme S,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G'}\}.$$

Let $c : G \to G$ be the inversion morphism. Since $c(\underline{G'}) = \underline{G'}$, we have $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, then $u \boxtimes v$ sends $\underline{S}$ into the subspace of $\underline{G \times_A G}$ consisting of the points whose two projections belong to $\underline{G'}$. This subspace is identified with the underlying space of $G' \times_A G'$, which is connected by Lemma 2.1.2.

[19] Editor's note: the original has been expanded in what follows.

[20] Editor's note: this paragraph has been added in order to make the connection with Exposé VI$_\mathrm{B}$, § 3.

Consequently, $\mu \circ (u \boxtimes v)$ sends $\underline{S}$ into $\underline{G'}$. This proves that G^0 is a group subfunctor of G over A.

If the connected component of e is open in G, then the subfunctor G^0 is represented by the subscheme induced by G on this open subset, which is therefore a group subscheme of G; this subscheme will also be denoted by G^0. In this case, with the notation of 2.1,

$$G' = (G^0)_{\text{red}},$$

and the topological spaces $\underline{G'}$ and $\underline{G^0}$ coincide.[21]

2.3. In accordance with our convention in 1.1, we again assume from now on that G is locally of finite type over A. Then G is locally Noetherian, hence locally connected;[22] thus every connected component of G is open.

We shall write G^0 for the subscheme induced by G on the connected component G' of the unit element. By 2.2.bis, G^0 is a group subscheme of G, which we shall call the *neutral component* of G. For every A-scheme S, therefore,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G^0} = \underline{G'}\}.$$

Let G^α be an arbitrary connected component of G, and let

$$\nu^\alpha : G^\alpha \times_A G^0 \longrightarrow G$$

be the morphism defined by

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1}$$

for every $S \in (\mathbf{Sch}_{/A})$, $g \in G^\alpha(S)$, and $\gamma \in G^0(S)$. If e is the origin of G, the restriction of ν^α to $G^\alpha \times_A \{e\}$ is the trivial morphism. Since $G^\alpha \times_A G^0$ is connected by 2.1.2, it follows that ν^α factors through G^0. Thus, for every A-scheme S, $G^0(S)$ is a normal subgroup of $G(S)$. We have proved the following proposition.[23]

Proposition 2.3.1. *Let G be an A-group locally of finite type. Then the neutral component G^0 is an open normal group subscheme of G.*

Proposition 2.4. *Let G be an A-group locally of finite type.*

(i) *G^0 is irreducible, and $G^0 \otimes_A k$ is geometrically irreducible over k.*

(ii) *G^0 is quasi-compact, and hence is of finite type over A.*

[24] *Proof.* (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to prove the second assertion. Let $\overline{k}$ be an algebraic closure of k. By 2.2, $(G \otimes_A \overline{k})_{\text{red}}$ is a reduced $\overline{k}$-group locally of finite type, and hence is smooth over $\overline{k}$ by 1.3.1. A fortiori, the local rings of $(G \otimes_A \overline{k})_{\text{red}}$ are integral domains. Therefore,[25] since $G \otimes_A \overline{k}$ is locally Noetherian, the connected components of $G \otimes_A \overline{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \overline{k}$ (cf. 2.1.1) is irreducible.

[21] Editor's note: in this exposé, the notation G^0 is reserved for the case in which the connected component of e is open; in Exposé VI$_B$, § 3, this connected component will be denoted by $\underline{G^0}$ in every case. This is a slight abuse of notation, but it is compatible with the preceding discussion when the connected component is the underlying topological space of an open group subscheme G^0 of G.

[22] Editor's note: indeed, in a Noetherian space there are only finitely many connected components, and each is therefore open; see also EGA I, 6.1.9.

[23] Editor's note: the number 2.3.1 has been added in order to emphasize this statement. Moreover, G^0 is even a characteristic subgroup of G; see 2.6.5(ii).

[24] Editor's note: in the statement, "G^0 is geometrically irreducible" has been replaced by "$G^0 \otimes_A k$ is geometrically irreducible over k," and the proof has been expanded.

[25] Editor's note: the original has been expanded here, and the reference to EGA I, 6.1.10, has been added.

(ii) We now show that G^0 is of finite type over A. Since G^0 is locally of finite type over A, it suffices to prove that G^0 is quasi-compact. Since G^0 is irreducible, this follows from 0.5.1.

Corollary 2.4.1. *Every connected component of G is irreducible,*[26] *of finite type over A,*[27] *and of the same dimension as G^0.*

Indeed, we may assume that A equals its residue field k. Let C be a connected component of G, let x be a closed point of C, let $\kappa(x)$ be the residue field of x, and let k' be a finite normal extension of k containing $\kappa(x)$. The canonical projection

$$\pi : C \otimes_k k' \longrightarrow C$$

is open and closed.[28] Consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \to C$ is surjective. Thus C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2); it follows that C' is the image of $G^0 \otimes_k k'$ under right translation r_y. Now $G^0 \otimes_k k'$ is of finite type over k', by 2.4, and $\pi^{-1}(x)$ is finite, of cardinality at most $[k' : k]$. Consequently, $C \otimes_k k'$ is of finite type over k', and hence C is of finite type over k.

Moreover, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same is true of C',[29] and hence also of C, because the projection $C' \to C$ is surjective.

Finally, we saw above that $C \otimes_k k'$ is the disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV$_2$, 4.1.4), it follows that C has the same dimension as G^0. (Moreover, by EGA IV$_2$, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5. This paragraph has been added. The results that follow appear in Exposé VI$_B$, but could (or should) have appeared in VI$_A$; it is useful to have them available already, in order to make Theorem 3.2 below more precise.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an Artinian local ring, and let $k = A/\mathfrak{m}$ be its residue field.*

(i) *If X is an A-scheme such that $X \otimes_A k$ is locally of finite type (respectively, of finite type) over k, then X has the same property over A.*

(ii) *Let $u : X \to Y$ be a morphism of A-schemes. If $u \otimes_A k$ is an immersion (respectively, a closed immersion), then so is u.*

Proof. (i) Suppose that $X \otimes_A k$ is locally of finite type over k. Let $U = \operatorname{Spec} B$ be an affine open subset of X. By hypothesis, there are elements $x_1, \dots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k-algebra, and the "nilpotent Nakayama lemma" then shows that the x_i generate B as an A-algebra. This proves that X is locally of finite type over A. If, in addition, $X \otimes_A k$ is quasi-compact, then so is X, which has the same underlying topological space; hence X is of finite type over A. This proves (i).

We prove (ii). Suppose that $u \otimes_A k$ is an immersion (respectively, a closed immersion). Then u is a homeomorphism of X onto a locally closed (respectively, closed) subset of Y, and, for every $x \in X$, the ring homomorphism

$$\varphi_x : \mathcal{O}_{Y,u(x)} \longrightarrow \mathcal{O}_{X,x}$$

[26] Editor's note: one should bear in mind that a nonneutral connected component is not geometrically connected in general. For example, if $k = \mathbb{R}$, the group $\mu_{3,\mathbb{R}}$, represented by $\mathbb{R}[X]/(X^3 - 1)$, has two connected components: $\{e\} = \operatorname{Spec} \mathbb{R}$ and $C = \operatorname{Spec} \mathbb{R}[X]/(X^2 + X + 1)$; and $C \otimes_{\mathbb{R}} \mathbb{C}$ has two components.

[27] Editor's note: the assertion that follows has been added; cf. Exposé VI$_B$, 1.5.

[28] Editor's note: the original has been expanded in what follows.

[29] Editor's note: the original has been simplified here.

is such that $\varphi_x \otimes_A k$ is surjective. The nilpotent Nakayama lemma therefore shows that φ_x is surjective, so u is an immersion (respectively, a closed immersion).

Proposition 2.5.2. *Let A be an Artinian local ring with residue field k, and let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type.*

(a) *The set $u(G)$ is closed in H, and its connected components are irreducible and all have the same dimension.*

(b) *One has*
$$\dim G = \dim u(G) + \dim \mathrm{Ker}(u).$$

(c) *If u is a monomorphism, then it is a closed immersion.*

Proof. By the preceding lemma, it suffices to prove the proposition when $A = k$. Moreover, the properties under consideration are stable under fpqc descent, and dimension is invariant under extension of the base field, so we may assume that k is algebraically closed.

We prove (a). Let C be the reduced subscheme of H whose underlying topological space is $u(G)$. Since $u(G)$ is stable under the inversion morphism of H, so is C. Furthermore, $u : G \to C$ is quasi-compact and dominant. Thus, by EGA IV_2, 2.3.7, the same is true of
$$u \times_k \mathrm{id}_G \qquad \text{and} \qquad \mathrm{id}_H \times_k u,$$
and hence of their composite
$$u \times_k u : G \times_k G \longrightarrow C \times_k C.$$
Consequently, multiplication on H sends $C \times_k C$ into C, and therefore C is a group subscheme of H.

Replacing H by C, we are thus reduced to the case in which u is dominant. Then $G(k)$ is dense in H, and hence meets every connected component of H; it therefore acts transitively on the set of those connected components. It is consequently enough to show that $u(G)$ contains H^0. Replacing G by $u^{-1}(H^0)$, we may therefore suppose that $H = H^0$. In that case, 2.4 shows that H is irreducible and of finite type over k, and hence Noetherian. Moreover, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, and is therefore of finite type. By Chevalley's constructibility theorem (cf. EGA IV_1, 1.8.5), $u(G)$ is consequently a constructible (and dense) subset of
$$H = \overline{u(G)},$$
and thus contains a dense open subset U of H (cf. EGA 0_{III}, 9.2.2). By 0.5, we then have
$$H = U \cdot U \subset u(G),$$
whence $u(G) = H$. Together with 2.4.1, this proves (a).

We prove (b). Recall first that the functor $\mathrm{Ker}(u)$ (cf. I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H. Since u is of finite type, $\mathrm{Ker}(u)$ is of finite type over k. On the other hand, replacing H by the reduced closed subscheme $u(G)$, we may assume that u is surjective. Let u^0 denote the restriction of u to G^0. Since G and $\mathrm{Ker}(u)$ are equidimensional, and since
$$\mathrm{Ker}(u)^0 \subset \mathrm{Ker}(u^0),$$
we are reduced to the case in which G, and hence also H, is irreducible.

By EGA IV_3, 9.2.6.2 and 10.6.1(ii), the set of $y \in H$ such that
$$\dim u^{-1}(y) = \dim G - \dim H$$

contains a nonempty open subset V. Since u is surjective, $U = u^{-1}(V)$ is then a nonempty open subset of G, and therefore contains a closed point x of G, because G is a Jacobson scheme (cf. EGA IV$_3$, 10.4.8). Right translation r_x is an isomorphism from $\mathrm{Ker}(u)$ onto $u^{-1}(u(x))$, whence

$$\dim \mathrm{Ker}(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

We prove (c), following [DG70], I, § 3.4. (Another proof is given in Exposé VI$_{\mathrm{B}}$, 1.4.2.) Suppose that u is a monomorphism. If C is a connected component of G, there is a closed point $x \in G$ such that $C = r_x(G^0)$. If u_C (respectively, u^0) denotes the restriction of u to C (respectively, to G^0), then

$$u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}.$$

It therefore suffices to show that u^0 is a closed immersion. We may thus suppose that $G = G^0$, so that G is irreducible and of finite type over k.

Let ξ be the generic point of G. Then $\mathcal{O}_{G,\xi}$ is an Artinian local ring; denote its maximal ideal by $\mathfrak{m}$. Let $h = u(\xi)$, let $\mathfrak{n}$ be the maximal ideal of $\mathcal{O}_{H,h}$, and set

$$A = \mathcal{O}_{G,\xi}/\mathfrak{n}\mathcal{O}_{G,\xi}.$$

Since u is a monomorphism, so is the morphism

$$u_h : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(\kappa(h))$$

obtained by base change. The multiplication morphism

$$A \otimes_{\kappa(h)} A \longrightarrow A$$

is therefore an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. Nakayama's lemma then shows that

$$\mathcal{O}_{H,h} \longrightarrow \mathcal{O}_{G,\xi}$$

is surjective, since $\mathfrak{n}\mathcal{O}_{G,\xi}$ is contained in $\mathfrak{m}$ and is therefore nilpotent.

Let V be an affine open subset of H containing h, let $U \neq \varnothing$ be an affine open subset of G contained in $u^{-1}(V)$, and let

$$\varphi : \mathcal{O}_H(V) \longrightarrow \mathcal{O}_G(U)$$

be the homomorphism of k-algebras induced by u. Let $\mathfrak{p}$ be the prime ideal of $\mathcal{O}_G(U)$ corresponding to ξ, and set $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Since G is of finite type over k, the k-algebra $\mathcal{O}_G(U)$ is generated by finitely many elements $a_1, \ldots, a_n$. By the preceding argument, there are elements $b_1, \ldots, b_n, s \in \mathcal{O}_H(V)$, with $s \notin \mathfrak{q}$, such that in $\mathcal{O}_G(U)_{\mathfrak{p}}$ one has

$$a_i/1 = \varphi(b_i)/\varphi(s).$$

There are therefore elements

$$t_1, \ldots, t_n \in \mathcal{O}_G(U) - \mathfrak{p}$$

such that

$$t_i(a_i\varphi(s) - \varphi(b_i)) = 0.$$

Set

$$t = t_1 \cdots t_n \varphi(s) \in \mathcal{O}_G(U) - \mathfrak{p}.$$

The equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ then already hold in $\mathcal{O}_G(U)_t$. Since

$$t \in \mathcal{O}_G(U) = k[a_1, \ldots, a_n],$$

there is an element $b \in \mathcal{O}_H(V)$ such that, for some $r \in \mathbb{N}$,

$$t/1 = \varphi(b)/\varphi(s)^r.$$

Consequently, φ induces a surjection

$$\mathcal{O}_H(V)_{sb} \longrightarrow \mathcal{O}_G(U)_t,$$

and hence u is a local immersion at ξ.

The open subset W of G consisting of the points at which u is a local immersion is therefore nonempty. Since G is a Jacobson scheme, W contains a closed point y. To prove that $W = G$, it is enough to show that every closed point of G belongs to W. But every closed point x is the image of y under the translation $r_x \circ r_y^{-1}$, and hence belongs to W. Thus $W = G$, which proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H, respectively, such that $x \in U_x$ and u induces a closed immersion of U_x into V_x. Since U_x is dense in G, the subset $u(U_x)$ is dense in $u(G) \cap V_x$. But $u(U_x)$ is closed in V_x, so

$$u(U_x) = V_x.$$

Moreover, since u is injective, $U_x = u^{-1}(V_x)$. Thus u induces a closed immersion of G into the open subscheme of H covered by the V_x. Hence $u : G \to H$ is an immersion. We have already seen that $u(G)$ is closed in H, so u is a closed immersion.

Lemma 2.5.3.[30] *Let A be an Artinian local ring, let k be its residue field, let G be a flat A-group, and let X be an A-scheme equipped with a left action*

$$\mu : G \times_A X \longrightarrow X$$

of G and with a section $s_0 : \operatorname{Spec} A \to X$. (This is the case, for example, if G' is a second A-group and one is given a morphism of A-groups $G \to G'$.)

Let φ be the morphism

$$\mu \circ (\mathrm{id}_G \times s_0) : G = G \times_A A \longrightarrow X.$$

If φ is flat at a point g of G, then φ is flat.

Proof. Since G is flat over A, the fiberwise criterion for flatness (EGA IV_3, 11.3.10.2) shows that it is enough to prove that $\varphi \otimes_A k$ is flat. We may therefore suppose that $A = k$. In that case, giving s_0 is equivalent to giving a k-point $x_0 \in X(k)$, and φ is the morphism $h \mapsto hx_0$.

Let $h \in G$; we show that φ is flat at h. Let K be an extension of k containing copies of both $\kappa(g)$ and $\kappa(h)$. We have a cartesian square

$$\begin{tikzcd} G_K \arrow[r, "\varphi_K"] \arrow[d] & X_K \arrow[d] \\ G \arrow[r, "\varphi"'] & X \end{tikzcd}$$

in which the two vertical arrows are faithfully flat. Thus, by Exposé V, 7.4(i), φ_K is flat at every point $g' \in G_K$ lying over g; and, in order to show that φ is flat at h, it is enough to show that φ_K is flat at one point h' lying over h. We are therefore reduced to the case in which g and h are rational.

Set $u = hg^{-1}$, and let ℓ_u (respectively, μ_u) be left translation on G (respectively, on X) defined by u. Since $\varphi \circ \ell_u = \mu_u \circ \varphi$, we obtain a commutative square

[30] Editor's note: no finiteness hypotheses are made in this statement; this will be useful later (cf. 6.2 and Exposé VI_B, § 12).

$$\begin{tikzcd}
\mathcal{O}_{G,h} \arrow[r, "\sim"] & \mathcal{O}_{G,g} \\
\mathcal{O}_{X,hx_0} \arrow[u] \arrow[r, "\sim"'] & \mathcal{O}_{X,gx_0} \arrow[u]
\end{tikzcd}$$

whose horizontal arrows are isomorphisms. Since the morphism

$$\mathcal{O}_{X,gx_0} \longrightarrow \mathcal{O}_{G,g}$$

is flat, the morphism

$$\mathcal{O}_{X,hx_0} \longrightarrow \mathcal{O}_{G,h}$$

is flat as well.

Proposition 2.5.4. *Let A be an Artinian local ring, let k be its residue field, let G be an A-group locally of finite type, and let $X \neq \varnothing$ be an A-scheme locally of finite type equipped with a left action of G. Suppose that the morphism*

$$\varphi : G \times_A X \longrightarrow X \times_A X, \qquad (g,x) \longmapsto (gx, x),$$

defined on underlying sets by the displayed rule, is surjective. Then:

(i) *The connected components of X are of finite type, irreducible, and all of the same dimension.*

(ii) *More precisely, let $\overline{k}$ be an algebraic closure of k, and let x be a closed point of $X \otimes_A \overline{k}$. Its stabilizer F is a closed group subscheme of $G \otimes_A \overline{k}$, and the dimension of the irreducible components of X is*

$$\dim G - \dim F.$$

In view of Lemma 2.5.1, we may suppose that $A = k$. Suppose first that k is algebraically closed. Then G_{red} is a k-group locally of finite type; hence, replacing G by G_{red} and X by X_{red}, we may suppose that G and X are reduced.

Since $G \times_k X$ is locally of finite type over k, the morphism φ is locally of finite type (cf. EGA I, 6.6.6(v)), and is therefore locally of finite presentation because $X \times_k X$ is locally Noetherian. Let x be a rational point of X. The morphism

$$\varphi_x : G \longrightarrow X$$

obtained from φ by base change is surjective and locally of finite presentation. If η is a maximal point of X, then $\mathcal{O}_{X,\eta}$ is a field, since X is reduced; consequently, φ_x is flat at every point of G lying over η. By Lemma 2.5.3, φ_x is therefore flat. Thus $\varphi_x : G \to X$ is faithfully flat and locally of finite presentation, and hence open (cf. EGA IV_2, 2.4.6).

Since G^0 is open in G, irreducible, and quasi-compact (by 2.4), every orbit

$$G^0 x = \varphi_x(G^0),$$

as x ranges over the rational points of X, is an open, irreducible, and quasi-compact subset of X, and is therefore of finite type over k, since X is locally of finite type over k.

Every nonempty open subset of X contains a rational point, so these open subsets cover X. Furthermore, any two of them are either disjoint or equal. Indeed, if

$$\varphi_x(G^0) \cap \varphi_y(G^0)$$

is nonempty, it contains a rational point z. There are then rational points $g, h \in G^0$ such that

$$gx = z = hy,$$

whence $x = g^{-1}z$ and $y = h^{-1}z$, and consequently

$$\varphi_x(G^0) = \varphi_z(G^0) = \varphi_y(G^0).$$

It follows that the orbits $\varphi_x(G^0)$ are also closed, and are therefore at once the connected components and the irreducible components of X.

Finally, let x, y be two rational points of X. Since φ_x is surjective, there is a rational point $g \in G$ such that $y = gx$. Since G^0 is a normal subgroup of G, the orbit G^0y is the image of G^0x under translation ℓ_g on X; hence G^0y and G^0x have the same dimension.

Moreover, by Exposé I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the following cartesian square:

$$\begin{tikzcd} F \arrow[r] \arrow[d] & G \arrow[d, "\varphi_x"] \\ \operatorname{Spec} k \arrow[r] & X \end{tikzcd}$$

Thus F is a k-group locally of finite type, while $F \cap G^0$ is a k-group of finite type containing F^0. By 2.4.1, the schemes F and $F \cap G^0$ are equidimensional, of the same dimension as F^0. Let

$$C = \varphi_x(G^0)$$

be the irreducible component of X containing x. Arguing as in the proof of part (b) of 2.5.2, we obtain

$$\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F.$$

In the general case, that is, when k is an arbitrary field, let $\overline{k}$ be an algebraic closure of k. Let C be a connected component of X, and let C' be a connected component of $C \otimes_k \overline{k}$. Then C' is a connected component of

$$X' = X \otimes_k \overline{k}.$$

The morphism $\pi : X' \to X$ is open (cf. EGA IV$_2$, 2.4.10), and, since it is integral, it is also closed; therefore $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, and hence is of finite type over k, because X is locally of finite type over k.

Finally, dimension is invariant under extension of the base field (cf. EGA IV$_2$, 4.1.4), so $\dim C = \dim C'$. Since all the irreducible components of X' have the same dimension, the same is true of those of X.

2.6. Complements.— This paragraph has been added; it is taken from [Per75], II, §§ 1–2, with complements due to O. Gabber.[31] It shows that the preceding results are valid for every group scheme G over a field k. (This will be used in Sections 5, 6, and 7 of Exposé VI$_B$.)

Fix a field k. We begin with the following lemma (loc. cit., II, 2.1.1), which does not appear explicitly in EGA IV$_2$, § 4.4 (although it can probably be read between the lines at the beginning of loc. cit., § 4.4.1).

Lemma 2.6.0. *Let X be an irreducible k-scheme, let K be an extension of k, and let X' be an irreducible component of X_K. The projection $X' \to X$ is surjective.*

[31] Editor's note: these results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in Section 5 of Exposé VI$_B$.

Proof. Let B be a transcendence basis of K over k, and set $L = k(B) \subset K$. By EGA IV_2, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L. Hence $X' \to X_L$ is integral and dominant, and therefore surjective. Since $X_L \to X$ is surjective (loc. cit., 4.4.1), so is $X' \to X$.

For the rest of 2.6, fix a k-group scheme G and an action $\mu : G \times X \to X$ of G on a k-scheme X, satisfying the following condition:

$$\Phi : G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x) \quad \text{is surjective.} \tag{$\star$}$$

(This holds in particular when G acts on itself by left translations.) For short, we shall say: "Let X be a G-scheme satisfying ($\star$)." Finally, write k' for the perfect closure of k.

Proposition 2.6.1. *Let X be a G-scheme satisfying ($\star$).*

(i) *X is geometrically pointwise irreducible over k: for every extension K of k, each point $x \in X_K$ belongs to a unique irreducible component of X_K.*

(ii) *Every local ring of $(X_{k'})_{\mathrm{red}}$ is normal.*

(iii) *Let η be a maximal point of $(X_{k'})_{\mathrm{red}}$, put $C = \overline{\{\eta\}}$, and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L-scheme and is geometrically irreducible over L; that is, $C \otimes_L K$ is irreducible for every extension K of L.*

(iv) *In particular, if x is a rational point of X, then the irreducible component of X containing x is geometrically irreducible over k.*

Proof. (i) Since hypothesis ($\star$) is preserved by every base change $k \to K$, it is enough to show that every $x \in X$ belongs to a unique irreducible component of X. Since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is a universal homeomorphism, we may further suppose that k is perfect. We may then suppose that G and X are reduced.

Let η be a maximal point of X, and let z be any point of $Z = \overline{\{\eta\}}$. Since X is reduced, $\mathcal{O}_{X,\eta} = \kappa(\eta)$; and, since k is perfect, $\kappa(\eta) \otimes_k K$ is normal for every extension K of k (cf. EGA IV_2, 6.14.2). Thus every point of X_K lying above η is normal.

Since Φ is surjective, there is a point γ of $G \times X$ such that the two projections of $\Phi(\gamma)$ are z and η. Set $K = \kappa(\gamma)$. There are then rational points $g \in G_K$ and $\eta' \in X_K$, with η' lying above η, such that $z' = g\eta'$ lies above z. By the preceding paragraph, η' is a normal point of X_K, and therefore so is z'. Since $\pi : X_K \to X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV_2, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) now follows from (ii). Since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L, by EGA IV_2, 4.5.9.

Finally, if X has a rational point x, it follows from (iii) that $L = k$, and hence the irreducible component C containing x is geometrically irreducible over k. This can also be seen directly as follows (cf. [Per75], II, 2.1). Let C be the irreducible component of X containing x, and let K be an extension of k. The scheme X_K has a unique point x_K above x, and by (i), x_K belongs to a unique irreducible component C' of X_K. On the other hand, by 2.6.0, every irreducible component of C_K contains x_K, and therefore equals C'.

Notation.— Write provisionally C^0 for the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the identity element e.

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is set-theoretically stable under the group law; that is, $(C^0_{k'})_{\mathrm{red}}$ is a subgroup of $(G_{k'})_{\mathrm{red}}$. Consequently, C^0 is quasi-compact.*
Proof. By 2.6.1, C^0 is geometrically irreducible over k, so $C^0 \times C^0$ is irreducible and

$$\nu(C^0 \times C^0) \subset C^0,$$

where ν denotes the morphism $(g,h) \mapsto gh^{-1}$. Since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a universal homeomorphism, the same conclusion holds for $C^0_{k'}$, and then for

$$H = (C^0_{k'})_{\mathrm{red}}.$$

Since $H \times_{k'} H$ is reduced, ν induces a morphism

$$H \times_{k'} H \longrightarrow H;$$

thus H is a subgroup of $(G_{k'})_{\mathrm{red}}$. Consequently, by 0.5.1, H, and hence C^0, is quasi-compact.

Recall 2.6.3.— Let Y be a scheme. Recall (EGA 0_{III}, 9.1.1) that a subset E of Y is called *retrocompact* if the inclusion $E \hookrightarrow Y$ is quasi-compact. Moreover, by EGA IV_1, 1.9.5(v) and 1.10.1, if U is a retrocompact open subset of Y, its closure $\overline{U}$ is the union of the closures $\overline{\{y\}}$ of the points $y \in U$. Of course, it is enough to let y run through the maximal points of U, that is, the maximal points of Y contained in U.

Consequently, if Y is quasi-separated and η is a maximal point of Y, then the intersection of the closed neighborhoods of η equals $\overline{\{\eta\}}$. Indeed, if $y \in Y - \overline{\{\eta\}}$, then y belongs to an affine open V not containing η. Since Y is quasi-separated, V is retrocompact, and therefore $\overline{V}$ is the union of the $\overline{\{\xi\}}$, for ξ running through the maximal points of Y belonging to V. Hence $\eta \notin \overline{V}$, so $Y - \overline{V}$ is a closed neighborhood of η not containing y.

Proposition 2.6.4. *Let X be a G-scheme satisfying $(\star)$, and let U be an open subset of X.*

(i) *C^0U is an open subset of X, equal to the union of the irreducible components of X whose generic point belongs to U.*

(i′) *Consequently, if X is irreducible, it is quasi-compact.*

(ii) *If, moreover, U is retrocompact in X, then $C^0U = \overline{U}$, and this subset is both open and closed in X.*

Proof. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open subsets

$$g \cdot U_{\kappa(g)} \subset X_{\kappa(g)},$$

and every projection $X_{\kappa(g)} \to X$ is open.

To prove the second assertion of (i), we may replace k by k' (since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a universal homeomorphism), and therefore suppose that k is perfect. We may then suppose that G and X are reduced, and hence geometrically reduced.

Let η be a maximal point of X contained in U, and put $Z = \overline{\{\eta\}}$. Consider

$$\mu' : C^0 \times Z \longrightarrow X.$$

Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; write γ for its generic point. The morphism μ' sends the point $\varepsilon(\eta)$, where ε denotes the identity section of C^0, to η. Hence γ is sent to a generization of η, and therefore to η. Thus μ' sends the underlying space of $C^0 \times Z$ into Z, and, since $C^0 \times Z$ is reduced, μ' factors through Z.

Now let $z \in Z$, and put $K = \kappa(z)$. The morphism

$$\mu_z : G_K \longrightarrow X_K, \qquad h \longmapsto h \cdot z,$$

is surjective. Let α be a maximal point of X_K. The local ring $\mathcal{O}_{X_K,\alpha}$ is a field, since X_K is reduced, so μ_z is flat at every point of G_K above α. It follows from Lemma 2.5.3 that μ_z is flat.

On the other hand, μ_z sends the generic point ω of C^0_K to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \overline{\{\beta\}}$. Since μ_z is flat, there is a generization ξ of ω such that $\mu_z(\xi) = \beta$. But ω is a maximal point of G_K, so necessarily $\xi = \omega$. Thus $\mu_z(\omega) = \beta$, and β lies above η, because $Z_K \to Z$ is flat.

Put $L = \kappa(\omega)$, and let ω_L and z_L be the L-points obtained from ω and z. Then

$$\omega_L \cdot z_L = \beta'$$

is a point of Z_L lying above β, and hence

$$z_L = \omega_L^{-1} \cdot \beta' \in C^0_L \cdot U_L.$$

Therefore $z \in C^0 \cdot U$, which proves (i).

Since C^0 is quasi-compact by 2.6.2, assertion (i′) follows: if X is irreducible and U is a nonempty affine open subset, then $X = C^0 U$, so X is the image of the morphism $C^0 \times U \to X$, and is therefore quasi-compact. Finally, if U is retrocompact in X, then by 2.6.3, $\overline{U}$ is the union of the $\overline{\{\eta\}}$, where η runs through the maximal points of X contained in U. Hence $\overline{U} = C^0 U$. This proves (ii).

One then obtains the following result ([Per75], II, Th. 2.4; see also [Per76], Prop. 4.1.1).

Theorem 2.6.5. *Let k be a field, and let G be a k-group scheme.*

(i) *There is a unique group subscheme G^0 of G, called the* identity component *of G, such that:*

   (a) *the underlying space of G^0 is the irreducible component of the identity element;*

   (b) *$G^0 \to G$ is a flat closed immersion; that is,*

$$\mathcal{O}_{G,g} = \mathcal{O}_{G^0,g} \qquad \text{for every } g \in G^0.$$

(ii) *Moreover, G^0 is quasi-compact, geometrically irreducible, and is a characteristic subgroup of G.*

(iii) *If G is connected, then $G = G^0$.*

Proof. (i) Recall first that G is separated (0.3), and hence, a fortiori, quasi-separated. Let U be an affine open subset of G containing the generic point ω of C^0. By 2.6.3 and 2.6.4,

$$\overline{U} = C^0 U$$

is both open and closed, and, set-theoretically,

$$C^0 = \overline{\{\omega\}}$$

is the intersection of these open-and-closed subsets.

As U runs through the affine open subsets containing ω, the G-schemes $\overline{U}$ form a projective system whose transition morphisms are affine, since they are closed immersions. We can therefore form its projective limit G^0 (cf. EGA IV_3, 8.2.2). Thus, for every affine open V of G, $G^0 \cap V$ is the spectrum of the algebra

$$\varinjlim_{U} \mathcal{O}_G(V \cap \overline{U}) = \mathcal{O}_G(V) \Big/ \sum_U I_{\overline{U}}(V),$$

where $I_{\overline{U}}(V)$ denotes the kernel of

$$\mathcal{O}_G(V) \longrightarrow \mathcal{O}_G(V \cap \overline{U}).$$

It follows that the underlying space of G^0 is C^0, and that $G^0 \to G$ is a closed immersion.

Moreover, for every $g \in G^0$, $\mathcal{O}_{G^0,g}$ is the inductive limit, as V runs through the affine open subsets of G containing g, of the k-algebras

$$\mathcal{O}_{G^0}(V \cap G^0) = \varinjlim_{U} \mathcal{O}_G(V \cap \overline{U}).$$

This double inductive limit is identified with

$$\varinjlim_{U} \varinjlim_{\substack{V \\ g \in V \subset \overline{U}}} \mathcal{O}_G(V) = \mathcal{O}_{G,g}.$$

Thus

$$\mathcal{O}_{G^0,g} = \mathcal{O}_{G,g}$$

(see also EGA IV$_2$, 5.13.3(ii)), and consequently $i : G^0 \to G$ is a flat closed immersion. Conversely, this condition implies that $i^*(\mathcal{O}_G) = \mathcal{O}_{G^0}$, so G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k-scheme and let ϕ be an automorphism of the S-group G_S. For every $s \in S$, ϕ_s sends $G^0_{\kappa(s)}$ into itself, and therefore $\phi(G^0_S) \subset G^0_S$. Moreover, the closed immersion

$$i_S : G^0_S \hookrightarrow G_S$$

obtained from i by base change is flat, so

$$\mathcal{O}_{G_S,\phi(z)} = \mathcal{O}_{G^0_S,\phi(z)} \qquad \text{for every } z \in G^0_S.$$

Hence $\phi \circ i_S$ factors through G^0_S. This proves that G^0 is a characteristic subgroup of G, and proves (ii). Finally, (iii) is a special case of part (i) of the following proposition.[32]

Proposition 2.6.6. *Let k be a field, and let G be a k-group acting on a k-scheme X in such a way that*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x)$$

is surjective. Suppose that X is quasi-separated. Then:

(i) *Every connected component C of X is irreducible.*

(ii) *Let η be the generic point of C, and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L-scheme, is geometrically irreducible over L, and the morphism*

$$G^0_L \times_L C \longrightarrow C \times_L C$$

is surjective.

(iii) *In particular, if C contains a rational point x, then C is geometrically irreducible over k, and*

$$\phi : G^0 \longrightarrow C, \qquad g \longmapsto gx,$$

is surjective.

[32]Editor's note: this proposition, as well as the preceding results, was communicated to us by O. Gabber. It will be used to correct the proof of Theorem 5.3 in Exposé VI$_B$.

Proof. (i) Let C be a connected component of X, let Y be an irreducible component of X contained in C, let η be the generic point of Y, and let U be an affine open subset of X containing η. Since X is quasi-separated, U is retrocompact in X. Hence, by 2.6.4,

$$\overline{U} = G^0 U$$

is an open-and-closed subset of X which meets C, and therefore contains C. By 2.6.3, the intersection of the $\overline{U}$, as U runs through a fundamental system of affine open neighborhoods of η, is $\overline{\{\eta\}}$. It follows that

$$C = \overline{\{\eta\}},$$

which proves (i).

The first assertion of (ii), and also that of (iii), follows from 2.6.1. We first prove the second assertion of (iii). Let ω, respectively η, denote the generic point of G^0, respectively C. Let $z \in C$, and set $K = \kappa(z)$. Since G^0, respectively C, is geometrically irreducible over k, the scheme G^0_K, respectively C_K, is irreducible; denote its generic point by ξ, respectively β. We saw in the proof of 2.6.4 that

$$\mu_z : G_K \longrightarrow C_K, \qquad h \longmapsto h \cdot z,$$

sends ξ to β. Likewise, $\phi(\omega) = \eta$.

Put $L = \kappa(\xi)$, and let ξ_L and z_L be the L-points obtained from ξ and z. Then

$$\xi_L \cdot z_L = \beta'$$

is a point of C_L lying above $\beta \in C_K$, and hence also above $\eta \in C$. Consider the cartesian square

$$\begin{array}{ccc} G^0_L & \xrightarrow{\varphi_L} & C_L \\ {\scriptstyle \pi_{G^0}}\big\downarrow & & \big\downarrow{\scriptstyle \pi_C} \\ G^0 & \xrightarrow[\varphi]{} & C \end{array}$$

Since

$$\phi_L(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$$

(cf. EGA I, 3.4.8), there is a $g \in G^0_L$ such that $\phi_L(g) = \beta'$. Therefore

$$z_L = \xi_L^{-1} \cdot \phi_L(g) = \phi_L(\xi_L^{-1} g),$$

whence

$$\phi(\pi_{G^0}(\xi_L^{-1} g)) = \pi_C(z_L) = z.$$

This proves that ϕ is surjective.

We now prove the second assertion of (ii). It is enough to show that, for every $z \in C$, the morphism

$$\mu_z : G^0_L \otimes_L \kappa(z) \longrightarrow C \otimes_L \kappa(z)$$

is surjective. But this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$.

Remark 2.6.7.— *Under the hypotheses of 2.6.6, if $C(k) = \varnothing$, the morphism*

$$G^0 \times_k C \longrightarrow C \times_k C$$

need not be surjective. For example, let $k = \mathbb{R}$ and $G = \{\pm 1\}_{\mathbb{R}}$. The G-torsor

$$X = \operatorname{Spec} \mathbb{R}[X]/(X^2 + 1)$$

is connected, but

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{R}} X$$

is not surjective. (Here $L = \mathbb{C}$, and

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{C}} X$$

is an isomorphism.)

3. Construction of quotients $F\backslash G$ (for G, F of finite type)

3.1.— Let A be an Artinian local ring, and let $u : F \to G$ be a homomorphism of A-groups. If

$$\mu : F \times_A F \longrightarrow F \qquad \text{and} \qquad \nu : G \times_A G \longrightarrow G$$

denote the multiplication morphisms, and if λ denotes the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\ \nu\ } G$$

recall that the *left quotient* $F\backslash G$ of G by F is the cokernel of the (Sch/A)-groupoid G_* described below:

$$\begin{tikzcd} F \times_A F \times_A G \arrow[r, shift left=2, "F\times\lambda"] \arrow[r, "\mu\times G" description] \arrow[r, shift right=2, "\mathrm{pr}_{2,3}"'] & F \times_A G \arrow[r, shift left, "\lambda"] \arrow[r, shift right, "\mathrm{pr}_2"'] & G \end{tikzcd}$$

Here pr_2 and $\mathrm{pr}_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$, respectively, onto the second factors. We shall say that G_* is the *groupoid with base G defined by u* (cf. Exposé V, § 2.a; as in Exposé V, we do not follow in this exposé the convention of IV, 4.6.15).

Since the unique A-morphism $F \to \operatorname{Spec} A$ is universally open (EGA IV$_2$, 2.4.9), pr_2 is an open morphism. The same therefore holds for λ, which is the composite of pr_2 with the automorphism σ of $F \times_A G$ defined by

$$\sigma(S)(x, y) = (x, u(S)(x) \cdot y),$$

where S is a variable A-scheme and $x \in F(S)$, $y \in G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A.

Let us also note, to conclude these preliminaries, that every A-morphism $s : \operatorname{Spec} A \to G$ defines an automorphism of the groupoid G_* which induces on G, $F \times_A G$, and $F \times_A F \times_A G$, respectively, the automorphisms

$$r_s, \qquad \mathrm{id}_F \times_A r_s, \qquad \mathrm{id}_F \times_A \mathrm{id}_F \times_A r_s.$$

We shall again write r_s for this automorphism of G_*, and shall say that r_s is the *right translation defined by s* (cf. 0.4).

Theorem 3.2. *Let F and G be flat groups locally of finite type over an Artinian local ring A. Let $u : F \to G$ be a quasi-compact homomorphism of A-groups whose kernel is finite over A. Then:*[33]

[33] Editor's note: we have added (ii′) and expanded point (iv), taking account of the additions made in 2.5.2, 2.5.4, and Exposé V, 6.1.

(i) *The left quotient $F\backslash G$ of G by F exists in* (Sch/A), *and the sequence*

$$F \times_A G \underset{\mathrm{pr}_2}{\overset{\lambda}{\rightrightarrows}} G \xrightarrow{\ p\ } F\backslash G$$

is exact in the category of all ringed spaces.

(ii) *The canonical morphism $p : G \to F\backslash G$ is surjective and open, and $F\backslash G$ is a direct sum of schemes of finite type over A.*

(ii′) *More precisely, $X = F\backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A, irreducible, and all of dimension* $\dim G - \dim F$.

(iii) *The canonical morphism*

$$F \times_A G \longrightarrow G \times_{(F\backslash G)} G$$

is surjective.

(iv) *If u is a monomorphism,*[34] *then:*

(a) *The morphism*

$$F \times_A G \xrightarrow{(\lambda,\mathrm{pr}_2)} G \times_{(F\backslash G)} G$$

is an isomorphism, and $G \to F\backslash G$ is faithfully flat and locally of finite presentation.

(a′) *$F\backslash G$ represents the* (fppf) *quotient sheaf $\widetilde{F\backslash G}$, and $G \to F\backslash G$ is an F-torsor locally trivial for the* (fppf) *topology.*

(b) *$F\backslash G$ is flat over A, and is smooth over A if G is so.*

(c) *$u : F \to G$ is a closed immersion, and $F\backslash G$ is separated.*

(d) *If, moreover, F is a normal subgroup of G, there is a unique A-group structure on $F\backslash G$ for which $p : G \to F\backslash G$ is a morphism of A-groups.*

In the proof of this theorem, A' will denote an A-algebra which is local, finite, and free over A. If R is a relation involving A', we shall say that "$R(A')$ is true when A' is sufficiently large" if there is an A-algebra A_1, local, finite, and free over A, such that the relation $R(A')$ holds for every A-algebra A' which is local, finite, and free over A_1.

We shall first prove the theorem when F and G are of finite type over A.

3.2.1.— Suppose for a moment that every point of G has an open saturated neighborhood W such that the groupoid induced by G_* on W has a quasi-section (cf. Exposé V, § 6). Then, by Exposé V, 6.1, assertions (i), (ii), (iii), and (iv)(a) hold, and $F\backslash G$ is of finite type over A. Moreover, under the hypothesis of (iv), since

$$G \longrightarrow F\backslash G$$

is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. Assertion (iv)(a′) follows from (iv)(a), by Exposé IV, 3.4.3.1, 5.2.2, and 5.1.6. Finally, assertion (iv)(c) will be proved in 3.2.5, while (ii′) and (iv)(d) will be proved in Section 5.

[34] Editor's note: in this case, the hypothesis that G be flat can be omitted; cf. subsection 3.3.

We now prove the following assertion:

$$\text{every finite subset of } F\backslash G \text{ is contained in an affine open subset.} \tag{†}$$

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If U is a quasi-section of the groupoid induced by G_* on a saturated open subset W of G, then $U \otimes_A A'$ is a quasi-section of the (Sch/A')-groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A)-groupoid induced by G_* on U, then $U_* \otimes_A A'$ is identified with the (Sch/A')-groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.[36]

It follows from the proofs in Exposé V that the construction of the quotient

$$X = F\backslash G$$

commutes with a base extension $A \to A'$ of the kind considered here.[37]

Let $x_1, \ldots, x_n$ be points of $X = F\backslash G$, which we may suppose to be closed,[38] and let $g_1, \ldots, g_n$ be closed points of G mapping to $x_1, \ldots, x_n$. Let V be an everywhere dense affine open subset of X,[39] and let U be its inverse image in G. By 1.2, there is an A-algebra A', local, finite, and free over A, such that the points $g'_1, \ldots, g'_p$ of

$$G' = G \otimes_A A'$$

lying above $g_1, \ldots, g_n$ are strictly rational over A'.[40] Since the morphisms $G' \to G$ and $G \to X$ are open,

$$U' = U \otimes_A A'$$

is dense in G'. Hence the open subset

$$\bigcap_{i=1}^{p} (U')^{-1} \cdot g'_i$$

is nonempty and therefore contains a closed point x. By 1.2 (and 0.4.1), after enlarging A' if necessary, we may suppose that x is strictly rational over A'. Since $x \in (U')^{-1} \cdot g'_i$, we have

$$g'_i \in U' \cdot x \qquad (1 \leq i \leq p).$$

[35] Editor's note: let us point out that if $A = k$ is a field, every quasi-compact open subset of $F\backslash G$ is quasi-projective (a result due to Chow for smooth algebraic groups); cf. [Ray70], VI, 2.6. By contrast, over the Artinian local ring $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, there exist abelian A-schemes G which are not projective (loc. cit., XII, 4.2).

[36] Editor's note: the preceding assertion is valid for every base change $A \to A'$.

[37] Editor's note: this is explained in detail in 4.6 below. One must show that formation of direct images by the morphisms p, λ, and pr_2 commutes with flat base changes $A \to A'$. Since F and G are of finite type over the Artinian ring A, the morphisms f in question are all quasi-compact and quasi-separated, and

$$f_*(\mathcal{O}_X) \otimes_A A' = f'_*(\mathcal{O}_{X'})$$

(with the evident notation) follows from EGA IV_1, 1.7.21.

[38] Editor's note: indeed, let $y_1, \ldots, y_n$ be arbitrary points of X. Since X is of finite type over A, the closure of each y_i contains a closed point x_i, and every open subset containing x_i contains y_i.

[39] Editor's note: such an open subset exists because X is of finite type over A: X has only finitely many irreducible components $C_1, \ldots, C_p$, and it is enough, for each i, to choose a nonempty affine open subset contained in

$$C_i - \bigcup_{j \neq i} C_j.$$

(Here, moreover, one knows from (ii′) that the C_i are disjoint.)

[40] Editor's note: the original has been expanded in what follows.

Write V' for the inverse image of V in

$$X' = X \otimes_A A'.$$

It is an affine open subset of X', and is also the image of U' under the projection $G' \to X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x, of the quotient X'. Consequently,

$$V' \cdot x = r_x(V'),$$

the image of $U' \cdot x$ in X', is an affine open subset of X' containing the images $x'_1, \dots, x'_p$ of $g'_1, \dots, g'_p$.

Consider the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \to X$:

$$X \otimes_A A' \otimes_A A' \underset{d_0}{\overset{d_1}{\rightrightarrows}} X \otimes_A A' \longrightarrow X$$

Here d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite free A-algebra, say of rank n, d_0 and d_1 are finite and locally free of rank n. We may therefore apply the argument of Exposé V, 5.b, taken from the proof of SGA 1, VIII.7.6. It follows that $x'_1, \dots, x'_p$ are contained in a saturated affine open subset W' contained in the affine open subset $V' \cdot x$. The image of W' in X then contains $x_1, \dots, x_n$, and is an affine open subset of X, by Exposé V, 4.1(ii).

3.2.2.— For every A-algebra A', local, finite, and free over A, let $U(A')$ denote the set of points of $G \otimes_A A'$ having an open saturated neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is clear that $U(A')$ is saturated under the operations of $G(\operatorname{Spec} A')$ on $G \otimes_A A'$. We shall show that, when A' is sufficiently large,

$$U(A') = G \otimes_A A'.$$

By Theorem V.8.1, $U(A)$ is nonempty, and therefore contains a closed point y. We argue by induction on

$$\dim(G - U(A)).$$

Let $g_1, \dots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there is an A-algebra A', local, finite, and free over A, such that the points

$$g'_1, \dots, g'_p \qquad \text{(respectively, } x = x_1, \dots, x_r)$$

of $G' = G \otimes_A A'$, lying above $g_1, \dots, g_n$ (respectively, above y), are strictly rational over A'. Then $U(A')$ contains

$$(U(A) \otimes_A A') \cdot x^{-1} g'_i \qquad \text{for every } i.$$

Consequently, $U(A')$ contains $g'_1, \dots, g'_p$, and

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis therefore gives an algebra A'', local, finite, and free over A', such that

$$U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''.$$

3.2.3.— We are now in a position to prove the existence of $F\backslash G$ when F and G are of finite type over A. Let A' be sufficiently large over A that $U(A')$ coincides with $G \otimes_A A'$ (cf. 3.2.2). Set

$$A'' = A' \otimes_A A',$$

and, for every A-scheme X, write X' and X'' for the fiber products $X \otimes_A A'$ and $X \otimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F'\backslash G'$ and $F''\backslash G''$ exist, and we have the following commutative diagram, in which the first two rows and columns are exact:

$$(*)\qquad \begin{tikzcd} F'' \times_{A''} G'' \arrow[r, shift left, "\mathrm{pr}_2''"] \arrow[r, shift right, swap, "\lambda''"] \arrow[d, shift right, swap, "w_1"] \arrow[d, shift left, "w_2"] & G'' \arrow[r, "p''"] \arrow[d, shift right, swap, "v_1"] \arrow[d, shift left, "v_2"] & F''\backslash G'' \arrow[d, shift right, swap, "u_1"] \arrow[d, shift left, "u_2"] \\ F' \times_{A'} G' \arrow[r, shift left, "\mathrm{pr}_2'"] \arrow[r, shift right, swap, "\lambda'"] \arrow[d, swap, "h"] & G' \arrow[r, "p'"] \arrow[d, "g"] & F'\backslash G' \\ F \times_A G \arrow[r, shift left, "\mathrm{pr}_2"] \arrow[r, shift right, swap, "\lambda"] & G & \end{tikzcd}$$

In this diagram, pr_2' and λ' (respectively, pr_2'' and λ'') are obtained from pr_2 and λ by the evident base changes; the morphisms g and h are induced by the canonical injection $A \to A'$. We write p' and p'' for the canonical morphisms, while v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A''.

Finally, formation of the quotient $F'\backslash G'$ commutes with the two base changes

$$f_1, f_2 : \operatorname{Spec} A'' \rightrightarrows \operatorname{Spec} A'.$$

Thus, if

$$\pi' : F'\backslash G' \longrightarrow \operatorname{Spec} A'$$

denotes the structure morphism, there are canonical isomorphisms, for $i = 1, 2$,

$$\tau_i : F''\backslash G'' \xrightarrow{\sim} (F'\backslash G') \times_{\pi', f_i} \operatorname{Spec} A'',$$

and u_i is the composite of τ_i with the projection

$$(F'\backslash G') \times_{\pi', f_i} \operatorname{Spec} A'' \longrightarrow F'\backslash G'.$$

Now, for a diagram of type $(*)$ in which the first two rows and columns are exact, one checks readily that $\operatorname{Coker}(\mathrm{pr}_2, \lambda)$ exists if and only if $\operatorname{Coker}(u_1, u_2)$ exists, and that these two cokernels are then identified. The existence of $F\backslash G$ therefore follows from that of $\operatorname{Coker}(u_1, u_2)$.

The compatibility of the formation of $F\backslash G$ with the base extensions considered here (cf. editor's note 37 in 3.2.1 and 4.6 below) shows that the composite

$$(F'\backslash G') \times_{\pi', f_1} \operatorname{Spec} A'' \xrightarrow{\tau_1^{-1}} F''\backslash G'' \xrightarrow{\tau_2} (F'\backslash G') \times_{\pi', f_2} \operatorname{Spec} A''$$

is a descent datum on $F'\backslash G'$ relative to

$$f : \operatorname{Spec} A' \longrightarrow \operatorname{Spec} A.$$

By 3.2.1 (†) and SGA 1, VIII.7.6, this descent datum is effective; that is, $\operatorname{Coker}(u_1, u_2)$ exists. (One could instead apply Exposé V, Theorem 4.1 directly.)

3.2.4.— To complete the proof of assertions (i), (ii), (iii), and (iv)(a) of 3.2 in the case where F and G are of finite type over A, it remains to study the quotient $F\backslash G$. By Exposé V, 6.1, assertions (ii), (iii), and (iv)(a) "become true" after the base change

$$f : \operatorname{Spec} A' \longrightarrow \operatorname{Spec} A.$$

By EGA IV_2, 2.6.1, 2.6.2, and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), namely that $F\backslash G$ is the cokernel of (pr_2, λ) in the category of all ringed spaces, it suffices to refer to Exposé V, §6(c).

3.2.5.—[41] We now prove assertion (iv)(c) of 3.2, following the proof of Exposé VI_B, 9.2.1. Write

$$X = F\backslash G$$

and let d be the morphism

$$F \times_A G \longrightarrow G \times_A G$$

whose components are λ and pr_2.

Since u is a closed immersion by 2.5.2, and since

$$d = \sigma \circ (u \times \mathrm{id}_G), \qquad \sigma(x, y) = (xy, y),$$

where σ is an automorphism of $G \times_A G$, the morphism d is a closed immersion. On the other hand, by (iv)(a), the following square is Cartesian:

$$\begin{tikzcd} F \times_A G \arrow[r, "d"] \arrow[d] & G \times_A G \arrow[d, "p\times p"] \\ X \arrow[r, "\Delta_X"'] & X \times_A X \end{tikzcd}$$

Moreover, p, and hence also $p \times p$, is faithfully flat and locally of finite presentation. Therefore, by fppf descent, since d is a closed immersion, so is Δ_X; that is, X is separated.

3.3.—[42] In Theorem 3.2, the hypothesis that G be flat can be omitted when $u : F \to G$ is a monomorphism. This generalization is mentioned in Exposé VI_B, Remark 9.3(b), and also in [Ray67a], Example a)(i), p. 82. The proof, given in [An73], Theorem 4, follows from Theorem 3.2 and from the following theorem of Grothendieck (mentioned in [Ray67a], Theorem 1(ii), and proved in [DG70], § III.2, 7.1). If X is a scheme and R an equivalence relation on X, we write $\widetilde{X/R}$ for the (fppf) quotient sheaf of X by R (cf. Exposé IV, 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, let X be an A-scheme, and let*

$$d_0, d_1 : R \rightrightarrows X$$

be an A-equivalence relation on X, with each d_i faithfully flat and of finite presentation. Let X_0 be a saturated closed subscheme of X, defined by a nilpotent ideal, and let R_0 be the equivalence relation induced by R on X_0. If the (fppf) *quotient sheaf $\widetilde{X_0/R_0}$ is representable by an A-scheme, then so is $\widetilde{X/R}$.*

For the proof, see [DG70], § III.2, 7.1. Let us now return to the case where A is an Artinian local ring. Let $u : F \to G$ be a quasi-compact morphism between A-groups locally

[41] Editor's note: we have added this paragraph.
[42] Editor's note: we have added this subsection.

of finite type, and suppose in addition that u is a monomorphism. Then u is a closed immersion by 2.5.2.

We may now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an Artinian local ring, let G be an A-group locally of finite type, and let F be a closed subgroup of G, flat over A.*[43] *Then:*

(i) *The* (fppf) *quotient sheaf $\widetilde{F\backslash G}$ is representable by a separated A-scheme $F\backslash G$, locally of finite type; moreover, the sequence*

$$F \times_A G \rightrightarrows G \xrightarrow{p} F\backslash G$$

is exact in the category of all ringed spaces.

(ii) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \mathrm{pr}_2)} G \times_{(F\backslash G)} G$$

is an isomorphism, and $p : G \to F\backslash G$ is faithfully flat and locally of finite presentation; hence p is an F-torsor locally trivial for the (fppf) *topology.*

(iii) *If G is flat (respectively, of finite type; respectively, smooth) over A, then so is $F\backslash G$.*

(iv) *$X = F\backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A, irreducible, and all of dimension* $\dim G - \dim F$.

(v) *If, in addition, F is a normal subgroup of G, there is a unique A-group structure on $F\backslash G$ for which $p : G \to F\backslash G$ is a morphism of A-groups.*

Assertions (i) and (ii) follow from 3.2 and 3.3.1. Since $G \to F\backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1, and 17.7.7. Assertions (iv) and (v) will be proved in Section 5. Let us record at once the following corollary.

Corollary 3.3.3. *Let A be an Artinian local ring, let G be an A-group locally of finite type, and let H be a closed subgroup of G, flat over A. Write p for the morphism $G \to G/H$, and write λ (respectively,* pr_1*) for the morphism $G \times H \to G$ defined by $\lambda(g, h) = gh$ (respectively, the projection $G \times H \to G$). Then, for every open subset U of G/H,*

$$\mathcal{O}(U) = \{\varphi \in \mathcal{O}(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ \mathrm{pr}_1\}.$$

In other words, $\mathcal{O}(U)$ is the set of $\varphi \in \mathcal{O}(p^{-1}(U))$ such that $\varphi(gh) = \varphi(g)$ for every A-scheme S and every $g \in G(S)$, $h \in H(S)$.

Indeed, $p : G \to G/H$ is faithfully flat and locally of finite presentation, hence a covering for the (fppf) topology; the result follows from Exposé IV, 3.3.3.2.

[43] Editor's note: for an example in which F is not flat and $\widetilde{F\backslash G}$ is not representable, see [DG70], § III.3, no. 3.3.

4. Construction of quotients $F\backslash G$ (general case)

We now assume the hypotheses of Theorem 3.2, without requiring F and G to be of finite type over A.

4.1.— Let G^α be a connected component of G. We first show that the saturation

$$\mathscr{S}(G^\alpha)$$

[44] of G^α under the equivalence relation defined by the groupoid G_* is an open-and-closed subset of G; in other words, it is a union of connected components of G.

This saturation is the image of $F \times_A G^\alpha$ under λ, and is therefore open in G (cf. § 3.1). Let k be the residue field of A, and let $\bar{k}$ be an algebraic closure of k. It remains to show that the image of

$$(F \times_A G^\alpha) \otimes_A k$$

under $\lambda \otimes_A k$ is closed in $G \otimes_A k$, or, equivalently by SGA 1, VIII.4.4, that the image of

$$(F \times_A G^\alpha) \otimes_A \bar{k}$$

under $\lambda \otimes_A \bar{k}$ is closed in $G \otimes_A \bar{k}$. Since $G^\alpha \otimes_A \bar{k}$ is a union of finitely many connected components of $G \otimes_A \bar{k}$, we are reduced to the case where A is an algebraically closed field, which we now assume. In this case, $\mathscr{S}(G^\alpha)$ is the union of the images of G^α under the left translations $\ell_{u(x)}$, where x runs through the closed points of F. The assertion follows because these images are connected components of G.

4.2.— Take G^α in particular to be the connected component G^0 of the identity in G. Then $\mathscr{S}(G^0)$ plainly contains the image of F under u, which is precisely the equivalence class of the identity. On the other hand, if F^β is a connected component of F, then $F^\beta \times_A G^0$ is connected by 2.1.2, so its image under λ is contained in the connected component of $u(F^\beta)$ in G. In other words, $\mathscr{S}(G^0)$ is the union of the connected components that meet the image of F.

Let us also observe that the open subscheme of G whose underlying space is $\mathscr{S}(G^0)$ is a subgroup of G; we continue to denote it by $\mathscr{S}(G^0)$. Indeed, inversion on G preserves the image of F and permutes the connected components of G that meet this image. It therefore suffices to prove that

$$\nu : G \times_A G \longrightarrow G$$

maps $\mathscr{S}(G^0) \times_A \mathscr{S}(G^0)$ into $\mathscr{S}(G^0)$. For this we may suppose that A is an algebraically closed field: with the notation of 4.1, $\mathscr{S}(G^0) \otimes_A \bar{k}$ is the saturation of $(G \otimes_A \bar{k})^0$ for the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$.

If G^γ and G^δ are connected components of $\mathscr{S}(G^0)$, then $G^\gamma \times_A G^\delta$ is connected and its image under ν meets the image of F. Consequently,

$$\nu(G^\gamma \times_A G^\delta)$$

is contained in a connected component of G that meets $u(F)$.

4.3.— It follows from the preceding discussion that the groupoid G_*, with base G, defined by u, is the direct sum of the groupoids $\mathscr{S}(G^\alpha_*)$ induced by G_* on the various open-and-closed parts of G of the form $\mathscr{S}(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of the groupoids $\mathscr{S}(G^\alpha_*)$, which we are thus led to study separately.

[44] Editor's note: we have written $\mathscr{S}(G^\alpha)$, rather than $\overline{G^\alpha}$, for the saturation of G^α.

First consider the groupoid $\mathscr{S}(G^0_*)$ induced by G_* on $\mathscr{S}(G^0)$. It is clear that $\mathscr{S}(G^0_*)$ is the groupoid with base $\mathscr{S}(G^0)$ defined by the homomorphism

$$F \longrightarrow \mathscr{S}(G^0)$$

induced by u (§ 3.1). The cokernel whose existence we wish to prove is thus identified with $F\backslash\mathscr{S}(G^0)$. Consider, on the other hand, the groupoid

$$G^0_2 \;\substack{\xrightarrow{\ell'_2} \\ \xrightarrow{\ell'_1} \\ \xrightarrow[\ell'_0]{}}\; G^0_1 \;\substack{\xrightarrow{\ell_1} \\ \xrightarrow[\ell_0]{}}\; G^0_0 = G^0$$

induced by $\mathscr{S}(G^0_*)$ on G^0. Referring to the explicit construction in Exposé V, § 3.b, the object denoted there by $Y_0 \times_{X_0} X_1$ is precisely $F \times_A G^0$. Hence G^0_1 is the inverse image of G^0 under the morphism

$$F \times_A G^0 \longrightarrow \mathscr{S}(G^0)$$

induced by λ.

I claim that this inverse image is $F_0 \times_A G^0$, where F_0 denotes the inverse image of G^0 under u. Indeed, if F^β is a connected component of F_0, then $F^\beta \times_A G^0$ is connected by 2.1.2 and $\lambda(F^\beta \times_A G^0)$ is contained in G^0.

Conversely, if F^β is a connected component of F not contained in F_0, then the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0, then $\lambda(F^\beta \times_A G^0)$ does not meet G^0.

It follows that the groupoid G^0_* induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism

$$F_0 \longrightarrow G^0$$

induced by u. Since G^0, and hence F_0, are of finite type over A, Section 3 shows that G^0_* has a cokernel, namely $F_0\backslash G^0$.

I now claim that $F_0\backslash G^0$ is identified with $F\backslash\mathscr{S}(G^0)$. The proof is analogous to that of the first part of assertion (i) of Exposé V, Lemma 6.1. Consider the diagram

$$\mathscr{S}(G^0) \xleftarrow[v]{} F \times_A G^0 \xrightarrow{\mathrm{pr}_2} G^0$$

where v is the morphism induced by λ. Since pr_2 has a section, it is a universal effective epimorphism. Consequently, $F_0\backslash G^0$ coincides with $\mathrm{Coker}(v_0, v_1)$, where

$$V_2 \;\substack{\xrightarrow{v'_2} \\ \xrightarrow{v'_1} \\ \xrightarrow[v'_0]{}}\; V_1 \;\substack{\xrightarrow{v_1} \\ \xrightarrow[v_0]{}}\; V = F \times_A G^0$$

is the inverse image under pr_2 of the groupoid G^0_* (cf. Exposé V, § 3.a), and hence also the inverse image of $\mathscr{S}(G^0_*)$ under the composite morphism

$$F \times_A G^0 \xrightarrow{\text{inclusion}} F \times_A \mathscr{S}(G^0) \xrightarrow{\mathrm{pr}_2} \mathscr{S}(G^0)$$

Similarly, since v is faithfully flat and quasi-compact, $F\backslash\mathscr{S}(G^0)$ coincides with the cokernel of the inverse image of $\mathscr{S}(G^0_*)$ under the base change v. This inverse image is isomorphic to V_* by Exposé V, § 3.c. It follows that the canonical inclusion of G^0_* in $\mathscr{S}(G^0_*)$ induces an isomorphism

$$F_0\backslash G^0 \xrightarrow{\sim} F\backslash\mathscr{S}(G^0).$$

Finally, formation of $F\backslash\mathscr{S}(G^0)$ commutes with finite locally free base changes, because the same holds for $F_0\backslash G^0$ (cf. editor's note 37 and 4.6 below).

4.4.— It remains to construct the cokernel of the groupoid $\mathscr{S}(G_*^\alpha)$ when G^α is an arbitrary connected component of G. Let A' be a sufficiently large local A-algebra, finite and free over A (cf. 3.2). Then $G^\alpha \otimes_A A'$ is the union of finitely many connected components

$$C^1, \dots, C^n$$

of $G \otimes_A A'$, each of which has a strictly rational point. For every i, there is therefore a right translation r_i of $G \otimes_A A'$ which maps $G^0 \otimes_A A'$ onto C^i. This translation induces an isomorphism

$$\mathscr{S}(G_*^0) \otimes_A A' \xrightarrow{\sim} \mathscr{S}(C_*^i).$$

Consequently, the groupoid induced by $G_* \otimes_A A'$ on the saturation of C^i has a cokernel.

Since $\mathscr{S}(G_*^\alpha) \otimes_A A'$ is the direct sum of some of the groupoids $\mathscr{S}(C_*^i)$, it too has a cokernel. This cokernel is a direct sum of some number of copies of

$$(F_0 \otimes_A A')\backslash(G^0 \otimes_A A').$$

Thus every finite subset of it is contained in an affine open subset. Moreover, formation of this cokernel commutes with finite locally free extensions of the base (cf. editor's note 37 and 4.6 below). As in 3.2.3, it follows that this cokernel has the form $Y \otimes_A A'$, where Y is a cokernel of $\mathscr{S}(G_*^\alpha)$.

4.5.— We have now constructed $F\backslash G$ and shown that it is a direct sum of schemes of finite type over A. The remaining assertions of Theorem 3.2 reduce directly to assertions about the groupoids $\mathscr{S}(G_*^\alpha)$. As in Exposé V, § 6, the second assertion in (i) follows from the first together with (ii) and (iii); it is therefore enough to prove (ii), (iii), and (iv)(a).

Since A' is a local A-algebra, finite and free over A, the morphism $A \to A'$ is faithfully flat and of finite presentation. By SGA 1, VIII (3.1, 4.6, 5.4), it is therefore enough to verify the corresponding assertions for the groupoid

$$\mathscr{S}(G_*^\alpha) \otimes_A A'.$$

This groupoid is isomorphic to a direct sum of finitely many copies of $\mathscr{S}(G_*^0) \otimes_A A'$ (cf. 4.4), so we are reduced to the groupoid $\mathscr{S}(G_*^0)$. For this last groupoid we continue to follow the proof given in Exposé V, § 6, as we began to do in 4.3.

4.6.— Let us conclude this section with some remarks concerning Lemma 6.1 and § 9(a) of Exposé V. Under the hypotheses and with the notation of Exposé V, § 9(a), we seek a condition under which formation of the cokernel of the (Sch/S)-groupoid X_* commutes with a base extension

$$\pi : S' \longrightarrow S.$$

The cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U', respectively. We are therefore reduced to the case of a (Sch/S)-groupoid satisfying the hypotheses of Exposé V, Theorem 4.1.

[45]Let Y be the cokernel of U_*, put

$$Y' = Y \times_S S',$$

and let Y_1 be the cokernel of U'_*; we saw in Exposé V, § 9(a), that the canonical morphism

$$Y_1 \longrightarrow Y'$$

[45]Editor's note: we have added the following sentence.

is a homeomorphism (indeed, a universal homeomorphism); we may thus identify Y_1 and Y' as topological spaces. If $p : U \to Y$ is the canonical morphism and $p' : U' \to Y'$ is obtained from it by base change, we want the following sequence of $\mathcal{O}_{Y'}$-modules to be exact:

$$\mathcal{O}_{Y'} \longrightarrow p'_*(\mathcal{O}_{U'}) \rightrightarrows p'_*u'_{1*}(\mathcal{O}_{U'_1}) = p'_*u'_{0*}(\mathcal{O}_{U'_1}). \tag{*}$$

[46]Since we are under the hypotheses of Exposé V, Theorem 4.1, u_0 and u_1 are finite and locally free; moreover, p is integral by Exposé V, 4.1(ii). Thus p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S, EGA III_1, 1.4.15 (taking account of correction $\mathrm{Err}_{\mathrm{III}}$ 25 in EGA III_2) identifies the sequence (*) with the inverse image of the sequence

$$\mathcal{O}_{Y} \longrightarrow p_*(\mathcal{O}_{U}) \rightrightarrows p_*u_{1*}(\mathcal{O}_{U_1}) = p_*u_{0*}(\mathcal{O}_{U_1}), \tag{**}$$

which is exact.[47] An analogous argument applies when the groupoid X_* has quasi-sections "locally" (cf. the proof of Exposé V, Theorem 7.1). We therefore obtain:

Proposition 4.6.1. *Formation of the cokernel of X_* commutes with flat base extensions when X_* has quasi-sections locally.*

4.7.— Consider the groupoid G_* of Theorem 3.2, supposing temporarily that F and G are of finite type over A. By 3.2.2 there is a local A-algebra A', finite and free over A, such that $G_* \otimes_A A'$ has quasi-sections "locally." For every base extension $T \to \operatorname{Spec} A$, the sequence

$$(F''\backslash G'') \times_{\operatorname{Spec} A} T \rightrightarrows (F'\backslash G') \times_{\operatorname{Spec} A} T \longrightarrow (F\backslash G) \times_{\operatorname{Spec} A} T$$

obtained from diagram $(*)$ of 3.2.3 is exact.

If, in addition, T is flat over $\operatorname{Spec} A$, then 4.6 identifies

$$(F''\backslash G'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (F'\backslash G') \times_{\operatorname{Spec} A} T$$

respectively with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\operatorname{Spec} A} T.$$

The diagram obtained from 3.2.3$(*)$ by the base change $T \to \operatorname{Spec} A$ then shows that

$$(F\backslash G) \times_{\operatorname{Spec} A} T$$

is the cokernel of $G_* \times_{\operatorname{Spec} A} T$. The same argument applies in the general case, when F and G are merely locally of finite type over A. Thus:

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A-scheme T, the scheme*

$$(F\backslash G) \times_{\operatorname{Spec} A} T$$

is identified with the left quotient of $G \times_{\operatorname{Spec} A} T$ by $F \times_{\operatorname{Spec} A} T$.[48]

[46]Editor's note: in what follows we have modified the original; the additional hypotheses imposed there on X_* are unnecessary.

[47]Editor's note: we have added both the following sentence and the numbering 4.6.1 below, in order to highlight the stated result.

[48]Editor's note: we have added the numbering 4.7.1 in order to highlight this result.

5. Connections with Exposé IV and consequences

5.1.—[49] We retain the notation of § 3 and the hypotheses of Theorem 3.2. We then have the following commutative diagram:

$$\begin{tikzcd} F \times_A G \times_A G \arrow[r, "F\times\nu"] \arrow[d, shift right, "\mathrm{pr}_2 \times G"'] \arrow[d, shift left, "\lambda\times G"] & F \times_A G \arrow[d, shift right, "\mathrm{pr}_2"'] \arrow[d, shift left, "\lambda"] \\ G \times_A G \arrow[r, "\nu"] \arrow[d, "p\times G"'] & G \arrow[d, "p"] \\ (F\backslash G) \times_A G \arrow[r, dashed, "\rho"] & F\backslash G \end{tikzcd}$$

It satisfies

$$\mathrm{pr}_2 \circ (F \times \nu) = \nu \circ (\mathrm{pr}_2 \times G), \qquad \lambda \circ (F \times \nu) = \nu \circ (\lambda \times G).$$

Moreover, since G is flat over A, the left-hand vertical sequence is exact by 4.7. Hence ν induces a morphism of A-schemes

$$\rho : (F\backslash G) \times_A G \longrightarrow F\backslash G.$$

The morphism ρ makes G act on the right on $F\backslash G$, as is immediately checked; moreover, the canonical morphism $G \to F\backslash G$ commutes with the right actions of G on G and on $F\backslash G$.[50] This proves the first assertion of Theorem 3.2(ii′). By 2.5.4, the connected components of

$$X = F\backslash G$$

are therefore of finite type, irreducible, and all of the same dimension.

To compute this dimension, we may suppose that $A = k$ and that k is algebraically closed. By Exposé I, 2.3.3.1, the stabilizer of the k-point $p(e)$ is represented by the fiber

$$H = p^{-1}(p(e)).$$

Since $F\backslash G$ is the quotient of G by F in the category of ringed spaces, the underlying space of this fiber is $u(F)$. As $\ker(u)$ is finite, we obtain

$$\dim H = \dim u(F) = \dim F.$$

It follows from 2.5.4(ii) that

$$\dim X = \dim G - \dim F.$$

This proves Theorem 3.2(ii′), and hence also Theorem 3.3.2(iv).

5.2.— When the homomorphism of A-groups $u : F \to G$ is a monomorphism, the result of 5.1 may also be recovered from the results of Exposé IV. Indeed, the canonical morphism

$$p : G \longrightarrow F\backslash G$$

is faithfully flat and open by 3.2; it is therefore a covering for the (fpqc) topology by Exposé IV, 6.3.1, and we may apply Exposé IV, Corollaries 5.2.2 and 5.2.4.

[49] Editor's note: we have changed the title of this section, which was called "Complements" in the original.
[50] Editor's note: we have added what follows.

In particular, suppose, in addition to the hypotheses of 3.2, that u is the inclusion in G of a normal subgroup F. Then there is a unique A-group structure on $F\backslash G$ for which the canonical morphism

$$p : G \longrightarrow F\backslash G$$

is a homomorphism of A-groups.[51] This proves Theorem 3.3.2(v).

5.3.— We now review several statements from Exposé IV.

5.3.1.— The statements of Exposé IV, 5.2.7 and 5.3.1 translate as follows. Let F and G be groups locally of finite type and flat over A, with F a closed normal subgroup of G. The maps

$$H \longmapsto F\backslash H, \qquad H' \longmapsto H' \times_{(F\backslash G)} G$$

define a bijection between the flat A-subgroups of G containing F and the flat A-subgroups of $F\backslash G$. Under this bijection, the closed (respectively, normal) subgroups of G containing F correspond to the closed (respectively, normal) subgroups of $F\backslash G$.[52]

5.3.2.— Exposé IV, Proposition 5.2.9, gives the following result. Let F, H, G be groups locally of finite type and flat over A, and suppose

$$F \subset H \subset G,$$

with F closed in G and normal in H. Then $F\backslash H$ acts freely on the left on $F\backslash G$, the quotient scheme

$$(F\backslash H)\backslash(F\backslash G)$$

exists, and there is a canonical isomorphism of G-schemes

$$(F\backslash H)\backslash(F\backslash G) \xrightarrow{\sim} H\backslash G.$$

5.3.3.— Finally, Exposé IV, 5.2.8, gives the following assertion. Let F, H, G be groups locally of finite type and flat over A. Suppose that F is contained in G, closed and normal in G, that H is contained in G, and that $F \cap H$ is flat over A. Let

$$F \overset{\tau}{\times}_A H$$

be the A-group whose underlying scheme is $F \times_A H$, with multiplication defined by

$$((x, h), (y, h')) \longmapsto (xhyh^{-1}, hh').$$

Likewise, let

$$u : H \cap F \longrightarrow F \overset{\tau}{\times}_A H$$

be the monomorphism $x \mapsto (x^{-1}, x)$, and let $F \cdot H$ be the quotient

$$(F \cap H)\backslash(F \overset{\tau}{\times}_A H).$$

Then there is a canonical isomorphism

$$F\backslash(F \cdot H) \xrightarrow{\sim} (F \cap H)\backslash H.$$

[51] Editor's note: we have added the following sentence.

[52] Editor's note: in addition to the cited statements of Exposé IV, we use the fact that, since $G \to F\backslash G$ is faithfully flat, an A-subgroup H of G is flat over A if and only if $F\backslash H$ is.

5.4.—[53] Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type, such that the kernel N of u is flat over A. In this case, by 3.3.2 and 5.2, the quotient A-group

$$C = N\backslash G$$

exists, and the morphism $p : G \to C$ is faithfully flat and locally of finite presentation. On the other hand, by Exposé IV, 5.2.6, u induces a monomorphism

$$v : C \longrightarrow H,$$

which is quasi-compact (because u is quasi-compact and $G \to C$ is surjective; cf. EGA IV_1, 1.1.3), and is therefore a closed immersion by 2.5.2. We have thus obtained the following proposition.

Proposition 5.4.1.— *Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type, such that $N = \ker u$ is flat over A. Then u admits the factorization*

$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "p"'] & H \\ N\backslash G \arrow[ur, "i"'] & \end{tikzcd}$$

where p is faithfully flat and locally of finite presentation, and i is a closed immersion.

Suppose in addition that G is flat over A. Then, by 3.3.2, $C = N\backslash G$ is flat over A, and hence the quotient

$$X = C\backslash H$$

exists in $(\mathbf{Sch}_{/A})$ and represents the quotient (fppf)-sheaf $\widetilde{C}\backslash H$, while

$$q : H \longrightarrow X$$

is a C-torsor. Consequently, if

$$e : \operatorname{Spec} A \longrightarrow H$$

denotes the unit section of H, then v induces an isomorphism of (fppf)-sheaves between $\widetilde{C}$ and the fiber product of q with

$$q \circ e : \operatorname{Spec} A \longrightarrow X,$$

which is represented by a closed subscheme of H. Consequently, v is an isomorphism of C onto a closed group subscheme K of H, equal to the stabilizer of the A-point $q \circ e$ of X. (This also gives another proof that every quasi-compact monomorphism $v : C \to H$ between A-groups locally of finite type is a closed immersion; cf. 2.5.2 and Exposé VI_B, 1.4.2.)

Suppose further that C is a normal subgroup of H. In this case, the A-group

$$\overline{H} = C\backslash H$$

is the cokernel, in the category of A-groups, of the morphism $u : G \to H$, and K is the kernel of the morphism

$$H \longrightarrow \overline{H}.$$

[53] Editor's note: we have expanded the original in what follows; in particular, we have added Proposition 5.4.1.

When G and H are abelian A-groups, K is the image of u in the category of abelian A-groups, whereas

$$C = (\ker u)\backslash G$$

is the coimage of u. Taking into account the isomorphism

$$C \xrightarrow{\sim} K$$

just established, we obtain the following.[53]

Theorem 5.4.2.— *Let k be a field. The category of commutative algebraic k-groups is abelian.*

Indeed, when k is a field, $\ker(u)$ is flat over k, whatever u may be.

[54] Note that the full subcategory of commutative affine algebraic k-groups is thick. Indeed, consider an exact sequence of commutative algebraic k-groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, then N is clearly affine, and G/N is also affine by a theorem of Chevalley; cf. Exposé VI$_B$, 11.17. Conversely, if N and G/N are affine, then G is also affine by Exposé VI$_B$, 9.2(viii). We therefore obtain the following corollary.

Corollary 5.4.3.— *Let k be a field. The category of commutative affine algebraic k-groups is abelian.*

Let us also point out that the category of all commutative affine k-groups (not necessarily of finite type) is abelian. This follows from Exposé VI$_B$, 11.17 and 11.18.2 (cf. [DG70], § III.3, 7.4); see also Exposé VII$_B$, 2.4.2 for a proof using formal groups.

5.5.— Let G be a group locally of finite type and flat over an Artinian local ring A. We know from 2.3 that the connected component G^0 of the identity is a normal open group subscheme of G, hence is also flat over A. It then follows from 3.2 and 5.2 that

$$G^0\backslash G$$

is an A-group scheme, flat over A. Moreover, since every connected component G^α of G is saturated for the equivalence relation defined by G^0, the quotient $G^0\backslash G$ is the disjoint union of the $G^0\backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the identity in $G^0\backslash G$ is precisely

$$G^0\backslash G^0 \simeq \operatorname{Spec} A,$$

and hence the morphism

$$G^0\backslash G \longrightarrow \operatorname{Spec} A$$

is a local isomorphism at the identity. Consequently, $G^0\backslash G$ is étale over $\operatorname{Spec} A$, by Exposé VI$_B$, 1.3.[55] We therefore obtain the following proposition (for assertion (ii), see [DG70], § II.5, 1.7–1.10).

Proposition 5.5.1.— *Let A be an Artinian local ring and G an A-group locally of finite type and flat.*

(i) *$G^0\backslash G$ is an étale A-group.*

[53] Editor's note: we have added the number 5.4.2 to this theorem.

[54] Editor's note: we have added what follows.

[55] Editor's note: we have expanded the preceding discussion from the original and added Proposition 5.5.1 in order to make this result explicit.

(ii) *Consequently, if $A = k$ is an algebraically closed field, $G^0\backslash G$ is a constant k-group acting simply transitively on the set of connected components of G. Thus, if G is algebraic, $G^0\backslash G$ is finite.*

5.6.— Now let k be a perfect field and G a k-group locally of finite type. We saw in 0.2 that G_{red} is then a group subscheme of G. Moreover, the equivalence class of the identity of G under the left action of G_{red} on G is the whole underlying space of G. Theorem 3.2 therefore gives the following result.

Proposition 5.6.1.— *Let k be a perfect field and G a k-group locally of finite type. Then the k-scheme*

$$G_{\mathrm{red}}\backslash G$$

is the spectrum of a finite local k-algebra with residue field k.

[56] Indeed, by 3.2, the quotient $G_{\mathrm{red}}\backslash G$ has a single point, with residue field k, and is a k-scheme of finite type. It is therefore the spectrum of a finite-dimensional local k-algebra (cf. EGA I, 6.4.4).

Proposition 5.6.2.— *Let $u : F \to G$ be a morphism between groups locally of finite type over a perfect field k. The following assertions are equivalent:*

(i) *u is flat.*

(ii) *The morphism*

$$u^0 : F^0 \longrightarrow G^0$$

is dominant, and the morphism

$$v : F_{\mathrm{red}}\backslash F \longrightarrow G_{\mathrm{red}}\backslash G$$

induced by u is flat.

[57] Indeed, consider the following commutative diagram:

$$\begin{tikzcd} F \arrow[r, "p"] \arrow[d, "u"'] & F_{\mathrm{red}}\backslash F \arrow[d, "v"] \\ G \arrow[r, "q"'] & G_{\mathrm{red}}\backslash G \end{tikzcd}$$

where p and q denote the canonical projections. By 3.2(iv), p and q are faithfully flat. Consequently, if u is flat, then

$$q \circ u = v \circ p$$

is flat, and hence v is flat as well.

Conversely, suppose that v is flat and u^0 is dominant. Since u^0 is quasi-compact (the scheme F^0 is of finite type over k by 2.4, and hence Noetherian), it maps the generic point ξ of F^0 to the generic point η of G^0. Let R be the finite local k-algebra such that

$$G_{\mathrm{red}}\backslash G = \operatorname{Spec} R,$$

and let $\mathfrak{m}$ be its maximal ideal. We have local homomorphisms of local rings

$$R \longrightarrow \mathcal{O}_{G,\eta} \longrightarrow \mathcal{O}_{F,\xi}.$$

Observe that the following square is Cartesian:

[56] Editor's note: we have added the number 5.6.1, as well as the proof that follows.
[57] Editor's note: we have expanded the proof of (ii) ⇒ (i) and simplified the diagram below.

$$\begin{tikzcd} G_{\mathrm{red}} \arrow[r] \arrow[d] & G \arrow[d, "q"] \\ \mathrm{Spec}(R/\mathfrak{m}) \arrow[r] & \mathrm{Spec}(R) \end{tikzcd}$$

It follows that

$$\mathscr{O}_{G,\eta}/\mathfrak{m}\mathscr{O}_{G,\eta} \simeq \mathscr{O}_{G_{\mathrm{red}},\eta} = \kappa(\eta),$$

so that

$$\mathscr{O}_{F,\xi}/\mathfrak{m}\mathscr{O}_{F,\xi}$$

is flat over $\mathscr{O}_{G,\eta}/\mathfrak{m}\mathscr{O}_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R. The local criterion for flatness (cf. EGA IV_3, 11.3.10.2) therefore shows that $\mathscr{O}_{F,\xi}$ is flat over $\mathscr{O}_{G,\eta}$; in other words, u is flat at ξ. It follows from 2.5.3 that u is flat.

6. Complements on k-groups not necessarily of finite type

[58] Let us also record the following results, which will be useful in the addendum, Exposé VI_B, § 12. We fix a base field k.

Lemma 6.1.— *Let G be a k-group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and both projections $p_1(u)$ and $p_2(u)$ are maximal points of G.*

Proof. Set $K = \kappa(x)$. Since the projection $G_K \to G$ sends maximal points to maximal points, we are reduced to the case in which x is rational. Left translation λ_x (respectively, right translation ρ_x) then gives a morphism

$$G \longrightarrow G \times G, \qquad g \longmapsto (\lambda_x(g^{-1}), g)$$

(respectively, $g \longmapsto (g, \rho_x(g^{-1}))$), which induces an isomorphism from G onto $\mu^{-1}(x)$, inverse to p_2 (respectively, p_1). Thus, if u is a maximal point of $\mu^{-1}(x)$, then $p_1(u)$ and $p_2(u)$ are maximal points of G, and u has the required properties.

Corollary 6.2.— *Let $f : G \to H$ be a quasi-compact dominant morphism of k-groups.*

(i) *f is surjective.*

(ii) *If H is reduced, f is faithfully flat.*

Proof. Write μ_H (respectively, μ_G) for the multiplication of H (respectively, G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and $\alpha = p_1(u)$, $\beta = p_2(u)$ are maximal points of H. Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are nonempty (cf. EGA IV_1, 1.1.5). Hence there exists $v \in G \times G$ such that

$$(f \times f)(v) = u$$

(cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This proves that f is surjective.

Suppose in addition that H is reduced. Then $\mathscr{O}_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \varnothing$. Thus f is flat at every point $\xi \in f^{-1}(\alpha)$, and consequently f is flat by Lemma 2.5.3.

[58] Editor's note: we have added the results that follow, taken from [Per75]. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is a product of two generic points."

Reminder 6.3.— Recall (cf. EGA IV_3, 11.10.1) that a morphism of schemes $f : X \to Y$ is said to be *schematically dominant* if it satisfies the following condition: for every open subset U of Y, if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \to U$ factors through Z, then $Z = U$. When f is quasi-compact and quasi-separated, this is equivalent to saying that the closed image of X under f is Y (cf. loc. cit., 11.10.3(iv), and EGA I, 9.5.8).

Proposition 6.4.— *Let $f : H \to G$ be a quasi-compact morphism of k-groups. Then the closed image of f is a group subscheme H' of G, and f factors as*

$$\begin{tikzcd} H \arrow[r, "f"] \arrow[d, "f'"'] & G \\ H' \arrow[ur, "i"'] & \end{tikzcd}$$

where f' is schematically dominant, quasi-compact, and surjective.

Proof. Since H is separated (0.3), f is quasi-compact and separated. Thus $f_*(\mathcal{O}_H)$ is a quasi-coherent $\mathcal{O}_G$-module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal

$$\mathcal{I} = \ker(\mathcal{O}_G \longrightarrow f_*(\mathcal{O}_H))$$

(cf. EGA I, § 9.5).

Write c_G, μ_G (respectively, c_H, μ_H) for the inversion and multiplication morphisms of G (respectively, H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence

$$H' \subset c_G(H') \qquad \text{and therefore} \qquad H' = c_G(H'),$$

since $c_G^2 = \mathrm{id}_G$. Similarly,

$$f \circ \mu_H = \mu_G \circ (f \times f)$$

factors through H', so $f \times f$ factors through the closed subscheme $\mu_G^{-1}(H')$ of $G \times G$. On the other hand, since formation of the closed image commutes with flat base change (EGA III, 1.4.15, and EGA IV_1, 1.7.21), the closed image of $f \times \mathrm{id}_H$ (respectively, $\mathrm{id}_{H'} \times f$) is $H' \times H$ (respectively, $H' \times H'$). Hence, by "transitivity of closed images" (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$. It is therefore contained in $\mu_G^{-1}(H')$; that is, the restriction of μ_G to $H' \times H'$ factors through H'. This proves that H' is a closed group subscheme of G. Write $i : H' \hookrightarrow G$ for the inclusion.

Now $f = i \circ f'$, where $f' : H \to H'$ is schematically dominant and quasi-compact (because f is quasi-compact and i is separated). By 6.2, f' is therefore surjective. This proves 6.4.

We can now state the following theorem ([Per75], V, 3.1 and 3.2; see also [Per76], 0.0 and 0.1).

Theorem 6.5 (D. Perrin).— *Let G be a quasi-compact k-group.*

(i) *G is the projective limit of a filtered system (G_i) of k-groups of finite type (whose transition morphisms $u_{ij} : G_j \to G_i$ are affine for i sufficiently large), and the morphisms $G \to G_i$ are faithfully flat (and affine for i sufficiently large).*

(ii) *Let H be a closed k-subgroup of G. Then the* fpqc *quotient sheaf $\widetilde{G/H}$ is a k-scheme in either of the following two cases:*

(1) *the immersion $H \to G$ is of finite presentation; in this case G/H is of finite type over k;*

(2) *H is normal in G.*

For the proof of this theorem, which rests on several intermediate theorems, we refer to [Per75]. For the reader's convenience, however, we prove the two corollaries below; cf. [Per75], V, 3.3–3.4, or [Per76], 4.2.3–4.2.5.

Corollary 6.6.— *Let $f : G \to H$ be a quasi-compact morphism of k-groups.*

(i) *If f is schematically dominant, then it is faithfully flat.*

(ii) *This is so, in particular, if H is affine and the morphism*

$$f^{\natural} : \mathcal{O}(H) \longrightarrow \mathcal{O}(G)$$

is injective.

Proof. Suppose that f is schematically dominant. By 6.2(i), f is surjective, so by 2.5.3 it is enough to prove that f is flat at the identity element of G. Since

$$\mathcal{O}_{H,e} = \mathcal{O}_{H^0,e}$$

(cf. 2.6.5), we may replace H by H^0, and hence suppose that H is irreducible. Then H is quasi-compact (loc. cit.); since f is quasi-compact, G is also quasi-compact. By 6.5(i),

$$H = \varprojlim_i H_i,$$

where each H_i is an algebraic k-group. Let N_i be the kernel of $G \to H_i$; it is a closed normal k-subgroup of G. Moreover, since the identity section $\{e\} \to H_i$ is of finite presentation, the immersion $N_i \to G$ is of finite presentation. Hence, by 6.5(ii), the fpqc quotient G/N_i is an algebraic k-group. We then have a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{\ f\ } & H \\ {\scriptstyle q_i}\big\downarrow & & \big\downarrow{\scriptstyle p_i} \\ G/N_i & \xrightarrow[f_i]{} & H_i \end{array}$$

in which f is schematically dominant and p_i, q_i are faithfully flat, hence schematically dominant. Thus

$$f_i \circ q_i = p_i \circ f$$

is schematically dominant, and therefore so is f_i. On the other hand, f_i is a monomorphism, hence a closed immersion because G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and consequently $G \to H_i$ is faithfully flat. By [BAC], I, § 2.7, Proposition 9, the morphism $G \to H$ is then flat; it is also surjective by 6.2(i), and is therefore faithfully flat. This proves (i). Assertion (ii) follows, because if H is affine and $f^{\natural}$ is injective, then the closed image of f is H, so f is schematically dominant (cf. 6.4).

Corollary 6.7.— *Let $u : G \to H$ be a morphism of k-groups and set $N = \ker(u)$. Suppose that u is quasi-compact.*

(i) *The* fpqc *quotient sheaf $\widetilde{G/N}$ is represented by a k-group scheme G/N, and u factors as*

$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "p"'] & H \\ G/N \arrow[ur, "i"'] & \end{tikzcd}$$

where p is faithfully flat and i is a closed immersion.

(ii) *In particular, if u is a monomorphism, then u is a closed immersion; and if u is schematically dominant, then u is faithfully flat.*

Proof. (i) By 6.4 and 6.6, the closed image G' of u is a closed group subscheme of H, and

$$p : G \longrightarrow G'$$

is faithfully flat and quasi-compact. Clearly $\ker(p) = \ker(u) = N$. Thus, by Exposé IV, 3.3.2.1 and 5.1.7, G' represents the fpqc quotient sheaf $\widetilde{G/N}$.

(ii) The second assertion is contained in 6.6; let us prove the first. If u is a monomorphism, then so is p. Thus p is both a monomorphism and an effective epimorphism, and is therefore an isomorphism (cf. Exposé IV, 1.14). This proves 6.7.

Let us finally record the following corollaries (cf. [Per76], 4.2.6–4.2.8).

Corollary 6.8.— *The category of commutative quasi-compact k-group schemes is abelian.*

Taking 6.7 into account, the proof is analogous to that of 5.4.2.

Corollary 6.9.— *If* $\mathrm{char}(k) = 0$*, every k-group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and K is an algebraic closure of $\kappa(g)$, then

$$\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K, g_K}.$$

It is therefore enough to prove that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. We are thus reduced to the case in which G is connected, hence quasi-compact by 2.6.5. The result then follows from 6.5 and Cartier's theorem for algebraic groups (cf. Exposé VI$_B$, 1.6.1, or [DG70], § II.6, Theorem 1.1).

Corollary 6.10.— *Let G be a quasi-compact k-group, and suppose that k is algebraically closed.*

(i) *If $f : G \to H$ is a faithfully flat morphism of k-groups, then the induced map $G(k) \to H(k)$ is surjective.*

(ii) *The set of rational points is dense in G.*

For the proof, we refer to [Per75], V, 3.7 and 3.9.

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Exposé $\mathrm{VI_B}$
Generalities on Group Schemes

by J.-E. Bertin

[0] This Exposé, which does not correspond to any oral lecture in the seminar, is intended to bring together a number of commonly used technical results concerning group schemes.*

1. Morphisms of groups locally of finite type over a field

1.1.— Let A be an Artinian local ring, let G and H be two A-groups,[2] and let $u : G \to H$ be a morphism of A-groups. Then u induces a group morphism

$$u(A) : G(A) \longrightarrow H(A).$$

[3] Since $H(A)$ acts on H by right translation, restriction along $u(A)$ defines an action of $G(A)$ on H. This action is compatible with u and with the action of $G(A)$ on G by right translation. Since $G(A)$ acts transitively on the strictly rational points of G (see Exposé VI_A, 0.4 for the definition), these points "all behave in the same way with respect to u." The following properties spring from this observation.

Proposition 1.2.— *Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type over A. Then the set $u(G)$ is closed in H, and*

$$\dim G = \dim u(G) + \dim \ker u.$$

[4]

Since u commutes with the inversion morphisms of G and H, the image $u(G)$ is invariant under inversion on H; the same is therefore true of its closure $\overline{u(G)}$ in H. On the other hand, let L be the set of points of $H \times_A H$ whose two projections belong to $u(G)$. It is clear that L is the image of

$$u \times_A u : G \times_A G \longrightarrow H \times_A H.$$

The multiplication morphism of H consequently sends L into $u(G)$; in other words,

$$u(G) \cdot u(G) = u(G).$$

[0]Editor's note: version of October 13, 2024.

*This Exposé was quite substantially revised after its mimeographed edition; in particular, Sections 10 and 11 were completely rewritten.[(1)]

[1]Editor's note: and also Section 5; cf. Editor's notes 34 and 46.

[2]Editor's note: recall the convention adopted at the beginning of Exposé VI_A: an A-group is a group scheme over A (and every such scheme is separated; cf. Exposé VI_A, 0.3).

[3]Editor's note: we have replaced "homomorphism" by "morphism."

[4]Editor's note: this statement also appears in Exposé VI_A, 2.5.2.

Lemma 1.2.1 below shows that the closure $\overline{L}$ of L in $H \times_A H$ is the set of points whose two projections belong to $\overline{u(G)}$. Hence

$$\overline{u(G)} \cdot \overline{u(G)} = \overline{u(G)}.$$

Thus the reduced subscheme of H whose underlying space is $\overline{u(G)}$ is naturally equipped with a group structure in the category $(\mathbf{Sch}_{/k})_{\mathrm{red}}$, where k is the residue field of A (cf. Exposé VI_A, 0.2).

Let us prove the first assertion of 1.2. After replacing A by an algebraic closure of its residue field k, we may suppose that A is an algebraically closed field k (cf. EGA IV_2, 2.3.12). Replacing u by

$$u_{\mathrm{red}} : G_{\mathrm{red}} \longrightarrow H_{\mathrm{red}},$$

we may suppose that G and H are reduced. In this case, as we have just seen, $\overline{u(G)}$ is the underlying space of a reduced group subscheme of H; we may therefore suppose that u is dominant. Then $G(k)$ acts transitively on the set of connected components of H, and it is enough to prove that $u(G) \cap H^0$ is closed. We are thus reduced to the case in which H is connected, hence irreducible and of finite type (Exposé VI_A, 2.4). The morphism u is then of finite type, since it is quasi-compact and locally of finite type. As H is Noetherian, $u(G)$ is constructible (EGA IV_1, 1.8.5), and therefore contains an open subset V of H (EGA 0_{III}, 9.2.2). By Exposé VI_A, 0.5, we then have

$$H = V \cdot V \subset u(G) \cdot u(G) = u(G).$$

Let us prove the second assertion. Recall first that the functor $\ker u$ (cf. Exposé I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H. When u is locally of finite type, $\ker u$ is therefore locally of finite type over A. As above, this time using EGA IV_2, 4.1.4, we reduce to the case in which A is an algebraically closed field k.

We may further suppose that G and H are irreducible and of finite type and that u is dominant. Indeed, since k is algebraically closed, the connected components of G, on whose set $G(k)$ acts transitively, clearly all have the same dimension; and if u^0 denotes the restriction of u to G^0, then

$$(\ker u)^0 \subset \ker u^0, \qquad \dim \ker u^0 = \dim \ker u.$$

Likewise, u is then of finite type over k. If η is the generic point of H, then

$$\dim u^{-1}(\eta) = \dim G - \dim H$$

(EGA IV_3, 10.6.1(ii)). By EGA IV_3, 9.2.3 and 9.2.6, the set of $y \in H$ such that

$$\dim u^{-1}(y) = \dim u^{-1}(\eta)$$

contains a nonempty open subset V. Since u is dominant, $U = u^{-1}(V)$ is a nonempty open subset of G; it contains a closed point x, because G is a Jacobson scheme (EGA IV_3, 10.4.7). Right translation r_x is an isomorphism from $\ker u$ onto $u^{-1}(u(x))$, and hence

$$\dim \ker u = \dim u^{-1}(u(x)) = \dim u^{-1}(\eta) = \dim G - \dim H.$$

Lemma 1.2.1.— *Let $f : X' \to X$ and $g : Y' \to Y$ be two quasi-compact dominant morphisms of schemes over an Artinian local ring A. Then*

$$f \times_A g : X' \times_A Y' \longrightarrow X \times_A Y$$

is dominant (and quasi-compact).

Indeed,

$$f \times_A g = (\mathrm{id}_X \times_A g) \circ (f \times_A \mathrm{id}_{Y'}).$$

It is therefore enough to prove that $f \times_A \mathrm{id}_{Y'}$ and $\mathrm{id}_X \times_A g$ are dominant. For this purpose we may replace A by its residue field k. Then X and Y' are flat over $A = k$, and $f \times_A \mathrm{id}_{Y'}$ (respectively, $\mathrm{id}_X \times_A g$) is obtained from f (respectively, g) by the flat base change $Y' \to A$ (respectively, $X \to A$). It is therefore dominant (and quasi-compact) by EGA IV_2, 2.3.7.

Counterexample 1.2.2.— *Let k be a field of characteristic zero, let G be the constant k-group $\mathbb{Z}$, and let H be the additive k-group $\mathbb{G}_{a,k}$. Let $u : G \to H$ be a morphism of k-groups. If $u \neq 0$, then $u(G)$ is not closed in H.*

Proposition 1.3.—[5] *Let A be an Artinian local ring with residue field k, let G be an A-group locally of finite type and flat, and let $u : G \to H$ be a morphism of A-groups. The following assertions are equivalent:*

(i) *u is flat (respectively, quasi-finite, unramified, smooth, or étale) at some point of G;*[6]

(ii) *u is flat (respectively, quasi-finite, unramified, smooth, or étale).*

Proof. It is enough to prove that (i) implies (ii). We first establish the following lemma.

Lemma 1.3.0.— *Let $A \to B \to C$ be morphisms of commutative rings, and let $\mathfrak{n}$ be a nilpotent ideal of A. Suppose that $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$-algebra of finite type.*

(i) *Then C is a B-algebra of finite type.*

(ii) *Moreover, if C is flat over A and $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$-algebra of finite presentation, then C is a B-algebra of finite presentation.*

Indeed, let $x_1, \dots, x_n$ be elements of C whose images generate $C/\mathfrak{n}C$ as a $(B/\mathfrak{n}B)$-algebra. By the nilpotent Nakayama lemma, the x_i generate C as a B-algebra. This proves (i). Let

$$\varphi : B[X_1, \dots, X_n] \longrightarrow C$$

be the resulting surjective morphism, and put $I = \ker \varphi$.

Suppose now that C is flat over A and that $C/\mathfrak{n}C$ is of finite presentation over $B/\mathfrak{n}B$. On the one hand, $I/\mathfrak{n}I$ identifies with the kernel of

$$\overline{\varphi} = \varphi \otimes_A (A/\mathfrak{n}).$$

On the other hand, by EGA IV_1, 1.4.4, $\ker \overline{\varphi}$ is an ideal of finite type. Choose $P_1, \dots, P_s \in I$ whose images generate $I/\mathfrak{n}I$ as an ideal. The nilpotent Nakayama lemma shows that the P_i generate I. This proves (ii).

[5] Editor's note: in the original, 1.3, 1.3.1, and 1.3.1.1 were stated with A a field. For completeness, we treat the case of an Artinian local ring, and for this purpose have added Lemma 1.3.0.

[6] Editor's note: recall that a morphism of schemes $f : X \to Y$ is said to be of finite type (respectively, of finite presentation or quasi-finite) at x if there are affine open subsets V, containing $f(x)$, and U, containing x, such that $f(U) \subset V$ and $\mathcal{O}_X(U)$ is an $\mathcal{O}_Y(V)$-algebra of finite type (respectively, of finite presentation, or with the additional condition that $f^{-1}(f(x'))$ is finite for every $x' \in U$); see also Editor's note 40 of Exposé V. The morphism is locally quasi-finite (respectively, locally of finite type or locally of finite presentation) if it has the property at every point x. It is smooth (respectively, unramified or étale) at x if there is an open neighborhood U of x such that $f|_U : U \to Y$ is smooth (respectively, unramified or étale). These definitions show that the loci where f is of finite presentation, of finite type, quasi-finite, smooth, unramified, or étale are open in X. Moreover, because the flatness locus of a morphism locally of finite presentation is open (EGA IV_3, 11.3.1), the locus where f is both of finite presentation and flat is open as well. These facts will be used repeatedly below.

We return to the proof of 1.3. Let x be an arbitrary point of G. Since G is flat over A, the fiberwise flatness criterion, in the form EGA IV$_3$, 11.3.10.2, shows that u is flat at x if $u \otimes_A k$ is. Likewise, the preceding lemma shows that u is of finite type (respectively, of finite presentation) at x if $u \otimes_A k$ is. The remaining properties can then be checked on the fibers (cf. EGA IV$_4$, 17.4.1, 17.5.1, and 17.6.1 for the unramified, smooth, and étale cases). We are therefore reduced to the case $A = k$.

Now let x be a point of G at which one of the conditions in 1.3(i) holds. The properties under consideration are preserved by fpqc descent (cf. EGA IV, 2.5.1, 2.7.1, and 17.7.1). Thus, after replacing k by an algebraic closure of $\kappa(x)$, we reduce to the case in which k is algebraically closed and $x \in G(k)$.

Since G is a Jacobson scheme (cf. EGA IV$_3$, 10.4.7), and the set W of points of G at which u is flat (respectively, quasi-finite, unramified, smooth, or étale) is stable under generization[7] (respectively, open), it is enough to show that every $y \in G(k)$ belongs to W. For such a point y, the translation

$$r_y \circ r_x^{-1}$$

sends x to y, so u has the required property at y. Thus $y \in W$, and the proof is complete.

Corollary 1.3.1.— *Let A be an Artinian local ring with residue field k, and let G be a flat A-group. The following assertions are equivalent:*[8]

(i) *G is locally quasi-finite (respectively, unramified, smooth, or étale) over A at some point;*

(ii) *G is locally quasi-finite (respectively, unramified, smooth, or étale) over A.*

[9] *Proof.* If G satisfies one of the conditions in (i) at a point x, there is an open neighborhood U of x that is of finite type over A. Consequently, it is enough to apply 1.3 in the case where H is the unit A-group, taking the following lemma into account.

Lemma 1.3.1.1.— *Let A be an Artinian local ring and G an A-group. If G contains a nonempty open subset of finite type over A, then G is locally of finite type over A.*

Proof. By Lemma 1.3.0, we may suppose that A equals its residue field k. Moreover, by fpqc descent, we may suppose that k is algebraically closed (cf. EGA IV$_2$, 2.7.1). Let V be the open subset of G consisting of the points at which G is of finite type over k; by hypothesis, $V \neq \varnothing$. Since G is a Jacobson scheme, V contains a closed point x, and to prove that $V = G$ it is enough to show that every closed point y of G belongs to V. For such a point y, the translation

$$r_y \circ r_x^{-1}$$

sends x to y, and hence $y \in V$.

Corollary 1.3.2.— *Let A be an Artinian local ring, and let $u : G \to H$ be a morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) *u is universally open;*

(ii) *u is open;*

[7] Editor's note: the original stated that the flat locus W is open, which does not seem evident a priori.

[8] Editor's note: it is not enough to suppose that G is flat over A at one point. For example, suppose that $A \neq k$, let H be the constant A-group $(\mathbb{Z}/2\mathbb{Z})_A$, and let G be the closed A-subgroup of H whose nonidentity component is reduced. Then the structural morphism $G \to \operatorname{Spec} A$ is a local isomorphism at the identity point e, but is not flat at the point $g \neq e$.

[9] Editor's note: we have added the sentence that follows and simplified the proof of Lemma 1.3.1.1 by invoking EGA IV, 2.7.1 instead of loc. cit., 17.7.5.

(iii) *u is open at some point of G;*

(iv) *the morphism $u^0 : G^0 \to H^0$ induced by u is dominant;*

(iv′) *u^0 is surjective;*

(v) *there is a connected component C of G such that, if D denotes the connected component of H containing $u(C)$, the induced morphism $u' : C \to D$ is dominant.*

Proof. The implications

$$\text{(i)} \Longrightarrow \text{(ii)} \Longrightarrow \text{(iii)} \qquad \text{and} \qquad \text{(iv}'\text{)} \Longrightarrow \text{(iv)} \Longrightarrow \text{(v)}$$

are clear. Since G^0 is of finite type over A (Exposé VI$_A$, 2.4), and hence Noetherian, u^0 is quasi-compact. Thus $u^0(G^0)$ is closed in H^0 by 1.2, and therefore (iv) implies (iv′). On the other hand, G^0 (respectively, C) is open in G (Exposé VI$_A$, 2.3), while H^0 (respectively, D) is irreducible (Exposé VI$_A$, 2.4.1). Hence (ii) implies (iv) (respectively, (iii) implies (v)). It remains to prove that (v) implies (i).

We endow the open subset C (respectively, D) of G (respectively, H) with its induced scheme structure, and let u' denote the induced morphism $C \to D$. Let k be the residue field of A.[10] Since the base change

$$\operatorname{Spec} k \longrightarrow \operatorname{Spec} A$$

is a universal homeomorphism, we may suppose that $A = k$. By hypothesis u' is dominant; and since C is of finite type over k (Exposé VI$_A$, 2.4.1), and hence Noetherian, u' is quasi-compact. By EGA IV$_2$, 2.3.7,

$$u' \otimes_k \bar{k}$$

is again quasi-compact and dominant, where $\bar{k}$ denotes an algebraic closure of k. Since $C \otimes_k \bar{k}$ is a union of connected components of $G \otimes_k \bar{k}$, the morphism

$$u \otimes_k \bar{k} : G \otimes_k \bar{k} \longrightarrow H \otimes_k \bar{k}$$

satisfies assertion (v). In view of EGA IV$_2$, 2.6.4, we are thus reduced to the case where $A = k$ is an algebraically closed field.

In this case we may further replace u by u_{red}, and are thereby reduced to the case where H is reduced. Let ξ (respectively, η) be the generic point of C (respectively, D). Since u' is quasi-compact and dominant,

$$u'(\xi) = \eta$$

(cf. EGA IV$_1$, 1.1.5). Since H is reduced, the local ring $\mathcal{O}_{H,\eta}$ is a field, and hence u' is flat at ξ.[11] It follows from 1.3 that u is flat. Moreover, since u is locally of finite type and H is locally Noetherian, u is locally of finite presentation. Therefore u is universally open by EGA IV$_2$, 2.4.6.

Proposition 1.4.— *Let A be an Artinian local ring, and let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) *u is proper;*

(ii) *there exists $h \in H$ such that the fiber $u^{-1}(h)$ is nonempty and proper over $\kappa(h)$;*

[10] Editor's note: we have expanded the original in what follows.

[11] Editor's note: the original invoked the generic flatness theorem (EGA IV$_2$, 6.9.1), which is not necessary here.

(iii) $\ker u$ *is proper over* A.

Proof. It is clear that (i) implies (iii), and that (iii) implies (ii). The hypotheses imply that u is of finite type; and since G is separated (Exposé VI_A, 0.3), u is separated as well (EGA I, 5.5.1). It remains to prove that (ii) implies that u is universally closed. We may therefore suppose that A equals its residue field k.[12] Let k' be an algebraic closure of $\kappa(h)$, let

$$u' : G' \longrightarrow H'$$

be obtained from u by base change, and let $h' \in H'$ be a point above h. Then

$$u'^{-1}(h') = u^{-1}(h) \times_{\kappa(h)} k'$$

is nonempty and proper, and it is enough to prove that u' is proper (EGA IV_2, 2.6.4). Thus we may suppose that k is algebraically closed and $h \in H(k)$.

By 1.2, $u(G)$ is then the underlying set of a closed reduced group subscheme of H. Since every closed immersion is proper (EGA II, 5.4.2), we may suppose that u is surjective and that H is reduced. The group $G(k)$ then acts transitively on the set of closed points of H. Consequently, for every closed point $y \in H$, the fiber $u^{-1}(y)$ is proper over $\kappa(y)$. By EGA IV_3, 9.6.1, the set of $y \in H$ for which $u^{-1}(y)$ is not proper over $\kappa(y)$ is locally constructible. It contains no closed point, and is therefore empty (cf. EGA IV_3, 10.3.1 and 10.4.7).

[13] Now consider the generic point η of H^0. By what precedes, the fiber

$$u^{-1}(\eta) = G \times_H \operatorname{Spec} \kappa(\eta)$$

is proper over $\kappa(\eta)$. Since H is reduced, $\kappa(\eta) = \mathcal{O}_{H,\eta}$. The latter ring is the direct limit of the rings $\mathcal{O}_H(V)$, as V ranges over the nonempty open subsets of H^0. It follows from EGA IV_3, 8.1.2(a) and 8.10.5(xii), that there exists a nonempty open subset $V \subset H^0$ such that the restriction of u over V is proper. The translates $g \cdot V$, for $g \in G(k)$, form an open covering of H; and for every $g \in G(k)$, the restriction of u over $g \cdot V$ is proper. Hence u is proper (cf. EGA II, 5.4.1).

Corollary 1.4.1.— *Let A be an Artinian local ring, and let $u : G \to H$ be a morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) *u is locally quasi-finite;*

(ii) *u is quasi-finite at some point;*

(iii) $\ker u$ *is discrete;*

(iv) *the restriction of u to each connected component of G is finite.*

If u is quasi-compact, these assertions are also equivalent to

(v) *u is finite.*

It is clear that (iv) implies (iii), that (iii) implies (ii) (EGA I, 6.4.4), and that, when u is quasi-compact, (iv) and (v) are equivalent. We have already seen in 1.3 that (i) and (ii) are equivalent.

It remains to prove that (i) implies (iv). Let C be a connected component of G. Since C is of finite type over A (Exposé VI_A, 2.4.1) and G, H are separated (Exposé VI_A, 0.3),

[12] Editor's note: we have added the following sentence.

[13] Editor's note: we have expanded the original in what follows.

EGA I, 5.5.1 and 6.3.4 show that the restriction u' of u to C is separated and of finite type.[14] Its fibers are discrete, so u' is quasi-finite (cf. EGA II, 6.2.2). Since every quasi-finite proper morphism is finite (cf. EGA III_1, 4.4.2), it is enough to prove that u' is universally closed.

For this, we may suppose that A equals its residue field k. By fpqc descent (cf. EGA IV_2, 2.6.4), it is enough to prove that

$$u' \otimes_k \overline{k}$$

is universally closed, where $\overline{k}$ is an algebraic closure of k. Since C is of finite type over k, the scheme $C \otimes_k \overline{k}$ is the disjoint union of finitely many connected components $C'_1, \ldots, C'_n$ of $G \otimes_k \overline{k}$, and it is enough to prove the assertion for each C'_i. We are thus reduced to the case in which k is algebraically closed.

Let g be a closed point of C, and let

$$u^0 : G^0 \longrightarrow H$$

be the restriction of u to G^0. Then

$$u' = r_{u(g)} \circ u^0 \circ r_{g^{-1}},$$

where r_g denotes right translation by g. Thus, to prove that u' is proper, it is enough to prove that u^0 is proper. By hypothesis u is locally quasi-finite, so the nonempty fiber $\ker u$ is discrete. We have seen that u^0 is of finite type, hence its fiber $\ker u^0$ is finite (cf. EGA II, 6.2.2), and is therefore nonempty and proper. Proposition 1.4 now shows that u^0 is proper.

Corollary 1.4.2.— *Let A be an Artinian local ring, and let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type. The following assertions are equivalent:*[15]

(i) *u is a closed immersion;*

(ii) *u is a monomorphism;*

(iii) $\ker u$ *is trivial, that is, isomorphic to the unit A-group.*

In particular, every group subscheme[16] *of H is closed.*

It is clear that (i) implies (ii). Considering the functors represented by G and H, one sees immediately that (ii) and (iii) are equivalent. Finally, if $\ker u$ is the unit A-group, it is a proper nonempty fiber. Proposition 1.4 therefore shows that u is a proper monomorphism; it is of finite presentation because H is locally Noetherian (EGA IV_1, 1.6.1), and hence is a closed immersion (EGA IV_3, 8.11.5).

The final assertion follows because H is locally Noetherian, so every immersion $G \to H$ is quasi-compact (EGA I, 6.6.4).

Counterexample 1.4.3.— *Let k be a field of characteristic zero, let G be the constant k-group $\mathbb{Z}$, and let H be the additive k-group $\mathbb{G}_{a,k}$. Let $u : G \to H$ be a morphism of k-groups. If $u \neq 0$, then* $\ker u = 0$, *but u is not a closed immersion (cf. 1.2.2).*

We shall use below the following two results, which should have appeared in Exposé VI_A.

Lemma 1.5.— *Let k be a field and let G be a k-group locally of finite type. Then:*

[14] Editor's note: we have expanded the original in what follows.

[15] Editor's note: the hypothesis that G and H be locally of finite type may be removed, because [Per76], 4.2.4, states that every quasi-compact monomorphism $u : G \to H$ between group schemes over a field k is a closed immersion. See also [DG70], III.3.7.2(b), for the case in which G and H are affine; in that case every morphism $G \to H$ is affine (EGA II, 1.6.2(v)), hence quasi-compact.

[16] Editor's note: in this particular case, see also Exposé VI_A, 0.5.2, which holds without any finiteness hypothesis.

(i) *all the irreducible components of G have the same dimension;*

(ii) *for every $g \in G$,*
$$\dim_g G = \dim G.$$

[17] Assertion (i) has already been proved in Exposé VI$_A$, 2.4.1, and (ii) follows. Indeed, let $g \in G$, and let C be the connected component of G containing g. By definition (EGA 0_{IV}, 14.1.2), $\dim_g G$ is the infimum of the integers $\dim U$, where U ranges over the open neighborhoods of g. Hence
$$\dim_g G = \dim U_0$$
for some U_0, which we may suppose is contained in C (because $\dim V \leq \dim U$ when $V \subset U$). Since C is irreducible and of finite type over k (Exposé VI$_A$, 2.4.1),
$$\dim U_0 = \dim C = \operatorname{trdeg}_k \kappa(\xi),$$
where ξ is the generic point of C (EGA IV$_2$, 5.2.1). Therefore
$$\dim_g G = \dim C = \dim G.$$

Proposition 1.6.—[18] *Let S be a scheme of characteristic zero, and let G be an S-group scheme locally of finite presentation over S at every point of the identity section $\epsilon(S)$. The scheme G is smooth over S at every point of the identity section if and only if the $\mathscr{O}_S$-module*
$$\omega_{G/S} = \epsilon^*(\Omega^1_{G/S}),$$
called the conormal module of the identity section of G, is locally free.

Recall that a scheme S is said to be of characteristic zero if the field $\kappa(x)$ has characteristic zero for every closed point $x \in S$.

Recall also (Exposé II, 4.11) that, if $\pi : G \to S$ is the structural morphism, then
$$\Omega^1_{G/S} = \pi^*(\omega_{G/S}).$$
Thus it is equivalent to say that the $\mathscr{O}_S$-module $\omega_{G/S}$ is locally free or that the $\mathscr{O}_G$-module $\Omega^1_{G/S}$ is locally free.

If there is an open neighborhood U of $\epsilon(S)$ that is smooth over S, then by EGA IV$_4$, 17.2.3, $\Omega^1_{U/S}$ is locally free of finite type, and so is
$$\omega_{G/S} = \epsilon^*(\Omega^1_{U/S}).$$

Conversely, if $\omega_{G/S}$ is locally free, then so is
$$\Omega^1_{G/S} = \pi^*(\omega_{G/S}).$$
Since S has characteristic zero, the Jacobian criterion (EGA IV$_4$, 16.12.2) shows that G is differentially smooth over S. It follows from EGA IV$_4$, 17.12.5, that G is smooth over S at every point of the identity section.

Corollary 1.6.1 (Cartier).— *Let k be a field of characteristic zero. Every k-group locally of finite type over k is smooth over k.*

Indeed, the k-module $\omega_{G/k}$ is then plainly free. By 1.6, G is smooth over k at the identity point e, and hence smooth over k by 1.3.1.

[17] Editor's note: the implication (ii) $\Rightarrow$ (i) is a general fact (cf. EGA 0_{IV}, 14.1.6), while (i) $\Rightarrow$ (ii) follows from the fact that if X is an irreducible scheme of finite type over a field, then $\dim X = \dim U$ for every nonempty open subset U of X. The essential point here is therefore assertion (i), which was already established in an addition to Exposé VI$_A$, 2.4.1. We have modified the statement and proof accordingly.

[18] Editor's note: in the statement, we have replaced "along the identity section" by "at every point of the identity section." At the end of the proof, we have also made explicit the cited results from EGA IV$_4$.

2. "Open properties" of groups and group morphisms locally of finite presentation

2.0.— In all that follows, S will denote an arbitrary scheme; an S-group scheme will be called an S-group. Given an S-group G, we shall denote by ϵ the identity section, by

$$c : G \longrightarrow G$$

the inversion morphism, and by

$$\mu : G \times_S G \longrightarrow G$$

the multiplication morphism. For every S-scheme X, we shall denote by π, or by π_X, the structural morphism

$$X \longrightarrow S.$$

Given a property $\mathscr{P}(u)$ of a morphism of S-schemes

$$u : X \longrightarrow Y,$$

we shall say that $\mathscr{P}(u)$ is *stable under base change* if, whenever u satisfies $\mathscr{P}(u)$, the same is true of the morphism $u_{(Y')}$, for every S-morphism $Y' \to Y$. We say that $\mathscr{P}(u)$ is *local in nature for the topology* $\mathcal{T}$ (cf. Exposé IV, §§ 4 and 6) if $\mathscr{P}(u)$ satisfies the following two conditions:

a) $\mathscr{P}(u)$ is stable under base change;

b) whenever there is a family of S-morphisms $Y_i \to Y$, covering for the topology $\mathcal{T}$, such that each morphism $u_{(Y_i)}$ satisfies $\mathscr{P}(u)$, then u satisfies $\mathscr{P}(u)$.

Proposition 2.1.— *Let $\mathscr{P}(u)$ be a property of a morphism of S-schemes, local in nature for the* fpqc *topology. Let $u : G \to H$ be a morphism of S-groups. Suppose that G is flat and universally open over S.*

Let W be the largest open subset of H over which u satisfies the property $\mathscr{P}(u)$, and let $V = u^{-1}(W)$. Then

$$U = \pi_G(V)$$

is open in S, and V is an open group subscheme of $G|_U$.

The existence of a largest open subset W of H over which u satisfies $\mathscr{P}(u)$ follows from the fact that $\mathscr{P}(u)$ is local in nature for the Zariski topology. Since π_G is universally open, $\pi_G(V)$ is an open subset U of S. It is enough to show that V is a group subscheme of $G|_U$. We may therefore suppose that $U = S$.

Set

$$G' = G \times_S V, \qquad H' = H \times_S V, \qquad V' = V \times_S V, \qquad W' = W \times_S V,$$

and $u' = u_{(V)}$. Let W'_1 be the largest open subset of H' over which u' satisfies $\mathscr{P}(u)$. Since V is flat and universally open over S, the same is true of H' over H, and Lemma 2.1.1 below shows that

$$W'_1 = W'.$$

Now consider the automorphism a (respectively, b) of the V-scheme G' (respectively, H'), namely right translation by the inverse of the diagonal section δ (respectively, by the inverse of $u(\delta)$), defined by

$$a(g, v) = (g \cdot v^{-1}, v), \qquad \text{(respectively, } b(h, v) = (h \cdot u(v^{-1}), v)\text{)},$$

for every morphism $T \to S$, every $g \in G(T)$, $v \in V(T)$, and $h \in H(T)$. It is clear that

$$u' \circ a = b \circ u',$$

which shows that W' is stable under b, hence that V' is stable under a, so that V is a group subscheme of G.

Lemma 2.1.1.— *Let $\mathscr{P}(u)$ be a property of an S-morphism u, local in nature for the* fpqc *topology. Consider a cartesian square of morphisms of S-schemes:*

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d] & Y' \arrow[d, "g"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

where g is flat and open. Let W (respectively, W_1') be the largest open subset of Y (respectively, Y') over which f (respectively, $f' = f_{(Y')}$) satisfies $\mathscr{P}(u)$. Then

$$W_1' = W \times_Y Y'.$$

Set $W' = W \times_Y Y'$. Since $\mathscr{P}(u)$ is stable under base change, we have

$$W' \subset W_1'.$$

Since g is open,

$$W_1 = g(W_1')$$

is an open subset of Y. Set

$$V_1 = f^{-1}(W_1), \qquad V_1' = V_1 \times_{W_1} W_1';$$

it is clear that $V_1' = f'^{-1}(W_1')$. Since g is flat and open, the morphism

$$W_1' \longrightarrow W_1 = g(W_1')$$

induced by g is faithfully flat and open, hence is a covering for the fpqc topology (cf. Exposé IV, 6.3.1(iv)). Since the morphism

$$V_1' \longrightarrow W_1'$$

induced by f' satisfies $\mathscr{P}(u)$, the same is true of the morphism $V_1 \to W_1$ induced by f. Thus

$$W_1 \subset W, \qquad W_1' \subset g^{-1}(W_1) \subset g^{-1}(W) = W',$$

and therefore $W' = W_1'$.

Remark 2.1.2.— *A great many properties of a morphism are local in nature for the* fpqc *topology; among them are the properties of being flat, (universally) open, (locally) of finite type, of finite presentation, quasi-finite (cf. EGA IV_2, 2.5.1, 2.6.1, and 2.7.1), smooth, étale, and unramified (EGA IV_4, 17.7.3).*

The proof of 2.1 in fact uses only base changes by flat morphisms. The proposition therefore applies to a property satisfying condition b) *of 2.0 relative to the* fpqc *topology and stable under base change by flat morphisms (for example, the property of being quasi-compact and dominant).*

Of course, one can state an analogous proposition concerning properties local in nature for a topology $\mathcal{T}$ finer than the Zariski topology, the condition to be verified by G then being that π_G be universally open and covering for the topology $\mathcal{T}$.

In particular, if G is flat and locally of finite presentation over S, there is an analogous statement for properties stable under base change by flat morphisms locally of finite presentation and satisfying condition b) *of 2.0 relative to the* fpqc *topology (for example, the properties of being regular, reduced, Cohen–Macaulay, and so on; cf. EGA IV$_2$, 6.8).*

Proposition 2.2.— *Let G and H be two S-groups, and let $u : G \to H$ be a morphism of S-groups. Then:*[19]

(i) *Suppose that G or H is flat over S, and that G or H is locally of finite presentation over S. Let V be the largest open subset of G such that the restriction of u to V is flat and locally of finite presentation (respectively, smooth, or étale). Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

(ii) *Suppose that G or H is universally open over S, and let V be the largest open subset of G such that the restriction of u to V is universally open. Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

We first prove (i). We show that the restriction π_V of π_G to V is flat and locally of finite presentation:

a) if π_G is flat (respectively, locally of finite presentation), then so is π_V;

b) if π_H is flat (respectively, locally of finite presentation), then so is π_V, as the composite of the restriction of u to V with π_H.

Thus, in the four cases contemplated in the statement, π_V is flat and locally of finite presentation, hence universally open (EGA IV$_2$, 2.4.6). Set $U = \pi_V(V)$; then U is open in S. It is therefore enough to show that V is an open group subscheme of $G|_U$, and we may suppose that $U = S$.

Set

$$G' = G \times_S V, \qquad H' = H \times_S V, \qquad V' = V \times_S V, \qquad \text{and} \qquad u' = u_{(V)}.$$

Since V is flat and locally of finite presentation over S, the same is true of H' over H. By EGA IV$_4$, 17.7.4, V' is then the largest open subset of G' such that the restriction of u' to V' is flat and locally of finite presentation (respectively, smooth, or étale). With the automorphisms a and b defined as in the proof of 2.1, it is clear that V' is stable under a, hence that V is a group subscheme of G.

We now prove (ii). The restriction π_V of π_G to V is universally open, either because π_G is universally open, or because it is the composite of the restriction of u to V with π_H, when π_H is universally open. Set $U = \pi_V(V)$; then U is open in S. It is enough to show that V is an open group subscheme of $G|_U$, and we may suppose that $U = S$.

Set, as before,

$$G' = G \times_S V, \qquad H' = H \times_S V, \qquad V' = V \times_S V, \qquad \text{and} \qquad u' = u_{(V)}.$$

[19] Editor's note: in case (i), the open subset V consists of all the points of G at which u is smooth, respectively étale, respectively of finite presentation and flat; cf. editor's note 6. By contrast, in case (ii), V is the largest open subset contained in the set E of points of G at which u is universally open, but E is not necessarily open, as the following example shows (EGA IV$_3$, 14.1.3(i)). Let k be a field, let $H = S = \operatorname{Spec} A$, where $A = k[x]$, and let G be the S-group $\operatorname{Spec} A[y]/(xy)$. Then E is the identity section of G, which is not open.

Then $\pi_V : V \to S$ is surjective and universally open; the same is therefore true of $G' \to G$. Consequently, V' is the largest open subset of G' such that the restriction of u' to V' is universally open, by EGA IV$_3$, 14.3.4(i),(ii). With the automorphisms a and b defined as before, it is clear that V' is stable under a, hence that V is a group subscheme of G.

Corollary 2.3.— *Let G be an S-group, and let V be the largest open subset of G that is flat and locally of finite presentation (respectively, smooth, étale, or universally open) over S. Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

It is enough to apply 2.2 when H is the identity S-group and u is the unique morphism of S-groups $G \to H$, since π_H is then an isomorphism and

$$\pi_G = \pi_H \circ u.$$

Corollary 2.4.— *Let G be an S-group. If there is a neighborhood X of the identity section having one (or more) of the following properties:*

X is flat and locally of finite presentation (respectively, smooth, étale, or universally open) over S,

then there is an open group subscheme of G having the same properties.

It is enough to apply 2.3, observing that here, with the notation of 2.2, we have

$$\epsilon(S) \subset V,$$

and hence $U = S$.

Proposition 2.5.—[20] *Let $u : G \to H$ be a morphism of S-groups.*

(i) *Suppose that G (respectively, H) is of finite presentation and flat (respectively, of finite type) over S at the points of its identity section. Then the sets*

$$U_{\text{plat}} \supset U_{\text{lisse}} \supset U_{\text{ét.}},$$

formed by the points $s \in S$ such that u_s is flat (respectively, smooth, or étale), are open in S.

If, moreover, G (respectively, H) is locally of finite presentation and flat (respectively, locally of finite type) over S, then the set V_{plat} (respectively, V_{lisse} or $V_{\text{ét.}}$) of points of G at which u is flat (respectively, smooth, or étale) is the inverse image under π_G of U_{plat} (respectively, U_{lisse} or $U_{\text{ét.}}$).

(ii) *Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$ (a condition satisfied if G is of finite type over S at the points of the identity section* (1.3.1.1)*), and that u is locally of finite type (respectively, locally of finite presentation) at the points of the identity section of G. Then the sets*

$$U_{\text{l.q.f.}} \supset U_{\text{n.r.}},$$

formed by the $s \in S$ such that u_s is locally quasi-finite (respectively, unramified), are open in S.

If, moreover, u is locally of finite type (respectively, locally of finite presentation), then the set $V_{\text{l.q.f.}}$ (respectively, $V_{\text{n.r.}}$) of points of G at which u is quasi-finite (respectively, unramified) is the inverse image under π_G of $U_{\text{l.q.f.}}$ (respectively, $U_{\text{n.r.}}$).

[20] Editor's note: we have expanded the statement and proof, and in (i) we have weakened the hypothesis on H, replacing "of finite presentation" by "of finite type."

We prove (i). First note (1.3.1.1) that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Let Y be the open subset of H consisting of the points at which π_H is of finite type, and let X_1 be the open subset of $u^{-1}(Y)$ consisting of the points at which π_G is of finite presentation. By EGA IV_3, 11.3.1, the set X of points of X_1 at which π_G is of finite presentation and flat is open in X_1, hence in G.

Let

$$u_X : X \longrightarrow Y$$

be the morphism induced by u. Since Y (respectively, X) contains the identity section of H (respectively, G), the restriction π_X of π_G to X is surjective, and the morphism

$$S \longrightarrow X$$

induced by ϵ_G is an S-section of X.

Consider the sets

$$W_{\text{plat}} \supset W_{\text{lisse}} \supset W_{\text{ét.}},$$

consisting of the points of X at which u_X is flat (respectively, smooth, or étale). It is clear that W_{lisse} and $W_{\text{ét.}}$ are open in X; cf. Editor's note 6. On the other hand, since π_Y is locally of finite type and

$$\pi_X = \pi_Y \circ u_X$$

is locally of finite presentation, EGA IV_1, 1.4.3(v), shows that u_X is locally of finite presentation. Consequently, the flatness locus W_X of u_X is open in X (EGA IV_3, 11.3.1).

Let $x \in X$ and set $s = \pi_X(x)$. Then, by EGA IV_2, 11.3.10, combined with EGA IV_4, 17.5.1 (respectively, 17.6.1), x belongs to W_{plat} (respectively, W_{lisse} or $W_{\text{ét.}}$) if and only if u_s is flat (respectively, smooth, or étale) at the point x, or, equivalently by 1.3, if and only if u_s is flat (respectively, smooth, or étale).

Consequently, for

$$\text{“}\flat = \text{flat, smooth, or étale,”}$$

we have

$$U_\flat = \epsilon_G^{-1}(W_\flat), \qquad W_\flat = \pi_X^{-1}(U_\flat).$$

Since $W_\flat$ is open in X, hence in G, the first equality shows that $U_\flat$ is open in S.

The second assertion of (i) follows from what we have just seen, for then

$$Y = H, \qquad X = G, \qquad V_\flat = W_\flat = \pi_G^{-1}(U_\flat).$$

We prove assertion (ii). Let

$$V_{\text{l.t.f.}} \supset V_{\text{l.q.f.}}$$

be the open subsets of G consisting of the points at which u is of finite type (respectively, quasi-finite). Set

$$X = V_{\text{l.t.f.}},$$

and let u_X denote the restriction of u to X. By hypothesis, X contains $\epsilon(S)$. Let $x \in X$ and set $s = \pi_X(x)$.

If u is quasi-finite at x, then, by base change, u_s is quasi-finite at x; and since G_s is assumed to be locally of finite type, u_s is locally quasi-finite by 1.3.1.

Conversely, if u_s is locally quasi-finite, then

$$u_X^{-1}(u_X(x)) \subset u_s^{-1}(u_s(x))$$

is a finite set. Since

$$u_X : X \longrightarrow H$$

is locally of finite type, Chevalley's semicontinuity theorem (EGA IV_3, 13.1.3) shows that there is an open neighborhood W of x in X such that the fiber

$$u_X^{-1}(u_X(x'))$$

is discrete for every $x' \in W$. Hence u_X is quasi-finite at x, by EGA II, 6.2.2 (and EGA III_2, Err_{III} 20).

Consequently,

$$U_{\text{l.q.f.}} = \epsilon_G^{-1}(V_{\text{l.q.f.}}), \qquad V_{\text{l.q.f.}} = \pi_X^{-1}(U_{\text{l.q.f.}}),$$

and the first equality shows that $U_{\text{l.q.f.}}$ is open in S.

If, moreover, u is locally of finite type over S, then $X = G$, and hence $V_{\text{l.q.f.}}$ is the inverse image under π_G of $U_{\text{l.q.f.}}$. This proves the first half of (ii).

Now consider the open subsets

$$V_{\text{l.p.f.}} \supset V_{\text{n.r.}},$$

consisting of the points at which u is of finite presentation (respectively, unramified), and suppose that

$$Z = V_{\text{l.p.f.}}$$

contains the identity section of G. Let $z \in Z$; set

$$s = \pi_Z(z), \qquad h = u_Z(z).$$

If u is unramified at z, then, by base change, u_s is unramified at z; and since G_s is assumed to be locally of finite type, u_s is unramified by 1.3.1.

Conversely, if u_s is unramified at the point z, then the fiber

$$u_s^{-1}(h) = u^{-1}(h)$$

is unramified over $\kappa(h)$ at the point z. Since Z is an open subset of G, the fiber

$$u_Z^{-1}(h) = u_s^{-1}(h) \cap Z$$

is unramified over $\kappa(h)$ at the point z. Since u_Z is locally of finite presentation, EGA IV_4, 17.4.1, implies that u_Z is unramified at the point z.

Thus

$$U_{\text{n.r.}} = \epsilon_G^{-1}(V_{\text{n.r.}}), \qquad V_{\text{n.r.}} = \pi_Z^{-1}(U_{\text{n.r.}}),$$

and the first equality shows that $U_{\text{n.r.}}$ is open in S.

If, moreover, u is locally of finite presentation over S, then $Z = G$, and hence $V_{\text{n.r.}}$ is the inverse image under π_G of $U_{\text{n.r.}}$. This completes the proof of assertion (ii).

Corollary 2.6.— *Let $u : G \to H$ be a morphism of S-groups that is radicial (as is the case if u is a monomorphism; EGA I, 3.5.4). Suppose that G (respectively, H) is locally of finite presentation and flat (respectively, locally of finite type) over S. Then the set U of $s \in S$ such that u_s is an open immersion is open in S, and the restriction of u over U is an open immersion.*

By 2.5(i), the set U' of points $s \in S$ such that u_s is étale is open in S. Since u is radicial, so is u_s, for every $s \in S$; hence EGA IV_4, 17.9.1, gives

$$U = U',$$

which shows that U is open. Finally, by 2.5(i), the restriction of u over U is étale; since u is radicial, this restriction is an open immersion (EGA IV_4, 17.9.1).

Proposition 2.7.— *Let G be an S-group. The following conditions are equivalent:*

(i) *G is unramified over S at the points of the identity section;*

(ii) *the identity section is an open immersion;*

(iii) *G is of finite presentation over S at the points of the identity section, and, for every $s \in S$, G_s is unramified over $\kappa(s)$.*

If, moreover, G is locally of finite presentation over S, then each of the three preceding conditions is equivalent to the following:

(iv) *G is unramified over S.*

The equivalence of assertions (i) and (ii) follows from the more general Lemma 2.7.1 below. Note (1.3.1.1) that either condition (i) or condition (iii) implies that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Thus, by EGA IV_4, 17.4.1, assertion (i) is equivalent to the assertion that, for every $s \in S$, G_s is unramified over $\kappa(s)$ at the point e_s, the identity of G_s, or, equivalently by 1.3.1, to the assertion that G_s is unramified over $\kappa(s)$. Consequently, assertions (i) and (iii) are equivalent. Finally, if G is locally of finite presentation over S, assertions (iii) and (iv) are equivalent (EGA IV_4, 17.4.1).

Lemma 2.7.1.— *Let G be an S-scheme equipped with a section ϵ. In order that G be unramified over S at the points of this section, it is necessary and sufficient that ϵ be an open immersion.*

The condition is necessary by EGA IV_4, 17.4.1, a)⇒b″). Conversely, if ϵ is an open immersion, then the restriction to $\epsilon(S)$ of the structural morphism

$$G \longrightarrow S$$

is an isomorphism; hence G is unramified over S at the points of $\epsilon(S)$.

Corollary 2.8.—[21] *Let $u : G \to H$ be a morphism of S-groups. Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$.*

a) *If u is locally of finite type, the following conditions are equivalent:*

(i) *u is locally quasi-finite;*

(ii) *for every $s \in S$, the morphism $u_s : G_s \to H_s$ is locally quasi-finite;*

(iii) $\operatorname{Ker} u$ *is locally quasi-finite over S;*

(iv) *the fibers of* $\operatorname{Ker} u$ *are discrete.*

b) *If u is locally of finite presentation, the following conditions are equivalent:*

(v) *u is unramified;*

(vi) *for every $s \in S$, the morphism $u_s : G_s \to H_s$ is unramified;*

(vii) $\operatorname{Ker} u$ *is unramified;*

(viii) *the identity section $S \to \operatorname{Ker} u$ is an open immersion.*

The equivalences

$$\text{(i)} \Longleftrightarrow \text{(ii)} \qquad \text{and} \qquad \text{(v)} \Longleftrightarrow \text{(vi)}$$

follow from 2.5(ii), and Reminder 2.8.1 below shows that

$$\text{(i)} \Longrightarrow \text{(iii)} \qquad \text{and} \qquad \text{(v)} \Longrightarrow \text{(vii)}.$$

[21] Editor's note: we have changed the presentation by separating the assertions concerning the "quasi-finite" case from those concerning the "unramified" case.

For every $s \in S$, denote by e_s the identity element of H_s. Then (iii) (respectively, (vii)) implies that, for every $s \in S$,

$$(\operatorname{Ker} u)_s = \operatorname{Ker} u_s = u_s^{-1}(e_s)$$

is locally quasi-finite (respectively, unramified) over $\kappa(s) = \kappa(e_s)$. It follows that u_s is quasi-finite (respectively, unramified) at the identity point of G_s, and hence, by 1.3, that u_s is locally quasi-finite (respectively, unramified). Thus

$$\text{(iii)} \Longrightarrow \text{(ii)} \qquad \text{and} \qquad \text{(vii)} \Longrightarrow \text{(vi)}.$$

Finally,

$$\text{(ii)} \Longleftrightarrow \text{(iv)} \quad \text{by 1.4.1,} \qquad \text{(vii)} \Longleftrightarrow \text{(viii)} \quad \text{by 2.7.}$$

Reminder 2.8.1.— Recall (cf. Exposé I, 2.3.6.1) that, given a morphism $u : G \to H$ of S-groups, the *kernel of* u, denoted $\operatorname{Ker} u$, is the group subfunctor of G defined, for every morphism $f : T \to S$, by

$$(\operatorname{Ker} u)(T) = \{a \in G(T) \mid u \circ a = \epsilon_H \circ f\}.$$

By loc. cit. (or EGA I, 4.4.1), this functor is represented by the S-group

$$G \times_H S = u^{-1}(\epsilon_H(S)),$$

which is denoted simply by $\operatorname{Ker} u$. In particular, the structural morphism

$$\operatorname{Ker} u \longrightarrow S$$

is obtained from u by base change.

Lemma 2.9.— *Let $\pi : X \to S$ be a morphism admitting an S-section ϵ.*

(i) *If π is injective, then it is integral.*(*)

(ii) *If π is locally of finite type and, for every $s \in S$, π_s is an isomorphism, then π is an isomorphism.*(**)

First observe that, by Lemma 2.9.1 below, π, being a homeomorphism, is an affine morphism.

If π is injective, then ϵ is surjective. Since ϵ is a surjective immersion, $\epsilon(S)$ is isomorphic to the closed subscheme of X defined by a nil ideal $\mathcal{I}$ of $\mathcal{O}_X$. Since ϵ is an S-section of π, there is a direct-sum decomposition

$$\mathcal{O}_X = \mathcal{O}_S \oplus \mathcal{I}$$

of $\mathcal{O}_S$-modules. Since $\mathcal{I}$ is a nil ideal of $\mathcal{O}_X$, it is plainly integral over $\mathcal{O}_S$. Thus $\mathcal{O}_X$ is integral over $\mathcal{O}_S$, and π is integral.

Now suppose that π is locally of finite type. Since $\pi \circ \epsilon = \mathrm{id}_S$, the section ϵ is locally of finite presentation (cf. EGA IV$_1$, 1.4.3(v)); hence $\mathcal{I}$ is an ideal of finite type in $\mathcal{O}_X$ (EGA IV$_1$, 1.4.1). For every $s \in S$, one has

$$\mathcal{O}_{X_s} = \kappa(s) \oplus \mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s).$$

(*) This is also a special case of EGA IV$_4$, 18.12.11, since π is plainly a universal homeomorphism.

(**) This is also an immediate consequence of EGA IV$_4$, 18.12.6.

By hypothesis, ϵ_s is an isomorphism, so

$$\mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s) = 0$$

for every $s \in S$, and therefore, a fortiori,

$$\mathcal{I} \otimes_{\mathcal{O}_X} \kappa(x) = 0$$

for every $x \in X$. Nakayama's lemma now gives $\mathcal{I} = 0$, and consequently π is an isomorphism.

Lemma 2.9.1.— *Let $f : X \to Y$ be a morphism of schemes that is a homeomorphism. Then f is an affine morphism.*(***)

It is enough to show that every point $y \in Y$ has an open neighborhood W such that the restriction of f over W is affine. Let $y \in Y$, and let V be an affine open neighborhood of y in Y. Set

$$V' = f^{-1}(V).$$

Then V' is an open neighborhood of $x = f^{-1}(y)$ in X. There is an affine open neighborhood W' of x in X contained in V'. Set

$$W = f(W').$$

Then W is an open neighborhood of y in Y, contained in the affine scheme V, and therefore W is separated. Since W' is affine, the restriction of f over W is an affine morphism (EGA II, 1.2.3).

Corollary 2.10.— *Let G be an S-group locally of finite type. Suppose that, for every $s \in S$, G_s is the identity $\kappa(s)$-group. Then G is the identity S-group.*

More generally:

Corollary 2.11.— *Let $f : X \to Y$ be an S-morphism locally of finite type. In order that f be a monomorphism, it is necessary and sufficient that f_s be a monomorphism for every $s \in S$.*

The condition is plainly necessary; we prove that it is sufficient. If f_s is a monomorphism for every $s \in S$, then, a fortiori, f_y is a monomorphism for every $y \in Y$. We may therefore assume that $Y = S$.

By EGA I, 5.3.8, to prove that f is a monomorphism it is enough to show that

$$\Delta_f : X \longrightarrow X \times_S X$$

is an isomorphism, or, equivalently, that the first projection

$$p : X \times_S X \longrightarrow X$$

is an isomorphism. If f_s is a monomorphism, EGA I, 5.3.8 likewise shows that the first projection

$$X_s \times_{\kappa(s)} X_s \longrightarrow X_s,$$

which is identified with p_s, is an isomorphism. But p has the S-section Δ_f; hence Lemma 2.9 shows that, if p_s is an isomorphism for every $s \in S$, then p itself is an isomorphism.

Corollary 2.12.— *Let G be an S-scheme admitting an S-section ϵ, and suppose that the structural morphism $\pi : G \to S$ is closed. Let $s \in S$ be such that π is of finite presentation at $\epsilon(s)$, and such that*

$$\epsilon_s : \operatorname{Spec}(\kappa(s)) \longrightarrow G_s$$

(***) Cf. EGA IV_4, 18.12.7.1 for a slightly more general result, proved by the same argument.

is an isomorphism (or, equivalently, such that

$$\pi_s : G_s \longrightarrow \operatorname{Spec} \kappa(s)$$

is an isomorphism). Then there exists an open neighborhood U of s in S such that

$$\epsilon|_U : U \longrightarrow G \times_S U$$

is an isomorphism.

Let V be the set of points of G at which π is unramified. This set is known to be open (cf. Editor's note 6), and it contains $\epsilon(s)$. Hence

$$W = \epsilon^{-1}(V)$$

is an open subset of S containing s, and π is unramified at $\epsilon(t)$ for every $t \in W$. Since π is closed, the same is true of $\pi|_W$, and we may therefore suppose that $W = S$.

By 2.7.1,

$$X = G - \epsilon(S)$$

is then a closed subset of G that does not meet G_s. Since π is closed,

$$F = \pi(X)$$

is a closed subset of S that does not contain s. Set

$$U = S - F.$$

Then U is an open subset of S, and $\epsilon|_U$ is an isomorphism from U onto $G|_U$.

3. Identity component of a group locally of finite presentation

3.0.— Given a subset A (respectively, B) of an S-scheme X (respectively, Y), by abuse of notation

$$A \times_S B$$

will denote the subset of $X \times_S Y$ consisting of the points whose first projection belongs to A and whose second projection belongs to B.

Given a subset A of an S-group G, we shall say that A is *stable under the group law of G* if

$$c(A) \subset A \qquad \text{and} \qquad \mu(A \times_S A) \subset A.$$

Definition 3.1.— Let G be an S-functor in groups satisfying the following condition:

$$\text{for every } s \in S, \text{ the functor } G_s = G \otimes_S \kappa(s) \text{ is representable.} \tag{+}$$

We then denote by $\underline{G_s^0}$ the connected component of the identity element of the $\kappa(s)$-group G_s.[22] We define a sub-S-functor in groups of G, called the *identity component of G* and denoted by G^0, by setting, for every morphism $T \to S$,

$$G^0(T) = \{u \in G(T) \mid \text{for every } s \in S,\ u_s(T_s) \subset \underline{G_s^0}\}.$$

[22]Editor's note: this is a slightly abusive notation, but it is compatible with the notation of Exposé VI$_A$, 2.3 when this connected component is the underlying topological space of an open group subscheme G^0 of G; cf. Exposé VI$_A$, 2.2.bis.

Thus we have defined the functor

$$G \longmapsto G^0$$

from

$$(\widehat{\mathbf{Sch}}_{/S})\text{-gr.} \quad \text{to} \quad (\widehat{\mathbf{Sch}}_{/S})\text{-gr.}$$

Remark 3.2.— (i) Let G be an S-functor in groups satisfying condition (+). Then

$$\operatorname{Lie}(G^0/S) = \operatorname{Lie}(G/S)$$

by Exposé II.[23] Indeed, for every S-scheme T, denote by I_T the T-scheme of dual numbers over T, and by

$$\tau : T \longrightarrow I_T$$

the zero section (cf. Exposé II, 2.1). By definition,

$$\operatorname{Lie}(G/S)(T) = \{u \in G(I_T) \mid u \circ \tau = e\},$$

where $e : T \to G$ denotes the composite of $T \to S$ with the identity section $S \to G$. Likewise,

$$\begin{aligned}\operatorname{Lie}(G^0/S)(T) &= \{u \in G^0(I_T) \mid u \circ \tau = e\} \\ &= \big\{u \in G(I_T) \bigm| u \circ \tau = e \text{ and } u_s\big((I_T)_s\big) \subset \underline{G^0_s} \text{ for every } s \in S\big\}.\end{aligned}$$

For every $s \in S$, the schemes T_s and

$$(I_T)_s = I_{T_s}$$

have the same underlying set. Hence, if $u \in \operatorname{Lie}(G/S)(T)$, then

$$u_s(I_{T_s}) = e_s,$$

where e_s denotes the identity point of G_s, and therefore $u \in \operatorname{Lie}(G^0/S)(T)$. Thus the inclusion

$$\operatorname{Lie}(G^0/S)(T) \subset \operatorname{Lie}(G/S)(T)$$

is an equality for every T, and consequently

$$\operatorname{Lie}(G^0/S) = \operatorname{Lie}(G/S).$$

(ii) Let G and G' be two S-functors in groups satisfying (+). Then:

a) if $G \subset G'$, then $G^0 \subset G'^0$;

b) if $G \subset G'$ and $G'^0 \subset G$, then $G^0 = G'^0$;

c) if, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$, then G^0 satisfies condition (+), by Exposé VI_A, 2.3, and

$$(G^0)^0 = G^0.$$

Proposition 3.3.— *Let G be an S-functor in groups satisfying condition* (+), *and let S' be an S-scheme. Then*

$$(G \times_S S')^0 = G^0 \times_S S'.$$

In other words, the functor $G \mapsto G^0$ commutes with base change; that is, the following diagram is commutative:

[23] Editor's note: we have added what follows.

$$\begin{tikzcd}
(\widehat{\mathrm{Sch}}_{/S})\text{-gr.} \arrow[r, "G\mapsto G^0"] \arrow[d] & (\widehat{\mathrm{Sch}}_{/S})\text{-gr.} \arrow[d] \\
(\widehat{\mathrm{Sch}}_{/S'})\text{-gr.} \arrow[r] & (\widehat{\mathrm{Sch}}_{/S'})\text{-gr.}
\end{tikzcd}$$

It is enough to check that, for every $s' \in S'$, whose image in S we denote by s,

$$\begin{aligned}\left((G \times_S S') \otimes_{S'} \kappa(s')\right)^0 &= \left(G_s \otimes_{\kappa(s)} \kappa(s')\right)^0 \\ &= G_s^0 \otimes_{\kappa(s)} \kappa(s').\end{aligned}$$

This follows from Exposé VI$_A$, 2.1.2. Note, for later use in 4.2, that we have not used the group structure of G_s, but only the fact that G_s^0 has a rational point, namely $\epsilon(s)$, and is therefore geometrically connected (see also EGA IV$_2$, 4.5.14).

Special Case 3.4.— Let G be an S-group scheme, and denote by $\underline{G^0}$ the subset of G equal to the union of the $\underline{G_s^0}$ as s runs through S. Then $\underline{G^0}$ is a subset of G stable under the group law of G (cf. 3.0), and, for every morphism $S' \to S$,

$$G^0(S') = \{u \in G(S') \mid u(S') \subset \underline{G^0}\}.$$

When $\underline{G^0}$ is an open subset of G, it is clear that G^0 is represented by the subscheme of G induced on the open subset $\underline{G^0}$.

Proposition 3.5.— *Let S be a quasi-compact and quasi-separated scheme, and let G be an S-group with fibers locally of finite type. Then there is a quasi-compact open subset U of G containing $\underline{G^0}$.*

Since the identity section ϵ is an immersion, $\epsilon(S)$ is a quasi-compact subspace of G. Thus there is a quasi-compact open subset V of G containing $\epsilon(S)$. Since S is quasi-separated and V is quasi-compact, V is quasi-compact over S (EGA IV$_1$, 1.2.4). Hence $V \times_S V$ is quasi-compact over S, and therefore quasi-compact. It follows that

$$V \cdot V = \mu(V \times_S V)$$

is quasi-compact. Set

$$V_s = V \cap G_s, \qquad V_s^0 = V \cap \underline{G_s^0}.$$

Then V_s^0 is open in $\underline{G_s^0}$, and is dense in $\underline{G_s^0}$ because G_s^0 is irreducible (Exposé VI$_A$, 2.4). Therefore

$$V_s^0 \cdot V_s^0 = \underline{G_s^0} \qquad \text{(Exposé VI}_A\text{, 0.5)},$$

which shows that

$$V_s \cdot V_s \supset \underline{G_s^0}, \qquad \text{and hence} \qquad V \cdot V \supset \underline{G^0}.$$

Finally, since $V \cdot V$ is quasi-compact, there is a quasi-compact open subset U of G containing $V \cdot V$, and therefore containing $\underline{G^0}$.

Corollary 3.6.— *Let G be an S-group whose fibers are locally of finite type and connected. Then G is quasi-compact over S.*

The assertion is local on S (EGA I, 6.6.1), so we are reduced to the case where S is affine. By 3.5, there is then a quasi-compact open subset U of G containing

$$\underline{G^0} = G.$$

Thus G is quasi-compact, and hence quasi-compact over the affine scheme S (EGA I, 6.6.4(v)).

Proposition 3.7.—[24] *Let G be an S-group locally of finite presentation. Then:*

(i) *$\underline{G}^0$ is ind-constructible in G.*

(ii) *If, moreover, G is quasi-separated over S, and S is quasi-compact and quasi-separated, then $\underline{G}^0$ is constructible.*

(iii) *Consequently, if G is quasi-separated over S, then $\underline{G}^0$ is locally constructible.*

We first prove the first assertion. Since

$$\pi : G \longrightarrow S$$

is locally of finite presentation, given $s \in S$, there are an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s such that

$$\pi(U) \subset V$$

and the induced morphism

$$\pi' : U \longrightarrow V$$

is of finite presentation. Then

$$T = \epsilon^{-1}(U)$$

is an open subset of S. Set

$$W = (\pi')^{-1}(T), \qquad \pi'' = \pi'|_W.$$

Thus $\pi'' : W \to T$ is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ.

For every $t \in T$, since $\underline{G}_t^0$ is irreducible (VI$_A$, 2.4), $W \cap \underline{G}_t^0$ is dense in $\underline{G}_t^0$, hence irreducible and therefore connected. It is thus the connected component of $(\pi'')^{-1}(t)$ containing $\epsilon''(t)$. It follows from EGA IV$_3$, 9.7.12, that the union W^0 of the $W \cap \underline{G}_t^0$, for $t \in T$, is locally constructible in W. On the other hand, VI$_A$, 0.5, gives

$$\underline{G}^0 \cap \pi^{-1}(T) = W^0 \cdot W^0;$$

that is, $\underline{G}^0 \cap \pi^{-1}(T)$ is the image of

$$W^0 \times_T W^0$$

under the morphism

$$\mu'' : W \times_T W \longrightarrow \pi^{-1}(T)$$

[24] Editor's note: we recall the following definitions and results (cf. EGA 0$_{\text{III}}$, §9.1, and EGA IV$_1$, §§1.8 and 1.9). Let X be a topological space. (i) An open subset U of X is called *retrocompact* if the inclusion $U \hookrightarrow X$ is quasi-compact, i.e. if $U \cap V$ is quasi-compact for every quasi-compact open subset $V \subset X$. (ii) A subset C of X is called *constructible* if it is a finite union of subsets of the form $U - V$, where U and V are retrocompact open subsets of X. (iii) A subset L of X is called *locally constructible* if, for every $x \in X$, there is an open neighborhood U of x in X such that $L \cap U$ is constructible in U. (If X is quasi-compact and quasi-separated, then L is constructible.) (iv) A subset I of X is called *ind-constructible* if, for every $x \in X$, there is an open neighborhood U of x in X such that $I \cap U$ is a union of locally constructible subsets of U. Now let $f : X \to Y$ be a morphism of schemes. By Chevalley's constructibility theorem (cf. EGA IV$_1$, 1.8.4 and 1.9.5(viii)), if f is of finite presentation (respectively, locally of finite presentation), then the image under f of every locally constructible subset is locally constructible (respectively, ind-constructible).

induced by μ.[25] Since $W \times_T W$ (respectively, $\pi^{-1}(T)$) is of finite presentation (respectively, locally of finite presentation) over T, the morphism μ'' is locally of finite presentation and quasi-separated, by EGA IV$_1$, 1.4.3 and 1.2.2. If, moreover, $\pi^{-1}(T)$ is quasi-separated over T, then μ'' is quasi-compact (loc. cit., 1.2.4), hence of finite presentation. Since $W^0 \times_T W^0$ is locally constructible in $W \times_T W$ (because W^0 is locally constructible in W), Chevalley's constructibility theorem (loc. cit., 1.8.4 and 1.9.5(viii)) shows that $\underline{G^0} \cap \pi^{-1}(T)$ is ind-constructible in $\pi^{-1}(T)$, and is locally constructible in $\pi^{-1}(T)$ if G, and hence $\pi^{-1}(T)$, is quasi-separated over T. This proves assertions (i) and (iii).

Now suppose that G is quasi-separated over S, and that S is quasi-compact and quasi-separated. By 3.5, there is a quasi-compact open subset U of G containing $\underline{G^0}$. Since G is quasi-separated over S, G is quasi-separated; hence the open subset U is retrocompact (EGA IV$_1$, 1.2.7), and it suffices to show that $\underline{G^0}$ is constructible in U (EGA 0_{III}, 9.1.8).

Moreover, U is quasi-compact, hence quasi-compact over S (EGA IV$_1$, 1.2.4), and is quasi-separated over S. Therefore the restriction of π to U is of finite presentation. By EGA IV$_3$, 9.7.12, $\underline{G^0}$ is locally constructible in U, hence constructible in U, since U is quasi-compact and quasi-separated (EGA IV$_1$, 1.8.1). This proves (ii), and it follows that, for every quasi-compact and quasi-separated open subset T of S (for example, every affine open subset of S),

$$\underline{G^0} \cap \pi^{-1}(T)$$

is constructible.

Corollary 3.8.— *Let S_0 be a quasi-compact and quasi-separated scheme, I an increasing filtered preordered set, $(\mathcal{A}_i)_{i \in I}$ a direct system of commutative quasi-coherent $\mathcal{O}_{S_0}$-algebras,*

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \qquad S_i = \operatorname{Spec} \mathcal{A}_i \quad (i \in I), \qquad S = \operatorname{Spec} \mathcal{A}$$

(cf. EGA II, 1.3.1). *Let G be an S_0-group scheme locally of finite presentation. Then the canonical map*

$$\varinjlim_i G^0(S_i) \longrightarrow G^0(S)$$

is bijective.

Since G is locally of finite presentation over S_0, the canonical map

$$\varinjlim_i G(S_i) \longrightarrow G(S)$$

is bijective, by EGA IV$_2$, 8.14.2(c). It follows immediately that the canonical map

$$\varinjlim_i G^0(S_i) \longrightarrow G^0(S)$$

is injective. We prove that it is surjective. Let

$$g \in G^0(S) \subset G(S).$$

There is an $i \in I$ such that g factors, by means of an element $g_i \in G(S_i)$, through $S \to S_i$. By hypothesis,

$$g^{-1}(\underline{G^0}) = S.$$

But $\underline{G^0}$ is ind-constructible in G, by 3.7, so

$$g_i^{-1}(\underline{G^0})$$

[25] Editor's note: we have expanded the original in what follows.

is ind-constructible in S_i. It follows from EGA IV$_2$, 8.3.4, that there is an index $j \geq i$ such that

$$g_j^{-1}(\underline{G^0}) = S_j,$$

where g_j is the map induced by g_i under the base change $S_j \to S_i$. Thus $g_j \in G^0(S_j)$, and hence g comes from an element of

$$\varinjlim_i G^0(S_i).$$

Proposition 3.9.— *Let G be an S-group locally of finite presentation. Suppose that G^0 is representable. Then the canonical morphism*

$$i : G^0 \longrightarrow G$$

is an open immersion; moreover, G^0 is quasi-compact over S.

Since G^0 is a subfunctor of the functor G, the morphism i is a monomorphism, hence is radicial. From the definition of the functor G^0, one checks immediately from the definition (EGA IV$_4$, 17.1.1) that i is formally étale (noting that $\underline{G^0}$ is the image of i in G). Finally, by the characterization (EGA IV$_3$, 8.14.2(c)) of S-schemes locally of finite presentation in terms of the functors they represent, and by 3.8, G^0 is locally of finite presentation over S, since G is. Hence i is locally of finite presentation (cf. EGA IV$_1$, 1.4.3). It is therefore a radicial étale morphism, and hence an open immersion (EGA IV$_4$, 17.9.1).

The last assertion follows from 3.6.

Theorem 3.10.— *Let G be an S-group. The following conditions are equivalent:*

(i) *G is smooth over S at the points of the identity section.*

(ii) *G is flat and locally of finite presentation over S at the points of the identity section, and, for every $s \in S$, G_s is smooth over $\kappa(s)$.*

(iii) *There is an open group subscheme G' of G that is smooth over S.*

(iv) *G^0 is representable by an open subscheme of G that is smooth over S.*

It is clear that

$$\text{(iv)} \Longrightarrow \text{(iii)} \Longrightarrow \text{(i)},$$

and, by 1.3.1 and 2.4, (i) implies both (ii) and (iii). Moreover, (ii) implies (i) by EGA IV$_4$, 17.5.1.

It remains to prove that (iii) implies (iv). Lemma 3.10.1 below shows that G' contains $\underline{G^0}$ and that

$$G'^0 = G^0.$$

It therefore suffices to show that $\underline{G^0}$ is open in G, for we have already seen in 3.4 that G^0 will then be represented by the smooth group subscheme induced by G' on the open subset

$$\underline{G'^0} = \underline{G^0}.$$

We may thus suppose that $G' = G$.

To show that $\underline{G^0}$ is open, it suffices to show that every $s \in S$ has a neighborhood T in S such that

$$\underline{G^0} \cap \pi^{-1}(T)$$

is open in $\pi^{-1}(T)$. Let $s \in S$. Since $G = G'$, the morphism π is locally of finite presentation. As in the proof of 3.7, we may therefore construct an open subset T of S containing s, and an open subset W of G containing $\epsilon(s)$, such that the induced morphism

$$\pi'' : W \longrightarrow T$$

is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ. For every $t \in T$,

$$W \cap \underline{G_t^0}$$

is then the connected component of $(\pi'')^{-1}(t)$ containing $\epsilon''(t)$. Since π is smooth, so is π'', which is therefore smooth and of finite presentation. By EGA IV$_3$, 15.6.5, the union W^0 of the $W \cap \underline{G_t^0}$, for $t \in T$, is open in W.

On the other hand, VI$_A$, 0.5, gives

$$W^0 \cdot W^0 = \underline{G^0} \cap \pi^{-1}(T),$$

and we must show that this is open in $\pi^{-1}(T)$. We may now suppose that $T = S$; it remains to show that $W^0 \cdot W^0$ is open in G. Since π is universally open, so is μ (VI$_A$, 0.1). Hence, since W^0 is open in G, so is

$$W^0 \cdot W^0 = \mu(W^0 \times_S W^0).$$

This result will be strengthened in 4.4.

Lemma 3.10.1.— *Let G be an S-group whose fibers are locally of finite type. Then every open group subscheme H of G contains $\underline{G^0}$ and satisfies*

$$G^0 = H^0.$$

Let $s \in S$, and set

$$G'_s = H_s \cap \underline{G_s^0}.$$

Then G'_s is an open subset of $\underline{G_s^0}$, and is dense in $\underline{G_s^0}$ since $\underline{G_s^0}$ is irreducible (VI$_A$, 2.4). Hence

$$G'_s \cdot G'_s = \underline{G_s^0}$$

by VI$_A$, 0.5, which shows that

$$\underline{G_s^0} = G'_s \cdot G'_s \subset H_s \cdot H_s = H_s.$$

Thus $\underline{G_s^0} = \underline{H_s^0}$ for every s, whence

$$\underline{G^0} \subset H, \qquad G^0 = H^0.$$

Proposition 3.11.—[26] *Let $u : G \to H$ be a morphism between S-groups locally of finite presentation. If u is flat, the induced map*

$$u^0 : G^0 \longrightarrow H^0$$

is surjective. The converse holds if G is flat over S and H has reduced fibers.

[26] Editor's note: compare VI$_A$, 5.6.

Suppose that u is flat. Then, for every $s \in S$, u_s is flat and locally of finite presentation, hence open (EGA IV_2, 2.4.6). Therefore

$$u^0_s : \underline{G^0_s} \longrightarrow \underline{H^0_s}$$

is surjective, by 1.3.2.[27] Thus the map

$$u^0 : G^0 \longrightarrow H^0$$

is surjective.

Conversely, suppose that $u^0 : G^0 \to H^0$ is surjective, that G is flat over S, and that H has reduced fibers. Then, for every $s \in S$, the morphism

$$u^0_s : \underline{G^0_s} \longrightarrow \underline{H^0_s}$$

is surjective, and hence is flat at every point lying above the generic point η of $\underline{H^0_s}$, since

$$\mathcal{O}_{\underline{H^0_s},\eta}$$

is a field. It follows from 1.3 that u_s is flat. Therefore u is flat, by the fiberwise flatness criterion (EGA IV_3, 11.3.11).

4. Dimension of the fibers of groups locally of finite presentation

Proposition 4.1.— *Let G be an S-scheme locally of finite type, equipped with an S-section ϵ, and suppose that, for every $s \in S$,*

$$\dim G_s = \dim_{\epsilon(s)} G_s$$

(as is the case when G is an S-group; 1.5).

(i) *The function*

$$s \longmapsto \dim G_s$$

is upper semicontinuous on S.

(ii) *If, moreover, G is locally of finite presentation over S, this function is locally constructible.*

Let

$$\pi : G \longrightarrow S$$

be the structural morphism. Chevalley's semicontinuity theorem (EGA IV_3, 13.1.3) asserts that the function

$$x \longmapsto \dim_x \pi^{-1}\big(\pi(x)\big)$$

is upper semicontinuous on G. For every $s \in S$, however,

$$\dim G_s = \dim \pi^{-1}(s) = \dim_{\epsilon(s)} \pi^{-1}\big(\pi(\epsilon(s))\big).$$

[27] Editor's note: we have shortened the original by referring here to 1.3.2. We have also simplified the argument below by omitting a reference to the generic flatness theorem (EGA IV_2, 6.9.1), which is not needed here.

Since the map $s \mapsto \epsilon(s)$ is continuous on S, the composite function

$$s \longmapsto \dim G_s$$

is upper semicontinuous on S.

Now suppose that G is locally of finite presentation over S. Arguing as above, to show that $s \mapsto \dim G_s$ is locally constructible, it is enough to show that the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x))$$

is locally constructible on G, which follows from EGA IV$_3$, 9.9.1.

Proposition 4.2.— *Let $\pi : G \to S$ be an S-scheme locally of finite presentation, equipped with an S-section ϵ, and satisfying the following two conditions* (which hold when G is an S-group, by 1.5 and VI$_A$, 2.4.1)*:*

(a) *For every $s \in S$ and every $x \in G_s$, one has*

$$\dim G_s = \dim_x G_s$$

(or equivalently (1.5), for every $s \in S$, all the irreducible components of G_s have the same dimension).

(b) *For every $s \in S$, if $\underline{G_s^0}$ denotes the connected component of G_s containing $\epsilon(s)$, then $\underline{G_s^0}$ is geometrically irreducible.*

Let $s \in S$. The following conditions are equivalent:

(i) *G is universally open over S at the points of $\underline{G_s^0}$.*

(i bis) *G is universally open over S at every point of a neighborhood of $\epsilon(s)$ in $\underline{G_s^0}$.*[28]

(ii) *The function*

$$t \longmapsto \dim G_t$$

is constant in a neighborhood of s in S.

(iii) *$\underline{G^0}$ is "universally open over S at the points of $\underline{G_s^0}$," that is, given $S' \to S$, $s' \in S'_s$, $g \in \underline{G_{s'}^0}$, and an open neighborhood V of g in $G' = G_{S'}$, then*

$$\pi(V \cap \underline{G'^0})$$

is an open neighborhood of s' in S'.[29]

Clearly (i) implies (i bis). By EGA IV$_3$, 14.3.3.1(ii), the set of points of $\underline{G_s^0}$ at which π_G is universally open is closed in $\underline{G_s^0}$. Since $\underline{G_s^0}$ is irreducible, it follows that (i bis) implies (i).

We show that (i) implies (ii). Let T be the set of $t \in S$ such that

$$\dim G_t = \dim G_s.$$

By 4.1, T is locally constructible; hence, by EGA IV$_1$, 1.10.1, to show that T is a neighborhood of s, it is enough to show that every generalization s' of s belongs to T.

[28] Editor's note: the original reads "at the points of $\epsilon(s)$." It is not enough to suppose that G is universally open over S at $\epsilon(s)$, as the example given in editor's note 19 shows. We have corrected the proof accordingly.

[29] Editor's note: we have added condition (iii), pointed out by O. Gabber; it will be useful later (5.7).

Let x be the generic point of $\underline{G^0_s}$, and let U be an open subset of G containing x. Since π_G is universally open at x, EGA IV$_3$, 14.3.13 gives, for every generalization s' of s,

$$\dim(U \cap G_{s'}) \geq \dim_x(U \cap G_s).$$

In view of hypothesis (a), this implies

$$\dim G_{s'} \geq \dim G_s.$$

On the other hand, by 4.1 the function $s \mapsto \dim G_s$ is upper semicontinuous, so that

$$\dim G_{s'} \leq \dim G_s.$$

Thus $s' \in T$, proving that (i) implies (ii).

It is clear that (iii) implies (i); we prove that (ii) implies (iii). The dimension of the fibers is unchanged by extension of the base field, and the formation of $\underline{G^0}$ commutes with base change (see the proof of 3.3). We may therefore assume that $S' = S$ and $s' = s$. Moreover, every open subset V of G meeting $\underline{G^0_s}$ contains the generic point η of $\underline{G^0_s}$, so we may assume that $g = \eta$.

We may further suppose that S is affine. Let U be an affine open neighborhood of $\epsilon(s)$; it is then of finite presentation over S. Replacing S by $\epsilon^{-1}(U)$, and then U by $U \cap \pi^{-1}(S)$, we reduce to the case in which

$$\pi : U \longrightarrow S$$

is of finite presentation and admits $\epsilon : S \to U$ as a section. By EGA IV$_3$, 9.7.12, $\underline{G^0} \cap U$ is constructible in U. Replacing V by an affine open subset contained in $V \cap U$, we find that $\underline{G^0} \cap V$ is constructible in V. Since $\pi : V \to S$ is of finite presentation, Chevalley's constructibility theorem (EGA IV$_1$, 1.8.4) shows that

$$\pi(\underline{G^0} \cap V)$$

is locally constructible in S.

Consequently, by *loc. cit.*, 1.10.1, to show that $\pi(\underline{G^0} \cap V)$ is an open neighborhood of s, it is enough to show that, for every generalization t of s, there is a generalization ξ of η belonging to $\underline{G^0}$, and hence to $\underline{G^0} \cap V$. The generic point ξ of $\underline{G^0_t}$ is such a generalization of η. Indeed, $\epsilon(s)$ belongs to the closure X of $\{\xi\}$ in G; hence, by Chevalley's semicontinuity theorem (EGA IV$_3$, 13.1.3),

$$\dim_{\epsilon(s)} X_s \geq \dim_\xi X_t.$$

On the other hand, hypothesis (ii) implies

$$\dim_\xi \underline{G^0_t} = \dim_{\epsilon(s)} G_s.$$

It follows that one of the irreducible components of X_s containing $\epsilon(s)$ is equal to $\underline{G^0_s}$, whence

$$\eta \in \overline{\{\xi\}}.$$

This proves that (ii) implies (iii), and hence proves the proposition.

[30]One can also prove that (ii) implies (i) as follows. Since π is locally of finite presentation, there exist an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s, such that $\pi(U) \subset V$ and the morphism

$$\pi' : U \longrightarrow V$$

[30]Editor's note: the preceding argument was communicated to us by O. Gabber. We have also retained the proof of the implication (ii) $\Rightarrow$(i) given in the original.

induced by π is of finite presentation. Set

$$T = \epsilon^{-1}(U), \qquad W = \pi'^{-1}(T) = U \cap \pi^{-1}(T).$$

The morphism

$$\pi'' : W \longrightarrow T$$

induced by π' is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ. Moreover, for every $t \in T$, since $\underline{G}_t^0$ is irreducible, $W \cap \underline{G}_t^0$ is dense in $\underline{G}_t^0$, hence irreducible and therefore connected. It is consequently the connected component of $\pi''^{-1}(t)$ containing $\epsilon''(t)$.

Since $W \cap \underline{G}_t^0$ is a dense open subset of $\underline{G}_t^0$, 1.5 and EGA IV$_2$, 5.2.1 give

$$\dim(W \cap \underline{G}_t^0) = \dim \underline{G}_t^0 = \dim G_t.$$

Thus the function

$$t \longmapsto \dim(W \cap \underline{G}_t^0)$$

is constant in a neighborhood of s in T. Finally, we show that $W \cap \underline{G}_t^0$ is geometrically irreducible for every $t \in T$. Let K be an extension of $\kappa(t)$. Then

$$(W \cap \underline{G}_t^0) \otimes_{\kappa(t)} K = (W \otimes_{\kappa(t)} K) \cap (\underline{G}_t^0 \otimes_{\kappa(t)} K)$$

is a nonempty open subset of $\underline{G}_t^0 \otimes_{\kappa(t)} K$, and is therefore irreducible because $\underline{G}_t^0 \otimes_{\kappa(t)} K$ is irreducible.

We are now in the situation of EGA IV$_3$, 15.6.6(ii), which asserts that π'', and hence π, is universally open at the points of $W \cap \underline{G}_s^0$. But, by EGA IV$_3$, 14.3.3.1(ii), the subset of G_s consisting of the points at which π is universally open is closed in G_s. Since it contains $W \cap \underline{G}_s^0$, it therefore contains its closure $\underline{G}_s^0$.

Corollary 4.3.— *Let G be a flat S-group locally of finite presentation. Then the function*

$$s \longmapsto \dim G_s$$

is locally constant on S.

This follows immediately from 4.2, since every flat morphism locally of finite presentation is universally open (EGA IV$_2$, 2.4.6).

Corollary 4.4.— *Let G be an S-group locally of finite presentation over S at the points of the identity section. Consider the following conditions:*

(i) *G is smooth over S at the points of the identity section* (cf. 3.10).

(ii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and the function*

$$s \longmapsto \dim G_s$$

is locally constant on S.

(iii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and there exists a neighborhood V of the identity section such that*

$$\pi_V : V \longrightarrow S$$

is universally open.

(iv) *For every $s \in S$, G^0_s is smooth over $\kappa(s)$, and G^0 is representable by an open group subscheme of G, universally open over S.*

Then the following implications hold:

$$\text{(i)} \Longrightarrow \text{(ii)} \Longleftrightarrow \text{(iii)} \Longleftrightarrow \text{(iv)}.$$

If, moreover, S is reduced, then conditions (i)–(iv) *are equivalent and imply that G^0 is smooth over S.*[31]

Let us show that (i) implies (ii). For every $x \in G$, by 1.5,

$$\dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x)).$$

Consequently, by EGA IV$_4$, 17.10.2, the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$$

is continuous in a neighborhood of the identity section. Therefore the function $s \mapsto \dim G_s$ is continuous on S, and hence locally constant on S. On the other hand, for every $s \in S$, G_s is smooth over $\kappa(s)$, by 1.3.1.

Let us show that (ii) implies (iv). It is enough to show that $\underline{G^0}$ is open in G, for then, by 3.4, G^0 will be representable by the group subscheme induced on $\underline{G^0}$, and the properties of G^0 stated above will follow from 2.4 and 4.2. Given $s \in S$, construct, as in the proof of 3.10,

$$W, \quad T, \quad \pi'', \quad \epsilon'', \quad \text{and} \quad W^0.$$

Then, by EGA IV$_3$, 15.6.7, W^0 is open in W. On the other hand, under hypothesis 4.4(ii), it follows from 4.2 that π is universally open at every point of W^0. Therefore, by VI$_A$, 0.1, μ is universally open at every point of

$$W^0 \times_S W^0,$$

which shows that $W^0 \cdot W^0$ is open; the proof is completed as in the proof of Theorem 3.10.

It is clear that

$$\text{(iv)} \Longrightarrow \text{(iii)},$$

and (iii) $\Longrightarrow$ (ii) follows by applying 4.2 to V.

Finally, suppose that (ii)–(iv) hold, and let us show that G^0 is smooth over S when S is reduced.[32] For this, we may suppose that $G = G^0$. Then G is of finite presentation over S, by 5.5 below. Thus π_G is of finite presentation, has geometrically integral fibers, and has locally constant relative dimension on S. It follows from EGA IV$_3$, 15.6.7 that the morphism

$$G \times_S S_{\text{red}} \longrightarrow S_{\text{red}}$$

induced by π_G is flat. Hence π_G is flat when S is reduced. In that case, $G = G^0$ is smooth over S, by Theorem 3.10.

[31]Editor's note: the hypothesis that S be reduced cannot be omitted here. Indeed, let k be a field, let $S = \operatorname{Spec} A$, where $A = k[\delta]$ with $\delta^2 = 0$, and let G be the closed subgroup

$$\operatorname{Spec} A[X]/(\delta X)$$

of $\mathbb{G}_{a,S}$ (which assigns to every A-algebra R the subgroup of those $r \in R$ such that $\delta r = 0$). Then G is an S-group satisfying (ii)–(iv), but it is not flat, and hence not smooth, over S.

[32]Editor's note: we have expanded the original in the argument that follows.

5. Separation of groups and homogeneous spaces

Proposition 5.1.— *For an S-group G to be separated, it is necessary and sufficient that the identity section of G be a closed immersion.*

The condition is necessary (EGA I, 5.4.6); it is sufficient by virtue of the following cartesian diagram (cf. VI$_A$, 0.3):

$$\begin{tikzcd} G \times_S G \arrow[r, "\mu\circ(\mathrm{id}_G \times c)"] & G \\ G \arrow[u, "\Delta_{G/S}"] \arrow[r, "\pi"] & S \arrow[u, "\varepsilon"'] \end{tikzcd} .$$

Proposition 5.2.— *If S is discrete, every S-group is separated.*

Indeed,

$$S = \coprod_{s\in S} \operatorname{Spec} \mathcal{O}_{S,s},$$

and, by EGA I, 5.5.5, it is enough to show that, for every $s \in S$,

$$G \times_S \operatorname{Spec} \mathcal{O}_{S,s}$$

is separated. This follows from VI$_A$, 0.3, since $\mathcal{O}_{S,s}$ is a local ring of dimension 0.[33]

Theorem 5.3.— *Let S be a scheme, let G be an S-group scheme locally of finite presentation over S and universally open over S in a neighborhood of the identity section, and let X be an S-scheme on which G acts in such a way that the morphism*

$$\Phi : G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

is surjective. Suppose moreover that, for every $s \in S$:

(i) *there exists an open subscheme U of X, separated over S, such that U_s is dense in X_s;*

(ii) *the fiber X_s is locally of finite type over $\kappa(s)$.*[34]

Then X is separated over S.

Corollary 5.4.— *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that X has fibers locally of finite type and connected. Then:*

(i) *X is separated over S.*

(ii) *If there exists an open subset V of X, quasi-compact over S, which meets every nonempty fiber X_s,*[35] *then X is quasi-compact over S.*

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[33]Editor's note: we have replaced "Artinian" by "of dimension 0" and mentioned this generalization in VI$_A$, 0.3.

[34]Editor's note: we have corrected the original by assuming that G is universally open over S in a neighborhood of the identity section and by adding hypothesis (ii); see below for examples, due to O. Gabber, showing that the statements 5.3 and 5.4 of the original are false without additional hypotheses. We also point out that Theorem 5.3 is a reworked version of Theorem 5.3A below, which appears in the 1965 edition of SGAD and is due to M. Raynaud; compare the notes $(*)$ in Exposé X, 8.5 and 8.8.

Theorem 5.3A (Raynaud).— *Let G be an S-group locally of finite presentation, universally open over S, and with connected fibers. Then G is separated over S. More generally, every S-scheme X locally of finite presentation over S, equipped with an action of G such that the morphism $\Phi : G \times_S X \to X \times_S X$, $(g, x) \mapsto (gx, x)$, is surjective, is separated over S.*

[35]Editor's note: without this hypothesis there is the following counterexample, pointed out by O. Gabber. Let $G = S$ be a local scheme with closed point s, such that $S^* = S - \{s\}$ is not quasi-compact, and let X be the disjoint union of $\{s\}$ and S^*.

Proof. (i) Let $s \in S$ be such that $X_s \neq \varnothing$. The morphism

$$\Phi_s : G_s \times_{\kappa(s)} X_s \longrightarrow X_s \times_{\kappa(s)} X_s$$

obtained from Φ by base change is surjective. Since X_s is connected, it is irreducible by VI_A, 2.5.4. Thus, if U is an affine open subset of X such that U_s is nonempty, then U_s is dense in X_s, and the theorem applies.

To prove (ii), we may suppose that S is affine. Then V is quasi-compact and, by 3.5, there exists a quasi-compact open subset U of G containing $\underline{G}^0$. Let $s \in S$ be such that $X_s \neq \varnothing$. Then X_s is irreducible (VI_A, 2.6.6), and, since U_s contains $\underline{G}^0_s$, the morphism

$$U_s \times_{\kappa(s)} V_s \longrightarrow X_s, \qquad (g, x) \longmapsto gx,$$

is surjective (VI_A, 2.6.4). Consequently the morphism

$$U \times_S V \longrightarrow X$$

is surjective, and X is therefore quasi-compact, since U and V are. As S was assumed affine, hence separated, it follows that X is quasi-compact over S (cf. EGA I, 6.6.4(v)).

Remark 5.4.1.—[36] We shall see in the course of the proof that the conclusion of Theorem 5.3 remains true if one assumes only:

> (i') there exists an open subscheme U of X, separated over S, such that U_s is dense in every irreducible component of X_s.

This follows from (i) if X_s locally has a finite number of irreducible components, as is the case under hypothesis (ii). We shall also see later that 5.4 remains true under the hypothesis that every fiber X_s is quasi-separated and connected.

Corollary 5.5.— *Let S be a scheme, and let G be an S-group scheme, locally of finite presentation, with connected fibers, and universally open over S. Then G is separated and of finite presentation over S.*[37]

Indeed, by 3.6 and 5.4, G is quasi-compact and separated over S. Since G is locally of finite presentation over S, it is therefore of finite presentation over S.

5.6.— *Proof of Theorem 5.3.* Before proving 5.3, let us establish a few lemmas.

Lemma 5.6.0.—[38]

(i) *Let $A \subset B$ be integral domains, with B integral over A, and let $\mathfrak{q} \in \mathrm{Spec}(B)$ be such that A is unibranch at $\mathfrak{p} = \mathfrak{q} \cap A$. Then the morphism*

$$\pi : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

is open at $\mathfrak{q}$.

[36] Editor's note: we have added this remark; compare editor's note 34.

[37] Editor's note: we point out here the following result ([Ray70a], VI, 2.5). If S is normal, G is smooth with connected fibers, and X is an (fppf) homogeneous G-space—that is, the morphisms $X \to S$ and $G \times_S X \to X \times_S X$ are coverings for the (fppf) topology—locally of finite type over S, then X is locally quasi-projective over S. In particular, G is quasi-projective over S. See also editor's note 35 in VI_A.

[38] Editor's note: we have added this lemma, communicated by O. Gabber. It improves EGA IV_3, 14.4.1.2 and corrects the proof of *loc. cit.*, 14.4.1.3 without changing its hypotheses (compare the erratum (ErrIV, 38) in EGA IV_4).

(ii) *Let X, Y be irreducible preschemes, let $f : X \to Y$ be a dominant morphism, and let $x \in X$ be a point at which f is quasi-finite, such that $y = f(x)$ is a unibranch point of Y. Then f is open at x. In particular, if*

$$Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$$

is a local prescheme with closed point y, then $f(U) = Y$ for every neighborhood U of x.

(i) Let K (respectively, L) be the fraction field of A (respectively, B), let A' be the normalization of A, and let B' be the subring of L generated by A' and B. Then B' is integral over A'. Set

$$Y = \mathrm{Spec}(A), \quad X = \mathrm{Spec}(B), \quad Y' = \mathrm{Spec}(A'), \quad X' = \mathrm{Spec}(B').$$

We then have the commutative diagram

$$\begin{tikzcd} X' \arrow[r, "\pi'"] \arrow[d] & Y' \arrow[d, "\phi"'] \\ X \arrow[r, "\pi"] & Y \end{tikzcd}$$

in which all morphisms are integral and surjective.

Since A is unibranch at $\mathfrak{p}$, Y' has a unique point $\mathfrak{p}'$ above $\mathfrak{p}$. Consequently, if U is an open neighborhood of $\mathfrak{p}'$ in Y', then

$$\phi(Y' - U)$$

is a closed subset not containing $\mathfrak{p}$, so that its open complement is contained in $\phi(U)$. This shows that $\phi : Y' \to Y$ is open at $\mathfrak{p}'$. It is therefore enough to show that π' is open, and we are reduced to the case where $A = A'$ is normal.

Let N be a quasi-Galois extension of K containing L, let

$$\widetilde{X} = \mathrm{Spec}(\widetilde{B}),$$

where $\widetilde{B}$ is the integral closure of B in N, and let $G = \mathrm{Aut}(N/K)$. Since $\widetilde{X} \to X$ is surjective, it is enough to show that

$$\widetilde{\pi} : \widetilde{X} \longrightarrow Y$$

is open. Let U be an open subset of $\widetilde{X}$, and set

$$U' = \bigcup_{g \in G} gU.$$

Because G acts transitively on the fibers of $\widetilde{X} \to Y$ (cf. [BAC], Chap. V, §2.3, Prop. 6), one has

$$\widetilde{\pi}(U) = \widetilde{\pi}(U'),$$

and the latter is the open complement of the closed subset

$$\widetilde{\pi}(\widetilde{X} - U').$$

This proves (i).

(ii) We may suppose that Y and X are reduced. By Zariski's Main Theorem (cf. EGA IV$_3$, 8.12.9), there exist affine open neighborhoods U of x and $V = \mathrm{Spec}(A)$ of y, such that $f(U) \subset V$, and a factorization

$$\begin{tikzcd} U \arrow[r, hook, "j"] \arrow[dr, "f"'] & V' \arrow[d, "u"] \\ & V \end{tikzcd},$$

where j is an open immersion and u is finite. Replacing V' by the closure of $j(U)$, we may suppose that V' is irreducible, hence that $V' = \mathrm{Spec}(B)$, where B is an integral A-algebra, finite over A. Moreover, since f is dominant, so is u, and the morphism $A \to B$ is therefore injective.

Since A is unibranch at y, it follows from (i) that u is open at $j(x)$, and hence that

$$f = u \circ j$$

is open at x. This proves the first assertion of (ii). The second follows because, if Y is a local prescheme with closed point y, every open subset containing y equals Y. In the case $Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$, one may also use, instead of EGA IV$_3$, 8.12.9, the local form of Zariski's Main Theorem found, for example, in [Pes66] or [Ray70b], Chap. IV, Th. 1.

Lemma 5.6.1.0.—[39] *Let $f : X \to S$ be a morphism locally of finite presentation, let $s \in S$, and let $x \in X_s$. Set $n = \dim_x(X_s)$, and let $q : \mathbb{A}^n_S \to S$ be the structure morphism.*

Suppose that a quasi-finite morphism $u : X \to \mathbb{A}^n_S$ is given such that $f = q \circ u$, and suppose that f is universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s, containing x and of dimension n. Then u is universally open at the point $x \in X$.

First observe that

(†) *the hypotheses are preserved under every base change $\pi : S' \to S$ covering s*

(that is, such that $\pi^{-1}(s) \neq \varnothing$). Indeed, let $S' \to S$ be such a morphism, let

$$\begin{tikzcd} X' \arrow[r, "u'"] \arrow[dr, "f'"'] & \mathbb{A}^n_{S'} \arrow[d, "q'"] \\ & S' \end{tikzcd}$$

be the diagram obtained by base change, and let s' be a point of S' above s and x' a point of $X'_{s'}$ above x. Since

$$X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s'),$$

the lifting of generizations and the invariance of dimension under field extension (EGA IV$_2$, 2.3.4(i) and 4.1.4) show that x' is contained in an irreducible component $\mathfrak{z}'$ of $X'_{s'}$ whose generic point z' lies above z, and

$$n \leq \dim \mathfrak{z}' \leq \dim_{x'} X'_{s'} \leq n.$$

Thus

$$\dim \mathfrak{z}' = n = \dim_{x'} X'_{s'}.$$

Since f is universally open at z, f' is universally open at z'.Set $Y = \mathbb{A}^n_S$. By EGA IV$_3$, 14.3.3.1(i), to prove that u is universally open at x, it is enough to prove that, for every integer $r \geq 0$ and every point x' of

$$X' = X[T_1, \dots, T_r]$$

[39] Editor's note: we have made explicit this lemma, which is used in the proof of Lemma 5.6.1.

above x, the morphism

$$u' : X' \longrightarrow Y' = \mathbb{A}^n_S[T_1, \dots, T_r]$$

is open at x'. Now set

$$S' = S[T_1, \dots, T_r],$$

and let q' be the projection

$$Y' \simeq \mathbb{A}^n_{S'} \longrightarrow S'.$$

This is the situation obtained by the base change $S' \to S$. Thus, by the preceding observation, we are reduced to showing that u is open at x.

Set $y = u(x)$. Since u is locally of finite presentation (because f and q are; cf. EGA IV$_1$, 1.4.3), EGA IV$_1$, 1.10.3 shows that it is enough to prove

$$u(\operatorname{Spec} \mathcal{O}_{X,x}) = \operatorname{Spec} \mathcal{O}_{Y,y}.$$

For this purpose we may assume that S is affine and integral. Let $\pi : S' \to S$ be its normalization, and denote by $u' : X' \to Y'$ the morphism obtained from u by base change. Since

$$\pi_Y : Y' \longrightarrow Y$$

is integral and surjective, one has

$$\operatorname{Spec} \mathcal{O}_{Y,y} = \bigcup_{y'} \pi_Y(\operatorname{Spec} \mathcal{O}_{Y',y'}),$$

where the union ranges over all points y' of Y' above y. It is therefore enough to show that, for every such y' and every $x' \in X'_{y'}$ above x,

$$u'(\operatorname{Spec} \mathcal{O}_{X',x'}) = \operatorname{Spec} \mathcal{O}_{Y',y'}.$$

Since the hypotheses are preserved under the base change $\pi : S' \to S$, we are thereby reduced to the case of S'; that is, we may assume that S is integral and normal.

The hypotheses on f now imply, by EGA IV$_3$, 14.3.13, that there is an irreducible component Z of X, containing $\mathfrak{z}$ (and hence x), dominating S, and such that

$$\dim_z(Z_s) = n = \dim(Z_\eta),$$

where η is the generic point of S. Let ξ be the generic point of Z. Since u, and hence also

$$u_\eta : Z_\eta \longrightarrow \mathbb{A}^n_\eta,$$

is quasi-finite, the closure in $\mathbb{A}^n_\eta$ of the point

$$u_\eta(\xi) = u(\xi)$$

has dimension n. Thus $u(\xi)$ is the generic point of $\mathbb{A}^n_\eta$, which is also the generic point of $Y = \mathbb{A}^n_S$. Consequently, if g denotes the restriction of u to Z, then

$$g : Z \longrightarrow Y$$

is quasi-finite and dominant. Since Y is normal, Lemma 5.6.0 shows that g is open. Hence u is open at every point of Z, and in particular at x. Consequently,

$$u(\operatorname{Spec} \mathcal{O}_{X,x}) = \operatorname{Spec} \mathcal{O}_{Y,y},$$

which completes the proof of Lemma 5.6.1.0.

The following lemma advantageously replaces EGA IV$_3$, 14.5.10,[40] in that it is independent of noetherian hypotheses.

Lemma 5.6.1.— *Let S be a scheme, let $f : X \to S$ be an S-scheme locally of finite presentation, let s be a point of S, and let x be a closed point of X_s. Suppose that f is universally open at the generic point z of an irreducible component Z of X_s containing x, and that*

$$\dim_x(X_s) = \dim Z.$$

Then there exists a commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] & S \\ S'' \arrow[u, "h"] \arrow[ur, "\phi"] \arrow[r, "\pi"] & S' \arrow[u, "w"'] \end{tikzcd},$$

where S' is affine, w is étale, π is finite, surjective, and of finite presentation, and $\phi^{-1}(s)$ consists of a single point s'', such that $h(s'') = x$ and $\kappa(s'') = \kappa(x)$.[41]

Proof. First, we may suppose that $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$, where B is an A-algebra of finite presentation. Let $\mathfrak{p}$ and $\mathfrak{q}$ be the prime ideals of A and B corresponding respectively to s and x, so that

$$\mathfrak{p} = A \cap \mathfrak{q}.$$

Set $n = \dim_x(X_s)$, and choose $t_1, \dots, t_n \in B$ whose images in

$$\mathcal{O}_{X_s,x} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$$

form a system of parameters. Then

$$\mathcal{O}_{X_s,x}/(t_1, \dots, t_n)$$

is finite-dimensional over $\kappa(x)$, and hence also over $\kappa(s)$, because x is a closed point of the algebraic $\kappa(s)$-scheme X_s. Consequently, the S-morphism

$$u : X \longrightarrow \mathbb{A}^n_S = \operatorname{Spec}(A[T_1, \dots, T_n]), \qquad T_i \longmapsto t_i,$$

is of finite presentation, x is isolated in the fiber $u^{-1}(u(x))$, and

$$u(x) = \tau_0(s),$$

where $\tau_0 : S \to \mathbb{A}^n_S$ denotes the zero section of $\mathbb{A}^n_S \to S$, corresponding to the A-algebra homomorphism

$$A[T_1, \dots, T_n] \longrightarrow A, \qquad T_i \longmapsto 0.$$

The set of points of X that are isolated in their fibers over $\mathbb{A}^n_S$ is open (EGA IV$_3$, 13.1.4). Thus, after shrinking X, we may suppose that u is quasi-finite and that $u^{-1}(u(x)) = \{x\}$. By Lemma 5.6.1.0, u is universally open at x.

Let

$$B_0 = B/(t_1, \dots, t_n), \qquad X_0 = X \times_{\mathbb{A}^n_S} \tau_0(S) = \operatorname{Spec}(B_0),$$

[40] Editor's note: we have corrected the original, which indicated 19.5.10.

[41] Editor's note: consequently, h induces a section σ of $X \times_S S'' \to S''$ such that $\sigma(s'')$ lies above x; compare EGA IV$_3$, 14.5.10.

and let $u_0 : X_0 \to S$ be obtained from u by the base change $\tau_0 : S \hookrightarrow \mathbb{A}^n_S$. Then u_0 is quasi-finite and of finite presentation, is universally open at x, and x is the unique point of X_0 above s.

Let A' be the henselization of the local ring

$$A_{\mathfrak{p}} = \mathcal{O}_{S,s},$$

and set

$$S' = \operatorname{Spec}(A'), \qquad B'_0 = B_0 \otimes_A A', \qquad X'_0 = X_0 \times_S S' = \operatorname{Spec}(B'_0).$$

The closed point s' of S' is the unique point of S' above s, and $\kappa(s') = \kappa(s)$. The scheme X'_0 has a unique point x' above s'; moreover $\kappa(x') = \kappa(x)$, and x' is also the unique point of X'_0 above both s and x. Since A' is henselian, EGA IV$_4$, 18.5.11 gives a disjoint decomposition into open and closed parts

$$X'_0 = V \amalg W, \qquad V = \operatorname{Spec}(\mathcal{O}_{X'_0,x'}). \tag{*}$$

The local ring $\mathcal{O}_{X'_0,x'}$ is finite and of finite presentation over A'. Hence the restriction π of u'_0 to V is finite and of finite presentation. Moreover, because $u'_0 : X'_0 \to S'$ is open at x',

$$u'_0(V) = \operatorname{Spec}(\mathcal{O}_{S',s'}) = S',$$

so π is surjective.

This proves the assertion when $S = S'$. In general, A' is the filtered direct limit of A-subalgebras A_i that are étale over A and for which

$$S_i = \operatorname{Spec}(A_i)$$

has a unique point s_i above s, with $\kappa(s_i) = \kappa(s)$. We have

$$B'_0 = \varinjlim_i B_i, \qquad B_i = B_0 \otimes_A A_i.$$

Set

$$X_i = \operatorname{Spec}(B_i) = X_0 \times_S S_i, \qquad C = \mathcal{O}_{X'_0,x'}.$$

By $(*)$, one has

$$C \simeq B'_0/fB'_0$$

for some idempotent $f \in B'_0$. There are an index i and an element $f_i \in B_i$ whose image in B'_0 is f. Put

$$C_i = B_i/f_iB_i, \qquad V_i = \operatorname{Spec}(C_i).$$

Then

$$C = C_i \otimes_{A_i} A', \qquad V = V_i \times_{S_i} S',$$

and C_i is an A_i-algebra of finite presentation, since the same is true of B_i. Consequently, the morphisms

$$\begin{tikzcd} X'_0 \arrow[dr, "u'_0"] & \\ V \arrow[u, "h'"] \arrow[r, "\pi"] & S' \end{tikzcd}$$

arise, by the base change $S' \to S_i$, from morphisms of finite presentation

$$\begin{tikzcd} X_i \arrow[dr, "u_i"] & \\ V_i \arrow[u, "h_i"] \arrow[r, "\pi_i"] & S_i \end{tikzcd}$$

For every $j \geq i$, let

$$V_j = V_i \times_{S_i} S_j$$

and let $\pi_j : V_j \to S_j$ be the morphism of finite presentation obtained from π_i by base change. Since $\pi : V \to S'$ is finite and surjective, EGA IV$_2$, 8.10.5 shows that, for some j, the morphism $\pi_j : V_j \to S_j$ is finite and surjective. Then $w : S_j \to S$ is affine and étale, S_j has a unique point s_j above s, and V_j has a unique point x_j above s_j, since x' is the unique point of

$$V = V_j \times_{S_j} S'$$

above s':

$$\begin{tikzcd} X_j \arrow[r] \arrow[dr, "u_j"] & X_0 \arrow[r] \arrow[dr, "u_0"] & X \arrow[d, "f"] \arrow[dr, "u"] & \\ V_j \arrow[u, "h_j"] \arrow[r, "\pi_j"] & S_j \arrow[r, "w"] & S \arrow[r, "\tau_0"] & \mathbb{A}^n_S \end{tikzcd}$$

Thus x_j is the unique point of V_j above s, its image under $V_j \to X_j \to X$ is x, and

$$\kappa(x_j) = \kappa(x).$$

Lemma 5.6.2.0.—[42] *Let k be a field and let G be a nonempty k-scheme of finite type.*

(i) *Let K be an extension of k, and let W be a dense open subset of G_K. Denote by π the projection $G_K \to G$. Then*

$$U = \{g \in G \mid W \text{ contains every maximal point of } \pi^{-1}(g)\}$$

is a dense open subset of G. (N.B. If g is a closed point of G belonging to U, then $\pi^{-1}(g) \subset W$.

(ii) *Suppose in addition that G is geometrically irreducible. Let*

$$\mu : G \times X \longrightarrow Y$$

be a morphism of k-schemes, let x be a point of X, and let Ω be an open subset of Y such that

$$\mu_x^{-1}(\Omega_{\kappa(x)}) \neq \varnothing,$$

where μ_x denotes the morphism

$$G_{\kappa(x)} \longrightarrow Y_{\kappa(x)}, \qquad g \longmapsto \mu(g, x).$$

Then

$$U = \{g \in G \mid \mu \text{ sends every maximal point of } g \times x \text{ into } \Omega\}$$

is a dense open subset of G. For every closed point g of G belonging to U, one has

$$\mu(g \times x) \subset \Omega,$$

[42] Editor's note: we have added this lemma, communicated by O. Gabber. It makes it possible to simplify the proof of 5.6.2 and to prove Theorem 5.3, as well as 5.4, in a more general form; see 5.7 and 5.8 below.

and hence

$$\mu(g', x) \in \Omega_{\kappa(x)} \quad (\text{respectively, } \mu(g, x') \in \Omega_{\kappa(g)}),$$

for every point $g' \in G_{\kappa(x)}$ above g (respectively, $x' \in X_{\kappa(g)}$ above x).

(iii) *Suppose that $G = H^0$, where H is a k-group scheme locally of finite type, acting on a nonempty k-scheme X in such a way that the morphism*

$$H \times_S X \longrightarrow X \times_S X, \qquad (h, x) \longmapsto (hx, x)$$

is surjective. Let U be an open subset of X. Then:

(a) *$G \cdot U$ is an open subset of X, equal to the union of the irreducible components of X whose generic point belongs to U.*

(b) *Suppose that U is dense in every irreducible component of X. Then, for every finite subset A of X, there exists a closed point $g \in G$ such that*

$$g \cdot A' \subset U_{\kappa(g)},$$

where A' is the inverse image of A in $X_{\kappa(g)}$. In particular, the morphism

$$G \times U \longrightarrow X$$

is surjective.

Proof. (i) Let $\bar{k}$ be a separable closure of k, and let L be an extension of k containing a copy of $\bar{k}$ and of K. Denote by π_L (respectively, ϕ) the projection

$$G_L \longrightarrow G \qquad (\text{respectively, } G_L \longrightarrow G_K).$$

For every $g \in G$, the map ϕ sends the maximal points of $\pi_L^{-1}(g)$ surjectively onto those of $\pi^{-1}(g)$. Consequently,

$$U = \{g \in G \mid W_L \text{ contains every maximal point of } \pi_L^{-1}(g)\}.$$

Thus, replacing K by L, we are reduced to the case in which K contains $\bar{k}$.

Since the projection

$$p : G_K \longrightarrow G_{\bar{k}}$$

is surjective and open, $V = p(W)$ is a dense open subset of $G_{\bar{k}}$. Moreover,

$$p^{-1}(g) = \operatorname{Spec}(\kappa(g) \otimes_{\bar{k}} K)$$

is irreducible for every $g \in G_{\bar{k}}$ (cf. EGA IV_2, 4.3.3). Hence, for every $g \in V$, the generic point of $p^{-1}(g)$ belongs to W.

On the other hand, set

$$\mathscr{G} = \operatorname{Gal}(\bar{k}/k).$$

The projection $q : G_{\bar{k}} \to G$ is surjective, and $\mathscr{G}$ acts transitively on the fibers of q. Therefore

$$U = q(V'),$$

where V' is the intersection of the $\mathscr{G}$-conjugates of V.

Let Z be the closed subset $G_{\bar{k}} - V$, endowed with its reduced closed subscheme structure. Since $G_{\bar{k}}$, and hence Z, is of finite presentation over $\bar{k}$, the scheme Z is obtained by base change from a reduced closed subscheme

$$Z_1 \subset G \otimes_k k_1$$

for some finite Galois extension k_1 of k. Thus the $\mathscr{G}$-conjugates of Z are finite in number, and their union is again a nowhere-dense closed subset F of $G_{\bar{k}}$. It follows that

$$V' = G_{\bar{k}} - F$$

is a dense open subset of $G_{\bar{k}}$. Since $q : G_{\bar{k}} \to G$ is surjective and open, $U = q(V')$ is therefore a dense open subset of G.

Moreover, for every closed point g of G, the fiber $\pi^{-1}(g)$ consists of finitely many closed points of G_K. Thus, if $g \in U$, then

$$\pi^{-1}(g) \subset W.$$

This proves (i), and (ii) follows from it.

Assertion (iii)(a) was proved in VI$_A$, 2.6.4. Finally, if U is dense in every irreducible component of X, then

$$X = G \cdot U.$$

Hence, for every $x \in X$,

$$\mu_x^{-1}(U_{\kappa(x)})$$

is a nonempty open subset of $G_{\kappa(x)}$, and (iii)(b) follows from (ii).

Corollary 5.6.2.— *Let S, G, and X be as in the preliminary hypotheses of 5.3, and let U be an open subset of X, let $s \in S$, and let A be a finite subset of X_s. Suppose that U_s is dense in X_s and that X_s is locally of finite type over $\kappa(s)$.*[43]

Then there are morphisms $f : S'' \to S$ and $h : S'' \to G$, where f factors as a finite surjective morphism $S'' \to S'$ followed by an étale morphism $S' \to S$, such that the inverse image A'' of A in $X \times_S S''$ (i.e., in $X_s \times_{\operatorname{Spec}\kappa(s)} S''_s$) *is contained in*

$$\ell_h^{-1}(U_{S''}),$$

where ℓ_h denotes the translation of $X_{S''} = X \times_S S''$ defined by the element $h \in G(S'')$.

Proof. Since X_s is locally of finite type over $\kappa(s)$, the connected components of X_s are open and irreducible (cf. VI$_A$, 2.5.4). Hence U_s is dense in every irreducible component of X_s.

By Lemma 5.6.2.0, there therefore exists a closed point

$$g \in G_s^0$$

such that

$$g \cdot a' \in U_s \otimes_{\kappa(s)} \kappa(g)$$

for every $a' \in X_{\kappa(g)}$ lying above a point of A.

By Lemma 5.6.1, there are morphisms $f : S'' \to S$ and $h : S'' \to G$, where f factors as a finite surjective morphism $S'' \to S'$ followed by an étale morphism $S' \to S$, such that $f^{-1}(s)$ consists of a single point s_0, and such that

$$h(s_0) = g, \qquad \kappa(s_0) = \kappa(g).$$

Denote by A'' the inverse image of A in

$$X_s \times_{\operatorname{Spec}\kappa(s)} S''_s = X_s \otimes_{\kappa(s)} \kappa(s_0) = X_s \otimes_{\kappa(s)} \kappa(g).$$

[43]Editor's note: we have added the hypothesis on X_s and simplified (and corrected) the proof, taking account of the addition of 5.6.2.0. Moreover, the proof shows that the conclusion remains valid if one assumes only that U_s is dense in every irreducible component of X_s.

Then the translation ℓ_h of $X_{S''}$ sends A'' into

$$U_s \otimes_{\kappa(s)} \kappa(s_0).$$

Lemma 5.6.3.— *Let X be an S-scheme. The following conditions are equivalent:*

(i) *X is separated over S.*

(ii) *For every S-scheme T, every section $\sigma : T \to X_T$ is a closed immersion.*

(iii) *For every reduced S-scheme T, two S-morphisms $f_1, f_2 : T \to X$ that agree on a dense open subset U of T are equal.*

(iv) *For every $s \in S$ and every pair of points x_1, x_2 of X_s, there exist a morphism*

$$\phi : S'' \longrightarrow S' \longrightarrow S$$

and an open subscheme V of $X_{S''}$, separated over S'', such that:

  (a) *$S' \to S$ is open, $S'' \to S'$ is closed and surjective, and $\phi^{-1}(s) \neq \varnothing$;*

  (b) *the inverse image of $\{x_1, x_2\}$ in $X_{S''}$ is contained in V.*

(iv′) *For every $s \in S$, every pair of points $x_1, x_2 \in X_s$ is contained in an open subset V of X that is separated over S.*[44]

Proof. The implication (ii)⇒(i) is clear (take $T = X$ and let σ be the diagonal section), as are

$$\text{(i)} \Rightarrow \text{(iv}'\text{)} \Rightarrow \text{(iv)}.$$

On the other hand, (i) ⇒ (ii) by EGA I, 5.4.6.

(iii)⇒(ii). Let $\sigma : T \to X_T$ be a section of $p : X_T \to T$. By EGA I, 5.3.13, σ is an immersion, that is, an isomorphism of T onto a locally closed subscheme E of X_T. To show that E is closed, we may suppose that T and E are reduced. Let $\overline{E}$ be the reduced closed subscheme of X_T whose underlying space is the closure of E, so that E is a dense open subscheme of $\overline{E}$. The immersion

$$i : \overline{E} \hookrightarrow X_T$$

and $\sigma \circ p \circ i$ agree on E, hence on $\overline{E}$ by hypothesis (iii). Thus every point of $\overline{E}$ belongs to $\sigma(T) = E$, and consequently $\overline{E} = E$.

(i)⇒(iii). Suppose that X is separated over S, and let T be a reduced S-scheme and $f_1, f_2 : T \to X$ two S-morphisms that agree on a dense open subset U of T. Let

$$g : T \longrightarrow X \times_S X$$

[44] Editor's note: we have simplified the formulation of condition (iv) and added condition (iv′). We have also added the proof of the implication (i)⇒(iii), which is used in the proof of (iv)⇒(iii). Observe moreover that, if

$$T = \operatorname{Spec} B, \qquad B = k[\varepsilon, x]/(\varepsilon^2, \varepsilon x)$$

(k a field) and

$$X = T_{\text{red}} = \operatorname{Spec} k[x],$$

then the morphisms

$$\phi_\lambda : T \longrightarrow X, \qquad x \longmapsto x + \lambda\varepsilon \quad (\lambda \in k),$$

agree on the dense open subset $\operatorname{Spec} B_x$, but are not equal.

be the morphism with components f_1 and f_2. Since

$$D = \Delta_{X/S}(X)$$

is closed, its inverse image under g is a closed subset of T containing U, and is therefore equal to T. Since T is reduced, g factors through D (cf. EGA I, 5.2.2); consequently

$$f_1 = p_1 \circ g = p_2 \circ g = f_2.$$

(iv)⇒(iii). Let T be a reduced S-scheme, and let $f_1, f_2 : T \to X$ be two S-morphisms that agree on a dense open subset U. Since T is reduced, in order to prove that $f_1 = f_2$, it is enough to prove that $f_1 = f_2$ set-theoretically. Indeed, suppose this has been established, let $t \in T$, let V be an affine open subset of X containing $f_1(t) = f_2(t)$, and let W be the open neighborhood of t that is the inverse image of V under the continuous map underlying both f_1 and f_2. Then the morphisms

$$f_i|_W : W \longrightarrow V$$

agree on the dense open subset $U \cap W$ of W. Since V is separated and W is reduced, this implies that $f_1|_W = f_2|_W$, and hence that $f_1 = f_2$.

Now let $t \in T$, let s be its image in S, and let us show that the points

$$x_1 = f_1(t), \qquad x_2 = f_2(t)$$

of X_s are equal. Let

$$\phi : S'' \longrightarrow S' \longrightarrow S$$

and an open subset V of $X \times_S S''$ be as in (iv). Set

$$T' = T \times_S S', \qquad T'' = T \times_S S'',$$

and denote by

$$g : T'' \longrightarrow T, \qquad f''_i : T'' \longrightarrow X_{S''} \quad (i = 1, 2)$$

the morphisms obtained by base change.

Since U is dense in T and $T' \to T$ is open, the inverse image U' of U in T' is dense in T'. Let U'' be the inverse image of U' in T'', and let F be the reduced subscheme of T'' whose underlying space is $\overline{U''}$. Since $T'' \to T'$ is surjective and closed, the image of F contains U' and is closed, and is therefore equal to T'. Consequently,

$$F \cap g^{-1}(t)$$

contains a point u.

For $i = 1, 2$, let h_i be the restriction of f''_i to F. Then $h_i(u)$ is a point of $X_{S''}$ above $f_i(t) = x_i$, and hence belongs to V, since V contains the inverse image of $\{x_1, x_2\}$ in $X_{S''}$.

Thus

$$W = h_1^{-1}(V) \cap h_2^{-1}(V)$$

is an open subset of F containing u, and the S''-morphisms

$$h_i|_W : W \longrightarrow V$$

agree on the dense open subset $U'' \cap W$ of W. Since V is separated over S'', we have

$$h_1|_W = h_2|_W.$$

It follows that $h_1(u) = h_2(u)$, and hence that $f_1(t) = f_2(t)$. This proves (iv) $\Rightarrow$ (iii).

Theorem 5.3 now follows from 5.6.2 and the implication (iv) $\Rightarrow$ (i) of 5.6.3.

Counterexample 5.6.4.— *Not every S-group G is separated. Let S be a scheme having a non-isolated closed point s, and let G be the scheme obtained by gluing two copies of S along the open subset $S - \{s\}$. It is easily seen that G is not separated over S, and that it carries a natural S-group structure whose fibers are all trivial, except for the fiber G_s, which is isomorphic to the two-element group $\mathbb{Z}/2\mathbb{Z}$.*[45]

Theorem 5.7.—[46] *Let S be a scheme, let G be an S-group locally of finite presentation over S, such that the function*

$$s \longmapsto \dim G_s$$

is locally constant on S, let X be an S-scheme on which G acts, and let U be an open subset of X, separated over S. Then $\underline{G^0} \cdot U$ is an open subset of X, separated over S.

Proof. Let

$$\mu : G \times_S X \longrightarrow X$$

denote the action of G on X. It is the composite of the automorphism

$$(g, x) \longmapsto (g, gx)$$

of $G \times_S X$ and the projection onto X. Since $G \times_S U$ is an open subset of $G \times_S X$, it follows from 4.2(iii) that

$$V = \underline{G^0} \cdot U$$

is open in X.

Then, arguing as in the proof of 5.6.2, we deduce from the implication (iv)$\Rightarrow$(i) of 5.6.3 that V is separated over S.

Corollary 5.7.1.— *Let S and G be as in 5.7, and let σ, τ be two S-sections of G^0, that is, $\sigma, \tau \in G^0(S)$. Then the coincidence subscheme $S(\sigma, \tau) \subset S$ of σ and τ* (that is, the inverse image of the diagonal of $G \times_S G$ under the morphism (σ, τ)) *is closed.*

Indeed, for every $s \in S$, let U be an affine open subset of G containing $\epsilon(s)$. Then

$$V = \epsilon^{-1}(U)$$

is an open neighborhood of s in S, and, since G^0U is separated, $S(\sigma, \tau) \cap V$ is closed in V.

Remark 5.7.2.— Gabber points out that one can show the following: if S is henselian local, with closed point s, then the intersection of the open subsets G^0U, where U ranges over the affine open neighborhoods of $\epsilon(s)$, is an open group subscheme of G, separated over S.

5.8. Complements

[47]We begin by recalling Proposition VI_A, 2.6.6:

[45]Editor's note: we reproduced this example in VI_A, 0.3, editor's note 5.

[46]Editor's note: on the one hand, we omitted Corollary 5.6.5, which repeated 5.5. On the other hand, the original stated Corollary 5.7.1 below as Remark 5.7, referring for its proof to 4.7, a number that does not exist in *Lecture Notes in Mathematics* 151 (but did appear in the 1965 edition of SGAD, whose numbers 4.5 and 4.6 became 5.6.1 and 5.6.2); we have added Theorem 5.7, communicated by O. Gabber, which makes the aforementioned statement of SGAD more precise.

[47]Editor's note: we have added this paragraph of complements and counterexamples, all communicated by O. Gabber.

Proposition 5.8.1.— *Let k be a field, let G be a k-group locally of finite type acting on a k-scheme X in such a way that the morphism*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is surjective. Suppose that X is quasi-separated. Then the connected components of X are irreducible.

Corollary 5.8.2.— *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that every fiber X_s is quasi-separated and connected. Then X is separated and quasi-compact over S.*

Indeed, by the preceding proposition, every fiber X_s is irreducible, and the rest of the proof is identical to that of 5.4.

Example 5.8.3.— Fix an algebraically closed field k. Recall first that every "locally Boolean" topological space X, that is, every space having a basis of compact open subsets, becomes a k-scheme when equipped with the sheaf of locally constant k-valued functions (cf. [DG70], I, §1, 2.12). Such an X will be called a *locally Boolean k-scheme.*

We also note that every topological space E having a basis of separated open subsets—and likewise every scheme X—has a dense separated open subset. Indeed, every increasing union of separated open subsets is separated (for a scheme X, this follows from 5.6.3), so there is a maximal such open subset U. It must be dense: otherwise there would be a nonempty open subset V with $U \cap V = \varnothing$; the open set V would contain a nonempty separated open subset W, and

$$U \cup W$$

would still be separated, contradicting the maximality of U.

Now let C be the Cantor space, viewed as the underlying space of the group

$$(\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$$

with its product topology. For every $p \in C$, let $C(p)$ be another copy of C, and let X be obtained by gluing each $C(p)$ to C along $C - \{p\}$. Then X is a nonseparated locally Boolean k-scheme, and C is a dense open subset of X.

Let G be the group of automorphisms of the k-scheme X, that is, of homeomorphisms of X. Then G acts transitively on X. Indeed, X is the union of C and, for each point $p \in C$, a second point p'. This second point is characterized as the unique $x \in X - \{p\}$ for which

$$(C - \{p\}) \cup \{x\}$$

is separated. Consequently, every automorphism ϕ of C extends to an automorphism ϕ_X of X satisfying

$$\phi_X(p') = \phi_X(p)'$$

for every p. Moreover, the map

$$\theta_p : X \longrightarrow X$$

that exchanges p and p' and fixes every other point is an automorphism of X. Finally, the automorphism group of C acts transitively on C: using the group structure of C, it is enough to consider translations.

Thus the discrete k-group G, which is locally of finite type, acts on the k-scheme X in such a way that

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is surjective, but X is not separated, even though C is a dense separated open subset.

Example 5.8.4.— Retain the notation of the preceding example. Using the description of C as the set of sequences
$$(u_n)_{n\in\mathbb{N}}, \qquad u_n \in \mathbb{Z}/2\mathbb{Z},$$
one sees that C with one point p removed is homeomorphic to a countable disjoint union of copies of C. Indeed, using the group structure of C, for example, translation reduces us to the case in which $p = 0$, the zero sequence. Then
$$C - \{0\} = \coprod_{i\in\mathbb{N}} C_i,$$
where
$$C_i = \{(u_n)_{n\in\mathbb{N}} \mid u_i = 1 \text{ and } u_j = 0 \text{ for } j < i\},$$
and every C_i is homeomorphic to C. It follows by induction on $|F|$ that, for every nonempty finite subset F of C, the space $C - F$ is homeomorphic to a countable disjoint union of copies of C, and hence to C with one point removed.

For each $F \subset C$ of cardinality 2, let $C(F)$ be another copy of C, denote by q_F the point 0 of $C(F)$, and choose a homeomorphism
$$\phi_F : C(F) - \{q_F\} \xrightarrow{\sim} C - F.$$
Let X' be obtained by gluing every $C(F)$ to C by means of ϕ_F. Then X' is a nonseparated locally Boolean k-scheme. Moreover, the construction shows that every locally constant function
$$f : X' \longrightarrow k$$
is constant. Indeed, if $x, y \in C$ and $F = \{x, y\}$, then every neighborhood of q_F meets every neighborhood of x and of y. Thus
$$f(x) = f(q_F) = f(y).$$
If $F' = \{z, t\}$, with $z, t \in C$, then likewise
$$f(q_{F'}) = f(z) = f(q_{\{z,x\}}) = f(x).$$
Consequently, X' is connected.

Every point $x \in X'$ also has a pointed neighborhood homeomorphic to $(C, 0)$. More precisely, for every F, fix a homeomorphism of pointed topological spaces
$$h_F : (C, 0) \xrightarrow{\sim} (C(F), q_F),$$
and let T be the group of translations of C. If $x = q_F$, use h_F; if $x \in C$, the translation $t_x \in T$ is a homeomorphism
$$t_x : (C, 0) \xrightarrow{\sim} (C, x),$$
and it is the unique element of T with this property.Let L be the free group generated by the h_F, and set
$$G = T * L,$$
the free product (coproduct) of T and L. For
$$h \in \{h_F\}_{|F|=2} \cup T,$$
let $\sigma(h)$ denote the corresponding generator of G, and let $S(h)$ and $B(h)$ be respectively the source and the target of h. It is convenient also to set
$$\sigma(h_F^{-1}) = \sigma(h_F)^{-1}, \qquad S(h_F^{-1}) = B(h_F), \qquad B(h_F^{-1}) = S(h_F),$$

and to let E be the set consisting of the elements of T and all h_F, h_F^{-1}.

On the product space

$$P = G \times X',$$

where G carries the discrete topology, consider the equivalence relation generated by

$$(g\sigma(h), x) \sim (g, h(x)) \qquad \text{when } x \in S(h),$$

for every $h \in E$, and let Z be the quotient space. Thus Z is obtained by gluing open subsets of the disjoint union

$$\coprod_{g \in G} \{g\} \times X'.$$

Consequently, the saturation $\widetilde{\Omega}$ of every open subset Ω of P is open (cf. [BTop], I, §5.1, Example 2). Explicitly, since every open subset of P is the union of its intersections with the slices $\{g\} \times X'$, it is enough to consider $\{1\} \times W$, with W open in X'. Its saturation is the union of $\{1\} \times W$ and the sets

$$\{\sigma(h_1)\} \times h_1^{-1}(W \cap B(h_1)),$$

$$\{\sigma(h_1)\sigma(h_2)\} \times h_2^{-1}\Big(h_1^{-1}(W \cap B(h_1)) \cap B(h_2)\Big),$$

and so on, for all finite sequences $h_1, \dots, h_n$ of elements of E. It is therefore open, and the projection

$$\pi : P \longrightarrow Z$$

is open.

Moreover, the word

$$\sigma(h_1) \cdots \sigma(h_n)$$

is reduced in G, unless one of the h_i is the identity of T, two consecutive h_i's belong to T, or one of the pairs (h_i, h_{i+1}) is

$$(h_F, h_F^{-1}) \quad \text{or} \quad (h_F^{-1}, h_F).$$

Suppose, therefore, that $x \in X'$ and that

$$\beta = (\sigma(h_1) \cdots \sigma(h_n), h_n^{-1} \cdots h_1^{-1}(x))$$

belongs to $\{1\} \times X'$. We may then suppose that every h_i is a translation t_i. In that case,

$$\sigma(t_1 \cdots t_n) = 1$$

implies $t_1 \cdots t_n = 1$, and hence $\beta = (1, x)$. The equivalence relation is compatible with the action of G on $P = G \times X'$ by left translation on the first factor. It follows that, for every $g \in G$, the restriction of π to $\{g\} \times X'$ is injective.

Now let $z \in Z$ and choose a representative $(g, x) \in P$. By the preceding discussion,

$$U = \pi\big(\{g\} \times X'\big)$$

is an open neighborhood of z, and the continuous map

$$\{g\} \times X' \longrightarrow U$$

induced by π is open and bijective, hence a homeomorphism. Thus Z is locally isomorphic to X', and hence also to C; it is therefore again a locally Boolean k-scheme.

Finally, G acts transitively on Z. Indeed, every $z \in Z$ is G-conjugate to an element of the form $\pi(1, x)$, so it is enough to show that every $(1, x) \in P$ is equivalent to some $(\sigma(h), 0)$. If $x \in C$, take $h = t_x$; if $x = q_F$, take $h = h_F$. Thus Z is a locally Boolean k-scheme endowed with a transitive action of the discrete group G, but Z is not separated.

6. Subfunctors and group subschemes*

Definition 6.1.— (i) Let X be an S-functor (that is, a functor from $(\mathbf{Sch}/S)^\circ$ to $(\mathbf{Ens})$), let G be an S-functor in groups, and let u and v be two S-morphisms from X to G. The *transporter of u into v*, denoted $\underline{\mathrm{Transp}}(u,v)$, is the sub-$S$-functor of G defined by

$$\begin{aligned}\underline{\mathrm{Transp}}(u,v)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g)\circ u_{S'} = v_{S'}\}\\ &= \left\{ g\in G(S') \;\middle|\; \begin{array}{l} g_{S''}u_{S''}(x)g_{S''}^{-1} = v_{S''}(x),\\ \text{for every } x\in X(S'') \text{ and every } S''\to S'\end{array}\right\}.\end{aligned}$$

In particular, $\underline{\mathrm{Transp}}(u,u)$ is a group sub-S-functor of G; it is called the *centralizer of* u and is denoted $\underline{\mathrm{Centr}}(u)$.

(ii) Let G be an S-functor in groups, and let X and Y be two sub-S-functors of G. The *transporter of X into Y* (respectively, the *strict transporter of X into Y*), denoted $\underline{\mathrm{Transp}}_G(X,Y)$ (respectively, $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y)$), is the sub-$S$-functor of G defined by

$$\begin{aligned}\underline{\mathrm{Transp}}_G(X,Y)(S') &= \{g\in G(S') \mid (\mathrm{int}\, g)(X_{S'})\subset Y_{S'}\}\\ &= \{g\in G(S')\mid g_{S''}X(S'')g_{S''}^{-1}\subset Y(S'') \text{ for every } S''\to S'\},\end{aligned}$$

and, respectively,

$$\begin{aligned}\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y)(S') &= \{g\in G(S') \mid (\mathrm{int}\, g)(X_{S'}) = Y_{S'}\}\\ &= \{g\in G(S')\mid g_{S''}X(S'')g_{S''}^{-1} = Y(S'') \text{ for every } S''\to S'\}.\end{aligned}$$

Note that

$$\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y) = \underline{\mathrm{Transp}}_G(X,Y)\cap c\big(\underline{\mathrm{Transp}}_G(Y,X)\big),$$

where c denotes the inversion morphism of G.[49]

(iii) Let G be an S-functor in groups, let H be a sub-S-functor of G, and let $i : H \to G$ be the canonical S-morphism. The *centralizer* and *normalizer of H in G* are the following group sub-S-functors of G:

$$\underline{\mathrm{Centr}}_G H = \underline{\mathrm{Centr}}(i) = \underline{\mathrm{Transp}}(i,i), \qquad \underline{\mathrm{Norm}}_G H = \underline{\mathrm{Transp}}_G^{\mathrm{str}}(H,H).$$

Finally, the *center of G* is the S-functor in groups

$$\underline{\mathrm{Centr}}(\mathrm{id}_G) = \underline{\mathrm{Centr}}_G G;$$

it is denoted $\underline{\mathrm{Centr}} G$.

*On the same theme, see also the results of Exposé XI, §6, whose natural place would be in the present Exposé VI$_B$. There one finds, in particular, representability criteria for certain group subfunctors of a given group scheme. (48) Editor's note: we have inserted these results below (6.5.2–6.5.5).

[49] Editor's note: if $u : X\to G$ and $v : Y \to G$ are two arbitrary morphisms of S-functors, let $u(X)$ be the image functor of u, defined by

$$u(X)(S') = u(S')\big(X(S')\big)\subset G(S');$$

it is a subfunctor of G, as is the image functor $v(Y)$. One may then consider the transporter of the image of u into the image of v, $\underline{\mathrm{Transp}}_G(u(X),v(Y))$. Thus, in part (ii) of the definition, it is not necessary to restrict to subfunctors, that is, to the case in which u and v are monomorphisms. In the original, this restriction sometimes imposed repetitions in the hypotheses, such as: "Let $u,v : X\to G$ be morphisms of S-functors and let $w : H\to G$ and $w' : K\to G$ be monomorphisms; then $\underline{\mathrm{Transp}}(u,v)$, $\underline{\mathrm{Centr}}_G(w) = \underline{\mathrm{Centr}}_G H$, and $\underline{\mathrm{Transp}}_G(H,K)$ satisfy ...." Such repetitions can be avoided by considering $\underline{\mathrm{Transp}}_G(u(X),v(Y))$. We have made such modifications in 6.2 and, later, in 10.11.

Remark 6.1.1.—[50] It follows from the definitions that the functors $\underline{\mathrm{Transp}}(u,v)$ and $\underline{\mathrm{Transp}}_G(X,Y)$ (and hence also $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y)$) commute with base change: for every $S' \to S$, if G', X', Y', u', v' are obtained from G, X, Y, u, v by base change, then

$$\underline{\mathrm{Transp}}(u,v)_{S'} = \underline{\mathrm{Transp}}(u',v'), \qquad \underline{\mathrm{Transp}}(X,Y)_{S'} = \underline{\mathrm{Transp}}(X',Y').$$

Proposition 6.2.—[51] *Let G be an S-group. For a subfunctor T of the functor G, consider the following property:*

$$(+_f) \qquad \text{for every } s \in S,\ T_s \text{ is representable by a closed subscheme of } G_s.$$

Let $u, w : X \to G$ and $v : Y \to G$ be morphisms of S-schemes. Then:

(i) $\underline{\mathrm{Transp}}(u,w)$ and $\underline{\mathrm{Centr}}(u) = \underline{\mathrm{Transp}}(u,u)$ satisfy condition $(+_f)$.

(ii) $\underline{\mathrm{Transp}}_G(u(X), v(Y))$ and $\underline{\mathrm{Norm}}_G v(Y)$ satisfy condition $(+_f)$ if, for every $s \in S$, v_s is a closed immersion.

(iii) $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(X,Y)$ satisfies condition $(+_f)$ if, for every $s \in S$, u_s and v_s are closed immersions.

This follows from Remark 6.1.1 and Corollary 6.2.5 below.[52]

Definition 6.2.1.— Let $f : X \to S$ be a morphism of schemes. One says that f is *essentially free*, or that X is *essentially free over S*, if one can find a covering of S by affine open subsets S_i, for every i an affine S_i-scheme S'_i faithfully flat over S_i, and a covering $(X'_{ij})_j$ of

$$X'_i = X \times_S S'_i$$

by affine open subsets X'_{ij}, such that, for every (i,j), the ring of X'_{ij} is a projective[53] module over the ring of S'_i.

Proposition 6.2.2.—

a) *If X is essentially free over S, it is flat over S; the converse holds if S is Artinian.*

b) *If S is the spectrum of a field, every S-scheme is essentially free over S.*

c) *If X is essentially free over S, then $X' = X \times_S S'$ is essentially free over S', for every $S' \to S$. The converse holds if $S' \to S$ is faithfully flat and quasi-compact.*

The proof is immediate, using for the converse in a) the fact that a flat module over an Artinian local ring is free.[54]

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[50] Editor's note: we have added this remark, which is used implicitly in Proposition 6.2.

[51] Editor's note: we have modified the statement as indicated in editor's note 49.

[52] Editor's note: the original referred to the results of Exposé VIII, §6. For the reader's convenience, we have reproduced these results (with the exception of VIII, 6.3 and 6.8) in 6.2.1–6.2.5 below. Moreover, this had been suggested by A. Grothendieck in a note at the beginning of VIII.6: "The present section is independent of the theory of diagonalizable groups; its natural place would be in VI_B."

[53] Editor's note: on the one hand, we have replaced the word "free" by "projective", as indicated in VIII, Remark 6.8. On the other hand, the notion of an essentially free S-scheme is closely related to the geometric notion of an S-scheme that is flat and pure, introduced and studied by M. Raynaud and L. Gruson ([RG71]); see the addition 6.9 below.

[54] Editor's note: indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A-module, and let $(x_i)_{i\in I}$ be elements of M whose images form a basis of $M/\mathfrak{m}M$ over k. Let F be the free A-module with basis $(e_i)_{i\in I}$, and let $\varphi : F \to M$ be the A-morphism defined by $\varphi(e_i) = x_i$. Then $Q = \operatorname{Coker}\varphi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose moreover that M is flat over A; then $K = \operatorname{Ker}\varphi$ satisfies $K \otimes_A k = 0$, that is, $K = \mathfrak{m}K$, whence $K = 0$.

The introduction of Definition 6.2.1 is justified by the following theorem.

Theorem 6.2.3.— *Let S be a scheme, let Z be an S-scheme that is essentially free, and let Y be a closed subscheme of Z. Consider the functor*

$$F = \prod_{Z/S} Y/Z : (\mathbf{Sch}_{/S})^{\circ} \longrightarrow \mathbf{Sets}$$

defined by the following condition: $F(S') = \varnothing$ when $Y_{S'} \neq Z_{S'}$, and $F(S')$ is a singleton otherwise.[55]

(i) *This functor is representable by a closed subscheme T of S.*

(ii) *If, moreover, $Y \to Z$ is of finite presentation, then so is $T \to S$.*

Proof. First observe that the functor in question is a sheaf for the fpqc topology. Since $F(S')$ is either $\varnothing$ or $\{\mathrm{pt}\}$ for every S', this amounts to checking the following assertions. If (S_i) is an open covering of S and every

$$Y_{S_i} \longrightarrow Z_{S_i}$$

is an isomorphism, then $Y \to Z$ is an isomorphism; and if $S' \to S$ is faithfully flat and quasi-compact and

$$Y_{S'} \longrightarrow Z_{S'}$$

is an isomorphism, then again $Y \to Z$ is an isomorphism. The first assertion is clear, and the second follows from SGA 1, VIII, 5.4, or EGA IV_2, 2.7.1.

Moreover, by SGA 1, VIII, 1.9, faithfully flat and quasi-compact morphisms are effective descent morphisms for the fibered category of closed-immersion arrows. With the notation of 6.2.1, this allows us to restrict to the case $S = S'_i$.

Let (Z_j) be a covering of Z by affine open subsets such that

$$\mathcal{O}(Z_j)$$

is a projective module over $A = \mathcal{O}(S)$. Put

$$Y_j = Y \cap Z_j,$$

and let

$$F_j : (\mathbf{Sch}_{/S})^{\circ} \longrightarrow \mathbf{Sets}$$

be the functor defined in terms of (Z_j, Y_j) as F was defined in terms of (Z, Y). It is a subfunctor of the final functor, and clearly

$$F = \bigcap_j F_j.$$

It is therefore enough to prove that every F_j is representable by a closed subscheme T_j of S, for then F is represented by the closed subscheme

$$T = \bigcap_j T_j$$

[55] Editor's note: cf. Exposé II, §1, where this functor is denoted $\prod_{Z/S} Y$. For every $S' \to S$,

$$F(S') = \Gamma(Y_{S'}/Z_{S'}),$$

which here is equal to $\{\mathrm{id}_{Z_{S'}}\}$ if $Y_{S'} = Z_{S'}$, and is empty otherwise. We have also added assertion (ii), which is used in the proof of 6.5.3.

of S.

We may consequently suppose that Z is itself affine, say

$$Z = \operatorname{Spec} B,$$

where B is a projective A-module. Let L be a free A-module with basis $(e_\lambda)_{\lambda\in\Lambda}$ of which B is a direct summand as an A-module, and let

$$\phi_\lambda : L \longrightarrow A$$

be the coordinate forms with respect to this basis. Let E be a set of generators for the ideal J of B defining the closed subscheme Y of Z, and let I be the ideal of A generated by the coordinates

$$\phi_\lambda(x), \qquad x \in E, \quad \lambda \in \Lambda.$$

For every A-algebra C, the morphism

$$B \otimes_A C \longrightarrow (B/J) \otimes_A C$$

is an isomorphism if and only if the image of $x \otimes 1$ in $L \otimes_A C$ is zero for every $x \in E$. This, in turn, is equivalent to saying that the kernel of $A \to C$ contains the ideal I. Hence

$$T = V(I) = \operatorname{Spec}(A/I)$$

has the required property, proving (i).

Finally, if $Y \to Z$ is of finite presentation, the set E may be chosen finite. Then I is a finitely generated ideal of A, that is, the closed immersion $T \to S$ is of finite presentation. This proves (ii).

Examples 6.2.4.— Let us give important examples of functors which reduce to functors

$$\prod_{Z/S} Y/Z$$

of the type considered in 6.2.3, and for which it will be useful to have representability criteria later on. We denote by S a scheme, and by $X, Y, Z, \ldots$ schemes over S.

a) Suppose given an S-morphism

$$q : X \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z), \tag{x}$$

("X acts on Y, with values in Z"), that is, a morphism

$$r : X \times_S Y \longrightarrow Z. \tag{xx}$$

Consider a subscheme Z' of Z, and hence a monomorphism

$$\underline{\operatorname{Hom}}_S(Y, Z') \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z),$$

which makes the first functor a subfunctor of the second. Let X' be the inverse image of this subfunctor under q. Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Z_T$$

factors through Z'_T.

This functor X' may be described as follows. Put

$$P = X \times_S Y,$$

and let P' be the inverse image of Z' under $r : P \to Z$. Then there is an evident isomorphism

$$X' \simeq \prod_{P/X} P'/P. \tag{xxx}$$

It follows that, if Y is essentially free over S and Z' is closed in Z, the subfunctor X' of X is representable by a closed subscheme of X.[56] If, moreover, $Z' \to Z$ is of finite presentation, then so is $X' \to X$.

b) Suppose given two ways of letting X act on Y, with values in Z, that is, two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\mathrm{Hom}}_S(Y, Z),$$

and put

$$X' = \mathrm{Ker}(q_1, q_2).$$

Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms

$$q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$$

are equal. The data of q_1, q_2 are equivalent to the data of a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z \times_S Z),$$

or, equivalently, of a morphism

$$r : X \times_S Y \longrightarrow Z \times_S Z.$$

Put $U = Z \times_S Z$, and let U' be the diagonal subscheme of $Z \times_S Z$. Then X' is precisely the inverse image under q of the subfunctor

$$\underline{\mathrm{Hom}}_S(Y, U') \subset \underline{\mathrm{Hom}}_S(Y, U).$$

It can therefore be put in the form (xxx), with $P = X \times_S Y$ and P' the inverse image of the diagonal under r, that is, the kernel of

$$r_1, r_2 : X \times_S Y \rightrightarrows Z.$$

We are thus under the hypotheses of (a).

Consequently, if Y is essentially free over S and Z is separated over S, the subfunctor X' of X is representable by a closed subscheme of X.[56] If, moreover, $Z \to S$ is locally of finite type, then $X' \to X$ is of finite presentation.

c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y),$$

that is, "X acts on Y." Let X' be the "kernel" of this morphism: X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Y_T$$

[56] Editor's note: we have added the sentence that follows.

is the identity. This functor falls under (b), as one sees by introducing a second morphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

by letting X act trivially on Y. Hence, if Y is essentially free and separated over S, the subfunctor X', the kernel of q, is representable by a closed subscheme of X.[56] If, moreover, $Y \to S$ is locally of finite type, then $X' \to X$ is of finite presentation.

d) Under the hypotheses of (c), consider the subfunctor Y' of Y "of the invariants under X." Thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism

$$\bar{q}(y) : X_T \longrightarrow Y_T$$

is the constant T-morphism with value y. Introducing $\bar{q}'$ as in (c), and the morphisms corresponding to $\bar{q}$ and $\bar{q}'$,

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that

$$Y' = \mathrm{Ker}(\bar{q}, \bar{q}').$$

It therefore falls again under (b), with the roles of X and Y reversed and with $Z = Y$.

Consequently, if X is essentially free over S and Y is separated over S, the subfunctor Y' of Y of the invariants under X is representable by a closed subscheme of Y.[56] If, moreover, $Y \to S$ is locally of finite type, then $Y' \to Y$ is of finite presentation.

e) Constructions of the kind described in the preceding examples occur especially often in group theory. Thus, when G is an S-group scheme acting on the S-scheme X,

$$q : G \longrightarrow \underline{\mathrm{Aut}}_S(X),$$

the kernel of q, the subgroup of G acting trivially, is a closed subscheme of G, provided X is essentially free and separated over S (example (c)). The subobject X^G of invariants is a closed subscheme of X, provided G is essentially free over S and X is separated over S (example (d)).

Let Y, Z be subschemes of X, and consider the subfunctor $\underline{\mathrm{Transp}}_G(Y, Z)$ of G, the transporter of Y into Z. Its T-valued points, for T over S, are the elements $g \in G(T)$ such that the corresponding automorphism of X_T satisfies

$$g(Y_T) \subset Z_T,$$

that is, induces a morphism $Y_T \to X_T$ which factors through $Y_T \to Z_T$. Hence, if Y is essentially free over S and Z is closed in X, then $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subscheme of G (example (a)).

One may also consider the strict transporter of Y into Z,[57] whose T-valued points, for T over S, are the elements $g \in G(T)$ such that $g(Y_T) = Z_T$. This functor is precisely

$$\underline{\mathrm{Transp}}_G(Y, Z) \cap \sigma\big(\underline{\mathrm{Transp}}_G(Z, Y)\big),$$

where σ is the inversion morphism of G. Consequently, if Y and Z are essentially free over S and closed in X, the strict transporter of Y into Z is a closed subscheme of G.

An important case is $X = G$, with G acting on itself by inner automorphisms. If H is a subscheme of G, the strict transporter of H into H is also called the *normalizer of H in*

[57] Editor's note: denoted $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y, Z)$.

G, and is denoted $\underline{\mathrm{Norm}}_G H$. Hence, if H is a closed group subscheme of G, essentially free over S, then $\underline{\mathrm{Norm}}_G H$ is representable by a closed group subscheme of G.

Finally, let Z be a subscheme of G. Its centralizer $\underline{\mathrm{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure of (d), when one regards Z as acting on G by the operations induced by those of G. Thus, if Z is essentially free over S and G is separated over S, then $\underline{\mathrm{Centr}}_G(Z)$ is a closed group subscheme of G. In particular, if G is essentially free and separated over S, then the center C of G, which is none other than $\underline{\mathrm{Centr}}_G(G)$, is a closed group subscheme of G.

When S is the spectrum of a field, 6.2.2(b) shows that the "essentially free" conditions in examples (a)–(e) above are automatically satisfied, leaving only the separation conditions. Recalling that a group scheme over a field is necessarily separated (VI_A, 0.3), we obtain, for example:

Corollary 6.2.5.— *Let G be a group scheme over a field k, and let Y, Y' be two subschemes of G. Then:*

(i) *The centralizer of Y in G is a closed group subscheme of G; in particular, this holds for the center $\underline{\mathrm{Centr}}_G(G)$ of G.*

(i′) *More generally, if $u, v : X \to G$ are morphisms of schemes, then $\underline{\mathrm{Transp}}_G(u, v)$ is representable by a closed subscheme of G.*

(ii) *If Y is closed, the transporter $\underline{\mathrm{Transp}}_G(Y', Y)$ is a closed subscheme of G. If Y' is also closed, the same conclusion holds for $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y', Y)$.*

(iii) *For every group subscheme[58] H of G, $\underline{\mathrm{Norm}}_G(H)$ is a closed group subscheme of G.*

Corollary 6.2.6.—[59] *Let k be a field and let G be a connected algebraic k-group. Then $\underline{\mathrm{Centr}}_G(G)$ is representable by a closed group subscheme Z of G, and G/Z is an affine algebraic k-group.*

Proof. The first assertion is, of course, contained in 6.2.5(i), but we shall see that it also follows from the proof of the second assertion. Indeed, G acts by the adjoint representation on the finite-dimensional k-vector spaces

$$V_n = \mathcal{O}_{G,e}/\mathfrak{m}_e^{n+1},$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$. Let K_n be the kernel of the morphism

$$\rho_n : G \longrightarrow \mathrm{GL}(V_n).$$

By VI_A, 5.4.1, ρ_n induces a closed immersion

$$G/K_n \hookrightarrow \mathrm{GL}(V_n),$$

so every G/K_n is affine. Since G is Noetherian, the intersection K of the K_n is equal to one of the K_n; hence G/K is affine.

On the other hand, if Z denotes the center of G, it is clear that

$$Z \subset K.$$

[58] Editor's note: indeed, over a field k, every group subscheme of G is closed; see VI_A, 0.5.2. Moreover, 6.2.5 completes the insertion of the results taken from VIII, §6.

[59] Editor's note: we have inserted this corollary here (cf. Exposé XII, 6.1), as it will be useful later. We return in 6.3 to the original text of Exposé VI_B.

We shall prove the reverse inclusion. Let $\widehat{\mathcal{O}}_{G,e}$ be the completion of $\mathcal{O}_{G,e}$ for the $\mathfrak{m}_e$-adic topology, and put

$$\widehat{S} = \operatorname{Spec}\widehat{\mathcal{O}}_{G,e}, \qquad S = \operatorname{Spec}\mathcal{O}_{G,e}.$$

Since $\widehat{S} \to S$ is faithfully flat, and since the two morphisms

$$K \times_k S \longrightarrow S, \qquad (g,x) \longmapsto gxg^{-1} \quad \text{and} \quad (g,x) \longmapsto x$$

coincide after the base change $\widehat{S} \to S$, they coincide. Thus K acts trivially on $\mathcal{O}_{G,e}$.

By 6.2.4(e), the subobject G^K of points of G fixed by K, which is precisely $\underline{\mathrm{Centr}}_G(K)$, is a closed subscheme of G. It is therefore defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_G$. Since G^K contains

$$S = \operatorname{Spec}\mathcal{O}_{G,e}$$

and $\mathcal{I}$ is of finite type (because G is Noetherian), there is an open neighborhood U of e such that

$$\mathcal{I}|_U = 0.$$

The subgroup

$$G^K = \underline{\mathrm{Centr}}_G(K)$$

therefore contains U, and hence also $U \cdot U$, which is equal to G because G is irreducible (VI_A, 0.5). Consequently,

$$\underline{\mathrm{Centr}}_G(K) = G.$$

Thus $K \subset Z$, and therefore $Z = K$, as required.

Remark 6.3.— *Let k be an algebraically closed field, let G be a k-group, and let H be a group subscheme of G. Suppose that G and H are of finite type over k and are reduced. Then $\underline{\mathrm{Norm}}_G H$ (respectively, $\underline{\mathrm{Centr}}_G H$) is representable by a group subscheme of G, whose associated reduced subscheme is precisely the normalizer (respectively, centralizer) of H in G in the sense of the* Bible.

Proposition 6.4.— *Let G be an S-group, and let $u : X \to G$ be a monomorphism of S-schemes. Set*

$$T = \underline{\mathrm{Transp}}_G(X,X).$$

The following conditions are equivalent:

(i) *T is a group sub-S-functor of G.*

(ii)

$$T = \underline{\mathrm{Transp}}^{\mathrm{str}}_G(X,X) = \underline{\mathrm{Norm}}_G X.$$

These conditions hold in each of the following two cases:

a) *X is of finite presentation over S.*

b) *T is representable by a scheme of finite presentation over S.*

The equivalence of (i) and (ii) follows from the fact that, for every morphism $S' \to S$ and all $t, t' \in T(S')$, one has

$$tt' \in T(S'),$$

and from the identity

$$\underline{\mathrm{Transp}}^{\mathrm{str}}_G(X,X) = T \cap c(T)$$

(cf. 6.1(ii)).

Suppose first that a) holds. If $t \in T(S)$, then $\mathrm{int}(t)$ is a monomorphism from X to itself, and therefore an S-automorphism of X by EGA IV$_4$, 17.9.6. Thus

$$t \in \underline{\mathrm{Transp}}^{\mathrm{str}}_G(X, X),$$

which proves the assertion in case a).

In case b), it is clear that

$$\mu(T \times_S T) \subset T,$$

and the assertion follows from the next lemma.

Lemma 6.4.2.—[60] *Let G be an S-scheme of finite presentation, equipped with an associative law (in the sense of EGA 0_{III}, 8.2.5). Suppose that, for every S-scheme S' and every $g \in G(S')$, right and left translation by g in the set $G(S')$ are injective, and suppose that $G(S) \neq \varnothing$. Then G is an S-group.*

Proof. It is enough to prove that $G(S')$ is a group for every S-scheme S'. The hypothesis immediately implies that right and left translation by every $g \in G(S')$ on $G_{S'}$ are S'-monomorphisms

$$G_{S'} \longrightarrow G_{S'}.$$

They are therefore S'-automorphisms, since G is of finite presentation over S (EGA IV$_4$, 17.9.6). Consequently, right and left translation by g in the set $G(S')$ are bijective, and the assertion reduces to the following lemma.

Lemma 6.4.3.— *Let G be a nonempty set with an associative law such that right and left translations are bijective. Then G is a group.*

The proof is left to the reader.

Definition 6.5.— *Let G be an S-group, let H be an S-functor, and let $u : H \to G$ be a monomorphism.*

(i) The *connected centralizer of H in G*, denoted

$$\underline{\mathrm{Centr}}^0_G H,$$

is the identity component of the functor $\underline{\mathrm{Centr}}_G H$ (cf. 3.1 and 6.1(iii)).

(i′) For every morphism $u : X \to G$, one similarly defines the functor

$$\underline{\mathrm{Centr}}^0(u)$$

(cf. 6.2(iv)).

(ii) When u_s is a closed immersion for every $s \in S$, the *connected normalizer of H in G*, denoted

$$\underline{\mathrm{Norm}}^0_G H,$$

is the identity component of the functor $\underline{\mathrm{Norm}}_G H$.

N.B. By 1.4.2, the hypothesis in (ii) is satisfied when H is an S-group scheme, G and H are locally of finite type over S, and u is a quasi-compact morphism of S-groups.

Proposition 6.5.1.— *Let G be an S-group locally of finite presentation and quasi-separated over S, let H be a smooth S-group with connected fibers, and let $u : H \to G$ be a monomorphism. Let N be the normalizer of H in G (cf. 6.1). By 6.5.5 below, N is representable by a closed group subscheme of G, of finite presentation over G. Under these circumstances, the following conditions are equivalent:*

[60] Editor's note: we have retained the numbering of the original; there is no number 6.4.1.

(i) *The canonical morphism $H \to N$ is an open immersion.*

(ii) $N^0 = H$ *(cf. 3.10).*

(iii) *For every $s \in S$, one has $H_s = (N_s)^0$.*

Condition (i) implies (ii) by Lemma 3.10.1, since $H^0 = H$. It is also clear that (ii) implies (iii), for

$$H_s = (N^0)_s = (N_s)^0.$$

Finally, let us prove that (iii) implies (i). Since $H_s = (N^0)_s$ for every $s \in S$, the morphism

$$H_s \longrightarrow N_s$$

is an open immersion. Moreover, H and N are locally of finite presentation over S, and H is flat over S. Consequently, $H \to N$ is an open immersion (EGA IV$_4$, 17.9.5).

For the reader's convenience, the results 6.8–6.11 of Exposé XI have been included below.[61]

Theorem 6.5.2.— *Let X be a smooth scheme over S, with geometrically irreducible fibers.*[62]

(i) *For every closed subscheme Y of X, the functor*

$$\prod_{X/S} Y/X$$

is representable by a closed subscheme T of S.

(ii) *Moreover, if $Y \to X$ is of finite presentation, then so is $T \to S$.*

Proof. Let $f : X \to S$ denote the structure morphism. Since f is faithfully flat and locally of finite presentation, it is a covering for the fpqc topology. Moreover,

$$T = \prod_{X/S} Y/X$$

is evidently a subsheaf of S for the fpqc topology. It follows that the representability of T by a closed subscheme of S is a question local on S for the fpqc topology,[63] and the same is true of the question whether T is of finite presentation over S (EGA IV$_2$, 2.7.1).

After the base change $S' \to S$ with $S' = X$, we are reduced to the case in which X admits a section

$$\epsilon : S \longrightarrow X.$$

We may also suppose that S is affine, and hence quasi-compact. We then have:

Corollary 6.5.3.— *Under the hypotheses of 6.5.2, suppose that S is quasi-compact, that $X \to S$ admits a section ϵ, and that $Y \to X$ is of finite presentation. Then there exists an integer $n \geq 0$ such that*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

[61] Editor's note: Sections 6.5.2–6.5.5, taken from Exposé XI, 6.8–6.11, have been inserted here. This was in fact suggested by A. Grothendieck in a note at the beginning of XI 6: "The present section does not use the results of Sections 3, 4, and 5 [of XI]; its natural place would be in VI$_B$."

[62] Editor's note: on the one hand, we have corrected the original by replacing "connected" with "irreducible" (see the proof). On the other hand, according to [RG71], I, 3.3.4(iii) and 4.1.1, it is enough to assume that X is flat over S, with geometrically irreducible fibers and without embedded components.

[63] Editor's note: see, for example, the addition 1.7 in Exposé VIII.

where X_n is the n-th infinitesimal neighborhood of the immersion $\epsilon : S \to X$, and $Y_n = Y \cap X_n$.

When Y is of finite presentation over X, this corollary does indeed imply 6.5.2. Since X is smooth over S, X_n is finite and locally free over S, and is therefore, a fortiori, essentially free over S (see 6.2.1). Hence

$$\prod_{X_n/S} Y_n/X_n$$

is representable by a closed subscheme T of S, of finite presentation over S, by 6.2.3.

Let us first prove 6.5.3, and hence 6.5.2, when S is Noetherian. Put

$$T_n = \prod_{X_n/S} Y_n/X_n.$$

The T_n form a decreasing sequence of closed subschemes of S, and since S is Noetherian this sequence is stationary. Let

$$R = \bigcap_{n\geq 0} \prod_{X_n/S} Y_n/X_n$$

be their common value for all sufficiently large n. Clearly $T \subset R$, and it is enough to prove $R \subset T$.

After the base change $R \to S$, we are reduced to the case

$$R = S,$$

that is,

$$Y_n = X_n \quad \text{for every } n, \qquad \text{or equivalently} \qquad Y \supset X_n \quad \text{for every } n.$$

We must then prove $T = S$, that is, $Y = X$.

Now $Y \supset X_n$ for every n, together with the fact that X is locally Noetherian, implies that in a neighborhood of each point of $\epsilon(S)$, Y is an induced open subscheme of X.[64] Consequently, there exists an induced open subscheme U of X, containing $\epsilon(S)$, such that $U \subset Y$. By EGA IV_3, 11.10.10, since the fibers of X/S are integral, U is schematically dense in X. Since Y is a closed subscheme containing U, it follows that $Y = X$. This proves 6.5.3, and hence 6.5.2, when S is Noetherian.

The general case is reduced to the preceding one. For every $s \in S$, there exist an affine open neighborhood U of s and an affine open neighborhood V of $\epsilon(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U. Let S' be an affine open neighborhood of s contained in

$$\epsilon^{-1}(V) \cap f(V),$$

and put

$$X' = V \cap f^{-1}(S').$$

Then X' and S' are affine open subsets of X and S, respectively, and X'/S' admits a section. By the local nature of 6.5.2 and 6.5.3, we may suppose that $S = S'$.

Thus X' is an affine open subset of X containing $\epsilon(S)$. Since every fiber X_s is irreducible, X'_s is schematically dense in X_s. Consequently, by EGA IV_3, 11.10.10, X' is schematically dense in X; likewise, after every base change $S_1 \to S$,

$$X' \times_S S_1$$

[64] Editor's note: see, for example, the proof of 6.2.6.

is schematically dense in $X \times_S S_1$.

It follows that

$$\prod_{X/S} Y/X = \prod_{X'/S} Y'/X', \qquad Y' = Y \cap X'.$$

This reduces us to the case $X = X'$, so we may suppose that both S and X are affine. Moreover, if

$$X = \operatorname{Spec} B$$

and J is the ideal of B defining Y, then J is the filtered colimit of its finitely generated subideals. Thus Y is the intersection of closed subschemes Y_i of X that are of finite presentation over X, and consequently

$$\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X.$$

To prove 6.5.2, we are therefore reduced to the case in which Y is of finite presentation over X, with S and X affine.

In that case X and Y over S are obtained by base change $S \to S_0$ from an analogous situation X_0, Y_0 over a Noetherian scheme S_0. This reduces us to the Noetherian case already treated, and completes the proof of 6.5.2 and 6.5.3.

Corollary 6.5.4.— *Let X be a smooth S-group scheme of finite presentation with connected fibers, let Y be a group scheme of finite presentation over S, and let $i : Y \to X$ be a monomorphism of S-group schemes.*

(i) *The functor*

$$\prod_{X/S} Y/X$$

is representable by a closed subscheme of S of finite presentation.

(ii) *If, moreover, S is quasi-compact, then for all sufficiently large n,*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n denotes the n-th infinitesimal neighborhood of the unit section $\epsilon : S \to X$, and $Y_n = X_n \cap Y$.

The proof is essentially that of 6.5.3.[65] On the one hand, i is locally of finite presentation (EGA IV$_1$, 1.4.3). On the other hand, the unit sections of Y and X induce bijective immersions

$$S \longrightarrow Y_n \qquad \text{and} \qquad S \longrightarrow X_n,$$

and hence isomorphisms of S_{red} with $(Y_n)_{\mathrm{red}}$ and $(X_n)_{\mathrm{red}}$. Consequently, i_n is quasi-compact, hence of finite type, and one has the commutative diagram

$$\begin{tikzcd} (Y_n)_{\mathrm{red}} \arrow[r, "\tau"] \arrow[dr, "\sigma"'] & Y_n \arrow[d, "i_n"] \\ & X_n \end{tikzcd}$$

[65]Editor's note: the original has been expanded in what follows, and references to EGA IV have been added.

where σ and τ are closed immersions and τ is surjective. Since i_n is separated (being a monomorphism), it follows that i_n is proper (EGA II, 5.4.3). Thus i_n is a proper monomorphism of finite presentation, and therefore a closed immersion (EGA IV$_3$, 8.11.5). It follows from 6.2.3 that

$$\prod_{X_n/S} Y_n/X_n$$

is representable by a closed subscheme T_n of S, of finite presentation over S. It remains to prove the last assertion of 6.5.4 when S is assumed affine as well.

We again reduce immediately to the case where S is Noetherian. With the notation of the proof of 6.5.3, it is enough to prove $R = T$, or equivalently that

$$Y \supset X_n \quad \text{for every } n \qquad \Longrightarrow \qquad Y = X.$$

The hypothesis implies that $i : Y \to X$ is étale at the points of the unit section of Y over S. Thus Y is smooth over S at those points, and the open subset U of points of Y at which Y is smooth over S is an induced open subgroup of Y (cf. 2.3). The morphism

$$\tau : U \longrightarrow X$$

is then an étale monomorphism by 2.5, and hence an open immersion. Since the fibers of X are connected and every open subgroup of an algebraic group is also closed, τ is a surjective open immersion, that is, an isomorphism. Therefore $U = X$, and a fortiori $Y = X$. This completes the proof of 6.5.4.

Proceeding as in 6.2.4(e), one deduces Corollary 6.5.5 from 6.5.4.

Corollary 6.5.5.— *Let G be an S-group scheme locally of finite type and quasi-separated over S, let H be a smooth group scheme of finite presentation over S with connected fibers, and let $i : H \to G$ be a monomorphism of S-groups. Then:*

a) $\underline{\mathrm{Centr}}_G(H)$ *and* $\underline{\mathrm{Norm}}_G(H)$ *are representable by closed subschemes of G, of finite presentation over G.*

a′) *Likewise, for every monomorphism $j : K \to G$ of S-group schemes that is of finite presentation, $\underline{\mathrm{Transp}}_G(H, K)$ is representable by a closed subscheme of G, of finite presentation over G.*

b) *If G is quasi-compact, there exists an integer $n \geq 0$ such that, if H_n denotes the n-th infinitesimal neighborhood of the unit section of H, then*

$$\underline{\mathrm{Centr}}_G(H) = \underline{\mathrm{Centr}}_G(H_n), \qquad \underline{\mathrm{Norm}}_G(H) = \underline{\mathrm{Norm}}_G(H_n),$$

$$\underline{\mathrm{Transp}}_G(H, K) = \underline{\mathrm{Transp}}_G(H_n, K) = \underline{\mathrm{Transp}}_G(H_n, K_n).$$

Proof.[66] First observe that the hypothesis on G implies that the monomorphism $H \to G$ is of finite presentation (EGA IV$_1$, 1.2.4 and 1.4.3), as is the diagonal immersion

$$\Delta_{G/S} : G \longrightarrow G \times_S G$$

(loc. cit., 1.4.3.1). The case of $\underline{\mathrm{Norm}}_G(H)$ therefore reduces to that of the transporter, by taking $H = K$.

[66] Editor's note: the argument that follows has been expanded. This also completes the insertion of XI, 6.8–6.11; at 6.6 below we return to the original text of VI$_B$.

In view of 6.2.4(e), we apply 6.5.4 to the group scheme

$$X = H_G = H \times_S G$$

over the base scheme G, and to the following group subscheme Y.

In the case of $\underline{\mathrm{Transp}}_G(H, K)$, take Y to be the inverse image of K_G under the morphism of G-groups

$$H \times_S G \longrightarrow G \times_S G, \qquad (h, g) \longmapsto (ghg^{-1}, g).$$

In the case of $\underline{\mathrm{Centr}}_G(H)$, take Y to be the inverse image of the diagonal subgroup

$$\Delta_{G/S}(G)_G$$

of $(G \times_S G)_G$ under the morphism of G-groups

$$H \times_S G \longrightarrow G \times_S G \times_S G, \qquad (h, g) \longmapsto (h, ghg^{-1}, g).$$

Definition 6.6.— Let G be an S-functor in groups and let H be a group sub-S-functor of G. One says that H is *invariant* (respectively, *central*, respectively, *characteristic*) in G if

$$\underline{\mathrm{Norm}}_G H = G$$

(respectively, if $\underline{\mathrm{Centr}}_G H = G$, respectively, if for every S-scheme T and every automorphism

$$a \in \mathrm{Aut}_{T\text{-gr}}(G_T)$$

one has $a(H_T) \subset H_T$). In other words, for every S-scheme T, the subgroup $H(T)$ of $G(T)$ is normal in $G(T)$ (respectively, central in $G(T)$, respectively, invariant under every automorphism of G_T).

N.B. If H is central (respectively, characteristic), then it is invariant.

Remark 6.7.— Let G and H be two S-groups and let $u : H \to G$ be a monomorphism. The group H is invariant (respectively, central) in G if and only if the morphism

$$\mu \circ (c \circ \mathrm{pr}_2,\ \mu \circ (u \times \mathrm{id}_G)) : H \times_S G \longrightarrow G$$

(defined by

$$(h, g) \longmapsto g^{-1}hg$$

for $g \in G(S')$ and $h \in H(S')$) factors through u (respectively, is equal to $u \circ \mathrm{pr}_1$). The group H is characteristic in G if and only if, for every S-scheme T and every T-group automorphism a of G_T, the morphism $a \circ u(T)$ factors through $u(T)$.

Example 6.8.— Let G be an S-functor in groups. Then $\underline{\mathrm{Centr}} G$ is characteristic and central. If G_s is representable for every $s \in S$, then G^0 is characteristic. This follows from the definitions and 3.3.

6.9.—[67] In [RG71], I, 3.3.3, the authors introduce the geometric notion of a *pure S-scheme*, which is local on S for the étale topology; we refer to *loc. cit.* for the precise definition. We merely record the following points:

a) (*loc. cit.*, Theorem 3.3.5) If B is a flat, finitely presented A-algebra, then B is a projective A-module if and only if $\mathrm{Spec}(B)$ is pure over $\mathrm{Spec}(A)$.

[67] Editor's note: this subsection has been added.

b) Consequently, if $X \to S$ is locally of finite presentation, flat, and pure, then X is essentially free over S.

c) (*loc. cit.*, 3.3.4(iii)) If $X \to S$ is locally of finite type, flat, has geometrically irreducible fibers, and has no embedded components, then X is pure over S.

Every group scheme locally of finite type over a field is Cohen–Macaulay (cf. VI_A, 1.1.1), and therefore has no embedded components. We obtain in particular:

d) Every S-group scheme G that is locally of finite presentation, flat, and has connected fibers is pure over S, and hence essentially free over S.

All the statements of 6.2.4(e) may therefore be used with the results of (b) and (d) above taken into account.

7. Generated subgroups; commutator group

Throughout this section, k denotes a fixed field.

Proposition 7.1.— *Let G be a k-group, let $(X_i)_{i\in I}$ be a family of geometrically reduced k-schemes,*[68] *and, for every $i \in I$, let $f_i : X_i \to G$ be a k-morphism.*

(i) *There exists a smallest closed k-group subscheme of G through which every f_i factors. It is denoted*

$$\Gamma_G((f_i)_{i\in I}).$$

It is a geometrically reduced k-scheme, and hence smooth when G is locally of finite type over k (1.3.1).

(ii) *Put*

$$X = \coprod_{i\in I} X_i,$$

and let $f : X \to G$ be the morphism whose restriction to X_i is f_i, for every $i \in I$. Put

$$X^1 = X \amalg X,$$

and let $f^1 : X^1 \to G$ be the morphism whose restrictions to the two copies of X are f and $c \circ f$, respectively. For every $n > 1$, put

$$X^n = X^1 \times_k X^{n-1}, \qquad f^n = \mu \circ (f^1 \times_k f^{n-1}) : X^n \longrightarrow G.$$

Then $\Gamma_G((f_i)_{i\in I})$ is the reduced subscheme of G whose underlying space is the closure of

$$\bigcup_{n\geq 1} f^n(X^n).$$

(iii) *For every k-scheme S, $\Gamma_G((f_i)_{i\in I})_S$ is the smallest closed group subscheme of G_S through which every*

$$f_{i,S} : (X_i)_S \longrightarrow G_S$$

factors.

[68] Editor's note: here and below, the little-used term "separable" has been replaced by the customary term "geometrically reduced"; see EGA IV_2, 4.6.2.

(iii′) *Moreover,* $\Gamma_G((f_i)_{i\in I})_S$ *is the smallest group subscheme of* G_S*, not necessarily closed, through which every* $f_{i,S}$ *factors.*[69]

First note that, to prove (i), (iii), and (iii′), after defining X and f as in (ii), we are reduced to the case in which I consists of a single element.

Let H be the reduced subscheme of G whose underlying set is

$$\overline{\bigcup_{n\geq 1} f^n(X^n)}.$$

The family of morphisms

$$f^n : X^n \longrightarrow H$$

is schematically dominant (cf. EGA IV$_3$, 11.10.4), and therefore every closed subscheme of G through which all the f^n factor also contains H. Moreover, by *loc. cit.*, 11.10.7, H is geometrically reduced. To prove (i) and (ii), it is therefore enough to show that H is a group subscheme of G, that is, that the restriction of c to H and the restriction of μ to $H \times_k H$ factor through the inclusion $H \to G$.

Since H is geometrically reduced, $H \times_k H$ is reduced, and it is enough to verify, set-theoretically, that

$$c(H) \subset H, \qquad \mu(H \times_k H) \subset H.$$

By EGA IV$_3$, 11.10.6, the union of the

$$f^n_{(H)}(X^n \times_k H)$$

is schematically dense in $H \times_k H$. Similarly, for every $n \geq 1$, the union of the

$$f^m_{(X^n)}(X^n \times_k X^m), \qquad m \geq 1,$$

is schematically dense in $X^n \times_k H$. It is therefore enough to prove

$$\mu\Big(f^n_{(H)}(f^m_{(X^n)}(X^n \times_k X^m))\Big) \subset H$$

and $c(f^n(X^n)) \subset H$. But

$$\begin{aligned}\mu\Big(f^n_{(H)}(f^m_{(X^n)}(X^n \times_k X^m))\Big) &= \mu((f^n \times_k f^m)(X^n \times_k X^m)) \\ &= f^{n+m}(X^{n+m}) \subset H.\end{aligned}$$

Also, since $c(f^1(X^1)) \subset f^1(X^1)$, for every n one has

$$c(f^n(X^n)) \subset f^n(X^n) \subset H.$$

This proves (i) and (ii).

We now prove (iii). Let G' be a closed group subscheme of G_S through which f_S factors. We must show that G' contains H_S, or equivalently that

$$H_S = H_S \times_{G_S} G'.$$

Put

$$H'_S = H_S \times_{G_S} G' = G' \cap H_S.$$

[69]Editor's note: the original stated (iii′) under the additional hypothesis that G be locally of finite type over k; this hypothesis may be omitted in view of VI$_A$, 0.5.2.

Every f_S^n factors through both G' and H_S, and hence through H'_S. Since the family

$$f^n : X^n \longrightarrow H$$

is schematically dominant, EGA IV$_3$, 11.10.6 implies that the base-changed family

$$f_S^n : X_S^n \longrightarrow H_S$$

is schematically dominant as well. Because every f_S^n factors through H'_S, EGA IV$_3$, 11.10.1(c) gives

$$H'_S = H_S.$$

This proves (iii).

Finally, we show that H_S is the smallest group subscheme, not necessarily closed, of G_S through which f_S factors. Let G' be such a group subscheme and again put

$$H'_S = H_S \times_{G_S} G'.$$

We must show $H'_S = H_S$. By (iii), it is enough to show that H'_S is closed in H_S. It is enough in turn to show that H_S and H'_S have the same underlying set, and for this it suffices to show, for every $s \in S$, that

$$H'_s := H'_S \times_S \kappa(s) = H_s \times_{G_s} G'_s = H_s.$$

By VI$_A$, 0.5.2, the $\kappa(s)$-group subscheme G'_s is closed in G_s. Hence H'_s is closed in H_s, and the preceding argument, applied to H'_s, H_s, and the f_s^n, shows that

$$H'_s = H_s.$$

This proves (iii$'$) and completes the proof.

Corollary 7.1.1.—[70] *Let G be a k-group locally of finite type, let A and B be two k-subgroups of G that are smooth and of finite type, and let i_A (respectively, i_B) be the inclusion of A (respectively, B) into G. Suppose that B normalizes A. Then*

$$A \cdot B = \Gamma_G(i_A, i_B).$$

Proof. Let

$$H = A \rtimes B$$

be the semidirect product of A and B (cf. I, 2.3.5); it is a smooth k-group of finite type. The group morphism

$$u : H \longrightarrow G, \qquad (a, b) \longmapsto ab,$$

is therefore quasi-compact. Hence, by 1.2,

$$u(H) = A \cdot B$$

is a closed reduced subscheme of G, and is a group in the category $(\mathrm{Sch}/k)_{\mathrm{red}}$. By EGA IV$_3$, 11.10.7, $A \cdot B = u(H)$ is geometrically reduced, since H is; it is therefore a closed subgroup of G. The evident inclusion

$$A \cdot B \subset \Gamma_G(i_A, i_B)$$

now proves the corollary.

[70] Editor's note: this corollary has been added to point out this special case of 7.1.

Definitions and Remarks 7.2.—[71]

(i) Let G be a k-group, let $(X_i)_{i\in I}$ be a family of geometrically reduced k-schemes, and, for every $i \in I$, let $f_i : X_i \to G$ be a k-morphism. The *closed group subscheme of* G *generated by the family* $(f_i)_{i\in I}$ is the smallest closed group subscheme of G through which every f_i factors; in this subsection it is denoted

$$\Gamma_G\big((f_i)_{i\in I}\big).$$

If X is a subscheme of G, geometrically reduced over k, and $f : X \hookrightarrow G$ is the immersion, we write $\Gamma_G(X)$ instead of $\Gamma_G(f)$.

(i′) With the notation of 7.1(ii), we shall sometimes put

$$\Gamma'_G(f) = \bigcup_{n\geq 1} f^n(X^n).$$

Notice that $\Gamma'_G(f)$ is a subset of G stable under the group law, in the sense of 3.0.

(ii) If X_1 and X_2 are geometrically reduced k-schemes and $f_1 : X_1 \to G$ and $f_2 : X_2 \to G$ are k-morphisms such that

$$f_1(X_1) = f_2(X_2)$$

as sets, then it is clear that

$$\Gamma_G(f_1) = \Gamma_G(f_2).$$

(iii) Let E be a subset of G such that the reduced subscheme $\overline{E}$ of G is geometrically reduced. The *closed group subscheme of* G *generated by* E is the group subscheme

$$\Gamma_G(E) := \Gamma_G(i),$$

where $i : \overline{E} \hookrightarrow G$ is the inclusion of that reduced subscheme.

(iv) Since every group subscheme of G is closed, by VI$_A$, 0.5.2,[72] we shall speak of the "generated group subscheme" rather than the "generated closed group subscheme."

(v) Let X be a geometrically reduced k-scheme and let $f : X \to G$ be a k-morphism. Suppose that $f(X)$ contains the unit element e of G. Put

$$X'^1 = X \times_k X, \qquad f'^1 = \mu \circ \big(f \times_k (c \circ f)\big),$$

and, for $n > 1$, put

$$X'^n = X'^1 \times_k X'^{n-1}, \qquad f'^n = \mu \circ \big(f'^1 \times_k f'^{n-1}\big).$$

Then $\Gamma_G(f)$ is the reduced subscheme of G whose underlying space is the closure of

$$\bigcup_{n\geq 1} f'^n(X'^n).$$

Indeed, recall the notation of 7.1(ii):

$$X^1 = X \amalg X, \qquad X^n = X^1 \times_k X^{n-1} \quad (n \geq 2).$$

[71] Editor's note: item (viii) of the original has been placed in (i′), and items (vi) and (vii) have been singled out as Corollary 7.2.1 and Definition 7.2.2 below.

[72] Editor's note: the hypothesis that G be locally of finite type over k has been removed here.

The hypothesis $e \in f(X)$ gives, for every $n \geq 1$,

$$f^n(X^n) \subset f'^n(X'^n) \subset f^{2n}(X^{2n}).$$

Consequently, there exists an integer n such that $f^n(X^n) = \Gamma_G(f)$ if and only if there exists an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

From Remark 7.2(v) we deduce the following result.

Corollary 7.2.1.— *Let X be a geometrically reduced and geometrically connected k-scheme, and let $f : X \to G$ be a k-morphism such that $f(X)$ contains the unit element of G. Then the k-group $\Gamma_G(f)$ is connected, and hence irreducible.*

Proof. Each X'^n is connected, so the union of the

$$f'^n(X'^n),$$

all of which contain the unit element, is connected. Its closure $\Gamma_G(f)$ is therefore connected as well. It follows that $\Gamma_G(f)$ is irreducible, by VI_A, 2.4 when G, and hence also $\Gamma_G(f)$, is locally of finite type over k, and by VI_A, 2.6.5(iii) in the general case.

Definition 7.2.2.— Let G be a k-group.

a) Let A and B be two geometrically reduced k-group subschemes of G. The *commutator group subscheme of A and B in G*, denoted

$$(A, B)_G \qquad \text{or simply} \qquad (A, B),$$

is the closed group subscheme of G generated by the morphism

$$\nu : A \times_k B \longrightarrow G$$

defined, for every k-scheme S, by

$$(a, b) \longmapsto aba^{-1}b^{-1}, \qquad a \in A(S), \quad b \in B(S).$$

b) Suppose that G is geometrically reduced over k. The *derived group of G*, denoted $\mathfrak{D}(G)$,[73] is the group

$$\mathfrak{D}(G) = (G, G).$$

N.B. The group G is commutative if and only if $\mathfrak{D}(G)$ is the unit k-group.

Reminders 7.3.0.—[74] Let $\varphi : Y \to Z$ be a morphism of S-schemes. Recall that the image presheaf $\varphi(Y)$ is the S-functor which assigns to every S' over S the subset

$$\varphi(Y(S')) \subset Z(S').$$

Suppose that T is a subscheme of Z. If the inclusion of presheaves

$$\varphi(Y) \hookrightarrow Z$$

factors through T, that is, if

$$\varphi \circ h \in T(S')$$

[73]Editor's note: the notation $\mathrm{Der}(G)$ of the original has been changed to avoid any risk of confusion with a space of derivations.

[74]Editor's note: these reminders have been added, and the statement of 7.3 has consequently been modified and expanded.

for every S-morphism $h : S' \to Y$, then, set-theoretically, $\varphi(Y) \subset T$: take $h = \mathrm{id}_Y$. Conversely, if Y is reduced and $\varphi(Y) \subset T$ set-theoretically, then φ factors through T.

Recall also that, by 6.2, if H is a closed k-group subscheme of G, then

$$\underline{\mathrm{Centr}}_G H \qquad \text{and} \qquad \underline{\mathrm{Norm}}_G H$$

are representable by closed k-group subschemes of G.

Corollary 7.3.— *Let G be a k-group, let X be a geometrically reduced k-scheme, and let $f : X \to G$ be a k-morphism.*

(i) *Let S be a k-scheme and let u be an endomorphism of the S-group G_S.*

  (a) *If*
$$u(f_S(X_S)) \subset \Gamma_G(f)_S$$
  set-theoretically, then the morphism
$$u : \Gamma_G(f)_S \longrightarrow G_S$$
  factors through $\Gamma_G(f)_S$.

  (b) *If u is an automorphism of the S-group G_S and*
$$u(f_S(X_S)) = f_S(X_S)$$
  set-theoretically, then u induces an automorphism of $\Gamma_G(f)_S$. In particular, if $g \in G(S)$ satisfies
$$\mathrm{int}(g)(f_S(X_S)) = f_S(X_S)$$
  set-theoretically, then
$$g \in \underline{\mathrm{Norm}}_{G_S}(\Gamma_G(f)_S)(S).$$

  (c) *If $u \circ f_S = f_S$, then the restriction of u to the subgroup $\Gamma_G(f)_S$ of G_S is the identity. In particular, if $g \in G(S)$ satisfies*
$$\mathrm{int}(g) \circ f_S = f_S,$$
  then
$$g \in \underline{\mathrm{Centr}}_{G_S}(\Gamma_G(f)_S)(S).$$

(ii) *Let H be a group subscheme of G. Then H centralizes (respectively, normalizes) $\Gamma_G(f)$ if and only if, for every k-scheme S and every $h \in H(S)$,*
$$\mathrm{int}(h) \circ f_S = f_S$$
(respectively,
$$\mathrm{int}(h)(f_S(X_S)) \subset \Gamma_G(f)_S$$
set-theoretically). Equivalently, for every $S' \to S$ and every $x \in X(S')$, the elements $h_{S'}$ and $f(x)$ of $G(S')$ commute (respectively,
$$h_{S'} f(x) h_{S'}^{-1} \in \Gamma_G(f)(S').$$
).

(iii) *In particular, let Y be a second geometrically reduced k-scheme and let $\varphi : Y \to G$ be a k-morphism. Suppose that, for every k-scheme S', every $x \in X(S')$, and every $y \in Y(S')$, the elements $\varphi(y)$ and $f(x)$ of $G(S')$ commute (respectively,*

$$\mathrm{int}(\varphi(y))(f(x)) \in \Gamma_G(f)(S').$$

). Then $\Gamma_G(\varphi)$ is a k-subgroup of

$$\underline{\mathrm{Centr}}_G(\Gamma_G(f)) \qquad \textit{(respectively, of)} \qquad \underline{\mathrm{Norm}}_G(\Gamma_G(f)).$$

(iv) *If g is a k-point of G, then the k-group $\Gamma_G(\{g\})$ is commutative.*

(v) *Let A and B be two group subschemes of G, geometrically reduced over k. If A and B are invariant (respectively, characteristic), then so is (A, B).*

Proof. We first prove (i)(a). Put

$$\Gamma_S = \Gamma_G(f)_S.$$

Then $u^{-1}(\Gamma_S)$ is a closed S-group subscheme of G_S. By 7.1(iii), it is therefore enough to show that f_S factors through $u^{-1}(\Gamma_S)$. Since

$$u(f_S(X_S)) \subset \Gamma_S$$

and X_S is reduced, $u \circ f_S$ factors through Γ_S; hence f_S factors through $u^{-1}(\Gamma_S)$. This proves (a).

The first assertion of (b) follows by applying (a) to u and u^{-1}. In fact, it is enough to assume

$$u(f_S(X_S)) \subset \Gamma_S \qquad \text{and} \qquad u^{-1}(f_S(X_S)) \subset \Gamma_S.$$

The second assertion is the special case $u = \mathrm{int}(g)$.

We now prove (c). Consider the morphism of S-groups

$$G_S \longrightarrow G_S \times_S G_S, \qquad g \longmapsto (u(g), g),$$

and let H be the inverse image of the diagonal. It is a group subscheme of G_S. Since G is separated over k, by VI_A, 0.3, G_S is separated over S, and hence H is closed in G_S. By hypothesis, f_S factors through H; therefore H contains $\Gamma_G(f)_S$, by 7.1(iii). This proves the first assertion of (c), and the second is the special case $u = \mathrm{int}(g)$.

We next prove (ii). Put

$$\Gamma = \Gamma_G(f)$$

and let $i : \Gamma \hookrightarrow G$ denote the inclusion. The group H is contained in

$$N = \underline{\mathrm{Norm}}_G(\Gamma)$$

if and only if, for every k-scheme S and every $h \in H(S)$,

$$\mathrm{int}(h)(\Gamma_S) \subset \Gamma_S.$$

Applying this condition to h and h^{-1} shows that the inclusion is then an equality. By (i)(a), this condition holds if and only if

$$\mathrm{int}(h)(f_S(X_S)) \subset \Gamma_S.$$

Similarly, H is contained in
$$C = \underline{\mathrm{Centr}}_G(\Gamma)$$
if and only if, for every k-scheme S and every $h \in H(S)$,

$$\mathrm{int}(h) \circ i_S = i_S.$$

By (i)(c), this holds if and only if

$$\mathrm{int}(h) \circ f_S = f_S.$$

This proves (ii).

We now prove (iii). In view of (ii), the hypothesis implies that φ factors through C (respectively, N). The latter is a closed subgroup of G, by 6.2, and therefore contains $\Gamma_G(\varphi)$, by 7.1(i).

Assertion (iv) follows from (iii) by taking both f and φ to be the closed immersion

$$\mathrm{Spec}\, k \hookrightarrow G$$

defined by g. Indeed, for every k-scheme S, the sets $X(S)$ and $Y(S)$ then each have a single element, which is sent by f, respectively by φ, to g.

Finally, we prove (v). Let

$$\nu : G \times_k G \longrightarrow G, \qquad (g, h) \longmapsto ghg^{-1}h^{-1},$$

and let ν' be its restriction to $A \times_k B$. By 7.2.2,

$$(A, B) = \Gamma_G(\nu').$$

Let S be a k-scheme, and let u be an inner automorphism, respectively an arbitrary S-group automorphism, of G_S. Then
$$u \circ \nu_S = \nu_S \circ (u \times_S u).$$

The hypothesis implies that u induces automorphisms u_1 of A_S and u_2 of B_S. Consequently,

$$u(\nu'_S(A_S \times_S B_S)) = \nu'_S(u_1(A_S) \times_S u_2(B_S)) = \nu'_S(A_S \times_S B_S).$$

By (i)(b), u therefore induces an automorphism of $(A, B)_S$. This proves (v).

Proposition 7.4.— *Let G be a k-group locally of finite type, let X be a finite-type k-scheme that is geometrically reduced and geometrically connected, and let $f : X \to G$ be a k-morphism such that $f(X)$ contains the unit element e of G. Then, with the notation of 7.1*(ii)*, there exists an integer N such that, set-theoretically,*

$$f^N(X^N) = \Gamma_G(f).$$

Proof. By 7.1(iii) and EGA IV$_2$, 2.6.1, we may assume that k is algebraically closed. By Corollary 7.2.1, we may assume that $G = G^0$. Finally, with the notation of 7.2(v), it is enough to show that there exists an integer N such that

$$f'^N(X'^N) = \Gamma_G(f).$$

First case. Suppose that X is irreducible. The closed irreducible subsets

$$\overline{f'^n(X'^n)}$$

then form an increasing sequence in the space G, which is Noetherian because $G = G^0$ is of finite type over k (VI$_A$, 2.4). Hence this sequence is stationary, and there exists an integer m such that

$$\overline{f'^m(X'^m)} = \Gamma_G(f).$$

Moreover, because X and G are of finite type over k, the morphisms f'^n are of finite type over k. Consequently, $f'^m(X'^m)$ is constructible in G (EGA IV$_1$, 1.8.5), and therefore contains an open subset U of its closure $\Gamma_G(f)$ (EGA 0_{IV}, 9.2.3). By VI$_A$, 0.5, one then has

$$\Gamma_G(f) \subset U \cdot U \subset f'^{2m}(X'^{2m}) \subset \Gamma_G(f),$$

and hence

$$f'^{2m}(X'^{2m}) = \Gamma_G(f).$$

Second case. Suppose that X has exactly two irreducible components A_1 and A_2. Since X is connected and k is algebraically closed, there exists

$$a \in (A_1 \cap A_2)(k).$$

The four irreducible subsets

$$A_i \times_k A_j \qquad (i, j = 1, 2)$$

therefore cover $X \times_k X$, and the image of each under the morphism

$$f'^1 = \mu \circ (f \times_k (c \circ f))$$

contains e. Let f'^1_{ij} denote the restriction of f'^1 to the reduced subscheme $A_i \times_k A_j$, and put

$$\begin{aligned} Y &= (A_1 \times_k A_1) \times_k (A_1 \times_k A_2) \times_k (A_2 \times_k A_2) \times_k (A_2 \times_k A_1), \\ g &= \mu \circ \Big(\mu \circ (f'^1_{11} \times_k f'^1_{12}) \times_k \mu \circ (f'^1_{22} \times_k f'^1_{21})\Big). \end{aligned}$$

The scheme Y is irreducible, reduced, and of finite type. By the first case, there is therefore an integer m such that

$$g'^m(Y'^m) = \Gamma_G(g).$$

For every $n \geq 1$, however,

$$f'^n(X'^n) \subset g'^n(Y'^n) \subset f'^{4n}(X'^{4n}).$$

It follows that

$$\Gamma_G(f) = \Gamma_G(g) \qquad \text{and} \qquad f'^{4m}(X'^{4m}) = \Gamma_G(f).$$

General case. We argue by induction on the number of irreducible components of X; this number is finite because X, being of finite type over k, is Noetherian. Suppose the proposition has been proved when X has at most $r-1$ irreducible components, and suppose that X has r such components, say $A_1, \ldots, A_r$. The unit e belongs to the image of one of the A_i; suppose, for example, that

$$e \in f(A_1).$$

Put

$$X_r = A_1 \cup \cdots \cup A_{r-1}, \qquad Y = \Gamma_G(f_r),$$

where f_r denotes the restriction of f to the reduced subscheme X_r. We choose the numbering of the A_i so that X_r is connected, which is always possible. Then Y is a closed, reduced, irreducible subgroup of G, by Corollary 7.2.1.

Put

$$Z = \overline{f(A_r)}$$

and let $T = Y \cup Z$, endowed with its closed reduced subscheme structure; let $i : T \hookrightarrow G$ be the inclusion. It follows immediately from 7.2(ii) that

$$\Gamma_G(i) = \Gamma_G(f).$$

Moreover, T is connected: since X is connected, X_r and A_r have a common point a, and Y and Z have the common point $f(a)$. Also, $e \in T$, and T has at most two irreducible components, because Y and Z are irreducible.

By the induction hypothesis, there exists an integer m such that

$$f_r'^m(X_r'^m) = \Gamma_G(f_r) = Y.$$

Since $X = X_r \cup A_r$, and since

$$Z = \overline{f(A_r)} \subset \overline{f'^m(X'^m)}$$

because $e \in f(A_1)$, we obtain

$$f(X) \subset Y \cup Z \subset \overline{f'^m(X'^m)}.$$

On the other hand, because T has at most two irreducible components, the second case gives an integer p such that

$$i'^p(T'^p) = \Gamma_G(i) = \Gamma_G(f).$$

Now

$$T \subset \overline{f'^m(X'^m)},$$

and hence

$$\overline{f'^{mp}(X'^{mp})} = \Gamma_G(f).$$

As in the first case, this last equality implies

$$f'^{2mp}(X'^{2mp}) = \Gamma_G(f).$$

This completes the proof.

Lemma 7.5.— *Let S be a scheme, let G be an S-group scheme, let X be an S-prescheme, and let $f : X \to G$ be an S-morphism. Suppose that X and G are locally of finite presentation over S, and that, for every $s \in S$ and every maximal point g of G_s, there exists a point x of X such that $f(x) = g$ and f is flat at x. Then the morphism*

$$\mu \circ (f \times_S f) : X \times_S X \longrightarrow G$$

is a covering for the (fppf) *topology.*

[75] By EGA IV_3, 11.3.1, the set V of points of X at which f is flat is open, and $f|_V$ is an open morphism. Thus

$$U = f(V)$$

is an open subset of G. Moreover, by hypothesis,

$$U \cap G_s$$

[75] Editor's note: the argument in the original has been expanded in what follows.

is dense in G_s for every $s \in S$. Let ϕ denote the restriction of $\mu \circ (f \times_S f)$ to $V \times_S V$.

It is enough to show that ϕ is a covering for the (fppf) topology, and for this it is enough to show that ϕ is faithfully flat and of finite presentation (Exposé IV, 6.3.1(iv)). Now ϕ is the composite

$$V \times_S V \xrightarrow{f|_V \times f|_V} U \times_S U \xrightarrow{\mu} G,$$

where the first morphism is faithfully flat and locally of finite presentation, since $f|_V$ is. It therefore remains only to prove the same assertions for the restriction of μ to $U \times_S U$.

Since $G \to S$ is locally of finite presentation and flat, so is

$$\mu : G \times_S G \longrightarrow G.$$

Indeed, this morphism is isomorphic to the morphism obtained from $G \to S$ by the base change $G \to S$ (VI$_A$, 0.1). The induced morphism

$$U \times_S U \longrightarrow G$$

therefore has the same properties. To prove that this last morphism is surjective, it is enough to check fiber by fiber. There the result follows from VI$_A$, 0.5, since $U \cap G_s$ is a dense open subset of G_s for every $s \in S$.

Remark 7.6.0.—[76] *Let S be a scheme, let H be an S-group, and let $f : X \to H$ be a morphism of S-schemes. The group subpresheaf of H generated by the image of f, denoted*

$$\langle \operatorname{Im} f \rangle,$$

is the group subfunctor of H which assigns to every S-scheme S' the subgroup of $H(S')$ generated by $f(X(S'))$. Since every element of this subgroup can be written as a finite product

$$f(x_1)^{\varepsilon_1} \cdots f(x_n)^{\varepsilon_n}, \qquad n \in \mathbb{N}^*, \quad x_i \in X(S'), \quad \varepsilon_i = \pm 1,$$

it follows that, if one puts

$$X^1 = X \amalg X$$

and defines the morphisms $f^n : X^n \to H$ as in 7.1, then $\langle \operatorname{Im} f \rangle$ is precisely the image presheaf of the morphism

$$f^\infty : X^\infty \longrightarrow H,$$

where

$$X^\infty = \coprod_{n \geq 1} X^n$$

and $f^\infty : X^\infty \to H$ is the S-morphism whose restriction to each X^n is f^n.

Proposition 7.6.—[77] *Let S be a scheme, let X be an S-scheme flat and locally of finite presentation over S, let H be an S-group locally of finite presentation over S and with reduced fibers, let $f : X \to H$ be a morphism of S-schemes, and let $f^\infty : X^\infty \to H$ be the S-morphism introduced above. The following conditions are equivalent:*

(i) *H represents the S-sheaf for the* (fppf) *topology associated with the presheaf $\langle \operatorname{Im} f \rangle$.*

(ii) *f^∞ is a covering for the* (fppf) *topology.*

[76] Editor's note: this definition has been added; in the original it was contained in the statement of Proposition 7.6.

[77] Editor's note: the original stated this result under the hypotheses of the special case, but the more general form was used implicitly in the proof of 10.12; the statement has been rewritten accordingly.

(iii) *f^∞ is surjective, that is,*

$$H = \bigcup_{n \geq 1} f^n(X^n).$$

If, in addition, H is quasi-compact, these conditions are also equivalent to the following one:

(iv) *There exists an integer n such that $f^n : X^n \to H$ is a covering for the* (fppf) *topology, and hence is surjective.*

This applies in particular when X is a geometrically reduced k-scheme locally of finite type, G is a k-group locally of finite type, $\varphi : X \to G$ is a morphism of k-schemes, $H = \Gamma_G(\varphi)$, and $f : X \to H$ is the morphism induced by φ.

Proof. The sheaf considered in (i) is the image sheaf of X^∞ under f^∞. Hence, by IV, 4.4.3, saying that H represents this sheaf is equivalent to saying that f^∞ is a covering; this in turn implies that f^∞ is surjective. Conversely, suppose that f^∞ is surjective. We shall show that it is then a covering for the (fppf) topology.

Let $s \in S$, let η be a maximal point of the fiber H_s, and let $x \in X^\infty_s$ be such that

$$f^\infty(x) = \eta;$$

such an x exists because f^∞ is surjective. Since the fiber H_s is reduced, $\mathcal{O}_{H_s,\eta}$ is a field, and therefore f^∞_s is flat at x. Since X^∞ and H are locally of finite presentation over S, and X^∞ is flat over S, the fiberwise criterion for flatness (EGA IV_3, 11.3.10) shows that f^∞ is flat at x. Thus, by Lemma 7.5, the morphism

$$\mu \circ (f^\infty \times_S f^\infty) : X^\infty \times_S X^\infty \longrightarrow H$$

is a covering for the (fppf) topology.

Now $X^n \times_S X^m$ is canonically isomorphic to X^{n+m}, and, under this isomorphism,

$$\mu \circ (f^n \times_S f^m)$$

corresponds to f^{n+m}. It is therefore clear that

$$\mu \circ (f^\infty \times_S f^\infty)$$

factors through f^∞, so f^∞ is a covering for the (fppf) topology. This proves (iii) $\Rightarrow$ (ii), and hence the equivalence of conditions (i), (ii), and (iii).

Furthermore, since the morphisms $X \to S$ and $H \to S$ are locally of finite presentation, so is $f : X \to H$ (EGA IV_1, 1.4.3(v)). The multiplication

$$\mu : H \times_S H \longrightarrow H$$

is also locally of finite presentation (VI_A, 0.1), and it follows that each $f^n : X^n \to H$ is locally of finite presentation. Consequently, by EGA IV_1, 1.9.5(viii), the sets

$$f^n(X^n)$$

are ind-constructible subsets of H.

Suppose, in addition, that H is quasi-compact; then S is also quasi-compact, since $H \to S$ is surjective. By EGA IV_1, 1.9.9, there exists an integer p such that

$$f^p(X^p) = H.$$

As above, the fact that the fibers of H are reduced, together with Lemma 7.5, shows that the morphism
$$\mu \circ (f^p \times_S f^p) : X^p \times_S X^p \longrightarrow H$$
is a covering for the (fppf) topology. Since this morphism is
$$f^{2p} : X^{2p} \longrightarrow H,$$
this completes the proof of Proposition 7.6.

Remark 7.6.1.— The equivalent conditions of 7.6 plainly imply that the sheaf F under consideration is representable. The converse is false in general:[78] for example, if k has characteristic 0, let $G = \mathbb{G}_{a,k}$, and let
$$f : \operatorname{Spec} k \longrightarrow G$$
be the morphism defined by the point 1 of $G(k)$. Then F is represented by the constant k-group $\mathbb{Z}_k$, whereas $\Gamma_G(f) = G$; hence the monomorphism
$$\mathbb{Z}_k \hookrightarrow G$$
is not surjective.

For simplicity, suppose now that the hypotheses of the special case in 7.6 hold, and suppose that F is representable. By EGA IV$_3$, 8.14.2, the sheaf F is locally of finite presentation over k. The question is therefore whether the dominant monomorphism
$$F \longrightarrow \Gamma_G(f)$$
is an isomorphism, or, equivalently, a closed immersion. By 1.4.2, this is so if F is quasi-compact; and by VI$_A$, 0.5.1, it is so if F is connected. In particular, by 7.2.1, it is so when X is connected and $f(X)$ contains the unit element of G.

Lemma 7.7.— *Let k be an algebraically closed field, let G be a k-group locally of finite type, let X be a geometrically reduced k-scheme locally of finite type, let $f : X \to G$ be a k-morphism, and let H be a group subscheme of G such that $H \subset f(X)$. Put*
$$\Gamma' = \bigcup_{n\geq 1} f^n(X^n), \qquad \Gamma'_0 = \Gamma' \cap G(k), \qquad H_0 = H(k),$$
and suppose that H_0 has finite index in Γ'_0. Then there exists an integer m such that
$$f^m(X^m) = \Gamma_G(f)$$
(cf. 7.6), and $\Gamma_G(f)$ is the union of finitely many translates of H.

For every $n \geq 1$, the subset $f^n(X^n)$ of G is ind-constructible (EGA IV$_1$, 1.9.5(viii)). The same is therefore true of Γ'. Since G is a Jacobson scheme, Γ'_0 is dense in Γ'. By hypothesis, there are finitely many points
$$a_1, \dots, a_r \in \Gamma'_0$$
such that
$$\Gamma'_0 = a_1 H_0 \cup \cdots \cup a_r H_0.$$
Consequently,
$$\begin{aligned}\Gamma_G(f) &= \overline{\Gamma'} = \overline{\Gamma'_0} = \overline{a_1 H_0 \cup \cdots \cup a_r H_0}\\ &= \overline{a_1 H_0} \cup \cdots \cup \overline{a_r H_0}\\ &= a_1 \overline{H_0} \cup \cdots \cup a_r \overline{H_0},\end{aligned}$$

[78] Editor's note: the argument in the original has been expanded in what follows.

where the last equality follows because translation by each a_i is a homeomorphism of G onto itself. Thus

$$\Gamma_G(f) = a_1 H \cup \cdots \cup a_r H.$$

It is also clear that there exists an integer p such that every a_i, for $1 \leq i \leq r$, belongs to $f^p(X^p)$. Finally, since $H \subset f(X)$, one has

$$a_i H \subset f^{p+1}(X^{p+1})$$

for every i. Hence

$$f^{p+1}(X^{p+1}) = \Gamma_G(f).$$

Proposition 7.8.— *Let G be a k-group locally of finite type, and let A and B be two geometrically reduced k-group subschemes of G (hence smooth at the generic points of their irreducible components, and therefore smooth by* 1.3*). Suppose that one of the following conditions holds:*

a) *A and B are invariant and of finite type over k;*

b) *A is connected and B is of finite type over k.*

Then (A, B) is of finite type over k, and represents the sheaf associated, for the (fppf) *topology (or the* (fpqc) *topology), with the group presheaf of commutators of A and B in G. Moreover,*[79] *the k-groups (A, B^0) and (A^0, B) are connected, and*

$$(A, B)^0 = (A, B^0) \cdot (A^0, B).$$

Proof. By 7.6, to show that (A, B) is the desired associated sheaf, it is enough to show that there exists an integer n such that

$$\nu^n((A \times_k B)^n) = (A, B),$$

with the notation of 7.2.2. To prove this and the other two assertions, we may suppose that k is algebraically closed.

Indeed, let $\overline{k}$ be an algebraic closure of k. By 7.1(iii) and VI_A, 2.4, one has, with the evident notation,

$$(A, B)_{\overline{k}} = (A_{\overline{k}}, B_{\overline{k}}),$$
$$((A, B)^0)_{\overline{k}} = (A_{\overline{k}}, B_{\overline{k}})^0,$$
$$(A, B^0)_{\overline{k}} = (A_{\overline{k}}, B^0_{\overline{k}}), \qquad \text{and so on.}$$

Consequently, suppose that $(A_{\overline{k}}, B_{\overline{k}})$ is of finite type over $\overline{k}$; respectively, suppose that $(A_{\overline{k}}, B^0_{\overline{k}})$ and $(A^0_{\overline{k}}, B_{\overline{k}})$ are connected and that the morphism

$$(A_{\overline{k}}, B^0_{\overline{k}}) \times_{\overline{k}} (A^0_{\overline{k}}, B_{\overline{k}}) \longrightarrow (A_{\overline{k}}, B_{\overline{k}})^0$$

is surjective. The analogous assertions then hold over k, by EGA IV_2, 2.7.1 and 2.6.1.

Let

$$B^1, \ldots, B^p$$

be the connected components of B other than the identity component B^0. There are only finitely many, since B is of finite type over k, hence Noetherian. In case a), likewise let

$$A^1, \ldots, A^q$$

[79] Editor's note: the assertions that follow have been made explicit, and the reduction to the algebraically closed case has been expanded in the proof.

be the connected components of A other than A^0. In case b), only $A^0 = A$ is considered. Let ν_{ij} be the restriction of ν to

$$A^i \times_k B^j.$$

Each A^i and B^j is irreducible, by VI$_A$, 2.4.1; hence so are $A^0 \times_k B^j$ and $A^i \times_k B^0$. Since the unit element of G belongs to A^0 and to B^0, it belongs to

$$\nu_{0j}(A^0 \times_k B^j) \qquad \text{and} \qquad \nu_{i0}(A^i \times_k B^0).$$

Thus every $\Gamma_G(\nu_{0j})$ and every $\Gamma_G(\nu_{i0})$ is connected, by 7.2.1.

Let u_{0j}, respectively u_{i0}, denote the inclusion of $\Gamma_G(\nu_{0j})$, respectively $\Gamma_G(\nu_{i0})$, in G. Then

$$(A^0, B) = \Gamma_G\big((u_{0j})_{j=0}^p\big), \qquad (A, B^0) = \Gamma_G\big((u_{i0})_{i=0}^q\big),$$

and these groups are connected. Moreover, it follows from 7.4 and the preceding constructions that there is an integer r such that both (A^0, B) and (A, B^0) are contained in

$$\nu^r\big((A \times_k B)^r\big).$$

In case b), one has $(A, B) = (A^0, B)$, and the proof is complete.

We now consider case a).[80] We already know that (A^0, B) and (A, B^0) are smooth connected k-subgroups of G, hence are of finite type over k (VI$_A$, 2.4). Since A^0 is a characteristic subgroup of A (VI$_A$, 2.6.5), it is an invariant subgroup of G; hence 7.3(v) shows that (A^0, B) is an invariant subgroup of G, and the same holds for (A, B^0).

By 7.1.1, the subgroup H of G generated by (A, B^0) and (A^0, B) is therefore

$$H = (A, B^0) \cdot (A^0, B).$$

In particular,

$$H \subset \nu^{2r}\big((A \times_k B)^{2r}\big).$$

If T is a subset of G stable under the group law (cf. 3.0), write T_0 for the group of k-points of G belonging to T. Put

$$\Gamma' = \bigcup_{q \geq 1} \nu^q\big((A \times_k B)^q\big).$$

By Proposition 7.9 below,

$$(A^0, B)_0 = (A_0^0, B_0), \qquad (A, B^0)_0 = (A_0, B_0^0), \qquad \Gamma'_0 = (A_0, B_0).$$

Therefore

$$H_0 = (A_0^0, B_0) \cdot (A_0, B_0^0)$$

has finite index in Γ'_0 (Bible, Exposé III, Appendix), since A_0 and B_0 are invariant and A_0^0, respectively B_0^0, has finite index in A_0, respectively B_0.

The hypotheses of Lemma 7.7 now hold. Since

$$H \subset \nu^{2r}\big((A \times_k B)^{2r}\big),$$

there exists an integer m such that

$$\nu^{2rm}\big((A \times_k B)^{2rm}\big) = (A, B),$$

[80] Editor's note: the argument that follows has been expanded to take account of the added Corollary 7.1.1.

and there are finitely many k-points $a_1, \dots, a_n$ of G such that

$$(A, B) = a_1 H \cup \cdots \cup a_n H.$$

Since H is of finite type over k, every $a_i H$ is quasi-compact. Their union (A, B) is therefore quasi-compact, hence of finite type over k. Finally, H is irreducible, so each $a_i H$ is irreducible; and because the unit e belongs to H, it follows that

$$H = (A, B)^0.$$

Proposition 7.9.—*Let k be an algebraically closed field, and let G be a k-group locally of finite type.*

(i) *Let A and B be two ind-constructible subsets of G. Denote by A_0 the set of rational points of G that belong to A. Then*

$$(A \cdot B)_0 = A_0 \cdot B_0,$$

where the product on the right is taken in the group $G(k)$.

(ii) *Let X be a geometrically reduced k-scheme locally of finite type, and let $f : X \to G$ be a k-morphism. Put*

$$\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n).$$

Then $\Gamma'_G(f)_0$ is the subgroup of $G(k)$ generated by $f(X)_0$.

(iii) *In particular, let A and B be two smooth group subschemes of G, and put*

$$\Gamma' = \bigcup_{n \geq 1} \nu^n((A \times_k B)^n)$$

with the notation of 7.2.2. Then Γ'_0 is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.

Proof. We first prove (i). It is clear that

$$A_0 \cdot B_0 \subset (A \cdot B)_0.$$

Conversely, let $z \in (A \cdot B)_0$. Then $\mu^{-1}(z)$ is a closed subset of $G \times_k G$, while $A \times_k B$ is an ind-constructible subset of $G \times_k G$ (cf. 3.0). Consequently,

$$\mu^{-1}(z) \cap (A \times_k B)$$

is a nonempty ind-constructible subset of $G \times_k G$. By EGA IV$_3$, 10.4.8, it therefore contains a rational point of $G \times_k G$. Its projections x and y are rational points of G such that $x \in A$, $y \in B$, and $x \cdot y = z$. Hence

$$(A \cdot B)_0 = A_0 \cdot B_0.$$

To prove (ii), observe that, since f^n is locally of finite type, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV$_1$, 1.9.5(viii)). Assertion (i) then shows by induction that, if

$$A = f(X)_0,$$

one has

$$f^n(X^n)_0 = (A \cup A^{-1})^n.$$

Consequently,

$$\Gamma'_G(f)_0 = \bigcup_{n\geq 1} f^n(X^n)_0 = \bigcup_{n\geq 1} (A \cup A^{-1})^n,$$

which is the subgroup of $G(k)$ generated by $A = f(X)_0$.

Finally, (iii) follows from (ii) and the definitions.

Corollary 7.10.— *Under the hypotheses of 7.8, if k is algebraically closed, then $(A, B)(k)$ is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.*

Indeed, it is enough to apply 7.9(iii), 7.8, and 7.6.

8. Solvable or nilpotent group schemes

8.1.— Let $\mathcal{C}$ be a category equipped with a topology T (cf. Exposé IV, §4). For every presheaf P on $\mathcal{C}$, write $P^\flat$ for the associated sheaf.

Let G be a presheaf of groups on $\mathcal{C}$, let A and B be two group subpresheaves of G, and let $\mathrm{Comm}(A, B)$ be the group presheaf of commutators of A and B in G. Thus, for every $S \in \mathrm{Ob}\,\mathcal{C}$, $\mathrm{Comm}(A, B)(S)$ is the subgroup of $G(S)$ generated by the commutators

$$aba^{-1}b^{-1}, \qquad a \in A(S), \quad b \in B(S).$$

We write

$$\mathrm{Comm}_T(A, B) = \mathrm{Comm}(A, B)^\flat.$$

In the proof of 8.2, we shall need the following results.[81]

Lemma 8.1.1.— *Let $A \subset G$ be sheaves of groups, with A invariant in G.*

(i) $\mathrm{Comm}_T(G, A)$ *is the smallest invariant group subsheaf C of G such that the sheaf $(A/C)^\flat$, associated with the quotient presheaf A/C, is central in $(G/C)^\flat$.*

(ii) *In particular,* $\mathrm{Comm}_T(G, G)$ *is the smallest invariant group subsheaf C of G such that the quotient sheaf $(G/C)^\flat$ is commutative.*

Clearly, (ii) is the special case $A = G$ of (i), so it is enough to prove (i). Let C be a group subsheaf of A, invariant in G, such that the quotient sheaf $(A/C)^\flat$ is central in $(G/C)^\flat$. By Lemma IV, 4.4.8.1, the presheaves A/C and G/C are separated. Hence, by IV, 4.3.11, all the morphisms in the following diagram are monomorphisms:

$$\begin{tikzcd} A/C \arrow[r, hook] \arrow[d, hook] & G/C \arrow[d, hook] \\ (A/C)^\flat \arrow[r, hook] & (G/C)^\flat \end{tikzcd}$$

Since $(A/C)^\flat$ is central in $(G/C)^\flat$, the presheaf A/C is central in G/C. It follows that

$$\mathrm{Comm}(G, A) \subset C,$$

and hence, by IV, 4.3.12, C contains $\mathrm{Comm}_T(G, A)$.

[81] Editor's note: these results were stated without proof in the original. We have highlighted them as Lemmas 8.1.1 and 8.1.2 and supplied the details of the proofs.

Conversely, $\mathrm{Comm}(G, A)$ is an invariant, separated group subpresheaf of A (cf. IV, 4.3.1, editor's note 24). Therefore, by IV, 4.4.8.2(i) and IV, 4.3.11,

$$C = \mathrm{Comm}_T(G, A)$$

is an invariant group subsheaf of A containing $\mathrm{Comm}(G, A)$. Consequently, the presheaf A/C is central in G/C, and hence, by IV, 4.4.8.2(ii), $(A/C)^\flat$ is central in $(G/C)^\flat$. This proves Lemma 8.1.1.

Lemma 8.1.2.— *Let G be a sheaf of groups, and let A, B be two group subpresheaves of G.*

(i) *The morphism*
$$\tau : \mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)$$
is a covering monomorphism.

(ii) *Consequently, there is an isomorphism*
$$\mathrm{Comm}_T(A, B) \xrightarrow{\sim} \mathrm{Comm}_T(A^\flat, B^\flat).$$

Proof. (i) Since A (respectively, B) is a subpresheaf of G, the associated sheaf $A^\flat$ (respectively, $B^\flat$) is a subpresheaf of G containing A (respectively, B). It follows that τ is a monomorphism.

We now show that τ is a covering. Let $S \in \mathrm{Ob}\,\mathcal{C}$ and let

$$g \in \mathrm{Comm}(A^\flat, B^\flat)(S).$$

There exist an integer $n \geq 1$ and, for $1 \leq i \leq n$, elements

$$a'_i \in A^\flat(S), \qquad b'_i \in B^\flat(S), \qquad \varepsilon_i \in \{\pm 1\},$$

such that
$$g = (a'_1, b'_1)^{\varepsilon_1} \cdots (a'_n, b'_n)^{\varepsilon_n},$$
where (a, b) denotes the commutator $aba^{-1}b^{-1}$. There is a refinement R of S such that

$$a'_i \in A(R) \qquad \text{and} \qquad b'_i \in B(R)$$

for every i. Then g is the composite morphism

$$R \xrightarrow{(a'_1,\ldots,b'_n)} (A \times B)^n \xrightarrow{\Phi^{\varepsilon_1,\ldots,\varepsilon_n}} \mathrm{Comm}(A, B)$$

where $\Phi = \Phi^{\varepsilon_1,\ldots,\varepsilon_n}$ is the morphism defined set-theoretically by

$$\Phi(a_1, b_1, \ldots, a_n, b_n) = (a_1, b_1)^{\varepsilon_1} \cdots (a_n, b_n)^{\varepsilon_n}$$

for every $T \in \mathrm{Ob}\,\mathcal{C}$, with $a_i \in A(T)$ and $b_i \in B(T)$. This shows that

$$g \in \mathrm{Comm}(A, B)(R).$$

It follows, as in the proof of IV, 4.3.11(i), that

$$\mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)$$

is a covering.

The morphism

$$\mathrm{Comm}(A^\flat, B^\flat) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)^\flat$$

is likewise a covering monomorphism (IV, 4.3.11(iv)). Hence the same is true of

$$\mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^\flat, B^\flat)^\flat.$$

By IV, 4.3.12, we therefore obtain an isomorphism

$$\mathrm{Comm}(A, B)^\flat \xrightarrow{\sim} \mathrm{Comm}(A^\flat, B^\flat)^\flat.$$

This proves Lemma 8.1.2.

Proposition 8.2.— *Let $\mathcal{C}$ be a category, let T be a topology on $\mathcal{C}$, let G be a sheaf of groups on $\mathcal{C}$, and let n be an integer with $n \geq 0$. The following conditions are equivalent:*

(i) *Put $K_0 = G$, and, for $p \geq 1$, put*

$$K_p = \mathrm{Comm}(K_{p-1}, K_{p-1}) \qquad (\text{respectively, } \; K_p = \mathrm{Comm}(G, K_{p-1})).$$

Then K_n is the unit presheaf of groups.

(ii) *Put $K'_0 = G$, and, for $p \geq 1$, put*

$$K'_p = \mathrm{Comm}_T(K'_{p-1}, K'_{p-1}) \qquad (\text{respectively, } \; K'_p = \mathrm{Comm}_T(G, K'_{p-1})).$$

Then K'_n is the unit sheaf of groups.

(iii) *There is a sequence*

$$G = H_0 \supset H_1 \supset \cdots \supset H_n$$

of invariant group subsheaves of G such that, for every i, the quotient sheaf H_i/H_{i+1} is commutative (respectively, central in G/H_{i+1})*, and H_n is the unit sheaf.*

It is clear that $K_n \subset K'_n$; consequently, (ii) implies (i). We show that (i) implies (ii). One has

$$K'_1 = \mathrm{Comm}_T(G, G) = K_1^\flat,$$

and induction using Lemma 8.1.2 gives

$$K'_p = K_p^\flat$$

for every p. Consequently, if K_n is the unit presheaf, then $K'_n = K_n^\flat$ is the unit sheaf.

Finally, conditions (ii) and (iii) are equivalent by Lemma 8.1.1.

Definition 8.2.1.— *When these conditions hold, the sheaf G is said to be* solvable of class n (respectively, nilpotent of class n)*. If there is an integer n for which these conditions hold, G is said to be* solvable (respectively, nilpotent)*.*

Observe that, by condition (i), this does not depend on the topology T.

Proposition 8.3.— *Let k be a field, let S be a nonempty k-scheme, let Ω be an algebraically closed extension of k, and let G be a smooth k-group of finite type. The following conditions are equivalent:*

(i) *G is solvable of class n* (respectively, nilpotent of class n)*.*

(ii) *$G \times_k S$ is solvable of class n* (respectively, nilpotent of class n)*.*

(iii) *The group $G(\Omega)$ is solvable of class n* (respectively, nilpotent of class n).

(iv) *Put $K_0 = G$, and, for $p \geq 1$, consider the k-groups*

$$K_p = (K_{p-1}, K_{p-1}) \qquad \text{(respectively, } \ K_p = (G, K_{p-1}))$$

(cf. 7.2.2). *Then K_n is the unit k-group.*

The equivalence of conditions (i) and (ii) follows from Proposition 8.2, since the formation of the group presheaf of commutators commutes with base change (IV, 4.1.3).

The equivalence of (i) and (iv) follows from the fact that, by 7.8, the k-group

$$K_p = (K_{p-1}, K_{p-1}) \qquad \text{(respectively, } \ K_p = (G, K_{p-1}))$$

represents the sheaf

$$\mathrm{Comm}_T(K_{p-1}, K_{p-1}) \qquad \text{(respectively, } \ \mathrm{Comm}_T(G, K_{p-1})),$$

where T is the (fppf) or (fpqc) topology.

To prove the equivalence of conditions (iii) and (iv), we may assume that $k = \Omega$; the result then follows from 7.10.

Proposition 8.4.— *Let G be an S-group of finite presentation such that, for every $s \in S$, the fiber G_s is smooth over $\kappa(s)$. Let T be the set of points $s \in S$ for which G_s is solvable* (respectively, nilpotent).

(i) *Then T is locally constructible in S.*

(ii) *If, in addition, G is assumed flat and separated over S* (that is, when G is smooth, quasi-compact, and separated over S)*, then T is closed in S.*

Proof. We may clearly suppose that S is affine, with ring A. By 10.1 and 10.10b),* there exist a Noetherian subring A' of A and an A'-group G' of finite type such that

$$G' \otimes_{A'} A \simeq G.$$

By EGA IV_3, 11.2.6 and 8.10.5,[82] if G is flat and separated over S, we may suppose that G' is flat and separated over $S' = \operatorname{Spec} A'$.[83] Since G is of finite presentation over S, the set of points $s \in S$ for which G_s is geometrically reduced —or, equivalently, smooth over $\kappa(s)$—is locally constructible (EGA IV_3, 9.7.7). Hence, by EGA IV_3, 9.3.3, we may suppose that $G'_{s'}$ is smooth over $\kappa(s')$ for every $s' \in S'$. On the other hand, if s' denotes the image of s in S', then

$$G'_{s'} \otimes_{\kappa(s')} \kappa(s) \simeq G_s.$$

It follows from 8.3 that G_s is solvable (respectively, nilpotent) if and only if the same is true of $G'_{s'}$. We are therefore reduced to the case where S is a Noetherian affine scheme.

We first show that T is constructible. Applying the criterion of EGA 0_{III}, 9.2.3, and arguing as above, we are reduced to showing that, when S is Noetherian and integral, either T or $S - T$ contains a nonempty open subset of S.

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* In the course of this proof we shall use results established in Section 10; like Section 9, they do not depend on the present Section 8.

[82]Editor's note: the reference 10.8.5 has been corrected to 8.10.5.

[83]Editor's note: what follows has been expanded.

Suppose, then, that S is integral and Noetherian, with generic point η. For every $s \in S$, put $K_s^0 = G_s$, and put, for $p \geq 1$,

$$K_s^p = (K_s^{p-1}, K_s^{p-1}) \qquad \text{(respectively, } K_s^p = (G_s, K_s^{p-1})). \qquad \dagger$$

We first show that the sequence of closed subschemes K_η^p is stationary. By 7.3(v), each K_η^p is invariant, and hence the sequence $(K_\eta^p)_p$ is decreasing. It is therefore stationary, since G_η is Noetherian. Thus there is an integer n such that

$$K_\eta^p = K_\eta^n \qquad \text{for every } p \geq n.$$

On the other hand, by 10.12.1 and 10.13, there exist a nonempty open subset S' of S and an S'-group D of finite presentation such that, for every $s \in S'$,

$$D_s = K_s^n, \qquad (D_s, D_s) = D_s \quad \text{(respectively, } (G_s, D_s) = D_s).$$

We may suppose that $S' = S$. Then, for every $s \in S$ and every $p \geq n$, one has $D_s = K_s^p$. Consequently, G_s is solvable (respectively, nilpotent) if and only if D_s is isomorphic to the unit $\kappa(s)$-group.

By EGA IV$_3$, 9.6.1(xi), the set of points $s \in S$ for which the structural morphism

$$D_s \longrightarrow \operatorname{Spec} \kappa(s)$$

is an isomorphism is constructible.[84] It is therefore either nowhere dense or contains a nonempty open subset of S. We have thus proved that T is locally constructible.

We now show that, if G is also flat and separated over S, then T is closed. Since T is locally constructible, it is closed if and only if it is stable under specialization (EGA IV$_1$, 1.10.1).

Let $s \in S$, and let s' be a specialization of s in S. Since we have reduced to the case where S is Noetherian, EGA II, 7.1.9 provides a discrete valuation ring A and a morphism

$$\operatorname{Spec} A \longrightarrow S$$

such that s, respectively s', is the image of the generic point α, respectively the closed point a, of $\operatorname{Spec} A$. It is enough to show that, if

$$G' = G \times_S \operatorname{Spec} A$$

and G'_α is solvable (respectively, nilpotent), then the same is true of G'_a. Since G is flat and separated over S, the A-group G' is flat and separated. We are thus reduced to the case where S is the spectrum of a discrete valuation ring A.

Since G_α is assumed solvable (respectively, nilpotent), there is an integer q such that K_α^q, with the notation introduced above, is isomorphic to the unit $\kappa(\alpha)$-group. For every n, let $\overline{K_\alpha^n}$ denote the schematic closure of K_α^n in G, in the sense of EGA IV$_2$, 2.8.5. We show by induction on p that

$$\big(\overline{K_\alpha^p}\big)_a \supset K_a^p.$$

First, since G is flat over A, one has $\overline{G_\alpha} = G$ by EGA IV$_2$, 2.8.5, and hence

$$\big(\overline{K_\alpha^0}\big)_a = K_a^0.$$

[84]Editor's note: here one uses the fact that S is assumed affine, and is therefore quasi-compact and quasi-separated (cf. EGA 0_{III}, 9.1.12).

Let $p \geq 0$, and suppose that

$$K_a^p \subset \left(\overline{K_\alpha^p}\right)_a.$$

Following 7.2.2, denote by $\nu_a, \nu_\alpha, \overline{\nu}, \overline{\nu}_a$ the following morphisms. In the solvable case they are

$$\begin{aligned}
\nu_a &: K_a^p \times_{\kappa(a)} K_a^p \longrightarrow G_a,\\
\nu_\alpha &: K_\alpha^p \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha,\\
\overline{\nu} &: \overline{K_\alpha^p} \times_A \overline{K_\alpha^p} \longrightarrow G,\\
\overline{\nu}_a &: \left(\overline{K_\alpha^p}\right)_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a \longrightarrow G_a;
\end{aligned}$$

respectively, in the nilpotent case they are

$$\begin{aligned}
\nu_a &: G_a \times_{\kappa(a)} K_a^p \longrightarrow G_a,\\
\nu_\alpha &: G_\alpha \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha,\\
\overline{\nu} &: G \times_A \overline{K_\alpha^p} \longrightarrow G,\\
\overline{\nu}_a &: G_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a \longrightarrow G_a.
\end{aligned}$$

Since ν_α factors through K_α^{p+1}, $\overline{\nu}$ factors through $\overline{K_\alpha^{p+1}}$. The latter is clearly a group subscheme of G, and therefore contains $\Gamma_G(\overline{\nu})$. By 7.1(iii),

$$\Gamma_G(\overline{\nu})_a = \Gamma_{G_a}(\overline{\nu}_a).$$

Moreover, the induction hypothesis gives

$$\begin{aligned}
K_a^p \times_{\kappa(a)} K_a^p &\subset \left(\overline{K_\alpha^p}\right)_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a \quad \text{in the solvable case,}\\
G_a \times_{\kappa(a)} K_a^p &\subset G_a \times_{\kappa(a)} \left(\overline{K_\alpha^p}\right)_a \quad \text{in the nilpotent case.}
\end{aligned}$$

Consequently,

$$K_a^{p+1} = \Gamma_{G_a}(\nu_a) \subset \Gamma_{G_a}(\overline{\nu}_a) = \Gamma_G(\overline{\nu})_a \subset \left(\overline{K_\alpha^{p+1}}\right)_a.$$

Finally, K_α^q is isomorphic to the unit $\kappa(\alpha)$-group. The unit A-group is flat over A and is isomorphic to a closed subscheme of G, since G is separated over A (cf. 5.1). It follows from EGA IV_2, 2.8.5 that $\overline{K_\alpha^q}$ is isomorphic to the unit A-group. Since we have proved that

$$K_a^q \subset \left(\overline{K_\alpha^q}\right)_a,$$

the group K_a^q is isomorphic to the unit $\kappa(a)$-group. Thus G_a is solvable (respectively, nilpotent).

9. Quotient sheaves

The present section is essentially limited to recalling, in the particular case of homogeneous spaces of groups, well-known general facts concerning passage to the quotient by flat equivalence relations (cf. Exposé IV).

Definition 9.1.— *Given a monomorphism $u : G' \to G$ of S-groups, we write G/G'* (respectively, $G'\backslash G$) *and call it the* right (respectively, left) quotient sheaf of G by G', *meaning the quotient sheaf, for the* (fpqc) *topology, of G by the equivalence relation defined by the monomorphism*

$$G \times_S G' \xrightarrow{\delta\circ(\mathrm{id}_G \times u)} G \times_S G,$$

(respectively,

$$G' \times_S G \xrightarrow{\gamma\circ(u\times \mathrm{id}_G)} G \times_S G),$$

where δ (respectively, γ) *denotes the automorphism of* $G \times_S G$ *defined, for* $g, h \in G(T)$, *by*

$$(g,h) \longmapsto (g,gh) \qquad (\text{respectively, } (h,g) \longmapsto (hg,g)).$$

Proposition 9.2.— *Let* $u : G' \to G$ *be a monomorphism of* S*-groups. Suppose that* G/G' *is representable by an* S*-scheme* G''. *Then:*

(i) *The canonical morphism* $p : G \to G''$ *is a covering for the* (fpqc) *topology.*

(ii) *If* $\varepsilon'' = p \circ \varepsilon$ (this morphism is called the unit section of G''), *the following diagrams are cartesian:*

$$\begin{tikzcd} G \times_S G' \arrow[r, "\mu\circ(\mathrm{id}_G \times u)"] \arrow[d, "\mathrm{pr}_1"'] & G \arrow[d, "p"] \\ G \arrow[r, "p"'] & G'' \end{tikzcd}, \qquad \begin{tikzcd} G' \arrow[r, "u"] \arrow[d, "\pi'"'] & G \arrow[d, "p"] \\ S \arrow[r, "\varepsilon''"'] & G'' \end{tikzcd}.$$

In particular, u *is an immersion.*

(iii) *There exists on* G'' *a unique structure of an* S*-scheme with left operator group* G *such that* p *is a morphism of* S*-schemes with operator group* G.

(iv) *If, in addition,* G' *is assumed invariant in* G, *there exists on* G'' *a unique* S*-group structure such that* p *is a morphism of* S*-groups.*

(v) *Let* S_0 *be an* S*-scheme, and put* $G_0 = G \times_S S_0$ *and* $G'_0 = G' \times_S S_0$. *Then* G_0/G'_0 *is representable by* $G''_0 = G'' \times_S S_0$.[85]

(vi) *Let* $\mathscr{P}$ *be a property of an* S*-morphism. Suppose that* $\mathscr{P}$ *is stable under base change. If* $p : G \to G''$ *has property* $\mathscr{P}$, *then the structural morphism* $\pi' : G' \to S$ *has property* $\mathscr{P}$ *as well.*

(vii) *Let* $\mathscr{P}$ *be a property of an* S*-morphism. Suppose that* $\mathscr{P}$ *is local for the* (fpqc) *topology* (cf. 2.0 and 2.1.2). *Then* $p : G \to G''$ *has property* $\mathscr{P}$ *if and only if* $\pi' : G' \to S$ *has property* $\mathscr{P}$.

(viii) *Let* $\mathscr{P}$ *be a property of an* S*-morphism. Suppose that* $\mathscr{P}$ *is local for the* (fpqc) *topology and stable under composition. If the structural morphisms* $G'' \to S$ *and* $G' \to S$ *have property* $\mathscr{P}$, *then the structural morphism* $G \to S$ *has property* $\mathscr{P}$ *as well.*

(ix) *If* G *is reduced, then* G'' *is reduced.*

(x) *For* G'' *to be separated over* S, *it is necessary and sufficient that* u (or, equivalently, ε'') *be a closed immersion.*

(xi) *For* G' *to be flat over* S, *it is necessary and sufficient that* p *be flat* (or, equivalently, faithfully flat).

In this case, for G'' *to be flat over* S, *it is necessary and sufficient that* G *be flat over* S.

[85] Editor's note: that is, the quotient is universal; cf. Exposé IV, §3.

(xii) *For G' to be flat and locally of finite presentation over S, it is necessary and sufficient that $p : G \to G''$ be faithfully flat and locally of finite presentation.*

In this case, for G'' to be locally of finite presentation (respectively, locally of finite type, of finite type, smooth, étale, unramified, locally quasi-finite, quasi-finite) *over S, it is enough that G have the same property over S* (and the condition is also necessary in the first two cases; cf. (viii)).

(xiii) *Suppose that G' is flat and of finite presentation over S.*

a) *Then p is of finite presentation and faithfully flat.*

b) *Moreover, for G to be of finite presentation over S, it is necessary and sufficient that G'' be so.*

Proof. Recall that the equivalence relation under consideration is universally effective (IV, 4.4.9). Assertions (i), (iii), (iv), (v), and the first assertion of (ii) then follow, respectively, from IV, 4.4.3, 5.2.2, 5.2.4, 3.4.5, and 3.3.2(iii). The second assertion of (ii) follows from the first, as shown by the following cartesian diagram, since

$$(G \times_S G') \times_G S$$

is isomorphic to G':

$$\begin{tikzcd} G' \arrow[r, "((\varepsilon\circ\pi'), \mathrm{id}_{G'})"] \arrow[d, "\pi'"'] & G \times_S G' \arrow[r, "\mu\circ(\mathrm{id}_G \times u)"] \arrow[d, "\mathrm{pr}_1"] & G \arrow[d, "p"] \\ S \arrow[r, "\varepsilon"'] & G \arrow[r, "p"'] & G'' \end{tikzcd}$$

Indeed, ε'' is an S-section of G'', hence an immersion (EGA I, 5.3.13); by the preceding cartesian diagram the same is true of u, which completes the proof of (ii). Moreover, (vi) is an immediate consequence of the second cartesian diagram in (ii).

We prove (vii). By (i), p is a covering for the (fpqc) topology. Hence, by (ii), in order to show that p has property $\mathscr{P}$, it is enough to show that the first projection

$$\mathrm{pr}_1 : G \times_S G' \longrightarrow G$$

has property $\mathscr{P}$. This follows from the stability of $\mathscr{P}$ under base change, since pr_1 is obtained from π' by base change.

It is clear that (viii) follows from (vii), since

$$\pi = \pi'' \circ p,$$

where $\pi'' : G'' \to S$ denotes the structural morphism.

We prove (ix). By (i), p is an epimorphism. Since G is reduced, p factors through the immersion

$$G''_{\mathrm{red}} \longrightarrow G'',$$

which is therefore also an epimorphism, and hence an isomorphism (IV, 4.4.4).

We prove (x). If G'' is separated over S, then ε'' is a closed immersion by EGA I, 5.4.6. On the other hand, we saw in (ii) that ε'' is a closed immersion if and only if u is. Finally, if u is a closed immersion, then so is

$$\delta \circ (\mathrm{id}_G \times u) : G \times_S G' \longrightarrow G \times_S G.$$

It follows from Lemma 9.2.1 below that G'' is separated over S.

Assertion (xi) follows from (vii) and EGA IV$_2$, 2.2.13.

Assertion (xii) follows from (vii), EGA IV$_4$, 17.7.5 and 17.7.7, and the fact that p is universally open: for every $s \in S$, if the underlying space of G_s is discrete, then the same is true of the underlying space of G''_s.

Finally, assertion (xiii) follows from (vii), (viii), and EGA IV$_4$, 17.7.5.

Lemma 9.2.1.— *Let X be an S-scheme, and let R be an equivalence relation on X defined by the monomorphism $v : R \to X \times_S X$. Suppose that R is effective. Then:*

(i) *The morphism v is an immersion; it is a closed immersion if $Y = X/R$ is separated.*

(ii) *Suppose, in addition, that Y represents the* (fpqc) *quotient sheaf of X by R,*[86] *and that v is a closed immersion. Then $Y = X/R$ is separated over S.*

Recall (IV, Definition 3.3.2) that the hypothesis "R is effective" means that there is a morphism of S-schemes

$$p : X \longrightarrow Y$$

such that the natural morphism

$$R \longrightarrow X \times_Y X$$

is an isomorphism. It follows (EGA I, 5.3.5) that the following diagram is cartesian:

$$\begin{tikzcd} R \arrow[r, "v"] \arrow[d] & X \times_S X \arrow[d, "p\times p"] \\ Y \arrow[r, "\Delta_{Y/S}"'] & Y \times_S Y \end{tikzcd}$$

Since $\Delta_{Y/S}$ is an immersion (EGA I, 5.3.9), the same is true of v. Likewise, if Y is separated over S, then $\Delta_{Y/S}$ is a closed immersion, and hence so is v.

Conversely, suppose that v is a closed immersion and that Y represents the (fpqc) quotient sheaf of X by R. Then p is a covering for the (fpqc) topology (IV, 4.4.3), and therefore $p \times p$ is also a covering: by base change, $p \times \mathrm{id}_X$ and $\mathrm{id}_Y \times p$ are coverings, and so is their composite $p \times p$. Thus, by (fpqc) descent (cf. EGA IV$_2$, 2.7.1), $\Delta_{Y/S}$ is a closed immersion; in other words, Y is separated over S.

Remark 9.2.2.— *Under the general hypotheses of 9.2, if G' is flat and locally of finite presentation over S, then p is a covering for the* (fppf) *topology,*[87] *by 9.2*(vii)*. Consequently, assertions 9.2*(vii) *and 9.2*(viii) *extend to properties $\mathscr{P}$ that are local for the* (fppf) *topology.*

Remark 9.3.— a) The question whether a quotient G/G' is representable is often delicate; in this seminar we prove the representability of certain particular quotients.

In general, in order to conclude that G/G' is representable, it is not enough to suppose that G and G' are of finite presentation over S and that G' is flat over S. Indeed, suppose in addition that G is smooth with connected fibers. In that case, if G/G' is a scheme, it is separated by Corollary 5.4; therefore $G' \hookrightarrow G$ is a closed immersion by 9.2(x). Consequently, if G' is not closed in G, then G/G' is not representable.

To obtain such a counterexample, take S to be the spectrum of a discrete valuation ring and put $G = \mathbb{G}_{m,S}$. Let $n > 1$ be an integer invertible on S. Then

$$\mu_n = \ker(G \xrightarrow{n} G)$$

[86] Editor's note: as O. Gabber pointed out, this condition is used in the proof and must be included among the hypotheses.

[87] Editor's note: (fpqc) has been corrected to (fppf).

is a closed subgroup of G, étale over S (cf. Exposé VII_A[88]). Let G' be the open subgroup of μ_n obtained by removing from μ_n the closed subset of its closed fiber that is complementary to the origin. Then G' is not closed in G, and hence G/G' is not representable. One can also construct such examples in which G' is smooth with connected fibers.

b) It remains possible, however, that G/G' is representable when G and G' are of finite presentation over S, and G' is flat over S and closed in G* [89]. Under these hypotheses, G/G' is known to be representable in the following particular cases:

1°— S is the spectrum of an Artinian ring (cf. VI_A, 3.2 and 3.3.2).

2°— G' is proper over S, and G is quasi-projective over S (cf. V, 7.1).

3°— S is locally Noetherian of dimension 1 (cf. [An73], Theorem 4.C).

10. Passage to the projective limit in group schemes and schemes with an operator group

10.0.— Recall the essential result of EGA IV_3, §8.8. Suppose that the following data are given: a quasi-compact and quasi-separated scheme S_0, an upward filtered preordered set I, a direct system $(\mathcal{A}_i)_{i\in I}$ of quasi-coherent commutative $\mathcal{O}_{S_0}$-algebras,

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \qquad S_i = \operatorname{Spec} \mathcal{A}_i \quad (i \in I), \qquad S = \operatorname{Spec} \mathcal{A}^{90}.$$

Then the category of S-schemes of finite presentation is determined, up to equivalence, by the categories of S_i-schemes of finite presentation, the functors between these categories

$$\rho_{ji} : X_i \longmapsto X_i \times_{S_i} S_j \qquad (i \le j),$$

and the transitivity isomorphisms

$$\rho_{kj} \circ \rho_{ji} \xrightarrow{\sim} \rho_{ki}.$$

More precisely, given $j \in I$ and an S_j-scheme X_j of finite presentation, put, for every $i \in I$ with $i \ge j$,

$$X_i = X_j \times_{S_j} S_i, \qquad X = X_j \times_{S_j} S.$$

Then EGA IV_3, 8.8.2 gives:

(i) Given $j \in I$ and two S_j-schemes X_j and Y_j of finite presentation, the canonical map

$$\varinjlim_{i\ge j} \operatorname{Hom}_{S_i}(X_i, Y_i) \longrightarrow \operatorname{Hom}_S(X, Y)$$

is bijective.

(ii) For every S-scheme X of finite presentation, there are an index $j \in I$, an S_j-scheme X_j of finite presentation, and an S-isomorphism

$$X \xrightarrow{\sim} X_j \times_{S_j} S.$$

[88]Editor's note: see Exposé VII_A, 8.4, or Exposé VIII, 2.1.

*This is too optimistic, as M. Raynaud shows in his thesis (loc. cit., X, 14).

[89]Editor's note: the starred remark refers to counterexample X, 14 in [Ray70a]. The base there is a regular local scheme of dimension 2.

[90]Editor's note: since S is affine over S_0, it is quasi-compact and quasi-separated; cf. Editor's note 92 below.

It follows from EGA IV_3, 8.8.3 that whenever a diagram $\mathcal{D}$ involves finitely many objects and arrows in the category of S-schemes of finite presentation, one can find an index $i \in I$ and a diagram $\mathcal{D}_i$ in the category of S_i-schemes of finite presentation such that $\mathcal{D}$, up to isomorphism, is obtained from $\mathcal{D}_i$ by the base change $S \to S_i$. One can even find i and $\mathcal{D}_i$ such that every cartesian square in $\mathcal{D}$ is obtained from a cartesian square in $\mathcal{D}_i$.

10.1.— Moreover, many familiar properties of a morphism that are stable under base change have the following property:

> *Let $u : X \to Y$ be an S-morphism between S-schemes of finite presentation. By 10.0, it is obtained by base change from an S_j-morphism $u_j : X_j \to Y_j$ between S_j-schemes of finite presentation. Then u has the property $\mathcal{P}$ if and only if there is an $i \geq j$ such that*
> $$u_i = u_j \times_{S_j} S_i$$
> *has the property $\mathcal{P}$.*

This applies when $\mathcal{P}$ is any of the following properties of a morphism: separated, surjective, radicial, affine, quasi-affine, finite, quasi-finite, proper, projective, quasi-projective, an isomorphism, a monomorphism, an immersion, an open immersion, a closed immersion (EGA IV_3, 8.10.5), flat (EGA IV_3, 11.2.6), smooth, unramified, or étale (EGA IV_4, 17.7.8).[91]

This also applies when $\mathcal{P}$ is the property of being a covering for the (fppf) topology. Indeed, given two S-schemes X and Y of finite presentation and an S-morphism $u : X \to Y$, it follows from Exposé IV, 6.3.1(i)[92] that u is a covering for the (fppf) topology if and only if there are an S-scheme Z and a faithfully flat S-morphism $v : Z \to Y$ of finite presentation that factors through u.

The purpose of this Section 10 is to give variants of results of this kind for the category of S-groups of finite presentation, for the category of S-schemes with an operator group, and for certain properties of group monomorphisms (being invariant, being central with representable quotient sheaf, and so on).

The first two preliminary results of this kind follow. In 10.2 through 10.9, we retain the notation introduced in 10.0.

Lemma 10.2.— *Let G_j and H_j be two S_j-groups of finite presentation. For every $i \geq j$, put*
$$G_i = G_j \times_{S_j} S_i, \qquad G = G_j \times_{S_j} S,$$
and define H_i and H similarly. Then the canonical map
$$\varinjlim_{i \geq j} \mathrm{Hom}_{S_i\text{-gr}}(G_i, H_i) \longrightarrow \mathrm{Hom}_{S\text{-gr}}(G, H)$$
is bijective.

Lemma 10.3.— *Let G be an S-group of finite presentation. Then there are an index $j \in I$, an S_j-group G_j of finite presentation, and an isomorphism of S-groups*
$$G \xrightarrow{\sim} G_j \times_{S_j} S.$$

Assertions 10.2 and 10.3 are easy consequences of 10.0 and 10.1, taking into account the interpretation[93] of the structure of an S-group given in EGA 0_{III}, 8.2.5 and 8.2.6.

[91] Editor's note: the word "flat" has been added, and the reference 17.7.6 has been corrected to 17.7.8.
[92] Editor's note: one also uses the fact that Y, being of finite presentation over S, is quasi-compact.
[93] Editor's note: in terms of commutative diagrams of S-morphisms.

Lemma 10.4.— *Let $u : G \to H$ be a morphism of S-groups between S-groups of finite presentation. By 10.3 and 10.2, u is obtained by base change from a morphism $u_j : G_j \to H_j$ between S_j-groups of finite presentation. Then u is a central monomorphism* (respectively, an invariant monomorphism) *if and only if there is an $i \geq j$ such that*

$$u_i = u_j \times_{S_j} S_i$$

has the same property.

This follows immediately from 10.0 and 10.1, in view of the characterization in 6.7 of central or invariant group monomorphisms.

Corollary 10.5.— *Let G_j be an S_j-group of finite presentation. Then $G_j \times_{S_j} S$ is commutative if and only if there is an $i \geq j$ such that $G_j \times_{S_j} S_i$ is commutative.*

Indeed, to say that an S-group is commutative is the same as saying that it is central when regarded as a group subscheme of itself.

Proposition 10.6.— *Let G_j be an S_j-group of finite presentation, and let G'_j be a group subscheme of G_j, flat and of finite presentation over S_j. Then the quotient*

$$(G_j \times_{S_j} S)/(G'_j \times_{S_j} S)$$

is representable for the (fpqc) *topology if and only if there is an $i \geq j$ such that*

$$(G_j \times_{S_j} S_i)/(G'_j \times_{S_j} S_i)$$

is representable.

This is a consequence of the following more general lemma.

Lemma 10.7.— *Let X_j be an S_j-scheme of finite presentation, and let R_j be an equivalence relation on X_j, flat and of finite presentation.*[94] *Then the quotient sheaf*

$$(X_j \times_{S_j} S)/(R_j \times_{S_j} S)$$

for the topology $\mathcal{T} =$ (fppf) *or* (fpqc) *is representable if and only if there is an $i \geq j$ such that the quotient sheaf*

$$(X_j \times_{S_j} S_i)/(R_j \times_{S_j} S_i)$$

for the topology $\mathcal{T}$ is representable.

In view of the statements EGA IV_2, 8.8.2, 8.8.3, 8.10.5, and 11.2.6 recalled in 10.0, this lemma follows from the next result.

Lemma 10.8.— *Let $\mathcal{T}$ be the* (fppf) *or* (fpqc) *topology. Let X be an S-scheme of finite presentation* (respectively, locally of finite presentation)*, and let R be an equivalence relation on X defined by a monomorphism*

$$v : R \longrightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ v$ *is flat and of finite presentation* (respectively, flat and locally of finite presentation)*. Then the following conditions are equivalent:*

(i) *The quotient sheaf X/R for the topology $\mathcal{T}$ is representable.*

[94] Editor's note: that is, such that the composite

$$R_j \hookrightarrow X_j \times_{S_j} X_j \xrightarrow{\mathrm{pr}_1} X_j$$

is flat and of finite presentation.

(ii) *There are an* S*-scheme* Y *of finite presentation* (respectively, locally of finite presentation) *and a faithfully flat morphism* $p : X \to Y$ *such that the diagram*

$$\text{(D)}\qquad \begin{tikzcd} R \arrow[r, "\mathrm{pr}_1 \circ v"] \arrow[d, "\mathrm{pr}_2 \circ v"'] & X \arrow[d, "p"] \\ X \arrow[r, "p"'] & Y \end{tikzcd}$$

is cartesian.

First note that, by IV, 3.3.2 and 4.4.3, the sheaf X/R for the topology $\mathcal{T}$ is representable by Y if and only if diagram (D) is cartesian and p is a covering for the topology $\mathcal{T}$.

Let us prove that (i) implies (ii). Hypothesis (i) implies that diagram (D) is cartesian. Thus $\mathrm{pr}_1 \circ v$ is obtained from p by base change along p, and p is a covering for the (fpqc) topology. Consequently, by (fpqc) descent (EGA IV_2, 2.7.1), since $\mathrm{pr}_1 \circ v$ is faithfully flat and (locally) of finite presentation, the same is true of p. Then EGA IV_4, 17.7.5 shows that, since X is (locally) of finite presentation over S, so is Y.

Let us prove that (ii) implies (i). It is enough to show that p is a covering for the (fppf) topology. But p is faithfully flat by hypothesis, and it is locally of finite presentation because X and Y are locally of finite presentation over S (EGA IV_1, 1.4.3(v)).

Lemma 10.9.— *Let* G_j *be an* S_j*-group of finite presentation, and put*

$$G = G_j \times_{S_j} S.$$

Then G^0 *is representable if and only if there is an* $i \geq j$ *such that*

$$(G_i)^0 = (G_j \times_{S_j} S_i)^0$$

is representable.

The condition is sufficient, since the functor $G \longmapsto G^0$ commutes with base change, by 3.3.

Conversely, suppose that G^0 is representable. Then, by 3.9, G^0 is open in G and quasi-compact over S, and hence is of finite presentation over S, since G is. By 10.3 and 10.1, there are an $i \geq j$ and an open group subscheme H_i of G_i such that

$$H_i \times_{S_i} S = G^0.$$

The structural morphism $G^0 \to S$ has geometrically connected fibers (VI_A, 2.1.1). Consequently, by EGA IV_3, 9.3.3 and 9.7.7, after increasing i we may suppose that the structural morphism $H_i \to S_i$ has geometrically connected fibers. It follows from 3.10.1 that the underlying space of H_i is precisely $(G_i)^0$; hence H_i represents $(G_i)^0$.

10.10.— Recall two very useful special cases of the situation stated in 10.0 (cf. EGA IV_3, 8.1.2(a) and (c)):

a) Given a point x of a scheme X, put $S_0 = \operatorname{Spec} \mathbb{Z}$, and consider the decreasing filtered inverse system $(S_i)_{i \in I}$ of affine open neighborhoods of x. Then

$$S = \operatorname{Spec} \mathcal{O}_{X,x}.$$

In particular, if x is the generic point of an integral scheme X, then

$$S = \operatorname{Spec} \kappa(x).$$

b) Put $S_0 = \operatorname{Spec} \mathbb{Z}$, and consider the family $(\mathcal{A}_i)_{i \in I}$, preordered by inclusion, of the finitely generated $\mathbb{Z}$-subalgebras of the ring of an affine scheme S. Since the $\mathcal{A}_i$ are Noetherian rings, this makes it possible in many cases to pass from the Noetherian case to the general case.

We shall now give two results concerning the special case considered in (a).[95]

Proposition 10.11.—[96] *Let S be an integral scheme with generic point η. Let G be an S-group and let Y, Z be S-schemes, all of finite presentation. Let*

$$u, v, i : Y \longrightarrow G, \qquad j : Z \longrightarrow G$$

be morphisms of S-schemes. Suppose that

$$i_\eta : Y_\eta \longrightarrow G_\eta, \qquad j_\eta : Z_\eta \longrightarrow G_\eta$$

are closed immersions.

Then there is a nonempty open subset S' of S such that the morphisms

$$i' : Y' \longrightarrow G', \qquad j' : Z' \longrightarrow G',$$

obtained by base change, are closed immersions, and such that the functors

$$\underline{\mathrm{Transp}}(u', v'), \qquad \underline{\mathrm{Transp}}_{G'}(i'(Y'), j'(Z')),$$
$$\underline{\mathrm{Transp}}^{\mathrm{str}}_{G'}(i'(Y'), j'(Z'))$$

respectively, the functors

$$\underline{\mathrm{Centr}}(u'), \qquad \underline{\mathrm{Norm}}_{G'} i'(Y'),$$

are representable by closed S'-subschemes (respectively, closed S'-subgroups) *of G', of finite presentation over S'.*

Proof. We shall apply the results of 10.1, first in the situation of 10.10(a), and then in that of 10.10(b). The schemes G_η, Y_η, Z_η are flat over the field $\kappa(\eta)$; the group G_η is separated over $\kappa(\eta)$ (VI_A, 0.3); and i_η, j_η are closed immersions. Therefore, by 10.1, there are an affine open subset $S' = \operatorname{Spec}(A')$ of S, a Noetherian subring A of A', A-schemes G_A, Y_A, Z_A, flat and of finite presentation over A, and morphisms

$$u_A, v_A, i_A : Y_A \longrightarrow G_A, \qquad j_A : Z_A \longrightarrow G_A,$$

such that G_A is an A-group separated over A, i_A and j_A are closed immersions, and

$$G \times_S S' = G_A \otimes_A A',$$

with analogous equalities for the remaining schemes and morphisms. Since the functors under consideration over S' are obtained by base change from the analogous functors over $\operatorname{Spec}(A)$, it is enough to prove the result for the latter.

By EGA IV_2, 6.9.2, after replacing A by a localization A_f (and hence S' by the affine open subset S'_f), we may suppose that G_A, Y_A, Z_A are essentially free over A, in the sense of 6.2.1.[97] It then follows from 6.2.4(b) and 6.2.4(e) that, under the hypotheses of the

[95]Editor's note: observe that the proof also uses case (b).

[96]Editor's note: the statement has been simplified, and the case of group subschemes has been treated separately in Corollary 10.11.1.

[97]Editor's note: indeed, since G_A is flat and of finite presentation over A, it is covered by affine open subsets $G_1, \dots, G_n$ such that each $\mathcal{O}(G_i)$ is a flat A-algebra of finite presentation. By EGA IV_2, 6.9.2, there is an $f_i \in A$ such that $\mathcal{O}(G_i)_{f_i}$ is a free A_{f_i}-module. One may therefore replace $\operatorname{Spec}(A)$ by the affine open subset $D(f)$, where $f = f_1 \cdots f_n$, and proceed in the same way for Y_A and Z_A.

proposition, the functors under consideration are representable by closed A-subschemes of G_A. These subschemes are of finite presentation over A, since A is Noetherian and G_A is of finite presentation over A; and in the cases of $\underline{\mathrm{Centr}}(u_A)$ and $\underline{\mathrm{Norm}}_{G_A} i_A(Y_A)$, they are closed A-subgroups of G_A.

Corollary 10.11.1.— *Let S be an integral scheme with generic point η. Let G, H, K be S-groups of finite presentation, and let*

$$i : H \longrightarrow G, \qquad j : K \longrightarrow G$$

be two quasi-compact monomorphisms of S-groups. Then there is a nonempty open subset S' of S such that the morphisms

$$i' : H' \longrightarrow G', \qquad j' : K' \longrightarrow G',$$

obtained by base change, are closed immersions, and such that the functors

$$\underline{\mathrm{Transp}}_{G'}(H', K'), \qquad \underline{\mathrm{Transp}}^{\mathrm{str}}_{G'}(H', K'),$$

respectively, the functors

$$\underline{\mathrm{Centr}}_{G'} H', \qquad \underline{\mathrm{Norm}}_{G'} H',$$

are representable by closed S'-subschemes (respectively, closed S'-subgroups) *of G', of finite presentation over S'.*

This follows from the preceding proposition, since the hypotheses imply, by 1.4.2, that i_η and j_η are closed immersions.

Proposition 10.12.— *Let S be an integral scheme, let G be an S-group of finite presentation, and let A and B be two group subschemes of G, of finite presentation over S and with smooth generic fibers. Suppose moreover that one of the following conditions holds:*

a) *A has connected generic fiber;*

b) *A and B are invariant in G.*

Then there are a nonempty open subset S' of S and a closed group subscheme D' of

$$G' = G|_{S'},$$

of finite presentation over S' and with smooth fibers, that represents the (fppf) *sheaf associated with the presheaf of groups of commutators of*

$$A' = A|_{S'} \qquad \textit{and} \qquad B' = B|_{S'}$$

in G'. Moreover, D' has connected fibers in case (a), *and is invariant in G' in case* (b).

In particular, for every $s \in S'$,

$$D'_s = (A_s, B_s),$$

with the notation of 7.2.2.

Proof. Let η be the generic point of S, and put

$$H_\eta = (A_\eta, B_\eta).$$

Since A_η and B_η are smooth, 7.8 in case (a) and 7.3(v) in case (b) show that H_η is connected, respectively invariant in G_η.

We are in the situation of 10.0 corresponding to 10.10(a). Consequently, by 10.3 and 10.1, there are a nonempty open subset S' of S and an S'-group subscheme D', of finite presentation and closed in G', such that

$$D'_\eta = D' \otimes_{S'} \kappa(\eta) = H_\eta.$$

Moreover, by EGA IV_3, 9.7.7 and 9.3.3, we may suppose that D' has geometrically reduced fibers. In case (a), another application of EGA IV_3, 9.7.7 and 9.3.3 allows us to suppose that D' has connected fibers, hence geometrically connected fibers (cf. VI_A, 2.1.1). In case (b), we may suppose, by 10.4, that D' is invariant in G'.

We also saw, in the course of the proof of 7.8, that there is an integer n such that

$$\nu_\eta^n\left((A_\eta \times_{\kappa(\eta)} B_\eta)^n\right) = D'_\eta,$$

where ν_η and ν_η^n are defined as in 7.2.2(a) and 7.1(ii). The same formulas define morphisms

$$\nu' : A' \times_{S'} B' \longrightarrow G', \qquad \nu'^n : (A' \times_{S'} B')^n \longrightarrow G',$$

and

$$\nu'^n \otimes_{S'} \kappa(\eta) = \nu_\eta^n.$$

It follows from 10.1 that we may choose S' so that the morphism ν'^n is flat and factors through D', and so that the resulting morphism

$$\nu'^n : (A' \times_{S'} B')^n \longrightarrow D'$$

is surjective. By 7.5, the morphism[98]

$$\nu'^{2n} = \mu \circ (\nu'^n \times_{S'} \nu'^n) : (A' \times_{S'} B')^{2n} \longrightarrow D'$$

is a covering for the (fppf) topology. Therefore, by 7.6, D' represents the (fppf) sheaf associated with the presheaf of commutators of A' and B' in G'.

Furthermore, for every $s \in S'$, ν'^n induces a surjective morphism

$$\nu_s^n : (A_s \times_{\kappa(s)} B_s)^n \longrightarrow D'_s.$$

[99] Thus D'_s is a closed subgroup of G_s containing

$$\nu_s(A_s \times_{\kappa(s)} B_s),$$

and hence containing (A_s, B_s). Since ν_s^n is surjective, $D'_s = (A_s, B_s)$; and by 7.6, it represents the (fppf) sheaf of commutators of A_s and B_s in G_s.

Corollary 10.12.1.—[100] *Let S be an integral scheme with generic point η, and let G be an S-group of finite presentation with smooth fibers. Put*

$$K_\eta^0 = G_\eta, \qquad K_\eta^p = (K_\eta^{p-1}, K_\eta^{p-1}) \quad (\text{respectively, } K_\eta^p = (G_\eta, K_\eta^{p-1}))$$

for every $p \in \mathbb{N}^*$.

Fix $n \in \mathbb{N}^*$. Then there are a nonempty open subset S' of S and a group subscheme D, invariant in $G|_{S'}$, of finite presentation and with smooth fibers, such that

$$D_s = K_s^n$$

[98] Editor's note: $n+1$ has been corrected below to $2n$.

[99] Editor's note: the argument that follows has been expanded from the original.

[100] Editor's note: this corollary, which is used in the proof of 8.4, has been added.

for every $s \in S'$.

This follows from 10.12, by induction on n.

Corollary 10.13.— *Let S be an integral scheme with generic point η. Let G be an S-group and let H be an invariant group subscheme of G. Suppose that G and H are of finite presentation over S and have smooth generic fibers.*[101]

If

$$(H_\eta, H_\eta) = H_\eta \qquad \text{(respectively, } (G_\eta, H_\eta) = H_\eta\text{)},$$

then there is a nonempty open subset S' of S such that, for every $s \in S'$,

$$(H_s, H_s) = H_s \qquad \text{(respectively, } (G_s, H_s) = H_s\text{)}.$$

Indeed, the proof of 10.12 gives a nonempty open subset S' of S and an S'-group subscheme D of $G|_{S'}$, of finite presentation and with smooth fibers, such that, for every $s \in S'$,

$$D_s = (H_s, H_s) \qquad \text{(respectively, } D_s = (G_s, H_s)\text{)}.$$

On the other hand, since $D_\eta = H_\eta$ and D and H are of finite presentation over S', EGA IV_3, 8.8.2.5 gives a nonempty open subset S'' of S' such that

$$D|_{S''} = H|_{S''}.$$

Consequently, for every $s \in S''$,

$$H_s = D_s = (H_s, H_s) \qquad \text{(respectively, } H_s = D_s = (G_s, H_s)\text{)}.$$

10.14.— Statements 10.2 and 10.3, concerning the category of S-groups of finite presentation, extend to the category of pairs consisting of an S-group of finite presentation and an S-scheme of finite presentation with operator group G. More precisely, in the situation recalled at the beginning of 10.0:

(i) let $j \in I$, and let G_j and G'_j be two S_j-groups of finite presentation; let H_j (respectively, H'_j) be an S_j-scheme of finite presentation with operator group G_j (respectively, G'_j). For $i \in I$, $i \geq j$, put

$$G_i = G_j \times_{S_j} S_i, \qquad G = G_j \times_{S_j} S,$$

and define G'_i, G', H_i, H, H'_i, and H' similarly. Denote by

$$\mathrm{Dihom}_{S\text{-gr.}}((G, H), (G', H'))$$

the set of di-morphisms of S-groups and S-schemes with operator groups from the pair (G, H) to the pair (G', H'). Then the canonical map

$$\varinjlim_{i \geq j} \mathrm{Dihom}_{S_i\text{-gr.}}((G_i, H_i), (G'_i, H'_i)) \longrightarrow \mathrm{Dihom}_{S\text{-gr.}}((G, H), (G', H'))$$

is bijective.

(ii) let G be an S-group of finite presentation and H an S-scheme of finite presentation with operator group G. Then there are an index $j \in I$, an S_j-group G_j of finite presentation, an S_j-scheme H_j of finite presentation with operator group G_j, and a di-isomorphism of S-groups and S-schemes with operator groups from

$$(G_j \times_{S_j} S,\ H_j \times_{S_j} S)$$

onto (G, H).

[101] Editor's note: the original assumed that G and H were of finite presentation and that H had smooth fibers; the hypothesis has been modified in order to apply 10.12. The proof has also been expanded.

Definition 10.15.—[102] *Let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. Given an S-group scheme G and an S-scheme X with operator group G, one says that X is a* formally homogeneous space under G (relative to the topology $\mathcal{T}$) *if the morphism*

$$\Phi : G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

defined for every $S' \to S$, $g \in G(S')$, and $x \in X(S')$, is an epimorphism in the category of sheaves for the topology $\mathcal{T}$; equivalently, Φ is a covering for the topology $\mathcal{T}$ (cf. IV, 4.4.3).

One says that X is a homogeneous space *if it is formally homogeneous and, in addition, the morphism $X \to S$ is also a covering for the topology $\mathcal{T}$.*

In particular, one says that X is a formally principal homogeneous space under G *if Φ is an isomorphism, and that X is a* principal homogeneous bundle (or a G-torsor) *if Φ is an isomorphism and, in addition, the morphism $X \to S$ is a covering for the topology $\mathcal{T}$* (cf. IV, 5.1.5 and 5.1.6(ii)).

Proposition 10.16.— *Retain the situation considered at the beginning of 10.0. Let $j \in I$, let G_j be an S_j-group, and let X_j be an S_j-scheme with operator group G_j. Suppose that G_j and X_j are of finite presentation over S_j.*

For

$$X = X_j \times_{S_j} S$$

to be a homogeneous space (respectively, a principal homogeneous bundle) *under*

$$G = G_j \times_{S_j} S$$

for the (fppf) *topology, it is necessary and sufficient that there be an index $i \geq j$ such that*

$$X_i = X_j \times_{S_j} S_i$$

is a homogeneous space (respectively, a principal homogeneous bundle) *under*

$$G_i = G_j \times_{S_j} S_i.$$

Taking into account 10.14 and EGA IV_3, 8.8.2, 8.8.3, and 8.10.5, the assertion follows from the property of covering morphisms for the (fppf) topology recalled in 10.1.[103]

11. Affine group schemes

11.0. Reminders.—[104] Let $q : X \to S$ be a quasi-compact and quasi-separated morphism of schemes (cf. EGA IV_1, 1.1 and 1.2), and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Recall that $q_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_S$-module (EGA I, 9.2.1). Moreover, by EGA III, 1.4.15 (completed by EGA IV_1, 1.7.21), one has (c) below, and the proof in *loc. cit.* also gives (a) and (b):

(a) If $\mathcal{F}$ is a filtered direct limit of quasi-coherent submodules $\mathcal{F}_\alpha$, then

$$q_*(\mathcal{F}) = \varinjlim_{\alpha} q_*(\mathcal{F}_\alpha).$$

[102] Editor's note: the original has been corrected by distinguishing the notions of a "formally homogeneous" object and a "homogeneous" object; see IV, §§5.1 and 6.7.

[103] Editor's note: for limit-passage properties of torsors expressed in terms of Čech cohomology, see also SGA 4, VII, 5.7 and its corollaries. This is developed in detail in [Ma07].

[104] Editor's note: this paragraph of reminders has been added.

(b) If $\mathcal{E}$ is a flat $\mathcal{O}_S$-module, the canonical morphism

$$\mathcal{E} \otimes_{\mathcal{O}_S} q_*(\mathcal{O}_X) \longrightarrow q_*q^*(\mathcal{E})$$

is an isomorphism.

(c) Let $p : S' \to S$ be a flat morphism, let $q' : X' \to S'$ be obtained from q by base change, and let $\mathcal{F}'$ be the inverse image of $\mathcal{F}$ on X'. Then the canonical morphism

$$p^*q_*(\mathcal{F}) \longrightarrow q'_*(\mathcal{F}')$$

is an isomorphism.

Indeed, let $U = \operatorname{Spec}(A)$ be an arbitrary affine open subset of S. By hypothesis, $q^{-1}(U)$ is the union of affine open subsets

$$V_i = \operatorname{Spec}(B_i), \qquad i = 1, \dots, n,$$

and every intersection $V_i \cap V_j$ is the union of finitely many affine open subsets

$$W_{ijk} = \operatorname{Spec}(C_{ijk}).$$

Then $\Gamma(U, q_*(\mathcal{F})) = \Gamma(q^{-1}(U), \mathcal{F})$ is the kernel of the morphism

$$\bigoplus_{i=1}^{n} \Gamma(V_i, \mathcal{F}) \longrightarrow \bigoplus_{i,j,k} \Gamma(W_{ijk}, \mathcal{F}).$$

Assertion (a) follows, since every term above commutes with filtered direct limits (the V_i and W_{ijk} being affine, hence quasi-compact).

Let us prove (b). The module $E = \Gamma(U, \mathcal{E})$ is flat over A, and $\Gamma(U, q_*q^*(\mathcal{E}))$ is the kernel $K(E)$ of the morphism

$$\bigoplus_{i=1}^{n} B_i \otimes_A E \longrightarrow \bigoplus_{i,j,k} C_{ijk} \otimes_A E.$$

Since E is flat over A, this kernel is identified with

$$K(A) \otimes_A E = \mathcal{O}_X(q^{-1}(U)) \otimes_A E.$$

Finally, if U' is an arbitrary affine open subset of S' lying over U, then $A' = \mathcal{O}(U')$ is a flat A-algebra, and the same argument gives

$$\mathcal{F}(q^{-1}(U)) \otimes_A A' \xrightarrow{\sim} \mathcal{F}'(q'^{-1}(U')).$$

Notation.— Let S be a scheme, let X be an S-scheme, and let $f : X \to S$ be the structural morphism. We shall put

$$\mathcal{A}(X) = f_*(\mathcal{O}_X).$$

Lemma 11.1.— *Let X and Y be two quasi-compact and quasi-separated S-schemes, and let $f : X \to S$ and $g : Y \to S$ be their structural morphisms. Then the canonical homomorphism*

$$\varphi : \mathcal{A}(X) \otimes_{\mathcal{O}_S} \mathcal{A}(Y) \longrightarrow \mathcal{A}(X \times_S Y)$$

is an isomorphism in each of the following cases:[105]

[105] Editor's note: if k is a field and X is an infinite disjoint union of copies of $S = \operatorname{Spec} k$ (so that X is not quasi-compact), then $\mathcal{A}(X) = k^X$, and the canonical morphism

$$k^X \otimes_k k^X \longrightarrow k^{X \times X}$$

is not surjective.

(a) *f and g are affine;*

(b) *g (or f) is flat and affine;*

(c) *g is flat and $f_*(\mathcal{O}_X)$ is a flat $\mathcal{O}_S$-module.*

In case (b), we shall suppose that it is g that is flat and affine. Put

$$S' = \operatorname{Spec}\mathcal{A}(X), \qquad Y' = Y \times_S S', \qquad g' = g \times_S S',$$

and write $v : S' \to S$ for the structural morphism:

$$\begin{tikzcd} Y' \arrow[r] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\ \operatorname{Spec}\mathcal{A}(X) = S' \arrow[r, "v"'] & S \end{tikzcd}$$

In cases (a) and (b), the morphism g is affine, and hence EGA II, 1.5.2 gives

$$g'_*(\mathcal{O}_{Y'}) = v^*g_*(\mathcal{O}_Y) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}. \tag{1}$$

The same equality holds in case (c), by 11.0(c), since S' is flat over S and g is quasi-compact and quasi-separated.

On the other hand, by EGA II, 1.2.7, the morphism $f : X \to S$ factors through v by means of a morphism $p : X \to S'$, and

$$p_*(\mathcal{O}_X) = \mathcal{O}_{S'}, \qquad X \times_S Y = X \times_{S'} Y'.$$

Since f is quasi-separated, so is p (EGA IV$_1$, 1.2.2); and since f is quasi-compact and v is quasi-separated, p is also quasi-compact (EGA IV$_1$, 1.2.4). Consider the cartesian square

$$\begin{tikzcd} X \times_S Y \arrow[r, "p'"] \arrow[d] & Y' \arrow[d, "g'"] \\ X \arrow[r, "p"'] & S' \end{tikzcd}$$

In cases (b) and (c), the scheme Y is flat over S, and hence Y' is flat over S'. Applying 11.0(c) once again, we obtain

$$p'_*(\mathcal{O}_{X\times_S Y}) = g'^*p_*(\mathcal{O}_X) = g'^*(\mathcal{O}_{S'}) = \mathcal{O}_{Y'}. \tag{2}$$

The same equality holds in case (a), because p and p' are then isomorphisms.

Finally, since v is affine, EGA II, 1.4.7 gives

$$v_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(X)$$

for every quasi-coherent $\mathcal{O}_S$-module $\mathcal{F}$. Combined with (2) and (1), this gives

$$\begin{aligned} \mathcal{A}(X \times_S Y) &= v_*g'_*p'_*(\mathcal{O}_{X\times_S Y}) \\ &\overset{(2)}{=} v_*g'_*(\mathcal{O}_{Y'}) \overset{(1)}{=} v_*(\mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \\ &= \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{A}(X). \end{aligned}$$

Corollary 11.2.— *The functor*

$$X \longmapsto \operatorname{Spec} \mathcal{A}(X),$$

from the full subcategory of $(\mathbf{Sch}/S)$ *formed by the* S*-schemes* X *that are flat, quasi-compact, and quasi-separated over* S*, and for which* $\mathcal{A}(X)$ *is a flat* $\mathcal{O}_S$*-module, to the category of flat affine* S*-schemes, commutes with finite products and therefore takes* S*-groups to* S*-groups.*

Definition 11.3.— *Given a flat* S*-group* G*, quasi-compact and quasi-separated over* S*, such that* $\mathcal{A}(G)$ *is flat over* $\mathcal{O}_S$*,*[106] *we write* G_{af}*, and call the* S*-group*

$$G_{\mathrm{af}} = \operatorname{Spec} \mathcal{A}(G)$$

the affine envelope *of* G.

Proposition 11.3.1.—[107] *The canonical morphism*

$$\tau_G : G \longrightarrow G_{\mathrm{af}}$$

is a morphism of S*-groups. Moreover, it has the following universal property:*

(i) *For every morphism of* S*-schemes* $\varphi : G \to H$*, where* H *is affine over* S*, there is a unique morphism of* S*-schemes* $\varphi' : G_{\mathrm{af}} \to H$ *such that*

$$\varphi = \varphi' \circ \tau_G.$$

(ii) *If, moreover,* H *is an* S*-group and* φ *is a morphism of* S*-groups, then the same is true of* φ'*.*

11.4.—[108] Let $\mathcal{E}$ and $\mathcal{F}$ be two quasi-coherent $\mathcal{O}_S$-modules. Consider the S-functor

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))$$

(cf. I, 3.1.4), that is, for every S-scheme $f : X \to S$,

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))(X) = \mathrm{Hom}_{\mathcal{O}_X}(\mathrm{W}(\mathcal{E})_X, \mathrm{W}(\mathcal{F})_X).$$

Moreover, by I, 4.6.2, one has $\mathrm{W}(\mathcal{E})_X = \mathrm{W}(f^*(\mathcal{E}))$ (and similarly for $\mathcal{F}$), and

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathrm{W}(f^*(\mathcal{E})), \mathrm{W}(f^*(\mathcal{F}))) = \mathrm{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})).$$

We therefore obtain (using the adjunction formula for the last equality)

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))(X) = \mathrm{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_*f^*(\mathcal{F})). \qquad (\dagger)$$

Proposition 11.5.— *Let* X *be an* S*-scheme that is quasi-compact and quasi-separated over* S*, let* $f : X \to S$ *be the structural morphism, and let* $\mathcal{E}$ *and* $\mathcal{E}'$ *be two quasi-coherent* $\mathcal{O}_S$*-modules. Suppose that one of the following two conditions holds:*

a) *f is affine;*

[106] Editor's note: if S is a regular locally Noetherian scheme of dimension at most 2, and X is an S-scheme that is flat, quasi-compact, and quasi-separated, then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$-module; cf. [Ray70a], VII, 3.2.

[107] Editor's note: this proposition has been added; see also the additional Section 12 below for a study of the morphism $G \to G_{\mathrm{af}}$ and of its kernel.

[108] Editor's note: in 11.4–11.6, we have considered $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))$ instead of $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{V}(\mathcal{F}), \mathrm{V}(\mathcal{E}))$ (cf. I, 4.6.3), and simplified the original by taking account of the addition 11.0(b).

b) *$\mathcal{E}$ is flat over $\mathcal{O}_S$.*

Then the canonical morphism

$$\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X) \longrightarrow f_* f^*(\mathcal{E})$$

is an isomorphism, and hence

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}'), \mathrm{W}(\mathcal{E}))(X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X)).$$

Indeed, the second assertion follows from 11.4 and the first; the latter follows from EGA II, 1.4.7 in case (a), and from 11.0(b) in case (b).[109]

11.6.—[110] Let G be an S-group and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$-module. Giving $\mathcal{E}$ a G-$\mathcal{O}_S$-module structure (that is, an $\mathcal{O}_S$-linear action of G on $\mathrm{W}(\mathcal{E})$, cf. I, 4.7.1) is equivalent to giving a morphism of S-functors in monoids

$$\rho : G \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}));$$

indeed, such a ρ necessarily takes G into $\underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$.

By 11.4, giving a morphism of S-functors

$$\rho : G \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$$

is equivalent to giving an element θ of

$$\mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{E}), f^*(\mathcal{E})),$$

which corresponds by adjunction to an element δ of

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{E})),$$

where f denotes the projection $G \to S$.

Let $m : G \times_S G \to G$ be multiplication, let δ_G be the morphism

$$\mathcal{O}_G \longrightarrow m_*(\mathcal{O}_{G\times_S G}),$$

and let $\varphi : G \times_S G \to S$ be the projection (which equals $f \circ m$). It is convenient to write $\boxtimes$ for the "external" tensor product. We thus obtain a morphism

$$\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \longrightarrow \mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G\times_S G}).$$

By abuse of notation, we again write $\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G$ for the composite of this morphism with the canonical morphism

$$\mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G\times_S G}) \longrightarrow m_* m^* f^*(\mathcal{E}) = m_* \varphi^*(\mathcal{E}).$$

On the other hand, let $h : G \to S$ denote a second copy of $f : G \to S$, and consider the following commutative diagram, in which p, q denote the two projections:

[109] Editor's note: we have simplified the original, which used the isomorphism

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}\big(\mathrm{W}(\mathcal{E}'), \mathrm{W}(\mathcal{E})\big) \simeq \underline{\mathrm{Hom}}_{\mathcal{O}_S}\big(\mathrm{V}(\mathcal{E}), \mathrm{V}(\mathcal{E}')\big),$$

then the inclusion of the right-hand term in

$$\underline{\mathrm{Hom}}_S\big(\mathrm{V}(\mathcal{E}), \mathrm{V}(\mathcal{E}')\big) = \mathrm{Hom}_{\mathcal{O}_S}\big(\mathcal{E}', f_* f^*(\mathrm{Sym}(\mathcal{E}))\big),$$

and applied EGA III, 4.1.15 to $\mathrm{V}(\mathcal{E}) = \mathrm{Spec}(\mathrm{Sym}(\mathcal{E}))$ to deduce 11.0(b).

[110] Editor's note: we have expanded 11.6 and displayed the results obtained in the form of Proposition 11.6.1.

$$\begin{tikzcd}
G \times_S G \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, "\varphi"] & G \arrow[d, "h"] \\
G \arrow[r, "f"'] & S
\end{tikzcd}$$

Writing δ again for the morphism $\mathcal{E} \to h_* h^*(\mathcal{E})$, we obtain the morphism

$$\delta \boxtimes \mathrm{id}_{\mathcal{O}_G} : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \longrightarrow h_* h^*(\mathcal{E}) \boxtimes_{\mathcal{O}_S} \mathcal{O}_G = f^* h_* h^*(\mathcal{E}).$$

By abuse of notation, we again write $\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}$ for the composite of this morphism with the canonical morphism

$$f^* h_* h^*(\mathcal{E}) \longrightarrow p_* \varphi^*(\mathcal{E}).$$

The condition that ρ be compatible with multiplication is then equivalent to saying that, for every open subset U of S, the following diagram is commutative:

$$(1) \qquad \begin{tikzcd}
\Gamma(U, \mathcal{E}) \arrow[r, "\delta"] \arrow[d, "\delta"'] & \Gamma\big(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G\big) \arrow[d, "\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G"] \\
\Gamma\big(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G\big) \arrow[r, "\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}"'] & \Gamma\big(U \times_S G \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \boxtimes_{\mathcal{O}_S} \mathcal{O}_G\big)
\end{tikzcd}$$

Furthermore, the unit section $\varepsilon : S \to G$ induces a morphism u from $\mathcal{O}_G$ to

$$\varepsilon_* \varepsilon^*(\mathcal{O}_G) = \varepsilon_*(\mathcal{O}_S),$$

and the condition that ρ preserve unit elements is equivalent to the commutativity of the diagram

$$(2) \qquad \begin{tikzcd}
\Gamma(U, \mathcal{E}) \arrow[rr, "\delta"] \arrow[dr, "\sim"'] & & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \arrow[dl, "\mathrm{id}_{\mathcal{E}} \boxtimes u"] \\
& \Gamma(U \times_S S, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_S) &
\end{tikzcd}$$

Thus, giving $\mathcal{E}$ a G-$\mathcal{O}_S$-module structure is equivalent to giving a morphism of $\mathcal{O}_S$-modules

$$\delta : \mathcal{E} \longrightarrow f_* f^*(\mathcal{E})$$

satisfying conditions (1) and (2) above. In that case, the morphism

$$\theta : f^*(\mathcal{E}) \longrightarrow f^*(\mathcal{E})$$

deduced from δ by adjunction is an isomorphism, because it corresponds to the isomorphism

$$G \times_S \mathrm{W}(\mathcal{E}) \xrightarrow{\sim} G \times_S \mathrm{W}(\mathcal{E})$$

defined set-theoretically by $(g, x) \longmapsto (g, gx)$; see also I, 6.5.4.

Now suppose that G is flat, quasi-compact, and quasi-separated over S, and that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module. Then, by 11.1(c), the canonical morphism

$$\mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow \mathcal{A}(G \times_S G)$$

is an isomorphism, and we write Δ for the morphism

$$\delta_G : \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

If, moreover,

$$\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G) = f_* f^*(\mathcal{E}),$$

as is the case by 11.5 if $G \to S$ is affine or if $\mathcal{E}$ is flat over $\mathcal{O}_S$, then conditions (1) and (2) are equivalent to the following conditions, which express that

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

makes $\mathcal{E}$ a right $\mathcal{A}(G)$-comodule (cf. I, 4.7.2):

(CM 1) Put $\mathcal{A} = \mathcal{A}(G)$. The following diagram is commutative:

$$\begin{tikzcd} \mathcal{E} \arrow[r, "\delta"] \arrow[d, "\delta"'] & \mathcal{E} \otimes \mathcal{A} \arrow[d, "\mathrm{id}_{\mathcal{E}} \otimes \Delta"] \\ \mathcal{E} \otimes \mathcal{A} \arrow[r, "\delta \otimes \mathrm{id}_{\mathcal{A}}"'] & \mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A} \end{tikzcd}$$

(CM 2) Write $\eta : \mathcal{A} \to \mathcal{O}_S$ for the morphism $\mathcal{A}(\varepsilon)$. The following diagram is commutative:

$$\begin{tikzcd} \mathcal{E} \arrow[rr, "\delta"] \arrow[dr, "\sim"'] & & \mathcal{E} \otimes \mathcal{A} \arrow[dl, "\mathrm{id}_{\mathcal{E}} \otimes \eta"] \\ & \mathcal{E} \otimes \mathcal{O}_S & \end{tikzcd}$$

Remark 11.6.A.— Recall that $\mathrm{V}(\mathcal{E})$ denotes the vector fibration over S representing the functor $\mathrm{V}(\mathcal{E})$; that is, for every $S' \to S$,

$$\mathrm{V}(\mathcal{E})(S') = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'}).$$

Since I, 4.6.2 gives an anti-isomorphism of S-functors in monoids

$$\underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E})) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{V}(\mathcal{E})),$$

if $\mathcal{E}$ is a left G-$\mathcal{O}_S$-module, then there is a right action

$$\mu : \mathrm{V}(\mathcal{E}) \times_S G \longrightarrow \mathrm{V}(\mathcal{E})$$

of G on $\mathrm{V}(\mathcal{E})$, defined set-theoretically by

$$(\phi g)(x) = \phi(gx)$$

for every $g \in G(S')$, $x \in \Gamma(S', \mathcal{E}_{S'})$, and $\phi \in \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. We therefore have the commutative diagrams

$$\begin{tikzcd} \mathrm{V}(\mathcal{E}) \times_S G \times_S G \arrow[r, "\mu \times \mathrm{id}_G"] \arrow[d, "\mathrm{id}_{\mathrm{V}(\mathcal{E})} \times m"'] & \mathrm{V}(\mathcal{E}) \times_S G \arrow[d, "\mu"] \\ \mathrm{V}(\mathcal{E}) \times_S G \arrow[r, "\mu"'] & \mathrm{V}(\mathcal{E}) \end{tikzcd} \qquad \begin{tikzcd} \mathrm{V}(\mathcal{E}) & \mathrm{V}(\mathcal{E}) \times_S G \arrow[l, "\mu"'] \\ & \mathrm{V}(\mathcal{E}) \times_S S \arrow[u, "\mathrm{id}_{\mathrm{V}(\mathcal{E})} \times \varepsilon"'] \arrow[ul, "\sim"] \end{tikzcd}$$

When G is flat, quasi-compact, and quasi-separated over S, when $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module, and when one of the conditions of 11.5 is satisfied, one similarly recovers conditions (CM 1) and (CM 2).

Consequently, we have proved:

Proposition 11.6.1.— *Let G be an S-group that is flat, quasi-compact, and quasi-separated over S, such that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module, and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$-module.*

(i) *Giving an $\mathcal{A}(G)$-comodule structure*

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

is equivalent to giving $\mathcal{E}$ a G_{af}-$\mathcal{O}_S$-module structure (that is, an $\mathcal{O}_S$-linear action of G_{af} on $\mathcal{E}$). *By composition with the morphism of S-groups $G \to G_{\mathrm{af}}$, this defines a G-$\mathcal{O}_S$-module structure on $\mathcal{E}$.*

(ii) *If, moreover, $\mathcal{E}$ is flat, every $\mathcal{O}_S$-linear action of G on $\mathcal{E}$ factors through G_{af} and corresponds to a unique $\mathcal{A}(G)$-comodule structure on $\mathcal{E}$.*

Lemma 11.7.— *Let G be an S-group that is flat, quasi-compact, and quasi-separated over S, such that $\mathcal{A} = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module. Let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$-module, let*

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{A}$$

be an $\mathcal{A}$-comodule structure, and let

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$$

be the associated action of G on $\mathcal{E}$. Let $\mathcal{E}_0$ be a quasi-coherent $\mathcal{O}_S$-submodule of $\mathcal{E}$ such that the restriction δ_0 of δ to $\mathcal{E}_0$

factors through

$$\mathcal{E}_0 \longrightarrow \mathcal{E}_0 \otimes \mathcal{A};$$

that is, suppose that the following diagram is commutative:

$$\begin{tikzcd} \mathcal{E}_0 \arrow[r, hook] \arrow[d, "\delta_0"'] & \mathcal{E} \arrow[d, "\delta"] \\ \mathcal{E}_0 \otimes \mathcal{A} \arrow[r, hook] & \mathcal{E} \otimes \mathcal{A} \end{tikzcd}$$

(N.B. The morphism $\mathcal{E}_0 \otimes \mathcal{A} \to \mathcal{E} \otimes \mathcal{A}$ is injective because $\mathcal{A}$ is flat over $\mathcal{O}_S$.) *Then δ_0 makes $\mathcal{E}_0$ an $\mathcal{A}(G)$-comodule, and hence defines an action ρ_0 of G on $\mathcal{E}_0$. This is called the action on $\mathcal{E}_0$ induced by ρ, and we say that $\mathcal{E}_0$ is stable under ρ.*

This follows immediately from the definitions and from 11.6. Notice, however, that in general the canonical map

$$\mathrm{W}(\mathcal{E}_0) \longrightarrow \mathrm{W}(\mathcal{E})$$

is not a monomorphism.

Remark 11.7.bis.—[111] Let G be a flat S-group and let $\mathcal{E}$ be a quasi-coherent G-$\mathcal{O}_S$-module. Write f for the morphism $G \to S$ and δ for the morphism

$$\mathcal{E} \longrightarrow f_* f^*(\mathcal{E})$$

defined in 11.6. Let $\mathcal{E}_0$ be a quasi-coherent $\mathcal{O}_S$-submodule of $\mathcal{E}$. Since f is flat, $f_* f^*(\mathcal{E}_0)$ is an $\mathcal{O}_S$-submodule of $f_* f^*(\mathcal{E})$, and the same holds for $\varphi = f \times f$. Consequently, if the restriction δ_0 of δ to $\mathcal{E}_0$ factors through $f_* f^*(\mathcal{E}_0)$, then it makes $\mathcal{E}_0$ a G-$\mathcal{O}_S$-module. In this case, we say that $\mathcal{E}_0$ is a G-stable submodule of $\mathcal{E}$.

[111] Editor's note: we have added this remark, which generalizes 11.7 and will be useful in 11.10.bis.

Definition 11.8.0.—[112] Let S be a scheme. An $\mathcal{O}_S$*-coalgebra* is an $\mathcal{O}_S$-module $\mathcal{C}$, together with two morphisms of $\mathcal{O}_S$-modules

$$\Delta : \mathcal{C} \longrightarrow \mathcal{C} \otimes \mathcal{C}, \qquad \varepsilon : \mathcal{C} \longrightarrow \mathcal{O}_S,$$

satisfying the following two axioms (cf. I, 4.2):

(CO 1) Δ is coassociative: the following diagram is commutative:

$$\begin{tikzcd} & \mathcal{C} \otimes \mathcal{C} \arrow[dr, "\mathrm{id}\otimes\Delta"] & \\ \mathcal{C} \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & \mathcal{C} \otimes \mathcal{C} \otimes \mathcal{C} \\ & \mathcal{C} \otimes \mathcal{C} \arrow[ur, "\Delta\otimes\mathrm{id}"'] & \end{tikzcd}$$

(CO 2) ε is a counit; that is, the following two composites are the identity:

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\mathrm{id}\otimes\varepsilon} \mathcal{C} \otimes \mathcal{O}_S \xrightarrow{\sim} \mathcal{C},$$

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\varepsilon\otimes\mathrm{id}} \mathcal{O}_S \otimes \mathcal{C} \xrightarrow{\sim} \mathcal{C}.$$

A $\mathcal{C}$*-comodule* (on the right) is an $\mathcal{O}_S$-module $\mathcal{E}$, together with a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{C}$$

satisfying axioms (CM 1) and (CM 2) of 11.6.

We say that $\mathcal{C}$ (respectively, $\mathcal{E}$) is a *quasi-coherent coalgebra* (respectively, a *quasi-coherent comodule*) if it is a quasi-coherent $\mathcal{O}_S$-module.

Let A be a commutative ring and let C be an A-coalgebra. Then

$$C^\vee = \mathrm{Hom}_A(C, A)$$

is an A-algebra. We denote by

$$\mathrm{ev} : C \otimes_A C^\vee \longrightarrow A$$

the natural evaluation map.

Lemma 11.8.— *Let C be an A-coalgebra, V a C-comodule, and M an A-submodule of V. Suppose that C is a projective A-module.*[113] *Let $c(M)$ be the image of the morphism of A-modules*

$$\theta : M \otimes_A C^\vee \xrightarrow{\mu\otimes\mathrm{id}} V \otimes_A C \otimes_A C^\vee \xrightarrow{\mathrm{id}\otimes\mathrm{ev}} V.$$

Then $c(M)$ is the smallest subcomodule of V containing M, and it is a finitely generated A-module if M is. We call $c(M)$ the subcomodule generated by *M.*

[112] Editor's note: the statements 11.8 and 11.9 concern only the notion of a comodule over a coalgebra. We have therefore introduced Definition 11.8.0 and reformulated 11.8 and 11.9 accordingly.

[113] Editor's note: in the original, C is assumed to be a free A-module. The generalization to the case where C is a projective A-module, pointed out by J.-P. Serre, is mentioned in Remark 11.10.1. We have incorporated this generalization here and in 11.9, and have expanded the proofs accordingly.

Moreover, for every homomorphism of rings $A \to A'$, if M' denotes the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$, then $c(M')$ is the image of $c(M) \otimes_A A'$ in V'. Thus the formation of $c(M)$ commutes with base change.

First, $M \subset c(M)$ by (CM 2). If N is a subcomodule of V containing M, then $\mu(M) \subset N \otimes C$, and hence $c(M) \subset N$.

By hypothesis, C is a direct summand of a free A-module L with basis $(e_i)_{i \in I}$. Let φ_i denote the restriction to C of the linear form e_i^*, defined by

$$e_i^*(e_j) = \delta_{ij}.$$

Let $x \in M$. We may write

$$\mu(x) = \sum_{i \in J} x_i \otimes e_i, \tag{1}$$

where $x_i \in V$ and J is a finite subset of I. Then

$$x_i = \theta(x \otimes \varphi_i) \in c(Ax),$$

and consequently

$$c(Ax) = \sum_{i \in J} Ax_i.$$

Since C is a direct summand of L, say $L = C \oplus R$, whence

$$V \otimes L = (V \otimes C) \oplus (V \otimes R),$$

we obtain

$$(c(Ax) \otimes L) \cap (V \otimes C) = c(Ax) \otimes C.$$

Consequently, $\mu(x)$ may also be written in the form

$$\mu(x) = \sum_{j \in J} x_j \otimes b_j, \tag{2}$$

with $b_j \in C$. Write

$$\Delta(b_j) = \sum_{i \in I} b_{ij} \otimes e_i, \qquad b_{ij} \in C.$$

Applying $\mu \otimes \mathrm{id}$ to (1) and $\mathrm{id} \otimes \Delta$ to (2), and using axiom (CM 1), we obtain, for every $i \in J$,

$$\mu(x_i) = \sum_{j \in J} x_j \otimes b_{ij} \in c(Ax) \otimes C.$$

This proves that $c(M)$ is a subcomodule of V, and therefore that it is the smallest subcomodule of V containing M.

It is clear that $c(M)$ is a finitely generated A-module whenever M is: if

$$M = Ax_1 + \cdots + Ax_n, \qquad \mu(x_k) = \sum_i x_{ik} \otimes e_i,$$

then the x_{ik}, for $k = 1, \ldots, n$ and i ranging over a finite subset of I, generate $c(M)$.

Finally, let $A \to A'$ be a homomorphism of rings, and let M' be the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$. Then $c(M')$, respectively the image of $c(M) \otimes_A A'$ in V', is the image of the morphism θ' below, respectively of the composite $\theta' \circ \tau$:

$$M \otimes_A A' \otimes_A C^\vee \xrightarrow{\tau} M \otimes_A \mathrm{Hom}_A(C, A') \xrightarrow{\theta'} V'.$$

These two morphisms have the same image. Indeed, let $\psi \in \mathrm{Hom}_A(C, A')$ and $x \in M$, and write

$$\mu(x) = \sum_{i \in J} x_i \otimes e_i.$$

Then

$$\theta'(x \otimes \psi) = \sum_{i \in J} \psi(e_i) x_i$$

is the image under $\theta' \circ \tau$ of the element

$$\sum_{i \in J} x \otimes \psi(e_i) \otimes \varphi_i \in M \otimes_A A' \otimes_A C^{\vee}.$$

This proves the lemma.

We also have the following proposition.

Proposition 11.8.bis.—[114] *Let A be a Noetherian ring, let C be an A-coalgebra flat over A, let V be a C-comodule, and let M be a finitely generated A-submodule of V. Then there is a subcomodule W of V, finitely generated over A, that contains M.*

Indeed, since M is finitely generated, so is $\Delta_V(M)$. Hence there is a finitely generated A-submodule M' of V such that

$$\Delta_V(M) \subset M' \otimes_A C.$$

Let $\pi : V \to V/M'$ be the projection, put

$$\overline{\Delta}_V = (\pi \otimes \mathrm{id}_C)\Delta_V,$$

and set

$$W = \{x \in V \mid \Delta_V(x) \in M' \otimes_A C\} = \ker \overline{\Delta}_V.$$

This is an A-submodule of V that contains M and is contained in M', because

$$x = (\mathrm{id}_V \otimes \varepsilon)\Delta_V(x).$$

It is therefore finitely generated over A. Moreover,

$$(\overline{\Delta}_V \otimes \mathrm{id}_C)\Delta_V = (\pi \otimes \Delta_C)\Delta_V$$

vanishes on W; in other words, $\Delta_V(W)$ is contained in the kernel K of $\overline{\Delta}_V \otimes \mathrm{id}_C$. Since C is flat over A, one has

$$K = W \otimes_A C.$$

Thus W is a subcomodule of V.

Lemma 11.8.1.—[115] *Let C be an A-coalgebra, let V be a C-comodule, let M be an A-submodule of V, and let $f : A \to A'$ be a faithfully flat homomorphism of rings. Suppose that $C' = C \otimes_A A'$ is a projective A'-module.*

(i) *There is a smallest subcomodule $t(M)$ of V containing M, and $t(M)$ is a finitely generated A-module if M is. Moreover, the formation of $t(M)$ commutes with base change.*

(ii) *More precisely, C is a projective A-module, and $t(M) = c(M)$.*

[114] Editor's note: this proposition, taken from [Se68], Section 1.5, Proposition 2, has been added.
[115] Editor's note: this lemma, which is used in the proof of 11.9, has been added.

Proof. For (ii), [RG71] (see Proposition 11.8.2 below) shows that C is a projective A-module. We may therefore apply Lemma 11.8: $c(M)$ is the smallest subcomodule of V containing M, it is a finitely generated A-module if M is, and its formation commutes with base change.

To avoid an anachronism ([RG71] postdates SGA 3), let us sketch a direct proof of (i). Since $A \to A'$ is flat,

$$M' = M \otimes_A A'$$

is an A'-submodule of $V' = V \otimes_A A'$. Since C' is a projective A'-module, $c(M')$ is the smallest subcomodule of V' containing M'. Denote by $V' \otimes_A A'$ and $A' \otimes_A V'$ the two $A' \otimes_A A'$-comodule structures on

$$V'' = V' \otimes_{A'} (A' \otimes_A A')$$

obtained by the two base changes

$$A' \rightrightarrows A' \otimes_A A', \qquad a' \longmapsto a' \otimes 1, \quad a' \longmapsto 1 \otimes a'.$$

The A'-comodule V' carries an isomorphism of $A' \otimes_A A'$-comodules

$$\varphi : V' \otimes_A A' \xrightarrow{\sim} A' \otimes_A V', \qquad (x \otimes a') \otimes b' \longmapsto b' \otimes (x \otimes a'),$$

which is a descent datum, that is,

$$\varphi_{31} = \varphi_{32} \circ \varphi_{21}.$$

Since $M' = M \otimes_A A'$, the isomorphism φ takes $M' \otimes_A A'$ onto $A' \otimes_A M'$, and hence takes $c(M' \otimes_A A')$ onto $c(A' \otimes_A M')$. Because the formation of $c(M')$ commutes with base change,

$$c(M' \otimes_A A') = c(M') \otimes_A A', \qquad c(A' \otimes_A M') = A' \otimes_A c(M').$$

Thus

$$\varphi\big(c(M') \otimes_A A'\big) = A' \otimes_A c(M'),$$

and φ equips $c(M')$ with a descent datum. By fpqc descent, there is a unique subcomodule $t(M)$ of V such that

$$c(M') = t(M) \otimes_A A'.$$

It contains M, since $t(M) \otimes_A A'$ contains M'. Moreover, if N is a subcomodule of V containing M, then N contains $t(M)$, because $N \otimes_A A'$ contains

$$c(M') = t(M) \otimes_A A'.$$

Hence $t(M)$ is the smallest subcomodule of V containing M.

Finally, let $A \to B$ be a homomorphism of rings. Put

$$B' = B \otimes_A A',$$

and let M_B, respectively $M'_{B'}$, be the image of $M \otimes_A B$ in $V_B = V \otimes_A B$, respectively of $M' \otimes_{A'} B'$ in $V' \otimes_{A'} B' = V \otimes_A B'$. Then

$$M_B \otimes_B B' = M'_{B'}.$$

On the one hand, the preceding construction, applied to C_B and $B \to B'$, gives

$$c(M_B \otimes_B B') = t(M_B) \otimes_B B' = t(M_B) \otimes_A A'.$$

On the other hand, because the formation of $c(M')$ commutes with base change, $c(M'_{B'})$ is the image in

$$V' \otimes_{A'} B' = V \otimes_A B \otimes_A A'$$

of

$$c(M') \otimes_{A'} B' = t(M) \otimes_A B \otimes_A A'.$$

It follows that $t(M_B)$ is the image of $t(M) \otimes_A B$ in V_B.

Proposition 11.8.2 (Gruson–Raynaud).— *Let $f : A \to A'$ be a faithfully flat homomorphism. Then f "descends projectivity": if M is an A-module and $M \otimes_A A'$ is a projective A'-module, then M is a projective A-module.*

Indeed, by [RG71], II, 2.5.1, f descends the Mittag–Leffler condition, and therefore, by *loc. cit.*, II, 3.1.3, f descends projectivity.[116]

Proposition 11.9.—[117] *Let $\mathcal{C}$ be an $\mathcal{O}_S$-coalgebra, let $\mathcal{E}$ be a $\mathcal{C}$-comodule, and let $\mathcal{F}$ be an $\mathcal{O}_S$-submodule of $\mathcal{E}$, all of them quasi-coherent. Suppose that S is covered by affine open subsets $U_\alpha = \operatorname{Spec} A_\alpha$, and that for each α there is a faithfully flat homomorphism of rings $A_\alpha \to A'_\alpha$ such that*

$$\Gamma(U_\alpha, \mathcal{C}) \otimes_{A_\alpha} A'_\alpha$$

is a projective A'_α-module.[(*)]

Then there is a smallest quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$, and $t(\mathcal{F})$ is a finitely generated $\mathcal{O}_S$-module if $\mathcal{F}$ is. Moreover, for every base change $S' \to S$, if $\mathcal{F}'$ denotes the image of $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ in

$$\mathcal{E}' = \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'},$$

then $t(\mathcal{F}')$ is the image of $t(\mathcal{F}) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ in $\mathcal{E}'$. In other words, the formation of $t(\mathcal{F})$ commutes with base change.

Proof. For each α, let T_α be the A_α-module

$$T_\alpha = t(\Gamma(U_\alpha, \mathcal{F})).$$

By 11.8.1(i), the associated $\mathcal{O}_{U_\alpha}$-module $\widetilde{T}_\alpha$ is the smallest quasi-coherent subcomodule of $\mathcal{E}|_{U_\alpha}$ containing $\mathcal{F}|_{U_\alpha}$; and it is a finitely generated $\mathcal{O}_{U_\alpha}$-module if $\mathcal{F}|_{U_\alpha}$ is.

For every α, β, put

$$U_{\alpha\beta} = U_\alpha \cap U_\beta.$$

Since the construction of $t(M)$ commutes with base change, there are canonical isomorphisms of $\mathcal{O}_{U_{\alpha\beta}}$-modules

$$\varphi_{\alpha\beta} : \widetilde{T}_\beta \otimes_{A_\beta} \mathcal{O}_{U_{\alpha\beta}} \xrightarrow{\sim} \widetilde{T}_\alpha \otimes_{A_\alpha} \mathcal{O}_{U_{\alpha\beta}}.$$

They satisfy the cocycle condition

$$\varphi'_{\alpha\gamma} = \varphi'_{\alpha\beta} \circ \varphi'_{\beta\gamma},$$

[116] Editor's note: let us also point out that assertion II, 2.5.2 of *loc. cit.*, which is more general than II, 2.5.1, is corrected in [Gr73]; this does not affect the case of faithfully flat homomorphisms.

[117] Editor's note: as indicated in editor's note 112, we have rewritten the statement for an $\mathcal{O}_S$-coalgebra $\mathcal{C}$ (rather than for an S-group G satisfying the hypotheses in Corollary 11.10). We have also expanded the proof (the original merely said: "$\cdots$ the proposition follows from Lemma 11.8").

[*] This holds, for example, when $\mathcal{C} = \mathcal{A}(G)$, where G is a reductive S-group, as we shall see in Exposé XXII, 5.7.8.

where $\varphi'_{\alpha\gamma}$, respectively the other maps, denotes the restriction of $\varphi_{\alpha\gamma}$, respectively the other maps, to $U_\alpha \cap U_\beta \cap U_\gamma$.

Consequently, the $\widetilde{T}_\alpha$ glue to a quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$. We leave it to the reader to verify that $t(\mathcal{F})$ is the smallest quasi-coherent subcomodule of $\mathcal{E}$ containing $\mathcal{F}$, and that its formation commutes with base change.

Definition 11.9.1.—[118] *Let S be a scheme and let $\mathcal{P}$ be a quasi-coherent $\mathcal{O}_S$-module. The following conditions are equivalent:*

(i) *For every affine open subset U of S, the module $\Gamma(U, \mathcal{P})$ is a projective $\mathcal{O}_S(U)$-module.*

(ii) *There is a covering (U_α) of S by affine open subsets such that every $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$-module.*

(iii) *There is a covering (U_α) of S by affine open subsets, and faithfully flat homomorphisms of rings*
$$A_\alpha = \mathcal{O}_S(U_\alpha) \longrightarrow A'_\alpha,$$
such that, for every α,
$$\Gamma(U_\alpha, \mathcal{P}) \otimes_{A_\alpha} A'_\alpha$$
is a projective A'_α-module.

It is clear that
$$\text{(i)} \Longrightarrow \text{(ii)} \Longrightarrow \text{(iii)}.$$
Conversely, if (iii) holds, 11.8.2 implies that every $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$-module, and hence (ii) holds.

Finally, suppose that (ii) holds, and let
$$V = \operatorname{Spec} A$$
be an arbitrary affine open subset. It is covered by finitely many affine open subsets
$$V_1, \dots, V_n, \qquad V_i = \operatorname{Spec} A_i,$$
each contained in at least one $V \cap U_\alpha$, so that $\Gamma(V_i, \mathcal{P})$ is a projective A_i-module. Put
$$A' = A_1 \times \cdots \times A_n.$$
Then $A \to A'$ is faithfully flat and
$$\Gamma(V, \mathcal{P}) \otimes_A A'$$
is a projective A'-module. Therefore 11.8.2 implies that $\Gamma(V, \mathcal{P})$ is a projective A-module.

When these equivalent conditions hold, we say that $\mathcal{P}$ is a *locally projective* $\mathcal{O}_S$-module.

Corollary 11.10.— *Let S be a quasi-compact and quasi-separated scheme, let G be an S-group, and let ρ be a linear action of G on a quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$. Suppose that:*

(i) *G satisfies one of the following conditions:*

(a) *G is affine and flat over S;*

[118] Editor's note: this definition, taken from the bottom of page 82 of [RG71], has been added. Thus the hypothesis in Proposition 11.9 is that the coalgebra $\mathcal{C}$ be a locally projective $\mathcal{O}_S$-module, and we have used this terminology in Corollary 11.10.

(b) *G is flat, quasi-compact, and quasi-separated over S, and $\mathcal{E}$ is flat;*

(ii) *$\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$-module.*

Then $\mathcal{E}$ is the direct limit of an increasing filtered family of finitely generated quasi-coherent $\mathcal{O}_S$-submodules of $\mathcal{E}$ that are stable under G.

By hypothesis (i) and 11.6.1, $\mathcal{E}$ is endowed with an $\mathcal{A}(G)$-comodule structure. On the other hand, since S is quasi-compact and quasi-separated, $\mathcal{E}$ is the direct limit of its finitely generated quasi-coherent submodules (EGA I, 9.4.9 and EGA IV_1, 1.7.7). Consequently, the corollary follows from Proposition 11.9, applied to the coalgebra $\mathcal{A}(G)$.

We also have the following proposition.

Proposition 11.10.bis.—[119] *Let S be a Noetherian scheme, let G be an S-group that is flat, quasi-compact, and quasi-separated over S, let $\mathcal{E}$ be a quasi-coherent G-$\mathcal{O}_S$-module, and let $\mathcal{M}$ be a coherent $\mathcal{O}_S$-submodule of $\mathcal{E}$. Then $\mathcal{M}$ is contained in a coherent $\mathcal{O}_S$-submodule stable under G.*

Proof. Write f for the morphism $G \to S$, and τ for the adjunction morphism

$$\mathcal{E} \longrightarrow f_* f^*(\mathcal{E}).$$

By 11.6, the G-$\mathcal{O}_S$-module structure on $\mathcal{E}$ is given by an automorphism θ of the $\mathcal{O}_G$-module $f^*(\mathcal{E})$, such that the morphism

$$\delta = \theta \circ \tau$$

satisfies conditions (1) and (2) of 11.6. (The situation considered in [Th87] is more general: the author considers a G-scheme X and a G-equivariant $\mathcal{O}_X$-module $\mathcal{E}$ (cf. Exposé I, Section 6); here $X = S$, endowed with the trivial action of G.)

Since S is Noetherian, $\mathcal{E}$ is the filtered direct limit of its coherent submodules $\mathcal{F}_\alpha$ (cf. EGA I, 9.4.9). Then $f^*(\mathcal{E})$ is the filtered direct limit of the $f^*(\mathcal{F}_\alpha)$, which are submodules of $f^*(\mathcal{E})$ because f is flat. Moreover, since f is quasi-compact and quasi-separated, by 11.0 the $\mathcal{O}_S$-module $f_*f^*(\mathcal{E})$ is quasi-coherent and is the filtered direct limit of the quasi-coherent submodules $f_*f^*(\mathcal{F}_\alpha)$. Consequently, $\mathcal{E}$ is the filtered direct limit of the quasi-coherent $\mathcal{O}_S$-submodules $\mathcal{E}_\alpha$, where $\mathcal{E}_\alpha$ denotes the inverse image under δ of $f_*f^*(\mathcal{F}_\alpha)$; equivalently, it is the kernel of the composite

$$\overline{\delta}_\alpha : \quad \mathcal{E} \xrightarrow{\delta} f_*f^*(\mathcal{E}) \longrightarrow f_*f^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Since $\mathcal{M}$ is coherent and S is Noetherian, every increasing sequence of submodules of $\mathcal{M}$ is stationary; hence $\mathcal{M}$ is contained in some $\mathcal{E}_\alpha$. We shall show that every $\mathcal{E}_\alpha$ is coherent and G-stable.

Let

$$u : f^*(\mathcal{E}) \longrightarrow \varepsilon_*(\mathcal{E})$$

be the morphism corresponding by adjunction to the identity morphism from $\mathcal{E}$ to $f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$. Then the composite

$$\mathcal{E} \xrightarrow{\tau} f_*f^*(\mathcal{E}) \xrightarrow{f_*(u)} f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$$

is the identity (cf. 11.6(2)). We have the commutative diagram

[119] Editor's note: this proposition has been added; it is a special case of [Th87], 1.4–1.5. The author there refers to an argument of Deligne (cf. [Kn71], III, Th. 1.1); one may also note its similarity with Serre's argument ([Se68], Prop. 2) recalled in 11.8.bis.

$$\begin{tikzcd}
\mathcal{E}_\alpha \arrow[r, "\delta"] \arrow[d, "i_\alpha"'] & f_*f^*(\mathcal{F}_\alpha) \arrow[r, "f_*(u)"] \arrow[d] & \mathcal{F}_\alpha \arrow[d, "j_\alpha"] \\
\mathcal{E} \arrow[r, "\tau"'] & f_*f^*(\mathcal{E}) \arrow[r, "f_*(u)"'] & \mathcal{E}
\end{tikzcd}$$

where i_α (respectively j_α) denotes the inclusion of $\mathcal{E}_\alpha$ (respectively $\mathcal{F}_\alpha$) in $\mathcal{E}$. It follows that i_α factors through j_α; thus $\mathcal{E}_\alpha$ is a submodule of $\mathcal{F}_\alpha$, and is therefore coherent.

Finally, let us show that $\mathcal{E}_\alpha$ is G-stable (cf. 11.7.bis). Let $h : G \to S$ denote a second copy of $f : G \to S$, and consider the following commutative diagram:

$$\begin{tikzcd}
G \times_S G \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, "\varphi"] & G \arrow[d, "h"] \\
G \arrow[r, "f"'] & S
\end{tikzcd}$$

The exact sequence

$$0 \longrightarrow \mathcal{E}_\alpha \longrightarrow \mathcal{E} \xrightarrow{\overline{\delta}_\alpha} h_*h^*(\mathcal{E}/\mathcal{F}_\alpha)$$

therefore gives, since f is flat, the exact sequence

$$0 \longrightarrow f_*f^*(\mathcal{E}_\alpha) \longrightarrow f_*f^*(\mathcal{E}) \xrightarrow{f_*f^*(\overline{\delta}_\alpha)} f_*f^*h_*h^*(\mathcal{E}/\mathcal{F}_\alpha). \tag{†}$$

Moreover, since h is quasi-compact and quasi-separated and f is flat, the canonical morphism

$$f^*h_*h^*(\mathcal{E}/\mathcal{F}_\alpha) \longrightarrow p_*q^*h^*(\mathcal{E}/\mathcal{F}_\alpha) = p_*\varphi^*(\mathcal{E}/\mathcal{F}_\alpha)$$

is an isomorphism. Thus the right-hand term in (†) is $\varphi_*\varphi^*(\mathcal{E}/\mathcal{F}_\alpha)$. Using the notation of 11.6 and writing π_α for the projection $\mathcal{E} \to \mathcal{E}/\mathcal{F}_\alpha$, we obtain the following commutative diagram, whose bottom row is exact:

$$\begin{tikzcd}
 & \mathcal{E}_\alpha \arrow[r, "\delta"] \arrow[d, "\delta"'] & f_*f^*(\mathcal{E}) \arrow[d, "f_*(f^*(\pi_\alpha)\boxtimes\delta_G)"] \\
f_*f^*(\mathcal{E}_\alpha) \arrow[r] & f_*f^*(\mathcal{E}) \arrow[r, "f_*f^*(\overline{\delta}_\alpha)"'] & \varphi_*\varphi^*(\mathcal{E}/\mathcal{F}_\alpha)
\end{tikzcd}$$

Since

$$f_*\big(f^*(\pi_\alpha) \boxtimes \delta_G\big) \circ \delta$$

vanishes on $\mathcal{E}_\alpha$, it follows that δ carries $\mathcal{E}_\alpha$ into $f_*f^*(\mathcal{E}_\alpha)$; in other words, $\mathcal{E}_\alpha$ is G-stable. This proves the proposition.

Remark 11.10.1.—[120] In 11.8, it is not enough to assume that C is a flat A-module, even when A is a principal ideal domain. Indeed, we have the following counterexamples, pointed out to us independently by O. Gabber and J.-P. Serre.

(a) Let $(A, \mathfrak{m})$ be a discrete valuation ring, let K be its field of fractions, and let G be the A-group obtained by "extension by zero" from the K-group $(\mathbb{Z}/2\mathbb{Z})_K$. The constant group $(\mathbb{Z}/2\mathbb{Z})_A$, and hence also its subgroup G, acts on the free A-module V with basis v_1, v_2 by

[120] Editor's note: the first part of the original Remark 11.10.1 has been incorporated into 11.8 and 11.9 (with "free" replaced by "projective"); the counterexamples below correct its second part.

interchanging v_1 and v_2. Then Av_1 is not a G-submodule of V, but it is the intersection of the G-submodules
$$Av_1 + \mathfrak{m}^n v_2, \qquad n \in \mathbb{N}^*.$$
Thus there is no smallest G-submodule of V containing v_1.

(b) Let A be an integral domain distinct from its field of fractions K, and let G be the flat affine A-group corresponding to the Hopf algebra

$$\mathcal{A}(G) = \{P \in K[T] \mid P(0) \in A\}.$$

Its comultiplication, counit, and antipode are respectively defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \qquad \varepsilon(T) = 0, \qquad \tau(T) = -T.$$

(Thus $G \otimes_A K = \mathbb{G}_{a,K}$.)

Let $V = Av_1 \oplus Av_2$ be free over A, and let u be the endomorphism of V defined by

$$u(v_1) = v_2, \qquad u(v_2) = 0,$$

so that $u^2 = 0$. Then V has the $\mathcal{A}(G)$-comodule structure

$$\mu(m) = 1 \otimes m + T \otimes u(m).$$

The G-submodules of V containing v_1 are precisely the A-submodules $Av_1 \oplus Iv_2$, where I is a nonzero ideal of A. Their intersection is Av_1, which is not a G-submodule. Hence there is no smallest G-submodule of V containing v_1. Moreover,

$$C = A \oplus K \cdot T$$

is a subcoalgebra of $\mathcal{A}(G)$, flat over A, and the coaction

$$\mu : V \longrightarrow V \otimes \mathcal{A}(G)$$

factors through $V \otimes C$. Thus the "very simple" coalgebra C also gives a counterexample.

Finally, the preceding two examples are special cases of the following construction. Let A be an integral domain distinct from its field of fractions K, and let B be a Hopf A-algebra, free over A. Write
$$\varepsilon : B \longrightarrow A, \qquad I = \ker(\varepsilon)$$
for the augmentation and the augmentation ideal. Since $B = A \cdot 1 \oplus I$, one readily sees that

$$B' = \{b \in B \otimes_A K \mid \varepsilon(b) \in A\}$$

is a Hopf subalgebra of B_K. Suppose that V is a B-comodule, free with basis $(v_1, \ldots, v_n)$ as an A-module, and that $\mu_V(v_1) \neq v_1 \otimes 1$. Then Av_1 is not a subcomodule of V, but it is the intersection of the subcomodules

$$Av_1 + J \cdot V$$

as J ranges over the nonzero ideals of A. Thus there is no smallest subcomodule of V containing v_1.

Proposition 11.11.— *Let G be an algebraic group over a field k. The following conditions are equivalent:*

(i) *G is affine.*

(ii) *G is quasi-affine.*

(iii) *G acts faithfully on a quasi-affine k-scheme X.*

(iv) *G acts linearly and faithfully on a k-vector space* (not necessarily finite-dimensional).

(v) *G is isomorphic to a closed subgroup of* $\mathrm{GL}(n)_k$.

Proof. The implication (i) ⇒ (ii) is immediate, and (ii) ⇒ (iii) because G acts faithfully on itself by translations.

Suppose that G acts faithfully on the right on a quasi-affine k-scheme X.[121] Since X is quasi-affine, it is separated and quasi-compact.[122] Similarly, G is separated (VI_A, 0.3) and quasi-compact (since it is of finite type over k). Hence, by 11.1(c), there are canonical isomorphisms

$$\mathcal{O}(X \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G), \qquad \mathcal{O}(X \times G \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G).$$

It follows that the morphism

$$\mu : \mathcal{O}(X) \longrightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$$

induced by $X \times G \to X$ gives $\mathcal{O}(X)$ the structure of a right $\mathcal{O}(G)$-comodule; that is, G acts linearly on the left on the k-algebra $\mathcal{O}(X)$. Consequently, G also acts on the right on the affine envelope

$$X_{\mathrm{af}} = \operatorname{Spec} \mathcal{O}(X)$$

of X, and the canonical morphism $X \to X_{\mathrm{af}}$ is G-equivariant.

Since X is quasi-affine, the morphism $X \to X_{\mathrm{af}}$ is an open immersion (EGA II, 5.1.2). Hence, a fortiori, G acts faithfully on X_{af}, and therefore the left linear action of G on the k-algebra $\mathcal{O}(X) = \mathcal{O}(X_{\mathrm{af}})$ is faithful. This proves (iii) ⇒ (iv).

Now suppose that G acts faithfully on a k-vector space V. By 11.10, V is the direct limit of finite-dimensional vector subspaces V_i stable under G. Let K_i be the kernel of the induced action of G on V_i; that is, the kernel of $G \to \mathrm{Aut}(V_i)$. Then K_i is a closed subscheme of G, and the faithfulness of the action of G means that the intersection of the K_i is the identity subgroup of G. Since G is Noetherian, one of the K_i is already the identity subgroup. Thus $G \to \mathrm{Aut}(V_i)$ is a monomorphism, and hence, by 1.4.2, a closed immersion. This proves (iv) ⇒ (v). Finally, (v) ⇒ (i) is immediate, and the proposition follows.

Remark 11.11.1.— Proposition 11.11 admits the following generalization. Let S be a regular, locally Noetherian scheme of dimension at most 1, and let G be a group scheme that is flat, quasi-compact, and quasi-separated over S. (In this case, $\mathcal{A}(G)$ is a torsion-free $\mathcal{O}_S$-module, and hence is flat.)

(a) The following conditions are equivalent:[123]

 (i) G is affine over S.

 (ii) G is quasi-affine over S.

 (iii) G acts faithfully on a quasi-affine flat S-scheme.

[121] Editor's note: the original has been expanded in what follows. We have also chosen a right action of G on X, so as to obtain a left linear action of G on $\mathcal{O}(X)$.

[122] Editor's note: recall that a k-scheme X is called quasi-affine if it is isomorphic to a quasi-compact open subscheme of an affine k-scheme (EGA II, 5.1.1).

[123] Editor's note: the equivalence of these conditions is proved in the added Section 12.

(iv) G acts linearly and faithfully on a flat quasi-coherent $\mathcal{O}_S$-module.

(b) If, moreover, G is of finite type over S and S is Noetherian, these conditions imply that G is isomorphic to a closed subgroup of $\mathrm{Aut}(\mathcal{E})$, where $\mathcal{E}$ is a finite-rank locally free $\mathcal{O}_S$-module.[124]

Lemma 11.12.— *Let k be a field and let G be an affine k-group. Put $A = \mathcal{A}(G)$. Given $x \in A$, there is a finitely generated k-subalgebra B of A such that*

$$x \in B, \qquad \Delta(B) \subset B \otimes_k B, \qquad u(B) \subset B,$$

where u denotes the involution of A corresponding to the inversion morphism of G.[125]

We may suppose that $x \neq 0$. Then $\Delta(x) \neq 0$, since

$$(\varepsilon \otimes \mathrm{id})\Delta(x) = x,$$

where ε denotes the augmentation (counit) of A. Write

$$\Delta(x) = \sum_{j=1}^{n} e_j \otimes a_j, \qquad n \text{ minimal}. \tag{1}$$

The e_j (respectively the a_j) are then linearly independent. Complete $(e_1, \dots, e_n)$ to a basis $(e_i)_{i \in I}$ of A, and, for

$$j \in J = \{1, \dots, n\},$$

write

$$\Delta(e_j) = \sum_{i \in I} e_i \otimes b_{ij}.$$

Applying $\Delta \otimes \mathrm{id}$ and $\mathrm{id} \otimes \Delta$ to (1), the axiom (CO 1) of 11.8.0 (see also (HA 1) in I, 4.2) gives

$$\sum_{j \in J} e_j \otimes \Delta(a_j) = \sum_{\ell \in J} \Delta(e_\ell) \otimes a_\ell = \sum_{i \in I} e_i \otimes \left(\sum_{\ell \in J} b_{i\ell} \otimes a_\ell \right).$$

Since the e_i are linearly independent, it follows that

$$\Delta(a_j) = \sum_{\ell \in J} b_{j\ell} \otimes a_\ell \qquad (j \in J). \tag{2}$$

Let B be the finitely generated k-subalgebra of A generated by the b_{ij} and the $u(b_{ij})$, for $i, j \in J$. It is clear that $u(B) = B$.

Applying $\Delta \otimes \mathrm{id}$ and $\mathrm{id} \otimes \Delta$ to (2), we obtain once again from (CO 1) the equalities

$$\sum_{\ell \in J} \Delta(b_{j\ell}) \otimes a_\ell = \sum_{i \in J} b_{ji} \otimes \Delta(a_i) = \sum_{i, \ell \in J} b_{ji} \otimes b_{i\ell} \otimes a_\ell.$$

Since the a_ℓ are linearly independent, this yields

$$\Delta(b_{j\ell}) = \sum_{i \in J} b_{ji} \otimes b_{i\ell} \qquad (j, \ell \in J). \tag{3}$$

[124]Editor's note: this is proved, with various generalizations, in the added Section 13.

[125]Editor's note: on the one hand, this is generalized in Section 13 to the case where G is affine and flat over a regular base of dimension at most 2. On the other hand, the original has been expanded and corrected in what follows.

Moreover,

$$\Delta \circ u = (u \otimes u) \circ v \circ \Delta, \qquad v(a \otimes b) = b \otimes a,$$

and hence

$$\Delta\big(u(b_{j\ell})\big) = \sum_{i \in J} u(b_{i\ell}) \otimes u(b_{ji}) \qquad (j, \ell \in J). \tag{4}$$

Since Δ is an algebra homomorphism, (3) and (4) imply

$$\Delta(B) \subset B \otimes_k B.$$

Finally, axiom (CO 2) of 11.8.0 (see also (HA 2) in I, 4.2), applied to (2) and (1), gives

$$a_j = \sum_{i \in J} b_{ji} \varepsilon(a_i), \qquad x = \sum_{j \in J} \varepsilon(e_j) a_j.$$

Thus $x \in B$, proving the lemma.

Proposition 11.13.— *Let k be a field and let G be an affine k-group with algebra A. Then G is the projective limit of an increasing filtered system of affine k-groups of finite type whose transition morphisms are faithfully flat.*

If B and B' are two finitely generated subalgebras of A stable under Δ and u, the subalgebra generated by B and B' has the same properties. Hence, by Lemma 11.12, A is the direct limit of an increasing filtered family $(B_i)_{i \in I}$ of finitely generated subalgebras stable under Δ and u. Each B_i, equipped with the restriction of u and with the morphism

$$B_i \longrightarrow B_i \otimes_k B_i$$

induced by Δ, is a Hopf algebra. Thus, by I, 4.2, it is the algebra of an affine k-group G_i of finite type over k. Finally, since

$$A = \varinjlim_i B_i,$$

we have

$$G = \varprojlim_i G_i$$

(cf. EGA IV_3, 8.2.3). The transition morphisms are faithfully flat by the following lemma.

Lemma 11.14.— *Let k be a field, let $u : G \to H$ be a morphism between affine k-groups of finite type, and let*

$$u^\natural : B \longrightarrow A$$

be the corresponding morphism of k-algebras.[126] *Then u is faithfully flat if and only if $u^\natural$ is injective.*

The condition is plainly necessary (cf. EGA 0_{I}, 6.6.1). Conversely, suppose that $u^\natural$ is injective, and put

$$N = \mathrm{Ker}(u).$$

By VI_A, 3.3.2 and VI_A, 5.4.1, G/N is a k-group of finite type and u factors as

$$G \xrightarrow{p} G/N \xrightarrow{v} H,$$

where p is faithfully flat and v is a closed immersion. Since H is affine, G/N is affine, and

$$v^\natural : B \longrightarrow \mathcal{O}(G/N)$$

[126]Editor's note: we write $u^\natural$ (instead of u°) for the morphism $B \to A$ corresponding to $u : \mathrm{Spec}(A) \to \mathrm{Spec}(B)$.

is surjective (cf. EGA I, 4.2.3). But

$$u^\natural = p^\natural \circ v^\natural$$

is injective, so $v^\natural$ is also injective. Hence $v^\natural$, and therefore v, is an isomorphism. Since p is faithfully flat, so is u.

Definition 11.15.—[127] *Let k be a field, let G be a quasi-compact k-group, and let V be a k-vector space endowed with a k-linear action of G, hence, by 11.6.1, with an $\mathcal{A}(G)$-comodule structure*

$$\delta : V \longrightarrow V \otimes \mathcal{A}(G).$$

Let $v \in V$ be nonzero. The following conditions are equivalent:

(i) *There is a $\lambda \in \mathcal{A}(G)$, necessarily unique, such that*

$$\delta(v) = v \otimes \lambda.$$

(ii) *For every k-algebra R and every $g \in G(R)$, one has $g \cdot v \in Rv$; that is, there is a necessarily unique $f \in R$ such that, in $V \otimes R$,*

$$g \cdot v = v \otimes f.$$

It is clear that (i) implies (ii). Conversely, if (ii) holds, apply it with

$$R = \mathcal{A}(G), \qquad g = \mathrm{id}_{\mathcal{A}(G)}\,.$$

There is then a unique $\lambda \in \mathcal{A}(G)$ such that $\delta(v) = v \otimes \lambda$.

When v satisfies these conditions, we call v a *semi-invariant vector under G*, and λ the *weight* of v. We also say that v is a *semi-invariant of weight λ*.

Let Δ denote the comultiplication of $\mathcal{A}(G)$. The equality

$$v \otimes \lambda \otimes \lambda = (\delta \otimes \mathrm{id})(\delta(v)) = (\mathrm{id} \otimes \Delta)(\delta(v)) = v \otimes \Delta(\lambda)$$

implies

$$\Delta(\lambda) = \lambda \otimes \lambda.$$

Consequently, λ defines a morphism of Hopf algebras

$$\mathcal{A}(\mathbb{G}_{m,k}) = k[T, T^{-1}] \longrightarrow \mathcal{A}(G), \qquad T \longmapsto \lambda,$$

and hence a morphism of k-groups

$$\lambda : G \longrightarrow \mathbb{G}_{m,k}.$$

Thus λ is a *character* of G, called the *character associated with the semi-invariant vector v*.

Lemma 11.16.0.—[128] *Let k be a field, let H be an affine k-group, let V be an H-module of dimension n, and let U be a d-dimensional vector subspace of V. Consider the line*

$$D = \bigwedge\nolimits^d U \subset \bigwedge\nolimits^d V.$$

Then U is stable under H if and only if D is stable under H.

[127] Editor's note: we have added the hypothesis that G be quasi-compact and have spelled out the equivalence of conditions (i) and (ii).

[128] Editor's note: this lemma, taken from [DG70], Section II.2, 3.5, has been added.

Necessity is clear. To prove sufficiency, we may assume $d < n$. Let $(e_1, \ldots, e_d)$ be a basis of U, and complete it to a basis $(e_1, \ldots, e_n)$ of V. For every k-algebra R, the module

$$V_R = V \otimes R$$

is free over R, and

$$U_R = \{\, v \in V_R \mid v \wedge (e_1 \wedge \cdots \wedge e_d) = 0 \,\}.$$

Indeed, for $i > d$, the elements

$$e_i \wedge e_1 \wedge \cdots \wedge e_d$$

are linearly independent in $\bigwedge_R^{d+1} V_R$. Since $H(R)$ acts on $\bigwedge_R^{\bullet} V_R$ by

$$h(x_1 \wedge \cdots \wedge x_s) = h(x_1) \wedge \cdots \wedge h(x_s),$$

it follows that, if $H(R)$ stabilizes

$$D_R = R(e_1 \wedge \cdots \wedge e_d),$$

then it also stabilizes U_R.

Theorem 11.16 (Chevalley).— *Let k be a field, let G be an affine algebraic k-group, and let H be a closed group subscheme of G.*[129] *Then there are a finite-dimensional G-module V and a line D in V such that*

$$H = \mathrm{Norm}_G(D);$$

that is, for every k-algebra R,

$$H(R) = \{\, g \in G(R) \mid g(D_R) = D_R \,\}.$$

Equivalently, there are finitely many elements $a_1, \ldots, a_n \in \mathcal{A}(G)$, all semi-invariant of the same weight λ for the "right" action of H; that is,

$$a_i(gh) = \lambda(h) a_i(g)$$

for every k-algebra R, $g \in G(R)$, and $h \in H(R)$, such that H is the largest closed group subscheme of G under which the a_i are semi-invariant.

Proof. Let Δ and ε denote, respectively, the comultiplication and the augmentation of

$$A = \mathcal{A}(G).$$

There is an ideal I of A, contained in $\operatorname{Ker} \varepsilon$, such that

$$H = \mathrm{Spec}(A/I), \qquad \Delta(I) \subset I \otimes A + A \otimes I.$$

Put $B = A/I$, and let $\pi : A \to B$ be the projection. Consider the left action of H on A given by

$$(h\phi)(g) = \phi(gh).$$

The corresponding B-comodule structure is

$$\overline{\Delta} : \quad A \xrightarrow{\Delta} A \otimes A \xrightarrow{\mathrm{id}_A \otimes \pi} A \otimes B.$$

[129] Editor's note: first, recall that every group subscheme of G is closed (1.4.2). Second, we have stated the result in its usual form, namely that "H is the stabilizer of a line in a representation of G," while retaining the original formulation in terms of a sequence $a_1, \ldots, a_n$ of semi-invariants in $\mathcal{A}(G)$.

Then I is an H-submodule of A, since

$$\overline{\Delta}(I) \subset I \otimes B.$$

The k-algebra A is finitely generated, and hence Noetherian, so I has a finite system of generators $(x_1, \dots, x_r)$. By 11.8, the x_i are contained in a finite-dimensional G-submodule V of A. Then

$$W = V \cap I$$

is a finite-dimensional H-module; write

$$d = \dim_k W.$$

Since V contains every x_i, W generates the ideal I.

Put

$$E = \bigwedge^d V,$$

choose a basis $(w_1, \dots, w_d)$ of W, and choose a basis $(e_0, \dots, e_n)$ of E containing

$$e_0 = w_1 \wedge \cdots \wedge w_d.$$

The action of G on V canonically induces an action of G on E, hence an A-comodule structure

$$\rho : E \longrightarrow E \otimes A.$$

For $j = 0, \dots, n$, write

$$\rho(e_j) = \sum_{i=0}^{n} e_i \otimes b_{ij}.$$

As in the proof of 11.12, one has

$$\Delta(b_{ij}) = \sum_{\ell=0}^{n} b_{i\ell} \otimes b_{\ell j}. \tag{$*$}$$

Put $a_i = b_{i0}$. Equivalently, if $(e_0^*, \dots, e_n^*)$ is the dual basis, then the a_i are the matrix coefficients

$$g \longmapsto e_i^*(ge_0).$$

The action of H on E corresponds to

$$\overline{\rho} : \quad E \xrightarrow{\rho} E \otimes A \xrightarrow{\mathrm{id}_E \otimes \pi} E \otimes B.$$

Since W is stable under H, the vector e_0 is semi-invariant under H. Hence

$$\overline{\rho}(e_0) = e_0 \otimes \pi(a_0), \qquad a_i = b_{i0} \in I \quad (1 \le i \le n).$$

Substitution in $(*)$ gives

$$\overline{\Delta}(a_i) = a_i \otimes \pi(a_0) \qquad (0 \le i \le n).$$

Thus the a_i are semi-invariant under H, all of weight $\pi(a_0)$. Moreover, after replacing E, if necessary, by the G-submodule generated by e_0 (cf. 11.8), we may suppose that the a_i are linearly independent.

Conversely, let

$$H' = \operatorname{Spec} B', \qquad B' = A/I',$$

be a closed group subscheme of G under which every a_i is semi-invariant, of weight $\lambda_i \in B'$. This includes, in particular, the case in which e_0 is semi-invariant under H' of weight λ'. We shall prove that $H' \subset H$. Let $\pi' : A \to B'$ be the projection. The hypothesis gives

$$a_i \otimes \lambda_i = (\mathrm{id}_A \otimes \pi')\Delta(b_{i0}) = \sum_{\ell=0}^{n} b_{i\ell} \otimes \pi'(a_\ell).$$

It follows that

$$\lambda_i = \pi'(a_0), \qquad a_\ell \in I' \quad (1 \leq \ell \leq n),$$

and hence e_0 is semi-invariant under H'. By Lemma 11.16.0, W is stable under H'. Since W generates I, the ideal I is also stable under H', and therefore

$$\Delta(I) \subset I \otimes A + A \otimes I'.$$

Finally,

$$I \subset \operatorname{Ker} \varepsilon, \qquad (\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A,$$

so $I \subset I'$, whence $H' \subset H$. This completes the proof.

Lemma 11.17.0.—[130] *Let k be an algebraically closed field, let G be a reduced affine algebraic k-group, let V be a finite-dimensional G-module over k, and let $v \in V$. Let E be the vector subspace of V spanned by the vectors gv, for $g \in G(k)$. Then E is the smallest G-submodule of V containing v. Consequently, the morphism*

$$G \times E \longrightarrow V, \qquad (g, x) \longmapsto gx,$$

factors through E.

Proof. By 11.8, there is a smallest G-submodule U of V containing v. Indeed, if

$$\mu : V \longrightarrow V \otimes \mathcal{A}(G)$$

denotes the comodule structure and

$$\mu(v) = \sum_{i=1}^{n} v_i \otimes a_i$$

with the a_i linearly independent, then

$$U = \mathrm{Span}_k(v_1, \ldots, v_n).$$

It is clear that U contains E, and that the morphism $G \times U \to V$ factors through U.

Conversely, the inverse image of E under the morphism

$$\mu_v : G \longrightarrow V, \qquad g \longmapsto gv,$$

is a closed subset of G containing all the rational points. Those points are dense in G, since G is of finite type over k (cf. EGA IV$_3$, 10.4.8). Hence

$$\mu_v^{-1}(E) = G.$$

Since G is reduced, μ_v therefore factors through E, whence $E = U$.

[130] Editor's note: we have added this lemma, drawn from the proof of Theorem 5.6 of [DG70], Section III.3. By an abuse of notation, the same letter E denotes a k-vector space and the k-module scheme $W(E) = \operatorname{Spec} S(E^*)$.

Theorem 11.17 (Chevalley).— *Let k be a field, let G be an affine k-group* (not necessarily of finite type)*, and let N be a closed group subscheme of G, invariant in G. Then the* (fpqc) *sheaf quotient G/N is representable by an affine k-group.*[131]

Proof. Suppose first that G is of finite type. By VI$_A$, 3.2 and 5.2, the (fpqc) sheaf quotient G/N is representable by a k-group Q; it remains to prove that Q is affine. We proceed in several steps, first supposing that k is algebraically closed.[132]

(a) Suppose, in addition, that G is reduced and connected, and that N is reduced. By 11.16, there are a finite-dimensional G-module V over k and a line $D = ke_0$ such that

$$\mathrm{Norm}_G(D) = N.$$

In particular, N acts on D through a character

$$\chi : N \longrightarrow \mathbb{G}_{m,k}.$$

Fix $h \in N(k)$. For every $g \in G(k)$, one has

$$hge_0 = g(g^{-1}hg)e_0 = \chi(g^{-1}hg)\, ge_0.$$

Thus $\chi(g^{-1}hg)$ is an eigenvalue of h. The continuous map

$$\varphi : G(k) \longrightarrow k, \qquad g \longmapsto \chi(g^{-1}hg),$$

therefore takes only finitely many values. Since $G(k)$ is irreducible (it is dense in G), it follows that

$$\varphi(g) = \varphi(e) = \chi(h) \qquad \text{for every } g \in G(k).$$

Consequently,

$$\chi(g^{-1}hg) = \chi(h) \qquad \text{for all } g \in G(k),\ h \in N(k).$$

Let E be the vector subspace of V spanned by the vectors ge_0, for $g \in G(k)$. By Lemma 11.17.0, this is the G-submodule of V generated by e_0.

By the preceding identity, the two morphisms

$$N \times E \longrightarrow E, \qquad (h,x) \longmapsto hx \quad \text{and} \quad (h,x) \longmapsto \chi(h)x,$$

agree on

$$(N \times E)(k) = N(k) \times E(k),$$

which is dense in $N \times E$. Since $N \times E$ is reduced (and E is separated), the two morphisms are equal. Thus N acts on E by scalar multiplications.

It follows that N is contained in the kernel K of the morphism

$$\rho : G \longrightarrow \mathrm{GL}(\mathrm{End}_k(E))$$

defined, for every k-algebra R, by

$$\rho(g)(u) = gug^{-1} \qquad (g \in G(R),\ u \in \mathrm{End}_R(E \otimes_k R)).$$

Conversely, if $g \in K(R)$, then

$$g(Re_0) = Re_0,$$

[131] Editor's note: for another proof of this theorem that does not use the results of Exposé VI$_A$, see [Ta72], Theorem 5.2 (see also Remark 11.18.5).

[132] Editor's note: in what follows, we have expanded the original and corrected the erroneous assertion $(G/N)_{\mathrm{red}} = G_{\mathrm{red}}/N_{\mathrm{red}}$, drawing on [DG70], Section III.3, 5.6.

and hence $g \in N(R)$. Thus $N = K$. By VI$_A$, 5.4.1, the morphism

$$G/N \longrightarrow \mathrm{GL}(\mathrm{End}_k(E))$$

is a closed immersion. Therefore G/N is affine.

(b) Suppose now that G and N are reduced, without assuming that G is connected. Put

$$N' = N \cap G^0.$$

Then G^0/N' is affine by (a). Moreover, NG^0 is an invariant subgroup of G, and G/NG^0, being a quotient of the finite constant group G/G^0 (cf. VI$_A$, 5.5.1), is itself a finite constant group. The scheme G/N is therefore the disjoint union of the fibers of

$$G/N \longrightarrow G/NG^0.$$

By VI$_A$, 5.3.3, all these fibers are isomorphic to NG^0/N, hence to G^0/N'. Thus G/N is affine.

(c) Suppose that G is reduced, with N arbitrary. The morphism

$$G \times N \longrightarrow N, \qquad (g, h) \longmapsto ghg^{-1},$$

induces a morphism

$$(G \times N)_{\mathrm{red}} \longrightarrow N_{\mathrm{red}}.$$

Since G is reduced and k is algebraically closed,

$$(G \times N)_{\mathrm{red}} = G \times N_{\mathrm{red}},$$

so N_{red} is an invariant subgroup of G. (This fails when G is not reduced; see VI$_A$, 0.2.)

Hence, by (b),

$$G' = G/N_{\mathrm{red}}$$

is affine. By VI$_A$, 5.6.1,

$$N' = N/N_{\mathrm{red}}$$

is a finite k-group. Hence, by Theorem 4.1 of Exposé V,

$$G/N = G'/N'$$

is affine.

(d) For arbitrary G and N, base change of the equivalence relation

$$G \times N \rightrightarrows G$$

along $G_{\mathrm{red}} \to G$ gives

$$G_{\mathrm{red}} \times N' \rightrightarrows G_{\mathrm{red}}, \qquad N' = N \cap G_{\mathrm{red}}.$$

Since the underlying spaces are the same (and the quotient is the quotient ringed space), the morphism

$$G_{\mathrm{red}}/N' \longrightarrow G/N$$

is a homeomorphism. The quotient G_{red}/N' is reduced, because

$$p : G_{\mathrm{red}} \longrightarrow G_{\mathrm{red}}/N'$$

is faithfully flat. It follows that

$$(G/N)_{\mathrm{red}} \simeq G_{\mathrm{red}}/N'.$$

The scheme on the right is affine by (c). Since G/N is of finite type over k (cf. VI$_A$, 3.3.2), EGA I, 5.1.10 implies that G/N is affine.

Finally, let k be arbitrary and let $\overline{k}$ be an algebraic closure. By 9.2(v), one has

$$(G \otimes_k \overline{k})/(N \otimes_k \overline{k}) \simeq (G/N) \otimes_k \overline{k}.$$

The first scheme is affine, hence so is the second. Therefore G/N is affine by (fpqc) descent (cf. EGA IV$_2$, 2.7.1). This proves Theorem 11.17 when G is of finite type. To pass to the general case, we need the following lemma.[133]

Lemma 11.17.1.— *Let $(C_i \to A_i)_{i\in I}$ be a filtered direct system of ring homomorphisms, all faithfully flat. Then*

$$A = \varinjlim_i A_i$$

is faithfully flat over

$$C = \varinjlim_i C_i.$$

Proof. By [BAC], Section I.3, Proposition 9, a ring homomorphism $B \to B'$ is faithfully flat if and only if it is injective and B'/B is a flat B-module. Since each $C_i \to A_i$ is faithfully flat, there are exact sequences

$$0 \longrightarrow C_i \longrightarrow A_i \longrightarrow A_i/C_i \longrightarrow 0,$$

and A_i/C_i is a flat C_i-module. Thus

$$(A_i/C_i) \otimes_{C_i} C$$

is a flat C-module. Since direct limits are exact and commute with tensor products, we obtain an exact sequence

$$0 \longrightarrow C \longrightarrow A \longrightarrow A/C \longrightarrow 0$$

and an isomorphism

$$\varinjlim_i ((A_i/C_i) \otimes_{C_i} C) = (A/C) \otimes_C C = A/C.$$

It follows that A/C is a flat C-module. Hence A is faithfully flat over C.

We now return to the proof of Theorem 11.17 in the general case, where G is not assumed to be of finite type. Put

$$\mathcal{A}(G) = A, \qquad \mathcal{A}(N) = B = A/J.$$

By 11.13, A is the direct limit of an increasing filtered family $(A_i)_{i\in I}$ of finitely generated sub-Hopf algebras. Thus

$$G = \varprojlim_i G_i, \qquad G_i = \mathrm{Spec}(A_i),$$

where the G_i are affine algebraic k-groups. Let Δ and τ denote, respectively, the comultiplication and antipode of A, and put

$$\Delta^2 = (\Delta \otimes \mathrm{id}_A) \circ \Delta.$$

[133] Editor's note: we have added this lemma, taken from [DG70], Section III.3, 7.1.

For each i,
$$B_i = A_i/(J \cap A_i)$$
is a quotient Hopf algebra of A_i, so
$$N_i = \mathrm{Spec}(B_i)$$
is a closed subgroup of G_i. Since N is invariant in G, the morphism
$$G \times N \longrightarrow G, \qquad (g, n) \longmapsto gng^{-1},$$
factors through N. Equivalently, the pair (A, J) satisfies
$$(m_{13} \circ (\Delta \otimes \tau) \circ \Delta)(J) \subset A \otimes_k J,$$
where
$$m_{13}(a_1 \otimes a_2 \otimes a_3) = a_1 a_3 \otimes a_2.$$
It follows that $(A_i, A_i \cap J)$ satisfies the analogous property, so N_i is invariant in G_i. Moreover,
$$\varinjlim_i B_i = B, \qquad \varprojlim_i N_i = N.$$

By the finite-type case already proved, each (fpqc) sheaf quotient G_i/N_i is represented by an affine k-group
$$Q_i = \mathrm{Spec}(C_i).$$
Put
$$C = \varinjlim_i C_i.$$
The Q_i form a filtered projective system of affine k-groups, and its projective limit
$$Q = \varprojlim_i Q_i$$
is the k-group $\mathrm{Spec}(C)$ (cf. EGA IV$_3$, 8.2.3). We have an exact sequence of k-groups
$$1 \longrightarrow N \longrightarrow G \longrightarrow Q.$$

We claim that Q represents the (fpqc) sheaf quotient of G by N. It is enough to verify that
$$G \longrightarrow Q$$
is a covering for the (fpqc) topology (cf. IV, 3.3.2.1 and 5.1.7.1). Every morphism
$$G_i \longrightarrow Q_i$$
is faithfully flat (cf. 9.2(xi)); equivalently, A_i is faithfully flat over C_i. Since
$$A = \varinjlim_i A_i, \qquad C = \varinjlim_i C_i,$$
Lemma 11.17.1 shows that A is faithfully flat over C. Thus $G \to Q$ is faithfully flat. It is also affine, hence quasi-compact, and is therefore an (fpqc) covering. This completes the proof of Theorem 11.17.

11.18. Complements.—[134] Theorem 11.17 and its proof also give the following results, taken from [DG70], Section III.3.7. Let k be a field. We begin with a lemma (cf. [An73], 2.3.3.2) that will be used later (see 12.10).

Lemma 11.18.1.— *Let $u : G \to G'$ be a morphism of k-groups and put $N = \mathrm{Ker}(u)$. Suppose that G' is affine and that G is of finite type.*

[134] Editor's note: we have added this subsection.

(i) *The morphism $\tau : G/N \to G'$ is a closed immersion. In particular, G/N is affine.*

(ii) *If, moreover, the morphism $u^\natural : \mathcal{O}(G') \to \mathcal{O}(G)$ is injective, then τ is an isomorphism. Thus G' is of finite type and u is faithfully flat.*

Indeed, by 11.13, G' is the projective limit of a filtered system of affine algebraic k-groups G'_i. Let

$$u_i : G \longrightarrow G' \longrightarrow G'_i$$

be the composite, and let N_i be its kernel. The N_i form a filtered decreasing system of closed subgroups of G, whose intersection is N. Since G is Noetherian, there is an index i such that

$$N = N_i.$$

The groups G and G'_i are of finite type. Therefore, by VI$_A$, 3.2 and 5.4.1, the quotient G/N is a k-group of finite type, and u_i is the composite of the faithfully flat projection

$$p : G \longrightarrow G/N$$

and a closed immersion

$$\tau_i : G/N \hookrightarrow G'_i.$$

Consider the commutative diagram

$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "p"'] & G' \arrow[d, "q_i"] \\ G/N \arrow[r, "\tau_i"'] \arrow[ur, "\tau"] & G'_i \end{tikzcd}$$

Since

$$q_i \circ \tau = \tau_i$$

is a closed immersion and q_i is separated (G' is separated; cf. VI$_A$, 0.3), τ is a closed immersion (cf. EGA I, 5.4.4). Hence G/N is affine and

$$\tau^\natural : \mathcal{O}(G') \longrightarrow \mathcal{O}(G/N)$$

is surjective, proving (i).

If $u^\natural : \mathcal{O}(G') \to \mathcal{O}(G)$ is also injective, then so is $\tau^\natural$. It follows that $\tau^\natural$ is an isomorphism, and hence so is τ, since G' and G/N are affine. This proves (ii).

Theorem 11.18.2.— *Let G and G' be affine k-groups, with algebras A and A', and let $u : G \to G'$ be a morphism of k-groups. Put $N = \mathrm{Ker}(u)$, and let $\varphi : A' \to A$ be the morphism induced by u.*

(i) *If φ is injective, then u is faithfully flat and identifies G' with G/N.*

(ii) *One has $G/N = \mathrm{Spec}(B)$, where $B = \varphi(A')$. Thus u is the composite of the faithfully flat morphism $G \to G/N$, corresponding to the inclusion $B \hookrightarrow A$, and the closed immersion $G/N \hookrightarrow G'$, corresponding to the surjection $A' \to B$. Moreover, N is defined in G by the ideal AB_+, where B_+ denotes the augmentation ideal of B.*

(iii) *In particular, if u is a monomorphism, it is a closed immersion.*

Proof. (i) Suppose that φ is injective and identify A' with a sub-Hopf algebra of A. By 11.13, A is the filtered union of finitely generated sub-Hopf algebras

$$A_i = \mathcal{O}(G_i).$$

Put

$$A'_i = A' \cap A_i, \qquad G'_i = \mathrm{Spec}(A'_i),$$

and let N_i be the kernel of the morphism

$$G_i \longrightarrow G'_i$$

induced by $A'_i \hookrightarrow A_i$. By Lemma 11.18.1,

$$G_i/N_i \simeq G'_i.$$

Thus, for each i, there is an exact sequence

$$1 \longrightarrow N_i \longrightarrow G_i \xrightarrow{p_i} G'_i \longrightarrow 1,$$

where p_i is faithfully flat. Since

$$G = \varprojlim_i G_i, \qquad G' = \varprojlim_i G'_i,$$

we obtain an exact sequence

$$1 \longrightarrow N \longrightarrow G \xrightarrow{p} G',$$

where $N = \varprojlim_i N_i$. By Lemma 11.17.1, p is faithfully flat (and affine). Therefore G' represents the (fpqc) sheaf quotient of G by N, proving (i).

In general, $B = \varphi(A')$ is a sub-Hopf algebra of A. Let

$$H = \mathrm{Spec}(B),$$

and let N' be the kernel of the morphism $G \to H$ induced by $B \hookrightarrow A$. By (i), H is identified with G/N', and u is the composite of the projection $G \to G/N'$ and the closed immersion

$$G/N' \hookrightarrow G'$$

induced by the surjection $A' \to B$. Hence $N' = N$. Moreover, by 9.2(ii), the square

$$\begin{tikzcd} N \arrow[r] \arrow[d] & G \arrow[d, "p"] \\ \mathrm{Spec}(k) \arrow[r, "\varepsilon"'] & G' \end{tikzcd}$$

is cartesian. Here ε is the unit section of G', corresponding to the augmentation $B \to k$. It follows that N is defined in G by the ideal AB_+. This proves (ii), and (iii) follows.

Remark 11.18.3.— Let G be an affine k-group and N an invariant k-subgroup. Since

$$p : G \longrightarrow G/N$$

is faithfully flat and quasi-compact, $\mathcal{O}(G/N)$ is, by IV, 3.3.3.2, the subalgebra of $\mathcal{O}(G)$ formed by the functions ϕ that are right N-invariant; that is,

$$\phi(gh) = \phi(g)$$

for every k-scheme S, every $g \in G(S)$, and every $h \in N(S)$. Let J be the ideal of

$$A = \mathcal{O}(G)$$

defining N. The preceding condition is equivalent to

$$\Delta(\phi) - \phi \otimes 1 \in \mathcal{O}(G) \otimes J,$$

where Δ is the comultiplication of A.

The preceding theorem may therefore be reformulated in terms of Hopf algebras as follows.

Corollary 11.18.4.— *Let k be a field, let A be a commutative k-Hopf algebra, and put $G = \operatorname{Spec}(A)$.*

(i) *If B is a sub-Hopf algebra of A, then A is faithfully flat over B.*

(ii) *The map*
$$N \longmapsto \mathcal{O}(G/N)$$
is a bijection from the set of invariant subgroups of G to the set of sub-Hopf algebras of A. Its inverse is
$$B \longmapsto \operatorname{Spec}(A/AB_+).$$
Moreover, if J is the ideal of A defining N, then
$$\mathcal{O}(G/N) = \{\, x \in A \mid \Delta(x) - x \otimes 1 \in A \otimes J \,\}.$$

Remarks 11.18.5.—

(a) One consequence of the preceding theorem is that the category of commutative affine k-groups is abelian. For this, as well as for other results on affine k-groups, we refer to [DG70], § III.3, 7.4–7.8.

(b) Finally, let us point out that M. Takeuchi gave another proof of the results of 11.17–11.18.4; see [Ta72], § 5. He moreover strengthened 11.18.4(i) above by showing that A is even a projective B-module; see [Ta79], Theorem 5 (see also [MW94], Theorem 3.6).

12. Complements on G_{af} and the "anti-affine" groups

[135] We begin with the following lemma, which extends 11.18.1 to the case where G is not assumed to be of finite type.[136]

Lemma 12.1.— *Let k be a field and let $u : G \to H$ be a monomorphism of k-groups, with H affine. Suppose that u is quasi-compact. Then u is a closed immersion.*

By (fpqc) descent, we may suppose that k is algebraically closed. By VI_A, 6.4, the closed image I of u is a closed group subscheme of H, hence is still affine. Replacing H by I, we may therefore suppose that u is schematically dominant. Since k is algebraically closed, $H' = H_{\mathrm{red}}$ is a group subscheme of H. Put $G' = G \times_H H'$. The morphism

$$u' : G' \longrightarrow H'$$

[135] Editor's note: this section has been added.

[136] Editor's note: this lemma was communicated to us by M. Raynaud; it will be used in the proof of Proposition 12.9.

deduced from u by base change is a quasi-compact monomorphism and is dominant (the underlying continuous map being the same for u and u'). Hence, by VI$_A$, 6.2, u' is faithfully flat. It is therefore a quasi-compact faithfully flat monomorphism, and thus an isomorphism (cf. IV, 1.14).

Consequently $u : G \to H$ is a homeomorphism, and hence is affine by 2.9.1. Thus G is affine, and u is a closed immersion by 11.18.2(iii).

Theorem 12.2.— *Let G be an algebraic k-group. Denote by $\rho : G \to G_{\mathrm{af}}$ the canonical morphism and by N its kernel.*

(i) *The canonical morphism $G/N \to G_{\mathrm{af}}$ is an isomorphism. Consequently G_{af} is an affine algebraic group and ρ is faithfully flat.*

(ii) *There is a canonical isomorphism $(G/N)_{\mathrm{af}} = G_{\mathrm{af}}$.*

(iii) *N is a characteristic subgroup of G.*

(iv) *$\mathcal{O}(N) = k$.*

(v) *N is smooth, connected, and commutative.*

Assertion (i) is a special case of 11.18.1, and (ii) follows from the universal properties of G_{af} and $(G/N)_{\mathrm{af}}$.

Let us prove (iii). For an arbitrary k-scheme $\pi : S \to \operatorname{Spec} k$, consider the cartesian square

$$\begin{tikzcd}
G_S \arrow[r] \arrow[d, "q"'] & G \arrow[d, "p"] \\
S \arrow[r, "\pi"'] & \operatorname{Spec} k
\end{tikzcd}$$

Since p is quasi-compact and separated and π is flat, EGA III, 1.4.15 and EGA IV$_1$, 1.7.21 give

$$q_*(\mathcal{O}_{G_S}) = \pi^*(\mathcal{O}(G)) = \pi^*(\mathcal{O}(G_{\mathrm{af}})).$$

Hence EGA II, 1.5.2 gives

$$(G_S)_{\mathrm{af}} = (G_{\mathrm{af}})_S.$$

Thus N_S, being the kernel of the canonical morphism $G_S \to (G_S)_{\mathrm{af}}$, is invariant under every automorphism of G_S; in other words, N is a characteristic subgroup of G.

To prove (iv), put

$$N' = \operatorname{Ker}(N \longrightarrow N_{\mathrm{af}}).$$

By (ii), this is an invariant subgroup of G. Since N is algebraic (being a closed subgroup of G), (i) gives $N/N' \simeq N_{\mathrm{af}}$. Moreover, by VI$_A$, 3.2 and 5.3.2, there is an isomorphism of k-groups

$$(G/N')/N_{\mathrm{af}} \simeq (G/N')/(N/N') \simeq G/N.$$

Since N_{af} is affine, the projection $G/N' \to G/N$ is affine by 9.2(vii); and since $G/N = G_{\mathrm{af}}$ is affine, G/N' is affine as well. The universal property of G_{af} therefore makes the projection

$$p' : G \longrightarrow G/N'$$

factor through $G_{\mathrm{af}} = G/N$. Hence $N \subset N'$, and consequently $N = N'$. Thus N_{af} is the trivial group, and $\mathcal{O}(N) = k$.

Finally, (v) follows from the next lemma.

Lemma 12.3.— *Let k be a field and let N be an algebraic k-group such that $\mathcal{O}(N) = k$. Then N is smooth, connected, and commutative.*

Indeed, we may suppose that k is algebraically closed. Then

$$H = N^0_{\mathrm{red}}$$

is a k-subgroup of N, and the quotient k-scheme $X = N/H$ is finite (hence affine) over k, by VI$_A$, 5.5.1 and 5.6.1. Since $p : N \to X$ is faithfully flat,

$$\mathcal{O}(X) \subset \mathcal{O}(N) = k.$$

It follows that $N = N^0_{\mathrm{red}}$, and hence that N is smooth (VI$_A$, 1.3.1) and connected.

Let Z be the center of N. By 6.2.6, N/Z is affine, and the preceding argument gives $\mathcal{O}(N/Z) = k$. Hence $N = Z$. This proves the lemma and completes the proof of 12.2.

We also record, without proof, the following theorem. Recall that an *abelian variety* over a field k is a proper, smooth, connected k-group scheme.

Theorem 12.4 (Chevalley).— *Let k be a perfect field and let G be a smooth, connected algebraic k-group. Then there exists a smooth, connected affine k-subgroup L, invariant in G, such that G/L is an abelian variety. Moreover, L is unique and its formation commutes with extension of the base field.*

Remarks 12.5.—

(1) This theorem was announced in 1953 by C. Chevalley, who published his proof in 1960 ([Ch60]). Meanwhile, other proofs were obtained independently by I. Barsotti and M. Rosenlicht ([Ba55, Ro56]); see [Se99] for historical comments.

(2) A modern version (that is, in the language of schemes) of Chevalley's proof was given by B. Conrad ([Co02]). (In *loc. cit.*, "algebraic group" means a smooth connected k-group scheme.)

(3) A modern version of Rosenlicht's proof, on the other hand, was given by Ngô B.-C. in a course at Orsay in 2005–2006.

(4) If the hypothesis that k be perfect is omitted, there still exists a smallest invariant connected affine subgroup L, not necessarily smooth, such that G/L is an abelian variety ([BLR], § 9.2, Theorem 1).

(5) One may also omit the hypothesis that G be smooth over k. Indeed, by VII$_A$, 8.3, there is an integer $n \geq 1$ such that the quotient

$$G' = G/(\mathrm{Fr}^n\, G)$$

is smooth. Then G' contains a subgroup L' as in (4) above, and the inverse image of L' in G again has the same properties. Thus, for every connected algebraic group G over a field k, there is an exact sequence

$$1 \longrightarrow H \longrightarrow G \longrightarrow A \longrightarrow 1,$$

where H is an affine k-group and A is an abelian k-variety. Moreover, by [Per76], Corollary 4.2.9, such an exact sequence exists for every connected k-group G, not necessarily algebraic.

(6) Let k be an algebraically closed field and let G be the semidirect product of an elliptic curve E by the constant k-group $\{\pm 1\}_k$, for the action defined by $(-1) \cdot x = -x$. In this case, if L is a closed invariant subgroup of G such that G/L is connected, then $L = G$.

Remark 12.6.— Following [Br09], a k-group N will be called *anti-affine* if $\mathcal{O}(N) = k$. By 12.3 and 12.4, if k is perfect, every anti-affine algebraic k-group is an extension of an abelian variety by a smooth, connected, commutative affine algebraic k-group. For the precise structure of anti-affine algebraic groups over a perfect field, and various consequences, see the recent articles of M. Brion and C. & F. Sancho de Salas ([Br09, SS09]).

To conclude this section, we shall prove two results due to M. Raynaud, the first being Remark 11.11.1, and the second Proposition 2.1 of Exposé XVII, Appendix III. We shall need the following lemma,[137] which improves, for a complete discrete valuation ring R, the flatness criteria given in [BAC], § III.5.

Lemma 12.7.— *Let R be a discrete valuation ring, K its fraction field, and π a uniformizer. Let A be a flat R-algebra and let M be an A-module flat over R. Suppose that*

(i) *$M/\pi M$ is flat over $\overline{A} = A/\pi A$,*

(ii) *$M \otimes_R K$ is flat over $A \otimes_R K$.*

Then M is flat over A.

By the flatness criterion in the nilpotent case (cf. [BAC], III § 5.2, Theorem 1),

$$M/\pi^n M$$

is flat over $A/\pi^n A$ for every $n \in \mathbb{N}^*$. It then follows from [RG71], II, Lemma 1.4.2.1, that M is flat over A. For the reader's convenience, let us briefly indicate the proof. Put

$$s = \pi^n, \qquad P = M/sM.$$

Since P is flat over A/sA, and A/sA has projective dimension 1 over A, the spectral sequence of composite functors shows that P has Tor-dimension at most 1 over A. Since M is flat over R, and thus has no π-torsion, M_K/M is the direct limit of the A-modules

$$\pi^{-n}M/M \simeq M/\pi^n M.$$

Consequently M_K/M also has Tor-dimension at most 1. From the exact sequence

$$0 \longrightarrow M \longrightarrow M_K \longrightarrow M_K/M \longrightarrow 0$$

and the hypothesis that M_K is flat over A_K, hence over A, it follows that M is flat.

For completeness, let us also give the following simpler proof, pointed out by O. Gabber. Let I be a finitely generated ideal of A. We must show that

$$u : M \otimes_A I \longrightarrow M$$

is injective. By (ii), $u \otimes_R K$ is injective, so $\operatorname{Ker}(u)$ is an R-module of π-torsion. It therefore suffices to show that the π-torsion submodule of $M \otimes_A I$ is zero. This submodule is a quotient of $\operatorname{Tor}_1^A(M, I/\pi I)$, as is seen by tensoring with M the exact sequence

$$0 \longrightarrow I \xrightarrow{\pi} I \longrightarrow I/\pi I \longrightarrow 0.$$

[137] Editor's note: this is a version improved by O. Gabber of a statement communicated by M. Raynaud.

On the other hand, since M has no π-torsion (being flat over R), one has

$$\mathrm{Tor}_i^A(M, \overline{A}) = 0 \qquad (i \geq 1).$$

Consequently, if $P_\bullet$ is a projective resolution of the A-module M, then

$$P_\bullet \otimes_A \overline{A}$$

is a projective resolution of the $\overline{A}$-module $\overline{M} = M/\pi M$. Thus, for every $\overline{A}$-module Q,

$$\mathrm{Tor}_i^A(M, Q) = \mathrm{Tor}_i^{\overline{A}}(\overline{M}, Q),$$

which vanishes for $i \geq 1$, since $\overline{M}$ is flat over $\overline{A}$. Hence $\mathrm{Tor}_1^A(M, I/\pi I) = 0$, proving the lemma.

Remark 12.8.— Let S be a regular locally noetherian scheme of dimension 1, and let X be an S-scheme that is flat, quasi-separated, and quasi-compact over S. Then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$-module. Indeed, we may suppose that S is local. Let s be its closed point, let $i : X_s \hookrightarrow X$ be the inclusion, and let π be a uniformizer of $R = \mathcal{O}(S)$. Since X is flat over S, there is an exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\pi} \mathcal{O}_X \longrightarrow i_*(\mathcal{O}_{X_s}) \longrightarrow 0.$$

Taking global sections shows that $\mathcal{A}(X)$ has no π-torsion as an R-module, and hence is flat.[138] Moreover, the morphism

$$\mathcal{A}((X_{\mathrm{af}})_s) = \mathcal{A}(X)/\pi\mathcal{A}(X) \longrightarrow \mathcal{A}(X_s),$$

induced by $X \to X_{\mathrm{af}}$, is injective.

We can now prove the following proposition (cf. Remark 11.11.1).

Proposition 12.9.— *Let S be a regular locally noetherian scheme of dimension at most 1, and let G be an S-group scheme that is flat, quasi-separated, and quasi-compact over S. Then the following conditions are equivalent:*

(i) *G is affine over S.*

(ii) *G is quasi-affine over S.*

(iii) *G acts faithfully on a quasi-affine and flat S-scheme X.*

(iv) *G acts linearly and faithfully on a quasi-coherent module $\mathcal{E}$ that is flat over S.*

(v) *The morphism $\rho : G \to G_{\mathrm{af}}$ is a monomorphism.*

The implication (i)$\Rightarrow$ (ii) is clear, as is (ii)$\Rightarrow$ (iii) (take $X = G$).

Suppose (iii) holds. Since $\mathcal{A}(G)$ and $\mathcal{A}(X)$ are flat $\mathcal{O}_S$-modules, the argument of 11.11 shows that G acts faithfully on the right on X_{af}, and hence acts linearly and faithfully on the left on the flat quasi-coherent $\mathcal{O}_S$-module $\mathcal{A}(X)$.

If (iv) holds, 11.6.1(ii) shows that the monomorphism

$$G \longrightarrow \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{E})$$

factors through G_{af}. Hence $G \to G_{\mathrm{af}}$ is a monomorphism.

[138] Editor's note: this is true more generally when S is regular, locally noetherian, and of dimension at most 2; see [Ray70a], VII 3.2.

Finally, suppose (v) holds, and let us prove that $\rho : G \to G_{\mathrm{af}}$ is an isomorphism. Replacing S by one of its connected components, we may suppose that S is irreducible, with generic point η. Since the formation of G_{af} commutes with flat base change,

$$(G_{\mathrm{af}})_\eta = (G_\eta)_{\mathrm{af}}.$$

Therefore $G_\eta \to (G_\eta)_{\mathrm{af}}$ is a monomorphism, hence a closed immersion by 12.1, and consequently an isomorphism because

$$\mathscr{O}((G_\eta)_{\mathrm{af}}) = \mathscr{O}(G_\eta).$$

If $S = \mathrm{Spec}(\kappa(\eta))$, the proof is complete; we may therefore suppose that $\dim S = 1$.

Let s be a closed point of S. We shall show that

$$G_s \longrightarrow (G_{\mathrm{af}})_s$$

is an isomorphism and that ρ is flat at every point of G_s. We may suppose that S is local, with closed point s. The base-change morphism

$$\rho_s : G_s \longrightarrow (G_{\mathrm{af}})_s$$

is a monomorphism, hence a closed immersion by 12.1. Thus the morphism

$$\mathscr{O}((G_{\mathrm{af}})_s) \longrightarrow \mathscr{O}(G_s)$$

induced by ρ_s is surjective. By the preceding remark it is also injective, and hence is an isomorphism. In particular, ρ is surjective.

It follows from Lemma 12.7 that $\rho : G \to G_{\mathrm{af}}$ is faithfully flat. Since G is quasi-compact over S and G_{af} is separated over S, ρ is also quasi-compact (cf. EGA I, 6.6.4). Therefore ρ is a quasi-compact faithfully flat monomorphism, and is thus an isomorphism. This proves the proposition.

Finally, we prove Proposition 2.1 of Exposé XVII, Appendix III. Taking [Per76], Corollary 4.2.5 into account, we have replaced "of finite type" in the hypotheses by "quasi-compact and quasi-separated". (If G is assumed to be of finite type, 12.1 may be used in place of *loc. cit.*)

Proposition 12.10.— *Let S be a regular locally noetherian scheme of dimension at most* 1*, and let G be a flat, quasi-compact, and quasi-separated S-group scheme.*

(i) *The canonical morphism $\rho : G \to G_{\mathrm{af}}$ is faithfully flat and quasi-compact. Consequently, G_{af} represents the* (fpqc) *sheaf quotient of G by $N = \mathrm{Ker}(\rho)$.*

(ii) *If moreover G is of finite type over S, then ρ is of finite presentation, G_{af} represents the* (fppf) *sheaf quotient of G by N, and G_{af} is of finite type over S.*

(iii) *Suppose moreover that G_η is affine for every maximal point η of S. Then N is an étale S-group and is the unit group if G is separated over S.*

Since G_{af} is affine, and therefore separated over S, the morphism ρ is quasi-compact (cf. EGA I, 6.6.4), and its kernel

$$N = \mathrm{Ker}(\rho)$$

is a closed subgroup of G. Replacing S by one of its connected components, we may suppose that S is irreducible, with generic point η.

To prove (i) and (ii), it is enough to show that ρ is faithfully flat. Indeed, Exposé IV, 5.1.7.1 then shows that G_{af} represents the (fpqc) sheaf quotient of G by N. If G is moreover of finite type over S, it is of finite presentation because S is locally noetherian; by 9.2(xiii), ρ is then of finite presentation, as is $G_{\mathrm{af}} \to S$, and ρ is therefore a covering for the (fppf) topology.

We may thus suppose that

$$S = \mathrm{Spec}(R),$$

where R is a discrete valuation ring if $\dim S = 1$, and is the field $\kappa(\eta)$ if $\dim S = 0$. Let s be the closed point of S. Since formation of G_{af} commutes with flat base change, the canonical morphism $G_\eta \to (G_\eta)_{\mathrm{af}}$ is identified with

$$\rho_\eta : G_\eta \longrightarrow (G_{\mathrm{af}})_\eta.$$

Since

$$\mathcal{O}((G_{\mathrm{af}})_\eta) = \mathcal{O}((G_\eta)_{\mathrm{af}}) = \mathcal{O}(G_\eta),$$

ρ_η is faithfully flat by [Per76], 4.2.5 (see also the addition VI$_A$, 6.6). If $\dim S = 1$, the same argument shows that ρ_s is faithfully flat, because the morphism

$$\mathcal{O}((G_{\mathrm{af}})_s) \longrightarrow \mathcal{O}(G_s)$$

is injective by Remark 12.8. Lemma 12.7 now shows that ρ is faithfully flat. This proves (i) and (ii). In particular, N is flat over S.

Suppose now that G_η is affine. Since ρ_η coincides with the canonical morphism

$$G_\eta \longrightarrow (G_\eta)_{\mathrm{af}},$$

its kernel N_η is the unit group. Let us prove that N is étale over S. Since N is flat over S, it remains to show that N_s is étale over $\kappa(s)$ for every point $s \neq \eta$ of S. There is nothing to prove when $S = \mathrm{Spec}(\kappa(\eta))$, so we may suppose that $S = \mathrm{Spec}(R)$, where R is a discrete valuation ring. Let s be the closed point of S, let K be the fraction field of R, and let π be a uniformizer.

Let $x \in N_s$, and let U_x be an affine open neighborhood of x in N. Since N is flat over S, the intersection $U_x \cap N_\eta$ is nonempty, and hence equals $\{\epsilon(\eta)\}$, where ϵ denotes the unit section. Thus

$$A = \mathcal{O}(U_x)$$

is a flat R-algebra such that $A_K = K$ and $\pi^{-1} \notin A$, since π belongs to the maximal ideal of $\mathcal{O}_{U_x,x}$. It follows that $A = R$, and hence the projection $U_x \to S$ is an isomorphism. Thus N is étale over S. If G is also separated over S, the inverse isomorphism $S \to U_x$ is the unit section, since the two coincide on the dense open subset $\{\eta\}$ of $S = \mathrm{Spec}(R)$. Hence N is the unit group. This proves the proposition.

In particular, we obtain the following corollary, two other proofs of which may be found in [An73], Proposition 2.3.1, and [PY06], Proposition 3.1.

Corollary 12.10.1.— *Let R be a discrete valuation ring, K its fraction field, and G a separated, flat R-group scheme of finite type over R. If G_K is affine, then G is affine.*

Remarks 12.10.2.—

(a) O. Gabber has pointed out to us examples in which G is a flat group of finite type over a discrete valuation ring, its generic fiber is an abelian variety, and the kernel N of $G \to G_{\mathrm{af}}$ is not smooth.

(b) M. Raynaud, on the other hand, gave an example, with S the affine plane of dimension 2 over a field k, of a smooth quasi-affine S-group scheme with affine, connected fibers that is not affine over S; see [Ray70a], § VII.3, p. 116.

13. Flat affine groups over a regular base of dimension at most 2

Let us begin by observing[139] that the familiar argument showing that every affine algebraic group over a field k is linear, as well as Lemma 11.12, extends to a group scheme G that is affine, flat, and of finite type over a Noetherian regular base scheme S of dimension at most 2. For S of dimension at most 1, this proves Remark 11.11.1(b). The extension to $\dim S = 2$ rests on the following lemma, communicated to us by O. Gabber.

Lemma 13.1.— *Let S be a normal Noetherian scheme, let $\mathcal{A}$ be a flat quasi-coherent $\mathcal{O}_S$-module, let $\mathcal{F}$ be a finite-type $\mathcal{O}_S$-submodule of $\mathcal{A}$, let $\mathcal{F}^{**}$ be its bidual, and let U be the flatness locus of $\mathcal{F}$, that is, the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$-module.*

(i) *U is open in S, and*
$$\mathcal{F}^{**} = j_*j^*(\mathcal{F}),$$
where $j : U \hookrightarrow S$ denotes the inclusion.

(ii) *For every quasi-coherent $\mathcal{O}_U$-module $\mathcal{E}$, the canonical morphism*
$$j_*(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow j_*(\mathcal{E} \otimes_{\mathcal{O}_U} j^*(\mathcal{A}))$$
is an isomorphism.

(iii) *In particular,*
$$\mathcal{V} = j_*j^*(\mathcal{F}) \subset \mathcal{A} = j_*j^*(\mathcal{A}),$$
and the canonical morphism
$$\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow j_*j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A})$$
is an isomorphism.

Proof. Replacing S by one of its connected components, we may suppose that S is integral. By EGA IV_2, 2.1.12, the flatness locus of $\mathcal{F}$, namely the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$-module, is an open subset U of S; write $j : U \hookrightarrow S$ for its inclusion. Since $\mathcal{A}_s$ is flat, hence torsion-free, the same is true of $\mathcal{F}_s$. Thus U contains every point of codimension at most 1. Consequently, by [BAC], VII, § 4.2, Corollary to Theorem 1,
$$\mathcal{F}^{**} = j_*j^*(\mathcal{F}),$$
and hence there is a monomorphism
$$\mathcal{F}^{**} \longrightarrow j_*j^*(\mathcal{A}).$$

The proof of (ii) is analogous to that of EGA III, 1.4.15, recalled in 11.0. Since S is normal, the morphism
$$\mathcal{O}_S \longrightarrow j_*j^*(\mathcal{O}_S)$$
is an isomorphism (cf. EGA IV_2, 5.8.6 and 5.10.5). Applying (ii) to $\mathcal{E} = \mathcal{O}_U$ therefore gives
$$\mathcal{A} = j_*j^*(\mathcal{A}).$$

Finally, since
$$j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}) = j^*(\mathcal{F}) \otimes_{\mathcal{O}_U} j^*(\mathcal{A}),$$

[139] Editor's note: this section has been added.

the last assertion of (iii) follows from (ii) applied to $\mathcal{E} = j^*(\mathcal{F})$. This proves the lemma.

Recall also that a finitely generated R-module M is called *reflexive* if the canonical morphism from M to its bidual M^{**} is an isomorphism. If R is a Noetherian regular ring of dimension at most 2, this implies that M is projective. Indeed, for every finitely generated R-module N, choose a resolution

$$L_1 \longrightarrow L_0 \longrightarrow N \longrightarrow 0,$$

where L_0 and L_1 are finite free R-modules. There is then an exact sequence

$$0 \longrightarrow N^* \longrightarrow L_0^* \longrightarrow L_1^* \longrightarrow Q \longrightarrow 0,$$

where Q is the cokernel of $L_0^* \to L_1^*$. Since R has homological dimension at most 2 ([BAC], X, § 4.2, Corollary 1 to Theorem 1), it follows that N^* is projective.

Proposition 13.2.— *Let S be a Noetherian regular scheme of dimension at most* 2, *let G be an affine flat S-group, and let $\mathcal{A}(G)$ be its affine algebra.*

(i) *If G is of finite type over S, it is isomorphic to a closed subgroup of*

$$H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

for some locally free $\mathcal{O}_S$-module $\mathcal{V}$ of finite rank. If, moreover, S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d, so that $H = \mathrm{GL}_{d,S}$.

(ii) *$\mathcal{A}(G)$ is the filtered direct limit of flat finite-type $\mathcal{O}_S$-sub-Hopf algebras.*

Proof. Let $\mathcal{B}$ be a finite-type $\mathcal{O}_S$-subalgebra of $\mathcal{A}(G)$. Since every coherent module on an open subset of S extends to a coherent module on S (EGA I, 9.4.5), there is a coherent $\mathcal{O}_S$-submodule $\mathcal{M}$ of $\mathcal{B}$ that generates $\mathcal{B}$ as an $\mathcal{O}_S$-algebra (*loc. cit.*, 9.6.5). By 11.10.bis, $\mathcal{M}$ is contained in a coherent G-stable $\mathcal{O}_S$-submodule $\mathcal{F}$. (N.B. Since G is affine over S, the proof of *loc. cit.* is simpler here: one may replace $f_*f^*(\mathcal{E})$ by $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$, and so on.)

Let $j : U \hookrightarrow S$ be the inclusion, where U is the flatness locus of $\mathcal{F}$. By Lemma 13.1 and the preceding observation,

$$\mathcal{V} = j_*j^*(\mathcal{F})$$

is a locally free $\mathcal{O}_S$-submodule of $\mathcal{A}(G)$, and the canonical morphism

$$\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow j_*j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G))$$

is an isomorphism. Thus $\mathcal{V}$ is a G-submodule of $\mathcal{A}(G)$. The action of G on $\mathcal{V}$ induces a morphism of affine S-groups

$$\rho_{\mathcal{V}} : G \longrightarrow H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

and therefore a morphism of $\mathcal{O}_S$-Hopf algebras

$$\varphi_{\mathcal{V}} : \mathcal{A}(H) \longrightarrow \mathcal{A}(G).$$

Write $\mathcal{A}_{\mathcal{V}}$ for the image of $\varphi_{\mathcal{V}}$. It is the affine algebra of a closed subgroup $G_{\mathcal{V}}$ of H, the closed image of $\rho_{\mathcal{V}}$. We shall show that $\mathcal{A}_{\mathcal{V}}$ contains $\mathcal{B}$.

The question is local on S, so we may suppose that $S = \mathrm{Spec}(R)$ and that

$$V = \Gamma(S, \mathcal{V})$$

is a free R-module with basis $v_1, \dots, v_n$. Then $H \simeq \mathrm{GL}_{n,R}$, and $\mathcal{A}(H)$ is generated as an R-algebra by the "matrix coefficients" c_{ij} and by d^{-1}, where d denotes the determinant. Let Δ (respectively, ε) denote the comultiplication (respectively, augmentation) of $A = \Gamma(S, \mathcal{A}(G))$. For $j = 1, \dots, n$, write

$$\Delta(v_j) = \sum_{i=1}^{n} v_i \otimes a_{ij}.$$

Then $a_{ij} = \varphi(c_{ij})$ belongs to $\mathrm{Im}(\varphi)$. Since V is an R-submodule of A, the identity

$$(\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A$$

gives

$$v_j = \sum_i \varepsilon(v_i) a_{ij} \in \mathrm{Im}(\varphi).$$

As V contains a set of generators of $B = \Gamma(S, \mathcal{B})$, it follows that

$$B \subset \mathrm{Im}(\varphi).$$

If G is of finite type over S, take $\mathcal{B} = \mathcal{A}(G)$. Then φ is surjective, so the morphism

$$G \longrightarrow H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

is a closed immersion.

If S is also affine, there is a locally free $\mathcal{O}_S$-module $\mathcal{V}'$ of finite rank such that

$$\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^{\oplus d}.$$

Regard $\mathcal{V}'$ as a trivial G-module and replace $\mathcal{V}$ by $\mathcal{O}_S^{\oplus d}$. This realizes G as a closed subgroup of $\mathrm{GL}_{d,S}$.

Finally, return to an arbitrary affine flat S-group G. By EGA I, 9.4.9, $\mathcal{A}(G)$ is the union of its coherent $\mathcal{O}_S$-submodules $\mathcal{M}$, hence also of the $\mathcal{O}_S$-sub-Hopf algebras $\mathcal{A}_{\mathcal{V}}$ constructed above. This proves (ii).

Example 13.3.— *Let R be a discrete valuation ring with uniformizer π and fraction field K. Consider the filtered projective system of R-groups*

$$\cdots \longrightarrow \mathbb{G}_{a,R} \xrightarrow{\times\pi} \mathbb{G}_{a,R} \xrightarrow{\times\pi} \mathbb{G}_{a,R}$$

corresponding to the direct system

$$R[X_0] \longrightarrow R[X_1] \longrightarrow R[X_2] \longrightarrow \cdots,$$

whose transition morphisms are given by $X_n = \pi X_{n+1}$. Its projective limit G is a flat affine R-group scheme, not of finite type, with trivial special fiber and generic fiber $\mathbb{G}_{a,K}$. The affine R-algebra $\mathcal{A}(G)$ is the subring of $K[X]$ consisting of the polynomials whose constant coefficient lies in R. (N.B. We already encountered this example in Remark 11.10.1.) *Furthermore, G represents the functor assigning to every R-algebra B the set of sequences $(x_n)_{n\in\mathbb{N}}$ of elements of B such that*

$$x_n = \pi x_{n+1} \qquad \text{for every } n.$$

(In particular, every x_n is indefinitely π-divisible.)

Now let S be a Noetherian scheme such that every coherent $\mathcal{O}_S$-module is a quotient of a locally free $\mathcal{O}_S$-module of finite type. This holds, for example, when S is a separated

Noetherian regular scheme; see SGA 6, Exposé II, 2.1.1 and 2.2.7. (One can show that every Noetherian regular scheme of dimension at most 1 also has this property. It does not hold, however, when S is the affine plane over a field k with the origin doubled; see *loc. cit.*, 2.2.7.2.)

Definition 13.4.— *Let G be an affine flat S-group. Following R. W. Thomason* ([Th87], 2.1)*, we say that the pair (G, S) has the* equivariant resolution property*, or satisfies* (RE)*, if, for every coherent G-$\mathcal{O}_S$-module $\mathcal{F}$, there are a locally free G-$\mathcal{O}_S$-module $\mathcal{E}$ of finite rank and a G-equivariant epimorphism*

$$\mathcal{E} \longrightarrow \mathcal{F}.$$

In *loc. cit.*, Theorem 3.1, Thomason proves the result below under the hypothesis that G be essentially free over S (see the remark below). Gabber pointed out to us the simpler proof given here, which does not use that hypothesis.

Proposition 13.5.— *Let S be a Noetherian scheme, and let G be an affine S-group, flat and of finite type, with affine algebra $\mathcal{A}(G)$. Suppose that (G, S) satisfies* (RE)*. Then:*

(i) *G is isomorphic to a closed subgroup of*

$$H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

for some locally free $\mathcal{O}_S$-module $\mathcal{V}$ of finite rank.

(ii) *If, moreover, S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d, so that $H = \mathrm{GL}_{d,S}$.*

The proof is analogous to that of 13.2. As there, there is a coherent G-stable $\mathcal{O}_S$-submodule $\mathcal{F}$ that generates $\mathcal{A} = \mathcal{A}(G)$ as an $\mathcal{O}_S$-algebra. Replacing S by one of its connected components, we may suppose that S is connected. By (RE), there are a locally free G-$\mathcal{O}_S$-module $\mathcal{E}$ of rank n and an epimorphism of $\mathcal{A}$-comodules

$$\pi : \mathcal{E} \longrightarrow \mathcal{F}.$$

Put $H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{E})$. This is an S-group scheme locally isomorphic to $\mathrm{GL}_{n,S}$. The action of G on $\mathcal{E}$ induces a morphism of affine S-groups

$$\rho : G \longrightarrow H,$$

corresponding to a morphism of $\mathcal{O}_S$-Hopf algebras

$$\varphi : \mathcal{A}(H) \longrightarrow \mathcal{A}(G).$$

We shall show that ρ is a closed immersion.

The question is local on S, so suppose that $S = \mathrm{Spec}(R)$ and that

$$V = \Gamma(S, \mathcal{E})$$

is a free R-module with basis $v_1, \dots, v_n$. Then $H \simeq \mathrm{GL}_{n,R}$, and

$$B = \Gamma(S, \mathcal{A}(H))$$

is generated as an R-algebra by the matrix coefficients c_{ij} and by d^{-1}, where d denotes the determinant. Let Δ (respectively, ε) be the comultiplication (respectively, augmentation) of

$$A = \Gamma(S, \mathcal{A}(G)),$$

let

$$\mu : V \longrightarrow V \otimes_R A$$

be the A-comodule structure on V, and put

$$F = \Gamma(S, \mathcal{F}).$$

For $j = 1, \dots, n$, write

$$\mu(v_j) = \sum_{i=1}^{n} v_i \otimes a_{ij}.$$

Then $\varphi(c_{ij}) = a_{ij}$. Since $\pi : V \to F$ is a morphism of A-comodules, one has

$$\sum_{i=1}^{n} \pi(v_i) \otimes a_{ij} = (\pi \otimes_R \mathrm{id}_A)(\mu(v_j)) = \Delta(\pi(v_j)),$$

and therefore

$$\begin{aligned}\pi(v_j) &= (\varepsilon \otimes_R \mathrm{id}_A)\Delta(\pi(v_j)) \\ &= \sum_{i=1}^{n} \varepsilon(\pi(v_i))a_{ij} \\ &= \varphi\left(\sum_{i=1}^{n} \varepsilon(\pi(v_i))c_{ij}\right).\end{aligned}$$

Thus $\varphi(B)$ contains $\pi(V) = F$, which generates A as an R-algebra. Hence φ is surjective, and ρ is a closed immersion. This proves (i).

If S is affine, there is a locally free $\mathcal{O}_S$-module $\mathcal{V}'$ of finite rank such that

$$\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^{\oplus d}.$$

Regard $\mathcal{V}'$ as a trivial G-module and replace $\mathcal{V}$ by $\mathcal{O}_S^{\oplus d}$. This realizes G as a closed subgroup of $\mathrm{GL}_{d,S}$ and completes the proof.

Remarks 13.6.—

(a) For completeness, let us briefly sketch Thomason's argument ([Th87], Theorem 3.1), retaining the preceding notation. The G-equivariant epimorphism

$$\pi : \mathcal{E} \longrightarrow \mathcal{F}$$

induces a closed immersion

$$\tau : G \longrightarrow V(\mathcal{E})$$

such that

$$\tau(g'g) = \tau(g') \cdot g.$$

(N.B. G acts on the right on $V(\mathcal{E})$.) There is an isomorphism

$$H \simeq \mathrm{Aut}_{\mathcal{O}_S}(V(\mathcal{E}))^{\mathrm{op}}$$

compatible with the left action of G on $\mathcal{E}$ and its right action on $V(\mathcal{E})$. Let

$$N = \mathrm{Transp}_H^{\mathrm{str}}(\tau(G), \tau(G))$$

be the strict transporter. If G is essentially free over S, it follows from 6.2.4(e) that N is a closed subgroup scheme of H, hence affine over S. Moreover, ρ factors through a morphism of S-groups

$$\rho' : G \longrightarrow N.$$

For every $S' \to S$ and $h \in N(S')$, put

$$\pi(h) = \tau(1) \cdot h,$$

where 1 is the unit element of $G(S')$. This defines a morphism of S-schemes

$$\pi : N \longrightarrow \tau(G)$$

that is a retraction of ρ', after identifying G with $\tau(G)$. Since N is separated over S, it follows that ρ' is a closed immersion.

(b) The proof of [Th87], Theorem 3.1, appears to require the hypothesis that G be essentially free over S, although that hypothesis is not stated there (the author invokes instead the fact that H is essentially free). It is, however, satisfied when G is reductive (cf. Exposé XXII, 5.7.8), and hence in all cases considered in *loc. cit.*, Corollary 3.2. In particular, Thomason proves in *loc. cit.*, 2.5, that if S is separated, Noetherian, and regular of dimension at most 2, if G is affine and of finite presentation, and if $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$-module, then (G, S) satisfies (RE). By 13.5, this gives 13.2 under a slightly more restrictive hypothesis.

To conclude, let us point out that the proof of [Th87], 2.5, can be simplified slightly as follows. (For brevity, we work in the affine situation $S = \operatorname{Spec}(R)$.)

Proposition 13.7.— *Let R be a Noetherian regular ring of dimension at most 2, let C be an R-coalgebra that is projective as an R-module, and let F be a C-comodule of finite type over R. Then F is a quotient of a C-comodule V that is projective and of finite type over R.*

Proof. Replacing $\operatorname{Spec}(R)$ by one of its connected components, we may suppose that R is integral, with fraction field K. Let Δ (respectively, ε) denote the comultiplication (respectively, augmentation) of C, and let

$$\rho : F \longrightarrow F \otimes C$$

be the comodule structure on F. Let

$$\pi : W \longrightarrow F$$

be a surjective morphism, with W a finite free R-module. Give $W \otimes C$ the comodule structure defined by $\mathrm{id}_W \otimes \Delta$, and do the same for $F \otimes C$. Then $\rho : F \to F \otimes C$ is a morphism of C-comodules with retraction $\mathrm{id}_F \otimes \varepsilon$.

Let W' be the C-comodule defined by the following cartesian square:

$$\begin{tikzcd} W' \arrow[r] \arrow[d] & F \arrow[d, "\rho"] \\ W \otimes C \arrow[r, "\pi\otimes\mathrm{id}_C"'] & F \otimes C \end{tikzcd}$$

Thus W' identifies with the kernel of the morphism

$$W \otimes C \longrightarrow (F \otimes C)/\rho(F),$$

and the projection

$$\pi' : W' \longrightarrow F, \qquad x \longmapsto (\pi \otimes \varepsilon)(x),$$

is surjective. Since F is a finitely generated R-module, there is a C-subcomodule V' of W', finitely generated over R, such that

$$\pi'(V') = F.$$

Since $W \otimes C$ is R-torsion-free, so is V'. Replacing F by V', we may therefore suppose from the outset that F is torsion-free.

Apply the preceding construction to this new F, obtaining V' as above. Consider the subcomodule V that is the kernel of the morphism

$$W \otimes C \longrightarrow E = \frac{W \otimes C \otimes K}{V' \otimes K}.$$

Then V contains V', and

$$Q = (W \otimes C)/V$$

is an R-submodule of the K-vector space E. Put

$$Q' = E/Q.$$

Since $W \otimes C$ and E are flat R-modules, for every R-module N one obtains

$$\mathrm{Tor}_1^R(V, N) \simeq \mathrm{Tor}_2^R(Q, N) \simeq \mathrm{Tor}_3^R(Q', N).$$

Because R is regular of dimension at most 2, the right-hand term vanishes. Hence V is a flat R-module.

It remains to prove that V is finitely generated. Put

$$M = W \otimes C.$$

By construction,

$$V/V' \simeq (M/V')_{\mathrm{tors}},$$

the R-torsion submodule of M/V'. Since M is a projective R-module, there is a projective R-module P such that

$$M \oplus P = L$$

is free. Then

$$(M/V')_{\mathrm{tors}} \simeq (L/V')_{\mathrm{tors}}.$$

Since V' is finitely generated, there is a direct summand

$$L' \simeq R^n$$

of L containing V'. Hence

$$(L/V')_{\mathrm{tors}} \simeq (L'/V')_{\mathrm{tors}},$$

and the latter is finitely generated because L'/V' is. Thus V is a finitely generated flat R-module, hence a finitely generated projective R-module, since R is Noetherian. This proves Proposition 13.7.

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Exposé $\mathrm{VII_A}$ Infinitesimal Study of Group Schemes

by P. Gabriel

In Exposé II we restricted ourselves to the study of first-order differential invariants, and we did not address certain phenomena special to characteristic $p > 0$ or to characteristic zero. Our purpose in Part A of this exposé is to fill that gap.

Moreover, the infinitesimal study of a group scheme to arbitrary order is related to that of the associated formal group; the purpose of the second part of this exposé is to present the first definitions and properties concerning formal groups.

A) Differential Operators and Restricted p-Lie Algebras*

1. Differential Operators

In this section, as well as in Sections 2 and 3, S denotes a fixed scheme, and all products under consideration are cartesian products in the category of S-schemes.[1] If X is an S-scheme, we write $p_{X/S}$, p_X, or simply p for the structural morphism from X to S.

1.1. Let $u : Y \to X$ be a morphism of S-schemes, and endow the direct image $u_*(\mathcal{O}_Y)$ of the structure sheaf of Y with the $\mathcal{O}_X$-module structure induced by u. The sheaf

$$\mathcal{H} = \mathcal{H}om_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y))$$

of homomorphisms of $p_X^{-1}(\mathcal{O}_S)$-modules from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$ is therefore naturally an $\mathcal{O}_X$-bimodule. If U is an open subset of X, and f and d are sections over U of $\mathcal{O}_X$ and $\mathcal{H}$, respectively, then fd and df are the morphisms

$$g \longmapsto f\,d(g) \qquad \text{and} \qquad g \longmapsto d(fg)$$

from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$. From now on we write $(\operatorname{ad} f)(d)$ in place of $fd - df$.

Definition 1.1.1. *An S-deviation of order $\leq n$ is a pair $D = (u, d)$, consisting of a morphism of S-schemes $u : Y \to X$ and a morphism of $p_X^{-1}(\mathcal{O}_S)$-modules*

$$d : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y),$$

*Part A of the present exposé was not treated seriously in the oral seminars.

[1] In particular, if X and Y are two S-schemes, $X \times_S Y$ is denoted simply by $X \times Y$. For the material in Sections 1 and 2, one may also consult *Groupes algébriques* by Demazure and Gabriel, II.4, nos. 5–6; see also Jantzen, *Representations of Algebraic Groups*, I.7.

such that, for every open subset U of X and every sequence of $n+1$ sections $f_0, \dots, f_n \in \mathcal{O}_X(U)$, one has

$$(\mathrm{ad}\, f_0)(\mathrm{ad}\, f_1)\cdots(\mathrm{ad}\, f_n)(d) = 0. \qquad (*_n)$$

[2]

In this case we also say that d is an S-deviation of u of order $\leq n$. In particular, an S-deviation of u of order ≤ 0 is a morphism of $\mathcal{O}_X$-modules from $\mathcal{O}_X$ to $u_*(\mathcal{O}_Y)$, that is, an element of $\Gamma(Y, \mathcal{O}_Y)$.

Definition 1.1.2. *A morphism of $p_X^{-1}(\mathcal{O}_S)$-modules*

$$d : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y)$$

is an S-deviation of u if, for every point $y \in Y$, there exist an open neighborhood U of $u(y)$ in X and an open neighborhood V of y in Y satisfying:

a) *$u(V) \subset U$;*

b) *if $v : V \to U$ is induced by u, there is an integer n such that the morphism $\mathcal{O}_U \to v_*(\mathcal{O}_V)$ induced by d is an S-deviation of v of order $\leq n$.*

[3]

If d is an S-deviation of u, we also call the pair $D = (u, d)$ an S-deviation, and we may write

$$Y \xrightarrow{D} X \qquad \text{or} \qquad Y \xrightarrow[u]{d} X.$$

When d is the homomorphism of algebras

$$u^\natural : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y)$$

corresponding to the morphism $u : Y \to X$, we also write u in place of D.

Remarks 1.1.3.[4] Let $\mathrm{Dev}(u)$ (respectively, $\mathrm{Dev}_{\leq n}(u)$) be the set of S-deviations of u (respectively, of order $\leq n$). It has a natural $\mathcal{O}_Y(Y)$-module structure: if $\lambda \in \mathcal{O}_Y(Y)$, then λd is the deviation that sends f to $\lambda d(f)$ for every local section f of $\mathcal{O}_X$.

For every open subset V of Y, set $\mathcal{D}\rceil\sqsubseteq(u)(V) = \mathrm{Dev}(u|_V)$. Thus $\mathcal{D}\rceil\sqsubseteq(u)(V)$ is the set of

$$\begin{aligned} d_V &\in \mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}\big(\mathcal{O}_X, (u|_V)_*(\mathcal{O}_V)\big) \\ &\simeq \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}\big((u|_V)^{-1}\mathcal{O}_X, \mathcal{O}_V\big) \\ &\simeq \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}\big(u^{-1}\mathcal{O}_X, \mathcal{O}_Y\big)(V) \end{aligned}$$

[2]Equivalently, for every $x \in X$ and $f_0, \dots, f_n, g \in \mathcal{O}_{X,x}$,

$$(\mathrm{ad}\, f_0)\cdots(\mathrm{ad}\, f_n)(d_x)(g) = 0.$$

Recall also that the adjunction isomorphism

$$\theta : \mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}\big(\mathcal{O}_X, u_*(\mathcal{O}_Y)\big) \xrightarrow{\sim} \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}\big(u^{-1}\mathcal{O}_X, \mathcal{O}_Y\big)$$

sends d to $d' = \varepsilon \circ u^{-1}(d)$, where $\varepsilon : u^{-1}u_*(\mathcal{O}_Y) \to \mathcal{O}_Y$ is canonical. Conversely, $\theta^{-1}(d') = u_*(d') \circ \eta$, with $\eta : \mathcal{O}_X \to u_* u^{-1}\mathcal{O}_X$ canonical. Hence d satisfies $(*_n)$ if and only if d' satisfies the analogous identity for all local sections of $u^{-1}\mathcal{O}_X$.

[3]If X and u are quasi-compact, every S-deviation of u is therefore of order $\leq n$ for some integer n.

[4]These remarks were added by the editors because they are useful in 1.3, 1.4, and 2.1.

such that, for every open $U \subset X$, the map

$$d_V(U) : \mathcal{O}_X(U) \longrightarrow \mathcal{O}_Y(u^{-1}(U) \cap V)$$

satisfies $(*_n)$. This defines a presheaf of $\mathcal{O}_Y$-modules on Y, and it is easily seen to be a sheaf (more precisely, a subsheaf of $\mathcal{H}om_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)$).

1.2. Consider two S-deviations $D = (u, d)$ and $E = (v, e)$:

$$Z \xrightarrow[v]{e} Y \xrightarrow[u]{d} X.$$

As U ranges over the open subsets of X, the composites

$$\Gamma(U, \mathcal{O}_X) \xrightarrow{d(U)} \Gamma(u^{-1}U, \mathcal{O}_Y) \xrightarrow{e(u^{-1}U)} \Gamma(v^{-1}u^{-1}U, \mathcal{O}_Z)$$

define an S-deviation of uv, denoted by de. If d has order $\leq m$ and e has order $\leq n$, then de has order $\leq m + n$. We also write

$$D \circ E = (uv, de) \qquad (\dagger)$$

[5] and call $D \circ E$, or DE, the *composite S-deviation.* When $d = u^\natural$, so that $D = u$ under the convention of 1.1, we also say that DE is the *image of E under u.*

The map $(D, E) \mapsto D \circ E$ allows us to speak from now on of the *category of S-deviations*, whose objects are S-schemes and whose morphisms are S-deviations.[6]

Definition 1.2.0.[7] *Let $w : Z \to X$ be an S-morphism. An S-derivation of w, or an S-derivation of $\mathcal{O}_X$ into $w_*(\mathcal{O}_Z)$, is a morphism of $p^{-1}(\mathcal{O}_S)$-modules*

$$d : \mathcal{O}_X \longrightarrow w_*(\mathcal{O}_Z)$$

such that, for every open $U \subset X$ and $f, g \in \mathcal{O}_X(U)$,

$$d(fg) = w^\natural(f)d(g) + w^\natural(g)d(f).$$

Then d is a deviation of w of order ≤ 1, vanishing on the unit section of $\mathcal{O}_X$. We denote the set of S-derivations of w by $\mathrm{Der}_S(w)$; it is an $\mathcal{O}(Z)$-module.

With the notation of 1.2, take

$$Y = I_Z = \operatorname{Spec} \mathcal{O}_Z[t], \qquad t^2 = 0,$$

let v be the zero section $\tau : Z \to I_Z$, defined by the homomorphism $\mathcal{O}_Z[t] \to \mathcal{O}_Z$ sending t to zero, and let e be the morphism of $\mathcal{O}_Z$-modules $\sigma : \mathcal{O}_Z[t] \to \mathcal{O}_Z$ defined by $\sigma(1) = 0$ and $\sigma(t) = 1$, which it is convenient to denote by ∂_t.[8]

If $u : I_Z \to X$ is a morphism satisfying $w = u \circ \tau$, then $\sigma \circ u^\natural$ is an S-derivation from $\mathcal{O}_X$ to $w_*(\mathcal{O}_Z)$. Conversely, an S-derivation d determines the morphism $u : I_Z \to X$ which equals w on the underlying spaces and satisfies

$$u^\natural(f) = w^\natural(f) + d(f)t$$

[5] With this notation, de means the composite "d followed by e."

[6] Often one considers only the S-deviations of id_X; they form the algebra of S-differential operators on X, cf. 1.4 below. The more general framework provides a convenient functorial language for statements such as the following: if G is an S-group, the algebra of left-translation-invariant S-differential operators on G is isomorphic to the algebra of S-deviations of the unit section $\varepsilon : S \to G$; cf. 2.1 and 2.4.

[7] The editors expanded this paragraph and assigned the number 1.2.0 to this definition and 1.2.1 to the lemma that follows.

[8] The editors added this notation ∂_t.

for every local section f of $\mathcal{O}_X$.

Lemma 1.2.1. *Let $E = (\tau, \partial_t)$ be the deviation of $\tau : Z \to I_Z$ defined above. For every S-morphism $w : Z \to X$, the map $u \mapsto u \circ E$ is a bijection between the S-morphisms $u : I_Z \to X$ such that $u \circ \tau = w$, and the S-derivations of w.*

1.2.2. Let d be an S-deviation of $u : Y \to X$. On the one hand, d is plainly an S'-deviation of u for every morphism $s : S \to S'$.

On the other hand, let $t : T \to S$ be a morphism with target S, let $u_T : Y_T \to X_T$ be obtained from u by base change, and let $t_Y : Y_T \to Y$ and $t_X : X_T \to X$ be the canonical projections. There exists a unique T-deviation of u_T, denoted d_T or $d \times T$, which satisfies

$$t_X d_T = d t_Y$$

in the sense of (†). Thus, for every open $U \subset X$, the following diagram is commutative:

$$\begin{tikzcd}
\mathcal{O}(U) \arrow[r, "t_X^\natural"] \arrow[d, "d(U)"'] & \mathcal{O}(U \times T) \arrow[d, "d_T(U \times T)"] \\
\mathcal{O}(u^{-1}U) \arrow[r, "t_Y^\natural"] & \mathcal{O}(u^{-1}U \times T) \, .
\end{tikzcd}$$

[9] If $D = (u, d)$, we also write $D_T = (u_T, d_T)$, and say that d_T and D_T are obtained from d and D by base change.

1.2.3. For example, let $u : Y \to X$ and $v : Z \to T$ be two S-morphisms, and let d and e be S-deviations of u and v. There is a commutative diagram

$$\begin{tikzcd}
X \times T & Y \times T \arrow[l, "u_T"'] \\
X \times Z \arrow[u, "v_X"] & Y \times Z \arrow[l, "u_Z"] \arrow[u, "v_Y"'] \arrow[ul, "u \times v"']
\end{tikzcd}$$

and we denote by $d \times e$, the *product of d and e*, the S-deviation of $u \times v$ equal to

$$d_T e_Y = e_X d_Z$$

under the convention of (†). More explicitly, for every open $U \subset X \times T$, set

$$W = v_Y^{-1} u_T^{-1} U = u_Z^{-1} v_X^{-1} U.$$

Then the following diagram is commutative:

$$\begin{tikzcd}
\mathcal{O}(U) \arrow[r, "d_T(U)"] \arrow[d, "e_X(U)"'] \arrow[dr, dashed, "(d \times e)(U)"] & \mathcal{O}(u_T^{-1}U) \arrow[d, "e_Y(u_T^{-1}U)"] \\
\mathcal{O}(v_X^{-1}U) \arrow[r, "d_Z(v_X^{-1}U)"] & \mathcal{O}(W) \, .
\end{tikzcd}$$

[9]Explicitly, if V is an affine open subset of S, and U (respectively, U') is an affine open subset of X (respectively, T) over V, then

$$\mathcal{O}_{X \times T}(U \times U') = \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U'),$$

and $d_T(U \times U')$ is the composite

$$\mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') \xrightarrow{d(U) \otimes \mathrm{id}} \mathcal{O}_Y(u^{-1}U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') \longrightarrow \mathcal{O}_{Y \times T}(u^{-1}U \times U').$$

The author left to the reader the verification that d_T is well-defined, and the editors do the same.

If $D = (u, d)$ and $E = (v, e)$, we also write

$$D \times E = (u \times v, d \times e).$$

1.3.[10] Let $u : Y \to X$ be a morphism of S-schemes. Recall that the adjunction isomorphism

$$\mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \xrightarrow{\sim} \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)$$

sends a morphism of $p^{-1}(\mathcal{O}_S)$-modules

$$d : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y)$$

to $d' = \varepsilon \circ u^{-1}(d)$, where $\varepsilon : u^{-1}u_*(\mathcal{O}_Y) \to \mathcal{O}_Y$ is the canonical morphism.

Write $\mathcal{J}_u$ (respectively, $\mathcal{I}_u$) for the kernel of the homomorphism of algebras

$$u^\natural : \mathcal{O}_X \longrightarrow u_*(\mathcal{O}_Y) \quad \Big(\text{respectively, } u^{\natural\prime} : u^{-1}\mathcal{O}_X \longrightarrow \mathcal{O}_Y\Big),$$

and let $d : \mathcal{O}_X \to u_*(\mathcal{O}_Y)$ be a morphism of $p^{-1}(\mathcal{O}_S)$-modules. If $U \subset X$ is open and $f_0, \ldots, f_n, g \in \mathcal{O}_X(U)$, induction on n shows that condition $(*_n)$ is equivalent to

$$0 = \sum_{I \subset \{0,\ldots,n\}} (-1)^{|I|} u^\natural(f_{\{0,\ldots,n\}\setminus I}) d(f_I g), \tag{$**_n$}$$

where f_I denotes the product of the f_i for $i \in I$. Consequently, if d satisfies $(*_n)$, then d vanishes on the ideal $\mathcal{J}_u^{n+1}$.

Now suppose that $Y = S$. Then $u : S \to X$ is a section of $p : X \to S$, hence an immersion. On the one hand,

$$\varepsilon : u^{-1}u_*\mathcal{O}_S \longrightarrow \mathcal{O}_S$$

is an isomorphism, and therefore $u^{-1}\mathcal{J}_u = \mathcal{I}_u$. On the other hand, there is an isomorphism

$$u^{-1}\mathcal{O}_X \simeq \mathcal{O}_S \oplus \mathcal{I}_u. \tag{$\star$}$$

Suppose that d vanishes on $\mathcal{J}_u^{n+1}$. Then $d' = \varepsilon \circ u^{-1}(d)$ vanishes on $\mathcal{I}_u^{n+1}$, and therefore satisfies the analogues $(**_n')$ and $(*_n')$ of $(**_n)$ and $(*_n)$ when

$$f_0, \ldots, f_n \in \mathcal{I}_u(u^{-1}(U)).$$

Moreover,

$$(\mathrm{ad}\, a)(\varphi) = 0$$

for every $a \in \mathcal{O}_S(u^{-1}U)$ and every morphism of $\mathcal{O}_{u^{-1}U}$-modules $\varphi : u^{-1}\mathcal{O}_U \to \mathcal{O}_{u^{-1}U}$. The splitting $(\star)$ therefore implies that d' satisfies $(*_n')$, and hence that d satisfies $(*_n)$.

Lemma. *If $u : S \to X$ is a section of $p : X \to S$, then d is an S-deviation of u of order $\leq n$ if and only if d' vanishes on $\mathcal{I}_u^{n+1}$.*

This interpretation generalizes as follows. Let $u : Y \to X$ be any S-morphism, and let $\Gamma_u : Y \to Y \times X$ be its graph, with components id_Y and u. Every S-deviation d of u of order $\leq n$ gives, by composition,

$$Y \xrightarrow{\mathrm{diag}} Y \times Y \xrightarrow[u_Y]{d_Y} Y \times X,$$

a Y-deviation of Γ_u of order $\leq n$, denoted Γ_d, the graph of d.

[10] The editors expanded the original in this paragraph; see also their note 2 to Definition 1.1.1.

Conversely, a Y-deviation e of Γ_u gives the composite S-deviation

$$e_X = \mathrm{pr}_2 \circ e : \qquad Y \xrightarrow[\Gamma_u]{e} Y \times X \xrightarrow{\mathrm{pr}_2} X.$$

One has immediately $(\Gamma_d)_X = d$. The identity $\Gamma_{e_X} = e$ follows from the $\mathcal{O}_Y$-linearity of e.[11] Thus there is an isomorphism of $\mathcal{O}_Y(Y)$-modules

$$\left\{ \begin{array}{c} S\text{-deviations of } u \\ \text{of order } \leq n \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{c} Y\text{-deviations of } \Gamma_u \\ \text{of order } \leq n \end{array} \right\}, \qquad d \longmapsto \Gamma_d.$$

Moreover, d is an S-derivation of u if and only if Γ_d is a Y-derivation of Γ_u.

Let $\mathcal{I}_{\Gamma_u}$ denote the kernel of the homomorphism of algebras

$$\Gamma_u^{-1}\mathcal{O}_{Y\times X} \longrightarrow \mathcal{O}_Y$$

corresponding to Γ_u. The preceding lemma yields:

Proposition. *Let $u : Y \to X$ be an S-morphism and $\Gamma_u : Y \to Y \times X$ its graph. The S-deviations of u of order $\leq n$ identify with the Y-deviations of Γ_u of order $\leq n$, and these are in bijection with*

$$\mathrm{Hom}_{\mathcal{O}_Y}\left(\frac{\Gamma_u^{-1}\mathcal{O}_{Y\times X}}{\mathcal{I}_{\Gamma_u}^{n+1}}, \mathcal{O}_Y \right).$$

1.3.1.[12] Return to the case in which $u : S \to X$ is a section of $p : X \to S$. The homomorphism

$$\varphi : u^{-1}\mathcal{O}_X \longrightarrow \mathcal{O}_S$$

has a section, denoted simply by $g \mapsto g \cdot 1$. Thus, with the notation of 1.3,

$$u^{-1}\mathcal{O}_X \simeq \mathcal{O}_S \oplus \mathcal{I}_u,$$

and, for every section f of $u^{-1}\mathcal{O}_X$,

$$f - \varphi(f) \cdot 1 \in \mathcal{I}_u.$$

Let d be an S-deviation of u of order ≤ 1, and let $d' : u^{-1}\mathcal{O}_X \to \mathcal{O}_S$ be the $\mathcal{O}_S$-morphism corresponding to d. If a, b are sections of $u^{-1}\mathcal{O}_X$, then

$$\begin{aligned} 0 &= d'\big((a - \varphi(a)\cdot 1)(b - \varphi(b)\cdot 1)\big) \\ &= d'(ab) - \varphi(a)d'(b) - \varphi(b)d'(a) + \varphi(ab)d'(1). \end{aligned}$$

Consequently, d is an S-derivation of u if and only if $d'(1) = 0$.

Lemma. *The S-derivations of u are precisely the S-deviations of u of order 1 which vanish on the unit section of $\mathcal{O}_X$. They correspond to the $\mathcal{O}_S(S)$-module*

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}_u/\mathcal{I}_u^2, \mathcal{O}_S),$$

and there is an isomorphism of $\mathcal{O}_S(S)$-modules

$$\mathrm{Dev}_{\leq 1}(u) \simeq \mathcal{O}_S(S) \oplus \mathrm{Der}_S(u).$$

[11] If λ and f are local sections of $\mathcal{O}_Y$ and $\mathcal{O}_X$, then

$$(\Gamma_{e_X})(\lambda \otimes f) = \lambda\, e(1 \otimes f) = e(\lambda \otimes f).$$

The French note prints g in the two right-hand tensor expressions; the variables just introduced require f.

[12] This paragraph was added by the editors.

Returning to the general case gives:

Corollary. *Let $u : Y \to X$ be an S-morphism and $\Gamma_u : Y \to Y \times X$ its graph. There is a canonical isomorphism of $\mathcal{O}_Y(Y)$-modules*

$$\mathrm{Der}_S(u) \simeq \mathrm{Der}_Y(\Gamma_u) \simeq \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_{\Gamma_u}/\mathcal{I}^2_{\Gamma_u}, \mathcal{O}_Y).$$

Definition 1.4. *Let X be an S-scheme. An S-differential operator (respectively, an S-differential operator of order $\leq n$) on X is an S-deviation (respectively, an S-deviation of order $\leq n$) of the identity morphism of X.*

By 1.1, an S-differential operator of order $\leq n$ is therefore an endomorphism of the $p^{-1}(\mathcal{O}_S)$-module $\mathcal{O}_X$ satisfying the identities $(*_n)$ of 1.1. We write $\mathrm{Dif}^n_{X/S}$ for the $\Gamma(\mathcal{O}_S)$-module[13] of S-differential operators of order $\leq n$, and $\mathrm{Dif}_{X/S}$ for the module of all S-differential operators.

As observed in 1.2, S-deviations of id_X can be composed. This makes $\mathrm{Dif}_{X/S}$ a $\Gamma(\mathcal{O}_S)$-algebra, called the *algebra of differential operators of X/S*.

Similarly, for every open subset $V \subset X$, set

$$\mathcal{D}\rangle\{_{X/S}(V) = \mathrm{Dif}_{V/S} = \mathrm{Dev}(\mathrm{id}_V).$$

By 1.1.3 this defines a sheaf of $\mathcal{O}_X$-modules, called the *sheaf of S-differential operators on X*.[14]

1.4.1. As in 1.3, differential operators on X/S may be interpreted using the graph of the identity morphism of X, namely the diagonal

$$\Delta = \Delta_{X/S} : X \longrightarrow X \times X.$$

Endow $\mathcal{O}_{X\times X}$ with the $\mathrm{pr}_1^{-1}(\mathcal{O}_X)$-algebra structure defined by pr_1. Then $\Delta^{-1}\mathcal{O}_{X\times X}$ is an algebra over

$$\mathcal{O}_X = \Delta^{-1}\,\mathrm{pr}_1^{-1}\,\mathcal{O}_X.$$

Let $\mathcal{I}_{X/S}$ be the kernel of the homomorphism

$$\Delta^{-1}\mathcal{O}_{X\times X} \xrightarrow{m} \mathcal{O}_X$$

adjoint to $\mathcal{O}_{X\times X} \to \Delta_*(\mathcal{O}_X)$, and set

$$\mathcal{P}^m_{X/S} = \Delta^{-1}\mathcal{O}_{X\times X}/\mathcal{I}^{m+1}_{X/S}.$$

If $V \subset S$ and $U \subset X$ are affine open subsets with U above V, and if

$$k = \Gamma(V, \mathcal{O}_S), \qquad A = \Gamma(U, \mathcal{O}_X),$$

then

$$\Gamma(U, \mathcal{P}^m_{X/S}) = (A \otimes_k A)/I^{m+1},$$

where I is generated by the elements $a \otimes 1 - 1 \otimes a$, for $a \in A$.

By 1.3 there is an isomorphism of $\mathcal{O}_X(X)$-modules

$$j_X : \mathrm{Dif}^m_{X/S} \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}^m_{X/S}, \mathcal{O}_X).$$

[13] In this exposé the ring $\Gamma(S, \mathcal{O}_S) = \mathcal{O}_S(S)$ is denoted by $\Gamma(\mathcal{O}_S)$.

[14] The editors modified the original here. It mentioned the sheaf $U \mapsto \mathrm{Dif}_{X_U/U}$, where U ranges over open subsets of S; that sheaf is the direct image of $\mathcal{D}\rangle\{_{X/S}$ under $p_X : X \to S$.

It is given explicitly as follows: if $d \in \mathrm{Dif}^m_{X/S}$ and a section c of $\mathcal{P}^m_{X/S}$ over U has the form

$$c = a \otimes b + I^{m+1},$$

then $j_X(d)(c) = a\, d(b)$.[15]

1.4.2. Let d be a differential operator and u a section of X over S. The *value of d at u* is the composite S-deviation

$$S \xrightarrow{u} X \xrightarrow[\mathrm{id}_X]{d} X.$$

By 1.3 and 1.4.1, if d has order $\leq m$, then du (respectively, d) is canonically associated with a morphism of $\mathcal{O}_S$-modules

$$d' : u^{-1}\mathcal{O}_X/\mathcal{I}_u^{m+1} \longrightarrow \mathcal{O}_S$$

(respectively, a morphism of $\mathcal{O}_X$-modules $d'' : \mathcal{P}^m_{X/S} \to \mathcal{O}_X$).

The morphism d' may be recovered from d'' as follows. The square

$$\begin{tikzcd} X \simeq S \times X \arrow[r, "u\times X"] \arrow[d, "p"'] & X \times X \arrow[d, "\mathrm{pr}_1"] \\ S \arrow[r, "u"] & X \end{tikzcd}$$

is cartesian. Hence X identifies with $S \times_X (X \times X)$, u with $S \times_X \Delta$, and $u^*(\mathcal{P}^m_{X/S})$ with $u^{-1}\mathcal{O}_X/\mathcal{I}_u^{m+1}$. Under these identifications, $u^*(d'')$ is the morphism

$$u^{-1}\mathcal{O}_X/\mathcal{I}_u^{m+1} \longrightarrow \mathcal{O}_S$$

which is precisely d'.

1.5. Set, as usual,

$$I_S = \operatorname{Spec} \mathcal{O}_S[T]/(T^2).$$

Let $\tau : S \to I_S$ be the zero section, and let σ be the canonical deviation of τ defined in 1.2.0: it is the $\mathcal{O}_S$-module homomorphism which vanishes on the unit section of $\mathcal{O}_S[T]/(T^2)$ and sends the class t of T modulo T^2 to the unit section of $\mathcal{O}_S$.

Let X be an S-scheme. Every I_S-automorphism u of $I_S \times X$ inducing the identity on X determines, by composition, a differential operator D_u on X:

$$X \simeq S \times X \xrightarrow{\sigma\times X} I_S \times X \xrightarrow{u} I_S \times X \xrightarrow{\mathrm{pr}_2} X\,.$$

By Exposé II, 3.14, the map $u \mapsto D_u$ is an isomorphism from the $\Gamma(\mathcal{O}_S)$-Lie algebra

$$\mathrm{Lie}(\operatorname{Aut} X) := \mathrm{Lie}(\operatorname{Aut} X)(S)$$

onto the $\Gamma(\mathcal{O}_S)$-Lie algebra of $p^{-1}(\mathcal{O}_S)$-derivations of $\mathcal{O}_X$. Its inverse sends a derivation D to the automorphism of $I_S \times X$ corresponding to

$$a + bt \longmapsto a + (Da + b)t$$

on $\mathcal{O}_X[T]/(T^2)$.

[15] Under this isomorphism, the X-derivations of $\Delta_{X/S}$ correspond, by 1.3.1, to the S-derivations of id_X, that is, to the $p^{-1}(\mathcal{O}_S)$-derivations of $\mathcal{O}_X$.

2. Invariant Differential Operators on Group Schemes

2.1. Let G be an S-group scheme; we denote its unit section by ε, or by $\varepsilon_G : S \to G$.

Definition. *Let $U(G)$ be the $\Gamma(\mathcal{O}_S)$-module of S-deviations of ε_G (or S-deviations of the origin), as in 1.1.*

If $d, e \in U(G)$, then $d \times e$ is an S-deviation of

$$\varepsilon \times \varepsilon : S \simeq S \times S \longrightarrow G \times G.$$

The image of $d \times e$ under the multiplication morphism $m : G \times G \to G$, as in 1.2, is called the product *of d and e and is denoted $d \cdot e$.*

The $\Gamma(\mathcal{O}_S)$-module $U(G)$ is thereby made into an associative $\Gamma(\mathcal{O}_S)$-algebra with unit ε_G. We call $U(G)$ the infinitesimal algebra *of G.*[16]

As T ranges over schemes above S, the infinitesimal algebra $U(G_T)$ of the T-group $G \times T$ plainly varies contravariantly with T; thus one may speak of the *infinitesimal-algebra functor*.

Restricting T to the open subsets of S gives a presheaf $T \mapsto U(G_T)$ of $\mathcal{O}_S$-algebras. By 1.1.3 it is a sheaf. We denote it by $\mathcal{U}(G)$ and call it the *sheaf of infinitesimal algebras of G*.

The algebra $U(G)$ is also covariantly functorial in G. Indeed, if $u : G \to H$ is a homomorphism of S-groups and d is an S-deviation of ε_G, its image under u is the element

$$U(u)(d) = ud$$

of $U(H)$. The resulting map $U(u) : U(G) \to U(H)$ is a homomorphism of $\Gamma(\mathcal{O}_S)$-algebras. Likewise one obtains a homomorphism $\mathcal{U}(u) : \mathcal{U}(G) \to \mathcal{U}(H)$.

2.2. Let $d \in U(G)$, that is, let d be an S-deviation of the origin of G. The base change of d is the S-deviation

$$d \times G$$

of $\varepsilon \times G : G \simeq S \times G \to G \times G$, as in 1.2.2. Its image under multiplication $m : G \times G \to G$ is an S-deviation of

$$m \circ (\varepsilon \times \mathrm{id}_G) = \mathrm{id}_G,$$

hence an element of $\mathrm{Dif}_{G/S}$, which we denote by d^G.

The map $d \mapsto d^G$ is plainly $\Gamma(\mathcal{O}_S)$-linear. The following commutative diagram shows that

$$(e \cdot d)^G = d^G \cdot e^G.$$

$$\begin{tikzcd}
 & & G \times G \times G \arrow[rr, "m\times G"] \arrow[dr, "G\times m"] & & G \times G \arrow[dd, "m"] \\
 & G \times G \arrow[ur, "G\times d\times G"] \arrow[dr, "m"] & & G \times G \arrow[dr, "m"] & \\
G \arrow[ur, shift left, "e\times G"] \arrow[ur, shift right, "\varepsilon\times G"'] \arrow[rr, shift left, "e^G"] \arrow[rr, shift right, "\mathrm{id}_G"'] & & G \arrow[ur, shift left, "d\times G"] \arrow[ur, shift right, "\varepsilon\times G"'] \arrow[rr, shift left, "d^G"] \arrow[rr, shift right, "\mathrm{id}_G"'] & & G
\end{tikzcd}$$

[16] This is now usually called the algebra of distributions (at the origin) of G; see Demazure–Gabriel, II.4, 6.1, and Jantzen, I.7.7.

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The two lower triangles commute by the definitions of d^G and e^G. The composite S-deviation of $e \times G$ and $G \times d \times G$ is $(e \times d) \times G$, by 1.2.2. Its image under $m \times G$ is $(e \cdot d) \times G$, and the image of the latter under m is therefore $(e \cdot d)^G$.

Thus one obtains an antihomomorphism

$$U(G) \longrightarrow \mathrm{Dif}_{G/S}$$

of $\Gamma(\mathcal{O}_S)$-algebras, called *right translation.*[18]

If $\mathcal{D}\rangle\{_{G/S}$ denotes the sheaf of S-differential operators on G, as in 1.4, and $p : G \to S$ is the structural morphism, one likewise defines a right translation

$$\mathcal{U}(G) \longrightarrow p_*(\mathcal{D}\rangle\{_{G/S}).$$

2.3. We now characterize the differential operators on G/S of the form d^G. Let $g : S \to G$ be a section of the structural morphism and let g_G denote right translation by g, namely

$$g_G : G \simeq G \times S \xrightarrow{G \times g} G \times G \xrightarrow{m} G.$$

For every differential operator D on G/S, the composite

$$g_G^{-1} D g_G$$

is again an S-deviation of id_G, hence an element of $\mathrm{Dif}_{G/S}$. We write

$$D^g = g_G^{-1} D g_G.$$

We call D *right-invariant* if for every base change $t : T \to S$ and every section $g : T \to G \times T$, one has

$$(D_T)^g = D_T.$$

Lemma. *For every differential operator D on G/S, the following assertions are equivalent, where m is the multiplication morphism of G:*

(i) D is right-invariant.

(ii) The following two deviations of m are equal:

$$Dm = m(D \times G).$$

(ii) $\Rightarrow$ *(i)*. Since condition (ii) is stable under base change, it suffices to show that it implies $D^g = D$ for every section $g : S \to G$. Let

$$h = G \times g : G \simeq G \times S \longrightarrow G \times G,$$

so that $m \circ h = g_G$. The equality $D^g = D$ is equivalent to $g_G \circ D = D \circ g_G$, and this follows from the commutative diagram

[17] The editors corrected the original diagram by replacing $d \times G \times G$ with $G \times d \times G$. The composite on the left side of the upper triangle is then $(e \times d) \times G$, and $d \mapsto d^G$ becomes an anti-isomorphism from $U(G)$ to the right-invariant differential operators (2.3–2.4). If instead ${}^G d$ is defined as the image of $G \times d$ under m, one similarly obtains an isomorphism onto the left-invariant differential operators. Sections 2.4 and 2.5 have been corrected accordingly.

[18] It would be preferable to call this a *left action.* For example, if d is an S-derivation of the origin, then by 1.2.1 it is the composite of $(\tau, \partial_t) : S \to I_S$ with a morphism $x : I_S \to G$ satisfying $x \circ \tau = \varepsilon$. The derivation d^G of $\mathcal{O}_G$ sends a local section φ to the section $g \mapsto \partial_t \varphi(xg)$. With this terminology one could say that the left action commutes with right translations.

$$\begin{tikzcd}
G & G \times G \arrow[l, "m"'] & G \arrow[l, "h"'] . \\
G \arrow[from=u, shift right, "D"'] \arrow[from=u, shift left, "\mathrm{id}_G"] & G \times G \arrow[l, "m"] \arrow[from=u, shift right, "D\times G"'] \arrow[from=u, shift left, "\mathrm{id}_{(G\times G)}"] & G \arrow[l, "h"] \arrow[from=u, shift right, "D"'] \arrow[from=u, shift left, "\mathrm{id}_G"]
\end{tikzcd}$$

$(i) \Rightarrow (ii)$. Take $t : T \to S$ to be the structural morphism $p : G \to S$, and take the section $g : T \to G \times T$ to be the diagonal $\Delta : G \to G \times G$. Right translation

$$\Delta_{G\times G} : G \times G \longrightarrow G \times G$$

then has components m and pr_2. The equality $(D_G)^\Delta = D_G$ is equivalent to commutativity of the first square in

$$\begin{tikzcd}
G \times G \arrow[r, "\Delta_{G\times G}"] \arrow[d, shift right, "D_G"'] \arrow[d, shift left, "\mathrm{id}_{G\times G}"] & G \times G \arrow[r, "\mathrm{pr}_1"] \arrow[d, shift right, "D_G"'] \arrow[d, shift left, "\mathrm{id}_{G\times G}"] & G . \arrow[d, shift right, "D"'] \arrow[d, shift left, "\mathrm{id}_G"] \\
G \times G \arrow[r, "\Delta_{G\times G}"] & G \times G \arrow[r, "\mathrm{pr}_1"] & G
\end{tikzcd}$$

because $m = \mathrm{pr}_1 \circ \Delta_{G\times G}$. This proves (ii).

Now consider $d \in U(G)$. Every square in the following diagram is commutative:

$$\begin{tikzcd}
G \times G \arrow[r, equal] \arrow[d, "m"'] & S \times G \times G \arrow[r, shift left, "d\times G\times G"] \arrow[r, shift right, "\varepsilon\times G\times G"'] \arrow[d, "S\times m"'] & G \times G \times G \arrow[r, "m\times G"] \arrow[d, "G\times m"] & G \times G \arrow[d, "m"] \\
G \arrow[r, equal] & S \times G \arrow[r, shift left, "d\times G"] \arrow[r, shift right, "\varepsilon\times G"'] & G \times G \arrow[r, "m"] & G
\end{tikzcd}$$

Since

$$m \circ (d \times G) = d^G \qquad \text{and} \qquad (m \times G) \circ (d \times G \times G) = d^G \times G,$$

one also has

$$d^G \circ m = m \circ (d^G \times G).$$

Thus, for every S-deviation d of the origin, d^G is a right-invariant differential operator.

Theorem 2.4.

(i) *The map $d \mapsto d^G$ is an anti-isomorphism of the infinitesimal algebra $U(G)$ onto the subalgebra $\mathrm{Dif}^G_{G/S}$ of $\mathrm{Dif}_{G/S}$ formed by the right-invariant differential operators.*

(ii) *Likewise, the map $d \mapsto {}^G d$ is an isomorphism from $U(G)$ onto the subalgebra of $\mathrm{Dif}_{G/S}$ formed by the left-invariant differential operators.*

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Let D be an arbitrary differential operator on G/S, and let D_0 denote its *value at the origin*, that is, the composite deviation

$$S \xrightarrow{\varepsilon} G \underset{\mathrm{id}_G}{\overset{D}{\rightrightarrows}} G.$$

The right-invariant differential operator $(D_0)^G$ is obtained by the composite

$$G \simeq S \times G \xrightarrow{\varepsilon\times G} G \times G \underset{\mathrm{id}_{G\times G}}{\overset{D\times G}{\rightrightarrows}} G \times G \xrightarrow{m} G\,.$$

[19] The editors corrected "isomorphism" to "anti-isomorphism" and added assertion (ii); see the preceding editorial note.

If D is right-invariant, then $Dm = m(D \times G)$, and hence

$$D = Dm(\varepsilon \times G) = m(D \times G)(\varepsilon \times G) = (D_0)^G.$$

Thus $d \mapsto d^G$ is surjective.

Conversely, for every S-deviation d of the origin one has the commutative square

$$\begin{tikzcd} G \times G & G \arrow[l, "d\times G"'] \\ G \times S \simeq G \arrow[u, "G\times\varepsilon"] & S \arrow[l, "d"'] \arrow[u, "\varepsilon"'] \end{tikzcd}$$

from which

$$d = m(G \times \varepsilon)d = m(d \times G)\varepsilon = (d^G)_0.$$

A fortiori, $d \mapsto d^G$ is injective. This proves the theorem.

As S varies, Theorem 2.4 also shows that right translation

$$\mathcal{U}(G) \longrightarrow p_*(\mathcal{D}\rangle\{_{G/S})$$

is an anti-isomorphism from $\mathcal{U}(G)$ onto the sheaf of $\mathcal{O}_S$-algebras

$$p_*(\mathcal{D}\rangle\{_{G/S})^G,$$

which assigns to every open $V \subset S$ the algebra $\mathrm{Dif}_{G_V/V}^{G_V}$.

Remark 2.4.1. Consider the commutative diagram

$$\begin{tikzcd} G & G \times G \arrow[l, "\eta"'] \arrow[d, shift right, "\mathrm{pr}_1"'] \\ S \arrow[u, shift right, "\varepsilon"'] & G, \arrow[l, "p"'] \arrow[u, shift right, "\Delta"'] \end{tikzcd}$$

with $G \xrightarrow{p} S$ downward, where η denotes the morphism $(x, y) \mapsto yx^{-1}$.[20] It induces morphisms

$$\eta' : \eta^{-1}(\mathcal{O}_G) \longrightarrow \mathcal{O}_{G\times G}$$

and

$$\Delta^{-1}(\eta') : p^{-1}\varepsilon^{-1}(\mathcal{O}_G) \longrightarrow \Delta^{-1}(\mathcal{O}_{G\times G}).$$

For every integer $n \geq 1$, set

$$\mathfrak{p}^n_{G/S} = \varepsilon^{-1}(\mathcal{O}_G)/\mathcal{I}_\varepsilon^{n+1}.$$

[21] Since the square formed by $\varepsilon, \eta, \Delta, p$ is cartesian, $\Delta^{-1}(\eta')$ induces an isomorphism of $\mathcal{O}_G$-modules

$$p^*(\mathfrak{p}^n_{G/S}) \xrightarrow{\sim} \mathcal{P}^n_{G/S}.$$

The differential operators on G/S of order $\leq n$ therefore correspond bijectively to the homomorphisms of $\mathcal{O}_G$-modules

$$p^*(\mathfrak{p}^n_{G/S}) \longrightarrow \mathcal{O}_G,$$

or equivalently to the homomorphisms of $\mathcal{O}_S$-modules

$$\mathfrak{p}^n_{G/S} \longrightarrow p_*(\mathcal{O}_G).$$

Under this bijection the right-invariant differential operators correspond to the composites

[20] Thus G acts on itself on the left by right translations.

[21] The editors corrected the original here: the relevant cartesian square is formed by $\varepsilon, \eta, \Delta, p$, not by $p, p, \eta, \mathrm{pr}_1$.

$$\mathfrak{p}^n_{G/S} \longrightarrow \mathcal{O}_S \xrightarrow{\text{can.}} p_*(\mathcal{O}_G)\,.$$

This recovers the isomorphism of Theorem 2.4.

2.5. [22] Let Lie(G) be the Lie algebra of G.[23] We shall define a homomorphism of $\Gamma(\mathcal{O}_S)$-Lie algebras

$$\alpha : \mathrm{Lie}(G) \longrightarrow U(G).$$

Let $s : S \to I_S$ be the zero section and let σ be the deviation of s defined in 1.2.0. Recall from II, 4.1 that Lie(G) is the set of morphisms $x : I_S \to G$ such that $x \circ s = \varepsilon_G$. The composite

$$S \underset{s}{\overset{\sigma}{\rightrightarrows}} I_S \xrightarrow{\quad x \quad} G$$

is an S-deviation of ε_G, hence an element of $U(G)$; in the notation of 1.2 it is denoted σx. Moreover, by 1.2.1 the map

$$\alpha : x \longmapsto \sigma x$$

is an isomorphism of $\mathcal{O}_S(S)$-modules from Lie(G) onto the submodule $\mathrm{Der}(\varepsilon_G) \subset U(G)$ formed by the S-derivations of ε_G. We show that α is a homomorphism of Lie algebras.[24]

Let

$$\rho' : U(G) \longrightarrow \mathrm{Dif}_{G/S}$$

be the algebra homomorphism which sends an S-deviation d of ε_G to the left-invariant differential operator ${}^G d$, as in 2.2.

Let

$$\rho : G \longrightarrow \mathrm{Aut}_S(G)$$

be the homomorphism of group functors which sends an S-morphism $g : T \to G$ to right translation by g on G_T, namely

$$G_T \simeq T \times_T G_T \xrightarrow{G_T \times g} G_T \times_T G_T \xrightarrow{m_T} G_T\,.$$

Recall also from 1.5 and II, 3.14 that

$$\mathrm{Lie}(\mathrm{Aut}\, G) = \mathrm{Lie}(\mathrm{Aut}_S(G)/S)(S)$$

identifies with the infinitesimal automorphisms of G, that is, the automorphisms of $I_S \times G$ which induce the identity on G.

Since ρ is a monomorphism, so is

$$\mathrm{Lie}(\rho) : \mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(\mathrm{Aut}_S(G)/S)$$

(see, for example, Exposé II, editorial note 50). Consequently

$$\mathrm{Lie}(\rho) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(\mathrm{Aut}\, G)$$

is injective.

On the other hand, by 1.5 the map β, which assigns to every infinitesimal automorphism u of G the differential operator D_u,

[22] The editors reordered this paragraph, beginning with the definition of $\alpha : \mathrm{Lie}(G) \to U(G)$, and corrected the original as indicated in the note to 2.2.

[23] In this exposé, for an S-group scheme G (respectively, an S-scheme X), Lie(G) (respectively, Lie(Aut X)) denotes, in the notation of Exposé II, $\mathrm{Lie}(G/S)(S)$ (respectively, $\mathrm{Lie}(\mathrm{Aut}_S(X)/S)(S)$). It is a $\Gamma(\mathcal{O}_S)$-Lie algebra by II, 4.11 and 3.14.

[24] See also Exposé II, 4.11.

$$G \simeq S \times G \xrightarrow{\sigma\times G} I_S \times G \xrightarrow{u} I_S \times G \xrightarrow{\mathrm{pr}_2} G$$

is an isomorphism from $\mathrm{Lie}(\mathrm{Aut}\, G)$ onto the Lie subalgebra of $\mathrm{Dif}_{G/S}$ formed by the $p^{-1}(\mathcal{O}_S)$-derivations of $\mathcal{O}_G$.

For every $x \in \mathrm{Lie}(G)$, the following commutative square determines the image of x under $\mathrm{Lie}(\rho)$:

$$\begin{tikzcd}
I_S \times G \arrow[r, "\mathrm{Lie}(\rho)(x)"] \arrow[d, "x\times G"'] & I_S \times G \arrow[d, "\mathrm{pr}_2"] \\
G \times G \arrow[r, "m"] & G\,.
\end{tikzcd}$$

It follows from this diagram that the image of $\mathrm{Lie}(\rho)(x)$ under β is the composite deviation

$$G \simeq S \times G \underset{s\times G}{\overset{\sigma\times G}{\rightrightarrows}} I_S \times G \xrightarrow{G\times x} G \times G \xrightarrow{m} G$$

which, by the editorial note to 2.2, is precisely

$$^{G}(\sigma x) = \rho'(\alpha(x)).$$

Thus one obtains the commutative diagram

$$\begin{tikzcd}
\mathrm{Lie}(G) \arrow[r, hook, "\mathrm{Lie}(\rho)"] \arrow[d, "\alpha"'] & \mathrm{Lie}(\mathrm{Aut}\, G) \arrow[d, "\beta"] \\
U(G) \arrow[r, hook, "\rho'"] & \mathrm{Dif}_{G/S}
\end{tikzcd}$$

in which $\mathrm{Lie}(\rho)$, β, and ρ' are homomorphisms of Lie algebras. Since ρ' is injective, α is also a homomorphism of Lie algebras. Hence:

Proposition. *The map α is an isomorphism of $\mathcal{O}_S(S)$-Lie algebras from* $\mathrm{Lie}(G)$ *onto the Lie algebra of S-derivations of ε_G; the latter is itself isomorphic, via* $\mathrm{Lie}(\rho)$*, to the Lie algebra of left-invariant S-derivations of G.* [25]

3. Coalgebras and Cartier Duality

3.1. Let S be a scheme (or, more generally, a ringed space). An $\mathcal{O}_S$-*coalgebra*[26] is a pair $(\mathcal{U}, \Delta_{\mathcal{U}})$ consisting of an $\mathcal{O}_S$-module $\mathcal{U}$ and a morphism of $\mathcal{O}_S$-modules

$$\Delta_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U},$$

called the *diagonal morphism* or *comultiplication*, such that:
(i) $\sigma \circ \Delta_{\mathcal{U}} = \Delta_{\mathcal{U}}$, where $\sigma(a \otimes b) = b \otimes a$.
(ii) The square

[25] There exist Lie algebras $\mathfrak{g}$ over a ring A for which the canonical map $\mathfrak{g} \to U(\mathfrak{g})$ is not injective; see Bourbaki, *Lie Groups and Lie Algebras*, I.2, Exercise 9. The result above shows, since α factors through

$$\mathrm{Lie}(G) \longrightarrow U(\mathrm{Lie}(G)) \longrightarrow U(G),$$

that this cannot happen for "algebraic" Lie algebras, namely those of the form $\mathrm{Lie}(G)$ for an A-group scheme G.

[26] The term *cogebra* is also used; see Bourbaki, *Algebra*, III, §11.1. Notice also that in this exposé, as well as in Exposé VII$_B$, we work in the category of cocommutative coalgebras (that is, coalgebras satisfying condition (i)). This is crucial for defining the product and the notion of a coalgebra in groups; see 3.1.0 and 3.2.

$$\begin{tikzcd}
\mathcal{U} \arrow[r, "\Delta_{\mathcal{U}}"] \arrow[d, "\Delta_{\mathcal{U}}"'] & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \arrow[d, "\mathrm{id}_{\mathcal{U}} \otimes \Delta_{\mathcal{U}}"] \\
\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \arrow[r, "\Delta_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}"] & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U}
\end{tikzcd}$$

is commutative.

(iii) There exists a morphism of $\mathcal{O}_S$-modules

$$\varepsilon_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{O}_S,$$

called the *augmentation*, such that the two composites

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \varepsilon_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{O}_S \simeq \mathcal{U}$$

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\varepsilon_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{U} \simeq \mathcal{U}$$

are the identity morphism of $\mathcal{U}$.

If $\varepsilon_{\mathcal{U}}$ and $\varepsilon'_{\mathcal{U}}$ are two augmentations, then

$$\varepsilon_{\mathcal{U}} = (\varepsilon_{\mathcal{U}} \otimes \varepsilon'_{\mathcal{U}}) \circ \Delta_{\mathcal{U}} = \varepsilon'_{\mathcal{U}};$$

the augmentation is therefore uniquely determined by (iii).

If $(\mathcal{U}, \Delta_{\mathcal{U}})$ and $(\mathcal{V}, \Delta_{\mathcal{V}})$ are two $\mathcal{O}_S$-coalgebras, a *morphism* from the first to the second is a morphism of $\mathcal{O}_S$-modules $f : \mathcal{U} \to \mathcal{V}$ for which the diagrams

$$\begin{tikzcd}
\mathcal{U} \arrow[r, "f"] \arrow[d, "\Delta_{\mathcal{U}}"] & \mathcal{V} \arrow[d, "\Delta_{\mathcal{V}}"] \\
\mathcal{U} \otimes \mathcal{U} \arrow[r, "f \otimes f"] & \mathcal{V} \otimes \mathcal{V}
\end{tikzcd}
\qquad \text{and} \qquad
\begin{tikzcd}
\mathcal{U} \arrow[rr, "f"] \arrow[dr, "\varepsilon_{\mathcal{U}}"'] & & \mathcal{V} \arrow[dl, "\varepsilon_{\mathcal{V}}"] \\
& \mathcal{O}_S &
\end{tikzcd}$$

are commutative. Morphisms of coalgebras compose like morphisms of $\mathcal{O}_S$-modules, so we may speak of the *category of $\mathcal{O}_S$-coalgebras*.

3.1.0. [27] This category has finite products: its final object is the $\mathcal{O}_S$-module $\mathcal{O}_S$, with the identity as comultiplication. The product of two coalgebras $(\mathcal{U}, \Delta_{\mathcal{U}})$ and $(\mathcal{V}, \Delta_{\mathcal{V}})$ is the tensor product $\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{V}$, whose comultiplication is the composite

$$\mathcal{U} \otimes \mathcal{V} \xrightarrow{\Delta_{\mathcal{U}} \otimes \Delta_{\mathcal{V}}} \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{V} \otimes \mathcal{V} \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \sigma \otimes \mathrm{id}_{\mathcal{V}}} \mathcal{U} \otimes \mathcal{V} \otimes \mathcal{U} \otimes \mathcal{V}$$

where $\sigma(a \otimes b) = b \otimes a$. The canonical projections of $\mathcal{U} \otimes \mathcal{V}$ onto the factors $\mathcal{U}$ and $\mathcal{V}$ are $\mathrm{id}_{\mathcal{U}} \otimes \varepsilon_{\mathcal{V}}$ and $\varepsilon_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{V}}$, respectively.[28] The "diagonal morphism" $\mathcal{U} \to \mathcal{U} \otimes \mathcal{U}$, corresponding to the pair $(\mathrm{id}_{\mathcal{U}}, \mathrm{id}_{\mathcal{U}})$, is precisely the comultiplication $\Delta_{\mathcal{U}}$.

3.1.1. Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$-algebra which, as an $\mathcal{O}_S$-module, is locally free and of finite type. Put

$$\mathcal{A}^* = \mathcal{H}om_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A}, \mathcal{O}_S).$$

[27] The editors added the numbering 3.1.0 for later references.

[28] The preceding assertion was added by the editors. Recall also that, to show that $\mathcal{U} \otimes \mathcal{V}$ is indeed the product of $\mathcal{U}$ and $\mathcal{V}$ in the category of cocommutative $\mathcal{O}_S$-cogebras, one argues as follows. Given an arbitrary $\mathcal{O}_S$-cogebra $\mathcal{E}$ and cogebra morphisms $f : \mathcal{E} \to \mathcal{U}$ and $g : \mathcal{E} \to \mathcal{V}$, any cogebra morphism $\varphi : \mathcal{E} \to \mathcal{U} \otimes \mathcal{V}$ satisfying $\mathrm{pr}_{\mathcal{U}} \circ \varphi = f$ and $\mathrm{pr}_{\mathcal{V}} \circ \varphi = g$ must equal $(f \otimes g) \circ \Delta_{\mathcal{E}}$; and this is a cogebra morphism if and only if it also equals $(g \otimes f) \circ \Delta_{\mathcal{E}}$.

The canonical morphism
$$\varphi : \mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}^* \longrightarrow (\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{A})^*$$
is invertible. If $m : \mathcal{A}\otimes\mathcal{A} \to \mathcal{A}$ is the morphism defining the multiplication of $\mathcal{A}$, composition gives a diagonal morphism

$$\Delta_{\mathcal{A}^*} : \quad \mathcal{A}^* \xrightarrow{m^*} (\mathcal{A} \otimes \mathcal{A})^* \xrightarrow{\varphi^{-1}} \mathcal{A}^* \otimes \mathcal{A}^* .$$

This diagonal morphism plainly makes $\mathcal{A}^*$ into an $\mathcal{O}_S$-coalgebra whose augmentation is the transpose of the morphism $\mathcal{O}_S \to \mathcal{A}$ defined by the unit section of $\mathcal{A}$. Moreover, it is clear that:

> *The functor $\mathcal{A} \mapsto \mathcal{A}^*$ is an anti-equivalence from the category of $\mathcal{O}_S$-algebras which are locally free and of finite type as $\mathcal{O}_S$-modules onto the category of $\mathcal{O}_S$-coalgebras which are locally free and of finite type as $\mathcal{O}_S$-modules.*

3.1.2. To every $\mathcal{O}_S$-coalgebra $\mathcal{U}$ there is canonically associated an S-functor

$$\operatorname{Spec}^* \mathcal{U} : (\mathbf{Sch}/S)^\circ \longrightarrow (\mathbf{Ens}) .$$

Indeed, for every S-scheme $q : T \to S$,

$$q^*(\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U}) \simeq q^*(\mathcal{U}) \otimes_{\mathcal{O}_T} q^*(\mathcal{U}),$$

so $q^*(\Delta_{\mathcal{U}})$ makes $\mathcal{U}_T = q^*(\mathcal{U})$ into an $\mathcal{O}_T$-coalgebra. Thus, by definition and with an obvious abuse of notation,[29]

$$(\operatorname{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1, \quad \Delta_{\mathcal{U}_T}(x) = x \otimes x\} .$$

Sections x of $\mathcal{U}_T$ plainly correspond to morphisms of $\mathcal{O}_T$-modules $\xi : \mathcal{O}_T \to \mathcal{U}_T$. The conditions $\varepsilon(x) = 1$ and $\Delta(x) = x \otimes x$ say exactly that ξ is a morphism of coalgebras. Hence also
$$(\operatorname{Spec}^* \mathcal{U})(T) = \operatorname{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{U}_T).$$
In particular:

Proposition 3.1.2.1. [30] *Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$-algebra which is locally free of finite type as an $\mathcal{O}_S$-module. Then the S-functor* $\operatorname{Spec}^* \mathcal{A}^*$ *is represented by* $\operatorname{Spec} \mathcal{A}$.

Indeed, for every S-scheme T, there are canonical isomorphisms

$$\begin{aligned}(\operatorname{Spec}^* \mathcal{A}^*)(T) &= \operatorname{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{A}^*_T)\\ &\simeq \operatorname{Hom}_{\mathcal{O}_T\text{-alg.}}(\mathcal{A}_T, \mathcal{O}_T) \simeq (\operatorname{Spec} \mathcal{A})(T).\end{aligned}$$

3.2. An $\mathcal{O}_S$-*coalgebra in groups*, that is, a group in the category of $\mathcal{O}_S$-coalgebras, consists of an $\mathcal{O}_S$-coalgebra $(\mathcal{U}, \Delta_{\mathcal{U}})$ together with three morphisms of $\mathcal{O}_S$-coalgebras

$$m_{\mathcal{U}} : \mathcal{U} \otimes \mathcal{U} \longrightarrow \mathcal{U}, \qquad \eta_{\mathcal{U}} : \mathcal{O}_S \longrightarrow \mathcal{U}, \qquad c_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U},$$

satisfying conditions (ii)*, (iii)*, and (vi) below. The requirement that $m_{\mathcal{U}}$ be a morphism of coalgebras is expressed by the commutativity of diagrams (iv) and (v):

[29] For $x \otimes y \in \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \Gamma(T, \mathcal{U}_T)$, its image in $\Gamma(T, \mathcal{U}_T \otimes_{\mathcal{O}_T} \mathcal{U}_T)$ is again denoted $x \otimes y$.

[30] The editors added the numbering 3.1.2.1 for later references. Notice also that the S-functor $\operatorname{Spec}^* \mathcal{U}$ is a sheaf for the Zariski topology; it is even an fpqc sheaf if $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$-module.

(iv)
$$\begin{tikzcd}
\mathcal{U}\otimes\mathcal{U} \arrow[rr, "m_{\mathcal{U}}"] \arrow[d, "\Delta_{\mathcal{U}}\otimes\Delta_{\mathcal{U}}"'] & & \mathcal{U} \arrow[dd, "\Delta_{\mathcal{U}}"] \\
\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U} \arrow[d, "\mathrm{id}_{\mathcal{U}}\otimes\sigma\otimes\mathrm{id}_{\mathcal{U}}"'] & & \\
\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U} \arrow[rr, "m_{\mathcal{U}}\otimes m_{\mathcal{U}}"] & & \mathcal{U}\otimes\mathcal{U}
\end{tikzcd}$$

(v)
$$\begin{tikzcd}
\mathcal{U}\otimes\mathcal{U} \arrow[rr, "m_{\mathcal{U}}"] \arrow[dr, "\varepsilon_{\mathcal{U}}\otimes\varepsilon_{\mathcal{U}}"'] & & \mathcal{U} \arrow[dl, "\varepsilon_{\mathcal{U}}"] \\
& \mathcal{O}_S &
\end{tikzcd}$$

(ii)* The square

$$\begin{tikzcd}
\mathcal{U}\otimes\mathcal{U}\otimes\mathcal{U} \arrow[r, "\mathrm{id}_{\mathcal{U}}\otimes m_{\mathcal{U}}"] \arrow[d, "m_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}"'] & \mathcal{U}\otimes\mathcal{U} \arrow[d, "m_{\mathcal{U}}"] \\
\mathcal{U}\otimes\mathcal{U} \arrow[r, "m_{\mathcal{U}}"] & \mathcal{U}
\end{tikzcd}$$

is commutative.
(iii)* The two composites

$$\mathcal{U} \simeq \mathcal{U}\otimes\mathcal{O}_S \xrightarrow{\mathrm{id}_{\mathcal{U}}\otimes\eta_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U},$$

$$\mathcal{U} \simeq \mathcal{O}_S\otimes\mathcal{U} \xrightarrow{\eta_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}.$$

are the identity morphism of $\mathcal{U}$.
(vi) The composite

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{c_{\mathcal{U}}\otimes\mathrm{id}_{\mathcal{U}}} \mathcal{U}\otimes\mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}\,.$$

is equal to $\eta_{\mathcal{U}}\circ\varepsilon_{\mathcal{U}}$.

3.2.1. The morphisms $\eta_{\mathcal{U}}$ and $c_{\mathcal{U}}$ are uniquely determined by $m_{\mathcal{U}}$. Conditions (ii)* and (iii)* simply say that $m_{\mathcal{U}}$ makes $\mathcal{U}$ into an $\mathcal{O}_S$-algebra whose unit section is the image under $\eta_{\mathcal{U}}$ of the unit section of $\mathcal{O}_S$. Condition (iv) also says that the diagonal morphism $\Delta_{\mathcal{U}}$ is compatible with multiplication. Indeed,

$$\Delta_{\mathcal{U}} : \mathcal{U} \longrightarrow \mathcal{U}\otimes\mathcal{U}$$

must be a homomorphism of coalgebras in groups, which also implies the commutativity of the triangle

(v)*
$$\begin{tikzcd}
& \mathcal{O}_S \arrow[dl, "\eta_{\mathcal{U}}"'] \arrow[dr, "\eta_{\mathcal{U}}\otimes\eta_{\mathcal{U}}"] & \\
\mathcal{U} \arrow[rr, "\Delta_{\mathcal{U}}"'] & & \mathcal{U}\otimes\mathcal{U}\,.
\end{tikzcd}$$

On the other hand, as in every category, the antipode $c_{\mathcal{U}}$ is an isomorphism from $\mathcal{U}$ onto the "opposite" group object;[31] in particular, $c_{\mathcal{U}}$ induces an algebra isomorphism from $\mathcal{U}$ onto the opposite algebra $\mathcal{U}^{\circ}$.

[31] That is, the object endowed with the multiplication $m'_{\mathcal{U}} = m_{\mathcal{U}}\circ\sigma$.

3.2.2. Since the functor $\mathcal{U} \mapsto \operatorname{Spec}^* \mathcal{U}$ commutes with finite products, it carries a coalgebra in groups to a group-valued S-functor. Indeed, for every S-scheme T, the elements

$$x \in \Gamma(T, \mathcal{U}_T)$$

belonging to $(\operatorname{Spec}^* \mathcal{U})(T)$ form a group under the multiplication of the algebra $\Gamma(T, \mathcal{U}_T)$, and the inverse of x is $c_{\mathcal{U}}(x)$. By 3.1.2.1:

Scholium 3.2.2.1. [32] *Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups, finite and locally free as an $\mathcal{O}_S$-module. Then the group-valued S-functor* $\operatorname{Spec}^* \mathcal{U}$ *is represented by the finite locally free S-group* $\operatorname{Spec} \mathcal{U}^*$.

Remark 3.2.2.2. Let $\mathcal{L}$ be an $\mathcal{O}_S$-Lie algebra and $\mathcal{U}(\mathcal{L})$ its enveloping algebra, that is, the sheaf on S associated with the presheaf assigning to every open subset V the enveloping algebra

$$U(\Gamma(V, \mathcal{L}))$$

of the Lie algebra $\Gamma(V, \mathcal{L})$.

Every homomorphism from $\mathcal{L}$ to the Lie algebra underlying an associative $\mathcal{O}_S$-algebra factors uniquely through the canonical morphism $\mathcal{L} \to \mathcal{U}(\mathcal{L})$. Besides the functoriality of $\mathcal{U}(\mathcal{L})$ in $\mathcal{L}$, this universal property also shows that the enveloping algebra of a product of Lie algebras is identified with the tensor product of their enveloping algebras.

In particular, the diagonal morphism $\delta : \mathcal{L} \to \mathcal{L} \times \mathcal{L}$ induces an algebra homomorphism

$$\Delta : \mathcal{U}(\mathcal{L}) \longrightarrow \mathcal{U}(\mathcal{L} \times \mathcal{L}) \simeq \mathcal{U}(\mathcal{L}) \otimes \mathcal{U}(\mathcal{L}).$$

The zero morphism $\mathcal{L} \to 0$ induces a homomorphism

$$\varepsilon : \mathcal{U}(\mathcal{L}) \longrightarrow \mathcal{U}(0) \simeq \mathcal{O}_S.$$

The isomorphism $x \mapsto -x$ from $\mathcal{L}$ onto the opposite Lie algebra $\mathcal{L}^\circ$ induces an anti-isomorphism c of the algebra $\mathcal{U}(\mathcal{L})$. One then checks readily that the multiplication m of $\mathcal{U}(\mathcal{L})$ makes $(\mathcal{U}(\mathcal{L}), \Delta)$ into an $\mathcal{O}_S$-coalgebra in groups with augmentation ε and antipode c.[33]

3.2.3. [34] Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups. We shall see that the group-valued S-functor

$$G = \operatorname{Spec}^* \mathcal{U}$$

is *very good*, in the sense of Exposé II, 4.6 and 4.10.

Let $\mathcal{M}$ be a free $\mathcal{O}_S$-module of rank r, and let $T \to S$ be an S-scheme. Since

$$I_T(\mathcal{M}) = \operatorname{Spec}(\mathcal{O}_T \oplus \mathcal{M}_T),$$

the morphism $\pi : I_T(\mathcal{M}) \to T$ is affine, and

$$\pi_*(\mathcal{U}_{I_T(\mathcal{M})}) = \mathcal{U}_T \otimes_{\mathcal{O}_T} \pi_*(\mathcal{O}_{I_T(\mathcal{M})}) = \mathcal{U}_T \otimes_{\mathcal{O}_T} (\mathcal{O}_T \oplus \mathcal{M}_T).$$

Consequently,

$$\Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}) \simeq \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} (\mathcal{O}(T) \oplus \Gamma(T, \mathcal{M}_T)). \tag{1}$$

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[32] The editors added this scholium, which is implicit in the original.

[33] The group-valued S-functor $\operatorname{Spec}^* \mathcal{U}(\mathcal{L})$ is not representable in general. Later, in 5.5, it will be shown that if S is a scheme of characteristic p, if $\mathcal{L}$ is finite locally free over $\mathcal{O}_S$, and if $\mathcal{U}_p(\mathcal{L})$ is its restricted enveloping algebra (see 5.3), then $\operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by a finite locally free S-group.

[34] The editors expanded this paragraph.

Let $(d_1, \dots, d_r)$ be a basis of $\mathcal{M}$. An element

$$u_0 + \sum_i u_i d_i \in \Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})})$$

belongs to $G(I_T(\mathcal{M}))$ if and only if

$$1 = \varepsilon\left(u_0 + \sum_i u_i d_i\right) = \varepsilon(u_0) + \sum_i \varepsilon(u_i) d_i$$

and

$$\left(u_0 + \sum_i u_i d_i\right) \otimes \left(u_0 + \sum_i u_i d_i\right) = \Delta\left(u_0 + \sum_i u_i d_i\right) = \Delta(u_0) + \sum_i \Delta(u_i) d_i.$$

Equivalently,

$$\begin{cases} \varepsilon(u_0) = 1, & \Delta u_0 = u_0 \otimes u_0 \qquad\qquad (\text{that is, } u_0 \in G(T)), \\ \varepsilon(u_i) = 0, & \Delta(u_i) = u_i \otimes u_0 + u_0 \otimes u_i \quad (i = 1, \dots, r). \end{cases} \tag{2}$$

Moreover, the morphism $G(I_T(\mathcal{M})) \to G(T)$ corresponding to the zero section of $I_T(\mathcal{M}) \to T$ is given by

$$u_0 + \sum_i u_i d_i \longmapsto u_0.$$

Combining this with (1) and (2), one sees that if $\mathcal{N}$ is a second free $\mathcal{O}_S$-module of finite rank, then the diagram of sets

$$\begin{tikzcd} G(I_T(\mathcal{M} \oplus \mathcal{N})) \arrow[r] \arrow[d] & G(I_T(\mathcal{N})) \arrow[d] \\ G(I_T(\mathcal{M})) \arrow[r] & G(T) \end{tikzcd}$$

is cartesian; that is, G satisfies condition (E) of Exposé II, 3.5.

Write

$$\operatorname{Prim}\Gamma(T, \mathcal{U}_T)$$

for the $\mathcal{O}(T)$-submodule of $\Gamma(T, \mathcal{U}_T)$ formed by the *primitive elements*, that is, the elements u satisfying, with the abuse of notation noted in 3.1.2,

$$\Delta u = u \otimes 1 + 1 \otimes u, \qquad \varepsilon(u) = 0.$$

[35]

Since $(\operatorname{Lie} G)(T)$ is the set of elements $u_0 + ud \in G(I_T)$ above the unit element $u_0 = 1$ of $G(T)$, there is an isomorphism of $\mathcal{O}(T)$-modules, functorial in T,[36]

$$(\operatorname{Lie} G)(T) \simeq \operatorname{Prim}\Gamma(T, \mathcal{U}_T).$$

On the other hand, (1) yields

$$\operatorname{Prim}\Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}) \simeq \operatorname{Prim}\Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})).$$

[35]The second condition follows from the first, since the first gives $u = (\mathrm{id} \otimes \varepsilon)\Delta(u) = u + \varepsilon(u)$, and hence $\varepsilon(u) = 0$.

[36]The $\mathcal{O}_S$-module structure on $\operatorname{Lie} G$ is defined in Exposé II, Proposition 3.6.

It follows that the natural morphism of $\mathcal{O}(I_T(\mathcal{M}))$-modules

$$(\mathrm{Lie}\, G)(T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})) \longrightarrow (\mathrm{Lie}\, G)(I_T(\mathcal{M}))$$

is an isomorphism; thus $\mathrm{Lie}\, G$ is a good $\mathcal{O}_S$-module (see Exposé II, Definition 4.4).

Therefore G is a good group-valued S-functor (see Exposé II, Definition 4.6), and by Exposé II, 4.7.2, $\mathrm{Lie}\, G$ carries an $\mathcal{O}_S$-bilinear Lie bracket satisfying the Jacobi identity. It remains to show that G is very good, that is, that the bracket on $(\mathrm{Lie}\, G)(T)$ satisfies $[u,u] = 0$ for every $u \in (\mathrm{Lie}\, G)(T)$; see Exposé II, 4.10.

Let u, v be two elements of $(\mathrm{Lie}\, G)(T)$, that is, two primitive elements of $\Gamma(T, \mathcal{U}_T)$. Put

$$I = \mathrm{Spec}\, \mathcal{O}_S[d]/(d^2), \qquad I' = \mathrm{Spec}\, \mathcal{O}_S[d']/(d'^2).$$

Since the composition law on $G(I \times I')$ is induced by the multiplication in $\mathcal{U}_{I \times I'}$, one has in $G(I \times I')$

$$\begin{aligned}(1+ud)&(1+vd')(1+ud)^{-1}(1+vd')^{-1}\\ &= (1+ud)(1+vd')(1-ud)(1-vd')\\ &= 1 + (uv - vu)dd'.\end{aligned}$$

By the description of the bracket $[u,v]$ given before Proposition 4.8 of Exposé II, it follows that

$$[u,v] = uv - vu,$$

where the right-hand side is the commutator of u and v in the algebra $\Gamma(T, \mathcal{U}_T)$. Hence $[u,u] = 0$, and we have proved:

Proposition. *Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups. The group-valued S-functor $G = \mathrm{Spec}^* \mathcal{U}$ is very good, and there is an isomorphism*

$$\mathrm{Lie}\, G \simeq \mathrm{Prim}\, \mathcal{W}(\mathcal{U})$$

of $\mathcal{O}_S$-Lie algebras, where $\mathrm{Prim}\, \mathcal{W}(\mathcal{U})$ *denotes the functor which assigns to every $T \to S$ the $\mathcal{O}(T)$-Lie algebra of primitive elements of $\mathcal{W}(\mathcal{U})(T) = \Gamma(T, \mathcal{U}_T)$.* [37]

3.3. Suppose finally that $\mathcal{U}$ is a *coalgebra in commutative groups*, that is, that the triangle

$$\text{(i)}^* \qquad \begin{tikzcd} \mathcal{U} \otimes \mathcal{U} \arrow[rr, "\sigma"] \arrow[dr, "m_{\mathcal{U}}"'] & & \mathcal{U} \otimes \mathcal{U} \arrow[dl, "m_{\mathcal{U}}"] \\ & \mathcal{U} & \end{tikzcd}$$

is commutative, or equivalently that $m_{\mathcal{U}}$ makes $\mathcal{U}$ into a commutative $\mathcal{O}_S$-algebra. Conditions (i), (ii), (iii), (iv), (v), (vi), (i)*, (ii)*, (iii)*, and (v)* then also say that $\mathcal{U}$ is a cogroup in the category of commutative $\mathcal{O}_S$-algebras. Thus, if in addition $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$-module, the affine S-scheme $\mathrm{Spec}\, \mathcal{U}$ is a commutative S-group scheme.

In this case the diagonal morphism Δ' of $\mathcal{O}_S[T, T^{-1}]$ sends T to $T \otimes T$. Consequently, morphisms of S-groups from $\mathrm{Spec}\, \mathcal{U}$ to $\mathbb{G}_{m,S}$ (Exposé I, 4.3.2) correspond bijectively to morphisms of unital $\mathcal{O}_S$-algebras

$$\varphi : \mathcal{O}_S[T, T^{-1}] \longrightarrow \mathcal{U}$$

such that

$$(\varphi \otimes \varphi) \circ \Delta' = \Delta_{\mathcal{U}} \circ \varphi.$$

[37] The editors added this proposition to summarize the preceding discussion.

In this situation, $\varepsilon_{\mathcal{U}} \circ \varphi$ is the neutral element of $\mathbb{G}_{m,S}(S)$, that is, the augmentation ε'. Such a morphism φ is determined by the image $\varphi(T)$, which must be an invertible element $x \in \mathcal{U}$ satisfying

$$\Delta_{\mathcal{U}} x = x \otimes x, \qquad \varepsilon_{\mathcal{U}}(x) = \varepsilon'(T) = 1.$$

Therefore

$$\mathrm{Hom}_{S\text{-gr.}}(\mathrm{Spec}\,\mathcal{U}, \mathbb{G}_{m,S}) \simeq (\mathrm{Spec}^* \mathcal{U})(S).$$

Since this formula remains valid after arbitrary base change, it gives

$$\mathrm{Spec}^* \mathcal{U} \simeq \mathrm{Hom}_{S\text{-gr.}}(\mathrm{Spec}\,\mathcal{U}, \mathbb{G}_{m,S}).$$

We have therefore proved:

Proposition 3.3.0. *If $\mathcal{U}$ is an $\mathcal{O}_S$-coalgebra in commutative groups, quasi-coherent as an $\mathcal{O}_S$-module, then the affine S-scheme $G = \mathrm{Spec}\,\mathcal{U}$ is a commutative S-group scheme, and there is an isomorphism of group-valued S-functors*

$$\mathrm{Spec}^* \mathcal{U} \simeq \mathrm{Hom}_{S\text{-}gr.}(G, \mathbb{G}_{m,S}).$$

If, moreover, $\mathcal{U}$ is locally free of finite type as an $\mathcal{O}_S$-module, then by 3.1.2.1 the group-valued S-functor $\mathrm{Spec}^* \mathcal{U}$ is represented by $\mathrm{Spec}\,\mathcal{U}^*$. Hence:

Proposition 3.3.1 (Cartier duality). *The functor*

$$\mathscr{A}(G) \longmapsto \mathscr{A}(G)^* = \mathcal{H}om_{\mathcal{O}_S\text{-}Mod.}(\mathscr{A}(G), \mathcal{O}_S)$$

induces a duality on the category of commutative finite locally free S-group schemes; it associates to G the S-group $\mathrm{Hom}_{S\text{-}gr.}(G, \mathbb{G}_{m,S})$.*

4. "Frobeniuseries"

Fix a prime number p, and let $(\mathbf{Sch}/\mathbb{F}_p)$ be the category of schemes of characteristic p, that is, schemes over the prime field $\mathbb{F}_p$. Following the general conventions of this seminar, we identify $(\mathbf{Sch}/\mathbb{F}_p)$ with a subcategory of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$ by means of the functor h of Exposé I, 1.1. We likewise use the isomorphism

$$\mathrm{Hom}(h_X, F) \xrightarrow{\sim} F(X)$$

defined there to identify these sets whenever X is an $\mathbb{F}_p$-scheme and F is an object of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$.

Notation 4.0. [38] *If T is an $\mathbb{F}_p$-scheme, a T-functor is a morphism $q : F \to T$ in $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$ with target T. For every T-scheme $r : X \to T$, the set of T-morphisms $X \to F$, that is, of $\mathbb{F}_p$-morphisms $s : X \to F$ satisfying $q \circ s = r$, is denoted*

$$q(r), \qquad q(X/T), \qquad F(r), \qquad \text{or} \qquad F(X/T).$$

It may also be denoted $F(X)$ when no confusion with $\mathrm{Hom}(h_X, F)$ can result.

*A duality of a category $\mathscr{C}$ is a pair (D, φ) consisting of a contravariant functor $D : \mathscr{C} \to \mathscr{C}$ and a natural isomorphism $\varphi : \mathrm{Id}_{\mathscr{C}} \to DD$, such that the isomorphisms $\varphi D : D \to DDD$ and $D\varphi^{-1} : DDD \to D$ are inverse to one another. The editors corrected $D\varphi$ to $D\varphi^{-1}$.

[38] The editors added the number 4.0 for later references.

4.1. For every scheme S of characteristic p, let $\mathrm{fr}(S)$, or simply fr, denote the endomorphism of S which is the identity on the underlying topological space and which sends a section x of $\mathcal{O}_S$ over an open subset to x^p.

The assignment

$$\mathrm{fr} : S \longmapsto \mathrm{fr}(S)$$

is an endomorphism of the identity functor of $(\mathbf{Sch}/\mathbb{F}_p)$.[39] This has the following consequences. If E is an $\mathbb{F}_p$-functor, the assignment which sends every $\mathbb{F}_p$-scheme S to the endomorphism $E(\mathrm{fr}(S))$ of $E(S)$ is a functorial endomorphism of E, denoted $\mathrm{fr}(E)$ or simply fr. This notation is compatible with the identification of $(\mathbf{Sch}/\mathbb{F}_p)$ with a subcategory of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$. Moreover, $E \mapsto \mathrm{fr}(E)$ is itself an endomorphism of the identity functor of $(\mathbf{Sch}/\mathbb{F}_p)^\wedge$, again denoted fr.[40]

For every $\mathbb{F}_p$-scheme S and every S-functor $q : X \to S$, let $X^{(p/S)}$, or $X^{(p)}$, denote the pullback of X along $\mathrm{fr}(S)$:

$$\begin{tikzcd} X^{(p/S)} \arrow[r, "\mathrm{pr}_X"] \arrow[d] & X \arrow[d, "q"] \\ S \arrow[r, "\mathrm{fr}(S)"] & S\,. \end{tikzcd}$$

The commutative square

$$\begin{tikzcd} X \arrow[r, "\mathrm{fr}(X)"] \arrow[d, "q"'] & X \arrow[d, "q"] \\ S \arrow[r, "\mathrm{fr}(S)"] & S \end{tikzcd}$$

induces an S-morphism

$$\mathrm{Fr}(X/S) : X \longrightarrow X^{(p/S)}$$

such that

$$\mathrm{fr}(X) = \mathrm{pr}_X \circ \mathrm{Fr}(X/S) :$$

$$\begin{tikzcd} X \arrow[dr, "\mathrm{Fr}(X/S)"] \arrow[drr, bend left, "\mathrm{fr}(X)"] \arrow[ddr, "q"'] & & \\ & X^{(p/S)} \arrow[r, "\mathrm{pr}_X"] \arrow[d] & X \arrow[d, "q"] \\ & S \arrow[r, "\mathrm{fr}(S)"] & S\,. \end{tikzcd}$$

We call $\mathrm{Fr}(X/S)$ the *Frobenius morphism of X relative to S.* Plainly the assignment

$$\mathrm{Fr} : X \longmapsto \mathrm{Fr}(X/S)$$

[39]Thus, for every morphism $f : Y \to X$ of $\mathbb{F}_p$-schemes, the following square commutes:

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[d, "\mathrm{fr}(Y)"'] & X \arrow[d, "\mathrm{fr}(X)"] \\ Y \arrow[r, "f"] & X\,. \end{tikzcd}$$

[40]The morphism $\mathrm{fr}(X)$ is called the *absolute* Frobenius morphism of X, to distinguish it from the relative Frobenius morphism $\mathrm{Fr}(X/S)$ introduced below.

is a functorial homomorphism.

[41] Let $r : T \to S$ be an S-scheme. For every

$$\varphi \in X(r) = \mathrm{Hom}_S(T, X),$$

one has the commutative diagram

$$\begin{tikzcd} X \arrow[r, "\mathrm{Fr}(X/S)"] \arrow[drr, "q"] & X^{(p/S)} \arrow[r, "\mathrm{pr}_X"] \arrow[d, "q^{(p/S)}"] & X \arrow[d, "q"] \\ T \arrow[u, "\varphi"] \arrow[r, "r"] & S \arrow[r, "\mathrm{fr}(S)"] & S\,. \end{tikzcd}$$

By the definition of $X^{(p/S)}$ as a fiber product, pr_X induces a bijection

$$X^{(p/S)}(r) = \mathrm{Hom}_S(T, X^{(p/S)}) \xrightarrow{\sim} \mathrm{Hom}_S(T, X) = X(\mathrm{fr}(S) \circ r).$$

On the other hand,

$$r \circ \mathrm{fr}(T) = \mathrm{fr}(S) \circ r,$$

because fr is an endomorphism of the identity functor. Hence

$$\mathrm{Fr}(X/S)(r) : X(r) \longrightarrow X^{(p/S)}(r)$$

is characterized by commutativity of

$$\begin{tikzcd} X(r) \arrow[r, "\mathrm{Fr}(X/S)(r)"] \arrow[d, "X(\mathrm{fr}(T))"'] & X^{(p/S)}(r) \arrow[d, "\wr"] \\ X(r \circ \mathrm{fr}(T)) \arrow[r, equal] & X(\mathrm{fr}(S) \circ r)\,. \end{tikzcd}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, then $X^{(p)}$ is the subscheme defined by the ideal $\mathcal{I}^{\{p\}}$ generated by the p-th powers of local sections of $\mathcal{I}$. Moreover, $\mathrm{Fr}(X/S)$ is the canonical immersion

$$\mathrm{Spec}(\mathcal{O}_S/\mathcal{I}) \longrightarrow \mathrm{Spec}(\mathcal{O}_S/\mathcal{I}^{\{p\}}).$$

4.1.1. [42] Let $t : T \to S$ be a base change and put

$$X_T = X \times_{q,t} T.$$

Consider the pullback of X_T along $\mathrm{fr}(T)$:

$$\begin{tikzcd} (X_T)^{(p/T)} \arrow[r] \arrow[d] & X_T \arrow[r] \arrow[d] & X \arrow[d, "q"] \\ T \arrow[r, "\mathrm{fr}(T)"] & T \arrow[r, "t"] & S\,. \end{tikzcd}$$

[41]The editors expanded the original in the remainder of this paragraph.
[42]The editors expanded the original in this paragraph.

Since
$$t \circ \mathrm{fr}(T) = \mathrm{fr}(S) \circ t,$$
$(X_T)^{(p/T)}$ identifies with the pullback of $X^{(p/S)}$ along t. Equivalently, there is a canonical isomorphism
$$X_T^{(p/T)} \xrightarrow{\sim} \left(X^{(p/S)}\right)_T.$$
Under this identification, $\mathrm{Fr}(X_T/T)$ identifies with the pullback $\mathrm{Fr}(X/S)_T$.

4.1.1.1. In particular, if $S = \operatorname{Spec} \mathbb{F}_p$, then $X^{(p/S)} = X$ and $\mathrm{Fr}(X/S) = \mathrm{fr}(X)$. Consequently, $X_T^{(p/T)}$ identifies with X_T, and $\mathrm{Fr}(X_T/T)$ with $\mathrm{fr}(X)_T$.

For example, if E is a set and E_T the constant T-scheme of type E, then
$$E_T^{(p/T)} \simeq E_T, \qquad \mathrm{Fr}(E_T/T) \simeq \mathrm{id}_{E_T}.$$

4.1.2. The functor $X \mapsto X^{(p/S)}$ plainly commutes with products. It therefore sends an S-group G to an S-group $G^{(p/S)}$. Since Fr is functorial,
$$\mathrm{Fr}(G/S) : G \longrightarrow G^{(p/S)}$$
is a homomorphism of S-groups. We denote its kernel by $\mathrm{Fr}\, G$.

If $r : T \to S$ is an S-scheme, the pointwise Frobenius diagram shows that the value of $\mathrm{Fr}\, G$ at r is the kernel of
$$G(\mathrm{fr}(T)) : G(r) \longrightarrow G(r \circ \mathrm{fr}(T)).$$

When $T = I_R$ is the scheme of dual numbers over an S-scheme R, the absolute Frobenius of I_R factors as
$$I_R, \xrightarrow{\text{can.}} R \xrightarrow{\mathrm{fr}(R)} R \xrightarrow{\quad s \quad} I_R$$
where s is the zero section. It follows that $(\mathrm{Fr}\, G)(I_R)$ contains the kernel $\mathrm{Lie}(G/S)(R)$ of
$$G(s) : G(I_R) \longrightarrow G(R),$$
and hence
$$\mathrm{Lie}(G/S) = \mathrm{Lie}(\mathrm{Fr}\, G/S).$$

4.1.3.

More generally, for every S-functor X, define the S-functor $X^{(p^n)}$ recursively in n by
$$X^{(p)} = X^{(p/S)} \qquad \text{and} \qquad X^{(p^n)} = \left(X^{(p^{n-1})}\right)^{(p)}.$$
Likewise, $\mathrm{Fr}^n(X/S)$, or simply Fr^n, denotes the composite functorial homomorphism
$$X \xrightarrow{\mathrm{Fr}(X/S)} X^{(p)} \xrightarrow{\mathrm{Fr}(X^{(p)}/S)} X^{(p^2)} \longrightarrow \cdots \longrightarrow X^{(p^{n-1})} \xrightarrow{\mathrm{Fr}(X^{(p^{n-1})}/S)} X^{(p^n)}.$$
By 4.1.1, $\mathrm{Fr}(X^{(p)}/S)$ coincides with $\mathrm{Fr}(X/S)^{(p)}$; in other words, the following diagram commutes:
$$\begin{tikzcd}
X^{(p)} \arrow[r] \arrow[d, "\mathrm{Fr}(X^{(p)}/S)"'] & X \arrow[d, "\mathrm{Fr}(X/S)"] \\
X^{(p^2)} \arrow[r] & X^{(p)}.
\end{tikzcd}$$

If G is a group S-functor, then $G^{(p^n)}$ is again a group S-functor and $\mathrm{Fr}^n(G/S)$ is a homomorphism of group S-functors.

Definition. *We denote the kernel of* $\mathrm{Fr}^n(G/S)$ *by* $\mathrm{Fr}^n G$*, and say that* G *has height* $\leq n$ *if* $\mathrm{Fr}^n(G/S)$ *is zero, that is, if* $\mathrm{Fr}^n G = G$.

Lemma. *The group subfunctor* $\mathrm{Fr}^n G$ *of* G *is characteristic: for every* S*-scheme* T*, every endomorphism* φ *of the group* T*-functor* G_T *induces an endomorphism of* $(\mathrm{Fr}^n G)_T$.

Proof. By 4.1.1, the constructions of $G^{(p^n)}$ and $\mathrm{Fr}^n(G/S)$ commute with base change. We may therefore suppose that $T = S$. The assertion then follows from the fact that $\mathrm{Fr}^n(G/S)$ is a functorial homomorphism. □

4.1.4. Examples.

a) Let M first be an "abstract" abelian group, and let

$$G = D_S(M)$$

be the diagonalizable group of type M (I, 4.4). For every S-scheme T,

$$G(T) = \mathrm{Hom}_{(\mathbf{Ab})}(M, \Gamma(T, \mathcal{O}_T)^\times).$$

Since G is the pullback of the diagonalizable group $D(M)$ over $\mathbb{F}_p$, $G^{(p)}$ identifies with G, and $\mathrm{Fr}(G/S)(T)$ identifies with the endomorphism $x \mapsto x^p$ of $G(T)$ (4.1.1). In particular, when $M = \mathbb{Z}$, one has $D_S(M) = \mathbb{G}_{m,S}$. Hence:

> $\mathrm{Fr}\,\mathbb{G}_{m,S}$ is the S-group $\boldsymbol{\mu}_{p,S}$ which associates to every S-scheme T the group of p-th roots of unity in $\Gamma(T, \mathcal{O}_T)^\times$.

b) Now let S be a scheme of characteristic p, and let $\mathcal{E}$ be a sheaf of modules on S. By I, 4.6.2 there is a canonical isomorphism

$$\mathcal{W}(\mathcal{E})^{(p)} \xrightarrow{\sim} \mathcal{W}(\mathcal{E}^{(p)}),$$

where $\mathcal{E}^{(p)}$ is the pullback of $\mathcal{E}$ along $\mathrm{fr}(S)$. For every S-scheme $\pi : T \to S$, the map $\mathrm{Fr}(\mathcal{W}(\mathcal{E})/S)(\pi)$ is determined, by 4.1 (†), by the commutative triangle

$$\begin{tikzcd} \Gamma(T, \pi^* \mathrm{fr}(S)^* \mathcal{E}) \arrow[rr, "\sim", "\text{can.}"'] & & \Gamma(T, \mathrm{fr}(T)^* \pi^* \mathcal{E}) \\ & \Gamma(T, \pi^* \mathcal{E}) \arrow[ul, "\mathrm{Fr}(\mathcal{W}(\mathcal{E})/S)(\pi)"] \arrow[ur, "f'"'] & \end{tikzcd},$$

where f' is the map induced by $\mathrm{fr}(T)$.

In particular, if $\mathcal{E} = \mathcal{O}_S$, then $\mathcal{W}(\mathcal{E})$ identifies with the additive group $\mathbb{G}_{a,S}$. In this case $\mathcal{E}^{(p)} = \mathcal{E} = \mathcal{O}_S$, and the Frobenius morphism $\mathrm{Fr}(\mathbb{G}_{a,S}/S)$ sends $x \in \Gamma(T, \mathcal{O}_T)$ to x^p. Thus:

> $\mathrm{Fr}\,\mathbb{G}_{a,S}$ is the S-group $\boldsymbol{\alpha}_{p,S}$ which associates to every S-scheme T the group
>
> $$\{x \in \Gamma(T, \mathcal{O}_T) \mid x^p = 0\}.$$

c) Similarly, for every quasi-coherent $\mathcal{O}_S$-algebra $\mathcal{A}$, the scheme $(\operatorname{Spec}\mathcal{A})^{(p)}$ identifies with $\operatorname{Spec}\mathcal{A}^{(p)}$, the spectrum of the pullback of $\mathcal{A}$ along $\mathrm{fr}(S)$. If π denotes the endomorphism $x \mapsto x^p$ of the sheaf of rings $\mathcal{O}_S$, then

$$\mathcal{A}^{(p)} = \mathcal{A} \otimes_\pi \mathcal{O}_S, \tag{44}$$

and $\mathrm{Fr}((\operatorname{Spec}\mathcal{A})/S)$ is induced by the morphism of $\mathcal{O}_S$-algebras

$$\mathcal{A} \otimes_\pi \mathcal{O}_S \longrightarrow \mathcal{A}, \qquad a \otimes_\pi x \longmapsto a^p x.$$

[43]

Finally, for every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, there are canonical isomorphisms

$$\mathcal{V}(\mathcal{E})^{(p)} \xrightarrow{\sim} \mathcal{V}(\mathcal{E}^{(p)}) \qquad \text{and} \qquad \mathcal{S}(\mathcal{E})^{(p)} \xrightarrow{\sim} \mathcal{S}(\mathcal{E}^{(p)}),$$

where $\mathcal{S}(\mathcal{E})$ denotes the symmetric algebra of the $\mathcal{O}_S$-module $\mathcal{E}$.

d) Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra (3.1), and let T be a scheme of characteristic p. If $\mathcal{U}^{(p/S)}$, or $\mathcal{U}^{(p)}$, denotes the pullback of the coalgebra $\mathcal{U}$ along $\mathrm{fr}(S)$, then, as in *b)*, there is a canonical isomorphism

$$(\operatorname{Spec}^* \mathcal{U})^{(p)} \xrightarrow{\sim} \operatorname{Spec}^* \mathcal{U}^{(p)}.$$

If $\mathcal{U}$ is a group coalgebra, then for an S-scheme T, the value of $\mathrm{Fr}(\operatorname{Spec}^* \mathcal{U})$, that is, of the kernel of the Frobenius morphism

$$\operatorname{Spec}^* \mathcal{U} \longrightarrow (\operatorname{Spec}^* \mathcal{U})^{(p)},$$

is the set of elements γ of

$$(\operatorname{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1,\ \Delta_{\mathcal{U}_T} x = x \otimes x\}$$

such that the image in $\Gamma(T, \mathcal{U}_T \otimes_{\mathrm{fr}(T)} \mathcal{O}_T)$ of the element

$$\gamma \otimes_{\mathrm{fr}(T)} 1 \in \Gamma(T, \mathcal{U}_T) \otimes_{\mathrm{fr}(T)} \mathcal{O}(T)$$

is equal to 1.

4.2. [44]

We now consider a construction closely related to the preceding one. Let S be a scheme of characteristic p, let X be an S-scheme, and let X^p_S be the product, in $(\mathbf{Sch}/S)$, of p copies of X.

Denote by $U^p(X)$ the open subscheme of X^p_S which is the union of the products U^p_S as U ranges over the affine open subsets of X. Thus a point x of X^p_S belongs to $U^p(X)$ if and only if its projections $\mathrm{pr}_i\, x$ on the factors of X^p_S belong to a common affine open subset of X. For example, if every finite subset of X is contained in an affine open subset, then

$$U^p(X) = X^p_S.$$

The symmetric group $\mathfrak{S}_p$ on p letters acts on X^p_S by permuting the factors and leaves $U^p(X)$ stable. We call the quotient of X^p_S by $\mathfrak{S}_p$ in the category of ringed spaces the *p-fold symmetric product of* X, and denote it by $\Sigma^p X$. Let

$$q(X), \text{ or simply } q : X^p_S \longrightarrow \Sigma^p X$$

[43] $\mathcal{A} \otimes_\pi \mathcal{O}_S$ denotes the $\mathcal{O}_S$-algebra obtained by extension of scalars along $\pi : \mathcal{O}_S \to \mathcal{O}_S$; thus $ax \otimes_\pi 1 = a \otimes_\pi x^p$ and $x \cdot (a \otimes_\pi 1) = a \otimes_\pi x$.

[44] For the material in 4.2 and 4.3, see also [DG70], IV, §3, nos. 4–6.

be the canonical projection.

The morphism q maps $U^p(X)$ onto an open subset $V^p(X)$ of the symmetric product, which may be described as follows (cf. V, 4.1). The structure sheaf of $\Sigma^p X$ induces a scheme structure on $V^p(X)$, and the morphism

$$q'(X) : U^p(X) \longrightarrow V^p(X)$$

induced by $q(X)$ is affine, indeed integral. As U ranges over the affine open subsets of X which map into a varying affine open V of S, the schemes $\Sigma^p U$ form an affine open covering of $V^p(X)$. If R is the affine algebra of V, and A that of U, the affine algebra of $\Sigma^p U$ is the subalgebra $\Sigma^p A$ of $\bigotimes_R^p A$ consisting of the symmetric tensors.

Now consider the diagonal morphism

$$\delta : X \longrightarrow U^p(X).$$

If $V = \operatorname{Spec} R$ is an affine open subset of S and $U = \operatorname{Spec} A$ is an affine open subset of X lying over V, the restriction of δ to U is defined by the algebra homomorphism

$$\eta : \bigotimes_R^p A \longrightarrow A, \qquad a_1 \otimes \cdots \otimes a_p \longmapsto a_1 a_2 \cdots a_p.$$

Consequently, if N denotes the symmetrization operator,

$$\begin{aligned} \eta(N(a_1 \otimes \cdots \otimes a_p)) &= \eta\left(\sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}\right) \\ &= p!\, a_1 \cdots a_p = 0. \end{aligned}$$

In other words, η vanishes on the subspace

$$N\left(\bigotimes_R^p A\right) \subseteq \Sigma^p A$$

consisting of symmetrized tensors. Moreover, if f is a symmetric tensor, then plainly $N(fa) = fN(a)$. Hence $N(\bigotimes_R^p A)$ is an ideal of $\Sigma^p A$. We put

$$U^{[p/S]} = \operatorname{Spec}\left(\Sigma^p A / N\left(\bigotimes_R^p A\right)\right).$$

This is a closed subscheme of $\Sigma^p(U) = V^p(U)$. As U ranges over the affine open subsets of X which map into a varying affine open V of S, the schemes $U^{[p/S]}$ glue to a closed subscheme of $V^p(X)$, denoted $X^{[p/S]}$.

If $i(X)$ denotes the inclusion of $X^{[p/S]}$ in $V^p(X)$, the preceding calculation shows that $q'(X) \circ \delta$ factors through $X^{[p/S]}$. We therefore obtain a morphism

$$F^{[p]}(X/S) : X \longrightarrow X^{[p/S]}.$$

[45] The factorization is displayed by the commutative diagram

[45] In the original, this morphism, respectively the relative Frobenius morphism, was denoted F', respectively F.

$$\begin{tikzcd}
X^p_S \arrow[d, "q(X)"'] & \supset & \mathrm{U}^p(X) \arrow[d, "q'(X)"'] & X \arrow[l, "\delta(X)"'] \arrow[d, "F^{[p]}(X/S)"] \\
\Sigma^p(X) & \supset & \mathrm{V}^p(X) & X^{[p/S]}. \arrow[l, "i(X)"']
\end{tikzcd}$$

It is clear that $X^{[p/S]}$ is functorial in X, and that

$$F^{[p]} : X \longmapsto F^{[p]}(X/S)$$

is a functorial homomorphism.

4.2.1.

The schemes $X^{[p/S]}$ and $X^{(p/S)}$ are naturally related. Let V be an affine open subset of S, with affine algebra R, and let U be an affine open subset of X lying over V, with affine algebra A. If π denotes the endomorphism $x \mapsto x^p$ of R, the affine algebra of $U^{(p/S)}$ is $A \otimes_\pi R$. The map

$$a \otimes_\pi \lambda \longmapsto \left(\lambda a \otimes \cdots \otimes a \bmod N\left(\bigotimes_R^p A\right)\right)$$

is a homomorphism of R-algebras

$$A \otimes_\pi R \longrightarrow \Sigma^p A/N\left(\bigotimes_R^p A\right).$$

It induces a morphism

$$\varphi(U) : U^{[p/S]} \longrightarrow U^{(p/S)}$$

satisfying

$$\varphi(U) \circ F^{[p]}(U/S) = \mathrm{Fr}(U/S).$$

Gluing these morphisms gives the commutative triangle

$$\begin{tikzcd}
& X \arrow[dl, "F^{[p]}(X/S)"'] \arrow[dr, "\mathrm{Fr}(X/S)"] & \\
X^{[p/S]} \arrow[rr, "\varphi(X)"] & & X^{(p/S)}.
\end{tikzcd}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, then $F^{[p]}(X/S)$ identifies with the identity morphism of X. Thus $\varphi(X)$ is the canonical immersion

$$\mathrm{Spec}(\mathcal{O}_S/\mathcal{I}) \longrightarrow \mathrm{Spec}(\mathcal{O}_S/\mathcal{I}^{\{p\}}).$$

This shows that $\varphi(X)$ need not be an isomorphism.

On the other hand, when M is a free R-module, the map

$$M \otimes_\pi R \longrightarrow \Sigma^p M/N\left(\bigotimes_R^p M\right),$$

$$m \otimes_\pi \lambda \longmapsto \left(\lambda m \otimes \cdots \otimes m \bmod N\left(\bigotimes_R^p M\right)\right)$$

is plainly bijective. It remains bijective when M is R-flat, because every flat module is a filtered direct limit of free modules.[46] It follows that

$$\boxed{\varphi(X) : X^{[p/S]} \longrightarrow X^{(p/S)} \text{ is an isomorphism if } X \text{ is flat over } S.}$$

4.3.

Finally, let G be an S-group scheme in abelian groups. The composite morphism

$$\mu(G) : U^p(G) \hookrightarrow G^p_S \longrightarrow G,$$

defined by multiplication, factors through $V^p(G)$. Thus there is a morphism

$$\nu(G) : V^p(G) \longrightarrow G$$

such that

$$\nu(G) \circ q'(G) = \mu(G).$$

The resulting commutative diagram is

$$\begin{tikzcd}
G & & \mathrm{U}^p(G) \arrow[ll, "\mu(G)"'] \arrow[dl, "q'(G)"] & & G \arrow[ll, "\delta(G)"'] \arrow[dl, "F^{[p]}(G/S)"'] \arrow[ddll, bend left, thick, "\mathrm{Fr}(G/S)"] \\
& \mathrm{V}^p(G) \arrow[ul, "\nu(G)"] & & G^{[p/S]} \arrow[ll, thick, "i(G)"] \arrow[dl, "\varphi(G)"] & \\
& & G^{(p/S)} \arrow[uull, dotted, bend left, "\mathrm{Ver}(G/S)"] & &
\end{tikzcd}.$$

When G is flat over S, $\varphi(G)$ is an isomorphism, and we may define the morphism called the *Verschiebung*

$$\mathrm{Ver}(G/S) : G^{(p/S)} \longrightarrow G$$

by

$$\mathrm{Ver}(G/S) = \nu(G) \circ i(G) \circ \varphi(G)^{-1}.$$

As G ranges over the flat commutative S-group schemes, the assignment

$$\mathrm{Ver} : G \longmapsto \mathrm{Ver}(G/S)$$

is plainly a functorial homomorphism. Consequently, $\mathrm{Ver}(G/S)$ is a group homomorphism.

For every S-scheme T, the composite

$$G(T) \xrightarrow{\delta(G)(T)} U^p(G)(T) \xrightarrow{\mu(G)(T)} G(T)$$

sends $x \in G(T)$ to $p \cdot x$. We may therefore write $p \cdot \mathrm{id}_G$ in place of $\mu(G) \circ \delta(G)$, obtaining the classical formula

$$\mathrm{Ver}(G/S) \circ \mathrm{Fr}(G/S) = p \cdot \mathrm{id}_G\,. \tag{*}$$

Examples 4.3.1.

[46] D. Lazard, *Sur les modules plats*, C. R. Acad. Sci. Paris **258** (1964), 6313–6316. See also D. Lazard, Bull. Soc. Math. France **97** (1969), 81–128, or [BAlg], X, §1.6, Th. 1.

a) If G is a constant S-scheme in abelian groups, then $\mathrm{Fr}(G/S)$ identifies with the identity morphism of G (cf. 4.1.1.1). Hence

$$\mathrm{Ver}(G/S) = p \cdot \mathrm{id}_G \,.$$

b) If G is the diagonalizable S-group of type M, then $\mathrm{Fr}(G/S) = p \cdot \mathrm{id}_G$ by 4.1.4 *a).* It follows at once that $\mathrm{Ver}(G/S)$ is the identity morphism of G.

c) If $\mathcal{E}$ is a flat $\mathcal{O}_S$-module and $G = \mathcal{V}(\mathcal{E})$, then $\mathrm{Ver}(G/S)$ is zero, as is $p \cdot \mathrm{id}_G$. In Exposé VII$_B$ we shall see that a commutative algebraic group G over a field k is "unipotent" if and only if, for some n, the composite homomorphism

$$G^{(p^n)} \xrightarrow{\mathrm{Ver}(G^{(p^{n-1})}/S)} G^{(p^{n-1})} \longrightarrow \cdots \longrightarrow G^{(p)} \xrightarrow{\mathrm{Ver}(G/S)} G$$

is zero. Here

$$G^{(p^n)} = \big(G^{(p^{n-1})}\big)^{(p)}$$

as in 4.1.3.[47]

4.3.2.

Since $\mathrm{Ver} : G \mapsto \mathrm{Ver}(G/S)$ is a functorial homomorphism on the category of flat commutative S-group schemes, the square

$$\begin{array}{ccc} G^{(p)} & \xrightarrow{\mathrm{Ver}(G/S)} & G \\ {\scriptstyle \mathrm{Fr}(G/S)^{(p)}}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{Fr}(G/S)} \\ G^{(p^2)} & \xrightarrow{\mathrm{Ver}(G^{(p)}/S)} & G^{(p)} \end{array}$$

commutes. Here $\mathrm{Fr}(G/S)^{(p)}$ denotes the pullback of $\mathrm{Fr}(G/S)$ along the base change $\mathrm{fr}(S)$. By 4.1.1,

$$\mathrm{Fr}(G/S)^{(p)} = \mathrm{Fr}(G^{(p)}/S).$$

Applying formula $(*)$ of 4.3 to $G^{(p)}$, we obtain

$$\mathrm{Fr}(G/S) \circ \mathrm{Ver}(G/S) = \mathrm{Ver}(G^{(p)}/S) \circ \mathrm{Fr}(G^{(p)}/S) = p \cdot \mathrm{id}_{G^{(p)}} \,. \qquad (**)$$

4.3.3.

Suppose finally that G is a commutative S-group which is finite and locally free. Let $\mathcal{A}$ be its affine $\mathcal{O}_S$-algebra, and let π be the endomorphism of the sheaf of rings $\mathcal{O}_S$ which sends a section x to x^p.[48] Denote by $\Sigma^p\mathcal{A}$ the subalgebra of $\bigotimes^p_{\mathcal{O}_S} \mathcal{A}$ consisting of the sections invariant under the symmetric-group action, and by

$$i(\mathcal{A}) : \Sigma^p\mathcal{A} \longrightarrow \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A}$$

its inclusion.

Let

$$\Delta^p(\mathcal{A}) : \mathcal{A} \longrightarrow \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A}$$

[47] This criterion is not stated explicitly in Exposé VII$_B$; see [DG70], IV, §3, Prop. 4.11.

[48] The editors changed the order of presentation, introducing first the objects which occur in the diagram below.

be the morphism obtained by iterating the diagonal morphism of the coalgebra $\mathcal{A}$. It corresponds to the multiplication morphism $U^p(G) = G^p_S \to G$. By the beginning of 4.3, $\Delta^p(\mathcal{A})$ factors through $\Sigma^p\mathcal{A}$; that is, it induces a morphism

$$a(\mathcal{A}) : \mathcal{A} \longrightarrow \Sigma^p\mathcal{A}$$

such that

$$i(\mathcal{A}) \circ a(\mathcal{A}) = \Delta^p(\mathcal{A}).$$

On the other hand, let $S^p(\mathcal{A})$ be the homogeneous component of degree p in the symmetric algebra of $\mathcal{A}$, and let

$$q(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A} \longrightarrow S^p(\mathcal{A})$$

be the canonical projection. The multiplication

$$m^p(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A} \longrightarrow \mathcal{A}$$

factors through $S^p(\mathcal{A})$; equivalently, it induces a map

$$b(\mathcal{A}) : S^p(\mathcal{A}) \longrightarrow \mathcal{A}$$

such that

$$b(\mathcal{A}) \circ q(\mathcal{A}) = m^p(\mathcal{A}).$$

Since $\Sigma^p\mathcal{A}$ is the affine algebra of $V^p(G)$, the composite $i(G) \circ \varphi(G)^{-1}$ induces, again by the beginning of 4.3, an algebra homomorphism

$$r(\mathcal{A}) : \Sigma^p\mathcal{A} \longrightarrow \mathcal{A} \otimes_\pi \mathcal{O}_S.$$

It vanishes on sections of the form

$$\sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}$$

and sends $a \otimes \cdots \otimes a$ to $a \otimes_\pi 1$. Similarly, $j(\mathcal{A})$ is the morphism of $\mathcal{O}_S$-modules

$$a \otimes_\pi 1 \longmapsto q(a \otimes \cdots \otimes a).$$

These maps form the commutative diagram

$$(\mathcal{A}) \qquad \begin{tikzcd} \mathcal{A} \arrow[rr, "\Delta^p(\mathcal{A})"] \arrow[dr, "a(\mathcal{A})"'] & & \bigotimes_{\mathcal{O}_S}^{p} \mathcal{A} \arrow[rr, "m^p(\mathcal{A})"] \arrow[dr, "q(\mathcal{A})"] & & \mathcal{A} \\ & \Sigma^p\mathcal{A} \arrow[ur, "i(\mathcal{A})"] \arrow[dr, "r(\mathcal{A})"'] & & \mathrm{S}^p(\mathcal{A}) \arrow[ur, "b(\mathcal{A})"'] & \\ & & \mathcal{A} \otimes_\pi \mathcal{O}_S \arrow[ur, "j(\mathcal{A})"'] & & \end{tikzcd}$$

.

The composite $r(\mathcal{A}) \circ a(\mathcal{A})$ corresponds to the Verschiebung morphism $\mathrm{Ver}(G/S)$, while $b(\mathcal{A}) \circ j(\mathcal{A})$ corresponds to the Frobenius morphism $\mathrm{Fr}(G/S)$.

The preceding commutative diagram $(\mathcal{A})$ is self-dual. Indeed, let D be the functor which associates to each $\mathcal{O}_S$-module $\mathcal{M}$ its dual

$$D\mathcal{M} = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{M}, \mathcal{O}_S).$$

The image of diagram $(\mathcal{A})$ under D is precisely diagram $(D\mathcal{A})$. The morphisms

$$Dr(\mathcal{A}), \quad Da(\mathcal{A}), \quad Dj(\mathcal{A}), \quad Db(\mathcal{A})$$

identify respectively with

$$j(D\mathcal{A}), \quad b(D\mathcal{A}), \quad r(D\mathcal{A}), \quad a(D\mathcal{A}).$$

By 3.3.1, it follows that:

> *In the category of commutative S-groups which are finite and locally free, Cartier duality interchanges the Frobenius morphism and the Verschiebung.*

[49]

5. Restricted p-Lie Algebras

We first recall some results from the *Séminaire Sophus Lie.*[50]

5.1.

Let p be a prime number, let R be a commutative ring of characteristic p, and let A be an associative R-algebra, not necessarily commutative. For $a, b \in A$, put

$$[a, b] = ab - ba, \qquad a \cdot b = L_a(b) = R_b(a).$$

Then

$$\begin{aligned}(\mathrm{ad}\, x^p)(y) &= [x^p, y] \\ &= (L_x^p - R_x^p)(y) \\ &= (L_x - R_x)^p(y) \\ &= (\mathrm{ad}\, x)^p(y),\end{aligned}$$

which gives Jacobson's first formula:

$$\mathrm{ad}(x^p) = (\mathrm{ad}\, x)^p. \tag{i}$$

If $a_1, \dots, a_p$ are arbitrary elements of A, and N is the symmetrization operator (cf. 4.2), then

$$\begin{aligned}N(a_1 \otimes \cdots \otimes a_p) &= \sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \cdots a_{\sigma(p)} \\ &= \sum_{\tau \in \mathfrak{S}_{p-1}} [a_{\tau(1)}, [a_{\tau(2)}, [\cdots, [a_{\tau(p-1)}, a_p] \cdots]]].\end{aligned} \tag{*}$$

Here σ ranges over the permutations of p letters and τ over those of $p-1$ letters.

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[49] See also [DG70], IV, §3, 4.9.

[50] See P. Cartier, *Exemples d'hyperalgèbres*, Séminaire Sophus Lie 1955/56, Exp. 3.

Indeed, the final expression is equal to

$$\sum_{\tau}\sum_{r=0}^{p-1}\sum_{i_1<\cdots<i_r}(-1)^s a_{\tau(i_1)}\cdots a_{\tau(i_r)}a_p a_{\tau(j_s)}\cdots a_{\tau(j_1)},$$

where τ ranges over the permutations of $p-1$ letters; $i_1,\ldots,i_r$ ranges over the strictly increasing sequences in $[1,p-1]$; and $j_1,\ldots,j_s$ is the strictly increasing sequence of those integers of $[1,p-1]$ which are different from $i_1,\ldots,i_r$. For fixed r, the sum of the terms

$$(-1)^s a_{\tau(i_1)}\cdots a_{\tau(i_r)}a_p a_{\tau(j_s)}\cdots a_{\tau(j_1)}$$

is plainly

$$(-1)^s\binom{p-1}{s}\sum_{\rho}a_{\rho(1)}\cdots a_{\rho(r)}a_p a_{\rho(r+1)}\cdots a_{\rho(p-1)},$$

where ρ ranges over the permutations of $p-1$ letters. But

$$(-1)^s\binom{p-1}{s}=1\quad\text{in }\mathbb{F}_p,$$

because in $\mathbb{F}_p[x]$,

$$(x-1)^p=x^p-1=(x-1)(x^{p-1}+\cdots+1),$$

and hence

$$(x-1)^{p-1}=x^{p-1}+\cdots+1.$$

This proves $(*)$.

On the other hand, if $x_0,x_1\in A$, then

$$(x_0+x_1)^p=x_0^p+x_1^p+\sum_z x_{z(1)}x_{z(2)}\cdots x_{z(p)},$$

where z ranges over the nonconstant maps $[1,p]\to\{0,1\}$. It follows that

$$\begin{aligned}(x_0+x_1)^p&=x_0^p+x_1^p\\&\quad+\sum_{0<r<p}\frac{1}{r!(p-r)!}N(\underbrace{x_0,\ldots,x_0}_{r},\underbrace{x_1,\ldots,x_1}_{p-r}).\end{aligned}$$

[51] By $(*)$,

$$\begin{aligned}&N(\underbrace{x_0,\ldots,x_0}_{r},\underbrace{x_1,\ldots,x_1}_{p-r})\\&\quad=r!(p-1-r)!\sum_t[x_{t(1)},[x_{t(2)},[\cdots,[x_{t(p-1)},x_1]\cdots]]],\end{aligned}$$

where t ranges over the maps $[1,p-1]\to\{0,1\}$ which take the value 0 exactly r times. We obtain Jacobson's second formula:

$$\begin{aligned}(x_0+x_1)^p&=x_0^p+x_1^p\\&\quad-\sum_{0<r<p}\sum_t\frac{1}{r}[x_{t(1)},[x_{t(2)},[\cdots,[x_{t(p-1)},x_1]\cdots]]],\end{aligned}\tag{ii}$$

where t has the same meaning.

[51] The editors inserted the following explanation from [DG70], III.7, 3.2.

5.2.

Let now $\mathfrak{g}$ be an R-Lie algebra. A map

$$x \longmapsto x^{(p)}$$

from $\mathfrak{g}$ to itself is said to make $\mathfrak{g}$ a *restricted p-Lie algebra over R* if the following conditions hold:

$$(0) \quad (\lambda x)^{(p)} = \lambda^p x^{(p)} \quad (\lambda \in R,\ x \in \mathfrak{g}),$$

$$(i) \quad \operatorname{ad} x^{(p)} = (\operatorname{ad} x)^p \quad (x \in \mathfrak{g}),$$

and

$$(ii) \quad (x_0 + x_1)^{(p)} = x_0^{(p)} + x_1^{(p)} - \sum_{0<r<p} \sum_t \frac{1}{r} [x_{t(1)}, [x_{t(2)}, [\cdots, [x_{t(p-1)}, x_1] \cdots]]],$$

where t ranges over the maps $[1, p-1] \to \{0, 1\}$ which take the value 0 exactly r times, and $x_0, x_1 \in \mathfrak{g}$. The map $x \mapsto x^{(p)}$ is called the *symbolic p-th power*.

For example, if A is an associative R-algebra, 5.1 shows that the R-module underlying A, endowed with

$$[x, y] = xy - yx, \qquad x^{(p)} = x^p,$$

is a p-Lie algebra, denoted A_{Lie}. We call it the p-Lie algebra underlying A.

We shall chiefly consider restricted p-Lie subalgebras of algebras of the form A_{Lie}. Here is one example. Let S be a scheme of characteristic $p > 0$, and let X be an S-scheme. Recall that a derivation of X over S is an endomorphism D of the sheaf of abelian groups $\mathcal{O}_X$ such that

$$D(\lambda s) = \lambda D(s), \qquad D(st) = (Ds)t + s(Dt),$$

whenever λ, respectively s, t, are sections of $\mathcal{O}_S$, respectively $\mathcal{O}_X$, on open subsets for which the formulas make sense. Leibniz's formula

$$D^n(st) = \sum_{i=0}^{n} \binom{n}{i} (D^i s)(D^{n-i} t)$$

shows that D^p is again a derivation of X over S, since

$$\binom{p}{i} \equiv 0 \pmod{p} \qquad (i \neq 0, p).$$

It follows that:

> *The algebra* $\mathrm{Der}_{X/S}$ *of derivations of X over S is a restricted p-Lie subalgebra of the* $\Gamma(S, \mathcal{O}_S)$*-algebra of differential operators of X over S.*

5.2.1. If $\mathfrak{g}$ and $\mathfrak{h}$ are two restricted p-Lie algebras, a *homomorphism*

$$h : \mathfrak{g} \longrightarrow \mathfrak{h}$$

is an R-linear map such that, for $x, y \in \mathfrak{g}$,

$$h([x, y]) = [h(x), h(y)], \qquad h(x^{(p)}) = h(x)^{(p)}.$$

The composite of two homomorphisms is again a homomorphism, so we may speak of the *category of restricted p-Lie algebras over R.*

If $(X, \mathcal{R})$ is a ringed space, we say that an $\mathcal{R}$-module $\mathfrak{g}$ is endowed with the structure of a restricted p-Lie algebra over $\mathcal{R}$ if, for every open subset U, the module $\Gamma(U, \mathfrak{g})$ is a restricted p-Lie algebra over $\Gamma(U, \mathcal{R})$, and if all restriction maps are homomorphisms.

5.3.

We are now interested in the left adjoint of the functor

$$A \longmapsto A_{\mathrm{Lie}}$$

of 5.2. Let $\mathfrak{g}$ be a restricted p-Lie algebra over a ring R of characteristic p, let $U(\mathfrak{g})$ be the enveloping algebra of the Lie algebra underlying $\mathfrak{g}$ (cf. [BLie], I, §2.1), and let

$$i_{\mathfrak{g}} : \mathfrak{g} \longrightarrow U(\mathfrak{g})$$

(or simply i) be the canonical map.

Let A be a unital associative R-algebra. For every Lie algebra homomorphism

$$\varphi : \mathfrak{g} \longrightarrow A_{\mathrm{Lie}},$$

there is a unique homomorphism of unital R-algebras

$$\psi : U(\mathfrak{g}) \longrightarrow A$$

such that $\psi \circ i = \varphi$. Moreover, φ is a homomorphism of restricted p-Lie algebras if and only if ψ vanishes on all the elements

$$i(x)^p - i(x^{(p)}), \qquad x \in \mathfrak{g}.$$

Definition. The quotient of $U(\mathfrak{g})$ by the two-sided ideal generated by the elements

$$i(x)^p - i(x^{(p)})$$

is denoted $U_p^R(\mathfrak{g})$, or simply $U_p(\mathfrak{g})$. We denote by $j_{\mathfrak{g}}$, or simply j, the composite

$$\mathfrak{g} \xrightarrow{i} U(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g}).$$

The algebra $U_p(\mathfrak{g})$ is called the *restricted enveloping algebra* of $\mathfrak{g}$.

The preceding discussion gives the following result.

Proposition. *For every unital associative R-algebra A and every homomorphism of restricted p-Lie algebras*

$$\varphi : \mathfrak{g} \longrightarrow A_{\mathrm{Lie}},$$

there is a unique homomorphism of unital algebras

$$\psi : U_p(\mathfrak{g}) \longrightarrow A$$

such that $\psi \circ j = \varphi$. In other words, the functor $\mathfrak{g} \mapsto U_p(\mathfrak{g})$ is left adjoint to the forgetful functor $A \mapsto A_{\mathrm{Lie}}$.

5.3.1.

With the notation of 5.3, put

$$\beta(x) = i(x)^p - i(x^{(p)}).$$

For every $y \in \mathfrak{g}$, formulas 5.1 (i) and 5.2 (i) give

$$\begin{aligned}\beta(x)i(y) &= i(y)\beta(x) + [\beta(x), i(y)]\\ &= i(y)\beta(x) + (\operatorname{ad} i(x))^p i(y) - i((\operatorname{ad} x)^p y)\\ &= i(y)\beta(x).\end{aligned}$$

Thus $\beta(x)$ belongs to the center of $U(\mathfrak{g})$; in particular, the left ideal generated by the elements $\beta(x)$ is already two-sided.

On the other hand,

$$\beta(\lambda x) = \lambda^p \beta(x) \qquad (\lambda \in R),$$

and formulas 5.1 (ii) and 5.2 (ii) give

$$\beta(x+y) = \beta(x) + \beta(y) \qquad (x, y \in \mathfrak{g}).$$

Consequently, if (x_α) is a family of generators of the R-module $\mathfrak{g}$, the left ideal generated by all $\beta(x)$ is already generated by the elements $\beta(x_\alpha)$.

5.3.2. Proposition.[52] *Let $\mathfrak{g}$ be an R-Lie algebra whose underlying R-module is free with basis (x_α). Then the restricted p-Lie algebra structures on $\mathfrak{g}$ correspond bijectively to the families (y_α) of elements of $\mathfrak{g}$ such that*

$$\operatorname{ad} y_\alpha = (\operatorname{ad} x_\alpha)^p.$$

Indeed, suppose that $\mathfrak{g}$ is endowed with a restricted p-Lie algebra structure $x \mapsto x^{(p)}$. By 5.2 (i), (0), and (ii), the elements

$$y_\alpha = x_\alpha^{(p)}$$

satisfy

$$\operatorname{ad} y_\alpha = (\operatorname{ad} x_\alpha)^p,$$

and they determine the restricted p-Lie algebra structure.

We prove the converse. Since $\mathfrak{g}$ is a free R-module, the canonical map

$$i : \mathfrak{g} \longrightarrow U(\mathfrak{g})$$

is injective by the Poincaré–Birkhoff–Witt theorem (cf. [BLie], I, §2.7). We may therefore identify $\mathfrak{g}$ with an R-submodule of $U(\mathfrak{g})$. Suppose that (y_α) is a family of elements of $\mathfrak{g}$ such that

$$\operatorname{ad} y_\alpha = (\operatorname{ad} x_\alpha)^p.$$

Let

$$\pi : R \longrightarrow R, \qquad r \longmapsto r^p,$$

and let $\mathfrak{g} \otimes_\pi R$ be the R-Lie algebra obtained by extension of scalars along π.[53]

There is an R-linear map

$$\gamma : \mathfrak{g} \otimes_\pi R \longrightarrow U(\mathfrak{g})$$

which sends $x_\alpha \otimes_\pi 1$ to $x_\alpha^p - y_\alpha$. Moreover, for every $x \in \mathfrak{g}$,

$$(\operatorname{ad} x_\alpha^p)(x) = (\operatorname{ad} x_\alpha)^p(x) = (\operatorname{ad} y_\alpha)(x).$$

[52] In this paragraph, the editors changed the order, stating the result first and then giving the detailed proof.

[53] Thus $xr \otimes_\pi 1 = x \otimes_\pi r^p$ and $r \cdot (x \otimes_\pi 1) = x \otimes_\pi r$, for $x \in \mathfrak{g}$ and $r \in R$.

It follows that γ maps $\mathfrak{g} \otimes_\pi R$ into the center of $U(\mathfrak{g})$. For $x \in \mathfrak{g}$, put

$$x^{(p)} = x^p - \gamma(x \otimes_\pi 1).$$

Then $x_\alpha^{(p)} = y_\alpha$ for every α. If

$$x = \sum_\alpha \lambda_\alpha x_\alpha,$$

formula 5.1 (ii), applied inductively to the number of indices α for which $\lambda_\alpha \neq 0$, gives

$$x^p - \sum_\alpha \lambda_\alpha^p x_\alpha^p \in \mathfrak{g}.$$

Denote this element by z. Then

$$x^{(p)} = \sum_\alpha \lambda_\alpha^p y_\alpha + z,$$

and hence $x^{(p)} \in \mathfrak{g}$.

It is clear that

$$(\lambda x)^{(p)} = \lambda^p x^{(p)}.$$

Since $\gamma(x \otimes_\pi 1)$ is central, we have $\operatorname{ad} x^{(p)} = \operatorname{ad} x^p$. Jacobson's first formula, 5.1 (i), therefore yields

$$\operatorname{ad} x^{(p)} = (\operatorname{ad} x)^p.$$

Finally, Jacobson's second formula, 5.1 (ii), shows that the map $x \mapsto x^{(p)}$ satisfies condition (ii) of 5.2. It thus makes $\mathfrak{g}$ into a restricted p-Lie algebra. This proves the proposition.

5.3.3. Proposition. *Let $\mathfrak{g}$ be a restricted p-Lie algebra over R whose underlying module is free with basis (x_α). Then*

$$j : \mathfrak{g} \longrightarrow U_p(\mathfrak{g})$$

is injective. If $z_\alpha = j(x_\alpha)$, then $U_p(\mathfrak{g})$ has as a basis the monomials

$$\prod_\alpha z_\alpha^{n_\alpha}, \qquad 0 \leq n_\alpha < p.$$

Here the n_α are zero except for finitely many α; the basis is assumed to be totally ordered, and the products are taken in increasing order.

Identify $\mathfrak{g}$ with a submodule of the enveloping algebra $U(\mathfrak{g})$ by means of the canonical map i. For every family

$$n = (n_\alpha)$$

of nonnegative integers, all but finitely many zero, put

$$|n| = \sum_\alpha n_\alpha, \qquad x^n = \prod_\alpha x_\alpha^{n_\alpha}.$$

Write

$$n_\alpha = m_\alpha + p\ell_\alpha, \qquad 0 \leq m_\alpha < p,$$

and put

$$T_n = \prod_\alpha x_\alpha^{m_\alpha} \beta(x_\alpha)^{\ell_\alpha},$$

where

$$\beta(x) = x^p - x^{(p)}$$

is the map $\mathfrak{g} \to U(\mathfrak{g})$ defined in 5.3.1.

For $r \in \mathbb{N}$, let U^r be the R-submodule of $U(\mathfrak{g})$ generated by the x^n for which $|n| \leq r$. Since the graded ring

$$\bigoplus_r U^r/U^{r-1}$$

is commutative (cf. [BLie], I, §2.6), for every n we have

$$T_n - \prod_\alpha x_\alpha^{n_\alpha} \in U^{|n|-1}.$$

For every $s \in \mathbb{N}$, the x^n with $|n| = s$ form a basis of U^s/U^{s-1} by the Poincaré–Birkhoff–Witt theorem (*loc. cit.*, §2.7); the same is therefore true of the T_n with $|n| = s$.

As $s = |n|$ varies, the T_n consequently form a basis of $U(\mathfrak{g})$. The kernel J of the canonical map

$$U(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g})$$

is the left ideal of $U(\mathfrak{g})$ generated by the central elements $\beta(x_\alpha)$, by 5.3.1. The T_n for which $\ell = (\ell_\alpha) \neq 0$ therefore form a basis of J, while the T_n for which $n_\alpha < p$ for every α form a basis of

$$U_p(\mathfrak{g}) = U(\mathfrak{g})/J.$$

5.3.3 bis.

Let $\mathfrak{g}$ be a restricted p-Lie algebra over R, and let $f : R \to R'$ be an extension of the base ring. We claim that the R'-module

$$R' \otimes_R \mathfrak{g}$$

admits one and only one restricted p-Lie algebra structure such that

$$[\lambda \otimes x, \mu \otimes y] = \lambda\mu \otimes [x, y], \qquad (\lambda \otimes x)^{(p)} = \lambda^p \otimes x^{(p)}. \tag{*}$$

In particular, it will follow that the functor

$$\mathfrak{g} \longmapsto R' \otimes_R \mathfrak{g}$$

is left adjoint to restriction of scalars from R' to R.

The uniqueness of the structure defined by $(*)$ is clear; we prove its existence. If $\mathfrak{g}$ is free with basis (x_α), Proposition 5.3.2 gives a unique restricted p-Lie algebra structure on the Lie algebra $R' \otimes_R \mathfrak{g}$ such that

$$(1 \otimes x_\alpha)^{(p)} = 1 \otimes x_\alpha^{(p)}.$$

This is the structure sought.

Suppose now that $\mathfrak{g}$ is arbitrary. There is a restricted p-Lie algebra L_0, free as an R-module, together with a surjective homomorphism

$$q_0 : L_0 \longrightarrow \mathfrak{g}.$$

For example, we may take

$$L_0 = R \otimes_{\mathbb{F}_p} \mathfrak{g}$$

and let q_0 be the homomorphism

$$\lambda \otimes x \longmapsto \lambda x;$$

here $\mathfrak{g}$ is free over the prime field $\mathbb{F}_p$. The kernel of q_0 is a p-ideal of L_0, that is, an ideal of the Lie algebra L_0 stable under $x \mapsto x^{(p)}$. Hence there is also a restricted p-Lie algebra L_1, free as an R-module, and a homomorphism $q_1 : L_1 \to L_0$ whose image is $\operatorname{Ker} q_0$. This gives the exact sequence

$$L_1 \xrightarrow{q_1} L_0 \xrightarrow{q_0} \mathfrak{g} \longrightarrow 0.$$

and therefore an exact sequence of R'-Lie algebras

$$R' \otimes_R L_1 \xrightarrow{R' \otimes_R q_1} R' \otimes_R L_0 \xrightarrow{R' \otimes_R q_0} R' \otimes_R \mathfrak{g} \longrightarrow 0.$$

Since $R' \otimes_R q_1$ is manifestly a homomorphism of restricted p-Lie algebras, the kernel of $R' \otimes_R q_0$ is a p-ideal. Thus the symbolic p-th power operation on $R' \otimes_R L_0$ induces, by passage to the quotient, a map from $R' \otimes_R \mathfrak{g}$ to itself (use formula (ii) of 5.2). This map equips $R' \otimes_R \mathfrak{g}$ with the required restricted p-Lie algebra structure.

5.3.4.

The canonical map

$$j_{\mathfrak{g}} : \mathfrak{g} \longrightarrow U_p(\mathfrak{g})$$

induces, for every extension $R \to R'$ of the base ring, a homomorphism

$$R' \otimes_R j_{\mathfrak{g}} : R' \otimes_R \mathfrak{g} \longrightarrow R' \otimes_R U_p(\mathfrak{g}).$$

It therefore induces a homomorphism

$$h : U_p(R' \otimes_R \mathfrak{g}) \longrightarrow R' \otimes_R U_p(\mathfrak{g})$$

such that

$$h \circ j_{R' \otimes_R \mathfrak{g}} = R' \otimes_R j_{\mathfrak{g}}.$$

The universal properties of $R' \otimes_R \mathfrak{g}$ and of the restricted enveloping algebra show immediately that h is an isomorphism. We may therefore identify

$$U_p(R' \otimes_R \mathfrak{g}) \simeq R' \otimes_R U_p(\mathfrak{g}).$$

In particular, let $r \in R$, and let $R' = R_r$ be the localized ring. Then

$$\mathfrak{g}_r = R_r \otimes_R \mathfrak{g}$$

has a canonical restricted p-Lie algebra structure over R_r. Consequently, the sheaf $\widetilde{\mathfrak{g}}$ on $\operatorname{Spec} R$ is a quasi-coherent restricted p-Lie algebra on $\operatorname{Spec} R$. Moreover,

$$U_p^{R_r}(\mathfrak{g}_r) \simeq U_p^{R}(\mathfrak{g})_r,$$

so the sheaf associated with the presheaf

$$V \longmapsto U_p(\Gamma(V, \widetilde{\mathfrak{g}}))$$

is quasi-coherent.

Definition. More generally, let S be a scheme of characteristic p, and let $\mathcal{G}$ be a quasi-coherent restricted p-Lie algebra over $\mathcal{O}_S$. The sheaf associated with the presheaf

$$V \longmapsto U_p(\Gamma(V, \mathcal{G}))$$

is quasi-coherent. It is denoted $\mathcal{U}_p(\mathcal{G})$ and is called the *restricted enveloping algebra* of $\mathcal{G}$. If V is affine, then

$$U_p(\Gamma(V, \mathcal{G})) \simeq \Gamma(V, \mathcal{U}_p(\mathcal{G})).$$

5.4.

The universal property of $U_p(\mathfrak{g})$ implies that $U_p(\mathfrak{g})$ is functorial in $\mathfrak{g}$. Every homomorphism of restricted p-Lie algebras

$$\varphi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

induces a unique homomorphism of unital algebras $U_p(\varphi)$ such that

$$j_{\mathfrak{h}} \circ \varphi = U_p(\varphi) \circ j_{\mathfrak{g}}.$$

Here are some examples.

a) If $\mathfrak{h} = 0$, then $U_p(\mathfrak{h})$ identifies with the base ring, and $U_p(\varphi)$ is an algebra homomorphism

$$\varepsilon_{\mathfrak{g}} : U_p(\mathfrak{g}) \longrightarrow R$$

called the *augmentation*.

b) Let now $\mathfrak{h} = \mathfrak{g}^\circ$, the Lie algebra opposite to $\mathfrak{g}$. Thus $\mathfrak{g}^\circ$ has the same underlying module and the same symbolic p-th power as $\mathfrak{g}$, while its bracket is the negative of the bracket of $\mathfrak{g}$. We may identify $U_p(\mathfrak{g}^\circ)$ with the algebra opposite to $U_p(\mathfrak{g})$. Moreover, the isomorphism

$$x \longmapsto -x$$

from $\mathfrak{g}$ onto $\mathfrak{g}^\circ$ induces an isomorphism

$$c_{\mathfrak{g}} : U_p(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g}^\circ) \simeq U_p(\mathfrak{g})^\circ.$$

We call $c_{\mathfrak{g}}$ the *antipode* of $U_p(\mathfrak{g})$.

c) Finally, let $\mathfrak{f}$ and $\mathfrak{g}$ be two restricted p-Lie algebras, and let $\mathfrak{h} = \mathfrak{f} \times \mathfrak{g}$ be their product. Its underlying R-module is the direct product, and

$$[(x,y),(x',y')] = ([x,x'],[y,y']), \qquad (x,y)^{(p)} = (x^{(p)}, y^{(p)}).$$

If

$$h_1 : \mathfrak{f} \to \mathfrak{k}, \qquad h_2 : \mathfrak{g} \to \mathfrak{k}$$

are restricted p-Lie algebra homomorphisms satisfying

$$[h_1(x), h_2(y)] = 0 \quad (x \in \mathfrak{f},\ y \in \mathfrak{g}),$$

then

$$h_1 + h_2 : (x,y) \longmapsto h_1(x) + h_2(y)$$

is a restricted p-Lie algebra homomorphism. Conversely, every homomorphism $\mathfrak{f} \times \mathfrak{g} \to \mathfrak{k}$ is of this form. This characterizes $\mathfrak{f} \times \mathfrak{g}$ by a universal property.

For example, the maps

$$h_1 : x \longmapsto j_{\mathfrak{f}}(x) \otimes 1, \qquad h_2 : y \longmapsto 1 \otimes j_{\mathfrak{g}}(y)$$

induce a homomorphism $h_1 + h_2$ from $\mathfrak{f} \times \mathfrak{g}$ to the restricted p-Lie algebra underlying

$$U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

The universal properties of $\mathfrak{f} \times \mathfrak{g}$ and of the restricted enveloping algebras show that this homomorphism extends to an isomorphism

$$\varphi : U_p(\mathfrak{f} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

Definition. If $\mathfrak{f} = \mathfrak{g}$, the diagonal map

$$\delta : x \longmapsto (x, x)$$

from $\mathfrak{g}$ to $\mathfrak{g} \times \mathfrak{g}$ induces a homomorphism from $U_p(\mathfrak{g})$ to $U_p(\mathfrak{g} \times \mathfrak{g})$. We denote by

$$\Delta_{\mathfrak{g}} : U_p(\mathfrak{g}) \longrightarrow U_p(\mathfrak{g}) \otimes U_p(\mathfrak{g})$$

the composite of this homomorphism with φ. [54]

It is then easy to see that $\Delta_{\mathfrak{g}}$, together with the multiplication of $U_p(\mathfrak{g})$, makes $U_p(\mathfrak{g})$ an R-coalgebra in groups (cf. 3.2), with augmentation $\varepsilon_{\mathfrak{g}}$ and antipode $c_{\mathfrak{g}}$.

5.5. [55]

Let S be a scheme of characteristic p. First let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups and put

$$G = \operatorname{Spec}^* \mathcal{U}.$$

We saw in 3.2.3 that, for every $T \to S$, $(\operatorname{Lie} G)(T)$ is the Lie subalgebra of $\Gamma(T, \mathcal{U}_T)$ consisting of primitive elements. If x is primitive, then

$$\Delta(x^p) = x^p \otimes 1 + 1 \otimes x^p,$$

since $\binom{p}{i} \equiv 0 \pmod p$ for $0 < i < p$. Thus x^p is again primitive. By 5.1 and 5.2, the map $x \mapsto x^p$ makes $(\operatorname{Lie} G)(T)$ into an $\mathcal{O}(T)$-restricted p-Lie algebra.

Now let $\mathcal{L}$ be a quasi-coherent restricted p-Lie algebra over $\mathcal{O}_S$. As V ranges over the open subsets of S, the group-coalgebra structures defined above on

$$U_p(\Gamma(V, \mathcal{L}))$$

induce on the associated sheaf $\mathcal{U}_p(\mathcal{L})$ a structure of $\mathcal{O}_S$-coalgebra in groups. Moreover, for every S-scheme T, there is an isomorphism

$$\mathcal{U}_p(\mathcal{L}_T) \xrightarrow{\sim} \mathcal{U}_p(\mathcal{L})_T.$$

Let $\operatorname{Prim} \mathcal{U}_p(\mathcal{L})$ denote the subpresheaf which assigns to every open subset V the primitive elements of $\mathcal{U}_p(\mathcal{L})(V)$; this is readily seen to be a sheaf. As V ranges over the open subsets of S, the composites

$$\Gamma(V, \mathcal{L}) \xrightarrow{j} \operatorname{Prim} \mathrm{U}_p(\Gamma(V, \mathcal{L})) \longrightarrow \operatorname{Prim} \mathcal{U}_p(\mathcal{L})(V)$$

define a morphism

$$\mathcal{L} \longrightarrow \operatorname{Prim} \mathcal{U}_p(\mathcal{L}),$$

again denoted j or $j_{\mathcal{L}}$. It in turn defines a morphism of $\mathcal{O}_S$-restricted p-Lie algebras

$$\mathcal{W}(\mathcal{L}) \longrightarrow \operatorname{Prim} \mathcal{W}(\mathcal{U}_p(\mathcal{L}))$$

(cf. 3.2.3).

Proposition 5.5.1. *Let $\mathcal{L}$ be an $\mathcal{O}_S$-restricted p-Lie algebra, locally free as an $\mathcal{O}_S$-module. Then $j_{\mathcal{L}}$ induces an isomorphism of $\mathcal{O}_S$-restricted p-Lie algebras*

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} \operatorname{Prim} \mathcal{W}(\mathcal{U}_p(\mathcal{L})).$$

[54] Thus $\Delta_{\mathfrak{g}}(x) = x \otimes 1 + 1 \otimes x$ for every $x \in \mathfrak{g}$; in particular, $\Delta_{\mathfrak{g}}$ is cocommutative.

[55] The editors reorganized §5.4.1 of the original into this §5.5. Proposition 5.5.1 combines results from Section 5 and Proposition 3.2.3 and incorporates the original Lemma 7.3; the proof of Proposition 5.5.3(ii) expands the proof of $(i) \Rightarrow (ii)$ in the original Theorem 7.4.

Proof. Let T be an S-scheme. In view of the identification

$$\mathcal{U}_p(\mathcal{L}_T) = \mathcal{U}_p(\mathcal{L})_T,$$

we must show that

$$\Gamma(T, \mathcal{L}_T) \longrightarrow \operatorname{Prim} \Gamma(T, \mathcal{U}_p(\mathcal{L}_T))$$

is bijective. Replacing S by T, we reduce to $T = S$; it then suffices to prove that the morphism of sheaves

$$j_{\mathcal{L}} : \mathcal{L} \longrightarrow \operatorname{Prim} \mathcal{U}_p(\mathcal{L})$$

is an isomorphism. The question is local on S, so suppose that $S = \operatorname{Spec} R$ and that $\mathcal{L}$ is the sheaf associated with an R-restricted p-Lie algebra L with a totally ordered basis (x_α).

As in 5.3.3, let z_α be the image of x_α in $U = U_p(L)$. For a family $n = (n_\alpha)$, with $0 \leq n_\alpha \leq p-1$ and only finitely many nonzero terms, put

$$z^{(n)} = \prod_\alpha \frac{z_\alpha^{n_\alpha}}{n_\alpha!},$$

the product being taken in increasing order. Since

$$\Delta(z_\alpha) = z_\alpha \otimes 1 + 1 \otimes z_\alpha,$$

one obtains

$$\Delta(z^{(n)}) = \sum_r z^{(n-r)} \otimes z^{(r)},$$

where the sum is over the finite set of r satisfying

$$0 \leq r_\alpha \leq n_\alpha \quad \text{for every } \alpha.$$

The $z^{(n)}$, respectively the $z^{(n)} \otimes z^{(m)}$, form a basis of U, respectively of $U \otimes U$. It follows that

$$\Delta(u) = u \otimes 1 + 1 \otimes u$$

holds for $u \in U$ if and only if u is a linear combination of the z_α. This proves the proposition. $\square$

Remark 5.5.2. Recall from 3.2.2 and 3.2.3 that the group S-functor

$$G = \operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$$

is very good and that

$$\operatorname{Lie} G = \operatorname{Prim} \mathcal{W}(\mathcal{U}_p(\mathcal{L})).$$

Thus Proposition 5.5.1 says that $j_{\mathcal{L}}$ induces an isomorphism

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} \operatorname{Lie} G.$$

If, moreover, $\mathcal{L}$ is locally free of finite rank, then $\mathcal{U}_p(\mathcal{L})$ is finite locally free over $\mathcal{O}_S$ by 5.3.3. Hence $\operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by the S-group

$$\mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{U}_p(\mathcal{L})^*$$

(cf. 3.2.2.1). We obtain the following more precise statement.

Proposition 5.5.3. *Let $\mathcal{L}$ be an $\mathcal{O}_S$-restricted p-Lie algebra, locally free of finite rank as an $\mathcal{O}_S$-module. Put*

$$\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*, \qquad G = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{A}.$$

Then:

(i) $j_{\mathcal{L}}$ induces an isomorphism

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} \operatorname{Lie} G$$

of $\mathcal{O}_S$-restricted p-Lie algebras.

(ii) Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$ and put $\omega_G = \mathcal{I}/\mathcal{I}^2$ (cf. II, 4.11.4). Then

$$\omega_G \simeq \mathcal{L}^* = \operatorname{Hom}_{\mathcal{O}_S}(\mathcal{L}, \mathcal{O}_S).$$

In particular, ω_G is locally free of finite rank, and $\omega^*_{G/S} \simeq \mathcal{L}$.

Proof. Part (i) follows from Remark 5.5.2. We prove (ii). Let $\eta_{\mathcal{U}}$ and $\varepsilon_{\mathcal{U}}$ denote the unit and augmentation of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, and let $\eta_{\mathcal{A}}$ and $\varepsilon_{\mathcal{A}}$ denote those of $\mathcal{A}$. Put

$$\mathcal{J} = \ker \varepsilon_{\mathcal{U}}.$$

Then

$$\mathcal{U} = \eta_{\mathcal{U}}(\mathcal{O}_S) \oplus \mathcal{J}. \tag{1}$$

Let δ be the morphism defined by the diagram

$$\begin{tikzcd} \mathcal{J} \arrow[r, "\delta"] \arrow[d, "\tau"'] & \mathcal{J} \otimes \mathcal{J} \\ \mathcal{U} \arrow[r, "\Delta"] & \mathcal{U} \otimes \mathcal{U} \arrow[u, "\pi"'] \end{tikzcd}$$

where τ and π are the inclusion and projection deduced from decomposition (1). There is an exact sequence

$$(*) \qquad 0 \longrightarrow \mathcal{L}^* \xrightarrow{j_{\mathcal{L}}} \mathcal{J} \xrightarrow{\delta} \mathcal{J} \otimes \mathcal{J}.$$

By 5.3.3, the $\mathcal{O}_S$-module $\mathcal{J}/\mathcal{L}$ is locally free, and by 5.5.1 the preceding sequence remains exact after every base change. Consequently, δ induces an isomorphism from $\mathcal{J}/\mathcal{L}$ onto a locally direct summand Q of $\mathcal{J} \otimes \mathcal{J}$. By duality, the exact sequence therefore becomes

$$(**) \qquad 0 \longleftarrow \mathcal{L} \xleftarrow{{}^t j_{\mathcal{L}}} \mathcal{I} \xleftarrow{{}^t\delta} \mathcal{I} \otimes \mathcal{I}.$$

Decomposition (1) corresponds by duality to

$$\mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{I}. \tag{2}$$

The transpose of Δ is the multiplication

$$m_{\mathcal{A}} : \mathcal{A} \otimes \mathcal{A} \longrightarrow \mathcal{A}.$$

Since $\mathcal{I}$ is an ideal of $\mathcal{A}$, $m_{\mathcal{A}}$ sends $\mathcal{I} \otimes \mathcal{I}$ into $\mathcal{I}$. More precisely, decomposition (2) gives the commutative square

$$\begin{tikzcd} \mathcal{I} & \mathcal{I} \otimes \mathcal{I} \arrow[l, "m'"'] \arrow[d, "{}^t\pi"] \\ \mathcal{A} \arrow[u, "{}^t\tau"] & \mathcal{A} \otimes \mathcal{A}. \arrow[l, "m_{\mathcal{A}}"'] \end{tikzcd}$$

This shows that the restriction m' of $m_{\mathcal{A}}$ to $\mathcal{I} \otimes \mathcal{I}$ is the transpose of δ. The dual exact sequence therefore yields

$$\mathcal{I}/\mathcal{I}^2 \simeq \mathcal{L}^*,$$

which proves the proposition. $\square$

6. The p-Lie Algebra of an S-Group Scheme

Let S be a scheme of characteristic $p > 0$. In 5.5 we associated with every quasi-coherent $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$ the group S-functor

$$\mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec}^* \mathcal{U}_p(\mathcal{L}).$$

We shall now see that, for every S-group scheme G, the $\mathcal{O}_S$-Lie algebra $\operatorname{Lie}(G/S)$ defined in II, 4.11 is naturally endowed with the structure of an $\mathcal{O}_S$-p-Lie algebra.

6.1.
First identify $\operatorname{Lie}(G/S)(S)$ and $\operatorname{Lie}(\operatorname{Aut} G/S)(S)$, respectively, with Lie subalgebras of $\mathrm{U}(G)$ and $\mathrm{Dif}_{G/S}$ by means of the injections α and β of 2.5. Thus $\operatorname{Lie}(\operatorname{Aut} G/S)(S)$ is identified with the $\Gamma(S, \mathcal{O}_S)$-Lie algebra of S-derivations of $\mathcal{O}_G$. By 5.2, the latter is a restricted p-Lie subalgebra of $\mathrm{Dif}_{G/S}$.

On the other hand, the image of

$$L = \operatorname{Lie}(G/S)(S)$$

under the injective algebra homomorphism

$$\ell : \mathrm{U}(G) \longrightarrow \mathrm{Dif}_{G/S}, \qquad d \longmapsto {}_G d,$$

consists of the left-invariant derivations (cf. 2.2, editor's note (17), 2.4, and 2.5). If $x \in L$, then $\ell(x)^p = \ell(x^p)$ by the cited passages. Since $\ell(x)^p$ is again a derivation, x^p belongs to $\operatorname{Lie}(G/S)(S)$. Hence

$\operatorname{Lie}(G/S)(S)$ *is a restricted p-Lie subalgebra of the infinitesimal algebra* $\mathrm{U}(G)$.

[56]

6.1.1.
Let $\varphi : G \to H$ be a homomorphism of S-group schemes. The homomorphisms $\operatorname{Lie}(\varphi/S)(S)$ and $\mathrm{U}(\varphi)$ are compatible with the identifications of $\operatorname{Lie}(G/S)(S)$ and $\operatorname{Lie}(H/S)(S)$ with restricted p-Lie subalgebras of $\mathrm{U}(G)$ and $\mathrm{U}(H)$. Since $\mathrm{U}(\varphi)$ is an algebra homomorphism, $\operatorname{Lie}(\varphi/S)(S)$ is a homomorphism of restricted p-Lie algebras.

Likewise, if $s : T \to S$ is a base change, the map from $\operatorname{Lie}(G/S)(S)$ to $\operatorname{Lie}(G/S)(T)$ induced by s is a homomorphism of restricted p-Lie algebras. Equivalently, the functor $\operatorname{Lie}(G/S)$ is endowed with a structure of $\mathcal{O}_S$-p-Lie algebra. In particular, as T ranges over the open subsets of S, the sheaf $\operatorname{Lie}(G/S)$ is endowed with a structure of $\mathcal{O}_S$-p-Lie algebra.

6.2.
Following an idea of Demazure, we shall generalize the preceding construction to certain group S-functors which need not be representable. We first give another definition of the symbolic p-th power in the Lie algebra of an S-group scheme G.

[56]One may also prove directly, without passing through $\mathrm{Dif}_{G/S}$, that the Lie algebra of derivations of G at the origin (isomorphic to $\operatorname{Lie}(G/S)(S)$ by 2.5) is stable under taking p-th powers in $\mathrm{U}(G)$. This is done in 6.2 below.

Let D be a derivation of G at the origin.[57] By 1.2.1, D is the composite of the S-derivation

$$\delta = (\tau, \partial_t)$$

of the zero section $\tau : S \to I_S$, and a morphism $x : I_S \to G$ such that $x \circ \tau = \varepsilon$, that is, $x \in \mathrm{Lie}(G/S)(S)$. By the definition in 2.1, D^p is the following composite deviation:

$$S \simeq \underbrace{S \times S \times \cdots \times S}_{p} \xrightarrow{\delta \times \cdots \times \delta} I_S \times \cdots \times I_S \xrightarrow{x \times \cdots \times x} G \times \cdots \times G \xrightarrow{m^{(p)}} G$$

Here $m^{(p)}$ is induced by the multiplication $m : G \times G \to G$. Since $I_S \times \cdots \times I_S$ is affine over S, with affine algebra

$$\mathcal{B} = \mathcal{O}_S[d_1, \ldots, d_p]/(d_1^2, \ldots, d_p^2),$$

the deviation $\delta \times \cdots \times \delta$ is defined by the homomorphism of $\mathcal{O}_S$-modules

$$\phi : \mathcal{B} \longrightarrow \mathcal{O}_S$$

which sends the monomial $d_1 d_2 \cdots d_p$ to 1, and all the other monomials $d_{i_1} \cdots d_{i_r}$, for $0 \leq r < p$, to 0.

Let pr_i denote the projection of I_S^p onto its i-th factor, and let x_i be the image of x in $G(I_S^p)$ under $G(\mathrm{pr}_i)$. Then

$$m^{(p)} \circ (x \times \cdots \times x)$$

is precisely the product $x_1 x_2 \cdots x_p$. Consequently, D^p is also the composite deviation

$$S \xrightarrow{\delta \times \cdots \times \delta} I_S \times \cdots \times I_S \xrightarrow{x_1 x_2 \cdots x_p} G.$$

This description gives another proof that D^p is a derivation of G at the origin. Since G is a very good group (II, 4.11), the images $G(\mathrm{pr}_1)(x)$ and $G(\mathrm{pr}_2)(x)$ of x in $G(I_S \times I_S)$ commute. It follows that the elements $x_i \in G(I_S^p)$ commute pairwise. Hence, for every permutation γ of the factors of I_S^p,

$$(x_1 \cdots x_p) \circ \gamma = x_1 \cdots x_p.$$

Thus $x_1 \cdots x_p$ factors through the canonical projection from I_S^p to the symmetric product $\Sigma^p I_S$ (cf. 4.2).

The affine algebra of $\Sigma^p I_S$ is the subalgebra $\mathcal{A}$ of $\mathcal{B}$ having as an $\mathcal{O}_S$-basis the elementary symmetric functions

$$1 = \sigma_0, \sigma_1, \ldots, \sigma_p$$

in $d_1, \ldots, d_p$. Let $\kappa : \mathcal{A} \hookrightarrow \mathcal{B}$ be the inclusion, and let

$$\pi : \mathcal{A} \longrightarrow \mathcal{O}_S[t]/(t^2)$$

be the homomorphism of $\mathcal{O}_S$-algebras which kills $\sigma_1, \ldots, \sigma_{p-1}$ and sends σ_p to t. Then

$$\phi \circ \kappa = \partial_t \circ \pi,$$

where $\partial_t : \mathcal{O}_S[t]/(t^2) \to \mathcal{O}_S$ kills 1 and sends t to 1. If

$$i : I_S \hookrightarrow \Sigma^p I_S$$

denotes the closed immersion defined by π, we obtain the commutative diagram

[57] The editors expanded the argument in the original text in what follows.

$$\begin{tikzcd}
S \arrow[r, "\delta\times\cdots\times\delta"'] \arrow[d, "\delta"'] \arrow[rr, bend left, "D^p"] & I_S^p \arrow[r, "x_1\cdots x_p"'] \arrow[d, "\text{can.}"] & G \arrow[d, equal] \\
I_S \arrow[r, "i"] \arrow[rr, bend right, "y"'] & \Sigma^p I_S \arrow[r] & G
\end{tikzcd}$$

which shows that D^p has the form $y \circ \delta$, and is therefore indeed a derivation of G at the origin.

6.3.

Let $\mathfrak{S}_p$ be the symmetric group of degree p, and let $I_S^p \times \mathfrak{S}_p$ denote the disjoint union of a family of copies of I_S^p indexed by $\mathfrak{S}_p$. Let

$$\pi : I_S^p \times \mathfrak{S}_p \longrightarrow I_S^p$$

be the canonical projection, and let

$$\mu : I_S^p \times \mathfrak{S}_p \longrightarrow I_S^p$$

be the morphism defining the action of $\mathfrak{S}_p$ on I_S^p: if $\rho \in \mathfrak{S}_p$, the restriction of μ to $I_S^p \times \rho$ has $\mathrm{pr}_{\rho(j)}$ as its j-th component.

Definition. A functor

$$X : (\mathrm{Sch}/S)^\circ \longrightarrow (\mathrm{Ens})$$

satisfies *condition (F)* if:

a) X takes finite disjoint unions to direct products;

b) for every S-scheme T, the following sequence is exact:

$$X(T \times \Sigma^p I_S) \longrightarrow X(T \times I_S^p) \underset{X(\mathrm{id}_T \times \mu)}{\overset{X(\mathrm{id}_T \times \pi)}{\rightrightarrows}} X(T \times I_S^p \times \mathfrak{S}_p)\,.$$

Every S-scheme satisfies (F). If $\mathcal{F}$ is an $\mathcal{O}_S$-module, then $\mathcal{W}(\mathcal{F})$ satisfies (F). Every projective limit of functors satisfying (F) again satisfies (F). Finally, if Y satisfies (F) and X is an arbitrary S-functor, then $\mathrm{Hom}_S(X, Y)$ satisfies (F).

Let X be a very good group (II, 4.10) satisfying condition (F). Let $x : I_S \to X$ be a morphism extending the unit section of X, and use the notation of 6.2. As above,

$$x_1 \cdots x_p : I_S^p \longrightarrow X$$

factors through $\Sigma^p I_S$:

$$\begin{tikzcd}
I_S^p \arrow[rr, "x_1\cdots x_p"] \arrow[dr, "\text{can.}"'] & & X \\
& \Sigma^p I_S \arrow[ur, "\Sigma^p(x)"'] &
\end{tikzcd}$$

and therefore defines, by composition, a morphism

$$x^{(p)}: \quad I_S \xrightarrow{i} \Sigma^p I_S \xrightarrow{\Sigma^p(x)} X$$

which we call the *symbolic p-th power* of x.

The endomorphism

$$x \longmapsto x^{(p)}$$

of $\mathrm{Lie}(G/S)(S)$ is plainly compatible with base change and functorial in G. It would be interesting to know for which G this endomorphism makes $\mathrm{Lie}(G/S)(S)$ into a restricted p-Lie algebra.

6.4.

The preceding definition of the symbolic p-th power is particularly well suited to computation. Here are some examples.

6.4.1.

Let M be an "abstract" abelian group, and let $D_S(M)$ be the diagonalizable S-group of type M (I, 4.4.2). For every S-scheme T,

$$D_S(M)(T) = \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(T)^\times).$$

Let $x \in \mathrm{Lie}(D_S(M)/S)(S)$, that is, an abelian-group homomorphism

$$x : M \longrightarrow \Gamma(S, \mathcal{O}_S + d\mathcal{O}_S)^\times$$

of the form

$$m \longmapsto 1 + d\xi(m), \qquad \xi \in \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S)).$$

With the notation of 6.2 and 6.3, the product $x_1 \cdots x_p$ associates with $m \in M$ the expression

$$(1 + d_1\xi(m)) \cdots (1 + d_p\xi(m)),$$

that is,

$$1 + \sigma_1\xi(m) + \sigma_2\xi(m)^2 + \cdots + \sigma_p\xi(m)^p.$$

This expression belongs to $\mathcal{O}(\Sigma^p I_S)$. Projecting it to $\mathcal{O}(S)[d]/(d^2)$, killing $\sigma_1, \ldots, \sigma_{p-1}$, and sending σ_p to d, we see that $x^{(p)}$ is the homomorphism

$$m \longmapsto 1 + d\xi(m)^p.$$

Thus, under the identification

$$\mathrm{Lie}(D_S(M)/S)(S) \simeq \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S))$$

of II, 5.1, the symbolic p-th power sends ξ to

$$\xi^{(p)} : m \longmapsto \xi(m)^p.$$

6.4.2.

Let $\mathcal{F}$ be an $\mathcal{O}_S$-module, and let $G = \mathcal{W}(\mathcal{F})$ be the group S-functor in abelian groups (cf. I, 4.6). Let

$$y \in \mathcal{W}(\mathcal{F})(S) = \Gamma(S, \mathcal{F}),$$

and let y' be its image in $\mathcal{W}(\mathcal{F})(I_S)$ under $\mathcal{W}(\mathcal{F})(I_S \to S)$.

By II, 4.4.2 and 4.5.1, the map

$$y \longmapsto dy'$$

is an isomorphism of $\mathcal{O}(S)$-modules from $\mathcal{W}(\mathcal{F})(S)$ onto $\mathrm{Lie}(\mathcal{W}(\mathcal{F})/S)(S)$. Put $x = dy'$. Then the element x_i of 6.2 is $d_i y''$, where y'' denotes the canonical image of y'[58] in $\mathcal{W}(\mathcal{F})(I_S^p)$. Hence

$$x_1 + \cdots + x_p = (d_1 + \cdots + d_p)y'' = \sigma_1 y'',$$

and this belongs to $\mathcal{W}(\mathcal{F})(\Sigma^p I_S)$. Since the homomorphism

$$\mathcal{O}(\Sigma^p I_S) \longrightarrow \mathcal{O}(I_S)$$

defining the morphism i kills σ_1, we have $x^{(p)} = 0$. Therefore

For every $\mathcal{O}_S$-module $\mathcal{F}$, the operation $x \mapsto x^{(p)}$ on $\mathrm{Lie}\,\mathcal{W}(\mathcal{F})$ is zero.

6.4.3.

Let X be an S-scheme, let $G = \mathrm{Aut}_S X$, and let D be an S-derivation of the structure sheaf $\mathcal{O}_X$. By 1.5, D may be identified with an I_S-automorphism x of X_{I_S} inducing the identity on X. It can be described as follows. If f is a section of $\mathcal{O}_S[d]/(d^2)$ of the form $a + bd$, put

$$D_{I_S} f = Da + (Db)d.$$

Thus D_{I_S} is obtained from D by the base change $I_S \to S$, and the automorphism of X_{I_S} is associated with the endomorphism

$$f \longmapsto f + (D_{I_S} f)d = a + (D(a) + b)d$$

of $\mathcal{O}_S[d]/(d^2)$.

Similarly, let $D_{I_S^p}$ be the differential operator on $X_{I_S^p}$ obtained from D by the base change $I_S^p \to S$. With the notation of 6.2, the automorphism x_i of $X_{I_S^p}$ is associated with the endomorphism

$$f \longmapsto f + (D_{I_S^p} f)d_i$$

of

$$\mathcal{O}_S[d_1, \ldots, d_p]/(d_1^2, \ldots, d_p^2).$$

The product $x_1 \cdots x_p$ is therefore associated with

$$(1 + d_1 D_{I_S^p})(1 + d_2 D_{I_S^p}) \cdots (1 + d_p D_{I_S^p}),$$

that is, with

$$1 + \sigma_1 D_{I_S^p} + \sigma_2 D_{I_S^p}^2 + \cdots + \sigma_p D_{I_S^p}^p.$$

The coefficient of σ_p is $D_{I_S^p}^p$. Consequently, the Lie-algebra isomorphism

$$\mathrm{Der}_S(\mathcal{O}_X) \xrightarrow{\sim} \mathrm{Lie}(\mathrm{Aut}_S X), \qquad D \longmapsto x$$

(cf. 1.5) is also an isomorphism of restricted p-Lie algebras.

6.4.4.

The same method shows that, for every $\mathcal{O}_S$-module $\mathcal{F}$, the Lie-algebra isomorphism

$$\mathrm{Lie}\big(\mathrm{Aut}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})/S\big)(S) \xrightarrow{\sim} \big(\mathrm{End}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})\big)(S)$$

(cf. II, 4.8) is also an isomorphism of restricted p-Lie algebras.

[58] The editors corrected x to y'.

6.4.5.[59]

Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups, and let

$$G = \operatorname{Spec}^* \mathcal{U}$$

be the group S-functor; suppose that G is representable. For every $T \to S$, 5.5 and 6.1.1 define two restricted p-Lie algebra structures on

$$L(T) = \operatorname{Lie}(G)(T).$$

There is a commutative diagram

$$\begin{tikzcd}
L(T) \arrow[r, hook, "\tau"] \arrow[d, hook, "\alpha"'] \arrow[dr, "i"] & \Gamma(T, \mathcal{U}_T) \\
\mathrm{U}(G_T) & \mathrm{U}(L(T)) \arrow[l, "\varphi"'] \arrow[u, "\psi"']
\end{tikzcd}$$

in which $\mathrm{U}(L(T))$ is the enveloping algebra of $L(T)$, and φ, ψ are the algebra homomorphisms induced by α, τ. It follows that the two restricted p-Lie algebra structures coincide: if $x \in L(T)$ is identified with its image in $\mathrm{U}(G_T)$, respectively in $\Gamma(T, \mathcal{U}_T)$, then $x^{(p)}$ is the image under φ, respectively under ψ, of the element $x^p \in \mathrm{U}(L(T))$.

7. Radicial Groups of Height 1

Let S be a scheme of characteristic $p > 0$.[60] We shall say that an $\mathcal{O}_S$-algebra $\mathcal{A}$ (respectively, an $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$) is *finite locally free* if its underlying $\mathcal{O}_S$-module is locally free and of finite type. If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, we know (cf. 5.5.2) that the group S-functor

$$\operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$$

is represented by a finite locally free S-group scheme $\mathfrak{G}_p(\mathcal{L})$. We shall see that this S-group scheme solves a universal problem (7.2), and we shall characterize the S-group schemes of the form $\mathfrak{G}_p(\mathcal{L})$ (7.4).

Definition 7.0.[61] *Let $H = \operatorname{Spec} \mathcal{A}$ be a finite locally free S-group scheme. We say that H is* infinitesimal *if its unit section $\varepsilon_H : S \to H$ is a homeomorphism; equivalently, the augmentation ideal of $\mathcal{A}$ is locally nilpotent.*

7.1.[62]

Let $\mathcal{L}$ be a finite locally free $\mathcal{O}_S$-p-Lie algebra, and let $\mathfrak{G}_p(\mathcal{L})$ be the affine S-group $\operatorname{Spec} \mathcal{U}_p(\mathcal{L})$. By 5.5, $\mathcal{L}$ is identified with $\operatorname{Lie} \mathfrak{G}_p(\mathcal{L})$.

Now let G be a very good group S-functor satisfying condition (F) of 6.3, and let

$$\varphi : \mathfrak{G}_p(\mathcal{L}) \longrightarrow G$$

be a morphism of group S-functors. By 6.3, the morphism of $\mathcal{O}_S$-Lie algebras

$$\operatorname{Lie} \varphi : \operatorname{Lie} \mathfrak{G}_p(\mathcal{L}) \longrightarrow \operatorname{Lie} G$$

[59] The editors supplied the number 6.4.5 and expanded the original text.

[60] For the results of this section, one may also consult [DG70], § II.7, nos. 3–4.

[61] The editors added this definition (cf. [DG70], § II.4, 7.1), which will be used in 7.2.1.

[62] The editors simplified the original, taking into account the additions made in 5.5.

is compatible with the symbolic p-th power. If $\operatorname{Hom}_p(\mathcal{L}, \operatorname{Lie} G)$ denotes the set of morphisms of $\mathcal{O}_S$-Lie algebras compatible with the symbolic p-th power, we therefore have a map

$$\operatorname{Lie} : \operatorname{Hom}_{S\text{-Gr.}}(\mathfrak{G}_p(\mathcal{L}), G) \longrightarrow \operatorname{Hom}_p(\mathcal{L}, \operatorname{Lie} G), \qquad \varphi \longmapsto \operatorname{Lie} \varphi.$$

7.2. Theorem.
If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the map

$$\operatorname{Hom}_{S\text{-gr.}}(\mathfrak{G}_p(\mathcal{L}), G) \longrightarrow \operatorname{Hom}_p(\mathcal{L}, \operatorname{Lie} G)$$

is bijective in each of the following cases:

(i) G is an S-group scheme;

(ii) G is of the form $\operatorname{Aut}_S X$, where X is an S-scheme;

(iii) G is of one of the forms $\mathcal{W}(\mathcal{F})$ or $\operatorname{Aut}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})$, where $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$-module.

The proof of the theorem rests on the following lemma.

Lemma.
If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the S-group

$$G = \mathfrak{G}_p(\mathcal{L})$$

is annihilated by the Frobenius morphism $\operatorname{Fr} : G \to G^{(p)}$. In particular, G is infinitesimal.

Indeed, let $\mathcal{U}$ be the restricted enveloping algebra of $\mathcal{L}$, let $\mathcal{A} = \mathcal{U}^*$ be the affine algebra of G, and let $\mathcal{I} = \ker \varepsilon_{\mathcal{A}}$ be the augmentation ideal of $\mathcal{A}$.[63] We have

$$\mathcal{A} = \mathcal{I} \oplus \eta_{\mathcal{A}}(\mathcal{O}_S), \tag{1}$$

where $\eta_{\mathcal{A}}$ denotes the unit section of $\mathcal{A}$. Since $\varepsilon_{\mathcal{A}}$ (respectively, $\eta_{\mathcal{A}}$) is the transpose of $\eta_{\mathcal{U}}$ (respectively, $\varepsilon_{\mathcal{U}}$), this decomposition corresponds by duality to

$$\mathcal{U} = \mathcal{J} \oplus \eta_{\mathcal{U}}(\mathcal{O}_S), \tag{2}$$

where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$; thus $\mathcal{J} = \mathcal{I}^*$.

Let π denote the endomorphism $x \mapsto x^p$ of $\mathcal{O}_S$. We must show that $\operatorname{Fr} : G \to G^{(p)}$ factors through the unit section of $G^{(p)}$. By 4.1.4 (c), this is equivalent to saying that the morphism

$$\Phi : \mathcal{I} \otimes_\pi \mathcal{O}_S \longrightarrow \mathcal{A}, \qquad a \otimes_\pi x \longmapsto a^p x,$$

is zero. Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, it suffices to show that the transpose ${}^t\Phi$ is zero.

Now Φ is precisely the following composite:

$$\mathcal{J} \otimes_\pi \mathcal{O}_S \xrightarrow{\tau} \mathcal{A} \otimes_\pi \mathcal{O}_S \xrightarrow{j(\mathcal{A})} \mathrm{S}^p\mathcal{A} \xrightarrow{b(\mathcal{A})} \mathcal{A},$$

Here τ is induced by the inclusion $\mathcal{I} \hookrightarrow \mathcal{A}$, while $b(\mathcal{A})$ and $j(\mathcal{A})$ are defined as in 4.3.3: $b(\mathcal{A})$ is induced by multiplication in $\mathcal{A}$, and $j(\mathcal{A})$ sends $a \otimes_\pi 1$ to the image of $a \otimes \cdots \otimes a$ in $\mathrm{S}^p\mathcal{A}$. The $\mathcal{O}_S$-dual of $\mathrm{S}^p\mathcal{A}$ is the submodule $\Sigma^p\mathcal{U}$ of $\bigotimes^p \mathcal{U}$ consisting of the sections invariant under the symmetric group of degree p. Hence ${}^t\Phi$ is the following composite:

[63] The editors supplied details in what follows. For another proof, see [DG70], § II.7, 3.9.

$$\mathcal{U} \xrightarrow{a(\mathcal{U})} \Sigma^p\mathcal{U} \xrightarrow{r(\mathcal{U})} \mathcal{U} \otimes_\pi \mathcal{O}_S \xrightarrow{q} \mathcal{J} \otimes_\pi \mathcal{O}_S,$$

Here q is induced by the projection $\mathcal{U} \to \mathcal{J}$ with kernel $\eta_{\mathcal{U}}(\mathcal{O}_S)$, $a(\mathcal{U})$ is induced by the comultiplication of $\mathcal{U}$, and $r(\mathcal{U})$ vanishes on symmetrized tensors and sends a section $x \otimes \cdots \otimes x$ to $x \otimes_\pi 1$ (cf. 4.3.3).

It is clear that ${}^t\Phi \circ \eta_{\mathcal{U}} = 0$; by (2), it remains to show that ${}^t\Phi$ annihilates the augmentation ideal $\mathcal{J}$. Since ${}^t\Phi$ is an algebra morphism and $\mathcal{J}$ is generated by $\mathcal{L}$, identified with its image in $\mathcal{U}$, it suffices to show that ${}^t\Phi(x) = 0$ for every section x of $\mathcal{L}$. But

$$-a(\mathcal{U})(x) = (p-1)!\, a(\mathcal{U})(x)$$

is the symmetrization of $x \otimes 1 \otimes \cdots \otimes 1$, so its image under $r(\mathcal{U})$ is zero. This proves the first assertion of the lemma.

The second follows. Every local section of $\mathcal{I}$ has zero p-th power, and $\mathcal{I}$ is an $\mathcal{O}_S$-module of finite type; hence $\mathcal{I}$ is locally nilpotent. Explicitly, if V is an affine open subset of S such that

$$I = \Gamma(V, \mathcal{I})$$

is generated by r elements, then

$$I^{r(p-1)+1} = 0.$$

Thus $G_{\mathrm{red}} = S_{\mathrm{red}}$, and the unit section $\varepsilon_G : S \to G$ is a homeomorphism.

7.2.1.[64]

We first prove assertion (ii) of Theorem 7.2. Let $\pi : X \to S$ be an S-scheme, and first consider an arbitrary infinitesimal S-group H.

Morphisms $\varphi : H \to \operatorname{Aut} X$ correspond bijectively to left actions $\mu : H \times X \to X$ of H on X. For such an action, if ε is the unit section of H, the composite

$$X \simeq S \times X \xrightarrow{\varepsilon \times X} H \times X \xrightarrow{\mu} X$$

must be the identity. Since $(H \times X)_{\mathrm{red}}$ is identified with X_{red}, the morphism μ must induce the identity on the associated reduced schemes. In particular, μ induces an action of H on every open subset of X. Thus, for every open $U \subset X$ affine over S, we obtain a morphism of unital associative algebras

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(H) \otimes \mathcal{A}(U)$$

which makes $\mathcal{A}(U)$ a left $\mathcal{A}(H)$-comodule, in such a way that the restriction maps

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(U'), \qquad U' \subset U,$$

are comodule morphisms. Conversely, every datum of this kind comes from a unique left action $\mu : H \times X \to X$.

We also have the following lemma.

Lemma.

Let $X = \operatorname{Spec} \mathcal{C}$ be an affine S-scheme, let $H = \operatorname{Spec} \mathcal{A}$ be an infinitesimal S-group, and put

$$\mathcal{U} = \mathcal{A}^* = \operatorname{Hom}_{\mathcal{O}_S}(\mathcal{A}, \mathcal{O}_S).$$

[64] The editors supplied details in what follows.

Left actions of H on X correspond bijectively to representations of the algebra $\mathcal{U}$ in the $\mathcal{O}_S$-module $\mathcal{C}$ such that

$$\begin{aligned} &\text{(a)} \quad u(1_{\mathcal{C}}) = \varepsilon(u)\, 1_{\mathcal{C}}, \\ &\text{(b)} \quad u(xy) = \sum_i v_i(x) w_i(y) \quad \textit{if} \quad \Delta u = \sum_i v_i \otimes w_i. \end{aligned}$$

Here u is an arbitrary section of $\mathcal{U}$ over an affine open subset V of S, while x and y are sections of $\mathcal{C}$ over V. The symbol $1_{\mathcal{C}}$ denotes the unit section of $\mathcal{C}$, and ε and Δ denote the augmentation and the diagonal morphism of $\mathcal{U}$, respectively.

Indeed, a left action μ of H on X is defined by a morphism of unital associative algebras

$$\lambda : \mathcal{C} \longrightarrow \mathcal{A} \otimes \mathcal{C}$$

which makes $\mathcal{C}$ a left $\mathcal{A}$-comodule. Let α be the composite

$$\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\mathcal{U} \otimes \lambda} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\gamma \otimes \mathcal{C}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{C} \simeq \mathcal{C}$$

where γ is the "contraction" from $\mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}$ to $\mathcal{O}_S$. Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, the map

$$\lambda \longmapsto (\gamma \otimes \mathcal{C})(\mathcal{U} \otimes \lambda)$$

is a bijection from

$$\operatorname{Hom}_{\mathcal{O}_S}(\mathcal{C}, \mathcal{A} \otimes \mathcal{C}) \quad \text{onto} \quad \operatorname{Hom}_{\mathcal{O}_S}(\mathcal{U} \otimes \mathcal{C}, \mathcal{C}).$$

Moreover, by duality, the condition that λ define a left $\mathcal{A}$-comodule structure (respectively, be a morphism of unital associative algebras) is equivalent to the condition that α be a representation of $\mathcal{U}$ in $\mathcal{C}$ (respectively, satisfy conditions (a) and (b)). This proves the lemma.

For any representation of $\mathcal{U}$ in the $\mathcal{O}_S$-module $\mathcal{C}$, it is also clear that the sections u of $\mathcal{U}$ satisfying conditions (a) and (b) form a subalgebra of $\mathcal{U}$.

In the particular case

$$H = \mathfrak{G}_p(\mathcal{L})$$

which concerns us, these conditions will therefore hold for every section u of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$ if they hold for the sections u of $\mathcal{L}$, identified with its image in $\mathcal{U}_p(\mathcal{L})$.

If u is a section of $\mathcal{L}$, conditions (a) and (b) simply mean

$$u(1_{\mathcal{C}}) = 0, \qquad u(xy) = u(x)y + xu(y);$$

that is, $\alpha(u)$ is an $\mathcal{O}_S$-derivation of $\mathcal{C} = \mathcal{A}(X)$. Assertion (ii) of 7.2 follows. Indeed, every morphism

$$\varphi : \mathfrak{G}_p = \mathfrak{G}_p(\mathcal{L}) \longrightarrow \operatorname{Aut} X$$

defines a morphism of p-Lie algebras $\operatorname{Lie}\varphi$ from $\mathcal{L} = \operatorname{Lie}\mathfrak{G}_p$ to

$$\pi_*(\operatorname{Der}_{\mathcal{O}_S} \mathcal{O}_X),$$

and conversely every datum of this kind comes, by what precedes, from a unique action $\mu : G \times X \to X$.

7.2.2.

We now show how assertion (i) of Theorem 7.2 follows from (ii). Let G be an S-group scheme. If T is an S-scheme and $x \in G(T)$, let ℓ_x^T (respectively, r_x^T) denote the left (respectively, right) translation of G_T defined by x. The maps

$$\ell^T : x \longmapsto \ell_x^T$$

therefore determine a homomorphism $\ell : G \to \operatorname{Aut} G$.

On the other hand, let f be a T-automorphism of G_T, and define

$${}^x f = (r_x^T)^{-1} f r_x^T;$$

thus, for every $T' \to T$ and $g \in G(T')$,

$$({}^x f)(g) = f(gx)x^{-1}.$$

In this way G acts on the left on the group S-functor $\operatorname{Aut} G$, and hence also on the functors

$$T \longmapsto \operatorname{Hom}_{T\text{-Gr.}}(\mathfrak{G}_p(\mathcal{L}_T), \operatorname{Aut} G_T)$$

and

$$T \longmapsto \operatorname{Hom}_p(\mathcal{L}_T, \operatorname{Lie}(\operatorname{Aut} G_T/T)).$$

The morphism $\ell^T : G_T \to \operatorname{Aut} G_T$ identifies G_T with the group of automorphisms of the T-scheme G_T which commute with right translations. Its derived morphism $\operatorname{Lie}(\ell^T)$ identifies $\operatorname{Lie} G_T$ with the p-Lie algebra of $\mathcal{O}_T$-derivations of $\mathcal{O}_{G_T}$ which commute with right translations (cf. II, 4.11.1). They therefore induce the commutative squares

$$\begin{tikzcd}
\operatorname{Hom}_{T\text{-Gr.}}(\mathfrak{G}_p(\mathcal{L}_T), G_T) \arrow[r, "\operatorname{Lie}"] \arrow[d, "\ell_T"'] & \operatorname{Hom}_p(\mathcal{L}_T, \operatorname{Lie}(G_T/T)) \arrow[d, "\operatorname{Lie} \ell_T"] \\
\operatorname{Hom}_{T\text{-Gr.}}(\mathfrak{G}_p(\mathcal{L}_T), \underline{\operatorname{Aut}}\, G_T) \arrow[r, "\operatorname{Lie}"] & \operatorname{Hom}_p(\mathcal{L}_T, \operatorname{Lie}(\underline{\operatorname{Aut}}\, G_T/T)).
\end{tikzcd}$$

The images of the two vertical arrows are the subfunctors formed by the invariants under the action of the S-group G. Since the lower horizontal arrow is invertible by 7.2.1 and is compatible with the action of G, the upper horizontal arrow is invertible as well. This proves 7.2 (i).

7.2.3.

Consider the case in which $G = \operatorname{Aut}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})$.[65] Put

$$\mathcal{U} = \mathcal{U}_p(\mathcal{L}), \qquad \mathcal{A} = \mathcal{U}^*, \qquad H = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{A}.$$

Since H is affine over S, VIB, 11.6.1 shows that a morphism of S-groups from H to $\operatorname{Aut}_{\mathcal{O}_S\text{-mod.}} \mathcal{W}(\mathcal{F})$ is the same as a right $\mathcal{A}$-comodule structure

$$\mu : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}.$$

Since $\mathcal{A}$ is finite locally free over $\mathcal{O}_S$, this is in turn equivalent to giving a representation

$$\alpha : \mathcal{U} \longrightarrow \operatorname{End}_{\mathcal{O}_S}(\mathcal{F})$$

[65]In what follows, the editors expanded and simplified the original, taking VIB, 11.6.1 into account.

of $\mathcal{U}$ in $\mathcal{F}$. Finally, by the universal property of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, giving such a morphism α is equivalent to giving its restriction ρ to $\mathcal{L}$, identified with its image in $\mathcal{U}$, and ρ is a morphism of p-Lie algebras from $\mathcal{L}$ to $\mathrm{End}_{\mathcal{O}_S}(\mathcal{F})$.

Finally, consider the case $G = \mathcal{W}(\mathcal{F})$, with the preceding notation.[66] To give a morphism of S-functors $\varphi : H \to \mathcal{W}(\mathcal{F})$ is first of all equivalent to giving an element θ of $\Gamma(H, \mathcal{F}\otimes\mathcal{O}_H)$. Since H is finite locally free over S, we have

$$\Gamma(H, \mathcal{F} \otimes \mathcal{O}_H) = \Gamma(S, \mathcal{F} \otimes \mathcal{A}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{U}, \mathcal{F}).$$

The condition that φ be a group morphism means that θ, regarded as a morphism of $\mathcal{O}_S$-modules $\mathcal{U} \to \mathcal{F}$, vanishes on $1_{\mathcal{U}}$ and on $\mathcal{J}^2$, where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$. Consequently,

$$\mathrm{Hom}_{S\text{-gr.}}(H, \mathcal{W}(\mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{J}/\mathcal{J}^2, \mathcal{F}). \tag{1}$$

Next consider the quasi-coherent sheaf $[\mathcal{L}, \mathcal{L}]$, the image of the morphism

$$\mathcal{L} \otimes \mathcal{L} \longrightarrow \mathcal{L}, \qquad x \otimes y \longmapsto [x, y].$$

For every affine open subset V of S, we have

$$[\mathcal{L}, \mathcal{L}](V) = [\mathcal{L}(V), \mathcal{L}(V)].$$

There is an exact sequence

$$(\dagger) \qquad 0 \longrightarrow [\mathcal{L}, \mathcal{L}] \longrightarrow \mathcal{L} \xrightarrow{\pi} \mathcal{J}/\mathcal{J}^2 \longrightarrow 0,$$

in which π is the composite of the inclusion $\mathcal{L} \hookrightarrow \mathcal{J}$ and the projection $\mathcal{J} \to \mathcal{J}/\mathcal{J}^2$.

Indeed, the question is local on S, so we may suppose that S is affine with ring R, and that

$$L = \mathcal{L}(S)$$

is free with basis $(x_1, \ldots, x_r)$. Identify L with its image in $U = U_p(L)$, and let K be the sub-R-module of U which is the direct sum of $[L, L]$ and the submodule with basis the monomials

$$x_1^{n_1} \cdots x_r^{n_r} \quad \text{such that} \quad n_1 + \cdots + n_r \geq 2.$$

One checks that K is a two-sided ideal of U. Since K is contained in J^2, where J is the augmentation ideal of U, and contains all the products $x_i x_j$, which generate J^2, it follows that $K = J^2$. Hence

$$J^2 \cap L = [L, L],$$

and the displayed sequence is exact.

By 6.4.2, $\mathrm{Lie}\,\mathcal{W}(\mathcal{F})$ is simply $\mathcal{W}(\mathcal{F})$, with zero Lie bracket and zero symbolic p-th power. From this and the preceding discussion, we obtain

$$\mathrm{Hom}_p(\mathcal{L}, \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{L}/[\mathcal{L}, \mathcal{L}], \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{J}/\mathcal{J}^2, \mathcal{F}). \tag{2}$$

Together with (1), this completes the proof of Theorem 7.2.

7.3. Lemma.

If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the morphism

$$j_{\mathcal{L}} : \mathcal{L} \longrightarrow \mathrm{Lie}\,\mathfrak{G}_p(\mathcal{L})$$

[66] The editors added what follows. The original said that "the case of $\mathcal{W}(\mathcal{F})$ is analogous."

of 5.5 is invertible.[67]

For the proof, see 5.5.1.

7.4.

To conclude this section, we give a characterization of the S-group schemes of the form $\mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra.

Let G be an S-group scheme, let ε_G be its unit section, and let $\mathcal{I}'$ be the kernel of the morphism

$$\varepsilon_G^{-1}(\mathcal{O}_G) \longrightarrow \mathcal{O}_S$$

corresponding to ε_G. By 2.5 and 1.3.1, the image of $\mathrm{Lie}(G/S)(S)$ in $\mathrm{U}(G)$ is identified with the morphisms of $\mathcal{O}_S$-modules from $\varepsilon_G^{-1}(\mathcal{O}_G)$ to $\mathcal{O}_S$ which vanish on the unit section of $\varepsilon_G^{-1}(\mathcal{O}_G)$ and on $\mathcal{I}'^2$.

We thereby recover the canonical isomorphism

$$\mathrm{Lie}(G/S)(S) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}'/\mathcal{I}'^2, \mathcal{O}_S)$$

of II, 3.3 and 4.11.4.[68] As in the cited passage, put

$$\omega_{G/S} = \mathcal{I}'/\mathcal{I}'^2.$$

Thus the sheaf $\mathrm{Lie}(G/S)$ is identified with

$$\omega^*_{G/S} = \mathrm{Hom}_{\mathcal{O}_S}(\omega_{G/S}, \mathcal{O}_S).$$

Moreover,[69] if $G = \mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, then 5.5.3 gives

$$\omega_{G/S} = \mathcal{L}^*.$$

Theorem.

If G is a group scheme over a scheme S of characteristic $p > 0$, the following assertions are equivalent:

(i) *There exists a finite locally free $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$ such that $G \simeq \mathfrak{G}_p(\mathcal{L})$.*

(i′) *The $\mathcal{O}_S$-p-Lie algebra* $\mathrm{Lie}(G)$ *is finite locally free, and $G \simeq \mathfrak{G}_p(\mathrm{Lie}(G))$.*

(ii) *G is affine over S, the $\mathcal{O}_S$-module $\omega_{G/S}$ is locally free of finite type, and the affine algebra of G is locally isomorphic to the quotient of the symmetric algebra $\mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})$ by the ideal generated by the p-th powers of the sections of $\omega_{G/S}$.*

(iii) *G is locally of finite presentation over S, is of height ≤ 1, and $\omega_{G/S}$ is locally free.*

(iii′) *G is locally of finite type over S, is of height ≤ 1, and $\omega_{G/S}$ is locally free.*

(iv) *G is locally of finite presentation and flat over S, and is of height ≤ 1.*[70]

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[67] To avoid changing the numbering, the editors retained statement 7.3, although they included it, together with its proof, in 5.5.1.

[68] If G is affine over S and $\mathcal{I}$ denotes the augmentation ideal of $\mathcal{A}(G)$, then $\mathcal{I}'/\mathcal{I}'^2$ is identified with $\varepsilon_G^*(\mathcal{I}/\mathcal{I}^2)$; cf. loc. cit.

[69] The editors added the following sentence.

[70] The editors added assertion (i′), implicit in the original, and assertions (iii′) and (iv), pointed out by O. Gabber. Assertion (iv) takes up a footnote in the original, which stated: "The condition on $\omega_{G/S}$ is in fact unnecessary, as is easily seen by reducing to the case where S is local with residue field k, and applying the theorem to G_k." As Gabber pointed out, this is incorrect without a flatness hypothesis. Indeed, let A

7.4.1.

The equivalence (i) $\Leftrightarrow$ (i′) follows from 5.5.3 (i), and the implications (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iii′) are clear. We also have (i) $\Rightarrow$ (iv), since $\mathfrak{G}_p(\mathcal{L})$ is finite locally free and of height ≤ 1, by 5.5.2 and the lemma in 7.2. We show that (i) implies (ii).

Let $\mathcal{I}$ be the augmentation ideal of

$$\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*.$$

We saw in 5.5.3 (ii) that $\omega_{G/S} = \mathcal{I}/\mathcal{I}^2$ is identified with $\mathcal{L}^*$, and hence is finite locally free.

Now suppose that S is affine. There is a section

$$\sigma : \omega_{G/S} \longrightarrow \mathcal{I}$$

of the projection $\mathcal{I} \to \mathcal{I}/\mathcal{I}^2$. It induces an algebra morphism

$$\sigma' : \mathrm{S}_{\mathcal{O}_S}(\omega_{G/S}) \longrightarrow \mathcal{A},$$

and, by the lemma in 7.2, σ' factors through a morphism

$$\varphi : \mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} \longrightarrow \mathcal{A},$$

where $\mathcal{K}$ denotes the ideal generated by the p-th powers of sections of $\omega_{G/S}$. Filter $\mathcal{A}$ by powers of $\mathcal{I}$, and filter $\mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K}$ by powers of the ideal generated by $\omega_{G/S}$. The morphism φ plainly induces an epimorphism on the associated graded modules. Hence φ is an epimorphism between locally free $\mathcal{O}_S$-modules of the same rank (cf. 5.3.3), and is therefore an isomorphism. This proves that (i) implies (ii).

7.4.2.

Suppose now that G is of height ≤ 1 and locally of finite presentation over S.[71] The Frobenius morphism $\mathrm{Fr} : G \to G^{(p)}$ is integral and factors through the unit section of $G^{(p)}$; consequently, G is integral, and hence affine, over S. Put $\mathcal{A} = \mathcal{A}(G)$. Since G is locally of finite presentation over S, it follows that G is finite and of finite presentation over S, and hence that $\mathcal{A}(G)$ is an $\mathcal{O}_S$-module of finite presentation (cf. EGA IV$_1$, 1.4.7).

Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$. Since

$$\mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{I},$$

where $\eta_{\mathcal{A}}$ is the unit section of $\mathcal{A}$, the module $\mathcal{I}$ is of finite presentation over $\mathcal{O}_S$, as is $\omega_G = \mathcal{I}/\mathcal{I}^2$. If instead G is assumed to be of height ≤ 1 and locally of finite type over S, the same argument shows that $\mathcal{A}(G)$, $\mathcal{I}$, and $\omega_G = \mathcal{I}/\mathcal{I}^2$ are $\mathcal{O}_S$-modules of finite type.

Thus, under hypothesis (iii′), $\omega_{G/S}$ is finite locally free over $\mathcal{O}_S$, as is

$$\mathcal{L} = \mathrm{Lie}(G/S) = \omega^*_{G/S}.$$

Put

$$\mathcal{B} = \mathcal{U}_p(\mathcal{L})^*, \qquad H = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec} \mathcal{B}.$$

---

be an Artinian local ring of characteristic $p > 0$, let J be a proper nonzero ideal of A, and let H be the subgroup

$$\operatorname{Spec} A[x]/(x^p, Jx)$$

of $\alpha_{p,A}$; that is, for every A-algebra R,

$$H(R) = \{x \in R \mid x^p = 0 \text{ and } Jx = 0\}.$$

Then H is not flat over A, and hence is not of the form $\mathfrak{G}_p(\mathcal{L})$ with $\mathcal{L}$ a finite-rank free p-Lie algebra over A.

[71] The editors supplied details and simplified the original in what follows.

By Theorem 7.2, the identity map of $\mathcal{L}$ corresponds to a group morphism from $H = \mathfrak{G}_p(\mathcal{L})$ to G, and hence to a morphism of $\mathcal{O}_S$-algebras

$$\theta : \mathcal{A} \longrightarrow \mathcal{B}.$$

By definition, θ induces an isomorphism from $\omega_{G/S}$ to $\omega_{H/S}$; we must show that θ itself is an isomorphism.

We may restrict to the case in which S is affine. There is then a section τ of the projection $\mathcal{I} \to \omega_{G/S}$, inducing an algebra morphism

$$\tau' : \mathrm{S}_{\mathcal{O}_S}(\omega_{G/S}) \longrightarrow \mathcal{A}.$$

Every local section of $\mathcal{I}$ has zero p-th power, because $\mathrm{Fr} : G \to G^{(p)}$ factors through the unit section of $G^{(p)}$. Thus τ' induces a morphism of $\mathcal{O}_S$-algebras ψ fitting into the commutative diagram

$$\begin{tikzcd}
\mathrm{S}_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} \arrow[r, "\psi"] \arrow[dr, "\varphi"'] & \mathcal{A} \arrow[d, "\theta"] \\
 & \mathcal{B}
\end{tikzcd}$$

where $\mathcal{K}$ is the ideal generated by the p-th powers of sections of $\omega_{G/S}$. As in 7.4.1, ψ is an epimorphism of $\mathcal{O}_S$-modules. On the other hand, 7.4.1 shows that $\varphi = \theta \circ \psi$ is an isomorphism. Hence θ is an isomorphism as well. This proves (iii′) $\Rightarrow$ (i).

7.4.3.[72]

We finally prove that (iv) implies (iii). It suffices to show that $\omega_{G/S}$ is locally free, so we may suppose that S is affine with ring R. As observed at the beginning of 7.4.2, hypothesis (iv) then implies that $G = \operatorname{Spec} A$, where A is an R-algebra which is an R-module of finite presentation, as is

$$\omega_{G/A} = I/I^2,$$

where I is the augmentation ideal of A. Since G is also assumed flat over S, the algebra A is a finite locally free R-module (cf. [BAC] II, § 5.2, Theorem 1 and Corollary 2). By the cited result, to show that $\omega_{G/A}$ is locally free of finite rank, it suffices to show that

$$(\omega_{G/A})_{\mathfrak{m}}$$

is flat for every maximal ideal $\mathfrak{m}$ of R. Thus we may suppose that R is local and that A is free of rank $n+1$; then I is free of rank n. Let $\mathfrak{m}$ be the maximal ideal of R, and put

$$k = R/\mathfrak{m}.$$

Let I_k be the augmentation ideal of A_k, and let r be the dimension of

$$\omega_{G_k/k} = I_k/I_k^2.$$

Choose a basis $(e_1, \dots, e_n)$ of I_k such that $(e_{r+1}, \dots, e_n)$ is a basis of I_k^2, and choose elements $x_1, \dots, x_n$ of I lifting the e_i. Nakayama's lemma shows that $(x_1, \dots, x_n)$ is a basis of I over R.

[72]The editors added this paragraph to prove (iv) $\Rightarrow$ (iii); cf. editor's note (71).

Let N be the sub-R-module of I with basis $(x_1, \ldots, x_r)$, and let B be the quotient of the symmetric algebra of N by the ideal generated by the elements x^p, for $x \in N$. Since every element of I has zero p-th power, we obtain a morphism of R-algebras

$$\psi : B \longrightarrow A.$$

By 7.4.2, $\psi \otimes k$ is an isomorphism. It follows that $\operatorname{Coker} \psi = 0$, and, if $K = \ker \psi$, the morphism

$$\tau : K \otimes k \longrightarrow B \otimes k$$

is zero. But ψ is surjective and A is flat over R, so τ is also injective. Hence $K \otimes k = 0$.

Since A is an R-module of finite presentation, K is an R-module of finite type (cf. [BAC] I, § 2.8, Lemma 9). Nakayama's lemma now gives $K = 0$. Thus ψ is an isomorphism of R-algebras. Since $\psi^{-1}(I)$ contains the augmentation ideal J of B, it follows that $\psi^{-1}(I) = J$. Therefore ψ^{-1} induces an isomorphism of R-modules

$$I/I^2 \xrightarrow{\sim} J/J^2 = N.$$

This proves that $\omega_{G/S}$ is finite locally free, and hence proves (iv) $\Rightarrow$ (iii). The proof of Theorem 7.4 is complete.

Remark 7.5.[73]

It follows immediately from Theorems 7.2 and 7.4 that the functors

$$G \longmapsto \operatorname{Lie}(G), \qquad \mathcal{L} \longmapsto \mathfrak{G}_p(\mathcal{L})$$

induce quasi-inverse equivalences between the category of S-groups which are locally of finite presentation, flat, and of height ≤ 1, and the full subcategory of the category of $\mathcal{O}_S$-p-Lie algebras consisting of the finite locally free $\mathcal{O}_S$-p-Lie algebras.

8. The Case of a Base Field

8.1.

We now summarize the results obtained when S is the spectrum of a field k of characteristic $p > 0$. We call a k-group scheme *algebraic* if its underlying scheme is of finite type over k. Theorem 7.2 then gives the following result.[74]

Theorem 8.1.1. *The functor $\mathfrak{G}_p$, which associates with every finite-dimensional restricted p-Lie algebra $\mathcal{L}$ over k the k-group $\mathfrak{G}_p(\mathcal{L})$, is left adjoint to the functor which associates* $\operatorname{Lie}(G)$ *with every algebraic k-group G.*

Combining this with Theorem 7.4 gives:

Theorem 8.1.2. *The functors*

$$\mathfrak{G}_p : \mathcal{L} \longmapsto \mathfrak{G}_p(\mathcal{L}), \qquad G \longmapsto \operatorname{Lie}(G),$$

induce quasi-inverse equivalences between the category of finite-dimensional restricted p-Lie algebras over k and the category of algebraic k-groups of height at most 1.

Since $\mathfrak{G}_p$ is a left adjoint, it commutes with inductive limits,[75] and in particular with cokernels. Let

$$\varphi : G \longrightarrow H, \qquad \varphi' : G' \longrightarrow H$$

[73]The editors added this remark.

[74]The editors supplied the numbering 8.1.1 through 8.1.3 in order to highlight the results stated here.

[75]The editors expanded the argument which follows, as well as the proof of Corollary 8.1.3.

be two morphisms between algebraic k-groups of height at most 1. The fiber product $G \times_H G'$ is again an algebraic k-group of height at most 1, because the Frobenius morphism

$$\mathrm{Fr} : G \longrightarrow G^{(p)}$$

commutes with fiber products. Thus the inclusion of the category of algebraic k-groups of height at most 1 into the category of all algebraic k-groups commutes with fiber products, and hence in particular with kernels. We obtain:

Corollary 8.1.3. *The functor $\mathfrak{G}_p$ is exact in the following sense. Let*

$$\pi : \mathcal{L}_1 \longrightarrow \mathcal{L}_2$$

be a surjective homomorphism between finite-dimensional restricted p-Lie algebras over k, and let i be the inclusion of $\mathcal{L}_0 = \ker \pi$ into $\mathcal{L}_1$. Then the following sequence of algebraic k-groups is exact:

$$1 \longrightarrow \mathfrak{G}_p(\mathcal{L}_0) \xrightarrow{\mathfrak{G}_p(i)} \mathfrak{G}_p(\mathcal{L}_1) \xrightarrow{\mathfrak{G}_p(\pi)} \mathfrak{G}_p(\mathcal{L}_2) \longrightarrow 1.$$

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Indeed, by the preceding discussion, $\mathfrak{G}_p(i)$ induces an isomorphism from $\mathfrak{G}_p(\mathcal{L}_0)$ onto $\ker(\mathfrak{G}_p(\pi))$. This kernel is the same whether it is taken in the category of all algebraic k-groups or in the subcategory of those of height at most 1. Furthermore,

$$\mathfrak{G}_p(\pi) : \mathfrak{G}_p(\mathcal{L}_1) \longrightarrow \mathfrak{G}_p(\mathcal{L}_2)$$

identifies $\mathfrak{G}_p(\mathcal{L}_2)$ with the quotient of $\mathfrak{G}_p(\mathcal{L}_1)$ by $\mathfrak{G}_p(\mathcal{L}_0)$ in the category of algebraic k-groups.

Remark 8.1.4.[77]

Let $\varphi : G \to H$ be a morphism of k-groups and put $K = \ker(\varphi)$. Suppose that φ is covering for the (fpqc) topology. This is in particular the case if it is covering for a coarser topology, such as the (fppf) topology. Then φ is a K-torsor over H (cf. IV, 5.1.7.1). Moreover, by IV, 6.3.1, there exists a covering of H by affine open subsets S_i, and for each i an affine faithfully flat morphism $T_i \to S_i$ factoring through φ. Now $G \times_H T_i$ is T_i-isomorphic to $K \times T_i$, hence faithfully flat over T_i. By (fpqc) descent, $G \times_H S_i \to S_i$ is faithfully flat, and therefore φ is faithfully flat.

Conversely, if φ is faithfully flat and quasi-compact, respectively faithfully flat and locally of finite presentation, then it is covering for the (fpqc), respectively (fppf), topology (cf. IV, 6.3.1). Finally, a morphism of sheaves is covering if and only if it is an epimorphism (cf. IV, 4.4.3). In particular, a quasi-compact morphism of k-groups is faithfully flat if and only if it is an epimorphism of (fpqc) sheaves.

8.2. Proposition. *Consider an exact sequence*[78] *of algebraic groups over a field k of characteristic $p > 0$:*

$$1 \longrightarrow G' \xrightarrow{\tau} G \xrightarrow{\pi} G'' \longrightarrow 1.$$

Consider the following assertions:

(i) *π is smooth;*

[76] Moreover, by VIA, 3.2, $\mathfrak{G}_p(\mathcal{L}_2)$ represents the (fppf) quotient sheaf of $\mathfrak{G}_p(\mathcal{L}_1)$ by $\mathfrak{G}_p(\mathcal{L}_0)$.

[77] The editors added this remark, communicated by O. Gabber, for use in 8.3.1.

[78] Thus π is faithfully flat and τ identifies G' with $\ker \pi$, so G'' represents the (fppf) quotient sheaf of G by G'; cf. VIA, 3.2 and 5.2.

(ii) *G' is smooth;*

(iii) *for every integer $n > 0$, the sequence induced by τ and π,*

$$1 \longrightarrow \mathrm{Fr}^n G' \longrightarrow \mathrm{Fr}^n G \longrightarrow \mathrm{Fr}^n G'' \longrightarrow 1.$$

is exact;

(iv) *the morphism*

$$\mathrm{Fr}\,\pi : \mathrm{Fr}\, G \longrightarrow \mathrm{Fr}\, G''$$

is an epimorphism of (fppf) sheaves;

(v) *the morphism*

$$\mathrm{Lie}(\pi) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(G'')$$

is surjective.

Then

$$\text{(i)} \Longleftrightarrow \text{(ii)} \Longrightarrow \text{(iii)} \Longrightarrow \text{(iv)} \Longleftrightarrow \text{(v)},$$

and all five assertions are equivalent when G is smooth over k.

Indeed, (i) is equivalent to (ii) by VIB, 9.2 (vii), and it is clear that (iii) implies (iv). The equivalence of (iv) and (v) follows from 8.1.3.

The implication (ii) ⇒ (iii) follows from the diagram

$$\begin{tikzcd}
1 \arrow[r] & G' \arrow[r, "\tau"] \arrow[d, "\mathrm{Fr}^n(G'/k)"'] & G \arrow[r, "\pi"] \arrow[d, "\mathrm{Fr}^n(G/k)"'] & G'' \arrow[r] \arrow[d, "\mathrm{Fr}^n(G''/k)"'] & 1 \\
1 \arrow[r] & G'^{(p^n)} \arrow[r, "\tau^{(p^n)}"'] & G^{(p^n)} \arrow[r, "\pi^{(p^n)}"'] & G''^{(p^n)} \arrow[r] & 1
\end{tikzcd}$$

whose two rows are exact. Since $\mathrm{Fr}^n(G'/k)$ is an epimorphism of (fppf) sheaves by Corollary 8.3.1 below, π induces an epimorphism from $\mathrm{Fr}^n G$ onto $\mathrm{Fr}^n G''$; here one uses the snake lemma for sheaves of groups, not necessarily commutative.

Likewise, suppose that G is smooth over k. Then $\mathrm{Fr}(G/k)$ is an epimorphism. If $\mathrm{Fr}\,\pi$ is also an epimorphism, the same snake lemma, applied to the preceding diagram with $n = 1$, shows that $\mathrm{Fr}(G'/k)$ is an epimorphism. Hence G' is smooth over k by 8.3.1 below.

8.3. Proposition.[79] *Let G be a group locally of finite type over a field k of characteristic $p > 0$. There exists an integer n_0 such that*

$$G/(\mathrm{Fr}^n G)$$

is smooth over k for every $n \geq n_0$.

The construction of $G/(\mathrm{Fr}^n G)$ commutes with extension of the base field (4.1.1 and VIA, 3.3.2), so we may suppose that k is perfect. Then G_{red} is a k-group locally of finite type (cf. VIA, 0.2), and, putting

$$H = G_{\mathrm{red}} \backslash G,$$

we have the following commutative exact diagram:

[79]The editors replaced "algebraic" by "locally of finite type."

$$\begin{tikzcd}
1 \arrow[r] & G_{\mathrm{red}} \arrow[r] \arrow[d, "\mathrm{Fr}^n(G_{\mathrm{red}}/k)"] & G \arrow[r] \arrow[d, "\mathrm{Fr}^n(G/k)"] & H \arrow[d, "\mathrm{Fr}^n(H/k)"] \\
1 \arrow[r] & G_{\mathrm{red}}^{(p^n)} \arrow[r] & G^{(p^n)} \arrow[r] & H^{(p^n)} .
\end{tikzcd}$$

The scheme H is the spectrum of a finite local k-algebra with residue field k (cf. VIA, 5.6.1). There is therefore an integer n_0 such that, for every $n \geq n_0$, $\mathrm{Fr}^n(H/k)$ factors through the unit section of $H^{(p^n)}$. It follows that, for $n \geq n_0$, $\mathrm{Fr}^n(G/k)$ factors through $G_{\mathrm{red}}^{(p^n)}$, and VIA, 5.4.1 gives the commutative diagram

$$\begin{tikzcd}
G \arrow[r, "\mathrm{Fr}^n(G/k)"] \arrow[d, "\pi"'] & G_{\mathrm{red}}^{(p^n)} \\
G/(\mathrm{Fr}^n G) \arrow[ur, "i"'] &
\end{tikzcd}$$

where i is a closed immersion and π is the canonical projection. Moreover, i induces a homeomorphism of the underlying topological spaces, so it is an isomorphism. Since k is perfect, $G_{\mathrm{red}}^{(p^n)}$ is smooth over k (VIA, 1.3.1). Thus $G/(\mathrm{Fr}^n G)$ is smooth over k for every $n \geq n_0$.

8.3.1. Corollary. *Let G be a group locally of finite type over a field k of characteristic $p > 0$, and let $n \geq 1$.*[80] *The following conditions are equivalent:*

(i) *G is smooth over k;*

(ii) $\mathrm{Fr}^n(G/k) : G \to G^{(p^n)}$ *is an epimorphism of (fppf) sheaves;*

(iii) $\mathrm{Fr}^n(G/k) : G \to G^{(p^n)}$ *is faithfully flat.*

Since G is locally of finite type over k, $\mathrm{Fr}^n(G/k)$ is of finite presentation, so the equivalence of (ii) and (iii) follows from 8.1.4. Suppose that G is smooth over k, and hence reduced. Since $\mathrm{Fr}^n(G/k)$ is surjective, it is faithfully flat (cf. VIA, 6.2 or VIB, 1.3).

Conversely, suppose that $\mathrm{Fr}^n(G/k)$ is faithfully flat. The morphism

$$\mathrm{Fr}^n(G^{(p^n)}/k)$$

is obtained from $\mathrm{Fr}^n(G/k)$ by base change (cf. 4.1.3), so it too is faithfully flat, as is the composite

$$\mathrm{Fr}^{2n}(G/k) : G \longrightarrow G^{(p^n)} \longrightarrow G^{(p^{2n})}.$$

It follows that, for every $m \in \mathbb{N}$,

$$\mathrm{Fr}^{mn}(G/k) : G \longrightarrow G^{(p^{mn})}$$

is faithfully flat. Hence it induces an isomorphism

$$G/(\mathrm{Fr}^{mn} G) \simeq G^{(p^{mn})}$$

[80] The editors made explicit the equivalence of (ii) and (iii) and expanded the proof.

(cf. VIA, 5.4.1). By Proposition 8.3, $G^{(p^{mn})}$ is smooth over k for m sufficiently large, and therefore G is smooth as well by (fpqc) descent (cf. EGA IV$_4$, 17.7.1).

8.4.

In the two statements which conclude this Exposé, we return to the case of a field k of arbitrary characteristic.

If k has characteristic 0, let $n \geq 1$. If k has characteristic $p > 0$, let $n \geq 1$ be prime to p. In both cases we simply say that n is *prime to the characteristic of* k. If G is a group scheme over k, we denote by

$$n_G : G \longrightarrow G$$

the morphism of k-schemes which sends $x \in G(T)$ to $x^n \in G(T)$, for every k-scheme T.

Proposition. *Let G be an algebraic group over a field k, and let n be prime to the characteristic of k. Then*

$$n_G : G \longrightarrow G$$

is an étale morphism.

By VIB, 1.3, it suffices to show that n_G is étale at the origin.[81] Let A be the local ring of G at the origin and I its maximal ideal. By II, 3.9.4, the map

$$\mathrm{Lie}(n_G) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(G)$$

induced by n_G is multiplication by n. It is therefore an isomorphism, as is the endomorphism induced by n_G on

$$I/I^2 = \mathrm{Lie}(G)^*.$$

Suppose first that $\operatorname{char} k = 0$. The group G is smooth over k (VIB, 1.6.1; see also VIIB, 3.3.1), and the canonical homomorphism

$$\mathrm{Sym}(I/I^2) \longrightarrow \mathrm{gr}_I(A)$$

is an isomorphism, where $\mathrm{gr}_I(A)$ is the graded ring associated with the I-adic filtration. Hence n_G induces an automorphism of $\mathrm{gr}_I(A)$, and therefore also of the completion $\widehat{A}$. Thus n_G is étale at the origin (cf. EGA IV$_4$, 17.6.3).

Suppose next that $\operatorname{char} k = p > 0$ and G has height at most 1. Then A is isomorphic to the quotient of the symmetric algebra on

$$\omega_{G/k} = I/I^2$$

by the ideal generated by the p-th powers of the elements of $\omega_{G/k}$ (cf. 7.4). The same argument as in characteristic 0 shows that n_G induces an automorphism of A.

Suppose that G has height at most r, and that the result has been proved for groups of height at most $r-1$. Let B, A, A' be the affine algebras of $\mathrm{Fr}\, G, G$, and

$$G' = \mathrm{Fr}\, G \backslash G,$$

and let $n_B, n_A, n_{A'}$ be the endomorphisms induced by $n_{\mathrm{Fr}\, G}, n_G, n_{G'}$, respectively. Put

$$I' = I \cap A',$$

the maximal ideal of A'. The cartesian square

[81] The editors changed "étale at the origin" to "étale" in the statement and added the preceding sentence.

$$\begin{tikzcd} \mathrm{Fr}\, G \arrow[r] \arrow[d] & G \arrow[d] \\ e \arrow[r] & G' \end{tikzcd}$$

gives

$$B = A/I'A.$$

[82] The endomorphism $n_{A'}$, respectively n_B, is the endomorphism induced by n_A on A', respectively on B. By VIA, 3.2, A is a faithfully flat A'-module. Since A' is an Artinian local ring, because G' is an algebraic k-group of height at most $r-1$, A is a free A'-module. The restriction of n_A to A' is $n_{A'}$, which is an isomorphism by the induction hypothesis. Nakayama's lemma therefore shows that n_A is an automorphism as soon as the endomorphism it induces on $A/I'A$ is an automorphism.

That endomorphism is n_B, which is an automorphism because $\mathrm{Fr}\, G$ has height at most 1. Hence n_A is an automorphism.

Finally, let G be any algebraic group over a field of characteristic $p > 0$. The preceding argument shows that n_G induces automorphisms of the k-schemes $\mathrm{Fr}^r G$. These schemes are affine over k, and their algebras are the quotients of the local algebra A by the ideal

$$I^{\{p^r\}}$$

generated by the p^r-th powers of the elements of I. Since n_G induces automorphisms of all the algebras $A/I^{\{p^r\}}$, passage to the projective limit shows that it induces an automorphism of $\widehat{A}$. Thus n_G is étale at the origin (EGA IV$_4$, 17.6.3).

8.5. Proposition. *Let G be a finite algebraic group of rank n over the field k. Then*

$$n_G : G \longrightarrow G$$

is the zero morphism of G.

Combining 8.4 and 8.5 immediately gives the following corollary.

Corollary 8.5.1. *Let G be a finite algebraic group of rank n over the field k. If n is prime to the characteristic of k, then G is étale over k.* [83]

We now prove 8.5. Let H be a normal subgroup of G, of rank m over k, and let

$$\lambda : H \times G \longrightarrow G$$

be induced by the multiplication of G. With the notation of VIA, 3.2, there is a cartesian square

$$\begin{tikzcd} H \times G \arrow[r, "\lambda"] \arrow[d, "\mathrm{pr}_2"'] & G \arrow[d, "\pi"] \\ G \arrow[r, "\pi"] & H\backslash G \end{tikzcd}$$

[82] The editors expanded the original argument in what follows.

[83] The editors added this corollary, which the original indicated implicitly by "cf. 8.4." For another proof which does not use 8.5, see [TO70], Lemma 5.

Since $\pi : G \to H\backslash G$ is faithfully flat and quasi-compact (VIA, 3.2), while pr_2 is locally free of rank m, EGA IV$_2$, 2.5.2 shows that $G \to H\backslash G$ is locally free of rank m. If

$$r = \mathrm{rk}_k(H\backslash G),$$

then

$$n = \mathrm{rk}_k(G) = rm.$$

On the other hand, for every k-scheme T there is an exact sequence of abstract groups

$$1 \longrightarrow H(T) \longrightarrow G(T) \longrightarrow (H\backslash G)(T).$$

It is therefore clear that n_G is zero whenever m_H and $r_{H\backslash G}$ are zero. Taking $H = G^0$, the identity component, the quotient $G^0\backslash G$ is étale (cf. VIA, 5.5.1). We may consequently assume that G is étale over k, or that it is infinitesimal (cf. 7.0).

If G is étale, extension of the base field reduces us to the case where k is algebraically closed. Then G is a constant group (cf. I, 4.1), and the assertion is classical.

If G is nonzero and infinitesimal, then k necessarily has characteristic $p > 0$ (cf. VIB, 1.6.1 or VIIB, 3.3.1). The subgroups $\mathrm{Fr}^r G$ form a composition series of G whose successive quotients have height at most 1.

We are thus reduced to the case where G has height at most 1. Let A, respectively L, be the affine algebra, respectively the Lie algebra, of G, and put

$$U = U_p(L).$$

By 7.4, $G = \mathfrak{G}_p(L)$, and hence $A = U^*$. Therefore, if $\dim_k L = r$, the rank of G over k is p^r (cf. 5.3.3). We shall study the morphism

$$p_G : G \longrightarrow G$$

defined by taking the p-th power. It induces an endomorphism p_A of A, and, by duality, an endomorphism p_U of U.

Let I be the augmentation ideal of A. We shall prove

$$p_A(I) \subset I^p. \tag{*}$$

Assuming this, one has

$$p_A^r(I) \subset I^{p^r}.$$

Moreover,

$$I^{r(p-1)+1} = 0,$$

because I is generated by r elements whose p-th powers vanish. Since $p^r > r(p-1)$, we obtain $p_A^r(I) = 0$, and therefore p_G^r is the zero morphism.

For every integer $s \geq 1$, let

$$m_A^{s-1} : A^{\otimes s} \longrightarrow A, \qquad \Delta_U^{s-1} : U \longrightarrow U^{\otimes s},$$

denote the maps induced by the multiplication m_A of A and the comultiplication Δ_U of U, respectively. Then p_A is the composite

$$A \xrightarrow{\Delta_A^{p-1}} A^{\otimes p} \xrightarrow{m_A^{p-1}} A.$$

Since the transpose of m_A, respectively of Δ_A, is Δ_U, respectively m_U, the endomorphism

$$p_U = {}^t p_A$$

is the composite

$$U \xrightarrow{\Delta_U^{p-1}} U^{\otimes p} \xrightarrow{m_U^{p-1}} U.$$

Let J be the augmentation ideal of U.[84] Then

$$U = k1_U \oplus J,$$

and we write $\pi : U \to J$ for the projection with kernel $k1_U$. For $s \geq 1$, let $(I^s)^\perp$ be the orthogonal of I^s in $A^* = U$. Thus $(I^s)^\perp$ consists of the elements $u \in U$ for which the composite

$$I^{\otimes s} \xrightarrow{m_A^{s-1}} I \xrightarrow{u} k$$

is zero. Since the transpose of m_A is Δ_U, $(I^s)^\perp$ is the vector subspace P_{s-1} consisting of those $u \in U$ for which $\Delta_U^{s-1}(u)$ vanishes on $I^{\otimes s}$. If

$$\overline{\Delta}_U^{s-1} = \pi^{\otimes s} \circ \Delta_U^{s-1},$$

then

$$(I^s)^\perp = P_{s-1} = \ker \overline{\Delta}_U^{s-1}$$

(see also VIIB, 1.3.6).

To prove $(*)$, it is enough to show that the transpose $p_U = {}^t p_A$ maps P_{p-1} into

$$I^\perp = k1_U.$$

Since $p_U(1_U) = 1_U$, it suffices to prove

$$p_U(P_{p-1}^+) = 0, \qquad P_{p-1}^+ = J \cap P_{p-1}. \tag{**}$$

An easy induction on s shows that P_{s-1}^+ is spanned by the products

$$x_1 \cdots x_t, \qquad 1 \leq t \leq s-1, \quad x_i \in L$$

(cf. VIIB, 4.3).

For $x_1, \ldots, x_t \in L$, one has

$$p_U(x_1 x_2 \cdots x_t) = m_U^{p-1}\left(\prod_{j=1}^{t} \sum_{i=1}^{p} 1 \otimes \cdots \otimes \underset{\text{position } i}{x_j} \otimes \cdots \otimes 1\right).$$

The expression inside parentheses is a sum of p^t terms x_h, indexed by the maps

$$h : \{1, \ldots, t\} \longrightarrow \{1, \ldots, p\}.$$

Such a map determines an ordered partition $\mathcal{P}_h$ of $\{1, \ldots, t\}$ into at most p parts.[85] Indeed, let

$$i_1 < \cdots < i_q$$

be the elements in the image of h, put

$$I_a = h^{-1}(i_a) \quad (1 \leq a \leq q), \qquad x_{I_a} = \prod_{j \in I_a} x_j,$$

[84] The editors expanded the original argument in this paragraph.

[85] The editors expanded the original argument and replaced the notion of a preorder by the equivalent notion of an ordered partition.

where the product is taken in increasing order. The map h corresponds to the p-tensor

$$1 \otimes \cdots \otimes x_{I_1} \otimes \cdots \otimes x_{I_q} \otimes \cdots \otimes 1,$$

where x_{I_a} occupies position i_a. Its image under the iterated multiplication is the product

$$x_{I_1} \cdots x_{I_q}.$$

This depends only on the ordered partition

$$\mathcal{P} = (I_1, \dots, I_q);$$

denote it by $x_{\mathcal{P}}$. For a fixed $\mathcal{P}$, the same term is obtained for every choice of $i_1 < \cdots < i_q$ in $\{1, \dots, p\}$. Consequently,

$$p_U(x_1 x_2 \cdots x_t) = \sum_{\mathcal{P}} \binom{p}{n(\mathcal{P})} x_{\mathcal{P}},$$

where $\mathcal{P}$ ranges over the ordered partitions of $\{1, \dots, t\}$ into at most p parts and $n(\mathcal{P})$ is the number of parts. Here

$$1 \leq n(\mathcal{P}) \leq \min(t, p).$$

If $t < p$, all the coefficients $\binom{p}{n(\mathcal{P})}$ vanish in k, and hence

$$p_U(x_1 \cdots x_t) = 0.$$

Thus p_U vanishes on P^+_{p-1}. This proves $(**)$, then $(*)$, and completes the proof of 8.5.

Corollary 8.5.2.[86] *Let S be a reduced scheme and G a finite locally free S-group of rank n. Then*

$$n_G : G \longrightarrow G$$

is the zero morphism of G.

Let S' be the disjoint union of the $\operatorname{Spec} \mathcal{O}_{S,\eta}$, where η ranges over the maximal points of S. Since S is reduced, the morphism $S' \to S$ is schematically dominant. The same is therefore true of

$$f : G_{S'} \longrightarrow G,$$

because G is finite locally free over S (cf. EGA IV_3, 11.10.5). Since $G \to S$ is affine, and hence separated, the coincidence locus of n_G and the zero morphism is a closed subscheme of G. By 8.5 it contains the image of f, and therefore equals G. Thus n_G is the zero morphism.

Remark 8.5.3.

A theorem of P. Deligne (see [TO70], p. 4) also says that if G is a commutative finite locally free S-group of rank n over an arbitrary base S, then $n_G = 0$.

[86] The editors added this corollary, which is noted in Exposé VIII, Remark 7.3.1.

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Exposé $\mathrm{VII}_{\mathrm{B}}$
Infinitesimal Study of Group Schemes

by P. Gabriel

B) Formal Groups

[0] The study of formal groups is usually extremely simple. If this does not appear clearly in the pages that follow, the responsibility lies with an arithmetician who claims to know formal groups over "something other than fields."[1] We have therefore developed, for formal groups "locally free over projective limits of Artinian rings," the generalities ordinarily stated for formal groups defined over a field. For a more detailed study of the latter, we refer to the 1964/65 Heidelberg–Strasbourg algebraic geometry seminar.[2]

0. Review of Pseudocompact Rings and Modules

This section contains a few technical preliminaries. We recall and complete here several results from [CA] (*Des catégories abéliennes*, Bull. Soc. Math. France **90** (1962)).

0.1. A *left pseudocompact ring* is a separated, complete topological ring with identity which has a basis of neighborhoods of 0 consisting of left ideals $\mathfrak{l}$ of finite colength, that is,

$$\mathrm{length}_A(A/\mathfrak{l}) < +\infty.$$

We shall assume here that A is commutative, so that no distinction need be made "between left and right."

In particular, the quotients $A/\mathfrak{l}$ are Artinian rings, and A identifies with the topological projective limit of these rings, each endowed with the discrete topology.

A complete Noetherian local ring $(A, \mathfrak{m})$ is plainly pseudocompact.[3]

[0]Editors' note: Version of 13 October 2024.

[1]The interest of formal groups over a complete Noetherian local ring appears, for example, in the work of Lubin and Tate (cf. [LT65]). The study of formal groups over an arbitrary base, and of lifting and deformation questions—in particular for Barsotti–Tate groups ("p-divisible groups")—has given rise to an abundant literature; see, for example, [LT66, Ta67, Gr74, Me72, La75, Fo77, Il85, Br00]. In particular, the results of the present exposé are largely taken up again in Chapter I of [Fo77].

[2]The editors found only such a seminar dated 1965/66 and entitled "Linear Algebraic Groups," in which the notion of a formal group does not appear; see instead [De72].

[3]When it is endowed with the $\mathfrak{m}$-adic topology.

0.1.1. Every closed ideal I of A is the intersection of the open ideals that contain it.[4] Every closed maximal ideal is therefore open. Moreover, if $\mathfrak{l}$ is an open ideal of A, the maximal ideals of $A/\mathfrak{l}$ correspond bijectively to the maximal ideals $\mathfrak{m}$ of A containing $\mathfrak{l}$; the latter are thus open and closed. Consequently, every closed maximal ideal is an open maximal ideal (and hence closed), the converse being clear. We denote the set of these ideals by $\Upsilon(A)$.

If $\mathfrak{l}$ is an open ideal of A and $\mathfrak{m} \in \Upsilon(A)$, the localization $(A/\mathfrak{l})_{\mathfrak{m}}$ is a local ring when $\mathfrak{m}$ contains $\mathfrak{l}$, and is zero otherwise. Since $A/\mathfrak{l}$ is Artinian, it is a direct product of finitely many local rings; thus

$$A/\mathfrak{l} \simeq \prod_{\mathfrak{m}\in\Upsilon(A)} (A/\mathfrak{l})_{\mathfrak{m}}.$$

It follows that there are "canonical" isomorphisms

$$A \simeq \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} \prod_{\mathfrak{m}} (A/\mathfrak{l})_{\mathfrak{m}} \simeq \prod_{\mathfrak{m}} \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_{\mathfrak{m}} \simeq \prod_{\mathfrak{m}} A_{\mathfrak{m}},$$

where

$$A_{\mathfrak{m}} = \varprojlim_{\mathfrak{l}} (A/\mathfrak{l})_{\mathfrak{m}}.$$

This *local component* $A_{\mathfrak{m}}$ is a filtered projective limit of Artinian local rings endowed with the discrete topology. It is therefore a local ring which is pseudocompact for the projective-limit topology.[5]

0.1.2. Let $r(A)$ be the intersection of the open maximal ideals of A, that is, the cartesian product of the ideals $\mathfrak{m}A_{\mathfrak{m}}$ under the identification

$$A = \prod_{\mathfrak{m}} A_{\mathfrak{m}}.$$

For every open ideal $\mathfrak{l}$ of A, the image of $r(A)$ in $A/\mathfrak{l}$ is contained in the radical of $A/\mathfrak{l}$. Some power of this image is therefore zero, so $r(A)^n \subset \mathfrak{l}$ for all sufficiently large n. Thus the sequence $r(A)^n$ tends to 0.

The same is true of x^n when $x \in r(A)$. In other words, every element of $r(A)$ is topologically nilpotent, and the converse is clear. It follows that the sequence with general term

$$1 + x + \cdots + x^n$$

converges, and converges to $1/(1-x)$, whenever $x \in r(A)$. This shows that $r(A)$ is the Jacobson radical of A, that is, the intersection of all maximal ideals of A (cf. Bourbaki, *Algèbre*, Chap. 8, §6, Th. 1).[6]

Remarks.[7]
a) If $\mathfrak{p}$ is an open prime ideal of A, then, since $A/\mathfrak{p}$ is Artinian, $\mathfrak{p}$ is maximal. Consequently, $\Upsilon(A)$ is the set of open prime ideals of A.

---

[4] Indeed, if $x \notin I$, there is an open ideal $\mathfrak{l}$ such that $(x + \mathfrak{l}) \cap I = \varnothing$; then $I + \mathfrak{l}$ is an open ideal that does not contain x. In what follows, the editors have also made explicit the fact that every "closed maximal" ideal is maximal and open.

[5] Notice that $A_{\mathfrak{m}}$ is the localization of A at the maximal ideal $\mathfrak{m}$. Indeed, the identity element $e_{\mathfrak{m}}$ of $A_{\mathfrak{m}}$ is an idempotent of A not belonging to $\mathfrak{m}$. Since $A = A_{\mathfrak{m}} \times B$, where $B = A(1 - e_{\mathfrak{m}})$, the ring $A_{\mathfrak{m}}$ identifies with the localization $A_{e_{\mathfrak{m}}}$, and hence also with $S^{-1}A$, where $S = A \setminus \mathfrak{m}$. Furthermore, $e_{\mathfrak{m}}(1 - e_{\mathfrak{m}}) = 0$, so $\mathfrak{m}$ contains $1 - e_{\mathfrak{m}}$, hence also B; thus $\mathfrak{m}$ identifies with $\mathfrak{m}A_{\mathfrak{m}} \times B$.

[6] Indeed, let $x \in r(A)$. If $\mathfrak{m}$ were a maximal ideal not containing x, there would be $y \in A$ such that $1 - xy \in \mathfrak{m}$. But $xy \in r(A)$, so $1 - xy$ is invertible, a contradiction. Notice the following consequence: if $\Upsilon(A) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$ is finite, then the $\mathfrak{m}_i$ are all the maximal ideals of A.

[7] The editors added these remarks in order to compare the definition of the formal spectrum $\mathrm{Spf}(A)$ given in 1.1 with those in EGA I, 10.1.2 and 10.4.2.

b) Each $\mathfrak{m}A_{\mathfrak{m}}$ is an ideal of definition of $A_{\mathfrak{m}}$, that is, an open ideal I for which I^n tends to 0 (cf. EGA 0_{I}, 7.1.2). Consequently,

$$\mathrm{Spec}(A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}),$$

endowed with the topological ring $A_{\mathfrak{m}}$, is an affine formal scheme in the sense of EGA I, 10.1.2.

c) The topological ring A is admissible in the sense of EGA 0_{I}, 7.1.2 if and only if $r(A)$ is open (and hence an ideal of definition), and this holds if and only if $\Upsilon(A)$ is finite. In that case the affine formal scheme

$$\mathrm{Spf}(A) = \mathrm{Spec}(A/r(A))$$

(cf. EGA I, 10.1.2) has $\Upsilon(A)$, endowed with the discrete topology, as its underlying space; the sections of its structure sheaf over a subset $E \subset \Upsilon(A)$ form the ring

$$\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}.$$

d) Let A be an arbitrary pseudocompact ring. In 1.1 below, the space $\Upsilon(A)$ is endowed with the discrete topology and with the sheaf of rings whose ring of sections over any subset E is

$$\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}.$$

By *b)*, every point then has an open neighborhood which is an affine formal scheme; this therefore defines a formal scheme (EGA I, 10.4.2), denoted by $\mathrm{Spf}(A)$. For this formal scheme to be affine, it must be quasi-compact, so $\Upsilon(A)$ must be finite; in that case $\mathrm{Spf}(A)$ agrees with the definition of EGA I, 10.1.2.

0.1.3. If A and B are pseudocompact rings, a *homomorphism* from A to B is, by definition, a continuous map compatible with addition, multiplication, and the identity elements. Such a homomorphism sends a topologically nilpotent element of A to a topologically nilpotent element of B; it therefore maps the radical $r(A)$ into the radical $r(B)$.

0.2. Let A be a commutative pseudocompact ring. A *pseudocompact A-module M* is a separated, complete topological A-module having a basis of neighborhoods of 0 consisting of submodules M' for which M/M' has finite length over A.

If M and N are pseudocompact A-modules, a *morphism* from M to N is, by definition, a continuous A-linear map. We write

$$\mathrm{Hom}_c(M, N)$$

for the group of these morphisms.

Proposition 0.2.B.[8]

(i) *The pseudocompact A-modules form an abelian category, denoted* $\mathrm{PC}(A)$*. In particular, for every morphism $f : M \to N$, the submodule* $\mathrm{Im}(f)$ *is complete and hence closed in N.*

(ii) *The pseudocompact A-modules of finite length, whose topology is therefore discrete, form a system of cogenerators of* $\mathrm{PC}(A)$*.*

[8]The editors have stated the results of this paragraph separately in the following proposition, and have then indicated the steps of the proof; cf. [CA], § IV.3, or [DG70], § V.2.

(iii) *Infinite products and filtered projective limits are exact; in other words,* $\mathrm{PC}(A)$ *satisfies axiom* (AB5*).[9]

For the reader's convenience, we briefly indicate the steps of the proof. First we use the following lemma ([CA], § IV.3, Lemma 1; for the proof, see [BEns], III, § 7.4, Theorem 1 and Example 2).

Lemma 0.2.C. *Let B be a ring, let I be a filtered ordered set, and let (M_i) and (N_i) be two projective systems of left B-modules indexed by I. Let*

$$(s_i) : (M_i) \longrightarrow (N_i)$$

be a morphism of projective systems such that s_i is surjective and has Artinian kernel for every i. Then the map

$$\varprojlim_i s_i : \varprojlim_i M_i \longrightarrow \varprojlim_i N_i$$

is surjective.

Corollary 0.2.D *(*[CA]*, § IV.3, Propositions 10 and 11). Let M be a pseudocompact A-module.*

(i) *If K is a closed submodule of M, then M/K, endowed with the quotient topology, is a pseudocompact A-module.*

(ii) *Let (M_i) be a filtered decreasing family of closed submodules of M.*

  (a) *The canonical map*

$$M \longrightarrow \varprojlim_i M/M_i$$

  is surjective and has kernel $\bigcap_i M_i$.

  (b) *For every closed submodule N of M, one has*

$$N + \bigcap_i M_i = \bigcap_i (N + M_i).$$

Proof. Let (L_j) be the filtered decreasing family of open submodules of M. We endow M/K with the quotient topology; that is, a basis of neighborhoods of 0 is formed by the open submodules $(K+L_j)/K$. Since K is closed, it is equal to the intersection of the $K+L_j$, and hence the map

$$\tau : M/K \longrightarrow \varprojlim_j M/(K + L_j)$$

is injective. It is also open, since the right-hand side is the inverse limit of the discrete modules $M/(K + L_j)$. Moreover, for each j, the map

$$t_j : M/L_j \longrightarrow M/(K + L_j)$$

is surjective and has Artinian kernel. Thus, by the preceding lemma, the map t in the following commutative diagram is surjective:

[9] Cf. [Gr57], I, § 1.5 and Proposition 1.8. One may also consult, for example, [Po73], § 2.8, or [We95], Appendix A.4.

$$\begin{tikzcd} M \arrow[r, "p", "\sim"'] \arrow[d] & \varprojlim_j M/L_j \arrow[d, "t"] \\ M/K \arrow[r, "\tau"] & \varprojlim_j M/(K+L_j). \end{tikzcd}$$

Since p is an isomorphism because M is complete, it follows that τ is surjective, and hence is an isomorphism. This proves (i).

Let us prove (ii)(a). By the foregoing, for every i there is an isomorphism

$$M/M_i \xrightarrow{\sim} \varprojlim_j M/(M_i + L_j),$$

so the two horizontal arrows in the following commutative diagram are isomorphisms:

$$\begin{tikzcd} M \arrow[r, "p", "\sim"'] \arrow[d, "g"] & \varprojlim_j M/L_j \arrow[d, "s"] \\ \varprojlim_i M/M_i \arrow[r, "\sim"'] & \varprojlim_{i,j} M/(M_i+L_j). \end{tikzcd}$$

Moreover, for each j, the family of submodules $(M_i + L_j)/L_j$ has a least element, since M/L_j is Artinian. Consequently, the morphism

$$s_j : M/L_j \longrightarrow \varprojlim_i M/(L_j + M_i)$$

is surjective; therefore, by the preceding lemma, s is surjective. It follows that g is surjective. Finally, the kernel of g is the inverse limit of the M_i, that is, their intersection. This proves (a).

Let us deduce (b) from this. Since N is a closed submodule, and hence separated and complete, it is a pseudocompact module for the topology induced by that of M. Therefore, by (a), the morphisms f and g in the following commutative exact diagram are surjective:

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r] \arrow[d, "f"] & M \arrow[r] \arrow[d, "g"] & M/N \arrow[r] \arrow[d, "h"] & 0 \\ 0 \arrow[r] & \varprojlim_i N/(N\cap M_i) \arrow[r] & \varprojlim_i M/M_i \arrow[r] & \varprojlim_i M/(N+M_i)\,. \end{tikzcd}$$

The "snake lemma" then shows that the sequence

$$0 \longrightarrow \ker f \longrightarrow \ker g \longrightarrow \ker h \longrightarrow 0$$

is exact, and the equality

$$N + \bigcap_i M_i = \bigcap_i (N + M_i)$$

follows.

We can now prove Proposition 0.2.B. Let $f : M \to N$ be a morphism of pseudocompact A-modules. Then $K = \ker(f)$ is a closed submodule of M, and hence is separated and complete; thus K, with the topology induced by that of M, is a pseudocompact module. By 0.2.D (i), the module M/K, endowed with the quotient topology, is pseudocompact.

Let us show that the continuous bijective morphism

$$M/K \longrightarrow \mathrm{Im}(f)$$

is bicontinuous. After identifying M/K with $\mathrm{Im}(f)$, it is enough to show that the topology T induced by that of N is finer than the quotient topology Q.Let (L_j), respectively (N_i), be the filtered decreasing families of open submodules of M, respectively N, and put

$$N'_i = N_i \cap \mathrm{Im}(f).$$

Let $P = (K + L_j)/K$ be a submodule of M/K open for Q. Since $M/(K + L_j)$ is Artinian, the family $N'_i + P$ has a least element $N'_{i_0} + P$. The N'_i are open, and hence closed, for T, and therefore also closed for Q. It follows from 0.2.D (ii)(b) that

$$N'_{i_0} + P = \bigcap_i (N'_i + P) = P + \bigcap_i N'_i = P,$$

whence $N'_{i_0} \subset P$. Thus P is open for T, and $M/K \to \mathrm{Im}(f)$ is consequently an isomorphism of pseudocompact modules.

In particular, $\mathrm{Im}(f)$ is complete for T, and hence closed in N. By 0.2.D (i) once more, $\mathrm{Coker}(f)$, endowed with the quotient topology, is pseudocompact. This proves that $\mathrm{PC}(A)$ is an abelian category.

Arbitrary inverse limits also exist in $\mathrm{PC}(A)$: if (M_i) is an inverse system of pseudocompact modules, its inverse limit has as underlying module the inverse limit of the underlying modules, endowed with the inverse-limit topology. Moreover, if one has a family of exact sequences in $\mathrm{PC}(A)$,

$$0 \longrightarrow K_i \longrightarrow M_i \longrightarrow Q_i \longrightarrow 0,$$

then the sequence

$$0 \longrightarrow \prod_i K_i \longrightarrow \prod_i M_i \longrightarrow \prod_i Q_i \longrightarrow 0$$

is exact. Assertion (iii) of 0.2.B follows, since in every abelian category in which arbitrary products exist, conditions (a) and (b) of 0.2.D are equivalent, and are equivalent to the exactness of filtered inverse limits (cf. [Mi65], III, 1.2–1.9, or [Po73], Chap. 2, Th. 8.6).

Finally, every pseudocompact module M is a submodule of the product

$$\prod_L M/L,$$

where L ranges over the open submodules of M. Thus the objects of finite length form a system of cogenerators of $\mathrm{PC}(A)$. (Moreover, every object of length n is isomorphic to a quotient A^n/L, where L is an open submodule of A^n of colength n; these quotients therefore form a *set* of cogenerators.) This completes the proof of 0.2.B.

Let (Ab)[10] be the category of abelian groups and let $\mathrm{LF}(A)$ be the full subcategory of $\mathrm{PC}(A)$ formed by the objects of finite length. For every object M of $\mathrm{PC}(A)$, let h_c^M denote the functor

$$\mathrm{LF}(A) \longrightarrow (\mathrm{Ab}), \qquad N \longmapsto \mathrm{Hom}_c(M, N).$$

By [CA], § II.4, Theorem 1, Lemma 4, and Corollary 1, we have the following results.[11]

[10] The editors inserted Proposition 0.2.E and Corollary 0.2.F into this paragraph; they will be useful in 0.2.2. In the original, these results appeared in 0.3.

[11] Indeed, $\mathrm{PC}(A)$ has a set of Artinian cogenerators, filtered projective limits are exact in it, and $\mathrm{LF}(A)$ is the subcategory of Artinian objects. The dual category $\mathrm{PC}(A)^\circ$ therefore has a set of Noetherian generators, and filtered inductive limits are exact in it. By the proof of [CA], § II.4, Theorem 1, the functor $M \mapsto \mathrm{Hom}_c(M, -)$ is an anti-equivalence from $\mathrm{PC}(A)$ to $\mathrm{Lex}(\mathrm{LF}(A), (\mathrm{Ab}))$. Likewise, Lemma 4 and Corollary 1 of *loc. cit.* state results "dual" to those of Corollary 0.2.F. For a "direct" proof, see [DG70], § V.2, Theorem 3.1, Lemma 3.5, and Corollaries 3.3 and 3.4.

Proposition 0.2.E. *The functor $M \mapsto h_c^M$ is an anti-equivalence from* $\mathrm{PC}(A)$ *to the category*

$$\mathrm{Lex}(\mathrm{LF}(A), (\mathrm{Ab}))$$

of left-exact functors $\mathrm{LF}(A) \longrightarrow (\mathrm{Ab})$.

Corollary 0.2.F.

(i) *An object P of* $\mathrm{PC}(A)$ *is projective if and only if the functor h_c^P is exact; that is, if and only if the functor* $\mathrm{Hom}_c(P, -)$ *is exact on* $\mathrm{LF}(A)$.

(ii) *Let (M_i) be a filtered projective system*[12] *of objects of* $\mathrm{PC}(A)$. *For every object N of* $\mathrm{LF}(A)$, *there is an isomorphism, functorial in N,*

$$\mathrm{Hom}_c\left(\varprojlim_i M_i, N\right) \simeq \varinjlim_i \mathrm{Hom}_c(M_i, N).$$

(iii) *Every filtered projective limit and every product*[12] *of projective objects of* $\mathrm{PC}(A)$ *is a projective object of* $\mathrm{PC}(A)$.

Finally, Corollary 0.2.F gives the following result.

Corollary 0.2.G. *Let $(M_i)_{i\in I}$ be a family of objects of* $\mathrm{PC}(A)$. *Then $\prod_{i\in I} M_i$ is projective if and only if each M_i is projective.*

Indeed, for every $N \in \mathrm{Ob}\,\mathrm{LF}(A)$, one has

$$\mathrm{Hom}_c\left(\prod_i M_i, N\right) \simeq \bigoplus_i \mathrm{Hom}_c(M_i, N).$$

0.2.1. Each local component $A_{\mathfrak{m}}$ of A is a direct factor of A, hence a projective object of $\mathrm{PC}(A)$ (A is manifestly projective). Moreover,

$$S_{\mathfrak{m}} = A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$$

is the unique simple quotient of $A_{\mathfrak{m}}$, so $A_{\mathfrak{m}}$ is indecomposable. On the other hand, every simple object of $\mathrm{PC}(A)$ is isomorphic to a unique $S_{\mathfrak{m}}$. By [CA], IV, § 3, Corollary 1 of Theorem 3,[13] we therefore have:

Proposition.

(i) *Every projective object of* $\mathrm{PC}(A)$ *is a direct product of indecomposable projective objects, uniquely determined up to isomorphism.*

(ii) *Every indecomposable projective object is isomorphic to a unique $A_{\mathfrak{m}}$, where $\mathfrak{m} \in \Upsilon(A)$.*

Definition. A pseudocompact A-module M is said to be *topologically free* if it is isomorphic to the product of a family (A_i) of copies of A.

In this case, a family (m_i) of elements of M is called a *pseudobasis* of M if the A-linear maps $A_i \longrightarrow M$ sending the unit element of A_i to m_i extend to an isomorphism

$$\prod_i A_i \xrightarrow{\ \sim\ } M.$$

[12]Since every infinite product is a filtered projective limit of finite products, every additive functor that commutes with filtered projective limits also commutes with infinite products.

[13]This refers to the "dual" statements established in *loc. cit.*, § 2, Theorem 2, and § 1, Proposition 2. For a "direct" proof, see [DG70], V, § 2, Theorem 4.5 and Example 4.6(b).

0.2.2. [14] If M is a pseudocompact A-module, we write $M^\dagger$ for the A-module, without topology,

$$\mathrm{Hom}_c(M, A).$$

Conversely, if N is an A-module, we write

$$N^* = \mathrm{Hom}_A(N, A)$$

for its dual, endowed with the topology of pointwise convergence. Thus a basis of neighborhoods of 0 in N^* is formed by the submodules

$$\mathcal{V}(x, \mathfrak{l}) = \{f \in N^* \mid f(x) \in \mathfrak{l}\},$$

where $x \in N$ and $\mathfrak{l}$ is an open ideal of A.

This makes N^* a pseudocompact A-module. Indeed, first suppose that $N = A$. Then $N^* = A$, endowed with its topology as a pseudocompact ring. If N is a free module $A^{(I)}$, then N^* is the product A^I, endowed with the product topology. Furthermore, for every morphism $\varphi : N_1 \longrightarrow N_2$, the transpose

$${}^t\varphi : N_2^* \longrightarrow N_1^*$$

is continuous, because the inverse image under ${}^t\varphi$ of a submodule $\mathcal{V}(x_1, \mathfrak{l})$ of N_1^* is precisely the submodule $\mathcal{V}(\varphi(x_1), \mathfrak{l})$ of N_2^*. For arbitrary N, choose a presentation

$$A^{(J)} \xrightarrow{\varphi} A^{(I)} \xrightarrow{\pi} N \longrightarrow 0.$$

Then N^* is the kernel of the continuous morphism

$${}^t\varphi : A^I \longrightarrow A^J,$$

and hence N^* is pseudocompact.

When A is Artinian (in which case one may take $\mathfrak{l} = 0$ above), Corollary 0.2.F gives:

Proposition. *When A is Artinian, the functors*

$$P \longmapsto P^\dagger \qquad \text{and} \qquad Q \longmapsto Q^*,$$

where P, respectively Q, is a projective object of $\mathrm{PC}(A)$*, respectively a projective A-module, establish an anti-equivalence between the category of projective pseudocompact A-modules and the category of projective A-modules.*[15]

In particular, when A is a field k, the functor

$$P \longmapsto P^\dagger$$

is an anti-equivalence between the category of all pseudocompact k-modules (also called *linearly compact k-vector spaces*) and the category of k-vector spaces.[16]

0.3.[17] Let L and M be two pseudocompact A-modules. The functor

$$\mathrm{LF}(A) \longrightarrow (\mathrm{Ab}), \qquad N \longmapsto \mathrm{Bil}_c(L \times M, N),$$

[14] The editors have spelled out the results of this paragraph, which play an important role below; cf. 1.2.3, 1.3.5, 2.2.1, and so on.

[15] Recall also that, over an Artinian ring, a module is projective if and only if it is flat; see, for example, Exposé VI$_\mathrm{B}$, editors' note (54), or 0.3.7 below.

[16] The direct sum in $\mathrm{PC}(k)$ of a family $(V_i)_{i\in I}$ of linearly compact k-vector spaces is $\left(\prod_{i\in I} V_i^\dagger\right)^*$.

[17] Editors' note: This paragraph has been modified to take account of the additions made in 0.2.

where $\mathrm{Bil}_c(L \times M, N)$ denotes the abelian group of continuous A-bilinear maps from $L \times M$ to N, is left exact.

By 0.2.E, there therefore exists a pseudocompact A-module $L\widehat{\otimes}_A M$, unique up to unique isomorphism, representing this functor; that is, for every object N of $\mathrm{LF}(A)$, there is a functorial isomorphism

$$\mathrm{Hom}_c(L\widehat{\otimes}_A M, N) \simeq \mathrm{Bil}_c(L \times M, N).$$

Moreover, $L\widehat{\otimes}_A M$ identifies with the projective limit

$$P = L \,\widehat{\otimes_A}\, M$$

of the discrete A-modules

$$(L/L') \otimes_A (M/M'),$$

where L' and M' range respectively over the open submodules of L and M.

Indeed, let $\varphi : L \times M \to N$ be a continuous bilinear map into a discrete A-module N of finite length. By Lemma 0.3.1 below, there exist open submodules L' and M' of L and M such that

$$\varphi(L' \times M) = \varphi(L \times M') = \{0\}.$$

This means that the map

$$\overline{\varphi} : L \otimes_A M \longrightarrow N$$

induced by φ has the form $\varphi' \circ p$, where p is the canonical projection from $L \otimes_A M$ onto $(L/L') \otimes_A (M/M')$. If $\widehat{\varphi}$ denotes the composite

$$P \longrightarrow (L/L') \otimes_A (M/M') \xrightarrow{\varphi'} N,$$

then the map $\varphi \mapsto \widehat{\varphi}$ is a functorial bijection from $\mathrm{Bil}_c(L \times M, N)$ onto $\mathrm{Hom}_c(P, N)$. Hence

$$P \simeq L\widehat{\otimes}_A M.$$

Thus the pseudocompact module $L\widehat{\otimes}_A M$ is the separated completion of $L \otimes_A M$ for the linear topology defined by the kernels of the canonical projections

$$L \otimes_A M \longrightarrow (L/L') \otimes_A (M/M'),$$

and it will be called the *completed tensor product* of L and M.

If $x \in L$ and $y \in M$, the image of $x \otimes_A y$ in $L\widehat{\otimes}_A M$ will be denoted by $x\widehat{\otimes}_A y$.

Lemma 0.3.1. *Let L, M, and N be pseudocompact A-modules, with N of finite length. If $\varphi : L \times M \to N$ is a continuous A-bilinear map, there exist open submodules L' and M' of L and M such that*

$$\varphi(L' \times M) = \varphi(L \times M') = \{0\}.$$

Proof. The inverse image $\varphi^{-1}(0)$ is an open neighborhood of $(0,0)$, and therefore contains an open set of the form $L_1 \times M_1$, where L_1 and M_1 are open submodules of L and M. Since L/L_1 has finite length, there are elements $x_1, \dots, x_r$ of L such that

$$L_1 + Ax_1 + \cdots + Ax_r = L.$$

If $M' \subset M_1$ is sufficiently small, then $\varphi(x_i, M') = 0$ for every i, since the map $y \mapsto \varphi(x_i, y)$ is continuous. It follows that $\varphi(L, M') = \{0\}$. Likewise, $\varphi(L', M) = \{0\}$ when L' is sufficiently small. □

Corollary 0.3.1.1.[18] *Let M be a pseudocompact A-module.*

[18] Editors' note: This corollary has been added.

(i) *For every open submodule M', there exists an open ideal $\mathfrak{l}$ of A such that*

$$\overline{\mathfrak{l}M} \subset M'.$$

(ii) *Consequently,*

$$M \simeq \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M},$$

where $\mathfrak{l}$ ranges over the filtered projective system of open ideals of A, and every $M/\overline{\mathfrak{l}M}$ is endowed with the quotient topology, which makes it a pseudocompact module; cf. 0.2.D.

Proof. Consider the map

$$\varphi : A \times M \longrightarrow M/M', \qquad (a, m) \longmapsto am + M'.$$

By 0.3.1, there is an open ideal $\mathfrak{l}$ of A such that $\mathfrak{l}M \subset M'$. Since M' is also closed, it contains $\overline{\mathfrak{l}M}$. Since the intersection of the open submodules of M is zero, it follows that

$$\bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M} = (0).$$

On the other hand, by 0.2.D the map

$$\phi : M \longrightarrow \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}$$

is surjective. By the proof of 0.2.B, ϕ therefore induces an isomorphism

$$M/\operatorname{Ker}(\phi) \xrightarrow{\sim} \varprojlim_{\mathfrak{l}} M/\overline{\mathfrak{l}M}.$$

But we have just seen that

$$\operatorname{Ker}(\phi) = \bigcap_{\mathfrak{l}} \overline{\mathfrak{l}M} = (0).$$

□

Remark 0.3.1.2.[19] The completed tensor product satisfies the usual associativity condition: if L, M, and P are pseudocompact A-modules, there is a canonical isomorphism

$$(L\widehat{\otimes}_A M)\widehat{\otimes}_A P \simeq L\widehat{\otimes}_A(M\widehat{\otimes}_A P).$$

Indeed, both objects represent the functor which associates to every object N of $\mathrm{LF}(A)$ the abelian group of continuous A-trilinear maps from $L \times M \times P$ to N.

0.3.2. Let

$$L' \xrightarrow{f} L \xrightarrow{g} L'' \longrightarrow 0$$

be an exact sequence, and let M be an object of $\mathrm{PC}(A)$. It is clear that for every object N of $\mathrm{LF}(A)$, the induced sequences

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Bil}_c(L'' \times M, N) & \longrightarrow & \mathrm{Bil}_c(L \times M, N) & \longrightarrow & \mathrm{Bil}_c(L' \times M, N) \\
 & & \| & & \| & & \| \\
0 & \longrightarrow & \mathrm{Hom}_c(L''\widehat{\otimes}_A M, N) & \longrightarrow & \mathrm{Hom}_c(L\widehat{\otimes}_A M, N) & \longrightarrow & \mathrm{Hom}_c(L'\widehat{\otimes}_A M, N)
\end{array}$$

[19] Editors' note: This remark has been added.

are exact. By 0.2.E, this is equivalent to saying that the sequence

$$L'\widehat{\otimes}_A M \xrightarrow{f\widehat{\otimes}M} L\widehat{\otimes}_A M \xrightarrow{g\widehat{\otimes}M} L''\widehat{\otimes}_A M \longrightarrow 0 \qquad (*)$$

is exact. Hence:

Corollary. *For every pseudocompact A-module M, the functor $-\widehat{\otimes}_A M$ is right exact.*

In particular, take $L = A$, let f be the inclusion of a closed ideal $\mathfrak{a}$ into A, and let g be the canonical projection of A onto $A/\mathfrak{a}$. We may then identify $A\widehat{\otimes}_A M$ with M by means of the map

$$x\widehat{\otimes}_A m \longmapsto xm.$$

Since the image of $\mathfrak{a}\widehat{\otimes}_A M$ is closed in M (cf. 0.2.B), and the image of $\mathfrak{a} \otimes_A M$ is dense in $\mathfrak{a}\widehat{\otimes}_A M$, the image of $f\widehat{\otimes}M$ is precisely the closure $\overline{\mathfrak{a}M}$ of $\mathfrak{a}M$ in M. The exact sequence $(*)$ therefore gives the isomorphism

$$(A/\mathfrak{a})\widehat{\otimes}_A M \xrightarrow{\sim} M/\overline{\mathfrak{a}M}.$$

0.3.3. Nakayama's Lemma. *Let A be a pseudocompact ring, M a pseudocompact A-module, and $\mathfrak{a}$ an ideal of A contained in the radical $r(A)$. Then*

$$\overline{\mathfrak{a}M} = M \qquad \Longrightarrow \qquad M = 0.$$

Proof. Suppose that $\overline{\mathfrak{a}M} = M$.[20] Let M' be an open submodule of M, and put $M'' = M/M'$. Since M'' is discrete, $\mathfrak{a}M''$ is closed in M'', and hence equals $\overline{\mathfrak{a}M''}$. By 0.3.2, the canonical map

$$M/\overline{\mathfrak{a}M} \longrightarrow M''/\overline{\mathfrak{a}M''}$$

is surjective, so

$$\mathfrak{a}M'' = \overline{\mathfrak{a}M''} = M''.$$

Since M'' is a finitely generated A-module and $\mathfrak{a} \subset r(A)$, the usual Nakayama lemma implies that $M'' = 0$. Consequently, every open submodule M' of M is equal to M, and therefore $M = 0$.[21] □

0.3.4. The usual consequences of Nakayama's lemma follow.

Corollary. *Let $\mathfrak{a}$ be a closed ideal contained in $r(A)$, and let $f : M \to N$ be a morphism of pseudocompact A-modules.*

(i) *The map f is surjective if the induced map*

$$f' : M/\overline{\mathfrak{a}M} \longrightarrow N/\overline{\mathfrak{a}N}$$

is surjective.[22]

(ii) *If N is projective, then f is an isomorphism whenever f' is an isomorphism.*

[20] Editors' note: Replacing $\mathfrak{a}$ by its closure if necessary, we may assume that $\mathfrak{a}$ is closed.

[21] Editors' note: Indeed, M is the projective limit of the quotients $M/M' = 0$.

[22] Editors' note: Consequently, every pseudocompact A-module is a quotient of a topologically free A-module. One first reduces to the case in which A is local; cf. the proof of 0.3.7.

Proof. Part (i) follows by applying Lemma 0.3.3 to $\operatorname{Coker}(f)$. For (ii), suppose that f' is an isomorphism. By (i), f is then surjective, and hence has a section. Apply Lemma 0.3.3 to $\operatorname{Ker}(f)$. □

When A is local with maximal ideal $\mathfrak{m}$, one may also deduce from 0.3.3 the following exchange theorem.

Theorem. *Let A be a local pseudocompact ring, let $\mathfrak{m}$ be its maximal ideal, let M be a topologically free A-module with pseudobasis $(m_i)_{i\in I}$ (0.2.1), and let N be a closed direct summand of M. There is a pseudobasis of M consisting of elements of N together with some of the m_i.*

Proof. This is clear when A is a field: use the duality of 0.2.2 and apply the usual exchange theorem. Consequently,

$$N/\overline{\mathfrak{m}N}$$

has a complementary topologically free $A/\mathfrak{m}$-module with pseudobasis $(\overline{m}_i)_{i\in J}$, where $\overline{m}_i$ is the image of m_i in

$$M/\overline{\mathfrak{m}M},$$

and J is a subset of I.[23] If

$$P = \prod_{i\in J} Am_i,$$

then the canonical map

$$N \oplus P \longrightarrow M$$

is "bijective modulo $\mathfrak{m}$." It is therefore bijective by what precedes. For another proof, see [CA], § IV.2, Proposition 8. □

0.3.5. Consider three pseudocompact A-modules L, M, N, where N has finite length. Give the A-module $\operatorname{Hom}_c(M, N)$ the discrete topology. Every element

$$\psi \in \operatorname{Hom}_c(L, \operatorname{Hom}_c(M, N))$$

defines a continuous bilinear map

$$\psi' : L \times M \longrightarrow N, \qquad (\ell, m) \longmapsto \psi(\ell)(m).$$

This gives a natural isomorphism

$$\operatorname{Hom}_c(L, \operatorname{Hom}_c(M, N)) \xrightarrow{\sim} \operatorname{Hom}_c(L \widehat{\otimes}_A M, N). \tag{1}$$

This is another characterization of $\operatorname{Hom}_c(L\widehat{\otimes}_A M, N)$, which we now use when M is the projective limit of a filtered projective system (M_i) of pseudocompact A-modules. By (1) and 0.2.F (ii), there are natural isomorphisms

$$\begin{aligned}\operatorname{Hom}_c\left(L\widehat{\otimes}_A \varprojlim_i M_i, N\right) &\simeq \operatorname{Hom}_c\left(L, \operatorname{Hom}_c(\varprojlim_i M_i, N)\right)\\ &\simeq \operatorname{Hom}_c\left(L, \varinjlim_i \operatorname{Hom}_c(M_i, N)\right).\end{aligned} \tag{2}$$

Moreover, because $\varinjlim_i \operatorname{Hom}_c(M_i, N)$ is discrete, every continuous map from L into this module factors through a finite-length quotient of L. Hence the natural map

$$\varinjlim_i \operatorname{Hom}_c(L, \operatorname{Hom}_c(M_i, N)) \xrightarrow{\sim} \operatorname{Hom}_c\left(L, \varinjlim_i \operatorname{Hom}_c(M_i, N)\right) \tag{3}$$

[23] Editors' note: We have corrected $N/\mathfrak{m}N$ to $N/\overline{\mathfrak{m}N}$, and likewise for $M/\overline{\mathfrak{m}M}$ below.

is an isomorphism. Finally, (1) and 0.2.F (ii) again give natural isomorphisms

$$\begin{aligned}\varinjlim_i \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) &\simeq \varinjlim_i \mathrm{Hom}_c(L\widehat{\otimes}_A M_i, N)\\ &\simeq \mathrm{Hom}_c\left(\varprojlim_i (L\widehat{\otimes}_A M_i), N\right).\end{aligned} \tag{4}$$

Combining (2), (3), and (4) yields the following result.

Proposition. *Let (M_i) be a filtered projective system of objects of $\mathrm{PC}(A)$, let $L \in \mathrm{PC}(A)$, and let $N \in \mathrm{LF}(A)$. There is an isomorphism, functorial in N,*

$$\mathrm{Hom}_c\left(L\widehat{\otimes}_A \varprojlim_i M_i, N\right) \simeq \mathrm{Hom}_c\left(\varprojlim_i (L\widehat{\otimes}_A M_i), N\right),$$

and consequently an isomorphism

$$L\widehat{\otimes}_A \varprojlim_i M_i \simeq \varprojlim_i (L\widehat{\otimes}_A M_i).$$

Thus the completed tensor product commutes with filtered projective limits.[24]

0.3.6. In particular,[25] the completed tensor product commutes with arbitrary products. For example, since A is the product of its local components $A_{\mathfrak{m}}$ (0.1.1), every pseudocompact A-module

$$M \simeq A\widehat{\otimes}_A M$$

identifies with the product of the modules

$$M_{\mathfrak{m}} = A_{\mathfrak{m}}\widehat{\otimes}_A M,$$

which are called the local components of M.

Likewise, let M and N be pseudocompact A-modules. Recall from 0.2.2 that

$$M^\dagger = \mathrm{Hom}_c(M, A).$$

Consider the map

$$\varphi : M^\dagger \otimes_A N^\dagger \longrightarrow (M\widehat{\otimes}_A N)^\dagger$$

which sends $f \otimes g$ to the map

$$m\widehat{\otimes} n \longmapsto f(m)g(n)$$

from $M\widehat{\otimes}_A N$ to A. The map φ is bijective when $M \simeq A$.

Corollary. *If A is Artinian, then φ is an isomorphism whenever M is topologically free, or more generally projective.*

Indeed, with N fixed, each of the functors

$$M \longmapsto (M\widehat{\otimes}_A N)^\dagger \qquad \text{and} \qquad M \longmapsto M^\dagger \otimes_A N^\dagger$$

takes arbitrary products to direct sums, by the preceding result and 0.2.F.

[24] This also recovers 0.3.1.1:

$$L = L\widehat{\otimes}_A A = L\widehat{\otimes}_A \varprojlim_{\mathfrak{l}} (A/\mathfrak{l}) \simeq \varprojlim_{\mathfrak{l}} L/\mathfrak{l}L,$$

where $\mathfrak{l}$ ranges over the open ideals of A.

[25] Cf. editor's note (12).

Remark 0.3.6.A.[26] *The same use of 0.2.F gives the following result. Let A be an Artinian ring, let $M, Q \in \mathrm{PC}(A)$, and let $N \in \mathrm{LF}(A)$. If Q is projective, then there are natural isomorphisms*

$$\mathrm{Hom}_c(M,Q) \xrightarrow{\sim} \mathrm{Hom}_A(Q^\dagger, M^\dagger), \qquad Q^\dagger \otimes_A N \xrightarrow{\sim} \mathrm{Hom}_c(Q,N).$$

0.3.7. For every $\mathfrak{m} \in \Upsilon(A)$, the functor

$$M \longmapsto A_{\mathfrak{m}} \widehat{\otimes}_A M$$

is plainly exact. Since every projective pseudocompact A-module P is a product of modules of the form $A_{\mathfrak{m}}$, it follows that

$$M \longmapsto P \widehat{\otimes}_A M$$

is exact whenever P is projective. The converse is also true.

Proposition. *Let A be a pseudocompact ring and P a pseudocompact A-module. The following conditions are equivalent:*

(i) *P is a projective object of* $\mathrm{PC}(A)$*;*

(ii) *every local component $P_{\mathfrak{m}}$ is a topologically free $A_{\mathfrak{m}}$-module;*

(iii) *the functor $M \mapsto P \widehat{\otimes}_A M$ is exact.*

The equivalence of (i) and (ii) follows from 0.2.F (iii) and 0.2.1, and we have just proved (ii) $\Rightarrow$ (iii). Suppose that the functor $M \mapsto P \widehat{\otimes}_A M$ is exact. Since $P \widehat{\otimes}_A M$ is the product of its local components,

$$(P \widehat{\otimes}_A M)_{\mathfrak{m}} \simeq P_{\mathfrak{m}} \widehat{\otimes}_{A_{\mathfrak{m}}} M_{\mathfrak{m}},$$

we reduce to the case in which A is local. We shall prove that P is then topologically free.

Let $\mathfrak{m}$ be the maximal ideal of A. The quotient $P/\overline{\mathfrak{m}}P$ is a linearly compact vector space over $A/\mathfrak{m}$, and hence a product of copies of $A/\mathfrak{m}$ (0.2.2). Therefore there is a family $(A_i)_{i \in I}$ of copies of A and an isomorphism

$$\phi : \prod_{i \in I} (A_i/\mathfrak{m}) \xrightarrow{\sim} P/\overline{\mathfrak{m}}P.$$

Since $\prod_{i \in I} A_i$ is projective, there is a commutative square

$$\begin{tikzcd} \prod_{i \in I} A_i \arrow[r, "\psi"] \arrow[d, "p"'] & P \arrow[d, "q"] \\ \prod_{i \in I} (A_i/\mathfrak{m}) \arrow[r, "\varphi", "\sim"'] & P/\mathfrak{m}P \end{tikzcd}$$

where p and q are the canonical projections. Apply Nakayama's lemma to $\operatorname{Coker}\psi$. Since

$$(A/\mathfrak{m}) \widehat{\otimes}_A \psi$$

is precisely ϕ, the map ψ is surjective.[27]

[26] The editors added this remark, which will be useful in 2.3.1.

[27] This proves the result announced in editor's note (22): every pseudocompact A-module M is a quotient of a topologically free A-module. Since products are exact in $\mathrm{PC}(A)$, one reduces to the local case treated here.

Put
$$B = \prod_{i \in I} A_i, \qquad N = \ker \psi.$$
There is then the following commutative exact diagram:
$$\begin{tikzcd}
 & & & 0 \arrow[d] \arrow[dddll, dashed, rounded corners] & \\
 & \mathfrak{m}\widehat{\otimes}_A N \arrow[r] \arrow[d] & \mathfrak{m}\widehat{\otimes}_A B \arrow[r] \arrow[d] & \mathfrak{m}\widehat{\otimes}_A P \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A\widehat{\otimes}_A N \arrow[r] \arrow[d] & A\widehat{\otimes}_A B \arrow[r, "\psi"] \arrow[d] & A\widehat{\otimes}_A P \arrow[r] \arrow[d] & 0 \\
 & (A/\mathfrak{m})\widehat{\otimes}_A N \arrow[r] & (A/\mathfrak{m})\widehat{\otimes}_A B \arrow[r, "\varphi"] & (A/\mathfrak{m})\widehat{\otimes}_A P \arrow[r] & 0
\end{tikzcd}.$$
The snake lemma, applied to the first two rows, shows that in the bottom row the morphism
$$(A/\mathfrak{m})\widehat{\otimes}_A N \longrightarrow (A/\mathfrak{m})\widehat{\otimes}_A B$$
is a monomorphism. But ϕ is an isomorphism, so $(A/\mathfrak{m})\widehat{\otimes}_A N = 0$. Hence $N = 0$ by 0.3.3, and ψ is an isomorphism.[28]

0.3.8. Corollary. *Let A be a complete Noetherian local ring and P a pseudocompact A-module. Then P is topologically free if and only if P is flat over A.*

Indeed, the canonical map
$$M \otimes_A P \longrightarrow M\widehat{\otimes}_A P$$
is bijective when $M = A$, and hence also when M is Noetherian: take a finite presentation of M and use the right exactness of both the tensor product and the completed tensor product.

Now P is flat if and only if
$$M \longmapsto M \otimes_A P$$
is exact on Noetherian modules. Likewise, the proof of Proposition 0.3.7 shows that P is topologically free provided the sequence
$$0 \longrightarrow \mathfrak{m}\widehat{\otimes}_A P \longrightarrow A\widehat{\otimes}_A P \longrightarrow (A/\mathfrak{m})\widehat{\otimes}_A P \longrightarrow 0$$
is exact. Thus P is topologically free if and only if the functor
$$M \longmapsto M\widehat{\otimes}_A P$$
is exact on Noetherian modules. The corollary follows from the equality
$$M \otimes_A P = M\widehat{\otimes}_A P$$
established above.

[28] A completely analogous proof, using the "nilpotent Nakayama lemma," shows that if A is Artinian and P is a flat A-module, then every local component of P is a free A-module (cf. [BAC], II, § 3.2, Corollary 2 to Proposition 5).

0.4. Profinite Algebras

Let k be a pseudocompact ring. A *topological k-algebra* is a commutative topological ring A, together with a morphism of topological rings $k \to A$. The algebra A is called a *profinite k-algebra* if its underlying topological k-module is pseudocompact.

Let $\mathfrak{l}$ be an open k-submodule of such an A. The composite map

$$\varphi : A \times A \xrightarrow{\text{mult}} A \longrightarrow A/\mathfrak{l}$$

is continuous. Lemma 0.3.1 therefore gives an open k-submodule $\mathfrak{n}$ of A such that $\varphi(A \times \mathfrak{n}) = 0$. Thus $\mathfrak{l}$ contains the open ideal $A\mathfrak{n}$, and A is a pseudocompact ring.

Likewise, let M be a topological A-module whose underlying k-module is pseudocompact. If M' is an open k-submodule of M, applying Lemma 0.3.1 to

$$A \times M \xrightarrow{\text{mult}} M \longrightarrow M/M'$$

shows that M' contains an open A-submodule. Hence M is also a pseudocompact A-module.[29] Conversely:

Lemma. *Let A be a profinite k-algebra and M a pseudocompact A-module. The k-module $M|_k$, obtained by restriction of scalars, is pseudocompact.*

Indeed, every finite-length pseudocompact A-module is isomorphic to a quotient

$$A^n/L,$$

where L is an open submodule of A^n, and is therefore a pseudocompact k-module. Since $M|_k$ is a projective limit of modules of this kind, it is a pseudocompact k-module.

0.4.1. If A and B are profinite k-algebras, a *morphism* from A to B is, by definition, a continuous homomorphism of k-algebras. We denote the category of profinite k-algebras by

$$\mathrm{Alp}\,/k.$$

It plainly has projective limits: the algebra underlying a projective limit is the projective limit of the underlying algebras, endowed with the projective-limit topology.

It also has finite direct limits.[30] For example, if

$$f : A \longrightarrow B, \qquad g : A \longrightarrow C$$

are morphisms of profinite k-algebras, the amalgamated sum of B and C over A has

$$B \widehat{\otimes}_A C$$

as its underlying topological A-module. Here 0.4 makes B and C pseudocompact A-modules via f and g, and the multiplication is characterized by

$$(b \widehat{\otimes} c)(b' \widehat{\otimes} c') = bb' \widehat{\otimes} cc'$$

for $b, b' \in B$ and $c, c' \in C$.

Definition 0.4.2. *A profinite k-algebra C is said to have* finite length *if its underlying k-module has finite length, and is therefore discrete. We denote by* $\mathrm{Alf}\,/k$ *the full subcategory of* $\mathrm{Alp}\,/k$ *formed by the k-algebras of finite length.*[31]

[29] The editors added the following lemma; cf. editor's note (36) in Proposition 0.5.

[30] In 0.4.2 we shall see that it has arbitrary direct limits.

[31] Caution: k is an object of $\mathrm{Alf}\,/k$ only when k is Artinian; cf. 1.2.2 below.

For every profinite k-algebra A, let h_A be the functor

$$h_A : \mathrm{Alf}\,/k \longrightarrow (\mathbf{Sets}), \qquad C \longmapsto \mathrm{Hom}_{\mathrm{Alp}\,/k}(A, C).$$

The functor h_A is left exact,[32] and the canonical projections $A \to A/\mathfrak{l}$, where $\mathfrak{l}$ ranges over the open ideals of A, induce for every $C \in \mathrm{Alf}\,/k$ an isomorphism

$$\varinjlim_{\mathfrak{l}} \mathrm{Hom}_{\mathrm{Alf}\,/k}(A/\mathfrak{l}, C) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Alp}\,/k}(A, C),$$

functorial in C. In other words, h_A is the direct limit of the representable functors $h_{A/\mathfrak{l}}$:

$$h_A \simeq \varinjlim_{\mathfrak{l}} h_{A/\mathfrak{l}}. \tag{$*$}$$

If B is another profinite k-algebra, the general properties of the bifunctor Hom, together with

$$\mathrm{Hom}(h_C, h_B) = h_B(C)$$

for C of finite length, give isomorphisms

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{Alp}\,/k}(B, A) &\simeq \varprojlim_{\mathfrak{l}} \mathrm{Hom}_{\mathrm{Alp}\,/k}(B, A/\mathfrak{l}) \\
&\simeq \varprojlim_{\mathfrak{l}} \mathrm{Hom}(h_{A/\mathfrak{l}}, h_B) \\
&\simeq \mathrm{Hom}\left(\varinjlim_{\mathfrak{l}} h_{A/\mathfrak{l}}, h_B\right).
\end{aligned}$$

Together with $(*)$, this proves that the contravariant functor $A \mapsto h_A$ is fully faithful. In fact:

Proposition. *The functor $A \mapsto h_A$ is an anti-equivalence from* $\mathrm{Alp}\,/k$ *onto the category of left-exact functors from* $\mathrm{Alf}\,/k$ *to* $(\mathbf{Sets})$.

By what precedes, it remains only to show that every left-exact functor

$$F : \mathrm{Alf}\,/k \longrightarrow (\mathbf{Sets})$$

is isomorphic to a functor h_A. The algebra A may be constructed as follows (cf. TDTE II, § 3).

Because F is left exact, for every finite-length k-algebra C and every $\xi \in F(C)$, there is a smallest subalgebra C' of C such that ξ belongs to the image of $F(C')$ in $F(C)$. If $C' = C$, the pair (C, ξ) is called *minimal*.

The minimal pairs form a category when a morphism from (C, ξ) to (D, η) is defined to be a homomorphism $\phi : C \to D$ such that

$$(F\phi)(\xi) = \eta.$$

Such a ϕ is necessarily surjective, and the category of minimal pairs is left filtered. Moreover, one may restrict to pairs (C, ξ) for which C belongs to a set containing one finite-length k-algebra of every isomorphism type.[33] The functor $(C, \xi) \mapsto C$, from the category of minimal

[32] That is, it commutes with finite projective limits.

[33] Indeed, every discrete k-module of length n is isomorphic to k^n/L, where L is an open submodule of k^n of colength n. One may then consider the set of isomorphism classes of profinite k-algebra structures on each such module.

pairs to the category of profinite k-algebras, therefore has a projective limit; take A to be this limit.

Corollary. *The category* Alp $/k$ *has arbitrary direct limits.*

Indeed, the category of left-exact functors from Alf $/k$ to (**Sets**) has projective limits, computed pointwise: if (F_i) is a projective system of such functors, then for every $C \in$ Alf $/k$,

$$(\varprojlim_i F_i)(C) = \varprojlim_i F_i(C).$$

[34]

0.5. Base Change

Let

$$\phi : k \longrightarrow \ell$$

be a homomorphism of pseudocompact rings (0.1.3). The construction of 0.3 admits the following generalization.[35]

Definition 0.5.A. *For every $M \in \mathrm{PC}(k)$ and $N \in \mathrm{PC}(\ell)$, let*

$$M \widehat{\otimes}_k N$$

be the separated completion of $M \otimes_k N$ for the linear topology defined by the kernels of the projections

$$M \otimes_k N \longrightarrow (M/M') \otimes_k (N/N'),$$

where M', respectively N', ranges over the open submodules of M, respectively N. Then $M\widehat{\otimes}_k N$ is a pseudocompact ℓ-module. If $m \in M$ and $x \in N$, the canonical image of $m \otimes_k x$ in $M\widehat{\otimes}_k N$ is denoted by

$$m \widehat{\otimes}_k x.$$

This applies in particular to $N = \ell$. In that case

$$M \widehat{\otimes}_k \ell$$

is called the pseudocompact ℓ-module obtained from M by the base change $k \to \ell$.

Remarks 0.5.B.

(i) In such a base change, ℓ need not be a profinite k-algebra. The typical example is that in which k is a field and $\ell = K$ is an arbitrary field extension of k.

(ii) If, however, the k-module underlying N is pseudocompact—for example, if ℓ is a profinite k-algebra—then 0.4 shows that every open k-submodule of N contains an open ℓ-submodule. In that case $M\widehat{\otimes}_k N$ agrees with the completed tensor product of 0.3 for the pseudocompact k-modules M and N, so the notation is unambiguous.

In every case, the k-module $N|_k$ obtained by restriction of scalars is a topological k-module: the map

$$k \times N \longrightarrow N, \qquad (t, n) \longmapsto \phi(t)n$$

[34] If $(A_i)_{i \in I}$ is a direct system in Alp $/k$, its direct limit in Alp $/k$ is the separated completion of the k-algebra B, the direct limit of the underlying k-algebras, for the topology defined by the ideals I such that $I \cap A_i$ is open in A_i for every i and $\mathrm{length}_k(B/I) < \infty$. Notice, for example, that if K/k is an algebraic extension of fields of infinite degree and (k_i) is the filtered direct system of finite-degree subextensions, then the direct limit of the system (k_i) in Alp $/k$ is the zero ring.

[35] The editors expanded the details of this paragraph.

is continuous. We write $\mathrm{Hom}_c(M, N|_k)$ for the abelian group of continuous k-linear maps from M to $N|_k$.

Proposition 0.5.C. *For every $M \in \mathrm{PC}(k)$ and $N \in \mathrm{PC}(\ell)$, there is a functorial isomorphism*

$$\mathrm{Hom}_{\mathrm{PC}(\ell)}(M\widehat{\otimes}_k\ell, N) \simeq \mathrm{Hom}_c(M, N|_k).$$

36

Indeed, let $u : M \to N|_k$ be a continuous homomorphism. The map

$$u' : M \times \ell \longrightarrow N, \qquad (m, \lambda) \longmapsto \lambda u(m)$$

is continuous and "bilinear," meaning k-linear in the first factor and ℓ-linear in the second. If N' is an open ℓ-submodule of N, the argument of Lemma 0.3.1 gives an open submodule $M' \subset M$ and an open ideal $\ell' \subset \ell$ such that

$$u'(M' \times \ell) \subset N', \qquad u'(M \times \ell') \subset N'.$$

Thus u induces a continuous ℓ-linear map

$$U : M\widehat{\otimes}_k\ell \longrightarrow N$$

satisfying

$$U(m\widehat{\otimes}_k\lambda) = \lambda u(m).$$

Conversely, every morphism

$$f : M\widehat{\otimes}_k\ell \longrightarrow N$$

gives the morphism

$$f' : M \longrightarrow N|_k, \qquad m \longmapsto f(m\widehat{\otimes}_k 1).$$

As in 0.3.2, 0.3.5, and 0.3.1.2, this yields:

Corollary 0.5.D. *The functor*

$$\mathrm{PC}(k) \longrightarrow \mathrm{PC}(\ell), \qquad M \longmapsto M\widehat{\otimes}_k\ell$$

is right exact and commutes with filtered projective limits. Thus, if (M_i) is a filtered projective system in $\mathrm{PC}(k)$, there is a canonical isomorphism

$$(\varprojlim_i M_i)\widehat{\otimes}_k\ell \simeq \varprojlim_i (M_i\widehat{\otimes}_k\ell).$$

Moreover, if M and N are pseudocompact k-modules, there is a canonical isomorphism

$$(M\widehat{\otimes}_k N)\widehat{\otimes}_k\ell \simeq (M\widehat{\otimes}_k\ell)\widehat{\otimes}_\ell(N\widehat{\otimes}_k\ell).$$

Definition 0.5.E. *Finally, if A is a profinite k-algebra, there is a unique profinite ℓ-algebra structure on $A\widehat{\otimes}_k\ell$ such that*

$$(a\widehat{\otimes}_k\lambda)(b\widehat{\otimes}_k\mu) = ab\widehat{\otimes}_k(\lambda\mu)$$

for $a, b \in A$ and $\lambda, \mu \in \ell$. The profinite ℓ-algebra $A\widehat{\otimes}_k\ell$ is said to be obtained from A by extension of scalars, or base change, along $k \to \ell$.

[36]Consequently, when ℓ is a profinite k-algebra, the functor $M \mapsto M\widehat{\otimes}_k\ell$ is left adjoint to the restriction-of-scalars functor.

1. Formal Varieties over a Pseudocompact Ring

1.1. To every pseudocompact ring A one may associate a formal scheme (EGA I, 10.4.2) as follows. Its underlying topological space is the set $\Upsilon(A)$ of open prime ideals of A, which are therefore maximal, endowed with the discrete topology. For every subset $E \subset \Upsilon(A)$, the space of sections of the structure sheaf over E is the Cartesian product

$$\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}.$$

The formal scheme so obtained is denoted by $\mathrm{Spf}(A)$, the *formal spectrum* of A.[37]

If A and B are two pseudocompact rings, a *morphism* from $\mathrm{Spf}(B)$ to $\mathrm{Spf}(A)$ consists of a map

$$f : \Upsilon(B) \longrightarrow \Upsilon(A)$$

and a family of continuous homomorphisms

$$f_y^\natural : A_{f(y)} \longrightarrow B_y, \qquad y \in \Upsilon(B).$$

Such a morphism induces a continuous homomorphism

$$f^\natural : A = \prod_{x\in\Upsilon(A)} A_x \longrightarrow B = \prod_{y\in\Upsilon(B)} B_y.$$

The converse is true:

Proposition. *The contravariant functor $A \mapsto \mathrm{Spf}(A)$ is fully faithful.*

Indeed, let $\phi : A \to B$ be a continuous homomorphism. The inverse image $\phi^{-1}(\mathfrak{n})$ of an open maximal ideal of B is an open prime ideal of A, and hence is maximal in A. We therefore obtain a map

$$\mathfrak{n} \longmapsto \phi^{-1}(\mathfrak{n})$$

from $\Upsilon(B)$ to $\Upsilon(A)$, and ϕ induces a continuous homomorphism

$$A_{\phi^{-1}(\mathfrak{n})} \longrightarrow B_{\mathfrak{n}}.$$

Thus ϕ induces a morphism of formal schemes

$$\mathrm{Spf}(\phi) : \mathrm{Spf}(B) \longrightarrow \mathrm{Spf}(A).$$

It is readily verified that

$$\mathrm{Spf}(\phi)^\natural = \phi, \qquad \mathrm{Spf}(f^\natural) = f$$

for every morphism $f : \mathrm{Spf}(B) \to \mathrm{Spf}(A)$, which proves the proposition.

Although we shall speak here only of formal schemes of the form $\mathrm{Spf}(A)$, we shall use the language of formal schemes rather than that of pseudocompact rings, so as to ground our assertions in geometric intuition.

1.2. Let k be a pseudocompact ring.

Definition 1.2.A.[38] A *formal variety over k* is a formal scheme X over $\mathrm{Spf}(k)$ which is isomorphic to a formal k-scheme $\mathrm{Spf}(A)$ for some profinite k-algebra A. The algebra A is then isomorphic to the *affine algebra* of X, that is, the algebra of sections of the structure sheaf $\mathcal{O}_X$ of X.

[37] Cf. Remarks 0.1.2.

[38] The editors added the numbering 1.2.A–1.2.E for later reference.

We denote by Vaf/k the full subcategory of the category of formal schemes over $\mathrm{Spf}(k)$ whose objects are the formal k-varieties.[39]

By 1.1, the functor $A \mapsto \mathrm{Spf}(A)$ is an anti-equivalence from Alp/k (0.4.1) onto Vaf/k. Hence, by the corollary of 0.4.2:

Proposition 1.2.B. *The category* Vaf/k *has projective and direct limits.*[40]

For example, let $X \in \mathrm{Ob}(\mathrm{Vaf}/k)$, let

$$f : Y \longrightarrow X, \qquad g : Z \longrightarrow X$$

be two formal k-varieties over X, and let A, B, C be the affine algebras of X, Y, Z, respectively. By 0.4.1, the affine algebra of the fiber product $Y \times_X Z$ identifies with $B \widehat{\otimes}_A C$,[41] Consequently, the inclusion of Vaf/k into the category of all formal k-schemes commutes with finite projective limits (cf. EGA I, 10.7).

Direct limits of formal k-varieties correspond to projective limits of their affine algebras.

Example 1.2.C (Cokernels). Let $d, e : X \rightrightarrows Y$ be a double arrow in Vaf/k. The affine algebra of $\mathrm{Coker}(d, e)$ is isomorphic to the kernel of the homomorphisms induced on the affine algebras of X and Y. There is also the following construction of $\mathrm{Coker}(d, e)$. Its underlying topological space is the cokernel of the underlying spaces.[42] Let p be the canonical projection from the set underlying Y to this cokernel, and let z be a point of the cokernel. The local algebra of $\mathrm{Coker}(d, e)$ at z is the kernel of the double arrow

$$d^\natural, e^\natural : \quad \prod_{p(y)=z} \mathcal{O}_{Y,y} \rightrightarrows \prod_{q(x)=z} \mathcal{O}_{X,x}$$

where

$$q = p \circ d = p \circ e,$$

and $d^\natural$ and $e^\natural$ are induced by the homomorphisms $d^\natural_x$ and $e^\natural_x$, in the notation of 1.1.

Definition 1.2.D. Let $\phi : k \to \ell$ be a homomorphism of pseudocompact rings and let X be a formal k-variety with affine algebra A. The formal scheme

$$X \times_{\mathrm{Spf}(k)} \mathrm{Spf}(\ell),$$

obtained by base change, is a formal ℓ-variety. We also denote it by

$$X \widehat{\otimes}_k \ell.$$

Its affine algebra is the completed tensor product $A \widehat{\otimes}_k \ell$ (cf. 0.5 and EGA I, § 10).

Remark 1.2.E. Since every formal variety over k decomposes into formal varieties over the local components of k, *in certain proofs we shall assume that k is a local pseudocompact ring.*

We now give some examples while fixing our terminology.

1.2.1. A *k-functor* is, by definition, a covariant functor from Alf/k to (**Sets**). By 1.1 and 0.4.2, we may identify

$$\mathrm{Vaf}/k \simeq (\mathrm{Alp}/k)^\circ$$

[39] By the definition of morphisms in 1.1, every $X \in \mathrm{Ob}(\mathrm{Vaf}/k)$ is the direct sum in Vaf/k of the one-point formal varieties $\mathrm{Spf}(\mathcal{O}_{X,x})$, for $x \in X$.

[40] In particular, cokernels exist in Vaf/k; see below. Moreover, a filtered projective limit in Vaf/k, all of whose transition morphisms are surjective, can be empty; cf. editor's note (34).

[41] The formal variety $Y \times_X Z$ is the direct sum, over $x \in X$, of the formal varieties $B_x \widehat{\otimes}_{\mathcal{O}_{X,x}} C_x$, where B_x is the product of the $\mathcal{O}_{Y,y}$ for $y \in Y$ such that $f(y) = x$, and C_x is defined analogously. This will be used in 2.5.1.

[42] That is, it is the quotient set of Y by the identifications $d(x) = e(x)$, for every $x \in X$, endowed with the quotient topology, which here is discrete.

with a full subcategory of the category of k-functors by associating to every formal k-variety X the functor

$$\mathrm{Alf}\,/k \longrightarrow (\mathbf{Sets}), \qquad C \longmapsto X(C) = \mathrm{Hom}_{\mathrm{Vaf}\,/k}(\mathrm{Spf}(C), X).$$

We shall later encounter k-functors $\mathbf{F}$ which associate to every object C of $\mathrm{Alf}\,/k$ a C-module $\mathbf{F}(C)$, and to every morphism $\phi : C \to D$ in $\mathrm{Alf}\,/k$ a k-linear map

$$\mathbf{F}(\phi) : \mathbf{F}(C) \longrightarrow \mathbf{F}(D)$$

such that, for $x \in \mathbf{F}(C)$ and $\lambda \in C$,

$$\mathbf{F}(\phi)(\lambda x) = \phi(\lambda)\mathbf{F}(\phi)(x).$$

By Exposé I, 3.1, such an $\mathbf{F}$ is endowed with the structure of an $\mathbf{O}_k$-module, where $\mathbf{O}_k$ denotes the k-functor in rings which assigns to every object C of $\mathrm{Alf}\,/k$ its underlying ring.

Definitions.
(i) An $\mathbf{O}_k$-module $\mathbf{F}$ is called *admissible* if every morphism $\phi : C \to D$ in $\mathrm{Alf}\,/k$ induces a bijection from $D \otimes_C \mathbf{F}(C)$ onto $\mathbf{F}(D)$.[43]
(ii) The module $\mathbf{F}$ is called *flat* if it is admissible and if, for every object C of $\mathrm{Alf}\,/k$, the C-module $\mathbf{F}(C)$ is flat.[44]

For example, if M is a k-module, not necessarily pseudocompact, we denote by $\mathbf{W}(M)$, as in Exposé I, 4.6.1, the $\mathbf{O}_k$-module

$$C \longmapsto C \otimes_k M.$$

Then $\mathbf{W}(M)$ is flat whenever M is flat over k.

Moreover, the functor $M \mapsto \mathbf{W}(M)$ has as right adjoint the functor which assigns to every $\mathbf{O}_k$-module $\mathbf{F}$ the k-module

$$\varprojlim_{\mathfrak{l}} \mathbf{F}(k/\mathfrak{l}),$$

where $\mathfrak{l}$ ranges over the open ideals of k.

1.2.2. In what follows, an $\mathbf{O}_k$-module will always be denoted by a boldface letter such as $\mathbf{F}$. When k is Artinian, we shall simply write F instead of $\mathbf{F}(k)$. In that case, it is clear that the functor

$$\mathbf{F} \longmapsto F$$

induces an equivalence from the category of flat $\mathbf{O}_k$-modules onto the category of flat k-modules! The purpose of this terminology is solely to allow us to reason "as if k were always Artinian."

In accordance with Exposé I, § 3.1, we shall use analogous conventions for other algebraic structures. Thus an $\mathbf{O}_k$-algebra, respectively an $\mathbf{O}_k$-coalgebra, an $\mathbf{O}_k$-Lie algebra, or an $\mathbf{O}_k$-p-Lie algebra, consists of an $\mathbf{O}_k$-module $\mathbf{M}$ together with, for every object C of $\mathrm{Alf}\,/k$, a structure of C-algebra, respectively C-coalgebra, C-Lie algebra, or C-p-Lie algebra, on $\mathbf{M}(C)$. We further require that for every morphism $\phi : C \to D$ in $\mathrm{Alf}\,/k$, the map

$$D \otimes_C \mathbf{M}(C) \longrightarrow \mathbf{M}(D)$$

induced by $\mathbf{M}(\phi)$ be a homomorphism of D-algebras, respectively of D-coalgebras, and so on.

[43] The editors introduced this terminology; cf. editor's note (47) in 1.2.3.
[44] It is therefore projective, since C is Artinian.

Finally, if $\mathbf{F}$ and $\mathbf{G}$ are two $\mathbf{O}_k$-modules, then $\mathbf{F} \otimes_k \mathbf{G}$ denotes the $\mathbf{O}_k$-module

$$C \longmapsto \mathbf{F}(C) \otimes_C \mathbf{G}(C).$$

1.2.3.[45] We begin with the following lemma.

Lemma 1.2.3.A. *Let $k \longrightarrow K$ be a morphism of pseudocompact rings, let B be a pseudocompact K-module, and let M be a projective k-module, without topology. There is a canonical isomorphism of pseudocompact K-modules*

$$\theta : \mathrm{Hom}_k(M, k)\widehat{\otimes}_k B \xrightarrow{\ \sim\ } \mathrm{Hom}_k(M, B). \tag{2}$$

Here $M^* = \mathrm{Hom}_k(M, k)$ is endowed with the topology defined in 0.2.2, which makes it a pseudocompact k-module. The topology on $\mathrm{Hom}_k(M, B)$ is defined analogously: a basis of neighborhoods of 0 is formed by the following K-submodules, where $x \in M$ and B' is an open K-submodule of B:

$$\mathscr{H}(x, B') = \{f \in \mathrm{Hom}_k(M, B) \mid f(x) \in B'\}.$$

Finally, $M^*\widehat{\otimes}_k B$ is the pseudocompact K-module obtained from M^* by base change; cf. 0.5.A.

With these conventions, θ is plainly an isomorphism when $M = k$. Moreover, both sides of (2), regarded as functors in M, carry direct sums to products; in particular, they commute with finite direct sums. It follows that (2) is an isomorphism first when M is a free k-module, and then when M is projective.

Definition 1.2.3.B. Let N be a pseudocompact k-module. We write $\mathbf{V}^f_k(N)$, or $\mathbf{N}^\dagger$,[46] for the $\mathbf{O}_k$-module which assigns to every $C \in \mathrm{Ob}(\mathrm{Alf}\,/k)$ the C-module

$$(C\widehat{\otimes}_k N)^\dagger,$$

the dual of $C\widehat{\otimes}_k N$ in the sense of 0.2.2; that is, the C-module

$$\mathrm{Hom}_c(C\widehat{\otimes}_k N, C) \simeq \mathrm{Hom}_c(N, C|_k),$$

where $C|_k$ denotes the k-module obtained from C by restriction of scalars. This $\mathbf{O}_k$-module $\mathbf{V}^f_k(N)$ is called the *dual* of N.

If $k_\mathfrak{m}$ is a local component of k, then for every object C of $\mathrm{Alf}\,/k$,

$$\mathrm{Hom}_c(k_\mathfrak{m}, C|_k) = C_\mathfrak{m} = C \otimes_k k_\mathfrak{m}$$

is a projective C-module. Moreover,

$$D \otimes_C C_\mathfrak{m} = D_\mathfrak{m}$$

for every morphism $C \longrightarrow D$ in $\mathrm{Alf}\,/k$. Thus $\mathbf{V}^f_k(k_\mathfrak{m})$ is a flat $\mathbf{O}_k$-module; cf. 1.2.1. Since every projective pseudocompact k-module is a product of modules $k_\mathfrak{m}$ (Proposition 0.2.1), Corollary 0.2.F gives:

Corollary 1.2.3.C. *The $\mathbf{O}_k$-module $\mathbf{V}^f_k(N)$ is flat whenever N is a projective pseudocompact k-module.*

[45] The editors added the following lemma and introduced the numbering 1.2.3.A through 1.2.3.F for later references.

[46] In fact, the second notation will not be used; see editors' note (47).

Definition 1.2.3.D. Conversely, let $\mathbf{M}$ be an admissible $\mathbf{O}_k$-module (cf. 1.2.1).[47] We call the following pseudocompact k-module the *dual* of $\mathbf{M}$, and denote it by $\Gamma^*(\mathbf{M})$. As $\mathfrak{l}$ ranges over the open ideals of k, endow each $k/\mathfrak{l}$-module

$$\mathbf{M}(k/\mathfrak{l})^* = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l})$$

with the topology described in 0.2.2; it is then a pseudocompact k-module. Since

$$\mathbf{M}(k/\mathfrak{l}) \simeq \mathbf{M}(k/\mathfrak{l}') \otimes_{k/\mathfrak{l}'} (k/\mathfrak{l}) \qquad (\mathfrak{l} \supset \mathfrak{l}'),$$

there are transition morphisms

$$\begin{aligned}\mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}') &\longrightarrow \mathrm{Hom}_{k/\mathfrak{l}'}(\mathbf{M}(k/\mathfrak{l}'), k/\mathfrak{l}) \\ &= \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), k/\mathfrak{l}).\end{aligned}$$

By definition, $\Gamma^*(\mathbf{M})$ is the projective limit of this filtered projective system of pseudocompact k-modules.

From now on, suppose in addition that $\mathbf{M}$ is a flat $\mathbf{O}_k$-module (cf. 1.2.1). Each $\mathbf{M}(k/\mathfrak{l}')$ is then a projective $k/\mathfrak{l}'$-module, so the transition morphisms above are surjective. Consequently the projections

$$\Gamma^*(\mathbf{M}) \longrightarrow \mathbf{M}(k/\mathfrak{l})^*$$

are also surjective, since filtered projective limits are exact in $\mathrm{PC}(k)$.

Let $\tau : k \longrightarrow K$ be a morphism of pseudocompact rings. We write $\mathbf{M} \otimes_k K$, or simply $\mathbf{M}_K$, for the $\mathbf{O}_K$-functor defined as follows. If C is a finite-length K-algebra, then the kernel of $k \longrightarrow C$ is an open ideal $\mathfrak{l}$, and we set

$$\mathbf{M}_K(C) = \mathbf{M}(k/\mathfrak{l}) \otimes_k C.$$

The same value is obtained from $\mathbf{M}(k/\mathfrak{l}') \otimes_k C$ for every open ideal $\mathfrak{l}' \subset \mathfrak{l}$. We then define $\Gamma^*_K(\mathbf{M}_K)$ as the projective limit, with I ranging over the open ideals of K, of the pseudocompact K-modules

$$\mathrm{Hom}_{K/I}(\mathbf{M}_K(K/I), K/I) = \mathrm{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), K/I),$$

where on the right $\mathfrak{l}$ is any open ideal of k such that $\tau(\mathfrak{l}) \subset I$.

By Lemma 1.2.3.A, the right-hand term identifies with

$$\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k (K/I).$$

Since the projections $\Gamma^*(\mathbf{M}) \longrightarrow \mathbf{M}(k/\mathfrak{l})^*$ are surjective, the projective limit of these latter modules is precisely the pseudocompact K-module

$$\Gamma^*(\mathbf{M}) \widehat{\otimes}_k K$$

(cf. 0.5.A). We have therefore proved that, for every flat $\mathbf{O}_k$-module $\mathbf{M}$, formation of $\Gamma^*(\mathbf{M})$ commutes with extension of the base; that is,

$$\Gamma^*_K(\mathbf{M} \otimes_k K) \simeq \Gamma^*(\mathbf{M}) \widehat{\otimes}_k K. \qquad (\star)$$

Proposition 1.2.3.E.

[47] The editors do not understand the definition that follows unless $\mathbf{M}$ is assumed to be admissible, which is why this hypothesis has been added. They also write $\Gamma^*(\mathbf{M})$, rather than $\mathbf{M}^*$, to avoid confusion between $\mathbf{M}^*$, the module dual to the functor $\mathbf{M}$, and $\mathbf{N}^\dagger$, the functor dual to the module N; cf. editors' note (46). Finally, the editors have spelled out the construction of $\Gamma^*(\mathbf{M})$.

(i) *The functors*

$$N \longmapsto \mathbf{V}_k^f(N), \qquad \mathbf{M} \longmapsto \Gamma^*(\mathbf{M})$$

define an anti-equivalence between the category of projective pseudocompact k-modules and the category of flat $\mathbf{O}_k$-modules.[48]

(ii) *Moreover, if $k \longrightarrow K$ is a morphism of pseudocompact rings, the preceding anti-equivalence "commutes with base change" in the following sense: if $N \simeq \Gamma^*(\mathbf{M})$, then*

$$N \widehat{\otimes}_k K \simeq \Gamma_K^*(\mathbf{M} \otimes_k K).$$

Proof. First, there is a natural transformation

$$\phi_N : N \longrightarrow \Gamma^*(\mathbf{V}_k^f(N)).$$

The functor $\Gamma^* \circ \mathbf{V}_k^f$ commutes with products by 0.3.5 and 0.2.F, so it is enough to verify that ϕ_N is an isomorphism when N is a local component $k_{\mathfrak{m}}$ of k. In this case, for every open ideal $\mathfrak{l}$ of k contained in $\mathfrak{m}$, the natural morphism

$$(k/\mathfrak{l})_{\mathfrak{m}} \longrightarrow \operatorname{Hom}_{k/\mathfrak{l}}\left(\operatorname{Hom}_c(k_{\mathfrak{m}} \widehat{\otimes}_k k/\mathfrak{l}, k/\mathfrak{l}), k/\mathfrak{l}\right)$$

is an isomorphism, which proves the assertion.

Conversely, let $\mathbf{M}$ be a flat $\mathbf{O}_k$-module. We show that

$$\Gamma^*(\mathbf{M}) = \varprojlim_{\mathfrak{l}} \mathbf{M}(k/\mathfrak{l})^*$$

is a projective object of $\mathrm{PC}(k)$. Let $N \longrightarrow N'$ be a surjective morphism between objects of $\mathrm{LF}(k)$. By 0.2.F (i) and (ii), it is enough to prove that the natural map

$$\varinjlim_{\mathfrak{l}} \operatorname{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N) \longrightarrow \varinjlim_{\mathfrak{l}} \operatorname{Hom}_c(\mathbf{M}(k/\mathfrak{l})^*, N')$$

is surjective. This is clear: N and N' are $k/\mathfrak{l}_0$-modules for some open ideal $\mathfrak{l}_0$, and if $\mathfrak{l} \subset \mathfrak{l}_0$, every morphism $\mathbf{M}(k/\mathfrak{l})^* \longrightarrow N'$ lifts to a morphism $\mathbf{M}(k/\mathfrak{l})^* \longrightarrow N$, since $\mathbf{M}(k/\mathfrak{l})^*$ is a projective object of $\mathrm{PC}(k/\mathfrak{l})$.

There is a morphism

$$\psi : \mathbf{M} \longrightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))$$

of functors from Alf/k to $(\mathbf{Sets})$. Let B be an object of Alf/k. We claim that

$$\begin{aligned} \psi(B) : \mathbf{M}(B) &\longrightarrow \mathbf{V}_k^f(\Gamma^*(\mathbf{M}))(B) \\ &= \operatorname{Hom}_c\left(\varprojlim_{\mathfrak{l}} (\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k B), B\right) \end{aligned} \tag{1}$$

is a bijection. In the equality above we used the fact that $\widehat{\otimes}$ commutes with filtered projective limits. Let $\mathfrak{l}_0$ be an open ideal of k contained in the kernel of $k \longrightarrow B$. By Lemma 1.2.3.A, for every $\mathfrak{l} \subset \mathfrak{l}_0$, there is a canonical isomorphism of pseudocompact B-modules

$$\mathbf{M}(k/\mathfrak{l})^* \widehat{\otimes}_k B \xrightarrow{\sim} \operatorname{Hom}_{k/\mathfrak{l}}(\mathbf{M}(k/\mathfrak{l}), B).$$

[48] The editors added assertion (ii) below and the proof that follows. The original said: "If $\mathbf{M}$ is a flat $\mathbf{O}_k$-module, it is clear that $\mathbf{M}$ is the dual of $\mathbf{M}^*$; hence the functors $N \mapsto N^*$ and $\mathbf{M} \mapsto \mathbf{M}^*$ define an anti-equivalence ..."

Since
$$\mathbf{M}(B) = \mathbf{M}(k/\mathfrak{l}) \otimes_{k/\mathfrak{l}} B,$$
the right-hand side is $\mathrm{Hom}_B(\mathbf{M}(B), B)$. Thus the projective system in (1) is constant for $\mathfrak{l} \subset \mathfrak{l}_0$, and (1) reduces to the canonical morphism
$$\mathbf{M}(B) \longrightarrow \mathrm{Hom}_c(\mathrm{Hom}_B(\mathbf{M}(B), B), B).$$
This is an isomorphism by 0.2.2, since B is Artinian and $\mathbf{M}(B)$ is a projective B-module. This proves assertion (i) of the proposition; assertion (ii) follows from the isomorphism $(\star)$ established above. □

Remark 1.2.3.F. Let us return to the statement of the proposition and suppose, in addition, that N is a topologically free pseudocompact k-module. In this case one can choose, "coherently," bases for the C-modules $\mathbf{V}^f_k(N)(C)$.

Indeed, let (n_i) be a pseudobasis of N (0.2.1), and let n_i^C be the canonical image of n_i in $C\widehat{\otimes}_k N$, for $C \in \mathrm{Alf}/k$. Define $\delta_i^C \in (C\widehat{\otimes}_k N)^\dagger$ by
$$\delta_i^C(n_i^C) = 1, \qquad \delta_i^C(n_j^C) = 0 \quad (i \neq j).$$
Then (δ_i^C) is a basis of $\mathbf{V}^f_k(N)(C)$. Moreover, for every morphism $\varphi : C \longrightarrow D$ in Alf/k, the map $\mathbf{V}^f_k(N)(\varphi)$ sends δ_i^C to δ_i^D.

1.2.4.[49] For every pseudocompact k-module E, let $\widehat{S}_k(E)$ denote the *completed symmetric algebra* of E, defined as follows. Let $T_k(E)$ be the direct sum of the pseudocompact k-modules
$$\widehat{\bigotimes}^n_k E = E\widehat{\otimes}_k \cdots \widehat{\otimes}_k E \qquad \left(n \geq 0;\quad \widehat{\bigotimes}^0_k E = k\right).$$
Give $T_k(E)$ its usual graded k-algebra multiplication. Let $S_k(E)$ be the quotient of $T_k(E)$ by the homogeneous ideal whose n-th component is the closed k-submodule of $\widehat{\bigotimes}^n_k E$ generated by the elements
$$\begin{aligned} &x_1\widehat{\otimes}\cdots\widehat{\otimes}x_i\widehat{\otimes}x_{i+1}\widehat{\otimes}\cdots\widehat{\otimes}x_n \\ &\quad - x_1\widehat{\otimes}\cdots\widehat{\otimes}x_{i+1}\widehat{\otimes}x_i\widehat{\otimes}\cdots\widehat{\otimes}x_n. \end{aligned}$$
If this quotient is denoted by $S^n_k(E)$, then $S_k(E)$ is a graded k-algebra whose n-th component is $S^n_k(E)$.

Endow $S_k(E)$ with the linear topology defined by the set Υ of ideals $\mathfrak{l}$ such that $S_k(E)/\mathfrak{l}$ is a k-module of finite length and
$$S^n_k(E) \cap \mathfrak{l}$$
is an open submodule of $S^n_k(E)$ for every n. The profinite algebra $\widehat{S}_k(E)$ is the separated completion of $S_k(E)$ for this topology.[50]

[49]Editors' note: In this paragraph the order has been changed: $\widehat{S}_k(E)$ is defined first, and it is then stated that $\mathbf{V}^f_k(E)$ represents $\mathbf{V}^f_k(E)$.

[50]Editors' note: For example, let k be a field, let E be a pseudocompact k-vector space, and put $V = \mathrm{Hom}_c(E, k)$; then $E \simeq V^*$ (cf. 0.2.2). For every finite-dimensional subspace W of V, let $\mathcal{F}(W)$ be the set of closed points of the k-scheme $\mathbf{V}(W) = \mathrm{Spec}\, S_k(W^*)$, and let $\mathcal{F}(V) = \varinjlim_W \mathcal{F}(W)$. Then $\widehat{S}_k(E)$ is the product, over $x \in \mathcal{F}(V)$, of the pseudocompact k-algebras
$$\widehat{S}_k(E)_x = \varprojlim_W \widehat{\mathcal{O}}_{W,x},$$
where W ranges over the finite-dimensional subspaces of V such that $x \in \mathcal{F}(W)$, and $\widehat{\mathcal{O}}_{W,x}$ is the separated completion of the local ring $\mathcal{O}_{\mathbf{V}(W),x}$ for the topology defined by the ideals of finite codimension (which here coincides with the $\mathfrak{m}$-adic topology). If k is algebraically closed and $(v_i)_{i\in I}$ is a basis of V, so that E has a pseudobasis $(e_i)_{i\in I}$, then every local component $\widehat{S}_k(E)_x$ is isomorphic to the formal power-series ring $k[[e_i,\ i \in I]]$, endowed with the topology defined by the ideals of finite codimension.

We write

$$\mathbf{V}_k^f(E) = \operatorname{Spf}(\widehat{S}_k(E))$$

for the corresponding formal k-variety. It represents the k-functor $\mathbf{V}_k^f(E)$; that is, for every object C of Alf $/k$, there is a canonical isomorphism

$$\begin{aligned}\mathbf{V}_k^f(E)(C) = \operatorname{Hom}_c(E, C|_k) &\xrightarrow{\sim} \operatorname{Hom}_{\mathrm{Alp}\,/k}(\widehat{S}_k(E), C)\\ &= \operatorname{Hom}_{\mathrm{Vaf}\,/k}(\operatorname{Spf}(C), \mathbf{V}_k^f(E)).\end{aligned}$$

In what follows, we identify the formal variety $\mathbf{V}_k^f(E)$ with the k-functor $\mathbf{V}_k^f(E)$.

1.2.5. By 1.2.4, the zero morphism from E to k corresponds to a morphism of profinite algebras

$$\pi : \widehat{S}_k(E) \longrightarrow k.$$

The morphism π induces the zero map on $S_k^n(E)$ for $n \geq 1$, and defines a section of the structure morphism

$$\mathbf{V}_k^f(E) \longrightarrow \operatorname{Spf}(k).$$

We denote by $\mathbf{V}_k^{f,0}(E)$ the formal variety whose points are the images of the points of $\operatorname{Spf}(k)$ under the section $\operatorname{Spf}(\pi)$, and whose local algebras at those points are the same as those of $\mathbf{V}_k^f(E)$. [51]

The affine algebra of $\mathbf{V}_k^{f,0}(E)$ is thus the separated completion of $S_k(E)$ for the topology defined by the ideals $\mathfrak{l} \in \Upsilon$ (cf. 1.2.4) which contain $S_k^n(E)$ for all sufficiently large n. Hence it is the infinite product

$$k[[E]] = k \times E \times S_k^2(E) \times S_k^3(E) \times \cdots .$$

Let C be an object of Alf $/k$, let

$$u \in \mathbf{V}_k^f(E)(C) = \operatorname{Hom}_c(E, C),$$

and let

$$\phi : \widehat{S}_k(E) \longrightarrow C$$

be the corresponding morphism of profinite k-algebras. Then $\operatorname{Ker}(\phi)$ is an open ideal—that is, ϕ belongs to $\mathbf{V}_k^{f,0}(E)(C)$—if and only if $\operatorname{Ker}(\phi)$ contains $S_k^n(E)$ for all sufficiently large n, or equivalently if and only if $u(E)$ is contained in the radical $r(C)$ of C. Consequently, for every object C of Alf $/k$, there are canonical isomorphisms

$$\mathbf{V}_k^{f,0}(E)(C) \simeq \operatorname{Hom}_{\mathrm{Alp}\,/k}(k[[E]], C) \simeq \operatorname{Hom}_c(E, r(C)).$$

1.2.6. A formal k-variety V is said to be *of finite length* if its affine algebra has finite length. Similarly, if S is a scheme, an S-scheme X is said to be *of finite length* if X is finite over S and the direct image of $\mathcal{O}_X$ on S is an $\mathcal{O}_S$-module of finite length.[52] Thus giving an S-scheme X of finite length is the same as giving a finite set

$$\{s_1, \ldots, s_n\}$$

of closed points of S, together with an $\mathcal{O}_{S,s_i}$-algebra of finite length at each such point.

[51] Editors' note: The remainder of this paragraph has been spelled out in detail.
[52] Editors' note: The following sentence has been added.

It follows that the S-schemes of finite length identify with the formal varieties of finite length over the following formal scheme $\widehat{S}$. The underlying topological space of $\widehat{S}$ is the set of closed points of S, endowed with the discrete topology. If s is such a closed point, the stalk $\mathcal{O}_{\widehat{S},s}$ of the structure sheaf is the separated completion

$$\widehat{\mathcal{O}}_{S,s}$$

of $\mathcal{O}_{S,s}$ for the linear topology defined by the ideals of finite colength. Thus

$$\widehat{S} = \operatorname{Spf} \mathcal{A}(\widehat{S}),$$

where $\mathcal{A}(\widehat{S})$ is the product of the $\widehat{\mathcal{O}}_{S,s}$, as s ranges over the closed points of S, endowed with the product topology.

Definition. *If X is an S-scheme, let $\widehat{X}/\widehat{S}$ denote the formal variety over*

$$k = \mathcal{A}(\widehat{S})$$

defined as follows. Its underlying topological space consists of the points $x \in X$ such that

$$[\kappa(x) : \kappa(s)] < \infty,$$

where s is the image of x in S. The local ring $\mathcal{O}_{\widehat{X}/\widehat{S},x}$ is the separated completion of $\mathcal{O}_{X,x}$ for the linear topology defined by the ideals I of $\mathcal{O}_{X,x}$ for which $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{S,s}$-module. Since $[\kappa(x) : \kappa(s)] < \infty$, this is equivalent to requiring that $\mathcal{O}_{X,x}/I$ have finite length as an $\mathcal{O}_{X,x}$-module.

Let $\mathrm{Vaf}^{\ell f}/\widehat{S}$ be the category of formal varieties of finite length over $\widehat{S}$, identified with the category of S-schemes of finite length. By 1.1 and 1.2.1, the category $\mathrm{Vaf}/\widehat{S}$ of formal varieties over $\widehat{S}$ is equivalent to the category of left-exact contravariant functors

$$\mathrm{Vaf}^{\ell f}/\widehat{S} \longrightarrow (\mathbf{Sets}).$$

In particular, for every S-scheme X, the functor

$$T \longmapsto \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, X)$$

from $\mathrm{Vaf}^{\ell f}/\widehat{S}$ to $(\mathbf{Sets})$ is left exact and therefore corresponds to a formal variety over $\widehat{S}$. This variety is precisely $\widehat{X}/\widehat{S}$.[53]

Proposition. *For every S-scheme X, the functors*

$$\mathrm{Hom}_{\mathrm{Vaf}/\widehat{S}}(-, \widehat{X}/\widehat{S}) \qquad \textit{and} \qquad \mathrm{Hom}_{(\mathrm{Sch}/S)}(-, X)$$

have the same restriction to $\mathrm{Vaf}^{\ell f}/\widehat{S}$. *This defines a functor*

$$X \longmapsto \widehat{X}/\widehat{S}$$

from (Sch/S) *to* $\mathrm{Vaf}/\widehat{S}$, *and this functor commutes with finite projective limits.*

Proof. It is immediate that the formal variety $\widehat{X}/\widehat{S}$ has the required property and that the correspondence $X \mapsto \widehat{X}/\widehat{S}$ is functorial. It remains to prove the second assertion.

[53] Editors' note: The following proposition has been expanded by adding the fact that the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with finite projective limits. In the original this appeared in the proof of 1.3.4; the proof given here is more direct than the original one. This result will also be used in Section 2 and in 3.3.1.

Put

$$\mathscr{S} = (\mathrm{Sch}\,/S), \qquad \mathscr{V} = \mathrm{Vaf}\,/\widehat{S},$$

and write $\widehat{X}$ for $\widehat{X}/\widehat{S}$. By 1.2.B, the category $\mathscr{V}$ has arbitrary projective limits. Let $(X_i)_{i\in I}$ be a projective system in $\mathscr{S}$, and suppose that

$$X = \varprojlim_i X_i$$

exists in $\mathscr{S}$, as it does when I is finite. The functor which associates with every object Y of $\mathscr{S}$ the functor

$$h_Y = \mathrm{Hom}_{\mathscr{S}}(-, Y)$$

and, respectively, with every object V of $\mathscr{V}$ the functor

$$h_V = \mathrm{Hom}_{\mathscr{V}}(-, V)$$

commutes with projective limits. Hence, for every S-scheme T of finite length, there are functorial isomorphisms

$$\begin{aligned} \mathrm{Hom}_{\mathscr{S}}(T, X) &\simeq \varprojlim_i \mathrm{Hom}_{\mathscr{S}}(T, X_i) \\ &\simeq \varprojlim_i \mathrm{Hom}_{\mathscr{V}}(T, \widehat{X}_i) \simeq \mathrm{Hom}_{\mathscr{V}}\left(T, \varprojlim_i \widehat{X}_i\right). \end{aligned}$$

Consequently, the functor $X \mapsto \widehat{X}/\widehat{S}$ commutes with projective limits whenever they exist in $(\mathrm{Sch}\,/S)$, and in particular with finite projective limits. □

1.3

Proposition 1.3. *Let $f : X \to Y$ be a morphism of formal varieties over k, let A and B be the affine algebras of X and Y, and let $g : B \to A$ be the morphism induced by f. Then f is a monomorphism in* Vaf $/k$ *if and only if g is surjective.*

By 1.1,[54] the functor $A \mapsto \mathrm{Spf}(A)$ is an anti-equivalence from Alp $/k$ to Vaf $/k$. Consequently, f is a monomorphism if and only if g is an epimorphism, which it certainly is when g is surjective.

Conversely, suppose that g is an epimorphism. For every $\mathfrak{n} \in \Upsilon(B)$, put

$$A_{\mathfrak{n}} = A \widehat{\otimes}_B B_{\mathfrak{n}}.$$

By 0.4, A is a pseudocompact B-module, and hence is the product of the $A_{\mathfrak{n}}$ (0.3.6). The morphism g is therefore the product of the morphisms

$$g_{\mathfrak{n}} : B_{\mathfrak{n}} \longrightarrow A_{\mathfrak{n}}$$

obtained from g by base change. These remain epimorphisms. We are thus reduced to the case in which B is local, with maximal ideal $\mathfrak{n}$; set $K = B/\mathfrak{n}$.

By Nakayama's lemma 0.3.3, it is enough to prove that

$$g \widehat{\otimes}_B K$$

is surjective. This map is obtained from g by base change and is therefore an epimorphism in Alp $/K$. We may thus suppose that $B = K$ is a field. Now f is a monomorphism if and only if the diagonal

$$X \longrightarrow X \times_Y X$$

[54] The editors expanded the beginning of the proof.

is an isomorphism; equivalently, the homomorphism

$$A\widehat{\otimes}_K A \longrightarrow A, \qquad x\widehat{\otimes}_K x' \longmapsto xx',$$

is an isomorphism. Since K is a field, this implies $A = K$.

Remark. *It follows from the proposition that every monomorphism $f : X \to Y$ of formal varieties is an isomorphism from X onto a necessarily closed formal subvariety of Y.*[55]

1.3.1

The preceding proposition implies, in particular, that every monomorphism $f : X \to Y$ in $\mathrm{Vaf}\,/k$ is effective (cf. IV 1.3).[56] The same is not true of epimorphisms, as one sees by slightly modifying the counterexample of Exposé V, § 2.c).[57] This is why we now consider a useful class of effective epimorphisms.

Lemma. [58] *Let $f : X \to Y$ be a morphism of formal k-varieties, let A and B be the affine algebras of X and Y, and let $f^\natural : B \to A$ be the morphism induced by f. The following conditions are equivalent:*

(i) *For every $x \in X$, the homomorphism*

$$f^\natural_x : \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x}$$

makes $\mathcal{O}_{X,x}$ a topologically free pseudocompact $\mathcal{O}_{Y,f(x)}$-module.

(ii) *For every $y \in Y$, the local component*

$$A_y = \prod_{f(x)=y} \mathcal{O}_{X,x}$$

is a topologically free pseudocompact B_y-module.

(iii) *The map $f^\natural : B \to A$ makes A a projective pseudocompact B-module.*

(iv) *The functor*

$$\mathrm{PC}(B) \longrightarrow \mathrm{PC}(A), \qquad M \longmapsto M\widehat{\otimes}_B A,$$

is exact.

When these conditions hold, f is said to be topologically flat*.*

The implications

$$\text{(i)} \Longrightarrow \text{(ii)} \Longleftrightarrow \text{(iii)} \Longleftrightarrow \text{(iv)}$$

[55] The editors added this remark.

[56] In the opposite category $(\mathrm{Vaf}\,/k)^\circ = \mathrm{Alp}\,/k$, the corresponding map $g : B \to A$ is an effective epimorphism: since it is surjective, it induces an isomorphism of profinite k-algebras $B/I \xrightarrow{\sim} A$, where $I = \ker g$. Hence every morphism $\phi : B \to C$ in $\mathrm{Alp}\,/k$ that vanishes on I descends to a morphism $\bar{\phi} : A \to C$ such that $\phi = \bar{\phi} \circ g$.

[57] Let k be a field, $X = \mathrm{Spf}(k[[T]])$, and $Y = \mathrm{Spf}(B)$, where B is the k-subalgebra of $A = k[[T]]$ topologically generated by T^3 and T^4; thus B consists of the series $\sum a_n T^n$ with $a_n = 0$ for $n = 1, 2, 5$. Then $X \to Y$ is an epimorphism that is not effective: the cokernel of $X \times_Y X \rightrightarrows X$ is $\mathrm{Spf}(B')$, where

$$B' = \{a \in A \mid a\widehat{\otimes}1 = 1\widehat{\otimes}a\},$$

and B' contains T^5.

[58] The editors added this lemma, which explains the term "topologically flat."

follow from 0.2.F (iii) and 0.3.7. Conversely, suppose that (ii) holds, and let $x \in X$, $y = f(x)$. Since $\mathcal{O}_{X,x}$ is a direct factor of A_y, it is a projective pseudocompact B_y-module, and therefore is topologically free by 0.2.1, because B_y is local.

A morphism $f : X \to Y$ of formal k-varieties is called *surjective* if it induces a surjection of the underlying sets.

Proposition. *Let $f : X \to Y$ be a surjective, topologically flat morphism of formal k-varieties. Then f is an effective epimorphism* (cf. IV 1.3).

Let A and B be the affine algebras of X and Y, and let $g : B \to A$ be the morphism induced by f. We must prove that Y identifies with the cokernel of

$$X \times_Y X \rightrightarrows X;$$

that is, for every $\mathfrak{n} \in \Upsilon(B)$, $B_{\mathfrak{n}}$ must identify with the subring of

$$A_{\mathfrak{n}} = A \widehat{\otimes}_B B_{\mathfrak{n}}$$

formed by the elements a such that

$$a \widehat{\otimes} 1 = 1 \widehat{\otimes} a.$$

We may suppose that B is local, with maximal ideal $\mathfrak{n}$. The hypothesis then says that g makes A a nonzero topologically free B-module. By Nakayama's lemma 0.3.3, $A/\overline{\mathfrak{n}A} \neq 0$, and hence the induced morphism

$$g' : B/\mathfrak{n} \longrightarrow A/\overline{\mathfrak{n}A}$$

is injective. By Lemma 1.3.2 below, B is a direct factor of A as a B-module; write $A = B \oplus A'$. It follows that B identifies with the set of $a \in A$ such that

$$a \widehat{\otimes}_B 1 = 1 \widehat{\otimes}_B a.$$

1.3.2

Lemma 1.3.2. *Let B be a local pseudocompact ring, let $\mathfrak{n}$ be its maximal ideal, let M, N be projective pseudocompact B-modules, and let $g : M \to N$ be a morphism. If*

$$(B/\mathfrak{n}) \widehat{\otimes}_B g$$

is injective, then g is an isomorphism from M onto a direct factor of N.

Indeed, suppose

$$g' = (B/\mathfrak{n}) \widehat{\otimes}_B g$$

is injective. Since $B/\mathfrak{n}$ is a field, g' has a retraction r'. Let p and q be the canonical projections from M and N onto

$$M/\overline{\mathfrak{n}M} \qquad \text{and} \qquad N/\overline{\mathfrak{n}N}.$$

Since N is projective, there exists $r : N \to M$ such that

$$p \circ r = r' \circ q;$$

thus r' is obtained from r by passage to the quotient. Since $r' \circ g'$ is an isomorphism, 0.3.4 shows that the same is true of $r \circ g$, because M is projective. If s denotes the inverse of $r \circ g$, then $s \circ r$ is a retraction of g.

1.3.3

Proposition 1.3.3. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of formal k-varieties.*

(i) *If f and g are topologically flat, so is $g \circ f$.*

(ii) *If f and $g \circ f$ are topologically flat and f is surjective, then g is topologically flat.*

(iii) *If f is topologically flat, then so is $f \times_Y Y'$ after every base change $Y' \to Y$.*

Assertions (i) and (iii) are clear. To prove (ii), let A, B, C be the affine algebras of X, Y, Z, and let $f' : B \to A$, $g' : C \to B$ be the homomorphisms induced by f, g. Since $g \circ f$ is topologically flat, $f' \circ g'$ makes A a projective pseudocompact C-module. Likewise, f' makes A a faithful projective pseudocompact B-module. Thus, as P ranges over pseudocompact C-modules and N over pseudocompact B-modules, the functors

$$P \longmapsto P \widehat{\otimes}_C A, \qquad N \longmapsto N \widehat{\otimes}_B A$$

are exact; the second is also faithful. Hence the functor

$$P \longmapsto P \widehat{\otimes}_C B$$

is exact, and 0.3.7 implies that B is a projective pseudocompact C-module.

1.3.4

Proposition 1.3.4. *Let S be a scheme, let Y be a locally Noetherian S-scheme, and let $f : X \to Y$ be a faithfully flat S-morphism locally of finite type. Thus f is an effective epimorphism; equivalently, the following sequence is exact* (cf. IV 6.3.1 (iv) and IV 1.3)*:*

$$(*) \qquad X \times_Y X \overset{\mathrm{pr}_1}{\underset{\mathrm{pr}_2}{\rightrightarrows}} X \xrightarrow{f} Y.$$

Then the morphism of formal $\widehat{S}$-varieties

$$\widehat{f} : \widehat{X}/\widehat{S} \longrightarrow \widehat{Y}/\widehat{S}$$

(cf. 1.2.6) *is surjective and topologically flat, and the following sequence, obtained from the preceding one, is exact:*

$$\widehat{(*)} \qquad \widehat{X \times_Y X}/\widehat{S} \overset{\widehat{\mathrm{pr}_1}}{\underset{\widehat{\mathrm{pr}_2}}{\rightrightarrows}} \widehat{X}/\widehat{S} \xrightarrow{\widehat{f}} \widehat{Y}/\widehat{S}.$$

Let y be a point of Y, with image s in S, such that $\kappa(y)$ is a finite extension of the residue field $\kappa(s)$. Since f is surjective and locally of finite type, $f^{-1}(y)$ is nonempty and locally of finite type over $\kappa(y)$. Its closed points are precisely the points of $\widehat{X}/\widehat{S}$ mapping to y. This proves that $\widehat{f}$ is surjective.

Let x be a closed point of $f^{-1}(y)$.[59] Since Y is locally Noetherian and f is locally of finite type, the local rings $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{X,x}$ are Noetherian. The powers of their maximal ideals therefore have finite colength, and the local rings of $\widehat{Y}/\widehat{S}$ at y and of $\widehat{X}/\widehat{S}$ at x are the completions

$$\widehat{\mathcal{O}}_{Y,y} \qquad \text{and} \qquad \widehat{\mathcal{O}}_{X,x}$$

for their adic topologies. Since f is flat, $\widehat{\mathcal{O}}_{X,x}$ is flat over $\widehat{\mathcal{O}}_{Y,y}$ by SGA 1, IV 5.8. By 0.3.8 it is therefore a topologically free $\widehat{\mathcal{O}}_{Y,y}$-module. Thus $\widehat{f}$ is topologically flat.

Proposition 1.3.1 now shows that $\widehat{f}$ is an effective epimorphism. In other words, the following sequence is exact, where $\widehat{X}$ is written for $\widehat{X}/\widehat{S}$:

[59] The editors expanded the argument that follows and then used the material added in 1.2.6.

$$\widehat{X} \times_{\widehat{Y}} \widehat{X} \underset{\widehat{\mathrm{pr}}_2}{\overset{\widehat{\mathrm{pr}}_1}{\rightrightarrows}} \widehat{X} \xrightarrow{\widehat{f}} \widehat{Y}.$$

Moreover, 1.2.6 gives a natural isomorphism, compatible in particular with the projections onto $\widehat{X}$:

$$\widehat{X \times_Y X} \simeq \widehat{X} \times_{\widehat{Y}} \widehat{X}.$$

Consequently, the completed sequence displayed in the proposition is exact.

1.3.5

Let k be a pseudocompact ring. A formal variety X over k is said to be *topologically flat* if its affine algebra A is a *projective* pseudocompact k-module, that is, if the structure morphism

$$X \longrightarrow \mathrm{Spf}(k)$$

is topologically flat.

Let us first note[60] that 0.2.2 and 0.3.6 imply the following result, analogous to VII$_{\mathrm{A}}$, 3.1.1.

Lemma 1.3.5.A. *Suppose that k is Artinian. The functors*

$$A \longmapsto A^\dagger = \mathrm{Hom}_c(A, k), \qquad C \longmapsto C^* = \mathrm{Hom}_k(C, k)$$

define an anti-equivalence between the category of topologically flat profinite k-algebras and the category of flat k-coalgebras.

Indeed, if A is a topologically flat profinite k-algebra, then, by 0.3.6, there is an isomorphism of k-modules

$$A^\dagger \otimes_k A^\dagger \xrightarrow{\sim} (A \widehat{\otimes} A)^\dagger.$$

Thus the multiplication

$$A \widehat{\otimes} A \longrightarrow A$$

induces by duality a k-coalgebra structure on $A^\dagger$. The rest then follows from Proposition 0.2.2.

Let us return to the case of an arbitrary pseudocompact ring k.

Definition 1.3.5.B. To every formal k-variety X whose affine algebra A is a projective pseudocompact k-module, one associates a flat $\mathbf{O}_k$-coalgebra $\mathbf{H}(X)$, defined as follows.

Denote by $\mathbf{H}(X)$ the $\mathbf{O}_k$-module $\mathbf{V}_k^f(A)$, the "dual of A". It is a flat $\mathbf{O}_k$-module, since the pseudocompact k-module underlying A is projective (cf. 1.2.3.C). Moreover, by 0.3.6, there is an isomorphism

$$\mathbf{V}_k^f(A \widehat{\otimes} A) \simeq \mathbf{V}_k^f(A) \otimes \mathbf{V}_k^f(A).$$

Consequently, the multiplication of A induces by transposition a diagonal morphism

$$\mathbf{H}(X) = \mathbf{V}_k^f(A) \longrightarrow \mathbf{V}_k^f(A \widehat{\otimes} A) = \mathbf{H}(X) \otimes \mathbf{H}(X),$$

which makes $\mathbf{H}(X)$ a flat $\mathbf{O}_k$-coalgebra. We shall say that $\mathbf{H}(X)$ is the *coalgebra of X over $\mathbf{O}_k$*.

[60]The editors added the following lemma, used implicitly in the original; they also introduced the numbering 1.3.5.A–1.3.5.D for later references.

Definition 1.3.5.C. Conversely, to every $\mathbf{O}_k$-coalgebra $\mathbf{C}$ one can associate a k-functor (cf. 1.2.1) $\mathrm{Spf}^*(\mathbf{C})$, defined as follows. For every object B of Alf $/k$, set, with the notation of VII$_\mathrm{A}$ 3.1,

$$\begin{aligned}\mathrm{Spf}^*(\mathbf{C})(B) &= \mathrm{Hom}_{B\text{-coalg.}}(B, \mathbf{C}(B)) \\ &= \{x \in \mathbf{C}(B) \mid \varepsilon_{\mathbf{C}(B)}(x) = 1 \text{ and } \Delta_{\mathbf{C}(B)}(x) = x \otimes x\}.\end{aligned} \tag{1}$$

Suppose in addition[61] that the $\mathbf{O}_k$-module underlying $\mathbf{C}$ is admissible (cf. 1.2.1), and put

$$A = \Gamma^*(\mathbf{C}) = \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^*. \tag{2}$$

The algebra structures on the modules $\mathbf{C}(k/\mathfrak{l})^*$ endow A with the structure of a profinite k-algebra. For every object B of Alf $/k$, one has

$$\begin{aligned}\mathrm{Hom}_{\mathrm{Vaf}\,/k}(\mathrm{Spf}(B), \mathrm{Spf}(A)) &= \mathrm{Hom}_{\mathrm{Alp}\,/k}(A, B) \\ &= \mathrm{Hom}_{\mathrm{Alp}\,/B}(A \widehat{\otimes} B, B).\end{aligned} \tag{3}$$

Suppose finally that $\mathbf{C}$ is a flat $\mathbf{O}_k$-module. Then, by 1.2.3.E, $A = \Gamma^*(\mathbf{C})$ is a projective pseudocompact k-module. Moreover, we saw in the proof of *loc. cit.* that if $\mathfrak{l}_0$ is an open ideal of k contained in the kernel of $k \to B$, there are isomorphisms

$$\begin{aligned}A \widehat{\otimes} B &= \varprojlim_{\mathfrak{l}} \mathbf{C}(k/\mathfrak{l})^* \widehat{\otimes}_k B \\ &\simeq \mathrm{Hom}_{k/\mathfrak{l}_0}(\mathbf{C}(k/\mathfrak{l}_0), B) \\ &\simeq \mathrm{Hom}_B(\mathbf{C}(B), B).\end{aligned} \tag{4}$$

We shall denote the term on the right by $\mathbf{C}(B)^*$. Finally, Lemma 1.3.5.A, applied to the Artinian ring B, gives a natural isomorphism

$$\mathrm{Hom}_{B\text{-coalg.}}(B, \mathbf{C}(B)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Alp}\,/B}(\mathbf{C}(B)^*, B). \tag{5}$$

Combining (1), (5), (4), (3), and (2), we obtain, when $\mathbf{C}$ is a flat $\mathbf{O}_k$-coalgebra, an isomorphism of functors

$$\mathrm{Spf}^*(\mathbf{C}) \simeq \mathrm{Spf}(A) = \mathrm{Spf}(\Gamma^*(\mathbf{C})). \tag{$\star$}$$

Consequently, if $\mathcal{A}(X)$ denotes the affine algebra of a formal k-variety X, then, taking 1.2.3.E into account, we obtain the following.

Proposition 1.3.5.D.

(i) *The functors*

$$X \longmapsto \mathbf{H}(X) = \mathbf{V}_k^f(\mathcal{A}(X)), \qquad \mathbf{C} \longmapsto \mathrm{Spf}^*(\mathbf{C}) = \mathrm{Spf}(\Gamma^*(\mathbf{C}))$$

define an equivalence between the category of topologically flat formal k-varieties and the category of flat $\mathbf{O}_k$-coalgebras.

(ii) *Moreover, this equivalence "commutes with base change": if $k \to K$ is a morphism of pseudocompact rings, then $X \widehat{\otimes}_k K$ corresponds to $\mathbf{H}(X) \otimes_k K$.*

[61] The editors supplied the details that follow.

1.3.6

In the remainder of this Exposé, we shall several times define a topologically flat formal k-variety X by exhibiting the coalgebra $\mathbf{H}(X)$. We shall then have to interpret, by means of $\mathbf{H}(X)$, certain geometric properties of X. Here is one example of this situation: suppose that a section σ of the structure morphism $X \to \mathrm{Spf}(k)$ is given, and ask under what condition σ induces an isomorphism of the underlying topological spaces.[62]

To begin, suppose that k is Artinian. Let (H, Δ, ε) be a flat k-coalgebra, put $H^+ = \ker(\varepsilon)$, and let $A = H^*$ be the profinite k-algebra dual to H. Suppose that a morphism of k-coalgebras $k \to H$ is given; that is, an element $\phi \in H$ such that

$$\varepsilon(\phi) = 1, \qquad \Delta(\phi) = \phi \otimes \phi.$$

On the one hand, ϕ defines a continuous algebra homomorphism $\Phi : A \to k$, and hence a section

$$\sigma : \mathrm{Spf}(k) \longrightarrow \mathrm{Spf}(A)$$

of the projection $\mathrm{Spf}(A) \to \mathrm{Spf}(k)$.

On the other hand, define k-submodules of H by setting $H_0 = k\phi$ and, for $n \geq 1$,

$$H_n = \{x \in H \mid \Delta(x) - x \otimes \phi \in H_{n-1} \otimes H^+\}.$$

This formula also applies when $n = 0$ if $H_{-1} = 0$. Induction on n shows that $H_{n-1} \subset H_n$. We call

$$H_0 \subset H_1 \subset \cdots$$

the *filtration of H defined by ϕ.*

Remark. Since

$$\Delta(H_n) \subset H_n \otimes H_0 \oplus H_{n-1} \otimes H^+,$$

we have $\Delta(H_n) \subset H_n \otimes H$. Since Δ is cocommutative—that is, $\tau \circ \Delta = \Delta$, where $\tau(a \otimes b) = b \otimes a$—we likewise have $\Delta(H_n) \subset H \otimes H_n$. When H/H_n is flat over k, it follows that H_n is a subcoalgebra of H (see also 1.3.6.A (iii) below). In general, however, the map

$$\Delta : H_n \longrightarrow H_n \otimes H$$

does not factor through $H_n \otimes H_n$.[63]

Lemma 1.3.6.A. *Let k be an Artinian ring, let H be a flat k-coalgebra, and let $A = H^*$ be its dual profinite k-algebra. Let $\phi \in H$ satisfy $\varepsilon(\phi) = 1$ and $\Delta(\phi) = \phi \otimes \phi$. Let $\Phi : A \to k$ be the continuous algebra homomorphism, let $\sigma : \mathrm{Spf}(k) \to \mathrm{Spf}(A)$ be the section of $\mathrm{Spf}(A) \to \mathrm{Spf}(k)$, and let (H_n) be the filtration of H defined by ϕ. Put $I = \ker \Phi$.*

(i) *For every $n \geq 1$, H_{n-1} is the orthogonal in H of the closure $\overline{I^n}$ of I^n.*

(ii) *Consequently, σ induces a bijection of the underlying sets if and only if*

$$H = \bigcup_n H_n.$$

[64]

[62] In Lemma 1.3.6.A below, the editors supplied detailed proofs of parts (i) and (ii) and added part (iii).

[63] For example, let k_0 be a field, let $k = k_0[T]/(T^n)$, where $n \geq 4$, and let $H = k\phi \oplus kx$, with $\varepsilon(x) = 0$ and

$$\Delta(x) = x \otimes \phi + \phi \otimes x + t\, x \otimes x,$$

where t is the image of T in k. Then $H_i = k\phi \oplus t^{n-i}x$ for $i = 0, \ldots, n$, but for $2 \leq i \leq n-2$, the element $\Delta(t^{n-i}x)$ does not belong to the image of $H_i \otimes H_i$ in $H \otimes H$.

[64] In this case the coalgebra H is said to be *connected*; cf. Addition 2.9.

(iii) *If, in addition, every H/H_n is flat over k, then for every $n \geq 0$,*

$$\Delta(H_n) \subset \sum_{i=0}^{n} H_i \otimes H_{n-i}. \tag{*}$$

In particular, every H_n is then a subcoalgebra of H.

Proof. For every $x \in H$, the map

$$A \longrightarrow k, \qquad f \longmapsto f(x),$$

is continuous. Thus, if I^n is annihilated by x, so is its closure $\overline{I^n}$. For $n \geq 1$, put

$$(\overline{I^n})^{\perp} = \{x \in H \mid f(x) = 0 \text{ for every } f \in \overline{I^n}\}.$$

Suppose first that

$$\sigma : \mathfrak{m} \longmapsto \Phi^{-1}(\mathfrak{m})$$

is a bijection from $\mathrm{Spf}(k)$ onto $\mathrm{Spf}(A)$. Since I is contained in the intersection of the ideals $\Phi^{-1}(\mathfrak{m})$, it follows from 0.1.2 that the sequence of ideals $(\overline{I^n})$ tends to zero. Let $x \in H$. The submodule

$$J(x) = \{f \in A \mid f(x) = 0\}$$

is open and closed in A; hence it contains $\overline{I^n}$ for all sufficiently large n. In other words, $x \in (\overline{I^n})^{\perp}$ for all sufficiently large n.

Conversely, suppose that

$$H = \bigcup_n (\overline{I^n})^{\perp}.$$

Let $\mathfrak{p}$ be an open prime ideal of A. By the definition of the topology on A (0.2.2), $\mathfrak{p}$ contains an open k-submodule of the form

$$\mathscr{V}(x_1, \ldots, x_s) = \{f \in A \mid f(x_1) = \cdots = f(x_s) = 0\}.$$

By hypothesis, there is an integer n such that

$$x_1, \ldots, x_s \in (\overline{I^n})^{\perp};$$

therefore $\overline{I^n} \subset \mathfrak{p}$. Moreover, since k is Artinian,

$$\mathrm{Spf}(k) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$$

is finite, and there is an integer $t \geq 1$ such that

$$\left(\prod_i \mathfrak{m}_i\right)^t = 0.$$

It follows that

$$\prod_i \Phi^{-1}(\mathfrak{m}_i)^t \subset I.$$

Thus $\mathfrak{p}$ contains the product of the ideals $\Phi^{-1}(\mathfrak{m}_i)^{tn}$. Since $\mathfrak{p}$ is prime, it contains, and hence equals, one of the $\Phi^{-1}(\mathfrak{m}_i)$. We have therefore proved that [65]

$$\sigma \text{ is a bijection} \quad \Longleftrightarrow \quad H = \bigcup_n (\overline{I^n})^{\perp}. \tag{65}$$

We also have $H = k\phi \oplus H^+$; let $\pi : H \to H^+$ be the projection with kernel $k\phi$. For $n \geq 0$, let

$$\Delta^n : H \longrightarrow H^{\otimes(n+1)}$$

be the iterated comultiplication, let $\overline{\Delta}^n$ be the composite of Δ^n with the projection

$$\pi^{\otimes(n+1)} : H^{\otimes(n+1)} \longrightarrow (H^+)^{\otimes(n+1)},$$

and put

$$H'_n = \ker(\overline{\Delta}^n) = \left\{ x \in H \;\middle|\; \Delta^n(x) \in \sum_{i=0}^{n} H^{\otimes(n-i)} \otimes H_0 \otimes H^{\otimes i} \right\}.$$

We set $\overline{\Delta}^0 = \mathrm{id}_H$, so that $H'_0 = H_0$. A straightforward induction on n gives

$$H_n \subset H'_n \subset (\overline{I^{n+1}})^\perp. \tag{*}$$

Up to this point we have not used the hypothesis that H is flat over k. Assume now that H is flat, and hence projective, over k. Then $A^\dagger = H$ by 0.2.2. We shall prove that

$$H_n = (\overline{I^{n+1}})^\perp.$$

This is clear for $n = 0$. Suppose it has been proved for $n < r$. Then H_{r-1} is the kernel of the homomorphism

$$H \longrightarrow (\overline{I^r})^\dagger.$$

Since H^+ is flat, the homomorphism

$$(H/H_{r-1}) \otimes H^+ \longrightarrow (\overline{I^r})^\dagger \otimes H^+$$

is injective. On the other hand, the hypothesis implies that I is a projective pseudocompact k-module, since it is a direct factor of $A = H^*$. Thus, by 0.3.6,

$$(\overline{I^r} \widehat{\otimes} I)^\dagger \simeq (\overline{I^r})^\dagger \otimes I^\dagger = (\overline{I^r})^\dagger \otimes H^+.$$

The exact sequence

$$\overline{I^r} \widehat{\otimes} I \longrightarrow A \longrightarrow A/\overline{I^{r+1}} \longrightarrow 0$$

therefore gives, by duality, the exact sequence

$$(\overline{I^r})^\dagger \otimes H^+ \xleftarrow{\delta} H \longleftarrow (A/\overline{I^{r+1}})^\dagger \longleftarrow 0, \tag{1}$$

where δ is obtained by composing Δ_H with the projection

$$H \otimes H \longrightarrow H \otimes H^+ \longrightarrow (H/H_{r-1}) \otimes H^+ \hookrightarrow (\overline{I^r})^\dagger \otimes H^+. \tag{2}$$

For every $u \in H$, the projection of $\Delta(u)$ onto $H \otimes H^+$ is $\Delta(u) - u \otimes \phi$. Equations (1) and (2) show that if

$$u \in (A/\overline{I^{r+1}})^\dagger = (\overline{I^{r+1}})^\perp,$$

[65] This remains valid without assuming that H is cocommutative, with k still a commutative Artinian ring. In that case a basis of neighborhoods of zero in $A = H^*$ is given by the two-sided ideals J such that A/J has finite length over k, and the preceding proof shows that $H = \bigcup_n (\overline{I^n})^\perp$ if and only if the $\Phi^{-1}(\mathfrak{m}_i)$ are the only open prime ideals of A.

then $\Delta(u) - u \otimes \phi$ belongs to the kernel of

$$H \otimes H^+ \longrightarrow (H/H_{r-1}) \otimes H^+,$$

namely $H_{r-1} \otimes H^+$. Thus $u \in H_r$. This completes the proof of (i) and (ii), and also proves that

$$H_n = \ker(\overline{\Delta}^n).$$

We now prove (iii). For $i \geq 0$, put

$$H_i^+ = H_i \cap H^+.$$

Let $n \geq 1$. For every $x \in H_n$, the element $\overline{x} = x - \varepsilon(x)\phi$ belongs to H_n^+, and

$$\Delta(x) = \varepsilon(x)\phi \otimes \phi + \overline{x} \otimes \phi + \phi \otimes \overline{x} + \overline{\Delta}(\overline{x}).$$

It is therefore enough to prove that

$$\overline{\Delta}(H_n^+) \subset \sum_{i=1}^{n-1} H_i^+ \otimes H_{n-i}^+.$$

For $i = 0, \ldots, n-1$, the map $\overline{\Delta}^n$ factors as follows:

$$\begin{tikzcd}
H^+ \arrow[r, "\overline{\Delta}"] & H^+ \otimes H^+ \arrow[r, "\overline{\Delta}^i \otimes \overline{\Delta}^{n-i-1}"] \arrow[d] & (H^+)^{\otimes(i+1)} \otimes (H^+)^{\otimes(n-i)} \\
& \dfrac{H^+}{H_i^+} \otimes \dfrac{H^+}{H_{n-i-1}^+} \arrow[r, "f"] & \dfrac{H^+}{H_i^+} \otimes (H^+)^{\otimes(n-i)} . \arrow[u, "g"']
\end{tikzcd}$$

Moreover, since H^+/H_i^+ and $(H^+)^{\otimes(n-i)}$ are flat, the maps f and g in this diagram are injective. It follows that

$$\overline{\Delta}(H_n^+) \subset H_i^+ \otimes H^+ + H^+ \otimes H_{n-i-1}^+$$

for every $i = 0, \ldots, n-1$. Part (iii) now follows from the next lemma, applied with $M = H^+$, $E_0 = 0$, and $E_i = H_{i-1}^+$ for $i \geq 1$.

Lemma 1.3.6.B. *Let k be a ring, and let*

$$0 = E_0 \subset E_1 \subset \cdots \subset E_n \subset M$$

be k-modules. Suppose that M/E_i is flat for every i. Then

$$\bigcap_{i=0}^{n} (E_i \otimes M + M \otimes E_{n-i}) = \sum_{i=1}^{n} E_i \otimes E_{n-i+1}.$$

Proof. Denote the left-hand and right-hand sides by K and S, respectively. Clearly $S \subset K$; we prove the reverse inclusion. For $i = 0, \ldots, n$, put

$$K_i = K \cap (E_i \otimes M).$$

For $i = 1, \ldots, n$, both M/E_{n-i+1} and E_i/E_{i-1} are flat. Consequently, the homomorphism τ_i below is injective, and the composite

$$(E_i/E_{i-1}) \otimes M \longrightarrow (E_i/E_{i-1}) \otimes (M/E_{n-i+1}) \xhookrightarrow{\tau_i} (M/E_{i-1}) \otimes (M/E_{n-i+1})$$

has kernel

$$(E_i/E_{i-1}) \otimes E_{n-i+1}.$$

The image of K_i in $(E_i \otimes M)/(E_{i-1} \otimes M)$ is both contained in and contains this kernel. Hence

$$K_i = K_{i-1} + E_i \otimes E_{n-i+1},$$

which proves the lemma.

To conclude this paragraph, return to an arbitrary pseudocompact ring k.[66] Let $(\mathbf{H}, \Delta, \varepsilon)$ be a flat $\mathbf{O}_k$-coalgebra, put $\mathbf{H}^+ = \ker(\varepsilon)$, and let

$$A = \Gamma^*(\mathbf{H})$$

be the dual profinite k-algebra. Put $X = \operatorname{Spf}(A)$, so that $\mathbf{H} = \mathbf{H}(X)$ (cf. 1.3.5). Suppose that a morphism of $\mathbf{O}_k$-coalgebras

$$\phi : \mathbf{O}_k \longrightarrow \mathbf{H}$$

is given. It defines a continuous homomorphism of k-algebras $A \to k$, and hence a section σ of the structure morphism $X \to \operatorname{Spf}(k)$.

For every object B of Alf/k, put

$$\mathbf{H}_0(B) = \phi(B) = B\phi_B, \qquad \phi_B = \phi(1_B) \in \mathbf{H}(B).$$

For $n \geq 1$, define $\mathbf{O}_k$-submodules $\mathbf{H}_n$ of $\mathbf{H}$ by

$$\mathbf{H}_n(B) = \Big\{ u \in \mathbf{H}(B) \;\Big|\; \Delta(u) - u \otimes \phi_B \in \mathbf{H}_{n-1}(B) \otimes \mathbf{H}^+(B) \Big\}.$$

This gives a filtration

$$\mathbf{H}_0 \subset \mathbf{H}_1 \subset \cdots$$

of $\mathbf{H}(X)$. The preceding result gives:

Proposition. *The section σ induces an isomorphism of the underlying topological spaces if and only if*

$$\mathbf{H}(X) = \bigcup_n \mathbf{H}_n.$$

1.4

Theorem 1.4. *Let k be a pseudocompact ring and let*

$$d_0, d_1 : X_1 \rightrightarrows X$$

be an equivalence pair in Vaf/k (cf. Exposé V, § 2.b) *such that d_1 is topologically flat.*

(i) *The canonical projection from X onto $X/X_1 =$* $\operatorname{Coker}(d_0, d_1)$ *is surjective and topologically flat, and the morphism*

$$X_1 \longrightarrow X \times_{X/X_1} X$$

with components d_0 and d_1 is an isomorphism.

(ii) *If X is topologically flat over k, then so is X/X_1.*

First note that (ii) follows from (i) by 1.3.3 (ii). The proof of (i) occupies 1.4.1, 1.4.2, and 1.4.3.

[66] The editors added what follows; the original treated only the case in which k is Artinian.

1.4.1

We first show that we may reduce to the case in which X has only one point. Since we are dealing with an equivalence pair, one sees, as in Exposé V, § 3.e), that an equivalence relation is defined on the underlying set of X by declaring two points x, y to be equivalent if there exists a point $z \in X_1$ such that

$$d_0(z) = y \qquad \text{and} \qquad d_1(z) = x.$$

We may plainly suppose, without loss of generality, that X contains a single equivalence class for this relation; in other words, that X/X_1 has only one point (see the construction of X/X_1 given in 1.2).

In this case, let x be any point of X, and let U be the formal variety having x as its only point and the same local ring as X at x. As in Exposé V, § 6, one then sees that the equivalence relation induced by (d_0, d_1) on U still satisfies the hypotheses of the theorem and that it is enough to prove the theorem for this latter equivalence relation (U is a "quasi-section").

Let us briefly recall the principle of the proof given in Exposé V, § 6. Put

$$V = d_0^{-1}(U) = U \times_{i,d_0} X_1,$$

where i is the inclusion of U into X, and let u and v be the morphisms with source V induced respectively by d_0 and d_1:

$$X \xleftarrow{\ v\ } V \xrightarrow{\ u\ } U.$$

It is clear that u and v are topologically flat and that u is surjective; since X contains a single equivalence class, v is surjective. If (v_0, v_1) is the inverse image of the equivalence pair (d_0, d_1) under v (cf. V, 3.a)), it follows from V, 3.c) and 3.d) that X/X_1 and the quotient of U by the equivalence relation induced by (d_0, d_1) both identify with $\operatorname{Coker}(v_0, v_1)$. As in the proof of V, 6.1, one then sees that if the conclusion of Theorem 1.4 holds for U, it also holds for X.

1.4.2

We are thus reduced to the case in which X has only one point.[67] Consider the following commutative diagram (cf. V, § 1, (0,1,2)):

$$\begin{tikzcd}
X_2 \arrow[r, shift left, "d'_1"] \arrow[r, shift right, "d'_0"'] \arrow[d, "d'_2"'] & X_1 \arrow[r, "d_0"] \arrow[d, "d_1"] & X \arrow[d] \\
X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X \arrow[r] & X/X_1
\end{tikzcd}$$

Here

$$X_2 = X_1 \times_{d_1,d_0} X_1,$$

and d'_0, d'_1, d'_2 are respectively the morphisms[68]

$$(x,y,z) \longmapsto (x,y), \qquad (x,y,z) \longmapsto (x,z), \qquad (x,y,z) \longmapsto (y,z).$$

If B, A, A_1, A_2 denote respectively the affine algebras of $X/X_1, X, X_1, X_2$, the preceding diagram induces the commutative diagram

[67] Editors' note: Thus one may suppose that k is local.

[68] Editors' note: Cf. Exposé V, § 2.b).

$$\begin{tikzcd} A_2 & A_1 \arrow[l, shift left, "j_1"'] \arrow[l, shift right, "j_0"] & A \arrow[l, "i_0"'] \\ A_1 \arrow[u, "j_2"] & A \arrow[l, shift left, "i_1"'] \arrow[l, shift right, "i_0"] \arrow[u, "i_1"'] & B \arrow[l, "i"'] \arrow[u, "i"'] \end{tikzcd}$$

in which both rows are exact and the squares determined by (i_0, j_0) and (i_1, j_1) are cocartesian. Since the morphism

$$X_1 \longrightarrow X \times X$$

with components d_0 and d_1 is a monomorphism by hypothesis, the morphism

$$A \widehat{\otimes}_k A \longrightarrow A_1$$

with components i_0 and i_1 is surjective by Proposition 1.3.

This means that i_1 makes A_1 a pseudocompact A-module, topologically free by hypothesis, generated by $i_0(A)$. Since A is local, Lemma 1.4.3 below gives a topologically free k-module V and a morphism of pseudocompact k-modules $f : V \to A$ such that the morphism

$$\alpha_1 : A \widehat{\otimes}_k V \longrightarrow A_1, \qquad a \widehat{\otimes} v \longmapsto i_1(a) \cdot i_0(f(v))$$

is an isomorphism. Let

$$\alpha : B \widehat{\otimes}_k V \longrightarrow A \qquad \text{and} \qquad \alpha_2 : A_1 \widehat{\otimes}_k V \longrightarrow A_2$$

be the morphisms

$$b \widehat{\otimes} v \longmapsto i(b) \cdot f(v) \qquad \text{and} \qquad a_1 \widehat{\otimes} v \longmapsto j_2(a_1) \cdot j_0 i_0(f(v)).$$

In the following commutative diagram, both rows are exact and the two left-hand squares are cocartesian. Since α_1 is an isomorphism, the same is true of α_2, and hence of α.

$$\begin{tikzcd} A_2 & A_1 \arrow[l, shift left, "j_1"'] \arrow[l, shift right, "j_0"] & A \arrow[l, "i_0"'] \\ A_1 \widehat{\otimes}_k V \arrow[u, "\alpha_2"] & A \widehat{\otimes}_k V \arrow[l, shift left, "i_1 \widehat{\otimes} V"'] \arrow[l, shift right, "i_0 \widehat{\otimes} V"] \arrow[u, "\alpha_1"'] & B \widehat{\otimes}_k V\,. \arrow[l, "i \widehat{\otimes} V"'] \arrow[u, "\alpha"'] \end{tikzcd}$$

This shows, on the one hand, that A is topologically free over B, with pseudobasis $f(V)$ (cf. 0.2.1), and that a pseudobasis of A_1 over A, where A_1 is regarded as an A-module through i_1, is obtained by taking the image under i_0 of $f(V)$. It follows that the morphism

$$A \widehat{\otimes}_B A \longrightarrow A_1$$

with components i_1 and i_0 is an isomorphism:

$$A \widehat{\otimes}_B A \simeq A \widehat{\otimes}_B B \widehat{\otimes}_k V \simeq A \widehat{\otimes}_k V \simeq A_1.$$

This proves Theorem 1.4, modulo the following lemma.

1.4.3

Lemma 1.4.3. *Let k be a pseudocompact ring, let A be a local profinite k-algebra, let A_1 be a topologically free A-module, and let*

$$i_0 : M \longrightarrow A_1$$

be a morphism of pseudocompact k-modules. Suppose that the map

$$A\widehat{\otimes}_k M \longrightarrow A_1, \qquad a\widehat{\otimes} m \longmapsto a \cdot i_0(m),$$

is surjective. Then there exist a topologically free k-module V and a morphism of pseudocompact k-modules

$$f : V \longrightarrow M$$

such that the map

$$A\widehat{\otimes}_k V \longrightarrow A_1, \qquad a\widehat{\otimes} v \longmapsto a \cdot i_0(f(v)),$$

is bijective.

Since every pseudocompact k-module is a quotient of a topologically free k-module (cf. Editors' note (27)), we may suppose, without loss of generality, that M is topologically free. Thus write

$$M = \prod_{i\in I} M_i,$$

where each M_i is a copy of k. In this case,

$$A\widehat{\otimes}_k M = \prod_{i\in I} A\widehat{\otimes}_k M_i.$$

Since the map

$$a\widehat{\otimes} m \longmapsto a \cdot i_0(m)$$

is surjective and A_1 is projective, its kernel is a direct factor of $A\widehat{\otimes}_k M$. As A is local, the exchange theorem (0.3.4) shows that this kernel has as a complement a partial product

$$\prod_{i\in J} A\widehat{\otimes}_k M_i,$$

where J is a subset of I. One may therefore take

$$V = \prod_{i\in J} M_i.$$

1.5

Let k be a pseudocompact ring.

Definition. We shall say that a family of morphisms

$$f_i : X_i \longrightarrow X$$

in Vaf $/k$ is a *surjective topologically flat family* if the morphism

$$\coprod_i X_i \longrightarrow X$$

induced by the f_i is surjective and topologically flat. This means that every f_i is topologically flat and that every point of X lies in the image of at least one X_i.

It follows from 1.3.3 that surjective topologically flat families define a pretopology on Vaf $/k$ (IV, 4.2.5). The corresponding topology will be called the *flat topology* on Vaf $/k$.

By IV, 4.3.5, a functor

$$F : (\mathrm{Vaf}\,/k)^{\mathrm{op}} \longrightarrow (\mathbf{Sets})$$

is a sheaf for the flat topology if and only if F carries every direct sum to a direct product and the sequence

$$F(Y) \xrightarrow{F(f)} F(X) \overset{F(\mathrm{pr}_1)}{\underset{F(\mathrm{pr}_2)}{\rightrightarrows}} F(X \times_Y X)$$

is exact for every surjective topologically flat morphism $f : X \to Y$.

By IV, 4.5,[69] Proposition 1.3.1 implies that the flat topology is coarser than the canonical topology. In other words, for every object X of Vaf $/k$, the functor

$$h_X : T \longmapsto \mathrm{Hom}_{\mathrm{Vaf}\,/k}(T, X)$$

is a sheaf for the flat topology. In what follows, as usual (cf. Exposé I), we identify X with h_X.

By IV, 4.6.5, Theorem 1.4 may be reformulated as follows.

Theorem. *Let k be a pseudocompact ring, let $d_0, d_1 : X_1 \rightrightarrows X$ be an equivalence pair in* Vaf $/k$, *and let X/X_1 be the quotient formal variety* (that is, $\mathrm{Coker}(d_0, d_1)$, cf. 1.2). *If d_1 is topologically flat, then X/X_1 represents the quotient sheaf for the flat topology.*

1.6

To complete these generalities on formal varieties, it remains to define briefly the formal varieties that are étale over k. [70]

Recollections 1.6.A. *(i)* Recall first (cf. EGA 0_{IV}, 19.10.2) that a topological k-algebra A is said to be *formally étale over k*—for the given topologies on k and A, respectively for the discrete topologies—if, for every discrete topological k-algebra C (not necessarily Artinian) and every nilpotent ideal J of C, every continuous morphism of k-algebras

$$A \longrightarrow C/J$$

lifts uniquely to a continuous morphism $A \to C$, where A is endowed with the given topology, respectively with the discrete topology.

This property is plainly preserved under base change. Thus, for every morphism $k \to k'$ of pseudocompact rings,

$$A \widehat{\otimes}_k k'$$

is formally étale over k'. It is also enough to verify the lifting condition for square-zero ideals J; cf. EGA IV$_4$, 17.1.2 (ii).

[69] The editors modified the original in what follows. In particular, they replaced the statement: "if $d_0, d_1 : X_1 \rightrightarrows X$ is an equivalence relation such that d_1 is topologically flat, formation of the quotient commutes with h," by the theorem below.

[70] The original continued as follows: "A formal variety X over k is said to be étale if the diagonal morphism $\Delta_X : X \to X \times X$ is a local isomorphism; that is, if Δ_X induces an isomorphism from $\mathcal{O}_{X\times X, \Delta_X(x)}$ onto $\mathcal{O}_{X,x}$ for every point x of X. It follows readily from SGA I that this formulation is equivalent to each of the following two: the formal variety X is topologically flat and, for every point $x \in X$ projecting to $s \in \mathrm{Spf}(k)$, $\mathcal{O}_{X,x}\widehat{\otimes}_k \kappa(s)$ is a finite separable extension of the residue field $\kappa(s)$; or, if A denotes the affine algebra of X, the local components (0.1) of $A\widehat{\otimes}_k(k/\mathfrak{l})$ are finite étale algebras over $k/\mathfrak{l}$, for every open ideal $\mathfrak{l}$ of k." In what follows, the editors corrected the omission of the flatness hypothesis from the first condition above and supplied details for the equivalence of the conditions.

The algebra A is said to be *étale over* k if it is formally étale over k for the discrete topologies and is, in addition, a k-algebra of finite presentation; cf. EGA IV$_4$, 17.3.2 (ii). In what follows, when k is a pseudocompact ring and A a profinite k-algebra, "formally étale" refers to the given topologies unless the contrary is stated.

(ii) Recall also that if $F \in k[T]$ is a separable monic polynomial of degree $d \geq 1$, meaning that the ideal generated by F and its derivative F' is all of $k[T]$, then

$$B = k[T]/(F),$$

which is free of rank d over k and is endowed with the product topology, is formally étale over k.

Indeed, let C be a discrete k-algebra, so that the kernel $\mathfrak{l}$ of $k \to C$ is an open ideal of k; let J be a square-zero ideal of C; and let $\phi : B \to C/J$ be a continuous morphism of k-algebras. Since B is a free k-module of finite rank, $\mathfrak{l}B$ is an open ideal of B, and hence every lift $\Phi : B \to C$ of ϕ is automatically continuous.

Let t be the image of T in B, and choose any lift $u_0 \in C$ of $\phi(t)$. Then $F(u_0) \in J$, since $\phi(t)$ is a root of F. There exist $G, H \in k[T]$ such that

$$GF + HF' = 1.$$

Consequently,

$$H(u_0)F'(u_0) = 1 - G(u_0)F(u_0).$$

The right-hand side is invertible because $F(u_0)$ is square-zero, and therefore $F'(u_0)$ is invertible. We seek $h \in J$ such that $u = u_0 + h$ is a root of F. This is equivalent to

$$0 = F(u) = F(u_0) + F'(u_0)h,$$

whose unique solution is

$$h = -F'(u_0)^{-1}F(u_0) \in J.$$

The same proof, with no continuity assumptions on the morphisms $k \to C$, ϕ, and Φ, also shows that B is an étale k-algebra.

(iii) Finally, if

$$A = A_1 \times \cdots \times A_n$$

is a finite product, then A is formally étale over k if and only if every A_i is.[71] It is enough to prove this for $n = 2$. Let $e = 1_{A_1}$ be the idempotent such that $A_1 = Ae$ and $A_2 = A(1-e)$, and suppose that a continuous morphism

$$\phi : A \longrightarrow C/J$$

is given, where J is square-zero. The polynomial $F = X^2 - X$ is separable, since

$$F' = 2X - 1 \qquad \text{and} \qquad (F')^2 - 4F = 1.$$

Therefore the idempotent $\phi(e)$ of C/J lifts uniquely to an idempotent f of C. Hence

$$C = Cf \oplus C(1-f),$$

and giving a lift Φ is equivalent to giving two morphisms

$$\Phi_1 : A_1 \longrightarrow Cf, \qquad \Phi_2 : A_2 \longrightarrow C(1-f),$$

[71] The editors point out that the proof given in EGA 0_{IV}, 19.3.5 (v), is erroneous.

lifting the respective restrictions of ϕ. The same argument shows that if e is an idempotent of k and $A = Ae$, then A is formally étale over k if and only if it is formally étale over the local component k_e, identified with ke.

Now let X be a formal k-variety with affine algebra A. If x is a point of X, that is, an open maximal ideal $\mathfrak{m}$ of A, and s is its image in $\mathrm{Spf}(k)$, we denote by $k_{\mathfrak{m}}$, or k_s, the local component of k corresponding to s.

Definition 1.6.B. *The following conditions are equivalent:*

(a) A is formally étale over k.

(b) For every $\mathfrak{m} \in \Upsilon(A)$, $A_{\mathfrak{m}}$ is formally étale over k, or equivalently over $k_{\mathfrak{m}}$.

(c) For every open ideal $\mathfrak{l}$ of k,
$$A\widehat{\otimes}_k(k/\mathfrak{l}) = A/A\mathfrak{l}$$
is formally étale over $k/\mathfrak{l}$.

(d) For every $\mathfrak{m} \in \Upsilon(A)$ and every open ideal $\mathfrak{l}$ of $k_{\mathfrak{m}}$, $A_{\mathfrak{m}}/A_{\mathfrak{m}}\mathfrak{l}$ is formally étale over $k_{\mathfrak{m}}/\mathfrak{l}$.

We say that X is *étale over k* if these conditions hold, and write
$$\mathrm{Vaf}^{\text{ét}}/k$$
for the full subcategory of Vaf/k consisting of formal varieties that are étale over k. [72]

Indeed, suppose that
$$\phi : A \longrightarrow C/J$$
is a continuous morphism of k-algebras, where C is a discrete k-algebra and J is square-zero. Then $I = \mathrm{Ker}(\phi)$ is an open ideal of A, so A/I is Artinian. Thus I is contained in only finitely many open maximal ideals
$$\mathfrak{m}_1, \ldots, \mathfrak{m}_r.$$
It therefore contains the product of the components $A_{\mathfrak{m}}$ for $\mathfrak{m} \notin \{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$. This product is $A(1-e)$, where e is the idempotent of A for which
$$Ae = \prod_{i=1}^{r} A_{\mathfrak{m}_i}.$$
Hence $\phi(e) = 1_{C/J}$, and giving a continuous lift of ϕ is equivalent to giving a continuous lift of the induced morphism
$$Ae \simeq A/A(1-e) \longrightarrow C/J.$$

On the other hand, by editors' note (24),
$$A \simeq \varprojlim_{\mathfrak{l}} A/A\mathfrak{l}.$$

The equivalence of the stated conditions follows readily from these observations and the preceding recollections.

Definitions 1.6.C. Put
$$\kappa(k) = \prod_{s \in \mathrm{Spf}(k)} \kappa(s),$$

[72]The category $\mathrm{Vaf}^{\text{ét}}/k$ is stable under finite projective limits.

endowed with the product topology. Thus the formal variety $\mathrm{Spf}(\kappa(k))$ is the direct sum of the schemes $\mathrm{Spec}\,\kappa(s)$, for $s \in \mathrm{Spf}(k)$. We write $S_{\kappa(k)}$ for the scheme that is the direct sum of the same $\mathrm{Spec}\,\kappa(s)$.

For every formal variety X over k, put

$$X_\kappa = X \widehat{\otimes}_k \kappa(k).$$

This is the formal variety over $\kappa(k)$ obtained by base change. It has the same points as X, and if $x \in X$ projects to $s \in \mathrm{Spf}(k)$, then

$$\mathcal{O}_{X_\kappa, x} = \mathcal{O}_{X,x} \widehat{\otimes}_k \kappa(s).$$

The base-change functor

$$\mathrm{Vaf}\,/k \longrightarrow \mathrm{Vaf}\,/\kappa(k)$$

sends $\mathrm{Vaf}^{\text{ét}}\,/k$ into $\mathrm{Vaf}^{\text{ét}}\,/\kappa(k)$, by 1.6.A (i).

One then has the following result (cf. SGA 1, I 6.2).

Lemma 1.6.D. *For every*

$$Y \in \mathrm{Ob}(\mathrm{Vaf}^{\text{ét}}\,/k) \qquad \textit{and} \qquad X \in \mathrm{Ob}(\mathrm{Vaf}\,/k),$$

the canonical map

$$\mathrm{Hom}_{\mathrm{Vaf}\,/k}(X, Y) \longrightarrow \mathrm{Hom}_{\mathrm{Vaf}\,/\kappa(k)}(X_\kappa, Y_\kappa)$$

is bijective. In particular, the functor

$$\mathrm{Vaf}^{\text{ét}}\,/k \longrightarrow \mathrm{Vaf}^{\text{ét}}\,/\kappa(k)$$

is fully faithful; below it will be shown to be an equivalence.

Indeed, let A, respectively B, be the affine algebra of X, respectively Y, and let r be the radical of k. Suppose that a morphism

$$B \widehat{\otimes}_k \kappa(k) \longrightarrow A \widehat{\otimes}_k \kappa(k)$$

is given, or equivalently a morphism

$$\phi : B \longrightarrow A/rA.$$

For every open ideal $\mathfrak{l}$ of k, there is an integer $n \geq 1$ such that

$$r^n \subset \mathfrak{l}.$$

Because the multiplication map

$$m^{n-1} : A^n \longrightarrow A$$

is continuous, it follows that

$$(rA)^n \subset \mathfrak{l}A.$$

Thus $rA/\mathfrak{l}A$ is a nilpotent ideal of $A/\mathfrak{l}A$. Consequently, ϕ lifts uniquely to a morphism

$$\phi_{\mathfrak{l}} : B \longrightarrow A/\mathfrak{l}A.$$

By uniqueness, these morphisms form a projective system and therefore give a continuous morphism

$$\Phi : B \longrightarrow \varprojlim_{\mathfrak{l}} A/\mathfrak{l}A = A.$$

Moreover, Φ is unique: if Φ' is another lift of ϕ, then Φ' and Φ agree modulo $\mathfrak{l}A$ for every $\mathfrak{l}$, and hence are equal.

Proposition 1.6.E. *(a) Let X be a formal variety over k, with affine algebra A. The following conditions are equivalent:*

(i) *X is étale over k.*

(ii) *X is topologically flat over k, and the diagonal morphism*

$$\Delta : X \longrightarrow X \times X$$

is a local isomorphism; that is, for every point x of X, Δ_X induces an isomorphism

$$\mathcal{O}_{X\times X,\Delta(x)} \xrightarrow{\sim} \mathcal{O}_{X,x}.$$

(iii) *For every $x \in X$ projecting to $s \in \mathrm{Spf}(k)$, there is an isomorphism*

$$\mathcal{O}_{X,x} \simeq k_s[T]/(F),$$

where $F \in k_s[T]$ is a separable monic polynomial (cf. 1.6.A (ii)).

(iv) *X is topologically flat over k, and, for every $x \in X$ projecting to $s \in \mathrm{Spf}(k)$,*

$$\mathcal{O}_{X,x}\widehat{\otimes}_k\kappa(s)$$

is a finite separable extension of $\kappa(s)$.

(v) *For every open ideal $\mathfrak{l}$ of k, each local component of*

$$A\widehat{\otimes}_k(k/\mathfrak{l})$$

is a finite étale algebra over the Artinian ring $k/\mathfrak{l}$.

(b) The category $\mathrm{Vaf}^{\text{ét}}/\kappa(k)$ *identifies with the category of étale schemes over $S_{\kappa(k)}$* (cf. 1.6.C)*, and the functor $X \mapsto X_\kappa$ induces an equivalence*

$$\mathrm{Vaf}^{\text{ét}}/k \xrightarrow{\sim} \mathrm{Vaf}^{\text{ét}}/\kappa(k)$$

(cf. SGA 1, I 6.1).

Proof. Let I be the kernel of the multiplication morphism

$$m : A\widehat{\otimes}_k A \longrightarrow A.$$

Suppose first that X is étale over k, so that A is formally étale over k. By EGA 0_{IV}, 20.7.4, the separated completion of I/I^2, for the quotient topology induced from I, is zero; hence

$$I = I^2.$$

For every $x \in X$, the ideal I is contained in the maximal ideal $\mathfrak{m}_{\Delta(x)}$ of $A\widehat{\otimes}_k A$. Thus its localization $I_{\mathfrak{m}_{\Delta(x)}}$ lies in the maximal ideal of $\mathcal{O}_{X\times X,\Delta(x)}$. Nakayama's lemma 0.3 therefore gives

$$I_{\mathfrak{m}_{\Delta(x)}} = 0.$$

Consequently, $\Delta : X \to X \times X$ is a local isomorphism.

Suppose next that Δ is a local isomorphism and that k is a field κ. We show that every $\mathcal{O}_{X,x}$ is a finite-dimensional étale κ-algebra. Replacing X by $\mathrm{Spf}(\mathcal{O}_{X,x})$, we may assume that

$$A = \mathcal{O}_{X,x}$$

is local. We proceed as in the proof of EGA IV$_4$, 17.4.1, (b)⇒(d”). Let K be a finite normal extension of κ containing the residue field $\kappa(x)$, and put

$$B = A \otimes_\kappa K = A \widehat{\otimes}_\kappa K, \qquad X_K = \mathrm{Spf}(B) = X \widehat{\otimes}_\kappa K.$$

The two tensor products agree because $[K : \kappa] < \infty$.

The morphism

$$\Delta_K : X_K \longrightarrow X_K \times_K X_K$$

is again a local isomorphism. Hence, for every $y \in X_K$, multiplication induces an isomorphism

$$\big(B_y \widehat{\otimes}_K B_y\big)_{\mathfrak{m}_{\Delta_K(y)}} \xrightarrow{\sim} B_y.$$

The residue field of B_y is K; cf., for example, VI$_A$, 1.1.1, editors’ note (11). Moreover,

$$C = B_y \widehat{\otimes}_K B_y$$

is already local. Indeed,

$$\mathfrak{n} = \mathfrak{m}_y \widehat{\otimes}_K B_y + B_y \widehat{\otimes}_K \mathfrak{m}_y$$

consists of topologically nilpotent elements and therefore lies in the radical of C, while

$$C/\mathfrak{n} = K \widehat{\otimes}_K K = K$$

is a field. It follows that multiplication itself is an isomorphism

$$B_y \widehat{\otimes}_K B_y \xrightarrow{\sim} B_y.$$

Choosing a pseudobasis of B_y over K that contains the unit element 1, we deduce that $B_y = K$. Since B is finite over A, the set X_K is finite. Thus

$$B = A \otimes_\kappa K$$

is a product of finitely many copies of K, and A is a finite étale κ-algebra.

We have therefore shown that, over a field κ, every profinite κ-algebra A that is étale over κ is a product of finite separable extensions K_i/κ, endowed with the product topology. Consequently, $\mathrm{Spf}(A)$ is the direct sum of the schemes

$$\mathrm{Spf}(K_i) = \mathrm{Spec}(K_i),$$

and $\mathrm{Vaf}^{\text{ét}}/\kappa$ identifies with the category of étale κ-schemes.

The preceding argument proves (ii)⇒(iv), because $\mathcal{O}_{X,x} \widehat{\otimes}_k \kappa(s)$ is formally étale over $\kappa(s)$, and also proves part (b). Indeed, return to an arbitrary pseudocompact ring k. The category $\mathrm{Vaf}^{\text{ét}}/\kappa(k)$ has just been identified with the category of étale schemes over $S_{\kappa(k)}$. By 1.6.D, to show that

$$\mathrm{Vaf}^{\text{ét}}/k \longrightarrow \mathrm{Vaf}^{\text{ét}}/\kappa(k)$$

is an equivalence, it remains only to prove essential surjectivity.

Let $s \in \mathrm{Spf}(k)$, and let $K/\kappa(s)$ be a finite separable extension. Choose a primitive element $\xi \in K$, let $n = [K : \kappa(s)]$, and choose a monic polynomial

$$F \in k_s[T]$$

of degree n whose image $\overline{F} \in \kappa(s)[T]$ is the minimal polynomial of ξ. Since $\overline{F}'$ is invertible in $\kappa(s)[T]/(\overline{F})$, Nakayama’s lemma shows that F' is invertible in $k_s[T]/(F)$. Thus F is separable, and 1.6.A (ii) shows that

$$A = k_s[T]/(F)$$

is an étale k_s-algebra satisfying

$$A\widehat{\otimes}_k \kappa(s) \simeq K.$$

It follows that every local profinite étale k_s-algebra is free of finite rank over k_s, and therefore is topologically free over k. Hence every profinite étale k-algebra is topologically free over k.

Condition (v) implies condition (d) of 1.6.B, and therefore implies (i). We have thus proved that (i), (iii), and (v) are equivalent and imply (ii), which in turn implies (iv).

Finally, let A be a profinite k-algebra satisfying (iv). We show that A is formally étale over k. We may suppose that A and k are local, and write κ for the residue field of k. By hypothesis,

$$K = A\widehat{\otimes}_k \kappa$$

is a finite separable extension of κ, say of degree n. By the preceding argument and Lemma 1.6.D, there is a k-algebra B, free of rank n and formally étale over k, together with a continuous morphism

$$\phi : B \longrightarrow A$$

such that

$$\phi\widehat{\otimes}_k \kappa$$

is an isomorphism. Since A is topologically flat over k, this implies

$$\operatorname{Coker}(\phi)\widehat{\otimes}_k \kappa = 0 \qquad \text{and} \qquad \operatorname{Ker}(\phi)\widehat{\otimes}_k \kappa = 0.$$

Nakayama's lemma 0.3 gives

$$\operatorname{Coker}(\phi) = 0 = \operatorname{Ker}(\phi).$$

Thus ϕ is an isomorphism (cf. the proof of 0.2.B), which completes the proof of 1.6.E.

Let X be a formal variety over k. If $x \in X$ projects to $s \in \operatorname{Spf}(k)$, then the residue field $\kappa(x)$ is a finite extension of $\kappa(s)$. We denote by $\kappa_e(x)$ the separable closure of $\kappa(s)$ in $\kappa(x)$.[73]

Proposition 1.6.F. *(i) The inclusion*

$$\operatorname{Vaf}^{\text{ét}}/k \longrightarrow \operatorname{Vaf}/k$$

has a left adjoint $X \mapsto X_e$. The formal variety X_e has the same points as X. For each $x \in X$ projecting to $s \in \operatorname{Spf}(k)$, choose a primitive element ξ of $\kappa_e(x)/\kappa(s)$, let n be its degree, choose any lift $u \in \mathcal{O}_{X,x}$ of ξ, and let $F \in k[T]$ be a monic polynomial of degree n that annihilates u. Then put

$$\mathcal{O}_{X_e,x} = k[T]/(F).$$

For every $Y \in \operatorname{Ob}(\operatorname{Vaf}^{\text{ét}}/k)$, the canonical maps in the following diagram are bijective:

$$\begin{array}{ccc} \operatorname{Hom}_{\operatorname{Vaf}/k}(X,Y) & \xrightarrow{\sim} & \operatorname{Hom}_{\operatorname{Vaf}/\kappa(k)}(X_\kappa, Y_\kappa) \\ & & \Big\downarrow \wr \\ \operatorname{Hom}_{\operatorname{Vaf}/k}(X_e,Y) & \xrightarrow{\sim} & \operatorname{Hom}_{\operatorname{Vaf}/\kappa(k)}((X_e)_\kappa, Y_\kappa)\,, \end{array}$$

[73] In what follows, the editors used the preceding additions to give details of the construction of the functor $X \mapsto X_e$ and to prove that it commutes with finite products; this is used in 2.5.1.

The vertical arrow is induced by the inclusions

$$\kappa_e(x) \hookrightarrow \kappa(x), \qquad x \in X.$$

In particular, this defines a morphism

$$p : X \longrightarrow X_e,$$

and every morphism of formal k-varieties $\phi : X \to Y$, with Y étale over k, factors uniquely through p.

(ii) The functor $X \mapsto X_e$ commutes with finite products.

Proof. Assertion (i) follows from 1.6.E (b) and 1.6.D. We prove (ii). By the equivalence of categories in 1.6.E (b), we may suppose that k is a field. In that case, if X is a semilocal formal k-variety, so that its affine algebra A is semilocal, the affine algebra of X_e is the largest subalgebra of A that is étale over k; denote it by A_e.

It is therefore enough to show that, if K and L are two finite extensions of k, then

$$K_e \otimes_k L_e = (K \otimes_k L)_e.$$

Let p be the characteristic exponent of k, so that $p = 1$ if $\mathrm{char}(k) = 0$, and $p = \mathrm{char}(k)$ otherwise. For every $x \in K \otimes_k L$, there is an integer $n \geq 1$ such that

$$x^{p^n} \in K_e \otimes_k L_e.$$

Thus $K \otimes_k L$ is radicial over $K_e \otimes_k L_e$, and its largest étale k-subalgebra is precisely $K_e \otimes_k L_e$. Hence

$$K_e \otimes_k L_e = (K \otimes_k L)_e.$$

Keeping this notation, observe that $\mathcal{O}_{X_e,x}$ is not necessarily a subalgebra of $\mathcal{O}_{X,x}$. It is a subalgebra when X is topologically flat over k, by the proposition that follows.

1.6.1

Let Y be an étale formal k-variety and let $f : X \to Y$ be a morphism in $\mathrm{Vaf}\,/k$. Then the following square is Cartesian, where Γ_f is the graph morphism $X \to X \times Y$, with components id_X and f:

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times Y \\ {\scriptstyle f}\big\downarrow & & \big\downarrow{\scriptstyle f\times \mathrm{id}_Y} \\ Y & \xrightarrow{\Delta_Y} & Y \times Y\,. \end{array}$$

It follows that Γ_f is a local isomorphism, and hence that

$$f = \mathrm{pr}_Y \circ \Gamma_f$$

is topologically flat if pr_Y is; this holds, for example, if X is topologically flat over k.

Conversely, since $Y \to k$ is topologically flat, $X \to k$ will also be topologically flat if f is (cf. 1.3.3). Taking in particular for f the canonical morphism

$$p : X \longrightarrow X_e$$

of 1.6, we obtain the following.

Proposition. *Let X be a formal variety over k. The morphism $X \to X_e$ is topologically flat if and only if X is topologically flat over k.*

Remark 1.6.2.[74] When k is a *perfect* field, the functor

$$X \longmapsto X_e$$

is also right adjoint to the inclusion

$$\mathrm{Vaf}^{\text{ét}}/k \hookrightarrow \mathrm{Vaf}/k.$$

Indeed, in this case

$$\mathcal{O}_{X_e,x} = \kappa(x)$$

for every $x \in X$, and the canonical projections

$$\mathcal{O}_{X,x} \longrightarrow \kappa(x)$$

define a morphism $s : X_e \to X$, which is a section of $p : X \to X_e$. For every morphism $f : X \to Y$, the following diagrams are commutative:

$$\begin{tikzcd} \mathcal{O}_{Y,f(x)} \arrow[r] \arrow[d] & \mathcal{O}_{X,x} \arrow[d] \\ \kappa(f(x)) \arrow[r] & \kappa(x) \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "f"] & Y \\ X_e \arrow[u, "s_X"] \arrow[r, "f_e"] & Y_e \arrow[u, "s_Y"'] \end{tikzcd}$$

Thus s is functorial in X, and it follows that $X \mapsto X_e$ is right adjoint to the inclusion

$$\mathrm{Vaf}^{\text{ét}}/k \hookrightarrow \mathrm{Vaf}/k.$$

Consequently, being a right adjoint, the functor $X \mapsto X_e$ commutes with projective limits whenever they exist in $\mathrm{Vaf}^{\text{ét}}/k$,[75] and hence, in particular, with finite products. (This can also be verified directly: for every formal k-variety X, with affine algebra A, the affine algebra of X_e is the quotient of A by its radical $r(A)$; and, since k is perfect, the quotient of $\mathcal{O}_{X,x}\widehat{\otimes}\mathcal{O}_{Y,y}$ by its radical is the algebra $\kappa(x) \otimes_k \kappa(y)$, because the latter is semisimple.)

2. Generalities on Formal Groups

2.1

Let k be a pseudocompact ring and let G be a formal k-group, that is, a group object in the category Vaf/k of formal varieties over k. Let A be the affine algebra of G. The composition law of G plainly defines a diagonal morphism, namely a homomorphism of profinite k-algebras

$$\Delta_A : A \longrightarrow A\widehat{\otimes}_k A.$$

This homomorphism has the following properties.

(i) The diagram

[74] The editors inserted this remark here; in the original it appeared in 2.5.2.

[75] This is the case for finite projective limits.

$$\begin{tikzcd} A \arrow[r, "\Delta_A"] \arrow[d, "\Delta_A"'] & A\widehat{\otimes}_k A \arrow[d, "\Delta_A\widehat{\otimes}\,\mathrm{id}_A"] \\ A\widehat{\otimes}_k A \arrow[r, "\mathrm{id}_A\widehat{\otimes}\Delta_A"] & A\widehat{\otimes}_k A\widehat{\otimes}_k A \end{tikzcd}$$

is commutative.

(ii) There is a necessarily unique *augmentation*, that is, a homomorphism of profinite k-algebras
$$\varepsilon_A : A \longrightarrow k,$$
such that the two composites displayed below are the identity maps of A:

$$A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{\varepsilon_A\widehat{\otimes}\,\mathrm{id}_A} k\widehat{\otimes}_k A \simeq A$$
$$\text{and} \qquad A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{\mathrm{id}_A\widehat{\otimes}\varepsilon_A} A\widehat{\otimes}_k k \simeq A$$

(iii) There is a necessarily unique *antipode*, that is, a homomorphism of profinite k-algebras
$$c_A : A \longrightarrow A,$$
such that the composite

$$A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{c_A\widehat{\otimes}\,\mathrm{id}_A} A\widehat{\otimes}_k A \xrightarrow{m_A} A$$

equals $\eta_A \circ \varepsilon_A$. Here m_A is the linear map that sends $a\widehat{\otimes}b$ to ab, and $\eta_A : k \to A$ sends λ to $\lambda 1_A$.

Conversely, data
$$(\Delta_A, \varepsilon_A, c_A)$$
satisfying (i)–(iii) endow G with the structure of a formal k-group.[76] Explicitly, for every profinite k-algebra B, the set
$$\mathrm{Hom}_c(A, B)$$
of continuous morphisms of k-modules $\phi : A \to B$ carries a group structure, functorial in B, defined by
$$\phi \cdot \phi' = m_B \circ (\phi\widehat{\otimes}\phi') \circ \Delta_A.$$
The identity element is $\eta_B \circ \varepsilon_A$, where m_B is the multiplication of B and $\eta_B : k \to B$ sends λ to $\lambda 1_B$; the inverse of ϕ is $\phi \circ c_A$. The set

$$\mathrm{Hom}_{\mathrm{Alp}\,/k}(A, B)$$

of continuous morphisms of k-algebras $A \to B$ is a subgroup, because B is commutative.

Definition. A *morphism of formal k-groups* $\theta : K \to G$ is, by definition, a morphism of formal k-varieties that respects the group structures. Let B, respectively A, be the affine

[76] One also says that A is a *cogroup* in the category of profinite k-algebras. The editors expanded the material that follows, in particular by making explicit what a morphism of formal groups $K \to G$ means; cf. Proposition 2.3.1.

algebra of K, respectively G, and let $f : A \to B$ be the morphism corresponding to θ. The condition is equivalently that f be compatible with comultiplication:

$$(f \widehat{\otimes} f) \circ \Delta_A = \Delta_B \circ f.$$

The identities

$$\varepsilon_B \circ f = \varepsilon_A, \qquad c_B \circ f = f \circ c_A$$

then follow automatically. We write $\mathrm{Grf}\,/k$ for the category of formal k-groups.

Notation. In what follows, the ideal

$$I_A = \mathrm{Ker}(\varepsilon_A)$$

will be called the *augmentation ideal* of A, and we write

$$\omega_{G/k} = I_A/\overline{I_A^2}$$

for the pseudocompact k-module obtained by quotienting I_A by the closed ideal generated by the products xy, with $x, y \in I_A$.

2.2

Let $\mathbf{H}$ be a group object in the category of coalgebras over $\mathbf{O}_k$. Thus, for every object C of $\mathrm{Alf}\,/k$, the object $\mathbf{H}(C)$ carries the structure of a C-coalgebra in groups (cf. VII$_\mathrm{A}$ 3.2). Following Manin, we shall say *bialgebra*[77] instead of "coalgebra in groups." Moreover, for every morphism

$$\phi : C \longrightarrow D$$

in $\mathrm{Alf}\,/k$, the map

$$D \otimes_C \mathbf{H}(C) \longrightarrow \mathbf{H}(D)$$

is a homomorphism of D-bialgebras.

Definition. These properties will be summarized by saying that $\mathbf{H}$ is a *bialgebra over* $\mathbf{O}_k$.

The functor

$$\mathbf{H} \longmapsto \mathrm{Spf}^*(\mathbf{H})$$

of 1.3.5 plainly commutes with finite products. It therefore sends a bialgebra over $\mathbf{O}_k$ to a k-group functor, that is, a covariant functor from $\mathrm{Alf}\,/k$ to the category of groups.

Indeed, for every k-algebra C of finite length, the elements of

$$\begin{aligned} \mathrm{Spf}^*(\mathbf{H})(C) &= \mathrm{Spf}^*(\mathbf{H}(C)) \\ &= \{x \in \mathbf{H}(C) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\} \end{aligned}$$

form a group under multiplication in the algebra $\mathbf{H}(C)$; cf. VII$_\mathrm{A}$ 3.2.2. Notice also that $\Delta(x) = x \otimes x$ implies

$$\varepsilon(x) = \varepsilon(x)^2.$$

Thus $\varepsilon(x) = 1$ whenever C is local and $x \neq 0$.[78]

[77] The term "bigebra" is also used (cf. [BAlg], III § 11.4). Recall (cf. VII$_\mathrm{A}$, 3.1, editors' note (26)) that all bialgebras considered here are assumed to be cocommutative and equipped with an antipode; they are therefore cocommutative Hopf algebras.

[78] Since $\Delta(x) = x \otimes x$, one has $x = \varepsilon(x)x$; hence $\varepsilon(x) = 0$ can occur only when $x = 0$.

2.2.1

A bialgebra $\mathbf{H}$ over $\mathbf{O}_k$ is said to be *flat* if its underlying module is flat (cf. 1.2.1).[79] If $\mathbf{H}$ is flat, then, by 1.3.5,

$$A = \Gamma^*(\mathbf{H})$$

is a topologically flat profinite k-algebra, and $\mathrm{Spf}^*(\mathbf{H})$, as a functor from

$$(\mathrm{Alf}\,/k)^{\mathrm{op}} = \mathrm{Vaf}\,/k$$

to the category of sets, is isomorphic to the functor

$$\mathrm{Spf}(A) : C \longmapsto \mathrm{Hom}_{\mathrm{Vaf}\,/k}(\mathrm{Spf}(C), \mathrm{Spf}(A)).$$

The group structure on $\mathrm{Spf}^*(\mathbf{H})$ therefore endows

$$\mathscr{G}(\mathbf{H}) = \mathrm{Spf}(A)$$

with the structure of a formal group, described explicitly as follows.

For every object C of $\mathrm{Alf}\,/k$, the C-module underlying $\mathbf{H}(C)$ is projective. Lemma 1.2.3.A therefore gives, by induction on n, natural isomorphisms

$$\begin{aligned}(\mathbf{H}(C)^*)^{\widehat{\otimes}(n+1)} &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^*)^{\widehat{\otimes} n}) \\ &\simeq \mathrm{Hom}_C(\mathbf{H}(C), (\mathbf{H}(C)^{\otimes n})^*) \\ &\simeq (\mathbf{H}(C)^{\otimes(n+1)})^*.\end{aligned}$$

For $n = 1, 2$, it follows that the C-algebra structure on $\mathbf{H}(C)$ endows $\mathbf{H}(C)^*$ with a diagonal map satisfying conditions 2.1(i)–(iii), functorially in C.

Consequently,

$$A = \Gamma^*(\mathbf{H}) = \varprojlim_{\mathfrak{l}} \mathbf{H}(k/\mathfrak{l})^*$$

is endowed with the structure of a cogroup in $\mathrm{Alp}\,/k$, which defines the announced formal-group structure on $\mathscr{G}(\mathbf{H})$.

Conversely, let G be a topologically flat formal k-group with affine algebra A, and write

$$\mathbf{H}(G) = \mathbf{V}_k^f(A)$$

for its $\mathbf{O}_k$-coalgebra (cf. 1.2.3). The diagonal morphism

$$\Delta_A : A \longrightarrow A \widehat{\otimes}_k A$$

then induces, for every finite-length k-algebra C, a C-linear map

$$\mathbf{V}_k^f(A)(C) \otimes_C \mathbf{V}_k^f(A)(C) \longrightarrow \mathbf{V}_k^f(A)(C),$$

which makes the coalgebra $\mathbf{V}_k^f(A)(C)$ into a C-bialgebra. We shall call $\mathbf{H}(G)$ the *covariant bialgebra* of the formal group G.[80] Thus, by Proposition 1.3.5.D, we obtain the following.

Proposition.

[79] The editors have expanded the original in what follows.

[80] The editors added the adjective "covariant" to make clear that the functor $G \mapsto \mathbf{H}(G)$ is covariant; this terminology is used in [Di73], I § 2.14.

(i) *The functors*

$$G \longmapsto \mathbf{H}(G), \qquad \mathbf{H} \longmapsto \mathscr{G}(\mathbf{H})$$

define an equivalence between the category of topologically flat formal k-groups and the category of flat $\mathbf{O}_k$-bialgebras.[81]

(ii) *This equivalence "commutes with base change": if $k \to K$ is a morphism of pseudo-compact rings, then*

$$\mathbf{H}(G\widehat{\otimes}_k K) = \mathbf{H}(G) \otimes_k K$$

and

$$\mathscr{G}(\mathbf{H} \otimes_k K) = \mathscr{G}(\mathbf{H})\widehat{\otimes}_k K.$$

When k is an Artinian ring and G is a topologically flat formal k-group, the functor $\mathbf{H}(G)$ is plainly determined by its value

$$\mathbf{H}(G) = \mathbf{H}(G)(k)$$

at k. We shall also call $\mathbf{H}(G)$ the *(covariant) bialgebra* of G.[82] Let $\mathcal{H}_k$ denote the category of k-Hopf algebras, and let

$$\mathcal{H}^{\text{cocom}}_{\text{flat}/k}$$

be its full subcategory formed by the k-Hopf algebras that are flat over k and cocommutative. We then obtain the following.

Corollary. *Let k be an Artinian ring.*

(i) *The functors*

$$G \longmapsto \mathbf{H}(G), \qquad \mathbf{H} \longmapsto \mathscr{G}(\mathbf{H}) = \operatorname{Spf}^*(\mathbf{H})$$

define an equivalence between the category of topologically flat formal k-groups and $\mathcal{H}^{\text{cocom}}_{\text{flat}/k}$.

(ii) *This equivalence "commutes with base change": if $k \to K$ is a morphism of Artinian rings, then*

$$\mathbf{H}(G\widehat{\otimes}_k K) = \mathbf{H}(G) \otimes_k K$$

and

$$\mathscr{G}(\mathbf{H} \otimes_k K) = \mathscr{G}(\mathbf{H})\widehat{\otimes}_k K.$$

On the other hand, let

$$\mathcal{H}^{\text{com}}_{\text{flat}/k}$$

denote the full subcategory of $\mathcal{H}_k$ formed by the k-Hopf algebras that are flat over k and commutative. Recall that the functor

$$T \longmapsto \mathscr{O}(T)$$

is an anti-equivalence from the category of affine k-group schemes onto $\mathcal{H}^{\text{com}}_{\text{flat}/k}$.

[81] Recall that in this Exposé, "bialgebra" means "cocommutative Hopf algebra."
[82] The editors have introduced here the notation $\mathcal{H}^{\text{cocom}}_{\text{flat}/k}$, which will be useful below.

2.2.2

Suppose, for simplicity, that k is Artinian, and let G be a topologically flat formal k-group.[83] Then G is commutative if and only if its affine algebra $\mathcal{A}(G)$ has a cocommutative comultiplication, which is equivalent to saying that the bialgebra $\mathbf{H}(G)$ has a commutative multiplication. In that case, $\mathbf{H}(G)$ is a commutative and cocommutative Hopf algebra, flat over k. Thus, if

$$D'(G) = \operatorname{Spec} \mathbf{H}(G),$$

then $D'(G)$ is a commutative k-group scheme, affine and flat over k.

Conversely, if T is such a k-group scheme, its affine algebra $\mathcal{O}(T)$ is a commutative group object in the category of cocommutative k-coalgebras flat over k. Hence 1.3.5.C gives a topologically flat formal k-group $D(T)$, defined by

$$D(T) = \operatorname{Spf}^* \mathcal{O}(T) = \operatorname{Spf}(\mathcal{A}), \qquad \mathcal{A} = \mathcal{O}(T)^*.$$

By 1.3.5.D, there are canonical isomorphisms

$$G = \operatorname{Spf}^* \mathbf{H}(G), \qquad \mathbf{H}(D(T)) = \mathcal{O}(T).$$

We therefore obtain canonical isomorphisms

$$\begin{aligned} D(D'(G)) &= \operatorname{Spf}^* \mathcal{O}(D'(G)) = \operatorname{Spf}^* \mathbf{H}(G) = G, \\ D'(D(T)) &= \operatorname{Spec} \mathbf{H}(D(T)) = \operatorname{Spec} \mathcal{O}(T) = T. \end{aligned}$$

Moreover, if k- Gr. denotes the category of k-group schemes, Corollary 2.2.1 gives functorial isomorphisms

$$\operatorname{Hom}_{\mathrm{Grf}\,/k}(G, D(T)) \simeq \operatorname{Hom}_{\mathcal{H}_k}(\mathbf{H}(G), \mathcal{O}(T)) \simeq \operatorname{Hom}_{k\text{-}\mathrm{Gr.}}(T, D'(G)).$$

This proves the following.

Proposition (Cartier duality). *Let k be an Artinian ring.*

(i) *The functors*

$$G \longmapsto D'(G) = \operatorname{Spec} \mathbf{H}(G), \qquad T \longmapsto D(T) = \operatorname{Spf}^* \mathcal{O}(T)$$

induce an anti-equivalence between the category $\mathcal{FC}_k$ of commutative topologically flat formal k-groups and the category $\mathcal{AC}_k$ of commutative affine flat k-group schemes. In other words, G and $D'(G)$, respectively T and $D(T)$, are related by the equalities [84]

$$\mathbf{H}(G) = \mathcal{O}(D'(G)), \qquad \mathcal{O}(T) = \mathbf{H}(D(T)).$$

(ii) *This anti-equivalence "commutes with base change": if $k \to K$ is a morphism of Artinian rings, then*

$$D'(G \widehat{\otimes}_k K) = D'(G) \otimes_k K$$

and

$$D(T \otimes_k K) = D(T) \widehat{\otimes}_k K.$$

(iii) *In particular, if k is a field, one obtains an anti-equivalence between the category of commutative formal k-groups and the category of commutative affine k-group schemes, and this anti-equivalence commutes with extension of the base field.*

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[83] The editors have expanded the original in what follows in order to introduce the notation $D'(G)$ and $D(K)$; cf. [Ca62], § 14.

[84] Let $\mathcal{FC}_k^f$, respectively $\mathcal{AC}_k^f$, denote the full subcategory of $\mathcal{FC}_k$, respectively $\mathcal{AC}_k$, consisting of the objects G, respectively T, for which $\mathbf{H}(G)$, respectively $\mathcal{O}(T)$, is a finite k-module (and hence finite locally free). Then $\mathcal{FC}_k^f$ and $\mathcal{AC}_k^f$ both have the same objects as the category $\mathcal{C}$ of commutative and cocommutative Hopf k-algebras that are finite and flat over k. The correspondence $G \mapsto \mathbf{H}(G)$, respectively $T \mapsto \mathcal{O}(T)$, is covariant, respectively contravariant. Thus one recovers the "Cartier duality" of the category $\mathcal{AC}_k$, already seen in VII$_{\mathrm{A}}$ 3.3.1.

2.3

Let G now be an arbitrary formal k-group,[85] with affine algebra A. We continue to write

$$\mathbf{H}(G) = \mathbf{V}_k^f(A)$$

for the $\mathbf{O}_k$-module dual to A, and denote by

$$\phi_G : \mathbf{H}(G) \otimes_k \mathbf{H}(G) \longrightarrow \mathbf{H}(G \times G)$$

the functorial homomorphism induced, for every object C of $\mathrm{Alf}\,/k$, by the natural map of 0.3.6:

$$\begin{aligned}&(A\widehat{\otimes}_k C)^\dagger \otimes_C (A\widehat{\otimes}_k C)^\dagger \\ &\qquad \longrightarrow \left((A\widehat{\otimes}_k C)\widehat{\otimes}_C (A\widehat{\otimes}_k C)\right)^\dagger .\end{aligned}$$

If

$$m : G \times G \longrightarrow G$$

is the multiplication of G, the composite

$$\mathbf{H}(G) \otimes_k \mathbf{H}(G) \xrightarrow{\phi_G} \mathbf{H}(G \times G) \xrightarrow{\mathbf{H}(m)} \mathbf{H}(G)$$

endows $\mathbf{H}(G)$ with an algebra structure over $\mathbf{O}_k$. For every $C \in \mathrm{Alf}\,/k$, the unit element of

$$\mathbf{H}(G)(C) = (A\widehat{\otimes}_k C)^\dagger$$

is the augmentation of $A\widehat{\otimes}_k C$; cf. 2.1. [86]

If G is not topologically flat over k, the morphism ϕ_G need not be an isomorphism. Consequently, the morphism

$$\delta_G : \mathbf{H}(G) \longrightarrow \mathbf{H}(G \times G)$$

induced by the diagonal $x \mapsto (x, x)$ of G need not factor through

$$\mathbf{H}(G) \otimes_k \mathbf{H}(G).$$

Thus $\mathbf{H}(G)$ need not be a bialgebra over $\mathbf{O}_k$. For this reason, in the general case we simply call $\mathbf{H}(G)$ the *covariant algebra* of the formal group G.

When G is topologically flat over k, of course, ϕ_G is an isomorphism, and one recovers the $\mathbf{O}_k$-bialgebra structure on $\mathbf{H}(G)$ defined in 2.2.1.

Proposition 2.3.1. *Let K and G be two formal k-groups, with affine algebras B and A, respectively. Suppose that K is topologically flat over k. Then there is a canonical bijection from*

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(K, G)$$

onto the set of homomorphisms of unital $\mathbf{O}_k$-algebras

$$h : \mathbf{H}(K) \longrightarrow \mathbf{H}(G)$$

for which the diagram

[85] That is, G is not necessarily topologically flat over k.

[86] The editors changed the order of the sentences that follow.

$$\begin{tikzcd}
\mathbf{H}(K) \otimes_k \mathbf{H}(K) \arrow[r, "h\otimes h"] & \mathbf{H}(G) \otimes_k \mathbf{H}(G) \arrow[dr, "\varphi_G"] & \\
 & & \mathbf{H}(G \times G) \\
\mathbf{H}(K) \arrow[uu, "\Delta_{\mathbf{H}(K)}"] \arrow[r, "h"] & \mathbf{H}(G) \arrow[ur, "\delta_G"'] &
\end{tikzcd}$$

is commutative.

Since K is topologically flat, $\mathbf{H}(K)$ is endowed with a bialgebra structure (cf. 2.2), and $\Delta_{\mathbf{H}(K)}$ is its diagonal morphism. In other words, with the notation of 2.3,

$$\Delta_{\mathbf{H}(K)} = \phi_K^{-1} \circ \delta_K.$$

When G too is topologically flat over k, the proposition follows from the equivalence of categories established in 2.2.1.

In the general case, we may suppose that k is Artinian and argue with the algebras

$$\mathbf{H}(K) = B^\dagger, \qquad \mathbf{H}(G) = A^\dagger.$$

Let $\mathrm{Hom}_c(A, B)$ denote the set of continuous k-linear maps from A to B, and let $\mathrm{Hom}_k(B^\dagger, A^\dagger)$ denote the set of k-linear maps from $B^\dagger$ to $A^\dagger$. [87]

By 0.3.6.A, if M and P are pseudocompact k-modules and P is projective, the canonical map

$$\mathrm{Hom}_c(M, P) \longrightarrow \mathrm{Hom}_k(P^\dagger, M^\dagger), \qquad f \longmapsto {}^t f,$$

where ${}^t f$ denotes the transpose of f, is bijective. We shall apply this either with

$$M = A \widehat{\otimes} A, \quad P = B,$$

or with

$$M = A, \quad P = B \widehat{\otimes} B.$$

Let $f \in \mathrm{Hom}_c(A, B)$. Consider the following diagrams. The squares marked (0) are commutative, and the two unnamed vertical arrows in the second diagram are ${}^t(f \widehat{\otimes} f)$.

$$\begin{tikzcd}
A \widehat{\otimes} A \arrow[r, "m_A"] \arrow[d, "f \widehat{\otimes} f"'] \arrow[dr, phantom, "(1)"] & A \arrow[r, "\Delta_A"] \arrow[d, "f"] \arrow[dr, phantom, "(2)"] & A \widehat{\otimes} A \arrow[d, "f \widehat{\otimes} f"] \\
B \widehat{\otimes} B \arrow[r, "m_B"] & B \arrow[r, "\Delta_B"] & B \widehat{\otimes} B
\end{tikzcd}$$

$$\begin{tikzcd}
A^\dagger \otimes A^\dagger \arrow[r, "\varphi_G"] & (A \widehat{\otimes} A)^\dagger & A^\dagger \arrow[l, "\delta_G = m_A^t"'] & (A \widehat{\otimes} A)^\dagger \arrow[l, "\Delta_A^t"'] & A^\dagger \otimes A^\dagger \arrow[l, "\varphi_G"'] \\
B^\dagger \otimes B^\dagger \arrow[u, "{}^t f \otimes {}^t f"] \arrow[r, equal] \arrow[ur, phantom, "(0)"] & (B \widehat{\otimes} B)^\dagger \arrow[u] \arrow[ur, phantom, "(1')"] & B^\dagger \arrow[l, "\delta_K = m_B^t"] \arrow[u, "{}^t f"'] \arrow[ur, phantom, "(2')"] & (B \widehat{\otimes} B)^\dagger \arrow[l, "\Delta_B^t"] \arrow[u] \arrow[r, equal] \arrow[ur, phantom, "(0)"] & B^\dagger \otimes B^\dagger \arrow[u, "{}^t f \otimes {}^t f"']
\end{tikzcd}$$

If $f : A \to B$ corresponds to a morphism of formal groups $K \to G$, then the squares (1) and (2) are commutative and

$$\varepsilon_B \circ f = \varepsilon_A.$$

[87] The editors supplied the details of the remainder of the proof.

Consequently, the squares (1′) and (2′) are commutative, and ${}^t f$ sends the unit of $B^\dagger = \mathbf{H}(K)$ to that of $A^\dagger = \mathbf{H}(G)$. Thus ${}^t f$ is a homomorphism of unital k-algebras

$$\mathbf{H}(K) \longrightarrow \mathbf{H}(G)$$

for which the diagram in the proposition is commutative.

Conversely, suppose that ${}^t f$ satisfies these conditions. Then

$$\varepsilon_B \circ f = \varepsilon_A$$

and the squares (1′) and (2′) are commutative. For

$$M = A\widehat{\otimes}A, \quad P = B \qquad \text{(respectively } M = A,\ P = B\widehat{\otimes}B\text{)},$$

the map $g \mapsto {}^t g$ is injective. Hence the squares (1) and (2) are commutative, so f is compatible with the multiplications and diagonal morphisms of A and B.

It remains to prove that $f(1_A) = 1_B$. What precedes gives

$$\varepsilon_B f(1) = 1, \qquad \Delta_B f(1) = f(1)\widehat{\otimes}f(1), \qquad f(1)\cdot f(1) = f(1).$$

By 2.1(iii), the first two conditions imply that $f(1)$ has inverse $c_B f(1)$ in B. The equality $f(1)\cdot f(1) = f(1)$ therefore implies $f(1) = 1$. Thus

$$f : A \longrightarrow B$$

is a morphism in $\mathrm{Alp}\,/k$, compatible with the comultiplications of A and B.

2.3.2

Let us now suppose, for simplicity, that the ring k is Artinian. When G is topologically flat over k, the algebra

$$\mathbf{H}(G) = \mathbf{H}(G)(k)$$

may be characterized by a universal property due to Cartier. Recall (cf. 1.2.1) that, if U is a k-module, we write $\mathbf{W}(U)$ for the functor that assigns to every finite-length k-algebra C the C-module

$$U \otimes_k C.$$

[88] If U is a k-algebra (associative and with a unit element), then so is $U \otimes_k C$. We shall write $\mathbf{W}(U)^\times$ for the k-group functor that assigns to every $C \in \mathrm{Ob}(\mathrm{Alf}\,/k)$ the multiplicative group of invertible elements of the algebra $U \otimes_k C$:

$$\mathbf{W}(U)^\times(C) = (U \otimes_k C)^\times.$$

Moreover, identify G with the k-group functor

$$C \longmapsto \mathrm{Hom}_{\mathrm{Vaf}\,/k}(\mathrm{Spf}(C), G),$$

and write

$$\mathrm{Hom}_{k\text{-}\mathrm{Gr.}}(G, \mathbf{W}(U)^\times)$$

for the set of homomorphisms of k-group functors from G to $\mathbf{W}(U)^\times$. We have the following.

[88] The editors added the preceding sentence and, in what follows, wrote $\mathbf{W}(U)^\times$ in place of U_m. Recall also (cf. 1.2.1) that a "k-functor" is a covariant functor from $\mathrm{Alf}\,/k$ to (**Sets**).

Proposition. *Let k be an Artinian ring. For every formal group G topologically flat over k, and every k-algebra U, there is a canonical isomorphism*

$$\mathrm{Hom}_{k\text{-}\mathrm{Gr.}}(G, \mathbf{W}(U)^\times) \xrightarrow{\sim} \mathrm{Hom}_{k\text{-}\mathrm{Alg.}}(\mathbf{H}(G), U).$$

Let A be the affine algebra of G. By hypothesis, A is a projective object of $\mathrm{PC}(k)$, and

$$\mathbf{H}(G) = A^\dagger.$$

For every object P of $\mathrm{PC}(k)$, write

$$U \widehat{\otimes}_k P$$

for the inverse limit of the k-modules

$$U \otimes_k (P/N),$$

where N runs through the open submodules of P. There are linear maps

$$U \otimes_k (P/N) \longrightarrow \mathrm{Hom}_k((P/N)^*, U)$$

that send $u \otimes x$ to the k-linear map

$$f \longmapsto f(x)u$$

and form a filtered inverse system. Passing to the inverse limit therefore gives a morphism

$$U \widehat{\otimes}_k P \xrightarrow{\psi_P} \mathrm{Hom}_k(P^\dagger, U). \tag{1}$$

When $P = k$, the morphism ψ_k is plainly an isomorphism. Moreover, the two sides of (1), considered as functors of P, commute with infinite products (every product being a filtered inverse limit of finite products). Thus (1) is an isomorphism when P is a product of copies of k, and then when P is a projective object of $\mathrm{PC}(k)$ (the two sides of (1) commute with finite direct sums).

Now let

$$\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$$

denote the set of morphisms of k-functors from G to $\mathbf{W}(U)$. Since

$$G = \mathrm{Spf}(A) = \varinjlim_{\mathfrak{l}} \mathrm{Spf}(A/\mathfrak{l}),$$

where $\mathfrak{l}$ runs through the open ideals of A, there are canonical isomorphisms

$$\begin{aligned} \mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) &= \varprojlim_{\mathfrak{l}} \mathrm{Hom}_{\mathcal{F}}(\mathrm{Spf}(A/\mathfrak{l}), \mathbf{W}(U)) \\ &= \varprojlim_{\mathfrak{l}} U \otimes_k (A/\mathfrak{l}) = U \widehat{\otimes}_k A. \end{aligned}$$

The preceding argument therefore gives a canonical isomorphism

$$\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)) = U \widehat{\otimes}_k A \xrightarrow{\psi_A} \mathrm{Hom}_k(\mathbf{H}(G), U). \tag{2}$$

For every finite-length k-algebra C, multiplication makes $U \otimes_k C$ a monoid with a unit element, and every homomorphism of monoids with unit

$$G(C) \longrightarrow U \otimes_k C$$

is necessarily a group homomorphism

$$G(C) \longrightarrow (U \otimes_k C)^\times.$$

It follows that

$$\mathrm{Hom}_{k\text{-}\mathrm{Gr.}}(G, \mathbf{W}(U)^\times)$$

is the part of $\mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U))$ consisting of morphisms of k-functors into monoids with a unit element.

It remains to show that these morphisms correspond to the k-linear maps from $\mathbf{H}(G)$ to U that preserve multiplication and unit elements. [89] To simplify the notation, write

$$H = \mathbf{H}(G) = A^\dagger$$

and write $\widehat{\otimes}$ instead of $\widehat{\otimes}_k$. Denote by

$$\Delta_A,\ m_A,\ \varepsilon_A \qquad \text{and} \qquad \Delta_H,\ m_H,\ \varepsilon_H$$

the comultiplication, multiplication, and augmentation of A and H, respectively. Let

$$\theta \in \mathrm{Hom}_{\mathcal{F}}(G, \mathbf{W}(U)),$$

let γ be its image in $U\widehat{\otimes}A$, and let

$$\phi : H \longrightarrow U$$

be the associated k-linear map. The morphism θ sends the unit section $s \in G(k)$ to an element u of U. Since s corresponds to the augmentation

$$\varepsilon : A \longrightarrow k,$$

which is the unit element 1_H of H, we have

$$\theta(s) = 1_U \quad \Longleftrightarrow \quad \phi(1_H) = 1_U.$$

The morphism

$$\theta \circ m_G : G \times G \longrightarrow \mathbf{W}(U)$$

corresponds to the element

$$(\mathrm{id}_U \widehat{\otimes} \Delta_A)(\gamma),$$

which, by duality, corresponds to the map

$$\phi \circ m_H : H \otimes H \longrightarrow U.$$

On the other hand, the morphism

$$\theta \circ \mathrm{pr}_1 : G \times G \longrightarrow \mathbf{W}(U)$$

corresponds to the element

$$\gamma \widehat{\otimes} 1_A \in U \widehat{\otimes} A \widehat{\otimes} A,$$

which, by duality, corresponds to

$$\phi \circ (\mathrm{id}_H \otimes \varepsilon_H) : H \otimes H \longrightarrow U.$$

[89] The original stated that this "follows from the functoriality of ψ_A." The editors gave the details that follow.

Similarly, $\theta \circ \mathrm{pr}_2$ corresponds to the element

$$\tau(\gamma \widehat{\otimes} 1_A) \in U \widehat{\otimes} A \widehat{\otimes} A,$$

where

$$\tau(u \widehat{\otimes} a \widehat{\otimes} b) = u \widehat{\otimes} b \widehat{\otimes} a,$$

and this element corresponds by duality to

$$\phi \circ (\varepsilon_H \otimes \mathrm{id}_H) : H \otimes H \longrightarrow U.$$

Finally, the multiplication map

$$\mu = m_{U \widehat{\otimes} A \widehat{\otimes} A}$$

given by

$$\begin{aligned} (U \widehat{\otimes} A \widehat{\otimes} A) \times (U \widehat{\otimes} A \widehat{\otimes} A) &\longrightarrow U \widehat{\otimes} A \widehat{\otimes} A, \\ (u \widehat{\otimes} a_1 \widehat{\otimes} a_2,\, u' \widehat{\otimes} a_3 \widehat{\otimes} a_4) &\longmapsto uu' \widehat{\otimes} a_1 a_3 \widehat{\otimes} a_2 a_4 \end{aligned}$$

may be viewed as the composite of the endomorphism σ_{23} of

$$(U \otimes U) \widehat{\otimes} A^{\widehat{\otimes} 4},$$

which "exchanges the factors a_2 and a_3," and the map

$$(U \otimes U) \widehat{\otimes} A^{\widehat{\otimes} 2} \widehat{\otimes} A^{\widehat{\otimes} 2} \xrightarrow{m_U \widehat{\otimes} m_A \widehat{\otimes} m_A} U \widehat{\otimes} A \widehat{\otimes} A.$$

It follows that the map

$$\mu \circ (\phi \circ \mathrm{pr}_1,\, \phi \circ \mathrm{pr}_2) : G \times G \longrightarrow \mathbf{W}(U)$$

corresponds to the composite map β below:

$$\begin{tikzcd}
 & H \otimes H \arrow[rr, "\beta"] \arrow[dl, "\mathrm{id}_H \otimes \sigma_{23} \otimes \mathrm{id}_H"'] & & U \\
H \otimes H \otimes H \otimes H \arrow[dr, "(\mathrm{id}_H \otimes \varepsilon_H) \otimes (\varepsilon_H \otimes \mathrm{id}_H)"'] & & & U \otimes U \arrow[u, "m_U"'] \\
 & H \otimes H \arrow[urr, "\phi \otimes \phi"'] & & .
\end{tikzcd}$$

Finally, θ is compatible with the laws of G and $\mathbf{W}(U)^\times$ if and only if

$$\theta \circ m_G = \mu \circ (\phi \circ \mathrm{pr}_1,\, \phi \circ \mathrm{pr}_2),$$

which, by the preceding argument, is equivalent to

$$\phi \circ m_H = \beta.$$

But plainly

$$(\phi \circ m_H)(x \otimes y) = \phi(xy),$$

whereas

$$\beta(x \otimes y) = \phi(x)\phi(y).$$

2.4

We now return to an arbitrary pseudocompact ring k, in order to apply to formal groups the results of 1.4–1.5 on passage to the quotient by a topologically flat equivalence relation.[90]

Let $u : H \to G$ be a monomorphism of formal k-groups, let $\mu : G \times G \to G$ be the multiplication morphism of G, and let λ be the composite

$$\lambda : \quad G \times H \xrightarrow{\mathrm{id}_G \times u} G \times G \xrightarrow{\mu} G.$$

Since u is a monomorphism, the pair

$$G \times H \underset{\lambda}{\overset{\mathrm{pr}_1}{\rightrightarrows}} G$$

is an equivalence pair in Vaf $/k$ (cf. V, 2.b). Recall (cf. 1.2.C) that the cokernel G/H of this pair is defined as follows.

Let $\mathcal{O}(G)$ and $\mathcal{O}(H)$ be the affine algebras of G and H, let

$$\Delta : \mathcal{O}(G) \longrightarrow \mathcal{O}(G)\widehat{\otimes}_k\mathcal{O}(G)$$

be the diagonal morphism of $\mathcal{O}(G)$, and let I be the kernel of

$$u^\natural : \mathcal{O}(G) \longrightarrow \mathcal{O}(H).$$

By Proposition 1.3, the morphism $u^\natural$ induces an isomorphism

$$\mathcal{O}(G)/I \xrightarrow{\sim} \mathcal{O}(H).$$

The affine algebra $\mathcal{O}(G/H)$ of G/H is then the kernel of the pair

$$\mathcal{O}(G) \underset{(\mathrm{id}\,\widehat{\otimes} u^\natural)\Delta}{\overset{\tau_1}{\rightrightarrows}} \mathcal{O}(G)\widehat{\otimes}_k\mathcal{O}(H),$$

where $\tau_1(x) = x\widehat{\otimes}1$. Thus

$$\mathcal{O}(G/H) = \{x \in \mathcal{O}(G) \mid \Delta(x) - x\widehat{\otimes}1 \in \mathcal{O}(G)\widehat{\otimes}I\}.$$

If, moreover, H is topologically flat over k, then pr_1 is topologically flat, and Theorem 1.4 gives the following result.

Theorem. *Let $u : H \to G$ be a monomorphism of formal k-groups. Suppose that H is topologically flat over k. Then the projection*

$$p : G \longrightarrow G/H$$

is surjective and topologically flat, there is an isomorphism

$$G \times H \xrightarrow{\sim} G \times_{G/H} G, \tag{*}$$

and G/H represents the quotient sheaf for the flat topology.

Consequently, G/H has a canonical structure of an object on which G acts, and $p : G \to G/H$ is an equivariant morphism. If, in addition, u identifies H with a normal subgroup of G, then G/H has a canonical structure of formal k-group for which $p : G \to G/H$ is a morphism of formal k-groups, and H is the kernel of p.

[90]The editors supplied the details of the original argument that follow.

Indeed, the first assertion follows from 1.4, and the other two from Exposé IV, Corollaries 5.2.2 and 5.2.4.

Corollary. [91] *Let G be a formal k-group and H a formal subgroup of G. Let A, A/J, and B be the affine algebras of G, H, and G/H, respectively; let I_A be the augmentation ideal of A, and put $I_B = B \cap I_A$. Suppose that H is topologically flat over k. Then*

$$J = AI_B,$$

the closed ideal generated by I_B.

Indeed, the projection $B \to B/I_B$ corresponds to the unit section

$$e : \mathrm{Spf}(k) \longrightarrow G/H$$

of G/H. By (*), H is identified with the fiber product

$$\mathrm{Spf}(k) \times_{G/H} G.$$

Its affine algebra A/J is therefore identified with

$$(B/I_B)\widehat{\otimes}_B A \simeq A/AI_B.$$

2.4.A

Let G and Q be formal k-groups that are topologically flat. Suppose that there are homomorphisms

$$\sigma : Q \longrightarrow G \qquad \text{and} \qquad \pi : G \longrightarrow Q$$

such that

$$\pi \circ \sigma = \mathrm{id}_Q\,.$$

[92] In particular, σ is a monomorphism, so Q is a formal subgroup of G (cf. Remark 1.3). Let

$$N = \mathrm{Ker}(\pi)$$

and let σ' be the inclusion $N \hookrightarrow G$. Then G is the semidirect product of N by Q (cf. I, 2.3.8). In other words, for every

$$B \in \mathrm{Ob}(\mathrm{Alf}\,/k),$$

identify $N(B)$ and $Q(B)$ with their images in $G(B)$ under σ' and σ. Then $N(B)$ is a normal subgroup of $G(B)$, and the map

$$\mu : N(B) \times Q(B) \longrightarrow G(B), \qquad (x, q) \longmapsto xq \tag{1}$$

is bijective.

The morphism of formal k-varieties

$$\theta : G(B) \longrightarrow N(B), \qquad g \longmapsto g \cdot \sigma\pi(g^{-1}) \tag{2}$$

is then a retraction of σ', and the inverse of μ is

$$g \longmapsto (\theta(g), \pi(g)). \tag{3}$$

[91] The editors made this corollary explicit; it is used, for example, in 5.2.1 and 5.2.3.
[92] The editors added paragraphs 2.4.A and 2.4.B, which will be useful in 5.1.3 and 2.9, respectively.

If $p : G \to G/Q$ denotes the quotient projection, then

$$p \circ \sigma' : N \longrightarrow G/Q$$

is an isomorphism of formal k-varieties.

In particular, N is topologically flat over k, by 1.4 (ii). Let

$$\alpha : Q \times N \longrightarrow N$$

and

$$\beta : G \times N = N \times Q \times N \longrightarrow N$$

be the maps defined on points by

$$\alpha(q, y) = qyq^{-1}, \qquad \beta(x, q, y) = x\alpha(q, y).$$

Then the following diagram is commutative:

$$\text{(4)} \qquad \begin{tikzcd} G \times G \arrow[r, "\mathrm{id} \times \theta"] \arrow[d, "m_G"'] & G \times N \arrow[r, "\theta \times \mathrm{id}"] \arrow[d, "\beta"] & N \times N \arrow[dl, "m_N"] \\ G \arrow[r, "\theta"] & N & \end{tikzcd} .$$

This can be expressed as follows in terms of the affine algebras A, A_0, and A' of G, Q, and N (cf. 5.1.3 below). Let

$$\rho' : A \longrightarrow A', \qquad \rho : A \longrightarrow A_0, \qquad \tau : A_0 \longrightarrow A$$

be the homomorphisms of k-bialgebras corresponding to σ', σ, and π, and put

$$I = \mathrm{Ker}(\rho).$$

By the preceding corollary, A' is identified with

$$A/\overline{A\,\tau(J_0)},$$

where J_0 is the augmentation ideal of A_0.

On the other hand, let B be the affine algebra of G/Q. It is the kernel of the pair of morphisms

$$A \underset{(\mathrm{id}_A \widehat{\otimes} \rho) \circ \Delta_A}{\overset{\tau_1}{\rightrightarrows}} A \widehat{\otimes}_k A_0,$$

where $\tau_1(x) = x \widehat{\otimes} 1$. Thus

$$B = \{x \in A \mid \Delta_A(x) - x \widehat{\otimes} 1 \in A \widehat{\otimes} I\}.$$

Let

$$\gamma = \theta^\natural : A' \longrightarrow A$$

be the corresponding continuous homomorphism of k-algebras. It is a section of ρ', and it is an isomorphism of profinite k-algebras from A' onto B. Moreover,

$$\tau = \pi^\natural$$

identifies A_0 with a sub-bialgebra of A, namely the affine algebra of the quotient $N \backslash G$.

It follows from equations (1) and (3) that there is an isomorphism of profinite k-algebras

$$A'\widehat{\otimes}_k A_0 \xrightarrow{\sim} A, \qquad a'\widehat{\otimes} a_0 \longmapsto \gamma(a')\tau(a_0), \tag{*}$$

whose inverse is the map

$$a \longmapsto (\rho'\widehat{\otimes}\rho)\Delta_A(a).$$

Finally, identify A' with its image in A under γ, so that the projection $A \to A'$ is then $\gamma\rho'$. Writing $\Delta_{A'}$ for the comultiplication of A', one deduces from (4) that

$$\Delta_A(A') \subset A\widehat{\otimes}A'$$

and that the following diagram is commutative:

$$\begin{tikzcd} A' \arrow[r, "\Delta_A"] \arrow[d, equal] & A\widehat{\otimes}A' \arrow[d, "\gamma\rho'\widehat{\otimes}\,\mathrm{id}"] \\ A' \arrow[r, "\Delta_{A'}"] & A'\widehat{\otimes}A' \end{tikzcd}$$

Consequently, one also has

$$\gamma\rho' \circ c_A = c_{A'},$$

where c_A and $c_{A'}$ are the antipodes of A and A', respectively. On the other hand, writing m_A for multiplication in A, equation (2) gives, for every $a \in A$,

$$\gamma\rho'(a) = (m_A \circ (\mathrm{id}_A \widehat{\otimes} \tau\rho c_A) \circ \Delta_A)(a).$$

2.4.B

Suppose, for simplicity, that k is Artinian. The preceding discussion can then be expressed more simply in terms of the covariant bialgebras of G, Q, and N. Indeed, since

$$G = N \times Q$$

as formal k-varieties, one has

$$\mathbf{H}(G) = \mathbf{H}(N) \otimes_k \mathbf{H}(Q)$$

as k-coalgebras. Moreover, since multiplication in G is given by

$$(x, q)\cdot(x', q') = (x\alpha(q, x'), qq'), \qquad \alpha(q, x') = qx'q^{-1},$$

multiplication in $\mathbf{H}(G)$ is given, for $x \in \mathbf{H}(N)$ and $q \in \mathbf{H}(Q)$, by

$$(x \otimes q)\cdot(x' \otimes q') = x\phi(q, x') \otimes qq',$$

where

$$\phi : \mathbf{H}(Q) \otimes_k \mathbf{H}(N) \longrightarrow \mathbf{H}(N)$$

is the morphism of k-coalgebras induced by α.

The morphism α is the following composite. Here δ_G and m_G denote, respectively, the diagonal and multiplication morphisms of G, c_Q is inversion in Q, and

$$v(q \otimes q' \otimes x) = q \otimes x \otimes q'.$$

$$Q \times N \xrightarrow{v\circ(\delta_G \times \mathrm{id})} Q \times N \times Q \xrightarrow{\mathrm{id}\times\mathrm{id}\times c_Q} Q \times N \times Q \xrightarrow{m_G} G,$$

Writing c_Q also for the antipode of $\mathbf{H}(Q)$, we obtain

$$\phi(q \otimes x') = \sum_i q_i x' c_Q(q'_i) \quad \text{if} \quad \Delta_{\mathbf{H}(Q)}(q) = \sum_i q_i \otimes q'_i. \tag{$\star$}$$

In particular, let M be an abstract group and suppose that $\mathbf{H}(Q)$ is the group algebra kM, so that

$$Q = \mathrm{Spf}^*\, kM$$

is the constant formal k-group M_k. Then for every $\gamma \in M$ and $x' \in \mathbf{H}(N)$,

$$\phi(\gamma \otimes x') = \gamma x' \gamma^{-1}.$$

This defines an action of M on $\mathbf{H}(N)$ by Hopf-algebra automorphisms.

Proposition 2.4.1. *Let $f : G \to K$ be a morphism of formal k-groups. If*

$$H = \ker(f)$$

is topologically flat over k, the homomorphism

$$f' : G/H \longrightarrow K$$

induced by f is a monomorphism.

This is a consequence of the results of Exposé IV,[93] but we give a direct proof. Let T be a formal variety of finite length over k, and let $t \in (G/H)(T)$ be such that $f' \circ t$ is the identity element of $K(T)$. We must show that t is the identity element of $(G/H)(T)$. Let

$$p : G \longrightarrow G/H$$

be the projection, and let

$$X = T \times_{G/H} G.$$

By 2.4, p is surjective and topologically flat. The same is therefore true of

$$\mathrm{pr}_1 : X \longrightarrow T.$$

By Proposition 1.3.1, pr_1 is consequently an epimorphism, so it is enough to show that $t \circ \mathrm{pr}_1$ is the identity element of $(G/H)(X)$.

Let $\mathrm{pr}_2 : X \to G$ be the second projection. We have

$$t \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2,$$

and hence

$$1 = f' \circ t \circ \mathrm{pr}_1 = f' \circ p \circ \mathrm{pr}_2 = f \circ \mathrm{pr}_2\,.$$

The exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{f} K$$

[93] See Exposé IV, 5.2.6.

therefore shows that pr_2 factors through H. Thus $p \circ \mathrm{pr}_2$ is the zero morphism. Since

$$p \circ \mathrm{pr}_2 = t \circ \mathrm{pr}_1,$$

this proves the proposition.

The following corollary follows. Write $\mathcal{O}(G)$, $\mathcal{O}(K)$, and $\mathcal{O}(G/H)$ for the affine algebras of G, K, and G/H, respectively. We saw in 2.4 that p induces an injection

$$\mathcal{O}(G/H) \hookrightarrow \mathcal{O}(G).$$

Moreover, since $f' : G/H \to K$ is a monomorphism, Proposition 1.3 shows that

$$\mathcal{O}(K) \longrightarrow \mathcal{O}(G/H)$$

is surjective. Hence:

Corollary. *Let $f : G \to K$ be a morphism of formal k-groups and let $H = \ker(f)$. If H is topologically flat over k, then*

$$\mathcal{O}(G/H)$$

is the image of $\mathcal{O}(K)$ in $\mathcal{O}(G)$.

2.4.2

Retain the preceding notation and suppose that H and G are topologically flat over k. Then G is topologically flat both over k and over G/H; hence, by 1.3.3, G/H is topologically flat over k. Consequently, by 2.4, the canonical projection

$$q : K \longrightarrow \mathrm{Coker}(f')$$

is topologically flat, and f' is an isomorphism from G/H onto $\ker(q)$. It is also clear that

$$\mathrm{Coker}(f) = \mathrm{Coker}(f').$$

Thus, under the hypothesis that G and $H = \ker(f)$ are topologically flat over k, we have obtained an isomorphism between $\ker(q)$, the image of f, and

$$G/\ker(f),$$

the coimage of f. This proves the following theorem.

Theorem. *Let k be a field. The commutative formal k-groups form an abelian category.*

Corollary. *Let k be a field. The commutative affine k-group schemes form an abelian category.* [94]

This follows from the theorem and the equivalence of categories in 2.2.2.

2.5

A formal k-group is said to be *étale* if its underlying formal variety is étale (cf. 1.6). Such formal groups have a very simple structure. Suppose that k is local, let κ be its residue field, let κ_s be a separable closure of κ, and let

$$\Gamma = \mathrm{Gal}(\kappa_s/\kappa)$$

[94] See also the remarks following Exposé VI$_A$, 5.4.3.

be the topological Galois group. A Γ-*set* means a set E with a continuous action of Γ; equivalently, the stabilizer of every element $x \in E$ is an open subgroup of Γ.

For every formal k-variety X, put

$$X(\kappa_s) = \varinjlim_{\ell} X(\ell),$$

where ℓ ranges over the finite extensions of κ contained in κ_s.[95] The group Γ acts continuously on every $X(\ell)$, and hence on $X(\kappa_s)$. Moreover, if

$$X_\kappa = X \widehat{\otimes}_k \kappa \qquad \text{(cf. 1.6.C)},$$

then $X(\ell) = X_\kappa(\ell)$ for every ℓ, and therefore

$$X(\kappa_s) = X_\kappa(\kappa_s).$$

Now suppose that X is étale over k. Then X_κ is the formal κ-variety that is the disjoint union of the $\operatorname{Spec} \kappa(x)$, for $x \in X$. If X'_κ denotes the κ-scheme given by the same disjoint union, then

$$\begin{aligned} X_\kappa(\kappa_s) &= X'_\kappa(\kappa_s) \\ &= \operatorname{Hom}_{\mathrm{Sch}/\kappa}(\operatorname{Spec} \kappa_s, X'_\kappa). \end{aligned}$$

Let $\mathcal{C}$ be the full subcategory

$$\mathrm{Sch}^{\text{ét}}/\kappa \subset \mathrm{Sch}/\kappa$$

of étale κ-schemes. It is known that

$$X' \longmapsto X'(\kappa_s)$$

is an equivalence from $\mathcal{C}$ onto the category $\mathcal{C}'$ of Γ-sets (cf. SGA 1, V, ğğ7–8, or [*DG*70], ğI.4, 6.4). It therefore induces an equivalence between the category of group objects in $\mathcal{C}$ and the category of group objects in $\mathcal{C}'$. A group object in $\mathcal{C}'$ is simply an abstract group endowed with a continuous action of Γ by group automorphisms; we call this a Γ-*group*.

Taking into account the equivalences of 1.6.E,

$$\mathrm{Vaf}^{\text{ét}}/k \xrightarrow{\sim} \mathrm{Vaf}^{\text{ét}}/\kappa \xrightarrow{\sim} \mathrm{Sch}^{\text{ét}}/\kappa,$$

we obtain the following.

Proposition. *Let k be a local pseudocompact ring, κ its residue field, κ_s a separable closure of κ, and $\Gamma = \mathrm{Gal}(\kappa_s/\kappa)$.*

(i) *The functor $X \mapsto X(\kappa_s)$ is an equivalence from the category of étale formal k-varieties onto the category of Γ-sets.*

(ii) *It induces an equivalence from the category of étale formal k-groups onto the category of Γ-groups.*

Remark 2.5.A. [96] Let k be a field, let k_s be a separable closure of k, let G be an étale formal k-group, and put

$$M = G(k_s).$$

[95] If $[\kappa_s : \kappa] = \infty$, then κ_s is not a profinite κ-algebra, so the preceding notation is an abuse. The editors also supplied the details that follow.

[96] The editors added this remark.

Choose a set X of representatives for the orbits of $\Gamma = \mathrm{Gal}(k_s/k)$ in M. For each $x \in X$, let Γ_x be its stabilizer, an open subgroup of finite index in Γ, and put

$$L_x = k_s^{\Gamma_x}.$$

Then L_x/k has degree $[\Gamma : \Gamma_x]$ (see, for example, [$BAlg$], V, ğ10.10).

By the preceding equivalence of categories, the affine algebra $\mathcal{A}(G)$ of G is the product of the L_x, endowed with the product topology. The L_x are therefore exactly the simple quotients

$$\mathcal{A}(G)/\mathfrak{m},$$

where $\mathfrak{m}$ ranges over the open maximal ideals of $\mathcal{A}(G)$. Since these quotients correspond by duality to the simple subcoalgebras of $\mathbf{H}(G)$, the coalgebra $\mathbf{H}(G)$ is pointed—that is,

$$\dim_k C = 1$$

for every simple subcoalgebra C—if and only if $L_x = k$ for every x. In that case, $\mathcal{A}(G)$ is the topological algebra k^M, so $\mathbf{H}(G)$ is the group algebra kM, and

$$M = \{x \in \mathbf{H}(G) \mid \varepsilon(x) = 1,\ \Delta(x) = x \otimes x\} = G(k).$$

2.5.1

Suppose again that k is an arbitrary pseudocompact ring. Since the functor

$$X \longmapsto X_e$$

of 1.6.F commutes with finite products, it sends every formal group G to an étale formal group G_e. The morphism $p : X \to X_e$ of that result is functorial in X, so in this case

$$p : G \longrightarrow G_e$$

is a morphism of formal groups.

Consider its kernel $\ker(p)$.[97] It is the fiber product of the diagram

$$\begin{array}{ccc} & & G \\ & & \big\downarrow{\scriptstyle p} \\ \mathrm{Spf}(k) & \xrightarrow{\ \varepsilon\ } & G_e \end{array}$$

where ε is the identity section of G_e. Since p induces a bijection on the underlying sets, it follows (cf. 1.2, editorial note 41) that the underlying set of $\ker(p)$ is the image of $\mathrm{Spf}(k)$ under ε, and that, for every point s of $\mathrm{Spf}(k)$, the local algebra of $\ker(p)$ at $\varepsilon(s)$ is

$$\mathcal{O}_{G,\varepsilon(s)} \widehat{\otimes}_{\mathcal{O}_{G_e,\varepsilon(s)}} \kappa(s).$$

At each point $\varepsilon(s)$, the residue field of $\mathcal{O}_{G,\varepsilon(s)}$ is $\kappa(s)$, and hence

$$\mathcal{O}_{G_e,\varepsilon(s)} = \kappa(s), \qquad \mathcal{O}_{\ker(p),\varepsilon(s)} = \mathcal{O}_{G,\varepsilon(s)}.$$

For these reasons, we call $\ker(p)$ the *infinitesimal neighborhood of the identity* of G, and write

$$\ker(p) = G_0.$$

We have thus obtained the exact sequence

[97] The editors supplied the details of the original argument that follow.

$$1 \longrightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e.$$

In what follows, we say that G is *infinitesimal* if

$$G = G_0.$$

[98] This is equivalent to saying that, for every $s \in \mathrm{Spf}(k)$, or equivalently for every continuous morphism $k \to \kappa$, where κ is a field with the discrete topology, the identity is the unique element of $G(\kappa(s))$, respectively of $G(\kappa)$.

Suppose in addition that G is topologically flat over k.[99] By 1.6.1, the morphism

$$p : G \longrightarrow G_e$$

is topologically flat. Since it is bijective, Proposition 1.3.1 shows that it is an effective epimorphism. Consequently, G_e is identified with the quotient G/G_0. We have proved:

Corollary. *Let G be a formal group topologically flat over k. Then G_e is identified with the quotient G/G_0; in other words, there is an exact sequence of formal groups*

$$1 \longrightarrow G_0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e \longrightarrow 1.$$

2.5.2

Suppose that k is a perfect field. In this case, we saw in 1.6.2 that the morphism

$$p : X \longmapsto X_e$$

has a section

$$s : X_e \longrightarrow X$$

that depends functorially on X. Consequently, when X is a formal group, this section is a morphism of formal groups. We thus obtain:

Proposition. *When k is a perfect field, every formal k-group G is canonically isomorphic to the semidirect product of an infinitesimal group G_0 and an étale group G_e acting on G_0.*[100]

If, moreover, G is commutative, then G is the product of G_0 and G_e. By 2.2.2, this canonical decomposition of commutative formal k-groups corresponds to an analogous decomposition of affine commutative k-group schemes; see 2.5.3 below.

Remark 2.5.2.A. [101] When k is not perfect, the exact sequence

$$1 \longrightarrow G_0 \longrightarrow G \longrightarrow G_e \longrightarrow 1$$

need not split. Consider the following example, taken from [*DG*70], ğIII.6, 8.6. Let k be a nonperfect field of characteristic $p > 0$, choose

$$\lambda \in k \setminus k^p,$$

[98] The editors added the sentence that follows.

[99] The editors supplied the details that follow.

[100] The same argument also shows that, for an arbitrary field k, if all the residue fields of G are equal to k, then G_e is the constant k-group $(M)_k$, where

$$M = G(k) = \mathrm{Spf}^* \mathbf{H}(G),$$

and $\mathbf{H}(G)$ is the semidirect product of $\mathbf{H}(G_0)$ by the group algebra kM; see the addition in 2.9 below.

[101] The editors added this remark.

and let L_λ be the abelian restricted p-Lie algebra with basis (x, y), whose symbolic p-th power (cf. VII$_A$, 5.2) is given by

$$x^{(p)} = x, \qquad y^{(p)} = \lambda x.$$

Let

$$U_p(L_\lambda) = k[x, y]/(x^p - x,\ y^p - \lambda x)$$

be the restricted enveloping algebra of L_λ (cf. VII$_A$, 5.3), and let G_λ be the commutative formal k-group whose affine algebra is $U_p(L_\lambda)$.

Then G_λ is a nonsplit extension of the constant étale k-group $(\mathbb{Z}/p\mathbb{Z})_k$ by the infinitesimal k-group

$$\alpha_{p,k} = \operatorname{Spf} k[x]/(x^p).$$

Thus there is a nonsplit exact sequence of commutative formal k-groups:[102]

$$0 \longrightarrow \boldsymbol{\alpha}_{p,k} \longrightarrow G_\lambda \longrightarrow (\mathbb{Z}/p\mathbb{Z})_k \longrightarrow 0.$$

By duality, this corresponds to a nonsplit exact sequence of commutative algebraic k-groups:

$$0 \longrightarrow \boldsymbol{\mu}_{p,k} \longrightarrow D'(G_\lambda) \longrightarrow \boldsymbol{\alpha}_{p,k} \longrightarrow 0$$

where

$$\mu_{p,k} = \operatorname{Spec} k[t]/(t^p - 1).$$

In this way one obtains all extensions of $\alpha_{p,k}$ by $\mu_{p,k}$; see [*DG*70], ğIII.6, 8.6.

2.5.3

Let k be a field.[103]

Definition 2.5.3.A. A commutative k-group scheme is said to be *unipotent* if it is isomorphic to

$$\operatorname{Spec} \mathbf{H}(G),$$

where G is an infinitesimal commutative formal k-group.[104]

On the other hand, following Exposé IX, Definition 1.1,[105] a k-group scheme H is said to be *of multiplicative type* if there exists a scheme S, faithfully flat and quasi-compact over $\operatorname{Spec} k$, such that H_S is a diagonalizable S-group; that is,

$$H_S \simeq D_S(M)$$

for some "abstract" abelian group M. By fpqc descent, this implies that H is affine and commutative. Moreover, since S may be replaced by the residue field of one of its points, H is of multiplicative type if and only if there exists an extension K of k such that H_K is a diagonalizable K-group.

Proposition 2.5.3.B. *Let T be an affine commutative k-group scheme, and let k_s be a separable closure of k.*

[102] The groups $\alpha_{p,k}$, $(\mathbb{Z}/p\mathbb{Z})_k$, and G_λ are also finite flat algebraic k-groups; see editorial note 84 in 2.2.2.

[103] The editors introduced the numbering 2.5.3 and 2.5.3.A–2.5.3.C.

[104] This agrees with the "usual" notion of a unipotent k-group scheme; cf. the addition in 2.8 at the end of Section 2.

[105] The original stated: "Let us say that a commutative k-group scheme is of multiplicative type if it is isomorphic to $\operatorname{Spec} \mathbf{H}(G)$, where G is an étale commutative formal k-group." The editors have given here the "usual" definition, taken from Exposé IX, 1.1, and proved its equivalence with the preceding condition; see also [*DG*70], ğIV.1, Theorem 2.2.

(i) *The scheme T is of multiplicative type if and only if its dual $D(T)$ is an étale commutative formal k-group.*

(ii) *Consequently, T is of multiplicative type if and only if*

$$T \otimes_k k_s$$

is diagonalizable.

Proof. Let $\mathcal{A}$ denote the affine algebra of $D(T)$, and let $\mathcal{A}_0$ be its local component at the identity. Then $D(T)$ is étale over k if and only if

$$\mathcal{A}_0 = k.$$

If K is an extension of k, then $\mathcal{A}_0 \widehat{\otimes}_k K$ is local, being an inverse limit of Artinian local rings, and hence coincides with the local component

$$\left(\mathcal{A}\widehat{\otimes}_k K\right)_0.$$

Moreover, since the formation of $D(T)$ commutes with base change (cf. 2.2.2), one also has

$$\mathcal{A}\widehat{\otimes}_k K \simeq \mathcal{A}(D(T_K)).$$

Suppose that T is of multiplicative type. Then there exists an extension K of k such that

$$\mathcal{O}(T) \otimes_k K$$

is isomorphic, as a Hopf K-algebra, to the group algebra KM for some abelian group M. It follows that $\mathcal{A}\widehat{\otimes}_k K$ is isomorphic to the algebra K^M, endowed with the product topology. Thus

$$K = \left(\mathcal{A}\widehat{\otimes}_k K\right)_0 = \mathcal{A}_0 \widehat{\otimes}_k K.$$

This implies that $\mathcal{A}_0 = k$, and hence that $D(T)$ is étale.

Conversely, suppose that $D(T)$ is étale. Then

$$D(T)\widehat{\otimes}_k k_s = D(T_{k_s})$$

is the constant k_s-group associated with a group M. Therefore its covariant bialgebra is the group algebra k_sM (cf. Remark 2.5.A), and consequently

$$\mathcal{O}(T) \otimes_k k_s = k_sM.$$

This proves that T is of multiplicative type and is split by the extension $k \to k_s$. The proposition follows.

By 2.2.2, 2.5.1, and 2.5, we obtain:

Corollary 2.5.3.C. *Let k be a field, and let G be an affine commutative k-group scheme.*

(i) *The group G contains a subgroup G_m of multiplicative type such that G/G_m is unipotent.*

(ii) *When k is perfect, there is in addition a canonical retraction of G onto G_m, so that G is the product of a unipotent group and a group of multiplicative type.*

(iii) [106] *Let k_s be a separable closure of k, and put*

$$\Gamma = \mathrm{Gal}(k_s/k).$$

The category of k-group schemes of multiplicative type is anti-equivalent to the category of continuous Γ-modules.

[106] The editors added item (iii), a consequence of Proposition 2.5.

2.6

We shall now study infinitesimal formal groups, to which the sections that follow are devoted. In this study, Lie algebras play a fundamental role.

Suppose first that the base ring k is Artinian, and let G be a formal group over k. The Lie algebra of G admits three different interpretations, all of which we shall use:

a) Let

$$D = k[d]/(d^2)$$

be the algebra of dual numbers over k, and let

$$\delta : D \longrightarrow k$$

be the homomorphism that annihilates d. For every formal group G over k, $\mathrm{Lie}(G)$ is the kernel of $G(\delta)$. Thus, by definition, there is an exact sequence of groups

$$1 \longrightarrow \mathrm{Lie}(G) \longrightarrow G(D) \xrightarrow{G(\delta)} G(k) \longrightarrow 1.$$

b) Let A be the affine algebra of G, and let

$$I_A = \ker \varepsilon_A$$

be its augmentation ideal. The elements of $G(D)$ are the morphisms of profinite k-algebras

$$f : A \longrightarrow D.$$

The condition $G(\delta)(f) = 1$ is equivalent to

$$\delta \circ f = \varepsilon_A.$$

Since

$$x - \varepsilon_A(x) \cdot 1_A \in I_A \qquad \text{for every } x \in A,$$

this is equivalent to $f(I_A) \subset k \cdot d$. Hence $\ker(f)$ contains I_A^2, and therefore also its closure $\overline{I_A^2}$. Thus f induces a continuous linear map

$$f' : I_A/\overline{I_A^2} = \omega_{G/k} \longrightarrow k$$

such that, for every $x \in A$,

$$f(x) = \varepsilon_A(x) \cdot 1_D + f'(\bar{x}) \cdot d,$$

where $\bar{x}$ denotes the image of $x - \varepsilon_A(x) \cdot 1_A$ in $I_A/\overline{I_A^2}$. It follows that the map

$$f \longmapsto f'$$

defines a bijection from $\mathrm{Lie}(G)$ onto the topological dual

$$\omega^{\dagger}_{G/k}$$

of $\omega_{G/k}$ (cf. 0.2.2).

This bijection respects the group structures. Indeed, let f and g be two elements of $\mathrm{Lie}(G)$. Their product $f \cdot g$ is the composite $h \circ \Delta_A$, where

$$h : A \widehat{\otimes} A \longrightarrow D$$

is defined by
$$h(b\widehat{\otimes}b') = f(b)g(b').$$
If $a \in I_A$, then, by 2.1(ii),
$$\Delta_A(a) - a\widehat{\otimes}1 - 1\widehat{\otimes}a \in I_A\widehat{\otimes}I_A.$$
Consequently,
$$(f \cdot g)(a) = f(a) + g(a)$$
(cf. also II, 3.10).

Henceforth we identify
$$\mathrm{Lie}(G) = \omega^{\dagger}_{G/k}$$
by means of the bijection $f \mapsto f'$ just described. The group Lie(G) is thereby endowed with a k-module structure.

c) Let $A^\dagger$ and $D^\dagger$ be the k-modules dual to A and D, and let
$$\{1^\dagger_D, d^\dagger\}$$
be the basis dual to the basis $\{1_D, d\}$ of D over k; thus $1^\dagger_D = \delta$.Since D is free of finite rank over k, the canonical map
$$\mathrm{Hom}_c(A, D) \longrightarrow \mathrm{Hom}_k(D^\dagger, A^\dagger), \qquad f \longmapsto {}^t f,$$
is bijective. On the other hand, ${}^t f$ is determined by the values
$${}^t f(1^\dagger_D) \qquad \text{and} \qquad {}^t f(d^\dagger) = x.$$
The condition $G(\delta)(f) = 1$ is equivalent to
$${}^t f(1^\dagger_D) = \varepsilon_A.$$
Moreover, one sees readily that f is compatible with multiplication if and only if (cf. 2.3)
$$\delta_G(x) = \phi_G(x \otimes 1 + 1 \otimes x). \qquad (*)$$

Finally, it is clear that a continuous linear map
$$f : A \longrightarrow D$$
that is compatible with multiplication and satisfies $\delta \circ f = \varepsilon_A$ sends the unit element of A to that of D.[107] The map $f \mapsto x$ therefore allows us to identify Lie(G) with the set of "primitive elements" of $\mathbf{H}(G)$, that is, the elements $x \in \mathbf{H}(G)$ satisfying relation (*). If x and y are two such elements, then
$$\begin{aligned}\delta_G(xy) &= \delta_G(x)\delta_G(y)\\ &= \phi_G\big((x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y)\big)\\ &= \phi_G\big(xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy\big),\end{aligned}$$
whence
$$\delta_G(xy - yx) = \phi_G\big((xy - yx) \otimes 1 + 1 \otimes (xy - yx)\big).$$
This shows that the k-module Lie(G) is identified with a Lie subalgebra of $\mathbf{H}(G)$; we call Lie(G) the *Lie algebra* of G.[108]

[107] Indeed, $\delta \circ f = \varepsilon_A$ implies $f(1) = 1 + \lambda d$, with $\lambda \in k$, and then
$$f(1) = f(1)^2 = 1 + 2\lambda d$$
gives $\lambda = 0$.

[108] Comparing with VII$_\mathrm{A}$, 2.5, we see that if K is a group scheme of finite type over k and G is the formal

2.6.1

Let k now be an arbitrary pseudocompact ring, and let G be a formal group over k. We call the functor $\mathrm{Lie}(G)$ that assigns to every object C of $\mathrm{Alf}\,/k$ the C-Lie algebra of the C-formal group

$$G' = G\widehat{\otimes}_k C$$

the $\mathbf{O}_k$-*Lie algebra of* G.[109] Put

$$A' = A\widehat{\otimes}_k C.$$

Since I_A is a direct factor of A, we have

$$I_{A'} = I_A\widehat{\otimes}_k C = I_A\widehat{\otimes}_A A'.$$

Moreover,

$$\omega_{G/k} = I_A\widehat{\otimes}_A k, \qquad \omega_{G'/C} = I_{A'}\widehat{\otimes}_{A'} C.$$

It follows that

$$\omega_{G'/C} = \omega_{G/k}\widehat{\otimes}_A C,$$

and hence

$$\mathrm{Lie}(G)(C) = \mathrm{Hom}_c(\omega_{G/k}\widehat{\otimes}_A C, C),$$

$$\text{i.e.} \qquad \mathrm{Lie}(G) = \mathbf{V}_k^f(\omega_{G/k}).$$

With the notation of 1.2.3.B, Proposition 1.2.3.E therefore shows that $\mathrm{Lie}(G)$ is flat over $\mathbf{O}_k$ if and only if $\omega_{G/k}$ is a projective pseudocompact k-module.

2.6.2

Conversely, every Lie algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a k-group functor. Indeed, let $U(\mathbf{L})$ denote the functor assigning to every object C of $\mathrm{Alf}\,/k$ the enveloping algebra

$$U(\mathbf{L}(C))$$

of the C-Lie algebra $\mathbf{L}(C)$. By VII$_{\mathrm{A}}$, 3.2.2, $U(\mathbf{L})$ is a bialgebra over $\mathbf{O}_k$, and hence, by 2.2, determines the k-group functor

$$\mathrm{Spf}^*\, U(\mathbf{L}),$$

which from now on we denote by $\mathscr{G}(\mathbf{L})$. Thus $\mathscr{G}(\mathbf{L})(C)$ is the group of elements $z \in U(\mathbf{L}(C))$ having augmentation 1 and satisfying

$$\Delta_{U(\mathbf{L}(C))}(z) = z \otimes z.$$

When $\mathbf{L}$ is flat over $\mathbf{O}_k$, we have the following proposition.

Proposition.[110] *Let* $\mathbf{L}$ *be a flat Lie algebra over* $\mathbf{O}_k$.

---

completion of K at the identity—that is, if $\mathcal{A}(G)$ is the completion of the local ring $\mathcal{O}_{K,e}$ for the $\mathfrak{m}$-adic topology, where

$$\mathfrak{m} = \ker\bigl(\varepsilon : \mathcal{O}_{K,e} \longrightarrow k\bigr)$$

—then $\mathbf{H}(G)$ is identified with the algebra of distributions $U(K)$, and $\mathrm{Lie}(G)$ with $\mathrm{Lie}(K)$. The hypothesis that K is of finite type over k is used to ensure that $\mathfrak{m}/\mathfrak{m}^2$ is a k-module of finite length, and hence discrete, so that its topological dual coincides with its ordinary dual. In particular, when G is a finite formal k-group—that is, when $\mathcal{A}(G)$ is a finite k-module—in which case G may also be regarded as the k-group scheme $\mathrm{Spec}\,\mathcal{A}(G)$, the two definitions of $\mathrm{Lie}(G)$ coincide.

[109] The editors expanded the original in what follows.

[110] The editors set off parts (i) and (ii) and added part (iii), which will be used in 2.6.3 and 3.3.2.

(i) *$\mathscr{G}(\mathbf{L})$ is a formal group topologically flat over k, whose $\mathbf{O}_k$-bialgebra is $U(\mathbf{L})$.*

(ii) *$\mathscr{G}(\mathbf{L})$ is infinitesimal.*

(iii) *For every object C of* Alf $/k$, $\mathrm{Lie}(\mathscr{G}(\mathbf{L}))(C)$ *is identified with the set*

$$\operatorname{Prim} U(\mathbf{L}(C)) = \{x \in U(\mathbf{L}(C)) \mid \varepsilon(x) = 0, \quad \Delta(x) = x \otimes 1 + 1 \otimes x\}$$

of primitive elements of $U(\mathbf{L}(C))$. In particular, there is a natural morphism of Lie algebras over $\mathbf{O}_k$,

$$\tau_{\mathbf{L}} : \mathbf{L} \longrightarrow \mathrm{Lie}(\mathscr{G}(\mathbf{L})).$$

Proof. The hypothesis that $\mathbf{L}$ is flat over $\mathbf{O}_k$ means that, for every morphism $C \to D$ in Alf $/k$,

$$\mathbf{L}(D) = \mathbf{L}(C) \otimes_C D,$$

and that, for every local component C' of C, the C'-module $\mathbf{L}(C')$ is free. The first condition, together with the universal properties of tensor products and enveloping algebras, gives

$$U(\mathbf{L}(D)) = U(\mathbf{L}(C)) \otimes_C D.$$

The second condition and the Poincaré–Birkhoff–Witt theorem (cf. Bourbaki, *Groupes et algèbres de Lie*, I, 2.7) show that $U(\mathbf{L}(C'))$ is a free C'-module. Consequently, the bialgebra $U(\mathbf{L})$ is flat over $\mathbf{O}_k$, proving (i).

To prove (ii) and (iii), we may suppose that k is Artinian. Put

$$L = \mathbf{L}(k), \qquad U = U(L), \qquad U_0 = k \cdot 1_U,$$

and let U^+ be the two-sided ideal of U generated by the image of L. For every $n \geq 1$, put

$$U_n = \left\{ u \in U \,\middle|\, \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+ \right\}.$$

By 1.3.6, it is enough to show that U is the union of the U_n. Identify L with its canonical image in U. Then $L \subset U_1$. If $x_1, \ldots, x_n \in L$, then

$$\Delta_U(x_1 \cdots x_n) = (x_1 \otimes 1 + 1 \otimes x_1) \cdots (x_n \otimes 1 + 1 \otimes x_n).$$

Induction on n therefore shows that $x_1 \cdots x_n \in U_n$, and hence

$$U = \bigcup_n U_n.$$

This proves (ii).

Finally, let

$$D = k[d]/(d^2)$$

be the algebra of dual numbers over k. By hypothesis,

$$\mathbf{L}(D) \simeq L \otimes_k D, \qquad U(\mathbf{L}(D)) \simeq U \otimes_k D.$$

It follows that $\mathrm{Lie}(\mathscr{G}(\mathbf{L}))(k)$ is identified with the set of elements

$$z = 1 + xd \in U \oplus Ud \qquad (x \in U)$$

such that $\varepsilon(z) = 1$ and $\Delta(z) = z \otimes z$. These conditions are equivalent to

$$\varepsilon(x) = 0, \qquad \Delta(x) = x \otimes 1 + 1 \otimes x,$$

that is, to $x \in \operatorname{Prim} U$. In particular,

$$\tau_{\mathbf{L}} : x \longmapsto 1 + dx$$

is a morphism of Lie algebras over $\mathbf{O}_k$ from $\mathbf{L}$ to $\mathrm{Lie}(\mathscr{G}(\mathbf{L}))$.

2.6.3

If $\mathbf{L}$ is a flat Lie algebra over $\mathbf{O}_k$, the formal group $\mathscr{G}(\mathbf{L})$ can be characterized by a universal property.[111] Indeed, every morphism

$$\phi : \mathscr{G}(\mathbf{L}) \longrightarrow G$$

to a formal group G induces a morphism

$$\mathrm{Lie}(\phi) : \mathrm{Lie}(\mathscr{G}(\mathbf{L})) \longrightarrow \mathrm{Lie}(G).$$

Composing it with

$$\tau_{\mathbf{L}} : \mathbf{L} \longrightarrow \mathrm{Lie}(\mathscr{G}(\mathbf{L}))$$

(cf. 2.6.2) gives a morphism

$$\phi' : \mathbf{L} \longrightarrow \mathrm{Lie}(G).$$

We then have:

Proposition. *Let G be a formal k-group and let $\mathbf{L}$ be a flat Lie algebra over $\mathbf{O}_k$. The map $\phi \mapsto \phi'$ defined above is a bijection*

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(\mathscr{G}(\mathbf{L}), G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Lie}}(\mathbf{L}, \mathrm{Lie}(G)),$$

where the right-hand side is the set of morphisms of Lie algebras over $\mathbf{O}_k$ from $\mathbf{L}$ to $\mathrm{Lie}(G)$*.*

Proof. We may immediately reduce to the case in which k is Artinian. Put $L = \mathbf{L}(k)$. By 2.3.1,

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(\mathscr{G}(\mathbf{L}), G)$$

is in bijection with the set of unital algebra morphisms

$$h : U(L) \longrightarrow \mathbf{H}(G)$$

for which the following diagram is commutative:

$$\begin{tikzcd} U(L) \arrow[r, "h"] \arrow[d, "\Delta_{U(L)}"'] & \mathbf{H}(G) \arrow[dr, "\delta_G"] & \\ & & \mathbf{H}(G \times G) \\ U(L) \otimes U(L) \arrow[r, "h\otimes h"] & \mathbf{H}(G) \otimes \mathbf{H}(G) \arrow[ur, "\varphi_G"'] & \end{tikzcd}.$$

Now h is determined by its restriction to L, which is a morphism from L to the Lie algebra underlying $\mathbf{H}(G)$. Commutativity of the diagram says precisely that h sends L into the subset of $\mathbf{H}(G)$ formed by the elements x satisfying

$$\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

This subset is exactly $\mathrm{Lie}(G)$; cf. 2.6(c). The proposition follows.

[111] The editors modified what follows to take advantage of the addition made in 2.6.2.

2.7

We conclude these generalities with a statement that goes back to S. Lie and that will be used in 5.1. A *formal monoid* over k is, by definition, a pair (M, m) consisting of a formal variety M and a morphism

$$m : M \times M \longrightarrow M$$

such that $m(C)$ makes $M(C)$ a monoid for every object C of Alf/k.[112] In particular, the "unit section," which assigns to every object C the identity element of $M(C)$, defines a section

$$\varepsilon_M : \mathrm{Spf}(k) \longrightarrow M$$

of the canonical projection $M \to \mathrm{Spf}(k)$. We shall say that the formal monoid M is *infinitesimal* if ε_M induces a bijection of the underlying sets.

Proposition. *Every topologically flat infinitesimal formal k-monoid M is a formal k-group.*[113]

Proof. We must show that $M(C)$ is a group for every object C of Alf/k. We therefore reduce immediately to the case in which k is Artinian. Let

$$U = \mathbf{H}(M)$$

be the coalgebra of M (1.3.5). The multiplication

$$m : M \times M \longrightarrow M$$

induces a morphism of coalgebras

$$m_U : U \otimes U \longrightarrow U$$

that makes U an associative algebra over k. The identity element of this algebra is the image of the identity element of k under the map $k \to U$ induced by the unit section ε_M of M. Likewise, the projection

$$M \longrightarrow \mathrm{Spf}(k)$$

induces a homomorphism

$$\varepsilon_U : U \longrightarrow k;$$

we write U^+ for the kernel of ε_U.

We must show that there is an antipode, that is, a morphism of coalgebras

$$c_U : U \longrightarrow U$$

such that, for every $u \in U$,

$$(m_U \circ (c_U \otimes \mathrm{id}_U) \circ \Delta_U)(u) = \varepsilon_U(u)\, 1_U. \tag{*}$$

[112]Here and in what follows, the editors have written "monoid" instead of "monoid with identity element" (recall that a monoid is, by definition, a set equipped with an associative composition law and possessing an identity element).

[113]By the equivalence of categories in 1.3.5.D, a monoid in the category of topologically flat formal k-varieties "is the same thing" as a monoid in the category of flat cocommutative $\mathbf{O}_k$-coalgebras, that is, a cocommutative $\mathbf{O}_k$-bialgebra (in the usual sense, hence not necessarily equipped with an antipode). Moreover, by 1.3.6, the hypothesis that M be infinitesimal is equivalent to saying that the corresponding bialgebra is connected. Thus, if k is an Artinian ring, the proposition is equivalent to the following statement: every connected cocommutative k-bialgebra that is flat over k is a k-Hopf algebra, that is, it possesses an antipode. In fact, the cocommutativity hypothesis is superfluous; cf. the proof.

Let (U_n) be the filtration of U defined in 1.3.6, and put

$$U_n^+ = U^+ \cap U_n.$$

Since M is infinitesimal, U^+ is the union of the submodules U_n^+.[114] Set

$$c_0(1) = 1$$

and

$$c_1(x) = -x \qquad \text{if } x \in U_1^+,$$

that is, if x is primitive. Suppose that

$$c_{n-1} : U_{n-1} \longrightarrow U$$

has been constructed so as to satisfy (*), with c_{n-1} in place of c_U, for every $x \in U_{n-1}$, and let $x \in U_n^+$. By the proof of Lemma 1.3.6.A,

$$\Delta_U(x) - x \otimes 1 \in U_{n-1} \otimes U^+;$$

this is where the hypothesis that U is flat over k enters. We may therefore write

$$\Delta_U(x) = x \otimes 1 + \sum_i y_i \otimes z_i, \qquad y_i \in U_{n-1},$$

and then set

$$c_n(x) = -\sum_i c_{n-1}(y_i) z_i.$$

Together with $c_n(1) = 1$, this prescription extends c_{n-1} coherently to U_n. In this way we obtain a k-linear map

$$c : U \longrightarrow U$$

that is the left inverse of id_U for the monoid law on $\mathrm{End}_k(U)$ defined by

$$f \cdot g = m_U \circ (f \otimes g) \circ \Delta_U;$$

the identity element is the map

$$\eta : u \longmapsto \varepsilon_U(u)\, 1_U.$$

It follows that c is uniquely determined and is also the right inverse of id_U; that is,

$$m_U \circ (\mathrm{id}_U \otimes c) \circ \Delta_U = \eta,$$

without any assumption that U be cocommutative. Thus $c_U := c$ is the required antipode, and M is a formal k-group.

[114] The original continued as follows: "one then shows easily, by induction on n, that there exists one and only one linear map $c_n : U_n \to U_n$ such that the composite

$$m_U \circ (c_n \otimes \mathrm{id}_U) \circ \Delta_U : U_n^+ \longrightarrow U$$

is zero." The editors have supplied the details of the proof, which rests on that of Lemma 1.3.6.A.

2.8. Unipotent group schemes over a field

115

Let k be a field. Recall that a k-group scheme G is called *unipotent* if it satisfies the following two conditions (cf. [DG70], § IV.2, Proposition 2.5):

(a) G is affine.

(b) Every simple $\mathcal{O}(G)$-comodule is trivial. Explicitly, if

$$\rho : V \longrightarrow V \otimes_k \mathcal{O}(G)$$

is an $\mathcal{O}(G)$-comodule structure on a nonzero k-vector space V, and there is no nonzero proper subspace $W \subset V$ such that

$$\rho(W) \subset W \otimes_k \mathcal{O}(G),$$

then V is one-dimensional and

$$\rho(v) = v \otimes 1 \qquad \text{for every } v \in V.$$

According to the cited result, when G is of finite type over k, this is equivalent to the definition given in Exposé XVII, § 1: G has a finite composition series whose successive quotients are isomorphic to k-subgroups of $\mathbb{G}_{a,k}$.

For every affine k-group scheme G, the comultiplication of $\mathcal{O}(G)$ endows the pseudo-compact k-module

$$A = \mathcal{O}(G)^*$$

with the structure of a profinite k-algebra, not necessarily commutative; its unit 1_A is the augmentation

$$\varepsilon : \mathcal{O}(G) \longrightarrow k.$$

Set

$$I = \{f \in A \mid f(1_{\mathcal{O}(G)}) = 0\}.$$

Then I is a two-sided ideal of A, and

$$A = k \cdot 1_A \oplus I;$$

cf. 1.3.6.

Let V be a finite-codimensional subspace of A, and consider the continuous k-bilinear map

$$\varphi : A \times A \longrightarrow A/V, \qquad (a, b) \longmapsto ab + V.$$

By Lemma 0.3.1, there are finite-codimensional subspaces $L_1, L_2 \subset A$ such that V contains AL_2 and $L_1 A$. Thus

$$L = L_1 \cap L_2$$

is finite-codimensional, and V contains the two-sided ideal ALA, which is itself finite-codimensional. Hence the finite-codimensional two-sided ideals form a neighborhood basis of zero in A.

It follows that an $\mathcal{O}(G)$-comodule is the same thing as a continuous A-module: an A-module V, equipped with the discrete topology, for which the map

$$A \times V \longrightarrow V$$

[115] The editors added this subsection.

is continuous. Every such module is the union of finite-dimensional A-submodules V_i, each of which is a module over a finite-dimensional quotient k-algebra A_i of A.

Consequently, a simple continuous A-module M is finite-dimensional over k and is a faithful simple module over the finite-dimensional algebra $A/\operatorname{Ann}(M)$. The latter is therefore a finite-dimensional simple k-algebra, so $\operatorname{Ann}(M)$ is an open maximal ideal of A.

Conversely, let P be an open prime ideal of A.[116] Then A/P is a finite-dimensional k-algebra in which the zero ideal is prime, and hence is a finite-dimensional simple k-algebra. Up to isomorphism, there is therefore a unique simple continuous A-module with annihilator P. Thus

$$M \longmapsto \operatorname{Ann}(M)$$

is a bijection from the isomorphism classes of simple continuous A-modules to the open prime ideals of A.

In particular, the A-module A/I, which is one-dimensional over k, is called the *trivial module.* It corresponds to the trivial one-dimensional $\mathcal{O}(G)$-comodule V, for which

$$\rho(v) = v \otimes 1_{\mathcal{O}(G)} \qquad \text{for every } v \in V.$$

We have proved:

Proposition 2.8.1. *Let k be a field, let G be an affine k-group scheme, and put $A = \mathcal{O}(G)^*$.*

(i) *G is unipotent if and only if I is the unique open prime ideal of A.*[117]

(ii) *In particular, suppose that G is commutative, so that*

$$\mathcal{O}(G) = \mathbf{H}(D(G)),$$

where $D(G) = \operatorname{Spf}(A)$ is the Cartier dual of G. Then G is unipotent if and only if $D(G)$ is infinitesimal.

2.9. Pointed cocommutative Hopf algebras over a field

[118]

Let k be a field, let C be a k-coalgebra, and endow

$$A = C^*$$

with the structure of a profinite k-algebra, not necessarily commutative, described in 2.8. By 0.2.2,

$$C = A^\dagger = \operatorname{Hom}_c(A, k).$$

The map

$$D \longmapsto D^\perp = \{f \in A \mid f(D) = 0\}$$

is therefore a bijection from the subcoalgebras of C to the closed two-sided ideals of A (which we shall simply call ideals below). Its inverse is

$$I \longmapsto I^\perp = \{x \in C = A^\dagger \mid x(I) = 0\}.$$

[116] Recall that a two-sided ideal P of a ring A is called prime if, in A/P, the product of any two nonzero two-sided ideals is nonzero.

[117] Equivalently, $k \cdot 1_{\mathcal{O}(G)}$ is the unique simple k-subcoalgebra of $\mathcal{O}(G)$; see, for example, [Ab80], 3.1.4. The editors also note a misprint in the cited work, p. 130, line 4: $M \simeq C^*/\operatorname{ann} M$ should be replaced by $C^*/\operatorname{ann} M \simeq \operatorname{End}_k(M)$.

[118] The editors added this subsection to express Proposition 2.5.2, together with the variant noted in editorial note 100, in the language of cocommutative Hopf algebras.

Since every maximal closed ideal is a maximal open ideal (cf. 0.2.1), every subcoalgebra contains a simple subcoalgebra, necessarily finite-dimensional.

Recall that a subcoalgebra $D \subset C$ is called *irreducible* if it contains a unique simple subcoalgebra S_0. Equivalently,

$$\mathfrak{m}_0 = S_0^\perp$$

is the unique open maximal ideal containing $D^\perp$, or, equivalently,

$$D^\perp + \mathfrak{m} = A \qquad \text{for every open maximal ideal } \mathfrak{m} \neq \mathfrak{m}_0.$$

This applies in particular when $D = S_0$.

Let Σ_0 be the sum of all irreducible subcoalgebras C_i containing S_0. It is plainly a subcoalgebra, and it is irreducible because, for every $\mathfrak{m} \neq \mathfrak{m}_0$, 0.2.D gives

$$\mathfrak{m} + \bigcap_i C_i^\perp = \bigcap_i (\mathfrak{m} + C_i^\perp) = A.$$

The coalgebra Σ_0 is called the *irreducible component of C corresponding to S_0. The French prints C_0 in this last phrase; the preceding definition makes S_0 the intended simple subcoalgebra.*

A coalgebra C is called *pointed* if every simple k-subcoalgebra of C is one-dimensional. Equivalently,

$$A/\mathfrak{m} = k$$

for every open maximal ideal $\mathfrak{m}$ of A. Recall also that C is called *connected* if it is irreducible and pointed. If C is a bialgebra, then it is connected if and only if it is irreducible, since $k \cdot 1_C$ is a simple subcoalgebra.

Suppose from now on that C is cocommutative. Then A is commutative and hence is the product of its local components

$$A_\mathfrak{m}, \qquad \mathfrak{m} \in \Upsilon(A);$$

cf. 0.1.1. Put

$$S_\mathfrak{m} = \mathfrak{m}^\perp \simeq (A/\mathfrak{m})^*,$$

and let $\Sigma_\mathfrak{m}$ be its irreducible component. This component can be described explicitly. Let

$$J_\mathfrak{m} = \ker(A \longrightarrow A_\mathfrak{m}).$$

Then $J_\mathfrak{m} \subset \mathfrak{m}$, and it is the smallest closed ideal $I \subset \mathfrak{m}$ satisfying

$$I + \mathfrak{m}' = A \qquad \text{for every open maximal ideal } \mathfrak{m}' \neq \mathfrak{m}.$$

Indeed, any such I contains $A_{\mathfrak{m}'}$ for every $\mathfrak{m}' \neq \mathfrak{m}$, and hence contains $J_\mathfrak{m}$. Since

$$A = J_\mathfrak{m} \oplus A_\mathfrak{m},$$

we obtain

$$\Sigma_\mathfrak{m} = J_\mathfrak{m}^\perp \simeq A_\mathfrak{m}^\dagger.$$

We can now prove:

Proposition 2.9.1. *Let k be a field.*

(i) *Let G be a formal k-group all of whose residue fields are equal to k. Then G_e is the constant k-group M_k, where*

$$M = G(k) = \{x \in \mathbf{H}(G) \,|\, \varepsilon(x) = 1, \quad \Delta(x) = x \otimes x\},$$

and $\mathbf{H}(G)$ is the semidirect product of $\mathbf{H}(G_0)$ by the group algebra kM (cf. 2.4.B).

(ii) *Equivalently, let H be a pointed cocommutative Hopf k-algebra. Then H is the semidirect product of the irreducible component H_0 of the unit 1_H by kM, where*

$$M = \{x \in H \mid \varepsilon(x) = 1,\ \Delta(x) = x \otimes x\}.$$

Proof. We first prove (i). Since every residue field of G is equal to k, the projection

$$\pi : G \longrightarrow G_e$$

has a section $s : G_e \to G$, defined at every point $g \in G$ by

$$\mathcal{O}_{G,g} \longrightarrow \kappa(g) = k.$$

Moreover, for all $g, h \in G$, the algebra

$$\mathcal{O}_{G,g} \widehat{\otimes}_k \mathcal{O}_{G,h}$$

is local with residue field k. Thus $s \times s$ is a section of

$$\pi \times \pi : G \times G \longrightarrow (G \times G)_e = G_e \times G_e.$$

Since π is a group morphism,

$$\pi \circ m_G \circ (s \times s) = m_{G_e} = \pi \circ s \circ m_{G_e}.$$

As π is an epimorphism, it follows that s is a group morphism. Hence

$$G = G_0 \rtimes G_e,$$

and $\mathbf{H}(G)$ is the semidirect product of $\mathbf{H}(G_0)$ by $\mathbf{H}(G_e)$.

Since every residue field of G is k, $\mathbf{H}(G_e)$ is the group algebra kM, where

$$M = G_e(k);$$

cf. 2.5.A. Finally, because G_0 is infinitesimal, the map

$$G(k) \longrightarrow G_e(k)$$

is injective. It is also surjective, since it has the section induced by s. Thus $M = G(k)$, proving (i).

For (ii), it remains only to show that

$$H_0 = \mathbf{H}(G_0).$$

The unit 1_H of H is the augmentation

$$\varepsilon_{\mathcal{A}} : \mathcal{A} \longrightarrow k,$$

which is the identity section of $G(k)$. Therefore the local component of the affine algebra $\mathcal{A}(G)$ corresponding to H_0 is precisely $\mathcal{A}(G_0)$. By the description above,

$$H_0 = \mathcal{A}(G_0)^\dagger = \mathbf{H}(G_0).$$

This proves the proposition.

Remarks 2.9.2.

(a) The preceding proposition, which is implicit in 2.5.2, was obtained independently by B. Kostant (cf. [Sw69], Preface). Combined with Cartier's Theorem 3.3 below (cf. [Ca62], § 12, Theorem 3), which was also obtained by Kostant (cf. [Sw69], loc. cit.), this result is often called the *Cartier–Gabriel–Kostant theorem.*

(b) In the form (ii), Proposition 2.9.1 was extended by R. G. Heyneman and M. E. Sweedler to the case in which H is assumed to be pointed and the direct sum of its irreducible components, but is not necessarily cocommutative; see [HS69], Theorem 3.5.8.

3. Phenomena specific to characteristic 0

Throughout Section 3, we assume that the pseudocompact ring k contains the field of rational numbers $\mathbb{Q}$.

3.1. Lemma

Lemma 3.1. *Let C be a commutative unital $\mathbb{Q}$-algebra, and let L be a Lie algebra over C whose underlying C-module is free. Then the canonical map*

$$L \longrightarrow U(L)$$

identifies L with the set of primitive elements of $U(L)$.

Proof. Identify L with its canonical image in $U(L)$. Let I be a totally ordered set, and let

$$(x_i)_{i\in I}$$

be a basis of L indexed by I. Denote by $\mathbb{N}^{(I)}$ the set of families

$$n = (n_i)_{i\in I}$$

of nonnegative integers for which $n_i = 0$ except possibly at finitely many indices

$$i_1 < i_2 < \cdots < i_s;$$

these indices depend on n. Put

$$x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \cdots x_{i_s}^{n_{i_s}}, \qquad n! = (n_{i_1}!)(n_{i_2}!)\cdots(n_{i_s}!).$$

By the Poincaré–Birkhoff–Witt theorem, the elements x^n form a basis of $U(L)$. Moreover,

$$\Delta_{U(L)}\left(\frac{x^n}{n!}\right) = \sum_{0\leq m\leq n} \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!}. \tag{*}$$

Here the sum ranges over all $m \in \mathbb{N}^{(I)}$ such that $0 \leq m \leq n$, that is,

$$0 \leq m_i \leq n_i \qquad \text{for every } i.$$

It follows immediately that

$$\Delta_{U(L)}(u) = u \otimes 1 + 1 \otimes u$$

if and only if u is a linear combination of the x_i. This proves the lemma.

3.2

Suppose now that C is Artinian, with radical $\mathfrak{r}$. For every C-algebra U (associative and unital), the ideal $\mathfrak{r}U$ therefore consists of nilpotent elements. If $x \in \mathfrak{r}U$, put

$$\exp_U(x) = 1 + x + \frac{x^2}{2!} + \cdots .$$

[119] This gives a bijection from $\mathfrak{r}U$ onto $1 + \mathfrak{r}U$; its inverse sends $1 + y \in 1 + \mathfrak{r}U$ to

$$\log_U(1+y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \cdots .$$

[119] If $x, x' \in \mathfrak{r}U$ commute, then

$$\exp_U(x + x') = \exp_U(x)\exp_U(x').$$

Moreover, the map $\exp_U$ is plainly functorial in U.[120]

Still assuming C Artinian, suppose that U is endowed with a bialgebra structure over C (cf. 2.2). For every primitive element $x \in \mathfrak{r}U$ (cf. VII$_{\mathrm{A}}$, 3.2.3), we then have

$$\begin{aligned}\Delta_U(\exp_U(x)) &= \exp_{U\otimes_C U}(\Delta_U(x))\\ &= \exp_{U\otimes_C U}(x\otimes 1 + 1\otimes x)\\ &= \exp_{U\otimes_C U}(x\otimes 1)\ \exp_{U\otimes_C U}(1\otimes x)\\ &= (\exp_U(x)\otimes 1)(1\otimes \exp_U(x))\\ &= \exp_U(x)\otimes\exp_U(x).\end{aligned}$$

Thus the bijection $\exp_U$ sends a primitive element of $\mathfrak{r}U$ to an element $z \in 1+\mathfrak{r}U$ satisfying

$$\Delta_U(z) = z\otimes z.$$

Conversely, if z satisfies these conditions, then, on putting $x = \log_U(z)$, the preceding calculation shows that

$$\exp_{U\otimes_C U}(x\otimes 1 + 1\otimes x) = z\otimes z = \Delta_U(\exp_U(x)) = \exp_{U\otimes_C U}(\Delta_U(x));$$

hence

$$x\otimes 1 + 1\otimes x = \Delta_U(x).$$

[121] Furthermore, if $z \in 1+\mathfrak{r}U$ satisfies $\Delta_U(z) = z\otimes z$, then

$$\varepsilon_U(z)^2 = \varepsilon_U(z).$$

Since $\varepsilon_U(z)$ is invertible (because z is invertible, and $\mathfrak{r}$ is nilpotent), it follows that $\varepsilon_U(z) = 1$.

In particular, let L be a Lie algebra flat over C, take for U the enveloping algebra $U(L)$ of L over C, and identify L with its canonical image in U. By Lemma 3.1, L is the set of primitive elements of U: indeed, on every local component of C, L is a free module. Consider the C-formal group

$$\mathscr{G}(L) = \mathrm{Spf}^*\, U(L),$$

whose covariant bialgebra is $U = U(L)$ (cf. 2.6.2). Let $\mathfrak{m}$ be a maximal ideal of C, and put

$$\overline{C} = C/\mathfrak{m}, \qquad \overline{U} = U\otimes_C \overline{C}.$$

Since $\mathscr{G}(L)$ is infinitesimal (loc. cit.), the unit element of $\overline{U}$ is the only element $\overline{z}\in\overline{U}$ such that[122]

$$\varepsilon_{\overline{U}}(\overline{z}) = 1, \qquad \Delta_{\overline{U}}(\overline{z}) = \overline{z}\otimes\overline{z}.$$

It follows that every $z\in U$ satisfying

$$\varepsilon_U(z) = 1, \qquad \Delta_U(z) = z\otimes z$$

necessarily belongs to $1+\mathfrak{r}U$.

Finally, since

$$L\cap\mathfrak{r}U = \mathfrak{r}L = L\otimes_C\mathfrak{r},$$

[120] That is, for every homomorphism $\phi: U\to V$ of C-algebras,

$$\phi\big(\exp_U(x)\big) = \exp_V\big(\phi(x)\big).$$

[121] The editors supplied the details of the preceding argument and added the sentence that follows.

[122] The editors added the condition $\varepsilon_{\overline{U}}(\overline{z}) = 1$, which was omitted in the original.

we conclude that $\exp_U$ defines a bijection from $L \otimes_C \mathfrak{r}$ onto the group $\mathscr{G}(L)(C)$. We summarize this as follows.

Proposition 3.2. *Let k be a pseudocompact ring containing $\mathbb{Q}$, and let $\mathbf{L}$ be a flat Lie algebra over $\mathbf{O}_k$.*

(i) *For every object C of* Alf $/k$, *let $\mathfrak{r}(C)$ denote its radical. Then the map*

$$\exp_{U(\mathbf{L}(C))} : \mathbf{L}(C) \otimes_C \mathfrak{r}(C) \longrightarrow \mathscr{G}(\mathbf{L})(C)$$

is bijective and functorial in C and $\mathbf{L}$.

(ii) *The natural morphism*

$$\mathbf{L} \longrightarrow \mathrm{Lie}(\mathscr{G}(\mathbf{L}))$$

(cf. 2.6.2) *is an isomorphism of Lie algebras over $\mathbf{O}_k$.*

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3.2.1

The bijection $\exp_{U(L(C))}$ allows us, by transport of structure, to define a group law on the set

$$L(C) \otimes_C \mathfrak{r}(C),$$

which, as in 3.2, we identify with a subset of $U(L(C))$. If x and y are two elements of $L(C) \otimes_C \mathfrak{r}(C)$, this law is given by

$$\begin{aligned} x \cdot y = \log\big((\exp x)(\exp y)\big) &= \log\left(1 + \sum_{p+q>0} \frac{x^p y^q}{p!\,q!}\right) \\ &= \sum_{m\geq 1}\sum_{p_i+q_i>0} \frac{(-1)^{m-1}}{m} \frac{x^{p_1}y^{q_1}}{p_1!\,q_1!} \cdots \frac{x^{p_m}y^{q_m}}{p_m!\,q_m!} \\ &= \sum_{\ell\geq 1} P_\ell(x,y). \end{aligned}$$

Here $P_\ell(x,y)$ denotes the sum of the monomials of total degree ℓ in x and y. For example,

$$P_1(x,y) = x + y,$$

and

$$\begin{aligned} P_2(x,y) &= \underbrace{\frac{x^2}{2} + xy + \frac{y^2}{2}}_{m=1} - \underbrace{\frac{1}{2}(x^2 + xy + yx + y^2)}_{m=2} \\ &= \frac{1}{2}(xy - yx) = \frac{1}{2}[x,y]. \end{aligned}$$

Moreover, $P_3(x,y)$ is the sum of the following three terms:

$$\begin{aligned} P_3(x,y) = &\underbrace{\frac{x^3}{6} + \frac{x^2y}{2} + \frac{xy^2}{2} + \frac{y^3}{6}}_{m=1} \\ &\underbrace{-\frac{1}{2}\left(x^3 + \frac{3}{2}x^2y + \frac{1}{2}yx^2 + xyx + yxy + \frac{1}{2}y^2x + \frac{3}{2}xy^2 + y^3\right)}_{m=2} \\ &\underbrace{+\frac{1}{3}\left(x^3 + x^2y + yx^2 + xyx + yxy + y^2x + xy^2 + y^3\right)}_{m=3}. \end{aligned}$$

[123] The editors added part (ii), which follows immediately from 3.1.

Consequently,

$$P_3(x,y) = \frac{1}{12}(x^2y + yx^2 - 2xyx - 2yxy + y^2x + xy^2) = \frac{1}{12}([[y,x],x] + [y,[y,x]]).$$

More generally, one can prove the Campbell–Hausdorff formula:[124]

$$\begin{aligned} P_\ell(x,y) = &\sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m\cdot\ell} \sum_{\substack{p_1,\dots,p_m\\ q_1,\dots,q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!}\frac{(\operatorname{ad} y)^{q_i}}{q_i!}\right)\frac{(\operatorname{ad} x)^{p_m}}{p_m!}(y) \\ &+ \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m\cdot\ell} \sum_{\substack{p_1,\dots,p_{m-1}\\ q_1,\dots,q_{m-1}}} \left(\prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}}{p_i!}\frac{(\operatorname{ad} y)^{q_i}}{q_i!}\right)(x). \end{aligned}$$

Here the $p_j, q_i \in \mathbb{N}$ satisfy

$$p_i + q_i \geq 1 \quad (1 \leq i \leq m-1), \qquad p_m + \sum_{i=1}^{m-1}(p_i + q_i) = \ell - 1.$$

In other words, in the sums above every nonzero "Lie monomial" has total degree ℓ. For a proof, see N. Jacobson, *Lie Algebras* (Interscience, 1962), §V.5, or [BLie], II, §6.4, Theorem 2.

3.3

If G is a formal k-group with affine algebra A, recall that I_A denotes the augmentation ideal of A, and that

$$\omega_{G/k} = I_A/I_A^2 \simeq I_A \widehat{\otimes}_A k$$

is a pseudocompact k-module.

Theorem 3.3 (Cartier). *Let k be a pseudocompact ring containing $\mathbb{Q}$,*[125] *and let G be a formal k-group. The following assertions are equivalent:*

(i) *There exists an $\mathscr{O}_k$-Lie algebra $\mathbf{L}$, flat over $\mathscr{O}_k$, such that G is isomorphic to $\mathscr{G}(\mathbf{L})$. In this case $\mathbf{L} = \mathrm{Lie}(G)$ by 3.2.*

(ii) *There exists a projective pseudocompact k-module ω such that the formal variety underlying G is isomorphic to*

$$\mathbf{V}_k^{f,0}(\omega)$$

(cf. 1.2.5)*, whose affine algebra is $k[[\omega]]$. In this case $\omega \simeq \omega_{G/k}$.*

(iii) *The group G is infinitesimal, and $\omega_{G/k}$ is a projective pseudocompact k-module.*

(iv) *The group G is infinitesimal and topologically flat over k.*

Proof of (i) $\Rightarrow$ (ii). Let

$$\omega = \Gamma^*(\mathbf{L})$$

be the pseudocompact k-module dual to $\mathbf{L}$ (cf. 1.2.3.D). For every $C \in \mathrm{Alf}/k$, we must construct, functorially in C, an isomorphism

$$\mathbf{V}_k^{f,0}(\omega)(C) \xrightarrow{\sim} \mathscr{G}(\mathbf{L})(C).$$

[124]The editors corrected the formula given in the original, which was erroneous, and added the reference [BLie].

[125]The editors removed the hypothesis that k be local; the proof reduces to that case.

By 1.2.5, the left-hand side identifies with

$$\mathrm{Hom}_c(\omega, \mathfrak{r}(C)),$$

where $\mathfrak{r}(C)$ is the radical of C. This set is naturally isomorphic to the set

$$\mathrm{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$$

of continuous C-linear maps. Since $\omega \widehat{\otimes}_k C$ is a projective pseudocompact C-module, the canonical map

$$(\omega \widehat{\otimes}_k C)^{\dagger} \otimes_C \mathfrak{r}(C) \longrightarrow \mathrm{Hom}_c(\omega \widehat{\otimes}_k C, \mathfrak{r}(C))$$

is bijective (cf. 0.3.6.A). By 1.2.3.E,

$$\mathbf{L}(C) \simeq \mathbf{V}_k^f(\Gamma^*(\mathbf{L}))(C) = (\omega \widehat{\otimes}_k C)^{\dagger}.$$

Thus $\mathbf{V}_k^{f,0}(\omega)(C)$ is canonically isomorphic to

$$\mathbf{L}(C) \otimes_C \mathfrak{r}(C),$$

which Proposition 3.2 identifies canonically with $\mathscr{G}(\mathbf{L})(C)$. This proves (i) $\Rightarrow$ (ii).

Proof of (ii) $\Rightarrow$ (iii). Let ω be a projective object of $\mathrm{PC}(k)$, and let

$$h : k[[\omega]] \xrightarrow{\sim} A$$

be an isomorphism onto the affine algebra of G. The composite

$$\varepsilon_A \circ h : k[[\omega]] \longrightarrow k$$

is determined by its restriction $\lambda : \omega \to \mathfrak{r}(k)$. The map

$$x \longmapsto x - \lambda(x)$$

from ω to the radical of $k[[\omega]]$ extends, by the universal property of $k[[\omega]]$, to an endomorphism ℓ_λ of $k[[\omega]]$. Since

$$\ell_\lambda \circ \ell_{-\lambda} = \ell_{-\lambda} \circ \ell_\lambda = \mathrm{id},$$

the map ℓ_λ is an automorphism. Replacing h by $h \circ \ell_\lambda$, we may assume that $\varepsilon_A \circ h$ vanishes on the closed ideal I of $k[[\omega]]$ generated by ω. Then h induces an isomorphism

$$I/I^2 \xrightarrow{\sim} I_A/I_A^2.$$

Since $I/I^2 \simeq \omega$, we obtain

$$\omega_{G/k} \simeq \omega,$$

so $\omega_{G/k}$ is projective. The formal variety $\mathbf{V}_k^{f,0}(\omega)$, and hence G, is infinitesimal.

Proof of (iii) $\Rightarrow$ (i). Assume that G is infinitesimal and $\omega_{G/k}$ is projective. Let $\mathbf{L}$ be the $\mathcal{O}_k$-Lie algebra of G. By 2.6.1, its underlying $\mathcal{O}_k$-module is

$$\mathbf{V}_k^f(\omega_{G/k}).$$

Consequently, $\mathbf{L}$ is flat over $\mathcal{O}_k$, and Proposition 1.2.3.E gives

$$\Gamma^*(\mathbf{L}) \simeq \omega_{G/k}.$$

The proof of (i) $\Rightarrow$ (ii) therefore identifies the affine algebra of $\mathscr{G}(\mathbf{L})$ with $k[[\omega_{G/k}]]$. By 2.6.3, the identity morphism of $\mathbf{L}$ corresponds canonically to a morphism

$$\mathscr{G}(\mathbf{L}) \longrightarrow G,$$

and hence to a continuous k-algebra morphism

$$\varphi : A \longrightarrow k[[\omega_{G/k}]].$$

Let I be the closed ideal generated by $\omega_{G/k}$. Filter $k[[\omega_{G/k}]]$ by the closures of the powers I^n, and A by the closures of the powers I_A^n. By definition, φ induces the identity on

$$\omega_{G/k} = I_A/I_A^2 = I/I^2.$$

Because $\omega_{G/k}$ is projective, the canonical projection

$$I_A \longrightarrow \omega_{G/k}$$

has a section τ. By the universal property of $k[[\omega_{G/k}]]$, this section defines a continuous algebra morphism

$$\psi : k[[\omega_{G/k}]] \longrightarrow A.$$

The composite $\varphi \circ \psi$ induces the identity on $\omega_{G/k}$, and hence on the associated graded algebra of $k[[\omega_{G/k}]]$. It follows from [CA], §V.7, Lemma 1 that $\varphi \circ \psi$ is an isomorphism.[126]

Moreover, ψ induces an isomorphism

$$I/I^2 \xrightarrow{\sim} I_A/I_A^2,$$

and hence a surjection on associated graded algebras. Since G is radicial, I_A is contained in the radical of A, so

$$\bigcap_n I_A^n = 0.$$

The cited lemma therefore shows that ψ is surjective. Since $\varphi \circ \psi$ is an isomorphism and ψ is surjective, both ψ and φ are isomorphisms. This proves (iii) $\Rightarrow$ (i).

Finally, (i) or (ii) plainly implies (iv), so it remains to prove (iv) $\Rightarrow$ (ii). We may assume that k is local, with residue field k_0. Put

$$G_0 = G\widehat{\otimes}_k k_0, \qquad \omega = \omega_{G/k}, \qquad \omega_0 = \omega\widehat{\otimes}_k k_0.$$

[127] The pseudocompact k_0-module ω_0 has a pseudobasis $(y_\lambda)_{\lambda\in\Lambda}$. Let

$$\omega' = \prod_{\lambda\in\Lambda} k_\lambda$$

be the topologically free k-module with that index set, and lift each y_λ to an element $x_\lambda \in \omega$. This gives a continuous k-linear map

$$f : \omega' \longrightarrow \omega$$

such that

$$f_0 = f\widehat{\otimes}_k k_0$$

[126] Compare 5.1.5 below, and also [BAC], III, §2.8, Theorem 1 and its corollaries. The editors supplied the details of the original argument that follow.

[127] The editors supplied the details of the original argument that follow.

is invertible.[128] Since ω' is projective, f lifts to a continuous k-linear map

$$g : \omega' \longrightarrow I_A$$

with $\pi \circ g = f$, where $\pi : I_A \to \omega$ is the projection. The universal property of $k[[\omega']]$ gives a morphism of topological algebras

$$\phi : k[[\omega']] \longrightarrow A.$$

Now

$$\omega' \widehat{\otimes}_k k_0 \simeq \omega \widehat{\otimes}_k k_0 = \omega_{G_0/k_0},$$

and consequently

$$k[[\omega']] \widehat{\otimes}_k k_0 \simeq k_0[[\omega_{G_0/k_0}]].$$

The pair k_0, G_0 satisfies (iii), and the proof of (iii) $\Rightarrow$ (i) shows that

$$\phi_0 = \phi \widehat{\otimes}_k k_0$$

is invertible. Both $k[[\omega']]$ and A are projective pseudocompact k-modules, so 0.3.4 implies that ϕ itself is invertible. In particular, if I is the augmentation ideal of $k[[\omega']]$, then ϕ induces an isomorphism

$$\omega' = I/I^2 \xrightarrow{\sim} I_A/I_A^2 = \omega.$$

This completes the proof.

3.3.1. Corollary

Let S be a locally Noetherian scheme over $\mathbb{Q}$, and let G be an S-group scheme that is flat and locally of finite type.[129] *If $\omega_{G/S}$ is a locally free $\mathcal{O}_S$-module, then G is smooth over S at every point of the unit section.*

Proof. Let x be a point of the unit section, let s be its image in S, and denote the local rings at x and s by $\mathcal{O}_x$ and $\mathcal{O}_s$, respectively.[130] By hypothesis, $\mathcal{O}_s$ and $\mathcal{O}_x$ are Noetherian local rings. The maximal-ideal-adic topology on each of them therefore agrees with the "pseudocompact" topology defined by the ideals of finite colength. Write

$$\widehat{\mathcal{O}}_x \qquad \text{and} \qquad \widehat{\mathcal{O}}_s$$

for their completions in this topology. By EGA IV$_4$, 17.5.3, G is smooth over S at x if and only if $\widehat{\mathcal{O}}_x$ is formally smooth over $\widehat{\mathcal{O}}_s$, both algebras being endowed with their maximal-ideal-adic topologies. It is therefore enough to prove this last assertion.

The rings $\widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_s$ are the local rings at x and s of the formal varieties $\widehat{G}/\widehat{S}$ and $\widehat{S}$ defined in 1.2.6. Put

$$k = \widehat{\mathcal{O}}_s, \qquad H = \operatorname{Spf}(\widehat{\mathcal{O}}_x).$$

Then H is an infinitesimal formal k-variety. Since the formation of $\widehat{G}/\widehat{S}$ commutes with products (loc. cit.), H is in fact an infinitesimal formal k-group.

Let I be the augmentation ideal of $\mathcal{O}_x$. By hypothesis,

$$\omega_{G/S,x} = I/I^2$$

[128] One does not know a priori that ω is a projective pseudocompact k-module; this will follow from the argument. Compare the proof of (iv) $\Rightarrow$ (iii) in VIIA 7.4.

[129] The editors added the flatness hypothesis, which was omitted from the original.

[130] The editors supplied the details of the argument that follows.

is a free $\mathcal{O}_s$-module of finite rank n. Since $\mathcal{O}_s$ is Noetherian, the module $\omega_{H/k}$, which is the completion of $\omega_{G/S,x}$, is identified with

$$\omega_{G/S,x} \otimes_{\mathcal{O}_s} \widehat{\mathcal{O}}_s.$$

It is therefore a topologically free k-module of rank n. By the implication (iv)⇒(ii) of Theorem 3.3,

$$\widehat{\mathcal{O}}_x \simeq k[[\omega_{H/k}]] \simeq k[[t_1, \dots, t_n]].$$

Finally, the latter formal power-series algebra is formally smooth over k, by EGA 0_{IV}, 19.3.3. This proves the corollary.

Thus we recover a result obtained independently for group schemes locally of finite presentation over an arbitrary scheme S; cf. VI$_B$, 1.6.

3.3.2 Corollary

Corollary 3.3.2. *Let k be a field of characteristic 0. The functor*

$$\mathbf{L} \longmapsto \mathscr{G}(\mathbf{L})$$

is an equivalence from the category of k-Lie algebras to the category of infinitesimal formal k-groups. [131]

Indeed, as G ranges over the infinitesimal formal k-groups, Theorem 3.3 identifies the functor

$$G \longmapsto \mathscr{G}(\operatorname{Lie} G)$$

with the identity functor. Likewise, 3.2(ii) identifies

$$\mathbf{L} \longmapsto \operatorname{Lie}(\mathscr{G}(\mathbf{L})) = \operatorname{Prim} U(\mathbf{L})$$

with the identity functor on k-Lie algebras.

3.3.3

Continue to suppose that k is a field of characteristic 0. Let $\overline{k}$ be an algebraic closure of k, and let Γ be the topological Galois group of $\overline{k}$ over k.

[131] By 2.7, 2.2.1, and 1.3.6, an infinitesimal formal k-group G is "the same thing" as a connected cocommutative k-bialgebra H (cf. 2.9). The corollary is therefore equivalent to the theorem below, obtained independently by Kostant (cf. 2.9.2). J. W. Milnor and J. C. Moore had earlier proved it in [MM65] under the additional hypothesis that H be generated as an algebra by its primitive elements. Although published in 1965, their text had circulated before 1960 (see the review [Ja65]); the result is consequently often called the Cartier–Kostant–Milnor–Moore theorem.

Cartier–Kostant–Milnor–Moore Theorem. Let k be a field of characteristic 0. The functors

$$\mathbf{L} \longmapsto U(\mathbf{L}), \qquad H \longmapsto \operatorname{Prim}(H),$$

define an equivalence between the category of k-Lie algebras and the category of connected cocommutative k-bialgebras.

If k is an Artinian ring containing $\mathbb{Q}$, then 3.2(ii) and 3.3, combined with 2.7, 2.2.1, and 1.3.6, likewise show that

$$\mathbf{L} \longmapsto \mathscr{G}(\mathbf{L})$$

(respectively, $\mathbf{L} \mapsto U(\mathbf{L})$) is an equivalence from the category of k-Lie algebras flat over k to the category of topologically flat infinitesimal formal k-groups (respectively, to the category of connected cocommutative k-bialgebras flat over k).

For every formal k-group G, denote by $\mathbf{Aut}_k(G)$ the k-group functor assigning to every finite-dimensional commutative k-algebra C the group of automorphisms of the formal C-group

$$G\widehat{\otimes}_k C.$$

[132] Since k is a field, G is topologically flat over k. Equivalently, its affine algebra

$$A = \mathcal{A}(G)$$

is topologically flat over k, or its covariant bialgebra

$$H = \mathbf{H}(G)$$

is flat over k. The functor $\mathbf{Aut}_k(G)$ is therefore a formal k-group.

Indeed, $\mathbf{Aut}_k(G)$ is the fiber product of the following diagram (cf. Exposé I, 1.7.3):

$$\begin{tikzcd} & \underline{\mathrm{End}}_k(G) \times \underline{\mathrm{End}}_k(G) \arrow[d] \\ \mathrm{Spf}(k) \arrow[r] & \underline{\mathrm{End}}_k(G) \times \underline{\mathrm{End}}_k(G) \end{tikzcd}$$

The vertical arrow sends

$$(\phi, \psi) \longmapsto (\phi \circ \psi, \psi \circ \phi),$$

whereas the horizontal arrow is the unit section. It is therefore enough to show that the k-functor $\mathbf{End}_k(G)$ is represented by an object of $\mathrm{Vaf}\,/k$; by 1.1 and 0.4.2, this amounts to showing that it is left exact, or, equivalently, that it commutes with fiber products of k-algebras.

To give an element of $\mathbf{End}_k(G)(C)$ is equivalent to giving a k-algebra morphism

$$\phi : H \longrightarrow H \otimes_k C$$

that also respects the coalgebra structure, that is, for which the usual compatibility diagrams commute. Since H is flat over k, the functor

$$H \otimes_k -$$

commutes with fiber products of k-algebras. Thus $\mathbf{End}_k(G)$ is left exact and can be treated as a formal k-group.

If L is a k-Lie algebra and $G = \mathscr{G}(L)$, Theorem 3.3 shows that $\mathbf{Aut}_k(G)$ is isomorphic to the k-group functor $\mathbf{Aut}_k(L)$ which assigns to each finite-dimensional k-algebra C the group of automorphisms of the C-Lie algebra

$$C \otimes_k L.$$

Now let G be an arbitrary formal k-group. We saw in 2.5.2 that G decomposes canonically as a semidirect product of an étale formal group G_e and an infinitesimal formal group G_0. This semidirect product is determined by the data of

$$L = \mathrm{Lie}(G) = \mathrm{Lie}(G_0),$$

[132] The original stated: "This k-functor is manifestly left exact (H is topologically flat over k!)." The editors supplied the details that follow.

the Γ-group

$$M = G_e(\overline{k}) = G(\overline{k}),$$

and a morphism of formal k-groups

$$\Phi : G_e \longrightarrow \mathbf{Aut}_k(G_0) \simeq \mathbf{Aut}_k(L).$$

Such a morphism sends G_e into the étale part $\mathbf{Aut}_k(L)_e$ of $\mathbf{Aut}_k(L)$ (cf. 2.5.1). It is consequently determined by a morphism of Γ-groups

$$\phi = \Phi(\overline{k}) : M \longrightarrow (\mathbf{Aut}_k L)(\overline{k}) = \mathrm{Aut}_{\overline{k}}(L \otimes_k \overline{k}).$$

Let Γ act on $L \otimes_k \overline{k}$ by

$$\gamma(\ell \otimes t) = \ell \otimes \gamma(t).$$

Then ϕ makes M act on $L \otimes_k \overline{k}$ by automorphisms of $\overline{k}$-Lie algebras in such a way that

$$\phi(\gamma(m)) = \gamma \circ \phi(m) \circ \gamma^{-1}.$$

Equivalently,

$$\gamma(m)(\ell \otimes t) = \gamma\Big(m\big(\ell \otimes \gamma^{-1}(t)\big)\Big)$$

for every $m \in M$, $\gamma \in \Gamma$, and $\ell \otimes t \in L \otimes_k \overline{k}$. We express this last condition by saying that M acts on $L \otimes_k \overline{k}$ *compatibly with* Γ.

The situation can be summarized in a "highbrow" statement. A Γ-*Lie algebra over* k is a triple

$$(L, M, \phi)$$

consisting of a k-Lie algebra L, a Γ-group M, and an action ϕ of M on $L \otimes_k \overline{k}$ that is compatible with the actions of Γ on M and on $L \otimes_k \overline{k}$.

If (L, M, ϕ) and (L', M', ϕ') are two such Γ-Lie algebras, a morphism from the first to the second is a pair (f, θ), where

$$f : L \longrightarrow L'$$

is a morphism of k-Lie algebras and

$$\theta : M \longrightarrow M'$$

is a morphism of Γ-groups, such that

$$(f \otimes 1)(m \cdot x) = \theta(m) \cdot (f \otimes 1)(x)$$

for every $x \in L \otimes_k \overline{k}$ and $m \in M$. We have therefore proved:

Theorem. *Let k be a field of characteristic* 0*. The functor that assigns to a formal k-group G the triple*

$$(\mathrm{Lie}(G), G(\overline{k}), \Phi(\overline{k}))$$

is an equivalence from the category of formal k-groups to the category of Γ*-Lie algebras.*[133]

4. Phenomena specific to characteristic $p > 0$

Throughout Section 4, p denotes a prime number, k a pseudocompact ring of characteristic p, and

$$\pi : k \longrightarrow k, \qquad x \longmapsto x^p,$$

the continuous Frobenius endomorphism of k.

[133] In particular, when $k = \overline{k}$, this recovers the Cartier–Gabriel–Kostant theorem mentioned in 2.9.2.

4.1

Let X be a formal k-variety with affine algebra A. We write $X^{(p/k)}$, or simply $X^{(p)}$, for the formal k-variety

$$X \widehat{\otimes}_\pi k$$

obtained by the base change $\pi : k \to k$ (cf. 1.2.D). Its affine algebra is the completed tensor product

$$A \widehat{\otimes}_\pi k.$$

The continuous morphism

$$A \widehat{\otimes}_\pi k \longrightarrow A, \qquad a \widehat{\otimes}_\pi \lambda \longmapsto a^p \lambda,$$

therefore induces a morphism of formal k-varieties

$$\mathrm{Fr}(X/k) : X \longrightarrow X^{(p/k)}.$$

We call $\mathrm{Fr}(X/k)$ the *Frobenius morphism of X relative to k*, and below we shall often write Fr in place of $\mathrm{Fr}(X/k)$.

4.1.1

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Let S be a scheme over the prime field $\mathbb{F}_p$, and let $\eta : X \to S$ be an S-scheme. Write $\mathrm{fr}(S)$ for the "absolute" Frobenius morphism of S: it induces the identity on the underlying topological space and sends every section f of the structure sheaf to f^p; cf. VII$_\mathrm{A}$, 4.1. Let

$$X^{(p)} = X \times_{\mathrm{fr}(S)} S$$

be the corresponding S-scheme (VII$_\mathrm{A}$, loc. cit.); thus its structural morphism to S is $\mathrm{fr}(S) \circ \eta$.

Let $\widehat{S}$ be the formal scheme whose underlying topological space is the set of closed points of S, endowed with the discrete topology, and whose local ring at a closed point s is the separated completion

$$\mathcal{O}_{\widehat{S},s} = \widehat{\mathcal{O}}_{S,s}$$

of $\mathcal{O}_{S,s}$ for the linear topology defined by ideals of finite colength (cf. 1.2.6). Its affine algebra is therefore

$$k = \mathcal{A}(\widehat{S}) = \prod_{s \in S_{\mathrm{cl}}} \widehat{\mathcal{O}}_{S,s}.$$

Recall (loc. cit.) that $\widehat{X}/\widehat{S}$ denotes the formal k-variety consisting of those points $x \in X$ for which

$$[\kappa(x) : \kappa(s)] < \infty,$$

where s is the image of x in S. Its local ring at x is the separated completion

$$\mathcal{O}_{\widehat{X}/\widehat{S},x} = \widehat{\mathcal{O}}_{X,x}$$

[134] The editors corrected the original, which gave the inclusion

$$(\widehat{X}/\widehat{S})^{(p)} \subset \widehat{X^{(p)}}/\widehat{S}$$

instead of the reverse inclusion. They also note that this paragraph is not used below.

of $\mathcal{O}_{X,x}$ for the linear topology defined by those ideals I such that $\mathcal{O}_{X,x}/I$ has finite length as an $\mathcal{O}_{X,x}$-module, and hence also as an $\mathcal{O}_{S,s}$-module. By base change we can therefore form the formal k-variety

$$(\widehat{X}/\widehat{S})^{(p)} = (\widehat{X}/\widehat{S})\widehat{\otimes}_\pi k.$$

We can also consider the formal k-variety $\widehat{X^{(p)}}/\widehat{S}$. Its underlying set consists of the points $x \in X^{(p)} = X$ such that, if s denotes the image of x in S, the composite

$$\kappa(s) \xrightarrow{\mathrm{fr}} \kappa(s) \xrightarrow{\eta^\natural} \kappa(x)$$

makes $\kappa(x)$ a finite extension of $\kappa(s)$. In that case the same is true of

$$\eta^\natural : \kappa(s) \longrightarrow \kappa(x),$$

so x is a point of $\widehat{X}/\widehat{S}$, and

$$\mathcal{O}_{\widehat{X^{(p)}}/\widehat{S},x} = \widehat{\mathcal{O}}_{X,x}\widehat{\otimes}_\pi\widehat{\mathcal{O}}_{S,s} = \widehat{\mathcal{O}}_{X^{(p)},x}.$$

The second equality follows because the construction $X \mapsto \widehat{X}/\widehat{S}$ commutes with products (cf. 1.2.6).

It follows that

$$\widehat{X^{(p)}}/\widehat{S} \;\subset\; (\widehat{X}/\widehat{S})^{(p)}$$

as a formal subvariety. Equality holds if and only if, for every closed point s of S, the field $\kappa(s)$ has finite degree over $\kappa(s)^p$. This is the case, for example, if S is locally of finite type over a field κ such that

$$[\kappa : \kappa^p] < \infty.$$

4.1.2

Let G be a formal k-group. Since the functor

$$X \longmapsto X^{(p)} = X\widehat{\otimes}_\pi k$$

commutes with finite products, it carries formal k-groups to formal k-groups. Moreover, $\mathrm{Fr}(X/k)$ is functorial in X, so

$$\mathrm{Fr} : G \longrightarrow G^{(p)}$$

is a homomorphism of formal k-groups.

Define recursively

$$G^{(p^{n+1})} = \big(G^{(p^n)}\big)^{(p)}.$$

Then the same is true of the composite

$$\mathrm{Fr}^n = \mathrm{Fr}^n(G/k) : G \xrightarrow{\mathrm{Fr}} G^{(p)} \xrightarrow{\mathrm{Fr}} G^{(p^2)} \xrightarrow{\mathrm{Fr}} \cdots \xrightarrow{\mathrm{Fr}} G^{(p^n)}.$$

Let A be the affine algebra of G, and let I_A be its augmentation ideal.

Definitions.

(a) The kernel of Fr^n is denoted

$$\mathrm{Fr}_n G.$$

It follows directly from the definitions that $\mathrm{Fr}_n G$ is infinitesimal and has affine algebra

$$A/I_A^{\{p^n\}},$$

where $I_A^{\{p^n\}}$ is the closed ideal of A generated by the p^n-th powers of the elements of I_A.

(b) The group G is said to have *height at most* n if

$$I_A^{\{p^n\}} = 0;$$

equivalently,

$$\mathrm{Fr}_n\, G = G.$$

4.2

It is clear that the Lie algebra $\mathrm{Lie}(G)$ of a formal k-group G is a p-Lie subalgebra of $\mathbf{H}(G)$ (cf. 2.3). Indeed, we may immediately reduce to the case in which k is Artinian. Then $\mathrm{Lie}(G)$ is the subset of $\mathbf{H}(G)$ consisting of the elements x such that

$$\varphi_G(x \otimes 1 + 1 \otimes x) = \Delta_G(x),$$

with the notation of 2.3 and 2.6(c). Hence

$$\begin{aligned}\varphi_G(x^p \otimes 1 + 1 \otimes x^p) &= \varphi_G\big((x \otimes 1 + 1 \otimes x)^p\big)\\ &= \varphi_G(x \otimes 1 + 1 \otimes x)^p\\ &= \Delta_G(x)^p = \Delta_G(x^p).\end{aligned}$$

4.2.1

Conversely, every p-Lie algebra $\mathbf{L}$ over $\mathbf{O}_k$ defines a k-group functor. Let $U_p(\mathbf{L})$ denote the functor assigning to every object C of Alf/k the restricted enveloping algebra

$$U_p(\mathbf{L}(C))$$

of the p-Lie algebra $\mathbf{L}(C)$ over C ($\mathrm{VII_A}$, 5.3). By $\mathrm{VII_A}$, 5.4, $U_p(\mathbf{L})$ is a bialgebra over $\mathbf{O}_k$, and by 2.2 it therefore determines the k-group functor

$$\mathrm{Spf}^*\, U_p(\mathbf{L}),$$

which we henceforth denote by $\mathscr{G}_p(\mathbf{L})$.

Thus $\mathscr{G}_p(\mathbf{L})(C)$ is the group of elements

$$z \in U_p(\mathbf{L}(C))$$

having augmentation 1 and satisfying

$$\Delta_{U_p(\mathbf{L}(C))}(z) = z \otimes z.$$

4.2.2

Suppose that $\mathbf{L}$ is a flat p-Lie algebra over $\mathbf{O}_k$. Taking $\mathrm{VII_A}$, 5.3.3 into account, the argument of part (i) of Proposition 2.6.2 shows that $U_p(\mathbf{L})$ is flat over $\mathbf{O}_k$. Consequently, $\mathscr{G}_p(\mathbf{L})$ is a formal k-group topologically flat over k, with $U_p(\mathbf{L})$ as its covariant $\mathbf{O}_k$-bialgebra.

[135] By the proof of 2.6.2(iii), for every finite-length k-algebra C, the Lie algebra $\mathrm{Lie}(\mathscr{G}_p(\mathbf{L}))(C)$ is the set of primitive elements of

$$U_p(\mathbf{L})(C) = U_p(\mathbf{L}(C));$$

[135] The editors supplied the details that follow.

see also VII$_\mathrm{A}$, 3.2.3. Moreover, by VII$_\mathrm{A}$, 5.5.3, the canonical morphism $\mathbf{L} \to U_p(\mathbf{L})$ induces an isomorphism of p-Lie algebras

$$\tau_{p,\mathbf{L}} : \mathbf{L} \xrightarrow{\sim} \mathrm{Lie}(\mathscr{G}_p(\mathbf{L})).$$

Compare this with 3.1 in characteristic 0.

The formal group $\mathscr{G}_p(\mathbf{L})$ can be characterized by a universal property. Every morphism

$$h : \mathscr{G}_p(\mathbf{L}) \longrightarrow G$$

to a formal group G induces a morphism

$$\mathrm{Lie}(h) : \mathrm{Lie}(\mathscr{G}_p(\mathbf{L})) \longrightarrow \mathrm{Lie}(G).$$

Composing this morphism with $\tau_{p,\mathbf{L}}$ gives

$$h' : \mathbf{L} \longrightarrow \mathrm{Lie}(G).$$

Proposition. *Let k be a pseudocompact ring of characteristic $p > 0$, let G be a formal k-group, and let $\mathbf{L}$ be a flat p-Lie algebra over $\mathbf{O}_k$. The map $h \mapsto h'$ defined above is a bijection*

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(\mathscr{G}_p(\mathbf{L}), G) \xrightarrow{\sim} \mathrm{Hom}_{p\text{-}\mathrm{Lie}}(\mathbf{L}, \mathrm{Lie}(G)),$$

where the right-hand side is the set of morphisms of p-Lie algebras from $\mathbf{L}$ to $\mathrm{Lie}(G)$*.*

Proof.[136] We may immediately reduce to the case in which k is Artinian. Put

$$L = \mathbf{L}(k).$$

By 2.3.1,

$$\mathrm{Hom}_{\mathrm{Grf}\,/k}(\mathscr{G}_p(\mathbf{L}), G)$$

is in bijection with the set of unital algebra morphisms

$$h : U_p(L) \longrightarrow \mathbf{H}(G)$$

for which the following diagram commutes:

$$\begin{tikzcd}
U_p(L) \arrow[r, "h"] \arrow[d, "\Delta_{U_p(L)}"'] & \mathbf{H}(G) \arrow[dr, "\delta_G"] & \\
& & \mathbf{H}(G \times G) \\
U_p(L) \otimes U_p(L) \arrow[r, "h \otimes h"] & \mathbf{H}(G) \otimes \mathbf{H}(G) \arrow[ur, "\varphi_G"'] &
\end{tikzcd} .$$

The morphism h is determined by its restriction to L, which is a morphism of p-Lie algebras from L to the underlying p-Lie algebra of $\mathbf{H}(G)$. The commutativity of the diagram says precisely that h sends L into the subset of $\mathbf{H}(G)$ consisting of those x such that

$$\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

This subset is $\mathrm{Lie}(G)$, by 2.6(c). The proposition follows.

[136] The proof is identical to that of 2.6.3.

4.3

We now study the formal k-group $\mathscr{G}_p(\mathbf{L})$ in greater detail when $\mathbf{L}$ is a flat p-Lie algebra over $\mathbf{O}_k$.

For this purpose, first let C be a ring of characteristic p, and let L be a p-Lie algebra over C whose underlying module is free with basis

$$(x_i)_{i\in I}.$$

Let $\mathcal{P}$ be the set of families

$$n = (n_i)_{i\in I}$$

of nonnegative integers such that

$$0 \leq n_i < p$$

and $n_i = 0$ except possibly for finitely many indices. Endow I with a total order, and let

$$i_1 < i_2 < \cdots < i_r$$

be the indices for which $n_i \neq 0$. Put

$$|n| = n_{i_1} + \cdots + n_{i_r},$$

and

$$x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \cdots x_{i_r}^{n_{i_r}}, \qquad n! = (n_{i_1}!) \cdots (n_{i_r}!).$$

By VII$_{\mathrm{A}}$, 5.3.3, the monomials $x^n/n!$ form a basis of $U_p(L)$, and

$$\Delta\left(\frac{x^n}{n!}\right) = \sum_{m\leq n} \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!}. \tag{*}$$

The sum ranges over all $m \in \mathcal{P}$ such that $m \leq n$, that is,

$$m_i \leq n_i \qquad \text{for every } i \in I.$$

Formula (*) gives an immediate description of the natural filtration of $U_p(L)$ (cf. 1.3.6). Put

$$U = U_p(L),$$

let U^+ be the two-sided ideal generated by L, and put

$$U_0 = C \cdot 1_U.$$

As in 1.3.6, define a filtration of U by C-subcoalgebras by setting, for $n \geq 1$,

$$U_n = \left\{ u \in U \,\middle|\, \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+ \right\}.$$

Formula (*) then shows that U_n is the free C-module with basis the monomials x^m satisfying

$$|m| \leq n.$$

4.3.1

With the notation of 4.3, let us determine the elements $\xi \in U = U_p(L)$ such that

$$\varepsilon_U(\xi) = 1, \qquad \Delta_U(\xi) = \xi \otimes \xi.$$

Every element $\xi \in U$ has a unique expression

$$\xi = \sum_{n \in \mathcal{P}} \xi_n \frac{x^n}{n!}, \qquad \xi_n \in C.$$

The condition $\varepsilon_U(\xi) = 1$ gives

$$\xi_0 = 1,$$

where 0 denotes the element of $\mathcal{P}$ all of whose components vanish. The condition $\Delta_{U_p(L)}(\xi) = \xi \otimes \xi$ gives

$$\sum_{\substack{n \in \mathcal{P} \\ m \leq n}} \xi_n \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!} = \sum_{q,r \in \mathcal{P}} \xi_q \xi_r \frac{x^q}{q!} \otimes \frac{x^r}{r!}.$$

Equivalently,

$$\xi_{q+r} = \xi_q \xi_r \quad \text{if } q_i + r_i < p \text{ for every } i,$$

and

$$\xi_q \xi_r = 0 \quad \text{otherwise.}$$

Let ξ_i denote the coefficient ξ_n for which $n_i = 1$ and $n_j = 0$ whenever $j \neq i$. The preceding conditions say precisely that

$$\xi_n = \prod_i \xi_i^{n_i} \quad (n \in \mathcal{P}), \qquad \xi_i^p = 0 \quad (i \in I).$$

We obtain:

Proposition. *Let k be a local pseudocompact ring*[137] *of characteristic $p > 0$, let $\mathbf{L}$ be a flat p-Lie algebra over $\mathbf{O}_k$, and let C be an object of* Alf $/k$. *Write*

$$\sqrt[p]{0_C} = \{x \in C \mid x^p = 0\}.$$

There is a bijection, "functorial in C,"

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathscr{G}_p(\mathbf{L})(C).$$

Proof. By Remark 1.2.3.F, one may choose a basis

$$({}^C x_i)_{i \in I}$$

of $\mathbf{L}(C)$ over C in such a way that, for every morphism $\phi : C \to D$ in Alf $/k$, the map $\mathbf{L}(\phi)$ sends ${}^C x_i$ to ${}^D x_i$. The preceding calculation shows that

$$\sum_i ({}^C x_i) \otimes \xi_i \longmapsto \sum_{n \in \mathcal{P}} \left(\prod_i \xi_i^{n_i} \right) \frac{({}^C x)^n}{n!}$$

is a functorial bijection from $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ onto $\mathscr{G}_p(\mathbf{L})(C)$.

[137] The editors added the hypothesis that k be local, so that every topologically flat pseudocompact k-module is topologically free; see the proof.

4.3.2

"There is no Campbell–Hausdorff formula in characteristic $p > 0$." Let us explain. The functorial isomorphism of 4.3.1 depends on the choice of the bases $({}^{C}x_i)_{i\in I}$. Unlike the situation in 3.2, in characteristic 0, there is in general no bijection

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \longrightarrow \mathscr{G}_p(\mathbf{L})(C)$$

that is "functorial both in C and in $\mathbf{L}$." The following argument shows, for example, that no such isomorphism exists when k is a field containing all (p^2-1)-st roots of unity.

Choose $\mathbf{L}$ so that, for every $C \in \mathrm{Alf}\,/k$, the algebra $\mathbf{L}(C)$ is the abelian p-Lie algebra over C generated by an element x subject to

$$x^{(p^2)} = 0.$$

Thus

$$\mathbf{L}(C) = Cx \oplus Cx^{(p)}, \qquad U_p(\mathbf{L}(C)) \simeq C[x]/(x^{p^2}).$$

We shall show that the only functorial morphism

$$\chi_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \longrightarrow U_p(\mathbf{L}(C))$$

which is compatible with the endomorphisms of $\mathbf{L}$ and sends $\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$ into $\mathscr{G}_p(\mathbf{L})(C)$ is the constant morphism with value 1.

Let

$$\psi_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\ \sim\ } \mathscr{G}_p(\mathbf{L})(C)$$

be the bijection of 4.3.1, namely

$$x \otimes a + x^{(p)} \otimes b \longmapsto \sum_{0\le i,j<p} a^i b^j x^{i+pj}.$$

Composing χ_C with ψ_C^{-1} gives a morphism, functorial in C,

$$\begin{aligned} \theta_C : \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} &\longrightarrow \mathbf{L}(C) \otimes_C \sqrt[p]{0_C}, \\ x \otimes a + x^{(p)} \otimes b &\longmapsto x \otimes P(a,b) + x^{(p)} \otimes Q(a,b). \end{aligned}$$

Functoriality in C implies that $P(a,b)$ and $Q(a,b)$ are the values at (a,b) of two polynomials

$$P, Q \in k[X,Y]$$

which may be assumed to have degree less than p in each of X and Y.[138]

Let $\alpha \in k$, and let ℓ_α be the endomorphism of the p-Lie algebra $\mathbf{L}$ sending

$$x \longmapsto \alpha x, \qquad x^{(p)} \longmapsto \alpha^p x^{(p)}.$$

Then $U_p(\ell_\alpha)$ sends x to αx, and hence sends x^i to $\alpha^i x^i$ for $i < p^2$. The following square commutes:

$$\begin{tikzcd} \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \arrow[r, "\psi_C"] \arrow[d, "\ell_\alpha(C)\otimes_C \mathrm{id}"] & U_p(\mathbf{L}(C)) \arrow[d, "U_p(\ell_\alpha)(C)"] \\ \mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \arrow[r, "\psi_C"] & U_p(\mathbf{L}(C)) \end{tikzcd}$$

[138] The editors expanded the original in the argument that follows.

The hypothesis

$$\chi_C \circ (\ell_\alpha(C) \otimes \mathrm{id}) = U_p(\ell_\alpha)(C) \circ \chi_C$$

therefore gives

$$P(\alpha a, \alpha^p b) = \alpha P(a,b), \qquad Q(\alpha a, \alpha^p b) = \alpha^p Q(a,b).$$

Write

$$P = \sum_{i,j<p} \lambda_{ij} X^i Y^j, \qquad Q = \sum_{i,j<p} \mu_{ij} X^i Y^j,$$

and take

$$C = k[X,Y]/(X^p, Y^p).$$

If $\lambda_{ij} \neq 0$, then

$$\alpha^{i-1+pj} = 1;$$

if $\mu_{ij} \neq 0$, then

$$\alpha^{i+p(j-1)} = 1.$$

Taking α to be a primitive root of unity of order $p^2 - 1$, we conclude that

$$P = \lambda X, \qquad Q = \mu Y$$

for some $\lambda, \mu \in k$. Consequently,

$$\chi_C(x \otimes a + x^{(p)} \otimes b) = \sum_{0 \leq i,j<p} (\lambda a)^i (\mu b)^j x^{i+pj}.$$

Finally, consider the endomorphism f of $\mathbf{L}$ sending

$$x \longmapsto x + x^{(p)}.$$

Take $C = k[a]/(a^3)$ and compare the coefficients of x^p and x^{p+1} in

$$(\chi_C \circ f(C))(x \otimes a) \qquad \text{and} \qquad (U_p(f)(C) \circ \chi_C)(x \otimes a).$$

This gives

$$\lambda = \mu, \qquad \lambda^2 a^2 = 0,$$

and hence $\lambda = \mu = 0$, as claimed.

4.4 Theorem

Theorem 4.4. *Let k be a pseudocompact ring of characteristic $p > 0$, and let G be a formal k-group. The following assertions are equivalent:*

(i) *There exists a p-Lie algebra $\mathbf{L}$, flat over $\mathcal{O}_k$, such that G is isomorphic to $\mathscr{G}_p(\mathbf{L})$. In this case, $\mathbf{L} = \mathrm{Lie}(G)$ by 4.2.2.*

(ii) *There exists a projective pseudocompact k-module ω such that the affine algebra of G is isomorphic to the quotient of $k[[\omega]]$ by the closed ideal generated by the elements x^p, $x \in \omega$. In this case, $\omega \simeq \omega_{G/k}$.*

(iii) *The group G has height at most 1, and $\omega_{G/k}$ is a projective pseudocompact k-module.*

(iv) *The group G has height at most 1, and is topologically flat over k.*

Proof of (i) $\Rightarrow$ (ii). Set

$$\omega = \Gamma^*(\mathbf{L})$$

(cf. 1.2.3.D), and let A be the quotient of $k[[\omega]]$ by the closed ideal generated by the elements x^p, $x \in \omega$. Every continuous morphism

$$h : A \longrightarrow C, \qquad C \in \mathrm{Alf}\,/k,$$

is determined by its restriction h' to ω, and this restriction maps ω into $\sqrt[p]{0_C}$. Thus there is a canonical bijection from

$$\mathrm{Hom}_{\mathrm{Alp}\,/k}(A, C)$$

onto the set

$$\mathrm{Hom}_c(\omega, \sqrt[p]{0_C})$$

of continuous k-linear maps from ω into $\sqrt[p]{0_C}$. The latter set is canonically isomorphic to

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C}$$

(see the proof of 3.3). The implication (i) $\Rightarrow$ (ii) therefore follows from the functorial bijection

$$\mathbf{L}(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\ \sim\ } \mathscr{G}_p(\mathbf{L})(C)$$

established in 4.3.1.

For the remaining implications, consult the analogous proofs of Theorems 3.3 and VIIA 7.4.1.

Remark 4.4.A. Let G be an infinitesimal formal k-group, with affine algebra A, such that

$$\omega_{G/k} = I_A / I_A^2$$

is a projective pseudocompact k-module. Then the projection

$$I_A \longrightarrow \omega_{G/k}$$

has a section

$$\tau : \omega_{G/k} \longrightarrow I_A,$$

and τ induces a surjective continuous morphism of algebras

$$\psi : k[[\omega_{G/k}]] \longrightarrow A;$$

compare the proof of the implication (iii) $\Rightarrow$ (i) in 3.3.[139]

4.4.1 Corollary

Corollary 4.4.1. *If k is a field of characteristic $p > 0$, the functor*

$$\mathbf{L} \longmapsto \mathscr{G}_p(\mathbf{L})$$

is an equivalence from the category of p-Lie algebras over k to the category of formal k-groups of height at most 1.

Indeed, as G ranges over the formal groups of height at most 1, Theorem 4.4 identifies the functor

$$G \longmapsto \mathscr{G}_p(\mathrm{Lie}\, G)$$

[139] The editors added this remark, which is used in 4.4.2.

with the identity functor. Likewise, 4.2.2 identifies the functor

$$\mathbf{L} \longmapsto \operatorname{Lie} \mathscr{G}_p(\mathbf{L})$$

with the identity functor (see also VIIA 5.5.3).[140]

4.4.2

Continue to assume that k is a field of characteristic p. Let G be an infinitesimal formal k-group, with affine algebra A. Since G is infinitesimal, every open ideal of A contains

$$I_A^{\{p^n\}}$$

for all sufficiently large n. Thus A is the projective limit of the affine algebras

$$A/I_A^{\{p^n\}}$$

of the groups $\operatorname{Fr}_n G$ (cf. 4.1.2). Equivalently, every infinitesimal formal k-group is an inductive limit of formal k-groups of finite height.

Suppose that G has height at most n, and write

$$H = G/\operatorname{Fr} G.$$

[141] By 2.4 and 2.4.1, the morphism

$$\operatorname{Fr} : G \longrightarrow G^{(p)}$$

factors as an epimorphism

$$\pi : G \longrightarrow H$$

followed by a monomorphism

$$i : H \longrightarrow G^{(p)}.$$

We therefore have the following commutative diagram:

$$\begin{tikzcd}
G \arrow[r, "\pi"] \arrow[dr, "\mathrm{Fr}"'] & H \arrow[r, "\mathrm{Fr}^{n-1}"] \arrow[d, "i"] & H^{(p^{n-1})} \arrow[d, "i^{(p^{n-1})}"] \\
& G^{(p)} \arrow[r, "\mathrm{Fr}^{n-1}"] & G^{(p^n)}
\end{tikzcd}$$

Since the functor $X \mapsto X^{(p)}$ commutes with fiber products, the morphism

$$i^{(p^{n-1})}$$

is again a monomorphism. Moreover, $\operatorname{Fr}^n(G/k)$ is zero and π is an epimorphism. Hence

$$\operatorname{Fr}^{n-1}(G^{(p)}/k) \circ i = i^{(p^{n-1})} \circ \operatorname{Fr}^{n-1}(H/k)$$

is zero. Because $i^{(p^{n-1})}$ is a monomorphism,

$$\operatorname{Fr}^{n-1}(H/k) = 0,$$

---

[140] If k is an Artinian ring of characteristic $p > 0$, the same proof gives an equivalence between the category of p-Lie algebras flat over k and the category of formal k-groups of height at most 1 that are topologically flat over k.

[141] The editors supplied the details of the original argument in what follows.

so H has height at most $n-1$.

It follows that every formal k-group of finite height has a composition series whose quotients have height at most 1, and therefore can be described by p-Lie algebras over k.

Finally, the affine algebra A of G is a quotient of

$$k[[\omega_{G/k}]]$$

by Remark 4.4.A. If $\omega_{G/k}$ is finite-dimensional over k, all the algebras

$$A/I_A^{\{p^n\}}$$

are finite-dimensional k-vector spaces. Consequently, every infinitesimal formal group G over a field of characteristic $p>0$ for which $\omega_{G/k}$ is finite-dimensional over k is an inductive limit of finite formal groups (that is, groups of finite length; cf. 1.2.6).

5. Homogeneous spaces of infinitesimal formal groups over a field

5.0

For simplicity, suppose that k is a field.[142] Let G be a formal k-group with affine algebra

$$A = \mathscr{A}(G).$$

Let A^+ be the augmentation ideal of A. For $a \in A$, write

$$\overline{a} = a - \varepsilon_A(a)1_A$$

for its projection onto A^+. Let F be a formal subgroup of G, defined by a closed ideal J of A, and let

$$\pi : A \longrightarrow A/J$$

be the projection. Write

$$\overline{\pi} : A \longrightarrow A^+/J$$

for the composite of the projections

$$A \longrightarrow A^+ \longrightarrow A^+/J.$$

For every $a \in A$, the projection of $\Delta_A(a)$ onto $A\widehat{\otimes}A^+$ is

$$\Delta_A(a) - a\widehat{\otimes}1.$$

By Theorem 2.4, the quotient formal k-variety $X = G/F$ exists. Its affine algebra is

$$\begin{aligned} B &= \mathrm{Eq}\left(A \underset{(\mathrm{id}_A \widehat{\otimes}\pi)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A\widehat{\otimes}_k(A/J) \right) \\ &= \{a \in A \mid (\mathrm{id}_A \widehat{\otimes}\pi)\Delta_A(a) = a\widehat{\otimes}\pi(1)\} \\ &= \ker\big((\mathrm{id}_A \widehat{\otimes}\overline{\pi})\Delta_A\big)\,, \end{aligned}$$

[142] The editors added subsection 5.0 in order to express Proposition 5.1 below in the language of cocommutative Hopf algebras and to cite results subsequently obtained in this direction.

where

$$\tau_1(a) = a\widehat{\otimes}\pi(1).$$

Equivalently, B is the subalgebra of A consisting of the elements φ such that

$$\varphi(gh) = \varphi(g)$$

for every $C \in \mathrm{Alf}\,/k$, $g \in G(C)$, and $h \in F(C)$. For $g, g' \in G(C)$ and $h \in F(C)$, one has

$$\varphi(gg'h) = \varphi(gg').$$

Thus $\Delta_A(\varphi)$ belongs to the kernel of

$$\mathrm{id}_A \widehat{\otimes}_k ((\mathrm{id}_A \widehat{\otimes} \overline{\pi})\Delta_A),$$

which equals $A\widehat{\otimes}_k B$, since A is topologically flat over k. Consequently,

$$\Delta_A(B) \subset A\widehat{\otimes}_k B;$$

that is, the closed subalgebra B is also a left coideal.

Conversely, B determines F, because Corollary 2.4.1 gives

$$J = AB^+, \qquad B^+ = B \cap A^+;$$

in other words, J is the closed ideal generated by B^+. We therefore obtain an injective map

$$\mathcal{Q} : \mathcal{F} \longrightarrow \mathcal{B}$$

from the set $\mathcal{F}$ of formal subgroups of G to the set $\mathcal{B}$ of closed subalgebras of A that are left coideals. Proposition 5.1 below shows that $\mathcal{Q}$ is bijective when G is infinitesimal. In fact, it is bijective for every formal k-group G.

Recall from 2.2.1 that the functor

$$G \longmapsto \mathbf{H}(G)$$

is an equivalence from formal k-groups to cocommutative k-Hopf algebras. If F is defined by the closed ideal $J \subset A$, then the Hopf subalgebra

$$\mathbf{H}(F) \subset H = \mathbf{H}(G)$$

is the orthogonal complement of J for the duality

$$A = H^*, \qquad H = A^\dagger$$

(cf. 0.2.2). Conversely, if B is a closed subalgebra of A that is also a left coideal, then

$$I = B^\perp$$

is a coideal and a left ideal of H; explicitly,

$$\Delta_H(I) \subset I \otimes H + H \otimes I, \qquad \varepsilon_H(I) = 0.$$

Let $\mathcal{H}$ be the set of Hopf subalgebras of H, and let $\mathcal{I}$ be the set of left ideals of H that are also coideals. For $I \in \mathcal{I}$, denote by

$$\pi_I : H \longrightarrow H/I$$

the projection, and by

$$\overline{\pi}_I : H \longrightarrow H^+/I$$

the composite

$$H \longrightarrow H^+ \longrightarrow H^+/I,$$

where H^+ is the augmentation ideal of H.

Let K be a Hopf subalgebra of H, and put

$$K^+ = K \cap H^+.$$

If F is the formal subgroup corresponding to K, then

$$J = K^\perp$$

and A^+/J identifies with the dual of K^+. Since $B = \mathcal{Q}(F)$ is the kernel of

$$A \xrightarrow{\Delta_A} A\widehat{\otimes}_k A \xrightarrow{\mathrm{id}\,\widehat{\otimes}\overline{\pi}} A\widehat{\otimes}_k(A^+/J),$$

duality identifies $\mathcal{Q}$ with the map

$$\Phi : \mathcal{H} \longrightarrow \mathcal{I}, \qquad K \longmapsto HK^+.$$

Indeed, HK^+ is the image of

$$H \otimes_k K^+ \xrightarrow{\mathrm{id} \otimes \mathrm{can}} H \otimes_k H \xrightarrow{m_H} H,$$

and is both a left ideal and a coideal.

Similarly, the map that assigns AB^+ to $B \in \mathcal{B}$ corresponds by duality to

$$\Psi : \mathcal{I} \longrightarrow \mathcal{H},$$

where

$$\Psi(I) = \{x \in H \mid (\mathrm{id}_H \otimes \pi_I)\Delta_H(x) = x \otimes \pi_I(1)\}.$$

Theorem 5.0.1. *Let k be a field and H a cocommutative Hopf algebra. The maps Φ and Ψ above are mutually inverse bijections between the Hopf subalgebras of H and the left ideals of H that are coideals.*

This theorem was first proved by K. Newman ([Ne75], Theorem 4.1, where the word "cocommutative" was omitted). Newman's proof uses the Cartier–Gabriel–Kostant theorem (cf. 2.9) to reduce to the case in which H is connected, and then uses the existence of a "Sweedler basis" in that case ([Sw67], Theorem 3), a result dual to the Dieudonné–Cartier Theorem 5.2.2 below. A second, shorter proof was given by H. J. Schneider ([Sch90], Theorem 4.15). A. Masuoka subsequently generalized the result to the case in which only the coradical H_0 of H, the sum of its simple subcoalgebras, is assumed commutative ([Ma91], Theorem 1.3 (3)).

Finally, for a commutative k-Hopf algebra corresponding to an affine k-group scheme G, one cannot expect an analogue of Theorem 5.0.1 without additional hypotheses: for a k-subgroup $F \subset G$, the quotient G/F need not be affine. Nevertheless, M. Takeuchi established analogous bijections. In [Tak72], Theorem 4.3, invariant k-subgroups F correspond to subalgebras $B \subset \mathcal{O}(G)$ satisfying

$$\Delta(B) \subset A \otimes B$$

and stable under the antipode. In [Tak79], Theorem 3, the k-subgroups F for which G/F is affine correspond to such subalgebras B for which $B \to A$ is faithfully flat.

5.1

Let k be a pseudocompact ring.[143] Let G be a topologically flat infinitesimal formal k-group, let A be its affine algebra, and let $B \subset A$ be a closed subalgebra. Put

$$X = \mathrm{Spf}(B),$$

and let

$$r : G \longrightarrow X$$

be the epimorphism induced by the inclusion $B \subset A$. We shall determine when r makes X the quotient of G on the right by a subgroup H (cf. 2.4).[144]

Proposition 5.1. *Let G be a topologically flat infinitesimal formal k-group, let A be its affine algebra, let I_A be the augmentation ideal of A, and let B be a closed subalgebra of A. Put*

$$I_B = B \cap I_A, \qquad J_B = AI_B.$$

Assume that A, and also

$$J_B^n / J_B^{n+1}$$

for every $n \geq 0$, are topologically flat over k. Then the following two assertions are equivalent:

(i) *For every $x \in B$,*

$$\Delta_A(x) - x \widehat{\otimes} 1 \in A \widehat{\otimes}_k J_B.$$

(ii) *Let*

$$\pi : A \longrightarrow A/J_B$$

be the projection, and put

$$\tau_1(a) = a \widehat{\otimes} 1.$$

The sequence

$$B \longrightarrow A \underset{(\mathrm{id}_A \widehat{\otimes} \pi)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A \widehat{\otimes}_k (A/J_B) \tag{*}$$

is exact. Equivalently, B is the set of all $x \in A$ for which

$$\Delta_A(x) - x \widehat{\otimes} 1 \in A \widehat{\otimes}_k J_B.$$

In this case,

$$H = \mathrm{Spf}(A/J_B)$$

is a formal subgroup of G, and the sequence

$$G \times H \underset{\lambda}{\overset{\mathrm{pr}_1}{\rightrightarrows}} G \longrightarrow \mathrm{Spf}(B) \tag{**}$$

is exact, where λ is the restriction of the multiplication of G to $G \times H$. Thus

$$\mathrm{Spf}(B) \simeq G/H.$$

[143] The original assumes in 5.1 that k is a field. In fact, that hypothesis can be replaced by flatness hypotheses; the editors modified 5.1–5.1.5 accordingly.

[144] The editors replaced "left" by "right" and modified the statement of Proposition 5.1 so as to display both the equivalent conditions (i), (ii) and the conclusion $\mathrm{Spf}(B) \simeq G/H$ more clearly.

Proof. Put
$$\overline{A} = A/J_B, \qquad H = \mathrm{Spf}(\overline{A}).$$
Then H is a formal subvariety of G. Since $J_B \subset I_A$, the augmentation ε_A induces a continuous k-algebra morphism
$$\overline{\varepsilon} : \overline{A} \longrightarrow k.$$

If (i) holds, then
$$\Delta_A(I_B) \subset I_B \widehat{\otimes} 1 + A \widehat{\otimes} J_B.$$
Therefore Δ_A induces, after passage to the quotient, a diagonal morphism
$$\overline{\Delta} : \overline{A} \longrightarrow \overline{A} \widehat{\otimes}_k \overline{A}.$$
The maps $\overline{\Delta}$ and $\overline{\varepsilon}$ make H a formal submonoid of G. Since G is infinitesimal, so is H; Proposition 2.7 then shows that H is a formal subgroup of G. By the definition of G/H (cf. 2.4), assertion (ii) implies the final assertion of the proposition.

It is clear that (ii) implies (i). The proof of the converse occupies 5.1.1–5.1.5.

5.1.1

Consider the following category $\mathcal{C}$. An object of $\mathcal{C}$ is a pair (A, J) consisting of a profinite k-algebra A and a closed ideal $J \subset A$. A morphism
$$\psi : (A, J) \longrightarrow (A', J')$$
in $\mathcal{C}$ is a continuous k-algebra homomorphism
$$A \longrightarrow A'$$
that maps J into J'.

The assignment
$$(A, J) \longmapsto (\mathrm{Spf}(A/J), \mathrm{Spf}(A))$$
is an antiequivalence from $\mathcal{C}$ to the category of pairs (Z, Y) consisting of a formal k-variety Y and a formal subvariety $Z \subset Y$. A morphism
$$\varphi : (Z, Y) \longrightarrow (Z', Y')$$
is a morphism of formal k-varieties
$$Y \longrightarrow Y'$$
that maps Z into Z'.

A cogroup structure on an object (A, J) of $\mathcal{C}$ consists of a formal-group structure on $\mathrm{Spf}(A)$ satisfying the following conditions (with the notation of 2.1):
$$\begin{aligned} &(1) \quad \Delta_A(J) \subset J \widehat{\otimes}_k A + A \widehat{\otimes}_k J, \\ &(2) \quad \varepsilon_A(J) = 0, \\ &(3) \quad c_A(J) \subset J. \end{aligned}$$
These conditions also say that
$$H = \mathrm{Spf}(A/J)$$
is a formal subgroup of
$$G = \mathrm{Spf}(A).$$

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Suppose in addition that A is local; equivalently, $\mathrm{Spf}(A)$ is an infinitesimal formal group. If $J \neq A$, then conditions (2) and (3) follow from (1). Indeed, a proper closed ideal J of A is contained in the augmentation ideal I_A, so (2) holds. Moreover,

$$M = \mathrm{Spf}(A/J)$$

is a formal submonoid of G. Since G is infinitesimal, Proposition 2.7 shows that M is a formal subgroup of G; hence (3) also holds.

5.1.2

Denote by $\mathbf{Alpg}/k$ the category of *graded profinite k-algebras*. An object of this category consists of a sequence

$$A_0, A_1, \ldots, A_n, \ldots$$

of pseudocompact k-modules, together with a profinite algebra structure on

$$\prod_{n \geq 0} A_n$$

such that

$$A_n A_m \subset A_{m+n}.$$

Here A_n is identified with a subset of $\prod_{i \geq 0} A_i$ by means of the canonical injection. A morphism

$$\psi : (A_n) \longrightarrow (B_n)$$

is a sequence of continuous linear maps

$$\psi_n : A_n \longrightarrow B_n$$

such that

$$\psi_{m+n}(aa') = \psi_m(a)\psi_n(a')$$

for $a \in A_m$ and $a' \in A_n$.

Definitions. Two graded profinite k-algebras (A_n) and (B_n) clearly have a coproduct[146] in $\mathbf{Alpg}/k$. Its n-th component is

$$\prod_{i=0}^{n} A_i \widehat{\otimes}_k B_{n-i},$$

the product of the pseudocompact k-modules

$$A_i \widehat{\otimes}_k B_{n-i}.$$

This coproduct will be denoted by

$$(A_n) \widehat{\otimes}_k (B_n).$$

[145] If (A', J') is a second cogroup of $\mathcal{C}$, corresponding to a pair $H' \subset G'$ of formal k-groups, then giving a cogroup morphism

$$(A', J') \longrightarrow (A, J)$$

is equivalent to giving a morphism of formal k-groups

$$G \longrightarrow G'$$

that maps H into H'.

[146] The editors replaced “direct sum” by “coproduct”.

A cogroup structure on an object (A_n) of $\mathbf{Alpg}/k$ is therefore the datum of continuous k-linear maps

$$\Delta_n : A_n \longrightarrow \prod_{i=0}^{n} A_i \widehat{\otimes}_k A_{n-i}, \qquad \varepsilon : A_0 \longrightarrow k,$$

which induce on $\prod_{n\geq 0} A_n$, upon setting $\varepsilon(A_i) = 0$ for $i \geq 1$, a cogroup structure in $\mathbf{Alp}/k$.

Finally, for every object (A, J) of $\mathcal{C}$, denote by

$$\mathrm{Gr}_J(A)$$

the graded profinite algebra associated with the filtration of A by the closures $\overline{J^n}$ of the powers of J. Thus

$$\mathrm{Gr}_J(A)_n = \overline{J^n}/\overline{J^{n+1}},$$

and the multiplication of $\mathrm{Gr}_J(A)$ is induced by that of A.

Lemma. [147] *Let U, V be two pseudocompact k-modules, with U topologically flat, and let*

$$U = U_0 \supset U_1 \supset \cdots, \qquad V = V_0 \supset V_1 \supset \cdots$$

be two decreasing sequences of closed k-submodules. Filter the completed tensor product

$$W = U \widehat{\otimes}_k V$$

by means of the closed submodules

$$W_n = U_n \widehat{\otimes}_k V_0 + U_{n-1} \widehat{\otimes}_k V_1 + \cdots + U_0 \widehat{\otimes}_k V_n.$$

Suppose that every U_i/U_{i+1} is topologically flat over k, so that U/U_n, and hence also U_n, is topologically flat for every n. Then, for every n, there is an isomorphism

$$W_n/W_{n+1} \simeq \bigoplus_{i+j=n} (U_i/U_{i+1}) \widehat{\otimes}_k (V_j/V_{j+1}).$$

Proof. For every $i, j \geq 0$, put

$$W_{i,j} = U_i \widehat{\otimes} V_j, \qquad \overline{W}_{i,j} = (U_i/U_{i+1}) \widehat{\otimes}_k (V_j/V_{j+1}).$$

We prove by induction on n that the natural map

$$\pi_n : W_n \longrightarrow \bigoplus_{i+j=n} \overline{W}_{i,j}$$

is surjective and that the inclusion

$$W_{n+1} \subset \ker(\pi_n)$$

is an equality. For $n = 0$, the projection

$$\pi_0 : U_0 \widehat{\otimes}_k V_0 \longrightarrow (U_0/U_1) \widehat{\otimes}_k (V_0/V_1)$$

is surjective. Since U_0, U_0/U_1, and hence U_1, are topologically flat over k, we have

$$\ker(\pi_0) = U_0 \widehat{\otimes}_k V_1 + U_1 \widehat{\otimes}_k V_0,$$

[147] In the original, the lemma is stated when k is a field, and in that case the proof is left to the reader.

and moreover
$$(U_0\widehat{\otimes}_k V_1) \cap (U_1\widehat{\otimes}_k V_0) = U_1\widehat{\otimes}_k V_1.$$

Suppose now that $n > 0$ and that the result has been proved for $n-1$. Put
$$M_0 = U_0\widehat{\otimes}_k V_n, \qquad S_0 = \sum_{i=1}^{n} U_i\widehat{\otimes}_k V_{n-i}.$$

We have $S_0 \subset U_1\widehat{\otimes}_k V_0$. Applying the preceding argument to $V_0 \supset V_n$ in place of $V_0 \supset V_1$, we obtain
$$M_0 \cap S_0 \subset (U_0\widehat{\otimes}_k V_n) \cap (U_1\widehat{\otimes}_k V) = U_1\widehat{\otimes}_k V_n.$$

It follows that
$$M_0 \cap S_0 = U_1\widehat{\otimes}_k V_n.$$

Since $W_n = M_0 + S_0$, we obtain a commutative diagram with exact rows, where
$$U'_i = U_{i+1}, \qquad W'_{i,n-1-i} = W_{i+1,n-1-i} \quad (0 \leq i \leq n-1):$$

$$\begin{tikzcd}
0 \arrow[r] & S_0 \arrow[r] \arrow[d, "\pi'_{n-1}"'] & W_n \arrow[r] \arrow[d, "\pi_n"'] & (U_0/U_1)\widehat{\otimes}_k V_n \arrow[r] \arrow[d, "p"'] & 0 \\
0 \arrow[r] & \bigoplus_{i=0}^{n-1} \overline{W}'_{i,n-1-i} \arrow[r] & \bigoplus_{i=0}^{n} \overline{W}_{i,n-1} \arrow[r] & \overline{W}_{0,n} \arrow[r] & 0
\end{tikzcd}$$

Then p is surjective, with kernel
$$(U_0/U_1)\widehat{\otimes}_k V_{n+1}.$$

Moreover, by the induction hypothesis applied to the sequence (U'_i), the map π'_{n-1} is surjective, with kernel
$$W'_n = \sum_{i=1}^{n} W_{i,n+1-i}.$$

It follows that π_n is surjective and that
$$W_{n+1} \subset \ker(\pi_n)$$
is an equality. This proves the lemma.

Let us return to an object (A, J) of $\mathcal{C}$. By 0.2.G, the hypothesis that every
$$\overline{J^n}\big/\overline{J^{n+1}}$$
is topologically flat over k is equivalent to saying that $\mathrm{Gr}_J(A)$ is topologically flat over k.

Corollary. *Let $\mathcal{P}$ be the full subcategory of $\mathcal{C}$ consisting of the objects (A, J) such that A and $\mathrm{Gr}_J(A)$ are topologically flat over k. Then the functor*
$$\mathcal{P} \longrightarrow \mathbf{Alpg}/k, \qquad (A, J) \longmapsto \mathrm{Gr}_J(A),$$
commutes with finite coproducts and hence carries a cogroup of $\mathcal{P}$ to a cogroup of $\mathbf{Alpg}/k$.

In particular, if k is a field, then for every cogroup (A, J) of $\mathcal{C}$, the algebra $\mathrm{Gr}_J(A)$ is a cogroup of $\mathbf{Alpg}/k$. [148]

[148]Thus, to a pair $H \subset G$ of formal k-groups one associates the "formal completion $\widehat{G}_H$ of G along H", which is a formal k-group. Moreover, we shall see in 5.1.3–5.1.4 that the inclusion $\sigma : H \hookrightarrow \widehat{G}_H$ has a retraction $\pi : \widehat{G}_H \to H$, and that the formal k-group $N = \ker(\pi)$ identifies, as a formal variety, with the completion of the homogeneous space G/H along the unit section. This will be useful in 5.2.2.

5.1.3

Identify every profinite k-algebra Γ with the graded profinite k-algebra $(\Gamma_n)_{n\geq 0}$ such that

$$\Gamma_0 = \Gamma, \qquad \Gamma_n = 0 \quad \text{if } n > 0.$$

In particular, if $(A_n)_{n\geq 0}$ is a graded profinite k-algebra, we shall regard A_0, without distinction, as either a profinite k-algebra or a graded profinite k-algebra. Denote by

$$\rho : (A_n) \longrightarrow A_0$$

the morphism of $\mathbf{Alpg}/k$ such that

$$\rho_0 = \mathrm{id}_{A_0}, \qquad \rho_n = 0 \quad \text{if } n > 0.$$

Similarly,

$$\tau : A_0 \longrightarrow (A_n)$$

will denote the section of ρ such that

$$\tau_0 = \mathrm{id}_{A_0}, \qquad \tau_n = 0 \quad \text{if } n > 0.$$

Every cogroup structure on $(A_n)_{n\in\mathbb{N}}$ induces a cogroup structure on A_0 for which ρ and τ are cogroup homomorphisms. In this case, let I_0 be the augmentation ideal of A_0, and put

$$A'_n = A_n/I_0A_n \qquad (n \geq 0),$$

so that $A'_0 = k$.

Then $(A'_n)_{n\in\mathbb{N}}$ is a cogroup in $\mathbf{Alpg}/k$. Indeed, since

$$A_0 = I_0 \oplus k \cdot 1,$$

the direct summand

$$I_0 \widehat{\otimes}_k A_n \simeq I_0 A_n$$

of A_n is defined for every n. Since τ is a section of ρ, it follows from 2.4.A that the cogroup $(A_n)_{n\in\mathbb{N}}$ is the "semidirect coproduct" of A_0 and the cogroup $(A'_n)_{n\in\mathbb{N}}$.

More precisely, $(A'_n)_{n\in\mathbb{N}}$ is isomorphic, as an object of $\mathbf{Alpg}/k$, to the kernel of the following pair:

$$(A_n) \underset{(\mathrm{id}\,\widehat{\otimes}\rho)\Delta}{\overset{\tau_1}{\rightrightarrows}} (A_n)\widehat{\otimes}_k A_0$$

Here

$$\Delta : (A_n) \longrightarrow (A_n)\widehat{\otimes}(A_n)$$

is the comultiplication of (A_n), and

$$\tau_1(x) = x\widehat{\otimes}1.$$

Identifying A'_n with its image in A_n, the map

$$A'_n\widehat{\otimes}_k A_0 \longrightarrow A_n, \qquad a'_n\widehat{\otimes}a_0 \longmapsto a'_n a_0,$$

is an isomorphism in $\mathbf{Alpg}/k$.

This is not an isomorphism of cogroups. Rather,

$$\Delta(A') \subset A\widehat{\otimes}A',$$

and

$$(\gamma'_\rho\widehat{\otimes}\,\mathrm{id}) \circ \Delta\big|_{A'} = \Delta',$$

where Δ' is the comultiplication of A' and $\gamma'_\rho : A \to A'$ is the projection (cf. 2.4.A).

5.1.4

Let (A, J) be an object of $\mathcal{C}$, and let

$$(A_n) = \mathrm{Gr}_J(A)$$

be the associated object of **Alpg**$/k$; thus

$$A_n = \overline{J^n}/\overline{J^{n+1}} \qquad (n \geq 0).$$

It is clear that the algebra

$$\mathscr{A} = \prod_{n \geq 0} A_n$$

is generated by A_0 and A_1. In other words, for $n \geq 1$, the multiplication map

$$A_1 \widehat{\otimes}_{A_0} A_1 \widehat{\otimes}_{A_0} \cdots \widehat{\otimes}_{A_0} A_1 \longrightarrow A_n \tag{1}$$

is surjective.

Suppose further that (A, J) is a cogroup of $\mathcal{C}$ and that A and the quotients

$$\overline{J^n}/\overline{J^{n+1}}$$

are topologically flat over k. By Corollary 5.1.2, $\mathrm{Gr}_J(A)$ is then a cogroup of **Alpg**$/k$. Hence, by 5.1.3, if one puts

$$A'_n = \left\{ x \in \overline{J^n}/\overline{J^{n+1}} \;\middle|\; \Delta(x) - x \widehat{\otimes} 1 \in \bigoplus_{i=1}^{n} (\overline{J^{\,n-i}}/\overline{J^{\,n-i+1}}) \widehat{\otimes}_k (\overline{J^i}/\overline{J^{i+1}}) \right\}, \tag{2}$$

then the map

$$A'_n \widehat{\otimes}_k A_0 \longrightarrow A_n, \qquad a'_n \widehat{\otimes} a_0 \longmapsto a'_n a_0,$$

is an isomorphism in **Alpg**$/k$.[149]

Let I_0 denote the augmentation ideal of A_0. It follows from (1) and the following commutative diagram, in which

$$A_1'^{\widehat{\otimes} n} = \underbrace{A'_1 \widehat{\otimes}_k \cdots \widehat{\otimes}_k A'_1}_{n \text{ factors}},$$

that:

$$\begin{tikzcd}
A_1 \widehat{\otimes}_{A_0} \cdots \widehat{\otimes}_{A_0} A_1 \arrow[d, "m"'] & A_1'^{\widehat{\otimes} n} \widehat{\otimes}_k A_0 \arrow[l, "\sim"'] \arrow[d, "m' \widehat{\otimes} \mathrm{id}"'] & (A_1'^{\widehat{\otimes} n} \widehat{\otimes}_k I_0) \oplus A_1'^{\widehat{\otimes} n} \arrow[l, "\sim"'] \arrow[d, "(m' \widehat{\otimes} \mathrm{id}) \oplus m'"] \\
A_n & A'_n \widehat{\otimes}_k A_0 \arrow[l, "\sim"] & (A'_n \widehat{\otimes}_k I_0) \oplus A'_n \arrow[l, "\sim"]
\end{tikzcd}$$

[149]Let G, H, and $\widehat{G}_H$ be the formal k-groups corresponding to A, $A_0 = A/J$, and $\mathrm{Gr}_J(A)$, respectively. Then $\tau : A_0 \hookrightarrow \mathrm{Gr}_J(A)$ corresponds to a retraction $\pi : \widehat{G}_H \to H$ of the inclusion $H \hookrightarrow \widehat{G}_H$, and the preceding assertion means that $\widehat{G}_H$ is the semidirect product of $N = \ker(\pi)$ by H.

the multiplication map

$$m' : A'_1 \widehat{\otimes}_k \cdots \widehat{\otimes}_k A'_1 \longrightarrow A'_n \tag{3}$$

is surjective. In other words, the profinite k-algebra

$$\mathscr{A}' = \prod_{n \geq 0} A'_n$$

is generated by its degree-one terms.

Let us now return to hypothesis (i) of Proposition 5.1.[150] Let G be a topologically flat infinitesimal formal k-group, A its affine algebra, I_A the augmentation ideal of A, B a closed subalgebra of A, and

$$J = AI_B, \qquad I_B = B \cap I_A.$$

Let H be the formal subgroup $\mathrm{Spf}(A/J)$, and let

$$\pi : A \longrightarrow A/J$$

be the projection. Suppose that

$$A/\overline{J^n}$$

is topologically flat over k for every $n \geq 1$, and that B is contained in the kernel $\widetilde{B}$ of the following pair:

$$A \underset{(\mathrm{id}_A \widehat{\otimes}_k \pi)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A \widehat{\otimes}_k (A/J).$$

Put

$$(A_n) = \mathrm{Gr}_J(A),$$

let

$$I_0 = I_A/J$$

be the augmentation ideal of $A_0 = A/J$, and define $(A'_n)_{n \in \mathbb{N}}$ as in (2) above. Let $(B_n)_{n \in \mathbb{N}}$, respectively $(\widetilde{B}_n)_{n \in \mathbb{N}}$, be the object of $\mathbf{Alpg}/k$ associated with the filtration of B, respectively of $\widetilde{B}$, induced by that of A; these filtrations are defined by the ideals

$$B \cap \overline{J^n}, \qquad \text{respectively} \qquad \widetilde{B} \cap \overline{J^n}.$$

It is clear that

$$B_n \subset \widetilde{B}_n \subset A'_n \qquad \text{for every } n,$$

and that

$$\mathscr{B} = \prod_{n \geq 0} B_n \subset \widetilde{\mathscr{B}} = \prod_{n \geq 0} \widetilde{B}_n$$

are subalgebras of

$$\mathscr{A}' = \prod_{n \geq 0} A'_n.$$

On the other hand, J, respectively $\overline{J^2}$, is the image in A of

$$I_B \widehat{\otimes}_k A, \qquad \text{respectively} \qquad \overline{I_B^2} \widehat{\otimes}_k A.$$

Consequently, the map

$$(I_B/\overline{I_B^2}) \widehat{\otimes}_k A \longrightarrow J/\overline{J^2} = A_1 \simeq A'_1 \widehat{\otimes}_k A_0$$

[150] The editors expanded the original in what follows and moved to the end the "supplement" $\overline{I_B^n} = B \cap \overline{J^n}$, which is not needed to prove Proposition 5.1.

is surjective, and it factors through

$$(I_B/\overline{I_B^2})\widehat{\otimes}_k A_0.$$

Since

$$A_0 = k \cdot 1 \oplus I_0,$$

the following commutative exact diagram shows that A'_1 is the image of $I_B/\overline{I_B^2}$:

$$\begin{tikzcd}
(I_B/\overline{I_B^2})\widehat{\otimes}_k A_0 \arrow[d] & ((I_B/\overline{I_B^2})\widehat{\otimes}_k I_0) \oplus (I_B/\overline{I_B^2}) \arrow[l, "\sim"'] \arrow[d] \\
J/\overline{J^2} \arrow[d] & (A'_1\widehat{\otimes}_k I_0) \oplus A'_1 \arrow[l, "\sim"'] \arrow[d] \\
0 & 0
\end{tikzcd}$$

Thus

$$B_1 = A'_1.$$

Since $\mathscr{A}'$ is generated by A'_1, it follows that, for every $n \geq 1$,

$$A'_n \subset B_n \subset \widetilde{B}_n \subset A'_n,$$

and hence

$$B_n = \widetilde{B}_n = A'_n.$$

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Finally, because G is infinitesimal,

$$\mathfrak{r}(A) = I_A,$$

and hence the ideals $\overline{I_A^n}$ tend to zero (cf. 0.1.2). A fortiori, the ideals $\overline{J^n}$ tend to zero, so the induced filtrations on $\widetilde{B}$ and B are separated. Moreover, since B is a closed subalgebra of A, it is complete for the topology defined by the ideals $B \cap \overline{J^n}$. It therefore follows from [CA], § V.7, Lemma 1 (see also 5.1.5 below) that

$$B = \widetilde{B}.$$

This completes the proof of Proposition 5.1.

There is also the following supplement. For every n,

$$\overline{J^n} = A\overline{I_B^n}$$

is the image in A both of

$$A\widehat{\otimes}_B \overline{I_B^n}$$

and of

$$A\widehat{\otimes}_B (B \cap \overline{J^n}).$$

[151] With the notation of editorial note 149, this implies that the formal completion of G/H along the unit section, whose affine algebra is $\prod_n B_n$, is isomorphic as a formal variety to the formal k-group N.

By hypothesis, the affine algebra A/J of the formal subgroup H is topologically flat over k. Thus, by Theorem 2.4, the morphism

$$G = \mathrm{Spf}(A) \longrightarrow G/H = \mathrm{Spf}(B)$$

is surjective and topologically flat. Consequently,

$$A\widehat{\otimes}_B\overline{I_B^n} = \overline{J^n} = A\widehat{\otimes}_B(B \cap \overline{J^n}),$$

which implies

$$\overline{I_B^n} = B \cap \overline{J^n} \qquad \text{for every } n.$$

This also follows from the surjectivity of the maps

$$\overline{I_B^n}/\overline{I_B^{n+1}} \longrightarrow A'_n = (B \cap \overline{J^n})/(B \cap \overline{J^{n+1}})$$

and from 5.1.5(ii) below, applied to

$$B'_n = \overline{I_B^n}, \qquad B_n = B \cap \overline{J^n}.$$

5.1.5

Lemma 5.1.5. [152]

(i) *Let M and N be two abelian groups filtered by decreasing sequences of subgroups*

$$(M_n)_{n\in\mathbb{Z}}, \qquad (N_n)_{n\in\mathbb{Z}}.$$

Suppose that

$$\bigcup_{n\in\mathbb{Z}} M_n = M, \qquad \bigcup_{n\in\mathbb{Z}} N_n = N,$$

that

$$\bigcap_{n\in\mathbb{Z}} M_n = 0, \qquad \bigcap_{n\in\mathbb{Z}} N_n = 0,$$

and that M is complete for the topology defined by the M_n. Let

$$f : M \longrightarrow N$$

be a homomorphism of filtered groups.

a) *If f induces a surjection on the associated graded groups, then f is surjective and N is complete for the topology defined by the N_n.*
b) *If f induces an injection on the associated graded groups, then f is injective.*

(ii) *Let B be an abelian group, and let*

$$B = B'_0 \supset B'_1 \supset \cdots, \qquad B = B_0 \supset B_1 \supset \cdots$$

be two separated filtrations of B by subgroups such that

$$B'_n \subset B_n \qquad \textit{for every } n.$$

Suppose that B is complete for the topology defined by the filtration (B'_n). If the map

$$B'_i/B'_{i+1} \longrightarrow B_i/B_{i+1}$$

is surjective for every i, then

$$B'_n = B_n \qquad \textit{for every } n.$$

[152]The original stated only part (ii). For the reader's convenience, the editors stated in part (i) Lemma 1 of [CA], § V.7.

Proof. Part (i) is Lemma 1 of [CA], § V.7 (see also [BAC], III, § 2.8). Part (ii) follows by taking

$$M = B'_n \supset B'_{n+1} \supset \cdots, \qquad N = B_n \supset B_{n+1} \supset \cdots.$$

5.2

Throughout the rest of Section 5, k denotes a perfect field of characteristic $p > 0$.

Put

$$\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}.$$

If B is a profinite k-algebra and $r \in \overline{\mathbb{N}}$, denote by

$$\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}$$

the closed ideal of B generated by the elements x^{p^r}, where x ranges over the radical $\mathfrak{r}(B)$ of B. If $r = \infty$, use the same notation with the convention that

$$\big((x^{p^\infty})\big)_{x \in \mathfrak{r}(B)}$$

is the zero ideal. In both cases, B_r denotes the quotient

$$B_r = B/\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}.$$

We say that B is *of height at most* r if

$$\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}$$

is the zero ideal. If this holds and r is finite, we say that B is *of finite height.*

Consider in particular the case

$$B = k[[\omega]],$$

where ω is a pseudocompact k-vector space (cf. 1.2.5).[153] We then say that B is a *formal power series algebra* and that B_r is a *truncated formal power series algebra* $(r \in \overline{\mathbb{N}})$; thus, by convention,

$$B = B_\infty$$

is also called "truncated." If $B = k[[\omega]]$, we also write

$$\big((x^{p^r})\big)_{x \in \omega}$$

instead of

$$\big((x^{p^r})\big)_{x \in \mathfrak{r}(B)}.$$

Notation. Let ω be a pseudocompact k-vector space filtered by an increasing sequence of closed vector subspaces

$$0 = \omega_0 \subset \omega_1 \subset \omega_2 \subset \omega_3 \subset \cdots.$$

(a) The closed ideal of $k[[\omega]]$ generated by the elements x^{p^r}, where r ranges over $\mathbb{N}$ and x ranges over ω_r, will be denoted by

$$\big((x^{p^r})\big)_{x \in \omega_r}.$$

[153] In the original, the author uses "linearly compact vector space," which is equivalent to "pseudocompact vector space" (cf. [BAC], §III.2, Exercises 15 a), 19 a), and 20 d)). The editors preferred to retain the terminology "pseudocompact," used up to this point.

(b) For $r \in \mathbb{N}$, denote by ${}^r\omega$ the filtered pseudocompact vector space defined by

$$ {}^r\omega_i = \omega_i \quad \text{if } i < r, \qquad {}^r\omega_i = \omega \quad \text{if } i \geq r. $$

Theorem (Dieudonné–Cartier). *Let*

$$ H \longrightarrow G $$

be a monomorphism of infinitesimal formal groups over a perfect field k of characteristic $p > 0$. Let B be the affine algebra of the homogeneous space G/H, and suppose that one of the following three conditions holds:[154]

(i) *B has finite height* (in particular, this holds if G has finite height).

(ii) *B is a complete Noetherian local ring.*

(iii) *B is a reduced ring.*

Then B is isomorphic to the completed tensor product of a finite family of truncated formal power series algebras.

The proof of this theorem occupies 5.2.1–5.2.5.

5.2.1

Let A be the affine algebra of G, let I_A be its augmentation ideal, and put

$$ I = B \cap I_A. $$

By 2.4,

$$ H = \operatorname{Spf}(A/\overline{AI}) $$

and

$$ B = \left\{ x \in A \;\middle|\; \Delta(x) - x\widehat{\otimes}1 \in A\widehat{\otimes}\overline{AI} \right\}. $$

Put

$$ \omega = I/\overline{I^2}. $$

For each r, denote by I_r the closed subideal of I consisting of the elements x such that

$$ x^{p^r} = 0, $$

and denote by ω_r the canonical image of I_r in ω. We shall prove the following proposition.

Proposition. *If there exists a continuous section*

$$ \sigma : \omega \longrightarrow I $$

of the projection $I \to I/\overline{I^2}$ such that

$$ \sigma(\omega_r) \subset I_r $$

for every r, then B is isomorphic to

$$ k[[\omega]]/\big((x^{p^r})\big)_{x \in \omega_r}. $$

[154] The editors replaced $H\backslash G$ by G/H, and did the same in the proof; they also added condition (iii).

Indeed, such a section extends to a continuous morphism

$$k[[\omega]] \longrightarrow B,$$

which factors through

$$B' = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r}.$$

In 5.2.2–5.2.5 we prove that the resulting morphism

$$\varphi : B' = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r} \longrightarrow B$$

is an isomorphism.

5.2.1.A. [155] For every $r \in \mathbb{N}$, put

$$\mathcal{I}_r = \overline{I^2} + I_r,$$

so that

$$\mathcal{I}_r/\overline{I^2} \simeq \omega_r.$$

There is then a commutative diagram with exact rows:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{I}_{r-1} & \longrightarrow & \mathcal{I}_r & \longrightarrow & \mathcal{I}_r/\mathcal{I}_{r-1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \wr & & \\
0 & \longrightarrow & \omega_{r-1} & \longrightarrow & \omega_r & \longrightarrow & \omega_r/\omega_{r-1} & \longrightarrow & 0
\end{array}$$

Since k is a field, the rows split. Thus a pseudobasis $\mathscr{B}_{r-1}$ of $\mathcal{I}_{r-1}$ can be completed to a pseudobasis

$$\mathscr{B}_{r-1} \cup \mathscr{B}'_r$$

of $\mathcal{I}_r$. The closed subspace S_r with pseudobasis $\mathscr{B}'_r$ is then a complement of $\mathcal{I}_{r-1}$ in $\mathcal{I}_r$, and the projection

$$\pi : I \longrightarrow \omega$$

induces an isomorphism from S_r onto a complement ω'_r of ω_{r-1} in ω_r.

Let $\mathcal{I}_\infty$ be the closed ideal

$$\mathcal{I}_\infty = \overline{\bigcup_r \mathcal{I}_r}.$$

It likewise has a complement S_∞ in I, and π induces an isomorphism from

$$\mathcal{I}_\infty/\overline{I^2}$$

onto the closure

$$\omega_\infty = \overline{\bigcup_r \omega_r}$$

of the union of the ω_r, and an isomorphism from S_∞ onto a complement ω'_∞ of ω_∞ in ω. Denote the latter isomorphism by

$$\eta : S_\infty \xrightarrow{\sim} \omega'_\infty.$$

We thus obtain continuous linear maps:

[155] The editors added 5.2.1.A and 5.2.1.B.

$$\begin{tikzcd} \overline{I^2} \times S_\infty \times \widehat{\bigoplus}_r S_r \arrow[r, "\phi"] \arrow[d, "\eta"', "\times\theta"] & I \\ \omega = \omega'_\infty \times \omega_\infty & \end{tikzcd}$$

Here

$$\widehat{\bigoplus_r} S_r$$

is the direct sum of the S_r in PC(k), that is,

$$\left(\prod_r S_r^\dagger\right)^*$$

(cf. editorial note (16) to 0.2.2), and

$$\theta : \widehat{\bigoplus_r} S_r \longrightarrow \omega_\infty$$

is induced by the maps

$$S_r \xrightarrow{\sim} \omega'_r \hookrightarrow \omega_\infty.$$

It follows that a sufficient condition (but not a necessary one, as seen below) for obtaining a section

$$\sigma : \omega \longrightarrow I$$

of the desired kind is that θ be an isomorphism. By duality (cf. 0.2.2), this is equivalent to saying that the linear map

$$\omega_\infty^\dagger \longrightarrow \prod_r S_r^\dagger$$

is bijective.

5.2.1.B. As above, put

$$\omega_\infty = \overline{\bigcup_{n\in\mathbb{N}} \omega_n}.$$

A second case in which a section

$$\sigma : \omega \longrightarrow I$$

of the desired kind exists is the case in which ω_∞ has a pseudobasis $\mathscr{B}_\infty$ that is the union, for $n \in \mathbb{N}^*$, of pseudobases of the spaces

$$\omega_n/\omega_{n-1}.$$

One can then complete it by a pseudobasis $\mathscr{B}'_\infty$ of ω/ω_∞, obtaining a pseudobasis of ω compatible with the filtration.

Put

$$V = \omega_\infty^\dagger,$$

and let V_n be the orthogonal complement of ω_n in V. The preceding condition is equivalent to saying that, in the category of ordinary k-vector spaces, the separated decreasing filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \cdots$$

splits: namely, V is the direct sum, over $n \in \mathbb{N}$, of subspaces F_n such that

$$F_n \simeq V_n/V_{n+1}.$$

This need not hold. For example, let

$$V = \mathscr{S} = k^{\mathbb{N}}$$

be the space of sequences of elements of k, and let $\mathscr{S}_n$ be the subspace of sequences (u_i) such that $u_i = 0$ for $i < n$. Then

$$\dim \mathscr{S}_n/\mathscr{S}_{n+1} = 1,$$

but $\mathscr{S}$ is not isomorphic to the direct sum of the $\mathscr{S}_n/\mathscr{S}_{n+1}$, because $\mathscr{S}$ is not countable-dimensional. In this example, however, $\mathscr{S}$ is the product of the $\mathscr{S}_n/\mathscr{S}_{n+1}$ (cf. 5.2.1.A).

The splitting does hold when V is countable-dimensional.[156] Indeed, let

$$(e_n)_{n\in\mathbb{N}}$$

be a basis of V. We shall construct inductively an increasing function

$$g : \mathbb{N} \longrightarrow \mathbb{N}$$

and subspaces F_i, for $i = 0, \dots, g(n)$, such that

$$F_i \simeq V_i/V_{i+1}$$

and

$$F_{\leq g(n)} = \bigoplus_{i=0}^{g(n)} F_i$$

is a complement of $V_{g(n)+1}$ containing

$$e_0, \dots, e_n.$$

It will then follow that

$$V = \bigoplus_{i\geq 0} F_i.$$

Let $n+1 \in \mathbb{N}$, and assume the assertion already proved for n (it is vacuous for $n = -1$). If

$$e_{n+1} \in F_{\leq g(n)},$$

put $g(n+1) = g(n)$. Otherwise write

$$e_{n+1} = f + x, \qquad f \in F_{\leq g(n)}, \qquad 0 \neq x \in V_{g(n)+1}.$$

Let j be the least integer such that

$$x \in V_j \setminus V_{j+1}.$$

For

$$i = g(n)+1, \dots, j,$$

choose a complement F_i of V_{i+1} in V_i, choosing F_j so that $x \in F_j$. Then

$$e_{n+1} \in F_{\leq j};$$

put $g(n+1) = j$.

[156] This paragraph arose from discussions with J.-M. Fontaine and E. Bouscaren; in particular, Bouscaren indicated the proof that follows.

5.2.1.C. [157] In particular, the two preceding conditions (5.2.1.A and 5.2.1.B) hold when the filtration of ω is stationary, that is, when there exists an integer n_0 such that

$$\omega_n = \omega_{n_0} \qquad \text{for } n_0 \leq n < +\infty.$$

In this case,

$$k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r}$$

is isomorphic to the completed tensor product

$$\frac{k[[\omega_1]]}{\big((x^p)\big)_{x\in\omega_1}} \widehat{\otimes}_k \frac{k[[\omega_2']]}{\big((x^{p^2})\big)_{x\in\omega_2'}} \widehat{\otimes}_k \cdots$$
$$\widehat{\otimes}_k \frac{k[[\omega_{n_0}']]}{\big((x^{p^{n_0}})\big)_{x\in\omega_{n_0}'}} \widehat{\otimes}_k k[[\omega_\infty']],$$

where ω_n' is a complement of ω_{n-1} in ω_n, and ω_∞' is a complement of

$$\omega_\infty = \omega_{n_0}$$

in ω.

The filtration of ω is plainly stationary in case (i), where $\omega_r = \omega$ for all sufficiently large r; in case (ii), where ω is finite-dimensional; and also in case (iii), where

$$I_r = 0$$

for every r. In the last case, B is isomorphic to the formal power series algebra $k[[\omega]]$. The preceding remarks therefore imply the theorem, subject to 5.2.2–5.2.5 below.

5.2.2

Suppose first that B has height at most 1; that is,

$$x^p = 0 \qquad \text{for every } x \in I.$$

By 5.1.4, the graded algebra

$$\mathrm{Gr}_I(B)$$

associated with the filtration

$$I \supset I^2 \supset I^3 \supset \cdots$$

of B is endowed with a cogroup structure in the category **Alpg**$/k$. In other words, the profinite algebra

$$C = \prod_{n\geq 0} \mathrm{Gr}_I(B)_n$$

is the affine algebra of a formal k-group N. Clearly,

$$\omega_{N/k} = I/I^2,$$

and N is infinitesimal of height at most 1.

By 4.4, the identity map of $\omega_{N/k}$ therefore induces an isomorphism

$$B' = k[[\omega_{N/k}]]/\big((x^p)\big)_{x\in\omega_{N/k}} \xrightarrow{\sim} C.$$

In particular, the map ϕ of 5.2.1 induces an isomorphism between the associated graded algebras of B' and B, when both are filtered by the powers of their augmentation ideals. It follows from [CA], §V.7, Lemma 1 (see also 5.1.5) that ϕ is an isomorphism.

[157] Here the editors return to the original, shortened to take account of the preceding additions.

5.2.3

Suppose now that B has finite height at most r. Let

$$\pi : k \xrightarrow{\sim} k, \qquad x \longmapsto x^p,$$

be the Frobenius automorphism. The linear map

$$B\widehat{\otimes}_\pi k \longrightarrow B, \qquad b\widehat{\otimes}_\pi x \longmapsto b^p x = (bx^{1/p})^p,$$

has closed image equal to the closed subalgebra

$$B^p = \{\, b^p \mid b \in B \,\}$$

of B. Put

$$J = B^p \cap I = B^p \cap I_A.$$

[158] Let G_1 be the kernel of the Frobenius morphism

$$\mathrm{Fr} : G \longrightarrow G^{(p)},$$

and let HG_1 be the formal subgroup of G that is the inverse image of the formal subgroup $H^{(p)} \subset G^{(p)}$. Then HG_1 is defined by the closed ideal generated by the p-th powers of the elements of AI; this ideal is AJ. Moreover, the formation of G/H commutes with base change, because G/H represents the quotient sheaf for the flat topology (cf. 2.4). Hence

$$(G/H)^{(p)} = G^{(p)}/H^{(p)},$$

and we have the following commutative diagrams:

$$\begin{tikzcd} G \arrow[r, "\mathrm{Fr}"] \arrow[d] & G^{(p)} \arrow[d] \\ G/H \arrow[r, "\mathrm{Fr}"'] & G^{(p)}/H^{(p)} \end{tikzcd} \qquad \begin{tikzcd} A & A\widehat{\otimes}_\pi k \arrow[l, "a\widehat{\otimes}1\mapsto a^p"'] \\ B \arrow[u] & B\widehat{\otimes}_\pi k \arrow[l, "b\widehat{\otimes}1\mapsto b^p"'] \arrow[u] \end{tikzcd}$$

It follows that B^p is the affine algebra of the quotient G/HG_1.[159] Temporarily denote by C the affine algebra of the quotient HG_1/H. Since the formation of G/H commutes with base change, the square

$$\begin{tikzcd} HG_1 \arrow[r] \arrow[d] & G \arrow[d] \\ HG_1/H \arrow[r] & G/H \end{tikzcd}$$

is Cartesian. Consequently,

$$A\widehat{\otimes}_B C \simeq A/AJ = A\widehat{\otimes}_B(B/BJ).$$

[158] The editors added what follows.

[159] As indicated in the original, this also follows from Proposition 5.1. The editors preferred to give the preceding argument, which does not use the implication (i) $\Rightarrow$ (ii) of that proposition.

Since A is topologically free over B (by 2.4, because A and B are local), the natural morphism

$$B/BJ \longrightarrow C$$

is an isomorphism. Thus B/BJ is the affine algebra of HG_1/H, and of course

$$B/BJ = B_1$$

has height at most 1, since

$$J = \big((x^p)\big)_{x\in\mathfrak{r}(B)}.$$

Let

$$B' = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r},$$

let $\phi : B' \to B$ be the morphism introduced in 5.2.1, let

$$B'^p = \{\, x^p \mid x \in B' \,\},$$

and let J' be the augmentation ideal of B'^p. We then have the commutative diagram

$$\begin{tikzcd} B' \arrow[r, "\psi"] \arrow[d] & B \arrow[d] \\ B'_1 = B'/\overline{B'J'} \arrow[r, "\phi_1"] & B_1 = B/\overline{BJ} \end{tikzcd}$$

and, by 5.2.2, ϕ_1 is an isomorphism.

On the other hand, by 2.4, A is topologically flat over

$$B = \mathscr{A}(G/H) \qquad \text{and over} \qquad B^p = \mathscr{A}(G/HG_1).$$

It follows from 1.3.3 that B is topologically flat over B^p. Moreover, 5.2.4 below shows that the morphism

$$B'^p \longrightarrow B^p$$

induced by ϕ is an isomorphism. We may therefore apply 0.3.4 to the pseudocompact ring $B'^p = B^p$ and the pseudocompact B^p-modules $M = B'$, $N = B$. By what precedes,

$$\phi_1 = \phi\widehat{\otimes}_{B^p} k$$

is an isomorphism, and it follows that ϕ itself is an isomorphism. This proves 5.2 when B has finite height, subject only to 5.2.4 below.

5.2.4

For every pseudocompact vector space V over k, denote by

$${}_{\pi}V = V\widehat{\otimes}_{\pi} k$$

the space obtained from V by extension of scalars along $x \mapsto x^p$.[160] We then have a commutative diagram with exact rows:

[160] Since k is perfect, one may identify ${}_{\pi}V$ with the abelian group V on which k acts by $\lambda \cdot v = \lambda^{1/p} v$.

$$\begin{tikzcd}
0 \arrow[r] & {}_{\pi}\overline{I^2} \arrow[r, "\alpha"] \arrow[d, "u"] & {}_{\pi}I \arrow[r, "\beta"] \arrow[d, "v"] & {}_{\pi}\omega \arrow[r] \arrow[d, "w"] & 0 \\
0 \arrow[r] & \overline{J^2} \arrow[r, "\gamma"] & J \arrow[r, "\delta"] & \overline{\omega} \arrow[r] & 0\,,
\end{tikzcd}$$

Here

$$\overline{\omega} = J/J^2,$$

and the maps u, v, w are induced by the linear map

$${}_{\pi}B \longrightarrow B^p, \qquad x\widehat{\otimes}a \longmapsto x^p a.$$

Since u and v are surjective, so is w, and its kernel is the image ${}_{\pi}\omega_1$ of

$${}_{\pi}I_1 = \ker(v).$$

Put

$$J_n = \{\, x \in J \mid x^{p^n} = 0 \,\}, \qquad \overline{\omega}_n = \delta(J_n).$$

Then, for every $n \geq 0$,

$$J_n = v({}_{\pi}I_{n+1}), \qquad \overline{\omega}_n = w({}_{\pi}\omega_{n+1}).$$

The section

$${}_{\pi}\sigma : {}_{\pi}\omega \longrightarrow {}_{\pi}I$$

induced by the section σ of 5.2.1 therefore descends to a section

$$\tau : \overline{\omega} \longrightarrow J$$

compatible with the filtrations of J and $\overline{\omega}$. Since B^p has height at most $r-1$, the induction hypothesis shows that this section induces an isomorphism

$$\psi : B'' = k[[\overline{\omega}]]/\big((x^{p^n})\big)_{x\in\overline{\omega}_n} \xrightarrow{\sim} B^p.$$

But B'' identifies with B'^p, and ψ identifies with the morphism $B'^p \to B^p$ induced by ϕ. Thus the theorem is proved when B has finite height.

Remark 5.2.4.A. [161] Suppose that B has height at most $r+1$, with $r \in \mathbb{N}^*$. Then $I_{r+1} = I$, and there is an isomorphism

$$B \simeq \frac{k[[S_1]]}{\big((x_1^p)\big)_{x_1\in S_1}} \widehat{\otimes}_k \cdots \widehat{\otimes}_k \frac{k[[S_r]]}{\big((x_r^{p^r})\big)_{x_r\in S_r}} \widehat{\otimes}_k \frac{k[[S_{r+1}]]}{\big((x_{r+1}^{p^{r+1}})\big)_{x_{r+1}\in S_{r+1}}}, \tag{1}$$

where each S_n is a complement of

$$I^2 + I_{n-1}$$

in $I^2 + I_n$. Thus $\omega = I/I^2$ identifies with

$$\prod_{i=1}^{r+1} S_i,$$

[161] The editors added this remark, which is used in 5.2.5.

and it is immediate that, for $n = 1, \dots, r+1$, the image ω_n of I_n in ω identifies with

$$\prod_{i=1}^{n} S_i.$$

This has the following consequence. Let

$$B_r = B/\check{J}_r, \qquad \check{J}_r = \big((x^{p^r})\big)_{x \in \mathfrak{r}(B)},$$

and let

$$\mathfrak{m} = I/\check{J}_r$$

be the augmentation ideal of B_r. Since $\check{J}_r \subset I^2$, the space

$$\omega(r) = \mathfrak{m}/\mathfrak{m}^2$$

identifies with ω. For $n = 1, \dots, r$, denote by $\omega(r)_n$ the image in $\omega(r)$ of

$$\mathfrak{m}_n = \{\, y \in \mathfrak{m} \mid y^{p^n} = 0 \,\}.$$

It is also the image in ω of

$$\{\, x \in I \mid x^{p^n} \in \check{J}_r \,\};$$

hence $\omega(r)_n$ contains ω_n.

On the other hand, isomorphism (1) gives

$$\check{J}_r = \big((x_{r+1}^{p^r})\big)_{x_{r+1} \in S_{r+1}},$$

and therefore

$$B_r \simeq \frac{k[[S_1]]}{\big((x_1^p)\big)_{x_1 \in S_1}} \widehat{\otimes}_k \cdots \widehat{\otimes}_k \frac{k[[S_r]]}{\big((x_r^{p^r})\big)_{x_r \in S_r}} \widehat{\otimes}_k \frac{k[[S_{r+1}]]}{\big((x_{r+1}^{p^r})\big)_{x_{r+1} \in S_{r+1}}}. \tag{2}$$

It follows that $\omega(r)_n$ identifies with $\prod_{i=1}^{n} S_i$ for $n = 1, \dots, r-1$. Thus the inclusion

$$\omega_n \subset \omega(r)_n$$

is an equality for $n = 1, \dots, r-1$.

5.2.5

It remains to consider the case in which B has infinite height and the projection $I \to \omega$ admits a section σ compatible with the filtrations of ω and I. Consider the morphism induced by σ:

$$\phi : B' = k[[\omega]]/\big((x^{p^n})\big)_{x \in \omega_n} \longrightarrow B.$$

It is enough to prove that, for every $r \in \overline{\mathbb{N}}$, the map

$$\phi_r : B'_r \longrightarrow B_r$$

induced by ϕ is an isomorphism.[162]

For every $r \in \mathbb{N}^*$, let G_r be the kernel of the iterated Frobenius morphism

$$G \longrightarrow G^{(p^r)},$$

[162]Indeed, B', respectively B, is the inverse limit of the B'_r, respectively the B_r. Moreover, the editors modified the original in what follows in order to take account of the addition made in 5.2.3.

and let HG_r be the formal subgroup of G that is the inverse image of $H^{(p^r)} \subset G^{(p^r)}$. Thus HG_r is defined by the closed ideal generated by the p^r-th powers of the elements of AI, namely

$$A\check{J}_r, \qquad \check{J}_r = \{\, x^{p^r} \mid x \in I \,\}.$$

Exactly as in 5.2.3, it follows that

$$B_r = B/B\check{J}_r$$

is the affine algebra of HG_r/H; in particular, it has height at most r.

Let

$$\mathfrak{m}(r) = I/B\check{J}_r$$

be the augmentation ideal of B_r. The canonical projection $B \to B_r$ induces an isomorphism

$$\omega = I/I^2 \xrightarrow{\sim} \omega(r) = \mathfrak{m}(r)/\mathfrak{m}(r)^2,$$

which we use to identify the two spaces. Let $\omega(r)_n$ be the image in $\omega(r)$ of the closed ideal

$$\mathfrak{m}(r)_n = \{\, y \in \mathfrak{m}(r) \mid y^{p^n} = 0 \,\}.$$

It is also the image in ω of the closed ideal

$$I(r)_n = \{\, x \in I \mid x^{p^n} \in B\check{J}_r \,\}.$$

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Clearly,

$$\omega(r)_n = \omega \qquad \text{if } n \geq r.$$

We shall prove that

$$\omega(r)_n = \omega_n \qquad \text{if } n < r.$$

For every r, n, the sequence

$$0 \longrightarrow I(r)_n \cap I^2 \longrightarrow I(r)_n \longrightarrow \omega(r)_n \longrightarrow 0$$

is exact. For fixed n,

$$\bigcap_r I(r)_n = I_n,$$

because $\bigcap_r B\check{J}_r = 0$. Filtered inverse limits are exact in PC(k) (cf. 0.2), so for every n,

$$\omega_n = \bigcap_r \omega(r)_n. \tag{*}$$

On the other hand, Remark 5.2.4.A gives

$$\omega(r)_n = \omega(r+1)_n \qquad \text{if } n < r.$$

Together with (*), this implies

$$\omega(r)_n = \omega_n \qquad \text{if } n < r.$$

Consequently, the vector space ω, filtered by the subspaces $\big(\omega(r)_n\big)_{n\geq 0}$, is precisely ${}^r\omega$ in the notation of 5.2. The composite

$$\sigma(r) : \omega \xrightarrow{\sigma} I \longrightarrow \mathfrak{m}(r)$$

is therefore compatible with the filtrations $\big(\omega(r)_n\big)$ and $\big(\mathfrak{m}(r)_n\big)$. Since

$$k[[{}^r\omega]]/\big((x^{p^n})\big)_{x\in {}^r\omega_n}$$

is exactly B'_r, and $\phi_r : B'_r \to B_r$ is induced by $\sigma(r)$, the result already proved for finite-height algebras shows that ϕ_r is an isomorphism.

[163]The editors expanded the original in what follows.

5.2.6

Definition 5.2.6. [164] Let $(A_\lambda)_{\lambda\in\Lambda}$ be a family of profinite k-algebras, each endowed with an augmentation

$$\varepsilon_\lambda : A_\lambda \longrightarrow k.$$

(This holds in particular if every A_λ is local with residue field k.) Define the infinite completed tensor product

$$\mathcal{A} = \widehat{\bigotimes}^{c}_{\lambda\in\Lambda} A_\lambda$$

to be the inverse limit in $\mathrm{Alp}\,/k$ of

$$\mathcal{A}_F = \widehat{\bigotimes}^{c}_{\lambda\in F} A_\lambda,$$

where F ranges through the finite subsets of Λ. For $F' = F \cup \{\lambda\}$, the transition morphism

$$\mathcal{A}_{F'} \longrightarrow \mathcal{A}_F$$

is

$$\mathrm{id}\,\widehat{\otimes}\varepsilon_\lambda.$$

In particular, suppose $\Lambda = \mathbb{N}^*$, and write

$$X_n = \mathrm{Spf}(A_n).$$

Then $\mathrm{Spf}(\mathcal{A})$ represents the functor that assigns to every $C \in \mathrm{Alf}\,/k$ the set of *finite* sequences in $\prod_{n\geq 1} X_n(C)$; that is, the sequences

$$(x_1, x_2, \ldots) \in \prod_{n\geq 1} X_n(C)$$

such that

$$x_n = \varepsilon_n$$

for all sufficiently large n. Here, by abuse of notation, ε_n denotes the composite of

$$\varepsilon_n : A_n \longrightarrow k$$

with the structural morphism $k \to C$.

If, in addition, every A_n is a quotient of an algebra $k[[\omega'_n]]$, one may write 0 for the unique morphism

$$A_n \longrightarrow C$$

that vanishes on ω'_n. The preceding description then becomes the set of sequences for which $x_n = 0$ for all sufficiently large n.

5.3. Remarks

(a) Call a profinite k-algebra *stationary* if it is the completed tensor product of a family of truncated formal power series algebras.[165]

[164] The editors added this number in order to define the infinite tensor products used in 5.3(a).

[165] The author doubtless had in mind a completed tensor product

$$\mathcal{A} = \widehat{\bigotimes}^{c}_{n\in\mathbb{N}^*} \frac{k[[\omega'_n]]}{\big((x^{p^n})\big)_{x\in\omega'_n}},$$

Let G be an infinitesimal formal k-group and let B be the affine algebra of a homogeneous space of G. It follows from Theorem 5.2 that, for every $r \in \mathbb{N}$, the algebra

$$B/\big((x^{p^r})\big)_{x\in\mathfrak{r}(B)}$$

is stationary. In particular, B is an inverse limit of stationary algebras.[166]

(b) I do not know whether, with the notation of 5.2.1, A and B can be chosen so that there exists no section

$$\sigma : \omega \longrightarrow I$$

compatible with the filtrations.[167] Nevertheless, ω may be any pseudocompact vector space filtered by an increasing sequence of closed subspaces. Indeed, given

$$\omega_1 \subset \omega_2 \subset \cdots \subset \omega,$$

one can define on

$$B = A = k[[\omega]]/\big((x^{p^r})\big)_{x\in\omega_r}$$

a diagonal morphism

$$\Delta_A : A \longrightarrow A\widehat{\otimes}_k A$$

satisfying conditions (i), (ii), and (iii) of 2.1. It suffices to put

$$\Delta_A(y) = y\widehat{\otimes}1 + 1\widehat{\otimes}y$$

when y is the image in A of an element of ω.

5.4. Corollary

Corollary 5.4. *Let G be an algebraic group over a perfect field k of characteristic $p > 0$, let H be an algebraic subgroup of G, let e be the image of the identity element of G in G/H, and let A be the local ring of G/H at e. Then $\widehat{A}$ is isomorphic to an algebra of the form*

$$k[[X_1, \ldots, X_r, \ldots, X_s]]/(X_1^{p^{n_1}}, \ldots, X_r^{p^{n_r}}).$$

Proof. Consider the infinitesimal formal groups

$$\widehat{G} = \mathrm{Spf}(\widehat{\mathcal{O}}_{G,e}), \qquad \widehat{H} = \mathrm{Spf}(\widehat{\mathcal{O}}_{H,e}).$$

By 1.3.4, the completion $\widehat{A}$ of

$$A = \mathcal{O}_{G/H,e}$$

is isomorphic to

$$\mathscr{A}(\widehat{G}/\widehat{H}),$$

and the corollary follows from Theorem 5.2(ii).[168]

where the ω'_n are arbitrary pseudocompact vector spaces. Then $\omega = \omega_{\mathcal{A}/k}$ identifies with $\prod_n \omega'_n$, and the filtration is $\omega_n = \prod_{i=1}^n \omega'_i$; this is the case of 5.2.1.B. For this reason these algebras would be better called *stable*, rather than *stationary*; cf. [Di73], II, §2.9, p. 75. If, for example, $\dim \omega'_n = 1$ for every $n \in \mathbb{N}^*$, then $\mathcal{A}$ represents the functor that assigns to $C \in \mathrm{Alf}/k$ the sequences $(x_n)_{n\in\mathbb{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ and $x_n = 0$ for all sufficiently large n. This case, where $\omega = \prod_{n\in\mathbb{N}^*} \omega'_n$, corresponds in the dual setting of connected cocommutative Hopf algebras to the case studied by M. E. Sweedler; cf. [Sw67], Theorem 3.

[166] Such an inverse limit need not itself be a stable profinite k-algebra in the sense of the preceding editor's note. For example, let S be the ordinary k-vector space of sequences $(u_1, u_2, \ldots)$ of elements of k, and put $\omega = S^*$. Then ω is the direct sum in $\mathrm{PC}(k)$ of copies kx_n of k, for $n \in \mathbb{N}^*$, where $x_n(u) = u_n$. Thus this is the case of 5.2.1.A. The algebra

$$\mathcal{A} = k[[\omega]]/\big((x_n^{p^n})\big)_{n\in\mathbb{N}^*}$$

has $\omega_{\mathcal{A}/k} = \omega$ and $\omega_n = \prod_{i=1}^n kx_i$, but it is not stable: $\mathrm{Spf}(\mathcal{A})$ represents the functor that assigns to every $C \in \mathrm{Alf}/k$ the infinite sequences $(x_n)_{n\in\mathbb{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ for every n.

[167] The editors do not know either, outside the cases considered in 5.2.1.A and 5.2.1.B.

[168] See also [DG70], §III.3, Theorem 6.1.

5.5. Complements

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Recall the following definitions. A Noetherian local ring A is called a *complete intersection* if its completion $\widehat{A}$ is the quotient of a complete regular Noetherian local ring B by an ideal I generated by a regular sequence of elements of B (cf. EGA IV$_4$, 19.3.1).

On the other hand, let

$$\tau : Y \hookrightarrow X$$

be a closed immersion of schemes. If $y \in Y$, then τ is called a *regular immersion at the point* y if the kernel of

$$\mathcal{O}_{X,y} \longrightarrow \mathcal{O}_{Y,y}$$

is generated by a regular sequence. If, moreover, X is locally Noetherian and τ is a regular immersion at every point, then τ is called a *regular immersion* (cf. *loc. cit.*, Proposition 16.9.10 and Definition 16.9.2).

Corollary 5.5.1. *If G is an algebraic group over a field k, then the local ring $\mathcal{O}_{G,e}$ is a complete intersection.*

Proof. By EGA IV$_4$, 19.3.4, we may assume that k is algebraically closed. If $\mathrm{char}(k) = 0$, then G is already known to be smooth (cf. 3.3.1 or VIB, 1.6.1); hence $\widehat{\mathcal{O}}_{G,e}$ is a k-algebra of formal power series, by EGA IV$_4$, 17.5.3 (d$''$). If $\mathrm{char}(k) = p > 0$, then 5.4, applied with $H = \{e\}$, shows that $\widehat{\mathcal{O}}_{G,e}$ is a complete intersection.

Remarks 5.5.2. *Let k be a field, let G be a smooth algebraic k-group, and let H be a closed subgroup of G.*

(a) We saw in Exposé III, 4.15, that the immersion

$$H \hookrightarrow G$$

is regular. This can also be deduced from 5.4 as follows. As in *loc. cit.*, we may assume that k is algebraically closed, and it is enough to show that the kernel I of

$$\mathcal{O}_{G,e} \longrightarrow \mathcal{O}_{H,e}$$

is generated by a regular sequence. Put

$$A = \mathcal{O}_{G,e}, \qquad \widehat{G} = \mathrm{Spf}(\widehat{A}), \qquad \widehat{H} = \mathrm{Spf}(\widehat{\mathcal{O}}_{H,e}).$$

Since A is Noetherian, there is an exact sequence

$$0 \longrightarrow I \otimes_A \widehat{A} \longrightarrow \widehat{A} \xrightarrow{\ \pi\ } \mathscr{A}(\widehat{H}) \longrightarrow 0,$$

and $\widehat{A}$ is faithfully flat over A. Therefore, by EGA IV$_4$, 16.9.10(ii) and 19.1.5(ii), it is enough to show that the kernel

$$\widehat{I} = I \otimes_A \widehat{A}$$

of π is generated by a regular sequence of elements of $\widehat{A}$.

Since G is smooth, $\widehat{A}$ is reduced. By 5.4, the subalgebra

$$B = \mathscr{A}(\widehat{G}/\widehat{H})$$

[169] The editors added this subsection in order to give some consequences of 5.4 mentioned in Exposés III and VIA.

is therefore isomorphic to a formal power series algebra

$$k[[x_1, \dots, x_n]].$$

Consequently, the identity section of $\widehat{G}/\widehat{H}$ is defined in B by the regular sequence $(x_1, \dots, x_n)$. Since $\widehat{A}$ is Noetherian, the ideal J of $\widehat{A}$ generated by $x_1, \dots, x_n$ is closed, and hence equals $\widehat{I}$, by Corollary 1.4. Moreover, $\widehat{A}$ is topologically flat, and hence flat over B (cf. 0.3.8). Thus $(x_1, \dots, x_n)$ is a regular sequence in $\widehat{A}$, by EGA IV$_4$, 19.1.5(ii).

(b) The following more precise result can also be deduced from 5.2(ii). Suppose that k is perfect. Applying 5.2(ii) to the algebra

$$C = \mathscr{A}(H),$$

there are a basis

$$(y_1, \dots, y_{r+s})$$

of ω_H and integers

$$1 \leq n_1 \leq \cdots \leq n_r$$

such that $\mathscr{A}(H)$ is isomorphic to the quotient of

$$k[[y_1, \dots, y_{r+s}]]$$

by the ideal generated by

$$y_i^{p^{n_i}}, \qquad i = 1, \dots, r.$$

Lift the y_i to elements x_i of ω_G, and extend $(x_1, \dots, x_{r+s})$ to a basis $(x_1, \dots, x_n)$ of ω_G. Since $\mathscr{A}(G)$ is reduced, the morphism

$$k[[x_1, \dots, x_n]] \longrightarrow \mathscr{A}(G)$$

is an isomorphism, by 5.2(iii).

Thus there exists a "coordinate system" $(x_1, \dots, x_n)$ on G, that is, an isomorphism

$$\mathscr{A}(G) \simeq k[[x_1, \dots, x_n]],$$

such that H is defined by the equations

$$x_i^{p^{n_i}} = 0 \quad (i = 1, \dots, r), \qquad x_i = 0 \quad (i > r + s).$$

This is "Dieudonné's theorem"; compare [Di55], §19, Theorem 6, and [Di73], II, §3.2, Proposition 3 and the material preceding it.

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Exposé VIII
Diagonalizable Groups

by A. Grothendieck[0]

1. Biduality

Let $\mathscr{C}$ be a category, which as usual we identify with a full subcategory of

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

(cf. Exposé I). Let I be a commutative group functor, that is, an object of $\widehat{\mathscr{C}}$ endowed with a commutative group structure (cf. I, 2.1).[1] For every $X \in \mathrm{Ob}(\widehat{\mathscr{C}})$, the object $\mathrm{Hom}(X, I)$ carries the commutative group structure induced by that of I. For every group G in $\widehat{\mathscr{C}}$, put

$$D(G) = \mathrm{Hom}_{\mathrm{gr}}(G, I),$$

the subobject of $\mathrm{Hom}(G, I)$ defined, for every $S \in \mathrm{Ob}(\mathscr{C})$, by

$$D(G)(S) = \mathrm{Hom}_{S\text{-gr}}(G_S, I_S), \tag{x}$$

where $G_S = G \times S$ and $I_S = I \times S$ are regarded as S-groups, that is, as groups in $\widehat{\mathscr{C}}_{/S}$. Thus $D(G)$ is a $\widehat{\mathscr{C}}$-subgroup of $\mathrm{Hom}(G, I)$. In this way we obtain a contravariant functor D from the category of $\widehat{\mathscr{C}}$-groups to the category of commutative $\widehat{\mathscr{C}}$-groups.

The right-hand side of (x) may also be interpreted as the subset of $\mathrm{Hom}(G \times S, I)$ consisting of morphisms $G \times S \to I$ that are "multiplicative in the first argument G." Moreover, the preceding formulas remain valid when S is an arbitrary object of $\widehat{\mathscr{C}}$, not necessarily one arising from $\mathscr{C}$.

Now take for S a group in $\widehat{\mathscr{C}}$, denoted G'. In the left-hand side $\mathrm{Hom}(G', D(G))$ of (x), consider the subset $\mathrm{Hom}_{\mathrm{gr}}(G', D(G))$ consisting of morphisms that respect the group structures of G' and $D(G)$.

It corresponds to the subset of $\mathrm{Hom}(G \times G', I)$ consisting of morphisms that are multiplicative in each of the two arguments—which one may call *bilinear morphisms from $G \times G'$ to I*, or *pairings of G and G' with values in I*. We therefore have

$$\mathrm{Hom}_{\mathrm{gr}}(G', D(G)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{bil}}(G \times G', I), \tag{xx}$$

[0]Editor's note: version 1.1 of November 8, 2009. Additions were made in 1.2, 1.4, 1.7, 3.1, 3.4, 4.5.1, 6.4, and 6.8; 1.5.1 and Section 7 remain to be revised.

[1]Editor's note: the original has been expanded in what follows.

functorially in the pair (G, G'). Since the right-hand side is symmetric in G and G', this gives a functorial bijection

$$\mathrm{Hom}_{\mathrm{gr}}(G', D(G)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(G, D(G')). \tag{xxx}$$

In other words, giving a group homomorphism $G' \to D(G)$ "amounts to the same thing" as giving a group homomorphism $G \to D(G')$: both amount to giving a pairing $G \times G' \to I$.

Apply this with $G' = D(G)$ and with the identity homomorphism $G' \to D(G)$. We obtain a canonical homomorphism

$$G \longrightarrow D(D(G)). \tag{xxxx}$$

Definition 1.0. [2] *The group G is called* reflexive *(relative to I) if the preceding homomorphism is an isomorphism. This implies, in particular, that G is commutative.*

We therefore have:

Proposition 1.0.1. *The functor D induces an anti-equivalence from the category of reflexive $\widehat{\mathscr{C}}$-groups to itself.*

In particular, if G and H are reflexive groups, D induces an isomorphism

$$\mathrm{Hom}_{\mathrm{gr}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(D(H), D(G)).$$

It is enough, in fact, that H be reflexive, as follows from (xxx).

Definition 1.0.2. *As usual, a $\mathscr{C}$-group G is called* reflexive *if it is reflexive as a $\widehat{\mathscr{C}}$-group, regardless of whether $D(G)$ is representable.*

Thus D gives an anti-equivalence from the category of reflexive $\mathscr{C}$-groups G for which $D(G)$ is representable to itself.

Remark 1.0.3. To conclude these generalities, note that formation of the dual $D(G)$ commutes with base extension, which therefore carries reflexive groups to reflexive groups.

We shall henceforth work in the following setting. Let $\mathscr{C} = (\mathbf{Sch})_{/S}$ be the category of preschemes over S, and let $I = \mathbf{G}_{m,S}$ be the "multiplicative group over S" (cf. Exposé I). For every ordinary group M, consider the constant S-group M_S. For every group prescheme J over S, there is a canonical isomorphism, functorial in M and J and compatible with base extension,

$$\mathrm{Hom}_{S\text{-gr}}(M_S, J) = \mathrm{Hom}_{\mathrm{gr}}(M, J(S)).$$

Applying this with $J = I = \mathbf{G}_{m,S}$ and with a variable S' over S, we obtain a functorial isomorphism

$$D(M_S)(S') \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(M, \mathbf{G}_m(S')). \tag{1.0.4}$$

We thus recover the functor already considered in I, 4.4, also denoted $D_S(M)$, which is representable when M is commutative, since

$$D_S(M) = D(M_S) = \mathrm{Spec}\,\mathscr{O}_S(M),$$

where $\mathscr{O}_S(M)$ denotes the group algebra of M over $\mathscr{O}_S$. In general, moreover, $D(M)$ is unchanged when M is replaced by its abelianization, so no information is lost by assuming that M is commutative.

[2] Editor's note: the numbering 1.0, 1.0.1, and so on has been added to make the definitions and statements contained here explicit.

Definition 1.1. *A group prescheme G over S is called* diagonalizable *if it is isomorphic to a scheme of the form*

$$D_S(M) = D(M_S) = \mathrm{Hom}_{S\text{-gr}}(M_S, \mathbf{G}_m)$$

for a suitable commutative group M. It is called locally diagonalizable *if every point of S has an open neighborhood U such that $G|_U$ is diagonalizable.*

Theorem 1.2. *Let Γ be a constant commutative group scheme over S; that is, suppose that it is isomorphic to a group scheme of the form M_S, where M is an ordinary commutative group. Then Γ is reflexive: the canonical homomorphism*

$$\Gamma \longrightarrow D(D(\Gamma))$$

is an isomorphism.[3] *Hence the diagonalizable group $D(M_S)$ is also reflexive.*

In view of the definitions, this follows from the next statement, applied to a prescheme S' over S.

Corollary 1.3. *Let $G = D(M_S)$. Every homomorphism of S-groups*

$$u : G \longrightarrow \mathbf{G}_{m,S}$$

is defined by a unique section of M_S over S, equivalently by a unique locally constant map from S to M.

Proof. By definition,

$$\mathbf{G}_{m,S} = \mathrm{GL}(1)_S = \mathrm{Aut}_{\mathscr{O}_S\text{-mod}}(\mathscr{O}_S).$$

Thus giving a group homomorphism $G \to \mathbf{G}_{m,S}$ is equivalent to giving $\mathscr{O}_S$ the structure of a G-$\mathscr{O}_S$-module compatible with its natural $\mathscr{O}_S$-module structure (cf. I, 4.7). By I, 4.7.3, this is also equivalent to giving an M-grading on $\mathscr{O}_S$, that is, a direct sum decomposition

$$\mathscr{O}_S = \bigoplus_{m \in M} \mathscr{L}_m.$$

It is well known that a direct summand of a locally free module of finite type is locally free of finite type. Hence, near each point of S, every $\mathscr{L}_m$ is either zero or free of rank one; in the latter case it identifies with $\mathscr{O}_S$ on that neighborhood. Let S_m be the open subset of S on which the latter alternative occurs. The direct-sum decomposition shows that the S_m cover S and are pairwise disjoint.

Consequently, giving a group homomorphism $G \to \mathbf{G}_{m,S}$ is equivalent to giving a decomposition of S as a union of pairwise disjoint open subsets S_m, $m \in M$, hence to giving a locally constant map from S to M. This proves Corollary 1.3 and therefore Theorem 1.2. □

Corollary 1.4. *A diagonalizable group is reflexive, and therefore so is a locally diagonalizable group. If M and N are ordinary commutative groups, the natural homomorphism*

$$\mathrm{Hom}_{S\text{-gr}}(M_S, N_S) \longrightarrow \mathrm{Hom}_{S\text{-gr}}(D(N_S), D(M_S))$$

is bijective.

[3]Editor's note: the sentence that follows has been added.

[4] Since the preceding isomorphism is compatible with base extension, it yields an isomorphism of S-functors in groups

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S)). \tag{1}$$

For every S-prescheme T,

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S)(T) = \mathrm{Hom}_{\mathrm{gr}}(M, \Gamma(N_T/T)),$$

and by I, 1.8, $\Gamma(N_T/T)$ is the abelian group of locally constant maps $T \to N$. On the other hand, let $\mathrm{Hom}_{\mathrm{gr}}(M,N)_S$ be the constant S-group associated with the ordinary abelian group $\mathrm{Hom}_{\mathrm{gr}}(M,N)$. There is an evident homomorphism of S-functors in commutative groups

$$\mathrm{Hom}_{\mathrm{gr}}(M,N)_S \xrightarrow{\theta} \underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, N_S), \tag{2}$$

which is always a monomorphism and is an isomorphism if M is of finite type.[5]

Part (a) of the following corollary follows from the preceding discussion; part (b) follows from it by the descent results recalled in 1.7. [6]

Corollary 1.5.

(a) *Let M and N be ordinary commutative groups, with M of finite type. Then there is an isomorphism*

$$\mathrm{Hom}_{\mathrm{gr}}(M,N)_S \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S));$$

consequently, $\underline{\mathrm{Hom}}_{S\text{-gr}}(D(N_S), D(M_S))$ is representable.

(b) *More generally, if G and H are locally diagonalizable and H is of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr}}(G,H)$ is representable.*

[7] Corollary 1.5 gives:

Corollary 1.6. *Under the hypotheses of 1.5, if S is connected, then*

$$\mathrm{Hom}_{S\text{-gr}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(M,N)$$

and

$$\mathrm{Isom}_{S\text{-gr}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Isom}_{\mathrm{gr}}(M,N).$$

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[4] Editor's note: the discussion that follows has been expanded.

[5] Editor's note: write $F(M)$ and $G(M)$ for the left- and right-hand sides of (2), respectively. If $M = M_1 \oplus M_2$, there are canonical isomorphisms $F(M) = F(M_1) \oplus F(M_2)$ and $G(M) = G(M_1) \oplus G(M_2)$. It is therefore enough to check that θ is an isomorphism for $M = \mathbf{Z}/r\mathbf{Z}$, $r \geq 0$. In that case $F(M) = ({}_rN)_S$, where ${}_rN$ is the kernel of $r\,\mathrm{id}_N$, and for every $T \to S$ the homomorphism

$$F(M)(T) = \Gamma({}_rN_T/T) \longrightarrow G(M)(T) = {}_r\Gamma(N_T/T)$$

is bijective.

[6] Editor's note: the original stated after 1.5 that, more generally, if G and H are locally diagonalizable and H is of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr}}(G,H)$ is representable. That assertion has been included in 1.5 and its proof made explicit in 1.7.

[7] Editor's note: a statement 1.5.1 treating the functor $\underline{\mathrm{Isom}}_{S\text{-gr}}(G,H)$, considered in X, 5.10 and 5.11, should be added.

1.7. Descent of representability. [8] We recall here several descent results that will be used frequently below.

Scholium 1.7.1. *Let S be a prescheme and let X, Y, T be S-preschemes. If (T_i) is an open covering of T, and if $T_{ij} = T_i \cap T_j = T_i \times_T T_j$, then, because giving a morphism of T-preschemes $X_T \to Y_T$ is local on T, there is an exact sequence of sets*

$$\mathrm{Hom}_T(X_T, Y_T) \longrightarrow \prod_i \mathrm{Hom}_{T_i}(X_{T_i}, Y_{T_i}) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{T_{ij}}(X_{T_{ij}}, Y_{T_{ij}}). \tag{1}$$

Thus $\underline{\mathrm{Hom}}_S(X, Y)$ is a local S-functor, that is, a sheaf on $(\mathbf{Sch})_{/S}$ for the Zariski topology.

More generally, by IV, 4.5.13, $\underline{\mathrm{Hom}}_S(X, Y)$ is a sheaf on $(\mathbf{Sch})_{/S}$ for every topology coarser than the canonical topology, for example the fpqc topology. If G and H are group preschemes over S, it follows that the subfunctor $\underline{\mathrm{Hom}}_{S\text{-gr}}(G, H)$ is an fpqc sheaf, hence a fortiori a local functor.

Lemma 1.7.2. [9] *Let F be a local S-functor.*

(i) *Suppose that there is an open covering (S_i) of S such that the restriction $F_i = F \times_S S_i$ of F to each S_i is representable by an S_i-prescheme X_i. Then F is representable by an S-prescheme X.*

(ii) *Suppose that F is an fpqc sheaf and that there is a faithfully flat quasi-compact morphism $S' \to S$ such that $F' = F \times_S S'$ is representable by an S'-prescheme X'. Then X' carries a descent datum relative to $S' \to S$ (cf. IV, 2.1). If this descent datum is effective—as it is when X' is affine over S'—then F is representable by an S-prescheme X.*

Proof. (i) The hypothesis implies that $X_i \times_S S_j$ and $X_j \times_S S_i$ both represent the restriction of F to $S_{ij} = S_i \times_S S_j$. The Yoneda lemma therefore gives a unique isomorphism of S_{ij}-preschemes

$$c_{ji} : X_i \times_S S_j \xrightarrow{\sim} X_j \times_S S_i.$$

Over $S_{ijk} = S_i \times_S S_j \times_S S_k$, we obtain the commutative diagram of isomorphisms

$$\begin{tikzcd}
X_i \times_S S_j \times_S S_k \arrow[r, "c_{ji} \times \mathrm{id}_{S_k}"] \arrow[d, equal] & X_j \times_S S_i \times_S S_k \arrow[r, equal] & X_j \times_S S_k \times_S S_i \arrow[d, "c_{kj} \times \mathrm{id}_{S_i}"] \\
X_i \times_S S_k \times_S S_j \arrow[r, "c_{ki} \times \mathrm{id}_{S_j}"'] & X_k \times_S S_i \times_S S_j \arrow[r, equal] & X_k \times_S S_j \times_S S_i.
\end{tikzcd}$$

Indeed, all of its objects represent the restriction of F to S_{ijk}. Thus the c_{ji} satisfy the usual cocycle relation $c_{kj} \circ c_{ji} = c_{ki}$.

It follows that the X_i glue to an S-prescheme X with $X \times_S S_i = X_i$ for every i. For every Y over S_i,

$$F(Y) = F_i(Y) = \mathrm{Hom}_{S_i}(Y, X \times_S S_i) = \mathrm{Hom}_S(Y, X) = h_X(Y). \tag{$*$}$$

For an arbitrary $Y \to S$, the $Y_i = Y \times_S S_i$ form an open covering of Y; put $Y_{ij} = Y_i \times_Y Y_j = Y \times_S S_{ij}$. Since F is local by hypothesis and h_X is local because the Zariski topology is

[8] Editor's note: this paragraph has been added to make explicit the "local on S" character of the representability of certain sheaves on S, which is used repeatedly below and was used implicitly in the original.

[9] Editor's note: see also Remark XI, 3.4.

coarser than the canonical topology, both $F(Y)$ and $h_X(Y)$, in view of $(*)$, identify with the kernels of the respective double arrows

$$\prod_i F(Y_i) \rightrightarrows \prod_{i,j} F(Y_{ij}), \qquad \prod_i h_X(Y_i) \rightrightarrows \prod_{i,j} h_X(Y_{ij}).$$

This proves (i).

(ii) Put $S'' = S' \times_S S'$, and let S''_1 and S''_2 denote S'' regarded as an S'-prescheme through the first and second projections. The functors $F''_1 = F' \times_{S'} S''_1$ and $F''_2 = F' \times_{S'} S''_2$ are represented respectively by $X''_1 = X' \times_{S'} S''_1$ and $X''_2 = X' \times_{S'} S''_2$. Since

$$F''_1 = F \times_S S'' = F''_2,$$

there is a unique S''-isomorphism

$$c : X''_1 \xrightarrow{\sim} X''_2.$$

Let q_i, respectively p_{ji}, be the projection from $S''' = S' \times_S S' \times_S S'$ onto its i-th factor, respectively onto its i-th and j-th factors. Let $X'''_i = X' \times_{S'} S'''_i$, where S'''_i means S''' viewed as an S'-prescheme through q_i, and let $p^*_{ji}(c) : X'''_i \xrightarrow{\sim} X'''_j$ be obtained from c by base change. We obtain the diagram

$$\begin{tikzcd} X'''_1 \arrow[r, "p^*_{21}(c)"] \arrow[dr, "p^*_{31}(c)"'] & X'''_2 \arrow[d, "p^*_{32}(c)"] \\ & X'''_3. \end{tikzcd}$$

All three objects represent the restriction of F to S''', so the diagram commutes:

$$p^*_{32}(c) \circ p^*_{21}(c) = p^*_{31}(c).$$

Thus c is a descent datum on X' relative to $S' \to S$ (cf. IV, 2.1).

Suppose that this descent datum is effective, so there is an S-prescheme X such that $X' \simeq X \times_S S'$. This is the case when X' is affine over S', by SGA 1, VIII, 2.1.[10] For every $Y \to S'$, one then has

$$(**) \qquad F(Y) = F'(Y) = \mathrm{Hom}_{S'}(Y, X \times_S S') = \mathrm{Hom}_S(Y, X) = h_X(Y).$$

For arbitrary $Y \to S$, put $Y' = Y \times_S S'$ and $Y'' = Y' \times_Y Y' \simeq Y \times_S S''$. Like $S' \to S$, the morphism $Y' \to Y$ is faithfully flat and quasi-compact, hence an $\mathcal{M}$-effective epimorphism, where $\mathcal{M}$ is the family of faithfully flat quasi-compact morphisms. Thus the equivalence relation

$$Y' \times_Y Y' \rightrightarrows Y'$$

is $\mathcal{M}$-effective and has quotient Y. Since F is an fpqc sheaf by hypothesis, while h_X is one because the fpqc topology is coarser than the canonical topology, both $F(Y)$ and $h_X(Y)$, in view of $(**)$, identify with the respective kernels of

$$F(Y') \rightrightarrows F(Y' \times_Y Y'), \qquad h_X(Y') \rightrightarrows h_X(Y' \times_Y Y').$$

This proves (ii). □

[10] Editor's note: for another effectivity criterion, see X, 5.4–5.6.

Corollary 1.7.3. *Let F be an fpqc sheaf on $(\mathbf{Sch})_{/S}$. Suppose there is an open covering (S_i) of S, and for each i a faithfully flat quasi-compact morphism $S'_i \to S_i$, such that $F'_i = F \times_S S'_i$ is representable by an S'_i-prescheme X'_i affine over S'_i. Then F is representable by an S-prescheme X affine over S, with $X \times_S S'_i = X'_i$ for every i. If, moreover, every $X'_i \to S'_i$ is a closed immersion (respectively, a finite étale morphism), then so is $X \to S$.*

The first assertion follows from 1.7.2. For the second, it is enough to check that every morphism $X \times_S S_i \to S_i$ is a closed immersion (respectively, finite and étale); this follows from EGA IV_2, 2.7.1 (respectively, together with IV_4, 17.7.3).

Remark 1.7.4. Assertion 1.5(b) follows, as announced, from 1.7.1 and 1.7.2(i).

2. Scheme-theoretic Properties of Diagonalizable Groups

These properties are summarized in the following proposition.

Proposition 2.1. *Let S be a nonempty prescheme, let M be an ordinary commutative group, and let $G = D(M_S)$ be the diagonalizable S-group defined by M. Then:*

(a) *G is faithfully flat over S and affine over S (a fortiori, quasi-compact over S);*

(b) *M is of finite type if and only if G is of finite type over S, if and only if G is of finite presentation over S;*

(c) *M is finite if and only if G is finite over S, if and only if G is of finite type over S and is annihilated by an integer $n > 0$. In that case,*

$$\deg(G/S) = \mathrm{Card}(M);$$

(c') *M is a torsion group if and only if G is integral over S;*

(d) *$M = 0$ if and only if G is the unit S-group;*

(e) *M is of finite type and the order of its torsion subgroup is prime to the residue characteristics of S if and only if G is smooth over S.*

The verification of (a) through (d) is immediate and is left to the reader. Let us prove (e). If G is smooth over S, it is locally of finite presentation over S, hence of finite presentation over S, since it is affine over S; thus M is of finite type. We may therefore assume from the outset that M is of finite type, and hence that G is of finite presentation over S. Then[11] G is smooth over S if and only if its geometric fibers are smooth, reducing us to the case in which S is the spectrum of an algebraically closed field k.

Write $M = T \times L$, where T is the torsion subgroup and L is free, with $L \simeq \mathbb{Z}^r$. Then

$$D(M) = D(T) \times D(L), \qquad D(L) \simeq \mathbb{G}_m^r,$$

and $D(L)$ is smooth over k. Thus $G = D(M)$ is smooth over k if and only if $D(T)$ is smooth. Since $D(T)$ is finite over k of degree equal to the order n of T, this means that $D(T)(k)$ has n elements. Now T is isomorphic to a sum of groups $\mathbb{Z}/n_i\mathbb{Z}$, where $n = \prod_i n_i$; consequently

$$D(T) = \prod_i D(\mathbb{Z}/n_i\mathbb{Z}) = \prod_i \mu_{n_i},$$

[11] Editor's note: this uses the fact that G is flat over S, by (a).

where μ_{n_i} is the group scheme of n_i-th roots of unity. Hence

$$\operatorname{card}(D(T)(k)) = \prod_i \operatorname{card}(\mu_{n_i}(k)).$$

Here $\operatorname{card}(\mu_{n_i}(k))$, the number of n_i-th roots of unity in k, is at most n_i, and equality holds if and only if n_i is prime to the characteristic p of k. It follows that

$$\operatorname{card}(D(T)(k)) = n, \qquad n = \prod_i n_i,$$

if and only if every n_i is prime to p, that is, if and only if n is prime to p. □

3. Exactness Properties of the Functor D_S

Theorem 3.1. *Let S be a prescheme, and let*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be an exact sequence of ordinary commutative groups. Consider the sequence of transposed homomorphisms

$$0 \longrightarrow D_S(M'') \xrightarrow{v^t} D_S(M) \xrightarrow{u^t} D_S(M') \longrightarrow 0.$$

(i) *The homomorphism v^t induces an isomorphism from $D_S(M'')$ onto the kernel of u^t, and u^t is faithfully flat and quasi-compact.*

(ii) [12] *$D_S(M')$ represents the quotient sheaf $D_S(M)/D_S(M'')$ for the* (fpqc) *topology.*

Let $\mathcal{M}$ denote the family of faithfully flat quasi-compact morphisms. First, (ii) follows from (i) (cf. IV, 4.6.5.1). Indeed, the equivalence relation on $D_S(M)$ defined by u^t is the same as that defined by the subgroup

$$\operatorname{Ker}(u^t) = D_S(M'').$$

Since $u^t \in \mathcal{M}$, this equivalence relation is $\mathcal{M}$-effective (cf. IV, 3.3.2.1); therefore $D_S(M')$ represents the quotient sheaf for the (fpqc) topology (cf. IV, 4.6.5).

The first assertion of (i) is an immediate consequence of the definition of the functors $D_S(-)$. More generally, every exact sequence

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with no zero on the left gives rise to an exact transposed sequence

$$0 \longrightarrow D_S(M'') \longrightarrow D_S(M) \longrightarrow D_S(M').$$

(This remains valid in the more general setting considered at the beginning of §1.) On the other hand, because $D_S(M)$ and $D_S(M')$ are affine over S, the morphism u^t is necessarily affine, and a fortiori quasi-compact, whatever the homomorphism $u : M' \to M$. The second assertion of (i) will therefore follow from part (a) of the following corollary.

Corollary 3.2. *Let S be a nonempty prescheme, let $u : M' \to M$ be a homomorphism of ordinary commutative groups, and let $u^t : G \to G'$ be the transposed homomorphism. Then:*

[12] Editor's note: what follows has been added, and the proof has been expanded accordingly.

(a) *u is a monomorphism if and only if u^t is faithfully flat;*

(b) *u is an epimorphism if and only if u^t is a monomorphism; in that case u^t is in fact a closed immersion.*

To prove (a), observe that if u is a monomorphism, then $\mathcal{O}_S(M)$, as a module over $\mathcal{O}_S(M')$, has a nonempty basis: namely, the system of sections defined by any system of representatives of M modulo M'. It is therefore faithfully flat. Conversely, if $\mathcal{O}_S(M)$ is faithfully flat over $\mathcal{O}_S(M')$, then

$$u^t : \mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$$

is injective, which, when $S \neq \varnothing$, implies that $u : M' \to M$ is injective.

To prove (b), observe that if u is an epimorphism, then $\mathcal{O}_S(M') \to \mathcal{O}_S(M)$ is surjective, so u^t is a closed immersion and, a fortiori, a monomorphism. Conversely, if u^t is a monomorphism, then $\operatorname{Ker} u^t$ is the unit group. Putting $M'' = \operatorname{Coker} u$, we have seen that

$$\operatorname{Ker} u^t \simeq D_S(M'').$$

Thus 2.1(d) gives $M'' = 0$, and u is an epimorphism.

The usual argument now gives the following consequence of 3.1.

Corollary 3.3. *Let*

$$M' \xrightarrow{u} M \xrightarrow{v} M''$$

be an exact sequence of ordinary commutative groups, and consider the transposed sequence

$$G'' \xrightarrow{v^t} G \xrightarrow{u^t} G'.$$

Then v^t induces a faithfully flat quasi-compact morphism from G'' to $\operatorname{Ker} u^t$, and this kernel is a diagonalizable group isomorphic to

$$D_S(v(M)) = D_S(\operatorname{Coker} u).$$

Corollary 3.4. *Let S be a prescheme, let $u : G \to H$ be a homomorphism of S-group preschemes that are locally diagonalizable, with H of finite type over S, and put $G' = \operatorname{Ker} u$. Then:*

(a) *G' is locally diagonalizable, and is of finite type over S if G is;*

(b) *the quotient G/G' "exists": more precisely, the equivalence relation on G defined by G' is $\mathcal{M}$-effective, where $\mathcal{M}$ is the family of faithfully flat quasi-compact morphisms (cf. IV, 3.4). Moreover, G/G' is locally diagonalizable and of finite type over S;*

(c) *the homomorphism $u : G \to H$ factors uniquely as*

$$G \xrightarrow{v} G/G' \xrightarrow{w} H,$$

where v is the canonical homomorphism, and is therefore faithfully flat and quasi-compact. Moreover, w is a closed immersion, and a fortiori a monomorphism.

Finally, the quotient

$$H' = H/\operatorname{Im} w = \operatorname{Coker} w = \operatorname{Coker} u$$

exists: more precisely, the equivalence relation on H defined by G/G' is $\mathcal{M}$-effective, and H' is of finite type over S.

The first assertion of (c) follows from (b), by the definition of the quotient G/G' (cf. IV, 3.2.3). [13] Let us show that the (fpqc) quotient sheaf $\widetilde{G}/\widetilde{G'}$ is representable. This can be checked locally on S, as can all the other assertions; we may therefore suppose that G and H are diagonalizable, of the form $D_S(M)$ and $D_S(N)$.

Since H is of finite type over S, the group N is of finite type by 2.1(b). Therefore, by 1.5, u is defined by a homomorphism $u' : N \to M$. It follows from 3.1 and 3.2 that

$$G' \simeq D_S(\operatorname{Coker} u')$$

and that $\widetilde{G}/\widetilde{G'}$ is represented by $D_S(\operatorname{Im} u')$. Moreover, on considering the exact sequence

$$0 \longrightarrow \operatorname{Ker} u' \longrightarrow N \xrightarrow{w'} \operatorname{Im} u' \longrightarrow 0,$$

we find that w is a closed immersion and that

$$H' = H/\operatorname{Im} w = D_S(\operatorname{Ker} u').$$

The latter is of finite type over S, since N, and hence $\operatorname{Ker} u'$, is of finite type.

Remarks. The existence result for quotients in 3.4 will be substantially generalized in §5.

On the other hand, note that in the present section and the preceding one, the hypothesis $S \neq \varnothing$ was used only to ensure the validity of certain converses, allowing properties of the corresponding ordinary groups to be deduced from certain hypotheses on diagonalizable S-groups. The results in the "forward" direction hold without any restriction on S, and the proofs given here apply in general.

Corollary 3.5. *Let G be a diagonalizable group prescheme over S, and let $n \neq 0$ be an integer. Then the subgroup ${}_nG$ of G, the kernel of the homomorphism*

$$n \cdot \mathrm{id}_G : G \longrightarrow G,$$

is integral over S, and is finite over S if G is of finite type over S.

Indeed, if $G = D_S(M)$, then

$${}_nG = D_S(M/nM)$$

by 3.1, and the result follows from 2.1(b), (c), and (c′).

4. Torsors under a Diagonalizable Group

Let S be a prescheme and let $G = D_S(M)$ be a diagonalizable group over S. We shall determine the G-torsors (or principal homogeneous G-bundles) over S for the faithfully flat quasi-compact topology; cf. Exposé IV, 5.1. Recall that a prescheme P over S, with S-operator group G, is called a *torsor* or a *principal homogeneous bundle* if every point of S has an open neighborhood U and a faithfully flat quasi-compact morphism $S' \to U$ such that

$$P' = P \times_S S'$$

is isomorphic, as an object with operators, to

$$G' = G \times_S S',$$

[13] Editor's note: the original has been expanded in what follows.

where G' acts on itself by right translations. Since G is affine over S, SGA 1, VIII 5.6 implies that P is necessarily affine over S. Also, because G itself is faithfully flat and quasi-compact over S, P is principal homogeneous under G if and only if it is formally principal homogeneous and is, in addition, faithfully flat and quasi-compact over S (cf. IV, 5.1.6).

Recall also (Exposé I, 4.7.3) that giving an S-prescheme P, affine over S, with S-operator group $G = D_S(M)$, amounts to giving a commutative quasi-coherent algebra graded by M over S. In other words, it amounts to giving a quasi-coherent algebra $\mathscr{A}$ over S, together with a direct-sum decomposition as a module

$$\mathscr{A} = \coprod_{m \in M} \mathscr{A}_m,$$

such that

$$\mathscr{A}_m \cdot \mathscr{A}_{m'} \subseteq \mathscr{A}_{m+m'} \qquad \text{for } m, m' \in M.$$

With this understood, the answer to the preceding problem is as follows.

Proposition 4.1. *Let P be the prescheme with S-operator group $G = D_S(M)$ defined by the M-graded algebra $\mathscr{A}$. Then P is a principal homogeneous bundle under G if and only if $\mathscr{A}$ satisfies the following conditions:*

(a) *for every $m \in M$, $\mathscr{A}_m$ is an invertible module over S;*

(b) *for $m, m' \in M$, the homomorphism*

$$\mathscr{A}_m \otimes_{\mathscr{O}_S} \mathscr{A}_{m'} \longrightarrow \mathscr{A}_{m+m'}$$

induced by multiplication in $\mathscr{A}$ is an isomorphism.

The necessity of these conditions follows immediately by descent, since they hold when P is the trivial principal homogeneous bundle, that is, when $\mathscr{A} = \mathscr{O}_S(M)$. For sufficiency, note that (a) already implies that P is faithfully flat over S; it is in any event quasi-compact over S, since it is affine over S. It therefore remains to check that P is formally principal homogeneous under G, that is, that the familiar morphism

$$P \times_S G \longrightarrow P \times_S P$$

is an isomorphism. On affine algebras this morphism is given explicitly by the homomorphism

$$\mathscr{A} \otimes \mathscr{A} \longrightarrow \mathscr{A}(M) = \mathscr{A} \otimes \mathscr{O}_S(M).$$

In bidegree (m, n), where $m, n \in M$, it is given by

$$x_m \otimes y_n \longmapsto x_m y_n \otimes e_n.$$

At the level of degrees, this homomorphism is compatible with the homomorphism $M \times M \to M \times M$ given by

$$(m, n) \longmapsto (m + n, n),$$

which is an isomorphism. This shows that (b) expresses precisely, and independently of (a), that P is formally principal homogeneous, and proves 4.1.

Faithfully flat descent also gives:

Corollary 4.2. *The conditions of 4.1 imply that the homomorphism*

$$\mathscr{O}_S \longrightarrow \mathscr{A}_0$$

is an isomorphism.

For example, if $M = \mathbb{Z}$, then, under the conditions of 4.1, the algebra $\mathscr{A}$ is essentially determined by $\mathscr{A}_1 = \mathscr{L}$, namely

$$\mathscr{A} \simeq \coprod_{n\in\mathbb{Z}} \mathscr{L}^{\otimes n}$$

as graded algebras. Thus we recover the familiar result:

Corollary 4.3. *The category of principal homogeneous bundles P over S under $\mathbb{G}_{m,S}$ is equivalent to the category of invertible modules $\mathscr{L}$ over S, where the morphisms in both categories are the isomorphisms of the structures in question. Quasi-inverse functors are obtained by assigning to P the degree-1 component of its $\mathbb{Z}$-graded affine algebra, and by assigning to $\mathscr{L}$ the spectrum of the $\mathbb{Z}$-graded algebra*

$$\coprod_{n\in\mathbb{Z}} \mathscr{L}^{\otimes n}.$$

In particular:

Corollary 4.4. *The group of isomorphism classes of principal homogeneous bundles over S under $\mathbb{G}_{m,S}$ is isomorphic to the group* $\mathrm{Pic}(S)$ *of isomorphism classes of invertible modules over S, that is, to*

$$H^1(S, \mathscr{O}_S^\times).$$

Since $\mathbb{G}_{m,S}$ is the scheme of automorphisms of the module $\mathscr{O}_S$, Corollary 4.4 is equivalent to the following statement, one of the forms of Hilbert's "Theorem 90."

Corollary 4.5. *Every principal homogeneous bundle over S under $\mathbb{G}_m$ is locally trivial for the Zariski topology.*

Remark 4.5.1. The preceding statement is no longer true in general for a group such as μ_n, or for a "twisted form" of $\mathbb{G}_m$. For example, the unique twisted form of $\mathbb{G}_m$ over the field $\mathbb{R}$ of real numbers has a first cohomology group equal to $\mathbb{Z}/2\mathbb{Z}$.

[14] Indeed, let $\mathbb{S}^1$ be the kernel of the norm morphism

$$N : \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \longrightarrow \mathbb{G}_{m,\mathbb{R}};$$

it is a $\mathbb{C}/\mathbb{R}$-twisted form of $\mathbb{G}_{m,\mathbb{R}}$. The equation $N(z) = -1$ in $\prod_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})$ defines an $\mathbb{S}^1$-torsor X over $\mathrm{Spec}(\mathbb{R})$, locally trivial for the étale topology, but not trivial, since $X(\mathbb{R}) = \varnothing$. Let us show that

$$H^1_{\text{ét}}(\mathbb{R}, \mathbb{S}^1) \simeq \mathbb{Z}/2\mathbb{Z}.$$

There is an exact sequence of smooth commutative algebraic $\mathbb{R}$-groups

$$1 \longrightarrow \mathbb{S}^1 \longrightarrow \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \longrightarrow \mathbb{G}_{m,\mathbb{R}} \longrightarrow 1,$$

which gives rise to a long exact sequence in étale cohomology (or in Galois cohomology):

$$\begin{aligned} 0 \longrightarrow \mathbb{S}^1(\mathbb{R}) \longrightarrow \mathbb{C}^\times \xrightarrow{N} \mathbb{R}^\times \longrightarrow H^1_{\text{ét}}(\mathbb{R}, \mathbb{S}^1) \\ \longrightarrow H^1_{\text{ét}}\left(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}\right) \longrightarrow \cdots . \end{aligned}$$

[14] Editor's note: what follows has been added.

Now (see, for example, XXIV, 8.4),

$$H^1_{\text{ét}}\left(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}\right) \simeq H^1_{\text{ét}}(\mathbb{C}, \mathbb{G}_{m,\mathbb{C}}),$$

and the latter group is zero by 4.5 (or, in this case, because $\mathbb{C}$ is algebraically closed). We therefore obtain an isomorphism

$$H^1_{\text{ét}}(\mathbb{R}, \mathbb{S}^1) \simeq \mathbb{R}^\times/N(\mathbb{C}^\times) \simeq \{\pm 1\}.$$

We shall need the following result in the next section.

Proposition 4.6. *Under the conditions of 4.1, conditions (a) and (b) are equivalent to the following conditions:*

(a') *the homomorphism $\mathcal{O}_S \to \mathscr{A}_0$ is an isomorphism;*

(b') *for every $m \in M$ (it is enough to require this for the members of a system of generators of M), one has*

$$\mathscr{A}_m \cdot \mathscr{A}_{-m} = \mathscr{A}_0.$$

The necessity is clear in view of 4.2.[15] We are thus reduced to proving the following corollary.

Corollary 4.7. *Let*

$$A = \coprod_{n \in \mathbb{Z}} A_n$$

be a $\mathbb{Z}$-graded ring such that

$$A_1 \cdot A_{-1} = A_0.$$

Then the A_n are invertible A_0-modules, and, for $n, n' \in \mathbb{Z}$, the homomorphism

$$A_n \otimes_{A_0} A_{n'} \longrightarrow A_{n+n'}$$

induced by multiplication in A is an isomorphism.

By hypothesis there exist $f_i \in A_1$ and $g_i \in A_{-1}$ such that

$$(*) \qquad \sum_i f_i g_i = 1.$$

Since the assertion to be proved is local on $\mathrm{Spec}(A_0)$,[16] and since $(*)$ shows that $\mathrm{Spec}(A_0)$ is covered by the affine open subsets $D(f_i g_i)$, we are reduced to the case in which there is an element $f \in A_1$ that is invertible in A. Then, for every $n \in \mathbb{Z}$, f^n is an invertible element of A_n, and therefore defines an isomorphism

$$A_0 \xrightarrow{\sim} A_n, \qquad h \longmapsto f^n h.$$

Moreover, this shows that one obtains an isomorphism

$$A_0[t, t^{-1}] \longrightarrow A$$

of graded A_0-algebras by sending t to f, which completes the proof of 4.7.

[15] Editor's note: "4.1" has been corrected to "4.2."
[16] Editor's note: the following passage has been expanded.

Under the hypotheses of 4.6, 4.7 already shows that the $\mathscr{A}_m$, for $m \in M$, are invertible. To prove condition 4.1(b), we may therefore suppose that $\mathscr{A}_m$ and $\mathscr{A}_{m'}$ have bases f_m and $f_{m'}$, with inverses

$$f_m^{-1} \in \Gamma(\mathscr{A}_{-m}) \qquad \text{and} \qquad f_{m'}^{-1} \in \Gamma(\mathscr{A}_{-m'}).$$

Multiplication by

$$f_m^{-1} f_{m'}^{-1} \in \Gamma(\mathscr{A}_{-m-m'})$$

then defines a homomorphism

$$\mathscr{A}_{m+m'} \longrightarrow \mathscr{A}_0 \simeq \mathscr{O}_S$$

that sends the image of $f_m \otimes f_{m'}$ to the unit section of $\mathscr{O}_S$. In the diagram

$$\begin{tikzcd} \mathscr{A}_m \otimes \mathscr{A}_{m'} \arrow[r, "u"] \arrow[rr, bend left, "w"] & \mathscr{A}_{m+m'} \arrow[r, "v"] & \mathscr{A}_0 \simeq \mathscr{O}_S, \end{tikzcd}$$

the maps w and v are therefore epimorphisms of invertible sheaves, hence isomorphisms; consequently u is an isomorphism. □

5. Quotient of an Affine Scheme by a Diagonalizable Group Acting Freely

[17] We denote by $\mathcal{M}$ the set of faithfully flat quasi-compact morphisms, and recall that torsors are understood with respect to the (fpqc) topology.

Theorem 5.1. *Let S be a prescheme, M an ordinary commutative group, $G = D_S(M)$ the diagonalizable group over S that it defines, and P an S-prescheme affine over S, on which G acts freely on the right.*

Then the equivalence relation defined by G on P is $\mathcal{M}$-effective (cf. IV, 3.4); that is, the quotient $X = P/G$ exists and P is a torsor over X under $G_X = D_X(M)$. Moreover, P/G is affine over S. More precisely, if P is defined by the M-graded algebra $\mathcal{A}$, then P/G is isomorphic to $\operatorname{Spec}(\mathcal{A}_0)$, *where $\mathcal{A}_0 = \mathcal{A}^G$ is the degree-zero component of $\mathcal{A}$.*

Proof. Put $X = \operatorname{Spec}(\mathcal{A}_0)$. The inclusion $\mathcal{A}_0 \to \mathcal{A}$ gives a natural morphism $P \to X$, which is invariant under the action of G. Thus P becomes an X-prescheme, affine over X, with operator group $G_X = D_X(M)$; the hypothesis that G acts freely on P/S implies that G_X acts freely on P/X. It remains only to show that P is a principal homogeneous bundle under G_X, using the fact that $\mathcal{B}_0 = \mathcal{O}_X$, where $\mathcal{B}$ is the M-graded algebra over X defining P/X.

We may therefore suppose that $X = S$ and that S is affine. Then P is affine, defined by an M-graded ring A; write A_m for its homogeneous part of degree m, so that $S = \operatorname{Spec}(A_0)$. In view of 4.6, it remains to verify

$$A_m \cdot A_{-m} = A_0 \qquad \text{for every } m \in M. \tag{x}$$

A direct calculation also shows that (x) is equivalent to saying that

$$P \times_S G \longrightarrow P \times_S P$$

[17] Editor's note: We have added the following sentence.

is a closed immersion, rather than merely a monomorphism (as follows from the hypothesis that G acts freely); equivalently, the homomorphism of affine rings

$$\theta : A \otimes_{A_0} A \longrightarrow A(M)$$

is surjective.[18] This proves 5.1 if one assumes that the equivalence relation defined by G on P is closed. We shall show, however, that this hypothesis already follows from the fact that G acts freely. This is also implicit in Theorem 5.1: the scheme $G \times_S P$ must be isomorphic to $P \times_X P$, which is closed in $P \times_S P$ because X is affine, hence separated, over S.

Let $R = P \times_S G$. The hypothesis that G acts freely, that is, that $R \to P \times_S P$ is a monomorphism, says that the diagonal morphism

$$R \longrightarrow R \times_{(P \times_S P)} R = R'$$

is an isomorphism. We have $R = \mathrm{Spec}(A(M))$ and

$$R' = \mathrm{Spec}\big(A(M \times M)/K\big),$$

[19] where K is the ideal generated by the elements

$$x_m(e_{m,0} - e_{0,m}), \qquad m \in M, \quad x_m \in A_m.$$

Let

$$\phi : A(M \times M) \longrightarrow A(M)$$

be the surjective ring homomorphism defined by

$$xe_{m,n} \longmapsto xe_{m+n} \qquad (m, n \in M,\ x \in A),$$

where the e_m, respectively $e_{m,n} = e_m \otimes e_n$, are the elements of the canonical basis of $A(M)$, respectively $A(M \times M)$. The diagonal morphism $R \to R'$ then corresponds to the homomorphism

$$\bar{\phi} : A(M \times M)/K \longrightarrow A(M)$$

obtained by passing to the quotient by K. The kernel of ϕ is the ideal K' generated by

$$d_m = e_{m,0} - e_{0,m}.$$

We have $K \subseteq K'$, and the hypothesis that G acts freely on P, that is, that $\bar{\phi}$ is an isomorphism, is equivalent to $K' = K$, expressed by the relations

$$\text{(xx)} \qquad d_m \in K = \sum_p A(M \times M)\, A_p\, d_p \qquad \text{for every } m \in M.$$

Using the natural trigrading of $A(M \times M)$, and the fact that the first degree of d_m is zero, this says that d_m can be written as a sum of elements of the form

$$fe_{r,s}(e_{p,0} - e_{0,p}), \qquad f \in A_{-p} \cdot A_p.$$

[18] Editor's note: Indeed, for every $b \in A$ and $a_m \in A_m$, one has $\theta(b \otimes a_m) = ba_m \otimes e_m$; hence the surjectivity of θ is equivalent to (x).

[19] Editor's note: We have denoted by K the ideal denoted by J in the original, in order to distinguish it from the ideals J_p of A_0 occurring in (xxx). We have also made explicit below that the relations (xx) are equivalent to the equality $K = K'$; see what follows.

Since the total degree of d_m is m, it is enough to use terms satisfying

$$r + s + p = m.$$

Thus, for every $m \in M$, there must be an expression

$$\text{(xxx)} \qquad \begin{cases} d_m = e_{m,0} - e_{0,m} = \sum_{r,s} \lambda_{r,s}(e_{m-s,s} - e_{r,m-r}), \\ \qquad\qquad \lambda_{r,s} \in J_p = A_p \cdot A_{-p} \subseteq A_0, \qquad p = m - (r+s). \end{cases}$$

We must deduce (x), that is, the relations

$$\text{(xxxx)} \qquad J_n = A_0 \qquad \text{for every } n \in M.$$

For this it is enough to establish the same relation modulo every maximal ideal of A_0. Since all the hypotheses are preserved under such a reduction, we may assume that A_0 is a field.

Lemma 5.2. *Under the preceding conditions, with A_0 a field, if $M \neq 0$, there exists $p \in M - \{0\}$ such that $J_p = A_0$.*

Indeed, otherwise the sum on the right-hand side of (xxx) would vanish for every $m \in M$, which is absurd.

Corollary 5.3. *Under the preceding conditions, but without assuming that A_0 is a field, there are finitely many elements $p_i \in M - \{0\}$ such that*

$$\sum_i J_{p_i} = A_0.$$

Indeed, apply 5.2 to the situations obtained from A/A_0 by reduction modulo the maximal ideals of A_0.[20]

Corollary 5.4. *Suppose again that A_0 is a field. For every subgroup N of M with $N \neq M$, there exists $p \in M - N$ such that $J_p = A_0$.*

Indeed, let $M' = M/N$, and consider the M'-graded ring A', whose underlying ring is A and whose grading is given by

$$A'_{m'} = \coprod_{m \in h^{-1}(m')} A_m,$$

where $h : M \to M' = M/N$ is the canonical homomorphism. Geometrically, this construction amounts to considering the subgroup $G' = D_S(M')$ of G, and the structure on P of a scheme with operator group G' induced by the action of G. It is then clear that G' acts freely on P ; that is, the pair (M', A') satisfies the hypotheses of 5.3. Thus

$$1 = \sum_i f_i g_i, \qquad f_i \in A'_{m'_i}, \quad g_i \in A'_{-m'_i}, \quad m'_i \in M' - \{0\}.$$

Taking the components of the right-hand side along A_0, and using the fact that A_0 is a field, immediately gives 5.4.

Now observe that

$$J_p J_q \subseteq J_{p+q} \qquad \text{and} \qquad J_p = J_{-p}.$$

[20] Editor's note: It follows from 5.2 that $\sum_{p \neq 0} J_p = A_0$; hence 1 can be written as a finite sum $\sum_i x_i$, with $x_i \in J_{p_i}$.

Thus, if N denotes the set of $m \in M$ such that $J_m = A_0$, then N is a subgroup of M. By 5.4, it equals M. This completes the proof of Theorem 5.1.

As noted in the course of the proof, Theorem 5.1 implies:

Corollary 5.5. *Under the hypotheses of 5.1, the graph morphism*

$$P \times_S G \longrightarrow P \times_S P$$

is a closed immersion.

It follows immediately that:

Corollary 5.6. *Let σ be a section of P over S. Then the morphism $g \mapsto \sigma \cdot g$ from G to P defined by σ is a closed immersion.*

Corollary 5.7. *Let G and H be two S-groups, with G diagonalizable and H affine over S, and let $u : G \to H$ be a homomorphism of S-groups that is a monomorphism. Then u is a closed immersion, $H/G = X$ exists, H is a principal homogeneous bundle over X under G_X, and X is affine over S.*

Corollary 5.8. *Under the hypotheses of 5.1, if P is of finite type (respectively, of finite presentation) over S, then the same is true of $X = P/G$.*

Indeed, the hypothesis implies that the fibers of G_X are of finite type. Hence G_X is of finite presentation over X, by 2.1(b); since P is a torsor under G_X, it is therefore of finite presentation over X.[21] It is also faithfully flat over X, so the conclusion follows from Exposé V, Proposition 9.1.

6. Essentially Free Morphisms and Representability of Certain Functors of the Form $\prod_{Y/S} Z/Y$ [(*)]

Definition 6.1. Let $f : X \to S$ be a morphism of preschemes. We say that f is *essentially free*, or that X is *essentially free over S*, if there are an affine open covering (S_i) of S, for each i an S_i-prescheme S'_i that is affine and faithfully flat over S_i, and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine opens X'_{ij}, such that, for every (i, j), the ring of X'_{ij} is a free module over the ring of S'_i.[22]

Proposition 6.2.

(a) *If X is essentially free over S, then it is flat over S. The converse holds if S is Artinian.*

(b) *If S is the spectrum of a field, every S-prescheme is essentially free over S.*

(c) *If X is essentially free over S, and if $S' \to S$ is a base-change morphism, then $X' = X \times_S S'$ is essentially free over S'. The converse holds if $S' \to S$ is faithfully flat and quasi-compact.*

[21] Editor's note: Cf. EGA II, 2.7.1(vi).

[(*)] This section is independent of the theory of diagonalizable groups; its natural place would be in Exposé $\mathrm{VI_B}$.

[22] Editor's note: "Free" may be replaced by "projective"; cf. 6.8 below. This notion should also be compared with that of an S-prescheme that is flat and pure, introduced and developed in [RG71]; see in particular *loc. cit.*, Part I, 3.3.12, and Part II, 3.1.4.1.

The proof is immediate, using for the converse in (a) the fact that a flat module over an Artinian local ring is free.[23]

Proposition 6.3. *Let H be a diagonalizable S-group prescheme (more generally, one that becomes diagonalizable after a suitable faithfully flat quasi-compact extension of every affine open of S; that is, H is "of multiplicative type," cf. IX, 1.1). Then H is essentially free over S.*

Indeed, if H is diagonalizable, it is affine over S and is defined by an algebra that is a free $\mathcal{O}_S$-module.

The introduction of Definition 6.1 is justified by the following theorem.

Theorem 6.4. *Let S be a prescheme, Z an S-prescheme essentially free over S, and Y a closed subprescheme of Z. Consider the functor*[24]

$$F = \prod_{Z/S} Y/Z : \ (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Set},$$

$$F(S') = \Gamma(Y_{S'}/Z_{S'}) = \begin{cases} \varnothing, & Z_{S'} \neq Y_{S'}, \\ \{\mathrm{id}_{Z_{S'}}\}, & Z_{S'} = Y_{S'}. \end{cases}$$

This functor is representable by a closed subprescheme of S.[25]

[26] First note that F is a sheaf for the (fpqc) topology. Since $F(S')$ is either $\varnothing$ or $\{\mathrm{pt}\}$ for every S', it is enough to check the following assertion: if (S_i) is an open covering of S (respectively, if $S' \to S$ is faithfully flat and quasi-compact), and every $Y_{S_i} \to Z_{S_i}$ is an isomorphism (respectively, $Y_{S'} \to Z_{S'}$ is an isomorphism), then $Y \to Z$ is an isomorphism. This is clear in the first case and follows from SGA 1, VIII, 5.4, or EGA IV$_2$, 2.7.1, in the second.

Moreover, by SGA 1, VIII, 2.1 and 5.5, faithfully flat quasi-compact morphisms are morphisms of effective descent for the fibered category of arrows that are closed immersions. Thus, with the notation of 6.1, we may reduce to the case $S = S'_i$.

Let (Z_j) be a covering of Z by affine opens such that $\mathcal{O}(Z_j)$ is a free module over $A = \mathcal{O}(S)$, put $Y_j = Y \cap Z_j$, and let

$$F_j : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Set}$$

be the functor defined from (Z_j, Y_j) as F is defined from (Z, Y). It is a subfunctor of the terminal functor, and clearly

$$F = \bigcap_j F_j.$$

It therefore suffices to prove that every F_j is representable by a closed subscheme T_j of S, for then F is represented by the closed subscheme $T = \bigcap_j T_j$. We may consequently suppose that Z is affine as well, say $Z = \mathrm{Spec}(B)$, where B is a free A-module. Let J be a subset of B defining the closed subprescheme Y of Z, and let K be the ideal of A

[23] Editor's note: Indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A-module, and $(x_i)_{i\in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k. Let F be the free A-module with basis $(e_i)_{i\in I}$, and let $\phi : F \to M$ be the A-morphism defined by $\phi(e_i) = x_i$. Then $Q = \mathrm{Coker}\,\phi$ satisfies $Q = \mathfrak{m}Q$, whence $Q = 0$, since $\mathfrak{m}$ is nilpotent. Suppose moreover that M is flat over A. Then $K = \mathrm{Ker}\,\phi$ satisfies $K \otimes_A k = 0$, that is, $K = \mathfrak{m}K$, whence $K = 0$.

[24] Editor's note: Cf. II.1, where this functor is denoted $\prod_{Z/S} Y$.

[25] Editor's note: We have corrected Z to S.

[26] Editor's note: We have expanded the original in what follows; see also 1.7.3.

generated by the $u_i(J) \subseteq A$, where the $u_i : B \to A$ are the coordinate forms for the chosen basis. It is immediate that

$$T = \mathcal{V}(K) = \operatorname{Spec}(A/K)$$

has the required property. This completes the proof.

Examples 6.5. We give some important examples of functors that reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.4, and for which criteria of representability will be useful below. Here S denotes a prescheme, and $X, Y, Z, \ldots$ denote preschemes over S.

(a) Suppose given an S-morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z), \tag{x}$$

("X acts on Y with values in Z"), that is, a morphism

$$r : X \times_S Y \longrightarrow Z. \tag{xx}$$

Let Z' be a subprescheme of Z. It gives a monomorphism

$$\underline{\mathrm{Hom}}_S(Y, Z') \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z),$$

making the first functor a subfunctor of the second. Let X' be the inverse image of this subfunctor under (x). Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Z_T$$

factors through Z'_T. This functor has the following description. Put $P = X \times_S Y$, and let P' be the inverse image of Z' under $r : P \to Z$. Then there is an evident isomorphism

$$X' \simeq \prod_{P/X} P'/P. \tag{xxx}$$

It follows that *if Y is essentially free over S and Z' is closed in Z, then the subfunctor X' of X is representable by a closed subprescheme of X.*

(b) Suppose given two actions of X on Y with values in Z, that is, two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\mathrm{Hom}}_S(Y, Z),$$

and put $X' = \operatorname{Ker}(q_1, q_2)$. This is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms

$$q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$$

are equal. Giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z \times_S Z),$$

or, equivalently, a morphism

$$r : X \times_S Y \longrightarrow Z \times_S Z.$$

Put $U = Z \times_S Z$, and let U' be the diagonal subprescheme of $Z \times_S Z$. Then X' is the inverse image under q of the subfunctor

$$\underline{\mathrm{Hom}}_S(Y, U') \longrightarrow \underline{\mathrm{Hom}}_S(Y, U).$$

It can therefore be written in the form (xxx), with $P = X \times_S Y$ and P' the inverse image of the diagonal under r, that is, the kernel of

$$X \times_S Y \underset{r_2}{\overset{r_1}{\rightrightarrows}} Z.$$

Thus we are in the situation of (a).

Consequently, *if Y is essentially free over S and Z is separated over S, then the subfunctor X' of X is representable by a closed subprescheme of X.*

(c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y);$$

that is, "X acts on Y." Let X' be the "kernel" of this morphism, namely the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \to Y_T$ is the identity. This functor falls under (b), as one sees by introducing a second morphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

that "makes X act trivially on Y." Hence, *if Y is essentially free and separated over S, then the kernel subfunctor of q is representable by a closed subprescheme of X.*

(d) Under the hypotheses of (c), consider the subfunctor Y' of Y of "invariants under X." Thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $\bar{q}(y) : X_T \to Y_T$ is the "T-constant morphism with value y." Introducing q' as in (c), and writing $\bar{q}$ and $\bar{q}'$ for the corresponding morphisms,

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that $Y' = \mathrm{Ker}(\bar{q}, \bar{q}')$. It therefore again falls under (b), with the roles of X and Y reversed and $Z = Y$. Consequently, *if X is essentially free over S and Y is separated over S, then the subfunctor Y' of Y of invariants under X is representable by a closed subprescheme of Y.*

(e) Constructions of the type described in the preceding examples occur especially often in group theory. Thus, when a group prescheme G over S acts on the S-prescheme X,

$$q : G \longrightarrow \underline{\mathrm{Aut}}_S(X),$$

the kernel of q, the "subgroup of G acting trivially," is a closed subprescheme of G provided that X is essentially free and separated over S, by (c). The subobject X^G of invariants is a closed subprescheme of X, provided that G is essentially free over S and X is separated over S,[27] by (d).

Let Y, Z be subpreschemes of X, and consider the subfunctor $\underline{\mathrm{Transp}}_G(Y, Z)$ of G, the "transporter of Y into Z." Its points with values in an S-prescheme T are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subseteq Z_T$, that is, induces a morphism $Y_T \to X_T$ factoring through Z_T. Hence, *if Y is essentially free over S, and Z is closed in X, then $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subprescheme of G*, by (a).

One may also consider the *strict transporter* of Y into Z,[28] whose T-valued points are the $g \in G(T)$ such that $g(Y_T) = Z_T$. It is precisely

$$\underline{\mathrm{Transp}}_G(Y, Z) \cap \sigma\big(\underline{\mathrm{Transp}}_G(Z, Y)\big),$$

[27] Editor's note: We have corrected S_X to S.

[28] Editor's note: It is denoted $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(Y, Z)$.

where σ is the symmetry of G. Consequently, if Y and Z are essentially free over S and closed in X, then the strict transporter of Y into Z is a closed subprescheme of G.

An important case is $X = G$, with G acting on itself by inner automorphisms. If H is a subprescheme of G, the strict transporter of H into itself is also called the *normalizer of H in G*, and is denoted $\underline{\mathrm{Norm}}_G H$. Hence, *if H is a closed subgroup prescheme of G, essentially free over S, then $\underline{\mathrm{Norm}}_G H$ is representable by a closed subgroup prescheme of G.*

Finally, let Z be a subprescheme of G. Its *centralizer* $\underline{\mathrm{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure in (d), regarding "Z acts on G" through the action induced by that of G. Thus, *if Z is essentially free over S and G is separated over S, then $\underline{\mathrm{Centr}}_G(Z)$ is a closed subgroup prescheme of G.* In particular, *if G is essentially free and separated over S, then the center C of G, which is none other than $\underline{\mathrm{Centr}}_G(G)$, is a closed subgroup prescheme of G.*

When S is the spectrum of a field, 6.2(b) shows that in examples (a)–(e) above the "essentially free" hypotheses are automatically satisfied; only separation hypotheses remain. Recalling that a group prescheme over a field is necessarily separated, one obtains, for example:

Corollary 6.7. [29] *Let G be a group prescheme over a field k. Then:*

- *For every subprescheme Z of G, the centralizer of Z in G is a closed subgroup prescheme of G. In particular, this holds for the center $\underline{\mathrm{Centr}}_G(G)$ of G.*
- *More generally, if $u, v : X \to G$ are morphisms of preschemes, then $\underline{\mathrm{Transp}}_G(u, v)$ is representable by a closed subprescheme of G.*
- *For two subpreschemes Y, Z of G, with Z closed, $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subprescheme of G. If Y is also closed, the same conclusion holds for $\underline{\mathrm{Transp}}^{\mathrm{str}}_G(Y, Z)$.*
- *For every subgroup prescheme[30] H of G, the normalizer $\underline{\mathrm{Norm}}_G(H)$ is a closed subgroup prescheme of G.*

Remark 6.8. [31] Let A be a commutative ring, let M be an A-module, and put

$$M^\vee = \mathrm{Hom}_A(M, A).$$

Endow the A-module $\mathrm{End}_A(M)$ with the topology of discrete pointwise convergence. Thus a basis of neighborhoods of zero is formed by the following A-submodules, where $n \in \mathbb{N}$ and $m_1, \dots, m_n \in M$:

$$K(m_1, \dots, m_n) = \{u \in \mathrm{End}_A(M) \mid u(m_i) = 0 \text{ for } i = 1, \dots, n\}.$$

We say that M is a *quasi-free A-module* if the image of the canonical morphism

$$\Theta : M \otimes_A M^\vee \longrightarrow \mathrm{End}_A(M)$$

contains id_M in its closure; that is, if the following condition holds:

[29] Editor's note: We have retained the numbering of the original; there is no 6.6.

[30] Editor's note: Indeed, over a field k, every subgroup prescheme of G is closed; cf. Exposé $\mathrm{VI_A}$, 0.6.1.

[31] Editor's note: We have expanded the original in what follows; in particular, we have added Lemma 6.8.1.

$$(*)\qquad \begin{cases}\text{for all } m_1,\dots,m_n \in M, \text{ there exist } x_1,\dots,x_r \in M \text{ and } f_1,\dots,f_r \in M^\vee\\ \text{such that}\quad m_i = \Theta\left(\sum_{s=1}^r x_s\otimes f_s\right)(m_i) = \sum_{s=1}^r f_s(m_i)x_s \quad \text{for } i=1,\dots,n.\end{cases}$$

(In this case $\operatorname{Im}\Theta$ is dense in $\operatorname{End}_A(M)$, since for every $u \in \operatorname{End}_A(M)$ one has $u(m_i) = \sum_{s=1}^r f_s(m_i)u(x_s) = \Theta(\sum_{s=1}^r u(x_s)\otimes f_s)(m_i)$.)

First note that this property is stable under base change. Indeed, let $\phi : A \to A'$ be a ring homomorphism, put $M' = M \otimes_A A'$, and let $m'_1,\dots,m'_n \in M'$. Write

$$m'_i = \sum_j m_{ij}\otimes b_{ij}, \qquad m_{ij}\in M, \quad b_{ij}\in A'.$$

By hypothesis there exist $x_1,\dots,x_r \in M$ and $f_1,\dots,f_r \in \operatorname{Hom}_A(M,A)$ such that

$$m_{ij} = \sum_s x_s f_s(m_{ij}) \qquad \text{for all } i,j.$$

Write $\phi\circ f_s$ for the image of f_s in

$$\operatorname{Hom}_A(M,A') = \operatorname{Hom}_{A'}(M',A').$$

Then, for $i=1,\dots,n$,

$$\begin{aligned}\left(\sum_s x_s\otimes\phi\circ f_s\right)(m'_i) &= \sum_{s,j} x_s\otimes\phi(f_s(m_{ij}))\,b_{ij}\\ &= \sum_j\left(\sum_s x_s f_s(m_{ij})\right)\otimes b_{ij}\\ &= m'_i.\end{aligned}$$

This proves that M' is quasi-free over A'.

Also, every projective A-module P is quasi-free. Indeed, there are A-morphisms

$$A^{(I)} \xrightarrow{\pi} P \xrightarrow{\tau} A^{(I)} \qquad \text{such that} \qquad \pi\circ\tau = \mathrm{id}_P\,.$$

Let (e_i) be the canonical basis of $A^{(I)}$, and put $f_i = e_i^*\circ\tau$, a linear form on P. Given $m_1,\dots,m_n\in P$, there is a finite subset J of I such that

$$m_k = \sum_{i\in J} f_i(m_k)\pi(e_i) \qquad \text{for } k=1,\dots,n.$$

Theorem 6.4 remains valid if, in the statement of Definition 6.1, "free" is replaced by "projective" or, more generally, by "quasi-free." Proceeding as in the proof of 6.4, one reduces to the following lemma.

Lemma 6.8.1. *Let M be a quasi-free A-module, N a submodule, and F the covariant functor*

$$F : (A\text{-algebras}) \longrightarrow \mathbf{Set}$$

such that $F(B) = \{\mathrm{pt}\}$ if $M_B = (M/N)_B$, and $F(B) = \varnothing$ otherwise. Then there is an ideal K of A such that $F(B) = \{\mathrm{pt}\}$ if and only if the morphism $A\to B$ factors through A/K; that is, there is an isomorphism, functorial in B,

$$F(B) \simeq \operatorname{Hom}_{A\text{-alg.}}(A/K, B).$$

Proof. Let (n_j) be a system of generators of N, and let F_j be the subfunctor of the terminal functor e, where $e(B) = \{\mathrm{pt}\}$ for every B, corresponding to the submodule An_j. We have $F(B) = \{\mathrm{pt}\}$ if and only if the image of every n_j in M_B is zero. Hence F is the intersection of the functors F_j, reducing us to the case in which N is generated by one element n.

Let B be an A-algebra, with structure map $\phi : A \to B$. If the image $n \otimes 1$ of n in M_B is zero, then for every $f \in M^\vee = \mathrm{Hom}_A(M, A)$,

$$0 = f(n) \otimes 1 = \phi(f(n)).$$

Conversely, because M is quasi-free, there exist $x_1, \ldots, x_r \in M$ and $f_1, \ldots, f_r \in M^\vee$ such that

$$n = \sum_s f_s(n) x_s,$$

whence

$$n \otimes 1 = \sum_s x_s \otimes \phi(f_s(n)).$$

It follows that $n \otimes 1 = 0$ if and only if ϕ factors through A/K, where K is the ideal generated by the $f_s(n)$, for $s = 1, \ldots, r$. This proves the lemma.

7. Appendix: On Monomorphisms of Group Preschemes

The result proved in this section is not needed in the remainder of the seminar, except in X, 8.8 and Exposés XV and XVI. It depends essentially on the existence theorem for quotient groups in Exposé VI_A. [32]

Corollary 5.7 raises the question of when a monomorphism $u : G \to H$ of S-groups can be asserted to be an immersion, or even a closed immersion.

We saw in VI_B, 1.4.2 that this is so when S is the spectrum of a field k, provided that G is of finite type over k and H is locally of finite type over k. It follows readily that the same result remains valid when S is merely assumed Artinian.[33]

On the other hand, examples of bijective monomorphisms that are not immersions are easy to give, for instance over the affine line over a field or over the spectrum of a discrete valuation ring. Take

$$H = (\mathbf{Z}/2\mathbf{Z})_S, \qquad G = G_1 \times_S G_2,$$

where G_1 is the open subgroup of H complementary to the closed point $x \neq 0$ in the fiber $H_s \simeq \mathbf{Z}/2\mathbf{Z}$, with s a fixed closed point of S, and G_2 is the closed subscheme of H which is the union of the subscheme given by the identity section and the reduced closed subscheme defined by x. One checks immediately that G_2 is stable under multiplication in H, and hence is a group scheme. The immersions $G_i \to H$, $i = 1, 2$, then define a homomorphism of S-groups

$$G = G_1 \times_S G_2 \longrightarrow H$$

which is visibly a bijective monomorphism, and even a local immersion, but is not an immersion. Indeed, G and H are reduced; G has three disjoint irreducible components, whereas H has only two.

Notice also that $G_1 \to H$ is an example of an open subgroup of H that is not closed, in contrast to the situation for algebraic groups over a field. The theory of degenerations

[32] Editor's note: cf. Exposé VI_A, Theorems 3.2 and 3.3.2.

[33] Editor's note: the additions made in Exposé VI_B should be taken into account.

of elliptic curves gives further examples of this phenomenon in which H is smooth over S of relative dimension one and G has connected fibers.

One might hope$^{(*)}$that once G is assumed flat over S, and, say, G and H are of finite presentation over S, every monomorphism $u : G \to H$ of S-groups is automatically an immersion. We shall prove a result of this kind under additional hypotheses.

First, we may assume that S is affine and, by the finite-presentation hypothesis on G and H, that S is Noetherian: one reduces to the spectrum of a ring of finite type over $\mathbf{Z}$.[34] Then G and H are Noetherian. Saying that $u : G \to H$ is a closed immersion (respectively, an immersion) is then equivalent to saying that u is a monomorphism—which holds by hypothesis—and is proper (respectively, proper at every point of $u(G)$).[35]

The valuative criterion for properness shows that it is enough to check, after every base change $S' \to S$ with S' the spectrum of a discrete valuation ring, complete if desired, that the morphism $u' : G' \to H'$ has the corresponding properness property. The case of local properness was omitted from EGA II, 7.3 and was to appear as an erratum in EGA IV.$^{(*)}$We are therefore reduced to the case in which S itself is the spectrum of a complete discrete valuation ring, subject to the proviso that any additional hypotheses imposed on G, H, u be stable under base change.

Let s, respectively s', be the closed, respectively generic, point of S. The induced homomorphisms on the fibers,

$$u_s : G_s \longrightarrow H_s, \qquad u_{s'} : G_{s'} \longrightarrow H_{s'},$$

are closed immersions because they are monomorphisms of group schemes of finite type over fields (VI_B, 1.4.2). We may therefore identify $G_{s'}$ with a closed subscheme of $H_{s'}$. We shall use the following result.

Lemma 7.1. *Let S be the spectrum of a discrete valuation ring, with generic point s'. Let H be an S-prescheme and let L' be a closed subprescheme of the generic fiber $H_{s'}$, hence also a subprescheme of H. Its schematic closure L in H—the smallest closed subprescheme of H containing L', cf. EGA I, 9.5—exists. It is also the unique closed subprescheme of H, flat over S, whose generic fiber is L'. Formation of L is functorial in the pair (H, L') and commutes with fiber products over S. In particular, if H is an S-group and L' is a subgroup of $H_{s'}$, then L is a subgroup of H.*

The proof is immediate and is left to the reader.$^{(**)}$Apply this to the pair $(H, G_{s'})$. The monomorphism $u : G \to H$ factors as

$$G \xrightarrow{u'} L \longrightarrow H,$$

where L is a subgroup of H, a closed subprescheme flat over S, and $u' : G \to L$ induces an isomorphism on generic fibers. Thus $u : G \to H$ is an immersion (respectively, a closed immersion) if and only if $u' : G \to L$ is one. We are reduced to the case in which H is flat over S and $u_{s'} : G_{s'} \to H_{s'}$ is an isomorphism, again subject to the requirement that any additional hypotheses remain valid when H is replaced by a closed subgroup prescheme.

Since H is then the schematic closure of $H_{s'}$, if u is an immersion its image $u(G)$ is a subprescheme[36] of H containing $H_{s'}$. Its schematic closure is therefore H, so $u(G)$ is an

$^{(*)}$In fact, M. Raynaud constructed a counterexample with G smooth and with connected fibers; cf. XVI, 1.1(c). Without the connected fiber hypothesis, one may take $S = \mathrm{Spec}(\mathbf{Z}_2)$, let $G = (\mathbf{Z}/2\mathbf{Z})_S$ with its nonidentity closed point removed, and let $H = (\mu_2)_S$.

[34] Editor's note: cf. EGA IV_3, ğ8 and Exposé VI_B, ğ10.

[35] Editor's note: cf. EGA IV_3, 8.11.5 and 15.7.1.

$^{(*)}$Cf. EGA IV_3, 15.7.

$^{(**)}$Cf. EGA IV_2, 2.8.

[36] Editor's note: the word "closed" has been removed.

open subprescheme of H. We must consequently prove that u is an open immersion, or an isomorphism if the desired conclusion is that it is a closed immersion.

Because G and H are flat over S, this is equivalent to the induced maps on the fibers being open immersions, respectively isomorphisms (cf. SGA 1, I, 5.7). It already holds for $u_{s'}$, so it remains to prove that u_s is an open immersion, respectively an isomorphism.

Since G and H are flat over S, their fiber dimensions are constant (VI_B, 4.3). Their generic fibers agree, hence these dimensions are the same for G and H. Thus $u_s : G_s \to H_s$ is a monomorphism of algebraic groups of the same dimension. It follows that G_s is open in H_s, and set-theoretically is a union of connected components of H_s. Indeed, after passing to an algebraically closed base field and reducing G and H, both are smooth over the field and the assertion is immediate.

The complement $H_s - G_s$ is closed in H_s, hence in H. Its complement H' in H is open and is stable under the group law. Upon replacing H by H', we may therefore reduce, with the usual proviso, to the case in which u is bijective, that is, $u_s : G_s \to H_s$ is bijective. In every case we are reduced to proving that u, equivalently u_s, is an isomorphism, possibly under this bijectivity hypothesis.

Assume first that u is bijective. If H_s is reduced, then u_s is an isomorphism, because G_s identifies with a closed subscheme of H_s having the same underlying set. In particular, if $k = \kappa(s)$ has characteristic zero, every algebraic group over k is reduced by Cartier's theorem (cf. VI_B, 1.6.1; VII_B, 3.3.1; or EGA IV_4, 16.12.2 and 17.12.5). We have proved:

Proposition 7.2. *Let $u : G \to H$ be a homomorphism of group preschemes of finite presentation over S. If u is a monomorphism, G is flat over S, and every residue field of S has characteristic zero, then u is an immersion.*

Suppose now that $\kappa(s)$ has characteristic $p > 0$, and restrict to the case in which G is commutative. Under the preceding reductions, H is also commutative, because it is the schematic closure of $H_{s'} \simeq G_{s'}$. For $n \geq 0$, set

$$S_n = \operatorname{Spec}(V/\mathfrak{m}^{n+1}), \qquad G_n = G \times_S S_n, \qquad H_n = H \times_S S_n,$$

where V is the valuation ring defining S and $\mathfrak{m}$ its maximal ideal. For $m > 0$, denote by ${}_mG$ and ${}_mH$ the kernels of multiplication by m in G and H. Define similarly

$${}_m(G_n) = ({}_mG)_n,$$

which we abbreviate to ${}_mG_n$, and likewise for H.

By VI_A, 3.2, the quotients

$$Q_n = H_n/G_n$$

exist. Each Q_n is a commutative group scheme over S_n, flat over S_n, and $H_n \to Q_n$ is faithfully flat with kernel G_n. Formation of the quotient commutes with base extension,[37] so

$$Q_n \simeq Q_m \times_{S_m} S_n \qquad (m \geq n).$$

In particular, $Q_n \times_{S_n} S_0 = Q_0 = H_0/G_0$. Since G_0 and H_0 have the same underlying set, Q_0 is set-theoretically a single point: it is a purely infinitesimal group.

It follows that every Q_n is finite and flat over S_n. Let it be defined by an algebra C_n over $V_n = V/\mathfrak{m}^{n+1}$, finite free as a V_n-module. For $m \geq n$,

$$C_n = C_m \otimes_{V_m} V_n,$$

[37]Editor's note: this should be made explicit in Exposés V and VI_A.

compatibly also with the comultiplication maps. Hence

$$C = \varprojlim C_n \quad \text{is a finite free module over} \quad V = \varprojlim V_n,$$

and the comultiplications on the C_n induce one on C. Thus $Q = \mathrm{Spec}(C)$ is a finite flat group scheme over S satisfying

$$Q \times_S S_n = Q_n$$

for every n.

Lemma 7.3. [38] *Let K be a field and let Q be a finite group scheme over K, of degree n. Then the n-th power morphism on Q is zero.*

See VII_A, 8.5.

Remark 7.3.1. The statement of Lemma 7.3 still makes sense for a finite locally free group scheme Q/S over an arbitrary base prescheme S. It would be interesting to find a proof in that generality.

Lemma 7.3 (that is, VII_A, 8.5) does show that the proposed statement holds whenever S is reduced: apply it to the fibers of Q at the maximal points of S, equivalently the generic points of its irreducible components.[39]

In the preceding situation, where S is the spectrum of a discrete valuation ring and Q is commutative, it follows that $n\,\mathrm{id}_Q = 0$. Here n is a power p^{ν_0} of the residue characteristic. Thus:

Corollary 7.4. *With Q and the Q_n as above, suppose that their common degree is p^{ν_0}. Then*

$$p^{\nu_0}\,\mathrm{id}_Q = 0, \qquad p^{\nu_0}\,\mathrm{id}_{Q_n} = 0 \quad \textit{for every } n.$$

Corollary 7.5. *Suppose in addition that G_0 is smooth over k and that $p\,\mathrm{id}_{G_0}$ is flat. Equivalently, by the structure theory of algebraic groups over an algebraically closed field $\overline{k}$, $G_{\overline{k}}$ contains no subgroup isomorphic to the additive group. Then, for every $\nu \geq \nu_0$ and every $n > 0$, the group ${}_{p^\nu}H_n$ is flat over S_n.*

Indeed, $p^{\nu_0}\,\mathrm{id}_{Q_n} = 0$ implies that $p^\nu\,\mathrm{id}_{H_n}$ factors through G_n, giving the commutative diagram

$$\begin{tikzcd} H_n \arrow[r, "p^\nu \mathrm{id}_{H_n}"] \arrow[dr, "v_n"'] & H_n \\ G_n \arrow[u, "u_n"] \arrow[r, "p^\nu \mathrm{id}_{G_n}"'] & G_n. \arrow[u, "u_n"'] \end{tikzcd}$$

We claim that v_n is flat. Since H_n and G_n are flat over S_n, it is enough to check that v_0 is flat (SGA 1, IV, 5.9). Thus we may take $n = 0$, so that $S_n = \mathrm{Spec}(k)$. Since $p\,\mathrm{id}_{G_0}$, and hence $p^\nu\,\mathrm{id}_{G_0}$, is flat, its image is an induced open subgroup G_0' of G_0. The map $u_0 : G_0 \to H_0$ is surjective, so v_0 takes its values in G_0' and may be regarded as a homomorphism to G_0'. Its composite with u_0 is an epimorphism; hence v_0 itself is an epimorphism, and therefore a flat homomorphism into G_0', and thus into G_0. Consequently v_n is flat and $\ker v_n = {}_{p^\nu}H_n$

is flat over S_n. □

[38] Editor's note: the hypothesis that Q be commutative has been removed here, and 7.3.1 modified accordingly. If Q is not commutative, the n-th power morphism need not be a group homomorphism.

[39] Editor's note: this should be added to Exposé VII_A. On the other hand, the statement is true for every S when Q is commutative; cf. Deligne's theorem in [TO70], p. 4.

Remark. We did not explicitly use the assumption that G_0 is smooth over k. It is, however, an easy consequence of the flatness of $p\,\mathrm{id}_{G_0}$; this is why the condition was stated explicitly in Corollary 7.5.

Lemma 7.6. *Let $u : G \to H$ be a surjective monomorphism of commutative algebraic groups over a field k of characteristic $p > 0$, and put $Q = H/G$, a purely infinitesimal group. There is an integer ν_1 such that, for every $\nu \geq \nu_1$, the sequence*

$$0 \longrightarrow {}_{p^\nu}G \longrightarrow {}_{p^\nu}H \longrightarrow Q \longrightarrow 0$$

is exact.

It is enough to establish exactness at Q, which in turn amounts to the injectivity of the homomorphism

$$\mathscr{O}_{Q,e} \longrightarrow \mathscr{O}_{{}_{p^\nu}H,e} \tag{x}$$

of local rings at the identity elements. Recall that Q is set-theoretically supported at the single point e. There is a natural homomorphism

$$\mathscr{O}_{{}_{p^\nu}H,e} \longrightarrow \mathscr{O}_{H,e}/\mathfrak{m}^{p^\nu}, \tag{xx}$$

where $\mathfrak{m}$ is the augmentation ideal, equivalently the maximal ideal, of $\mathscr{O}_{H,e}$. Indeed, $p^\nu\,\mathrm{id}_H$ vanishes on the kernel of the iterated Frobenius homomorphism F^ν. The composite of (x) and (xx) is the natural composite

$$\mathscr{O}_{Q,e} \longrightarrow \mathscr{O}_{H,e} \longrightarrow \mathscr{O}_{H,e}/\mathfrak{m}^{p^\nu}. \tag{xxx}$$

Now $\mathscr{O}_{Q,e} \to \mathscr{O}_{H,e}$ is injective, because $H \to Q$ is an epimorphism and therefore flat. Moreover, $\mathscr{O}_{Q,e}$ is Artinian, while $\bigcap_\nu \mathfrak{m}^{p^\nu} = 0$ in $\mathscr{O}_{H,e}$, so the intersection of the corresponding ideals in $\mathscr{O}_{Q,e}$ is also zero. One of them is therefore zero. Thus (xxx) is injective, and a fortiori so is (x).

Lemma 7.7. *Under the hypotheses of 7.5, there is an integer ν_1 such that, for every $\nu \geq \nu_1$ and every n, the sequence of S_n-groups*

$$0 \longrightarrow {}_{p^\nu}G_n \longrightarrow {}_{p^\nu}H_n \xrightarrow{w_n} Q_n \longrightarrow 0$$

is exact. More precisely, w_n is faithfully flat and its kernel is ${}_{p^\nu}G_n$.

Take ν_1 as in Lemma 7.6, applied to $u_0 : G_0 \to H_0$, and require $\nu_1 \geq \nu_0$, where $p^{\nu_0} = \operatorname{rank} Q_0$. It remains only to check that w_n is faithfully flat. By Corollary 7.5, ${}_{p^\nu}H_n$ is flat over S_n, as is Q_n; it is therefore enough to check that

$$w_0 : {}_{p^\nu}H_0 \longrightarrow Q_0$$

is faithfully flat, equivalently an epimorphism. This follows from the choice of ν_1.

Corollary 7.8. *Suppose in addition that H is separated over S, or more generally that ${}_{p^\nu}H$ is separated over S for every ν, and that ${}_{p^\nu}G$ is finite over S for every ν. Then, for $\nu \geq \nu_1$, the group schemes ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are finite and flat over S, and there is an exact sequence*

$$0 \longrightarrow {}_{p^\nu}G \longrightarrow {}_{p^\nu}H \xrightarrow{w} Q \longrightarrow 0.$$

Because $u : G \to H$ is surjective, the induced morphism ${}_{p^\nu}G \to {}_{p^\nu}H$ is also surjective. The source is finite over S and the target is separated over S, hence the target is finite over S. To prove that ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are flat over S, it is enough to prove this after every base change to S_n, where it is contained in Corollary 7.5. Finally, the exact sequence above is obtained from the exact sequences in Lemma 7.7 as n varies.

Taking generic fibers in the sequence of Corollary 7.8, and recalling that $u_{s'} : G_{s'} \to H_{s'}$ is an isomorphism, gives that $Q_{s'}$ is the unit group. Since Q is flat over S, Q itself is the unit group; therefore so is Q_s, and $u_s : G_s \to H_s$ is an isomorphism. We have proved:

Proposition 7.9. *Let $u : G \to H$ be a homomorphism of group preschemes of finite presentation over a prescheme S. Assume:*

(a) *u is a monomorphism;*

(b) *G is flat over S;*

(c) *for every $s \in S$ whose residue field has characteristic $p > 0$, let $S' = \mathrm{Spec}(\mathscr{O}_{S,s})$, and let $u' : G' \to H'$ be obtained from u by base change to S'. Then G' is commutative, the special fiber $G'_0 = G_s$ is smooth over $\kappa(s)$, and, for every $\nu > 0$, the group ${}_{p^\nu}G'$ is finite over S' while ${}_{p^\nu}H'$ is separated over S'.*

Under these assumptions, u is an immersion.

Indeed, the finiteness of ${}_pG'$ over S' implies the finiteness of its special fiber over $\kappa(s)$. Hence $G'_0 \otimes_{\kappa(s)} \overline{\kappa(s)}$ contains no subgroup isomorphic to the additive group, and the hypotheses of Corollary 7.5 apply.

Remark 7.10. Examples of M. Raynaud (XVI, 1.1(a),(b)) show that in condition (c) one cannot omit either the finiteness of the ${}_{p^\nu}G'$ over S' or the separatedness of the ${}_{p^\nu}H'$ over S'.

We now seek conditions under which u is a closed immersion. Retain the hypotheses preceding Proposition 7.9, but no longer suppose that u is surjective; assume only that it is an isomorphism on generic fibers. We already know that $u : G \to H$ is an open immersion. The same is therefore true of $u_0 : G_0 \to H_0$, whose image contains the identity component $(H_0)^0$. Since G_0 is smooth over k and has no additive component after extending to $\overline{k}$, the same is true of H_0. We use:

Lemma 7.11. *Let H be a commutative algebraic group over a field k, and suppose that $H \otimes_k \overline{k}$ contains no subgroup isomorphic to $\mathbf{G}_a$. If $n = \deg(H/H^0)$, then*

$$ {}_nH \longrightarrow H/H^0 $$

is surjective.

We may assume that k is algebraically closed, in which case this follows from the familiar fact that $H^0(k)$ is a divisible group.

Return to $u : G \to H$, and suppose that the ${}_nG$, $n > 0$, are proper over S, while the ${}_nH$ are separated over S. The ${}_nH$ are also flat over S. Indeed, it is enough to check flatness at the points above s, and hence to check that $n\,\mathrm{id}_{H_0}$ is flat. As noted above, this is equivalent to $(H_0)^0$ being smooth[40] over k and having no $\mathbf{G}_a$-component over $\overline{k}$. Since $(H_0)^0 = (G_0)^0$, this follows from the corresponding hypothesis on G_0.

Because H is separated over S, so are the ${}_nH$. The morphism ${}_nG \to {}_nH$ is proper and therefore has closed image. Its image contains the generic fiber of ${}_nH$, because $u_{s'} : G_{s'} \to H_{s'}$ is an isomorphism. Since ${}_nH$ is flat over S, it is the schematic closure of its generic fiber. Hence ${}_nG \to {}_nH$, and therefore ${}_nG_0 \to {}_nH_0$, is surjective. Together with

[40] Editor's note: the terminology has been updated, replacing "simple" by "smooth"; see, for example, A. Grothendieck's note (1) in SGA 1, Exposé II.

$G_0 \supset (H_0)^0$, Lemma 7.11 now implies that $G_0 \to H_0$ is surjective. Hence u is surjective. We obtain:

Proposition 7.12. *With the notation of 7.9, suppose that conditions (a) and (b) hold, together with the stronger condition (c′) obtained from (c) by requiring, for every integer $n > 0$, not only those of the form p^ν, that ${}_nG'$ be finite over S' and ${}_nH'$ separated over S'. Then u is a closed immersion.*

Remark 7.13.

(a) The separatedness hypothesis on the ${}_nH$ easily implies that H itself is separated over S. This is also formally contained in Proposition 7.12, by taking G to be the unit S-group. If S is locally Noetherian, in Proposition 7.9 it is enough to assume that H is locally of finite type over S, rather than of finite type; the proof applies verbatim. For Proposition 7.12, one also assumes that the fibers of H are of finite type.

(b) Using Proposition 7.12, it is not difficult to prove that if $u : G \to H$ is a monomorphism of group preschemes of finite presentation over S, G is diagonalizable—or, more generally, of multiplicative type—and H is separated over S, then u is a closed immersion. This is a satisfactory generalization of the first conclusion in 5.7. If G is smooth over S, apply Proposition 7.12 directly; the general case reduces readily to this one. If H is not assumed separated over S, one can still show that u is an immersion; for a torus G, this also follows from the material that comes next.

(c) If G is assumed to have connected fibers in Proposition 7.9, the separatedness of ${}_{p^\nu}H'$ over S' may be omitted from condition (c). Indeed, after the reductions in the proof, H may be assumed flat over S and u bijective. Thus H has connected fibers, and Raynaud's theorem (VI_B, 5.5) implies that H is separated over S.[41]

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[42]

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[TO70] J. Tate and F. Oort, *Group schemes of prime order*, Ann. Sci. Éc. Norm. Supér. (4) **3** (1970), 1–21.

[41] Editor's note: the attribution, which appeared in SGAD 1965, should be made explicit in Exposé VI_B, ğ5, together with the changes between SGAD and Lecture Notes 151.

[42] Editor's note: additional references cited in this exposé.

Exposé IX
Groups of Multiplicative Type: Homomorphisms into a Group Scheme

by A. Grothendieck[0]

1. Definitions

Definition 1.1. — *Let S be a prescheme and let G be a group prescheme over S. The group G is said to be* of multiplicative type *if it is locally diagonalizable for the faithfully flat quasi-compact topology (cf. IV 6.3); that is, if for every $s \in S$, there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that*

$$G' = G \times_S S'$$

is a diagonalizable S'-group (I 4.4 and VIII 1.1).

The group G is said to be of quasi-isotrivial multiplicative type *if it is locally diagonalizable even for the étale topology (IV 6.3); that is, if in the preceding definition one can take $S' \to U$ to be étale and surjective. Equivalently—as one sees by taking the disjoint union of the schemes S' attached to the various $s \in S$—there is an étale surjective morphism $S' \to S$ such that $G' = G \times_S S'$ is a locally diagonalizable S'-group.*

If one can moreover choose $S' \to S$ to be finite, étale, and surjective, then G is said to be of isotrivial multiplicative type*.*

Finally, G is said to be a group of locally trivial multiplicative type *(respectively,* of locally isotrivial multiplicative type*) if every $s \in S$ has an open neighborhood U such that $G \times_S U$ is a diagonalizable U-group (respectively, is of isotrivial multiplicative type; in other words, there is a finite étale surjective morphism $S' \to U$ such that $G \times_S S'$ is a diagonalizable S'-group).*

Comments 1.2. — The five preceding notions are all obtained from the notion of a diagonalizable group by localization, with respect to five different "topologies" associated with (**Sch**).

It is understood in general that, when the word "locally" is not qualified by naming the topology under consideration, the Zariski topology is meant, in accordance with established terminology. In the terminology introduced here, "of locally trivial multiplicative

[0]Editor's note: version 1.0 of November 8, 2009. Additions were made in the proof of 3.6 bis, in 4.4–7, 5.0–6, 6.1, 7.1, and 8.2. Sections 8.1 and 4.5 remain to be revised.

type" is equivalent to "locally diagonalizable" in VIII 1.1; likewise, "trivial" translates as "diagonalizable." The five notions introduced above fit into the following implication diagram, which follows from the corresponding relations among the topologies involved:

$$\begin{tikzcd}
& \text{general multiplicative type} & \\
& \text{quasi-isotrivial} \arrow[u, Rightarrow] & \\
& \text{locally isotrivial} \arrow[u, Rightarrow] & \\
\text{isotrivial} \arrow[ur, Rightarrow] & & \text{locally trivial} \arrow[ul, Rightarrow] \\
& \text{trivial} \arrow[ul, Rightarrow] \arrow[ur, Rightarrow] &
\end{tikzcd}$$

For practical purposes, let us point out at once that every group of multiplicative type that we shall encounter will be quasi-isotrivial. Thus we shall see in the next exposé (X 4.5) that, when G is of finite type over S, it is automatically quasi-isotrivial; we shall nevertheless give examples in which it is not locally isotrivial. We shall likewise see that G may be locally trivial without being isotrivial, and hence *a fortiori* without being trivial. This readily implies that the inclusions in the preceding diagram are strict.

On the other hand, we shall also see there (X 5.16) that, when S is locally noetherian and normal—or, more generally, geometrically unibranch—every group of multiplicative type and finite type over S is necessarily isotrivial; moreover, it is trivial as soon as it is locally trivial (or even merely trivial over a dense open subset). This explains why most groups of multiplicative type encountered in practice will doubtless be isotrivial, all the more so because we shall later see that the maximal tori of semisimple group schemes are automatically isotrivial.

Definition 1.3. — *Let S and G be as in 1.1. The group G is called a* torus *if, for the faithfully flat quasi-compact topology, it is locally isomorphic to a group of the form $\mathbf{G}_m^r$, where r is an integer with $r \geq 0$.*

With the notation of 1.1, this says that S' may be chosen so that G' is isomorphic to a group of the form $(\mathbf{G}_{m,S'})^r$. The integer r depends on $s \in S$: it is the dimension of the fiber

$$G_s = G \otimes_S \kappa(s).$$

It is a locally constant function of s, as is seen at once. This observation should be generalized as follows.

Definition 1.4. — *Let G be a diagonalizable group scheme over a field k. Thus G is isomorphic to a group $D_k(M)$, where M is an ordinary commutative group determined up to isomorphism by this condition—more precisely,*

$$M \simeq \operatorname{Hom}_{k\text{-gr}}(G, \mathbf{G}_{m,k})$$

(VIII 1.3). The isomorphism class of M is called the type *of the diagonalizable group G; it is plainly invariant under extension of the base field.*

Now let G be a group scheme of multiplicative type over an arbitrary prescheme S. For every $s \in S$, there is an extension k of $\kappa(s)$ such that $G \times_S \operatorname{Spec}(k)$ is a diagonalizable k-group.[1] *Its type is independent of the chosen extension by the preceding observation and*

[1] Editor's note: indeed, by hypothesis there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G_{S'}$ is diagonalizable. For every $s' \in S'$ above s, the field $\kappa(s')$ is an extension of $\kappa(s)$, and $G_{s'} = G \times_S \operatorname{Spec}(\kappa(s'))$ is diagonalizable.

is called the type of G at s*, or the* type of G_s*. In particular, if G itself is diagonalizable, and hence isomorphic to a group of the form $D_S(M)$, then for every $s \in S$ the type of G at s is the isomorphism class of M.*

More generally, if G is a group of multiplicative type over S and M is an ordinary commutative group, then G is said to be of type M *if it is of type M at every point $s \in S$; equivalently, if G is locally isomorphic to $D_S(M)$ for the faithfully flat quasi-compact topology.*

Remark 1.4.1. [2] *a) For a group G of multiplicative type over S, the function*

$$s \longmapsto \textit{the type of } G_s$$

is locally constant on S. Indeed, with the notation of 1.1, if G' is of type M, then G is of type M over the neighborhood U of s. Consequently S has a canonical decomposition as a disjoint union of preschemes S_i such that, for every i, G_{S_i} is of type M_i, where the commutative groups M_i are pairwise nonisomorphic.

b) In particular, if S is connected, the type of the fibers of G is constant; that is, there is a commutative group M such that G is of type M. Finally, if G is a torus, the type of G at s is determined by the integer $\dim G_s$*: indeed, G_s is of type $\mathbb{Z}^n$, where $n = \dim G_s$.*

Remark 1.5. — *a) It follows immediately from Definitions 1.1 and 1.3 that these notions are "compatible with base extension." Thus, if G is a group prescheme over S and $S' \to S$ is a base-change morphism, then whenever G is of multiplicative type (respectively, of isotrivial multiplicative type, and so on), the same is true of G'.*

b) If, moreover, $S' \to S$ is faithfully flat and quasi-compact, then G' being of multiplicative type, respectively a torus, implies the same for G. If $S' \to S$ is also étale (respectively, finite étale), then G' being quasi-isotrivial (respectively, isotrivial) implies the same for G.

c) Finally, return to an arbitrary base-change morphism $f : S' \to S$. If $s' \in S'$ and $s = f(s')$, then the type of G' at s' is the same as that of G at s, since

$$G'_{s'} = G_s \otimes_{\kappa(s)} \kappa(s').$$

2. Extension of Certain Properties of Diagonalizable Groups to Groups of Multiplicative Type

The extensions in question are essentially trivial consequences of the results of the preceding Exposé, in view of the definitions in 1.1 and the "local" nature (for the faithfully flat and quasi-compact topology) of the results under consideration.

Definition 2.0. [3] *We denote by $\mathcal{M}$ the set of faithfully flat and quasi-compact morphisms.*

Proposition 2.1. *Let G be a group of multiplicative type over a prescheme S. Then:*

(a) *G is faithfully flat over S, and affine over S (a fortiori, quasi-compact over S);*

(b) *G is of finite type over S if and only if G is of finite presentation over S, if and only if, for every $s \in S$, the type of G at s is given by a commutative group of finite type;*

[2] Editor's note: the number 1.4.1 has been added to the remarks that follow, for use in later references.

[3] Editor's note: This definition has been added; it occurs in Propositions 2.3 and 2.7.

(c) *G is finite over S if and only if, for every $s \in S$, the type of G at s is given by a finite commutative group, if and only if (when S is quasi-compact) G is of finite type over S and is annihilated by an integer $n > 0$;*

(c') *G is integral over S if and only if, for every $s \in S$, the type of G at s is given by a torsion commutative group;*

(d) *G is the unit group if and only if, for every $s \in S$, the type of G at s is given by the zero commutative group;*

(e) *G is smooth over S if and only if, for every $s \in S$, the type of G at s is given by a commutative group of finite type whose torsion subgroup has order prime to the characteristic of $\kappa(s)$.*

These statements follow from VIII 2.1, taking into account that the properties in question descend under faithfully flat and quasi-compact morphisms (cf. SGA 1, VIII, or EGA IV_2, §2).

Using VIII 3.5, one likewise obtains:

Proposition 2.2. *Let G be a group of multiplicative type and of finite type over S. For every integer $n \neq 0$, the kernel ${}_nG = \ker(n \cdot \mathrm{id}_G)$ is a group of multiplicative type, finite over S.*

Proposition 2.3. *Let G be a group of multiplicative type over the prescheme S, acting freely on the right on an S-prescheme X that is affine over S. Then:*

(a) *the equivalence relation defined by G in X is $\mathcal{M}$-effective (IV 3.4), and $Y = X/G$ is affine over S;*

(b) *if, moreover, X is of finite presentation (respectively, of finite type) over S, then the same is true of Y.*

The first assertion follows from VIII 5.1, which treats the case in which G is diagonalizable, and from IV 3.5.2, which allows one to reduce to that case. Indeed, faithfully flat and quasi-compact morphisms $S' \to S$ are effective descent morphisms for the fibered category of affine schemes over schemes: if Y' is affine over S' and is equipped with a descent datum relative to $S' \to S$, then this datum is effective, that is, Y' comes from a scheme Y affine over S (cf. SGA 1, VIII 2.1).

For the second assertion, one is likewise reduced to the diagonalizable case of VIII 5.8, since the finiteness conditions under consideration descend under faithfully flat and quasi-compact morphisms (SGA 1, VIII 3.3 and 3.6[4]). Proceeding as in VIII, Corollaries 5.5–5.7, one deduces from 2.3:

Corollary 2.4. *Under the hypotheses of 2.3, the graph morphism*

$$X \times_S G \longrightarrow X \times_S X$$

is a closed immersion. For every section σ of X over S, the corresponding morphism

$$G \longrightarrow X, \qquad g \longmapsto \sigma \cdot g,$$

is a closed immersion.

In particular:

[4]Editor's note: See also V, 9.1, or EGA IV_2, 2.7.1.

Corollary 2.5. *Let $u : G \to H$ be a monomorphism of S-preschemes in groups, with G of multiplicative type and H affine over S. Then u is a closed immersion, $H/G = Y$ exists and is affine over S, and H is a principal homogeneous bundle over Y under G_Y.*

Remark 2.6. Let $u : G \to H$ be a monomorphism of S-preschemes in groups, where G is of multiplicative type and of finite type over S, and H is of finite presentation and separated over S. One can then show that u is a closed immersion by using VIII 7.12 (see Remark VIII 7.13(b)).

Proposition 2.7. *Let $u : G \to H$ be a homomorphism of S-groups of multiplicative type, with H of finite type over S. Then:*

(i) *$G_0 = \ker u$ is an S-group of multiplicative type; the equivalence relation defined by G_0 in G is $\mathcal{M}$-effective, and hence (IV 3.4) u factors as*

$$G \longrightarrow G/G_0 = I \longrightarrow H,$$

where $I \to H$ is a monomorphism[5] *of S-groups. The group I is of multiplicative type, the equivalence relation in H defined by I is $\mathcal{M}$-effective, and consequently $H_0 = H/I$ exists and is also of multiplicative type;*

(ii) *H_0 and I are of finite type, and so is G_0 when G is.*

Proof. Proceeding as for 2.3, one is reduced to the case in which G and H are diagonalizable; then 2.7 reduces to VIII 3.4.

We record the following corollaries.

Corollary 2.8.

(a) *Let S be a prescheme. The category of S-groups of multiplicative type and of finite type is an abelian category.*

(b) *Let $u : G \to H$ be a homomorphism in this category. It is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in this category if and only if u is a monomorphism of preschemes (respectively, u is faithfully flat, respectively, u is an isomorphism of preschemes).*

It is enough to note that the product of two S-groups of multiplicative type is again of multiplicative type (which is immediate), and is plainly of finite type over S if both factors are. The rest of 2.8 follows immediately from 2.7; the details are left to the reader.

Corollary 2.9. *Let $u : G \to H$ be a homomorphism of S-groups of multiplicative type and of finite type. Let U be the set of points $s \in S$ for which $u_s : G_s \to H_s$ is a monomorphism (respectively, faithfully flat, respectively, an isomorphism). Then U is both open and closed, and the induced homomorphism $u_U : G_U \to H_U$ is a monomorphism (respectively, faithfully flat, respectively, an isomorphism).*

Let P (respectively, Q) be the kernel (respectively, cokernel) of u. By 2.7, Q exists and P and Q are of multiplicative type; moreover, formation of P and Q commutes with every base change $S' \to S$, in particular with the formation of fibers. On the other hand, u is a monomorphism (respectively, faithfully flat, respectively, an isomorphism) if and only if $P = 0$ (respectively, $Q = 0$, respectively, $P = Q = 0$). These remarks reduce us to checking the following assertion: if R is an S-group of multiplicative type, then the set U

[5] Editor's note: Indeed, even a closed immersion.

of points $s \in S$ for which $R_s = 0$ is open and closed, and $R_U = 0$. This is contained in Remark 1.4.1(a).

Corollary 2.10. *Let G be an S-group of multiplicative type and of finite type, and let H and H' be two subgroups of multiplicative type.*

(a) *$H'' = H \cap H'$ is a subgroup of G of multiplicative type.*

(b) *Let U be the set of $s \in S$ for which $H_s \subseteq H'_s$ (respectively, $H_s = H'_s$). Then U is open and closed, and $H_U \subseteq H'_U$ (respectively, $H_U = H'_U$).*

Here, of course, the intersection sign in $H \cap H'$ denotes the intersection in the functorial sense, namely $H \times_G H'$, which is plainly an S-subgroup of G. Likewise, the inclusion sign (respectively, the equality sign) denotes the order relation (respectively, equality) between subfunctors of H, not inclusion (respectively, equality) of the underlying sets.

Applying 2.7 first to the inclusion morphism $H \to G$, one finds that H is of finite type; the same holds for H'. It then follows from 2.8 that $H \cap H'$ is of multiplicative type and of finite type. (Note that the canonical functor from the category considered in 2.8 to the category $(\mathbf{Sch})/S$ commutes with finite inverse limits.)

Formation of $H \cap H'$ commutes with base extension, in particular with formation of the fibers. Moreover, $H \subseteq H'$ (respectively, $H = H'$) is equivalent to $H'' = H$ (respectively, to $H'' = H$ and $H'' = H'$). Consider the canonical homomorphisms $H'' \to H$ and $H'' \to H'$. Then U is the set of points $s \in S$ for which the induced homomorphism $H''_s \to H_s$ is an isomorphism (respectively, for which $H''_s \to H_s$ and $H''_s \to H'_s$ are isomorphisms).

By 2.9, it follows that U is open and closed, and that $H''_U \to H_U$ (respectively, $H''_U \to H_U$ and $H''_U \to H'_U$) are isomorphisms. This gives the desired conclusion.

Proposition 2.11. *Let S be a prescheme, let G be an S-group of multiplicative type and of finite type over S, let H be a subgroup of G of multiplicative type, and put $K = G/H$, a quotient group of G of multiplicative type.*

(i) *Suppose G is trivial, that is, diagonalizable. There is a partition of S into open and closed subsets S_i such that, for every i, H_{S_i} and K_{S_i} are diagonalizable. In particular, if S is connected, H and K are diagonalizable.*

(ii) *The same assertion holds with "diagonalizable" replaced by "isotrivial," provided that S is connected, locally connected, or quasi-compact.*

(iii) *If G is locally trivial (respectively, locally isotrivial, respectively, quasi-isotrivial), then the same is true of K and H.*

Proof. (i) By hypothesis, $G = D_S(M)$, where M is a commutative group of finite type. For each quotient group M_i of M, let $H_i = D_S(M_i)$ be the corresponding diagonalizable subgroup of G (VIII 3.1). Let S_i be the set of points $s \in S$ for which $H_s = (H_i)_s$. By 2.10, S_i is open and closed, and $H_{S_i} = (H_i)_{S_i}$; thus H_{S_i} is diagonalizable, and hence so is K_{S_i} (VIII 3.1). The S_i are plainly pairwise disjoint, and we claim that they cover S. Indeed, for every s, the group H_s, being a subgroup of the diagonalizable group G_s, is diagonalizable (cf. 8.1 below[6]). Hence H_s has the form $D_{\kappa(s)}(M_i)$, by VIII 1.5 and 3.2(b). Restricting to the family of nonempty S_i, assertion (i) follows.

(ii) By hypothesis, there exists a finite surjective étale morphism $S' \to S$ such that $G_{S'}$ is diagonalizable. Every point of S' therefore has an open and closed neighborhood U'

[6]Editor's note: The erroneous reference to IV 4.7.5 has been corrected. Note that Section 8 of the present Exposé is independent of Sections 3–7.

such that $H_{U'}$ and $K_{U'}$ are diagonalizable. The image U of U' in S is open and closed, and $S' \to S$ still induces a finite surjective étale morphism $U' \to U$. Thus every point $s \in S$ has an open and closed neighborhood U such that H_U and K_U are isotrivial. Assertion (ii) follows at once.

(iii) This follows at once from (i), (ii), and the definitions. Moreover, the quasi-isotrivial case will also follow from the more general fact "finite type $\Longrightarrow$ quasi-isotrivial," announced in 1.2 (cf. X 4.5).

3. Infinitesimal Properties: Lifting and Conjugation Theorems

The fundamental infinitesimal properties of groups of multiplicative type follow from the following theorem.

Theorem 3.1. *Let S be an affine scheme, let H be an S-group of multiplicative type, and let $\mathscr{F}$ be a quasi-coherent sheaf on S on which H acts (I 4.7). Then*

$$\mathrm{H}^i(H, \mathscr{F}) = 0 \qquad \text{for } i > 0,$$

where H^i *denotes the Hochschild cohomology studied in Exposé I, §5.*

Indeed, by loc. cit., 5.3, Hochschild cohomology is expressed, in terms of the affine rings A and B of S and G, and the A-module M defining $\mathscr{F}$, as the cohomology of a complex of A-modules $C^\bullet(H, M)$, whose formation commutes with every base change $A \to A'$. Consequently, for a base change $A \to A'$ with A' flat over A, one has

$$\mathrm{H}^i(H', \mathscr{F}') = \mathrm{H}^i(H, \mathscr{F}) \otimes_A A'.$$

If A' is moreover faithfully flat over A, it is therefore enough, in order to prove that $\mathrm{H}^i(H, \mathscr{F})$ vanishes, to prove that $\mathrm{H}^i(H', \mathscr{F}')$ vanishes. This observation reduces us to checking 3.1 in the case where G is diagonalizable; in that case it was proved in I 5.3.3.

Using the results of Exposé III, we shall derive from 3.1 several consequences of a geometric nature.

Theorem 3.2. *Let S be an affine scheme, let S_0 be an affine subscheme defined by an ideal J, let H be a group of multiplicative type over S, and let G be an arbitrary group prescheme over S. Let $u, v : H \to G$ be two group homomorphisms, and let $u_0, v_0 : H_0 \to G_0$ be the homomorphisms obtained from u, v by the base change $S_0 \to S$. If $J^2 = 0$ and $u_0 = v_0$, then there exists $g \in G(S)$ such that*

$$v = \mathrm{int}(g) \circ u \qquad \textit{and} \qquad g_0 = e.$$

This follows from III 2.1(ii) and 3.1 (the vanishing of H^1).

Corollary 3.3. *Under the preliminary hypotheses of 3.2, suppose in addition that G is smooth over S, but assume only that J is nilpotent (not necessarily square-zero). If there exists $g_0 \in G_0(S_0)$ such that $v_0 = \mathrm{int}(g_0) \circ u_0$, then g_0 lifts to an element $g \in G(S)$ such that $v = \mathrm{int}(g) \circ u$.*

By induction on an integer n such that $J^n = 0$, one is reduced to the case in which J is square-zero. Since G is smooth over S, one may lift g_0 to an element g' of $G(S)$. Replacing v by $v' = \mathrm{int}(g') \circ u$, one is then reduced to the situation of 3.2.

Corollary 3.4. *Under the preliminary hypotheses of 3.2, with J nilpotent, suppose in addition that u is central (for example, that G is commutative). Then $u_0 = v_0$ implies $u = v$.*

Indeed, the reduction to the case in which J is square-zero is again immediate, and then it is enough to apply 3.2. In particular:

Corollary 3.5. *Let S, S_0, H, G be as in 3.2, with J nilpotent. Let $u : H \to G$ be a homomorphism of S-groups such that $u_0 : H_0 \to G_0$ is the unit homomorphism. Then u is the unit homomorphism.*

Theorem 3.6. *Let S be an affine scheme, let S_0 be a subscheme defined by a nilpotent ideal J, let H be a group of multiplicative type over S, let G be a group prescheme smooth over S, and let $u_0 : H_0 \to G_0$ be a homomorphism of the S_0-groups obtained by base change $S_0 \to S$. Then there exists a homomorphism $u : H \to G$ of S-groups that lifts u_0. Moreover, by 3.3, any two such liftings u, u' are conjugate by an element of $G(S)$ whose reduction in $G(S_0)$ is the identity element.*

This follows from III 2.1 and 2.3 and from 3.1: the vanishing of H^2 gives the existence of the lifting of u_0, and the vanishing of H^1 gives uniqueness up to conjugation.

One may prove in the same way the following variants of 3.2–3.6. In fact, these variants imply the preceding statements when applied to the graph subgroups of the homomorphisms considered there.

Theorem 3.2 bis. *Let S be an affine scheme, let S_0 be a subscheme defined by an ideal J, let G be an S-group prescheme, and let H, H' be subgroup preschemes of multiplicative type. Let G_0, H_0, H'_0 be the groups over S_0 obtained by base change $S_0 \to S$. If $J^2 = 0$ and $H_0 = H'_0$, then there exists $g \in G(S)$ such that*

$$H' = \mathrm{int}(g)(H) \qquad \textit{and} \qquad g_0 = e.$$

Corollary 3.3 bis. *Under the preliminary hypotheses of 3.2 bis, suppose in addition that G is smooth over S, and that J is nilpotent (not necessarily square-zero). If $g_0 \in G_0(S_0)$ satisfies $H'_0 = \mathrm{int}(g_0)(H_0)$, then g_0 lifts to an element $g \in G(S)$ such that $H' = \mathrm{int}(g)(H)$.*

Corollary 3.4 bis. *Under the preliminary hypotheses of 3.2 bis, suppose that J is nilpotent and that H is central (for example, that G is commutative). Then $H_0 = H'_0$ implies $H = H'$.*

Theorem 3.6 bis. *Let S be an affine scheme, let S_0 be a subscheme defined by a nilpotent ideal J, let G be an S-group prescheme smooth over S, and let H_0 be a subgroup prescheme of $G_0 = G \times_S S_0$, of multiplicative type. Then:*

(a) *there exists a subgroup prescheme H of G, flat over S, such that $H \times_S S_0 = H_0$;*

(b) *such an H is necessarily of multiplicative type;*

(c) *finally, by 3.2 bis, any two such liftings H, H' of H_0 are conjugate by an element $g \in G(S)$ whose reduction in $G(S_0)$ is the identity element.*

Let us indicate how one can prove 3.2 bis (of which 3.3 bis and 3.4 bis are immediate consequences) and assertion (a) of 3.6 bis. The latter follows from 3.1 (the vanishing of H^2) and III 4.37(ii). Similarly, 3.2 bis follows from 3.1 (the vanishing of H^1) and III 4.37(i), at least when G and H are smooth over S.

Using the results of the following Exposé, however, one can very simply deduce 3.2 bis and 3.6 bis, in the general form stated here, from 3.2 and 3.6, respectively. For 3.2 bis, it is enough to note that, by X 2.1, H and H' are isomorphic, which reduces us to 3.2.

For 3.6 bis, note that H_0 is necessarily of finite type over S_0:[7] indeed, as a subgroup prescheme of G_0, which is smooth over S_0, the group H_0 is locally of finite type over S_0. But H_0 is affine over S_0 by 2.1(a), and is therefore of finite type over S_0. By X 4.5, H_0 is consequently quasi-isotrivial, and hence comes, by X 2.1, from a group H of multiplicative type over S.[8] By 3.6, there exists a homomorphism of S-groups $u : H \to G$ lifting the immersion $H_0 \hookrightarrow G_0$. Since H and H_0 (respectively, G and G_0) have the same underlying topological space, and since, for every $h \in H$, the morphism

$$\mathscr{O}_{G,u(h)} \longrightarrow \mathscr{O}_{H,h}$$

is surjective (because it is so after reduction modulo the nilpotent ideal J), the morphism u is also an immersion (cf. EGA I, 4.2.2).

Finally, every lifting H of H_0 to a flat subgroup of G is necessarily of multiplicative type by X 2.3,[9] which also proves assertion (b) of 3.6 bis. The reader may verify that the results 3.2 bis–3.6 bis are not used in Exposé X, so there is no vicious circle.

Proposition 3.8. [10] *Let S be a prescheme, let S_0 be a closed subscheme defined by a locally nilpotent ideal J, let G be an S-group of multiplicative type, and let X be an S-prescheme locally of finite presentation. Suppose that G acts on X in such a way that $G_0 = G \times_S S_0$ acts trivially on $X_0 = X \times_S S_0$. Then G acts trivially on X.*

The proof is the proof of III 2.4(b). One may also reduce to loc. cit. by observing that 3.8 immediately reduces to the case in which G is diagonalizable, which is the case treated there.

Corollary 3.9. *Let S, S_0 be as above, and let $u : G \to H$ be a homomorphism of S-groups, with G of multiplicative type and H locally of finite presentation over S. Suppose that the homomorphism $u_0 : G_0 \to H_0$ obtained by base change $S_0 \to S$ is central. Then u is central.*

Indeed, it is enough to apply 3.8 with $X = H$, letting G act on H by

$$(g, h) \longmapsto \operatorname{int}(u(g)) \cdot h = u(g)h\, u(g)^{-1}.$$

4. The Density Theorem

This theorem (cf. 4.7 below)*, together with the theorem[11] of Section 7, will be the essential tool in the present exposé and the two following ones for passing from the infinitesimal properties of groups of multiplicative type developed above to their "finite" properties.

Definition 4.1. *Let X be a prescheme. A family $(Z_i)_{i\in I}$ of subpreschemes of X is said to be* schematically dense *if, for every open subset U of X and every closed subprescheme U_0 of U containing all the $Z_i \cap U$, one has $U_0 = U$.*

A subprescheme Z of X is said to be schematically dense in X *if the one-member family consisting of Z is schematically dense.*

[7]Editor's note: The original said, "by 2.1(b), since its fibers are." The editors did not understand this argument and have replaced it by the argument that follows.

[8]Editor's note: The sentence that follows has been expanded.

[9]Editor's note: X 2.3 depends essentially on Theorem 3.6 of the present Exposé.

[10]Editor's note: The numbering of the original has been retained; there is no 3.7.

*All the results from 4.1 through 4.6 are contained in EGA IV_3, 11.10, to which we refer the reader for a systematic account of the notion of schematic density.

[11]Editor's note: the "algebraicity" theorem 7.1.

It is immediate (cf. EGA IV_3, 11.10.1) that the definition is equivalent to saying that, for every open subset U of X, every section f of $\mathscr{O}_U$ that vanishes on each $Z_i \cap U$ is zero. Equivalently, the intersection of the kernels of the canonical homomorphisms

$$u_i : \mathscr{O}_X \longrightarrow (v_i)_*(\mathscr{O}_{Z_i})$$

is zero, where $v_i : Z_i \to X$ is the canonical immersion.[12] When X lies over a prescheme S, this is also equivalent to the following assertion: for every open subset U of X, every S-prescheme Y separated over S, and every pair of morphisms $u, v : U \to Y$ that coincide on the $Z_i \cap U$, one has $u = v$. Indeed, $u = v$ is equivalent to $U_0 = U$, where U_0 is the inverse image of the diagonal in $Y \times_S Y$ under the S-morphism $U \to Y \times_S Y$ defined by (u, v). The diagonal is a closed subprescheme of $Y \times_S Y$, so U_0 is a closed subprescheme of U containing all the $Z_i \cap U$. Thus schematic density gives $U_0 = U$, hence $u = v$. Conversely, take $Y = \operatorname{Spec} \mathscr{O}_S[T]$.

In the terminology introduced at the end of EGA I, 9.5, saying that the subprescheme Z is schematically dense in X also means that X is the *closure subprescheme* of Z in X.

Lemma 4.2. *Let X be a flat S-prescheme, and let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, flat over S. Let S_0 be a subprescheme of S defined by a nilpotent ideal $\mathscr{J}$, and suppose that the modules $\mathscr{J}^n/\mathscr{J}^{n+1}$ are locally free over S_0. Let X_0, Z_{i0} be obtained from X, Z_i by the base change $S_0 \to S$. If $(Z_{i0})_{i\in I}$ is schematically dense in X_0, then $(Z_i)_{i\in I}$ is schematically dense in X.*

Proof. Suppose $\mathscr{J}^{n+1} = 0$, where $n \geq 0$, and argue by induction on n; the assertion is trivial for $n = 0$. Define S_m, X_m, Z_{im} by reduction modulo $\mathscr{J}^{m+1}$, as usual. The induction hypothesis already implies that $(Z_{i,n-1})_{i\in I}$ is schematically dense in X_{n-1}. After replacing X by an induced open subset, it is enough to prove that every section f of $\mathscr{O}_X$ that vanishes on all the Z_i is zero.

The section f_{n-1} of

$$\mathscr{O}_{X_{n-1}} = \mathscr{O}_X/\mathscr{J}^n\mathscr{O}_X$$

defined by f vanishes on the $Z_{i,n-1}$, hence is zero. Consequently, f is a section of $\mathscr{J}^n\mathscr{O}_X$. Since X is flat over S, there is an isomorphism

$$\mathscr{J}^n\mathscr{O}_X \xleftarrow{\sim} \mathscr{E} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{X_0}, \qquad \mathscr{E} = \mathscr{J}^n = \mathscr{J}^n/\mathscr{J}^{n+1}.$$

Similarly, since each Z_i is flat over S, the restriction f_i of f to Z_i may be regarded as a section of

$$\mathscr{J}^n\mathscr{O}_{Z_i} \xleftarrow{\sim} \mathscr{E} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Z_{i0}}.$$

By hypothesis, all the f_i are zero. The module $\mathscr{E}$ is locally free by assumption, and therefore so is

$$\mathscr{F} = \mathscr{E} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{X_0}.$$

Thus f is a section of the locally free module $\mathscr{F}$ on X_0 whose restriction to every Z_{i0} is zero. Since the family $(Z_{i0})_{i\in I}$ is schematically dense in X_0, it follows at once that $f = 0$. □

Corollary 4.3. *Let X be a locally Noetherian S-prescheme, flat over S, and let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, flat over S. Suppose that, for every $s \in S$, the family of fibers $(Z_{is})_{i\in I}$ is schematically dense in X_s. Then $(Z_i)_{i\in I}$ is schematically dense in X.*

Proof. After replacing X by an induced open subset, it is enough to prove that every section f of $\mathscr{O}_X$ whose restriction to all the Z_i is zero is itself zero.[13] For this, it suffices to

[12]Editor's note: all the results on schematic density stated in EGA IV_3, §11.10 follow from the results on "separating families of homomorphisms" proved in *loc. cit.*, §11.9, to which some of the editor's notes below will refer.

[13]Editor's note: the original has been expanded in what follows.

show that the image of f in $\mathscr{O}_{X,x}$ is zero for every $x \in X$. Let $\mathfrak{m}_x$ be the maximal ideal of $\mathscr{O}_{X,x}$, and let $\mathfrak{m}_s$ be that of $\mathscr{O}_{S,s}$, where s is the image of x in S. Since $\mathscr{O}_{X,x}$ is Noetherian,

$$\bigcap_{n\in\mathbb{N}} \mathfrak{m}_x^n = 0, \qquad \text{and hence, a fortiori,} \qquad \bigcap_{n\in\mathbb{N}} \mathscr{O}_{X,x}\mathfrak{m}_s^n = 0.$$

It is therefore enough to show, for every integer n, that the section induced by f on

$$X \times_S \operatorname{Spec}(\mathscr{O}_{S,s}/\mathfrak{m}_s^{n+1})$$

is zero. By hypothesis,

$$Z_{is} = Z_i \otimes_{\mathscr{O}_{S,s}} \kappa(s)$$

is a schematically dense family in

$$X_s = X \otimes_{\mathscr{O}_{S,s}} \kappa(s).$$

We are thus reduced to the case in which S is the spectrum of a local ring whose maximal ideal $\mathscr{J}$ is nilpotent, $S_0 = \operatorname{Spec}\kappa(s)$, and the hypotheses of Lemma 4.2 hold. The conclusion follows. ☐

Lemma 4.4. *Let S be a locally Noetherian prescheme, let X be a locally Noetherian S-prescheme, flat over S, and let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, flat over S. Suppose that, for every $s \in S$, the family of fibers $(Z_{is})_{i\in I}$ is schematically dense in X_s. Then, for every base change $S' \to S$, with S' not necessarily locally Noetherian, the family $(Z_i')_{i\in I}$ is schematically dense in X'. In other words, $(Z_i)_{i\in I}$ is universally schematically dense in X relative to S (cf. EGA IV$_3$, Definition 11.10.8).*

Proof.[14] Let f' be a section of $\mathscr{O}_{X'}$ over an open subset U' of X', vanishing on the Z_i'; we must show that $f' = 0$. Fix $s' \in S'$. It is enough to prove that f' vanishes at every point of U' above s'. If s is the image of s' in S, then, after replacing S, S' by the spectra of their local rings at s, s', we may suppose that S, S' are local and that s, s' are their closed points. We may also suppose that X is affine, so that

$$S = \operatorname{Spec}(A), \qquad S' = \operatorname{Spec}(A'),$$
$$X = \operatorname{Spec}(B), \qquad X' = \operatorname{Spec}(B'), \qquad B' = B \otimes_A A',$$

where A, A' are local, $A \to A'$ is a local homomorphism, and A, B are Noetherian. We may further suppose that U' is affine, of the form $X'_{g'}$, with $g' \in B'$.

For every A-subalgebra A'' of A', put

$$S'' = \operatorname{Spec}(A'')$$

and let X'', Z_i'' be obtained from X, Z_i by the base change $S'' \to S$. We obtain the diagram

$$\begin{array}{ccccc}
Z^i & \longleftarrow & Z_n^{i''} & \longleftarrow & Z_n^{i'} \\
\downarrow & & \downarrow & & \downarrow \\
X & \longleftarrow & X'' & \longleftarrow & X' \\
\downarrow & & \downarrow & & \downarrow \\
S & \longleftarrow & S'' & \longleftarrow & S'.
\end{array}$$

[14] Editor's note: for another proof, reducing to the case in which S' is locally Noetherian and using 4.3 and 4.5 as here, see EGA IV$_3$, 11.9.16 and 11.9.12. In the last line of the proof of 11.9.16, replace 11.9.5 by 11.9.12.

In what follows, we restrict to those A'' that are localizations of A-subalgebras A''_1 of A', of finite type over A, at $\mathfrak{m}' \cap A''_1$, where $\mathfrak{m}'$ is the maximal ideal of A'. Thus the homomorphisms $A \to A'' \to A'$ are local. These A'' form an increasing filtered family of subalgebras whose union, and hence whose direct limit, is A'. Likewise,

$$B' = \varinjlim_{A''} (B \otimes_A A'').$$

There therefore exist such an A'' and an element $g'' \in B'' = B \otimes_A A''$ whose image in B' is g'.

By Lemma 4.5 below, the property that $(Z_{is})_{i \in I}$ be schematically dense in X_s is preserved by every base change $S'' \to S$.[15] Consequently, because each S'' is locally Noetherian, replacing S by a suitable S'' and X, Z_i by X'', Z''_i preserves the hypotheses of Lemma 4.4.

We may therefore suppose that g' comes from an element $g \in B$. After replacing X by the open subset $X_g = U$, we may suppose that $U' = X'$. In the same way, we may suppose that f' comes from a section f of $\mathscr{O}_X$ over X.

Let $Y = V(f)$ be the subscheme of X defined by f, and put

$$Y_i = Z_i \times_X Y.$$

Thus Y_i is the closed subscheme $V(f_i)$ of Z_i, where f_i is the section of $\mathscr{O}_{Z_i}$ induced by f. Denoting by Y', Y'_i the S'-schemes obtained by base change, we have

$$Y' = V(f'), \qquad Y'_i = Z'_i \times_{X'} Y' = V(f'_i),$$

and analogous relations hold for Y'', Y''_i. The assertion that f' vanishes on the Z'_i is equivalent to

$$Y'_i = Z'_i \qquad \text{for every } i.$$

[16]

Let $\mathfrak{m}$ be the maximal ideal of A. For every $n \geq 0$, introduce the closed subscheme

$$S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1})$$

of S, and let X_n, Y_n, Z^i_n, Y^i_n be obtained from X, Y, Z_i, Y_i by the base change $S_n \to S$. More generally, for every S-prescheme P, write $P_n = P \times_S S_n$.

For every n and $i \in I$, consider the functor

$$F^i_n = \prod_{Z^i_n/S_n} Y^i_n/Z^i_n : (\mathbf{Sch})^\circ_{/S_n} \longrightarrow \mathbf{Set},$$

defined, for every P over S_n, by

$$F^i_n(P) = \Gamma\big((Y^i_n)_P/(Z^i_n)_P\big) = \begin{cases} \varnothing, & (Y^i_n)_P \neq (Z^i_n)_P, \\ \{\mathrm{id}\}, & (Y^i_n)_P = (Z^i_n)_P. \end{cases}$$

Since S_n is local Artinian and Z^i_n is flat, hence essentially free, over S_n, VIII, 6.4 shows that each F^i_n is represented by a closed subprescheme T^i_n of S_n.

[15] Editor's note: the preceding sentence has been added, and the original has been expanded in what follows.

[16] Editor's note: the original added "at least at every point x' of X' above s'," but this restriction seems unnecessary.

The completion $\widehat{A}$ of the local ring A is Noetherian because A is, and it is the projective limit of the rings A_n.[17] Let K_n^i be the ideal of

$$A_n = A/\mathfrak{m}^{n+1}$$

defining T_n^i, and let $K^i = \varprojlim_n K_n^i$; this is an ideal of $\widehat{A}$.

For fixed i, integers $m \geq n$, and any S_n-prescheme P,

$$(Y_n^i)_P = Y_i \times_S S_n \times_{S_n} P = Y_i \times_S P = (Y_m^i)_P,$$

and similarly $(Z_n^i)_P = (Z_m^i)_P$. It follows that F_n^i is the restriction $F_m^i \times_{S_m} S_n$ of F_m^i to $(\mathbf{Sch})_{/S_n}$, and therefore

$$T_n^i = T_m^i \times_{S_m} S_n.$$

We thus have a commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & K_m^i \arrow[r] \arrow[d, two heads] & A_m \arrow[r] \arrow[d, two heads] & \mathscr{O}(T_m^i) \arrow[r] \arrow[d, two heads] & 0 \\
0 \arrow[r] & K_n^i \arrow[r] & A_n \arrow[r] & \mathscr{O}(T_n^i) \arrow[r] & 0.
\end{tikzcd}$$

All the vertical arrows are surjective. It follows, first, that the projective system $(K_n^i)_{n\in\mathbb{N}}$ satisfies the Mittag–Leffler condition, so that

$$\varprojlim_n \mathscr{O}(T_n^i) \simeq \widehat{A}/K^i$$

as topological rings (cf. EGA 0_{III}, §13.2). Second, the map $K^i \to K_n^i$ is surjective (cf. [BEns], III, §7.4, Proposition 5). Hence

$$K_n^i \simeq \frac{K^i + \mathfrak{m}^{n+1}\widehat{A}}{\mathfrak{m}^{n+1}\widehat{A}}, \qquad \mathscr{O}(T_n^i) \simeq (\widehat{A}/K^i) \otimes_A (A/\mathfrak{m}^{n+1}).$$

In other words, $(T_n^i)_n$ is an inductive system of affine Artinian schemes, and its inductive limit

$$T^i = \varinjlim_n T_n^i$$

is a closed formal subscheme of

$$\widehat{S} = \mathrm{Spf}(\widehat{A})$$

(cf. EGA I, §10), whose reduction modulo $\mathfrak{m}^{n+1}$ is T_n^i.

Let T be the closed formal subscheme of $\widehat{S}$ given by the intersection of the T^i, that is, defined by

$$K = \sum_i K^i.$$

Since $\widehat{A}$ is Noetherian, there is a finite subset J of I such that

$$K = \sum_{i\in J} K^i, \qquad \text{equivalently} \qquad T = \bigcap_{i\in J} T^i.$$

[17] Editor's note: the original has been expanded and corrected to show that the projective limit of the rings $\mathscr{O}(T_n^i)$ defines a closed formal subscheme of $\mathrm{Spf}(\widehat{A})$.

For every n, therefore,

$$K_n = \sum_{i \in J} K_n^i,$$

where K_n is the image of K in A_n.

Recall that f_i is the image of $f \in B = \mathscr{O}(X)$ in $\mathscr{O}(Z_i)$, and that f'_i is its image in

$$\mathscr{O}(Z'_i) = \mathscr{O}(Z_i) \otimes_A A'.$$

The relation $Y'_i = Z'_i$ is equivalent to $f'_i = 0$. Now $\mathscr{O}(Z_i) \otimes_A A'$ is the direct limit of the subalgebras $\mathscr{O}(Z_i) \otimes_A A''$, for A'' as above. Consequently, there is an A'' for which $Y''_i = Z''_i$. A priori A'' depends on i, but, because J is finite, one A'' can be chosen that works for every $i \in J$. Put

$$S'' = \operatorname{Spec}(A''), \qquad S''_{(n)} = S'' \times_S S_n.$$

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For every $i \in J$, the equality

$$Y''_i \times_{S''} S''_{(n)} = (Y^i_n)_{S''_{(n)}} = Z''_i \times_{S''} S''_{(n)} = (Z^i_n)_{S''_{(n)}}$$

shows, by the definition of T^i_n, that $S''_{(n)} \to S_n$ factors through T^i_n. It therefore factors through

$$T_n = \bigcap_{i \in J} T^i_n,$$

and hence through T^i_n for every $i \in I$. Let f'' be the image of f in $B'' = B \otimes_A A''$, and let $f''_{(n)}$ be its image in

$$B''_{(n)} = B''/\mathfrak{m}^{n+1}B''.$$

The preceding factorization means that, for every n, the section $f''_{(n)}$ vanishes on all the $Z_i \times_S S''_{(n)}$.

Because $A \to A'' \to A'$ are local homomorphisms, the image s'' of s' in S'' is the closed point of S'', corresponding to the maximal ideal $\mathfrak{m}''$ of A'', and its image in S is s. Fix n, and now let S''_n denote the n-th infinitesimal neighborhood of s'' in S'':

$$S''_n = \operatorname{Spec}(A''/\mathfrak{m}''^{n+1}).$$

Put

$$B''_n = B''/\mathfrak{m}''^{n+1}B'', \qquad Z^{i''}_n = Z_i \times_S S''_n.$$

Then the image f''_n of $f''_{(n)}$ in B''_n vanishes on every $Z^{i''}_n$. By Lemma 4.5 below, the family of fibers

$$Z^{i'''}_0 = Z^{i''}_n \times_{S''_n} \kappa(s'') = Z_{is} \times_{\kappa(s)} \kappa(s'')$$

is schematically dense in

$$X'''_0 = X''_n \times_{S''_n} \kappa(s'') = X_s \times_{\kappa(s)} \kappa(s'').$$

Thus $S''_n, X''_n, (Z^{i''}_n)_{i \in I}$, and

$$S'''_0 = \operatorname{Spec} \kappa(s'')$$

[18]Editor's note: the original has been expanded and corrected, distinguishing $S''_{(n)}$ from the subscheme $S''_n = S''_n(s'')$ introduced below.

satisfy the hypotheses of Lemma 4.2. It follows that $f''_n = 0$, that is,

$$f'' \in \mathfrak{m}''^{n+1}B'' \qquad \text{for every } n.$$

Since B is Noetherian and $B'' = B \otimes_A A''$ is a localization of a finite-type B-algebra, B'' is Noetherian. Its local rings are therefore separated for their usual topology. Hence f'' is zero at every point x'' of X'' for which $\mathfrak{m}''B''$ is contained in the maximal ideal of $\mathscr{O}_{X'',x''}$, that is, at every point of X'' above s''. A fortiori, f' is zero at every point of X' above s'. ☐

Lemma 4.5.0.[19] *Let X be an S-prescheme, let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, and let $S' \to S$ be faithfully flat. If $(Z'_i)_{i\in I}$ is schematically dense in X', then $(Z_i)_{i\in I}$ is schematically dense in X.*

This follows from EGA IV_3, 11.10.5(i) and 11.9.10(i). It remains to give the proof of the following lemma.

Lemma 4.5. *Let X be a locally Noetherian prescheme*[20] *over a field k, let $(Z_i)_{i\in I}$ be a family of subpreschemes of X, let k'/k be a field extension, and let X', Z'_i be obtained from X, Z_i by base change. Then $(Z_i)_{i\in I}$ is schematically dense in X if and only if $(Z'_i)_{i\in I}$ is schematically dense in X'.*

Proof. The "if" direction follows from Lemma 4.5.0; let us prove the converse.[21] First, we may suppose $X = \mathrm{Spec}(B)$ affine. For every i, let $(Z_{ij})_{j\in J_i}$ be an affine open covering of Z_i. Replacing $(Z_i)_{i\in I}$ by $(Z_{ij})_{(i,j)\in J}$, where

$$J = \coprod_{i\in I} J_i,$$

we may also suppose that every Z_i is affine.

Let

$$g \in B' = B \otimes_k k', \qquad t' \in B'_g$$

be a section of $\mathscr{O}_{X'}$ on the affine open $U' = X'_g$ whose restriction to each $Z'_i \cap U'$ is zero. Thus its image in every

$$\big(\mathscr{O}(Z_i) \otimes_k k'\big)_g$$

is zero. There exists a finite-type k-subalgebra A of k' such that

$$g \in B_A = B \otimes_k A, \qquad t' \in (B_A)_g.$$

The map

$$\mathscr{O}(Z_i) \otimes_k A \longrightarrow \mathscr{O}(Z_i) \otimes_k k'$$

is injective, and therefore so is

$$\big(\mathscr{O}(Z_i) \otimes_k A\big)_g \longrightarrow \big(\mathscr{O}(Z_i) \otimes_k k'\big)_g.$$

The hypothesis consequently implies that the image of t' in every $\big(\mathscr{O}(Z_i) \otimes_k A\big)_g$ is zero. It remains to show that $(Z_{iA})_{i\in I}$ is schematically dense in X_A, which follows from EGA IV_3, 11.9.10(b).[22] ☐

[19] Editor's note: this lemma has been inserted here because it is used in the proofs of 4.5 and 4.7.

[20] Editor's note: this hypothesis is in fact superfluous; cf. EGA IV_3, 11.10.6 and 11.9.13.

[21] Editor's note: the original has been expanded in what follows.

[22] Editor's note: the original continued: "Using 4.3 for the $X_{A'}$, where A' is a quotient of A by a power of a maximal ideal, and Hilbert's Nullstellensatz, one is reduced to the case where k' is a finite extension of k." This reduction could be expanded and that argument pursued, since the case of a finite extension k'/k is somewhat simpler than the more general case treated in EGA IV_3, 11.9.10(b).

Let us record in passing the following result, which will be used in a later exposé.[23]

Corollary 4.6. *Let X be a flat S-prescheme and U an open subset of X. Suppose that U_s is schematically dense in X_s for every $s \in S$, and that either X is locally Noetherian or X is locally of finite presentation over S. Then U is schematically dense in X.*

Proof. For simplicity, suppose U is retrocompact in X (cf. EGA IV$_3$, 11.10.10 and 11.9.17 for the general case). The locally Noetherian case is included only for reference; it is contained in 4.3.

In the second case, we may plainly suppose S and X affine. Then X and U are of finite presentation over $S = \operatorname{Spec}(A)$. The ring A is the direct limit of its finite-type $\mathbb{Z}$-subalgebras A_i. The "patented procedure" already used (cf. EGA IV$_3$, 8.8.2 and 8.10.5(iii)) shows that there are an index i, an affine scheme X_i over

$$S_i = \operatorname{Spec}(A_i),$$

and an open subset U_i of X_i, from which X, U are obtained by the base change $S \to S_i$. Let E_i be the subset of S_i consisting of those $s \in S_i$ for which $(U_i)_s$ is schematically dense in $(X_i)_s$.

By EGA IV$_2$, 5.9.9 and 5.10.2, E_i is the set of points $s \in S_i$ for which $(U_i)_s$ contains the set

$$\operatorname{Ass} \mathscr{O}_{(X_i)_s}$$

of points associated with the structure sheaf of $(X_i)_s$. By EGA IV$_3$, 11.9.17.1, this is a constructible condition; hence E_i is constructible.[24]

By 4.5, the inverse image of E_i under $S_j \to S_i$ is E_j, and its inverse image under $S \to S_i$ is the set E of those $s \in S$ for which U_s is schematically dense in X_s.[24] By hypothesis, $E = S$, which is also the inverse image of S_i under every $S \to S_i$. EGA IV$_3$, 8.3.11 therefore implies that there is a $j \geq i$ such that $E_j = S_j$; that is, $(U_j)_s$ is schematically dense in $(X_j)_s$ for every $s \in S_j$.

Apply 4.4 to (S_j, X_j, U_j) and the base change $S \to S_j$. It follows that U is schematically dense in X. Notice that here 4.4 is used only for a family with a finite index set (in fact, a singleton); in this case its proof simplifies considerably. □

Theorem 4.7. *Let S be a prescheme, and let G be a group of multiplicative type and of finite type over S. For every integer $n > 0$, let ${}_nG$ be the kernel of $n \cdot \mathrm{id}_G$. Then the family of subpreschemes $({}_nG)_{n>0}$ is schematically dense in G in the sense of Definition 4.1.*

Proof. We distinguish two cases.

(a) S locally Noetherian. By 4.3, we are reduced to the case in which S is the spectrum of a field k. By 4.5.0, we may suppose that k is algebraically closed and that G is diagonalizable, say

$$G = D_k(M),$$

where M is a commutative group of finite type. Then

$$M \simeq \Gamma \times \mathbb{Z}^r$$

with Γ finite, and consequently

$$G \simeq G_1 \times G_2, \qquad G_1 = D(\Gamma), \qquad G_2 = \mathbf{G}_m^r.$$

[23] Editor's note: this reference remains to be specified.

[24] Editor's note: the original has been expanded in this paragraph and the next.

For n sufficiently divisible—namely, for n a multiple of the order of Γ—one has

$$ {}_nG = G_1 \times {}_nG_2, \qquad \text{since} \qquad {}_nG_1 = G_1. $$

Applying 4.3 again to the projection $G \to G_1$, we are reduced to G_2, hence to the case $G = \mathbf{G}_m^r$. Since G is then reduced, schematic density of $({}_nG)_{n>0}$ is equivalent to density of $\bigcup_n {}_nG$ in the ordinary topology. As

$$ {}_nG = ({}_n\mathbf{G}_m)^r, $$

we are reduced to $G = \mathbf{G}_m$, an irreducible group of dimension one. The assertion now follows because $\bigcup_n {}_nG$, the set of roots of unity in k, is infinite.

(b) The general case. For every $s \in S$, there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G' = G_{S'}$ is diagonalizable, say $G' = D_{S'}(M)$. By 4.5.0 we may reduce to G', and therefore suppose that G itself is diagonalizable. It is obtained by base change

$$ S \longrightarrow \operatorname{Spec}(\mathbb{Z}) $$

from the absolute group $H = D_{\mathbb{Z}}(M)$. Part (a) shows that, for every $s \in \operatorname{Spec}(\mathbb{Z})$, the family $({}_nH_s)_{n>0}$ is schematically dense in H_s. It remains only to apply 4.4.[25] □

Corollary 4.8.

(a) *Let $u, v : H \to G$ be two homomorphisms of S-preschemes in groups, with H of multiplicative type*[26] *and of finite type. Suppose that, for every integer $n > 0$, the restrictions of u and v to ${}_nH$ are equal. Then $u = v$.*

(b) *Let H_1, H_2 be two subgroups of multiplicative type and of finite type of a prescheme in groups G. If ${}_nH_1 = {}_nH_2$ for every integer $n > 0$, then $H_1 = H_2$.*

Proof. Part (a) follows from part (b) by considering the subgroups H_1, H_2 of $H \times G$ that are the graphs of u, v. For part (b), put

$$ H = H_1 \cap H_2. $$

This is a subgroup prescheme of H_i, for $i = 1, 2$, and we must show that it equals H_i. The hypothesis says precisely that H contains all the ${}_nH_i$. We are therefore reduced to the next corollary. □

Corollary 4.9. *Let G be an S-group of multiplicative type and of finite type, and let H be a subgroup prescheme containing ${}_nG$ for every $n > 0$. Then $H = G$.*

Proof. By 4.7, it is enough to prove that H is closed, or equivalently that it equals G set-theoretically. This reduces us to the case in which S is the spectrum of a field. Every subgroup prescheme of G is then closed, by VI_B, 1.4.2. □

Remark 4.10. Under the hypotheses of 4.7, let $m > 0$ be an integer with the following properties: for every $s \in S$, m is not a power of the characteristic of $\kappa(s)$; and if G_s has type M, every prime divisor of the torsion subgroup of M divides m. The second condition is automatically satisfied if G is a torus. The proof shows that in Theorem 4.7 and Corollary 4.8 and Corollary 4.9 one may restrict to the subgroups

$$ {}_{m^r}G, \qquad r > 0. $$

[25] Editor's note: this result will be needed in X, 4.3 for a non-locally-Noetherian base, namely $S = \operatorname{Spec}(\widehat{A} \otimes_A \widehat{A})$, where $\widehat{A}$ is the completion of the Noetherian local ring A.

[26] Editor's note: "diagonalizable" has been replaced by "of multiplicative type."

Moreover, if G is smooth over S, then for every $s \in S$ there are an open neighborhood U of s and an integer $m > 0$, prime to every residue characteristic on U, that satisfy the preceding conditions for G_U. For instance, take m to be the order of the torsion subgroup of the type of G at s (cf. 1.4.1(a) and 2.1(e)). After restriction to U, the groups ${}_{m^r}G$ are then finite and étale over S, and their family is schematically dense. In certain applications—in particular those involving Theorem 3.2 and Theorem 3.2 bis (cf. XI, 6), though not those involving Theorem 3.6 and Theorem 3.6 bis—this observation allows one to avoid Theorem 3.1 and its use of Hochschild cohomology.

5. Central Homomorphisms of Groups of Multiplicative Type

Lemma 5.0.[27] *Let $(A, \mathfrak{m})$ be a Noetherian local ring, put $S = \operatorname{Spec}(A)$, and let H be a finite scheme over S, say $H = \operatorname{Spec}(B)$, where B is finite over A. Let K be a subscheme of H such that reduction modulo $\mathfrak{m}^{n+1}$ gives $K_n = H_n$ for every n. Then $K = H$.*

Proof. Let s be the closed point of S. First, K is a closed subscheme of H. Indeed, a priori it is a closed subscheme of an open subset U of H. But K, and hence U, contains the fiber

$$K_s = H_s.$$

Since $H \to S$ is finite, and therefore closed, the complement of U is empty; thus $U = H$. Hence K is defined by an ideal I of B. The hypothesis implies

$$I \subset \mathfrak{m}^n B \qquad \text{for every } n.$$

Since B is a finite A-module, it is separated for the $\mathfrak{m}$-adic topology. Therefore $I = 0$. □

Theorem 5.1. *Let $u, v : H \to G$ be two homomorphisms of S-preschemes in groups, where H is of multiplicative type and of finite type. Suppose either that S is locally Noetherian or that G is of finite presentation over S. Let $s \in S$ satisfy $u_s = v_s$, and suppose that u_s is central. Then there is an open neighborhood U of s such that $u_U = v_U$.*

Proof. We distinguish the two cases.

(a) S locally Noetherian.[28] Let

$$K = \operatorname{Ker}(u, v)$$

be the inverse image of the diagonal of $G \times_S G$ under $(u, v) : H \to G \times_S G$. It is a subgroup prescheme of H. We seek an open U for which $K_U = H_U$. Because S is locally Noetherian and H is of finite type over S, the prescheme H is locally Noetherian. Hence the immersion $K \hookrightarrow H$ is of finite type (cf. EGA I, 6.3.5), so K is of finite type, and therefore of finite presentation, over S. By EGA IV$_3$, 8.8.2.4, it is enough to prove that

$$K_{S_0} = H_{S_0}, \qquad S_0 = \operatorname{Spec}(\mathscr{O}_{S,s}).$$

We may thus suppose that S is local with closed point s. Replacing the Noetherian local ring A by its completion $\widehat{A}$, if desired, gives a faithfully flat quasi-compact base change $\widehat{S} \to S$, and allows us to suppose that A is complete.[29]

[27] Editor's note: this lemma has been made explicit because it is used several times below; it was implicit in the original.

[28] Editor's note: the original has been expanded in what follows.

[29] Editor's note: this is because $K = H$ if $K_{\widehat{S}} = H_{\widehat{S}}$; cf. SGA 1, VIII, 5.4. The argument below does not use completeness of A.

By 3.4, one has $u_n = v_n$ for every n, where the subscript denotes reduction modulo $\mathfrak{m}^{n+1}$, with $\mathfrak{m}$ the maximal ideal of A. For every $m > 0$, let

$$ {}_m u, {}_m v : {}_m H \longrightarrow G $$

be the homomorphisms induced by u, v. Then

$$ ({}_m u)_n = ({}_m v)_n \qquad \text{for every } n. $$

The group ${}_m H$ is finite over S, by 2.2, so 5.0 gives ${}_m u = {}_m v$. This holds for every m, and therefore 4.7 gives $u = v$.

(b) G of finite presentation.[30] The group H is also of finite presentation over S. By EGA IV$_3$, 8.8.2, we may again suppose that S is local with closed point s, and it suffices to prove $u = v$. Let $f : S' \to S$ be a faithfully flat quasi-compact morphism. Write

$$ f_H : H' \to H, \qquad f_G : G' \to G, \qquad u', v' : H' \to G' $$

for the resulting morphisms. If $u' = v'$, then

$$ u \circ f_H = f_G \circ u' = f_G \circ v' = v \circ f_H, $$

and $u = v$ because f_H is an epimorphism. After a faithfully flat quasi-compact extension of the base, we may therefore suppose that H is diagonalizable:

$$ H = D_S(M) $$

for a commutative group M of finite type.

As in the proof of 4.6, introduce the increasing filtered family of $\mathbb{Z}$-subalgebras A_i of $A = \mathscr{O}_{S,s}$ that are of finite type, and put

$$ S_i = \operatorname{Spec}(A_i). $$

[30] For every i, the group $H = D_S(M)$ comes from

$$ H_i = D_{S_i}(M). $$

Since G is of finite presentation over S, EGA IV$_3$, 8.8.2 (see also VI$_B$, 10.2 and 10.3) gives an index i, an S_i-group prescheme G_i of finite presentation, and homomorphisms

$$ u_i, v_i : H_i \longrightarrow G_i $$

whose base changes are u, v. Let s_i be the image of s in S_i, and define

$$ \rho_i : H_i \times_{S_i} G_i \longrightarrow G_i, \qquad \rho_i(h, g) = u_i(h) g\, u_i(h)^{-1}. $$

Because u_s is central,

$$ \rho_s = \rho_i \times_{\kappa(s_i)} \kappa(s) $$

is the second projection. The extension $\kappa(s_i) \to \kappa(s)$ is faithfully flat, so ρ_i is also the second projection over s_i; hence $(u_i)_{s_i}$ is central. Likewise $u_s = v_s$ implies

$$ (u_i)_{s_i} = (v_i)_{s_i}. $$

[30]Editor's note: the original has been expanded in what follows.

Part (a), applied over S_i, now gives the conclusion. □

Corollary 5.2. *Let $u : H \to G$ be a homomorphism of S-preschemes in groups, where H is of multiplicative type and of finite type, and suppose either that S is locally Noetherian or that G is of finite presentation over S. If $s \in S$ and u_s is the unit homomorphism, then there is an open neighborhood U of s on which u_U is the unit homomorphism.*

Corollary 5.3. *Let u, H, G be as in 5.2, and suppose in addition that G is separated over S. Let U be the set of points $s \in S$ for which u_s is the unit homomorphism. Then U is open and closed in S, and $u_U : H_U \to G_U$ is the unit homomorphism.*

Proof. Only closedness remains to be proved. Put $K = \mathrm{Ker}(u)$. Since G is separated over S, K is a closed subprescheme of H, and U is the set of points $s \in S$ for which $K_s = H_s$. It follows readily that U is closed, for example by VIII, 6.4, since H is essentially free over S by VIII, 6.3. Alternatively, we may suppose that S is reduced and that U is dense in S, hence schematically dense. Then H_U is schematically dense in H because H is flat over S,[31] and u and the unit homomorphism, being equal on H_U, are equal on H. □

Corollary 5.4. *Let S be a prescheme; let H, G be S-groups, with H of multiplicative type and of finite type and G separated and of finite presentation over S; and let $\pi : S' \to S$ be a*[32] *universal effective epimorphism—for example, a faithfully flat quasi-compact morphism or a morphism admitting a section (cf. IV, 1.12)—whose fibers are geometrically connected.*[33]

Let $u' : H' \to G'$ be a central homomorphism of the S'-groups obtained from H, G by base change. Then there is a unique homomorphism $u : H \to G$ of S-groups such that $u \times_S S' = u'$. If S'/S admits a section g, then u is the morphism obtained from u' by the base change $g : S \to S'$.

Proof. Because π is a universal effective epimorphism, so is $H' \to H$; uniqueness follows as at the beginning of the proof of 5.1(b). If π has a section g, then

$$u' = \pi^* u \quad \Longrightarrow \quad u = g^* \pi^* u = g^* u'.$$

For existence, IV, 2.3 reduces us to showing that the two homomorphisms

$$u_1'', u_2'' : H'' \longrightarrow G''$$

of S''-groups obtained from u' by the two base changes

$$\mathrm{pr}_1, \mathrm{pr}_2 : S'' = S' \times_S S' \longrightarrow S'$$

are equal. Their pullbacks along the diagonal $S' \to S''$ are equal, since both are u'.[34] Because u_1'', u_2'' are central, we may apply 5.3 to

$$u_1''(u_2'')^{-1}.$$

There is therefore an open-and-closed subset U of S'', containing the diagonal, over which u_1'' and u_2'' coincide. The fibers of S'/S are geometrically connected, hence so are those

[31] Editor's note: and of finite presentation over S; cf. EGA IV_3, 11.10.5(ii) and 11.9.10(ii).

[32] Editor's note: "covering morphism for the faithfully flat quasi-compact topology" has been replaced by "universal effective epimorphism," so that the result applies to the morphism $K \to S$ in 5.5; the beginning of the proof has been modified accordingly.

[33] Editor's note: this holds, for example, when $S = \mathrm{Spec}(k)$ and $S' = \mathrm{Spec}(k')$, where k is a field and k'/k is radicial; cf. X, Proposition 1.4.

[34] Editor's note: the preceding sentence has been added.

of S''/S, and in particular they are connected. Since U contains the diagonals of these fibers, it contains every fiber and therefore equals S''. This proves existence. □

Corollary 5.5. *Let S be a prescheme, let K be an S-group prescheme locally of finite type*[35] *and with connected fibers, and let H be an invariant subgroup of K, of multiplicative type and of finite type. Then H is a central subgroup of K.*

Proof.[36] First, $\pi : K \to S$ has geometrically connected fibers, because for a group scheme locally of finite type over a field, connectedness implies geometric connectedness (cf. VI_A, 2.4). Apply 5.4 with $G = H$ and $S' = K$ to the homomorphism of K-groups

$$u' : H_K \longrightarrow H_K, \qquad (h,k) \longmapsto (khk^{-1}, k).$$

It is central because H is commutative. Its pullback along the unit section $\epsilon : S \to K$ is the identity homomorphism of H. Corollary 5.4 therefore implies that u' itself is the identity on H_K. Thus H is central in K. □

Let us state the variants of the preceding results for central subgroups of multiplicative type. Arguing as above, and using 3.2 bis, gives:

Theorem 5.1 bis. *Let G be an S-group prescheme, and let H_1, H_2 be two subgroup preschemes of multiplicative type and of finite type, with $(H_1)_s$ central. Suppose either that S is locally Noetherian or that G is of finite presentation over S. For every $s \in S$ such that $(H_1)_s = (H_2)_s$, there is an open neighborhood U of s for which $H_1|_U = H_2|_U$.*

Corollary 5.3 bis. *Under the preceding hypotheses, suppose moreover that G is separated over S. Let U be the set of points $s \in S$ for which $(H_1)_s = (H_2)_s$. Then U is open and closed in S, and $H_1|_U = H_2|_U$.*

Corollary 5.4 bis. *Let S be a prescheme, let G be a group prescheme separated and of finite presentation over S, let $S' \to S$ be a covering morphism for the faithfully flat quasi-compact topology with geometrically connected fibers, and let H' be a subgroup of multiplicative type and of finite type of $G' = G \times_S S'$. Then there is a unique subgroup H of G such that $H' = H \times_S S'$. The subgroup H is of multiplicative type and of finite type over S, by 1.1 and 2.1(b). If $S' \to S$ admits a section g, then $H = g^*H'$.*

Theorem 5.6. *Let S be a prescheme and let $u : H \to G$ be a homomorphism of S-preschemes in groups, where H is of multiplicative type and of finite type and G is of finite presentation over S.*

(a) *Suppose that G has connected fibers. If $s \in S$ and $u_s : H_s \to G_s$ is central, then there is an open neighborhood U of s such that the homomorphism $H|_U \to G|_U$ induced by u is central.*

(b) *If $u_s : H_s \to G_s$ is central for every $s \in S$, then u is central.*

Proof. We first prove (a).[37] Arguing exactly as in the proof of 5.1(b), we may suppose first that S is local with closed point s, then that $H = D_S(M)$ is diagonalizable, and finally that there exist a finite-type $\mathbb{Z}$-subalgebra A_i of $A = \mathscr{O}_{S,s}$ and a homomorphism

$$u_i : D_{S_i}(M) \longrightarrow G_i$$

[35] Editor's note: the hypothesis "locally of finite type" has been added to agree with VI_A, 2.4. In fact, over a field every connected group scheme is geometrically connected.

[36] Editor's note: the original has been expanded in what follows.

[37] Editor's note: the original has been expanded in the proof of (a).

whose base change is u and for which $(u_i)_{s_i}$ is central, where s_i is the image of s in S_i. We are thereby reduced to the case in which S is Noetherian local with closed point s, and we must prove that u is central.

Let $K = \mathrm{Ker}(v, w)$, where

$$v, w : H \times_S G \rightrightarrows G$$

are defined by

$$v(h, g) = g, \qquad w(h, g) = \mathrm{int}(u(h)) \cdot g = u(h) g\, u(h)^{-1}.$$

Then K is a subgroup over G of

$$H_G = H \times_S G,$$

and we must prove that $K = H_G$. By 4.9, it is enough to show that K contains

$${}_m(H_G) = ({}_m H) \times_S G$$

for every $m > 0$. We are thus reduced to the case $H = {}_m H$, so that H is finite over S.

Let e be the unit element of the fiber G_s, and put

$$S^0 = \mathrm{Spec}(\mathscr{O}_{G,e}), \qquad S^0_n = S^0 \times_S S_n, \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}).$$

The scheme S^0 is Noetherian local because G is of finite presentation over the Noetherian scheme S. The base change

$$K_{S^0} = K \times_G S^0$$

is a subscheme of

$$H_{S^0} = H \times_S S^0,$$

and 3.9 gives

$$K_{S^0_n} = H_{S^0_n} \qquad \text{for every } n.$$

[38] Since H_{S^0} is finite over S^0, 5.0, applied to the Noetherian local ring $\mathscr{O}_{G,e}$, gives

$$K_{S^0} = H_{S^0}.$$

On the other hand, $H \times_S G$ is Noetherian, being of finite presentation over the Noetherian scheme S. Thus the immersion

$$K \hookrightarrow H \times_S G$$

is of finite type (cf. EGA I, 6.3.5), and K is of finite presentation over G. The equality

$$K_{S^0} = H_G \times_G S^0$$

therefore implies, by EGA IV$_3$, 8.8.2.4, that there is an open neighborhood W of e in G such that

$$K \times_G W = H_G \times_G W = H \times_S W.$$

Hence K contains the open neighborhood

$$V = H \times_S W$$

of the unit section of $G_H = G \times_S H$ over H.

[38] Editor's note: the original has been expanded and corrected, replacing "the unit section of $(G_H)_n$ over H_n" by $H_{S^0_n}$.

For every $t \in H$, the fiber G_t is an algebraic group over $\kappa(t)$, hence Cohen–Macaulay by VI_A, 1.1.1 and therefore has no embedded components. It is also connected, hence irreducible by VI_A, 2.4, so its generic point is its unique associated point. EGA IV_2, 3.1.8 therefore shows that V_t is schematically dense in G_t. By 4.3, V is schematically dense in G_H.

Moreover, since K is an H-subgroup of G_H, it induces on each fiber G_h a subgroup K_h. This subgroup contains an open neighborhood of the unit element; because G_h is connected, it follows that $K_h = G_h$. Thus $K = G_H$ set-theoretically. Finally, K is a closed subprescheme of G_H containing the schematically dense subprescheme V, and hence $K = G_H$. This proves (a).

Part (b) follows directly from 3.9: to verify that a subprescheme K of $P = G \times_S H$ equals P, it is enough to verify, after every base change $S' \to S$ in which S' is the spectrum of an Artinian quotient of a local ring of S, that $K' = P'$. □

6. Monomorphisms of Groups of Multiplicative Type, and Canonical Factorization of a Homomorphism from Such a Group

Lemma 6.1. *Let S be a quasi-compact prescheme, and let G be a commutative S-group prescheme, of finite presentation and quasi-finite over S. Then there is an integer $n > 0$ such that*

$$n \cdot \mathrm{id}_G = 0, \qquad \textit{that is,} \qquad G = {}_nG.$$

Proof. If S is the spectrum of a field k, then G is finite over k, and by $\mathrm{VII_A}$ 8.5 it is enough to take

$$n = \deg(G/k).$$

Suppose now that $S = \mathrm{Spec}(A)$, where A is local Artinian. Let k be the residue field of A, put

$$G_0 = G \otimes_A k, \qquad n_0 = \deg(G_0/k),$$

and distinguish two cases.

(a) k has characteristic zero. Then G_0 is separable over k, and hence G is unramified over S. The unit section of G is therefore an open immersion, so

$${}_nG = \mathrm{Ker}(n \cdot \mathrm{id}_G)$$

is an open subscheme of G. For it to equal G, it is consequently enough that it do so set-theoretically; thus one may take $n = n_0$.

(b) k has characteristic $p > 0$. Let $\mathfrak{m}$ be the maximal ideal of A, and put

$$S_r = \mathrm{Spec}(A/\mathfrak{m}^{r+1}) \qquad (r \geq 0).$$

Choose an integer m such that $\mathfrak{m}^{m+1} = 0$. We claim that one may take $n = p^m n_0$. Indeed, arguing by induction on m and putting $n' = p^{m-1} n_0$, we may suppose that $n' \cdot \mathrm{id}_G$ induces the zero endomorphism on

$$G_{m-1} = G \times_S S_{m-1}.$$

[39] Let E be the set of endomorphisms u of G having this property, and let K be the kernel of the morphism of group functors

$$G \longrightarrow \prod_{S_{m-1}/S} G$$

(cf. III 0.1.2). Then

$$E = \mathrm{Hom}_{S\text{-gr}}(G, K);$$

in particular, the abelian group law on E is induced by that on K. By III 0.9, K is the S-group functor which, to every $g : T \to S$, assigns the k-vector space

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(g_0^*(\Omega^1_{G_0/S_0}), \mathfrak{m}^m \mathscr{O}_T\big).$$

Thus $pu = 0$ for every $u \in E$. It follows that

$$pn' \cdot \mathrm{id}_G = 0, \qquad \text{that is,} \qquad p^m n_0 \cdot \mathrm{id}_G = 0.$$

Suppose now that S is Noetherian; this is the case to which the general assertion reduces by the usual reduction to the Noetherian case.[40] It is enough to combine the foregoing with the following lemma.

Lemma 6.2. *Let X be a quasi-finite prescheme over a Noetherian prescheme S, and let $(X_i)_{i \in I}$ be an increasing filtered family of subpreschemes having the following property:*

$$\begin{cases} \textit{for every } s \in S \textit{ and every } n \geq 0, \textit{ putting} \\ S_{s,n} = \mathrm{Spec}(\mathscr{O}_{S,s}/\mathfrak{m}^{n+1}), \textit{ there is an } i \in I \textit{ such that} \\ X_i \times_S S_{s,n} = X \times_S S_{s,n}. \end{cases}$$

Then there is an $i \in I$ such that $X_i = X$.

Proof. Since S is Noetherian, there is a maximal open subset U on which

$$X|_U = X_i|_U$$

for all sufficiently large i. We shall prove that $U = S$. In other words, if $U \neq S$, we shall find an open subset U' strictly larger than U, and an i, such that

$$X|_{U'} = X_i|_{U'}.$$

Localizing at a maximal point s of $S - U$, we reduce to the case in which S is local with closed point s, and

$$U = S - \{s\}.$$

Indeed, put $S' = \mathrm{Spec}(\mathscr{O}_{S,s})$. If

$$X \times_S S' = X_i \times_S S'$$

for some i,[41] there is an open neighborhood V of s such that

$$X|_V = X_i|_V.$$

[39] Editor's note: the original has been expanded in what follows, with the results of Exposé III that are used made explicit.

[40] Editor's note: since S is quasi-compact, it is covered by finitely many affine open subsets, so one reduces to $S = \mathrm{Spec}(A)$. Since G is of finite presentation over S, EGA IV$_3$, 8.8.2 may then be applied.

[41] Editor's note: $X - \{s\}$ has been corrected above to $S - \{s\}$. Recall also that a quasi-finite morphism is assumed to be of finite type; cf. EGA II, 6.2.3, and EGA III$_2$, Err$_{\mathrm{III}}$ 20 for the definition of "locally quasi-finite." Thus here, since S is Noetherian, X is of finite presentation over S; every immersion $X_i \hookrightarrow X$ is of finite type by EGA I, 6.3.5, so X_i is of finite presentation over S, and EGA IV$_3$, 8.8.2 applies.

Taking i large enough that $X_i|_U = X|_U$, we then have

$$X|_W = X_i|_W, \qquad W = U \cup V.$$

For all sufficiently large i, the equality $X|_U = X_i|_U$ shows that X_i is a closed subprescheme of X, defined by an ideal $\mathscr{I}^{(i)}$ whose support is contained in

$$X_s = X_0 = X \times_S S_0,$$

where

$$S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1}), \qquad S = \operatorname{Spec}(A).$$

Since X_0 is quasi-finite over S_0, it is a finite closed subset of the Noetherian prescheme X. Hence $\mathscr{I}^{(i)}$ is a module of finite length. There is therefore an integer $n \geq 0$ such that

$$\mathscr{I}^{(i)} \cap \mathfrak{m}^{n+1}\mathscr{O}_X = 0.$$

On the other hand, by the hypothesis of the lemma, after increasing i if necessary we may suppose that the image of $\mathscr{I}^{(i)}$ in

$$\mathscr{O}_{X_n} = \mathscr{O}_X/\mathfrak{m}^{n+1}\mathscr{O}_X$$

is zero. It follows that $\mathscr{I}^{(i)} = 0$, and therefore $X_i = X$. □

Lemma 6.3. *Let S be a prescheme, let K be a group prescheme over S, locally of finite presentation, and let $s \in S$. Suppose that K_s is quasi-finite (respectively, unramified) over $\kappa(s)$ at the unit element. Then there is an open neighborhood U of s such that $K|_U$ is locally quasi-finite (respectively, unramified) over U.**

Proof. Let V be the set of points $x \in K$ having the following property: if t is the image of x in S, then K_t is quasi-finite (respectively, unramified) over $\kappa(t)$ at x; that is, x is isolated in K_t (respectively, it is isolated and its local ring in K_t is a separable extension of $\kappa(t)$). The set V is open because K is locally of finite presentation over S.[42] If ϵ denotes the unit section of K, then $\epsilon^{-1}(V)$ is open. By hypothesis it contains s, and is therefore an open neighborhood U of s. This U has the required property. Indeed, if $t \in U$, then K_t, being a group locally of finite type over $\kappa(t)$, is locally quasi-finite (respectively, unramified) over $\kappa(t)$ because it has that property at $\epsilon(t)$; cf. $\mathrm{VI_B}$ 1.3. □

Combining 6.1 and 6.3, one obtains:

Theorem 6.4. *Let S be a prescheme, let H be an S-group of multiplicative type and of finite type, and let K be a closed group subprescheme of H, of finite presentation over S. Let $s \in S$ be such that K_s is finite. Then there are an open neighborhood U of s and an integer $n > 0$ such that*

$$K|_U \subset {}_nH|_U.$$

In particular, since ${}_nH$ is finite over S, the group $K|_U$ is finite over U.

Nakayama's lemma now gives:

Corollary 6.5. *With the preceding notation, suppose that K_s is the unit group. Then there is an open neighborhood U of s such that $K|_U$ is the unit group.*

Corollary 6.6. *Let $u : H \to G$ be a homomorphism of S-group preschemes, where H is of multiplicative type and of finite type and G is separated over S. Suppose in addition*

*Cf. $\mathrm{VI_B}$ 2.5(ii) for a more general statement.

[42] Editor's note: cf. EGA IV_3, 13.1.4 and EGA IV_4, 17.4.1. In addition, t, rather than s, has been used for the image of x.

that either S is locally Noetherian or G is locally of finite presentation over S.[43] *If $s \in S$ and*

$$u_s : H_s \longrightarrow G_s$$

is a monomorphism, then there is an open neighborhood U of s such that u induces a monomorphism

$$H|_U \longrightarrow G|_U.$$

Proof. Put $K = \mathrm{Ker}(u)$. The hypothesis on u_s says that K_s is the unit group, and the desired conclusion says that $K|_U$ is the unit group for a suitable U. Since G is separated over S, the group K is a closed subgroup of H. If S is not assumed locally Noetherian, but G is assumed locally of finite presentation over S, then K is locally of finite presentation over S.[43] It is therefore of finite presentation over S, since it is separated over S (as H is) and quasi-compact over S (being closed in H, which is quasi-compact over S). Corollary 6.5 now applies. □

Remark 6.6.1.[44] *Under the hypotheses of 6.6, note that if, in addition, G is affine over S (respectively, of finite presentation over S), then*

$$H|_U \longrightarrow G|_U$$

is even a closed immersion, as was observed in 2.5 (respectively, 2.6).

Corollary 6.7. *Let $u : H \to G$ be a homomorphism of S-group preschemes, where H is of multiplicative type and of finite type. Suppose that either S is locally Noetherian or G is locally of finite presentation over S. If every homomorphism on fibers*

$$u_s : H_s \longrightarrow G_s$$

is a monomorphism, then u is a monomorphism.

Proof. The argument is the same as in 6.6. Here the hypothesis that G be separated over S is unnecessary in order to ensure that $K = \mathrm{Ker}(u)$ is closed in H, since the assumption that all the u_s are monomorphisms implies that K is set-theoretically reduced to the unit section. □

Theorem 6.8. *Let $u : H \to G$ be a homomorphism of S-group preschemes, where H is of multiplicative type and of finite type and G is separated over S. Suppose in addition that either S is locally Noetherian or G is of finite presentation over S. Then*

$$K = \mathrm{Ker}(u)$$

is a subgroup of H of multiplicative type and of finite type, and u factors as

$$H \xrightarrow{\quad u' \quad} H/K \xrightarrow{\quad u'' \quad} G,$$

where H/K is of multiplicative type and of finite type, u' is the canonical homomorphism and is faithfully flat (and affine), and u'' is a monomorphism.

N.B. As noted in 6.6.1, the homomorphism u'' is a closed immersion if G is affine over S, or if G is of finite presentation over S.

[43] Editor's note: it is enough to suppose that G is locally of finite type over S. By EGA IV_4, 1.4.3(v), this implies, since $H \to S$ is of finite presentation, that $H \to G$ is locally of finite presentation; the same is then true of $K \to S$, which is obtained from it by base change.

[44] Editor's note: the number 6.6.1 has been added for use in later references.

Proof. It is enough to prove that K is of multiplicative type; the rest then follows from 2.7 and IV 5.2.6.

Suppose first that G is of finite presentation over S. Since this hypothesis is stable under base change, in proving that K is of multiplicative type we may reduce to the case in which H is diagonalizable:

$$H = D_S(M),$$

where M is a commutative group of finite type. Let $s \in S$. The group K_s is a closed subgroup of

$$H_s = D_{\kappa(s)}(M).$$

By 8.1,[45] it is therefore of the form $D_{\kappa(s)}(N)$, where N is a quotient of M. Put

$$K' = D_S(N).$$

Then K' is a subgroup of H of multiplicative type. Let

$$v' : K' \longrightarrow G$$

be the homomorphism induced by u.[46] By construction, v'_s is the unit homomorphism. Corollary 5.2 therefore gives an open neighborhood U of s such that

$$v'|_U : K'|_U \longrightarrow G|_U$$

is the unit homomorphism. Replacing S by U, we may suppose that v' itself is the unit homomorphism, and hence that u factors as

$$H \xrightarrow{\ u'\ } H/K' \xrightarrow{\ u''\ } G.$$

The homomorphism u''_s is obtained from u_s by factoring through

$$H_s \longrightarrow H_s/\operatorname{Ker}(u_s),$$

and is therefore a monomorphism by IV 5.2.6. By 6.6, after restricting to a suitable open neighborhood U of s, the homomorphism

$$u''|_U : (H/K')|_U \longrightarrow G|_U$$

is a monomorphism. Thus, after this restriction,

$$\operatorname{Ker}(u) = \operatorname{Ker}(u') = K',$$

which proves that $\operatorname{Ker}(u)$ is of multiplicative type.

The same proof applies if, instead of assuming G of finite presentation over S, one assumes S locally Noetherian, at least when H is diagonalizable. Without this additional hypothesis on H, one must show that there is a covering base change

$$S' \longrightarrow S,$$

with S' locally Noetherian, which trivializes H. This will be proved in the next exposé, X 4.6. □

[45]Editor's note: the erroneous reference to IV 4.7.5 has been corrected. Note that Section 8 of the present exposé is independent of Sections 2 through 7.

[46]Editor's note: "induced by G" has been corrected to "induced by u."

7. Algebraicity of Formal Homomorphisms into an Affine Group

Theorem 7.1. *Let A be a Noetherian ring equipped with an ideal I for which A is separated and complete in the I-adic topology. Put*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/I^{n+1}),$$

and let H, G be S-group preschemes, with H of isotrivial multiplicative type and G affine. Then the canonical map

$$\theta : \mathrm{Hom}_{S\text{-gr}}(H, G) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}(H_n, G_n) \tag{x}$$

is bijective, where H_n, G_n are the S_n-groups obtained from H, G by the base change $S_n \to S$.

Proof. Suppose first that H is diagonalizable, hence of the form

$$H = \mathrm{Spec}\, A(M),$$

where $B = A(M)$ is the algebra of the commutative group M with coefficients in A. We may also write

$$G = \mathrm{Spec}(C),$$

where C is an A-algebra equipped with a diagonal map satisfying the usual axioms. Homomorphisms of S-groups $H \to G$ correspond bijectively to homomorphisms of A-algebras

$$\varphi : C \longrightarrow B$$

compatible with the diagonal maps; that is, such that, for every $f \in C$,

$$\Delta_H(\varphi(f)) = (\varphi \otimes \varphi)(\Delta_G(f)).$$

There is an analogous description of the homomorphisms of S_n-groups $H_n \to G_n$, given by homomorphisms of A_n-algebras

$$\varphi_n : C_n \longrightarrow B_n,$$

where

$$A_n = A/I^{n+1}, \qquad B_n = B \otimes_A A_n, \qquad C_n = C \otimes_A A_n.$$

Put

$$\widehat{B} = \varprojlim_n B_n, \qquad \widehat{C} = \varprojlim_n C_n.$$

Then

$$\widehat{B} = \varprojlim_n A_n(M)$$

identifies with a submodule of the product A^M, namely the submodule of families $(a_m)_{m \in M}$ of elements of A that tend to zero for the I-adic topology along the filter of complements of finite subsets of M. In particular, the canonical homomorphism

$$B \longrightarrow \widehat{B}$$

is injective.[47] The projective system

$$\theta(\varphi) = (\varphi_n)_{n \geq 0}$$

[47] Editor's note: the argument that follows has been expanded.

defines a homomorphism

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B}$$

such that the following diagram is commutative:

$$\begin{tikzcd} C \arrow[r, "\varphi"] \arrow[d] & B \arrow[d] \\ \widehat{C} \arrow[r, "\widehat{\varphi}"'] & \widehat{B}. \end{tikzcd}$$

We prove that θ is surjective. Let $(\varphi_n)_{n\geq 0}$ be a projective system; we must show that it is obtained by reduction from some φ in the left-hand side of (x). A priori, the system (φ_n) defines a homomorphism on the completed algebras,

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B}.$$

It is enough to prove that the composite

$$\Phi : C \longrightarrow \widehat{C} \xrightarrow{\widehat{\varphi}} \widehat{B}$$

sends C into B. Indeed, in that case it gives a homomorphism of A-algebras

$$\varphi : C \longrightarrow B$$

whose reductions are the φ_n. This homomorphism is immediately seen to be compatible with the diagonal maps, since the φ_n are so, and since

$$B \otimes_A B \longrightarrow \widehat{B \otimes_A B}$$

is injective, as follows from the same argument after replacing M by $M \times M$.

Passing to the completions, the diagonal homomorphism Δ_H defines

$$\widehat{\Delta}_H : \widehat{B} \longrightarrow \widehat{B \otimes_A B}.$$

Taking the projective limit of the analogous diagrams defined by the φ_n, one obtains the commutative diagram

$$\begin{tikzcd} C \arrow[r, "\Phi"] \arrow[d, "\Delta_G"'] & \widehat{B} \arrow[d, "\widehat{\Delta}_H"] \\ C \otimes_A C \arrow[r, "\Psi"'] & \widehat{B \otimes_A B}, \end{tikzcd}$$

where Ψ is the composite

$$C \otimes_A C \xrightarrow{\Phi\otimes\Phi} \widehat{B} \otimes_A \widehat{B} \longrightarrow \widehat{B \otimes_A B};$$

the last arrow is the evident canonical homomorphism. Consequently, for every $f \in C$, the element $\Phi(f) \in \widehat{B}$ has the property that its image under $\widehat{\Delta}_H$ is a "decomposable" element of $\widehat{B \otimes_A B}$, namely an element in the image of $\widehat{B} \otimes_A \widehat{B}$.

Let $(e_m)_{m\in M}$ be the canonical basis of $A(M)$, and let $(e_{m,m'})$ be that of

$$A(M \times M) = A(M) \otimes_A A(M).$$

Since

$$\Delta_H(e_m) = e_m \otimes e_m = e_{m,m} \qquad (m \in M),$$

it remains only to apply the following lemma.

Lemma 7.2. *Let A be a Noetherian ring, let M be a set, and let $(a_{m,m'})$ be a family of elements of A, indexed by $M \times M$, satisfying:*

(i) $a_{m,m'} = 0$ if $m \neq m'$; that is, the support of the family is contained in the diagonal of $M \times M$;

(ii) one has

$$a_{m,m'} = \sum_i b^i_m c^i_{m'},$$

where the b^i, c^i are finitely many elements of A^M; in other words, $(a_{m,m'})$ belongs to the image of the canonical homomorphism

$$A^M \otimes_A A^M \longrightarrow A^{M \times M}.$$

Then the support of $(a_{m,m'})$ is finite.

Proof. By (i), the family $(a_{m,m'})$ is determined by the elements

$$a_m = a_{m,m}.$$

For $x = (x_n)_{n \in M} \in A(M)$, put

$$(u \cdot x)_m = \sum_{m'} a_{m,m'} x_{m'}.$$

Thus $(a_{m,m'})$ is viewed as the matrix of a homomorphism

$$u : A(M) \longrightarrow A^M.$$

Condition (i) gives simply

$$(u \cdot x)_m = a_m x_m.$$

On the other hand, condition (ii) gives

$$u \cdot x \in \sum_i A \cdot b^i.$$

Hence $u(A(M))$ lies in a finitely generated A-module. If $(e_m)_{m \in M}$ denotes the canonical basis of

$$A(M) \subset A^M,$$

then all the elements $a_m e_m$ lie in a finitely generated A-module. Since A is Noetherian, the module generated by these elements is itself finitely generated. Because the e_m are linearly independent, all but finitely many of the a_m are zero. This proves the lemma, and hence Theorem 7.1 when H is diagonalizable. □

We now prove the general case of 7.1, in which H is only assumed isotrivial. There is then a finite étale surjective morphism

$$S' \longrightarrow S$$

such that

$$H' = H \times_S S'$$

is diagonalizable. We shall use only the fact that $S' \to S$ is finite and a covering for the faithfully flat quasi-compact topology, or simply for the canonical topology on (**Sch**). Thus "étale" could be replaced by "flat."

Put

$$S'' = S' \times_S S',$$

and similarly introduce

$$S'_n \quad \text{and} \quad S''_n = S'' \times_S S_n = S'_n \times_{S_n} S'_n,$$

together with $H', G', H'', G'', H'_n, G'_n, H''_n, G''_n$, obtained from H, G by the evident base changes. The groups H' and H'' are diagonalizable. Moreover, S', and hence S'', is affine. If

$$S' = \operatorname{Spec}(A'), \qquad S'' = \operatorname{Spec}(A''),$$

then A' and A'' are separated and complete for the topologies defined by IA' and IA'', respectively, because both are finite over A.

Since $S' \to S$ and $S'_n \to S_n$ are covering morphisms, we obtain a commutative diagram of maps of sets whose two rows are exact:

$$\begin{tikzcd}
\operatorname{Hom}_{S\text{-gr}}(H,G) \arrow[r] \arrow[d, "u"'] & \operatorname{Hom}_{S'\text{-gr}}(H',G') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, "u'"'] & \operatorname{Hom}_{S''\text{-gr}}(H'',G'') \arrow[d, "u''"] \\
\varprojlim_n \operatorname{Hom}_{S_n\text{-gr}}(H_n,G_n) \arrow[r] & \varprojlim_n \operatorname{Hom}_{S'_n\text{-gr}}(H'_n,G'_n) \arrow[r, shift left] \arrow[r, shift right] & \varprojlim_n \operatorname{Hom}_{S''_n\text{-gr}}(H''_n,G''_n).
\end{tikzcd}$$

The second row is exact because it is the projective limit of the exact diagrams associated with the various $S'_n \to S_n$. By the diagonalizable case already proved, u' and u'' are bijective. Thus u is bijective as well, completing the proof of 7.1. □

Corollary 7.3. *Under the hypotheses of 7.1, suppose in addition that G is smooth over S, and let*

$$u_0 : H_0 \longrightarrow G_0$$

be a homomorphism of S_0-groups. Then there is a homomorphism of S-groups

$$u : H \longrightarrow G$$

lifting u_0. Any two such liftings u, u' are conjugate by an element $g \in G(S)$ whose reduction is the unit element of $G_0(S_0)$.

Proof. This follows by combining 3.6 and 7.1. To construct u, construct the u_n successively, as is possible by 3.6; Theorem 7.1 then shows that the system (u_n) comes from an S-homomorphism u. Given two liftings u, u', construct successively elements g_n such that

$$g_0 = 1, \qquad g_n \text{ is the reduction of } g_{n+1}, \qquad u'_n = \operatorname{int}(g_n)u_n.$$

Again, this is possible by 3.6. Since A is separated and complete, the g_n come from an element $g \in G(S)$. Finally, the injectivity in 7.1 gives

$$u' = \operatorname{int}(g)u.$$

□

Remark 7.4. *Compare 7.1 with EGA III, 5.4.1, which implies that the statement of 7.1 remains true if, instead of assuming H of multiplicative type and G affine, one assumes* *that H is proper over S, and that G is separated and locally of finite type over S. A statement such as 7.1 without a properness hypothesis is rather exceptional, and should be understood here as one aspect of the great "rigidity" of the structure of a group of multiplicative type.*

The analogous statement with

$$G = H = \mathbf{G}_a$$

is false in general. Indeed, take A of characteristic $p > 0$, and define the u_n, by reduction modulo I^{n+1}, from an additive formal power series

$$u(T) = \sum_{n\geq 0} a_n T^{p^n},$$

where the $a_n \in A$ tend to zero for the I-adic topology, but not for the discrete topology.

The statement of 7.1 also becomes false if the hypothesis that G be affine is omitted, even when $H = \mathbf{G}_m$. An example, with A a complete discrete valuation ring, is obtained from an elliptic curve over the fraction field K of A whose reduction, in the Néron–Kodaira theory, say, is the group $\mathbf{G}_m$ over the residue field k.[48] *One then has a smooth commutative group scheme G over S, with special fiber $\mathbf{G}_{m,k}$. By 3.6 this permits the construction of a projective system of isomorphisms*

$$u_n : H_n \xrightarrow{\sim} G_n, \qquad H = \mathbf{G}_{m,A},$$

while the generic fiber of G is an abelian variety. Consequently, there is no homomorphism of S-groups $H \to G$ other than the zero homomorphism.

8. Subgroups, Quotient Groups, and Extensions of Groups of Multiplicative Type over a Field[*]

Scholium 8.0. [49] — *Let k be a field and let H be an affine k-group scheme. Write $k[H]$ for its affine algebra and*

$$X(H) = \mathrm{Hom}_{k\text{-gr}}(H, \mathbf{G}_{m,k})$$

for the group of characters of H. By the independence lemma for characters, $X(H)$ is a linearly independent subset of $k[H]$. Consequently, H is diagonalizable if and only if $X(H)$ spans $k[H]$ as a k-vector space.

Proposition 8.1. — *Let G be a diagonalizable group scheme (respectively, a group scheme of multiplicative type) over a field k.*[50] *Then, for every subgroup scheme H of G, the schemes H and G/H are diagonalizable (respectively, of multiplicative type).*

Definition 1.1 immediately reduces the assertion to the case without the parenthetical qualification. An easy passage to the limit, using $\mathrm{VI_B}$ 11.13 and VIII 3.1, reduces us to

[48] Editor's note: such an example could be made explicit.

[*] Added in July 1969. This afterthought section is independent of Sections 3 through 7.

[49] Editor's note: this scholium has been added.

[50] Editor's note: the words "over a field k," implicit in the original, have been added. In addition, "algebraic group" has been replaced by "group scheme," since G is not assumed to be of finite type.

the case in which G is of finite type over k.[51] By $\mathrm{VI_B}$ 11.16,[52] one can find a finite family of nonzero elements

$$f_i = \sum_m a_{im} m, \qquad a_{im} \in k, \tag{8.1.1}$$

of the affine ring $k(M)$ of $G = D_k(M)$, such that the points of H, with values in an arbitrary k-algebra k', are precisely the points $g \in G(k')$ for which

$$\tau_g f_i = \lambda_i(g) f_i, \qquad \lambda_i(g) \in k', \tag{8.1.2}$$

where τ_g denotes translation by g. But

$$\tau_g f_i = \sum_m a_{im} \chi_m(g) m, \tag{8.1.3}$$

where $\chi_m : G \to \mathbf{G}_{m,k}$ is the character associated with m. Thus (8.1.2) is equivalent to

$$\chi_m(g) = \chi_{m'}(g) \qquad \text{if } m, m' \in Z_i, \tag{8.1.4}$$

where Z_i denotes the set of $m \in M$ for which $a_{im} \neq 0$. The same relation may be written

$$\chi_{m'-m}(g) = 1 \qquad \text{if } m, m' \in Z_i.$$

Let N be the subgroup of M generated by all the elements $m' - m$, where i varies and $m, m' \in Z_i$. It follows from the definition of $D_k(M) \simeq G$ that H identifies with $D_k(M/N)$. Hence VIII 3.1 identifies G/H with $D_k(N)$. ☐

Proposition 8.2. [53] — *Let k be a field; let H and K be k-group schemes of multiplicative type and finite type; and let G be a k-group scheme fitting into an exact sequence*

$$1 \longrightarrow H \longrightarrow G \longrightarrow K \longrightarrow 1$$

(which implies that G is of finite type over k).

(i) If G is commutative or K is connected, then G is of multiplicative type.

(ii) If K and H are diagonalizable and K is a torus, then G is diagonalizable.

For the proof of (i), the reader is referred to XVII 7.1.1, of which 8.2(i) is a special case. The case of a field is treated in part (i) of the proof of XVII 7.1.1, without using the results of the later exposés.

For (ii), now suppose that K and H are diagonalizable:

$$K \simeq D_k(M) \qquad \text{and} \qquad H \simeq D_k(N),$$

and suppose that G is of multiplicative type, as follows from (i) when K is a torus. By X 1.4, G is isotrivial over k; that is, there is a finite separable extension k'/k such that

$$G' = G \times_k k'$$

[51] Editor's note: this "passage to the limit" should be spelled out. It uses 8.0 and the equality

$$G/H = \varprojlim_i G_i/H_i$$

(cf. $\mathrm{VI_B}$, proof of 11.17). To see that $H = \varprojlim_i (H \cap G_i)$, does one use the fact that H is closed in G? Using this fact, one can give a direct proof.

[52] Editor's note: $\mathrm{VI_B}$ 11.16 should doubtless be rewritten in the usual, more agreeable form. In particular, a single f_i suffices below.

[53] Editor's note: both the statement and its proof have been expanded.

is diagonalizable. Hence $G' = D_{k'}(E)$ for a commutative group E, and there is an exact sequence

$$0 \longrightarrow D_{k'}(N) \longrightarrow D_{k'}(E) \longrightarrow D_{k'}(M) \longrightarrow 0. \tag{1}$$

By VIII 3.1 and 3.2, M is therefore a subgroup of E, and one has an exact sequence

$$0 \longrightarrow M \longrightarrow E \longrightarrow N \longrightarrow 0. \tag{2}$$

For a given extension E_0 of N by M, consider the diagonalizable k-group $G_0 = D_k(E_0)$. The k-group functor A of automorphisms of the extension

$$1 \longrightarrow H \longrightarrow G_0 \longrightarrow K \longrightarrow 1, \tag{3}$$

that is, the group subfunctor of $\mathrm{Aut}_{k\text{-gr}}(G_0)$ whose points on a k-prescheme T are the

$$\phi \in \mathrm{Aut}_{T\text{-gr}}(G_T)$$

inducing the identity on H_T and K_T, identifies with $\mathrm{Hom}_{k\text{-gr}}(K, H)$. By VIII 1.5, this is the constant k-group with value

$$L = \mathrm{Hom}_{\mathrm{gr}}(N, M).$$

It follows that the extensions G of K by H which become isomorphic to (3) over a separable closure k_s of k are classified in the same way as the k-torsors, for the étale topology, under the constant group L_k. Put $\Gamma = \mathrm{Gal}(k_s/k)$. Since L is a trivial Γ-module, these torsors are classified by

$$H^1_{\text{ét}}(k, L_k) = H^1(\Gamma, L) = \mathrm{Hom}_{\mathrm{gr.\,top.}}(\Gamma, L).$$

If K is moreover a torus, then M is torsion-free, and hence so is L. Therefore

$$\mathrm{Hom}_{\mathrm{gr.\,top.}}(\Gamma, L) = 0,$$

and every extension of K by H is already diagonalizable.

Remark 8.3. — *As already noted in the proof of 8.2, the first assertion is stated and proved over an arbitrary base in XVII 7.1.1. The second assertion generalizes, with essentially the same proof, to the case of an integral normal base scheme—or, more generally, a geometrically unibranch one—by using X 5.13 below.*

Finally, 8.1 may also be regarded as a corollary of the distinctly less trivial result 6.8, which is likewise valid over an arbitrary base scheme.[54]

Bibliography

[55]

[BEns] N. Bourbaki, *Théorie des ensembles*, Chapters I–IV, Hermann, 1970.

[54] Editor's note: the proof of 6.8 nevertheless uses 8.1.

[55] Editor's note: additional references cited in this exposé.

Exposé X
Characterization and Classification of Groups of Multiplicative Type

by A. Grothendieck

1. Classification of Isotrivial Groups. Case of a Base Field

Recall (IX 1.1) that a group of multiplicative type H over the prescheme S is said to be *isotrivial* if there exists a finite étale surjective morphism $S' \to S$ such that $H' = H \times_S S'$ is diagonalizable. When S is connected, S' decomposes as a finite disjoint sum of connected components S'_i, and hence, after replacing S' by one of the S'_i, we may suppose that S' is connected. Finally, S' can be dominated by a connected finite étale S'_1 that is Galois, that is, a principal homogeneous bundle over S with group Γ_S, where Γ is an ordinary finite group (cf. SGA 1, V, nos. 7 and 3 when S is locally noetherian, and EGA IV_4, 18.2.9 in general).[1] We suppose S' chosen in this way and propose to determine the groups of multiplicative type H over S that are "trivialized" by S', that is, those for which $H' = H \times_S S'$ is diagonalizable.[2] By descent theory (cf. SGA 1, VIII, 5.4), the category of such H is equivalent to the category of diagonalizable groups H' over S', equipped with actions of Γ on H' compatible with the actions of Γ on S'. (N.B. Since the groups under consideration are affine over the base, every descent datum here is effective; cf. SGA 1, VIII, 2.1.) Since S' is connected, the contravariant functor

$$M \longmapsto D_{S'}(M)$$

is an anti-equivalence from the category of ordinary commutative groups to the category of diagonalizable groups over S' (cf. VIII 1.6); a quasi-inverse is[3]

$$H \longmapsto \mathrm{Hom}_{S'\text{-gr}}(H, \mathbb{G}_{m,S'}).$$

Proposition 1.1. — *Let S be a connected prescheme and let S' be a connected principal cover of S with finite group Γ. Then the category of groups of multiplicative type over S*

[0] xy version of November 6, 2009: Addenda placed in Section 9; reviewed through 8.8.

[1] Editor's note: More precisely, this follows from the "axiomatic conditions of a Galois theory," cf. SGA 1, V, no. 4 g); when S is locally noetherian, the axioms are verified in *loc. cit.*, nos. 7 and 3, especially 3.7, which rests on SGA 1, I, 10.9. The latter result is proved without noetherian hypotheses in EGA IV_4, 18.2.9.

[2] Editor's note: In what follows, an S-group of multiplicative type will be called "split" if it is "trivial" in the sense of IX 1.2, that is, if it is a diagonalizable S-group.

[3] Editor's note: S has been corrected to S'.

split by S'[4] *is anti-equivalent to the category of Γ-modules, that is, of ordinary commutative groups M equipped with a homomorphism*

$$\Gamma \longrightarrow \mathrm{Aut}_{\mathrm{gr}}(M).$$

The following standard consequence results.

Corollary 1.2. — *Let S be a connected prescheme, and let $\xi : \mathrm{Spec}(\Omega) \to S$ be a "geometric point" of S, that is, a morphism into S from the spectrum of an algebraically closed field Ω. Consider the fundamental group of S at ξ (cf. SGA 1, V, no. 7):*

$$\pi = \pi_1(S, \xi).$$

Then the category of isotrivial groups of multiplicative type H over S is anti-equivalent to the category of "Galois modules" under π, that is, of π-modules M such that the stabilizer in π of every element of M is an open subgroup.

Under this correspondence, the group associated with an isotrivial group of multiplicative type H is

$$M = \mathrm{Hom}_{\Omega\text{-gr}}(H_\xi, \mathbb{G}_{m,\Omega}),$$

where H_ξ is the fiber of H at ξ; this group is naturally endowed with an action of $\pi_1(S, \xi)$.

Remark 1.3. — We shall see below (cf. 5.16) that if S is normal, or more generally geometrically unibranch, then every group of multiplicative type and of finite type over S is necessarily isotrivial. Consequently, the classification principle of 1.2 applies to groups of multiplicative type and of finite type over S; they correspond to Galois π-modules that are of finite type over $\mathbb{Z}$. For the moment, let us confine ourselves to the following result.

Proposition 1.4. — *Let k be a field and H a group of multiplicative type and of finite type over k. Then H is isotrivial; that is, there exists a finite separable extension k' of k that splits H.*

Consequently, by 1.2, if π is the topological Galois group of an algebraic closure Ω of k, the category of groups H of multiplicative type and of finite type over k is anti-equivalent to the category of Galois modules M under π that are of finite type as $\mathbb{Z}$-modules.

Since H is of finite type over k, the "principle of finite extension" (cf. EGA IV$_3$, 9.1.4) first gives a finite extension k' of k that splits H. Recall the argument. By hypothesis, there exist a diagonalizable group G of finite type over k, a faithfully flat S' over $S = \mathrm{Spec}(k)$, and an isomorphism of S'-groups

$$H_{S'} \simeq G_{S'}.$$

After replacing S' by the residue field of one of its points, we may suppose that S' is the spectrum of an extension K of k. The field K is the filtered direct limit of its finitely generated k-subalgebras A_i; it follows readily (cf. EGA IV$_3$, 8.8.2.4) that the given isomorphism u comes, for sufficiently large i, from an A_i-isomorphism

$$u_i : H_{A_i} \simeq G_{A_i}.$$

By the Nullstellensatz, A_i has a quotient ring k' that is a finite extension of k. Thus k' splits H.

Now k' is a radicial extension of a separable extension k'_s of k. By IX 5.4, the isomorphism

$$u' : H_{k'} \simeq G_{k'}$$

[4]Editor's note: Here and below, "splittés" has been replaced in the French text by "déployés," and "splitte" by "déploie," etc. Both are rendered here by "split."

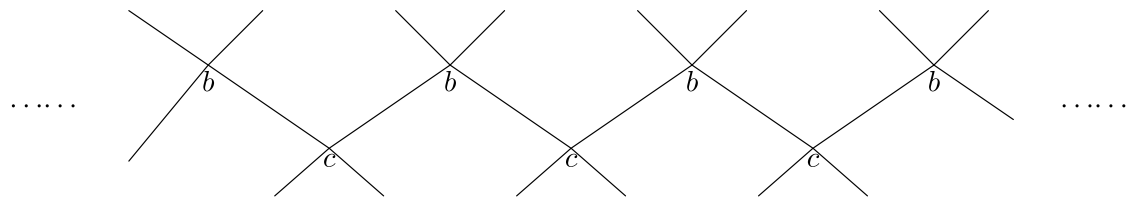
comes from an isomorphism $H_{k'_s} \simeq G_{k'_s}$. Thus k'_s splits H, proving 1.4.

Remark 1.5. — Statement 1.2 gives, in particular, a characterization of the isotrivial tori over S of relative dimension n: setting $\pi = \pi_1(S, \xi)$, they correspond to classes, up to "equivalence," of representations

$$\pi \longrightarrow \mathrm{GL}(n, \mathbb{Z})$$

whose kernel is an open subgroup of π.

1.6. Even when S is an algebraic curve, there may exist tori over S (of relative dimension 2) that are not locally isotrivial (and *a fortiori* not isotrivial); there may also exist locally trivial tori that are not isotrivial. (Observe, however, that such phenomena can occur only when S is not normal, as already noted in 1.3.) For example, let S be an irreducible algebraic curve (over an algebraically closed field, to fix ideas) with an ordinary double point a, let S' be its normalization, and let b and c be the two points of S' above a. A connected principal homogeneous bundle P over S, with structural group $\mathbb{Z}$, is obtained by attaching infinitely many copies of S' according to the diagram



(N.B. This is a principal bundle for the étale topology.) There is a homomorphism

$$\varphi : \mathbb{Z} \longrightarrow \mathrm{GL}(2, \mathbb{Z}), \qquad \varphi(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

which makes it possible to construct a torus T over S, of relative dimension 2, from the trivial torus $\mathbb{G}_m^2$ over P and the descent datum on it deduced from φ. (N.B. The projection $P \to S$ is a covering for the étale topology and *a fortiori* for the canonical topology of (**Sch**); the descent datum in question is necessarily effective because $\mathbb{G}_{m,P}^2$ is affine over P.) It is not difficult to prove that T is not isotrivial in a neighborhood of a[5] (although it is trivial on $S - \{a\}$).

A variant of this construction is obtained by taking for S a curve with two irreducible components S_1 and S_2 meeting in two points a' and a''. This makes it possible to construct a connected, locally trivial principal homogeneous bundle P over S with group $\mathbb{Z}_S$, and hence an associated torus T that is locally trivial but not isotrivial.

2. Infinitesimal Variations of Structure

[6] Let us begin by recalling the following result (cf. SGA 1, I, 8.3 when S is locally noetherian, and EGA IV$_4$, 18.1.2 in general).

Recall 2.0. — *Let S be a prescheme and S_0 a subprescheme with the same underlying set. Then the functor*

$$X \longmapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale preschemes over S and the analogous category over S_0.

[5] Editor's note: See 7.3 below.

[6] Editor's note: This "recall" has been added for later references.

Proposition 2.1. — *Let S be a prescheme and S_0 a subprescheme with the same underlying set (that is, defined by a nilideal $\mathscr{I}$). Then the functor*

$$H \longmapsto H_0 = H \times_S S_0$$

from the category of group preschemes of multiplicative type over S to the analogous category over S_0 is fully faithful.

Moreover, it induces an equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the analogous category over S_0.

We first prove full faithfulness, that is, if H and G over S are of multiplicative type, then

$$\mathrm{Hom}_{S\text{-gr}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(H_0, G_0)$$

is bijective. The question is local on S, so we may suppose that S is affine. There then exists a faithfully flat quasi-compact morphism $S' \to S$ that splits H and G. Let $S'' = S' \times_S S'$; write H', G', respectively H'', G'', for the groups obtained from H, G by the base changes $S' \to S$, respectively $S'' \to S$. Define S'_0 and S''_0 similarly; the latter is also isomorphic to $S'_0 \times_{S_0} S'_0$. We obtain a commutative diagram with exact rows:

$$\begin{tikzcd}
\mathrm{Hom}_{S\text{-gr.}}(H,G) \arrow[r] \arrow[d, "u"] & \mathrm{Hom}_{S'\text{-gr.}}(H',G') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, "u'"] & \mathrm{Hom}_{S''\text{-gr.}}(H'',G'') \arrow[d, "u''"] \\
\mathrm{Hom}_{S_0\text{-gr.}}(H_0,G_0) \arrow[r] & \mathrm{Hom}_{S'_0\text{-gr.}}(H'_0,G'_0) \arrow[r, shift left] \arrow[r, shift right] & \mathrm{Hom}_{S''_0\text{-gr.}}(H''_0,G''_0),
\end{tikzcd}$$

Thus, to prove that u is bijective, it is enough to prove that u' and u'' are bijective. This reduces us to the case where H and G are diagonalizable, hence of the form $D_S(M)$ and $D_S(N)$, where M and N are ordinary commutative groups. Likewise,

$$H_0 = D_{S_0}(M), \qquad G_0 = D_{S_0}(N).$$

We then have a commutative diagram[7]

$$\begin{tikzcd}
\mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) \arrow[r, "\sim"] \arrow[d] & \mathrm{Hom}_{S\text{-gr.}}\big(D_S(M), D_S(N)\big) \arrow[d] \\
\mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0}) \arrow[r, "\sim"] & \mathrm{Hom}_{S_0\text{-gr.}}\big(D_{S_0}(M), D_{S_0}(N)\big),
\end{tikzcd}$$

whose horizontal arrows are isomorphisms by VIII 1.4. We are therefore reduced to proving that the homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(N_S, M_S) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(N_{S_0}, M_{S_0}) \tag{x}$$

is bijective; in other words, the functor $M_S \mapsto M_{S_0}$, from constant commutative group preschemes over S to constant commutative group preschemes over S_0, is fully faithful. Now (x) is also identified with the natural map

$$\mathrm{Hom}_{\mathrm{gr}}(N, \Gamma(M_S)) \longrightarrow \mathrm{Hom}_{\mathrm{gr}}(N, \Gamma(M_{S_0}))$$

deduced from $\Gamma(M_S) \to \Gamma(M_{S_0})$. The latter map is plainly bijective, since $\Gamma(M_S)$, the set of locally constant maps from S to M, depends only on the underlying topological space of S. This proves the desired assertion.

[7]Editor's note: The original has been corrected by interchanging M and N in the right-hand terms.

To prove the second assertion of 2.1, it remains to show that every quasi-isotrivial group H_0 of multiplicative type over S_0 comes from a quasi-isotrivial group H of multiplicative type over S. Let $S'_0 \to S_0$ be an étale surjective morphism that splits H_0.

By 2.0, there exist an étale morphism $S' \to S$ and an S_0-isomorphism

$$S' \times_S S_0 \simeq S'_0.$$

Thus we may suppose that S'_0 is obtained from S' by reduction. Since H'_0 is diagonalizable, it is immediately seen to be isomorphic to the group obtained by the base change $S'_0 \to S'$ from a diagonalizable group H' over S'. (N.B. If $H'_0 = D_{S'_0}(M)$, take $H' = D_{S'}(M)$.) As usual, set

$$S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S',$$

and similarly define S''_0 and S'''_0 by base change from S to S_0. They are also isomorphic to the fibered square and fibered cube, respectively, of S'_0 over S_0. Applying the full faithfulness already proved to (S'', S''_0) and (S''', S'''_0), we see that the natural descent datum on H'_0 relative to $S'_0 \to S_0$ (cf. IV 2.1) comes from a uniquely determined descent datum on H' relative to $S' \to S$. This descent datum is effective because H' is affine over S' (SGA 1, VIII, 2.1). Hence there exists an S-group H such that

$$H \times_S S' = H' = D_{S'}(M),$$

and H is therefore a quasi-isotrivial group of multiplicative type.

Using the full-faithfulness result for (S', S'_0), one now checks readily that the given isomorphism between H'_0 and $H' \times_{S'} S'_0$ comes from an isomorphism between H_0 and $H \times_S S_0$. (For a more formal presentation of results of this kind, see Giraud's article in preparation on descent theory.*)

Corollary 2.2. — *Let H be an S-group of multiplicative type, and set $H_0 = H \times_S S_0$. Then H is quasi-isotrivial (respectively, locally isotrivial, isotrivial, locally trivial, or trivial) if and only if H_0 has the corresponding property.*

The "only if" direction is immediate. The "if" direction has already been proved in the quasi-isotrivial case: by full faithfulness, it is enough to know that every quasi-isotrivial group over S_0 lifts to a quasi-isotrivial group over S. The same argument applies to "trivial." For "isotrivial," repeat the argument establishing the second assertion of 2.1, but take $S'_0 \to S_0$ to be finite étale and surjective. The locally isotrivial and locally trivial cases follow at once from the isotrivial and trivial cases.

When $\mathscr{I}$ is nilpotent, 2.2 can be generalized slightly without assuming *a priori* that H is of multiplicative type.[8]

Corollary 2.3. — *Suppose that the ideal $\mathscr{I}$ defining S_0 is locally nilpotent. Let H be a group prescheme over S, flat over S, and set $H_0 = H \times_S S_0$. Then H is of quasi-isotrivial multiplicative type if and only if H_0 is so.*

Suppose that H_0 is of quasi-isotrivial multiplicative type. We prove that the same is true of H. The question is local for the étale topology, and by 2.0 the category of étale preschemes over S is equivalent to the category of étale preschemes over S_0 by the functor $S' \mapsto S' \times_S S_0$. We are therefore immediately reduced to the case where H_0 is diagonalizable, hence isomorphic to $D_{S_0}(M)$. Let $G = D_S(M)$. We then have an isomorphism

$$u_0 : H_0 \xrightarrow{\sim} G_0.$$

*Cf. J. Giraud, "Méthode de la descente," *Bull. Soc. Math. France*, Mémoire 2 (1964).

[8] Editor's note: Recall that 2.3 is used in proving Theorem IX 3.6 bis.

I claim that u_0 comes from a unique homomorphism $u : H \to G$, which is necessarily an isomorphism because u_0 is one, H and G are flat over S, and $\mathscr{I}$ is locally nilpotent.[9] This will prove 2.3. We have (cf. VIII 1 (xxx))

$$\mathrm{Hom}_{S\text{-gr}}(H, G) \simeq \mathrm{Hom}_{S\text{-gr}}\big(M_S, \underline{\mathrm{Hom}}_{S\text{-gr}}(H, \mathbb{G}_{m,S})\big),$$

and the right-hand side is also identified with

$$\mathrm{Hom}_{\mathrm{gr}}\big(M, \mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_{m,S})\big).$$

Thus the homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(H_0, G_0) \tag{x}$$

is isomorphic to the homomorphism

$$\mathrm{Hom}_{\mathrm{gr}}\big(M, \mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_{m,S})\big) \longrightarrow \mathrm{Hom}_{\mathrm{gr}}\big(M, \mathrm{Hom}_{S_0\text{-gr}}(H_0, \mathbb{G}_{m,S_0})\big)$$

deduced from the restriction homomorphism

$$\mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_{m,S}) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(H_0, \mathbb{G}_{m,S_0}). \tag{xx}$$

It is therefore enough to prove that (xx) is bijective. The question is local on S, so we may suppose that S is affine. Since $\mathbb{G}_{m,S}$ is commutative and smooth over S, IX 3.6 applies and completes the proof.

Corollary 2.4. — *Let A be an artinian local ring with residue field k, $S = \mathrm{Spec}(A)$, and $S_0 = \mathrm{Spec}(k)$.*

(i) [10] *Let H be an S-group prescheme, flat and locally of finite type, such that $H_0 = H \times_S S_0$ is of multiplicative type. Then H is of multiplicative type, of finite type, and isotrivial. In particular, every S-group prescheme H of multiplicative type and of finite type is isotrivial.*

(ii) *The functor $H \mapsto H_0$ is an equivalence between the category of groups of multiplicative type and of finite type over A and the analogous category over k.*

Indeed, let H be as in (i). Then H_0 is of multiplicative type and locally of finite type, hence of finite type, and therefore isotrivial by 1.4. By 2.3 and 2.2, H is consequently of multiplicative type (and hence of finite type) and isotrivial. Assertion (ii) now follows from 2.1.

Corollary 2.5. — *Let $S' \to S$ be a faithfully flat radicial morphism.*

(i) *The functor*

$$H \longmapsto H' = H \times_S S'$$

from the category of groups of multiplicative type over S to the analogous category over S' is fully faithful. Moreover, it induces an equivalence between the subcategories of quasi-isotrivial groups of multiplicative type.

[9]Editor's note: Since u_0 is an isomorphism, it is enough to see that for every $h \in H$, the morphism $\varphi : \mathscr{O}_{G,u(h)} \to \mathscr{O}_{H,h}$ is bijective. It is bijective after reduction modulo $\mathscr{I} = \mathscr{I}_s$, where s is the image of h in S; hence its cokernel C satisfies $C = \mathscr{I}C$, so $C = 0$ because $\mathscr{I}$ is nilpotent. Since $\mathscr{O}_{H,h}$ is flat over $\mathscr{O}_{S,s}$, the kernel K of φ likewise satisfies $K = \mathscr{I}K$, and hence $K = 0$.

[10]Editor's note: The statement has been rewritten, retaining only the nontrivial implication in order to highlight it.

(ii) *If H is of multiplicative type, then H is quasi-isotrivial (respectively, locally isotrivial, isotrivial, locally trivial, or trivial) if and only if H' has the corresponding property.*

Indeed, let
$$S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S'.$$
The assumption that $S' \to S$ is radicial implies that the diagonal immersions $S' \to S''$ and $S' \to S'''$ are surjective, so base change by either immersion is governed by 2.1 and 2.2. Taking into account that $S' \to S$ is a morphism of effective descent for the fibered category of groups of multiplicative type over preschemes (because it is faithfully flat and quasi-compact[11]), the assertion follows formally from 2.1 and 2.2 (cf. the already cited article of Giraud for a formal argument).

3. Finite Variations of Structure: Complete Base Ring

Lemma 3.1. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology; set $S = \operatorname{Spec}(A)$ and $S_0 = \operatorname{Spec}(A/I)$. Let G and H be two S-group preschemes, with G of multiplicative type and isotrivial, and with H affine over S, flat over S at the points of $H_0 = H \times_S S_0$, where H_0 is of quasi-isotrivial multiplicative type. Then the natural map*
$$\operatorname{Hom}_{S\text{-gr}}(G, H) \longrightarrow \operatorname{Hom}_{S_0\text{-gr}}(G_0, H_0)$$
is bijective.

For every integer $n \geq 0$, let
$$S_n = \operatorname{Spec}(A/I^{n+1}),$$
and let G_n, H_n be obtained from G, H by the base change $S_n \to S$. Since G/S is of isotrivial multiplicative type and H is affine over S, IX 7.1 shows that the natural homomorphism
$$\operatorname{Hom}_{S\text{-gr}}(G, H) \longrightarrow \varprojlim_n \operatorname{Hom}_{S_n\text{-gr}}(G_n, H_n)$$
is bijective. On the other hand, 2.3 shows that the H_n are of quasi-isotrivial multiplicative type, and 2.1 shows that all transition homomorphisms in the inverse system
$$\big(\operatorname{Hom}_{S_n\text{-gr}}(G_n, H_n)\big)_n$$
are isomorphisms. Lemma 3.1 follows at once.

Theorem 3.2. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, and set $S = \operatorname{Spec}(A)$ and $S_0 = \operatorname{Spec}(A/I)$. Then:*

(i) *The functor*
$$H \longmapsto H_0 = H \times_S S_0$$
is an equivalence between the category of isotrivial groups of multiplicative type over S and the analogous category over S_0.

(ii) *Let H be an S-group of multiplicative type and of finite type. Then H is isotrivial if and only if H_0 is isotrivial.*

[11] Editor's note: Spell out this point . . .

Proof. For (i), one may either repeat the proof of 2.1 or use 1.2. In either case one uses the fact that the functor

$$S' \longmapsto S'_0 = S' \times_S S_0$$

from the category of finite étale schemes over S to the category of finite étale schemes over S_0 is an equivalence (SGA 1, I, 8.4). Equivalently, after reducing to the case where S is connected (that is, S_0 is connected) and choosing a geometric point ξ of S_0, the canonical homomorphism

$$\pi_1(S_0, \xi) \longrightarrow \pi_1(S, \xi)$$

is an isomorphism.

We prove (ii), namely that if H_0 is isotrivial, then H is isotrivial. By (i), there exist an isotrivial group G of multiplicative type over S and an S_0-isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

[12] Since H is of finite type, so are H_0 and G_0. Therefore, by IX 2.1 b), the type of G at every point of S is a finitely generated abelian group, and hence G is of finite type over S. Moreover, by 3.1, u_0 comes from a homomorphism of S-groups

$$u : G \longrightarrow H.$$

Finally, G and H are groups of multiplicative type and of finite type over S, and u_0 is an isomorphism. It follows from IX 2.9 that u is an isomorphism, since every neighborhood of S_0 in S is equal to S.

Corollary 3.3. — *Let A be a complete noetherian local ring with residue field k.*

(i) *Every group of multiplicative type and of finite type over A is isotrivial.*

(ii) *The functor*

$$H \longmapsto H \otimes_A k = H_0$$

is an equivalence between the categories of groups of multiplicative type and of finite type over A and over k.

[12] First, (i) follows from 3.2 (ii) and 1.4. Then (ii) follows from 3.2 (i), taking into account that H is of finite type if and only if H_0 is of finite type (cf. the proof of 3.2 (ii)).

Remark 3.3.1. — By 1.4, statement 3.3 gives a complete classification of groups of multiplicative type and of finite type over A in terms of the topological Galois group of an algebraic closure Ω of k.

Remarks 3.4. — Under the hypotheses of 3.2 (that is, A noetherian, separated and complete for the I-adic topology, with no further hypothesis on A/I), it will follow from no. 5 that the functor

$$H \longmapsto H_0$$

from the category of groups of multiplicative type and of finite type over S to the analogous category over S_0 is still fully faithful, without any isotriviality hypothesis.[13]

It is not, however, essentially surjective in general. Indeed, there may exist a group H_0 of multiplicative type and of finite type over S_0, even locally trivial if desired (but not isotrivial), that does not arise by reduction from a group H of multiplicative type over S.

[12] Editor's note: The original has been expanded in what follows.

[13] Editor's note: Specify this reference, possibly adding a corollary in no. 5 ...

To see this, take either example from 1.6 of a non-isotrivial group of multiplicative type over a nonnormal curve. The curve may plainly be taken affine, say S_0, and embedded in affine 2-space; it is therefore defined by an ideal J in $k[X, Y]$. Take A to be the completion of $k[X, Y]$ for the J-adic topology. Then A is a regular ring of dimension 2, and *a fortiori* normal. We shall see in 5.16 that every group of multiplicative type and of finite type over $S = \mathrm{Spec}(A)$ is consequently isotrivial. Thus H_0, being of finite type and non-isotrivial, cannot come from a group H of multiplicative type over S, since such an H would necessarily be of finite type and hence isotrivial.

Lemma 3.5.* — *Let S be a prescheme and $u : G \to H$ a homomorphism of S-group preschemes that are locally of finite presentation and flat over S. Let U be the set of $s \in S$ such that the homomorphism induced on the fibers,*

$$u_s : G_s \longrightarrow H_s,$$

is flat (respectively, smooth, unramified, étale, or quasi-finite). Then U is open, and the restriction

$$u|_U : G|_U \longrightarrow H|_U$$

is flat (respectively, smooth, unramified, étale, or quasi-finite).

Let V be the set of points at which u is flat (respectively, ...). It is known that V is open (cf. SGA 1, I through IV in the locally noetherian case, and EGA IV in general[14]), and for $x \in G$ above $s \in S$, one has $x \in V$ if and only if $u_s : G_s \to H_s$ is flat (respectively, ...) at x (same reference; in the flat, smooth, and étale cases, the flatness of G and H over S is used here). Since u_s is a homomorphism of locally algebraic groups, this is also equivalent to u_s being flat (respectively, ...) everywhere ($\mathrm{VI_B}$, 1.3), that is, to $s \in U$. Therefore U is the inverse image of the open set V under the unit section of G, and hence is open. Moreover, $V = G|_U$, so $G|_U \to H|_U$ is flat (respectively, ...), completing the proof. (N.B. In the unramified and quasi-finite cases, the flatness hypothesis on G and H is unnecessary.)

Lemma 3.6. — *Let H be a commutative algebraic group over a field k that admits an open subgroup G of multiplicative type. Then the family of subschemes ${}_nH$, $n > 0$, is schematically dense in H. In particular, if ${}_nH = {}_nG$ for every $n > 0$, then $H = G$.*

Here ${}_nH$ denotes the kernel of $n \cdot \mathrm{id}_H$.[15] Let $\bar{k}$ be an algebraic closure of k. It is enough to show that the family $({}_nH_{\bar{k}})_{n>0}$ is schematically dense in $H_{\bar{k}}$, for then $({}_nH)_{n>0}$ is schematically dense in H (cf. IX 4.5). We may therefore suppose that k is algebraically closed, so that $G = D_k(M)$, where M is an ordinary finitely generated commutative group.

Let

$$M_0 = M/\,\mathrm{Torsion}(M).$$

Then $T = D_k(M_0)$ is the largest torus contained in G, and H/T is finite. Hence $H(k)/T(k)$ is annihilated by an integer $\nu > 0$. There exist finitely many $g_i \in H(k)$ such that

$$H = \coprod_i g_i \cdot G,$$

and $g_i^\nu \in T(k)$. Since k is algebraically closed, the map $\nu \cdot \mathrm{id}_T$ is surjective on

$$T(k) \simeq (k^\times)^d.$$

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*Cf. also $\mathrm{VI_B}$, 2.5, for more systematic developments of this kind.

[14] Editor's note: Specify these references. Since G, H are locally of finite presentation, u is locally of finite presentation (IV_1, 1.4.3 (v)); one then applies IV_3, 11.3.10 and 13.1.4 for flat and quasi-finite, and IV_4, 17.4.1, 17.5.1, and 17.6.1 for unramified, smooth, and étale; compare $\mathrm{VI_B}$, 2.5.

[15] Editor's note: The preceding sentence has been added.

After replacing g_i by $g_i t_i^{-1}$, where $t_i \in T(k)$ satisfies $t_i^\nu = g_i^\nu$, we may therefore suppose that $g_i^\nu = 1$. If n is now a multiple of ν, then

$$_nH \supset g_i \cdot {}_nG.$$

As $n > 0$ varies, the family $_nG$ is schematically dense in G by IX 4.7, and the conclusion of 3.6 follows.

Theorem 3.7. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology; set $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/I)$. Let H be an affine S-group scheme of finite type, flat over S at the points of H_0, such that $H_0 = H \times_S S_0$ is of multiplicative type and isotrivial. Then there exists an open and closed subgroup G of H, of isotrivial multiplicative type (and of finite type), such that $G_0 = H_0$.*

By 3.2 (i), there exist an isotrivial group G of multiplicative type over S and an isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

By 3.1, u_0 comes from a unique homomorphism of S-groups

$$u : G \longrightarrow H.$$

Using IX 6.6, we see that u is a monomorphism: if U is the set of $s \in S$ such that $u_s : G_s \to H_s$ is a monomorphism, then U is an open neighborhood of S_0, hence equals S, and $G|_U \to H|_U$ is a monomorphism. By IX 2.5, u is even a closed immersion.

Thus G is of finite type, and therefore of finite presentation, over S.[16] Applying Lemma 3.5 in the étale case, there is an open neighborhood U of S_0, hence $U = S$, such that $G|_U \to H|_U$ is étale. Thus u is étale and therefore an open immersion, since it is an étale monomorphism.[17] This completes the proof.

Corollary 3.8. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology; set $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/I)$. Let H be an S-group prescheme of finite type, affine and flat over S.*

Then H is of multiplicative type and isotrivial if and only if H_0 is so and H satisfies either one of the following conditions (a) and (b), which are therefore equivalent under the preceding hypotheses:

(a) *The fibers of H are of multiplicative type and have constant type on each connected component of S.*

(b) *H is commutative and the $_nH$, $n > 0$, are finite over S.*

Both of these conditions are implied by the following one:

(c) *The fibers of H are connected.*

Of course, if H is of multiplicative type (and isotrivial), then (a) and (b) hold by IX 1.4.1 a) and 2.1 c). Thus only the "if" direction requires proof. We use the subgroup G provided by 3.7. When (c) holds, plainly $G = H$, and there is nothing to prove.

In case (b), the immersion $u : G \to H$ induces an open immersion

$$_nu : {}_nG \longrightarrow {}_nH$$

[16] Editor's note: The preceding sentence has been added.

[17] Editor's note: Cf. EGA IV_4, 17.9.1.

and an isomorphism

$$({}_nG)_0 \xrightarrow{\sim} ({}_nH)_0.$$

Since ${}_nH$ is finite over S, it follows at once that ${}_nu$ is an isomorphism: the complement of its image is finite over S and reduces to the empty set. By 3.6, the induced morphisms on fibers

$$u_s : G_s \longrightarrow H_s$$

are isomorphisms. Hence u is surjective and therefore an isomorphism.

Finally, in case (a), we may suppose that S is connected.[18] It follows that, for every $s \in S$,

$$u_s : G_s \longrightarrow H_s$$

is a monomorphism of algebraic groups of multiplicative type and of the same type over $\kappa(s)$.[19] Such a homomorphism is necessarily an isomorphism, which again completes the proof.[20] After extending the base field, we may suppose that both groups over $\kappa(s)$ are diagonalizable. The assertion then follows from VIII 3.2 b) and the fact that a surjective homomorphism between isomorphic finitely generated $\mathbb{Z}$-modules

$$M \longrightarrow N$$

is necessarily bijective.

Corollary 3.9. — *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, set $S = \operatorname{Spec}(A)$, and let H be an S-group prescheme of finite type, affine and flat over S, with connected fibers. Then H is an isotrivial torus if and only if $H_0 = H \times_S \operatorname{Spec}(A/I)$ is an isotrivial torus.*

4. Case of an arbitrary base. Quasi-isotriviality theorem

Let A be a local ring. Recall, following Nagata, that A is said to be *henselian* if every algebra B finite over A is a product of local algebras B_i finite over A.

Recall 4.0.[21] *Let A be a henselian local ring, k its residue field, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$, and ξ a geometric point of S_0. Then the functor*

$$X \longmapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale coverings of S and the analogous category over S_0 (cf. EGA IV$_4$, § 18.5). Consequently (cf. SGA 1, V),

$$\pi_1(S_0, \xi) = \pi_1(S, \xi).$$

Suppose in addition that A is noetherian, and write A' for its completion and $S' = \operatorname{Spec}(A')$. Then A' is a complete noetherian local ring, hence henselian (*loc. cit.*, 18.5.14), and the functor

$$X \longmapsto X' = X \times_S S'$$

[18] Editor's note: The original said "that is, S_0 connected." Since A is separated for the I-adic topology, S_0 meets every connected component of S; and since A is complete, the connected components of S_0 and S are in bijection.

[19] Editor's note: This is because G_s and H_s have the same type for every $s \in S_0$, and hence for every s.

[20] Editor's note: Indeed, u will again be a surjective open immersion.

[21] *Editors' note.* This recall has been spelled out, and Remark 4.0.1 has been added.

from the category of étale coverings of S to the analogous category over S' fits into a commutative diagram

$$\begin{tikzcd} \text{ÉtCov}(S) \arrow[rr] \arrow[dr, "\sim"'] & & \text{ÉtCov}(S') \arrow[dl, "\sim"] \\ & \text{ÉtCov}(S_0) & \end{tikzcd}$$

and is therefore also an equivalence of categories, whence $\pi_1(S_0, \xi) = \pi_1(S', \xi)$.

Remark 4.0.1. Since S is connected (A being local), it follows from 1.2 that the category of isotrivial groups of multiplicative type over S is equivalent to the analogous category over S_0 (and also over S' if A is noetherian).

Lemma 4.1. *Let A be a henselian local ring with residue field k, $S = \operatorname{Spec}(A)$, and $S_0 = \operatorname{Spec}(k)$.*

(i) *The functor*

$$H \longmapsto H_0 = H \times_S S_0$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S_0.

(ii) *If, in addition, A is noetherian, writing A' for its completion and $S' = \operatorname{Spec}(A')$, the functor*

$$H \longmapsto H' = H \times_S S'$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S'.

As in 4.0, the second assertion follows from the first; let us prove the first. We already know that the functor under consideration is essentially surjective: every group of multiplicative type H_0 over $S_0 = \operatorname{Spec}(k)$, being finite and hence of finite type over k, is isotrivial by 1.4, and therefore comes from an isotrivial group of multiplicative type over S, by 4.0.1.

It remains to prove full faithfulness, that is, that for two finite groups G, H of multiplicative type over S, the following map is bijective:

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \operatorname{Hom}_{S_0\text{-gr.}}(G_0, H_0).$$

In other words,[22] if F denotes the S-functor $\operatorname{Hom}_{S\text{-gr.}}(G, H)$, the natural map

$$\operatorname{Hom}_S(S, F) \longrightarrow \operatorname{Hom}_{S_0}(S_0, F_0)$$

induced by the base change $S_0 \to S$ is bijective. In view of Recall 4.0, it is enough to prove the following lemma.

Lemma 4.2. *Let G, H be two finite groups of multiplicative type over a prescheme S. Then $F = \operatorname{Hom}_{S\text{-gr.}}(G, H)$ is representable by a finite étale scheme over S.*

[23] Let $f : S' \to S$ be a faithfully flat quasi-compact morphism such that G' and H' are diagonalizable. It is enough to show that $F_{S'}$ is represented by a prescheme X' étale and finite (hence affine) over S'. Indeed, the descent datum on X' relative to f (cf. VIII 1.7.2) is then effective (by SGA 1, VIII 2.1), so there is an S-prescheme X such that $X \times_S S' = X'$, representing F, and X is étale and finite over S (cf. SGA 1, V 5.7 and EGA IV_4, 17.7.3 (ii)).

[22] *Editors' note.* What follows has been added.

[23] *Editors' note.* The original has been expanded in what follows.

We may therefore suppose that $G = D_S(M)$ and $H = D_S(N)$, where M and N are finite commutative groups (cf. VIII 2.1 c)). Then $K = \mathrm{Hom}_{\text{gr.}}(N, M)$ is a finite abelian group and, by VIII 1.5, there is an isomorphism

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \simeq K_S,$$

which completes the proof of 4.2 and hence of 4.1.

Proposition 4.3.0.[24] *Let S be a henselian local scheme, s its closed point, $g : X \to S$ a morphism locally of finite type, and x an isolated point of the fiber X_s.*

(i) *Then $\mathcal{O}_{X,x}$ is finite over $\mathcal{O}_{S,s}$. (In particular, if the residue-field extension $\kappa(x)/\kappa(s)$ is trivial, then $\mathcal{O}_{S,s} \to \mathcal{O}_{X,x}$ is surjective, by Nakayama's lemma.)*

(ii) *If g is also separated, then $X' = \mathrm{Spec}(\mathcal{O}_{X,x})$ is an open and closed subscheme of X; that is, there is a decomposition into a disjoint sum $X = X' \amalg X''$.*

Proof. By the local form of Zariski's Main Theorem found in [Pes66], or in [Ray70, Ch. IV, Th. 1], x has an affine open neighborhood $U = \mathrm{Spec}(B)$, of finite type and quasi-finite over $A = \mathcal{O}_{S,s}$, and there is an open immersion $U \hookrightarrow Y = \mathrm{Spec}(C)$, where C is a finite A-algebra. (N.B. To obtain this conclusion one may also use Chevalley's semicontinuity theorem (EGA IV_3, 13.1.4), followed by the form of Zariski's Main Theorem given in *loc. cit.*, 8.12.8.)

Since A is henselian, Y is the disjoint sum of local schemes $Y_1, \dots, Y_n$, each finite over S, and the points of Y above s are their closed points $y_1, \dots, y_n$. Thus $x = y_i$ for some i, and $\mathcal{O}_{X,x} = \mathcal{O}_{U,x} = C_i$ is finite over A. Moreover, $X' = \mathrm{Spec}(C_i)$ is an open subscheme of U, hence of X.

Suppose in addition that g is separated. Since $X' \to S$ is finite (C_i being finite over A), so is the immersion $X' \to X$ (cf. EGA II, 6.1.5 (v)); therefore X' is also closed in X. $\square$

Lemma 4.3. *Let A be a noetherian henselian local ring, A' its completion, $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, and s the closed point of S. Let G, H be two S-preschemes in groups, with G of multiplicative type and of finite type over S, and with H locally of finite type and separated over S, H_s of multiplicative type, and H flat over S at the points of H_s.*

Let G', H' be obtained from G, H by the base change $S' \to S$. Then the following natural map is bijective:

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{S'\text{-gr.}}(G', H').$$

Since S' is faithfully flat and quasi-compact over S, SGA 1, VIII 5.2 shows that this map identifies the first member with the subset of the second formed by the $u' : G' \to H'$ for which the two inverse images $u_1'', u_2'' : G'' \to H''$ of u' on $S'' = S' \times_S S'$, under the projections $\mathrm{pr}_1, \mathrm{pr}_2 : S'' \to S'$, are equal.

It therefore remains to prove that $u_1'' = u_2''$ for every homomorphism of S'-groups $u' : G' \to H'$. By the density theorem IX 4.8, it is enough to prove that u_1'' and u_2'' agree on ${}_nG''$ for every integer $n > 0$. (N.B. Here the density theorem is needed essentially in a case where the base prescheme, namely S'', is not locally noetherian.)

Replacing G by ${}_nG$, this reduces us to the case in which $n \cdot \mathrm{id}_G = 0$ for some integer $n > 0$, and hence G is finite over S. Likewise let ${}_nH$ be the kernel of the n-th power morphism φ_n on H. (N.B. We have not assumed H commutative, so φ_n, respectively ${}_nH$, need not be a group homomorphism, respectively a subgroup of H.) Since ${}_nH$ is defined as the fiber product of $\varphi_n : H \to H$ and the unit section $\varepsilon : S \to H$, its formation commutes with every base change $T \to S$:

$$({}_nH)_T = {}_n(H_T).$$

[24] *Editors' note.* This recall, which is used in the proof of 4.3, has been added (the reference to EGA IV_4, 18.5.11 was not entirely satisfactory).

[25] We use a right subscript m for reduction modulo $\mathfrak{m}^{m+1}$, where $\mathfrak{m}$ is the maximal ideal of A. Then H_m is flat over S_m (because H is flat over S at the points of H_0); hence, by 2.4, H_m is of multiplicative type and of finite type, since H_0 is. Thus every $({}_nH)_m = {}_n(H_m)$ is a subgroup of H_m, of multiplicative type, finite and flat over S_m.

In particular, $({}_nH)_0 = {}_n(H_0)$ is finite over S_0. Since S is henselian local and ${}_nH$, like H, is locally of finite type and separated over S, Proposition 4.3.0 shows that the local rings of ${}_nH$ at the points of $({}_nH)_0$ are finite over S, and that there is a decomposition as a disjoint sum of open subsets

$$ {}_nH = {}_nH^+ \amalg {}_nH^-, \tag{*}$$

where $Z = {}_nH^+$ is finite over S, while ${}_nH^-$ lies above $S - \{s\}$.

Observe that for every finite S-scheme Y (such as G), every S-morphism $Y \to {}_nH$ factors through Z, and that the formation of the decomposition $(*)$ commutes with the base change $S' \to S$, where $S' = \operatorname{Spec}(A')$ and A' is the completion of A.

Then $Z' = Z \times_S S'$ is finite over S', as is $P' = Z' \times_{S'} Z'$. Let ν denote the restriction to P' of the multiplication on H', and let σ be the automorphism of P' interchanging the two factors. Since P' is finite over S', while H' is separated and locally of finite type over S', EGA II 5.4.3 and IV_1 1.1.3 show that $Y = \nu(P')$ is a closed subscheme of H', universally closed and quasi-compact, hence of finite type and therefore proper over S'. Moreover, $Y \to S'$ has finite fibers (because $P' \to S'$ does). Since S' is noetherian, $Y \to S'$ is therefore finite (cf. EGA III, 4.4.2). Since $Z' \subseteq Y$ and $Z'_m = Y_m$ for every m, Lemma IX 5.0 gives $Z' = Y$; thus Z' is a subgroup of H'. Similarly, the kernel $K = \operatorname{Ker}(\nu, \nu\circ\sigma)$ is a closed subscheme of P' such that $K_m = P'_m$ for every m (because $Z'_m = {}_nH_m$ is commutative). Hence $K = P'$, that is, Z' is a commutative subgroup of H'.

On the other hand, every reduction $Z'_m = {}_nH_m$ is flat over S_m. The "local criterion for flatness" (cf. EGA 0_{III}, 10.2.2, or [BAC], III § 5, Example 1 and Th. 1) therefore shows that Z' is flat over S'. Moreover, $Z'_0 = {}_nH_0$ is a finite group of multiplicative type over S_0, and hence isotrivial by 1.4. It follows from 3.8 b) that Z' is of multiplicative type (and isotrivial) over S'. Since $S' \to S$ is faithfully flat and quasi-compact, the multiplication $Z \times_S Z \to H$ factors through Z and makes Z a subgroup of H, finite over S and of multiplicative type (because Z' is).

Finally, by the observations made above concerning $(*)$, Z' is the "finite part" of ${}_nH'$, and therefore $u' : G' \to {}_nH'$ takes its values in Z'. Since Z, as well as $G = {}_nG$, is finite over S and of multiplicative type, Lemma 4.1 shows that u' comes from a $u : G \to Z$, and hence $u''_1 = u''_2$. This completes the proof of 4.3.

Lemma 4.4.0.[26] *Let A be a ring, $S = \operatorname{Spec}(A)$, and let H be an S-group scheme of multiplicative type and of finite type, quasi-isotrivial (respectively isotrivial). Let (A_i) be a filtered family of subrings of A whose inductive limit is A, and put $S_i = \operatorname{Spec}(A_i)$.*

Then there are an index i and an S_i-group scheme H_i, of multiplicative type and of finite type, quasi-isotrivial (respectively isotrivial), such that $H = H_i \times_{S_i} S$.

Theorem 4.4. *Let S be a prescheme, let H be an affine S-prescheme in groups of finite presentation over S, and let $s \in S$. Suppose:*

α) H is flat over S at the points of H_s;

β) H_s is of multiplicative type.

Then there exist an étale morphism $S' \to S$, a point $s' \in S'$ above s such that $\kappa(s')/\kappa(s)$ is trivial, and an open and closed subgroup G' of $H' = H \times_S S'$, of multiplicative type and finite type, and isotrivial, such that $G'_{s'} = H'_{s'}$.

[25] *Editors' note.* The original has been expanded in what follows, on the basis of indications by M. Raynaud.

[26] *Editors' note.* This lemma, used several times below, has been added.

[27] **a)** Write provisionally $T = \mathrm{Spec}(\mathcal{O}_{S,s})$, and let us show, by "descent of the conclusions", that one may reduce to the case $S = T$. Suppose indeed that we have found an étale morphism $g : T' \to T$, a point $s' \in T'$, and a subgroup G' of $H' = H_{T'}$ satisfying the conditions of the statement. Replacing T' by an affine open neighborhood of s', we may assume that T' is of finite presentation over T.

Since G' is isotrivial, there is a finite étale surjective morphism $f : \widetilde{T} \to T'$ such that $\widetilde{G} = G' \times_{T'} \widetilde{T}$ is isomorphic to $D_{\widetilde{T}}(M)$, where M is a finitely generated abelian group. Since T', H, $\widetilde{T}$, and G' are of finite presentation over $T = \mathrm{Spec}(\mathcal{O}_{S,s})$, and $\mathcal{O}_{S,s}$ is the inductive limit of the subalgebras $A_i = \mathcal{O}_S(S_i)$, where S_i runs through the affine open neighborhoods of s in S, EGA IV$_3$, 8.8.2 (and Exposé VI$_\mathrm{B}$, 10.2 and 10.3) gives an index i, S_i-preschemes $S'_i, \widetilde{T}_i$ of finite presentation (respectively an S_i-prescheme in groups G'_i of finite presentation), and morphisms

$$g_i : S'_i \to S_i, \qquad f_i : \widetilde{T}_i \to S'_i, \qquad u_i : G'_i \to H \times_S S'_i,$$

from which g, f, respectively $u : G' \to H'$, are obtained by base change. Taking i large enough, g_i is étale, f_i is finite étale surjective, and u_i is an open and closed immersion (cf. EGA IV, 8.10.5 and 17.7.8).

Now $\widetilde{G}$ comes from both $\widetilde{G}_i = G'_i \times_{S'_i} \widetilde{T}_i$ and $D_{\widetilde{T}_i}(M)$. By EGA IV$_3$, 8.8.2 (i) (and VI$_\mathrm{B}$, 10.2), there is an index $j \geq i$ such that $\widetilde{G}_j \simeq D_{\widetilde{T}_j}(M)$, so G'_j is an isotrivial group of multiplicative type. Let s'_j be the image of s' in S'_j. Then the étale morphism $S'_j \to S_j \hookrightarrow S$, the point s'_j, and the open and closed subgroup G'_j of $H \times_S S'_j$ satisfy the conditions of the statement. Thus we may assume that $S = \mathrm{Spec}(A)$ is local, with closed point s.

b) The ring A is the inductive limit of local rings A_i which are localizations of finitely generated $\mathbb{Z}$-algebras; put $S_i = \mathrm{Spec}(A_i)$. Let us show that hypotheses α and β "descend" to some A_i. Since $H \to S$ is of finite presentation, the set of $x \in H$ at which H is flat over S is an open set W. It contains H_s by hypothesis, and hence contains a quasi-compact open set V containing H_s (since the affine scheme H_s can be covered by finitely many affine open subsets contained in W). The open immersion $\tau : V \hookrightarrow H$ is of finite presentation, so $V \to S$ is also of finite presentation.

By EGA IV$_3$, 8.8.2 (and Exposé VI$_\mathrm{B}$, 10.2 and 10.3), there are an index i, an S_i-prescheme V_i of finite presentation (respectively an S_i-prescheme in groups H_i of finite presentation), and an S_i-morphism $\tau_i : V_i \to H_i$, from which V, H, τ are obtained by base change $S \to S_i$. Taking i large enough, τ_i is an open immersion and V_i is flat over S_i, by EGA IV$_3$, 8.10.5 and 11.2.6. Let s_i be the image of s in S_i. Since the open immersion $(V_i)_{s_i} \hookrightarrow (H_i)_{s_i}$, after base change $\kappa(s_i) \to \kappa(s)$, gives the equality $V_s = H_s$, one already has $(V_i)_{s_i} = (H_i)_{s_i}$. Thus H_i is flat over S_i at the points of $(H_i)_{s_i}$. Finally, $(H_i)_{s_i}$ is of multiplicative type, because

$$H_s = (H_i)_{s_i} \otimes_{\kappa(s_i)} \kappa(s)$$

is. Hence (S_i, H_i, s_i) satisfies the hypotheses of 4.4, and if the desired assertion holds for this triple, it also holds for (S, H, s) by base change. We are reduced to the case in which A is local and noetherian. We now distinguish two cases.

1° *A is local noetherian and henselian.* Let $\widehat{A}$ be its completion, $\widehat{S} = \mathrm{Spec}(\widehat{A})$, and $\widehat{H} = H \times_S \widehat{S}$. Applying Theorem 3.7, we find an isotrivial $\widehat{S}$-group $\widehat{G}$ of multiplicative type and of finite type, together with a homomorphism

$$\widehat{u} : \widehat{G} \longrightarrow \widehat{H}$$

[27] *Editors' note.* The reductions that follow have been spelled out (the original said: "One reduces at once to the case where S is the spectrum of a noetherian local ring A, and where s is the closed point of S.")

which is both an open and a closed immersion, and which induces an isomorphism

$$\widehat{u}_0 : \widehat{G}_0 \xrightarrow{\sim} \widehat{H}_0.$$

By Remark 4.0.1, base change along $\widehat{S} \to S$ induces an equivalence between the categories of isotrivial groups of multiplicative type over S and over $\widehat{S}$. In particular, $\widehat{G}$ comes from an isotrivial S-group G of multiplicative type and of finite type. By 4.3, $\widehat{u}$ comes from a homomorphism

$$u : G \longrightarrow H.$$

Moreover, u is an open and closed immersion and induces an isomorphism $u_0 : G_0 \to H_0$, since this is true after the faithfully flat quasi-compact base change $\widehat{S} \to S$. This proves 4.4 in this case (taking $S' = S$ and $s' = s$ in its conclusion).

2° *A is local noetherian.* The reduction to case 1° is immediate, by applying that case to the henselization A^h of A. More precisely, one sees easily (using SGA 1, I § 5 *) that the local rings $\mathcal{O}_{S',s'}$ of étale S-preschemes S', equipped with a point s' above s for which $\kappa(s')/\kappa(s)$ is trivial, form a filtered inductive system whose inductive limit is a noetherian henselian local ring A^h (called the henselization of A); for details of this construction, due to Nagata in the normal case, see SGA 4, VIII § 4. The standard arguments of EGA IV$_3$, § 8 then allow one, as in part a), to deduce from the known result over the inductive limit A^h of the $\mathcal{O}_{S',s'}$ an analogous result over one of the pairs (S', s'). This proves 4.4.

Corollary 4.5. *Let S be a prescheme and H an S-prescheme in groups of multiplicative type and of finite type. Then H is quasi-isotrivial; that is, H is split by an étale surjective morphism $S' \to S$.*

Indeed, let $s \in S$. By 4.4, there are an étale morphism $S' \to S$, a point $s' \in S'$ above s with $\kappa(s) = \kappa(s')$, and an isotrivial subgroup G' of H', of multiplicative type and of finite type, such that $G'_{s'} = H'_{s'}$. Since G' and H' are of multiplicative type and finite type, IX 2.9 gives an open neighborhood U' of s' such that $G'|_{U'} = H'|_{U'}$.

[28] Suppose in addition that S is henselian local, with closed point s. By EGA IV$_4$, 18.5.11 b), there is a section σ of $S' \to S$ such that $\sigma(s) = s'$. (N.B. One can see directly that $B = \mathcal{O}_{S',s'}$ equals $A = \mathcal{O}_{S,s}$ as follows. By 4.3.0 (i), $B \simeq A/I$. Since B is a flat A-algebra of finite presentation, I is a finitely generated ideal (cf. EGA IV$_1$, 1.4.7), and $I = I\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal of A; hence $I = 0$.) Thus H is already isotrivial. We obtain:

Corollary 4.6. *Let A be a henselian local ring, k its residue field, and π the topological Galois group of an algebraic closure of k.*

(i) *Every group of multiplicative type and of finite type over $S = \mathrm{Spec}(A)$ is isotrivial.*

(ii) *The category of these groups over S is equivalent, through the functor $H \mapsto H \otimes_A k$, to the analogous category over $S_0 = \mathrm{Spec}(k)$. It is therefore anti-equivalent, by 1.4, to the category of Galois modules under π which are finitely generated as $\mathbb{Z}$-modules.*

Corollary 4.7. *Under the hypotheses of 4.4, suppose in addition that one of the following conditions holds:*

a) for every generization t of s, H_t is of multiplicative type and has the same type as H_s;

b) H is commutative and the ${}_nH$, $n > 0$, are finite over S in a neighborhood of s;

c) for every generization t of s, the fiber H_t is connected.

Then there is an open neighborhood U of s such that $H|_U$ is of multiplicative type.

*Or EGA IV$_4$, § 18.6.

[28] *Editors' note.* The original has been expanded in what precedes and what follows.

Since an étale morphism is open, it is enough to show, with the notation of the conclusion of 4.4, that there is an open neighborhood U' of s' such that $G'|_{U'} = H'|_{U'}$. Put $S'' = \mathrm{Spec}(\mathcal{O}_{S',s'})$. Since G' and H' are of finite presentation over S', EGA IV$_3$, 8.8.2 reduces the matter to showing that $G'' = H''$. We may therefore assume that $S = S''$; then conditions a), b), c) become those of the following lemma, whose proof is the same as that of 3.8.

Lemma 4.7.1. *Let S be a local scheme with closed point s, let H be an S-group scheme of finite type, and let G be an open and closed subgroup of H, of multiplicative type, such that $G_s = H_s$. Suppose in addition that one of the following conditions holds:*

a) the fibers of H are of multiplicative type and have the same type as H_s;

b) H is commutative and the ${}_nH$, $n > 0$, are finite over S;

c) the fibers of H are connected.

Then $G = H$.

Corollary 4.8. *Let S be a prescheme and let H be an affine S-prescheme in groups, flat and of finite presentation over S, with fibers of multiplicative type.*

For H to be of multiplicative type, it is necessary and sufficient that it satisfy one of the following two conditions (which are equivalent under the preceding hypotheses):

a) the type of H_s (cf. IX 1.4) is a locally constant function of $s \in S$;

b) H is commutative and the ${}_nH$, $n > 0$, are finite over S.

Moreover, a) and b) are implied by:

c) the fibers of H are connected.

In particular:

Corollary 4.9. *Let S be a prescheme and H an S-prescheme in groups, flat and of finite presentation over S. Suppose in addition that H is affine over S and has connected fibers.*[29] *If $s \in S$ is such that H_s is a torus, there is an open neighborhood U of s such that $H|_U$ is a torus.*

In particular, if all the fibers of H are tori, H is a torus.

5. Scheme of homomorphisms from one group of multiplicative type to another. Twisted constant groups and groups of multiplicative type

Definition 5.1. *a) Let S be a prescheme and R a prescheme in groups over S. We say that R is a* twisted constant group *over S if it is locally isomorphic, for the faithfully flat quasi-compact topology, to a constant group scheme, that is, one of the form M_S for an ordinary group M.*

b) A twisted constant group R over S is said to be quasi-isotrivial*, respectively* isotrivial*, respectively* locally isotrivial*, respectively* locally trivial*, respectively* trivial*, if in the preceding definition one can replace the faithfully flat quasi-compact topology by the étale topology, respectively the global finite étale topology, respectively the finite étale topology, respectively the Zariski topology, respectively the coarsest (or "chaotic") topology; cf. IX 1.1 and 1.2, and IV 6.6.*

Thus, to say that R is quasi-isotrivial (respectively isotrivial) means that there is an étale surjective (respectively also finite) morphism $S' \to S$ such that $R' = R \times_S S'$ is a constant group over S'; to say that R is trivial means that R is a constant group.

[29] *Editors' note.* This result is generalized in 8.1: it is enough to suppose that the fibers of H are affine and connected.

c) As in VIII 1.4, one defines the type *of a twisted constant group R over S at a point $s \in S$. It is an isomorphism class of ordinary groups and, as s varies, is a locally constant function of s, hence a constant function if S is connected. We also say that R is "of type M" if all the fibers of R are of type M.*

One should beware that R is quasi-compact over S only if it is finite over S, that is, only if its type at every $s \in S$ is a finite group.[30]

d) The case of greatest interest to us is that in which R is commutative. We then say that R is finitely generated *if its type at every point $s \in S$ is given by a finitely generated $\mathbb{Z}$-module. This notion must not be confused with the scheme- theoretic notion "R of finite type over S" (cf. above).*

Remark 5.2. We shall also have to consider S-preschemes X which are locally isomorphic, for the faithfully flat quasi-compact topology, to constant preschemes, independently of any group structure. We then call X a *twisted constant bundle* over S, and extend to these preschemes the terminology introduced in 5.1. Of course, when X is endowed with an S-group structure, the meaning of the expressions "twisted constant", "isotrivial", and so on changes according as one does or does not take the group structure over S into account. The same applies when X carries any other kind of algebraic structure (for example, that of a Galois principal bundle, to be considered in the next section).

Proposition 5.3. *Let R be a commutative twisted constant group over S.*

(i) *The functor $H = D_S(R)$ (cf. VIII 1) is representable and is a group of multiplicative type over S.*

(ii) *For every $s \in S$, the type of R at s equals the type of H at s.*

(iii) *The group R is quasi-isotrivial (respectively isotrivial, respectively trivial, respectively locally isotrivial, respectively locally trivial) if and only if H is so.*

Remark 5.3.1.[31] One may be concerned that in 5.3 the word "type" is being used in two different senses, according as it refers to R or to H. Fortunately, when an S-prescheme in groups G is both a twisted constant group and a group of multiplicative type, its type in either sense is the same, thanks to the fact that an ordinary finite commutative group is isomorphic to its dual.

Proof. Covering families for the faithfully flat quasi-compact topology are families of effective descent for the fibered category of affine preschemes in groups over variable base preschemes (SGA 1, VIII 2.1). Consequently H is representable (and affine over S), since this is true "locally"[32] (when R is constant, H is a diagonalizable group).

That H is of multiplicative type is then immediate from the definition, as is the equality of the types of R and H at $s \in S$. Finally, since

$$H_{S'} = D_{S'}(R_{S'}),$$

the last assertion reduces to the "trivial" case, namely to checking that R is trivial if and only if H is trivial; this follows at once from the biduality theorem VIII 1.2. □

To complete the description of the correspondence between twisted constant groups and groups of multiplicative type, one must start with a group of multiplicative type H and study $R = D_S(H)$. If R is representable, it is plainly a twisted constant group and $H \simeq D_S(R)$. In other words:

[30] *Editors' note.* See Lemma 5.12 below.

[31] *Editors' note.* This remark, which appeared after the proof in the original, has been moved here.

[32] *Editors' note.* Cf. VIII § 1.7.

Scholium 5.4.0. *The functor $R \mapsto D_S(R)$ is an anti-equivalence*[33] *between the category of twisted constant groups over S and the category of groups of multiplicative type H over S for which $D_S(H)$ is representable.*

I do not know whether this condition on H is always satisfied. We shall see, however, that it is satisfied when H is quasi-isotrivial, and in particular when H is of finite type.

Lemma 5.4. *Let $S' \to S$ be a faithfully flat morphism locally of finite presentation, and let X' be an S'-prescheme separated, locally of finite presentation, and locally quasi-finite over S'.*

Then every descent datum on X' relative to $S' \to S$ is effective: there are an S-prescheme X and an S'-isomorphism $X \times_S S' \simeq X'$ compatible with the descent datum.

We have already observed that the hypothesis on $S' \to S$ implies that it is a covering morphism for the faithfully flat quasi-compact topology (IV 6.3.1 (iv)), and hence *a fortiori* a morphism of effective descent.

When $X' \to S'$ is quasi-compact, and hence of finite presentation and quasi-finite, it is a quasi-affine morphism (cf. SGA 1, VIII 6.2[34]), and effectivity follows in this case from SGA 1, VIII 7.9. In the general case, one reduces at once to the case where S and S' are affine. Cover X' by affine open subsets U'_i, and let V'_i be the saturation of U'_i under the equivalence relation on X' defined by the descent datum:

$$V'_i = q_2(q_1^{-1}(U'_i)).$$

Here q_1, q_2 are the two projections to X' of

$$X''_1 = X' \times_{S'} (S'', p_1) \quad \text{and} \quad X''_2 = X' \times_{S'} (S'', p_2),$$

respectively: $q_1 = \mathrm{pr}_1$, and q_2 is obtained from the first projection of X''_2 by means of the given descent isomorphism $X''_1 \simeq X''_2$. Since $S' \to S$ is faithfully flat, quasi-compact, and locally of finite presentation, the same is true of

$$p_1 : S'' = S' \times_S S' \longrightarrow S',$$

and hence of q_1, q_2. These are consequently open morphisms (SGA 1, IV 6.6[35]). Therefore V'_i is an open quasi-compact part of X'. By the case already proved, the induced descent data on the V'_i are effective; hence so is the descent datum on X' (SGA 1, VIII 7.2).

Corollary 5.5. *A faithfully flat morphism of finite presentation $S' \to S$ is a morphism of effective descent for the fibered category of twisted constant groups over variable base preschemes.*

Indeed, this amounts to the effectivity assertion of 5.4 when X' is a constant S'-prescheme.

Theorem 5.6. *Let S be a prescheme, and let G, H be two quasi-isotrivial S-preschemes in groups of multiplicative type, with G*[36] *of finite type.*

Then $\mathrm{Hom}_{S\text{-}gr.}(H, G)$ is representable and is a quasi-isotrivial twisted constant group over S;[37] *for every $s \in S$, if the type of G at s is M and the type of H at s is N, then the type of $\mathrm{Hom}_{S\text{-}gr.}(H, G)$ is $\mathrm{Hom}_{\mathrm{gr.}}(M, N)$.*

[33] *Editors' note.* "Equivalence" has been corrected to "anti-equivalence".
[34] *Editors' note.* When S' is locally noetherian; EGA IV_3, 8.11.2 in general.
[35] *Editors' note.* When S' is locally noetherian; EGA IV_2, 2.4.6 in general.
[36] *Editors' note.* H has been corrected to G.
[37] *Editors' note.* What follows has been added.

The proof proceeds as in 4.2, using the fact that the assertion is known (VIII 1.5) when G and H are trivial. The required effectivity criterion is provided by 5.5, applied to an étale surjective morphism $S' \to S$.

Taking $G = \mathbb{G}_{m,S}$, in particular, gives:

Corollary 5.7. (i) *Let H be a quasi-isotrivial S-group of multiplicative type. Then the S-group $D_S(H)$ is representable and is a quasi-isotrivial twisted constant group over S.*

(ii) *The functors $R \mapsto D_S(R)$ and $H \mapsto D_S(H)$ are mutually quasi-inverse anti-equivalences between the category of quasi-isotrivial twisted constant S-groups and the category of quasi-isotrivial S-groups of multiplicative type.*

(iii) *These functors induce anti-equivalences between the subcategories of isotrivial, respectively locally isotrivial, respectively locally trivial, respectively trivial groups.*

The last assertion is included only for reference, since it is already contained in 5.3.

Moreover, by 4.5 every S-group of multiplicative type and of finite type is quasi-isotrivial. Theorem 5.6 therefore gives:

Corollary 5.8. *Let S be a prescheme, and let G, H be two S-preschemes in groups of multiplicative type and of finite type. Then* $\mathrm{Hom}_{S\text{-}gr.}(H, G)$ *is representable and is a finitely generated quasi-isotrivial twisted constant S-group.*

Also, in 5.3, R is finitely generated if and only if $H = D_S(R)$ is of finite type (IX 2.1 b)). By 4.5, H is then quasi-isotrivial, and hence so is R. Thus:

Corollary 5.9. *The functors of 5.7 induce mutually quasi-inverse anti-equivalences between the category of S-groups H of multiplicative type and of finite type and the category of finitely generated twisted constant S-groups R. Moreover, every such group R is quasi-isotrivial.*

Corollary 5.10. *Let H, G be two S-preschemes in groups of multiplicative type and of finite type.*

(i) $\mathrm{Isom}_{S\text{-}gr.}(H, G)$ *is representable by an open and closed subprescheme of* $\mathrm{Hom}_{S\text{-}gr.}(H, G)$*, and is a twisted constant S-prescheme.*

(ii) *In particular,* $\mathrm{Aut}_{S\text{-}gr.}(H)$ *is representable and is a twisted constant S-group, in general noncommutative.*

This follows from 5.8 and VIII 1.6.[38]

Recall 5.11.0.[39] Recall that if X is a locally noetherian prescheme, its connected components (which are always closed) are open. Indeed, let C be a connected component of X, let $x \in C$, and let U be a noetherian open neighborhood of x. The prescheme U has only finitely many connected components, so the connected component U_x of x in U is open in U, hence in X. Since $U_x \subseteq C$, this shows that C is open in X.

Proposition 5.11. *Let S be a prescheme, let R be a commutative twisted constant group over S, and let $H = D_S(R)$ be the group of multiplicative type that it defines. Consider the following conditions:*

(i) *H is isotrivial (that is, R is isotrivial);*

(ii) *R is a union of open and closed subpreschemes R_i which are quasi-compact over S (and then necessarily finite over S);*

(iii) *the connected components of R are finite over S.*

[38] *Editors' note.* This reference should be corrected by treating the functor $\mathrm{Isom}_{S\text{-gr.}}(H, G)$ in an addition 1.5.1 to VIII ...

[39] *Editors' note.* This recall, used several times below, has been added. See also EGA I, 6.1.9; note, however, that the proof in *loc. cit.* seems unnecessarily complicated.

a) Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii).[40]

b) If S is locally noetherian, then (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii).

c) Finally, if R is finitely generated (that is, if H is of finite type over S), then (i) $\Leftrightarrow$ (ii), *at least when S is quasi-compact or when its connected components are open.*

First decompose S as a prescheme sum of preschemes S_i on each of which H has constant type (cf. IX 1.4.1). We are reduced to the case where H, and therefore R, has constant type M. We shall use:

Lemma 5.12. *Let S be a prescheme and R a twisted constant prescheme over S. Every closed subprescheme Z of R which is quasi-compact over S is finite over S.*

Indeed, we reduce to the case where R is constant, hence of the form I_S for a set I. It is the filtered increasing union of the J_S, where J runs through the finite subsets of I. We may also assume that S is affine. Then Z is quasi-compact, and is therefore contained in some J_S. Since J_S is finite over S, so is Z.

Lemma 5.12 already proves the parenthetical assertion in 5.11 (ii).[41] The implication (ii) $\Rightarrow$ (iii) is then clear, since the connected components of R are closed. If S is locally noetherian, they are open and closed by 5.11.0 (R being étale and hence locally of finite type over S); this gives (iii) $\Rightarrow$ (ii) in that case.

Let us prove (i) $\Rightarrow$ (ii). Choose a finite étale surjective morphism $S' \to S$ which splits H, and hence R (cf. 5.3), so that

$$R' \simeq M_{S'} = \coprod_{m \in M} R'_m,$$

where the R'_m are disjoint open subsets of R', each S'-isomorphic to S'. Let $g : R' \to R$ be the projection. It is a finite étale surjective morphism, and hence both open and closed. The subsets $R_m = g(R'_m)$ are therefore open and closed parts of R, and are plainly quasi-compact over S, since the R'_m are.

Finally, suppose that H is of finite type over S, and let us prove (ii) $\Rightarrow$ (i). If the connected components of S are open, the matter reduces at once to the case where S is connected; thus we may assume that S is quasi-compact or connected. Since M is finitely generated, write

$$M = \mathbb{Z}^r \times N,$$

where $r \geq 0$ and N is a finite abelian group. Let $G = D_S(M)$, and consider the preschemes

$$P = \mathrm{Isom}_{S\text{-gr.}}(H, G) \subset \mathrm{Hom}_{S\text{-gr.}}(H, G) = Q$$

(cf. 5.8 and 5.10). There are isomorphisms

$$\begin{aligned} Q &\simeq \mathrm{Hom}_{S\text{-gr.}}(M_S, R) \\ &\simeq \mathrm{Hom}_{S\text{-gr.}}(\mathbb{Z}^r_S, R) \times \mathrm{Hom}_{S\text{-gr.}}(N_S, R) \\ &\simeq R^r \times E, \end{aligned}$$

where

$$E = \mathrm{Hom}_{S\text{-gr.}}(N_S, R)$$

is finite over S.[42] It follows that Q is a union of open and closed subpreschemes Q_i finite over S. Hence P is a union of the open and closed subpreschemes $P_i = P \cap Q_i$, finite over

[40] *Editors' note.* Compare the examples of 1.6 . . .

[41] *Editors' note.* The original has been expanded in what follows.

[42] *Editors' note.* Indeed, it is a twisted constant group of type $\mathrm{End}_{\mathrm{gr.}}(N)$; cf. 5.6 and 5.8.

S. Since they are étale over S, their images in S are open and closed parts S_i, and these cover S. If S is connected or quasi-compact, there is a finite set of indices i for which the S_i cover S. Let S' be the union of the corresponding P_i. Then $S' \to S$ is finite étale surjective. Putting

$$P' = P \times_S S' = \mathrm{Isom}_{S'\text{-gr.}}(H', G'),$$

we see that P' has a section over S'; hence there is an isomorphism of S'-groups

$$H' = H \times_S S' \xrightarrow{\sim} G' = G \times_S S' = D_{S'}(M).$$

Thus S' splits H, completing the proof of 5.11.[43]

Remark 5.11 bis. If H is of finite type over S and of constant type, one can also give the following isotriviality criterion, with no restriction on S: $H = D_S(R)$ is isotrivial if and only if R is the union of a sequence of open and closed parts finite over S.[44]

Lemma 5.13. *Let S be a locally noetherian connected prescheme, let P be a quasi-isotrivial twisted constant S-prescheme, and let Z be an open and closed part of P. If Z_s is finite for some $s \in S$, then Z is finite over S.*

First, do not assume S connected. Let U be the set of $s \in S$ for which Z_s is finite. We shall prove that U is both open and closed, and that $Z|_U$ is finite over U. This statement is essentially equivalent to 5.13, but has the advantage of being local in S for the étale topology, say. We are therefore reduced to the case where P is constant, of the form I_S for a set I. (N.B. The locally noetherian hypothesis is preserved by étale base change; this is where the quasi-isotriviality of P over S is used.)

We may also suppose that S is connected, since its connected components are open (S being locally noetherian). Then necessarily $Z = J_S$ for a subset $J \subseteq I$, and consequently $U = \varnothing$ or $U = S$ according as J is infinite or finite. This proves the assertion.

Recalls 5.14.0.[45] Let S be a locally noetherian prescheme and let $\widetilde{S}$ be the normalization of S_{red} (cf. EGA II, 6.3.8). Recall that S is said to be *geometrically unibranch* (cf. EGA 0_{IV}, § 23.2 and IV_2, § 6.15) if the canonical morphism $\widetilde{S} \to S$ is radicial, and hence a universal homeomorphism. In particular, the connected components of S are irreducible.

Suppose now that S is connected, and hence irreducible. Let η be its generic point and let $f : P \to S$ be a flat, locally quasi-finite morphism. Let P_i be the irreducible components of P, with generic points ξ_i. Since P is flat over S, every ξ_i lies above η (cf. EGA IV_2, 2.3.4), and

$$(P_i)_\eta = P_i \cap P_\eta$$

is the closure of ξ_i in P_η. Since the fiber P_η is discrete, one has $(P_i)_\eta = \{\xi_i\}$. This applies in particular when f is étale. In that case P is also locally noetherian and geometrically unibranch (cf. EGA IV_4, 17.5.7), so its irreducible components are its connected components and are open (and closed).

[43] *Editors' note.* This is modulo verifying (ii) $\Rightarrow$ (i) when S is locally noetherian and R is not finitely generated ...

[44] *Editors' note.* To spell this out: since M is a finitely generated $\mathbb{Z}$-module, it is countable, and the proof of 5.11 (i) $\Rightarrow$ (ii) shows that R is the union of a countable family of open and closed parts finite over S. The converse would have to be checked ...

[45] *Editors' note.* The original treated in 5.14 the case where S is locally noetherian and normal, and observed in Remark 5.15 that the argument applies more generally when S is merely geometrically unibranch rather than normal. The statement of 5.14 (and also 5.16) has accordingly been modified, and these "recalls", drawn from EGA IV_4, 18.10.6 and 18.10.7, have been added; they show that the proof of 5.14 applies verbatim to the geometrically unibranch case.

Corollary 5.14. *Let S be a locally noetherian geometrically unibranch prescheme, and let P be a quasi-isotrivial twisted constant S-prescheme. Then the connected components of P are finite over S.*

We may plainly suppose that S is connected and hence irreducible, and write η for its generic point. By the preceding recalls, every connected component P_i of P is open and closed and meets the fiber P_η in a single point. Lemma 5.13 therefore applies and shows that every P_i is finite over S.

Theorem 5.16.[46] *Let S be a locally noetherian geometrically unibranch prescheme. Then every S-group of multiplicative type and of finite type is isotrivial.*

Indeed, we may suppose S connected, and hence H of constant type M. It is enough to apply 5.14 to

$$P = \mathrm{Isom}_{S\text{-gr.}}(H, G), \qquad G = D_S(M),$$

and then argue as in the proof of 5.11 (ii) $\Rightarrow$ (i). Alternatively, one may apply 5.14 to $P = R = D_S(H)$ (cf. 5.9), and then use 5.11.

6. Infinite Galois principal bundles and the enlarged fundamental group

(The results of the present section and of the next will not be used again in the remainder of this Seminar.)

Let S be a prescheme. We propose to determine the principal homogeneous bundles P over S whose structural group is of the form G_S, the constant S-group defined by an ordinary group G, not necessarily finite. We shall also call them *Galois principal bundles over S with group G*. We use "principal bundle" in the sense of the faithfully flat quasi-compact topology (cf. Exposé IV, Def. 5.1.5). For such a P, however, the structural morphism $P \to S$ is necessarily étale and surjective, hence a covering for the étale topology; consequently P is also locally trivial for the étale topology (cf. IV, Prop. 5.1.6).[47]

We assume that S is a sum of connected preschemes, that is, that its connected components are open; this reduces us at once to the case in which S is connected. Choose a "geometric point" ξ of S, that is, an S-scheme ξ which is the spectrum of an algebraically closed field $\Omega = \kappa(\xi)$. For every Galois principal bundle P over S with group G, P_ξ is a Galois principal bundle over the algebraically closed field $\kappa(\xi)$, and is therefore trivial. We refine the initial problem by asking for the category of Galois principal bundles over S *pointed* above ξ, that is, endowed with an S-morphism $\xi \to P$, or equivalently with a trivialization of P_ξ. For fixed G, let $\bar\pi^1(S,\xi;G)$ denote the set of classes of such bundles, up to an isomorphism respecting the base point. Then the set $\bar\pi^1(S;G)$ of isomorphism classes of Galois principal bundles over S with group G, without a specified base point, is the set of orbits of G in $\bar\pi^1(S,\xi;G)$. Here G acts naturally by translating the marked point of a pointed Galois principal bundle:

$$\bar\pi^1(S;G) = \bar\pi^1(S,\xi;G)/G.$$

Let $S' \to S$ be any morphism which is a morphism of universal effective descent for the fibered category of twisted constant preschemes over a variable base (for example, $S' \to S$

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[46] *Editors' note.* Remark 5.15, rendered obsolete by the addition of 5.14.0 (cf. the preceding editors' note), has been deleted, and in 5.16 "normal" has been replaced by "geometrically unibranch".

[47] *Editors' note.* Notice that P is assumed to be a prescheme. In the *a priori* more general case of an fpqc sheaf P which is a G_S-torsor, is P necessarily representable?

faithfully flat and locally of finite presentation, cf. 5.4; for other examples, cf. SGA 1 IX). We propose to determine the subsets

$$\bar{\pi}^1(S'/S, \xi; G) \quad \text{and} \quad \bar{\pi}^1(S'/S; G)$$

of the preceding sets, formed by the Galois principal bundles over S which become trivial over S', or, as one says, are "split" by S'. In fact, we shall determine the category of Galois principal bundles P over S split by S'. Of course,

$$\bar{\pi}^1(S, \xi; G) = \varinjlim_{S'} \bar{\pi}^1(S'/S, \xi; G),$$

where S' runs through a cofinal system among the S'/S which cover for the étale topology (for example, if S is quasi-compact, among the quasi-compact S' over S whose structural morphism is étale and surjective). Likewise, the category of Galois principal bundles over S is the inductive limit of the subcategories defined by the S', formed by the bundles split over S'.

By the assumption on S'/S, the category of Galois principal bundles over S split by S' is equivalent to the category of trivial Galois principal bundles over S', hence of the form $G_{S'}$, with G acting by right translations, equipped with descent data relative to $S' \to S$. In terms of the corresponding trivial bundle P' over S' and its descent datum, a base point on a Galois principal bundle P over S split by S' amounts to a trivialization of $P' \times_{S'} S'_\xi$ compatible with the induced descent datum relative to $S'_\xi \to \xi$, where $S'_\xi = S' \times_S \xi$; equivalently, it is a section σ of P'_ξ over S'_ξ compatible with the descent datum.

It is therefore useful, for an arbitrary S-prescheme S', without now assuming that $S' \to S$ is a morphism of universal effective descent for the fibered category of twisted constant bundles, to define $\bar{\pi}^1(S'/S; G)$ and $\bar{\pi}^1(S'/S, \xi; G)$ as the sets of isomorphism classes of the structures with descent data just specified. These functors in G then have a very simple simplicial description in terms of the fibered square and cube S'' and S''' of S' over S; we shall sketch it below (cf. 6.3).

The important conclusion is the following.

Proposition 6.1. *Suppose that the connected components of S' and S'' are open, and, for example, that the quotient of $\pi_0(S')$ by the equivalence relation induced by the two projections $\pi_0(S'') \to \pi_0(S')$ is a single point.*[48]

(i) *The functor*

$$G \longmapsto \bar{\pi}^1(S'/S, \xi; G)$$

from the category of groups to the category of sets is representable by a group, denoted $\pi_1(S'/S, \xi)$ and called the fundamental group of S at ξ relative to $S' \to S$. *Thus there is a functorial bijection*

$$\bar{\pi}^1(S'/S, \xi; G) \simeq \mathrm{Hom}_{\mathrm{gr.}}(\pi_1(S'/S, \xi), G).$$

(ii) *This group has a set of generators in bijection with $\pi_0(S'')$, and is described in terms of those generators by relations in bijection with the elements of $\pi_0(S''')$.*[49] *In particular,*

[48] *Editors' note.* The hypothesis that the connected components of S' be open, as well as the second hypothesis, has been added. The latter says that the simplicial set $K_\bullet = \pi_0(S_\bullet)$ defined in 6.3 is connected. In fact a weaker hypothesis suffices: the cone $\widetilde{K}_\bullet$ of a certain morphism of simplicial sets $k_\bullet \to K_\bullet$ need only be connected (cf. *loc. cit.*).

[49] *Editors' note.* The superfluous hypothesis that the connected components of S''' be open has been deleted. On the other hand, the later description (cf. ...) gives the natural set of generators $\pi_0(S'') \amalg \pi_0(S'_\xi)$; the relations between these generators arising from the 2-cells then reduce it to $\pi_0(S'')$. See, for example, [Kan58], § 19 for a finer description.

$\pi_1(S'/S,\xi)$ is finitely generated (respectively finitely presented) if $\pi_0(S'')$ is finite (respectively if both $\pi_0(S'')$ and $\pi_0(S''')$ are finite).

(iii) *The category of Galois principal bundles over S split by S', with base point above ξ, is equivalent to the category of ordinary groups G equipped with a homomorphism $\pi_1(S'/S,\xi) \to G$.*

The proof is given below; cf. ...

6.2. When S is a connected locally noetherian prescheme, every étale prescheme S' over S is locally noetherian and therefore has open connected components. It follows from the preceding discussion[50] that the functor

$$G \longmapsto \bar{\pi}^1(S,\xi;G)$$

from ordinary groups to sets is *strictly pro-representable* (cf. *Séminaire Bourbaki*, February 1960, No. 195, §§ A.2 and A.3). Thus there is a projective system

$$\Pi = \Pi_1(S;\xi) = (\pi_i)_{i\in I}$$

of ordinary groups over an increasing filtered index set I, which is "strict", meaning that its transition morphisms $\pi_j \to \pi_i$ are surjective, together with an isomorphism of functors in G

$$\bar{\pi}^1(S,\xi;G) \simeq \varinjlim_i \mathrm{Hom}_{\text{gr.}}(\pi_i, G).$$

The second member is also denoted $\mathrm{Hom}_{\text{pro-gr.}}(\Pi, G)$ (cf. *loc. cit.*).

If the projective limit

$$\pi = \varprojlim_i \pi_i$$

is "large enough", more precisely if the canonical homomorphisms $\pi \to \pi_i$ are surjective, it is natural to equip π with the projective-limit topology of the discrete topologies on the π_i. The preceding isomorphism can then also be written

$$\bar{\pi}^1(S,\xi;G) \simeq \mathrm{Hom}_{\text{gr. top.}}(\pi, G),$$

where the right-hand side is the set of homomorphisms of topological groups, G being given the discrete topology.

The hypothesis just formulated on the projective system Π holds, as is well known, when the π_i are finite groups (cf. [BEns], III § 7.4, Th. 1). This last condition plainly also means that every Galois principal bundle over S is isotrivial, that is, split by a finite étale surjective morphism. This is the case when S is geometrically unibranch (for example normal), as follows at once from 5.14.[51] When the π_i are finite, π also coincides with the fundamental group $\pi_1(S,\xi)$ introduced in SGA 1, V.

Thus, in the favorable case in which the maps $\pi \to \pi_i$ are surjective, one might call π the *enlarged fundamental group of S at ξ*. Outside this favorable case, π itself is of little interest, and the role of the usual fundamental group is played by the projective system Π itself, which we call the *enlarged fundamental pro-group of S at ξ*. (Any better terminological suggestion than "enlarged" is welcome![52]) Knowledge of this pro-group is more precise than knowledge of the usual fundamental group $\pi_1(S,\xi)$ of SGA 1, V; more precisely, the latter is the projective limit of the projective system formed by the finite quotients of the π_i.

[50] *Editors' note.* This deduction could be spelled out: I is in bijection with a cofinal set of morphisms $S' \to S$ covering for the étale topology ...

[51] *Editors' note.* 5.13 has been corrected to 5.14.

[52] *Editors' note.* Some authors speak of the "true" fundamental group.

6.3. Let us indicate briefly the "calculation" of $\pi_1(S'/S, \xi)$. Let S_i be the $(i+1)$-st fibered power of S' over S, so that $S_0 = S'$, $S_1 = S''$, and so on. The evident simplicial operations between the S_i make $(S_i)_{i\in\mathbb{N}}$ a simplicial object of $(\mathrm{Sch})/S$.

Applying to this simplicial object the functor "set of connected components"

$$\pi_0 : (\mathrm{Sch})/S \longrightarrow (\mathrm{Ens}),$$

we obtain a simplicial set $K_\bullet = (K_i)_{i\in\mathbb{N}}$, with

$$K_i = \pi_0(S_i).$$

Likewise, the $S_{i,\xi}$, the $(i+1)$-st fibered powers of S'_ξ over ξ, form a simplicial object of $(\mathrm{Sch})/\xi$, hence of $(\mathrm{Sch})/S$, together with a natural morphism of simplicial objects into $(S_i)_{i\in\mathbb{N}}$. Thus we obtain a simplicial set $k_\bullet$, with

$$k_i = \pi_0(S_{i,\xi}),$$

and a canonical morphism

$$k_\bullet \longrightarrow K_\bullet.$$

Taking the cone of this morphism (cf. 9.5.1), we form a new simplicial set

$$\widetilde{K}_\bullet = \mathrm{Cone}(k_\bullet \longrightarrow K_\bullet).$$

[53] In this way we obtain a "pointed simplicial set" $\widetilde{K}_\bullet$, that is, a simplicial set equipped with a morphism

$$\widetilde{\xi} : e_\bullet \longrightarrow \widetilde{K}_\bullet,$$

where $e_\bullet$ is the final simplicial set. We may form its familiar combinatorial invariants

$$\pi_0(\widetilde{K}_\bullet, \widetilde{\xi}) \quad \text{and} \quad \pi_1(\widetilde{K}_\bullet, \widetilde{\xi}),$$

whose constructions involve only the components of degree at most 1 and at most 2, respectively. These invariants are defined without any restriction on S or S'. When the connected components of S_0 and S_1 are open and $\widetilde{K}_\bullet$ is connected,[54] one checks without difficulty that $\pi_1(\widetilde{K}_\bullet, \widetilde{\xi})$ represents the functor $G \mapsto \bar{\pi}^1(S'/S, \xi; G)$; that is,

$$\pi_1(S'/S; \xi) \simeq \pi_1(\widetilde{K}_\bullet, \widetilde{\xi}).$$

We also note that if $S' \to S$ is "universally submersive" (cf. SGA 1, IV 2.1) and the connected components of S' are open, then the simplicial set $K_\bullet$, and hence also $\widetilde{K}_\bullet$, is connected.[55]

Examples 6.4. It remains to give examples of enlarged fundamental groups. Return to the examples of 1.6. Let k be an algebraically closed field; let C_1 be a complete rational curve over k with exactly one singular point, that point being an ordinary double point; and let C_2 be a curve which is the union of two irreducible components, each isomorphic to

[53] *Editors' note.* The original, which takes $\widetilde{K}_\bullet = K_\bullet/k_\bullet$, has been corrected: on the one hand $k_\bullet$ is already contractible (cf. 9.7.1), and on the other hand $k_\bullet \to K_\bullet$ is not in general injective. It is an epimorphism if S'/S is finite étale (cf. 9.7.3).

[54] *Editors' note.* The hypothesis that $\widetilde{K}_\bullet$ be connected has been added; for the proof, see the addendum below (Section 9).

[55] *Editors' note.* The sequel has been modified to take account of the correction made above; cf. Editors' note (53). (The original read: "$\pi_0(\widetilde{K}_\bullet, \widetilde{\xi})$ is also canonically isomorphic to the set $\pi_0(S, \xi)$ of connected components of S, pointed by the connected component of ξ in S.")

$\mathbb{P}^1_k$, meeting in exactly two points, which are ordinary double points of C_2. In either case, the enlarged fundamental group of the curve is isomorphic to $\mathbb{Z}$.[56]

More generally, one should take up, simplifying and correcting them, the results of SGA 1, IX 5 in the setting of the enlarged fundamental group. The examples of *loc. cit.*, 5.5 would give as many examples of enlarged fundamental pro-groups which are not profinite. Thus, if in the first example the ordinary double point were replaced by a double point with n distinct branches,[57] the enlarged fundamental group would be the free (discrete!) group on $n-1$ generators.

7. Classification of twisted constant preschemes and of groups of multiplicative type of finite type in terms of the enlarged fundamental group

7.0.

Let S be a prescheme, which we continue to assume locally noetherian, in order to ensure that S, and certain preschemes over S that we shall consider (notably those étale over S, and more generally those locally of finite type over S), are locally connected.

Proposition 7.0.1.— *Every twisted constant prescheme X over S that is locally trivial for the* (fppf) *topology (that is, that is split by a faithfully flat morphism $S' \to S$ locally of finite presentation) is quasi-isotrivial (that is, one may even choose $S' \to S$ étale and surjective).*

Indeed, we may suppose that S is connected, and hence that X is of type I, where I is a fixed set. Thus

$$X' = X \times_S S'$$

is isomorphic to $I_{S'}$, so $I_{S'}$ is endowed with descent data relative to $S' \to S$; in other words, there is an isomorphism

$$I_{S''} \xrightarrow{\sim} I_{S''}$$

satisfying the usual transitivity condition. Now $S'' = S' \times_S S'$ is locally noetherian and hence locally connected. It follows that the automorphisms of $I_{S''}$ correspond to the sections of $G_{S''}$, where $G = \mathrm{Aut}(I)$ is the group of permutations of I.

In this way one obtains descent data on $G_{S'}$ (regarded as a trivial Galois principal bundle) relative to $S' \to S$. By 5.4 these descent data are effective, and therefore give a Galois principal bundle P over S with group G. By construction, P represents the functor $\mathrm{Isom}_S(I_S, X)$ in the category of preschemes over S that are locally noetherian. Consequently, the étale surjective base change $P \to S$ splits X, so X is indeed quasi-isotrivial.

Remark 7.0.2.— One should bear in mind that, even when S is the spectrum of a field, it is not true in general that every twisted constant bundle over S is quasi-isotrivial. For example, one may take X to be the disjoint union of a sequence of schemes of the form $\mathrm{Spec}(k_i)$, where the k_i are separable extensions of k of strictly increasing degrees.

[56] *Editors' note.* This could be spelled out. First, by 5.14, the enlarged fundamental group of the projective line $\mathbb{P}^1_k$ is zero; that is, $\mathbb{P}^1_k$ is "truly" simply connected. One would then have to extend Corollary 5.4 of SGA 1, IX, and the discussion following it, to the enlarged fundamental group and to the category of principal bundles not necessarily finite. Let Γ_1, Γ_2 be two copies of $\mathbb{P}^1_k$, and let a_i, b_i be two distinct points of Γ_i. Let C''_2 be the disjoint union of Γ_1 and Γ_2, and let C'_2 be the curve obtained by identifying a_1 with a_2; then C_2 is obtained from C'_2 by also identifying b_1 with b_2. The discussion following *loc. cit.*, extended to the enlarged case, shows that the enlarged fundamental group of C'_2, respectively C_2, is zero, respectively $\mathbb{Z}$.

[57] *Editors' note.* This means an n-fold point with n distinct tangents, for example the curve $X^n - Y^n = X^{n+1}$.

The proof just given shows at the same time that the classification of quasi-isotrivial twisted constant bundles X over S of type I is equivalent to that of Galois principal bundles over S with group $G = \mathrm{Aut}(I)$. Indeed, this is an equivalence of categories.

It can be put into a more convenient form, as in SGA 1, V. To this end, suppose that S is connected and equipped with a geometric point ξ. The enlarged fundamental pro-group $\Pi = \Pi_1(S, \xi)$ is then defined. For every quasi-isotrivial twisted constant bundle X over S, let $I = X(\xi)$ be its set-theoretic fiber at ξ. Thus X is of type I, and is consequently associated, as above, with a Galois principal bundle

$$P = \mathrm{Isom}_S(I_S, X)$$

over S, with group $G = \mathrm{Aut}(I)$.

By the definition of Π, one therefore obtains a canonical homomorphism from Π to G, that is, from one of the Π_i to G. Since G is the group of permutations of $I = X(\xi)$, this says that Π "acts continuously on $I = X(\xi)$", meaning that the π_i, for i sufficiently large, act on I compatibly with the transition morphisms.

We leave it to the reader to verify that every S-morphism $X \to Y$ between quasi-isotrivial twisted constant bundles over S induces a map $X(\xi) \to Y(\xi)$ compatible with the actions of Π, and that the resulting functor is an equivalence of categories:

Proposition 7.0.3.— *Let S be a locally noetherian connected prescheme, let ξ be a geometric point of S, and let $\Pi = \Pi_1(S, \xi)$ be the enlarged fundamental pro-group of S at ξ. Then the functor*

$$X \longmapsto X(\xi)$$

is an equivalence between the category of quasi-isotrivial twisted constant bundles over S and the category of sets on which Π acts continuously.

This functor is compatible with finite sums and finite inverse limits. It follows, for example, that quasi-isotrivial twisted constant groups (or rings, and so forth) over S correspond to ordinary groups (respectively rings, and so forth) on which the pro-group Π acts continuously. In particular:

Corollary 7.0.4.— *The category of quasi-isotrivial twisted constant commutative groups over S is equivalent to the category of "Π-modules", that is, commutative groups M on which Π acts continuously.*

Using 5.7, we now obtain:

Theorem 7.1.— *Let S be a locally noetherian connected prescheme, let ξ be a geometric point of S, and let $\Pi = \Pi_1(S, \xi)$ be the enlarged fundamental pro-group of S at ξ. Then the functor*

$$G \longmapsto \mathrm{Hom}_{\kappa(\xi)\text{-gr}}(G_\xi, \mathbf{G}_{m,\xi})$$

induces an anti-equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the category of Π-modules.

Using 4.5, we obtain:

Corollary 7.2.— *The preceding functor induces an anti-equivalence between the category of groups of multiplicative type and of finite type over S, and the category of Π-modules that are of finite type over $\mathbb{Z}$.*

Example 7.3.— Take for S, for example, a complete rational curve over an algebraically closed field, having exactly one multiple point with $n + 1$ distinct branches. By 6.4, the enlarged fundamental group $\Pi(S, \xi)$ is a free group on n generators. Thus, by 7.2, the classification of tori of relative dimension m over S is equivalent to the classification of systems of n endomorphisms $A_1, \ldots, A_n$ of the $\mathbb{Z}$-module $\mathbb{Z}^m$, up to automorphism of $\mathbb{Z}^m$. Except when $n \leq 1$ or $m \leq 1$, an explicit classification of such systems seems hopeless. One can at least define a multitude of nontrivial invariants for such a system, such as the characteristic polynomials of the A_i.[58]

Remark 7.4.— Without any hypothesis on S, it remains true that, for a given ordinary commutative group M of finite type over $\mathbb{Z}$, the category of groups of multiplicative type of type M over S is anti-equivalent to the category of Galois principal bundles over S with group $G = \mathrm{Aut}_{\mathrm{gr}}(M)$. This follows easily from 5.9 and 5.10.

Remark 7.5.— The theory of the fundamental pro-group sketched in the present two sections is more advantageously formulated in the setting of general sites. In that form it applies equally well, for example, to ordinary topological spaces, and gives a satisfactory theory at least for a locally connected topological space (not necessarily locally simply connected). Here too it seems that one cannot make do with a fundamental group, and that a pro-group is needed. Finally, once one has at one's disposal the language of topologies and descent (which really lies at the heart of these questions), the account sketched here is also technically simpler than that of SGA 1, V, and should therefore in principle replace it.

8. Appendix: Elimination of certain affineness hypotheses

Our aim is to prove the following generalization of 4.9.

Theorem 8.1.— *Let S be a prescheme, and let H be a flat S-prescheme in groups of finite presentation, with connected affine fibers.*[59] *Let $s \in S$, and suppose that H_s is a torus. Then there exists an open neighborhood U of s such that $H|_U$ is a torus.*

We immediately obtain:

Corollary 8.2.— *Let H be an S-prescheme in groups, flat and of finite presentation over S. Then H is a torus if and only if its fibers are tori.*

Remark 8.3.— Even when S is the spectrum of a discrete valuation ring, the hypothesis in 8.1 that the fibers of H (here, the generic fiber) be affine cannot be dropped: there are examples of smooth groups over S whose generic fiber is an elliptic curve and whose special fiber is $\mathbf{G}_m$.

Proof of 8.1. We may clearly suppose that $S = \mathrm{Spec}(A)$ is affine. By the standard procedure (cf. EGA IV_2, 8.8.2), this reduces us to the case in which S is moreover noetherian. We begin by proving 8.2 in this case.

By 4.9, we are reduced to proving that H is affine over S. We may therefore suppose[60] that A is noetherian and local. Since its completion $\widehat{A}$ is faithfully flat over A, descent reduces us to the case in which A is a complete noetherian local ring. Let S' be the normalization of S_{red}. Nagata's theorem says that it is finite over S (EGA 0_{IV}, 23.1.5). Moreover, $S' \to S$ is surjective, so

$$H' = H \times_S S' \longrightarrow H$$

is finite and surjective. Hence, to prove that H is affine, it suffices to show that H' is affine (EGA II, 6.7.1).

[58] *Editors' note.* In particular, this shows that the tori of relative dimension 2 considered in 1.6 are non-isotrivial.

[59] *Editors' note.* Observe that the hypotheses imply that H is separated over S, by $\mathrm{VI_B}$, Theorem 5.3 or Corollary 5.5.

[60] *Editors' note.* By EGA IV_2, 8.8.2 once again.

Replacing S by a connected component of S', we are reduced to the case in which A is a noetherian (semilocal), normal, integral ring.[61] Moreover, after possibly replacing A by its normalization in a finite separable extension of its field of fractions, we may suppose that the generic fiber H_η of H is diagonalizable, that is, that there is an isomorphism

$$u_\eta : H_\eta \xrightarrow{\sim} T_\eta, \qquad T = \mathbf{G}^r_{m,S}.$$

We now have:

Lemma 8.4.— *Let S be a locally noetherian normal irreducible prescheme with generic point η. Let H be a smooth prescheme in groups over S, with connected fibers, let T be a prescheme in groups of multiplicative type and of finite type over S, and let $u_\eta : H_\eta \to T_\eta$ be a homomorphism of algebraic groups over $\kappa(\eta)$. Then u_η extends to a homomorphism of groups $u : H \to T$.*

After possibly replacing S by its normalization in a finite separable extension of its function field, we may suppose that T is diagonalizable (which is already the case in the application we have in mind). Then T is a closed subgroup of a group of the form $\mathbf{G}^r_{m,S}$, reducing us to the case $T = \mathbf{G}_{m,S}$.

It remains only to prove that u_η, regarded as a rational map from H to $T = \mathbf{G}_{m,S}$, is everywhere defined (for then the morphism $u : H \to T$ extending it is necessarily a group homomorphism). We may regard u_η as an invertible section f of the structure sheaf of H_η, and must show that it extends to an invertible section of the structure sheaf of H. Since H is smooth over the normal prescheme S, it is normal (SGA 1, II 3.1). It therefore suffices to find a closed subset Z of H, of codimension at least 2, such that f extends to an invertible section of the structure sheaf of $H - Z$. By localizing at the codimension-one points of S, this at once reduces us to the case in which S is the spectrum of a discrete valuation ring A.

Let t be a uniformizer of A, and let t' be the section of $\mathcal{O}_H$ that it defines, so that the special fiber H_0 equals $V(t')$. By hypothesis, H_0 is smooth and connected over the residue field k. The function f is then a rational function on H with neither zeros nor poles in $H - H_0$. Since $H_0 = V(t')$ is an irreducible divisor, there is an integer $n \in \mathbb{Z}$ such that

$$t'^n f = t^n f$$

has neither zeros nor poles, and hence is an invertible section of $\mathcal{O}_H$. It therefore defines a morphism $v : H \to \mathbf{G}_{m,S}$. Since $v_\eta = t'^n u_\eta$, and u_η is a group homomorphism, v sends the identity section of H to a section of $\mathbf{G}_{m,S}$ whose value at the generic point of S is t^n. Because this is a section of $\mathbf{G}_{m,S}$, t^n must be a unit; thus $n = 0$. Hence v extends u_η, completing the proof of 8.4.

Applying this lemma in the present situation, we obtain a group homomorphism

$$u : H \longrightarrow T = \mathbf{G}^r_{m,S}$$

that induces an isomorphism on the generic fibers. We shall prove that u is an isomorphism.

Lemma 8.5.— *For every integer $n > 0$ prime to the residual characteristic of A, the group*

$${}_nH = \ker(n \cdot \mathrm{id}_H)$$

is finite over S.

If n is prime to the residual characteristic of A, then it is prime to all the residual characteristics of the points of S. Consequently, $n \cdot \mathrm{id}_H$ induces an étale morphism on every

[61] *Editors' note.* The original has been expanded in what follows.

fiber of H, and hence is itself étale (SGA 1, I 5.9); its kernel ${}_nH$ is therefore étale over S. On the other hand, ${}_nH$ is separated over S, because H is. [*] [62] Moreover, all its fibers have the same rank n^r, since the fibers of H are tori, all of the same dimension r (as H is smooth over S). We conclude that ${}_nH$ is finite over S (SGA 1, I 10.9, or EGA IV$_4$, 18.2.9).

By the preceding lemma, $u({}_nH)$ is therefore a closed subset of T. On the generic fiber T_η, this subset is identical to ${}_n(T_\eta)$, so it contains the closure of that subset, namely ${}_nT$. For every $s \in S$, the groups ${}_n(T_s)$, as n varies, are dense in T_s; since $u_s(H_s)$ is a closed subset (VI$_\mathrm{B}$, 1.4.2) containing them, we have $u_s(H_s) = T_s$. Thus u is surjective. A surjective homomorphism between tori of the same dimension over a field is flat,[63] and hence u induces a flat morphism on every fiber. Therefore u is flat (SGA 1, I 5.9). Consequently $K = \ker(u)$ is flat over S,[64] and hence equals the closure of its generic fiber K_η. By construction, K_η is the identity group. Since K is separated over S (because H is), its identity section is closed, and it follows that K is the identity group. Thus u is a monomorphism; since we have seen that it is faithfully flat, it is an isomorphism (cf. SGA 1, I 5.1, or EGA IV$_4$, 17.9.1). This proves that H is a torus, and completes the proof of 8.2.[65]

Remark 8.5.1.— Instead of invoking 8.5, one can also invoke Zariski's "Main Theorem", which directly implies that u is an open immersion and hence an isomorphism.[66]

To prove 8.1, the quasi-isotriviality theorem reduces us to the case in which S is local and s is its closed point,[67] and it remains to prove that, under the other hypotheses, H is then a torus. By 8.2, already proved, we are reduced to showing that the fibers of H are tori. We may suppose that S is the spectrum of a complete noetherian local ring A. By 3.3, for every $n > 0$ there exists a group of multiplicative type T_n, finite over S, and an isomorphism

$$(T_n)_s \simeq {}_n(H_s),$$

where s is the closed point of S. Arguing as in 3.1 and using the fact that T_n is finite over S,[68] we see that the preceding isomorphism comes from a homomorphism $u_n : T_n \to H$, which is moreover uniquely determined. (Pass to the limit over the artinian quotients of A.)

Moreover, by the uniqueness properties, when m is a multiple of n, the homomorphism $u_n : T_n \to H$ is obtained from $u_m : T_m \to H$ by restriction to ${}_n(T_m) \simeq T_n$. It follows from IX 6.6 that the u_n are monomorphisms, so the T_n are subgroups of H, and for m a multiple of n one has

$$T_n = {}_n(T_m).$$

Thus, for every $t \in S$, the groups $(T_n)_t$ are subgroups of H_t, of type $(\mathbb{Z}/n\mathbb{Z})^r$, where $r = \dim H_s = \dim H_t$, and when m is a multiple of n one has

$$(T_n)_t = {}_n(T_m)_t.$$

The fact that H_t is a torus now follows from:

Lemma 8.6.— *Let H be a smooth affine algebraic group over an algebraically closed field k, and let $(T_n)_{n>0}$ be a family of subgroups of multiplicative type of H, such that for every*

[*] By Raynaud's theorem, VI$_\mathrm{B}$, 5.3.

[62] *Editors' note.* This point should be expanded in connection with the modifications in VI$_\mathrm{B}$, §5.

[63] *Editors' note.* This follows, for example, from VIII 3.2(a). More generally, if $f : G \to H$ is a surjective morphism of algebraic groups over a field k, and H is reduced, then f is flat (cf. VI$_\mathrm{B}$, 1.3).

[64] *Editors' note.* The original has been expanded in what follows.

[65] *Editors' note.* At least in the case considered up to this point, namely S locally noetherian.

[66] *Editors' note.* The editors did not understand this remark, since they did not see why u would *a priori* have finite and surjective fibers.

[67] *Editors' note.* Expand this reduction.

[68] *Editors' note.* Expand this point.

integer $n > 0$, T_n is of type $(\mathbb{Z}/n\mathbb{Z})^r$, and for every multiple m of n, one has $T_n = {}_n(T_m)$. Under these conditions, H contains a torus of dimension at least r containing the T_n. Consequently, if H is connected and has dimension at most r, then H is a torus of dimension r.

This is an exercise on affine algebraic groups, which we shall handle by repeated reference to *Bible*. We restrict attention to the T_n for n prime to the characteristic. Let K be the closure of the union of the T_n in H, endowed with the induced reduced structure. Standard arguments then show that K is a commutative algebraic subgroup of H. By *Bible*, §4.5, Theorem 4, K is isomorphic to a product $K_u \times K_s$, with K_u "unipotent" and K_s diagonalizable. Every diagonalizable subgroup of K is contained in K_s, so the T_n are subgroups of K_s, and therefore $K = K_s$. Write $K = D(M)$, where M is an ordinary commutative group of finite type over $\mathbb{Z}$. The inclusion $T_n \subseteq K$ means that M has a quotient group isomorphic to $(\mathbb{Z}/n\mathbb{Z})^r$. Since this holds for every integer n prime to the characteristic of k (powers of a single fixed prime would suffice), M has rank at least r. Thus K contains a torus T of dimension r. When H is connected of dimension r, it follows that $T = H$, completing the proof of 8.6. Theorem 8.1 is now proved.

Remarks 8.7.— Using 8.1, it should not be difficult to give analogous generalizations of 4.7 and 4.8. A more interesting problem would be to study the situation of 8.1 after dropping the hypothesis that the fibers of H be affine. One can show that there is then an open neighborhood U of s such that $H|_U$ is commutative and, for every $t \in U$, the geometric fiber H_t is an extension of an abelian variety by a torus.[69] Of course, in questions of this kind one may restrict to the case in which S is the spectrum of a discrete valuation ring, s its closed point, and t its generic point.

This result can be generalized as follows. For every smooth connected algebraic group G over an algebraically closed field k, a well-known theorem of Chevalley says that G is, in a unique way, an extension of an abelian variety A by a smooth connected affine group V. Define the *abelian rank* (respectively, *reductive rank*, *nilpotent rank*, and *semisimple dimension*) of G, denoted $\rho_{\mathrm{ab}}(G)$ (respectively, $\rho_r(G)$, $\rho_n(G)$, and $d_s(G)$), to be the dimension of A (respectively, the dimension of the maximal tori of G, the dimension of the Cartan subgroups[70] of G, and the dimension of the quotient of G, or equivalently of V, by its radical); see *Bible* for all these notions. Also define the *unipotent rank* by

$$\rho_u(G) = \rho_u(V) = \rho_n(G) - \rho_r(G) - \rho_{\mathrm{ab}}(G).$$

When the ground field is not algebraically closed, the same names and notations designate the corresponding invariants of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k.

Now let G be a smooth group scheme over the spectrum S of a discrete valuation ring. Let ρ_{ab}, and so on (respectively, ρ'_{ab}, and so on), denote the invariants associated with the special fiber (respectively, the generic fiber). Then one has the inequalities

$$\begin{cases} \rho_{\mathrm{ab}} \leq \rho'_{\mathrm{ab}}, \\ \rho_r + \rho_{\mathrm{ab}} \leq \rho'_r + \rho'_{\mathrm{ab}}, \\ d_s \leq d'_s, \end{cases} \qquad \begin{cases} \rho_n \geq \rho'_n, \\ \rho_u \geq \rho'_u. \end{cases}$$

Equivalently, if G is smooth of finite type over an arbitrary base S, the functions

$$s \longmapsto \rho_{\mathrm{ab}}(s), \qquad s \longmapsto \rho_{\mathrm{ab}}(s) + \rho_r(s), \qquad s \longmapsto d_s(s)$$

[69] *Editors' note.* Give a reference here?

[70] *Editors' note.* Recall the definition of "Cartan subgroup".

are lower semicontinuous, while the functions

$$s \longmapsto \rho_n(s), \qquad s \longmapsto \rho_u(s)$$

are upper semicontinuous.[71]

The same results probably remain valid without assuming G smooth over S, but assuming only that it is flat and of finite presentation over S, provided that, for an algebraic group G over an algebraically closed field k, $\rho_{\mathrm{ab}}(G)$, and so on, are understood to mean the corresponding invariants of G^0_{red}.

In this Seminar we present some results of this kind when G is affine over S, or more generally has affine fibers. In this case we shall verify the semicontinuity properties for ρ_r and ρ_n, hence for $\rho_u = \rho_n - \rho_r$, and the continuity of ρ_s in a neighborhood of a point s whose fiber is a reductive group.[71]

When G is already assumed commutative, 8.2 can be generalized as follows.

Theorem 8.8.— [72]*Let G be a commutative S-prescheme in groups, flat and of finite presentation over S, with affine connected fibers. Let $s \in S$, and suppose that:*

a) if $\bar{k}$ denotes an algebraic closure of $\kappa(s)$, then $(G_{\bar{k}})_{\mathrm{red}}$ is a torus;

b) there is a generization t of s such that G_t is smooth over $\kappa(t)$.

Under these conditions, there exists an open neighborhood U of s such that $G|_U$ is a torus.

(Observe moreover that, if one assumes only that G_t is affine, respectively connected, for every generization t of s, it follows easily that the same conclusion holds for t in an open neighborhood of s.)

Proof of 8.8. It suffices to prove that G_s is smooth over $\kappa(s)$. Indeed, since G is flat and of finite presentation over S, it then follows that G is smooth over S above a neighborhood of s (cf. 3.5), and we are then in the situation of 8.1.

To prove that G_s is smooth over $\kappa(s)$, the usual procedure reduces us to the case in which S is affine and noetherian. Choose a morphism $S' \to S$, where S' is the spectrum of a discrete valuation ring, sending the closed point to s and the generic point η to t. We are thereby reduced to the case in which S is itself the spectrum of a discrete valuation ring; we may moreover suppose that it is complete and has algebraically closed residue field, and that s and $t = \eta$ are respectively the closed point and the generic point of S.

Thus G is flat, separated, and of finite type over S; the generic fiber G_η is smooth and connected, and the special fiber G_0 is such that

$$T_0 = (G_0)_{\mathrm{red}}$$

is a torus. Let $m > 0$ be an integer prime to the residual characteristic at s, and hence also to that at η. Then $m \cdot \mathrm{id}_G$ is étale on every fiber ($\mathrm{VII_A}$, 8.4), and is therefore étale because G is flat over S (SGA 1, I 5.9). Its kernel ${}_mG$ is thus étale over S; since G is separated * [73] and of finite type over S, the same holds for ${}_mG$. Since

$$({}_mG)_0 = {}_m(T_0),$$

[71]*Editors' note.* Give a reference for these results?

[72]*Editors' note.* G. Prasad and J.-K. Yu generalized this result (subject to some additional hypotheses) without assuming G commutative, and with "torus" replaced by "reductive group" in hypothesis (a) and in the conclusion; cf. [PY06], Theorem 6.2.

*By Raynaud's theorem, $\mathrm{VI_B}$, 5.3.

[73]*Editors' note.* See Editors' note (62) in 8.5.

its degree is m^r, where
$$r = \dim T_0 = \dim G_0 = \dim G.$$
It follows that the rank of $({}_mG)_\eta = {}_m(G_\eta)$ is at least m^r, which already proves, using 8.6, that G_η is a torus of dimension r, since the two fibers of ${}_mG$ have the same rank, and hence, as in 8.5, that ${}_mG$ is finite over S.[74]

Since S is complete with algebraically closed residue field, the finite étale cover ${}_mG$ splits completely. Thus, through every point of ${}_mG_0$ there passes a section of G over S; in particular, the set of points of G_0 through which there passes a section of G over S is everywhere dense. We now use the following lemma.

Lemma 8.9.— *Let S be a locally noetherian regular prescheme of dimension* 1*, and let G be an S-prescheme in groups, flat and locally of finite type, such that G_η is smooth over $\kappa(\eta)$ for every maximal point η of S (which implies that G is reduced). Suppose that the normalization X of G is finite over G (this is the case, by Nagata, when S is the spectrum of a complete discrete valuation ring; cf. EGA IV$_2$, 7.7.4), and let G' be the open subset of X formed by the points at which X is smooth over S. With these notations:*

a) If the projection $G' \to S$ is surjective, then G' carries a unique structure of an S-prescheme in groups for which the canonical morphism $G' \to G$ is a group homomorphism.

b) Suppose that, for every closed point s of S, the set of points of G_s^0 through which an étale quasi-section passes is dense in G_s^0 for the Zariski topology. Then X is regular, the hypotheses of (a) *hold, and the map*
$$\Gamma(G'/S) \longrightarrow \Gamma(G/S)$$
is bijective.

In the case that concerns us, this lemma applies and gives a group homomorphism $u : G' \to G$, where G' is smooth over S and
$$u_\eta : G'_\eta \xrightarrow{\sim} G_\eta$$
is an isomorphism. After possibly replacing G' by an open subgroup, we may suppose that G'_0 is connected. Since $u_0 : G'_0 \to G_0$ induces a surjective morphism $G'_0 \to T_0$, where T_0 is a torus of the same dimension as G'_0, it follows easily that G'_0 is a torus (for example, by using 8.6, or by referring to *Bible*, §7.3, Theorem 3(a)). Consequently, by 8.2, G' is a torus. By IX 6.8, $\ker u$ is then a subgroup of multiplicative type of G'; since its generic fiber is the identity group, it is itself the identity group, and therefore u is a monomorphism. It follows from VIII 7.9 that u is an immersion. Since it is surjective and G is reduced, u is an isomorphism, completing the proof of 8.8.

It remains to prove 8.9. For part (a), the uniqueness of the group law on G' for which $u : G' \to G$ is a group homomorphism is clear, since the group law on the generic fiber of G' is known (we may suppose S irreducible). For existence, we reduce easily to the case in which S is local, the spectrum of a discrete valuation ring A. By uniqueness, and because integral closure commutes with an étale extension of the base, we may make étale base extensions on S. This reduces us to the case in which A is "strictly local", that is, henselian with separably closed residue field. The same reduction applies to part (b); under the hypothesis made there, "étale quasi-section" can now be replaced by "section".

[74] *Editors' note.* Revise the preceding sentence.

Since G' is smooth over S and X is normal, the prescheme $G' \times_S X$ is normal (being smooth over the normal prescheme X). Thus the composite morphism

$$G' \times_S X \longrightarrow G \times_S G \longrightarrow G$$

factors as

$$p : G' \times_S X \longrightarrow X.$$

We shall prove that the restriction of this morphism to the open subset $G' \times_S G'$ of $G' \times_S X$ induces a morphism

$$G' \times_S G' \longrightarrow G'.$$

It is therefore enough to show that $G'_0 \times_k G'_0$ maps into the open subset G'_0 of X_0. It suffices to check that, for every k-valued point g'_0 of G'_0, the morphism

$$h'_0 \longmapsto p_0(g'_0, h'_0) \tag{$\times$}$$

from G'_0 to X_0 takes its values in G'_0. Since G' is smooth over S and S is henselian, every such g'_0 is induced by a section g' of G' over S. The morphism $(\times)$ is then induced by the morphism

$$h' \longmapsto p(g', h')$$

from G' to X, which itself is obtained by transport of structure from the automorphism $h \mapsto g \cdot h$ of G, namely left translation by the section g of G that is the image of g'. Thus $h' \mapsto p(g', h')$ is itself an automorphism of X, and hence maps G' into G'. This proves the assertion.

It remains to prove that the composition law thereby obtained on G' is a group law. Associativity follows at once from associativity on the generic fiber (which is isomorphic to that of G). Moreover, the inversion automorphism of the S-prescheme G induces an automorphism of X, which therefore preserves G' and induces an automorphism σ of G'. One verifies that σ has the properties of an inverse for the composition law on G'. Indeed, this again amounts to verifying the commutativity of certain diagrams involving fiber powers of G' over S; since these are smooth over S, it suffices to check commutativity on the generic fiber, where it is clear. This proves part (a) of 8.9.

We prove part (b). Let Z' be the set of $x \in X$ for which $\mathcal{O}_{X,x}$ is not regular. It is closed by a theorem of Nagata (EGA IV_2, 6.12.6), and is clearly contained in X_0. Let Z be its image in G, which is therefore a closed subset of G_0. Then Z is nowhere dense in G_0, that is, it contains no maximal point y of G_0. Indeed, G_0 is defined in G by an equation $t = 0$, where t is a uniformizer of the discrete valuation ring A. By the Hauptidealsatz, $\mathcal{O}_{G,y}$ has dimension 1. Hence, for every $x \in X$ above y, $\mathcal{O}_{X,x}$ has dimension 1, and is therefore a discrete valuation ring, since X is normal and hence regular in codimension 1.

On the other hand, for every section g of G over S, the set Z' is plainly stable under the automorphism of X obtained by transport of structure from left translation by g on G. Thus Z is stable under the left translation on G_0 defined by g_0. By hypothesis, the set of these g_0 is dense in G^0_0. Since Z is stable under translation by these g_0 and is a nowhere-dense closed subset, it follows at once that $Z = \varnothing$, whence $Z' = \varnothing$, so X is regular. Consequently X is smooth over S at every point through which a section passes. Every section of G over S lifts uniquely to a section of X over S, and hence to a section of G' over S. This applies in particular to the identity section, proving that the image of G' in S is all of S, that is, that the hypothesis of (a) holds. This completes the proof of 8.9, and hence of 8.8.

9. Addenda

[75]

0.9.1 Simplicial sets, topoi, groupoids, and topological spaces

Notations 9.1.1.— *(1) Let E be a simplicial set. The following objects can be associated with it:*

(2) a topos $T = E^{\sim}$, obtained from the topoi naturally associated with the sets E_i, by the procedure described in [Del74]*, 6.3.1 (see also* [Ill72]*, VI.5.2 and SGA 4, VI.7);*

(3) a groupoid $G = \Pi(E)$, whose objects are the elements of E_0 (the "vertices") and whose arrows are defined in [GZ67]*, II.7;*

(4) a topological space $X = |E|$, a cellular complex called the "geometric realization" (or "topological realization") (cf. loc. cit.*, III.1).*

Observe that a sheaf on T is nothing other than a simplicial set over E.

0.9.2 Locally constant sheaves; descent data

Definitions 9.2.1.— A *locally constant sheaf* on

(1) a simplicial set E is a morphism of simplicial sets $E' \to E$ such that, for every $i \in \mathbb{N}$ and every $e \in E_i$, the face operators d induce isomorphisms

$$E'_e \xrightarrow{\sim} E'_{d(e)}$$

between the fibers (cf. [AM69], §10);

(2) a topos T is an object F of T such that there exist an epimorphism $U \to 1$ and an isomorphism

$$F \times U \simeq f^*I \times U,$$

where I is a set and $f : T \to \mathbf{Ens}$ is the final morphism (cf. SGA 4, IX.2);

(3) a groupoid G is a presheaf on G, that is, a contravariant functor from G to the category of sets (cf. [GZ67], Appendix I.1.2);

(4) a topological space X is a sheaf of sets on X that is locally constant in the usual sense.

Finally:

(5) a *descent datum* on a simplicial set E consists of a sheaf F on $E_0^{\sim}$ (that is, a set-valued function on E_0) and an isomorphism

$$\alpha : p_1^*F \xrightarrow{\sim} p_2^*F,$$

where $p_1, p_2 : E_1^{\sim} \to E_0^{\sim}$ are the morphisms induced by the faces, satisfying the usual cocycle relation (cf. Exposé IV, 2.1 [1] and below).

The morphisms between these five kinds of objects, as well as the associated inverse-image functors, are defined in the evident way.

[75] *Editors' note.* This additional section was written by Fabrice Orgogozo (following indications by Ofer Gabber).

[1] PP: I have replaced "FGA, *Technique de descente* I, 1.6" by "Exposé IV, 2.1".

0.9.3 Some equivalences of categories

Let E be a simplicial set, and let T, G, and X be, respectively, the associated topos, groupoid, and topological space.

Proposition 9.3.1.— *The categories of locally constant sheaves on E, T, G, and X, as well as the category of descent data on E, are equivalent.*

Sketch of proof. Denote by (1) through (5) the categories of objects defined in the preceding paragraph.

(1) $\Longleftrightarrow$ (5). This is a special case of [AM69], 10.6 (see also [GZ67], Appendix I.2.3, [Fri82], ?.5.6, or [Ill72], VI.8.1.6). An equivalence of categories is given by the functor that associates with the object $E' \to E$ the pair (E'_0, α), where E'_0 is regarded as a sheaf on E_0 and α is the unique isomorphism

$$p_1^* E'_0 \xrightarrow{\sim} p_2^* E'_0$$

whose fiber at each $y \in E_1$, with images denoted x_1 and x_2 under the two projections, is given by the isomorphisms

$$(E'_0)_{x_1} \xleftarrow{\sim} (E'_1)_y \xrightarrow{\sim} (E'_0)_{x_2}.$$

(5) $\Longleftrightarrow$ (3). This is evident: one of the two relations defining the morphisms in the groupoid G associated with E is a cocycle relation.

(1) $\Longleftrightarrow$ (4). Cf. [GZ67], Appendix I.3.2.1.

(1) $\Longleftrightarrow$ (2). One must show that an object over E is a locally constant sheaf in the simplicial sense if and only if it is one as a sheaf on $T = E^{\sim}$. This follows from the fact that the locally constant sheaves $F_\bullet$ on T are precisely the cartesian sheaves, that is, those for which the arrows

$$([n] \to [m])^* F_m \longrightarrow F_n$$

are isomorphisms. This last assertion is a special case of a general fact about simplicial topoi, together with the fact that every sheaf on $E_0^{\sim}$ is locally constant. (See also [Ill72], VI.8.1.6.)

Q.E.D.

For every group H, the preceding equivalences of categories induce equivalences between the categories of H-torsors. Here these are locally constant sheaves equipped with an action of H, whose fibers are isomorphic to H acting on itself by translation.

0.9.4 Fundamental groups and groupoids

It follows from the preceding equivalences of categories that, for every group H and every simplicial set E, the sets of isomorphism classes of H-torsors

$$H^1(E, H), \qquad H^1(E^{\sim}, H), \qquad \pi_0(\mathrm{Hom}(\Pi(E), BH)), \qquad H^1(|E|, H)$$

are naturally in bijection. Recall that BH denotes the one-object groupoid associated with H, and that π_0 is the functor that associates with a category the set of isomorphism classes of its objects.

Likewise, if e is a point of E, the preceding equivalences induce bijections between the sets of isomorphism classes of H-torsors trivialized over e, denoted respectively by

$$H^1(E \operatorname{rel} e, H), \qquad H^1(E^{\sim} \operatorname{rel} e^{\sim}, H), \qquad \mathrm{Hom}(\pi_1(\Pi(E), e), H), \qquad H^1(|E| \operatorname{rel} |e|, H).$$

Recall that

$$\pi_1(\Pi(E), e) = \mathrm{Isom}_{\Pi(E)}(e, e).$$

For variable H, these functors are represented, in the connected case, by a group denoted $\pi_1(E, e)$. The group $\pi_1(E, e)$ is isomorphic to $\pi_1(\Pi(E), e)$ and to $\pi_1(|E|, |e|)$, and hence also to the fundamental group of a simplicial set as defined by Kan (cf., for example, [May67], 16.1, or [Ill72], I.2.1.1). Recall also that $H^1(E, H)$ is in turn isomorphic to the set $H^1(\pi_1(E, e), H)$ of homomorphisms to H modulo conjugation, also denoted

$$\mathrm{Hom\,ext}(\pi_1(E, e), H).$$

0.9.5 Cones

Definitions 9.5.1.— *(1) Let $f : E' \to E$ be a morphism of simplicial sets. Recall (cf., for example,* [Del74]*, 6.3.1) that $C(f)$ denotes the pointed simplicial set whose underlying set in degree $n \geq 0$ is*

$$E_n \amalg \left(\coprod_{i<n} E'_i \right) \amalg \star,$$

where $\star$ is a singleton. We leave it to the reader to define the simplicial maps; the faces in degrees at most two are recalled below. (See also [GZ67]*, VI.2, for a pointed variant.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on E equipped with a trivialization of the inverse image on E'. The simplicial set $C(f)$ is naturally pointed by $\star \to C(f)$.*

(2) Let $f : T' \to T$ be a morphism of topoi. Denote by $C(f)$ the topos whose objects are quintuples

$$(F, F', A, \alpha : f^*F \to F', \beta : A \to \Gamma(T', F')),$$

where F (respectively F') is an object of T (respectively T'), A is a set, and α (respectively β) is a morphism in T' (respectively in **Ens***). (See also* [Del80]*, 4.3.4 and* [Ill72]*, III.4 for a variant of this construction.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on T equipped with a trivialization of the inverse image on T'. The topos $C(f)$ is naturally pointed by the fiber functor that sends the quintuple above to the set A.*

(3) Let $f : G' \to G$ be a morphism of groupoids. Denote by $C(f)$ the colimit of the diagram

$$G \longleftarrow G' \longrightarrow B1,$$

where $B1$ is the one-object category (one object, one arrow). The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on G equipped with a trivialization of the inverse image on G'.

(4) Let $f : X' \to X$ be a morphism of topological spaces. Denote by $C(f)$ the colimit of the diagram

$$\star \longleftarrow X' \times \{0\} \longrightarrow X' \times [0,1] \longleftarrow X' \times \{1\} \longrightarrow X.$$

The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on X equipped with a trivialization of the inverse image on X'. The topological space $C(f)$ is naturally pointed by $\star \to C(f)$.

9.5.2. Descent data on the cone of a simplicial map

Let $f : E' \to E$ be a morphism of simplicial sets, and let $C(f)$ be its cone (cf. above, (1)). Use p (respectively q, r) to denote the face maps of E' (respectively $E, C(f)$). Before stating the proposition below, we describe the faces r in degrees at most two in terms of p and q. By convention, p_{ij} (respectively p_i) is the face map

$$E'_2 = E'_{\{1,2,3\}} \longrightarrow E'_1 = E'_{\{1,2\}} \quad \Big(\text{respectively } E'_1 = E'_{\{1,2\}} \longrightarrow E'_0 = E'_{\{1\}}\Big)$$

corresponding to the increasing map

$$\{1,2\} \longrightarrow \{1,2,3\} \quad (\text{respectively } \{1\} \longrightarrow \{1,2\})$$

whose image is $\{i,j\}$ (respectively $\{i\}$). [2] The same convention is adopted for the faces of E and $C(f)$.

The morphism

$$r_1 : C(f)_1 = E_1 \amalg E'_0 \amalg \star \longrightarrow C(f)_0 = E_0 \amalg \star$$

(respectively r_2) is:

- $q_1 : E_1 \to E_0$ (respectively $q_2 : E_1 \to E_0$) on E_1;
- $E'_0 \to \star$ (respectively $f_0 : E'_0 \to E_0$) on E'_0;
- $\star \to \star$ on $\star$.

Likewise, the morphism

$$r_{21} : C(f)_2 = E_2 \amalg E'_1 \amalg E'_0 \amalg \star \longrightarrow C(f)_1 = E_1 \amalg E'_0 \amalg \star$$

(respectively r_{32}, r_{31}) is:

- $q_{21} : E_2 \to E_1$ (respectively q_{32}, q_{31}) on E_2;
- $p_1 : E'_1 \to E'_0$ (respectively $f_1 : E'_1 \to E_1$, $p_2 : E'_1 \to E'_0$) on E'_1;
- $E'_0 \to \star$ (respectively $\mathrm{Id} : E'_0 \to E'_0$, $\mathrm{Id} : E'_0 \to E'_0$) on E'_0.

Let H be a descent datum on $C(f)$, that is, a sheaf H_0 on

$$C(f)_0 = E_0 \amalg \star$$

equipped with an isomorphism

$$\gamma : r_1^* H_0 \xrightarrow{\sim} r_2^* H_0$$

between its inverse images on

$$C(f)_1 = E_1 \amalg E'_0 \amalg \star,$$

satisfying the cocycle relation

$$r_{31}^* \gamma = r_{32}^* \gamma \circ r_{21}^* \gamma.$$

The restriction of γ to E_1 (respectively E'_0) is a descent datum

$$\beta : q_1^* G_0 \xrightarrow{\sim} q_2^* G_0,$$

[2] This convention departs from the present simplicial convention but is closer to the notation used by Grothendieck in his theory of descent.

where G_0 is the restriction of H_0 to E_0 (respectively, a trivialization

$$t : H_0(\star) \xrightarrow{\sim} f^*G_0 =: F_0,$$

where $H_0(\star)$ is the constant sheaf with stalk $H_0(\star)$). The restriction of the cocycle relation satisfied by γ to E_2 (respectively E'_1) is the cocycle relation for β (respectively,

$$p_2^*(t) = f_1^*(\beta) \circ p_1^*(t)).$$

This latter relation says that the trivialization t is compatible with the descent datum $f_1^*(\beta)$ induced by β on F_0. It follows that:

Proposition 9.5.3.— *Let $f : E' \to E$ be a morphism of simplicial sets, and let $C(f)$ be its cone, pointed by $\star$. The category of descent data on $C(f)$ trivialized over $\star$ is equivalent to the category of descent data on E equipped with a trivialization of the descent datum induced on E'.*

0.9.6 Representability of the functor $\overline{\pi}^1(S'/S, \xi; -)$

The notation is that of page 90, and from now on we suppose that the connected components of

$$S_0 = S' \qquad \text{and} \qquad S_1 = S' \times_S S'$$

are open. Under this hypothesis we have the following explicit combinatorial description, which follows immediately from the preceding recall.

Proposition 9.6.1.— *For every group G, the set $\overline{\pi}^1(S'/S, \xi; G)$ of isomorphism classes of descent data equipped with a trivialization over ξ is naturally in bijection with the relative nonabelian cohomology set*

$$H^1(K \operatorname{rel} k, G) := H^1(\widetilde{K} \operatorname{rel} \star, G).$$

It follows that this functor is representable by a group, denoted $\pi_1(K, k)$, the *relative fundamental group*, as soon as the simplicial set

$$\widetilde{K} = C(k \to K)$$

is connected. One verifies immediately that this holds if and only if the map $\pi_0(k) \to \pi_0(K)$ is surjective. It is enough, in particular, that the simplicial set K be connected. As indicated in the text, this is the case when the scheme S is connected and the morphism $S' \to S$ is universally submersive.

0.9.7 Contractibility of k; counterexample to the injectivity of $k_\bullet \to K_\bullet$

The following two propositions make Editors' note (53) more precise.

Proposition 9.7.1.— *Let κ be an algebraically closed field, and let X be a connected κ-scheme. For every integer $i \geq 0$, let X_i denote the $(i+1)$-st fiber power of X over κ. For every integer $i \geq 0$, the canonical map*

$$\pi_0(X_i) \longrightarrow \pi_0(X)^{i+1}$$

is a bijection. In particular, the simplicial set $\pi_0(X_\bullet)$ is contractible.

It suffices to prove the following lemma, which follows by passage to the limit from the Künneth formula. [Expand?]

Lemma 9.7.2.— *Let κ be an algebraically closed field, and let X and Y be two connected κ-schemes. Then the fiber product $X \times_\kappa Y$ is connected.*

Proposition 9.7.3.— *Let X be a connected scheme, let $X' \to X$ be a finite étale morphism, and let $\bar{x}$ be a geometric point of X localized at a point x. The canonical morphism*

$$\pi_0(X'_{\bar{x}}) \longrightarrow \pi_0(X')$$

is surjective.

Indeed, the image of every connected component of X' is open and closed in X, and therefore meets the fiber X'_x. Thus

$$\pi_0(X'_x) \longrightarrow \pi_0(X')$$

is surjective, as is

$$\pi_0(X'_{\bar{x}}) \longrightarrow \pi_0(X'_x).$$

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[76] *Editors' note.* Additional references cited in this Exposé; the references [AM69] and following concern the Addenda.

Exposé XI
Representability Criteria
Applications to
Multiplicative-Type Subgroups
of Affine Group Schemes

by A. Grothendieck[0]

0. Introduction

As we have already seen in Exposé X, nos. 4 and 5, the representability of certain functors—for example, functors of the form $\mathrm{Hom}_S(X, Y)$ and various related constructions—plays an important role in many questions concerning group preschemes.

Among the results that are particularly useful in this theory, besides the representability of quotients studied in Exposés V and $\mathrm{VI_B}$ and in Exposé VIII, no. 5, let us mention the representability of functors of the form

$$\prod_{X/S} Y/X,$$

where Y is a subobject of X. This was studied in a very elementary case in Exposé VIII, no. 6; variants will be given in no. 6 of the present exposé. These results provide the representability of various centralizers, normalizers, and transporters.

Less elementary representability criteria, using results that will appear in EGA VI, are indicated in 6.12 and in Exposés XV and XVI. There we shall give a criterion for the representability of quotients G/H in cases not covered by the preceding exposés; this criterion was not developed in the oral lectures.

Our principal purpose in the present exposé is to prove Theorems 4.1 and 4.2, which provide a typical example of a nonprojective construction technique, close to the technique that will be developed in EGA VI. Since the oral lecture and the drafting of the present text, it has moreover become apparent that the affine hypotheses in 4.1 and 4.2 can be removed to a large extent (cf. Exposé XV), and that the main part of the theory developed in the following exposé can dispense with 4.1 and 4.2. Finally, it might be interesting to prove analogues of these results for a general reductive group prescheme (for example, a semisimple one) in place of a group of multiplicative type; in that case 4.1 and 4.2 would doubtless be the key result in the proof.

[0]Editor's note: xy version of December 5, 2008.

1. Review of Smooth, Étale, and Unramified Morphisms

The reader is referred to EGA IV, ğğ17 and 18, and, pending its publication, to SGA 1, Exposés I, II, and III. Certain noetherian hypotheses used there, which are inconvenient in applications, should however be replaced by finite-presentation hypotheses.

Definition 1.1. — *Let S be a prescheme and let*

$$F : (\mathbf{Sch})^{\circ}/S \longrightarrow (\mathbf{Ens})$$

be a functor. The functor F is said to be formally smooth *(respectively,* formally unramified*, respectively,* formally étale*) if, for every S-prescheme S' that is affine in the absolute sense and every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal $\mathscr{J}$, the map*

$$F(S') \longrightarrow F(S'_0)$$

is surjective (respectively, injective, respectively, bijective).

An S-prescheme X is said to be formally smooth over S (respectively, formally unramified over S, respectively, formally étale over S) if the corresponding functor has that property. It is said to be smooth over S (respectively, unramified over S, respectively, étale over S) if it satisfies the preceding condition and is, in addition, locally of finite presentation over S.

The usefulness of these definitions for X lies, on the one hand, in their remarkably simple expression in terms of the functor represented by X (and in practice X is often given as the object over S representing an explicitly described functor), and, on the other hand, in the remarkable local structural properties that they express. We recall those properties in the following statements; for their proofs, see the references just cited.

Proposition 1.2. — *Let X be a prescheme locally of finite presentation over S. Then:*

(i) *X is smooth over S if and only if it is flat over S and its geometric fibers*

$$X \otimes_S \operatorname{Spec} \kappa(s)$$

are regular schemes. More generally, X is smooth over S in a neighborhood of $x \in X$ (one then says that X is smooth over S at x) if and only if X is flat over S at x and X_s is smooth over $\kappa(s)$ at x; that is,

$$X_s \otimes_{\kappa(s)} \overline{\kappa(s)}$$

is regular at the points (or merely at one point) lying over x.

(ii) *Suppose that S, and hence X, is locally noetherian. Let $x \in X$, and let $s \in S$ be its image. The smoothness of X over S at x can then be detected from the local homomorphism*

$$A = \mathscr{O}_{S,s} \longrightarrow B = \mathscr{O}_{X,x}$$

of noetherian local rings (or even from the local homomorphism $\widehat{A} \to \widehat{B}$ of their completions), by the following characteristic property: B is flat over A, and $B \otimes_A k$ is geometrically regular over k, the residue field of A. Equivalently, for every finite extension k' of k,

$$(B \otimes_A k) \otimes_k k' = B \otimes_A k'$$

is a regular semilocal ring. When the residue-field extension $k(B)/k(A)$ is trivial, these conditions are also equivalent to the following one: $\widehat{B}$ is isomorphic, as an $\widehat{A}$-algebra, to an algebra of formal power series

$$\widehat{A}[[t_1, \ldots, t_n]].$$

Thus, from the "formal" point of view, the structure of X over S is that of the standard affine space $S[t_1, \dots, t_n]$ over S.

Proposition 1.3. — *Let X be a prescheme locally of finite presentation over S. Then:*

(i) *The following conditions are equivalent:*

(a) *X is unramified over S.*

(b) *The diagonal morphism $X \to X \times_S X$ is an open immersion.*

(c) $\Omega^1_{X/S} = 0$.

(d) *The geometric fibers of X/S are discrete and reduced; that is, they are isomorphic to disjoint sums of copies of the base field.*

(e) *For every $s \in S$, the fiber*

$$X \otimes_S \operatorname{Spec} \kappa(s) = X_s$$

is unramified over $\kappa(s)$, or equivalently is isomorphic to a disjoint sum of spectra of finite separable extensions of $\kappa(s)$.

(ii) *There are analogous pointwise conditions for X to be unramified over S at a given point x, that is, in a neighborhood of that point. For example, it is necessary and sufficient that $\mathscr{O}_{X_s,x}$ be a finite separable extension of $\kappa(s)$. In terms of the local homomorphism*

$$A = \mathscr{O}_{S,s} \longrightarrow B = \mathscr{O}_{X,x},$$

this says that $B \otimes_A k$ is a finite separable extension of the residue field k of A; equivalently,

$$\mathfrak{r}(A)B = \mathfrak{r}(B) \quad \textit{and} \quad k(B)/k(A) \textit{ is finite separable.}$$

When A, and hence B, is noetherian and $k(A) \xrightarrow{\sim} k(B)$, this also means that $\widehat{A} \to \widehat{B}$ is surjective.

Thus, from the "formal" point of view, saying that X is unramified over S means that X is essentially a subprescheme of S. Observe also that conditions (d) and (e) are expressed solely in terms of the fibers of X/S.

Proposition 1.4. — *Let X be a prescheme locally of finite presentation over S. Then X is étale over S if and only if it is smooth and unramified over S (which is immediate from the definition), so the criteria in 1.2 and 1.3 apply. In particular:*

(i) *X is étale over S if and only if it is flat and unramified over S. There is an analogous local criterion for X to be étale over S at a given point x.*

(ii) *Suppose in addition that S, and hence X, is locally noetherian. Whether X is étale over S at x can then be detected from the local homomorphism*

$$A = \mathscr{O}_{S,s} \longrightarrow B = \mathscr{O}_{X,x}$$

(and even from $\widehat{A} \to \widehat{B}$) by the following characteristic property: B is flat over A, and $B \otimes_A k$ is a finite separable extension of k, the residue field[1] *of A. When $k(A) \xrightarrow{\sim} k(B)$, this condition simply says that $\widehat{A} \to \widehat{B}$ is an isomorphism.*

[1] Editor's note: "residue extension" was replaced by "residue field."

Thus, from the "formal" point of view, saying that X is étale over S simply means that X is locally isomorphic to S.

Remark 1.5. — When X is locally of finite presentation over S and a point $x \in X$ is given, the fact that X is smooth (respectively, unramified, respectively, étale) over S at x, that is, in a neighborhood of x, can also be detected from the functor

$$X : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

by the following property. For every S-scheme S' that is the spectrum of a local ring, every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal, and every S-morphism $u_0 : S'_0 \to X$ sending the closed point s' of S' to x, there exists at least one (respectively, at most one, respectively, exactly one) S-morphism $u : S' \to X$ extending u_0.

This statement shows in particular that, in Definition 1.1, one may restrict to local schemes S', provided the functor under consideration is representable by a prescheme locally of finite presentation over S. Moreover, when S is locally noetherian, one may even restrict the preceding pointwise criterion to local artinian schemes S', and the same restriction may therefore be made in Definition 1.1, provided that S is locally noetherian and that the functor in question is represented by a prescheme locally of finite type over S.

Remark 1.6. — Of course, given a morphism $f : X \to Y$ of preschemes, one says that f is smooth (respectively, unramified, respectively, étale) if f makes X a smooth (respectively, unramified, respectively, étale) Y-prescheme. When X and Y are S-preschemes and f is an S-morphism of finite presentation, these properties of f can be expressed immediately in terms of the morphism of functors

$$F, G : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

defined by f. Namely, f is smooth (respectively, unramified, respectively, étale) if, for every S-prescheme S' that is affine in the absolute sense and every subscheme $S'_0 \subset S'$ defined by a nilpotent ideal $\mathscr{J}$, the map

$$F(S') \longrightarrow F(S'_0) \times_{G(S'_0)} G(S')$$

deduced from the commutative square

$$\begin{tikzcd} F(S') \arrow[r] \arrow[d] & G(S') \arrow[d] \\ F(S'_0) \arrow[r] & G(S'_0) \end{tikzcd}$$

is surjective (respectively, injective, respectively, bijective). Equivalently, for every commutative square of morphisms of functors over S, where S' and S'_0 are as above and $i : S'_0 \to S'$ is the canonical immersion,

$$\text{(Q)} \qquad \begin{tikzcd} S'_0 \arrow[r, "i"] \arrow[d, "u_0"'] & S' \arrow[d, "v"] \\ F \arrow[r, "f"'] & G, \end{tikzcd}$$

there exists at least one (respectively, at most one, respectively, exactly one) morphism

$$\begin{tikzcd} & S' \arrow[dl, "u"'] \\ F & \end{tikzcd}$$

that makes the two corresponding triangles commute:

$$u_0 = u \circ i \qquad \text{and} \qquad v = f \circ u.$$

This property for a homomorphism of functors $f : F \to G$ over S retains its meaning even when the functors under consideration are not representable. If it holds, the homomorphism $f : F \to G$ is said to be *formally smooth* (respectively, *formally unramified*, respectively, *formally étale*). Notice that this property depends only on the homomorphism of the underlying functors

$$(\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens})$$

defined by F and G (cf. Exposé I, 1.4.1), and not on the structure morphisms $F \to S$ and $G \to S$. An equivalent formulation of the preceding definition, somewhat easier to use in applications, is the following: a morphism of functors $F \to G$ over S is formally smooth (respectively, formally unramified, respectively, formally étale) if, for every S' over S and every morphism $S' \to G$, the functor

$$F' = F \times_G S'$$

over S' is formally smooth (respectively, formally unramified, respectively, formally étale).

Remark 1.7. — Under the hypotheses of 1.6, suppose that we are given a "point of F with values in a field k," that is, an element $x \in F(\operatorname{Spec} k)$, or equivalently a morphism

$$x : \operatorname{Spec} k \longrightarrow F.$$

We likewise say that f is *formally smooth* (respectively, *formally unramified*, respectively, *formally étale*) *at* x if the condition in the preceding definition is satisfied whenever S' is a local scheme and the morphism $S'_0 \to F$ in diagram (Q) above is "compatible with the marked points" in the following sense. Let k' denote the residue field of S' and S'_0 at their closed point. Then the diagram

$$\begin{tikzcd} \operatorname{Spec}(k') \arrow[d, "j"'] & \operatorname{Spec}(k) \arrow[d, "x"] \\ S'_0 \arrow[r, "u_0"'] & F \end{tikzcd}$$

in which $j : \operatorname{Spec}(k') \to S'_0$ is the canonical immersion, must admit a completion to a commutative diagram

$$\begin{tikzcd} & \operatorname{Spec}(k'') \arrow[dl] \arrow[dr] & \\ \operatorname{Spec}(k') \arrow[d] & & \operatorname{Spec}(k) \arrow[d] \\ S'_0 \arrow[rr] & & F, \end{tikzcd}$$

where k'' is a field.When F and G are represented by S-preschemes X and Y, and the S-morphism $f : X \to Y$ is locally of finite presentation, this condition says precisely, by 1.5, that $f : X \to Y$ is smooth at the point of X that is the image of $\operatorname{Spec}(k)$ under $x : \operatorname{Spec}(k) \to X$.

Remark 1.8. — When the condition in the preceding definition is satisfied after restricting to local artinian schemes S', we say that $F \to G$ is *infinitesimally smooth* (respectively, *infinitesimally unramified*, respectively, *infinitesimally étale*) *at* x. We say that $F \to G$ is infinitesimally smooth (respectively, infinitesimally unramified, respectively, infinitesimally étale) if it has the corresponding property at every point x; equivalently, if the condition considered in 1.6 is satisfied whenever S' is local artinian.

This variant of the preceding notions is technically useful. Since it is weaker, it is often easier to verify, while it is nevertheless frequently sufficient to imply the stronger condition; this is the case, for example, when $F \to G$ is a morphism locally of finite presentation and G is represented by a locally noetherian prescheme.

Proposition 1.9. — *Let $f : X \to S$ be a smooth morphism of preschemes. Then $\Omega^1_{X/S}$ is a locally free module of finite type on X, and its rank at a point $x \in X$ is equal to the dimension of the fiber X_s, where $s = f(x)$, in a neighborhood of x.*

This dimension is called the *relative dimension of X over S at x*. Notice that it is zero (when f is smooth at x) if and only if f is étale at x. It can also be computed directly in terms of the functor F represented by X, as follows. Let ξ be a point of F with values in a field k, "localized at x," that is, a morphism

$$\operatorname{Spec}(k) \longrightarrow F = X$$

whose image is x. Consider the algebra

$$D(k) = k[t]/(t^2)$$

of dual numbers over k, regarded as an S-prescheme, and the map

$$F(D(k)) \longrightarrow F(k)$$

induced by the augmentation $D(k) \to k$. Finally, let $F(D(k), \xi)$ be the inverse image of $\xi \in F(k)$ under this map. This set is naturally endowed with the structure of a vector space over k. In fact, it is the vector-space dual of

$$\Omega^1_{X/S} \otimes_{\mathscr{O}_{X,x}} k,$$

and its dimension is the relative dimension of X over S at x.

To describe the vector-space law on $F(D(k), \xi)$ explicitly, it is useful to introduce, more generally and as in Exposé II, for every vector space V over k, the algebra

$$D_k(V) = k + V,$$

where V is an ideal of square zero, and to regard $F(D_k(V), \xi)$, the inverse image of ξ under

$$F(D_k(V)) \longrightarrow F(k),$$

as a covariant functor of V with values in (**Ens**). It is then enough that this functor commute with products of two factors. This means that F carries certain amalgamated sums of a very special kind to fiber products; compare Exposé II. This condition always holds when F is representable. It follows that the sets $F(D_k(V), \xi)$, and in particular

$F(D(k), \xi)$, are endowed with vector-space structures over k. Thus one can define the relative dimension of F over X at the "point" ξ under conditions appreciably more general than the representability of F.

In the present Exposé, the fact that certain functors to be described explicitly are representable by preschemes smooth over S will be used chiefly through the following result. It will be our technical means of passing from constructions over the completion of the local ring A of a point s of a noetherian scheme S to a local ring A' over A that is étale over A, and hence, in particular, of passing from s to neighboring points.

Proposition 1.10. — *Let $f : X \to S$ be a smooth morphism of preschemes, let $s \in S$, and let $x \in X$ lie over s, with $\kappa(x)$ a finite separable extension of $\kappa(s)$. Then there exists a subscheme $S' \subset X$, étale over S, passing through x. Thus one can find an étale morphism $S' \to S$, a point $s' \in S'$ above s whose residue-field extension is $\kappa(x)/\kappa(s)$, and an S-morphism $S' \to X$ carrying s' to x.*

To construct S', simply choose sections $g_1, \dots, g_n$ of $\mathscr{O}_X$ on a neighborhood of x whose images at x form a regular system of parameters for the local ring of the fiber X_s at x. The subscheme S' defined by the g_i is then étale over S at x, and, after shrinking S', it is étale over S.

We shall use 1.10 when $\kappa(x) = \kappa(s)$, that is, when x is rational over $\kappa(s)$, or equivalently when x may be regarded as a section of X_s over $\operatorname{Spec} \kappa(s)$. In that case, 1.10 has the character of an extension theorem for sections, after an étale extension of the base. It takes a particularly simple form in the following special case.

Corollary 1.11. — *Under the hypotheses of 1.10, suppose that S is the spectrum of a henselian local ring and that $\kappa(x) = \kappa(s)$. Then there exists a section of X over S passing through x. It is uniquely determined if X is in fact étale over S at x.*

Indeed, since S is henselian, under the hypotheses of 1.10 the prescheme S' contains an open subscheme that is finite over S and whose fiber at s is reduced to x. Since this open subscheme is étale over S, it is isomorphic to S, which proves the assertion. Notice that when S is the spectrum of a complete local ring, 1.10 or 1.11 is more or less equivalent to the classical "Hensel's lemma," and one sometimes refers to the result by that name.

2. Examples of Formally Smooth Functors from the Theory of Groups of Multiplicative Type

We shall interpret, in the language introduced in the preceding number, the results stated in Exposé IX, no. 3, concerning infinitesimal extensions of a homomorphism from a group of multiplicative type. Those results are consequences of the vanishing of the Hochschild cohomology of such a group, established in Exposé I.

Proposition 2.1. — *Let S be a prescheme, let H be a group of multiplicative type over S, and let G be an S-group prescheme smooth over S. Consider the functor over S*

$$M_H = \operatorname{Hom}_{S\text{-gr}}(H, G).$$

Its value at an S' over S is the set of homomorphisms of S'-groups from $H_{S'}$ to $G_{S'}$. This functor is formally smooth over S.

Cf. Exposé IX, 3.6. More generally:

Corollary 2.2. — *Let S and G be as above, and consider a homomorphism $u : H_1 \to H_2$ of group schemes of multiplicative type over S. With the preceding notation this induces* *a morphism of functors over S*

$$M_{H_2} \longrightarrow M_{H_1}, \qquad M_{H_i} = \mathrm{Hom}_{S\text{-gr}}(H_i, G),$$

given by $w \mapsto w \circ u$. This morphism is formally smooth.

Indeed, by the definitions in 1.6, this is equivalent to the following assertion. Suppose that S is affine and that $S_0 \subset S$ is a subscheme defined by a nilpotent ideal. Let

$$v : H_1 \longrightarrow G$$

be a homomorphism of S-groups, and let

$$w_0 : (H_2)_{S_0} \longrightarrow G_{S_0}$$

be a homomorphism of S_0-groups such that

$$w_0 u_{S_0} = v_{S_0}.$$

Then there exists a homomorphism of S-groups

$$w : H_2 \longrightarrow G$$

extending w_0 and satisfying $wu = v$.

To see this, first extend w_0 to a homomorphism of S-groups

$$w' : H_2 \longrightarrow G,$$

which is possible by 2.1. Set

$$v' = w'u : H_1 \longrightarrow G.$$

By the hypothesis on w_0, one has $v'_{S_0} = v_{S_0}$. Consequently, by Exposé VIII, 3.6, there exists an element $g \in G(S)$ whose image in $G(S_0)$ is the identity element and such that

$$v = \mathrm{int}(g)v'.$$

It follows that

$$v = \big(\mathrm{int}(g)w'\big)u,$$

and it is therefore enough to take

$$w = \mathrm{int}(g)w'.$$

Corollary 2.3. — *With the notation of 2.1, regard $M_H = M$ as a functor with operator group G, where G acts by*

$$(v, g) \longmapsto \mathrm{int}(g) \circ v.$$

Then the corresponding morphism

$$R : G \times_S M \longrightarrow M \times_S M,$$

defined by

$$(g, v) \longmapsto \big(\mathrm{int}(g) \circ v, v\big),$$

is formally smooth.

After a base change $S' \to S$, this is equivalent to the following assertion.

Corollary 2.4. — *With the notation of 2.1, let $v_1, v_2 : H \to G$ be two morphisms of S-groups, and let* $\mathrm{Transp}(v_1, v_2)$ *be the subfunctor of G consisting of the g such that*

$$\mathrm{int}(g) \circ v_1 = v_2.$$

Then this functor is formally smooth over S. In particular, if $v_1 = v_2 = v$, the functor $\mathrm{Centr}(v)$*, the subgroup of G consisting of the h such that* $\mathrm{int}(h)v = v$*, is formally smooth over S.*

N.B. The pair (v_1, v_2) may be regarded as a section over S of the second factor in the codomain of the morphism R in 2.3, and $\mathrm{Transp}(v_1, v_2)$ as the inverse-image functor of that section under R.

The assertion in 2.4 is itself equivalent to the following. Suppose that S is affine and that $S_0 \subset S$ is a subscheme defined by a nilpotent ideal. If $g_0 \in G(S_0)$ satisfies

$$\mathrm{int}(g_0)(v_1)_{S_0} = (v_2)_{S_0},$$

then g_0 extends to an element $g \in G(S)$ such that

$$\mathrm{int}(g)v_1 = v_2.$$

To prove this, first extend g_0 to a section g' of G over S, which is possible because G is smooth over S, and put

$$v_2' = \mathrm{int}(g')v_1.$$

The morphisms v_2 and v_2' have the same restriction over S_0. Hence, by Exposé IX, 3.6, there exists $g'' \in G(S)$ inducing the identity section over S_0 and such that

$$v_2 = \mathrm{int}(g'')v_2'.$$

Therefore

$$v_2 = \mathrm{int}(g'')\,\mathrm{int}(g')v_1 = \mathrm{int}(g''g')v_1,$$

and it is enough to take $g = g''g'$.

Proposition 2.1 bis. — *Let S be a prescheme and let G be an S-group prescheme smooth over S. Consider the functor*

$$M : (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens}),$$

defined by

$$M(S') = \left\{ \begin{array}{c} \text{multiplicative-type subgroups} \\ \text{of } G_{S'} \end{array} \right\}.$$

Then M is formally smooth over S.

Cf. Exposé IX, 3.6 bis.

Corollary 2.2 bis. — *Let n be an integer, and consider the morphism of functors*

$$\varphi_n : M \longrightarrow M$$

defined by

$$\varphi_n(H) = {}_nH = \ker\bigl(n \cdot \mathrm{id}_H\bigr).$$

Then φ_n is formally smooth. For every integer $p > 0$, let M_p denote the subfunctor of M for which $M_p(S')$ is the set of multiplicative-type subgroups H of $G_{S'}$ satisfying ${}_pH = H$. Then the morphism induced by φ_n

$$M_{np} \longrightarrow M_n$$

is formally smooth.

The second assertion is trivially contained in the first and is included only for the convenience of a later reference. The proof of the first assertion is analogous to that of 2.2, now invoking Exposé IX, 3.6 bis.

Corollary 2.3 bis. — *With the notation of 2.1 bis, regard M as a functor with operator group G, where G acts by*

$$(g, H) \longmapsto \operatorname{int}(g)(H).$$

Then the corresponding morphism

$$R : G \times_S M \longrightarrow M \times_S M,$$

defined by

$$(g, v) \longmapsto (\operatorname{int}(g)v, v),$$

is formally smooth.

This is equivalent to the following assertion.

Corollary 2.4 bis. — *With the notation of 2.1 bis, let H_1 and H_2 be two multiplicative-type subgroups of G, and let $\operatorname{Transp}_G(H_1, H_2)$ be the subfunctor of G consisting of the g such that*

$$\operatorname{int}(g)H_1 = H_2.$$

Then this functor is formally smooth over S. In particular, if $H_1 = H_2 = H$, the subfunctor $\operatorname{Norm}_G(H)$ of G, the normalizer of H, is formally smooth over S.

The proof is analogous to that of 2.4, again invoking Exposé IX, 3.6 bis.

Proposition 2.5. — *Let S be a prescheme, let G be an S-group prescheme smooth over S, let K be an S-group subprescheme that is smooth over S or of multiplicative type, let H be an S-group prescheme of multiplicative type, and let $u : H \to G$ be a homomorphism of S-groups. Denote by $\operatorname{Transp}_G(u, K)$ the subfunctor of G whose value at an S' over S consists of the elements $g \in G(S')$ for which*

$$\operatorname{int}(g)u_{S'} : H_{S'} \longrightarrow G_{S'}$$

factors through $K_{S'}$. Then this functor is formally smooth over S.

The proof is analogous to that of 2.2, invoking Exposé IX, 3.6 and Exposé X, 2.1, the latter in the case where K is of multiplicative type.

3. Auxiliary Representability Results

Proposition 3.1. — *Let $F \to S$ be a functor over the prescheme S. The following conditions are equivalent:*

(i) *F is representable and $F \to S$ is an open immersion (one also says simply that $F \to S$ is an open immersion).*

(ii) *F is a sheaf for the faithfully flat quasi-compact topology, "commutes with filtered direct limits of rings," and $F \to S$ is a monomorphism. Moreover, the following condition holds: for every local prescheme S' over S, with residue field k, and every S-morphism*

$$\operatorname{Spec}(k) = S'_0 \longrightarrow F,$$

there exists an S-morphism $S' \to F$ extending it.

(iii) *Suppose that S is locally noetherian. Then F is a sheaf for the faithfully flat quasi-compact topology, commutes with filtered direct limits of rings, commutes with adic inverse limits of local artinian rings, and $F \to S$ is a monomorphism and is infinitesimally étale (cf. 1.8).*

Let us first make two points of terminology precise.

Remark 3.2. — A functor[2] F over S is said to *commute with direct limits of rings* (understood to mean filtered direct limits) if the following holds. Let $(S'_i)_{i \in I}$ be any filtered inverse system of affine S-schemes lying over an affine open subscheme of S, and let A'_i be their rings. Then the natural homomorphism

$$(*) \qquad \varinjlim_i F(S'_i) \longrightarrow F(S'), \qquad S' = \operatorname{Spec}(A'), \quad A' = \varinjlim_i A'_i$$

is bijective.

Notice that S' is precisely the inverse limit of the S'_i in the category of preschemes, and even in the category of all ringed spaces. Thus the condition under consideration has the character of a right-exactness condition—commutation with certain direct limits in $(\mathbf{Sch})^\circ/S$—just as does the condition of being a sheaf for a topology.

Care must be taken that this condition is essentially relative: it involves the morphism $F \to S$, not merely the functor

$$F : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens}).$$

More precisely, in $(*)$, $F(S'_i)$ and $F(S')$ mean $\operatorname{Hom}_S(S'_i, F)$ and $\operatorname{Hom}_S(S', F)$. Consequently, when F is representable, the condition says that F is locally of finite presentation over S. We used several times in the preceding two exposés the fact that a functor represented by an S-prescheme locally of finite presentation commutes with direct limits of rings.

Remark 3.3. — A functor F over S is said to *commute with adic inverse limits of local artinian rings* if the following condition holds. For every S' over S that is the spectrum of a complete noetherian local ring A', set

$$S'_n = \operatorname{Spec}\big(A'/\operatorname{rad}(A')^{n+1}\big).$$

Then the natural map

$$(**) \qquad F(S') \longrightarrow \varprojlim_n F(S'_n)$$

is required to be bijective.

This condition has the character of a left-exactness condition and is satisfied whenever F is representable. In contrast to commutation with direct limits of rings, it is intrinsic to F as an object of $\widehat{\mathbf{Sch}}$; that is, it does not involve the morphism $F \to S$.

Remark 3.4. — Let F be a functor over S that is a sheaf for the Zariski topology, or, as one also says, is "of local nature." It is enough for this that F be a sheaf for a finer topology, such as the faithfully flat quasi-compact topology. Let (S_i) be an open covering of S. A straightforward gluing argument shows that F is representable if and only if each

$$F_i = F \times_S S_i$$

[2]Editor's note: here and in Remark 3.3, the functor is contravariant.

is representable; this allows one, for example, to reduce to the case where S is affine.

Suppose that the functor F, of local nature, commutes with direct limits of rings. Then F is representable if and only if its restriction to the category of preschemes locally of finite presentation over S is representable. Necessity was noted in 3.2. Sufficiency amounts to the following assertion: if X is a prescheme locally of finite presentation over S, and $X \to F$ is a morphism such that, for every S' locally of finite presentation over S, the induced map

$$\mathrm{Hom}_S(S', X) \longrightarrow \mathrm{Hom}_S(S', F)$$

is bijective, then $X \to F$ is an isomorphism. This follows readily from the fact that X and F are both functors of local nature that commute with direct limits of rings.

Proof of Proposition 3.1. The implications (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii) are immediate. We prove the converse implications.

Assume (ii). Let U be the set of points $s \in S$ for which the canonical monomorphism

$$\operatorname{Spec} \kappa(s) \longrightarrow S$$

factors through F. By the last condition in (ii), for every $s \in U$, the canonical map $\operatorname{Spec} \mathscr{O}_{S,s} \to S$, equivalently its factorization through U, factors through F. The local ring $\mathscr{O}_{S,s}$ is the direct limit of the rings of affine neighborhoods of s. Since F commutes with direct limits of rings, for every $s \in U$there is an open neighborhood U^s such that the canonical immersion $U^s \to S$ factors through F. It follows that $U^s \subset U$, and hence that U is open.

Because $F \to S$ is a monomorphism and F is of local nature, the S-morphisms $U^s \to F$ agree on the intersections $U^s \cap U^{s'}$, for $s, s' \in U$, and therefore glue to an S-morphism $U \to F$. It remains to prove that this morphism is an isomorphism, or equivalently that every S-morphism $S' \to F$ factors uniquely through U, where S' is an S-prescheme. Since both $F \to S$ and $U \to S$ are monomorphisms, it is equivalent to say that the structure morphism $S' \to S$ factors through U.

By locality, we are reduced to the case in which S' is the spectrum of a field and hence has a single point s'. Let s be the point of S below s'. We claim that the given S-morphism $S' \to F$ factors through

$$\operatorname{Spec} \kappa(s) = S_0 \longrightarrow F.$$

This implies $s \in U$ and proves the assertion. Since $S' \to S_0$ is an fpqc covering and F is a sheaf for the fpqc topology, it is enough that the two composites

$$S'' = S' \times_{S_0} S' \rightrightarrows S' \longrightarrow F$$

coincide. They do coincide because $F \to S$ is a monomorphism. This proves (ii) $\Rightarrow$ (i).

Now assume (iii), with S locally noetherian. It is enough to prove the last condition of (ii); by the preceding proof, it is enough to do so when

$$S' = \operatorname{Spec} \mathscr{O}_{S,s}$$

for some $s \in S$. Put

$$A = \mathscr{O}_{S,s}, \qquad A_n = A/\mathfrak{m}^{n+1}, \qquad S_n = \operatorname{Spec}(A_n).$$

Because $F \to S$ is infinitesimally smooth, the given morphism $S'_0 \to F$ extends to morphisms $S_n \to F$. Since $F \to S$ is a monomorphism, these morphisms form an element of

$\varprojlim_n F(S_n)$. Since F commutes with adic inverse limits of local artinian rings, they arise from a morphism

$$\operatorname{Spec}(\widehat{A}) = \widehat{S}' \longrightarrow F.$$

Finally, $F \to S$ is a monomorphism, F is an fpqc sheaf, and $\widehat{S}' \to S'$ is an fpqc covering. The last morphism therefore descends to $S' \to F$, completing the proof of (iii) $\Rightarrow$ (ii). ☐

Proposition 3.5. — *Let S be a locally noetherian prescheme, let $F \to S$ be a functor over S, and let $(X_i, u_i)_{i \in I}$ be a family of S-morphisms $u_i : X_i \to F$, where the X_i are preschemes locally of finite type over S. Suppose that the following conditions hold:*

(a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with direct limits of rings, and commutes with adic inverse limits of local artinian rings.*

(b) *The morphisms $u_i : X_i \to F$ are monomorphisms and are infinitesimally étale (cf. 1.8).*

(c) *The family (u_i) is "set-theoretically surjective."*

Under these conditions, F is representable by a prescheme locally of finite type over S. Moreover, the u_i are open immersions and the family (X_i) is an open covering of F.

Remark 3.6. — Proceeding as at the end of Remark 1.7, one defines, for every functor

$$F : (\mathbf{Sch})^\circ \longrightarrow (\mathbf{Ens}),$$

an "underlying set" $\operatorname{ens}(F)$: it is the quotient of the set of field-valued points of F by the equivalence relation described in 1.7. When F is represented by X, this recovers the underlying set of X.

Plainly, $\operatorname{ens}(F)$ depends functorially on F. Thus, if $G \to F$ is a morphism of functors, we say that it is *set-theoretically surjective* if the induced map

$$\operatorname{ens}(G) \longrightarrow \operatorname{ens}(F)$$

is surjective. Equivalently, every point of F with values in a field k "comes from" a point of G with values in a suitable extension of k. This definition extends immediately to a family of morphisms $G_i \to F$, and thereby specifies the meaning of condition (c).

Proof of Proposition 3.5. For $(i, j) \in I \times I$, put

$$X_{i,j} = X_i \times_F X_j$$

and consider the projections

$$v_{i,j} : X_{i,j} \longrightarrow X_i, \qquad w_{i,j} : X_{i,j} \longrightarrow X_j.$$

We claim that these morphisms are representable by open immersions. Apply criterion 3.1(iii). The functor $X_{i,j}$ satisfies the three exactness conditions: it is a sheaf, it commutes with direct limits of rings, and it commutes with adic inverse limits of local artinian rings. Indeed, F, X_i, and X_j satisfy these conditions, and the conditions are stable under finite inverse limits, in particular under fiber products. Since every $X_i \to F$ is a monomorphism, both projections are monomorphisms, each being obtained from one of these maps by base change. Finally, being infinitesimally étale is also preserved by base change. Thus the hypotheses of 3.1(iii) are satisfied.

We may now use the X_i, $X_{i,j}$, $v_{i,j}$, and $w_{i,j}$ to construct in the usual way an S-prescheme X, in which the X_i are identified with open subschemes, the $X_{i,j}$ with the intersections $X_i \cap X_j$, and $v_{i,j}$, $w_{i,j}$ with the canonical immersions.

Put

$$Y = \coprod_i X_i.$$

Then X is also the quotient of Y by the equivalence relation

$$R = \coprod_{i,j} X_{i,j} \rightrightarrows Y,$$

whose two projections v, w are defined by the $v_{i,j}$ and $w_{i,j}$, respectively.

More precisely, because F is an fpqc sheaf, the morphisms $u_i : X_i \to F$ induce a morphism $u : Y \to F$, and

$$R = Y \times_F Y$$

is exactly the equivalence relation defined by u. The quotient $X = Y/R$ is also a quotient in the category of fpqc sheaves, and even in the category of Zariski sheaves; this follows directly from the definitions of quotient and sheaf. Consequently, u factors uniquely through a morphism

$$\overline{u} : X \longrightarrow F,$$

and this morphism is a monomorphism.

It remains to show that $\overline{u}$ is an isomorphism. Since F is of local nature, we may assume that S is affine. Since F also commutes with direct limits of rings, it is enough, by 3.4, to verify that, for every affine T of finite type over S, every morphism $T \to F$ factors through X. Consider

$$G = X \times_F T \longrightarrow T.$$

It is enough to prove that this morphism is an isomorphism. Since T is noetherian, the argument above shows that it is an open immersion. Here one uses that $X \to F$ is infinitesimally étale, which follows immediately because the induced morphisms $u_i : X_i \to F$ are infinitesimally étale. By hypothesis, $X \to F$ is set-theoretically surjective, and this condition is stable under base change. Hence $G \to T$ is set-theoretically surjective. Being also an open immersion, it is an isomorphism. □

Proposition 3.7. — *Let S be a locally noetherian prescheme, let I be a directed partially ordered index set, let $(T_i)_{i\in I}$ be a projective system of S-preschemes locally of finite type, let*

$$T = \varprojlim_{i\in I} T_i$$

be the projective-limit functor, let F be a functor over S, and let $u : F \to T$ be an S-morphism. Suppose that the following conditions hold:

(a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, and commutes with adic projective limits of local artinian rings.*

(b) *The morphism $u : F \to T$ is a monomorphism.*

b' *The morphism $u : F \to T$ is infinitesimally étale.*

(c) *For every point ξ of F with values in the spectrum of a field k, let $\xi_i \in T_i(\operatorname{Spec} k)$ be its image and let t_i be the corresponding point of T_i. There exists $i \in I$ such that, for every $j \geq i$, the transition morphism*

$$p_{i,j} : T_j \longrightarrow T_i$$

is étale at t_j.

(d) *For every prescheme X locally of finite type over S and every S-morphism $X \to F$, the set of points $x \in X$ at which this morphism is infinitesimally étale is open.*

Under these conditions, F is representable by a prescheme locally of finite type over S.

Let us note at once that, in the case to be considered in the next section, conditions (c) and (d) will be verified by means of the following corollary.

Corollary 3.8. — *Assuming (a), (b), and (b′), conditions (c) and (d) are implied by the following conditions:*

c' *The T_i are smooth over S, and the transition morphisms $p_{i,j} : T_j \to T_i$ are smooth.*

d' *For every point ξ of F with values in the spectrum of a field k, let $t_i(\xi)$ be the point of T_i defined by ξ, let $d_i(\xi)$ be the relative dimension of T_i over S at $t_i(\xi)$, and put*

$$d(\xi) = \sup_i d_i(\xi).$$

Then:

1° *for every ξ as above, $d(\xi) < +\infty$; and*

2° *for every prescheme X locally of finite type over S and every S-monomorphism $v : X \to F$, the function*

$$x \longmapsto d(\xi_x)$$

on X is locally constant, where, for $x \in X$, ξ_x denotes the point of F with values in $\kappa(x)$ induced by v.

Proof of Proposition 3.7. Work under condition (c). Let t be the element of $\operatorname{ens}(F)$ defined by ξ (cf. 3.6), and put

$$\mathscr{O}_t = \varinjlim_i \mathscr{O}_{T_i,t_i}.$$

Using condition (c), one sees readily that $\mathscr{O}_t$ is a noetherian local ring (EGA 0_{IV}, 10.3.1.3). Its residue field $\kappa(t)$ is the inductive limit of the residue fields $\kappa(t_i)$, and k is an extension of it. If η denotes $\operatorname{Spec} \kappa(t)$, and ξ denotes $\operatorname{Spec} k$, there is a commutative diagram

$$\begin{tikzcd} \xi \arrow[r] \arrow[d] & F \arrow[d] \\ \eta \arrow[r] & T, \end{tikzcd}$$

whose definition is evident and is left to the reader. Since $\xi \to \eta$ is a covering for the fpqc topology, F is a sheaf for that topology by (a), and $F \to T$ is a monomorphism by (b), the morphism $\eta \to T$ factors uniquely through a morphism $\eta \to F$. We now record the following lemma.

Lemma 3.9. — *Under the hypotheses of 3.7, let*

$$T' = \operatorname{Spec}(A) \quad \textit{and} \quad T'_0 = \operatorname{Spec}(k(A))$$

with T' a noetherian local scheme, and let $i : T'_0 \to T'$ be the canonical immersion. Suppose that a commutative square of morphisms is given:

$$\begin{tikzcd} T'_0 \arrow[r] \arrow[d, "i"'] & F \arrow[d] \\ T' \arrow[r] & T. \end{tikzcd}$$

Then there exists a unique T-morphism $T' \to F$.

The proof is the proof of 3.1, (iii) ⇒ (ii), in which the representability of the S occurring in that statement was not used. One uses the facts that F is a sheaf for the fpqc topology, that it commutes with adic projective limits of local artinian rings, here the rings

$$A/\operatorname{rad}(A)^{n+1},$$

and that $F \to T$ is both a monomorphism and infinitesimally étale.

We apply Lemma 3.9 with

$$T' = \operatorname{Spec}(\mathscr{O}_t), \qquad T'_0 = \operatorname{Spec}(\kappa(t)).$$

We have just constructed $T'_0 \to F$, and the canonical homomorphisms

$$\mathscr{O}_{T_i,t_i} \longrightarrow \mathscr{O}_t$$

define a projective system of morphisms $\operatorname{Spec}(\mathscr{O}_t) \to T_i$, hence a canonical morphism $\operatorname{Spec}(\mathscr{O}_t) \to T$. The corresponding square (x) plainly commutes: by the definition of T, it is enough to check this after replacing T by T_i. We therefore obtain a unique T-morphism

$$v' : T' = \operatorname{Spec}(\mathscr{O}_t) \longrightarrow F.$$

Since F commutes with inductive limits of rings, this morphism factors, for sufficiently large i, through

$$v'_i : T'_i = \operatorname{Spec}(\mathscr{O}_{T_i,t_i}) \longrightarrow F.$$

For such an i, one has

$$T' \xrightarrow{\sim} T'_i, \qquad \text{that is,} \qquad \mathscr{O}_{T_i,t_i} \xrightarrow{\sim} \mathscr{O}_t.$$

Indeed, $T' \to T'_i$ is faithfully flat and quasi-compact, hence an effective epimorphism, so it is enough to show that it is a monomorphism. Suppose that two morphisms $R \rightrightarrows T'$, with R representable, have the same composite with $T' \to T'_i$. Their composites with $T' \to T$, which factors through $T' \to T'_i$, are then equal. Hence their composites with $T' \to T_j$, and therefore with $T' \to T'_j$, are equal for every j. The two original morphisms are equal because T' is the projective limit of the T'_j in the category (**Sch**).

Consider the diagram

$$\begin{tikzcd} T' \arrow[r, "v'"] \arrow[d] & F \arrow[d] \\ T'_i \arrow[r] \arrow[ur, "v'_i"'] & T_i, \end{tikzcd}$$

where the unlabeled arrows are the evident ones. The square and the upper triangle commute. Since $T' \to T'_i$ is an epimorphism, the lower triangle also commutes.

The ring $\mathscr{O}_{T_i,t_i}$ is the inductive limit of the rings of affine open neighborhoods of t_i in T_i. Since F commutes with inductive limits of rings, v'_i comes from a morphism

$$v_t : U_t \longrightarrow F$$

defined on an open neighborhood U_t of t_i in T_i. After shrinking this neighborhood if necessary, v_t is a T_i-morphism. Indeed, T_i, being locally of finite type over S, also commutes with inductive limits of rings. The composite of v_t with $F \to T_i$ is a monomorphism, and therefore v_t is a monomorphism.

We claim that v_t is infinitesimally étale at t_i. Since $F \to T$ is infinitesimally étale, this is equivalent to the composite

$$T'_i \xrightarrow{v'_i} F \longrightarrow T$$

being infinitesimally étale at t_i, or to the induced morphism $T'_i \to T$ being infinitesimally étale. Since $T' \xrightarrow{\sim} T'_i$, it is in turn equivalent to $T' \to T$ being infinitesimally étale. This is immediate, because that morphism is the projective limit of the infinitesimally étale morphisms $T'_j \to T_j$.

Apply condition (d), which has not yet been used. It follows that v_t is infinitesimally étale in a neighborhood of t_i. After replacing U_t by a smaller open, we may therefore suppose that v_t is a monomorphism that is infinitesimally étale.

As t varies in $\mathrm{ens}(F)$, the family of morphisms v_t satisfies Proposition 3.5, which gives the conclusion of Proposition 3.7.

Proof of Corollary 3.8. Assume conditions (c') and (d'). With the notation of (d'), $d_i(\xi)$ is constant for all sufficiently large i. Consequently, the relative dimension of T_j over T_i at t_j, which is

$$d_j(\xi) - d_i(\xi),$$

is zero. Thus $T_j \to T_i$ is étale at t_j, which proves condition (c).

The preceding proof now gives, for each $t \in \mathrm{ens}(F)$, an index i, an open neighborhood U_t of t_i, and a morphism $U_t \to F$ that is a monomorphism and is infinitesimally étale at t_i. It remains to show that this morphism is infinitesimally étale throughout a neighborhood of t_i. In condition (d'), 2°, take $X = U_t$. Replacing U_t, if necessary, by the connected component of t_i in U_t, we may suppose that

$$d(\xi_x) = d(\xi_{t_i}) \qquad \text{for every } x \in U_t.$$

Now $d(\xi_{t_i})$ is the relative dimension of T_i over S at t_i. Since that relative dimension remains constant on U_t, we also have, for every $x \in U_t$,

$$d(\xi_x) = \text{the relative dimension of } U_t \text{ over } S \text{ at } x. \tag{*}$$

On the other hand, using the notation of the proof of 3.7, consider any local noetherian prescheme

$$R' = \mathrm{Spec}(A), \qquad R'_0 = \mathrm{Spec}(k(A)).$$

Every morphism $R' \to F$ whose induced point $R'_0 \to F$ represents $t \in \mathrm{ens}(F)$, that is, sends the closed point of R' to t, factors uniquely through T'. The proof is the proof of 3.1 or of Lemma 3.9: one uses that $T' \to F$ is a monomorphism infinitesimally étale at t, and that T' is an fpqc sheaf commuting with adic projective limits.

Apply this result to the morphism

$$\mathrm{Spec}(\mathscr{O}_{U_t,x}) = R' \longrightarrow F$$

induced by v_t, and to the point $t_x \in \mathrm{ens}(F)$ that is the image of the closed point of R', or equivalently the image of x under v_t. We obtain a factorization

$$\mathrm{Spec}(\mathscr{O}_{U_t,x}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{t_x}) \longrightarrow F.$$

The second arrow is infinitesimally étale at t_x. To prove that the composite is infinitesimally étale at x, it is therefore enough to prove the same for the first arrow at x.

By (x) above and formula $(*)$, the first arrow is induced by a local S-homomorphism between the localizations, at x and t_x, of the smooth S-preschemes U_t and U_{t_x}, which have the same relative dimension d over S at those points. By EGA I, 6.5.1, this local homomorphism is induced by an S-morphism

$$w : U \longrightarrow V,$$

where U is an open neighborhood of x in U_t and $V = U_{t_x}$. It remains to prove that w is étale at x. Moreover, U and V carry monomorphisms to F, and, after shrinking U once more, w is an F-morphism. Hence w is a monomorphism. It is now enough to prove the following lemma.

Lemma 3.10. — *Let U and V be two smooth S-preschemes of the same relative dimension d over S, and let $w : U \to V$ be an S-morphism that is a monomorphism. Then w is an open immersion, and hence is étale.*

By SGA 1, I, 5.7, we reduce to the case in which S is the spectrum of a field, which may be assumed algebraically closed. By SGA 1, I, 5.1, it is enough to prove that w is étale, and it is enough to do so at the closed points of U. Let x be such a point and put $y = w(x)$. Choose a regular system of parameters $f_1, \dots, f_d$ of $\mathscr{O}_{V,y}$. Then

$$A = \mathscr{O}_{U,x} / \sum_i f_i \mathscr{O}_{U,x}$$

is the trivial extension of

$$k = \mathscr{O}_{V,y} / \sum_i f_i \mathscr{O}_{V,y};$$

that is, $A \simeq k$. Indeed, since w is a monomorphism, so is the structural morphism

$$\mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(k)$$

obtained from it by base change. Since $\mathscr{O}_{U,x}$ is a regular local ring of dimension d, the f_i form a regular system of parameters of that ring. It follows immediately that w is étale at x. This proves Lemma 3.10, and hence Corollary 3.8.

Corollary 3.11. — *Under the hypotheses of 3.8, for every quasi-compact open subfunctor U of F that is separated over S, there exists $i \in I$ such that, for every $j \geq i$, the morphism*

$$u_j|_U : U \longrightarrow T_j$$

is an open immersion. In particular, if the T_i are quasi-affine over S, then every open subfunctor U of F that is quasi-compact over S, that is, of finite type over S, is quasi-affine over S.

The proof of Proposition 3.7 shows that for every $t \in \mathrm{ens}(F)$ there exists $i \in I$ such that $u_i : F \to T_i$ is a local isomorphism at t, and then u_j is a local isomorphism at that point for every $j \geq i$. By quasi-compactness, i may be chosen independently of $x \in U$. It remains to prove that, for sufficiently large i,

$$u_i|_U : U \longrightarrow T_i$$

is a monomorphism.

Since $U \to T$ is a monomorphism, the intersection of the equivalence relations

$$U \times_{T_i} U \ \subset \ U \times_S U$$

is the diagonal. The prescheme $U \times_S U$ is noetherian, and the $U \times_{T_i} U$ are closed subpreschemes. It follows that one of them is already the diagonal; equivalently, $u_i|_U$ is a monomorphism. This proves the first assertion of Corollary 3.11, and the second follows immediately.

Proposition 3.12. — *Let S be a prescheme and G an affine S-group prescheme.*

(a) *Let*

$$F : (\mathbf{Sch})^{\circ}/S \longrightarrow (\mathbf{Ens})$$

be the functor such that, for every T over S,

$$F(T) = \left\{ \begin{array}{c} \text{multiplicative-type subgroups of } G_T \\ \text{that are finite over } T \end{array} \right\}.$$

Suppose that S is locally noetherian or that G is of finite presentation over S. Then F is representable and affine over S. If G is of finite presentation over S, then F is locally of finite presentation over S.

(b) *Let H be a finite S-group prescheme of multiplicative type. Then $\mathrm{Hom}_{S\text{-gr}}(H, G)$ is representable and affine over S. If G is of finite type (respectively, of finite presentation) over S, then the same is true of $\mathrm{Hom}_{S\text{-gr}}(H, G)$.*

Remark 3.13. — Apart from the assertion that $\mathrm{Hom}_{S\text{-gr}}$ is affine, and from the case in which G is of finite presentation over S, which is the case that will suffice for us, 3.12 is an immediate consequence of the theory of Hilbert schemes (A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique: IV. Les Schémas de Hilbert*, Séminaire Bourbaki, May 1961, no. 221). It is enough, in fact, that G be quasi-projective over S. In case *(a)* one can also represent the larger functor

$$F_0(T) = \left\{ \begin{array}{c} \text{group subpreschemes of } G_T \text{ that are flat, proper,} \\ \text{and of finite presentation over } T \end{array} \right\}.$$

The canonical monomorphism $F \to F_0$ is an "open immersion," as follows from the criterion in 3.1 and Exposé X, 4.7 *(b)*. Thus the representability of F_0 implies that F is representable by an open subprescheme of F_0. In case *(b)* it is enough to assume that H is projective and of finite presentation over S. In both cases one obtains a functor locally of finite presentation over S.

In the present Exposé, 3.12 is only a technical lemma used to prove a key result in the next section. We shall therefore sketch an easy direct proof that does not use Hilbert schemes.

We first prove *(b)*. It is enough to prove that $\mathrm{Hom}_S(H,G)$ is representable, independently of any group structure on H and G, and that it has the additional properties asserted for $\mathrm{Hom}_{S\text{-gr}}(H,G)$. Indeed, applying the same result to $\mathrm{Hom}_S(H \times_S H, G)$, the subfunctor $\mathrm{Hom}_{S\text{-gr}}(H,G)$ of $\mathrm{Hom}_S(H,G)$ can be expressed by a finite inverse limit—in fact, by fiber products—involving G, $\mathrm{Hom}_S(H,G)$, and $\mathrm{Hom}_S(H \times_S H, G)$. We leave this expression to the reader.

Write

$$H = \mathrm{Spec}(B),$$

where B is a sheaf of algebras on S that is locally free as a sheaf of modules; this is the only hypothesis on H that the argument uses. Since the representability question is local on S, we may suppose that S is affine, with ring A. We may also write $G = \mathrm{Spec}(C)$, where C is an A-algebra.

When $G = S[t] = G_0$, with t an indeterminate, the Hom functor is simply

$$T \longmapsto \Gamma(T, B_T).$$

Since B is locally free, this functor is represented by the vector bundle $V(B^\vee)$, where $B^\vee$ is the dual module sheaf. When $G = S[(t_i)_{i\in I}] = G_0^I$, for a not necessarily finite family of indeterminates, it follows that

$$\mathrm{Hom}_S(H,G) = \mathrm{Hom}_S(H,G_0)^I = V(B^\vee)^I = V(B^{\vee(I)}).$$

In general G is isomorphic to a closed subscheme of a scheme of the form $S[(t_i)_{i\in I}]$; equivalently, C is a quotient of an A-algebra $A[(t_i)_{i\in I}]$. Let (F_j) be a family of generators of the ideal by which one takes the quotient. After covering S by smaller affine open subsets, we may assume that B is free of rank n. Choose a basis $(e_k)_{1\le k\le n}$ of B. Writing

$$x_i = \sum_{k=1}^{n} x_{ik} e_k$$

with indeterminate coefficients x_{ik} in an unspecified A-algebra A', expand $F_j((x_i))$ in this basis. For every F_j this gives n polynomials

$$F_{j,k} \qquad (1 \le k \le n)$$

in the variables $(x_{i,k})_{i\in I,\,1\le k\le n}$, with coefficients in A. It follows at once that $\mathrm{Hom}_S(H,G)$ is represented by the spectrum of the quotient of the polynomial ring $A[(x_{i,k})]$ by the ideal generated by the $F_{j,k}$. This proves *(b)*.

We now prove *(a)*. For every ordinary finite commutative group M, let F_M be the subfunctor of F obtained by restricting to those subgroups of G_T that are of multiplicative type and of type M. A gluing argument like the one used in 3.5 (which we should have stated after decomposing the problem a little further) shows that it is enough to prove that each F_M is representable. Then F is represented by the disjoint sum of the F_M, with M ranging over the isomorphism classes of finite commutative groups. Indeed, F is the corresponding sum in the category of sheaves.

Fix M and write F in place of F_M. Put $H = D_S(M)$, and let

$$F' = \mathrm{Imm}_{S\text{-gr}}(H,G) \subset \mathrm{Hom}_{S\text{-gr}}(H,G)$$

be the subfunctor whose value at T is the set of homomorphisms of T-groups $H_T \to G_T$ that are closed immersions. By *(b)*, $\mathrm{Hom}_{S\text{-gr}}(H,G)$ is represented by an affine S-prescheme.

Using Exposé IX, 6.8, one sees immediately that F' is represented by an open-and-closed subprescheme of it; in particular, F' is affine over S.

Consider the canonical morphism $F' \to F$, which assigns to every monomorphism $H_T \to G_T$ its image group. Directly from the definitions, F' is thereby a principal homogeneous bundle over F, in the category of fpqc sheaves, under

$$\Gamma_F = \Gamma_S \times_S F, \qquad \Gamma = \mathrm{Aut}_{\mathrm{gr}}(M).$$

It follows by descent that $F' \to F$ is representable: for every morphism $T \to F$ with T representable, the prescheme $T \times_F F'$ is representable over T, in fact by a principal Galois bundle under Γ.

By Exposé IV, 4.6.6, F is therefore representable if and only if the quotient F'/F'', where $F'' = F' \times_F F'$, exists in $(\mathbf{Sch})$ and is universally effective for faithfully flat quasi-compact morphisms. Equivalently, the quotient F'/Γ must exist and be universally effective for such morphisms. Since F' is affine over S, Exposé V, 4.1 shows that this condition is satisfied. This proves the representability of F in *(a)*.

Finally, suppose that G is of finite presentation over S. The additional assertion follows immediately from the preceding proof: by *(b)*, F' is then locally of finite presentation over S, so that Exposé V may be applied.

4. The Scheme of Multiplicative-Type Subgroups of a Smooth Affine Group

The principal result of the present Exposé is the following.

Theorem 4.1. — *Let S be a prescheme, let G be an S-group prescheme smooth and affine over S, and let*

$$F\colon (\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

be the functor defined by

$$F(T) = \{\text{multiplicative-type subgroups of } G_T\}.$$

Then F is representable, smooth, and separated over S.

Let us record at once the following variant.

Corollary 4.2. — *Let G and H be two S-group preschemes, with G smooth and affine over S, and H of multiplicative type and of finite type. Then* $\mathrm{Hom}_{S\text{-gr}}(H, G)$ *is representable, smooth, and separated over S.*

Indeed, when H is smooth over S, so is $H \times_S G$, and one may apply 4.1 to the latter. By considering graph subgroups, $\mathrm{Hom}_{S\text{-gr}}(H, G)$ becomes a subfunctor F' of the functor F of multiplicative-type subgroups of $H \times_S G$: namely, $F'(T)$ is the set of multiplicative-type subgroups K of

$$(H \times_S G)_T = H_T \times_T G_T$$

such that the homomorphism induced by the first projection,

$$K \longrightarrow H_T,$$

is an isomorphism. By Exposé IX, 2.9, the canonical morphism $F' \to F$ is an open immersion. Since F is representable, the same is true of F', which is represented by an open subprescheme of F; and since F is separated and smooth over S, so is F'.

When H is finite over S, it is enough to apply 3.12(b) to obtain the representability of $\mathrm{Hom}_{S\text{-gr}}(H, G)$. When $H = H_1 \times_S H_2$, with H_1 smooth over S and H_2 finite over S, then $\mathrm{Hom}_{T\text{-gr}}(H_T, G_T)$ is identified with the subset of

$$\mathrm{Hom}_{T\text{-gr}}\big((H_1)_T, G_T\big) \times \mathrm{Hom}_{T\text{-gr}}\big((H_2)_T, G_T\big)$$

formed by the pairs (u_1, u_2) such that $u_1 \cdot u_2 = u_2 \cdot u_1$. It follows that $\mathrm{Hom}_{S\text{-gr}}(H, G)$ is represented by a closed subprescheme of

$$\mathrm{Hom}_{S\text{-gr}}(H_1, G) \times_S \mathrm{Hom}_{S\text{-gr}}(H_2, G) = X.$$

Indeed, apply Exposé VIII, 6.5(b), with $Y = H$, $Z = G$, and with q_1 and q_2 defined respectively by

$$(u_1, u_2) \longmapsto u_1 \cdot u_2 \qquad \text{and} \qquad (u_1, u_2) \longmapsto u_2 \cdot u_1.$$

In the general case, the question is local on S for the Zariski topology, so we may suppose that S is affine and that H has constant type over S. Since H is quasi-isotrivial (Exposé X, 4.5), there is a surjective étale morphism $S' \to S$, with S' affine, that splits H, that is, such that $H' = H_{S'}$ is diagonalizable. The preceding result then applies, because a diagonalizable group is the product of a diagonalizable torus and a finite diagonalizable group. It remains to see that the descent datum obtained on the S'-prescheme

$$\mathrm{Hom}_{S'\text{-gr}}(H', G') = X'$$

is effective. This follows from the argument of Exposé X, 5.4, noting that, there, the hypothesis that $X' \to S'$ be separated and locally quasi-finite was used only to ensure that every open part of X' quasi-compact over S' was quasi-affine over S'. That property still holds here, as follows readily from the fact that it holds for the functor F of 4.1 (as will be seen during the proof of 4.1). This argument—or any similar variant of this small dévissage—establishes the representability of $\mathrm{Hom}_{S\text{-gr}}(H, G)$, and at the same time shows that it is separated over S. It is locally of finite presentation over S, as follows, for example, from the fact already noted in 3.2 that this functor "commutes with filtered direct limits of rings." Finally, the functor is formally smooth over S by 2.1, and is therefore smooth over S.

Remark 4.3. — We have deduced 4.2 from 4.1; this is genuinely immediate only when H is also smooth over S. To make the deduction without contortions in the general case, one would need the representability assertion in 4.1 without assuming that G is smooth over S, assuming only that it is affine and of finite presentation over S. Of course, F would then no longer be smooth over S in general. There is little doubt that 4.1 remains true under these more general hypotheses, but the proof appears likely to be more delicate, because one can no longer invoke 3.8.[3][4]

Let us nevertheless point out that when G is a closed subgroup of an affine smooth group G_0 over S, the functor F of multiplicative-type subgroups of G is represented by a closed subprescheme of the prescheme representing the analogous functor F_0 for G_0, which

[3] This is indeed proved for G flat and quasi-affine over S, with connected fibres, provided one restricts to the central subtori of G (Exposé XV, 8.8).

[4] Editor's note: when H is smooth (not necessarily affine) over a normal locally noetherian base S, M. Raynaud showed that the largest representable open subfunctor of the functor of subtori of H is a disjoint sum of opens smooth and affine over S. This is Theorem IX.9.26 in *Faisceaux amples sur les schémas en groupes et sur les espaces homogènes*, Lecture Notes in Mathematics 119 (1970).

is covered by 4.1; this follows readily by applying Exposé VIII, 6.4. This also raises the question whether an affine S-group scheme G, affine and of finite presentation over S, is isomorphic to a subgroup scheme of $\mathrm{GL}(n)_S$ for some suitable n. This is true when S is the spectrum of a field (cf. Exposé $\mathrm{VI_B}$, 11.11), but is unfortunately false in general, even for tori; see 4.6. Finally, one could also prove 4.2 directly by exactly the same method as 4.1.

We now prove 4.1. Since the functor F is evidently local in nature, we may suppose that S is affine, say $S = \mathrm{Spec}(A)$, where A is a ring. Regarding A as the filtered direct limit of its subrings of finite type over $\mathbb{Z}$, and noting that G descends from a smooth affine group over one such subring (EGA IV, 8), we are reduced to the case in which S is noetherian.

For every integer $n > 0$, let T_n be the functor defined like F, but restricted to multiplicative-type subgroups H of G_T such that $n \cdot \mathrm{id}_H = 0$, that is, such that ${}_nH = H$. Order the set I of positive integers by divisibility. When m is a multiple of n, define

$$p_{n,m} \colon T_m \longrightarrow T_n$$

by

$$p_{n,m}(H) = {}_nH = \mathrm{Ker}(n \cdot \mathrm{id}_H).$$

In this way the T_n form an inverse system of functors over S. By 3.12(a), the functors T_n are representable, affine, and of finite type over S. Similarly define morphisms

$$u_n \colon F \longrightarrow T_n$$

by $u_n(H) = {}_nH$. Thus one obtains a morphism

$$u \colon F \longrightarrow T = \varprojlim_n T_n,$$

where the inverse limit is taken in the category of functors over S. Since the T_n are representable and affine over S, so is T: it is the spectrum of the filtered direct limit of the quasi-coherent algebras on S defining the T_n. Of course, T is not in general of finite type over S.

We shall apply 3.7. It is enough to verify conditions *(a)–(d)* there; these will imply that F is represented by a prescheme locally of finite type over S. It then follows from 2.1 bis that F is even smooth over S. Moreover, since F is a subfunctor of T, which is affine over S, F is separated over S: it is separated over T, and T is separated over S. We can also prove at once the supplementary assertion used above: every open $U \subset F$ that is quasi-compact over S is quasi-affine over S. This follows from 3.11 and the fact that the T_n are affine over S.

Let us now verify the conditions of 3.7.

(a) F is a sheaf for the faithfully flat quasi-compact topology, by the descent theory of SGA 1, Exposé VIII, which applies here because groups of multiplicative type over T are affine over T.It commutes with filtered direct limits of rings by the standard machinery of EGA IV, 8. Let us show that it commutes with adic inverse limits of local artinian rings. For the analogous functor considered in 4.2, this is precisely Exposé IX, 7.1, in the special case where A is a complete noetherian local ring endowed with an ideal of definition for its usual topology. Notice that this is exactly where the hypothesis that G be affine enters essentially. In the present case we are reduced to the following lemma.

Lemma 4.4. — *Let A be a complete noetherian local ring with maximal ideal $\mathfrak{m}$, let G be a group scheme affine over $S = \mathrm{Spec}(A)$, and, for every integer $n \geq 0$, put*

$$S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}), \qquad G_n = G \times_S S_n.$$

For every n, let H_n be a multiplicative-type subgroup of finite type of G_n, and suppose that, whenever $m \geq n$, H_n is obtained from H_m by reduction. Then there exists a unique multiplicative-type subgroup H of G whose reductions are the H_n.

By Exposé X, 3.2, there exists a multiplicative-type group H over S, necessarily of finite type and isotrivial and determined up to unique isomorphism, together with an isomorphism

$$H \times_S S_0 \simeq H_0.$$

Here one uses the fact that H_0 is isotrivial, since it is of finite type over a field; see Exposé X, 1.4. By Exposé X, 2.1, for every n the isomorphism $H \times_S S_0 \simeq H_0$ lifts to a unique isomorphism $H \times_S S_n \simeq H_n$. Exposé IX, 7.1, already cited, now shows that the homomorphisms $H_n \to G_n$ arise from a unique homomorphism of S-groups

$$u \colon H \longrightarrow G.$$

By Exposé IX, 6.6, this homomorphism is a monomorphism because $u_0 \colon H_0 \to G_0$ is one. This proves 4.4.

(b) The morphism $u \colon F \to T$ is a monomorphism. This follows from the density theorem, Exposé IX, 4.7, in the form of Corollary 4.8(b). It is essential for the present application that this result is available over a base that need not be noetherian. Since the functor T does not commute with filtered direct limits of rings, one cannot reduce to the noetherian case a priori.

(b′) The morphism $u \colon F \to T$ is infinitesimally étale; in other words, we have the following lemma.

Lemma 4.5. — *Let A be an artinian local ring with residue field k, let $S = \operatorname{Spec}(A)$, let $I \subset \operatorname{rad}(A)$ be an ideal, and put $S_0 = \operatorname{Spec}(A/I)$. Let G be an S-group prescheme smooth over S, and put $G_0 = G \times_S S_0$. Suppose given a multiplicative-type subgroup H_0 of G_0 and, for every integer $n > 0$, a multiplicative-type subgroup $H(n)$ of G, in such a way that*

1° *for every multiple m of n, $H(n) = {}_nH(m)$; and*

2° *$H(n)_0 = {}_nH_0$.*

Then there exists a unique multiplicative-type subgroup H of G such that ${}_nH = H(n)$ for every n.

Uniqueness is already contained in *(b)*. For existence, an immediate induction reduces us to the case in which $I\mathfrak{m} = 0$, where $\mathfrak{m}$ is the maximal ideal of A.Put

$$k = A/\mathfrak{m}, \qquad S_0 = \operatorname{Spec}(k), \qquad G_0 = G \times_S S_0,$$

and let $\mathfrak{g}_0$ and $\mathfrak{h}_0$ be the Lie algebras of G_0 and H_0, respectively. There is a canonical isomorphism of groups

$$\mathfrak{g}_0 \otimes_k I \simeq \operatorname{Ker}\bigl(G(S) \longrightarrow G(S_0)\bigr);$$

see Exposé III. By Exposé IX, 3.6 bis and 3.7, there exists a multiplicative-type subgroup H of G reducing to H_0, and such an H is determined, up to an inner automorphism by an element of

$$N = \operatorname{Ker}\bigl(G(S) \longrightarrow G(S_0)\bigr).$$

Thus the set P of liftings H of H_0 is a principal homogeneous set under N/M, where

$$M = \{g \in N \mid \operatorname{int}(g) \cdot H = H\}.$$

One sees readily (Exposé III) that this subgroup is precisely the vector subspace of $\mathfrak{g}_0 \otimes_k I$ consisting of the invariants under H_0, where H_0 acts on $\mathfrak{g}_0 \otimes_k I$ through the representation induced by the adjoint representation of G_0. Likewise, the set $P(n)$ of liftings of $H(n)_0 = {}_nH_0$ is a principal homogeneous set under $N/M(n)$, where $M(n)$ is the vector subspace of $\mathfrak{g}_0 \otimes_k I = N$ consisting of the invariants under ${}_nH_0$. The density theorem, Exposé IX, 4.7, shows readily that $M = M(n)$ for sufficiently large n, where "large" is understood with respect to the divisibility order on the positive integers. Consequently, the natural map

$$H \longmapsto {}_nH$$

from P to $P(n)$, which is compatible with the actions of N, and hence with the homomorphism

$$N/M \longrightarrow N/M(n)$$

between the acting groups, is bijective for sufficiently large n. The conclusion of 4.5 follows immediately.

To verify conditions *(c)* and *(d)* of 3.7, we use 3.8, which reduces us to checking *(c′)* and *(d′)* below.

(c′) The T_n are smooth over S, and the transition morphisms $p_{n,m} \colon T_m \to T_n$ are smooth.

This is precisely 2.2 bis.

(d′) With the notation of 3.8, a point ξ of F with values in a field k over S is just a multiplicative-type subgroup H_0 of $G_0 = G_k$. Repeating the argument in *(b′)* above, we see that the integer $d(\xi)$ considered in 3.8 is

$$d(\xi) = \dim_k(\mathfrak{g}_0/\mathfrak{g}_0^{H_0}),$$

where $\mathfrak{g}_0$ is the Lie algebra of G_0 and $\mathfrak{g}_0^{H_0}$ is the vector subspace of invariants under H_0. This is a finite integer, so condition *(d')*1°*)* of 3.8 holds.

With the notation of *(d')*2°*)*, giving a morphism $v \colon X \to F$ amounts to giving a multiplicative-type subgroup H of G_X. For $x \in X$, the integer $d(\xi_x)$is then the dimension of

$$(\mathfrak{g} \otimes \kappa(x))/(\mathfrak{g} \otimes \kappa(x))^{H_x}.$$

where $\mathfrak{g}$ is the sheaf of Lie algebras of G_X. It is locally free of finite type over X, because G_X is smooth over X, and $\mathfrak{g} \otimes \kappa(x)$ is the Lie algebra of the fibre G_x of G_X at x; here H_x is the fibre of H at x. Now $\mathfrak{g}$ is a locally free module on Xon which the multiplicative-type group H acts, and the invariant subfunctor $\mathfrak{g}^H$ is therefore given by a locally direct-summand, hence locally free, subsheaf of $\mathfrak{g}$. By descent, one reduces to the case in which H is diagonalizable, and then applies Exposé I, 4.7.3, noting that the invariant subsheaf corresponds to the degree-zero component. Consequently,

$$d(\xi_x) = \operatorname{rank}_x(\mathfrak{g}/\mathfrak{g}^H),$$

so this is a locally constant function of x. This proves condition *(d′)*.

We have thus verified the conditions of 3.7, completing the proof of 4.1.

Remark 4.6. — When $G = \mathrm{GL}(n)_S$, one can give a much simpler and more explicit direct proof of 4.1, using Exposé I, 4.7.3. The proof moreover shows that, in this case, the moduli scheme is a disjoint sum of affine S-schemes. Proceeding as indicated in 4.3, one obtains the same result whenever G is a closed subgroup of a group of the form $\mathrm{GL}(n)_S$.

One must not, however, suppose that the preschemes representing the functors in 4.1 or 4.2 are always disjoint sums of affine S-schemes. For example, let H be a multiplicative-type group of finite type over a locally noetherian prescheme S. By Exposé X, 5.11, H is isotrivial if and only if

$$\mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_m)$$

is a disjoint sum of preschemes affine over S. But, as noted in Exposé X, 1.6, there may exist nonisotrivial tori H of relative dimension 2. For such a torus, $\mathrm{Hom}_{S\text{-gr}}(H, \mathbb{G}_m)$ is not a disjoint sum of S-preschemes affine over S, and likewise the "dual" twisted constant group

$$\mathrm{Hom}_{S\text{-gr}}(\mathbb{G}_m, H)$$

is not such a sum. Indeed, if two commutative twisted constant groups R, R' of finite presentation are dual to one another,

$$R' = \mathrm{Hom}_{S\text{-gr}}(R, \mathbb{Z}_S),$$

then one readily sees that one is isotrivial if and only if the other is.

This last point also shows that such an H is not isomorphic to a subgroup of a group of the form $\mathrm{GL}(n)_S$. More precisely, one can show that a multiplicative-type group of finite type over a connected locally noetherian S is isomorphic to a subgroup of a group of the form $\mathrm{Aut}_{\mathrm{Mod}}(E)$, where E is a locally free module of finite type over S, if and only if it is isotrivial. Finally, taking $G = H \times \mathbb{G}_m$ in the two preceding examples gives an example in which the prescheme representing the functor F of 4.1 is not a disjoint sum of affine S-preschemes; if desired, G may be taken to be a torus of relative dimension 3.

5. First Corollaries of the Representability Theorem

Let H and G be as in 4.2. Put

$$M = \mathrm{Hom}_{S\text{-gr}}(H, G),$$

which, by 4.2, is a smooth S-prescheme separated over S. Notice that G acts on the functor $M = \mathrm{Hom}_{S\text{-gr}}(H, G)$ by

$$(g, u) \longmapsto \mathrm{int}(g) \circ u.$$

Hence there is a canonical morphism

$$G \times_S M \longrightarrow M \times_S M, \tag{x}$$

whose components are the preceding morphism $G \times_S M \to M$ and the second projection $G \times_S M \to M$.

Corollary 5.1. — *The preceding morphism* (x) *is smooth.*

This follows from 4.2 and 2.3. This statement is equivalent to the following one.

Corollary 5.2. — *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S-groups. Then the subfunctor* $\mathrm{Transp}(u_1, u_2)$ *of G (cf. 2.4) is representable by a closed subprescheme of G, smooth over S.*

It remains only to prove that $\mathrm{Transp}(u_1, u_2) \to G$ is indeed a closed immersion. This follows because it is the equalizer of a pair of morphisms $G \rightrightarrows M$, namely

$$g \longmapsto \mathrm{int}(g)u_1 \qquad \text{and} \qquad g \longmapsto u_2,$$

the latter being the "constant" morphism, and because M is separated over S. In particular:

Corollary 5.3. — *Let $u : H \to G$ be a morphism of S-groups. Then*

$$\mathrm{Centr}_G(u) = \mathrm{Transp}(u, u)$$

is representable by a closed group subprescheme of G, smooth over S. Moreover, the quotient of G by $\mathrm{Centr}_G(u)$ *is representable by an open subprescheme of M.*

It remains to prove the last assertion. The morphism

$$g \longmapsto \mathrm{int}(g) \circ u$$

from G to M is smooth and of finite type by 5.2, hence is an open morphism (EGA IV, 6.6). Let U denote its image, equipped with the structure induced from M. The induced morphism $G \to U$ is smooth, surjective, and of finite type, and is therefore a covering for the faithfully flat quasi-compact topology. Moreover, it is clear that the preceding morphism $G \to M$ makes G a formally principal homogeneous sheaf under $\mathrm{Centr}_G(u)_M$. It follows that the sheaf $G/\mathrm{Centr}_G(u)$ is represented by U.

Corollary 5.4. — *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S-groups. The following conditions are equivalent:*

(i) *u_1 and u_2 are conjugate locally for the étale topology.*

(i bis) *u_1 and u_2 are conjugate locally for the faithfully flat quasi-compact topology.*

(ii) *For every $s \in S$, let $\overline{s}$ denote the spectrum of an algebraic closure of $\kappa(s)$. The morphisms*

$$u_{1,\overline{s}}, u_{2,\overline{s}} : H_{\overline{s}} \rightrightarrows G_{\overline{s}}$$

are conjugate by an element of $G(\kappa(\overline{s}))$.

(ii bis) *The structural morphism* $\mathrm{Transp}(u_1, u_2) \to S$ *is surjective.*

(iii) $\mathrm{Transp}(u_1, u_2)$ *is a torsor under the action of the smooth S-group prescheme of finite type* $\mathrm{Centr}_G(u_1)$.

The implications (i) ⇒ (i bis) and (ii) ⇒ (ii bis) are immediate, the latter by the Nullstellensatz. The implication (i bis) ⇒ (ii) follows from the "principle of finite extension" (EGA IV, 9). Moreover, (ii bis) ⇒ (iii): since $\mathrm{Transp}(u_1, u_2)$ is smooth over S, it is flat over S, and since it is of finite type, it is quasi-compact over S. Thus it is faithfully flat and quasi-compact over S if and only if its structural morphism is surjective. On the other hand, it is formally principal homogeneous under $\mathrm{Centr}_G(u_1)$, which is faithfully flat and quasi-compact over S. The last condition is therefore also equivalent to saying that $\mathrm{Transp}(u_1, u_2)$ is a torsor under $\mathrm{Centr}_G(u_1)$, in the faithfully flat quasi-compact topology. Finally, (iii) ⇒ (i) follows from the smoothness of $\mathrm{Transp}(u_1, u_2)$ over S and from Hensel's lemma in the form 1.10.

Remark 5.5. — Fix $u_1 = u : H \to G$. Consider the functor

$$(\mathbf{Sch})^\circ/S \longrightarrow (\mathbf{Ens})$$

that assigns to every T over S the set of homomorphisms of T-groups $u_2 : H_T \to G_T$ that are locally conjugate, for the étale topology, to $u_T : H_T \to G_T$. This functor is represented precisely by the open subprescheme of M, isomorphic to $G/\mathrm{Centr}_G(u)$, considered in 5.3.

Let us sketch the variants of the preceding results obtained by applying 4.1 instead of 4.2. Thus let G be a group prescheme smooth and affine over S, and now let M denote the smooth S-prescheme separated over S that represents the functor considered in 4.1. Again G acts on M:

$$(g, H) \longmapsto \operatorname{int}(g)(H).$$

Hence, as above, there is a morphism

$$G \times_S M \longrightarrow M \times_S M. \tag{x bis}$$

Corollary 5.1 bis. — *The preceding morphism is smooth.*

This follows from 4.1 and 2.3 bis. We again conclude:

Corollary 5.2 bis. — *Let H_1, H_2 be two multiplicative-type subgroups of G. Then the subfunctor* $\operatorname{Transp}_G(H_1, H_2)$ *of G is representable by a closed subprescheme of G, smooth over S.*

In particular:

Corollary 5.3 bis. — *Let H be a multiplicative-type subgroup of G. Then the group subfunctor* $\operatorname{Norm}_G(H)$ *of G is representable by a closed subprescheme of G, smooth over S. Moreover, the quotient $G/\operatorname{Norm}_G(H)$ is representable by an open subprescheme of M.*

Corollary 5.4 bis. — *Let H_1, H_2 be two multiplicative-type subgroups of G. The following conditions are equivalent:*

(i) *H_1 and H_2 are conjugate locally for the étale topology.*

(i bis) *H_1 and H_2 are conjugate locally for the faithfully flat quasi-compact topology.*

(ii) *For every $s \in S$, let $\overline{s}$ denote the spectrum of an algebraic closure of $\kappa(s)$. The subgroups $H_{1,\overline{s}}$ and $H_{2,\overline{s}}$ of $G_{\overline{s}}$ are conjugate by an element of $G(\kappa(\overline{s}))$.*

(ii bis) *The structural morphism* $\operatorname{Transp}(H_1, H_2) \to S$ *is surjective.*

(iii) $\operatorname{Transp}(H_1, H_2)$ *is a principal homogeneous bundle under the action of the smooth S-group prescheme of finite type* $\operatorname{Norm}_G(H_1)$.

Remark 5.5 bis. — Remark 5.5 carries over to the present case as well.

Remark 5.6. — To prove 5.2, and hence also the first assertion of 5.3 and the assertions of 5.4, the reference to 4.2 may be replaced by a reference to VIII, 6.4, whose proof is much easier. This also shows that the hypothesis that G be affine over S is unnecessary there. Moreover, in Section 6 we shall show how a variant of this method extends these results to certain groups H more general than groups of multiplicative type. The same observations apply to the variants 5.2 bis and following. By contrast, the result 5.8 below makes essential use of all the hypotheses imposed—in particular, that G be affine and smooth over S, and that H be of multiplicative type—and the lecturer knows no proof other than one using the representability theorems 4.1 or 4.2.[5]

5.7. Since the morphism (x), respectively (x bis), is smooth and hence open, its image is open. Let R be this image, equipped with the structure induced from $M \times_S M$. One checks readily that R is an equivalence relation on M, namely the relation made explicit in 5.4, respectively 5.4 bis. It would be interesting to know whether the quotient sheaf M/R,

[5] The situation has changed since this text was written; cf. Exposés XV and XIX, no. 6.

which is formally étale over S, is representable; if so, it is represented by a prescheme étale over S. This is the case, for example, when S is the spectrum of a field. Notice also that R need not be closed in $M \times_S M$, even when G is quasi-finite over S. Thus, when M/R is representable, it may be nonseparated over S.

Theorem 5.8. — *Let S be a prescheme, let G be a smooth affine S-prescheme, and let $s \in S$. Then:*

(a) *For every multiplicative-type subgroup H_0 of G_s, there exist an étale morphism $S' \to S$, a point $s' \in S'$ lying over s such that the residue-field extension $\kappa(s')/\kappa(s)$ is trivial, and a multiplicative-type subgroup H' of $G' = G \times_S S'$, such that*

$$H'_{s'} = H_0 \otimes_{\kappa(s)} \kappa(s').$$

(b) *For every group homomorphism $u_0 : H_s \to G_s$, where H is a multiplicative-type S-group prescheme of finite type, there exist an étale morphism $S' \to S$, a point $s' \in S'$ lying over s such that $\kappa(s')/\kappa(s)$ is trivial, and a group homomorphism*

$$u' : H' \longrightarrow G', \qquad H' = H \times_S S', \quad G' = G \times_S S',$$

such that

$$u'_{s'} : H'_{s'} \longrightarrow G'_{s'} \quad \text{equals} \quad u_0 \otimes_{\kappa(s)} \kappa(s').$$

This follows from 4.1, respectively 4.2, and from Hensel's lemma in the form 1.10.

Proposition 5.9. — *Let S be a prescheme, let G be a smooth affine S-prescheme, and let H be a multiplicative-type subgroup of G. Consider*

$$\mathrm{Centr}_G(H) \hookrightarrow \mathrm{Norm}_G(H),$$

which, by 5.3 and 5.3 bis, are closed group subpreschemes of G, smooth over S. The first is then an open and closed subprescheme of the second, and the quotient sheaf

$$W_G(H) = \mathrm{Norm}_G(H)/\,\mathrm{Centr}_G(H)$$

is represented by an open group subprescheme of $\mathrm{Aut}_{S\text{-gr}}(H)$. *It is therefore a quasi-finite, étale, and separated S-group prescheme.*

Indeed, consider the evident homomorphism

$$\theta : \mathrm{Norm}_G(H) \longrightarrow \mathrm{Aut}_{S\text{-gr}}(H),$$

whose kernel is, by definition, $\mathrm{Centr}_G(H)$. Since $\mathrm{Aut}_{S\text{-gr}}(H)$ is represented by an étale S-group prescheme separated over S (X, 5.10), its unit section is an open and closed immersion. Its inverse image under θ is therefore an open and closed subgroup of $\mathrm{Norm}_G(H)$.

Moreover, θ is smooth. This follows formally from the definitions and from the facts that $\mathrm{Norm}_G(H)$ is smooth over S and that $\mathrm{Aut}_{S\text{-gr}}(H)$ is étale over S. As in 5.3, the image of θ is consequently an open subprescheme U of $\mathrm{Aut}_{S\text{-gr}}(H)$, and, with the induced structure, U represents the quotient sheaf $\mathrm{Norm}_G(H)/\,\mathrm{Centr}_G(H)$. The latter is therefore étale and separated over S. It is also quasi-finite over S, since it is quasi-compact over S, being the image of $\mathrm{Norm}_G(H)$, which is quasi-compact. This proves 5.9.

Corollary 5.10. — *For every $s \in S$, put*

$$w(s) = \mathrm{rank}(\mathrm{Norm}_{G_s}(H_s)/\,\mathrm{Centr}_{G_s}(H_s)).$$

This is also the index of $\mathrm{Centr}_{G(\overline{k})}(H(\overline{k}))$ *in* $\mathrm{Norm}_{G(\overline{k})}(H(\overline{k}))$*, where* $\overline{k}$ *is an algebraic closure of* $\kappa(s)$*. Then the function* $s \mapsto w(s)$ *is lower semicontinuous. It is constant in a neighborhood of* s *if and only if* $W_G(H)$ *is finite over* S *in a neighborhood of* s*.*

Indeed, for every S-prescheme W that is étale, of finite type, and separated over S, the function

$$s \longmapsto \mathrm{rank}[W_s : \kappa(s)]$$

is lower semicontinuous, and it is constant in a neighborhood of s if and only if W is finite over S in a neighborhood of s. This fact was noted in SGA 1, I, 10.9; its proof, which presents no difficulty, is given in EGA IV.[6]

Remark 5.11.[7] — Let G be a group prescheme affine and smooth over S, and let H be a multiplicative-type subgroup. Then 5.3 and 5.3 bis imply that the quotients

$$G/\mathrm{Centr}_G(H) \qquad \text{and} \qquad G/\mathrm{Norm}_G(H)$$

are preschemes smooth and quasi-affine over S. Indeed, in both cases the moduli prescheme M in which the quotient is embedded has the property that every open $U \subset M$ quasi-compact over S is quasi-affine over S, by 3.11.

We shall moreover see in the next Exposé[8] that for every such U, the schematic closure $\overline{U}$ of U in M is even affine over S. Thus the quotients under consideration are affine over S, provided that the open subprescheme U of the corresponding moduli prescheme—of group homomorphisms from H to G, respectively of multiplicative-type subgroups of G—is closed in M. This is the case, for example, when H is a "maximal torus" in the second case, or when S is the spectrum of a field; cf. XII.5.4.[9] I do not know whether the quotients under consideration are affine over S in general.

6. A Rigidity Property for Homomorphisms of Certain Group Schemes, and the Representability of Certain Transporters[10]

Proposition 6.1. — *Let* S *be a locally noetherian prescheme, let* H *be a commutative* S*-group prescheme of finite type over* S*, and let* E *be a set of positive integers stable under multiplication. Suppose that the following conditions hold:*

(a) *for every* $n \in E$*, the subgroup*

$$_nH = \mathrm{Ker}(n \cdot \mathrm{id}_H)$$

is finite and flat over S*;*

(b) *for every* $s \in S$*, the family of the* $_nH_s$*,* $n \in E$*, is schematically dense in the fibre* H_s*.*

[6] EGA IV, 15.5.1 and 18.10.7.

[7] Editor's note: For a generalization to the nonaffine case, see M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes in Mathematics 119 (1970), IX.2.8.

[8] Editor's note: Cf. Remark XII.5.7. We mention here the following generalization. Without assuming G affine, if H is a torus and S is locally noetherian, U is affine and smooth by a theorem of Raynaud (*loc. cit.*, IX.2.6 and IX.2.8). Indeed, $G/\mathrm{Norm}_G(H)$ is an open subprescheme of M, and the largest representable open subfunctor of M is a disjoint sum of open subpreschemes smooth and affine over S.

[9] Editor's note: This assertion assumes that G has locally constant reductive rank.

[10] The present section does not use the results of Sections 3, 4, and 5; its natural place would be in Exposé $\mathrm{VI_B}$.

Let Y be a closed subprescheme of H. Then the subfunctor $\prod_{H/S}(Y/H)$ of S (compare Exposé VIII, 6) is representable by a closed subprescheme T of S. If S is noetherian, there exists $n \in E$ such that

$$\prod_{H/S}(Y/H) = \prod_{{}_nH/S}((Y \cap {}_nH)/{}_nH). \tag{x}$$

Indeed, by Exposé VIII, 6.4, condition (a) implies that ${}_nH$ is finite and flat, and hence a fortiori "essentially free" over S. Thus the right-hand side of (x) is representable by a closed subprescheme T_n of S.Order E by divisibility. The T_n then form a decreasing family of closed subpreschemes of S; consequently, when S is noetherian—as we may now suppose—this family is stationary for sufficiently large n. Let T be the stable value of T_n. We claim that T represents the left-hand side of (x), which will complete the proof.

It is enough to prove the following. Let S' be a prescheme over S such that

$$(Y \cap {}_nH)_{S'} = ({}_nH)_{S'} \qquad \text{for every } n \in E,$$

or, equivalently, such that $Y_{S'} \supset ({}_nH)_{S'}$ for every $n \in E$. Then $Y_{S'} = H_{S'}$. This follows at once from Exposé IX, 4.4: in view of conditions (a) and (b), the family of the $({}_nH)_{S'}$ is schematically dense in $H_{S'}$.

Theorem 6.2. — *Let S be a prescheme, let H, E be as in 6.1 and satisfy conditions (a) and (b), let $u : H \to G$ be a homomorphism of S-groups from H to an S-group prescheme locally of finite type over S, and let K be a closed subgroup prescheme of G.Consider the subfunctor* $\mathrm{Transp}_G(u, K)$ *of G (cf. 2.5). It is representable by a closed subprescheme of G. If G is noetherian—for example, if S is noetherian and G is of finite type over S—there exists $n \in E$ such that*

$$\mathrm{Transp}_G(u, K) = \mathrm{Transp}_G(u|_{{}_nH}, K).$$

Suppose, finally, that G is smooth over S, that K is smooth over S or is of multiplicative type, and that H, E satisfy the following condition, stronger than (a):

a' *for every $n \in E$, the subgroup ${}_nH$ of H is of multiplicative type.*

Then $\mathrm{Transp}_G(u, K)$ *is smooth over S.*

Consider the G-group

$$H_G = H \times_S G,$$

which plainly satisfies the conditions of 6.1, with S replaced by G and H by H_G. Let Y be the closed subprescheme of H_G that is the inverse image of $K \subset G$ under the map

$$H_G = H \times_S G \longrightarrow G, \qquad (h, g) \longmapsto \mathrm{int}(g) \cdot u(h).$$

Then

$$\mathrm{Transp}_G(u, K) = \prod_{H_G/G}(Y/H_G)$$

(compare Exposé VIII, Example 6.5(e)). The first assertions therefore follow from 6.1. Moreover, for every quasi-compact open $U \subset G$, there exists $n \in E$ for which the restrictions to U of the two functors

$$\mathrm{Transp}_G(u, K) \qquad \text{and} \qquad \mathrm{Transp}_G(u|_{{}_nH}, K)$$

coincide. To verify the last assertion of 6.2, we may therefore replace H by some ${}_nH$, after which it is enough to apply 2.5, since ${}_nH$ is assumed to be of multiplicative type over S.

Remark 6.3. — The preceding proof uses only the very elementary result of Exposé VIII, 6.4, together, for its last part, with 2.5; when K is smooth over S, the latter is the infinitesimal result of Exposé IX, 3.6, and hence the vanishing of the cohomology of multiplicative-type groups. In the most important cases (cf. 6.7), the integers $n \in E$ may even be taken prime to the residue characteristics of S, that is, invertible in $\mathscr{O}_S$. The ${}_nH$ are then finite étale over S, and the cohomological result invoked is practically trivial. Thus, in that situation, 6.2 is independent of the theory of multiplicative-type groups.

6.4. — As usual, one sees that 6.1 extends to the following situation. Let S' be an S-prescheme (not necessarily locally noetherian), put $H' = H_{S'}$, and let Y' be a closed subprescheme of H', provided that $Y' \to H'$ is of finite presentation, that is, that the ideal defining Y' is of finite type. Then $\prod_{H'/S'}(Y'/H')$ is representable by a closed subprescheme T' of S', the morphism $T' \to S'$ is of finite presentation, and, if S' is quasi-compact, there exists $n \in E$ for which the analogue of (x) holds. It follows as well that the first assertion of 6.2 remains valid without assuming that G is locally of finite type over S, provided that the immersion $K \to G$ is of finite presentation.

6.5. — As announced in 5.6, Theorem 6.2 makes it possible to extend to group preschemes satisfying (a') and (b) above certain results proved by another method, under more restrictive hypotheses, for groups of multiplicative type. This applies to 5.2, the beginning of 5.3, and 5.4 and their corresponding "bis" variants, and also to 5.9 and 5.10. It also applies to the results of Exposé IX, Sections 5 and 6, with the exception of Exposé IX, 6.8. The latter assertion is already false for a homomorphism of abelian schemes $u : H \to G$ over the spectrum S of a discrete valuation ring of residue characteristic $p > 0$: it can happen that $\operatorname{Ker} u$ has the trivial group as its generic fibre and, as its special fibre, a radicial group that is not reduced to the trivial group.

6.6. — We have just given an example of a rigidity property of multiplicative-type groups that is not shared by abelian schemes. Another, extremely important, example is that the existence theorem for infinitesimal extensions of homomorphisms in Exposé IX, 3.6 no longer holds when H is an abelian scheme. Thus a nonzero abelian scheme over a field admits nontrivial infinitesimal variations, in contrast to a multiplicative-type group. This is the infinitesimal aspect of the fact that there is a theory of moduli—one that, moreover, is far from complete—for abelian varieties, whereas the theory of moduli of multiplicative-type groups is empty.

Another, "global," aspect of this infinitesimal difference is the following. Let H be an abelian scheme over a locally noetherian prescheme S, and let G be a commutative S-group prescheme locally of finite type over S. One can show that $\operatorname{Hom}_{S\text{-gr}}(H, G)$ is representable by a prescheme locally of finite type over S. Unlike the corresponding prescheme when H is of multiplicative type, however, it is not étale over S, but only unramified. Thus, for example, if S is the spectrum of a complete discrete valuation ring and H, G are abelian schemes over S, there may exist homomorphisms $H_0 \to G_0$ on the special fibres that do not arise "by specialization" from a homomorphism on the generic fibres.

6.7. — Theorem 6.2 applies whenever H is an abelian scheme over S, or, more generally, an extension of such a scheme by a torus. Indeed, the question is local on S, so one may suppose that there is a prime number ℓ prime to the residue characteristics of S. Taking E to be the set of powers of ℓ then suffices to satisfy conditions (a') and (b).

An argument analogous to that of 6.1 gives the following result.

Theorem 6.8. — *Let X be a prescheme smooth over S, with nonempty connected geometric fibres. For every closed subprescheme Y of X, the functor $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme T of S. If Y is of finite presentation over X, then T is of finite presentation over S.*

Since $f : X \to S$ is faithfully flat and locally of finite presentation, it is a covering for the fpqc topology. On the other hand, $T = \prod_{X/S}(Y/X)$ is plainly a subsheaf of S for the fpqc topology. Hence both the representability of T by a closed subprescheme of S and the question whether T is of finite presentation over S are local on S for the fpqc topology. After the base change $S' \to S$ with $S' = X$, we are reduced to the case in which X admits a section e over S. We may also suppose that S is affine, and hence in particular quasi-compact. We then have:

Corollary 6.9. — *Under the hypotheses of 6.8, suppose that S is quasi-compact and that X admits a section e over S. For every integer $n \geq 0$, let X_n be the order-n infinitesimal neighbourhood of the section e in X. Suppose that Y is of finite presentation over X. Then there exists an integer $n \geq 0$ such that*

$$\prod_{X/S}(Y/X) = \prod_{X_n/S}(Y_n/X_n), \qquad Y_n = Y \cap X_n.$$

This corollary implies 6.8 when Y is of finite presentation over X, by Exposé VIII, 6.4. Indeed, since X is smooth over S, X_n is finite and locally free over S, and a fortiori "essentially free" over S in the sense of Exposé VIII, 6.1. Thus $\prod_{X_n/S}(Y_n/X_n)$ is representable by a closed subprescheme of S. Furthermore, either the proof of the cited result or reduction to the noetherian case shows at once that this closed subprescheme is of finite presentation over S.

Let us first prove 6.9, and hence 6.8, when S is noetherian. Put

$$T_n = \prod_{X_n/S}(Y_n/X_n).$$

The T_n form a decreasing sequence of closed subpreschemes of S, and this sequence is stationary because S is noetherian. Let

$$R = \bigcap_{n\geq 0} \prod_{X_n/S}(Y_n/X_n)$$

be their common value for large n. Plainly $T \subset R$, and it remains to show that $R \subset T$. After the base change $R \to S$, we are reduced to the case $R = S$, that is, $Y_n = X_n$ for every n, or equivalently $Y \supset X_n$ for every n. We must then prove $T = S$, that is, $Y = X$.

The inclusions $Y \supset X_n$ for all n, together with the fact that X is locally noetherian, imply that in a neighbourhood of each point of $e(S)$, Y is an induced open subprescheme of X. Hence there is an induced open subprescheme U of X such that $e(S) \subset U \subset Y$. By Exposé IX, 4.3, since the fibres of X/S are integral, U is schematically dense in X. As Y is a closed subprescheme containing U, it follows that $Y = X$. This proves 6.9, and hence 6.8, in the noetherian case.

The general case is reduced to the preceding one. For every $s \in S$, there is an affine open neighbourhood U of s and an affine open neighbourhood V of $e(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighbourhood of s contained in U. Choose an affine open neighbourhood S' of s contained in $e^{-1}(V) \cap f(V)$, and put $X' = V \cap f^{-1}(S')$. Then X' and S' are affine open subpreschemes of X and S, respectively, and X'/S' admits a section. By the local nature of 6.8 and 6.9, we may suppose that $S = S'$.

We claim that

$$\prod_{X/S}(Y/X) = \prod_{X'/S'}(Y'/X'), \qquad Y' = Y \cap X'.$$

Indeed, by Exposé IX, 4.6, X' is schematically dense in X, at least when X is quasi-separated over S, so that X' is retrocompact in X. In fact, one can readily prove that Exposé IX, 4.6 remains valid without the retrocompactness hypothesis. Likewise, after every base change $S_1 \to S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$, and the claimed equality follows immediately. We are thus reduced to the case $X = X'$, so we may suppose that both S and X are affine.

Now let $X = \operatorname{Spec}(B)$, and let J be the ideal of B defining Y. The ideal J is the filtered colimit of its finite-type subideals. Consequently, Y is the intersection of closed subpreschemes Y_i of X that are of finite presentation over S, and

$$\prod_{X/S}(Y/X) = \bigcap_i \prod_{X/S}(Y_i/X).$$

To prove 6.8, it is therefore enough to treat the case in which Y is of finite presentation over X. It remains to prove 6.9 with S and X affine. But X and Y over S then arise by base change $S \to S_0$ from an analogous situation X_0, Y_0 over a noetherian prescheme S_0. This reduces us to the already treated case in which S is noetherian, and completes the proof of 6.8 and 6.9.

Corollary 6.10. — *Let X be an S-group prescheme, smooth and of finite presentation, with connected fibres. Let Y be a group prescheme of finite presentation over S, and let $i : Y \to X$ be a monomorphism of S-group preschemes, so that Y is a subgroup of X. Then $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation. If S is quasi-compact, let X_n denote, for every $n \geq 0$, the order-n infinitesimal neighbourhood of the unit section of X, and put $Y_n = X_n \cap Y$. Then, for all sufficiently large n,*

$$\prod_{X/S}(Y/X) = \prod_{X_n/S}(Y_n/X_n).$$

The proof is essentially that of 6.9. The unit sections of X and Y induce bijective immersions $S \to X_n$ and $S \to Y_n$, and hence isomorphisms

$$S_{\mathrm{red}} \xrightarrow{\sim} (X_n)_{\mathrm{red}}, \qquad S_{\mathrm{red}} \xrightarrow{\sim} (Y_n)_{\mathrm{red}}.$$

It follows that the injection $Y_n \to X_n$ is proper. Since it is a monomorphism of finite presentation, it is therefore a closed immersion. Exposé VIII, 6.4, already used above, now shows that $\prod_{X_n/S}(Y_n/X_n)$ is representable by a closed subprescheme of S of finite presentation over S. It remains to prove the last assertion of 6.10 under the additional assumption that S is affine.

We reduce once more to the case in which S is noetherian. With the notation of the proof of 6.9, it is enough to prove $R = T$, or, equivalently, that $Y \supset X_n$ for every n implies $Y = X$. This hypothesis implies that $i : Y \to X$ is étale at the points of the unit section of Y over S, and therefore that Y is smooth over S at those points. Hence the open subprescheme Y' consisting of the points where Y is smooth over S is an induced open subgroup of Y. By Exposé X, 3.5, the morphism $Y' \to X$ is an étale monomorphism and hence an open immersion. Since the fibres of X are connected and every open subgroup of an algebraic group is also closed, this open immersion is surjective, and therefore an isomorphism. Thus $Y' = X$, and a fortiori $Y = X$, completing the proof of 6.10.

Arguing as in Exposé VIII, 6.5, we deduce:

Corollary 6.11. — *Let G be an S-group prescheme locally of finite type and quasi-separated over S, let H be a group prescheme smooth and of finite presentation over S, with connected fibres, and let $i : H \to G$ be a monomorphism of S-groups, so that H is a subgroup of G. Then:*

(a) $\mathrm{Centr}_G(H)$ *and* $\mathrm{Norm}_G(H)$ *are representable by closed subpreschemes of G of finite presentation over G. Likewise, for every monomorphism $j : K \to G$ of finite presentation between S-group preschemes,* $\mathrm{Transp}_G(H, K)$ *is representable by a closed subprescheme of G of finite presentation over G.*

(b) *If G is quasi-compact, then in each of the cases in (a) there exists an integer $n \geq 0$ such that, if H_n denotes the order-n infinitesimal neighbourhood of the unit section of H (and K_n is defined similarly in the transporter case),*

$$\begin{aligned}\mathrm{Centr}_G(H) &= \mathrm{Centr}_G(H_n),\\ \mathrm{Norm}_G(H) &= \mathrm{Norm}_G(H_n),\\ \mathrm{Transp}_G(H, K) &= \mathrm{Transp}_G(H_n, K) = \mathrm{Transp}_G(H_n, K_n).\end{aligned}$$

Apply 6.10 over the base prescheme G to the group prescheme

$$X = H_G = H \times_S G.$$

For the centralizer, take as Y the inverse image of the diagonal subgroup of $(G \times_S G)_G$ under a suitable homomorphism of G-groups from X to $(G \times_S G)_G$. For the transporter, take the inverse image of K_G under a suitable homomorphism of G-groups from X to G_G. The normalizer case reduces to the transporter case by taking $K = H$. The hypothesis on G ensures that $H \to G$ is of finite presentation, and hence that $Y \to X$ is of finite presentation. In the centralizer case, the same hypothesis ensures that the diagonal subgroup of $G \times_S G$ is of finite presentation over $G \times_S G$; again it follows that Y is of finite presentation over X.

Remark 6.12. — One can prove, using rather delicate results of EGA VI,[11] that if X is a prescheme flat and of finite presentation over S, and is either proper over S or has nonempty connected geometric fibres, then for every closed subprescheme Y of X of finite presentation over S, the functor $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation over S. Likewise, if X is an S-group prescheme flat and of finite presentation with connected fibres, and $i : Y \to X$ is a monomorphism of S-groups, with Y an S-group prescheme of finite presentation, then $\prod_{X/S}(Y/X)$ is representable by a closed subprescheme of S of finite presentation over S. In particular, 6.11(a) remains valid if the hypothesis "H smooth over S" is replaced by "H flat over S."

[11] Cf. also J. P. Murre, "Representation of unramified functors. Applications," *Séminaire Bourbaki*, no. 294 (May 1965), Theorem 3, p. 13.

Exposé XII
Maximal Tori, the Weyl Group, Cartan Subgroups, and the Reductive Center

by A. Grothendieck

From the present Exposé onwards, in contrast to the preceding Exposés, we shall use the known results on the structure of smooth algebraic groups over an algebraically closed field k, and above all Borel's theory of affine algebraic groups, expounded in the 1956/57 Chevalley Seminar, *Classification of Algebraic Lie Groups.*[0] Following the usage in the theory of algebraic groups, we shall refer to that seminar by the siglum *BIBLE*. In the following Exposés we shall need, in particular, the results of *BIBLE* 4, 5, 6, and 7 (the number following *BIBLE* is the number of the Exposé). Moreover, it appears that scheme theory brings no notable simplification to Borel's theory as it is presented in *BIBLE*. For this reason it did not seem useful to reproduce that theory in the present Seminar: our purpose here is to deduce from results known over algebraically closed fields analogous results valid over an arbitrary base prescheme. (The situation will be different for Chevalley's structure theory of semisimple groups, which, it appears, is more advantageously treated *ab ovo* over an arbitrary base prescheme.)

In the oral Exposés (which we followed in Nos. 1 to 4), we restricted ourselves to affine group preschemes over S, relying essentially on the representability theorems of Exposé XI, No. 4. In the present notes (cf. Nos. 6 to 8), we show quickly how the affineness hypotheses can be eliminated by a simpler method that does not use the most delicate results of Exposé XI. For further generalizations of the results contained in the present Exposé, see also Exposés XV and XVI. (It is clear that the contents of Exposés XI, XII, XV, and XVI ought to be completely rewritten.)

0.1 Maximal tori

1.0. Let first G be an algebraic group over an algebraically closed field k. A *maximal torus of* G is an algebraic subgroup T of G which is a torus (here, since k is algebraically closed, this means that it is isomorphic to a group of the form $\mathbf{G}_m^r$) and which is maximal for this property. Notice that, since k is perfect, G_{red} is a subgroup of G, smooth over k, and hence is essentially an algebraic group in the sense of *BIBLE*. Moreover, every reduced

[-1] Version xy of 2 December 2008.

[0] Editor's note: this seminar has since been published, in a form revised by P. Cartier, as volume 3 of Chevalley's collected works (Springer, 2005).

subgroup of G (such as a torus) is automatically a subgroup of G_{red}. Consequently the maximal tori of G are precisely the maximal tori of G_{red}. When G is affine, and therefore G_{red} is affine, a fundamental theorem of Borel says that any two maximal tori of G are conjugate by an element of

$$G(k) = G_{\text{red}}(k)$$

(*BIBLE* 6, Theorem 4(c)); in particular, they have the same dimension. We call the common dimension of the maximal tori of G the *reductive rank of* G. The assumption that G be affine is in fact unnecessary. This follows from a theorem of Chevalley asserting that every connected smooth algebraic group over k is an extension of an abelian variety by an affine group; cf. No. 6. In Nos. 1 to 4 we shall nevertheless most often restrict ourselves to affine group preschemes over the base.

Let now G be a smooth algebraic group over k, and let T be a maximal torus of G. The centralizer $C(T)$ of T in G will be called the *Cartan subgroup of* G *associated with* T. For us it is a group subprescheme defined by VIII 6.7. Since G is smooth over k, however, its Cartan subgroup C is also smooth over k, by XI 5.3. Thus C is in this case the unique group subprescheme of G, smooth over k (that is, an algebraic subgroup of G in the sense of *BIBLE*), for which $C(k)$ is the subgroup of $G(k)$ centralizing $T(k)$; in other words, it is essentially the centralizer in the sense of *BIBLE*.

By the conjugacy theorem just cited, the Cartan subgroups associated with the various maximal tori are conjugate to one another and therefore have the same dimension. We call their common dimension the *nilpotent rank of* G; it is equal to that of G_{red}. If $\rho_r(G)$ and $\rho_n(G)$ denote the reductive and nilpotent ranks of G, then

$$\rho_r(G) \leq \rho_n(G),$$

and the difference

$$\rho_u(G) = \rho_n(G) - \rho_r(G) = \dim(C/T)$$

might be called, when G is affine, the *unipotent rank of* G. If G is smooth, affine, and connected, then C is a connected nilpotent algebraic group (*BIBLE* 6, Theorem 6(a),(c)). Hence, by *BIBLE* 6, Theorem 2, it is isomorphic to the product

$$C_s \times C_u,$$

where $C_s = T$ is the original maximal torus (and *a fortiori* a maximal torus of C), while C_u is a smooth unipotent subgroup, that is, a successive extension of groups isomorphic to $\mathbf{G}_a$ (*BIBLE* 6, Theorem 1, Corollary 1, and *BIBLE* 7, Theorem 4). In this case one also has

$$\rho_u(G) = \dim C_u.$$

Remark 1.1. Besides the three notions of rank just specified for an affine algebraic group, two others are useful. The first is the *semisimple rank* $\rho_s(G)$, defined as the reductive rank of the quotient G/R, where R is the radical of G. The second is the *infinitesimal rank* $\rho_i(G)$, defined as the nilpotent rank of the Lie algebra of G, which will be defined and studied in the following Exposé. We shall not use them in the present Exposé. We record only the inequalities

$$\rho_s \leq \rho_r \leq \rho_n \leq \rho_i.$$

Lemma 1.2. *Let G be an algebraic group over the algebraically closed field k, let T be an algebraic subgroup of G, and let k' be an algebraically closed extension of k. Let G' and T' be the groups obtained from G and T by base extension. Then T is a maximal torus of G if and only if T' is a maximal torus of G'.*

This is an immediate consequence of the "principle of finite extension" (EGA IV, 9.1.1).

Definition 1.3. Let S be a prescheme, let G be an S-group prescheme of finite type, and let T be a group subprescheme of G. One says that T is a *maximal torus of G* if:

a) T is a torus (IX 1.3); and

b) for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure of $\kappa(s)$, then $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Remarks 1.4. It follows from 1.2 that, when S is the spectrum of an algebraically closed field, this recovers the usual definition, and that property 1.3 is stable under arbitrary base change. Notice that, by X 8.8, condition (a) in Definition 1.3 may be replaced by:

a') T is of finite presentation and flat over S.

One should take care: a maximal torus in the sense of Definition 1.3 is indeed maximal among the subtori of G, as follows immediately from IX 2.9, but the converse cannot hold—say when the base S is connected—unless G actually admits a maximal torus in the sense of 1.3. This is not true in general, even when $S = \operatorname{Spec}(\mathbb{Z})$ and G is "semisimple"; see also 1.6. We shall see in Exposé XIV, however, that the converse is true when S is artinian, or when S is a local scheme and G is "reductive": in that case every torus of G is contained in a maximal torus.

Definition 1.5. Let G be an algebraic group over a field k. The *reductive rank* (respectively the *nilpotent rank*, the *unipotent rank*, and so on) of G is the reductive rank (respectively, and so on) of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k.

By 1.2 and the compatibility of the formation of $\operatorname{Centr}_G(T)$ with base extension, the notions of rank introduced in 1.5 are invariant under extension of the base field. For algebraically closed k, they coincide with the notions introduced at the beginning of the present number.

Remark 1.6. It is not difficult to construct a smooth affine group scheme G over the spectrum S of a discrete valuation ring whose generic fiber is isomorphic to $\mathbf{G}_m$ and whose special fiber is isomorphic to $\mathbf{G}_a$. Such a G contains no torus other than the trivial torus T, reduced to the identity subgroup, and T is plainly not a maximal torus. More precisely, in the special fiber $G_0 = \mathbf{G}_{a,k}$, the torus T_0 is indeed maximal, whereas in the generic fiber $G_1 = \mathbf{G}_{m,K}$, the torus T_1 is no longer maximal; here k is the residue field and K the fraction field of the valuation ring. This example also shows that the reductive rank of G_s, for $s \in S$, is not a continuous function of s. Nevertheless, one has the following results.

Theorem 1.7. *Let S be a prescheme and let G be a smooth affine S-group prescheme. For every $s \in S$, set*

$$\rho_r(s) = \rho_r(G_s), \qquad \rho_n(s) = \rho_n(G_s),$$

the reductive and nilpotent ranks of G_s as in 1.5. Then:

a) *The function ρ_r on S is lower semicontinuous, while ρ_n is upper semicontinuous. Hence $\rho_u = \rho_n - \rho_r$ is upper semicontinuous.*

b) *The following conditions, each stable under arbitrary base change, are equivalent:*

(i) *the function ρ_r on S is locally constant;*

(ii) *G admits a maximal torus locally for the étale topology;*

(ii bis) *G admits a maximal torus locally for the faithfully flat quasi-compact topology.*

c) *Let T_1, T_2 be two maximal tori of G, which in particular places us under the conditions of (b). Then T_1 and T_2 are locally conjugate for the étale topology: there exists a surjective étale morphism $S' \to S$ such that the subgroups $(T_1)_{S'}$ and $(T_2)_{S'}$ of $G_{S'}$ are conjugate by a section of $G_{S'}$ over S'.*

d) *Under the conditions of (b), that is, when ρ_r is locally constant, ρ_n is locally constant as well, and hence so is $\rho_u = \rho_n - \rho_r$.*

Proof. a) Observe that for every morphism $S' \to S$, if $G' = G \times_S S'$, then the functions ρ'_r, and so on, on S', defined in terms of G' as ρ_r, and so on, are defined in terms of G, are obtained simply by composing the latter functions with $S' \to S$. When $S' \to S$ is faithfully flat and quasi-compact, it follows that ρ' is continuous, respectively upper semicontinuous, respectively lower semicontinuous, if and only if ρ is, since the Zariski topology on S is the quotient of that on S' (SGA 1, VIII 4.3). Consequently, the assertions in (a) are local for the faithfully flat quasi-compact topology.

Let $s \in S$. We wish to show that the set U of $t \in S$ such that

$$\rho_r(t) \geq \rho_r(s) \qquad (\text{respectively } \rho_n(t) \leq \rho_n(s))$$

is a neighborhood of s. By the principle of finite extension, there exists a finite extension k of $\kappa(s)$ such that G_k admits a maximal torus. There then exist an open neighborhood U of s and a finite, surjective, locally free morphism $S' \to U$ such that the fiber S'_s is $\kappa(s)$-isomorphic to $\operatorname{Spec}(k)$ (cf. EGA III, 10.3.2, where the noetherian hypothesis is plainly unnecessary). Since $S' \to S$ is an open morphism (SGA 1, IV 6.6), we are reduced to the case $S' = S$, that is, to the case in which G_s admits a maximal torus T_s. Moreover, by XI 5.8(a), after possibly replacing S once more by an étale S' over S, equipped with a point s' above s, we may suppose that T_s is the fiber at s of a subtorus T of G. Then, for every $t \in S$,

$$\rho_r(t) = \rho_r(G_t) \geq \dim T_t = \dim T_s = \rho_r(G_s) = \rho_r(s),$$

which proves that ρ_r is lower semicontinuous.

On the other hand, by XI 5.3, the functor

$$\operatorname{Centr}_G(T)$$

is represented by

$$C = \operatorname{Centr}_G(T),^{1}$$

a closed group subprescheme of G, smooth over S. There is therefore a neighborhood U of s such that $t \in U$ implies

$$\dim C_t = \dim C_s = \rho_n(s).$$

[1] Editor's note: the text has been modified here to introduce the prescheme $C = \operatorname{Centr}_G(T)$ representing the functor $\operatorname{Centr}_G(T)$.

The upper semicontinuity of ρ_n now follows from the relation

$$\rho_n(t) \leq \dim C_t \qquad \text{for every } t \in S,$$

which is itself contained in the following purely geometric lemma.

Lemma 1.8. *Let k be a field, let G be a smooth affine algebraic group over k, let T be a torus in G, and let C be its centralizer. Then*

$$\rho_n(G) \leq \dim C.$$

Indeed, we may suppose k algebraically closed, so that T is contained in a maximal torus T'. If C' is the centralizer of T', then $C' \subset C$, and hence $\dim C' \leq \dim C$. This proves the lemma.

b) If ρ_r is locally constant, then for every torus T in G and every $s \in S$, if T_s is a maximal torus in G_s, there exists an open neighborhood U of s such that $T|_U$ is a maximal torus in $G|_U$. Applying the argument of (a), we see that (i) implies (ii bis). Conversely, (ii bis) implies (i), because when G admits a maximal torus T, it is clear that

$$\rho_r(s) = \dim T_s$$

is a locally constant function of s; and we observed at the beginning of the proof of (a) that continuity of ρ_r is local for the faithfully flat topology. Thus (i) is equivalent to (ii bis), while (ii) plainly implies (ii bis); it remains to prove the converse implication (i) $\Rightarrow$ (ii).

For this purpose, introduce the functor F of XI 4.1,[2] which is a smooth separated prescheme over S, and consider the subfunctor $\mathcal{T}$ of F whose value on an S' over S is the set of maximal tori in $G_{S'}$. By the preceding observation that, under (i), a torus in $G_{S'}$ which is maximal in the fiber over a point $s \in S'$ is maximal over an open neighborhood of s, the functor $\mathcal{T}$ is represented by an open subprescheme of F, and is therefore smooth and separated over S. Since the structural morphism $\mathcal{T} \to S$ is plainly surjective, it admits a section locally for the étale topology by XI 1.10. This proves (i) $\Rightarrow$ (ii).

c) This is an immediate consequence of XI 5.4 bis, taking account of Borel's conjugacy theorem recalled at the beginning of this number.

d) By the observation at the beginning of the proof of (a), and in view of (b), we may suppose that G admits a maximal torus T. If C is its centralizer, then C is representable and smooth over S by XI 5.3, so the function

$$s \longmapsto \rho_n(s) = \dim C_s$$

is indeed locally constant.

This completes the proof of 1.7. We shall refer to the conditions in 1.7(b) by saying that, in this case, G is *of locally constant reductive rank.* We record the following consequence.

Corollary 1.9. *Let G be as in 1.7, and let $s \in S$ be such that $\rho_u(s) = 0$, that is, $\rho_r(s) = \rho_n(s)$ (equivalently, the Cartan subgroups of $G_{\bar{k}}$ are tori, where $\bar{k}$ is an algebraic closure of $\kappa(s)$). Then there exists an open neighborhood U of s on which ρ_r and ρ_n are constant. In particular, for every $t \in U$, the unipotent rank $\rho_u(t)$ of G_t is zero.*

Indeed, this follows immediately from 1.7(a) and the inequality $\rho_r(t) \leq \rho_n(t)$ for every $t \in S$.

Notice also that, in the course of proving (b), we proved the following result.

[2] Editor's note: this is the functor of multiplicative-type subgroups of G.

Corollary 1.10. *Let G be as in 1.7, and suppose that G has locally constant reductive rank. Consider the functor*

$$\mathcal{T} : (\mathbf{Sch})^{\circ} \longrightarrow (\mathbf{Ens})$$

such that, for every S' over S,

$$\mathcal{T}(S') = \{\textit{maximal tori of } G_{S'}\}.$$

Then $\mathcal{T}$ is represented by a smooth, separated prescheme of finite type over S.

It remains only to verify that $\mathcal{T}$ is of finite type over S. Since the question is local for the faithfully flat quasi-compact topology, we may suppose that G admits a maximal torus T. By XI 5.3 bis and the bis version of XI 5.5, $\mathrm{Norm}_G(T)$ and $G/\mathrm{Norm}_G(T)$ are represented by preschemes $\mathrm{Norm}_G(T)$ and $G/\mathrm{Norm}_G(T)$, and $\mathcal{T}$ is isomorphic to

$$G/\mathrm{Norm}_G(T).^{3}$$

The morphism $G \to \mathcal{T}$ defined by $g \mapsto \mathrm{int}(g)(T)$ is surjective. Since G is quasi-compact over S, so is $\mathcal{T}$, which completes the proof.

Remarks 1.11. a) The prescheme $\mathcal{T}$ of 1.10 will properly be called the *prescheme of maximal tori of G*. We shall see in No. 5 that it is in fact affine over S. This can be checked directly when G is isomorphic to a closed group subprescheme of a prescheme of the form $\mathrm{GL}(n)$, using XI 4.6 and observing that, by the argument in the proof of 1.7(b), $\mathcal{T}$ identifies with a subprescheme both open and closed in the prescheme F representing the multiplicative-type subgroups of G.

b) It is possible to prove 1.7, and hence 1.9, without using the delicate results of Exposé XI, using only the elementary results XI 3.12 and XI 6.2 and working solely with multiplicative-type groups finite over the base (morally, the groups ${}_nT$, where T is a maximal torus); compare No. 7. The same observation applies to the proof of 1.10.

We end this number with examples in which there is a unique maximal torus.

Proposition 1.12. *Let G be an S-group prescheme of multiplicative type and finite type. Then G admits a unique maximal torus, and every torus in G is contained in this maximal torus.*

Uniqueness plainly follows from the latter assertion, which characterizes the maximal torus as the largest subtorus of G. Uniqueness also makes existence a question local for the faithfully flat quasi-compact topology. We may therefore suppose that G is diagonalizable, say $G = D_S(M)$, where M is a finitely generated commutative group. Let M_0 be the quotient of M by its torsion subgroup. We claim that the torus

$$T = D_S(M_0)$$

in G is a maximal torus and the largest subtorus. Indeed, a subtorus T' of G is locally diagonalizable for the fpqc topology. Thus, in order to prove $T' \subset T$, we may suppose that T' is diagonalizable, and hence of the form $D_S(N)$, where N is a free quotient of M (VIII 1.4 and 3.2(b)). It follows that N is a quotient of M_0, and therefore that $T' \subset T$. Since the construction of $T = D_S(M_0)$ is compatible with arbitrary base extension, this proves at the same time that T is a maximal torus of G, and completes the proof. When G is smooth over S, Proposition 1.12 has the following generalization.

[3] Editor's note: the text has been modified here to introduce the preschemes $\mathrm{Norm}_G(T)$ and $G/\mathrm{Norm}_G(T)$.

Proposition 1.13. *Let G be an S-group prescheme of finite presentation over S. Suppose that, locally for the faithfully flat quasi-compact topology, G admits a central maximal torus. Then G admits globally a unique maximal torus,*[4] *and it is the largest subtorus of G.*

Here again, uniqueness follows trivially from the last assertion and makes existence a local question for the fpqc topology. We may therefore suppose that G admits a central maximal torus T. We shall show that every torus R of G is contained in T. This follows from the next lemma.

Lemma 1.14. *Let G be an S-group prescheme of finite presentation over S, let T be a maximal torus of G, let R be a subtorus of G, and suppose that T and R commute. Then $R \subset T$.*

Indeed, since R and T commute, the morphism

$$R \times_S T \longrightarrow G, \qquad (r, t) \longmapsto r \cdot t,$$

is a group homomorphism. By IX 6.8 it therefore has an image subgroup T' in G, which is a multiplicative-type group quotient of $R \times_S T$, hence a torus, and plainly contains T. Since T is a maximal torus, $T' = T$ by 1.4, and therefore $R \subset T$. This proves 1.14 and hence 1.13. In particular, using 1.7(b), we obtain:

Corollary 1.15. *Let G be a commutative, smooth, affine S-group prescheme of locally constant reductive rank. Then G admits a unique maximal torus, and this torus contains every subtorus of G.*

Corollary 1.16. *Let G be a smooth affine S-group prescheme. Suppose that, for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure $\bar{k}$ of $\kappa(s)$, the geometric fiber $G_{\bar{s}}$ is a connected nilpotent algebraic group (in the sense of* BIBLE, *that is, the group $G(\bar{k})$ of its $\bar{k}$-valued points is nilpotent). Suppose moreover that G has locally constant reductive rank. Then G admits a unique maximal torus T; moreover, T is central and is the largest subtorus of G.*

By 1.7(b), G admits a maximal torus locally for the fpqc topology, and by 1.13 we are reduced to proving that a maximal torus of G is central. By IX 5.6(b), we are reduced to the case in which S is the spectrum of a field, which we may suppose algebraically closed. Then $T(k)$ lies in the center of $G(k)$ by *BIBLE* 6, Theorem 2. This implies that T lies in the center of G, because $\mathrm{Centr}_G(T)$ is a closed subscheme of G (VIII 6.6) containing all the points of $G(k)$, and is therefore equal to G by the Nullstellensatz, since G is reduced.

Finally, for later reference, we record the following elementary proposition, which we have already used implicitly.

Proposition 1.17. *Let $G \supset H \supset T$ be group preschemes of finite type over S. The following conditions are equivalent:*

(i) *T is a maximal torus of G;*

(ii) *T is a maximal torus of H, and for every $s \in S$ one has equality of reductive ranks*

$$\rho_r(G_s) = \rho_r(H_s).$$

[4] Which is central!

0.2 The Weyl group

Let first k be an algebraically closed field, and let G be a smooth affine algebraic group over k. If T is a maximal torus, C its centralizer, and N its normalizer, then by XI 5.9 these are smooth closed subgroups of G, and C is an open subgroup of N. Thus

$$W = N/C$$

is a finite étale group over k, and is consequently determined by the group $W(k)$ of its k-valued points:

$$W = W(k)_k$$

(the constant group defined by the ordinary finite group $W(k)$). The finite group $W(k)$ will be called the *geometric Weyl group*, or simply the *Weyl group* when no confusion is likely, of G relative to T. By Borel's conjugacy theorem, the Weyl groups relative to the various maximal tori are isomorphic to one another; this is why one sometimes speaks of "the" Weyl group of G, without specifying a maximal torus. Since the formation of C, N, and N/C commutes with arbitrary base extension, if k' is an algebraically closed extension of k, the geometric Weyl group of $G_{k'}$ relative to $T_{k'}$ is canonically isomorphic to that of G relative to T. Consequently, "the" geometric Weyl group of G, properly speaking an isomorphism class of ordinary finite groups, coincides with that of $G_{k'}$.

This permits us, when G is a smooth affine algebraic group over an arbitrary field k, to speak of the *geometric Weyl group of* G as the isomorphism class of the geometric Weyl group of $G_{k'}$, where k' is any algebraically closed extension of k. When G admits a maximal torus T, we can form

$$C = \mathrm{Centr}_G(T), \qquad N = \mathrm{Norm}_G(T), \qquad W = N/C$$

as above. The latter is a finite étale group over k, called the *Weyl group of* G *relative to* T. The geometric Weyl group is then simply the class of the group of points of W with values in any algebraically closed extension k' of k. In general, knowledge of the geometric Weyl group $W(k')$ plainly no longer suffices to reconstruct the algebraic group W: one must also know the action on $W(k')$ of the Galois group of the separable algebraic closure of k in k'.

Finally, let G be a group prescheme over an arbitrary base S, smooth and affine over S, and let T be a maximal torus of G. By XI 5.9 we may again form the group

$$W(T) = N(T)/C(T),$$

which is an étale, separated, quasi-finite S-group prescheme. Its geometric fibers, relative to algebraic closures of the residue fields $\kappa(s)$, $s \in S$, are the geometric Weyl groups of the fibers G_s. Thus XI 5.10 gives information about the variation of these groups with $s \in S$. We can make this information more precise and generalize it as follows.

Theorem 2.1. *Let S be a prescheme and let G be a smooth affine S-group prescheme. For every $s \in S$, let $w(s)$ be the geometric Weyl group of G_s, hence an isomorphism class of finite groups. On the set E of such classes, introduce the following preorder: $w \leq w'$ if and only if w and w' can be represented by finite groups W and W' such that W is isomorphic to a quotient of a subgroup of W'. With this convention:*

a) *The function $s \mapsto w(s)$ from S to E is lower semicontinuous.*

b) *Suppose that G has locally constant reductive rank. Then the following conditions are equivalent:*

(i) *the function $s \mapsto w(s)$ is locally constant;*

(i bis) *the function $s \mapsto \operatorname{card} w(s)$ is locally constant;*

(ii) *locally for the étale topology (or merely for the fpqc topology), there exists a maximal torus T such that $W(T)$ is finite over its base;*

(ii bis) *for every S' over S and every maximal torus T of $G_{S'}$, the associated Weyl group $W(T)$ is finite over S'.*

Proof. a) Arguing as in 1.7(a), in order to prove that for every $s \in S$ there exists an open neighborhood U of s such that $t \in U$ implies $w(t) \geq w(s)$, we are reduced to the case in which there is a torus R in G such that R_s is a maximal torus in G_s. Put

$$W(R) = \operatorname{Norm}_G(R)/\operatorname{Centr}_G(R).$$

As in XI 5.9, this is an étale, separated, quasi-finite group prescheme over S. For $t \in S$, let $w'(t) \in E$ be its geometric fiber at t. Since R_s is a maximal torus in G_s, and the formation of Norm, Centr, and Norm / Centr commutes with arbitrary base extension, in particular with passage to fibers, one has

$$w(s) = w'(s).$$

We claim moreover that for t near s,

$$w(t) \geq w'(t) \qquad \text{and} \qquad w'(t) \geq w'(s),$$

which will suffice to prove (a). These two inequalities are contained in the following two lemmas.

Lemma 2.2. *Let S be a prescheme, and let W be an étale, separated, quasi-finite S-group prescheme. For every $s \in S$, let $f(s)$ be the class of the geometric fiber of W at s, an element of the ordered set E of isomorphism classes of finite groups introduced in 2.1. Then the function $f : S \to E$ is lower semicontinuous. It is constant in a neighborhood of $s \in S$ if and only if the function $s \mapsto \operatorname{card} f(s)$ is constant there, and this holds if and only if W is finite over S above some open neighborhood U of s.*

This result refines, in group-theoretic terms, the result used in the proof of XI 5.10. We give only a sketch of the proof, which is entirely standard. As usual, we are reduced to the case in which S is affine and noetherian. The function f is immediately seen to be constructible (EGA 0_{III}, 9.3.1 and 9.3.2, together with the sorites of EGA IV, 9), and EGA 0_{III}, 9.3.4 reduces semicontinuity to proving that if t is a generalization of s, then

$$f(t) \geq f(s).$$

By EGA II, 7.1.7, this reduces us to the case in which S is the spectrum of a discrete valuation ring, which may be supposed complete with algebraically closed residue field. Let s be the closed point of S. Since W is étale and separated over S, it contains a subscheme W', both open and closed and finite over S, such that $W'_s = W_s$ (EGA II, 6.2.6); it is immediate here that W' is a subgroup of W. Moreover, since W' is finite étale over $S = \operatorname{Spec}(V)$, with V complete and with algebraically closed residue field, it is a constant group, hence of the form A_S, where

$$A = W'(\kappa(s)) = W(\kappa(s))$$

has class $f(s)$. If B is the geometric fiber of W at the generic point t of S, there is therefore a canonical monomorphism $A \to B$, proving $f(s) \leq f(t)$. Notice that this argument actually proves semicontinuity for an order on E finer than the one specified in 2.1.

The equivalence between constancy of f at s and constancy of $s \mapsto \operatorname{card} f(s)$ follows from the fact that, for $w, w' \in E$,

$$w \leq w' \quad \text{and} \quad \operatorname{card} w = \operatorname{card} w' \quad \Longrightarrow \quad w = w'.$$

The equivalence of this condition with finiteness of W over a neighborhood of s is independent of the group structure on W; it was noted after XI 5.10, and also follows readily from the preceding argument using the valuative criterion of properness (EGA II, 7.3.8).

Lemma 2.3. *Let G be a smooth affine algebraic group over an algebraically closed field k, and let $R \subset T$ be two subtori. Let $W(R)$ and $W(T)$ be the two associated finite groups of XI 5.9, obtained as the quotient of the normalizer by the centralizer. Then $W(R)$ is isomorphic to a quotient of a subgroup of $W(T)$.*

Indeed, consider the diagram

$$\begin{tikzcd}
T \arrow[r, hook] & C(T) \arrow[r, hook] \arrow[d, hook] & C(R) \arrow[d, hook] \\
& N(T) \cap N(R) \arrow[r, hook] \arrow[d, hook] & N(R) \\
& N(T) &
\end{tikzcd}$$

Then $(N(T) \cap N(R))/C(T)$ is a subgroup of

$$W(T) = N(T)/C(T),$$

and there is an evident homomorphism

$$(N(T) \cap N(R))/C(T) \longrightarrow W(R) = N(R)/C(R).$$

Everything reduces to proving that this last homomorphism is surjective, that is, that for every k-valued point g of $N(R)$, there exists a k-valued point c of $C(R)$ such that cg normalizes T, or equivalently such that

$$\operatorname{int}(c)\bigl(\operatorname{int}(g)T\bigr) = T.$$

For this, it suffices to observe that $\operatorname{int}(g)T$ is a torus of $N(R)$, hence of $C(R)$, which is an open subgroup of $N(R)$. The tori T and $\operatorname{int}(g)T$ are maximal tori of $C(R)$, since they are maximal in G, and the assertion follows from Borel's conjugacy theorem.

This proves 2.3, and hence 2.1(a).

b) We have already observed that (i) and (i bis) are plainly equivalent. They imply (ii bis), by the converse part of 2.2 or, equivalently, XI 5.10. Condition (ii bis) implies (ii) by 1.7(b). Finally, (ii) implies (i bis): as in 1.7(a), condition (i) is local for the fpqc topology, so we may suppose that G admits a maximal torus T such that $W(T)$ is finite over S, and XI 5.10 again gives the conclusion. This completes the proof of 2.1.

0.3 Cartan subgroups

Definition 3.1. Let G be a smooth group prescheme of finite type over a prescheme S. A *Cartan subgroup of* G is a group subprescheme C of G, smooth over S, such that for every $s \in S$, if $\bar{s}$ denotes the spectrum of an algebraic closure of $\kappa(s)$, then $C_{\bar{s}}$ is a Cartan subgroup, in the sense of No. 1, of $G_{\bar{s}}$.

It is immediate that if C is a Cartan subgroup of G, then for every S' over S, $C_{S'}$ is a Cartan subgroup of $G_{S'}$. If S is the spectrum of an algebraically closed field, this recovers the notion recalled in No. 1. Finally, the property that a group subprescheme C of G be a Cartan subgroup is plainly local for the faithfully flat quasi-compact topology.

Theorem 3.2. *Let G over S be as in 3.1, and suppose that G is affine over S and has locally constant reductive rank. Then the map*

$$T \longmapsto \mathrm{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups[5] *of G. If C corresponds to T, then T is the unique maximal torus of C.*

If T is a maximal torus of G, the functor $\mathrm{Centr}_G(T)$ is represented, by XI 5.3 or 6.2, by a closed smooth subprescheme

$$C = \mathrm{Centr}_G(T)$$

of G, and the definitions show that C is a Cartan subgroup of G. Moreover, T is plainly a maximal torus of C, and, being central, it is the unique maximal torus of C by 1.13. Thus the map $T \mapsto \mathrm{Centr}_G(T)$ from the set of maximal tori to the set of Cartan subgroups is injective; it remains to prove surjectivity.

Let C be a Cartan subgroup of G. We shall prove that $C = \mathrm{Centr}_G(T)$ for a maximal torus T of G. It is enough to find a maximal torus T of C, for then T is a maximal torus of G: for every $s \in S$, the groups G_s and C_s have the same reductive rank. By IX 5.6(b), T lies in the center of C, so

$$C \subset C' = \mathrm{Centr}_G(T).$$

The smooth subgroup C of the smooth group C' over S coincides with C' fiber by fiber; hence $C = C'$.

Since G has locally constant reductive rank, so does C. Therefore C admits a maximal torus locally for the étale topology by 1.7(b); that torus is central by the preceding argument, and 1.13 then shows that C itself admits a maximal torus. This completes the proof.

Corollary 3.3. *Let G be a smooth affine group prescheme over S, of locally constant reductive rank, and let*

$$\mathcal{C} : (\mathrm{Sch})^{\circ}_{/S} \longrightarrow (\mathrm{Ens})$$

be the functor defined by

$$\mathcal{C}(S') = \{\textit{Cartan subgroups of } G_{S'}\}.$$

Then $\mathcal{C}$ is isomorphic to the functor $\mathcal{T}$ of 1.10, and is therefore represented by the same smooth separated prescheme of finite type over S.

[5] The proof given here in fact proves Theorem 3.2 only for closed Cartan subgroups of G. However, 7.1(a) establishes Theorem 3.2 as stated, and implies that the Cartan subgroups of G are closed.

Corollary 3.4. *Under the hypotheses of 3.2, if* $C = \mathrm{Centr}_G(T)$*, then*

$$\mathrm{Norm}_G(C) = \mathrm{Norm}_G(T).$$

Remark 3.5. When G does not have locally constant reductive rank, it may nevertheless admit Cartan subgroups. For example, if G has connected nilpotent fibers, then G is a Cartan subgroup of itself, although it need not have locally constant reductive rank; see 1.6. In Exposé XV we develop the theory of Cartan subgroups without assuming either that G is affine over S or that its reductive rank is locally constant, using the theory of regular elements from the next Exposé. See also Nos. 6 and 7 below for the removal of the affineness hypothesis.

0.4 The reductive center

Definition 4.1. Let S be a prescheme, let G be an S-group of finite presentation over S with affine fibers, and let Z be a group subprescheme of G. We say that Z is a *reductive center* of G if:

(i) Z is central and of multiplicative type;

(ii) for every base change $S' \to S$ and every central homomorphism

$$u : H \longrightarrow G_{S'},$$

where H is a group of multiplicative type and of finite type over S', the homomorphism u factors through $Z_{S'}$.

(For a variant of this notion when the fibers of G are not assumed affine, see 8.6.)

Such a Z is necessarily of finite type over S, since its fibers are, and is therefore uniquely determined as the largest central subgroup of multiplicative type in G. It is easy to give examples in which G, smooth and affine over S, admits a largest central subgroup Z of multiplicative type but Z is not a reductive center—that is, G admits no reductive center; see, for example, 1.6. It follows readily, however, from IX 6.8 that a subgroup Z of G is a reductive center if and only if it is a largest central subgroup of multiplicative type and retains this property after every base change. See also 4.3 below.

It is clear from 4.1 that if Z is the reductive center of G, then $Z_{S'}$ is the reductive center of $G_{S'}$ after every base change $S' \to S$. This fact and the uniqueness of the reductive center, together with faithfully flat quasi-compact descent (SGA 1, VIII), give:

Proposition 4.2. *Let G be an S-group prescheme of finite presentation over S, with affine fibers. If G admits a reductive center locally for the faithfully flat quasi-compact topology, then it admits a reductive center. For Z to be a reductive center of G, it is necessary and sufficient that it be so locally for the fpqc topology.*

We also record:

Proposition 4.3. *Let G be an S-group prescheme of finite presentation with affine fibers, and let Z be a group subprescheme. For Z to be a reductive center of G, it is necessary and sufficient that it be of multiplicative type and that, for every $s \in S$, Z_s be a reductive center of G_s.*

Indeed, IX 5.6(b) first shows that Z is central. Since the conditions under consideration are stable under base change, it remains to prove that every central homomorphism

$$u : H \longrightarrow G,$$

with H of multiplicative type and of finite type, factors through Z. Since Z is central, u and the canonical immersion $v : Z \to G$ define a group homomorphism

$$w : H \times_S Z \longrightarrow G.$$

By IX 6.8, this homomorphism has a group image K, which is a subgroup of multiplicative type in G. Everything reduces to showing that $K = Z$. This holds fiber by fiber by the hypothesis on Z, and IX 5.1 bis now applies, completing the proof of 4.3.

In criterion 4.3, the hypothesis that Z_s be the reductive center of G_s for every $s \in S$ is in fact purely geometric: it is enough to verify it after passing to an algebraic closure of $\kappa(s)$, as follows from the second assertion of 4.2.

Theorem 4.4. *Let G be an affine algebraic group over a field k. Then G admits a reductive center.*

Since the center of G is represented by a closed subgroup of G by VIII 6.7, we are immediately reduced to the case in which G is commutative. Moreover, by 4.2 we may suppose that k is algebraically closed. We shall see in Exposé XVII that in this case G can be written as a product $Z \times U$, where Z is of multiplicative type and U is "unipotent", from which it follows at once that Z is a reductive center of G. We give here a proof independent of the general structure theorem for commutative algebraic groups just indicated.

By IX 6.8, the set of subgroups H of multiplicative type in G is filtered upward. We show that it has a largest element.

When G is smooth over k, the structure theorem BIBLE 4, Th. 4 gives

$$G = G_s \times G_u,$$

where G_s is of multiplicative type and G_u is "unipotent": here this means that G_u has a composition series whose factors are subgroups of $\mathbb{G}_a$, smooth if desired. Indeed, G_u/G_u^0 is a unipotent group by BIBLE 4, Corollary to Th. 3, hence a p-group, where p is the characteristic exponent, by BIBLE 4, Prop. 4. On the other hand, G_u^0 has a composition series with smooth connected one-dimensional quotients by BIBLE 6, Th. 1, Cor. 1, and these quotients are isomorphic to $\mathbb{G}_a$ by BIBLE 7, Th. 4. Every homomorphism from a group of multiplicative type to $\mathbb{G}_a$, and therefore also to G_u, is trivial, as follows from the lemma below. Thus every subgroup of multiplicative type in G is contained in G_s.

In the general case, consider the subgroup

$$G_0 = G_{\mathrm{red}}$$

of G. It is smooth over k, and hence by the preceding case has a largest subgroup Z_0 of multiplicative type. The subgroups of multiplicative type in G containing Z_0 correspond to the subgroups of multiplicative type in $G' = G/Z_0$. We have an exact sequence

$$0 \longrightarrow G_0/Z_0 \longrightarrow G/Z_0 \longrightarrow G/G_0 \longrightarrow 0,$$

where G_0/Z_0 has no subgroup of multiplicative type other than the unit group. Since every subgroup of a group of multiplicative type is again of multiplicative type by IX 8, every subgroup H of multiplicative type in G/Z_0 satisfies

$$H \cap (G_0/Z_0) = 0;$$

thus $H \to G/G_0$ is injective. The algebraic group G/G_0 is radicial, hence finite over k, so H is itself radicial and its rank is bounded above by that of G/G_0. Consequently the family of subgroups of multiplicative type in G has a largest element, say Z.

We claim that G/Z has no subgroup of multiplicative type other than the unit subgroup. Indeed, a commutative extension of two algebraic groups of multiplicative type is of multiplicative type (7.1.1). If Z' is the inverse image in G of a subgroup of multiplicative type in G/Z, then Z' is of multiplicative type. The maximality of Z gives $Z' = Z$, and hence $Z'/Z = 0$.

The "principle of finite extension" now readily shows that for every extension K of k,

$$(G/Z)_K = G_K/Z_K$$

likewise has no subgroup of multiplicative type other than the unit group.

We can now prove that Z is a reductive center of G. Let

$$u : H \longrightarrow G_S$$

be a homomorphism of S-groups, where S is a prescheme over k and H is of multiplicative type and of finite type over S. We must show that u factors through Z_S, or equivalently that the composite

$$H \longrightarrow G_S \longrightarrow (G/Z)_S = G_S/Z_S$$

is trivial. Put $U = G/Z$. Every homomorphism $H \to U_S$ is trivial: by IX 5.2 one reduces to the case where S is the spectrum of a field K, and IX 6.8 then reduces the claim to the fact that U_K has no nontrivial subgroup of multiplicative type. This proves 4.4, modulo the following lemma.

Lemma 4.4.1. *Let H be an S-group prescheme of multiplicative type. Then every homomorphism of S-groups*

$$u : H \longrightarrow \mathbb{G}_a$$

is trivial.

Regard $E = \mathcal{O}_S^2$ as an extension with submodule $E' \simeq \mathcal{O}_S$ and quotient $E'' \simeq \mathcal{O}_S$. Then $\mathbb{G}_a$ identifies with the prescheme of automorphisms of this extension. A homomorphism $u : H \to \mathbb{G}_a$ therefore amounts to an H-module structure on E preserving the extension, with E' stable under H and with the induced actions on E' and E'' trivial. By I 4.7.3, H acts trivially on E; hence u is trivial.

Remark 4.4.2. If G is a nonzero abelian variety over k, then G admits no reductive center in the sense of 4.1 with the restriction "G affine" omitted. Indeed, if n is prime to the characteristic, then ${}_nG$ is étale over k, of order prime to the characteristic, and hence of multiplicative type. The family of the ${}_nG$ is schematically dense in G. A reductive center, if it existed, would therefore equal G, which is impossible because G is not of multiplicative type. This explains the requirement in 4.1 that G have affine fibers.

Lemma 4.5. *Let S be a prescheme, let G be a smooth affine S-group prescheme with connected fibers, let T be a maximal torus of G, and let $u : H \to G$ be a central homomorphism, where H is an S-group prescheme of multiplicative type and of finite type. Then u factors through T.*

Let $C = \operatorname{Centr}_G(T)$. By XI 5.3 it is a closed smooth group subprescheme of G, hence affine over S. Since T is central in C, it is invariant, and the quotient group

$$U = C/T$$

is represented by VIII 5.1. Because u is central it factors through C, and it remains only to show that the composite

$$H \longrightarrow C \longrightarrow U = C/T$$

is trivial. By IX 5.2 we reduce to the case where S is the spectrum of a field, which we may suppose algebraically closed. Then BIBLE 6, Th. 2 shows that U is a connected algebraic group, smooth over k, which is "unipotent"; as recalled above, it has a composition series with quotients isomorphic to $\mathbb{G}_a$. Every homomorphism from the finite-type group H of multiplicative type to U is therefore trivial. This proves 4.5.

Corollary 4.6. *Let G be as in 4.5. If G admits a reductive center, then that center is contained in every maximal torus of G.*

Theorem 4.7. *Let S be a prescheme and let G be an S-group prescheme, smooth and affine over S, with connected fibers.*

(a) *For every $s \in S$, let $z(s)$ denote the type of the reductive center of G_s, which is defined by 4.4. Order the set E of isomorphism classes of finite-type $\mathbb{Z}$-modules by declaring the class of M to be greater than that of M' when M' is isomorphic to a quotient of M. Then the map*

$$S \longrightarrow E, \qquad s \longmapsto z(s),$$

is lower semicontinuous.

(b) *The group G admits a reductive center Z if and only if the preceding function $z : S \to E$ is locally constant. If this holds, then G/Z is representable* (cf. VIII 5.1), *and its reductive center is the unit subgroup.*

(c) *Suppose that the reductive rank of G is locally constant* (cf. 1.7(b)). *Then G admits a reductive center Z. If G/Z is representable—for example, if G is affine over S— then the maximal tori T of G* (respectively, the Cartan subgroups C of G) *and the maximal tori T' of $G' = G/Z$* (respectively, the Cartan subgroups C' of G') *are in bijective correspondence:*

$$T' = T/Z, \qquad C' = C/Z,$$

and conversely

$$T = \varphi^{-1}(T'), \qquad C = \varphi^{-1}(C'),$$

where $\varphi : G \to G'$ is the canonical homomorphism.

(d) *Let T be a maximal torus of G, let $\mathfrak{g}$ denote the Lie algebra of G, and consider the homomorphism*

$$\theta : T \longrightarrow \mathrm{GL}(\mathfrak{g})$$

induced by the adjoint representation of G (II 4). *Then* $\ker\theta$ *is a reductive center of G.*

Proof. (a) and (b). The proof of (a) and of the first assertion of (b) is essentially identical to that of 1.7(a) and (b), and we shall not repeat the argument. We point out only that in (a) one must appeal to IX 5.6, using the fact that G has connected fibers.

We prove the second assertion of (b). Suppose that Z is a reductive center of G, and put $G' = G/Z$. By VIII 5.1 the quotient is representable. It is affine over S, of finite presentation over S by VIII 5.8, and even smooth over S: this may be checked on the fibers because G' is flat and of finite presentation over S, and $\mathrm{VI_B}$ 9.2(xii) states that a quotient of a smooth algebraic group over a field is smooth. Since G has connected fibers, so does G'. Thus G' satisfies the same hypotheses as G.

By 4.3 we may reduce to the case in which S is the spectrum of a field. Let Z' be a central subgroup of multiplicative type in G'; it remains to prove that Z' is the unit subgroup. Let Z_1 be the inverse image of Z' in G, and consider the action of Z_1 on G by inner automorphisms. Since Z is central, it acts trivially, and hence Z_1 acts through an action of Z' on G. Moreover Z' acts trivially both on Z and on $G/Z = G'$. Since Z is central, the action of Z' on G therefore comes from a group homomorphism

$$u : Z' \longrightarrow \mathrm{Hom}_{S\text{-gr}}(G', Z),$$

characterized by

$$r(z') \cdot g = g \cdot u(z')(\varphi(g)),$$

where $r : Z' \to \mathrm{Aut}_{S\text{-gr}}(G)$ is the representation just considered and $\varphi : G \to G'$ is the canonical homomorphism. Giving u is equivalent to giving a group homomorphism

$$v : G' \longrightarrow \mathrm{Hom}_{S\text{-gr}}(Z', Z).$$

By X 5.8 the group on the right is representable and étale over S. Its unit section is therefore an open immersion, so $\ker v$ is an open subgroup of G'. Since G' has connected fibers, $\ker v = G'$. Thus $v = 0$, hence $u = 0$, and Z', and therefore Z_1, acts trivially on G. Consequently Z_1 is central in G. It is a commutative extension of the group Z' of multiplicative type by the group Z of multiplicative type, and hence, over a field, is itself of multiplicative type by 7.1.1. Being central in G, it is contained in the reductive center Z. Thus $Z_1 = Z$, whence Z' is the unit subgroup. This proves (b).

(d) Put $Z = \ker \theta$, a subgroup of multiplicative type in T, for example by IX 6.8. By 4.5, every central homomorphism $u : H \to G$, with H of multiplicative type and of finite type, factors through T, and therefore through Z. Since formation of Z commutes with every base change, it remains to show that Z is central, or equivalently that its centralizer C is equal to G. By XI 5.3, C is a closed smooth subgroup of G. Since Z acts trivially on $\mathfrak{g}$, we have

$$\mathrm{Lie}(C) = \mathfrak{g}.$$

It follows that $C = G$. Indeed, the immersion $C \to G$ is étale, as may be checked fiberwise: it is an unramified homomorphism of smooth algebraic groups of the same dimension ($\mathrm{VI_B}$ 1.3), so X 3.5 applies. Hence $C \to G$ is an étale closed immersion, and therefore an open immersion by SGA 1, I 5.1. Since it is both open and closed and G has connected fibers, it is an isomorphism.

(c) By 1.7(b), G admits a maximal torus locally for the fpqc topology. Proposition 4.2 and part (d) therefore show that G admits a reductive center Z. Part (d) also shows that every maximal torus T of G contains Z. Clearly $T' = T/Z$ is a torus. To prove that it is a maximal torus of G', Definition 1.3 reduces us to the case in which S is the spectrum of an algebraically closed field; the assertion then follows from *BIBLE* 7, Th. 3(a).

Conversely, let T' be a maximal torus of G', and set $T = \varphi^{-1}(T')$. We prove that T is a maximal torus of G. The question is local for the fpqc topology, so we may suppose that G already admits a maximal torus T_0. By what precedes, $T_0' = T_0/Z$ is a maximal torus of G'. By the conjugacy theorem 1.7(c), T' and T_0' are locally conjugate for the fpqc topology. Since $\varphi : G \to G'$ is a covering for this topology, sections of G' lift locally to sections of G; thus T and T_0 are likewise locally conjugate. Hence T is a maximal torus. The argument for Cartan subgroups is identical. This completes the proof.

Here is a useful reformulation of (d) when T is diagonalizable, say $T = D_S(M)$, where M is a finite-type free $\mathbb{Z}$-module. By I 4.7.3, the T-module $\mathfrak{g}$ decomposes into the direct sum of its T-submodules $\mathfrak{g}_m$, $m \in M$:

$$\mathfrak{g} = \bigoplus_{m \in M} \mathfrak{g}_m.$$

These submodules are necessarily locally free. Suppose that for every $m \in M$ the rank of $\mathfrak{g}_m$ is constant, as is the case in particular when S is connected. Then the set

$$R = \{m \in M \mid \mathfrak{g}_m \neq 0\}$$

of "roots" is finite. We obtain:

Corollary 4.8. *Under the preceding hypotheses and with the preceding notation, the reductive center of G is the intersection of the kernels in T of the root characters $m \in R$. Thus there is an isomorphism*

$$Z \simeq D_S(N),$$

where N is the quotient of M by the subgroup generated by R.

Corollary 4.9. *Let G be an algebraic group over an algebraically closed field k. Suppose that G is smooth, connected, and affine over k, that its reductive center is the unit subgroup, and that its Lie algebra $\mathfrak{g}$ is nilpotent. Then G is "unipotent," that is, it admits a composition series whose quotients are isomorphic to $\mathbb{G}_a$.*

By *BIBLE* 6, Th. 4, Cor. 3, it is enough to prove that a maximal torus T of G is the unit subgroup, or equivalently that its Lie algebra $\mathfrak{t}$ is zero. The T-module $\mathfrak{g}$ decomposes according to the characters of T by I 4.7.3, so for every $t \in \mathfrak{t}$, the endomorphism $\mathrm{ad}_{\mathfrak{g}}(t)$ of $\mathfrak{g}$ is semisimple. If $\mathfrak{g}$ is nilpotent, this endomorphism is also nilpotent, hence zero. By 4.7(d), the hypothesis on the reductive center makes the homomorphism

$$T \longrightarrow \mathrm{GL}(\mathfrak{g})$$

a monomorphism, and it therefore induces an injective map on Lie algebras. By II 4.5, this says that $\mathrm{ad}_{\mathfrak{g}}(t) = 0$ implies $t = 0$. Consequently $\mathfrak{t} = 0$, as required.

Proposition 4.10. *Let G be a smooth connected affine algebraic group over an algebraically closed field k. Then the reductive center of G is the intersection of the maximal tori of G.*

The intersection is meant schematically, or equivalently in the sense of subfunctors of G: it is the largest closed subprescheme of G contained in every maximal torus. Since G is noetherian, it is already the intersection of a suitable finite family of maximal tori.

Let Z denote this intersection. It is a closed subgroup of a torus, hence is of multiplicative type, and by 4.5 it contains the reductive center of G. To prove equality it remains to show that Z is central. Since G is connected, IX 5.5 reduces this to showing that Z is invariant. By construction Z is invariant under every inner automorphism $\mathrm{int}(g)$, with $g \in G(k)$. Its normalizer N, represented by VIII 6.7, is therefore a closed subgroup of G containing all rational points of G. Since G is reduced, $N = G$, which proves the result.

Proposition 4.11. *Let S be a prescheme and let G be a smooth affine S-group prescheme with connected fibers and unipotent rank zero. Then G has locally constant reductive rank by 1.9, and hence the "prescheme of maximal tori of G," denoted by $\mathcal{T}$, is defined by 1.10 and is smooth, separated, and of finite type over S. Let G act on $\mathcal{T}$ through inner automorphisms, giving a homomorphism of group functors*

$$u : G \longrightarrow \mathrm{Aut}_S(\mathcal{T}).$$

Under these hypotheses the following three subfunctors of G are identical:

(i) *the reductive center Z of G;*

(ii) *the center Z' of G;*

(iii) *the kernel $Z'' = \ker u$ of the preceding homomorphism.*

In particular, the center of G is represented by a subgroup of multiplicative type in G.

Proof. Clearly $Z \subset Z' \subset Z''$. It remains to prove $Z'' \subset Z$, or, since the hypotheses are stable under base change, that every section g of G over S which acts trivially on $\mathcal{T}$ is a section of Z. Put $G' = G/Z$. By 4.7(b) and (c), the prescheme $\mathcal{T}'$ of maximal tori of G' is canonically isomorphic to $\mathcal{T}$. We are therefore reduced to the case $G = G'$, in which the reductive center Z is the unit subgroup. Notice that 4.7(c) also shows that the unipotent rank of G' equals that of G, and hence is zero.

We must prove that g is the unit section of G. The usual reduction process reduces us first to the case in which S is noetherian, and then to the case in which S is local artinian: when S is locally noetherian, it is enough to check the assertion after every base change $S' \to S$ with S' local artinian. In this case $Z'' = \ker u$ is representable by VIII 6.2(a) and 6.5(c); alternatively, representability follows from XI 6.8. By Nakayama's lemma it is enough to show that the fiber Z''_0 is the unit subgroup. We may therefore suppose that S is the spectrum of a field k, which we may take to be algebraically closed.

The subgroup Z'' lies in the stabilizer of every point of $\mathcal{T}(k)$, that is, in the normalizer $N(T)$ of every maximal torus T of G. Since the unipotent rank of G is zero, XI 5.9 says that T is an open subgroup of $N(T)$. Thus

$$\mathrm{Lie}\, N(T) = \mathrm{Lie}\, T,$$

and $\mathrm{Lie}\, Z''$ is contained in $\mathrm{Lie}\, T$ for every maximal torus T. Proposition 4.10 shows that the intersection of the Lie algebras of all maximal tori is the Lie algebra of the reductive center Z, which is zero in the present reduction. Hence $\mathrm{Lie}\, Z'' = 0$, so Z'' is étale over k.

Moreover, Z'' is invariant in G. Since G is connected, it follows that Z'' is central in G. Thus it is contained in

$$T = \mathrm{Centr}_G(T)$$

for every maximal torus T, and therefore in their intersection. By 4.10 this intersection is $Z = 0$, completing the proof.

0.5 Application to the prescheme of subgroups of multiplicative type[6]

Theorem 5.1. *Let S be a prescheme, let G be a smooth affine S-group prescheme, and let M be the "prescheme of subgroups of multiplicative type in G," representing the functor*

[6]For a generalization of the results of this section to the case in which G is not assumed affine over S, see M. Raynaud, "Ample sheaves on group schemes and homogeneous spaces," Chapter IX.

described in XI 4.1. For every integer $n > 0$, let T_n be the subfunctor of M for which $T_n(S')$ is the set of subgroup preschemes H of multiplicative type in $G_{S'}$ satisfying

$$n\,\mathrm{id}_H = 0.$$

Thus T_n is representable and affine over S by XI 3.12(a). Let

$$u_n : M \longrightarrow T_n$$

be the morphism defined by $u_n(H) = {}_nH$, where ${}_nH = \ker(n\,\mathrm{id}_H)$. Then:

(a) *Every subprescheme U of M of finite type over S is contained in a closed subprescheme of finite type over S. Every closed subprescheme X of M of finite type over S is affine over S.*

(b) *Suppose that S is quasi-compact, and let X be a closed subprescheme of M of finite type over S. Then there is an integer $n > 0$ such that, for every multiple m of n, the induced morphism*
$$u_m|_X : X \longrightarrow T_m$$
is a closed immersion.

Proof. (a) For the first assertion, take for X the schematic closure of U (EGA I, 9.5.1 and 9.5.3). This closure is defined because the immersion $i : U \to M$ is quasi-compact: U is of finite type over S, and M is separated over S by XI 4.1. It remains to prove that such an X is affine over S, which will prove the second assertion at the same time. The question is local on S, so we may suppose that S is affine. Since U is of finite type over S, it is contained in a quasi-compact open of M. We are therefore reduced to the case in which U itself is an open subprescheme of finite type over S, hence quasi-compact. Notice that a closed subscheme of a scheme affine over S is affine over S.

It is enough to show that such a U is contained in a closed affine subprescheme of M. In this form the usual reduction process reduces us immediately to the case in which S is noetherian. The same argument applies to (b), reducing it to the case in which S is affine noetherian.

Recall the inverse limit T of the T_n used in the proof of XI 4.1. It is an affine prescheme over S, though not of finite type, and its local rings are noetherian, as was observed at the beginning of the proof of IX 3.7. More explicitly, if $t = (t_n) \in T$, then the transition morphisms $T_m \to T_n$ are smooth and the dimensions of the T_m are bounded. Hence there is an n_0 such that $T_n \to T_{n_0}$ is étale at t_n for every multiple n of n_0. The proof just cited, or XI 3.11, shows that the canonical morphism

$$u : M \longrightarrow T$$

is an immersion and induces isomorphisms on local rings. It need not be an open immersion or a quasi-compact morphism.

Let U be a quasi-compact open of M. We shall show that its closure in T is contained in M. If X is the schematic closure of U in M, this will make

$$u|_X : X \longrightarrow T$$

a closed immersion. Then X is affine, proving (a). Since U is noetherian it has only finitely many irreducible components. Every $t \in T$ in the closure of U is a specialization of a point $x \in U$. The local ring of t in T is noetherian, and so is the local ring A of t in $\overline{\{x\}}$, endowed with the induced reduced structure.

The canonical morphism from the noetherian integral local scheme

$$S' = \operatorname{Spec}(A)$$

to T sends its closed point to t and its generic point to $x \in M$. We must show that $t \in M$. Replacing A, if necessary, by the quotient by a minimal prime of a suitable complete local A-algebra flat over A (EGA 0_{III}, 10.3.1), we may suppose that A is complete and has algebraically closed residue field. One could even reduce further to a discrete valuation ring by EGA II, 7.1.7. Renaming S' as S, we are reduced to the following lemma.

Lemma 5.2. *Let S be the spectrum of a complete noetherian integral local ring with algebraically closed residue field. Let G be a smooth affine S-group prescheme, let $\big(H(n)\big)_{n>0}$ be a system of subgroups of multiplicative type in G, indexed by the positive integers, and let η be the generic point of S. Suppose that:*

(a) *if m is a multiple of n, then*
$$H(n) = {}_nH(m);$$

(b) *there is a subgroup H_η of multiplicative type in G_η such that*
$$H(n)_\eta = {}_n(H_\eta)$$
for every $n > 0$.

Then there is a subgroup H of multiplicative type in G such that $H(n) = {}_nH$ for every $n > 0$.

Proof. For every n, let
$$C_n = \operatorname{Centr}_G\big(H(n)\big).$$
This is a closed subgroup prescheme of G, smooth over S, by XI 5.3. The positive integers being ordered by divisibility, the C_n form a decreasing family of closed subpreschemes of G. Since G is noetherian, this family is stationary. Let C be the common value of the C_n for sufficiently divisible n. Then C satisfies the same conditions as G, and all the $H(n)$ are in fact subgroups of C. Replacing G by C, we may therefore suppose that all the $H(n)$ are central subgroups of G.

Let s be the closed point of S, and let Z_s be the reductive center of G_s, defined in 4.4. By the definition in 4.1, Z_s contains every $H(n)_s$. Since A is complete, Z_s lifts to a subgroup prescheme Z of multiplicative type in G, by XI 5.8. We claim that Z contains every $H(n)$. Indeed, $H(n)$ is central and hence commutes with Z in G. The resulting homomorphism
$$Z \times H(n) \longrightarrow G$$
has, by IX 6.8, an image K which is a subgroup of multiplicative type in G containing Z. It remains only to prove $K = Z$; this follows from $K_s = Z_s$ and IX 5.1 bis.

We are thus reduced to proving the analogue of the lemma with G replaced by a group Z of multiplicative type and of finite type over S, not necessarily smooth over S. Since A is complete with algebraically closed residue field, Z is diagonalizable by X 3.3 and 1.4, hence is of the form $D_S(M)$, where M is a finitely generated commutative group. Every subgroup of multiplicative type H of Z is therefore diagonalizable by IX 2.11(i), and so is of the form $D_S(N)$, where N is a quotient group of M, by VIII 3.2. Thus the $H(n)$

correspond to quotient groups $M(n)$ of M, and the existence of a subgroup H such that $H(n) = {}_nH$ for every n is equivalent to the existence of a quotient group N of M such that

$$M(n) = N/nN$$

for every n. This follows at once from the existence of the subgroup H_η of G_η such that $H(n)_\eta = {}_n(H_\eta)$ for every n. This proves Lemma 5.2 and hence Theorem 5.1(a).

(b) We already know that

$$u|_X : X \longrightarrow T$$

is a closed immersion. It follows readily that, for all sufficiently large m, the composite

$$u_m|_X : X \longrightarrow T_m$$

is also a closed immersion. Since we shall not need this fact below, we omit the details of the proof.

Corollary 5.3. *With the notation of 5.1, let U be a subprescheme of M, both open and closed, and of finite type over S. Then U, with the structure induced by M, is affine over S. If S is quasi-compact, there is an integer $n > 0$ such that, for every multiple m of n, the induced morphism*

$$u_m|_U : U \longrightarrow T_m$$

is both an open and a closed immersion; equivalently, it is an isomorphism onto an open-and-closed subprescheme of T_m, endowed with the induced structure.

Proof. The first assertion follows from 5.1(a), and the second from 5.1(b), since $u_m : M \to T_m$ is smooth by XI 2.2 bis and a smooth immersion, that is, an étale immersion, is an open immersion.

Corollary 5.4. *Let S be a prescheme, let G be a smooth affine S-group prescheme whose reductive rank is locally constant* (1.7(b)), *and let $\mathcal{T}$ be the "prescheme of maximal tori of G"* (1.10). *Then $\mathcal{T}$ is smooth and affine over S. If T is a maximal torus of G and $N(T)$ its normalizer, then $G/N(T)$* (XI 5.3 bis) *is affine over S. The same is true of $G/C(T)$, where $C(T)$ is the centralizer of T, provided that*

$$W(T) = N(T)/C(T)$$

is finite over S; cf. 2.1(b).

Proof. The second assertion is contained in the first, since the conjugacy theorem identifies $G/N(T)$ with $\mathcal{T}$, by XI 5.5 bis. For the first assertion, observe that by construction $\mathcal{T}$ is isomorphic to an open-and-closed subprescheme of M. Indeed, one may suppose that the reductive rank of G is constant and equal to r; then $\mathcal{T}$ is the subprescheme of M corresponding to the subtori of relative dimension r, that is, the largest subprescheme of M over which the universal subgroup of multiplicative type

$$H \subset G_M$$

is a torus of relative dimension r. Corollary 5.3 therefore applies. Finally, $G/C(T)$ is finite over

$$G/N(T) \simeq (G/C(T))/W(T),$$

and hence is affine because $G/N(T)$ is affine.

Corollary 5.5. *Let G be a smooth affine algebraic group over a field k. Then the scheme M of subgroups of multiplicative type in G is a disjoint union of affine k-schemes. For every subgroup H of multiplicative type in G, if C and N denote respectively its centralizer and its normalizer, then the quotients G/C and G/N are affine.*

Proof. By XI 5.1 bis, the saturation under the action of G on M of every finite closed subset of M is open. Indeed, one reduces to the case in which k is algebraically closed, and hence to the orbit of a k-rational point x; then, by the cited result, the morphism

$$G \longrightarrow M, \qquad g \longmapsto g \cdot x$$

is smooth and therefore open, so its image is open.

Let U be the union of the equivalence classes of the closed points of M under the action of G. Then U is open and contains every closed point of M; by the Nullstellensatz, $U = M$. Thus M is a disjoint union of open subpreschemes, which are necessarily also closed. Hence M is the sum prescheme of subpreschemes M_i, each of which is the G-orbit of a closed point. Each M_i is quasi-compact, hence of finite type, and is therefore affine by 5.3.

If H is a subgroup of multiplicative type in G, it corresponds to a k-rational point x of M, and G/N identifies with the orbit of x under G, by XI 5.5 bis. Thus G/N is affine. Since C is an open subgroup of N, by XI 5.9, the quotient G/C is finite over G/N; indeed,

$$G/N \simeq (G/C)/(N/C).$$

Consequently G/C is affine as well.

This proof also gives the following result.

Corollary 5.6. *Under the hypotheses of 5.5 on G and H, the subprescheme U of M consisting of the subgroups of multiplicative type in G which are locally conjugate to H is an open-and-closed subprescheme of M.*

In picturesque language, every subgroup H' of multiplicative type in G which is a limit of subgroups conjugate to H is itself conjugate to H.

Remarks 5.7. Let S be a prescheme, let G be a smooth affine S-group prescheme, let H be an S-prescheme of multiplicative type and of finite type, and put

$$M = \mathrm{Hom}_{S\text{-gr}}(H, G).$$

By XI 4.2, M is representable, smooth over S, and separated over S. One can prove for M a result entirely analogous to 5.1, either by reducing to 5.1 through an argument analogous to that of XI 4.2, or directly by an argument modeled on the proof of 5.1. The corresponding variants of 5.3, 5.5, and 5.6 follow; we leave their formulation to the reader.

0.6 Maximal tori and Cartan subgroups of not necessarily affine algebraic groups (algebraically closed base field)

Lemma 6.1. *Let G be a connected algebraic group over a field k, and let Z be an algebraic subgroup of finite index in the center of G. Then G/Z is affine.*

Proof. We may suppose that Z is the center of G, since a scheme finite over an affine scheme is affine. Consider the vector spaces

$$P^n = P^n(G) = \mathcal{O}_{G,e}/\mathfrak{m}_{G,e}^{n+1} \qquad (n \geq 0),$$

where $\mathfrak{m}_{G,e}$ is the maximal ideal at the identity. The group G acts on the $P^n(G)$ through the adjoint representation. Let Z_n be the kernel of the corresponding homomorphism

$$\mathrm{ad}_n : G \longrightarrow \mathrm{GL}(P^n).$$

Using the connectedness of G, one readily verifies that Z is the intersection of the Z_n. Since G is noetherian, this intersection is equal to one of the Z_n. Passing to the quotient, ad_n defines a monomorphism

$$G/Z_n \longrightarrow \mathrm{GL}(P^n),$$

which is a closed immersion. Hence G/Z_n is affine, and so is G/Z.

Lemma 6.2. *Let G be a smooth algebraic group over a field k, let Z be a central algebraic subgroup, put $G' = G/Z$, and let*

$$u : G \longrightarrow G'$$

be the canonical homomorphism. Let T be a subgroup of multiplicative type in G, let $T' = u(T)$, let C be the centralizer of T in G, and let C' be the centralizer of T' in G'. Then

$$C'^0 \subset u(C).$$

Proof. We may suppose that k is algebraically closed. Put

$$C_1 = \left(u^{-1}(C'^0)\right)_{\mathrm{red}}.$$

It is enough to prove that $C_1 \subset C$. Indeed, $u(C_1)$ is connected and of finite index in C'^0, hence is equal to it set-theoretically and therefore scheme-theoretically, since C', and consequently C'^0, is smooth.

The commutator morphism

$$(g, t) \longmapsto gtg^{-1}t^{-1}$$

from $G \times G$ to G induces a morphism

$$\varphi : C_1 \times T \longrightarrow Z_1, \qquad Z_1 = Z_{\mathrm{red}}.$$

Indeed, the source is reduced, and for $g \in C_1(k)$ and $t \in T(k)$, the element $gtg^{-1}t^{-1}$ belongs to $Z(k)$ because g centralizes T modulo Z.

A calculation on k-rational points shows that $\varphi(g, t)$ is additive in g and in t, hence "bilinear." We claim that this homomorphism is identically zero; this proves $C_1 \subset C$. By the density theorem for T, we reduce to the case in which T is finite, so that $n\,\mathrm{id}_T = 0$ for some integer $n > 0$. The morphism φ is then defined by a group homomorphism

$$C_1 \longrightarrow \mathrm{Hom}_{k\text{-gr}}(T, Z_1).$$

Since Z_1 is commutative and smooth, the group on the right is represented by an étale algebraic group over k. Since C_1 is connected, every group homomorphism from C_1 to that group is zero.

Corollary 6.3. *Under the preceding hypotheses, suppose that C' is connected. Then*

$$C' = u(C), \qquad C = u^{-1}(C').$$

Proof. In this case $C' = C'^0$. Moreover, C plainly contains Z, and hence

$$C = u^{-1}(u(C)).$$

Lemma 6.4. *Let G be an algebraic group over a field k, and let Z be a central algebraic subgroup such that*

$$G/Z = G'$$

is a torus. Then G is commutative. If k is algebraically closed, there is a torus T in G such that $u(T) = G'$, where

$$u : G \longrightarrow G/Z = G'$$

is the canonical homomorphism.

Proof. We may suppose that k is algebraically closed. Consider again the commutator morphism

$$G \times G \longrightarrow G, \qquad (g, h) \longmapsto ghg^{-1}h^{-1}.$$

Since Z is central, this morphism factors through a morphism $G' \times G' \to G$. Since G' is commutative, the latter takes its values in Z, and even in Z_{red}, because $G' \times G'$ is reduced. As above, the resulting morphism

$$\varphi : G' \times G' \longrightarrow Z_{\mathrm{red}}$$

is bilinear and hence zero. Thus G is commutative.

To find a torus T in G with $u(T) = G'$, we may, after replacing G by G^0_{red}, suppose that G is smooth and connected. Replacing Z by Z^0_{red}, we may also suppose that Z is smooth and connected. By a well-known theorem of Chevalley,[7] Z is an extension of an abelian variety by an affine group. We are therefore reduced to the two cases in which Z is affine or Z is an abelian variety. In the first case G is affine and the result is classical (BIBLE 7, Theorem 3(a)).

Suppose then that Z is an abelian variety. Every homomorphism from the additive group $\mathbb{G}_a$ to the torus $G' = R$, or to the abelian variety Z, is trivial. The same is therefore true of every homomorphism from $\mathbb{G}_a$ to G; thus G contains no subgroup isomorphic to $\mathbb{G}_a$. Chevalley's structure theorem gives an exact sequence

$$(*) \qquad 0 \longrightarrow T \longrightarrow G \longrightarrow A \longrightarrow 0,$$

where A is an abelian variety and T is a smooth connected affine group. Since T is commutative and contains no additive subgroup, T is a torus; it is plainly the unique maximal torus of G. It remains to prove that every epimorphism

$$u : G \longrightarrow R$$

to a torus R satisfies $u(T) = R$.

Put

$$\mathrm{Hom}_{\mathrm{gr}}(T, \mathbb{G}_m) = M, \qquad \mathrm{Hom}_{\mathrm{gr}}(R, \mathbb{G}_m) = P.$$

These are finitely generated free $\mathbb{Z}$-modules, and

$$T = D_k(M), \qquad R = D_k(P).$$

Let

$$B = \mathrm{Ext}^1_{k\text{-gr}}(A, \mathbb{G}_m);$$

this is also the set of k-rational points of the abelian variety dual to A. We have

$$(**) \qquad \mathrm{Ext}^1(A, T) = \mathrm{Hom}_{\mathrm{gr}}(M, B), \qquad \mathrm{Ext}^1(A, R) = \mathrm{Hom}_{\mathrm{gr}}(P, B).$$

[7] Editor's note: references should be supplied here.

In particular, the extension G of A by T is determined by a homomorphism

$$\theta : M \longrightarrow B.$$

The exact sequence (*) also gives

$$0 \longrightarrow \mathrm{Hom}(A,R) \longrightarrow \mathrm{Hom}(G,R) \longrightarrow \mathrm{Hom}(T,R) \longrightarrow \mathrm{Ext}^1(A,R).$$

Moreover, $\mathrm{Hom}(A,R) = 0$, and the last map identifies with the homomorphism

$$\mathrm{Hom}(P,M) \longrightarrow \mathrm{Hom}(P,B)$$

induced by $\theta : M \to B$. Setting

$$N = \ker\theta,$$

we obtain canonical bijections

$$\mathrm{Hom}(G,R) \simeq \mathrm{Hom}(P,N) \simeq \mathrm{Hom}(S,R), \qquad S = D_k(N).$$

This has the following immediate geometric description.

Lemma 6.5. *Let G be an extension of an abelian variety A by a torus $T = D_k(M)$, defined by a homomorphism*

$$\theta : M \longrightarrow B = \mathrm{Ext}^1(A, \mathbb{G}_m),$$

over an algebraically closed field k. Put $N = \ker\theta$, and let $S = D_k(N)$ be the corresponding torus, isomorphic to T/U, where $U = D_k(M/N)$. Consider the extension $H = G/U$ of A by S. This extension splits,[8] *so there is a unique projection from H to S, and hence a unique homomorphism*

$$v : G \longrightarrow S$$

extending the canonical homomorphism $T \to S$. For every torus R and every homomorphism $u : G \to R$, there is a unique homomorphism $u' : S \to R$ such that $u = u'v$. In particular,

$$\mathrm{Im}(u) = \mathrm{Im}(u') = u(T).$$

It follows in particular that if u is an epimorphism, then its restriction to T is also an epimorphism. This completes the proof of 6.4.

Theorem 6.6. *Let G be a smooth connected algebraic group over an algebraically closed field k.*

(a) *The maximal tori of G are conjugate.*

(b) *If T is a torus in G, then its centralizer is smooth and connected.*

(c) *The assignment*

$$T \longmapsto \mathrm{Centr}_G(T)$$

gives a bijection between the maximal tori T of G and the Cartan subgroups of G defined in No. 1. An algebraic subgroup C of G is a Cartan subgroup if and only if it

[8] Editor's note: the extension is split.

is smooth, nilpotent, and set-theoretically equal to its connected normalizer; in that case it is even scheme-theoretically equal to its connected normalizer. Moreover,

$$C = \mathrm{Centr}_G(T),$$

where T is the unique maximal torus of C, and

$$\mathrm{Norm}_G(C) = \mathrm{Norm}_G(T).$$

(d) *Let $v : G \to H$ be an epimorphism of smooth connected algebraic groups. The maximal tori, respectively the Cartan subgroups, of H are the images of the maximal tori, respectively the Cartan subgroups, of G. If T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$.*

(e) *Under the hypotheses of* (d)*, suppose that* $\ker v$ *is a central subgroup of G. Then*

$$T \longmapsto v(T), \qquad C \longmapsto v(C)$$

give bijections between the maximal tori, respectively the Cartan subgroups, of G and those of H. The Cartan subgroups of G contain the center of G, and hence $\ker v$*; they are precisely the groups $v^{-1}(C')$, where C' is a Cartan subgroup of H.*

We also record the following immediate consequence of Theorem 6.6(c).

Corollary 6.7. *The group G is nilpotent, that is, $G(k)$ is nilpotent, if and only if every torus in G is central, or equivalently if G has a unique maximal torus. In that case this torus is the largest subtorus of G.*

Proof of Theorem 6.6. Let Z be a central algebraic subgroup of G, put $G' = G/Z$, and let

$$u : G \longrightarrow G'$$

be the canonical homomorphism. Then G' is smooth and connected. If T' is a subtorus of G', it follows from 6.4 that $u^{-1}(T')$ is commutative and that T' is the image of a subtorus of $u^{-1}(T')$, hence of a subtorus of G. Since $u^{-1}(T')$ is commutative, it plainly has a largest subtorus T: the sum of two subtori is a third subtorus containing both. Thus

$$u(T) = T'.$$

It follows immediately that, for every maximal torus T of G, its image $T' = u(T)$ is a maximal torus of G', and that

$$T \longmapsto u(T)$$

is a bijection between the maximal tori of G and those of G'.

Now take

$$Z = \mathrm{Centr}(G)^0_{\mathrm{red}}.$$

Then G' is affine by 6.1. Its maximal tori are conjugate, and therefore so are those of G, which proves (a). Moreover, G is nilpotent, respectively has a unique maximal torus, if and only if G' has the same property. Since G' is affine, these two conditions on G' are equivalent (BIBLE 6, Th. 4, Cor. 2); hence they are also equivalent for G. If G has a unique maximal torus T, then T is invariant and therefore central; and because every torus of G is contained in a maximal torus, every torus is central. Conversely, if every torus is central, then so are the maximal tori, and the conjugacy theorem (a) shows that there is only one. This proves 6.7.

Let T be any torus of G, let $T' = u(T)$, and put

$$C' = \mathrm{Centr}_{G'}(T').$$

The group C' is connected (BIBLE 6, Th. 6(a)). By 6.3 the centralizer C of T is $u^{-1}(C')$, and is therefore connected because Z is connected. This proves (b). If T is maximal, then T' is maximal; hence C' is nilpotent, and so is C, which is a central extension of C'. Moreover, T is a maximal torus of C, and by 6.7 it is the unique maximal torus of C. Consequently

$$T \longmapsto \mathrm{Centr}_G(T)$$

is a bijection from the maximal tori of G to the Cartan subgroups of G. We also have

$$\mathrm{Centr}_G(T) = C \subset \mathrm{Norm}_G(C) \subset \mathrm{Norm}_G(T).$$

The centralizer of T has finite index in its normalizer (cf. XI 5.9); the argument there remains valid without the affineness hypothesis, because it uses only the representability of the two functors in question, as noted in XI 6.5. Since C is smooth and connected, it follows that

$$C = \mathrm{Norm}_G(C)^0.$$

Furthermore, the bijection between maximal tori and Cartan subgroups shows that the normalizers of C and T have the same k-valued points. Since the latter normalizer is smooth,

$$\mathrm{Norm}_G(T) = \mathrm{Norm}_G(C).$$

To finish the proof of (c), it remains to show that a smooth, connected, nilpotent subgroup C of G which has finite index in its normalizer is a Cartan subgroup. Since Z is central, the normalizer of C contains Z; and since Z is smooth and connected, it follows that $Z \subset C$. Hence

$$C = u^{-1}(C'), \qquad C' = u(C),$$

and

$$\mathrm{Norm}_G(C) = u^{-1}(\mathrm{Norm}_{G'}(C')).$$

Thus C' is connected and nilpotent and has finite index in its normalizer. By Theorem 1 of BIBLE 7, it is a Cartan subgroup of G', and it follows at once that C is a Cartan subgroup of G.

We next prove (e). We are in the situation considered at the beginning of the proof, with $\ker u = Z$ and $G' = H$, and we have already seen that $T \mapsto u(T)$ is a bijection between the maximal tori of G and those of G'. Using the bijection just proved between maximal tori and Cartan subgroups, we obtain a bijection between Cartan subgroups C of G and Cartan subgroups C' of G': to

$$C = \mathrm{Centr}_G(T)$$

we associate

$$C' = \mathrm{Centr}_{G'}(T'), \qquad T' = u(T).$$

Since C' is connected by (b), 6.3 gives

$$C' = u(C), \qquad C = u^{-1}(C'),$$

which proves (e).

It remains to prove (d). Let T be a maximal torus of G. We first prove that $v(T)$ is a maximal torus of H; by the conjugacy theorem (a), this will imply that all maximal tori of H arise in this way. Let R be a torus of H containing $v(T)$. Replacing H by R and G by $v^{-1}(R)^0_{\mathrm{red}}$, we may suppose that $R = H$, so that H is a torus; it remains to prove $v(T) = H$. Put again

$$Z = \mathrm{Centr}(G)^0_{\mathrm{red}}, \qquad G' = G/Z, \qquad H' = H/v(Z).$$

We have a commutative diagram with exact rows

$$\text{(D)} \qquad \begin{tikzcd} e \arrow[r] & Z \arrow[r] \arrow[d, "v''"'] & G \arrow[r, "u"] \arrow[d, "v"'] & G' \arrow[r] \arrow[d, "v'"'] & e \\ e \arrow[r] & v(Z) \arrow[r] & H \arrow[r, "u'"'] & H' \arrow[r] & e. \end{tikzcd}$$

We already know that $u(T)$ is a maximal torus of G'. Because G' is affine, the epimorphism

$$v' : G' \longrightarrow H'$$

is an epimorphism of smooth connected affine groups. Therefore $v'(u(T))$ is a maximal torus of H' (BIBLE 7, Th. 3(a)), and hence equals H'. Thus $u'(v(T)) = H'$, so in order to prove $v(T) = H$ it is enough to show that $v(T) \supset v(Z)$. Now $v(Z)$ is a smooth connected subgroup of H, hence a torus, and

$$v'' : Z \longrightarrow v(Z)$$

is an epimorphism. By 6.4,

$$v(Z) = v''(S)$$

for a torus S of Z. Since S is central in G, it is contained in the maximal torus T, and therefore $v(Z) \subset v(T)$. This proves (d) for maximal tori.

In view of the bijection between maximal tori and Cartan subgroups, it remains to prove that, if T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$. Return to diagram (D), now without assuming that H is a torus, and put

$$T' = u(T), \qquad C' = u(C).$$

By (e), C' is the centralizer of the maximal torus T'. Since G', and hence H', is affine, $v'(C')$ is the centralizer of the maximal torus $v'(T')$ of H' (BIBLE 7, Th. 3(a)). Equivalently, $u'(v(C))$ is the centralizer of $u'(v(T))$. Because C contains Z, the group $v(C)$ contains $v(Z)$. It is therefore the inverse image under u' of $u'(v(C))$, that is, of the centralizer of $u'(v(T))$. By (e), applied to $u' : H \to H'$ and the maximal torus $v(T)$ of H, this inverse image is precisely the centralizer of $v(T)$. This completes the proof of 6.6. □

0.7 Application to smooth group preschemes not necessarily affine

Theorem 7.1. *Let S be a prescheme and let G be an S-group prescheme that is smooth, separated, and of finite type over S. Suppose that G admits a maximal torus locally for the faithfully flat quasi-compact topology. Then:*

(a) *The map*

$$T \longmapsto \mathrm{Centr}_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups of G. If C corresponds to T, then T is the unique maximal torus of C.

(b) *Let T, T' be two maximal tori of G, and let C, C' be the corresponding Cartan subgroups. Then*

$$\mathrm{Transp}_G(T, T') = \mathrm{Transp}_G(C, C') = \mathrm{Transp}_G(T, C').$$

The first two terms are also identical to the strict transporters. Moreover, the functor in question is representable by a closed subprescheme of G, smooth over S. The tori T, T' and the Cartan subgroups C, C' are locally conjugate for the étale topology.

(c) *A maximal torus of G, and a Cartan subgroup of G, exist locally for the étale topology.*

(d) *Suppose that every finite subset of a fiber G_s is contained in an affine open of G—for example, that G is quasi-projective over S, or that S is artinian. Then the functor*

$$\mathcal{T} : (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets})$$

defined in 1.10, the functor of maximal tori of G, is isomorphic to the functor

$$\mathcal{C} : (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets})$$

of Cartan subgroups of G, and is representable by a smooth, separated prescheme of finite type over S. This prescheme is quasi-projective over S when G is, and is affine over S when G is affine over S, or when S is artinian.

(e) *Let $u : G \to G'$ be a morphism of S-group preschemes, where G' is smooth, separated, and of finite type over S, and suppose that*

$$u_s(G_s) = G'_s$$

for every $s \in S$, that is, that u_s is faithfully flat.[9] *For every maximal torus T of G, $u(T)$ is a maximal torus of G'. If G' has connected fibers, then for every Cartan subgroup C of G, $u(C)$ is a Cartan subgroup of G'; and if C is the centralizer of T, then $u(C)$ is the centralizer of $u(T)$. In either case the induced morphism $T \to u(T)$, respectively $C \to u(C)$, is faithfully flat.*

(f) *Under the hypotheses of* (e), *suppose in addition that* $\ker u$ *is a central subgroup of G. Then*

$$T \longmapsto u(T)$$

is a bijection from the set of maximal tori of G onto the set of maximal tori of G'. If G' has connected fibers, then

$$C \longmapsto u(C)$$

is likewise a bijection from the set of Cartan subgroups of G onto the set of Cartan subgroups of G'; if $C' = u(C)$, then $C = u^{-1}(C')$.

[9] The editors did not locate the printed reference in Exposé VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), Proposition II.5.1(c).

Remarks 7.2. We shall see in Exposé XV that the conclusion of Theorem 7.1(d) remains valid without the restrictive condition on the finite subsets of the G_s. We shall also prove there the conclusions concerning Cartan subgroups alone that occur in (b), (c), (d), and (e), under the weaker hypothesis that G admit a Cartan subgroup locally for the faithfully flat quasi-compact topology, but not necessarily a maximal torus. For that purpose we shall moreover need to use 7.1 when S is artinian. The method of Exposé XV also considerably simplifies the proof of (c) and (d) when S is not artinian.

Proof of Theorem 7.1. (a) Proceeding as in 3.2, one sees that for every maximal torus T of G,

$$C = \mathrm{Centr}_G(T)$$

is indeed a Cartan subgroup of G, and that T is determined by C as the unique maximal torus of C. It remains only to show that every Cartan subgroup C of G is defined by a maximal torus of G, or equivalently that C admits a central maximal torus.

The question is still local for the fpqc topology, so we may suppose that G admits a maximal torus T. By XI 6.2,

$$\mathrm{Transp}_G(T, C)$$

is representable by a closed subprescheme S' of G, smooth over S; moreover $S' \to S$ is surjective by 6.6(c). After the base change $S' \to S$, we may therefore suppose that there is a section g of G such that

$$\mathrm{ad}(g) \cdot T \subset C.$$

Then $\mathrm{ad}(g) \cdot T$ is a maximal torus of C. It is central fiber by fiber by 6.6(c), and hence central by IX 5.6(b). This proves (a).

(b) Let T, T' be two maximal tori of G, and C, C' their corresponding Cartan subgroups. The following conditions are equivalent:

$$(1)\ T \subset T', \quad (2)\ T = T', \quad (3)\ T \subset C', \quad (4)\ C \subset C', \quad (5)\ C = C'.$$

This follows immediately from (a). Applying the same result after arbitrary base change gives the asserted identities among the transporters and strict transporters.

We have already observed in (a) that $\mathrm{Transp}_G(T, C')$ is representable by a closed subprescheme of G, smooth over S, whose structural morphism is surjective. By Hensel's lemma this prescheme has a section locally for the étale topology. Thus T, T', on the one hand, and C, C', on the other, are locally conjugate for the étale topology. This proves (b).

(c) Suppose first that S is artinian and local. If G admits a maximal torus T, the conjugacy theorem just proved shows that the functor $\mathcal{T}$ of maximal tori of G is represented by the homogeneous space G/N, where N is the normalizer of T in G; it is smooth by (b). Moreover, as observed in VI_A 3.2.1, every finite subset of $\mathcal{T}$ is contained in an affine open, because $\mathcal{T}$ is a homogeneous space under G.

In general there is a finite extension k'/k of the residue field of S such that $G_{k'}$ has a maximal torus. The extension k'/k is obtained by a finite flat base change $S' \to S$, and the maximal torus of $G_{k'}$ lifts to a maximal torus of $G_{S'}$ by XI 2.1 bis. Thus $\mathcal{T}_{S'}$ is represented by a smooth, separated prescheme of finite type over S', every finite subset of which is contained in an affine open. Its natural descent datum is effective. Hence $\mathcal{T}$ is representable, and descent shows that it is smooth, separated, and of finite type over S. By smoothness and Hensel's lemma, $\mathcal{T}$ has a section locally for the étale topology. This proves (c) and (d) when S is artinian. We shall prove below that in this case $\mathcal{T}$ is in fact affine over S.

Now let S be arbitrary. To prove (c) and (d), which are local assertions on S for the Zariski topology, we may suppose that S is affine, that there is a prime ℓ prime to all residue characteristics of S, and that the reductive rank of the fibers of G is constant, say equal to r. The latter rank is visibly locally constant because of the assumed fpqc-local existence of a maximal torus.

Choose a faithfully flat quasi-compact morphism $S' \to S$, with S' affine, such that $G_{S'}$ admits a maximal torus T_0, and let C_0 be its centralizer. I claim that there is a suitable power n of ℓ such that

$$C_0 = \mathrm{Centr}_{G_{S'}}({}_nT_0).$$

Indeed, this reduces immediately to the case where S' is noetherian, which was treated in XI 6.2. Fix such an n, and put

$$M = (\mathbb{Z}/n\mathbb{Z})^r, \qquad P = \mathrm{Hom}_{S\text{-gr}}(M_S, G).$$

Then P is represented by a closed subprescheme of finite presentation of $({}_nG)^r$, where

$${}_nG = \mathrm{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S, G).$$

This is also the kernel of the n-th-power morphism on G, and is therefore represented by a closed subprescheme of finite presentation of G. Moreover, P is smooth over S by XI 2.1.

Let $s \in S$. From what precedes, there is a finite separable extension $k''/\kappa(s)$ for which $G_{k''}$ has a maximal torus T''. Since ${}_nT''$ is étale over k'', we may, after replacing k'' by a finite separable extension, suppose that

$${}_nT'' \simeq M_{k''}.$$

Choose an étale morphism $S'' \to S$ and a point $s'' \in S''$ above s, inducing the residue-field extension $\kappa(s'') \simeq k''$. We obtain a section of $P_{S''} \otimes \kappa(s'')$ over $\operatorname{Spec}\kappa(s'')$. By Hensel's lemma, after replacing the pointed prescheme (S'', s'') by an étale neighborhood if necessary, this section extends to a section of $P_{S''}$. Equivalently, there is a homomorphism

$$M_{S''} \longrightarrow G_{S''}$$

inducing the given isomorphism

$$M_{\kappa(s'')} \simeq {}_nT''.$$

By IX 6.4—which here reduces to Nakayama's lemma—this homomorphism is a closed immersion above an open neighborhood of s'', which we may take to be all of S''. Let H'' be its image, and let C'' be its centralizer in $G_{S''}$; by XI 6.2, C'' is a closed smooth subprescheme of $G_{S''}$.

Lemma 7.3. *Under the preceding hypotheses on ℓ, n, r, for every prescheme S'' over S and every maximal torus T'' of $G_{S''}$, one has*

$$\mathrm{Centr}_{G_{S''}}(T'') = \mathrm{Centr}_{G_{S''}}({}_nT'').$$

Indeed, after faithfully flat descent from $S' \times_S S''$, we reduce to the case in which $G_{S''}$ admits a maximal torus T''_1 for which the displayed equality holds. By (b), T'' and T''_1 are locally conjugate for the étale topology, so the same equality holds for T''.

Apply this result over $\operatorname{Spec}\kappa(s'')$. It shows that the fiber $C''_{s''}$ is a Cartan subgroup of $G_{s''}$. Now, since G is smooth over S, the union of the identity components of the fibers G_s is an open subprescheme of G; this is the general EGA IV result for smooth morphisms recorded by the editors without a precise reference. This open is visibly stable under the group law and hence is a subgroup, denoted G^0. It satisfies the same preliminary hypotheses as G, and maximal tori of G correspond bijectively to maximal tori of G^0. Thus, for the proof of (c) and (d), we may and do suppose that G has connected fibers. Its Cartan subgroups then also have connected fibers by 6.6(a).

I claim that C''^0 is a Cartan subgroup of $G_{S''}$ above an open neighborhood of s''. This will complete the proof of (c). The claim follows from the next lemma.

Lemma 7.4. *Under the hypotheses of 7.1, suppose that G has connected fibers. Let C be a closed group subprescheme of G, smooth over S, and let $s \in S$ be such that C_s is a Cartan subgroup of G_s. Then C^0 is a Cartan subgroup of G over an open neighborhood of s.*

We reduce easily to the case where S is local and s is its closed point, and to proving there that C^0 is a Cartan subgroup of G. By flat descent we may further suppose that G admits a maximal torus T. By XI 6.2,

$$\mathrm{Transp}_G(T, C)$$

is represented by a closed subprescheme of G, smooth over S. The fiber of this transporter at s is nonempty by the hypothesis on C_s and the conjugacy theorem 6.6(a). Faithfully flat descent therefore reduces us to the case where the transporter has a section over S, hence to the case where C contains a maximal torus T of G. Then T is central in C^0 by IX 5.6(a), so

$$C^0 \subset \mathrm{Centr}_G(T).$$

This is an inclusion of smooth group preschemes over S, with connected fibers and the same relative dimension—namely, the dimension of their common fiber at s. It is therefore an equality, proving the lemma and completing (c).

(d) Retain the preceding notation and hypotheses on ℓ, n, r, and the assumption that G has connected fibers. Let

$$Q : (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets})$$

be the functor defined by

$$Q(S') = \left\{ \begin{array}{c} \text{subgroups of multiplicative type in } G_{S'} \\ \text{of type } M = (\mathbb{Z}/n\mathbb{Z})^r \end{array} \right\}$$

in the sense of IX 1.4. The rule

$$T \longmapsto {}_nT$$

defines a morphism

$$\varphi : \mathcal{T} \longrightarrow Q,$$

which is a monomorphism by Lemma 7.3.

I claim that Q is representable by a separated prescheme of finite presentation over S. As noted in the proof of XI 3.12(a), there is an isomorphism

$$Q \simeq P_0/\Gamma,$$

where P_0 is the open and closed subprescheme of

$$P = \mathrm{Hom}_{S\text{-gr}}(M_S, G)$$

corresponding to monomorphisms $M_{S'} \to G_{S'}$, as in IX 6.8, and

$$\Gamma = (\mathrm{Aut}_{\mathrm{gr}}(M))_S.$$

The hypothesis that every finite subset of a fiber G_s lies in an affine open is stable under passage to a closed subprescheme and under fiber products. It is therefore inherited by $({}_nG)^r$, then by P, and then by P_0. Hence Q is represented by a prescheme of finite presentation over S.[10] The same argument shows that Q is quasi-projective, respectively affine, over S when G is. In every case Q is separated over S.

Lemma 7.5. *The preceding morphism $\varphi : \mathcal{T} \to Q$ is representable by an open immersion.*

In other words, let H be a subgroup of multiplicative type in G, of type M. We must prove that there is an open subset $U \subset S$ with the following property: for every $S' \to S$, the group $H_{S'}$ is of the form ${}_nT_{S'}$, for a suitable maximal torus $T_{S'}$ of $G_{S'}$, if and only if $S' \to S$ factors through U.

We may suppose that the nilpotent rank of the fibers of G is constant, say equal to r'; it is locally constant by (b). Put

$$C = \mathrm{Centr}_G(H),$$

a closed group subprescheme smooth over S by XI 6.2. Replacing S by the open and closed subset where C has relative dimension r', we may suppose that this is its relative dimension everywhere. Then $H = {}_nT$ for a maximal torus T of G if and only if C^0 admits a central maximal torus T, everywhere of relative dimension r, and $H = {}_nT$. This gives another description of the subfunctor U.

By flat descent, we may suppose that G admits a maximal torus T_1. Consider the subfunctor

$$R = \mathrm{Transp}_G(T_1, \mathrm{Centr}_{C^0}(C^0)) \subset \mathrm{Transp}_G(T_1, C).$$

The transporter on the right is represented by a smooth prescheme over S by XI 6.2. The one on the left is represented by the induced open subprescheme, as follows immediately from IX 5.6(a); in particular it too is smooth over S. Its structural morphism is therefore open. Replacing S by its image, equipped with the induced structure, we may suppose that this morphism is surjective. Applying 1.13 to C^0, we find that C^0 admits a central maximal torus T, because it admits one fpqc-locally. Its relative dimension is that of T_1, namely r. The remaining condition on H is $H = {}_nT$, and by IX 2.10 this again cuts out a suitable open and closed subset of S. This proves the lemma.

Lemma 7.5 shows that $\mathcal{T}$ is represented by a separated prescheme locally of finite presentation over S, and in fact of finite presentation. For the latter assertion, one may either inspect the proof of Lemma 7.5 to see that φ is a quasi-compact open immersion, or descend fpqc-locally to the case where G has a maximal torus T, when

$$\mathcal{T} \simeq G/\mathrm{Norm}_G(T).$$

This expression, or XI 2.1, also shows that $\mathcal{T}$ is smooth over S. If G is quasi-projective over S, then so are Q and $\mathcal{T}$. If G is affine over S, the affineness of $\mathcal{T}$ over S was proved in 5.4. It is unknown here whether, without the hypothesis that G/S is affine, n can be chosen so that the open immersion of Lemma 7.5 is also closed.

[10]The editors left the printed reference to Exposé V unspecified.

It remains to show that $\mathcal{T}$ is affine over S when S is artinian. By EGA I 6.1.7, we reduce to the case where S is the spectrum of a field, which we may suppose algebraically closed. In view of (f), proved below, it suffices to prove the assertion for

$$G/\operatorname{Centr}(G).$$

This quotient is affine by 6.1, so the preceding affine case applies. This proves (d).

(e) By IX 6.8, there is a subtorus T' of G' such that u induces a faithfully flat morphism

$$T \longrightarrow T'.$$

This characterizes T' as the subsheaf $u(T)$ of G'. Let C and C' be the centralizers of T and T', respectively. We prove that $C \to C'$ is flat, and faithfully flat when G' has connected fibers. Since C and C' are flat and of finite presentation over S, SGA 1, I 5.9 reduces us to a base field, which we may suppose algebraically closed. Replacing G, G' by their identity components does not change C^0, C'^0, so we may suppose G, G' connected. The assertion then follows from 6.6(d), together with (a).

(f) By (a) and (e), it is enough to prove the assertion about Cartan subgroups. A Cartan subgroup C of G, being the centralizer of a maximal torus, contains the center of G, and therefore contains $\ker u$. Thus

$$C = u^{-1}(C'), \qquad C' = u(C),$$

where C' is the Cartan subgroup of G' supplied by (e). Hence $C \mapsto u(C)$ is injective. To prove surjectivity, it suffices to show that $u^{-1}(C')$ is a Cartan subgroup of G for every Cartan subgroup C' of G'.

The question is fpqc-local. We may suppose that G admits a Cartan subgroup C_1. Then $C'_1 = u(C_1)$ is a Cartan subgroup of G', and is fpqc-locally conjugate to C' by (b). Since

$$u^{-1}(C'_1) = C_1$$

is a Cartan subgroup of G, the same is true of $u^{-1}(C')$. This proves (f) and completes the proof of the theorem. □

Alternatively, to prove that $u^{-1}(C')$ is a Cartan subgroup, observe that it is flat over S, since u is; SGA 1, I 5.9 then reduces the assertion, by definition, to the case of a base field, where 6.6(e) applies.

Corollary 7.6. *Under the hypotheses of 7.1* (e)*, G' admits a maximal torus locally for the étale topology, and therefore satisfies the preliminary hypotheses imposed on G. If moreover* $\ker u$ *is central, then the functors $\mathcal{T}_G, \mathcal{C}_G$ of maximal tori and Cartan subgroups of G are isomorphic to the analogous functors $\mathcal{T}_{G'}, \mathcal{C}_{G'}$ for G'. Consequently, whenever these functors are representable, they are represented by isomorphic S-preschemes.*

Remark 7.7. *(a)* In contrast with the case where G is affine over S—in fact it is enough that G have affine fibers, as will be seen in Exposé XVI—local constancy of the reductive rank of G does not imply that G admits a maximal torus fpqc-locally. For example, let S be the spectrum of a discrete valuation ring, and let G_1, G_2 be smooth separated group schemes of finite type over S, such that the generic fiber of G_1 is an elliptic curve and its special fiber is $\mathbb{G}_m$, while the generic fiber of G_2 is $\mathbb{G}_m$ and its special fiber is $\mathbb{G}_a$. Put

$$G = G_1 \times_S G_2.$$

Both fibers of G have reductive rank 1, but G plainly does not admit a maximal torus fpqc-locally.

It is, on the other hand, very plausible that the following condition on a smooth separated group of finite type over a prescheme S is sufficient for the étale-local existence of a maximal torus: the reductive rank and the abelian rank of the fibers of G are locally constant functions.

(b) In the proof of 7.1, particularly in (a), we used XI 6.2 in cases where S was not assumed locally noetherian. In the instances at hand, however, the reduction to affine noetherian S is immediate.

Here is a variant of 7.1(b).

Proposition 7.8. *Let G be a smooth S-group prescheme of finite presentation with connected fibers, let H be a smooth S-group prescheme, and let*

$$i : H \longrightarrow G$$

be a monomorphism of S-groups, making H an S-subgroup of G. For every Cartan subgroup C of G,[11] *the transporter* $\mathrm{Transp}_G(C, H)$ *is represented by a closed subprescheme of G, smooth over S.*

The representability of the transporter by a closed subprescheme of G, of finite presentation, follows from XI 6.11, because C has connected fibers when G does, by 6.6(b). To prove smoothness, the standard procedure reduces first to affine noetherian S, then to artinian local S, and finally, by descent, to an algebraically closed residue field. In that case C has a maximal torus T, which is also a maximal torus of G. Unless the reductive and nilpotent ranks of the fiber H_0 equal those of G_0, the transporter is empty. Under this equality one sees, using the connectedness of C and the smoothness of the centralizer in H of a maximal torus of H, that

$$\mathrm{Transp}_G(C, H) = \mathrm{Transp}_G(T, H).$$

The right-hand side is smooth by XI 2.5, and therefore so is the left-hand side.

The same argument proves the more general assertion 7.9(b) below.

Proposition 7.9. *Let G and $i : H \to G$ be as in 7.8. Suppose moreover that, for every $s \in S$, H_s is connected and has the same reductive rank and the same nilpotent rank as G_s; equivalently, H_s contains a Cartan subgroup of G_s. Then:*

(a) $\mathrm{Norm}_G(H)$ *is represented by a closed subprescheme of G, of finite presentation over S, and the canonical monomorphism*

$$H \longrightarrow \mathrm{Norm}_G(H)$$

is an open immersion. Consequently i is an immersion and

$$H = \mathrm{Norm}_G(H)^0.$$

(b) For every Cartan subgroup C of G, $\mathrm{Transp}_G(C, H)$ *is a closed subprescheme of G, smooth over S, with surjective structural morphism. If C is the centralizer of a maximal torus T of G, then moreover*

$$\mathrm{Transp}_G(T, H) = \mathrm{Transp}_G(C, H).$$

[11] The proof only requires C to be a smooth subgroup of G whose geometric fibers are connected centralizers of subgroups of maximal tori.

(c) *Let C be a group subprescheme of H. Then C is a Cartan subgroup of H if and only if it is a Cartan subgroup of G.*

(d) *If G admits a Cartan subgroup, respectively a maximal torus, locally for the étale topology or for the fpqc topology, then so does H.*

Proof. *(a)* The representability of $\mathrm{Norm}_G(H)$ by a closed subprescheme of G, of finite presentation over S, is contained in XI 6.11. Since H is smooth, hence flat and locally of finite presentation over S, VI_B 2.6 reduces the assertion that

$$H \longrightarrow \mathrm{Norm}_G(H)$$

is an open immersion to the fibers. We therefore reduce to the case where $S = \operatorname{Spec} k$, with k algebraically closed.

The assertion then reduces to showing that the corresponding map on Lie algebras is an isomorphism,[12] or equivalently that

$$(\mathfrak{g}/\mathfrak{h})^H = 0,$$

where $\mathfrak{g}, \mathfrak{h}$ are the Lie algebras of G, H and the superscript denotes invariants; compare II 5.2.3(i). By hypothesis H contains a Cartan subgroup C of G, the centralizer of a maximal torus T of G. It therefore suffices to prove

$$(\mathfrak{g}/\mathfrak{h})^T = 0.$$

By complete reducibility of representations of T (I 4.7.3), this follows from

$$(\mathfrak{g}/\mathfrak{c})^T = 0, \qquad \mathfrak{c} = \mathrm{Lie}(C).$$

The latter equality is equivalent to

$$\mathfrak{g}^T = \mathfrak{c}.$$

It says that the centralizer C and the normalizer of T have the same Lie algebra, and follows from the fact that C is an open subgroup of that normalizer by XI 5.9. This proves (a).

(b) was proved in 7.8.

(c) Since H is a subprescheme of G, the assertion reduces immediately to an algebraically closed base field, where it follows directly from the hypothesis on H.

(d) follows from (b), (c), and the Hensel lemma XI 1.10. □

Corollary 7.10. *Let G, H be as in 7.9, and suppose that for every algebraically closed field k over S, every element of $G_k(k)$ normalizing H_k belongs to $H_k(k)$. Then H is a closed subprescheme of G and is its own normalizer.*

This follows immediately from 7.9(a). We shall apply it in particular to Borel subgroups, and more generally to parabolic subgroups, of G.

[12] The editors did not locate the printed reference in Exposé VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), Corollary II.5.6.

Corollary 7.11. *Let G, H be as in 7.9. Then H contains every group subprescheme Z of G which is central in G, flat, and of finite presentation over S.*

As usual, we reduce first to affine noetherian S, then to artinian S; the hypotheses of 7.1 then hold. By 7.9(c), we may suppose that H contains a Cartan subgroup C of G, and it remains to show $C \supset Z$. Since S is artinian, the quotient

$$G' = G/Z$$

is represented by a group prescheme of finite type over S; the canonical morphism $u : G \to G'$ is faithfully flat and has kernel Z by VI_A 3.2. The group G' is smooth over S. Theorem 7.1(f) therefore shows that $C = u^{-1}(C')$ for a Cartan subgroup C' of G', and hence that C contains Z.

Corollary 7.12. *Let G, G' be smooth S-group preschemes of finite presentation, and let $u : G \to G'$ be a faithfully flat group homomorphism, meaning that u_s is faithfully flat for every $s \in S$. Suppose that* $\ker u$ *is central and that G has connected fibers. Then*

$$H' \longmapsto H = u^{-1}(H')$$

defines a bijection between:

- *the group subpreschemes H' of G', smooth and of finite presentation over S, having at every $s \in S$ the same reductive rank and the same nilpotent rank as G'; and*
- *the group subpreschemes H of G, smooth and of finite presentation over S, having at every $s \in S$ the same reductive rank and the same nilpotent rank as G.*

Moreover, H has connected fibers if and only if H' does.

Let H be a group subprescheme of G with the stated properties. By 7.11 it contains $Z = \ker u$, so faithfully flat descent gives

$$H = u^{-1}(H')$$

for a uniquely determined group subprescheme H' of G'. Using 6.6(e), one checks immediately that H' has the asserted properties. If H has connected fibers, so does $H' = H/Z$.

Conversely, start with $H' \subset G'$ having the stated properties, and put $H = u^{-1}(H')$. In view of 6.6(e), it remains to show that H is smooth over S, and that it has connected fibers when H' does. Since u is flat and H' is flat over S, H is already flat over S. It therefore suffices to check smoothness, respectively connectedness, on geometric fibers, reducing to an algebraically closed base field.

Then H' contains a Cartan subgroup C' of G', and H contains its inverse image C. By 6.6(e), C is a Cartan subgroup of G, hence is smooth and connected. The conclusion now follows from the next lemma.

Lemma 7.13. *Let G, G' be group preschemes flat and of finite presentation over S, let $u : G \to G'$ be a faithfully flat group homomorphism, and let C' be a group subprescheme of G', of finite presentation over S, such that $C = u^{-1}(C')$ is smooth over S, respectively has connected fibers. For every group subprescheme H' of G', of finite presentation over S, containing C', and smooth over S, respectively with connected fibers, its inverse image $H = u^{-1}(H')$ is smooth over S, respectively has connected fibers.*

As noted above, this reduces at once to the case where S is the spectrum of a field. The group H is then a principal C-bundle over H/C, by VI_B 9, while

$$H/C \simeq H'/C'$$

by Exposé IV. Since H' is smooth, respectively connected, so is H'/C', hence so is H/C. The same property holds for C by hypothesis and therefore for the bundle H. This proves the lemma and Corollary 7.12.

0.8 Semisimple elements; unions and intersections of maximal tori in group schemes not necessarily affine

Throughout this section, G denotes an S-group prescheme that is smooth and of finite presentation over S, with connected fibers.

Suppose first that S is the spectrum of an algebraically closed field k. When G is affine, the notion of a semisimple element of $G(k)$ was defined in *BIBLE* 4, No. 4; one checks at once that this notion is invariant under an algebraically closed extension k'/k of the base field. Moreover, by *BIBLE* 6, Theorem 5(c), an element $g \in G(k)$ is semisimple if and only if it is contained in a maximal torus of G.

When G is no longer assumed affine, Chevalley's theorem gives a canonical expression of G as an extension of an abelian variety by a smooth connected affine algebraic group:

$$e \longrightarrow V \longrightarrow G \longrightarrow A \longrightarrow e. \tag{*}$$

We shall say that $g \in G(k)$ is *semisimple* if it is a semisimple element of $V(k)$. Since the maximal tori of V are precisely the maximal tori of G, this is equivalent to saying that g belongs to a maximal torus of G. This notion is again invariant under algebraically closed extensions of the base field.

Suppose next that $S = \operatorname{Spec} k$ for an arbitrary field k, and let $g \in G$. After choosing an algebraically closed extension K of $\kappa(g)$, the point g is the image of a geometric point g' of G with values in an algebraically closed extension K of k. We call g semisimple when g' is semisimple. What precedes shows that this is independent of the choice of g'. If k'/k is any extension, the set of semisimple elements of $G_{k'}$ is the inverse image of the set G^{ss} of semisimple elements of G.

Finally, for arbitrary S, a point $g \in G$ is called semisimple when it is semisimple in its fiber G_s. Thus, for every morphism $S' \to S$, the subset $G^{\mathrm{ss}}_{S'}$ is the inverse image of G^{ss}. Suppose that the functor $\mathcal{T}$ of maximal tori defined in 1.10 is representable by a prescheme of finite presentation over S. This is so, for example, if G admits a maximal torus locally for the fpqc topology, by 7.1(d), at least when G is quasi-projective over S. Let T be the canonical maximal torus of $G_{\mathcal{T}}$, the "universal maximal torus of G", and consider the morphism

$$u : T \longrightarrow G$$

induced by the projection $G_{\mathcal{T}} \to G$. It follows immediately from the definition that G^{ss} is the image of this morphism. We obtain:

Proposition 8.1. *The set G^{ss} of semisimple elements of G is locally constructible. It is therefore constructible when S, and hence G, is quasi-compact and quasi-separated.*

As usual, we reduce to affine noetherian S. By the standard noetherian criterion for constructibility (EGA 0_{III}, 9.2.3), we may further suppose that S is integral and need only show that there is a nonempty open $U \subset S$ such that $G^{\mathrm{ss}}|_U$ is constructible. After shrinking U, and replacing it if necessary by a finite covering, we may suppose that G is separated over S and contains a maximal torus. By 7.1(d), $\mathcal{T}$ is then represented by a prescheme of finite presentation over S; the same is true of its universal torus, whose image in G is therefore constructible.

Suppose again that k is a field, and let H be the subgroup of G generated by the preceding morphism (VI_B 1.2). It is a smooth group subscheme of G, connected because the universal torus is connected, and its formation commutes with every extension of the base field, as does the formation of $\mathcal{T}$.

When k is algebraically closed, H is also the algebraic subgroup of G generated by the maximal tori of G, or, equivalently, by all the tori of G. It is likewise the smallest algebraic subgroup of G containing all semisimple elements of $G(k)$. These characterizations remain valid whenever k is infinite, because, as will be proved in Exposé XIV, the set of k-rational points of $\mathcal{T}$ is dense in $\mathcal{T}$.

The subgroup H is invariant in G. Indeed, it is enough to check this after passing to an algebraically closed field; since H and G are then smooth, it suffices to check stability under the inner automorphisms $\mathrm{int}(g)$, $g \in G(k)$, which is immediate. One can even show that H is characteristic in G, that is, stable under $\mathrm{Aut}_{k\text{-gr}}(G)$.

It follows from 6.6(d) that G/H has reductive rank zero. More precisely, if K is an invariant algebraic subgroup of G, then, over an algebraically closed field, the maximal tori of G/K are the direct images of the maximal tori of G. Hence G/K has reductive rank zero if and only if K contains all maximal tori of G, or, over an arbitrary field, if and only if $K \supset H$. Thus H is the smallest invariant algebraic subgroup of G for which G/H has reductive rank zero. It is also the smallest algebraic subgroup of G having the same reductive rank as G.

Finally, H is affine. This may again be checked after passing to an algebraic closure. In the exact sequence $(*)$, every maximal torus of G is contained in V; consequently $H \subset V$, and V is affine. We have proved:

Proposition 8.2. *Let G be a smooth connected algebraic group over a field k, and let $\bar{k}$ be an algebraic closure of k. There exists an algebraic subgroup H of G such that $H_{\bar{k}}$ is the algebraic subgroup of $G_{\bar{k}}$ generated by the maximal tori, or equivalently by all the tori, of $G_{\bar{k}}$. The group H is also characterized as the smallest algebraic subgroup of G having the same reductive rank as G, or as the smallest invariant algebraic subgroup of G such that G/H has reductive rank zero. It is a smooth, connected, invariant, affine subgroup of G, and its formation commutes with every extension of the base field.*

The characterization in terms of G/H gives the following consequences.

Corollary 8.3. *Let G be a smooth connected algebraic group over an algebraically closed field k. Then G has reductive rank zero —equivalently, in the notation of 8.2, $H = (e)$— if and only if G is an extension of an abelian variety by a smooth connected unipotent algebraic group, that is, a successive extension of groups isomorphic to the additive group $\mathbb{G}_a$.*

Indeed, Chevalley's exact sequence $(*)$ reduces the assertion to the fact that a smooth connected affine group V has reductive rank zero if and only if it is unipotent; this is contained in *BIBLE* 6.4, Theorem 4, Corollary 3. Thus, in the notation of 8.2, H is the smallest invariant algebraic subgroup of G such that $(G/H)_{\bar{k}}$ is an extension of an abelian variety by a smooth connected affine unipotent algebraic group.

Corollary 8.4. *In the notation of 8.2, $G = H$, or equivalently $G_{\bar{k}}$ is generated by its maximal tori, if and only if G is affine and every homomorphism $G_{\bar{k}} \to \mathbb{G}_a$ is trivial.*

Remarks 8.5.

(a) Let V be the largest smooth connected affine algebraic subgroup of $G_{\bar{k}}$, so that $G_{\bar{k}}$ is an extension of an abelian variety by V. It is well known, by Rosenlicht,[13] that when k is not perfect, V is not in general defined over k: there need not exist an algebraic subgroup V_0 of G such that $(V_0)_{\bar{k}} = V$. If, however, V is generated by its maximal tori—equivalently, if V has no quotient isomorphic to $\mathbb{G}_{a,\bar{k}}$—then such a group does exist, namely the group H of 8.2. Thus, in this rationality question as in many others (compare Exposé XIV, No. 6), all the difficulties arise from unipotent groups, hence from the additive group, whereas the presence of sufficiently many groups of multiplicative type makes descent work.

(b) In the notation of 8.2, let H' be the inverse image in G of the commutator subgroup of G/H. Then H' is the smallest invariant algebraic subgroup of G such that $(G/H')_{\bar{k}}$ is a commutative extension of an abelian variety by a smooth connected affine unipotent algebraic group. It is again a smooth connected affine subgroup of G.

Assume that the characteristic p is positive, and let H'' be the inverse image in G of $p^n(G/H')$ for n sufficiently large. Then H''/H' is an abelian variety, and H'' is the smallest invariant algebraic subgroup of G such that G/H'' is a commutative smooth connected affine unipotent algebraic group. It follows that $V = H_{\bar{k}}$, in the notation of (a), if and only if $H'' = G$, or equivalently if and only if every homomorphism $G_{\bar{k}} \to \mathbb{G}_a$ is trivial.

We now generalize to smooth groups with connected fibers the notion of reductive center developed in Section 4, following the idea of 4.10. Let $\mathcal{Z}$ be the subfunctor of G defined by

$$\begin{aligned}\mathcal{Z}(S') = \{\, g' \in G_{S'}(S') :\ & \text{for every } S'' \to S' \text{ and every maximal torus } T'' \subset G_{S''},\\ & \text{the pullback } g'' \text{ of } g' \text{ is a section of } T'' \text{ over } S'' \,\}.\end{aligned}$$

Let $\mathcal{T}(S')$ be the set of maximal tori of $G_{S'}$, and let $\widetilde{\mathcal{T}}(S')$ be the set of pairs (T', g') where T' is a maximal torus of $G_{S'}$ and g' is a section of T' over S'. Then $\widetilde{\mathcal{T}}$ is a subfunctor of

$$\mathcal{T} \times_S G = \mathcal{T}_G,$$

and in this notation one may also write

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} \widetilde{\mathcal{T}}/\mathcal{T}_G.$$

Using XI 6.8 and 7.1(d), which under suitable hypotheses represents $\mathcal{T}$ by a smooth S-prescheme, one could deduce representability of $\mathcal{Z}$ under corresponding hypotheses. We shall instead obtain it more directly below.

[13] The editors ask that references be supplied here.

Definition 8.6. *Let G be an S-group prescheme, smooth and of finite presentation with connected fibers. We say that G admits a* reductive center *if the preceding functor $\mathcal{Z}$, which is evidently a subgroup of G, is representable by a group of multiplicative type. In that case $\mathcal{Z}$ is called the reductive center of G.*

If Z is a reductive center of G, then $Z_{S'}$ is a reductive center of $G_{S'}$ for every base change $S' \to S$. The existence of a reductive center is consequently local for the fpqc topology. The terminology is justified because $\mathcal{Z}$ is invariant under

$$\mathrm{Aut}_S(G),$$

and hence is an invariant subgroup of G; by IX 5.5, it is central.

Lemma 4.5 is replaced here by the following statement.

Lemma 8.7. *Let $u : H \to G$ be a central homomorphism, where H is of multiplicative type and of finite type over S. Suppose that for every algebraically closed field k over S, the subgroup $u_k(H_k)$ is contained in the largest smooth connected affine subgroup of G_k. Then u factors through every maximal torus T of G. In particular, u factors through the subfunctor $\mathcal{Z}$ defined above.*

An easy variant of IX 5.1 bis, with equality replaced by inclusion, reduces the assertion to the case in which S is the spectrum of an algebraically closed field: first reduce to affine noetherian S, then to artinian S, and use IX 3.6. Since T is contained in the largest smooth connected affine subgroup V of G, we are reduced to $G = V$, hence to the affine case proved in 4.5.

Proposition 8.8. *Let G be an S-group prescheme, smooth and of finite presentation with connected fibers.*

(a) If $S = \operatorname{Spec} k$, then G admits a reductive center Z. When G is an extension of an abelian variety by a smooth connected affine algebraic group V —for example, when k is algebraically closed—then Z is also the reductive center of V, and it is the largest central subgroup of multiplicative type of V.

(b) Let Z be a group subprescheme of multiplicative type in G. Then Z is a reductive center of G if and only if Z_s is a reductive center of G_s for every $s \in S$. It is then the largest subgroup of multiplicative type K in G such that K_s is contained in the reductive center of G_s for every $s \in S$. More generally, let $u : H \to G$ be a homomorphism with H of multiplicative type and of finite type over S, and suppose that for every algebraically closed field k over S, u_k factors through the largest smooth connected affine subgroup of G_k. Then u factors through Z, and in particular is central.

(c) If G admits a maximal torus locally for the fpqc topology, then G admits a reductive center.

(d) If T is a maximal torus of G, then

$$T \cap \mathrm{Centr}(G) = \ker(T \longrightarrow \mathrm{GL}(\mathfrak{g})),$$

and this group is a reductive center of G.

(e) *Let Z be a reductive center of G, and suppose that $G' = G/Z$ is representable, for example when S is artinian. Then the unit subgroup is a reductive center of G', and*

$$T' \longmapsto T = u^{-1}(T')$$

is a bijection from the set of maximal tori of G' to the set of maximal tori of G.

Proof. *(a)* Let $S = \operatorname{Spec} k$. To prove existence, we may pass to an algebraic closure of k, and thus suppose that G is an extension of an abelian variety by a smooth connected affine algebraic group V. For every S' over k, the maximal tori of $G_{S'}$ are those of $V_{S'}$: this follows from IX 5.2 and from the fact that a homomorphism from a torus to an abelian variety over an algebraically closed field is trivial. Consequently the functor $\mathcal{Z}$ defined using G is the same as the one defined using V, and we reduce to affine G.

The functor $\mathcal{T}$ is representable and is necessarily "essentially free" over k (VIII 6.1); hence $\mathcal{T}_G$ is essentially free over G. Since the universal torus is a closed subscheme of $\mathcal{T}_G = G_{\mathcal{T}}$ (for example by VIII 5.7), VIII 6.4 shows that

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} \widetilde{\mathcal{T}}/\mathcal{T}_G$$

is represented by a closed subscheme of G. It is therefore a group subscheme. It is of multiplicative type: after passing to an algebraic closure, G admits a maximal torus T, and by definition $Z \subset T$; a subgroup of a group of multiplicative type is of multiplicative type (IX 8). Thus Z is a reductive center. Lemma 8.7 shows that it is the largest central subgroup of multiplicative type of V.

(b) Only sufficiency requires proof. Suppose that Z is of multiplicative type and that Z_s is the reductive center of G_s for every $s \in S$. Lemma 8.7 first shows that Z is contained in every maximal torus of G, and this remains true after every base change. It remains to show that if g is a section of G over S whose pullback to every S' over S is a section of every maximal torus of $G_{S'}$, then g is a section of Z.

As usual, since the fpqc sheaf

$$\mathcal{Z}_0 = \prod_{\mathcal{T}_G/G} \widetilde{\mathcal{T}}/\mathcal{T}_G$$

commutes with filtered direct limits of rings, we reduce successively to affine, noetherian, and artinian S. By 7.1(d), $\mathcal{T}$ is then represented by a smooth prescheme over S. Arguing as in (a), $\mathcal{Z}_0$ is represented by a closed subscheme Z_0 of G. We already know $Z \subset Z_0$; by hypothesis $Z_k = (Z_0)_k$ over the residue field k, and Z is flat over S. It follows that $Z = Z_0$, proving that Z is the reductive center. The remaining assertions of (b) follow from 8.7.

(c) follows immediately from (d).

(d) Here $\mathfrak{g}$ denotes the Lie algebra of G, and $T \to \mathrm{GL}(\mathfrak{g})$ is induced by the adjoint representation. Trivially,

$$T \cap \operatorname{Centr}(G) \subset \ker\bigl(T \longrightarrow \mathrm{GL}(\mathfrak{g})\bigr).$$

The proof of the reverse inclusion is the same as in 4.7(d). Put

$$Z = \ker\bigl(T \longrightarrow \mathrm{GL}(\mathfrak{g})\bigr).$$

Then Z is of multiplicative type (IX 6.8). By (b), we may check that it is a reductive center over a field. Let Z_0 be the reductive center supplied by (a). Plainly

$$Z_0 \subset T \cap \operatorname{Centr}(G) = Z.$$

Conversely, Z is a central subgroup of multiplicative type contained in the smooth connected affine group T, so (a) gives $Z \subset Z_0$. Thus $Z = Z_0$.

(e) Under the hypotheses of 7.1(f), put $Z = \ker u$, and suppose that for every algebraically closed field k over S, Z_k is contained in a maximal torus of G_k. One checks readily that

$$T' \longmapsto u^{-1}(T')$$

is a bijection from the maximal tori of G' to those of G. Applying this observation to the situation of (e) proves the assertion. □

Remarks 8.9.

(a) The proof of 8.8 is independent of the results of Section 4; in particular, the proof of 8.8(a) does not use 4.4, whose proof is somewhat laborious.

(b) The subfunctor $\mathscr{Z}$ considered in 8.6 is always central, whether or not it is representable. It may therefore be tempting to call it the reductive center of G in all cases.

(c) One may likewise generalize 4.9. If G is a smooth connected algebraic group over an algebraically closed field, then G is an extension of an abelian variety by a smooth connected unipotent algebraic group—equivalently, G has reductive rank zero, by 8.3—if and only if the reductive center of G is the unit group and the Lie algebra of G is nilpotent.

0.9 Complement: actions of group schemes and fixed points

The purpose of this section, added in January 2008, is to study fixed points under the action of a group scheme.

0.9.1 Representability of the fixed-point functor

Let S be a scheme, let G be an S-group scheme, and suppose that G acts on an S-scheme X. Define the subfunctor X^G of fixed points of X under G by requiring, for every S-scheme T, that $X^G(T)$ be the subset of $X(T)$ consisting of the points fixed under G (VIII 6(e)).

Recall the notion of an S-pure scheme introduced in Raynaud–Gruson, §3.3, and in Raynaud. Let G be a flat S-group scheme of finite presentation. Since the fibers of G have no embedded components, purity takes the following form: G is *S-pure* if, for every strictly henselian local S-scheme T with closed point t, every generic point $\widetilde{x}$ of a fiber of $G \times_S T$ specializes to a point of G_t; equivalently, the closure of $\widetilde{x}$ in $G \times_S T$ meets G_t.

Examples.

(1) If G is quasi-finite and separated over S, then G is S-pure if and only if it is finite over S.

(2) If $S = \operatorname{Spec} R$ and $G = \operatorname{Spec} A$, then G is S-pure if and only if A is a projective R-module (Raynaud–Gruson, 3.3.5). In particular, a diagonalizable G is S-pure.

(3) Every group scheme of multiplicative type is S-pure, since purity is local for the étale topology on S.

(4) The scheme G is S-pure if it is proper over S, if its fibers are irreducible, or if S is semilocal artinian.

(5) In particular, every reductive S-group scheme is S-pure.

Proposition 9.2. *Let G be an S-group scheme of finite presentation, flat and S-pure over S, acting on an S-scheme X of finite presentation. Then the fixed-point functor X^G is represented by a closed subscheme of X, of finite presentation over S.*

Indeed, let $a \in X(S)$, and let H be the group subscheme of G fixing a. The point a is fixed under G if and only if $H = G$. The coincidence functor of H and G—whose value on an S-scheme T is a singleton when

$$H \times_S T \longrightarrow G \times_S T$$

is bijective and is empty otherwise—is represented by a closed subscheme of S, defined by a finite-type sheaf of ideals (Raynaud–Gruson, 4.1.1). Apply this result with $S = X$ and with a the universal point of X, that is, the identity element of

$$X(X) = \mathrm{Hom}_S(X, X).$$

0.9.3 The infinitesimal obstruction

Suppose that G is a flat S-group scheme of finite presentation acting on the left on a smooth S-prescheme X. We shall study the formal smoothness of X^G.

Let S_0 be a closed subscheme of S, defined by a quasi-coherent sheaf of ideals $\mathcal{I}$ with $\mathcal{I}^2 = 0$, and write

$$j : S_0 \longrightarrow S, \qquad G_0 = G \times_S S_0, \qquad X_0 = X \times_S S_0.$$

Let $\epsilon_0 \in X^G(S_0)$ be a point that lifts to $X(S)$. We shall study the obstruction to lifting ϵ_0 to $X^G(S)$.

Since X is smooth, the set of lifts of ϵ_0 to $X(S)$ is a trivial principal homogeneous space under the abelian group

$$\mathrm{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^*\Omega^1_{X_0/S_0}, \mathcal{I})$$

(III 0.2, 0.3). Put

$$L_0 = \mathrm{Hom}_{\mathcal{O}_{S_0}}(\epsilon_0^*\Omega^1_{X_0/S_0}, \mathcal{I}).$$

This is an $\mathcal{O}_{S_0}$-module. Since ϵ_0 is fixed under G, it is naturally a G_0-$\mathcal{O}_{S_0}$-module. Let

$$\rho_0 : G_0 \longrightarrow \mathrm{Aut}_{S_0}(L_0), \qquad \rho : G \longrightarrow \mathrm{Aut}_S(j_*L_0)$$

be the associated representations.

Lemma 9.4. *There is a canonically defined class*

$$c(\epsilon_0) \in H^1(G, j_*L_0) \simeq H^1(G_0, L_0)$$

such that ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

The displayed adjunction is the one in III 1.1.2. We work on the small fppf site of S. For every S-scheme T, flat and of finite presentation, write

$$G_T = G \times_S T, \qquad X_T = X \times_S T.$$

Consider the sheaf $\mathcal{A}$ on this site defined by

$$\mathcal{A}(T) = \{ \text{lifts of } \epsilon_0 \times_S T \text{ to } X(T) \}.$$

Since the formation of L_0 commutes with flat base change, $\mathcal{A}(T)$ is a trivial principal homogeneous space under

$$H^0(T, j_* L_0).$$

Moreover, because ϵ_0 is fixed, every $g \in G(T)$ acts by affine automorphisms on $\mathcal{A}(T)$, compatibly with the action of G on $j_* L_0$:

$$g(a_T + v) = g(a_T) + \rho(g)(v), \qquad v \in H^0(T, j_* L_0).$$

Since G is flat and of finite type, we may take $T = G$ and g equal to the universal point id_G. We thereby obtain an action of the fppf sheaf G on $\mathcal{A}$.

Let a be a lift of ϵ_0 to $X(S)$, and let $g^\sharp$ denote the universal point of G. Define

$$v^\sharp \in H^0(G, j_* L_0)$$

by the relation

$$\rho(g^\sharp)(v^\sharp) = g^\sharp \cdot a_G - a_G.$$

For every S-prescheme Y, set

$$z(Y) : G(Y) \longrightarrow H^0(Y, j_* L_0), \qquad g \longmapsto v^\sharp(g).$$

This defines a 1-cocycle $z \in Z^1(G, j_* L_0)$ (Exposé I, §5). Its class $c(\epsilon_0)$ does not depend on the choice of a. In particular, if ϵ_0 lifts to $X^G(S)$, then $c(\epsilon_0) = 0$.

Conversely, if $c(\epsilon_0) = 0$, there is a section

$$w \in H^0(S, j_* L_0)$$

such that, for every S-prescheme Y and every $g \in G(Y)$,

$$z(Y)(g) = g \cdot w_Y - w_Y.$$

Apply this with $Y = G$ and $g = g^\sharp$. It follows that $a - w \in X^G(S)$. Thus ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

Remark 9.5. When S is affine and G is affine, flat, and of finite type over S, this obstruction can be refined to a class

$$\widetilde{c}(\epsilon_0) \in H^1_{G_0}(X_0, L_0),$$

where the right-hand side is the G_0-equivariant cohomology group defined by Wewers (Appendix C). In this case it is unnecessary to assume that ϵ_0 lifts to $X(S)$.

0.9.6 Smoothness of fixed points

Theorem 9.7. *Let G be a flat S-group scheme of finite type over a noetherian base S, acting on a smooth S-scheme X. Suppose that G is S-pure and that, for every geometric point s of S with algebraically closed residue field,*

$$H^1(G_s, (\Omega^1_{X_s})^*) = 0.$$

Then the fixed-point functor X^G is represented by a closed subscheme of X, smooth over S.

By 9.2, X^G is represented by a closed subscheme of X, of finite presentation over S. We may suppose $S = \operatorname{Spec} A$ with A local. By the smoothness criterion of SGA 1, III 3.1(iii bis), it is enough to verify formal smoothness for

$$S' = \operatorname{Spec} A'$$

and its closed subscheme

$$S'' = \operatorname{Spec}(A'/I'),$$

where A' is an artinian local ring and I' is an ideal of square zero. By dévissage, we may assume that I' is annihilated by the maximal ideal of A'. If k is the residue field of A', then I' is therefore a k-vector space.

Let $\epsilon'' \in X^G(S'')$ be a point that we wish to lift to $X^G(S')$, and put

$$X' = X \times_S S', \qquad G' = G \times_S S',$$
$$X'' = X \times_S S'', \qquad G'' = G \times_S S''.$$

Since S' is affine and X' is smooth over S', ϵ'' lifts to $X(S')$. By Lemma 9.4, the obstruction to lifting ϵ'' to $X^G(S')$ is a class

$$c(\epsilon'') \in H^1(G'', L''),$$

where

$$L'' = \operatorname{Hom}_{\mathcal{O}_{S''}}((\epsilon'')^*\Omega^1_{X''/S''}, I').$$

Because L'' is a k-vector space, there is a canonical isomorphism

$$H^1(G'', L'') \simeq H^1(G'' \times_{A'/I'} k, L'').$$

Let $\bar{k}$ be an algebraic closure of k. By IX 3.1 in the affine case, and by Lemma 9.11 in general, there is an isomorphism

$$H^1(G'' \times_{A'/I'} k, L'') \otimes_k \bar{k} \xrightarrow{\sim} H^1(G'' \times_{A'/I'} \bar{k}, L'' \otimes_k \bar{k}).$$

The representation $L'' \otimes_k \bar{k}$ is a direct sum of copies of

$$(\Omega^1_{X_{\bar{k}}})^*.$$

The group on the right therefore vanishes by hypothesis. It follows that $H^1(G'', L'') = 0$, so the obstruction vanishes and X^G is formally smooth.

Corollary 9.8. *Let G be a flat S-group scheme of finite presentation acting on a smooth S-scheme X of finite presentation. Suppose that G admits a composition series whose factors are of one of the following types:*

(1) an abelian S-scheme, that is, a smooth S-group scheme whose fibers are abelian varieties;

(2) an S-group scheme of multiplicative type;

(3) a finite étale S-group scheme of degree invertible on S;

(4) a reductive S-group scheme, when S is a $\mathbb{Q}$-scheme.

Then the fixed-point functor X^G is represented by a closed subscheme of X, of finite presentation and smooth over S.

Indeed, suppose that

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1$$

is an exact sequence of S-groups satisfying the hypotheses. The group G'' acts on $X^{G'}$, and X^G is the fixed-point functor of $X^{G'}$ under G''. We are therefore reduced to the case of a group of one of the four types listed above. By passage to the limit, we may suppose S noetherian. We verify the cohomological criterion of Theorem 9.7 for a geometric fiber G_s and a finite-dimensional linear representation V of G_s.

For an abelian scheme, every morphism from G_s^n to V is constant; consequently

$$H^i(G_s, V) = 0 \qquad (i > 0).$$

For a group scheme of multiplicative type, G_s is diagonalizable, and I 5.3.3 again gives

$$H^i(G_s, V) = 0 \qquad (i > 0).$$

In the remaining two cases the assertion follows from semisimplicity of the linear representations of G_s. If G_s is finite étale of degree invertible in $\kappa(s)$, it is a constant finite group of invertible order, and every linear representation of G_s is semisimple. In the reductive case, $\kappa(s)$ is algebraically closed of characteristic zero, and every linear representation of G_s is semisimple (Tauvel–Yu, Theorem 27.3.3).

Remark 9.9. This applies in particular to an action of G/S on a smooth S-group scheme H. If G is of multiplicative type and acts by inner automorphisms on H, with H smooth and affine over S, one recovers XI 5.3: the centralizer of G in H is smooth.

Remark 9.10. When $S = \operatorname{Spec} k$ and G is a linearly reductive algebraic group over an algebraically closed field k, the result is due independently to Fogarty (Theorem 5.4) and Iversen (Proposition 1.3).

The proof of Theorem 9.7 used the following lemma, which is well known for affine group schemes (IX 3.1).

Lemma 9.11. *Let G be a group scheme over an affine scheme $S = \operatorname{Spec} A$, and let $f : S' = \operatorname{Spec} A' \to S$ be flat. Put $G' = G \times_S S'$. Then, for every quasi-coherent G-$\mathcal{O}_S$-module M, there are canonical isomorphisms*

$$H^i(G, M) \otimes_A A' \xrightarrow{\sim} H^i(G', M \otimes_A A') \qquad (i \geq 0).$$

For $n \geq 1$, the degree-n Hochschild cochains are determined by their value at

$$(\mathrm{id}, \dots, \mathrm{id}) \in G(G) \times \cdots \times G(G).$$

Thus there is an isomorphism of A-modules

$$C^n(G, M) \simeq \Gamma\big(G^n, M \otimes_A \mathcal{O}_{G^n}\big).$$

The group $H^n(G, M)$ is the n-th cohomology group of the complex

$$M \longrightarrow \Gamma(G, M \otimes_A \mathcal{O}_G) \longrightarrow \Gamma(G^2, M \otimes_A \mathcal{O}_{G^2}) \longrightarrow \Gamma(G^3, M \otimes_A \mathcal{O}_{G^3}) \longrightarrow \cdots .$$

Since A' is flat over A, EGA IV_1, 1.7.21 gives

$$\begin{aligned}\Gamma\big(G^n, M \otimes_A \mathcal{O}_{G^n}\big) \otimes_A A' \xrightarrow{\ \sim\ } \Gamma\big(&(G')^n,\\ &(M \otimes_A A') \otimes_{A'} \mathcal{O}_{(G')^n}\big).\end{aligned}$$

Taking cohomology proves the assertion.

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Exposé XIII
Regular Elements of Algebraic Groups and Lie Algebras

by A. Grothendieck

0.10 An auxiliary lemma on varieties with operators

Let S be a prescheme, let G be an S-group prescheme acting on the left on an S-prescheme V, let W be a closed S-subprescheme of V, and let N be its stabilizer in G. Thus N is the subgroup of G whose points with values in an S' over S are the elements $g \in G(S')$ such that

$$g \cdot W_{S'} = W_{S'}.$$

We endow (Sch/S) with the faithfully flat quasi-compact topology and identify G, V, and W with the corresponding sheaves.We shall therefore reason in the category of sheaves on (Sch/S); throughout this section, "locally" refers to the topology just specified on (Sch/S).

Since N is a sheaf, consider the quotient sheaf G/N. It is immediately seen to be isomorphic to the following functor: to every S' over S, it assigns the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ under G. Let X be the subsheaf of $G/N \times_S V$ whose value on an S' over S is the set of pairs (W', v), where W' is as above and v is a section of W' over S' (hence also a section of $V_{S'}$ over S'). Let Z be the inverse image of X in $G \times_S V$. We then have the cartesian diagram

$$\text{(x)} \qquad \begin{tikzcd} Z \arrow[r] \arrow[d, hook] & X \arrow[d, hook, "i"] \\ G \times_S V \arrow[r] & G/N \times_S V, \end{tikzcd}$$

where i is the canonical immersion, while the lower horizontal arrow is induced by the canonical morphism $G \to G/N$.

This morphism sends $g \in G(S')$ to the subsheaf $g \cdot W_{S'}$ of $V_{S'}$. Consequently, $Z(S')$ is the set of pairs $(g, v) \in G(S') \times V(S')$ such that $v \in (g \cdot W_{S'})(S')$. It follows that Z is isomorphic to the sheaf $G \times_S W$, via the isomorphism

$$G \times_S W \xrightarrow{\sim} Z, \qquad (g, w) \longmapsto (g, g \cdot w).$$

Thus the preceding cartesian diagram yields the cartesian diagram

(xx)
$$\begin{tikzcd} G \times_S W \arrow[r, "q"] \arrow[d, "\lambda"'] & X \arrow[d, "i"] \\ G \times_S V \arrow[r] & G/N \times_S V \end{tikzcd}$$

where
$$\lambda(g, w) = (g, g \cdot w), \qquad q(g, w) = (\bar{g}, g \cdot w),$$
where $\bar{g}$ denotes the image of g under the canonical map $G(S') \to (G/N)(S')$.

Diagram (x) also shows that $Z \to X$ makes Z a principal bundle over X, with N acting on the right by
$$(g, v) \cdot n = (gn, v).$$
Accordingly, in diagram (xx), the morphism $q : G \times_S W \to X$ makes $G \times_S W$ a principal bundle over X, with right N-action
$$(g, w) \cdot n = (gn, n^{-1} \cdot w).$$

The principal morphisms just introduced are summarized by

(D)
$$\begin{tikzcd} & G \times_S W \arrow[d, "q"] \arrow[dr, "\varphi"] & \\ G/N & X \arrow[l, "p"'] \arrow[r, "\psi"] \arrow[d, hook, "i"] & V \\ & G/N \times_S V \arrow[ul, dashed, "\mathrm{pr}_1"] \arrow[ur, dashed, "\mathrm{pr}_2"'] & \end{tikzcd}$$

where
$$\psi = \mathrm{pr}_2 \circ i, \qquad \varphi = \psi \circ q, \qquad \varphi(g, w) = g \cdot w.$$

If v is a section of V over S, let X_v be its inverse image in X under ψ. For every S' over S, $X_v(S')$ is the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ under G and contain the section $v_{S'}$ of $V_{S'}$. The inverse image of v in $G \times_S W$ under φ is isomorphic to the subsheaf M_v of G defined by
$$M_v(S') = \{g \in G(S') \mid v_{S'} \in (g \cdot W_{S'})(S')\} = \{g \in G(S') \mid g^{-1} v_{S'} \in W(S')\}.$$
If v is a section of W, and not merely of V, then M_v evidently contains N.

None of these explicit descriptions used the representability of G, V, or W (nor even the fact that the site in question is defined in terms of preschemes). Suppose now, however, that N is representable, faithfully flat, and quasi-compact over S, and that G/N is representable. When S is the spectrum of a field and G is of finite type over that field, this hypothesis is necessarily satisfied (VIII 6 and $\mathrm{VI_B}$.11.18). Using the cartesian diagram (xx), faithfully flat quasi-compact descent, and the fact that $Z \to G \times_S V$ is a closed immersion, one sees that X is representable: it is obtained by descending the closed subprescheme Z of $G \times_S V$ along the faithfully flat quasi-compact morphism
$$G \times_S V \longrightarrow G/N \times_S V.$$

Thus diagram (D) is a diagram of morphisms of preschemes over S.

We henceforth assume that S is the spectrum of a field k, and that G, V, and W are of finite type over k. Let $\mathfrak{n}$ be the Lie algebra of N. Then $\dim N \leq \operatorname{rank}_k \mathfrak{n}$, with equality if and only if N is smooth over k (Exposé VI[14]). Let $a \in W(k)$, and consider the subscheme M_a of G defined above; it contains N and is isomorphic to $\varphi^{-1}(a)$. We denote by $\mathfrak{m}_a$ its Zariski tangent space at the identity element e of N. Thus

$$\mathfrak{n} \subset \mathfrak{m}_a, \qquad \dim N \leq \operatorname{rank}_k \mathfrak{n} \leq \operatorname{rank}_k \mathfrak{m}_a. \tag{1}$$

Lemma 1.1. — *With the preceding notation:*

a) Consider the following conditions:

(i) $\mathfrak{n} = \mathfrak{m}_a$, and N is smooth over k.

(i bis) $\dim N = \operatorname{rank}_k \mathfrak{m}_a$.

(ii) The morphism $\psi : X \to V$ is unramified at $(\bar{e}, a)$.

(iii) M_a and N coincide in a neighborhood of e.

Then
$$\text{(i)} \Longleftrightarrow \text{(i bis)} \Longrightarrow \text{(ii)} \Longleftrightarrow \text{(iii)}.$$

b) Suppose that $\varphi : G \times_S W \to V$ is smooth at (e, a). Then M_a is smooth over k at e, and ψ is smooth at $(\bar{e}, a)$.

Proof. *a)* The equivalence of (i) and (i bis) follows immediately from (1) and from the fact noted above that N is smooth over k if and only if $\dim N = \operatorname{rank}_k \mathfrak{n}$. Consider the inclusion morphism $N \to M_a$. It is well known[15] that if N is smooth over k at e and the tangent map at e is surjective, then $N \to M_a$ is smooth at e; since it is an immersion, it is therefore an isomorphism at e. This proves that (i) implies (iii).

To prove the equivalence of (ii) and (iii), let $X_a = \psi^{-1}(a)$, as above. From the isomorphisms
$$M_a \simeq \varphi^{-1}(a) \simeq q^{-1}(X_a)$$
we obtain a morphism $p_a : M_a \to X_a$ making M_a a principal homogeneous bundle with group N_{X_a}. Consider the diagram

$$\begin{tikzcd} M_a \arrow[d, "p_a"'] & N \arrow[l, "j_a"'] \arrow[d] \\ X_a \arrow[r, "j'_a"] & S = \operatorname{Spec}(k). \end{tikzcd}$$

Here $\operatorname{Spec}(k) \to X_a$is defined by the point $\bar{a} = (\bar{e}, a)$ of X, and $j_a : N \to M_a$ is the canonical immersion. To say that ψ is unramified at $(\bar{e}, a)$ means that j'_a is an open immersion, or equivalently that it induces an isomorphism

$$S \xrightarrow{\sim} \operatorname{Spec}(\mathcal{O}_{X_a, \bar{a}}).$$

[14]Editor's note in the French source: since this reference could not be identified, see Theorem II.5.2.1 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson (1970).

[15]Editor's note in the French source: see, for example, EGA IV_4, Theorem 17.11.1(d).

Since p_a is flat, this is equivalent to saying that the morphism obtained from the preceding one by the base change

$$\mathrm{Spec}(\mathcal{O}_{M_a,e}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{X_a,\bar{a}})$$

is an isomorphism. The resulting morphism is precisely

$$\mathrm{Spec}(\mathcal{O}_{N,e}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{M_a,e}),$$

which proves the equivalence of (ii) and (iii).

Notice moreover that the proof shows that (ii) and (iii) imply the following condition, apparently stronger than (iii):

(iii bis) N is an open and closed subscheme of M_a.

b) The first assertion follows from the isomorphism $M_a \simeq \varphi^{-1}(a)$, and the second from the fact that q is flat and $q(e,a) = (\bar{e}, a)$.

0.11 Density theorem and theory of the regular points of G

We apply the constructions and notation of the preceding section in the following situation. Let G be a smooth connected algebraic group over k; take $V = G$, with G acting on V by inner automorphisms; and let $W = H$, where H is a smooth connected algebraic subgroup of G. We denote the Lie algebras of G and H by $\mathfrak{g}$ and $\mathfrak{h}$, the normalizer of H in G by N, and its Lie algebra by $\mathfrak{n}$. For $a \in G(k)$, we again denote by M_a, as in Section 1, the inverse of the transporter of a into H. Thus, if $a \in H(k)$, then $N \subset M_a$; in that case $\mathfrak{m}_a$ denotes the Zariski tangent space of M_a at the identity element e of G. Note that

$$\mathfrak{h} \subset \mathfrak{n} \subset \mathfrak{m}_a \subset \mathfrak{g} \qquad \text{for } a \in H(k).$$

We shall use the following lemma.

Lemma 2.0. — *The equality $H = N^0$ holds if and only if*

$$(\mathfrak{g}/\mathfrak{h})^H = 0,$$

where the left-hand side denotes the subspace of invariants for the action of H induced by the adjoint representation. When this condition holds, N is smooth and $\dim X = \dim G$*. In all cases,* $\dim X \leq \dim G$*, and equality holds if and only if H has finite index in N.*

Indeed, by II 5.2.3(i), $\mathfrak{n}$ is the inverse image of $(\mathfrak{g}/\mathfrak{h})^H$ under $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$. Hence $(\mathfrak{g}/\mathfrak{h})^H = 0$ is equivalent to $\mathfrak{h} = \mathfrak{n}$, which in turn is equivalent to $H = N^0$, since H is a smooth connected algebraic subgroup of the algebraic group N (cf. VI.2). This also shows that N is smooth. Furthermore,

$$\dim X = (\dim G - \dim N) + \dim H = \dim G - (\dim N - \dim H),$$

and therefore

$$\dim X \leq \dim G,$$

with equality if and only if $\dim H = \dim N$, that is, if and only if H has finite index in N. This holds in particular when $H = N^0$, and proves the lemma.

Theorem 2.1. — *Let G be a smooth connected algebraic group over an algebraically closed field k, let H be a smooth connected algebraic subgroup, and let N be its normalizer. Write $\mathfrak{g}, \mathfrak{h}, \mathfrak{n}$ for their Lie algebras, let $X = G \times^N H$ be the scheme introduced in Section*

1 (fibered over G/N with typical fiber H), and let $\psi : X \to G$ be the canonical morphism. Its image is also the image of $\varphi : G \times H \to G$, where

$$\varphi(g, h) = \operatorname{int}(g)h = ghg^{-1}.$$

The following conditions are equivalent:

(i) H contains a Cartan subgroup C of G (XII 1).

(i bis) H has the same reductive rank and the same nilpotent rank (XII 1) as G.

(ii) H contains a maximal torus T of G, and $(\mathfrak{g}/\mathfrak{h})^T = 0$.

(iii) The set of conjugates of H containing a given maximal torus is finite and nonempty, and H has finite index in its normalizer.

(iv) There exists $a \in H(k)$ that belongs to only finitely many conjugates of H (or merely such that $\psi^{-1}(a)$ has an isolated point), and H has finite index in its normalizer.

(iv bis) The morphism $\psi : X \to G$ is generically quasi-finite (that is, there is a dense open subset of X on which ψ is quasi-finite), and H has finite index in its normalizer.

(v) There is a dense open subset U of G such that, for every $x \in U(k)$, the set of conjugates of H containing x is finite and nonempty; equivalently, $\psi : X \to G$ is dominant and generically quasi-finite.

(vi) There is a dense open subset U of G such that every $x \in U(k)$ is contained in a conjugate of H; equivalently, $\psi : X \to G$ is dominant.

(vii) There exists $a \in H(k)$ such that the fixed subspace of $\mathfrak{g}/\mathfrak{h}$ under $\operatorname{ad}_{\mathfrak{g}/\mathfrak{h}}(a)$ is zero.

Moreover, these conditions imply that H is its own connected normalizer: N is smooth and $\dim H = \dim N$. *They also imply that $\psi : X \to G$ is generically étale.*

Proof. By 2.0, $\dim X \leq \dim G$, with equality if and only if $\dim H = \dim N$, or equivalently if and only if H has finite index in N. The inequality $\dim X \leq \dim G$ shows that ψ is dominant if and only if it is dominant and generically quasi-finite, and also if and only if it is generically quasi-finite and $\dim X = \dim G$. Since the last equality is equivalent, by 2.0, to H having finite index in N, conditions (vi), (v), and (iv bis) are equivalent. The equivalence of (iv) and (iv bis) is immediate.

The equivalence of (i) and (i bis) follows directly from the definitions and is left to the reader. If H contains a Cartan subgroup C of G, it contains the maximal torus T of C, which is a maximal torus of G. Since C is the centralizer of T, its Lie algebra $\mathfrak{c}$ is

$$\mathfrak{c} = \mathfrak{g}^T$$

(II 5.2.3(ii)). Hence $\mathfrak{g}^T = \mathfrak{c} \subset \mathfrak{h}$, and I 4.7.3 gives the relation

$$(\mathfrak{g}/\mathfrak{h})^T = 0. \tag{x}$$

Conversely, suppose that H contains the maximal torus T and that (x) holds, so that $\mathfrak{h} \supset \mathfrak{c}$. We claim that H contains the centralizer C of T, which will prove the equivalence of (i) and (ii). This follows from the next lemma.

Lemma 2.1.1. — *Let G be a smooth algebraic group over a field k, let T be a subgroup of multiplicative type of G, let C be its connected centralizer (equal to the centralizer of*

T when G is connected and T is a torus, by XII 6.6(b)), and let H be a smooth subgroup of G containing T. Then H contains C if and only if its Lie algebra $\mathfrak{h}$ contains the Lie algebra $\mathfrak{c}$ of C.

Indeed, $\mathrm{Centr}_G(T)$ is smooth over k by XI 2.4, and hence C is smooth over k. Likewise, $\mathrm{Centr}_H(T)$ is smooth, and

$$\mathrm{Centr}_H(T) = \mathrm{Centr}_G(T) \cap H$$

has Lie algebra $\mathfrak{h} \cap \mathfrak{c}$. The hypothesis therefore implies that this smooth subgroup has the same Lie algebra as the smooth group $\mathrm{Centr}_G(T)$, so it contains the connected component C of the latter. Thus H contains C. $\square$

We now prove the equivalence of (i) and (iii). If H contains the maximal torus T of G, we must show that $H \supset C$ (equivalently, that (x) holds) if and only if H has finite index in its normalizer and the set of conjugates of H containing T is finite. If $H \supset C$, then (x) implies, a fortiori,

$$(\mathfrak{g}/\mathfrak{h})^H = 0. \tag{xx}$$

The Lie algebra $\mathfrak{n}$ is the inverse image of the left-hand side under the canonical homomorphism $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ (II 5.2.3(i)). Consequently, by 2.0, (xx) says that $H = N^0$; in particular, H has finite index in its normalizer. Consider the diagram of subgroups

$$\begin{tikzcd}
T \arrow[r] & N(T) \cap H \arrow[r] \arrow[d] & H \arrow[d] \\
 & N(T) \cap N(H) \arrow[r] \arrow[d] & N(H) \\
 & N(T). &
\end{tikzcd}$$

By the conjugacy theorem for maximal tori in H (XII 6.6(a)), every conjugate of H containing T is obtained from H by conjugation by an element of $N(T)(k)$. Thus the set of such conjugates is in bijection with

$$(N(T)/(N(T) \cap N(H)))(k).$$

Since $H \supset C$, we have $N(T) \cap H \supset C$, so the preceding set is a quotient of $(N(T)/C)(k)$, which is finite. This proves (i) $\Rightarrow$ (iii).

Conversely, suppose (iii), so that $N(T)/(N(T) \cap N(H))$ and $N(H)/H$ are finite. The conjugacy theorem in H shows that the homomorphism

$$(N(T) \cap N(H))/(N(T) \cap H) \longrightarrow N(H)/H$$

induced by the preceding diagram is bijective on k-valued points (indeed, it is an isomorphism). Since the latter quotient is finite, so is the former. Hence $N(T) \cap H$ has finite index in $N(T)$ and therefore contains $C = N(T)^0$. Thus $H \supset C$, and (i), (i bis), (ii), and (iii) are equivalent.

We next prove (ii) $\Rightarrow$ (vii). Both conditions are invariant under an extension $k \to k'$ with k' algebraically closed. We may therefore assume that k has infinite transcendence degree over its prime subfield. There is then an element $a \in T(k)$ such that the subgroup of $T(k)$ generated by a is Zariski dense in T. It follows that

$$(\mathfrak{g}/\mathfrak{h})^T = (\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(a)}.$$

The left-hand side is zero by hypothesis, which proves (vii).

The implication (vii) $\Rightarrow$ (vi) is contained in the following result, which sharpens 2.1.

Corollary 2.2. — *Let G be a smooth algebraic group over a field k, let H be a smooth algebraic subgroup, let N be its normalizer in G, and let*

$$\varphi : G \times H \longrightarrow G, \qquad \varphi(g,h) = \mathrm{ad}(g)h = ghg^{-1}.$$

Let $\psi : X = G \times^N H \to G$ be the morphism obtained from φ by passage to the quotient (cf. Section 1), and let $a \in H(k)$. The following conditions are equivalent:

(i) φ is smooth at (e,a).

(ii) ψ is étale at $(\bar{e},a)$, and N is smooth over k.

(iii) $(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(a)} = 0$, where $\mathfrak{g}$ and $\mathfrak{h}$ are the Lie algebras of G and H.

These conditions imply $H^0 = N^0$.

The smoothness at a k-rational point of φ, a morphism between smooth k-preschemes, is equivalent to the surjectivity of the tangent map at that point. Under the usual identifications of the tangent spaces of G and H with their Lie algebras, a direct calculation gives

$$d\varphi(\xi,\eta) = \big(\mathrm{id} - \mathrm{ad}(a)\big)\xi + \eta,$$

as a map from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Its surjectivity is therefore equivalent to the surjectivity of $\mathrm{id} - \mathrm{ad}(a)$ on $\mathfrak{g}/\mathfrak{h}$, which is precisely (iii).

Condition (iii) implies, a fortiori, $(\mathfrak{g}/\mathfrak{h})^H = 0$, and hence $H^0 = N^0$ by 2.0 (the connectedness hypothesis made at the beginning of the section is not needed there). It follows that N is smooth and that $\dim H = \dim N$, whence $\dim X = \dim G$. Since $q : G \times H \to X$ is flat and $\psi \circ q = \varphi$is smooth at (e,a), the morphism ψ is smooth at $q(e,a) = (\bar{e},a)$, and hence is étale there by dimension. We have proved (i) $\Longleftrightarrow$ (iii) $\Longrightarrow$ (ii). Conversely, (ii) implies (i), because the smoothness of N implies that of q. ☐

In particular, under condition (vii) of 2.1 the morphism φ is smooth at (e,a), so its image contains a nonempty open subset of G. Thus ψ is dominant, proving (vii) $\Rightarrow$ (vi).

It remains to prove (vi) $\Rightarrow$ (i). Suppose first that G is affine. Let U be a nonempty open subset of G such that every $x \in U(k)$ is contained in a conjugate of H, and let C be a Cartan subgroup of G. Applying the already proved implication (i) $\Rightarrow$ (iv) with C in place of H (Borel's "density theorem"), we find a conjugate of C meeting U. We may therefore assume that $U \cap C \neq \varnothing$; equivalently, there is a nonempty open subset V of C such that every $x \in V(k)$ belongs to a conjugate of H.

Write

$$C = T \cdot C_u,$$

where T is the maximal torus of C, hence a maximal torus of G, and C_u is the unipotent part of C; T lies in the center of C (*BIBLE* 6, Theorem 2). We may again assume that k has infinite transcendence degree over its prime subfield. Choose $t \in T(k)$ in the projection of V onto T, a nonempty open subset of T, so that $tC_u \cap V \neq \varnothing$, and choose it so that it generates a Zariski-dense subgroup of T. Every algebraic subgroup of G containing a product tu, with $t \in T(k)$ and $u \in C_u(k)$, contains both factors (*BIBLE* 4, Theorem 3). Taking $tu \in V(k)$, we conclude that a conjugate of H contains t, and therefore contains T. After replacing H by that conjugate, we may assume

$$T \subset H.$$

Let W be the inverse image in C_u of V under $u \mapsto tu$. For every $x \in (T \cdot W)(k)$, there is then a conjugate of H containing both T and x. As observed above, every such conjugate is of the form $\mathrm{int}(g)H$ for some $g \in N(T)(k)$. Consider

$$f : N(T) \times H \longrightarrow G, \qquad f(g,h) = \mathrm{int}(g)h = ghg^{-1}.$$

The image of f contains TW. Since C has finite index in $N(T)$, the latter is a finite union of translates Cg_i, with $g_i \in N(T)(k)$. It follows that some dense open subset V' of $C = TC_u$ is contained in the image under f of $(Cg_i) \times H$. Replacing H by $\mathrm{int}(g_i)H$, we may assume $g_i = e$, so that

$$f(C \times H) \supset V'.$$

Thus for every $u \in V'(k)$ there are $v \in C(k)$ and $h \in H(k)$ such that

$$v^{-1}hv = u, \qquad \text{hence} \qquad vuv^{-1} \in H(k).$$

Set

$$C' = C \cap H = \mathrm{Centr}_H(T).$$

Since vuv^{-1} also belongs to C, it lies in C', and hence $u \in \mathrm{int}(v^{-1})C'$. Therefore the union of the conjugates of C' in C, by elements of $C(k)$, is dense. As in the preceding argument for the pair (G,H), now applied to (C,C'), this implies that C' has finite index in its normalizer in C. By the lemma of *BIBLE* 7, Section 1, we obtain $C' = C$, and consequently $C \subset H$.This proves (vi) $\Rightarrow$ (i) when G is affine.

In the general case we argue by induction on $n = \dim G$, the assertion being trivial for $n = 0$. Let Z be the center of G. There are two cases.

1° If $\dim(Z \cap H) > 0$, set $G' = G/(Z \cap H)$. Then $\dim G' < n$, and condition (vi) for H implies the same condition for its image H' in G'. By induction, H' contains a Cartan subgroup C' of G'. Hence H contains the inverse image C of C', which is a Cartan subgroup by XII 6.6(e).

2° If $\dim(Z \cap H) = 0$, the canonical morphism $H \to G/Z$ is finite. Since G/Z is affine by XII 6.1, H is affine. Every homomorphism from H to an abelian variety is therefore zero (indeed, every morphism of preschemes from H to an abelian variety is constant). This follows from the fact that a smooth connected affine algebraic group over an algebraically closed field is a rational variety, or simply from the fact that it is the union of its Borel subgroups (*BIBLE* 6, Theorem 5(b)); the structure theorems *BIBLE* 6.2 and 6.3 show readily that a smooth connected solvable affine group is a rational variety.

Apply Chevalley's structure theorem to G: the group G is an extension of an abelian variety A by a smooth affine group. Since H is affine, its image in A is zero. On the other hand, that image must be all of A, because the union of its conjugates in A is dense. Hence $A = 0$, so G is affine, reducing us to the case already proved.

This completes the proof of 2.1.

Corollary 2.3. — *Assume that the equivalent conditions of 2.1 hold.*

a) Let $k(X)$ and $k(G)$ be the fields of rational functions of X and G, respectively. Then $k(X)$ is a finite separable extension of $k(G)$. Denote its degree by d.

b) Let T be a maximal torus of G contained in H, which exists by 2.1(ii), and let C be the corresponding Cartan subgroup of G. Then $C \subset H$. Moreover, $\mathrm{Norm}_G(T)$ is a smooth subgroup of G, and

$$\mathrm{Norm}_G(T) \cap \mathrm{Norm}_G(H) = \mathrm{Norm}_{\mathrm{Norm}_G(H)}(T)$$

is a smooth subgroup of it, of finite index equal to the degree d defined in (a). The number of conjugates of H containing a given maximal torus, or a given Cartan subgroup, is d.

c) Let U be the largest open subset of G such that $\psi : X \to G$ induces a finite étale morphism

$$\psi^{-1}(U) \longrightarrow U.$$

Then U is dense. For $g \in G(k)$, the following are equivalent:

(a) $g \in U(k)$;

(b) exactly d conjugates of H contain g;

(c) there are at least d distinct conjugates H_i of H containing g, and for every i,

$$(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0, \qquad \mathfrak{h}_i = \mathrm{Lie}(H_i).$$

Assertion (a) follows from the generic étaleness of ψ, stated at the end of 2.1. This also shows that the open set U in (c) is nonempty, hence dense, and gives the two stated characterizations of its k-points: here one uses that ψ is separated, that X is integral, that G is integral and normal (SGA 1, I 10.11), and that $(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0$ means that ψ is étale at the point $x_i \in \psi^{-1}(g)$ corresponding to H_i.

If H contains the maximal torus T of G, the centralizers of T in H and G have the same dimension and are smooth and connected (XII 6.6(b)); they are therefore equal, proving that $C \subset H$.The normalizer of T in any smooth group containing it is smooth (XI 2.4 bis), so $\mathrm{Norm}_G(T)$ and $\mathrm{Norm}_{\mathrm{Norm}_G(H)}(T)$ are smooth. Recall also that $N = \mathrm{Norm}_G(H)$ is smooth by the last assertion of 2.1. Furthermore, $\mathrm{Norm}_N(T)$ contains C, which has finite index in $\mathrm{Norm}_G(T)$; hence $\mathrm{Norm}_N(T)$ itself has finite index in $\mathrm{Norm}_G(T)$.

The conjugacy theorem for maximal tori in H shows that this index is the number of conjugates of H containing T, equivalently the number containing C. The union of the conjugates of C in G is dense, by 2.1(i) $\Rightarrow$ (vi) applied to C in place of H, and U is stable under inner automorphisms. Hence $C \cap U \neq \varnothing$. Arguing as in the proof of 2.1(vi) $\Rightarrow$ (ii), and making a harmless extension of the base field if necessary, one finds $g \in (C \cap U)(k)$ such that every conjugate of H containing g contains T, and hence C. Thus the conjugates of H containing C are precisely those containing g; since $g \in U(k)$, their number is d. This proves (b).

In fact, the argument shows that the set of conjugates of H containing T is a homogeneous set under the group of rational points of

$$W_G(T) = \mathrm{Norm}_G(T)/\,\mathrm{Centr}_G(T).$$

In particular,

$$d \leq \text{the order of the Weyl group of } G.$$

Corollary 2.4. — *With the notation of 2.1, the following conditions are equivalent:*

(i) $\psi : X \to G$ is birational.

(ii) Exactly one conjugate of H contains a given Cartan subgroup of G.

(iii) H contains a Cartan subgroup C of G, and $\mathrm{Norm}_G(H) \supset \mathrm{Norm}_G(C)$.

(iv) There is a nonempty open subset V of G such that every $g \in V(k)$ belongs to exactly one conjugate of H.

This follows immediately from 2.1 and 2.3.

Corollary 2.5. — *Assume that the conditions of 2.4 hold, and let $g \in G(k)$. The following conditions are equivalent:*

(i) $g \in U(k)$, where U is defined in 2.3(c); that is, g is contained in exactly one conjugate of H.

(ii) The set of conjugates of H containing g is finite and nonempty.

(iii) The scheme $\psi^{-1}(g)$, the "scheme of conjugates of H containing g," has an isolated point.

(iv) There is a conjugate H' of H containing g such that

$$(\mathfrak{g}/\mathfrak{h}')^{\mathrm{ad}(g)} = 0, \qquad \mathfrak{h}' = \mathrm{Lie}(H').$$

Finally, U is also the largest open subset of G on which ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

The equivalence of (i), (ii), and (iii), together with the last assertion, follows from the "Main Theorem"[16] applied to the birational morphism $\psi : X \to G$, since G is normal. Their equivalence with (iv) follows immediately from Corollary 2.2, which characterizes the points of X at which ψ is étale.**Theorem 2.6.** — *Let G be a smooth connected algebraic group over an algebraically closed field k, let C be a Cartan subgroup associated with a maximal torus T, and put*

$$N = \mathrm{Norm}_G(C) = \mathrm{Norm}_G(T) \qquad \text{(cf. XII 8.4).}$$

Let $X = G \times^N C$, where N acts on the left factor G by right translations and on the right factor C by inner automorphisms, and let $\psi : X \to G$ be the canonical morphism.

a) ψ is birational.

b) Let U be the largest open subset of G such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U$$

(cf. 2.5), and let

$$\rho = \rho_\nu(G) = \dim C$$

be the nilpotent rank of G. For every $g \in G(k)$, the multiplicity of the eigenvalue 1 in the action of $\mathrm{ad}(g)$ on $\mathfrak{g}$ is at least ρ; it equals ρ if and only if $g \in U(k)$.

[16] Editor's note in the French source: Zariski's Main Theorem.

Proof. Condition 2.1(i) holds, so 2.4(iii) $\Rightarrow$ (i) proves (a). In *BIBLE* 7, when G is affine, the points of $U(k)$ are called the *regular points* of $G(k)$. We shall use the same terminology and call U the *open subset of regular points* of G. Notice that the proof in *BIBLE* that this set is itself open is incorrect; the present Exposé establishes its openness under more general hypotheses.

To prove (b), for each $g \in G(k)$ introduce the characteristic polynomial

$$P(\mathrm{ad}(g), t) = t^n + c_1(g)t^{n-1} + \cdots + c_n(g).$$

Replacing k by an arbitrary k-algebra shows immediately that the functions $c_i(g)$ arise from uniquely determined sections

$$c_i \in \Gamma(G, \mathcal{O}_G).$$

Suppose that $g \in G(k)$ belongs to a Cartan subgroup, which we may take to be C; this includes the case of a regular element. By 2.5(iv),

$$(\mathfrak{g}/\mathfrak{c})^{\mathrm{ad}(g)} = 0$$

if and only if g is regular, where $\mathfrak{c} = \mathrm{Lie}(C)$. Since C is nilpotent, $\mathrm{ad}_{\mathfrak{c}}(g)$ has only the eigenvalue 1. It follows that the multiplicity of the eigenvalue 1 in $\mathrm{ad}_{\mathfrak{g}}(g)$ is at least ρ, and is exactly $\dim C = \rho$ if and only if g is regular.

In particular, the characteristic polynomial above is divisible by $(t-1)^\rho$. Divisibility by $(t-1)^\rho$ is expressed by linear relations, with integer coefficients, among the coefficients of the polynomial. These relations hold for $g \in U(k)$, where U is dense; since G is reduced, they therefore hold everywhere. Thus

$$(\dagger) \qquad t^n + c_1 t^{n-1} + \cdots + c_n = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho})$$

in the polynomial ring over $\Gamma(G, \mathcal{O}_G)$. In particular, for every field extension K/k and every $g \in G(K)$, the eigenvalue 1 of $\mathrm{ad}(g)$ has multiplicity at least ρ.

We have already seen that equality holds when g is regular; let us prove the converse. Suppose first that G is affine, and write

$$g = g_s g_u$$

as the product of its semisimple and unipotent parts (*BIBLE* 4, Section 4). Then

$$\mathrm{ad}(g) = \mathrm{ad}(g_s)\,\mathrm{ad}(g_u)$$

is the corresponding decomposition of $\mathrm{ad}(g)$ (loc. cit., corollary to Theorem 3). Hence $\mathrm{ad}(g)$ and $\mathrm{ad}(g_s)$ have the same eigenvalues, counted with multiplicity; in particular, the eigenvalue 1 has the same multiplicity in both.

By *BIBLE* 7, Theorem 2, Corollary 1, the element g is regular if and only if g_s is regular. We may therefore assume that $g = g_s$ is semisimple. It is then contained in a maximal torus by *BIBLE* 6, Theorem 5(c), and hence in a Cartan subgroup, a case already treated. This proves (b) when G is affine.

In general, let $Z = \mathrm{Centr}(G)_{\mathrm{red}}$. By XII 6.6(e), the Cartan subgroups of G are the inverse images of those of $G' = G/Z$. Thus g is regular in G if and only if its image g' is regular in G'. Since Z is smooth, the Lie algebra of G' is $\mathfrak{g}' = \mathfrak{g}/\mathfrak{z}$, where $\mathfrak{z} = \mathrm{Lie}(Z)$, and $\mathrm{ad}(g')$ is the endomorphism induced by $\mathrm{ad}(g)$ on $\mathfrak{g}/\mathfrak{z}$. Consequently, the multiplicity of the eigenvalue 1 in $\mathrm{ad}(g)$ is

$$d = \dim Z$$

plus its multiplicity in $\mathrm{ad}(g')$. The former therefore equals the nilpotent rank of G if and only if the latter equals the nilpotent rank of G'. We are reduced to G', which is affine by XII 6.1, and that case has already been proved. This completes the proof of 2.6.

Corollary 2.7. — *With the notation of the preceding proof,*[17] *let*

$$b = 1 + b_1 + \cdots + b_{n-\rho} \in \Gamma(G, \mathcal{O}_G).$$

Then the open subset of regular points of G is

$$U = G_b,$$

the set of points of G at which b is invertible. In particular, U is an affine open subset when G is affine.

Corollary 2.8. — *Let H be a smooth connected algebraic subgroup of G containing a Cartan subgroup of G.*

a) Let C be an algebraic subgroup of H. Then C is a Cartan subgroup of H if and only if it is a Cartan subgroup of G.

b) Let $g \in G(k)$, and let d be the integer introduced in 2.3. Then g is a regular point of G if and only if there are exactly d conjugates H_i of H containing g, and g is a regular element of every H_i. Equivalently, there are at most d conjugates of H containing g, and g is regular in one of them. When these conditions hold, if C is the unique Cartan subgroup of G containing g, then the conjugates of H containing g are precisely the conjugates of H containing C.

c) Let $g \in H(k)$. Then g is regular in G if and only if it is regular in H and

$$(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(g)} = 0.$$

Proof. a) Under either hypothesis on C, its unique maximal torus T is a maximal torus of both H and G, since H has the same reductive rank as G. The smooth connected groups $\mathrm{Centr}_H(T) \subset \mathrm{Centr}_G(T)$ have the same dimension, and therefore are equal. Hence C is equal to one of these two centralizers if and only if it is equal to the other, proving (a).

b) Suppose first that g is regular in G, and let C be the unique Cartan subgroup of G containing g. By 2.3(b), exactly d conjugates H_i of H contain C. Since

$$(\mathfrak{g}/\mathfrak{c})^{\mathrm{ad}(g)} = 0,$$

the endomorphism $\mathrm{ad}(g)$ has no eigenvalue 1 on $\mathfrak{g}/\mathfrak{c}$, and a fortiori

$$(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0.$$

By 2.3(c), exactly d conjugates of H contain g, namely the H_i. A Cartan subgroup of such an H_i containing g is a Cartan subgroup of G by (a), and hence is C. Thus g is regular in every H_i.

Conversely, suppose that at most d conjugates H_i of H contain g, and that g is regular in one of them, which we may take to be H. Since g is regular in H, it belongs to a unique Cartan subgroup C of H; by (a), C is a Cartan subgroup of G. Let C' be any Cartan subgroup of G containing g. By 2.3(b), exactly d conjugates of H contain C'. They all

[17] Editor's note in the French source: cf. (†) above.

contain g, so they are precisely the H_i; in particular, H contains C'. Thus C and C' are two Cartan subgroups of H containing the same regular element g of H, and therefore $C = C'$. Hence g is regular in G. $\square$

c) For an endomorphism u of a finite-dimensional vector space, let $\nu(u)$ denote the nullity[18] of $\mathrm{id} - u$. Then

$$\nu(\mathrm{ad}(g)_{\mathfrak{g}}) = \nu(\mathrm{ad}(g)_{\mathfrak{h}}) + \nu(\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}).$$

The two terms on the right are respectively at least the nilpotent rank of H, which equals the nilpotent rank ρ of G by (a), and at least 0. Hence $\nu(\mathrm{ad}(g)_{\mathfrak{g}}) = \rho$ if and only if

$$\nu(\mathrm{ad}(g)_{\mathfrak{h}}) = \rho, \qquad \nu(\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}) = 0.$$

This says precisely that g is regular in G if and only if it is regular in H and $\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}$ has no nonzero invariants. $\square$

Remarks 2.9. — In the statement of 2.1, condition (iii) cannot be weakened by assuming only that H contains a maximal torus and has finite index in its normalizer, even if one also requires that normalizer to be smooth, that is, that $H = N^0$, and even when G is affine and solvable. An example is the group G of matrices of the form

$$g = \begin{pmatrix} t & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

and the subgroup H formed by the matrices above for which $b = c = 0$. The Cartan subgroup of G here consists of the matrices g for which $a = c = 0$.

Remarks 2.10. — Let G be a smooth algebraic group over k, and let H be a smooth algebraic subgroup; we no longer assume that G and H are connected. Suppose that H^0 contains a Cartan subgroup of G^0. Then

$$(\mathfrak{g}/\mathfrak{h})^H \subset (\mathfrak{g}/\mathfrak{h})^{H^0} = 0,$$

so $H^0 = N^0$, where N is the normalizer of H; in particular, N is smooth.

Nevertheless, even with G connected, examples are easily constructed in which H has a connected component H_i such that

$$(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(h)} \neq 0 \qquad \text{for every } h \in H_i(k).$$

Equivalently, the morphism

$$G \times H_i \longrightarrow G, \qquad (g, h) \longmapsto \mathrm{ad}(g) \cdot h$$

is nowhere étale, or even quasi-finite. One may take, for example, H to be the normalizer of a maximal torus in $\mathrm{SL}(2)_k$.

Similarly, even if the image H'' of H in the finite group $G'' = G/G^0$ is all of G'', the union of the conjugates of H in G need not be dense. For example, take G to be the semidirect product of $\mathbb{Z}/2\mathbb{Z}$ with $G^0 = \mathrm{SL}(2)_k$, on which it acts by “symmetry,” and take H to be the semidirect product $\mathbb{Z}/2\mathbb{Z} \cdot T$, where T is a maximal torus of G^0.

On the other hand, if one does not assume in advance that H^0 contains a Cartan subgroup of G^0, but assumes that the union of the conjugates of H in G is dense, then H^0 necessarily contains a Cartan subgroup of G^0. To see this, one may plainly assume that G is connected and repeat the proof of 2.1(vi) $\Rightarrow$ (i), which does not require H to be connected.

[18] Editor's note in the French source: that is, the dimension of the nullspace of $\mathrm{id} - u$.

0.12 The case of an arbitrary base prescheme

Suppose first that the base is a field k, not necessarily algebraically closed. Since conditions 2.1 (i bis), (iv bis), (v), and (vi) are invariant under extension of the base field, passage to the algebraic closure $\overline{k}$ of k shows that they are equivalent to one another, and equivalent to the assertion that $H_{\overline{k}}$ contains a Cartan subgroup of $G_{\overline{k}}$. When this condition is satisfied, it is still true (with the notation of 2.3) that $k(X)$ is a finite separable extension of $k(G)$, of degree d independent of every extension of the base. If U is the largest open subset of G such that ψ induces a finite étale morphism

$$\psi^{-1}(U) \longrightarrow U,$$

then the formation of U commutes with extension of the base field. If ψ is birational, then U is also the largest open subset of G such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U;$$

and if $g \in U(k)$, there is one and only one subgroup H' of G, conjugate to H over $\overline{k}$, such that $H' \ni g$.

A point $g \in G(k)$ is called *regular* if it is regular as an element of

$$G(\overline{k}) = G_{\overline{k}}(\overline{k}).$$

More generally, the construction of 2.7 gives an open subset of G, whose formation commutes with every extension of the base field, called the *open subset of regular points* of G. It is also characterized by the following property: for every algebraically closed extension K of k and every $g \in G(K)$, the point g is regular in G_K if and only if $g \in U(K)$. If $g \in U(k)$[19], then g is contained in one and only one Cartan subgroup of G, as we shall prove below under more general hypotheses.

Let G be a smooth, separated group prescheme of finite type, with connected fibers over the prescheme S, and consider the functor

$$\mathcal{C} : (\mathrm{Sch})^{\circ}_{/S} \longrightarrow (\mathrm{Ens})$$

defined by

$$\mathcal{C}(S') = \Big\{ \text{ Cartan subgroups (XII 3.1) of } G_{S'} \Big\}.$$

Suppose that this functor is representable by a prescheme smooth over S. In Exposé XV we give a necessary and sufficient condition for this to hold, but we already know that the hypothesis is satisfied if G is affine over S with locally constant reductive rank (XII 3.3), or, more generally, if G locally admits a maximal torus for the fpqc topology (XII 7.1 a)); this is the case, for example, if S is the spectrum of a field.

Let X be the Cartan subgroup of the $\mathcal{C}$-group prescheme $G_{\mathcal{C}}$, the "universal Cartan subgroup" of G. Thus, as a prescheme over S, X represents the functor

$$X(S') = \left\{ \begin{array}{ll} (C, g) & \mid\ C \text{ is a Cartan subgroup of } G_{S'}, \\ & \ \ g \text{ is a section of } C \text{ over } S' \end{array} \right\}.$$

Consider the canonical projection morphism $(C, g) \mapsto g$,

$$\psi : X \longrightarrow G.$$

We then have the following theorem.

[19] Editor's note: $G(k)$ has been corrected to $U(k)$.

Theorem 0.12.1. *Under the preceding hypotheses on G, and with the preceding notation, let U be the set of $g \in G$ such that g is a regular element of its fiber G_s. Then U is open, and it is also the largest open subset U of G such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Let us first prove that U is open. By the representability of $\mathcal{C}$ as a prescheme smooth over S, and since its structure morphism is plainly surjective, it follows at once that G locally admits a Cartan subgroup for the étale topology and that the nilpotent rank of the fibers of G is locally constant. The same holds for the dimension of the fibers of G. After localizing on S, we may therefore suppose that these two integers are constant, say ρ and n. Consider the Killing polynomial

$$P_G(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t], \qquad A = \Gamma(G, \mathcal{O}_G).$$

The restriction of this polynomial to the fibers G_s of G, and in particular to the fibers at the maximal points of S, is divisible by $(t-1)^\rho$. This says that certain linear combinations of the c_i, with integer coefficients, vanish on the fibers G_s. When S is reduced, as we may assume in order to prove that U is open, it follows that these combinations themselves vanish. Hence the Killing polynomial itself is divisible by $(t-1)^\rho$, say

$$P_G(t) = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho}).$$

Let b be the sum of the coefficients $b_0 = 1, b_1, \ldots, b_{n-\rho}$ of the second factor. By 2.7, applied to the fibers of G, we have

$$U = G_b,$$

which proves that U is open.

To prove that $\psi^{-1}(U) \to U$ is an isomorphism, SGA 1, I 5.7 reduces us to checking this fiber by fiber. We are thereby reduced to the case of a base field, which may be assumed algebraically closed. There is then a Cartan subgroup C of G. If N is its normalizer, the conjugacy theorem XII 7.1 a) and b) identifies $\mathcal{C}$ with G/N, and the morphism $\psi : X \to G$ considered here is precisely the one defined in Section 2. The conclusion follows from 2.6 b). The same argument also shows that U is the largest open subset of G for which ψ induces an isomorphism $\psi^{-1}(U) \to U$.

Corollary 0.12.2. *Under the hypotheses of 3.1, let g be a regular section of G, that is, suppose that $g(s)$ is a regular point of G_s for every $s \in S$. Then there is one and only one Cartan subgroup C of G such that g is a section of C.*

Indeed, the hypothesis on g says that g is a section of U, while the conclusion says that it admits a unique lift to a section of X. This is simply another way of saying that $\psi^{-1}(U) \to U$ is an isomorphism.

Observe now that the open subset $\psi^{-1}(U)$ of the Cartan subgroup X of $G_{\mathcal{C}}$ is precisely the open subset of X consisting of the points regular in $G_{\mathcal{C}}$, by which we mean regular in their fibers. We thus obtain a natural "fibration" of the dense open subset U of regular points of G over the prescheme $\mathcal{C}$; its fibers are dense open subsets of Cartan subgroups of the fibers of $G_{\mathcal{C}}$, namely their open subsets of points regular in G. For example, we obtain the following result, which will be considerably sharpened in the next Exposé.

Corollary 0.12.3. *Let G be a smooth connected algebraic group over the field k, and let $\mathcal{T}$ be the scheme of maximal tori of G, which is isomorphic to the scheme of Cartan subgroups of G. Then the function field $k(G)$ of G is isomorphic to the function field of* *a smooth connected affine nilpotent algebraic group C over the function field $k(\mathcal{T})$ of $\mathcal{T}$, namely*

$$C = \text{“the generic Cartan subgroup of } G\text{”}.$$

If G is affine and has unipotent rank zero, that is, if the Cartan subgroups of $G_{\overline{k}}$ are tori, then $k(G)$ is a unirational extension of $k(\mathcal{T})$.

Of course, by the generic Cartan subgroup of G we mean, by an abuse of language, the Cartan subgroup of $G_{k(\mathcal{T})}$ that is the generic fiber of X over $\mathcal{T}$. It remains only to prove the last assertion of 3.3, which is contained in the following well-known result, due to Chevalley.

Lemma 0.12.4. *Let k be a field, T a torus over k, and $k(T)$ the field of rational functions on T. Then $k(T)$ is a unirational extension of k, that is, it is contained in a purely transcendental extension of k.*

Indeed, let k' be a finite separable extension of k that splits[20] T (X 1.4). Then $T \otimes_k k'$ is a rational variety: it contains a dense open subset isomorphic to a dense open subset of affine n-space $\mathbb{A}^n_{k'}$. Hence

$$T' = \prod_{\operatorname{Spec}(k')/\operatorname{Spec}(k)} T_{k'}/\operatorname{Spec}(k')$$

is a rational variety, since it has a dense open subset isomorphic to a dense open subset of

$$\prod_{\operatorname{Spec}(k')/\operatorname{Spec}(k)} \mathbb{A}^n_{k'},$$

which is isomorphic to affine mn-space over k, where $m = [k' : k]$. Consider the norm homomorphism from T' to T, which is defined whenever T is a commutative group scheme over k. The composite

$$T \longrightarrow T' \longrightarrow T$$

is the m-th power map on T, and is therefore dominant. Consequently $T' \to T$ is dominant, proving that T is unirational.

Return to the hypotheses of 3.1, but suppose in addition that G locally admits a maximal torus for the fpqc topology (XII 7.1). Let Y be the maximal torus of the Cartan subgroup X of $G_{\mathcal{C}}$. The morphism $\psi : X \to G$ then induces a morphism $Y \to G$ whose set-theoretic image consists of the semisimple elements of the fibers of G (XII 8). Finally, 3.1 shows that the restriction of ψ to the open subset Y^{reg} of regular points of Y induces a closed immersion

$$Y^{\mathrm{reg}} \longrightarrow U = G^{\mathrm{reg}}.$$

Making explicit the functorial meaning of $Z = Y^{\mathrm{reg}}$, we obtain the following.

Corollary 0.12.5. *Let G be a smooth, separated S-group prescheme of finite type, with connected fibers over the prescheme S, and suppose that G locally admits a maximal torus for the fpqc topology. Let*

$$Z : (\mathrm{Sch})^{\circ}_{/S} \longrightarrow (\mathrm{Ens})$$

be the functor defined by

$$Z(S') = \left\{ \begin{array}{c} \text{regular sections of } G_{S'} \text{ over } S' \text{ that are} \\ \text{contained in a maximal torus of } G_{S'} \end{array} \right\}.$$

Then Z is representable by a closed subprescheme, smooth over S, of the open subset $U = G^{\mathrm{reg}}$ of G introduced in 3.1.

[20]Editor's note: “splits” (?), or: “over which T is split”.

To conclude, let us record the following result, which sharpens the density theorem 2.1 (i) $\Rightarrow$ (vi).

Corollary 0.12.6. *Under the hypotheses of 3.5, let C be a Cartan subgroup of G, and consider the morphism*

$$\varphi : Z \times C \longrightarrow G$$

defined by

$$\varphi(g, h) = \mathrm{ad}(g)h = ghg^{-1}.$$

Then φ is dominant.

It plainly suffices to prove the assertion fiber by fiber, which reduces us to the case where S is the spectrum of an algebraically closed field k. Let T be the maximal torus of C, let $t_0 \in T(k)$ be regular in G, and let $c_0 \in C(k)$ be regular in G. Consider $\varphi^{-1}(\varphi(t_0, c_0))$. Its k-rational points are the pairs (t, c), with $t \in Z(k)$ and $c \in C(k)$, such that

$$\mathrm{ad}(t)c = \mathrm{ad}(t_0)c_0, \qquad \text{that is,} \qquad c = \mathrm{ad}(t^{-1}t_0)c_0.$$

They are therefore in bijection with the $t \in Z(k)$ such that $\mathrm{ad}(t^{-1}t_0)c_0 \in C$, or equivalently, since c_0 is regular, such that $t^{-1}t_0 \in N$, where N is the normalizer of C; in other words, $t \in N$. Restricting to those $t \in Z(k)$ that belong to $C(k)$ gives an open and closed part of this fiber. We have therefore found a connected component of $\varphi^{-1}(\varphi(t_0, c_0))$ isomorphic to T. The preceding set-theoretic argument indeed gives an isomorphism of schemes: one simply replaces k-valued points by points with values in an arbitrary k-prescheme.

It follows that the generic fiber of φ has dimension at most $\dim T$, and therefore

$$\dim \mathrm{Im}(\varphi) \geq \dim(Z \times C) - \dim T = \dim Z + \dim C - \dim T.$$

But

$$\dim Z = \dim Y = \dim \mathcal{C} + \dim T = \dim G - \dim C + \dim T.$$

Thus $\dim \mathrm{Im}(\varphi) \geq \dim G$, and φ is dominant.

Q.E.D.

Remarks 0.12.7. The argument shows in addition that the connected component at (t_0, c_0) of the fiber $\varphi^{-1}(\varphi(t_0, c_0))$ is isomorphic to T. In particular, it is smooth over k and has the same dimension as the generic fiber. It follows that φ is in fact smooth at (t_0, c_0), something that one should also be able to verify by calculating the tangent map. Consequently, under the hypotheses of 3.6, the induced morphism

$$Z \times_S C^{\mathrm{reg}} \longrightarrow G^{\mathrm{reg}}, \qquad C^{\mathrm{reg}} = C \cap G^{\mathrm{reg}},$$

is smooth. Likewise, the analogous morphism

$$Z \times T^{\mathrm{reg}} \longrightarrow Z,$$

where T is a maximal torus of G, is smooth. More generally, for every smooth connected normal algebraic subgroup H of C containing an element $c_0 \in G(k)$ that is regular in G, the image of $Z \times H \to G$ is dense in the image of $G \times H \to G$.

0.13 Lie algebras over a field: rank, regular elements, Cartan subalgebras

In the remainder of this Exposé we take up the theory developed by Chevalley in his book *Théorie des Groupes de Lie III* (Act. Sc. Ind. 1226, Paris 1955); the methods of schemes allow us to dispense with the hypothesis of characteristic zero. We begin by recalling in this section several well-known notions and results.

Let $\mathfrak{g}$ be a Lie algebra over a ring k. For every $a \in \mathfrak{g}$, denote by $\mathrm{ad}(a)$ the endomorphism

$$\mathrm{ad}(a) \cdot x = [a, x]$$

of $\mathfrak{g}$; it is a derivation of the Lie algebra $\mathfrak{g}$. For every derivation D of $\mathfrak{g}$, the nil-space of D, that is, the union of the kernels of the iterates of D, is a Lie subalgebra of $\mathfrak{g}$, as follows from the Leibniz formula

$$D^m([x, y]) = \sum_{0 \leq p \leq m} \binom{m}{p} [D^p x, D^{m-p} y].$$

We set

$$\mathrm{Nil}(a, \mathfrak{g}) = \text{nil-space of } \mathrm{ad}(a) = \bigcup_{m \geq 0} \mathrm{Ker}\, \mathrm{ad}(a)^m.$$

When no confusion can arise, we write simply $\mathrm{Nil}(a)$, and call it the *nil-space of a in $\mathfrak{g}$*.

Proposition 0.13.1. *For every $a \in \mathfrak{g}$, its nil-space $\mathrm{Nil}(a, \mathfrak{g})$ is a Lie subalgebra of $\mathfrak{g}$, equal to its own normalizer.*

It remains only to prove that it is its own normalizer. In other words, every element of $\mathfrak{g}/\mathrm{Nil}(a, \mathfrak{g})$ annihilated by the adjoint representation of $\mathrm{Nil}(a, \mathfrak{g})$ on this quotient must be zero. This is immediate, since every element of the quotient annihilated by $\mathrm{ad}(a)$is zero.

For the remainder of this section, suppose that k is a field and that $\mathfrak{g}$ is finite-dimensional over k. We denote by $W(\mathfrak{g})$ the scheme over k defined by $\mathfrak{g}$; its points in a k-algebra A are the elements of $\mathfrak{g} \otimes_k A$. If $a \in \mathfrak{g}$, the characteristic polynomial of $\mathrm{ad}(a)$ is also called the *characteristic polynomial*, or the *Killing polynomial*, of a in $\mathfrak{g}$:

$$P_{\mathfrak{g}}(a, t) = t^n + c_1(a) t^{n-1} + \cdots + c_n(a),$$

where $n = \mathrm{rank}_k\, \mathfrak{g}$ and $c_i(a) \in k$. Taking the same polynomial for $a \in \mathfrak{g} \otimes_k A$[21], where A is any k-algebra, we see that the $c_i(a)$ arise from uniquely determined sections c_i of the structure sheaf of $W(\mathfrak{g})$, that is, from elements of the symmetric algebra

$$A = \mathrm{Sym}_k(\mathfrak{g}^\vee),$$

where $\mathfrak{g}^\vee$ is the dual of the k-module $\mathfrak{g}$. When k is infinite, the c_i are determined by the corresponding polynomial functions $\mathfrak{g} \to k$, but this is no longer true when k is finite.

Let r be the largest integer such that the Killing polynomial

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t]$$

is divisible by t^r; thus

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_{n-r} t^r, \qquad c_{n-r} \neq 0.$$

The integer r is called the *nilpotent rank* of the Lie algebra $\mathfrak{g}$. It is invariant under extension of the base field.

[21] Editor's note: $W \otimes_k A$ has been corrected to $\mathfrak{g} \otimes_k A$.

Proposition 0.13.2. *Let r be the nilpotent rank of $\mathfrak{g}$, and let $a \in \mathfrak{g}$. Then*

$$\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g}) \geq r,$$

with equality if and only if

$$c_{n-r}(a) \neq 0.$$

In this case $\operatorname{Nil}(a, \mathfrak{g})$ *is a nilpotent Lie algebra. We shall see the converse in 5.7 b) when* $\mathfrak{g}$ *is the Lie algebra of an algebraic group G smooth over k.*

The first assertion is immediate, since by definition

$$\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g}) = \text{the multiplicity of the zero root in } P_{\mathfrak{g}}(a, t).$$

Suppose now that $c_{n-r}(a) \neq 0$. We prove that $\operatorname{Nil}(a)$ is nilpotent, which is equivalent to saying that $\operatorname{ad}_{\operatorname{Nil}(a)}(x)$ is a nilpotent endomorphism for every $x \in \operatorname{Nil}(a)$. We may suppose that k is algebraically closed. Since

$$\operatorname{ad}(a)_{\mathfrak{g}/\operatorname{Nil}(a)}$$

is injective, there is a nonempty open subset $U \subset W(\operatorname{Nil}(a))$ such that, for every $x \in U(k)$, the endomorphism $\operatorname{ad}(x)_{\mathfrak{g}/\operatorname{Nil}(a)}$ is injective and hence $\operatorname{Nil}(x) \subset \operatorname{Nil}(a)$. We may further suppose that U is contained in the open subset on which c_{n-r} does not vanish; that open subset is nonempty because $c_{n-r}(a) \neq 0$. Then $\operatorname{Nil}(x)$ and $\operatorname{Nil}(a)$ have the same dimension, so

$$\operatorname{Nil}(x) = \operatorname{Nil}(a).$$

It follows that $\operatorname{ad}(x)_{\operatorname{Nil}(a)}$ is nilpotent for every $x \in U(k)$. By the principle of extension of algebraic identities, this remains true for every $x \in \operatorname{Nil}(a)$, and hence $\operatorname{Nil}(a)$ is nilpotent.

An element $a \in \mathfrak{g}$ is called *regular* if $c_{n-r}(a) \neq 0$, that is, if

$$\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g}) = r.$$

When k is infinite, this also says that $\operatorname{rank}_k \operatorname{Nil}(a, \mathfrak{g})$ is as small as possible as a varies in $\mathfrak{g}$. In every case, the notion of a regular element of $\mathfrak{g}$ is invariant under extension of the base field. The set of regular points of $W(\mathfrak{g})$, meaning those that arise from regular points with values in a suitable extension of k, is open, since it is

$$W(\mathfrak{g})_{c_{n-r}},$$

the set of points at which c_{n-r} is invertible.

Corollary 0.13.3. *Let a be a regular element of $\mathfrak{g}$, and let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ containing a. Then $\mathfrak{h}$ is nilpotent if and only if*

$$\mathfrak{h} \subset \operatorname{Nil}(a, \mathfrak{g}).$$

In particular, $\operatorname{Nil}(a, \mathfrak{g})$ *is a maximal nilpotent subalgebra of* $\mathfrak{g}$.

Since $\operatorname{Nil}(a)$ is nilpotent, the inclusion $\mathfrak{h} \subset \operatorname{Nil}(a)$ implies that $\mathfrak{h}$ is nilpotent. Conversely, if $\mathfrak{h}$ is nilpotent, it is contained in the nil-space of its element a; that is, $\mathfrak{h} \subset \operatorname{Nil}(a)$.

Proposition 0.13.4. *Suppose that k is infinite. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. Consider the following conditions:*

(i) $\mathfrak{d}$ *is maximal nilpotent and contains a regular element of* $\mathfrak{g}$.

(i bis) $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ *for some regular element* a *of* $\mathfrak{g}$.

(ii) $\mathfrak{d}$ *is nilpotent and* $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ *for some* $a \in \mathfrak{g}$.

(ii bis) $\mathfrak{d}$ *is nilpotent, and there exists* $a \in \mathfrak{d}$ *such that* $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ *is injective.*

(iii) $\mathfrak{d}$ *is nilpotent and equal to its own normalizer.*

Then

$$\text{(i)} \iff \text{(i bis)} \Longrightarrow \text{(ii)} \iff \text{(ii bis)} \iff \text{(iii)}.$$

We shall see in 5.7 a) that if $\mathfrak{g}$ *is the Lie algebra of a smooth algebraic group, then all the preceding conditions are equivalent.*

The equivalence of (i) and (i bis) follows immediately from 4.3, and these conditions plainly imply (ii). The equivalence of (ii) and (ii bis) is also immediate, as is (ii bis) $\Rightarrow$ (iii) (cf. 4.1). It remains to prove (iii) $\Rightarrow$ (ii bis). This is the only implication that uses the infinitude of k, and it follows at once from the next lemma.

Lemma 0.13.5. *Let* $\mathfrak{d}$ *be a nilpotent Lie algebra over an infinite field* k*, acting on a finite-dimensional vector space* V*. Suppose that the endomorphism* $u(x)$ *is not injective for every* $x \in \mathfrak{d}$*. Then there is a nonzero element* $v \in V$ *annihilated by* $\mathfrak{d}$*.*

We may suppose that k is algebraically closed and that $\mathfrak{d}$ is finite-dimensional. Then V is known to be a direct sum of finitely many nonzero stable vector subspaces V_i, $1 \leq i \leq n$, such that, for every i and every $x \in \mathfrak{d}$, the restriction $u(x)|_{V_i}$ has a single eigenvalue $\lambda_i(x)$ (cf. Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Exercise 22). Let $c_i(x)$ be the constant term of the characteristic polynomial of $u(x)|_{V_i}$, so that

$$\lambda_i(x) = 0 \quad \iff \quad c_i(x) = 0.$$

Each c_i is a polynomial function on $\mathfrak{d}$, and the hypothesis says that $\mathfrak{d}$ is the union of their zero sets. Thus one of the c_i is identically zero. Replacing V by V_i, we are reduced to the case in which all $u(x)$, $x \in \mathfrak{d}$, are nilpotent. Engel's theorem (Bourbaki, loc. cit., Th. 1) then implies that some nonzero $v \in V$ is annihilated by $\mathfrak{d}$.

Q.E.D.

It is easy to see, still assuming that k is infinite, that conditions (i) and (i bis) of 4.4 are invariant under every extension of the base field. When they hold, $\mathfrak{d}$ is called a *Cartan subalgebra* of $\mathfrak{g}$. In general, with k not necessarily infinite, $\mathfrak{d}$ is called a Cartan subalgebra of $\mathfrak{g}$ if it becomes one after one, and hence every, extension $k \to k'$ with k' infinite. It follows in particular that $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Proposition 0.13.6. a) *Let* $a \in \mathfrak{g}$*. If* a *is regular, then it is contained in one and only one Cartan subalgebra of* $\mathfrak{g}$*. We shall see the converse in 6.1 d), when* k *is algebraically closed and* $\mathfrak{g}$ *is the Lie algebra of a smooth algebraic group.*

b) *Let* $\mathfrak{d}$ *be a Cartan subalgebra of* $\mathfrak{g}$*, and let* $a \in \mathfrak{d}$*. Then* a *is regular in* $\mathfrak{g}$ *if and only if* $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ *is injective.*

For a), if a is regular, then $\mathrm{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$, since this is true after an infinite extension k' of k. It follows at once from 4.3 that every Cartan subalgebra of $\mathfrak{g}$ containing a is equal to this one. For b), observe that the nullity[22] of $\mathrm{ad}(a)_{\mathfrak{g}}$ is the sum of the nullities of $\mathrm{ad}(a)_{\mathfrak{d}}$ and $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$. Since the first is r, their sum is r if and only if $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.

Q.E.D.

[22] Editor's note: that is, the dimension of the nil-space.

Corollary 0.13.7. *Let a be a regular element of $\mathfrak{g}$, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$ containing a, let A be a k-algebra, and let $\mathfrak{g}_A$ and $\mathfrak{d}_A \subset \mathfrak{g}_A$ be the A-Lie algebras obtained from $\mathfrak{g}$ and $\mathfrak{d}$ by base change; write a_A for the image of a in $\mathfrak{g}_A$. Let u be an automorphism of $\mathfrak{g}_A$. Then*

$$u(\mathfrak{d}_A) = \mathfrak{d}_A$$

if and only if

$$u(a_A) \in \mathfrak{d}_A.$$

The condition is plainly necessary; let us prove that it is sufficient. Suppose it holds. Then

$$\mathfrak{d}' = u^{-1}(\mathfrak{d}_A)$$

is a Lie subalgebra containing a_A, and every $b \in \mathfrak{d}'$ is such that $\operatorname{ad}(b)_{\mathfrak{d}'}$ is nilpotent.Indeed, $\mathfrak{d}'$ is isomorphic to $\mathfrak{d}_A$, which has this property, as follows immediately from the definition of "nilpotent" in Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Def. 1. Taking $b = a_A$, we see that $\operatorname{Nil}(b, \mathfrak{g}_A)$ contains $\mathfrak{d}'$. On the other hand,

$$\operatorname{Nil}(b, \mathfrak{g}_A) = \mathfrak{d}_A.$$

Moreover, $\mathfrak{d}'$ is locally a direct summand of $\mathfrak{g}_A$, since $\mathfrak{d}$ is, and hence is locally a direct summand of $\mathfrak{d}_A$. It is a projective module of the same rank r as $\mathfrak{d}_A$, and therefore equals $\mathfrak{d}_A$.

Q.E.D.

Proposition 0.13.8. *Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$.*

a) *If k is infinite, the following conditions are equivalent:*

(i) $\mathfrak{h}$ *contains a Cartan subalgebra* $\mathfrak{d}$ *of* $\mathfrak{g}$*.*

(ii) $\mathfrak{h}$ *contains a regular element a of* $\mathfrak{g}$*, and an element b such that* $\operatorname{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ *is injective.*

(iii) $\mathfrak{h}$ *has the same nilpotent rank as* $\mathfrak{g}$*, and contains a regular element of* $\mathfrak{g}$*.*

These conditions are invariant under extension of the base field k.

b) *Suppose that these conditions hold after a suitable infinite extension k' of k. For $a \in \mathfrak{h}$, the element a is regular in $\mathfrak{g}$ if and only if it is regular in $\mathfrak{h}$ and* $\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ *is injective; equivalently, if and only if*

$$\operatorname{Nil}(a, \mathfrak{h}) = \operatorname{Nil}(a, \mathfrak{g})$$

and this common Lie algebra is a Cartan subalgebra of $\mathfrak{h}$*.*

c) *Under the hypotheses of b), let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$. In order that $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{h}$, it is sufficient that it be a Cartan subalgebra of $\mathfrak{g}$. We shall see in 5.8 that this condition is also necessary when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G, and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G.*

Conditions (ii) and (iii) of a) are visibly invariant under extension of the base field k, assumed infinite. In b) and c), we may suppose that k is infinite, and shall do so. If $\mathfrak{h}$ contains the Cartan subalgebra

$$\mathfrak{d} = \operatorname{Nil}(a, \mathfrak{g}),$$

then a is a regular element of $\mathfrak{g}$ such that $\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective. Thus (i) $\Rightarrow$ (ii). Conversely, if (ii) holds, then a "sufficiently general" element a of $\mathfrak{h}$ simultaneously satisfies both conditions in (ii). Hence $\operatorname{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ contained in $\mathfrak{h}$, proving (i). Conditions (i) and (ii) are therefore equivalent.

Suppose they hold, and let a vary in $\mathfrak{h}$. Then

$$\text{(x)} \qquad \operatorname{rank}_k \operatorname{Nil}(a,\mathfrak{g}) = \operatorname{rank}_k \operatorname{Nil}(a,\mathfrak{h}) + \operatorname{rank}_k \operatorname{Nil}\big(\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}\big).$$

The two terms on the right are respectively at least r', the nilpotent rank of $\mathfrak{h}$, and at least 0; moreover, both equalities are attained[23] for a sufficiently general element of $\mathfrak{h}$. We also have

$$\operatorname{rank}_k \operatorname{Nil}(a,\mathfrak{g}) \geq r,$$

where r is the nilpotent rank of $\mathfrak{g}$. Equality is attained for a sufficiently general element of $\mathfrak{h}$, and holds if and only if a is regular in $\mathfrak{g}$.

It follows that $r = r'$, and that a is regular if and only if the two terms on the right-hand side of (x) are respectively r' and 0; that is, if and only if a is regular in $\mathfrak{h}$ and $\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective. This proves b), and c) follows immediately by taking an element $a \in \mathfrak{d}$ regular in $\mathfrak{g}$, so that $\operatorname{Nil}(a,\mathfrak{g}) = \mathfrak{d}$.

The preceding result also proves (i) $\Rightarrow$ (iii). Finally, (iii) $\Rightarrow$ (i): under (iii), a sufficiently general $a \in \mathfrak{h}$ is regular both in $\mathfrak{h}$ and in $\mathfrak{g}$. Thus

$$\operatorname{Nil}(a,\mathfrak{h}) \subset \operatorname{Nil}(a,\mathfrak{g})$$

are respectively Cartan subalgebras of $\mathfrak{h}$ and $\mathfrak{g}$. Since they have the same rank over k, they are equal, proving (i). This completes the proof of 4.8.

0.14 The case of the Lie algebra of a smooth algebraic group: the density theorem

Let G be a smooth algebraic group over the field k, and let $\mathfrak{g}$ be its Lie algebra. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Let G act on $W(\mathfrak{g})$ through the adjoint representation, and consider the subscheme $W(\mathfrak{h})$. The construction of Section 1 leads us to introduce

$$N = \operatorname{Norm}_G(\mathfrak{h}) = \operatorname{Norm}_G(W(\mathfrak{h})),$$

which is an algebraic subgroup of G (not necessarily smooth), and

$$\mathfrak{n} = \operatorname{Lie}(N),$$

together with the scheme

$$X = G \times^N W(\mathfrak{h}),$$

the quotient of $\widetilde{X} = G \times W(\mathfrak{h})$ by the right action of N given by

$$(g,x)\cdot n = (gn, \operatorname{Ad}(n^{-1})x).$$

We consider the canonical morphisms

$$\begin{tikzcd} G \times W(\mathfrak{h}) \arrow[d, "q"'] \arrow[dr, "\varphi"] & \\ X \arrow[r, "\psi"] & W(\mathfrak{g}). \end{tikzcd}$$

[23] Editor's note: "the equality" has been replaced by "the equalities".

Theorem 0.14.1. *With the preceding notation, suppose that k is infinite. Consider the following conditions:*

(i) $\mathfrak{h}$ *contains a Cartan subalgebra* $\mathfrak{d}$ *of* $\mathfrak{g}$.

(ii) There exists $a \in \mathfrak{h}$ *such that* $\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ *is injective.*

(iii) The morphism $\varphi : G \times W(\mathfrak{h}) \to W(\mathfrak{g})$ *is generically smooth.*

(iv) The preceding morphism φ *(or, equivalently,* $\psi : X \to W(\mathfrak{g})$*) is dominant, and* $\mathfrak{h}$ *has the same nilpotent rank as* $\mathfrak{g}$.

(v) The morphism $\psi : X \to W(\mathfrak{g})$ *is generically smooth and* $\mathfrak{h} = \mathfrak{n}$.

(vi) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant, and* $\mathfrak{h} = \mathfrak{n}$.

(vii) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant.*

(viii) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant, and* N *is smooth.*

(ix) The morphism $\psi : X \to W(\mathfrak{g})$ *is dominant,* $\mathfrak{h} = \mathfrak{n}$*, and* $\mathfrak{h}$ *is the Lie algebra of a smooth algebraic subgroup* H *of* G.

(x) The morphism $\psi : X \to W(\mathfrak{g})$ *is generically quasi-finite, and* $\mathfrak{n} = \mathfrak{h}$.

(xi) The morphism $\psi : X \to W(\mathfrak{g})$ *is generically étale, and* $\mathfrak{n} = \mathfrak{h}$.

(xii) There exists a smooth algebraic subgroup H *of* G*, with Lie algebra* $\mathfrak{h}$*, such that* $\mathfrak{h}$ *contains a Cartan subalgebra* $\mathfrak{d}$ *of* $\mathfrak{g}$.

(xiii) There exists $a \in \mathfrak{h}$ *such that* N *and the transporter* M_a *of* a *into* $\mathfrak{h}$ *(cf. Section 1) coincide in a neighborhood of* e*, and* $\mathfrak{n} = \mathfrak{h}$.

One then has the following diagram of implications:

$$\begin{tikzcd}
\text{(xi)} \arrow[rr, Leftrightarrow] & & \text{(xii)} \arrow[r, Leftrightarrow] \arrow[d, Rightarrow] & \text{(xiii)} \arrow[r, Rightarrow] & \text{(viii)} \arrow[r, Leftrightarrow] & \text{(ix)} \arrow[r, Leftrightarrow] \arrow[d, Rightarrow] & \text{(x)} \\
\text{(i)} \arrow[r, Leftrightarrow] & \text{(ii)} \arrow[r, Leftrightarrow] & \text{(iii)} \arrow[r, Leftrightarrow] & \text{(iv)} \arrow[r, Rightarrow] & \text{(v)} \arrow[r, Rightarrow] & \text{(vi)} \arrow[r, Rightarrow] & \text{(vii)}.
\end{tikzcd}$$

When k has characteristic zero, all the conditions under consideration are equivalent. Finally,

$$\text{(xi)} \iff [\text{(i)} \textit{ and } \text{(viii)}] \iff [\text{(v)} \textit{ and } \text{(viii)}].$$

Let us first record the trivial implications

$$\text{(v)} \Rightarrow \text{(vi)} \Rightarrow \text{(vii)}, \qquad \text{(ix)} \Rightarrow \text{(vi)}, \qquad \text{(xi)} \iff [\text{(x) and (v)}].$$

We prove the equivalence of conditions (i) through (iv), and that they imply (v). The implication (i)⇒(ii) is immediate. On the other hand, when k is algebraically closed, condition (iii) says that there is a k-rational point of $G \times W(\mathfrak{h})$ at which the tangent map of φ is surjective. It is clear that, after translating by a suitable element of $G(k)$, this point may be taken to have the form (e, a), with $a \in \mathfrak{h}$. It follows that when k is infinite, though not necessarily algebraically closed, this condition—which is plainly sufficient for generic smoothness—is also necessary.

The tangent map is readily computed. Upon identifying the tangent space of $W(\mathfrak{h})$ at a with $\mathfrak{h}$, it is the map

$$(\xi, x) \longmapsto [\xi, a] + x$$

from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Its surjectivity is equivalent to the surjectivity of $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$, or, what amounts to the same thing, to its injectivity. This proves the equivalence of (ii) and (iii). Moreover, (ii) plainly implies $\mathfrak{h} = \mathfrak{n}$, and (iii) implies that ψ is generically smooth: if φ is smooth at a point u, then, since q is flat, ψ is smooth at $q(u)$. Thus (ii) and (iii) imply (v).

We now show that they imply (i). Since ψ is dominant and the set of regular points of $\mathfrak{g}$ is dense and open, $\mathfrak{h}$ contains regular elements of $\mathfrak{g}$. Hence a sufficiently general element b of $\mathfrak{h}$ is regular in $\mathfrak{g}$ and makes $\mathrm{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ injective. Consequently $\mathrm{Nil}(b, \mathfrak{g}) \subset \mathfrak{h}$, so $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \mathrm{Nil}(b, \mathfrak{g})$. Conditions (i), (ii), and (iii) are therefore equivalent. Finally, (i)$\Longleftrightarrow$(iv): we have already observed that if $\mathfrak{h}$ contains a Cartan subalgebra, then it has the same nilpotent rank as $\mathfrak{g}$ (4.6), so (i) implies (iv). Conversely, if (iv) holds, then $\mathfrak{h}$ contains a regular element of $\mathfrak{g}$, and since it has the same rank as $\mathfrak{g}$, it contains a Cartan subalgebra by 4.6.

We next prove the equivalence of conditions (viii) through (x). We first note the following facts.

Lemma 0.14.2. *a) If N is smooth, then*

$$\dim X \leq \dim G,$$

with equality if and only if $\mathfrak{h} = \mathfrak{n}$.

b) If $\mathfrak{h} = \mathfrak{n}$, then

$$\dim X \geq \dim G,$$

with equality if and only if N is smooth.

These assertions follow at once from the formula

$$\dim X = \dim G - \dim N + \dim_k \mathfrak{h}$$

and from the fact that $\dim N = \dim_k \mathfrak{n}$ is equivalent to the smoothness of N.

Now (viii) implies (x): by dominance, (viii) gives $\dim X \geq \dim G$, so Lemma 5.2(a) gives equality and $\mathfrak{h} = \mathfrak{n}$, hence (x). The same argument in reverse, using Lemma 5.2(b), shows that (x) implies (viii). Moreover, (viii) implies (ix), since it implies $\mathfrak{h} = \mathfrak{n}$, so $\mathfrak{h}$ is the Lie algebra of the smooth algebraic subgroup N of G. Conversely, (ix) implies (viii): since H normalizes its Lie algebra $\mathfrak{h}$, it is contained in N; and since H is smooth and has the same Lie algebra as N, it follows that N is smooth.

Finally, we prove the equivalence of (xi), (xii), and (xiii), and show that they imply (iii); this will complete the implication diagram. We have (xi)$\Longleftrightarrow$(xiii). Indeed, if $\mathfrak{n} = \mathfrak{h}$, then Lemma 5.2(b) gives $\dim X \geq \dim G$. Since $W(\mathfrak{g})$ is normal, condition (xi) is therefore equivalent to ψ being generically unramified, and this in turn is equivalent to (xiii) by 1.1, (ii)$\Longleftrightarrow$(iii), arguing as above in the proof of (ii)$\Longleftrightarrow$(iii).

Since (xi)$\Rightarrow$(x)$\Rightarrow$(viii), what we have already proved shows that (xi) implies that N is smooth. Thus $q : G \times W(\mathfrak{h}) \to X$ is smooth, and the composite $\varphi = \psi \circ q$ is generically smooth; this is (iii). Since (iii) implies (i), it also follows that (xi)$\Rightarrow$(xii).[24] Finally, (xii) implies (xi). Indeed, (xii) plainly implies (i), and we have seen that (i)$\Rightarrow$(iii)$\Rightarrow$(v); hence

[24]Editor's note: because $H = N$ will do.

(xii) implies (v). It also implies (ix), and (ix) implies (x). Thus (xii) implies both (v) and (x), and therefore (xi), since

$$\text{generically étale} = \text{generically smooth} + \text{generically quasi-finite}.$$

If k has characteristic zero, then (vii) implies (viii), because Cartier's theorem makes N automatically smooth ($\mathrm{VI_B}$ 1.6.1). Moreover, [(viii) and (x)] implies (xi), because in characteristic zero, for a morphism of integral preschemes,

$$\text{generically étale} = \text{dominant and generically quasi-finite}.$$

Thus in this case all conditions (i) through (xiii) are equivalent.

Corollary 0.14.3. *Under the equivalent conditions (viii) through (x), there exists a unique smooth connected algebraic subgroup H of G whose Lie algebra is $\mathfrak{h}$, and*

$$\mathrm{Norm}_G(H) = \mathrm{Norm}_G(\mathfrak{h}) = N, \qquad H = N^0.$$

Indeed, $H = N^0$ has the required properties. Conversely, if H has these properties, then $H \subset N$, since H normalizes its Lie algebra $\mathfrak{h}$. This is an inclusion of smooth groups with the same Lie algebra, and H is connected, so $H = N^0$. To prove $\mathrm{Norm}_G(H) = \mathrm{Norm}_G(\mathfrak{h})$, we may assume that k is algebraically closed. What we have just proved then shows immediately that the two groups have the same k-points. On the other hand, the inclusions

$$H \subset \mathrm{Norm}_G(H) \subset N$$

show that $\mathrm{Norm}_G(H)$ and N have the same Lie algebra; hence they are equal.

Corollary 0.14.4. *Under the equivalent conditions (i) through (iv), let $a \in \mathfrak{h}$. Then the following conditions are equivalent, and they hold whenever a is regular in $\mathfrak{g}$:*

(i) φ *is smooth at* (e, a).

(ii) $M_a = \mathrm{Transp}_G(a, \mathfrak{h})$ *is smooth at* e, *and* $\dim_e(M_a) = \mathrm{rank}_k\,\mathfrak{h}$.

(iii) $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ *is injective (equivalently, bijective).*

Under the equivalent conditions (xi) through (xiii), let H be the algebraic subgroup of G considered in Corollary 5.3. The preceding conditions are also equivalent to:

(iv) ψ *is étale at* $(\overline{e}, a)$.

(v) If M_a^0 denotes the connected component of e in the transporter M_a of a into $\mathfrak{h}$, endowed with the structure induced by M_a, then

$$H = M_a^0.$$

Plainly (i) implies (ii), since M_a is isomorphic to the fiber $\varphi^{-1}(a)$, with e corresponding to (e, a). Conversely, (ii) implies (i): condition (ii) says that φ is "equidimensional" at (e, a), in the sense that the dimension of the fiber through this point is the dimension of the generic fiber. Since $G \times W(\mathfrak{h})$ and $W(\mathfrak{g})$ are regular, this implies that φ is flat at (e, a), hence smooth there because its fiber is smooth at that point. The equivalence of (i) and (iii) was obtained in the proof of Theorem 5.1 by the elementary calculation of the tangent map. We also saw in 4.8(b) that regularity of a in $\mathfrak{g}$ implies (iii).

Under conditions (xi) through (xiii), the morphism $q : G \times W(\mathfrak{h}) \to X$ is smooth, since N is smooth. Thus (i) is equivalent to the smoothness of ψ at $(\bar{e}, a)$; and since ψ is generically étale, this is equivalent to (iv). Finally, as was noted at the end of the proof of 1.1, condition (iv) implies that N is the prescheme induced on M_a by an open and closed subset of M_a, which gives (v). Conversely, (v) trivially implies (ii). Alternatively, (v) implies (iv) by 1.1, since ψ is dominant and $W(\mathfrak{g})$ is normal, and "unramified" is equivalent here to "étale". This completes the proof of Corollary 5.4.

Corollary 0.14.5. *Let G be a smooth algebraic group over a field k, and let H be a smooth algebraic subgroup such that its Lie algebra $\mathfrak{h}$ contains, at least after a suitable extension of k, a Cartan subalgebra of the Lie algebra $\mathfrak{g}$ of G. Let K be a connected algebraic subgroup of G, not necessarily smooth, with Lie algebra $\mathfrak{k}$, and suppose that $\mathfrak{k}$ contains a regular element a of $\mathfrak{g}$, again at least after a suitable extension of k. Then H contains K if and only if $\mathfrak{h}$ contains $\mathfrak{k}$.*

Indeed, Corollary 5.4, (iii)⇒(v), applied to the connected subgroup supplied by Corollary 5.3, gives $H^0 = M_a^0$.This identity is invariant under extension of the base field, so it remains valid without assuming that the base field is infinite. Moreover, $\mathfrak{k} \subset \mathfrak{h}$ plainly implies $K \subset M_a$. Since K is connected, $K \subset M_a^0 \subset H$. □

Corollary 0.14.6. *Let G and H be as in Corollary 5.5, and let K be an algebraic subgroup of G. Suppose that H is connected and K is smooth. Then K contains H if and only if $\mathfrak{k}$ contains $\mathfrak{h}$.*

Indeed, if $\mathfrak{k} \supset \mathfrak{h}$, then K satisfies, with respect to H, the hypothesis imposed on H in Corollary 5.5. Meanwhile H plainly satisfies, with respect to K, the condition imposed on K there. The conclusion follows from Corollary 5.5.

Corollary 0.14.7. *Let $\mathfrak{g}$ be the Lie algebra of a smooth algebraic group G over a field k. Then:*

- *a) A Lie subalgebra $\mathfrak{d}$ of $\mathfrak{g}$ is a Cartan subalgebra if and only if it is nilpotent and equal to its own normalizer.*
- *b) An element $a \in \mathfrak{g}$ is regular if and only if* $\mathrm{Nil}(a, \mathfrak{g})$ *is nilpotent.*

After extending the base field, we may assume that k is infinite. In view of 4.4, to prove (a) it remains only to show that if $\mathfrak{d}$ is nilpotent and contains an element a for which $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective, then $\mathfrak{d}$ is a Cartan subalgebra. By Theorem 5.1, (ii)⇒(i), $\mathfrak{d}$ contains a Cartan subalgebra

$$\mathfrak{d}' = \mathrm{Nil}(a, \mathfrak{g}).$$

Since $\mathfrak{d}$ is nilpotent, 4.3 now gives $\mathfrak{d} = \mathfrak{d}'$. For (b), one notes that $\mathrm{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ by (a), and therefore a is regular.

Corollary 0.14.8. *Let G be a smooth algebraic group over a field k, and let H be a smooth algebraic subgroup, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Suppose that after a suitable extension of the base field, $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$. Then $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$ if and only if it is a Cartan subalgebra of $\mathfrak{g}$.*

In view of 4.8(c), it remains to show that if $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$, then it is one of $\mathfrak{g}$. It is enough to show that $\mathfrak{d}$ contains an element a regular in $\mathfrak{g}$, and we may assume that k is algebraically closed. But there is a dense open subset of $\mathfrak{h}$ consisting of points regular in $\mathfrak{g}$, so the assertion follows from Theorem 5.1, (i)⇒(vii), applied to $(\mathfrak{h}, \mathfrak{d})$.

0.15 Cartan subalgebras and subgroups of type (C) associated with a smooth algebraic group

For simplicity, in the following theorem we restrict ourselves to an algebraically closed base field; the case of an arbitrary base prescheme will be treated in the next Exposé.

Theorem 0.15.1. *Let G be a smooth algebraic group over an algebraically closed field k, and let $\mathfrak{g}$ be its Lie algebra. Then:*

a) The Cartan subalgebras of $\mathfrak{g}$ are conjugate.

b) Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Its normalizer N in G is smooth, and $D = N^0$ is the unique smooth connected subgroup of G whose Lie algebra is $\mathfrak{d}$. One has

$$\mathrm{Norm}_G(\mathfrak{d}) = \mathrm{Norm}_G(D) = N, \qquad \textit{and hence} \qquad D = \mathrm{Norm}_G(D)^0.$$

c) With $\mathfrak{d}$ as in (b), set, as in Section 5,

$$X = G \times^N W(\mathfrak{d}),$$

and consider the canonical morphism

$$\psi : X \longrightarrow W(\mathfrak{g}).$$

Its fiber over $a \in \mathfrak{g}$ has as its k-valued points the set of Cartan subalgebras of $\mathfrak{g}$ containing a. Then ψ is birational.

d) With the notation of (c), let U be the largest open subset of $W(\mathfrak{g})$ for which ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Then, for $a \in \mathfrak{g}$, the following conditions are equivalent:

(i) $a \in U(k)$.

(ii) The element a belongs to one and only one Cartan subalgebra of $\mathfrak{g}$.

(iii) The set of Cartan subalgebras of $\mathfrak{g}$ containing a is finite and nonempty.

(iv) The fiber $\psi^{-1}(a)$ has an isolated point.

(v) If $a \in \mathfrak{d}$, the morphism ψ is étale—or merely quasi-finite—at the point $(\bar{e}, a)$.

(vi) The element a is regular in $\mathfrak{g}$.

(vii) With the notation of Corollary 5.4(iii), one has $N = M_a$.

Proof. Apply Theorem 5.1 with the Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$ in place of $\mathfrak{h}$. We show that the strongest conditions, (xi) through (xiii), hold. This is clear either in the form (xiii), by 4.7—which says that if a regular element $a \in \mathfrak{g}$ lies in $\mathfrak{d}$, then $N = M_a$—or in the form

$$\text{(xi)} = \text{(x)} + \text{(i)}.$$

Indeed, (i) is automatic, while (x) follows from the fact that a regular point of $\mathfrak{g}$ belongs to a unique Cartan subalgebra of $\mathfrak{g}$ (4.6(a)), and therefore, a fortiori, to a unique conjugate of $\mathfrak{d}$.

Part (b) now follows from Corollary 5.3. Part (c) follows because ψ is generically étale and because a sufficiently general point of $\mathfrak{g}$—more precisely, a regular point—belongs to a unique Cartan subalgebra of $\mathfrak{g}$. Under these conditions, the equivalence of conditions (i) through (v) on a follows immediately from Zariski's Main Theorem, applied to the separated birational morphism ψ, since $W(\mathfrak{g})$ is normal and X is integral. The equivalence of (v) and (vi) is the special case of Corollary 5.4, (iii)$\Longleftrightarrow$(iv), obtained after carrying a into $\mathfrak{d}$ by a suitable element $g \in G(k)$, in view of 4.6(b). Moreover, Corollary 5.4 shows that (v) and (vi) are also equivalent to

$$N \supset M_a^0.$$

By 4.7 this even implies $N = M_a$.This proves (d).

Of course, (b), (c), and the equivalence of (i), (iv), (v), (vi), and (vii) remain valid over an arbitrary base field. We now prove (a), using the assumption that k is algebraically closed. By Theorem 5.1, (i)$\Rightarrow$(vii), the morphism $\psi : X \to W(\mathfrak{g})$ is dominant. Hence there is a dense open subset V of $W(\mathfrak{g})$ such that every $a \in V(k)$ is the image of a point of $X(k)$, or equivalently, belongs to a conjugate of $\mathfrak{d}$. Apply the same result to a second Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}$. We may choose V so that every element of $V(k)$ is conjugate both to an element of $\mathfrak{d}$ and to an element of $\mathfrak{d}'$. Choose a regular element of $V(k)$. Some conjugate $\mathfrak{d}''$ of $\mathfrak{d}'$ then contains an element of $\mathfrak{d}$ regular in $\mathfrak{g}$, and 4.6(a) gives $\mathfrak{d}'' = \mathfrak{d}$. This proves (a) and completes the proof. ☐

Definition 0.15.2. Let G be a smooth algebraic group over a field k. A *subgroup of type* (C) of G is a smooth connected algebraic subgroup whose Lie algebra is a Cartan subalgebra of $\mathfrak{g}$. The *infinitesimal rank* of G is the nilpotent rank of its Lie algebra $\mathfrak{g}$; it is equal to the dimension of every subgroup of type (C) of G.

Theorem 6.1(b) immediately gives:

Corollary 0.15.3. *Under the hypotheses of Definition 6.2, the map*

$$D \longmapsto \mathfrak{d} = \mathrm{Lie}(D)$$

establishes a bijection between the subgroups of type (C) *of* G *and the Cartan subalgebras* $\mathfrak{d}$ *of* $\mathfrak{g}$*. If* D *is a subgroup of type* (C) *of* G*, then* D *is its own connected normalizer:*

$$D = \mathrm{Norm}_G(D)^0.$$

Combining Theorem 6.1(a) and Corollary 6.3 gives:

Corollary 0.15.4. *If* k *is algebraically closed, the subgroups of type* (C) *of* G *are conjugate to one another.*

Corollary 0.15.5. *Let* G *be a smooth algebraic group over an algebraically closed field* k*, and let* H *be a smooth algebraic subgroup of* G*, with Lie algebras* $\mathfrak{g}$ *and* $\mathfrak{h}$*, respectively. Then* $\mathfrak{h}$ *contains a Cartan subalgebra of* $\mathfrak{g}$ *if and only if* H *contains a subgroup* D *of type* (C) *of* G*.*

The condition is plainly sufficient. It is also necessary, since by Corollary 5.5,

$$H \supset D \quad \Longleftrightarrow \quad \mathfrak{h} \supset \mathfrak{d}.$$

Remarks 0.15.6.

a) The connected subgroups H of G described in Corollary 6.5 correspond bijectively to the Lie subalgebras $\mathfrak{h}$ of $\mathfrak{g}$ satisfying the strongest condition, (xii), of Theorem 5.1. By Corollary 5.5, inclusion relations among such subgroups can already be read from their Lie algebras.

b) Continue to suppose that k is algebraically closed, and let D be a subgroup of type (C) of G. It is then easy to show that D contains a Cartan subgroup C of G. Indeed, put $V = \mathfrak{g}/\mathfrak{d}$. For a general element $a \in \mathfrak{d}$, the endomorphism $\mathrm{ad}(a)$ of V is injective; it suffices to take a regular in $\mathfrak{g}$. From this one readily deduces that, for a general $g \in D(k)$, the action of $\mathrm{Ad}(g)$ on V has no nonzero invariant vector. Thus 2.1, (vii)$\Rightarrow$(i), applies. In fact, in the next Exposé we shall prove the more precise result that every Cartan subgroup C of G is contained in one and only one subgroup of type (C) of G.

c) In general one must not confuse Cartan subgroups with subgroups of type (C). The subgroups of type (C) of G are Cartan subgroups if and only if they are nilpotent, since Cartan subgroups are maximal nilpotent subgroups. It may happen that $\mathfrak{g}$ is nilpotent although G is not; an example is $\mathrm{SL}(2)_k$ when k has characteristic 2. Then the Cartan subalgebras of $\mathfrak{g}$ are all equal to $\mathfrak{g}$. Thus, if G is connected, G is a subgroup of type (C) of itself, but it is not a Cartan subgroup of itself.

   On the other hand, we shall see in Exposé XIV that if G is an adjoint semisimple group, its subgroups of type (C) are precisely its Cartan subgroups. The same conclusion holds in characteristic zero for every smooth algebraic group G over k. This follows at once from the fact that, in characteristic zero, a connected algebraic group is nilpotent if and only if its Lie algebra is nilpotent. In turn, this follows from the fact that in characteristic zero the center Z of a connected algebraic group G has Lie algebra equal to the center of $\mathfrak{g}$. Indeed, one obtains a priori the space of invariants in $\mathfrak{g}$ under the adjoint representation of G. Since G is connected and k has characteristic zero, these are also the infinitesimal invariants of the adjoint representation of $\mathfrak{g}$, namely the center of $\mathfrak{g}$. Finally, Cartier's theorem (cf. $\mathrm{VI_B}$ 1.6.1) shows that Z is smooth.

Exposé XIV
Regular Elements: Continuation Applications to Algebraic Groups

1 Construction of Cartan subgroups and maximal tori for a smooth algebraic group

Theorem 1.1. *Let G be a smooth algebraic group over a field k. Then G admits a maximal torus T, and hence a Cartan subgroup $C = \mathrm{Centr}_G(T)$.*

By XII 3.2, finding a maximal torus T of G is equivalent to finding a Cartan subgroup C of G. Moreover, since the maximal tori of G are those of G^0, we may suppose that G is connected. We distinguish two cases.

1°) *The field k is finite.* Let $\mathcal{T}$ be the scheme of maximal tori of G (XII 1.10); it is smooth over k. The group G acts on $\mathcal{T}$ through inner automorphisms, and the conjugacy theorem (XII 6.6(a)) says that any two $\bar{k}$-rational points of $\mathcal{T}_{\bar{k}}$ are conjugate under $G_{\bar{k}}(\bar{k})$. Since $\mathcal{T}$ is smooth over k, $\mathcal{T}_{\bar{k}}$ is smooth over $\bar{k}$; it follows that $\mathcal{T}_{\bar{k}}$ is isomorphic to

$$G_{\bar{k}}/N,$$

where N is the stabilizer of an element of $\mathcal{T}_{\bar{k}}(\bar{k})$, that is, the normalizer of a maximal torus $\bar{T}$ of $G_{\bar{k}}$. Consequently $\mathcal{T}$ is a homogeneous space under the action of G. A well-known theorem of Lang (*Amer. J. Math.* **78** (1956), 555–563) says that every homogeneous space under a smooth connected algebraic group over a finite field k has a rational point. In particular, $\mathcal{T}$ has a rational point; equivalently, G has a maximal torus T.

2°) *The field k is infinite.* We shall use the following lemma.

Lemma 1.2. *Let G be a smooth algebraic group over a field k. Then G admits a subgroup of type* (C) (XIII 6.2).

By XIII 6.3, this is equivalent to saying that $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{d}$. When k is infinite this is immediate: $\mathfrak{g}$ contains a regular element a, and one takes

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}).$$

The case of finite k is treated exactly as in the proof of 1°) above, but it requires first constructing the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ and proving that this scheme is smooth over k; this will be done below in 2.16.

To prove 1.1 in case 2°), which is the case now under consideration, it is therefore enough to know 1.2 when k is infinite. We also record the following result for later use.

[0] Version xy of 1 December 2008.

Lemma 1.3. *Let G be a smooth connected affine algebraic group whose reductive center* (XII 4.1 and 4.4) *is reduced to the identity group. Then G is nilpotent if and only if its Lie algebra $\mathfrak{g}$ is nilpotent.*

This is contained in XII 4.9.

We can now give a procedure for constructing Cartan subgroups of G. The same procedure applies when k is finite, provided Lemma 1.2 is granted in that case. Suppose first that G is affine. We argue by induction on $n = \dim G$, the assertion being trivial when $n = 0$. Suppose therefore that $n > 0$, and that the assertion has been proved in dimensions $n' < n$. Let Z be the reductive center of G, and let

$$u \colon G \longrightarrow G' = G/Z$$

be the canonical homomorphism. By XII 4.7(c), the rule

$$C' \longmapsto C = u^{-1}(C')$$

is a bijection from the Cartan subgroups of G' to those of G. Replacing G by G', we may thus suppose that the reductive center Z of G is reduced to the identity element (the same is true for the reductive center of G', by XII 4.7(b)).

By Lemma 1.2,G admits a subgroup D of type (C). Over the algebraic closure $\bar{k}$ of k, the group $D_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$ (XIII 6.6(b)); hence every Cartan subgroup of D is a Cartan subgroup of G (XIII 2.8(a)). We are therefore reduced to finding a Cartan subgroup of D.

If $\dim D = \dim G$, that is, if $D = G$, then the Lie algebra of G is a Cartan subalgebra of itself and is therefore nilpotent. Lemma 1.3 then shows that G is nilpotent, so G is a Cartan subgroup of itself. If $\dim D < \dim G$, the induction hypothesis gives a Cartan subgroup of D, which is consequently a Cartan subgroup of G. This proves the assertion when G is affine.

In the general case, let Z be the center of G. Then $G/Z = G'$ is affine (XII 6.1), and the inverse image in G of every Cartan subgroup C' of G' is a Cartan subgroup of G (XII 6.6(e)). We are thus reduced to finding a Cartan subgroup of the smooth affine algebraic group G', which is the case already treated.

Corollary 1.4. *Let G be a smooth group scheme of finite type over an Artinian scheme S. Then G admits a maximal torus T, and hence a Cartan subgroup $C = \mathrm{Centr}_G(T)$. Every torus in G is contained in a maximal torus.*

Indeed, we may suppose that S is local, with residue field k. By 1.1, the group

$$G_0 = G \times_S \mathrm{Spec}(k)$$

admits a maximal torus T_0. By IX 3.6 bis and X 2.3, T_0 comes from a torus T of G, which is plainly maximal. The last assertion follows by applying the preceding result to the centralizer of the given torus, which is smooth over k (XI 2.4).

Remarks 1.5. (a) We shall give below, in 3.20, 3.21, and Exposé XV, variants of 1.4 in which S is not assumed Artinian.

(b) I do not know whether every algebraic group G, not necessarily smooth, over a field k admits a maximal torus. The question arises only in characteristic $p > 0$. Applying 1.1 to a smooth quotient group of the form $G' = G/I$, where I is a suitable radicial subgroup of G (for example, the kernel of a sufficiently high power of the Frobenius homomorphism), and then taking in G the inverse image of a maximal torus of G', reduces the question to the case in which $(G_{\bar{k}})_{\mathrm{red}}$ is a torus; here, as always, $\bar{k}$ denotes the algebraic closure of k. The answer is easily seen to be affirmative when G is commutative, or more generally nilpotent: in that case G has a unique maximal torus, which may for example be constructed by descent from the maximal torus of $G_{\bar{k}}$.*

(c) When G is affine and either k is perfect or G is solvable, 1.1 was already known, by a theorem of Rosenlicht; his proof is quite different from the one given here.

(d) When k is infinite, 1.1 follows from the much more precise result, proved below in 6.1, that the scheme $\mathcal{T}$ of maximal tori of G is a rational variety. The method essentially combines the proof of 1.1 with an explicit description of the structure of the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$. To obtain the desired result, we must first extend to an arbitrary base prescheme certain results of XIII 4–6 (this is the purpose of the next two sections), and then refine the preceding construction proving 1.1, using the fact that every Cartan subgroup of G is contained in one and only one subgroup of type (C) in G (Nos. 4 and 5).

2 Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras

Lemma 2.1. *Let A be a ring, let $\mathfrak{d}$ be a Lie algebra over A, and for each $s \in \operatorname{Spec}(A)$ let*

$$\mathfrak{d}(s) = \mathfrak{d} \otimes_A k(s)$$

be the corresponding Lie algebra over the residue field $k(s)$. Suppose that the A-module $\mathfrak{d}$ is of finite presentation. The following conditions are equivalent:

(i) *for every $s \in \operatorname{Spec}(A)$, the Lie algebra $\mathfrak{d}(s)$ is nilpotent;*

(ii) *for every $x \in \mathfrak{d}$, the endomorphism $\operatorname{ad}(x)$ is nilpotent;*

(iii) *there is an integer $N \geq 0$ such that, for every sequence $x_1, \ldots, x_N$ of elements of $\mathfrak{d}$, one has*

$$\operatorname{ad}(x_1)\operatorname{ad}(x_2)\cdots\operatorname{ad}(x_N) = 0.$$

When A is a field, the equivalence of (i) and (iii) is the definition of nilpotence, while the equivalence of (ii) and (iii) is the familiar consequence of Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, No. 2). In the general case, (iii) plainly implies (ii), and (ii) implies (i) by the preceding result and the fact that (ii) is preserved under passage to quotients and under localization. It remains to prove that (i) implies (iii).

Suppose first that A is local Artinian, with maximal ideal $\mathfrak{m}$. Choose $n > 0$ such that $\mathfrak{m}^n = 0$, and choose N such that condition (iii) holds for

$$\mathfrak{d}(s) = \mathfrak{d} \otimes_A A/\mathfrak{m}.$$

Taking $N' = nN$, one sees at once that N' satisfies (iii) for $\mathfrak{d}$.

If A is Noetherian, there are finitely many points $s_i \in \operatorname{Spec}(A)$ and ideals of definition $\mathfrak{q}_i \subset A_{s_i}$ such that the natural map

$$\mathfrak{d} \longrightarrow \prod_i (\mathfrak{d} \otimes_A A_i), \qquad A_i = A_{s_i}/\mathfrak{q}_i,$$

is injective. By the preceding case, for each i there is an integer N_i satisfying (iii) for the Lie algebra $\mathfrak{d} \otimes_A A_i$. Taking N to be the largest of the N_i, condition (iii) follows for $\mathfrak{d}$. Finally, the general case reduces to the Noetherian case by the usual procedure described in EGA IV 8.

*M. Raynaud gave a negative answer to the question raised here; see XVII, Example 5.9(c).

Definition 2.2. Let S be a prescheme and let $\mathfrak{d}$ be a quasi-coherent Lie algebra over S whose underlying module is of finite presentation. We say that $\mathfrak{d}$ is *locally nilpotent* if, for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ over $k(s)$ is nilpotent. We say that $\mathfrak{d}$ is *strictly locally nilpotent* if it is locally free and, on every open subset U of S over which it has constant rank r, its Killing polynomial reduces to

$$P_{\mathfrak{d}}(t) = t^r.$$

Of course, when $\mathfrak{d}$ is a Lie algebra over S that is locally free as a module, its Killing polynomial is defined as a polynomial

$$P_{\mathfrak{d}} \in A[t], \qquad A = \Gamma\Big(\mathrm{Sym}_{\mathcal{O}_S}(\mathfrak{d})\Big) \simeq \Gamma(W(\mathfrak{d}))\,,$$

where A is the ring of sections of the structure sheaf of the vector bundle $W(\mathfrak{d})$ defined by $\mathfrak{d}$.

The two notions introduced in 2.2 are plainly stable under base change and local for the fpqc topology. We record:

Proposition 2.3. *If $\mathfrak{d}$ is strictly locally nilpotent, then it is locally nilpotent. The converse holds when $\mathfrak{d}$ is locally free and S is reduced.*

The proof is immediate.

Definition 2.4. Let S be a prescheme, let $\mathfrak{g}$ be a Lie algebra over S whose underlying module is locally free of finite type, and let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. We call $\mathfrak{d}$ a *Cartan subalgebra* if it satisfies the following conditions:

(i) $\mathfrak{d}$ is locally a direct-factor submodule (and hence is itself locally free of finite type);

(ii) for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ is a Cartan subalgebra of $\mathfrak{g}(s)$.

Definition 2.5. Let S and $\mathfrak{g}$ be as in 2.4. A section a of $\mathfrak{g}$ is called *quasi-regular* if, for every $s \in S$, the element $a(s) \in \mathfrak{g}(s)$ is regular in the Lie algebra $\mathfrak{g}(s)$ over $k(s)$. We call a a *regular section* if it is quasi-regular and is contained in a Cartan subalgebra of $\mathfrak{g}$.

The notions in 2.4 and 2.5 are again stable under base change and local for the fpqc topology. Only the last assertion, and only for the notion of a regular section, requires proof. It follows from the fact that the Cartan subalgebra containing a given regular section is uniquely determined. More precisely:

Proposition 2.6. *Let S and $\mathfrak{g}$ be as in 2.4, and let a be a quasi-regular section of $\mathfrak{g}$. There is at most one Cartan subalgebra of $\mathfrak{g}$ containing a. Such a Cartan subalgebra exists—that is, a is a regular section—if and only if a satisfies the following condition:*

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$$

is locally a direct-factor submodule of $\mathfrak{g}$, and $\mathrm{ad}(a)$ *induces an automorphism of $\mathfrak{g}/\mathfrak{d}$. In that case $\mathfrak{d}$ is the unique Cartan subalgebra of $\mathfrak{g}$ containing a.*

Suppose first that a is contained in a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$. Then $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is bijective on every fiber. Since $\mathfrak{g}/\mathfrak{d}$ is locally free of finite type, it follows that this endomorphism is an automorphism of $\mathfrak{g}/\mathfrak{d}$. On the other hand, by 2.1 the endomorphism $\mathrm{ad}(a)_{\mathfrak{d}}$ is locally nilpotent. Hence

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}),$$

which proves both the uniqueness of $\mathfrak{d}$ and the necessity of the stated regularity criterion.

For sufficiency, observe that the hypothesis on a implies that the formation of $\mathrm{Nil}(a, \mathfrak{g})$ commutes with base extension, and in particular with passage to fibers. Thus the fibers $\mathfrak{d}(s)$ of $\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$ are Cartan subalgebras of the $\mathfrak{g}(s)$. Moreover, $\mathfrak{d}$ is a Lie subalgebra of $\mathfrak{g}$ by XIII 4.1. It is therefore a Cartan subalgebra.

Corollary 2.7. *Under the hypotheses of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, let a be a section of $\mathfrak{d}$ that is regular in $\mathfrak{g}$, and let u be an automorphism of $\mathfrak{g}$. Then u preserves $\mathfrak{d}$ if and only if $u(a)$ is a section of $\mathfrak{d}$.*

Indeed, by transport of structure $u(a)$ is then a regular section of $\mathfrak{g}$, contained in both Cartan subalgebras $\mathfrak{d}$ and $u(\mathfrak{d})$; Proposition 2.6 therefore shows that these two subalgebras are equal.

Corollary 2.8. *Under the hypotheses of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. For every $s \in S$ such that $k(s)$ is infinite, there exist an open neighborhood V of s and a regular section a of $\mathfrak{g}$ over V such that*

$$\mathfrak{d}|_V = \mathrm{Nil}(a, \mathfrak{g}|_V)\,;$$

equivalently, a is a section of $\mathfrak{d}|_V$.

Indeed, since $k(s)$ is infinite, there is a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}(s)$. After replacing S by an open neighborhood U of s, we may extend a_0 to a section a of $\mathfrak{d}$ over U. The endomorphism induced by $\mathrm{ad}(a)$ on the fiber

$$\mathfrak{g}(s)/\mathfrak{d}(s)$$

is an automorphism. Shrinking U to a smaller open neighborhood V, we may therefore suppose that $\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is an automorphism. It follows that a is a quasi-regular section of $\mathfrak{g}|_V$ satisfying the required condition.

Let $\mathfrak{g}$ be as in 2.4. Inspection of its Killing polynomial immediately shows that the function

$$s \longmapsto \text{the nilpotent rank of } \mathfrak{g}(s)$$

on S is upper semicontinuous. We shall be especially interested in the case in which this function is continuous, that is, locally constant. Here are several variants of this property.

Proposition 2.9. *Let $S, \mathfrak{g}$ be as in 2.4. Consider the following conditions:*

(C_0) *The nilpotent rank of $\mathfrak{g}(s)$, for $s \in S$, is a locally constant function of s.*

(C_1) *Locally for the fpqc topology, there exists a Cartan subalgebra of $\mathfrak{g}$.*

(C_1') *The same as (C_1), with "fpqc topology" replaced by "étale topology".*

(C_2) *Condition (C_0) holds, and for every S' over S, every quasi-regular section of*

$$\mathfrak{g}_{S'} = \mathfrak{g} \otimes_S S'$$

is regular.

(C_3) *Every $s \in S$ has an open neighborhood V on which the Killing polynomial of $\mathfrak{g}$ has the form*

$$t^r\big(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}\big),$$

where, for every $s \in V$,

$$c_{n-r}(s) \in \mathrm{Sym}\big(\mathfrak{g}(s)^\vee\big)$$

is nonzero.

With this notation, one has the implications

$$(\mathrm{C}_3) \Longrightarrow (\mathrm{C}_2) \Longrightarrow (\mathrm{C}'_1) \Longrightarrow (\mathrm{C}_1) \Longrightarrow (\mathrm{C}_0),$$

and, when S is reduced, the five conditions above are equivalent.

Notice also that these conditions are manifestly stable under base change and are local for the fpqc topology.

The implications

$$(\mathrm{C}'_1) \Longrightarrow (\mathrm{C}_1) \Longrightarrow (\mathrm{C}_0)$$

are trivial, and the implication $(\mathrm{C}_0) \Longrightarrow (\mathrm{C}_3)$ when S is reduced is immediate. We note moreover:

Corollary 2.10. *Suppose that condition* (C_0) *holds. Let U be the set of elements of $W(\mathfrak{g})$ that are regular in their fiber. Then U is open. In particular, for every section a of $\mathfrak{g}$ over S, the set V of $s \in S$ for which $a(s) \in \mathfrak{g}(s)$ is regular is open.*

Indeed, the first assertion follows from the second, applied to $\mathfrak{g}_{S'}$ for every base change S' over S. For the second, we may suppose that S is reduced; then (C_3) holds, and it is enough to inspect the Killing polynomial of a in $\mathfrak{g}$.

The implication $(\mathrm{C}_2) \Longrightarrow (\mathrm{C}'_1)$ now follows easily from part b) of the following more precise statement.

Corollary 2.11. *Suppose that condition* (C_2) *holds. Then:*

a) *For every $s \in S$ and every Cartan subalgebra $\mathfrak{d}_0$ of $\mathfrak{g}(s)$ such that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$* (a condition automatically satisfied when $k(s)$ is infinite)*, there exist an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ whose fiber at s is $\mathfrak{d}_0$. If S_1 is a subscheme of S containing s, and if $\mathfrak{d}_0$ has already been extended to a Cartan subalgebra $\mathfrak{d}_1$ of $\mathfrak{g} \otimes_S S_1$, then one can find an open neighborhood V of s in S and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ such that*

$$\mathfrak{d} \otimes_V (S_1 \cap V) = \mathfrak{d}_1|_{S_1 \cap V}.$$

b) *For every $s \in S$ such that $\mathfrak{g}(s)$ contains a regular element* (a condition automatically satisfied when $k(s)$ is infinite)*, there exist an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$.*

Part b) follows from part a) by taking

$$\mathfrak{d}_0 = \mathrm{Nil}(a_0, \mathfrak{g}(s))\,,$$

where a_0 is a regular element of $\mathfrak{g}(s)$. To prove part a), in its second formulation, choose a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$. Extend it near s in S_1 to a section of $\mathfrak{d}_1$, and extend that section in turn near s to a section of $\mathfrak{g}$. By 2.10 this section is quasi-regular on an open neighborhood V of s, hence regular by (C_2). Therefore, by 2.6,

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}|_V)$$

has the required properties.

Remark 2.12. I do not know whether the assertions of 2.11 a) and b) remain valid without the hypothesis that regular points exist, when $k(s)$ is finite.

It remains to prove

$$(\mathrm{C}_3) \Longrightarrow (\mathrm{C}_2).$$

We also record the following equivalent form of (C_3):

(C'_3) Condition (C_0) holds, that is, the nilpotent rank of $\mathfrak{g}(s)$, for $s \in S$, is locally constant; and on every open subset V of S on which this rank has the value r, the Killing polynomial of $\mathfrak{g}|_V$ is divisible by t^r.

We must show that this condition implies that every quasi-regular section a of $\mathfrak{g}$ is regular. In view of 2.6, this follows from the implication (iv) $\Longrightarrow$ (iii) in the following lemma, applied to the endomorphism $\mathrm{ad}(a)$ of $\mathfrak{g}$.

Lemma 2.13. *Let A be a ring, let M be a projective A-module of finite type, and let u be an endomorphism of M. The following conditions are equivalent:*

(i) *M is the direct sum of two u-stable submodules M' and M'', such that $u|_{M'}$ is nilpotent and $u|_{M''}$ is an automorphism of M''.*

(ii) *There exists an integer $n > 0$ such that*

$$\operatorname{Im} u^n + \operatorname{Ker} u^n = M.$$

(iii) *The nilspace*

$$\operatorname{Nil}(u) = \bigcup_{n>0} \operatorname{Ker} u^n$$

is a direct factor of M, and

$$M = \operatorname{Nil}(u) + u(M).$$

These conditions are implied by the following one, and are equivalent to it when A is reduced:

(iv) *Locally on $\operatorname{Spec}(A)$, for the Zariski topology, the characteristic polynomial $P_u(t)$ of u can be written in the form*

$$t^r\bigl(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}\bigr),$$

where c_{n-r} is invertible.

The equivalence of (i), (ii), and (iii) is immediate and is recalled for reference. When A is reduced, (i) implies (iv) because a nilpotent endomorphism of a projective module of rank r then has characteristic polynomial t^r. On the other hand, in all cases the constant term of the characteristic polynomial of an automorphism of a projective module of finite type is, up to sign, the determinant of u, locally on $\operatorname{Spec}(A)$; it is therefore an invertible element of A.

Finally, to prove (iv) $\Longrightarrow$ (i), regard M as a module over the polynomial ring $A[t]$, with t acting as u. The familiar identity

$$P(u) = 0$$

shows that M is annihilated by $PA[t]$, and hence may be regarded as a module over $A[t]/PA[t]$. Write

$$P = t^r Q,$$

where the constant term of Q is invertible. Then

$$PA[t] = t^r A[t] \cap QA[t],$$

so that $A[t]/PA[t]$ decomposes as the product of the rings

$$A[t]/t^r A[t] \qquad \text{and} \qquad A[t]/QA[t].$$

This gives a corresponding decomposition of M into the direct sum of two $A[t]$-modules, or equivalently two u-stable A-submodules M' and M''; this is precisely the decomposition of (i).

This completes the proof of 2.13, and hence that of 2.9.

Corollary 2.14. *When condition* (C$_3$) *of 2.9 holds, the Cartan subalgebras of* $\mathfrak{g}$ *are strictly nilpotent* (2.2)*.*

The proof is immediate.

Remark 2.15. a) One can prove a converse to 2.14: condition (C$_3$) is equivalent to the assertion that, for every S' over S, every quasi-regular section of $\mathfrak{g}_{S'}$ is regular and every Cartan subalgebra of $\mathfrak{g}_{S'}$ is strictly nilpotent; equivalently, every quasi-regular section of $\mathfrak{g}_{S'}$ is contained in a strictly nilpotent Cartan subalgebra of $\mathfrak{g}_{S'}$.

b) Unlike the other conditions (C$_0$) through (C$_2$), condition (C$_3$) is of "infinitesimal nature." More precisely, when S is locally noetherian, $\mathfrak{g}$ satisfies (C$_3$) if and only if, for every base change $S' \to S$ with S' local Artinian, $\mathfrak{g}_{S'}$ satisfies the same condition. If desired, one may take S' to be the spectrum of an Artinian quotient of a local ring of S. Similarly, when (C$_0$) holds, the condition for a section of $\mathfrak{g}$ to be regular is of infinitesimal nature.

c) When $\mathfrak{g}$ is the Lie algebra of a smooth group prescheme of finite presentation over S, we shall see that the conditions

$$(\mathrm{C}_0), \quad (\mathrm{C}_1), \quad (\mathrm{C}'_1), \quad (\mathrm{C}_2)$$

on $\mathfrak{g}$ are equivalent (5.2 a). I do not know what happens in general, except that even when S is local Artinian, (C$_0$) does not imply (C$_1$). However, even when $\mathfrak{g}$ comes from a group G and S is local Artinian, it is not true in general that (C$_2$) implies (C$_3$), since $\mathfrak{g}$ may be nilpotent without being strictly nilpotent.

An example is obtained from a smooth affine group scheme G over the spectrum S of a discrete valuation ring, such that the Lie algebra of the generic fiber is not nilpotent—for example, the generic fiber may be an adjoint semisimple group—whereas that of the special fiber is nilpotent—for example, the special fiber may be a vector group. Then, for sufficiently large n, the Lie algebra of

$$G_n = G \times_S S_n$$

is not strictly nilpotent, although it is nilpotent.

We continue under the hypotheses of 2.4. Let

$$\mathcal{D}\colon \left((\mathrm{Sch})_{/S}\right)^{\circ} \longrightarrow \mathrm{Sets}$$

be the functor defined by

$$\mathcal{D}(S') = \{\text{Cartan subalgebras of } \mathfrak{g}_{S'}\}\,.$$

Introduce also the functor

$$X(S') = \left\{ (\mathfrak{d}, a) \;\middle|\; \begin{array}{l} \mathfrak{d} \text{ is a Cartan subalgebra of } \mathfrak{g}_{S'}, \\ a \text{ is a section of } \mathfrak{d} \end{array} \right\}.$$

The two projections

$$(\mathfrak{d}, a) \longmapsto \mathfrak{d} \qquad \text{and} \qquad (\mathfrak{d}, a) \longmapsto a$$

give morphisms

$$p\colon X \longrightarrow \mathcal{D} \qquad \text{and} \qquad \psi\colon X \longrightarrow W(\mathfrak{g}).$$

When (C$_0$) holds, let U also denote the open subset of regular points of $W(\mathfrak{g})$ (2.10). Condition (C$_2$) is then equivalent to saying that the morphism

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism; a priori it is a monomorphism by 2.6.

It is immediate that

$$p\colon X \longrightarrow \mathcal{D}$$

is representable by a projection of vector bundles. More explicitly, for every S-morphism $S' \to \mathcal{D}$ corresponding to a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}_{S'}$, the fiber product

$$X \times_{\mathcal{D}} S'$$

is represented by the vector bundle $W(\mathfrak{d})$ over S'. Thus, if $\mathcal{D}$ is representable, then so is X. We have:

Theorem 2.16. *Let S be a prescheme and let $\mathfrak{g}$ be a Lie algebra over S whose underlying $\mathcal{O}_S$-module is locally free of finite type. Suppose that condition* (C_0) *of 2.9 holds.*

a) *The functor $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$, defined above, is represented by a quasi-projective prescheme of finite presentation over S. The same holds for the functor X.*

b) *If condition* (C_2) *of 2.9 holds, then $\mathcal{D}$ and X are smooth over S, and the morphism*

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism.

c) *Continue to suppose that* (C_2) *holds. Let $s \in S$, and let $\mathfrak{d}_0$ be a Cartan subalgebra of $\mathfrak{g}(s)$, corresponding to a $k(s)$-rational point d of $\mathcal{D}$. Suppose that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$, a condition automatically satisfied when $k(s)$ is infinite. If r is the infinitesimal rank of $\mathfrak{g}(s)$ and n its rank over $k(s)$, then there exists an open neighborhood V of d in $\mathcal{D}$ that is S-isomorphic to an open subset V' of*

$$S[t_1, \dots, t_{n-r}].$$

Proof. We may suppose that $\mathfrak{g}$ has constant rank n and constant nilpotent rank r. The assertions concerning X in a) and b) follow immediately from those concerning $\mathcal{D}$ and from the fact that X is a vector bundle over $\mathcal{D}$, defined by a locally free module; they were included only for completeness.

a) The functor $\mathcal{D}$ is a subfunctor of $\mathrm{Grass}_{n-r}(\mathfrak{g})$, whose value at S' is the set of locally free quotient modules of rank $n-r$ of $\mathfrak{g}_{S'}$. It is well known that the latter functor is represented by a projective prescheme smooth over S; see, for example, Cartan Seminar 1960/61, Exposé 12, Nos. 2 and 3, whose constructions carry over unchanged to preschemes.[*] [3]

We are therefore reduced to the following relative problem. Given a locally free quotient module of rank $n-r$ of $\mathfrak{g}$, or, equivalently, a locally free submodule $\mathfrak{d}$ of rank r that is locally a direct factor, represent the functor

$$F(S') = \begin{cases} \varnothing, & \text{if } \mathfrak{d}_{S'} \text{ is not a Cartan subalgebra of } \mathfrak{g}_{S'}, \\ \{\varnothing\}, & \text{otherwise.} \end{cases}$$

We shall show that F is represented by a subprescheme of finite presentation over S. This will show that

$$\mathcal{D} \longrightarrow \mathrm{Grass}_{n-r}(\mathfrak{g})$$

[*]See also EGA I, second edition (announced as forthcoming from North-Holland Publishing Company).

[3]Editors' note: See §I.1.3 of M. Demazure and P. Gabriel, *Groupes algébriques*, Masson (1970).

is represented by an immersion of finite presentation and will complete the proof of a).

First express the condition that $\mathfrak{d}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$. This condition says precisely that $S' \to S$ factors through a certain closed subprescheme S_1 of finite presentation over S; locally on S, its equations are obtained at once from a basis of $\mathfrak{g}$ adapted to $\mathfrak{d}$. We may therefore suppose that $S = S_1$.

Next we must express that $\mathfrak{d}_{S'}$ contains, fpqc-locally, a quasi-regular section of $\mathfrak{g}_{S'}$. For this purpose consider

$$V = W(\mathfrak{d}) \cap U,$$

where U is the open subset of regular points of $W(\mathfrak{g})$ (2.10). Since the structure morphism $V \to S$ is smooth and quasi-compact, its image S_1 is an open subset of S, and the immersion $S_1 \to S$ is quasi-compact, hence of finite presentation. The condition on S' is exactly that $S' \to S$ factor through S_1. We are thus reduced to the case $S = S_1$, and, by descent, to the case in which $\mathfrak{d}$ has a section a that is quasi-regular in $\mathfrak{g}$.

It remains to express that, for the induced section $a_{S'}$, the endomorphism

$$\operatorname{ad}(a_{S'})_{\mathfrak{g}_{S'}/\mathfrak{d}_{S'}}$$

is bijective. This again says that $S' \to S$ factors through an open subprescheme of finite presentation, namely S_D, where

$$D = \det\Big(\operatorname{ad}(a)_{\mathfrak{g}/\mathfrak{d}}\Big)\,.$$

But $\mathfrak{d}|_{S_D}$ is then a Cartan subalgebra of $\mathfrak{g}|_{S_D}$. Consequently S_D represents F, which proves a).

b) This follows immediately from 2.11(a) and XI 1.5. It is also, of course, a consequence of the more precise assertion c).

c) Let a_0 be a regular point of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$, and extend it to a section a of $\mathfrak{g}$ over an open neighborhood of s. We may plainly suppose that this neighborhood is all of S. Choose a complement M_0 of $\mathfrak{d}_0$ in the vector space $\mathfrak{g}(s)$. After again replacing S by an open neighborhood of s, there is a submodule M of $\mathfrak{g}$, locally a direct factor, such that $M(s) = M_0$; we again suppose that the neighborhood is S.

Let $\mathcal{V}$ be the subfunctor of $\mathcal{D}$ for which $\mathcal{V}(S')$ consists of the Cartan subalgebras $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ satisfying:

1°) $\mathfrak{d}'$ is a complement of $M_{S'}$;

2°) the unique section in $(a_{S'} + M_{S'}) \cap \mathfrak{d}'$ is a regular section of $\mathfrak{g}_{S'}$.

Condition 1°) defines an open V_1 of $\mathcal{D}$, induced by the open of $\mathrm{Grass}_{n-r}(\mathfrak{g})$ given by the same condition. By 2.10 and (C_2), the conjunction of 1°) and 2°) defines an open of V_1. Thus $\mathcal{V}$ is represented by an open subprescheme V of $\mathcal{D}$, plainly containing the point d.

Now let $\mathcal{V}'$ be the subfunctor of $W(M)$ defined by

$$\mathcal{V}'(S') = \left\{ u' \in \Gamma(S', M_{S'}) \;\middle|\; \begin{array}{l} \text{(i) } a_{S'} + u' \text{ is a regular section of } \mathfrak{g}_{S'}, \\ \text{(ii) the unique Cartan subalgebra } \mathfrak{d}' \text{ of } \mathfrak{g}_{S'} \\ \quad \text{containing } a_{S'} + u' \text{ is a complement of } M_{S'} \end{array} \right\}.$$

Condition (i) defines an open subprescheme V_1' of $W(M)$: it is the inverse image of the open U of regular points of $W(\mathfrak{g})$ (2.10) under the translation morphism $m \mapsto a + m$. The conjunction of (i) and (ii) defines an open V' of V_1', namely the inverse image of V under the evident morphism

$$V_1' \longrightarrow \mathcal{D}$$

that associates with u' the unique Cartan subalgebra $\mathfrak{d}'$ containing $a_{S'} + u'$. Its restriction gives a morphism

$$V' \longrightarrow V,$$

which is plainly an isomorphism. This proves c).

Corollary 2.17. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k. Then the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$* (2.16(a)) *is quasi-projective, smooth, and irreducible. If $\mathfrak{g}$ contains a regular element* (for example, if k is infinite)*, then $\mathcal{D}$ is a rational variety; that is, its function field is a purely transcendental extension of k.*

Indeed, $\mathcal{D}$ is irreducible because there is a surjective morphism

$$\psi^{-1}(U) \longrightarrow \mathcal{D},$$

and $\psi^{-1}(U)$ is irreducible, being isomorphic to the open subset U of $W(\mathfrak{g})$. The assertion concerning the function field follows immediately from 2.16(c).

Remarks 2.18. I do not know whether this conclusion remains true when k is finite without assuming that $\mathfrak{g}$ contains a regular point; compare 2.12. It can be proved when $\mathfrak{g}$ is the Lie algebra of an algebraic group G smooth over k, at least when $G/\operatorname{radical}(G)$ is an "adjoint" semisimple group, by using a result of Chevalley cited below (see the Appendix). It is plausible that the result remains valid without restriction on G; for this it would suffice that the cited result of Chevalley be proved for every semisimple algebraic group, not necessarily an adjoint group.

3 Subgroups of type (C) of group preschemes over an arbitrary prescheme

Theorem 3.1. *Let S be a prescheme, let G be a smooth S-group prescheme, and let $\mathfrak{g}$ be its Lie algebra, which is a locally free module of finite type over S. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ that, as a module, is locally a direct factor in $\mathfrak{g}$, and suppose that, for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let a be a quasi-regular section of $\mathfrak{g}$* (2.5)*. Then*

$$M_a = \operatorname{Transp}_G(a, \mathfrak{h}),$$

the subfunctor of G whose S'-valued points are the $g \in G(S')$ such that

$$\operatorname{ad}(g) \cdot a_{S'} \in \Gamma(S', \mathfrak{h}_{S'}),$$

is represented by a closed subprescheme of G, smooth over S, whose structure morphism to S is surjective. [4]

Proof. Consider the canonical morphism

$$\varphi\colon G \times_S W(\mathfrak{h}) \longrightarrow W(\mathfrak{g}), \qquad (g, x) \longmapsto \operatorname{ad}(g) \cdot x.$$

Then M_a is S-isomorphic to $\varphi^{-1}(a)$, the inverse image under φ of a, viewed as a section of $W(\mathfrak{g})$ over S. To prove that M_a is smooth, it is enough to show that φ is smooth at the points of $G \times_S W(\mathfrak{h})$ lying over $\operatorname{Im}(a)$. More generally, φ is smooth at every point lying over a point of $W(\mathfrak{g})$ that is regular in its fiber $W(\mathfrak{g}(s))$ over S.

[4] Editors' note: This transporter is denoted $\operatorname{Transp}_G(a, \mathfrak{h})$.

Indeed, the source and target of φ are smooth over S, and hence are flat and locally of finite presentation over S. We may therefore check the assertion fiber by fiber. This reduces us to the case in which S is the spectrum of an algebraically closed field k, G is a locally algebraic group over k, $\mathfrak{h}$ is a Lie subalgebra of its Lie algebra $\mathfrak{g}$ containing a Cartan subalgebra of $\mathfrak{g}$, and a is a regular point of $\mathfrak{g}$.

Since φ is a G-morphism, we may suppose that the point of $G \times W(\mathfrak{h})$ under consideration is of the form (e, a). We may also suppose that G is connected, and hence of finite type over k. The assertion is then precisely XIII 5.4.

It is also immediate that M_a is a closed subprescheme of G of finite presentation over S: it is the inverse image of $W(\mathfrak{h})$ under the morphism

$$G \longrightarrow W(\mathfrak{g}), \qquad g \longmapsto \mathrm{ad}(g) \cdot a.$$

Finally, the surjectivity of $M_a \to S$ likewise reduces to the case of an algebraically closed base field. By hypothesis, $\mathfrak{h}$ then contains a Cartan subalgebra $\mathfrak{d}'$. By the conjugacy theorem XIII 6.1(a), $\mathfrak{d}'$ is conjugate to

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g}).$$

It follows that $\mathfrak{h}$ contains a conjugate of a. This completes the proof.

Corollary 3.2. *Let G and $\mathfrak{g}$ be as in 3.1, with G of finite type over S, and let $\mathfrak{k}$ and $\mathfrak{h}$ be two Lie subalgebras of $\mathfrak{g}$ that are locally direct factors as modules. Suppose that one of the following two hypotheses holds:*

a) *For every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent and contains a regular element of $\mathfrak{g}_{\bar{s}}$, while the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

b) *$\mathfrak{k}$ is a Cartan subalgebra of $\mathfrak{g}$.*

Then $\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h})$ *is a closed subprescheme of G, smooth over S. Moreover, in case* a)*, its structure morphism to S is surjective.*

The representability of the transporter by a closed subprescheme of G of finite presentation is immediate and is left to the reader. We first prove smoothness in case a). Suppose first that $\mathfrak{k}$ has a section a that is quasi-regular in $\mathfrak{g}$. It is then enough to apply 3.1 and the following lemma.

Lemma 3.3. *Under the hypotheses of 3.2*(a)*, if a is a section of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$, then*

$$\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}) = \mathrm{Transp}_G(a, \mathfrak{h}).$$

Indeed, by the definitions it is enough to show that, if a is also a section of $\mathfrak{h}$, then $\mathfrak{k} \subset \mathfrak{h}$. Since $\mathfrak{k}$ is locally nilpotent by hypothesis, 2.1 gives

$$\mathfrak{k} \subset \mathrm{Nil}(a, \mathfrak{g}).$$

On the other hand,

$$\mathrm{Nil}(a, \mathfrak{g}) \subset \mathfrak{h},$$

because

$$\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$$

is injective; this can be checked fiber by fiber by XIII 4.8(b). The lemma follows.

In the general case, the usual standard argument first reduces us to the case where S is affine noetherian, and then to the case where S is local artinian, smoothness being an infinitesimal property. By flat descent we may further suppose that the residue field is infinite. The fiber of $\mathfrak{k}$ then contains an element regular in the corresponding fiber of $\mathfrak{g}$. Lifting this element to a section of $\mathfrak{k}$ reduces us to the preceding case.

Thus the transporter is smooth in case a). Its structure morphism is surjective because this may be checked after reducing to the case where S is the spectrum of an algebraically closed field; $\mathfrak{k}$ then contains a regular point of $\mathfrak{g}$, and 3.3 and 3.1 apply.

For case b), the definition of smoothness (XI 1.1) reduces the assertion to the following lifting statement. Let S be affine, let S_0 be a closed subscheme defined by a quasi-coherent nilpotent ideal J, and let $g_0 \in G(S_0)$ transport $\mathfrak{k}_0$ into $\mathfrak{h}_0$. Then g_0 lifts to an element $g \in G(S)$ that transports $\mathfrak{k}$ into $\mathfrak{h}$. The hypothesis on g_0 places this lifting problem under case a), already proved. This completes the proof.

Of course, if in 3.2(b) the subalgebra $\mathfrak{h}$ satisfies the stronger hypothesis of 3.1, then, and only then, the structure morphism

$$\mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}) \longrightarrow S$$

is surjective. Using Hensel's lemma XI 1.10, one obtains from 3.1 and 3.2 the following corollary.

Corollary 3.4. *Under the hypotheses of 3.1 for G and $\mathfrak{h}$, and assuming that G is of finite type over S, the following assertions hold:*

a) *For every quasi-regular section a of $\mathfrak{g}$, étale locally there is a conjugate of a that is a section of $\mathfrak{h}$.*

b) *For every Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$, étale locally $\mathfrak{d}$ is conjugate to a subalgebra of $\mathfrak{h}$.*

In particular, when $\mathfrak{h}$ is itself a Cartan subalgebra of $\mathfrak{g}$, one obtains the following.

Corollary 3.5. *Let G be a smooth S-prescheme in groups of finite type, let $\mathfrak{g}$ be its Lie algebra, and let $\mathfrak{d}$ and $\mathfrak{d}'$ be two Cartan subalgebras of $\mathfrak{g}$. Then* $\mathrm{Transp}_G(\mathfrak{d}, \mathfrak{d}')$ *is identical to the strict transporter of $\mathfrak{d}$ into $\mathfrak{d}'$, and is a closed subprescheme of G, smooth over S, whose structure morphism is surjective. Étale locally, $\mathfrak{d}$ and $\mathfrak{d}'$ are conjugate.*

Here the transporter is identical to the strict transporter simply because $\mathfrak{d}$ and $\mathfrak{d}'$ are locally direct factors in $\mathfrak{g}$ and have the same rank at every point. Thus 3.5 is a special case of 3.4. Taking $\mathfrak{d} = \mathfrak{d}'$ gives the following.

Corollary 3.6. *Let G be as in 3.5, and let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then* $\mathrm{Norm}_G(\mathfrak{d})$ *is a closed subprescheme in groups of G, smooth over S, and its Lie algebra is $\mathfrak{d}$.*

Indeed, the last assertion says that $\mathfrak{d}$ is its own normalizer in $\mathfrak{g}$, and this follows at once because it is true fiber by fiber.

Corollary 3.7. *Let G and $\mathfrak{g}$ be as in 3.5. Then the conditions* (C_2), (C'_1), *and* (C_1) *of 2.9 are equivalent. In other words, if $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally, then every quasi-regular section of $\mathfrak{g}$ is regular.*

Indeed, let a be a quasi-regular section; we prove that it is regular. Since the question is fpqc local, we may suppose that $\mathfrak{g}$ admits a Cartan subalgebra $\mathfrak{d}$. By 3.4(a), étale locally a is conjugate to a section of $\mathfrak{d}$. We are thereby reduced to the case in which a is a section of $\mathfrak{d}$, where the conclusion follows immediately from the definition.

Definition 3.8. *Let G be a smooth prescheme in groups over a prescheme S. A* subgroup of type (C) *of G is a subprescheme in groups $D \subset G$, smooth over S and with connected fibers, such that*

$$\mathfrak{d} = \mathrm{Lie}(D)$$

is a Cartan subalgebra (2.4) *of* $\mathfrak{g} = \mathrm{Lie}(G)$*. Equivalently, for every* $s \in S$*, the fiber* D_s *is a subgroup of type* (C) *of the algebraic group* G_s (XIII 6.2)*.*

Theorem 3.9. *Let* G *be a smooth* S*-prescheme in groups of finite presentation, and let* $\mathfrak{g}$ *be its Lie algebra. Then:*

a) *The map*

$$D \longmapsto \mathfrak{d} = \mathrm{Lie}(D)$$

establishes a bijective correspondence between the subgroups of type (C) *of* G *and the Cartan subalgebras of* $\mathfrak{g}$*.*

b) *If* D *and* $\mathfrak{d}$ *correspond, then*

$$\mathrm{Norm}_G(D) = \mathrm{Norm}_G(\mathfrak{d}).$$

This is a closed subprescheme of G*, smooth over* S*, and*

$$D = \mathrm{Norm}_G(D)^0 = \mathrm{Norm}_G(\mathfrak{d})^0.$$

c) *Any two subgroups* D *and* D' *of type* (C) *in* G *are conjugate étale locally.*

Proof. Let D be a subgroup of type (C) of G, and let $\mathfrak{d}$ be its Lie algebra. Then

$$D \subset \mathrm{Norm}_G(\mathfrak{d}).$$

By Definition 3.8 and Corollary 3.6, this is an inclusion of smooth preschemes in groups over S that induces an isomorphism on Lie algebras. Since D has connected fibers, it follows that

$$D = \mathrm{Norm}_G(\mathfrak{d})^0.$$

Thus the map in a) is injective. We prove that it is surjective. Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. By 3.6,

$$N = \mathrm{Norm}_G(\mathfrak{d})$$

is a closed subprescheme in groups of G, smooth over S, and has Lie algebra $\mathfrak{d}$. Since G is of finite presentation over S, so is N. Consequently, as noted in XII after 7.3, the union of the connected components of the fibers of N is the underlying set of an open subprescheme in groups $N^\circ \subset N$. Evidently N° is a subgroup of type (C) of G with Lie algebra $\mathfrak{d}$. This proves a). The first assertion of b) follows at once, and the formula

$$D = \mathrm{Norm}_G(\mathfrak{d})^0$$

has already been proved. Finally, c) follows from a) and 3.5.

Corollary 3.10. *Suppose that* $\mathfrak{g}$ *admits a Cartan subalgebra fpqc locally, or equivalently by 3.9*(a)*, that* G *admits a subgroup of type* (C) *fpqc locally. Consider the functor*

$$\mathcal{D}\colon (\mathrm{Sch})^\circ/S \longrightarrow (\mathrm{Sets}),$$

where $\mathcal{D}(S')$ *is the set of subgroups of type* (C) *of* $G_{S'}$*. This functor is representable by a quasi-projective prescheme, smooth over* S*, with connected geometric fibers. If* $S = \mathrm{Spec}(k)$*, so that* G *is a smooth algebraic group over* k*, and if* $\mathfrak{g}$ *admits a regular point—a condition automatically satisfied when* k *is infinite—then* $\mathcal{D}$ *is a rational variety over* k*.*

Indeed, by 3.9(a) the functor $\mathcal{D}$ is canonically isomorphic to the functor considered in 2.16. On the other hand, by 3.7 condition (C_2) is satisfied. Therefore 3.10 follows from 2.16 and 2.17.

Corollary 3.11. *Let D be a subgroup of type* (C) *of G, and let N be its normalizer in G. Then the quotient sheaf G/N is canonically isomorphic to the functor $\mathcal{D}$ of 3.10. It is therefore representable by a quasi-projective prescheme, smooth over S, with connected geometric fibers.*

Proposition 3.12. *Let G be a smooth S-prescheme in groups of finite presentation over S. Let H and K be two subpreschemes in groups, smooth and of finite presentation over S, with K having connected fibers, and let $\mathfrak{g}, \mathfrak{h}, \mathfrak{k}$ be the corresponding Lie algebras. Suppose that one of the following conditions holds on their geometric fibers:*

a) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$, and $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

b) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

Under these conditions, $H \supset K$ if and only if $\mathfrak{h} \supset \mathfrak{k}$.

It remains only to prove that $\mathfrak{h} \supset \mathfrak{k}$ implies $H \supset K$. In case b), the inclusion $\mathfrak{h} \supset \mathfrak{k}$ places us under the hypotheses of case a), so it is enough to prove a). Proceeding as in 3.2(a), by reduction to the case where S is local artinian, we reduce to the case in which there exists a section a of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$. Arguing then as in XIII 5.5 reduces us to the following statement.

Corollary 3.13. *Let G be a smooth S-prescheme in groups of finite presentation over S, let $H \subset G$ be a subprescheme in groups smooth and of finite presentation over S, and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. Let a be a section of $\mathfrak{h}$ that is quasi-regular in $\mathfrak{g}$. Suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Put*

$$M_a = \mathrm{Transp}_G(a, \mathfrak{h}).$$

This is a closed subprescheme of G, smooth over S (cf. 3.1)*. Hence M_a^0, the union in the fibers of M_a of the connected components containing the identity, is an open subset of M_a, endowed with the structure induced from M_a. Then*

$$H^0 = M_a^0.$$

Clearly $H \subset M_a$, and therefore $H^0 \subset M_a^0$. Since this is an inclusion of preschemes smooth over S, it is enough to check equality on the fibers. This reduces us to the case where S is the spectrum of an algebraically closed field, which was treated in XIII 5.4.

Corollary 3.14. *Let G be a smooth S-prescheme in groups of finite presentation, let D be a subgroup of type* (C) *of G, and let $H \subset G$ be a subprescheme in groups smooth and of finite presentation over S. If $\mathfrak{d}$ and $\mathfrak{h}$ are their Lie algebras, then*

$$\mathrm{Transp}_G(D, H) = \mathrm{Transp}_G(\mathfrak{d}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G, smooth over S.

Indeed, the identity between the two transporters follows from 3.12(b), after which 3.2 applies.

Corollary 3.15. *Let G and H be as in 3.14, and suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Suppose also that $\mathfrak{g}$ admits a Cartan subalgebra fpqc locally. Then, étale locally, H contains a subgroup of type* (C) *of G.*

By 3.7 and 3.9(a), G admits a subgroup of type (C) étale locally. We may therefore suppose that G admits such a subgroup D. The hypothesis on $\mathfrak{h}$ then also says that the structure morphism of the transporter in 3.14 is surjective, by the conjugacy theorem XIII 6.1(a). The conclusion follows from Hensel's lemma XI 1.10.

Corollary 3.16. *Let G, H, K be as in 3.12*(a)*, and suppose in addition that every geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent, that is, that $\mathfrak{k}$ is locally nilpotent. Then*

$$\mathrm{Transp}_G(K, H) = \mathrm{Transp}_G(\mathfrak{k}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G, smooth over S, whose structure morphism to S is surjective. Étale locally, H contains a subgroup conjugate to K.

The identity between the two transporters is again contained in 3.12(a). The assertion about its structure is precisely 3.2(a), and the last assertion of 3.16 then follows from Hensel's lemma.

Remarks 3.17. a) In 3.12 and 3.16, the hypothesis that K be smooth over S may be replaced by the weaker hypothesis that the sheaf

$$e_K^*(\Omega^1_{K/S}) = \omega^1_K$$

of relative 1-differentials along the identity section be locally free. In this form, 3.12 includes XIII 5.5.

b) Let $G, \mathfrak{g}, \mathfrak{h}$ be as in 3.1, with G of finite presentation over S. Then

$$N = \mathrm{Norm}_G(\mathfrak{h})$$

need not be smooth over S along the identity section. Equivalently, there need not exist a subprescheme in groups $H \subset G$, smooth over S, whose Lie algebra is $\mathfrak{h}$, even when S is the spectrum of a field. When such an H exists, taking H with connected fibers gives

$$N = \mathrm{Norm}_G(H);$$

I do not know whether N is then smooth over S. In this question one may plainly reduce to the case where S is local artinian.

To conclude this section, we examine the case in which G is "semisimple" over S.

Theorem 3.18. *Let S be a prescheme and let G be a smooth S-prescheme in groups whose geometric fibers are semisimple "adjoint" groups, that is, semisimple groups whose reductive center* (XII 4.1 and 4.4) *is the identity group. Then the subgroups of type* (C) *of G are precisely its maximal tori, and hence also precisely its Cartan subgroups* (XII 3.1).

In view of the definitions, it is enough to treat the case where S is the spectrum of an algebraically closed field. We must then show that, for a maximal torus T of G, its Lie algebra $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$. Put $r = \dim T = \operatorname{rank} \mathfrak{t}$. Taking into account the inequality

$$\operatorname{nil rank}(\mathfrak{g}) \geq \operatorname{nil rank}(G) = \dim T = \operatorname{rank} \mathfrak{t} = r,$$

it is enough to find $x \in \mathfrak{t}$ such that

$$\dim \mathrm{Nil}(x, \mathfrak{g}) = r.$$

Since $\mathfrak{t}$ is abelian, and therefore nilpotent, this is equivalent to finding $a \in \mathfrak{t}$ for which

$$\mathrm{ad}(a)_{\mathfrak{g}/\mathfrak{t}}$$

is injective (XIII 5.7(a)).

Consider the characters α of T occurring in the representation of T induced by the adjoint representation of G. The structure theory of the semisimple group G *(Bible)*, 13, Theorem 1(a), and Theorem 3, Corollary 2, or more precisely the "big cell" of G, shows that the eigenspace of $\mathfrak{g}$ for the trivial character is exactly $\mathfrak{t}$. The other eigenspaces are one-dimensional, and their characters are precisely the roots of G relative to T. Here the big cell is built from T and the subgroups P_α, each isomorphic to the additive group $\mathbf{G}_a$, stable under T, and corresponding to the root characters.

The reductive center of G is the intersection of the kernels of the characters of T occurring in the adjoint representation (XII 4.8). Thus the assertion that G is adjoint says that its roots generate the lattice

$$M = \mathrm{Hom}(T, \mathbf{G}_m).$$

A familiar lemma from root theory says that every root belongs to a system of simple roots, hence to a basis of the group generated by the roots, and therefore to a basis of the dual lattice of T.We obtain the following.

Corollary 3.19. *Let G be an adjoint semisimple algebraic group over an algebraically closed field k, and let T be a maximal torus of G. For every root*

$$\alpha\colon T \longrightarrow \mathbf{G}_m$$

of G relative to T, the corresponding homomorphism

$$\alpha'\colon \mathfrak{t} \longrightarrow k$$

is nonzero.

This assertion is essentially equivalent to Theorem 3.18. Indeed, for $t \in \mathfrak{t}$, the endomorphism $\mathrm{ad}(t)$ is semisimple, and its eigenvalues on $\mathfrak{g}/\mathfrak{t}$ are precisely the numbers $\alpha'(t)$. Hence $\mathrm{ad}(t)_{\mathfrak{g}/\mathfrak{t}}$ is injective if and only if all the $\alpha'(t)$ are nonzero; and such a $t \in \mathfrak{t}$ exists if and only if every α' is nonzero.

Corollary 3.20. *Let S be a prescheme and let G be a smooth S-prescheme in groups of finite presentation over S, whose geometric fibers are connected reductive algebraic groups, that is, extensions of a semisimple group by a torus. For every $s \in S$ there is an open neighborhood U of s such that $G|_U$ admits a maximal torus.*[5]

We shall see in Exposé XVI that the hypothesis on G implies that G is affine over S and has locally constant reductive rank. Hence, by XII 4.7(c), G admits a reductive center Z, the quotient

$$G' = G/Z$$

is a smooth affine group over S whose reductive center is the identity group, and the maximal tori of G correspond bijectively to those of G'. Moreover, the geometric fibers of G' are plainly connected semisimple groups, and are adjoint by definition because their reductive center is trivial.

It is therefore enough to treat the case where G is semisimple and adjoint. By 3.18, the problem reduces to finding an open neighborhood U of s and a subgroup of type (C) of $G|_U$, or equivalently by 3.9(a), a Cartan subalgebra of $\mathfrak{g}|_U$. This is possible when $k(s)$ is infinite: by 3.7, $\mathfrak{g}$ satisfies condition (C_2) of 2.9, and 2.11(b) applies.

In fact, 3.20 remains true without assuming that $k(s)$ is infinite. By the preceding argument, it is enough to know that, for every adjoint semisimple group G over a finite field k, the Lie algebra $\mathfrak{g}$ contains a regular element. Chevalley proved this assertion using the properties of a Coxeter element of the Weyl group; see the appendix below by J.-P. Serre.

[5]This result, as well as 2.11 on which it rests, immediately generalizes to the case in which s is replaced by a finite subset of S contained in an affine open.

Remarks 3.21. a) Assertion 3.20 remains valid, with essentially the same proof, if G is replaced by a closed subprescheme in groups H, smooth over S, whose rank is everywhere the same as that of G (for example, a "parabolic subgroup" of G), provided that $k(s)$ is infinite. I do not know whether the infinitude hypothesis is again superfluous. It can at least be removed when H is parabolic: constructing the radical U of H and the quotient H/U reduces the question to the semisimple case. Unfortunately, the method of regular elements seems powerless here. One readily constructs examples—for instance, the projective group with its "standard" Borel subgroup—in which the Lie algebra $\mathfrak{h}$ of H contains no regular element.

b) When $k(s)$ is infinite, the proof of 3.20 in fact gives the following sharper statement, using 3.9(b): every maximal torus T_0 of G_s is induced by a maximal torus T over an open neighborhood of s. I do not know whether this remains true without the infinitude hypothesis. The evident difficulty is that the Lie algebra $\mathfrak{t}_0$ of T_0 need not contain a regular element of the Lie algebra $\mathfrak{g}_0$ of the fiber G_s.

An affirmative answer would imply the following assertion, which is known only when the residue field is infinite or when A is separated and complete. Let A be a local ring with residue field k, and let M be an "Azumaya algebra" over A, meaning an algebra that is a finite free A-module and whose reduction

$$M_0 = M \otimes_A k$$

is a central simple algebra over k. Let $D_0 \subset M_0$ be a commutative subalgebra separable over k, with

$$[M_0 : k] = [D_0 : k]^2.$$

Then there should exist a commutative subalgebra $D \subset M$ that is a direct summand as an A-module and satisfies

$$D \otimes_A k = D_0.$$

The datum of M is equivalent to that of a principal homogeneous bundle under the projective group $\mathrm{PGL}(n)_A$, and hence to an inner twisted form G of $\mathrm{PGL}(n)$. The maximal tori of G correspond bijectively to the commutative subalgebras $D \subset M$ that are étale of rank n over A.

c) Applying 3.20 to the centralizer of a subtorus Q of a reductive S-prescheme in groups G, one sees that every such Q is contained, Zariski locally, in a maximal torus of G.

4 A Digression on Borel Subgroups

Definition 4.1. *Let G be a smooth algebraic group over an algebraically closed field. A* Borel subgroup *of G is a smooth, solvable, connected algebraic subgroup that is maximal with these properties.*

When G is affine, this agrees with the terminology of *(Bible)*, 6, Definition 1. Observe at once that if Z is a connected smooth subgroup of G contained in its center (or, more generally, if Z is solvable and normal), then, for every Borel subgroup B of G, the image BZ of $B \times Z$ under the morphism

$$(b, z) \longmapsto bz$$

from $B \times Z$ to G is a smooth, solvable, connected subgroup of G containing B. It is therefore equal to B. Thus B contains Z, and consequently B is the inverse image of an algebraic subgroup B' of $G' = G/Z$; it is immediate that B' is a Borel subgroup of G'.

Taking

$$Z = \mathrm{Centr}(G^0)^0_{\mathrm{red}},$$

the quotient $G' = G/Z$ is affine by XII 6.1. Hence the Borel subgroups of G' are conjugate, and G'/B' is a projective variety for every such B' *(Bible)*, 6, Theorem 4. Consequently:

Proposition 4.2. *Let G be as in 4.1. Then the Borel subgroups of G are conjugate. If B is a Borel subgroup, then G/B is a projective variety. The maximal tori of B are maximal tori of G; and, when G is connected, the Cartan subgroups of B are Cartan subgroups of G.*

It remains to prove the last assertion, and we may plainly suppose that $G = G^0$. For Cartan subgroups, it follows from the analogous assertion in G' *(Bible)*, 6, Theorem 4, Corollary 4 and XII 6.6(e). The assertion for maximal tori follows from this one, since by XII 6.6(c) the maximal tori of a smooth algebraic group are precisely the maximal tori of its Cartan subgroups.

Corollary 4.3. *Suppose that G is connected. Then every element of G is contained in a Borel subgroup of G.*

This reduces to the same assertion in G', where it is well known *(Bible)*, 6, Theorem 5(d).

Corollary 4.4. *Let B be a Borel subgroup of G, let C be a Cartan subgroup of B, and let N be its normalizer in G. Then*

$$N \cap B = C.$$

Indeed,

$$N \cap B = \mathrm{Norm}_B(C),$$

so it remains to show that, when G is connected and solvable, a Cartan subgroup C is its own connected normalizer. With the preceding notation, C is the inverse image of a Cartan subgroup C' of G', and we are therefore reduced to the case in which G is affine. The normalizer of a Cartan subgroup is smooth (C being its own connected normalizer; see, for example, XII 6.6(c)). It is thus enough to check that C and N have the same k-valued points, which is exactly *(Bible)*, Theorem 6(d).

Definition 4.5. *Let G be a smooth prescheme in groups of finite presentation over a prescheme S. A* Borel subgroup *of G is a smooth subprescheme in groups B of G, of finite presentation, such that for every $s \in S$ the geometric fiber $B_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$.*

As is immediately verified, this notion is stable under base change and is local for the fpqc topology. Indeed, if k' is an algebraically closed extension of an algebraically closed field k, then an algebraic subgroup B of a smooth algebraic group G over k is a Borel subgroup of G if and only if $B_{k'}$ is a Borel subgroup of $G_{k'}$.

It follows from the definition that, if G is a smooth algebraic group over an arbitrary field k and B is a Borel subgroup of G, then G/B is a projective variety; every maximal torus T of B is a maximal torus of G; and the normalizer of T in B equals its centralizer C, which is a Cartan subgroup of G when G is connected.

Remarks 4.6. Unfortunately, it is no longer true in general that two Borel subgroups of G are locally conjugate for the fpqc topology. This can fail even when G is affine over S and S is the spectrum of the algebra of dual numbers over an algebraically closed field k.

As a consequence of this regrettable fact, let us point out that if G is a smooth, affine, connected algebraic group over a nonperfect field k, it is in general impossible to define naturally a homogeneous space D under G that plays the role of a flag variety, that is, of the variety of Borel subgroups of G. Over the algebraic closure $\bar{k}$ of k, such a space would have to be isomorphic to $G_{\bar{k}}/B$, where B is a Borel subgroup of $G_{\bar{k}}$.

Indeed, suppose that the quotient of $G_{\bar{k}}$ by its radical $\bar{R}$ is an adjoint semisimple group. The kernel of

$$G_{\bar{k}} \longrightarrow \operatorname{Aut}_{\bar{k}}(\bar{D})$$

is then the radical of $G_{\bar{k}}$. Consequently, if $\bar{D}$ came from a homogeneous space D under G, the radical $\bar{R}$ would come from a subgroup R of G. But examples are easily constructed for which $G_{\bar{k}}/\bar{R}$ is adjoint while $\bar{R}$ is not defined over k.

Under these conditions, the functor

$$\mathfrak{B}\colon (\mathrm{Sch}/S)^{\circ} \longrightarrow \mathrm{Ens}, \qquad \mathfrak{B}(S') = \{\text{Borel subgroups of } G_{S'}\},$$

is not representable by a smooth S-prescheme. From the infinitesimal point of view (III, §3), the failure of the conjugacy theorem is expressed by the fact that, if B is a Borel subgroup of the smooth algebraic group G, the cohomology group

$$H^1(B, \mathfrak{g}/\mathfrak{b})$$

may be nonzero.[6]

We shall see, on the other hand, in a later exposé that when G is semisimple, or more generally reductive, such unpleasant phenomena do not occur. These phenomena, together no doubt with the absence of good existence theorems, explain why Borel subgroups play only a relatively unobtrusive role in the scheme-theoretic study of general group schemes, whereas they dominate the theory of semisimple group schemes in the later exposés.

Proposition 4.8. *Let G be a smooth S-prescheme in groups of finite presentation with connected fibers, and let B be a Borel subgroup of G. Then B is its own normalizer and is a closed subprescheme of G.*

By XII 7.10, it is enough to prove that, over an algebraically closed field k, every element of $G(k)$ that normalizes B belongs to $B(k)$. When G is affine this is a fundamental result of Chevalley *(Bible)*, 9, Theorem 1; the general case reduces immediately to it by the reduction already used in 4.2.

Remarks 4.8.1. Definition 4.5 may be generalized by also introducing the notion of a *parabolic subgroup* of G. This means a subprescheme in groups P of G, smooth and of finite presentation over S, such that for every $s \in S$ the geometric fiber $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$, that is, contains a Borel subgroup of $G_{\bar{s}}$.

Proposition 4.8 extends to a parabolic subgroup P of G, by the same reduction to the known set-theoretic assertion. We note the following consequence (compare Exposé XVI). If P is a parabolic subgroup of G, then G/P is representable by a quasi-projective prescheme of finite presentation over S; here G is assumed to have connected fibers. Moreover, G/P is plainly smooth over S, and its geometric fibers are connected and proper. It follows readily from EGA III, 5.5.1 that

$$D = G/P$$

is proper, hence projective, over S. If its relative dimension is n, it is known that the inverse of the invertible sheaf $\Omega^n_{D/S}$ induces ample sheaves on the geometric fibers of D/S; hence, by EGA III, 4.7.1,

$$(\Omega^n_{D/S})^{-1}$$

is ample on D relative to S.

[6] Here $\mathfrak{b}$ denotes the Lie algebra of B.

It is readily seen, by reduction to the affine case and to an algebraically closed base field, that if

$$u\colon G \longrightarrow G'$$

is an epimorphism of smooth algebraic groups, then $u(B) = B'$ is a Borel subgroup of G' for every Borel subgroup B of G. We now consider the case in which this gives a bijective correspondence between the Borel subgroups of G and those of G'.

Proposition 4.9. *Let G and G' be smooth S-preschemes in groups of finite presentation with connected fibers, and let*

$$u\colon G \longrightarrow G'$$

be a faithfully flat, that is, surjective, homomorphism of groups.[7] *Put $N = \ker u$, and suppose that one of the following two conditions holds:*

a) *N is central in G.*

b) *$S = \operatorname{Spec}(k)$ for a field k, and, if $\bar{k}$ is an algebraic closure of k, then*

$$N_{\bar{k}} = \ker(u_{\bar{k}})$$

is contained in the radical of $G_{\bar{k}}$, that is, in its largest smooth, connected, solvable, normal subgroup.

Then the map

$$B' \longmapsto u^{-1}(B')$$

induces a bijection from the set of Borel subgroups of G' onto the set of Borel subgroups of G.

Case b) follows immediately from the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N, together with the fact that, over an algebraically closed field, the Borel subgroups of G contain its radical. The latter fact follows at once from the argument preceding 4.2.

We prove case a). By XII 7.12, the map

$$H' \longmapsto H = u^{-1}(H')$$

gives a bijection between the subpreschemes in groups H' of G' that are smooth and of finite presentation over S, have connected fibers, and at every $s \in S$ have the same reductive rank and the same nilpotent rank as G', and the subpreschemes in groups H of G having the analogous properties. Borel subgroups of G' and of G have these properties.

It remains to show that, when H' and H correspond, H' is a Borel subgroup of G' if and only if H is a Borel subgroup of G. By definition, this question reduces to the case where S is the spectrum of an algebraically closed field. Since N is central in G, and hence in H, the group H is solvable if and only if H' is solvable. Finally, the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N shows immediately that H' has the maximality property of Definition 4.1 if and only if H does. This completes the proof.

[7] The editors ask whether a reference should be added here.

Corollary 4.10. *With the notation of 4.9, if B' and B are corresponding Borel subgroups of G' and G, then*

$$\mathfrak{b} = \mathrm{Lie}(u)^{-1}(\mathfrak{b}'),$$

where $\mathfrak{g}, \mathfrak{g}', \mathfrak{b}, \mathfrak{b}'$ are the Lie algebras of G, G', B, B', respectively, and

$$\mathrm{Lie}(u)\colon \mathfrak{g} \longrightarrow \mathfrak{g}'$$

is the homomorphism induced by u.

This follows immediately from the definitions and the identity

$$u^{-1}(B') = B.$$

We can now prove the principal result of this section.

Theorem 4.11. *Let G be a smooth algebraic group over an algebraically closed field k, and let $\mathfrak{g}$ be its Lie algebra. Then $\mathfrak{g}$ is the union of the Lie algebras $\mathfrak{b}$ of the Borel subgroups B of G.*

We may plainly suppose that G is connected. Let R be the radical of G, and put $G' = G/R$. By 4.9(b) and 4.10, it is enough to prove the theorem for G'; thus we may suppose that G is semisimple.

Let Z be the center of G, which is its reductive center, and put $G' = G/Z$. The same argument, now using 4.9(a), reduces the theorem to G'. We may therefore suppose that G is semisimple and adjoint.

Let B be a Borel subgroup of G, let T be a maximal torus of B, hence of G, and let $\mathfrak{b}$ and $\mathfrak{t}$ be their Lie algebras. By 3.18, T is a subgroup of type (C) of G, that is, $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$. The union of the conjugates of $\mathfrak{t}$ is consequently dense in $\mathfrak{g}$ (XIII 5.1, (i)⇒(vii)). *A fortiori*, the union of the conjugates of $\mathfrak{b}$ is dense in $\mathfrak{g}$.

Let X be the closed subscheme of

$$G/B \times W(\mathfrak{g})$$

whose k-valued points are the pairs (gB, x) such that

$$x \in \mathrm{Ad}(g)\mathfrak{b}$$

(compare XIII 1). The morphism

$$\psi\colon X \longrightarrow W(\mathfrak{g})$$

induced by the second projection is proper because G/B is proper over k. We have just seen that it is dominant; it is therefore surjective. This proves 4.11.

The only result of this section that we shall use in the remainder of this exposé is the following corollary.

Corollary 4.12. *Let k be an infinite field, let G be a smooth algebraic group over k, and let T be a maximal torus of G. Write $\mathfrak{g}$ and $\mathfrak{t}$ for their Lie algebras. Let*

$$u\colon G \longrightarrow \mathrm{GL}(V)$$

be a linear representation, where V is finite-dimensional over k, and let

$$u'\colon \mathfrak{g} \longrightarrow \mathfrak{gl}(V)$$

be the induced representation of Lie algebras. Then the minimum of the nullity of $u'(x)$, for $x \in \mathfrak{g}$, is attained at an element $x \in \mathfrak{t}$.

We reduce immediately to the case where k is algebraically closed. We may plainly suppose that G is connected, and, after dividing G by $(\ker u)_{\mathrm{red}}$, that G is affine. Using 4.11 and the conjugacy theorem for the maximal tori of G, we reduce to the further case in which G is solvable.

Then G is a semidirect product $T \cdot U$, where U is the unipotent part of G, a smooth connected unipotent group *(Bible)*, 6, Theorem 3. Thus, as a vector space,

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{u}, \qquad \mathfrak{u} = \mathrm{Lie}(U).$$

By the Lie–Kolchin theorem *(Bible)*, 6, Theorem 1, the representation space V has a composition series by stable subspaces V_i, with one-dimensional quotients

$$W_i = V_i/V_{i+1}.$$

For every i there is therefore an induced representation

$$u_i \colon G \longrightarrow \mathrm{GL}(W_i) = \mathbb{G}_m$$

and a corresponding homomorphism of Lie algebras

$$u_i' \colon \mathfrak{g} \longrightarrow k.$$

For every $x \in \mathfrak{g}$, the nullity of $u'(x)$ is the number of indices i for which $u_i'(x) = 0$.

Since U is unipotent, every u_i is trivial on U, and hence every u_i' vanishes on $\mathfrak{u}$. If

$$x = t + v, \qquad t \in \mathfrak{t}, \quad v \in \mathfrak{u},$$

then $u_i'(x) = u_i'(t)$ for every i. Thus $u'(x)$ and $u'(t)$ have the same nullity, and 4.12 follows.

5 Relations Between Cartan Subgroups and Cartan Subalgebras

Applying 4.12 to the adjoint representation of G, we obtain:

Theorem 5.1. *Let G be a smooth algebraic group over an infinite field, let T be a maximal torus of G, and let $\mathfrak{g} \supset \mathfrak{t}$ be their Lie algebras. Then $\mathfrak{t}$ contains a regular element of $\mathfrak{g}$.*

Corollary 5.2. *Let G be a smooth S-prescheme in groups of finite presentation, and let $\mathfrak{g}$ be its Lie algebra.*

a) *Conditions* (C_0) *through* (C_2) *of 2.9 for $\mathfrak{g}$ are equivalent. In particular, if the infinitesimal rank of the fibers of $\mathfrak{g}$ at the points of S is locally constant, then, étale locally, $\mathfrak{g}$ admits a Cartan subalgebra and hence, by 3.9*(a)*, G admits a subgroup of type* (C)*.*

b) *Let H be a subprescheme in groups of G, smooth and of finite presentation over S, whose fibers are connected and at every $s \in S$ have the same reductive rank as G; for example, H may be a maximal torus or a Cartan subgroup of G. Let D be a subgroup of type* (C) *of G. Then*

$$H \subset D \iff \mathfrak{h} \subset \mathfrak{d}.$$

c) *Suppose that condition* (C_0) *holds, that is, the infinitesimal rank of G is locally constant. Let H be a subprescheme in groups of G, smooth and of finite presentation over S, with* *connected nilpotent fibers, and having at every point the same reductive rank as G; for example, H may be a maximal torus or a Cartan subgroup of G. Then H is contained in a unique subgroup D of type* (C) *of G.*

d) *If G admits a Cartan subgroup fpqc locally, then so does every subgroup D of type* (C) *of G.*

Proof. a) Assume (C_0), and let us prove (C_2), that is, that every quasi-regular section a of $\mathfrak{g}$ is regular. As usual, we reduce first to S affine noetherian and then, since the question is infinitesimal by 2.15(b), to S local Artinian. In this last case (C_0) is automatic. We may moreover suppose that the residue field k of S is infinite.

By 3.7, it is enough to show that $\mathfrak{g}$ admits a Cartan subalgebra. Let T be a maximal torus of G. By 5.1 there is a quasi-regular element t in the Lie algebra $\mathfrak{t}$ of T. We shall show that t is regular.

Consider the linear representation of T on $\mathfrak{g}$ induced by the adjoint representation of G. There is a finite set of characters $(u_i)_{i\in I}$ of T and a direct-sum decomposition

$$\mathfrak{g} = \bigoplus_{i\in I} \mathfrak{g}_i$$

into T-stable submodules such that T acts on $\mathfrak{g}_i$ through u_i (I, §4.7.3). Let

$$u_i' \colon \mathfrak{t} \longrightarrow A$$

be the homomorphism of Lie algebras induced by $u_i \colon T \to \mathbb{G}_m$, where A is the ring of S. Let u_{i0} and u_{i0}' be their reductions under $A \to k$. Put

$$I' = \{i \in I \mid u_{i0}' \neq 0\}, \qquad I'' = I \setminus I'.$$

The regularity of t_0 is expressed by

$$u_{i0}'(t_0) \neq 0 \qquad (i \in I'),$$

and therefore $u_i'(t)$ is invertible for $i \in I'$.

The Cartan subalgebra of $\mathfrak{g}_0$ defined by t_0, that is, the kernel of the semisimple endomorphism $\operatorname{ad}(t_0)$, is

$$\bigoplus_{i\in I''} (\mathfrak{g}_i)_0.$$

Set

$$\mathfrak{d} = \bigoplus_{i\in I''} \mathfrak{g}_i.$$

Then $\operatorname{ad}(t)$ is nilpotent on $\mathfrak{d}$, and is an automorphism of

$$\mathfrak{g}/\mathfrak{d} \simeq \bigoplus_{i\in I'} \mathfrak{g}_i.$$

By 2.6, t is regular.

b) This follows from 3.12(a). By 5.1, H satisfies the required hypothesis that every geometric fiber of $\mathfrak{h}$ contains a regular element of the corresponding fiber of $\mathfrak{g}$.

c) By b), it is enough to show that $\mathfrak{h}$ is contained in a unique Cartan subalgebra of $\mathfrak{g}$. By a), $\mathfrak{g}$ satisfies (C_2). We are therefore reduced to the following lemma.

Lemma 5.3. *Let $\mathfrak{g}$ be a Lie algebra over S, locally free of finite type as a module, and satisfying condition* (C_2) *of 2.9. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ such that $\mathfrak{h}$ is locally a direct summand, is locally nilpotent, and, for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$. Then $\mathfrak{h}$ is contained in a unique Cartan subalgebra of $\mathfrak{g}$.*

In the situation at hand, $\mathfrak{h}$ has these properties: it is locally nilpotent because H has nilpotent fibers, and the condition on regular elements follows from 5.1.

Since 5.3 is fpqc local, it is enough, at a point $s \in S$ with $\kappa(s)$ infinite, to find an open neighborhood U of s on which existence and uniqueness hold after every base change $S' \to S$ factoring through U. Choose a regular element of $\mathfrak{g} \otimes \kappa(s)$ lying in $\mathfrak{h} \otimes \kappa(s)$, and extend it to a section a of $\mathfrak{h}$ over an open neighborhood U of s. Condition (C_2) and 2.10 allow us to shrink U so that a is regular. We may thus suppose $U = S$.

Every Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ contains a, and hence equals

$$\mathfrak{d} = \mathrm{Nil}(a, \mathfrak{g})$$

by 2.6; this proves uniqueness. Conversely, $\mathfrak{h} \subset \mathfrak{d}$ because $\mathfrak{h}$ is locally nilpotent, which proves existence.

d) This is a special case of XII 7.9(d).

Corollary 5.4. *Let G be a smooth connected algebraic group over an algebraically closed field k. Let H be a connected algebraic subgroup such that $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$, or equivalently such that H contains a subgroup of type* (C) *of G. Then the number of conjugates of H containing a regular element of $G(k)$ equals the number of conjugates of $\mathfrak{h}$ containing a regular element of $\mathfrak{g}$.*

Indeed, let g be a regular element of $G(k)$, and let C be the unique Cartan subgroup of G containing g (XIII 2). The conjugates of H that contain g are exactly those that contain C (XIII 2.8(b)).

Similarly, let x be a regular element of $\mathfrak{g}$. If a conjugate $\mathfrak{h}'$ of $\mathfrak{h}$ contains x, then

$$\mathrm{ad}(x)_{\mathfrak{g}/\mathfrak{h}'}$$

is injective by XIII 5.4, and therefore

$$\mathfrak{h}' \supset \mathrm{Nil}(x, \mathfrak{g}) = \mathfrak{d}.$$

Thus the number of conjugates of $\mathfrak{h}$ containing x equals the number containing the Cartan subalgebra $\mathfrak{d}$.

The subalgebra $\mathfrak{d}$ is the Lie algebra of a subgroup D of type (C) of G. We may suppose that $C \subset D$. It remains to show that

$$H \supset C \iff \mathfrak{h} \supset \mathfrak{d}.$$

By XIII 5.5, the inclusion $\mathfrak{h} \supset \mathfrak{d}$ implies $H \supset D$, and *a fortiori* $H \supset C$. Conversely, by 5.1 the Lie algebra $\mathfrak{t}$ of C contains a regular element x of $\mathfrak{g}$. Hence $H \supset C$ implies $x \in \mathfrak{h}$, and, as observed above, this implies $\mathfrak{h} \supset \mathfrak{d}$. This completes the proof.

Let G be as in 5.2, and suppose that the infinitesimal rank of its fibers is locally constant, that is, condition (C_0) holds. By 5.2(c), there is a morphism of functors on $(\mathrm{Sch}/S)^\circ$,

$$\mathcal{C} \longrightarrow \mathcal{D},$$

where

$$\mathcal{C}(S') = \{\text{Cartan subgroups of } G_{S'}\},$$
$$\mathcal{D}(S') = \{\text{subgroups of type (C) of } G_{S'}\}.$$

By 3.10 and 5.2(a), the functor $\mathcal{D}$ is represented by a smooth quasi-projective prescheme over S. Let D be the universal subgroup of type (C) of $G_{\mathcal{D}}$. One may then define the functor

$$\mathcal{C}_D\colon (\mathrm{Sch}\,/\mathcal{D})^\circ \longrightarrow \mathrm{Ens}$$

for the $\mathcal{D}$-group D, exactly as $\mathcal{C}$ was defined for the S-group G.

Proposition 5.5. *Under the preceding hypotheses, regard $\mathcal{C}$ as a functor over $\mathcal{D}$. Then $\mathcal{C}$ is $\mathcal{D}$-isomorphic to the functor $\mathcal{C}_D$ of Cartan subgroups of D.*

This follows immediately from the next corollary.

Corollary 5.6. *Let G be a smooth S-prescheme in groups of finite presentation, and let D be a subgroup of type* (C) *of G. There is a bijective correspondence between the Cartan subgroups of G contained in D and the Cartan subgroups of D. More precisely, for a subgroup $H \subset D$, H is a Cartan subgroup of G if and only if it is a Cartan subgroup of D.*

This is a special case of XII 7.9(c), taking into account that, over an algebraically closed field, every subgroup of type (C) of G contains a Cartan subgroup of G.

For the next section, the principal result obtained here is 5.2(c) with H a Cartan subgroup of G. It permits the formulation of 5.5 and thus gives a useful dévissage of $\mathcal{C}$.

6 Applications to the Structure of Algebraic Groups

Theorem 6.1. *Let G be a smooth algebraic group over a field k. Consider the scheme $\mathcal{T}$ of maximal tori of G, isomorphic to the scheme $\mathcal{C}$ of Cartan subgroups of G. This is a homogeneous space under G^0 and a smooth affine connected algebraic scheme* (XII 7.1(d))*. Then $\mathcal{C}$ is a rational variety: its field of rational functions is a purely transcendental extension of k.*

We first prove the theorem when k is infinite. We may plainly suppose that G is connected, since $\mathcal{T}$, and hence $\mathcal{C}$, is unchanged when G is replaced by G^0. Moreover, by XII 7.6, $\mathcal{C}$ is unchanged when G is divided by a central subgroup. We may therefore first divide by the center of G and suppose that G is affine (XII 6.1), and then divide by its reductive center (XII 4.1 and 4.4) and suppose that the reductive center of G is trivial (XII 4.7(b)).

We proceed by induction on $n = \dim G$, assuming the result for dimensions $n' < n$. If G is nilpotent, then $\mathcal{C}$ consists of a single k-rational point and the assertion is trivial. Otherwise, the Lie algebra of G is not nilpotent (1.3). Its Cartan subalgebras, and therefore the subgroups of type (C) of G, have some dimension $n' < n$. Consider the morphism

$$\mathcal{C} \longrightarrow \mathcal{D}$$

of 5.5. By 3.10, since the infinite-field Lie algebra $\mathfrak{g}$ contains a regular element, $\mathcal{D}$ is a rational variety. Thus its function field K is purely transcendental over k.

Let x be the generic point of $\mathcal{D}$. By 5.5, the fiber of $\mathcal{C}$ at x is the scheme of Cartan subgroups of a smooth connected algebraic group D_x over $K = \kappa(x)$, namely the generic subgroup of type (C) of G. The function field L of $\mathcal{C}$ is therefore the function field of $\mathcal{C}_{D_x}$. Since $\dim D_x = n' < n$, the induction hypothesis makes L purely transcendental over K, and hence, by transitivity, purely transcendental over k.

Suppose now that k is finite. A different argument is required. We may again suppose that G is affine and connected. Since k is perfect, the radical of $G_{\bar{k}}$ descends to a subgroup R of G. Suppose first that $R \neq G$, so that G is not solvable, and let

$$u\colon G \longrightarrow G' = G/R$$

be the canonical morphism. By XII 7.1(e), it induces

$$v\colon \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \qquad C \longmapsto u(C).$$

Let x be the generic point of $\mathcal{C}_{G'}$, and put $K = \kappa(x)$. The fiber $v^{-1}(x)$ is the scheme of Cartan subgroups of G_K whose image in G'_K is the generic Cartan subgroup C'_x of G'. By XII 7.9(c), it is also the scheme of Cartan subgroups of

$$H = u_K^{-1}(C'_x).$$

The field K is infinite. The infinite-field case just proved therefore shows that the function field L of $\mathcal{C}_G$, which is also that of $\mathcal{C}_H$, is purely transcendental over K.It remains to prove that K is purely transcendental over k, and we are thus reduced to the case in which G is semisimple. Dividing by its reductive center, we may further suppose that G is adjoint. Then 3.18 gives $\mathcal{C}_G \simeq \mathcal{D}_G$, and by 3.10 it is enough to show that $\mathfrak{g}$ has a regular point. As noted in 3.20, this is an unpublished result of Chevalley.[8]

It remains only to treat the case in which k is finite and G is connected, affine, and solvable. In fact, the next result holds over an arbitrary field.

Corollary 6.2. *Let G be a smooth solvable algebraic group over a field k. Then the variety $\mathcal{C}$ of Cartan subgroups of G is isomorphic to an affine space* $\operatorname{Spec} k[t_1, \dots, t_N]$.

We may suppose that G is connected and affine. Let G_∞ be the terminal group in the descending central series

$$G_0 = G, \qquad G_{i+1} = [G, G_i].$$

Thus G_∞ is the smallest invariant algebraic subgroup of G for which G/G_∞ is nilpotent. Let C be a Cartan subgroup of G, which exists by 1.1. Its image in G/G_∞ is a Cartan subgroup and hence is all of G/G_∞. Consequently the morphism

$$G_\infty \longrightarrow \mathcal{C}, \qquad g \longmapsto \operatorname{Ad}(g)C$$

is an epimorphism and identifies

$$\mathcal{C} \simeq G_\infty/(N \cap G_\infty),$$

where N is the normalizer of C in G. In fact $N = C$, as recalled in 4.4, but that is immaterial here.

The group $U = G_\infty$ is smooth, connected, and unipotent: over $\bar{k}$ it lies in the unipotent part of G, by the known structure of smooth affine solvable groups *Bible* 6, Theorem 3. If k is perfect, as in the finite-field case needed for 6.1, it follows easily that U is k-unipotent: it has a composition series by algebraic subgroups U_i with $U_i/U_{i+1} \simeq \mathbb{G}_a$.

Rosenlicht proved that a group of the form G_∞ above is k-unipotent without any restriction on k (M. Rosenlicht, *Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica, 1963, pp. 97–120, Theorem 4, Corollary 2). We admit this substantially more delicate result and apply the following lemma.

Lemma 6.3. *Let U be a smooth connected algebraic group over a field k, and let $X = U/V$ be a homogeneous space under U having a k-rational point. If U is k-unipotent, then, as a k-scheme, X is isomorphic to an affine space* $\operatorname{Spec} k[t_1, \dots, t_N]$.

Choose a composition series

$$U = U_0 \supset U_1 \supset \cdots \supset U_n = (e)$$

[8] See the Appendix by J.-P. Serre.

by smooth connected normal subgroups of U, with $U_i/U_{i+1} \simeq \mathbb{G}_a$. The groups

$$K_i = U_i V$$

are algebraic subgroups of U. They need not be smooth or connected when V is not, but this does not matter, and K_{i+1} is normal in K_i. The canonical epimorphism

$$U_i/U_{i+1} \longrightarrow K_i/K_{i+1}$$

shows that K_i/K_{i+1} is either the unit group or isomorphic to $\mathbb{G}_a/H_i$, where H_i is a finite subgroup of $\mathbb{G}_a$. Rosenlicht's Theorem 2 in the cited paper gives

$$K_i/K_{i+1} \simeq \mathbb{G}_a;$$

this last assertion is immediate when k is perfect.

Put $X_i = U/K_i$. We prove inductively that every X_i is an affine space. If K_i/K_{i+1} is trivial, then $X_i = X_{i+1}$. Otherwise X_{i+1} is a principal $\mathbb{G}_a$-bundle over X_i. Since X_i is affine, the bundle is trivial, so

$$X_{i+1} \simeq X_i \times \mathbb{G}_a.$$

The induction proves the lemma. It proves 6.2 and thereby completes the proof of 6.1.

Corollary 6.4. *Let G be a smooth algebraic group over an infinite field k. Then the set of k-rational points of $\mathcal{C}$, in the notation of 6.1, is Zariski dense. The union of the Cartan subgroups of G is dense in G.*

The first assertion holds for every rational variety over an infinite field and is, for us, the most important arithmetic consequence of the rationality result. The second follows from the first and the density theorem XIII 2.1.

Corollary 6.5. *Let G be a smooth connected algebraic group over k. Then the variety Z of regular semisimple points of G* (XIII 3.5) *is unirational. In particular, if k is infinite, its k-rational points are dense in Z.*

Indeed, Z is an open subscheme of a torus over $\mathcal{C}$. Its function field L is therefore the function field of a torus over the function field K of $\mathcal{C}$, namely the generic maximal torus of G. Thus L is unirational over K by XIII 3.4. Since 6.1 makes K purely transcendental over k, L is unirational over k.

Corollary 6.6. *Let G be a smooth connected algebraic group over k, and let H be the subgroup generated by the subvariety Z of regular semisimple points. Equivalently* (XII 8.2)*, H is the smallest invariant algebraic subgroup of G for which G/H has reductive rank zero; over $\bar{k}$, G/H is an extension of an abelian variety by a smooth connected unipotent group. Then H is unirational.*

In particular, suppose $G = H$; equivalently (XII 8.4)*, suppose G is affine and there is no nontrivial homomorphism $G_{\bar{k}} \to \mathbb{G}_a$. Then G is unirational. If, in addition, k is infinite, $G(k)$ is dense in G.*

This follows immediately from 6.5. Indeed, if Z_i are smooth connected unirational k-preschemes and $u_i \colon Z_i \to G$ are morphisms, then the algebraic subgroup of G generated by the u_i is unirational ($\mathrm{VI_B}$ 7.1).

An interesting special case of either 6.5 or 6.6 is the following.

Corollary 6.7. *Let G be a smooth connected affine algebraic group of unipotent rank zero. Then G is unirational.*

The conclusion of 6.4 may be sharpened as follows.

Corollary 6.8. *Let G be a smooth connected algebraic group over an infinite field k, and let C be a Cartan subgroup of G. Then the union of the conjugates of C by regular semisimple elements of $G(k)$ is dense in G.*

This follows at once from 6.4 and XIII 3.6, which says that the morphism

$$\varphi\colon Z \times C \longrightarrow G, \qquad \varphi(t,c) = \mathrm{Ad}(t)c,$$

is dominant. The same result also implies the next corollary, without any assumption that k is infinite.

Corollary 6.9. *Let G be a smooth connected algebraic group over k, and let H be a smooth connected algebraic subgroup having the same reductive rank and the same nilpotent rank as G. Equivalently, over $\bar{k}$, the group $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. If H is unirational, so is G. If $H(k)$ is dense in H, then $G(k)$ is dense in G.*

The morphism

$$\varphi\colon Z \times H \longrightarrow G, \qquad \varphi(t,h) = \mathrm{Ad}(t)h,$$

is dominant. By 6.5, Z is unirational, and by hypothesis so is H, hence so are $Z \times H$ and G. The density assertion is proved in the same way.

We recover the following well-known result, due to Chevalley in characteristic zero and to Rosenlicht in characteristic $p > 0$.

Corollary 6.10. *Let G be a smooth connected affine algebraic group over a perfect field k. Then G is unirational. If k is also infinite, $G(k)$ is dense in G.*

By 1.1, G has a Cartan subgroup C. By 6.9, it is enough to prove that C is unirational. Since k is perfect, Galois descent from the algebraically closed case *Bible* 6, Theorem 2 gives

$$C = T \times C_u,$$

where T is the maximal torus of C and C_u is smooth, connected, and unipotent. The torus T is unirational by XIII 3.4. Moreover, C_u is k-unipotent because k is perfect, so 6.3 applies to C_u.

Remarks 6.11. (a) Rosenlicht gave examples of twisted forms of $\mathbb{G}_a$ over a nonperfect field having only finitely many rational points; *a fortiori*, they are not unirational. Chevalley, on the other hand, gave an example of a torus over a field of characteristic zero that is not rational. By contrast, Chevalley's theory of semisimple groups shows that, over an algebraically closed field, every smooth connected affine algebraic group is rational. In any case, the question of unirationality arises only for affine algebraic groups: by Chevalley's structure theorem, a unirational algebraic group is necessarily affine.

(b) With the notation of 6.6, it is tempting to seek a criterion for the unirationality of G in terms of G/H, which is unipotent when G is affine. Clearly G/H must be unirational; is that condition also sufficient? An example of Rosenlicht shows that a smooth connected unipotent algebraic group U may be rational without being k-unipotent.

(c) Over a finite field k, it would be interesting to study questions of "density" such as the following one, raised by Rosenlicht: if G is a smooth connected algebraic group over k, is G generated by its Cartan subgroups?[9]

[9] This question was subsequently answered in the affirmative by Steinberg.

Editors' note. The editors did not identify this result in Steinberg's *Collected Papers*; here is a proof based on the Bruhat decomposition.

Let G/k be semisimple over the finite field k, and let $G^\sharp$ be the subgroup generated by its k-subtori. The question is stable under central isogeny and restriction of scalars, so, as in Lemma 1 of the Appendix below, it is enough to treat the geometrically simple case. The group G is quasi-split by Lang's theorem and admits a Killing pair (T, B). By definition $T \subset G^\sharp$, hence T normalizes $G^\sharp$. The big cell

$$R_u(B^-)\, T\, R_u(B)$$

shows that it suffices to prove $R_u(B) \subset G^\sharp$. Let S be the maximal split k-torus of T. For the relative root system $\Phi(G, S)$, choose a basis Δ_k, and denote by $U(\alpha)$ the unipotent subgroup associated with α (A. Borel, *Linear Algebraic Groups*, second edition, 1991, Proposition 21.9). Since $R_u(B)$ is generated by the $U(\alpha)$, $\alpha \in \Delta_k$, it is enough to show $U(\alpha) \subset G^\sharp$.

The classification supplies a quasi-split semisimple subgroup G_α of type A_1, ${}^2A_1$, ${}^3A_1$, or ${}^2A_2$, with

$$U(\alpha) \subset G_\alpha \subset G.$$

Thus it is enough to consider $G = \mathrm{PGL}_2$ or $G = \mathrm{SU}_3(K)$, where K/k is the unique quadratic extension.

For PGL_2, since $G^\sharp$ contains the standard split torus T, the possibilities up to $G(k)$-conjugacy are

$$G^\sharp = T, \qquad G^\sharp = B, \qquad G^\sharp = \mathrm{PGL}_2\,.$$

The first is excluded because $G^\sharp$ contains the k-torus $R_{K/k}(\mathbb{G}_m)/\mathbb{G}_m$. The preceding discussion also shows that $G^\sharp = B$ forces $G^\sharp = \mathrm{PGL}_2$. Hence $G^\sharp = \mathrm{PGL}_2$.

For $G = \mathrm{SU}_3(K)$, one has $\Delta_k = \{\alpha\}$, and the possibilities up to conjugacy are

$$\begin{aligned} G^\sharp = T, \quad B, \quad U(2\alpha) \rtimes T,& \\ \langle U(2\alpha), U(-2\alpha), T\rangle = \mathrm{SL}_2 \cdot T, \quad \text{or } G.& \end{aligned}$$

The case B is excluded as for PGL_2. The case $\mathrm{SL}_2 \cdot T$ is excluded because $G^\sharp$ contains the k-torus

$$R^1_{K/k}\Big(R^1_{K_3/K}(\mathbb{G}_m)\Big)\,,$$

where K_3/K is the degree-three extension, and the isogeny class of this torus is irreducible. The same torus excludes T and $U(2\alpha) \rtimes T$. Hence $G^\sharp = G$.

For Rosenlicht's question one may reduce to the affine case by dividing by the center. In the semisimple case, an affirmative answer would also follow from a strengthening of Chevalley's existence result for regular points mentioned in 3.20: one would need a regular element of $\mathfrak{g}$ outside $\mathfrak{h} = \mathrm{Lie}(H)$, where $H \neq G$ is a prescribed smooth algebraic subgroup of an adjoint semisimple group G over the finite field k.

7 Appendix: Existence of Regular Elements over Finite Fields

by J.-P. Serre

Throughout the Appendix, k denotes a finite field, $\bar{k}$ its algebraic closure, and $\mathcal{G} = \mathrm{Gal}(\bar{k}/k)$. If $q = \mathrm{Card}(k)$, then $\mathcal{G}$ is topologically generated by the Frobenius element

$$F\colon x \longmapsto x^q.$$

We shall prove the following theorem.[10]

[10] The theorem is due to Chevalley. The editor wishes to express his gratitude to American Express which, by misplacing a trunk of Chevalley's manuscripts, obliged him to reconstruct the proof.

Theorem. *Let G be an adjoint semisimple group defined over k, and let $\mathfrak{g}$ be its Lie algebra. Then the k-Lie algebra $\mathfrak{g}(k)$ contains a regular element.*

Remarks. (1) Recall that "adjoint" means that the center of G, as a subgroup scheme, is trivial. In view of the Chevalley seminar, this also means that, for a maximal torus T of G, the character group $X(T)$ of $T_{\bar{k}}$ is generated by the set R of roots. In particular, the rank of $\mathfrak{g}$ equals $\dim T$, the rank of G.

(2) The editor does not know whether the adjoint hypothesis is indispensable.

Lemma 1. *It is enough to prove the theorem when G is geometrically simple.*

Here geometrically simple means that $G_{\bar{k}}$ has no smooth normal subgroup scheme other than $G_{\bar{k}}$ and (e); equivalently, the associated root system is irreducible.

We may first suppose that G is indecomposable over k. Then $G_{\bar{k}}$ is a product of geometrically simple components S, permuted transitively by $\mathcal{G}$. If $\mathcal{H} \subset \mathcal{G}$ fixes one component S, and K is its fixed field, then S is defined over K, and the standard restriction-of-scalars argument gives

$$G = R_{K/k}(S).$$

Similarly,

$$\mathfrak{g} = R_{K/k}(\mathfrak{s}),$$

where $\mathfrak{s}$ is the Lie algebra of S. If the theorem holds for S, there is an element

$$x \in \mathfrak{s}(K) = \mathfrak{g}(k)$$

regular in $\mathfrak{s}$, and one checks immediately that it is regular in $\mathfrak{g}$. This proves the lemma.

Henceforth assume that G is geometrically simple, and choose a maximal torus T of G. Let X be the character group of $T_{\bar{k}}$, let R be its root system, W its Weyl group, and E the group of automorphisms of X preserving R. Then W is normal in E.

For another maximal torus T', write X', R', W', E' for the corresponding objects. If $y \in G(\bar{k})$ satisfies

$$yT_{\bar{k}}y^{-1} = T'_{\bar{k}},$$

then $\operatorname{int}(y)$ identifies (X', R', W', E') with (X, R, W, E). Replacing y changes this identification by an element of W.

Frobenius acts on X' while preserving R', and thus defines an element $f_{T'} \in E'$. Put $f = f_T$. After the above identification $E' \simeq E$, the element $f_{T'}$ becomes an element $f' \in E$, well defined up to W-conjugacy. We now compare f' with f.

Lemma 2. *One has $f' \equiv f \pmod{W}$. Conversely, every $f' \in E$ satisfying this condition arises from a maximal torus T' of G.*

Put $\bar{T} = T_{\bar{k}}$ and $\bar{T}' = T'_{\bar{k}}$, and choose $y \in G(\bar{k})$ such that

$$y\bar{T}y^{-1} = \bar{T}'.$$

Since $y^q\bar{T}y^{-q} = \bar{T}'$, the element

$$n = y^{-1}y^q$$

belongs to the normalizer $N(T)(\bar{k})$. The action of f' on the points of $\bar{T}$ is

$$\begin{aligned} f'(t) &= y^{-1}(yty^{-1})^q y \\ &= y^{-1}y^q\, t^q\, y^{-q}y = nt^qn^{-1}. \end{aligned}$$

If $w \in W$ is represented by n, then f' and $f \circ w$ have the same action on $\bar{T}(\bar{k})$, hence on its characters. Thus $f' = f \circ w$, and $f' \equiv f \pmod{W}$.

Conversely, represent a given $w \in W$ by $n \in N(T)(\bar{k})$. Lang's theorem gives

$$n = y^{-1} y^q$$

for some $y \in G(\bar{k})$. Then $y\bar{T}y^{-1}$ is defined over k, and the preceding calculation shows that its corresponding element is $f \circ w$.

Lemma 3. *Let X, R, W, E be as above, with R irreducible, and let $\varphi \in E/W$. There exist a representative $f \in E$ of φ and roots $\theta_1, \dots, \theta_n$ such that:*

1. *$(\theta_1, \dots, \theta_n)$ is a basis of X;*
2. *R is the union of the orbits of the θ_i under the powers of f: every $a \in R$ has the form $a = f^m \theta_i$ for suitable m and i.*

The proof will be given below. We first need a lemma of linear algebra.

Lemma 4. *Let V be a vector space over k, and let $(\theta_1, \dots, \theta_n)$ be a basis of the dual of $V \otimes_k \bar{k}$. Then there is an $x \in V$ such that $\theta_i(x) \neq 0$ for every i.*

Let V^* be the dual of V, and let W_i be the $\bar{k}$-subspace of $V^* \otimes_k \bar{k}$ generated by the Galois conjugates of θ_i. It descends to a k-subspace, again denoted W_i. The natural map

$$W_1 \otimes \cdots \otimes W_n \longrightarrow \bigwedge^n V^*$$

is nonzero: after extension to $\bar{k}$, it contains $\theta_1 \wedge \cdots \wedge \theta_n \neq 0$. Hence V^* has a basis $u_1, \dots, u_n$ with $u_i \in W_i$. Choose $x \in V$ with $u_i(x) \neq 0$ for all i, for example with $u_i(x) = 1$. If $\theta_i(x) = 0$, every Galois conjugate of θ_i vanishes at x, and so does u_i, a contradiction.

End of the proof of the theorem. By Lemmas 2 and 3, choose a maximal torus T whose element f has the properties of Lemma 3. Let Y be the dual of X. The Lie algebra of T satisfies

$$\mathfrak{t}(\bar{k}) \simeq Y \otimes_{\mathbb{Z}} \bar{k},$$

compatibly with the Galois action. Put $V = \mathfrak{t}(k)$. An element $x \in V$ is regular precisely when it is annihilated by no root $\alpha \in R$, viewed as a linear form in $V^* \otimes_k \bar{k}$.

Lemma 4 supplies $x \in V$ annihilated by none of the θ_i. Every root is a Galois conjugate of one of the θ_i, by property (2) of Lemma 3. Hence no root annihilates x, and x is regular. This proves the theorem.

Proof of Lemma 3. The proof uses Coxeter transformations. Let $a_1, \dots, a_n$ be a simple system of roots of R, and let r_i be the reflection corresponding to a_i. Put

$$c = r_1 \cdots r_n.$$

Then $c \in W$. Although c depends on the simple system and on its ordering, its conjugacy class does not. It is called the Coxeter element. We shall use, without proof, the fact that 1 is not an eigenvalue of c.

Lemma 5. Put

$$\theta_i = r_n \cdots r_{i+1}(a_i).$$

Then:

(a) the θ_i form a basis of the group X generated by the roots;

(b) $\theta_i > 0$ and $c(\theta_i) < 0$ for every i;

(c) every root $a \in R$ such that $a > 0$ and $c(a) < 0$ is one of the θ_i;

(d) R is the union of the orbits of the θ_i under the powers of c.

Indeed,

$$\theta_i = a_i + \sum_{j>i} m_{ij} a_j, \qquad m_{ij} \in \mathbb{Z},$$

which proves (a). Assertions (b) and (c) follow from the observation that r_i preserves the sign of every root other than $\pm a_i$, and reverses the sign of $\pm a_i$.

For (d), an orbit of c cannot consist entirely of positive roots, nor entirely of negative roots. Otherwise the sum of the roots in that orbit would give a nonzero element of X fixed by c, contrary to the fact that 1 is not an eigenvalue. Hence every orbit contains a root $a > 0$ for which $c(a) < 0$, and (c) applies.

Remark. The proof was recalled only for the reader's convenience; one could instead refer to the standard texts on Coxeter transformations, for example Koszul, Séminaire Bourbaki 1959/1960, Exposé 191. They also show that all orbits of c have the same number h of elements and each contains exactly one θ_i. In particular,

$$h = \frac{\mathrm{Card}(R)}{n}.$$

We return to Lemma 3 and distinguish three cases.

(1) Suppose that $\varphi \in E/W$ is the identity. We must choose $f \in W$; take $f = c$. Lemma 5 shows that this works.

(2) Suppose that R has type A_n, where $n \geq 2$ is even, and that φ is the unique nontrivial element of E/W. Then $-1 \notin W$, and we take

$$f = -c.$$

The Coxeter element c has order $h = n+1$, which is odd. If $a \in R$, Lemma 5 gives $a = c^m \theta_i$ for suitable m, i. Adding h to m if necessary, we may suppose that m is even. Then

$$a = (-c)^m \theta_i = f^m \theta_i.$$

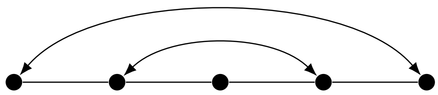
Thus the orbits of the θ_i under f fill R.

(3) Suppose that $\varphi \in E/W$ is nontrivial and that R has one of the following types:

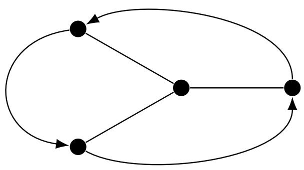
$$A_n \ (n \geq 3 \text{ odd}), \qquad D_4, \qquad D_n \ (n \geq 5), \qquad E_6.$$

These, together with the even A_n case treated separately above, are exactly the cases in which E/W is nontrivial. The following diagrams record the corresponding Dynkin-diagram automorphisms.

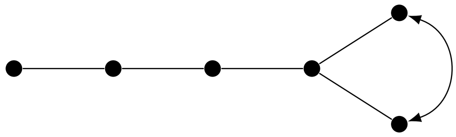
$A_n,\ n$ odd ≥ 3



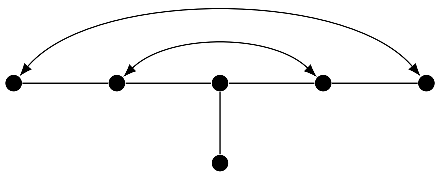
D_4



$D_n,\ n \geq 5$



E_6.



Let S be a simple system of roots, not yet ordered. The group E is the semidirect product of W with the group Ψ of permutations of S preserving the Cartan matrix, or equivalently the Dynkin diagram. Thus $E/W \simeq \Psi$, and φ corresponds to some $\psi \in \Psi$.

Inspection of the diagrams shows that each ψ-orbit in S consists of pairwise orthogonal roots. This fails for A_n with n even, which is why that case was separated. If σ is such an orbit, the reflections r_a, $a \in \sigma$, commute; write

$$\rho_\sigma = \prod_{a \in \sigma} r_a.$$

Plainly ρ_σ commutes with ψ.

Choose a total order on S for which every ψ-orbit is a segment, and write $S = \{a_1, \dots, a_n\}$. Let $\theta_1, \dots, \theta_n$ and $c = r_1 \cdots r_n$ be defined as above. The element c is a product of the ρ_σ, with the orbits in some order, so it commutes with ψ. Put

$$f = c\psi.$$

The automorphism ψ permutes the θ_i. Indeed, $\psi(\theta_i) > 0$, and, since c commutes with ψ,

$$c(\psi(\theta_i)) = \psi(c(\theta_i)) < 0.$$

Lemma 5(c) therefore gives $\psi(\theta_i) = \theta_j$ for some j.

Finally, if $a \in R$, Lemma 5 gives $a = c^m \theta_i$ for some m, i. Hence

$$a = f^m(\psi^{-m}\theta_i) = f^m\theta_j$$

for a suitable j. Thus f and the θ_i satisfy both conditions of Lemma 3, completing its proof.

Remark. Except in case (2), one can prove that every orbit of f has exactly

$$h = \frac{\mathrm{Card}(R)}{n}$$

elements and contains exactly one θ_i. In case (2), some of the θ_i are superfluous.

Exposé XV
Complements on Sub-tori of a Group Prescheme
Application to Smooth Groups

by M. Raynaud[11,12]

0. Introduction

This Exposé supplements and partly recasts Exposés XI and XII; knowledge of Exposés XIII and XIV is not indispensable. Continuing the effort begun in XII, we shall work with S-group preschemes that are not necessarily affine and not necessarily separated over S.

Sections 1, 2, 3, and 4 are devoted to the study of sub-tori of a group prescheme. We obtain infinitesimal lifting theorems (ğ2) and a global lifting theorem (ğ4), in which points of finite order (ğ1) play an essential role.

Sections 5, 6, and 7 are independent of the preceding sections. The consideration of infinitesimal neighbourhoods leads to the representability of the functor of smooth subgroups equal to their connected normalizer (ğ5). In ğğ6 and 7 we turn more particularly to Cartan subgroups.

Finally, in ğ8 we give a necessary and sufficient condition for the functor of sub-tori of a smooth group, or the functor of maximal tori, to be representable.

1. Lifting finite subgroups

1.1. Finite, smooth, central subgroups of multiplicative type

Proposition 1.1.— *Let S be an affine scheme, let S_0 be a closed subscheme of S defined by a square-zero ideal, let G be an S-group prescheme, and let H_0 be a subgroup scheme of $G_0 = G \times_S S_0$ that is smooth and of finite type over S_0, and of multiplicative type. A subgroup scheme of G, of multiplicative type, lifting H_0 exists if and only if there exists a subscheme H' of G, flat over S, lifting H_0.*

The necessity of the condition is clear; let us prove its sufficiency. The group H_0 of multiplicative type is quasi-isotrivial (X 4.5). By Exposé X 2.1, there exist an S-group H

[11] Source-edition note: xy version dated 1 December 2008.

[12] This Exposé and the two following ones (Exposés XVI and XVII) do not correspond to oral lectures delivered at the seminar. With some additional material, they develop the substance of concise manuscript notes written by A. Grothendieck in 1964 for the present seminar.

of multiplicative type and an S_0-isomorphism of groups

$$u_0 : H \times_S S_0 \xrightarrow{\sim} H_0.$$

Since H' is flat over S and $H' \times_S S_0$ is of finite presentation over S_0 (Exposé IX 2.1(b)), H' is of finite presentation over S. Its fibres are smooth, so H' is smooth over S (EGA IV 17.5.1). Since S is affine, u_0 therefore lifts to an S-morphism of preschemes

$$u : H \longrightarrow H'.$$

It then follows from Exposé III 2.1 and Exposé IX 3.1 that the composite

$$v_0 : H \times_S S_0 \xrightarrow{\sim} H_0 \longrightarrow G_0$$

also lifts to an S-homomorphism

$$v : H \longrightarrow G.$$

Since v_0 is an immersion, so is v. The image of H under v is therefore a subgroup scheme of G, of multiplicative type, lifting H_0.

Proposition 1.2.— *Let S be a prescheme, let S_0 be a subprescheme of S defined by a locally nilpotent sheaf of ideals, let G be an S-group prescheme flat and of finite presentation over S, and let H_0 be a subgroup scheme of $G_0 = G \times_S S_0$ that is smooth, finite over S_0, of multiplicative type, and central. Then there exists a unique subgroup scheme H of G, of multiplicative type, lifting H_0. Moreover, H is central. (See XVII, Appendix III, 1.)*

Proposition 1.2 bis.— *Let S, G, S_0, G_0 be as above, let H be an S-group scheme of multiplicative type, smooth and finite over S, and let $u_0 : H_0 = H \times_S S_0 \to G_0$ be a central homomorphism. Then u_0 lifts uniquely to a homomorphism $u : H \to G$. Moreover, u is central.*

The existence of the lifting u in Proposition 1.2 bis follows readily from Proposition 1.2 by considering the graph of u_0. The lifting u is unique and central by Exposé IX 3.4 and Exposé IX 5.1.

Proof of Proposition 1.2. The uniqueness of H, and the fact that H is central, follow from Exposé IX 5.6(b) and Exposé IX 3.4 bis. In view of uniqueness, to prove existence we may assume that S is affine and that S_0 is defined by a square-zero ideal; by Proposition 1.1 it is enough to find a subscheme of G, flat over S, lifting H_0.

Since H_0 is smooth and finite over S_0, after possibly restricting S we may assume that there is an integer $n > 0$, invertible on S, such that $H_0 = {}_nH_0$. Consider the n-th power morphism on G:

$$u : G \longrightarrow G, \qquad x \longmapsto x^n.$$

We continue to denote by ${}_nG$ the "kernel of u", that is, the inverse image under u of the identity section of G. Notice that u is not, in general, a group homomorphism. We first admit the following lemma.

Lemma 1.3.— *Let k be a field, let G be a group scheme locally of finite type over k, let n be an integer prime to the characteristic of k, and let u be the n-th power morphism on G. Then u is étale at every point x of G that belongs to the centre of G.*

Since G is flat and of finite presentation over S, the preceding lemma and EGA IV 17.8.2 show that, if $x \in G$ lies above $s \in S$ and belongs to the centre of G_s, then u is étale at x. If in addition $x \in {}_nG$, then ${}_nG$ is étale over S at x. By hypothesis, H_0 is central and is contained in ${}_nG_0$; it is therefore contained in the largest open subset V of ${}_nG$ that is étale over S. Since H_0 and $V \times_S S_0 = V_0$ are étale over S_0, H_0 is an open subscheme of V_0 (EGA IV 17.8.7 and 17.9.1). The open subprescheme of V having the same underlying space as H_0 is consequently a subscheme of G, flat over S, lifting H_0.

It remains to prove Lemma 1.3. The largest open subset of G on which u is étale is invariant under extension of the base field (EGA IV 17.8.2), so we may reduce to the case in which x is k-rational. Let t_x denote translation by x, a k-automorphism of G. Since x belongs to the centre of G, the diagram

$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "t_x"'] & G \arrow[d, "t_x^n"] \\ G \arrow[r, "u"'] & G \end{tikzcd}$$

is commutative. It is therefore enough to prove that u is étale at the identity; this was established in $\mathrm{VII_A}$ 8.4.

1.2. Global lifting of finite groups

Lemma 1.4.— *Let A be a local ring, separated and complete for the topology defined by its maximal ideal $\mathfrak{m}$, let*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^n).$$

Then, for every prescheme X (respectively, for every S-prescheme X), the canonical map

$$\mathrm{Hom}(S, X) \longrightarrow \varprojlim_n \mathrm{Hom}(S_n, X) \tag{$*$}$$

(respectively,

$$\Gamma(X/S) \longrightarrow \varprojlim_n \Gamma(X_n/S_n), \qquad X_n = X \times_S S_n) \tag{$**$}$$

is bijective.

Statement $(**)$ is an easy consequence of $(*)$. Let us prove $(*)$. Let

$$u_n : S_n \longrightarrow X \qquad (n \in \mathbb{N})$$

be a coherent system of morphisms. The image y of the closed point of S_n is independent of n, and u_n factors through $\mathrm{Spec}\,\mathcal{O}_y$. On passing to the inverse limit, the morphisms u_n define a ring homomorphism

$$\widetilde{u} : \mathcal{O}_y \longrightarrow \varprojlim_n (A/\mathfrak{m}^n) = A.$$

This proves that $(*)$ is surjective; it is injective as soon as A is separated for the $\mathfrak{m}$-adic topology.

Corollary 1.5.— *Let A be a complete noetherian local ring, let $\mathfrak{m}$ be its maximal ideal, let*

$$S = \mathrm{Spec}(A), \qquad S_n = \mathrm{Spec}(A/\mathfrak{m}^n),$$

let X be a finite scheme over S, and let Y be an S-prescheme. Then the canonical map

$$\mathrm{Hom}_S(X, Y) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n}(X_n, Y_n)$$

(where $X_n = X \times_S S_n$, and similarly for Y_n) is bijective.

Indeed, it follows from EGA II 6.2.5 that X is a finite disjoint sum of local S-schemes finite over S. We are thus reduced to the case in which X itself is the spectrum of a complete noetherian local ring. But

$$\mathrm{Hom}_S(X,Y) = \Gamma(Z/X), \qquad Z = Y \times_S X,$$

and the preceding lemma applies.

Proposition 1.6.— *Let A, S, S_n be as above, and let G and M be two S-group preschemes, with M finite over S. Then:*

a) *The canonical map*

$$\mathrm{Hom}_{S\text{-gr}}(M,G) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}(M_n, G_n)$$

is bijective.

b) *If M is of multiplicative type and G is smooth over S, the canonical map*

$$\varphi : \mathrm{Hom}_{S\text{-gr}}(M,G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr}}(M_0, G_0)$$

is surjective. Moreover, if $\varphi(u) = \varphi(u') = u_0$, then u and u' are conjugate by an element of $G(S)$ that reduces to the identity element of $G(S_0)$.

c) *If M is of multiplicative type and smooth over S, if G is flat and of finite type over S, and if $u_0 : M_0 \to G_0$ is a central homomorphism, then u_0 lifts uniquely to a homomorphism*

$$u : M \longrightarrow G.$$

Moreover, u is central if G has connected fibres.

Proof. *a)* This follows from 1.5, from the fact that $M \times_S M$ is finite over S, and from the characterization of group morphisms as the morphisms for which the familiar diagram

$$\begin{tikzcd} M \times_S M \arrow[r, "u\times u"] \arrow[d] & G \times_S G \arrow[d] \\ M \arrow[r, "u"'] & G \end{tikzcd}$$

is commutative.

b) By Exposé IX 3.6, one can construct a coherent system of homomorphisms $u_n : M_n \to G_n$ lifting a homomorphism $u_0 : M_0 \to G_0$. In view of part a), this proves the first assertion of b).

If u and u' are two liftings of u_0, then u_n and u'_n are conjugate by an element $g_n \in G(S_n)$ lifting the identity element of $G(S_0)$ (Exposé IX 3.6). The same reference also implies that the g_n may be chosen coherently; by 1.4 they therefore arise from a section $g \in G(S)$. The morphisms u and $\mathrm{Int}(g)u'$ then agree modulo $\mathfrak{m}^n$ for every n, and hence agree by 1.5.

c) The existence and uniqueness of u follow from part a) and 1.2 bis. If G has connected fibres, u is central by Exposé IX 5.6(a).

2. Infinitesimal lifting of sub-tori

2.1. Statement of the theorem

We shall give an infinitesimal lifting theorem for sub-tori of a group prescheme that does not require smoothness hypotheses (in contrast with Exposé IX 3.6 bis) and that also answers a very natural question:[13] is it enough to be able to lift "sufficiently many" points of finite order of a sub-torus T_0 in order to ensure that T_0 itself can be lifted (infinitesimally, of course)?

Theorem 2.1.— *Let S be a noetherian affine scheme, let S_0 be a closed subscheme of S defined by a square-zero ideal J, let G be an S-group prescheme of finite type, let $G_0 = G \times_S S_0$, let T_0 be a sub-torus of G_0, and let $q > 0$ be an integer invertible on S. Suppose that, for every integer n equal to a power of q, there exists a subscheme M_n of G, flat over S, such that*

$$M_n \times_S S_0 = {}_nT_0.$$

Then there exists a sub-torus T of G such that $T \times_S S_0 = T_0$.

Theorem 2.1 will be useful to us through the following two corollaries.

Corollary 2.2.— *Let S be a locally noetherian prescheme, let S_0 be a closed subprescheme of S defined by a locally nilpotent sheaf of ideals, let G be an S-group prescheme of finite type, let T_0 be a sub-torus of $G_0 = G \times_S S_0$, and let $q > 0$ be an integer invertible on S. As n ranges over the powers of q, let (M_n) be a coherent system of S-subgroup schemes of G, of multiplicative type, lifting the ${}_nT_0$. (The system (M_n) is said to be coherent if*

$$M_m = {}_m(M_n)$$

whenever m divides n.) Then there exists a unique sub-torus T of G such that

$$T \times_S S_0 = T_0 \qquad \textit{and} \qquad {}_nT = M_n$$

for every n.

Corollary 2.3.— *Let G be an S-group prescheme, flat and of finite presentation over S, let S_0 be a closed subprescheme of S defined by a locally nilpotent sheaf of ideals of finite type, and let T_0 be a central torus of $G_0 = G \times_S S_0$. Then there exists a unique sub-torus T of G lifting T_0. Moreover, T is central.*

Remark 2.4.— *We leave it to the reader to formulate the analogues of statements 2.1, 2.2, and 2.3 in which, instead of lifting a sub-torus of G_0, one is given a torus T over S and asks to lift a morphism*

$$u_0 : T_0 \longrightarrow G_0.$$

One reduces to the preceding cases by considering the graph of u_0.

Let us show how Corollaries 2.2 and 2.3 follow from Theorem 2.1.

Proof of Corollary 2.2. The uniqueness of T follows from Exposé IX 4.8(b) and Exposé IX 4.10. To prove existence, we may therefore reduce to the case in which S is affine, hence noetherian, and S_0 is defined by a square-zero ideal.

[13] Editor's note: The idea of approximating a torus by its finite subgroups occurs in Grothendieck's proof of the connectedness of centralizers of tori; see ğ4.6 of *BIBLE*.

Lemma 2.5.— *Let G be an S-group prescheme of finite presentation, and let H be a subgroup scheme of G, of multiplicative type and smooth over S. Then* $\mathrm{Centr}_G(H)$ *is representable by an S-group subprescheme of G, of finite presentation.*

When G is separated over S, the lemma follows from Exposé VIII 6.5(e), without any smoothness hypothesis on H. In the present case, observe that the assertion is local for the fpqc topology. We may therefore assume that H is diagonalizable, hence of the form $D_S(M)$. We may also assume that S is affine and then, by EGA IV 8, noetherian. Since H is smooth and of finite type over S, the order q of the torsion subgroup of M is invertible on S (Exposé VIII 2.1(e)). It is then immediate (cf. Exposé IX 4.10) that the subgroups ${}_nH$, with n ranging over the powers of q, are schematically dense in H (Exposé IX 4.1). But ${}_nH$ is a completely split covering of S, that is, it is isomorphic to a finite disjoint sum of copies of S. Consequently

$$\mathrm{Centr}_G({}_nH) = Z_n$$

is representable as the intersection of the centralizers in G of the sections of ${}_nH$ over S. It remains to apply the following lemma.

Lemma 2.5 bis.— *Let S be a noetherian prescheme, let G be an S-group prescheme of finite type, let H be a subgroup of G of multiplicative type, and let (H_i) be an increasing filtered family of subgroups of G of multiplicative type. Suppose that*

$$Z_i = \mathrm{Centr}_G(H_i)$$

is representable by a group subprescheme of G. Then the family (Z_i) is stationary. If, in addition, the H_i are schematically dense in H, then

$$Z_i = \mathrm{Centr}_G(H)$$

for all sufficiently large i.

To see that the family (Z_i) is stationary, it is enough to prove that the family of underlying sets $\mathrm{ens}(Z_i)$ is stationary. Indeed, the stationary value will then be a closed subset of an open subset U of G. After replacing G by U, we are reduced to studying a decreasing filtered family of closed subpreschemes of a noetherian prescheme. An elementary constructibility argument reduces us to the case in which S is integral. We then have to show that $\mathrm{ens}(Z_i)$ is stationary over some nonempty open subset of S. The generic fibre of G is separated (Exposé $\mathrm{VI_A}$ 0.3); after restricting S, we may therefore assume that G is separated over S (EGA IV 8). Then Z_i is closed in G (Exposé VIII 6.5(e)).

To prove the last assertion of the lemma, let Z denote the stationary value of the family (Z_i). It is clear that $\mathrm{Centr}_G(H)$ is a subfunctor of Z; we show that Z centralizes H. Let E be the subprescheme of $H \times_S Z$ that is the kernel of the pair of morphisms

$$\begin{aligned} H \times_S Z &\rightrightarrows G, \\ (h, c) &\longmapsto c, \\ (h, c) &\longmapsto hch^{-1}. \end{aligned}$$

The prescheme E contains $H_i \times_S Z$ for every i. The H_i are flat over S, so EGA IV 11.10.9 shows that, for every point s of S, the $(H_i)_s$ are schematically dense in H_s, and $H_i \times_S Z$ is schematically dense in $H \times_S Z$. Since G_s is separated, E_s is closed in $(H \times_S Z)_s$, and is therefore equal to it. Thus E is closed in $H \times_S Z$, and hence equals $H \times_S Z$. This says that Z centralizes H.

Returning to the proof of 2.2, Lemma 2.5 shows that

$$Z_n = \mathrm{Centr}_G(M_n)$$

is representable, and Lemma 2.5 bis shows that the decreasing family of subpreschemes (Z_n) is stationary; let Z be its stationary value. The group Z contains T_0 and all the M_n. After replacing G by Z, we may therefore suppose that the M_n are central.

The hypotheses of Theorem 2.1 are now satisfied, so there exists a sub-torus T of G lifting T_0. The groups ${}_nT$ and M_n are two liftings of ${}_nT_0$; they are therefore conjugate (Exposé IX 3.2 bis), and consequently coincide because M_n is central. This torus T has the required properties.

Proof of Corollary 2.3. The uniqueness of T follows from Exposé IX 5.1 bis, and its centrality follows from Exposé IX 5.6(b). By the usual procedure, this observation reduces us first to the case in which S is affine (and hence S_0 is defined by a nilpotent ideal of finite type), and then to the noetherian case. After restricting S if necessary, we may suppose that there exists an integer q invertible on S. Corollary 2.3 then follows from 2.2 and 1.2.

Remark 2.6.— *Corollary 2.3 remains true if the torus T_0 is replaced by a smooth central subgroup of multiplicative type of G_0.*

2.2. Proof of Theorem 2.1

a) Reduction to the case $T_0 = \mathbb{G}_{m,S_0}$.

By 1.1, we may assume that M_n is a subgroup of multiplicative type. Using Exposé IX 3.2 bis, we may suppose that the family (M_n) is coherent (2.2). The centralizer Z_n of M_n in G is representable (2.5), and the filtered family (Z_n) is stationary (2.5 bis). After replacing G by Z_n for sufficiently large n, we may therefore suppose that T_0 and the M_n are central. The uniqueness of a lifting of T_0 is then guaranteed by Exposé IX 5.1 bis.

Proceeding as in the proof of 1.1, we may suppose that there exist an S-torus T and an S_0-isomorphism

$$u_0 : T \times_S S_0 \xrightarrow{\sim} T_0.$$

Lifting u_0 is equivalent to lifting T_0. By uniqueness, it is enough to prove the existence of a lifting of u_0 after an affine faithfully flat extension $S' \to S$ of finite type (fpqc descent). This allows us to suppose that

$$T = \mathbb{G}_{m,S}^r$$

(Exposé X 4.5). If the restriction of u_0 to each factor $\mathbb{G}_m$ lifts to an S-morphism—necessarily central—then these liftings immediately give a lifting of u_0. In short, we may suppose that $T_0 = \mathbb{G}_{m,S_0}$.

b) Definition of the obstruction to the existence of a lifting of T_0.

By 1.1, to prove Theorem 2.1 it is enough to find a subscheme of G, flat over S, lifting T_0. We shall see that the obstruction to the existence of such a lifting can be defined as an element of a certain Ext^1 group of sheaves of modules.

Let U be an open subset of G in which T_0 is closed, and again denote by U (respectively, U_0) the open subscheme of G (respectively, G_0) having this open subset as its underlying space. The sheaf $\mathcal{O}_{T_0}$, regarded as a sheaf on U, is therefore a quotient of $\mathcal{O}_{U_0}$. Let

$$h : \mathcal{O}_{U_0} \longrightarrow \mathcal{O}_{T_0}$$

be the canonical epimorphism.

Lemma 2.7.— *The canonical map*

$$h' = \mathrm{id}_J \otimes h : J \otimes_{S_0} \mathcal{O}_{U_0} \longrightarrow J \otimes_{S_0} \mathcal{O}_{T_0}$$

factors, necessarily uniquely, as $i \circ j_U$, *where* j_U *is the canonical epimorphism*

$$J \otimes_{S_0} \mathcal{O}_{U_0} \longrightarrow J\mathcal{O}_U \simeq J\mathcal{O}_{U_0}.$$

We must show that $h'(K) = 0$, where K is the kernel of j_U. For every integer n equal to a power of q, there is an epimorphism

$$h_n : \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0}.$$

Since ${}_nT_0$ lifts to a scheme M_n flat over S, the canonical morphism

$$j_n : J \otimes_{S_0} \mathcal{O}_{{}_nT_0} \xrightarrow{\sim} J\mathcal{O}_{M_n}$$

is an isomorphism. The commutative diagram

$$\begin{tikzcd}
K \arrow[r] \arrow[dr, phantom, "\circlearrowright"] & J \otimes_{S_0} \mathcal{O}_{U_0} \arrow[r, "j_U"] \arrow[d, "h'"'] & J\mathcal{O}_U \arrow[r, hook] \arrow[dl, "i"] & \mathcal{O}_U \\
 & J \otimes_{S_0} \mathcal{O}_{T_0} \arrow[d, "h'_n"'] & & \\
 & J \otimes_{S_0} \mathcal{O}_{{}_nT_0} \arrow[r, "j_n", "\sim"'] & J\mathcal{O}_{M_n} \arrow[r, hook] & \mathcal{O}_{M_n}
\end{tikzcd}$$

shows that $h'(K)$ is contained in $\ker h'_n$ for every n. It is therefore contained in $\bigcap_n \ker h'_n$, and it remains to prove that this intersection is zero. Now

$$\mathcal{O}_{T_0} = \mathcal{O}_{S_0}[\mathbb{Z}], \qquad \mathcal{O}_{{}_nT_0} = \mathcal{O}_{S_0}[\mathbb{Z}/n\mathbb{Z}].$$

Let

$$a = \sum_{m \in \mathbb{Z}} a_m \otimes m \in J \otimes_{S_0} \mathcal{O}_{T_0}.$$

The a_m are sections of J, almost all zero. Choose n large enough that the indices m for which $a_m \neq 0$ have distinct images in $\mathbb{Z}/n\mathbb{Z}$. If $a \in \ker h'_n$, then necessarily $a = 0$. Thus $\bigcap_n \ker h'_n = 0$, proving 2.7.

Let K_0 be the kernel of h. Consider the diagram

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
 & J\mathcal{O}_U \arrow[d, "\sim"'] & & K_0 \arrow[d] & \\
0 \arrow[r] & J\mathcal{O}_{U_0} \arrow[r] \arrow[d, "i"'] & \mathcal{O}_U \arrow[r] & \mathcal{O}_{U_0} \arrow[r] \arrow[d, "h"] & 0 \\
 & J \otimes_{S_0} \mathcal{O}_{T_0} & & \mathcal{O}_{T_0} \arrow[d] & \\
 & & & 0. &
\end{tikzcd}$$

The sheaf $\mathcal{O}_U$ defines an element

$$\Phi \in \operatorname{Ext}^1_{\mathcal{O}_U}(\mathcal{O}_{U_0}, J\mathcal{O}_{U_0}).$$

Let Ψ be the element of

$$\operatorname{Ext}^1_{\mathcal{O}_U}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

obtained from Φ, by the bifunctoriality of Ext^1, through the morphisms

$$K_0 \longrightarrow \mathcal{O}_{U_0}, \qquad i : J\mathcal{O}_{U_0} \longrightarrow J \otimes_{S_0} \mathcal{O}_{T_0}.$$

It follows from Exposé III 4.1 and the infinitesimal criterion of flatness (cf. Exposé III 4.3) that a subscheme of U, flat over S and lifting T_0, exists if and only if $\Psi = 0$.

Since T_0 is affine, Exposé III 4.5 and 4.6 show that it is enough to construct, locally on U, a subscheme flat over S lifting T_0. It therefore suffices to show that the image of Ψ in the sheaf

$$\mathscr{E} = \mathcal{E}xt^1_{\mathcal{O}_U}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

is zero. We again denote this element of $\Gamma(U, \mathscr{E})$ by Ψ.

c) Reduction to the case in which S is local artinian with algebraically closed residue field.

Since K_0 and $J \otimes_{S_0} \mathcal{O}_{T_0}$ are coherent sheaves, so is $\mathscr{E}$; moreover, $\mathscr{E}$ is supported on T_0, because the same is true of $J \otimes_{S_0} \mathcal{O}_{T_0}$. To show that the section Ψ of $\mathscr{E}$ is zero, it is enough to show that, for every point u of T_0, its image in the fibre of $\mathscr{E}$ at u is zero. Formation of the $\mathcal{E}xt^i$ of coherent sheaves commutes with flat extensions of the base,[14] so we are reduced to proving the existence of a lifting of

$$T_0 \cap \operatorname{Spec} \mathcal{O}_u$$

flat over S. Let s be the image of u in S. We may replace S by $\operatorname{Spec} \mathcal{O}_s$ and G by $G \times_S \operatorname{Spec} \mathcal{O}_s$.

After another faithfully flat extension if necessary, we may suppose that $\mathcal{O}_s$ has an algebraically closed residue field (EGA 0_{III}, 10.3.1). Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_s$, and set

$$S_n = \operatorname{Spec}(\mathcal{O}_s/\mathfrak{m}^n).$$

Suppose that, for every $n > 0$, the obstruction to lifting

$$T_0 \times_S S_n = (T_0)_n$$

to a subscheme of $G_n = G \times_S S_n$, flat over S_n, has been shown to vanish, and let T_n be the unique flat lifting of $(T_0)_n$ over S_n that is a sub-torus of G_n. If $n > n'$, it is clear that

$$T_n \times_{S_n} S_{n'} = T_{n'}.$$

If u is a point of T_0 lying over s, the lemma below, applied to the coherent system of liftings

$$T_n \cap \operatorname{Spec} \mathcal{O}_u$$

of $(T_0)_n \cap \operatorname{Spec} \mathcal{O}_u$, shows that $T_0 \cap \operatorname{Spec} \mathcal{O}_u$ indeed has a lifting flat over $\mathcal{O}_s$. We are therefore reduced to proving that $\Psi = 0$ when $S = S_n$ is the spectrum of an artinian local ring with algebraically closed residue field, and to proving the following lemma.

[14]Editor's note: cf. EGA 0_{III}, 12.3.5.

Lemma 2.8.— *Let $A \to B$ be a local homomorphism of noetherian local rings, let $\mathfrak{m}$ be the maximal ideal of A, let J be a square-zero ideal of A, let M be a finite-type B-module, and set*

$$A_0 = A/J, \qquad B_0 = B/JB, \qquad M_0 = M/JM.$$

Let N_0 be a quotient B_0-module of M_0, flat over A_0. For every $n > 0$, let $A_n, A_{0,n}$, and so forth denote the objects obtained by the base extension

$$A \longrightarrow A/\mathfrak{m}^n = A_n,$$

and let $\overline{J}_n$ be the image of J in A_n. For every $n > 0$, let N_n be a quotient B_n-module of M_n, flat over A_n, lifting $N_{0,n}$, and suppose that, for $n \geq n'$, $N_{n'}$ is obtained from N_n by the base extension $A_n \to A_{n'}$. Then there exists a quotient B-module N of M, flat over A, lifting N_0.

Proof of 2.8. Let P_0 be the kernel of the epimorphism $M_0 \to N_0$. For every $n > 0$, there is a commutative diagram with exact horizontal rows:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
 & & & P_{0,n} \arrow[d] & \\
0 \arrow[r] & \overline{J}_n M_n \arrow[r] & M_n \arrow[r] \arrow[dd] & M_{0,n} \arrow[r] \arrow[dd] & 0 \\
 & \overline{J}_n \otimes_{A_{0,n}} M_{0,n} \arrow[u] \arrow[d] & & & \\
0 \arrow[r] & \overline{J}_n \otimes_{A_{0,n}} N_{0,n} \arrow[r] & N_n \arrow[r] \arrow[d] & N_{0,n} \arrow[r] \arrow[d] & 0 \\
 & & 0 & 0 &
\end{tikzcd} \qquad (*_n)$$

Moreover, by hypothesis, the diagram $(*)_{n+1}$ reduces modulo $\mathfrak{m}^n$ to $(*)_n$.

The Artin–Rees lemma (Bourbaki, *Algèbre commutative*, Chapter III, ğ3, Corollary 1) shows that the filtrations on JM, $J \otimes_{A_0} M_0$, and $J \otimes_{A_0} N_0$, defined by the kernels of the respective morphisms

$$\begin{aligned} JM &\longrightarrow \overline{J}_n M_n, \\ J \otimes_{A_0} M_0 &\longrightarrow \overline{J}_n \otimes_{A_{0,n}} M_{0,n}, \\ J \otimes_{A_0} N_0 &\longrightarrow \overline{J}_n \otimes_{A_{0,n}} N_{0,n}, \end{aligned}$$

are $\mathfrak{m}B$-good. Passing to the inverse limit in the diagrams $(*)_n$, and writing $\widehat{Q}$ for the separated completion of a B-module Q for the $\mathfrak{m}B$-adic topology, we obtain a commutative diagram in which the two horizontal rows remain exact:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
 & & & \widehat{P}_0 \arrow[d] & \\
0 \arrow[r] & \widehat{JM} \arrow[r] & \widehat{M} \arrow[r] \arrow[dd] & \widehat{M}_0 \arrow[r] \arrow[dd] & 0 \\
 & J \widehat{\otimes_{A_0}} M_0 \arrow[u] \arrow[d] & & & \\
0 \arrow[r] & J \widehat{\otimes_{A_0}} N_0 \arrow[r] & \varprojlim_n N_n \arrow[r] \arrow[d] & \widehat{N}_0 \arrow[r] \arrow[d] & 0 \\
 & & 0 & 0 &
\end{tikzcd} \tag{$\widehat{*}$}$$

Now $(B, \mathfrak{m}B)$ is a Zariski ring and $J \otimes_{A_0} N_0$ is of finite type over B, hence separated for the $\mathfrak{m}B$-adic topology. The diagram $(\widehat{*})$ then shows that the morphism

$$J \otimes_{A_0} M_0 \longrightarrow J \otimes_{A_0} N_0$$

induced by the epimorphism $M_0 \to N_0$ factors through JM:

$$\begin{tikzcd}
J \otimes_{A_0} M_0 \arrow[rr, "\text{can.}"] \arrow[dr] & & JM \arrow[dl] \\
 & J \otimes_{A_0} N_0 &
\end{tikzcd}$$

Under these conditions, Exposé III 4.1 and 4.3 show that a quotient B-module N of M, flat over A and lifting N_0, exists if and only if a certain element

$$g \in E = \mathrm{Ext}^1_B(P_0, J \otimes_{A_0} N_0)$$

vanishes. More precisely, the exact sequence

$$0 \longrightarrow JM \longrightarrow M \longrightarrow M_0 \longrightarrow 0$$

defines an element $f \in F$, where

$$F = \mathrm{Ext}^1_B(M_0, JM),$$

and g is the image of f under the natural morphism $F \to E$ arising by bifunctoriality from

$$P_0 \longrightarrow M_0, \qquad JM \longrightarrow J \otimes_{A_0} N_0.$$

The diagram $(\widehat{*})$ and Exposé III 4.1 show that the image of g in

$$\widehat{E} \simeq \mathrm{Ext}^1_{\widehat{B}}(\widehat{P}_0, J \widehat{\otimes_{A_0}} N_0)$$

is zero. But E is of finite type over B, and hence separated for the $\mathfrak{m}B$-adic topology. Consequently $g = 0$, which completes the proof of 2.8.

d) Study of $\mathscr{E}$.

We now suppose that S is the spectrum of an artinian local ring A with algebraically closed residue field k. Let A_0 be the ring of S_0, and let $\mathfrak{m}_0$ be the maximal ideal of A_0. Since S_0 is artinian, T_0 is closed in G (Exposé $\mathrm{VI_B}$ 1.4.2). We may therefore take $U = G$, so that

$$\mathscr{E} = \mathcal{E}xt^1_{\mathcal{O}_G}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0}).$$

Let B_0 be the ring of the affine S_0-scheme $T_0 = \mathbb{G}_{m,S_0}$, and let $\overline{B}_0$ be the ring of its special fibre

$$T_0 \times_{S_0} \mathrm{Spec}(k) = \mathbb{G}_{m,k}.$$

The sheaf $\mathscr{E}$ is a coherent $\mathcal{O}_{T_0}$-module, hence is defined by a finite-type B_0-module, which we denote by E.

Consider the graded module associated with E and the $\mathfrak{m}_0 B_0$-adic filtration:

$$E_r = \mathfrak{m}_0^r E / \mathfrak{m}_0^{r+1} E.$$

Each E_r is a finite-type $\overline{B}_0$-module, and $E_r = 0$ for all sufficiently large r, since S_0 is artinian.

Let g be a section of G over S whose restriction to S_0 is a section g_0 of T_0. Left translation by g defines an "automorphism of the situation" from the point of view of the obstruction problem. In particular, g induces an S_0-automorphism of the sheaf $\mathscr{E}$ that fixes the obstruction Ψ. More precisely, g defines a semilinear automorphism of the B_0-module E, relative to the A_0-automorphism of B_0 defined by translation by g_0 in T_0. Reducing modulo $\mathfrak{m}_0^{r+1}$, g therefore defines a semilinear automorphism of E_r, relative to the k-automorphism of $\overline{B}_0$ defined by translation by

$$g_0 \times_{S_0} \mathrm{Spec}(k)$$

in $T_0 \times_{S_0} \mathrm{Spec}(k)$.

Lemma 2.9.— *For every integer $r \geq 0$, E_r is a locally free $\overline{B}_0$-module.*

Let x be a point of T_0, let $\kappa(x)$ be its residue field, and let

$$(E_r)_x = E_r \otimes_{\overline{B}_0} \kappa(x)$$

be the "fibre" of E_r at x. Denote by $\ell(x)$ the rank of $(E_r)_x$ over $\kappa(x)$, and by ℓ the maximum of $\ell(x)$ as x ranges over the points of T_0. Let L be the largest closed subscheme of

$$\mathrm{Spec}\,\overline{B}_0 = \mathbb{G}_{m,k}$$

over which E_r is locally free of rank ℓ (TDTE IV, Lemma 3.6). Let $\beta \in L(k)$, which exists because L is of finite type over the algebraically closed field k, and let $\alpha \in \mathbb{G}_{m,k}(k)$ be a point whose order is a power n of q. The point α is k-rational. Since, by hypothesis, ${}_nT_0$ lifts to a subscheme M_n étale over S, there is a section a of G over S lifting α and whose restriction to S_0 is a section of T_0.

The preceding observations show that the fibres of E_r at the points β and $\alpha\beta$ of $\mathbb{G}_{m,k}(k)$ are k-isomorphic. The points whose order is a power of q are dense in $\mathbb{G}_{m,k}$, as are their translates by β. Since L is closed in $\mathbb{G}_{m,k}$ and $\mathbb{G}_{m,k}$ is reduced, $L = \mathbb{G}_{m,k}$, and E_r is locally free of rank ℓ over $\overline{B}_0$.

e) Completion of the proof of Theorem 2.1.

Let $K_0(n)$ be the sheaf of ideals of $\mathcal{O}_{G_0}$ defining the closed subscheme ${}_nT_0$. The sheaf K_0[15] is therefore a subsheaf of $K_0(n)$. For brevity, set

$$R = J \otimes_{S_0} \mathcal{O}_{T_0}, \qquad R(n) = J \otimes_{S_0} \mathcal{O}_{{}_nT_0}.$$

The sheaf $R(n)$ is a quotient of R, and we have the diagram

$$\begin{tikzcd}
 & & K_0 \arrow[d] & & \\
 & & K_0(n) \arrow[d] & & \\
0 \arrow[r] & J\mathcal{O}_G \arrow[r] \arrow[d] & \mathcal{O}_G \arrow[r] & \mathcal{O}_{G_0} \arrow[r] & 0 \\
 & R \arrow[d] & & & \\
 & R(n). & & &
\end{tikzcd}$$

Using the bifunctoriality of $\operatorname{Ext}^1_{\mathcal{O}_G}$, we obtain the commutative diagram

$$\begin{tikzcd}
\operatorname{Ext}^1_{\mathcal{O}_G}(\mathcal{O}_{G_0}, J\mathcal{O}_G) \arrow[dr] & & \\
 & \operatorname{Ext}^1_{\mathcal{O}_G}(K_0(n), R) \arrow[r] \arrow[d] & \operatorname{Ext}^1_{\mathcal{O}_G}(K_0, R) = \mathscr{E} \arrow[d] \\
 & \operatorname{Ext}^1_{\mathcal{O}_G}(K_0(n), R(n)) \arrow[r] & \operatorname{Ext}^1_{\mathcal{O}_G}(K_0, R(n)).
\end{tikzcd}$$

Again let Φ denote the element of $\operatorname{Ext}^1_{\mathcal{O}_G}(\mathcal{O}_{G_0}, J\mathcal{O}_G)$ defined by the exact sequence

$$0 \longrightarrow J\mathcal{O}_G \longrightarrow \mathcal{O}_G \longrightarrow \mathcal{O}_{G_0} \longrightarrow 0,$$

so that Ψ is the image of Φ in $\mathscr{E}$. Since ${}_nT_0$ lifts to a subscheme M_n of G, flat over S, the image of Φ in

$$\operatorname{Ext}^1_{\mathcal{O}_G}(K_0(n), R(n))$$

is zero (Exposé III 4.1). A fortiori, the image of Φ in $\operatorname{Ext}^1_{\mathcal{O}_G}(K_0, R(n))$, which is also the image of Ψ, is zero.

Lemma 2.10.— *For every integer n, the canonical morphism*

$$\operatorname{Ext}^1_{\mathcal{O}_G}(K_0, R) \otimes_{B_0} \mathcal{O}_{{}_nT_0} \longrightarrow \operatorname{Ext}^1_{\mathcal{O}_G}(K_0, R(n))$$

is injective.

[15] Editor's note: Introduced after Lemma 2.7.

Indeed, the affine scheme $T_0 = \mathbb{G}_{m,S_0}$ has ring

$$B_0 = A_0[T, T^{-1}].$$

The subscheme ${}_nT_0$ is defined by the vanishing of

$$h(n) = T^n - 1.$$

This section is regular (EGA 0_{IV}, 15.2.2) and remains regular after every base change $S_0' \to S_0$. Hence there is an exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_{T_0} \xrightarrow{h(n)} \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0} \longrightarrow 0.$$

Since ${}_nT_0$ is flat over S, tensoring over $\mathcal{O}_{S_0}$ with J gives the exact sequence

$$0 \longrightarrow R \xrightarrow{h(n)} R \longrightarrow R(n) \longrightarrow 0,$$

and then the exact sequence of Ext groups

$$\cdots \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R) \xrightarrow{h(n)} \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R) \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_G}(K_0, R(n)),$$

which proves the lemma.

The foregoing shows that, for every integer n equal to a power of q, the image of Ψ in $\mathscr{E}/h(n)\mathscr{E}$ is zero. To prove that $\Psi = 0$, it is enough to show that

$$\Psi \in \mathfrak{m}_0^r\mathscr{E} \quad \Longrightarrow \quad \Psi \in \mathfrak{m}_0^{r+1}\mathscr{E}.$$

Let $\overline{\Psi}$ be the image of Ψ in E_r. There is an element $\Psi(n) \in \mathscr{E}$ such that

$$\Psi = \Psi(n)h(n) \qquad (n \text{ a power of } q).$$

We observed that the image $\overline{h(n)}$ of $h(n)$ in $\overline{B}_0$ is again $\overline{B}_0$-regular. Since E_s is locally free over $\overline{B}_0$ for every s (2.9), multiplication by $\overline{h(n)}$ on E_s is injective. It follows immediately that $\Psi(n) \in \mathfrak{m}_0^r\mathscr{E}$. Let $\overline{\Psi(n)}$ be its image in E_r; then

$$\overline{\Psi} = \overline{h(n)}\,\overline{\Psi(n)} \qquad (n \text{ a power of } q).$$

Thus the set F of points of $\mathbb{G}_{m,k}$ at which $\overline{\Psi}$ vanishes contains the dense set of points whose order is a power of q. Moreover, F is closed because E_r is locally free over $\overline{B}_0$, and $\mathbb{G}_{m,k}$ is reduced. Therefore $\overline{\Psi} = 0$.

This completes the proof of Theorem 2.1.

3. Characterization of a sub-torus by its underlying set

3.1. Statement of the theorem

Notation.—If X is a prescheme, $\mathrm{ens}(X)$ denotes the underlying set of X. If X and S' are two S-preschemes,

$$X_{S'} = X \times_S S', \qquad u : X_{S'} \longrightarrow X$$

is the canonical morphism, and if E is a subset of $\mathrm{ens}(X)$, we denote by $E_{S'}$ (or $E \times_S S'$) the subset $u^{-1}(E)$ of $\mathrm{ens}(X_{S'})$.

Theorem 3.1.— *Let S be a locally noetherian prescheme, let G be an S-group prescheme of finite type, let $q > 0$ be an integer invertible on S, and let E be a subset of* $\mathrm{ens}(G)$*. Consider the following assertions about E:*

(i) *The set E is the underlying set of a sub-torus T of G.*

(ii) a) *For every point s of S, there is a sub-torus T_s of G_s such that*

$$E \cap \operatorname{ens}(G_s) = E_s$$

is the underlying set of T_s.

b) *As n ranges over the powers of q, there is a coherent family (cf. 2.2) M_n of subgroup schemes of G, of multiplicative type, such that for every point s of S,*

$$(M_n)_s = {}_nT_s.$$

(iii) a) *As in* (ii) a).

b) *The set E is locally closed in* $\operatorname{ens}(G)$, *and the dimension of the fibers of E over S is locally constant.*

(iv) a) *As in* (ii) a).

b) *For every S-scheme S' that is the spectrum of a complete discrete valuation ring with algebraically closed residue field, $E_{S'}$ is the underlying set of a sub-torus of $G_{S'}$.*

Then the following implications hold:

A) (i) $\Longleftrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (iv).

B) *If G is separated over S, then* (iii) $\Longleftrightarrow$ (iv), *and moreover E is closed.*

C) *Conditions* (i), (ii), *and* (iii) *(and also* (iv) *if G is separated over S) are equivalent in the following two cases:*

First case: a) *The prescheme S is reduced, or G is flat over S; and*
b) *for every point s of S, T_s is a central torus of G_s.*

Second case: *S is normal.*

Moreover, in the two cases above, the torus T with underlying set E is unique.

Remarks 3.2.— a) *When S is reduced, there is no need in* (ii) *to assume that the family M_n is coherent, since this condition follows from the other hypotheses. Indeed, if m divides n, the subgroups ${}_mM_n$ and M_m are étale over S, hence reduced, and have the same underlying space; they therefore coincide.*

b) *If S is no longer assumed normal, it is in general no longer true that* (iii) $\Rightarrow$ (i), *even when S is reduced and geometrically unibranch and G is a smooth group scheme over S. Indeed, consider the Borel subgroup of* $\mathrm{SL}_{2,S}$ *consisting of matrices of the form*

$$\begin{pmatrix} t & u \\ 0 & t^{-1} \end{pmatrix},$$

where S is the affine curve over a field k with ring

$$k[x, y]/(y^2 - x^3).$$

Define E as follows. Above the cusp of S, where $x = y = 0$, take the diagonal torus $u = 0$. Above the complementary open subset $x \neq 0$, take the torus obtained from the diagonal torus by conjugation by

$$\begin{pmatrix} 1 & y/x \\ 0 & 1 \end{pmatrix}.$$

The resulting set E satisfies (iii) a)*; it is also closed in G, and the reduced subscheme having E as its underlying set has equations*

$$\begin{cases} xu + y(t - t^{-1}) = 0, \\ u^2 - x(t - t^{-1})^2 = 0. \end{cases}$$

It is therefore not a sub-torus of G, since its fiber above the cusp is not reduced.

Plan of the proof of 3.1. In a first part we shall establish the following "easy" implications:

$$\begin{tikzcd}
 & \text{(ii)} \arrow[dr] & \\
\text{(i)} \arrow[ur] \arrow[dr] & & \text{(iv)} \\
 & \text{(iii)} \arrow[ur] &
\end{tikzcd}$$

and

$$\text{(iv)} \Longrightarrow [\,\text{(iii) and } E \text{ is closed}\,] \quad \text{if } G \text{ is separated over } S.$$

The more delicate implications will be proved in three stages:

I) Reduction of (iii) ⇒ (i), under the hypotheses of C), first case, to the case in which S is normal.

II) (iii) ⇒ (ii) when S is normal.

III) (ii) ⇒ (i).

3.2. Proof of the "easy" assertions in 3.1

The implications (i) ⇒ (ii) and (i) ⇒ (iii) are trivial.

(iii) ⇒ (iv). After replacing S by S', we may assume that S is the spectrum of a discrete valuation ring. Let t be the generic point of S, and let s be its closed point.

Since E is locally closed, there is a reduced subprescheme of G with underlying space E; we also denote it by E. Its generic fiber E_t is therefore the sub-torus T_t of G_t, of rank r, by (iii) a). Let E' be the schematic closure of E_t in E. The prescheme E' is irreducible, and its closed fiber E'_s is nonempty (it contains the unit section of G_s); hence E'_s has dimension r (EGA IV 14.3.10). But E'_s is a closed subscheme of E_s, while E_s has dimension r by (iii) b) and is irreducible because it has the same underlying space as T_s. Thus E'_s and E_s have the same underlying space, and consequently $E' = E$. This proves that E is flat over S.

Let E'' now be the schematic closure of E_t in G. Then E'' is a subgroup prescheme of G, flat over S (Exposé VIII 7.1), which contains E', and therefore E. Its closed fiber E''_s is an algebraic subgroup of G_s of dimension r (*loc. cit.*). Since T_s is a closed irreducible subset of G_s of dimension r, E_s has the same underlying set as the identity component $(E''_s)^0$ of

E''_s. Let $(E'')^0$ denote the "identity component" of E'', that is, the open subgroup of E'' complementary to the union of the irreducible components of E''_s that do not contain the origin. Then

$$E = \mathrm{ens}\big((E'')^0\big).$$

Since E and E'' are reduced, one even has $E = (E'')^0$. Finally, E is a subgroup prescheme of G, flat and of finite type over S, with connected fibers, hence separated (Exposé $\mathrm{VI_B}$ 5.2); its generic fiber is the torus T_t, and its reduced closed fiber is the torus T_s. It follows that E is a torus (Exposé X 8.8).

(ii) $\Rightarrow$ (iv). Again we may assume that S is the spectrum of a discrete valuation ring, and we retain the notation introduced above. The schematic closure T of T_t in G is a subgroup prescheme of G, flat over S, and it contains M_n for every n that is a power of q. Consequently, its closed fiber is a closed subscheme of G_s containing $(M_n)_s$ for every n, hence containing T_s. For dimension reasons, the "identity component" T^0 of T has E as its underlying set, and the preceding argument shows that T^0 is a sub-torus of G.

(iv) $\Rightarrow$ [(iii) and E is closed], if G is separated over S. Let us show that E is closed.[16] We first prove the following lemma.

Lemma 3.3.— *If the conditions stated in 3.1* (iv) *are satisfied, E is a locally constructible subset of* ens(G).

By the usual method, we are reduced to the case in which S is noetherian and integral, with generic point η. After shrinking S, we may assume (Exposé $\mathrm{VI_B}$ 10.10) that there is a subgroup scheme H of G, flat over S and with connected fibers, whose generic fiber H_η is T_η. To prove Lemma 3.3 it is enough to show that $\mathrm{ens}(H) = E$.

Let s be a point of S. By EGA II 7.1.9 there is an S-scheme S', the spectrum of a discrete valuation ring which we may assume complete with algebraically closed residue field, whose generic point t' maps to η and whose closed point s' maps to s. By (iv) b), there is a sub-torus $T_{S'}$ of $G_{S'}$ having $E_{S'}$ as its underlying space. The two subgroup preschemes $T_{S'}$ and $H_{S'} = H \times_S S'$ of $G_{S'}$ are flat over S', have connected fibers, and have the same generic fiber $T_{t'}$. Therefore both coincide with the identity component of the schematic closure of $T_{t'}$ in $G_{S'}$. Hence $\mathrm{ens}(H_s) = E_s$, and therefore $\mathrm{ens}(H) = E$, proving the lemma.

Now that E is known to be locally constructible, to see that it is closed it is enough to prove that every specialization in G of a point of E again belongs to E. By the usual technique, we are reduced to the case in which S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. In that case the sub-torus of G with underlying space E, supplied by (iv) b), is closed in G, because G is separated (Exposé VIII 7.12).

3.3. Continuation of the proof of 3.1

I) Reduction of (iii) $\Rightarrow$ (i), case C, first case, to the case in which S is normal.

a) Reduction to the case in which S is affine and reduced. We therefore assume that, for every point s of S, E_s is the underlying space of a central sub-torus of G_s. The uniqueness of T then follows from Exposé IX 5.1 bis, and moreover T will necessarily be central (*loc. cit.*). In view of this uniqueness, in order to prove the existence of T we may suppose that S is noetherian and affine, with ring A. If S is not reduced, G is flat over S by hypothesis. By 2.3 it is then enough to solve the problem for S_{red} and $G \times_S S_{\mathrm{red}}$. We may therefore suppose in addition that S is reduced.

[16] Editor's note: The implication (iv) $\Rightarrow$ (iii) will appear in the course of the proof.

b) Reduction to the case in which A is an algebra of finite type over $\mathbb{Z}$. We shall use the following lemma.

Lemma 3.4.— *Let k be a field, G an algebraic k-group, and E a subset of G. Let k'/k be a field extension, and suppose that $E_{k'}$ is the underlying set of a sub-torus $T_{k'}$ of $G_{k'}$. If $T_{k'}$ is central in $G_{k'}$, or if k is perfect, then E is the underlying set of a sub-torus T of G.*

Proof. By fpqc descent it is enough to prove that the two inverse images of $T_{k'}$ in $G_{k''}$, where

$$k'' = k' \otimes_k k',$$

coincide. They have the same underlying set. If $T_{k'}$ is central, they therefore coincide by Exposé IX 5.1 bis. If k is perfect, k'' is reduced; the two inverse images, being smooth over k'', are reduced and consequently coincide.

Remarks 3.5.— (1) *Lemma 3.4 applies in particular when the residue fields of S are perfect.*

(2) *Under the hypotheses of the first case in Theorem 3.1 C, for every S-scheme S' as in 3.1* (iv) b)*, the sub-torus of $G_{S'}$ having $E_{S'}$ as its underlying space is central.*

Lemma 3.6.— *Let S be the projective limit of a filtered inverse system of affine schemes S_i, and let H be an S-group of multiplicative type and of finite type. There exist an index i, a group H_i of multiplicative type over S_i, and an S-isomorphism*

$$H_i \times_{S_i} S \xrightarrow{\sim} H.$$

If H is isotrivial, H_i may be chosen isotrivial.

Proof. Since H is of finite presentation over S, there are an index i, an S_i-group prescheme H_i, of finite presentation over S_i, and an S-isomorphism

$$H_i \times_{S_i} S \xrightarrow{\sim} H$$

(Exposé $\mathrm{VI_B}$ 10.16). After increasing i, we may suppose that H_i is of multiplicative type (Exposé IX 2.11). It remains to prove that H_i can be chosen quasi-isotrivial, and isotrivial when H is isotrivial.

We may thus suppose that H is trivialized by an étale, surjective morphism $S' \to S$ of finite presentation. For i sufficiently large, there are an S_i-prescheme S'_i, an étale, surjective morphism $S'_i \to S_i$ of finite presentation, and an S-isomorphism

$$S'_i \times_{S_i} S \xrightarrow{\sim} S'$$

(EGA IV 17.16). For j sufficiently large set

$$S'_j = S'_i \times_{S_i} S_j, \qquad H'_j = H_j \times_{S_j} S'_j, \qquad H' = H \times_S S'.$$

By the choice of S', there are a finitely generated abelian group M and an S'-isomorphism of group schemes

$$D_{S'}(M) \xrightarrow{\sim} H'.$$

Since the S'_i are quasi-compact and $S' = \varprojlim S'_i$, Exposé $\mathrm{VI_B}$ 10 shows that there are an index j and an S'_j-isomorphism of group schemes

$$D_{S'_j}(M) \xrightarrow{\sim} H'_j.$$

This says precisely that H_j is a quasi-isotrivial group of multiplicative type. When H is isotrivial, one argues similarly using a trivialization of H by a finite étale morphism $S' \to S$.

We can now carry out reduction b). The ring A of S is the filtered direct limit of its subrings A_i of finite type over $\mathbb{Z}$. Put $S_i = \operatorname{Spec} A_i$, and let

$$u_j : S \longrightarrow S_j, \qquad u_{jk} : S_k \longrightarrow S_j \quad (k \geq j)$$

be the transition morphisms. Since S is noetherian and G is of finite presentation over S, there are an index i and an S_i-group prescheme G_i, of finite type over S_i, such that G is S-isomorphic to $G_i \times_{S_i} S$. Likewise, since E is locally closed in G, we may suppose that there is a locally closed subset E_i of G_i such that

$$E = E_i \times_{S_i} S$$

(EGA IV 8.3.11). For $j > i$, put

$$G_j = G_i \times_{S_i} S_j, \qquad E_j = E_i \times_{S_i} S_j,$$

and let Q_j be the set of points s of S_j for which $(E_j)_s$ is the underlying set of a central sub-torus of $(G_j)_s$.

It follows from 3.4 that

$$Q_k = u_{jk}^{-1}(Q_j) \quad (k \geq j),$$

and by hypothesis

$$\operatorname{ens}(S) = u_j^{-1}(Q_j) \quad (j \geq i).$$

I further claim that Q_j is ind-constructible (EGA IV 1.9.4). Indeed, since S_j is noetherian, it is enough (EGA IV 1.9.10) to see the following. Let S be a noetherian integral scheme with generic point η, and suppose that E_η is the underlying set of a central sub-torus T_η of G_η. Then there is a neighborhood U of η such that E_s has the same property for every $s \in U$. After shrinking S, we may suppose that the center Z of G is representable and that there is a subgroup scheme T of G whose generic fiber is T_η (Exposé $\mathrm{VI_B}$, §10). The two locally closed subsets $\operatorname{ens}(T)$ and E of $\operatorname{ens}(G)$ coincide on the generic fiber. We can therefore find a neighborhood U of η such that, above U,

$$\operatorname{ens}(T) = E$$

(EGA IV 8.3.11). For the same reasons, we may suppose that T is central above U; it is a torus by 3.6.

Now that Q_j is known to be ind-constructible, EGA IV 8.3.4 implies that $Q_j = \operatorname{ens}(S_j)$ for j sufficiently large. We may then replace S, G, E by S_j, G_j, E_j, which reduces us to the case in which S is an affine reduced scheme of finite type over $\mathbb{Z}$.

c) Reduction to the case in which S is the spectrum of a complete reduced noetherian local ring. Because the torus with underlying space E is unique, the usual technique (EGA IV 8) and Lemma 3.6 allow us to replace S by the spectrum of the local ring A at a point of S. Let $\widehat{S}$ be the spectrum of the completion $\widehat{A}$ of A for the topology defined by its maximal ideal. Since A is the localization of an algebra of finite type over $\mathbb{Z}$, $\widehat{S}$ is still reduced (EGA IV 7.6.5).

I claim that it is enough to solve the problem after the base extension $\widehat{S} \to S$. Indeed, let $\widehat{T}$ be the sub-torus of $G_{\widehat{S}}$ with underlying space $E_{\widehat{S}}$. Its two inverse images in $G_{\widehat{S} \times_S \widehat{S}}$ are central sub-tori with the same underlying space; they therefore coincide (Exposé IX 5.1 bis). By fpqc descent, $\widehat{T}$ comes from a torus T of G that solves the problem (cf. 3.4).

d) A descent lemma. Recall the following properties of finite morphisms, noted in TDTE I. Let S, S' be two preschemes and $u : S' \to S$ a finite morphism. Then:

(1) The morphism u is an epimorphism if and only if the canonical morphism of sheaves

$$\mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'})$$

is injective.

(2) The morphism u is an effective epimorphism (Exposé IV 1.3) if and only if the canonical diagram

$$0 \longrightarrow \mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'}) \rightrightarrows u_*(\mathcal{O}_{S'}) \otimes_{\mathcal{O}_S} u_*(\mathcal{O}_{S'})$$

is exact.

(3) If in addition S is noetherian and u is an epimorphism, u is the composite of a finite sequence of finite effective epimorphisms.

We can now prove the following lemma, whose proof uses a non-flat descent technique.

Lemma 3.7.— *Let S be a locally noetherian prescheme, G an S-group prescheme of finite type, S' a prescheme, and $u : S' \to S$ a finite morphism. For every S-prescheme T, let* $\mathrm{M}(T)$ *be the set of subgroups of G_T of multiplicative type* (respectively, the set of central subgroups of G_T of multiplicative type)*, so that* M *is naturally a contravariant functor on* Sch/S*. If u is an effective epimorphism* (respectively, an epimorphism)*, then the diagram of sets*

$$\mathrm{M}(S) \longrightarrow \mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_S S') \tag{*}$$

is exact.

Proof. *i) Reduction to the case in which u is an effective epimorphism.* Here we consider the functor of central subgroups of G of multiplicative type. The injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$ is local on S; once this injectivity is granted, the exactness of $(*)$ is also local on S. We may therefore suppose that S is affine and noetherian.

Consider the case in which $u : S' \to S$ is the composite of two finite epimorphisms

$$S' \xrightarrow{v} S'' \xrightarrow{w} S.$$

I claim that if both diagrams

$$\mathrm{M}(S'') \longrightarrow \mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_{S''} S') \tag{*'}$$

and

$$\mathrm{M}(S) \longrightarrow \mathrm{M}(S'') \rightrightarrows \mathrm{M}(S'' \times_S S'') \tag{*''}$$

are exact, then so is $(*)$.

Indeed, the injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$ is clear. If T' belongs to

$$\mathrm{Ker}\big(\mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_S S')\big),$$

then, a fortiori, T' belongs to

$$\mathrm{Ker}\big(\mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_{S''} S')\big).$$

By the exactness of $(*')$, it therefore comes from a unique element T'' of $\mathrm{M}(S'')$. It is enough to show that T'' belongs to

$$\mathrm{Ker}\big(\mathrm{M}(S'') \rightrightarrows \mathrm{M}(S'' \times_S S'')\big),$$

for then the exactness of $(*'')$ descends T'' to S. Let T''_1, T''_2 be the two inverse images of T'' in $\mathrm{M}(S'' \times_S S'')$. These are central subgroups of $G_{S''\times_S S''}$ of multiplicative type, so to prove that $T''_1 = T''_2$ it is enough to see that they have the same fibers (Exposé IX 5.1 bis). Consider the commutative diagram

$$\begin{tikzcd} S' & S' \times_S S' \arrow[l] \arrow[d, "v\times v"] \\ S'' \arrow[u, "v"] & S'' \times_S S'' \arrow[l] \end{tikzcd}$$

The morphism v is a finite epimorphism, hence dominant by property (1) above, and closed; it is therefore surjective, and so is $v \times v$. Let x'' be a point of $S'' \times_S S''$, and let x' be a point of $S' \times_S S'$ lying above x''. The commutativity of the diagram shows that the two inverse images of $(T''_1)_{x''}$ and $(T''_2)_{x''}$ in $G_{x'}$ coincide. Thus

$$(T''_1)_{x''} = (T''_2)_{x''}$$

(EGA IV 2.2.15).

The preceding argument and an immediate induction on the number of factors in a factorization of $u : S' \to S$ into effective epimorphisms, supplied by property (3) above, show that in order to prove the exactness of $(*)$ for central subgroups of multiplicative type it is enough to treat the case in which u is an effective epimorphism. Finally, another application of Exposé IX 5.1 bis shows that it is enough to prove 3.7 when M is the functor of all subgroups of G of multiplicative type and u is an effective epimorphism.

ii) Injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$. Since $u : S' \to S$ is an epimorphism, the canonical morphism

$$\mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'})$$

is injective. Moreover, an S-group of multiplicative type is flat over S. The injectivity of $\mathrm{M}(S) \to \mathrm{M}(S')$ is therefore a consequence of the following more general lemma.

Lemma 3.8.— *Let $f : X \to S$ and $g : S' \to S$ be morphisms of preschemes, let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, and put*

$$X' = X \times_S S', \qquad \mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

Let $\mathrm{Q}(\mathcal{F})$ be the set of quotient $\mathcal{O}_X$-modules $\mathcal{G}$ of $\mathcal{F}$ that are quasi-coherent and flat over S, and define $\mathrm{Q}(\mathcal{F}')$ analogously relative to $\mathcal{F}', X'$, and S'. Suppose that g is quasi-compact and that

$$\mathcal{O}_S \longrightarrow g_*(\mathcal{O}_{S'})$$

is injective. Then the canonical map

$$\begin{aligned} \mathrm{Q}(\mathcal{F}) &\longrightarrow \mathrm{Q}(\mathcal{F}'), \\ \mathcal{G} &\longmapsto \mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \end{aligned}$$

is injective.

Proof. Indeed, we may suppose that S is affine, and then that X is affine. Since S' is quasi-compact, it has a finite covering by affine open subsets. Replacing S' by the disjoint union of these open subsets, we may therefore suppose that S' is affine. We are then reduced to the following lemma, whose proof is immediate.

Lemma 3.9.— *Let $A \to A'$ be an injective homomorphism of rings, let M be an A-module, let $N = M/P$ be a quotient A-module flat over A, and put*

$$M' = M \otimes_A A', \qquad N' = N \otimes_A A' = M'/P'.$$

Then P is the inverse image of P' under the canonical homomorphism $M \to M'$. Thus N and P are determined by N' and P'.

iii) Exactness of 3.7 $(*)$ *at* $\mathrm{M}(S')$. Let

$$T' \in \mathrm{Ker}\big(\mathrm{M}(S') \rightrightarrows \mathrm{M}(S' \times_S S')\big).$$

Suppose that iii) has been proved when S is the spectrum of a noetherian local ring. For every point s of S, there then exists a subgroup T_s of multiplicative type in $G \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$, obtained by descent from

$$T' \times_{S'} u^{-1}(\mathrm{Spec}(\mathcal{O}_{S,s})).$$

By Exposé $\mathrm{VI_B}$, §10, and 3.6, there is an open neighborhood U of s in S and a subgroup scheme T of G over U, of multiplicative type, that extends T_s. Let U' be the inverse image of U in S'. We then have two subschemes of $G'|_{U'}$, namely $T'|_{U'}$ and $T \times_U U'$, which coincide over $u^{-1}(\mathrm{Spec}(\mathcal{O}_{S,s}))$.

Regarding the latter scheme as the projective limit of the $u^{-1}(V)$, where V runs through the open neighborhoods of s in U, EGA IV 8 shows that there is an open subset $V \subset U$, containing s, over whose inverse image $T \times_U U'$ and T' coincide. Thus, under the stated hypotheses, T' descends locally on S; by the uniqueness proved in ii), it consequently descends globally. In short, it is enough to prove iii) when S is the spectrum of a noetherian local ring.

Let $\widehat{S}$ be the spectrum of the completion of the ring of S, and put

$$S'' = S' \times_S S', \qquad \widehat{S}' = \widehat{S} \times_S S', \qquad \widehat{S}'' = \widehat{S} \times_S S'' = \widehat{S}' \times_S S'.$$

I claim that it is enough to show that

$$\mathrm{M}(\widehat{S}) \longrightarrow \mathrm{M}(\widehat{S}') \rightrightarrows \mathrm{M}(\widehat{S}'') \tag{$\widehat{*}$}$$

is exact at $\mathrm{M}(\widehat{S}')$. This follows from the commutative diagram

$$\begin{tikzcd}
\mathrm{M}(S) \arrow[r] \arrow[d] & \mathrm{M}(S') \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & \mathrm{M}(S'') \arrow[d] \\
\mathrm{M}(\widehat{S}) \arrow[r] \arrow[d, shift left] \arrow[d, shift right] & \mathrm{M}(\widehat{S}') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left] \arrow[d, shift right] & \mathrm{M}(\widehat{S}'') \\
\mathrm{M}(\widehat{S} \times_S \widehat{S}) \arrow[r, "f"] & \mathrm{M}(\widehat{S}' \times_{S'} \widehat{S}'). &
\end{tikzcd}$$

Here the second row is exact at $\mathrm{M}(\widehat{S}')$ by hypothesis, the first two columns are exact by fpqc descent, and f is injective by ii), applied to the finite epimorphism

$$\widehat{S}' \times_{S'} \widehat{S}' \longrightarrow \widehat{S} \times_S \widehat{S}$$

deduced from $S' \to S$ by the flat base change $\widehat{S} \times_S \widehat{S} \to S$. The diagram chase is left to the reader. The characterization of finite effective epimorphisms recalled in d)(2) shows that

$\widehat{S}' \to \widehat{S}$, obtained from $S' \to S$ by the flat base change $\widehat{S} \to S$, is again a finite effective epimorphism. We are therefore reduced to the case in which S is, in addition, the spectrum of a complete noetherian local ring.

Let S_0 be the reduced closed subscheme of S whose underlying space is the closed point. Let T' be as above and write T'' for its common image in $\mathrm{M}(S'')$. Since

$$S'_0 = S' \times_S S_0$$

is faithfully flat over S_0, there is a subgroup T_0 of G_{S_0}, of multiplicative type, whose inverse image in $G_{S'_0}$ is $T'_{S'_0}$, by fpqc descent. Since S is complete noetherian local, there exist an S-group T of multiplicative type and an S_0-morphism

$$u_0 : T \times_S S_0 \longrightarrow T_0$$

(Exposé X 3.3). The inverse image of u_0 over S'_0 extends uniquely to an S'-isomorphism

$$u' : T_{S'} \xrightarrow{\sim} T',$$

again by Exposé X 3.3. Notice here that, because S' is finite over the complete local scheme S, it is a finite disjoint sum of complete local schemes.

The two inverse images of u' over S'' are morphisms from $T_{S''}$ to T'' that coincide over $S_0 \times_S S''$; hence they coincide everywhere by the same result. Since T is flat over S and $S' \to S$ is a finite effective epimorphism, TDTE I, page 8, gives an exact diagram

$$\mathrm{Hom}_S(T, G) \longrightarrow \mathrm{Hom}_{S'}(T_{S'}, G_{S'}) \rightrightarrows \mathrm{Hom}_{S''}(T_{S''}, G_{S''}).$$

Thus u' comes from a morphism of preschemes $u : T \to G$. Likewise, $T \times_S T$ is flat over S, and therefore

$$\mathrm{Hom}_S(T \times_S T, G) \longrightarrow \mathrm{Hom}_{S'}(T_{S'} \times_{S'} T_{S'}, G_{S'})$$

is injective. It follows at once that u is a homomorphism of groups, since u' is one.

Moreover, u is a monomorphism. First observe that $\mathrm{Ker}(u)$ is flat over S. To prove this we may suppose that S is artinian local (EGA 0_{III}, 10.2.2), hence that G is separated (Exposé $\mathrm{VI_A}$, 0.3); then $\mathrm{Ker}(u)$ is of multiplicative type (Exposé IX 6.8), and is therefore flat over S. Since

$$\mathrm{Ker}(u) \times_S S' = \mathrm{Ker}(u') = 0,$$

Lemma 3.8 gives $\mathrm{Ker}(u) = 0$. Finally, a monomorphism u is an immersion (Exposé VIII, Remarks 7.13), and its image $u(T)$ is an element of $\mathrm{M}(S)$ whose image in $\mathrm{M}(S')$ is T'. This proves Lemma 3.7.

e) End of the proof of I). By reduction c), we are reduced to the case in which S is the spectrum of a complete reduced noetherian local ring A. Let S' be the spectrum of the normalization A' of A. Nagata's theorem says that A' is finite over A (EGA 0_{IV}, 23.1.5), and $S' \to S$ is an epimorphism because $A \to A'$ is injective. Suppose that $G_{S'}$ has a torus T' whose underlying space is $E_{S'}$. The two inverse images of T' in $G_{S' \times_S S'}$ are central sub-tori with the same underlying space, so they coincide (Exposé IX 5 bis). By Lemma 3.7, T' therefore comes from a central sub-torus T of G_S, which plainly has E as its underlying space. It is thus enough to prove the existence of T'. This reduces us to the normal case and completes the proof of I).

II) Proof that (iii) *implies* (ii) *when S is normal.* We may restrict ourselves to the case in which S is integral; let t be its generic point. For every integer n that is a power of q, let ${}_nG$ be the "kernel" in G of raising to the n-th power. Since E is locally closed in G, by (iii)(b), the intersection of $\mathrm{ens}({}_nG)$ with E is locally closed in $\mathrm{ens}(G)$. Let $E(n)$ denote the reduced subscheme of G whose underlying space is $\mathrm{ens}({}_nG) \cap E$.

We show that the structural morphism

$$E(n) \longrightarrow S$$

is separated and universally open. Both properties have a valuative criterion (EGA II 7.2.3 and EGA IV 14.5.8). Let S' therefore be an S-scheme that is the spectrum of a complete discrete valuation ring with algebraically closed residue field. We have proved that 3.1(iii) implies 3.1(iv), so there is a sub-torus T' of $G_{S'}$ whose underlying space is $E_{S'}$. Now ${}_nT'$ is finite étale over S', hence is separated and universally open over S'; the same is therefore true of $E(n) \times_S S'$, which has the same underlying space.

The fibers of $E(n)$, moreover, all have the same number of geometric points, namely n^r if r is the rank of the torus T_t.Finally, the generic fiber $E(n)_t$, being reduced, is equal to ${}_nT_t$, and hence is étale over $\mathrm{Spec}(t)$. Since S is normal, SGA I 10.11 now shows that $E(n)$ is an étale covering of S.

If s is a point of S, then $E(n)_s$ is étale over $\mathrm{Spec}\,\kappa(s)$, hence reduced, and consequently coincides with the group ${}_nT_s$ of multiplicative type. We show that $E(n)$ is a group subscheme of G. Let

$$m : E(n) \times_S E(n) \longrightarrow G$$

be the morphism induced by multiplication in G. The underlying map $\mathrm{ens}(m)$ factors through $E(n)$, so m itself factors through the scheme $E(n)$, because its source $E(n) \times_S E(n)$ is étale over S, and hence reduced. It then follows from Exposé X 4.8(a) that $E(n)$ is a subgroup of multiplicative type. As observed already in 3.2(a), the family of subgroups $E(n)$ is necessarily coherent. This proves that (iii) implies (ii) when S is normal.

III) Proof that (ii) *implies* (i). In fact, we shall show that there is a unique sub-torus T of G, with underlying space E, such that

$${}_nT = M_n$$

for every n that is a power of q. Uniqueness follows immediately from Exposé IX 4.8(b). In proving existence, in view of this uniqueness, we may successively suppose that:

a) S is noetherian.

b) The M_n are central subgroups. Indeed, it is enough to replace G by

$$Z_n = \mathrm{Centr}_G(M_n)$$

for sufficiently large n, by 2.5 and 2.5 bis.

c) S is the spectrum of a local ring. Suppose that the problem has been solved after every base change $\mathrm{Spec}(\mathcal{O}_{S,s}) \to S$, for $s \in S$, and let T_s be the resulting sub-torus of $G \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$. There is then an open neighborhood U of s and a sub-torus T of $G|_U$ extending T_s, by Exposé $\mathrm{VI_B}$, §10, and 3.6. We have proved that 3.1(ii) implies 3.1(iv); since S is noetherian, E is therefore constructible by 3.3. After shrinking U, we may consequently suppose that

$$\mathrm{ens}(T) = E \times_S U$$

(EGA IV 9.5.2). For every n that is a power of q, the groups ${}_nT$ and $M_n \times_S U$ are then two subgroups of multiplicative type in $G|_U$, with the same fibers; because M_n is central, they coincide by Exposé IX 5.3 bis. Thus the problem has a local solution on S, and uniqueness makes these local solutions into a global one.

d) S is the spectrum of a complete noetherian local ring and, if s is the closed point, T_s is trivial. This follows from EGA 0_{III}, 10.3.1, and fpqc descent.

e) S is reduced. Apply 2.2.

f) S is normal. Apply Nagata's finiteness theorem (EGA 0_{IV}, 23.1.5) and Lemma 3.7 for central sub-tori.

After these reductions, T_s is trivial, and hence $(M_n)_s$ is trivial. It follows from Exposé X 3.3 that M_n itself is trivial. If t is any point of S, all the subgroups ${}_nT_t$ of T_t, for n a power of q, are therefore trivial; Exposé X 1.4 then shows at once that T_t itself is trivial. It remains only to prove the following lemma.

Lemma 3.10.— *Under the hypotheses of 3.1(ii), suppose in addition that S is noetherian and normal and that, for every point s of S, T_s is a trivial torus. Then there exists a unique sub-torus T of G, with underlying space E, such that*

$$ {}_nT = M_n $$

for every n that is a power of q. Moreover, T is trivial.

Proof. Uniqueness follows from the fact that T, being smooth over S, is reduced. To prove existence, we may suppose that S is irreducible, with generic point η. Let r be the rank of T_η, put

$$ T' = \mathbb{G}^r_{m,S}, $$

and choose an isomorphism

$$ u_\eta : T'_\eta \xrightarrow{\sim} T_\eta. $$

Write ${}_nu_\eta$ for its restriction to ${}_nT'_\eta$. Since ${}_nT'$ and M_n are trivial, ${}_nu_\eta$ has a unique extension to an S-isomorphism

$$ {}_nu : {}_nT' \xrightarrow{\sim} M_n. $$

I claim that, for every point s of S, there is a necessarily unique group isomorphism

$$ u_s : T'_s \xrightarrow{\sim} T_s $$

that extends $({}_nu)_s$ for every n that is a power of q.

Indeed, let S^1 be an S-scheme that is the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point t_1 maps to η and whose closed point s_1 maps to s (EGA II 7.1.9). The proof that 3.1(ii) implies 3.1(iv) gives a sub-torus T^1 of G_{S^1} such that

$$ (M_n)_{S^1} = {}_nT^1 $$

for every n that is a power of q. Since S^1 is normal, T^1 is isotrivial by Exposé X 5.16. The classification of isotrivial tori (Exposé X 1.2) and SGA I, Exposé V 8.2, together with the fact that the generic fiber of T^1 is trivial, show that T^1 is trivial. We may therefore extend

$$ u_\eta \times_\eta t_1 $$

to an S^1-isomorphism of group schemes

$$ u^1 : T' \times_S S^1 \xrightarrow{\sim} T^1. $$

On ${}_nT' \times_S S^1$, the morphisms u^1 and ${}_nu \times_S S^1$ agree on the generic fiber, and hence agree everywhere. Restricting u^1 to the closed fiber gives the required u_s after the field extension $\kappa(s) \to \kappa(s_1)$; it descends to $\kappa(s)$ by Exposé IX 4.8(a) and fpqc descent.

The scheme T' is smooth over the normal scheme S, hence is normal. The extension criterion for rational maps in EGA IV_4, 20.4.6, now shows that there is an S-morphism

$$u : T' \longrightarrow G$$

whose restriction to each fiber T'_s is the composite

$$T'_s \xrightarrow{u_s} T_s \longrightarrow G_s.$$

Indeed, for every base change $S^1 \to S$ of the kind used above, the preceding construction gives precisely the required extension over S^1.

It remains to show that u is a homomorphism. Let $m_{T'}$ and m_G denote multiplication in T' and G, respectively. The coincidence subscheme of

$$m_G \circ (u \times_S u) \quad \text{and} \quad u \circ m_{T'}$$

contains every fiber, since every u_s is a homomorphism. It therefore has the same underlying space as $T' \times_S T'$, and so equals $T' \times_S T'$, because the latter is smooth over S and hence reduced. Finally, u is a monomorphism because it is one on every fiber; it is therefore an immersion (Exposé VIII 7.9). The image of T' under u is a sub-torus of G whose underlying set is E.

This completes the proof of Theorem 3.1.

4. Characterization of a sub-torus T by the subgroups ${}_nT$

4.1. Statement of the main theorem

Theorem 4.1.— *Let S be a connected locally noetherian prescheme, let G be an S-group prescheme of finite type over S, let $q > 1$ be an integer invertible on S, and let r be a positive integer. For every integer n that is a power of q, let $M(n)$ be a subgroup scheme of G, of multiplicative type and of type $(\mathbb{Z}/n\mathbb{Z})^r$. Suppose that:*

a) *the family $M(n)$ is coherent: if $m \mid n$, then*

$${}_mM(n) = M(m);$$

b) *there exist a point $s \in S$ and a sub-torus T_s of G_s such that*

$$M(n)_s = {}_nT_s \qquad \textit{for every } n;$$

c) *for every point $t \in S$, there exists a closed affine subscheme F_t of G_t containing $M(n)_t$ for every n.*

Then there exists a unique sub-torus T of G such that ${}_nT = M(n)$ for every n that is a power of q.

There is an analogous theorem for lifting morphisms.

Theorem 4.1 bis.— *Let S, G, q be as above, let T be an S-torus, and, for every n that is a power of q, let $u(n) : {}_nT \to G$ be an S-homomorphism. Suppose that:*

a) *the family $u(n)$ is coherent: if $m \mid n$, then*

$$u(m) = u(n)|_{{}_mT};$$

b) *there exist a point $s \in S$ and a homomorphism $u_s : T_s \to G_s$ such that*

$$u_s|_{{}_nT_s} = u(n)_s \qquad \text{for every } n \text{ that is a power of } q;$$

c) *for every point $t \in S$, there exists a closed affine subscheme F_t of G_t containing $u(n)_t({}_nT_t)$ for every n.*

Then there exists a unique homomorphism $u : T \to G$ whose restriction to ${}_nT$ equals $u(n)$ for every n that is a power of q.

Remark 4.2.— *Using the lower semicontinuity of the abelian rank of a flat group prescheme of finite type over the spectrum of a discrete valuation ring (cf. Exposé X 8.7), condition (c) in Theorems 4.1 and 4.1 bis may be weakened by requiring the closed affine subscheme F_t only for every maximal point t of S.*

Let us deduce Theorem 4.1 bis from Theorem 4.1. Put $G' = G \times_S T$. For every n that is a power of q, consider the homomorphism

$$v(n) : {}_nT \longrightarrow G'$$

whose projections to G and T are respectively $u(n)$ and the canonical immersion ${}_nT \to T$. Thus $v(n)$ is an immersion; let $M(n)$ be its image subgroup. The family $M(n)$ is coherent in the sense of Theorem 4.1. The group $M(n)_s$ equals ${}_nT'_s$, where T'_s is the sub-torus of G'_s that is the graph of u_s, and, for every $t \in S$, the closed affine subscheme $F_t \times_t T_t$ of G'_t contains all the groups $M(n)_t$. By Theorem 4.1 there is therefore a sub-torus T' of G' such that ${}_nT' = M(n)$ for every n.

Let $f : T' \to T$ be the restriction of the projection $G' \to T$, and let $f(n)$ be its restriction to $M(n)$. Then

$$f(n) \circ v(n) = \mathrm{id}_{{}_nT}\,.$$

The fibre T'_s is the graph of u_s (Exposé IX 4.8(b)), so f_s is an isomorphism. The groups $\ker f$ and $\operatorname{coker} f$ are of multiplicative type (Exposé IX 2.7) and of constant type, since S is connected. They are therefore trivial, and f is an isomorphism. If $v = f^{-1}$, then

$$v|_{{}_nT} = v(n).$$

Composing v with the projection $G' \to G$ gives the required homomorphism $u : T \to G$. This proves existence; uniqueness follows from Exposé IX 4.8(a).

4.2. Application

We shall generalize Theorem IX 7.1. Let A be a complete noetherian local ring, let I be its maximal ideal, and put

$$S = \operatorname{Spec} A, \qquad S_m = \operatorname{Spec}(A/I^m).$$

For every prescheme X, put $X_m = X \times_S S_m$.

Let G be an S-group prescheme of finite type, let T be an S-torus, let q be an integer invertible on S, and let

$$u_m : T_m \longrightarrow G_m \qquad (m \in \mathbb{N})$$

be a coherent family of homomorphisms. For n ranging over the powers of q, denote by $u_m(n)$ the restriction of u_m to ${}_nT_m$, and by

$$u(n) : {}_nT \longrightarrow G$$

the unique homomorphism extending the morphisms $u_m(n)$ for all m (Proposition 1.6(a)). We call the family $(u_m)_{m\in\mathbb{N}}$ *admissible* if, for every point $t \in S$, there exists a closed affine subprescheme F_t of G_t containing $u(n)_t({}_nT_t)$ for every n that is a power of q. As will be seen, this property does not depend on q.

Proposition 4.3.— *With the preceding notation, the canonical map*

$$\mathrm{Hom}_{S\text{-gr}}(T, G) \longrightarrow \varprojlim_m \mathrm{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

identifies its source with the subset of the target consisting of admissible coherent families.

Indeed, apply Theorem 4.1 bis with s the closed point of S.

Corollary 4.4.— *With the preceding notation, suppose in addition that G has affine fibres. Then the canonical map*

$$\mathrm{Hom}_{S\text{-gr}}(T, G) \longrightarrow \varprojlim_m \mathrm{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

is an isomorphism.

Indeed, when G has affine fibres every coherent family is admissible.

Remark 4.5.— *When G is separated, the ring A in Proposition 4.3 may be replaced by a noetherian I-adic ring: one uses EGA III 5.4.1 in place of Proposition 1.6(a).*

4.3. Proof of Theorem 4.1

Lemma 4.6.— *Let k be a field, let G be a k-algebraic group, let q be an integer prime to the characteristic of k, let $r > 0$, and let $M(n)$, for n ranging over the powers of q, be a coherent family of subgroups of G, of multiplicative type and of type $(\mathbb{Z}/n\mathbb{Z})^r$. There exists a smallest closed subscheme H of G containing all the $M(n)$. Moreover, H is a smooth, connected, commutative algebraic subgroup of G, and its formation commutes with extension of the base field.*

Let $M \subseteq \mathrm{ens}(G)$ be the union of the underlying sets of all the $M(n)$, and let H be the reduced closed subscheme of G whose underlying space is the closure of M. Each $M(n)$ is étale over k, hence reduced, and is therefore contained in H. Thus H is the smallest closed subscheme containing all the $M(n)$.

Let $\bar{k}$ be an algebraically closed extension of k. By construction the $M(n)$ are schematically dense in H (Exposé IX 4.1), hence the $M(n)_{\bar{k}}$ are schematically dense in $H_{\bar{k}}$ (Exposé IX 4.5). Since every $M(n)_{\bar{k}}$ is reduced, $H_{\bar{k}}$ is the reduced closed subscheme of $G_{\bar{k}}$ whose underlying space is the closure of $M_{\bar{k}}$. Thus H is geometrically reduced (EGA IV 4.6.1).

Coherence of the family implies that M is stable under the group law. The subset $M \times_k M$ is dense in $\mathrm{ens}(H \times_k H)$, and $H \times_k H$ is reduced (EGA IV 4.6.5); it follows that H is an algebraic subgroup of G. It is smooth over k, since it is geometrically reduced (Exposé $\mathrm{VI_A}$ 1.3.1), and it is commutative because all the $M(n)$ are commutative.

It remains to prove connectedness. Let H^0 be the identity component of H, let m be the number of geometric points of H/H^0, and let q^s be the q-primary part of m. For every n that is a power of q, the subgroup ${}_{q^s}M(n)$ is contained in H^0. Coherence and the prescribed type of $M(n)$ give

$${}_{q^s}M(nq^s) = M(n).$$

Hence H^0 contains every $M(n)$, and therefore $H^0 = H$.

We can now sharpen condition (c) of Theorem 4.1. By Lemma 4.6 there is, for each $t \in S$, a smallest closed subscheme H_t of G_t containing all the $M(n)_t$. Its formation commutes with extension of the residue field. Thus $H_t \otimes_t \bar{t}$ is contained in the affine closed subset F_t, and is affine; hence H_t itself is affine. Lemma 4.6 also says that H_t is a smooth, connected, commutative algebraic group. The structure theorem for such affine groups over an algebraically closed field (*Bible*, 4, Theorem 4) expresses H_t as a direct product of a torus T_t and a unipotent group U_t. Since every $M(n)_t$ is contained in T_t, minimality gives $H_t = T_t$. Consequently H_t is a torus.

We begin the proof of Theorem 4.1.

a) Uniqueness. This is Exposé IX 4.8(b).

b) The case in which S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field k. Let K be the fraction field of A, let s and t be the closed and generic points of S, let J be the maximal ideal of A, and put

$$S_m = \operatorname{Spec}(A/J^m), \qquad X_m = X \times_S S_m.$$

First case: the point in Theorem 4.1(b) is the generic point t. Let T' be the schematic closure of T_t in G. Its closed fibre T'_s is an algebraic subgroup of G_s, of dimension r, containing every $M(n)_s$, hence containing H_s. The torus H_s contains every $M(n)_s$, so it has rank at least r. Consequently H_s has the same underlying space as the identity component of T'_s.

The identity component $(T')^0$ of T' is then a subgroup prescheme of G, flat and separated (Exposé $\mathrm{VI_B}$ 5.2), whose generic fibre T_t is a torus and whose reduced closed fibre H_s is a torus. Hence $(T')^0$ is a torus (Exposé X 8.8); denote it by T. The groups ${}_nT$ and $M(n)$ are smooth over S, hence reduced, and have the same underlying space. They therefore coincide, and T is the required solution.

Second case: the point in Theorem 4.1(b) is the closed point s. After replacing G by the schematic closure in G of the smallest algebraic subgroup H_t containing the family $M(n)_t$ (Lemma 4.6), we may suppose that G_t is affine.

For every $m \geq 0$, Proposition 2.2 gives a unique sub-torus T_m of G_m lifting T_s and satisfying

$${}_nT_m = M(n)_m \qquad \text{for every } n \text{ that is a power of } q.$$

Put $T' = \mathbb{G}^r_{m,S}$. Since k is algebraically closed, T_s is split, and there is a k-isomorphism

$$u_s : T'_s \xrightarrow{\sim} T_s.$$

It lifts uniquely to an S_m-isomorphism $u_m : T'_m \to T_m$ (Exposé IX 3.3). The coherent family $(u_m)_{m\in\mathbb{N}}$ defines a morphism of formal completions $\widehat{u} : \widehat{T'} \to \widehat{G}$, fitting into

$$\begin{tikzcd} \widehat{T'} \arrow[r, "\widehat{u}"] \arrow[d, "i"'] & \widehat{G} \arrow[d, "j"] \\ T' \arrow[r, dashed] & G. \end{tikzcd}$$

We shall prove that $\widehat{u}$ is algebraizable, reducing first to the case in which G is affine.

Lemma 4.7.— *Let $S = \operatorname{Spec} A$, where A is a discrete valuation ring, and let X, Y be S-preschemes that are quasi-compact, quasi-separated, and flat over S. Then the canonical map*

$$\Gamma(X, \mathcal{O}_X) \otimes_A \Gamma(Y, \mathcal{O}_Y) \longrightarrow \Gamma(X \times_S Y, \mathcal{O}_{X\times_S Y})$$

is an isomorphism.

Let $f : X \to S$ and $g : Y \to S$ be the structural morphisms. Because X and Y are quasi-compact and quasi-separated, EGA I 9.2.2 and EGA IV 1.7.4 show that $f_*\mathcal{O}_X$ and $g_*\mathcal{O}_Y$ are quasi-coherent $\mathcal{O}_S$-algebras. Let $\widetilde{X}$ and $\widetilde{Y}$ be the corresponding affine S-schemes. Since Y is flat over S, EGA III 1.4.15 and EGA IV 1.7.21 give an isomorphism

$$\Gamma(\widetilde{X} \times_S Y, \mathcal{O}_{\widetilde{X}\times_S Y}) \xrightarrow{\sim} \Gamma(X \times_S Y, \mathcal{O}_{X\times_S Y}).$$

The scheme $\widetilde{X}$ is flat over S, because over A flatness is equivalent to torsion-freeness. Applying EGA III 1.4.15 again, with X and Y interchanged, yields

$$\begin{aligned}\Gamma(\widetilde{X}, \mathcal{O}_{\widetilde{X}}) \otimes_A \Gamma(\widetilde{Y}, \mathcal{O}_{\widetilde{Y}}) &\xrightarrow{\sim} \Gamma(\widetilde{X} \times_S \widetilde{Y}, \mathcal{O}_{\widetilde{X}\times_S \widetilde{Y}}) \\ &\xrightarrow{\sim} \Gamma(\widetilde{X} \times_S Y, \mathcal{O}_{\widetilde{X}\times_S Y}),\end{aligned}$$

which proves the lemma.

The S-group G is flat and of finite type over S, hence quasi-compact and quasi-separated. Applying the lemma to $G \times_S G$ gives

$$\Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G) \xrightarrow{\sim} \Gamma(G \times_S G, \mathcal{O}_{G\times_S G}).$$

Thus the multiplication $m_G : G \times_S G \to G$ determines a homomorphism

$$\Gamma(G, \mathcal{O}_G) \longrightarrow \Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G),$$

and hence an S-morphism

$$m_{\widetilde{G}} : \widetilde{G} \times_S \widetilde{G} \longrightarrow \widetilde{G},$$

where $\widetilde{G}$ is the affine S-scheme corresponding to $\Gamma(G, \mathcal{O}_G)$. The usual identities show that $m_{\widetilde{G}}$ makes $\widetilde{G}$ an S-group scheme and that the canonical morphism $v : G \to \widetilde{G}$ is a homomorphism.

Remark 4.8.— *The group $\widetilde{G}$ plays the role of a largest affine quotient of G. In fact it is an fpqc quotient of G, and is therefore of finite type over S (XVII, Appendix III, 2).*

In our situation the generic fibre G_t is affine, so $v_t : G_t \to \widetilde{G}_t$ is an isomorphism (EGA I 9.3.3). Since $\widetilde{G}$ is affine, the coherent family

$$w_m = v_m \circ u_m : T'_m \longrightarrow \widetilde{G}_m \qquad (m \in \mathbb{N})$$

comes from a unique S-homomorphism (Exposé IX 7.1)

$$w : T' \longrightarrow \widetilde{G}.$$

Let T_t be the sub-torus

$$T_t = v_t^{-1}(w_t(T'_t))$$

of G_t. Its rank is at most r. We show that it contains every $M(n)_t$. Let

$$u(n)_m : ({}_nT')_m \xrightarrow{\sim} M(n)_m$$

be the restriction of u_m. This coherent family comes from a unique S-isomorphism

$$u(n) : {}_nT' \xrightarrow{\sim} M(n),$$

because $M(n)$ is finite over S. For every $m \geq 0$,

$$w_m|_{({}_nT')_m} = (v_m \circ u_m)|_{({}_nT')_m} = (v \circ u(n))_m.$$

It follows from Proposition 1.6(a) that

$$w|_{{}_nT'} = v \circ u(n).$$

On the generic fibre,

$$v_t^{-1}(w_t(({}_nT')_t)) = u(n)_t(({}_nT')_t) = M(n)_t.$$

Thus T_t contains every $M(n)_t$, and consequently has rank r.

We conclude as in the first case. Let T'' be the schematic closure of T_t in G. Since T_t contains every $M(n)_t$, the closed fibre T''_s contains every $M(n)_s$, and hence contains T_s by the density theorem. On the other hand T'' is flat over S and T_t has dimension r, so T''_s has dimension r (Exposé $\mathrm{VI_B}$ 4). Therefore T_s has the same underlying space as the identity component of T''_s, and the identity component of T'' is the required sub-torus of G.

c) End of the proof. To prove existence of T, we may assume that S is reduced (Proposition 2.2). By Theorem 3.1, (ii)⇒(i), it is then enough to show that the set U of points $x \in S$ for which there exists a sub-torus T_x of G_x satisfying

$${}_nT_x = M(n)_x \qquad \text{for every } n \text{ that is a power of } q$$

equals $\mathrm{ens}(S)$. Such a torus is necessarily unique, and fpqc descent reduces its existence to existence after an extension of the residue field of x (Exposé IX 4.8(b)).

Since S is locally noetherian and connected and U is nonempty (it contains the point s of Theorem 4.1(b)), it is enough to prove the following: if $x, x' \in S$, if x is a specialization of x', and if one of these two points lies in U, then both lie in U. The usual method (EGA II 7.1.9) reduces this to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, which was treated in part (b). This completes the proof of Theorem 4.1.

5. Representability of the functor of smooth subgroups equal to their connected normalizer

Proposition 5.1.— *Let S be a prescheme, let G be an S-group prescheme of finite presentation, and let H be a subgroup prescheme of G, smooth with connected fibres. Then:*

a) *the normalizer N of H in G is represented by a closed subprescheme of G, of finite presentation over S;*

b) *the following conditions are equivalent:*

   i) *the canonical immersion $H \to N$ is an open immersion;*

   ii) *N is smooth along the identity section, and its identity component, which is then representable (Exposé VI_B 3.10), equals H;*

   iii) *for every $s \in S$, one has $H_s = (N_s)^0$.*

Proof. The group prescheme H is locally of finite presentation over S, since it is smooth, and has connected fibres; it is therefore of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3). Part (a) follows from Exposé XI 6.11. The equivalence in (b), recalled here for reference, was proved in Exposé $\mathrm{VI_B}$ 6.5.1.

We can now state the main theorem of this section.

Theorem 5.2.— *Let S be a prescheme and G an S-group prescheme of finite presentation over S. Let $\mathcal{L}_G$, or simply $\mathcal{L}$, be the S-functor defined, for every S-prescheme S', by*

$$\mathcal{L}_G(S') = \left\{ \begin{array}{c} \textit{subgroup preschemes of } G_{S'},\ \textit{smooth over } S', \\ \textit{with connected fibres, equal to their connected normalizer} \end{array} \right\}.$$

Then $\mathcal{L}$ is represented by an S-prescheme that is the union of an increasing family $(U_i)_{i\in\mathbb{N}}$ of open subpreschemes, each quasi-projective and of finite presentation over S, and hence in particular separated over S.

Initial reductions. For every $r \geq 0$, let $\mathcal{L}_r$ be the subfunctor of $\mathcal{L}$ defined by

$$\mathcal{L}_r(S') = \left\{ \begin{array}{c} H \subseteq G_{S'}:\ H \text{ is smooth with connected fibres,} \\ H \text{ equals its connected normalizer, and } \dim(H/S') = r \end{array} \right\}.$$

The fibre dimension of a smooth group is locally constant. Hence $\mathcal{L}_r \to \mathcal{L}$ is represented by an immersion that is both open and closed. It is therefore enough to prove the asserted representability and finiteness properties for every $\mathcal{L}_r$, for then

$$\mathcal{L} = \coprod_{r\in\mathbb{N}} \mathcal{L}_r.$$

For $n \geq 0$, an S-prescheme S', and a subgroup prescheme H of $G_{S'}$, denote by $H^{(n)}$ the n-th normal invariant of the identity section $S' \to H$ (EGA IV 16.1.2). Thus $H^{(n)}$ is a finite-type $\mathcal{O}_{S'}$-module corresponding to the n-th infinitesimal neighbourhood of the identity section of H. If H is smooth of relative dimension r, then $H^{(n)}$ is locally free of a rank $\varphi(n,r)$ depending only on n, r. Since H is a subprescheme of $G_{S'}$, there is a canonical epimorphism, compatible with base change,

$$G^{(n)} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \xrightarrow{\sim} G^{(n)}_{S'} \longrightarrow H^{(n)}.$$

Introduce the projective S-scheme

$$P_{\varphi(n,r)} = \mathrm{Grass}_{\varphi(n,r)}(G^{(n)})$$

(EGA I, 2nd ed., 9.7; see also Cartan Seminar 1960/61, Exposé 14 by A. Grothendieck). The assignment

$$H \longmapsto H^{(n)}$$

therefore defines a canonical morphism

$$u_{n,r} : \mathcal{L}_r \longrightarrow P_{\varphi(n,r)}.$$

The group G acts naturally on $G^{(n)}$, and hence on $P_{\varphi(n,r)}$, through

$$\mathrm{int} : G \longrightarrow \mathrm{Aut}_{S\text{-gr}}(G), \qquad g \longmapsto \mathrm{int}(g).$$

If S' is quasi-compact over S and $H \in \mathcal{L}_r(S')$, then for sufficiently large n one has (Exposé XI 6.11)

$$\mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(n)}).$$

For each $n \geq 0$, define the subfunctor $\mathcal{L}_r^n$ of $\mathcal{L}_r$ by

$$\mathcal{L}_r^n(S') = \left\{ H \in \mathcal{L}_r(S') \;\middle|\; \mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(n)}) \right\}.$$

Representability of $\mathcal{L}_r^n$. The integer r remains fixed through the proof, so we write $\mathcal{L}, \mathcal{L}^n, P_n, u_n$ in place of $\mathcal{L}_r, \mathcal{L}_r^n, P_{\varphi(n,r)}, u_{n,r}$. Let

$$v_n : \mathcal{L}^n \longrightarrow P_n$$

be the restriction of u_n. The definition of $\mathcal{L}^n$ and Proposition 5.1(b)(ii) show that v_n is a monomorphism. More precisely:

Lemma 5.3.— *The morphism v_n is an immersion of finite presentation. Consequently $\mathcal{L}^n$ is represented by an S-prescheme, quasi-projective and of finite presentation over S.*

After replacing S by P_n, the usual method reduces the assertion to the following. Let $Q \in P_n(S)$, and define a subfunctor F of the final functor h_S by

$$F(S') = \begin{cases} h_S(S'), & \text{if there exists } H \in \mathcal{L}^n(S') \text{ with } H^{(n)} = Q_{S'}, \\ \varnothing, & \text{otherwise.} \end{cases}$$

We must prove that $F \to S$ is an immersion of finite presentation.

Let N be the normalizer of Q in G. It is a subgroup prescheme of finite presentation over S, namely the inverse image of Q under

$$G \longrightarrow P_n, \qquad g \longmapsto g \cdot Q.$$

We claim that F is equivalently described by

$$F(S') = \begin{cases} h_S(S'), & \begin{array}{l} N_{S'} \text{ is smooth along the identity section, of} \\ \text{relative dimension } r, \text{ and } (N_{S'})^{(n)} = Q_{S'}, \end{array} \\ \varnothing, & \text{otherwise.} \end{cases}$$

Call the first and second descriptions F_1 and F_2. Suppose $F_1(S') = h_S(S')$, and let $H \in \mathcal{L}^n(S')$ with $H^{(n)} = Q_{S'}$. Then

$$\begin{aligned} \mathrm{Norm}_{G_{S'}}(H) &= \mathrm{Norm}_{G_{S'}}(H^{(n)}) \\ &= \mathrm{Norm}_{G_{S'}}(Q_{S'}) = N_{S'}. \end{aligned}$$

By Proposition 5.1(b)(ii), $N_{S'}$ is smooth along the identity section and its identity component is H. Thus it has relative dimension r; since H is open in $N_{S'}$,

$$(N_{S'})^{(n)} = H^{(n)} = Q_{S'}.$$

Hence $F_2(S') = h_S(S')$.

Conversely, suppose $F_2(S') = h_S(S')$. The identity component of $N_{S'}$ is then represented (Exposé $\mathrm{VI_B}$ 3.10) by a subgroup prescheme H, smooth over S', with connected fibres of dimension r. Since H is invariant and open in $N_{S'}$,

$$\begin{aligned} N_{S'} \subseteq \mathrm{Norm}_{G_{S'}}(H) &\subseteq \mathrm{Norm}_{G_{S'}}(H^{(n)}) \\ &= \mathrm{Norm}_{G_{S'}}((N_{S'})^{(n)}) = \mathrm{Norm}_{G_{S'}}(Q_{S'}) = N_{S'}. \end{aligned}$$

All inclusions are equalities. The first says that H equals its connected normalizer, and the second says $H \in \mathcal{L}^n(S')$. Hence $F_1 = F_2$.

We retain the second description. Condition (a) is represented by an immersion of finite presentation, by the following lemma.

Lemma 5.4.— *Let S be a prescheme, let X be locally of finite presentation over S, let $\sigma : S \to X$ be a section, let $r \geq 0$, and define the subfunctor L of S by*

$$L(S') = \begin{cases} h_S(S'), & \begin{array}{l} X_{S'} \textit{ is smooth along } \sigma_{S'}, \textit{ of relative} \\ \textit{dimension } r \textit{ at the points of } \sigma_{S'}(S'), \end{array} \\ \varnothing, & \textit{otherwise.} \end{cases}$$

Then:

a) *$L \to S$ is an immersion of finite presentation;*

b) *let $\mathcal{J}$ be the conormal sheaf of $S \to X$. If for every $s \in S$ the vector space $\mathcal{J} \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ has rank at most r, then $L \to S$ is a closed immersion.*

Proof. The functor L is local on S, so we may suppose S affine. After replacing X by a neighbourhood of $\sigma(S)$, we may suppose X of finite presentation, and then reduce to noetherian S by EGA IV 8.9. In part (b), use also compatibility of the conormal sheaf with base change (EGA IV 16.6.4) and constructibility of the rank of the fibres of $\mathcal{J}$.

For an S-prescheme S', let $\mathcal{J}' = \mathcal{J}_{S'}$, let $S^\bullet(\mathcal{J}')$, canonically isomorphic to $S^\bullet(\mathcal{J})_{S'}$, be the symmetric algebra of $\mathcal{J}'$ over $\mathcal{O}_{S'}$, let $\mathrm{Gr}^\bullet(\sigma_{S'})$ be the graded algebra associated with $\sigma_{S'}$ (EGA IV 16), and let

$$\sigma_{n,S'} : S^n(\mathcal{J}') \longrightarrow \mathrm{Gr}^n(\sigma_{S'})$$

be the canonical epimorphism. By EGA IV 17.12.3 and EGA 0_{IV} 19.5.4, $X_{S'}$ is smooth along $\sigma_{S'}$ of relative dimension r if and only if:

i) $\mathcal{J}'$ is locally free of rank r;

ii) $\sigma_{n,S'}$ is an isomorphism for every $n \geq 1$.

TDTE IV, Lemma 3.6, shows that the functor making $\mathcal{J}$ locally free of rank r is represented by a subprescheme S_1, closed in S in case (b). Replacing S by S_1, assume that $\mathcal{J}$ is locally free of rank r. Proceed by induction on n. Suppose a closed subprescheme S_{n-1} represents the functor making $\sigma_{q,S'}$ injective for $q \leq n-1$. The subfunctor[17] making $\sigma_{n,S'}$ injective is represented by a closed subprescheme S_n of S_{n-1}.

Indeed, after replacing S by S_{n-1}, all $\sigma_{q,S}$ for $q \leq n-1$ are isomorphisms. The $(n-1)$-st normal invariant $X^{(n-1)}$ of σ_S has a filtration with successive quotients $\mathrm{Gr}^0(\sigma_S), \ldots, \mathrm{Gr}^{n-1}(\sigma_S)$, all locally free; hence $X^{(n-1)}$ is flat over S. Compatibility of the normal invariants with base change gives

$$\begin{aligned} \mathrm{Gr}^n(\sigma_{S'}) &= \ker\Big(X_{S'}^{(n)} \to X_{S'}^{(n-1)}\Big) = \mathrm{Gr}^n(\sigma)_{S'}, \\ \sigma_{n,S'} &= (\sigma_{n,S})_{S'}. \end{aligned}$$

Thus we must represent the functor making $\sigma_{n,S}$ injective. Locally we may suppose S affine and $S^n(\mathcal{J})$ free. The required functor is represented by the closed subscheme defined by the ideal generated by the coordinates of $\ker \sigma_{n,S}$ in a basis of $S^n(\mathcal{J})$.

Since S is noetherian, the descending chain (S_n) stabilizes. Its stable value represents L, proving the lemma.

[17]The editors replaced "functor" by "subfunctor" here, and ask whether the intended wording is instead "which makes $\sigma_{q,S'}$ injective for $q \leq n$".

Return to $\mathcal{L}^n$. After replacing S by the subprescheme provided by Lemma 5.4, assume N smooth along the identity section of relative dimension r. Then $\mathcal{L}^n$ is the coincidence functor of two sections $h, g : S \to P_n$, corresponding to Q and $N^{(n)}$. It is therefore represented by the inverse image of the diagonal of $P_n \times_S P_n$ under the finite-presentation morphism $h \times_S g$. This proves Lemma 5.3.

The transition morphisms $\mathcal{L}^n \to \mathcal{L}^m$. If $H \in \mathcal{L}^n(S')$, then $H \in \mathcal{L}^m(S')$ for every $m \geq n$. Thus there are natural monomorphisms

$$u_n^m : \mathcal{L}^n \longrightarrow \mathcal{L}^m \qquad (m \geq n).$$

Lemma 5.5.— *The morphism u_n^m is an open immersion.*

After a base change, it is enough to consider $H \in \mathcal{L}^m(S)$ and

$$N = \mathrm{Norm}_G(H) = \mathrm{Norm}_G(H^{(m)}), \qquad N' = \mathrm{Norm}_G(H^{(n)}).$$

Let D be the coincidence functor of N, N': $D(S') = h_S(S')$ if $N_{S'} = N'_{S'}$, and $D(S') = \varnothing$ otherwise. We must show that $D \to S$ is an open immersion.

The same functor is obtained by requiring that the immersion $H \to N'$ become open. One implication follows from Proposition 5.1. Conversely, if $H_{S'} \to N'_{S'}$ is open, then, because H has connected fibres, $H_{S'}$ is the identity component of $N'_{S'}$ (Exposé $\mathrm{VI_B}$ 3.10), hence is invariant in $N'_{S'}$. Thus $N'_{S'} \subseteq N_{S'}$. The opposite inclusion always holds, so equality follows. Since H, N' are of finite presentation and H is flat over S, Exposé $\mathrm{VI_B}$ 2.6 shows that $D \to S$ is open.

End of the proof of Theorem 5.2. The functors $\mathcal{L}^n$ are represented, and the transition morphisms u_n^m are compatible open immersions. There is therefore an S-prescheme X, the union of an increasing sequence of open subpreschemes $(X_i)_{i \in \mathbb{N}}$, such that X_i represents $\mathcal{L}^i$ and $X_i \to X_j$ corresponds to u_i^j. In the category of sheaves for the Zariski topology on Sch/S,

$$X = \varinjlim_i X_i, \qquad \mathcal{L} = \varinjlim_i \mathcal{L}^i.$$

Thus X represents $\mathcal{L}$, completing the proof.

Remark 5.6.— *Suppose all residue characteristics of S are zero. Then, for every $r \geq 0$, $\mathcal{L}_r = \mathcal{L}_r^1$, so $\mathcal{L}_r$ is represented by an S-prescheme quasi-projective and of finite presentation over S.*

Indeed, for $H \in \mathcal{L}_r(S')$, it suffices to prove that

$$H \longrightarrow N = \mathrm{Norm}_{G_{S'}}(H^{(1)})$$

is an open immersion. Because H is flat and both H, N are of finite presentation, this may be checked on the fibres (Exposé $\mathrm{VI_B}$ 2.6). Cartier's theorem (Exposé $\mathrm{VI_B}$ 1.6) gives, for a connected algebraic subgroup H of an algebraic group in characteristic zero,

$$\mathrm{Norm}_G(H) = \mathrm{Norm}_G(\mathrm{Lie}\, H) = \mathrm{Norm}_G(H^{(1)}).$$

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[18] For example, one may apply Proposition II.6.1 of M. Demazure and P. Gabriel, *Groupes algébriques* I, Masson–North Holland, 1970.

In nonzero residue characteristic the subfunctors $\mathcal{L}_r^n$ can form a strictly increasing sequence, even when S is quasi-compact, and then $\mathcal{L}_r$ is not represented by an S-prescheme quasi-compact over S. For example, over a field k of characteristic $p > 0$, let

$$T = \mathbb{G}_m \times \mathbb{G}_m, \qquad U = \mathbb{G}_a \times \mathbb{G}_a,$$

and let $G = T \ltimes U$, where $(t, t') \cdot (u, u') = (tu, t'u')$. For $n > 0$, let $U_n \subset U$ be defined by $u' = u^{p^n}$, and let $T_n \subset T$ be defined by $t' = t^{p^n}$. Then T_n acts on U_n, and $G_n = T_n U_n$ is smooth, connected, and equal to its normalizer in G. For every m, the distinct groups G_n, $n \geq m$, have the same infinitesimal neighbourhood of order p^m.

Remark 5.7.— *There is a canonical invertible sheaf L on $\mathcal{L}$ whose restriction to every open subset of $\mathcal{L}$ quasi-compact over S is S-ample. One may take*

$$L = (\det(\operatorname{Lie} H))^{-1},$$

where $H \subseteq G_{\mathcal{L}}$ is the universal subgroup.

Recall that for a finite-rank locally free sheaf $\mathcal{F}$, its determinant restricts on the open-and-closed locus where $\operatorname{rank} \mathcal{F} = r$ to $\bigwedge^r \mathcal{F}$. It is enough to restrict to $\mathcal{L}_r^n$. Let $v_n : \mathcal{L}_r^n \to P_n$ be the canonical immersion, and let Q be the universal locally free quotient on the Grassmannian P_n. By construction,

$$v_n^* Q = H^{(n)}.$$

Since $\det Q$ is the canonical ample sheaf on P_n (EGA I, 2nd ed., 9.7), $\det H^{(n)}$ is ample relative to S (EGA II 4.6.13(i) bis).

Let $\mathcal{J} = H^{(1)}$ be the conormal sheaf, and let $S^q(\mathcal{J})$ be the homogeneous degree-q part of its symmetric algebra. Smoothness of H gives

$$\det H^{(n)} \xrightarrow{\sim} \bigotimes_{1 \leq q \leq n} \det S^q(\mathcal{J}).$$

For every locally free $\mathcal{J}$ of rank r and every $q > 0$ there is a canonical isomorphism

$$\det S^q(\mathcal{J}) \xrightarrow{\sim} (\det \mathcal{J})^{\otimes s},$$

where $s > 0$ depends only on r, q. Thus, for some $s > 0$,

$$\det H^{(n)} \simeq (\det \mathcal{J})^{\otimes s}.$$

Since $\det \mathcal{J} = (\det \operatorname{Lie} H)^{-1}$, EGA II 4.5.6 proves that L is S-ample.

Remark 5.8.— *Let G, H be S-group preschemes of finite presentation and let $i : H \to G$ be a monomorphism. Suppose H is smooth over S with connected fibres. The normalizer $N =$* $\operatorname{Norm}_G(H)$ *is then represented by a closed subgroup prescheme of G, of finite presentation over S (Exposé XI 6.11). If, in addition, N is smooth along the identity section and has the same relative dimension as H, then $H = N^0$; in particular H is a subgroup prescheme of G.*

Indeed, N^0 is represented by an open subgroup prescheme of N, smooth over S (Exposé $\mathrm{VI_B}$ 3.10), and i factors through N^0. For every $s \in S$, the connected smooth algebraic groups $H_s, (N^0)_s$ have the same dimension and hence coincide. Since H is flat, $H \to N^0$ is an isomorphism (EGA IV 17.9.5). Thus the functor $\mathcal{L}$ of this section may equivalently be described as the functor of subgroup preschemes H of G that are smooth over S, have connected fibres, and equal their connected normalizer.

6. Functor of Cartan subgroups and functor of parabolic subgroups

When G is a smooth connected algebraic group defined over an algebraically closed field k, sub-tori of G, maximal sub-tori, Cartan subgroups (Exposé XII 1), Borel subgroups (Exposé XIV 4.1), and parabolic subgroups (Exposé XIV 4.8 bis) have already been defined. We extend these notions to a group prescheme over an arbitrary base as follows.

Definition 6.1. *Let S be a prescheme, let G be an S-group prescheme of finite presentation, let H be an S-group prescheme, and let $i : H \to G$ be an S-monomorphism making H a subgroup of G. We say that H is a sub-torus of G (respectively, a maximal sub-torus, a Cartan subgroup, a Borel subgroup, or a parabolic subgroup) if:*

i) *i) H is smooth over S.*

ii) *ii) For every geometric point $\bar{s}$ above S, $H_{\bar{s}}$ is a sub-torus of $(G_{\bar{s}})^0_{\mathrm{red}}$ (respectively, a maximal sub-torus, a Cartan subgroup, a Borel subgroup, or a parabolic subgroup).*

Remarks 6.1 bis.

a) If the S-group H is a sub-torus of G (respectively, …), its fibres are connected, and consequently H is of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3).

b) If H is a sub-torus of G, then H is a torus in the sense of Exposé IX, as follows immediately from Exposé X 8.1. Moreover, the monomorphism $i : H \to G$ is an immersion (see 8.3 below).

c) If G is smooth over S, with connected fibres, and if H is a Cartan subgroup of G (respectively, a Borel subgroup or a parabolic subgroup), the monomorphism $i : H \to G$ is an immersion. Thus our definitions coincide with those introduced in Exposés XII and XIV. Indeed, H is then equal to its connected normalizer (by XII 6.6(c), XIV 4.8 and 4.8 bis), and it suffices to apply 5.8.

Definition 6.1 ter. *Let S be a prescheme, let G be an S-group prescheme locally of finite type, and let s be a point of S. The* ***nilpotent rank*** *of G at s, denoted by $\rho_n(s)$, is the dimension of the Cartan subgroups of $(G_s)^0_{\mathrm{red}}$. The reductive rank $\rho_r(s)$, unipotent rank $\rho_u(s)$, and abelian rank $\rho_{\mathrm{ab}}(s)$ are defined similarly (see Exposé X 8.7).*

If G is now a smooth connected algebraic group defined over an algebraically closed field k, recall that the **radical** of G, denoted by $\mathrm{rad}(G)$, is the largest invariant smooth connected solvable algebraic subgroup of G; $G/\mathrm{rad}(G)$ is then semisimple (use Exposé XII 6.1 to reduce to the affine case). If G is moreover affine, the **unipotent radical** $\mathrm{rad}^u(G)$ is defined as the largest invariant smooth connected unipotent algebraic subgroup of G; $G/\mathrm{rad}^u(G)$ is then reductive.

Proposition 6.2. *Let S be a prescheme, let G and H be two S-group preschemes of finite presentation, and let $i : H \to G$ be an S-monomorphism making H a subgroup of G. Let $P(s)$ be one of the following properties concerning a point s of S:*

i) *$(G_{\bar{s}})^0_{\mathrm{red}}$ is an abelian variety (respectively, is affine, is a torus, or is unipotent).*

ii) *$(H_{\bar{s}})^0_{\mathrm{red}}$ is a maximal torus of $G_{\bar{s}}$.*

iii) *$(H_{\bar{s}})^0_{\mathrm{red}}$ is the centralizer in $(G_{\bar{s}})^0_{\mathrm{red}}$ of a torus of $G_{\bar{s}}$ (respectively, is a Cartan subgroup of $G_{\bar{s}}$).*

iv) $(H_{\bar{s}})^0_{\mathrm{red}}$ *is a Borel subgroup (respectively, a parabolic subgroup) of* $G_{\bar{s}}$.

v) $(H_{\bar{s}})^0_{\mathrm{red}}$ *is the radical of* $G_{\bar{s}}$ *(respectively,* $(G_{\bar{s}})^0_{\mathrm{red}}$ *is semisimple).*

vi) $G_{\bar{s}}$ *is affine and* $(H_{\bar{s}})^0_{\mathrm{red}}$ *is the unipotent radical of* $G_{\bar{s}}$ *(respectively,* $(G_{\bar{s}})^0_{\mathrm{red}}$ *is reductive).*

Then the set E *of points* s *of* S *for which* $P(s)$ *holds is locally constructible (EGA* 0_{III}*, 9.1.11).*

Remark 6.2.1. This proposition complements Exposé $\mathrm{VI_B}$, §10. The structure of E can be described more precisely by using semicontinuity theorems (see Exposé X 8.7); we shall see an example below.

Proof of 6.2. Note that if S is the spectrum of a field, E is invariant under extension of that field. A standard reduction (EGA IV 9) then allows us to reduce to the case where S is noetherian and integral, with generic point η. One must show that E or $\mathrm{ens}(S) \setminus E$ contains a neighbourhood of η (EGA IV 9.2.1). One may assume S affine of ring A and field of fractions K. If L is a finite extension of K, it is immediate that there exists an A-subalgebra B of L, finite over A, having L as field of fractions. The canonical morphism $S' \to S$, where $S' = \operatorname{Spec} B$, is dominant, of finite presentation, so the image of a non-empty open subset of S' contains a non-empty open subset of S (EGA IV 1.8.4). From the viewpoint that interests us, we may therefore replace S by S', hence replace K by a finite extension L. Thus we may choose L so that $(G_L)_{red}$ and $(H_L)_{red}$ are smooth over L (EGA IV 4.6.6). Possibly restricting S', we may assume that G_{red} and H_{red} are group preschemes smooth over S (Exposé $\mathrm{VI_B}$, §10, and EGA IV 17). In view of the properties to be proved, we may replace G and H by their reduced identity components (Exposé $\mathrm{VI_B}$ 10.9), and hence assume that G and H are smooth over S with connected fibres. Finally, we may assume that H is a closed subgroup prescheme of G (Exposé $\mathrm{VI_B}$ 10.4).

Proof of i). Possibly after a finite extension of K, we may assume that G_η admits a "Chevalley decomposition", i.e. is an extension of an abelian variety D_η by a smooth connected linear algebraic group F_η (Séminaire Bourbaki 1956/57, no. 145). By Exposé $\mathrm{VI_B}$ 10.16, there is an open neighbourhood U of η over which this generic extension comes from an extension

$$1 \longrightarrow F \longrightarrow G|_U \longrightarrow D \longrightarrow 1.$$

One may moreover assume that F and D are smooth over U with connected fibres, that F is affine over U, and that D is proper over U (EGA IV 8, 9 and 17). For every point s of U, (D_s, F_s) is then the Chevalley decomposition of G_s. Consequently, G_s is an abelian variety (respectively, is affine) if and only if F_s (respectively, D_s) is the unit group; this is a constructible condition (EGA IV 9.2.6.1).

To establish the last two assertions of (i), we may, by the preceding, assume that G is affine over S. Let q be a prime number invertible on S, and let ${}_qG$ be the "kernel" of the q-th-power morphism in G. It follows easily from the structure of affine algebraic groups that G_s is a torus (respectively, is unipotent) if and only if

a) ${}_qG_s$ is quasi-finite, which is a constructible property (EGA IV 9.3.2), and

b) ${}_qG_s$ has q^r geometric points, where r is the relative dimension of G over S (respectively, ${}_qG_s$ has a single point).

The function

$$s \longmapsto (\text{number of geometric points of } {}_qG_s)$$

is constructible (EGA IV 9.7.9). This completes the proof of (i).

Proof of iii). **a) Case of the centralizer of a torus.** Suppose

$$H_\eta = \mathrm{Centr}_{G_\eta}(T_\eta),$$

where T_η is a torus of G_η. We shall show that H_s is the centralizer of a sub-torus of G_s for s in a neighbourhood of η. By (i), after possibly restricting S, we may assume that T_η comes from a sub-torus T of G. Then $Z = \mathrm{Centr}_G(T)$ is represented (Exposé XI 6.11) by a subgroup prescheme of G. Since H and Z coincide generically, they coincide over a neighbourhood of η. Thus the set E of points s of S for which H_s is the centralizer in G_s of a sub-torus of G_s is ind-constructible (EGA IV 1). This will suffice in Lemma 6.6 below to prove that E is open in S, and hence, *a fortiori*, locally constructible.

b) Case of a Cartan subgroup. Suppose that H_η is a Cartan subgroup of G_η. We shall show that H_s is a Cartan subgroup of G_s for every s in a neighbourhood of η. The group H_η is the centralizer in G_η of a torus of G_η, and is nilpotent (Exposé XII 6.6). By (a) and Exposé $\mathrm{VI_B}$ 8.4, H_s has the same properties for every s in some neighbourhood U of η. For every $s \in U$, the group H_s therefore has the same reductive rank as G_s, and its unique maximal torus is central (Exposé XII 6.7). Hence H_s is the centralizer of a maximal torus of G_s, that is, a Cartan subgroup of G_s.

Suppose now that H is not a Cartan subgroup of G, and let us show that H_s is not a Cartan subgroup of G_s for s in a neighborhood U of η. In view of the assertion provisionally admitted in (a) above, we may restrict to the case where H is the centralizer in G of a sub-torus T. But then H_η contains a Cartan subgroup C_η of H_η. We have just seen that, possibly restricting S, C_η extends to a Cartan subgroup C of G, which one may assume contained in H. By hypothesis H_η strictly contains C_η, so H_s strictly contains C_s for s in a neighborhood U of η (EGA IV 9.5.2); *a fortiori*, H_s is not

a Cartan subgroup of G_s for s in U.

Proof of ii). Suppose H_η is a maximal torus of G_η and let C_η be its centralizer in G_η. By i) and iii), H is a torus over a neighborhood U of η and $C = \mathrm{Centr}_{G|U}(H|U)$ is a Cartan subgroup of $G|U$. To prove that H is a maximal torus of G over a neighborhood of η, we may then replace G by C, then by the linear component F of a Chevalley decomposition of C (cf. i)). Let q be an integer invertible on S, ${}_qF$ the kernel of the q-th-power morphism in F. Since F_s is affine, nilpotent, smooth and connected, F_s is the direct product of its maximal torus T_s by a unipotent group (*Bible* 6-04), so ${}_qF_s = {}_qT_s$. Since H_η is a maximal torus, ${}_qH_\eta = {}_qF_\eta$, and consequently ${}_qH = {}_qF$ over a neighborhood V of η. For every point s of V, ${}_qH_s = {}_qT_s$, so $H_s = T_s$ is a maximal torus.

Suppose now that H_η is not a maximal torus of G_η. By i), we may restrict to the case where H_η is a torus, then assume that it is contained in a strictly larger torus T_η. The latter extends to a torus T strictly containing H over a neighborhood U of η. *A fortiori*, H_s is not a maximal torus for $s \in U$.

Proof of iv). Possibly restricting S, we may assume that the center Z of G is representable (Exp. $\mathrm{VI_B}$ 10.11) and flat over S, as well as the quotient G/Z (*loc. cit.*). The property "H_s contains Z_s" is constructible (EGA IV 9.5.2) and every parabolic subgroup of G_s contains Z_s (Exp. XIV 4.9 a)); this allows us to replace G by G/Z, hence assume G affine over S (Exp. XII 6.1 and i)). We may further

assume that G/H is representable, but then H_s is a parabolic subgroup of G_s if and only if $(G/H)_s$ is proper (*Bible* 6 Th. 4 b)), which is an ind-constructible property (EGA IV 9.3.5). So E is ind-constructible, and this will suffice for us to prove that E is open (Lemma 6.6), hence locally constructible.

Let us now examine the case of Borel subgroups. If H_η is a Borel subgroup of G_η, i.e. a solvable parabolic subgroup of G_η, what precedes and Exposé $\mathrm{VI_B}$, 8.4, entail that these properties remain true at every point s of a neighborhood of η. If now H_η is not a

Borel subgroup of G_η, to prove that the same holds at points s of a neighborhood of η, we may restrict (in view of what precedes) to the case where H_η is a parabolic subgroup, then assume that H_η contains a Borel subgroup B_η. We have just shown that the latter extends to a Borel subgroup B of G over a neighborhood U of η. Since H_η strictly contains B_η, H_s strictly contains B_s at every point of an open subset V, and H_s is not a Borel subgroup of G_s for $s \in V$.

Proof of v). Suppose H_η is the radical of G_η. The group H_η is then invariant in G_η, solvable (smooth and connected); the same is therefore true for H_s for s belonging to a neighborhood U of η (Exp. $\mathrm{VI_B}$ 8 and 10), so for $s \in U$, H_s is contained in the radical of G_s. Replacing G by G/H (Exp. $\mathrm{VI_B}$ 10), we must

prove that if G_η is semisimple, then G_s is semisimple at every point of a neighborhood V of η. Using i) and ii), one may assume that G is affine over S and possesses a maximal torus T. Let W be the Weyl group of T (Exp. XII 2), which is quasi-finite and étale over S, hence finite and étale over an open subset V. It then follows from the elementary properties of roots (Exp. XIX 1.12) that G is semisimple over V.

Suppose now that H_η is not the radical of G_η. Possibly replacing K by a finite extension L, we may assume that G_η admits a radical R_η. By what precedes, R_η extends to a subgroup prescheme R of $G|_U$ such that for every $s \in U$, R_s is the radical of G_s. By hypothesis, $R_\eta \neq H_\eta$. So $R_s \neq H_s$ for $s \in V$. It remains to prove that if G_η is not semisimple, neither is G_s at neighboring points, but this is a particular case of what precedes (take H = unit group).

Proof of vi). The proof is entirely analogous to that of v), in view of i), and is left to the care of the reader.

Corollary 6.3. *Let S_0 be a quasi-compact prescheme, let $(S_i)_{i\in L}$ be a projective system of S_0-preschemes affine over S_0, and let $S = \varprojlim S_i$ (EGA IV 8.2). Let G_0 be a group prescheme of finite presentation over S_0, set*

$$G_i = G_0 \times_{S_0} S_i, \qquad G = G_0 \times_{S_0} S,$$

and let H be a subgroup of G. If H is a sub-torus of G (respectively, a maximal sub-torus, a Cartan subgroup, a Borel subgroup, or a parabolic subgroup), then there exist an index $i \in L$ and a subgroup H_i of G_i such that

$$H = H_i \times_{S_i} S$$

and H_i is a sub-torus of G_i (respectively, ...).

Indeed, H is smooth with connected fibres, hence of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3). By Exposé $\mathrm{VI_B}$, §10, there exist $i \in L$ and a subgroup H_i of G_i, smooth over S_i, such that $H = H_i \times_{S_i} S$. Corollary 6.3 then follows from Definition 6.1, Proposition 6.2 and EGA IV 9.3.3.

Corollary 6.3 bis. *Let S be a prescheme and let G be an S-group prescheme of finite presentation. Then the functions $\rho_n, \rho_r, \rho_u, \rho_{\mathrm{ab}}$ (cf. 6.1 ter) are locally constructible on S.*

It suffices to show (EGA IV 9.) that if S is a noetherian integral scheme with generic point η, the functions in question are constant on a neighborhood of η. Possibly replacing S by a scheme S' finite over S and dominating S, we may assume that G_η admits a Cartan subgroup C_η with a Chevalley decomposition $1 \to L_\eta \to C_\eta \to A_\eta \to 1$. The argument made in 6.2 i) proves that this decomposition extends to a Chevalley decomposition over a neighborhood of η:

$$1 \longrightarrow L \longrightarrow C \longrightarrow A \longrightarrow 1.$$

Moreover, one may assume that C is a Cartan subgroup of G (6.3) and that the maximal torus T_η of L_η extends to a maximal torus T of L (6.3). The corollary follows immediately from this and from the definitions.

6.4.0. Let S be a prescheme, G an S-prescheme in groups of finite presentation, and $\mathcal{L}$ the functor of subgroups of G that are smooth, have connected fibres, and are equal to their connected normalizer (cf. ğ5). The functor $\mathcal{L}$ is representable (5.2 and 5.8). The rest of this section is devoted to certain subfunctors of $\mathcal{L}$. More precisely, define $\mathcal{L}_C$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$) as follows: for every S-prescheme S', the set $\mathcal{L}_C(S')$ (resp. $\mathcal{CT}(S')$, resp. $\mathcal{P}(S')$) consists of the subgroups H of $G_{S'}$ that are smooth over S', have connected fibres, are equal to their connected normalizer, and are such that, for every point s' of S', $H_{s'}$ contains a Cartan subgroup (6.1) of $G_{s'}$ (resp. is the centralizer in $(G_{s'})^0_{\text{red}}$ of a subtorus of $G_{s'}$, resp. is a parabolic subgroup (6.1) of $G_{s'}$).

Theorem 6.4. *Let S be a prescheme and G an S-prescheme in groups of finite presentation. The S-functors $\mathcal{L}_C$, $\mathcal{CT}$, and $\mathcal{P}$ defined in 6.4.0 are representable by S-preschemes that are quasi-projective and of finite presentation over S.*

Remark 6.5. If G has smooth fibres—for example, if G is smooth over S, or if the residue characteristics of S are zero (Exposé VI_B, 1.6.1)—then every subgroup H of G, smooth over S with connected fibres, such that H_s contains a Cartan subgroup of $(G_s)^0$ for every point s of S, is necessarily equal to its connected normalizer (and hence is an element of $\mathcal{L}_C(S)$). Indeed, by 5.1 iii) we may assume that S is the spectrum of a field, in which case the property was noted at the end of the statement of Exposé XIII, 2.1.

There are natural monomorphisms

$$\begin{tikzcd}
\mathcal{CT} \arrow[dr, hook] & & \\
& \mathcal{L}_C \arrow[r, hook] & \mathcal{L}. \\
\mathcal{P} \arrow[ur, hook] & &
\end{tikzcd}$$

Let us show that these monomorphisms are open immersions of finite presentation (which will already prove, in view of 5.2, that $\mathcal{L}_C$, $\mathcal{CT}$, $\mathcal{P}$ are representable by S-preschemes, union of an increasing sequence of open subpreschemes, quasi-projective and of finite presentation over S). Now this will follow, by the usual technique, from the following lemma:

Lemma 6.6. *Let S be a prescheme, G an S-prescheme in groups of finite presentation, and H a subgroup of G, smooth with connected fibres. The set of points s of S such that H_s contains a Cartan subgroup of G_s (resp. is equal to the centralizer in $(G_s)^0_{\text{red}}$ of a torus of G_s) is open in S. If, moreover, H is equal to its connected normalizer,*[19] *then the set of points s of S such that H_s is a parabolic subgroup of G_s is also open in S.*

The assertion to be proved is local on S, which allows us to assume S affine, then, by EGA IV 8.1 and 6.3, S noetherian. Denote by E the set of points of S having the property in question. It then follows from the assertions effectively proved in 6.2 that E is ind-constructible. But S is noetherian, so to prove that E is open, it suffices to show that E is stable under generizations (EGA IV 1.10.1). Using EGA II 7.1.9, we are finally led to prove that if S is the spectrum of a discrete valuation ring, and if the closed point s belongs to E, then so does the generic point t.

We shall need the following lemma:

[19] This hypothesis is in fact superfluous; see Exposé XVII, Appendix III, 3.

Lemma 6.7. *Let A be a complete noetherian local ring, $S = \operatorname{Spec} A$, s the closed point of S, H an S-prescheme in groups, smooth with connected fibres, and T_s a subtorus of H_s. Then:*

i) there exists a closed subgroup prescheme C of H, smooth with connected fibres, such that $C_s = \operatorname{Centr}_{H_s}(T_s)$;

ii) for every point t of S, C_t is the centralizer in H_t of a subtorus T_t of H_t.

Proof of 6.7. Let T' be an S-torus for which there is an isomorphism $u_0 : T'_s \xrightarrow{\sim} T_s$ (Exposé X, 4.6). Let $\mathfrak{m}$ be the maximal ideal of A, and put $A_n = A/\mathfrak{m}^n$, $S_n = \operatorname{Spec} A_n$, $H_n = H \times_S S_n$, and so on. Since H is smooth over S, for every integer $n > 0$ there is an S_n-group morphism

$$u_n : T'_n \longrightarrow H_n$$

lifting u_0 (Exposé IX, 3.6), and by induction on n we may assume that u_n lifts u_{n-1}. Let q be a prime invertible on S. For every integer ℓ that is a power of q, write ${}_\ell u_n$ for the restriction of u_n to ${}_\ell T'_n$. For fixed ℓ and varying n, the morphisms ${}_\ell u_n$ form a projective system and therefore arise from a unique S-group morphism

$${}_\ell u : {}_\ell T' \longrightarrow H$$

(1.6 a)). Since H is separated (Exposé $\mathrm{VI_B}$, 5.2) and u_0 is a monomorphism, ${}_\ell u$ is a monomorphism (Exposé IX, 6.8), and even a closed immersion because it is finite, ${}_\ell T'$ being finite over S. Denote its image by $M(\ell)$. The family of subgroups of multiplicative type $M(\ell)$ is clearly coherent in the sense of 4.1. Put

$$C_\ell = \operatorname{Centr}_H(M(\ell)).$$

This is represented by a closed subgroup prescheme (2.5), since H is separated, and is smooth over S (Exposé XI, 2.4). The C_ℓ form a filtered decreasing family of closed subschemes and are therefore stationary, since H is noetherian. Their stationary value is a smooth closed subgroup C such that $C_s = \operatorname{Centr}_{H_s}(T_s)$, by the density theorem (Exposé IX, 4.7). It remains to show that for every point t of S, C_t is the centralizer in H_t of a subtorus T_t; this will also imply that C_t is connected. This follows from the more precise lemma below, applied to the family $M(\ell)_t$ of subgroups of H_t.

Lemma 6.8. *Let G be a connected algebraic group over a field k, let $r > 0$ be an integer, and let q be an integer prime to the characteristic*[20] *of k. Let $M(\ell)$, where ℓ ranges over the powers of q, be a coherent family of subgroups of G, of multiplicative type and of type $(\mathbb{Z}/\ell\mathbb{Z})^r$ (cf. 4.6). Let M be the algebraic subgroup of G generated by the $M(\ell)$ (*loc. cit.*), and let T be the unique maximal torus of M (cf. 3.4). Then, for all sufficiently large ℓ,*

$$\operatorname{Centr}_G(T) = \operatorname{Centr}_G(M) = \operatorname{Centr}_G(M(\ell)).$$

The last equality is clear. To prove the first, introduce the centre Z of G, put $G' = G/Z$, let M' (resp. T') be the image of M (resp. T) in G', and let K be the inverse image of T' in G, that is, the algebraic subgroup of G generated by T and Z. It plainly suffices to prove $K \supset M$, or equivalently $T' = M'$. Now M is smooth and connected (4.6), while G' is affine (Exposé XII, 6.1); hence M' is the direct product of its maximal torus T' (Exposé XII, 6.6 d)) and a unipotent group (*Bible* 4, Th. 4), after assuming k algebraically closed. The image of $M(\ell)$ in M' is therefore contained in T'. Thus the inverse image of T' in M

[20] Editor's note: the word "residual" has been removed.

contains $M(\ell)$ for every ℓ, and consequently is equal to M. Hence $M' = T'$. This proves 6.8, and therefore 6.7.

We now prove 6.6. We have reduced to the case where S is the spectrum of a discrete valuation ring A, which we may moreover assume complete with algebraically closed residue field. After replacing A, if necessary, by its normalization in a finite extension of its fraction field, we may assume that $(G_t)_{\text{red}}$ is smooth.[21] To prove 6.6 we may clearly replace G by the identity component of the schematic closure in G of $(G_t)^0_{\text{red}}$, and therefore assume that G is flat over S, has connected fibres, and has smooth generic fibre G_t.

a) Suppose that H_s is the centralizer in $(G_s)_{\text{red}}$ of a torus T_s. We show that H_t is then the centralizer in G_t of a subtorus of G_t. By Lemma 6.7 there is a subgroup scheme C of H, smooth over S, whose closed fibre is H_s, such that $C_t = \mathrm{Centr}_{H_t}(T_t)$ for a subtorus T_t of H_t. Since H is smooth over S with connected fibres, dimension considerations give $C = H$. With the notation of 6.7, $H = C = C_\ell$ for all sufficiently large ℓ. Similarly, put $C'_\ell = \mathrm{Centr}_G(M(\ell))$ (2.5), and let C' be the stationary value of the C'_ℓ for large ℓ (2.5 bis). The group scheme C' contains H, and

$$C'_t = \mathrm{Centr}_{G_t}(T_t) \quad (6.8), \qquad C'_s = \mathrm{Centr}_{G_s}(T_s).$$

The hypothesis on H_s gives $\dim H_s = \dim C'_s$. Moreover, $\dim H_s = \dim H_t$, since H is smooth over S, and $\dim C'_t \leq \dim C'_s$ (Exposé $\mathrm{VI_B}$, 4.1). Hence $\dim H_t = \dim C'_t$. Since G_t is smooth and connected, $C'_t = \mathrm{Centr}_{G_t}(T_t)$ is smooth and connected; finally,

$$H_t = C'_t = \mathrm{Centr}_{G_t}(T_t).$$

b) Suppose that H_s contains a Cartan subgroup C_s of G_s, that is, the centralizer in $(G_s)_{\text{red}}$ of a maximal torus T_s of G_s. By 6.7 there is a subgroup scheme C of H, smooth over S with connected fibres, lifting C_s. It suffices to prove that C_t contains a Cartan subgroup of G_t. By part a), applied with $H = C$, C_t is the centralizer in G_t of a subtorus of G_t, and hence contains a Cartan subgroup of G_t.

c) Suppose that H_s is a parabolic subgroup of G_s. Let $N = \mathrm{Norm}_G(H)$. By hypothesis H is equal to its connected normalizer, so N_s is smooth and consequently

$$N_s = \mathrm{Norm}_{(G_s)_{\text{red}}}(H_s) = H_s \qquad \text{(Exposé XII, 8 bis)}.$$

Thus N is flat over S. We shall see in Exposé XVI that under these conditions G/N is representable. Since $H_s = N_s$ is a parabolic subgroup, $(G/N)_s$ is proper. Since G/N is flat over S with connected fibres, EGA III, 5.5.1 implies that G/N is proper over S. Thus $(G/N)_t = G_t/N_t$ is proper over t, and so is G_t/H_t, because N_t/H_t is finite. It remains only to apply the following lemma.

Lemma 6.9. *Let k be a field, G a k-algebraic group, H a smooth connected algebraic subgroup of G, and $N = \mathrm{Norm}_G(H)$. If $\dim H = \dim N$ and G/H is proper, then H is a parabolic subgroup of G.*

Indeed, we may assume that k is algebraically closed and that G is smooth and connected. The centre Z of G is contained in N, and $\dim H = \dim N$ implies that $Z' = (Z)^0_{\text{red}}$ is contained in H. Hence $G' = G/Z'$ is affine (Exposé XII, 6.1). Replacing G by G' and H by its image in G', we reduce to the case in which G is affine (Exposé XIV, 4.9), and Lemma 6.9 then follows from *Bible* 6, Th. 4. This proves Lemma 6.6.

[21] Editor's note: details or references should be supplied here.

To complete the proof of 6.4, we must show that the S-preschemes representing $\mathcal{L}_C$ (respectively $\mathcal{CT}$ and $\mathcal{P}$) are of finite presentation over S. This assertion is local on S, so we may assume first that S is affine and then, since G is of finite presentation, that S is noetherian (EGA IV, 8.9). We have just seen that the natural inclusions $\mathcal{CT} \to \mathcal{L}$ and $\mathcal{P} \to \mathcal{L}$ are immersions. The same is therefore true of $\mathcal{CT} \to \mathcal{L}_C$ and $\mathcal{P} \to \mathcal{L}_C$. Consequently, it is enough to prove that $\mathcal{L}_C$ is represented by an S-prescheme of finite presentation.

Resume the notation introduced in 5.2. For every integer $n \geq 0$, let $\mathcal{L}^n$ be the subfunctor of $\mathcal{L}$ defined by

$$\mathcal{L}^n(S') = \left\{ H \in \mathcal{L}(S') \;\middle|\; \mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(n)}) \right\}.$$

The S-functor $\mathcal{L}^n$ is thus represented by an open subprescheme of $\mathcal{L}$, the disjoint union of the $\mathcal{L}^n_r$. Each $\mathcal{L}^n_r$ is of finite presentation over S (5.3), and is empty for

$$r > \sup_{s \in S} \dim G_s,$$

which is a finite number because S is quasi-compact. Hence $\mathcal{L}^n$ is of finite presentation over S. It remains to prove that $\mathcal{L}_C$ is contained in $\mathcal{L}^n$ for all sufficiently large n.

For each point s of S, let $d(s)$ be the least integer n, finite or infinite, such that

$$\mathcal{L}_{C,G_s} \subset \mathcal{L}^n_{G_s}.$$

It is enough to show that d is bounded on S. Indeed, if M is an upper bound, then $\mathcal{L}_C$ is set-theoretically contained in $\mathcal{L}^M$, and hence is contained in $\mathcal{L}^M$, since the latter is open in $\mathcal{L}$. A direct constructibility argument reduces us to showing that, when S is noetherian and integral with generic point t, the function d is bounded on a neighbourhood of t.

a) **Reduction to the case in which G is smooth over S.** Proceeding as in 6.2, we see that, after possibly changing S, we may assume that G_{red} is a group prescheme smooth over S; denote it by G'. Put

$$X = \mathcal{L}_{C,G}, \qquad X' = \mathcal{L}_{C,G'},$$

and let H (respectively H') be the subgroup prescheme of G_X (respectively $G'_{X'}$) universal for the functor $\mathcal{L}_{C,G}$ (respectively $\mathcal{L}_{C,G'}$). Since H is smooth over X, the prescheme $H \times_X X_{\mathrm{red}}$ is reduced, and is therefore contained in $G'_{X_{\mathrm{red}}}$. It is an element of $\mathcal{L}_{C,G'}(X_{\mathrm{red}})$, and hence defines a canonical morphism

$$p : X_{\mathrm{red}} \longrightarrow X'.$$

It is clear that p is a monomorphism; we show that it is in fact an immersion. Let N' be the normalizer of H' in $G_{X'}$. The set of points $s \in X'$ for which $H'_s \to N'_s$ is an open immersion is an open subset U, and

$$H'|_U \longrightarrow N'|_U$$

is an open immersion (Exposé $\mathrm{VI_B}$, 2.5, and EGA IV, 17.9.5). It follows from 5.1(iii) that U is the largest open subset of X' over which H' equals its connected normalizer in $G_{X'}$. Thus $H'|_U \in \mathcal{L}_{C,G}(U)$, and it follows at once that p is an isomorphism of X_{red} onto U_{red}. If $\mathcal{L}_{C,G}$ is known to be of finite type when G is smooth, then X' is of finite type over S; consequently X_{red} is of finite type (since S is noetherian), and is therefore contained in $\mathcal{L}^M_G$ for all sufficiently large M. The same then holds for X.

b) **The case in which S is the spectrum of an algebraically closed field k of characteristic $p > 0$, and G is a smooth algebraic group over k.** Instead of the infinitesimal neighbourhoods $H^{(n)}$ of a subgroup prescheme H of G, we use the radicial subgroups $F_n(H)$, the kernels of the iterates of the Frobenius morphism of H (Exposé $\mathrm{VII_A}$, 4). This is legitimate here because $F_n(H)$ is contained in the infinitesimal neighbourhood of order p^n of the unit section of H. If T is a subtorus of G, then

$$F_n(T) = {}_{p^n}T,$$

and duality immediately shows that $\mathrm{Centr}_G({}_{p^n}T) = \mathrm{Centr}_G(T)$ for all sufficiently large n. We have the following more precise proposition.

Proposition 6.10. *Let k be a field of characteristic $p > 0$, let G be a smooth k-algebraic group, let T be a maximal torus of $G_{\bar{k}}$, and let m be the least integer such that*

$$\mathrm{Centr}_G({}_{p^m}T)^0 = \mathrm{Centr}_G(T)^0.$$

Then, for every k-prescheme S and every $H \in \mathcal{L}_C(S)$, one has

$$\mathrm{Norm}_{G_S}(H) = \mathrm{Norm}_{G_S}(F_m(H)).$$

In particular, $\mathcal{L}_C$ is contained in $\mathcal{L}^{p^m}$.

Since H is smooth and of finite presentation over S, and has constant nilpotent rank, namely that of G, we shall see in the next section (7.3) that the functor $\mathcal{C}_H$ of Cartan subgroups of H is represented by an S-prescheme smooth over S. The reader will verify that the proof given there does not use the finite-type property of $\mathcal{L}_{C,H}$. It follows from Exposé XIII, 3.1, that we may consider the open subset U of regular points of H.

Let S' be an S-prescheme and let $g \in G(S')$ normalize $F_m(H)_{S'}$. To prove that g normalizes $H_{S'}$, it is enough to prove that

$$\mathrm{int}(g)U_{S'} \subset H_{S'}.$$

Indeed, $H_{S'} \cap \mathrm{int}(g)H_{S'}$ would then contain an open subgroup of $\mathrm{int}(g)H_{S'}$, and hence would equal it, since the latter has connected fibres. After replacing S' by a suitable covering S'', and then replacing S'' by S, we are reduced to proving the following: if $g \in G(S)$ normalizes $F_m(H)$ and $u \in U(S)$, then $\mathrm{int}(g)u \in H(S)$.

Let C be the unique Cartan subgroup of H "containing" u (Exposé XIII, 3.2). It is enough to show that

$$C' = \mathrm{int}(g)C \subset H.$$

Since $H \in \mathcal{L}_{C,G}(S)$, the group C is also a Cartan subgroup of G. The latter admits maximal tori, so C is the centralizer in G_S^0 of its unique maximal torus T (Exposé XII, 7.1(a)). By the definition of m, and because any two maximal tori of G are locally conjugate for the fpqc topology (Exposé XII, 7.1),

$$C = \mathrm{Centr}_G(T)^0 = \mathrm{Centr}_G({}_{p^m}T)^0.$$

Conjugating by g, we obtain

$$C' = \mathrm{Centr}_G\big(\mathrm{int}(g)({}_{p^m}T)\big)^0.$$

Now $\mathrm{int}(g)({}_{p^m}T)$ is a subgroup of multiplicative type of

$$\mathrm{int}(g)(F_m(H)) = F_m(H),$$

because g normalizes $F_m(H)$; it is therefore contained in H. Exposé XIII, 2.1 (proved there when the base is a field, but extending at once to an arbitrary base) then shows that, in order to prove $C' \subset H$, it is enough to prove

$$\operatorname{Lie} C' \subset \operatorname{Lie} H.$$

But

$$\operatorname{Lie} C' = \operatorname{int}(g)(\operatorname{Lie} C) \subset \operatorname{int}(g)(\operatorname{Lie} H).$$

On the other hand, we may assume $m \geq 1$, and Exposé $\mathrm{VII_A}$, 4.1.2, gives

$$\operatorname{Lie} H = \operatorname{Lie}(F_m(H)) = \operatorname{Lie}(\operatorname{int}(g)(F_m(H))) = \operatorname{int}(g)(\operatorname{Lie} H).$$

Thus $\operatorname{Lie} C' \subset \operatorname{Lie} H$, which completes the proof of 6.10.

We shall need another description of the integer m introduced in 6.10.

Lemma 6.11. *Let G be a smooth algebraic group over an algebraically closed field k of characteristic $p > 0$, let T be a maximal torus of G, put $\mathfrak{g} = \operatorname{Lie} G$, and let R be the set of nonzero characters of T occurring in the representation of T on $\mathfrak{g}$ induced by the adjoint representation of G. For $r \in R$, let e_r be the greatest integer n such that p^n divides r in the character group of T. Then*

$$m = \sup_{r\in R}(e_r + 1)$$

if $R \neq \varnothing$, and $m = 0$ otherwise.

Indeed, $\operatorname{Centr}_G(T)$ is smooth and is contained in $\operatorname{Centr}_G({}_{p^m}T)$. Hence

$$\begin{aligned}\operatorname{Centr}_G(T)^0 = \operatorname{Centr}_G({}_{p^m}T)^0 &\\ \iff \operatorname{Lie}(\operatorname{Centr}_G(T)) &= \operatorname{Lie}(\operatorname{Centr}_G(F_m(T)))\\ \iff \mathfrak{g}^T &= \mathfrak{g}^{p^m T} \qquad \text{(Exposé II, 5.2.3).}\end{aligned}$$

With the usual notation,

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{r\in R} \mathfrak{g}_r.$$

Thus $\mathfrak{g}^T = \mathfrak{g}_0$, whereas

$$\mathfrak{g}^{p^m T} = \mathfrak{g}_0 \oplus \bigoplus_{r\in R'} \mathfrak{g}_r,$$

where R' is the subset of R formed by the characters whose restriction to ${}_{p^m}T$ is zero. A nonzero character r of T has trivial restriction to ${}_{p^m}T$ if and only if $m \leq e_r$, which proves the lemma.

c) **Return to the proof of 6.4.** By part a) and the paragraph preceding it, we have reduced to the case in which S is a noetherian integral scheme and G is smooth over S. We must show that d is bounded on a neighbourhood of the generic point t of S. After possibly changing S, we may assume that G has a split maximal torus (6.2)

$$T = D_S(M).$$

Let

$$\mathfrak{g} = \bigoplus_{\lambda\in M} \mathfrak{g}^{\lambda}$$

be the decomposition of the Lie algebra of G according to the characters of T, and let R be the finite set of nonzero $\lambda \in M$ for which $\mathfrak{g}^{\lambda} \neq 0$. We distinguish two cases.

First case: t, and hence every point of S, has positive residue characteristic p. It is clear from the foregoing that d is then bounded by p^m, where m is defined as in 6.11.

Second case: t has residue characteristic zero. For each $\lambda \in R$, let n_λ be the greatest integer dividing λ in the group M, and put

$$n = \prod_{\lambda \in R} n_\lambda.$$

For every prime number q dividing n, let S_q be the closed subset of S formed by the points of residue characteristic q, and let U be the nonempty open subset, containing t, complementary to the union of the S_q. If $s \in U$, then either s has residue characteristic zero, in which case $d(s) = 1$ by 5.6, or s has residue characteristic $p > 0$ not dividing n. In the latter case the integer m attached to G_s in 6.11 is at most one. Moreover, Exposé VII[22] gives

$$\mathrm{Norm}_{G_{S'}}(F_1(H)) = \mathrm{Norm}_{G_{S'}}(\mathrm{Lie}\, H) = \mathrm{Norm}_{G_{S'}}(H^{(1)}).$$

Finally, 6.10 shows that, for $H \in \mathcal{L}_{C,G_s}(S')$,

$$\mathrm{Norm}_{G_{S'}}(H) = \mathrm{Norm}_{G_{S'}}(H^{(1)}).$$

Consequently $d(s) \leq 1$, so d is bounded by 1 on U. This completes the proof of 6.4.

Corollary 6.12. *Let S be a prescheme and G an S-prescheme in groups of finite presentation. Suppose that the nilpotent rank of the fibres of G (respectively the dimension of their Borel subgroups) is locally constant on S. Then the functor $\mathcal{C}$ of Cartan subgroups of G (respectively the functor $\mathcal{B}$ of Borel subgroups of G) is represented by an S-prescheme of finite presentation over S.*

Indeed, after restricting S, we may assume that the nilpotent rank ν of the fibres is constant. Then $\mathcal{C}$ is represented by the open-and-closed subprescheme of the prescheme X representing $\mathcal{L}_C$ (6.4) over which the universal subgroup of G_X for the functor $\mathcal{L}_C$ has relative dimension ν. The proof is analogous for $\mathcal{B}$, using the representability of $\mathcal{P}$.

7. Cartan subgroups of a smooth group

Proposition 7.1.— *Let S be a prescheme and let G be an S-group prescheme of finite presentation, smooth over S, with connected fibres.*

i) *Let $\mathcal{CT}$ be the S-functor defined by*

$$\mathcal{CT}(S') = \left\{ \begin{array}{c} \textit{subgroup preschemes } H \textit{ of } G_{S'}, \textit{ smooth over } S', \\ \textit{such that every } H_{s'} \textit{ is the centralizer in } G_{s'} \\ \textit{of a sub-torus of } G_{s'} \end{array} \right\}.$$

Then $\mathcal{CT}$ is represented by an S-prescheme of finite presentation, smooth and quasi-projective over S.

ii) *If S is artinian and $H \in \mathcal{CT}(S)$, then H is the centralizer in G of a sub-torus of G. If S is the spectrum of a henselian local ring and $H \in \mathcal{CT}(S)$, then H is the centralizer in G of a subgroup of multiplicative type that is étale over S.*

[22]Editor's note: the argument should be made explicit.

iii) *Let L be an S-group prescheme smooth and of finite presentation, let $i : L \to G$ be an S-group monomorphism, and let $H \in \mathcal{CT}(S)$. Then* $\mathrm{Transp}_G(H, L)$ *and* $\mathrm{Transpstr}_G(H, L)$ *(Exposé VIII 6.5(e)) are represented by closed subpreschemes of G, smooth over S.*

iv) *If $H \in \mathcal{CT}(S)$, then H is closed in G; $N = \mathrm{Norm}_G(H)$ is represented by a closed subgroup prescheme of G, smooth over S; N/H is represented by an S-group prescheme, separated, étale, and of finite type over S; and G/N is represented by an S-prescheme smooth and quasi-projective over S.*

v) *Let G' be an S-group prescheme of finite presentation and let $u : G \to G'$ be faithfully flat. Then G' satisfies the same hypotheses as G (Exposé $\mathrm{VI_B}$ 9). If $H \in \mathcal{CT}_G(S)$, its image is represented by a subgroup prescheme $H' \in \mathcal{CT}_{G'}(S)$, and $H \to H'$ is faithfully flat. If $H = \mathrm{Centr}_G(T)$ for a torus T, then*

$$H' = \mathrm{Centr}_{G'}(T'), \qquad T' = u(T).$$

vi) *Under the hypotheses of (v), the S-morphism*

$$\widetilde{u} : \mathcal{CT}_G \longrightarrow \mathcal{CT}_{G'}, \qquad H \longmapsto H' = u(H),$$

is quasi-compact and faithfully flat. If, moreover, $\ker u$ *is central, then $\widetilde{u}$ is an isomorphism, with inverse $H' \mapsto u^{-1}(H')$.*

Proof of (ii). For the first assertion we may suppose that S is local artinian, with closed point s. Let $H \in \mathcal{CT}(S)$, and let T_s be the maximal central torus of H_s, which is defined over $\kappa(s)$ (cf. Lemma 3.4). Since

$$H_s = \mathrm{Centr}_{G_s}(T_s),$$

and H is smooth, T_s lifts uniquely to a sub-torus T of H, central in H (Exposé IX 3.6 bis and 5.6). Then $H' = \mathrm{Centr}_G(T)$ contains H and has the same fibre as H. Because H is flat, $H = H'$ (Exposé $\mathrm{VI_B}$ 2.5).

Suppose now that S is the spectrum of a henselian local ring, which the usual reductions allow us to take noetherian. Let s be its closed point, let T_s be the maximal central torus of H_s, let q be invertible on S, let ℓ be a power of q, and let T be an S-torus whose closed fibre is isomorphic to T_s (Exposé X 4.6). Let ${}_\ell H$ denote the "kernel" of the ℓ-th power morphism in H, and let U_ℓ be the largest open subset of ${}_\ell H$ étale over S. Lemma 1.3 and flatness of H show that U_ℓ contains ${}_\ell T_s$. Hensel's lemma gives a unique S-morphism

$$u_\ell : {}_\ell T \longrightarrow U_\ell$$

inducing the canonical immersion ${}_\ell T_s \to (U_\ell)_s$ on the closed fibre. Uniqueness shows that u_ℓ is a central homomorphism (Exposé IX 5.6(a)). Proceeding as in 6.6 and 6.7, one proves that u_ℓ is an immersion and, if $M_\ell = u_\ell({}_\ell T)$, that

$$\mathrm{Centr}_G(M_\ell) = H$$

for all sufficiently large ℓ. This is where noetherianity of S is used.

Proof of (i). If $H \in \mathcal{CT}(S)$, then H has connected fibres (Exposé XII 6.6(b)) and equals its connected normalizer (6.5). Hence the present functor $\mathcal{CT}$ is the functor of 6.4.0. Theorem

6.4 therefore represents it by an S-prescheme of finite presentation, quasi-projective over S. It remains to prove smoothness.

EGA IV 8 reduces us to affine noetherian S. By Exposé XI 1.5 it then suffices to prove the following infinitesimal lifting property: if S is local artinian, S_0 is defined by a nilpotent ideal, and $H_0 \in \mathcal{CT}(S_0)$, then H_0 lifts to a smooth subgroup prescheme H of G. By part (ii),

$$H_0 = \mathrm{Centr}_{G_0}(T_0)$$

for a sub-torus T_0 of G_0. Since G is smooth, T_0 lifts to a sub-torus T of G (Exposé IX 3.6 bis), and $H = \mathrm{Centr}_G(T)$ works; it is smooth by Exposé XI 2.4 and Lemma 2.5.

Proof of (iii). Because H is smooth with connected fibres, Exposé XI 6.11 represents $\mathrm{Transp}_G(H, L)$ by a closed subprescheme of G, of finite presentation over S. Smoothness reduces to the following lifting problem. Let S be local artinian, S_0 a closed subprescheme, and $g_0 \in G(S_0)$ such that

$$\mathrm{int}(g_0)H_0 \subseteq L_0.$$

We must lift g_0 to $g \in G(S)$ with $\mathrm{int}(g)H \subseteq L$.

Smoothness of G gives $g_1 \in G(S)$ lifting g_0. Put $H' = \mathrm{int}(g_1)H$. Then $H' \in \mathcal{CT}(S)$ and $H'_0 \subseteq L_0$. By (ii), $H' = \mathrm{Centr}_G(T')$ for a torus $T' \subseteq G$. Since L is smooth, $T'_0 \subseteq L_0$ lifts to a torus $T'' \subseteq L$ (Exposé IX 3.6 bis). The group $\mathrm{Centr}_L(T'')$ lies in $\mathrm{Centr}_G(T'')$, has the same closed fibre as the latter, namely H'_0, and is smooth. Hence

$$\mathrm{Centr}_L(T'') = \mathrm{Centr}_G(T'').$$

The sub-tori T' and T'' of G lift the same T'_0, so they are conjugate by an element $h \in G(S)$ reducing to the identity (Exposé IX 3.3 bis). Their centralizers are therefore conjugate. Then $g = hg_1$ lifts g_0 and satisfies $\mathrm{int}(g)H \subseteq L$.

For $g \in \mathrm{Transp}_G(H, L)(S)$, the equality $\mathrm{int}(g)H = L$ holds if and only if $\dim H_s = \dim L_s$ for every $s \in S$. If U is the open-and-closed subprescheme over which these dimensions agree, then

$$\mathrm{Transpstr}_G(H, L) = U \times_S \mathrm{Transp}_G(H, L).$$

Proof of (iv). To prove H closed in G, reduce first to affine noetherian S, then, by EGA IV 8, to the spectrum of a complete local ring. Part (ii) writes H as the centralizer in G of a subgroup of multiplicative type, so H is closed because G is separated (Exposé $\mathrm{VI_B}$ 5.2).

By (iii),

$$N = \mathrm{Norm}_G(H) = \mathrm{Transpstr}_G(H, H)$$

is represented by a subgroup prescheme smooth over S and closed in G. The morphism

$$G \longrightarrow \mathcal{CT}, \qquad g \longmapsto \mathrm{int}(g)H$$

is smooth by (iii), so its image is an open subset U of $\mathcal{CT}$. As in Exposé XI 5.3, G/N is represented by U; it is therefore quasi-projective.

We now treat N/H. By EGA IV 8, assume S affine noetherian and then $S = \operatorname{Spec} A$, with A local of finite dimension[23]. We use induction on $\dim S$. When $\dim S = 0$, the assertion is Exposé $\mathrm{VI_A}$, ğ4. If N/H is representable, it is separated because H is closed in N, of finite type and étale because N is smooth of finite type and H is open. It is therefore quasi-affine (SGA 1, VIII 6.2). Effective descent for quasi-affine schemes (*loc. cit.*, 7.9) permits replacement of A by its completion.

[23] Krull dimension.

Thus assume S complete noetherian local, with closed point s, and put $U = S \setminus \{s\}$. By induction, $(N|_U)/(H|_U)$ is represented by a U-group K. Let T_s be the maximal central torus of H_s, choose q invertible on S, let ℓ be a power of q, and let M_ℓ be the unique central subgroup of multiplicative type of H lifting ${}_\ell T_s$, as in (ii). Choose ℓ so large that

$$\operatorname{Centr}_G(M_\ell) = H,$$

and put $N' = \operatorname{Norm}_G(M_\ell)$.

The torus T_s, and hence $(M_\ell)_s$, is characteristic in H_s. Therefore N_s normalizes $(M_\ell)_s$, and $N_s = N'_s$. The proof of Exposé XI 5.9 represents

$$N'/H = \operatorname{Norm}_G(M_\ell)/\operatorname{Centr}_G(M_\ell)$$

by an S-group K'. Since N' is smooth and $N'_s = N_s$, it is an open subgroup of N containing H (Exposé $\mathrm{VI_B}$ 2.5). Its image over U is an open subgroup of K, isomorphic to $K'|_U$. Glue K and K' along this open subgroup to obtain L, and glue the projections $N|_U \to K$ and $N' \to K'$ to obtain $p : N \to L$. Then (L, p) represents N/H.

Proof of (v). First suppose $S = \operatorname{Spec} k$. The image H' of H is a smooth subgroup of G'. The required assertion follows from:

Lemma 7.2.— *Let $u : G \to G'$ be an epimorphism of smooth connected algebraic groups over a field k, let T be a torus of G, and let T' be its image in G'. Then*

$$u(\operatorname{Centr}_G(T)) = \operatorname{Centr}_{G'}(T').$$

Put $H = \operatorname{Centr}_G(T)$, $H' = \operatorname{Centr}_{G'}(T')$, and $H'' = u(H)$. We have $H'' \subseteq H'$. After extending the base field, assume k algebraically closed. It suffices to prove that every Cartan subgroup of H' lies in H'': then H'' contains the open subset of regular points of H', so is open, and equals the connected group H'.

Let C' be a Cartan subgroup of H'. It is also a Cartan subgroup of G', since H', being the centralizer of the torus T', has the same reductive and nilpotent ranks as G'. Put

$$K = (u^{-1}(C'))^0_{\mathrm{red}}.$$

Because T' is central in H', C' contains T', and K contains T. Let C be a Cartan subgroup of K containing T. The torus T lies in the unique maximal torus of C, central in C (Exposé XII 6.6(c)); hence $C \subseteq H$. Any two Cartan subgroups of G are conjugate, and the image of a Cartan subgroup of G is a Cartan subgroup of G' (Exposé XII 6.6). It follows that C is a Cartan subgroup of G, and its image is a Cartan subgroup of G' contained in C'. Thus $u(C) = C'$, proving $C' \subseteq H''$.

This proves (v) over a field. In general, finite presentation and the usual reductions permit us to assume S affine noetherian, then local, and finally the spectrum of a complete noetherian local ring A, by fpqc descent of subpreschemes of G'.

Use the notation of (ii). Let T_s be the maximal central torus of H_s, and let $M_\ell \subset H$ lift ${}_\ell T_s$, with $H = \operatorname{Centr}_G(M_\ell)$. Let $T'_s = u_s(T_s)$. Since G' is separated, $u(M_\ell)$ is represented by a subgroup M'_ℓ of multiplicative type (Exposé IX 6.8). Put

$$H' = \operatorname{Centr}_{G'}(M'_\ell),$$

which is smooth. For every power ℓ' of q, a sufficiently large ℓ makes $(M'_\ell)_s$ contain ${}_{\ell'}T'_s$. We may therefore choose ℓ such that

$$H'_s = \operatorname{Centr}_{G'_s}(T'_s) = u_s(H_s),$$

the last equality being Lemma 7.2.

The restriction $v : H \to H'$ of u is flat. Indeed, H, H' are flat over S, $H_s \to H'_s$ is flat, and hence v is flat near H_s (EGA IV 11.3.10 and 11.3.1). It is therefore flat on an open subgroup of H (Exposé $\mathrm{VI_B}$ 2.2), and thus everywhere because the fibres of H are connected. The set-theoretic image of H is consequently an open subset of H', necessarily $(H')^0$, and with its induced structure represents the fpqc image sheaf $u(H)$. Lemma 7.2 shows that it belongs to $\mathcal{CT}_{G'}(S)$.

If $H = \mathrm{Centr}_G(T)$, then $T' = u(T)$ is a sub-torus of G' (Exposé IX 6.8). The image $u(H)$ is contained in $\mathrm{Centr}_{G'}(T')$, agrees with it fibrewise by Lemma 7.2, and is smooth. Hence

$$u(H) = \mathrm{Centr}_{G'}(T').$$

Proof of (vi). Since $\mathcal{CT}_G, \mathcal{CT}_{G'}$ are smooth over S, it is enough to prove faithful flatness on geometric fibres. Thus take $S = \operatorname{Spec} k$, with k algebraically closed. Let $H' \in \mathcal{CT}_{G'}(k)$, let T' be its maximal central torus, choose a sub-torus $T \subset G$ mapping onto T', and put

$$H = \mathrm{Centr}_G(T), \qquad N = \mathrm{Norm}_G(H), \qquad N' = \mathrm{Norm}_{G'}(H').$$

By Lemma 7.2, $H' = u(H)$. Part (iv) identifies G/N and G'/N' with open neighbourhoods of H and H' in the two $\mathcal{CT}$-spaces. On these neighbourhoods, $\widetilde{u}$ is

$$w : G/N \longrightarrow G'/N',$$

obtained from u by passage to quotients. This is an epimorphism of homogeneous spaces under G, hence faithfully flat. Therefore $\widetilde{u}$ is flat and its image is an open subset of $\mathcal{CT}_{G'}$ containing all k-points. Since $\mathcal{CT}_{G'}$ is of finite type, $\widetilde{u}$ is surjective and hence faithfully flat. If $\ker u$ is central, the remaining assertions are Exposé XII 7.12.

The next theorem generalizes Exposé XII 7.1.

Theorem 7.3.— *Let S be a prescheme and let G be an S-group prescheme of finite presentation, smooth over S, with connected fibres. Define the functor*

$$\mathcal{C}(S') = \{\textit{Cartan subgroups of } G_{S'}\}.$$

i) *The following are equivalent:*

   a) *$\mathcal{C}$ is representable;*

   b) *$\mathcal{C}$ is represented by an S-prescheme smooth, quasi-projective, and of finite presentation over S, with affine fibres;*

   c) *G admits a Cartan subgroup locally for the étale topology;*

   d) *G admits a Cartan subgroup locally for the faithfully flat topology;*

   e) *the nilpotent rank of the fibres of G is locally constant on S.*

ii) *If these conditions hold, any two Cartan subgroups of G are locally conjugate for the étale topology. The regular locus of the fibres of G (Exposé XIII 2.7) is an open subprescheme G_{reg}, of finite presentation over S, and every section of G_{reg} over S is contained in a unique Cartan subgroup of G.*

iii) *Let G' be an S-group prescheme of finite presentation and $u : G \to G'$ faithfully flat. Then, for every Cartan subgroup C of G, its image $C' = u(C)$ is represented by a Cartan subgroup of G', and $C \to C'$ is faithfully flat.*

iv) *Under the conditions of (i) and (iii), the morphism*

$$\widetilde{u} : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \qquad C \longmapsto u(C),$$

is faithfully flat. If $\ker u$ *is central, it is an isomorphism.*

v) *Further information on transporters and relations with maximal tori is given in Proposition 7.1 and Exposé XII 7.1.*

Proof. For (i) we prove

$$\text{(b)} \Rightarrow \text{(c)} \Rightarrow \text{(d)} \Rightarrow \text{(e)} \Rightarrow \text{(b)} \Rightarrow \text{(a)} \Rightarrow \text{(d)}.$$

Assume (b), and let $s \in S$. Since $\mathcal{C}_s$ is smooth over $\kappa(s)$, it has points with residue field a finite separable extension of $\kappa(s)$. Exposé XI 1.10 gives an open neighbourhood U of s and an étale surjective morphism $U' \to U$ such that $G_{U'}$ admits a Cartan subgroup. Thus (b)⇒(c), while (c)⇒(d) is clear.

Assume (d), and let $s \in S$. There is a flat S-prescheme S', whose image contains s, such that $G_{S'}$ admits a Cartan subgroup. For $s' \in S'$ above s, the nilpotent rank of the fibres of $G_{S'}$ is constant on $\operatorname{Spec} \mathcal{O}_{S',s'}$. Hence the nilpotent rank of the fibres of G is constant on $\operatorname{Spec} \mathcal{O}_{S,s}$, which is the image of that local spectrum (EGA IV 2.3.4(ii)). Let the value be r. By 6.3 bis, the set E_r where the nilpotent rank equals r is ind-constructible; it contains a neighbourhood of s by EGA IV 1.10.1. Thus (d)⇒(e).

Assume (e). Locally on S, the nilpotent rank is a constant r. For every S', the Cartan subgroups of $G_{S'}$ are exactly the smooth subgroup preschemes of relative dimension r whose geometric fibres are centralizers of tori. By Proposition 7.1, $\mathcal{C}$ is the open-and-closed subprescheme of $\mathcal{C}\mathcal{T}$ representing the relative-dimension-r subfunctor. It has all the properties in (b); affineness of its fibres is Exposé XII 7.1(d). This proves (e)⇒(b), and (b)⇒(a) is immediate.

Finally assume (a). By 6.2(iii), $\mathcal{C}$ commutes with filtered inductive limits of rings, so its representing prescheme is locally of finite presentation (EGA IV 8.14.2). To prove it smooth, it is enough to show that, when S is affine and $S_0 \subseteq S$ is defined by a nilpotent ideal, every Cartan subgroup C_0 of G_0 lifts to a Cartan subgroup of G. Existence of C_0 implies (e) on S_0, hence on S, which has the same underlying space. Since (e)⇒(b), $\mathcal{C}$ is smooth. It is also surjective over S, so (a)⇒(d).

For (ii), if C, C' are Cartan subgroups, then $\operatorname{Transp}_G(C, C')$ is represented by a smooth S-prescheme (Proposition 7.1(iii)) with nonempty fibres (Exposé XII 6.6(a),(c)). Hensel's lemma (Exposé XI 1.10) gives local conjugacy for the étale topology. The assertions about regular points follow from Exposé XIII 3.1 and 3.2.

For (iii), Proposition 7.1(v) represents $u(C)$ by a smooth subgroup C' of G'. Its fibres are Cartan subgroups of the fibres of G' (Exposé XII 6.6(d)), so C' is a Cartan subgroup.

For (iv), faithful flatness is proved as in Proposition 7.1(vi). If $\ker u$ is central and C' is a Cartan subgroup of G', then $C = u^{-1}(C')$ is smooth with connected fibres (Proposition 7.1(vi)), and its fibres are Cartan subgroups (Exposé XII 6.6(f)). Thus C is a Cartan subgroup, and Proposition 7.1(vi) completes the proof.

8. Representability criterion for the functor of sub-tori of a smooth group

8.0. In this section, if S is a prescheme and G an S-group prescheme, $\mathcal{T}_G$ (or simply $\mathcal{T}$, if there is no ambiguity) denotes the S-functor

$$(\mathrm{Sch}\,/S)^\circ \longrightarrow \mathrm{Ens}$$

such that, for every S-prescheme S',

$$\mathcal{T}(S') = \{\text{sub-tori of } G_{S'}\}.$$

The functor $\mathcal{T}_C$ of central sub-tori of G is defined similarly.

Proposition 8.1.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type. Then the following conditions are equivalent:*

i) *The functor $\mathcal{T}_G$ "commutes with adic limits of local artinian rings": for every S-prescheme $S' = \operatorname{Spec}(A)$, where A is a complete noetherian local ring for the topology defined by its maximal ideal $\mathfrak{m}$, the canonical map*

$$\mathcal{T}(S') \longrightarrow \varprojlim_n \mathcal{T}(S'_n), \qquad S'_n = \operatorname{Spec}(A/\mathfrak{m}^n),$$

is bijective.

ii) *The assertion in* i)*, restricted to the case in which A is a complete discrete valuation ring with algebraically closed residue field.*

iii) *The assertion in* ii)*, restricted to the subfunctor $\mathcal{T}^{(1)}$ of $\mathcal{T}$ formed by sub-tori of G of relative dimension* 1.

i bis) *For every S-prescheme S' as in* i) *and every S'-torus T, the canonical map*

$$\operatorname{Hom}_{S'\text{-}gr}(T, G_{S'}) \longrightarrow \varprojlim_n \operatorname{Hom}_{S'_n\text{-}gr}(T_{S'_n}, G_{S'_n})$$

is bijective.

ii bis) *The assertion in* i bis)*, restricted to the case in which A is a complete discrete valuation ring with algebraically closed residue field.*

iii bis) *The assertion in* ii bis)*, restricted to $T = \mathbb{G}_{m,S'}$.*

Remark 8.2. There is an analogous proposition in which one restricts to central sub-tori of G and to central homomorphisms from a torus into G.

Proof. We shall use the following lemma.

Lemma 8.3.— *Let S be a prescheme, G an S-group prescheme, T an S-torus, and $u : T \to G$ an S-group morphism. Suppose, moreover, either that G is of finite presentation over S, or that S is locally noetherian and G is locally of finite type. Then:*

a) $\ker u$ *is a subgroup of multiplicative type of T;*

b) *the quotient $T' = T/\ker u$ is a torus;*

c) *the canonical monomorphism $T' \to G$ induced by u after passage to the quotient is an immersion.*

When G is separated over S, this lemma follows from Exposé IX 6.8. In the general case, the usual reduction allows us to suppose that S is noetherian. We first show that $K = \ker u$ is flat. We may suppose that S is the spectrum of a local artinian ring (EGA 0_{III}, 10.2.6); then G is separated (Exposé VI_{B} 5.2), so K is of multiplicative type (Exposé IX 6.8), and hence is flat over S. We next show that $\ker u$ is closed in T. This reduces us to the case in which S is the spectrum of a discrete valuation ring (EGA II 7.2.1). Since T is flat with connected fibres, u factors through the identity component of the schematic closure in G of the generic fibre of G. We may thus suppose that G is flat with connected fibres; it is then separated (Exposé $\mathrm{VI_B}$ 5.2), and K is closed. It follows from Exposé X 4.8(b) that K is a subgroup of multiplicative type of T. The quotient $T' = T/K$ is therefore representable and is a group of multiplicative type (Exposé IX 2.7) whose fibres are tori; hence T' is a torus. Finally, the monomorphism $T' \to G$ is an immersion by Exposé VIII 7.9.

Proof of Proposition 8.1. i)⇒i bis). Put $T_n = T_{S'_n}$, $G_n = G_{S'_n}$, and let

$$(u_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathrm{Hom}_{S'_n\text{-gr}}(T_n, G_n).$$

For every n, $u_n(T_n)$ is a sub-torus T'_n of G_n (Lemma 8.3). By hypothesis, there is a unique sub-torus T' of G lifting T'_n for every n. Since T' has affine fibres, the assertion follows from 4.4.

i bis)⇒i). Let $(T_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathcal{T}(S'_n)$. By Exposé X 4.6 there are an S'-torus T^0 and an S'_0-isomorphism

$$u_0 : T^0_0 \xrightarrow{\sim} T_0.$$

Since T_n is smooth over S'_n, u_0 lifts to an S'_n-morphism $u_n : T^0_n \to T_n$ (Exposé IX 3.6), and the family $(u_n)_{n\in\mathbb{N}}$ may be chosen coherent. It therefore comes from a morphism $u : T^0 \to G$. By Lemma 8.3, the image of T^0 under u is a sub-torus of G, and it lifts T_n for every n.

The implications

$$\text{i)} \Rightarrow \text{ii)} \Rightarrow \text{iii)} \quad \text{and} \quad \text{i bis)} \Rightarrow \text{ii bis)} \Rightarrow \text{iii bis)}$$

are immediate. The implication iii) ⇒ iii bis) is proved in the same way as i) ⇒ i bis). It remains to prove

$$\text{iii bis)} \Rightarrow \text{ii bis)} \Rightarrow \text{i bis)}.$$

ii bis)⇒i bis). In the terminology of 4.3, assertion i bis) holds if and only if every element

$$(u_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathrm{Hom}_{S'_n\text{-gr}}(T_n, G_n)$$

is admissible. For every point t of S' distinct from its closed point s, there is an S-scheme S'', the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point maps to t and whose closed point maps to s (EGA II 7.1.9). This gives at once a valuative criterion for a family of morphisms to be admissible, and proves the implication.

iii bis)⇒ii bis). Let T be an S-torus, where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. The torus T is split (Exposé X 4.6), say

$$T \simeq \mathbb{G}^r_{m,S}.$$

Let

$$(u_n)_{n\in\mathbb{N}} \in \varprojlim_n \mathrm{Hom}_{S_n\text{-gr}}\big((\mathbb{G}_m)^r_{S_n}, G_n\big).$$

By hypothesis, the restriction of u_n to each factor of $\mathbb{G}^r_{m,S}$ comes from a group morphism $\mathbb{G}_{m,S} \to G$. Their product, followed by the composition law $G^r \to G$, gives a morphism

$$v : \mathbb{G}^r_{m,S} \longrightarrow G.$$

Clearly $u_n = v_n$ for every n. It remains to prove that v is a group morphism. Equivalently, two evident morphisms

$$f, g : X = \mathbb{G}^r_{m,S} \times_S \mathbb{G}^r_{m,S} \longrightarrow G$$

must coincide. Let Z be their coincidence subscheme. Since $\mathbb{G}^r_{m,S}$ has connected fibres and its image under v contains the unit section, the argument of Lemma 8.3 shows that v factors through the identity component of G. We may therefore suppose that G is separated (Exposé $\mathrm{VI_B}$ 5.2), so Z is closed. On the other hand, v_n is a group morphism, hence $f_n = g_n$ for every n. Thus Z contains a neighbourhood of the closed fibre of X (EGA I 10.9.4), which is schematically dense in X, since X has smooth irreducible fibres (Exposé IX 4.6). Consequently $Z = X$, and $f = g$.

Definition 8.4.— *Let S be a locally noetherian prescheme and G an S-group prescheme of finite type. Let $\mathcal{S}$ be the set of S-schemes S' that are spectra of complete discrete valuation rings with algebraically closed residue field. Write s and t for the closed and generic points of S', and require that $(G_t)^0_{\mathrm{red}}$ be smooth and admit a Chevalley decomposition: an extension of an abelian variety by a smooth connected linear algebraic subgroup L_t. This decomposition is then unique. For $S' \in \mathcal{S}$, let G', respectively L, denote the schematic closure in $G_{S'}$ of G_t, respectively of L_t.*

We say that "the abelian part of G does not degenerate into a toric part", or briefly that G has property AT*, if, for every $S' \in \mathcal{S}$, L_s has the same reductive rank as G'_s. Intuitively, suppose that the quotient $A = G'/L$ is representable. Then A is a flat group prescheme and $(A_t)^0_{\mathrm{red}}$ is an abelian variety; property* AT *says that A_s has reductive rank zero, so that $(A_s)^0_{\mathrm{red}}$ is an extension of an abelian variety by a unipotent group.*

Similarly, if G has connected fibres, we say that G has property ATC *if, for every $S' \in \mathcal{S}$, the schematic closure Z of the centre Z_t of G_t has property* AT.

These technical definitions are justified by the following proposition.

Proposition 8.5.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type.*

a) *The functor $\mathcal{T}$ of sub-tori of G commutes with adic limits of local artinian rings (cf. Proposition 8.1) if and only if G has property* AT.

b) *If G has connected fibres, the functor $\mathcal{T}_C$ of central sub-tori of G commutes with adic limits of local artinian rings if and only if G has property* ATC.

We need the following technical lemma.

Lemma 8.6.— *Let S be the spectrum of a discrete valuation ring, with closed point s, let G be an S-group prescheme flat and of finite type, and let T_s be a sub-torus of G_s. There are an S-scheme S', the spectrum of a discrete valuation ring faithfully flat over S, with closed point s', and a subgroup scheme C of $G_{S'}$, flat over S', commutative, and with connected fibres, such that $C_{s'}$ contains $T_s \times_s s'$.*

After replacing S by the spectrum S' of a discrete valuation ring faithfully flat over it, we may suppose that $T_s = \mathbb{G}^r_{m,s}$ and that the transcendence degree of $\kappa(s)$ over its prime field is at least r (EGA 0_{III}, 10.3.1, and EGA II 7.1.9). There is then an element $x \in T_s(s)$ such that every algebraic subgroup of G_s containing x contains T_s (cf. Exposé XIII, proof of 2.1, ii) $\Rightarrow$ vii)). Since G is flat over S, a quasi-section passes through x (EGA IV 14.5.8). After another such replacement of S, we may therefore suppose that x lifts to a section $\widetilde{x}$ of G over S. Let C_t be the commutative algebraic subgroup of G_t generated by $\widetilde{x}_t$ (Exposé $\mathrm{VI_B}$ 7), and let C be its schematic closure in G. Clearly C_s contains x, hence contains T_s.

The identity component of C is therefore a flat commutative group scheme with connected fibres having the required property.

Proof of Proposition 8.5(a). Suppose first that $\mathcal{T}$ commutes with adic limits of artinian rings. Let $S' \in \mathcal{S}$, and let T_s be a maximal torus of G'_s. We must show that $T_s \subset L_s$. The formation of L and G' plainly commutes with faithfully flat extensions $S'' \to S'$ of discrete valuation rings. By Lemma 8.6, we may therefore replace S' so that there is a flat commutative subgroup scheme C of $G'_{S'}$ for which C_s contains T_s. The torus T_s is central in C_s, and hence lifts infinitesimally to a central sub-torus (Corollary 2.3). By the hypothesis on G, T_s lifts to a sub-torus T of G. Clearly T_t lies in the linear part L_t of G_t, and therefore $T \subset L$.

Conversely, suppose that G has property AT. We prove condition 8.1 iii bis). Let $S = \operatorname{Spec}(A)$, where A is a complete discrete valuation ring with algebraically closed residue field, let $\mathfrak{m}$ be its maximal ideal, put $S_n = \operatorname{Spec}(A/\mathfrak{m}^n)$, and let

$$u_n : (\mathbb{G}_{m,S})_n \longrightarrow G_n = G \times_S S_n, \qquad n \in \mathbb{N},$$

be a coherent system of group morphisms. Let q be a prime invertible on S. As ℓ ranges over the powers of q, there is a unique S-group morphism

$$_\ell u : {}_\ell\mathbb{G}_{m,S} \longrightarrow G$$

lifting the restriction of u_n to $_\ell(\mathbb{G}_{m,S})_n$ for every n (Proposition 1.6(a)). Consequently, if an S-group morphism $u : \mathbb{G}_{m,S} \to G$ lifting every u_n exists, its restriction to $_\ell\mathbb{G}_{m,S}$ is uniquely determined. The density theorem (Exposé IX 4.8(a)) proves both the uniqueness of u and that, for its existence, we may make a faithfully flat extension of the base.

Now $(G_t)^0_{\mathrm{red}}$ admits a Chevalley decomposition, already defined over a finite extension L of the fraction field K of A. After replacing S by its normalization in L, we may suppose that $S \in \mathcal{S}$. The morphism $_\ell u_t$ factors through $(G_t)_{\mathrm{red}}$, and $_\ell u$ therefore factors through the schematic closure G'' of $(G_t)_{\mathrm{red}}$ in G'. The density theorem then shows that every u_n factors through G''_n. Since G has property AT, every sub-torus of G'_s, and hence of G''_s, is contained in L_s; thus u_s factors through L_s. In fact every u_n factors through L_n. Indeed, L_t is invariant in G''_t, so L is invariant in G''; moreover L is flat over S, and the quotient

$$H_n = G''_n / L_n$$

is representable for every n (Exposé $\mathrm{VI_A}$, §4). The image of $(\mathbb{G}_{m,S})_n$ in H_n is a sub-torus (Exposé IX 6.8) whose closed fibre is zero. It is therefore zero, and u_n factors through L_n. Since L_t is affine, Proposition 4.3 now shows that the family (u_n) is admissible.

The proof of Proposition 8.5(b) is entirely analogous, in view of Remark 8.2 and Exposé IX 5.6(a).

Proposition 8.7.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type.*

i) *If G has connected fibres and has property* AT*, then it has property* ATC*.*

ii) *If G has property* AT*, every subgroup prescheme of G has property* AT*.*

iii) *If G is flat over S and the abelian rank of its fibres is a locally constant function on S, then G has property* AT*.*

Assertion (i) follows at once from Proposition 8.5 and Exposé IX 5.6(a).

For (ii), let H be a subgroup prescheme of G. To prove that H has property AT, we may suppose that S is the spectrum of a complete discrete valuation ring. We must show that every coherent family of S_n-group morphisms

$$u_n : (\mathbb{G}_{m,S})_n \longrightarrow H_n$$

is admissible (Proposition 8.5 and 8.1 iii bis)). By hypothesis, G has property AT, so there is an S-group morphism

$$u : \mathbb{G}_{m,S} \longrightarrow G$$

lifting every u_n. As in the proof of Proposition 8.5, the restriction of u to ${}_\ell\mathbb{G}_{m,S}$, with ℓ ranging over powers of a prime q invertible on S, factors through H. By density, u_t factors through H_t on the generic fibre. Since $\mathbb{G}_{m,S}$ is reduced, u itself factors through H.

For (iii), let $S' \in \mathcal{S}$, with the notation of Definition 8.4. The group L is flat over S, and its generic fibre L_t is affine. Under these conditions L_s is necessarily affine (Exposé XVII, Appendix III 2). Since G and L are flat, $G = G'$, and the dimensions of the fibres of G and L are constant on S (Exposé $\mathrm{VI_B}$ 4). Consequently

$$\mathrm{rk_{ab}}\, G_s \leq \dim G_s - \dim L_s = \dim G_t - \dim L_t = \mathrm{rk_{ab}}\, G_t.$$

By hypothesis these inequalities are equalities. Thus $(G_s/L_s)^0_{\mathrm{red}}$ is an abelian variety and has reductive rank zero. It follows that L_s has the same reductive rank as G_s.

We can now prove the principal theorems of this section.

Theorem 8.8.— *Let G be a group prescheme of finite type over a locally noetherian prescheme S. Suppose that G is flat over S and has connected fibres.*

a) *The functor $\mathcal{T}_C$ of central sub-tori of G is representable if and only if G has property* ATC. *In that case, $\mathcal{T}_C$ is represented by an S-prescheme étale and separated over S.*

b) *Under the hypotheses of* (a), *for every S-torus T, the functor* $\mathrm{Hom}^{\mathrm{cent}}_{S\text{-}gr}(T, G)$ *of central homomorphisms from T into G is represented by an S-prescheme étale and separated over S.*

Theorem 8.9.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme of finite type, smooth over S.*

a) *The functor $\mathcal{T}$ of sub-tori of G is representable if and only if G has property* AT. *In that case, $\mathcal{T}$ is represented by an S-prescheme smooth and separated over S. More precisely, its structural morphism admits a canonical factorization*

$$\mathcal{T} \xrightarrow{u} Y \xrightarrow{v} S,$$

where v is smooth and quasi-projective (hence of finite type) and u is étale and separated.

b) *Under the hypotheses of* (a), *for every S-torus T, the functor* $\mathrm{Hom}_{S\text{-}gr}(T, G)$ *of homomorphisms from T into G is represented by an S-prescheme smooth and separated over S.*

Proof of Theorem 8.8(a). If the functor $\mathcal{T}_C$ is representable, it commutes with adic limits of artinian rings. Consequently, by 8.2 and Proposition 8.5(b), G has property ATC. To prove the converse, we shall use the following result, which will be proved in EGA VI and may also be found in Murre's exposé, *Séminaire Bourbaki*, May 1965, no. 294, Theorem 1, Corollary 2.

Lemma 8.10.— *Let S be a locally noetherian prescheme and let F be a contravariant functor from* Sch/S *to the category of sets. In order that F be representable by an S-prescheme étale and separated over S, it is necessary and sufficient that F satisfy the following five properties:*

i) *F is a sheaf for the fpqc topology (Exposé IV).*

ii) *F commutes with filtered inductive limits of rings (Exposé XI 3.2).*

iii) *F commutes with adic limits of local artinian rings (8.1*(i)*).*

iv) *F satisfies the valuative criterion for separatedness: for every S-prescheme S' that is the spectrum of a discrete valuation ring, with generic point t, the canonical map*

$$F(S') \longrightarrow F(t)$$

is injective.

v) *F is infinitesimally étale (Exposé XI 1.8).*

We show that the functor $\mathcal{T}_C$ of Theorem 8.8 satisfies the five conditions of Lemma 8.10.

i) The functor $\mathcal{T}$, and likewise $\mathcal{T}_C$, is a sheaf for the fpqc topology whenever G is of finite presentation over S. Indeed, every monomorphism from a torus into G is then an immersion (Exposé VIII 7.9), and the assertion follows from fpqc descent for sub-preschemes.

ii) Corollary 6.3 proves that $\mathcal{T}$ commutes with filtered inductive limits of rings when G is of finite presentation over S; the same property for $\mathcal{T}_C$ follows immediately.

iii) By Proposition 8.5, condition (iii) is precisely equivalent to property ATC.

iv) If S is the spectrum of a discrete valuation ring, two sub-tori of G with the same generic fibre coincide. More precisely, each is the identity component of the schematic closure in G of that common generic fibre.

v) This follows from Proposition 2.3 because G is flat over S.

Proof of Theorem 8.8(b). Proceeding as in Exposé XI 4.2, it is enough to prove that the product group $T \times_S G$ again has property ATC, which is immediate from the definition.

Proof of Theorem 8.9(a). After replacing G, if necessary, by its identity component (Exposé $\mathrm{VI_B}$ 3.10), we may suppose that G has connected fibres. If T is a sub-torus of G, its centralizer $\mathrm{Centr}_G(T)$ is represented (Exposé XI 6.11) by a subgroup prescheme of G, smooth over S (Exposé XI 2.4), with connected fibres (Exposé XII 6.6(b)). It is therefore an element of $\mathcal{CT}(S)$, where $\mathcal{CT}$ is the functor defined in 7.1(i). The rule

$$T \longmapsto \mathrm{Centr}_G(T)$$

clearly defines an S-morphism

$$u : \mathcal{T} \longrightarrow \mathcal{CT}.$$

Since $\mathcal{CT}$ is represented by an S-prescheme smooth and quasi-projective over S (7.1(i)), it remains only to prove that u is representable by a separated étale morphism.

After a suitable base change $S' \to S$, with $S' = \mathcal{CT}$ and hence S' locally noetherian, we are reduced to the following problem. Let S be a locally noetherian prescheme, let G be an S-group prescheme smooth and of finite type over S, with connected fibres, and let H be a subgroup of G, smooth over S, with connected fibres. Consider the subfunctor X of $\mathcal{T}_{C,H}$ such that, for every S-prescheme S', $X(S')$ is the set of central sub-tori T of $H_{S'}$ for which

$$\mathrm{Centr}_{G_{S'}}(T) = H_{S'}.$$

We shall show that X is represented by an S-prescheme étale and separated over S.

Indeed, G has property AT by hypothesis, and so does H by Proposition 8.7(ii). Since H has connected fibres, it also has property ATC by Proposition 8.7(i). Moreover, H is smooth over S, hence flat. Theorem 8.8(a) therefore represents $\mathcal{T}_{C,H}$ by an S-prescheme étale and separated over S. It is enough to show that the canonical monomorphism

$$X \longrightarrow \mathcal{T}_{C,H}$$

is represented by an open immersion.

Put $S' = \mathcal{T}_{C,H}$, and let K be the centralizer in $G_{S'}$ of the universal central torus of $H_{S'}$. The group K is a smooth group scheme over S', with connected fibres, and contains $H_{S'}$. By definition,

$$X = \prod_{K/S'} H_{S'}/K,$$

which is represented by the open-and-closed subprescheme of S' over which $H_{S'}$ and K have the same relative dimension.

The proof of Theorem 8.9(b) is analogous to that of Theorem 8.8(b).

Corollary 8.11.— *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S. If the abelian rank of the fibres of G is a locally constant function on S—in particular, if the fibres of G are affine—then the functor of sub-tori of G is represented by an S-prescheme smooth and separated over S. The same holds for the functor* $\mathrm{Hom}_{S\text{-}gr}(T, G)$*, for every S-torus T.*

The assertion is local on S, so we may suppose that S is affine and that the abelian rank of the fibres of G is constant. There are then a noetherian affine scheme S_0 and an S_0-group prescheme G_0, smooth and of finite type over S_0, such that

$$G \simeq G_0 \times_{S_0} S$$

as S-group preschemes (Exposé $\mathrm{VI_B}$, §10). The abelian rank of the fibres of an S-group prescheme of finite presentation is an ind-constructible function (6.3 bis). A standard argument (EGA IV 8) therefore allows us, in the present situation, to choose G_0 so that the abelian rank of its fibres is constant on S_0. Proposition 8.7(iii) gives property AT for G_0, and Theorem 8.9 then represents its functor of sub-tori by an S_0-prescheme smooth and separated over S_0. Base change gives the assertion for G. The argument for $\mathrm{Hom}_{S\text{-}\mathrm{gr}}(T, G)$ is analogous.

Generalization of Theorem 8.9. The functor $\mathcal{T}$ of sub-tori of a smooth group G need not be representable. We now give sufficient conditions for a subfunctor of $\mathcal{T}$ to be representable.

Proposition 8.12.— *Let S be a locally noetherian prescheme, let G be an S-group prescheme smooth over S with connected fibres, and let F be an S-subfunctor of the functor $\mathcal{T}$ of sub-tori of G. Suppose that:*

i) *F is a sheaf for the fpqc topology (Exposé IV);*

ii) *F commutes with filtered inductive limits of rings (Exposé XI 3.2);*

iii) *F commutes with adic limits of local artinian rings (Exposé XI 3.3);*

iv) *F is infinitesimally smooth over S (Exposé XI 1.8);*

v) *F is stable under inner automorphisms of G: for every S-prescheme S',*

$$T \in F(S'), \quad g \in G(S') \quad \Longrightarrow \quad \operatorname{int}(g)T \in F(S').$$

Then F is represented by an S-prescheme smooth and separated over S.

Proposition 8.12 bis.— *Let S and G be as in Proposition 8.12, let T be an S-torus, and let F be a subfunctor of* $\mathrm{Hom}_{S\text{-}gr}(T, G)$. *Suppose that F satisfies conditions* (i)–(iv) *of Proposition 8.12 and is stable under inner automorphisms of G: for every S-prescheme S',*

$$u \in F(S'), \quad g \in G(S') \quad \Longrightarrow \quad \operatorname{int}(g)u \in F(S').$$

Then F is represented by an S-prescheme smooth and separated over S.

Proof of Proposition 8.12. The proof of Proposition 8.12 bis is entirely analogous and is left to the reader. Let

$$u : F \longrightarrow \mathcal{CT}$$

be the S-morphism which assigns to every $T \in F(S')$ the element $\mathrm{Centr}_{G_{S'}}(T)$ of $\mathcal{CT}(S')$ (7.1(i)). Since $\mathcal{CT}$ is represented by an S-prescheme smooth and quasi-projective over S, we are reduced to proving that u is representable.

After the base change $\mathcal{CT} \to S$, the problem is the following. Given S and G as above and a smooth subgroup H of G with connected fibres, represent the functor F_H defined by

$$F_H(S') = \Big\{ T \in F(S') \;\Big|\; H_{S'} = \mathrm{Centr}_{G_{S'}}(T) \Big\}.$$

We show that F_H is étale and separated over S by verifying the five conditions of Lemma 8.10.

The first three conditions follow directly from Proposition 8.12(i)–(iii). We have already observed that $\mathcal{T}_{C,H}$ is separated and infinitesimally étale. Thus F_H, being a subfunctor of $\mathcal{T}_{C,H}$, is separated and infinitesimally unramified (Exposé XI 1.8). It remains to prove that F_H is infinitesimally smooth.

Let S be the spectrum of a local artinian ring, let $S_0 \subseteq S$ be defined by a nilpotent ideal, and let $T_0 \in F_H(S_0)$. By Proposition 8.12(iv), T_0 lifts to an element $T' \in F(S)$. Since H is smooth, T_0 also lifts to a sub-torus T'' of H_S (Exposé IX 3.6 bis). This sub-torus is conjugate to T' by an element $g \in G(S)$ (*loc. cit.*), and hence $T'' \in F(S)$ by Proposition 8.12(v). The centralizer $\mathrm{Centr}_{G_S}(T'')$ is smooth over S and agrees with H_{S_0} over S_0. Therefore T'' lies in the centre of H_S (Exposé IX 5.6(a)), and its centralizer in G equals H_S. Thus $T'' \in F_H(S)$, and we take $T = T''$.

Corollary 8.13.— *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S, with connected fibres. Let $\mathcal{T}_D$ be the subfunctor of $\mathcal{T}$ for which $\mathcal{T}_D(S')$ is the set of sub-tori T of $G_{S'}$ such that, for every $s' \in S'$, $T_{s'}$ is contained in the derived subgroup of $G_{s'}$ (Exposé VI$_B$ 7). Then $\mathcal{T}_D$ is represented by an S-prescheme smooth and separated over S.*

Corollary 8.13 bis.— *Let S and G be as in Corollary 8.13, and let T be an S-torus. Let*

$$\mathrm{Hom}^{\mathrm{der}}_{S\text{-}gr}(T, G)$$

be the subfunctor of $\mathrm{Hom}_{S\text{-}gr}(T,G)$ *whose* S'*-points are the* S'*-morphisms* $u : T_{S'} \to G_{S'}$ *such that, for every* $s' \in S'$*, the morphism* $u_{s'}$ *factors through the derived group of* $G_{s'}$*. Then this functor is represented by an* S*-prescheme smooth and separated over* S*.*

The two corollaries have analogous proofs; we prove Corollary 8.13 bis. By the usual procedure, we reduce to the case in which S is noetherian. We verify the five conditions of Proposition 8.12 bis.

Conditions (i) and (iv) follow immediately from the corresponding properties of the functor

$$\mathrm{Hom}_{S\text{-gr}}(T,G).$$

Condition (v) holds because the derived group of an algebraic group is invariant (Exposé $\mathrm{VI_B}$, §7).

For condition (ii), a standard reduction (EGA IV 8) reduces us to the following assertion. Let S be an integral noetherian scheme with generic point η, and let $u : T \to G$ be an S-group morphism whose generic fibre factors through the derived group of G_η. Then there is a neighbourhood U of η such that, for every $s \in U$, u_s factors through the derived group of G_s. This is an immediate consequence of Exposé $\mathrm{VI_B}$ 10.12.

For condition (iii), retain the notation of Proposition 4.3. We must show that an element

$$(u_m)_{m\in\mathbb{N}} \in \varprojlim_m \mathrm{Hom}^{\mathrm{der}}_{S_m\text{-gr}}(T_m, G_m)$$

is admissible in the sense of Proposition 4.3 and comes from an element of $\mathrm{Hom}^{\mathrm{der}}_{S\text{-gr}}(T,G)$. As in 8.1, we reduce immediately to the case in which S is the spectrum of a complete discrete valuation ring. Let t be its generic point, s its closed point, and let D be the schematic closure in G of the derived group D_t of G_t.

a) The group D_s contains the derived group of G_s. Indeed, D_t is invariant in G_t, and G is flat over S, so D is invariant in G. Moreover, the morphism

$$G \times_S G \longrightarrow G, \qquad (x,y) \longmapsto xyx^{-1}y^{-1},$$

factors through D_t on the generic fibre and therefore factors through D. Consequently G_s/D_s is commutative.

b) If $u_m \in \mathrm{Hom}^{\mathrm{der}}_{S_m\text{-gr}}(T_m, G_m)$, then u_m factors through D_m. Its closed fibre u_s factors through the derived group of G_s, hence through D_s by (a). Since D_m is flat over S_m and invariant in G_m, the quotient

$$H_m = G_m/D_m$$

is representable (Exposé $\mathrm{VI_A}$, §4). The image of T_m in H_m is a torus (Exposé IX 6.8) with trivial closed fibre, hence is trivial. Thus u_m factors through D_m.

c) The family $(u_m)_{m\in\mathbb{N}}$ is admissible and lifts to a morphism $T \to G$ belonging to $\mathrm{Hom}^{\mathrm{der}}_{S\text{-gr}}(T,G)$. By what precedes and Proposition 4.1 bis, it is enough to prove that D_t is affine. This follows from the next lemma.

Lemma 8.14.— *Let* G *be a smooth connected algebraic group over a field* k*. Then the derived group* D *of* G *is affine.*

Formation of D commutes with extension of the base field (Exposé $\mathrm{VI_B}$ 7), so we may suppose that k is algebraically closed. The group G is then an extension of an abelian variety A by a linear group L. Since A is commutative, D is necessarily contained in L, and is therefore affine.

Maximal tori.

Theorem 8.15.— *Let S be a locally noetherian prescheme and let G be an S-group prescheme smooth and of finite type over S. The following conditions are equivalent:*

i) *The S-functor $\mathcal{T}_M$, whose S'-points are the maximal tori of $G_{S'}$ (Exposé XII 1.3), is representable.*

ii) *The functor $\mathcal{T}_M$ is represented by an S-prescheme smooth and quasi-projective over S, with affine fibres.*

iii) *The group G admits a maximal torus locally for the étale topology.*

iv) *The group G admits a maximal torus locally for the faithfully flat topology.*

v) *The group G has property* AT*, and the reductive rank of its fibres is a locally constant function on S.*

Proof. (ii)⇒(i) is clear.

(i)⇒(iii). Since G is of finite presentation over S, Corollary 6.3 shows that $\mathcal{T}_M$ commutes with filtered inductive limits of rings. If representable, it is therefore locally of finite presentation (EGA IV 8.14). Moreover, $\mathcal{T}_M$ is formally smooth (Exposé XI 2.1 bis). Hence its representing prescheme is smooth over S, and (iii) follows from Hensel's lemma (Exposé XI 1.10).

(iii)⇒(iv) is clear.

(iv)⇒(v). Let $s \in S$. There is an S-prescheme S', flat over S and with image containing s, such that $G_{S'}$ has a maximal torus T'. If $s' \in S'$ lies over s, the reductive rank of the fibres of $G_{S'}$ is constant on $\operatorname{Spec} \mathcal{O}_{S',s'}$. Thus the reductive rank of the fibres of G is constant on its image $\operatorname{Spec} \mathcal{O}_{S,s}$ (EGA IV 2.3.4(ii)). Since this rank is a locally constructible function on S (6.3 bis), it is constant on a neighbourhood of s (EGA IV 1.10.1).

To prove property AT, let S_1 be an S-scheme which is the spectrum of a discrete valuation ring, whose closed point s_1 maps to s. The prescheme

$$S_1' = S_1 \times_S S'$$

is faithfully flat over S_1, and $G_{S_1'}$ has a maximal torus. Let A and A' be the rings of S_1 and S_1', respectively. Writing A' as the filtered inductive limit of its finitely generated A-subalgebras and applying Corollary 6.3, we obtain an S_1-scheme S_1'' such that $G_{S_1''}$ has a maximal torus and $S_1'' \to S_1$ is of finite type and surjective. By EGA II 7.1.9, we may reduce to the case in which S_1' is the spectrum of a discrete valuation ring. Then $G_{S_1'}$, and hence G_{S_1}, has property AT. Since this holds for every such S_1, G has property AT.

(v)⇒(i). By Theorem 8.9 the functor $\mathcal{T}$ of sub-tori of G is representable. If the reductive rank of G is the constant function r, then $\mathcal{T}_M$ is represented by the open-and-closed subprescheme of $\mathcal{T}$ representing the subfunctor of tori of rank r.

(iii)⇒(ii). Under condition (iii), the results of Exposé XII 7.1 identify $\mathcal{T}_M$ canonically with the functor of Cartan subgroups of G. The assertion follows from Theorem 7.3(i).

Remark 8.16. The prescheme $\mathcal{T}_M$ of maximal tori of G can in fact be shown to be affine over S,[24] generalizing Exposé XII 5.4.

[24]Cf. M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes* (thesis, forthcoming). Editor's note: see *Lecture Notes in Mathematics* 119 (1970), Springer, especially IX 2.9.

Corollary 8.17.— *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S. If the abelian and reductive ranks of the fibres of G are locally constant functions on S, then G satisfies the equivalent conditions* (i)–(iv) *of Theorem 8.15.*

We may suppose that the abelian and reductive ranks of the fibres of G are constant. Proceeding as in Corollary 8.11 and using 6.3 bis, we reduce to the case in which S is noetherian. Then G has property AT by Proposition 8.7, and the conclusion follows from Theorem 8.15(v).

Corollary 8.18.— *Let S be a prescheme and let G be an S-group prescheme smooth over S. The following conditions are equivalent:*

a) *The unipotent rank ρ_u and the abelian rank ρ_{ab} of the fibres of G are locally constant functions on S.*

b) *The unipotent rank ρ_u is locally constant, and G admits maximal tori locally for the fpqc topology.*

c) *The reductive rank ρ_r and the abelian rank ρ_{ab} of the fibres of G are locally constant functions on S.*

Remark 8.19. Under the hypotheses of Corollary 8.18, a more precise argument using lower semicontinuity of the abelian rank, announced in Exposé X 8.7, shows that if two of the three ranks $\rho_u, \rho_r, \rho_{\mathrm{ab}}$ are locally constant, then so is the third.

Proof of Corollary 8.18. After replacing G, if necessary, by its identity component, we may suppose that G is of finite presentation over S (Exposé $\mathrm{VI_B}$ 5.3.3).

(a)⇒(c). Let $s \in S$. Since ρ_{ab} is locally constant, Corollary 8.11 allows us, after an étale extension covering S, to suppose that G has a sub-torus T whose fibre T_s is a maximal torus of G_s. Put

$$C = \mathrm{Centr}_G(T).$$

This is a subgroup prescheme of G, smooth over S, with connected fibres. For every $t \in S$, the group C_t contains a Cartan subgroup C'_t of G_t. After shrinking S, we may suppose that ρ_u, ρ_{ab}, and the relative dimension of C over S are constant. We then have

$$\begin{aligned} \dim C_s = \dim C_t &\geq \dim C'_t \\ &\geq \rho_u(t) + \rho_{\mathrm{ab}}(t) + \dim T_t \\ &= \rho_u(s) + \rho_{\mathrm{ab}}(s) + \dim T_s \\ &= \rho_\nu(s) = \dim C_s. \end{aligned}$$

It follows that $C_t = C'_t$. Thus T is a maximal torus of G, and in particular $\rho_r(t) = \rho_r(s)$.

(c)⇒(b) follows from Corollary 8.17.

(b)⇒(a). Since G admits maximal tori locally for the fpqc topology, it also admits Cartan subgroups locally for that topology. Hence, by Exposé XII 7.3, the nilpotent rank

$$\rho_\nu = \rho_u + \rho_r + \rho_{\mathrm{ab}}$$

is locally constant. The existence of maximal tori makes ρ_r locally constant, while ρ_u is locally constant by hypothesis; therefore ρ_{ab} is locally constant.

Exposé XVI
Groups of Unipotent Rank Zero

by M. Raynaud[(*)]

1. An Immersion Criterion

1.1. Examples of Monomorphisms of Group Preschemes That Are Not Immersions

We shall construct a prescheme S, two S-group preschemes G and H, and an S-group monomorphism

$$u : G \longrightarrow H$$

that is not an immersion. The groups G and H will be of finite presentation over S, flat over S, and G will even be smooth over S (cf. Exposé VIII, the paragraph and note $(*)$ preceding 7.1; see also XVII, Appendix III, 4).

First take S to be the spectrum of a discrete valuation ring A of unequal characteristic and residue characteristic 2. Let t be the generic point of S and s the closed point.

a) Take for G the open subgroup of the constant group

$$G' = (\mathbb{Z}/2\mathbb{Z})_S$$

obtained by removing the point of the closed fiber $(\mathbb{Z}/2\mathbb{Z})_s$ distinct from the identity element, and take for H the group of multiplicative type $(\mu_2)_S$ of square roots of unity (Exposé I, 4.4.4). By the choice of S, H_t is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_t$, whereas H_s is a radicial group. There is an evident morphism $G' \to H$, defined by the section (-1) of H, hence a morphism $u : G \to H$ which is a monomorphism but not an immersion (otherwise it would necessarily be an isomorphism; cf. Exposé VIII, 7). Here Exposé VIII, 7.9 does not apply, since ${}_2G = G$ is not finite over S.

b) Let K denote the étale, nonseparated S-group prescheme obtained from the identity group by "doubling" the unique point of the closed fiber. Take for H the product group

$$(\mu_2)_S \times_S K.$$

Let a (respectively b) be the unique section of μ_2 (respectively K) over S distinct from the identity section, and let $h = (a, b)$ be the corresponding section of H. The datum of h defines an S-group morphism

$$u : G = (\mathbb{Z}/2\mathbb{Z})_S \longrightarrow H.$$

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[0]xy version of December 1, 2008.

[*]Cf. the note on page 1 of Exposé XV.

It is clear that u is a monomorphism and is not an immersion. Here Exposé VIII, 7.9 does not apply, since ${}_2H = H$ is not separated over S.

c) More interesting is the following example, in which G is a smooth group with connected fibers (and even $G = (\mathbb{G}_a)_S$); cf. Exposé VIII, 7.10.

Take for S the spectrum of a discrete valuation ring A of equal characteristic 2, and let π be a uniformizer of A and $\mathfrak{m}$ the maximal ideal of A.

Consider the additive group $(\mathbb{G}_a)_S$, the product group $(\mathbb{G}_a \times \mathbb{G}_a)_S$ with ring $A[X, Y]$, and let G_1 be the closed subgroup defined by

$$X^2 + X - \pi Y = 0. \tag{1}$$

The group G_1 is smooth over S; its generic fiber is isomorphic to $(\mathbb{G}_a)_t$, and its closed fiber is isomorphic to the product of $(\mathbb{G}_a)_s$ with $(\mathbb{Z}/2\mathbb{Z})_s$. For the $(\mathbb{Z}/2\mathbb{Z})_s$-factor one may take the subgroup N_s of $(G_1)_s$ defined by

$$Y = X^2 + X = 0. \tag{2}$$

Let N' and N'' be two group subschemes of G_1, isomorphic to $(\mathbb{Z}/2\mathbb{Z})_S$, whose closed fibers coincide with N_s and whose generic fibers are distinct. Thus N' and N'' are defined by two sections of G_1 over S, with coordinates (a', b') and (a'', b''), respectively, such that

$$\begin{cases} a'^2 + a' - \pi b' = a''^2 + a'' - \pi b'' = 0, \\ a' \text{ and } a'' \equiv 1 \pmod{\mathfrak{m}}, \qquad a' \neq a'', \\ b' \text{ and } b'' \equiv 0 \pmod{\mathfrak{m}}. \end{cases} \tag{3}$$

(For example, one may take $(1, 0)$ and $(1 + \pi^2, \pi + \pi^3)$.)

The quotient groups

$$G' = G_1/N' \qquad \text{and} \qquad G'' = G_1/N''$$

are representable (Exposé V, 4.1). Let u be the canonical morphism

$$u : G_1 \longrightarrow G' \times_S G'',$$

and let H be the group subscheme of $G' \times_S G''$ equal to the schematic closure in $G' \times_S G''$ of $u_t((G_1)_t)$, so that u factors through H. The kernel of u is

$$K = N' \cap N'',$$

whose generic fiber is the identity group and whose closed fiber is equal to N_s; in particular, K is not flat over S. On the generic fiber, u_t is therefore an isomorphism from $(G_1)_t$ onto H_t. On the other hand, H_s is not smooth. Indeed, if H_s were smooth, then, since u_s is a morphism with finite kernel and $(G_1)_s$ and H_s have the same dimension, namely 1, u_s would be flat. Since G_1 and H are flat, u would then be flat, and consequently $\operatorname{Ker} u$ would be flat over S, which it is not. It is clear that the restriction of u to the identity component G of G_1 is a monomorphism: the fibers of $K \cap G$ are identity groups, so $K \cap G$ is the identity group (Exposé $\mathrm{VI_B}$, 2.9). But it is not an immersion, for otherwise it would be an open immersion and H_s would be smooth. Here Exposé VIII, 7.9 does not apply, since ${}_2G = G$ is not finite over S.

For lovers of equations, in the preceding example one may take for H the closed subgroup of $(\mathbb{G}_a \times_S \mathbb{G}_a)_S$ with ring $A[V, W]/(F)$, where

$$F = (a''b' - b''a')(a''^2 V - a'^2 W) - (a''V - a'W)^2$$

(where a', a'', b', b'' satisfy (3)). For G one takes the group $(\mathbb{G}_a)_S$ with ring $A[T]$, and for the morphism $u : G \to H$ the S-morphism defined by

$$V \longmapsto a'(a'T + \pi T^2), \qquad W \longmapsto a''(a''T + \pi T^2).$$

Remark 1.2. — The preceding construction is inspired by the method of Koizumi–Shimura.[1] It does not apply when S is the spectrum of a ring of unequal characteristic, because the points of finite order of G are then "too rigid."

1.3. Statement of the Immersion Criterion

Theorem 1.3. — *Let S be a locally noetherian prescheme, let G be an S-group prescheme, smooth and of finite type over S, possessing Cartan subgroups locally for the fpqc topology (Exposé XV, 6.1 and 7.3 (i)), let H be an S-group prescheme locally of finite type, and let $u : G \to H$ be an S-group monomorphism. Then:*

a) If G has connected fibers, then u is an immersion if and only if, for every S-scheme S' that is the spectrum of a complete discrete valuation ring with algebraically closed residue field, and every Cartan subgroup C of $G_{S'}$, the restriction of $u_{S'}$ to C is an immersion.

b) The morphism u is an immersion (respectively a closed immersion) if and only if, for every S' as above and every Cartan subgroup C of the identity component $(G_{S'})^0$ of $G_{S'}$, the restriction of $u_{S'}$ to the normalizer N of C in $G_{S'}$ (Exposé XI, 6.11) is an immersion (respectively a closed immersion).

Before proving 1.3, let us state some applications.

Corollary 1.4. — *Let S be a prescheme, let G be an S-group prescheme, smooth and of finite presentation over S, of unipotent rank zero (Exposé XV, 6.1 ter), possessing maximal tori locally for the fpqc topology, let H be an S-group prescheme, and let $u : G \to H$ be an S-group monomorphism. Suppose in addition either that H is of finite presentation over S, or that S is locally noetherian and H is locally of finite type. Then:*

a) If G has connected fibers, u is an immersion.

b) If H is separated over S, and if for every S' over S and every maximal torus T of $G_{S'}$, the Weyl group

$$W = \mathrm{Norm}_{G_{S'}}(T)/\,\mathrm{Centr}_{G^0_{S'}}(T)$$

is representable (a condition always satisfied when G has connected fibers; Exposé XV, 7.1 (iv)) and finite over S, then u is a closed immersion.

Corollary 1.5. — *Let S be a prescheme, let G be an S-group prescheme, smooth and of finite presentation over S, with connected fibers, let H be an S-group prescheme, and let $u : G \to H$ be an S-group monomorphism. Suppose either that H is of finite presentation over S, or that S is locally noetherian and H is locally of finite type. Then:*

a) If G is reductive, that is, affine over S with reductive fibers (cf. Exposé XIX), u is an immersion, and a closed immersion if H is separated.

[1] Editor's note: S. Koizumi and G. Shimura, "Specialization of abelian varieties," *Sci. Papers Coll. Gen. Ed. Univ. Tokyo* 9 (1959), 187–211.

b) *If G has affine, solvable, connected fibers of unipotent rank zero, u is an immersion, and a closed immersion if H is separated.*

Proof of 1.4 from 1.3.

Reduction to the noetherian case. If S is locally noetherian, the reduction is immediate, since the properties to be proved are local on S. In the second case G and H are of finite presentation over S. After shrinking S, we may suppose that S is affine. By Exposé $\mathrm{VI_B}$, §10, there exist a noetherian scheme S_0, an S_0-group prescheme G_0, smooth over S_0 (with connected fibers in case a)) and of finite type over S_0, an S_0-group prescheme H_0 of finite type (separated in case b)), and an S_0-group morphism

$$u_0 : G_0 \longrightarrow H_0$$

such that G, H, u are obtained from G_0, H_0, u_0 by the base extension $S \to S_0$. The assertion that u is a monomorphism means that $\operatorname{Ker} u$ is the identity group; we may therefore suppose that u_0 is a monomorphism.

Since G has unipotent rank $\rho_u = 0$ and possesses maximal tori locally for the fpqc topology, the abelian rank ρ_{ab} of the fibers of G is locally constant (Exposé XV, 8.18). But ρ_u and ρ_{ab} are locally constructible functions (Exposé XV, 6.3 bis). A standard argument (cf. EGA IV, 8.3.4) shows that one may choose S_0 and G_0 so that the unipotent rank (respectively the abelian rank) of the fibers of G_0 is zero (respectively locally constant). Then G_0 possesses maximal tori locally for the fpqc topology (Exposé XV, 8.18), and the functor $\mathcal{TM}_{G_0}$ of maximal tori of G_0 is representable by an S_0-scheme X_0 of finite type over S_0 (Exposé XV, 8.15). Let T_0 be the "universal" maximal torus of $(G_0)_{X_0}$, let $X = X_0 \times_{S_0} S$, and let $T = T_0 \times_{S_0} S$, the universal maximal torus for G. By hypothesis, in case b), the Weyl group relative to T is representable and finite over X. These two properties are compatible with projective limits of preschemes (Exposé $\mathrm{VI_B}$, 10.1 iii, and EGA IV, 8.10.5). We may therefore choose S_0 so that the Weyl group of T_0 is finite over X_0. Under these conditions, to prove 1.4 we may replace S, G, u, H by S_0, G_0, u_0, H_0, and hence suppose that S is noetherian.

Use the valuative criterion furnished by 1.3. We are reduced to the case where S is the spectrum of a discrete valuation ring and where G^0 possesses a Cartan subgroup C.

Proof of 1.4 a). We must show that the restriction of u to C is an immersion (1.3 a)), which reduces us to the case where $G = C$ has connected, nilpotent fibers. Standard reductions (cf. the proof of Exposé VIII, 7.1) reduce us to the case where H is flat over S, u is an isomorphism on the generic fiber, and u is an isomorphism of the underlying spaces on the closed fiber. The group H then has connected fibers and is therefore separated over S (Exposé $\mathrm{VI_B}$, 5.2). Grant for a moment the following lemma.

Lemma 1.6. — *Let S be a prescheme and let G be an S-group prescheme, smooth over S, with connected, nilpotent fibers and unipotent rank zero. Then:*

i) *G is commutative.*

ii) *For every integer $n > 0$, ${}_nG$ is an S-group prescheme, flat and quasi-finite over S, and it is finite over S if and only if the abelian rank, or the reductive rank, of G is locally constant on S.*

iii) *For every integer $q > 0$ invertible on S, the family of subgroups ${}_{q^n}G$ is universally schematically dense in G relative to S (EGA IV, 11.10.8).*

Lemma 1.6 applies to G, and since G possesses maximal tori locally, the reductive rank of the fibers of G is locally constant. Thus, by 1.6 ii), ${}_nG$ is finite over S for every integer $n > 0$. It follows from Exposé VIII, 7.9 that u is an immersion.

Proof of 1.6. Let us first examine the case where S is the spectrum of a field.

Lemma 1.7. — *Let k be a field of characteristic p, and let G be a smooth, connected, nilpotent algebraic k-group of unipotent rank zero. Then G is a commutative group, an extension of an abelian variety A by a torus T. For every integer $n > 0$, ${}_nG$ is a finite group, étale if $(n, p) = 1$, defined by a k-algebra of rank $n^{\rho_r + 2\rho_{ab}}$. If $(q, p) = 1$, the family of subgroups ${}_{q^n}G$, $n \in \mathbb{N}$, is schematically dense in G.*

Proof of 1.7. Let Z be the center of G. The group G/Z is affine (Exposé XII, 6.1), smooth, connected, and of unipotent rank zero; hence it is a torus (BIBLE 4, th. 4). It follows that G is commutative (Exposé XII, 6.4). If T is the unique maximal torus of G (Exposé XV, 3.4), Chevalley's theorem (Séminaire Bourbaki 1956/57, no. 145) immediately shows that G is an extension of an abelian variety A by T. For every integer $n > 0$, raising to the n-th power is an epimorphism on T, whence the exact sequence

$$0 \longrightarrow {}_nT \longrightarrow {}_nG \longrightarrow {}_nA \longrightarrow 0.$$

A classical theorem of Weil (A. Weil, *Variétés abéliennes et courbes algébriques*, §IX, th. 33, cor. 1) says that ${}_nA$ is a finite group defined by a k-algebra of rank $n^{2\rho_{ab}}$. Since ${}_nT$ is a finite group of rank n^{ρ_r}, the asserted structure of ${}_nG$ follows.

Now let H be the smallest closed subscheme of G containing ${}_{q^n}G$ for every n. It follows from the preceding discussion and Exposé XV, 4.6 that H is a smooth, connected subgroup, hence of the same type as G. Raising to the q-th power on H is an epimorphism (since ${}_qH$ is finite), so there is an exact sequence

$$0 \longrightarrow {}_qH \longrightarrow {}_qG \longrightarrow {}_q(G/H) \longrightarrow 0.$$

It follows that ${}_q(G/H) = 0$, and therefore $G/H = 0$. Thus the subgroups ${}_{q^n}G$ are schematically dense in G.

Continuation of the proof of 1.6 i). To show that G is commutative, the usual procedure reduces us first to the case where S is noetherian, and then to the case where S is the spectrum of a local ring with closed point s. The center of G is represented by a closed group subscheme Z of G (Exposé XI, 6.11). To prove that $Z = G$, it suffices to prove this after reduction modulo every power of the maximal ideal of the ring of S: Z will then be an open subgroup of G, and hence equal to G, since G has connected fibers. This reduces us to the case where S is local Artinian.

Let q be an integer invertible on S. For every integer n, ${}_{q^n}G_s$ is an étale, central subgroup of multiplicative type of G_s (1.7). Since G is smooth, ${}_{q^n}(G_s)$ lifts to an étale, central S-group subscheme M_n of G (Exposé XV, 1.2). The family of flat subgroups M_n, $n \in \mathbb{N}$, is then schematically dense in G (1.7 and EGA IV, 11.10.9). Since Z is closed in G and contains every M_n, we have $Z = G$.

Proof of 1.6 ii). To see that ${}_nG$ is flat and quasi-finite over S, it suffices to show that the n-th power map on G is flat and quasi-finite. Since G is flat over S, we are reduced to the case where S is the spectrum of a field (EGA IV, 11.3.10), and we observed in the proof of 1.7 that the n-th power map is an epimorphism. Moreover, ${}_nG$ is separated over S, since G is separated over S (Exposé $\mathrm{VI_B}$, 5.2). For ${}_nG$ to be finite over S, it is necessary and sufficient that the fibers of ${}_nG$ be spectra of finite algebras of locally constant rank. This is seen immediately by reducing to the case where S is the spectrum of a henselian local ring.

By 1.7, this condition is equivalent to saying that the abelian rank, or the reductive rank, of the fibers of G is locally constant.

Proof of 1.6 iii). To see that the family of subgroups ${}_{q^n}G$, $n \in \mathbb{N}$, is universally schematically dense in G, we again reduce to the noetherian case. In view of 1.7 and 1.6 ii), it suffices to apply EGA IV, 11.10.9.

Proof of 1.4 b). We have reduced to the case where S is the spectrum of a discrete valuation ring and where G^0 possesses a Cartan subgroup C. Let

$$N = \mathrm{Norm}_G(C) = \mathrm{Norm}_G(T),$$

where T is the unique maximal torus of C (Exposé XII, 7.1 a) and b)). Since H is separated over S, it follows from 1.6 ii) and Exposé VIII, 7.12 that the restriction of u to C is a closed immersion. On the other hand, to prove that u is a closed immersion, it suffices to prove this for $u|_N$ (1.3 b)). Since, by hypothesis, $W = N/C$ is represented by a finite group scheme, the assertion follows from the next lemma, applied to the exact sequence

$$0 \longrightarrow C \longrightarrow N \longrightarrow W \longrightarrow 1.$$

(Recall that a proper immersion is a closed immersion.)

Lemma 1.8. — *Let S be a prescheme and let G be an S-group prescheme of finite presentation over S, an extension of an S-group prescheme G'', proper and of finite presentation over S, by an S-group prescheme G', flat and of finite presentation over S:*

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1.$$

Let H be an S-group prescheme of finite presentation over S (or locally of finite presentation if S is locally noetherian), and let $u : G \to H$ be an S-group morphism. If the restriction of u to G' is proper, then u is proper.

Proof of 1.8. As usual, we reduce to the case where S is noetherian (Exposé $\mathrm{VI_B}$, §10, and Exposé XV, 6.2), and we may then apply the valuative criterion of properness (EGA II, 7.3.8). We therefore suppose that S is the spectrum of a discrete valuation ring A, with closed point s and generic point t. Let $\bar{x} \in G(t)$ and $h \in H(S)$ satisfy

$$u_t(\bar{x}) = h(t).$$

We must show that $\bar{x}$ comes from a unique element $x \in G(S)$. It is enough to prove the existence and uniqueness of x after a faithfully flat extension of the discrete valuation ring A. Let $\bar{y}$ be the image of $\bar{x}$ in $G''(t)$. Since G'' is proper over S, $\bar{y}$ comes from a unique element $y \in G''(S)$. Let X be the inverse image of y in G. The prescheme X is faithfully flat over S, since G' is faithfully flat over S, as is the morphism $G \to G''$ (Exposé $\mathrm{VI_B}$, §9). After a faithfully flat change of discrete valuation ring, we may suppose that X has a section ℓ over S (EGA IV, 14.5.8). Replacing $\bar{x}$ by $\bar{x}\ell_t^{-1}$, we may suppose that $\bar{x} \in G'(t)$. The existence and uniqueness of x now follow from the properness of $u|_{G'}$.

Proof of 1.5.

a) We shall see in Exposé XIX that, if G is reductive, then G possesses maximal tori locally for the faithfully flat topology, has a finite Weyl group, and has unipotent rank zero. Assertion a) therefore follows from 1.4.

b) If G has affine fibers and unipotent rank zero, then G possesses maximal tori locally for the fpqc topology (Exposé XV, 8.18). It remains only to apply 1.4, observing that, since G

has connected, smooth, affine, solvable fibers, its Weyl group is the identity group (BIBLE 6, th. 1, cor. 3).

Proof of 1.3 a). Since u is already a monomorphism, to prove that u is an immersion it suffices to prove that u is proper at the points of $u(G)$ (EGA IV, 15.7.1); for this we may use the valuative criterion of local properness (EGA IV, 15.7.5). We are therefore reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, closed point s, and generic point t. Since G is flat over S, standard reductions (cf. the proof of Exposé VIII, 7.1) reduce us to the case where H is flat over S, u_t is an isomorphism, and u_s is an isomorphism of underlying spaces. To say that u is an immersion is then equivalent to saying that u is an isomorphism.

By hypothesis, G possesses Cartan subgroups locally for the fpqc topology, so we may speak of the open subset U of regular points of G (Exposé XV, 7.3 i) and ii)).

Lemma 1.9. — *Under the preceding hypotheses, for u to be an immersion it suffices that $u|_U$ be an immersion.*

Indeed, to say that $u|_U$ is an immersion means that there exist an open subscheme V of H and a closed subscheme F of V such that $u|_U$ factors through F and induces an isomorphism from U onto F. Since H_t, and therefore V_t, is irreducible, one necessarily has $V_t = F_t$. Then F contains the schematic closure of V_t in V, which equals V because H is flat over S. Thus $V = F$. It follows that $u|_U$ is an open immersion, and hence that u is an open immersion (Exposé $\mathrm{VI_B}$, 2.6).

It remains to prove that $u|_U$ is an immersion. For this we apply the valuative criterion of local properness. After replacing S by the spectrum of a discrete valuation ring faithfully flat over S, we must show the following: if h is a section of H over S whose image $h(S)$ is contained in $u(U)$, then h is the image of a section g of U over S. It suffices to show that h is contained in the image of a Cartan subgroup C of G. Indeed, by hypothesis $u|_C$ is an immersion, so h is the image of a section g of C, necessarily a section of U.

Let $a \in U(s)$ and $b \in U(t)$ satisfy, respectively,

$$u_s(a) = h(s), \qquad u_t(b) = h(t).$$

Since u_t is an isomorphism, $h(t)$ is a regular point of H_t, and is therefore contained in a unique Cartan subgroup D_t of H_t (Exposé XIII, 3.2). Let D be the schematic closure of D_t in H.

Lemma 1.10. —

i) D_s is a nilpotent algebraic group (Exposé VI_B, §8).

ii) $\dim D_s = \nu$, where ν is the nilpotent rank of G, and also the nilpotent rank of $(H_s)_{\mathrm{red}}$.

iii) $h(s) \in D_s(\kappa(s))$.

i) Since H is separated (Exposé $\mathrm{VI_B}$, 5.2), so is D. Moreover, D is flat over S and its generic fiber is nilpotent. It follows from Exposé $\mathrm{VI_B}$, 8.4 that D_s is nilpotent.

ii) Since $(H_s)_{\mathrm{red}} \simeq G_s$, these two groups have the same nilpotent rank. Since G possesses Cartan subgroups locally for the fpqc topology, G_s and G_t have the same nilpotent rank ν. Finally, since D is flat over S,

$$\dim D_s = \dim D_t = \nu$$

(Exposé $\mathrm{VI_B}$, §4).

iii) Since h_t factors through D_t, the section h plainly factors through D, which proves iii).

The next lemma now shows that $(D_s)_{\mathrm{red}}$ is a Cartan subgroup of

$$(H_s)_{\mathrm{red}} \simeq G_s.$$

Lemma 1.11. — *Let G be a smooth, connected algebraic group over an algebraically closed field k, and let D be a smooth, nilpotent algebraic subgroup containing a regular element $a \in G(k)$. Then D^0 is contained in a Cartan subgroup of G. If, in addition,* $\dim D$ *equals the nilpotent rank of G, then D is a Cartan subgroup of G, and in particular is connected.*

Let Z be the center of G, let $G' = G/Z$, which is affine (Exposé XII, 6.1), and let a' and D' be the images of a and D in G'. The correspondence between Cartan subgroups of G and Cartan subgroups of G' (Exposé XII, 6.6 e)) shows at once that a' is a regular element of G' and that it suffices to prove the lemma for the pair (D', G'). Thus we may suppose that G is affine.

Let s be the semisimple component of a. It belongs to $D(k)$ (BIBLE 4, th. 3) and is regular (BIBLE 7, th. 2, cor. 1). By BIBLE 6, th. 2, s centralizes D^0; hence D^0 is contained in the connected centralizer of s, which is a Cartan subgroup of G (BIBLE 7, th. 2). If now $\dim D$ equals the nilpotent rank of G, then D^0 is a Cartan subgroup of G, equal to $\mathrm{Centr}_G(T)$, where T is the unique maximal torus of D^0 (Exposé XII, 6.6). But D is nilpotent, and therefore centralizes T (BIBLE 6, th. 2), so $D = D^0$.

Return now to the group G. The group

$$C_t = u_t^{-1}(D_t)$$

is the unique Cartan subgroup of G_t containing b. Let C be the schematic closure of C_t in G.

By functoriality of schematic closure, $u|_C$ factors through D, and therefore $u_s(C_s) \subset D_s$. Since u_s is an isomorphism from G_s onto $(H_s)_{\mathrm{red}}$, since

$$\dim C_s = \dim C_t = \nu = \dim D_s$$

by 1.10, and since D_s is connected by 1.11, u_s induces an isomorphism from $(C_s)_{\mathrm{red}}$ onto $(D_s)_{\mathrm{red}}$. Thus $(C_s)_{\mathrm{red}}$ is a Cartan subgroup of G_s. The next lemma shows that C itself is a Cartan subgroup of G.

Lemma 1.12. — *Let S be the spectrum of a discrete valuation ring, let G be a smooth S-group prescheme of finite type, and let C be a group subscheme of G, flat over S, such that the generic fiber C_t is a Cartan subgroup of G_t and the geometric reduced closed fiber $(C_{\bar{s}})_{\mathrm{red}}$ is a Cartan subgroup of $G_{\bar{s}}$. Then C is a Cartan subgroup of G.*

We must prove that C is smooth over S, and it suffices to establish this after a faithfully flat extension of the base. The hypotheses on C imply that the nilpotent rank of the fibers of G is constant; hence we may speak of the open subset U of regular points of G (Exposé XV, 7.3). Since $(C_{\bar{s}})_{\mathrm{red}}$ is a Cartan subgroup of $G_{\bar{s}}$, $C_{\bar{s}}$ contains a regular point $a_{\bar{s}}$ of $G_{\bar{s}}$. Since C is flat over S, after a faithfully flat extension of the base if necessary (EGA IV, 14.5.8), we may suppose that $a_{\bar{s}}$ is an element of $G(s)$ which lifts to a section $a \in C(S)$. Since U is open and contains $a(s)$, it contains a. Let C' be the unique Cartan subgroup of G containing a (Exposé XV, 7.3). Necessarily $C_t = C'_t$, and consequently C' and C coincide with the identity component of the schematic closure of C_t in G (observe that both C' and C are flat over S).

End of the proof of 1.3 a). By hypothesis, the restriction of u to the Cartan subgroup C of G is an immersion. It is clear that $h(S)$ is contained in $u(C) = D$, and hence h is the image of a section $g \in C(S)$. $\square$

Proof of 1.3 b). We shall again use the valuative criterion of local properness (EGA IV, 15.5) in the case of an immersion (respectively, the valuative criterion of properness (EGA II, 7.3.6) in the case of a closed immersion). This reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. Let s be the closed point and t the generic point of S. Replacing H by the schematic closure in H of $u_t(G_t)$, we may suppose that H is flat over S and that u_t is an isomorphism.

Let G^0 be the connected component of G. By 1.2 a), the restriction $u|_{G^0}$ is an immersion; denote by H^0 the group subprescheme of H that is the image of G^0. Thus

$$H^0_s = (H_s)^0_{\mathrm{red}}.$$

To verify the valuative criterion, we must show that every section h of H over S such that $h(S)$ is contained in $u(G)$, in the case of an immersion (respectively, every section h of H over S, in the case of a closed immersion), is the image of a section g of G over S.

Let C be a Cartan subgroup of G, and let $D = u(C)$ be the Cartan subgroup of H^0 that is the image of C. The S-group

$$D' = \mathrm{int}(h)D$$

has connected fibers, so its underlying space is contained in that of H^0; on the other hand, since it is smooth over S, it is reduced, and hence D' is a smooth group subprescheme of H^0. The fibers of D' are Cartan subgroups of the fibers of H^0; therefore D' is a Cartan subgroup of H^0, and

$$C' = u^{-1}(D')$$

is a Cartan subgroup of G. But $\mathrm{Transpstr}_G(C, C')$ is smooth and surjective over S (Exposé XII, 7.1 b)). Since S is henselian with algebraically closed residue field, there exists $g \in G(S)$ such that $\mathrm{int}(g)C = C'$. Replacing h by $u(g)^{-1}h$, we may suppose that $h(t)$ normalizes D_t (and that $h(s)$ is the image of an element of the normalizer of C_s in G_s, in the case of an immersion). Let

$$N = \mathrm{Norm}_G(C).$$

By hypothesis, $u|_N$ is an immersion (respectively, a closed immersion); hence h is indeed the image of a section of N over S, which completes the proof. ☐

2. A Representability Theorem for Quotients

Let us “recall” the following result.

Theorem 2.1. — *Let S be a prescheme, let X and Y be two S-preschemes, and let $f : X \to Y$ be an S-morphism. Suppose that one is in one of the following two cases:*

a) The morphism f is locally of finite presentation.

b) The prescheme S is locally noetherian and X is locally of finite type over S.

Then the following conditions are equivalent:

i) There exist an S-prescheme X' and a factorization of f

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism, locally of finite presentation, and f'' is a monomorphism.

ii) The first projection

$$p_1 : X \times_Y X \longrightarrow X$$

is a flat morphism.

Moreover, if the preceding conditions hold, then (X', f') is a quotient of X, for the fpqc topology, by the equivalence relation defined by f. Consequently the factorization $f = f'' \circ f'$ in i) is unique up to isomorphism.

The proof of this delicate theorem will be given in EGA V. One may also consult the lecture by J.-P. Murre, *Séminaire Bourbaki*, May 1965, no. 294, theorem 2, corollary 2, which treats the case where Y is locally noetherian and X is of finite type over Y. We shall see that one can reduce to that case.

Let us first make a few remarks.

a) The implication i) $\Rightarrow$ ii) is trivial. Indeed, the first projection

$$p_1' : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p_1' : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism because f'' is a monomorphism, and p_1' is flat because f' is flat; therefore p_1 is flat.

b) The assertions of 2.1 are local on Y (hence local on S); they are also local on X, as follows readily from the fact that a flat morphism locally of finite presentation is open (EGA IV, 11.3.1).

c) Under the hypotheses of 2.1 a), the foregoing reduces us to the case where X and Y are affine and f is of finite presentation. Replacing S by Y, we may suppose that X and Y are of finite presentation over S. We then reduce to the noetherian case for S by EGA IV, 11.2.6.

d) Under the hypotheses of 2.1 b), we may suppose that S, X, and Y are affine, that S is noetherian, and that X is of finite type over S. Regard Y as a filtered projective limit of affine schemes Y_i of finite type over S. The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has

$$X \times_{Y_i} X = X \times_Y X$$

for all sufficiently large i. Thus

$$f_i : X \xrightarrow{f} Y \longrightarrow Y_i$$

satisfies the hypotheses of 2.1 ii) if f does. Since the equivalence relation on X defined by f coincides with the one defined by f_i, it is clear that it suffices to prove ii) $\Rightarrow$ i) for f_i. This reduces us to the case where Y is of finite type over S.

Application to Group Preschemes

Theorem 2.2. — *Let S be a prescheme and let G be an S-group prescheme locally of finite presentation over S, acting on an S-prescheme X. Let $\xi \in X(S)$ be a section of X over S, such that the stabilizer H of ξ in G is an S-group subprescheme of G, flat over S. If X is locally of finite type over S, or if S is locally noetherian, then the homogeneous quotient* *space G/H is representable by an S-prescheme locally of finite presentation over S, and the S-morphism*

$$f : G \longrightarrow X, \qquad g \longmapsto g \cdot \xi$$

factors as

$$\begin{tikzcd} G \arrow[d, "p"'] \arrow[dr, "f"] & \\ G/H \arrow[r, "i"] & X. \end{tikzcd}$$

where p is the canonical projection, a faithfully flat morphism locally of finite presentation, and i is a monomorphism. (For definiteness, we have supposed that G acts on the left on X.)

Proof. The morphism f makes G into an X-prescheme. By the definition of the stabilizer of ξ, the morphism

$$G \times_S H \longrightarrow G \times_X G, \qquad (g, h) \longmapsto (g, gh)$$

is an isomorphism. Since H is flat over S, $G \times_S H$ is flat over G; hence the first projection

$$p_1 : G \times_X G \longrightarrow G$$

is flat. Moreover, if X is locally of finite type over S, then f is locally of finite presentation (EGA IV, 1.4.3 v)); otherwise S is locally noetherian by hypothesis. It therefore suffices to apply 2.1 to f.

It remains only to show that G/H is locally of finite presentation over S, and this follows immediately from Exposé V, 9.1. □

Corollary 2.3. — *Let S be a prescheme, let G and H be two S-group preschemes, and let $u : G \to H$ be an S-homomorphism. Suppose that G is locally of finite presentation over S, and that either H is locally of finite type over S, or S is locally noetherian. If $K = \mathrm{Ker}(u)$ is flat over S, then the quotient group G/K is representable by an S-group prescheme locally of finite presentation over S, and u factors as*

$$\begin{tikzcd} G \arrow[rr, "u"] \arrow[dr, "p"'] & & H \\ & G/K \arrow[ur, "i"'] & \end{tikzcd}$$

where p is the canonical projection and i is a monomorphism.

Proof. Apply 2.2 with $X = H$ and with ξ the unit section of H. □

Corollary 2.4. — *Let S be a prescheme, let G be an S-group prescheme of finite presentation over S, let H be an S-group prescheme smooth over S with connected fibers (and hence of finite presentation over S, by Exposé VI_B, 5.5), and let $i : H \to G$ be a monomorphism of S-groups, so that H is a subgroup of G. Suppose that*

$$N = \mathrm{Norm}_G(H),$$

which is representable by a closed group subprescheme of G, of finite presentation over S (Exposé XI, 6.11), is flat over S. Then G/N is representable by an S-prescheme of finite presentation over S and quasi-projective over S.

All the assertions to be proved, except the last one, are local on S. To establish them, we may therefore suppose that S is quasi-compact and that the relative dimension of H over S is constant, equal to r. Proceed as in Exposé XV, §5. For every integer $n > 0$, let $G^{(n)}$ (respectively $H^{(n)}$) be the n-th normal invariant of the unit section of G (respectively of H) (EGA IV, 16). The sheaf of $\mathscr{O}_S$-modules $H^{(n)}$ is a quotient of $G^{(n)}$. Since H is smooth over S of dimension r, $H^{(n)}$ canonically defines an element ξ_n of

$$X_n = \mathrm{Grass}_{\varphi(n,r)} G^{(n)}$$

(EGA I, second edition, §9), for a suitable integer $\varphi(n, r)$. On the other hand, G acts naturally on $G^{(n)}$, and hence also on X_n, through the representation

$$G \longrightarrow \mathrm{Aut}_{S\text{-gr}}(G), \qquad g \longmapsto \mathrm{int}(g).$$

Since S is quasi-compact, there exists an integer m such that, for $n \geq m$,

$$N = \mathrm{Norm}_G(H) = \mathrm{Norm}_G H^{(n)} \qquad \text{(Exposé XI, 6.11 b))}.$$

In other words, N is the stabilizer of ξ_n for sufficiently large n. The representability of G/N therefore follows from 2.2. The fact that G/N is of finite presentation over S is a consequence of Exposé V, 9.1. It remains to prove that G/N is quasi-projective over S. For this, we shall exhibit a canonical invertible sheaf on G/N that is ample relative to S. Consider the functor

$$\mathscr{F} : (\mathbf{Sch}/S)^{\circ} \longrightarrow \mathbf{Ens}$$

such that, for every S-prescheme T,

$$\mathscr{F}(T) = \left\{ \begin{array}{c} T\text{-subgroups } H \text{ of } G_T\text{, representable and smooth over } T, \\ \text{with connected fibers} \end{array} \right\}.$$

Such a group H is of finite presentation over S (Exposé $\mathrm{VI_B}$, 5.5), and $H \to G_T$ is a quasi-affine morphism (EGA IV, 8.11.2). By effective descent for quasi-affine morphisms (SGA 1, VIII, 7.9), it follows that $\mathscr{F}$ is a sheaf for the fpqc topology. Since G/N is the sheaf associated with a subfunctor of $\mathscr{F}$, there is a canonical monomorphism

$$G/N \longrightarrow \mathscr{F}.$$

Consequently there exists a subgroup H' of $G \times_S (G/N)$, representable and smooth over G/N, with connected fibers, that is "universal" for the functor G/N. I claim that the invertible sheaf

$$\mathscr{L} = \big(\det(\mathrm{Lie}\, H')\big)^{-1}$$

is ample relative to S. In this form the assertion is local on S, and its proof is analogous to the one given in Exposé XV, 5.8.

The following corollary was announced in Exposé XIV, 4.8 bis.

Corollary 2.5. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let P be a parabolic subgroup of G (Exposé XIV, 4.8 bis). Then G/P is representable by an S-prescheme smooth and projective over S.*

It remains only to show (*loc. cit.*) that G/P is representable by an S-prescheme that is quasi-projective and of finite presentation. This follows from 2.4, since

$$P = \mathrm{Norm}_G(P)$$

(*loc. cit.*) and is therefore flat over S. □

3. Groups with Flat Center

Proposition 3.1. — *Let S be a prescheme and let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers. Suppose that the center Z of G, which is representable by Exposé XI, 6.11, is flat over S. For every integer $n > 0$, let $G^{(n)}$ denote the locally free $\mathcal{O}_S$-module equal to the n-th normal invariant of G along the unit section, and let ρ_n be the natural "adjoint representation" of G in $\mathrm{GL}_S(G^{(n)})$. Then:*

a) The quotient $G' = G/Z$ is representable by an S-group prescheme that is smooth, of finite presentation over S, quasi-affine, and has affine connected fibers.

b) The representation ρ_n factors through G':

$$\begin{tikzcd} G \arrow[rr, "\rho_n"] \arrow[dr] & & \mathrm{GL}_S\Big(G^{(n)}\Big) \\ & G/Z \arrow[ur, "i_n"'] & \end{tikzcd}$$

If S is quasi-compact, then i_n is a monomorphism for all sufficiently large n.

c) The functor $\mathscr{T}_{G'}$ of subtori of G' is representable by an S-prescheme smooth over S, and, if S is quasi-compact, this prescheme is a sum of a family of preschemes affine over S.

Proof. The assertions in a) are local on S, which allows us to suppose that S is quasi-compact. By Exposé XI, 6.11 b), one has

$$Z = \operatorname{Ker} \rho_n$$

for all sufficiently large n; this kernel is therefore flat over S. The representability of G/Z and the fact that i_n is a monomorphism now follow from 2.3. Since G is smooth and of finite presentation over S, the same is true of G' (Exposé $\mathrm{VI_B}$, §9). The group $\mathrm{GL}_S(G^{(n)})$ is affine over S, and a monomorphism of finite presentation is quasi-affine (EGA IV, 8.11.2). Hence G' is quasi-affine over S and has affine fibers. Since G' is smooth over S, the functor $\mathscr{T}_{G'}$ is formally smooth (Exposé XI, 2.1 bis). Taking Exposé XI, 4.6 and 4.3 into account, the assertions of 3.1 c) will therefore follow from the next lemma.

Lemma 3.2. — *Let S be a prescheme, let G and H be two S-group preschemes of finite presentation over S, let $u : G \to H$ be an S-group monomorphism, and let $\mathscr{T}_G$ (respectively $\mathscr{T}_H$) be the functor of subtori of G (respectively of H) (cf. Exposé XV, §8). Then the map $T \mapsto u(T)$ defines a monomorphism (Exposé XV, 8.3 c))*

$$\widetilde{u} : \mathscr{T}_G \longrightarrow \mathscr{T}_H$$

that is representable by a closed immersion of finite presentation.

By the usual procedure, we reduce to the following problem. Let T be a subtorus of H, let

$$T' = G \times_H T$$

be its inverse image in G, and let $u_T : T' \to T$ be the S-group monomorphism induced by u. One must show that the S-functor

$$\prod_{T/S}(T'/T)$$

is representable by a closed subprescheme of S, of finite presentation. But T, being a torus, is smooth over S with connected fibers, and it suffices to apply Exposé XI, 6.10. □

Theorem 3.3. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let Z be the center of G. Suppose that Z is flat over S and that the unipotent rank of G (Exposé XV, 6.1 ter) equals that of Z. Then:*

a) The group $G' = G/Z$ is representable; if S is quasi-compact, the canonical morphism

$$i_n : G' \longrightarrow \mathrm{GL}_S(G^{(n)})$$

(cf. 3.1 b)) is an immersion for all sufficiently large n.

b) The group G' is quasi-affine over S, with affine fibers; its center is the unit group, and G' has unipotent rank zero.

c) Locally for the étale topology, G' possesses maximal tori, and these are also Cartan subgroups of G'. The functor $\mathscr{T}\mathscr{M}_{G'}$ of maximal tori of G' (Exposé XV, §8) is representable by an S-prescheme smooth and affine over S.

d) Locally for the étale topology, G possesses Cartan subgroups, and the functor $\mathscr{C}_G$ of Cartan subgroups of G is representable by an S-prescheme smooth and affine over S.

Proof. By 3.1, the group G' is represented by an S-prescheme that is smooth and quasi-affine over S, with affine fibers. In view of the correspondence between Cartan subgroups of the fibers of G and those of the fibers of G' (Exposé XII, 6.6 e)), the hypothesis on the unipotent rank of Z implies that G' has unipotent rank zero. Exposé XV, 8.18 now shows that, locally for the étale topology, G' possesses maximal tori. The fact that i_n is an immersion for all sufficiently large n then follows from 3.1 b) and 1.4 a). This proves a).

We now prove c). Since G' has unipotent rank zero, every maximal torus of G' is plainly also a Cartan subgroup of G'. The functor of maximal tori of G' is represented by a subprescheme that is both open and closed in the functor of subtori of G'; moreover it is of finite type over S (Exposé XV, 8.15). It follows from 3.1 c) that this functor is represented by an S-prescheme smooth and affine over S.

Since, locally for the étale topology, G' possesses Cartan subgroups, the same is true of G; and the functor of Cartan subgroups of G is canonically isomorphic to that of G' (Exposé XV, 7.3 iv)). Thus c) implies d).

It remains to prove b), and more precisely to prove that the center Z' of G' is the unit group. Since Z' is representable (Exposé XI, 6.11) and of finite presentation over S, it suffices to show that the fibers of Z' are reduced to the unit group (Exposé $\mathrm{VI_B}$, 2.9). We are thereby reduced to the case where S is the spectrum of an algebraically closed field. The center Z' is evidently contained in every Cartan subgroup of G', hence, by c), in every maximal torus of G'; it is therefore of multiplicative type. On the other hand, we shall see in Exposé XVII that, if G is a connected algebraic group with center Z, the center Z' of G/Z is unipotent. In the present case, Z', being at once of multiplicative type and unipotent, is reduced to the unit group (cf. Exposé XVII). □

Examples of Groups Whose Center Is Flat

Proposition 3.4. — *Let S be a prescheme, let G be an S-group prescheme of finite presentation over S, smooth, with connected fibers, and let Z be the center of G. Then Z is flat over S in either of the following two cases:*

a) The unipotent rank ρ_u of the fibers of G (Exposé XV, 6.1 ter) is zero.

b) i) S is reduced; ii) the dimension of the fibers of Z is a locally constant function on S; iii) the unipotent rank of G equals that of Z.

Proof of 3.4 a). We shall prove at the same time the following proposition.

Proposition 3.5. — *Under the hypotheses of 3.4 a), suppose moreover that G possesses a maximal torus T, and let*

$$C = \mathrm{Centr}_G(T)$$

be the Cartan subgroup of G associated with T (Exposé XII, 7.1). Then:

(i) $Z \cap T$ is a subgroup of multiplicative type of G.

(ii) C is commutative and equals $T \cdot Z$.

(iii) If the quotients $Z/(Z \cap T)$ and C/T are representable, then they are represented by abelian preschemes (that is, S-group preschemes smooth over S, whose fibers are abelian varieties), and the canonical monomorphism

$$Z/(Z \cap T) \longrightarrow C/T$$

is an isomorphism.

Using the general properties of passage to the limit established in Exposé $\mathrm{VI_B}$, §10 and Exposé XV, 6.2, 6.3, and 6.3 bis, we reduce as usual to the case where S is noetherian (observe that the assertions of 3.4 and 3.5 are local on S). We noted in 3.1 that the hypotheses on G imply that Z is representable. To prove that Z is flat over S, EGA 0_{III}, 10.2.6 reduces us to the case where S is local artinian. Since G is then smooth, it possesses maximal tori locally for the étale topology (Exposé XV, 8.17). After a finite flat extension of S, if necessary (which is permissible when proving that Z is flat), we may therefore suppose that G possesses a maximal torus T.

Let us show, in the same way, that to establish 3.5 we may reduce to the case where S is artinian.

i) Since Z is closed in G (Exposé XI, 6.11), $Z \cap T$ is a closed group subscheme of T. It follows from Exposé X, 4.8 b), that $Z \cap T$ is of multiplicative type if and only if it is flat over S. As above, it suffices to establish that $Z \cap T$ is flat when S is artinian.

ii) Since G has unipotent rank zero, every Cartan subgroup C of G satisfies the hypotheses of Lemma 1.6, and is therefore commutative. The equality $C = T \cdot Z$ will follow from iii).

iii) If C/T is representable, then C/T is smooth over S (Exposé $\mathrm{VI_B}$, §9), and its fibers are abelian varieties (1.7); thus C/T is an abelian prescheme. To show that the canonical monomorphism

$$i : Z/(Z \cap T) \longrightarrow C/T$$

is an isomorphism, it suffices to verify this when S is local artinian. Indeed, one then deduces successively that $Z/(Z \cap T)$ is flat over S (EGA 0_{III}, 10.2.6), that i is flat (EGA IV, 11.3.10), that i is an open immersion (EGA IV, 17.9.1), and finally that i is an isomorphism (because C/T has connected fibers).

We henceforth suppose that S is local artinian, with closed point s, and that G possesses a maximal torus T. Put

$$C = \mathrm{Centr}_G(T).$$

The group C is a Cartan subgroup of G (Exposé XII, 7.1) and contains Z.

The algebraic group $M_s = Z_s \cap T_s$ is a central subgroup of multiplicative type of G_s. Since T is smooth over S, M_s lifts to a group subscheme M of T, of multiplicative type (Exposé IX, 3.6 bis), and contained in the center of G (Exposé IX, 3.9), hence contained in $Z \cap T$. Since S is artinian and M and T are flat over S, the quotient groups

$$(Z \cap T)/M, \qquad Z' = Z/M, \qquad A = C/T$$

are representable (Exposé $\mathrm{VI_A}$, §§4 and 5). By construction, $\big((Z\cap T)/M\big)_s$ is the unit group, so $Z \cap T = M$ (Exposé $\mathrm{VI_B}$, 2.9); in particular it is flat over S, which proves 3.5 i).

Passing to the quotient gives a canonical monomorphism

$$i : Z' \longrightarrow A,$$

which is therefore a closed immersion (Exposé $\mathrm{VI_B}$, 1.4.2). We have already observed that A is an abelian prescheme. It remains to show that i is an isomorphism. This will prove 3.5 iii) and will imply that Z' is flat over S, and hence that Z is flat over S (as an extension of Z' by the flat group M; Exposé $\mathrm{VI_B}$, §9). Since Z_s has the same abelian rank as G_s (Exposé XII, 6.1), i_s is an epimorphism, hence an isomorphism. Let $q > 0$ be an integer invertible on S. By the density theorem of 1.6 iii), to see that $i(Z') = A$, it suffices to show that, for every integer n equal to a power of q, $i(Z')$ contains ${}_nA$. Now let M_s^0 be the identity component of M_s. It follows immediately by duality that raising to the q-th power in M_s^0 is an epimorphism. Consequently, if m_0 is the exponent of q in the prime factorization of $\mathrm{card}(M_s/M_s^0)$, the image of ${}_{nm_0}Z_s$ in

$$A_s \simeq Z_s/M_s$$

contains ${}_nA_s$. There is therefore a subgroup $M_s(n)$ of multiplicative type of Z_s whose image in A_s is ${}_nA_s$. As above, $M_s(n)$ lifts to a central subgroup $M(n)$ of multiplicative type of G. The image of $M(n)$ in A is a subgroup of multiplicative type (Exposé IX, 6.8), and is necessarily equal to ${}_nA$, since it is so on the reduced fiber (Exposé IX, 5.1 bis). This completes the proof of 3.4 a) and of 3.5.

Proof of 3.4 b). The assertion to be proved is local on S, so we may suppose that S is affine, with ring A. Regarding A as the inductive limit of its finitely generated $\mathbb{Z}$-subalgebras, we reduce as above to the case where S is noetherian and reduced.

To prove that Z is flat, we may use the valuative criterion for flatness (EGA IV, 11.8.1), which reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with generic point t and closed point s. Let Z' be the schematic closure of Z_t in Z. We must show that $Z' = Z$, and it even suffices to show that $Z_s = Z'_s$ (Exposé $\mathrm{VI_B}$, 2.6). By Exposé $\mathrm{VI_B}$, §4, one has the inequalities

$$\dim Z_t = \dim Z'_t = \dim Z'_s \leq \dim Z_s.$$

By hypothesis, however, $\dim Z_t = \dim Z_s$; hence $\dim Z'_s = \dim Z_s$, and consequently Z'_s contains $(Z_s)^0_{\mathrm{red}}$. It follows that

$$G'_s = G_s/Z'_s$$

is affine (Exposé XII, 6.1). In view of the correspondence between Cartan subgroups of G and G' (Exposé XII, 6.6 e)), hypothesis 3.4 b) iii) implies that the unipotent rank of G'_s is zero, and consequently its Cartan subgroups are also its maximal tori. The image Z''_s of Z_s in G'_s is a central algebraic subgroup of G'_s, hence is contained in every Cartan subgroup of G'_s, that is, in every maximal torus; Z''_s is therefore a subgroup of multiplicative type of G'_s. Under these conditions, we shall show in Exposé XVII, §7, that there exists a finite subgroup M_s of multiplicative type of Z_s whose image in Z_s/Z'_s is Z''_s (we used this fact in the proof of 3.4 a) when Z''_s is étale). Since G is smooth over S and S is the spectrum of a complete local ring, M_s lifts to an S-subgroup M of multiplicative type of G (Exposé IX, 3.6 bis and Exposé XV, 1.6 b)), which is central (Exposé IX, 5.6 a)). Thus M_t is contained in $Z'_t = Z_t$, and, since M is flat, M is contained in Z'. The group M_s is therefore contained in Z'_s, but this implies that Z''_s is the unit group, that is, $Z_s = Z'_s$. This completes the proof of 3.4 b).

Example of a Smooth Group Prescheme with Connected Fibers Whose Center Is Not Flat

Let S be the spectrum of a discrete valuation ring A, let π be a uniformizer of A, and let t and s be the generic and closed points of S, respectively. Let G be the smooth affine S-group whose ring is

$$B = A[T, T^{-1}, U]/F, \qquad F = 1 - T + \pi U,$$

the composition law being defined by

$$(t, u), (t', u') \longmapsto (tt', \pi uu' + u + u').$$

The generic fiber G_t is therefore isomorphic to the multiplicative group $(\mathbb{G}_m)_t$, while the closed fiber G_s is isomorphic to the additive group $(\mathbb{G}_a)_s$. The function T is invertible in $\Gamma(G, \mathcal{O}_G)$ and defines an S-group morphism

$$\varphi : G \longrightarrow (\mathbb{G}_m)_S$$

which is an isomorphism on the generic fiber and the trivial morphism on the closed fiber. The datum of φ permits one to construct the semidirect-product group

$$H = G \cdot (\mathbb{G}_a)_S$$

(note that $(\mathbb{G}_m)_S$ acts on $(\mathbb{G}_a)_S$). The center Z of H is the unit group on the generic fiber and is equal to H_s on the closed fiber, and is therefore not flat over S.

4. Groups with Affine Fibers, of Unipotent Rank Zero

Theorem 4.1. — *Let S be a prescheme and let G be an S-group prescheme smooth over S, of finite presentation, with affine, connected fibers, of unipotent rank zero (Exposé XV, 6.1 ter). Then:*

a) The center Z of G is a group of multiplicative type (and is therefore a reductive center of G; Exposé XII, 4.1).

b) G satisfies conditions a) through d) of 3.3.

c) *Locally for the étale topology, G possesses maximal tori, and these are also Cartan subgroups of G. The functor $\mathscr{T}\mathscr{M}_G$ of maximal tori of G is representable by an S-prescheme smooth and affine over S.*

d) *G is quasi-affine over S. Moreover, if T is a maximal torus of G, if*

$$N = \mathrm{Norm}_G(T)$$

is its normalizer in G, and if $W = N/T$ is the Weyl group relative to T, the following conditions are equivalent:

- *(i) G is affine over S;*
- *(ii) $G' = G/Z$ is affine over S;*
- *(iii) N is affine over S;*
- *(iv) W is affine over S.*

These conditions are always satisfied if S is locally noetherian of dimension at most 1.

Proof. Since G has affine fibers (and hence abelian rank zero) and unipotent rank zero, G possesses maximal tori locally for the étale topology (Exposé XV, 8.18), and every maximal torus of G is plainly a Cartan subgroup of G. This proves the first part of c). To see that Z is of multiplicative type, we may suppose, by what precedes, that G possesses a maximal torus T. Since T is also a Cartan subgroup, it contains Z (because $T = \mathrm{Centr}_G(T)$ by Exposé XII, 7.1), and

$$Z = Z \cap T$$

is of multiplicative type by 3.5 i). This proves a). Assertion b) is clear in view of a). On the other hand, the functor $\mathscr{T}\mathscr{M}_G$ is isomorphic to the functor of Cartan subgroups of G (Exposé XII, 7.1), and is therefore smooth and affine over S by 3.3 c). This finishes the proof of c). It remains to prove d).

Proof of d). Since G' is quasi-affine (3.3 b)) and Z is of multiplicative type, hence affine over S, G is quasi-affine (Exposé $\mathrm{VI_B}$, §9).

i) $\Longleftrightarrow$ ii). If G is affine, then G' is affine by Exposé IX, 2.3. If G' is affine, then G is affine as an extension of an affine group by an affine group (Exposé $\mathrm{VI_B}$, §9).

i) $\Longleftrightarrow$ iii). If G is affine, then N is affine, since it is closed in G (Exposé XI, 6.11 a)). Moreover, G/N is isomorphic to the functor $\mathscr{T}\mathscr{M}_G$ (conjugacy of maximal tori; cf. Exposé XII, 7.1 b)), and is therefore affine over S by c). Thus if N is affine over S, then G is affine, as an "extension" of an affine homogeneous space by an affine group (Exposé $\mathrm{VI_B}$, §9).

iii) $\Longleftrightarrow$ iv). If N is affine, then $W = N/T$ is affine (Exposé IX, 2.3), and the converse follows from Exposé $\mathrm{VI_B}$, §9.

Moreover, N/T is representable by an S-group prescheme that is étale, separated, and of finite type over S (Exposé XV, 7.1 iv)), hence quasi-finite over S. The last assertion of d) therefore follows from the next lemma, applied with $X = W$ and $Y = S$.

Lemma 4.2. — *Let S be a locally noetherian prescheme of dimension 1, let X and Y be two S-preschemes locally of finite type over S, and let $u : X \to Y$ be a quasi-finite and separated S-morphism. Suppose that, for every point s of S, the morphism*

$$u_s : X_s \longrightarrow Y_s$$

induced from u by the base change $\operatorname{Spec}\kappa(s) \to S$ *is finite. Then u is an affine morphism.*

The assertion to be proved is local on Y, which allows us to suppose that Y (and hence also X) is of finite type over S. By EGA IV, §8, it suffices to prove 4.2 when S is the spectrum of a local ring A. By fpqc descent we may suppose that A is complete, then that A is reduced (EGA II, 1.6.4), and then that A is normal (Nagata's theorem, EGA 0_{IV}, §22, and Chevalley's theorem, EGA II, 6.7.1). If A is a field, then u is finite by hypothesis, hence affine. Otherwise A is a discrete valuation ring; let s and t be the closed and generic points of S, and let π be a uniformizer of A. Let y be a point of Y_s. Applying EGA IV, §8, once more, we may replace Y by the spectrum Y' of $\mathcal{O}_{Y,y}$, and X by

$$X' = X \times_Y Y';$$

we may even suppose that $\mathcal{O}_{Y,y}$ is complete. Since u is quasi-finite and separated, EGA II, 6.2.6, shows that X' is the disjoint union of two schemes X_1 and X_2, where X_1 is finite over Y' (hence affine), while $u(X_2)$ does not contain the closed point y of Y'. I claim that, under these conditions, X_2 does not meet the closed fiber X'_s. Indeed, by hypothesis,

$$X'_s \longrightarrow Y'_s$$

is finite, so the restriction of this morphism to the closed subset $(X_2)_s$ is finite, and its image in $(Y')_s$ is closed. Since this image does not contain the closed point of the local scheme $(Y')_s$, $(X_2)_s$ is empty. Since u_t is finite, the restriction to $(X_2)_t = X_2$ of the morphism

$$X'_t \longrightarrow Y'_t$$

is finite. Moreover $Y'_t = Y'_\pi$, so the open immersion $Y'_t \to Y'$ is affine. In short, the composite morphism

$$X_2 \longrightarrow Y'_t \longrightarrow Y'$$

is affine, and it follows that

$$X' = X_1 \amalg X_2 \longrightarrow Y'$$

is affine.

Corollary 4.3. — *Let S be a prescheme, and let G be an S-group prescheme smooth and of finite presentation over S, with affine, solvable, connected fibers, of unipotent rank zero. Then G is affine over S. If, moreover, the center of G is the unit group and S is quasi-compact, the canonical morphism*

$$i_n : G \longrightarrow \mathrm{GL}_S(G^{(n)}) \qquad \textit{(cf. 3.1 b))}$$

is a closed immersion for all sufficiently large n.

To prove that G is affine over S, we may suppose that G possesses a maximal torus T (4.1 c)). Since G has solvable fibers, the Weyl group relative to T is the unit group (BIBLE, 6, Th. 1, Cor. 3), and condition 4.1 d) iv) is satisfied. If the center of G is the unit group, i_n is a monomorphism for all sufficiently large n (3.1), and is therefore a closed immersion (1.5 b)).

5. Application to Reductive and Semisimple Groups

Definition 5.1. — *An S-group prescheme G is said to be* reductive *(respectively* semisimple*) if G is smooth and affine over S, with reductive (respectively semisimple) fibers.*

5.1.1. Reductive groups will be studied systematically beginning with Exposé XIX. In this section we shall need the following properties, which will be proved in Exposé XIX (without using the developments of the present Exposé).

a) *Let S be a prescheme, let G be an S-group prescheme smooth and affine over S, with connected fibers, and let s be a point of S such that G_s is reductive (respectively semisimple). Then there exists a neighborhood U of s such that $G|_U$ is reductive (respectively semisimple).*

b) *If G is reductive, then locally for the étale topology G possesses maximal tori; and if T is a maximal torus of G, the Weyl group*

$$W = \mathrm{Norm}_G(T)/T$$

is finite over S.

c) *A reductive group has unipotent rank zero.*

Admitting these facts, we propose to strengthen assertion a) above.

Theorem 5.2. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let s be a point of S. Then:*

(i) *If the fibers of G are affine and G_s is reductive, there exists an open neighborhood U of s such that $G|_U$ is reductive.*

(ii) *If G_s is semisimple, there exists an open neighborhood U of s such that $G|_U$ is semisimple.*

Using Exposé XV, 6.2 i), and Exposé $\mathrm{VI_B}$, §10, we reduce to the case where S is noetherian.

In case ii), consider the center Z of G, which is representable (Exposé XI, 6.11 a)). Since G_s is semisimple, it is well known that Z_s is finite. Consequently (Exposé $\mathrm{VI_B}$, §4), there exists a neighborhood U of s such that Z is quasi-finite over U. For every point t of U, G_t is then affine (Exposé XII, 6.1). Shrinking S if necessary, we may therefore suppose that the fibers of G are affine, in case ii) as well as in case i).

In view of a), to prove 5.2 it suffices to prove that G is affine over S. Since G has affine fibers, we know that the functor of subtori of G is representable by a smooth S-prescheme (Exposé XV, 8.11 and 8.9). After making an étale extension covering s, which is permissible, we may therefore suppose that there is a subtorus T of G such that T_s is a maximal torus of G_s. But then

$$C = \mathrm{Centr}_G(T)$$

is a smooth subgroup of G, with connected fibers, containing T, such that $C_s = T_s$ (because C_s is a Cartan subgroup of G_s, and G_s has unipotent rank zero by c)). It follows that $C = T$, so T is a Cartan subgroup and a maximal torus of G; in particular, the unipotent rank of G is zero, and 4.1 applies.

By 4.1 d), it suffices to show that the Weyl group W relative to T is affine over a neighborhood U of s. In fact, we shall show that W is even finite over a neighborhood of s. Since W is quasi-finite over S (Exposé XV, 7.1 iv)), saying that W is finite over a neighborhood U of s is equivalent to saying that the morphism

$$W \longrightarrow S$$

is proper at s (EGA IV, 15.7, and EGA III, 4.4.2). To establish this, the valuative criterion of local properness (EGA IV, 15.7) reduces us to the case where S is the spectrum of a discrete valuation ring, with closed point s. But then, by the last assertion of 4.1 d), G is affine over S, and we may apply property a) recalled above to conclude that G is reductive (respectively semisimple). Applying property b), we conclude that W is indeed finite over S, which completes the proof of 5.2.

6. Applications: Extension of Certain Rigidity Properties of Tori to Groups of Unipotent Rank Zero

Proposition 6.1. — *Let S be a prescheme and let G be an S-group prescheme smooth over S, of finite presentation, with connected affine fibers, and whose center is of multiplicative type (for example, G of unipotent rank zero; cf. 4.1 a)). Then every closed normal subgroup K of G, of finite presentation over S and quasi-finite over S, is finite over S.*

Indeed, for every geometric point $\bar{s}$ above S, $(K_{\bar{s}})_{\mathrm{red}}$ is a finite étale subgroup of $G_{\bar{s}}$, normal and therefore central (since $G_{\bar{s}}$ is connected). Consequently

$$K' = Z \cap K,$$

where Z denotes the center of G, has the same underlying space as K, and it suffices to show that K' is finite over S. Now K' is a closed quasi-finite subgroup of the group of multiplicative type Z, and is therefore finite, as is immediate (locally on S, K' is contained in ${}_nZ$ for a suitable integer n, and ${}_nZ$ is finite over S).

Corollary 6.2. — *Let S and G be as above, let K be an S-group subprescheme of G, of finite presentation over S, normal and closed in G, and let s be a point of S. If K_s is finite (respectively is the unit group), there exists an open neighborhood U of s such that $K|_U$ is finite over S (respectively is the unit group).*

In view of the upper semicontinuity of the dimension of the fibers of K (Exposé $\mathrm{VI_B}$, §4), after shrinking S we may already suppose that K is quasi-finite over S, and hence finite by 6.1. If, moreover, K_s is the unit group, then Nakayama's lemma easily implies that K is the unit group over $\operatorname{Spec} \mathcal{O}_s$, and hence over a neighborhood of s.

Proposition 6.3. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with affine, connected fibers, of unipotent rank zero, let H be an S-group prescheme, let s be a point of S, and let*

$$u : G \longrightarrow H$$

be an S-group morphism such that $\operatorname{Ker} u_s$ is central. Suppose, moreover, that H is of finite presentation over S, or that S is locally noetherian and G is locally of finite type over S. Then there exists an open neighborhood U of s such that, if $K = \operatorname{Ker} u$, $K|_U$ is central and of multiplicative type. Moreover,

$$G' = (G|_U)/(K|_U)$$

is representable, and the morphism

$$u' : G' \longrightarrow H|_U$$

induced from u by passage to the quotient is an immersion.

Proof. Let Z be the center of G, and put $K' = K \cap Z$.

a) K' is a subgroup of multiplicative type over a neighborhood U of s. Indeed, after making an étale extension over a neighborhood U of s, which is legitimate, we may suppose that G possesses a maximal torus T (4.1 c)). But then $T \cap K$ is a group of multiplicative type (Exposé XV, 8.3) whose fiber at s is central by hypothesis; therefore $T \cap K$ is central over a neighborhood of s (Exposé IX, 5.6 a)), and consequently coincides with K'.

b) Let us show that $K' = K$ over a neighborhood U of s. By a), we may already suppose that K' is of multiplicative type, and hence flat over S. Since $K'_s = K_s$, the natural immersion

$$K' \longrightarrow K$$

is then open over a neighborhood U of s (Exposé $\mathrm{VI_B}$, 2.6). If $t \in U$, the image of K_t in the group

$$G'_t = G_t/Z_t$$

is a finite étale group, normal in G'_t (since K_t is normal in G_t), hence central, and therefore reduced to the unit group (3.3 b)). This says that K_t is contained in Z_t, and hence is equal to K'_t, whence $K = K'$ over U.

c) The representability of G' now follows from 2.3; the fact that u' is an immersion is contained in 1.3 a), in view of 4.1 c) applied to G'.

Proposition 6.4. — *Let S be a prescheme, let G be an S-group prescheme smooth and of finite presentation over S, with connected fibers, and let s be a point of S such that G_s is generated by its subtori (Exposé XII, 8.2) (for example, G_s affine of unipotent rank zero). Let H be an S-group prescheme, and let*

$$u, v : G \longrightarrow H$$

be two S-homomorphisms such that $u_s = v_s$ and u_s is central. Suppose, moreover, that H is of finite presentation over S, or that S is locally noetherian. Then there exists a neighborhood U of s such that

$$u|_U = v|_U.$$

We reduce as usual to the case where S is noetherian (to study the condition "G_s is generated by its subtori," one uses Exposé $\mathrm{VI_B}$, 7.4). Since G has connected fibers, to prove that $u = v$ over a neighborhood U of s, it suffices to prove that $u = v$ after reduction modulo every power of the maximal ideal of the local ring $\mathcal{O}_{S,s}$, which reduces us to the case where S is local artinian.

But then the functor of maximal subtori of G is representable by an S-scheme X, smooth and of finite type over S (Exposé XV, 8.17). Let T be the maximal subtorus of G_X universal for the functor X. The hypothesis on G_s means that the algebraic subgroup of G_s generated (Exposé $\mathrm{VI_B}$, §7) by the $\kappa(s)$-morphism

$$f : T_s \longrightarrow G_s$$

(the composite of the immersion $T_s \to G_{X_s}$ and the canonical projection $G_{X_s} \to G_s$) is all of G_s. But X_s is geometrically connected (for if $\overline{T}$ is a maximal torus of $G_{\bar{s}}$, and N its normalizer in $G_{\bar{s}}$, then $X_{\bar{s}}$ is isomorphic to $G_{\bar{s}}/N$), and the image of T_s under f contains the unit section of G_s. It follows from Exposé $\mathrm{VI_B}$, 7.4, that there exist an integer $N > 0$ and an S-morphism

$$f^N : T^N \longrightarrow G, \qquad T^N = T \times_S \cdots \times_S T \quad (N \text{ factors}),$$

where f^N depends only on f and the multiplication law of G, such that $(f^N)_s$ is surjective.

Consider the base change $X \to S$ and the restrictions of u_X and v_X to the subtorus T of G_X. By hypothesis,

$$u_{X_s} = v_{X_s},$$

and

$$u_{X_s} : T_{X_s} \longrightarrow H_{X_s}$$

is a central homomorphism. It then follows from Exposé IX, 5.1, that

$$u_X|_T = v_X|_T.$$

Making the definition of f^N explicit, and taking account of the fact that u and v are homomorphisms, one immediately deduces that

$$uf^N = vf^N.$$

But the $\kappa(s)$-morphism f_s^N is surjective and G_s is smooth, hence reduced; consequently f_s^N is generically flat. Since T is smooth over S, hence flat, there exists a nonempty open subset V of T^N such that

$$f^N|_V$$

is flat (EGA IV, 11.3.10).The image of V is an open subset W of G (EGA IV, 11.3.1), and

$$f^N : V \longrightarrow W$$

is faithfully flat and of finite presentation, hence covering for the fpqc topology. The equality $uf^N = vf^N$ then implies

$$u|_W = v|_W.$$

Since W is schematically dense in G (EGA IV, 11.10.10), and H is separated over S (Exposé $\mathrm{VI_A}$, 0.3), it follows immediately that $u = v$.

EXPOSÉ XVII

UNIPOTENT ALGEBRAIC GROUPS. EXTENSIONS BETWEEN UNIPOTENT GROUPS AND GROUPS OF MULTIPLICATIVE TYPE

by M. Raynaud[*]

0. Some notation[0]

In this chapter we shall be concerned chiefly with algebraic groups defined over a field k. The number $p \geq 0$ will always denote the characteristic of k; $\mathbf{F}_p$ will denote the prime field with p elements when $p > 0$; $\bar{k}$ will denote an algebraically closed extension of k; and q will denote a prime number different from p.

For every S-prescheme, $(\mathbb{G}_a)_S$ (respectively $(\mathbb{G}_m)_S$) denotes the additive group (respectively the multiplicative group) over S (cf. Exposé I, 4.3). For every integer $n > 0$, $(\boldsymbol{\mu}_n)_S$ (respectively $(\mathbb{Z}/n\mathbb{Z})_S$) denotes the group of n-th roots of unity (Exposé I, 4.4.4) (respectively the constant group over S associated with the abstract group $\mathbb{Z}/n\mathbb{Z}$ (Exposé I, 4.1)). The group $(\mathbb{Z}/n\mathbb{Z})_S$ is finite and étale over S; the group $(\boldsymbol{\mu}_n)_S$ is flat and finite over S, and is étale over S if and only if n is invertible on S (Exposé VIII, 2.1).

Suppose that S is a prescheme of characteristic $p > 0$. For every integer $n > 0$ and every S-group prescheme G, we denote by $F^n(G)$ the radicial S-subgroup prescheme of G equal to the kernel of the n-th iterate of the relative Frobenius morphism of G (Exposé VII$_{\mathrm{A}}$). In particular, if $G = (\mathbb{G}_a)_S$, we set

$$F^n(G) = (\boldsymbol{\alpha}_{p^n})_S.$$

This is a radicial S-group, flat and finite over S, representing the following functor: for every S-prescheme S', $(\boldsymbol{\alpha}_{p^n})_S(S')$ is the set of $x' \in \Gamma(S', \mathcal{O}_{S'})$ such that $x'^{p^n} = 0$.

The already defined group $(\boldsymbol{\mu}_{p^n})_S$ is canonically isomorphic to $F^n((\mathbb{G}_m)_S)$.

When there is no ambiguity about the base scheme S, we shall simply write $\mathbb{G}_a$, $\mathbb{G}_m$, $\boldsymbol{\alpha}_{p^n}$, and so on, in place of $(\mathbb{G}_a)_S$, $(\mathbb{G}_m)_S$, $(\boldsymbol{\alpha}_{p^n})_S$, and so on.

[*]Cf. the note on page 1 of Exposé XV.

[0]Version x.y of 18 November 2008.

If G is a commutative S-group prescheme, then for every integer $n > 0$, ${}_nG$ denotes the subgroup prescheme of G equal to the kernel of raising to the n-th power in G.

For the reader's convenience, we have collected in an appendix some properties of algebraic groups proved in Exposés VI and VII, together with elementary properties of Hochschild cohomology that will be useful in this chapter.

1. Definition of unipotent algebraic groups

Definition 1.1.— *An algebraic group G defined over an algebraically closed field k is called* unipotent *if G admits a composition series whose successive quotients are isomorphic to algebraic subgroups of $(\mathbb{G}_a)_k$.*

Proposition 1.2.— *Let k be an algebraically closed field, let K be an algebraically closed extension of k, and let G be an algebraic group defined over k. Then G is unipotent if and only if G_K is unipotent.*

The necessity is clear, since a composition series remains a composition series after extension of the base.

The proof of sufficiency is standard. The field K is the filtered direct limit of its finitely generated k-subalgebras. By Exposé $\mathrm{VI_B}$, §10, one can find a finitely generated k-subalgebra A of K, a composition series G_i of G_S, where $S = \operatorname{Spec} A$, and immersions

$$u_i : H_i = G_i/G_{i+1} \longrightarrow (\mathbb{G}_a)_S.$$

To prove that G is unipotent, it therefore suffices to make a k-valued base change $A \to k$. This is possible because $\mathrm{Hom}_{k\text{-alg}}(A, k)$ is nonempty: A is a nonzero finitely generated algebra over the algebraically closed field k.

Definition 1.3.— *Let G be an algebraic group defined over a field k. We say that G is* unipotent *if there exists an algebraically closed extension $\bar{k}$ of k such that $G_{\bar{k}}$ is unipotent in the sense of Definition 1.1.*

By Proposition 1.2, this property is independent of the chosen algebraically closed extension $\bar{k}$.

Definition 1.4.— *Let G be an algebraic group defined over a field k, and let H be an algebraic group defined over an extension k' of k. We say that H is a* form *of G over k' if the algebraic groups $G_{\bar{k}'}$ and $H_{\bar{k}'}$ become isomorphic over an algebraic closure $\bar{k}'$ of k'. As above, this property is independent of the choice of $\bar{k}'$. We shall also say that H is a* twisted form *of G.*

We can now describe the algebraic subgroups of $\mathbb{G}_a$.

Proposition 1.5.— *Let k be a field of characteristic $p \geq 0$. An algebraic subgroup H of $(\mathbb{G}_a)_k$ is of one of the following types:*

(i) *$H = 0$;*

(ii) *$H = \mathbb{G}_a$;*

(iii) *if $p > 0$, H is an extension of a twisted constant group $(\mathbb{Z}/p\mathbb{Z})^r$ by a radicial group $\boldsymbol{\alpha}_{p^n}$, where $r, n \geq 0$ and $r + n > 0$. If, moreover, k is perfect, this extension is necessarily trivial.*

Proof. If $\dim H = 1$, then clearly $H = \mathbb{G}_a$. Otherwise H has dimension zero and is therefore an extension[4] of an étale group H'' by its identity component H', which is radicial.

[4]Editor's note: reference to Exposé $\mathrm{VII_A}$?

To describe $H''_{\bar{k}}$, it is enough to know the abstract group $H''(\bar{k})$, which is isomorphic to $H(\bar{k})$.

The latter is a finite subgroup of $\mathbb{G}_a(\bar{k})$. It is therefore zero when $p = 0$, and otherwise is of the form $(\mathbb{Z}/p\mathbb{Z})^r$, since it is annihilated by p. The group H' is closed in $\mathbb{G}_a$ and is defined by a single equation (the coordinate ring $\bar{k}[T]$ of $\mathbb{G}_a$ is principal). Since this equation has 0 as its only root over $\bar{k}$, it is of the form $T^n = 0$. Compatibility with the group law implies that $(T+T')^n$ belongs to the ideal generated by T^n and T'^n in $\bar{k}[T,T']$. Thus, if $p = 0$, then $n = 1$ and H' is the trivial group; while if $p > 0$, then $n = p^m$ and $H' = \boldsymbol{\alpha}_{p^m}$.

The last assertion of Proposition 1.5 follows more generally from the following lemma.

Lemma 1.6.— *If k is a perfect field, every extension H of an étale algebraic group H'' by a radicial group H' is trivial. Moreover, H'' has a unique lifting in H, namely H_{red}.*

Indeed, since k is perfect, the reduced k-scheme H_{red} is an algebraic subgroup of H (Exposé $\mathrm{VI_A}$, 0.2), is geometrically reduced and hence smooth over k (Exposé $\mathrm{VI_A}$, 1.3.1), and is therefore étale because H has dimension zero. To see that the canonical projection $H_{\mathrm{red}} \to H''$ is an isomorphism, it is enough to check this after the base-field extension $k \to \bar{k}$; there it is enough to check that it induces an isomorphism on $\bar{k}$-valued points, which is clear. The final assertion follows because every lifting of H'' in H, being étale over k, is reduced and hence necessarily contained in H_{red}.

Notice that $\boldsymbol{\alpha}_{p^n}$ is an iterated extension of groups isomorphic to $\boldsymbol{\alpha}_p$. Proposition 1.5 therefore gives the following corollary.

Corollary 1.7.— *Let G be an algebraic group defined over an algebraically closed field k. Then G is unipotent if and only if it has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$ when $p = 0$, and to one of the groups $\mathbb{G}_a, \mathbb{Z}/p\mathbb{Z}, \boldsymbol{\alpha}_p$ when $p > 0$. We shall call these the* elementary unipotent groups.

2. First properties of unipotent groups

Proposition 2.1.— *A unipotent algebraic group defined over a field k is affine over k.*

By fpqc descent for affine morphisms, it suffices to prove the assertion when k is algebraically closed. In that case, G is by definition an iterated extension of affine algebraic groups, and is therefore affine by Exposé $\mathrm{VI_B}$, 9.2 (viii), applied to affine morphisms.[5]

Proposition 2.2.—

(i) *The property that an algebraic group is unipotent is invariant under extension of the base field.*

(ii) *Every algebraic subgroup of a unipotent group is unipotent.*

(iii) *Every algebraic quotient group of a unipotent group is unipotent.*

(iv) *Every extension of a unipotent algebraic group by a unipotent algebraic group is likewise unipotent.*

Proof. Assertion (i) follows immediately from Definitions 1.3 and Proposition 1.2. To prove the remaining assertions, we may suppose that k is algebraically closed. Assertion (iv) is then immediate from the definition.

[5]Editor's note: verify this reference.

Let G be a unipotent algebraic group, G' an algebraic subgroup of G, G'' an algebraic quotient group of G, and let G_i, $1 \le i \le n$, be a composition series of G such that

$$H_i = G_i/G_{i+1}$$

is an elementary unipotent group (Corollary 1.7).

For (ii), consider the composition series of G' induced by that of G: $G'_i = G_i \cap G'$. The group G'_i/G'_{i+1} identifies with an algebraic subgroup of H_i, hence is isomorphic to a subgroup of $\mathbb{G}_a$. It follows that G' is unipotent.

For (iii), consider the composition series of G'' obtained as the image of that of G: let G''_i be the image of G_i in G''. Then G''_i/G''_{i+1} is a quotient of H_i, so it remains only to prove the following lemma.

Lemma 2.3.— *Let H be an elementary unipotent group (Corollary 1.7) defined over a field k. Every algebraic quotient group H'' of H is either zero or isomorphic to H over k.*

Proof. If $H = \mathbb{G}_a$ in characteristic zero, or if $H = \boldsymbol{\alpha}_p$ or $\mathbb{Z}/p\mathbb{Z}$ in characteristic $p > 0$, it is enough to observe from Proposition 1.5 that H has no algebraic subgroups other than 0 and H. Indeed, a nonzero algebraic subgroup of $\boldsymbol{\alpha}_p$ is defined by a k-algebra of rank at least p over k, by Proposition 1.5(iii), and is therefore $\boldsymbol{\alpha}_p$.

Now let $H = \mathbb{G}_a$ in characteristic $p > 0$, so that there is an exact sequence

$$0 \longrightarrow N \longrightarrow \mathbb{G}_a \longrightarrow H'' \longrightarrow 0.$$

If $N = \mathbb{G}_a$, then $H'' = 0$. Otherwise, arguing by induction on the length of a composition series of N, we may suppose that $N = \boldsymbol{\alpha}_p$, or that N is a form of $(\mathbb{Z}/p\mathbb{Z})^r$.

(a) If $N \simeq \boldsymbol{\alpha}_p$, the proof of Proposition 1.5(iii) shows that N is necessarily the kernel of the Frobenius morphism F of $\mathbb{G}_a$, and the assertion follows from the exact sequence

$$0 \longrightarrow \boldsymbol{\alpha}_p \longrightarrow \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 0.$$

(b) If $N \simeq \mathbb{Z}/p\mathbb{Z}$, there is plainly an $a \in k^\times$ such that N is the closed subscheme of $\mathbb{G}_a = \operatorname{Spec} k[X]$ defined by $X^p - aX = 0$. The assertion then follows from the exact sequence

$$0 \longrightarrow N \longrightarrow \mathbb{G}_a \xrightarrow{P} \mathbb{G}_a \longrightarrow 0,$$

where P is the Artin–Schreier morphism $x \mapsto x^p - ax$.

(c) If N is a form of $(\mathbb{Z}/p\mathbb{Z})^r$, there is a finite Galois extension k'/k that trivializes N. By (b) and an evident induction on r, the quotient $\mathbb{G}_a/N$ is a form of $\mathbb{G}_a$ trivialized by k'. It remains to apply the next lemma.

Lemma 2.3 bis.— *Let k be a field, and let G be a k-algebraic group that is a form of $\mathbb{G}_a$, trivialized by a finite separable extension k'/k. Then G is isomorphic to $\mathbb{G}_a$.*

Proof of Lemma 2.3 bis. Indeed, the group of k'-automorphisms of the algebraic group $(\mathbb{G}_a)_{k'}$ is the group of nonzero homotheties $(k')^\times$ (*Bible*, Exposé 9, Lemma 1), and the Galois cohomology group $H^1(k'/k, \mathbb{G}_m)$ is zero (we may suppose that k' is a Galois extension of k), by Hilbert's Theorem 90. Lemma 2.3 bis then follows from the classification of the k-forms of an algebraic group (cf. J.-P. Serre, *Cohomologie galoisienne*, Chapter III, 1.3).

This completes the proof of 2.2.

Proposition 2.4.— *Let k be a field, let M be a k-group of multiplicative type (Exposé VIII) and of finite type, and let U be a unipotent k-algebraic group. Then:*

(i) $\underline{\mathrm{Hom}}_{k\text{-}gr}(M,U) = \underline{e}$ *(a fortiori* $\mathrm{Hom}_{k\text{-}gr}(M,U) = e$*);*

(ii) $\mathrm{Hom}_{k\text{-}gr}(U,M) = e$.

To prove (i), we must establish that, for every prescheme S over k,

$$\mathrm{Hom}_{S\text{-gr}}(M_S, U_S) = e.$$

This follows from the next lemma.

Lemma 2.5.— *Let S be a prescheme, let M be an S-group of multiplicative type and of finite type over S, and let U be an S-group prescheme, of finite presentation over S, whose fibers are unipotent. Then* $\mathrm{Hom}_{S\text{-}gr}(M,U) = e$.

Proof. Under these hypotheses, for an S-group morphism $u : M \to U$ to be the unit homomorphism, it suffices that the restriction of u to the fibers of M over the points of S be the zero morphism (Exposé IX, 5.2). We are therefore reduced to the case in which S is the spectrum of a field, which we may moreover suppose to be algebraically closed. In view of Definition 1.1, it is enough to take $U = \mathbb{G}_a$, in which case the assertion has already been proved (Exposé XII, 4.4.1).

We now prove Proposition 2.4(ii). Let $u : U \to M$ be a k-group morphism. The image $u(U)$ is represented by an algebraic subgroup U'' of M (Exposé $\mathrm{VI_B}$, 5.4).[6] The group U'' is unipotent as a quotient of a unipotent group (2.2(iii)), and is of multiplicative type as a subgroup of a group of multiplicative type (cf. *Bible*, Exposé 4, Theorem 2, Corollary 1, or Exposé IX, 6.8). Thus U'' is the unit group by Proposition 2.4(i).

Remark 2.6.— *Keeping the notation of Proposition 2.4, it is no longer true in general that the functor* $\underline{\mathrm{Hom}}_{k\text{-}gr}(U,M)$ *is equal to* $\underline{e}$. *Indeed, let S be a prescheme such that $\Gamma(S, \mathcal{O}_S)$ contains a nonzero element ε with $\varepsilon^2 = 0$ (for example, the spectrum of the algebra of dual numbers over a ring A). For every S' over S, the map*

$$u \longmapsto 1 + \varepsilon_{S'} u$$

defines, functorially in S', a homomorphism from the additive group $\Gamma(S', \mathcal{O}_{S'})$ to the multiplicative group $\Gamma(S', \mathcal{O}_{S'}^{\times})$. It therefore defines an S-group morphism

$$(\mathbb{G}_a)_S \longrightarrow (\mathbb{G}_m)_S,$$

and, since $\varepsilon \neq 0$, this morphism is not zero.

Recall (Exposé $\mathrm{VII_A}$, §3) that when G is a commutative S-group prescheme, finite and flat over S, Cartier duality is available and G is reflexive in the sense of Exposé VIII, §1. More precisely, the functor

$$G \longmapsto \underline{\mathrm{Hom}}_{S\text{-gr}}(G, \mathbb{G}_m)$$

is represented

by a commutative S-group prescheme $D(G)$, finite and flat over S, and the canonical morphism

$$G \longrightarrow D(D(G))$$

is an isomorphism. In particular:

(i) $D(\boldsymbol{\mu}_n) \simeq \mathbb{Z}/n\mathbb{Z}$, and consequently $D(\mathbb{Z}/n\mathbb{Z}) \simeq \boldsymbol{\mu}_n$;

(ii) if S has characteristic $p > 0$, then $D(\boldsymbol{\alpha}_p) \simeq \boldsymbol{\alpha}_p$.

[6]Editor's note: verify this reference.

3. Unipotent groups acting on a vector space

Recall (Exposé I, 4.6.1) that if S is a prescheme and M is a sheaf of $\mathcal{O}_S$-modules, $\mathbf{W}(M)$ denotes the S-functor

$$(\mathrm{Sch}\,/S)^{\circ} \longrightarrow (\mathrm{Ens})$$

defined, for every S-prescheme S', by

$$\mathbf{W}(M)(S') = \Gamma(S', M \otimes_{\mathcal{O}_S} \mathcal{O}_{S'})\,.$$

Recall also that if an S-group G acts on an S-functor V, the S-functor V^G of invariants of V under G is the subfunctor of V whose S'-valued points, for S' over S, are those $x \in V(S')$ such that $x_{S''}$ is fixed under $G(S'')$ for every S'' over S'.

We then have the following lemma.

Lemma 3.1.— *Let S be a prescheme, and let G be an S-group prescheme, affine over S, defined by the quasi-coherent $\mathcal{O}_S$-algebra A. Suppose that G acts on a quasi-coherent sheaf M of $\mathcal{O}_S$-modules. Let*

$$\mu : M \longrightarrow A \otimes_{\mathcal{O}_S} M$$

be the comorphism defining the action of G on M (Exposé I, 4.7.2), and let

$$\nu : M \longrightarrow A \otimes_{\mathcal{O}_S} M$$

be the morphism $x \mapsto \mu(x) - 1 \otimes x$. Then:

(i) $\underline{M}^G(S) = \Gamma\,\mathrm{Ker}(\nu)$;

(ii) if S is the spectrum of a field k, then $\underline{M}^G$ is of the form $\mathbf{W}(N)$, where N is a vector subspace of M;

(iii) if S is the spectrum of a field k, every element x of M is contained in a finite-dimensional k-vector subspace of M stable under the action of G.

Proof. Assertion (iii) is recalled here for reference and was already proved in Exposé $\mathrm{VI_B}$, 11.2.

For (i), it is clear that $\underline{M}^G(S)$ contains $\Gamma\,\mathrm{Ker}(\nu)$. To prove the reverse inclusion, we may suppose that S is affine, with ring B. Let $m \in \underline{M}^G(S)$. For every B-algebra B' and every

$$u \in \mathrm{Hom}_{B\text{-alg}}(A, B')$$

(where u corresponds to an element of $G(\mathrm{Spec}\,B')$), we have

$$(u \otimes 1_M)\nu(m) = 0 \qquad \text{in } B' \otimes_B M.$$

Taking $B' = A$ and u to be the identity of A, we obtain $\nu(m) = 0$.

For (ii), let N be the kernel of ν, equal to $\underline{M}^G(k)$ by (i). Every k-prescheme S is flat over k; hence

$$\underline{M}^G(S) = \Gamma\,\mathrm{Ker}(\nu \otimes_k S) = \Gamma(N \otimes_k S).$$

Thus $\underline{M}^G$ is isomorphic to the functor $\mathbf{W}(N)$.

Proposition 3.2.— *Let G be a unipotent algebraic group, defined over a field k, acting on a k-vector space V. If $V \neq 0$, then $V^G \neq 0$.*

In view of 3.1(ii) and (iii), we may suppose that k is algebraically closed and that V is finite-dimensional over k.

Let
$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$
be an exact sequence of algebraic groups, and suppose that G acts on a sheaf V (for the fpqc topology). Of course $V^G \subset V^{G'}$, and the quotient presheaf G/G' acts naturally on $V^{G'}$. But $V^{G'}$ is a sheaf, being the kernel of the familiar pair of morphisms
$$V \rightrightarrows \underline{\mathrm{Hom}}(G', V).$$
Consequently the sheaf associated with G/G', namely G'', acts on $V^{G'}$, and it is immediate that
$$\left(V^{G'}\right)^{G''} = V^G = \left(V^{G'}\right)^{G/G'}.$$

This remark reduces the proof of 3.2 to the case in which G is an elementary unipotent group (1.7).

(a) $G = \mathbb{G}_a$, $p = 0$. It follows from (BIBLE 4, Proposition 4) that a morphism from $\mathbb{G}_a$ to the linear group $\mathrm{GL}(V)$ is given by an exponential map
$$T \longmapsto \sum_{q=0}^{\infty} T^q \frac{n^q}{q!},$$
where n is a nilpotent endomorphism of V. But then $V \neq 0$ implies $\operatorname{Ker} n \neq 0$, and it is clear that every vector of V annihilated by n is fixed by G.

Suppose now that $p > 0$.

(b) $G = \boldsymbol{\alpha}_p$. Since $\boldsymbol{\alpha}_p$ is a radicial group of height 1 (Exposé $\mathrm{VII_A}$, §7), giving a representation of $\boldsymbol{\alpha}_p$ in V amounts to giving a representation of the p-Lie algebra $\boldsymbol{\alpha}_p$ in $\mathfrak{gl}(V)$ (Appendix II, 2.2), that is, in this case, an element $X \in \mathrm{End}(V)$ such that $X^p = 0$ (Appendix II, 2.1). But then
$$V \neq 0 \quad \Longrightarrow \quad W = \mathrm{Ker}(X) \neq 0,$$
and, again by Appendix II, 2.2, one has $W = V^{\boldsymbol{\alpha}_p}$.

(c) $G = \mathbb{Z}/p\mathbb{Z}$. A representation of G in V is equivalent to the choice of an element $x \in \mathrm{Aut}(V)$ such that $x^p = 1$, that is, $(1-x)^p = 0$. Thus x is of the form $1 + n$, with n nilpotent, and $W = \operatorname{Ker} n$ is fixed by x.

(d) $G = \mathbb{G}_a$. Let G_i, $i \in I$, be the increasing filtered family of étale algebraic subgroups of $\mathbb{G}_a$, hence groups isomorphic to $(\mathbb{Z}/p\mathbb{Z})^{r_i}$ (Proposition 1.5), and put $V_i = V^{G_i}$. Since V has finite nonzero dimension and each V_i is nonzero by (c), the decreasing filtered family of the V_i is stationary, and
$$W = \bigcap_{i \in I} V_i \neq 0.$$
We shall use the following lemma.

Lemma 3.3.— *The family of étale subgroups of $(\mathbb{G}_a)_S$, where S is a prescheme of characteristic $p > 0$, is schematically dense in G (Exposé IX, 4.1).*

By Exposé IX, 4.4, it suffices to prove the lemma when S is the spectrum of the prime field $\mathbb{F}_p$. In that case it is enough to consider the family of étale subgroups G_n, $n \geq 1$, defined by
$$X^{p^n} - X = 0.$$
This family is schematically dense in $(\mathbb{G}_a)_{\mathbb{F}_p}$, since it contains every closed point.

With this established, let us return to the proof of 3.2(d). If $w \in W$, its stabilizer in G is an algebraic subgroup of $\mathbb{G}_a$ containing G_i for every $i \in I$; it is therefore equal to $\mathbb{G}_a$ by 3.3. Consequently $W = V^G$.

Proposition 3.2 immediately gives the following.

Corollary 3.4.— *Let k be a field, and let G be a unipotent k-algebraic group acting on a finite-dimensional k-vector space V. Then V has a sequence of vector subspaces V_i, defined over k and stable under G,*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V,$$

such that G acts trivially on V_{i+1}/V_i. Moreover, the V_i may be chosen so that V_{i+1}/V_i has dimension 1*.*

We shall now summarize and supplement the properties of unipotent groups already proved in the following theorem.

Theorem 3.5.— *Let G be an algebraic group defined over a field k. The following properties are equivalent:*

(i) G is unipotent.

(ii) G has a composition series defined over k, whose successive quotients are isomorphic to $\mathbb{G}_a$ if $p = 0$, and, respectively, to $\boldsymbol{\alpha}_p$, $\mathbb{G}_a$, or a twisted form of $(\mathbb{Z}/p\mathbb{Z})^r$ (1.4) if $p > 0$.

(iii) As in (ii), with the additional requirement that the composition series be central.

(iv) G has a characteristic composition series (Exposé VI_B) defined over k, whose successive quotients are isomorphic to $(\mathbb{G}_a)^r$ if $p = 0$, and, respectively, to $(\boldsymbol{\alpha}_p)^r$, a twisted form of $(\mathbb{G}_a)^s$, or a twisted form of $(\mathbb{Z}/p\mathbb{Z})^t$, taken in this order, if $p > 0$.

(v) G is isomorphic to an algebraic subgroup of the group $\mathrm{Trigstr}(n)_k$ *of strictly upper-triangular matrices in the linear group* $\mathrm{GL}(n)_k$*, for a suitable integer $n \geq 0$.*

(vi) G is affine, and for every linear representation of G on a nonzero finite-dimensional k-vector space V, one has $V^G \neq 0$.

Proof. The implication (i)⇒(vi) follows from 2.1 and 3.2.

For (vi)⇒(v), since the algebraic group G is affine, it is an algebraic subgroup of a suitable linear group $\mathrm{GL}(V)$ (Exposé VI_B, 11.3). Applying 3.4 to the representation of G on V defined by this embedding gives (v).

For (v)⇒(iii), the algebraic group $\mathrm{Trigstr}(n)$ is known to have a central composition series whose successive quotients are isomorphic to $\mathbb{G}_a$. The composition series induced on G gives property (iii), in view of 1.5.

The implications

$$\text{(iii)} \Rightarrow \text{(ii)} \Rightarrow \text{(i)} \qquad \text{and} \qquad \text{(iv)} \Rightarrow \text{(i)}$$

are clear.

We shall prove (i)⇒(iv) shortly, but first note some consequences of what has already been proved.

Definition 3.6.— *A p-Lie algebra $\mathfrak{g}$, $p > 0$ (cf. Exposé VII_A, §5), is called* unipotent *if the map*

$$x \longmapsto x^{(p)}$$

is nilpotent; that is, if for every $x \in \mathfrak{g}$ there is an integer $n > 0$ such that $x^{(p^n)} = 0$.

Corollary 3.7.— *A unipotent algebraic group G is nilpotent (Exposé VI_B, §8); its Lie algebra $\mathfrak{g}$ is nilpotent (Bourbaki,* Groupes et algèbres de Lie*, Chap. I, §4) and is isomorphic to a Lie algebra of nilpotent endomorphisms of a finite-dimensional vector space. In characteristic $p > 0$, $\mathfrak{g}$ is a unipotent p-Lie algebra (3.6).*

Since (i) implies (v), it is enough to prove 3.7 when $G = \mathrm{Trigstr}(n)$. We have already used the fact that $\mathrm{Trigstr}(n)$ is a nilpotent algebraic group. Moreover, the Lie algebra $\mathfrak{h}$ of $\mathrm{Trigstr}(n)$ consists of the upper-triangular endomorphisms of V that have zeros on the main diagonal. These endomorphisms are therefore nilpotent, and consequently $\mathfrak{h}$ is nilpotent (Bourbaki, *loc. cit.*, Chap. I, §4, Cor. 3). If $p > 0$, the p-th power in the p-Lie algebra $\mathfrak{gl}(V) = \mathrm{End}(V)$ coincides with the p-th power of endomorphisms of V (Exposé $\mathrm{VII_A}$, 6.4.4); hence $\mathfrak{h}$ is unipotent.

Corollary 3.8.— *Let k be an algebraically closed field, and let G be a smooth affine algebraic group over k. Then the following properties are equivalent:*

(i) G is unipotent.

(ii) $G(k)$ consists of unipotent elements (BIBLE 4, Prop. 4, Cor. 1); that is, G is unipotent in the sense of BIBLE.

The implication (i)⇒(ii) follows because, by 3.2, (i)⇒(v), G is isomorphic to an algebraic subgroup of a group $\mathrm{Trigstr}(n)$.

For (ii)⇒(i), suppose that G is a unipotent algebraic group in the sense of BIBLE, and write G^0 for its identity component. The maximal tori of G^0, being made up of unipotent elements, are trivial; thus G^0 is equal to each of its Cartan subgroups. It follows that G^0 is solvable (BIBLE 6, Th. 6), and hence triangularizable (BIBLE 6, Th. 1). In short, G^0 is an algebraic subgroup of a group $\mathrm{Trigstr}(n)$, and is therefore unipotent in the sense of this Exposé.

The group $(G/G^0)(k)$ is a finite group consisting of unipotent elements. It is therefore trivial if $p = 0$, and is a finite p-group if $p > 0$ (BIBLE 4, Prop. 4). But then G/G^0 is unipotent in the sense of this Exposé, since it is an iterated extension of groups isomorphic to $\mathbb{Z}/p\mathbb{Z}$. This proves that G is unipotent.

Completion of the proof of 3.5. It remains to prove (i)⇒(iv).

(a) $p > 0$. Consider the increasing sequence of algebraic subgroups of G

$$\{e\} \subset F(G) \subset F^2(G) \subset \cdots \subset F^n(G) \subset G^0 \subset G.$$

This gives a characteristic composition series of G (Appendix II, 1), and for n sufficiently large, $G/F^n(G)$ is smooth (Appendix II, 3.1). Thus its successive quotients are, in order:

(1) radicial groups of height 1;

(2) a smooth connected group;

(3) an étale group.

To prove (i)⇒(iv), it is therefore enough to prove the following lemma.

Lemma 3.9.— *Let G be a unipotent algebraic group defined over a field k of characteristic $p > 0$. Then G has a characteristic composition series, defined over k, whose successive quotients are isomorphic to:*

(i) $(\boldsymbol{\alpha}_p)^r$*, if* G *is radicial;*

(ii) a twisted form of $(\mathbb{G}_a)^r$*, if* G *is smooth and connected;*

(iii) a twisted form of $(\mathbb{Z}/p\mathbb{Z})^r$*, if* G *is étale.*

Proof of 3.9(i). The group G is radicial. Filtering G by the $F^n(G)$, we reduce to the case in which G is radicial of height 1. Since G is nilpotent (3.5, (i)⇒(iii)), and since the center of an algebraic group is representable (Exposé VIII, 6.5(e)), we may consider the ascending central series of G. This series is plainly characteristic in G, and reduces us to the case in which G is also commutative.

Put $\mathfrak{g} = \mathrm{Lie}(G)$. The p-th-power morphism π is then additive on $\mathfrak{g}$ (Exposé $\mathrm{VII_A}$); we shall reduce to the case in which it is zero. For every prescheme S over k, set

$$\mathfrak{g}_S = \mathfrak{g} \otimes_k \mathcal{O}_S,$$

and let $\mathfrak{h}_S$ be the subsheaf of abelian groups of $\mathfrak{g}_S$ that is the image of $\mathfrak{g}_S$ under π_S. Finally, let $\overline{\mathfrak{h}}_S$ be the subsheaf of $\mathcal{O}_S$-modules of $\mathfrak{g}_S$ generated by $\mathfrak{h}_S$. It is clear that

$$\overline{\mathfrak{h}}_S = \overline{\mathfrak{h}}_k \otimes_k \mathcal{O}_S$$

and that $\overline{\mathfrak{h}}_S$[4] is a characteristic p-Lie subalgebra of $\mathfrak{g}_S$; that is, it is stable under the S-functor $\mathrm{Aut}_{p\text{-Lie}}(\mathfrak{g})$. It follows from Appendix II, 2.2 that $\overline{\mathfrak{h}}_k$ is the Lie algebra of an algebraic subgroup H of G that is characteristic in G.

Moreover, in view of 3.7 and Lemma 3.9 bis below, if $G \neq \{e\}$, then $H \neq G$, since $\overline{\mathfrak{h}}_k \neq \mathfrak{g}$. On the other hand, if $G/H = G''$, then

$$\mathrm{Lie}\, G'' = \mathfrak{g}/\overline{\mathfrak{h}}_k \qquad \text{(Appendix II, 2.2)},$$

and consequently the p-th power is zero in $\mathrm{Lie}\, G''$. Induction on $\dim \mathrm{Lie}\, G$ therefore reduces us to the case in which $\mathrm{Lie}\, G$ is a p-Lie algebra whose p-th power is zero. But then $\mathrm{Lie}\, G''$ is isomorphic to $\mathrm{Lie}((\boldsymbol{\alpha}_p)^r)$ for some suitable integer $r \geq 0$ (Appendix II, 2.1), and hence G'' is isomorphic to $(\boldsymbol{\alpha}_p)^r$ (Appendix II, 2.2). It remains to prove the following lemma.

Lemma 3.9 bis.— *Let k be a field of characteristic $p > 0$, let $\mathfrak{g}$ be a commutative unipotent p-Lie algebra (3.6) of finite dimension over k, and let $\mathfrak{h}$ be the p-Lie subalgebra of $\mathfrak{g}$ generated by the image of the p-th power in $\mathfrak{g}$. If $\mathfrak{g} \neq 0$, then $\mathfrak{h} \neq \mathfrak{g}$.*

Indeed, since $\mathfrak{g}$ is commutative, $\mathfrak{h}$ is simply the k-vector subspace of $\mathfrak{g}$ generated by the $X^{(p)}$, $X \in \mathfrak{g}$. If $\mathfrak{g} \neq 0$ and is unipotent, there exists an $X \in \mathfrak{g}$, $X \neq 0$, such that $X^{(p)} = 0$. Let $X_1, \ldots, X_n$ be a basis of a complement in $\mathfrak{g}$ to the line kX. The Lie algebra $\mathfrak{h}$ is then the k-vector subspace of $\mathfrak{g}$ generated by

$$X_1^{(p)}, \ldots, X_n^{(p)};$$

it therefore has dimension at most $n = \dim \mathfrak{g} - 1$.

Proof of 3.9(ii). The group G is smooth and connected. In this case the descending central series of G is represented by smooth, connected, characteristic algebraic subgroups G_i (Exposé $\mathrm{VI_B}$, 8.3 and 7.4), and $G_i = 0$ for sufficiently large i, since G is nilpotent (3.5, (i)⇒(iii)). It is enough to prove 3.9 for the groups G_i/G_{i+1}, which reduces us to the case in which G is also commutative.

[4] Editor's note: $\mathfrak{h}$ has been changed to $\overline{\mathfrak{h}}$.

For every integer $n > 0$, let G_n be the algebraic subgroup of G that is the image of G under the p^n-th-power morphism. The group G_n is therefore smooth, connected, and characteristic; and it follows from Definition 1.1 of unipotent groups that $G_n = 0$ for sufficiently large n. Replacing G by G_n/G_{n+1}, we may further suppose that G is annihilated by raising to the p-th power. But then, by J.-P. Serre, *Groupes algébriques et corps de classes*, Chap. VII, Prop. 11, G is a form of $(\mathbb{G}_a)^r$ for a suitable integer r.

Proof of 3.9(iii). The group G is étale. Proceeding as in (ii), we reduce first to the case in which G is commutative, and then to the case in which G is annihilated by p. But then

$$G_{\overline{k}} \simeq (\mathbb{Z}/p\mathbb{Z})^r.$$

Proof of 3.5, (i)⇒(iv), in case (b), $p = 0$. The group G is then smooth and connected; proceeding as in 3.9(ii), we reduce to the case in which G is also commutative. We then have the following more precise result.

Lemma 3.9 ter.— *Let k be a field of characteristic 0, let G be a commutative unipotent k-algebraic group, and put $\mathfrak{g} = \operatorname{Lie} G$. There is a canonical isomorphism*

$$\exp : \mathbf{W}(\mathfrak{g}) \xrightarrow{\sim} G.$$

The morphism exp *is the unique homomorphism $\mathbf{W}(\mathfrak{g}) \to G$ that induces the identity on Lie algebras.*

Since G is unipotent, it can be realized as an algebraic subgroup of $\operatorname{Trigstr}(n)$, for a suitable integer n (3.5, (i)⇒(v)). Choosing such an embedding identifies $\mathfrak{g}$ with a Lie subalgebra of $\mathfrak{gl}(n)$ consisting of nilpotent endomorphisms. We thereby obtain a k-morphism

$$\begin{aligned} \exp : \mathbf{W}(\mathfrak{g}) &\longrightarrow \mathrm{GL}(n), \\ T &\longmapsto \sum_{i \geq 0} \frac{T^i}{i!}. \end{aligned}$$

Since G is commutative, the morphism exp is a homomorphism. Let G' be the algebraic group that is the image of $\mathbf{W}(\mathfrak{g})$ under exp. Under the canonical identification $\operatorname{Lie} \mathbf{W}(\mathfrak{g}) = \mathfrak{g}$, the tangent linear map of exp is simply the inclusion $\mathfrak{g} \hookrightarrow \mathfrak{gl}(n)$. Hence

$$\operatorname{Lie}(G \cap G') = \mathfrak{g} \cap \mathfrak{g} = \mathfrak{g}.$$

Since $G \cap G'$ is smooth (Exposé $\mathrm{VI_B}$, 1.6.1) and G is connected—being an iterated extension of groups $\mathbb{G}_a$ (3.5, (i)⇒(ii))—one necessarily has $G = G'$. The kernel ker(exp) is an étale unipotent group, and hence is the trivial group. Consequently exp is an isomorphism from $\mathbf{W}(\mathfrak{g})$ onto G.

If $h : \mathbf{W}(\mathfrak{g}) \to G$ is another homomorphism such that $\operatorname{Lie}(h)$ is the identity map of $\mathfrak{g}$, then $h - \exp$ is a homomorphism, since G is commutative, and its tangent linear map is zero. Because k has characteristic 0 and $\mathbf{W}(\mathfrak{g})$ is connected, Cartier's theorem again implies that $h = \exp$.

Proposition 3.10.— *Let G be an algebraic group defined over an algebraically closed field k. The following properties are equivalent:*

(i) Every homomorphism from a group M of multiplicative type into G is trivial.

(i bis) G has no nontrivial subgroup of multiplicative type.

(ii) (a) If $p = 0$, then $G(k)$ has no point of finite order other than e.

(b) *If $p \neq 0$, then:*
  - *for every prime number $q \neq p$, if $x \in G(k)$ and $x^q = e$, then $x = e$;*
  - *for every $X \in \mathfrak{g} = \operatorname{Lie} G$ such that $X^{(p)} = X$, one has $X = 0$.*

(ii bis) (a) *If $p = 0$, the condition is the same as in (ii)(a).*

(b) *If $p \neq 0$, then:*
  - *every finite subgroup of $G(k)$ is a p-group;*
  - *for every $X \in \mathfrak{g} = \operatorname{Lie} G$ such that $X^{(p)} = X$, one has $X = 0$.*

Proof. The equivalence (i)⇔(i bis) follows because the image of a group of multiplicative type is of multiplicative type (Exposé IX, 2.7).

The equivalence (ii)⇔(ii bis) follows from the following observation. If an ordinary finite group H has order that is not a power of p, then there exist a prime number $q \neq p$ and an element $x \in H$, distinct from e, such that $x^q = e$ (Sylow's theorem; cf. J.-P. Serre, *Corps locaux*, Chapter IX, §2).

The implication (i)⇒(ii) follows from the next lemma.

Lemma 3.11.— *Let G be an algebraic group defined over a field k.*

(a) *If k contains the n-th roots of unity, where n is prime to p, then*

$$\operatorname{Hom}_{k\text{-gr}}(\mu_n, G) \simeq \operatorname{Hom}_{k\text{-gr}}(\mathbb{Z}/n\mathbb{Z}, G) \simeq {}_nG(k),$$

where ${}_nG(k)$ denotes the points of order n in $G(k)$.

(b) *If $p > 0$, then*

$$\operatorname{Hom}_{k\text{-gr}}(\mu_p, G) \simeq \{X \in \mathfrak{g} = \operatorname{Lie} G \mid X^{(p)} = X\}.$$

Indeed, for (a), note that μ_n is then isomorphic over k to $\mathbb{Z}/n\mathbb{Z}$; assertion (b) follows from Appendix II, 2.1.

For (ii)⇒(i bis), Lemma 3.11 shows that (ii) is equivalent to the assertion that, for every prime number r, the group G contains no subgroup μ_r. The implication now follows from:

Lemma 3.12.— *Let G be a diagonalizable S-group, of finite type over S, and distinct from the trivial group. There exist a prime number r and a subgroup of G isomorphic to $(\mu_r)_S$.*

Write $G = D_S(M)$, where M is a finitely generated abelian group, hence an extension of a free group M'' by a finite group M'. If $M'' \neq 0$, it is clear that M has quotients isomorphic to $\mathbb{Z}/r\mathbb{Z}$ for every integer r. If $M'' = 0$, then M' has a quotient isomorphic to $\mathbb{Z}/r\mathbb{Z}$ for every prime number r dividing the order of M. The lemma follows by duality.

We saw in Proposition 2.4 that a unipotent group satisfies the equivalent conditions of 3.10. The purpose of the next section is to prove the converse.

4. A Characterization of Unipotent Groups

As announced, we shall prove that an algebraic group G defined over an algebraically closed field k, and containing no nontrivial subgroup of multiplicative type, is unipotent. In fact, it is enough that G contain none of certain particular "elementary" subgroups of multiplicative type, which depend on the hypotheses imposed on G. Before stating the general theorem, let us study some special cases in detail.

4.1. Smooth, Connected, Affine Algebraic Groups

Proposition 4.1.1.— *Let k be a field, let G be a smooth, connected, affine k-algebraic group, and put $\mathfrak{g} = \mathrm{Lie}\, G$. The following properties are equivalent:*

(i) G is unipotent.

(ii) G has a central composition series whose successive quotients are forms of $\mathbb{G}_a$.

(iii) G has a central, characteristic composition series whose successive quotients are forms of $(\mathbb{G}_a)^r$.

(iv) There is an integer $n > 1$ such that $G_{\overline{k}}$ contains no subgroup isomorphic to μ_n.

(v) Every maximal torus of G is the trivial group.

Suppose, moreover, that G is an algebraic subgroup of a linear group $\mathrm{GL}(n)$. The preceding conditions are then also equivalent to:

(vi) $\mathfrak{g} \subset \mathfrak{gl}(n)$ consists of nilpotent endomorphisms.

(vii) $\mathfrak{g}$ is nilpotent, and its center contains no nonzero semisimple endomorphism.

Proof. The implication (ii)⇒(i) is clear, and (i)⇒(iii) was proved in 3.9. The implication (iii)⇒(ii) will follow from the next lemma.

Lemma 4.1.2.— *Let k be a field, and let G be a k-algebraic group that is a form of $(\mathbb{G}_a)^r$. Then:*

(a) G can be realized as an algebraic subgroup of $(\mathbb{G}_a)^n$, for a suitable integer n.

(b) G has a composition series whose successive quotients are forms of $\mathbb{G}_a$.

Indeed, by hypothesis there is an extension k' of k such that $G_{k'}$ is isomorphic to $(\mathbb{G}_{a,k'})^r$. By the principle of finite extension (EGA IV, 9.1.1), we may assume that k'/k is finite. For (a), it then suffices to consider the canonical closed immersion (EGA V[5])

$$G \longrightarrow \prod_{k'/k} (G_{k'})_{/k'} \xrightarrow{\sim} (\mathbb{G}_{a,k})^n, \qquad n = r \deg(k'/k).$$

To prove (b), in view of (a), we may assume that G is a closed subgroup of $G' = (\mathbb{G}_a)^n$. If $G \neq 0$, there is a hyperplane $\mathfrak{h}$ in $\mathfrak{g}' = \mathrm{Lie}\, G'$ that does not contain $\mathfrak{g}$. Let H be the vector subgroup $\mathbf{W}(\mathfrak{h})$ of $\mathbf{W}(\mathfrak{g}') = G'$. Since H is defined by one equation in G', the intersection $G \cap H$ is defined by one equation in G, and

$$\begin{aligned} \dim G - 1 &\leq \dim(G \cap H) \\ &\leq \dim \mathrm{Lie}(G \cap H) = \dim_k(\mathfrak{g} \cap \mathfrak{h}) \\ &= \dim_k \mathfrak{g} - 1 = \dim G - 1. \end{aligned}$$

Thus $\dim(G \cap H) = \dim \mathrm{Lie}(G \cap H)$, and consequently $G \cap H$ is smooth. The group

$$G_1 = (G \cap H)^0$$

[5]Editor's note: provide another reference here...

is a smooth, connected algebraic subgroup of G such that G/G_1 is smooth, connected, and one-dimensional, hence is a form of $\mathbb{G}_a$ (4.1, (i)⇒(iii)). Induction on the dimension of G completes the proof.

Before continuing the proof of 4.1, note that the equivalence (i)⇔(ii), together with 2.3 bis, gives the following corollary.

Corollary 4.1.3.— *If k is a perfect field, a smooth, connected k-algebraic group is unipotent if and only if it has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.*

Continuation of the proof of 4.1. The implication (i)⇒(iv) follows from 2.4(i).

For (iv)⇒(v), by Exposé XIV, 4.1, the group G has a maximal torus T defined over k. If $r = \dim T$, then

$$({}_nT)_{\overline{k}} \simeq (\mu_n)^r.$$

It follows that $r = 0$.

The implication (v)⇒(i) follows from the observation made in the proof of 3.8.

For (i)⇒(vi), by 3.4 the group G is in fact contained in an algebraic subgroup of $\mathrm{GL}(n)$ isomorphic to $\mathrm{Trigstr}(n)$. Hence $\mathfrak{g}$ consists of nilpotent endomorphisms.

For (vi)⇒(vii), the Lie algebra $\mathfrak{g}$ is nilpotent by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, Corollary 3).

For (vii)⇒(v), let T be a maximal subtorus of G (Exposé XIV, 1.1), and let $\mathfrak{t}$ be its Lie algebra. The embedding of G in $\mathrm{GL}(n)$ defines a representation of T, which is necessarily semisimple; this can be seen after passing to an algebraic closure $\overline{k}$ of k and applying Exposé I, 4.7.3. Thus, if $X \in \mathfrak{t}$, then X is a semisimple endomorphism in $\mathfrak{gl}(n)$. It follows at once that

$$\mathrm{ad}\, X : \; Y \longmapsto [X, Y]$$

is a semisimple endomorphism of $\mathfrak{gl}(n)$, and hence of $\mathfrak{g}$. This endomorphism is also nilpotent, since $\mathfrak{g}$ is nilpotent. Therefore $\mathrm{ad}\, X = 0$, so X is central. Consequently $\mathfrak{t}$ is central and consists of semisimple endomorphisms; by hypothesis it is zero. A fortiori, T is the trivial group.

Remark 4.1.4.—

(a) *We shall give below (4.3.1) an infinitesimal characterization of unipotent groups in characteristic $p > 0$ that is independent of an embedding in* $\mathrm{GL}(n)$.

(b) *When k is perfect, conditions (ii) and (iii) of 4.1.1 simplify by virtue of the following lemma.*

Lemma 4.1.5.— *If k is a perfect field, every k-algebraic group G that is a form of $(\mathbb{G}_a)^r$ is isomorphic to $(\mathbb{G}_a)^r$.*

The lemma follows from 3.9 ter when the characteristic p of k is zero, and from 2.3 bis when $r = 1$. In the general case ($p > 0$), realize G as an algebraic subgroup of $(\mathbb{G}_a)^n$, for a suitable integer n (4.1.2), and argue by induction on $n - r$. If $r = n$, then indeed $G = (\mathbb{G}_a)^r$. Otherwise, the quotient $(\mathbb{G}_a)^n/G$ is a nonzero smooth, connected, unipotent group. By 4.1.1, (i)⇒(ii), and 2.3 bis, it has a composition series whose quotients are isomorphic to $\mathbb{G}_a$. It follows that there is a smooth, connected algebraic subgroup G_1 of $(\mathbb{G}_a)^n$, containing G, such that

$$(\mathbb{G}_a)^n/G_1 = \mathbb{G}_a.$$

By induction, it suffices to show that G_1 is isomorphic to $(\mathbb{G}_a)^{n-1}$.

It is immediate that a homomorphism

$$\operatorname{Spec} k[X_1,\dots,X_n] = (\mathbb{G}_a)^n \longrightarrow \mathbb{G}_a = \operatorname{Spec} k[T]$$

is defined by an additive polynomial of the form

$$\sum_{i,j} a_{i,j} X_i^{p^j}.$$

Since G_1 is smooth, the linear part of this polynomial is nonzero. After a linear change of the coordinates X_i, we may assume that G_1 is the algebraic subgroup of $(\mathbb{G}_a)^n$ defined by

$$P(X) = -X_1 + \sum_{j=1}^{q} a_j X_1^{p^j} + Q(X_2,\dots,X_n) = 0, \tag{$*$}$$

where

$$Q(X_2,\dots,X_n) = \sum_{\substack{i>1\\ j>0}} b_{i,j} X_i^{p^j}.$$

We now argue by induction on the degree of P. If $\deg P = 1$, it is clear that

$$G_1 \xrightarrow{\sim} (\mathbb{G}_a)^{n-1}.$$

Otherwise, since k is perfect, one has $Q(X) = Q_1(X)^p$, and we may define an endomorphism v of $(\mathbb{G}_a)^n$ by

$$\begin{aligned} X_i &\longmapsto X_i && \text{for } i>1,\\ X_1 &\longmapsto \sum_{j=1}^{q} a_j^{1/p} X_1^{p^{j-1}} + Q_1(X). \end{aligned}$$

It is clear that v induces an isomorphism on G_1, and that $v(G_1)$ is defined in $(\mathbb{G}_a)^n$ by the equation

$$P_1(X) = -X_1 + \sum_{j=1}^{q} a_j^{1/p} X_1^{p^j} + Q_1(X_2,\dots,X_n) = 0. \tag{$*_1$}$$

We distinguish two cases.

First case. $\deg Q = \deg P > p^q$. Then $\deg P_1 < \deg P$, and the induction hypothesis applies.

Second case. $\deg P = p^q$, with $a_q \neq 0$. Then

$$\deg P_1 = \deg P = p^q \qquad \text{and} \qquad \deg Q_1 < p^q.$$

We cannot have $Q = 0$, for otherwise G_1 would not be connected. If the polynomial Q_1 has no linear part, the preceding transformation may be repeated. Continuing in this fashion, we eventually obtain an equation of the form

$$P_s(X) = -X_1 + \sum_{j=1}^{q} a_j^{1/p^s} X_1^{p^j} + Q_s(X), \tag{$*_s$}$$

where

$$Q_s(X) = Q(X_2,\dots,X_n)^{1/p^s}$$

is an additive polynomial with a nonzero linear part, and moreover $\deg Q_s < p^q$. Suppose, for example, that the coefficient of X_2 in Q_s is nonzero, and let $-L$ be the linear part of $P_s(X)$. After a linear change of coordinates, the equation of G_1 becomes

$$P'(X) = -L + \sum_{j=1}^{q} a_j^{1/p^s} X_1^{p^j} + Q'(L, X_3, \dots, X_n),$$

where Q' is an additive polynomial without a linear part and $\deg Q' < p^q$. We are thereby reduced to the first case.[6]

4.2. Radicial Groups

Proposition 4.2.1.— *Let G be a radicial algebraic group defined over a field k of characteristic $p > 0$. The following conditions are equivalent:*

(i) G is unipotent.

(ii) G has a central composition series whose successive quotients are isomorphic to $\boldsymbol{\alpha}_p$.

(iii) G has a central, characteristic composition series whose successive quotients are isomorphic to $(\boldsymbol{\alpha}_p)^r$.

(iv) $G_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$.

(v) $\mathfrak{g} = \operatorname{Lie} G$ is a unipotent p-Lie algebra (3.6).

Proof. The implications (iii)⇒(ii)⇒(i) are clear; (i)⇒(iii) is 3.9(i), and (i)⇒(iv) follows from 2.4(i).

We shall need the following lemma on abelian p-Lie algebras.

Lemma 4.2.2.— *Let $\mathfrak{g}$ be an abelian, finite-dimensional p-Lie algebra over a perfect field k. Then $\mathfrak{g}$ can be written uniquely as a direct sum*

$$\mathfrak{g} = \mathfrak{r} \oplus \mathfrak{u}$$

of a p-Lie subalgebra $\mathfrak{r}$ on which the p-th power is bijective and a unipotent p-Lie subalgebra $\mathfrak{u}$ (3.6). The algebra $\mathfrak{r}$ is called the reductive part *of $\mathfrak{g}$, and $\mathfrak{u}$ its* unipotent part. *The formation of $\mathfrak{r}$ and $\mathfrak{u}$ is compatible with extension of the field k. If, moreover, k is algebraically closed, then $\mathfrak{r}$ has a basis e_i such that*

$$e_i^{(p)} = e_i.$$

The proof of this lemma is easy and is left to the reader; see Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §1, Exercise 23. We merely note that $\mathfrak{u}$ is the kernel of a suitable iterate of the map $X \mapsto X^{(p)}$, while $\mathfrak{r}$ is the image of that same iterate.

We now prove (iv)⇒(v). After extending the base field if necessary, we may suppose that k is algebraically closed. Let $X \in \mathfrak{g}$, and let $\mathfrak{h}$ be the p-Lie subalgebra of $\mathfrak{g}$ generated by X. The algebra $\mathfrak{h}$ is plainly commutative, and its reductive part (4.2.2) is zero. Otherwise, by the cited result, $\mathfrak{h}$ would contain an element $Y \neq 0$ such that $Y^{(p)} = Y$, and consequently, by Appendix II, 2.1 and 2.2, G would contain a subgroup isomorphic to $\boldsymbol{\mu}_p$, contrary to the hypothesis. Thus $\mathfrak{h}$, and hence $\mathfrak{g}$, is a unipotent p-Lie algebra.

It remains to prove (v)⇒(i), the least immediate implication in 4.2.1.

[6]Editor's note: replace X_1 by L.

(a) The case in which G has height 1 (Exposé $\mathrm{VII_A}$, 4.1.3). Since G is radicial, it is affine and hence isomorphic to an algebraic subgroup of a linear group $\mathrm{GL}(V)$ (Exposé $\mathrm{VI_B}$, §11). This embedding identifies $\mathfrak{g}$ with a p-Lie subalgebra of $\mathrm{End}(V)$, and the p-th power of X in $\mathfrak{g}$ agrees with the p-th power of the endomorphism X (Exposé $\mathrm{VII_A}$, 6.4.4). Since $\mathfrak{g}$ is unipotent by hypothesis, it is therefore an algebra of nilpotent endomorphisms of V. By Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chapter I, §4, Theorem 1), it is a Lie subalgebra of the Lie algebra $\mathfrak{h}$ of the group of strictly upper-triangular matrices $\mathrm{Trigstr}(n)$, relative to a suitable basis of V. Since G has height 1, Appendix II, 2.2 then shows that G itself is an algebraic subgroup of $\mathrm{Trigstr}(n)$, and hence is unipotent by 3.5(v)⇒(i).

(b) The general case. We argue by induction on the height h of G.* The case $h = 1$ has just been treated. Suppose that $h > 1$, and put

$$G' = {}_F G, \qquad G'' = G/G'.$$

The group G' has height 1, and $\mathrm{Lie}\, G' = \mathrm{Lie}\, G$ is unipotent. Consequently G' is unipotent by (a). To prove that G is unipotent, it is therefore enough to show that G'' is unipotent (2.2). But G'' has height $h-1$, so the induction hypothesis reduces us to proving that $\mathrm{Lie}\, G''$ is unipotent.

In view of (iv)⇒(v), it suffices to prove that $G''_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$. Suppose, then, that such a subgroup has been given, and let H be its inverse image in $G_{\overline{k}}$. Since G' is unipotent, we shall prove in §5 that the extension

$$e \longrightarrow G'_{\overline{k}} \longrightarrow H \longrightarrow \boldsymbol{\mu}_p \longrightarrow e$$

is necessarily trivial; the proof of that fact is independent of the results of the present section. In short, $\boldsymbol{\mu}_p$ lifts to H. Being of height 1, it is then necessarily contained in

$$G'_{\overline{k}} = {}_F(G_{\overline{k}}),$$

which is impossible because G' is unipotent.

4.3. Connected Affine Groups in Characteristic $p > 0$

Proposition 4.3.1.— *Let G be a connected affine algebraic group over a field k of characteristic $p > 0$. The following conditions are equivalent:*

(i) G is unipotent.

(ii) G has a composition series whose successive quotients are isomorphic to $\boldsymbol{\alpha}_p$ and $\mathbb{G}_a$, taken in that order.

(iii) G has a characteristic composition series whose successive quotients are isomorphic to $(\boldsymbol{\alpha}_p)^r$ and $(\mathbb{G}_a)^s$, taken in that order.

(iv) $G_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$.

(v) $\mathfrak{g} = \mathrm{Lie}\, G$ is unipotent (3.6).

(vi) $\mathfrak{g}$ is nilpotent, and the reductive part of the center of $\mathfrak{g}$ (4.2.2) is trivial.

(vi bis) $\mathfrak{g}$ is nilpotent, and every subgroup of multiplicative type of the identity component of the center of G is trivial.

*That is, the least integer h such that $F^h = \mathrm{id}_G$; see Appendix II, 1.

(vi ter) *G is nilpotent, and every subgroup of multiplicative type in the identity component of the center of G is trivial.*

Proof. The implications (iii)⇒(ii)⇒(i) are clear. To prove (i)⇒(iii), we shall need the following lemma.

Lemma 4.3.2.— *Let k be a field of characteristic $p > 0$, let $n \in \mathbb{N}$, and let k' be a radicial extension of k such that $(k')^{p^n}$ is contained in k. For every k-prescheme X (respectively, every k'-prescheme X'), denote by $X^{(p^n)}$ (respectively, $(X')^{\phi}$) the k-prescheme obtained from X (respectively, from X') by the base change*

$$F^n : k \longrightarrow k, \qquad x \longmapsto x^{p^n}, \qquad (\textit{respectively, } \phi : k' \longrightarrow k, \qquad x' \longmapsto (x')^{p^n}).$$

Then, for every k-prescheme X, there is a functorial isomorphism

$$(X_{k'})^{\phi} \xrightarrow{\sim} X^{(p^n)}.$$

Consequently, if X and Y are two k-preschemes for which there exists a k'-isomorphism

$$u' : X_{k'} \xrightarrow{\sim} Y_{k'},$$

then there exists a k-isomorphism

$$v : X^{(p^n)} \xrightarrow{\sim} Y^{(p^n)}.$$

If, moreover, X and Y are endowed with structures of k-group preschemes and u' is a k'-homomorphism, then v is a k-homomorphism.

The lemma follows immediately from the transitivity of base changes and from the fact that the composite

$$k \longrightarrow k' \xrightarrow{\phi} k$$

is F^n.

Continuation of the proof of 4.3.1.

(i)⇒(iii). We argue by induction on $\dim G$. If $\dim G = 0$, then, because G is connected, it is radicial, and 3.9(i) applies. If $\dim G > 0$, there exists an integer $m \geq 0$ such that the quotient G/F_mG is a smooth group (Appendix II, 3.1), plainly connected and nonzero. Applying 4.1.1(i)⇒(iii) to this quotient, we see that G has an algebraic subgroup G' which is characteristic and connected, and for which the quotient

$$G'' = G/G'$$

is a form of $(\mathbb{G}_a)^r$, where $r > 0$. By 4.1.5, if K is a perfect closure of k, then

$$G''_K \xrightarrow{\sim} (\mathbb{G}_{a,K})^r.$$

Since G'' is of finite type over k, there exists a finite radicial extension k' of k such that

$$G''_{k'} \xrightarrow{\sim} (\mathbb{G}_{a,k'})^r$$

(Exposé $\mathrm{VI_B}$, §10). Choose $n > 0$ such that $(k')^{p^n} \subset k$. Retaining the notation of 4.3.2, we deduce a k-isomorphism of algebraic groups

$$(\mathbb{G}_{a,k})^r = \big((\mathbb{G}_{a,k'})^r\big)^{\phi} \xrightarrow{\sim} (G'')^{(p^n)}.$$

Now consider the Frobenius homomorphism relative to G'' (Exposé $\mathrm{VII_A}$, §4):

$$F^n : G'' \longrightarrow (G'')^{(p^n)}.$$

Since G'', and hence also $(G'')^{(p^n)}$, is smooth over k, while F^n is radicial, F^n is an epimorphism for the fpqc topology. Thus $(G'')^{(p^n)}$ is identified with $G''/F_n(G'')$, and we have shown that

$$G''/F_n(G'') \xrightarrow{\sim} (\mathbb{G}_a)^r$$

as algebraic groups. The inverse image G'_n of $F_n(G'')$ in G is a connected, characteristic subgroup of G, of dimension strictly less than that of G; the induction hypothesis therefore applies to G'_n.

(i)⇒(iv). This follows from 2.4(i).

(iv)⇒(i). Regard G as an extension of a smooth connected group G'' by a radicial group G' (Appendix II, 3.1). The group G' is unipotent by 4.2.1(iv)⇔(i). It is therefore enough to prove that G'' is unipotent, and for this it is enough to show that $G''_{\overline{k}}$ contains no subgroup isomorphic to $\boldsymbol{\mu}_p$, by 4.1.1(i)⇔(iv). If $G''_{\overline{k}}$ contained such a subgroup, it would lift to $G_{\overline{k}}$ by the result 5.1, already used above, whose proof is given in §5. This would contradict (iv).

Continuation of the proof of 4.3.1.

(i)⇒(v). This follows from 3.7.

(v)⇒(vi). Indeed, since

$$(\operatorname{ad} X)^{p^r} = \operatorname{ad}\bigl(X^{(p^r)}\bigr) \qquad \text{(Exposé VII}_\mathrm{A}\text{, 5.2)},$$

$\operatorname{ad} X$ is nilpotent when $\mathfrak{g}$ is unipotent; hence $\mathfrak{g}$ is nilpotent by Engel's theorem. Moreover, if $\mathfrak{g}$ is unipotent, then its center plainly is as well, and the reductive part of that center is therefore trivial (4.2.2).

(vi)⇒(iv). Indeed, if $G_{\overline{k}}$ contained a subgroup isomorphic to $\boldsymbol{\mu}_p$, its Lie algebra would contain a nonzero element X such that

$$X^{(p)} = X \qquad \text{(Appendix II, 2.1)},$$

and hence $X^{(p^r)} = X$ for every $r > 0$. Since $\operatorname{ad} X$ is nilpotent because $\mathfrak{g}$ is nilpotent, and

$$(\operatorname{ad} X)^{p^r} = \operatorname{ad}\bigl(X^{(p^r)}\bigr),$$

we necessarily have $\operatorname{ad} X = 0$. Thus X belongs to the reductive part of the center of $\mathfrak{g}_{\overline{k}}$, contradicting (vi).

(i)⇒(vi ter). This follows from 2.4(i) and 3.5(i)⇒(iii).

(vi ter)⇒(vi bis). Indeed, if G is nilpotent, then so is its subgroup ${}_FG$. Appendix II, 2.2 shows that ${}_FG$ is nilpotent if and only if

$$\operatorname{Lie}({}_FG) = \operatorname{Lie} G \qquad \text{(Exposé VII}_\mathrm{A}\text{)}$$

is nilpotent.

(vi bis)⇒(vi). Let Z be the identity component of the center of G, and let $\mathfrak{r}$ be the reductive part of the center of $\mathfrak{g}$. We must prove that $\mathfrak{r} = 0$. It is immediate that $\mathfrak{r}$ is a characteristic p-Lie subalgebra of

$$\mathfrak{g} = \operatorname{Lie}({}_FG);$$

that is, it is stable under the functor $\operatorname{Aut}_{p\text{-Lie}}(\mathfrak{g})$. Consequently $\mathfrak{r}$ is the Lie algebra of a characteristic radicial subgroup R of ${}_FG$ (Appendix II, 2.2). The last assertion of 4.2.2 and Appendix II, 2.1 show moreover that R is a form of $(\boldsymbol{\mu}_p)^r$. Since R is characteristic in ${}_FG$, which is itself characteristic in G (Appendix II, 1), it is invariant in G, and hence central because G is connected (Exposé IX, 5.5). By (vi bis), $R = 0$, and therefore $\mathfrak{r} = 0$.

4.4. Étale Groups

The following proposition is an easy consequence of the Sylow theorems and of the structure of finite q-groups (cf. J.-P. Serre, *Corps locaux*, Chapter IX, §1).

Proposition 4.4.1.— *Let G be a finite étale algebraic group over an algebraically closed field. Then G is unipotent if and only if, for every prime number q distinct from the characteristic p of the field, G contains no subgroup isomorphic to $\boldsymbol{\mu}_q$.*

4.5. Abelian Varieties

Let G be an abelian variety over an algebraically closed field k. The following conditions are equivalent:

(i) G is unipotent.

(ii) $G = 0$.

(iii) There exists an integer n, relatively prime to the characteristic p of k, such that G contains no subgroup isomorphic to $\boldsymbol{\mu}_n$.

Indeed, if G is an abelian variety of dimension d, it is known (cf. S. Lang, *Abelian Varieties*, Chapter IV, §3, Theorem 6) that the group ${}_nG(k)$, for n prime to p, is isomorphic to

$$(\mathbb{Z}/n\mathbb{Z})^{2d},$$

and hence to $(\boldsymbol{\mu}_n)^{2d}$. Thus (iii)$\Rightarrow$(ii), while (ii)$\Rightarrow$(i)$\Rightarrow$(iii) are clear.

4.6. The General Case

If G and H are two algebraic groups defined over an algebraically closed field k, we shall denote by $P(G, H)$ the property

"there is no algebraic subgroup of G isomorphic to H."

We then obtain the following characterizations of unipotent groups.

Theorem 4.6.1.— *Let G be an algebraic group defined over an algebraically closed field k of characteristic p. Then:*

(i) If G is smooth, affine, and connected, then

$$\begin{aligned} G \text{ is unipotent} &\Longleftrightarrow \text{there exists } n > 1 \text{ such that } P(G, \boldsymbol{\mu}_n) \text{ holds} \\ &\Longleftrightarrow P(G, \mathbb{G}_m) \text{ holds.} \end{aligned}$$

(ii) If G is smooth and connected, then

$$G \text{ is unipotent} \Longleftrightarrow \text{there exists } n \text{ prime to } p \text{ such that } P(G, \boldsymbol{\mu}_n) \text{ holds.}$$

(iii) If G is smooth, then

$$G \text{ is unipotent} \Longleftrightarrow P(G, \boldsymbol{\mu}_n) \text{ holds for every prime } n \neq p.$$

(iv) If G is affine and connected and $p > 0$, then

$$G \text{ is unipotent} \Longleftrightarrow P(G, \boldsymbol{\mu}_p) \text{ holds.}$$

(v) *If G is connected and $p > 0$, then*

$$G \text{ is unipotent} \iff \text{there exists } n \text{ prime to } p \text{ such that } P(G, \boldsymbol{\mu}_n) \text{ and } P(G, \boldsymbol{\mu}_p) \text{ both hold.}$$

(vi) *For an arbitrary algebraic group G,*

$$G \text{ is unipotent} \iff P(G, \boldsymbol{\mu}_n) \text{ holds for every prime } n.$$

Proof. Assertion (i) follows from 4.1.1, and (iv) from 4.3.1. We shall prove (vi); assertions (ii), (iii), and (v) are proved in the same way and are left to the reader.

Let G be an algebraic group. If G is unipotent, then $P(G, \boldsymbol{\mu}_n)$ holds for every $n > 1$ by 2.4(i). Conversely, suppose that $P(G, \boldsymbol{\mu}_n)$ holds for every prime n, and let us prove that G is unipotent. Write G^0 for the identity component of G. If G^0 is smooth, a classical theorem of Chevalley* shows that G^0 is an extension of an abelian variety A by a smooth connected affine group L. If G is not smooth—which entails $p > 0$ (Exposé $\mathrm{VI_B}$, 1.6.1)—there is an integer $n > 0$ such that

$$G'' = G^0/{}_{F^n}(G)$$

is smooth (Appendix II, 3.1). Thus G'' is an extension of an abelian variety A by a smooth connected linear group L''. Let L be the inverse image of L'' in G^0. It is still affine and connected, since ${}_{F^n}(G)$ is radicial. In every case, therefore, G has a composition series

$$0 \subset L \subset G^0 \subset G$$

such that L is affine and connected, $G^0/L = A$ is an abelian variety, and G/G^0 is an étale group.

If $P(G, \boldsymbol{\mu}_n)$ holds, then a fortiori $P(L, \boldsymbol{\mu}_n)$ holds, and hence L is unipotent (4.1.1 and 4.3.1). If $A \neq 0$, there are a prime number n and a subgroup of A isomorphic to $\boldsymbol{\mu}_n$ (4.5). By 5.1 below, this subgroup lifts to G, contradicting $P(G, \boldsymbol{\mu}_n)$. Hence $A = 0$. Finally, if G/G^0 is not unipotent, there are an integer q and a subgroup of G/G^0 isomorphic to $\boldsymbol{\mu}_q$ (4.4.1). As above, it follows that $P(G, \boldsymbol{\mu}_q)$ does not hold. Thus G/G^0 is unipotent, and consequently so is G.

5. Extensions of Groups of Multiplicative Type by Unipotent Groups

5.1. Statement of the Theorem

Definition 5.1.0.— *Let k be a field and G an algebraic k-group. Following the terminology introduced by Rosenlicht (*Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica 61 (1963)), we shall say that G is k-*solvable *if it satisfies the following equivalent conditions:*

(i) *G has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.*

(ii) *G has a characteristic composition series (G_i) such that the successive quotients G_i/G_{i+1} are commutative and themselves have composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.*

* *Séminaire Bourbaki*, no. 145, 1956/57.

The implication (i)⇒(ii) is proved in *loc. cit.* In fact, for the composition series (G_i) one may take the series introduced in the proof of 3.9(ii).

Theorem 5.1.1.— *Let k be a field, U a unipotent algebraic k-group, H a k-group of multiplicative type, and E an algebraic k-group which is an extension of H by U, so that there is an exact sequence*

$$1 \longrightarrow U \longrightarrow E \longrightarrow H \longrightarrow 1.$$

Then:

(i) The extension E is trivial in each of the following cases:

  (a) k is algebraically closed;

  (b) k is perfect and one of the groups U or H is connected;

  (c) U is k-solvable;

  (d) U is smooth and H is connected.

(ii) If H' and H'' are two lifts of H in E, then they are conjugate by an element of $U(k)$ in each of the following cases:

  (a) k is algebraically closed and H is smooth;

  (b) k is algebraically closed and U is smooth.

(During the proof we shall indicate other cases in which the conclusion of (ii) is true without assuming that k is algebraically closed.)

If U is an algebraic group (respectively, a commutative algebraic group) defined over a field k, we denote by $H^1(k, U)$ (respectively, by $H^i(k, U)$, $i \geq 0$) the pointed set (respectively, the i-th group) of Galois cohomology of k with values in U; see J.-P. Serre, *Cohomologie galoisienne*, Lecture Notes in Mathematics no. 5, Springer.

If S is a prescheme, H a commutative S-group prescheme, and G an S-group prescheme acting on H, we denote by $H^i(G, H)$[8] the i-th Hochschild cohomology group of G with values in H (Appendix I, 1).

To prove 5.1.1, we shall proceed in several steps.

5.2. Proof of 5.1.1(i) and (ii) when U is smooth and H is étale

Lemma 5.2.1.— *With the notation of 5.1.1, suppose that H is étale. The canonical morphism $E \to H$ has a section $s : H \to E$, defined over k, in each of the following cases:*

(a) k is algebraically closed;

(b) k is perfect and U is smooth and connected;

(c) U is k-solvable.

Assertion (a) follows simply from the surjectivity of $E(k) \to H(k)$. We have (b)⇒(c) by 4.1.2(b) and 2.3 bis, so it is enough to treat case (c). Let x be a point of H. It is an open and closed subset of H, and the induced subscheme is isomorphic to $\operatorname{Spec} K$, where K is a finite separable extension of k. Let X be the inverse image in E of the scheme x. The

[8] Editor's note: H^i_0 has been replaced by H^i, in agreement with the notation of Appendix I and Exposé I.

K-scheme X is a principal homogeneous K-space under U_K. But U_K has a composition series whose successive quotients are isomorphic to $(\mathbb{G}_a)_K$; hence

$$H^1(K, U_K) = 0$$

(J.-P. Serre, *Cohomologie galoisienne*, Chapter III, Proposition 6). Consequently X is trivial and has a K-rational point. Thus, over every point x of H, we obtain a k-section of $E \to H$, and these give a section $H \to E$.

Lemma 5.2.3.— [9] *With the notation of 5.1.1, suppose that H is étale. Then:*

(i) The extension E is trivial in each of the following cases:

(a) U is commutative and $E \to H$ has a section;

(b) k is algebraically closed;

(c) k is perfect and U is connected;

(d) U is k-solvable.

Moreover, in each of the preceding cases, two lifts H' and H'' of H in E are conjugate by an element of $U(k)$.

(ii) Let R be the k-functor

$$(\mathrm{Sch}\,/k)^\circ \longrightarrow \mathrm{Ens}$$

for which, for every k-prescheme S, $R(S)$ is the set of lifts of H_S in E_S. If U is commutative, then R is representable by a nonempty affine k-scheme. The group U acts on R by inner automorphisms, and this action makes R a homogeneous space under U for the fpqc topology (cf. Exposé IV).

Proof of (i). We shall reduce cases (b), (c), and (d) to case (a).

Case (b). Since U has a characteristic composition series with commutative successive quotients (3.5, (i)⇔(iv)), we immediately reduce to the case in which U is commutative. Moreover, $E \to H$ has a section by 5.2.1(a), so we are reduced to case (a).

Case (d). We argue in the same way, using 5.1.0(ii) and 5.2.1(c).

Case (c). Let E_{red} be the reduced subgroup associated with E. Since H is smooth, E_{red} is an extension of H by

$$U_0 = U \cap E_{\mathrm{red}},$$

and we may replace E by E_{red}. The group U, and therefore U_0, is connected, and U_0 is plainly the identity component of E_{red}; in particular it is smooth. Thus U_0 is smooth and connected and k is perfect, so U_0 is k-solvable by 4.1.4(b), reducing us to case (d).

The preceding reductions show that in cases (b), (c), and (d) we may suppose that U has a characteristic composition series (U_i) such that U_i/U_{i+1} is commutative and the maps

$$U_i(k) \longrightarrow (U_i/U_{i+1})(k)$$

are surjective. (In cases (c) and (d), the latter assertion follows from $H^1(k, \mathbb{G}_a) = 0$.) An immediate dévissage then shows that it is enough to prove the conjugacy of two lifts H' and H'' in E when U is commutative. In short, it is enough to prove (i)(a). In that case the

[9] Editor's note: there is no item numbered 5.2.2.

extension E is trivial provided that $H^2(H,U)=0$ (Appendix I, 3.1), and H' and H'' are conjugate provided that $H^1(H,U)=0$. We now use the following lemma.

Lemma 5.2.4.— *Let S be a prescheme, U a commutative S-group prescheme, and H an étale finite S-group prescheme of rank n acting on U. Then the groups $H^i(H,U)$, for $i>0$, are annihilated by n in each of the following cases:*

(a) H is a constant S-group (Exposé I, 4.1);

(b) S is the spectrum of a field.

Proof of (a). By hypothesis, H is the constant group associated with an ordinary group $\{H\}$ of order n. It is clear[10] that $H^i(H,U)$ is isomorphic to the i-th cohomology group

$$H^i(\{H\},U(S))$$

of the group $\{H\}$ with values in the ordinary group $U(S)$. It is classical that these groups are annihilated by n (J.-P. Serre, *Corps locaux*, Chapter VIII, Proposition 4, Corollary 1).

Proof of (b). Let $x\in H^i(H,U)$, where $i>0$, and represent it by an i-cocycle

$$f : H^{(i)} \longrightarrow U,$$

where $H^{(i)}$ denotes the product over k of i copies of H. If K is a finite Galois extension of k which splits H, assertion (a) shows that nf_K is a coboundary. More precisely, a direct computation shows that, if one defines the K-morphism

$$F_K : H_K^{(i-1)} \longrightarrow U_K$$

by

$$F_K(h_1,\dots,h_{i-1}) = \sum_{h\in H(K)} f_K(h_1,\dots,h_{i-1},h),$$

then, up to sign,

$$d(F_K)=nf_K, \qquad d \text{ denoting the coboundary operator.}$$

An immediate Galois argument shows that F_K descends to a k-morphism $F : H^{(i-1)}\to U$, and consequently

$$d(F)=nf.$$

Corollary 5.2.4 bis.— *With the notation of 5.2.4, suppose in addition that U is flat and of finite presentation over S, with unipotent fibres, and that n is prime to the residue characteristics of S. Then, in cases (a) and (b) above,*

$$H^i(H,U)=0 \qquad (i>0).$$

Proof. It is enough to show that the n-th power map on U is an isomorphism, for it will then follow that multiplication by n on $H^i(H,U)$ is both an isomorphism and the zero morphism, and hence that $H^i(H,U)=0$. Under the hypotheses made on U, it is enough to check that the n-th power map is an isomorphism on the fibres of U (EGA IV, 17.9.5). We are thus reduced to the case in which S is the spectrum of a field k of characteristic p. Since

[10] Editor's note: this is Proposition III.6.4.2 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

$(n, p) = 1$, the n-th power map on U is étale (Exposé VII); it is also a monomorphism (2.4(i)), and hence an open immersion (EGA IV, 17.9.1). This already shows that its restriction to the identity component U^0 is an isomorphism. Since U/U^0 is a finite p-group, the proof is complete.

This completes the proof of 5.2.3(i)(a), since an étale group of multiplicative type H has order prime to p.

Proof of 5.2.3(ii). It is clear that R is a sheaf for the fpqc topology. Moreover, in view of descent for affine schemes, the assertions of 5.2.3(ii) are local for the fpqc topology. We may therefore suppose that k is algebraically closed. The group H is then split and the extension E is trivial by (i)(b); let H' be a lift of H in E. For every k-prescheme S and every lift H'' of H_S in E_S, the groups H'_S and H'' are conjugate by an element of $U(S)$, since

$$H^1(H_S, U_S) = 0$$

by 5.2.4 bis.

Let $U^{H'}$ be the sheaf of invariants of U under H'. It is representable by an algebraic subgroup of U (Exposé VIII, 6.5(d)). The preceding remarks show that the k-morphism

$$U \longrightarrow R, \qquad u \longmapsto \mathrm{int}(u)H' \quad (u \in U(S))$$

induces a k-isomorphism

$$U/U^{H'} \xrightarrow{\sim} R.$$

This proves that R is representable and affine (2.1).

Remark 5.2.5.— *The assertions of 5.2.3(ii) can also be proved when U is not commutative, but we shall not need this in order to prove 5.1.1.*

5.3. The Case Where H Is Smooth

Proposition 5.3.1.— *The assertions of 5.2.3(i) remain true when the hypothesis "H is étale" is replaced by "H is smooth."*

Proceeding as in the proof of 5.2.3(i), we reduce to the case in which U is also commutative.

Let N' be the set of positive integers prime to p, ordered by divisibility. For every $n \in N'$, the group ${}_nH$ is étale, and the family of groups ${}_nH$, $n \in N'$, is schematically dense in H, since H is smooth (Exposé IX, 4.10). Let E_n be the inverse image of ${}_nH$ in E, so that E_n is an extension of ${}_nH$ by U, and let R_n be the k-functor of lifts of ${}_nH$ in E_n (cf. 5.2.3(ii)). If n divides m, there is plainly a natural k-morphism

$$R_m \longrightarrow R_n,$$

so the R_n form an inverse system of k-functors.

By 5.2.3(ii), R_n is representable by a nonempty affine k-scheme. Since a filtered direct limit of nonzero rings is nonzero, the functor

$$R = \varprojlim_n R_n$$

is representable by a nonempty affine k-scheme (EGA IV, 8 and 1.9.1). There are therefore an extension K of k and a point $u \in R(K)$. The image u_n of u in $R_n(K)$ corresponds to a lift H'_n of $({}_nH)_K$ in $(E_n)_K$. By construction,

$$H'_n = {}_n(H'_m) \qquad \text{if } n \text{ divides } m.$$

Put

$$U_n = (U_K)^{H'_n}.$$

The choice of H'_n identifies $(R_n)_K$ with U_K/U_n. The family of groups H'_n is filtered increasing, and hence the family U_n is filtered decreasing. It is therefore stationary for sufficiently large n, since U_K is noetherian. It follows that the family $(R_n)_K$ is stationary, and consequently so is the family R_n. In short,

$$R_n = R$$

for sufficiently large n.

Under the hypotheses in force, 5.2.3(i) shows that $R_n(k)$ is nonempty. We may therefore find a coherent system of lifts H'_n of ${}_nH$, for $n \in N'$. Let H' be the smallest closed subscheme of E containing H'_n for every n (Exposé $\mathrm{VI_B}$, §7). The argument of Exposé XV, 4.6 shows that H' is a smooth commutative algebraic group whose formation commutes with extension of the base field. To prove that H' is a lift of H in E, we may therefore suppose that k is algebraically closed. According to BIBLE 4, Theorem 4, H' is then the direct product of a smooth group M of multiplicative type and a unipotent group V. The groups H'_n are necessarily contained in M by 2.4; the definition of H' then gives $H' = M$. Thus H' is of multiplicative type, and hence

$$H' \cap U = 0.$$

The morphism $H' \to H$ is consequently a monomorphism, and the density theorem of Exposé IX, 4.10 shows that it is an epimorphism. It is therefore an isomorphism.

Now let H' and H'' be two lifts of H in E. For each $n \in N'$, let

$$T_n = \mathrm{Transp}({}_nH', {}_nH'')$$

be the transporter of ${}_nH'$ into ${}_nH''$. It is representable by a closed subscheme of U (Exposé VIII, 6.5(e)). The T_n form a filtered decreasing family of closed subschemes of U, and they are nonempty by 5.2.3(i)(a). Let T be their stationary value. Under the hypotheses of 5.2.3(i), $T_n(k)$ is nonempty. Thus there is an element $u \in U(k)$ such that

$${}_nH'' = \mathrm{int}(u)({}_nH') \qquad \text{for every } n \in N'.$$

It follows that

$$H'' = \mathrm{int}(u)H'$$

by Exposé IX, 4.8(b).

5.4. The Case Where U Is Radicial

Proposition 5.4.1.— *If k is a perfect field of characteristic $p > 0$, and if U is radicial, then the extension E of 5.1.1 is trivial.*

Using a characteristic composition series of U, we may restrict ourselves to the case

$$U = (\boldsymbol{\alpha}_p)^r$$

by 3.9.

It follows from Appendix II, 2.2 and 2.1 that there are isomorphisms of k-functors

$$\mathrm{Aut}_{k\text{-gr}}((\boldsymbol{\alpha}_p)^r) \xrightarrow{\sim} \mathrm{Aut}_{p\text{-Lie}}(\mathrm{Lie}((\boldsymbol{\alpha}_p)^r)) \xrightarrow{\sim} \mathrm{GL}(\mathrm{Lie}((\boldsymbol{\alpha}_p)^r)).$$

Let V be a k-vector space of rank r, let $\mathbf{W}(V)$ be the corresponding vector k-group (Exposé I, 4.6), and identify $(\boldsymbol{\alpha}_p)^r$ with ${}_FV$. The preceding remarks show that the action of H on ${}_FV$ defined by the extension E comes from a linear representation v of H on V. Consider the exact sequence

$$0 \longrightarrow {}_FV \longrightarrow V \xrightarrow{F} V^{(p)} \longrightarrow 0, \tag{*}$$

where F is the Frobenius morphism.

The sequence $(*)$ is an exact sequence of H-groups, provided that H acts on $V^{(p)}$ through the linear representation

$$H \xrightarrow{v} \mathrm{GL}(V) \xrightarrow{F} \mathrm{GL}(V^{(p)}).$$

Since k is perfect, the morphism $F : V \to V^{(p)}$ induces a surjective map

$$V(k) \longrightarrow V^{(p)}(k).$$

It follows from Appendix I, 2.1 that the exact sequence $(*)$ defines an exact sequence

$$H^1(H, V^{(p)}) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(H, {}_FV) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(H, V). \tag{**}$$

We shall show that

$$\mathrm{Ext}_{\mathrm{alg}}(H, V) = 0.$$

Let E' be an algebraic group which is an extension of H by V:

$$1 \longrightarrow V \longrightarrow E' \xrightarrow{h} H \longrightarrow 1.$$

The scheme E' is a torsor over H under $\mathbb{G}_a^r$, and hence defines an element of

$$H^1(H, \mathcal{O}_H^r)$$

in coherent-sheaf cohomology. Since H is affine,

$$H^1(H, \mathcal{O}_H^r) = 0$$

by EGA III, §1. Thus $E' \to H$ has a section. Consequently, $\mathrm{Ext}_{\mathrm{alg}}(H, V)$ is isomorphic to $H^2(H, V)$ by Appendix I, 3.1. But

$$H^i(H, V) = H^i(H, \mathbf{W}(V)) = 0 \qquad (i > 0)$$

by Exposé IX, 3.1. The exact sequence $(**)$ now gives

$$\mathrm{Ext}_{\mathrm{alg}}(H, {}_FV) = 0,$$

and hence E is a trivial extension.

5.5. Proof of 5.1.1(i)

If k has characteristic 0, then U is k-solvable by 4.1.3 and H is smooth. It follows from 5.3.1 and 5.2.3(d) that E is a trivial extension. The same argument shows that two lifts of H in E are conjugate by an element of $U(k)$.

Henceforth we suppose that k is a field of characteristic $p > 0$.

Proof of (i)(b): the case where k is perfect and U is connected. We shall reduce to the case in which U is radicial. Since k is perfect, H_{red} is smooth; let E' be its inverse image in E. By 5.3.1 and 5.2.3(i)(c), the extension

$$1 \longrightarrow U \longrightarrow E' \longrightarrow H_{\mathrm{red}} \longrightarrow 1$$

is trivial. Let H_1 be a lift of H_{red} in E. By Appendix II, 3.1, there is an integer $n > 0$ such that

$$E^{(n)} = E/{}_{F^n}(E)$$

is smooth. Let E'' be the algebraic subgroup of E generated by H_1 and ${}_{F^n}(E)$; equivalently, E'' is the inverse image in E of the image of H_1 in $E^{(n)}$. Let H'' be the image of E'' in H. We claim that $H'' = H$.

Indeed, let R be the image of ${}_{F^n}(E)$ in H, so that H'' is generated by R and H_{red}. The group H/R is a quotient of $E^{(n)}$, and is therefore smooth. Consequently the canonical morphism

$$H_{\mathrm{red}} \longrightarrow H/R$$

is an epimorphism. Thus H is generated by R and H_{red}, and hence equals H''. We have obtained an exact sequence

$$1 \longrightarrow U'' = U \cap E'' \longrightarrow E'' \longrightarrow H \longrightarrow 1. \tag{†}$$

But E'' has the same underlying space as H_1, so U'' is radicial. It is plainly enough to prove that the extension (†) is trivial, and this follows from 5.4.1.

Proof of (i)(b): the case where k is perfect and H is connected. The group U is an extension of an étale group by its identity component U^0. The case where U is connected having just been treated, it is enough to consider the case where U is étale. We shall use the following more precise lemma.

Lemma 5.5.1.— *With the notation of 5.1.1, suppose in addition that U is étale. Then:*

(i) If H is connected, it has a unique lift in E, namely the identity component E^0 of E.

(ii) If k is algebraically closed, then E is trivial, and any two lifts of H in E are conjugate by an element of $U(k)$.

Proof of (i). Since formation of the identity component commutes with extension of the base field, we may restrict ourselves to the case where k is algebraically closed. If H is a torus, then E is trivial by 5.3.1 and 5.2.3(i)(b), and E^0 is plainly the unique lift of H. This already proves that in the general case $E^0 \cap U$ is radicial. It is also étale, since U is étale, and is therefore zero. The morphism

$$E^0 \longrightarrow H$$

is consequently a monomorphism; it is flat because E^0 is open in E, and it is surjective because H is connected. It is therefore an isomorphism. If H' is any other lift of H, then H' is connected, hence contained in E^0, and thus equal to E^0.

Proof of (ii). Let H^0 be the identity component of H. By (i), E^0 is the unique lift of H^0 in E. Put

$$E' = E/E^0, \qquad H' = H/H^0.$$

There is then an extension

$$1 \longrightarrow U \longrightarrow E' \longrightarrow H' \longrightarrow 1.$$

Since H' is étale, this extension is trivial by 5.2.3(i)(b). If H'_1 is a lift of H' in E', and H_1 is its inverse image in E, then H_1 plainly lifts H in E. If H_2 is a second lift of H in E,

then it contains E^0. By 5.2.3(i)(b), the image of H_2 in E' is conjugate to H_1' by an element $u \in U(k)$, and hence
$$H_2 = \mathrm{int}(u)H_1.$$

Remark 5.5.2.— *Under the hypotheses of 5.5.1(i), the group E^0 centralizes U.*

Proof of 5.1.1(i)(a). Using the composition series
$$1 \longrightarrow U^0 \longrightarrow U \longrightarrow U/U^0 \longrightarrow 1,$$
assertion (i)(a) follows by combining (i)(b) with 5.5.1(ii).

Before proving 5.1.1(i)(c) and (d), we shall first establish 5.1.1(ii).

Proof of 5.1.1(ii)(a). For want of a satisfactory general statement, we shall describe a number of cases in which, when H is smooth, two lifts of H in E are conjugate.

Proposition 5.6.1.— *With the notation of 5.1.1, suppose in addition that H is smooth, and let H' and H'' be two lifts of H in E. Then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:*

(a) k is algebraically closed.

(b) U is commutative.

(c) k is perfect and U is connected.

(d) U is k-solvable.

(e) The centralizer of H' in U is k-solvable.

(f) The group of multiplicative type H is split by a finite Galois extension K/k of degree prime to p.

Proof. Parts (a), (b), (c), and (d) follow from 5.3.1.

(e) Let Z be the centralizer of H' in U, and let T be the transporter of H' into H''. By (a), T is nonempty; hence T is a principal homogeneous space under Z. The hypothesis on Z therefore implies that T is trivial.

(f) Proceeding as in 5.3.1, we see that it is enough to consider the case where H is étale. Suppose first that H is diagonalizable, defined by an abstract group M of order m prime to p. The two lifts H' and H'' determine a one-cocycle of M with values in $U(k)$, that is, a map
$$h : M \longrightarrow U(k)$$
such that
$$h(mn) = h(m)\,{}^{m}h(n)$$
for every pair m, n of elements of M. The groups H' and H'' are conjugate by an element of $U(k)$ if and only if there exists $a \in U(k)$ such that
$$h(m) = a^{-1}({}^{m}a).$$

Now the abstract group $U(k)$ has a composition series whose successive quotients are commutative and annihilated by a power of p (in view of 5.6.1(c), we may suppose that $p > 0$). It follows at once in this case that h is a coboundary.

Consider now the general case. Again let Z denote the centralizer of H' in U, and T the transporter of H' into H''; thus T is a torsor under Z. By hypothesis there is a finite

Galois extension K/k, with Galois group G of order prime to p, which splits H. By the preceding case, H'_K and H''_K are conjugate by an element of $U(K)$, and therefore T_K is trivial. Consequently T is defined by an element of

$$H^1(G, Z).$$

For the same reasons as above, the hypothesis on G implies that

$$H^1(G, Z) = e;$$

hence T is trivial.

5.7. Proof of 5.1.1(ii)(b)

Lemma 5.7.1.— *Let S be a prescheme, let G be an S-group prescheme, separated and smooth over S, and let H be an S-group of multiplicative type acting on G. Then the S-functor*

$$Z = G^H$$

of invariants of G under H is representable by a closed smooth S-group subprescheme of G.

The representability of G^H by a closed group subprescheme of G follows from Exposé VIII, 6.5(d). Consider the semidirect product

$$K = G \rtimes_S H.$$

The centralizer of H in K is then $Z \times_S H$. To prove that Z is smooth, we must verify the following. Suppose that S is affine, that S_0 is a closed subscheme of S defined by a square-zero ideal, and that $u_0 \in Z(S_0)$. We must lift u_0 to an element of $Z(S)$.

Since G is smooth over S, there is an element $u \in G(S)$ lifting u_0. Let i be the canonical immersion of H into K, and consider the two S-group morphisms

$$H \rightrightarrows K, \qquad i \quad \text{and} \quad j = \operatorname{int}(u)i.$$

Since $u_0 \in Z(S_0)$, we have $i_{S_0} = j_{S_0}$. By Exposé IX, 3.2, there exists $v \in K(S)$, lifting the identity section in $K(S_0)$, such that

$$i = \operatorname{int}(v)j = \operatorname{int}(vu)i.$$

Thus

$$vu = (u', v') \in (Z \times_S H)(S).$$

It follows that $u' \in Z(S)$, and u' lifts u_0.

Lemma 5.7.2.— *Let k be a field and H a k-algebraic group, and let $P(H)$ denote the following property:*

> *For every smooth unipotent k-group U, and every extension E of H by U, any two lifts of H in E are conjugate by an element of $U(k)$.*

Suppose that H is an algebraic group which is an extension of a group H'' of multiplicative type by a group H' of multiplicative type. If $P(H')$ and $P(H'')$ hold, then $P(H)$ holds.

Indeed, let U be a smooth unipotent k-group, let E be an extension of H by U, and let H_1, H_2 be two lifts of H in E. Let H'_1, H'_2 be the corresponding lifts of H'. Since $P(H')$ holds, there exists $u \in U(k)$ such that

$$H'_2 = \operatorname{int}(u)H'_1.$$

After replacing H_1 by $\operatorname{int}(u)H_1$, we may suppose that $H_1' = H_2'$; denote this common group simply by H'. Put

$$E' = \operatorname{Centr}_E(H') = U^{H'}H_1 = U^{H'}H_2.$$

By 5.7.1, $U^{H'} = U'$ is smooth. We then have an extension

$$1 \longrightarrow U' \longrightarrow E' \longrightarrow H \longrightarrow 1.$$

By construction H' is central in E', and hence invariant. Passing to the quotient gives an exact sequence

$$1 \longrightarrow U' \longrightarrow E'' \longrightarrow H'' \longrightarrow 1.$$

Since U' is smooth and $P(H'')$ holds, the images of H_1 and H_2 in E'' are conjugate by an element $u \in U'(k)$. It follows that

$$H_1 = \operatorname{int}(u)H_2.$$

To prove 5.1.1(ii)(b), observe that, since k is algebraically closed, H has a composition series whose successive quotients are smooth or, when $p > 0$, isomorphic to $\boldsymbol{\mu}_p$. Repeated application of 5.7.2 reduces us to the case where H is smooth or $H = \boldsymbol{\mu}_p$. In the first case it is enough to apply 5.1.1(ii)(a). It remains to consider $H = \boldsymbol{\mu}_p$. Since U is smooth, U has a characteristic composition series whose successive quotients are étale or isomorphic to $\mathbb{G}_a^r$, by 3.9. If U is étale, apply 5.5.1. We are finally reduced to

$$H = \boldsymbol{\mu}_p, \qquad U = \mathbb{G}_a^r.$$

We must prove that

$$H^1(\boldsymbol{\mu}_p, U) = 0.$$

The method used in 5.4.1 no longer applies, because $\boldsymbol{\mu}_p$ does not in general act linearly on $\mathbb{G}_a^r$. Fix the following notation: H' denotes a lift of H in E, and

$$\mathfrak{e} = \operatorname{Lie} E, \qquad \mathfrak{u} = \operatorname{Lie} U, \qquad \mathfrak{h} = \operatorname{Lie} H, \qquad \mathfrak{h}' = \operatorname{Lie} H'.$$

Let X be a nonzero element of $\mathfrak{h}$ such that

$$X^{(p)} = X$$

(Appendix II, 3.1), and let X' be its lift in $\mathfrak{h}'$.

Since $\boldsymbol{\mu}_p$ is a radicial group of height 1, the set of lifts of H in E is in bijective correspondence with the set A of elements $Y \in \mathfrak{e}$ such that

$$Y^{(p)} = Y$$

and whose image is X (Appendix II, 2.2). Likewise, if $Y \in A$ corresponds to a lift H'' of H in E, and if $u \in U(k)$, then

$$\operatorname{int}(u)H' = H'' \quad \Longleftrightarrow \quad \operatorname{Ad}(u)X' = Y.$$

Let B be the subset of $\mathfrak{e}$ consisting of the elements $\operatorname{Ad}(u)X'$, with $u \in U(k)$. Plainly $B \subset A$, and it remains only to prove $A = B$ when k is algebraically closed.

(a) *Study of A.* Since $\mathfrak{u}$ is commutative and normalized by $\mathfrak{h}'$, Jacobson's formula (Exposé $\mathrm{VII_A}$, 5.2) gives, for $u \in \mathfrak{u}$,

$$(X' + u)^{(p)} = X'^{(p)} + u^{(p)} + (\operatorname{ad} X')^{p-1}(u) = X' + (\operatorname{ad} X')^{p-1}(u).$$

Consequently,

$$X' + u \in A \iff u = (\operatorname{ad} X')^{p-1}(u).$$

Let

$$\mathfrak{u} = \bigoplus_{n \in \mathbb{Z}/p\mathbb{Z}} \mathfrak{u}_n = \mathfrak{u}_0 \oplus \bigoplus_{n \in (\mathbb{Z}/p\mathbb{Z})^\times} \mathfrak{u}_n$$

be the canonical decomposition of $\mathfrak{u}$ under the action of H'. If $u \in \mathfrak{u}_0$, then $\operatorname{ad} X'(u) = 0$. If $u \in \mathfrak{u}_n$, with $n \in (\mathbb{Z}/p\mathbb{Z})^\times$, then

$$(\operatorname{ad} X')^{p-1}(u) = u.$$

Thus $Y = X' + u$ belongs to A if and only if

$$u \in \bigoplus_{n \in (\mathbb{Z}/p\mathbb{Z})^\times} \mathfrak{u}_n.$$

In particular, A is the set of k-rational points of an irreducible subscheme of $\mathbf{W}(\mathfrak{e})$, of dimension

$$\operatorname{rk} \mathfrak{u} - \operatorname{rk} \mathfrak{u}_0 = \operatorname{rk} \mathfrak{u} - \operatorname{rk} \operatorname{Centr}_{\mathfrak{u}}(X').$$

(b) *Study of B.* We shall need the following lemma.

Lemma 5.7.3 (Rosenlicht).— *Let U be a unipotent algebraic group over a field k, acting on a quasi-affine k-scheme X. Then the orbit of every point $x \in X(k)$ is closed in X. Here, by the orbit of x, we mean the subset of* $\operatorname{ens}(X)$ *which is the image of U under the morphism $g \mapsto g \cdot x$.*

By functoriality, U acts on the affine envelope of X, that is,

$$\operatorname{Spec} \Gamma(X, \mathcal{O}_X),$$

so we may suppose that X is affine. We may then suppose that k is algebraically closed, X is reduced, and U is smooth (observe that U_{red} acts on X_{red} when k is perfect). Let Y be the schematic image of U (EGA I, 9.5.1) under the morphism $g \mapsto g \cdot x$. It is a closed reduced subscheme of X, on which U acts. It follows readily from EGA IV, 1.8.6 that the orbit of x is an open subset Z of Y, dense in Y. We must prove that

$$Z = \operatorname{ens}(Y).$$

Let F be the closed reduced subscheme of Y whose underlying space is $Y \setminus Z$. Thus $F = V(J)$, where J is a nonzero ideal of $\Gamma(Y, \mathcal{O}_Y)$. Since U is smooth, U acts on F, hence on J, and therefore

$$J^U \neq 0$$

by 3.2. If a is a nonzero element of J^U, then a is constant on the orbit Z, and hence on Y, because Z is dense in Y. The ideal J consequently contains k, and $F = \varnothing$.

Apply the preceding lemma to the group U acting on the affine space $\mathbf{W}(\mathfrak{e})$ through the adjoint representation. The orbit of X' is the underlying set of a closed subscheme of $\mathbf{W}(\mathfrak{e})$. Moreover, the stabilizer Z of X' is its centralizer in U, and there is a closed immersion

$$U/Z \longrightarrow \mathbf{W}(\mathfrak{e}).$$

Thus the orbit of X' is the underlying space of a closed subscheme of $\mathbf{W}(\mathfrak{e})$ of dimension

$$\dim U - \dim Z.$$

(c) *End of the proof of 5.1.1(ii)(b).* When k is algebraically closed, the canonical map

$$U(k) \longrightarrow (U/Z)(k)$$

is surjective. By (b), B is therefore the set of k-rational points of a closed subscheme of $\mathbf{W}(\mathfrak{e})$ of dimension $\dim U - \dim Z$. In view of (a), to prove $A = B$ it is enough to show that

$$\operatorname{rk}\mathfrak{u} - \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X') \leq \dim U - \dim \operatorname{Centr}_U(X').$$

Since U is smooth,

$$\dim U = \operatorname{rk}\mathfrak{u}.$$

On the other hand, Exposé II, 5.3.3 gives

$$\dim \operatorname{Centr}_U(X') \leq \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X'),$$

which proves the assertion. Notice in fact that

$$\dim \operatorname{Centr}_U(X') = \operatorname{rk}\operatorname{Centr}_{\mathfrak{u}}(X');$$

this also proves once more that Z is smooth, as in 5.7.1.

5.8. End of the proof of 5.1.1(i)

It remains to prove (i)(c) and (i)(d).

Proof of 5.1.1(i)(d): U smooth and H connected. We shall need the following lemma.

Lemma 5.8.1.— *With the notation of 5.1.1, suppose that U is smooth and H is radicial, and let H_1 be an algebraic subgroup of H which has a lift H_1' in E. Then H has a lift H' in E which contains H_1'.*

Proof. By induction on the height of H/H_1, we may suppose that H/H_1 has height 1. Put

$$C' = \operatorname{Centr}_E(H_1').$$

We claim that the canonical morphism $C' \longrightarrow H$ is an epimorphism. To prove this, we may suppose that k is algebraically closed. The extension E is then trivial, by 5.1.1(i)(a). Let H'' be a lift of H in E, and let H_1'' be the inverse image of H_1 in H''. The groups H_1' and H_1'' are two lifts of H_1 in E; hence they are conjugate by an element of $U(k)$, since k is algebraically closed and U is smooth, by 5.1.1(ii)(b). Consequently, to prove the assertion about C', it is enough to prove it for

$$C'' = \operatorname{Centr}_E(H_1'').$$

But C'' contains H'', so the assertion is clear.

It also follows from 5.7.1 that $C' \cap U$ is smooth. We may therefore replace E by C', and thus suppose that H_1' is central. Passing then to the quotient by H_1', we may suppose that $H_1 = 0$ and that H has height 1.

Since U is smooth, we have the following exact sequence of p-Lie algebras (Appendix II, 3.2):

$$(*) \qquad 0 \longrightarrow \operatorname{Lie} U \longrightarrow \operatorname{Lie} E \longrightarrow \operatorname{Lie} H \longrightarrow 0.$$

In view of Appendix II, 2.2, to say that E is trivial is equivalent to saying that $(*)$ is a trivial extension of p-Lie algebras. Suppose that $H \neq 0$, and hence that $\operatorname{Lie} H \neq 0$. Suppose also that we have found a nonzero p-Lie subalgebra

$$\mathfrak{h}_1 \subset \mathfrak{h} = \operatorname{Lie} H$$

which lifts to a p-Lie subalgebra $\mathfrak{h}_1' \subset \operatorname{Lie} E$. By the cited result, there is a subgroup H_1 of H, with $\operatorname{Lie} H_1 = \mathfrak{h}_1$, and a lift H_1' of H_1 in E, with $\operatorname{Lie} H_1' = \mathfrak{h}_1'$. Applying once more the reduction just described, we are brought back to the same problem with H replaced by H/H_1. Since H has height 1,

$$\operatorname{Lie}(H/H_1) = \operatorname{Lie} H/\operatorname{Lie} H_1$$

(loc. cit.), and consequently

$$\operatorname{rk}\operatorname{Lie}(H/H_1) \leq \operatorname{rk}\operatorname{Lie} H - 1.$$

Thus, by induction on the rank of $\operatorname{Lie} H$, it is enough, when $\mathfrak{h} \neq 0$, to find a nonzero Lie subalgebra of $\operatorname{Lie} H$ which lifts to $\operatorname{Lie} E$. This follows from the next lemma.

Lemma 5.8.2.— *Let k be a field of characteristic $p > 0$, and let*

$$\varphi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

be a surjective morphism of finite-rank p-Lie algebras over k. Then:

(i) If $\mathfrak{h}_{\overline{k}}$ is reductive (4.2.2) and nonzero, there exists a reductive Lie subalgebra $\mathfrak{h}_1$ of $\mathfrak{g}$ whose image in $\mathfrak{h}$ is nonzero.

(ii) If k is perfect and u is a unipotent element of $\mathfrak{h}$, that is, if $u^{(p^n)} = 0$ for some n, then u lifts to a unipotent element of $\mathfrak{g}$.

Proof. Let X be a nonzero element of $\mathfrak{h}$ in case (i), and let $X = u$ in case (ii); choose a lift X' of X in $\mathfrak{g}$. The p-Lie subalgebra $\mathfrak{h}'$ of $\mathfrak{g}$ generated by X' is abelian (Exposé VII).

Case (i). The description given in 4.2.2 shows that the reductive part (loc. cit.) of $\mathfrak{h}'_{\overline{k}}$ is already defined over k; denote it by $\mathfrak{r}'$. We claim that the image of $\mathfrak{r}'$ in $\mathfrak{h}$ is nonzero. We may suppose that k is algebraically closed, so that

$$\mathfrak{h}' = \mathfrak{r}' \oplus \mathfrak{u}',$$

where $\mathfrak{u}'$ is the unipotent part of $\mathfrak{h}'$. If the image of $\mathfrak{r}'$ in $\mathfrak{h}$ were zero, then the image of $\mathfrak{h}'$ in $\mathfrak{h}$ would be unipotent, and hence zero because $\mathfrak{h}$ is reductive (2.4(ii)); but by construction that image contains X, a contradiction.

Case (ii). The same argument applies, with the roles of $\mathfrak{r}'$ and $\mathfrak{u}'$ interchanged.

End of the proof of 5.8.1. Suppose that $H \neq 0$. By 5.8.2 there exists a nonzero reductive p-Lie subalgebra $\mathfrak{h}_1' \subset \operatorname{Lie} E$. Since $\operatorname{Lie} U$ is unipotent, by 4.3(i), we necessarily have

$$(\operatorname{Lie} U) \cap \mathfrak{h}_1' = 0.$$

Thus $\mathfrak{h}_1'$ is a lift of a p-Lie subalgebra of $\operatorname{Lie} H$.

End of the proof of 5.1.1(i)(d). By 5.8.1 there is a family of algebraic subgroups H_n' ($n \in \mathbb{N}$) of E such that H_{n+1}' contains H_n', and such that H_n' lifts ${}_{F^n}H$. The descending sequence of subgroups

$$\operatorname{Centr}_E(H_n')$$

is stationary; let C be its stationary value. The center Z of C contains H_n' for every n. Hence the image of Z in H contains ${}_{F^n}(H)$ for every n, and is therefore an open subgroup of H (Exposé $\mathrm{VII_A}$, §4). It is consequently equal to H, since H is connected. To prove that E is trivial, we may thus replace E by Z, and hence suppose that E is commutative.

We shall see in 7.2.1 that E then contains a maximal subgroup M of multiplicative type, whose formation commutes with extension of the base field. Since $E_{\overline{k}}$ is a trivial extension, by 5.1.1(i)(a), and U is unipotent, it is clear that M is the unique lift of H in E.

Proof of 5.1.1(i)(c): U k-solvable. Since $(H/H^0)(\overline{k})$ has order prime to p, duality immediately shows that there exists an integer n such that ${}_nH$ is an étale subgroup and the canonical morphism

$$ {}_nH \longrightarrow H/H^0 $$

is an epimorphism. Thus $H/{}_nH$ is connected. By 5.2.3(d), there exists a lift H' of ${}_nH$ in E. As at the beginning of the proof of 5.8.1, one sees that

$$ C = \mathrm{Centr}_E(H') $$

is a subgroup of E such that $C \longrightarrow H$ is an epimorphism and $C \cap U$ is smooth. Replacing E by C and passing to the quotient by H', we are reduced to the case where U is smooth and H is connected, namely case (i)(d).

This completes the proof of 5.1.1.

5.9. Counterexamples

Let us first indicate a procedure for obtaining nontrivial extensions of a k-group H of multiplicative type by a unipotent k-group U. Suppose that an action of H on U is given, and let

$$ E = U \rtimes H $$

be the semidirect product. Let, moreover, a class in $\mathrm{H}^1(k, U)$ be represented by a 1-cocycle a. The group U acts on E by inner automorphisms, and a therefore defines a k-form of E, denoted by E_a. Suppose in addition that U is commutative. Then U acts trivially on U and on the quotient $E/U = H$, so E_a is again an extension of H by U.

For simplicity, suppose that H is an étale diagonalizable group, so that the Galois group $\mathscr{G} = \mathrm{Gal}(\overline{k}/k)$ acts trivially on $H(\overline{k})$. Its action on $E_a(\overline{k})$ is then given by

$$ {}^g(u, h) = ({}^gu + a(g) - {}^ha(g), h) \qquad (g \in \mathscr{G},\ u \in U(\overline{k}),\ h \in H(\overline{k})). $$

If $h \in H(k)$, and X is its inverse image in E_a, then X is a torsor under U defined by the class of the 1-cocycle

$$ g \longmapsto a(g) - {}^ha(g) $$

of $\mathscr{G}$ with values in U. It follows that if there is a point $h \in H(k)$ for which this cocycle is nontrivial, then the extension E_a is nontrivial. We apply this construction in the following two particular cases.

(a) A nontrivial extension of an étale diagonalizable group H by $\mathbb{Z}/p\mathbb{Z}$. Take

$$ H = (\mathbb{Z}/p\mathbb{Z})^\times, \qquad U = \mathbb{Z}/p\mathbb{Z}, $$

with H acting on U by multiplication. Let k be a field of characteristic p, let K/k be a Galois extension whose Galois group $\mathscr{G}$ is isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and take a nonzero element

$$ a \in \mathrm{H}^1(\mathscr{G}, U) = \mathrm{Hom}(\mathscr{G}, U). $$

Then E_a is the required example.

(b) A nontrivial extension of an étale diagonalizable group H by a smooth connected unipotent group U. Choose a nonperfect field k admitting a k-form U of $\mathbb{G}_a$ such that $\mathrm{H}^1(k, U) \neq$

0. For example (see J.-P. Serre, *Cohomologie Galoisienne*), one may take for k the fraction field of a discrete valuation ring of equal characteristic $p > 0$, and for U the algebraic subgroup of $\mathbb{G}_a \times_k \mathbb{G}_a$ defined by

$$X^p + X + tY^p = 0,$$

where t is a uniformizer of k. Indeed, suppose for simplicity that k contains the $(p-1)$-st roots of unity. There is then an exact sequence of algebraic groups

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow U \longrightarrow \mathbb{G}_a \longrightarrow 0, \qquad (x, y) \longmapsto y,$$

and hence an exact cohomology sequence

$$\mathbb{G}_a(k) \xrightarrow{d} \mathrm{H}^1(k, \mathbb{Z}/p\mathbb{Z}) \longrightarrow \mathrm{H}^1(k, U) \longrightarrow 0,$$

where d associates to x the homogeneous space under $\mathbb{Z}/p\mathbb{Z}$ defined by

$$X^p + X + tx^p = 0.$$

Moreover,

$$\mathrm{H}^1(k, \mathbb{Z}/p\mathbb{Z}) \simeq k/P(k), \qquad P(x) = x^p + x,$$

and therefore

$$\mathrm{H}^1(k, U) \simeq k/\big(P(k) + tk^p\big).$$

If, in addition, $p \neq 2$, then t^2 does not belong to $P(k) + tk^p$; consequently $\mathrm{H}^1(k, U) \neq 0$.

The group $\boldsymbol{\mu}_{p-1}$ acts on U by

$$(h, x, y) \longmapsto (hx, hy).$$

Let $\mathscr{G}$ be the Galois group of the extension K/k defined by

$$X^p + X + t^2 = 0,$$

and let $a \in \mathrm{H}^1(\mathscr{G}, U)$ be the nonzero class described above. It follows immediately that E_a is a nontrivial extension of $\boldsymbol{\mu}_{p-1}$ by U.

(c) A nontrivial extension of $\mathbb{G}_m$ *by* $\boldsymbol{\alpha}_p$. By 5.1.1(i)(b), such an extension can exist only over a nonperfect field. Let k therefore be nonperfect, and let G be the semidirect product of $U = \mathbb{G}_a$ by $H = \mathbb{G}_m$, where $\mathbb{G}_m$ acts on $\mathbb{G}_a$ by homotheties. Since U is invariant in G, so is ${}_FU$. Put

$$G' = G/{}_FU.$$

The group G' is isomorphic to $\mathbb{G}_a \rtimes \mathbb{G}_m$, where $\mathbb{G}_m$ acts on $\mathbb{G}_a$ by

$$(h, u) \longmapsto h^p u.$$

The functor $\mathscr{T}_G$ (respectively $\mathscr{T}_{G'}$) of subtori of G (respectively G') (see Exposé XV) is isomorphic to $\mathbb{G}_a$, and the morphism

$$\mathscr{T}_G \longrightarrow \mathscr{T}_{G'}$$

induced by $G \longrightarrow G'$ identifies with

$$u \longmapsto u^p.$$

It follows that if T' is a subtorus of G' corresponding to a point $x \in k \simeq \mathbb{G}_a(k)$ such that $x^{1/p} \notin k$, then its inverse image E in G is an extension of the torus T' by ${}_FU = \boldsymbol{\alpha}_p$. It has no maximal torus defined over k, and hence is nontrivial. As a subgroup of

$$G = \operatorname{Spec} k[U, T, T^{-1}],$$

the group E is defined by

$$U^p = x - xT^p.$$

Remark 5.9.1.— *This last example shows that a nonsmooth algebraic group defined over a nonperfect field need not possess a maximal torus, and thus answers the question posed in Exposé XIV, 1.5(b).*

(d) A trivial extension with nonconjugate lifts. We now give an example of a trivial extension E of a group H of multiplicative type by a unipotent group U, and of two lifts H' and H'' of H which are not conjugate by an element of $U(k)$.

Take for E the semidirect product of $U = \mathbb{G}_a$ by

$$\boldsymbol{\mu}_p = \operatorname{Spec} k[T]/(T^p - 1),$$

where the action of $\boldsymbol{\mu}_p$ on $\mathbb{G}_a = \operatorname{Spec} k[U]$ is defined by the comorphism

$$U \longmapsto (U + U^p)T - U^p.$$

The centralizer of $\boldsymbol{\mu}_p$ in U is then the étale group Z defined by

$$U + U^p = 0.$$

It follows that if k is not algebraically closed, the canonical map

$$U(k) \longrightarrow (U/Z)(k)$$

is not in general surjective. In view of 5.1.1(ii)(b), two lifts of $\boldsymbol{\mu}_p$ in E therefore need not be conjugate by an element of $U(k)$.

Here is another example, with k algebraically closed of characteristic $p > 0$. Let G be the radicial semidirect product in which $\boldsymbol{\mu}_p$ acts on $\boldsymbol{\alpha}_p$ by homotheties. There is an exact sequence of algebraic groups with $\boldsymbol{\mu}_p$-operators

$$0 \longrightarrow \boldsymbol{\alpha}_p \longrightarrow \mathbb{G}_a \xrightarrow{x \longmapsto x^p} \mathbb{G}_a \longrightarrow 0,$$

where $\boldsymbol{\mu}_p$ acts by homotheties on the first $\mathbb{G}_a$ and trivially on the second. The exact cohomology sequence (Appendix I, Proposition 11) gives

$$0 \longrightarrow \mathbb{G}_a(k) \longrightarrow \mathrm{H}^1(\boldsymbol{\mu}_p, \boldsymbol{\alpha}_p) \longrightarrow \mathrm{H}^1(\boldsymbol{\mu}_p, \mathbb{G}_a).$$

Since the last term is zero by I, 5.3.3, the group $\mathrm{H}^1(\boldsymbol{\mu}_p, \boldsymbol{\alpha}_p)$ is nonzero. Thus two lifts of $\boldsymbol{\mu}_p$ in G need not be conjugate.

6. Extensions of a Unipotent Group by a Group of Multiplicative Type

6.1. Statement of the Theorem

Theorem 6.1.1.— *Let k be a field, U a unipotent algebraic k-group, H a k-group of multiplicative type, and E an algebraic k-group which is an extension of U by H, so that there is an exact sequence*

$$1 \longrightarrow H \longrightarrow E \longrightarrow U \longrightarrow 1.$$

Then the extension E is trivial and there exists a unique lift of U in E in each of the following cases:

(A) The group U is smooth and one of the following conditions holds:

- *(i) U is connected and the canonical morphism $E \longrightarrow U$ admits a section.*
- *(ii) U has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$.*
- *(iii) H is étale.*
- *(iv) k is perfect.*

(B) $U = \boldsymbol{\alpha}_p$ and k is perfect.

(C) E is commutative and k is perfect.

6.2. Proof of 6.1.1(A)

We first establish three lemmas.

Lemma 6.2.1.— *Let S be a prescheme, E an S-group prescheme which is an extension of an S-group prescheme U, with connected fibres, by an S-group H of multiplicative type and finite type (that is, U is the quotient of E by H for the fpqc topology). Then E is a central extension.*

Proof. Since H is commutative, U acts on H by inner automorphisms, through an S-morphism of groups

$$u : U \longrightarrow \mathrm{Aut}_{S\text{-gr}}(H).$$

The functor $\mathrm{Aut}_{S\text{-gr}}(H)$ is representable by an étale S-scheme (Exposé X, 5.10), and consequently its identity section is both an open and a closed immersion. Since U has connected fibres, it follows that u is the unit morphism.

Lemma 6.2.2.— *With the notation of 6.1.1, if E is trivial, there exists a unique lift of U in E.*

Proof. Let U' and U'' be two lifts of U in E. To prove that $U' = U''$, we may suppose that k is algebraically closed, and it is enough to prove that $\mathrm{H}^1(U, H) = 0$. If U is connected, then U centralizes H by 6.2.1, and hence

$$\mathrm{H}^1(U, H) = \mathrm{Hom}_{k\text{-gr}}(U, H) = 0$$

by 2.4(ii).

In the general case, let U_1' denote the unique lift in E of the identity component U^0 of U, and put

$$N = \mathrm{Norm}_E(U_1').$$

If $g \in E(k)$, then $\operatorname{int}(g)U'_1$ is a lift of U^0, and is therefore equal to U'_1; thus $N(k) \supset E(k)$. Moreover, N contains H, by 6.2.1, as well as U'_1, and it follows at once that $N = E$. On passing to the quotient by U'_1, we are reduced to the case in which U is étale. Since k is algebraically closed, in this case $\mathrm{H}^1(U, H)$ identifies with the ordinary cohomology group[11]

$$\mathrm{H}^1(U(k), H(k)),$$

and is therefore zero, since $U(k)$ is a finite p-group and $H(k)$ is uniquely p-divisible.

Corollary 6.2.3.— *To prove that E is trivial, it is enough to prove that it becomes trivial after a finite separable extension of the base field. In particular, we may suppose that H is diagonalizable.*

Lemma 6.2.4.— *In order that every central extension E of U by a diagonalizable k-group H be trivial, it is enough that every central extension of U by $\mathbb{G}_m$ be trivial.*

Proof. Induction on r first shows that the hypothesis implies that E is trivial when $H = \mathbb{G}_m^r$. In the general case, H embeds in $\mathbb{G}_m^r$ for a suitable integer r (this is immediate by duality); put

$$H'' = \mathbb{G}_m^r / H.$$

We obtain the exact sequence (Appendix I, 2.1)

$$\mathrm{Z}^1(U, H'') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(U, H) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(U, \mathbb{G}_m^r) = 0,$$

where U acts trivially on H, H'', and $\mathbb{G}_m^r$. But

$$\mathrm{Z}^1(U, H'') = \mathrm{Hom}_{k\text{-gr}}(U, H'') = 0$$

by 2.4(ii), and hence $\mathrm{Ext}_{\mathrm{alg}}(U, H) = 0$.

Proof of 6.1.1(A)(i). Since $E \longrightarrow U$ admits a section, E defines an element f of $\mathrm{H}^2(U, H)$ (Appendix I, 3.1). We must prove that f is zero. For this it is enough to prove that a 2-cocycle

$$f : U \times_k U \longrightarrow H$$

is a constant morphism, which follows from the next lemma.

Lemma 6.2.5.— *Let U be a smooth connected unipotent algebraic k-group, and H a k-group of multiplicative type. Then every k-morphism of preschemes*

$$f : U \longrightarrow H$$

is constant.

Proof. We may suppose that k is algebraically closed. We argue by induction on $\dim U$. If $\dim U > 0$, then U has a composition series (see 3.9)

$$1 \longrightarrow U' \longrightarrow U \xrightarrow{\pi} U'' \longrightarrow 1, \qquad U' \simeq \mathbb{G}_a, \quad \dim U'' < \dim U.$$

It is enough to show that f factors through U''. Since the graph of the equivalence relation defined by π is smooth over k, and hence reduced, it is enough to show that if $x, y \in U(k)$ have the same image z in $U''(k)$, then $f(x) = f(y)$. Now $\pi^{-1}(z)$ is isomorphic to $\mathbb{G}_a$, so the restriction of f to $\pi^{-1}(z)$ factors through a reduced irreducible component of H, and hence

[11] Editor's note: see, for example, Proposition III.6.4.2 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

through a k-scheme isomorphic to $\mathbb{G}_m^r$. It remains only to note that every morphism from $\mathbb{G}_a$ to $\mathbb{G}_m^r$ is constant, since every invertible regular function on $\mathbb{G}_a$ is constant.

Proof of 6.1.1(A)(ii). By 6.2.1, 6.2.3, and 6.2.4, we may suppose that $H = \mathbb{G}_m$. By hypothesis, U has a composition series

$$U \supset U_1 \supset \cdots$$

such that U_i/U_{i+1} is isomorphic to $\mathbb{G}_a$. Let E_1 be the inverse image of U_1 in E. By induction on $\dim U$, we may suppose that the extension E_1 is trivial; let U_1' be the unique lift of U_1. Arguing as in the normalizer argument of 6.2.2, we find that U_1' is invariant in E. After taking the quotient by U_1', we are reduced to the case $U = \mathbb{G}_a$. The S-scheme E, where $S = U$, is then a torsor under the S-group $\mathbb{G}_m \times_k S$, and therefore admits a section, since $\mathrm{Pic}(S) = 0$ (Exposé VIII, 4.3). The extension E is then trivial by (A)(i).

Proof of 6.1.1(A)(iii) (U smooth, H étale). Suppose first that U is connected. The group $U_{\overline{k}}$ then has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$, so $E_{\overline{k}}$ is trivial by (A)(ii). Since H is étale, the unique lift of $U_{\overline{k}}$ in $E_{\overline{k}}$ is plainly the identity component of $E_{\overline{k}}$; this lift is therefore already defined over k. In the general case, after quotienting by the identity component of E, we are reduced to the case where E is étale, and then to the case where E is completely split (6.2.3). Since $U(k)$ is a p-group and $H(k)$ has order prime to p, we may take the Sylow p-subgroup of $E(k)$ as a lift of $U(k)$.

Proof of 6.1.1(A)(iv) (U smooth, k perfect). If U is connected, it has a composition series whose successive quotients are isomorphic to $\mathbb{G}_a$ (4.1.2(b)), and we apply (A)(ii). In the general case, the preceding argument reduces us to the case in which U is étale, and then to the case in which U is completely split and H is diagonalizable (6.2.3). Using a characteristic composition series of H,

$$H \supset H^0 \supset {}_{F^n}(H) \supset 0,$$

we are reduced to the case in which H is of one of the following three types:

(a) H is étale;

(b) $H = \mathbb{G}_m^r$;

(c) H is radicial.

In case (a) we apply (A)(iii), and in case (c) we apply 1.6. Finally, in case (b), Hilbert's Theorem 90 shows that $E \longrightarrow U$ admits a section, so it is enough to show that

$$\mathrm{H}^2(U, \mathbb{G}_m^r) = \mathrm{H}^2(U(k), \mathbb{G}_m^r(k)) = 0.$$

But $U(k)$ is a finite p-group, whereas $\mathbb{G}_m^r(k)$ is uniquely p-divisible, since k is perfect.

6.3. Proof of 6.1.1(B) and (C)

By 6.2.1, 6.2.3, and 6.2.4, it is enough to prove (B) when $H = \mathbb{G}_m$. We therefore have an exact sequence

$$1 \longrightarrow \mathbb{G}_m \longrightarrow E \longrightarrow \boldsymbol{\alpha}_p \longrightarrow 1.$$

Since $\mathbb{G}_m$ is smooth, it gives an exact sequence of restricted Lie algebras (Appendix II, 3.2)

$$0 \longrightarrow \mathrm{Lie}\,\mathbb{G}_m \longrightarrow \mathrm{Lie}\,E \longrightarrow \mathrm{Lie}\,\boldsymbol{\alpha}_p \longrightarrow 0. \tag{*}$$

The group $\boldsymbol{\alpha}_p$ has height 1, so the extension E is trivial if and only if the exact sequence $(*)$ of restricted Lie algebras splits (Appendix II, 2.2). The algebra $\mathrm{Lie}\,\boldsymbol{\alpha}_p$ is generated by an element $X \neq 0$ such that $X^{(p)} = 0$ (Appendix II, 2.1). It is therefore enough to show that X lifts to an element Z of $\operatorname{Lie} E$ for which $Z^{(p)} = 0$. By 5.8.2(ii), there is a unipotent element Z of $\operatorname{Lie} E$ which lifts X. Since the unipotent part of $\operatorname{Lie} E$ plainly has dimension at most 1, it follows necessarily that $Z^{(p)} = 0$.

Proof of 6.1.1(C). If U_1' is a lift in E of an algebraic subgroup U_1 of U, then U_1' is invariant in E, since E is assumed commutative, and we may pass to the quotient by U_1'. Using a composition series of U (3.5(ii)), the preceding remark reduces us to the case in which U is smooth or equal to $\boldsymbol{\alpha}_p$. But then E is trivial by (A)(iv) and (B).

6.4. Examples of nontrivial extensions of a unipotent group U by a group H of multiplicative type

In view of 6.1.1(A)(iv), the question arises only in characteristic $p > 0$.

(a) $H = \mathbb{G}_m$, and U is a nontrivial form of $\mathbb{G}_a$ (cf. Appendix III, §5).

Let k be a nonperfect field of characteristic 2, and let $u \in k$ be such that $u^{1/2} \notin k$. Consider the affine group E with coordinate ring

$$k\big[X, Y, (X^2 + uY^2)^{-1}\big],$$

whose multiplication is given by the comorphism

$$(X, Y) \longmapsto (XX' + uYY',\ XY' + YX').$$

The group E is smooth, connected, commutative, and of dimension 2; the subscheme $Y = 0$ defines a subgroup $H \simeq \mathbb{G}_m$. The kernel K of the squaring map on E has equation

$$X^2 + uY^2 = 1,$$

and hence has dimension 1. The group K contains the unipotent radical of E, defined over $\bar{k}$, as well as the contribution of H, which is isomorphic to $\boldsymbol{\mu}_2$. Since K is reduced over $\bar{k}$, the unipotent radical of E is not defined over k, and $U = E/H$ does not lift in E.

One checks immediately that U is the form of $\mathbb{G}_a$ with coordinate ring

$$k[V, W]/(V + uV^2 + W^2),$$

the morphism $E \longrightarrow U$ corresponding to the comorphism

$$V \longmapsto \frac{Y^2}{X^2 + uY^2}, \qquad W \longmapsto \frac{XY}{X^2 + uY^2}.$$

(b) $H = \mathbb{G}_m$, $U = \mathbb{Z}/2\mathbb{Z}$, and k is nonperfect of characteristic 2.

Choose k and u as in (a), and consider the subgroup E of GL_2 generated by the element

$$X = \begin{pmatrix} 0 & 1 \\ u & 0 \end{pmatrix} = \begin{pmatrix} u^{1/2} & 0 \\ 0 & u^{1/2} \end{pmatrix} \begin{pmatrix} 0 & u^{-1/2} \\ u^{1/2} & 0 \end{pmatrix}.$$

The group E is an extension of $\mathbb{Z}/2\mathbb{Z}$ by $\mathbb{G}_m$, but this extension is not trivial because the unipotent part of X is not defined over k.

(c) $H = \boldsymbol{\mu}_p$, $U = \boldsymbol{\alpha}_p$, and k is nonperfect.

Let $\mathfrak{e}$ be the commutative restricted Lie algebra generated by two elements X and Y such that

$$X^{(p)} = X, \qquad Y^{(p)} = aX.$$

By Appendix II, 2.2, $\mathfrak{e}$ is the restricted Lie algebra of an algebraic group E which is an extension of $\boldsymbol{\alpha}_p$ by $\boldsymbol{\mu}_p$. This extension is trivial if and only if there exists $b \in k$ such that $b^p = a$, for then

$$(Y - bX)^{(p)} = 0.$$

(d) $H = \boldsymbol{\mu}_2$, $U = \boldsymbol{\alpha}_2 \times \boldsymbol{\alpha}_2$, E noncommutative, and k a field of characteristic 2.

Consider the special linear group $\mathrm{SL}_{2,k}$, and put $E = F(\mathrm{SL}_{2,k})$. The group E is a radicial group of height 1, whose Lie algebra is generated by three elements X, Y, Z satisfying

$$[X, Y] = Z, \qquad [X, Z] = [Y, Z] = 0,$$
$$X^{(p)} = Y^{(p)} = 0, \qquad Z^{(p)} = Z.$$

Consequently, E is a central extension of $U \simeq \boldsymbol{\alpha}_2 \times \boldsymbol{\alpha}_2$ by $\boldsymbol{\mu}_2$. Each factor $\boldsymbol{\alpha}_2$ of U lifts uniquely in E, but U itself does not lift in E, since $[X, Y] \neq 0$.

7. Nilpotent Affine Algebraic Groups

7.1. Extensions of Groups of Multiplicative Type

Proposition 7.1.1.— *Let S be a prescheme, let H and K be two S-group preschemes of multiplicative type and finite type, and let E be an S-group prescheme which is an extension of K by H (that is, K is the quotient of E by H for the fpqc topology). Then E is of multiplicative type in either of the following two cases:*

(a) E is commutative.

(b) The fibres of K are connected.

Proof.

(i) Suppose first that S is the spectrum of a field k. The assertion is local for the fpqc topology, so we may suppose that k is algebraically closed and hence that H and K are diagonalizable. Notice that K acts trivially on H by inner automorphisms: this is clear in case (a), and follows from 6.2.1 in case (b). Induction on the length of a suitable composition series of K reduces us to the case where K is of one of the following three types:

$$\text{(a) } K = \mathbb{G}_m, \qquad \text{(b) } K = \boldsymbol{\mu}_p, \qquad \text{(c) } K = \boldsymbol{\mu}_q, \quad (q, p) = 1,$$

and in case (c), E is commutative. Now embed H in $\mathbb{G}_m^r$. It follows that E embeds in an extension of K by $\mathbb{G}_m^r$, and we may therefore suppose that $H = \mathbb{G}_m^r$.

(a) If $K = \mathbb{G}_m$, then E is a smooth, connected, affine algebraic group (Exposé $\mathrm{VI_B}$, 9.2) of unipotent rank zero, and is therefore a torus.

(b) Let $K = \boldsymbol{\mu}_p$. In this case E is a trivial extension. Indeed, since H is smooth, the canonical morphism

$$\operatorname{Lie} E \longrightarrow \operatorname{Lie} \boldsymbol{\mu}_p$$

is surjective (Appendix II, 3.2), and the assertion follows from 5.8.2(i), in view of Appendix II, 2.2.

(c) Let $K = \boldsymbol{\mu}_q$, where $(q,p) = 1$, and suppose that E is commutative. Again the extension E is trivial. Indeed, let $x \in E(\bar{k})$ lift a generator $\bar{x}$ of $\boldsymbol{\mu}_q(\bar{k})$. The element x^q belongs to $H(\bar{k})$, and hence has the form y^q, with $y \in H(\bar{k})$ (notice that $\mathbb{G}_m^r(\bar{k})$ is q-divisible). Since E is commutative, $y^{-1}x$ is a lift of $\bar{x}$ having order q.

(ii) Consider now the general case. The groups H and K are flat, affine, and of finite presentation over S (Exposé IX, 2.1), and consequently so is E (Exposé $\mathrm{VI_B}$, 9.2). Using the general technique of Exposé $\mathrm{VI_B}$, §10, we reduce to the case where S is noetherian. To prove that E is of multiplicative type, it then suffices to prove that E is commutative and that ${}_nE$ is finite over S for every n (Exposé X, 4.8(b)).

(a) E is commutative. We must verify that the morphism

$$E \times_S E \longrightarrow E, \qquad (x,y) \longmapsto [x,y] = xyx^{-1}y^{-1}$$

factors through the unit section of E, and it is enough to verify this when S is the spectrum of an artinian local ring. But then E is of multiplicative type by (i) and Exposé X, 2.3.

(b) The group ${}_nE$ is finite over S. Indeed, there is an exact sequence

$$0 \longrightarrow {}_nH \longrightarrow {}_nE \longrightarrow {}_nK \xrightarrow{u} H_n,$$

where H_n is the cokernel of the n-th power map on H. We know that H_n is of multiplicative type (Exposé IX, 2.7), and hence separated, and that ${}_nH$ and ${}_nK$ are finite over S. The group $\ker u$ is a closed subgroup of ${}_nK$, and hence is finite over S; consequently ${}_nE$ is finite over S, as an extension of one finite group by another (Exposé $\mathrm{VI_B}$, 9.2).

7.2. Structure of Commutative Affine Algebraic Groups

Theorem 7.2.1.— *Let k be a field and G a commutative affine algebraic k-group. Then:*

(a) G contains a largest subgroup M of multiplicative type. The group M is characteristic in G, the quotient G/M is unipotent, and the formation of M commutes with extension of the ground field k.

(b) If k is perfect, G is the direct product of M and a unipotent algebraic subgroup U, uniquely so.

Proof.

(i) Suppose that k is algebraically closed. When G is smooth, 7.2.1(b) is well known (BIBLE 4, Theorem 4). If G is radicial of height 1, the decomposition of $\operatorname{Lie} G$ described in 4.2.2 corresponds, in view of 4.3.1(v) and Appendix II, 2.2, to a decomposition of G of the kind asserted in 7.2.1(b). In general, G has a composition series whose successive quotients are smooth or radicial of height 1 (Appendix II, 3.1). To prove 7.2.1(b), it therefore suffices to observe the following. Suppose there is an exact sequence of commutative algebraic groups

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0,$$

in which G', respectively G'', is the product of a group of multiplicative type and a unipotent group,

$$G' = M' \cdot U', \qquad G'' = M'' \cdot U''.$$

Then the same is true of G. Indeed, consider the exact sequence

$$0 \longrightarrow G'/U' \longrightarrow G/U' \longrightarrow G'' \longrightarrow 0.$$

By 7.1.1(a), the inverse image of M'' in G/U' is a subgroup M_1 of multiplicative type. The group M_1 lifts to a subgroup M of G by 5.1.1(i)(a). Likewise, now using 6.1.1(C), one proves that there is a unipotent subgroup U of G which is an extension of U'' by U'. Clearly $G = M \cdot U$.

(ii) Let k now be arbitrary. By (i), $G_{\bar{k}}$ is the direct product of a group $M_{\bar{k}}$ of multiplicative type and a unipotent group $U_{\bar{k}}$. Put

$$S = \bar{k} \times_k \bar{k},$$

and let M_1 and M_2 be the two inverse images of $M_{\bar{k}}$ under the two projections $S \rightrightarrows \bar{k}$. The group

$$G_S/M_2 = (G/M_{\bar{k}})_S$$

has unipotent fibres; hence the image of M_1 in G_S/M_2 is trivial by 2.4(i), and $M_1 \subset M_2$. The converse inclusion follows in the same way, so $M_1 = M_2$. By fpqc descent, $M_{\bar{k}}$ therefore descends to an algebraic subgroup M of G. It is clear that M is of multiplicative type, that G/M is unipotent, and that the formation of M is compatible with every extension of k. For every k-prescheme S, every subgroup H of multiplicative type in G_S is contained in M_S. Indeed, its image in the group with unipotent fibres

$$(G/M)_S = G_S/M_S$$

is trivial by 2.5.

Taking H, in particular, to be the transform of M_S under an automorphism of G_S, we conclude that M is characteristic in G.

Finally, if k is perfect, G/M lifts in G to a unipotent group, uniquely so by 6.1.1(C).

Remark 7.2.2.—

(i) If k is not perfect, the unipotent component of $G_{\bar{k}}$ need not be defined over k, as Example 6.4(a) shows.

(ii) In contrast to the multiplicative-type component M, the unipotent component U is not in general characteristic in G, whatever the characteristic of k. Of course, the uniqueness of the decomposition in 7.2.1(b) implies that U is invariant under every k-automorphism of G. But if U and M are such that there exist a k-prescheme S and a nonzero S-homomorphism

$$h : U_S \longrightarrow M_S$$

(cf. 2.6), then one obtains an S-automorphism of G_S,

$$(u, m) \longmapsto (u, h(u) + m),$$

which does not leave U_S invariant.

(iii) If G is finite over k, then under Cartier duality (2.6) the quotient G/M corresponds to the connected component of the identity in the dual $D(G)$ of G.

7.3. Structure of Nilpotent Affine Algebraic Groups

Theorem 7.3.1.— *Let k be a field and G a nilpotent (Exposé VI_B, §8), affine, connected algebraic k-group. Then G has a largest subgroup M of multiplicative type. The group M is central and characteristic, and G/M is a unipotent algebraic group.*

Let Z be the centre of G, and let M be the largest subgroup of multiplicative type in Z, as supplied by 7.2.1. Since Z is characteristic in G, and M is characteristic in Z, the group M is characteristic in G. It remains to show that G/M is unipotent. By induction on the length of the ascending central series of G, this follows from the following more general lemma.

Lemma 7.3.2 (Rosenlicht).— *Let G be a connected algebraic k-group and let Z be its centre. Then the centre Z' of $G' = G/Z$ is unipotent.*

Proof. We may suppose that k is algebraically closed. It is then enough to show that Z' contains no subgroup isomorphic to $\boldsymbol{\mu}_\ell$, for any prime number ℓ, by 4.6.1(vi). Let $\boldsymbol{\mu}_\ell$ be such a subgroup of Z', and let N be its inverse image in G. Since $\boldsymbol{\mu}_\ell$ is central in G', the group N is invariant in G.

(i) Suppose that $(\ell, p) = 1$. We can find an element $x \in N(k)$ and an integer n with the following properties:

  (a) x lifts a generator $\bar{x}$ of $\boldsymbol{\mu}_\ell$;

  (b) $x^{\ell^n} \in Z^0(k)$.

  Indeed, it suffices to choose a lift of $\bar{x}$ whose image in $(N/Z^0)(k)$ belongs to the ℓ-Sylow subgroup. Raising to the ℓ-th power in the commutative group Z^0 is an étale morphism, so $Z^0(k)$ is ℓ-divisible. Consequently, after multiplying x by an element of $Z^0(k)$, we may suppose that $x^{\ell^n} = 0$. The group N is then generated by two commuting commutative groups, namely Z and the group generated by x, and is therefore commutative. The group ${}_{\ell^n}N$ is of multiplicative type and is characteristic in N; hence it is invariant in G, and consequently central because G is connected (Exposé IX, 5.5). Thus ${}_{\ell^n}N$ is contained in Z, contradicting the fact that its image in G' contains $\boldsymbol{\mu}_\ell$.

(ii) Suppose that $\ell = p$. There is then an integer n such that the image of $F_n N$ in G' contains $\boldsymbol{\mu}_p$, and hence an exact sequence

$$1 \longrightarrow K \longrightarrow F_n N \longrightarrow \boldsymbol{\mu}_p \longrightarrow 1. \tag{*}$$

  The group K is contained in Z, and is therefore commutative. It follows from 7.2.1, 7.1.1(b), and 5.5.1 that there is a subgroup of multiplicative type in $F_n N$ whose image in the quotient by K is $\boldsymbol{\mu}_p$. As in (i), we deduce that $F_n N$ is commutative. Its multiplicative-type component (7.2.1) is characteristic in $F_n N$, hence invariant in G, and therefore central (Exposé IX, 5.5). Since its image in G' contains $\boldsymbol{\mu}_p$, this is a contradiction.

A. Appendix I. Hochschild Cohomology and Extensions of Algebraic Groups

A.1. Definition of the Cohomology Groups

Let k be a field, let G be an algebraic k-group, and let $(\mathrm{Ab})_G$ be the abelian category of commutative algebraic k-groups on which G acts. If $A \in \mathrm{Ob}(\mathrm{Ab})_G$, the functor

$$h_A : (\mathrm{Sch}\,/k)^\circ \longrightarrow \mathrm{Ens}$$

canonically defined by A is a G-$\mathbb{Z}$-module in the sense of Exposé I, 3.2. We may therefore consider the standard complex $C^\bullet(G, A)$ of algebraic cochains of G with values in A (Exposé I, 5.1), together with the group $Z^i(G, A)$ of i-cocycles, the group $B^i(G, A)$ of i-coboundaries, and the i-th cohomology group $H^i(G, A)$. As usual, $H^0(G, A)$ is identified with the group $A^G(k)$, where A^G is the k-functor of invariants of A under G. The group $H^1(G, A)$ classifies the trivial principal homogeneous spaces under A on which G acts.

In general, the functor

$$A \longmapsto H^\bullet(G, A)$$

from $(\mathrm{Ab})_G$ to Ab is not a cohomological functor. Nevertheless, we have the following proposition.

Proposition A.1.1.— *Let*

$$0 \longrightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \longrightarrow 0$$

be an exact sequence in $(\mathrm{Ab})_G$. *Then:*

(a) if v has a section—that is, if there is a k-morphism of preschemes $s : A'' \to A$ such that $vs = 1_{A''}$—then there is the usual exact cohomology sequence

$$\cdots \longrightarrow H^i(G, A) \longrightarrow H^i(G, A'') \xrightarrow{d} H^{i+1}(G, A') \longrightarrow \cdots ;$$

(b) if $A(k) \to A''(k)$ is surjective, then the sequence

$$0 \longrightarrow A'^G(k) \longrightarrow A^G(k) \longrightarrow A''^G(k) \xrightarrow{d} H^1(G, A') \longrightarrow H^1(G, A) \longrightarrow H^1(G, A'') \quad (1)$$

is exact.

Proof. For (a), observe that the existence of a section implies the exactness of the sequence of complexes

$$0 \longrightarrow C^\bullet(G, A') \longrightarrow C^\bullet(G, A) \longrightarrow C^\bullet(G, A'') \longrightarrow 0.$$

For (b), if $x'' \in A''^G(k)$, its inverse image in A is a trivial principal homogeneous space under A' (because $A(k) \to A''(k)$ is assumed to be surjective) on which G acts, and therefore defines an element $d(x'') \in H^1(G, A')$. The exactness of the sequence (1) is then immediate.

A.2. The Group $\mathrm{Ext}_{\mathrm{alg}}(G, A)$

Let A and G be two algebraic k-groups, and let E and E' be two algebraic k-groups that are extensions of G by A. These two extensions are said to be *isomorphic* if there is a

k-morphism of groups $u : E \to E'$ that makes the following diagram commutative:

$$\begin{tikzcd} & E \arrow[dd, "u"] \arrow[dr] & \\ A \arrow[ur] \arrow[dr] & & G \\ & E' \arrow[ur] & \end{tikzcd}.$$

The group E acts on A by inner automorphisms. If A is commutative, this action factors through G, so that G acts on A. Conversely, if $A \in \mathrm{Ob}(\mathrm{Ab})_G$, we denote by $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ the set of classes of algebraic extensions E of G by A for which the action of G on A defined by E agrees with the action coming from the structure of A as an object of $(\mathrm{Ab})_G$.

The set $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ is naturally a bifunctor, covariant in A and contravariant in G. More precisely:

(a) if $f : A \to B$ is a morphism in $(\mathrm{Ab})_G$, and if E represents an element of $\mathrm{Ext}_{\mathrm{alg}}(G, A)$, then $f_*(E) \in \mathrm{Ext}_{\mathrm{alg}}(G, B)$ is defined to be the class of the extension of G by B equal to the quotient of the semidirect product $B \cdot E$ (where E acts on B through G) by the algebraic subgroup that is the image of A under the morphism (f, i). This quotient is representable by Exposé $\mathrm{VI_A}$, §5, and there is therefore a commutative diagram

$$\begin{tikzcd} 1 \arrow[r] & A \arrow[r, "i"] \arrow[d, "f"] & E \arrow[r] \arrow[d] & G \arrow[r] \arrow[d, "1_G"] & 1 \\ 1 \arrow[r] & B \arrow[r] & f_*(E) \arrow[r] & G \arrow[r] & 1 \end{tikzcd}.$$

(b) if $g : H \to G$ is a k-morphism of algebraic k-groups, and if E is an extension of G by A, the fiber product $E \times_G H$ is naturally an extension of H by A, denoted by $g^*(E)$. Thus there is a commutative diagram

$$\begin{tikzcd} 1 \arrow[r] & A \arrow[r] \arrow[d, "1_A"] & g^*(E) \arrow[r] \arrow[d] & H \arrow[r] \arrow[d, "g"] & 1 \\ 1 \arrow[r] & A \arrow[r] & E \arrow[r] & G \arrow[r] & 1 \end{tikzcd}.$$

By adapting the proofs given in J.-P. Serre, *Groupes algébriques et corps de classe*, Chapter VII, one endows $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ with a natural abelian-group structure, functorial in A and G.

Proposition A.2.1.— *Let*

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

be an exact sequence in $(\mathrm{Ab})_G$. *Then:*

(a) there is a canonical exact sequence of abelian groups

$$Z^1(G, A) \longrightarrow Z^1(G, A'') \xrightarrow{d} \mathrm{Ext}_{\mathrm{alg}}(G, A') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A'');$$

(b) if $A(k) \to A''(k)$ is surjective, one deduces from (a) the exact sequence

$$H^1(G, A) \longrightarrow H^1(G, A'') \xrightarrow{d} \mathrm{Ext}_{\mathrm{alg}}(G, A') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A''). \quad (2)$$

The exact sequence in (a) generalizes the usual exact $\mathrm{Hom}(\cdot, \cdot)$ sequence valid for commutative extensions (*loc. cit.*), and is proved in the same way. We recall only the definition of the coboundary

$$d : Z^1(G, A'') \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(G, A').$$

For this purpose, consider the extension

$$1 \longrightarrow A' \longrightarrow A \cdot G \longrightarrow A'' \cdot G \longrightarrow 1,$$

obtained in the evident way from the exact sequence $0 \to A' \to A \to A'' \to 0$. If $u \in Z^1(G, A'')$, then u defines in the usual way a section homomorphism

$$u : G \longrightarrow A'' \cdot G.$$

One then has

$$d(u) = u^*(A \cdot G).$$

A.3. Comparison of $H^2(G, A)$ and $\mathrm{Ext}_{\mathrm{alg}}(G, A)$

It is well known that, for abstract groups, there is a functorial isomorphism between the abelian groups $H^2(G, A)$ and $\mathrm{Ext}(G, A)$. Likewise in the present setting, if A is an object of $(\mathrm{Ab})_G$, one can associate with every 2-cocycle $u \in Z^2(G, A)$ an algebraic-group structure on the prescheme $A \times_k G$, making it an element of $\mathrm{Ext}_{\mathrm{alg}}(G, A)$. Moreover, this extension is trivial if and only if $u \in B^2(G, A)$ (see Exposé III, 1.2.2). Recall that the composition law on $A \times G$ is defined by

$$(a, g)(a', g') = a + {}^g a' + u(g, g').$$

It is clear that the extensions of G by A obtained in this way are not arbitrary, since they possess a section. Conversely, however, if $E \in \mathrm{Ext}_{\mathrm{alg}}(G, A)$ possesses a section s, then E is isomorphic to the extension of G by A associated with the 2-cocycle u such that

$$u(g, g') = s(g)s(g')s(gg')^{-1}.$$

We thus obtain the following proposition.

Proposition A.3.1.— *There is a functorial isomorphism between the bifunctors with values in abelian groups*

$$(G, A) \longmapsto H^2(G, A) \qquad \textit{and} \qquad (G, A) \longmapsto \mathrm{Ext}_s(G, A),$$

where $\mathrm{Ext}_s(G, A)$ denotes the subgroup of $\mathrm{Ext}_{\mathrm{alg}}(G, A)$ formed by the classes of extensions of G by A that possess a section.

B. Appendix II. Review and Supplements on Radicial Groups

Let $p > 1$ be a prime number, and let S be an $\mathbb{F}_p$-prescheme.

B.1. The Frobenius morphism

For every S-prescheme X and every integer $n > 0$, let

$$P^n : X \longrightarrow X$$

denote the $\mathbb{F}_p$-endomorphism corresponding to raising sections of $\mathcal{O}_X$ to the p^n-th power, and let $X^{(n)}$ denote the S-prescheme obtained from X by base change along $P^n : S \to S$. There is then a unique S-morphism

$$F^n : X \longrightarrow X^{(n)}$$

that makes the following diagram commutative:

$$\begin{tikzcd}
 & & X \arrow[dll, "P^n"'] \arrow[dl, "F^n"] \arrow[ddl] \\
X \arrow[d] & X^{(n)} \arrow[l] \arrow[d] & \\
S & S \arrow[l, "P^n"] &
\end{tikzcd}$$

It is clear that F^n identifies with the "n-th iterate" of $F^1 = F$, called the Frobenius morphism of X/S.

If G is an S-group prescheme, then $G^{(n)}$ is an S-group prescheme and

$$F^n : G \longrightarrow G^{(n)}$$

is an S-group morphism. Its kernel ${}_{F^n}(G)$ is a characteristic subgroup prescheme of G (that is, it is stable under the functor $\mathrm{Aut}_{S\text{-gr}}(G)$), and is radicial over S. If G is a radicial S-group prescheme, we say that G has height at most h if ${}_{F^h}(G) = G$.

B.2. Groups and restricted p-Lie algebras

If G is an S-group prescheme, then $\mathrm{Lie}(G)$ (Exposé II) is naturally a restricted p-Lie algebra (Exposé $\mathrm{VII_A}$, ğğ5 and 6). In particular, one has the following result (cf. Exposé $\mathrm{VII_A}$).

Proposition B.2.1.—

(i) $\mathrm{Lie}(\boldsymbol{\alpha}_p)_S = \mathrm{Lie}(\mathbb{G}_a)_S$[12] *is a sheaf of $\mathcal{O}_S$-modules, free of rank* 1 *over $\mathcal{O}_S$, generated by an element X such that $X^{(p)} = 0$.*

(ii) $\mathrm{Lie}(\boldsymbol{\mu}_p)_S = \mathrm{Lie}(\mathbb{G}_m)_S$ *is a sheaf of $\mathcal{O}_S$-modules, free of rank* 1 *over $\mathcal{O}_S$, with a canonical basis X such that $X^{(p)} = X$.*

We now recall the fundamental result proved in Exposé $\mathrm{VII_A}$, ğ7.

Theorem B.2.2.— *Suppose that S is affine, with ring A. Then the functor*

$$G \longmapsto \mathrm{Lie}\, G$$

[12] Editor's note: the definition of $\boldsymbol{\alpha}_p$, already given at the beginning of this Exposé, has been omitted here.

defines an equivalence between, on the one hand, the category of S-group preschemes G that are of finite presentation and flat over S, of height 1, and whose Lie algebra is locally free over $\mathcal{O}_S$, and, on the other hand, the category of restricted p-A-Lie algebras that are locally free of finite rank.

Moreover, if G is as above and H is any S-group prescheme of finite presentation, the canonical morphism

$$\mathrm{Hom}_{S\text{-gr}}(G, H) \longrightarrow \mathrm{Hom}_{p\text{-}A\text{-Lie}}(\mathrm{Lie}\, G(S), \mathrm{Lie}\, H(S))$$

is an isomorphism.

B.3. Radicial groups and smooth groups

We now suppose that S is the spectrum of a field k of characteristic p. Recall that Exposé $\mathrm{VI_A}$, ğ5 proves that, if G is an algebraic k-group and H an algebraic subgroup of G, then the sheaf G/H (for the fpqc topology) is representable. We also recall ($\mathrm{VII_A}$, 8.3):

Proposition B.3.1.— *Let G be an algebraic k-group. There exists an integer m such that, for every $n \geq m$, the algebraic group*

$$G/_{F^n}(G)$$

is smooth over k.

Proposition B.3.2.— *Consider an exact sequence of algebraic k-groups*

$$1 \longrightarrow G' \longrightarrow G \xrightarrow{u} G'' \longrightarrow 1$$

and the following assertions:

(i) the morphism u is smooth;

(ii) G' is smooth over k;

(iii) for every integer $n > 0$, the sequence

$$1 \longrightarrow {}_{F^n}(G') \longrightarrow {}_{F^n}(G) \longrightarrow {}_{F^n}(G'') \longrightarrow 1$$

is exact;

(iv) the morphism ${}_F G \to {}_F G''$ is an epimorphism;

(v) the morphism

$$(\mathrm{Lie}\, G)(k) \longrightarrow (\mathrm{Lie}\, G'')(k)$$

is surjective.

Then

$$\text{(i)} \iff \text{(ii)} \implies \text{(iii)} \implies \text{(iv)} \iff \text{(v)}.$$

Moreover, if G is smooth, all five assertions are equivalent.

The equivalence (i)$\iff$(ii) follows from Exposé $\mathrm{VI_B}$, 9.2(vii).

For (ii)$\implies$(iii), if G' is smooth, then

$$F^n : G' \longrightarrow G'^{(n)}$$

is an epimorphism, and (iii) follows from the snake diagram. The implication (iii)$\implies$(iv) is clear, while (iv)$\iff$(v) follows from Theorem B.2.2. Finally, suppose that G is smooth and that (v) holds. Then G'' is smooth, and (v) implies

$$\dim G' = \dim_k (\mathrm{Lie}\, G')(k),$$

so G' is smooth.

C. Appendix III. Remarks and Supplements Concerning Exposés XV, XVI, and XVII

C.1.

It may be that Propositions 1.2 and 1.2 bis of Exposé XV remain valid if the hypothesis that H_0 is smooth over S_0 is removed. This is the case, in particular, if G is finite, flat, and commutative.

C.2. Complement to Exposé XV, 4.8

The following proposition, as well as Theorems C.3.1 and C.4.1 below, will appear in a forthcoming article by M. Raynaud on group schemes over a discrete valuation ring.[13]

Proposition C.2.1.— *Let S be the spectrum of a discrete valuation ring, let t be its generic point, let G be a flat S-group prescheme of finite type, and put*

$$\widetilde{G} = \operatorname{Spec} \Gamma(G, \mathcal{O}_G).$$

Let $u : G \to \widetilde{G}$ be the canonical morphism. Then:

(1) $\widetilde{G}$ is naturally an S-group scheme, and u is a homomorphism.

(2) $\ker(u)$ is flat over S, and $(\widetilde{G}, u)$ is the fpqc quotient of G by $\ker(u)$. Thus $\widetilde{G}$ is the largest affine quotient of G.

(3) If G_t is affine, then $\ker(u)$ is an étale group over S, and it is the trivial group if and only if G is separated over S. In particular, a flat, separated S-group scheme of finite type whose generic fiber is affine is affine.

C.3.

In the statement of Exposé XV, 6.6, the hypothesis that H be equal to its connected normalizer is unnecessary in order that the set of points s of S for which H_s is a parabolic subgroup of G_s be open. Indeed, repeat the proof given in paragraph c) preceding Lemma 6.9, and let N denote the scheme-theoretic closure of N_t in G. Then N is a closed subgroup scheme of G, flat over S, which contains H and is contained in $\mathcal{N}$. The theorem below shows that G/N is representable. Since G_s/N_s is proper and connected, the proof concludes as in *loc. cit.*

Theorem C.3.1[14].— *Let S be the spectrum of a henselian discrete valuation ring, let G be an S-group prescheme locally of finite type, and let H be a closed subgroup prescheme of G, flat over S. Then G/H is representable.*

C.4. Complement to Exposé XVI, 1.1

The following theorem refines Exposé VIII, 7.9.

Theorem C.4.1.— *Let S be the spectrum of a discrete valuation ring with residue characteristic $p > 0$, and let G be a commutative, smooth, separated S-group scheme of finite type. Then the following conditions are equivalent:*

[13]Editor's note: comments are to be added here, including references to Exposé $\mathrm{VI_B}$, ...

[14]Editor's note: see Theorem 4C in S. Anantharaman, "Schémas en groupes, espaces homogènes et espaces algébriques sur une base de dimension 1," *Bull. Soc. Math. France, Mém.* 33 (1976), 5–79.

(i) $_pG$ *is finite over* S.

(ii) *For every* S*-prescheme* S', *every separated* S'*-group prescheme* H' *of finite presentation over* S', *and every* S'*-monomorphism*

$$u : G_{S'} \longrightarrow H',$$

the morphism u *is an immersion.*

C.5.

Example XVII, 6.4(a), gives a smooth group over a field k whose unipotent radical is not defined over k. The following proposition gives a general method for obtaining such groups.

Proposition C.5.1.— *Let* k *be a field, let* K *be a finite radicial extension of* k *of degree greater than* 1, *and let* H *be a smooth, connected* K*-algebraic group of dimension* r. *Put*

$$G = \prod_{K/k} H/K,$$

which is a smooth, connected k*-algebraic group of dimension* $r[K : k]$. *Let*

$$u : G_K \longrightarrow H$$

be the canonical homomorphism, and put $R = \ker(u)$. *Then:*

(i) u *is an epimorphism, and* R *is a smooth, connected, unipotent algebraic group.*

(ii) *If* U *is a smooth algebraic subgroup of* G *such that* U_K *contains* R, *then* $U = G$.

Corollary.— *Retain the notation of the preceding proposition.*

(a) *If* H *is not unipotent, then the unipotent radical of* $G_{\bar{k}}$ *is not defined over* k.

(b) *If* H *is a nonzero abelian variety, then* G *is not an extension of an abelian variety by a smooth linear group.*

(c) *If* H *is not solvable, then the solvable radical of* $G_{\bar{k}}$ *is not defined over* k, *and* G *has no Borel subgroup defined over* k.

Let us first show how the corollary follows from the proposition.

(a) Let U be a unipotent radical of G. Then U_K is the unipotent radical of G_K, and hence contains R, since R is smooth, connected, and unipotent by (i). Thus $U = G$ by (ii). But G_K has H as a quotient, and H is not unipotent by hypothesis. Therefore G is not unipotent, a contradiction.

(b) Suppose that G is an extension of an abelian variety by a linear group L that is smooth over k. Then L_K necessarily contains R, so $L = G$. But G cannot be a linear group, because G_K has a nonzero abelian variety as a quotient.

(c) Let S be a solvable radical of G, and let B be a Borel subgroup of G. Then S_K and B_K contain R, so $S = G$ and $B = G$. But then G is solvable, contradicting the fact that G_K has a quotient that is not solvable.

Proof of Proposition C.5.1. *(i)* We begin by describing the morphism u. A K-scheme

$$f : S \longrightarrow \mathrm{Spec}(K)$$

gives rise to the diagram

$$\begin{tikzcd}
\mathrm{Spec}(K) \arrow[d, "j"'] & & T \arrow[ll, "h"'] \arrow[d, bend right, "j_T"'] \\
\mathrm{Spec}(k) & & S \arrow[ll, "g"] \arrow[ull, "f"] \arrow[u, bend right, "s"']
\end{tikzcd}$$

where

$$g = j \circ f, \qquad T = \mathrm{Spec}(K) \times_{\mathrm{Spec}(k)} S,$$

h and j_T are the two projections, and s is the section of T over S such that $h \circ s = f$. The map

$$u(S) : G_K(S) \longrightarrow H(S)$$

is simply the composite

$$G_K(S) \xrightarrow{\sim} H(T) \longrightarrow H(S),$$

where the last arrow is defined by the section s.

Take, in particular, S to be the spectrum of an algebraic closure $\bar{k}$ of k, and let f be the unique k-morphism induced by $K \to \bar{k}$, so that T is a local Artinian scheme. To prove (i), it is enough to do so after extending the base field from K to $\bar{k}$. It is clear that

$$G_S = \prod_{T/S} H_T/T$$

represents the Greenberg functor of H_T relative to S (M. J. Greenberg, "Schemata over local rings," *Ann. of Math.* 73 (1961), 624–648). Under this identification, the description above shows that u_S is the canonical transition morphism

$$\mathrm{Green}(H_T) \longrightarrow H_S = H_T \times_T S.$$

Assertion (i) now follows from the smoothness of H over K and M. J. Greenberg, "Schemata over local rings II," *Ann. of Math.* 78 (1963), 256–266.

(ii) To prove (ii), we may suppose that k is separably closed. Let U be a smooth algebraic subgroup of G such that $U_K \supset R$, and let us prove that $U = G$. Since G is connected, we may suppose that U is connected. Let $V = u(U_K)$, a smooth, connected algebraic subgroup of H, and put

$$V' = \prod_{K/k} V/K,$$

a smooth, connected algebraic subgroup of G. The group V'_K is an algebraic subgroup of G_K, and the canonical morphism $V'_K \to V$ is the restriction of u to V'_K. By hypothesis, U_K contains R, hence *a fortiori* contains $\ker(V'_K \to V)$; by construction, moreover, the image of U_K in H is V. It follows that U_K contains V'_K, and hence that U contains V'.

On the other hand, the canonical isomorphism

$$G(k) \hookrightarrow G(K) \xrightarrow{u(K)} H(K)$$

plainly sends $U(k)$ into $V(K) = V'(k)$. Thus $U(k) \subset V'(k)$. Since U is smooth and k is separably closed, it follows that $U \subset V'$, whence $U = V'$. We then have

$$([K : k] - 1) \dim V = \dim \ker(V'_K \to V) = \dim R = ([K : k] - 1) \dim H.$$

Since $K \neq k$, we conclude that $\dim V = \dim H$, hence that $V = H$, and finally that $U = V' = G$.

Exposé XVIII
Weil's Theorem on the Construction of a Group from a Rational Law

by Michael Artin

0. Introduction

This Exposé is devoted to the well-known theorem of Weil [1], which constructs a group variety from a birational law. It seems that the generalization of this result to the case where the base is the spectrum of a discrete valuation ring was already known to several people; see, for example, [2]. Here we shall prove the theorem for a flat group of finite presentation over an arbitrary base prescheme S. Since the hypotheses must be formulated somewhat more carefully, the statement is not in the form given by Weil; but when S is the spectrum of a field, the form given here is seen to be essentially equivalent to Weil's. We use the following suggestion of Grothendieck: given a "group germ" X over a prescheme S, one should construct the group as the quotient of $X \times_S X$ by a suitable equivalence relation.

1. "Recollections" on Rational Maps

Let X/S be a relative prescheme and let $U \subseteq X$ be an open subset. We say that U is *schematically dense in X relative to S*, or that $U \hookrightarrow X$ is *relatively schematically dense*, if, for every base change $S' \to S$, the open subset $U \times_S S'$ is schematically dense in $X \times_S S'$. For the definition and properties of this notion, we refer to Exposé IX, §4.*

Proposition 1.1. *(i) A finite intersection, as well as a union of a nonempty family, of open subsets schematically dense relative to S, is schematically dense relative to S.*

(ii) If $U \subseteq X$ is schematically dense relative to S, and if $S \to T$ is a morphism, then $U_T \subseteq X_T$ is schematically dense relative to T.

(iii) Let $U \subseteq V \subseteq X$ be open immersions. For U to be schematically dense in X relative to S, it is necessary and sufficient that it be so in V, and that V be so in X.

(iv) If $U \subseteq X$ and $V \subseteq Y$ are relatively schematically dense, then $U \times_S V$ is schematically dense in $X \times_S Y$ relative to S.

[0] Version of May 8, 2009.

*See also EGA IV_3, 11.9 and 11.10 (especially 11.10.8), where the term "universally schematically dense relative to S" is used.

In Exposé IX one finds the following criterion:

Proposition 1.2. *Let X/S be locally of finite presentation and flat, and let U be an open subset of X. If, for every $s \in S$, the fiber U_s is schematically dense in X_s, then U is schematically dense in X relative to S.*

In particular:

Corollary 1.3. *Suppose that X/S is locally of finite presentation and flat, and that every fiber X_s is "without embedded components." Then U is schematically dense in X relative to S if and only if, for every $s \in S$, U_s is dense in X_s in the topological sense.*

One easily deduces from the definition the following proposition.

Proposition 1.4. *Let $U \subseteq X$ be schematically dense relative to S, and let $f, g : X \to Y$ be two morphisms of functors over S, where Y satisfies one of the following conditions:*

(i) *Y is a prescheme over S, and every fiber Y_s is separated.*

(ii) *Y is a presheaf*[1] *over S, such that the diagonal morphism $Y \to Y \times_S Y$ is representable by a closed immersion.*[2]

Then, if $f = g$ on U, one has $f = g$.

Proof. In case (i), the standard argument following IX, 4.1 shows that f and g are equal on every fiber; hence the subprescheme $\operatorname{Ker}(f, g)$ of X is set-theoretically equal to X, and is therefore closed. Since it contains U, it is equal to X, that is, $f = g$. Case (ii) is proved as in *loc. cit.*[3]

Definition 1.5. [*] *Let X be a prescheme over S, and let Y be a presheaf over S. A* rational map *$f : X \dashrightarrow Y$ over S is an equivalence class of morphisms $f : U \to Y$ over S, where U is an open subset of X that is schematically dense relative to S. One sets*

$$(f : U \to Y) \sim (f' : U' \to Y)$$

if and only if there exists an open subset $U'' \subseteq U' \cap U$, schematically dense relative to S, such that $f = f'$ on U''.

This definition is made in such a way that, for every base change $S' \to S$, the rational map $f \times_S S'$ may be defined in the obvious way.

If U is an open subset of X, we say that the rational map f is *defined on U* if there exists a morphism representing f whose set of definition contains U.

Definition 1.5.1. [4] *If Y satisfies one of conditions (i), (ii) of 1.4, it is clear that there exists a largest open subset U of X on which f is defined, and this U is schematically dense relative to S. It is called the* domain of definition of f over S, *and is denoted* $\operatorname{Dom}(f)$.

This notion does not in general commute with base change, but one has:

[1] Editor's note: specify for which topology; a priori, the fpqc topology.

[2] Editor's note: Let us recall here the definition of a closed immersion. A morphism of S-functors $F \to G$ is *relatively representable* if, for every S-morphism $T \to G$, where T is an S-prescheme, the T-functor $F_T = F \times_G T$ is representable by a T-prescheme. The French source prints $F \times_S T$ here; the fiber product over G is the one dictated by the preceding morphism $T \to G$. A morphism of S-functors $F \to G$ is an open (respectively, closed) immersion if it is relatively representable and if, for every S-morphism $T \to G$, where T is an S-prescheme, the T-morphism $F_T \to T$ is an open (respectively, closed) immersion.

[3] Editor's note: Give details on this point…

[*] See EGA IV_4, 20.5, where the term "pseudo-morphism from X to Y relative to S" is used in order not to conflict with EGA I, 7.12.

[4] Editor's note: The number 1.5.1 has been added.

Proposition 1.6. *Let X be an S-prescheme and Y an S-functor satisfying one of hypotheses (i), (ii) of 1.4. Let $f : X \dashrightarrow Y$ be a rational map over S, let $S' \to S$ be a flat morphism locally of finite presentation, and put $f' = f \times_S S'$. Then*

$$\mathrm{Dom}(f') = \mathrm{Dom}(f) \times_S S'.$$

Proof. Put $U = \mathrm{Dom}(f)$. It is clear that $V' = \mathrm{Dom}(f')$ contains $U \times_S S'$. Let V be the image of V'; it is an open subset of X, because $X' \to X$ is open. We must prove that $V = U$, that is, we must find a morphism $V \to Y$ representing f. Put $S'' = S' \times_S S'$, and

$$X'' = X \times_S S'', \qquad U'' = U \times_S S'', \qquad V'' = V' \times_V V'.$$

Then U'' is schematically dense in X'' relative to S''; hence U'' is schematically dense in V'' relative to S'', since $U'' \subseteq V'' \subseteq X''$. The restriction of $f' : V' \to Y'$ to U' is obtained from $f : U \to Y$ by base change. The two morphisms $V'' \to Y$ obtained from f' by the two projections agree on U'', and hence agree. Since $V' \to V$ is flat and locally of finite presentation, it is an fppf covering (Exposé IV, 6.3), and the morphism $V \to Y$ is obtained by descent.

We shall frequently use the following triviality:

Proposition 1.7. *Let X/S be faithfully flat and locally of finite presentation, and let $U \subseteq X$ be a relatively schematically dense open subset. Then there exists an fppf covering base change $S' \to S$, and a section $x \in X(S')$ contained in $U(S')$.*

Indeed, U/S is an fppf covering. Take $S' = U$, and as the section take the S-graph of the inclusion $U \hookrightarrow X$.

2. Local Determination of a Group Homomorphism

Let G and H be groups, and let $U \subseteq G$ be a subset such that $U \cdot U = G$. If f and g are homomorphisms from G to H such that $f = g$ on U, then $f = g$. Likewise:

Proposition 2.1. *Let S be a site (cf. Exposé IV[5]), let G be a sheaf of groups on S, and let $U \subseteq G$ be a subsheaf of sets such that the morphism $U \times U \to G$ induced by multiplication is an epimorphism. If f and g are homomorphisms from G to a sheaf of groups H and are equal on U, then $f = g$.*

Corollary 2.2. *Let G be an S-group prescheme that is locally of finite presentation and flat, let $U \subseteq G$ be a relatively schematically dense open subset, and let H be a sheaf of groups on S for the fppf topology. Then every homomorphism $f : G \to H$ is determined by its restriction to U.*

Indeed, since G is flat over S, the composition law on G is a flat morphism ($\mathrm{VI_B}$, 9.2(xi)); it follows that $U \times_S U \to G$ is faithfully flat and locally of finite presentation, hence an fppf covering, and therefore an epimorphism.

Proposition 2.3. *Let G be a group prescheme locally of finite presentation and flat over S, let $U \subseteq G$ be a relatively schematically dense open subset, and let H be a sheaf of groups for the fppf topology. Write m_G for the multiplication on G and m_H for that on H.*

Let $f : U \to H$ be an S-morphism, and suppose that there exists a relatively schematically dense open subset V of $m_G^{-1}(U) \cap (U \times_S U)$ such that the following diagram is commutative:

[5] Editor's note: that is, a category endowed with a topology; cf. IV, §4.2.

$$\begin{tikzcd} V \arrow[r, "(\bar f\times\bar f)|_V"] \arrow[d, "m_G"] & H\times_S H \arrow[d, "m_H"] \\ U \arrow[r, "\bar f"] & H \end{tikzcd}$$

Then there exists a morphism of S-groups $f : G \to H$, necessarily unique, extending $\bar f$, in each of the following situations:

(i) *H is representable.*

(ii) *The diagonal morphism $H \to H \times_S H$ is representable by a closed immersion.*[6]

(iii) *For every section $a \in U(S)$, the open subset $\mathrm{pr}_2((a\times_S U)\cap V)$ is relatively schematically dense in U^*, and this assertion remains true after every base change $S' \to S$.*

Proof. First note that V is relatively schematically dense in $G\times_S G$. Indeed, V is relatively schematically dense in

$$m_G^{-1}(U) \cap (U \times_S G) \cap (G \times_S U),$$

and each of the three factors is obtained from $U \subseteq G$ by an evident base change $G\times_S G \to G$. The assertion therefore follows from 1.1.

To construct $f : G \to H$, since $U \times_S U \to G$ is an fppf covering, it is enough to construct a morphism $U \times_S U \to H$ whose two pullbacks to

$$(U \times_S U) \times_G (U \times_S U)$$

are equal. Take the morphism

$$m_H \circ (\bar f \times \bar f) : U \times_S U \longrightarrow H.$$

For sections $a, b, c, d \in U(S')$, where $S' \to S$ is arbitrary, such that $ab = cd$, we must prove

$$\bar f(a)\bar f(b) = \bar f(c)\bar f(d).$$

If (a, b) and (c, d) belong to $V(S')$, the desired equality follows at once. Thus it holds on $(V \times_G V)(S')$.

I claim that $V \times_G V$ is a relatively schematically dense open subset of

$$(U \times_S U) \times_G (U \times_S U).$$

This will complete the proof in cases (i) and (ii) by 1.4, because the morphism f constructed in this way extends $\bar f$, and its being a homomorphism is then evident, again by 1.4. Now

$$V \times_G V = (V \times_G (G \times_S G)) \cap ((G \times_S G) \times_G V).$$

By symmetry and 1.1, it is enough to check that $V \times_G (G \times_S G)$ is relatively schematically dense in $(G\times_S G)\times_G (G\times_S G)$. The latter prescheme is isomorphic over S to $G\times_S G\times_S G$, the isomorphism being given by

$$(a, b, c, d) \longmapsto (a, b, c).$$

[6] Editor's note: See the note at 1.4.

[*] Or in G, which is equivalent by 1.1(iv).

What remains is therefore the relative schematic density of $V \times_S G$ in $G \times_S G \times_S G$, which follows from that of V in $G \times_S G$.

It remains to treat case (iii). To prove $\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d)$, we may make an fppf covering base change. Suppose that, after such a base change $S' \to S$, there is a section $x \in G(S')$ such that

$$(b, x), \quad (a, bx), \quad (d, x), \quad (c, dx)$$

all belong to V. Then

$$\begin{aligned} \bar{f}(a)\bar{f}(b)\bar{f}(x) &= \bar{f}(a)\bar{f}(bx) = \bar{f}(abx) = \bar{f}(cdx) \\ &= \bar{f}(c)\bar{f}(dx) = \bar{f}(c)\bar{f}(d)\bar{f}(x), \end{aligned}$$

which gives the desired equality.

To find such an x[7], set, for every $z \in U(S')$,

$$V_z = \mathrm{pr}_{2/S'}\big((z \times_{S'} U_{S'}) \cap V_{S'}\big).$$

The conditions imposed on x say exactly that $x \in W$, where

$$W = V_b(S') \cap V_d(S') \cap b^{-1}V_a(S') \cap d^{-1}V_c(S').$$

By hypothesis (iii), W is relatively schematically dense in G. Proposition 1.7 therefore supplies such a section x after an fppf covering base change.

Let us verify that the morphism thus constructed is multiplicative. Let $a, b \in G(S')$, where $S' \to S$ is arbitrary; after this base change, rename S' as S. Choose an fppf covering base extension $S' \to S$ and a section $x \in G(S')$ such that

$$x \in U(S') \qquad \text{and} \qquad ax^{-1} \in U(S'),$$

that is, $x \in (a^{-1}U)^{-1}(S')$. Choose in addition another fppf covering extension $S'' \to S'$ and a section $y \in G(S'')$ such that

$$y, \quad by^{-1}, \quad aby^{-1} \in U(S''),$$

and

$$by^{-1} \in V_x, \qquad xby^{-1} \in V_{ax^{-1}}.$$

Then, by the definition of f, over S'' we have

$$f(a) = \bar{f}(ax^{-1})\bar{f}(x), \qquad f(b) = \bar{f}(by^{-1})\bar{f}(y), \qquad f(ab) = \bar{f}(aby^{-1})\bar{f}(y).$$

Furthermore,

$$\begin{aligned} f(a)f(b) &= \bar{f}(ax^{-1})\bar{f}(x)\bar{f}(by^{-1})\bar{f}(y) \\ &= \bar{f}(ax^{-1})\bar{f}(xby^{-1})\bar{f}(y) \\ &= \bar{f}(aby^{-1})\bar{f}(y) = f(ab), \end{aligned}$$

which proves multiplicativity.

It is now easy to see that f extends $\bar{f}$. Let $a \in U(S')$; after base change, rename S' as S, and choose an fppf covering $S' \to S$ and sections $x, y \in G(S')$ such that

$$(x, y), \qquad (ax, y), \qquad (a, xy)$$

[7]Editor's note: The beginning of the following sentence has been added.

belong to $V(S')$. Then

$$f(xy) = \bar{f}(x)\bar{f}(y) = \bar{f}(xy), \qquad f(axy) = \bar{f}(ax)\bar{f}(y) = \bar{f}(axy).$$

Consequently,

$$\begin{aligned} f(a) &= f(axy)f\big((xy)^{-1}\big) = f(axy)f(xy)^{-1} \\ &= \bar{f}(axy)\bar{f}(xy)^{-1} = \bar{f}(a). \end{aligned}$$

□

Remark 2.3.1. In many cases hypothesis (iii) holds, for it will still hold if U is replaced by a smaller open subset U', still relatively schematically dense in G. For example:

Proposition 2.4. *In the situation of 2.3, suppose that every geometric fiber of G/S is irreducible. Put*

$$U' = \mathrm{pr}_1 V \qquad \textit{and} \qquad V' = V \cap m_G^{-1}(U') \cap (U' \times_S U').$$

Then $U' \subseteq U$ is an open subset relatively schematically dense in G, and U', $\bar{f}|_{U'}$, and V' satisfy hypothesis (iii).

Proof. The subset U' is open because $G \times_S G \to G$ is flat and locally of finite presentation. All the other verifications are immediate except for hypothesis (iii).

Let $a \in U(S')$; after base change, rename S' as S. To verify that

$$\mathrm{pr}_2\big((a \times_S U') \cap V'\big)$$

is relatively schematically dense in U', it suffices by Corollary 1.3 to do so fiber by fiber. Thus it is enough to treat the case where S is the spectrum of a field, and in that case it is enough to prove that U' is nonempty, since G is irreducible and has no embedded components (cf. $\mathrm{VI_A}$, 1.1.1). Now

$$\begin{aligned} \mathrm{pr}_2\big((a \times_S U') \cap V'\big) = {} & \mathrm{pr}_2\big((a \times_S U') \cap V\big) \\ & \cap \mathrm{pr}_2\big((a \times_S U') \cap m_G^{-1}(U')\big). \end{aligned}$$

The second factor on the right is dense in G. It therefore remains only to prove that

$$\mathrm{pr}_2\big((a \times_S U') \cap V\big)$$

is dense in G, that is, nonempty; and this is clear because $a \in U' = \mathrm{pr}_1 V$. □

3. Construction of a Group from a Rational Law

3.0. Suppose given an S-prescheme X/S and a rational map

$$X \times_S X \dashrightarrow X$$

over S. We seek an S-group G/S and a birational map relative to S[8]

$$X \dashrightarrow G$$

[8]Editor's note: The notion of a birational map (relative to S) should be made explicit, perhaps in an addition to 1.5.2.

that commutes with the composition laws. We treat only the case in which X/S satisfies the following hypothesis:

$$\begin{gathered} X/S \text{ is faithfully flat and of finite presentation,} \\ \text{with separated fibers and without embedded components.} \end{gathered} \tag{$\diamond$}$$

(The last two conditions hold for a group prescheme.[9])

We shall often omit the symbol S from fiber products.

Let X/S be a prescheme having the properties ($\diamond$) above, and let W be a subprescheme of finite presentation of $X \times X \times X$ having the following property:

$$\begin{gathered} \text{the three morphisms } W \to X \times X \text{ induced by the projections} \\ \text{from } X^3 \text{ onto } X^2 \text{ are relatively schematically dense open immersions over } S. \end{gathered} \tag{*}$$

Notation. We shall use the following terminology. Given sections $a, b, c \in X(S')$, where $S' \to S$ is arbitrary, such that $(a, b, c) \in W(S')$, we write

$$c = ab, \qquad b = a^{-1}c, \qquad a = cb^{-1}.$$

Definition 3.0.1. [10] Given a section $(a, b) \in X^2(S')$, we say that ab is *defined* if, and only if, there exists a section $c \in X(S')$ such that $(a, b, c) \in W(S')$; equivalently, if and only if $(a, b) \in \mathrm{pr}_{12} W(S')$. Similarly, to say that $a^{-1}b$ or ab^{-1} is defined has the analogous meaning, and we extend this terminology also to products of several factors.

Remark 3.0.2. Let us note at once the following fact. By (*), W defines a rational map $X^2 \dashrightarrow X$ over S, namely the one given by $(a, b) \mapsto ab$. This rational map may well have a domain of definition larger than $\mathrm{pr}_{12} W$. Nevertheless, we say that ab is defined only if $(a, b) \in \mathrm{pr}_{12} W(S')$.

Definition 3.1. A *group germ* is an S-prescheme X/S having the properties ($\diamond$) above, together with a subprescheme W of finite presentation of $X \times X \times X$, having the following properties:

(i) W satisfies property (*) above.

(ii) For every section $a \in X(S)$, the sets

$$\begin{aligned} &\mathrm{pr}_i\big((a \times X \times X) \cap W\big), \quad i = 2, 3, \\ &\mathrm{pr}_i\big((X \times a \times X) \cap W\big), \quad i = 3, 1, \\ &\mathrm{pr}_i\big((X \times X \times a) \cap W\big), \quad i = 1, 2, \end{aligned}$$

are schematically dense in X relative to S, and this assertion remains true after every base change $S' \to S$. (Condition (i) implies that these sets are open in X.) Intuitively, this says that ax is defined for sufficiently general x, and similarly for the other products.

(iii) The law is associative: if $(a, b, c) \in X^3(S')$, for an arbitrary $S' \to S$, and both $(ab)c$ and $a(bc)$ are defined, then

$$(ab)c = a(bc).$$

[9] Editor's note: Cf. $\mathrm{VI_A}$; the reference is to be specified.

[10] Editor's note: The numbering 3.0.1 and 3.0.2 has been introduced.

Remarks. (a) In (i), the condition that W be schematically dense in X^2 relative to S may be replaced by the condition that, for every $s \in S$, the fiber W_s be topologically dense in X_s^2, by the hypotheses on X and by 1.2.

(b) Condition (iii) is equivalent to the following. Let V_1 (respectively V_2) be the open subset of X^3 on which the rational map

$$(a, b, c) \longmapsto a(bc) \qquad \text{(respectively } (a, b, c) \longmapsto (ab)c)$$

is defined* . Then there exists an open subset $V \subseteq V_1 \cap V_2$, schematically dense in X^3 relative to S, on which the two maps agree. This follows from 1.4, since V_1 and V_2 are schematically dense in X^3 relative to S.

(c) Hypothesis (ii) will be used below to ensure that X is a subobject of the group prescheme that it defines. In many cases one can deduce (ii) from (i), provided that X is replaced by a relatively schematically dense open subset X' and W by a relatively schematically dense open subset of $W \cap X'^3$. More precisely:

Proposition 3.2. *Suppose that every geometric fiber of X/S is irreducible, and let W be a subprescheme of X^3 satisfying condition (i) of 3.1. Then there exist a relatively schematically dense open subset $X' \subseteq X$ and a relatively schematically dense open subset $W' \subseteq W \cap X'^3$ such that the pair (X', W') satisfies conditions (i) and (ii). If (iii) holds for (X, W), then it holds for (X', W').*

Proof. Set

$$X' = \bigcap_{i=1}^{3} \operatorname{pr}_i W.$$

Each $\operatorname{pr}_i W$ is open in X, because $W \to X^2$ is an open immersion and the projections $X^2 \to X$ are flat and of finite presentation. Moreover, $\operatorname{pr}_i W$ is schematically dense in X relative to S, because W is so in X^2 and the projections $X^2 \to X$ are surjective; it is enough here to check topological density. Put

$$W' = W \cap X'^3.$$

To verify (i), note that

$$\begin{aligned} W' = (W \cap (X' \times X' \times X)) &\cap (W \cap (X \times X' \times X')) \\ &\cap (W \cap (X' \times X \times X')). \end{aligned}$$

Now

$$W \cap (X' \times X' \times X) \simeq \operatorname{pr}_{12} W \cap (X' \times X'),$$

so this term is schematically dense in W relative to S. The other two terms on the right are likewise schematically dense in W, and hence W' is schematically dense in W. Consequently $\operatorname{pr}_{ij} W'$ is schematically dense in $\operatorname{pr}_{ij} W$, hence in $X \times X$, and therefore in $X' \times X'$.

To verify (ii), let $a \in X'(S')$. After this base change, rename S' as S. We must show, for example, that

$$\operatorname{pr}_2((a \times X' \times X') \cap W')$$

is schematically dense in X' relative to S. By 1.3 it is enough to check this fiber by fiber. Thus it is enough to treat the case in which S is the spectrum of a field, and in that case it

*In the sense explained in 3.0.1; one could also replace these open subsets by the domains of definition (cf. 1.5) of the rational maps under consideration.

is enough to show that the open subset is nonempty, since the fibers of X/S are irreducible and have no embedded components. Since

$$\mathrm{pr}_2((a \times X \times X) \cap W)$$

is nonempty, because a is a section of X', this open subset is dense in X. We have

$$\mathrm{pr}_2((a \times X \times X) \cap W) = \mathrm{pr}_2((a \times X) \cap \mathrm{pr}_{12} W).$$

Therefore

$$\begin{aligned} &\mathrm{pr}_2((a \times X) \cap \mathrm{pr}_{12} W) \cap X' \\ &\quad = \mathrm{pr}_2((a \times X') \cap \mathrm{pr}_{12} W) = \mathrm{pr}_2((a \times X' \times X) \cap W) \end{aligned}$$

is dense in X, and hence in X'.

Similarly, $\mathrm{pr}_3((a \times X \times X') \cap W)$ is dense in X, hence dense in $\mathrm{pr}_3((a \times X \times X) \cap W)$. Equivalently, $(a \times X \times X') \cap W$ is dense in $(a \times X \times X) \cap W$, and therefore

$$\mathrm{pr}_2((a \times X \times X') \cap W)$$

is dense in $\mathrm{pr}_2((a \times X \times X) \cap W)$, hence in X, and thus in X'.

Finally,

$$\begin{aligned} \mathrm{pr}_2((a \times X' \times X') \cap W') &= \mathrm{pr}_2((a \times X' \times X') \cap W) \\ &= \mathrm{pr}_2((a \times X' \times X) \cap W) \cap \mathrm{pr}_2((a \times X \times X') \cap W), \end{aligned}$$

which is indeed dense in X'. This is what was required. The other assertions in (ii) follow by symmetry, and preservation of condition (iii) is immediate.[11] This proves Proposition 3.2. □

3.2.1. [12] Fix a group germ (X, W) over S. We first make some preliminary observations about the situation, collected below as rules. These rules will often be used without explicit mention in what follows.

Let $a \in X(S')$, and after base change rename S' as S. There is then a birational map φ over S from X to itself, sending a section x to ax whenever the latter is defined. By 3.1(ii), the domain of definition of φ contains the relatively schematically dense open subset

$$\mathrm{pr}_2((a \times X \times X) \cap W),$$

and φ defines an isomorphism from this open subset onto the open subset

$$\mathrm{pr}_3((a \times X \times X) \cap W),$$

on which $a^{-1}x$ is defined. This observation is generalized in the evident way in Rule 1 below.

Rule 1. Let $P = P(x, t_1, \ldots, t_n)$ be a "product" of the symbols $x, t_1, \ldots, t_n$, obtained recursively as follows: $P_0 = x$, and P_{i+1} is one of the expressions

$$P_i t, \qquad t P_i, \qquad P_i^{-1} t, \qquad t P_i^{-1}, \qquad P_i t^{-1}, \qquad t^{-1} P_i,$$

where t is one of the t_j; and $P_r = P$. Let $a_1, \ldots, a_n \in X(S')$. Then there exists a relatively schematically dense open subset U of $X_{S'}$ such that the product $P(x, a_1, \ldots, a_n)$ is defined (in the sense of Remark 3.0.2) for a section $x \in X(S'')$ if and only if $x \in U(S'')$, and the map

$$x \longmapsto P(x, (a))$$

gives an isomorphism from U onto another relatively schematically dense open subset of $X_{S'}$, denoted $P(U, (a))$.

[11] Editor's note: The following sentence has been added.

[12] Editor's note: The numbering 3.2.1, as well as 3.2.2, has been introduced.

Rule 2. Let $a, b \in X(S')$. Then:

$$\text{if } ab \text{ is defined, so is } a^{-1}(ab), \text{ and } a^{-1}(ab) = b;$$
$$\text{if } a^{-1}b \text{ is defined, so is } a(a^{-1}b), \text{ and } a(a^{-1}b) = b;$$
$$\text{if } ba^{-1} \text{ is defined, so is } (ba^{-1})a, \text{ and } (ba^{-1})a = b.$$

Rule 3. Let $a, b, b' \in X(S')$. If

$$ab = ab', \qquad ba = b'a, \qquad a^{-1}b = a^{-1}b', \qquad \text{or} \qquad ba^{-1} = b'a^{-1},$$

then $b = b'$. It is understood here that an equality entails that both of its sides are defined.

Rule 4. Let $a, b, c \in X(S')$. Whenever both sides are defined, one has

$$a((ba)^{-1}c) = b^{-1}c.$$

Similarly,

$$(c(ab)^{-1})a = cb^{-1}, \qquad a^{-1}((ab^{-1})c) = b^{-1}c, \qquad (c(b^{-1}a))a^{-1} = cb^{-1}.$$

Rule 5. All the following associativity laws hold whenever both sides are defined:

$$(a^{-1}b)c = a^{-1}(bc), \qquad (ab^{-1})c = a(b^{-1}c), \qquad (ab)c^{-1} = a(bc^{-1}),$$
$$(ab)^{-1}c = b^{-1}(a^{-1}c), \quad (a^{-1}b)c^{-1} = a^{-1}(bc^{-1}), \quad (ab^{-1})c^{-1} = a(cb)^{-1}.$$

3.2.2. Verification of the rules. (1) This follows by an evident induction on the length r of P, the case $r = 1$ being a direct consequence of 3.1(ii).

(2) This is immediate from the definition.

(3) Indeed, by Rule 2, in the first case for example,

$$b = a^{-1}(ab) = a^{-1}(ab') = b'.$$

(4) Let us verify the first relation, for example. On the right-hand side, left multiplication by b is defined and gives c, by Rule 2. Suppose that it is also defined on the left-hand side. Then

$$b(a((ba)^{-1}c)) = (ba)((ba)^{-1}c) = c.$$

Indeed, ba is defined by hypothesis, since it occurs in the expression. The middle term is therefore defined and equal to c by Rule 2, and it is equal to the left-hand term by associativity (cf. 3.1(iii)). Rule 3 thus implies that the desired equality holds whenever this multiplication by b is defined.

Now fix a and b. Rule 1 implies that $b(a((ba)^{-1}c))$ is defined for c "in" a relatively schematically dense open subset U of X. Hence, on this relatively schematically dense open subset, the two rational maps

$$c \longmapsto b^{-1}c \qquad \text{and} \qquad c \longmapsto a((ba)^{-1}c)$$

are equal. By 1.4, they are equal on every common domain of definition, which proves the desired result.

(5) The argument is of the same kind as the preceding one. For example, let us verify

$$(ab^{-1})c^{-1} = a(cb)^{-1}.$$

If right multiplication by c is defined on the right-hand side, equality follows from Rule 4. Since it is enough to verify such a formula on a relatively schematically dense open subset, we are reduced to the case in which this multiplication is defined.

3.2.3. Consider the relation R on $X \times X$ defined as follows. For $a, b, a', b' \in X(S')$, one has

$$(a, b) \sim (a', b') \pmod{R(S')}$$

if and only if there exist an fppf covering $S'' \to S'$ and a section $x \in X(S'')$ such that $(xa)b$ and $(xa')b'$ are defined and equal. Then R is an equivalence relation.

Indeed, this relation is evidently symmetric. By Rule 1, the product $(xa)b$ is defined when x is "in" a suitable relatively schematically dense open subset. Thus 1.7 asserts that there exist an fppf covering $S'' \to S'$ and an $x \in X(S'')$ for which $(xa)b$ is defined. The relation is therefore reflexive. Its transitivity follows from the next lemma.

Lemma 3.3. *Let x, y, a, b, a', b' be sections of $X(S')$ such that $(xa)b$, $(xa')b'$, $(ya)b$, and $(ya')b'$ are defined. If*

$$(xa)b = (xa')b',$$

then

$$(ya)b = (ya')b'.$$

Indeed, the lemma says that one may test $(a, b) \sim (a', b')$ using any $x \in X(S')$, with $S' \to S$ an fppf covering, for which both products are defined. Given

$$a, b, a', b', a'', b'' \in X(S),$$

Rules 1, 1.1, and 1.7 allow us to find an fppf covering $S' \to S$ and a section $x \in X(S')$ such that the three products in question are defined; transitivity follows.

Proof of Lemma 3.3. Let us begin by writing formally

$$(za)b = (((zx^{-1})x)a)b = ((zx^{-1})(xa))b = (zx^{-1})((xa)b),$$
$$(za')b' = (((zx^{-1})x)a')b' = ((zx^{-1})(xa'))b' = (zx^{-1})((xa')b').$$

The appropriate rules show that these equalities do hold whenever their terms are defined.[13] It follows that

$$(za)b = (za')b'$$

whenever all these expressions are defined. Moreover, by Rule 1 and the hypotheses already made, these expressions are defined when $z \in X(S'')$ belongs to $V(S'')$, where V is a certain relatively schematically dense open subset of X (we have taken $S' = S$). Therefore the two rational self-maps of X given by

$$z \longmapsto (za)b \qquad \text{and} \qquad z \longmapsto (za')b'$$

are equal, and consequently

$$(ya)b = (ya')b'.$$

□

[13] Editor's note: To be made precise...

Lemma 3.4. *Consider the rational map*

$$\varphi\colon X^3 \dashrightarrow X$$

over S, defined by

$$(a, b, c) \longmapsto c^{-1}(ab).$$

Let U be the domain of definition of φ, and let Γ be the graph of the morphism $f\colon U \to X$ induced by φ, regarded as a subprescheme of X^4. Then a section $(a, b, c, d) \in X^4(S)$ belongs to $\Gamma(S)$ if and only if

$$(a, b) \sim (c, d).$$

Proof. First observe that the rational map φ is also given by the formula

$$(a, b, c) \longmapsto (xc)^{-1}((xa)b)$$

for an arbitrary section $x \in X(S)$. Equivalently,

$$c^{-1}(ab) = (xc)^{-1}((xa)b)$$

whenever both sides are defined. We leave the verification to the reader.

We now show that φ is defined at a section $(a, b, c) \in X^3(S)$ if and only if there exists a d such that $(a, b) \sim (c, d)$, and that in this case

$$d = \varphi(a, b, c).$$

Suppose first that $(a, b) \sim (c, d)$. To verify that φ is defined, we may make an fppf-covering base change (Proposition 1.6), and may therefore suppose that there exists a section $x \in X(S')$ such that $(xa)b$ and $(xc)d$ are defined and equal. It follows from Rule 2 that

$$(xc)^{-1}((xa)b)$$

is defined and equal to d.

Conversely, suppose that φ is defined at (a, b, c), and put $d = \varphi(a, b, c)$. Choose an fppf covering $S' \to S$ and a section $x \in X(S')$ such that $(xa)b$ and $(xc)d$ are defined. We wish to show that they are equal. It is enough to show that the two rational self-maps of X over S, given by

$$b \longmapsto (xa)b \qquad \text{and} \qquad b \longmapsto (xc)\varphi(a, b, c),$$

are equal; this follows from the observation in the first paragraph. $\square$

Proposition 3.5. *The equivalence relation*

$$R \rightrightarrows X^2$$

is representable, and it is a flat relation of finite presentation; that is, the projections of R onto X^2 are flat morphisms of finite presentation.

Proof. We may suppose that S is affine. Since (X, W) is of finite presentation over S, we may descend the entire situation to an S of finite type over $\operatorname{Spec}\mathbb{Z}$, hence to a noetherian S. We may therefore suppose that S is noetherian. It is then immediate that the graph Γ of 3.4 is of finite presentation over X^2. The projection $\mathrm{pr}_{12}|_\Gamma$ is flat because

$$\Gamma \xrightarrow{\sim} \mathrm{pr}_{123}\,\Gamma,$$

the latter being an open subset of X^3, and because the projection

$$\mathrm{pr}_{12}\colon X^3 \longrightarrow X^2$$

is flat, since X is flat over S.

I claim that Γ represents R. There is something to prove here, because the domain of definition of a rational map does not in general commute with base change. Let $S'' \to S$. What is clear from 3.4 applied over S'' is that

$$R(S'') \supseteq \Gamma(S''),$$

because $\varphi \times_S S''$ is certainly defined on $U \times_S S''$. Now let

$$(a, b, c, d) \in R(S'').$$

We must prove that $(a, b, c, d) \in \Gamma(S'')$. This can be checked locally, say for the étale topology. We may therefore suppose that S'' is strictly local, that is, the spectrum of a henselian ring with separably closed residue field. Moreover, by applying 1.6 and the usual passage-to-the-limit arguments, we are reduced to the case in which S is strictly local and $S'' \to S$ is local.

Suppose that there is a section $x \in X(S)$ such that over S'' the products $(xa)b$ and $(xc)d$ are defined. This implies that

$$(xc)^{-1}((xa)b)$$

is defined and equal to d. Now there is a relatively schematically dense open subset V of X^3 such that $(xc)^{-1}((xa)b)$ is defined if and only if $(a, b, c) \in V(S'')$, and $V \subseteq U$. Consequently, $(a, b, c, d) \in \Gamma(S'')$ if such an x exists. By 1.6, we may make an fppf-covering base change $S' \to S$ in order to find such an x. Since S is strictly local, we can find* a faithfully flat, local, finite morphism $S' \to S$ and a section $x \in X(S')$ that passes through an arbitrary closed point of the closed fiber of X/S. To say that $(xa)b$ and $(xc)d$ are defined means that, over S'', x lies in a certain relatively schematically dense open subset; this can be checked on the closed fiber of $X_{S''}$. Thus the required section exists. □

Let G now be the quotient of X^2 by R, as a sheaf for the fppf topology. We define a composition law on G as follows. Let

$$(g, g') \in G(S')$$

be represented by a section

$$((a, b), (c, d)) \in (X^2 \times_S X^2)(S''),$$

where $S'' \to S'$ is an fppf covering. Suppose moreover that there is a section $x \in X(S'')$ such that $a(b(cx))$ and $x^{-1}d$ are defined; this is permitted by Rule 1 and 1.7. We define gg' to be the class in $G(S')$ represented by the section

$$(a(b(cx)), x^{-1}d) \in X^2(S'').$$

Let us verify that gg' is independent of the choice of the section x and of the representative $((a, b), (c, d))$. Let

$$(a', b') \sim (a, b), \qquad (c', d') \sim (c, d),$$

*Combining Exposé $\mathrm{VI_A}$, 1.1.1 and EGA IV_4, 17.16.2.

and let x' be such that $a'(b'(c'x'))$ and ${x'}^{-1}d'$ are defined. We may suppose that all these are sections over S''. We must prove that

$$(a(b(cx)), x^{-1}d) \sim (a'(b'(c'x')), {x'}^{-1}d');$$

that is, for a suitable fppf covering $S''' \to S''$ and a suitable section $z \in X(S''')$, we must have

$$(z(a(b(cx))))(x^{-1}d) = (z(a'(b'(c'x'))))({x'}^{-1}d').$$

Whenever all the products involved are defined,

$$\begin{aligned}(z(a(b(cx))))(x^{-1}d) &= ((za)(b(cx)))(x^{-1}d)\\ &= (((za)b)(cx))(x^{-1}d)\\ &= ((((za)b)c)x)(x^{-1}d)\\ &= (((za)b)c)\big(x(x^{-1}d)\big)\\ &= (((za)b)c)d,\end{aligned}$$

and the same identities hold with primes. By Rule 1 and 1.7, such a z exists. It therefore remains to prove that

$$(((za)b)c)d = (((za')b')c')d'.$$

But $(za)b = (za')b'$, since $(a,b) \sim (a',b')$ by 3.3; the desired equality now follows from $(c,d) \sim (c',d')$.

Consider the natural morphism

$$i\colon X \longrightarrow G$$

defined as follows. For $a \in X(S')$, choose an fppf covering $S'' \to S'$ and a section $b \in X(S'')$ such that ab^{-1} is defined, and put

$$i(a) = \text{the class in } G \text{ of } \big(ab^{-1}, b\big).$$

It is readily verified that this class, which a priori lies in $G(S'')$, is independent of the choice of b, and therefore defines a well-determined element of $G(S')$.

The reader may enjoy verifying the following proposition.

Proposition 3.6. *The morphism i commutes with the composition laws on X and G. In other words, if $a, b \in X(S')$ are sections for which ab is defined, then*

$$i(a)i(b) = i(ab).$$

The goal of this section is the following theorem.

Theorem 3.7. [14] *Let (X,W) be a group germ over S, with X/S faithfully flat and of finite presentation, and with separated fibers having no embedded component. With the notation above:*

(i) *G is a sheaf of groups.*

(ii) *$i\colon X \to G$ is representable by an open immersion.*

[14] Editor's note: If X is smooth and separated over S, faithfully flat and of finite presentation over S, then G/S is representable by a smooth S-group scheme of finite type over S. This is Theorem 6.6.1 of *Néron Models*, by Bosch–Lütkebohmert–Raynaud, Springer (1990).

(iii) *G is representable fppf-locally over S.*

(iv) *If G/S is representable, then it is a flat group of finite presentation, and $i\colon X \to G$ is schematically dense relative to S.*

Observe that G is plainly characterized by properties (i)–(iv); one may therefore forget its explicit construction.

The proof proceeds in several stages.

Lemma 3.8.

(i) *Let c be a section of $X(S)$. Then the morphism from the prescheme $c \times X$, which is S-isomorphic to X, into G, given by*

$$c \times X \hookrightarrow X \times X \longrightarrow G,$$

is a monomorphism.

(ii) *Let $\{c_i\}_{i\in I}$ be sections of $X(S)$, put*

$$Z = \coprod_i c_i \times X,$$

and let $R' \rightrightarrows Z$ be the equivalence relation induced on Z by the evident morphism $Z \to X^2$. Then R' is a "gluing" of the $c_i \times X \simeq X$.

Proof. (i) For two sections (c, a) and (c, a') of $(c \times X)(S')$ to have the same image in G, one must have $(c, a) \sim (c, a')$; that is, for a suitable section $x \in X(S'')$, where $S'' \to S'$ is an fppf covering,

$$(xc)a = (xc)a'$$

. It follows from Rule 3 that $a = a'$.

(ii) Let c_i, c_j be two sections, and consider the birational self-map ψ_{ji} of X over S, given by

$$x \longmapsto c_j^{-1}(c_i x).$$

For $y \in X(S)$, this is the same map as

$$x \longmapsto (yc_j)^{-1}((yc_i)x),$$

as is readily seen. Moreover, ψ_{ji} is defined at a section $b \in X(S')$ if and only if there exists a section b' such that

$$(c_i, b) \sim (c_j, b'),$$

and then $b' = \psi_{ji}(b)$. Let U_{ji} be the domain of definition of ψ_{ji} over S. It remains to prove that this domain of definition is universal: if $b \in X(S'')$, for an arbitrary $S'' \to S$, and ψ_{ji} is defined at b, then $b \in U_{ji}(S'')$. Equivalently, one must prove that if $b, b' \in X(S'')$ satisfy

$$(c_i, b) \sim (c_j, b'),$$

then $b \in U_{ji}(S'')$. We leave to the reader the verification of this fact, which is analogous to that in 3.5. □

Lemma 3.9. *Suppose that $\{c_i\}_{i\in I}$ are sections of $X(S)$ such that*

$$\coprod_i c_i \times X \longrightarrow G$$

is surjective as a morphism of sheaves. Then G is representable, flat, and of finite presentation over S, and the structural morphism

$$X^2 \longrightarrow G$$

is flat and of finite presentation.

Proof. The representability of G is an immediate consequence of Lemma 3.8, and it follows that

$$\coprod_i c_i \times X \longrightarrow G$$

is an open covering. To prove that $X^2 \to G$ is flat, it is enough to do so locally, hence to prove that the rational map

$$X^2 \dashrightarrow c_i \times X$$

induces a flat morphism on its domain of definition. This rational map is given by

$$(a,b) \longmapsto (c_i, ((xc_i)^{-1}(xa))b),$$

where x is an arbitrary section. If it is defined at $(a,b) \in X^2(S')$, one can find an fppf covering $S'' \to S'$ and a section $x \in X(S'')$ such that $\big((xc_i)^{-1}(xa)\big)b$ is defined. It is then readily seen that this morphism is flat. The same argument shows that it is locally of finite presentation, and hence is fppf-covering.

By construction the relation R is effective. Proposition 3.5 therefore shows that $X^2 \to G$ becomes a morphism of finite presentation after the fppf-covering base change $G \leftarrow X^2$. Thus $X^2 \to G$ is of finite presentation.

Let us prove that $G \to S$ is flat and of finite presentation. It is flat and locally of finite presentation because G is covered by the $c_i \times X \simeq X$. Moreover, X^2/S is quasi-compact and $X^2 \to G$ is surjective, so G/S is quasi-compact. To prove that $G \to S$ is quasi-separated, note the following cartesian diagram:

$$\begin{tikzcd} R \arrow[r, "\alpha"] \arrow[d, "\gamma"'] & X^2 \times_S X^2 \arrow[d, "\beta"] \\ G \arrow[r, "\Delta"] & G \times_S G. \end{tikzcd}$$

Here γ is surjective and α, β are quasi-compact. Hence $\beta\alpha$ is quasi-compact, and therefore so is Δ.[15] □

[15] Editor's note: Another way to conclude here is by faithfully flat descent (EGA IV_2, 2.7.1), since β is an fppf covering.

Lemma 3.10. *Let $\{c_i\}_{i\in I}$ be sections of $X(S)$. For*

$$\coprod_i c_i \times X \longrightarrow G$$

to be surjective as a morphism of fppf sheaves, it is enough that the following condition hold: for every $S' \to S$ and every $(a,b) \in X^2(S')$, there exist an open covering $\{S'_\nu\}_{\nu\in N}$ of S' and a function

$$N \longrightarrow I, \qquad \nu \longmapsto i(\nu),$$

such that $(c_{i(\nu)}^{-1}a)b$ is defined over S'_ν.

Proof. Let $S'' \to S$ be arbitrary, and let $g \in G(S'')$. Choose an fppf covering $S' \to S''$ and a section $(a,b) \in X^2(S')$ representing g. Take the open covering $\{S'_\nu\}$ supplied by the hypothesis. Over each S'_ν one has

$$(a,b) \sim (c_{i(\nu)}, (c_{i(\nu)}^{-1}a)b).$$

Thus, locally over S'_ν, the element g is represented by a section of $c_{i(\nu)} \times X$. Since the family $\{S'_\nu \to S''\}$ is fppf-covering, the desired surjectivity follows. □

Lemma 3.11. *Let Y/S be a prescheme of finite presentation, and let $\{a_i\}_{i\in I}$ be sections of $Y(S)$. Let s_0, s_1 be points of S, with s_0 a specialization of s_1, and let Y_j be the fiber of Y/S at s_j. Let C_j be the closure in Y_j of the set of points $\{a_i(s_j)\}_{i\in I}$. Then*

$$\dim C_1 \geq \dim C_0.$$

Proof. It is enough to verify the assertion after a base change $S' \to S$ together with chosen points s'_0, s'_1 such that $s'_j \mapsto s_j$ and s'_0 is a specialization of s'_1. By EGA II, 7.1.4, we are therefore reduced to the case in which S is the spectrum of a valuation ring A, s_0 is its closed point, and s_1 is its generic point.

Let V be the closure of C_1 in Y. Clearly $C_0 \subseteq V$. The lemma is consequently a special case of the "well-known" fact that an irreducible closed subprescheme V of a prescheme Y/S of finite presentation satisfies

$$\dim(V \times_S s_1) \geq \dim(V \times_S s_0)$$

when S is the spectrum of a valuation ring and $V \times_S s_1 \neq \varnothing$ (EGA IV, 13.1.6). □

Lemma 3.12. *Suppose that S is the spectrum of a local ring, with closed point s_0, and let $\{c_i\}_{i\in I}$ be sections such that the closure C_0, in the closed fiber X_0, of the set $\{c_i(s_0)\}_{i\in I}$ has dimension*

$$\dim C_0 = \dim X_0 = n.$$

Then the condition of Lemma 3.9 is satisfied.

Proof. First note that all the fibers of X/S have the same dimension. This follows from EGA IV, 12.1.1(i), and from the fact that X has a rational composition law. Lemma 3.11 therefore implies that, for every morphism $S_1 \to S$, with S_1 the spectrum of a field, the closure of $\{c_i \times_S S_1\}_{i\in I}$ in X_{S_1} has dimension n.

Let us verify the condition of Lemma 3.10. Let $(a,b) \in X^2(S')$. For $(c_i^{-1}a)b$ to be defined, it is necessary and sufficient that c_i lie in a certain open subset $U \subseteq X_{S'}$ that is schematically dense relative to S' (Rule 1). We must show that this holds for a suitable i, locally on S'. It is therefore enough to treat the case in which S' is the spectrum of a local ring; membership of c_i in $U(S')$ can then be checked on the closed fiber. We are thus reduced to $S' = \operatorname{Spec} k$, where k is a field.

With the notation above, take $S' = S_1$. Then

$$\dim C_1 = \dim X_{S_1},$$

and U is relatively schematically dense in X_{S_1}. Hence $U \cap C_1 \neq \varnothing$, and therefore

$$U \cap \{c_i \times_S S_1\}_{i \in I} \neq \varnothing.$$

This proves the assertion. □

Proof of Theorem 3.7. The proof is now easy. First note the following consequence of the finiteness assertion in Lemma 3.9. Let $\{A_i\}$ be a directed system of rings over S, put

$$\widetilde{A} = \varinjlim_i A_i,$$

and suppose that the hypotheses of Lemma 3.9 hold over $S = \operatorname{Spec} \widetilde{A}$. Then the object representing the quotient G of

$$R \rightrightarrows X^2$$

descends to some $S_i = \operatorname{Spec} A_i$, together with the finiteness and flatness properties stated in Lemma 3.9. This is the usual passage to the limit (EGA IV, 8 and 11).

It follows that, in proving Theorem 3.7(iii) and (iv), we may restrict ourselves to the case

$$S = \operatorname{Spec} A,$$

where A is a strictly local ring. Let x_0 be a closed point of the closed fiber X_0 of X/S. There exists* a local, finite, free extension A' of A, and a section of

$$X' = X \times_S S'$$

passing through the unique point x'_0 of X' lying over x_0. Notice that $X'_0 \to X_0$ is radicial because S is strictly local, and hence the residue field of A is separably closed.

It follows that there exists a directed system $\{A_i\}$ of local rings, flat and finite over A, such that, on putting

$$\widetilde{A} = \varinjlim_i A_i, \qquad \widetilde{X} = X \times_S \operatorname{Spec} \widetilde{A},$$

the prescheme $\widetilde{X}$ has a set of sections inducing a dense set on its closed fiber $\widetilde{X}_0$. Indeed, for each closed point x_0 of X_0, choose an extension $A(x_0)$ such that the corresponding base change of X admits a section "passing through x_0", and take for the directed system the finite tensor products of the $A(x_0)$. Notice that $\widetilde{A}$ is local, being a direct limit of local rings. Thus Lemmas 3.12 and 3.10 give the quotient G over $\operatorname{Spec} \widetilde{A}$, hence, by the preceding remarks, over one of the $\operatorname{Spec} A_i$, and therefore fppf-locally.

In fact, it follows from the constructions that fppf-locally one can find a finite set of sections $\{c_i\}$, $i = 1, \ldots, r$, such that G is covered by the $c_i \times X$. Assertions (ii) and (iv) follow easily from this fact, and we leave the verification of (i) to the reader.

*Cf. the note at the foot of page 15 in Exposé VII.

Corollary 3.13. *The group sheaf G determined by a group germ (X, W) over S is representable in each of the following situations:*

(i) *S is Artinian.*

(ii) *For every local scheme $S' \to S$ at a closed point s of S, the scheme $X_{S'}$ has a set of sections whose restrictions to the closed fiber form a set with closure of dimension $\dim X_s$.*

(iii) *S is strictly local and X/S is smooth.*

(iv) *There exists an fppf covering $S' \to S$ such that $G_{S'}$ is representable and affine over S.*

Indeed, (ii) follows from Lemmas 3.10 and 3.12; (iii) follows directly from (ii) and Hensel's lemma; (iv) follows from descent for affine schemes; and (i) follows from descent for group schemes, which is possible here because every finite subset of a group over a field is known to be contained in an affine open subset (Exposé VI).

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EXPOSÉ XIX

REDUCTIVE GROUPS—GENERALITIES

by M. Demazure

The remainder of this Seminar is devoted to the study of reductive groups. Its principal aim is to generalize Chevalley's classical results (the *Bible* and *Tôhoku*)[16] to arbitrary base schemes. The two central results are the uniqueness theorem (Exposé XXIII, Theorem 4.1 and Corollaries 5.1–5.10) and the existence theorem (Exposé XXV, Theorem 1.1) for pinned reductive group schemes corresponding to prescribed root data. The proofs used here are inspired by Chevalley's; the language of schemes makes it possible to give them greater effectiveness.

The results of the first volume of the *Bible* (Exposés 1–13) will be used systematically. By contrast, we shall prove directly over an arbitrary scheme the results of the second volume (in particular the isomorphism theorem); knowledge of the proofs over an algebraically closed field is therefore not absolutely indispensable.

In proving these two fundamental results, we shall use only the most elementary results from the theory of groups of multiplicative type, found essentially in Exposés VIII and IX; on the other hand, we shall make essential use of the results of Exposé XVIII.[17] The reader interested specifically in the existence and uniqueness theorems may, on a first reading, skip Exposés X–XVII.

1. Review of groups over an algebraically closed field

1.1. Throughout this subsection, k will denote an algebraically closed field. As announced above, the only results from the *Bible* used below are contained in its first volume (Exposés 1–13).

All the results of the *Bible* (*loc. cit.*) concerning semisimple groups hold more generally for reductive groups (defined below), and their proofs are identical after the following harmless modifications:

- Exposé 9, ğ4, Definition 3: see 1.6.1 below;

[0]Editor's note: version of 13 October 2024.

[16]Editor's note: see the bibliographical references at the end of this Exposé. In particular, the 2005 re-edition of the Chevalley Seminar 1956–1958, cited as *Bible*, was revised by P. Cartier; see [Ch05].

[17]Editor's note: more precisely, Proposition XVIII 2.3 (extension of a "generic homomorphism" between groups) is used in XXII 4.1.11 and then in the proof of uniqueness in XXIII, as well as in XXIV. Theorem XVIII 3.7 (construction of a group from a group germ) is used only in XXV.

- Exposé 12, ğ4, Theorem 2(e): omit "finite";
- Exposé 13, ğ3, Theorem 2: replace "rank" by "semisimple rank";
- Exposé 13, ğ4, Corollary 2 to Theorem 3: replace "rank" by "reductive rank".

1.2. Let G be a smooth affine connected k-group. The *radical* of G (*Bible*, ğ9.4, Proposition 2)[18] is the reduced subgroup associated with the identity component of the intersection of the Borel subgroups of G; it is also the largest smooth connected solvable normal subgroup of G. We denote it by $\mathrm{rad}(G)$.

The *unipotent radical* of G is the unipotent part of the radical of G; it is also the largest smooth connected unipotent normal subgroup of G. We denote it by $\mathrm{rad}^u(G)$.

1.3. Let G be a smooth affine connected k-group, and let Q be a torus in G. Then the centralizer $\underline{\mathrm{Centr}}_G(Q)$ of Q in G is a closed subgroup of G (Exposé VIII, 6.7), smooth (Exposé XI, 2.4), and connected (cf. *Bible*, ğ6.6, Theorem 6(a), or [Ch05], ğ6.7, Theorem 7(a)).

We have the fundamental relation

$$\mathrm{rad}^u(\underline{\mathrm{Centr}}_G(Q)) = \mathrm{rad}^u(G) \cap \underline{\mathrm{Centr}}_G(Q). \tag{1.3.1}$$

First,[19]

$$U = \mathrm{rad}^u(G) \cap \underline{\mathrm{Centr}}_G(Q)$$

is a normal unipotent subgroup of $\underline{\mathrm{Centr}}_G(Q)$; let us show that it is smooth and connected. Let Q act on $\mathrm{rad}^u(G)$ by inner automorphisms. Then U is precisely the scheme of fixed points of this action. This scheme is smooth and connected by Lemma 1.4 below.

Consequently, U is a closed subgroup of $\mathrm{rad}^u(\underline{\mathrm{Centr}}_G(Q))$. On the other hand, by the *Bible*, Exposé 12, ğ3, Corollary to Theorem 1,

$$\mathrm{rad}^u(\underline{\mathrm{Centr}}_G(Q))(k) \subset \mathrm{rad}^u(G)(k).$$

Equality (1.3.1) follows.

Let us record one special case of the preceding result. If G is a smooth affine connected k-group and T is a maximal torus of G, then

$$\underline{\mathrm{Centr}}_G(T) = T \cdot \big(\underline{\mathrm{Centr}}_G(T) \cap \mathrm{rad}^u(G)\big). \tag{1.3.2}$$

Indeed (cf. *Bible*, ğ6.6, Theorem 6(c), or [Ch05], ğ6.7, Theorem 7(c)), $\underline{\mathrm{Centr}}_G(T)$ is solvable and hence is the semidirect product of its maximal torus T and its unipotent radical.

By the density theorem (cf. *Bible*, ğ6.5, Theorem 5(a), or [Ch05], ğ6.6, Theorem 6(a)), the union of $\underline{\mathrm{Centr}}_G(T)$, as T ranges over the maximal tori of G, contains a dense open subset of G. It therefore follows from (1.3.2) that:

Corollary 1.3.3.— *If G is a smooth affine connected k-group, then the union of $T \cdot \mathrm{rad}^u(G)$, where T ranges over the maximal tori of G, contains a dense open subset of G.*

Notation 1.4.0.—[20] *We shall systematically use the following notation. If S is a scheme and s is a point of S, then $\bar{s}$ denotes the spectrum of an algebraic closure $\overline{\kappa(s)}$ of $\kappa(s)$.*

Lemma 1.4.—[5] *Let S be a scheme and let Q be an S-torus acting on an S-group scheme H, separated and smooth over S.*

---

[18]Editor's note: the reference 9-06 (that is, Exposé 9, p. 6) has been replaced by ğ9.4 (that is, Exposé 9, ğ4), which applies both to the *Bible* and to [Ch05]. When the numbering in [Ch05] differs from that in the *Bible*, as happens in Exposé 6, both references will be given explicitly.

[19]Editor's note: the argument that follows has been expanded.

[20]Editor's note: this notation, which appeared in 2.3 of the original, has been inserted here; moreover, the statement and proof of 1.4 have been expanded.

(i) The fixed-point functor H^Q is representable by a closed subscheme of H, smooth over S.

(ii) If, moreover, H is affine over S and has connected fibers, then the same holds for H^Q.

Proof. First, by VIII 6.5(d),[21] since H is separated over S and Q is essentially free over S, H^Q is representable by a closed subscheme of H. In particular, if H is affine over S, then so is H^Q.

Now consider the semidirect product $G = H \cdot Q$; it is smooth and separated over S. The centralizer $\underline{\mathrm{Centr}}_G(Q)$ is then representable by a closed subscheme of G, of finite presentation over G, by Exposé XI 6.11(a),[6] and it is smooth over S by Exposé XI 2.4.

Since

$$\underline{\mathrm{Centr}}_G(Q) = H^Q \cdot Q,$$

it follows that H^Q is smooth over S. Indeed, using the section

$$\sigma : H^Q \longrightarrow \underline{\mathrm{Centr}}_G(Q)$$

deduced from the identity section ϵ_Q of Q, one sees at once that H^Q is formally smooth over S; and since ϵ_Q is of finite presentation, H^Q is locally of finite presentation over S. This proves (i).

Finally, suppose that H is affine over S and has connected fibers. Then every geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$-group. Hence, by the first assertion of 1.3, the group

$$\underline{\mathrm{Centr}}_{G_{\bar{s}}}(Q_{\bar{s}}) = \big(\underline{\mathrm{Centr}}_G(Q)\big)_{\bar{s}} = (H^Q)_{\bar{s}} \cdot Q_{\bar{s}}$$

is also connected, and this implies that $(H^Q)_{\bar{s}}$ is connected. □

1.5. Recall that the *reductive rank* of a smooth affine k-group G is the common dimension of its maximal tori. We denote it by $\mathrm{rgred}(G/k)$ or $\mathrm{rgred}(G)$. The equality $\mathrm{rgred}(G/k) = 0$ holds if and only if G is unipotent (that is, $G = \mathrm{rad}^u(G)$), by the *Bible*, ğ6.4, Corollary 1 to Theorem 4, or [Ch05], ğ6.5, Corollary 1 to Theorem 5.

If H is a normal subgroup of the smooth affine k-group G, then the quotient G/H is affine and smooth (Exposé $\mathrm{VI_B}$, 11.17 and $\mathrm{VI_A}$, 3.3.2(iii)). Moreover (*Bible*, ğ7.3, Theorem 3(a),(c)),

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/H) + \mathrm{rgred}(H).$$

Definition 1.6.1.—[22] *A k-group G is said to be* reductive *if it is smooth, affine, and connected, and if* $\mathrm{rad}(G)$ *is a torus; that is, if*

$$\mathrm{rad}^u(G) = \{e\}.$$

Lemma 1.6.2.— *Let G be a reductive k-group.*

(i) If Q is a torus of G, then $\underline{\mathrm{Centr}}_G(Q)$ is reductive.

(ii) In particular, if T is a maximal torus of G, then $\underline{\mathrm{Centr}}_G(T) = T$.

(iii) The center of G is contained in every maximal torus and is therefore diagonalizable.

[21] Editor's note: see also the additions made in $\mathrm{VI_B}$, 6.2.4(d) and 6.5.5(a).

[22] Editor's note: the former number 1.6 has been replaced by numbers 1.6.1 through 1.6.3. Note that in this Seminar every reductive k-group (respectively, every semisimple k-group; cf. 1.8) is connected by definition.

(iv) The radical of G is the unique maximal torus of $\underline{\mathrm{Centr}}(G)$.

Proof. Indeed, (i) follows from (1.3.1), (ii) from (1.3.2), and (iii) follows from (ii). Finally, the maximal torus of $\underline{\mathrm{Centr}}(G)$, that is, the neutral component $\underline{\mathrm{Centr}}(G)^0$, is a smooth connected solvable subgroup normal in G, and hence is contained in $\mathrm{rad}(G)$. Conversely, $\mathrm{rad}(G)$ is a torus normal in G, and hence central (*Bible*, ğ4.3, Corollary to Proposition 2); this proves (iv). □

Remark 1.6.3.— *If G is reductive and* $\dim(G) \neq \mathrm{rgred}(G)$, *then*

$$\dim(G) - \mathrm{rgred}(G) \geq 2.$$

Indeed, this difference is always even (cf. 1.10 below).

1.7. Let G be a smooth affine connected k-group, and let H be a smooth connected normal subgroup. Then

$$\mathrm{rad}(H) = \big(\mathrm{rad}(G) \cap H\big)^0_{\mathrm{red}}, \qquad \mathrm{rad}^u(H) = \big(\mathrm{rad}^u(G) \cap H\big)^0_{\mathrm{red}}, \tag{1.7.1}$$

as is seen immediately. In particular, if G is reductive, then so is H.

Let $f : G \longrightarrow G'$ be a surjective morphism of smooth affine connected k-groups. Then

$$f\big(\mathrm{rad}^u(G)\big) = \mathrm{rad}^u(G'). \tag{1.7.2}$$

In particular, if G is reductive, then so is G'.

Let us prove (1.7.2). First, $f\big(\mathrm{rad}^u(G)\big)$ is contained in $\mathrm{rad}^u(G')$. On introducing

$$H = \big(f^{-1}(\mathrm{rad}^u(G'))\big)^0_{\mathrm{red}},$$

which contains $\mathrm{rad}^u(G)$, we have $\mathrm{rad}^u(H) = \mathrm{rad}^u(G)$; thus we are reduced to the case $G = H$, that is, to the case in which G' is unipotent. As T runs over the maximal tori of G, the union of the groups $T\,\mathrm{rad}^u(G)$ is dense in G. Hence the union of the groups $f(T)\,f\big(\mathrm{rad}^u(G)\big)$ is dense in G'. But $f(T)$ consists of semisimple elements, so $f(T) = \{e\}$, since G' is unipotent. It follows that $f\big(\mathrm{rad}^u(G)\big)$ is dense in G'. Therefore, by *Bible*, ğ5.4, Lemma 4, or [Ch05], ğ6.1, Lemma 1,[23] $f\big(\mathrm{rad}^u(G)\big)$ is an open subgroup of G'. Since G' is connected, it follows that

$$f\big(\mathrm{rad}^u(G)\big) = G'.$$

N.B. Under the same hypotheses one can prove that

$$f\big(\mathrm{rad}(G)\big) = \mathrm{rad}(G').$$

1.8. A k-group G is said to be *semisimple* if it is smooth, affine, and connected, and if

$$\mathrm{rad}(G) = \{e\},$$

that is, if G is reductive and $\underline{\mathrm{Centr}}(G)$ is finite. If G is an arbitrary smooth affine connected k-group, then $G/\mathrm{rad}(G)$ is semisimple (*Bible*, ğ9.4, Proposition 2), and $G/\mathrm{rad}^u(G)$ is reductive. The *semisimple rank of* G, denoted by $\mathrm{rgss}(G/k)$, or simply by $\mathrm{rgss}(G)$, is the reductive rank of $G/\mathrm{rad}(G)$.

Thus, if G is reductive,

$$\mathrm{rgred}(G) = \mathrm{rgss}(G) + \dim\big(\mathrm{rad}(G)\big).$$

[23] Editor's note: the following argument has been supplied in detail.

If G is a smooth affine connected k-group and Q is a central subtorus of G, then G/Q is semisimple if and only if G is reductive and $Q = \mathrm{rad}(G)$. Indeed, if G/Q is semisimple, then $Q \supset \mathrm{rad}(G)$; hence $\mathrm{rad}(G)$ is a torus, so G is reductive, and $\mathrm{rad}(G)$ is the maximal torus of $\underline{\mathrm{Centr}}(G)$, whence $\mathrm{rad}(G) = Q$. If G is reductive and Q is a central subgroup, then G/Q is reductive and

$$\mathrm{rgss}(G) = \mathrm{rgss}(G/Q).$$

1.9. Let K be an algebraically closed extension of k, and let G be a smooth affine connected k-group. Then G is reductive (respectively, semisimple) if and only if G_K is, and

$$\mathrm{rgred}(G/k) = \mathrm{rgred}(G_K/K), \qquad \mathrm{rgss}(G/k) = \mathrm{rgss}(G_K/K).$$

1.10. Let G be a smooth connected k-group and let T be a torus of G.[24] Write $\mathfrak{g}$ for the k-Lie algebra of G, that is, $\mathfrak{g} = \mathrm{Lie}(G) = \mathrm{Lie}(G/k)(k)$, and similarly write $\mathfrak{t} = \mathrm{Lie}(T)$. Under the action of T, through $T \longrightarrow G$ and the adjoint representation of G, the Lie algebra $\mathfrak{g}$ decomposes as

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha, \tag{1.10.1}$$

where the elements $\alpha \in R$ are nontrivial characters of T and the spaces $\mathfrak{g}^\alpha$ are nonzero. The characters $\alpha \in R$ are called the *roots of G relative to T*. By Exposé II, 5.2.3(i),

$$\mathfrak{g}^0 = \mathrm{Lie}(\underline{\mathrm{Centr}}_G(T)). \tag{1.10.2}$$

In particular,[25] since $\underline{\mathrm{Centr}}_G(T)$ is connected (by 1.3 when G is affine, and by Exposé XII, 6.6(b), in general), T is its own centralizer if and only if $\mathfrak{g}^0 = \mathrm{Lie}(T)$. This condition implies that T is maximal and that $\underline{\mathrm{Centr}}(G) \subset T$; hence, by Exposé XII, 8.8(d),

$$\underline{\mathrm{Centr}}(G) = \mathrm{Ker}(T \longrightarrow \mathrm{GL}(\mathfrak{g})) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha). \tag{1.10.3}$$

Moreover, $\underline{\mathrm{Centr}}(G)$ is then affine, and therefore the morphism $G \longrightarrow G/\underline{\mathrm{Centr}}(G)$ is affine. Since the latter group is affine (Exposé XII, 6.1),[26] G is affine.

When G is reductive and T is maximal, the roots in the preceding sense coincide with the roots in the sense of *Bible*, ğ12.2, Definition 1. The latter are indeed roots in the sense of this number (*Bible*, ğ13.2, Theorem 1(c)), and their number is $\dim(G) - \dim(T)$ (*Bible*, ğ13.4, Corollary 2 to Theorem 3). Moreover, if α is a root, then so is $-\alpha$ (*Bible*, ğ12.2, Corollary to Proposition 1). As usual, we write the group law of $\mathrm{Hom}_{k\text{-gr}}(T, \mathbb{G}_{m,k})$ indifferently additively or multiplicatively. It follows that, for reductive G,

$$\dim(G) - \mathrm{rgred}(G) = \mathrm{Card}(R) \tag{1.10.4}$$

is always even.

The semisimple rank of the reductive group G is the rank of the subset R of the $\mathbb{Q}$-vector space

$$\mathrm{Hom}_{k\text{-gr}}(T, \mathbb{G}_{m,k}) \otimes_{\mathbb{Z}} \mathbb{Q}$$

(*Bible*, ğ13.3, Theorem 2).

[24] Editor's note: the following sentence has been added.
[25] Editor's note: the following argument has been given in detail.
[26] Editor's note: see also Addition $\mathrm{VI_B}$, 6.2.6.

Lemma 1.11.— *Let k be an algebraically closed field, G a smooth affine connected k-group, and T a torus of G. Put*

$$W(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T),$$

the Weyl group of G relative to T. The following conditions are equivalent:

(i) G *is reductive,* T *is maximal, and* $\mathrm{rgss}(G) = 1$.

(ii) G *is reductive,* T *is maximal, and* $G \neq T$*; there exists a subtorus* Q *of* T*, of codimension* 1 *in* T*, that is central in* G.

(iii) G *is not solvable and* $\dim(G) - \dim(T) \leq 2$.

(iv) $W(T) \neq \{e\}$ *and* $\dim(G) - \dim(T) \leq 2$.

(v) $W(T) = (\mathbb{Z}/2\mathbb{Z})_k$ *and* $\dim(G) - \dim(T) = 2$.

Moreover, under these conditions there are exactly two roots of G relative to T, and they are opposite. Under the conditions of (ii)*,* $Q = \mathrm{rad}(G)$.

Proof. The implication (v)⇒(iv) is clear, and (iv)⇒(iii) follows from *Bible*, ğ6.1, Corollary 3 to Theorem 1, or [Ch05], ğ6.2, Corollary 3 to Theorem 2.

Let us prove (iii)⇒(ii). Let U be the unipotent radical of G. The group G/U is reductive and is not a torus (otherwise G would be solvable); hence, by (1.10.4),

$$\dim(G/U) - \mathrm{rgred}(G/U) = \mathrm{Card}(R) \geq 2.$$

On the other hand,

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/U) + \mathrm{rgred}(U) = \mathrm{rgred}(G/U),$$

and therefore

$$\begin{aligned}\dim(G) - \dim(U) - \mathrm{rgred}(G) &= \dim(G/U) - \mathrm{rgred}(G/U)\\ &\geq 2 \geq \dim(G) - \dim(T) \geq \dim(G) - \mathrm{rgred}(G).\end{aligned}$$

It follows that $\dim(U) = 0$, so G is reductive, and that $\mathrm{rgred}(G) = \dim(T)$; thus T is maximal and $\dim(G) - \dim(T) = 2$. By *Bible*, ğ10.4, Proposition 8, there exists a singular torus Q of codimension 1 in T. Then $\underline{\mathrm{Centr}}_G(Q)$ is reductive (1.6.2(i)) and not solvable (by the definition of a singular torus), and hence

$$\dim(\underline{\mathrm{Centr}}_G(Q)) - \mathrm{rgred}(G) \geq 2.$$

This proves that $G = \underline{\mathrm{Centr}}_G(Q)$, and that Q is central in G.

Let us prove (ii)⇒(i). We know that G/Q is reductive by 1.7 and that $\mathrm{rgred}(G/Q) = 1$; hence $\mathrm{rgss}(G/Q)$ is either 0 or 1. The first case is impossible, since it would imply $\mathrm{rgss}(G) = 0$, and therefore $G = T$. Thus

$$\mathrm{rgred}(G/Q) = \mathrm{rgss}(G/Q) = 1,$$

which proves that G/Q is semisimple. Consequently Q is the radical of G and $\mathrm{rgss}(G) = 1$.

Let us finally prove (i)⇒(v). If Q is the radical of G, then $\dim(T) - \dim(Q) = 1$, and Q is central in G. Hence $G = \underline{\mathrm{Centr}}_G(Q)$, which proves that Q is a singular torus. By *Bible*, ğ11.3, Theorem 2,

$$W_G(T) = (\mathbb{Z}/2\mathbb{Z})_k,$$

and, by *Bible*, ğ12.1, Lemma 1, $\dim(G) - \dim(T) = 2$. Thus G has at most two roots relative to T, while it has at least two, opposite to one another, by 1.10.

Proposition 1.12.— *Let k be an algebraically closed field, G a smooth connected k-group, T a torus of G, and R the set of roots of G relative to T. Write*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha, \qquad \mathfrak{g}^\alpha \neq 0,$$

for the decomposition of $\mathfrak{g}$ under $\mathrm{Ad}(T)$. For every $\alpha \in R$, put

$$T_\alpha = \mathrm{Ker}(\alpha)^0, \qquad Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

[27] *The following conditions are equivalent:*

(i) G is affine and reductive, and T is maximal.

(ii) $\mathfrak{g}^0 = \mathfrak{t}$, and every Z_α, for $\alpha \in R$, is reductive.

(iii) $\mathfrak{g}^0 = \mathfrak{t}$; every $\mathfrak{g}^\alpha$, for $\alpha \in R$, has dimension 1; if $\alpha, \beta \in R$ and $\beta = q\alpha$ for some $q \in \mathbb{Q}$, then $q = \pm 1$; and, for every $\alpha \in R$, there exists $w_\alpha \in G(k)$ which normalizes T, centralizes T_α, but does not centralize T.

Moreover, under these conditions every Z_α has semisimple rank 1, and

$$\mathrm{Lie}(Z_\alpha) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}.$$

Proof. Suppose first that G is affine and reductive and that T is maximal. Every Z_α is then affine and reductive by 1.6.2(i), with maximal torus T; furthermore, T is its own centralizer by 1.6.2(ii). Thus $\mathfrak{g}^0 = \mathfrak{t}$, proving (i)⇒(ii).

Conversely, the equality $\mathfrak{g}^0 = \mathfrak{t}$ implies that T is maximal and that G is affine, by 1.10. Hence every Z_α is affine, smooth, and connected, by 1.3. In all cases, Exposé II, 5.2.2 gives

$$\mathrm{Lie}(Z_\alpha) = \mathfrak{g}^{T_\alpha} = \mathfrak{g}^0 \oplus \bigoplus_{\beta \in R \cap \mathbb{Q}\alpha} \mathfrak{g}^\beta.$$

Consequently,

$$\mathrm{Lie}(Z_\alpha) \supset \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

so $Z_\alpha \neq T$. Since T_α is a central subtorus of Z_α, of codimension 1 in T, Lemma 1.11 applied to Z_α proves the equivalence of (ii) and (iii), together with the additional assertions.

It remains to prove (ii)⇒(i). We already know that (ii) implies that T is maximal and that G is affine. Let U be the unipotent radical of G. It is invariant in G, and its Lie algebra is invariant under $\mathrm{Ad}(T)$; hence

$$\mathrm{Lie}(U) = \big(\mathrm{Lie}(U) \cap \mathfrak{g}^0\big) \oplus \bigoplus_{\alpha \in R} \big(\mathrm{Lie}(U) \cap \mathfrak{g}^\alpha\big).$$

By (1.3.1),

$$U \cap T = U \cap \underline{\mathrm{Centr}}_G(T) = \mathrm{rad}^u\big(\underline{\mathrm{Centr}}_G(T)\big) = \mathrm{rad}^u(T) = \{e\},$$
$$U \cap Z_\alpha = U \cap \underline{\mathrm{Centr}}_G(T_\alpha) = \mathrm{rad}^u\big(\underline{\mathrm{Centr}}_G(T_\alpha)\big) = \mathrm{rad}^u(Z_\alpha) = \{e\}.$$

[27] Editor's note: thus T_α is the identity component of $\mathrm{Ker}(\alpha)$.

It follows[28] that

$$\begin{aligned}\mathrm{Lie}(U) \cap \mathfrak{g}^0 &= \mathrm{Lie}(U) \cap \mathfrak{t} = \mathrm{Lie}(U \cap T) = 0,\\ \mathrm{Lie}(U) \cap \mathfrak{g}^\alpha &\subset \mathrm{Lie}(U) \cap \mathrm{Lie}(Z_\alpha) = \mathrm{Lie}(U \cap Z_\alpha) = 0.\end{aligned}$$

Thus $\mathrm{Lie}(U) = 0$, and therefore $U = \{e\}$; in other words, G is reductive.

Corollary 1.13.— *Let k, G, T, R, and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha$$

be as in Proposition 1.12. For every $\alpha \in R$, let T_α be the maximal torus of $\mathrm{Ker}(\alpha)$ *and put $Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha)$. The following conditions are equivalent:*

(i) G is affine and semisimple, and T is maximal.

(ii) $\mathfrak{g}^0 = \mathfrak{t}$, every Z_α is reductive, and $\bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$ is finite.

29

2. Reductive group schemes: definitions and first properties

Scholium 2.1.— *For an S-group scheme G, the following properties are equivalent:*

(i) G is affine and smooth over S, with connected fibers.

(ii) G is affine and flat over S, of finite presentation over S, with geometrically integral fibers.

These properties are stable under base change and are local for the fpqc topology.

[30] Indeed, suppose that (i) holds. Since G is affine and smooth over S, it is of finite presentation over S; and since its fibers are smooth and connected, they are geometrically integral, by Exposé $\mathrm{VI_A}$, 2.4.

Conversely, if (ii) holds, the fibers of G are connected and geometrically reduced, and hence smooth (Exposé $\mathrm{VI_A}$, 1.3.1). It follows that G is smooth over S, by EGA IV_4, 17.5.2.

These properties are plainly stable under base change: see EGA II, 1.5.1 for "affine", EGA IV_1, 1.6.2(iii) for "of finite presentation", EGA IV_2, 2.1.4 for "flat", and EGA IV_2, 4.6.5(i) for "with geometrically integral fibers".

They are also clearly local for the Zariski topology. It is therefore enough to check that, if $S' \longrightarrow S$ is faithfully flat and quasi-compact and $G_{S'} \longrightarrow S'$ has the indicated properties, then so does $G \longrightarrow S$. This follows from EGA IV_2, 2.5.1 for "flat", 2.7.1(vi) and (xiii) for "of finite presentation" and "affine", and 4.6.5(i) for "with geometrically integral fibers"; for "smooth", see also EGA IV_4, 17.7.3(ii).

2.2. Let G be an S-group scheme satisfying the preceding conditions, and let Q be a torus of G in the sense of Exposé IX, Definition 1.3.[31] By Exposé XI, 6.11(a) and 2.4, $\underline{\mathrm{Centr}}_G(Q)$ is represented by a closed subgroup scheme of G; it is therefore affine over S, and it is of

[28] Editor's note: if U and L are two subgroup schemes of G, then $\mathrm{Lie}(U) \cap \mathrm{Lie}(L) = \mathrm{Lie}(U \cap L)$.

[29] Editor's note: this follows from the equalities $\mathrm{rad}(G) = \underline{\mathrm{Centr}}(G)^0$ and $\underline{\mathrm{Centr}}(G) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$, established in 1.6.2(iv) and (1.10.3).

[30] Editor's note: the following details have been added.

[31] Editor's note: the following argument has been given in detail.

finite presentation and smooth over S. Moreover, every geometric fiber $G_{\bar{s}}$ is a smooth affine connected $\kappa(\bar{s})$-group. The first assertion of 1.3 therefore shows that the same is true of

$$\underline{\mathrm{Centr}}_{G_{\bar{s}}}(Q_{\bar{s}}) = \big(\underline{\mathrm{Centr}}_G(Q)\big)_{\bar{s}}.$$

Lemma 2.3.— *Let S be a scheme, let G be an S-group scheme that is smooth and affine over S, with connected fibers, and let T be a torus of G. The set of points $s \in S$ for which $G_{\bar{s}}$ is reductive, of semisimple rank* 1*, with maximal torus $T_{\bar{s}}$, is an open subset U of S.*

Proof. Since G and T are flat over S, the function

$$s \longmapsto \dim(G_{\bar{s}}/\bar{s}) - \dim(T_{\bar{s}}/\bar{s})$$

is locally constant. Let U_1 be the open subset of S on which this function has the value 2.

[32] By 6.3 the Weyl group

$$W_G(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T)$$

is represented by an S-group scheme that is étale and separated over S, and the function

$$s \longmapsto \text{the number of points of } W_G(T)_{\bar{s}}$$

is lower semicontinuous. Let U_2 be the locus on which this number is greater than 1; it is open. By Lemma 1.11, the locus on which $G_{\bar{s}}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}}$, is

$$U = U_1 \cap U_2.$$

Furthermore, $W_G(T)_{\bar{s}}$ has exactly two points for every $s \in U$. Consequently, by SGA 1, I, 10.9 and EGA IV$_3$, 15.5.1 and IV$_4$, 18.12.4, $W_G(T)_U$ is finite and étale over U.

Remark 2.4.— *The group $W_G(T)_U$ is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_U$.*

Indeed, the preceding argument shows that it is finite and étale over U. Since the automorphism functor of $(\mathbb{Z}/2\mathbb{Z})_U$ is the trivial group, it is enough to verify the assertion locally. It is immediate whenever the algebra $\mathscr{A}$ defining $W_G(T)_U$ is a free $\mathscr{O}_U$-module.[33]

Notations and reminders 2.5.0.—[34] Let S be a scheme, let G be an S-group scheme, and let $\epsilon\colon S \to G$ be the unit section of G. We saw in II, ğ4.11 that the functor $\underline{\mathrm{Lie}}(G/S)$ is represented by the vector bundle $\mathbb{V}(\omega^1_{G/S})$, where $\omega^1_{G/S} = \epsilon^*(\Omega^1_{G/S})$, and we write

$$\mathfrak{g} = \mathrm{Lie}(G/S) = \big(\omega^1_{G/S}\big)^\vee$$

for the sheaf of sections of this vector bundle. Suppose in addition that G is smooth over S. Then $\omega^1_{G/S}$, and hence $\mathrm{Lie}(G/S)$, are finite locally free $\mathscr{O}_S$-modules, and (cf. I, 4.6.5.1)

$$\underline{\mathrm{Lie}}(G/S) = \mathbb{W}\big(\mathrm{Lie}(G/S)\big);$$

that is, for every S-scheme S',

$$\underline{\mathrm{Lie}}(G/S)(S') = \Gamma\big(S', \mathrm{Lie}(G/S) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}\big)\,.$$

[32]Editor's note: the following details have been added. Note also that ğ6 is independent of the rest of this Exposé.

[33]Editor's note: the hypothesis implies that $\mathscr{A}$ is locally free of rank 2. Since the augmentation ideal $\mathscr{I}$ is a direct summand of $\mathscr{A}$, after replacing U by a sufficiently small affine open $\mathrm{Spec}(R)$, one may suppose that $I = \Gamma(U, \mathscr{I})$ is a free R-module of rank 1. If x is a generator of I, then $x^2 = ax$ for some $a \in R$; the assumption that $A = \Gamma(U, \mathscr{A})$ is étale implies that a is invertible, which gives the asserted local identification.

[34]Editor's note: this paragraph of notations and reminders has been added.

By II, 4.1.1.1, the adjoint action of G gives $\underline{\mathrm{Lie}}(G/S) = \mathbb{W}(\mathrm{Lie}(G/S))$ the structure of a G-$\mathscr{O}_S$-module; thus $\mathrm{Lie}(G/S)$ is a G-$\mathscr{O}_S$-module (cf. I, 4.7.1). If G is moreover affine over S, this says, by I, 4.7.2, that $\mathrm{Lie}(G/S)$ is an $A(G)$-comodule.

If T is a torus over S (IX, Definition 1.3), we say that T is *split* ("trivial" in the original) if it is isomorphic to $\mathbb{G}^r_{m,S}$ for some integer $r \geq 0$, and that T is *trivialized* if such an isomorphism has been fixed, or more generally if an isomorphism

$$T \simeq D_S(M)$$

has been fixed, where M is a free abelian group of rank r.

Finally, recall (XII, Definition 1.3) that a torus T of G is called *maximal* if, for every $s \in S$, $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Theorem 2.5.— *Let S be a scheme, let G be an S-group scheme that is affine and of finite presentation over S, with connected fibers, and let s_0 be a point of S. Suppose that G is flat over S at $\epsilon(s_0)$, and that the geometric fiber $G_{\bar{s}_0}$ is (reduced and) reductive* (resp. semisimple). *Then there exist an open subset $U \subset S$, containing s_0, and an étale surjective morphism $S' \to U$ such that:*

(i) G_U is smooth, with reductive (resp. semisimple) *geometric fibers, and its reductive rank and semisimple rank are constant.*

(ii) $G_{S'}$ has a split[35] *maximal torus T, and the Weyl group* (cf. 6.3)

$$W_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T)/\underline{\mathrm{Centr}}_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T)/T$$

is finite over S'.

Proof. [36] Write $\pi\colon G \to S$ for the structure morphism and $\epsilon\colon S \to G$ for the unit section. Since G is flat over S at $\epsilon(s_0)$, and since $G_{\bar{s}_0}$ is reduced (hence smooth over $\bar{s}_0$, by $\mathrm{VI_A}$, 1.3.1), G is smooth over S at $\epsilon(s_0)$ (EGA IV_4, 17.5.1). Thus there is an open neighborhood V of $\epsilon(s_0)$ on which π is smooth. Then $S' = \epsilon^{-1}(V)$ is open in S, and $G_{S'}$ is smooth over S' at the points of $\epsilon(S')$. Since G has connected fibers, $\mathrm{VI_B}$, 3.10 shows that $G_{S'}$ is smooth over S'. Replacing S by S', we may therefore suppose that G is smooth over S.

By Exposé XI, Theorem 4.1, the functor of subgroups of multiplicative type of G is represented by an S-scheme $\mathscr{M}$, smooth and separated over S. Let r_0 be the rank of $G_{\bar{s}_0}$, and consider the open subscheme $\mathscr{M}_{r_0} \subset \mathscr{M}$ representing the functor of subtori of rank r_0 of G (IX, 1.4). Smoothness implies that $\mathscr{M}_{r_0}$ has a rational point over a finite separable extension of $\kappa(s_0)$ (EGA IV_4, 17.15.10(iii)). Consequently there is an étale morphism $S' \to S$, together with a point s'_0 above s_0, such that $G_{s'_0}$ has a torus of rank r_0. Replacing S by S', we may therefore suppose that G_{s_0} has a torus of rank r_0.

By "Hensel's lemma" (XI, 1.10), the smoothness of $\mathscr{M}_{r_0}$ allows this torus to be lifted to an S'-torus T of G, where $S' \to S$ is étale and is equipped with a point s'_0 above s_0. By Exposé X, 4.5 (see also 6.1[37]), there is an étale morphism $f\colon S'' \to S'$ that splits T and for which $f^{-1}(s'_0) \neq \varnothing$. Because an étale morphism is open, and the assertions in (i) are local for the étale topology, we may thus suppose that G has a split torus T,[38] maximal at s_0. Write $T = D_S(M)$, and let

$$\mathfrak{g} = \prod_{m \in M} \mathfrak{g}^m$$

[35] Editor's note: the word "split" has been added.
[36] Editor's note: the proof has been expanded.
[37] Editor's note: ğ6.1 is independent of the rest of this Exposé.
[38] Editor's note: throughout this volume, "trivial torus" has been replaced by "split torus".

be the decomposition of $\mathfrak{g} = \mathrm{Lie}(G/S)$ under $\mathrm{Ad}(T)$ (Exposé I, 4.7.3).

Put $\mathfrak{t} = \mathrm{Lie}(T)$, and, for every $m \in M$, write

$$\mathfrak{g}^m(s_0) = \mathfrak{g}^m \otimes_{\mathcal{O}_S} \kappa(s_0).$$

[39] Let R be the set of $m \in M$, $m \neq 0$, such that $\mathfrak{g}^m(s_0) \neq 0$.[40] Since $G_{\bar{s}_0}$ is reductive, $\mathfrak{g}^0(s_0) = \mathfrak{t}(s_0)$, and hence

$$\mathfrak{g}(s_0) = \mathfrak{t}(s_0) \oplus \prod_{\alpha \in R} \mathfrak{g}^\alpha(s_0).$$

Since the modules in question are locally free, after shrinking S we may suppose that the $\mathfrak{g}^\alpha$ are free and that

$$\mathfrak{g} = \mathfrak{t} \oplus \prod_{\alpha \in R} \mathfrak{g}^\alpha.$$

Recall from Exposé XII, 1.12 that a group of multiplicative type has a unique maximal torus (this also follows almost immediately by descent, the diagonalizable case being evident). Let T_α be the maximal torus of $\ker(\alpha)$, and put

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

[41] By 2.2, Z_α is affine and smooth over S, with connected fibers; and by 1.12, its fiber $(Z_\alpha)_{\bar{s}_0}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}_0}$. By 2.3 there is therefore an open subset $U_\alpha \subset S$, containing s_0, on which Z_α has reductive fibers.

Set

$$U = \bigcap_{\alpha \in R} U_\alpha.$$

By 1.12, for every $s \in U$, $G_{\bar{s}}$ is reductive, with maximal torus $T_{\bar{s}}$, and the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$ is identified with R. Consequently

$$\mathrm{rgred}(G_{\bar{s}}) = \dim(T) = \mathrm{rg}(M), \qquad \mathrm{rgss}(G_{\bar{s}}) = \mathrm{rg}(R) \quad (\text{cf. } 1.10).$$

This proves (i) and the first assertion of (ii). It remains to prove that $W_{G_U}(T_U)$ is finite over U, that is, that it has the same number of points in every geometric fiber (SGA 1, I, 10.9; EGA IV_3, 15.5.1; and EGA IV_4, 8.12.4).

For this it is enough to observe that, for $s \in U$, the geometric fiber of this group is determined by the configuration $R \subset M$: it is the constant group associated with the abstract Weyl group of this root system. In particular it is independent of s; see the *Bible*, ğ11.3, Theorem 2 (and also Exposé XXII, no. 3). □

Corollary 2.6.— *Let G be an S-group scheme that is affine and smooth over S, with connected fibers. The set of points $s \in S$ for which $G_{\bar{s}}$ is reductive* (resp. semisimple) *is an open subset U of S, and the functions*

$$s \longmapsto \mathrm{rgred}(G_{\bar{s}}/\bar{s}), \qquad s \longmapsto \mathrm{rgss}(G_{\bar{s}}/\bar{s})$$

are locally constant on U.

Definition 2.7.— *An S-group* (that is, an S-group scheme) *G is called* reductive (resp. *semisimple*) *if it is affine and smooth over S, with connected reductive* (resp. semisimple) *geometric fibers.*

[39] Editor's note: the preceding sentence has been added.
[40] Editor's note: since $\mathfrak{g}$ is a finite locally free $\mathcal{O}_S$-module, R is finite.
[41] Editor's note: the following argument has been expanded.

For an S-group scheme G, the property of being reductive (resp. semisimple) is stable under base change and local for the fpqc topology.

2.8.— Let G be a reductive S-group. For every torus (resp. maximal torus) Q of G,

$$Z(Q) = \underline{\mathrm{Centr}}_G(Q)$$

is reductive (resp. is equal to Q). This follows at once from 2.2 and 1.6. Applying 2.5 to $\underline{\mathrm{Centr}}_G(Q)$, we deduce that Q is contained, locally for the étale topology, in a maximal torus.

Remark 2.9.— *Using, as usual, the technique of EGA IV$_3$, ğ8, the finite-presentation hypotheses, and Theorem 2.5, one sees that if G is a reductive group over S, there is an open cover $\{U_i\}$ of S such that each G_{U_i} is obtained by base change from a reductive group over a noetherian ring* (in fact, over a finitely generated $\mathbb{Z}$-algebra). *Likewise, if G has a split maximal torus T, one may further suppose that T_{U_i} comes from a split maximal torus of the preceding group, ...*

3. Roots and root systems of reductive group schemes

3.1.— Let S be a scheme, and let T be an S-torus acting linearly on a finite locally free $\mathcal{O}_S$-module F (cf. I, ğ4.7). For every character α of T, that is,

$$\alpha \in \mathrm{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}),$$

define a subfunctor of $\mathbb{W}(F)$ by

$$\mathbb{W}(F)^\alpha(S') = \{x \in \mathbb{W}(F)(S') \mid t \cdot x = \alpha(t)x \text{ for every } t \in T(S''),\ S'' \to S'\}.$$

Lemma.—[42] *Then $\mathbb{W}(F)^\alpha = \mathbb{W}(F^\alpha)$, where F^α is a submodule of F, locally a direct summand of F, and hence also locally free.*

Indeed, the assertion is local for the fpqc topology, and we may suppose that $T = D_S(M)$, where M is a finite-type (free) abelian group. Then α is identified with a locally constant function from S to M (Exposé VIII, 1.3), and, after restricting S if necessary, we may suppose that this function is constant. We are then reduced to Exposé I, 4.7.3.

Definition 3.2.— *Let S be a scheme, let G be an S-group scheme smooth over S and with connected fibers,*[43] *and let T be a torus of G. Put $\mathfrak{g} = \mathrm{Lie}(G/S)$, and let T act on $\mathfrak{g}$ through the adjoint representation of G.*

A character α of T is called a root *of G with respect to T if the following equivalent conditions are satisfied:*

(i) For every $s \in S$, α_s is a root of G_s with respect to T_s (1.10).

(ii) α is nontrivial on every fiber, and $\mathfrak{g}^\alpha(s) \neq 0$ for every $s \in S$.

Lemma 3.3.— *Let S be a scheme, let T be an S-torus, and let α be a character of T. The following conditions are equivalent:*

[42] Editor's note: this statement has been made into a lemma in order to bring it out.

[43] Editor's note: the original also assumed that G was of finite presentation over S, an assumption that does not appear to be used below. In any case, since G is smooth over S and has connected fibers, it is quasi-compact and separated over S ($\mathrm{VI_B}$, 5.5), and hence of finite presentation over S.

(i) *α is nontrivial on every fiber; that is, for every $s \in S$, α_s is distinct from the unit character of T_s.*

(ii) *For every $S' \to S$ with $S' \neq \varnothing$, $\alpha_{S'}$ is distinct from the unit character of $T_{S'}$.*

(iii) *The morphism α is faithfully flat.*

[44] It is clear that (ii) implies (i), and it is easy to see that (iii) implies (i). Condition (i) implies (ii): if $s' \in S'$ lies over s, and $\alpha_{s'}$ is the unit character, then so is α_s. Finally, to prove that (i) implies (iii), we reduce to the case in which T is diagonalizable and conclude by Exposé VIII, 3.2(a).

3.4.—[45] Let G be a reductive S-group scheme, let T be a maximal torus of G, and let α be a root of G with respect to T. Then, by 2.5.0 and 1.12, $\mathfrak{g}^\alpha$ is a locally free $\mathcal{O}_S$-module of rank one. Moreover, by 1.10, $-\alpha$ is also a root of G with respect to T. In particular, if G has semisimple rank one, 1.11 and 1.12 give the following statement.

Lemma 3.5.— *Let S be a scheme, let G be a reductive S-group of semisimple rank one, and let T be a maximal torus of G. If α is a root of G with respect to T, then so is $-\alpha$, and*

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank one.

Definition 3.6.— *Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let R be a set of roots of G with respect to T. We say that R is a* root system *of G with respect to T if the following equivalent conditions are satisfied:*

(i) *For every $s \in S$, the map $\alpha \mapsto \alpha_s$ is a bijection from R onto the set of roots of G_s with respect to T_s.*

(ii) *The elements of R are distinct on every fiber* (that is, if $\alpha, \alpha' \in R$ and $\alpha \neq \alpha'$, then $\alpha - \alpha' = \alpha\alpha'^{-1}$ is nontrivial on every fiber)*, and, for every $s \in S$,*

$$\dim(G_s/\kappa(s)) - \dim(T_s/\kappa(s)) = \mathrm{Card}(R).$$

(iii) *One has*

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha.$$

The equivalence of these conditions is immediate.

Lemma 3.7.— *Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let R be a root system of G with respect to T. Every root of G with respect to T is locally on S equal to an element of R.*

Proof. This follows from condition (iii).

Put

$$\mathcal{M} = \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S});$$

this is a twisted constant S-group scheme (Exposé X, 5.6). If G has a root system R with respect to T, then the inclusion $R \hookrightarrow \mathcal{M}(S)$ defines a morphism

$$R_S \longrightarrow \mathcal{M},$$

[44]Editor's note: the original has been simplified; it invoked Exposé IX, 5.2. Note, however, that loc. cit., 5.3 shows that, if S is connected, the conditions of the lemma hold as soon as (i) holds at one point s of S.

[45]Editor's note: what follows has been modified so as to recall the hypotheses of 3.2 and to add that T is a maximal torus.

where R_S is the constant S-scheme defined by R. By condition (ii), this morphism is an open and closed immersion whose image is precisely

$$\bigcup_{\alpha\in R} \alpha(S),$$

where each $\alpha \in R$ is regarded as a section $S \to \mathcal{M}$.

Let $\mathcal{R}$ be the functor of roots of G with respect to T. By definition, $\mathcal{R}(S')$ is the set of roots of $G_{S'}$ with respect to $T_{S'}$, for every $S' \to S$. If $S' = \varnothing$, set $\mathcal{R}(\varnothing) = \{e\}$; if $S' \neq \varnothing$, the inclusion $R \hookrightarrow \mathcal{M}(S')$ identifies R with a root system of $G_{S'}$ with respect to $T_{S'}$. Thus, by 3.7,

$$\mathcal{R}(S') = \mathrm{Hom}_{\text{loc. const.}}(S', R),$$

which shows that $\mathcal{R}$ is represented by R_S.

Now drop the assumption that G necessarily has a root system with respect to T. The functor $\mathcal{R}$ is in any case an fpqc subsheaf of $\mathcal{M}$. Locally for this topology, G does have a root system with respect to T (for example, make T split and use the proof of Theorem 2.5). By Exposé IV, 4.6.8 and the descent theory of open (respectively closed) subschemes,[46] we obtain the following proposition.

Proposition 3.8.— *Let S be a scheme, let G be an S-group, and let T be a maximal torus of G. The functor $\mathcal{R}$ of roots of G with respect to T is represented by a finite twisted constant S-scheme* (Exposé X, 5.1) *which is an open and closed subscheme of*

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}).$$

A subset $R \subset \mathrm{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S})$ is a root system of G with respect to T if and only if the morphism

$$R_S \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}),$$

defined by the preceding inclusion, induces an isomorphism

$$R_S \xrightarrow{\sim} \mathcal{R}.$$

3.9.— Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let α be a root of G with respect to T, that is, a section of $\mathcal{R}$. Consider the kernel $\mathrm{Ker}(\alpha)$ of α, its unique maximal torus T_α, and the centralizer of the latter,

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

This is a closed S-subgroup of G, reductive (2.8), and of semisimple rank one (1.12). Moreover,

$$\mathrm{Lie}(Z_\alpha/S) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha};$$

hence $\{\alpha, -\alpha\}$ is a root system of Z_α with respect to T.

3.10.— Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let α be a root of G with respect to T. If $q \in \mathbb{Q}$ and $q\alpha$ is a root of G with respect to T, then $q = 1$ or $q = -1$. This follows at once from 1.12.

[46] Editor's note: see SGA 1, VIII, 4.4 or EGA IV_2, 2.7.1.

4. Roots and vector group schemes

4.1.— Let S be a scheme and let F be a finite locally free $\mathscr{O}_S$-module. The S-scheme $\mathbb{W}(F)$ is smooth over S. Its Lie algebra is canonically isomorphic to F. Indeed, there is a canonical isomorphism

$$\mathbb{W}(F) \xrightarrow{\sim} \mathrm{Lie}(\mathbb{W}(F)/S) = \mathbb{W}\big(\mathrm{Lie}(\mathbb{W}(F)/S)\big)$$

(Exposé II, 4.4.1 and 4.4.2). We shall always identify F with $\mathrm{Lie}(\mathbb{W}(F)/S)$.

Lemma 4.2.— *Let S be a scheme and let V be a vector bundle over S, smooth over S. There is then a unique isomorphism of $\mathscr{O}_S$-modules*

$$\exp\colon \mathbb{W}\big(\mathrm{Lie}(V/S)\big) \xrightarrow{\sim} V$$

which induces the identity on Lie algebras. [47]

Proof. We have $V = \mathbb{V}(\mathscr{F})$ for some quasi-coherent $\mathscr{O}_S$-module $\mathscr{F}$. Since V is smooth over S, the module $\mathscr{F} \simeq \omega^1_{V/S}$ is finite locally free, and hence

$$\mathrm{Lie}(V/S) = \mathbb{V}(\omega^1_{V/S}) \simeq \mathbb{W}\big(\mathrm{Lie}(V/S)\big).$$

Moreover, by Exposé II, loc. cit., there is a canonical isomorphism

$$V \xrightarrow{\sim} \mathrm{Lie}(V/S) \simeq \mathbb{W}\big(\mathrm{Lie}(V/S)\big),$$

and the uniqueness of exp follows at once because $\mathbb{W}$ is fully faithful. □

4.3. Notation.— If V is a vector bundle over S, let $V^\times$ denote the open subset obtained from V by removing the zero section. Write the group law of V multiplicatively. The action of $\mathscr{O}_S$ on V which defines its module structure will then be written exponentially:

$$\mathscr{O}_S \times_S V \longrightarrow V, \qquad (x, v) \longmapsto v^x.$$

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[47] Editor's note: the proof in the original edition has been retained; it may also be expanded as follows.

Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_S$-module and put $V = \mathbb{V}(\mathscr{F})$. Write $\pi\colon V \to S$ for the projection and $\epsilon\colon S \to V$ for the zero section. Then $\Omega^1_{V/S} = \pi^*\mathscr{F}$, whence

$$\omega^1_{V/S} = \epsilon^*\Omega^1_{V/S} \simeq \epsilon^*\pi^*\mathscr{F} \simeq \mathscr{F}, \tag{1}$$

and therefore $\mathrm{Lie}(V/S) = (\omega^1_{V/S})^\vee \simeq \mathscr{F}^\vee$. If V is smooth over S, then by (1) $\mathscr{F}$ is finite locally free, and hence

$$V = \mathbb{V}(\mathscr{F}) \simeq \mathbb{W}(\mathscr{F}^\vee) \simeq \mathbb{W}\big(\mathrm{Lie}(V/S)\big). \tag{2}$$

Now let U be an S-scheme equipped with a section $\tau\colon S \to U$, and suppose that an isomorphism $\phi\colon V \xrightarrow{\sim} U$ of pointed S-schemes has been given, that is, $\phi \circ \epsilon = \tau$ (for example, when U is an S-group). Then ϕ induces an isomorphism

$$d\phi\colon \mathscr{F}^\vee \xrightarrow{\sim} \mathrm{Lie}(U/S),$$

and therefore an isomorphism ψ displayed by

$$\begin{tikzcd}
\mathbb{W}(\mathscr{F}^\vee) \arrow[r, equal] & \mathbb{V}(\mathscr{F}) \arrow[r, "\phi", "\sim"'] & U \\
& \mathbb{W}\big(\mathrm{Lie}(U/S)\big) \arrow[ul, "\mathbb{W}(d\phi^{-1})", "\sim"'] \arrow[ur, "\sim", "\psi"'] &
\end{tikzcd}$$

which identifies U with $\mathbb{W}(\mathrm{Lie}(U/S))$. Since the functor $\mathbb{W}$ is fully faithful, there is a unique isomorphism

$$\exp\colon \mathbb{W}\big(\mathrm{Lie}(U/S)\big) \xrightarrow{\sim} U$$

such that $\mathrm{Lie}(\psi^{-1} \circ \exp) = \mathrm{id}$. In fact, one sees directly that $\mathrm{Lie}(\psi) = \mathrm{id}$, so that $\exp = \psi$.

Thus
$$(vv')^x = v^x(v')^x, \qquad v^{x+x'} = v^x v^{x'}, \qquad v^{xx'} = (v^x)^{x'}.$$

In particular, upon restricting the operator action to $\mathbb{G}_{m,S}$, the open subset $V^\times$ is stable and is therefore an object with operator group $\mathbb{G}_{m,S}$. We shall continue to write this action as $(z, v) \longmapsto v^z$.

Definition 4.4.1.[48]— *Let L be an invertible module on S, and let $\mathbb{W}(L)$ be the associated vector bundle. Then $\mathbb{W}(L)^\times$ is a* (locally trivial) *principal homogeneous bundle under $\mathbb{G}_{m,S}$. We write*
$$\Gamma(S, L)^\times = \mathbb{W}(L)^\times(S).$$

Scholium 4.4.2.— *There are natural bijections among the isomorphisms of $\mathcal{O}_S$-modules $\mathcal{O}_S \simeq L$, the isomorphisms of $\mathcal{O}_S$-modules $\mathcal{O}_S \simeq \mathbb{W}(L)$,*[49] *and the sections $S \to \mathbb{W}(L)^\times$.*

The correspondence is
$$f \longmapsto \mathbb{W}(f) \longmapsto \mathbb{W}(f)(1).$$

It is compatible with base change. Thus $\mathbb{W}(L)^\times$ may be regarded as the "scheme of trivializations of $\mathbb{W}(L)$".

4.5.— Let S be a scheme, let T be an S-torus, let P be an S-group with operator group $\mathbb{G}_{m,S}$ (for example, a vector bundle), and let α be a character of T. We write $T \cdot_\alpha P$ for the semidirect product of P by T, where T acts on P through the composite morphism
$$T \xrightarrow{\alpha} \mathbb{G}_{m,S} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(P).$$

Definition 4.6.— *Let S be a scheme, let G be an S-group scheme, let T be a subgroup of G, let α be a character of T, and let L be an $\mathcal{O}_S$-module. Let*
$$p\colon \mathbb{W}(L) \longrightarrow G$$
be a morphism of S-group functors.[50] *We say that p is* normalized by T with multiplier α *if it satisfies the following equivalent conditions:*

(i) p is a morphism of objects with operator group T, where T acts on $\mathbb{W}(L)$ through α, and on G by inner automorphisms. In other words, for every $S' \to S$, every $t \in T(S')$, and every
$$x \in \mathbb{W}(L)(S') = \Gamma(S', L \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$
one has
$$\mathrm{int}(t)p(x) = p\big(\alpha(t)x\big).$$

(ii) The morphism
$$T \cdot_\alpha \mathbb{W}(L) \longrightarrow G, \qquad (t, x) \longmapsto t\,p(x),$$
defined by multiplication in G, is a group morphism.

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[48] Editor's note: for purposes of later references, paragraph 4.4 has been divided into 4.4.1 and 4.4.2.

[49] Editor's note: here the vector bundle $\mathbb{W}(L)$ is identified with the $\mathcal{O}_S$-module functor which it represents.

[50] Editor's note: the edition uses a bold $\mathbf{W}(L)$ here because, for an arbitrary $\mathcal{O}_S$-module L, the functor need not be representable. Below, however, L is assumed finite locally free (indeed, invertible), in which case the functor is represented by the vector bundle $\mathbb{V}(L^\vee)$, and the notation $\mathbb{W}(L)$ is used without this distinction.

Lemma 4.7.— *Under the hypotheses of 4.6, suppose in addition that G is smooth with connected fibers,*[51] *that T is a maximal torus of G, and that L is invertible. If p is a monomorphism and α is nonzero on every fiber, then α is a root of G with respect to T.*

Indeed,

$$\mathrm{Lie}(p)\colon L \longrightarrow \mathfrak{g}$$

is a monomorphism which factors through $\mathfrak{g}^\alpha$.

Proposition 4.8.— *Under the hypotheses of 4.7, suppose that G is reductive and that p is a monomorphism. Then α is a root of G with respect to T, and* $\mathrm{Lie}(p)$ *induces an isomorphism*

$$\mathrm{Lie}(p)\colon L \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

Indeed, by 4.7 and the fact that $\mathfrak{g}^\alpha$ is invertible, it is enough to prove that α is nonzero on every fiber. Let $s \in S$ be such that $\alpha_{\bar{s}} = 0$ ($= 1$ in multiplicative notation). If X is a nonzero section of $L \otimes_S \kappa(\bar{s})$, then $p(X)$ is a unipotent element different from the identity in $G(\bar{s})$ which centralizes $T_{\bar{s}}$. This is impossible, since $T_{\bar{s}}$ is its own centralizer.

Corollary 4.9.— *Under the hypotheses of 4.8, there exists a monomorphism of groups with operator group T* (that is, normalized by T with multiplier α)

$$\mathbb{W}(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^\alpha \to \mathfrak{g}$.

We shall soon see that 4.9 in fact holds whenever G is a reductive group and α is a root of G with respect to T, and that such a morphism is unique.

Reminder 4.10.— *Let k be an algebraically closed field, let G be a reductive k-group, let T be a maximal torus of G, and let α be a root of G with respect to T. There exists a monomorphism*

$$p\colon \mathbb{G}_{a,k} \longrightarrow G$$

normalized by T with multiplier α.

See the *Bible*, ğ13.1, Theorem 1.

4.11.— We finish this section with a technical result which will be useful to us. Let S be a scheme, and let L be an invertible $\mathcal{O}_S$-module. Let $q > 0$ be an integer such that $x \longmapsto x^q$ is an endomorphism of the S-group $\mathbb{G}_{a,S}$. If $S \neq \varnothing$, then $q = 1$, or $q = p^n$, where p is a prime number which vanishes on S. This follows at once from the elementary fact that the greatest common divisor of the binomial coefficients $\binom{q}{i}$, for $i \neq 0, q$, is p if $q = p^n$ with p prime, and is 1 otherwise.

The morphism defined by taking the q-th power,

$$L \longrightarrow L^{\otimes q},$$

is a morphism of sheaves of abelian groups. By base change it defines a morphism of S-group schemes

$$\mathbb{W}(L) \longrightarrow \mathbb{W}(L^{\otimes q}).$$

In particular, if L' is another invertible module and an $\mathcal{O}_S$-linear morphism

$$h\colon L^{\otimes q} \longrightarrow L'$$

[51] Editor's note: compare editor's note (28).

is given, one obtains a morphism of S-group schemes

$$\mathbb{W}(L) \longrightarrow \mathbb{W}(L'), \qquad x \longmapsto h(x^q).$$

With this notation, we have the following result.

Proposition 4.12.— *Let S be a scheme, let T* (resp. *T'*) *be an S-torus, let L* (resp. *L'*) *be an invertible $\mathcal{O}_S$-module, and let α* (resp. *α'*) *be a character of T* (resp. *T'*). *Let $f\colon T \to T'$ be a group morphism and let*

$$g\colon \mathbb{W}(L) \longrightarrow \mathbb{W}(L')$$

be a morphism of S-schemes (not necessarily a morphism of S-groups) *satisfying*

$$g(\alpha(t)x) = \alpha'(f(t))g(x) \tag{$\star$}$$

for every $S' \to S$, every $x \in \mathbb{W}(L)(S')$, and every $t \in T(S')$. Let $s_0 \in S$ be such that $\alpha_{\bar{s}_0} \neq 0$. [52]

(a) *Suppose that g sends the zero section to the zero section and that, for every integer $n > 0$,*

$$(\alpha' \circ f)_{\bar{s}_0} \neq n\alpha_{\bar{s}_0}.$$

Then $g = 0$ in a neighborhood of s_0.

(b) *Suppose that g is a group morphism and that $g_{\bar{s}_0} \neq 0$. There then exist an open subset $U \subset S$ containing s_0 and an integer $q > 0$ such that $x \longmapsto x^q$ is an endomorphism of $\mathbb{G}_{a,U}$ and*

$$(\alpha' \circ f)_U = q\alpha_U.$$

(c) *Suppose that*

$$(\alpha' \circ f)_{\bar{s}_0} = q\alpha_{\bar{s}_0},$$

where $q > 0$ is an integer such that $x \longmapsto x^q$ is an endomorphism of $\mathbb{G}_{a,S}$. There then exist an open subset $U \subset S$ containing s_0 and a unique morphism of $\mathcal{O}_S$-modules

$$h\colon L^{\otimes q}|_U \longrightarrow L'|_U$$

such that g_U is the composite

$$\mathbb{W}(L)_U \xrightarrow{x \mapsto x^q} \mathbb{W}(L^{\otimes q})_U \xrightarrow{\mathbb{W}(h)} \mathbb{W}(L')_U.$$

Proof of Proposition 4.12.. Let us prove (a). Since the conclusion is local on S, we may suppose that $\mathbb{W}(L) = \mathbb{W}(L') = \mathbb{G}_{a,S}$, and hence that g is given by a polynomial

$$g(X) = \sum_{n \geq 0} a_n X^n, \qquad a_n \in \Gamma(S, \mathcal{O}_S).$$

[52] Editor's note: the phrases "consider the semidirect product $T \cdot_\alpha \mathbb{W}(L)$, respectively $T' \cdot_{\alpha'} \mathbb{W}(L')$" have been deleted. On the other hand, in part (b) the hypothesis that g be a group morphism, together with ($\star$), is equivalent to requiring

$$(t, x) \longmapsto \big(f(t), g(x)\big)$$

to be a group morphism from $T \cdot_\alpha \mathbb{W}(L)$ to $T' \cdot_{\alpha'} \mathbb{W}(L')$.

Condition $(\star)$, which relates f and g, is the identity in $\Gamma(S', \mathcal{O}_{S'})[X]$

$$\sum_{n\geq 0} a_n \alpha'(f(t)) X^n = \sum_{n\geq 0} a_n \alpha(t)^n X^n.$$

Thus, for every $n \geq 0$, every $S' \to S$, and every $t \in T(S')$,

$$a_n(\alpha'(f(t)) - \alpha(t)^n) = 0.$$

For every $n \geq 0$, let S_n be the set of points $s \in S$ such that

$$(\alpha' \circ f)_{\bar{s}} = n\alpha_{\bar{s}}.$$

It is known (Exposé IX, 5.3) that the S_n are both open and closed and that

$$(\alpha' \circ f)_{S_n} = n\alpha_{S_n}.$$

Moreover, since $\alpha_{\bar{s}_0} \neq 0$, after shrinking S we may suppose that α is nonzero on every fiber (same reference); it follows that the S_n are pairwise disjoint. After shrinking S once more, we may therefore suppose that one of the following two cases holds: $S = S_n$ for some n, or every S_n is empty.

Let $m \geq 0$ be such that $S_m = \varnothing$. We claim that $a_m = 0$. Indeed, $\alpha' \circ f$ and $m\alpha$ are distinct on every fiber of S, and we have:

Lemma 4.13.— *Let S be a scheme, let T be an S-torus, and let α, α' be two characters of T which are distinct on every fiber. There exist an fpqc covering family $\{S_i \to S\}$ and, for every i, an element $t_i \in T(S_i)$ such that*

$$\alpha(t_i) - \alpha'(t_i) = 1.$$

The lemma is immediate by reduction first to the diagonalizable case, and then to $T = \mathbb{G}_{m,S}$.

We return to the proof of Proposition 4.12. We have just proved (a). In cases (b) and (c), there is an integer n such that $S = S_n$ ($n = q$ in (c)). The preceding argument gives $a_m = 0$ for $m \neq n$, and therefore

$$g(X) = a_n X^n, \qquad a_n \in \Gamma(S, \mathcal{O}_S).$$

This proves (c) at once. In case (b), we know that $a_n(s_0) \neq 0$; we may therefore suppose that a_n is invertible on S. It follows that $x \longmapsto x^n$ is an endomorphism of $\mathbb{G}_{a,S}$, by the hypothesis in (b), and this completes the proof.

5. An instructive example

5.1.— Let k be an algebraically closed field of characteristic 0. Put $A = k[t]$, the polynomial ring in one variable over k, and let $S = \operatorname{Spec}(A)$. Consider the following Lie algebra $\mathfrak{g}$ over $\mathcal{O}_S$. As a module it is free of rank 3, with basis $\{X, Y, H\}$, and its multiplication table is

$$[X, Y] = 2tH, \qquad [H, X] = X, \qquad [H, Y] = -Y.$$

For $s \in S$, $s \neq s_0$, where s_0 is the point defined by $t = 0$, the fiber

$$\mathfrak{g}(s) = \mathfrak{g} \otimes_A \kappa(s)$$

is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$. For $s = s_0$, it is a solvable Lie algebra.

5.2.— Let G_1 be the group scheme of automorphisms of $\mathfrak{g}$: for every $S' \to S$, $G_1(S')$ is the group of automorphisms of the $\mathscr{O}_{S'}$-Lie algebra $\mathfrak{g} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$. It is a closed subscheme of the group $\mathrm{GL}(\mathfrak{g})$ of automorphisms of the $\mathscr{O}_S$-module $\mathfrak{g}$.

Let $S' \to S$, and let

$$u \in M_3(\Gamma(S', \mathscr{O}_{S'}))$$

be viewed as an endomorphism of the $\mathscr{O}_{S'}$-module $\mathfrak{g} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$:

$$\begin{aligned} u(X) &= aX + bY + eH, \\ u(Y) &= b'X + a'Y + e'H, \\ u(H) &= cX + c'Y + dH. \end{aligned}$$

It is immediate that u is a section of G_1 if and only if $\det(u)$ is invertible and the following relations hold:

$$\begin{array}{llll} (1) & a(d-1) = ec, & (1') & a'(d-1) = e'c', \\ (2) & b(d+1) = ec', & (2') & b'(d+1) = e'c, \\ (3) & e = 2t(bc - ac'), & (3') & e' = 2t(b'c' - a'c), \\ (4) & 2tc = eb' - ae', & (4') & 2tc' = be' - ea', \\ (5) & 2t(aa' - bb') = 2td. & & \end{array}$$

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Lemma 5.3.— *Relations* (1), (1'), (2), *and* (2') *imply*

$$\det(u) = aa'(2-d) + bb'(d+2), \qquad aa' - bb' = d\det(u).$$

Indeed, the first assertion follows at once by substituting relations (1), (1'), (2), and (2') in the expansion of $\det(u)$:

$$\begin{aligned} \det(u) &= aa'd + be'c + b'c'e - a'ec - ae'c' - bb'd \\ &= aa'd + bb'(d+1) + bb'(d+1) - aa'(d-1) - bb'd - aa'(d-1) \\ &= aa'(d - d + 1 - d + 1) + bb'(d + 1 + d + 1 - d) \\ &= aa'(2-d) + bb'(d+2). \end{aligned}$$

Multiplying this relation by d, we obtain

$$d\det(u) = aa'(2d - d^2) + bb'(d^2 + 2d).$$

On the other hand, the relation $(1) \times (1') = (2) \times (2')$ gives at once

$$aa'(d-1)^2 = bb'(d+1)^2.$$

Combining the last two relations immediately yields the second formula of Lemma 5.3.

5.4.— Now consider

$$G = G_1 \cap \mathrm{SL}(\mathfrak{g}).$$

This is the closed subgroup of G_1 defined by the equation $\det(u) = 1$; hence it is an affine group over S.

[53] Editor's note: the equality $[u(X), u(Y)] = 2t\,u(H)$ gives relations (4), (4'), and (5); the equality $[u(H), u(X)] = u(X)$ gives (1)–(3); and the equality $[u(H), u(Y)] = -u(Y)$ gives (1')–(3').

Proposition 5.5.— *The group G is smooth over S.*

To prove the proposition, we shall need the following lemmas.

Lemma 5.6.— *Let U be the open subset of*

$$\mathrm{End}_A(\mathfrak{g}) \simeq \mathbb{W}(M_3(\mathcal{O}_S))$$

defined by the condition that ad be invertible; that is, the principal open subset $\mathrm{End}_A(\mathfrak{g})_f$, where $f(u) = ad$. Then $U \cap G$ is the closed subscheme of U defined by the six equations

$$(1), \quad (2), \quad (2'), \quad (3), \quad (3'), \quad (\mathrm{D})\colon aa' - bb' = d.$$

It is clear that these six relations are satisfied by every point of G, by Lemma 5.3, and therefore by every point of $U \cap G$. Conversely, suppose that $u \in U(S')$, for some $S' \to S$, satisfies the six conditions in the statement. We must show that $\det(u) = 1$, and that u also satisfies (1'), (4), (4'), and (5).

First, (D) implies (5). Relations (2) and (2') give

$$bb'(d+1) = bce' = b'c'e.$$

On the other hand, from (3) and (3'), writing $2t(bc - ac')(b'c' - a'c)$ in two ways, we obtain

$$(bc - ac')e' = (b'c' - a'c)e.$$

Together with the preceding relation, this gives

$$ac'e' = a'ce.$$

By (1), however,

$$a'ce = a'a(d-1).$$

Thus

$$a\big(a'(d-1) - e'c'\big) = 0,$$

and this implies (1'), since a is invertible.

Consequently, (1), (2), (2'), and (1') hold. Lemma 5.3 and (D) now give

$$d\big(\det(u) - 1\big) = 0.$$

Since d is invertible, it follows that $\det(u) = 1$.

It remains to prove (4) and (4'). We prove (4'); the other calculation follows by symmetry. From (3), (3'), and (D), we have

$$a'e + be' = -2t(aa' - bb')c' = -2tdc'.$$

Combining (1') and (2), we also have

$$a'e + be' = -d(be' - ea'),$$

[54] which proves (4'), since d is invertible.

Lemma 5.7.— *The group G is smooth over S along the identity section.*

[54]Editor's note: the sign has been corrected.

Proof. By 5.6 and SGA 1, II 4.10, it is enough to prove that the differentials of the functions

$$a(d-1)-ec, \qquad b(d+1)-ec', \qquad b'(d+1)-e'c,$$
$$e-2t(bc-ac'), \qquad e'-2t(b'c'-a'c), \qquad aa'-bb'-d$$

at the points of the identity section of G are linearly independent. If the differential of a lower-case letter is denoted by the corresponding capital letter, these differentials are

$$D, \qquad 2B, \qquad 2B', \qquad E+2tC', \qquad E'+2tC, \qquad A+A'-D,$$

[55] and they are indeed linearly independent modulo every $(t-\lambda)$, $\lambda \in k$. [56]

Lemma 5.8.— *For $s \in S$, $s \neq s_0$, the fiber G_s is connected and semisimple.*

Proof. Indeed, since $s \neq s_0$, $\mathfrak{g}(s)$ is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$, and, on the other hand, $G_s = (G_1)_s$. The automorphism group of the Lie algebra of PGL_2 over a field of characteristic 0 is known to be PGL_2 itself, which is connected and semisimple. [57]

Lemma 5.9.— *The fiber G_{s_0} is solvable and has two connected components, of the following form:*

$$G^0_{s_0} = \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ c & c' & 1 \end{pmatrix} \right\}, \quad G^-_{s_0} = \left\{ \begin{pmatrix} 0 & b & 0 \\ b^{-1} & 0 & 0 \\ c & c' & -1 \end{pmatrix} \right\}.$$

Proof. We have $e = e' = 0$, since $t = 0$ at s_0. The equations (1), (1'), …, (5), and (D) can then be solved immediately.

Lemma 5.10.— *The matrix*

$$w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

is a section of G over S, such that $w(s_0) \in G^-_{s_0}$.

Let us now prove Proposition 5.5. Denote by G^0 the union of the identity components of the fibers of G, that is, the complement of $G^-_{s_0}$. Since G is smooth over S along the identity section by Lemma 5.7, G^0 is an open subgroup of G, smooth over S, by Exposé $\mathrm{VI_B}$, 3.10. By translation, G is also plainly smooth at the points of $w(S)$; hence G is smooth over S. [58]

5.11.— Consider the morphism

$$\mathbb{G}_{m,S} \longrightarrow G^0, \qquad z \longmapsto \begin{pmatrix} z & 0 & 0 \\ 0 & z^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is a monomorphism defining a torus T of G^0. We claim that

$$T = \mathrm{Centr}_G(T) = \mathrm{Centr}_{G^0}(T).$$

It is enough to verify the first equality. Since the two sides are smooth S-subgroups of G, it is enough to check that they have the same geometric points. In the fibers over points $s \neq s_0$,

[55] Editor's note: $A+A'+B$ has been corrected to $A+A'-D$.
[56] Editor's note: "$(t-a)$, $a \in A$" has been corrected to "$(t-\lambda)$, $\lambda \in k$".
[57] Editor's note: the preceding sentence has been slightly modified.
[58] Editor's note: the preceding sentence has been modified, and the reference to Exposé $\mathrm{VI_B}$, 3.10 has been added.

this follows because $\mathrm{PGL}_{2,\kappa(s)}$ is reductive and T_s is a maximal torus in it for dimensional reasons (cf. 1.11). In the fiber over s_0, the verification is immediate. In particular, T is a maximal torus of both G and G^0.

5.12.— The section w of G defined in Lemma 5.10 normalizes T. It follows immediately (cf. 2.4) that the Weyl group of G is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_S$, and is therefore finite over S:

$$W_G(T) = \mathrm{Norm}_G(T)/T = (\mathbb{Z}/2\mathbb{Z})_S.$$

By contrast, $W_{G^0}(T)$ is not finite over S: it "lacks one point" above s_0.

5.13.— The open immersion $G^0 \to G$ is not a closed immersion, since G^0 is dense in G; it is nevertheless an affine morphism (and hence G^0 is affine over S). Indeed, $G^0_{s_0}$ is closed in G_{s_0}, which is closed in G, so its complement U in G is open. The open subsets G^0 and U cover G, and it is enough to verify that the immersions

$$G^0 \longrightarrow G^0 \qquad \text{and} \qquad G^0 \cap U \longrightarrow U$$

are affine. This is trivial for the first; for the second, observe that $U \cap G^0$ is defined in U by the condition $b \neq 0$.

We have thus constructed a smooth affine S-group G^0, with connected fibers, possessing a self-centralizing maximal torus T whose Weyl group $W_{G^0}(T)$ is not finite (compare Theorem 2.5).

6. Local existence of maximal tori. The Weyl group

In the proof of Theorem 2.5 we used a result from Exposé XI on the local existence, for the étale topology, of maximal tori. The proof in Exposé XI uses a deep representability result (XI 4.1). In the particular case under consideration, one can give another, far more elementary proof, based on the ideas of Exposé XII, no. 7.

Proposition 6.1.— *Let S be a scheme, let G be a smooth affine S-group scheme with connected fibers over S, and let $s_0 \in S$ be a point such that the maximal tori of the geometric fiber $G_{\bar{s}_0}$ are self-centralizing. There exist an étale morphism $S' \to S$ whose image contains s_0 and a split maximal torus T of $G_{S'}$.*

Proof. First, we may suppose that S is affine. [59] Since G is of finite presentation over S, we may successively suppose that S is noetherian, local, and then henselian with separably closed residue field (cf. EGA IV, 8.12, §§8.8 and 18.8). Thus let

$$S = \mathrm{Spec}(A), \qquad k = \kappa(s_0),$$

where A is henselian and k is separably closed. Choose a maximal torus T_0 of $G_0 = G_k$. Such a torus exists, for example because the scheme of maximal tori of G_0 is smooth over k (Exposé XII, 7.1(c)). Since k is separably closed, T_0 is split (cf. Exposé X, 1.4), and is therefore given by a group monomorphism

$$f_0 \colon \mathbb{G}^r_{m,k} \longrightarrow G_0.$$

Let $m > 1$ be an integer prime to the characteristic of k. By Exposé VIII, 6.7, for every $h > 0$, $\mathrm{Centr}_{G_0}({}_{m^h}T_0)$ is represented by a closed subscheme of G_0. The subgroup schemes

[59] Editor's note: the preceding sentence and the reference to EGA IV below have been added.

${}_{m^h}T_0$ are schematically dense in T_0 (cf. Exposé IX, 4.10), and G_0 is noetherian. Consequently there is an h such that

$$\operatorname{Centr}_{G_0}({}_{m^h}T_0) = \operatorname{Centr}_{G_0}(T_0) = T_0.$$

Put $n = m^h$. Since n is invertible on S, ${}_n\mathbb{G}_{m,S}$ is isomorphic to $(\mathbb{Z}/n\mathbb{Z})_S$, so f_0 defines a group monomorphism

$$u_0 \colon (\mathbb{Z}/n\mathbb{Z})_k^r \longrightarrow G_0$$

such that $\operatorname{Centr}_{G_0}(u_0) = T_0$.

Now the S-functor

$$P = \operatorname{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S^r, G)$$

is represented by an S-scheme of finite type: it is a closed subscheme of r copies of

$${}_nG = \operatorname{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S, G) = \operatorname{Ker}(G \xrightarrow{n} G).$$

But P is smooth over S (Exposé IX, 3.6). Therefore $u_0 \in P(k)$ lifts to a section $u \in P(S)$, by Hensel's lemma (Exposé XI, 1.11):

$$u \colon (\mathbb{Z}/n\mathbb{Z})_S^r \longrightarrow G.$$

Consider $H = \operatorname{Centr}_G(u)$. It is a closed subgroup scheme of G, by Exposé VIII, 6.5(e), and its special fiber is $H_0 = T_0$. Moreover, H is smooth over S. Indeed, let $S' = \operatorname{Spec}(A')$ be an affine S-scheme, let

$$u' \colon ({}_n\mathbb{G}_{m,S'})^r \longrightarrow G_{S'}$$

be obtained from u by base change, put $S'_J = \operatorname{Spec}(A'/J)$, where J is a square-zero ideal, and let $x \in H(S'_J)$. Since G is smooth, x lifts to an element $g \in G(S')$. Then

$$v = \operatorname{int}(g)(u')$$

satisfies $v_J = u'_J$. By Exposé IX, 3.2, there is therefore an element $g' \in G(S')$ such that $g'_J = e$ and $\operatorname{int}(g')(v) = u'$. It follows that $h = g'g$ belongs to $H(S')$ and satisfies $h_J = x$. [60]

Let H^0 be the identity component of H. It is a subgroup scheme of G, smooth and with connected fibers, whose special fiber is a torus. By Exposé X, 8.1, it is a torus, necessarily split by Exposé X, 4.6. Put $T = H^0$, and let

$$C = \operatorname{Centr}_G(T).$$

This is a closed subgroup of G (Exposé VIII, 6.5(e)) and is smooth (Exposé XI, 2.4). Consider C^0 (in fact $C^0 = C$, but we do not need this). Then $C^0 \supset T$, and both are smooth groups with connected fibers. They coincide at s_0, hence in a neighborhood of s_0. After shrinking S, we may therefore suppose that $C^0 = T$; then, a fortiori, T is maximal.

Remark 6.2.— *The proof shows in particular that the reductive rank of $G_{\bar{s}}$ is constant in a neighborhood of $s = s_0$.*

Proposition 6.3.— *Let S be a scheme, let G be an S-group scheme that is smooth and of finite presentation over S, and let Q be a subtorus of G.*

(i) $\operatorname{Centr}_G(Q)$ *and* $\operatorname{Norm}_G(Q)$ *are represented by closed subgroup schemes, smooth* (and hence of finite presentation) *over S.*

[60] Editor's note: the reference to Exposé VIII, 6.5(e) was added above, and the preceding smoothness argument has been expanded.

(ii) $\mathrm{Centr}_G(Q)$ *is an open and closed subscheme of* $\mathrm{Norm}_G(Q)$. *The quotient*

$$W_G(Q) = \mathrm{Norm}_G(Q)/\,\mathrm{Centr}_G(Q)$$

is represented by an open subgroup scheme of $\mathrm{Aut}_{S\text{-gr}}(Q)$*; it is therefore a quasi-finite, étale, separated S-group scheme.*

(iii) *For every $s \in S$, put*

$$w(s) = \left|\mathrm{Norm}_{G(\bar{s})}(Q(\bar{s}))/\,\mathrm{Centr}_{G(\bar{s})}(Q(\bar{s}))\right|.$$

Then $s \longmapsto w(s)$ is lower semicontinuous, and is constant in a neighborhood of s if and only if $W_G(Q)$ is finite over S in a neighborhood of s.

Proof. By Exposé XI, 6.11, $\mathrm{Centr}_G(Q)$ and $\mathrm{Norm}_G(Q)$ are represented by closed subschemes of G of finite presentation. They are smooth by Exposé XI, 2.4 and 2.4 bis, which proves (i). Assertions (ii) and (iii) are then proved as in Exposé XI, 5.9 and 5.10; those proofs in fact use only (i), and not the deeper theorems of Exposé XI, 4.1 and 4.2.

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Exposé XX
Reductive Groups of Semisimple Rank 1

by M. Demazure

1 Elementary systems. The groups U_α and $U_{-\alpha}$

Recollection 1.1. *Let $S = \mathrm{Spec}(k)$, where k is an algebraically closed field, and let G be a reductive S-group of semisimple rank one and T a maximal torus of G, necessarily split. Then*

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha},$$

where α and $-\alpha$ are the roots of G with respect to T. Moreover, there exist two group monomorphisms

$$p_\alpha \colon \mathbb{G}_{a,S} \longrightarrow G, \qquad p_{-\alpha} \colon \mathbb{G}_{a,S} \longrightarrow G$$

such that

$$t\, p_\alpha(x)\, t^{-1} = p_\alpha(\alpha(t)x), \qquad t\, p_{-\alpha}(x)\, t^{-1} = p_{-\alpha}(\alpha(t)^{-1}x)$$

for every $S' \to S$, $t \in T(S')$, and $x \in \mathbb{G}_a(S')$, and such that the morphism

$$\mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \qquad (y, t, x) \longmapsto p_{-\alpha}(y)\, t\, p_\alpha(x),$$

is radicial and dominant (*Bible*, Section 13.4, Corollary 2 to Theorem 3).

Since its tangent map at the identity is bijective, this morphism is also separable, hence birational. Zariski's Main Theorem (EGA III_1, 4.4.9) therefore shows that it is an open immersion.

Lemma 1.2. *Let S be a scheme, G an S-group scheme, T a torus of G, Q a subtorus of T, and α a character of T whose restriction to Q is nontrivial on every fiber. Let*

$$p_\alpha \colon \mathbb{G}_{a,S} \longrightarrow G \quad \text{and} \quad p_{-\alpha} \colon \mathbb{G}_{a,S} \longrightarrow G$$

be group morphisms normalized by T, with multipliers α and $-\alpha$, respectively. Suppose that the morphism

$$u \colon \mathbb{G}_{a,S} \times_S T \times_S \mathbb{G}_{a,S} \longrightarrow G, \qquad u(y, t, x) = p_{-\alpha}(y)\, t\, p_\alpha(x),$$

[0]Editors' note: version of 13 October 2024.

defined on points by the displayed formula, is an open immersion. Finally, let $q \geq 0$ be an integer and let

$$p\colon \mathbb{G}_{a,S} \longrightarrow G$$

be a group morphism such that, for every $S' \to S$, $t \in Q(S')$, and $x \in \mathbb{G}_a(S')$,

$$\mathrm{int}(t)^q(p(x)) = p(\alpha(t)x).$$

Then there is a unique $\nu \in \mathbb{G}_a(S)$ such that

$$p(x) = p_\alpha(\nu x^q).$$

Indeed, let Ω be the image of u, and put $U = p^{-1}(\Omega)$. This is an open subset of $\mathbb{G}_{a,S}$ containing the zero section. For every section t of Q, multiplication by $\alpha(t)$ leaves U globally invariant. We have $U = \mathbb{G}_{a,S}$. It is enough to check this when S is the spectrum of an algebraically closed field k; then

$$\alpha\colon Q(k) \longrightarrow k^\times$$

is surjective, so $U(k) \supset k^\times$, and hence $U = \mathbb{G}_{a,k}$.

There are consequently three morphisms

$$a\colon \mathbb{G}_{a,S} \longrightarrow \mathbb{G}_{a,S}, \qquad b\colon \mathbb{G}_{a,S} \longrightarrow T, \qquad c\colon \mathbb{G}_{a,S} \longrightarrow \mathbb{G}_{a,S}$$

such that

$$p(x) = p_{-\alpha}\big(a(x)\big)\, b(x)\, p_\alpha\big(c(x)\big).$$

The condition on p becomes

$$a\big(\alpha(t)x\big) = \alpha(t)^{-q}a(x), \qquad b\big(\alpha(t)x\big) = b(x), \qquad c\big(\alpha(t)x\big) = \alpha(t)^q c(x).$$

For the same reason, for every $S' \to S$ and $z \in \mathbb{G}_m(S')$,

$$a(zx) = z^{-q}a(x), \qquad b(zx) = b(x), \qquad c(zx) = z^q c(x),$$

and therefore

$$z^q a(z) = a(1), \qquad b(z) = b(1), \qquad c(z) = z^q c(1).$$

Since $\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}$, for every $S' \to S$ and $x \in \mathbb{G}_a(S')$ we obtain

$$\begin{aligned} x^q a(x) &= a(1) = a(0) = 0, && \text{hence } a = 0,\\ c(x) &= x^q c(1) = \nu x^q, && \text{for some } \nu \in \mathbb{G}_a(S),\\ b(x) &= b(1) = b(0) = e, && \text{hence } b = e. \end{aligned}$$

This completes the proof.

Definition 1.3. *An S-elementary system is a triple (G, T, α) in which*

(i) *G is a reductive S-group of semisimple rank one* (Exposé XIX, 2.7)*;*

(ii) *T is a maximal torus of G;*

(iii) *α is a root of G with respect to T* (Exposé XIX, 3.2).

Thus there is a direct-sum decomposition (Exposé XIX, 3.5)

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank one.[1]

[1] As $\mathcal{O}_S$-modules.

1.4. If (G, T, α) is an S-elementary system, then $(G_{S'}, T_{S'}, \alpha_{S'})$ is an S'-elementary system for every $S' \to S$. If (G, T, α) is an S-elementary system, then so is $(G, T, -\alpha)$.

If S is a scheme, G a reductive S-group, T a maximal torus of G, and α a root of G with respect to T, then (Z_α, T, α) is an S-elementary system (Exposé XIX, 3.9).

Let (G, T, α) be an S-elementary system. The invertible module $\mathfrak{g}^\alpha$ is canonically endowed with a T-module structure. Hence the vector bundle $W(\mathfrak{g}^\alpha)$ is also a T-module. On the other hand, the inner automorphisms by elements of T make G a group equipped with a group of operators T.

Theorem 1.5. *Let (G, T, α) be an S-elementary system.*

(i) *There exists a unique morphism of groups with operator group T*

$$\exp\colon W(\mathfrak{g}^\alpha) \longrightarrow G$$

whose induced morphism on Lie algebras is the canonical map $\mathfrak{g}^\alpha \to \mathfrak{g}$.[2] *Equivalently,* exp *is the unique morphism such that, for every $S' \to S$, $t \in T(S')$, and $X, X' \in W(\mathfrak{g}^\alpha)(S')$,*

$$\begin{aligned} \exp(X + X') &= \exp(X)\exp(X'), \\ \mathrm{int}(t)(\exp(X)) &= \exp(\alpha(t)X), \\ \mathrm{Lie}(\exp)(X) &= X. \end{aligned}$$

(ii) *Define analogously, in the S-elementary system $(G, T, -\alpha)$,*

$$\exp\colon W(\mathfrak{g}^{-\alpha}) \longrightarrow G.$$

Then the morphism

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^\alpha) \longrightarrow G, \qquad (Y, t, X) \longmapsto \exp(Y)\, t\, \exp(X),$$

defined on points by the displayed formula, is an open immersion.

Suppose for the moment that the required exponential morphisms exist, and prove the other assertions. We first prove (ii). Both sides are of finite presentation and flat over S, so it is enough to work when S is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Put $S = \mathrm{Spec}(k)$, and choose

$$X \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \qquad Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times.$$

It is enough to prove that

$$\mathbb{G}_{a,k} \times_k T \times_k \mathbb{G}_{a,k} \longrightarrow G, \qquad (y, t, x) \longmapsto \exp(yY)\, t\, \exp(xX),$$

is an open immersion. By 1.1 and 1.2, there exist $a, b \in k$ such that

$$\exp(yY) = p_{-\alpha}(ay), \qquad \exp(xX) = p_\alpha(bx).$$

Since $\exp\colon W(\mathfrak{g}^{-\alpha}) \to G$ induces a monomorphism on Lie algebras, $a \neq 0$; likewise $b \neq 0$. The assertion is therefore reduced to 1.1.

The uniqueness of exp may be proved locally on S. We reduce to the case in which $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free, and then apply 1.2 with $Q = T$ and $q = 1$.

[2] It will be seen below, in Corollary 5.9, that exp is an isomorphism from $W(\mathfrak{g}^\alpha)$ onto a closed subgroup of G.

It remains to prove existence. Faithfully flat descent, together with the uniqueness just proved, shows that it is enough to prove existence locally on S for the fpqc topology. The usual finite-presentation arguments reduce first to S noetherian and then to S noetherian local. We may therefore assume that

$$S = \mathrm{Spec}(A),$$

where A is a complete noetherian local ring with algebraically closed residue field k. Let

$$p_0 \colon \mathbb{G}_{a,k} \longrightarrow G_k$$

be a monomorphism of k-groups normalized by T_k, with multiplier $\alpha_0 = \alpha \otimes_A k$; such a morphism exists by 1.1. By 1.1 and 1.2, the associated morphism

$$T_k \cdot_{\alpha_0} \mathbb{G}_{a,k} \longrightarrow G_k$$

is an immersion and hence, in particular, a monomorphism. We provisionally admit the following two lemmas.

Lemma 1.6. *Let S be a scheme, G an S-group of finite presentation, T an S-torus, α a character of T that is nontrivial on every fiber, and $s_0 \in S$. Let*

$$f \colon T \cdot_\alpha \mathbb{G}_{a,S} \longrightarrow G$$

be an S-group morphism such that f_{s_0} is a monomorphism and the restriction of f to T is a monomorphism. There exists an open neighborhood U of s_0 such that $f|_U$ is a monomorphism.

Lemma 1.7. *Let A be a complete noetherian local ring with algebraically closed residue field k, let (G, T, α) be an A-elementary system, and let*

$$p_0 \colon \mathbb{G}_{a,k} \longrightarrow G_k$$

be a morphism of k-groups normalized by T_k, with multiplier $\alpha \otimes_A k$. There exists a group morphism

$$p \colon \mathbb{G}_{a,A} \longrightarrow G$$

normalized by T, with multiplier α.

Let p be the morphism asserted by 1.7, and let

$$f \colon T \cdot_\alpha \mathbb{G}_{a,S} \longrightarrow G$$

be the corresponding morphism. It satisfies the hypotheses of 1.6, so it is a monomorphism; in particular, p is a monomorphism. The theorem now follows from Exposé XIX, 4.9.

Proof of 1.6. Let

$$\epsilon \colon S \longrightarrow T \cdot_\alpha \mathbb{G}_{a,S}$$

be the identity section. Since f is unramified at $\epsilon(s_0)$, it is unramified at $\epsilon(s)$ for all s in an open neighborhood U of s_0. Hence $f|_U$ is unramified (Exposé X, 3.5),[3] and its kernel $\mathrm{Ker}(f)_U$ is unramified over U. To prove that $f|_U$ is a monomorphism, it is therefore enough[4] to prove that $\mathrm{Ker}(f)_U$ is radicial over U, which is a set-theoretic question. We are reduced to the following assertion.

[3]See also Exposé VI_B, 2.5.

[4]By EGA IV_4, 17.9.1.

Lemma 1.8. *Let k be an algebraically closed field, and let N be an invariant subgroup of $T \cdot_\alpha \mathbb{G}_{a,k}$, where α is a nontrivial character of the torus T. Suppose that N is étale over k and that $N \cap T = \{e\}$. Then $N = \{e\}$.*

We have
$$\mathrm{int}(t')(t,x) = (t, \alpha(t')x).$$
If (t,x) is a point of N with $x \neq 0$, then (t, zx) is also a point of N for every $z \in k^\times$. Thus (t,x) is not isolated, contradicting the quasi-finiteness of N. Set-theoretically, therefore, $N \subset T$, and the hypothesis $N \cap T = \{e\}$ completes the proof.

Proof of 1.7. Let $\mathfrak{m}$ be the radical of A, and put
$$S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1}), \qquad n \geq 0.$$
We first show, by induction on n, that p_0 can be extended, for every n, to a morphism of S_n-groups
$$p_n \colon \mathbb{G}_{a,S_n} \longrightarrow G_{S_n},$$
normalized by T_{S_n} with multiplier α_{S_n}, in such a way that
$$p_{n+1} \times_{S_{n+1}} S_n = p_n.$$

Let $H = T \cdot_\alpha \mathbb{G}_{a,S}$. Denote by f_n the morphism $H_{S_n} \to G_{S_n}$ defined by p_n. We provisionally admit the following lemma.

Lemma 1.9. *Let (G, T, α) be a k-elementary system, where k is algebraically closed, and let*
$$p \colon \mathbb{G}_{a,k} \longrightarrow G$$
be a monomorphism normalized by T, with multiplier α. Then
$$\mathrm{H}^2(T \cdot_\alpha \mathbb{G}_{a,k}, \mathfrak{g}) = 0.$$
Here $T \cdot_\alpha \mathbb{G}_{a,k}$ acts on $\mathfrak{g}$ through the morphism $T \cdot_\alpha \mathbb{G}_{a,k} \to G$ defined by p and through the adjoint representation of G.

By Exposé III, 2.8, f_n therefore extends to a morphism of S_{n+1}-groups
$$f'_{n+1} \colon H_{S_{n+1}} \longrightarrow G_{S_{n+1}}.$$
The morphism f'_{n+1} and the canonical immersion of $T_{S_{n+1}}$ into $G_{S_{n+1}}$ have the same restriction to T_{S_n}. By Exposé III, 2.5, there is an element $g \in G(S_{n+1})$ such that
$$g \times_{S_{n+1}} S_n = e$$
and such that
$$f_{n+1} = \mathrm{int}(g) \circ f'_{n+1}$$
restricts to $T_{S_{n+1}}$ by the canonical immersion. Let p_{n+1} be the restriction of f_{n+1} to $\mathbb{G}_{a,S_{n+1}}$. It is normalized by $T_{S_{n+1}}$, has multiplier $\alpha_{S_{n+1}}$, and extends p_n.

We have thus constructed a coherent system (f_n); it remains to algebraize it. For this we use:

Lemma 1.10. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal,*

$$S = \operatorname{Spec}(A), \qquad S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1}),$$

let T be an S-torus, let α be a nonzero character of T, and let X be an affine S-scheme on which T acts. Let T act on $\mathbb{G}_{a,S}$ through α. Let $q \geq 0$, and let $(f_n)_{n\geq 0}$ be a coherent system of morphisms

$$f_n \colon \mathbb{G}_{a,S_n}^q \longrightarrow X_{S_n}$$

of objects with operators T_{S_n}. There exists a unique morphism of objects with operators T

$$f \colon \mathbb{G}_{a,S}^q \longrightarrow X$$

that induces the f_n (compare Exposé IX, 7.1).

Corollary 1.11. *If X is a group with operator group T, and if the f_n are group morphisms, then f is also a group morphism.*

Indeed, apply the uniqueness assertion of the lemma to the two morphisms $\mathbb{G}_{a,S}^{2q} \to X$ deduced from f in the usual way.

Proof of 1.10. Suppose that T is split, as it is in the application of 1.10 to the proof of 1.5. By Exposé I, 4.7.3.1, the functor

$$X \longmapsto \mathscr{A}(X)$$

defines an equivalence between the category of affine S-schemes equipped with an action of T and the category opposite to that of M-graded S-algebras, where

$$M = \operatorname{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}).$$

Thus there are gradings

$$B = \mathscr{A}(X) = \bigoplus_{m\in M} B_m, \qquad C = \mathscr{A}(\mathbb{G}_{a,S}^q) = \bigoplus_{m\in M} C_m.$$

Each C_m is visibly a finite free A-module. Indeed, $C_m = 0$ if m is not a multiple of α, whereas, if $m = d\alpha$, then C_m is isomorphic to the A-module of homogeneous polynomials of degree d in q variables. Put

$$\begin{aligned} \widehat{B}_m &= \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{C}_m &= \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{B} &= \bigoplus_{m\in M} \widehat{B}_m, \qquad \widehat{C} = \bigoplus_{m\in M} \widehat{C}_m. \end{aligned}$$

There are canonical morphisms of M-graded algebras

$$g_B \colon B \longrightarrow \widehat{B}, \qquad g_C \colon C \longrightarrow \widehat{C}.$$

The preceding observation shows that g_C is an isomorphism. Giving a coherent system (f_n) as in the statement is equivalent to giving a morphism of graded A-algebras

$$\widehat{F} \colon \widehat{B} \longrightarrow \widehat{C}.$$

Finding a morphism f as in the statement is equivalent to finding a morphism of graded A-algebras $F\colon B \to C$ making the diagram

$$\begin{tikzcd} B \arrow[r, "F"] \arrow[d, "g_B"'] & C \arrow[d, "g_C"] \\ \widehat{B} \arrow[r, "\widehat{F}"] & \widehat{C} \end{tikzcd}$$

commute. Since g_C is an isomorphism, the existence and uniqueness of F are immediate. This proves 1.10.

To complete the proof of 1.5, it remains only to prove 1.9.

1.12. Proof of 1.9. We have

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha}.$$

As explained in 1.9, regard $\mathfrak{g}$ as a $T \cdot_\alpha \mathbb{G}_{a,k}$-module. Clearly $\mathfrak{t} \oplus \mathfrak{g}^{\alpha}$ is a submodule of $\mathfrak{g}$, and the quotient is isomorphic to $\mathfrak{g}^{-\alpha}$ as a k-vector space and even as a T-module. The group $\mathbb{G}_{a,k}$ acts trivially on this one-dimensional quotient, since every group morphism $\mathbb{G}_{a,k} \to \mathbb{G}_{m,k}$ is trivial. Likewise, $\mathfrak{g}^{\alpha}$ is a submodule of $\mathfrak{t} \oplus \mathfrak{g}^{\alpha}$; the quotient is isomorphic to $\mathfrak{t}$ as a T-module, and $\mathbb{G}_{a,k}$ acts trivially on it. In summary:

Lemma 1.13. *Under the hypotheses of 1.9, $\mathfrak{g}$, as a $T \cdot_\alpha \mathbb{G}_{a,k}$-module, has a composition series whose successive quotients are*

$$\mathfrak{g}^{-\alpha}, \qquad \mathfrak{t}, \qquad \mathfrak{g}^{\alpha},$$

regarded as $T \cdot_\alpha \mathbb{G}_{a,k}$-modules through the projection $T \cdot_\alpha \mathbb{G}_{a,k} \to T$.

We are thus reduced to computing the cohomology of $T \cdot_\alpha \mathbb{G}_{a,k}$ acting on $W(k)$ through the projection $T \cdot_\alpha \mathbb{G}_{a,k} \to T$ and a character β of T (here $\beta = 0, \alpha$, or $-\alpha$). [5] Write $k[x_1, \dots, x_n]$ for the polynomial algebra over k in n variables, and $k_q[x_1, \dots, x_n]$ for its subspace of homogeneous polynomials of degree q.

Lemma 1.14. *With the preceding notation,*

$$\mathrm{H}^n(T \cdot_\alpha \mathbb{G}_{a,k}, k) = \mathrm{H}^n(C^*_{\alpha,\beta}),$$

where the complex $C^*_{\alpha,\beta}$ is defined by

$$C^n_{\alpha,\beta} = \begin{cases} k_q[x_1, \dots, x_n], & \text{if } \beta = q\alpha, \text{ with } q \in \mathbb{N}^*, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\begin{aligned} (\delta f)(x_1, x_2, \dots, x_{n+1}) = f(x_2, \dots, x_{n+1}) & \\ + \sum_{i=1}^{n} (-1)^i f(x_1, \dots, x_i x_{i+1}, \dots, x_{n+1}) & \\ + (-1)^{n+1} f(x_1, \dots, x_n). & \end{aligned}$$

Indeed, by Exposé I, 5.3.2, the functor $M \mapsto \mathrm{H}^0(T, M)$ is exact on the category of T-modules, and the $\mathrm{H}^q(T, -)$ vanish. It follows, as in the usual theory of group cohomology,

[5] The editors added the sentence that follows.

that $\mathrm{H}^n(T \cdot_\alpha \mathbb{G}_{a,k}, k)$ may be computed as the n-th cohomology group of the complex of cochains of $\mathbb{G}_{a,k}$ in k that are invariant under T, that is, those satisfying

$$f(\alpha(t)x_1, \ldots, \alpha(t)x_n) = \beta(t)f(x_1, \ldots, x_n).$$

This is precisely the complex just displayed.

To prove 1.9, it is therefore enough to show that $\mathrm{H}^2(C^*_{\alpha,\beta}) = 0$ for $\beta = 0, \alpha, -\alpha$, which is immediate.

Remark 1.15. The groups $\mathrm{H}^n(C^*_{\alpha,\beta})$ can be computed explicitly when $\beta = q\alpha$; see M. Lazard, *Lois de groupes et analyseurs*, Annales E.N.S., 1955. In particular,

$$\mathrm{H}^n(C^*_{\alpha,q\alpha}) = 0 \qquad \text{for } n > q.$$

Notations 1.16. The image of the canonical immersion

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^{\alpha}) \longrightarrow G$$

will be denoted by Ω. It is an open subset of G containing the identity section. The images of

$$W(\mathfrak{g}^{-\alpha}), \qquad W(\mathfrak{g}^{\alpha}), \qquad W(\mathfrak{g}^{-\alpha}) \times_S T, \qquad T \times_S W(\mathfrak{g}^{\alpha})$$

will be denoted, respectively,[6]

$$U_{-\alpha}, \qquad U_\alpha, \qquad U_{-\alpha} \cdot T, \qquad T \cdot U_\alpha.$$

Then U_α (respectively $U_{-\alpha}$) is a subgroup of G, canonically endowed with a vector-bundle structure, and

$$\mathrm{int}(t)(x) = x^{\alpha(t)} \qquad (\text{respectively } x^{-\alpha(t)})$$

for every $S' \to S$, $t \in T(S')$, and $x \in U_\alpha(S')$ (respectively $x \in U_{-\alpha}(S')$).

There are canonical isomorphisms

$$T \cdot U_\alpha \simeq T \cdot_\alpha U_\alpha, \qquad T \cdot U_{-\alpha} \simeq T \cdot_{-\alpha} U_{-\alpha}.$$

The open subset Ω is stable under $\mathrm{int}(T)$, and

$$\mathrm{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

Corollary 1.17. *We have*

$$\mathrm{Lie}(U_\alpha/S) = \mathfrak{g}^{\alpha}, \qquad \mathrm{Lie}(U_{-\alpha}/S) = \mathfrak{g}^{-\alpha}.$$

The isomorphisms

$$W(\mathfrak{g}^{\alpha}) \xrightarrow[\sim]{\exp} U_\alpha, \qquad W(\mathfrak{g}^{-\alpha}) \xrightarrow[\sim]{\exp} U_{-\alpha}$$

are those of Exposé XIX, 4.2.

Corollary 1.18. *The open subset Ω is relatively schematically dense in G* (cf. Exposé XVIII, §1).

This follows immediately from Exposé XVIII, 1.3.[7]

[6] The editors replaced $P_{-\alpha}$ and P_α by $U_{-\alpha}$ and U_α.

[7] See also EGA IV_3, 11.10.10.

Corollary 1.19. *The center of G is*

$$\mathrm{Centr}(G) = \mathrm{Ker}(\alpha).$$

It is therefore a closed subgroup of G, of multiplicative type and of finite type.

The second assertion follows from the first by Exposé IX, 2.7. Let us prove the first. The inner automorphism defined by a section of $\mathrm{Ker}(\alpha)$ acts trivially on Ω, by the last formula of 1.16, and hence on G, by 1.18. Conversely, if $g \in G(S)$ centralizes G, then it centralizes T and U_α. It is therefore a section of T, by Exposé XIX, 2.8, and annihilates α. Since the same argument applies after every base change, we indeed have $\mathrm{Centr}(G) = \mathrm{Ker}(\alpha)$.

Corollary 1.20. *A monomorphism*

$$p_\alpha \colon \mathbb{G}_{a,S} \longrightarrow G$$

normalized by T, with multiplier α, exists if and only if the $\mathscr{O}_S$-module $\mathfrak{g}^\alpha$ is free. More precisely, the assignments

$$X_\alpha \longmapsto (x \longmapsto \exp(xX_\alpha)), \qquad p_\alpha \longmapsto \mathrm{Lie}(p_\alpha)$$

give a bijection between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of such monomorphisms p_α, which is also the set of isomorphisms of vector group schemes

$$\mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha.$$

8

Corollary 1.21. *The subgroups U_α and $U_{-\alpha}$ of G commute on no fiber.*

Indeed, if $(U_\alpha)_s$ and $(U_{-\alpha})_s$ commute, then Ω_s is a subgroup of G_s, and hence $\Omega_s = G_s$.[9] It follows that G_s is solvable, contrary to the hypothesis that G_s is reductive of semisimple rank 1.

2 Structure of elementary systems

Theorem 2.1. *Let S be a scheme and (G, T, α) an S-elementary system. There exist a morphism of $\mathscr{O}_S$-modules*

$$\mathfrak{g}^\alpha \otimes_{\mathscr{O}_S} \mathfrak{g}^{-\alpha} \longrightarrow \mathscr{O}_S, \qquad (X, Y) \longmapsto \langle X, Y\rangle,$$

and a morphism of S-groups

$$\alpha^* \colon \mathbb{G}_{m,S} \longrightarrow T$$

such that, for every $S' \to S$ and all

$$X \in \Gamma(S', \mathfrak{g}^\alpha \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}), \qquad Y \in \Gamma(S', \mathfrak{g}^{-\alpha} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}),$$

one has

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \iff 1 + \langle X, Y\rangle \in \mathbb{G}_m(S').$$

[8] Indeed, $\mathrm{Lie}(\mathbb{G}_{a,S}) = \mathscr{O}_S$, and $\mathrm{Lie}(p_\alpha)$ is an element of $\mathrm{Hom}_{\mathscr{O}_S}(\mathscr{O}_S, \mathfrak{g}^\alpha) = \Gamma(S, \mathfrak{g}^\alpha)$.

[9] By 1.18.

Under these conditions,

$$\begin{aligned}\exp(X)\cdot\exp(Y) = {} & \exp\left(\frac{Y}{1+\langle X,Y\rangle}\right)\alpha^*(1+\langle X,Y\rangle) \\ & \exp\left(\frac{X}{1+\langle X,Y\rangle}\right).\end{aligned} \tag{F}$$

Moreover, the morphisms $(X,Y)\mapsto\langle X,Y\rangle$ *and* α^* *are uniquely determined. The first is an isomorphism, and therefore places* $\mathfrak{g}^\alpha$ *and* $\mathfrak{g}^{-\alpha}$ *in duality; furthermore,*

$$\alpha\circ\alpha^* = 2,$$

where 2 *denotes squaring on* $\mathbb{G}_{m,S}$*.*

In view of the uniqueness assertions, it is enough to prove the theorem locally on S. We may therefore assume that $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free on S. Choose

$$X\in\Gamma(S,\mathfrak{g}^\alpha)^\times,\qquad Y\in\Gamma(S,\mathfrak{g}^{-\alpha})^\times,$$

and put

$$p_\alpha(x)=\exp(xX),\qquad p_{-\alpha}(y)=\exp(yY)$$

for $x,y\in\mathbb{G}_a(S')$, $S'\to S$. By 1.5 and 1.21, it is enough to prove the following proposition.

Proposition 2.2. *Let* S *be a scheme,* G *an* S*-group,* T *a torus of* G*, and* α *a character of* T *that is nontrivial on every fiber. Let*

$$p_\alpha\colon\mathbb{G}_{a,S}\longrightarrow G,\qquad p_{-\alpha}\colon\mathbb{G}_{a,S}\longrightarrow G$$

be group monomorphisms normalized by T*, with multipliers* α *and* $-\alpha$*, respectively. Suppose that:*

(i) *the morphism*

$$\mathbb{G}_{a,S}\times_S T\times_S\mathbb{G}_{a,S}\longrightarrow G,\qquad (y,t,x)\longmapsto p_{-\alpha}(y)\,t\,p_\alpha(x),$$

is an open immersion; denote its image by Ω*;*

(ii) *for every* $s\in S$*, the groups*

$$(p_\alpha)_s(\mathbb{G}_{a,\kappa(s)})\quad\text{and}\quad(p_{-\alpha})_s(\mathbb{G}_{a,\kappa(s)})$$

do not commute.

Then there exist unique elements

$$a\in\mathbb{G}_a(S),\qquad \alpha^*\in\operatorname{Hom}_{S\text{-gr.}}(\mathbb{G}_{m,S},T)$$

with the following properties: for every $S'\to S$ *and all* $x,y\in\mathbb{G}_a(S')$*,*

$$p_\alpha(x)p_{-\alpha}(y)\in\Omega(S')\quad\Longleftrightarrow\quad 1+axy\in\mathbb{G}_m(S'),$$

and, under this condition,

$$p_\alpha(x)p_{-\alpha}(y)=p_{-\alpha}\left(\frac{y}{1+axy}\right)\alpha^*(1+axy)p_\alpha\left(\frac{x}{1+axy}\right).$$

Moreover, a *is invertible, that is,* $a\in\mathbb{G}_m(S)$*, and* $\alpha\circ\alpha^*=2$*.*

Proof. A) Consider the morphism

$$\mathbb{G}_{a,S}^2 \longrightarrow G, \qquad (x,y) \longmapsto p_\alpha(x)p_{-\alpha}(y).$$

Let U be the inverse image of Ω. It is an open subset of $\mathbb{G}_{a,S}^2$ containing $0 \times_S \mathbb{G}_{a,S}$ and $\mathbb{G}_{a,S} \times_S 0$. There are therefore unique morphisms of S-schemes

$$A\colon U \longrightarrow \mathbb{G}_{a,S}, \qquad C\colon U \longrightarrow \mathbb{G}_{a,S}, \qquad B\colon U \longrightarrow T$$

satisfying the set-theoretic identity

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(A(u,v))\, B(u,v)\, p_\alpha(C(u,v)).$$

Immediately,

$$A(0,v) = v, \qquad A(u,0) = 0, \qquad C(u,0) = u, \qquad C(0,v) = 0,$$
$$B(u,0) = B(0,v) = e.$$

Let S' be a separated S-scheme and let $t \in T(S')$. Since $\Omega_{S'}$ is stable under $\operatorname{int}(t)$, the last formula of 1.16 shows that $U_{S'}$ is stable under the automorphism

$$(x,y) \longmapsto (\alpha(t)x, \alpha(t)^{-1}y)$$

of $\mathbb{G}_{a,S'}^2$, and

$$\begin{aligned} A(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)^{-1}A(u,v), \\ C(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)C(u,v), \\ B(\alpha(t)u, \alpha(t)^{-1}v) &= B(u,v). \end{aligned}$$

Because α is faithfully flat, it follows that for every $S' \to S$ and every $z \in \mathbb{G}_m(S')$, the open subset $U_{S'}$ is stable under $(x,y) \mapsto (zx, z^{-1}y)$, and

$$\begin{aligned} A(zu, z^{-1}v) &= z^{-1}A(u,v), \\ C(zu, z^{-1}v) &= zC(u,v), \\ B(zu, z^{-1}v) &= B(u,v). \end{aligned}$$

Suppose first that v is invertible. Taking $z = v$, we see that if (u,v) is a section of U, then $(uv,1)$ is also a section of U, and

$$A(uv,1) = v^{-1}A(u,v), \qquad B(uv,1) = B(u,v).$$

Let V be the open subset of $\mathbb{G}_{a,S}^2$ defined by[10]

$$(u,v) \in V(S') \iff (u,v),\ (uv,1),\ \text{and } (1,uv) \text{ belong to } U(S').$$

Since U contains $0 \times_S \mathbb{G}_{a,S}$ and $\mathbb{G}_{a,S} \times_S 0$, the open subset V is a neighborhood of the zero section of $\mathbb{G}_{a,S}^2$. The preceding calculation shows that, on $V \cap (\mathbb{G}_{a,S} \times_S \mathbb{G}_{m,S})$, the two morphisms

$$(u,v) \longmapsto A(u,v), \qquad (u,v) \longmapsto vA(uv,1)$$

coincide, as do

$$(u,v) \longmapsto B(u,v), \qquad (u,v) \longmapsto B(uv,1).$$

Since $\mathbb{G}_{a,S} \times_S \mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}^2$, these morphisms coincide on V.

[10]The editors added the condition "$(1,uv) \in U(S')$".

We have $A(0,1) = 1$. Hence there is an open subset W_1 of $\mathbb{G}_{a,S}$, containing the zero section, such that $A(x,1)$ is invertible for every section x of W_1. Put

$$A(x,1)^{-1} = F(x).$$

If $(u,v) \in V(S')$[11] and $uv \in W_1(S')$, then

$$A(u,v) = vA(uv,1) = vF(uv)^{-1}.$$

The same argument applied to C gives an open subset W_2 of $\mathbb{G}_{a,S}$, containing the zero section, and an element $E \in \mathscr{O}(W_2)^{\times}$[12] such that

$$C(u,v) = uC(1,uv) = uE(uv)^{-1}$$

whenever $(u,v) \in V(S')$ and $uv \in W_2(S')$.

Consequently, with $W = W_1 \cap W_2$, there are S-morphisms

$$\begin{aligned} F\colon W &\to \mathbb{G}_{m,S}, & F(0) &= 1,\\ H\colon W &\to T, & H(0) &= e,\\ E\colon W &\to \mathbb{G}_{m,S}, & E(0) &= 1, \end{aligned}$$

such that, if $(u,v) \in V(S')$ and $uv \in W(S')$, then

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\big(vF(uv)^{-1}\big)\,H(uv)\,p_\alpha\big(uE(uv)^{-1}\big). \tag{+}$$

B) We now use the associativity of G to write

$$p_\alpha(u)p_{-\alpha}(v)p_{-\alpha}(w) = p_\alpha(u)p_{-\alpha}(v+w).$$

There is an open subset L of $\mathbb{G}^3_{a,S}$, containing the zero section, such that $(u,v,w) \in L(S')$ is equivalent to

$$\begin{aligned} &(u,v) \in V(S'), && (uE(uv)^{-1},w) \in V(S'), && (u,v+w) \in V(S'),\\ &uv \in W(S'), && uwE(uv)^{-1} \in W(S'), && u(v+w) \in W(S'). \end{aligned}$$

Using formula (+), we immediately obtain, for $(u,v,w) \in L(S')$,

$$E(uv+uw) = E\big(uwE(uv)^{-1}\big)E(uv), \tag{1}$$

$$H(uv+uw) = H\big(uwE(uv)^{-1}\big)H(uv), \tag{2}$$

and

$$\begin{aligned}(v+w)F(uv+uw)^{-1} = \alpha\big(H(uv)^{-1}\big)\,wF\big(uwE(uv)^{-1}\big)^{-1}\\ + vF(uv)^{-1}.\end{aligned} \tag{3}$$

It follows immediately from the definition of L that $(1,0,0) \in L(S)$. Consider therefore

$$L \cap_T \big(1 \times_S \mathbb{G}_{a,S} \times_S \mathbb{G}_{a,S}\big) = 1 \times_S M.$$

[11] Here and below, the editors replaced $U(S')$ by $V(S')$.

[12] The editors use E for the element denoted G in the original, since G already denotes the S-group under consideration.

The scheme M is an open subset of $\mathbb{G}_{a,S}^2$ containing the section $(0,0)$. For $(v,w) \in M(S')$, the elements

$$v, \qquad wE(v)^{-1}, \qquad v+w$$

belong to $W(S')$, and

$$E(v+w) = E\big(wE(v)^{-1}\big)E(v), \tag{1'}$$

$$H(v+w) = H\big(wE(v)^{-1}\big)H(v), \tag{2'}$$

$$\begin{aligned}(v+w)F(v+w)^{-1} = \alpha\big(H(v)\big)^{-1}&wF\big(wE(v)^{-1}\big)^{-1} \\ &+ vF(v)^{-1}.\end{aligned} \tag{3'}$$

Finally, consider the morphism from M to $\mathbb{G}_{a,S}^2$ defined set-theoretically by

$$(v,w) \longmapsto \big(v, wE(v)^{-1}\big).$$

[13] It preserves the section $(0,0)$ and induces an isomorphism of M onto an open subset N of $\mathbb{G}_{a,S}^2$ containing the zero section; the inverse isomorphism is

$$(x,y) \longmapsto \big(x, yE(x)\big).$$

[14] We have therefore proved the following assertion:

There is an open subset N of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, if $(x,y) \in N(S')$, then x, y, and $x+yE(x)$[15] belong to $W(S')$, and

$$E\big(x+yE(x)\big) = E(x)E(y), \tag{1''}$$

$$H\big(x+yE(x)\big) = H(x)H(y), \tag{2''}$$

$$\begin{aligned}\big(x+yE(x)\big)F\big(x+yE(x)\big)^{-1} = {}& xF(x)^{-1} \\ &+ \alpha\big(H(x)\big)^{-1}yE(x)F(y)^{-1}.\end{aligned} \tag{3''}$$

C) Arguing in the same way with left associativity, we obtain the following assertion.[16]

There is an open subset N' of $\mathbb{G}_{a,S}^2$, containing the zero section, such that, if $(x,y) \in N'(S')$, then x, y, and $x+yF(x)$[17] belong to $W(S')$, and

$$F\big(x+yF(x)\big) = F(x)F(y), \tag{4''}$$

$$H\big(x+yF(x)\big) = H(x)H(y), \tag{5''}$$

$$\begin{aligned}\big(x+yF(x)\big)E\big(x+yF(x)\big)^{-1} = {}& xE(x)^{-1} \\ &+ \alpha\big(H(x)\big)^{-1}yF(x)E(y)^{-1}.\end{aligned} \tag{6''}$$

We are thus led to solve the "functional equation" $(1'')$.

[13] The editors corrected what follows.

[14] Thus the editors made the change of variables $x=v,\ y=wE(v)^{-1}$, or equivalently $v=x,\ w=yE(x)$.

[15] The editors corrected $yE(x)$ to $x+yE(x)$.

[16] That is, one writes the equalities resulting from $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t+u)p_{-\alpha}(v)$, and sets $v = 1,\ x=u,\ t=yF(u)$, or equivalently $y=tF(u)^{-1}$.

[17] The editors corrected $yF(x)$ to $x+yF(x)$.

Lemma 2.3. *Let S be a scheme, let W be an open subset of $\mathbb{G}_{a,S}$ containing the zero section, and let*

$$F\colon W \longrightarrow \mathbb{G}_{m,S}$$

be an S-morphism. Suppose that $F(0) = 1$ and that there is an open subset N of $\mathbb{G}^2_{a,S}$, containing the zero section, such that, for $(x, y) \in N(S')$, the elements x, y, and $x + yF(x)$ belong to $W(S')$, and

$$F(x + yF(x)) = F(x)F(y). \tag{†}$$

(i) *If $S = \mathrm{Spec}(k)$ for a field k, there is an $a \in k$ such that $F(x) = 1 + ax$.*

(ii) *If $a = F'(0) \in \Gamma(S, \mathscr{O}_S)$ is invertible, then $F(x) = 1 + ax$.*

Proof. Under the hypotheses, we may differentiate the given equation at $x = 0$, respectively at $y = 0$, and obtain

$$F'(y)(1 + yF'(0)) = F'(0)F(y) \qquad \text{for } (0, y) \in N(S'), \tag{$*$}$$

respectively

$$F'(x)F(x) = F(x)F'(0) \qquad \text{for } (x, 0) \in N(S').$$

Since F takes its values in $\mathbb{G}_m$, the second relation gives

$$F'(x) = F'(0) \qquad \text{for } (x, 0) \in N(S'). \tag{$*'$}$$

The first relation therefore gives

$$F'(0)(1 + yF'(0)) = F'(0)F(y) \qquad \text{for } (y, 0), (0, y) \in N(S').$$

If $a = F'(0)$ is invertible, it follows that

$$F(y) = 1 + ay$$

for y a section of an open subset of W containing the zero section, hence schematically dense in W. This proves (ii), and also proves (i) when $F'(0) \neq 0$.

Suppose $F'(0) = 0$. Then $(*')$ shows that $F'(x) = 0$ for x near 0, and hence $F' = 0$ by schematic density. If k has characteristic zero, F is a rational function with zero derivative, and is therefore constant and equal to $F(0) = 1$.

Suppose now that k has characteristic p, and that F is not constant.[18] There are an integer $n > 0$ and a rational function $F_1 \in k(X)$ such that $F_1'(X) \neq 0$ and

$$F(X) = F_1(X^{p^n}) = F_1(X)^{p^n}.$$

Substitution in the functional equation gives

$$F_1(x + yF_1(x)^{p^n}) = F_1(x)F_1(y). \tag{$†_1$}$$

Differentiating at $x = 0$ gives

$$F_1'(y) = F_1'(0)F_1(y), \tag{$*_1$}$$

whereas differentiation of $(†_1)$ at $y = 0$ gives

$$F_1'(x)F_1(x)^{p^n} = F_1(x)F_1'(0). \tag{$*_1'$}$$

[18]The editors corrected what follows.

By hypothesis, $F_1'(X)$ is invertible in $k(X)$; these two equalities therefore imply

$$F_1(X)^{p^n} = 1.$$

Thus F_1 is constant, contrary to the choice of F_1. Consequently F is constant and equal to $1 = F(0)$. □

D) Suppose first that S is the spectrum of a field. If $F'(0) = 0$, then $F = 1$. Formula $(5'')$ then gives

$$H(x+y) = H(x)H(y),$$

which shows that H extends to a group morphism

$$\mathbb{G}_{a,S} \longrightarrow T$$

(Exposé XVIII, 2.3), necessarily the constant morphism with value e. On the other hand, Lemma 2.3 also gives

$$E(x) = 1 + bx$$

for some $b \in k$. Formula $(6'')$ therefore gives, for $(x, y) \in N(S')$,

$$(x+y)E(x+y)^{-1} = xE(x)^{-1} + yE(y)^{-1}.$$

Thus,[19] again by Exposé XVIII, 2.3, the map

$$x \longmapsto xE(x)^{-1}$$

extends to a morphism of k-groups $\mathbb{G}_{a,k} \to \mathbb{G}_{a,k}$. Consequently

$$\frac{x}{1+bx} = cx$$

for some $c \in k$, whence $b = 0$ and $c = 1$.

Thus F, H, E are constant with values $(1, e, 1)$ in a neighborhood of the zero section, and hence everywhere. Formula (+) then shows that U_α and $U_{-\alpha}$ commute, contrary to hypothesis (ii).

For arbitrary S, we have therefore proved that $F'(0)$ is nonzero on every fiber, and hence invertible. The same is plainly true of $E'(0)$. Lemma 2.3 now shows that there are $a, b \in \mathbb{G}_m(S)$ such that

$$F(x) = 1 + ax, \qquad E(x) = 1 + bx \qquad \text{for } x \in W(S'). \tag{$\diamondsuit_1$}$$

E) The rest is now easy. Substitution of the preceding results in $(3'')$ gives

$$y\,\alpha\big(H(x)\big)(1+ay) = y\big(1 + ax + ay(1+bx)\big)(1+ax).$$

This formula holds for every section (x, y) of N. Since $\mathbb{G}_{a,S} \times_S \mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}^2_{a,S}$, it follows that

$$(1+ay)\alpha\big(H(x)\big) = \big(1 + ax + ay(1+bx)\big)(1+ax).$$

Setting $y = 0$ gives

$$\alpha\big(H(x)\big) = (1+ax)^2.$$

[19]The editors added the sentence that follows. One may also see by direct calculation that the preceding equality implies $0 = xyb\big(2 + (x+y)b\big)$, hence $0 = b\big(2 + (x+y)b\big)$, and finally $b = 0$.

Substitution in the preceding equality gives[20]

$$a^2xy = abxy.$$

Since $\mathbb{G}_{m,S}$ is schematically dense in $\mathbb{G}_{a,S}$, we obtain $a^2 = ab$. As a is invertible,

$$a = b, \qquad \alpha(H(x)) = (1+ax)^2. \tag{$\diamondsuit_2$}$$

Because a is invertible, the map $x \mapsto 1+ax$ is an automorphism of $\mathbb{G}_{a,S}$. There are therefore an open subset W' of $\mathbb{G}_{a,S}$, containing the section 1, and a morphism

$$P\colon W' \longrightarrow T$$

such that

$$P(1+ax) = H(x).$$

[21] Substitution in $(2')$ gives immediately, for $(x,y) \in N(S')$,

$$P(1+ax+ay) = P\left(\frac{1+ax+ay}{1+ax}\right) P(1+ax).$$

Thus there is an open neighborhood of 1 in $\mathbb{G}_{m,S}$ on which

$$P(x)P(y) = P(xy).$$

By Exposé XVIII, 2.3, there is a group morphism

$$\alpha^*\colon \mathbb{G}_{m,S} \longrightarrow T \tag{$\diamondsuit_3$}$$

extending P. Since $\alpha(H(x)) = (1+ax)^2$ near the zero section, we have $\alpha(\alpha^*(z)) = z^2$ near the section 1, and hence

$$\alpha \circ \alpha^* = 2. \tag{$\diamondsuit_4$}$$

F)[22] Combining $(+)$ and $(\diamondsuit_1)$–$(\diamondsuit_4)$, we see that there are

$$a \in \mathbb{G}_m(S), \qquad \alpha^* \in \mathrm{Hom}_{S\text{-gr.}}(\mathbb{G}_{m,S}, T)$$

such that $\alpha \circ \alpha^* = 2$, and such that, whenever $(u,v) \in V(S')$ and $uv \in W(S')$, the element $1+auv$ is invertible and

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\left(\frac{v}{1+auv}\right)\alpha^*(1+auv)p_\alpha\left(\frac{u}{1+auv}\right).$$

Let V' be the open subset of $\mathbb{G}^2_{a,S}$ defined by the invertibility of $1+auv$; thus

$$V' = (\mathbb{G}^2_{a,S})_f, \qquad f(u,v) = 1+auv.$$

The two sides of the preceding formula define morphisms $V' \to G$ which agree in a neighborhood of the zero section, and hence agree on V'. Thus the formula holds for every

[20] In the three preceding equalities, the editors corrected the original by replacing the factor $1+bx$ on the right by $1+ax$. Substitution of the third equality in the second, and use of the invertibility of $1+ax$, gives precisely $a^2xy = abxy$.

[21] This is the change of variables $x' = 1+ax$, or $x = (x'-1)/a$.

[22] The editors slightly modified what follows, since an open subset V had already been introduced in part A.

section (u,v) of V', and in particular $V' \subset U$, where U is the open subset introduced at the beginning of part A.

We prove that $U = V'$. Returning to the notation of part A, there is a morphism

$$A\colon U \longrightarrow \mathbb{G}_{a,S}$$

whose restriction to V' is

$$A(u,v) = v(1+auv)^{-1}.$$

The equality $U = V'$ is a set-theoretic question and may therefore be checked when $S = \operatorname{Spec}(k)$ for a field k. It then reduces to the evident assertion that the domain of definition of the rational map

$$\mathbb{G}_{a,k}^2 \dashrightarrow \mathbb{G}_{a,k}, \qquad (X,Y) \longmapsto \frac{Y}{1+aXY},$$

is the open subset defined by $1+aXY$.

G) We have proved the existence of a and α^*, as well as the two asserted supplementary properties. It remains to prove uniqueness. Let a' and $\alpha^{*\prime}$ also satisfy the required conditions. For $u,v \in \mathbb{G}_a(S')$, we immediately have

$$1+auv \text{ invertible} \implies 1+a'uv \text{ invertible},$$
$$\frac{v}{1+auv} = \frac{v}{1+a'uv}.$$

Hence, for every $u \in \mathbb{G}_a(S')$,

$$1+au \text{ invertible} \implies 1+au = 1+a'u,$$

which proves $a = a'$. With the same notation,

$$1+au \text{ invertible} \implies \alpha^*(1+au) = \alpha^{*\prime}(1+au),$$

and therefore $\alpha^* = \alpha^{*\prime}$. □

Corollary 2.4. *Let*

$$\exp(Y)t\exp(X) \quad \text{and} \quad \exp(Y')t'\exp(X')$$

be two elements of $\Omega(S')$*. Their product belongs to* $\Omega(S')$ *if and only if*

$$u = 1 + \langle X, Y'\rangle$$

is invertible; in that case,

$$\begin{aligned}&\exp(Y)t\exp(X)\cdot\exp(Y')t'\exp(X')\\ &\quad = \exp(Y + u^{-1}\alpha(t)^{-1}Y')\, tt'\alpha^*(u)\, \exp(u^{-1}\alpha(t')^{-1}X + X').\end{aligned} \tag{F'}$$

Remark 2.5. *Formula* (F) *of Theorem 2.1 may also be written without using the morphisms* exp*. Transporting the duality* $\mathfrak{g}_\alpha \otimes \mathfrak{g}_{-\alpha} \to \mathscr{O}_S$ *through those morphisms gives a canonical pairing of vector bundles*

$$U_\alpha \times_S U_{-\alpha} \longrightarrow \mathbb{G}_{a,S},$$

which we continue to denote by $(x,y) \mapsto \langle x,y\rangle$*. Thus*

$$\langle \exp X, \exp Y\rangle = \langle X, Y\rangle.$$

If $x \in U_\alpha(S')$*,* $y \in U_{-\alpha}(S')$*, and* $1+\langle x,y\rangle \in \mathbb{G}_m(S')$*, then*

$$x\cdot y = y^{(1+\langle x,y\rangle)^{-1}}\, \alpha^*\big(1+\langle x,y\rangle\big)\, x^{(1+\langle x,y\rangle)^{-1}}.$$

Corollary 2.6. *The pairing*

$$W(\mathfrak{g}_\alpha) \times_S W(\mathfrak{g}_{-\alpha}) \longrightarrow \mathbb{G}_{a,S}$$

induces a pairing of principal $\mathbb{G}_{m,S}$*-bundles*

$$W(\mathfrak{g}_\alpha)^\times \times_S W(\mathfrak{g}_{-\alpha})^\times \longrightarrow \mathbb{G}_{m,S}.$$

We denote this pairing by $(X, Y) \mapsto \langle X, Y\rangle$, *or simply* $(X, Y) \mapsto XY$.

For every $X \in \Gamma(S, \mathfrak{g}_\alpha)^\times$, there is therefore a unique

$$X^{-1} \in \Gamma(S, \mathfrak{g}_{-\alpha})^\times$$

such that $XX^{-1} = 1$. Moreover,

$$(zX)^{-1} = z^{-1}X^{-1}.$$

The morphism

$$s\colon W(\mathfrak{g}_\alpha)^\times \longrightarrow W(\mathfrak{g}_{-\alpha})^\times$$

so defined is an isomorphism of schemes, compatible with the isomorphism $s\colon z \mapsto z^{-1}$ of their operator groups.

Definition 2.6.1. *The elements* X *and* $s(X) = X^{-1}$ *are said to be paired.*

Apply Corollary 2.4 with $Y = 0 = X'$ and

$$Y' = aX^{-1}, \qquad a \in \mathscr{O}_S(S).$$

Then $u = 1 + a$, and

$$u^{-1}Y' = u^{-1}(u - 1)X^{-1} = (1 - u^{-1})X^{-1}.$$

Hence:

Corollary 2.7. *Let* $X \in \Gamma(S, \mathfrak{g}_\alpha)^\times$ *and* $u \in \Gamma(S, \mathscr{O}_S)^\times$. *Then*

$$\alpha^*(u) = \exp\big((u^{-1} - 1)X^{-1}\big)\, \exp(X)\, \exp\big((u - 1)X^{-1}\big)\, \exp(-u^{-1}X).$$

Definition 2.8. *The morphism* α^* *is called the coroot associated with the root* α.

Remark 2.9. *If* (G, T, α) *is an* S*-elementary system, then so is* $(G, T, -\alpha)$. *Theorem 2.1 therefore gives a duality between* $\mathfrak{g}_{-\alpha}$ *and* $\mathfrak{g}_\alpha$, *and a coroot* $(-\alpha)^*$. *Taking the inverse of formula* (F) *gives immediately*

$$\langle X, Y\rangle = \langle Y, X\rangle, \qquad (-\alpha)^* = -\alpha^*.$$

We now pass to the Lie algebra of G. The root α and the coroot α^* define linear forms

$$\mathscr{O}_S \xrightarrow{\overline{\alpha^*}} \mathfrak{t} \xrightarrow{\bar\alpha} \mathscr{O}_S.$$

Put

$$H_\alpha = \overline{\alpha^*}(1).$$

We call $\bar\alpha$ the infinitesimal root associated with α, and H_α the corresponding infinitesimal coroot.

Lemma 2.10. *Let $S' \to S$, and let*

$$X, X' \in W(\mathfrak{g}_\alpha)(S'), \qquad H \in W(\mathfrak{t})(S'), \qquad Y, Y' \in W(\mathfrak{g}_{-\alpha})(S'), \qquad t \in T(S').$$

Then

$$\mathrm{Ad}(t)H = H, \qquad \mathrm{Ad}(t)X = \alpha(t)X, \qquad \mathrm{Ad}(t)Y = \alpha(t)^{-1}Y; \tag{1}$$

$$\begin{aligned} &\mathrm{Ad}(\exp X)H = H - \bar{\alpha}(H)X, \qquad \mathrm{Ad}(\exp X)X' = X', \\ &\mathrm{Ad}(\exp X)Y = Y + \langle X, Y\rangle H_\alpha - \langle X, Y\rangle X; \end{aligned} \tag{2}$$

$$\begin{aligned} &\mathrm{Ad}(\exp Y)H = H + \bar{\alpha}(H)Y, \qquad \mathrm{Ad}(\exp Y)Y' = Y', \\ &\mathrm{Ad}(\exp Y)X = X + \langle X, Y\rangle H_{-\alpha} - \langle X, Y\rangle Y; \end{aligned} \tag{2'}$$

$$\begin{gathered} [H, X] = \bar{\alpha}(H)X, \qquad [H, Y] = -\bar{\alpha}(H)Y, \\ [X, Y] = \langle X, Y\rangle H_\alpha; \end{gathered} \tag{3}$$

$$H_{-\alpha} = -H_\alpha, \tag{4}$$

$$\bar{\alpha}(H_\alpha) = 2. \tag{5}$$

The proof of these formulas is either immediate or an immediate consequence of formula (F) of 2.1.

Corollary 2.11. *Suppose that H_α is nonzero on every fiber; by* (5)*, this holds in particular if* 2 *is invertible on S. Then*

$$X_\alpha \in \Gamma(S, \mathfrak{g}_\alpha)^\times, \qquad X_{-\alpha} \in \Gamma(S, \mathfrak{g}_{-\alpha})^\times$$

are paired if and only if

$$[X_\alpha, X_{-\alpha}] = H_\alpha.$$

2.12. Let (G, T, α) be an S-elementary system. By 1.19,

$$\mathrm{Centr}(G) = \ker(\alpha)$$

is a group of multiplicative type and finite type. If Q is a subgroup of multiplicative type of $\mathrm{Centr}(G)$, then G/Q is affine over S (Exposé IX, 2.5), smooth over S (Exposé $\mathrm{VI_B}$, 9.2), and has connected reductive fibers of semisimple rank 1 (Exposé XIX, 1.8).

Put $G' = G/Q$. It is a reductive S-group of semisimple rank 1, and $T' = T/Q$ is a maximal torus of G'. The open subset

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

of G is stable under Q, and its quotient is isomorphic to

$$U_{-\alpha} \times_S (T/Q) \times_S U_\alpha.$$

If α' denotes the character of T' induced by α, the canonical morphism $G \to G'$ induces isomorphisms

$$\mathfrak{g}_\alpha \xrightarrow{\sim} \mathfrak{g}'_{\alpha'}, \qquad \mathfrak{g}_{-\alpha} \xrightarrow{\sim} \mathfrak{g}'_{-\alpha'}.$$

In particular, α' is a root of G' relative to T'. Writing α/Q for the character

$$T/Q \longrightarrow \mathbb{G}_{m,S}$$

induced by α, we obtain:

Lemma 2.13. *If Q is a subgroup of multiplicative type of* $\ker(\alpha)$*, then*

$$(G/Q, T/Q, \alpha/Q)$$

is an elementary system.

Lemma 2.14. *Under the preceding conditions, the following diagrams are commutative:*

$$\begin{tikzcd}
\mathrm{W}(\mathfrak{g}^{\alpha}) \arrow[r, "\exp"] \arrow[d, "\mathrm{can}"', "\sim"] & G \arrow[d, "\mathrm{can}"'] & \mathrm{W}(\mathfrak{g}^{-\alpha}) \arrow[l, "\exp"'] \arrow[d, "\mathrm{can}"', "\sim"] \\
\mathrm{W}(\mathfrak{g}'^{\alpha'}) \arrow[r, "\exp"] & G' & \mathrm{W}(\mathfrak{g}'^{-\alpha'}) \arrow[l, "\exp"']
\end{tikzcd}$$

$$\begin{tikzcd}
\mathfrak{g}^{\alpha} \otimes \mathfrak{g}^{-\alpha} \arrow[r, "\sim"] \arrow[d, "\mathrm{can}"', "\sim"] & \mathscr{O}_S \arrow[d, "\mathrm{id}"] \\
\mathfrak{g}'^{\alpha'} \otimes \mathfrak{g}'^{-\alpha'} \arrow[r, "\sim"] & \mathscr{O}_S
\end{tikzcd}$$

$$\begin{tikzcd}
 & T \arrow[dr, "\alpha"] \arrow[dd, "\mathrm{can}"] & \\
\mathbb{G}_{m,S} \arrow[ur, "\alpha^*"] \arrow[dr, "\alpha'^*"'] & & \mathbb{G}_{m,S} \\
 & T' \arrow[ur, "\alpha'"'] &
\end{tikzcd}$$

3 The Weyl group

Notations 3.0. [23] If (G, T, α) is an S-elementary system, write

$$N = \mathrm{Norm}_G(T), \qquad W = \mathrm{Norm}_G(T)/T$$

(cf. Exposé XIX, 6.3). The group N is a closed subgroup of G, smooth over S. Write

$$N^{\times} = N - T$$

for the open subscheme of N induced on the complement of T.[24]

Let R be the unique maximal torus of $\mathrm{Ker}(\alpha)$, and let T' be the image of

$$\alpha^* \colon \mathbb{G}_{m,S} \longrightarrow T;$$

it is a one-dimensional subtorus of T. The morphism

$$T' \times_S R \longrightarrow T$$

induced by multiplication in T is surjective, hence faithfully flat. Indeed, it is enough to check this on the geometric fibers, where it follows at once from the formula

$$\alpha \circ \alpha^* = 2.$$

[23] The editors added the number 3.0 for later references.
[24] The editors replaced Q by the more suggestive notation $N^{\times}$.

Theorem 3.1. *With the preceding notation:*

(i) *W is isomorphic to the constant group $(\mathbb{Z}/2\mathbb{Z})_S$.*

(ii) *$N^\times$ is a locally trivial principal homogeneous bundle under T, on the left by the law $(t,q) \mapsto tq$, and on the right by the law $(q,t) \mapsto qt$.*

(iii) *One has*

$$\operatorname{int}(w)t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for $w \in N^\times(S')$, $t \in T(S')$, and $S' \to S$. In the decomposition $T_{S'} = T'_{S'} \cdot R_{S'}$, $\operatorname{int}(w)$ induces the identity on $R_{S'}$ and inversion on $T'_{S'}$. One has the relations

$$\alpha \circ \operatorname{int}(w) = \alpha^{-1}, \qquad \operatorname{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

(iv) *For $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \to S$, set*

$$w_\alpha(X) = \exp(X)\exp(-X^{-1})\exp(X).$$

Then $w_\alpha(X) \in N^\times(S')$, and the morphism $w_\alpha\colon W(\mathfrak{g}^\alpha)^\times \to N^\times$ so defined satisfies

$$w_\alpha(zX) = \alpha^*(z)w_\alpha(X) = w_\alpha(X)\alpha^*(z)^{-1}$$

for $z \in \mathbb{G}_m(S')$, $X \in W(\mathfrak{g}^\alpha)^\times(S')$, and $S' \to S$.

(v) *For $X, Y \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \to S$, one has, with the notation of 2.6,*

$$w_\alpha(X)w_\alpha(Y) = \alpha^\vee(-XY^{-1}).$$

[25] *In particular,*

$$w_\alpha(X)^2 = \alpha^*(-1) \in {}_2T(S) \cap \operatorname{Centr}(G)(S),$$
$$w_\alpha(X)^{-1} = w_\alpha(-X) = \alpha^*(-1)w_\alpha(X).$$

(vi) *If, similarly, one defines for $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$*

$$w_{-\alpha}(Y) = \exp(Y)\exp(-Y^{-1})\exp(Y),$$

then, in addition to the formulas analogous to the preceding ones,

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X), \qquad w_\alpha(X)w_{-\alpha}(Y) = \alpha^*(XY).$$

Proof. Assertion (i) was already proved in Exposé XIX, 2.4. It follows at once that $N^\times$ is indeed a principal homogeneous bundle under T for the laws defined in (ii); its local triviality for the Zariski topology follows in particular from (iv).

Let us prove (iii). If $w \in N^\times(S)$, it is clear that $\alpha \circ \operatorname{int}(w)$ is a root of G relative to T, and hence is locally equal to α or $-\alpha$. Since it is $-\alpha$ on every fiber (Bible, 12-05, proof of the corollary to Proposition 1), one has

$$\alpha \circ \operatorname{int}(w) = -\alpha.$$

By transport of structure, it follows that

$$-\alpha^* = \operatorname{int}(w)^{-1} \circ \alpha^* = \operatorname{int}(w) \circ \alpha^*,$$

since $\operatorname{int}(w)^2 = \operatorname{int}(w^2)$ and w^2 is a section of T. Thus $\operatorname{int}(w)$ induces inversion on T'; since R is central, it induces the identity on R. The formula in (iii) defines a morphism $T \to T$ having the same properties, and therefore coincides with $\operatorname{int}(w)$.

[25] The editors corrected the first formula of the original.

Let us prove (iv). Successively,

$$\begin{aligned} w_\alpha(X) t w_\alpha(X)^{-1} = &\ \exp(X)\exp(-X^{-1})\exp(X)t\exp(-X)\exp(X^{-1})\exp(-X) \\ = &\ \exp(X)\exp(-X^{-1})\exp(X-\alpha(t)X)\exp(\alpha(t)^{-1}X^{-1})\exp(-\alpha(t)X)t. \end{aligned}$$

Applying formula (F) gives

$$\begin{aligned} \exp(-X^{-1})\exp\bigl((1-\alpha(t))X\bigr) = &\ \exp\bigl((\alpha(t)^{-1}-1)X\bigr)\alpha^*\bigl(\alpha(t)^{-1}\bigr) \\ &\ \cdot \exp\bigl(-\alpha(t)^{-1}X^{-1}\bigr). \end{aligned}$$

Substitution in the preceding relation yields

$$\begin{aligned} \operatorname{int}(w_\alpha(X))t = &\ \exp\bigl(\alpha(t)^{-1}X\bigr)\alpha^*\bigl(\alpha(t)^{-1}\bigr)\exp\bigl(-\alpha(t)X\bigr)t \\ = &\ \exp(aX)\alpha^*\bigl(\alpha(t)^{-1}\bigr)t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha\circ\alpha^*)\bigl(\alpha(t)^{-1}\bigr)\alpha(t).$$

But $\alpha\circ\alpha^* = 2$, which gives $a = 0$ at once and shows that $w_\alpha(X) \in N^\times(S')$.

We now prove the second assertion of (iv). One has[26]

$$\begin{aligned} \alpha^*(z)w_\alpha(X) = &\ \exp(z^2X)\exp(-z^{-2}X^{-1})\exp(z^2X)\alpha^*(z) \\ = &\ \exp(zX)\exp\bigl((z^2-z)X\bigr)\exp(-z^{-2}X^{-1})\exp(z^2X)\alpha^*(z) \\ = &\ \exp(zX)\exp(-z^{-1}X^{-1})\alpha^*(z)^{-1}\exp\bigl((z^3-z^2)X\bigr)\exp(z^2X)\alpha^*(z) \\ = &\ \exp(zX)\exp(-z^{-1}X^{-1})\exp(zX) = w_\alpha(zX). \end{aligned}$$

Let us prove (v). By the preceding result, the first formula in (v) follows at once from the second; we prove the latter:

$$\begin{aligned} w_\alpha(X)^2 = &\ \exp(X)\exp(-X^{-1})\exp(2X)\exp(-X^{-1})\exp(X) \\ = &\ \exp(X)\exp(-X^{-1})\exp(X^{-1})\alpha^*(-1)\exp(-2X)\exp(X) \\ = &\ \exp(X)\alpha^*(-1)\exp(-X) = \alpha^*(-1), \end{aligned}$$

since

$$\alpha(\alpha^*(-1)) = (-1)^2 = 1.$$

This also proves that $\alpha^*(-1) \in \operatorname{Centr}(G)(S)$.

Finally, let us prove (vi). The first assertion is a special case of the second, so we prove the latter. Its two sides define morphisms

$$W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times \longrightarrow G.$$

To prove that they coincide, it suffices to do so on a nonempty open subset of every fiber

[26] The first equality follows from 1.5 (i), which, together with $\alpha\circ\alpha^* = 2$, gives $\alpha^*(z)\exp(X)\alpha^*(z)^{-1} = \exp(z^2X)$ and $\alpha^*(z)\exp(X^{-1})\alpha^*(z)^{-1} = \exp(z^{-2}X)$. The third equality follows from formula (F), and the fourth again from these formulas. Finally, an analogous calculation gives $w_\alpha(X)\alpha^*(z^{-1}) = w_\alpha(zX)$.

(Exposé XVIII, 1.4). We may therefore suppose that $1+XY$ is invertible. Then, successively,

$$\begin{aligned}
w_\alpha(X)w_{-\alpha}(Y) =& \exp(X)\exp(-X^{-1})\exp(X)\exp(Y) \\
& \exp(-Y^{-1})\exp(Y) \\
=& \exp(X)\exp(-X^{-1})\exp\left(\frac{Y}{1+XY}\right)\alpha^*(1+XY) \\
& \exp\left(\frac{X}{1+XY}\right)\exp(-Y^{-1})\exp(Y) \\
=& \exp(X)\exp\left(\frac{-X^{-1}}{1+XY}\right)\alpha^*(1+XY)\exp\left(\frac{-Y^{-1}}{1+XY}\right)\exp(Y) \\
=& \exp(-X^{-2}Y^{-1})\alpha^*\left(\frac{XY}{1+XY}\right)\exp(X+Y^{-1})\alpha^*(1+XY) \\
& \cdot\exp\left(\frac{-Y^{-1}}{1+XY}\right)\exp(Y) \\
=& \exp(-X^{-2}Y^{-1})\alpha^*(XY)\exp\left(\frac{Y^{-1}+X}{(1+XY)^2}\right) \\
& \exp\left(\frac{-Y^{-1}}{1+XY}\right)\exp(Y) \\
=& \alpha^*(XY)\exp(-Y)\exp(Y) = \alpha^*(XY).
\end{aligned}$$

Corollary 3.2. *Let $n \in \mathbb{Z}$, $n \neq 0$. For every $w \in G(S)$, the following conditions are equivalent:*

(i) *$w \in N^\times(S)$;*

(ii) *$\mathrm{int}(w) \circ n\alpha^* = -n\alpha^*$, where $(n\alpha^*)(z) = \alpha^*(z)^n$.*

One has (i)⇒(ii) by Theorem 3.1 (iii). Conversely, one may suppose that $N^\times$ has a section, and one is reduced to proving:

Lemma 3.3. *For $n \neq 0$, one has*

$$\mathrm{Centr}_G(n\alpha^*) = T.$$

Indeed, the image T' of $n\alpha^*$ is a subtorus of G. It follows from Exposé XIX, 2.8 that $\mathrm{Centr}_G(n\alpha^*)$ is a reductive subgroup of G containing T. On every fiber,

$$\mathrm{Centr}_{G_s}(n\alpha^*_s) \neq G_s,$$

and hence

$$\mathrm{Centr}_{G_s}(n\alpha^*_s) = T_s$$

by Exposé XIX, 1.6.3.[27] Therefore

$$\mathrm{Centr}_G(n\alpha^*) = T,$$

since these are smooth subgroups of G.

[27] The hypothesis $\mathrm{Centr}_{G_s}(n\alpha^*_s) \neq G_s$ implies

$$\dim \mathrm{Centr}_{G_s}(n\alpha^*_s) - \dim T_s < 2;$$

this difference is even by the cited result.

Remark 3.4. *The construction of w_α, and the fact that $w_\alpha(X)$ normalizes T, depend only on formula* (F). *In particular, if G is an S-group satisfying the conditions of 2.2, then* $\mathrm{Norm}_G(T)$ *differs from T on every fiber. Consequently, if G is an affine S-group with connected fibers satisfying the conditions of 2.2, then G is reductive of semisimple rank* 1. *Indeed, it is smooth in a neighborhood of the identity section, hence smooth, and the criterion of Exposé XIX, 1.11 applies.*

3.5. Before stating the following theorem, let us make a few remarks. As usual, we identify $\mathfrak{g}^{-\alpha}$ with $(\mathfrak{g}^\alpha)^{\otimes -1}$. Similarly, we identify

$$\mathrm{Hom}_{\mathscr{O}_S}(\mathfrak{g}^{-\alpha}, \mathfrak{g}^\alpha) \quad \text{with} \quad (\mathfrak{g}^\alpha)^{\otimes 2},$$

and hence

$$\mathrm{Isom}_{\mathscr{O}_S\text{-mod.}}\big(W(\mathfrak{g}^{-\alpha}), W(\mathfrak{g}^\alpha)\big) \simeq W\big((\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times.$$

If $w \in N^\times(S)$, then $\mathrm{Ad}(w)$ permutes $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ (3.1 (iii)), and therefore defines an isomorphism

$$a_\alpha(w)\colon \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

We thus identify it with a section

$$a_\alpha(w) \in \Gamma\big(S, (\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times.$$

This construction is compatible with base change and therefore defines a morphism

$$a_\alpha\colon N^\times \longrightarrow W\big((\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times$$

such that

$$a_\alpha(w)Y = \mathrm{Ad}(w)Y$$

for every $w \in N^\times(S')$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})^\times$, and $S' \to S$.

Theorem 3.6. (i) *For every $S' \to S$, $w \in N^\times(S')$, and $Y \in W(\mathfrak{g}^{-\alpha})(S')$, one has*

$$\mathrm{int}(w)\exp(Y) = \exp\big(a_\alpha(w)Y\big).$$

(ii) *One has*

$$a_\alpha(tw) = \alpha(t)a_\alpha(w), \qquad a_\alpha(wt) = \alpha(t)^{-1}a_\alpha(w).$$

(iii) *If one similarly defines*

$$a_{-\alpha}\colon N^\times \longrightarrow W\big((\mathfrak{g}^{-\alpha})^{\otimes 2}\big)^\times,$$

then

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

[28]

(iv) *For every $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \to S$, one has*

$$a_\alpha(w_\alpha(X)) = -X^2.$$

Proof. Assertion (i) is immediate from the characterization of the morphisms exp given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv). Let

$$X \in \Gamma(S', \mathfrak{g}^\alpha)^\times, \qquad Z \in \Gamma(S', \mathfrak{g}^\alpha).$$

[28] That is, $a_{-\alpha}(w)$ and $a_\alpha(w)$ are paired; cf. 2.6.1.

By definition,[29]

$$\begin{aligned} a_\alpha\big(w_\alpha(X)\big)^{-1}(Z) &= \operatorname{Ad}(w_\alpha(X))(Z) \\ &= \operatorname{Ad}(\exp(X))\operatorname{Ad}(\exp(-X^{-1}))\operatorname{Ad}(\exp(X))(Z). \end{aligned}$$

Applying formulas (2′) and (2) of Lemma 2.10, together with

$$H_{-\alpha} = -H_\alpha, \qquad \bar{\alpha}(H_\alpha) = 2$$

(loc. cit., (4) and (5)), and $\langle X, X^{-1}\rangle = 1$ (2.6), the expression on the right is, successively,

$$\begin{aligned} &\operatorname{Ad}(\exp(X))\operatorname{Ad}(\exp(-X^{-1}))(Z) \\ &\quad= \operatorname{Ad}(\exp(X))(Z + \langle X^{-1}, Z\rangle(H_\alpha - X^{-1})) \\ &\quad= Z + \langle X^{-1}, Z\rangle(H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &\quad= Z - \langle X^{-1}, Z\rangle X - \langle X^{-1}, Z\rangle X^{-1}. \end{aligned}$$

But

$$Z = \langle X^{-1}, Z\rangle X, \qquad \langle X^{-1}, Z\rangle X^{-1} = X^{-2}Z.$$

Thus

$$a_\alpha\big(w_\alpha(X)\big)^{-1} = -X^{-2},$$

and consequently

$$a_\alpha\big(w_\alpha(X)\big) = -X^2.$$

Corollary 3.7. *In particular, one has*

$$\operatorname{int}\big(w_\alpha(X)\big)\exp(X) = \exp(-X^{-1}).$$

Hence, by the definition of $w_\alpha(X)$,

$$w_\alpha(X)\exp(X)w_\alpha(X)^{-1} = \exp(-X)w_\alpha(X)\exp(-X),$$

and therefore, by an immediate calculation,

$$\big(w_\alpha(X)\exp(X)\big)^3 = e.$$

Corollary 3.8. *Let $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, and let $n \in \mathbb{Z}$, $n \neq 0$. Then $w_\alpha(X)$ is the unique section $w \in G(S)$ satisfying*
(i) $\operatorname{int}(w) \circ n\alpha^* = -n\alpha^*$*;*
(ii) $\big(w\exp(X)\big)^3 = e$.

We know that $w_\alpha(X)$ satisfies these conditions. Conversely, let $w \in G(S)$ satisfy (i) and (ii). By 3.2 and 3.1 (ii), there exists $t \in T(S)$ such that

$$w = w_\alpha(X)t.$$

Set $u = \exp(X)$. Then

$$\begin{aligned} wuw^{-1} &= w_\alpha(X)t\exp(X)t^{-1}w_\alpha(X)^{-1} \\ &= \exp(-\alpha(t)X^{-1}), \end{aligned}$$

[29]The editors supplied the details that follow.

whereas

$$\begin{aligned} u^{-1}wu^{-1} &= \exp(-X)w_\alpha(X)t\exp(-X) \\ &= \exp(-X)w_\alpha(X)\exp(-X)\exp(X-\alpha(t)X)t \\ &= \exp(-X^{-1})\exp(X-\alpha(t)X)t \\ &= \exp(-X^{-1})t\exp(H). \end{aligned}$$

Now

$$(wu)^3 = e \quad\Longleftrightarrow\quad wuw^{-1} = u^{-1}wu^{-1}.$$

Comparing the two decompositions of this element in

$$U_{-\alpha}\cdot T\cdot U_\alpha$$

gives $t = e$.

Remark 3.9. We may summarize a number of the results of this section by the following diagram of principal homogeneous bundles (on the left):

$$\begin{array}{ccccc} \mathrm{W}(\mathfrak{g}^\alpha)^\times & \xrightarrow{\quad w_\alpha \quad} & N^\times & \xrightarrow{\quad a_\alpha \quad} & \mathrm{W}\big((\mathfrak{g}^\alpha)^{\otimes 2}\big)^\times \\ \mathbb{G}_{m,S} & \xrightarrow{\quad \alpha^* \quad} & T & \xrightarrow{\quad \alpha \quad} & \mathbb{G}_{m,S} \end{array}$$

Notice that a_α is faithfully flat (since α is), and that w_α is a monomorphism if and only if α^* is a monomorphism. We leave to the reader the task of writing down the corresponding diagrams for right principal-bundle structures, analogous diagrams for the root $-\alpha$, and of studying the relations among these various diagrams.

Lemma 3.10. *Let S be a scheme, and let $q > 0$ be an integer such that $x \mapsto x^q$ defines an endomorphism of $\mathbb{G}_{a,S}$. Let (G,T,α) and (G',T',α') be two elementary S-systems, and let $f\colon G \to G'$ be a morphism of S-groups. Let*

$$h\colon (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\ \sim\ } \mathfrak{g}'^{\,\alpha'}$$

be an isomorphism of $\mathcal{O}_S$-modules, and let

$$h^\vee\colon (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\ \sim\ } \mathfrak{g}'^{-\alpha'}$$

be the contragredient isomorphism. Suppose that, for every $S' \to S$ and every $X \in W(\mathfrak{g}^\alpha)(S')$,

$$f\big(\exp(X)\big) = \exp\big(h(X^q)\big).$$

Then the following conditions are equivalent:

(i) $f\big(\alpha^*(z)\big) = \alpha'^*(z)^q$.

(ii) $f\big(w_\alpha(Z)\big) = w_{\alpha'}\big(h(Z^q)\big)$.

(iii) $f\big(\exp(Y)\big) = \exp\big(h^\vee(Y^q)\big)$.

Each condition is to be read as holding for every $S' \to S$ and every

$$z \in \mathbb{G}_m(S'), \qquad Z \in W(\mathfrak{g}^\alpha)^\times(S'), \qquad Y \in W(\mathfrak{g}^{-\alpha})(S'),$$

respectively. Indeed, (i) implies (ii) by 3.8, (ii) implies (iii) by 3.7, and (iii) implies (i) by 2.7.

Proposition 3.11. *Let S be a scheme, and let $q \in \mathbb{Z}$, $q > 0$, be such that $x \mapsto x^q$ defines an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S-systems, and let $f\colon G \to G'$ be a morphism of S-groups. The following conditions on f are equivalent:*

(i) *The restriction of f to T factors through a morphism $f_T\colon T \to T'$ making the diagram*

$$\begin{tikzcd} \mathbb{G}_{m,S} \arrow[r, "\alpha^*"] \arrow[d, "q"'] & T \arrow[r, "\alpha"] \arrow[d, "f_T"'] & \mathbb{G}_{m,S} \arrow[d, "q"'] \\ \mathbb{G}_{m,S} \arrow[r, "\alpha'^*"] & T' \arrow[r, "\alpha'"] & \mathbb{G}_{m,S} \end{tikzcd}$$

commutative.

(ii) *There exists a unique isomorphism of $\mathcal{O}_S$-modules*

$$h\colon (\mathfrak{g}^{\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

such that

$$f\big(\exp(X)\big) = \exp\big(h(X^q)\big), \qquad f\big(\exp(Y)\big) = \exp\big(h^{\vee}(Y^q)\big)$$

for every $S' \to S$, $X \in W(\mathfrak{g}^{\alpha})(S')$, and $Y \in W(\mathfrak{g}^{-\alpha})(S')$. Consequently, f also satisfies the equivalent conditions of 3.10.

We first prove that (ii) implies (i). By 3.10, condition (ii) gives

$$f \circ \alpha^* = q\alpha'^*;$$

hence, by 3.3, $f|_T$ factors through T'. It remains to prove

$$\alpha'(f(t)) = \alpha(t)^q,$$

which follows at once from the fact that f induces a morphism of groups

$$T \cdot U_{\alpha} \longrightarrow T' \cdot U_{\alpha'}.$$

We now prove that (i) implies (ii). Let

$$X \in \Gamma(S, \mathfrak{g}^{\alpha}), \qquad Y \in \Gamma(S, \mathfrak{g}^{-\alpha}),$$

and set

$$p_+(x) = f\big(\exp(xX)\big), \qquad p_-(y) = f\big(\exp(yY)\big).$$

These are morphisms of groups

$$p_+, p_-\colon \mathbb{G}_{a,S} \longrightarrow G.$$

Now

$$\begin{aligned} \operatorname{int}\big(\alpha'^*(z)\big)^q\big(p_+(x)\big) &= \operatorname{int}\big(f_T(\alpha^*(z))\big)\big(f(\exp(xX))\big) \\ &= f\Big(\operatorname{int}\big(\alpha^*(z)\big)\big(\exp(xX)\big)\Big) \\ &= f\big(\exp(z^2xX)\big) = p_+(z^2x). \end{aligned}$$

Applying Lemma 1.2 with $Q = \alpha'^*(\mathbb{G}_{m,S})$, we obtain a section

$$X' \in \Gamma(S, \mathfrak{g}'^{\alpha'})$$

such that

$$f\big(\exp(xX)\big) = p_+(x) = \exp(x^q X').$$

Similarly, there exists a section

$$Y' \in \Gamma(S, \mathfrak{g}'^{-\alpha'})$$

such that

$$f\big(\exp(yY)\big) = \exp(y^q Y').$$

Since f is a group morphism, it respects formula (F); hence

$$X^q Y^q = (XY)^q = X'Y'.$$

It follows readily that $X^q \mapsto X'$ and $Y^q \mapsto Y'$ define the announced isomorphisms h and $h^\vee$.

Proposition 3.12. *Let (G, T, α) be an elementary S-system and let $w \in N^\times(S)$. Put*

$$\Omega_0 = \Omega \cap \operatorname{int}(w^{-1})(\Omega).$$

Let d be the function on Ω defined by

$$d\big(\exp(Y) \cdot t \cdot \exp(X)\big) = \alpha(t)^{-1} + XY.$$

Then $\Omega_0 = \Omega_d$, and, for $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$, one has the following formula, where $z = d(\exp(Y) \cdot t \cdot \exp(X))$:

$$\operatorname{int}(w)\big(\exp(Y) \cdot t \cdot \exp(X)\big) = \exp\big(z^{-1} a_\alpha(w)^{-1} X\big) \cdot t\alpha^*(z) \cdot \exp\big(z^{-1} a_\alpha(w) Y\big). \qquad (\star)$$

Moreover,

$$d \circ \operatorname{int}(w) = d^{-1}.$$

Indeed, one immediately has[30]

$$\begin{aligned}\operatorname{int}(w)\big(\exp(Y) \cdot t \cdot \exp(X)\big) &= \exp\big(a_\alpha(w)Y\big) \cdot t\alpha^*\big(\alpha(t)^{-1}\big) \cdot \exp\big(a_\alpha(w)^{-1}X\big) \\ &= \exp\big(a_\alpha(w)Y\big) \cdot \exp\big(\alpha(t)a_\alpha(w)^{-1}X\big) \cdot t\alpha^*\big(\alpha(t)^{-1}\big).\end{aligned}$$

By 2.1, this is a section of Ω if and only if $1 + \alpha(t)XY$ is invertible, which proves $\Omega_0 = \Omega_d$. Applying formula (F) of the same paragraph then gives $(\star)$ by an immediate computation. Finally, $(\star)$ gives

$$\begin{aligned}\big(d \circ \operatorname{int}(w)\big)\big(\exp(Y) \cdot t \cdot \exp(X)\big) &= \alpha\big(t\alpha^*(z)\big)^{-1} + z^{-2}XY \\ &= z^{-2}\big(\alpha(t)^{-1} + XY\big) = z^{-1},\end{aligned}$$

which proves the last assertion.

N.B. The function d is independent of the choice of w.

[30] The editors corrected the original by interchanging $a_\alpha(w)$ and $a_\alpha(w)^{-1}$, and supplied the details of the proof of $d \circ \operatorname{int}(w) = d^{-1}$.

4 The Isomorphism Theorem

Theorem 4.1. *Let S be a scheme and let $q \in \mathbb{Z}$, $q > 0$, be such that $x \mapsto x^q$ is an endomorphism of $\mathbb{G}_{a,S}$. Let (G, T, α) and (G', T', α') be two elementary S-systems. Let*

$$h\colon (\mathfrak{g}^{\alpha})^{\otimes q} \xrightarrow{\ \sim\ } \mathfrak{g}'^{\alpha'}, \qquad h^{\vee}\colon (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\ \sim\ } \mathfrak{g}'^{-\alpha'}$$

be mutually contragredient isomorphisms. Let $f_T\colon T \to T'$ be a morphism of S-groups making the following diagram commute:

$$\begin{tikzcd}
\mathbb{G}_{m,S} \arrow[r, "q"] \arrow[d, "\alpha^*"'] & \mathbb{G}_{m,S} \arrow[d, "\alpha'^*"] \\
T \arrow[r, "f_T"] \arrow[d, "\alpha"'] & T' \arrow[d, "\alpha'"] \\
\mathbb{G}_{m,S} \arrow[r, "q"] & \mathbb{G}_{m,S}
\end{tikzcd}$$

There exists a unique morphism of S-groups $f\colon G \to G'$ extending f_T and satisfying

$$f\big(\exp(X)\big) = \exp\big(h(X^q)\big)$$

for every $S' \to S$ and $X \in W(\mathfrak{g}^{\alpha})(S')$. Moreover, this morphism also satisfies

$$f\big(\exp(Y)\big) = \exp\big(h^{\vee}(Y^q)\big), \qquad f\big(w_{\alpha}(Z)\big) = w_{\alpha'}\big(h(Z^q)\big)$$

for every $S' \to S$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})$, and $Z \in \Gamma(S', \mathfrak{g}^{\alpha})^{\times}$.

If $f\colon G \to G'$ extends f_T, then

$$f \circ \alpha^* = (\alpha'^*)^q.$$

If, in addition, f satisfies the second displayed condition in the statement, then it satisfies the other two conditions as well by 3.10. It follows that f is determined on Ω by

$$f\big(\exp(Y)t\exp(X)\big) = \exp\big(h^{\vee}(Y^q)\big)\, f_T(t)\, \exp\big(h(X^q)\big).$$

Since Ω is schematically dense in G, this already proves the uniqueness of f. To prove existence, it is enough, by Exposé XVIII, 2.3, to show that the preceding formula defines a "generically multiplicative" morphism from Ω to G'. By 2.4, this amounts to checking

$$\alpha' \circ f = \alpha^q,$$

which follows from the fact that f extends f_T.

Scholium 4.2. Theorem 4.1 may also be interpreted as follows. Let $\mathcal{E}$ be the category of elementary S-systems, and let $\mathcal{D}$ be the category of tuples

$$\big(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\ L\big),$$

where T is a torus, α and α^* are group morphisms such that $\alpha \circ \alpha^* = 2$, and L is an invertible $\mathcal{O}_S$-module. The reader may specify the morphisms in the two categories under consideration. Define a functor

$$\mathcal{E} \longrightarrow \mathcal{D},$$

$$(G, T, \alpha) \longmapsto \big(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\ \mathfrak{g}^{\alpha}\big).$$

The preceding theorem says that this functor is fully faithful. In fact, it is an equivalence of categories, as will be seen in the next section. We already have:

Corollary 4.3. *If $q = 1$ and f_T is an isomorphism, then f is an isomorphism.*

Corollary 4.4. *If $q = 1$ and f_T is faithfully flat with kernel Q* (cf. Exposé IX, 2.7)*, then f is faithfully flat and quasi-compact with kernel Q, and hence identifies G' with G/Q.*

Indeed, if f_T is faithfully flat with kernel Q, then

$$Q = \mathrm{Ker}(f_T) \subset \mathrm{Ker}(\alpha' \circ f_T) = \mathrm{Ker}(\alpha).$$

Introduce the elementary S-system $(G/Q, T/Q, \alpha/Q)$ of 2.13. By 2.14, we are reduced to proving that f/Q induces an isomorphism from G/Q onto G', which follows immediately from 4.3.

5 Examples of Elementary Systems; Applications

5.1. Let S be a scheme and L an invertible $\mathscr{O}_S$-module. Consider the group GL_L over S defined by

$$\mathrm{GL}_L(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; \begin{array}{ll} a, d \in \mathbb{G}_a(S'), & b \in W(L)(S'), \\ c \in W(L^{-1})(S'), & ad - bc \in \mathbb{G}_m(S') \end{array} \right\},$$

equipped with the usual matrix-multiplication law. It is locally isomorphic to $\mathrm{GL}_{2,S}$. It is therefore an affine smooth S-group scheme with connected fibers.

Remark. Let L' and L'' be invertible sheaves on S such that

$$L = L' \otimes L''^{-1}.$$

Then there is an isomorphism of S-groups

$$\mathrm{GL}_L \xrightarrow{\sim} \mathrm{GL}(L' \oplus L'')$$

defined as follows: if x (respectively y) is a section of L' (respectively L'') on an open subset V of S, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

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5.2. Denote by SL_L the closed subgroup of GL_L defined by $ad - bc = 1$. It too is an affine smooth S-group scheme with connected fibers; under the preceding isomorphism it is isomorphic to $\mathrm{SL}(L' \oplus L'')$.

Likewise, consider the morphism

$$\mathbb{G}_{m,S} \longrightarrow \mathrm{GL}_L, \qquad z \longmapsto \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}.$$

It is a central monomorphism. Passage to the quotient gives a smooth affine S-group P_L with connected fibers (cf. Exposé VIII, 5.7). By passing to the quotient in the isomorphism of the preceding remark, one sees that P_L identifies with the automorphism group of the projective bundle $\mathbb{P}(L' \oplus L'')$ (cf. EGA II, 4.2.7). Write i and p for the canonical morphisms

$$\mathrm{SL}_L \xrightarrow{i} \mathrm{GL}_L \xrightarrow{p} \mathrm{P}_L.$$

Here i is a closed immersion, while p is faithfully flat and affine.

[31]The editors corrected $L'' \otimes L'^{-1}$ to $L' \otimes L''^{-1}$ and expanded the sentence that follows.

5.3. Consider the group morphisms

$$t_G\colon \mathbb{G}^2_{m,S} \longrightarrow \mathrm{GL}_L, \quad t_G(z,z') = \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix},$$

$$t_S\colon \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_L, \qquad t_S(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix},$$

$$t_P\colon \mathbb{G}_{m,S} \longrightarrow \mathrm{P}_L, \qquad t_P(z) = p\big(t_G(z,1)\big).$$

They are group monomorphisms and define in each group a split torus of relative codimension 2. For $s \in S$, let

$$X \in \Gamma(s, L \otimes s)^{\times}.$$

The section $\left(\begin{smallmatrix} 0 & X \\ -X^{-1} & 0 \end{smallmatrix}\right)$ of $\mathrm{GL}_{L,s}$ normalizes $t_G(\mathbb{G}^2_{m,s})$ and does not centralize it. Exposé XIX, 1.6 therefore shows that GL_L is reductive of semisimple rank 1, with maximal torus $t_G(\mathbb{G}^2_{m,S})$. The same argument shows that SL_L and P_L are reductive of semisimple rank 1, with respective maximal tori $t_S(\mathbb{G}_{m,S})$ and $t_P(\mathbb{G}_{m,S})$.

5.4. The Lie algebras of these groups, and the adjoint actions of the chosen maximal tori, are determined in the usual way. For GL_L, Exposé II, 4.8 immediately gives

$$\mathrm{Lie}(\mathrm{GL}_L/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a, d \in \mathcal{O}_S,\ b \in L,\ c \in L^{-1} \right\},$$

with the usual bracket, and

$$\mathrm{Ad}\big(t_G(z,z')\big) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & zz'^{-1}b \\ z'z^{-1}c & d \end{pmatrix}.$$

Put $\mathfrak{g} = \mathrm{Lie}(\mathrm{GL}_L/S)$, and define

$$\alpha_G\colon t_G(\mathbb{G}^2_{m,S}) \longrightarrow \mathbb{G}_{m,S}, \qquad \alpha_G\big(t_G(z,z')\big) = zz'^{-1}.$$

The preceding identity shows that α_G is a root of GL_L relative to $t_G(\mathbb{G}^2_{m,S})$, and that

$$u\colon L \longrightarrow \mathfrak{g}, \quad X \longmapsto \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \qquad u^-\colon L^{-1} \longrightarrow \mathfrak{g}, \quad Y \longmapsto \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix}$$

identify L with $\mathfrak{g}^{\alpha_G}$ and L^{-1} with $\mathfrak{g}^{-\alpha_G}$, respectively. Thus

$$(\mathrm{GL}_L, t_G(\mathbb{G}^2_{m,S}), \alpha_G)$$

is an elementary S-system.

Similarly, setting

$$\alpha_S(t_S(z)) = z^2, \qquad \alpha_P(t_P(z)) = z,$$

one finds elementary S-systems

$$(\mathrm{SL}_L, t_S(\mathbb{G}_{m,S}), \alpha_S), \qquad (\mathrm{P}_L, t_P(\mathbb{G}_{m,S}), \alpha_P),$$

and corresponding identifications of L and L^{-1} with the root summands of their Lie algebras.

5.5. Set
$$\exp\begin{pmatrix}0 & X\\ 0 & 0\end{pmatrix}=\begin{pmatrix}1 & X\\ 0 & 1\end{pmatrix}.$$
This defines a morphism
$$W(\mathfrak{g}^{\alpha_G})\longrightarrow \mathrm{GL}_L$$
whose tangent map is the canonical inclusion; it is therefore the unique morphism of this kind by 1.5. Likewise, set
$$\exp\begin{pmatrix}0 & 0\\ Y & 0\end{pmatrix}=\begin{pmatrix}1 & 0\\ Y & 1\end{pmatrix}.$$
Direct calculation of formula (F) gives
$$\left\langle\begin{pmatrix}0 & X\\ 0 & 0\end{pmatrix},\begin{pmatrix}0 & 0\\ Y & 0\end{pmatrix}\right\rangle = XY,\qquad \alpha^*_G(z)=\begin{pmatrix}z & 0\\ 0 & z^{-1}\end{pmatrix}=t_G(z,z^{-1}).$$

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The open subset $N^\times = N^\times_G$ defined before 3.1 is
$$N^\times_G(S')=\left\{\begin{pmatrix}0 & P\\ Q & 0\end{pmatrix}\ \middle|\ P\in W(\mathfrak{g}^{\alpha})^\times(S'),\ Q\in W(\mathfrak{g}^{-\alpha})^\times(S')\right\}.$$
For $X\in W(\mathfrak{g}^{\alpha})^\times(S')$,
$$w_{\alpha_G}(X)=\begin{pmatrix}0 & X\\ -X^{-1} & 0\end{pmatrix}.$$
Moreover, if
$$w=\begin{pmatrix}0 & P\\ Q & 0\end{pmatrix}\in N^\times_G(S'),$$
then
$$a_{\alpha_G}(w)=PQ^{-1}\in W\big((\mathfrak{g}^{\alpha})^{\otimes 2}\big)^\times(S');$$
that is, for $Y\in W(\mathfrak{g}^{-\alpha})^\times(S')$,
$$a_{\alpha_G}(w)(Y)=PQ^{-1}Y\in W(\mathfrak{g}^{\alpha})^\times(S').$$

5.6. We leave the analogous calculations for SL_L and P_L to the reader. One obtains the same duality formula and the coroots
$$\alpha^*_S(z)=t_S(z),\qquad \alpha^*_P(z)=t_P(z^2).$$
Let p_T be the morphism induced by $p\colon \mathrm{GL}_L\to\mathrm{P}_L$ on $t_S(\mathbb{G}_{m,S})$, namely
$$p_T(t_S(z))=t_P(z^2).$$
Thus the following diagram is commutative: [33]

[32]The editors expanded the calculations that follow.
[33]Here t_S and t_P are isomorphisms.

$$\begin{tikzcd}
& & \mathbb{G}_{m,S} \arrow[dll, "\mathrm{id}"'] \arrow[dl, "\alpha_S^*"'] \arrow[dr, "\alpha_P^*"] \arrow[drr, "2"] & & \\
\mathbb{G}_{m,S} \arrow[r, "t_S"] \arrow[drr, "2"'] & t_S(\mathbb{G}_{m,S}) \arrow[rr, "p_T"] \arrow[dr, "\alpha_S"] & & t_P(\mathbb{G}_{m,S}) \arrow[dl, "\alpha_P"'] & \mathbb{G}_{m,S} \arrow[l, "t_P"'] \arrow[dll, "\mathrm{id}"] \\
& & \mathbb{G}_{m,S} & &
\end{tikzcd}$$

Its central portion is the commutative diagram of 4.1 [34] associated with the canonical morphism $p \circ i \colon \mathrm{SL}_L \to \mathrm{P}_L$, which induces a morphism between the preceding elementary S-systems.

5.7. Let (G, T, α) now be an arbitrary elementary S-system. Consider the commutative diagram

$$\begin{tikzcd}
& \mathbb{G}_{m,S} \arrow[dl, "\mathrm{id}"'] \arrow[d, "\alpha^*"] \arrow[dr, "2"] & \\
\mathbb{G}_{m,S} \arrow[r, "\alpha^*"] \arrow[dr, "2"'] & T \arrow[r, "\alpha"] \arrow[d, "\alpha"] & \mathbb{G}_{m,S} \arrow[dl, "\mathrm{id}"] \\
& \mathbb{G}_{m,S} &
\end{tikzcd}$$

Combining this diagram with that of 5.6 gives a commutative diagram

[34] With $q = 1$.

$$\begin{tikzcd}
 & \mathbb{G}_{m,S} \arrow[dl, "\alpha_S^*"'] \arrow[d, "\alpha^*"] \arrow[dr, "\alpha_P^*"] & \\
t_S(\mathbb{G}_{m,S}) \arrow[r, "\alpha^* \circ t_S^{-1}"] \arrow[dr, "\alpha_S"'] & T \arrow[r, "t_P \circ \alpha"] \arrow[d, "\alpha"] & t_P(\mathbb{G}_{m,S}) \arrow[dl, "\alpha_P"] \\
 & \mathbb{G}_{m,S} &
\end{tikzcd}$$

Using 4.1, one obtains:

Proposition 5.8. *Let S be a scheme and let (G, T, α) be an elementary S-system. Put $L = \mathfrak{g}^\alpha$, so that $L^{-1} = \mathfrak{g}^{-\alpha}$.*

(i) *There exists a unique group morphism $f \colon \mathrm{SL}_L \to G$ satisfying the following equivalent conditions:*

$$\text{(a)}\quad f\begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), \qquad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X);$$

$$\text{(b)}\quad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), \qquad f\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y);$$

$$\text{(c)}\quad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), \qquad f\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_\alpha(X).$$

(ii) *There exists a unique group morphism $g \colon G \to \mathrm{P}_L$ satisfying*

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \qquad g(\exp(X)) = p\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

Moreover,

$$g(\exp(Y)) = p\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \qquad g(w_\alpha(X)) = p\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism g is faithfully flat and quasi-compact, with

$$\mathrm{Ker}(g) = \mathrm{Ker}(\alpha) = \mathrm{Centr}(G),$$

and $g \circ f$ is the canonical morphism $\mathrm{SL}_L \to \mathrm{P}_L$.

Notice that condition (b) in (i) gives an explicit description of the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$.

Corollary 5.9. *Let (G, T, α) be an elementary S-system. The subgroups*

$$T \cdot U_\alpha, \qquad T \cdot U_{-\alpha}, \qquad U_\alpha, \qquad U_{-\alpha}$$

are closed.

Since U_α is a closed subgroup scheme of $T \cdot U_\alpha$, it suffices to prove the assertion for the latter. By Noether's theorem (Exposé IV, 5.3.1 and 6.4.1), it is enough to show that

$$(T \cdot U_\alpha)/\,\mathrm{Ker}(\alpha)$$

is a closed subgroup of $G/\operatorname{Ker}(\alpha)$. By 5.8, this reduces to the evident fact that the subgroup of P_L defined by $c = 0$ is closed; one may equivalently work in GL_L, by another application of Noether's theorem. Consequently, the exponential morphisms of Theorem 1.5(i) are closed immersions.

N.B. The corollary also follows from the fact that $T \cdot U_\alpha$ and $T \cdot U_{-\alpha}$ are "Borel subgroups" of G (cf. Exposé XII, 7.10).

5.10. Let L be an invertible $\mathscr{O}_S$-module, and let

$$\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S}$$

be a diagram of groups, where T is a torus,[35] such that $\alpha \circ \alpha^* = 2$. Let R be the maximal torus of $\operatorname{Ker}(\alpha)$, and put $K = \alpha^{*-1}(R)$. Then K is a subgroup of multiplicative type of $\mathbb{G}_{m,S}$; since $\alpha \circ \alpha^* = 2$, it is even a subgroup of $\mu_{2,S}$. In particular,

$$K \longrightarrow \mathrm{SL}_L, \qquad z \longmapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. Hence there is a central group monomorphism

$$K \longrightarrow R \times \mathrm{SL}_L, \qquad z \longmapsto \left(\alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} \right).$$

Let

$$G = (R \times \mathrm{SL}_L)/K.$$

This is an affine smooth group over S, with connected fibers. The sequence

$$1 \longrightarrow K \longrightarrow R \times t_S(\mathbb{G}_{m,S}) \xrightarrow{u} T \longrightarrow 1, \qquad u(x, t_S(z)) = x\alpha^*(z),$$

is exact. The image of $R \times t_S(\mathbb{G}_{m,S})$ in G is therefore a torus T' isomorphic to T. If α' is the character of T' transported from α, one readily checks that (G, T', α') is an elementary S-system, that $\mathfrak{g}^{\alpha'}$ is isomorphic to L, and that α'^* is obtained from α^* under the isomorphism $T \xrightarrow{\sim} T'$. Thus the object

$$\left(\mathbb{G}_{m,S} \xrightarrow{\alpha'^*} T' \xrightarrow{\alpha'} \mathbb{G}_{m,S},\ \mathfrak{g}^{\alpha'} \right)$$

of the category $\mathcal{D}$ defined in 4.2 is isomorphic to

$$\left(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\ L \right).$$

We have therefore proved:

Theorem 5.11. *With the notation of 4.2, the functor*

$$(G, T, \alpha) \longmapsto \left(\mathbb{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbb{G}_{m,S},\ \mathfrak{g}^{\alpha} \right)$$

is an equivalence of categories between $\mathcal{E}$ and $\mathcal{D}$.

[35] This qualification is supplied by the editors.

6 Generators and Relations for an Elementary System

6.1. Let S be a scheme and let (G, T, α) be an elementary S-system. Let

$$X \in W(\mathfrak{g}^{\alpha})^{\times}(S), \qquad u = \exp(X).$$

As observed in 3.8, the element $w = w_\alpha(X)$ satisfies, in particular,

$$(wu)^3 = e.$$

[36] Let s_α denote the automorphism of T induced by $\mathrm{int}(w)$. By Theorem 3.1(iii), for every $S' \to S$ and every $t \in T(S')$,

$$s_\alpha(t) = \mathrm{int}(w)(t) = t\,\alpha^*\big(\alpha(t)^{-1}\big).$$

Theorem 6.2. *Let H be a sheaf of groups over S for the fppf topology. Let*

$$f_T \colon T \longrightarrow H, \qquad f_\alpha \colon U_\alpha \longrightarrow H$$

be group morphisms, and let $h \in H(S)$. There exists a (necessarily unique) *group morphism*

$$f \colon G \longrightarrow H$$

extending f_T and f_α and satisfying $f(w) = h$ if and only if the following conditions hold:

(i) *For every $S' \to S$, $t \in T(S')$, and $x \in U_\alpha(S')$,*

$$f_T(t) f_\alpha(x) f_T(t)^{-1} = f_\alpha(txt^{-1}) = f_\alpha(x^{\alpha(t)}). \tag{1}$$

Equivalently, f_T and f_α extend to a group morphism $T \cdot U_\alpha \to H$.

(ii) *For every $S' \to S$ and $t \in T(S')$,*

$$h f_T(t) h^{-1} = f_T(s_\alpha(t)) = f_T\Big(t\,\alpha^*\big(\alpha(t)^{-1}\big)\Big). \tag{2}$$

(iii) *The following two relations hold in $H(S)$:*

$$h^2 = f_T(\alpha^*(-1)), \tag{3}$$

$$\big(h f_\alpha(u)\big)^3 = e. \tag{4}$$

Proof. Write the laws on U_α and $U_{-\alpha}$ additively, while writing their vector-space structures multiplicatively. If f has the asserted properties, then for $y \in U_{-\alpha}(S')$ one necessarily has

$$f(y) = f(w^{-1}wyw^{-1}w) = h f_\alpha(w^{-1}yw)h^{-1}.$$

Define a morphism $f_{-\alpha} \colon U_{-\alpha} \to H$ by

$$f_{-\alpha}(y) = h f_\alpha(w^{-1}yw)h^{-1}. \tag{$*_1$}$$

It is a group morphism. On the big cell Ω, the morphism f is forced to satisfy

$$f(ytx) = f_{-\alpha}(y) f_T(t) f_\alpha(x).$$

This proves uniqueness. The conditions in the statement are manifestly necessary; let us prove that they are sufficient.

[36] The editors added the sentence that follows.

Relation (4) gives
$$hf_\alpha(u)h^{-1}h^2 = f_\alpha(-u)h^{-1}f_\alpha(-u).$$
By (3) and (1), $h^2 = f_T(\alpha^*(-1))$ commutes with $f_\alpha(-u)$, and hence
$$hf_\alpha(u)h^{-1} = f_\alpha(-u)hf_\alpha(-u).$$
By definition,
$$hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1}),$$
whereas 3.7 gives
$$wuw^{-1} = -\widetilde{u}, \tag{$*_2$}$$
with $\widetilde{u}$ the element paired with u. Therefore
$$f_{-\alpha}(-\widetilde{u}) = f_\alpha(-u)hf_\alpha(-u). \tag{$*_3$}$$

Let t be a section of T over a variable $S' \to S$. Apply $\mathrm{int}(f_T(t))$ to this identity. On the left one obtains[37]
$$\begin{aligned} f_T(t)f_{-\alpha}(-\widetilde{u})f_T(t)^{-1} &= f_T(t)hf_\alpha(u)h^{-1}f_T(t)^{-1} \\ &= h\big(h^{-1}f_T(t)h\big)f_\alpha(u)\big(h^{-1}f_T(t)^{-1}h\big)h^{-1} \\ &= hf_T(s_\alpha(t))f_\alpha(u)f_T(s_\alpha(t))^{-1}h^{-1} \\ &= hf_\alpha\big(\alpha(s_\alpha(t))u\big)h^{-1} \\ &= hf_\alpha\big(\alpha(t)^{-1}u\big)h^{-1} \\ &= f_{-\alpha}\big(-\alpha(t)^{-1}\widetilde{u}\big). \end{aligned}$$
Here we used (2), (1), $s_\alpha(t) = t\alpha^*(\alpha(t)^{-1})$, and $\alpha \circ \alpha^* = 2$.

The right-hand side of $(*_3)$, after conjugation, becomes
$$f_\alpha(-\alpha(t)u)\, f_T(t)hf_T(t)^{-1}h^{-1}\, hf_\alpha(-\alpha(t)u).$$
Since
$$hf_T(t)^{-1}h^{-1} = f_T(s_\alpha(t^{-1})) = f_T\big(t\,\alpha^*(\alpha(t))\big),$$
this equals
$$f_\alpha(-\alpha(t)u)\, f_T\big(\alpha^*(\alpha(t))\big)\, hf_\alpha(-\alpha(t)u).$$
Comparison gives
$$f_{-\alpha}\big(-\alpha(t)^{-1}\widetilde{u}\big) = f_\alpha(-\alpha(t)u)\, f_T\big(\alpha^*(\alpha(t))\big)\, hf_\alpha(-\alpha(t)u).$$
Since $\alpha\colon T \to \mathbb{G}_{m,S}$ is faithfully flat and H is a separated presheaf, $-\alpha(t)^{-1}$ may be replaced by an arbitrary section of $\mathbb{G}_{m,S}$. Thus:

Lemma 6.2.1. *For every $S' \to S$ and every $z \in \mathbb{G}_m(S')$,*
$$f_{-\alpha}(z\widetilde{u}) = f_\alpha(z^{-1}u)\, f_T\big(\alpha^*(-z^{-1})\big)\, hf_\alpha(z^{-1}u).$$

Let $x, y \in \mathbb{G}_a(S')$, where $S' \to S$, and suppose that y and $1+xy$ are invertible. Applying Lemma 6.2.1 first with $z = y$, one obtains[38]
$$f_\alpha(xu)f_{-\alpha}(y\widetilde{u}) = f_\alpha\big((x+y^{-1})u\big)\, f_T\big(\alpha^*(-y^{-1})\big)\, hf_\alpha(y^{-1}u).$$

[37] The editors corrected the calculation that follows.
[38] The editors expanded the calculations that follow.

Now $x + y^{-1} = y^{-1}(1 + xy)$. Applying the lemma with $z = y/(1 + xy)$ gives

$$f_\alpha\big((x + y^{-1})u\big) = f_{-\alpha}\Big(\frac{y}{1 + xy}\widetilde{u}\Big)\, f_\alpha\big(-(x + y^{-1})u\big) \\ \cdot h\, f_T\Big(\alpha^*\Big(-\frac{y}{1 + xy}\Big)\Big).$$

Substitution in the preceding identity yields

$$f_\alpha(xu)f_{-\alpha}(y\widetilde{u}) = f_{-\alpha}\Big(\frac{y}{1 + xy}\widetilde{u}\Big)\, f_\alpha\big(-(x + y^{-1})u\big)h^{-1} \\ \cdot f_T(\alpha^*(1 + xy)^{-1})hf_\alpha(y^{-1}u).$$

Since $h^{-1}f_T(t)h = f_T(s_\alpha(t))$ by (2) (recall that $s_\alpha^2 = \mathrm{id}$), and $s_\alpha \circ \alpha^* = -\alpha^*$, this is

$$f_{-\alpha}\Big(\frac{y}{1 + xy}\widetilde{u}\Big)\, f_\alpha\big(-(x + y^{-1})u\big)f_T\big(\alpha^*(1 + xy)\big)f_\alpha(y^{-1}u).$$

For $x' \in U_\alpha(S')$ and $z \in \mathbb{G}_m(S')$, one has

$$f_\alpha(x')f_T(\alpha^*(z)) = f_T(\alpha^*(z))f_\alpha(z^{-2}x').$$

Consequently,

$$\begin{aligned} f_\alpha(xu)f_{-\alpha}(y\widetilde{u}) &= f_{-\alpha}\Big(\frac{y}{1 + xy}\widetilde{u}\Big)\, f_T\big(\alpha^*(1 + xy)\big) \\ &\quad \cdot f_\alpha\bigg(\bigg[-\frac{y^{-1}(1 + xy)}{(1 + xy)^2} + y^{-1}\bigg]u\bigg) \\ &= f_{-\alpha}\Big(\frac{y}{1 + xy}\widetilde{u}\Big)\, f_T\big(\alpha^*(1 + xy)\big)f_\alpha\Big(\frac{x}{1 + xy}u\Big). \end{aligned}$$

We have proved:

Lemma 6.2.2. *Let $S' \to S$. If $a \in U_\alpha(S')$, $b \in U_{-\alpha}^\times(S')$, and $1 + ab \in \mathbb{G}_m(S')$, then*

$$f_\alpha(a)f_{-\alpha}(b) = f_{-\alpha}\Big(\frac{b}{1 + ab}\Big)\, f_T\big(\alpha^*(1 + ab)\big)f_\alpha\Big(\frac{a}{1 + ab}\Big).$$

By schematic density, this formula remains valid for $b \in U_{-\alpha}(S')$, with $1 + ab$ invertible. Define

$$f\colon \Omega \longrightarrow H, \qquad f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x).$$

It follows at once from Lemma 6.2.2, condition 6.2(i), and formula (F') of 2.4 that whenever $g, g' \in \Omega(S')$ and $gg' \in \Omega(S')$, one has

$$f(gg') = f(g)f(g').$$

By Exposé XVIII, 2.3(iii) and 2.4,[39] there is therefore a group morphism $G \to H$ extending f. Denote it again by f. It remains only to prove that $f(w_\alpha) = h$. Since

$$w_\alpha = u \cdot (-\widetilde{u}) \cdot u,$$

formula $(*_3)$ gives[40]

$$f(w_\alpha) = f_\alpha(u)f_{-\alpha}(-\widetilde{u})f_\alpha(u) = h.$$

This completes the proof of Theorem 6.2.

Remark 6.3. *These results will be completed in Exposé XXIII, 3.5.*

[39] Each geometric fiber of G is connected, for example by 1.1.

[40] The editors simplified the original argument by invoking $(*_3)$.

Exposé XXI. Root Data

This Exposé[41] collects, in the absence of a suitable reference,[42] known results on root data (that is, "abstract" root systems), most of which will be used below.

Notation. We denote by $\mathbb{Q}_+$ the set of positive (or zero) rational numbers; $\mathbb{Z} \cap \mathbb{Q}_+ = \mathbb{N}$. Let V be a $\mathbb{Q}$-vector space. If A (respectively B) is a subset of $\mathbb{Q}$ (respectively of V), we denote by $A \cdot B$ the image of $A \otimes B$ under the morphism $\mathbb{Q} \otimes V \to V$, in other words the set of linear combinations of elements of B with coefficients in A. We write $-B = \{-1\}B$. We denote by $E - F$ the set of elements of E that do not belong to F.

1 Generalities

1.1 Definitions and first properties

Definition 1.1.1. Let M and M^* be two free finite-type $\mathbb{Z}$-modules in duality. Write

$$V = M \otimes \mathbb{Q}, \qquad V^* = M^* \otimes \mathbb{Q};$$

these are two $\mathbb{Q}$-vector spaces in duality. We identify M (respectively M^*) with a subset of V (respectively V^*). The canonical bilinear form on $M^* \times M$ (respectively $V^* \times V$) is denoted $(\ ,\ )$.

Let R be a finite subset of M. Suppose given a map $\alpha \mapsto \alpha^*$ from R to M^*; the set of α^*, for $\alpha \in R$, is denoted R^*. To every $\alpha \in R$ associate the endomorphism s_α (respectively s^*_α) of M and V (respectively of M^* and V^*) given by

$$s_\alpha(x) = x - (\alpha^*, x)\alpha, \qquad s_\alpha = \mathrm{id} - \alpha^* \otimes \alpha, \tag{1}$$

$$s^*_\alpha(u) = u - (u, \alpha)\alpha^*, \qquad s^*_\alpha = \mathrm{id} - \alpha \otimes \alpha^*. \tag{1*}$$

The pair (R, R^*) (more precisely, the pair $(R, R \to M^*)$) is called a *root datum* in (M, M^*), and (M, M^*, R, R^*) is called a root datum, if the following axioms hold:

$$\text{(DR I)} \quad (\alpha^*, \alpha) = 2 \qquad \text{for every } \alpha \in R,$$

$$\text{(DR II)} \quad s_\alpha(R) \subset R, \qquad s^*_\alpha(R^*) \subset R^* \qquad \text{for every } \alpha \in R.$$

The set R is called the *root system* of the root datum $\mathcal{R} = (M, M^*, R, R^*)$. The elements of R (respectively R^*) are called the *roots* (respectively *coroots*) of the root datum.

Remark 1.1.2. Axiom (DR I) is equivalent to any one of the following properties:

$$s_\alpha s_\alpha = \mathrm{id}, \qquad s^*_\alpha s^*_\alpha = \mathrm{id}, \tag{2,2*}$$

$$s_\alpha(\alpha) = -\alpha, \qquad s^*_\alpha(\alpha^*) = -\alpha^*. \tag{3,3*}$$

[41] Editors' note: Version of 13 October 2024.

[42] Editors' note: For results on root systems (Sections 1–5), one may consult *Groupes et algèbres de Lie*, Chapter VI.

Remark 1.1.3. Axioms (DR I) and (DR II) imply

$$R = -R, \qquad R^* = -R^*, \qquad 0 \notin R, \qquad 0 \notin R^*.$$

Lemma 1.1.4. *The map $R \to R^*$ is a bijection. More generally, if $\alpha, \beta \in R$ and $(\alpha^*, x) = (\beta^*, x)$ for every $x \in R$, then $\alpha = \beta$.*

Proof. Indeed, one then has

$$s_\beta(\alpha) = \alpha - 2\beta, \qquad s_\alpha(\beta) = \beta - 2\alpha.$$

It follows at once that

$$s_\beta s_\alpha(\alpha) = 2\beta - \alpha = \alpha + 2(\beta - \alpha), \qquad s_\beta s_\alpha(\beta - \alpha) = s_\beta(\beta - \alpha) = \beta - \alpha.$$

Hence, by (DR II),

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha) \in R.$$

Since R is finite, $\beta - \alpha = 0$. □

Corollary 1.1.5. *The inverse map $R^* \to R$ defines a root datum*

$$\mathcal{R}^* = (M^*, M, R^*, R),$$

called the dual *of* $\mathcal{R}$.[43]

Definition 1.1.6. We denote by $\Gamma_0(R)$ the subgroup of M generated by R. We denote by $\mathcal{V}(R)$ the vector subspace of V generated by R, that is,

$$\mathcal{V}(R) = \Gamma_0(R) \otimes \mathbb{Q}.$$

Applying these definitions to $\mathcal{R}^*$, one likewise obtains $\Gamma_0(R^*)$ and $\mathcal{V}(R^*)$.

The *reductive rank* of $\mathcal{R}$ is the number

$$\operatorname{rgred}(\mathcal{R}) = \operatorname{rank}(M) = \dim(V) = \dim(V^*) = \operatorname{rank}(M^*) = \operatorname{rgred}(\mathcal{R}^*).$$

The *semisimple rank* of $\mathcal{R}$ is the number

$$\operatorname{rgss}(\mathcal{R}) = \operatorname{rank}(R) = \operatorname{rank}(\Gamma_0(R)) = \dim(\mathcal{V}(R)).$$

Thus $\operatorname{rgss}(\mathcal{R}) \le \operatorname{rgred}(\mathcal{R})$.

We shall see below that $\operatorname{rgss}(\mathcal{R}) = \operatorname{rgss}(\mathcal{R}^*)$, that is, that $\mathcal{V}(R)$ and $\mathcal{V}(R^*)$ have the same dimension.

Definition 1.1.7.

The root datum $\mathcal{R}$ is called *semisimple* (respectively *trivial*) if $\operatorname{rgss}(\mathcal{R}) = \operatorname{rgred}(\mathcal{R})$ (respectively $\operatorname{rgss}(\mathcal{R}) = 0$). Thus $\mathcal{R}$ is trivial if and only if R is empty. The trivial root datum of reductive rank zero is denoted

$$0 = (\{0\}, \{0\}, \varnothing, \varnothing).$$

[43] Editors' note: $(\alpha^*)^* = \alpha$.

Definition 1.1.8. We denote by $W(\mathcal{R})$ the group of transformations of M generated by the s_α, $\alpha \in R$. It is called the *Weyl group* of $\mathcal{R}$. We write

$$W^*(\mathcal{R}) = W(\mathcal{R}^*).$$

Then $W(\mathcal{R})$ acts on R, $\Gamma_0(R)$, $\mathcal{V}(R)$, M, and V. If $w \in W(\mathcal{R})$ and $x \in M$ (respectively $x \in V$), then

$$wx - x \in \Gamma_0(R) \quad \text{(respectively } wx - x \in \mathcal{V}(R));$$

this follows immediately from formula (1). Likewise for $W^*(\mathcal{R})$.

Lemma 1.1.9. *For every $\alpha \in R$, $x \in V$, and $u \in V^*$, one has*

$$(s^*_\alpha(u), s_\alpha(x)) = (u, x). \tag{4}$$

Proof. Taking the scalar product of (1) and (1*), the left-hand side is

$$(u, x) + (u, \alpha)(\alpha^*, x)\big((\alpha^*, \alpha) - 2\big) = (u, x).$$

□

Remark 1.1.10. If one assumes $0 \notin R$ and $0 \notin R^*$, then Lemma 1.1.9 is equivalent to (DR I).

Corollary 1.1.11. *Let $h \mapsto h^\vee$ denote the isomorphism from* $\mathrm{GL}(M)$ *onto* $\mathrm{GL}(M^*)$ *that associates to h its contragredient. Formula* (4) *may then also be written*

$$(s_\alpha)^\vee = s^*_\alpha. \tag{5}$$

Corollary 1.1.12. *The preceding isomorphism induces an isomorphism from $W(\mathcal{R})$ onto $W^*(\mathcal{R})$.*

Scholium 1.1.13. By the preceding result, we shall identify W and W^*, and regard W as a group of transformations of

$$R,\ R^*,\ M,\ M^*,\ \Gamma_0(R),\ \Gamma_0(R^*),\ V,\ V^*,\ \mathcal{V}(R),\ \mathcal{V}(R^*).$$

We shall write s_α for s^*_α.

1.2 The map p

Lemma 1.2.1. *Let $p : M \to M^*$ (respectively $V \to V^*$) be the linear map defined by*

$$p(x) = \sum_{u \in R^*} (u, x)u. \tag{6}$$

Write $\ell(x) = (p(x), x)$. Then:

$$\begin{aligned}
\ell(x) \geq 0, \qquad \ell(\alpha) > 0 \qquad & \textit{for } \alpha \in R, && (7)\\
\ell(wx) = \ell(x) \qquad & \textit{for } w \in W, && (8)\\
(p(x), y) = (p(y), x) \qquad & \textit{for } x, y \in V, && (9)\\
\ell(\alpha)\alpha^* = 2p(\alpha) \qquad & \textit{for } \alpha \in R. && (10)
\end{aligned}$$

Proof. The first three relations are immediate.[44] We prove the last one. By (1), for $u \in V^*$,

$$\begin{aligned}(u,\alpha)^2\alpha^* &= (u,\alpha)u - (u,\alpha)s_\alpha(u)\\ &= (u,\alpha)u + (u, s_\alpha(\alpha))s_\alpha(u)\\ &= (u,\alpha)u + (s_\alpha(u),\alpha)s_\alpha(u) \qquad (s_\alpha^2 = \mathrm{id}).\end{aligned}$$

Since s_α permutes R^* by (DR II), summing over $u \in R^*$ gives the result. □

Scholium 1.2.2. Relation (10) says that the map $\alpha \mapsto \ell(\alpha)\alpha^*$ is the restriction to R of a linear map from M to M^*. In particular,

$$(-\alpha)^* = -\alpha^*.$$

Corollary 1.2.3. *The map p induces an isomorphism from $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$.*

Proof. The map p sends $\mathcal{V}(R)$ onto $\mathcal{V}(R^*)$. Hence

$$\dim \mathcal{V}(R) \geq \dim \mathcal{V}(R^*).$$

Applying this inequality to the dual root datum gives

$$\dim \mathcal{V}(R) = \dim \mathcal{V}(R^*),$$

so the surjective map p is also bijective. □

Corollary 1.2.4. *One has* $\dim \mathcal{V}(R) = \dim \mathcal{V}(R^*)$, *and therefore*

$$\mathrm{rgss}(\mathcal{R}) = \mathrm{rgss}(\mathcal{R}^*).$$

Corollary 1.2.5. *The bilinear form (u, x) is nondegenerate on $\mathcal{V}(R^*)\times\mathcal{V}(R)$, and thus puts these $\mathbb{Q}$-vector spaces in duality.*

Proof. If $(u, x) = 0$ for every $u \in R^*$, then $p(x) = 0$. □

Corollary 1.2.6. *The symmetric bilinear form $(p(x), y)$ is positive and nondegenerate on $\mathcal{V}(R)$.*

Corollary 1.2.7. *The group W acts faithfully on R (and hence on the other sets listed in Scholium 1.1.13).*

Proof. Let $u \in V^*$, and suppose $w \in W$ satisfies $w(\alpha) = \alpha$ for every $\alpha \in R$. We prove $w(u) = u$. For every $\alpha \in R$,

$$(w(u) - u, \alpha) = (w(u), \alpha) - (u, \alpha) = (u, w^{-1}(\alpha)) - (u, \alpha) = 0.$$

But $w(u) - u \in \mathcal{V}(R^*)$. Since it is orthogonal to every root, it vanishes by Corollary 1.2.5. □

Corollary 1.2.8. *The group W is finite.*

Proposition 1.2.9. *The operations of W respect the correspondence between roots and coroots. In other words, for $\alpha \in R$ and $w \in W$,*

$$w(\alpha^*) = w(\alpha)^*.$$

[44] Editors' note: Formula (8) follows from (4), (2,2*), and (DR II).

Proof. It suffices to verify this for $w = s_\beta$, $\beta \in R$, that is, to verify

$$s_\beta(\alpha^*) = s_\beta(\alpha)^*.$$

Now $\ell(s_\beta(\alpha))s_\beta(\alpha)^*/2$ equals

$$\begin{aligned} p(s_\beta(\alpha)) &= \sum_{u\in R^*} (u, s_\beta(\alpha))u \\ &= \sum_{u\in R^*} (s_\beta(u), \alpha)u \\ &= \sum_{u\in R^*} (u, \alpha)s_\beta(u) = s_\beta(p(\alpha)). \end{aligned}$$

Since $\ell(s_\beta(\alpha)) = \ell(\alpha)$, we obtain

$$s_\beta(\alpha)^* = s_\beta\left(\frac{2p(\alpha)}{\ell(\alpha)}\right) = s_\beta(\alpha^*).$$

□

Corollary 1.2.10. *If $\alpha \in R$ and $w \in W$, then*

$$ws_\alpha w^{-1} = s_{w(\alpha)}.$$

Proof. For $x \in V$,

$$ws_\alpha w^{-1}(x) = x - (\alpha^*, w^{-1}(x))w(\alpha) = x - (w(\alpha^*), x)w(\alpha).$$

By Proposition 1.2.9, the last expression is

$$x - (w(\alpha)^*, x)w(\alpha) = s_{w(\alpha)}(x).$$

□

Corollary 1.2.11.
Let $R' \subset R$. The quadruple (M, M^*, R', R'^*) is a root datum if and only if $\alpha, \beta \in R'$ implies $s_\alpha(\beta) \in R'$.

2 Relations between two roots

2.1 Proportional roots

Proposition 2.1.1. *Let α and β be two roots. The following conditions are equivalent:*

(i) there exists $k \in \mathbb{Q}$ such that $\alpha = k\beta$;

(ii) $s_\alpha = s_\beta$.

Under these conditions, $\alpha^* = k^{-1}\beta^*$, and k is one of

$$1,\ -1,\ 2,\ -2,\ \tfrac{1}{2},\ -\tfrac{1}{2}.$$

Proof. Assume first (i). Then

$$\alpha^* = \ell(\alpha)^{-1} 2p(\alpha) = k^{-2}\ell(\beta)^{-1} 2k\, p(\beta) = k^{-1}\beta^*.$$

This immediately gives $s_\alpha = s_\beta$. Conversely, if $s_\alpha = s_\beta$, then

$$\alpha = s_\alpha(-\alpha) = s_\beta(-\alpha) = (\beta^*, \alpha)\beta - \alpha,$$

so

$$2\alpha = (\beta^*, \alpha)\beta, \qquad (\beta^*, \alpha) \in \mathbb{Z};$$

thus (ii) implies (i). Finally, if $\alpha = k\beta$, then $\alpha^* = k^{-1}\beta^*$, whence

$$(\alpha^*, \beta) = 2k^{-1}, \qquad (\beta^*, \alpha) = 2k.$$

Thus $2k$ and $2k^{-1}$ are integers, which gives the stated list. □

Application 2.1.2. Root data $\mathcal{R}$ with $\mathrm{rgss}(\mathcal{R}) = 1$ are of one of the following two types:

(i) Type A_1: there exists $\alpha \in M$ such that the roots are α and $-\alpha$. The coroots are then α^* and $-\alpha^*$.

(ii) Type A_1': there exists $\alpha \in M$ such that the roots are $\alpha, -\alpha, 2\alpha, -2\alpha$. The coroots are then $\alpha^*, -\alpha^*, \alpha^*/2, -\alpha^*/2$.

Definition 2.1.3. A root $\alpha \in R$ is called *indivisible* if $\alpha/2 \notin R$. The root datum $\mathcal{R}$ is called *reduced* if every root is indivisible.

The root datum $\mathcal{R}$ is reduced if and only if $\mathcal{R}^*$ is reduced. If α is indivisible and $w \in W$, then $w(\alpha)$ is indivisible.

Definition 2.1.4. Let $\alpha \in R$. If α is indivisible, set $\mathrm{ind}(\alpha) = \alpha$; otherwise set $\mathrm{ind}(\alpha) = \alpha/2$.

Corollary 2.1.5. *If $\alpha \in R$ is indivisible and $k\alpha \in R$, where $k \in \mathbb{Q}$, then $k \in \mathbb{Z}$.*

Proposition 2.1.6. *Let $\mathcal{R}$ be a root datum. Then*

$$\mathrm{ind}(\mathcal{R}) = \big(M, M^*, \mathrm{ind}(R), \mathrm{ind}(R)^*\big)$$

is a reduced root datum, and

$$W\big(\mathrm{ind}(\mathcal{R})\big) = W(\mathcal{R}).$$

Proof. The quadruple $\mathrm{ind}(\mathcal{R})$ is a root datum by Corollary 1.2.11, since the Weyl group permutes the indivisible roots. The second assertion follows from Proposition 2.1.1. □

Remark 2.1.7. If $\mathcal{R}$ is not reduced, then

$$\mathrm{ind}(R)^* \neq \mathrm{ind}(R^*), \qquad \mathrm{ind}(\mathcal{R}^*) \neq \mathrm{ind}(\mathcal{R})^*.$$

2.2 Orthogonal roots

Lemma 2.2.1. *Let α and β be two roots. Then*

$$\ell(\alpha)(\alpha^*, \beta) = \ell(\beta)(\beta^*, \alpha). \tag{11}$$

Proof. This follows immediately from Lemma 1.2.1, formulas (9) and (10). □

Corollary 2.2.2. *Let $\alpha, \beta \in R$. The following conditions are equivalent:*

(i) $(\alpha^*, \beta) = 0$;

(i bis) $(\beta^*, \alpha) = 0$;

(ii) $(p(\alpha), \beta) = 0$;

(iii) $s_\alpha(\beta) = \beta$;

(iii bis) $s_\beta(\alpha) = \alpha$;

(iv) $s_\alpha(\beta^*) = \beta^*$;

(iv bis) $s_\beta(\alpha^*) = \alpha^*$;

(v) $s_\alpha \neq s_\beta$*, and* s_α *and* s_β *commute.*

Proof. All the equivalences are immediate except those involving (v). We first show that (i), together with its equivalent condition (i bis), implies (v). One has

$$s_\alpha s_\beta(x) = x - (\beta^*, x)\beta - (\alpha^*, x)\alpha + (\beta^*, x)(\alpha^*, \beta)\alpha.$$

If $(\alpha^*, \beta) = 0$, then $(\beta^*, \alpha) = 0$, and consequently

$$s_\alpha s_\beta(x) = s_\beta s_\alpha(x).$$

Conversely, suppose (v). Then

$$s_\alpha = s_\beta s_\alpha s_\beta = s_{s_\beta(\alpha)} \qquad \text{by Corollary 1.2.10.}$$

By Proposition 2.1.1, there is a $k \in \mathbb{Q}^*$ such that

$$\alpha = k\, s_\beta(\alpha) = k\alpha - k(\beta^*, \alpha)\beta.$$

Since $s_\alpha \neq s_\beta$, the roots α and β are not proportional, again by Proposition 2.1.1. It follows that $(\beta^*, \alpha) = 0$. □

Definition 2.2.3. Two roots satisfying the equivalent conditions of Corollary 2.2.2 are called *orthogonal*.

Remark 2.2.4. The roots α and β are orthogonal if and only if the coroots α^* and β^* are orthogonal.

Lemma 2.2.5. *If α and β are two orthogonal roots, then $\alpha+\beta \in R$ if and only if $\alpha-\beta \in R$.*

Proof. Indeed,

$$s_\beta(\alpha + \beta) = s_\beta(\alpha) + s_\beta(\beta) = \alpha - \beta.$$

□

Lemma 2.2.6. *Let α and β be two nonorthogonal roots. For a coroot γ^*, define $\ell(\gamma^*)$ to be the value of ℓ on the root γ^* of $\mathcal{R}^*$. Then*

$$\ell(\alpha)\ell(\alpha^*) = \ell(\beta)\ell(\beta^*). \tag{12}$$

Proof. Multiply formula (11) by the corresponding formula for $\mathcal{R}^*$, and use $(\gamma^*)^* = \gamma$ for every $\gamma \in R$. This gives

$$(\beta^*, \alpha)(\alpha^*, \beta)\ell(\alpha)\ell(\alpha^*) = (\beta^*, \alpha)(\alpha^*, \beta)\ell(\beta)\ell(\beta^*).$$

The common integer factor is nonzero because the roots are not orthogonal, so it may be cancelled. □

2.3 General case

Proposition 2.3.1. *If α and β are any two roots, then*

$$0 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 4. \tag{13}$$

If α and β are neither proportional nor orthogonal, then

$$1 \leq (\alpha^*, \beta)(\beta^*, \alpha) \leq 3.$$

Proof. One has

$$4(p(\alpha), \beta)^2 = \ell(\alpha)\ell(\beta)(\alpha^*, \beta)(\beta^*, \alpha).$$

On the other hand, by Corollary 1.2.6, the symmetric bilinear form $(p(x), y)$ is positive and nondegenerate on $\mathcal{V}(R)$. Hence

$$(p(x), y)^2 \leq \ell(x)\ell(y).$$

[45] The first inequality follows. In the nonproportional, nonorthogonal case, the product is a nonzero integer and cannot equal 4, which gives the second inequality. □

Corollary 2.3.2. *Let α and β be two nonorthogonal roots. If $\ell(\alpha) = \ell(\beta)$, then there is a $w \in W$ such that $w(\alpha) = \beta$.*

Proof. If α and β are proportional, equality of their ℓ-values gives $\alpha = \beta$ or $\alpha = -\beta$; take $w = 1$ or $w = s_\alpha$, respectively. If they are neither proportional nor orthogonal, formula (11) and Proposition 2.3.1 give

$$(\beta^*, \alpha) = (\alpha^*, \beta) = \pm 1.$$

If both integers are 1, take $w = s_\beta s_\alpha s_\beta$. If both are -1, take $w = s_\alpha s_\beta$. □

Corollary 2.3.3. *Let α and β be roots. If $\alpha \neq \beta$ and $(\beta^*, \alpha) > 0$, then $\alpha - \beta$ is a root. Respectively, if $\alpha \neq -\beta$ and $(\beta^*, \alpha) < 0$, then $\alpha + \beta$ is a root.*

Proof. The second case follows from the first by replacing β by $-\beta$. Suppose first that α and β are proportional and that $(\beta^*, \alpha) > 0$. Then

$$\beta = \alpha, \qquad 2\beta = \alpha, \qquad \text{or} \qquad \beta = 2\alpha.$$

The first case is excluded. In the other two cases, $\alpha - \beta = \beta \in R$ and $\alpha - \beta = -\alpha \in R$, respectively.

[45] Editors' note: equality holds if and only if x and y are proportional.

If α and β are not proportional, then (α^*, β) and (β^*, α) are strictly positive integers whose product is at most 3. Thus one of them equals 1. If $(\beta^*, \alpha) = 1$, then

$$\alpha - \beta = s_\beta(\alpha) \in R;$$

if $(\alpha^*, \beta) = 1$, then

$$\alpha - \beta = -s_\alpha(\beta) \in R.$$

□

Lemma 2.3.4. *Let α and β be two nonproportional roots. If $\beta - \alpha$ is not a root, then $\beta + k\alpha \in R$ for $k = -(\alpha^*, \beta)$, but not for $k = -(\alpha^*, \beta) + 1$.*

Proof. Indeed,

$$\beta - (\alpha^*, \beta)\alpha = s_\alpha(\beta) \in R,$$

whereas

$$\beta + (-(\alpha^*, \beta) + 1)\alpha = s_\alpha(\beta - \alpha) \notin R.$$

□

Proposition 2.3.5. *Let α and β be two nonproportional roots. The set of integers k for which $\beta + k\alpha \in R$ is an interval $[p, q]$, with $p \leq 0$ and $q \geq 0$, and*

$$p + q = -(\alpha^*, \beta).$$

Proof. For the first assertion it suffices, for example, to show that if $\beta + k\alpha \in R$ and $k > 0$ is an integer, then $\beta + (k-1)\alpha \in R$. This is trivial if $k = 1$. If $k \geq 2$, then

$$(\alpha^*, \beta + k\alpha) = (\alpha^*, \beta) + 2k \geq -3 + 4 > 0,$$

and the assertion follows from Corollary 2.3.3.

Let $[p, q]$ be the interval in question. Applying Lemma 2.3.4 to $\beta + p\alpha$, one obtains

$$q - p = -(\alpha^*, \beta + p\alpha) = -(\alpha^*, \beta) - 2p.$$

[46] The asserted formula follows. □

Remark 2.3.6. The preceding formula contains the qualitative statements of Lemma 2.2.5 and Corollary 2.3.3.

Complements 2.3.7. [47] By formulas (9) and (10), and by Corollary 1.2.6, the bilinear form on $\mathcal{V}(R)$ defined by

$$\langle x, y \rangle = (p(x), y)$$

is symmetric and positive definite. For every $\alpha \in R$ and $y \in \mathcal{V}(R)$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y), \qquad \ell(\alpha) = \langle \alpha, \alpha \rangle.$$

Consequently, Corollary 2.3.3 gives the following corollary.

Corollary. Let $\alpha \neq \beta$ in R. If $\alpha - \beta \notin R$, then

$$\langle \alpha, \beta \rangle \leq 0.$$

[46] Editors' note: in this equality, $+2p$ was corrected to $-2p$.

[47] Editors' note: these complements, useful for the proof of Proposition 3.1.5, were added by the editors.

3 Simple roots, positive roots

3.1 Systems of simple roots

Lemma 3.1.1. *Let α and α_i, $i = 1, \dots, n$, be roots, and suppose that α is distinct from the α_i. If*

$$\alpha = \sum_{i=1}^{n} q_i \alpha_i, \qquad q_i \in \mathbb{Q}_+,$$

then there is an index i such that

$$q_i \neq 0, \qquad (\alpha^*, \alpha_i) > 0, \qquad \alpha - \alpha_i \in R.$$

Proof. Write

$$2 = (\alpha^*, \alpha) = \sum_{i=1}^{n} q_i (\alpha^*, \alpha_i).$$

This proves the first two assertions. The third then follows from Corollary 2.3.3. □

Proposition 3.1.2. *Let α and α_i, $i = 1, \dots, n$, be roots. If*

$$\alpha = \sum_{i=1}^{n} m_i \alpha_i, \qquad m_i \in \mathbb{N},$$

then there is a sequence $\beta_1, \dots, \beta_m$ of roots chosen from among the α_i, not necessarily pairwise distinct, such that, on setting

$$\gamma_p = \sum_{i=1}^{p} \beta_i, \qquad p = 1, \dots, m,$$

one has $\gamma_p \in R$ and $\gamma_m = \alpha$.

Proof. We argue by induction on the integer $m' = \sum_i m_i$. If α equals one of the α_i, say α_{i_0}, which necessarily happens when $m' = 1$, take

$$m = 1, \qquad \beta_1 = \alpha_{i_0}.$$

Otherwise, Lemma 3.1.1 gives an index i for which $m_i \neq 0$ and $\alpha - \alpha_i$ is a root. Then $m_i - 1 \in \mathbb{N}$ and

$$\alpha - \alpha_i = (m_i - 1)\alpha_i + \sum_{j \neq i} m_j \alpha_j.$$

Apply the induction hypothesis to $\alpha - \alpha_i$, and append α_i to the resulting sequence. □

Corollary 3.1.3. *Let $R' \subset R$. The following conditions are equivalent:*

(i) if $\alpha, \beta \in R'$ and $\alpha + \beta \in R$, then $\alpha + \beta \in R'$;

(ii) $(\mathbb{N} \cdot R') \cap R \subset R'$.

Proof. Clearly (ii) implies (i). The converse follows immediately from Proposition 3.1.2. □

Definition 3.1.4. A set of roots satisfying the equivalent conditions of Corollary 3.1.3 is called *closed*.

Proposition 3.1.5. *Let $\Delta \subset R$ be a set of roots. The following conditions are equivalent:*

(i) *the elements of* Δ *are indivisible and linearly independent, and*

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta);$$

(ii) *the elements of* Δ *are linearly independent, and*

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta);$$

(iii) *every root can be written uniquely as a linear combination of the elements of* Δ*, with integer coefficients all of the same sign.*

Proof. [48] It is clear that (ii) implies (iii). Write $\alpha_1, \dots, \alpha_n$ for the distinct elements of Δ.

Assume (i), and let $\alpha \in R$. There is a unique expression

$$\varepsilon\alpha = \sum_i q_i \alpha_i, \qquad q_i \in \mathbb{Q}_+, \qquad \varepsilon = \pm 1.$$

We show that the q_i are integers. This is immediate when all but one of them vanish, by Corollary 2.1.5. Otherwise, $\varepsilon\alpha$ is distinct from the α_i, and Lemma 3.1.1 gives an i_0 such that

$$\alpha' = \varepsilon\alpha - \alpha_{i_0} \in R, \qquad q_{i_0} \neq 0.$$

Thus

$$\alpha' = (q_{i_0} - 1)\alpha_{i_0} + \sum_{i \neq i_0} q_i \alpha_i.$$

At least one q_i, with $i \neq i_0$, is nonzero, so (i) forces $q_{i_0} - 1 \geq 0$. Repeating this operation with α', after finitely many steps we find that all q_i are integers. This proves (ii).

Assume now (iii). We first prove that $\alpha_1, \dots, \alpha_n$ are linearly independent over $\mathbb{Q}$. Otherwise there would be an equality

$$x = \sum_{i \in I} a_i \alpha_i = \sum_{j \in J} b_j \alpha_j,$$

where I and J are disjoint subsets of $\{1, \dots, n\}$, at least one of them, say I, is nonempty, and $a_i, b_j \in \mathbb{N}^*$. By the corollary in Complements 2.3.7, one has $\langle \alpha_i, \alpha_j \rangle \leq 0$ for $i \in I$ and $j \in J$. Therefore $\langle x, x \rangle \leq 0$, and hence $x = 0$.

Choose $i_0 \in I$. The two expressions

$$\alpha_{i_0} = \alpha_{i_0} + \sum_{i \in I} a_i \alpha_i = \alpha_{i_0} + \sum_{j \in J} b_j \alpha_j$$

then imply, by (iii), that $a_i = 0 = b_j$ for all i, j, a contradiction. Thus the elements of Δ are linearly independent over $\mathbb{Q}$.

It remains to show that they are indivisible. Suppose that $\beta \in \Delta$ is divisible. Then $\beta/2 \in R$, so

$$\frac{\beta}{2} = \varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha, \qquad m_\alpha \in \mathbb{N}, \qquad \varepsilon = \pm 1.$$

Consequently,

$$\beta = 2\varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha.$$

Uniqueness gives $m_\alpha = 0$ for $\alpha \neq \beta$ and $2\varepsilon m_\beta = 1$, a contradiction. □

[48] Editors' note: in the argument that follows, the order was changed and the proof of the implication (iii)⇒(i) was spelled out.

Definition 3.1.6. A set Δ of roots satisfying the conditions of Proposition 3.1.5 is called a *system of simple roots*, or a *base* of $\mathcal{R}$.

If $w \in W$ and Δ is a system of simple roots, then $w(\Delta)$ is a system of simple roots.

Remark 3.1.7. This definition involves only R, not $\mathcal{R}$; indeed, it involves only $\mathrm{ind}(R)$.

Remark 3.1.8. If Δ is a system of simple roots, then Δ is a basis of the free abelian group $\Gamma_0(R)$. Hence

$$\mathrm{Card}(\Delta) = \mathrm{rgss}(\mathcal{R}).$$

Remark 3.1.9. The conditions of Proposition 3.1.5 are therefore equivalent to the following:

(i′) the elements of Δ are indivisible, their number is at most $\mathrm{rgss}(\mathcal{R})$, and

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta);$$

(ii′) $\mathrm{Card}(\Delta) \leq \mathrm{rgss}(\mathcal{R})$, and

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta).$$

Corollary 3.1.10. *If Δ is a system of simple roots, then* $\mathrm{ind}(\Delta^*)$ *is a system of simple coroots* (i.e. *a system of simple roots of $\mathcal{R}^*$*).

Proof. If

$$\beta = \sum_{\alpha\in\Delta} a_\alpha \alpha, \qquad a_\alpha \in \mathbb{N},$$

then

$$p(\beta) = \sum_{\alpha\in\Delta} a_\alpha p(\alpha).$$

Consequently, by (10),

$$\beta^* = \sum_{\alpha\in\Delta} a_\alpha \frac{\ell(\alpha)}{\ell(\beta)} \alpha^*,$$

which shows that $\mathrm{ind}(\Delta^*)$ satisfies condition (i′) of Remark 3.1.9. □

By Proposition 2.1.1, if α^* is not indivisible, then

$$\alpha^* = 2u^*, \qquad u^* \in \mathrm{ind}(\Delta^*) \quad \text{and} \quad u = 2\alpha.$$

It follows that:[49]

Corollary 3.1.11. *Let Δ be a system of simple roots, and let*

$$\beta = \sum_{\alpha\in\Delta} a_\alpha \alpha, \qquad a_\alpha \in \mathbb{Z},$$

be the expression of β relative to Δ. Then $\ell(\beta)$ divides $2a_\alpha\ell(\alpha)$; it even divides $a_\alpha\ell(\alpha)$ if α^* is indivisible.

Before continuing to state the properties of systems of simple roots, we show that such systems exist.

[49] Editors' note: The preceding sentence was added, and in Corollary 3.1.11 the expression $4a_\alpha\ell(\alpha)$ was replaced by $2a_\alpha\ell(\alpha)$.

3.2 Systems of positive roots

Definition 3.2.1. A subset $R_+ \subset R$ is called a *system of positive roots* of $\mathcal{R}$ *(or of R; see Remark 3.2.2*) if it satisfies:

(P 1) $R_+ \cap (-R_+) = \varnothing$;

(P 2) $R_+ \cup (-R_+) = R$;

(P 3) $(\mathbb{Q}_+ \cdot R_+) \cap R \subset R_+$.

In particular, such a set is closed. We shall see later, in Corollary 3.3.8, that a closed set satisfying (P 1) and (P 2) also satisfies (P 3), and hence is a system of positive roots. If $w \in W$ and R_+ is a system of positive roots, then $w(R_+)$ is a system of positive roots.

Remark 3.2.2. This definition involves only R. We shall also say that R_+ is a system of positive roots of R.

Remark 3.2.3. Conditions (P 1) and (P 2) immediately give

$$\mathrm{Card}(R_+) = \frac{\mathrm{Card}(R)}{2}.$$

It follows that if R_+ and R'_+ are two systems of positive roots and $R_+ \subset R'_+$, then $R_+ = R'_+$.

Remark 3.2.4. If R_+ is a system of positive roots, then R^*_+ is a system of positive coroots *(i.e.* a system of positive roots of R^**).*

This follows immediately from Lemma 1.1.4 and Scholium 1.2.2.

Definition 3.2.5. Let Δ be a system of simple roots. Set

$$\mathscr{P}(\Delta) = (\mathbb{Q}_+ \cdot \Delta) \cap R = (\mathbb{N}_+ \cdot \Delta) \cap R.$$

Proposition 3.2.6. *If Δ is a system of simple roots, then $\mathscr{P}(\Delta)$ is a system of positive roots. If Δ is a system of simple roots and R_+ is a system of positive roots, then*

$$\Delta \subset R_+ \quad\Longleftrightarrow\quad R_+ = \mathscr{P}(\Delta).$$

Proof. The first assertion is immediate. If $\Delta \subset R_+$, then $\mathscr{P}(\Delta) \subset R_+$ by (P 3), and therefore $\mathscr{P}(\Delta) = R_+$ by Remark 3.2.3. The rest is trivial. □

Remark 3.2.7. Systems of positive roots exist: endow $\mathcal{V}(R)$ with the structure of a totally ordered vector space. The set of roots that are ≥ 0 for this order is a system of positive roots.

Theorem 3.2.8. *Let R_+ be a system of positive roots. There exists a unique system of simple roots $\mathscr{S}(R_+)$ such that*

$$\mathscr{S}(R_+) \subset R_+, \qquad \textit{i.e.} \qquad R_+ = \mathscr{P}(\mathscr{S}(R_+)).$$

Uniqueness follows immediately from:

Lemma 3.2.9. *Let Δ be a system of simple roots. An element $\alpha \in \mathscr{P}(\Delta)$ belongs to Δ if and only if α is not the sum of two elements of $\mathscr{P}(\Delta)$.*

Proof. This follows immediately from the definitions and Proposition 3.1.2. □

We now prove the existence of $\mathscr{S}(R_+)$. Consider the set of subsets T of R_+ such that

$$T = \operatorname{ind}(T) \qquad \text{and} \qquad \mathbb{Q}_+ \cdot T \supset R_+.$$

This set is nonempty, since it contains $\operatorname{ind}(R_+)$. Let Δ be a minimal element of this set under inclusion. We show that Δ is a system of simple roots; by Proposition 3.1.5(i), it is enough to show that Δ is a linearly independent subset of $M \otimes \mathbb{Q}$.

Lemma 3.2.10. *If $\alpha, \beta \in \Delta$ and*

$$q\alpha + q'\beta \in R, \qquad q, q' \in \mathbb{Q},$$

then $qq' \geq 0$.

Proof. If $qq' < 0$, then, after interchanging α and β if necessary, we may write

$$a\alpha - b\beta \in R_+, \qquad a, b \in \mathbb{Q}_+^*.$$

By the defining property of Δ, there is a relation

$$a\alpha - b\beta = \sum_{\gamma \in \Delta} c(\gamma)\gamma, \qquad c(\gamma) \in \mathbb{Q}_+.$$

If $a \leq c(\alpha)$, this becomes

$$-\beta = \frac{c(\alpha) - a}{b}\alpha + \sum_{\gamma \neq \alpha} \frac{c(\gamma)}{b}\gamma.$$

Thus $-\beta \in (\mathbb{Q}_+ \cdot \Delta) \cap R$, which is contained in R_+ by (P 3). This gives $\beta \in R_+ \cap (-R_+)$, contradicting (P 1).

If instead $a > c(\alpha)$, then

$$(a - c(\alpha))\alpha = b\beta + \sum_{\gamma \neq \alpha} c(\gamma)\gamma,$$

which shows that

$$\Delta \subset \mathbb{Q}_+ \cdot (\Delta \setminus \{\alpha\}),$$

contrary to the minimality of Δ. □

Recall from Complements 2.3.7 that the bilinear form on $\mathcal{V}(R)$ defined by

$$\langle x, y \rangle = (p(x), y)$$

is a Euclidean inner product. Moreover, for $\alpha \in R$,

$$\langle \alpha, y \rangle = \frac{\ell(\alpha)}{2}(\alpha^*, y),$$

so $\langle \alpha, y \rangle$ and (α^*, y) have the same sign.

Lemma 3.2.11. *If $\alpha, \beta \in \Delta$, then*

$$(\beta^*, \alpha) \leq 0 \qquad \textit{and hence} \qquad \langle \beta, \alpha \rangle \leq 0.$$

Proof. Indeed,

$$s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta \in R,$$

and therefore $(\beta^*, \alpha) \leq 0$ by Lemma 3.2.10. □

We now show that Δ is linearly independent. Otherwise, Lemma 3.2.11 would imply, exactly as in the proof of Proposition 3.1.5, the existence of a nontrivial relation

$$\sum_{\alpha\in\Delta} m(\alpha)\alpha = 0, \qquad m(\alpha) \in \mathbb{N}.$$

If $m(\alpha_0) \geq 1$, then

$$-\alpha_0 = (m(\alpha_0) - 1)\alpha_0 + \sum_{\alpha\neq\alpha_0} m(\alpha)\alpha.$$

By (P 3), this gives $\alpha_0 \in R_+ \cap (-R_+)$, contradicting (P 1). Thus Δ is a base of $\mathcal{R}$, completing the proof of Theorem 3.2.8.

Corollary 3.2.12. *Let R_+ be a system of positive roots, let Δ be a base of $\mathcal{R}$, and let $w \in W$. Then*

$$\Delta \subset R_+ \iff R_+ = \mathscr{P}(\Delta) \iff \Delta = \mathscr{S}(R_+),$$

$$\mathscr{P}(\mathrm{ind}(\Delta^*)) = \mathscr{P}(\Delta)^*, \qquad \mathscr{S}(R_+^*) = \mathrm{ind}(\mathscr{S}(R_+)^*),$$

and

$$\mathscr{S}(w(R_+)) = w(\mathscr{S}(R_+)), \qquad \mathscr{P}(w(\Delta)) = w(\mathscr{P}(\Delta)).$$

Definition 3.2.13. Once a system of simple roots Δ has been chosen, the elements of $\mathscr{P}(\Delta)$ are called *positive*. Once a system of positive roots R_+ has been chosen, the elements of $\mathscr{S}(R_+)$ are called *simple*.

Corollary 3.2.14. *Let R_+ be a system of positive roots, and let $\alpha \in R_+$. The following conditions are equivalent:*

(i) α is simple, that is, $\alpha \in \mathscr{S}(R_+)$;

(ii) α is not the sum of two elements of R_+;

(iii) $R_+ \setminus \{\alpha\}$ is closed.

Proof. The equivalence of (i) and (ii) follows immediately from Lemma 3.2.9; the equivalence of (ii) and (iii) is immediate from Definition 3.1.4. □

Definition 3.2.15. Let Δ be a system of simple roots. The sum of the coefficients in the expression of a root α relative to Δ is called the *order of α relative to* Δ and is denoted $\mathrm{ord}_\Delta(\alpha)$.

One has

$$\alpha \in \mathscr{P}(\Delta) \iff \mathrm{ord}_\Delta(\alpha) > 0, \qquad \alpha \in \Delta \iff \mathrm{ord}_\Delta(\alpha) = 1.$$

Lemma 3.2.16. *Let Δ be a system of simple roots and $\alpha \in \mathscr{P}(\Delta)$. There is a sequence $\alpha_i \in \Delta$, $i = 1, \ldots, m$, such that*

$$\alpha_1 + \alpha_2 + \cdots + \alpha_p \in \mathscr{P}(\Delta) \quad \textit{for } p = 1, \ldots, m,$$

and

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \alpha.$$

Moreover, for every sequence satisfying these conditions,

$$m = \mathrm{ord}_\Delta(\alpha).$$

Proof. This is immediate from Proposition 3.1.2. □

3.3 Characterization and conjugacy of systems of positive roots

Lemma 3.3.1. *If $\alpha \in \mathscr{P}(\Delta)$, $\beta \in \Delta$, and α and β are not proportional, then*

$$s_\beta(\alpha) \in \mathscr{P}(\Delta).$$

Proof. One has

$$s_\beta(\alpha) = \alpha - (\beta^*, \alpha)\beta.$$

At least one simple root other than β occurs with a nonzero, and hence strictly positive, coefficient in the expression of α. It occurs with the same coefficient in the expression of $s_\beta(\alpha)$; therefore $s_\beta(\alpha)$ is also positive. □

Corollary 3.3.2. *If $\beta \in \Delta$, then the reflection s_β permutes the elements of $\mathscr{P}(\Delta)$ that are not proportional to β.*

Lemma 3.3.3. *Let*

$$\alpha \in \mathscr{P}(\Delta) \setminus \Delta$$

be indivisible. There exists $\beta \in \Delta$ such that

$$s_\beta(\alpha) \in \mathscr{P}(\Delta) \qquad \textit{and} \qquad \operatorname{ord}_\Delta(s_\beta(\alpha)) < \operatorname{ord}_\Delta(\alpha).$$

Proof. By Lemma 3.1.1, there is a $\beta \in \Delta$ such that $(\beta^*, \alpha) > 0$. Since $\alpha \notin \Delta$ and α is indivisible, it cannot be proportional to β. Thus $s_\beta(\alpha) \in \mathscr{P}(\Delta)$ by Lemma 3.3.1, and

$$\operatorname{ord}_\Delta(s_\beta(\alpha)) = \operatorname{ord}_\Delta(\alpha) - (\beta^*, \alpha) < \operatorname{ord}_\Delta(\alpha).$$

□

Corollary 3.3.4. *If $\alpha \in \mathscr{P}(\Delta)$ is indivisible, then there is a sequence $\beta_p \in \Delta$, $p = 1, \ldots, q$, such that*

$$\alpha_p = s_{\beta_p} \cdots s_{\beta_1}(\alpha) \in \mathscr{P}(\Delta) \quad \textit{for } p = 1, \ldots, q,$$

and $\alpha_q \in \Delta$.

Proof. Apply Lemma 3.3.3 repeatedly, inducting on $\operatorname{ord}_\Delta(\alpha)$. □

Proposition 3.3.5. *The Weyl group is generated by the s_α, for $\alpha \in \Delta$. Every indivisible root is conjugate under an element of the Weyl group to a simple root.*

Proof. The second assertion follows immediately from Corollary 3.3.4. The first then follows from Corollary 1.2.10 and Proposition 2.1.1. □

Proposition 3.3.6. *Let R_+ be a system of positive roots, and let $P' \subset R$ be closed and satisfy (P 2). Then there exists $w \in W$ such that*

$$w(R_+) \subset P'.$$

We first state the corollaries.

Corollary 3.3.7. *The Weyl group acts transitively on the set of systems of positive roots, and on the set of systems of simple roots.*

Corollary 3.3.8. *A subset of R is a system of positive roots if and only if it is closed and satisfies (P 1) and (P 2).*

Corollary 3.3.9. *Suppose that $\Gamma_0(R)$ is endowed with an ordered-group structure for which every root is either > 0 or < 0. Then the set of positive roots for this order is a system of positive roots.*

Proof of Proposition 3.3.6. Choose $w \in W$ such that $\mathrm{Card}(w(R_+) \cap P')$ is maximal. Replacing R_+ by $w(R_+)$, we may assume that

$$\mathrm{Card}(R_+ \cap P') \geq \mathrm{Card}\big(R_+ \cap s_\alpha(P')\big) \quad \text{for every } \alpha \in \mathscr{S}(R_+) = \Delta. \tag{$*$}$$

We show that $R_+ \subset P'$. Otherwise, since P' is closed, there is an $\alpha \in \Delta$ such that $\alpha \notin P'$. As P' satisfies (P 2), one then has $-\alpha \in P'$, and also $-2\alpha \in P'$ if 2α is a root.

For every $\beta \in R_+ \cap P'$, one has $\beta \neq \alpha$. If 2α is not a root, then $s_\alpha(\beta) \in R_+$ by Lemma 3.3.1; hence

$$s_\alpha(R_+ \cap P') \subset R_+ \cap s_\alpha(P').$$

But $R_+ \cap s_\alpha(P')$ also contains $\alpha = s_\alpha(-\alpha)$, contradicting $(*)$. If 2α is a root, the same argument applies. □

Thus, to study closed sets of roots satisfying (P 2), it is enough to consider those that contain a system of positive roots.

Proposition 3.3.10. *Let R_+ be a system of positive roots, and let P' be a closed subset of R containing R_+. Put*

$$\Delta = \mathscr{S}(R_+), \qquad \Delta' = \Delta \cap (-P').$$

Then P' is the union of R_+ and the set of roots that are linear combinations of the elements of Δ' with negative coefficients.

Proof. We argue by induction on the order of a root $\gamma \in P' \setminus R_+$. If $\mathrm{ord}_\Delta(\gamma) = -1$, then $-\gamma \in \Delta$, and hence $\gamma \in -\Delta'$.

Suppose that $\mathrm{ord}_\Delta(\gamma) < -1$. By Proposition 3.1.2, there exists $\beta \in \Delta$ such that $-\gamma - \beta \in R$. Then

$$0 < \mathrm{ord}_\Delta(-\gamma - \beta) = \mathrm{ord}_\Delta(-\gamma) - 1 < \mathrm{ord}_\Delta(-\gamma).$$

The first inequality shows that $-\gamma - \beta \in R_+$. Consequently, since $R_+ \cap (-R_+) = \varnothing$, and since $\gamma + \beta$ is the sum of two roots of P', the root $\gamma + \beta$ belongs to $P' \setminus R_+$ and satisfies

$$\mathrm{ord}_\Delta(\gamma + \beta) > \mathrm{ord}_\Delta(\gamma).$$

By induction, $\gamma + \beta$ is a linear combination of the elements of Δ' with negative coefficients. Since $\gamma = (\gamma + \beta) - \beta$, it remains only to show that $\beta \in -P'$. But $\gamma \in P'$ and $-(\gamma + \beta) \in R_+ \subset P'$, so

$$-\beta = \gamma - (\gamma + \beta) \in P'.$$

□

Definition 3.3.10.1. [50] A subset R' of R is called *symmetric* if

$$-R' = R'.$$

[50] Editors' note: This definition was inserted here.

Scholium 3.3.11. Let P' be a set of roots satisfying (P 2) and closed. There exist a system of simple roots Δ and a subset Δ' of Δ such that

$$P' = R \cap (\mathbb{N} \cdot \Delta \cup -\mathbb{N} \cdot \Delta').$$

If

$$R_{\Delta'} = (\mathbb{Z} \cdot \Delta') \cap R,$$

which is a closed and symmetric subset of R, then P' is the disjoint union of $R_{\Delta'}$ and the closed subset of $\mathscr{P}(\Delta)$ formed by the

$$\alpha \in \mathscr{P}(\Delta) \setminus \mathbb{N} \cdot \Delta',$$

that is, the positive roots whose decomposition with respect to Δ "contains at least one root of $\Delta \setminus \Delta'$".

3.4 Closed and symmetric sets of roots

Proposition 3.4.1. *Let*

$$\mathcal{R} = (M, M^*, R, R^*)$$

be a root datum, and let R' be a closed and symmetric subset of R. Then:

(i) $\mathcal{R}' = (M, M^*, R', R'^*)$ is a root datum;

(ii) for every system of positive roots R_+ of R, $R'_+ = R_+ \cap R'$ is a system of positive roots of R';

(iii) the Weyl group $W(\mathcal{R}')$ of $\mathcal{R}'$ is the subgroup of $W(\mathcal{R})$ generated by the s_α, for $\alpha \in R'$.

Proof. The first assertion is immediate from Corollary 1.2.11; the second follows from Corollary 3.3.8; and the third is evident. □

Corollary 3.4.2. *Let $w \in W(\mathcal{R}')$. The order of w is the smallest integer $n > 0$ such that*

$$w^n(\alpha') = \alpha' \qquad \text{for every } \alpha' \in R'.$$

Proof. Apply Corollary 1.2.7 to the root datum $\mathcal{R}'$. □

Corollary 3.4.3. *Let α and β be two nonproportional roots. Let n be the smallest integer > 0 such that*

$$(s_\alpha s_\beta)^n(\alpha) = \alpha, \qquad (s_\alpha s_\beta)^n(\beta) = \beta.$$

Then the subgroup $W_{\alpha,\beta}$ of W generated by s_α and s_β is defined by the relations

$$s_\alpha^2 = 1, \qquad s_\beta^2 = 1, \qquad (s_\alpha s_\beta)^n = 1.$$

Proof. In view of Corollary 3.4.2, it remains only to verify the following lemma. □

Lemma 3.4.4. *Let G be the group generated by two elements x and y subject to the relations $x^2 = y^2 = 1$. Every normal subgroup of G containing neither x nor y is generated, as a normal subgroup, by an element of the form $(xy)^n$, where $n > 0$.*

Proof. [51] Every element of G can be written as $(xy)^n$, as $(yx)^n = (xy)^{-n}$, or in one of the forms

$$(a) \qquad x(yx)^{2n} \qquad \text{or } y(xy)^{2n+1},$$
$$(b) \qquad y(xy)^{2n} \qquad \text{or } x(yx)^{2n+1},$$

where $n \in \mathbb{N}$. The elements of type (a), respectively of type (b), are conjugate to x, respectively to y. □

Remark 3.4.5. The integer n is computed immediately. Put

$$(\alpha^*, \beta) = p, \qquad (\beta^*, \alpha) = q.$$

Then[52]

$$(s_\alpha s_\beta)(\alpha) = (pq-1)\alpha - q\beta, \qquad (s_\alpha s_\beta)(\beta) = p\alpha - \beta.$$

Thus n is the order of the matrix

$$\begin{pmatrix} pq-1 & p \\ -q & -1 \end{pmatrix}.$$

If $pq = 0$, then $p = q = 0$ by Corollary 2.2.2, whence $n = 2$. Otherwise, Proposition 2.3.1 shows that pq equals 1, 2, or 3, and one finds respectively $n = 3$, 4, or 6.

N.B. Expressing the finiteness of the order of the preceding matrix recovers inequality (13) of Proposition 2.3.1.

Definition 3.4.6. Let Δ be a system of simple roots and let $\Delta' \subset \Delta$. We write

$$R_{\Delta'} = R \cap (\mathbb{Q} \cdot \Delta') = R \cap (\mathbb{Z} \cdot \Delta').$$

Lemma 3.4.7. *The set $R_{\Delta'}$ is closed and symmetric, and Δ' is a system of simple roots of the root datum*

$$(M, M^*, R_{\Delta'}, R^*_{\Delta'}).$$

Its Weyl group is the subgroup $W_{\Delta'}$ of W generated by the s_α, for $\alpha \in \Delta'$. Moreover,

$$\Delta \cap R_{\Delta'} = \Delta'.$$

Proof. Trivial. □

Proposition 3.4.8. *Let $R' \subset R$. The following conditions are equivalent:*

(i) There exists a vector subspace V' of V, or of $\mathcal{V}(R)$, such that $R' = R \cap V'$.

(ii) There exist a system of simple roots Δ of R and a subset Δ' of Δ such that $R' = R_{\Delta'}$.

More precisely, under these conditions, every system of simple roots Δ' of (M, M^*, R', R'^*) is contained in a system of simple roots Δ of R, and one has $R' = R_{\Delta'}$.

Proof. The implication (ii) ⇒ (i) is clear. Suppose (i). Then R' is closed and symmetric, so (M, M^*, R', R'^*) is a root datum. Let Δ' be a system of simple roots of this datum, and put

$$R'_+ = \mathscr{P}(\Delta').$$

[51] Editors' note: The original has been corrected in what follows.
[52] Editors' note: The original has been corrected in what follows.

If $V' = V$, then Δ' is a system of simple roots of $\mathcal{R}$, and there is nothing to prove. Otherwise, there exists $x \in V^*$ such that

$$(x, V') = \{0\}, \qquad (x, \alpha) \neq 0 \quad \text{for every } \alpha \in R \setminus R'.$$

Put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}, \qquad R_+ = R_+(x) \cup R'_+.$$

For every $\alpha \in R$, one has

$$\begin{aligned} (x, \alpha) > 0 &\Longleftrightarrow \alpha \in R_+(x), \\ (x, \alpha) < 0 &\Longleftrightarrow \alpha \in -R_+(x), \\ (x, \alpha) = 0 &\Longleftrightarrow \alpha \in R'. \end{aligned}$$

It follows at once from Corollary 3.3.8 that R_+ is a system of positive roots of R. Put $\Delta = \mathscr{S}(R_+)$. It is enough to prove that $\Delta' \subset \Delta$. Otherwise, let $\alpha \in \Delta' \setminus \Delta$. By Corollary 3.2.14, there exist $\beta, \gamma \in R_+$ such that

$$\alpha = \beta + \gamma.$$

If $\beta, \gamma \in R_+(x)$, then $\alpha \in R_+(x)$, which is impossible because $(x, \Delta') = 0$. If one of β, γ, say β, belongs to R'_+, then, since R' is symmetric and closed,

$$\gamma = \alpha - \beta \in R_+ \cap R' = R'_+.$$

But then α is not simple in R'_+, again a contradiction. □

Lemma 3.4.9. *Under the preceding conditions, let*

$$\alpha \in \mathscr{P}(\Delta) \setminus R'.$$

For every $w \in W_{\Delta'}$, one has

$$w(\alpha) \in \mathscr{P}(\Delta) \setminus R'.$$

Proof. It suffices to verify the assertion for $w = s_\beta$, $\beta \in \Delta'$. It then follows from Lemma 3.3.1 and from $s_\beta(R') = R'$. □

Lemma 3.4.10. *Let $w \in W$. Under the conditions of Proposition 3.4.8, the following conditions are equivalent:*

(i) $w \in W_{\Delta'}$.

(ii) For every $m \in M$, one has $w(m) - m \in V'$.

(iii) For every $\alpha \in R$, one has $w(\alpha) - \alpha \in V'$.

Proof. Clearly (i) implies (ii), and (ii) implies (iii). Suppose (iii). Write

$$w = s_{\alpha_n} \cdots s_{\alpha_1}, \qquad \alpha_i \in \Delta,$$

and argue by induction on n that every α_i belongs to Δ'. We may assume that

$$w' = s_{\alpha_{n-1}} \cdots s_{\alpha_1} \in W_{\Delta'}.$$

Then, for every $\alpha \in R$,

$$w(\alpha) - \alpha = w'(\alpha) - \alpha - (\alpha_n^*, w'(\alpha))\alpha_n.$$

Taking $\alpha = w'^{-1}(\alpha_n)$, one obtains $2\alpha_n \in V'$, hence $\alpha_n \in \Delta'$. Therefore

$$w = s_{\alpha_n} w' \in W_{\Delta'}.$$

□

3.5 Miscellaneous remarks

Proposition 3.5.1. *Let R_+ be a system of positive roots. Put*

$$\rho_{R_+} = \frac{1}{2} \sum_{\alpha \in \operatorname{ind}(R_+)} \alpha.$$

Then

$$(\beta^*, \rho_{R_+}) = 1 \qquad \text{for every } \beta \in \mathscr{S}(R_+).$$

Proof. One can write

$$2\rho_{R_+} = \beta + \sum_{\substack{\alpha \in \operatorname{ind}(R_+) \\ \alpha \neq \beta}} \alpha.$$

Hence

$$s_\beta(2\rho_{R_+}) = 2\rho_{R_+} - 2\beta$$

by Lemma 3.3.2. □

Corollary 3.5.2. *Put*

$$\rho^*_{R_+} = \frac{1}{2} \sum_{\alpha^* \in \operatorname{ind}(R^*_+)} \alpha^*.$$

Then:

(i) $(\rho^*_{R_+}, \alpha) > 0$ for every $\alpha \in R_+$; equivalently, $\rho^*_{R_+} \in \mathcal{C}(R_+)$, in the notation of Definition 3.6.8.

(ii) For every $\alpha \in R$,

$$\operatorname{ord}_{\mathscr{S}(R_+)}(\alpha) = (\rho^*_{R_+}, \alpha).$$

[53]

Remark 3.5.3. If $w \in W$, then

$$\rho_{w(R_+)} = w(\rho_{R_+}), \qquad \rho^*_{w(R_+)} = w(\rho^*_{R_+}).$$

Proposition 3.5.4. *Let α and γ be two nonproportional roots, with α indivisible. There exist a system of simple roots containing α and a root β such that*

$$\gamma = a\alpha + b\beta, \qquad a, b \in \mathbb{N}.$$

Proof. Choose a basis of the vector space $\mathcal{V}(R)$ containing $\alpha_1 = \alpha$ and $\alpha_2 = \gamma$, and consider the lexicographic order with respect to this basis. Let R_+ be the set of roots > 0. It is clear that R_+ is a system of positive roots and[54] that the least element of R_+ not proportional to α is simple. This element has the form

$$\beta = p\alpha + q\gamma, \qquad 0 < q \leq 1.$$

Consequently

$$\gamma = q^{-1}\beta - q^{-1}p\alpha.$$

Since $q^{-1} > 0$, one has

$$q^{-1} \in \mathbb{N}^*, \qquad -q^{-1}p \in \mathbb{N}.$$

□

[53] Editors' note: By Lemma 3.1.5, it is enough to verify the formula for $\alpha \in \Delta$. Proposition 3.5.1, applied to R^*, then gives $(\rho^*_{R_+}, \alpha) = 1 = \operatorname{ord}_{\mathscr{S}(R_+)}(\alpha)$.

[54] Editors' note: The original was modified here to take account of the case in which 2α is a root.

We conclude with two remarks on the group $\Gamma_0(R)$.

Proposition 3.5.5. *Let G be an abelian group, and let $f : R \to G$ be a map satisfying:*

(i) if $\alpha \in R$, then $f(-\alpha) = -f(\alpha)$;

(ii) if $\alpha, \beta, \alpha+\beta \in R$, then
$$f(\alpha+\beta) = f(\alpha) + f(\beta).$$

There exists a unique group homomorphism

$$\overline{f} : \Gamma_0(R) \longrightarrow G$$

such that $\overline{f}(\alpha) = f(\alpha)$ for every $\alpha \in R$.

Proof. Let Δ be a system of simple roots of R, and write

$$\beta = \sum_{\alpha\in\Delta} a(\alpha)\alpha.$$

Lemma 3.2.16 gives at once
$$f(\beta) = \sum_{\alpha\in\Delta} a(\alpha) f(\alpha).$$

Now Δ is a basis of $\Gamma_0(R)$. □

Proposition 3.5.6. *Let Δ be a system of simple roots. There exists on $\Gamma_0(R)$ a totally ordered group structure such that the roots > 0 are the elements of $\mathscr{P}(\Delta)$, and such that*

$$\alpha \longmapsto \mathrm{ord}_\Delta(\alpha)$$

is increasing.

Proof. Let
$$\alpha_1, \alpha_2, \ldots, \alpha_n, \qquad n = \mathrm{rgss}(\mathcal{R}),$$
be the elements of Δ. Every $x \in \Gamma_0(R)$ has a decomposition

$$x = \sum_{i=1}^{n} m_i(x)\alpha_i.$$

Take the lexicographic order with respect to the functions

$$\sum_i m_i, \quad m_n, \quad m_{n-1}, \quad \ldots, \quad m_2.$$

□

Remark 3.5.7. The first roots, in order, are

$$\alpha_1, \alpha_2, \ldots, \alpha_n;$$

then come, when they are roots,

$$2\alpha_1, \quad \alpha_1+\alpha_2, \quad \alpha_1+\alpha_3, \quad \ldots.$$

3.6 Weyl chambers

Lemma 3.6.1. *Let V be a finite-dimensional real vector space.*[55] *Let f_i be independent linear forms. Put*

$$C = \{x \in V \mid f_i(x) > 0\}.$$

Then C is a maximal convex subset of

$$X = V \setminus \bigcup_i f_i^{-1}(0).$$

Proof. Trivial. □

Definition 3.6.2. A subset $C \subset V$ described by the construction of Lemma 3.6.1 is called, here, a *chamber* of V.

Definition 3.6.3. A hyperplane H of V is called a *wall* of C if

$$H \cap (\overline{C} \setminus C)$$

contains a nonempty open subset of H.

Remark 3.6.4. For a convex subset, the closure can be described without reference to the topology of V: it is the set of endpoints of all open segments contained in the given subset.

Lemma 3.6.5. *Under the conditions of Lemma 3.6.1,*

$$\overline{C} = \{x \in V \mid f_i(x) \geq 0\}.$$

The walls of C are the hyperplanes $f_i^{-1}(0)$.

Proof. The first assertion is clear. The second follows from

$$\overline{C} \setminus C \subset \bigcup_i f_i^{-1}(0)$$

and from the fact that the $f_i^{-1}(0)$ are visibly walls of C. □

Proposition 3.6.6. *Let C be a chamber of V. If*

$$H_i, \qquad i = 1, 2, \ldots, n,$$

are the distinct walls of C, then, for every system of linear forms $\{u_i\}$ such that $H_i = u_i^{-1}(0)$, there exist $\varepsilon_i \in \{-1, +1\}$ for which

$$C = \{x \in V \mid \varepsilon_i u_i(x) > 0\}.$$

For every wall H of C,

$$H \cap C = \varnothing, \qquad H \cap \overline{C} = \overline{C_H},$$

where C_H is a chamber in H. The walls of C_{H_i} are the $H_j \cap H_i$, for $j \neq i$.

Proof. This follows immediately from the lemma. □

Definition 3.6.7. The C_{H_i} are called the *faces* of C.

[55] Editors' note: The original $\mathbb{Q}$ has been replaced by $\mathbb{R}$.

Let now

$$\mathcal{R} = (M, M^*, R, R^*)$$

be a root datum, and put

$$V^*_{\mathbb{R}} = M^* \otimes \mathbb{R}.$$

Definition 3.6.8. [56] For every $\alpha \in R$, put

$$\mathcal{H}_\alpha = \{x \in V^*_{\mathbb{R}} \mid (x, \alpha) = 0\}.$$

Write

$$X = V^*_{\mathbb{R}} \setminus \bigcup_{\alpha \in R} \mathcal{H}_\alpha.$$

For every $x \in X$, put

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}.$$

For every system of positive roots R_+, put

$$\mathcal{C}(R_+) = \{x \in V^*_{\mathbb{R}} \mid (x, \alpha) > 0 \text{ for every } \alpha \in R_+\}.$$

Proposition 3.6.9. *(i) For every $x \in X$, $R_+(x)$ is a system of positive roots. For every system of positive roots R_+, $\mathcal{C}(R_+)$ is a chamber in $V^*_{\mathbb{R}}$. The $\mathcal{C}(R_+)$ are the maximal convex subsets of X.*

(ii) Let Δ be a system of simple roots. One has

$$\mathcal{C}(\mathscr{P}(\Delta)) = \{x \in V^*_{\mathbb{R}} \mid (x, \alpha) > 0 \textit{ for every } \alpha \in \Delta\}.$$

The walls of $\mathcal{C}(\mathscr{P}(\Delta))$ are the hyperplanes

$$\mathcal{H}_\alpha = \alpha^{-1}(0), \qquad \alpha \in \Delta;$$

its faces are

$$C_\alpha = \{x \in V^*_{\mathbb{R}} \mid (x, \alpha) = 0,\ (x, \beta) > 0 \textit{ for } \beta \in \Delta,\ \beta \neq \alpha\}.$$

(iii) One has the equivalence

$$R_+(x) = R_+ \iff x \in \mathcal{C}(R_+).$$

Proof. It is first clear that $R_+(x)$ is a system of positive roots. Since $x \in \mathcal{C}(R_+(x))$, the union of the $\mathcal{C}(R_+(x))$ is X. Property (iii) is immediate; it follows that the $\mathcal{C}(R_+)$ form a partition of X. The first assertion of (ii) is evident. It follows at once that $\mathcal{C}(R_+)$ is a chamber in $V^*_{\mathbb{R}}$, which proves the remainder of (i). Put

$$C = \mathcal{C}(\mathscr{P}(\Delta)).$$

It remains only to note that

$$\overline{C} \cap \left(V^*_{\mathbb{R}} \setminus \bigcup_{\alpha \in R} \alpha^{-1}(0) \right) = \overline{C} \cap \left(V^*_{\mathbb{R}} \setminus \bigcup_{\alpha \in \Delta} \alpha^{-1}(0) \right)$$

to complete the proof of (ii) by Lemma 3.6.1. □

[56] Editors' note: The definition of the hyperplanes $\mathcal{H}_\alpha$ was added here.

Definition 3.6.10. The $\mathcal{C}(R_+)$ are called the *Weyl chambers* of the root datum. Conversely, for every Weyl chamber C, put

$$R_+(C) = R_+(x)$$

for any $x \in C$.

Corollary 3.6.11. *The maps*

$$R_+ \longmapsto \mathcal{C}(R_+), \qquad C \longmapsto R_+(C)$$

give a bijective correspondence between systems of positive roots and Weyl chambers.

This correspondence is invariant under the Weyl group:

Lemma 3.6.12. *If $w \in W(\mathcal{R})$, then*

$$\mathcal{C}(w(R_+)) = w(\mathcal{C}(R_+)).$$

Corollary 3.6.13. *The correspondences*

$$\Delta \longleftrightarrow R_+ \longleftrightarrow C$$

are isomorphisms of homogeneous spaces under $W(\mathcal{R})$.

We shall see later that these homogeneous spaces are principal (5.5).

Remark 3.6.14. If C is a Weyl chamber, then $-C$ is also one; it is called the *opposite* of C. Consequently there exists $w_0 \in W$—and in fact exactly one, cf. 5.5—such that

$$w_0(C) = -C.$$

It is called the *symmetry* of the root datum relative to the Weyl chamber C (or relative to $R_+(C)$, or to $\mathscr{S}(R_+(C))$, and so on).

4 Reduced root data of semisimple rank 2

4.0.[57] Let $\mathcal{R}$ be a root datum of semisimple rank 2. Let $\{\alpha, \beta\}$ be a system of simple roots. Suppose that $\ell(\alpha) \leq \ell(\beta)$. By 2.3.1 and 3.2.1, there are then four possibilities:

Type	$\ell(\beta)/\ell(\alpha)$	$\ell(\beta^*)/\ell(\alpha^*)$	(β^*, α)	(α^*, β)
$A_1 \times A_1$	–	–	0	0
A_2	1	1	-1	-1
B_2	2	1/2	-1	-2
G_2	3	1/3	-1	-3

It follows from Remark 3.4.5 that the order of $s_\alpha s_\beta$ is respectively $2, 3, 4, 6$.

We study each of these systems separately and give the list of indivisible roots.

Type $A_1 \times A_1$. The indivisible roots are

$$\alpha, \quad \beta, \quad -\alpha, \quad -\beta.$$

The corresponding coroots are

$$\alpha^*, \quad \beta^*, \quad -\alpha^*, \quad -\beta^*.$$

[57] Editors' note: The numbering 4.0 was added for later references.

Type A_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha+\beta$
$\ell(\gamma)/\ell(\alpha)$	1	1	1
coroot γ^*	α^*	β^*	$\alpha^*+\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	1	1	1

The half-sum of the indivisible positive roots is

$$\rho = \alpha + \beta.$$

Type B_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha+\beta$	$2\alpha+\beta$
$\ell(\gamma)/\ell(\alpha)$	1	2	1	2
coroot γ^*	α^*	β^*	$2\alpha^*+\beta^*$	$\alpha^*+\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	2	1	2	1

The half-sum of the indivisible positive roots is

$$\rho = \frac{4\alpha+3\beta}{2}.$$

Type G_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha+\beta$	$2\alpha+\beta$	$3\alpha+\beta$	$3\alpha+2\beta$
$\ell(\gamma)/\ell(\alpha)$	1	3	1	1	3	3
coroot γ^*	α^*	β^*	$\alpha^*+3\beta^*$	$2\alpha^*+3\beta^*$	$\alpha^*+\beta^*$	$\alpha^*+2\beta^*$
$\ell(\gamma^*)/\ell(\beta^*)$	3	1	3	3	1	1

The half-sum of the indivisible positive roots is

$$\rho = 5\alpha + 3\beta.$$

Proposition 4.1. [58] *Let n be the order of $s_\alpha s_\beta$. Put $u_0 = 0$, and, for $p \in \mathbb{N}$, define*

$$\begin{cases} u_{2p+1} = u_{2p} + (s_\alpha s_\beta)^p(\alpha), \\ u_{2p+2} = u_{2p+1} + (s_\alpha s_\beta)^p s_\alpha(\beta). \end{cases}$$

Then: [59]

(i) $u_{k+2n} = u_k$, *for every* $k \in \mathbb{N}$.

(ii) $u_0 = 0$, $u_1 = \alpha$, $u_{2n-1} = \beta$, *and* $u_{2n} = 0$.

(iii) If $1 < k < 2n-1$, *then*

$$u_k = a_k\alpha + b_k\beta, \qquad a_k, b_k \in \mathbb{N}^*.$$

[58] The following numbers 4.1 to 4.4 are used in the proof of Theorem 5.1. There are now simpler proofs of that theorem; see *Groupes et algèbres de Lie*, Chapter V, §3, Theorem 1. Editors' note: The reference was made explicit and this note, which occurred at 4.2 in the original, was moved here.

[59] Editors' note: Let ρ be the half-sum of the positive roots, cf. Proposition 3.5.1. If, as below, one puts $w_0 = 1$, $w_1 = s_\alpha$, $w_2 = s_\beta s_\alpha$, $w_3 = s_\alpha s_\beta s_\alpha$, and so on, then $u_k = \rho - w_k^{-1}(\rho)$. This proves (i), since $w_{2n} = 1$.

Proof. Assertion (i) follows from $(s_\alpha s_\beta)^n = 1$ and $u_{2n} = 0$.

We prove (ii) and (iii). Direct calculation gives, in the four cases, the following sequences of values: [60]

$$\begin{array}{r|l} A_1 \times A_1 & 0, \alpha, \beta+\alpha, \beta, 0, \\ A_2 & 0, \alpha, \beta+2\alpha, 2\beta+2\alpha, 2\beta+\alpha, \beta, 0, \\ B_2 & 0, \alpha, \beta+3\alpha, 2\beta+4\alpha, 3\beta+4\alpha, 3\beta+3\alpha, 2\beta+\alpha, \beta, 0, \\ G_2 & 0, \alpha, \beta+4\alpha, 2\beta+6\alpha, 4\beta+9\alpha, 5\beta+10\alpha, 6\beta+10\alpha, \\ & \quad 6\beta+9\alpha, 5\beta+6\alpha, 4\beta+4\alpha, 2\beta+\alpha, \beta, 0. \end{array}$$

□

Lemma 4.2. *Put*

$$w_{2p} = (s_\beta s_\alpha)^p, \qquad w_{2p+1} = s_\alpha (s_\beta s_\alpha)^p,$$

so that $w_0, \dots, w_{2n-1}$ are the distinct elements of W. Let $u_0, \dots, u_{2n-1}$ be the elements of V defined in Proposition 4.1. For every $x \in V^*$, one has

$$x - w_k(x) = n_k \alpha^* + m_k \beta^*,$$

where $n_k, m_k \in \mathbb{Q}$ and

$$n_k + m_k = (x, u_k).$$

[61]

Proof. The proof is by induction on k. If $k = 0$, the formula is trivial. We carry out, for example, the passage from w_{2p} to w_{2p+1}. Since $w_{2p+1} = s_\alpha w_{2p}$,

$$\begin{aligned} x - w_{2p+1}(x) &= x - w_{2p}(x) + w_{2p}(x) - s_\alpha w_{2p}(x) \\ &= n_{2p}\alpha^* + m_{2p}\beta^* + (w_{2p}(x), \alpha)\alpha^*. \end{aligned}$$

Consequently

$$\begin{aligned} n_{2p+1} + m_{2p+1} &= n_{2p} + m_{2p} + \big((s_\beta s_\alpha)^p(x), \alpha\big) \\ &= (x, u_{2p}) + \big(x, (s_\alpha s_\beta)^p(\alpha)\big) \\ &= (x, u_{2p+1}). \end{aligned}$$

□

Corollary 4.3. [62] *Let $x \in V^*$. For every $w \in W$, put*

$$x - w(x) = a_w \alpha^* + b_w \beta^*.$$

If $(x, \alpha) \geq 0$ and $(x, \beta) \geq 0$, then

$$a_w + b_w \geq 0.$$

If moreover $(x, \alpha) > 0$ (respectively $(x, \beta) > 0$), then

$$a_w + b_w > 0$$

for $w \neq 1, s_\beta$ (respectively for $w \neq 1, s_\alpha$).

[60] Editors' note: The original was corrected here to agree with the convention $\ell(\alpha) \leq \ell(\beta)$ of 4.0.

[61] Editors' note: In view of the preceding editor's note, this follows immediately from Propositions 3.5.1 and 1.1.9: one has

$$n_k + m_k = (x - w_k(x), \rho) = (x, \rho - w_k^{-1}(\rho)) = (x, u_k).$$

[62] Editors' note: The statement was corrected.

Proof. This follows at once from Proposition 4.1 and Lemma 4.2. □

Corollary 4.4. *Let $\mathcal{R}$ be any root datum, and let Δ be a system of simple roots. Let γ be a positive root, let α, β be two simple roots, and let $W_{\alpha,\beta}$ be the subgroup of W generated by s_α and s_β. If*

$$\mathrm{ord}_\Delta(s_\alpha(\gamma)) < \mathrm{ord}_\Delta(\gamma), \qquad \mathrm{ord}_\Delta(s_\beta(\gamma)) \leq \mathrm{ord}_\Delta(\gamma),$$

then, for every $w \in W_{\alpha,\beta}$,

$$\mathrm{ord}_\Delta(w(\gamma)) \leq \mathrm{ord}_\Delta(\gamma).$$

Moreover,

$$\mathrm{ord}_\Delta(w(\gamma)) < \mathrm{ord}_\Delta(\gamma)$$

if $w \neq 1, s_\beta$.

Proof. Consider the dual root datum $\mathcal{R}^*$, and then the datum

$$(M^*, M, R'^*, R'),$$

where R' is the set of roots that are rational linear combinations of α and β. Applying Corollary 4.3 to this datum gives the stated result. [63] □

5 The Weyl group: generators and relations

Let $\mathcal{R}$ be a root datum. Since the Weyl group is the same for this datum and for the corresponding reduced datum, we may assume $\mathcal{R}$ reduced when studying the Weyl group.

Let

$$\Delta = \{\alpha_1, \ldots, \alpha_n\}$$

be a system of simple roots, where $n = \mathrm{rgss}(\mathcal{R})$. Let n_{ij} be the order of $s_{\alpha_i} s_{\alpha_j}$ in W. In particular $n_{ii} = 1$, and Remarks 3.4.2 and 3.4.3 show that the subgroup W_{ij} of W generated by s_{α_i} and s_{α_j} is defined by

$$s_{\alpha_i}^2 = s_{\alpha_j}^2 = (s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

Theorem 5.1. *The group W is the group generated by the elements*

$$s_{\alpha_i}, \qquad i = 1, 2, \ldots, n,$$

subject to the relations

$$(s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

Proof. We have already seen that the theorem is true when $n = 2$; we shall use this fact in the proof. Introduce the group $\widetilde{W}$ generated by elements

$$T_i, \qquad i = 1, 2, \ldots, n,$$

subject to the relations

$$(T_i T_j)^{n_{ij}} = 1.$$

In particular $n_{ii} = 1$, whence $T_i^2 = 1$. Let

$$p : \widetilde{W} \longrightarrow W$$

be the group homomorphism sending T_i to s_{α_i}. We know that p is surjective; we shall prove that it is injective. □

[63] Editors' note: The hypotheses say that $(\gamma, \alpha^*) > 0$ and $(\gamma, \beta^*) \geq 0$. Consequently $\gamma - w(\gamma) \in \mathbb{N}\alpha + \mathbb{N}\beta$ for every $w \in W_{\alpha,\beta}$, and it is nonzero if $w \notin \{1, s_\beta\}$.

Lemma 5.2. *For every $\alpha \in \mathscr{P}(\Delta)$, one can define uniquely an element $T_\alpha \in \widetilde{W}$ with the following properties:*

(i) $p(T_\alpha) = s_\alpha$;

(ii) $T_{\alpha_i} = T_i$;

(iii) if β and α are positive roots such that $s_{\alpha_i}(\alpha) = \beta$, then

$$T_i T_\alpha T_i = T_\beta.$$

Proof. First observe that it follows from 1.2.10 and 3.3.6 that (i) is a consequence of (ii) and (iii), and that (ii) and (iii) determine the T_α completely. We construct them by induction on $\mathrm{ord}_\Delta(\alpha)$. If $\mathrm{ord}_\Delta(\alpha) = 1$, then $\alpha \in \Delta$, and we put $T_\alpha = T_i$ if $\alpha = \alpha_i$.

Consider the hypothesis

(H_p) there exist T_α, for $\alpha \in \mathscr{P}(\Delta)$ with $\mathrm{ord}_\Delta(\alpha) \leq p$, satisfying (ii), and satisfying (iii) whenever $\mathrm{ord}_\Delta(\alpha) \leq p$ and $\mathrm{ord}_\Delta(\beta) \leq p$.

It holds for $p = 1$. Indeed, if α and $s_{\alpha_i}(\alpha) = \beta$ are simple, then α_i and α are orthogonal. Thus, if $\alpha_j = \alpha = \beta$, one has $n_{ij} = 2$, whence

$$T_i T_j T_i = T_j.$$

Suppose $p > 1$ and (H_{p-1}) holds.

A) Construction of the T_α for $\mathrm{ord}_\Delta(\alpha) \leq p$. It is enough to do this when $\mathrm{ord}_\Delta(\alpha) = p$. There exists $\alpha_i \in \Delta$ such that

$$s_{\alpha_i}(\alpha) \in \mathscr{P}(\Delta), \qquad \mathrm{ord}_\Delta(s_{\alpha_i}(\alpha)) < p$$

by Lemma 3.3.3. Put

$$T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i. \tag{$\star$}$$

We verify that T_α depends only on α. Let $\alpha_j \in \Delta$ also satisfy

$$s_{\alpha_j}(\alpha) \in \mathscr{P}(\Delta), \qquad \mathrm{ord}_\Delta(s_{\alpha_j}(\alpha)) < p.$$

We must prove

$$T_i T_{s_{\alpha_i}(\alpha)} T_i = T_j T_{s_{\alpha_j}(\alpha)} T_j. \tag{+}$$

We distinguish two cases.

(1) Suppose that α is a linear combination of α_i and α_j. The same is then true of $s_{\alpha_i}(\alpha)$ and $s_{\alpha_j}(\alpha)$; by (H_{p-1}), the elements $T_{s_{\alpha_i}(\alpha)}$ and $T_{s_{\alpha_j}(\alpha)}$ are words in T_i, T_j. Since the projection of (+) to W holds and the theorem is true for $n = 2$, the restriction of p to the subgroup of $\widetilde{W}$ generated by T_i, T_j is injective. Thus (+) holds.

(2) Suppose that α is not a linear combination of α_i and α_j. Then, for $w \in W_{\alpha_i,\alpha_j}$, all the $w(\alpha)$ are positive (cf. Lemma 3.4.9). The relation to be verified can also be written

$$(T_i T_j)^{n_{ij}-1} T_{s_{\alpha_i}(\alpha)} (T_j T_i)^{n_{ij}-1} = T_{s_{\alpha_j}(\alpha)}. \tag{++}$$

Corollary 4.4 shows that all the $w(\alpha)$ have order $< p$ for $w \in W_{\alpha_i,\alpha_j}$, $w \neq 1$. We may therefore apply (H_{p-1}) $2(n_{ij} - 1)$ times, which proves the relation.

B) Verification of (H_p)**.** [64] We must verify that, if $\alpha_j \in \Delta$ and $\beta = s_{\alpha_j}(\alpha)$ satisfies $\mathrm{ord}_\Delta(\beta) \leq p$, then

$$T_j T_\alpha T_j = T_\beta.$$

If $\mathrm{ord}_\Delta(\beta) < p$, this follows from the preceding construction, since $T_j^2 = 1$. We may therefore suppose

$$\mathrm{ord}_\Delta(\beta) = p = \mathrm{ord}_\Delta(\alpha).$$

In this case α and α_j are orthogonal, so $\beta = \alpha$, and it remains to prove

$$T_j T_\alpha T_j = T_\alpha. \tag{$\dagger$}$$

By $(\star)$,

$$T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i,$$

so it remains to verify

$$T_j T_i T_{s_{\alpha_i}(\alpha)} T_i T_j = T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i. \tag{+++}$$

□

Proof of Lemma 5.2, continued. Put $m = n_{ij}$, $s_i = s_{\alpha_i}$, and $s_j = s_{\alpha_j}$. We have

$$T_j T_i = (T_i T_j)^{m-1}.$$

By Corollary 4.4, $\mathrm{ord}_\Delta(w(\alpha)) < p$ for every $w \in W_{ij}$ distinct from

$$1 = (s_i s_j)^m \quad \text{and} \quad s_j = s_i (s_i s_j)^{m-1}.$$

The induction hypothesis therefore gives

$$\begin{aligned} &T_j (T_i T_j)^{m-2} T_{s_i(\alpha)} (T_j T_i)^{m-2} T_j \\ &\qquad = T_{s_j (s_i s_j)^{m-2} s_i(\alpha)} = T_{s_i s_j(\alpha)} = T_{s_i(\alpha)}. \end{aligned}$$

The last equality holds because $s_j(\alpha) = \alpha$. Hence

$$(T_i T_j)^{m-1} T_{s_i(\alpha)} (T_j T_i)^{m-1} = T_i T_{s_i(\alpha)} T_i,$$

which proves (+++). □

Lemma 5.3. *Let $h \in \widetilde{W}$. Write*

$$h = T_{\alpha_1} \cdots T_{\alpha_m},$$

where the $\alpha_i \in \Delta$, not necessarily distinct, are chosen so that m is minimal. Then

$$p(h)(\alpha_m) \in -\mathscr{P}(\Delta).$$

Proof. Since

$$p(T_{\alpha_m})(\alpha_m) = s_{\alpha_m}(\alpha_m) = -\alpha_m,$$

if $p(h)(\alpha_m)$ were positive, there would be an index k, $1 \leq k \leq m-1$, such that

$$u = s_{\alpha_{k+1}} \cdots s_{\alpha_m}(\alpha_m) = -s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) \in -\mathscr{P}(\Delta)$$

and $s_{\alpha_k}(u) \in \mathscr{P}(\Delta)$. Necessarily $u = -\alpha_k$, by Lemma 3.3.1, whence

$$s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) = \alpha_k.$$

Property (iii) of Lemma 5.2 then gives

$$T_{\alpha_k} T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}} T_{\alpha_m} = T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}},$$

contradicting the minimality of m. □

[64] Editors' note: The argument in the original has been expanded in what follows.

Let now $h \in \widetilde{W}$ satisfy

$$p(h)(\mathscr{P}(\Delta)) \subset \mathscr{P}(\Delta).$$

Lemma 5.3 gives $p(h) = 1$. This proves Theorem 5.1 and also the following two corollaries.

Corollary 5.4. *If R_+ is a system of positive roots and $w \in W$ satisfies $w(R_+) = R_+$, then $w = 1$.*

Corollary 5.5. *The Weyl group acts freely and transitively on the set of systems of positive roots, respectively on the set of systems of simple roots, respectively on the set of Weyl chambers.*

Choose a system of simple roots Δ, and put

$$R^+ = \mathscr{P}(\Delta).$$

[65] For every pair of simple roots $(\alpha, \beta) \in \Delta \times \Delta$, let $R_{\alpha,\beta}$ be the set of roots that are linear combinations of α and β. Put

$$R^+_{\alpha,\beta} = R^+ \cap R_{\alpha,\beta},$$

and let $W_{\alpha,\beta}$ be the Weyl group of $R_{\alpha,\beta}$, that is, the subgroup of W generated by s_α and s_β.

Theorem 5.6 (Tits)**.** *Let α and β be two simple roots, and let $w \in W$ satisfy $w(\alpha) = \beta$. There exist a sequence of simple roots $\alpha_0, \dots, \alpha_m$ and a sequence $w_0, \dots, w_{m-1}$ of elements of W satisfying:*

(i) $\alpha_0 = \alpha$ *and* $\alpha_m = \beta$;

(ii) $w = w_{m-1} w_{m-2} \cdots w_0$;

(iii) $w_i(\alpha_i) = \alpha_{i+1}$ *for* $0 \le i \le m-1$;

(iv) for every i, $0 \le i \le m-1$, such that $\alpha_i \ne \alpha_{i+1}$, one has $w_i \in W_{\alpha_i,\alpha_{i+1}}$;

(v) for every i, $0 \le i \le m-1$, such that $\alpha_i = \alpha_{i+1}$, there exists a simple root β_i such that $w_i \in W_{\alpha_i,\beta_i}$.

Proof. Put[66]

$$M(w) = \mathrm{Card}(R^+ \cap w^{-1}(-R^+)) = \mathrm{Card}\{\gamma \in R^+ \mid w(\gamma) \in -R^+\}.$$

If $M(w) = 0$, then $w(R^+) = R^+$, so $w = 1$ by Corollary 5.4; the theorem is then trivial: $m = 0$, and assertions (iii)–(v) are empty.

We argue by induction on $M(w)$. Suppose $M(w) > 0$. There exists $\beta_0 \in \Delta$ such that $w(\beta_0) \in -R^+$. Put $\alpha_0 = \alpha$, and consider

$$A = w^{-1}(R^+) \cap R_{\alpha_0,\beta_0}.$$

This is a system of positive roots of R_{α_0,β_0}. Hence there exists $w_0 \in W_{\alpha_0,\beta_0}$ such that

$$w_0^{-1}(R^+_{\alpha_0,\beta_0}) = A.$$

[65] Editors' note: The superscript notation R^+, rather than R_+, is used here so that $R^+_{\alpha,\beta} = R_{\alpha,\beta} \cap R^+$.

[66] Editors' note: In what follows, the original argument has been expanded, and equality (1) has been corrected.

Put $w' = ww_0^{-1}$. Lemma 3.4.9 gives

$$R^+ - R^+_{\alpha_0,\beta_0} = w_0(R^+ - R^+_{\alpha_0,\beta_0}),$$

and therefore

$$\begin{aligned}(R^+ - R^+_{\alpha_0,\beta_0}) &\cap w^{-1}(-R^+)\\ &= w_0\Big((R^+ - R^+_{\alpha_0,\beta_0}) \cap w'^{-1}(-R^+)\Big).\end{aligned} \tag{1}$$

On the other hand,

$$\beta_0 \in R^+_{\alpha_0,\beta_0} \cap w^{-1}(-R^+). \tag{2}$$

Since $w_0(R_{\alpha_0,\beta_0}) = R_{\alpha_0,\beta_0}$, one has

$$w_0(-A) = R_{\alpha_0,\beta_0} \cap w'^{-1}(-R^+),$$

and hence

$$\begin{aligned}R^+_{\alpha_0,\beta_0} \cap w'^{-1}(-R^+) &= R^+_{\alpha_0,\beta_0} \cap w_0(-A)\\ &= R^+_{\alpha_0,\beta_0} \cap (-R^+_{\alpha_0,\beta_0}) = \varnothing.\end{aligned} \tag{2$'$}$$

It follows from (1), (2), and (2′) that

$$M(w') < M(w).$$

Put $\alpha_1 = w_0(\alpha_0)$. We show that $\alpha_1 \in \Delta$, or equivalently that $\alpha_0 \in w_0^{-1}(\Delta)$. Since $w(\alpha_0) \in \Delta$, we have

$$\alpha_0 \in w^{-1}(\Delta) \cap R_{\alpha_0,\beta_0}.$$

Thus α_0 is a simple root of

$$A = w^{-1}(R^+) \cap R_{\alpha_0,\beta_0} = w_0^{-1}(R^+_{\alpha_0,\beta_0}),$$

and consequently belongs to

$$w_0^{-1}(\Delta \cap R^+_{\alpha_0,\beta_0}) = w_0^{-1}\{\alpha_0, \beta_0\}$$

by Lemma 3.4.8. Hence $\alpha_1 = w_0(\alpha_0)$ equals α_0 or β_0. If $\alpha_1 \neq \alpha_0$, then $\alpha_1 = \beta_0$ and $w_0 \in W_{\alpha_0,\alpha_1}$.

Finally,

$$\beta = w'(\alpha_1) \qquad \text{and} \qquad M(w') < M(w),$$

so the induction hypothesis completes the proof. □

6 Morphisms of root data

6.1 Definition

Let

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two root data. Let $f : M' \to M$ be a linear map and ${}^t f : M^* \to M'^*$ its transpose.

Definition 6.1.1. The map f is called a *morphism* from $\mathcal{R}'$ to $\mathcal{R}$, and is denoted

$$f : \mathcal{R}' \longrightarrow \mathcal{R},$$

if f induces a bijection from R' onto R, and ${}^t f$ induces a bijection from R^* onto R'^*.

Then ${}^t f$ is a morphism of the dual root data:

$$ {}^t f : \mathcal{R}^* \longrightarrow \mathcal{R}'^*. $$

If $f : \mathcal{R}' \to \mathcal{R}$ is a morphism and $\alpha = f(\alpha')$ for $\alpha' \in R'$, then

$$ \alpha'^* = {}^t f(\alpha^*). $$

Indeed, if p and p' denote the maps of 1.2.1 for $\mathcal{R}$ and $\mathcal{R}'$, respectively, then

$$ p' = {}^t f \circ p \circ f, $$

which gives the assertion at once. The reader may verify the statements below, almost all of which are immediate.

Proposition 6.1.2. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism of root data. If $\alpha' \in R'$ and $\alpha = f(\alpha')$, then*

$$ \alpha'^* = {}^t f(\alpha^*). $$

Moreover, f induces isomorphisms

$$ R' \xrightarrow{\sim} R, \qquad \Gamma_0(R') \xrightarrow{\sim} \Gamma_0(R), \qquad \mathcal{V}(R') \xrightarrow{\sim} \mathcal{V}(R), $$

and ${}^t f$ induces isomorphisms

$$ R^* \xrightarrow{\sim} R'^*, \qquad \Gamma_0(R^*) \xrightarrow{\sim} \Gamma_0(R'^*), \qquad \mathcal{V}(R^*) \xrightarrow{\sim} \mathcal{V}(R'^*). $$

The last is the transpose of the corresponding morphism induced by f. The map

$$ s_{\alpha'} \longmapsto s_{f(\alpha')} $$

extends to an isomorphism

$$ W(\mathcal{R}') \xrightarrow{\sim} W(\mathcal{R}) $$

compatible with the actions of these two groups on the sets of 1.1.13.

Proposition 6.1.3. *The maps*

$$ \Delta' \longmapsto f(\Delta'), \qquad R'_+ \longmapsto f(R'_+), \qquad C' \longmapsto ({}^t f \otimes \mathbb{R})^{-1}(C') $$

define bijective correspondences between the systems of simple roots, the systems of positive roots, and the Weyl chambers for $\mathcal{R}'$ and $\mathcal{R}$. These correspondences are compatible with the actions of the Weyl groups and with the correspondences

$$ \mathscr{S}(R_+) \longleftrightarrow R_+ \longleftrightarrow \mathscr{C}(R_+). $$

Lemma 6.1.4. *Morphisms compose. A morphism $f : \mathcal{R}' \to \mathcal{R}$ is an isomorphism if and only if the linear map $f : M' \to M$ is bijective.*

6.2 Isogenies

Definition 6.2.1. A morphism $f : \mathcal{R}' \to \mathcal{R}$ of root data is called an *isogeny* if $f : M' \to M$ is injective with finite cokernel.

If f is an isogeny, then ${}^t f$ is an isogeny.

Definition 6.2.2. Let $f : \mathcal{R}' \to \mathcal{R}$ be an isogeny. Put

$$K(f) = \operatorname{Coker}(M' \xrightarrow{f} M).$$

Lemma 6.2.3. *There is a natural pairing*

$$K(f) \times K({}^t f) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

that puts these two finite groups in duality.

Proof. This is classical. □

Lemma 6.2.4. *If $f : \mathcal{R}' \to \mathcal{R}$ is a morphism, then*

$$\operatorname{rgss}(\mathcal{R}') = \operatorname{rgss}(\mathcal{R}).$$

If f is moreover an isogeny, then

$$\operatorname{rgred}(\mathcal{R}') = \operatorname{rgred}(\mathcal{R}).$$

Proof. This is immediate. □

Lemma 6.2.5. *Every morphism of semisimple root data is an isogeny.*

Proof. The map f must induce an isomorphism from $V' = \mathcal{V}(R')$ onto $V = \mathcal{V}(R)$. □

If $\mathcal{R}'$ and $\mathcal{R}$ are semisimple, every isogeny $f : \mathcal{R}' \to \mathcal{R}$ defines a commutative diagram:

$$\begin{tikzcd}
\Gamma_0(R') \arrow[r, "\sim"] \arrow[d] & \Gamma_0(R) \arrow[d] \\
M' \arrow[r, "f"'] & M
\end{tikzcd}$$

If $M = \Gamma_0(R)$, then f is necessarily an isomorphism.

Definition 6.2.6. A root datum is called *adjoint*, respectively *simply connected*, if

$$M = \Gamma_0(R), \qquad \text{respectively} \qquad M^* = \Gamma_0(R^*).$$

An adjoint or simply connected root datum is therefore semisimple. Moreover, $\mathcal{R}$ is adjoint, respectively simply connected, if and only if $\mathcal{R}^*$ is simply connected, respectively adjoint. The preceding result gives the following proposition.

Proposition 6.2.7. *Let $\mathcal{R}$ be a semisimple root datum. The following conditions are equivalent:*

(i) $\mathcal{R}$ is adjoint, respectively simply connected;

(ii) every isogeny $\mathcal{R}' \to \mathcal{R}$, respectively $\mathcal{R} \to \mathcal{R}'$, is an isomorphism.

Proposition 6.2.8. *Let $\mathcal{R}$ be an adjoint, respectively simply connected, root datum. Every indivisible root, respectively coroot, is an indivisible element of M, respectively of M^*.*

Proof. Every indivisible root belongs to a basis of $\Gamma_0(R)$, by 3.3.5. □

6.3 Radical and coradical

Let $\mathcal{R}$ be a root datum. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha^* \in R^*\}, \qquad N^* = M^*/\big(\mathcal{V}(R^*) \cap M^*\big).$$

Lemma 6.3.1. *Consider the canonical morphisms*

$$N \longrightarrow M, \qquad M^* \longrightarrow N^*.$$

They are transposes of one another, and N^* is identified with the dual of N.

Proof. This follows immediately from 1.2.5. □

Definition 6.3.2. The *coradical* of $\mathcal{R}$, denoted $\operatorname{corad}(\mathcal{R})$, is the trivial root datum

$$\operatorname{corad}(\mathcal{R}) = (N, N^*, \varnothing, \varnothing).$$

If

$$\mathcal{R}^0 = (M, M^*, \varnothing, \varnothing),$$

which is a trivial root datum, there is therefore a morphism

$$\operatorname{corad}(\mathcal{R}) \longrightarrow \mathcal{R}^0.$$

Definition 6.3.3. The *radical* of $\mathcal{R}$, denoted $\operatorname{rad}(\mathcal{R})$, is the trivial root datum [67]

$$\operatorname{rad}(\mathcal{R}) = \operatorname{corad}(\mathcal{R}^*)^*.$$

Thus there is a diagram

$$\begin{tikzcd} \operatorname{corad}(\mathcal{R}) \arrow[rr] \arrow[dr] & & \operatorname{rad}(\mathcal{R}) \\ & \mathcal{R}^0 \arrow[ur] & \end{tikzcd}$$

whose transpose is the corresponding diagram for $\mathcal{R}^*$.

Lemma 6.3.4. *The canonical morphism*

$$u : \operatorname{corad}(\mathcal{R}) \longrightarrow \operatorname{rad}(\mathcal{R})$$

is an isogeny.

Definition 6.3.5. Put

$$N(\mathcal{R}) = K(u) = M \Big/ \left((\mathcal{V}(R) \cap M) + \bigcap_{\alpha \in R} \ker(\alpha^*) \right).$$

There is then a canonical pairing

$$N(\mathcal{R}) \times N(\mathcal{R}^*) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

[67] Editors' note: Equivalently, if

$$P = \{y \in M^* \mid (y, \alpha) = 0 \text{ for every } \alpha \in R\}, \qquad P^* = M/\big(\mathcal{V}(R) \cap M\big),$$

then $\operatorname{rad}(\mathcal{R}) = (P^*, P, \varnothing, \varnothing)$.

Lemma 6.3.6. *One has*

$$\operatorname{rgred}\bigl(\operatorname{rad}(\mathcal{R})\bigr) = \operatorname{rgred}\bigl(\operatorname{corad}(\mathcal{R})\bigr) = \operatorname{rgred}(\mathcal{R}) - \operatorname{rgss}(\mathcal{R}),$$

and the following conditions are equivalent:

(i) $\mathcal{R}$ *is semisimple;*

(ii) $\operatorname{rad}(\mathcal{R}) = 0$*;*

(iii) $\operatorname{corad}(\mathcal{R}) = 0$.

6.4 Products of root data

Definition 6.4.1. Let

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two root data. The *product root datum* of $\mathcal{R}$ and $\mathcal{R}'$, denoted

$$\mathcal{R}'' = \mathcal{R} \times \mathcal{R}',$$

is the root datum (M'', M''^*, R'', R''^*), where

$$\begin{aligned} M'' &= M \times M', & M''^* &= M^* \times M'^*, \\ R'' &= (R \times 0) \cup (0 \times R'), & R''^* &= (R^* \times 0) \cup (0 \times R'^*), \end{aligned}$$

and the map $\alpha \mapsto \alpha^*$ is the evident one.

Proposition 6.4.2. *Under the preceding hypotheses there are canonical isomorphisms*

$$\Gamma_0(R'') \simeq \Gamma_0(R) \times \Gamma_0(R'), \qquad \mathcal{V}(R'') \simeq \mathcal{V}(R) \times \mathcal{V}(R'),$$

$$W(\mathcal{R}'') \simeq W(\mathcal{R}) \times W(\mathcal{R}'),$$

and similarly for the other associated constructions. Moreover,

$$\operatorname{rgred}(\mathcal{R}'') = \operatorname{rgred}(\mathcal{R}) + \operatorname{rgred}(\mathcal{R}'), \qquad \operatorname{rgss}(\mathcal{R}'') = \operatorname{rgss}(\mathcal{R}) + \operatorname{rgss}(\mathcal{R}').$$

There is also a canonical isomorphism of root data

$$(\mathcal{R} \times \mathcal{R}')^* \simeq \mathcal{R}^* \times \mathcal{R}'^*.$$

The preceding definitions extend immediately to products of any finite number of factors.

Proposition 6.4.3. *Let*

$$\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2 \times \cdots \times \mathcal{R}_n$$

be a product of root data. The following conditions are equivalent:

(i) $\mathcal{R}$ *is semisimple, respectively simply connected, respectively adjoint, respectively reduced;*

(ii) every $\mathcal{R}_i$ *is semisimple, respectively simply connected, respectively adjoint, respectively reduced.*

Consider the following special case. Let $\mathcal{R}_0$ be a trivial root datum and $\mathcal{R}_1$ a semisimple root datum. There is then a commutative diagram

$$\begin{tikzcd}[column sep=small] & \mathcal{R}_0 \times \mathcal{R}_1 \arrow[dr] & \\ \mathcal{R}_1 \arrow[ur] \arrow[rr, "\mathrm{id}"] & & \mathcal{R}_1 \end{tikzcd}$$

Lemma 6.4.4. *There are canonical isomorphisms*

$$\begin{tikzcd}[column sep=small] \operatorname{corad}(\mathcal{R}_0 \times \mathcal{R}_1) \arrow[rr, "\sim"] \arrow[dr, "\sim"'] & & \operatorname{rad}(\mathcal{R}_0 \times \mathcal{R}_1) \\ & \mathcal{R}_0 \arrow[ur, "\sim"'] & \end{tikzcd}$$

In particular,

$$N(\mathcal{R}_0 \times \mathcal{R}_1) = 0.$$

We shall see later that, conversely, if $N(\mathcal{R}) = 0$, then $\mathcal{R}$ is the product of a semisimple root datum and a trivial root datum.

6.5 Induced and coinduced root data

Let $\mathcal{R} = (M, M^*, R, R^*)$ be a root datum, and let $N \subset M$ be a subgroup containing the roots, that is,

$$\Gamma_0(R) \subset N \subset M.$$

The canonical linear map $i_N : N \to M$ gives, by transposition, a linear map

$${}^t i_N : M^* \longrightarrow N^*.$$

Put

$$R_N = R, \qquad R_N^* = {}^t i_N(R^*).$$

Lemma 6.5.1. *The quadruple*

$$\mathcal{R}_N = (N, N^*, R_N, R_N^*)$$

is a root datum, and i_N is a morphism.

Proof. First, ${}^t i_N$ induces an isomorphism from R^* onto R_N^*. Indeed, if $\alpha, \beta \in R$ and

$${}^t i_N(\alpha^*) = {}^t i_N(\beta^*),$$

then

$$(\alpha^*, x) = (\beta^*, x)$$

for every $x \in N$, in particular for $x \in R$. Hence $\alpha = \beta$ by 1.1.4. The rest follows without difficulty. □

Definition 6.5.2. The root datum $\mathcal{R}_N$ is called the root datum *induced by $\mathcal{R}$ on N*.

Lemma 6.5.3. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism, and put*

$$N = f(M') \subset M.$$

Then f factors uniquely through i_N.

In particular, up to isomorphism, the isogenies $\mathcal{R}' \to \mathcal{R}$ correspond bijectively to the finite-index subgroups of
$$M/\Gamma_0(R).$$
This sharpens 6.2.7.

Now let $N^* \subset M^*$ be a subgroup containing R^*. Define the root datum *coinduced by* $\mathcal{R}$ *on* N^* by
$$\mathcal{R}^{N^*} = (\mathcal{R}^*_{N^*})^*.$$
There is a canonical morphism
$$p^{N^*} : \mathcal{R} \longrightarrow \mathcal{R}^{N^*}.$$

Lemma 6.5.4. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism. There exist subgroups $N \subset M$ and $N'^* \subset M'^*$ such that f factors as*

$$\begin{tikzcd} \mathcal{R}' \arrow[r, "f"] \arrow[d, "p^{N'^*}"'] & \mathcal{R} \\ \mathcal{R}'^{N'^*} \arrow[r, "f_0", "\sim"'] & \mathcal{R}_N \arrow[u, "i_N"'] \end{tikzcd}$$

where f_0 is an isomorphism.

Proof. Take $N = f(M')$, as in 6.5.3. The resulting morphism $M' \to N$ is surjective, so its transpose is injective. Take its image for N'^*. □

We now treat some special cases. If $N = \Gamma_0(R)$, write
$$\mathcal{R}_N = \mathrm{ad}(\mathcal{R}).$$
If $N = \mathcal{V}(R) \cap M$, write
$$\mathcal{R}_N = \mathrm{ss}(\mathcal{R}).$$
Thus there is a diagram
$$\mathrm{ad}(\mathcal{R}) \longrightarrow \mathrm{ss}(\mathcal{R}) \longrightarrow \mathcal{R}.$$
Put
$$\mathrm{der}(\mathcal{R}) = \mathrm{ss}(\mathcal{R}^*)^*, \qquad \mathrm{sc}(\mathcal{R}) = \mathrm{ad}(\mathcal{R}^*)^*.$$
By duality there is a diagram
$$\mathcal{R} \longrightarrow \mathrm{der}(\mathcal{R}) \longrightarrow \mathrm{sc}(\mathcal{R}).$$

Proposition 6.5.5. *(i) In the first row of the diagram*

$$\begin{tikzcd} \mathrm{ad}(\mathcal{R}) \arrow[r] & \mathrm{ss}(\mathcal{R}) \arrow[rr] \arrow[dr] & & \mathrm{der}(\mathcal{R}) \arrow[r] & \mathrm{sc}(\mathcal{R}) \\ & & \mathcal{R} \arrow[ur] & & \end{tikzcd}$$

the four root data are semisimple and the three morphisms are isogenies.

(ii) The datum $\mathrm{ad}(\mathcal{R})$ *is adjoint, and* $\mathcal{R}$ *is adjoint if and only if* $\mathrm{ad}(\mathcal{R}) \to \mathcal{R}$ *is an isomorphism.*

(iii) *The datum* $\mathrm{sc}(\mathcal{R})$ *is simply connected, and* $\mathcal{R}$ *is simply connected if and only if* $\mathcal{R} \to \mathrm{sc}(\mathcal{R})$ *is an isomorphism.*

(iv) *The following conditions are equivalent:*

   (a) $\mathcal{R}$ *is semisimple;*

   (b) $\mathrm{ss}(\mathcal{R}) \to \mathcal{R}$ *is an isomorphism;*

   (c) $\mathcal{R} \to \mathrm{der}(\mathcal{R})$ *is an isomorphism.*

Consider the morphism

$$\mathrm{ss}(\mathcal{R}) \longrightarrow \mathrm{der}(\mathcal{R}).$$

The definitions immediately give the following result.

Lemma 6.5.6. *Let*

$$h : \mathrm{ss}(\mathcal{R}) \longrightarrow \mathrm{der}(\mathcal{R})$$

be the canonical isogeny. Then

$$K(h) \simeq N(\mathcal{R}).$$

Consider other special cases of induced root data. Put

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for every } \alpha \in R\} \times \Gamma_0(R).$$

The sum is direct by 1.2.5. It follows that $\mathcal{R}_N$ is identified with

$$\mathrm{ad}(\mathcal{R}) \times \mathrm{corad}(\mathcal{R}).$$

One may replace $\Gamma_0(R)$ by $\mathcal{V}(R) \cap M$, and then dualize both constructions. This gives the following diagram of root data:

$$\begin{tikzcd}
\mathrm{ad}(\mathcal{R}) \arrow[r] \arrow[d] & \mathrm{ss}(\mathcal{R}) \arrow[rr] \arrow[d] & & \mathrm{der}(\mathcal{R}) \arrow[r] \arrow[d] & \mathrm{sc}(\mathcal{R}) \arrow[d] \\
\mathrm{ad}(\mathcal{R}) \times \mathrm{corad}(\mathcal{R}) \arrow[r] \arrow[d] & \mathrm{ss}(\mathcal{R}) \times \mathrm{corad}(\mathcal{R}) \arrow[r] \arrow[d] & \mathcal{R} \arrow[r] & \mathrm{der}(\mathcal{R}) \times \mathrm{rad}(\mathcal{R}) \arrow[r] \arrow[d] & \mathrm{sc}(\mathcal{R}) \times \mathrm{rad}(\mathcal{R}) \arrow[d] \\
\mathrm{ad}(\mathcal{R}) \arrow[r] & \mathrm{ss}(\mathcal{R}) \arrow[rr] & & \mathrm{der}(\mathcal{R}) \arrow[r] & \mathrm{sc}(\mathcal{R})
\end{tikzcd}$$

It is commutative, as is checked at once, and is self-dual in the evident sense. The horizontal morphisms are isogenies, and the composites of the vertical arrows are the identity.

Lemma 6.5.7. *Let* h_1 *and* h_2 *be the canonical isogenies*

$$\mathrm{ss}(\mathcal{R}) \times \mathrm{corad}(\mathcal{R}) \xrightarrow{h_1} \mathcal{R} \xrightarrow{h_2} \mathrm{der}(\mathcal{R}) \times \mathrm{rad}(\mathcal{R}).$$

Then

$$K(h_1) \simeq K(h_2) \simeq N(\mathcal{R}).$$

Proof. This is immediate from the definitions. □

Corollary 6.5.8. *For a root datum* $\mathcal{R}$*, the following conditions are equivalent:*

(i) *$N(\mathcal{R}) = 0$, that is,* $\operatorname{corad}(\mathcal{R}) \to \operatorname{rad}(\mathcal{R})$ *is an isomorphism;*

(ii) *the canonical map* $\operatorname{ss}(\mathcal{R}) \to \operatorname{der}(\mathcal{R})$ *is an isomorphism;*

(iii) *h_1 is an isomorphism;*

(iv) *h_2 is an isomorphism;*

(v) *$\mathcal{R}$ is the product of a semisimple root datum and a trivial root datum.*

Corollary 6.5.9. *For every root datum $\mathcal{R}$, there are isogenies*

$$\operatorname{ad}(\mathcal{R}) \times \mathcal{R}_0 \longrightarrow \mathcal{R} \longrightarrow \operatorname{sc}(\mathcal{R}) \times \mathcal{R}_0,$$

where $\mathcal{R}_0$ is the trivial root datum of rank $\operatorname{rgred}(\mathcal{R}) - \operatorname{rgss}(\mathcal{R})$.

Lemma 6.5.10. *Let $\mathcal{R}$ be a root datum, let Δ be a system of simple roots, and let $\Delta' \subset \Delta$. Consider the root datum*

$$\mathcal{R}_{\Delta'} = (M, M^*, R_{\Delta'}, R^*_{\Delta'})$$

of 3.4.7.

(i) *If $\mathcal{R}$ is simply connected, then* $\operatorname{der}(\mathcal{R}_{\Delta'})$ *is simply connected.*

(ii) *If $\mathcal{R}$ is adjoint, then* $\operatorname{ss}(\mathcal{R}_{\Delta'})$ *is adjoint.*

Proof. The two assertions are equivalent by duality. The second reduces to

$$M \cap \mathcal{V}(R_{\Delta'}) = \Gamma_0(R_{\Delta'}).$$

When $M = \Gamma_0(R)$, both sides are the subgroup of M generated by Δ'. □

6.6 Weights

Definition 6.6.1. For a root datum $\mathcal{R}$, put [68]

$$\Lambda(\mathcal{R}) = \{x \in \mathcal{V}(R) \mid (\alpha^*, x) \in \mathbb{Z} \text{ for every } \alpha^* \in R^*\}.$$

The elements of $\Lambda(\mathcal{R})$ are called the *weights* of $\mathcal{R}$. The weights of $\mathcal{R}^*$ are called the *coweights* of $\mathcal{R}$.

One has

$$\Gamma_0(R) \subset \Lambda(\mathcal{R}),$$

and $\Lambda(\mathcal{R})$ is stable under $W(\mathcal{R})$.

Lemma 6.6.2. *The bilinear map*

$$V^* \times V \longrightarrow \mathbb{Q}$$

induces a duality

$$\Gamma_0(R^*) \times \Lambda(\mathcal{R}) \longrightarrow \mathbb{Z}.$$

Proof. Immediate. □

[68] Editors' note: The notation $\Gamma(\mathcal{R})$ has been replaced by $\Lambda(\mathcal{R})$ to avoid any possible confusion with $\Gamma_0(R)$.

Corollary 6.6.3. *Let*

$$\Delta^* = (\alpha_1^*, \dots, \alpha_n^*)$$

be a system of simple coroots, and let $p_i \in \mathcal{V}(R)$, $i = 1, \dots, n$, be defined by

$$(\alpha_i^*, p_j) = \delta_{ij}.$$

Thus

$$s_{\alpha_i}(p_i) = p_i - \alpha_i, \qquad s_{\alpha_i}(p_j) = p_j \quad (i \neq j).$$[69]

Then $\Lambda(\mathcal{R})$ is the free abelian group generated by the p_i.

The p_i are called the *fundamental weights* corresponding to the system of simple coroots Δ^*.

Corollary 6.6.4. *For every $\alpha^* \in \Delta^*$,*

$$\left(\alpha^*, \sum_i p_i\right) = 1.$$

Consequently,

$$\sum_i p_i = \rho_{R_+}, \qquad R_+ = \mathscr{P}(\mathrm{ind}(\Delta)),$$

in the notation of 3.5.1.

Corollary 6.6.5. *For every $x \in \mathcal{V}(R)$,*

$$x = \sum_i (\alpha_i^*, x) p_i.$$

Since

$$R^* \subset \Gamma_0(R^*), \qquad R \subset \Lambda(\mathcal{R}),$$

the quadruple

$$(\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*)$$

is a root datum.

Corollary 6.6.6. *The canonical morphism*

$$\Gamma_0(R^*) \longrightarrow M^*$$

is the transpose of

$$x \longmapsto \sum_i (\alpha_i^*, x) p_i.$$

This map defines a morphism of root data, and there is a commutative diagram

$$\begin{tikzcd}
\mathcal{R} \arrow[rr] \arrow[dr] & & \mathrm{sc}(\mathcal{R}) \arrow[dl, "\wr"'] \\
& (\Lambda(\mathcal{R}), \Gamma_0(R^*), R, R^*) &
\end{tikzcd}$$

Thus we have an explicit description of $\mathrm{sc}(\mathcal{R})$ in terms of the weights of $\mathcal{R}$. Likewise, one obtains a commutative diagram

[69] Caution: when the datum is not reduced, the α_i need not form a system of simple roots.

$$\begin{tikzcd}
\mathrm{ad}(\mathcal{R}) \arrow[rr] \arrow[d, "\wr"] & & \mathcal{R} \\
(\Gamma_0(R), \Lambda(\mathcal{R}^*), R, R^*) \arrow[urr] & &
\end{tikzcd}$$

Corollary 6.6.7. *The datum $\mathcal{R}$ is simply connected if and only if*

$$M = \Lambda(\mathcal{R}).$$

Remark 6.6.8. One has

$$\Lambda(\mathcal{R}) \cap M = \mathcal{V}(R) \cap M.$$

Consequently, $\mathcal{R}$ is semisimple if and only if

$$M \subset \Lambda(\mathcal{R}).$$

The results of §6.5 also give:

Corollary 6.6.9. *The datum $\mathcal{R}$ is the product of a simply connected datum and a trivial datum if and only if*

$$M \supset \Lambda(\mathcal{R}).$$

Consider now the canonical isogeny

$$f : \mathrm{ad}(\mathcal{R}) \longrightarrow \mathrm{sc}(\mathcal{R}),$$

and put $Z(\mathcal{R}) = K(f)$. One has

$$Z(\mathcal{R}) \simeq Z(\mathrm{sc}\,\mathcal{R}) \simeq Z(\mathrm{ad}\,\mathcal{R}).$$

Corollary 6.6.10. *There is a canonical isomorphism*

$$Z(\mathcal{R}) = \Lambda(\mathcal{R})/\Gamma_0(R).$$

More precisely, there is an exact sequence of $W(\mathcal{R})$-modules

$$0 \longrightarrow \Gamma_0(R) \longrightarrow \Lambda(\mathcal{R}) \longrightarrow Z(\mathcal{R}) \longrightarrow 0.$$

Corollary 6.6.11. *There is a canonical pairing*

$$Z(\mathcal{R}^*) \times Z(\mathcal{R}) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

which puts these groups in duality.

Remark 6.6.12. One has

$$Z(\mathcal{R} \times \mathcal{R}') \simeq Z(\mathcal{R}) \times Z(\mathcal{R}').$$

In particular, consider simply connected data $\mathcal{R}_i$, $i = 1, 2, \ldots, n$, and a trivial datum $\mathcal{R}_0$. Put

$$\mathcal{R} = \mathcal{R}_0 \times \mathcal{R}_1 \times \cdots \times \mathcal{R}_n.$$

If

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}_0 = (M_0, M_0^*, \varnothing, \varnothing),$$

then

$$M/\Gamma_0(R) \simeq M_0 \times Z(\mathcal{R}_1) \times \cdots \times Z(\mathcal{R}_n).$$

6.7 Automorphisms

By 6.1.4, an automorphism of $\mathcal{R}$ is an automorphism u of M such that

$$u(R) = R, \qquad {}^t u(R^*) = R^*.$$

In particular, every $w \in W(\mathcal{R})$ defines an automorphism of $\mathcal{R}$.

Lemma 6.7.1. *The group $W(\mathcal{R})$ is a normal subgroup of $\mathrm{Aut}(\mathcal{R})$. More precisely, if $u \in \mathrm{Aut}(\mathcal{R})$ and $\alpha \in R$, then*

$$u s_\alpha u^{-1} = s_{u(\alpha)}.$$

Proof. The proof is the same as that of 1.2.10. □

Proposition 6.7.2. *Let Δ be a system of simple roots, and put*

$$E_\Delta(\mathcal{R}) = \{u \in \mathrm{Aut}(\mathcal{R}) \mid u(\Delta) = \Delta\}.$$

Then $\mathrm{Aut}(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E_\Delta(\mathcal{R})$.

Proof. This follows at once from the fact that $W(\mathcal{R})$ acts simply transitively on the systems of simple roots and from the fact that, if Δ is a system of simple roots of $\mathcal{R}$, then $u(\Delta)$ is a system of simple roots for every automorphism u of $\mathcal{R}$. □

We shall later give a simpler description of $E_\Delta(\mathcal{R})$ in the case of reduced irreducible root data.[70]

Definition 6.7.3. We denote by $\mathrm{Aut}^s(\mathcal{R})$[71] the set of $u \in \mathrm{Aut}(\mathcal{R})$ for which the following diagram is commutative:

$$\begin{tikzcd} \mathcal{R} \arrow[rr, "u"] \arrow[dr] & & \mathcal{R} \arrow[dl] \\ & \mathrm{rad}(\mathcal{R}) & \end{tikzcd}$$

We write

$$E^s_\Delta(\mathcal{R}) = E_\Delta(\mathcal{R}) \cap \mathrm{Aut}^s(\mathcal{R}).$$

Remark 6.7.4. If $u \in \mathrm{Aut}(\mathcal{R})$, then $u \in \mathrm{Aut}^s(\mathcal{R})$ if and only if

$$(u - \mathrm{id})(M) \subset \mathcal{V}(R).$$

In particular,

$$W(\mathcal{R}) \subset \mathrm{Aut}^s(\mathcal{R}).$$

Proposition 6.7.5. *For every system of simple roots Δ, the group $\mathrm{Aut}^s(\mathcal{R})$ is the semidirect product of $W(\mathcal{R})$ by $E^s_\Delta(\mathcal{R})$.*

By functoriality, every automorphism of $\mathcal{R}$ induces an automorphism of $\mathrm{ad}(\mathcal{R})$. Thus there is a canonical homomorphism

$$\mathrm{Aut}(\mathcal{R}) \longrightarrow \mathrm{Aut}(\mathrm{ad}(\mathcal{R})).$$

[70] Editors' note: When $\mathcal{R}$ is simply connected or adjoint, see 7.4.5.

[71] Editors' note: The superscript s is intended to suggest "semisimple."

Lemma 6.7.6. *The homomorphism*

$$\mathrm{Aut}^s(\mathcal{R}) \longrightarrow \mathrm{Aut}(\mathrm{ad}(\mathcal{R}))$$

is injective.

Proof. Let u be an automorphism of M such that

$$(u - \mathrm{id})(M) \subset \mathcal{V}(R)$$

and

$$^t u(\alpha^*) = \alpha^* \qquad (\alpha^* \in R^*).$$

For every $x \in M$, one has

$$(\alpha^*, u(x) - x) = (^t u(\alpha^*) - \alpha^*, x) = 0.$$

Hence $u(x) - x = 0$, by 1.2.5. □

Lemma 6.7.7. *The group* $\mathrm{Aut}^s(\mathcal{R})$ *is finite.*

Proof. It suffices to prove that $\mathrm{Aut}(\mathcal{R})$ is finite when $\mathcal{R}$ is adjoint. In that case M is generated by R, so every automorphism of $\mathcal{R}$ is determined by the permutation it induces on R. □

Remark 6.7.8. It is immediate that $\mathrm{Aut}(\mathcal{R})$ (respectively $E_\Delta(\mathcal{R})$) is finite if and only if

$$\mathrm{rgred}(\mathcal{R}) - \mathrm{rgss}(\mathcal{R}) \leq 1.$$

6.8 p-morphisms of reduced root data

Throughout this subsection, p is a fixed positive integer.

Definition 6.8.1. Let

$$\mathcal{R} = (M, M^*, R, R^*), \qquad \mathcal{R}' = (M', M'^*, R', R'^*)$$

be two reduced root data. A group homomorphism

$$f : M' \longrightarrow M$$

is called a *p-morphism from $\mathcal{R}'$ to $\mathcal{R}$* if there exist a bijection

$$u : R \xrightarrow{\sim} R'$$

and a map

$$q : R \longrightarrow \{p^n \mid n \in \mathbb{N}\}$$

such that:

(i) $f(u(\alpha)) = q(\alpha)\alpha$ for every $\alpha \in R$;

(ii) $^t f(\alpha^*) = q(\alpha) u(\alpha)^*$ for every $\alpha \in R$.[72]

[72]Editors' note: These two conditions imply

$$q(\alpha)\big(u(\alpha)^*, u(\beta)\big) = q(\beta)(\alpha^*, \beta)$$

for all $\alpha, \beta \in R$.

Corollary 6.8.2. *A* 1*-morphism is precisely a morphism.*

Corollary 6.8.3. *The transpose of a* p*-morphism is a* p*-morphism.*

Lemma 6.8.4. *If* $w \in W(\mathcal{R})$ *and* $\alpha \in R$*, then*

$$q(w(\alpha)) = q(\alpha).$$

The map

$$s_\alpha \longmapsto s_{u(\alpha)}$$

extends to an isomorphism

$$u : W(\mathcal{R}) \xrightarrow{\sim} W(\mathcal{R}')$$

such that

$$u(w(\alpha)) = u(w)(u(\alpha)).$$

Proof. It suffices to prove, for $\alpha, \beta \in R$, that

$$u(s_\alpha(\beta)) = s_{u(\alpha)}u(\beta), \qquad q(s_\alpha(\beta)) = q(\beta).$$

Successively, one has

$$\begin{aligned} f\big(s_{u(\alpha)}u(\beta)\big) &= f(u(\beta)) - \big(u(\alpha)^*, u(\beta)\big) f(u(\alpha)) \\ &= q(\beta)\beta - q(\beta)q(\alpha)^{-1}(\alpha^*, \beta)q(\alpha)\alpha \\ &= q(\beta)(\beta - (\alpha^*, \beta)\alpha) \\ &= q(\beta)s_\alpha(\beta). \end{aligned}$$

If

$$\gamma = u^{-1}\big(s_{u(\alpha)}u(\beta)\big),$$

then

$$q(\gamma)\gamma = f(u(\gamma)) = q(\beta)s_\alpha(\beta).$$

The roots γ and $s_\alpha(\beta)$ are proportional over $\mathbb{Q}$, hence equal or opposite. Since $q(\gamma)$ and $q(\beta)$ are positive, it follows that

$$q(\gamma) = q(\beta), \qquad \gamma = s_\alpha(\beta).$$

□

Definition 6.8.5. The numbers $q(\alpha)$ are called the *root exponents* of f.

Example 6.8.6. Let $\mathcal{R}$ be a reduced root datum and let $q = p^n$, $n \in \mathbb{N}$. Multiplication by q,

$$M \longrightarrow M, \qquad x \longmapsto qx,$$

is a p-morphism whose root exponents are all equal to q (and for which $u = \mathrm{id}$). It is denoted

$$q : \mathcal{R} \longrightarrow \mathcal{R}.$$

Proposition 6.8.7. *In the notation of 6.8.1,* u *induces an isomorphism from the set of systems of simple roots (respectively positive roots) of* R *onto the corresponding set for* R'*.*

Proof. This follows immediately from 3.1.5 (respectively 3.2.1). □

7 Structure

7.1 Decomposition of a root datum

Proposition 7.1.1. *Let $\mathcal{R}$ be a root datum and let Δ be a system of simple roots.*

(i) Let R' and R'' be two closed symmetric sets of roots forming a partition of R. If

$$\Delta' = \Delta \cap R', \qquad \Delta'' = \Delta \cap R'',$$

then

$$R' = R_{\Delta'}, \qquad R'' = R_{\Delta''},$$

and every root of Δ' is orthogonal to every root of Δ''.

(ii) Let Δ' and Δ'' be two orthogonal subsets forming a partition of Δ. Then $R_{\Delta'}$ and $R_{\Delta''}$ form a partition of R.

Proof. We first prove (i).

Lemma 7.1.2. *Under the hypotheses of* (i)*, if α, β, and $\alpha+\beta$ are roots, then all three belong to R', or all three belong to R''.*

Suppose, for example, that $\alpha+\beta \in R'$. One cannot have $\alpha, \beta \in R''$, since R'' is closed. Suppose therefore that $\alpha \in R'$. Then $-\alpha \in R'$, and

$$\beta = (\beta+\alpha) - \alpha \in R'.$$

We now prove $R' = R_{\Delta'}$ by induction on the order of a positive root $\alpha \in R' \cap \mathscr{P}(\Delta)$. If $\operatorname{ord}_\Delta(\alpha) = 1$, then

$$\alpha \in R' \cap \Delta = \Delta'.$$

If $\operatorname{ord}_\Delta(\alpha) > 1$, there is $\beta \in \Delta$ such that $\alpha - \beta \in R$. By the lemma,

$$\beta \in \Delta', \qquad \alpha - \beta \in R'.$$

Hence $\alpha - \beta \in R_{\Delta'}$ by induction, and finally

$$\alpha = (\alpha - \beta) + \beta \in R_{\Delta'}.$$

We next prove that Δ' and Δ'' are orthogonal. If $\alpha \in \Delta'$ and $\beta \in \Delta''$, then

$$(\beta^*, \alpha) \leq 0.$$

If $(\beta^*, \alpha) \neq 0$, then $\beta + \alpha$ is a root, contrary to the lemma.

We now prove (ii). If Δ' or Δ'' is empty, the assertion is immediate. Otherwise, if $R_{\Delta'}$ and $R_{\Delta''}$ do not form a partition of R, there is a root α of the form

$$\alpha = \sum_i m'_i \alpha'_i + \sum_j m''_j \alpha''_j, \qquad m'_i, m''_j \in \mathbb{Z}_+,$$

where the α'_i (respectively α''_j) belong to Δ' (respectively Δ''). Applying 3.1.2, and interchanging Δ' and Δ'' if necessary, yields a relation

$$\delta = \gamma + \beta, \qquad \gamma \in R_{\Delta'}, \quad \beta \in \Delta'', \quad \delta \in R.$$

But $(\beta^*, \gamma) = 0$, so $\gamma - \beta$ is also a root by 2.2.5, which is impossible. □

Proposition 7.1.3. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

(i) There is no nontrivial partition of R into two closed symmetric subsets.

(ii) For some (respectively every) system of simple roots Δ of R, there is no partition of Δ into two nonempty orthogonal subsets.

(iii) The natural representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible.

(iv) For every pair (α, β) of roots, there is a sequence of roots

$$\alpha_0, \alpha_1, \dots, \alpha_n, \qquad \alpha_0 = \alpha, \quad \alpha_n = \beta,$$

such that α_i and α_{i+1} are nonorthogonal for $i = 0, \dots, n-1$.

Proof. The equivalence (i)$\Leftrightarrow$(ii) follows from 7.1.1, and (iv)$\Rightarrow$(ii) is immediate. Conversely, if (ii) holds for Δ, then (iv) holds when $\alpha, \beta \in \Delta$. Every root is nonorthogonal to some simple root, by 3.1.1, for example. Moreover, (iii)$\Rightarrow$(i): under the hypotheses of 7.1.1, the space $\mathcal{V}(R')$ is stable under $W(\mathcal{R})$. It remains to prove (i)$\Rightarrow$(iii).

Let H be a vector subspace of $\mathcal{V}(R)$, stable under $W(\mathcal{R})$. For every $\alpha \in R$, the equality $s_\alpha(H) = H$ immediately gives

$$\alpha \in H \qquad \text{or} \qquad \alpha^* \in H^\perp,$$

where $H^\perp$ denotes the orthogonal complement of H in $\mathcal{V}(R^*)$, which is in duality with $\mathcal{V}(R)$. Put

$$R' = \{\alpha \in R \mid \alpha \in H\}, \qquad R'' = \{\alpha \in R \mid \alpha^* \in H^\perp\}.$$

This gives a partition of R into two closed symmetric subsets. □

Definition 7.1.4. A root datum (respectively root system) satisfying the equivalent conditions of 7.1.3 and having nonzero semisimple rank is called *irreducible*.

Corollary 7.1.5. *For every root datum $\mathcal{R}$, there is a unique partition, up to order, of R into closed symmetric irreducible subsets.*

Corollary 7.1.6. *Every adjoint (respectively simply connected) root datum is a product of irreducible adjoint (respectively simply connected) root data.*

Proof. It suffices to consider the adjoint case. The assertion then follows from the fact that, under the hypotheses of 7.1.1,

$$\Gamma_0(R) = \Gamma_0(R') \times \Gamma_0(R'').$$

□

Corollary 7.1.7. *For every root datum (respectively reduced root datum) $\mathcal{R}$, there is an isogeny*

$$\mathcal{R} \longrightarrow \mathcal{R}',$$

where $\mathcal{R}'$ is a product of a trivial root datum and irreducible simply connected root data (respectively irreducible reduced simply connected root data).

7.2 Properties of irreducible root data

Definition 7.2.1. Let $\mathcal{R}$ be an irreducible root datum. For every $\alpha \in R$, put

$$\operatorname{long}(\alpha) = \ell(\alpha)/\ell(\alpha_0),$$

where $\alpha_0 \in R$ is such that $\ell(\alpha_0)$ is minimal. The number $\operatorname{long}(\alpha)$ is called the *length* of α.

Lemma 7.2.2. *Let $\mathcal{R}$ be an irreducible root datum. The Weyl group acts transitively on the set of roots of a given length.*

Proof. Let $\alpha, \beta \in R$. Since the representation of $W(\mathcal{R})$ on $\mathcal{V}(R)$ is irreducible, α cannot be orthogonal to all $w(\beta)$, $w \in W(\mathcal{R})$. Hence some $w(\beta)$ is nonorthogonal to α. Since

$$\ell(w(\beta)) = \ell(\beta),$$

the assertion follows from 2.3.2. □

Lemma 7.2.3. *If $\mathcal{R}$ is irreducible and reduced, then*

$$\operatorname{long}(R) \in \{\{1\}, \{1,2\}, \{1,3\}\}.$$

Proof. By the observation used above, for every $\alpha, \beta \in R$ there is $w \in W(\mathcal{R})$ such that $w(\beta)$ is nonorthogonal to α. Therefore, by 2.3.1,

$$\frac{\ell(\alpha)}{\ell(\beta)} \in \left\{1, 2, 3, \frac{1}{2}, \frac{1}{3}\right\}.$$

Thus $\operatorname{long}(\alpha)$ is 1, 2, or 3. But lengths 2 and 3 cannot both occur, since their ratio 2/3 is not in the displayed set. □

Remark 7.2.4. Reasoning similarly gives the following result. If $\mathcal{R}$ is irreducible and nonreduced, with $\operatorname{rgss}(\mathcal{R}) > 1$, then

$$\operatorname{long}(R) = \{1, 2, 4\}.$$

If $R_i = \operatorname{long}^{-1}(i)$, then

$$\operatorname{ind}(R) = R_1 \cup R_2, \qquad R_4 = 2R_1,$$

and two nonproportional roots of R_1 are orthogonal. Conversely, suppose that R is an irreducible reduced root system such that $\operatorname{long}(R) = \{1,2\}$. Put $R_i = \operatorname{long}^{-1}(i)$, and suppose that two nonproportional roots of R_1 are orthogonal. Then $R \cup 2R_1$ is irreducible and nonreduced, and

$$\operatorname{ind}(R \cup 2R_1) = R.$$

Lemma 7.2.5. *If $\mathcal{R}$ is an irreducible root datum, then $\mathcal{R}^*$ is irreducible as well, and the product*

$$\operatorname{long}(\alpha)\operatorname{long}(\alpha^*)$$

is constant as α runs through R.

Proof. This follows immediately from 7.1.3 (iv) and 2.2.6. □

Definition 7.2.6. Let $\mathcal{R}$ be any root datum. The *length* of $\alpha \in R$, denoted $\operatorname{long}(\alpha)$, is the length of α in its irreducible component.

Lemma 7.2.7. *There is a unique group homomorphism*

$$u : \Gamma_0(R) \longrightarrow \Gamma_0(R^*)$$

such that

$$u(\alpha) = \mathrm{long}(\alpha)\alpha^* \qquad (\alpha \in R).$$

Proof. By 3.5.5, it is enough to verify that, whenever $\alpha, \beta, \alpha + \beta \in R$,

$$\mathrm{long}(\alpha)\alpha^* + \mathrm{long}(\beta)\beta^* = \mathrm{long}(\alpha + \beta)(\alpha + \beta)^*.$$

But $\alpha, \beta, \alpha + \beta$ belong to the same irreducible component of R, by 7.1.2; the assertion is therefore reduced to 1.2.2. □

Remark 7.2.8. Let u be as in 7.2.7. For $\alpha, \beta \in R$,

$$(u(\alpha), \beta) = (u(\beta), \alpha).$$

Indeed, this amounts to

$$\mathrm{long}(\alpha)(\alpha^*, \beta) = \mathrm{long}(\beta)(\beta^*, \alpha),$$

which is clear when α and β are orthogonal. If they are not orthogonal, they lie in the same irreducible component of R, and the assertion reduces to 1.2.1, formula (9).

Remark 7.2.9. The symmetric bilinear form

$$(x, y) \longmapsto (u(x), y)$$

is positive and nondegenerate on $\Gamma_0(R)$.

Proof. Let R_i be the irreducible components of R. Then

$$\Gamma_0(R) = \prod_i \Gamma_0(R_i),$$

and the bilinear form $(u(x), y)$ is the product, on the groups $\Gamma_0(R_i)$, of the forms

$$2^{-1}\ell(\alpha_i)^{-1}(p(x), y),$$

where $\ell(\alpha_i)$ is the minimum of $\ell(\alpha)$ for $\alpha \in R_i$. These symmetric bilinear forms are positive and nondegenerate by 1.2.6. □

7.3 Cartan matrix

Let $\mathcal{R}$ be a root datum. If Δ is a system of simple roots, the *Cartan matrix* of $\mathcal{R}$ relative to Δ is the square matrix indexed by Δ and defined by

$$a_{\alpha,\beta} = (\alpha^*, \beta), \qquad \alpha, \beta \in \Delta.$$

If Δ' is another system of simple roots and $w \in W(\mathcal{R})$ satisfies $w(\Delta) = \Delta'$, then

$$(w(\alpha)^*, w(\beta)) = (\alpha^*, \beta).$$

Thus the Cartan matrix relative to Δ' is obtained from that relative to Δ by the isomorphism $\Delta \to \Delta'$ of index sets defined by w. Consequently, up to isomorphism of its index set, the Cartan matrix depends only on $\mathcal{R}$.

Proposition 7.3.1. *The Cartan matrix has the following properties.*

(i) $a_{\alpha,\alpha} = 2$, *and* $a_{\alpha,\beta} \leq 0$ *for* $\alpha \neq \beta$.

(ii) $a_{\alpha,\beta} = 0$ *implies* $a_{\beta,\alpha} = 0$.

(iii) There are strictly positive integers $m_\alpha = \operatorname{long}(\alpha)$ *such that the matrix*
$$(m_\alpha a_{\alpha,\beta})$$
is symmetric, positive, and nondegenerate.

(iv) The principal minors of $(a_{\alpha,\beta})_{\alpha,\beta\in\Delta}$*, namely*
$$\det(a_{\alpha,\beta})_{\alpha,\beta\in\Delta'} \qquad (\Delta' \subset \Delta),$$
are strictly positive.

(v) One has
$$s_\alpha(\beta) = \beta - a_{\alpha,\beta}\alpha, \qquad s_\alpha(\beta^*) = \beta^* - a_{\beta,\alpha}\alpha^*.$$

Proof. Part (v) is a definition; (i) follows from 3.2.11, (ii) from 2.2.2, and (iii) from 7.2.9. Part (iv) follows immediately from (iii) and
$$\det(m_\alpha a_{\alpha,\beta})_{\alpha,\beta\in\Delta'} = \left(\prod_{\alpha\in\Delta'} m_\alpha\right) \det(a_{\alpha,\beta})_{\alpha,\beta\in\Delta'}.$$
□

Proposition 7.3.2. *Let* $\mathcal{R}$ *and* $\mathcal{R}'$ *be two reduced simply connected (respectively adjoint) root data, let* Δ *and* Δ' *be systems of simple roots of* $\mathcal{R}$ *and* $\mathcal{R}'$*, and let* $u : \Delta \to \Delta'$ *be an isomorphism. Denote the corresponding Cartan matrices by* $(a_{\alpha,\beta})$ *and* $(a'_{\alpha',\beta'})$*, and suppose that*
$$a'_{u(\alpha),u(\beta)} = a_{\alpha,\beta}.$$
Then there is a unique isomorphism from $\mathcal{R}$ *onto* $\mathcal{R}'$ *that induces* u *on* Δ.

Proof. It is enough to give the proof in the adjoint case. Then $M = \Gamma_0(R)$ and $M' = \Gamma_0(R')$ are the free abelian groups generated by Δ and Δ'. Hence there is a unique group isomorphism $M \to M'$ inducing u on Δ; denote it again by u.

We show that $u(R) \subset R'$. Every root α of R can be written
$$\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}), \qquad \alpha_i \in \Delta.$$
By the hypothesis on u and the relations 7.3.1 (v),
$$u(\alpha) = s_{u(\alpha_1)} \cdots s_{u(\alpha_n)}(u(\alpha_{n+1})).$$
It remains to prove that ${}^t u(R'^*) \subset R^*$. This follows from the fact that the elements of M^* (respectively M'^*) are determined by their duality with R or Δ (respectively R' or Δ'), by 1.2.5. □

Corollary 7.3.3. *A reduced simply connected or adjoint root datum is determined up to isomorphism by its Cartan matrix.*

Corollary 7.3.4. *Let* $\mathcal{R}$ *be a reduced simply connected (respectively adjoint) root datum and let* Δ *be a system of simple roots. The group* $E_\Delta(\mathcal{R})$ *is identified with the group of automorphisms of the set* Δ *that preserve the Cartan matrix.*

Remark 7.3.5. The existence of a root datum corresponding to a given Cartan matrix satisfying, for example, (i), (ii), and (iv) is not easily settled directly without using the classification.

7.4 Dynkin diagrams

Definition 7.4.1. A *Dynkin-diagram structure* on a finite set Δ (the word "scheme" being banished for obvious reasons) consists of a set of pairs of distinct elements of Δ, called *linked pairs*, together with a map

$$\Delta \longrightarrow \{1, 2, 3\}.$$

The notion of isomorphism between such structures is evident.

Definition 7.4.2. Let $\mathcal{R}$ be a root datum and let Δ be a system of simple roots. The *Dynkin diagram* of $\mathcal{R}$ relative to Δ is the set Δ, in which two simple roots are linked if and only if they are nonorthogonal, and each root is assigned its length.

Proposition 7.4.3. *The Dynkin diagram and the Cartan matrix determine one another uniquely.*

Proof. The equivalence

$$\alpha \text{ and } \beta \text{ are not linked} \quad \Longleftrightarrow \quad a_{\alpha,\beta} = 0$$

and the relation

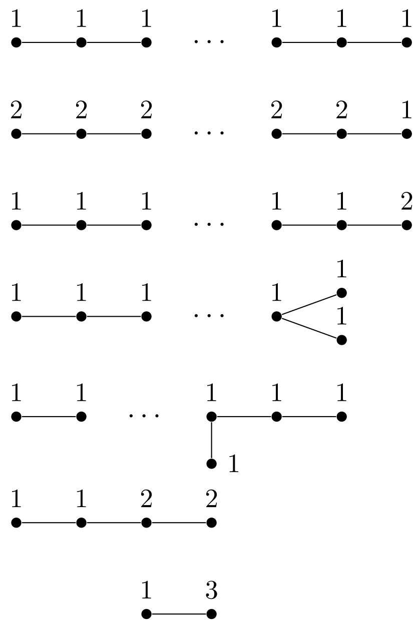
$$\operatorname{long}(\alpha) a_{\alpha,\beta} = \operatorname{long}(\beta) a_{\beta,\alpha},$$

with $\inf \operatorname{long}(\alpha) = 1$ in each connected component of the diagram, determine the $a_{\alpha,\beta}$ from the links and lengths, and conversely. The details are left to the reader. □

Corollary 7.4.4. *A reduced simply connected or adjoint root datum is determined by its Dynkin diagram.*

Corollary 7.4.5. *Let $\mathcal{R}$ be a reduced simply connected or adjoint root datum and let Δ be a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the Dynkin diagram of $\mathcal{R}$ relative to Δ; that is, with the group of permutations of Δ preserving lengths and links.*

Remark 7.4.6. The usual method[73] [74] classifies the connected Dynkin diagrams and shows that each actually corresponds to an irreducible reduced simply connected root datum. One obtains the familiar types:



A_n, $n \geq 1$.

B_n, $n \geq 2$.

C_n, $n \geq 3$.

D_n, $n \geq 4$.

E_n, $n = 6, 7, 8$.

F_4.

G_2.

[73] See Bourbaki, *Groupes et algèbres de Lie*, Chapter VI, no. 4.2, or the Sophus Lie Seminar.
[74] Editors' note: for a slightly different proof, see also [De80].

By 7.4.5, the corresponding group $E_\Delta(\mathcal{R})$ is obtained at once:

$$\begin{aligned} &E_\Delta(\mathcal{R}) = \{e\}, && A_1,\ B_n,\ C_n,\ E_7,\ E_8,\ F_4,\ G_2,\\ &E_\Delta(\mathcal{R}) = \mathbb{Z}/2\mathbb{Z}, && A_n\ (n \geq 2),\ D_n\ (n \geq 5),\ E_6,\\ &E_\Delta(\mathcal{R}) = \mathfrak{S}_3, && D_4. \end{aligned}$$

7.5 Complements on p-morphisms

Let $f : \mathcal{R} \to \mathcal{R}'$ be a p-morphism (cf. 6.8). It is clear from the definitions that the bijection $u : R \xrightarrow{\sim} R'$ associated with f makes the systems of simple roots, the systems of positive roots, the irreducible components, and so forth of R and R' correspond. We may therefore suppose, for simplicity, that R and R' are irreducible.

Lemma 7.5.1. *If R and R' are irreducible, there is a $k \in \mathbb{Q}$ such that, for every $\alpha \in R$,*

$$k \operatorname{long}(u(\alpha)) = q(\alpha)^2 \operatorname{long}(\alpha).$$

Proof. One has

$$\operatorname{long}(\alpha)(\alpha^*, \beta) = \operatorname{long}(\beta)(\beta^*, \alpha)$$

and, similarly,

$$\operatorname{long}(u(\alpha))(u(\alpha)^*, u(\beta)) = \operatorname{long}(u(\beta))(u(\beta)^*, u(\alpha)).$$

It follows immediately that, for nonorthogonal α, β,

$$\frac{q(\alpha)^2 \operatorname{long}(\alpha)}{\operatorname{long}(u(\alpha))} = \frac{q(\beta)^2 \operatorname{long}(\beta)}{\operatorname{long}(u(\beta))}.$$

The conclusion then follows from 7.1.3 (iv). □

Remark 7.5.2. It follows from 7.2.2 and 6.8.4 that $q(\alpha)$ depends only on $\operatorname{long}(\alpha)$. One then sees readily that, if $q(\alpha)$ is not constant, the product $q(\alpha)\operatorname{long}(\alpha)$ is constant; hence $p = 2$ or 3. Inspection of the preceding diagrams gives the four possible cases (the same letter denotes a Dynkin diagram and the corresponding reduced simply connected root datum):

$$\begin{aligned} &p = 2, \quad B_n \xrightarrow{f_1} C_n, \quad C_n \xrightarrow{f_2} B_n, \quad (C_2 = B_2),\\ &p = 2, \quad F_4 \xrightarrow{g} F_4,\\ &p = 3, \quad G_2 \xrightarrow{h} G_2. \end{aligned}$$

The reader will observe that $f_1 \circ f_2$, $f_2 \circ f_1$, $g \circ g$, and $h \circ h$ are p-morphisms of the form described in 6.8.6.

7.5.3. The preceding description shows immediately that, if $\mathcal{R}$ and $\mathcal{R}'$ are two reduced root data of semisimple rank at most 2 and there is a p-morphism from $\mathcal{R}'$ to $\mathcal{R}$, then $\mathcal{R}$ and $\mathcal{R}'$ have the same type. More precisely, one has the following table.

Notation. Let $f : \mathcal{R}' \to \mathcal{R}$ be a p-morphism. The symbols q and q_1 denote arbitrary positive powers of p. For rank-two systems we use the notation of Section 4, with α, β the simple roots and $\ell(\alpha) \leq \ell(\beta)$.

Type	p	**values of** f	**values of** ${}^t f$
Trivial	arbitrary	—	—
A_1	arbitrary	$f(\alpha') = q\alpha$	${}^t f(\alpha^*) = q\alpha'^*$
$A_1 \times A_1$	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q_1\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q_1\beta'^*$
A_2, B_2, G_2	arbitrary	$f(\alpha') = q\alpha$ $f(\beta') = q\beta$	${}^t f(\alpha^*) = q\alpha'^*$ ${}^t f(\beta^*) = q\beta'^*$
B_2	$p = 2$	$f(\alpha') = q\beta$ $f(\beta') = 2q\alpha$	${}^t f(\alpha^*) = 2q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$
G_2	$p = 3$	$f(\alpha') = q\beta$ $f(\beta') = 3q\alpha$	${}^t f(\alpha^*) = 3q\beta'^*$ ${}^t f(\beta^*) = q\alpha'^*$

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Exposé XXII
Reductive Groups: Splittings, Subgroups, Quotient Groups

by Michel Demazure

[0] This Exposé consists of two parts. The first (§§1–5.5) gathers the technical results needed for the proofs of the uniqueness and existence theorems. The second (§5.6 to the end) will not be used in that proof; the end of §5 will be used in particular in Exposé XXVI, which is devoted to parabolic subgroups; §6 establishes, in the scheme-theoretic setting, the classical results concerning the derived group of a reductive group.

1. Roots and Coroots. Split Groups and Root Data

Theorem 1.1. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and α a root of G relative to T.*

(i) *There exists a unique T-equivariant homomorphism of group schemes*

$$\exp_\alpha : W(\mathfrak{g}^\alpha) \longrightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^\alpha \to \mathfrak{g}$. This homomorphism is a closed immersion. The corresponding homomorphism

$$T \cdot_\alpha W(\mathfrak{g}^\alpha) \longrightarrow G$$

is likewise a closed immersion.

If $p_\alpha : \mathbb{G}_{a,S} \to G$ is a monomorphism normalized by T, with multiplier α, there exists a unique $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$[1] such that $p_\alpha(x) = \exp_\alpha(xX_\alpha)$; one has $\mathrm{Lie}(p_\alpha)(1) = X_\alpha$, and the preceding two formulas establish a bijective correspondence between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of monomorphisms $\mathbb{G}_{a,S} \to G$ normalized by T, with multiplier α.

(ii) *There exists a unique duality, denoted $(X, Y) \longmapsto XY$,*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathcal{O}_S,$$

and a unique homomorphism of group schemes

$$\alpha^* : \mathbb{G}_{m,S} \longrightarrow T,$$

[0] Editor's note: Version dated October 13, 2024.

[1] Editor's note: The set $\Gamma(S, \mathfrak{g}^\alpha)^\times$ is defined in Exposé XIX, 4.4.1.

such that formula (F) *of Exposé XX, 2.1 holds. One has*

$$\alpha \circ \alpha^* = 2, \qquad (-\alpha)^* = -\alpha^*,$$

and α^* *is given by the formula of Exposé XX, 2.7.*

Indeed, a homomorphism normalized by T, with multiplier α, necessarily factors through the closed subgroup

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha)$$

of G (cf. Exposé XIX, 3.9). Now (Z_α, T, α) is an elementary S-system (Exposé XX, 1.4), and one is reduced to the results of Exposé XX (1.5, 2.1, and 5.9).

Remark 1.2. Part (i) of Theorem 1.1 remains valid if one assumes only that α is a character of T, nontrivial on every fiber. Indeed, there is then a decomposition

$$S = S' \amalg S'',$$

such that $\alpha|_{S'}$ is a root of $G_{S'}$ relative to $T_{S'}$, and $\mathfrak{g}^\alpha|_{S''} = 0$. If $S = S'$, one is reduced to 1.1; if $S = S''$, the result is trivial; the general case follows at once.

Notations 1.3. As in Exposé XX, let U_α denote the image of $W(\mathfrak{g}^\alpha)$; it is a closed subgroup of G, canonically endowed with a vector-group structure. We call it the *vector group associated with the root* α. We say that α^* is the *coroot associated with* α. Sections

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha) \quad \text{and} \quad X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})$$

are said to be *paired* if $X_\alpha X_{-\alpha} = 1$. Then $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, and likewise for $X_{-\alpha}$. The corresponding homomorphisms p_α and $p_{-\alpha}$ are contragredient to one another in the sense of Exposé XX, 1.5[2], and one has

$$p_\alpha(x) p_{-\alpha}(y) = p_{-\alpha}\left(\frac{y}{1+xy}\right) \alpha^*(1+xy) p_\alpha\left(\frac{x}{1+xy}\right).$$

Proposition 1.4. *Under the hypotheses of 1.1, let* $w \in \underline{\mathrm{Norm}}_G(T)(S)$*. Then*

$$\beta = \alpha \circ \mathrm{int}(w)^{-1} : T \longrightarrow \mathbb{G}_{m,S}$$

is a root of G *relative to* T*,* $\beta^* = \mathrm{int}(w) \circ \alpha^*$ *is the corresponding coroot, and the following diagram is commutative:*

$$\begin{tikzcd} W(\mathfrak{g}^\alpha) \arrow[r, "\exp_\alpha"] \arrow[d, "\mathrm{Ad}(w)"'] & G \arrow[d, "\mathrm{int}(w)"] \\ W(\mathfrak{g}^\beta) \arrow[r, "\exp_\beta"] & G. \end{tikzcd}$$

Trivial: transport of structure.

[2] Editor's note: That is, if $\langle\, ,\, \rangle$ denotes the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$, and if $\mathfrak{g}^\alpha$ is identified with U_α via $p_\alpha(x) = \exp(xX_\alpha)$, and likewise for $-\alpha$, then $\langle p_\alpha(x), p_{-\alpha}(y)\rangle = xy$.

Definitions 1.5. (a) Under the hypotheses of 1.1, let s_α denote the automorphism of T defined by

$$s_\alpha(t) = t \cdot \alpha^*(\alpha(t))^{-1}.$$

Let (,) denote the canonical pairing

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T) \times \underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S}) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, \mathbb{G}_{m,S}) = \mathbb{Z}_S.$$

Then s_α acts on $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S})$, respectively on $\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T)$, by the following formulas, where χ, respectively u, denotes an arbitrary section of the corresponding scheme:

$$s_\alpha(\chi) = \chi - (\alpha^*, \chi)\alpha, \qquad s_\alpha(u) = u - (u, \alpha)\alpha^*.$$

One has

$$s_\alpha \circ s_\alpha = \mathrm{id}, \qquad s_{-\alpha} = s_\alpha.$$

(b) If $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then the inner automorphism $w_\alpha(X)$ of T, defined by

$$w_\alpha(X) = \exp_\alpha(X) \exp_{-\alpha}(-X^{-1}) \exp_\alpha(X)$$

(cf. Exposé XX, 3.1), coincides with s_α (loc. cit.). Proposition 1.4 therefore gives the following.

Corollary 1.6. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and α, β two roots of G relative to T. Then*

$$s_\alpha(\beta) = \beta - (\alpha^*, \beta)\alpha$$

is a root of G relative to T, and its corresponding coroot is

$$s_\alpha(\beta)^* = s_\alpha(\beta^*) = \beta^* - (\beta^*, \alpha)\alpha^*.$$

Corollary 1.7. *Under the preceding conditions, $\alpha^* = \beta^*$ implies $\alpha = \beta$.*

Indeed, if $\alpha^* = \beta^*$, one has (cf. Exposé XXI, 1.4)

$$s_\beta(\alpha) = \alpha - 2\beta, \qquad s_\alpha(\beta) = \beta - 2\alpha,$$

and hence immediately

$$(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha).$$

If $\beta \neq \alpha$, there exists $s \in S$ such that $\alpha_s \neq \beta_s$. The preceding formula would then give infinitely many distinct roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$, which is impossible.

Definitions 1.8.0. [3] If $u : \mathbb{G}_{m,S} \to T$ is a homomorphism of groups, we say that u is a *coroot* of G relative to T if there exists a root α of G relative to T such that $\alpha^* = u$. Consider the functor $\mathcal{R}^*$ of coroots of G relative to T, defined by

$$\mathcal{R}^*(S') = \{\text{coroots of } G_{S'} \text{ relative to } T_{S'}\}.$$

If $\mathcal{R}$ is the functor of roots of G relative to T (Exposé XIX, 3.8), there is a canonical morphism $\mathcal{R} \to \mathcal{R}^*$. By 1.7 and Exposé XIX, 3.8, one has:

[3] Editor's note: We added the numbering 1.8.0 to highlight these definitions.

Corollary 1.8. *The canonical morphism $\mathcal{R} \to \mathcal{R}^*$ is an isomorphism. In particular, $\mathcal{R}^*$ is represented by a twisted finite constant S-scheme that is an open and closed subscheme of*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T).$$

This leads to the following definition.

Definition 1.9. *Let S be a scheme and T an S-torus. A* twisted root datum *in T consists of*

(i) *a finite subscheme $\mathcal{R}$ of*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S});$$

(ii) *a finite subscheme $\mathcal{R}^*$ of*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T);$$

(iii) *an isomorphism $\mathcal{R} \xrightarrow{\sim} \mathcal{R}^*$, denoted $\alpha \mapsto \alpha^*$,*

satisfying the following conditions:

(DR 1) *For every $S' \to S$ and every $\alpha \in \mathcal{R}(S')$, one has $\alpha \circ \alpha^* = 2$.*

(DR 2) *For every $S' \to S$ and all $\alpha, \beta \in \mathcal{R}(S')$, one has*

$$\alpha - (\beta^*, \alpha)\beta \in \mathcal{R}(S'), \qquad \alpha^* - (\alpha^*, \beta)\beta^* \in \mathcal{R}^*(S').$$

Moreover, the twisted root datum is said to be reduced *if, for every nonempty $S' \to S$ and every $\alpha \in \mathcal{R}(S')$, one has $2\alpha \notin \mathcal{R}(S')$.*

Proposition 1.10. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and let $\mathcal{R}$, respectively $\mathcal{R}^*$, be the scheme of roots, respectively coroots, of G relative to T. Then $(\mathcal{R}, \mathcal{R}^*)$ is a reduced twisted root datum in T.*

It remains only to verify that this twisted root datum is reduced. This was done in Exposé XIX, 3.10.

1.11. Let $T = \mathrm{D}_S(M)$ be a *trivialized* torus. If M^* denotes the abelian group dual to M, there are canonical isomorphisms (cf. Exposé VIII, 1.5)

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S}) \xrightarrow{\sim} M_S, \qquad \underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbb{G}_{m,S}, T) \xrightarrow{\sim} M^*_S,$$

and hence group isomorphisms

$$\mathrm{Hom}_{S\text{-gr}}(T, \mathbb{G}_{m,S}) \xrightarrow{\sim} \mathrm{Hom}_{\text{loc.const.}}(S, M),$$

$$\mathrm{Hom}_{S\text{-gr}}(\mathbb{G}_{m,S}, T) \xrightarrow{\sim} \mathrm{Hom}_{\text{loc.const.}}(S, M^*).$$

A character of T, respectively a group homomorphism $\mathbb{G}_{m,S} \to T$, is said to be *constant* relative to the given trivialization if the preceding isomorphism carries it to a constant map from S to M, respectively to M^*.

1.12. With the same notation, let (M, M^*, R, R^*) be a root datum (Exposé XXI). Then (R_S, R_S^*) is a twisted root datum in T. Conversely, if $(\mathcal{R}, \mathcal{R}^*)$ is a twisted root datum in a torus T, a *splitting* of this root datum means the data of an ordinary root datum (M, M^*, R, R^*) and an isomorphism

$$T \simeq \mathrm{D}_S(M)$$

that carries $(\mathcal{R}, \mathcal{R}^*)$ to (R_S, R_S^*).

Definition 1.13. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. A* splitting of G relative to T *consists of*

(i) *an abelian group M and an isomorphism $T \simeq \mathrm{D}_S(M)$;*

(ii) *a system of roots R of G relative to T* (Exposé XIX, 3.6)*,*

satisfying the following two conditions:

(D 1) *S is nonempty, and the roots $\alpha \in R$* (respectively, the corresponding coroots) *identify with constant functions from S to M* (respectively, to M^*)*.*

(D 2) *The $\mathfrak{g}^\alpha$, $\alpha \in R$, are free $\mathcal{O}_S$-modules.*

We say that G is *splittable relative to T* if a splitting of G relative to T exists. A *splitting of G* consists of a maximal torus T of G together with a splitting of G relative to T. We say that G is *splittable* if it admits a splitting. A *split S-group* is a reductive S-group equipped with a splitting; we denote it by a symbol of the form (G, T, M, R), or simply by G when there is no risk of confusion.

Condition (D 1) canonically identifies R, respectively R^*, with a subset of M, respectively of M^*.

Proposition 1.14. *Let S be a nonempty scheme and let (G, T, M, R) be a split S-group. Then*

$$\mathcal{R}(G, T, M, R) = (M, M^*, R, R^*)$$

is a reduced root datum (Exposé XXI, 1.1 and 2.1.3)*; it is a splitting of the twisted root datum of 1.10.*

This follows trivially from 1.10 and Exposé XIX, 3.7.

For brevity we shall sometimes write $\mathcal{R}(G, T, M, R) = \mathcal{R}(G)$. We shall systematically use the notation V, $V(\mathcal{R})$, $W, \ldots$ of Exposé XXI.

Remark 1.15. (a) If S is nonempty and connected, respectively if $\mathrm{Pic}(S) = 0$, condition (D 1), respectively condition (D 2), is automatically satisfied.

(b) If (G, T, M, R) is a split S-group, then for every nonempty $S' \to S$, $(G_{S'}, T_{S'}, M, R)$ is a split S'-group and

$$\mathcal{R}(G, T, M, R) = \mathcal{R}(G_{S'}, T_{S'}, M, R).$$

1.16. Let $T = \mathrm{D}_S(M)$ be a trivialized torus. The Lie algebra $\mathfrak{t}$ of T identifies canonically (Exposé II, 5.1.1) with

$$\mathfrak{t} \simeq M^* \otimes \mathcal{O}_S.$$

For every homomorphism of groups $u : T \to \mathbb{G}_{m,S}$, $\mathrm{Lie}(u)$ is a linear form

$$\mathrm{Lie}(u) : \mathfrak{t} \longrightarrow \mathcal{O}_S = \mathrm{Lie}(\mathbb{G}_{m,S}/S).$$

In particular, if u is defined by an element $\alpha \in M$, then $\mathrm{Lie}(u)$ is the linear form $\bar{\alpha}$ on $M^* \otimes \mathcal{O}_S$ defined by

$$\bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Symmetrically, for every homomorphism of groups $h : \mathbb{G}_{m,S} \to T$, $\mathrm{Lie}(h)$ is an $\mathcal{O}_S$-linear morphism

$$\mathcal{O}_S = \mathrm{Lie}(\mathbb{G}_{m,S}/S) \longrightarrow \mathfrak{t},$$

canonically determined by the section

$$H = \mathrm{Lie}(h)(1) \in \Gamma(S, \mathfrak{t}).$$

In particular, if h is defined by an element $m \in M^*$, then

$$H = \mathrm{Lie}(h)(1) = m \otimes 1.$$

Comparing the two definitions, we find in particular

$$\bar{\alpha}(H) = (h, \alpha) \cdot 1 \in \Gamma(S, \mathcal{O}_S).$$

1.17. These definitions apply in particular when T is the maximal torus of a split group. Every root $\alpha \in R$ defines an *infinitesimal root*

$$\bar{\alpha} \in \mathrm{Hom}_{\mathcal{O}_S}(\mathfrak{t}, \mathcal{O}_S), \qquad \bar{\alpha}(m \otimes x) = (m, \alpha)x.$$

Every coroot α^*, $\alpha \in R$, defines an *infinitesimal coroot*

$$H_\alpha \in \Gamma(S, \mathfrak{t}), \qquad H_\alpha = \alpha^* \otimes 1.$$

For $\alpha, \beta \in R$, we have

$$\bar{\alpha}(H_\beta) = (\beta^*, \alpha) \cdot 1,$$

and in particular

$$\bar{\alpha}(H_\alpha) = 2.$$

Consequently, if 2 is invertible on S, then $\bar{\alpha}$ and H_α are nonzero on every fiber.

2. Existence of a Splitting. Type of a Reductive Group

Proposition 2.1. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. Suppose that T is split. Then G is locally splittable relative to T: for every $s_0 \in S$, there exists an open neighborhood U of s_0 such that the U-group G_U is splittable relative to T_U.*

Indeed, write $T \simeq \mathrm{D}_S(M)$ and

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}^m.$$

Let
$$R = \{m \in M \mid m \neq 0,\ \mathfrak{g}^m(s_0) \neq 0\}.$$
After restricting S and replacing it by an open neighborhood of s_0, we may suppose that the $\mathfrak{g}^\alpha$, $\alpha \in R$, are free and that the $\mathfrak{g}^m$, for $m \neq 0$ and $m \notin R$, vanish. We then have
$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha,$$
where the $\mathfrak{g}^\alpha$ are free of rank 1. It follows that R is a system of roots of G relative to T (Exposé XIX, 3.6). The coroots α^* corresponding to the $\alpha \in R$ then identify with locally constant functions on S with values in M^*. Shrinking S once more, we may suppose that they are constant, and the proof is complete.

Observe that the proof immediately gives:

Proposition 2.2. *Let S be a nonempty connected scheme such that* $\mathrm{Pic}(S) = 0$*, for example* $\mathrm{Spec}(\mathbb{Z})$ *or a local scheme* (in particular, the spectrum of a field)*. If G is a reductive S-group possessing a split maximal torus T, then G is splittable relative to T.*

We deduce immediately from 2.1 and from the fact that a reductive group locally possesses maximal tori for the étale topology (Exposé XIX, 2.5):

Corollary 2.3. *Let S be a scheme and G a reductive S-group* (respectively, let T also be a maximal torus of G)*. Then G is locally splittable* (respectively, locally splittable relative to T) *for the étale topology on S.*

Corollary 2.4. *Let k be a field and G a reductive k-group. There exists a finite separable extension K/k such that G_K is splittable.*

Remark 2.5. Using 2.1 and Exposé XIX, Remark 2.9, one immediately proves the following result: let $G = (G, T, M, R)$ be a split S-group. There exists an open covering (U_i) of S such that each split group G_{U_i} is obtained by base change from a split group over a noetherian ring (and in fact over a finitely generated $\mathbb{Z}$-algebra). We shall moreover prove that every split group over S is already obtained from a split $\mathbb{Z}$-group (Exposé XXV).

2.6. Let k be an algebraically closed field and G a reductive k-group. We know (for example by 2.4) that splittings of G exist. Let (G, T, M, R) and (G, T', M', R') be two splittings of G. Then the root data
$$\mathcal{R}(G, T, M, R) \qquad \text{and} \qquad \mathcal{R}(G, T', M', R')$$
are isomorphic.

Indeed, we first see that it is enough to reduce to the case $T = T'$: there exists $g \in G(k)$ such that $T' = \mathrm{int}(g)T$, and one readily verifies that transporting a splitting by an automorphism of G produces a root datum isomorphic to the original one. But since $S = \mathrm{Spec}(k)$ is connected, the isomorphism
$$\mathrm{D}_k(M) \xrightarrow{\sim} T \xrightarrow{\sim} \mathrm{D}_k(M')$$
comes from a unique isomorphism $M \simeq M'$. For the same reason, there exists at most one root system of G relative to T.

Definition 2.6.1. [4] If G is a reductive k-group, where k is algebraically closed, the *type of* G is the isomorphism class of the root datum defined by any splitting of G. If G is a torus of type M in the sense of Exposé IX, 1.4, then the type of G as a reductive group is given by the trivial root datum

$$(M, M^*, \varnothing, \varnothing).$$

By 1.15(b)[5], the type is invariant under an (algebraically closed) extension of the base field.

Definition 2.7. If G is a reductive S-group and $s \in S$, the *type of* G *at* s is the type of the reductive $\bar{s}$-group $G_{\bar{s}}$.

For every $S' \to S$ and every $s' \in S'$ mapping to $s \in S$, the type of $G_{S'}$ at s' is equal to the type of G at s.

If G is splittable and (G, T, M, R) is a splitting of G, then the type of G at s is the isomorphism class of $\mathcal{R}(G, T, M, R)$, by 1.15(b)[5]. The following then follows immediately from 2.3:

Proposition 2.8. *Let G be a reductive S-group, where $S \neq \varnothing$. The function*

$$s \longmapsto \textit{the type of } G \textit{ at } s$$

is locally constant on S. In particular, there exists a partition of S into nonempty open subschemes on each of which G has constant type. More precisely, let E be the set of types of the fibers of G, and, for every $t \in E$, let S_t be the set of points $s \in S$ where G has type t. Then $(S_t)_{t \in E}$ is a partition of S, and each S_t is open and closed (and nonempty).

3. The Weyl Group

3.1. Let S be a scheme, G a reductive S-group, and T a maximal torus of G. Then

$$W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$$

is a finite étale S-group (Exposé XIX, 2.5). The morphism $n \mapsto \mathrm{int}(n)$ induces, upon passage to the quotient, a canonical monomorphism (which is moreover an open immersion)

$$W_G(T) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(T).$$

3.2. Suppose now that G is splittable relative to T. Choose a splitting, say (G, T, M, R). We then have a canonical isomorphism (Exposé VIII, 1.5)

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(T) \simeq \big(\mathrm{Aut}_{\mathrm{gr.}}(M)\big)_S.$$

In particular, if W is the Weyl group of the root datum $\mathcal{R}(G)$ (Exposé XXI, 1.1.8), we have a monomorphism

$$W_S \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(T).$$

[4]Editor's note: The number 2.6.1 was added for later references.
[5]Editor's note: We have corrected the original, which referred to 1.17.

3.3. For every root $\alpha \in R$, the reflection $s_\alpha \in W$ acts on M by

$$s_\alpha(x) = x - (\alpha^*, x)\alpha,$$

and hence acts on T, by the preceding morphism, according to

$$s_\alpha(t) = t \cdot \alpha^*\big(\alpha(t)\big)^{-1}.$$

On the other hand, since $\mathfrak{g}^\alpha$ is assumed to be free, there exists

$$X \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

Consider

$$w_\alpha(X) \in \underline{\mathrm{Norm}}_G(T)(S) \qquad \text{(Exposé XX, 3.1)}.$$

We have (loc. cit.)

$$\mathrm{int}\big(w_\alpha(X)\big)(t) = s_\alpha(t).$$

Since W is generated by the s_α, $\alpha \in R$, it follows from the preceding remarks that, if W and $\underline{\mathrm{Norm}}_G(T)(S)/T(S)$ are regarded as groups of automorphisms of T, then

$$W \subset \underline{\mathrm{Norm}}_G(T)(S)/T(S) \subset W_G(T)(S).$$

By definition of the constant group W_S associated with W (cf. Exposé I, 1.8), we therefore have a commutative diagram:

$$\begin{tikzcd} W_S \arrow[rr] \arrow[dr] & & W_G(T) \arrow[dl] \\ & \underline{\mathrm{Aut}}_{S\text{-gr.}}(T). & \end{tikzcd}$$

Proposition 3.4. *Let S be a scheme, let (G, T, M, R) be a split S-group, and let W be the Weyl group of the root datum $\mathcal{R}(G)$. Then the canonical monomorphism*

$$W_S \longrightarrow W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$$

is an isomorphism.

Indeed, these are étale groups over S; it therefore suffices to verify that, for every $s \in S$, the morphism

$$W_S(\bar{s}) \longrightarrow W_G(T)(\bar{s})$$

is an isomorphism.[6] The latter assertion follows, for example, from *Bible*, §11.3, Theorem 2.

Remark 3.5. Using 2.3, the preceding proposition gives a new proof of the fact that the Weyl group of a maximal torus of a reductive S-group G is finite over S (Exposé XIX, 2.5(ii)). [7]

[6]Editor's note: Indeed, since W_S and $W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$ are étale over S, the morphism $f : W_S \to W_G(T) = \underline{\mathrm{Norm}}_G(T)/T$ is étale (EGA IV_4, 17.3.4). If, moreover, each f_s is an isomorphism, then, by loc. cit., 17.9.1, f is a surjective open immersion, hence an isomorphism.

[7]Editor's note: Indeed, let T be a maximal torus of G. The fact that $W_G(T)$ is finite over S is local for the fpqc topology (EGA IV_2, 2.7.1), hence *a fortiori* for the étale topology. By 2.3, we may therefore suppose that G is split relative to T, in which case the assertion follows from 3.4.

3.6. Under the hypotheses of 3.1, for every $w \in W_G(T)(S)$, let N_w[8] denote the fiber product in the following diagram:

$$\begin{tikzcd} N_w \arrow[r] \arrow[d] & \underline{\mathrm{Norm}}_G(T) \arrow[d] \\ S \arrow[r, "w"] & W_G(T). \end{tikzcd}$$

It is an open and closed subscheme of $\underline{\mathrm{Norm}}_G(T)$, and is a principal homogeneous bundle under T on the left (respectively, on the right) for the law

$$(t, q) \longmapsto tq \qquad \text{(respectively, } (q, t) \longmapsto qt).$$

If $n \in N_w(S)$, then

$$N_{ww'} = n \cdot N_{w'}, \qquad N_{w'w} = N_{w'} \cdot n.$$

3.7. In particular, if α is a root of G with respect to T, then N_{s_α} is none other than what was denoted by $N^\times$ in Exposé XX, 3.0. If $\mathfrak{g}^\alpha$ is free over S, we therefore have

$$N_{s_\alpha}(S) \neq \varnothing.$$

By 3.4 and condition (D 2) of the splitting, we deduce:

Corollary 3.8. *Under the hypotheses of 3.4, the morphism*

$$\underline{\mathrm{Norm}}_G(T)(S) \longrightarrow W_G(T)(S) = \mathrm{Hom}_{\mathrm{loc.const.}}(S, W)$$

is surjective. In particular, for every $w \in W$, there exists $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ such that

$$\mathrm{int}(n_w)|_T = w.$$

4. Homomorphisms of Split Groups

4.1. The "Big Cell"

4.1.1. Let (G, T, M, R) be a split reductive S-group. Choose a system of positive roots R_+ of the root datum $\mathcal{R}(G)$ (Exposé XXI, 3.2.1). Put $R_- = -R_+$.

Choose a total ordering on R_+ (respectively, R_-) and consider the morphism induced by multiplication in G:

$$u : \left(\prod_{\alpha \in R_-} U_\alpha \right) \times_S T \times_S \left(\prod_{\alpha \in R_+} U_\alpha \right) \longrightarrow G.$$

This is an open immersion. Indeed, since both sides are flat and of finite presentation over S, it suffices to verify this on every geometric fiber (SGA 1, I, 5.7 and VIII, 5.4). We are thus reduced to the case in which S is the spectrum of an algebraically closed field. But by *Bible*, §13.4, Corollary 2 to Theorem 3, u is radicial and dominant. Since the tangent map

[8] Editor's note: We have replaced Q_w by N_w, just as in Exposé XX, 3.0, Q was replaced by $N^\times$.

of u at the origin is an isomorphism (definition of a root system), u is birational. As G is normal, Zariski's "Main Theorem" (EGA III_1, 4.4.9) applies, and u is an open immersion.

Let us show that the image Ω of this open immersion is independent of the ordering chosen on R_+ (respectively, R_-). Since this amounts to comparing open subsets of G, we are reduced to proving that they have the same geometric points, and may therefore again suppose that S is the spectrum of an algebraically closed field. The assertion is then precisely *Bible*, §13, Proposition 1(c) and Theorem 1(a).

We have therefore proved:

Proposition 4.1.2. *Let (G, T, M, R) be a split S-group, and let R_+ be a system of positive roots of R. There exists an open subset Ω_{R_+} of G such that, for every total ordering on R_+* (respectively, R_-)*, the morphism induced by multiplication in G*

$$\left(\prod_{\alpha\in R_-} U_\alpha\right) \times_S T \times_S \left(\prod_{\alpha\in R_+} U_\alpha\right) \longrightarrow G$$

is an open immersion with image Ω_{R_+}.

Remark 4.1.3. Proposition 4.1.2 may be restated as follows: for every $\alpha \in R$, choose an isomorphism of vector groups

$$p_\alpha : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha \qquad \text{(cf. 1.19)}.$$

Then the morphism (where $N = \mathrm{Card}(R_+) = \mathrm{Card}(R_-)$)

$$\mathbb{G}_{a,S}^N \times_S T \times_S \mathbb{G}_{a,S}^N \longrightarrow G$$

defined on points by

$$((x_\alpha)_{\alpha\in R_-}, t, (x_\alpha)_{\alpha\in R_+}) \longmapsto \prod_{\alpha\in R_-} p_\alpha(x_\alpha)\cdot t \cdot \prod_{\alpha\in R_+} p_\alpha(x_\alpha)$$

is an open immersion whose image depends only on R_+, and not on the choice of the p_α or of the orderings on R_+ and R_-.

Notation 4.1.4. We write[9]

$$\Omega_{R_+} = \prod_{\alpha\in R_-} U_\alpha \cdot T \cdot \prod_{\alpha\in R_+} U_\alpha.$$

Proposition 4.1.5. *The scheme Ω_{R_+} is of finite presentation over S* (hence quasi-compact in G) *and is universally schematically dense in G relative to S* (cf. Exposé XVIII, 1).

The first assertion is trivial. Thus,[10] Ω_{R_+} is flat and of finite presentation over S, and contains the identity section; it therefore meets every fiber of G in a nonempty, hence dense, open subset. The second assertion thus follows from Exposé XVIII, 1.3.[11]

[9]Editor's note: It is called the "big cell" corresponding to R_+.
[10]Editor's note: We have expanded the original in what follows.
[11]Editor's note: See also EGA IV_3, 11.10.10.

Corollary 4.1.6. *Let (G, T, M, R) be a split reductive S-group. Then*

$$\underline{\mathrm{Centr}}(G) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha).$$

Consequently, $\underline{\mathrm{Centr}}(G)$ is representable by a closed diagonalizable subgroup of G. [12]

The second assertion follows immediately from the first. To prove the first, one may invoke Exposé XII, 4.8 and 4.11; alternatively, one may argue directly as follows.[13] Let $S' \to S$. If $t \in T(S')$ and $\alpha(t) = 1$ for every $\alpha \in R$, then $\mathrm{int}(t)$ induces the identity on $T_{S'}$ and on every $(U_\alpha)_{S'}$, $\alpha \in R$, and therefore also on $(\Omega_{R_+})_{S'}$. By schematic density it consequently induces the identity on $G_{S'}$, whence

$$t \in \underline{\mathrm{Centr}}(G)(S').$$

Conversely, since

$$\underline{\mathrm{Centr}}_{G_{S'}}(T_{S'}) = T_{S'} \qquad \text{(cf. Exposé XIX, 2.8)},$$

if $g \in G(S')$ centralizes $T_{S'}$ and the $(U_\alpha)_{S'}$, then g is a section of $T_{S'}$ on which every $\alpha \in R$ vanishes.

Corollary 4.1.7. *Let S be a scheme and G a reductive S-group. Then the center of G is representable by a closed subgroup of G of multiplicative type. It is also "the intersection of the maximal tori of G" in the following sense: for every $S' \to S$, $\underline{\mathrm{Centr}}(G)(S')$ is the set of $g \in G(S')$ such that, for every $S'' \to S'$, the inverse image of g in $G(S'')$ belongs to $T(S'')$ for every maximal torus T of $G_{S''}$.*

In view of 2.3, the first assertion follows from 4.1.6 by descent.[14] Let us prove the second assertion. Let H denote "the intersection of the maximal tori of G" in the preceding sense. Clearly,[15]

$$\underline{\mathrm{Centr}}(G) \subset H.$$

By descent, it is therefore enough to prove $\underline{\mathrm{Centr}}(G) = H$ when G is split. Since H is contained in the intersection of the maximal tori of G in the usual sense, this follows from the following remark: if (G, T, M, R) is a splitting, $\alpha \in R$, and $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, then

$$\mathrm{int}\big(\exp_\alpha(X)\big)(T) \cap T = \mathrm{Ker}(\alpha),$$

as a trivial calculation shows. See also Exposé XII, 8.6 and 8.8 for a more general statement.

Remark 4.1.8. In what follows, in the split case we shall systematically identify T with $D_S(M)$. Then

$$\underline{\mathrm{Centr}}(G) = D_S\big(M/\Gamma_0(R)\big),$$

where $\Gamma_0(R)$ is the subgroup of M generated by R (cf. Exposé XXI, 1.1.6).
If $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is a system of simple roots of R, then one immediately has (cf. Exposé XX, 1.19)

$$\underline{\mathrm{Centr}}(G) = \bigcap_i \mathrm{Ker}(\alpha_i) = \bigcap_i \underline{\mathrm{Centr}}(Z_{\alpha_i}).$$

[12] Editor's note: We have added the preceding sentence.

[13] Editor's note: We have expanded the original in what follows.

[14] Editor's note: Indeed, representability of the center by a closed subscheme of G is local for the fpqc topology (SGA 1, VIII, 5.2 and 5.4), hence *a fortiori* for the étale topology; the same is true of the property of being of multiplicative type.

[15] Editor's note: We have expanded the original in what follows.

Proposition 4.1.9. *Let S be a scheme, let (G,T,M,R) be a split S-group, let Q be an S-torus, let α_0 be a character of Q, let $\mathcal{L}$ be an invertible $\mathcal{O}_S$-module, and let*

$$f : Q \longrightarrow T, \qquad p : W(\mathcal{L}) \longrightarrow G$$

be group homomorphisms satisfying the following set-theoretic relation:

$$p(\alpha_0(q)x) = \operatorname{int}(f(q)) \cdot p(x)$$

for every $q \in Q(S')$, $x \in W(\mathcal{L})(S')$, and $S' \to S$. Suppose that f separates the elements of R in the following sense: if $\alpha, \alpha' \in R$ and $m, m' \in \mathbb{Z}$, then[16]

$$m\alpha \circ f = m'\alpha' \circ f \quad \Longrightarrow \quad m\alpha = m'\alpha'.$$

Finally, let $s \in S$ be such that $(\alpha_0)_{\bar{s}} \neq e$ and $p_{\bar{s}} \neq e$.

There then exist an open subset U of S containing s, an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $\mathbb{G}_{a,U}$, a root $\alpha \in R$, and an isomorphism of $\mathcal{O}_U$-modules

$$h : (\mathcal{L}|_U)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}^\alpha|_U$$

such that

$$\text{(i)} \quad (\alpha \circ f)|_U = (q\alpha_0)|_U,$$

$$\text{(ii)} \quad p(X) = \exp_\alpha(h(X^q)) \quad \text{for every } X \in W(\mathcal{L})(S'),\ S' \to U.$$

Moreover, once U has been chosen, q, α, and h are uniquely determined.

After shrinking S, we may suppose that α_0 is nonzero on every fiber of S. Choose a system of positive roots R_+ of R, and put

$$V = p^{-1}(\Omega_{R_+}).$$

This is an open subset of $W(\mathcal{L})$ containing the zero section and stable under multiplication by every $\alpha_0(q)$, where $q \in Q(S')$ and $S' \to S$. Since α_0 is nontrivial on every fiber, it follows immediately that $V = W(\mathcal{L})$, and hence that p factors through Ω_{R_+}. Choose arbitrary orderings of R_+ and R_-; every product below is understood to be taken in the chosen ordering. There are therefore unique morphisms

$$a_\alpha : W(\mathcal{L}) \longrightarrow U_\alpha, \qquad \alpha \in R, \qquad b : W(\mathcal{L}) \longrightarrow T$$

such that

$$p(x) = \prod_{\alpha \in R_-} a_\alpha(x) \cdot b(x) \cdot \prod_{\alpha \in R_+} a_\alpha(x).$$

Writing out the covariance condition under Q, one immediately obtains

$$a_\alpha(\alpha_0(q)x) = \alpha(f(q))a_\alpha(x), \qquad \alpha \in R,$$

$$b(\alpha_0(q)x) = b(x)$$

for every $x \in W(\mathcal{L})(S')$, $q \in Q(S')$, and $S' \to S$. The second condition immediately gives $b = e$.

[16]Editor's note: This is equivalent to the following hypothesis: if $\alpha, \alpha' \in R$, $m, m' \in \mathbb{Z}$, and

$$(m\alpha \circ f)_{\bar{s}} = (m'\alpha' \circ f)_{\bar{s}}$$

for every geometric point $\bar{s}$ of S, then $m\alpha = m'\alpha'$. In particular, this separation hypothesis is stable under base change.

Now let $\alpha \in R$ be such that $(a_\alpha)_{\bar{s}} \neq e$; such an α exists because $p_{\bar{s}} \neq e$ by hypothesis. Applying Exposé XIX, 4.12(a), we find an integer $n > 0$ such that

$$(\alpha \circ f)_{\bar{s}} = (n\alpha_0)_{\bar{s}}.$$

After shrinking S, we may suppose that $\alpha \circ f = n\alpha_0$ (Exposé IX, 5.3). But then, for every $\alpha' \in R$ with $\alpha' \neq \alpha$, one has

$$(\alpha' \circ f)_{\bar{s}} \neq m\alpha_0 \qquad \text{for every integer } m > 0,$$

by the hypothesis on f and the fact that the only roots proportional to α are α and $-\alpha$. Applying Exposé XIX, 4.12(a), now to $a_{\alpha'}$, we find that $a_{\alpha'}$ vanishes in a neighborhood of s. Since R is finite, after shrinking S once more we may suppose that $a_{\alpha'} = 0$ for every $\alpha' \in R$, $\alpha' \neq \alpha$. Thus $p = a_\alpha$, and we may apply Exposé XIX, 4.12(b), followed by 4.12(c), which gives the asserted result. The uniqueness assertions are clear.

Remark 4.1.10. The condition imposed on f in 4.1.9 is satisfied in particular when f is surjective (that is, faithfully flat).

Proposition 4.1.11. *Let (G, T, M, R) be a split S-group, let R_+ be a system of positive roots of R, and let Ω_{R_+} be the corresponding "big cell."*

(i) *Let H be a group-valued S-functor that is separated*[17] *for the fppf topology. If $f, g : G \rightrightarrows H$ are two group homomorphisms that coincide on Ω_{R_+}, then $f = g$.*

(ii) *Let H be an S-sheaf of groups for the fppf topology, and let $f : \Omega_{R_+} \to H$ be an S-morphism with the following property: for every $S' \to S$ and every $x, y \in \Omega_{R_+}(S')$ such that $xy \in \Omega_{R_+}(S')$, one has $f(xy) = f(x)f(y)$. Then there exists a group homomorphism $\bar{f} : G \to H$, unique by* (i), *extending f.*

Indeed, by 4.1.5, (i) (respectively (ii)) follows immediately from Exposé XVIII, 2.2 (respectively 2.3 and 2.4).

Remark 4.1.12. If $\alpha \in R_+$, then

$$\Omega_{R_+} \cap Z_\alpha = U_{-\alpha} \cdot T \cdot U_\alpha. \tag{†}$$

Indeed,[18] for every $S' \to S$, if

$$g = \prod_{\beta \in R_-} p_\beta(x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(x_\beta)$$

is an element of $\Omega_{R_+}(S')$, and if $t' \in T_\alpha(S')$, then

$$t' g t'^{-1} = \prod_{\beta \in R_-} p_\beta(\beta(t') x_\beta) \cdot t \cdot \prod_{\beta \in R_+} p_\beta(\beta(t') x_\beta).$$

Since α and $-\alpha$ are the only elements of R whose restriction to T_α is 1, it follows that g belongs to $Z_\alpha = \mathrm{Centr}(T_\alpha)$ if and only if $x_\beta = 0$ for $\beta \neq \pm\alpha$.

From (†) and Exposé XX, 2.1, one deduces that if $X \in \Gamma(S, \mathfrak{g}^\alpha)$ and $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$, then

$$\exp_\alpha(X) \exp_{-\alpha}(Y) \in \Omega_{R_+}(S) \iff 1 + XY \text{ is invertible.}$$

[17]Editor's note: Recall (cf. Exposé IV, 4.3.5) that an S-presheaf H is separated for a topology $\mathcal{T}$ if, for every $S' \to S$ and every family of S-morphisms $(S'_i \to S')_{i \in I}$ covering for $\mathcal{T}$, the map

$$H(S') \longrightarrow \prod_i H(S'_i)$$

is injective.

[18]Editor's note: We have expanded the original in what follows.

4.2. Morphisms of Split Groups

Definition 4.2.1. Let S be a nonempty scheme, and let (G, T, M, R) and (G', T', M', R') be two split S-groups. An S-group homomorphism $f : G \to G'$ is said to be *compatible with the splittings*, or to define a *morphism of split groups*, if the restriction of f to T factors as a morphism $f_T : T \to T'$ of the form $f_T = D_S(h)$, where $h : M' \to M$ is a group homomorphism satisfying the following condition: there exist a bijection $d : R \xrightarrow{\sim} R'$[19] and, for every $\alpha \in R$, an integer $q(\alpha) > 0$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbb{G}_{a,S}$ and

$$h(d(\alpha)) = q(\alpha)\alpha, \qquad {}^t h(\alpha^*) = q(\alpha)d(\alpha)^*.$$

Remark 4.2.2. It is immediate that h, d, and the $q(\alpha)$, for $\alpha \in R$, are uniquely determined by f. We write $h = \mathscr{R}(f)$. The $q(\alpha)$ are the radicial exponents of f (or of h).

Let p be the prime number, if one exists, that vanishes on S; put $p = 1$ if no prime number vanishes on S. Then $\mathscr{R}(f)$ is a p-morphism of reduced root data in the sense of Exposé XXI, 6.8. We have thus defined a functor $\mathscr{R}$ from the category of split S-groups to the category of reduced root data (equipped with p-morphisms).

Proposition 4.2.3. *Under the hypotheses of 4.2.1, the following properties hold:*

(i) *For every $\alpha \in R$, there exists a unique isomorphism of $\mathcal{O}_S$-modules*

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)} \xrightarrow{\sim} \mathfrak{g}'^{\,d(\alpha)}$$

such that

$$f(\exp_\alpha(X)) = \exp_{d(\alpha)}(f_\alpha(X^{q(\alpha)}))$$

for every $X \in W(\mathfrak{g}^\alpha)(S')$ and every $S' \to S$.

(ii) *For every $\alpha \in R$, one has $q(-\alpha) = q(\alpha)$, and f_α and $f_{-\alpha}$ are contragredient to one another.*

(iii) *For every $\alpha \in R$, every $Z \in W(\mathfrak{g}^\alpha)^*(S')$, and every $S' \to S$, one has*

$$f(w_\alpha(Z)) = w_{d(\alpha)}(Z^{q(\alpha)}).$$

By hypothesis, the following diagram is commutative:

$$\begin{tikzcd}
\mathbb{G}_{m,S} \arrow[r, "\alpha^*"] \arrow[d, "q(\alpha)"'] & T \arrow[r, "\alpha"] \arrow[d, "f_T"'] & \mathbb{G}_{m,S} \arrow[d, "q(\alpha)"] \\
\mathbb{G}_{m,S} \arrow[r, "d(\alpha)^*"] & T' \arrow[r, "d(\alpha)"] & \mathbb{G}_{m,S}
\end{tikzcd}$$

It follows that f maps $\mathrm{Ker}(\alpha)$ into $\mathrm{Ker}(d(\alpha))$, hence T_α into $T'_{d(\alpha)}$, and hence Z_α into $Z'_{d(\alpha)}$. It remains only to apply Exposé XX, 3.10 and 3.11, to the groups Z_α and $Z'_{d(\alpha)}$.

Proposition 4.2.4. *The morphism f induces a morphism f_N from $\underline{\mathrm{Norm}}_G(T)$ to $\underline{\mathrm{Norm}}_{G'}(T')$, hence a morphism f_W from $W_G(T)$ to $W_{G'}(T')$; the latter is an isomorphism. More precisely, let*

$$\bar{d} : W(\mathscr{R}(G)) = W \longrightarrow W' = W(\mathscr{R}(G'))$$

be the isomorphism extending $s_\alpha \mapsto s_{d(\alpha)}$ (Exposé XXI, 6.8.4). *Then there is a commutative diagram of isomorphisms:*

[19] Editor's note: We have replaced the notation $u : R \to R'$ by $d : R \to R'$.

$$\begin{tikzcd} W_G(T) \arrow[r, "f_W", "\sim"'] & W_{G'}(T') \\ W_S \arrow[u, "\sim"] \arrow[r, "\bar{d}_S", "\sim"'] & W'_S. \arrow[u, "\sim"'] \end{tikzcd}$$

This follows immediately from 3.4, Exposé XXI, 6.8.4, and (iii) above.

Remark 4.2.5. With the notation of 4.2.3, the restriction of f to Ω_{R_+}, for a system of positive roots R_+, can be written explicitly: it maps Ω_{R_+} into $\Omega'_{d(R_+)}$ ($d(R_+)$ is a system of positive roots of R' by Exposé XXI, 6.8.7), and is given by the set-theoretic formula

$$f\Big(\prod_{\alpha\in R_-}\exp_\alpha(X_\alpha)\cdot t\cdot\prod_{\alpha\in R_+}\exp_\alpha(X_\alpha)\Big)=\prod_{\alpha\in R_-}\exp_{d(\alpha)}\Big(f_\alpha(X_\alpha^{q(\alpha)})\Big)\cdot f_T(t)\cdot \\ \prod_{\alpha\in R_+}\exp_{d(\alpha)}\Big(f_\alpha(X_\alpha^{q(\alpha)})\Big).$$

Proposition 4.2.6. (i) *The morphism f is surjective* (= faithfully flat in the present case; cf. Exposé $\mathrm{VI_B}$, 3.11) *if and only if f_T is surjective.*

(ii) *One has* $\mathrm{Ker}(f)\subset\Omega_{R_+}$.

We prove (i). If f is surjective, then $f_T(T)=f(T)$ is a maximal torus of G'. Indeed, $f(T)$ is a subtorus of a maximal torus T' (Exposé IX, 6.8); to verify that $f(T)=T'$, one reduces to the case of an algebraically closed field, where this is *Bible*, §7.3, Theorem 3(a).

If f_T is surjective, the preceding formula shows that f induces a surjection from $\Omega=\Omega_{R_+}$ onto $\Omega'=\Omega'_{d(R_+)}$.[20] Since the fibers of G' are connected, it follows (cf. Exposé $\mathrm{VI_A}$, 0.5) that f is surjective.

We prove (ii), and for this purpose admit a result that will be proved below (5.7.4). For each $w\in W$, choose an $n_w\in\underline{\mathrm{Norm}}_G(T)(S)$ representing it. Then the open subschemes $n_w\Omega$, for $w\in W$, cover G. It is therefore enough to prove that

$$\mathrm{Ker}(f)\cap n_w\Omega\neq\varnothing\quad\Longrightarrow\quad w=1.$$

If $x\in\Omega(S')$, where $S'\to S$, and $f(n_wx)=1$, then $f(x)=f(n_w)^{-1}$. But $f(x)\in\Omega'(S')$, whereas

$$f(n_w)^{-1}\in\underline{\mathrm{Norm}}_{G'}(T')(S').$$

By 4.2.4, we are reduced to proving the following.

Lemma 4.2.7. *Under the hypotheses of 4.1.2, one has*

$$\Omega\cap\underline{\mathrm{Norm}}_G(T)=T.$$

Let

$$x=\prod_{\alpha\in R_-}p_\alpha(x_\alpha)\cdot t\cdot\prod_{\alpha\in R_+}p_\alpha(x_\alpha)=vtu\in\Omega(S').$$

If x normalizes $T_{S'}$, then, for every $t'\in T(S')$,

$$xt'x^{-1}=t''\in T(S'),$$

[20] Editor's note: We have given the reference to Exposé VI in greater detail below.

that is, $xt' = t''x$, which may be written

$$v(tt')(t'^{-1}ut') = (t''vt''^{-1})(t''t)u.$$

This gives $t'^{-1}ut' = u$, hence

$$u \in \mathrm{Centr}_G(T)(S') = T(S'),$$

and therefore $u = 1$. Similarly, $v = 1$.

Corollary 4.2.8. *One has*

$$\mathrm{Ker}(f) = \prod_{\alpha \in R_-} K_\alpha \cdot \mathrm{Ker}(f_T) \cdot \prod_{\alpha \in R_+} K_\alpha,$$

where, for each $\alpha \in R$, K_α denotes the finite S-group

$$K_\alpha = \mathrm{Ker}\Big(U_\alpha \longrightarrow U_\alpha^{\otimes q(\alpha)}\Big) \simeq \boldsymbol{\alpha}_{q(\alpha),S}.$$

To apply this corollary, we make the following definition.

Definition 4.2.9. Let S be a scheme and let G and G' be two reductive S-groups. A faithfully flat and finite morphism of S-groups $f : G \to G'$ (i.e. one that is surjective and has finite kernel over S) is called an *isogeny.* If, moreover, $\mathrm{Ker}(f)$ is a central subgroup of G, then f is called a *central isogeny.*

Proposition 4.2.10. *Let $f : G \to G'$ be a morphism of split groups. For f to be an isogeny* (respectively, a central isogeny)*, it is necessary and sufficient that f_T be an isogeny, i.e. that $\mathscr{R}(f)$ be injective with finite cokernel* (respectively, and that $q(\alpha) = 1$ for every $\alpha \in R$)*.*

Indeed, by 4.2.8, $\mathrm{Ker}(f)$ is finite over S if and only if $\mathrm{Ker}(f_T)$ is finite over S, and $\mathrm{Ker}(f) \subset T$ if and only if every $q(\alpha)$ is equal to 1. In that case $\mathrm{Ker}(f)$ is central, since it is of multiplicative type and invariant; cf. Exposé IX, 5.5.

Remark 4.2.11. (a) Thus $f : G \to G'$ is a central isogeny if and only if

$$\mathscr{R}(f) : \mathscr{R}(G') \longrightarrow \mathscr{R}(G)$$

is an isogeny in the sense of Exposé XXI, 6.2. Moreover, in this case, with the notation of *loc. cit.*, one has

$$\mathrm{Ker}(f) = D_S\big(K(\mathscr{R}(f))\big), \qquad K\big(\mathscr{R}(f)\big) = \mathrm{Coker}\big(\mathscr{R}(f)\big).$$

(b) If G and G' are semisimple, every morphism of split groups $f : G \to G'$ is an isogeny.[21]

(c) If $f : G \to G'$ is faithfully flat and finite and G is reductive (respectively, semisimple), then G' is so as well. Indeed, G' is of finite presentation over S (Exposé V, 9.1), affine over S (EGA II, 6.7.1), smooth over S (Exposé VI, 9.2), and has connected reductive (respectively, semisimple) geometric fibers by Exposé XIX, 1.7.

Definition 4.2.1 may seem arbitrary. It is justified by the following proposition, which, for simplicity, we shall state for semisimple groups.

A morphism $f : G \to G'$ of reductive S-groups is said to be *splittable* if there exist splittings of G and G' with which f is compatible. We then have the following.

[21] Editor's note: Indeed, since G and G' are semisimple, f_T is surjective with finite kernel; hence f is faithfully flat with finite kernel by 4.2.6(i) and 4.2.8.

Proposition 4.2.12. *Let S be a scheme, let G and G' be two semisimple S-groups, and let $f : G \to G'$ be a morphism of groups. Let $s \in S$. The following conditions are equivalent:*

(i) $\mathrm{Ker}(f_{\bar{s}})$ is finite ($\Longleftrightarrow$ e is isolated in $\mathrm{Ker}(f)(\bar{s})$), and $f_{\bar{s}}$ is surjective, i.e. $f_{\bar{s}}$ is an isogeny.

(ii) $f_{\bar{s}}$ is splittable.

(iii) There exists an étale morphism $S' \to S$ covering s such that $f_{S'} : G_{S'} \to G'_{S'}$ is splittable.

Clearly, (iii) is equivalent to (ii), and (ii) implies (i) by 4.2.11(b). This is where the hypothesis that G and G' are semisimple enters; the other implications are valid for reductive groups.

We now prove that (i) implies (iii). We may suppose that G and G' are split in such a way that f induces a morphism
$$f_T : T \longrightarrow T'$$
(2.3 and Exposé XIX, 2.8). After restricting S, we may suppose that $f_T = D_S(h)$, where $h : M' \to M$ is a morphism of groups. Let $\alpha \in R$, and consider the composite morphism

$$p : W(\mathfrak{g}^\alpha) \xrightarrow{\exp_\alpha} G \xrightarrow{f} G'.$$

Since $\mathrm{Ker}(p_{\bar{s}})$ is finite, $p_{\bar{s}} \neq e$. On the other hand, $f_{T_{\bar{s}}}$ is surjective. We may therefore apply 4.1.9: there exist an open subset V_α of S containing s, a root $\alpha' \in R'$, and an integer $q(\alpha)$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of $\mathbb{G}_{a,V_\alpha}$, and an isomorphism of $\mathcal{O}_{V_\alpha}$-modules

$$f_\alpha : (\mathfrak{g}^\alpha)^{\otimes q(\alpha)}\Big|_{V_\alpha} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}\Big|_{V_\alpha},$$

such that
$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(f_\alpha(X_\alpha^{q(\alpha)}))$$
and
$$\alpha' \circ f_T = h(\alpha') = q(\alpha)\alpha.$$
We may replace S by the intersection of the V_α, for $\alpha \in R$. Put $\alpha' = d(\alpha)$. It is clear that

$$d : R \longrightarrow R'$$

is a bijection, since the kernel of h is finite ($f_{T_{\bar{s}}}$ being surjective). It remains only to prove that
$$f_T \circ \alpha^* = q(\alpha)\alpha'^*,$$
which follows by a trivial modification of the argument used in Exposé XX, 3.11.

In any event, as we have seen in the course of the proof, (i) implies (iii). We have therefore proved the following.

Corollary 4.2.13. *Let S be a scheme and let $f : G \to G'$ be an isogeny of reductive groups. Then f is locally splittable for the étale topology.*

4.3. Central Quotients of Reductive Groups

We first consider a special case.

Proposition 4.3.1. *Let S be a scheme, let (G, T, M, R) be a split S-group, let N be a subgroup of M containing R, and put*

$$Q = D_S(M/N) \subset \mathrm{Centr}(G).$$

Then:

(i) *$G' = G/Q$ is a reductive S-group, and $T' = T/Q$ is a maximal torus of G';*

(ii) *if T' is identified with $D_S(N)$, then $R \subset N$ is a root system of G' relative to T', (G', T', N, R) is a splitting of G', and $\mathscr{R}(G')$ is canonically identified with the induced root datum* (Exposé XXI, 6.5) *$\mathscr{R}(G)_N$;*

(iii) *the canonical morphism $G \to G'$ is compatible with the splittings, has root exponents equal to 1, and, by functoriality, induces the canonical morphism* (loc. cit.)

$$\mathscr{R}(G)_N \longrightarrow \mathscr{R}(G).$$

We know that $G' = G/Q$ is representable by an affine group scheme over S (Exposé VIII, 5.7), smooth over S (Exposé $\mathrm{VI_B}$, 9.2), and with connected reductive geometric fibers (as quotients of reductive groups; cf. Exposé XIX, 1.7). Thus G' is a reductive S-group.

It is clear that

$$T' = T/Q \simeq D_S(N)$$

is a maximal torus of G'. Next, choose a system R_+ of positive roots of R. The open subset Ω_{R_+} of 4.2 is stable under Q, and there is a canonical isomorphism

$$\Omega_{R_+}/Q \simeq \left(\prod_{\alpha \in R_-} U_\alpha\right) \underset{S}{\times} (T/Q) \underset{S}{\times} \left(\prod_{\alpha \in R_+} U_\alpha\right).$$

Moreover, Ω_{R_+}/Q is an open subset of G' containing the identity section (cf. Exposé IV, 4.7.2 and 6.4.1).

It follows that, if $\mathfrak{g}'$ denotes the Lie algebra of G' and $\bar{\alpha}$ the character of T/Q induced by α (or, equivalently, defined by $\alpha \in N$ under the identification $T/Q = D_S(N)$), then the canonical morphism $\mathfrak{g} \to \mathfrak{g}'$ induces, for every $\alpha \in R$, an isomorphism

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\bar{\alpha}}.$$

We have thus proved that R is a root system of G' relative to T', and the proof is completed without difficulty.

Corollary 4.3.2. *Let S be a scheme, let G be a reductive S-group, and let Q be a normal subgroup of G of multiplicative type. Then Q is central in G, the quotient G/Q is representable by a reductive S-group, and the canonical morphism $G \to G/Q$ is locally splittable for the étale topology* (with root exponents equal to 1).

The first assertion follows from Exposé IX, 5.5; the others are local for the étale topology and reduce to 4.3.1.

Definition 4.3.3. Let G be a reductive S-group. We say that G is *adjoint* (respectively, *simply connected*) if, for every $s \in S$, the type of G at s is given by an adjoint (respectively, simply connected) root datum, i.e. (Exposé XXI, 6.2.6) one for which M is generated by R (respectively, M^* is generated by R^*).

Proposition 4.3.4. (i) *An adjoint* (respectively, simply connected) *reductive group is semisimple.*

(ii) *If T is a maximal torus of the adjoint* (respectively, simply connected) *reductive group G, and if α is a root of G relative to T, then the infinitesimal root $\bar{\alpha}$ is nonzero on every fiber* (respectively, α^* is a monomorphism and the infinitesimal coroot H_α is nonzero on every fiber).

Indeed, (i) is trivial; (ii) may be checked on the geometric fibers and follows at once from Exposé XXI, 6.2.8.

Proposition 4.3.5. (i) *The reductive group G is adjoint if and only if*

$$\mathrm{Centr}(G) = \{e\}_S.$$

(ii) *For every reductive group G, the quotient group $G/\mathrm{Centr}(G)$ is an adjoint reductive group.*

Indeed, we may suppose that G is split. Then (i) is immediate, since

$$\mathrm{Centr}(G) = D_S(M/\Gamma_0(R)),$$

and (ii) follows from 4.3.1.

Definition 4.3.6. Let G be a reductive S-group. The *adjoint group* of G, denoted by $\mathrm{ad}(G)$, is the group $G/\mathrm{Centr}(G)$. The *radical* of G, denoted by $\mathrm{rad}(G)$, is the maximal torus (unique by Exposé XII, 1.12) of $\mathrm{Centr}(G)$. The *semisimple group associated with G* is the quotient $G/\mathrm{rad}(G)$.

The preceding definitions are compatible with base change. If $s \in S$, then $\mathrm{rad}(G)_{\bar{s}}$ is indeed the radical of $G_{\bar{s}}$ in the usual sense (Exposé XIX, 1.6).

4.3.7. If (G, T, M, R) is a split group, then

$$\mathrm{rad}(G) = D_S(M/N), \qquad N = M \cap V(R).$$

Thus the semisimple group associated with G (as also the adjoint group of G) is equipped with a canonical splitting (4.3.1), and we have a diagram of split groups

$$G \longrightarrow \mathrm{ss}(G) \longrightarrow \mathrm{ad}(G)$$

corresponding to the canonical diagram of root data (Exposé XXI, 6.5.5)

$$\mathrm{ad}(\mathscr{R}(G)) \longrightarrow \mathrm{ss}(\mathscr{R}(G)) \longrightarrow \mathscr{R}(G).$$

Remark 4.3.8. *Let (G, T, M, R) be a split adjoint* (respectively, simply connected) *S-group, and let Δ be a system of simple roots of R. Then the family $\{\alpha\}_{\alpha\in\Delta}$, respectively $\{\alpha^*\}_{\alpha\in\Delta}$, induces an isomorphism*

$$T \xrightarrow{\sim} (\mathbb{G}_{m,S})^\Delta, \qquad \textit{respectively} \qquad (\mathbb{G}_{m,S})^\Delta \xrightarrow{\sim} T.$$

Indeed,

$$M = \Gamma_0(R) \qquad (\text{respectively, } M^* = \Gamma_0(R^*)),$$

and Δ (respectively, $\{\alpha^*\}_{\alpha\in\Delta}$) is a basis of the free abelian group $\Gamma_0(R)$ (respectively, $\Gamma_0(R^*)$); see Exposé XXI, 3.1.8.

Remark 4.3.9. The radical of a reductive group is a *characteristic* subgroup (i.e. stable under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$), by its definition.

5. Subgroups of Type (R)

We are especially interested in reductive groups, but some of the results that we shall establish hold more generally for a wider class of groups: groups of type (RR).

5.1. Groups of Type (RR)

Definition 5.1.1. Let S be a scheme and G an S-group scheme. We say that G is *of type* (RR) if it satisfies the following conditions:

(i) G is smooth and of finite presentation over S, with connected fibers.

(ii) G locally possesses maximal tori for the fpqc topology.

(iii) For every $s \in S$, every maximal torus T of $G_{\bar{s}}$, and every root α of $G_{\bar{s}}$ relative to $T_{\bar{s}}$ (Exposé XIX, 1.10),
$$\mathrm{Lie}(G_{\bar{s}})^{\alpha}$$
has rank 1 (as a vector space over $\overline{\kappa(s)}$).

(iv) For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, let R denote the set of roots of $G_{\bar{s}}$ relative to T, and let $\Gamma_0(R)$ be the subgroup of
$$\mathrm{Hom}_{\bar{s}\text{-gr.}}(T, \mathbb{G}_{m,\bar{s}})$$
generated by R. Then the content[22] of every root $\alpha \in R$ in the free abelian group $\Gamma_0(R)$ (and hence a positive integer) is invertible on S.

Reminder 5.1.1.1. [23] Let us recall that, if G is a smooth connected algebraic group over an algebraically closed field k, a *Cartan subgroup* of G is the centralizer of a maximal torus of G (Exposé XII, 1.0), and such a subgroup is smooth and connected. For this, as well as for other characterizations of Cartan subgroups, see *Bible*, §7.1, Theorem 1 in the case where G is affine, and Exposé XII, Theorem 6.6 in the general case.

If S is an arbitrary scheme and G is a smooth S-group of finite type, a *Cartan subgroup* of G is a smooth S-subgroup C of G such that, for every $s \in S$, $C_{\bar{s}}$ is a Cartan subgroup of $G_{\bar{s}}$ (Exposé XII, Definition 3.1).

Remark 5.1.2. (a) By Exposé XII, 7.1 (where the hypothesis that G is separated is satisfied here because G has connected fibers; cf. Exposé $\mathrm{VI_B}$, 5.5), conditions (i) and (ii) imply that G locally possesses, for the étale topology, maximal tori (respectively, Cartan subgroups), which are locally conjugate for the étale topology.

(b) The Cartan subgroups of G have connected fibers (Exposé XII, 6.6).

(c) If G is affine over S, conditions (i) and (ii) are respectively equivalent to:

(i′) G is smooth over S and has connected fibers;

[22] Editor's note: The content of the root α is the positive generator of the ideal $\{f(\alpha) \mid f \in \Gamma_0(R)^*\}$ of $\mathbb{Z}$; it is the greatest integer $c > 0$ such that $\alpha/c \in \Gamma_0(R)$.

[23] Editor's note: This reminder was added.

(ii′) the reductive rank of the fibers of G is locally constant (Exposé XII, 1.7).

(d) By Exposé XII, 8.8 (c) and (d), G has a reductive center Z, and for every $s \in S$, in the notation of (iv),[24]

$$Z_{\bar{s}} = \bigcap_{\alpha \in R} \operatorname{Ker}(\alpha),$$

whence

$$\operatorname{Hom}_{\bar{s}\text{-gr.}}\big((T/Z)_{\bar{s}}, \mathbb{G}_{m,\bar{s}}\big) \simeq \Gamma_0(R).$$

(e) Condition (iv) holds in particular in the following two cases:

(1) S has characteristic 0;

(2) every root is an indivisible element of the group generated by the roots.

(f) The property of being of type (RR) is stable under base change and is local for the fpqc topology.

Remarks (c) and (e) immediately give the following proposition.

Proposition 5.1.3. *A reductive group is of type* (RR)*.*

Proposition 5.1.4. *Let S be a scheme, let G be an S-group of type* (RR)*, and let Q be a central subgroup of G, of finite presentation over S, such that the quotient G/Q is representable* (for example, G affine over S and Q of multiplicative type (Exposé IX, 2.3), or S artinian (Exposé $\mathrm{VI_A}$, 3.3.2))*. Then G/Q is an S-group of type* (RR)*.*

Indeed, G/Q is smooth over S (Exposé $\mathrm{VI_B}$, 9.2), of finite presentation over S (Exposé V, 9.1), and has connected fibers, so condition (i) holds. On the other hand, condition (ii) follows from Exposé XII, 7.6; it remains to verify conditions (iii) and (iv).

Put $G' = G/Q$, let

$$u : G \longrightarrow G'$$

be the canonical morphism, and let $T' = u(T)$ be the maximal torus of G' that is the image of T (cf. Exposé XII, 7.1 (e)). For each $\alpha \in R$, again denote by α the character of T' induced by α; indeed,

$$Q \cap T \subset \bigcap_{\alpha \in R} \operatorname{Ker}(\alpha)$$

by 5.1.2 (d). We first prove:

Lemma 5.1.5. *Under the hypotheses of 5.1.4, let $T = D_S(M)$ be a trivialized maximal torus of G, and suppose that the decomposition of $\mathfrak{g} = \operatorname{Lie}(G)$ under $\operatorname{Ad}(T)$ has the form*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha, \qquad R \subset M - \{0\},$$

where, for every $s \in S$, $\mathfrak{g}^\alpha(s) \neq 0$ for $\alpha \in R$ (thus $\mathfrak{g}^\alpha$ is an invertible $\mathcal{O}_S$-module for every $\alpha \in R$, and R is the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$, for every $s \in S$)*. Then the Lie algebra $\mathfrak{g}'$ of G' decomposes under $\operatorname{Ad}(T')$ as follows:*

$$\mathfrak{g}' = \mathfrak{g}'^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}'^\alpha,$$

[24] Editor's note: The original said "under the conditions of (iv)," but the last condition of (iv) does not appear to be used here.

and $\mathrm{Lie}(u)$ *induces an isomorphism from* $\mathfrak{g}^\alpha$ *onto* ${\mathfrak{g}'}^\alpha$.

Indeed, let

$$p = \mathrm{Lie}(u) : \mathfrak{g} \longrightarrow \mathfrak{g}'.$$

One immediately has $p(\mathfrak{g}^\alpha) \subset {\mathfrak{g}'}^\alpha$ for every $\alpha \in R$, and $p(\mathfrak{g}^0) \subset {\mathfrak{g}'}^0$. Since

$$\mathrm{Ker}(p) = \mathrm{Lie}(Q) \subset \mathrm{Lie}\big(\mathrm{Centr}_G(T)\big) = \mathfrak{g}^0,$$

p induces a monomorphism from $\mathfrak{g}^\alpha$ into ${\mathfrak{g}'}^\alpha$, for every $\alpha \in R$.

To prove the lemma, it is enough to do so when S is the spectrum of an algebraically closed field; in view of the preceding remarks, it is then enough to prove that

$$\mathrm{rk}(\mathfrak{g}') = \mathrm{rk}({\mathfrak{g}'}^0) + \mathrm{Card}(R).$$

Set

$$C = \mathrm{Centr}_G(T), \qquad C' = \mathrm{Centr}_{G'}(T').$$

By Exposé XII, 7.1 (e), u induces a faithfully flat morphism $C \to C'$ with kernel Q. Hence

$$\dim C' + \dim Q = \dim C.$$

But G, G', C, and C' are smooth, and therefore

$$\begin{aligned}\dim G = \mathrm{rk}(\mathfrak{g}) &= \mathrm{rk}(\mathfrak{g}^0) + \mathrm{Card}(R) = \dim C + \mathrm{Card}(R)\\ &= \dim Q + \dim C' + \mathrm{Card}(R) = \dim Q + \mathrm{rk}({\mathfrak{g}'}^0) + \mathrm{Card}(R),\\ \mathrm{rk}(\mathfrak{g}') &= \dim G' = \dim G - \dim Q.\end{aligned}$$

It follows that

$$\mathrm{rk}(\mathfrak{g}') = \mathrm{rk}({\mathfrak{g}'}^0) + \mathrm{Card}(R),$$

which is the desired relation.

We return to the proof of 5.1.4. We may suppose that S is the spectrum of an algebraically closed field. Let T be a maximal torus of G. Applying 5.1.5, we immediately obtain (iii) and (iv) for G/Q.

To use the preceding proposition, we introduce a definition.

Definition 5.1.6. We say that the S-group scheme G is *of type* (RA) if it is of type (RR) and also satisfies condition (iv′) below, which is stronger than (iv):

(iv′) For every $s \in S$ and every maximal torus T of $G_{\bar{s}}$, the content of every root of $G_{\bar{s}}$ relative to T, in

$$\mathrm{Hom}_{\bar{s}\text{-gr.}}(T, \mathbb{G}_{m,\bar{s}}),$$

is invertible on S.

Remarks 5.1.7. (a) An adjoint reductive S-group is of type (RA).

(b) If S has characteristic 0, every group of type (RR) is of type (RA).

(c) The property of being of type (RA) is stable under base change and is local for the fpqc topology.

Remark (a) above admits the following generalization.

Proposition 5.1.8. *Let S be a scheme, let G be an S-group of type* (RR)*, and let Z be its reductive center. Suppose that the quotient G/Z is representable* (for example, if G is affine over S, or if S is artinian)*. Then G/Z is of type* (RA).

Indeed, this follows immediately from 5.1.4, 5.1.5, and 5.1.2 (d).

Remark 5.1.9. If G is of type (RR) and T is a maximal torus of G, Exposé XIX, 6.3 applies. In particular, $W_G(T)$ is étale, quasi-finite, and separated over S.

5.2. Subgroups of Type (R)

Definition 5.2.1. Let S be a scheme, let G be a smooth S-group scheme of finite presentation with connected fibers,[25] and let H be a group subscheme of G. We say that H is *of type* (R) if:

(i) H is smooth and of finite presentation over S, and has connected fibers;[25]

(ii) for every $s \in S$, $H_{\bar{s}}$ contains a Cartan subgroup of $G_{\bar{s}}$.

This notion is stable under base change and is local for the fpqc topology.

Recall 5.2.2. (cf. Exposé XII, 7.9) Under the preceding conditions:

(a)
$$H = \underline{\mathrm{Norm}}_G(H)^0.$$

(b) If G locally possesses Cartan subgroups (respectively maximal tori) for the étale topology, then so does H, and the Cartan subgroups (respectively maximal tori) of H are Cartan subgroups (respectively maximal tori) of G.

Examples 5.2.3. (a) *Borel subgroups.* A Borel subgroup of G is a subgroup H of type (R) whose geometric fibers are Borel subgroups of the corresponding fibers of G.[26]

(b) *Parabolic subgroups.* A parabolic subgroup of G is a subgroup of type (R) whose geometric fibers contain Borel subgroups.

Other examples are given by the following propositions.

Proposition 5.2.4. *Let G be as in 5.2.1, let $K \subset H$ be two group subschemes of G, and suppose that H is of type* (R)*. Then K is a subgroup of type* (R) *of H if and only if it is a subgroup of type* (R) *of G.*

Indeed,[27] let $s \in S$. Since H is of type (R), every maximal torus of $H_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$, and the same therefore holds for Cartan subgroups.

Proposition 5.2.5. *Let G be as in 5.2.1, let T be a maximal torus of G, let Q be a subtorus of T, and put $Z = \underline{\mathrm{Centr}}_G(Q)$. If H is a subgroup of type* (R) *of G containing T, then $H \cap Z$ is a subgroup of type* (R) *of Z.*

First recall that Z is a closed group subscheme of G, of finite presentation (Exposé XI, 6.11), with connected fibers (Exposé XII, 6.6), and smooth over S (Exposé XI, 2.4), so it satisfies the conditions imposed in Definition 5.2.1. Likewise, $H \cap Z$ is of finite presentation, smooth, and has connected fibers (since $H \cap Z = \underline{\mathrm{Centr}}_H(Q)$); moreover,

$$H \cap Z \supset \underline{\mathrm{Centr}}_G(T).$$

[25] Editor's note: The hypothesis that G (respectively H) is of finite presentation over S is automatically satisfied: since G (respectively H) is smooth over S with connected fibers, it is quasi-compact and separated over S (Exposé $\mathrm{VI_B}$, 5.5), and hence of finite presentation over S.

[26] Editor's note: Equivalently, H is a smooth subgroup of G such that every geometric fiber $H_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$ (since every Borel subgroup of $G_{\bar{s}}$ is connected and contains a Cartan subgroup of $G_{\bar{s}}$).

[27] Editor's note: We have expanded the argument that follows.

Proposition 5.2.6. *Let S be a scheme, let G be an S-group of type* (RR) (respectively (RA))*, and let H be a subgroup of type* (R) *of G. Then H is an S-group of type* (RR) (respectively (RA))*.*

Indeed, (i) is clear, (ii) follows from 5.2.2 (b), and (iii) and (iv) (respectively (iv′)) are to be verified when S is the spectrum of an algebraically closed field. Then H contains a maximal torus T of G, and therefore also

$$C = \underline{\mathrm{Centr}}_G(T),^{28}$$

and the assertions to be proved follow immediately from:

Lemma 5.2.7. *Let S be a scheme, let G be an S-group of type* (RR)*, and let T be a maximal torus of G equipped with a trivialization $T \simeq D_S(M)$. Suppose that*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha\in R} \mathfrak{g}^\alpha$$

(the $\mathfrak{g}^\alpha$ are then invertible $\mathscr{O}_S$-modules)*. Let H be a subgroup of type* (R) *containing $C = \underline{\mathrm{Centr}}_G(T)$* (that is, containing T)*. Then $\mathfrak{h} = \mathrm{Lie}(H/S)$ is locally on S of the form*

$$\mathfrak{g}^0 + \coprod_{\alpha\in R'} \mathfrak{g}^\alpha = \mathfrak{g}_{R'}.$$

More precisely, for each $s \in S$, put

$$R'(s) = \{\alpha \in R \mid \mathfrak{g}^\alpha(s) \subset \mathfrak{h}(s)\}.$$

Then $R'(s)$ is a locally constant function of s. If U is an open subset of S on which $R'(s) = R'$, then

$$\mathfrak{h}_U = \mathfrak{g}^0_U \oplus \coprod_{\alpha\in R'} \mathfrak{g}^\alpha_U.$$

Indeed, $\mathfrak{h}$ is a submodule of $\mathfrak{g}$, locally a direct factor, containing $\mathfrak{g}^0$, and stable under T.

5.3. Strict Transporter of Two Subgroups of Type (R). Applications

Recall 5.3.0. [29] Let S be a scheme, let G be a smooth S-group, put $\mathfrak{g} = \mathrm{Lie}(G/S)$, and let $\mathfrak{h}$ be an $\mathscr{O}_S$-submodule of $\mathfrak{g}$ that is locally a direct factor. The $\mathscr{O}_S$-algebra

$$\mathscr{A} = \mathrm{Sym}(\omega^1_{G/S})$$

is locally free; hence the S-scheme

$$\underline{\mathrm{Lie}}(G/S) = W(\mathfrak{g}) = \mathrm{Spec}(\mathscr{A})$$

is essentially free in the sense of Exposé VIII, 6.1. Since $W(\mathfrak{h})$ is a closed subscheme of $\underline{\mathrm{Lie}}(G/S)$, of finite presentation over $\underline{\mathrm{Lie}}(G/S)$, it follows from Exposé VIII, 6.5 (a) that

$$N = \underline{\mathrm{Norm}}_G(\mathfrak{h})$$

is represented by a closed group subscheme of G, of finite presentation over G. (See also the additions 6.2.3 and 6.2.4 (a) in Exposé $\mathrm{VI_B}$.) On the other hand, by Exposé II, 5.3.1,

$$\underline{\mathrm{Lie}}(N/S) = \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(\mathfrak{h}).$$

Finally, by Exposé $\mathrm{VI_B}$, 3.10, if N is smooth over S at the points of the unit section, then the group subfunctor N^0 (defined in Exposé $\mathrm{VI_B}$, 3.1) is represented by an open group subscheme of N, smooth over S.

[28] Editor's note: By hypothesis, H contains $\underline{\mathrm{Centr}}_G(T')$ for some maximal torus T' of G. The tori T and T' are then conjugate in H, and hence H also contains $C = \underline{\mathrm{Centr}}_G(T)$.

[29] Editor's note: This reminder was added; it is used in the proofs of 5.3.1 and 5.3.4.

Proposition 5.3.1. *Let S be a scheme, let G be an S-group of type* (RA) (5.1.6)*, let H be a subgroup of type* (R) *of G, and let $\mathfrak{g} \supset \mathfrak{h}$ be their Lie algebras. Then* $\underline{\mathrm{Norm}}_G(\mathfrak{h})$ (which, by 5.3.0, is representable by a closed subscheme of G, of finite presentation over S) *is smooth over S at every point of the unit section, and*

$$\underline{\mathrm{Norm}}_G(\mathfrak{h})^0 = H.$$

[30] *Proof.* Put

$$N = \underline{\mathrm{Norm}}_G(\mathfrak{h}), \qquad \mathfrak{n} = \mathrm{Lie}(N/S).$$

We have $H \subset N$, and by Exposé II, 5.3.1, for every $s \in S$,

$$\mathfrak{h}(s) \subset \mathfrak{n}(s) = \mathrm{Norm}_{\mathfrak{g}(s)}(\mathfrak{h}(s)).$$

Now, by 5.3.2 below,

$$\mathfrak{h}(s) = \mathrm{Norm}_{\mathfrak{g}(s)}(\mathfrak{h}(s)),$$

and since H is smooth over S,

$$\dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s$$

(cf. [DG70], §II.5, Theorem 2.1). Hence

$$\dim_{\kappa(s)} \mathfrak{n}(s) = \dim_{\kappa(s)} \mathfrak{h}(s) = \dim H_s \leq \dim N_s.$$

It follows that

$$N_s^0 = H_s^0 = H_s,$$

since H has connected fibers. Consequently the group subfunctor N^0 (defined in Exposé $\mathrm{VI_B}$, 3.1) is represented by the smooth S-group H. This proves 5.3.1, modulo the following lemma.

Lemma 5.3.2. *Under the conditions of 5.2.7, if G is of type* (RA)*, then, for every $s \in S$,*

$$\mathrm{Norm}_{\mathfrak{g}(s)}(\mathfrak{h}(s)) = \mathfrak{h}(s).$$

Indeed, we are reduced to the case where S is the spectrum of a field, and hence where

$$\mathfrak{h} = \mathfrak{g}_{R'}$$

for some $R' \subset R$. But we already have

$$\mathrm{Transp}_{\mathfrak{g}}(\mathfrak{t}, \mathfrak{h}) = \mathfrak{h}.$$

Indeed, if $H \in \mathfrak{t}$ and $X \in \mathfrak{g}^{\alpha}$, then

$$[H, X] = \bar{\alpha}(H)X,$$

where $\bar{\alpha} : \mathfrak{t} \to \mathcal{O}_S$ is the morphism derived from α. Condition (iv$'$) says precisely that $\bar{\alpha} \neq 0$ for every $\alpha \in R$.

Corollary 5.3.3. *Let S be a scheme, let G be an S-group of type* (RA)*, let H and H' be two subgroups of type* (R) *of G, and let $\mathfrak{h}$ and $\mathfrak{h}'$ be their Lie algebras. Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

[30] Editor's note: We have expanded the original in the argument that follows.

Corollary 5.3.4. *Under the conditions of 5.2.7, with G of type* (RA)*, the maps*

$$H \longmapsto \mathrm{Lie}(H/S), \qquad \mathfrak{h} \longmapsto \underline{\mathrm{Norm}}_G(\mathfrak{h})^0$$

give a bijective correspondence between the set of subgroups of type (R) *of G containing T, and the set of Lie subalgebras of $\mathfrak{g}$ containing $\mathfrak{g}^0$, stable under T, and whose normalizer in G is smooth over S at every point of the unit section.*

Indeed,[31] let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the properties above. By 5.3.0,

$$H = \underline{\mathrm{Norm}}_G(\mathfrak{h})^0$$

is a smooth S-group scheme. Moreover, since

$$C = \underline{\mathrm{Centr}}_G(T)$$

stabilizes every $\mathfrak{g}^\alpha$ and has connected fibers (Exposé XII, 6.6), we have $C \subset H$. Therefore H is a subgroup of G of type (R). By Exposé II, 5.3.1,

$$\underline{\mathrm{Lie}}(H) = \underline{\mathrm{Norm}}_{\mathfrak{g}}(\mathfrak{h}).$$

Finally, by the proof of 5.3.2,

$$\underline{\mathrm{Norm}}_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

Corollary 5.3.5. *Let S be a scheme, let G be an S-group of type* (RR) (5.1.1)*, let T be a maximal torus of G, and let H and H' be two subgroups of type* (R) *of G, both containing T. Then*

$$H = H' \iff \mathfrak{h} = \mathfrak{h}'.$$

By virtue of the finite-presentation hypotheses, we reduce as usual (cf. EGA IV_3, §8, and Exposé $\mathrm{VI_B}$, §10) to the case where S is noetherian. It is then enough to verify that $\mathfrak{h} = \mathfrak{h}'$ implies

$$H_{S'} = H'_{S'}$$

for every S' that is the spectrum of an artinian quotient of a local ring of S;[32] we are therefore reduced to the case where S is artinian, in which case we may apply 5.1.8.

Let

$$u : G \longrightarrow G' = G/Z$$

be the canonical morphism, and let $T' = T/Z$ be the maximal torus of G' corresponding to T. By Exposé XII, 7.12, there exist subgroups H_1 and H'_1 of type (R) of G', containing T', such that

$$H = u^{-1}(H_1), \qquad H' = u^{-1}(H'_1).$$

It is enough to prove that $H_1 = H'_1$. But by 5.2.7 and 5.1.5,

$$\mathrm{Lie}(H_1) = \mathrm{Lie}(H'_1),$$

and we are reduced to 5.3.3.

[31] Editor's note: The proof that follows was added.

[32] Editor's note: Indeed, let $g \in G$, let s be its image in S, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{G,g}$, let $\mathfrak{n}$ be that of $\mathcal{O}_{S,s}$, and let I (respectively I') be the kernel of the morphism from $\mathcal{O}_{G,g}$ to $\mathcal{O}_{H,g}$ (respectively $\mathcal{O}_{H',g}$), the latter being the zero ring if $g \notin H$ (respectively $g \notin H'$). Since $\mathcal{O}_{G,g}$ is noetherian, I and I' are closed for the $\mathfrak{m}$-adic topology, hence *a fortiori* for the $\mathfrak{n}$-adic topology. It is therefore enough to prove

$$I + \mathfrak{n}^n\mathcal{O}_{G,g} = I' + \mathfrak{n}^n\mathcal{O}_{G,g}$$

for every $n \in \mathbb{N}$.

Remark 5.3.6. The fact that H and H' contain the same maximal torus is essential for the validity of 5.3.5 when G is not of type (RA). For example, consider the maximal tori of $\mathrm{SL}_{2,k}$, where k has characteristic 2.[33]

Corollary 5.3.7. *Let S be a scheme, let G be an S-group of type* (RR)*, let T be a maximal torus of G, and let H and H' be two subgroups of type* (R) *of G, both containing T. The set U of $s \in S$ such that $H_s = H'_s$ is open and closed in S, and*

$$H_U = H'_U.$$

Indeed, this follows immediately from 5.3.5 and 5.2.7.

Corollary 5.3.8. *The "functor of subgroups of type* (R) *containing T", where T is a given maximal torus in a group G of type* (RR)*, is formally unramified* (Exposé XI, 1.1).

Theorem 5.3.9. *Let G be an S-group of type* (RR) (5.1.1)*, and let H and H' be two subgroups of type* (R) (5.2.1)*. Let $\underline{\mathrm{Transt}}_G(H, H')$ be the strict transporter of H into H', defined by*

$$\underline{\mathrm{Transt}}_G(H, H')(S') = \{g \in G(S') \mid \mathrm{int}(g)H_{S'} = H'_{S'}\}.$$

Then $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G, smooth and of finite presentation over S.

The fact that $\underline{\mathrm{Transt}}_G(H, H')$ is representable by a closed subscheme of G, of finite presentation over S, follows from Exposé XI, 6.11 (a). To prove that it is smooth over S, it is necessary to prove that if S is affine, if S_0 is the closed subscheme defined by a nilpotent ideal J, and if

$$g_0 \in G(S_0), \qquad \mathrm{int}(g_0)H_0 = H'_0,$$

then there exists $g \in G(S)$, lifting g_0, such that

$$\mathrm{int}(g)H = H'.$$

Since smoothness is local for the étale topology, we may suppose that H contains a maximal torus T of G.

Then T_0 is a maximal torus of H_0, and hence $\mathrm{int}(g_0)T_0$ is a maximal torus of H'_0. By Exposé IX, 3.6 bis, there exists a torus T' of H' such that

$$T'_0 = \mathrm{int}(g_0)T_0;$$

by Exposé IX, 3.3 bis, there therefore exists $g \in G(S)$, lifting g_0, such that

$$\mathrm{int}(g)T = T'.$$

After replacing H by $\mathrm{int}(g)H$, we may therefore suppose that H and H' contain the same maximal torus T and that $H_0 = H'_0$. But then $H = H'$ by 5.3.7. □

Corollary 5.3.10. *Let G be an S-group of type* (RR)*, and let H be a subgroup of type* (R) *of G. Then $\underline{\mathrm{Norm}}_G(H)$ is representable by a closed group subscheme of G, of finite presentation and smooth over S.*

Using now the reasoning employed in Exposé XI to deduce 5.4 bis from 5.2 bis, we obtain:

[33] Editor's note: Indeed, if k is algebraically closed, all maximal tori of $G = \mathrm{SL}_{2,k}$ are conjugate under $G(k)$, and they all have as Lie algebra the line $k\,\mathrm{id} \subset M_2(k)$, which is invariant under the adjoint action.

Corollary 5.3.11. *Under the hypotheses of 5.3.9, the following conditions are equivalent:*

(i) *H and H' are locally conjugate in G for the étale topology;*

(i bis) *the same holds for the fpqc topology;*

(ii) *for every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$;*

(ii bis) *the structural morphism*

$$\underline{\mathrm{Transt}}_G(H, H') \longrightarrow S$$

is surjective;

(iii) *$\underline{\mathrm{Transt}}_G(H, H')$ is a principal homogeneous bundle under the action of the smooth S-group scheme of finite presentation $\underline{\mathrm{Norm}}_G(H)$.*

We merely note that the nontrivial implication (iii)⇒(i) is Hensel's lemma.

Using now *Bible*, §6.4, Theorem 4 (= [Ch05], §6.5, Theorem 5) and §9.3, Theorem 1, we obtain from 5.3.10 and 5.3.11:

Corollary 5.3.12. *Let G be an S-group of type* (RR). *The Borel subgroups of G are closed in G, equal to their normalizers, and locally conjugate for the étale topology.*

Definition 5.3.13. *Let S be a scheme, and let G be a smooth S-group scheme of finite presentation with connected fibers.*[34] *A* Killing pair *of G is a pair $T \subset B$, where T is a maximal torus of G and B is a Borel subgroup of G containing T.*

Using now the conjugacy of maximal tori in B (cf. 5.1.2 (a) and 5.2.6, for example), we have:

Corollary 5.3.14. *Let G be an S-group of type* (RR). *The Killing pairs of G are locally conjugate for the étale topology.*

Corollary 5.3.15. *Let G be an S-group of type* (RR). *Let T be a maximal torus of G, and let*

$$W_G(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T)$$

be the corresponding Weyl group (Exposé XIX, 6.3). *The "functor of Borel subgroups of G containing T" is formally principal homogeneous under $W_G(T)$.*

This follows immediately from 5.3.14 and from the fact that, if B is a Borel subgroup of G containing T, then

$$\underline{\mathrm{Norm}}_G(T) \cap B = \underline{\mathrm{Centr}}_G(T),$$

cf. Exposé XIV, 4.4.

Proposition 5.3.16. *Let G be an S-group of type* (RR), *let H be a subgroup of type* (R), *and let $N = \underline{\mathrm{Norm}}_G(H)$ be its normalizer* (5.3.10). *Let T be a maximal torus of H, and let $W_H(T)$ and $W_N(T)$ be the corresponding Weyl groups* (étale, quasi-finite, and separated by Exposé XIX, 6.3). *There is the following exact sequence of sheaves* (for the étale topology), *whose morphisms are induced by*

$$\underline{\mathrm{Norm}}_H(T) \longrightarrow \underline{\mathrm{Norm}}_N(T) \longrightarrow N/H :$$

[34] Editor's note: See note 25.

$$1 \longrightarrow W_H(T) \longrightarrow W_N(T) \longrightarrow N/H \longrightarrow 1.$$

The only nontrivial point is that the last arrow is an epimorphism. Let $n \in N(S')$, where $S' \to S$. The two maximal tori T and $\mathrm{int}(n)T$ of H are locally conjugate in H for the étale topology. Thus there is a covering family $\{S'_i \to S'\}$ and, for every i, an $h_i \in H(S'_i)$ such that

$$\mathrm{int}(h_i)T = \mathrm{int}(n)T.$$

Consequently

$$nh_i^{-1} \in \underline{\mathrm{Norm}}_N(T),$$

which gives the desired result.

Remark 5.3.17. The group $W_N(T)$ may be described as follows. Suppose that we have reduced to the situation of 5.2.7, with $\mathfrak{h} = \mathfrak{g}_{R'}$. Then $W_N(T)$ is equal to $\underline{\mathrm{Norm}}_W(R')$, the sheaf of sections of $W = W_G(T)$ which, acting on R, normalize R'. Indeed, by 5.3.5,

$$\underline{\mathrm{Norm}}_N(T) = \underline{\mathrm{Norm}}_G(H) \cap \underline{\mathrm{Norm}}_G(T) = \underline{\mathrm{Norm}}_G(\mathfrak{h}) \cap \underline{\mathrm{Norm}}_G(T).$$

Corollary 5.3.18. *Let G be an S-group of type* (RR)*, and let H be a subgroup of type* (R)*. Suppose that "the Weyl groups of G are finite", that is, that for every $S' \to S$ and every maximal torus T of $G_{S'}$, the étale S'-scheme*

$$\underline{\mathrm{Norm}}_{G_{S'}}(T)/\underline{\mathrm{Centr}}_{G_{S'}}(T)$$

is finite (cf. Exposé XIX, 6.3 (iii))*. Then the following conditions are equivalent:*

(i) *H is closed in G;*

(ii) *$\underline{\mathrm{Norm}}_G(H)/H$ is representable by a finite étale S-scheme;*

(iii) *"the Weyl groups of H are finite".*

Indeed, we may suppose that H has a maximal torus T. By 5.3.10, $N = \underline{\mathrm{Norm}}_G(H)$ is closed in G; therefore $W_N(T)$ is closed in $W_G(T)$, and hence is finite over S. Clearly (i) implies (iii), and (iii) implies (ii) by the exact sequence of 5.3.16. Finally, (ii) implies (i), since if N/H is finite, it is separated, so H is closed in N, and hence in G.

Remark 5.3.19. When G is reductive, the preceding conditions on H seem always to be satisfied. We prove them below in most cases.

5.4. Subgroups of Type (R) of a Split Reductive Group (Generalities)

5.4.1. If H is a subgroup of type (R) of the reductive group G, then H locally for the étale topology contains a maximal torus of G (5.2.2). By 2.3, locally for the étale topology we may suppose that G is split relative to this torus. Thus let (G, T, M, R) be a split S-group, and let H be a subgroup of type (R) of G containing T. By 5.3.5, such a subgroup is characterized by its Lie algebra, which (5.2.7) is locally on S of the form $\mathfrak{g}_{R'}$:

$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \prod_{\alpha \in R'} \mathfrak{g}^\alpha.$$

Definition 5.4.2. *Let (G, T, M, R) be a split S-group. A subset R' of R is said to be of type* (R) *if $\mathfrak{g}_{R'}$ is the Lie algebra of a subgroup of type* (R) *of G containing T. This subgroup, uniquely determined by R', is denoted $H_{R'}$.*

Lemma 5.4.3. *Under the preceding conditions, the following equivalences hold:*

$$\begin{aligned} H \cap Z_\alpha = T \quad &\Longleftrightarrow \quad \alpha \notin R',\ -\alpha \notin R',\\ H \supset U_\alpha \quad &\Longleftrightarrow \quad \alpha \in R',\\ H \cap U_\alpha = e \quad &\Longleftrightarrow \quad \alpha \notin R',\\ H \supset Z_\alpha \quad &\Longleftrightarrow \quad \alpha \in R',\ -\alpha \in R'. \end{aligned}$$

Indeed, $H \cap Z_\alpha$ is a subgroup of type (R) of Z_α, by 5.2.5; but a subgroup of type (R) of Z_α containing T is locally equal to one of the following subgroups:

$$T, \qquad T \cdot U_\alpha, \qquad T \cdot U_{-\alpha}, \qquad Z_\alpha,$$

by 5.3.5.

Lemma 5.4.4. *Under the conditions of 5.4.2, let R_+ be a system of positive roots; choose orderings on $R' \cap R_+$ and $R' \cap -R_+$. The morphism*

$$\Omega_{R_+,R'} = \prod_{\alpha \in R' \cap -R_+} U_\alpha \underset{S}{\times} T \underset{S}{\times} \prod_{\alpha \in R' \cap R_+} U_\alpha \longrightarrow G,$$

induced by multiplication in G, induces an open immersion

$$\Omega_{R_+,R'} \longrightarrow H_{R'}.$$

Indeed, by 5.4.3 this morphism factors through $H_{R'}$, and therefore induces an immersion $\Omega_{R_+,R'} \to H_{R'}$. We then argue as in 4.1.1.

Proposition 5.4.5. *Let (G, T, M, R) be a split S-group. Let R' and R'' be two subsets of R of type* (R).

(i) *$H_{R'} \cap H_{R''}$ is smooth at every point of the unit section, $R' \cap R''$ is of type* (R), *and*

$$\left(H_{R'} \cap H_{R''}\right)^0 = H_{R' \cap R''}.$$

(ii) *The following equivalence holds:*

$$H_{R'} \subset H_{R''} \quad \Longleftrightarrow \quad R' \subset R''.$$

Indeed, (ii) follows immediately from (i). To prove (i), it suffices to prove that $H_{R'} \cap H_{R''}$ is smooth at every point of the unit section: its neutral component (cf. Exposé $\mathrm{VI_B}$, 3.10) will then be a subgroup of type (R) containing T, and hence equal to $H_{R' \cap R''}$. But

$$\Omega_{R_+,R'} \cap \Omega_{R_+,R''} = \Omega_{R_+,R' \cap R''}$$

is an open subset of $H_{R'} \cap H_{R''}$, contains the unit section, and is smooth over S.

Corollary 5.4.6. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, and s a point of S. If H and H' are two subgroups of type* (R) *containing T, and if $H_s \subset H'_s$, then there is an open subset U of S containing s such that $H_U \subset H'_U$.*

Indeed, we may suppose that G is split relative to T, and the assertion then follows immediately from 5.4.5 (ii).

We are thus led to ask which subsets R' of R are of type (R). We may suppose that the group is adjoint; one must then verify that $\mathfrak{g}_{R'}$ is a Lie algebra and that its normalizer is smooth at every point of the unit section. The most important case is given by the following theorem.

Theorem 5.4.7. *Every closed subset R' of R is of type* (R)*.*

Recall (Exposé XXI, 3.1.4) that a subset R' of R is said to be *closed* if

$$\alpha, \beta \in R', \quad \alpha + \beta \in R \quad \Longrightarrow \quad \alpha + \beta \in R'.$$

Remark 5.4.8. We shall see later (Exposé XXIII, 6.6) that, if $6 \cdot 1_S \neq 0$[35]—for example, if S has residue characteristics different from 2 and 3—the fact that $\mathfrak{g}_{R'}$ is a Lie algebra already implies that R' is closed. In that case a subset R' is of type (R) if and only if it is closed. Thus Theorem 5.4.7 gives all the subsets of type (R) that are "independent of the characteristic."

Let us first prove the following lemma.

Lemma 5.4.9. *For each $\alpha \in R$, choose an element*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

Let $\alpha, \beta \in R$, with $\alpha + \beta \neq 0$, and let q be the largest integer i such that $\beta + i\alpha \in R$. There exist uniquely determined sections

$$M_{\alpha,\beta,i} \in \Gamma(S, \mathcal{O}_S)$$

such that

$$\mathrm{Ad}\big(\exp_\alpha(xX_\alpha)\big)(X_\beta) = X_\beta + \sum_{i=1}^{q} M_{\alpha,\beta,i} x^i X_{\beta+i\alpha}$$

for every $x \in \mathbb{G}_a(S')$, where $S' \to S$.

The map

$$x \longmapsto \mathrm{Ad}\big(\exp_\alpha(xX_\alpha)\big)(X_\beta)$$

defines a morphism

$$\mathbb{G}_{a,S} \longrightarrow \mathbf{W}(\mathfrak{g}) \simeq \mathbb{G}_{a,S}^m.$$

There are therefore uniquely determined sections $Y_n \in \Gamma(S, \mathfrak{g})$ such that

$$\mathrm{Ad}\big(\exp_\alpha(xX_\alpha)\big)(X_\beta) = \sum_{n \geq 0} x^n Y_n.$$

Applying the inner automorphism defined by a section t of T, we immediately find

$$\mathrm{Ad}(t)(Y_n) = \beta(t)\alpha(t)^n Y_n,$$

and hence $Y_n \in \Gamma(S, \mathfrak{g}^{\beta+n\alpha})$. Since α and β are not proportional, none of the $\beta + n\alpha$ is zero. Thus $Y_n = 0$ for $n > q$, and

$$Y_n = M_{\alpha,\beta,n} X_{\beta+n\alpha}$$

for $0 \leq n \leq q$, where $M_{\alpha,\beta,n} \in \mathbb{G}_a(S)$ is uniquely determined. Setting $x = 0$ in the formula obtained gives $Y_0 = X_\beta$, which completes the proof.

[35] Editor's note: In fact, it suffices (cf. loc. cit.) that 2 and 3 be nonzero on S.

Remark 5.4.10. Differentiating the preceding formula at $x = 0$, we find

$$[X_\alpha, X_\beta] = \begin{cases} N_{\alpha,\beta}X_{\alpha+\beta}, & \text{where } N_{\alpha,\beta} = M_{\alpha,\beta,1}, \text{ if } \alpha+\beta \in R, \\ 0, & \text{if } \alpha+\beta \notin R \text{ and } \alpha+\beta \neq 0. \end{cases}$$

Let us now prove 5.4.7. If R' is a closed subset of R, then $\mathfrak{g}_{R'}$ is a Lie subalgebra of $\mathfrak{g}$, by 5.4.10 and Exposé XX, 2.10, formula (3). By 5.4.9 and Exposé XX, 2.10, formula (2), U_α normalizes $\mathfrak{g}_{R'}$ for every $\alpha \in R'$. Choose a system of positive roots R_+, and consider the open subset Ω_{R_+} of 4.1.2. Let $\Omega_{R_+,R'}$ be the closed subscheme of Ω_{R_+} defined by

$$\Omega_{R_+,R'} = \left(\prod_{\alpha\in R'\cap -R_+} U_\alpha\right) \cdot T \cdot \left(\prod_{\alpha\in R'\cap R_+} U_\alpha\right).$$

The canonical immersion $\Omega_{R_+,R'} \to G$ factors through

$$i : \Omega_{R_+,R'} \longrightarrow \underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'}).$$

Suppose that G is adjoint. The tangent map of i at the points of the unit section is bijective by 5.3.2. In particular, i is étale at every point of the unit section, and hence is a local immersion[36] at every point of the unit section. Therefore $\underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the unit section, as was to be proved.

5.5. Borel Subgroups of a Split Reductive Group

Proposition 5.5.1. *Let (G, T, M, R) be a split S-group. For every system of positive roots R_+ of R, the subgroup H_{R_+}* (which exists by 5.4.7) *is a Borel subgroup of G. Moreover, for every ordering on R_+, the morphism induced by multiplication in G,*

$$T \underset{S}{\times} \prod_{\alpha\in R_+} U_\alpha \longrightarrow G,$$

is a closed immersion with image H_{R_+}. We write $B_{R_+} = H_{R_+}$.

By the definition of Borel subgroups, the first assertion may be verified after replacing S by the spectrum of an algebraically closed field. Let B be the Borel subgroup of G containing T and corresponding to the system of positive roots R_+ (*Bible*, §10.4, Proposition 9). The Lie algebra of B is $\mathfrak{g}_{R_+}$; therefore $B = H_{R_+}$ by 5.3.5.

For the second assertion, the morphism in the statement induces an open immersion

$$i : T \underset{S}{\times} \prod_{\alpha\in R_+} U_\alpha \longrightarrow H_{R_+}$$

by 5.4.4. But i is surjective (*Bible*, §15.1, Corollary 1 to Proposition 1).

Corollary 5.5.2. *Choose an arbitrary ordering on R_+, and for each $\alpha \in R_+$ choose an*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

Let $\alpha, \beta \in R_+$. For every pair $(i, j) \in \mathbb{N}^* \times \mathbb{N}^*$ such that $i\alpha + j\beta \in R$, there is a unique section

$$C_{i,j,\alpha,\beta} \in \Gamma(S, \mathcal{O}_S)$$

[36] Editor's note: By EGA IV_4, 17.11.2, i is étale at every point of the unit section (and $\underline{\mathrm{Norm}}_G(\mathfrak{g}_{R'})$ is smooth at every point of the unit section). Moreover, let V be the largest open subset of $\Omega_{R_+,R'}$ on which i is étale. Since i is a monomorphism, i_V is an open immersion (ibid., 17.9.1).

such that, for all $x, y \in \mathbb{G}_a(S')$*, where* $S' \to S$*, one has*

$$\exp_\alpha(xX_\alpha)\exp_\beta(yX_\beta)\exp_\alpha(xX_\alpha)^{-1} = \exp_\beta(yX_\beta)\prod_{\substack{i,j\in\mathbb{N}^*\\ i\alpha+j\beta\in R}}\exp_{i\alpha+j\beta}(C_{i,j,\alpha,\beta}x^iy^jX_{i\alpha+j\beta}).$$

If $\alpha = \beta$, the assertion is trivial. Suppose therefore that $\alpha \neq \beta$. By the proposition there are unique morphisms

$$F_0 : \mathbb{G}^2_{a,S} \longrightarrow T, \qquad F_\gamma : \mathbb{G}^2_{a,S} \longrightarrow \mathbb{G}_{a,S} \quad (\gamma \in R_+)$$

such that

$$\exp(xX_\alpha)\exp(yX_\beta)\exp(xX_\alpha)^{-1} = F_0(x,y)\prod_{\gamma\in R_+}\exp(F_\gamma(x,y)X_\gamma).$$

Let $t \in T(S')$, where $S' \to S$. Applying $\mathrm{int}(t)$ to this formula, we immediately obtain

$$F_0(\alpha(t)x, \beta(t)y) = F_0(x,y), \tag{1}$$

$$F_\gamma(\alpha(t)x, \beta(t)y) = \gamma(t)F_\gamma(x,y). \tag{2}$$

Since α and β are two linearly independent characters of T over $\mathbb{Q}$, the first relation implies, as usual, that F_0 is constant, and hence $F_0(x,y) = e$. Write next

$$F_\gamma(x,y) = \sum a_{ij}x^iy^j, \qquad a_{ij} \in \Gamma(S, \mathcal{O}_S).$$

Substituting in relation (2) and comparing the polynomials on both sides gives

$$a_{ij}\big(\alpha(t)^i\beta(t)^j - \gamma(t)\big) = 0.$$

If $\gamma \neq i\alpha + j\beta$, we know (Exposé XIX, 4.13) that there exist a faithfully flat quasi-compact morphism $S' \to S$ and an element $t \in T(S')$ such that

$$\alpha(t)^i\beta(t)^j - \gamma(t) = 1.$$

Thus $a_{ij} = 0$ over S', and hence over S. If $\gamma = i\alpha + j\beta$, set $a_{ij} = C_{i,j,\alpha,\beta}$. Setting $x = 0$ (respectively, $y = 0$), one obtains

$$C_{0,1,\alpha,\beta} = 1 \qquad (\text{respectively, } C_{1,0,\alpha,\beta} = 0).$$

Remark 5.5.3. Differentiating with respect to y at $y = 0$ and comparing with 5.4.9, one finds

$$C_{i,1,\alpha,\beta} = M_{\alpha,\beta,i}.$$

Corollary 5.5.4. *Let S be a scheme, let G be a reductive S-group, let T be a maximal torus of G, and let $\alpha \neq \beta$ be two roots of G relative to T such that $\alpha + \beta$ is nontrivial on every fiber. Order arbitrarily the set of roots $i\alpha + j\beta$, where $(i,j) \in \mathbb{N}^* \times \mathbb{N}^*$. For all $i, j \in \mathbb{N}^*$ such that $i\alpha + j\beta \in R$, there is a unique morphism of $\mathcal{O}_S$-modules*

$$f_{\alpha,\beta,i,j} : (\mathfrak{g}^\alpha)^{\otimes i} \otimes (\mathfrak{g}^\beta)^{\otimes j} \longrightarrow \mathfrak{g}^{i\alpha+j\beta}.$$

such that, for every $S' \to S$ and all $X \in \mathbf{W}(\mathfrak{g}^\alpha)(S')$, $Y \in \mathbf{W}(\mathfrak{g}^\beta)(S')$, one has (the exponentials in the right-hand side being understood in the sense of 1.2)[37]

$$\exp_\alpha(X)\exp_\beta(Y)\exp_\alpha(-X) = \exp_\beta(Y)\prod_{(i,j)}\exp_{i\alpha+j\beta}(f_{\alpha,\beta,i,j}(X^{\otimes i}\otimes Y^{\otimes j})).$$

The assertion is local for the fpqc topology. By §2 we may therefore suppose that G is split relative to T, with α and β constant in the splitting. Since $\alpha + \beta \neq 0$, there is a system of positive roots R_+ containing α and β (Exposé XXI, 3.5.4), and we are reduced to 5.5.2.

[37] Editor's note: That is, if $\mathfrak{g}^{i\alpha+j\beta} = 0$ on a connected component of S, the corresponding exponential is 1 on that component.

Corollary 5.5.5. *Let S be a scheme and G a reductive S-group.*

(i) *G has Borel subgroups locally for the étale topology. If T is a maximal torus of G, then, locally for the étale topology, G also has Borel subgroups containing T.*

(ii) *If T is a maximal torus of G, the "functor of Borel subgroups of G containing T" is represented by a principal homogeneous bundle under $W_G(T)$.*

(iii) *If (G, T, M, R) is split, every Borel subgroup B of G containing T is locally on S of the form B_{R_+}, where R_+ is a system of positive roots of R.*

(iv) *If $T \subset B$ is a Killing pair of G, there is an étale covering family $\{S_i \to S\}$, and for every i a splitting $(G_{S_i}, T_{S_i}, M_i, R_i)$ and a system of positive roots R_{i+} of R_i, such that $B_{S_i} = B_{R_{i+}}$.*

Indeed, (i) follows from 2.3 and 5.5.1, (ii) from (i) and 5.3.15, (iii) from (ii) and 5.5.1, and (iv) from (iii) and 2.3.

Lemma 5.5.6. *Choose on the group $\Gamma_0(R)$ generated by the roots a totally ordered group structure such that the roots > 0 are the elements of R_+* (cf. Exposé XXI, 3.5.6)[38] *Let $\alpha_1 < \cdots < \alpha_N$ be the elements of R_+. Consider the isomorphism*

$$f : T \underset{S}{\times} U_{\alpha_1} \underset{S}{\times} \cdots \underset{S}{\times} U_{\alpha_N} \longrightarrow B_{R_+}$$

induced by multiplication in G. For $i = 1, \dots, N$, put

$$U_{\geq i} = f\left(U_{\alpha_i} \underset{S}{\times} \cdots \underset{S}{\times} U_{\alpha_N}\right).$$

(i) *Each $U_{\geq i}$ is an invariant subgroup of B_{R_+}.*

(ii) *For $1 \leq i \leq N-1$, $U_{\geq i}$ identifies with the semidirect product*

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1}.$$

(iii) *B_{R_+} identifies with the semidirect product*

$$B_{R_+} = T \cdot U_{\geq 1}.$$

(iv) *For $1 \leq i \leq N-1$, the inner automorphisms of $U_{\geq 1}$ act trivially on $U_{\geq i}/U_{\geq i+1}$, identified with U_{α_i} by* (ii).

We first prove, by descending induction on i, the following assertion:

$U_{\geq i}$ is an invariant subgroup of B_{R_+}, the semidirect product of U_{α_i} and $U_{\geq i+1}$.

This is true for $i = N$; suppose it has been proved for $i+1$, and let us prove it for i. As schemes, we have

$$U_{\geq i} = U_{\alpha_i} \cdot U_{\geq i+1}.$$

We must show that $U_{\geq i}$ is invariant under the inner automorphisms of B_{R_+}. This is clear for $\mathrm{int}(t)$, where $t \in T(S)$; it is therefore enough to consider $\mathrm{int}(x)$, where $x \in U_\alpha(S)$ and $\alpha \in R_+$. Since $U_{\geq i+1}$ is invariant, it is enough to prove that

$$\mathrm{int}(x)(U_{\alpha_i}) \subset U_{\geq i}.$$

[38] Editor's note: Such an ordering is necessarily compatible with ord_Δ, where $\Delta = \mathscr{S}(R_+)$ (cf. Exposé XXI, 3.2.15).

By 5.5.2, if $y \in U_{\alpha_i}(S')$, then
$$y^{-1}xyx^{-1} \in U_{\geq i+1}(S'),$$
which implies that
$$\mathrm{int}(x)(y) \in U_{\geq i}(S').$$

It remains to prove that $U_{\geq i}$ is a subgroup. Let $x, y \in U_{\geq i}(S)$, and write
$$x = px', \qquad y = qy',$$
where $p, q \in U_{\alpha_i}(S)$ and $x', y' \in U_{\geq i+1}(S)$. We then have
$$xy = px'qy' = pq(q^{-1}x'q)y' \in U_{\alpha_i}(S')U_{\geq i+1}(S'),$$
and likewise
$$x^{-1} = p^{-1}(px'^{-1}p^{-1}) \in U_{\alpha_i}(S')U_{\geq i+1}(S').$$
We have thus proved (i) and (ii), as well as (iv) along the way. Assertion (iii) is a trivial consequence of 5.5.1.

Lemma 5.5.7. *With the preceding notation, choose, for each $1 \leq i \leq N$, an element*
$$X_i \in \Gamma(S, \mathfrak{g}^{\alpha_i})^\times,$$
and consider the isomorphism
$$a : \mathbb{G}_{a,S}^N \longrightarrow U_{\geq 1}$$
defined on underlying sets by
$$a(x_1, \ldots, x_N) = \exp_{\alpha_1}(x_1X_1) \cdots \exp_{\alpha_N}(x_NX_N).$$
There is a unique family $(Q_i)_{1 \leq i \leq N}$ of polynomials
$$Q_i = Q_i(x_1, \ldots, x_N, y_1, \ldots, y_N)$$
with coefficients in $\Gamma(S, \mathcal{O}_S)$, such that
$$\begin{aligned} &a(x_1, \ldots, x_N)a(y_1, \ldots, y_N) \\ &\qquad = a\big(Q_1(x_1, \ldots, x_N, y_1, \ldots, y_N), \ldots, Q_N(x_1, \ldots, x_N, y_1, \ldots, y_N)\big). \end{aligned}$$
Moreover, the coefficients of these polynomials belong to the subring of $\Gamma(S, \mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$ of 5.5.2 $(\alpha, \beta \in R_+,\ i, j \in \mathbb{N}^*)$, and each Q_i is of the form
$$\begin{aligned} Q_i(x_1, \ldots, x_N, y_1, \ldots, y_N) &= x_i + y_i \\ &\qquad + Q_i'(x_1, \ldots, x_{i-1}, y_1, \ldots, y_{i-1}). \end{aligned}$$

The existence and uniqueness of the Q_i are immediate from the fact that a is an isomorphism of schemes. Let z, z', z'' denote sections of $U_{\geq i+1}$. We have
$$\begin{aligned} &a(x_1, \ldots, x_N)a(y_1, \ldots, y_N) \\ &\quad = a(x_1, \ldots, x_{i-1}, 0, \ldots, 0) \exp_{\alpha_i}(x_iX_i)z \\ &\qquad \cdot a(y_1, \ldots, y_{i-1}, 0, \ldots, 0) \exp_{\alpha_i}(y_iX_i)z'. \end{aligned}$$
Using 5.5.6 (i) and (iv), the right-hand side can be written
$$\begin{aligned} &a(x_1, \ldots, x_{i-1}, 0, \ldots, 0)a(y_1, \ldots, y_{i-1}, 0, \ldots, 0) \\ &\qquad \cdot \exp_{\alpha_i}((x_i + y_i)X_i)z'', \end{aligned}$$

and, applying 5.5.6 once again, this gives

$$Q_i(x_1,\dots,x_N,y_1,\dots,y_N) = x_i + y_i + Q'_i(x_1,\dots,x_{i-1},y_1,\dots,y_{i-1}),$$

where

$$Q'_i(x_1,\dots,x_{i-1},y_1,\dots,y_{i-1}) = Q_i(x_1,\dots,x_{i-1},0,\dots,0,y_1,\dots,y_{i-1},0,\dots,0);$$

this is precisely the required form.[39]

It remains to prove the assertion concerning the coefficients of the polynomials Q_i. Let A be the subring of $\Gamma(S,\mathcal{O}_S)$ generated by the $C_{i,j,\alpha,\beta}$, where $\alpha,\beta\in R_+$ and $i,j\in\mathbb{N}^*$. We prove by descending induction on i that, if

$$x_1=\cdots=x_{i-1}=0 \qquad\text{and}\qquad y_1=\cdots=y_{i-1}=0,$$

that is, if $a(x_1,\dots,x_N)$ and $a(y_1,\dots,y_N)$ are sections of $U_{\geq i}$, then the polynomials

$$R_j(x_i,\dots,x_N,y_i,\dots,y_N) = Q_j(x_1,\dots,x_N,y_1,\dots,y_N)$$

have coefficients in A. This is trivial for $i=N$, and also for $j<i$, since $R_j=0$ when $j<i$. Let $i<N$; suppose the assertion proved for $i+1$, and let us prove it for i and $j\geq i$. We have

$$\begin{aligned} a(0,\dots,0,x_i,\dots,x_N) &= \exp(x_iX_i)a(0,\dots,0,x_{i+1},\dots,x_N)\\ &= \exp(x_iX_i)Z_i. \end{aligned}$$

Similarly, write

$$a(0,\dots,y_i,\dots,y_N) = \exp(y_iX_i)T_i.$$

Then

$$\begin{aligned} &a(0,\dots,x_i,\dots,x_N)a(0,\dots,y_i,\dots,y_N)\\ &\qquad = \exp\big((x_i+y_i)X_i\big)\operatorname{int}\big(\exp(-y_iX_i)\big)(Z_i)\cdot T_i. \end{aligned}$$

Now[40]

$$\operatorname{int}\big(\exp(-y_iX_i)\big)(Z_i) = \operatorname{int}\big(\exp(-y_iX_i)\big)\big(\exp(x_{i+1}X_{i+1})\cdots\exp(x_NX_N)\big)$$

is a product of $N-i-1$ sections of $U_{\geq i+1}$, whose coefficients in the decomposition

$$U_{\geq i+1} = U_{\alpha_{i+1}}\cdots U_{\alpha_N}$$

are polynomials in $y_i,x_{i+1},\dots,x_N$ with coefficients in A, by 5.5.2. Applying the induction hypothesis, we deduce that the coefficients of

$$\operatorname{int}\big(\exp(-y_iX_i)\big)(Z_i)\cdot T_i$$

are likewise polynomials with coefficients in A, which completes the proof.

The preceding induction immediately gives a proof of the following.

[39] Editor's note: The French source prints x_i as the first argument on the right-hand side of this formula. We have written x_1, as required by the defining specialization and by the preceding calculation.

[40] Editor's note: In the second conjugating exponential the French source prints the undefined symbol Y_i. We have written X_i, as required by the preceding identity and by the definition of Z_i.

Lemma 5.5.8. *With the notation of 5.5.6, let, for each $i = 1, \dots, N$,*

$$f_i : U_{\alpha_i} \longrightarrow H$$

be a group morphism, where H is an S-group functor. For the morphism

$$f : U_{\geq 1} \longrightarrow H$$

defined by

$$f\big(\exp(x_1X_1)\cdots\exp(x_NX_N)\big) = f_1\big(\exp(x_1X_1)\big)\cdots f_N\big(\exp(x_NX_N)\big)$$

to be a group morphism, it is necessary and sufficient that, for every pair $i < j$,

$$\begin{aligned} &f_j\big(\exp(x_jX_j)\big)f_i\big(\exp(x_iX_i)\big)f_j\big(\exp(-x_jX_j)\big)\\ &\qquad = f\big(\exp(x_jX_j)\exp(x_iX_i)\exp(-x_jX_j)\big). \end{aligned}$$

5.6. Subgroups of Type (R) with Solvable Fibers

Proposition 5.6.1. *Let (G, T, M, R) be a split S-group, let R' be a subset of R of type* (R) (5.4.2)*, and let $H_{R'}$ be the corresponding subgroup of G. The following conditions are equivalent:*

(i) *$H_{R'}$ has solvable geometric fibers.*

(ii) *There exists a system of positive roots R_+ such that $R' \subset R_+$, and hence $H_{R'} \subset B_{R_+}$* (cf. 5.4.5).

(iii) $R' \cap (-R') = \varnothing$.

(iv) *For every ordering of R', the morphism induced by multiplication in G*

$$T \times_S \prod_{\alpha \in R'} U_\alpha \longrightarrow H_{R'}$$

is an isomorphism.

(v) $H_{R'} \cap \underline{\mathrm{Norm}}_G(T) = T$.

(vi) *For every subset R'' of R of type* (R)*, one has* (cf. 5.4.5)

$$H_{R'} \cap \underline{\mathrm{Norm}}_G(H_{R''}) = H_{R' \cap R''}.$$

We shall prove these equivalences according to the following logical scheme.

$$\begin{tikzcd}
\text{(iii)} \arrow[dd, Rightarrow] & \text{(ii)} \arrow[l, Rightarrow] \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{(vi)} \arrow[dd, Rightarrow] \\
 & \text{(iv)} \arrow[dr, Rightarrow] & \\
\text{(i)} \arrow[ur, Rightarrow] & & \text{(v).} \arrow[ll, Rightarrow]
\end{tikzcd}$$

Clearly (ii) implies (iii), and (vi) implies (v) (take $R'' = \varnothing$). By 5.4.6, it is enough to verify that (i) implies (ii) on the geometric fibers. If S is the spectrum of an algebraically closed field, then $H_{R'}$ is contained in a Borel subgroup containing T, and this subgroup is therefore of the form H_{R_+} (5.5.5(iii)).

Likewise, the implication from (iii) to (i) may be checked on the geometric fibers. Suppose that (iii) holds. If $H_{R'}$ were not solvable, there would exist a subtorus Q of T, of codimension 1 in T, such that

$$\underline{\mathrm{Centr}}_{H_{R'}}(Q)$$

is not solvable (*Bible*, §10.4, Proposition 8). But $\underline{\mathrm{Centr}}_{H_{R'}}(Q)$ has Lie algebra $\mathfrak{g}_{R''}$, where R'' is the set of roots in R' which vanish on Q. Thus, by (iii),

$$R'' = \varnothing \qquad \text{or} \qquad R'' = \{\alpha\}.$$

Consequently, $\underline{\mathrm{Centr}}_{H_{R'}}(Q)$, which is a subgroup of G of type (R), is T or $T \cdot U_\alpha$, and is therefore solvable, contrary to the hypothesis.

Likewise, the implication from (ii) to (iv) may be checked on the geometric fibers (since the schemes in question are flat S-schemes of finite presentation). By *Bible*, §13.2, Theorem 1(d), the morphism in question is bijective. It induces an isomorphism on the tangent spaces at the origin, and the conclusion follows as usual (cf. 4.1.1).

The implication from (iv) to (v) follows from 4.2.7. To prove that (v) implies (i), we may again reduce to the case in which S is the spectrum of an algebraically closed field; the conclusion then follows from *Bible*, §10.3, the corollary to Proposition 6, and §9.3, Corollary 3 to Theorem 1.

It remains only to prove the implication from (ii) to (vi). We may reduce to the case in which G is adjoint. Then, by 5.3.3,

$$\underline{\mathrm{Norm}}_G(H_{R''}) = \underline{\mathrm{Norm}}_G(\mathfrak{g}_{R''}) \subset \underline{\mathrm{Transp}}_G(\mathfrak{t}, \mathfrak{g}_{R''}). \tag{39}$$

[41] By 5.4.5, it is enough to prove

$$H_{R'}(S) \cap \underline{\mathrm{Transp}}_{G(S)}(\mathfrak{t}, \mathfrak{g}_{R''}) \subset H_{R' \cap R''}(S). \tag{x}$$

We first prove a lemma.

Lemma 5.6.2. *With the notation of 5.5.7, let*

$$u = \exp(x_1 X_1) \cdots \exp(x_N X_N),$$

where $x_i \in \mathbb{G}_a(S)$. Let m, with $1 \leq m \leq N$, be such that $x_i = 0$ for $i < m$.

(a) *If $H \in \Gamma(S, \mathfrak{t})$, the component of $\mathrm{Ad}(u)H$ in $\mathfrak{g}^{\alpha_m}$ is*

$$-\overline{\alpha_m}(H) x_m X_m.$$

(b) *If $Y \in \Gamma(S, \mathfrak{g}^{-\alpha_m})$, the component of $\mathrm{Ad}(u)Y$ in $\mathfrak{t}$ is, with the notation of Exposé XX, 2.6,*

$$x_m \langle X_m, Y \rangle H_{\alpha_m}.$$

[41] Editor's note: The inclusion follows from $\mathfrak{t} \subset \mathfrak{g}_{R''}$.

Proof. Set
$$\mathfrak{u} = \mathfrak{g}^{\alpha_{m+1}} + \cdots + \mathfrak{g}^{\alpha_N}.$$
By 5.4.9, one has
$$\mathrm{Ad}\big(\exp(x_i X_i)\big)\mathfrak{u} \subset \mathfrak{u} \qquad \text{for } i > m.$$
By Exposé XX, 2.10,
$$\mathrm{Ad}\big(\exp(x_i X_i)\big)H = H - \overline{\alpha_i}(H)x_i X_i.$$
It follows immediately that, modulo $\mathfrak{u}$,
$$\mathrm{Ad}(u)H = \mathrm{Ad}\big(\exp(x_m X_m)\big)H = H - \overline{\alpha_m}(H)x_m X_m,$$
which proves the first assertion.

Likewise, set[42]
$$\mathfrak{n} = \mathfrak{g}^{\alpha_1} + \cdots + \mathfrak{g}^{\alpha_N}, \qquad u_1 = \exp(x_{m+1}X_{m+1})\cdots\exp(x_N X_N).$$
For $i > m$, one has $\alpha_i > \alpha_m$; hence, by 5.4.9, modulo $\mathfrak{n}$,
$$\mathrm{Ad}(u_1)Y \equiv Y, \qquad \text{and therefore} \qquad \mathrm{Ad}(u)Y \equiv \mathrm{Ad}\big(\exp(x_m X_m)\big)Y.$$
Applying Exposé XX, 2.10, one therefore obtains, modulo $\mathfrak{n}$,
$$\mathrm{Ad}(u)Y - Y \equiv x_m\langle X_m, Y\rangle H_{\alpha_m},$$
which proves the second assertion.

Let us return to the proof of inclusion (x). Suppose that there exists
$$h \in H_{R'}(S), \qquad h \notin H_{R'\cap R''}(S),$$
such that
$$\mathrm{Ad}(h)\mathfrak{t} \subset \mathfrak{g}_{R''}.$$
We may write
$$h = t\exp(x_1X_1)\cdots\exp(x_NX_N).$$
Since $h \notin H_{R'\cap R''}(S)$, there is a least m such that
$$t\exp(x_1X_1)\cdots\exp(x_{m-1}X_{m-1}) \in H_{R'\cap R''}(S), \qquad \alpha_m \notin R'', \qquad x_m \neq 0.$$
Then
$$h' = \exp(x_mX_m)\cdots\exp(x_NX_N)$$
still satisfies the conditions imposed above on h. By Lemma 5.6.2, for every $H \in \Gamma(S,\mathfrak{t})$, the component of $\mathrm{Ad}(h')H$ in $\mathfrak{g}^{\alpha_m}$ is
$$-\overline{\alpha_m}(H)x_mX_m.$$
[43] It therefore follows from the hypothesis on h and from the definition of m that
$$\overline{\alpha_m}(H) = 0 \qquad \text{for every } H \in \Gamma(S,\mathfrak{t}).$$
This is impossible, since G is adjoint and $\overline{\alpha_m}$ is nonzero on every fiber.

[42]Editor's note: The original has been corrected in what follows.

[43]Editor's note: In this sentence the French source prints $\overline{\alpha_m}(X)$; Lemma 5.6.2(a) and the following sentence require H.

Remark 5.6.2 bis. With the notation of Lemma 5.6.2, suppose that $\mathrm{Ad}(u)$ induces the identity on $\mathfrak{t}$ and on $\mathfrak{g}^{-\alpha_m}$. Then $x_m = 0$. Indeed,

$$x_m \overline{\alpha_m} = 0, \qquad x_m H_{\alpha_m} = 0.$$

If $\alpha_m \notin 2M$, then $\overline{\alpha_m}$ is nonzero on every fiber; if $\alpha_m \in 2M$, then $\alpha_m^* \notin 2M^*$, and H_{α_m} is nonzero on every fiber. Thus $x_m = 0$. It follows that $u = e$ if $\mathrm{Ad}(u)$ acts trivially on $\mathfrak{g}$.

Remark 5.6.3. Let H be a subgroup of type (R) of a reductive S-group G, and suppose that its geometric fibers are solvable. Then H is closed in G, and

$$\underline{\mathrm{Norm}}_G(H)/H$$

is represented by a separated finite étale S-scheme. This follows from 5.3.18 and either 3.5 or Exposé XIX, 2.5(ii).

Corollary 5.6.4. *Let (G, T, M, R) be a split reductive S-group. If $R' \subset R$ is closed and $R' \cap (-R') = \varnothing$, then R' is contained in a system of positive roots.* [44]

Proof. By 5.4.7, R' is of type (R), so the result follows from Proposition 5.6.1.

Corollary 5.6.5. *Under the hypotheses of Proposition 5.6.1, multiplication in G induces an isomorphism*

$$\prod_{\alpha \in R'} U_\alpha \xrightarrow{\sim} U_{R'},$$

where $U_{R'}$ is a closed subgroup scheme of G, smooth over S, with connected unipotent geometric fibers, and is independent of the chosen ordering of R'. Moreover, $H_{R'}$ is the semidirect product $T \cdot U_{R'}$, with $U_{R'}$ invariant in $H_{R'}$.

Proof. If $R' \subset R_+$, then

$$U_{R'} = H_{R'} \cap U_{\geq 1},$$

with the notation of 5.5.6, is a closed subgroup of G, of finite presentation, and invariant in $H_{R'}$. By Proposition 5.6.1(iv), one has $H_{R'} = T \cdot U_{R'}$, from which all the assertions follow.

Remark 5.6.6. In particular, U_{R_+} is the group $U_{\geq 1}$ of 5.5.6.

Let us draw some corollaries from the preceding results concerning groups of type $U_{R'}$.

Corollary 5.6.7. *Let (G, T, M, R) be a split reductive S-group, and let R' and R'' be two subsets of R of type* (R)*, with $R' \cap (-R') = \varnothing$.*

(i) *One has*

$$U_{R'} \cap \underline{\mathrm{Norm}}_G(H_{R''}) = U_{R' \cap R''}.$$

(ii) *Suppose that R' is closed. If the conditions $\alpha \in R'$, $\beta \in R''$, and $\alpha + \beta \in R$ imply that $\alpha + \beta \in R'$, then $H_{R''}$ normalizes $U_{R'}$.*

[44] Editor's note: This is in fact a statement about root systems, supplementing Exposé XXI (cf. *Bible*, VI, §1.7, Proposition 22), and it is proved here indirectly.

Proof. Assertion (i) follows at once from 5.6.5 and 5.6.1(vi). To prove (ii), it is enough, by 5.4.4, to show that T and each U_β, for $\beta \in R''$, normalize $U_{R'}$.[45] For T this is trivial; for U_β it follows from 5.5.2 and Exposé XXI, 3.1.2.[46]

Corollary 5.6.8. *Let (G, T, M, R) be a split S-group, let R_+ be a system of positive roots, and let α be a simple root of R_+* (that is, an element of R_+ such that $R_+ \setminus \{\alpha\}$ is closed). *Set*

$$U_{\widehat{\alpha}} = U_{R_+ \setminus \{\alpha\}}.$$

Then:

(i) *$U_{\widehat{\alpha}}$ is a normal subgroup of B_{R_+};*

(ii) *U_{R_+} is the semidirect product of $U_{\widehat{\alpha}}$ by U_α;*

(iii) *$U_{-\alpha}$ normalizes $U_{\widehat{\alpha}}$;*

(iv) *Z_α normalizes $U_{\widehat{\alpha}}$.*

If one similarly defines

$$U_{-\widehat{\alpha}} = U_{R_- \setminus \{-\alpha\}}, \qquad R_- = -R_+,$$

then

$$\Omega_{R_+} = U_{-\widehat{\alpha}} \cdot U_{-\alpha} \cdot T \cdot U_\alpha \cdot U_{\widehat{\alpha}}.$$

Proof. Assertion (ii) follows from 5.6.5, and (i) from 5.6.7(ii). Likewise, (iii) follows from 5.5.2: indeed, if $\beta \in R_+$, with $\beta \neq \alpha$, no combination

$$i(-\alpha) + j\beta, \qquad i, j > 0,$$

can be negative, since β contains at least one simple root different from α. Assertion (iv) then follows from (i) and (iii), because

$$U_{-\alpha} \cdot T \cdot U_\alpha$$

is schematically dense in Z_α. Finally, the last assertion follows from (ii) and its analogue for U_{R_-}.

Let us return to the general situation.

Proposition 5.6.9. *Let S be a scheme, let G be a reductive S-group, and let H be a subgroup of type* (R) *with solvable geometric fibers.*

(i) *The group*

$$\mathrm{D}_S(H) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(H, \mathbb{G}_{m,S})$$

[45] Editor's note: Indeed, since G has connected fibers, it is separated over S by Exposé $\mathrm{VI_B}$, 5.5; hence, by Exposé XI, 6.11 (see also Addendum 6.5.5 in Exposé $\mathrm{VI_B}$), $\underline{\mathrm{Norm}}_G(U_{R'})$ is represented by a closed subgroup scheme N of G. If N contains T and the U_β, for $\beta \in R''$, it contains the big cell $\Omega_{R_+, R''}$. This cell is schematically dense in $H_{R''}$ by 5.4.4 and EGA IV_3, 11.10.10 (the fibers of $H_{R''}$ are integral, and $\Omega_{R_+,R''}$ contains a nonempty open subset of every fiber). It follows that $H_{R''} \subset N$, and therefore that $H_{R''}$ normalizes $U_{R'}$.

[46] Editor's note: One must show that, under the hypotheses of (ii), if $\alpha \in R'$ and $\beta \in R''$, then every root of the form $i\alpha + j\beta$, with $i, j \in \mathbb{N}^*$, belongs to R'. For this purpose the reference Exposé XXI, 2.3.5, has been replaced by Exposé XXI, 3.1.2. This may also be seen directly by inspecting the root systems of rank 2.

is represented by a twisted constant S-group whose type at $s \in S$ is

$$\mathbb{Z}^{\mathrm{rgred}(G_s)}.$$

The biduality morphism (Exposé VIII, §1)

$$f : H \longrightarrow \mathrm{D}_S(\mathrm{D}_S(H))$$

is smooth and surjective.

(ii) *The kernel H^u of f is the largest closed normal subgroup scheme of H, smooth over S, with connected unipotent geometric fibers. It is called the unipotent part of H, and is also denoted*

$$H^u = \mathrm{rad}^u(H).$$

Moreover, H^u is the sheaf of commutators of H: every morphism of groups from H to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on H^u, and therefore factors through

$$H/H^u = \mathrm{D}_S(\mathrm{D}_S(H)).$$

(iii) *If T is a maximal torus of H, the morphism $T \longrightarrow H$ induces isomorphisms*

$$\mathrm{D}_S(H) \xrightarrow{\sim} \mathrm{D}_S(T), \qquad T \xrightarrow{\sim} \mathrm{D}_S(\mathrm{D}_S(H)).$$

Moreover, H is identified with the semidirect product of H^u by T.

(iv) *In the situation of 5.6.1, if $H = H_{R'}$, then*

$$H^u = U_{R'}.$$

Proof. The assertions of the proposition are local for the étale topology (Exposé X, 5.5). We may therefore reduce to the situation of 5.6.1. We then know from 5.6.5 that $H_{R'}$ is the semidirect product of $U_{R'}$ by T. Let us show that $U_{R'}$ is the sheaf of commutators of $H_{R'}$. Since $H_{R'}/U_{R'} = T$ is commutative, it suffices to prove that every morphism of groups

$$\phi : H_{R'} \longrightarrow V$$

as in (ii) vanishes on $U_{R'}$. It is enough to prove that ϕ vanishes on each U_α, for $\alpha \in R'$. If $t \in T(S')$ and $X \in W(\mathfrak{g}^\alpha)(S')$, then

$$1 = \phi\big(t \exp_\alpha(X) t^{-1} \exp_\alpha(-X)\big) = \phi\big(\exp_\alpha((\alpha(t) - 1)X)\big).$$

Since $\alpha : T \longrightarrow \mathbb{G}_{m,S}$ is faithfully flat, it follows at once that ϕ vanishes on $U_\alpha^\times$. But every section of U_α is locally the sum of two sections of $U_\alpha^\times$. Thus

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, V) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(H/U_{R'}, V)$$

for every V as above. Applying this result to $V = \mathbb{G}_{m,S}$, we immediately obtain (i) and (iii), then (iv) and the second assertion of (ii).

It remains to prove the first assertion of (ii). The only nontrivial point is that every closed normal subgroup U of H, smooth over S, with connected unipotent geometric fibers, is a subgroup of H^u. We first have:

Lemma 5.6.10. *Let G be a reductive S-group, T a maximal torus, and U a subgroup scheme of G, smooth over S, with unipotent geometric fibers, and normalized by T. Then*

$$U \cap T = e.$$

Proof. Indeed, since

$$T = \underline{\mathrm{Centr}}_G(T),$$

one has $U \cap T = U^T$, where the right-hand side denotes the invariants under $\mathrm{int}(T)$. By Exposé XIX, 1.4, $U \cap T$ is therefore smooth over S. It is also radicial over S: for every $s \in S$, the group

$$U(\overline{s}) \cap T(\overline{s})$$

consists of elements that are both unipotent and semisimple. This proves the lemma.

We return to the proof of 5.6.9(ii). If U is a normal subgroup of H as above, then the semidirect product $T \cdot U$ is a subgroup of G of type (R), with solvable geometric fibers. We may therefore suppose it to be of the form $H_{R''}$, with $R'' \subset R'$. It is enough to prove that $U = U_{R''}$, and we are thus reduced to the case $H = T \cdot U$. Since H/U is commutative, the French text says that U is a subsheaf of the sheaf of commutators of H, which is H^u.[47] □

Let us note that we have in fact just proved:

Proposition 5.6.11. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. The maps*

$$H \longmapsto H^u, \qquad U \longmapsto T \cdot U$$

are mutually inverse bijections between the set of subgroups H of G of type (R)*, containing T, and having solvable geometric fibers, and the set of subgroups U of G, smooth over S, normalized by T, and having connected unipotent geometric fibers.*[48]

In particular, when (G, T, M, R) is split, the groups $H_{R'}$ and $U_{R'}$ correspond.

Corollary 5.6.12. *Let S be a scheme, let (G, T, M, R) be a split S-group* (respectively, and let R_+ be a system of positive roots of R defining the Borel subgroup B)*. Every smooth group subscheme of G with connected unipotent geometric fibers* (respectively, every smooth group subscheme of B^u)*, normalized by T, is locally on S of the form $U_{R'}$, where R' is a subset of R contained in a system of positive roots* (respectively, a subset of R_+) *and is of type* (R)*.*

Proof. For the "respectively" case, it is enough to observe that the geometric fibers of the given group are unipotent and connected by the *Bible*, §13.2, Theorem 1(d). □

Proposition 5.6.9 also has the following corollary:

Corollary 5.6.13. *Let S be a scheme, G a reductive S-group, and H a subgroup of type* (R) *with solvable geometric fibers. Let $\underline{\mathrm{Tor}}(H)$ be the functor of maximal tori of H:*

$$\underline{\mathrm{Tor}}(H)(S') = \{\, \text{maximal tori of } H_{S'} \,\}.$$

[47] Editor's note: The implication in the French is stated in the wrong direction: commutativity of H/U gives $H^u \subset U$. The desired inclusion follows because H/H^u is a torus whereas U has connected unipotent geometric fibers.

[48] Editor's note: The hypothesis that U be closed has been omitted, since it is automatic. Indeed, for such a U, one has $U \cap T = e$ by 5.6.10, so the semidirect product $H = T \cdot U$ is a subgroup of G of type (R) (cf. 5.2.1), with solvable geometric fibers. Thus, by 5.6.3, H is closed in G; and since U is closed in H, it is closed in G.

Then $\underline{\mathrm{Tor}}(H)$ *is represented by an affine smooth* S*-scheme, which is a principal homogeneous bundle under* H^u *for the action*

$$(h, T) \longmapsto \mathrm{int}(h)T.$$

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Proof. Indeed, if T and T' are two maximal tori of $H_{S'}$, there exists a unique section $h \in H^u(S')$ such that $\mathrm{int}(h)T = T'$. The uniqueness of h follows at once from the equality

$$\underline{\mathrm{Norm}}_G(T) \cap H^u = e$$

(cf., for example, 5.6.1); it is therefore enough to prove the existence of h locally for the étale topology. By 5.2.6 and 5.1.2(a), we may suppose that T and T' are conjugate by a section of H, whence the desired conclusion because

$$H = H^u \cdot T$$

by 5.6.9(iii). It follows that $\underline{\mathrm{Tor}}(H)$ is a principal homogeneous sheaf under H^u, which is affine and smooth over S; the assertion follows at once.[50] □

5.7. Bruhat's Theorem

Recall 5.7.1. *Let* k *be an algebraically closed field,* G *a reductive* k*-group,* B *a Borel subgroup of* G*,* T *a maximal torus of* B*, and* $N = \underline{\mathrm{Norm}}_G(T)$*. Then*

$$G(k) = B(k)N(k)B(k);$$

this is Bruhat's theorem (*Bible*, §13.4, Corollary 1 to Theorem 3). *More precisely, with the notation of 3.6, the sets*

$$B(k)N_w(k)B(k) = B^u(k)N_w(k)B^u(k)$$

form a partition of $G(k)$ *as* w *ranges over* $(N/T)(k)$*. If* B' *is another Borel subgroup of* G *containing* T*, the sets* $B'(k)N_w(k)B(k)$ *likewise form a partition of* $G(k)$*. Indeed, if* $y \in N(k)$ *is such that* $\mathrm{int}(y)B = B'$*, then*

$$yB(k)N_w(k)B(k) = B'(k)N_{yw}(k)B(k).$$

Definition 5.7.2. *Let* (G, T, M, R) *be a split* S*-group, let* R_- *be a system of positive roots of* R*, and let* $B' = B_{R_-}$ *be the Borel subgroup it defines. For* $w \in W$*, put* (cf. 5.6.5)

$$R_-^w = R_- \cap w(R_-), \qquad B'^u_w = U_{R_-^w} = \prod_{\alpha \in R_-^w} U_\alpha.$$

If $n_w \in \underline{\mathrm{Norm}}_G(T)(S)$ *is a representative of* w (3.8)*, one may also write*

$$B'^u_w = B'^u \cap \mathrm{int}(n_w)B'^u.$$

[49] Editor's note: This reproves and sharpens Exposé XII, 1.10 (for G reductive).
[50] Editor's note: By SGA 1, VIII, 2.1, and EGA IV_4, 17.7.1.

Lemma 5.7.3. *Let (G, T, M, R) be a split S-group, let R_+ be a system of positive roots of R, let $R_- = -R_+$, and let B* (respectively B') *be the Borel subgroup of G defined by R_+* (respectively R_-). *Let $w \in W$, and let N_w and B'^u_w be the corresponding subschemes of G* (3.8 and 5.7.2).

(i) *The sheaf $B' \cdot N_w \cdot B$, the image of the morphism*

$$B' \underset{S}{\times} N_w \underset{S}{\times} B \longrightarrow G$$

induced by multiplication in G, is represented by a subscheme of G (in fact, by a closed subscheme of the open subscheme $n_w \Omega_{R_+}$).

(ii) *The morphism*

$$B'^u_w \underset{S}{\times} N_w \underset{S}{\times} B^u \longrightarrow G,$$

induced by multiplication in G, is an immersion whose image is the preceding subscheme.

Proof. Let us first show that the morphism in (ii) is an immersion. By definition, $\mathrm{int}(n_w)^{-1}$ induces a closed immersion of B'^u_w into B'^u; hence the morphism

$$(u, b) \longmapsto n_w^{-1} u\, n_w b$$

induces a closed immersion

$$B'^u_w \underset{S}{\times} B \longrightarrow \Omega_{R_+}.$$

It follows immediately that the morphism in (ii) induces a closed immersion of its source into the open subscheme $n_w \Omega_{R_+}$. To prove (i), it is enough to see that

$$B'(S) N_w(S) B(S) = B'^u_w(S) N_w(S) B^u(S).$$

Now, if $\alpha \in R$, then $\mathrm{int}(n_w) U_\alpha(S) = U_{w(\alpha)}(S)$. Thus, if $w^{-1}(\alpha) \in R_+$,

$$\begin{aligned} U_\alpha(S) N_w(S) B(S) &= U_\alpha(S) n_w T(S) B^u(S) \\ &= n_w U_{w^{-1}(\alpha)}(S) T(S) B^u(S) \\ &= n_w B(S) = N_w(S) B^u(S). \end{aligned}$$

In view of the definition of R_-^w, this immediately gives the desired assertion. ☐

Theorem 5.7.4. *Let S be a scheme, let (G, T, M, R) be a split S-group, let B be the Borel subgroup defined by the system of positive roots R_+, and let B' be the Borel subgroup defined by $R_- = -R_+$.*

(i) (Bruhat's theorem) *The schemes $B'^u_w \cdot N_w \cdot B$, as w ranges over W, form a partition of the underlying set of G.*

(ii) *For each $w \in W$, let n_w be a representative of w in $\underline{\mathrm{Norm}}_G(T)(S)$* (3.8). *Then the open subschemes*

$$n_w \Omega = n_w B'^u \cdot T \cdot B^u$$

as w ranges over W, form an open covering of G.

Proof. Both assertions may be checked on the geometric fibers, where they follow from 5.7.1 and 5.7.3. ☐

Remark 5.7.5. Assertion (i) implies that if S is the spectrum of a field, then $G(S)$ is the disjoint union of the sets

$$B'^u_w(S) \cdot T(S) \cdot B^u(S).$$

The corresponding assertion for an arbitrary S, even a local or Artinian one, is plainly false. Note, however, that (ii) implies that if S is local, then $G(S)$ is indeed the union of the $n_w\Omega(S)$. In fact:

Corollary 5.7.6. *Let Δ be a system of simple roots of the split group G over the local scheme S.*

(i) *Then $G(S)$ is generated by $T(S)$ and the groups $U_\alpha(S)$, for $\alpha \in \Delta \cup -\Delta$.*

(ii) *If G is simply connected* (4.3.3)*, then $G(S)$ is already generated by the groups $U_\alpha(S)$, for $\alpha \in \Delta \cup -\Delta$.*

Proof. Let H be the subgroup of $G(S)$ generated by the $U_\alpha(S)$, for $\alpha \in \Delta \cup -\Delta$. First observe that H contains a representative of every s_α, for $\alpha \in \Delta$, in $\underline{\mathrm{Norm}}_G(T)(S)$ (Exposé XX, 3.1), and hence a representative n_w of every $w \in W$. Since every $\alpha \in R$ can be written as $w(\alpha_0)$, with $w \in W$ and $\alpha_0 \in \Delta$, we have

$$U_\alpha(S) = \mathrm{int}(n_w)U_{\alpha_0}(S) \subset H.$$

The subgroup generated by H and $T(S)$ therefore contains $\Omega(S)$, and hence, by the preceding observation, is all of $G(S)$.

Now suppose that G is simply connected. Let us prove that $H \supset T(S)$. By Exposé XX, 2.7, H contains $\alpha^*(\mathbb{G}_m(S))$ for every $\alpha \in \Delta$, and it remains only to apply 4.3.8. □

Remark 5.7.6.1. Instead of taking, for each $\alpha \in \Delta$, both $U_\alpha(S)$ and $U_{-\alpha}(S)$, it is enough to take $U_\alpha(S)$ together with a representative w_α of the reflection s_α, or else $U_\alpha(S)$ together with a section of $U^\times_{-\alpha}$, ...

Corollary 5.7.7. *Suppose that G has semisimple rank* 1*, and choose $u_\alpha \in U^\times_\alpha(S)$. Then Ω and $u_\alpha\Omega$ form an open covering of G.*

Proof. If $u_{-\alpha}$ is the section of $U_{-\alpha}$ paired with u_α (cf. 1.3), then 5.7.4 (ii) gives

$$G = \Omega \cup u^{-1}_{-\alpha}u_\alpha u^{-1}_{-\alpha}\Omega,$$

and hence

$$G = u_{-\alpha}G = u_{-\alpha}\Omega \cup u_\alpha u^{-1}_{-\alpha}\Omega = \Omega \cup u_\alpha\Omega.$$

□

Corollary 5.7.8. *Let S be a scheme and let G be a reductive S-group. Then G is essentially free over S in the sense of Exposé VIII, 6.1.*[51]

Proof. The assertion is local for the fpqc topology, so we may assume that G is split. It then admits an open covering by open subschemes isomorphic to

$$\mathbb{G}^N_{a,S} \underset{S}{\times} \mathbb{G}^n_{m,S},$$

and these are essentially free. □

[51]Editor's note: See also the additions made in Exposé $\mathrm{VI_B}$, 6.2.1–6.2.6 and 6.5.2–6.5.5.

Lemma 5.7.9. *Under the hypotheses of 5.7.4, let α be a simple root of R_+, and let $u_\alpha \in U_\alpha^\times(S)$. For every $v \in U_{-\alpha}(S)$, one has*

$$\Omega \cdot v \subset \Omega \cup u_\alpha \cdot \Omega.$$

Proof. We are comparing two open subschemes of G, so it is enough to do so when S is the spectrum of a field k. We must therefore prove

$$\Omega(k)v \subset \Omega(k) \cup u_\alpha\Omega(k).$$

Now

$$\begin{aligned}\Omega(k)v &= B'^u(k)T(k)B^u(k)v \\ &= U_{-\widehat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\widehat{\alpha}}(k)v \\ &\subset U_{-\widehat{\alpha}}(k)Z_\alpha(k)U_{\widehat{\alpha}}(k)v.\end{aligned}$$

Here we use the decomposition of 5.6.8. Applying 5.6.8 (iii) and then 5.7.7 to the group Z_α, we obtain

$$\begin{aligned}\Omega(k)v &\subset U_{-\widehat{\alpha}}(k)Z_\alpha(k)vU_{\widehat{\alpha}}(k) \\ &\subset U_{-\widehat{\alpha}}(k)Z_\alpha(k)U_{\widehat{\alpha}}(k) \\ &\subset U_{-\widehat{\alpha}}(k)U_{-\alpha}(k)T(k)U_\alpha(k)U_{\widehat{\alpha}}(k) \\ &\qquad \cup U_{-\widehat{\alpha}}(k)u_\alpha U_{-\alpha}(k)T(k)U_\alpha(k)U_{\widehat{\alpha}}(k).\end{aligned}$$

Applying 5.6.8 (iii) once more, now with R_- in place of R_+, gives the result. $\square$

Proposition 5.7.10. *Under the hypotheses of 5.7.4, choose, for each simple root α, an element $u_\alpha \in U_\alpha^\times(S)$. Let U_1 be the submonoid of $B^u(S)$ generated by the u_α. Then the open subschemes $u\Omega$, for $u \in U_1$, form an open covering of G.*

Proof. Once again, we may assume that S is the spectrum of a field k. In view of 5.7.6, it is enough to prove that

$$\bigcup_{u\in U_1} u\Omega(k)$$

is stable under right multiplication by $T(k)$, $U_\alpha(k)$, and $U_{-\alpha}(k)$, for every simple root α. This is trivial in the first two cases, and in the last it follows from the lemma. $\square$

Remark 5.7.11. Let us point out a special case of 5.7.2. If $w = s_\alpha$ is the reflection with respect to the simple root α, then

$$R_- \cap s_\alpha(R_-) = R_- \setminus \{-\alpha\}$$

(Exposé XXI, 3.3.1), and hence, in the notation of 5.6.8,

$$B'^u_{s_\alpha} = U_{-\widehat{\alpha}}.$$

Remark 5.7.12. In fact, the proof of 5.7.10 immediately gives the following statement. Under the hypotheses of 5.7.10, let Γ be a submonoid of $G(S)$. The open subschemes $g\Omega$, for $g \in \Gamma$, form an open covering of G if and only if, for every $s \in S$ and every simple root α, one has

$$(u_\alpha)_{\bar{s}}B'^u(\bar{s}) \subset \Gamma \cdot B'^u(\bar{s}) \cdot T(\bar{s}) \cdot B^u(\bar{s}).$$

Remark 5.7.13. By 5.5.5 (iii), arguing as in 5.7.1, one immediately obtains the following variant of 5.7.4. Let (G, T, M, R) be a split S-group, and let B and B' be two Borel subgroups of G containing T. For every $w \in W$, the sheaf $B' \cdot N_w \cdot B$ is representable by a subscheme of G; these subschemes, as w ranges over W, form a partition of the underlying set of G. One can also give the analogue of 5.7.3 (ii): one must put

$$B'^u_w = B'^u \cap \operatorname{int}(n_w)\widetilde{B}^u,$$

where $\widetilde{B}$ is the Borel subgroup opposite to B with respect to T (cf. 5.9.2).

Proposition 5.7.14. *Let S be a scheme, let G be a reductive S-group, and let*

$$\mathrm{Ad}\colon G \longrightarrow GL_{\mathcal{O}_S}(\mathfrak{g})$$

be its adjoint representation. Then

$$\mathrm{Ker}(\mathrm{Ad}) = \underline{\mathrm{Centr}}(G).$$

Equivalently, the canonical homomorphism induced by Ad *upon passage to the quotient,*

$$\overline{\mathrm{Ad}}\colon G/\underline{\mathrm{Centr}}(G) = \mathrm{ad}(G) \longrightarrow GL_{\mathcal{O}_S}(\mathfrak{g}),$$

is a monomorphism.[52]

Proof. We may assume that G is split. Choose a total order on $\Gamma_0(R)$ compatible with its group structure, and let R_+ be the set of positive roots. By 5.7.4 (ii) and 4.1.6, it is enough to prove that, if n_w represents $w \in W$, if

$$u \in U(S), \qquad t \in T(S), \qquad v \in U_-(S),$$

and if $\mathrm{Ad}(n_w v t u) = \mathrm{id}$, then $w = e$, $v = e$, and $u = e$. For each $m \in R \cup \{0\}$, put

$$\mathfrak{g}^{>m} = \bigoplus_{n>m} \mathfrak{g}^n, \qquad \mathfrak{g}^{<m} = \bigoplus_{n<m} \mathfrak{g}^n.$$

Let $X \in \Gamma(S, \mathfrak{g}^m)$; write

$$\mathrm{Ad}(tu)X = \mathrm{Ad}(v^{-1}n_w^{-1})X.$$

Now

$$\begin{aligned} \mathrm{Ad}(t)\,\mathrm{Ad}(u)X - m(t)X &\in \Gamma(S, \mathfrak{g}^{>m}),\\ \mathrm{Ad}(v^{-1}n_w^{-1})X - \mathrm{Ad}(n_w^{-1})X &\in \Gamma(S, \mathfrak{g}^{<w^{-1}(m)}). \end{aligned}$$

If $w \neq e$, there is an $\alpha \in R$ such that $w^{-1}(\alpha) < \alpha$. On taking $m = \alpha$, this gives a contradiction, since

$$\begin{aligned} \mathrm{Ad}(tu)X &\in \Gamma(S, \mathfrak{g}^{\alpha} + \mathfrak{g}^{>\alpha}) \cap \Gamma(S, \mathfrak{g}^{w^{-1}(\alpha)} + \mathfrak{g}^{<w^{-1}(\alpha)})\\ &= 0. \end{aligned}$$

Thus $w = e$, and we may choose $n_w = e$. We then have

$$\mathrm{Ad}(v^{-1})X - X \in \Gamma(S, \mathfrak{g}^{<m} \cap (\mathfrak{g}^m + \mathfrak{g}^{>m})) = 0,$$

whence $\mathrm{Ad}(v)X = X$ for every $X \in \Gamma(S, \mathfrak{g}^m)$, and therefore $\mathrm{Ad}(v) = \mathrm{id}$. Similarly, $\mathrm{Ad}(u) = \mathrm{id}$. The result now follows from 5.6.2 bis. □

[52] Editor's note: It is even a closed immersion, by Exposé XVI, 1.5(a).

5.8. Schemes Associated with a Reductive Group

Theorem 5.8.1. *Let S be a scheme and let G be a reductive S-group. Let $\mathcal{H}$ be the functor of subgroups of type* (R) *of G: for every $S' \to S$, $\mathcal{H}(S')$ is the set of subgroups of type* (R) *of $G_{S'}$* (cf. 5.2.1)*. Then $\mathcal{H}$ is representable by a quasi-projective S-scheme of finite presentation over S.*

Proof.[53] Let

$$G' = G/\underline{\mathrm{Centr}}(G)$$

be the adjoint group of G (4.3.6), and let $u\colon G \to G'$ be the quotient morphism. By Exposé XII, 7.12, the map

$$H' \longmapsto u^{-1}(H')$$

is a bijection from the set of subgroups of type (R) of G' onto the set of subgroups of type (R) of G; this remains true after every base change. Thus, replacing G by G', we may assume that G is adjoint. Consider the morphism

$$u\colon \mathcal{H} \longrightarrow \underline{\mathrm{Grass}}(\mathfrak{g})$$

that assigns to every subgroup of type (R) its Lie algebra (a submodule of $\mathfrak{g}$ that is locally a direct summand[54]). Then u is a monomorphism by 5.3.3. It is enough to prove that it is representable by an immersion of finite presentation; in other words, it is enough to prove the following assertion. For every $S_1 \to S$, given a locally direct-summand submodule $\mathfrak{h}$ of $\mathfrak{g}_{S_1}$, the morphisms $S' \to S_1$ for which $\mathfrak{h}_{S'}$ is the Lie algebra of a subgroup of type (R) of $G_{S'}$ are exactly those that factor through a certain subscheme Σ of finite presentation over S_1. Replacing S_1 by S, we reduce to $S_1 = S$, and we may moreover assume that S is affine; then there exists an affine Noetherian scheme S_0 such that G (respectively, $\mathfrak{h}$) is obtained by base change from an adjoint reductive S_0-group G_0 (respectively, from a locally direct-summand submodule $\mathfrak{h}_0$ of $\mathfrak{g}_0 = \mathrm{Lie}(G_0/S_0)$). It is enough to show that there exists a subscheme Σ_0 of S_0 having the required properties, for then $\Sigma = \Sigma_0 \times_{S_0} S$. Replacing S by S_0, we may therefore suppose that S is affine and Noetherian (note that then every subscheme of S is of finite presentation over S). Finally, after replacing S by a sufficiently small open subscheme, we may suppose that $\mathfrak{g}$ is free of rank n and that $\mathfrak{h}$ is a direct summand, free of rank r.

We must first express the condition that $\mathfrak{h}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$, that is, that the morphism induced by the Lie bracket

$$\varphi\colon \mathfrak{h} \otimes \mathfrak{h} \xrightarrow{[\,,]} \mathfrak{g}$$

factor through $\mathfrak{h}$. If $(e_1, \dots, e_n)$ is a basis of $\mathfrak{g}$ such that $(e_1, \dots, e_r)$ is a basis of $\mathfrak{h}$, then φ is given by sections a_k^{ij} of $\mathcal{O}_S$, where $i, j = 1, \dots, r$ and $k = 1, \dots, n$. The preceding condition is equivalent to requiring that $S' \to S$ factor through the closed subscheme of S defined by the equations

$$a_k^{ij} = 0 \qquad \text{for } k = r+1, \dots, n \text{ and } i, j = 1, \dots, r.$$

We replace S by this closed subscheme.

Then, by 5.3.0,

$$N = \underline{\mathrm{Norm}}_G(\mathfrak{h})$$

is a closed subgroup scheme of G, of finite presentation over G. We must now express (cf. 5.3.1) the condition that $N_{S'}$ be smooth at every point of the unit section, of relative dimension $r = \mathrm{rank}(\mathfrak{h})$, and that the inclusion of $\mathfrak{h}_{S'}$ in $\mathrm{Lie}(N_{S'}/S')$ be an equality.

[53] Editor's note: The original is expanded in what follows.

[54] Editor's note: Cf. Exposé II, 4.11.8.

Since N is affine over S[55] (being closed in G), the unit section $\varepsilon\colon S \to N$ is a closed immersion. Thus $\varepsilon(S)$ is defined by a quasi-coherent ideal $\mathcal{J}$ of $\mathcal{A}(N)$. Note that

$$\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2 \simeq \varepsilon^*(\Omega^1_{N/S}),$$

so its formation commutes with every base change $S' \to S$. By the equivalence $(c') \iff (a)$ in EGA IV_4, 17.12.1 (applied to $f\colon N \to S$ and $j = \varepsilon$), the morphism $N_{S'} \to S'$ is smooth, of relative dimension r, at every point of $\varepsilon(S')$ if and only if

$$\mathfrak{n}_{S'/N'} = \mathfrak{n}_{S/N} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$$

is locally free of rank r, and the morphism

$$\varphi_n(S')\colon\ \mathrm{Sym}^n(\mathfrak{n}_{S'/N'}) \longrightarrow \mathcal{J}^n_{S'}/\mathcal{J}^{n+1}_{S'}$$

is an isomorphism for every $n \geq 1$. Put

$$K_n(S') = \ker \varphi_n(S').$$

By TDTE I, Lemma 3.6, $\mathfrak{n}_{S/N} \otimes \mathcal{O}_{S'}$ is locally free of rank r if and only if $S' \to S$ factors through a certain subscheme Z of S. Replacing S by Z, we may therefore suppose that $\mathfrak{n}_{S/N} = \mathcal{J}/\mathcal{J}^2$ is locally free of rank r. Then, for every $S' \to S$,

$$(\mathcal{J}^2/\mathcal{J}^3) \otimes \mathcal{O}_{S'} = \mathcal{J}^2_{S'}/\mathcal{J}^3_{S'},$$

and consequently

$$K_2(S') = K_2(S) \otimes \mathcal{O}_{S'}.$$

It follows that $\varphi_2(S')$ is an isomorphism if and only if $S' \to S$ factors through the closed subscheme S_2 of S defined by the ideal generated by the image of

$$\mathrm{Sym}^2(\mathfrak{n}_{S/N})^* \otimes K_2(S) \longrightarrow \mathcal{O}_S.$$

Over S_2, the module $\mathcal{J}^2/\mathcal{J}^3$ is then isomorphic to $\mathrm{Sym}^2(\mathfrak{n}_{S/N})$, hence is locally free; the same argument shows that $\varphi_3(S')$ is an isomorphism if and only if $S' \to S$ factors through a certain closed subscheme S_3 of S, and so on. We thus obtain that $N_{S'} \to S'$ is smooth, of relative dimension r, at every point of $\varepsilon(S')$ if and only if $S' \to S$ factors through the closed subscheme

$$Z = \bigcap_n S_n.$$

But then, for every $S' \to Z$, $\mathrm{Lie}(N_{S'}/S')$ is locally a direct summand of rank r of $\mathfrak{g}_{S'}$, and hence the inclusion

$$\mathfrak{h}_{S'} \subset \mathrm{Lie}(N_{S'}/S')$$

is an equality. We then put $H = N^0$.

Replacing S by Z, it remains only to express the condition that $H_{s'}$ have the same reductive rank as $G_{s'}$ at every point $s' \in S'$, or, equivalently, that H_s have the same reductive rank as G_s at every point s in the set-theoretic image of S' in S. This condition defines an open subset of S (Exposé XIX, 6.2). □

[55]Editor's note: The original then indicated that, denoting by $\mathfrak{n}_{N/G}$ the inverse image of the conormal sheaf $\mathcal{N}_{N/G}$ under the unit section $S \to N$, the condition that $\mathfrak{n}_{N_{S'}/G_{S'}} \to \omega^1_{G_{S'}/S'}$ be universally injective is equivalent to the condition that $S' \to S$ factor through a certain open subscheme of S. We were unable to justify this point, since formation of $\mathcal{N}_{N/G}$ does not commute with base change, and we have replaced that argument by the one that follows, suggested by O. Gabber.

Remark. In general, the scheme $\mathcal{H}$ is not smooth over S. It is nevertheless smooth if 6 is invertible on S, or if there exists a prime number p such that $p \cdot 1_S = 0$ (that is, if S has characteristic $p > 0$).

Corollary 5.8.2. *Let S be a scheme, let G be a reductive S-group, and let H be a subgroup of type* (R) *of G. Recall from 5.3.10 that $\underline{\mathrm{Norm}}_G(H)$ is representable by a closed subgroup scheme of G, smooth over S. Then the quotient sheaf*

$$G/\underline{\mathrm{Norm}}_G(H)$$

is representable by a quasi-projective S-scheme, smooth and of finite presentation over S (in fact, by an open subscheme of $\mathcal{H}$).

Indeed, consider the morphism

$$f\colon G \longrightarrow \mathcal{H}$$

defined set-theoretically by

$$f(g) = \mathrm{int}(g)H.$$

By 5.3.9, this morphism is smooth and of finite presentation, hence open. Let $V = f(G)$, equipped with its open-subscheme structure in $\mathcal{H}$. The morphism $f\colon G \to V$ is a covering morphism with kernel $\underline{\mathrm{Norm}}_G(H)$. Consequently $G/\underline{\mathrm{Norm}}_G(H)$ is representable by V (cf. Exposé IV, 4.6.5). $\square$

Corollary 5.8.3. *Let S be a scheme and let G be a reductive S-group. Consider the functors $\underline{\mathrm{Tor}}(G)$, $\underline{\mathrm{Bor}}(G)$, and $\underline{\mathrm{Kil}}(G)$, defined by*

$$\begin{aligned} \underline{\mathrm{Tor}}(G)(S') &= \{\textit{maximal tori of } G_{S'}\}, \\ \underline{\mathrm{Bor}}(G)(S') &= \{\textit{Borel subgroups of } G_{S'}\}, \\ \underline{\mathrm{Kil}}(G)(S') &= \{\textit{Killing pairs of } G_{S'} \quad \text{(cf. 5.3.13)}\}. \end{aligned}$$

(i) *$\underline{\mathrm{Tor}}(G)$, $\underline{\mathrm{Bor}}(G)$, and $\underline{\mathrm{Kil}}(G)$ are representable by smooth S-schemes of finite presentation, with geometrically integral fibres, and are respectively affine, projective, and affine over S.*

(ii) *The canonical morphism*

$$\underline{\mathrm{Kil}}(G) \longrightarrow \underline{\mathrm{Tor}}(G)$$

(respectively, $\underline{\mathrm{Kil}}(G) \to \underline{\mathrm{Bor}}(G)$) *is finite étale and surjective* (respectively, affine, smooth, and surjective).

(iii) *Let T be a maximal torus of G* (respectively, let B be a Borel subgroup of G, or let $B \supset T$ be a Killing pair of G). *The morphism*

$$G \longrightarrow \underline{\mathrm{Tor}}(G), \quad \textit{resp.} \quad G \longrightarrow \underline{\mathrm{Bor}}(G), \quad \textit{resp.} \quad G \longrightarrow \underline{\mathrm{Kil}}(G),$$

defined by

$$g \longmapsto \mathrm{int}(g)T, \quad \textit{respectively} \quad g \longmapsto \mathrm{int}(g)B, \quad \textit{respectively} \quad g \longmapsto \big(\mathrm{int}(g)B, \mathrm{int}(g)T\big)$$

induces an isomorphism

$$G/\underline{\mathrm{Norm}}_G(T) \xrightarrow{\sim} \underline{\mathrm{Tor}}(G), \quad \text{resp.} \quad G/B \xrightarrow{\sim} \underline{\mathrm{Bor}}(G), \quad \text{resp.} \quad G/T \xrightarrow{\sim} \underline{\mathrm{Kil}}(G).$$

One first sees that (iii) follows from the conjugacy theorem for maximal tori (respectively, Borel subgroups, respectively, Killing pairs) and from the facts that

$$\underline{\mathrm{Norm}}_G(B) = B, \qquad \underline{\mathrm{Norm}}_G(B) \cap \underline{\mathrm{Norm}}_G(T) = T,$$

all of which were established above (5.1.2, 5.3.12, 5.3.14, 5.6.1).

It follows first that the canonical morphisms

$$\underline{\mathrm{Tor}}(G) \longrightarrow \mathcal{H}, \qquad \underline{\mathrm{Bor}}(G) \longrightarrow \mathcal{H}$$

are representable, locally for the étale topology, by open immersions (5.8.2 and 5.1.2, respectively 5.5.5). Hence, by descent, $\underline{\mathrm{Tor}}(G)$ and $\underline{\mathrm{Bor}}(G)$ are representable by open subschemes of $\mathcal{H}$. Likewise, $\underline{\mathrm{Kil}}(G)$ is locally, for the étale topology, representable by a scheme affine over the base (Exposé IX, 2.3), and hence is representable by an affine S-scheme, by descent for affine schemes.[56]

The assertions of (ii) follow at once from 5.5.5(ii) and 5.6.13. It already follows that $\underline{\mathrm{Tor}}(G)$ is affine over S (EGA II, 6.7.1). It remains only to prove that $\underline{\mathrm{Bor}}(G)$ is projective over S. We already know that it is quasi-projective, so it remains to prove that it is proper;[57] but it has connected fibres, and therefore, by EGA IV_3, 15.7.10, one is reduced to proving this on geometric fibres. If S is the spectrum of an algebraically closed field, then $\underline{\mathrm{Bor}}(G) = G/B$ by (iii), and the conclusion follows from *Bible*, §6.4, Theorem 4 (or [Ch05], §6.5, Theorem 5). □

Remark 5.8.4. Under the hypotheses of 5.8.3, let Q be a central subgroup of multiplicative type of G. The evident morphisms define isomorphisms

$$\underline{\mathrm{Tor}}(G) \simeq \underline{\mathrm{Tor}}(G/Q), \qquad \underline{\mathrm{Bor}}(G) \simeq \underline{\mathrm{Bor}}(G/Q), \qquad \underline{\mathrm{Kil}}(G) \simeq \underline{\mathrm{Kil}}(G/Q).$$

Corollary 5.8.5. *Let S be a scheme, G a reductive S-group, and P a subgroup scheme of G, smooth and of finite presentation over S. The following conditions are equivalent:*

(i) *For every $s \in S$, $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$* (that is, the quotient scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper over $\bar{s}$, or equivalently $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$; cf. *Bible*, §6.4, Theorem 4, or [Ch05], §6.5, Theorem 5).

(ii) *The quotient sheaf G/P is representable by an S-scheme smooth and projective over S.*

Moreover, under these conditions, P is closed in G, has connected fibres, and

$$P = \underline{\mathrm{Norm}}_G(P).$$

Clearly (ii) implies (i). If (i) holds, then

$$P(\bar{s}) = \underline{\mathrm{Norm}}_{G(\bar{s})}(P_{\bar{s}})$$

and $P_{\bar{s}}$ is connected (for the first assertion, cf. *Bible*, §12.3, Lemma 4;[58] the second assertion follows from this, since $P' = P^0_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$ normalized by $P_{\bar{s}}$, whence $P'(\bar{s}) = P(\bar{s})$ and therefore $P' = P_{\bar{s}}$). It follows that P is of type (R) and that

$$P = \underline{\mathrm{Norm}}_G(P),$$

so that P is closed in G. By 5.8.2,

$$G/P = G/\underline{\mathrm{Norm}}_G(P)$$

is representable by a quasi-projective S-scheme. Its fibres are connected and proper, and it is therefore projective by the argument of 5.8.3. □

[56] Editor's note: Cf. SGA 1, VIII, 2.1.

[57] Editor's note: We may suppose that S is affine and, since $\underline{\mathrm{Bor}}(G)$ is of finite presentation over S by (i), reduce to the case in which S is Noetherian; we are then under the hypotheses of EGA IV_3, 15.7.10.

[58] Editor's note: We have supplied the details that follow.

Remark 5.8.6. The statements 5.8.1, 5.8.2, and 5.8.5 are valid for an S-group of type (RA), or for an S-group of type (RR) satisfying 5.1.8.[59]

Remark 5.8.7. Through the inner automorphisms of G, there are canonical actions

$$G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Tor}}(G)), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Bor}}(G)), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G)),$$

which, in the situation of 5.8.3(iii), are identified with the canonical actions

$$G \longrightarrow \underline{\mathrm{Aut}}_S(G/\underline{\mathrm{Norm}}_G(T)), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(G/B), \quad G \longrightarrow \underline{\mathrm{Aut}}_S(G/T).$$

In particular, one obtains

$$\begin{aligned}\ker(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Tor}}(G))) &= \ker(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Bor}}(G)))\\ &= \ker(G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G)))\\ &= \underline{\mathrm{Centr}}(G).\end{aligned}$$

Indeed, it is clear that $\underline{\mathrm{Centr}}(G)$ acts trivially on each of the three schemes. Conversely, the kernel of

$$G \longrightarrow \underline{\mathrm{Aut}}_S(\underline{\mathrm{Kil}}(G))$$

is "the intersection of the maximal tori of G" in the sense of 4.1.7, and hence equals $\underline{\mathrm{Centr}}(G)$ (loc. cit.). For $\underline{\mathrm{Bor}}(G)$, one observes that "the intersection of the Borel subgroups of G" is also "the intersection of its maximal tori" (see the following section). For $\underline{\mathrm{Tor}}(G)$, one uses Exposé XII, 4.11.

5.9. Properties Peculiar to Borel Subgroups

Most of these properties will be generalized in Exposé XXVI to parabolic subgroups.

Definition 5.9.1. Let S be a scheme, G a reductive S-group, and B, B' two Borel subgroups of G. We say that B and B' are *in general position* (or that B' is in general position relative to B) if $B \cap B'$ is a torus (necessarily maximal) of G.

If T is a maximal torus of G contained in B and B', we say that B and B' are *opposite* (relative to T) if

$$B \cap B' = T.$$

Proposition 5.9.2. *Let S be a scheme, G a reductive S-group, B a Borel subgroup of G, and T a maximal torus of B. There exists a unique Borel subgroup B' of G, opposite to B relative to T.*

If (G, T, M, R) is a splitting of G relative to T, and if $B = B_{R_+}$ (5.5.1)*, then*

$$B' = B_{-R_+}.$$

By faithfully flat descent, it is enough to prove the proposition in the split case, with $B = B_{R_+}$ (5.5.5(iv)). Then B_{-R_+} is indeed opposite to B (4.1.2). Let us show that it is the only Borel subgroup of G containing T that is opposite to B. If B' is a Borel subgroup of G containing T, then locally on S it has the form $B_{R'_+}$, where R'_+ is another system of positive roots of R (5.5.5(iii)). If $R'_+ \neq -R_+$, there exists $\alpha \in R'_+ \cap R_+$, and hence

$$U_\alpha \subset B_{R_+} \cap B_{R'_+}.$$

□

[59]Editor's note: Indeed, the proof of 5.8.1 uses only 5.3.3 (which is valid for a group of type (RA)) and Exposé XIX, 6.2, which, by Exposé XII, 1.7(b), is also valid for groups of type (RR).

Proposition 5.9.3. *Let S be a scheme, G a reductive S-group, and B a Borel subgroup of G.*

(i) *If B' is a Borel subgroup of G, the following conditions are equivalent:*

(a) *B' is in general position relative to B* (5.9.1)*.*

(b) *$B'^u \cap B^u = e$.*

(b′) *$B'^u \cap B = e$.*

(c) *Multiplication in G induces an open immersion*

$$B'^u \times_S B \longrightarrow G.$$

(c′) *The canonical morphism*

$$B'^u \longrightarrow G/B$$

is an open immersion.

(ii) *The functor $\underline{\mathrm{Opp}}(B)$,*

$$S' \longmapsto \left\{ \begin{array}{c} \textit{Borel subgroups of } G_{S'} \textit{ in general position} \\ \textit{relative to } B_{S'} \end{array} \right\},$$

is representable by an open subscheme of $\underline{\mathrm{Bor}}(G)$ (5.8.3)*. The morphism*

$$\underline{\mathrm{Opp}}(B) \longrightarrow \underline{\mathrm{Tor}}(B)$$

defined by $B' \mapsto B \cap B'$ is an isomorphism. In particular (5.6.13)*, the inner automorphisms of B^u equip $\underline{\mathrm{Opp}}(B)$ with the structure of a principal homogeneous bundle under B^u.*

Let us first examine (i). We have (a) ⇒ (c). Indeed, (c) is local for the étale topology; by 5.5.5(iv), one reduces to the case in which G is split relative to $B \cap B'$ and B has the form B_{R_+}. By 5.9.2,

$$B'^u = U_{-R_+},$$

and one is reduced to 4.1.2.

Trivially,

$$(\mathrm{c}') \Longleftrightarrow (\mathrm{c}) \Longrightarrow (\mathrm{b}') \Longrightarrow (\mathrm{b}).$$

It remains to prove (b) ⇒ (a). Let us first prove (ii). The second assertion is a formal consequence of 5.9.2, and the third follows at once from it by 5.6.13. Let us prove the first. It is local for the étale topology, and we may therefore suppose that B contains a maximal torus T. Let B'_0 be the Borel subgroup opposite to B relative to T (5.9.2).

By what precedes, the morphism

$$B^u \longrightarrow \underline{\mathrm{Bor}}(G)$$

induced by the canonical morphism

$$G \longrightarrow G/B'_0 \longrightarrow \underline{\mathrm{Bor}}(G)$$

(5.8.3) induces an isomorphism

$$B^u \xrightarrow{\sim} \underline{\mathrm{Opp}}(B).$$

Thus there is a commutative diagram

$$\begin{tikzcd} B^u \arrow[r] \arrow[d, "\sim"] & G/B'_0 \arrow[d, "\sim"] \\ \underline{\mathrm{Opp}}(B) \arrow[r] & \underline{\mathrm{Bor}}(G). \end{tikzcd}$$

Now the morphism $B^u \to G/B'_0$ is an open immersion (by (i), (a) $\Rightarrow$ (c′)), which completes the proof of (ii). We record at once the following corollary.

Corollary 5.9.4. *Let G be a reductive S-group and let B, B' be two Borel subgroups of G. If $s \in S$ is such that $B_{\bar{s}}$ and $B'_{\bar{s}}$ are in general position, there exists an open subset V of S, containing s, such that B_V and B'_V are in general position.*

It therefore remains only to prove (b) $\Rightarrow$ (a). By the preceding corollary, it is enough to do so when S is the spectrum of an algebraically closed field k. We may suppose that G is split relative to a maximal torus T of B. Let B'_0 be the Borel subgroup opposite to B. Since the Borel subgroups of G are conjugate under $G(k)$, there exists $g \in G(k)$ such that $\mathrm{int}(g)B'_0 = B'$. By the Bruhat theorem (5.7.4), we may write

$$g = bnb', \qquad b \in B(k), \quad b' \in B'_0(k), \quad n \in \underline{\mathrm{Norm}}_G(T)(k).$$

Thus

$$B' = \mathrm{int}(b)\,\mathrm{int}(n)B'_0$$

and

$$B' \cap B = \mathrm{int}(b)\big(\mathrm{int}(n)B'_0 \cap B\big).$$

If $n \notin T(k)$, then

$$\mathrm{int}(n)(B'_0)^u \cap B^u \neq e$$

(cf. the proof of 5.9.2); it follows that (b) implies

$$B' \cap B = \mathrm{int}(b)(B'_0 \cap B) = \mathrm{int}(b)T.$$

□

Proposition 5.9.5. *Let S be a scheme, G a reductive S-group, B a Borel subgroup of G, and B^u its unipotent part. There exists a sequence of subgroups of B,*

$$U_0 = B^u \supset U_1 \supset \cdots \supset U_n \supset \cdots,$$

having the following properties:

(i) *Each U_i is smooth, has connected fibers, and is characteristic in B; the inner automorphisms of B^u act trivially on the* (sheaf) *quotients U_i/U_{i+1}.*

(ii) *For each $i \geq 0$, there exists a locally free $\mathcal{O}_S$-module $\mathcal{E}_i$ and an isomorphism of sheaves of groups on S,*

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i).$$

(iii) *For every $s \in S$,*

$$(U_n)_s = e \qquad \text{for} \qquad n \geq \dim(B^u_s).$$

Suppose first that there exists a splitting (G, T, M, R) of G and a system R_+ of positive roots of R such that $B = B_{R_+}$. Denote by Δ the set of simple roots of R_+. For each $\alpha \in R_+$, denote by $\mathrm{ord}(\alpha)$ the sum of the coefficients of α in the basis Δ of $\Gamma_0(R)$; this is the *order* of α relative to R_+. We have

$$\mathrm{ord}(\alpha) \leq \mathrm{Card}(R_+).$$

For every $i > 0$, let $R^{(i)}$ be the set of roots of order $> i$. This is a closed set of positive roots, and we may therefore construct (5.6.5)

$$U_i = U_{R^{(i)}}.$$

If $\alpha \in R_+$ and $\beta \in R^{(i)}$, then $\alpha + \beta \in R^{(i+1)}$. It follows from 5.5.2 that each U_i is an invariant subgroup of B, and that the inner automorphisms of B^u act trivially on U_i/U_{i+1}. Moreover, this group identifies with

$$\prod_{\mathrm{ord}(\alpha)=i+1} U_\alpha$$

and is therefore endowed with a vector-group structure.

If B is of the form $B_{R'_+}$ for another splitting (G, T', M', R') of G, let us show that the groups U'_i constructed as above using the new splitting coincide with the U_i, and that the vector-group structures on the successive quotients likewise coincide. By 5.6.13, there exists $b \in B^u(S)$ such that $T' = \mathrm{int}(b)T$; the assertion to be proved is local on S, and we may therefore suppose that the isomorphism

$$T \xrightarrow{\sim} T'$$

induced by $\mathrm{int}(b)$ arises by duality from an isomorphism of root data

$$h : (M', M'^*, R', R'^*) \xrightarrow{\sim} (M, M^*, R, R^*).$$

The roots in R'_+ are then the roots

$$\alpha \circ \mathrm{int}(b) = h(\alpha), \qquad \alpha \in R_+,$$

and its simple roots are the $h(\alpha)$, with $\alpha \in \Delta$. Consequently

$$\mathrm{ord}\big(h(\alpha)\big) = \mathrm{ord}(\alpha).$$

By transport of structure, the vector groups $U'_{h(\alpha)}$ are the groups $\mathrm{int}(b)U_\alpha$. It follows that

$$\mathrm{int}(b)U_i = U'_i.$$

But U_i is invariant in B, so that $U_i = U'_i$. Likewise, the isomorphism

$$\mathrm{int}(b) : U_i/U_{i+1} \xrightarrow{\sim} U'_i/U'_{i+1}$$

is the identity.

We now pass to the general case. Let $\{S_i \to S\}$ be an étale covering family such that each (G_i, T_i, M_i, R_i) is a splitting, and choose a system R_{i+} of positive roots such that

$$B \times_S S_i = B_{R_{i+}}$$

(5.5.5(iii)). For every i, the preceding construction gives a sequence

$$B^u_{S_i} = U_{i,0} \supset U_{i,1} \supset \cdots \supset U_{i,j} \supset \cdots$$

and vector-group structures on its successive quotients. By descent, it is enough to verify, for every i, i' and every j, that

$$U_{i,j} \times_{S_i} S_{ii'} = U_{i',j} \times_{S_{i'}} S_{ii'}, \qquad S_{ii'} = S_i \times_S S_{i'},$$

and that the vector-group structures on

$$(U_{i,j}/U_{i,j+1}) \times_{S_i} S_{ii'} \quad \text{and} \quad (U_{i',j}/U_{i',j+1}) \times_{S_{i'}} S_{ii'}$$

coincide. If $S_{ii'} = \varnothing$, there is nothing to prove. Otherwise we are in the situation considered above: $B \times_S S_{ii'}$ is defined by the system of positive roots R_{i+}, respectively $R_{i'+}$, in the splittings

$$(G_{S_{ii'}}, T_i \times_{S_i} S_{ii'}, M_i, R_i), \qquad (G_{S_{ii'}}, T_{i'} \times_{S_{i'}} S_{ii'}, M_{i'}, R_{i'}).$$

This proves the proposition. □

Corollary 5.9.6. *If S is affine, then*

$$H^1(S, B^u) = e;$$

that is, every principal homogeneous bundle under B^u has a section.

Indeed, S is a disjoint union of subschemes on each of which B^u has constant relative dimension. In view of 5.9.5(iii), we may therefore suppose that $U_n = e$. By TDTE I, B.1.1,[56] one has

$$H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0.$$

It follows by induction that $H^1(S, B^u) = e$.

Corollary 5.9.7. *If S is affine, then B has maximal tori. If T is a maximal torus of B, then*

$$H^1(S, T) = H^1(S, B).$$

The first assertion follows at once from 5.9.6 and 5.6.13; the second follows from it in the standard way.[57]

Corollary 5.9.8. *If G is a reductive S-group, the canonical morphism* (cf. 5.8.3)

$$\underline{\mathrm{Kil}}(G) \longrightarrow \underline{\mathrm{Bor}}(G)$$

has sections over every affine open subset.

[56] Editor's note: This is the corollary on page 18 of TDTE I.

[57] Editor's note: Indeed, there is an exact sequence

$$H^1(S, B^u) \longrightarrow H^1(S, B) \xrightarrow{\pi} H^1(S, B/B^u) = H^1(S, T);$$

see [Se64], I §5.5, Proposition 38, or [Gi71], III Proposition 3.3.1. Now $H^1(S,T) \to H^1(S,B)$ is a section of π, so that π is surjective; on the other hand, $H^1(S, B^u) = 0$ by 5.9.6.

Corollary 5.9.9. *Under the hypotheses of 5.9.5, suppose that S is affine. Then there exists a locally free $\mathcal{O}_S$-module $\mathcal{E}$ such that B^u, as a scheme, is S-isomorphic to $W(\mathcal{E})$.*

We prove by induction on i that B^u/U_i is S-isomorphic to

$$W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

This is clear for $i = 0$; suppose that $i \geq 1$. Then B^u/U_i is a principal homogeneous bundle with base

$$X = B^u/U_{i-1}$$

under the group $(U_{i-1}/U_i)_X$. Since B^u/U_{i-1} is affine by the induction hypothesis, and since

$$U_{i-1}/U_i = W(\mathcal{E}_{i-1}),$$

this bundle is trivial. We therefore have, at least, an isomorphism of S-schemes

$$B^u/U_i \xrightarrow{\sim} (B^u/U_{i-1}) \times_S W(\mathcal{E}_{i-1}) = W(\mathcal{E}_0 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

The conclusion now follows at once from condition 5.9.5(iii).

Corollary 5.9.10. *Let S be a semilocal scheme, let $\{s_i\}$ be its closed points, and let B be a Borel subgroup of the reductive S-group G. The canonical map*

$$B^u(S) \longrightarrow \prod_i B^u(\operatorname{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \operatorname{Spec}(A)$, $\kappa(s_i) = A/\mathfrak{p}_i$, and $\mathcal{E}$ is defined by the A-module E, then

$$B^u(S) = E \otimes A, \qquad B^u(\operatorname{Spec} \kappa(s_i)) = E \otimes_A (A/\mathfrak{p}_i).$$

The assertion follows at once from the surjectivity of

$$A \longrightarrow \prod_i A/\mathfrak{p}_i.$$

5.10. Subgroups of Type (R) with Reductive Fibers

Proposition 5.10.1. *Let (G, T, M, R) be a split S-group, let R' be a subset of R of type* (R) (5.4.2)*, and let $H_{R'}$ be the corresponding subgroup of G. The following conditions are equivalent:*

(i) *$H_{R'}$ is reductive* (that is, its geometric fibers are reductive).

(ii) *$R' = -R'$, that is, R' is symmetric.*

Moreover, under these conditions, $(H_{R'}, T, M, R')$ is a splitting of $H_{R'}$; for every system R_+ of positive roots of R,

$$R'_+ = R' \cap R_+$$

is a system of positive roots of R', and

$$B_{R_+} \cap H_{R'} = H_{R'_+}.$$

This is a Borel subgroup of $H_{R'}$, whose unipotent part is

$$U_{R_+} \cap H_{R'} = U_{R'_+}.$$

Clearly (i) implies (ii): it is enough to check this fiber by fiber, and R' is the root system of $H_{R'}$ relative to T. To prove that (ii) implies (i), observe, by 5.4.3, that

$$H_{R'} \cap Z_\alpha = \underline{\mathrm{Centr}}_{H_{R'}}(T_\alpha) = Z_\alpha$$

for every $\alpha \in R'$, and apply the criterion of Exposé XIX, 1.12.

If R_+ is a system of positive roots of R, then $R'_+ = R_+ \cap R'$ is plainly a closed subset of R' such that

$$R'_+ \cup (-R'_+) = R', \qquad R'_+ \cap (-R'_+) = \varnothing;$$

it is therefore a system of positive roots of R'. The other two assertions follow respectively from 5.6.1(vi) and 5.6.7(i). □

Corollary 5.10.2. *Let S be a scheme, G a reductive S-group, and H a reductive S-subgroup such that, for every $s \in S$, $G_{\bar{s}}$ and $H_{\bar{s}}$ have the same reductive rank. Then H is closed in G, $\underline{\mathrm{Norm}}_G(H)$ is smooth over S, and*

$$\underline{\mathrm{Norm}}_G(H)/H$$

is representable by a finite étale S-scheme.

If T is a maximal torus of H, and B a Borel subgroup of G containing T, then $B \cap H$ is a Borel subgroup of H, whose unipotent part is

$$(B \cap H)^u = B^u \cap H.$$

The first assertions follow at once from 5.3.10 and 5.3.18, using the fact that the Weyl groups of G and H are finite (Exposé XIX, 2.5). The other assertions are local for the étale topology and reduce to the case studied in 5.10.1. □

Proposition 5.10.3. *Let S be a scheme and G a reductive S-group.*

(a) *If Q is a torus of G, then $\underline{\mathrm{Centr}}_G(Q)$ is a subgroup of G of type* (R) *with reductive fibers. If $Q \subset Q'$ are two tori of G, then*

$$\underline{\mathrm{Centr}}_G(Q) \supset \underline{\mathrm{Centr}}_G(Q').$$

(b) *If H is a subgroup of G of type* (R) *with reductive fibers, then* $\mathrm{rad}(H)$ (4.3.6) *is a torus of G. If $H \subset H'$ are two subgroups of G of type* (R) *with reductive fibers, then*

$$\mathrm{rad}(H) \supset \mathrm{rad}(H').$$

(c) *If Q is a torus of G, then*

$$\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q)) \supset Q \quad \textit{and} \quad \underline{\mathrm{Centr}}_G(\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q))) = \underline{\mathrm{Centr}}_G(Q).$$

(d) *If H is a subgroup of G of type* (R) *with reductive fibers, then*

$$\underline{\mathrm{Centr}}_G(\mathrm{rad}(H)) \supset H \quad \textit{and} \quad \mathrm{rad}(\underline{\mathrm{Centr}}_G(\mathrm{rad}(H))) = \mathrm{rad}(H).$$

Indeed, (a) follows at once from Exposé XIX, 2.8. To prove (b), it is enough to observe that $\mathrm{rad}(H') \subset H$, since H contains, fpqc-locally, a maximal torus of G, hence of H'. The first assertion of (c) (respectively, of (d)) is trivial, and the second follows by the usual argument. □

This proposition leads to the following definition:

Definition 5.10.4. Let S be a scheme, G a reductive S-group, H a reductive subgroup of G of type (R), and Q a subtorus of G.[58]

1) H is called a *critical subgroup* if it is the centralizer of its radical.

2) Q is called a *C-critical torus* if it is the radical of its centralizer.

Proposition 5.10.3 then gives:

Corollary 5.10.5.

(i) For every subtorus Q of G, $\underline{\mathrm{Centr}}_G(Q)$ is critical.

(ii) For every subgroup H of G of type (R) with reductive fibers, $\mathrm{rad}(H)$ is a C-critical torus of G.

(iii) The maps
$$Q \longmapsto \underline{\mathrm{Centr}}_G(Q), \qquad H \longmapsto \mathrm{rad}(H)$$
are mutually inverse bijections between the set of C-critical tori of G and the set of its critical reductive subgroups of type (R).

(iv) If Q is a torus of G, then
$$\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q))$$
is the smallest C-critical torus of G containing Q.

(v) If H is a reductive subgroup of G of type (R), then
$$\underline{\mathrm{Centr}}_G(\mathrm{rad}(H))$$
is the smallest critical reductive subgroup of G of type (R) containing H.

Remark 5.10.5.1. (58)

1) A torus T of G is a critical subgroup of G if and only if it is a maximal torus.

2) Henceforth, "critical torus" means "C-critical torus".

Proposition 5.10.6. *Let (G, T, M, R) be a split S-group, and let R' be a subset of R. The following conditions are equivalent:*

(i) *R' is of type* (R)*, and $H_{R'}$ is reductive and critical.*

(ii) *There exist a system Δ of simple roots of R and a subset Δ' of Δ such that R' is the set of those elements of R that are linear combinations of elements of Δ'.*

(iii) *R' is closed and symmetric, and every system of simple roots of R' is the intersection with R' of a system of simple roots of R.*

[58] Editor's note: We have modified the original by introducing the term "C-critical torus" in place of "critical torus", in order to avoid confusion in later references (cf. Exposé XXVI, 3.9). We have also expanded the statement of 5.10.5 and added Remark 5.10.5.1.

Indeed, by Exposé XXI, 3.4.8, conditions (ii) and (iii) are equivalent; they are also equivalent to the condition that R' be the intersection of R with a $\mathbf{Q}$-vector subspace of $M \otimes \mathbf{Q}$. This last condition follows from (i): if

$$H_{R'} = \underline{\mathrm{Centr}}_G(Q),$$

then R' is the set of elements of R that vanish on Q (Exposé II, 5.2.3(ii)). Conversely, this condition implies (i), because $\mathrm{rad}(H_{R'})$ is the maximal torus of

$$\bigcap_{\alpha \in R'} \mathrm{Ker}(\alpha).$$

Consequently,

$$\underline{\mathrm{Centr}}_G(\mathrm{rad}(H_{R'})) = H_{R''},$$

where R'' is the intersection of R with the vector subspace spanned by R'; hence $R'' = R'$. $\square$

5.10.7. Let us summarize some of the preceding results. Let (G, T, M, R) be a split S-group, let Δ be a system of simple roots of R, and let R_+ be the corresponding system of positive roots. Choose a subset Δ' of Δ, denote by R' the set of those elements of R which are linear combinations of elements of Δ', and put

$$R'_+ = R' \cap R_+.$$

Let $T_{\Delta'}$ be the maximal torus of

$$\bigcap_{\alpha \in \Delta'} \mathrm{Ker}(\alpha)$$

and put

$$Z_{\Delta'} = \underline{\mathrm{Centr}}_G(T_{\Delta'}).$$

Then $Z_{\Delta'}$ is a reductive subgroup of G, with radical $T_{\Delta'}$; $(Z_{\Delta'}, T, M, R')$ is a split S-group; $B_{R_+} \cap Z_{\Delta'}$ is the Borel subgroup of $Z_{\Delta'}$ defined by the system R'_+ of positive roots (or, equivalently, by the system Δ' of simple roots), and its unipotent part is

$$U_{R_+} \cap Z_{\Delta'} = U_{R'_+}.$$

Remark 5.10.8. Under the conditions of 5.10.4, let Q be a critical torus of G, and let

$$L = \underline{\mathrm{Centr}}_G(Q)$$

be its centralizer. Since $Q = \mathrm{rad}(L)$, Q is a characteristic subgroup of L; it follows at once that

$$\underline{\mathrm{Norm}}_G(L) = \underline{\mathrm{Norm}}_G(Q),$$

and hence also that

$$\underline{\mathrm{Norm}}_G(L)/L = \underline{\mathrm{Norm}}_G(Q)/\underline{\mathrm{Centr}}_G(Q) = W_G(Q).$$

By 5.10.2, we deduce:

Proposition 5.10.9. *Let S be a scheme, let G be a reductive S-group, and let Q be a critical torus of G. The Weyl group $W_G(Q)$ is finite* (étale) *over S.*

Remark 5.10.10. Under the conditions of 5.10.7, one may describe explicitly

$$W_G(T_{\Delta'}) = \underline{\mathrm{Norm}}_G(Z_{\Delta'})/Z_{\Delta'}.$$

This is the constant group associated with the quotient W_1/W_2, where W_1 is the subgroup of W consisting of those elements which normalize the subgroup of M generated by Δ', and W_2 is the subgroup of W generated by the s_α, for $\alpha \in \Delta'$.

5.11. Subgroups of Type (RC)

Definition 5.11.1. Let S be a scheme and let G be a reductive S-group. A group subscheme H of G is said to be *of type* (RC) if it is of type (R), that is, if it satisfies the following two conditions from 5.2.1:

(i) H is smooth over S, with connected fibers;

(ii) for every $s \in S$, $H_{\bar{s}}$ contains a maximal torus of $G_{\bar{s}}$;

and if it also satisfies the following condition:

(iii) for every $s \in S$ and every maximal torus T of $H_{\bar{s}}$, the set of roots of $H_{\bar{s}}$ relative to T is a closed subset of the set of all roots of $G_{\bar{s}}$ relative to T.

Remark 5.11.2. As we already observed in 5.4.8, condition (iii) follows from the other two when 6 is invertible on S.[59]

Lemma 5.11.3. *Let (G, T, M, R) be a split S-group, and let R' be a closed subset of R. Put*

$$R_1 = \{\alpha \in R' \mid -\alpha \in R'\}, \qquad R_2 = \{\alpha \in R' \mid -\alpha \notin R'\}.$$

Then R_1 and R_2 are closed. Consider the groups $H_{R'}$, H_{R_1}, and U_{R_2} (5.4.7 and 5.6.5), *which are smooth and have connected fibers.*

(i) *U_{R_2} is normal in $H_{R'}$, and $H_{R'}$ is the semidirect product of U_{R_2} by H_{R_1}.*

(ii) *The group H_{R_1} is reductive, and U_{R_2} has connected unipotent geometric fibers. Every normal subgroup scheme of $H_{R'}$ that is smooth over S and has connected unipotent geometric fibers is contained in U_{R_2}. Every reductive subgroup of $H_{R'}$ that contains T is contained in H_{R_1}.*

(iii)

$$U_{R_2} \cap \underline{\mathrm{Norm}}_G(H_{R_1}) = e.$$

Let us first prove (iii), which follows from 5.6.7(i). The first assertion of (i) follows from 5.6.7(ii). Since

$$U_{R_2} \cap H_{R_1} = e$$

by (iii), the semidirect product $H_{R_1} \cdot U_{R_2}$ is a subgroup of $H_{R'}$. But these are two subgroups of G of type (R), containing T, and having the same Lie algebra $\mathfrak{g}_{R'}$; they coincide by 5.3.5, which proves (i).

For (ii), the first two assertions are merely 5.10.1 and 5.6.5. Let U be a subgroup scheme of $H_{R'}$, smooth and of finite presentation over S, normal in $H_{R'}$ (and hence normalized by

[59] Editor's note: That is, when every residue characteristic of S is greater than 3.

T), and with connected unipotent geometric fibers. By 5.6.12, locally one has $U = U_{R''}$, where $R'' \subset R'$ and

$$R'' \cap (-R'') = \varnothing.$$

If U were not contained in U_{R_2}, then R'' would not be contained in R_2; there would therefore exist $\alpha \in R''$ such that $-\alpha \in R'$. Then $Z_\alpha \subset H_{R'}$ by 5.4.3, so Z_α normalizes U. But U contains U_α, and Z_α has a section w such that

$$\operatorname{int}(w)U_\alpha = U_{-\alpha};$$

thus $-\alpha \in R''$, a contradiction.

Finally, if L is a reductive subgroup of $H_{R'}$ containing T, locally one has $L = H_{R'''}$, where R''' is symmetric and contained in R'; hence $R''' \subset R_1$. □

Proposition 5.11.4. *Let S be a scheme, let G be a reductive S-group, and let H be a subgroup scheme of G of type* (RC)*.*

(i) *H is closed in G, and $\underline{\mathrm{Norm}}_G(H)/H$ is representable by a finite étale S-group scheme.*

(ii) *H has a largest normal subgroup scheme that is smooth and of finite presentation over S, with connected unipotent geometric fibers. It is called the* unipotent radical *of H and is denoted* $\operatorname{rad}^u(H)$*. The quotient sheaf*

$$H/\operatorname{rad}^u(H)$$

is representable by a reductive S-group.

(iii) *If T is a maximal torus of H, then H has a reductive subgroup L, containing T, of type* (RC)*, with the following properties:*

  (a) *every reductive subgroup of H containing T is contained in L;*

  (b) *H is the semidirect product*

$$H = L \cdot \operatorname{rad}^u(H);$$

  in other words, the canonical morphism

$$L \longrightarrow H/\operatorname{rad}^u(H)$$

  is an isomorphism.

Moreover, L is the unique reductive subgroup of H that contains T and satisfies either one of the two preceding conditions. Finally,

$$\underline{\mathrm{Norm}}_H(L) = L, \qquad \underline{\mathrm{Norm}}_H(T) = \underline{\mathrm{Norm}}_L(T), \qquad W_H(T) = W_L(T);$$

in particular, $W_H(T)$ is finite over S.

Proof. First observe that (i) is local for the étale topology. Thus, by Corollary 5.3.18, (i) follows from the last assertion of (iii).

The assertions of (ii) are local for the étale topology. We may therefore suppose ourselves in the situation of 5.11.3, where they follow immediately from (i) and (ii).

By virtue of the uniqueness assertions contained there, (iii) is likewise local for the étale topology, and we may again reduce to the situation of 5.11.3, where properties (a) and (b) have been verified. The uniqueness of an L satisfying (a) is immediate; in view of (a), the uniqueness of an L satisfying (b) is clear. The equality

$$\underline{\mathrm{Norm}}_H(L) = L$$

is precisely 5.11.3(iii). If a section of H normalizes T, then it normalizes L, by the uniqueness of L, and hence is a section of L by what we have just proved. This proves the second equality; the third is then immediate.

Proposition 5.11.5. *Let S be a scheme, let G be a reductive S-group, and let $\mathcal{H}_c$ be the functor of subgroups of G of type* (RC)*, which is a subfunctor of the functor $\mathcal{H}$ of 5.8.1.*

(i) *$\mathcal{H}_c$ is representable by an open subscheme of $\mathcal{H}$, smooth, quasi-projective, and of finite presentation over S.*

(ii) *There exist a finite étale S-scheme $\mathcal{C}\ell_c$ and a morphism*

$$\mathrm{cl}\colon \mathcal{H}_c \longrightarrow \mathcal{C}\ell_c$$

that is smooth, quasi-projective, of finite presentation, surjective, and has connected geometric fibers, with the following property:

For every $S' \to S$ and all $H, H' \in \mathcal{H}_c(S')$, one has $\mathrm{cl}(H) = \mathrm{cl}(H')$ if and only if H and H' are conjugate in G locally for the étale topology (or, equivalently by 5.3.11, if for every $s \in S$, $H_{\bar{s}}$ and $H'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$).

(iii) *$\mathcal{C}\ell_c$ and* cl *are determined* (up to a unique isomorphism) *by the preceding conditions.*

(iv) *If (G, T, M, R) is a splitting of G, let E be the set of conjugacy classes modulo W of closed subsets of R. Then there is an isomorphism*

$$\mathcal{C}\ell_c \xrightarrow{\sim} E_S$$

such that, for every closed subset R' of R, $\mathrm{cl}(H_{R'})$ corresponds to the canonical image of R' in

$$E_S(S) = \mathrm{Hom}_{\mathrm{loc.const.}}(S, E).$$

It is first of all clear that $\mathcal{H}_c$ is a sheaf for the étale topology, and that (ii) implies that $\mathcal{C}\ell_c$ is simply the quotient sheaf of $\mathcal{H}_c$ by the equivalence relation defined by conjugation.

This first implies (iii), as well as the fact that it suffices to verify (i) and (ii) locally for the étale topology. We are therefore reduced to the situation of (iv); let us first construct a morphism

$$f\colon \mathcal{H}_c \longrightarrow E_S.$$

It suffices to construct a map $\mathcal{H}_c(S) \to E_S(S)$, functorial in S; let H be a subgroup of G of type (RC). Since H locally for the étale topology has maximal tori, and since the maximal tori of G are locally conjugate for the étale topology, there exist a covering family $(S_i \to S)$, elements $g_i \in G(S_i)$, and closed subsets R_i of R such that

$$\mathrm{int}(g_i)(H \times_S S_i) = H_{R_i} \times_S S_i.$$

Each R_i defines a section η_i of E_{S_i}, that is, an element of $E_S(S_i)$. It remains to prove that the family (η_i) comes from a section $\eta = f(H)$ of E_S over S, and that it depends only on H.

For this it is enough to prove that $H_{R'}$ and $H_{R''}$ are locally conjugate for the étale topology if and only if R' and R'' are conjugate by an element of the Weyl group W, which is clear.

For every $\eta \in E$, there exists $H_0 \in \mathcal{H}_c(S)$ such that $f(H_0) = \eta$: take $H_0 = H_{R'}$, where R' is a closed subset whose image is η. If $H \in \mathcal{H}_c(S')$, for $S' \to S$, then H is locally conjugate to H_0 if and only if $f(H) = \eta$. Consequently,

$$f^{-1}(\eta) \simeq G/\underline{\mathrm{Norm}}_G(H_0),$$

which, by 5.8.2, is an open subscheme of $\mathcal{H}$, smooth, quasi-projective, and of finite presentation over S, with nonempty connected fibers. Since E_S is the disjoint union of the open subschemes that are the images of the sections corresponding to the elements $\eta \in E$, $\mathcal{H}_c$ identifies with the disjoint union of the $f^{-1}(\eta)$. This proves (i) and (ii); assertion (iv) follows from the construction.

Corollary 5.11.6. *Let $u \in \mathcal{C}\ell_c(S')$, where $S' \to S$. Then $\mathrm{cl}^{-1}(u)$ is a smooth, quasi-projective S'-scheme of finite presentation, with nonempty connected fibers. It is an open subscheme of $\mathcal{H}_c$ and a "homogeneous" scheme under $G_{S'}$* (acting by inner automorphisms)*.*

In particular, if $H \in \mathrm{cl}^{-1}(u)(S')$, the morphism

$$G_{S'} \longrightarrow (\mathcal{H}_c)_{S'}, \qquad g \longmapsto \mathrm{int}(g)H$$

identifies

$$G_{S'}/\underline{\mathrm{Norm}}_{G_{S'}}(H) \quad \textit{with} \quad \mathrm{cl}^{-1}(u).$$

Examples 5.11.7. There are two canonical sections u_t and u_b of $\mathcal{C}\ell_c$. The section u_t corresponds to maximal tori, for which $R' = \varnothing$. The section u_b corresponds to Borel subgroups, for which R' is a system of positive roots. The schemes $\mathrm{cl}^{-1}(u_t)$ and $\mathrm{cl}^{-1}(u_b)$ are respectively $\underline{\mathrm{Tor}}(G)$ and $\underline{\mathrm{Bor}}(G)$, introduced in 5.8.3. Other examples will be given in Exposé XXVI.

Remark 5.11.8. One can construct an S-scheme $\mathcal{C}\ell$, of finite presentation and unramified over S, and a smooth surjective morphism

$$\mathcal{H} \longrightarrow \mathcal{C}\ell$$

with connected geometric fibers, having properties analogous to those of 5.11.5(ii) and (iii).

6. The Derived Group

6.1. Preliminaries

In this section, we fix a scheme S, a split S-group (G, T, M, R), and a system R_+ of positive roots of R. We write

$$B = B_{R_+}, \qquad B^- = B_{R_-}, \qquad U = B^u, \qquad U^- = (B^-)^u,$$

and

$$\Omega = \Omega_{R_+} = U^- \cdot T \cdot U.$$

6.1.1. Let T' denote the subtorus of T that is the "image of the family α^*, $\alpha \in R$"; in other words, T' is the image of the homomorphism

$$(\mathbb{G}_{m,S})^R \longrightarrow T$$

defined by

$$(z_\alpha)_{\alpha\in R} \longmapsto \prod_{\alpha\in R} \alpha^*(z_\alpha).$$

It follows at once that, if Δ denotes the set of simple roots of R_+, the homomorphism

$$(\mathbb{G}_{m,S})^\Delta \longrightarrow T'$$

defined in the same way is surjective and has finite kernel. If we identify T with $D_S(M)$, then T' identifies with $D_S(M/N)$, where

$$N = M \cap V(R^*)^\perp$$

(we write $V(R^*)^\perp$ for the orthogonal complement of $V(R^*)$ under the duality between V and V^*).

Lemma 6.1.2. *The homomorphism defined by multiplication in T,*

$$\operatorname{rad}(G) \times_S T' \longrightarrow T,$$

is an isogeny (cf. 4.2.9).

Indeed, the canonical homomorphism

$$\operatorname{rad}(T) \longrightarrow T/T'$$

comes by duality from the homomorphism of commutative groups

$$M \cap V(R^*)^\perp \longrightarrow M/(M \cap V(R)),$$

which is immediately seen to be injective with finite cokernel (cf. Exposé XXI, 6.3).

Definition 6.1.3. We put

$$\Omega' = U^- \cdot T' \cdot U;$$

this is a closed subscheme of $\Omega = U^- \cdot T \cdot U$.

Lemma 6.1.4. *Let α be a simple root and let $w_\alpha \in \underline{\mathrm{Norm}}_G(T)(S)$ lift s_α. Then*

$$\operatorname{int}(w_\alpha)\Omega' \cap \Omega \subset \Omega'.$$

It is enough to prove that, if $g \in \Omega'(S)$ and $\operatorname{int}(w_\alpha)g \in \Omega(S)$, then $\operatorname{int}(w_\alpha)g \in \Omega'(S)$. By 5.6.8, write

$$g = a \exp_{-\alpha}(Y) t \exp_\alpha(X) b,$$

where

$$a \in U_{-\widehat{\alpha}}(S), \qquad Y \in \Gamma(S, \mathfrak{g}^{-\alpha}), \qquad t \in T'(S), \qquad X \in \Gamma(S, \mathfrak{g}^\alpha), \qquad b \in U_{\widehat{\alpha}}(S).$$

We then have

$$\operatorname{int}(w_\alpha)g = \operatorname{int}(w_\alpha)a \cdot \operatorname{int}(w_\alpha)(\exp_{-\alpha}(Y) t \exp_\alpha(X)) \cdot \operatorname{int}(w_\alpha)b.$$

By 5.6.8(iv),

$$\operatorname{int}(w_\alpha)a \in U_{-\widehat{\alpha}}(S), \qquad \operatorname{int}(w_\alpha)b \in U_{\widehat{\alpha}}(S).$$

It follows that, on putting

$$h = \exp_{-\alpha}(Y)t \exp_\alpha(X),$$

we have the equivalences

$$\operatorname{int}(w_\alpha)g \in \Omega(S) \quad \Longleftrightarrow \quad \operatorname{int}(w_\alpha)h \in \Omega(S),$$

and

$$\operatorname{int}(w_\alpha)g \in \Omega'(S) \quad \Longleftrightarrow \quad \operatorname{int}(w_\alpha)h \in \Omega'(S).$$

We are therefore reduced to the case $g = h$. Since, by (4.1.12),

$$Z_\alpha \cap \Omega = U_{-\alpha} \cdot T \cdot U_\alpha, \qquad Z_\alpha \cap \Omega' = U_{-\alpha} \cdot T' \cdot U_\alpha,$$

we are reduced to proving the following assertion:

$$\operatorname{int}(w_\alpha)h \in (U_{-\alpha} \cdot T \cdot U_\alpha)(S) \quad \Longrightarrow \quad \operatorname{int}(w_\alpha)h \in (U_{-\alpha} \cdot T' \cdot U_\alpha)(S).$$

But this follows at once from Exposé XX, 3.12, which shows that the T-component of $\operatorname{int}(w_\alpha)h$ is of the form

$$t \cdot \alpha^*(z) \in T'(S).$$

Lemma 6.1.5. *For every $w \in \underline{\mathrm{Norm}}_G(T)(S)$, there exists an open subscheme V_w of G, containing the unit section, such that*

$$\operatorname{int}(w)\Omega' \cap V_w \subset \Omega'.$$

For each simple root α, choose an element $n_\alpha \in \underline{\mathrm{Norm}}_G(T)(S)$ lifting s_α. For every point $s \in S$, there exist an open subscheme V of S containing s, an element $t \in T(V)$, and, over V, a relation

$$w = n_{\alpha_1} \cdots n_{\alpha_p} t,$$

where the α_i are simple. We may plainly restrict ourselves to proving the result when $V = S$; we argue by induction on p. If $p = 0$, then $w \in T(S)$, and we take $V_w = G$. Suppose, therefore, that $w = n_\alpha w'$, where w' satisfies the conclusion of the lemma. There is then an open subscheme $V_{w'}$ of G, containing the unit section, such that

$$\operatorname{int}(w')\Omega' \cap V_{w'} \subset \Omega'.$$

We may then write

$$\begin{aligned}\operatorname{int}(w)\Omega' \cap \operatorname{int}(n_\alpha)V_{w'} \cap \Omega &= \operatorname{int}(n_\alpha)\big(\operatorname{int}(w')\Omega' \cap V_{w'}\big) \cap \Omega \\ &\subset \operatorname{int}(n_\alpha)\Omega' \cap \Omega \subset \Omega',\end{aligned}$$

by 6.1.4. We now take

$$V_w = \operatorname{int}(n_\alpha)V_{w'} \cap \Omega,$$

and the proof is complete.

Lemma 6.1.6. *There exists an open subscheme V_0 of G, containing the unit section, such that for every $S' \to S$,*

$$U(S')U^-(S') \cap V_0(S') \subset \Omega'(S').$$

Let n_0 be an element of $\underline{\mathrm{Norm}}_G(T)(S)$ lifting the symmetry w_0 of the Weyl group,[60] that is, such that

$$\mathrm{int}(n_0)U = U^- \qquad \text{(cf. Exposé XXI, 3.6.14).}$$

Then $n_0^2 \in T(S)$. We show that the open subscheme $V_0 = V_{n_0}$ of 6.1.5 has the required property. Indeed,

$$\begin{aligned} U(S')U^-(S') &= \mathrm{int}(n_0)\big(\mathrm{int}(n_0)^{-1}U(S') \cdot \mathrm{int}(n_0)^{-1}U^-(S')\big) \\ &= \mathrm{int}(n_0)\big(U^-(S') \cdot U(S')\big) \subset \mathrm{int}(n_0)\Omega'(S'). \end{aligned}$$

Consequently,

$$U(S')U^-(S') \cap V_0(S') \subset \mathrm{int}(n_0)\Omega'(S') \cap V_0(S') \subset \Omega'(S').$$

Lemma 6.1.7. *Consider the morphism*

$$f : \Omega = U^- \cdot T \cdot U \longrightarrow T/T'$$

obtained by composing the second projection with the canonical morphism from T to T/T'. Then f is "generically multiplicative": there exists an open subscheme V of $\Omega \times_S \Omega$, containing the unit section (and hence relatively schematically dense, Exposé XVIII, 1.3)*, such that for every $S' \to S$ and every $(x, y) \in V(S')$, one has $xy \in \Omega(S')$ and*

$$f(xy) = f(x)f(y).$$

Let x, y be sections of Ω over S'. Write

$$x = utv, \qquad y = u't'v',$$

where

$$u, u' \in U^-(S'), \qquad t, t' \in T(S'), \qquad v, v' \in U(S').$$

Let V_0 be the open subscheme of 6.1.6, and let V be the open subscheme of $\Omega \times_S \Omega$ defined by the condition

$$vu' \in V_0(S').$$

This open subscheme has the required property. Indeed, if $(x, y) \in V(S')$, then

$$xy = (utv)(u't'v') = (ut)(vu')(t'v').$$

Since $vu' \in \Omega'(S')$, we obtain

$$xy \in U^-(S')\, t\Omega'(S')t'\, U(S') \subset U^-(S')\, tt'T'(S')\, U(S'),$$

and therefore $xy \in \Omega(S')$ and

$$f(xy) = f(tt') = f(t)f(t') = f(x)f(y).$$

[60] Editor's note: The symmetry w_0 is defined in Exposé XXI, 3.6.14.

Proposition 6.1.8. *There exists a group morphism*

$$f : G \longrightarrow T/T'$$

inducing the canonical projection on T. The kernel $\mathrm{Ker}(f)$ *of f is a closed subgroup scheme of G, smooth over S, with connected fibers. Every group morphism from G to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on* $\mathrm{Ker}(f)$.

The first assertion follows at once from 4.1.11. We immediately have

$$\mathrm{Ker}(f) \cap \Omega = \Omega',$$

which proves that $\mathrm{Ker}(f)$ is smooth over S at every point of the unit section.[61] By 5.6.9(ii), every morphism ϕ from G to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on U and on U^-. By Exposé XX, 2.7, ϕ also vanishes on T', and hence on Ω'. Using the notation of 5.7.10, we see that the monoid U_1 is contained in $\mathrm{Ker}(f)(S)$, which shows that

$$\mathrm{Ker}(f) = \bigcup_{u \in U_1} u\Omega'.$$

It follows, on the one hand, that every ϕ as above vanishes on $\mathrm{Ker}(f)$, and, on the other hand, that $\mathrm{Ker}(f)$ has connected fibers and is therefore smooth over S, by Exposé $\mathrm{VI_B}$, 3.10.

6.2. The Derived Group of a Reductive Group

Theorem 6.2.1. *Let S be a scheme and G a reductive S-group.*

(i) *The group*

$$\mathrm{D}_S(G) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, \mathbb{G}_{m,S})$$

is represented by a twisted constant S-group whose type at $s \in S$ is

$$\mathbb{Z}^{\mathrm{rgred}(G_s) - \mathrm{rgss}(G_s)}.$$

(ii) *Write*

$$\mathrm{corad}(G) = \mathrm{D}_S(\mathrm{D}_S(G)),$$

which is therefore an S-torus. The biduality morphism (cf. Exposé VIII, §1)

$$f_0 : G \longrightarrow \mathrm{corad}(G)$$

is smooth and surjective.

(iii) *The composite morphism*

$$\mathrm{rad}(G) \longrightarrow G \longrightarrow \mathrm{corad}(G)$$

is an isogeny (cf. 4.2.9).

(iv) *The kernel of f_0, denoted*

$$\mathrm{der}(G) = \mathrm{Ker}(f_0),$$

is a closed subgroup scheme of G, semisimple over S, called the derived group of G. If G is semisimple, then $\mathrm{der}(G) = G$.

[61] Editor's note: We have added "at every point of the unit section," as well as the reference to Exposé $\mathrm{VI_B}$, 3.10, at the end of the proof. Moreover, in the following sentence "prescheme" has been replaced by "presheaf."

(v) *Every group morphism from G to an S-presheaf of commutative groups that is separated for the fppf topology vanishes on* $\mathrm{der}(G)$*, and therefore factors through* f_0*.*

Proof. All the assertions of the theorem are local for the étale topology; we may therefore reduce to the case in which G is split over S. Consider the morphism f of 6.1.8. By the last assertion of 6.1.8, we immediately have an isomorphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, \mathbb{G}_{m,S}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(T/T', \mathbb{G}_{m,S}),$$

which proves (i), then (ii), and yields a commutative diagram

$$\begin{tikzcd} G \arrow[r, "f_0"] \arrow[dr, "f"'] & \mathrm{corad}(G) \arrow[d, "\sim"] \\ & T/T'. \end{tikzcd}$$

Assertion (v) now follows from 6.1.8, and (iii) from 6.1.2. We also have $\mathrm{Ker}(f) = \mathrm{Ker}(f_0)$, which, by 6.1.8, implies that $\mathrm{der}(G)$ is smooth over S with connected fibers. It remains to verify that its fibers are semisimple. They are reductive by Exposé XIX, 1.7, as invariant subgroups of reductive groups. By (iii), $\mathrm{rad}(G) \cap \mathrm{der}(G)$ is finite, which indeed implies that the fibers of $\mathrm{der}(G)$ are semisimple.

Remark 6.2.2. a) By construction, when G is split, $\mathrm{der}(G)$ is the fppf subsheaf of G generated by the U_α, $\alpha \in R$. It even suffices to take the U_α, $\alpha \in \Delta \cup -\Delta$, where Δ is a base of R.

b)[62] Let C be the presheaf of commutators of G, that is, the S-functor in groups that assigns to every $S' \to S$ the commutator subgroup of $G(S')$ (that is, the subgroup of $G(S')$ generated by the elements $xyx^{-1}y^{-1}$, for $x, y \in G(S')$), and let $\widetilde{C}$ be the associated fppf sheaf. Since the quotient $G/\mathrm{der}(G) = T/T'$ is commutative, $\mathrm{der}(G)$ contains C, and hence $\widetilde{C}$ (cf. Exposé IV, 4.3.12).

On the other hand, the quotient presheaf $G/\widetilde{C}$ is separated (Exposé IV, 4.4.8.1), and therefore, by (v), we have $\mathrm{der}(G) \subset \widetilde{C}$. Thus $\mathrm{der}(G) = \widetilde{C}$; in other words, $\mathrm{der}(G)$ is the fppf sheaf of commutators of G.

Let us finally note that, since C is a subpresheaf of G, it is separated, but it is not equal to $\mathrm{der}(G)$ in general: for example, $\mathrm{der}(\mathrm{SL}_2) = \mathrm{SL}_2$, whereas $\mathrm{SL}_2(\mathbb{F}_2) \simeq \mathfrak{S}_3$ is not equal to its commutator subgroup.

c) When S is the spectrum of an algebraically closed field k, $\mathrm{der}(G)(k)$ is the commutator subgroup of $G(k)$ (Exposé $\mathrm{VI_B}$, 7.10).

6.2.3. Consider now the two exact sequences

$$1 \longrightarrow \mathrm{rad}(G) \longrightarrow G \longrightarrow \mathrm{ss}(G) \longrightarrow 1,$$
$$1 \longrightarrow \mathrm{der}(G) \longrightarrow G \longrightarrow \mathrm{corad}(G) \longrightarrow 1.$$

Since $\mathrm{rad}(G)$ is central in G, multiplication in G defines a group morphism

$$u : \mathrm{rad}(G) \times_S \mathrm{der}(G) \longrightarrow G,$$

[62] Editor's note: In what follows, we have expanded the original and removed the assertion that "$\mathrm{der}(G)$ is the fppf-separated presheaf of commutators of G."

which is covering by 6.2.1(iii), and hence surjective and flat (Exposé $\mathrm{VI_B}$, 9.2(xi)). [63] Its kernel is isomorphic to $\mathrm{rad}(G) \cap \mathrm{der}(G)$, which is also the kernel of

$$\mathrm{rad}(G) \longrightarrow \mathrm{corad}(G);$$

it is a finite subgroup of multiplicative type of $\mathrm{rad}(G)$. Similarly, the kernel of the morphism

$$G \longrightarrow \mathrm{corad}(G) \times_S \mathrm{ss}(G)$$

is isomorphic to $\mathrm{der}(G) \cap \mathrm{rad}(G)$.

Proposition 6.2.4. *Let G be a reductive S-group. The three morphisms*

$$\mathrm{rad}(G) \times_S \mathrm{der}(G) \longrightarrow G,$$
$$G \longrightarrow \mathrm{corad}(G) \times_S \mathrm{ss}(G),$$
$$\mathrm{rad}(G) \longrightarrow \mathrm{corad}(G)$$

are central isogenies, and their kernels are isomorphic.

Corollary 6.2.5. *The following conditions are equivalent:*

(i) *G is the product of a semisimple group and a torus.*

(ii) *The morphism*

$$\mathrm{rad}(G) \times_S \mathrm{der}(G) \xrightarrow{\sim} G$$

is an isomorphism.

(iii) *The morphism*

$$G \xrightarrow{\sim} \mathrm{corad}(G) \times_S \mathrm{ss}(G)$$

is an isomorphism.

(iv) $\mathrm{rad}(G) \cap \mathrm{der}(G) = e$.

6.2.6. Let us return temporarily to the case of a split group. We retain the notation of 6.1. Put

$$N = M \cap V(R^*)^{\perp}, \qquad \text{so that} \qquad T' = \mathrm{D}_S(M/N).$$

Then $U^- \cdot T' \cdot U$ is an open neighborhood of the unit section in $\mathrm{der}(G)$. Consequently,

$$\mathrm{Lie}\big(\mathrm{der}(G)/S\big) = \mathfrak{t}' \oplus \coprod_{\alpha \in R} \mathfrak{g}^{\alpha}.$$

Since the characters induced on T' by the $\alpha \in R$ are nonzero and distinct (Exposé XXI, 1.2.5—which we already used in 6.1.2), R is a root system of $\mathrm{der}(G)$ relative to T'. It is then immediate (since $U_\alpha \subset \mathrm{der}(G)$) that the exp-morphisms of $\mathrm{der}(G)$ "are" those of G, and likewise for the coroots.

It follows that:

[63] Editor's note: Indeed, G is the fppf quotient of $\mathrm{rad}(G) \times_S \mathrm{der}(G)$ by $\mathrm{Ker}(u)$, which is a group of multiplicative type and hence is flat over S. Therefore, by Exposé $\mathrm{VI_B}$, 9.2(xi), u is flat.

Proposition 6.2.7. *With the preceding notation,*

$$(\mathrm{der}(G), T', M/N, R)$$

is a split group with root datum $\mathrm{der}(\mathcal{R}(G))$. *The canonical morphism* $\mathrm{der}(G) \to G$ *induces, by functoriality, the canonical morphism of root data*

$$\mathcal{R}(G) \longrightarrow \mathrm{der}(\mathcal{R}(G))$$

of Exposé XXI, 6.5.

N.B. As an exercise, the reader may construct the diagram of split groups corresponding to the three left-hand columns of the diagram of root data in Exposé XXI, 6.5.7.

Proposition 6.2.8. *Let S be a scheme, G a reductive S-group, and* $\mathrm{der}(G)$ *its derived group.*

(i) *For every maximal torus T of G, $T \cap \mathrm{der}(G)$ is a maximal torus of* $\mathrm{der}(G)$. *For every maximal torus T' of* $\mathrm{der}(G)$,

$$\underline{\mathrm{Centr}}_G(T') = \mathrm{rad}(G) \cdot T'$$

is a maximal torus of G. These two constructions are inverse to one another and establish a bijective correspondence between the maximal tori of G and those of $\mathrm{der}(G)$.

(ii) *For every Borel subgroup B of G, $B \cap \mathrm{der}(G)$ is a Borel subgroup B' of* $\mathrm{der}(G)$. *We have $B'^u = B^u$. For every Borel subgroup B' of* $\mathrm{der}(G)$,

$$\underline{\mathrm{Norm}}_G(B') = \mathrm{rad}(G) \cdot B'$$

is a Borel subgroup of G. These constructions are inverse to one another and establish a bijective correspondence between the Borel subgroups of G and those of $\mathrm{der}(G)$.

By the local conjugacy theorem for maximal tori and the construction of the derived group, the only assertion that remains to be proved in (i) is the following: if T is a maximal torus of G, then

$$T = (T \cap \mathrm{der}(G)) \cdot \mathrm{rad}(G) = \underline{\mathrm{Centr}}_G(T \cap \mathrm{der}(G)).$$

The first equality is immediate (since we may reduce to the split case); the second follows at once, because $\mathrm{rad}(G)$ is central in G, and hence

$$T = \underline{\mathrm{Centr}}_G(T) = \underline{\mathrm{Centr}}_G(T \cap \mathrm{der}(G)).$$

The argument for (ii) is the same.

6.3. Subgroups with Commutative Quotients

6.3.1. Let G be a reductive S-group. If H is a subsheaf in groups of G, the following conditions are equivalent:

- H contains $\mathrm{der}(G)$;
- H is normal and G/H is commutative.

In this case, the canonical morphism

$$f_0 : G \longrightarrow \mathrm{corad}(G)$$

maps H onto a subsheaf $f_0(H)$ of $\mathrm{corad}(G)$, and we have

$$G/H \simeq \mathrm{corad}(G)/f_0(H), \qquad H/\mathrm{der}(H) \simeq f_0(H),$$
$$\mathrm{der}(G) = \mathrm{der}(H), \qquad H = f_0^{-1}(f_0(H)).$$

Since $\mathrm{der}(G)$ is smooth over S with connected fibers,[64] it follows from Exposé IV, 5.3.1 and 6.3.1, and Exposé $\mathrm{IV_B}$, 9.2, that the assignment

$$H \longmapsto f_0(H)$$

establishes a bijective correspondence between the group subschemes (respectively, the closed group subschemes) of G that contain $\mathrm{der}(G)$, are smooth over S, and have connected fibers, and the group subschemes (respectively, the closed group subschemes) of $\mathrm{corad}(G)$ that are smooth over S and have connected fibers.

Now, if H' is a group subscheme of $\mathrm{corad}(G)$, smooth over S with connected fibers, then H' is of finite presentation over S (Exposé $\mathrm{VI_B}$, 5.5), and its fibers are tori (because those of $\mathrm{corad}(G)$ are). Hence, by Exposé X, 8.2, H' is a subtorus of $\mathrm{corad}(G)$, and is therefore closed in $\mathrm{corad}(G)$ (Exposé IX, 2.6).

Consequently, every subgroup of G that is smooth with connected fibers and contains $\mathrm{der}(G)$ is closed in G.

6.3.2. If H is a closed group subscheme of G, smooth over S, with connected fibers, and normal in G, then H is reductive. If, moreover, $H \supset \mathrm{der}(G)$, then

$$\mathrm{der}(H) = \mathrm{der}(G) \qquad \text{and} \qquad f_0(H) \simeq \mathrm{corad}(H).$$

We have therefore proved the following proposition.

Proposition 6.3.3. *Let G be a reductive S-group. Every group subscheme H of G that is normal in G, has commutative quotient* (that is, contains $\mathrm{der}(G)$)*, is smooth over S, and has connected fibers*[65] *is closed and reductive. We have*

$$\mathrm{der}(H) = \mathrm{der}(G), \qquad f_0(H) \simeq \mathrm{corad}(H),$$

and

$$G/H \simeq \mathrm{corad}(G)/\mathrm{corad}(H), \qquad H = \big(H \cap \mathrm{rad}(G)\big) \cdot \mathrm{der}(G).$$

Moreover, the assignment $H \mapsto f_0(H)$ defines a bijection between the set of subgroups H of G having the preceding properties and the set of subtori of $\mathrm{corad}(G)$.[66]

By another application of Noether's theorem (Exposé IV, 5.3.1 and 6.3.1), we deduce the following.

[64] Editor's note: We have expanded the original in what follows. In particular, we have added the conclusion, implicit in the original, that every subgroup of G that is smooth with connected fibers and contains $\mathrm{der}(G)$ is closed in G.

[65] Editor's note: We have deleted the hypothesis that H be retrocompact in G, which is automatic: by Exposé $\mathrm{VI_B}$, 5.5, both G and H are separated and quasi-compact over S; hence $H \hookrightarrow G$ is quasi-compact by EGA $\mathrm{IV_1}$, 1.1.2(v).

[66] Editor's note: The French source prints $\mathrm{corad}(H)$ here; the codomain of f_0 and the preceding correspondence require $\mathrm{corad}(G)$.

Proposition 6.3.4. *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. For every subgroup H of G as above, $T \cap H$ is a maximal torus of H,*[67] *and*

$$G/H \simeq T/(T \cap H), \qquad H = (T \cap H) \cdot \mathrm{der}(G).$$

Moreover, the assignment $H \mapsto T \cap H$ is a bijection between the set of subgroups H of G as above and the set of subtori of T containing $T \cap \mathrm{der}(G)$.

[67] Editor's note: The French source prints G here; since $T \cap H \subset H$, and by the formulas that follow, the intended group is H.

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Exposé XXIII
Reductive Groups: Uniqueness of Pinned Groups

by M. Demazure

[68] The aim of this Exposé is to prove the uniqueness theorem (Theorem 4.1). It was proved by Chevalley in the case of an algebraically closed field; the reduction-to-rank-two method used here is likewise due to Chevalley (see *Bible*, Exposés 23 and 24). Along the way, we obtain an explicit description of reductive groups by generators and relations (3.5).

1. Pinnings

Definition 1.1.— *Let S be a scheme and let (G, T, M, R) be an S-split group (XXII, 1.13). A* pinning [69] *of this split group is the datum of a system Δ of simple roots of R, together with, for each $\alpha \in \Delta$, a section*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

In other words, a pinning of the reductive group G over the nonempty scheme S is the datum of:

(i) a maximal torus T;

(ii) an abelian group M and an isomorphism $T \simeq D_S(M)$;

(iii) a system of roots R of G with respect to T;

(iv) a system Δ of simple roots of R;

(v) an element $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ for every $\alpha \in \Delta$, that is, an element

$$u_\alpha = \exp_\alpha(X_\alpha) \in U_\alpha^\times(S) \qquad \text{for every } \alpha \in \Delta,$$

satisfying condition (D1) of Exposé XXII, 1.13 (indeed, condition (D2) of *loc. cit.* is automatic[70]).

[68] *Editors' note.* Version of 13 October 2024.

[69] *Editors' note.* Demazure tells us that behind this terminology lies the image of the butterfly (supplied to him by Grothendieck): the body is a maximal torus T, the wings are two Borel subgroups opposite with respect to T; one spreads the butterfly by laying out the wings and then fixes elements in the additive groups (pins) to rigidify the situation, that is, to eliminate the automorphisms.

[70] *Editors' note.* It is implied by condition (v), namely the existence of a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

Every split group admits a pinning; in particular, every reductive group is locally pinnable for the étale topology.

1.2.— If G is a pinned S-group, that is, an S-split group equipped with a pinning, it is canonically endowed with the system of positive roots R^+ defined by Δ, the corresponding Borel subgroup $B = B_{R^+}$, the opposite Borel subgroup $B^- = B_{R^-}$, the unipotent groups $U = B_u$ and $U^- = (B^-)_u$, the open subscheme $U^- \cdot T \cdot U$, and so forth. Likewise, for each $\alpha \in \Delta$, there is a canonical isomorphism of vector groups

$$p_\alpha : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha, \qquad x \longmapsto \exp_\alpha(xX_\alpha) = u_\alpha^x,$$

normalized by T with multiplier α, and whose datum is equivalent to that of X_α (Exposé XXII, 1.1).

By duality one obtains an $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ and an isomorphism

$$p_{-\alpha} : \mathbb{G}_{a,S} \xrightarrow{\sim} U_{-\alpha}$$

which is contragredient to the preceding one (Exposé XXII, 1.3). We set (Exposé XX, 3.1)

$$w_\alpha = w_\alpha(X_\alpha) = p_\alpha(1)p_{-\alpha}(-1)p_\alpha(1) = p_{-\alpha}(-1)p_\alpha(1)p_{-\alpha}(-1).$$

Then (*loc. cit.*, 3.1 and 3.7)

$$w_\alpha^2 = \alpha^*(-1), \qquad \mathrm{int}(w_\alpha)t = s_\alpha(t) = t \cdot \alpha^*(\alpha(t)^{-1}),$$

$$\mathrm{int}(w_\alpha)p_\alpha(x) = p_{-\alpha}(-x) = p_{-\alpha}(x)^{-1}, \qquad \mathrm{Ad}(w_\alpha)X_\alpha = -X_{-\alpha},$$
$$\mathrm{int}(w_\alpha)p_{-\alpha}(x) = p_\alpha(-x) = p_\alpha(x)^{-1}, \qquad \mathrm{Ad}(w_\alpha)X_{-\alpha} = -X_\alpha.$$

We shall systematically use the preceding notation in what follows.

Definition 1.3.— *Let S be a scheme, and suppose that*

$$(G, T, M, R, \Delta, (X_\alpha)) \quad \textit{and} \quad (G', T', M', R', \Delta', (X'_{\alpha'}))$$

are two pinned S-groups. The morphism of split groups (Exposé XXII, 4.2.1)

$$f : G \longrightarrow G'$$

is said to be compatible with the pinnings, *or to define a* morphism of pinned groups, *if the associated bijection $d : R \to R'$ (cf. loc. cit.) satisfies $d(\Delta) = \Delta'$ and if, for every $\alpha \in \Delta$,*

$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(X'_{d(\alpha)}), \qquad \textit{i.e.} \qquad f(u_\alpha) = u'_{d(\alpha)}.$$

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1.4.— If $q(\alpha)$ denotes the integer of *loc. cit.*, then

$$f(p_\alpha(x)) = p'_{d(\alpha)}(x^{q(\alpha)}) \qquad (\alpha \in \Delta),$$

and hence

$$f(p_{-\alpha}(x)) = p'_{-d(\alpha)}(x^{q(\alpha)}), \qquad f(w_\alpha) = w'_{d(\alpha)}.$$

Recall (Exposé XXII, 4.2) that, for every $\alpha \in R$, every $z \in \mathbb{G}_m(S')$, and every $t \in T(S')$,

$$f(\alpha^*(z)) = d(\alpha)^*(z)^{q(\alpha)} = d(\alpha)^*(z^{q(\alpha)}), \qquad d(\alpha)(f(t)) = \alpha(t)^{q(\alpha)}.$$

[71] *Editors' note.* The bijection $R \xrightarrow{\sim} R'$ is denoted by d (rather than by u) in order to avoid the notation $u'_{u(\alpha)}$.

1.5.— We call a root datum equipped with a system of simple roots a *pinned root datum*, and a p-morphism of root data (Exposé XXI, 6.8) that sends simple roots to simple roots a *p-morphism of pinned root data.*

If G is a pinned S-group, we denote by $\mathscr{R}(G)$ the corresponding pinned root datum (the root datum of Exposé XXII, 1.14 equipped with Δ). Let p be the integer defined in Exposé XXII, 4.2.2. Then:

Scholium 1.6.— *The correspondence $G \longmapsto \mathscr{R}(G)$ defines a functor from the category of pinned reductive S-groups to the category of pinned root data (whose morphisms are the p-morphisms).*

1.7. The pinned groups Z_{Δ_1}

Let Δ_1 be a subset of the system Δ of simple roots of the pinned group G. Let T_{Δ_1} be the maximal torus of $\bigcap_{\alpha\in\Delta_1}\ker(\alpha)$, and set

$$Z_{\Delta_1} = \mathrm{Centr}_G(T_{\Delta_1}).$$

Put $R' = \mathbb{Z}\Delta_1 \cap R$. It is known (Exposé XXII, 5.10.7) that Z_{Δ_1} is a reductive S-group with radical T_{Δ_1}, that (T, M, R') is a splitting of this group, and that Δ_1 is a system of simple roots. Thus

$$(Z_{\Delta_1}, T, M, R', \Delta_1, (X_\alpha)_{\alpha\in\Delta_1})$$

is a pinned S-group. We shall always give Z_{Δ_1} this pinning. In particular, we shall consider

$$Z_\alpha = Z_{\{\alpha\}}, \qquad Z_{\alpha\beta} = Z_{\{\alpha,\beta\}}.$$

Write $B_{\Delta_1} = B \cap Z_{\Delta_1}$. It is known (*loc. cit.*) that this is the canonical Borel subgroup [72] of Z_{Δ_1}, and that its unipotent part is $U_{\Delta_1} = U \cap Z_{\Delta_1}$. In particular,

$$B_\alpha = T \cdot U_\alpha.$$

We write

$$U_{\alpha\beta} = U_{\{\alpha,\beta\}} = U \cap Z_{\alpha\beta} = \prod_{\gamma\in R^+_{\alpha\beta}} U_\gamma,$$

where $R^+_{\alpha\beta}$ is the set of positive roots that are linear combinations of α and β.

Now let $f : G \to G'$ be a morphism of pinned S-groups. If $d : R \to R'$ is the corresponding bijection and $\Delta_1 \subset \Delta$, then $d(\Delta_1) = \Delta'_1 \subset \Delta'$. Clearly f maps T_{Δ_1} into $T'_{\Delta'_1}$, and hence Z_{Δ_1} into $Z'_{\Delta'_1}$. The corresponding S-morphism

$$f_{\Delta_1} : Z_{\Delta_1} \longrightarrow Z'_{\Delta'_1}$$

is compatible with the canonical pinnings. It defines a morphism of pinned root data

$$\mathscr{R}(f_{\Delta_1}) : \mathscr{R}(Z_{\Delta_1}, T, M, \ldots) \longrightarrow \mathscr{R}(Z'_{\Delta'_1}, T', M', \ldots),$$

and the underlying morphisms $M' \to M$ of $\mathscr{R}(f)$ and $\mathscr{R}(f_{\Delta_1})$ coincide.

[72] *Editors' note.* That is, the Borel subgroup of Z_{Δ_1} corresponding to $R'^+ = R' \cap R^+$.

1.8. Study of the group $\mathrm{Norm}_G(T)$

For every pair (α, β) of simple roots, let $n_{\alpha\beta}$ be the order of $s_\alpha s_\beta$ in the Weyl group W. In particular $n_{\alpha\alpha} = 1$, and therefore $(w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S)$.

Definition 1.8.1.— *For every $(\alpha, \beta) \in \Delta \times \Delta$, set*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S).$$

Moreover, set (Exposé XX, 3.1)

$$t_\alpha = t_{\alpha\alpha} = w_\alpha^2 = \alpha^*(-1) \in T(S).$$

Proposition 1.8.2.— *Let H be an S-functor in groups that carries direct sums of schemes to products (for example, a sheaf for the Zariski topology). Let*

$$f_T : T \longrightarrow H$$

be a morphism of groups, and let $h_\alpha \in H(S)$ $(\alpha \in \Delta)$. There exists a necessarily unique morphism of groups

$$f_N : \mathrm{Norm}_G(T) \longrightarrow H$$

inducing f_T on T and satisfying $f_N(w_\alpha) = h_\alpha$ for $\alpha \in \Delta$ if and only if the following two conditions hold:

(i) for every $\alpha \in \Delta$,

$$f_T(s_\alpha(t)) = h_\alpha f_T(t) h_\alpha^{-1}$$

for every $t \in T(S')$, $S' \to S$; equivalently, $f_T \circ s_\alpha = \mathrm{int}(h_\alpha) \circ f_T$;

(ii) for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$f_T(t_{\alpha\beta}) = (h_\alpha h_\beta)^{n_{\alpha\beta}}.$$

Proof. Equip (Sch) with the topology $\mathcal{T}$ generated by the pretopology whose covering families are direct sums. The hypothesis says that H is a $\mathcal{T}$-sheaf. Let L be the free group on the generators $(m_\alpha)_{\alpha\in R}$, and let L_1 be the invariant subgroup generated by

$$(m_\alpha m_\beta)^{n_{\alpha\beta}}, \qquad (\alpha, \beta) \in \Delta \times \Delta.$$

Let $g : L \to W$ be defined by $g(m_\alpha) = s_\alpha$. It is known (Exposé XXI, 5.1) that g induces an isomorphism $\overline{g} : L/L_1 \xrightarrow{\sim} W$. Let L act on T through g, equivalently by $m_\alpha \cdot t = s_\alpha(t)$. Writing L_S for the constant group defined by L, form the semidirect product $N = T \cdot L_S$. There is a unique morphism of S-groups

$$h : N = T \cdot L_S \longrightarrow \mathrm{Norm}_G(T)$$

such that $h(m_\alpha) = w_\alpha$ and $h(t) = t$ for every $t \in T(S')$, $S' \to S$.

Let N_1 be the invariant subgroup sheaf of N generated by

$$t_{\alpha\beta}^{-1}(m_\alpha m_\beta)^{n_{\alpha\beta}}, \qquad (\alpha, \beta) \in \Delta \times \Delta.$$

Clearly $N_1 \subset \ker h$, so h induces

$$h_1 : N/N_1 \longrightarrow \mathrm{Norm}_G(T).$$

We prove that h_1 is an isomorphism. Since $h|_T$ is the canonical immersion, hence a monomorphism, the canonical morphism $T \to N/N_1$ is also a monomorphism and induces an isomorphism $T \xrightarrow{\sim} TN_1/N_1$. Thus h_1 induces an isomorphism $TN_1/N_1 \xrightarrow{\sim} T$. It remains only to show that

$$h_2 : N/TN_1 \longrightarrow \mathrm{Norm}_G(T)/T$$

is an isomorphism. But TN_1 is the invariant $\mathcal{T}$-subgroup sheaf of N generated by T and the $(m_\alpha m_\beta)^{n_{\alpha\beta}}$; hence it is the subgroup sheaf generated by T and L_1, namely $T \cdot (L_1)_S$. Therefore h_2 identifies with

$$\overline{g} : L_S/(L_1)_S \longrightarrow W_S,$$

which is an isomorphism by construction.

The proof of the proposition is now immediate. Necessity is clear. Conversely, condition (i) gives a morphism

$$u : N \longrightarrow H$$

such that $u(m_\alpha) = h_\alpha$ for $\alpha \in \Delta$ and $u|_T = f_T$. Condition (ii) says that u vanishes on N_1, and the result follows.

1.9. Faithfulness of the functor $\mathscr{R}$

Proposition 1.9.1.— *The functor $\mathscr{R}$ of 1.6 is faithful: if*

$$f, g : G \rightrightarrows G'$$

are two morphisms of pinned groups such that $\mathscr{R}(f) = \mathscr{R}(g)$, then $f = g$.

Indeed, f and g agree on T, on every U_α ($\alpha \in \Delta$), and on every $U_{-\alpha}$ ($\alpha \in \Delta$); it therefore suffices to apply:

Lemma 1.9.2.— *Let S be a scheme, let G be a pinned reductive S-group, and let H be an S-presheaf in groups separated for the fppf topology. Let*

$$f, g : G \rightrightarrows H$$

be two morphisms of S-groups. The following conditions are equivalent:

(i) $f = g$;

(ii) f and g agree on T, on every U_α ($\alpha \in \Delta$), and on every $U_{-\alpha}$ ($\alpha \in \Delta$);

(iii) f and g agree on T and on every U_α ($\alpha \in \Delta$), and $f(w_\alpha) = g(w_\alpha)$ for every $\alpha \in \Delta$.

Proof. The implication (i)⇒(ii) is trivial, while (ii)⇒(iii) follows immediately from the definition of w_α in 1.2. It remains to prove (iii)⇒(i). If $\alpha \in R$, there is a sequence $\{\alpha_i\} \subset \Delta$ such that

$$\alpha = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_n}(\alpha_{n+1}),$$

and hence

$$U_\alpha = \mathrm{int}(w_{\alpha_1} \cdots w_{\alpha_n}) U_{\alpha_{n+1}}.$$

Thus f and g agree on every U_α. They consequently agree on Ω, and therefore agree everywhere (Exposé XXII, 4.1.11).

Remark 1.9.3.— *If G is semisimple, the hypothesis that f and g agree on T may be omitted in (ii) and (iii). Indeed, G, as an fppf sheaf, is generated by the U_α, $\alpha \in R$ (Exposé XXII, 6.2.2(a)).*

2. Generators and relations for a pinned group

Throughout this section G is a fixed pinned S-group. For $\alpha, \beta \in \Delta$, we use the notation $Z_\alpha, Z_{\alpha\beta}, U_{\alpha\beta}, R^+_{\alpha\beta}$ of 1.7.

Theorem 2.1.— *Let S be a scheme, let G be a pinned S-group, and let H be an fppf S-sheaf in groups. Suppose given morphisms of groups*

$$f_N : \mathrm{Norm}_G(T) \longrightarrow H, \qquad f_\alpha : U_\alpha \longrightarrow H \quad (\alpha \in R).$$

There exists a necessarily unique morphism of groups

$$f : G \longrightarrow H$$

extending f_N and all the f_α, $\alpha \in R$, if and only if the following conditions hold:

(i) for every $\alpha \in \Delta$ and $\beta \in R$,

$$\mathrm{int}(f_N(w_\alpha)) \circ f_\beta = f_{s_\alpha(\beta)} \circ \mathrm{int}(w_\alpha);$$

(ii) for every $\alpha \in \Delta$, there exists a morphism of groups

$$F_\alpha : Z_\alpha \longrightarrow H$$

extending f_α, $f_{-\alpha}$, and $f_N|_{\mathrm{Norm}_{Z_\alpha}(T)}$;

(iii) for every $(\alpha, \beta) \in \Delta \times \Delta$, there exists a morphism of groups $U_{\alpha\beta} \to H$ inducing f_γ on U_γ for every $\gamma \in R^+_{\alpha\beta}$ (so $U_\gamma \subset U_{\alpha\beta}$).

2.1.1. Proof

The conditions are plainly necessary. Choose a totally ordered group structure on $\Gamma_0(R)$ for which the positive elements of R are precisely R^+ (Exposé XXI, 3.5.6); every product indexed by a subset of R will be taken in this order. Let f_T denote the restriction of f_N to T, and define a morphism of sets

$$f : \Omega \longrightarrow H$$

by

$$f\left(\prod_{\alpha \in R^-} y_\alpha \cdot t \cdot \prod_{\alpha \in R^+} x_\alpha\right) = \prod_{\alpha \in R^-} f_\alpha(y_\alpha) \cdot f_T(t) \cdot \prod_{\alpha \in R^+} f_\alpha(x_\alpha).$$

Every morphism satisfying the theorem must extend this map. Conversely, any morphism of groups $f : G \to H$ extending it also extends f_N: since it extends f_α and $f_{-\alpha}$, one has

$$f(w_\alpha) = F_\alpha(w_\alpha) = f_N(w_\alpha),$$

and it extends f_T by hypothesis. By Exposé XXII, 4.1.11(ii), it therefore remains to prove:

Proposition 2.1.2.— *The morphism $f : \Omega \to H$ defined above is* generically multiplicative*: for every $S' \to S$ and every $x, y \in \Omega(S')$ such that $xy \in \Omega(S')$, one has $f(xy) = f(x)f(y)$.*

Lemma 2.1.3.— *Let $n \in \mathrm{Norm}_G(T)(S')$, $S' \to S$, and let $\alpha, \beta \in R$ satisfy* $\mathrm{int}(n)U_\alpha = U_\beta$*, equivalently $n(\alpha) = \beta$. Then*

$$\mathrm{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \mathrm{int}(n).$$

It suffices to verify this formula for a system of generators of $\mathrm{Norm}_G(T)$. It holds for every w_α, $\alpha \in \Delta$, by 2.1(i), so it remains to treat $n \in T(S')$. This is immediate from 2.1(ii) when α is simple. Otherwise choose $w \in W$ with $w^{-1}(\alpha) \in \Delta$. Writing w as a product of simple reflections[73], one reduces to showing that if the formula holds for α and every n, it also holds for $w_{\alpha_0}(\alpha)$ and $t \in T(S')$, where $\alpha_0 \in \Delta$. By 2.1(i),

$$\begin{aligned}\mathrm{int}(f_N(t)) \circ f_{w_{\alpha_0}(\alpha)} &= \mathrm{int}(f_N(tw_{\alpha_0})) \circ f_\alpha \circ \mathrm{int}(w_{\alpha_0}^{-1})\\ &= f_{w_{\alpha_0}(\alpha)} \circ \mathrm{int}(tw_{\alpha_0}) \circ \mathrm{int}(w_{\alpha_0}^{-1}).\end{aligned}$$

Lemma 2.1.4.— *The restriction of f to U (respectively U^-) is a morphism of groups. In particular, this restriction is independent of the chosen ordering of the roots.*

It suffices to prove the assertion for U. By Exposé XXII, 5.5.8, it is enough to show that, for every pair $\alpha < \beta$ of positive roots and every $x_\alpha \in U_\alpha(S')$, $x_\beta \in U_\beta(S')$, $S' \to S$,

$$f_\beta(x_\beta) f_\alpha(x_\alpha) f_\beta(x_\beta^{-1}) = f(x_\beta x_\alpha x_\beta^{-1}).$$

By Exposé XXII, 5.5.2, there are elements $x_\gamma \in U_\gamma(S')$, where $\gamma = i\alpha + j\beta \in R$, $i > 0$, $j \geq 0$ [74], such that

$$x_\beta x_\alpha x_\beta^{-1} = \prod_\gamma x_\gamma.$$

It remains to verify

$$f_\beta(x_\beta) f_\alpha(x_\alpha) f_\beta(x_\beta^{-1}) = \prod_{\substack{\gamma = i\alpha + j\beta \\ i>0,\ j\geq 0}} f_\gamma(x_\gamma).$$

By Exposé XXI, 3.5.4, there is a $w \in W$ such that $w(\alpha) = \alpha_0 \in \Delta$ and $w(\beta) \in A_{\alpha_0\beta_0}$, in the notation of 1.7, for some $\beta_0 \in \Delta$. Lift w to $n \in \mathrm{Norm}_G(T)(S)$ by Exposé XXII, 3.8. It suffices to verify the relation after conjugation by $f_N(n)$; by 2.1.3, this reduces to $\alpha, \beta \in R^+_{\alpha_0\beta_0}$, where condition 2.1(iii) applies.

Lemma 2.1.5.— *Let $n \in \mathrm{Norm}_G(T)(S)$ satisfy $\mathrm{int}(n)U = U^-$, so that n represents the symmetry of the Weyl group*[75] *(Exposé XXI, 3.6.14). For every $u \in U(S')$, $S' \to S$ (respectively $u \in U^-(S')$), one has*

$$f(nun^{-1}) = f_N(n) f(u) f_N(n^{-1}).$$

This follows immediately from 2.1.3 and 2.1.4.

Lemma 2.1.6.— *Let $u \in B(S')$, $v \in B^-(S')$, and $g \in \Omega(S')$, where $S' \to S$. Then*

$$f(vgu) = f(v)f(g)f(u).$$

Write

$$v = v_1 t_1, \qquad g = v_2 t_2 u_2, \qquad u = t_3 u_3,$$

with $v_i \in U^-(S')$, $t_i \in T(S')$, and $u_i \in U(S')$. On the one hand,

$$f(v)f(g)f(u) = f(v_1) f_T(t_1) f(v_2) f_T(t_2) f(u_2) f_T(t_3) f(u_3).$$

[73] *Editors' note.* "Fundamental symmetries" was replaced by "simple reflections."

[74] *Editors' note.* The printed $i, j > 0$ was corrected to $i > 0, j \geq 0$, since x_α occurs in the product on the right.

[75] *Editors' note.* Relative to Δ.

On the other hand,

$$\begin{aligned} f(vgu) &= f(v_1 t_1 v_2 t_1^{-1}\, t_1 t_2 t_3\, t_3^{-1} u_2 t_3 u_3) \\ &= f(v_1 \cdot t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3 \cdot u_3). \end{aligned}$$

Using 2.1.4 on the two outer factors gives

$$f(vgu) = f(v_1) f(t_1 v_2 t_1^{-1}) f_T(t_1 t_2 t_3) f(t_3^{-1} u_2 t_3) f(u_3).$$

It remains only to use

$$\begin{aligned} f(t_1 v_2 t_1^{-1}) &= f_T(t_1) f(v_2) f_T(t_1)^{-1}, \\ f(t_3^{-1} u_2 t_3) &= f_T(t_3)^{-1} f(u_2) f_T(t_3), \end{aligned}$$

which follow from the definition of f and 2.1.3.

Lemma 2.1.7.— *Let $\alpha \in \Delta$ and*

$$g \in \Omega(S') \cap \mathrm{int}(w_\alpha)^{-1}\Omega(S'), \qquad S' \to S.$$

Then

$$f(w_\alpha g w_\alpha^{-1}) = f_N(w_\alpha) f(g) f_N(w_\alpha)^{-1}.$$

By Exposé XXII, 5.6.8, write

$$g = a x_{-\alpha} t x_\alpha b,$$

where $a \in U^-_{-\widehat{\alpha}}(S')$, $x_{-\alpha} \in U_{-\alpha}(S')$, $t \in T(S')$, $x_\alpha \in U_\alpha(S')$, and $b \in U_{\widehat{\alpha}}(S')$. By 2.1.3 and 2.1.4, and because s_α permutes the positive roots other than α (Exposé XXI, 3.3.2),

$$\mathrm{int}(w_\alpha) a \in U^-_{-\widehat{\alpha}}(S'), \qquad \mathrm{int}(w_\alpha) b \in U_{\widehat{\alpha}}(S').$$

The desired formula holds for $g = a$ and $g = b$. By 2.1.6 it therefore suffices to treat $g = x_{-\alpha} t x_\alpha \in Z_\alpha(S')$; in that case everything takes place in Z_α, and condition 2.1(ii) applies.

Lemma 2.1.8.— *Let $n \in \mathrm{Norm}_G(T)(S)$. For every $S' \to S$ and every $g \in \Omega(S')$ such that $\mathrm{int}(n)g \in \Omega(S')$,*

$$f(ngn^{-1}) = f_N(n) f(g) f_N(n)^{-1}.$$

This is immediate for $n \in T(S)$, by 2.1.3. The two sides define morphisms from $\Omega \cap \mathrm{int}(n)^{-1}\Omega$ to H. It is enough to find an open $V_n \subset \Omega$, containing the unit section, such that $\mathrm{int}(n)V_n \subset \Omega$, and on which $f \circ \mathrm{int}(n)$ and $\mathrm{int}(f_N(n)) \circ f$ agree. By the structure of $\mathrm{Norm}_G(T)$, it suffices to show that if this is true for n', then it is true for $n = n'w_\alpha$, $\alpha \in \Delta$. Put

$$V_n = \Omega \cap \mathrm{int}(w_\alpha)^{-1} V_{n'}.$$

Then $\mathrm{int}(n)V_n \subset \mathrm{int}(n')V_{n'} \subset \Omega$. If $g \in V_n(S')$, then

$$\mathrm{int}(n)g = \mathrm{int}(n')\,\mathrm{int}(w_\alpha) g.$$

Now $\mathrm{int}(w_\alpha)g \in V_{n'}(S')$, so the induction hypothesis gives

$$f(\mathrm{int}(n')\,\mathrm{int}(w_\alpha)g) = \mathrm{int}(f_N(n')) f(\mathrm{int}(w_\alpha)g).$$

Lemma 2.1.7 gives

$$f(\mathrm{int}(w_\alpha)g) = \mathrm{int}(f_N(w_\alpha)) f(g),$$

and the conclusion follows.

We now prove Proposition 2.1.2. Let $x, x' \in \Omega(S')$, and write

$$x = vtu, \qquad x' = v't'u',$$

so $xx' = vt(uv')t'u'$. By 2.1.6 and $U^-(S')\Omega(S')U(S') = \Omega(S')$, it suffices to show that $f(uv') = f(u)f(v')$ whenever $uv' \in \Omega(S')$. Choose $n \in \mathrm{Norm}_G(T)(S')$ as in 2.1.5. Then

$$f(u) = f_N(n)^{-1} f(nun^{-1}) f_N(n),$$
$$f(v') = f_N(n)^{-1} f(nv'n^{-1}) f_N(n),$$

and hence

$$f(u)f(v') = f_N(n)^{-1} f(nun^{-1}) f(nv'n^{-1}) f_N(n).$$

Since $nun^{-1} \in U^-(S')$ and $nv'n^{-1} \in U(S')$,

$$f(nun^{-1})f(nv'n^{-1}) = f\big((nun^{-1})(nv'n^{-1})\big) = f(nuv'n^{-1}).$$

Thus

$$f(u)f(v') = f_N(n)^{-1} f(nuv'n^{-1}) f_N(n).$$

If $uv' \in \Omega(S')$, Lemma 2.1.8 finishes the proof.

Remark 2.2.— *Instead of prescribing f_N, one may prescribe a morphism $f_T : T \to H$ and sections $h_\alpha \in H(S)$, $\alpha \in \Delta$, satisfying the conditions of 1.8.2. Condition (ii) of Theorem 2.1 must then be replaced by:*

> (ii′) There exists a morphism of groups $F_\alpha : Z_\alpha \to H$ extending $f_\alpha, f_{-\alpha}, f_T$ and satisfying $F_\alpha(w_\alpha) = h_\alpha$.

We now give the preceding theorem a more explicit form, retaining its notation.

Theorem 2.3.— *Let S be a scheme, let G be a pinned S-group, and let H be an fppf S-sheaf in groups. Suppose given morphisms of groups*

$$f_T : T \longrightarrow H, \qquad f_\alpha : U_\alpha \longrightarrow H \quad (\alpha \in \Delta),$$

and sections $h_\alpha \in H(S)$, $\alpha \in \Delta$. There exists a necessarily unique morphism of groups $f : G \to H$ inducing f_T and the f_α and satisfying $f(w_\alpha) = h_\alpha$ for all $\alpha \in \Delta$ if and only if:

(i) for every $S' \to S$, $t \in T(S')$, $\alpha \in \Delta$, and $x \in U_\alpha(S')$,

$$f_T(t) f_\alpha(x) f_T(t)^{-1} = f_\alpha(\mathrm{int}(t)x); \tag{1}$$

(ii) for every $\alpha \in \Delta$, $S' \to S$, and $t \in T(S')$,

$$h_\alpha f_T(t) h_\alpha^{-1} = f_T(s_\alpha(t)) = f_T\big(t\alpha^*(\alpha(t)^{-1})\big); \tag{2}$$

(iii) for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$(h_\alpha h_\beta)^{n_{\alpha\beta}} = f_T(t_{\alpha\beta}); \tag{3}$$

(iv) for every $\alpha \in \Delta$, recalling that $u_\alpha = \exp_\alpha(X_\alpha)$,

$$(h_\alpha f_\alpha(u_\alpha))^3 = e; \tag{4}$$

(v) for every distinct $\alpha, \beta \in \Delta$, there exists a morphism $f_{\alpha\beta} : U_{\alpha\beta} \to H$ such that

$$f_{\alpha\beta}|_{U_\alpha} = f_\alpha, \qquad f_{\alpha\beta}|_{U_\beta} = f_\beta, \tag{5}$$

and, for every $\gamma \in R^+_{\alpha\beta}$ with $\gamma \neq \alpha$ (respectively $\gamma \neq \beta$) and every $x \in U_\gamma(S')$, $S' \to S$,

$$\mathrm{int}(h_\alpha) f_{\alpha\beta}(x) = f_{\alpha\beta}(\mathrm{int}(w_\alpha)x) \tag{6}$$

(respectively $\mathrm{int}(h_\beta) f_{\alpha\beta}(x) = f_{\alpha\beta}(\mathrm{int}(w_\beta)x)$*).*

Proof. Uniqueness follows from 1.9.2. We prove existence.

Lemma 2.3.1.— *There exists a morphism*

$$f_N : \mathrm{Norm}_G(T) \longrightarrow H$$

extending f_T and satisfying $f_N(w_\alpha) = h_\alpha$.

This is precisely 1.8.2, in view of conditions (2) and (3).

Lemma 2.3.2.— *For every $\alpha \in \Delta$, there is a morphism of groups $F_\alpha : Z_\alpha \to H$ extending f_T and f_α, satisfying $F_\alpha(w_\alpha) = h_\alpha$, and hence extending $f_N|_{\mathrm{Norm}_{Z_\alpha}(T)}$.*

This follows from Exposé XX, 6.2 and conditions (1), (3), and (4).

Lemma 2.3.3.— *Let $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$. Let $n \in \mathrm{Norm}_{Z_{\alpha\beta}}(T)(S)$ satisfy $n(\alpha) = \alpha$ (respectively $n(\alpha) = \beta$), equivalently $\mathrm{int}(n)U_\alpha = U_\alpha$ (respectively U_β). Then, for every $S' \to S$ and $x \in U_\alpha(S')$,*

$$\mathrm{int}(f_N(n)) f_\alpha(x) = f_\alpha(\mathrm{int}(n)x)$$

(respectively the right-hand side is $f_\beta(\mathrm{int}(n)x)$).

There exist $t \in T(S)$ and a sequence $\{\alpha_i\} \subset \{\alpha, \beta\}$ such that

$$n = t w_{\alpha_k} \cdots w_{\alpha_1}.$$

Condition (1) treats $n = t$, so we may omit t. Proceed by induction on k, and set

$$\gamma_i = s_{\alpha_i} \cdots s_{\alpha_1}(\alpha).$$

If every γ_i is positive, condition (v) yields

$$\mathrm{int}\big(f_N(w_{\alpha_i} \cdots w_{\alpha_1})\big) f_\alpha(x) = f_{\alpha\beta}\big(\mathrm{int}(w_{\alpha_i} \cdots w_{\alpha_1})x\big),$$

and $i = k$ gives the result. Otherwise there is a $j < k$ such that $\gamma_j \in R^+$ but $\gamma_{j+1} \notin R^+$. Since $\gamma_{j+1} = s_{\alpha_j}(\gamma_j)$, Exposé XXI, 3.3.2 gives $\gamma_j = \alpha_j$, hence γ_j is α or β. Decompose

$$n = n'n'', \qquad n' = w_{\alpha_k} \cdots w_{\alpha_{j+1}}, \qquad n'' = w_{\alpha_j} \cdots w_{\alpha_1}.$$

Both n' and n'' satisfy the lemma's hypotheses and involve fewer than k reflections, so induction applies.

Lemma 2.3.4.— *Let $\alpha \in R$. Suppose $n, n' \in \mathrm{Norm}_G(T)(S)$ and $\beta, \beta' \in \Delta$ satisfy $n(\alpha) = \beta$ and $n'(\alpha) = \beta'$. Then, for every $x \in U_\alpha(S')$, $S' \to S$,*

$$\mathrm{int}(f_N(n)^{-1}) f_\beta(nxn^{-1}) = \mathrm{int}(f_N(n')^{-1}) f_{\beta'}(n'xn'^{-1}).$$

It suffices to prove that if $n(\alpha) = \beta$, with $\alpha, \beta \in \Delta$, then

$$\operatorname{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \operatorname{int}(n).$$

By Tits's lemma (Exposé XXI, 5.6), there are simple roots

$$\alpha_0 = \alpha, \alpha_1, \ldots, \alpha_m = \beta$$

and elements $w_i \in W$, $0 \leq i < m$, such that

$$n = w_{m-1} \cdots w_0, \qquad w_i(\alpha_i) = \alpha_{i+1},$$

and for each i there is a simple root β_i such that

$$w_i \in W_{\alpha_i \beta_i}, \qquad \alpha_{i+1} = \alpha_i \text{ or } \beta_i.$$

Induction reduces the assertion to Lemma 2.3.3.

Lemma 2.3.5.— *There is a family of morphisms of groups $f'_\gamma : U_\gamma \to H$, $\gamma \in R$, such that:*

(i) for $\alpha \in \Delta$, $f'_\alpha = f_\alpha$ and $f'_{-\alpha} = F_\alpha|_{U_{-\alpha}}$;

(ii) if $\alpha, \beta \in \Delta$ and $\gamma \in R^+_{\alpha\beta}$, then $f'_\gamma = f_{\alpha\beta}|_{U_\gamma}$;

(iii) if $n \in \operatorname{Norm}_G(T)(S)$ and $\alpha, \beta \in R$ satisfy $n(\alpha) = \beta$, then

$$\operatorname{int}(f_N(n)) f'_\alpha(x) = f'_\beta(\operatorname{int}(n)x)$$

for every $x \in U_\alpha(S')$, $S' \to S$.

For every root α, choose $n \in \operatorname{Norm}_G(T)(S)$ with $n(\alpha) \in \Delta$, and define $f'_\alpha(x)$ by the expression in Lemma 2.3.4. This is independent of n, and f'_α is a morphism of groups. Property (iii) holds by construction. The first part of (i) follows by taking $n = e$, and the second by taking $n = w_\alpha$ and using 2.3.2. If $\gamma \in R^+_{\alpha\beta}$, choose $n \in \operatorname{Norm}_{Z_{\alpha\beta}}(T)(S)$ with $n(\gamma) = \alpha$ or β; then (iii), together with conditions (5) and (6), proves (ii).

2.3.6.— We finish the proof by verifying the conditions of Theorem 2.1 for f_N and f'_α, $\alpha \in R$:

- 2.1(i) follows from 2.3.5(iii);
- 2.1(ii) follows from 2.3.5(i) and 2.3.2;
- 2.1(iii) follows from 2.3.5(ii) and the fact that $f_{\alpha\beta}$ is a morphism of groups.

An immediate corollary is:

Corollary 2.4.— *Let S be a scheme, let G be a pinned S-group of semisimple rank at least 1, and let H be an fppf S-sheaf in groups. For each $(\alpha, \beta) \in \Delta \times \Delta$, let*

$$F_{\alpha\beta} : Z_{\alpha\beta} \longrightarrow H$$

be a morphism of groups. There exists a morphism $f : G \to H$ inducing the $F_{\alpha\beta}$ if and only if, for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$F_{\alpha\beta}|_{Z_\alpha} = F_{\alpha\alpha}.$$

Necessity is clear. Conversely, put

$$f_T = F_{\alpha\alpha}|_T;$$

this is independent of α, since

$$F_{\alpha\alpha}|_T = F_{\alpha\beta}|_T = F_{\beta\beta}|_T.$$

Set

$$f_\alpha = F_{\alpha\alpha}|_{U_\alpha}, \qquad h_\alpha = F_{\alpha\alpha}(w_\alpha), \qquad f_{\alpha\beta} = F_{\alpha\beta}|_{U_{\alpha\beta}}.$$

The conditions of Theorem 2.3 hold: (1), (2), and (4) hold in Z_α, while (3), (5), and (6) hold in $Z_{\alpha\beta}$. Thus there is a morphism $f : G \to H$ extending f_T and all f_α, and satisfying $f(w_\alpha) = h_\alpha$. It agrees with $F_{\alpha\beta}$ on the generators of $Z_{\alpha\beta}$; hence $f|_{Z_{\alpha\beta}} = F_{\alpha\beta}$.

Corollary 2.5.— *Let S be a scheme; let G and G' be two pinned S-groups of semisimple rank 2; let $q > 0$ be an integer such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$; and let*

$$h : R(G') \longrightarrow R(G)$$

be a q-morphism of pinned root data. For each $\alpha \in R^+$, choose

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \qquad X'_{d(\alpha)} \in \Gamma(S, \mathfrak{g}'^{\,d(\alpha)})^\times,$$

extending the canonical choices for $\alpha \in \Delta$. Suppose:

(i) if $\Delta = \{\alpha, \beta\}$, then

$$D_S(h)(t_{\alpha\beta}) = t'_{d(\alpha)d(\beta)};$$

(ii) for every $\alpha \in \Delta$ and every $\beta \in R^+$, $\beta \neq \alpha$, if $z \in \mathbf{G}_m(S)$ is defined by

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{s_\alpha(\beta)},$$

then

$$\mathrm{Ad}(w'_{d(\alpha)})X'_{d(\beta)} = z^{q(\beta)}X'_{d(s_\alpha(\beta))};$$

(iii) there is a morphism of groups $f : U \to U'$ such that, for every $\alpha \in R^+$, every $S' \to S$, and every $x \in \mathbf{G}_a(S')$,

$$f(\exp(xX_\alpha)) = \exp\bigl(x^{q(\alpha)}X'_{d(\alpha)}\bigr).$$

Then there is a morphism of pinned groups $g : G \to G'$ such that $R(g) = h$.

Indeed, define $f_\alpha : U_\alpha \to G'$ by

$$f_\alpha(\exp(xX_\alpha)) = \exp\bigl(x^{q(\alpha)}X'_{d(\alpha)}\bigr),$$

and put

$$f_T = D_S(h), \qquad h_\alpha = w'_{d(\alpha)}.$$

The conditions of Theorem 2.3 hold: note that

$$q(s_\alpha(\beta)) = q(\beta)$$

by Exposé XXI, 6.8.4, and that

$$D_S(h)(t_\alpha) = t'_{d(\alpha)}$$

always holds. The assertion follows at once.

Remark 2.6.— *Condition (v) of Theorem 2.3 may be made more explicit as follows. One must first verify:*

(a) *for every word in w_α, w_β whose corresponding word carries α or β to α or β, the corresponding relation of the type occurring in 2.3.3 holds.*

The proof of 2.3.3 shows that it suffices to check the words that are minimal, in the sense that no nontrivial initial subword satisfies the imposed conditions.

If (a) holds, define $f_\gamma : U_\gamma \to H$ for every $\gamma \in R^+_{\alpha\beta}$ as in 2.3.5. One must then require:

(b) *the map defined by the f_γ,*

$$U_{\alpha\beta} = \prod_{\gamma\in R^+_{\alpha\beta}} U_\gamma \longrightarrow H,$$

is a morphism of groups.

By Exposé XXII, 5.5.8, condition (b) follows from:

(b′) *the preceding map respects the commutation relations between U_γ and U_δ, for $\gamma, \delta \in R^+_{\alpha\beta}$, $\gamma > \delta$; namely, the relations involving $C_{i,j,\gamma,\delta}$ in Exposé XXII, 5.5.2.*

Conversely, the conjunction of (a) and (b′) is equivalent to condition (v).

These conditions may even be reduced to conditions involving only

$$h_\alpha, \quad h_\beta, \quad f_\alpha(u_\alpha), \quad f_\beta(u_\beta)$$

in H. For example, if $s_\alpha s_\beta s_\alpha(\beta) = \alpha$, a condition of type (a) reads

$$\mathrm{int}(h_\alpha h_\beta h_\alpha) f_\beta(x) = f_\alpha(\mathrm{int}(w_\alpha w_\beta w_\alpha)x)\,. \tag{1}$$

Here $x \in U_\beta(S')$, $S' \to S$. In particular, for $x = u_\beta$,

$$\mathrm{int}(w_\alpha w_\beta w_\alpha) u_\beta = u_\alpha^z$$

for a certain $z \in \mathbf{G}_m(S)$, and

$$\mathrm{int}(h_\alpha h_\beta h_\alpha) f_\beta(u_\beta) = f_\alpha(u_\alpha)^z. \tag{1′}$$

Conversely, taking conditions (i)–(iv) of 2.3 into account, (1′) implies (1). If $t \in T(S')$, conjugating (1′) by t, and using (i) and (ii), gives (1) for

$$x = \mathrm{int}(t) u_\beta = u_\beta^{\beta(t)}.$$

The morphism $\beta : T \to \mathbf{G}_{m,S}$ is faithfully flat, so (1) holds for $x \in U_\beta(S')^\times$. It is additive in x, and every section of U_β is locally the sum of two invertible sections. Thus

$$(1’) + (\mathrm{i}) + (\mathrm{ii}) \quad \Longrightarrow \quad (1).$$

The same argument applies to conditions of type (b). One uses the fact that, for two distinct positive roots γ, γ', hence linearly independent over $\mathbf{Q}$, the morphism

$$T \longrightarrow \mathbf{G}^2_{m,S}$$

with components γ, γ' is faithfully flat. We leave the details of this transposition to the reader.

3. Groups of Semisimple Rank 2

3.1. Generalities

Lemma 3.1.1.— *Let S be a scheme, let G be a pinned S-group, and let α, β be two roots of G such that $\alpha+\beta \neq 0$.*

(i) If $\alpha+\beta \notin R$, then

$$\exp(X_\alpha)\exp(X_\beta) = \exp(X_\beta)\exp(X_\alpha)$$

for every $S' \to S$, $X_\alpha \in W(\mathfrak{g}^\alpha)(S')$, and $X_\beta \in W(\mathfrak{g}^\beta)(S')$.

(ii) If neither $\alpha+\beta$ nor $\alpha-\beta$ is a root, then

$$w_\alpha(z_\alpha)\exp(X_\beta)w_\alpha(z_\alpha)^{-1} = \exp(X_\beta)$$

for every $X_\beta \in W(\mathfrak{g}^\beta)(S')$ and $z_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$, and

$$w_\alpha(z_\alpha)w_\beta(z_\beta) = w_\beta(z_\beta)w_\alpha(z_\alpha)$$

for every $z_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$ and $z_\beta \in W(\mathfrak{g}^\beta)^\times(S')$.

(iii) Let $X_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$, $X_\beta \in W(\mathfrak{g}^\beta)^\times(S')$, and $w \in \mathrm{Norm}_G(T)(S')$ satisfy $w(\alpha)=\beta$. Define $z \in \mathbf{G}_m(S')$ by

$$\mathrm{Ad}(w)X_\alpha = zX_\beta.$$

Then

$$\begin{aligned}\mathrm{int}(w)\exp(xX_\alpha) &= \exp(xzX_\beta),\\ \mathrm{int}(w)\exp(yX_\alpha^{-1}) &= \exp(yz^{-1}X_\beta^{-1}),\\ \mathrm{int}(w)w_\alpha(X_\alpha) &= \beta^*(z)w_\beta(X_\beta).\end{aligned}$$

In particular, if $z=\pm 1$, then

$$\mathrm{int}(w)w_\alpha(X_\alpha) = w_\beta(X_\beta)^z.$$

(iv) If $t_\alpha = \alpha^*(-1)$ and $t_\beta = \beta^*(-1)$, then

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{(\beta^*,\alpha)}, \qquad \beta(t_\alpha) = (-1)^{(\alpha^*,\beta)}.$$

Proof. Part (i) follows immediately from Exposé XXII, 5.5.2. Part (ii) follows from Exposé XX, 3.1 and from (i) applied to

$$(\beta,\alpha), \quad (\beta,-\alpha), \quad (-\beta,\alpha), \quad (-\beta,-\alpha).$$

Part (iii) is immediate from the definitions; for its last assertion, observe that

$$\beta^*(-1) = w_\beta(X_\beta)^{-2}.$$

Finally, (iv) is tautological.

Proposition 3.1.2 (Groups of type $A_1 \times A_1$).— *Let S be a scheme and G a pinned S-group of type $A_1 \times A_1$. Write $\Delta = R^+ = \{\alpha,\beta\}$.*

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^2 = t_\alpha t_\beta = (w_\beta w_\alpha)^2 = t_{\beta\alpha}.$$

(ii)

$$\mathrm{Ad}(w_\alpha)X_\beta = X_\beta, \qquad \mathrm{Ad}(w_\beta)X_\alpha = X_\alpha.$$

(iii) The groups U_α and U_β commute; in other words, U is commutative.

Proof. Lemma 3.1.1(ii) gives $w_\alpha w_\beta = w_\beta w_\alpha$, whence

$$(w_\alpha w_\beta)^2 = w_\alpha^2 w_\beta^2 = t_\alpha t_\beta,$$

which proves (i). The same lemma gives (ii), and (iii) is Lemma 3.1.1(i).

3.1.3.— Condition (v) of Theorem 2.3 takes the following explicit form. Put

$$v_\alpha = f_\alpha(u_\alpha) \qquad (\alpha \in \Delta).$$

Then the two groups of conditions are

$$\text{(A)} \quad \begin{cases} h_\alpha v_\beta h_\alpha^{-1} = v_\beta, \\ h_\beta v_\alpha h_\beta^{-1} = v_\alpha, \end{cases} \qquad \text{(B)} \quad v_\alpha v_\beta = v_\beta v_\alpha.$$

3.2. Groups of Type A_2

Proposition 3.2.1.— *Let S be a scheme and G a pinned S-group of type A_2. Write*

$$\Delta = \{\alpha, \beta\}, \qquad R^+ = \{\alpha, \beta, \alpha+\beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^3 = e = (w_\beta w_\alpha)^3 = t_{\beta\alpha}.$$

(ii) Put $\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$. Then

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_\beta &= -X_{\alpha+\beta}, \\ \mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_\beta, \\ \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha. \end{aligned}$$

(iii) Put

$$p_{\alpha+\beta}(x) = \exp(xX_{\alpha+\beta}) = \mathrm{int}(w_\beta)p_\alpha(x).$$

Then

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy).$$

3.2.2.— The proof occupies 3.2.2–3.2.7. First,

$$(\beta^*, \alpha) = (\alpha^*, \beta) = -1,$$

so $\alpha(t_\beta) = \beta(t_\alpha) = -1$. Put

$$\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}.$$

Then

$$\mathrm{Ad}(w_\beta)X_{\alpha+\beta} = \alpha(t_\beta)X_\alpha = -X_\alpha.$$

Write

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{\alpha+\beta}, \qquad z \in \mathbf{G}_m(S).$$

It follows that

$$\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} = \beta(t_\alpha)z^{-1}X_\beta = -z^{-1}X_\beta.$$

By Exposé XXII, 5.5.2, there is a unique $A \in \mathbf{G}_a(S)$ such that

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy). \tag{$+$}$$

Thus (ii) and (iii) reduce to proving $A = -z = 1$.

3.2.3.— Conjugating $(+)$ by w_β gives

$$p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy). \tag{$++$}$$

3.2.4.— By definition,

$$w_\beta p_\alpha(x)w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Expanding w_β, this is

$$\begin{aligned} p_\beta(1)p_{-\beta}(-1)p_\beta(1)p_\alpha(x)p_\beta(-1)p_{-\beta}(1)p_\beta(-1)& \\ = p_{\alpha+\beta}(x).& \end{aligned}$$

Since p_β commutes with $p_{\alpha+\beta}$, this may be written

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using $(+)$ on the left and $(++)$ on the right gives

$$\begin{aligned} p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_\beta(-1) = p_{\alpha+\beta}(x)p_{-\beta}(1)& \\ \cdot\, p_\alpha(Ax)p_{-\beta}(-1).& \end{aligned}$$

Because $\alpha+2\beta$, $\alpha-\beta$, and $2\alpha+\beta$ are not roots, this reduces to

$$p_\alpha(x)p_{\alpha+\beta}(Ax) = p_\alpha(Ax)p_{\alpha+\beta}(x).$$

Exposé XXII, 4.1.3 now gives $A = 1$.

3.2.5.— Conjugating $(+)$ by w_α, and using $A = 1$, gives

$$p_{\alpha+\beta}(zy)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(zy)p_\beta(-z^{-1}xy). \tag{$+++$}$$

3.2.6.— As before,

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{\alpha+\beta}(zy).$$

Since p_α and $p_{\alpha+\beta}$ commute,

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{\alpha+\beta}(zy)p_{-\alpha}(-1).$$

Using $(+)$ and $(+++)$, this becomes

$$p_\beta(y)p_{\alpha+\beta}(-y) = p_{\alpha+\beta}(zy)p_\beta(-z^{-1}y).$$

The two factors commute, and hence $z = -1$.

3.2.7.— It remains to prove (i). One has

$$w_\alpha w_\beta w_\alpha = w_\alpha w_\beta w_\alpha^{-1}w_\alpha^2 = w_{\alpha+\beta}^{-1}t_\alpha,$$

and therefore

$$\begin{aligned} w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{\alpha+\beta}^{-1} t_\alpha w_\beta \\ &= w_\beta w_{\alpha+\beta}^{-1} w_\beta^{-1}\, s_\beta(t_\alpha) t_\beta \\ &= w_\alpha t_\alpha t_\beta t_\beta = w_\alpha t_\alpha = w_\alpha^{-1}. \end{aligned}$$

Thus

$$(w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 = e.$$

Remark 3.2.8.— *Put $v_\alpha = f_\alpha(u_\alpha)$. Condition (v) of 2.3 becomes*

$$\text{(A)} \quad \begin{cases} \operatorname{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \operatorname{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$

and

$$\text{(B)} \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta\, (h_\beta v_\alpha h_\beta^{-1}), \\ v_\beta (h_\beta v_\alpha h_\beta^{-1}) = (h_\beta v_\alpha h_\beta^{-1}) v_\beta, \\ v_\alpha (h_\beta v_\alpha h_\beta^{-1}) = (h_\beta v_\alpha h_\beta^{-1}) v_\alpha. \end{cases}$$

Writing $v_{\alpha+\beta} = \operatorname{int}(h_\beta) v_\alpha$, the latter conditions are equivalently

$$\text{(B)} \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta}, \\ v_\alpha v_{\alpha+\beta} = v_{\alpha+\beta} v_\alpha, \\ v_\beta v_{\alpha+\beta} = v_{\alpha+\beta} v_\beta. \end{cases}$$

3.3. Groups of Type B_2

Proposition 3.3.1.— *Let S be a scheme and G a pinned S-group of type B_2. Write*

$$\Delta = \{\alpha, \beta\}, \qquad R^+ = \{\alpha, \beta, \alpha+\beta, 2\alpha+\beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^4 = t_\alpha = (w_\beta w_\alpha)^4 = t_{\beta\alpha}.$$

(ii) If

$$\operatorname{Ad}(w_\beta) X_\alpha = X_{\alpha+\beta}, \qquad \operatorname{Ad}(w_\alpha) X_\beta = X_{2\alpha+\beta},$$

then

$$\begin{aligned} &\operatorname{Ad}(w_\alpha) X_{\alpha+\beta} = -X_{\alpha+\beta}, && \operatorname{Ad}(w_\alpha) X_{2\alpha+\beta} = X_\beta, \\ &\operatorname{Ad}(w_\beta) X_{\alpha+\beta} = -X_\alpha, && \operatorname{Ad}(w_\beta) X_{2\alpha+\beta} = X_{2\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(x X_{\alpha+\beta}) = \operatorname{int}(w_\beta) p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(x X_{2\alpha+\beta}) = \operatorname{int}(w_\alpha) p_\beta(x). \end{aligned}$$

Then

$$\begin{aligned} p_\beta(y) p_\alpha(x) &= p_\alpha(x) p_\beta(y) p_{\alpha+\beta}(xy) p_{2\alpha+\beta}(x^2 y), \\ p_{\alpha+\beta}(y) p_\alpha(x) &= p_\alpha(x) p_{\alpha+\beta}(y) p_{2\alpha+\beta}(2xy). \end{aligned}$$

3.3.2.— The proof occupies 3.3.2–3.3.6. We have

$$(\beta^*, \alpha) = -1, \qquad (\alpha^*, \beta) = -2,$$

so $\alpha(t_\beta) = -1$ and $\beta(t_\alpha) = 1$. In particular, $\beta(t_\alpha) = \alpha(t_\alpha) = 1$, so t_α is a section of $\mathrm{Centr}(G)$. Put

$$\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \qquad \mathrm{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta}.$$

Then

$$\begin{aligned} \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= \beta(t_\alpha)X_\beta = X_\beta, \\ \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}. \end{aligned}$$

For some $k \in \mathbf{G}_m(S)$,

$$\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} = kX_{\alpha+\beta}.$$

By Exposé XXII, 5.5.2, there are $A, B, C \in \mathbf{G}_a(S)$ such that

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y), \tag{1}$$

$$p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Cxy). \tag{2}$$

It remains to prove

$$A = B = 1, \qquad C = 2, \qquad k = -1.$$

3.3.3.— Conjugating (1) by w_α gives

$$\begin{aligned} p_{2\alpha+\beta}(y)p_{-\alpha}(-x) = p_{-\alpha}(-x)&p_{2\alpha+\beta}(y) \\ &\cdot p_{\alpha+\beta}(Akxy)p_\beta(Bx^2y). \end{aligned} \tag{3}$$

Transforming (2) in the same way gives

$$p_{\alpha+\beta}(ky)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(ky)p_\beta(Cxy). \tag{4}$$

Conjugating (1) by w_β gives

$$\begin{aligned} p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)&p_{-\beta}(-y) \\ &\cdot p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y). \end{aligned} \tag{5}$$

3.3.4.— Write

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Because $\alpha + 2\beta$ is not a root,

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) on the left and (5) on the right yields

$$\begin{aligned} &p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_\beta(-1) \\ &\quad = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{-\beta}(-1). \end{aligned}$$

After commuting the indicated factors and applying (2), this becomes

$$\begin{aligned} p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) = p_\alpha(Ax)&p_{\alpha+\beta}(x) \\ &\cdot p_{2\alpha+\beta}((AC-B)x^2). \end{aligned}$$

Hence
$$A = 1, \qquad C = 2B.$$

3.3.5.— Now write
$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y).$$
Since $3\alpha + \beta$ is not a root,
$$p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{2\alpha+\beta}(y) p_{-\alpha}(-1).$$
Using (1) on the left and (3) on the right gives
$$p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) = p_{2\alpha+\beta}(y) p_{\alpha+\beta}(Aky) p_\beta(By).$$
The three groups commute, so
$$B = 1, \qquad -A = Ak.$$
Consequently
$$A = B = 1, \qquad C = 2, \qquad k = -1.$$

3.3.6.— We have proved (ii) and (iii). Since
$$s_\beta(t_\alpha) = t_\alpha t_\beta^{(\alpha^*,\beta)} = t_\alpha,$$
one obtains successively
$$\begin{aligned} w_\alpha w_\beta w_\alpha &= w_{2\alpha+\beta} t_\alpha, \\ w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_{2\alpha+\beta} t_\alpha t_\beta, \\ w_\alpha w_\beta w_\alpha w_\beta w_\alpha w_\beta w_\alpha &= w_\beta^{-1} t_\alpha. \end{aligned}$$
[76] It follows that
$$(w_\beta w_\alpha)^4 = t_\alpha, \qquad (w_\alpha w_\beta)^4 = s_\beta(t_\alpha) = t_\alpha.$$

Remark 3.3.7.— *Put*
$$v_{\alpha+\beta} = \operatorname{int}(h_\beta) v_\alpha, \qquad v_{2\alpha+\beta} = \operatorname{int}(h_\alpha) v_\beta.$$
Condition (v) of 2.3 becomes
$$(A) \quad \begin{cases} \operatorname{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \operatorname{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$
and
$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta}, \\ v_{\alpha+\beta} v_\alpha = v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2, \\ v_{\alpha+\beta} v_\beta = v_\beta v_{\alpha+\beta}, \\ v_{2\alpha+\beta} v_\alpha = v_\alpha v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_\beta = v_\beta v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_{\alpha+\beta} = v_{\alpha+\beta} v_{2\alpha+\beta}. \end{cases}$$

[76] *Editors' note.* The French prints $(\alpha^*, \beta) = 2$; the convention used immediately above gives -2. The conclusion is unchanged because $t_\beta^2 = e$.

3.4. Groups of Type G_2

Proposition 3.4.1.— *Let S be a scheme and G a pinned S-group of type G_2. Write*

$$\Delta = \{\alpha, \beta\}, \qquad R^+ = \{\alpha, \beta, \alpha+\beta, 2\alpha+\beta, 3\alpha+\beta, 3\alpha+2\beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^6 = e = (w_\beta w_\alpha)^6 = t_{\beta\alpha}.$$

(ii) If

$$\begin{aligned} \mathrm{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, \\ \mathrm{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \mathrm{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \end{aligned}$$

then

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \mathrm{Ad}(w_\alpha)X_{3\alpha+\beta} &= X_\beta, \\ \mathrm{Ad}(w_\alpha)X_{3\alpha+2\beta} &= X_{3\alpha+2\beta}, & \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha, \\ \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \mathrm{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \mathrm{int}(w_\beta)p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \mathrm{int}(w_\alpha w_\beta)p_\alpha(x), \\ p_{3\alpha+\beta}(x) &= \exp(xX_{3\alpha+\beta}) = \mathrm{int}(w_\alpha)p_\beta(-x), \\ p_{3\alpha+2\beta}(x) &= \exp(xX_{3\alpha+2\beta}) = \mathrm{int}(w_\beta w_\alpha)p_\beta(-x). \end{aligned}$$

Then

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy)p_{3\alpha+\beta}(3x^2y)p_{3\alpha+2\beta}(3xy^2), \\ p_{2\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(3xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy), \\ p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(3xy). \end{aligned}$$

3.4.2.— The proof occupies 3.4.2–3.4.9. Since

$$(\beta^*, \alpha) = -1, \qquad (\alpha^*, \beta) = -3,$$

we have $\beta(t_\alpha) = \alpha(t_\beta) = -1$. Define the four nonsimple root vectors as in (ii). Then

$$\begin{aligned} \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= \alpha(t_\alpha)\beta(t_\alpha)X_{\alpha+\beta} = -X_{\alpha+\beta}, \\ \mathrm{Ad}(w_\alpha)X_{3\alpha+\beta} &= -\beta(t_\alpha)X_\beta = X_\beta, \\ \mathrm{Ad}(w_\beta)X_{3\alpha+2\beta} &= \alpha(t_\beta)^3\beta(t_\beta)X_{3\alpha+\beta} = -X_{3\alpha+\beta}. \end{aligned}$$

Since $(3\alpha+2\beta) \pm \alpha$ and $(2\alpha+\beta) \pm \beta$ are not roots,

$$\mathrm{Ad}(w_\alpha)X_{3\alpha+2\beta} = X_{3\alpha+2\beta}, \qquad \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta}.$$

3.4.3.— By Exposé XXII, 5.5.2, there are sections

$$A, B, C, D, E, F, G, H, J \in \mathbf{G}_a(S)$$

such that

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y)p_{3\alpha+\beta}(Cx^3y)p_{3\alpha+2\beta}(Dx^3y^2), \tag{1}$$

$$p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Exy)p_{3\alpha+\beta}(Fx^2y)p_{3\alpha+2\beta}(Gxy^2), \tag{2}$$

$$p_{2\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hxy), \tag{3}$$

$$p_{3\alpha+\beta}(y)p_\beta(x) = p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(Jxy). \tag{4}$$

3.4.4.— Conjugating (1), (3), and (4) by w_β gives

$$\begin{aligned} p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y)& \\ \cdot p_{3\alpha+2\beta}(Cx^3y)p_{3\alpha+\beta}(-Dx^3y^2),& \end{aligned} \tag{5}$$

$$p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(Hxy), \tag{6}$$

$$p_{3\alpha+2\beta}(y)p_{-\beta}(-x) = p_{-\beta}(-x)p_{3\alpha+2\beta}(y)p_{3\alpha+\beta}(-Jxy). \tag{7}$$

Conjugating (1) by w_α gives

$$\begin{aligned} p_{3\alpha+\beta}(-y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{3\alpha+\beta}(-y)p_{2\alpha+\beta}(Axy)p_{\alpha+\beta}(-Bx^2y)& \\ \cdot p_\beta(Cx^3y)p_{3\alpha+\beta}(Dx^3y^2).& \end{aligned} \tag{8}$$

3.4.5.— The identity

$$w_\beta p_\alpha(x)w_\beta^{-1} = p_{\alpha+\beta}(x)$$

may be written, since $\alpha + 2\beta$ is not a root,

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1). \tag{9}$$

Using (1) and then (4), its left side becomes

$$p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D-CJ)x^3). \tag{10}$$

Using (5) and then (7), its right side becomes

$$p_{\alpha+\beta}(x)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ-D)x^3).$$

Applying (2) transforms this into

$$\begin{aligned} p_\alpha(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE-B)x^2)p_{3\alpha+\beta}((A^2F+CJ-D)x^3)& \\ \cdot p_{3\alpha+2\beta}((AG-C)x^3).& \end{aligned} \tag{12}$$

Comparing (10) and (12) gives

$$A = 1, \qquad E = 2B, \qquad C + D = F + CJ, \qquad F = G.$$

3.4.6.— Write

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{3\alpha+\beta}(-y).$$

Since $4\alpha + \beta$ is not a root,

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{3\alpha+\beta}(-y)p_{-\alpha}(-1). \tag{13}$$

Using (1), the left side is

$$p_\beta(y)p_{\alpha+\beta}(-Ay)p_{2\alpha+\beta}(By)p_{3\alpha+\beta}(-Cy)p_{3\alpha+2\beta}(-Dy^2).$$

Using (8), (6), and (4), the right side is

$$p_\beta(Cy)p_{\alpha+\beta}(-By)p_{2\alpha+\beta}(Ay)p_{3\alpha+\beta}(-y)p_{3\alpha+2\beta}((D-CJ-ABH)y^2).$$

Thus

$$C=1, \qquad A=B, \qquad D-CJ-ABH=-D.$$

Together with the preceding identities, this yields

$$A=B=C=1, \quad E=2, \quad F=G, \quad D+1=F+J, \quad 2D=H+J.$$

3.4.7.— From

$$w_\beta p_{3\alpha+\beta}(x)w_\beta^{-1}=p_{3\alpha+2\beta}(x)$$

one obtains

$$p_{3\alpha+\beta}(x)p_{3\alpha+2\beta}(-Jx)=p_{3\alpha+2\beta}(x)p_{3\alpha+\beta}(-Jx),$$

and therefore $J=-1$.

3.4.8.— Finally,

$$w_\alpha p_{\alpha+\beta}(y)w_\alpha^{-1}=p_{2\alpha+\beta}(y),$$

or equivalently

$$p_\alpha(1)p_{\alpha+\beta}(y)p_\alpha(-1)=p_{-\alpha}(1)p_\alpha(-1)p_{2\alpha+\beta}(y)p_\alpha(1)p_{-\alpha}(-1).$$

Using (2) on the left and (3) on the right gives

$$\begin{aligned}p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-Ey)p_{3\alpha+\beta}(Fy)p_{3\alpha+2\beta}(-Gy^2)\\ =p_{-\alpha}(1)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hy)p_{-\alpha}(-1).\end{aligned}$$

Commuting $p_{-\alpha}(-1)$ first past $p_{3\alpha+\beta}(Hy)$, and then past $p_{2\alpha+\beta}(y)$, introduces no new $p_{3\alpha+\beta}$-factor. Comparing those factors gives $F=H$. Hence

$$2D=D+1, \qquad D=1, \qquad F=G=H=2D-J=3.$$

This determines all the coefficients and proves (iii).

3.4.9.— For (i), one obtains successively

$$\begin{aligned}w_\alpha w_\beta w_\alpha &= w_{3\alpha+\beta}^{-1}t_\alpha,\\ w_\beta(w_\alpha w_\beta)^2 &= w_{3\alpha+2\beta}^{-1}t_\alpha,\\ w_\alpha(w_\beta w_\alpha)^3 &= w_{3\alpha+2\beta}^{-1},\\ w_\beta(w_\alpha w_\beta)^4 &= w_{3\alpha+\beta}t_\beta,\\ w_\alpha(w_\beta w_\alpha)^5 &= w_\beta^{-1}.\end{aligned}$$

Consequently

$$(w_\alpha w_\beta)^6=(w_\beta w_\alpha)^6=e.$$

Remark 3.4.10.— *Condition (v) of 2.3 consists first of*

$$\textit{(A)} \quad \begin{cases}\operatorname{int}(h_\alpha h_\beta h_\alpha h_\beta h_\alpha)v_\beta=v_\beta,\\ \operatorname{int}(h_\beta h_\alpha h_\beta h_\alpha h_\beta)v_\alpha=v_\alpha,\end{cases}$$

and, after setting

$$v_{\alpha+\beta} = \operatorname{int}(h_\beta)v_\alpha, \qquad v_{2\alpha+\beta} = \operatorname{int}(h_\alpha h_\beta)v_\alpha,$$
$$v_{3\alpha+\beta} = \operatorname{int}(h_\alpha)v_\beta^{-1}, \quad v_{3\alpha+2\beta} = \operatorname{int}(h_\beta h_\alpha)v_\beta^{-1},$$

of the commutation relations

$$(B)\qquad \begin{aligned}
v_\beta v_\alpha &= v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta} v_{3\alpha+\beta} v_{3\alpha+2\beta},\\
v_{\alpha+\beta} v_\alpha &= v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2 v_{3\alpha+\beta}^3 v_{3\alpha+2\beta}^3,\\
v_{2\alpha+\beta} v_\alpha &= v_\alpha v_{2\alpha+\beta} v_{3\alpha+\beta}^3,\\
v_{3\alpha+\beta} v_\alpha &= v_\alpha v_{3\alpha+\beta}, & v_{3\alpha+2\beta} v_\alpha &= v_\alpha v_{3\alpha+2\beta},\\
v_{\alpha+\beta} v_\beta &= v_\beta v_{\alpha+\beta}, & v_{2\alpha+\beta} v_\beta &= v_\beta v_{2\alpha+\beta},\\
v_{3\alpha+\beta} v_\beta &= v_\beta v_{3\alpha+\beta} v_{3\alpha+2\beta}^{-1},\\
v_{3\alpha+2\beta} v_\beta &= v_\beta v_{3\alpha+2\beta},\\
v_{2\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{2\alpha+\beta} v_{3\alpha+2\beta}^3,\\
v_{3\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{3\alpha+\beta},\\
v_{3\alpha+2\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{3\alpha+2\beta},\\
v_{3\alpha+\beta} v_{2\alpha+\beta} &= v_{2\alpha+\beta} v_{3\alpha+\beta},\\
v_{3\alpha+2\beta} v_{2\alpha+\beta} &= v_{2\alpha+\beta} v_{3\alpha+2\beta},\\
v_{3\alpha+2\beta} v_{3\alpha+\beta} &= v_{3\alpha+\beta} v_{3\alpha+2\beta}.
\end{aligned}$$

3.5. Explicit Form of the Generators-and-Relations Theorem

Theorem 3.5.1.— *Let S be a scheme and G a pinned S-group, with maximal torus T, system of simple roots Δ, and elements*

$$u_\alpha \in U_\alpha(S)^\times, \qquad w_\alpha \in \operatorname{Norm}_G(T)(S) \qquad (\alpha \in \Delta)$$

defined by the pinning. Let H be an fppf S-sheaf in groups, let

$$f_T : T \to H, \qquad f_\alpha : U_\alpha \to H \quad (\alpha \in \Delta)$$

be morphisms of groups, and let $h_\alpha \in H(S)$. Put

$$v_\alpha = f_\alpha(u_\alpha).$$

There exists a morphism of groups $f : G \to H$ extending f_T and all the f_α, and satisfying $f(w_\alpha) = h_\alpha$, if and only if:

(i) for every $S' \to S$, $\alpha \in \Delta$, $t \in T(S')$, and $x \in U_\alpha(S')$,

$$\operatorname{int}(f_T(t))f_\alpha(x) = f_\alpha(\operatorname{int}(t)x) = f_\alpha(x^{\alpha(t)}); \tag{1}$$

(ii) for every $\alpha \in \Delta$, every $S' \to S$, and every $t \in T(S')$,

$$\operatorname{int}(h_\alpha)f_T(t) = f_T(s_\alpha(t)) = f_T\Big(t\alpha^*(\alpha(t)^{-1})\Big)\,; \tag{2}$$

(iii) for every $\alpha \in \Delta$,

$$h_\alpha^2 = f_T(\alpha^*(-1)), \tag{3_1}$$

and

$$(h_\alpha v_\alpha)^3 = e; \tag{4}$$

(iv) for distinct $\alpha, \beta \in \Delta$:

(a) the appropriate braid relation is

$$\begin{cases} (h_\alpha h_\beta)^2 = f_T(\alpha^*(-1)\beta^*(-1)), & (\alpha^*, \beta) = 0, \\ (h_\alpha h_\beta)^3 = e, & (\alpha^*, \beta) = -1, \\ (h_\alpha h_\beta)^4 = f_T(\alpha^*(-1)), & (\alpha^*, \beta) = -2, \\ (h_\alpha h_\beta)^6 = e, & (\alpha^*, \beta) = -3; \end{cases} \tag{3_2}$$

(b) the corresponding relations (A) and (B) of 3.1.3, 3.2.8, 3.3.7, or 3.4.10 hold.

When it exists, f is necessarily unique.

This follows immediately from Theorem 2.3, Remark 2.6, and the calculations in each rank-two case.

Remark 3.5.2.— *The result may equivalently be presented by giving morphisms*

$$a_\alpha : T\cdot U_\alpha \longrightarrow H, \qquad b_\alpha : \mathrm{Norm}_{Z_\alpha}(T) \longrightarrow H \qquad (\alpha \in \Delta),$$

and putting

$$h_\alpha = b_\alpha(w_\alpha), \qquad v_\alpha = a_\alpha(u_\alpha).$$

The required conditions are:

(1) all the a_α and b_α have the same restriction to T;

(2) condition (4) and all the conditions in (iv) of Theorem 3.5.1 hold.

3.5.3.— Further applications will be given in the next Exposé. Here is one. Theorem 3.5.1 describes G by generators and relations in the category of fppf S-sheaves. More explicitly, for every $S' \to S$, let $H(S')$ be the group generated by

$$T(S'), \qquad U_\alpha(S')\ (\alpha \in R), \qquad w_\alpha\ (\alpha \in R),$$

subject to relations analogous to

$$(1),\ (2),\ (3_1),\ (4),\ (3_2),\ (A),\ (B).$$

Then G is the sheaf associated with the presheaf $S' \mapsto H(S')$.

In particular, if S' is the spectrum of an algebraically closed field k, then $G(S') = H(S')$. This follows immediately from the Nullstellensatz in the form: an fppf-covering sieve over an algebraically closed field is trivial. Thus Theorem 3.5.1 gives an explicit presentation by generators and relations of the abstract group $G(k)$.

4. Uniqueness of Pinned Groups: Fundamental Theorem

Theorem 4.1.— *Let S be a nonempty scheme. The functor R of 1.6 is fully faithful. More precisely, let G, G' be pinned S-groups, let $p > 0$ be an integer such that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$, and let*

$$h : R(G') \longrightarrow R(G)$$

be a p-morphism of pinned root data. There is a unique morphism of pinned groups $f : G \to G'$ such that $R(f) = h$.

Uniqueness was proved in 1.9, so only existence remains. By hypothesis there are a bijection $d : R \xrightarrow{\sim} R'$ and a map

$$q : R \longrightarrow \{p^n \mid n \in \mathbf{N}\}$$

such that, for every $\alpha \in R$,

$$h(d(\alpha)) = q(\alpha)\alpha, \qquad {}^t h(\alpha^*) = q(\alpha)d(\alpha)^*.$$

In particular, G and G' have the same semisimple rank.

4.1.1.— Suppose that both semisimple ranks are 0. Then G, G' are tori:

$$G = T = D_S(M), \qquad G' = T' = D_S(M'),$$

and $h : M' \to M$ is an ordinary homomorphism. Take $f = D_S(h)$.

4.1.2.— Suppose that both semisimple ranks are 1, and put

$$f_T = D_S(h) : T \longrightarrow T'.$$

Writing $\alpha' = d(\alpha)$, the diagram

$$\begin{array}{ccccc}
\mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\
\Big\downarrow{\scriptstyle q(\alpha)} & & {\scriptstyle f_T}\Big\downarrow & & \Big\downarrow{\scriptstyle q(\alpha)} \\
\mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S}
\end{array}$$

commutes. Apply Exposé XX, 4.1.

4.1.3.— Suppose that both semisimple ranks are 2. Exposé XXI, 7.5.3 lists every possible h; we verify the conditions of Corollary 2.5 in each case. Write

$$\Delta = \{\alpha, \beta\}, \qquad \Delta' = \{\alpha', \beta'\}, \qquad d(\alpha) = \alpha', \quad d(\beta) = \beta'.$$

4.1.4. G and G' of type $A_1 \times A_1$.— Here

$$h(\alpha') = q\alpha, \qquad h(\beta') = q_1\beta.$$

Conditions 2.5(ii) and (iii) follow from Proposition 3.1.2(ii),(iii). For (i),

$$\begin{aligned}
D_S(h)t_{\alpha\beta} &= D_S(h)(t_\alpha t_\beta) \\
&= {}^t h(\alpha^*)(-1)\, {}^t h(\beta^*)(-1) \\
&= \alpha'^*((-1)^q)\, \beta'^*((-1)^{q_1}).
\end{aligned}$$

The assumption on $x \mapsto x^p$, with $S \neq \varnothing$, implies that $p = 1$ or S has characteristic p. Thus $(-1)^q = -1$: if q is even, then $p = 2$ and $1 = -1$. Consequently

$$D_S(h)t_{\alpha\beta} = \alpha'^*(-1)\beta'^*(-1) = t'_{\alpha'\beta'}.$$

4.1.5. ***G* and *G′* of type A_2.—** Here

$$h(\alpha') = q\alpha, \qquad h(\beta') = q\beta.$$

Put

$$X_{\alpha+\beta} = \mathrm{Ad}(w_\beta)X_\alpha, \qquad X'_{\alpha'+\beta'} = \mathrm{Ad}(w'_{\beta'})X'_{\alpha'}.$$

Condition (i) follows as above from Proposition 3.2.1(i), and (ii) follows immediately from Proposition 3.2.1(ii). For (iii), the map

$$p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z) \longmapsto p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(z^q)$$

must be a homomorphism. The only nontrivial relation is

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy),$$

which maps to

$$p'_{\beta'}(y^q)p'_{\alpha'}(x^q) = p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(x^q y^q).$$

4.1.6.— The same argument treats types B_2 and G_2 when the radicial exponents are equal, using Propositions 3.3.1 and 3.4.1. It remains to treat the two exceptional cases in Exposé XXI, 7.5.3.

4.1.7. Type B_2, characteristic 2, $q(\alpha) = 2q$, $q(\beta) = q$.— The positive roots are

$$\{\alpha, \beta, \alpha+\beta, 2\alpha+\beta\}$$

and

$$\{\alpha', \beta', \alpha'+\beta', \alpha'+2\beta'\},$$

where α and β' are the short simple roots. We have

$$\begin{aligned} h(\alpha') &= 2q\alpha, & h(\beta') &= q\beta, \\ h(\alpha'+\beta') &= q(2\alpha+\beta), & h(\alpha'+2\beta') &= 2q(\alpha+\beta), \end{aligned}$$

so

$$\begin{aligned} d(\alpha+\beta) &= \alpha'+2\beta', & q(\alpha+\beta) &= 2q, \\ d(2\alpha+\beta) &= \alpha'+\beta', & q(2\alpha+\beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \mathrm{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \mathrm{Ad}(w_\alpha)X_\beta, \\ X'_{\alpha'+\beta'} &= \mathrm{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \mathrm{Ad}(w'_{\beta'})X'_{\alpha'}. \end{aligned}$$

For (i), characteristic 2 gives $-1 = 1$, hence

$$t_{\alpha\beta} = t_\alpha = \alpha^*(-1) = e = \beta'^*(-1) = t'_{\beta'} = t'_{\alpha'\beta'}.$$

For (ii), the two defining identities are

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_\beta &= X_{2\alpha+\beta}, & \mathrm{Ad}(w'_{\alpha'})X'_{\beta'} &= X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}, \\ \mathrm{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \mathrm{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}. \end{aligned}$$

Proposition 3.3.1(ii), together with $-1 = 1$, gives the four remaining identities:

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{\alpha+\beta}, & \mathrm{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= X'_{d(\alpha+\beta)}, \\ \mathrm{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, & \mathrm{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= X'_{d(\beta)}, \\ \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= X_\alpha, & \mathrm{Ad}(w'_{\beta'})X'_{d(\alpha+\beta)} &= X'_{d(\alpha)}, \\ \mathrm{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \mathrm{Ad}(w'_{\beta'})X'_{d(2\alpha+\beta)} &= X'_{d(2\alpha+\beta)}. \end{aligned}$$

For (iii), the sole nontrivial relations are

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y),$$
$$p'_{\alpha'}(y')p'_{\beta'}(x') = p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(x'y')p'_{\alpha'+2\beta'}(x'y'^2).$$

The proposed map is

$$\begin{aligned} &p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z)p_{2\alpha+\beta}(t) \\ &\qquad\longmapsto p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q)p'_{\alpha'+2\beta'}(z^{2q})p'_{\alpha'+\beta'}(t^q). \end{aligned}$$

It respects the first relation because the second, with $y' = x^{2q}$ and $x' = y^q$, reads

$$\begin{aligned} p'_{\beta'}(y^q)p'_{\alpha'}(x^{2q}) = p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q)&p'_{\alpha'+2\beta'}((xy)^{2q}) \\ &\cdot p'_{\alpha'+\beta'}((x^2y)^q). \end{aligned}$$

4.1.8. Type G_2, characteristic 3, $q(\alpha) = 3q$, $q(\beta) = q$.— The positive roots are

$$\{\alpha, \beta, \alpha+\beta, 2\alpha+\beta, 3\alpha+\beta, 3\alpha+2\beta\}$$

and

$$\{\alpha', \beta', \alpha'+\beta', \alpha'+2\beta', \alpha'+3\beta', 2\alpha'+3\beta'\},$$

where again α and β' are short. We have

$$\begin{aligned} h(\alpha') &= 3q\alpha, & h(\beta') &= q\beta, \\ h(\alpha'+\beta') &= q(3\alpha+\beta), & h(\alpha'+2\beta') &= q(3\alpha+2\beta), \\ h(\alpha'+3\beta') &= 3q(\alpha+\beta), & h(2\alpha'+3\beta') &= 3q(2\alpha+\beta), \end{aligned}$$

and hence

$$\begin{aligned} d(\alpha+\beta) &= \alpha'+3\beta', & q(\alpha+\beta) &= 3q, \\ d(2\alpha+\beta) &= 2\alpha'+3\beta', & q(2\alpha+\beta) &= 3q, \\ d(3\alpha+\beta) &= \alpha'+\beta', & q(3\alpha+\beta) &= q, \\ d(3\alpha+2\beta) &= \alpha'+2\beta', & q(3\alpha+2\beta) &= q. \end{aligned}$$

Define

$$\begin{aligned} X_{\alpha+\beta} &= \operatorname{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \operatorname{Ad}(w_\alpha)X_{\alpha+\beta}, \\ X_{3\alpha+\beta} &= -\operatorname{Ad}(w_\alpha)X_\beta, & X_{3\alpha+2\beta} &= \operatorname{Ad}(w_\beta)X_{3\alpha+\beta}, \\ X'_{\alpha'+\beta'} &= -\operatorname{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \operatorname{Ad}(w'_{\beta'})X'_{\alpha'+\beta'}, \\ X'_{\alpha'+3\beta'} &= \operatorname{Ad}(w'_{\beta'})X'_{\alpha'}, & X'_{2\alpha'+3\beta'} &= \operatorname{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'}. \end{aligned}$$

Condition (i) follows from Proposition 3.4.1(i). Four instances of (ii) hold by definition:

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{d(\alpha+\beta)}, \\ \operatorname{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{\beta'} &= -X'_{d(3\alpha+\beta)}, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{d(3\alpha+\beta)} &= X'_{d(3\alpha+2\beta)}, \\ \operatorname{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= X'_{d(2\alpha+\beta)}. \end{aligned}$$

The remaining instances follow from Proposition 3.4.1(ii); for example,

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= -X'_{d(\alpha+\beta)}, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{d(3\alpha+2\beta)} &= -X'_{d(3\alpha+\beta)}. \end{aligned}$$

For (iii), in characteristic 3, the nontrivial relations in U are

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy). \end{aligned}$$

The corresponding relations in U' are

$$\begin{aligned} p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(-x'y')p'_{\alpha'+2\beta'}(-x'^2y') \\ &\quad \cdot p'_{\alpha'+3\beta'}(-x'^3y')p'_{2\alpha'+3\beta'}(-x'^3y'^2), \\ p'_{\alpha'+\beta'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'+\beta'}(y')p'_{\alpha'+2\beta'}(x'y'), \\ p'_{\alpha'+3\beta'}(y')p'_{\alpha'}(x') &= p'_{\alpha'}(x')p'_{\alpha'+3\beta'}(y')p'_{2\alpha'+3\beta'}(x'y'). \end{aligned}$$

Define $\phi : U \to U'$ by

$$\begin{aligned} &\phi(p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(t)p_{2\alpha+\beta}(u)p_{3\alpha+\beta}(v)p_{3\alpha+2\beta}(w)) \\ &\quad = p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}(t^{3q})p'_{2\alpha'+3\beta'}(u^{3q})p'_{\alpha'+\beta'}(v^q)p'_{\alpha'+2\beta'}(w^q). \end{aligned}$$

Substituting x^{3q}, y^q in the three displayed relations for U' yields exactly the images of the three relations for U; thus ϕ is a homomorphism.

4.1.9. Semisimple rank greater than 2.— For $\alpha \in \Delta$, put $\alpha' = d(\alpha)$. For every (α, β), consider the pinned rank-at-most-two groups

$$Z_{\alpha\beta}, \qquad Z'_{\alpha'\beta'}.$$

The homomorphism $M' \to M$ underlying h induces a p-morphism

$$h_{\alpha\beta} : R(Z'_{\alpha'\beta'}) \longrightarrow R(Z_{\alpha\beta}).$$

By the preceding cases, there is a morphism of pinned groups

$$f_{\alpha\beta} : Z_{\alpha\beta} \longrightarrow Z'_{\alpha'\beta'}$$

with $R(f_{\alpha\beta}) = h_{\alpha\beta}$. The restrictions

$$f_{\alpha\beta}|_{Z_\alpha}, \qquad f_{\alpha\alpha} : Z_\alpha \to Z'_{\alpha'}$$

correspond to the same morphism of pinned root data, and therefore coincide by uniqueness. Corollary 2.4 now glues the $f_{\alpha\beta}$ to a morphism $f : G \to G'$. It is a morphism of pinned groups and satisfies $R(f) = h$, completing the proof.

5. Corollaries of the Fundamental Theorem

The most important is the following.

Corollary 5.1.— *Let S be a nonempty scheme, let G, G' be pinned S-groups, and let*

$$h : R(G') \xrightarrow{\sim} R(G)$$

be an isomorphism of pinned root data. There is a unique isomorphism of pinned S-groups $f : G \xrightarrow{\sim} G'$ such that $R(f) = h$.

This also follows from Theorem 3.5.1, because its relations can be written using only $R(G)$. It also follows from the elementary part of the proof of Theorem 4.1, without the exceptional isogenies of 4.1.7 and 4.1.8.

Corollary 5.2 (Uniqueness theorem).— *Let S be a scheme and let G, G' be two splittable S-groups. If they have the same type, then they are isomorphic.*

Corollary 5.3.— *Let S be a scheme and G, G' two splittable S-groups. The following are equivalent:*

(i) G and G' are isomorphic;

(ii) they are locally isomorphic for the fpqc topology;

(iii) there is $s \in S$ such that G_s and G'_s have the same type.

Clearly (i) implies (ii), which implies (iii). Conversely, (iii) gives

$$R(G) = R(G_s) = R(G'_s) = R(G'),$$

so Corollary 5.2 applies.

Corollary 5.4 (Uniqueness of Chevalley schemes).— *Let G, G' be reductive groups over $\mathbf{Z}$, each possessing a split maximal torus. The following are equivalent:*

(i) G and G' are isomorphic;

(ii) there is $s \in \operatorname{Spec} \mathbf{Z}$ such that G_s and G'_s have the same type;

(iii) $G_{\mathbf{C}}$ and $G'_{\mathbf{C}}$ have the same type.

Indeed, G and G' are splittable by Exposé XXII, 2.2. [77]

Corollary 5.5 (Existence of outer automorphisms).— *Let S be a scheme, G a pinned S-group, and h an automorphism of the pinned root datum $R(G)$. There is a unique automorphism u of G preserving the pinning and satisfying $R(u) = h$.*

Corollary 5.5 bis.— *Let S be a scheme, G a split reductive S-group, and R^+ a system of positive roots. For every simple root α, choose an isomorphism of vector groups*

$$p_\alpha : \mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha.$$

Let h be an automorphism of M permuting the positive roots and the corresponding coroots: if $\alpha \in R^+$, then

$$h(\alpha) \in R^+, \qquad h^\vee(\alpha^*) = h(\alpha)^*.$$

There is a unique automorphism u of G inducing $D_S(h)$ on T and satisfying $u \circ p_\alpha = p_{h(\alpha)}$ for every simple root α.

Corollary 5.6.— *Let G, G' be reductive S-groups. The following are equivalent:*

(i) G, G' are locally isomorphic for the fpqc topology;

(i bis) they are locally isomorphic for the étale topology;

[77] *Editors' note.* In fact every $\mathbf{Z}$-torus is split: every $\mathbf{Z}$-torus is isotrivial, and every étale cover of $\operatorname{Spec} \mathbf{Z}$ is trivial.

(ii) G_s, G'_s *are isomorphic for every* $s \in S$*;*

(iii) the functions $s \mapsto \mathrm{type}(G_s)$ *and* $s \mapsto \mathrm{type}(G'_s)$ *are equal.*

The implication (i bis)⇒(i) is immediate; (i)⇒(ii) follows from the principle of finite extension (EGA IV_3, 9.1.4); and (ii)⇒(iii) is immediate. For (iii)⇒(i bis), one may suppose G, G' splittable (Exposé XXII, 2.3), and then apply Corollary 5.3.

Corollary 5.7.— *Let* S *be a scheme,* G *a reductive* S*-group, and* G' *an affine smooth* S*-group with connected fibers. If* $s \in S$ *and* $G_s \simeq G'_s$*, then there is an étale morphism* $S' \to S$*, covering* s*, such that* $G_{S'} \simeq G'_{S'}$*.*

By Exposés XIX, 2.5 and XXII, 2.1, one may suppose G, G' reductive and splittable, and Corollary 5.3 applies.

Corollary 5.8.— *Let* k *be a field and* G, G' *reductive* k*-groups. The following are equivalent:*

(i) G, G' *have the same type;*

(ii) $G \otimes_k \bar{k}$ *and* $G' \otimes_k \bar{k}$ *are isomorphic;*

(iii) for some finite separable extension K/k*,* $G \otimes_k K$ *and* $G' \otimes_k K$ *are isomorphic.*

Corollary 5.9.— *Let* S *be a nonempty scheme and* R *a root datum. The following are equivalent:*

(i) there is a pinned S*-group of type* R*;*

(ii) there is an S*-group of type* R*;*

(iii) fpqc-locally, there is a reductive S*-group of type* R*.*

Only (iii)⇒(i) needs proof. Suppose, for simplicity, that there are a faithfully flat quasi-compact morphism $S' \to S$ and a reductive S'-group G' of type R. We may suppose G' splittable and choose a pinning E', writing $R = R(G', E')$. The two pullbacks of (G', E') to

$$S'' = S' \times_S S'$$

are pinned groups (G''_1, E''_1), (G''_2, E''_2), with canonical isomorphisms

$$p_i : R(G''_i, E''_i) \xrightarrow{\sim} R.$$

Thus

$$p = p_2^{-1} \circ p_1 : R(G''_1, E''_1) \xrightarrow{\sim} R(G''_2, E''_2).$$

The uniqueness theorem gives a unique isomorphism

$$f : (G''_1, E''_1) \xrightarrow{\sim} (G''_2, E''_2)$$

with $R(f) = p$. This is a descent datum on G'. Its cocycle condition may be checked after applying R, since R is fully faithful, and then follows from the construction of p. Because G' is affine, the datum is effective. The pinning is stable under descent, so it descends to a pinned S-group (G, E) of type R.

N.B. In the language of fibered categories, the preceding proof becomes both shorter and more transparent.

Corollary 5.10.— *Let S be a nonempty scheme and R a pinned root datum for which there exists a reductive S-group of type R. Then there is a pinned S-group of type R, unique up to unique isomorphism.*

Definition 5.11.— *Under these hypotheses, denote the unique pinned S-group of type R by $\text{Ép}_S(R)$, its canonical maximal torus by $T_S(R)$, its canonical Borel subgroup by $B_S(R)$, and so on.*

For every morphism $S' \to S$, with $S' \neq \varnothing$, one may identify

$$\text{Ép}_{S'}(R) = \text{Ép}_S(R) \times_S S'.$$

If $\text{Ép}_{\operatorname{Spec} \mathbf{Z}}(R)$ exists, write it $\text{Ép}(R)$; then

$$\text{Ép}_S(R) = \text{Ép}(R) \times_{\operatorname{Spec} \mathbf{Z}} S.$$

The group $\text{Ép}(R)$ is called the *Chevalley group scheme of type R.*

5.12.— Thus an S-sheaf in groups G is a reductive S-group of type R if and only if it is locally isomorphic, for the étale or fpqc topology, to $\text{Ép}_S(R)$. Similarly, by the conjugacy theorems, giving a reductive S-group of type R with a maximal torus is locally equivalent to

$$(\text{Ép}_S(R), T_S(R)).$$

The analogous statements hold for Borel subgroups and Killing pairs.

6. Chevalley Systems

The explicit calculations of Section 3 have important numerical consequences.

Definition 6.1.— *Let S be a scheme and (G, T, M, R) a split S-group. A* Chevalley system *of G is a family*

$$(X_\alpha)_{\alpha \in R}, \qquad X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times,$$

satisfying:

(SC) *for every pair $\alpha, \beta \in R$,*

$$\operatorname{Ad}(w_\alpha(X_\alpha))X_\beta = \pm X_{s_\alpha(\beta)}.$$

Recall that

$$w_\alpha(X_\alpha) = \exp(X_\alpha)\exp(-X_\alpha^{-1})\exp(X_\alpha).$$

Condition (SC) implies, in particular,

$$X_{-\alpha} = \pm X_\alpha$$

by the identity

$$\operatorname{Ad}(w_\alpha(X_\alpha))X_\alpha = -X_\alpha^{-1}.$$

Proposition 6.2.— *Every split group admits a Chevalley system. More precisely, if*

$$(\Delta, (X'_\alpha)_{\alpha \in \Delta})$$

is a pinning of (G, T, M, R), there is a Chevalley system $(X_\alpha)_{\alpha \in R}$ with $X_\alpha = X'_\alpha$ for $\alpha \in \Delta$.

First, it suffices to verify (SC) for $\alpha \in \Delta$. Indeed, for every $\alpha \in R$, choose a sequence $\alpha_i \in \Delta$ such that

$$\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}).$$

Repeated use of (SC) gives

$$X_\alpha = \pm \operatorname{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))\, X_{\alpha_{n+1}}.$$

By Lemma 3.1.1(iii),

$$\begin{aligned} w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}) w_{\alpha_{n+1}}(X_{\alpha_{n+1}}) w_{\alpha_n}(X_{\alpha_n})^{-1} \cdots w_{\alpha_1}(X_{\alpha_1})^{-1} \\ = \alpha^*(\pm 1) w_\alpha(X_\alpha). \end{aligned}$$

Now

$$w_{\alpha_i}(X_{\alpha_i})^{-1} = \alpha_i^*(-1) w_{\alpha_i}(X_{\alpha_i}),$$

and

$$\beta(\gamma^*(-1)) = (-1)^{(\gamma^*, \beta)} = \pm 1.$$

It follows that (SC) for every pair (α_i, γ) implies (SC) for every (α, β).

For each $\alpha \in R$, choose such a sequence and define

$$X_\alpha = \operatorname{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))\, X'_{\alpha_{n+1}}.$$

It remains to prove the following lemma.

Lemma 6.3.— *Let*

$$(G, T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$$

be a pinned S-group. If α_i, $0 \leq i \leq n+1$, are simple roots such that

$$s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}) = \alpha_0,$$

then

$$\operatorname{Ad}(w_{\alpha_1} \cdots w_{\alpha_n}) X_{\alpha_{n+1}} = \pm X_{\alpha_0}.$$

Arguing as in 2.3.4, using Tits's lemma, reduces the proof to:

(a) G has semisimple rank at most 2; or

(b) $w_{\alpha_1} \cdots w_{\alpha_n}$ is a section of T.

Case (a) is exactly the calculation in 3.1.2(i), 3.2.1(i), 3.3.1(i), or 3.4.1(i).

In case (b), it suffices to show that if

$$s_{\alpha_1} \cdots s_{\alpha_n} = 1,$$

then $t = w_{\alpha_1} \cdots w_{\alpha_n}$ satisfies $\alpha(t) = \pm 1$ for every root α. By the structure of the Weyl group, the word in the free group generated by the s_α lies in the normal subgroup generated by

$$(s_\alpha s_\beta)^{n_{\alpha\beta}}.$$

Thus one reduces to a word

$$s_{\alpha_1} \cdots s_{\alpha_i} (s_{\alpha_{i+1}} s_{\alpha_{i+2}})^{n_{\alpha_{i+1}\alpha_{i+2}}} s_{\alpha_i} \cdots s_{\alpha_1},$$

for which

$$t = s_{\alpha_1} \cdots s_{\alpha_i}(t_{\alpha_{i+1}\alpha_{i+2}}).$$

The assertion is now immediate from the values of $t_{\alpha_1\alpha_2}$ computed in 3.1.2(i), 3.2.1(i), 3.3.1(i), and 3.4.1(i).

Proposition 6.4.— *Let (G, T, M, R) be a split S-group with Chevalley system $(X_\alpha)_{\alpha\in R}$. Let α, β be nonproportional roots, with*

$$\mathrm{length}(\alpha) \le \mathrm{length}(\beta), \qquad |(\beta^*, \alpha)| \le |(\alpha^*, \beta)|.$$

Let q and $p-1$ be the nonnegative integers for which the roots of the form $\beta + k\alpha$ are

$$\{\beta - (p-1)\alpha, \ldots, \beta, \ldots, \beta + q\alpha\}.$$

Thus $-(\alpha^*, \beta) = q - p + 1$. Writing $p_\gamma(x) = \exp(xX_\gamma)$, the commutation relation between U_α and U_β is given by:

(p, q)	$p_\beta(y)p_\alpha(x)p_\beta(-y)p_\alpha(-x)$
$(-, 0)$	e
$(1, 1)$	$p_{\alpha+\beta}(\pm xy)$
$(1, 2)$	$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$
$(1, 3)$	$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)p_{3\alpha+\beta}(\pm x^3y)p_{3\alpha+2\beta}(\pm x^3y^2)$
$(2, 1)$	$p_{\alpha+\beta}(\pm 2xy)$
$(2, 2)$	$p_{\alpha+\beta}(\pm 2xy)p_{2\alpha+\beta}(\pm 3x^2y)p_{\alpha+2\beta}(\pm 3xy^2)$
$(3, 1)$	$p_{\alpha+\beta}(\pm 3xy)$

These are all possibilities, since the root string has length at most 3.

Proof. By Exposé XXI, 3.5.4, there is a system of simple roots Δ such that $\alpha \in \Delta$, and there are $\alpha' \in \Delta$, $a, b \in \mathbf{Q}_+$, with

$$\beta = a\alpha + b\alpha'.$$

The desired relation lies in $U_{\alpha\alpha'}$, so it follows from the rank-two calculations of Section 3 and condition (SC).

Corollary 6.5 (Chevalley's rule).— *Let $(X_\alpha)_{\alpha\in R}$ be a Chevalley system of the split S-group G. If $\alpha, \beta, \alpha + \beta \in R$, then*

$$[X_\alpha, X_\beta] = \pm pX_{\alpha+\beta},$$

where p is the least positive integer for which $\beta - p\alpha$ is not a root.

The assertion is symmetric in α, β; after arranging $\mathrm{length}(\alpha) \le \mathrm{length}(\beta)$, it follows from Proposition 6.4.

Corollary 6.6.— *Let S be a scheme such that $6 \cdot 1_S \ne 0$, and let (G, T, M, R) be a split S-group. If $R' \subset R$ is such that*

$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \bigoplus_{\alpha\in R'} \mathfrak{g}^\alpha \tag{*}$$

is a Lie subalgebra of $\mathfrak{g}$, then R' is a closed subset of R.

Choose a Chevalley system and let $\alpha, \beta \in R'$ with $\alpha + \beta \in R$. By Corollary 6.5 and Proposition 6.4,

$$[X_\alpha, X_\beta] = \pm pX_{\alpha+\beta}, \qquad p \in \{1, 2, 3\}.$$

Since neither 2 nor 3 is zero on S, (*) implies $\mathfrak{g}^{\alpha+\beta} \subset \mathfrak{g}_{R'}$, and hence $\alpha + \beta \in R'$. [78]

[78] *Editors' note.* It suffices that 2 and 3 are nonzero on S; for example the result applies to $S = \mathrm{Spec}(\mathbf{Z}/6\mathbf{Z})$. In the excluded characteristics, short-root subsystems furnish standard counterexamples.

6.7.— The exact signs in this section can be determined by studying $\mathrm{Norm}_G(T)$, and more precisely the extended Weyl group

$$W^* = N(\mathbf{Z}), \qquad N = \mathrm{Norm}_{\mathrm{\acute{E}p}_{\mathbf{Z}}(R)}(T_{\mathbf{Z}}(R)).$$

It is an extension of $W(R)$ by an abelian group of type $(2, 2, \ldots, 2)$, which accounts for the signs.

Remark 6.7.1.— *By 3.1.2(i) and 3.n.1, $n = 2, 3, 4$, the elements $w_\alpha, w_\beta \in N(\mathbf{Z})$ satisfy the braid relations*

$$w_\alpha w_\beta \cdots = w_\beta w_\alpha \cdots$$

with $n_{\alpha\beta}$ factors on each side.

See also Tits [Ti66], Bourbaki [BLie], IX.4, Ex. 12, and Springer [Sp98, 9.3.2].

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Exposé XXIV
Automorphisms of Reductive Groups

by Michel Demazure

[0] The first part of this Exposé (nos. 1 to 5) is a direct consequence of the existence, for a reductive group, of "sufficiently many outer automorphisms", a result which is itself a consequence of the weakest form of the isomorphism theorem for pinned groups. The second part (nos. 6 and 7) sets out two applications of the more precise results of the preceding Exposé; in particular, no. 7 uses the generators-and- relations theorem in its explicit form. Finally, we have included in an appendix (no. 8) results on "Galois" cohomology used in the text.

Let us specify our cohomological notation. If S is a scheme and G an S-group scheme, we write $H^1(S,G)$ for the first cohomology set of S with coefficients in G, computed for the (fpqc) topology; this is also the set of isomorphism classes of principal homogeneous (fpqc) sheaves under G. We write $H^1_{\text{ét}}(S,G)$ for the corresponding set for the étale topology; it is therefore the part of $H^1(S,G)$ formed by classes of homogeneous sheaves under G which are quasi-isotrivial (that is, locally trivial for the étale topology). We write $\mathrm{Fib}(S,G)$ for the part of $H^1(S,G)$ formed by classes of representable sheaves (principal homogeneous bundles). Thus we have the inclusions

$$H^1_{\text{ét}}(S,G) \subset H^1(S,G), \qquad \mathrm{Fib}(S,G) \subset H^1(S,G).$$

If every principal homogeneous sheaf under G is representable (for example, if G is quasi-affine over S; cf. SGA 1, VIII, 7.9), then

$$\mathrm{Fib}(S,G) = H^1(S,G).$$

If $S' \to S$ is a covering morphism for the (fpqc) topology, we write $H^1(S'/S,G)$ for the kernel of the canonical map

$$H^1(S,G) \longrightarrow H^1(S',G_{S'}).$$

It is known that $H^1(S'/S,G)$ can be computed simplicially (TDTE I, §A.4), which implies that, when $S' \to S$ is a covering for the étale topology, $H^1(S'/S,G)$ is also the kernel of

$$H^1_{\text{ét}}(S,G) \longrightarrow H^1_{\text{ét}}(S',G_{S'}).$$

Finally, following Exp. VIII, 4.5, the name "Theorem 90" is given to the following assertion: "every principal homogeneous sheaf under $G_{m,S}$ is representable and locally trivial." This assertion is equivalent to

$$H^1(S,G_{m,S}) = \mathrm{Pic}(S),$$

[0] Editor's note: Version dated October 13, 2024.

or again to

$$H^1(S, G_{m,S}) = 0$$

when S is local (or, more generally, semilocal).

1. Scheme of Automorphisms of a Reductive Group

1.0.

We must first make precise certain definitions from the preceding Exposé. Let S be a nonempty scheme, let G be a reductive S-group, and let

$$\mathscr{R} = (M, M^*, R, R^*, \Delta)$$

be a pinned reduced root datum (Exp. XXIII, 1.5). A *pinning of G of type $\mathscr{R}$*, or an *$\mathscr{R}$-pinning of G*, consists of:

(i) an isomorphism of $D_S(M)$ onto a maximal torus T of G (or, equivalently, a monomorphism $D_S(M) \to G$ whose image is a maximal torus T of G), identifying R with a root system of G relative to T (Exp. XIX, 3.6) and R^* with the corresponding set of coroots;

(ii) for each $\alpha \in \Delta$, an element
$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

The group G possesses an $\mathscr{R}$-pinning if and only if it is split and of type $\mathscr{R}$ (Exp. XXII, 2.7).

If $u : G \to G'$ is an isomorphism of reductive S-groups, then every $\mathscr{R}$-pinning $\mathcal{E}$ of G determines, by "transport of structure", an $\mathscr{R}$-pinning $u(\mathcal{E})$ of G'. If $v : \mathscr{R}' \to \mathscr{R}$ is an isomorphism of pinned root data, then every $\mathscr{R}'$-pinning $\mathcal{E}'$ of G' determines, by transport of structure, an $\mathscr{R}$-pinning $v(\mathcal{E}')$ of G.

Call a *pinned group* a triple $(G, \mathscr{R}, \mathcal{E})$, where G is a reductive S-group, $\mathscr{R}$ is a pinned reduced root datum, and $\mathcal{E}$ is a pinning of G of type $\mathscr{R}$. An isomorphism from the pinned group $(G, \mathscr{R}, \mathcal{E})$ onto the pinned group $(G', \mathscr{R}', \mathcal{E}')$ is a pair (u, v), where $u : G \to G'$ is an isomorphism and $v : \mathscr{R}' \to \mathscr{R}$ is an isomorphism of pinned root data, such that

$$u(\mathcal{E}) = v(\mathcal{E}').$$

[1]

N.B. If S is nonempty, v is uniquely determined by u, and by an abuse of language u will also be called an isomorphism of the pinned groups. In particular, if $(G, \mathscr{R}, \mathcal{E})$ is a pinned group, an automorphism of $(G, \mathscr{R}, \mathcal{E})$ is therefore an automorphism u of G for which there exists an automorphism v of $\mathscr{R}$ such that $u(\mathcal{E}) = v(\mathcal{E})$. It is thus an automorphism of G which normalizes T, induces on T an automorphism of the form $D_S(h)$, where h is an automorphism of M, and permutes the elements X_α, $\alpha \in \Delta$. (As is readily seen, the preceding conditions in fact characterize the automorphisms of $(G, \mathscr{R}, \mathcal{E})$.)

There is an evident contravariant functor

$$\Phi : \quad (G, \mathscr{R}, \mathcal{E}) \longmapsto \mathscr{R}, \qquad (u, v) \longmapsto v,$$

and the principal result of the preceding Exposé (Exp. XXIII, 4.1) shows that this functor is fully faithful (we shall moreover see in the following Exposé that it is an equivalence of categories). In particular, it follows that the automorphism group of $(G, \mathscr{R}, \mathcal{E})$ is canonically isomorphic to the automorphism group of the pinned root datum $\mathscr{R}$ (cf. Exp. XXIII, 5.5).

[1] Editor's note: Thus $\mathrm{Lie}(u)(X_{v(\alpha)}) = X'_\alpha$ for every $\alpha \in R'$.

1.1.

Let S be a scheme; endow $(\mathbf{Sch})/S$ with the (fpqc) topology, and consider the S-sheaf of groups $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$, where G is an S-group scheme. There is an exact sequence of S-sheaves of groups

$$1 \longrightarrow \underline{\mathrm{Centr}}(G) \longrightarrow G \xrightarrow{\mathrm{int}} \underline{\mathrm{Aut}}_{S\text{-gr.}}(G),$$

which defines a monomorphism

$$j : G/\underline{\mathrm{Centr}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G).$$

The image sheaf of j is the sheaf of inner automorphisms of G. An automorphism u of G is inner if and only if there is a covering family $\{S_i \to S\}$ and, for every i, an element $g_i \in G(S_i)$ such that

$$\mathrm{int}(g_i) = u_{S_i}.$$

In this case, if v is another automorphism of G, one sees at once that

$$\mathrm{int}(v)u = vuv^{-1}$$

is the inner automorphism defined by the family $g_i' = v(g_i)$. It follows that the image of j is normal in $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$. The quotient sheaf of groups, denoted $\underline{\mathrm{Autext}}(G)$, is the sheaf of outer automorphisms of G. Thus there is an exact sequence

$$1 \longrightarrow G/\underline{\mathrm{Centr}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

All the preceding definitions are compatible with base change. They are naturally valid in any site.

1.2.

Let S be a scheme and let $(G, \mathscr{R}, \mathscr{E})$ be a pinned reductive S-group. Let E be the (abstract) automorphism group of the pinned root datum $\mathscr{R}$, that is, the group of automorphisms of $\mathscr{R}$ which normalize Δ. By Exp. XXIII, 5.5, there is a canonical monomorphism

$$E \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$$

which associates with $h \in E$ the unique automorphism u of the pinned group G such that $\Phi(u) = h$. This monomorphism canonically defines a monomorphism of sheaves

$$a : E_S \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G). \tag{*}$$

An automorphism u of G is a section of the image sheaf of a if and only if the following conditions hold:

(i) u normalizes T. It is then known that u permutes the roots of G relative to T. If $\alpha \in R$, then

$$u(\alpha) : t \longmapsto \alpha(u^{-1}(t))$$

is therefore locally on S of the form $t \longmapsto \beta(t)$, with $\beta \in R$. The second condition may then be written as follows:

(ii) If $\alpha \in \Delta$ and U is an open subset of S such that $u(\alpha)_U \in R$, then $u(\alpha)_U \in \Delta$ and

$$\mathrm{Lie}(u_U)(X_\alpha)_U = (X_{u(\alpha)})_U.$$

It follows at once from the definitions that the sections of $a(E_S)$ normalize the subgroups of G defined by the pinning:

$$T, \quad B, \quad B^-, \quad U, \quad U^-.$$

With these definitions in place, we have:

Theorem 1.3. *Let S be a scheme and G a reductive S-group. Consider the canonical exact sequence of S-sheaves of groups* [2]

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \xrightarrow{p} \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

Then:

(i) $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable by a smooth separated S-scheme;

(ii) $\underline{\mathrm{Autext}}(G)$ is representable by a twisted constant S-scheme of finite generation (Exp. X, 5.1);

(iii) if G is split, the preceding exact sequence is split. More precisely, for every pinning of G, the morphism (cf. 1.2 $(*)$)

$$p \circ a : E_S \longrightarrow \underline{\mathrm{Autext}}(G)$$

is an isomorphism.

Let us first show how the theorem follows from the following lemma.

Lemma 1.4. *Under the hypotheses of* (iii)*, $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is the semidirect product*

$$a(E_S) \cdot \mathrm{ad}(G).$$

[3]

The lemma immediately implies the theorem when G is split. Since G is locally split for the étale topology (Exp. XXII, 2.3), hence also for the (fppf) topology, and since the latter is “of effective descent” for the fibred category of twisted constant morphisms (Exp. X, 5.5), assertion (ii) follows in the general case (cf. Exp. IV, 4.6.8). To deduce (i), observe that $\mathrm{ad}(G)$ is affine over S; thus p is affine whenever $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is representable, and the result follows by descent for affine schemes. [4]

It remains only to prove 1.4. For this, it suffices to prove:

Lemma 1.5. *If $(\mathscr{R}, \mathcal{E})$ and $(\mathscr{R}', \mathcal{E}')$ are two pinnings of the reductive S-group G, there exists a unique inner automorphism u of G over S carrying one pinning to the other; that is, such that there exists an isomorphism*

$$v : \mathscr{R}' \xrightarrow{\sim} \mathscr{R}$$

for which $u(\mathcal{E}) = v(\mathcal{E}')$; cf. 1.0.

1.5.1. Uniqueness.

It suffices to prove that, if G is a pinned S-group and $\mathrm{int}(g)$ is an automorphism of the pinned group ($g \in G(S)$), then $\mathrm{int}(g) = \mathrm{id}$. First,

$$\mathrm{int}(g)T = T, \qquad \mathrm{int}(g)B = B,$$

and hence

$$g \in \underline{\mathrm{Norm}}_G(T)(S) \cap \underline{\mathrm{Norm}}_G(B)(S) = T(S)$$

[2] Editor’s note: Recall that $\mathrm{ad}(G) = G/\underline{\mathrm{Centr}}(G)$ denotes the adjoint group of G; cf. Exp. XXII, 4.3.6.
[3] Editor’s note: Here and below, the notation $\mathrm{int}(G)$ has been replaced by $\mathrm{ad}(G)$.
[4] Editor’s note: Cf. SGA 1, VIII, 2.1.

(cf., for example, Exp. XXII, 5.6.1). It follows that $\mathrm{int}(g)$ normalizes every U_α and that

$$\mathrm{Lie}(\mathrm{int}(g))X_\alpha = \mathrm{Ad}(g)X_\alpha = \alpha(g)X_\alpha$$

for every $\alpha \in \Delta$. Thus $\alpha(g) = 1$ for $\alpha \in \Delta$, and therefore

$$g \in \bigcap_{\alpha\in\Delta} (\ker\alpha)(S) = \underline{\mathrm{Centr}}(G)(S)$$

(Exp. XXII, 4.1.8). □

1.5.2. Existence.

It suffices to prove existence locally for the (fpqc) topology. Let

$$(T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta}) \quad \text{and} \quad (T', M', R', \Delta', (X'_{\alpha'})_{\alpha'\in\Delta'})$$

be the two pinnings. By conjugacy of maximal tori, we may assume $T = T'$. After restricting S, we may assume that the isomorphism

$$D_S(M) \simeq D_S(M')$$

comes from an isomorphism $M \simeq M'$ carrying R onto R'; we are thus reduced to the situation

$$T = T', \qquad M = M', \qquad R = R'.$$

Since the systems of simple roots are conjugate under the Weyl group (Exp. XXI, 3.3.7), we may also assume $\Delta = \Delta'$. For every $\alpha \in \Delta$, there is then a scalar $z_\alpha \in \mathbb{G}_m(S)$ such that

$$X'_\alpha = z_\alpha X_\alpha,$$

and it is enough to construct, locally for the (fpqc) topology, a section t of T such that $\alpha(t) = z_\alpha$ for every $\alpha \in \Delta$. But the morphism

$$T \longrightarrow (\mathbb{G}_{m,S})^\Delta$$

whose components are the α, $\alpha \in \Delta$, is dual to an injection $\mathbb{Z}^\Delta \to M$, and is therefore faithfully flat. This completes the proof of 1.5.2, and hence of 1.3.

Corollary 1.6. *Let S be a scheme and G a reductive S-group. The following conditions are equivalent:*

(i) $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is affine (respectively of finite type, respectively of finite presentation, respectively quasi-compact) over S;

(ii) $\underline{\mathrm{Autext}}(G)$ is finite over S;

(iii) for every $s \in S$,

$$\mathrm{rg}_{\mathrm{red}}(G_s) - \mathrm{rg}_{\mathrm{ss}}(G_s) \leq 1.$$

Indeed, since $\mathrm{ad}(G)$ is affine, flat, and of finite presentation over S, the morphism

$$p : \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\mathrm{Autext}}(G)$$

is affine, faithfully flat, and of finite presentation.

If $\underline{\mathrm{Autext}}(G)$ is finite over S, it is affine over S, and hence so is $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$; this proves (ii)⇒(i). If $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$ is quasi-compact over S, it is of finite presentation over S, since it is in any case locally of finite presentation and separated over S. By Exp. V, 9.1, $\underline{\mathrm{Autext}}(G)$ is then of finite presentation over S, and hence finite; this proves (i)⇒(ii). Finally, to prove the equivalence of (ii) and (iii), one may suppose that G is split, and the assertion reduces to Exp. XXI, 6.7.8.

Corollary 1.7. *Let S be a scheme and G a reductive S-group. Then*

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)^0 \simeq \mathrm{ad}(G).$$

Corollary 1.8. *Let S be a scheme, G a reductive S-group, and H an S-group scheme that is smooth, affine, and has connected fibres over S. Then the S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H)$$

is representable by a smooth separated S-scheme, which is affine over S if G is semisimple.

Indeed, let U be the set of points $s \in S$ for which H_s is reductive; this is an open subset (Exp. XIX, 2.6). If S' is an S-scheme, $H_{S'}$ is reductive if and only if $S' \to S$ factors through U. It follows that the canonical morphism

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H) \longrightarrow S$$

factors through U. We may therefore assume $S = U$, and we are reduced to the following corollary.

Corollary 1.9. *Let S be a scheme, and let G and G' be two reductive S-groups. Then*

$$F = \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$$

is representable by a smooth separated S-scheme, affine if G or G' is semisimple. Moreover, S is the disjoint union of two open subschemes S_1 and S_2 such that $F_{S_1} = \varnothing$, while F_{S_2} is a principal homogeneous bundle on the left (respectively, on the right) *under* $\underline{\mathrm{Aut}}_{S_2\text{-gr.}}(G'_{S_2})$ (respectively, $\underline{\mathrm{Aut}}_{S_2\text{-gr.}}(G_{S_2})$).

Indeed, let S_2 be the set of points of S at which G and G' have the same type, and let S_1 be its complement. Since the type of a reductive group is a locally constant function, S_1 and S_2 are open. Clearly $F_{S_1} = \varnothing$, and we may assume $S = S_2$. By Exp. XXIII, 5.6, F is a principal homogeneous sheaf under $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$, locally trivial for the étale topology. It follows that

$$F_0 = F/\,\mathrm{ad}(G)$$

is a principal homogeneous sheaf under $\underline{\mathrm{Autext}}(G)$, locally trivial for the étale topology, and hence representable (Exp. X, 5.5). Since $\mathrm{ad}(G)$ is affine, F is therefore representable as well.

We note that the proof also gives the following result.

Corollary 1.10. *Let S be a scheme, and let G and G' be two reductive S-groups having the same type at every $s \in S$. Then* $\mathrm{ad}(G)$ *acts freely on the right on* $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$. *The quotient sheaf*

$$\underline{\mathrm{Isomext}}(G, G') = \underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')/\,\mathrm{ad}(G)$$

is representable by a twisted constant S-scheme; it is a principal homogeneous bundle under $\underline{\mathrm{Autext}}(G)$, *and is therefore finite over S if G is semisimple. Moreover, the isomorphism*

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \simeq \underline{\mathrm{Isom}}_{S\text{-gr.}}(G', G), \qquad u \longmapsto u^{-1},$$

induces an isomorphism

$$\underline{\mathrm{Isomext}}(G, G') \simeq \underline{\mathrm{Isomext}}(G', G).$$

Remark 1.11. If $\underline{\mathrm{Isomext}}(G, G')(S) \neq \varnothing$, one says that G is an *inner twisted form* of G'. Then G' is an inner twisted form of G, and the structure group of $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ can be reduced to $\mathrm{ad}(G)$. More precisely, let

$$u \in \underline{\mathrm{Isomext}}(G, G')(S)$$

and regard u as a section $u : S \to \underline{\mathrm{Isomext}}(G, G')$. Denote by

$$\underline{\mathrm{Isomint}}_u(G, G')$$

the inverse image, under the canonical morphism

$$\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G') \longrightarrow \underline{\mathrm{Isomext}}(G, G'),$$

of the closed subscheme of $\underline{\mathrm{Isomext}}(G, G')$ that is the image of u. The natural action of $\mathrm{ad}(G)$ on $\underline{\mathrm{Isomint}}_u(G, G')$ makes this scheme a principal homogeneous bundle. By extension of the structure group[5]

$$\mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}(G),$$

one recovers $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, G')$ from $\underline{\mathrm{Isomint}}_u(G, G')$.

By Hensel's lemma (Exp. XI, 1.11), Corollary 1.8 immediately gives:

Corollary 1.12. *Let S be a henselian local scheme, let G be a reductive S-group, let G' be a smooth affine S-group with connected fibres, and let s be the closed point of S. If G_s and G'_s are isomorphic algebraic groups over $\kappa(s)$, then G and G' are isomorphic. More precisely, every $\kappa(s)$-isomorphism*

$$G_s \simeq G'_s$$

comes from an S-isomorphism $G \simeq G'$.

Applying Corollary 1.7 (respectively, Corollary 1.12) to the scheme of dual numbers over a field, Exp. III, 2.10 (respectively, 3.10) yields part (i) (respectively, part (ii)) of the next corollary.

Corollary 1.13. *Let k be a field and G a reductive k-group.*

(i) *If G is adjoint, then*

$$H^1(G, \mathrm{Lie}(G/k)) = 0.$$

[6]

(ii) *One has*

$$H^2(G, \mathrm{Lie}(G/k)) = 0.$$

[5] The source's editorial note corrects $\mathrm{ad}(G) \to \underline{\mathrm{Autext}}(G)$ to $\mathrm{ad}(G) \to \underline{\mathrm{Aut}}(G)$.

[6] The source's editorial note corrects the original, which stated (i) without assuming G adjoint. By the description of $H^i(G, -)$ as the derived functors of $\mathrm{Hom}_G(k, -)$ (Exp. I, 5.3.1; see also [Ja87], I, 4.16), one has $H^i(G, V) = \mathrm{Ext}^i_G(k, V)$ for every G-module V. If $\mathrm{char}(k) = p > 0$ and $G = \mathrm{SL}_{p,k}$, then $\mathrm{Lie}(\mathrm{GL}_p/k)$ is a nontrivial extension of k by $\mathfrak{g} = \mathrm{Lie}(\mathrm{SL}_p/k)$, so $H^1(G, \mathfrak{g}) \neq 0$. See also the addition 1.15.1 below.

Remark 1.14.

(i) The assertion concerning H^1 was known to Chevalley; the assertion concerning H^2 had been proved in most cases of the classification by Chevalley.

(ii) In fact, the conjunction of Corollary 1.13 and the uniqueness theorem over an algebraically closed field is essentially equivalent to the uniqueness theorem. A direct proof of Corollary 1.13 would therefore give a way to deduce the general uniqueness theorem from Chevalley's uniqueness theorem over a field.

The existence of reductive groups of every type over every scheme (Exp. XXV) shows that the obstructions to lifting a reductive k-group G over artinian rings with residue field k, which by Exp. III, 3.8 are elements of

$$H^3(G, \mathrm{Lie}(G/k) \otimes V) \simeq H^3(G, \mathrm{Lie}(G/k)) \otimes V,$$

where V is a certain k-vector space endowed with the trivial action of G, vanish. This suggests that

$$H^3(G, \mathrm{Lie}(G/k)) = 0.$$

A direct proof, if the assertion is true, would likewise doubtless give a way to deduce the general existence theorem from the existence theorem over a field (Chevalley's Tôhoku).

Corollary 1.15. *Let k be a field,*[7] *and let G be a reductive k-group. Regard k as a trivial G-module. Then*

$$H^1(G, k) = H^2(G, k) = 0.$$

[8] Since

$$H^i(G_{\bar{k}}, \bar{k}) = H^i(G, k) \otimes_k \bar{k},$$

we may assume k algebraically closed. An element of $H^1(G, k)$ is simply a morphism of k-groups

$$\varphi : G \longrightarrow \mathbb{G}_{a,k}.$$

Then

$$\varphi(G) = G/\ker(\varphi)$$

is a smooth connected reductive subgroup of $\mathbb{G}_{a,k}$ (cf. XIX, 1.7), and is therefore trivial. Thus $H^1(G, k) = 0$. Now consider the reductive k-group

$$H = G \times_k \mathbb{G}_{m,k}.$$

There is an H-stable decomposition

$$\mathrm{Lie}(H/k) = \mathrm{Lie}(G/k) \oplus k.$$

For every H-module V,

$$H^i(H, V) = H^i(G, H^0(\mathbb{G}_{m,k}, V)).$$

Indeed, the $H^i(H, -)$ are the derived functors of

$$H^0(H, -) = H^0(G, H^0(\mathbb{G}_{m,k}, -)),$$

[7] The source's editorial note replaces "scheme" with "field."

[8] Because of the correction made in 1.13, the source's editors supply another proof for H^1. Moreover, a theorem of G. Kempf gives $H^i(G, k) = 0$ for every $i > 0$; cf. [Ja87], II, 4.5 and 4.11.

and the functor $H^0(\mathbb{G}_{m,k}, -)$ is exact (cf. Exp. I, 5.3.1 and 5.3.3). In particular, for every i,

$$H^i(H, \mathrm{Lie}(H/k)) = H^i(G, \mathrm{Lie}(G/k)) \oplus H^i(G, k).$$

Applying 1.13(ii) to the reductive group H gives $H^2(G, k) = 0$.

Corollary 1.15.1. — [9] *Let k be a field, G a reductive k-group, Z its center, and $\mathscr{R} = (M, M^*, R, R^*)$ the type of G. Then*

$$H^1(G, \mathrm{Lie}(G/k)) \simeq \mathrm{Ext}^1_{\mathbb{Z}}(M/\Gamma_0(R), k).$$

In particular, $H^1(G, \mathrm{Lie}(G/k)) = 0$ if and only if Z is smooth over k.

Let $\mathfrak{g}$, $\mathfrak{z}$, and $\mathfrak{g}_{\mathrm{ad}}$ be the Lie algebras of G, Z, and $G_{\mathrm{ad}} = G/Z$, respectively. It follows from 1.15 and its proof that

$$H^1(G, \mathfrak{g}) = H^1(G_{\mathrm{ad}}, \mathfrak{g}) \simeq H^1(G_{\mathrm{ad}}, \mathfrak{g}/\mathfrak{z}).$$

Set $C = \mathrm{Coker}(\mathfrak{g} \to \mathfrak{g}_{\mathrm{ad}})$. There is an exact sequence

$$0 \longrightarrow \mathfrak{g}/\mathfrak{z} \longrightarrow \mathfrak{g}_{\mathrm{ad}} \longrightarrow C \longrightarrow 0.$$

Since $H^1(G_{\mathrm{ad}}, \mathfrak{g}_{\mathrm{ad}}) = 0$ by 1.13 and

$$H^0(G_{\mathrm{ad}}, \mathfrak{g}_{\mathrm{ad}}) = 0$$

(cf. Exp. II, 5.2.3), we obtain

$$H^1(G_{\mathrm{ad}}, \mathfrak{g}/\mathfrak{z}) \simeq H^0(G_{\mathrm{ad}}, C).$$

To calculate the right-hand side, we may assume k algebraically closed. Let (G, T, M, R) be a splitting of G. Then

$$Z = D_k(M/\Gamma_0(R))$$

and C is identified with

$$\mathrm{Coker}(\mathrm{Hom}_{\mathbb{Z}}(M, k) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\Gamma_0(R), k)) \simeq \mathrm{Ext}^1_{\mathbb{Z}}(M/\Gamma_0(R), k),$$

with trivial G_{ad}-action. The corollary follows, since Z is not smooth over k exactly when $\mathrm{char}(k) = p > 0$ and $M/\Gamma_0(R)$ has p-torsion (Exp. IX, 2.1).

Definition 1.16. — *Let S be a scheme and G a reductive S-group. A* form *of G over S is an S-group scheme G' locally isomorphic to G for the* (fpqc) *topology. It is equivalent to require that G' be locally isomorphic to G for the étale topology (cf. Exp. XXIII, 5.6), or that G' be a reductive S-group of the same type as G at every point of S.*

Corollary 1.17. — *Let S be a scheme and G a reductive S-group.*

(i) The functor

$$G' \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$$

is an equivalence between the category of forms of G over S and the category of principal homogeneous bundles under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$.

(ii) If $S' \to S$ is a covering morphism, forms of G trivialized by S' correspond to bundles trivialized by S'.

[9] Editor's note: This corollary, drawn from remarks of O. Gabber, has been added to make 1.13(i) more precise.

(iii) *Every principal homogeneous sheaf under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$ is representable and quasi-isotrivial, that is, locally trivial for the étale topology.*

The first assertion is formal in the category of sheaves, for example for the (fpqc) topology. Every sheaf locally isomorphic to G for the (fpqc) topology is representable, since G is affine over S, and is locally isomorphic to G for the étale topology. Finally, for every form G' of G, the S-sheaf $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ is representable by 1.8. The corollary follows.

Corollary 1.18. — *The set of isomorphism classes of forms of the reductive group G over S is isomorphic to*

$$H^1\Big(S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big) = H^1_{\text{ét}}\Big(S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big) = \mathrm{Fib}\Big(S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big).$$

If $S' \to S$ is a covering morphism, the subset consisting of forms trivialized by S' is isomorphic to

$$H^1\Big(S'/S, \underline{\mathrm{Aut}}_{S\text{-gr}}(G)\Big).$$

Corollary 1.19. — *Let S be a scheme and $\mathscr{R}$ a pinned reduced root datum such that $\mathrm{\acute{E}p}_S(\mathscr{R})$ exists,*[10] *a condition which is automatically satisfied by Exp. XXV. Put*

$$A_S(\mathscr{R}) = \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{\acute{E}p}_S(\mathscr{R})) = \mathrm{ad}(\mathrm{\acute{E}p}_S(\mathscr{R})) \cdot E(\mathscr{R})_S.$$

(i) *The set of isomorphism classes of reductive S-groups of type $\mathscr{R}$ (Exp. XXII, 2.7) is isomorphic, by Exp. XXIII, 5.12, to*

$$H^1(S, A_S(\mathscr{R})) = H^1_{\text{ét}}(S, A_S(\mathscr{R})) = \mathrm{Fib}(S, A_S(\mathscr{R})).$$

(ii) *If $S' \to S$ is a covering morphism, the subset consisting of classes of groups splittable over S' is isomorphic to $H^1(S'/S, A_S(\mathscr{R}))$.*

Remark 1.20. — With the preceding notation, every reductive S-group G of type $\mathscr{R}$ has a canonically associated right principal homogeneous bundle under $A_S(\mathscr{R})$:

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(\mathrm{\acute{E}p}_S(\mathscr{R}), G) = P.$$

The scheme P may be interpreted as the "scheme of pinnings of G of type $\mathscr{R}$" (cf. 1.0). It is also a principal homogeneous bundle on the left under $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$, a structure visible immediately from the description above.

Proposition 1.21. — *Let S be a henselian local scheme and s its closed point. The functor*

$$G \longmapsto G_s$$

induces a bijection from the set of isomorphism classes of reductive S-groups onto the set of isomorphism classes of reductive $\kappa(s)$-groups. In particular, for every reductive S-group G, there exists a finite étale surjective morphism $S' \to S$ such that $G_{S'}$ is splittable.

Using the existence of $\mathrm{\acute{E}p}_S(\mathscr{R})$ (Exp. XXV), it is enough to prove that, with $H = A_S(\mathscr{R})$, the canonical map

$$\mathrm{Fib}(S, H) \longrightarrow \mathrm{Fib}(\kappa(s), H_s)$$

is bijective, and that every element of $\mathrm{Fib}(S, H)$ has the property just stated. Every finite subset of H is contained in an affine open subscheme: this is clear for a constant group, and H is affine over a constant group. The result proved in the Appendix, 8.1, therefore applies.

[10] Editor's note: Cf. XXIII, Definition 5.11.

2. Automorphisms and Subgroups

Introduce the following notation. If

$$\mathcal{H} = \underline{\mathrm{Aut}}_{S\text{-gr}}(G)$$

and X is a subfunctor of G, set

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G, X) = \mathrm{Norm}_{\mathcal{H}}(X), \qquad \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_X) = \underline{\mathrm{Centr}}_{\mathcal{H}}(X).$$

If Y is a second subfunctor of G, define

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G, X, Y) = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, X) \cap \underline{\mathrm{Aut}}_{S\text{-gr}}(G, Y).$$

If G' is a second S-group and X' a subfunctor of G',write

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, X; G', X')$$

for the subfunctor of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ defined, for every $S' \to S$, by

$$\Big\{ u \in \underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')(S') \,\Big|\, u(X_{S'}) = X'_{S'} \Big\}.$$

The notation $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, X, Y; G', X', Y')$, and so on, is defined similarly.[11]

Proposition 2.1. — *Let S be a scheme, G a reductive S-group, and T a maximal torus of G (respectively, B a Borel subgroup of G, or $B \supset T$ a Killing couple of G). Let T_{ad} (respectively, B_{ad}) denote the corresponding maximal torus (respectively, Borel subgroup) of* $\mathrm{ad}(G)$*:*

$$B_{\mathrm{ad}} \simeq B/\underline{\mathrm{Centr}}(G) = B/\underline{\mathrm{Centr}}(B), \qquad T_{\mathrm{ad}} \simeq T/\underline{\mathrm{Centr}}(G).$$

Then $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$ (respectively, $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B)$, or $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)$) is represented by a closed subscheme of $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$, smooth over S, and the exact sequence of Theorem 1.3 induces the exact sequences

$$1 \longrightarrow \mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}}) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, T) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1,$$

$$1 \longrightarrow B_{\mathrm{ad}} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1,$$

$$1 \longrightarrow T_{\mathrm{ad}} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T) \longrightarrow \underline{\mathrm{Autext}}(G) \longrightarrow 1.$$

By descent of closed subschemes,[12] we reduce immediately to the case in which G is pinned and $B \supset T$ is its canonical Killing couple (cf. Exp. XXII, 5.5.5(iv)). Since the group E of 1.2 normalizes B and T, the result follows at once from the normalization theorems in $\mathrm{ad}(G)$ (Exp. XXII, 5.3.12 and 5.6.1).

Using now the conjugacy theorems (cf. Exp. XXIII, 5.12), and arguing as in §1, one obtains:

Corollary 2.2. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, and let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G').*

(i) *The S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')$$

[11] Editor's note: The preceding definitions have been made explicit. The original only said that $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, X, Y), \ldots$ and $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, X; G', X'), \ldots$ were defined similarly.

[12] Editor's note: Cf. SGA 1, VIII, 1.9.

is represented by a closed smooth subscheme of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ *which is a principal homogeneous space under* $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$. *Moreover,* $\mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}})$ *acts freely on this scheme, and there is a canonical isomorphism*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')/\,\mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}}) \simeq \underline{\mathrm{Isomext}}(G, G').$$

(ii) *The S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B; G', B')$$

is represented by a closed smooth subscheme of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ *which is a principal homogeneous space under* $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B)$. *Moreover,* B_{ad} *acts freely on this scheme, and there is a canonical isomorphism*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B; G', B')/B_{\mathrm{ad}} \simeq \underline{\mathrm{Isomext}}(G, G').$$

(iii) *The S-functor*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T')$$

is represented by a closed smooth subscheme of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$, *which is a principal homogeneous space under* $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)$. *Moreover,* T_{ad} *acts freely on this scheme, and there is a canonical isomorphism*

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T')/T_{\mathrm{ad}} \simeq \underline{\mathrm{Isomext}}(G, G').$$

Arguing once again as in §1, one obtains:

Corollary 2.3. — *Let S be a scheme, let G be a reductive S-group, and let $B \supset T$ be a Killing couple of G. The functor*

$$(G', T') \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T'),$$

respectively

$$(G', B') \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, B; G', B'),$$

respectively

$$(G', B', T') \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T'),$$

is an equivalence between the category of pairs (G', T') (respectively, pairs (G', B'), respectively, triples (G', B', T')), where G' is a form of G and T' is a maximal torus of G' (respectively, B' is a Borel subgroup of G', respectively, $B' \supset T'$ is a Killing couple of G'), and the category of principal homogeneous bundles under the S-group H, where $H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$ (respectively, $H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B)$, respectively, $H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)$). Moreover, every principal homogeneous sheaf under H is representable and quasi-isotrivial, so that

$$H^1(S, H) = H^1_{\text{ét}}(S, H) = \mathrm{Fib}(S, H).$$

Remark 2.4. — Under the hypotheses of 2.2, the morphism denoted set-theoretically by $u \mapsto u(T)$ (respectively, $u \mapsto u(B)$, respectively, $u \mapsto (u(B), u(T))$) induces an isomorphism

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')/\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T) \simeq \mathrm{Tor}(G'),$$

respectively,

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')/\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B) \simeq \mathrm{Bor}(G'),$$

respectively,

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')/\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T) \simeq \mathrm{Kil}(G').$$

The proof is immediate: it suffices to prove the assertion locally for (fpqc), so one may assume $G \simeq G'$, and the result then reduces to Exp. XXII, 5.8.3(iii).

Remark 2.5. — The preceding results are immediately interpreted in terms of reduction of the structure group. If G' is a form of G, corresponding to the principal bundle $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$, to give a reduction of the structure group of this bundle to $\underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)$ amounts to giving a maximal torus T' of G'. The indicated bijections are, on the one hand, that of 2.4 and, on the other hand, the map

$$T' \longmapsto \underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T').$$

The same applies to Borel subgroups and Killing couples.

Proposition 2.6. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, and let T (respectively, T') be a maximal torus of G (respectively, G'). Then T_{ad} acts freely on $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')$, and the quotient*

$$P = \underline{\mathrm{Isom}}_{S\text{-gr}}(G, T; G', T')/T_{\mathrm{ad}}$$

is representable; it is a principal homogeneous bundle under

$$A = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, T)/T_{\mathrm{ad}},$$

where A is represented by a twisted constant S-scheme and is an extension of $\underline{\mathrm{Autext}}(G)$ by

$$W_{\mathrm{ad}(G)}(T_{\mathrm{ad}}) = \mathrm{Norm}_{\mathrm{ad}(G)}(T_{\mathrm{ad}})/T_{\mathrm{ad}}.$$

Moreover, if A is made to act on T in the evident way, the bundle associated with P is precisely T'.[13]

The first part of the proposition follows immediately from the preceding results. To prove the second, observe that there is an evident morphism

$$P \times_S T \longrightarrow T', \qquad (u, t) \longmapsto u(t).$$

To show that, after passage to the quotient by A, it induces an isomorphism, one may once again assume $(G, T) \simeq (G', T')$, in which case the assertion is immediate.

In an entirely analogous way, one has:

Proposition 2.7. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, and let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G'). If*

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G, B, T)/T_{\mathrm{ad}} \simeq \underline{\mathrm{Autext}}(G)$$

is made to act on T in the evident way, the bundle associated with $\underline{\mathrm{Isomext}}(G, G')$ is precisely T'.

Corollary 2.8. — *Let G and G' be two reductive S-groups which are inner twisted forms of one another, and let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G'). Then T and T' are isomorphic.*[14]

Remark 2.9. — It is not true in general that B and B' are isomorphic; they are nevertheless inner twisted forms of one another (cf. §5).

[13] Editor's note: Consequently, if P has a section, then T and T' are isomorphic; in particular, T_P and T'_P are isomorphic. See the addition 4.5.1, where this observation is developed in the case $G = \mathrm{\acute{E}p}_S(\mathscr{R})$.

[14] Editor's note: This follows from 2.7, since $\underline{\mathrm{Isomext}}(G, G')$ has a section (cf. the preceding editor's note).

One may develop "Isomint" variants[15] of the preceding results. We mention one:

Proposition 2.10. — *Let S be a scheme, let G and G' be two reductive S-groups having the same type at every point, let $B \supset T$ (respectively, $B' \supset T'$) be a Killing couple of G (respectively, G'), and let $u \in \underline{\mathrm{Isomext}}(G, G')(S)$. Consider*

$$\begin{aligned}&\underline{\mathrm{Isomint}}_u(G, B, T; G', B', T')\\ &\qquad = \underline{\mathrm{Isom}}_{S\text{-gr}}(G, B, T; G', B', T') \cap \underline{\mathrm{Isomint}}_u(G, G').\end{aligned}$$

It is a smooth affine S-scheme which is a principal homogeneous bundle under T_{ad}. In particular,

$$\underline{\mathrm{Isomint}}_u(G, B, T; G', B', T')(S) \neq \varnothing$$

if and only if the corresponding element of $H^1(S, T_{\mathrm{ad}})$ is zero.

To conclude this section, we prove two results that will be useful later.

Proposition 2.11. — *Let S be a scheme, let G be a reductive S-group, and let T be a maximal torus of G. The evident morphism*

$$T_{\mathrm{ad}} \xrightarrow{\sim} \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_T)$$

is an isomorphism.

This follows from the preceding statements. A direct proof was also given in the course of the proof of 1.5.2.

Corollary 2.12. — *Under the preceding hypotheses, there is an equivalence between the category of pairs (G', f), where G' is a form of G and f is an isomorphism from T onto a maximal torus of G', and the category of principal homogeneous bundles under T_{ad}.*

Corollary 2.13. — *If $H^1(S, T_{\mathrm{ad}}) = 0$, and if G' is a form of G having a maximal torus isomorphic to T, then G' is isomorphic to G.*

Corollary 2.14. — *Let S be a scheme such that $\mathrm{Pic}(S) = 0$, and let G be a reductive S-group of constant type. Then G is splittable if and only if G has a split maximal torus.*

Let G be a reductive group and let $\mathrm{rad}(G)$ be its radical (Exp. XXII, 4.3.9). Since $\mathrm{rad}(G)$ is central and characteristic in G, there is a canonical morphism

$$q : \underline{\mathrm{Autext}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{rad}(G)).$$

Proposition 2.15. — *Let G be a reductive S-group. The following sequence is exact:*

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_{\mathrm{rad}(G)}) \xrightarrow{p} \underline{\mathrm{Autext}}(G) \xrightarrow{q} \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{rad}(G)),$$

and $\ker(q) = \mathrm{im}(p)$ is an open and closed subscheme of $\underline{\mathrm{Autext}}(G)$, finite over S.

One may assume that G is split. The first assertion is immediate, and the second follows from Exp. XXI, 6.7.5 and 6.7.7.

Writing

$$H = \underline{\mathrm{Aut}}_{S\text{-gr}}(G, \mathrm{id}_{\mathrm{rad}(G)}),$$

one obtains:

Corollary 2.16. — *There is an equivalence between the category of pairs (G', f), where G' is a form of G and f is an isomorphism from $\mathrm{rad}(G)$ onto the radical of G', and the category of principal bundles under a certain S-group scheme H, where H is such that there is an exact sequence*

$$1 \longrightarrow \mathrm{ad}(G) \longrightarrow H \longrightarrow F \longrightarrow 1,$$

in which the S-group F is étale and finite over S.

[15] Editor's note: Recall that $\underline{\mathrm{Isomint}}_u(G, G')$ is defined in 1.11.

3. Dynkin Scheme of a Reductive Group. Quasi-Split Groups

3.1. — Recall (Exp. XXI, 7.4.1) that a Dynkin diagram is a finite set endowed with the structure defined by a set of pairs of distinct elements (bonds) and a map to $\{1, 2, 3\}$ (lengths). To every pinned reduced root datum $\mathscr{R}$ there is associated a Dynkin diagram $\Delta(\mathscr{R})$, whose underlying set is the set of simple roots.

3.2. — Let S be a scheme. A Dynkin S-scheme is a finite twisted constant S-scheme Δ, endowed with the structure defined by a subscheme L of $\Delta \times_S \Delta$ having empty intersection with the diagonal, and by a morphism

$$\Delta \longrightarrow \{1, 2, 3\}_S.$$

If S' is a connected S-scheme,[16] $\Delta(S')$ is naturally endowed with the structure of a Dynkin diagram.

The following notions are defined immediately: an isomorphism of two Dynkin schemes, the base extension of a Dynkin scheme, and the constant Dynkin scheme associated with a Dynkin diagram.

Every étale descent datum on a Dynkin scheme is effective.

3.3. — We shall associate a Dynkin S-scheme with every reductive S-group G. First suppose that G is splittable over S. For every pinning E of G, let $\Delta(E)$ be the constant Dynkin scheme associated with the pinned root datum defined by E. If E and E' are two pinnings of G, then by 1.5 there is a unique inner automorphism of G over S carrying E to E'. This automorphism of G defines an isomorphism

$$a_{EE'} : \Delta(E) \xrightarrow{\sim} \Delta(E').$$

The $a_{EE'}$ plainly form a transitive system, so the $\Delta(E)$ may be identified (that is, one takes their inductive limit); the result is a constant Dynkin scheme denoted by $\mathrm{Dyn}(G)$. If now G is an arbitrary reductive S-group, there is an étale covering family $\{S_i \to S\}$ such that

G_{S_i} is splittable. Arguing as above, one therefore obtains a canonical descent datum on the $\mathrm{Dyn}(G_{S_i})$, and by descent this constructs a Dynkin S-scheme $\mathrm{Dyn}(G)$.

3.4. — This construction has the following properties (which, moreover, essentially characterize it):

(i) With every reductive S-group G is associated a Dynkin scheme $\mathrm{Dyn}(G)$; with every isomorphism

$$u : G \xrightarrow{\sim} G'$$

is functorially associated an isomorphism

$$\mathrm{Dyn}(u) : \mathrm{Dyn}(G) \xrightarrow{\sim} \mathrm{Dyn}(G').$$

(ii) If S' is an S-scheme and G is a reductive S-group, then

$$\mathrm{Dyn}(G \times_S S') \simeq \mathrm{Dyn}(G) \times_S S'.$$

(iii) If E is a pinning of G defining the pinned root datum with Dynkin diagram Δ, then

$$\mathrm{Dyn}(G) \simeq \Delta_S.$$

[16] Editor's note: The hypothesis that S' be connected has been added.

(iv) If u is an inner automorphism of G, then $\mathrm{Dyn}(u)$ is the identity automorphism of $\mathrm{Dyn}(G)$.

3.5. — Let S be a scheme and G a reductive S-group. It is clear that the functor

$$\underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G))$$

of automorphisms of $\mathrm{Dyn}(G)$ as a Dynkin scheme is represented by a finite twisted constant S-scheme. By 3.4(i) and (ii), there is a canonical morphism

$$\underline{\mathrm{Aut}}_{S\text{-gr}}(G) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G)),$$

which, by (iv), factors through a morphism

$$\underline{\mathrm{Autext}}(G) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G)).$$

More generally, if G and G' are two reductive S-groups, there is a canonical morphism

$$\underline{\mathrm{Isomext}}(G, G') \longrightarrow \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G), \mathrm{Dyn}(G')).$$

In particular, if G' is an inner twisted form of G (1.11), the Dynkin schemes $\mathrm{Dyn}(G)$ and $\mathrm{Dyn}(G')$ are isomorphic.

3.6. — If G is semisimple (respectively, adjoint or simply connected), the morphism

$$\underline{\mathrm{Autext}}(G) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G))$$

is a monomorphism (respectively, an isomorphism). Indeed, one may assume that G is pinned, and the assertion reduces to the corresponding result for pinned reduced root data (cf. Exp. XXI, 7.4.5).

There is an analogous result for the Isom functors. It follows in particular that two semisimple adjoint (respectively, simply connected) S-groups are inner twisted forms of one another if and only if their Dynkin schemes are isomorphic.

3.7. — One can give a different construction of the Dynkin scheme associated with a reductive group. Let $\mathscr{R}$ be a pinned reduced root datum and G a reductive S-group of type $\mathscr{R}$, and write $\Delta(\mathscr{R})$ for the Dynkin diagram defined by $\mathscr{R}$. There is, by 3.5, a canonical morphism

$$A_S(\mathscr{R}) = \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{\acute{E}p}_S(\mathscr{R})) \longrightarrow \underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\Delta(\mathscr{R})_S).$$

The reductive S-group G corresponds (1.17) to a bundle

$$\underline{\mathrm{Isom}}_{S\text{-gr}}(\mathrm{\acute{E}p}_S(\mathscr{R}), G),$$

principal homogeneous under $A_S(\mathscr{R})$. The associated bundle under $\underline{\mathrm{Aut}}_{\mathrm{Dyn}}(\Delta(\mathscr{R})_S)$ corresponds to a form over S of $\Delta(\mathscr{R})_S$: it is $\mathrm{Dyn}(G)$. In other words, this associated bundle is precisely

$$\underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\Delta(\mathscr{R})_S, \mathrm{Dyn}(G)).$$

In this form the proof is immediate.

3.8. Dynkin Scheme and Killing Couples. — Let S be a scheme, G a reductive S-group, and $B \supset T$ a Killing couple of G. There is a canonical morphism

$$i : \mathrm{Dyn}(G) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr}}(T, \mathbb{G}_{m,S})$$

which identifies $\mathrm{Dyn}(G)$ with the "scheme of simple roots of B relative to T." This morphism is defined immediately by descent from the pinned case. Observe also that the data of T and i allow one to reconstruct B (the "one-to-one correspondence between systems of simple roots and systems of positive roots").

It follows from the preceding description of $D = \mathrm{Dyn}(G)$ that there is a canonical root of B_D relative to T_D: this root α_D is the image, under $i(D)$, of the identity morphism of D. One obtains from it a canonical invertible $\mathscr{O}_D$-module $\mathfrak{g}_D$:

$$\mathfrak{g}_D = (\mathfrak{g} \otimes_{\mathscr{O}_S} \mathscr{O}_D)^{\alpha_D}.$$

In the pinned case,

$$D = \coprod_{\alpha\in\Delta} S_\alpha,$$

where every S_α is a copy of S, and $\mathfrak{g}_D$ is then the $\mathscr{O}_D$-module inducing $\mathfrak{g}_\alpha$ on S_α, for every $\alpha \in \Delta$.

3.9. Quasi-Pinnings. Quasi-Pinned Groups. — If G is a reductive S-group, a *quasi-pinning of G* means the data

(i) of a Killing couple $B \supset T$ of G;

(ii) of a section

$$X \in \Gamma(\mathrm{Dyn}(G), \mathfrak{g}_D)^\times.$$

A reductive S-group is called *quasi-splittable* if it has a quasi-pinning. A *quasi-pinned group* is a reductive group equipped with a quasi-pinning.

Let $B \supset T$ be a Killing couple of the reductive S-group G. Then G is quasi-pinnable relative to this Killing couple if and only if $\mathfrak{g}_D$ has a section nonzero at every point, that is, if the element of $\mathrm{Pic}(\mathrm{Dyn}(G))$ defined by $\mathfrak{g}_D$ is zero. Suppose in particular that S is semilocal. Then $\mathrm{Dyn}(G)$ is also semilocal, and hence $\mathrm{Pic}(\mathrm{Dyn}(G)) = 0$. Thus:

Proposition 3.9.1. — *Let S be a semilocal scheme and G a reductive S-group. Then G is quasi-splittable if and only if it has a Borel subgroup.*

[17] Indeed, S is affine, so by the first assertion of Exp. XXII, 5.9.7, if G has a Borel subgroup B, it also has a Killing couple $B \supset T$. Then, because $\mathrm{Pic}(D) = 0$, the module $\mathfrak{g}_D$ has a nowhere-zero section X.

Continue to let A be a semilocal ring and $S = \mathrm{Spec}(A)$. For every reductive S-group G, the morphism

$$\mathrm{Bor}(G) \longrightarrow S$$

is surjective (because G_s has Borel subgroups for every $s \in S$), smooth, and projective (Exp. XXII, 5.8.3). It therefore has sections after a finite, surjective, étale extension of the base.[18] It follows that:

[17] Editor's note: The reference to Exp. XXII, 5.9.7 has been given in detail.

[18] Editor's note: The original referred to EGA IV, §24, which did not appear. The point is to use "Bertini's theorem." We give the details of the argument, which were indicated to us by O. Gabber. Let $X \to S$ be a surjective smooth projective morphism. Replacing S by a connected component, one may assume that X/S has relative dimension d at every point (cf. EGA IV_4, 17.10.2). One may also assume that X is a closed subscheme of a projective space $\mathbb{P}^n_S = \mathbb{P}(E)$, where E is a free A-module of rank $n+1$ (cf. EGA II, 5.3.3). Let $s_1, \ldots, s_r$ be the closed points of S. By Bertini's theorem (see, for example, [Jou83], I, 6.10), there is an open subset U of

$$P = \mathbb{P}(E^*)^d$$

with nonempty fibers such that, for every point $u = (f_1, \ldots, f_d)$ of U_{s_i}, the intersection of $X_{\kappa(u)}$ with the d hyperplanes of $\mathbb{P}^n_{\kappa(u)}$ defined by u is étale over $\kappa(u)$. After shrinking U, one may assume that U is the

Corollary 3.9.2. — *Let S be a semilocal scheme and G a reductive S-group. There is an étale, finite, and surjective morphism $S' \to S$ such that $G_{S'}$ is quasi-splittable.*

Remark 3.9.3. — Under the preceding hypotheses, let T be a maximal torus of G (cf. Exp. XIV, 3.20). One may moreover assume that $G_{S'}$ is quasi-splittable relative to $T_{S'}$: it suffices to apply the preceding argument to the "scheme of Borel subgroups containing T," which is finite and étale over S (Exp. XXII, 5.5.5(ii)).

3.10. — Let E and E' be two quasi-pinnings of the reductive S-group G. There is a unique inner automorphism of G carrying E to E'. Indeed, one reduces immediately to the split case, where the assertion has already been proved (1.5; it suffices to observe that for an inner automorphism of G, preserving a pinning is equivalent to preserving the underlying quasi-pinning). One concludes, as in §1, that a

quasi-pinning of the reductive S-group G defines a splitting h of the exact sequence:

$$1 \longrightarrow \underline{\mathrm{ad}}(G) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \underset{p}{\overset{h}{\rightleftarrows}} \underline{\mathrm{Autext}}(G) \longrightarrow 1$$

the image of h being the subgroup of $\underline{\mathrm{Aut}}_{S\text{-gr}}(G)$ that preserves the quasi-pinning.

Likewise, if G and G' are two quasi-pinned S-groups, one defines naturally the subfunctor

$$\underline{\mathrm{Isom}}_{S\text{-gr},\, q\text{-}\acute{e}p}(G, G')$$

of $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$. The projection from $\underline{\mathrm{Isom}}_{S\text{-gr}}(G, G')$ to $\underline{\mathrm{Isomext}}(G, G')$ induces an isomorphism

$$(*) \qquad \underline{\mathrm{Isom}}_{S\text{-gr},\, q\text{-}\acute{e}p}(G, G') \xrightarrow{\ \sim\ } \underline{\mathrm{Isomext}}(G, G').$$

Theorem 3.11. — *Let S be a scheme, let $\mathscr{R}$ be a pinned reduced root datum such that $\mathrm{\acute{E}p}_S(\mathscr{R})$ exists (cf. Exp. XXV), and let E be its group of automorphisms. Consider the following three categories:*

(i) the category Rev *of principal Galois coverings of S with group E (the morphisms are isomorphisms of E-bundles);*

(ii) the category Redext*, whose objects are the reductive S-groups of type $\mathscr{R}$ (Exp. XXII, 2.7), the morphisms from G to G' being the elements of* $\underline{\mathrm{Isomext}}(G, G')(S)$*;*

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complement in affine space $\mathbb{A}^{nd}_S$ of the zero locus $V(Q)$ of a polynomial Q of degree $m > 0$. An easy induction on the number of variables then shows that $V(Q)$ cannot contain every rational point of $\mathbb{A}^{nd}_{\kappa(s_i)}$ if $|\kappa(s_i)| > m$. Since the morphism

$$A \longrightarrow \prod_i \kappa(s_i)$$

is surjective, one can find a monic polynomial $R \in A[X]$ of degree m whose image in $\kappa(s_i)[X]$ is irreducible when $\kappa(s_i)$ is finite and has m distinct roots when $\kappa(s_i)$ is infinite. Put

$$A' = A[X]/(R), \qquad S' = \mathrm{Spec}(A').$$

Then $S' \to S$ is étale, finite, and surjective, and the residue field at each of its closed points $s'_1, \ldots, s'_t$ has cardinality at least $2m$. Thus U has a rational point u_i above every closed point of S', and, since

$$A' \longrightarrow \prod_i \kappa(s'_i)$$

is surjective,these points lift to a section u of $P_{S'}$. Let Z be the intersection of $X_{S'}$ with the d hyperplanes of $\mathbb{P}^n_{S'}$ defined by u, and let V be the open subset of Z consisting of the points at which Z is étale over S'. By EGA IV$_3$, 11.3.8, V contains the fibers $Z_{s'_i}$ for every i. Since $\pi : Z \to S'$ is proper, it follows that the closed set $\pi(Z - V)$ is empty, whence $V = Z$. Thus $Z \to S'$ is surjective, étale, and proper, hence finite, as is the composite $Z \to S$; this gives the desired section of $X \to S$.

(iii) the category $\mathsf{Q\acute{e}p}$ *of quasi-pinned reductive S-groups of type $\mathscr{R}$ (the morphisms are the isomorphisms preserving the quasi-pinnings).*

These three categories are equivalent. More precisely, there is a diagram of functors, commutative up to functorial isomorphisms:

$$\begin{tikzcd}
\mathsf{Rev} \arrow[rr, "q\acute{e}p"] & & \mathsf{Q\acute{e}p} \arrow[dl, "i"] \\
& \mathsf{Redext} \arrow[ul, "\mathrm{rev}"] &
\end{tikzcd}$$

We describe these three functors below, leaving to the reader the verification that the diagram commutes.[19]

3.11.1. The functor i**.** — This is the evident functor:

$$i(G) = G, \qquad i(f) = \text{the image of } f \text{ under the isomorphism 3.10 } (*).$$

3.11.2. The functor *qép***.** — Let S' be a principal Galois covering of S with group E. Let

$$\big(G, T, M, \mathscr{R}, \Delta, (X_\alpha)_{\alpha\in\Delta}\big)$$

be a pinning of $G = \mathrm{\acute{E}p}_S(\mathscr{R})$, and let $(X'_\alpha)_{\alpha\in\Delta}$ be the corresponding pinning of

$$G' = G_{S'} = \mathrm{\acute{E}p}_{S'}(\mathscr{R}).$$

By Exp. XXIII, 5.5 bis, there is a unique group homomorphism

$$\theta : E \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(\mathrm{\acute{E}p}_S(\mathscr{R})) = A(\mathscr{R})(S)$$

such that, for every $h \in E$, the automorphism $\theta(h)$ induces $D_S(h)$ on T, and

$$\mathrm{Lie}(\theta(h))(X_\alpha) = X_{h(\alpha)} \qquad (\alpha \in \Delta).$$

Taking into account the action of E on S', this yields an action of E on G', compatible with the action of E on S' and preserving the canonical quasi-pinning of G'.[20] Since $S' \to S$ is a covering for the (fpqc) topology, it is an effective descent morphism for the fibered category of affine morphisms. Write

$$q\acute{e}p(S') = Q\text{-}\,\mathrm{\acute{E}p}_{S'/S}(\mathscr{R})$$

for the quasi-pinned S-group obtained by Galois descent.[21]

[19] Editor's note: In 3.11.4 we have sketched the verification that the functors i, rev, and *qép* are fully faithful and that the composite $\mathrm{rev}\circ i \circ q\acute{e}p$ is isomorphic to the identity functor of Rev.

[20] Editor's note: The preceding discussion expands the original in order to show that the action of E on G' is obtained by combining the natural action of E on $\mathrm{\acute{E}p}_{S'}(\mathscr{R})$ with the given action on S'. The canonical Killing couple of G' is preserved by this action, as is the quasi-pinning given by the element $\widetilde{X}'$ of

$$\Gamma\big(\Delta_{S'}, \mathrm{Lie}(G'/S')_{\Delta_{S'}}\big),$$

equal to X'_α on the copy of S' indexed by α. Indeed, since E acts on $\Delta_{S'}$ by permuting the copies of S', one has $h(\widetilde{X}') = \widetilde{X}'$ for every $h \in E$.

[21] Editor's note: See, for example, TDTE I, p. 22, Example 1. One may also describe $q\acute{e}p(S')$ as the twist of $G = \mathrm{\acute{E}p}_S(\mathscr{R})$ by the E-torsor S'/S, that is, the (fpqc) sheaf quotient of $S' \times_S G$ by the right action of E defined by

$$(s', g)\cdot h = (s'h, h^{-1}(g)).$$

3.11.3. The functor rev. — Let G be a reductive S-group of type $\mathscr{R}$. Write

$$\mathrm{rev}(G) = \underline{\mathrm{Isomext}}(\mathrm{\acute{E}p}_S(\mathscr{R}), G).$$

This is a principal homogeneous bundle under E_S (cf. 1.10 and 1.19), that is, an object of Rev.

3.11.4. — [22] It is clear from the isomorphism 3.10 $(*)$ that the functor i is fully faithful. On the other hand, if G_0, G, G' are reductive S-groups, the natural morphism

$$\underline{\mathrm{Isom}}(G_0, G) \times_S \underline{\mathrm{Isom}}(G, G') \longrightarrow \underline{\mathrm{Isom}}(G_0, G')$$

induces a morphism

$$(\star) \qquad \underline{\mathrm{Isomext}}(G_0, G) \times_S \underline{\mathrm{Isomext}}(G, G') \longrightarrow \underline{\mathrm{Isomext}}(G_0, G'),$$

since, for every $S' \to S$, if

$$g_0 \in G_0(S'), \quad g \in G(S'), \quad u \in \underline{\mathrm{Isom}}(G_0, G)(S'), \quad v \in \underline{\mathrm{Isom}}(G, G')(S'),$$

one has

$$v \circ \mathrm{int}(g) \circ u \circ \mathrm{int}(g_0) = v \circ u \circ \mathrm{int}\big(u^{-1}(g)g_0\big).$$

Taking $G_0 = \mathrm{\acute{E}p}_S(\mathscr{R})$, one obtains a map

$$\begin{aligned}\underline{\mathrm{Isomext}}(G, G')(S) \longrightarrow \mathrm{Hom}_{\mathsf{Rev}}\big(&\underline{\mathrm{Isomext}}(\mathrm{\acute{E}p}_S(\mathscr{R}), G),\\ &\underline{\mathrm{Isomext}}(\mathrm{\acute{E}p}_S(\mathscr{R}), G')\big),\end{aligned}$$

and, to show that this map is a bijection, one may assume by (fpqc) descent that G and G' are pinned, in which case it is evident.

Similarly, if S_1, S_2 are two objects of Rev, and if one writes

$$S_i \times^E \mathrm{\acute{E}p}_S(\mathscr{R})$$

for the S-group $q\acute{e}p(S_i)$ obtained by twisting $\mathrm{\acute{E}p}_S(\mathscr{R})$ by the E-torsor S_i (cf. editor's note (21)), there is a natural map

$$\begin{aligned}\mathrm{Hom}_{\mathsf{Rev}}(S_1, S_2) \longrightarrow \mathrm{Hom}_{\mathsf{Q\acute{e}p}}\big(&S_1 \times^E \mathrm{\acute{E}p}_S(\mathscr{R}),\\ &S_2 \times^E \mathrm{\acute{E}p}_S(\mathscr{R})\big).\end{aligned}$$

Since G is affine over S, and since E acts on G by group automorphisms, this sheaf is represented by an S-group $G^\sharp$, which is a "twisted" form of G, and $D^\sharp = \mathrm{Dyn}(G^\sharp)$ is the twist of the constant Dynkin scheme Δ_S by the torsor S'/S. Since E normalizes B and T, one obtains likewise a Killing couple $(B^\sharp, T^\sharp)$ of $G^\sharp$. On the other hand, put $\mathfrak{g} = \mathrm{Lie}(G/S)$ and $\mathfrak{g}^\sharp = \mathrm{Lie}(G^\sharp/S)$. For every $U \to S$, the sections of $\mathfrak{g}^\sharp$ over U are the E-equivariant S-morphisms

$$U \times_S S' \longrightarrow W(\mathfrak{g}).$$

Since $D^\sharp \times_S S'$ is E-isomorphic to $\Delta \times S'$, endowed with the action

$$(\alpha, s') \cdot h = (h^{-1}(\alpha), s'h),$$

the morphism $(\alpha, s') \mapsto X_\alpha$ gives a section of $\mathfrak{g}^\sharp$ over $D^\sharp$ which is a quasi-pinning, that is, a nowhere-zero section of

$$\big(\mathfrak{g}^\sharp \otimes_{\mathcal{O}_S} \mathcal{O}_{D^\sharp}\big)^{D^\sharp}$$

(cf. 3.8).

[22] Editor's note: This subsection number has been added.

To show that this map is a bijection, one may assume by (fpqc) descent that S_1 and S_2 are trivial E-bundles, in which case the assertion is evident.

Thus the three functors in the diagram are fully faithful. Moreover, for every object S' of Rev, there is a natural morphism

$$S' \longrightarrow \underline{\mathrm{Isomext}}\Big(\mathrm{\acute{E}p}_S(\mathscr{R}), S' \times^E \mathrm{\acute{E}p}_S(\mathscr{R})\Big),$$

and to see that it is an isomorphism, one may assume by (fpqc) descent that S' is a trivial E-bundle, in which case the assertion is evident.

There is therefore an isomorphism of functors

$$\mathrm{Id}_{\mathsf{Rev}} \longrightarrow \mathrm{rev} \circ i \circ q\acute{e}p.$$

Since rev is fully faithful, for every reductive S-group H (not necessarily quasi-split) one obtains a bijection

$$\underline{\mathrm{Isomext}}\Big(Q\text{-}\,\mathrm{\acute{E}p}_{\mathrm{rev}(H)/S}(\mathscr{R}), H\Big)\,(S) \simeq \mathrm{Aut}_{\mathsf{Rev}}(\mathrm{rev}(H)),$$

functorial in H. In particular, the identity morphism of $\mathrm{rev}(H)$ corresponds to a "canonical" element

$$u_0 \in \underline{\mathrm{Isomext}}\Big(Q\text{-}\,\mathrm{\acute{E}p}_{\mathrm{rev}(H)/S}(\mathscr{R}), H\Big)\,(S).$$

Moreover, every element u of this same set corresponds to an automorphism φ_u of $\mathrm{rev}(H)$, and $q\acute{e}p(\varphi_u)$ is the unique automorphism of the pinned S-group

$$Q\text{-}\,\mathrm{\acute{E}p}_{\mathrm{rev}(H)/S}(\mathscr{R})$$

such that

$$u_0 \circ q\acute{e}p(\varphi_u) = u.$$

This proves Theorem 3.11.

We develop one of the corollaries of 3.11.

Corollary 3.12. — *For every reductive S-group G, there is a quasi-pinned S-group $G_{q\text{-}\acute{e}p}$ and an "outer isomorphism"*

$$u \in \underline{\mathrm{Isomext}}(G_{q\text{-}\acute{e}p}, G)(S).$$

The pair $(G_{q\text{-}\acute{e}p}, u)$ is unique up to a unique isomorphism.

Indeed, one may assume that G has constant type $\mathscr{R}$, and one takes

$$G_{q\text{-}\acute{e}p} = Q\text{-}\,\mathrm{\acute{E}p}_{\mathrm{rev}(G)/S}(\mathscr{R}).$$

3.12.1. — Observe that the data of the pair $(G_{q\text{-}\acute{e}p}, u)$ canonically define the S-scheme (cf. 1.11)

$$Q = \underline{\mathrm{Isomint}}(G_{q\text{-}\acute{e}p}, G),\,[23]$$

which is a principal homogeneous bundle under $\mathrm{ad}(G_{q\text{-}\acute{e}p})$, and whose data are "equivalent" to those of the inner twisted form G of $G_{q\text{-}\acute{e}p}$. Moreover, Q is precisely the "scheme of quasi-pinnings of G," a definition that explains its structure as a principal homogeneous bundle (on the left) under $\mathrm{ad}(G)$ (3.10); compare 1.20.

[23] Editor's note: We write $\underline{\mathrm{Isomint}}(G_{q\text{-}\acute{e}p}, G)$ instead of $\underline{\mathrm{Isomint}}_u(G_{q\text{-}\acute{e}p}, G)$ (cf. 1.11), since the pair $(G_{q\text{-}\acute{e}p}, u)$ is unique up to a unique isomorphism.

Proposition 3.13. *Let S be a scheme, G a semisimple adjoint (resp. simply connected) S-group, $B \supset T$ a Killing couple of G,* $\mathrm{Dyn}(G)$ *the Dynkin S-scheme of G. There exists a canonical isomorphism of S-group schemes*

$$T \xrightarrow{\sim} \prod_{\mathrm{Dyn}(G)/S} \mathbb{G}_{m,\mathrm{Dyn}(G)}.$$

(Recall (cf. Exp. II, ğ 1) that the right-hand side is by definition the S-functor which to $S' \to S$ associates $G_m(\mathrm{Dyn}(G) \times_S S')$, or, what amounts to the same, $G_m(Dyn(G_{S'}))$.)

First proof. Let us do it for simplicity in the adjoint case. Consider the composite morphism

$$T \longrightarrow \prod_{D/S} T_D \longrightarrow \prod_{D/S} \mathbb{G}_{m,D}.$$

where the first morphism is the canonical morphism, the second is $\prod_{D/S} \alpha_D$ (we have written $D = \mathrm{Dyn}(G)$). To verify that this morphism is an isomorphism, one may assume G split; now, in this case, this is none other than the morphism $T \to (G_{m,S})^{\Delta}$ with components α, for $\alpha \in \Delta$, and the latter is an isomorphism (Exp. XXII, 4.3.8).

Second proof. By 2.8, 3.5 and 3.11, one may assume that $G = Q - \acute{E}p_{S'/S}(R)$, T being the canonical maximal torus. On S', one has by Exp. XXII 4.3.8, an isomorphism $T_{S'} \xrightarrow{\sim} (G_{m,S'})^{\Delta}$, defined by the simple roots (resp. the simple coroots). The group E operates on the right-hand side by permutation of Δ. Now the bundle associated to S'/S by $E \to \mathrm{Aut}(\Delta)$ is $\mathrm{Dyn}(G)$ (3.7), and one concludes at once.

Using Prop. 8.4 of the appendix, one deduces:

Corollary 3.14. *Under the preceding conditions, one has*

$$H^1(S,T) \xrightarrow{\sim} H^1(\mathrm{Dyn}(G), \mathbb{G}_m) \xrightarrow{\sim} \mathrm{Pic}(\mathrm{Dyn}(G)).$$

In particular, $H^1(S,T) = 0$ when S is semi-local.

Remark 3.15. Let S be a scheme, G (resp. G') an S-reductive group, $B \supset T$ (resp. $B' \supset T'$) a Killing couple of G (resp. G'), $u \in Isomext(G, G')(S)$. Set (cf. 2.10)

$$P = \underline{\mathrm{Isomint}}_u(G, B, T; G', B', T').$$

this is a principal homogeneous bundle under T_{ad} (by $(f,t) \mapsto f \circ int(t)$).

Let on the other hand $D = \mathrm{Dyn}(G) = \mathrm{Dyn}(G')$ (identified via u (3.5)), and let $L = Isom_D(g_D, g'_D)$ be the principal homogeneous bundle under $G_{m,D}$ defined by the invertible $\mathcal{O}_D$-module

$$\mathrm{Hom}_{\mathcal{O}_D}(\mathfrak{g}_D, \mathfrak{g}'_D) = \mathfrak{g}'_D \otimes (\mathfrak{g}_D)^{\vee}.$$

Each $f \in P(S')$ defines an isomorphism of $g_D \otimes_{O_S} O_{S'}$ onto $g'_D \otimes_{O_S} O_{S'}$, whence a canonical morphism

$$P \to \prod_{D/S} L.$$

This morphism is an isomorphism, compatible with the isomorphism of operator groups

$$T_{\mathrm{ad}} \xrightarrow{\sim} \prod_{D/S} \mathbb{G}_{m,D}.$$

defined above. Indeed, it suffices to verify it in the case where the two groups are pinned, where it is easy.

It follows in particular that in the isomorphism

$$H^1(S, T_{ad}) \xrightarrow{\sim} \mathrm{Pic}(\mathrm{Dyn}(G))$$

of 3.14, the class of the bundle P is transformed into $c\ell(g'^D) - c\ell(g_D)$. The bundle P is therefore trivial if and only if g'^D and g_D are isomorphic.

If (G, B, T) is quasi-splittable, for example if one takes for G the group $G'_{q-ép.}$, with its canonical Killing couple, it follows that the image of the class of P is none other than the obstruction to the quasi-splitting of G' defined in 3.9.

3.16. Symmetries

3.16.1.

Let S be a scheme, G an S-reductive group, $B \supset T$ a Killing couple of G. Write $D = \mathrm{Dyn}(G)$. Recall (Exp. XXII, 5.9.1) that there exists a unique Borel subgroup B^- of G such that $B \cap B^- = T$. If $X \in \Gamma(D, g_D)^\times$ defines a quasi-pinning of G relative to (B, T) (3.9), then $Y = -X^{-1} \in \Gamma(D, (g_D)^\vee)^\times$ defines a quasi-pinning of G relative to (B^-, T); one says that this is the *opposite quasi-pinning.*

If R is a pinned reduced root datum and if E is an R-pinning of the S-reductive group G, one defines an R-pinning E^- said to be *opposite to* E in the following way: one keeps the same maximal torus T, one takes the opposite of the isomorphism $D_S(M) \xrightarrow{\sim} T$, and one "pins" by $Y_\alpha = -X_\alpha^{-1} \in \Gamma(S, g^{-\alpha})^\times$, for $\alpha \in \Delta$. The quasi-pinning underlying E^- is the quasi-pinning opposite to the quasi-pinning underlying E.

Remark. In the notation of Exp. XIX, 3.1, if one sets

$$w_\alpha(X_\alpha) = \exp(X_\alpha)\exp(-X_\alpha^{-1})\exp(X_\alpha).$$

one has $w_\alpha(X_\alpha) = w_{-\alpha}(Y_\alpha)$ (loc. cit., 3.1 (vi)).

Proposition 3.16.2. *Let S be a scheme, G an S-reductive group.*

(i) Let T be a maximal torus of G; there exists a unique

$$i_T \in \left(\underline{\mathrm{Aut}}_{S\text{-gr.}}(G, T)/T_{\mathrm{ad}}\right)(S) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(T).$$

such that $i_T(t) = t^{-1}$ for every section t of T.

(ii) Let (B, T) be a Killing couple of G; there exists a unique section

$$w_{B,T} \in (\mathrm{Norm}_G(T)/T)(S) = W_G(T)(S).$$

such that $int(w_{B,T})(B) = B^-$ (with the obvious abuse of language).

(iii) Let $Q = (B, T, X)$ be a quasi-pinning of G, $Q^- = (B^-, T, Y)$ the opposite quasi-pinning; there exists a unique inner automorphism of G

$$n_Q \in Norm_{ad(G)}(T_{ad})(S) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(G)$$

such that $n_Q(Q) = Q^-$, that is, $n_Q(T) = T$, $n_Q(B) = B^-$, $n_Q(X) = Y$.

(iv) Let (R, E) be a pinning of G, (R, E^-) the opposite pinning; there exists a unique automorphism of G

$$u_E \in \underline{\mathrm{Aut}}_{S\text{-gr.}}(G, T) \subset \underline{\mathrm{Aut}}_{S\text{-gr.}}(G).$$

such that $u_E(E) = E^-$, i.e. $u_E(t) = t^{-1}$ for every section t of T, and $Ad(u_E)X_\alpha = Y_\alpha$ for every $\alpha \in \Delta$.

Proof. (ii) follows from Exp. XXII, 5.5.5 (ii), then (iii) follows from 3.10, and (iv) follows from Exp. XXIII, 4.1. Finally to prove (i), one may assume G pinned. Existence follows from (iv) for example, uniqueness from the fact that an automorphism of G inducing the identity on T is given by a section of T_{ad} (2.11).

Corollary 3.16.3. *One has*

$$i_T^2 = w_{B,T}^2 = n_Q^2 = u_E^2 = e.$$

Moreover, i_T (resp. u_E) is $\neq e$ if $G \neq e$, and $w_{B,T}$ (resp. n_Q) is $\neq e$ if G is not a torus.

Corollary 3.16.4. *In the situation of (iii) (resp. (iv)), n_Q projects onto $w_{B,T}$ (resp. u_E projects onto i_T) by the canonical morphism*

$$Norm_{ad(G)}(T_{ad}) \to W_{ad(G)}(T_{ad}) \simeq W_G(T),$$

resp.

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(G, T) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G, T)/T_{\mathrm{ad}}.$$

Corollary 3.16.5. *The preceding definitions are compatible with base change, and are functorial under isomorphism (in an obvious sense).*

Proposition 3.16.6. *(i) One can define uniquely for each reductive group G over a scheme S an element*

$$s_G \in Autext(G)(S)$$

in such a way that this construction is functorial in G under isomorphism, is compatible with base change, and is such that whenever T is a maximal torus of the S-reductive group G, s_G is the image of i_T under the canonical morphism

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(G, T)/T_{\mathrm{ad}} \longrightarrow \underline{\underline{\mathrm{Autext}}}(G).$$

(ii) One has $s_G^2 = e$, and s_G is a central element of $Autext(G)$.

(iii) Under the conditions of 3.16.2 (ii), if one identifies $\underline{\mathrm{Aut}}_{S\text{-gr.}}(G, B, T)/T_{ad}$ with $Autext(G)$ (2.2), one has

$$w_{B,T}i_T = i_T w_{B,T} = s_G.$$

(iv) Under the conditions of 3.16.2 (iv), if one identifies $\mathrm{Aut}_{S-gr.-ép}(G, R, E)$ with $Autext(G)$ (1.3 (iii)), one has

$$n_E u_E = u_E n_E = s_G.$$

Proof. (i) is proved without difficulty by descent. On the other hand, since i_T is evidently a central section of square e in $\underline{\mathrm{Aut}}_{S\text{-gr.}}(T)$, (ii) follows immediately; (iii) is a consequence of (iv) by descent. Finally, under the conditions of (iv), it is clear that $n_E u_E = u_E n_E$ and that this automorphism of G respects the pinning; modulo the identification made, it is therefore equal to its image in $Autext(G)$; but n_E is inner and u_E projects onto s_G.

Remark 3.16.7. (i) One determines s_G explicitly in each case of the classification thanks to (iii): for each irreducible pinned root datum it suffices to compose the symmetry through the origin with the symmetry in the Weyl group (i.e. the element w_0 of the Weyl group such that $w_0(\Delta) = -\Delta$). One finds the following results: one has $s_G = e$ except for A_n ($n \geqslant 2$), D_n (n odd) and E_6, in which case s_G is the unique non-trivial "outer automorphism".

(ii) The automorphism u_E is the one used to construct "the compact real forms" in the theory of semisimple Lie algebras.

Remark 3.16.8. We defined in 3.16.1 an involution in the S-scheme

$$Q = Isomint(G_{q-\acute{e}p.}, G)$$

of quasi-pinnings of G (cf. 3.12.1); by transport of structure from $G_{q-\acute{e}p.}$ to G, one sees at once that this involution is given by the action of an element of $ad(G_{q-\acute{e}p.})(S)$: the element n_0 defined (3.16.2 (iii)) by the canonical quasi-pinning of $G_{q-\acute{e}p.}$.

In the same way, one has defined an involution in the S-scheme

$$P = Isom_{S-gr.}(\acute{E}p_S(R), G)$$

of R-pinnings of G (cf. 1.20). Arguing as before, one sees that this involution is given by the action of the automorphism u_0 of $\acute{E}p_S(R)$ defined (3.16.2 (iv)) by the canonical pinning of $\acute{E}p_S(R)$.

4. Isotriviality of reductive groups and of principal bundles under reductive groups

4.1. Definitions. Isotriviality theorem

Definition 4.1.1. *Let S be a scheme, G an S-group scheme, P a principal homogeneous bundle under G. One says that P is* locally isotrivial *(resp.* semi-locally isotrivial) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open set) there exist an open set U of S containing s (resp. F) and a finite surjective étale morphism $S' \to U$ such that $P_{S'}$ is trivial.

Definition 4.1.2. *Let S be a scheme, G an S-reductive group. One says that G is* locally isotrivial *(resp.* semi-locally isotrivial) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open set) there exist an open set U of S containing s (resp. F) and a finite surjective étale morphism $S' \to U$ such that $G_{S'}$ is splittable.*

Remark 4.1.3. (i) The equivalence of categories 4.1.1 of 1.17 respects by definition local (resp. semi-local) isotriviality.

(ii) Add to the conditions of 4.1.1: G locally of finite presentation over S. Then the principal bundle P (or the reductive group G) is locally isotrivial (resp. semi-locally isotrivial) if and only if for every $S' \to S$, S' local (resp. semi-local), $P_{S'}$ is isotrivial (or $G_{S'}$ isotrivial), that is to say, if there exists $S'' \to S'$ finite surjective étale such that $P_{S''}$ is trivial (or $G_{S''}$ split).

(iii) In the case of tori, definition 4.1.2 coincides with that of Exp. IX, 1.1.

4.1.4.

Recall (Exp. XXII, 4.3 and 6.2) that for every reductive group G, we have introduced the groups $rad(G)$, $corad(G)$ and $d\acute{e}r(G)$. The groups $rad(G)$ and $corad(G)$ are tori, which are split when G is split; moreover, there exists an isogeny $rad(G) \to corad(G)$. The S-group $d\acute{e}r(G)$ is semisimple, one has $G/d\acute{e}r(G) = corad(G)$; it follows that for every principal homogeneous bundle P under G, $P/d\acute{e}r(G)$ is a principal homogeneous bundle under $corad(G)$. This said, one has:

Theorem 4.1.5. *Let S be a scheme, G an S-reductive group of constant type.*

(i) The following conditions are equivalent:

(a) G is locally (resp. semi-locally) isotrivial.

(b) The torus $rad(G)$ is.

(b') The torus $corad(G)$ is.

(c) The Galois covering of S associated to G (3.11) is.

If T is a maximal torus of G, these conditions are also equivalent to

(d) The torus T is locally (resp. semi-locally) isotrivial.

(ii) Let P be a principal homogeneous bundle under G; for P to be locally (resp. semi-locally) isotrivial, it is necessary and sufficient that the $corad(G)$-principal bundle $P/d\acute{e}r(G)$ be so.

Corollary 4.1.6. *The conditions of (i) are satisfied when G is semisimple or when S is locally noetherian and normal (or more generally geometrically unibranch). The conditions of (ii) are satisfied when G is locally (resp. semi-locally) isotrivial.*

For (i), the first assertion is trivial from (b), the second follows from (c) and Exp. X, 5.14 and 5.15. For (ii), it suffices to note that by virtue of theorem 90, a principal bundle under a split torus is semi-locally isotrivial.

4.2. Proof: the semisimple case

Let us first prove, for later reference:

Proposition 4.2.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G (resp. B a Borel subgroup, resp. $B \supset T$ a Killing couple of G), P a principal homogeneous bundle under G, G' the twisted form of G associated to P (via the morphism $int : G \to \mathrm{Aut}_{S-gr.}(G)$). One has canonical isomorphisms*

$$P/\operatorname{Norm}_G(T) \xrightarrow{\sim} \operatorname{Tor}(G'), \qquad P/B \xrightarrow{\sim} \operatorname{Bor}(G'), \qquad P/T \xrightarrow{\sim} \operatorname{Kil}(G').$$

By construction, G' is the quotient of $P \times_S G$ by a certain action of G ($((p, g'))g = (pg, g^{-1}g'g)$); one therefore has a morphism $P \times_S G \to G'$, that is, a morphism

$$P \to \operatorname{Hom}_S(G, G'),$$

which, as one sees at once, factors through a morphism

$$f : P \longrightarrow \operatorname{Isom}_{S\text{-gr.}}(G, G').$$

(to verify this assertion, one may assume $G = P$, in which case one has $G' = G$, $f = int$). The set-theoretic map $p \mapsto f(p)(T)$ defines a morphism

$$P \to Tor(G').$$

To verify that this morphism induces an isomorphism $P/Norm_G(T) \xrightarrow{\sim} Tor(G')$ as announced, one may again assume $P = G$ in which case one is reduced to Exp. XXII, 5.8.3 (iii). (In fact, *loc. cit.* should be replaced by the statement above.) One argues similarly for Bor and Kil.

Proposition 4.2.2. *Let S be a semi-local scheme, G an S-semisimple group of constant type.*

(i) G is isotrivial.

(ii) Every principal bundle under G is isotrivial.

Let us prove (i). Up to making a finite surjective étale extension of the base, one may, by 3.9.2, assume G quasi-split. But then G is isotrivial by construction (3.11, the group E being finite). To prove (ii), one may, by (i), assume G split; one is then reduced to:

Lemma 4.2.3. *Let S be a semi-local scheme. Every principal bundle under a split reductive group is isotrivial.*

Indeed, with the notation of 4.2.1, where $B \supset T$ denotes the canonical Killing couple of the split G, the S-scheme $Kil(G')$ possesses a section, after finite surjective étale extension of the base, by 3.9.2. One may therefore reduce the structure group of G to T; but T is split, so $H^1(S, T) = 0$ (theorem 90).

Corollary 4.2.4. *Let S be a scheme and*

$$1 \to G \to G' \to G'' \to 1$$

an exact sequence of S-group schemes (for (fpqc)), G being semisimple of constant type. Let P be a principal homogeneous bundle under G', suppose the sheaf[24] *P/G associated representable (for example G'' affine over S). For P to be locally isotrivial (resp. semi-locally isotrivial), it is necessary and sufficient that P/G be so (as bundle under G'').*

If P is trivial, P/G is too, which shows that the condition is necessary. Conversely, suppose S local (resp. semi-local) and P/G isotrivial, so trivialized by a finite surjective étale extension S' of S. Extending the base to S', one may reduce the structure group of $P_{S'}$ to $G_{S'}$. But S' is still semi-local and $G_{S'}$ semisimple of constant type, so the obtained bundle is isotrivial (4.2.2).

4.3. Proof: general case

Let us first note that 4.1.5 (ii) follows at once from the application of 4.2.4 to the exact sequence

$$1 \to d\acute{e}r(G) \to G \to corad(G) \to 1.$$

Let us therefore prove (i). If G is split, $rad(G)$ and $corad(G)$ are split, as is $rev(G)$; so (a) implies (b), (b'), and (c).

4.3.1.

One has (c) $\Rightarrow$ (a). Let R be the type of G; one has an exact sequence

$$1 \to ad(\acute{E}p_S(R)) \to A_S(R) \to E_S \to 1.$$

Applying 4.2.4 to the bundle $P = Isom_{S-gr.}(\acute{E}p_S(R), G)$ and to the associated bundle $rev(G) = P/ad(\acute{E}p_S(R))$, one has (c) $\Rightarrow$ (a).

4.3.2.

One has (b) $\Rightarrow$ (a). It suffices to prove that if $rad(G)$ is split, G is semi-locally isotrivial. Now let $G_0 = \acute{E}p_S(R)$; consider the category of pairs (G', f), where G' is a form of G_0 and f is an isomorphism of $rad(G_0)$ onto $rad(G)$.

One knows (2.16) that this category is equivalent to the category of principal homogeneous bundles under a certain S-group H extension of a finite twisted constant group by a semisimple group. It suffices to prove that every principal bundle under H is semi-locally isotrivial; this follows at once from 4.2.4.

[24] N.D.E.: We have replaced "bundle" by "sheaf", and then G' by G''.

4.3.3.

One has (b') ⇒ (a). One can argue as previously (it will moreover be the same group H that arises). One can also see that (b) and (b') are equivalent: a torus isogenous to a locally split torus is also locally split (cf. Exp. IX, 2.11 (iii)).

4.3.4.

One has (d) ⇒ (a). It suffices to prove that a group of constant type possessing a split maximal torus is semi-locally isotrivial; this follows at once from 2.14.

4.3.5.

One has (a) ⇒ (d). It suffices to prove that a maximal torus of a split group is semi-locally isotrivial. More precisely:

Lemma 4.3.6. *Let S be a semi-local scheme, G an S-reductive group, T_0 a split maximal torus of G, $W_0 = Norm_G(T_0)/T_0$ (this is a locally constant S-group, constant if G is of constant type, by 2.14), T a maximal torus of G.*

There exists a morphism $S' \to S$ which is principal homogeneous under W_0 (hence étale finite and surjective, and even principal Galois if G is of constant type) such that $T_{S'}$ is conjugate to $(T_0)_{S'}$ by an element of $G(S')$ (and so in particular split).

Indeed, one knows that $Transp_G(T_0, T)$ is a principal homogeneous bundle under $Norm_G(T_0)$ (cf. for example Exp. XI, 5.4 bis). Set $S' = Transp_G(T_0, T)/T_0$; this is a principal homogeneous bundle under W_0. Extending the base from S to S', one can reduce the structure group of $Transp_G(T_0, T)$ to T_0. Now S' is semi-local and T_0 split, so $Transp_G(T_0, T)$ possesses a section over S'.[25] QED

4.4. Use of the existence of maximal tori

Using Grothendieck's existence theorem for maximal tori (Exp. XIV, 3.20), one can considerably sharpen the preceding results. Let us state at once:

Theorem 4.4.1. *Let S be a semi-local scheme, R a pinned reduced root datum, W its Weyl group, E the group of its automorphisms. (Recall that E operates naturally on W and that the semi-direct product $A = W \cdot E$ is identified with the group of automorphisms of R non-pinned, cf. Exp. XXI, 6.7.2).*

(i) Every principal homogeneous bundle under $\acute{E}p_S(R)$ is trivialized by a principal Galois covering $S' \to S$ of group W.

(ii) Let G be an S-reductive group of type R; let $rev(G) = Isomext(\acute{E}p_S(R), G)$ be the associated Galois covering of S with group E. Let W_0 be the form of W_S associated to $rev(G)$. There exists a morphism $S' \to S$, which is a principal homogeneous bundle under W_0, such that $G_{S'}$ is quasi-splittable (i.e. possesses a Borel subgroup).

(iii) Every S-reductive group of type R is split by a principal Galois covering $\tilde{S} \to S$ with group A.

Let us first state:

Proposition 4.4.2. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, P a principal homogeneous bundle under G, G' the twisted form of G associated to P (one then has, cf. 4.2.1, a canonical isomorphism $P/Norm_G(T) \xrightarrow{\sim} Tor(G')$). Let T' be a maximal torus of G', defining a composite morphism*

$$S \longrightarrow \mathrm{Tor}(G') \xrightarrow{\sim} P/\mathrm{Norm}_G(T).$$

[25] N.D.E.: by "theorem 90".

Consider the canonical morphisms

$$P \longrightarrow P/T \longrightarrow P/\operatorname{Norm}_G(T).$$

and take their inverse images by the preceding morphism:

$$\begin{tikzcd} P \arrow[r] & P/T \arrow[r] & P/\operatorname{Norm}_G(T) \\ H \arrow[u] \arrow[r] & S' \arrow[u] \arrow[r] & S \arrow[u] \end{tikzcd}$$

Then S' (resp. H) is a principal homogeneous bundle over S (resp. S') under $W_G(T)$ (resp. $T_{S'}$). Moreover, if one makes $W_G(T)$ operate on T in the obvious way, the bundle associated to S' is isomorphic to T'.

The first two assertions are trivial, the last is proved like the corresponding assertion of 2.6.

Corollary 4.4.3. *Let S be a semi-local scheme, G an S-reductive group, T a maximal torus of G. Suppose one of the two following conditions is satisfied:*

(i) T is split.

(ii) T is contained in a Borel subgroup of G, and G is either adjoint, or simply connected.

Let moreover P be a principal homogeneous bundle under G. There exists an S-scheme S', which is a principal homogeneous bundle under $W_G(T)$, such that $P_{S'}$ is trivial.

Indeed, if G' is the form of G associated to P, then G' possesses a maximal torus T' (Exp. XIV, 3.20). Resuming the notation of the preceding proposition, one sees that $H^1(S', T_{S'}) = 0$ (by theorem 90 for (i), by 3.14 for (ii)). The morphism $H \to S'$ possesses a section, so $P_{S'}$ also possesses a section over S'. QED

Let us now prove the theorem. Assertion (i) is a particular case of the preceding corollary (take $G = \acute{E}p_S(R)$, endowed with its canonical split torus). Let us prove (ii). One knows (3.12) that G is an inner twisted form of

$$G_0 = Q - \acute{E}p_{rev(G)/S}(R).$$

If T_0 is the canonical maximal torus of G_0, $W_{G_0}(T_0)$ is the group W_0 described in the statement. The form G of G_0 corresponds to a principal homogeneous bundle P under $ad(G_0)$ ($P = Isomint(G_0, G)$). The group $W_{ad(G_0)}(T_0^{ad})$ is canonically isomorphic to W_0, and one obtains the wanted result by applying 4.4.3 to the situation $(ad(G_0), T_0^{ad}, P)$, hypothesis (ii) of 4.4.3 being well verified. Let us finally prove (iii). Resume the notation of (ii); one has a diagram

$$\begin{tikzcd} & \operatorname{rev}(G) \arrow[d, "E_S"'] \\ S' \arrow[r, "W_0"'] & S \end{tikzcd}$$

One knows that $G_{S'}$ is isomorphic to $(G_0)_{S'}$ and that $(G_0)_{rev(G)}$ is splittable. If one sets $\tilde{S} = S' \times_S rev(G)$, $G_{\tilde{S}}$ is splittable, and it only remains to verify that $\tilde{S}$ is indeed a principal Galois covering of S with group A, which follows from the following more general lemma (naturally valid in any site):

Lemma 4.4.4. *Let S be a scheme, G and H two S-group schemes, $G \to \operatorname{Aut}_{S-gr.}(H)$ an action of G on H, E a G-principal homogeneous bundle, F an H_0-principal homogeneous* *bundle, where H_0 is the form of H associated to E. Then $E \times_S F$ is endowed naturally with a structure of principal homogeneous bundle under the semi-direct product $H \cdot G$.*

Write $(e, g) \mapsto eg$ (resp. $(f, u) \mapsto fu$) for the action (on the right) of G on E (resp. of H_0 on F). Write

$$p : E \times_S H \longrightarrow H_0.$$

for the canonical projection (H_0 is by definition the quotient of $E \times_S H$ by G acting on it by the formula $(e, h)g = (eg, g^{-1}hg)$). Consider the morphism

$$r : E \times_S F \times_S H \times_S G \longrightarrow E \times_S F.$$

defined set-theoretically by

$$r(e, f, h, g) = \big(eg, f \cdot p(e, h)\big).$$

The morphism r indeed defines an action of the semi-direct product $H \cdot G$ on $E \times_S F$. Indeed, one has set-theoretically

$$r\big(r(e, f, h, g), h', g'\big) = \big(egg', f\, p(e, h)\, p(eg, h')\big).$$

but

$$p(e, h)p(eg, h') = p(e, h)p(e, gh'g^{-1}) = p(e, hgh'g^{-1}).$$

whence

$$r\big(r(e, f, h, g), h', g'\big) = r(e, f, hgh'g^{-1}, gg').$$

which had to be proved.

To prove now that this law is indeed a law of principal homogeneous bundle, one may assume that E and F are trivial, in which case one sees at once that $E \times_S F$ is also a trivial bundle.

4.5. Isotriviality of maximal tori of semisimple groups

[26]

Let us bring out here the following result, contained implicitly in 2.6 (cf. the N.D.E. (13)) and which was mentioned at the end of Exp. IX 1.2. Recall (XXIII 5.11) that, for every scheme S and every reduced root datum R, one writes $\acute{E}p_S(R)$ for the unique pinned S-group of type R (which exists by Exp. XXV) and $T_S(R)$ its canonical maximal torus.

Proposition 4.5.1. *Let G be an S-semisimple group of constant type R; assume that G possesses a maximal torus T (which is the case if S is semi-local, by XIV 3.20).*

(i) Then $P = Isom_{S-gr.}(\acute{E}p_S(R), T_S(R); G, T)/T_S(R)_{ad}$ is a principal S-bundle under the finite group $\mathrm{Aut}(R)$, *and T_P is a split maximal torus of G_P. Consequently, T is isotrivial.*

(ii) Moreover, for every finite set F of points of P contained in an affine open set, there exists an open set U of P containing F such that G_U is split.

Proof. The exact sequence

$$1 \longrightarrow T_S(\mathscr{R})_{\mathrm{ad}} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(\mathrm{\acute{E}p}_S(\mathscr{R}), T_S(\mathscr{R})) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(T_S(\mathscr{R})).$$

[26] *Editors' note.* This no. 4.5 has been added.

induces for every $S' \to S$ a morphism of bundles $P_{S'} \to Isom_{S'-gr.}(T_{S'}(R), T_{S'})$, so if $P_{S'}$ possesses a section, then $T_{S'}(R)$ and $T_{S'}$ are isomorphic. This is in particular the case for $S' = P$.

Note moreover that $\mathrm{Aut}_{S-gr.}(\acute{E}p_S(R), T_S(R))/T_S(R)_{ad}$ is none other than the constant S-group $\mathrm{Aut}(R)_S$, and that P is identified with the $\mathrm{Aut}(R)_S$-torsor $Isom(R, \tilde{R}(G,T))$, where $\tilde{R}(G,T)$ denotes the twisted root datum associated to (G,T) (cf. XXII 1.10).

Then (ii) follows from (i) and from 2.14, taking into account the fact that the g_α (α ranging over the roots of R) are free $\mathcal{O}_U$-modules of rank 1 on a sufficiently small affine open set U containing F.

5. Canonical decomposition of an adjoint or simply connected group

In this section, we shall use the results of no 1 to generalize to the case of schemes a classical decomposition of adjoint (resp. simply connected) groups. So as not to overburden the exposition indefinitely, the proofs are sketched and the details are left to the reader; in fact it is always a matter of absolutely standard proofs in the theory of principal bundles: reduction of the structure group, twisting, etc.

5.1.

Recall (Exp. XXI, 7.4) that a Dynkin diagram is a disjoint union of its connected components, which are Dynkin diagrams. Moreover, every non-empty connected Dynkin diagram corresponding to a root datum is isomorphic to one of the standard diagrams (A_n, B_n, ..., G_2) which were exhibited in Exp. XXI, 7.4.6. In the sequel, we shall be interested only in Dynkin diagrams whose connected components are of one of the preceding types. Let T be the set of these standard diagrams. For every Dynkin diagram D, let $n(t) = n_D(t)$ be the number of connected components of D isomorphic to t, where $t \in T$. The *type* of D is by definition $\sum_{t\in T} n_D(t)\cdot t$.

A Dynkin diagram of type t is said to be *simple of type* t, a Dynkin diagram of type $n\cdot t$ is said to be *isotypic of type* t. Let D_0 be the set of connected components of D and let

$$a : D_0 \to T$$

be the obvious map. The type of D is none other than $\sum_{x\in D_0} a(x)$.

5.2.

Let S be a scheme, D an S-Dynkin scheme (verifying the restrictive condition stated above). The cokernel of the pair of morphisms $L \rightrightarrows D$ ($L =$[27] scheme of bonds of D) is written D_0. This is the "scheme of connected components" of D (it

exists trivially when D is constant; the general case is deduced from this by descent); it is a finite twisted constant S-scheme. One has a canonical morphism

$$a : D_0 \to T_S.$$

For $t \in T$, set $a^{-1}(t) = D_{0,t}$; this is a subscheme of D_0, whose inverse image in D, written D_t, is the *isotypic component of type* t of the Dynkin scheme D. Each D_t is a subscheme of D, and one has

[27]N.D.E.: We have replaced R by L, as in 3.2.

$$D = \coprod_{t \in T} D_t.$$

Note that the degree of $D_{0,t}$ at $s \in S$ is $n(s,t)$, if the type of D at s is $\sum_t n(s,t) \cdot t$.

5.3.

In what follows, we shall consider only semisimple adjoint (resp. simply connected) groups. To simplify the language, we shall state the results for adjoint groups; all statements will remain valid if one substitutes everywhere "simply connected" for "adjoint".

Recall that an adjoint reduced root datum is determined up to isomorphism by the type of its Dynkin diagrams. We shall therefore say that an adjoint root datum R (resp. a semisimple adjoint group G) is *of type* $\sum n(t) \cdot t$ if its Dynkin diagrams are (resp. if its type is given by an adjoint root datum of type $\sum n(t) \cdot t$). We shall say that R or G is *simple of type* t (resp. *isotypic of type* t) if its type is t (resp. $n \cdot t$, $n > 0$).

If G is a semisimple adjoint group, we shall use the symbols $Dyn_0(G)$ and $Dyn_{0,t}(G)$ in the sense defined in 5.2.

5.4.

Let t_i, $i = 1, 2, \ldots, n$, be distinct elements of T, and let R_i be a pinned adjoint root datum isotypic of type t_i. Consider the product $R = R_1 \times \cdots \times R_n$ (Exp. XXI, 6.4.1). Let S be a scheme such that the different $\acute{E}p_S(R_i)$ exist (cf. Exp. XXV). One verifies at once that there exists a canonical isomorphism

$$\mathrm{\acute{E}p}_S(\mathscr{R}) = \mathrm{\acute{E}p}_S(\mathscr{R}_1) \times_S \cdots \times_S \mathrm{\acute{E}p}_S(\mathscr{R}_n). \tag{*}$$

Proposition 5.5.— *Let S be a scheme and G an adjoint semisimple S-group. There is a unique decomposition*

$$G \simeq \prod_{t \in \mathrm{T}} G_t,$$

where G_t is an adjoint semisimple S-group, isotypic of type t. Moreover, the preceding decomposition induces isomorphisms

$$\mathrm{Dyn}_t(G) \simeq \mathrm{Dyn}(G_t), \qquad \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \simeq \prod_{t \in \mathrm{T}} \underline{\mathrm{Aut}}_{S\text{-gr.}}(G_t).$$

This was proved above when G is split. In the general case one may suppose that G has constant type $\mathscr{R}$. Using the preceding decomposition of $A_S(\mathscr{R})$ and 1.17, one obtains the desired decomposition of G. The other assertions then follow by descent.

Remark 5.6.— More generally, if G and H are two adjoint semisimple groups, there are canonical isomorphisms in the following commutative diagram:

$$\begin{tikzcd}
\underline{\mathrm{Isom}}_{S\text{-gr.}}(G,H) \arrow[r, "\sim"] \arrow[d] & \prod_{t\in\mathrm{T}} \underline{\mathrm{Isom}}_{S\text{-gr.}}(G_t,H_t) \arrow[d] \\
\underline{\mathrm{Isomext}}(G,H) \arrow[r, "\sim"] \arrow[d, "\wr"] & \prod_{t\in\mathrm{T}} \underline{\mathrm{Isomext}}(G_t,H_t) \arrow[d, "\wr"] \\
\underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G),\mathrm{Dyn}(H)) \arrow[r, "\sim"] & \prod_{t\in\mathrm{T}} \underline{\mathrm{Isom}}_{\mathrm{Dyn}}(\mathrm{Dyn}(G_t),\mathrm{Dyn}(H_t)).
\end{tikzcd}$$

Remark 5.7.— One can give the following intrinsic characterization of G_t, which we state without proof: it is the largest reductive subgroup of G that is invariant (and indeed characteristic) and isotypic of type t.

The preceding proposition reduces the study of adjoint semisimple groups to that of isotypic adjoint semisimple groups. We now undertake this study.

5.8.

Let $\mathscr{R}$ be a pinned reduced adjoint simple root datum of type t, let I be a nonempty finite set, and let $\mathscr{R}^I$ be the product root datum of copies $\mathscr{R}_i$ of $\mathscr{R}$, for $i \in I$. Denote by E the group of automorphisms of $\mathscr{R}$, identified with the group of automorphisms of its Dynkin diagram D. The Dynkin diagram of $\mathscr{R}^I$ is identified with D^I, and hence there is an exact sequence

$$1 \longrightarrow \mathrm{Aut}(D)^I \longrightarrow \mathrm{Aut}(D^I) \longrightarrow \mathrm{Aut}(I) \longrightarrow 1,$$

where $\mathrm{Aut}(I)$ is the group of permutations of I. It follows that the canonical isomorphism

$$\mathrm{\acute{E}p}_S(\mathscr{R})^I \simeq \mathrm{\acute{E}p}_S(\mathscr{R}^I)$$

induces an exact sequence

$$1 \longrightarrow A_S(\mathscr{R})^I \longrightarrow A_S(\mathscr{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S) \longrightarrow 1.$$

The last group is the constant S-group associated with $\mathrm{Aut}(I)$. Notice also that I is canonically identified with the set of connected components of D^I.

Let G be an S-semisimple group of type $\mathscr{R}^I$, thus defining (cf. 1.17) a principal homogeneous bundle P under $A_S(\mathscr{R}^I)$. The definitions by descent of $\mathrm{Dyn}(G)$ (3.7) and of $\mathrm{Dyn}_0(G)$ (5.2) show that the bundle associated with P by

$$A_S(\mathscr{R}^I) \longrightarrow \underline{\mathrm{Aut}}(I_S)$$

is precisely $\underline{\mathrm{Isom}}_S(I_S, \mathrm{Dyn}_0(G))$, corresponding to the form $\mathrm{Dyn}_0(G)$ of I_S. Using again the equivalence of categories 1.17 and the preceding exact sequence, one deduces formally that there is a reductive $\mathrm{Dyn}_0(G)$-group G_0 of type $\mathscr{R}$, and an S-isomorphism

$$G \simeq \prod_{\mathrm{Dyn}_0(G)/S} G_0.$$

Proposition 5.9.— *Let S be a scheme and G an adjoint semisimple S-group isotypic of type t. There exist an adjoint semisimple $\mathrm{Dyn}_0(G)$-group G_0, simple of type t, and a unique S-isomorphism*

$$G \simeq \prod_{\mathrm{Dyn}_0(G)/S} G_0.$$

Moreover, this isomorphism induces an exact sequence

$$1 \longrightarrow \prod_{\mathrm{Dyn}_0(G)/S} \underline{\mathrm{Aut}}_{\mathrm{gr.}}(G_0) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \longrightarrow \underline{\mathrm{Aut}}_S(\mathrm{Dyn}_0(G)) \longrightarrow 1.$$

One may plainly suppose that G has constant type $n\,t$. The first assertion has already been proved (and uniqueness is clear). The second then follows from the split case by descent.

Combining 5.5 and 5.9 gives:

Proposition 5.10.— *Let S be a scheme, let G be an adjoint semisimple S-group, and put $\mathfrak{D} = \mathrm{Dyn}(G)$.*

(i) There is a canonical decomposition

$$G \simeq \prod_{t\in \mathrm{T}} \prod_{\mathfrak{D}_{0,t}/S} G_{0,t} \simeq \prod_{\mathrm{Dyn}_0(G)/S} G_0,$$

where each $G_{0,t}$ is a simple adjoint $\mathrm{Dyn}_{0,t}(G)$-group of type t; equivalently, G_0 is an adjoint semisimple $\mathrm{Dyn}_0(G)$-group whose type at $x \in \mathrm{Dyn}_0(G)$ is $a(x) \in \mathrm{T}$, where $a : \mathrm{Dyn}_0(G) \to \mathrm{T}_S$ was defined in 5.2.

(ii) The preceding decompositions induce isomorphic exact sequences, written vertically below. Here $\underline{\mathrm{Aut}}_{S,a}(\mathfrak{D}_0)$ denotes the scheme of automorphisms of $\mathrm{Dyn}_0(G)$ commuting with $a : \mathrm{Dyn}_0(G) \to \mathrm{T}_S$:

$$\begin{tikzcd}
1 \arrow[d] & 1 \arrow[d] \\
\prod_{t\in\mathrm{T}} \prod_{\mathfrak{D}_{0,t}/S} \underline{\mathrm{Aut}}_{\mathrm{gr.}}(G_{0,t}) \arrow[r, "\sim"] \arrow[d] & \prod_{\mathrm{Dyn}_0(G)/S} \underline{\mathrm{Aut}}_{\mathrm{gr.}}(G_0) \arrow[d] \\
\underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \arrow[r, equal] \arrow[d] & \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) \arrow[d] \\
\prod_{t\in\mathrm{T}} \underline{\mathrm{Aut}}_S(\mathrm{Dyn}_{0,t}(G)) \arrow[r, "\sim"] \arrow[d] & \underline{\mathrm{Aut}}_{S,a}(\mathrm{Dyn}_0(G)) \arrow[d] \\
1 & 1
\end{tikzcd}$$

Corollary 5.11.— *Under the preceding hypotheses, the following three categories are equivalent:*

(i) the category of principal homogeneous bundles under G;

(ii) the category of principal homogeneous bundles under G_0;

(iii) the product, for $t \in \mathrm{T}$, of the categories of principal homogeneous bundles under the $G_{0,t}$.

This follows formally from the preceding decompositions and 8.4.

Corollary 5.12.— *There are canonical isomorphisms*

$$\underline{\mathrm{Tor}}(G) \simeq \prod_{t\in \mathrm{T}} \prod_{\mathfrak{D}_{0,t}/S} \underline{\mathrm{Tor}}(G_{0,t}) \simeq \prod_{\mathrm{Dyn}_0(G)/S} \underline{\mathrm{Tor}}(G_0),$$

and likewise with $\underline{\mathrm{Tor}}$ *replaced by* $\underline{\mathrm{Bor}}$*, respectively by* $\underline{\mathrm{Kil}}$*.*

This is immediate from the split case.

Remark 5.13.— The canonical morphism

$$\mathrm{Dyn}(G) \longrightarrow \mathrm{Dyn}_0(G)$$

allows one to regard $\mathrm{Dyn}(G)$ as a Dynkin scheme over $\mathrm{Dyn}_0(G)$; in fact, it is the Dynkin scheme $\mathrm{Dyn}(G_0)$ of G_0.

Likewise, if $T \subset B$ is a Killing couple of G, corresponding canonically to a Killing couple $T_0 \subset B_0$ of G_0, then the obstructions to the quasi-splitting of G and G_0, which lie by 3.9 in

$$\mathrm{Pic}(\mathrm{Dyn}(G)) = \mathrm{Pic}(\mathrm{Dyn}(G_0)),$$

coincide. Hence:

Corollary 5.14.— *The following conditions are equivalent:*

(i) G is quasi-splittable;

(ii) G_0 is quasi-splittable;

(iii) each $G_{0,t}$, $t \in \mathrm{T}$, is quasi-splittable.

6. Automorphisms of Borel Subgroups of Reductive Groups

Lemma 6.1.— *Let S be a scheme, let (G, T, M, R) be a split S-group, let Δ be a system of simple roots of R, and let B be the corresponding Borel subgroup. Then B^u is generated, as an* (fppf) *sheaf of groups, by the U_α, $\alpha \in \Delta$.*

Let H be the subgroup sheaf of B^u generated by the U_α, $\alpha \in \Delta$. We prove $H \supset U_\beta$, for $\beta \in R_+$, by induction on $\mathrm{ord}(\beta) = \mathrm{ord}_\Delta(\beta) > 0$ (cf. Exp. XXI, 3.2.15). The assertion holds by hypothesis when $\mathrm{ord}(\beta) = 1$. Suppose $\mathrm{ord}(\beta) > 1$, and suppose $U_\gamma \subset H$ whenever $\mathrm{ord}(\gamma) < \mathrm{ord}(\beta)$. There is an $\alpha \in \Delta$ such that $\beta - \alpha \in R_+$ (Exp. XXI, 3.1.1). Let p be the largest integer such that $\beta - p\alpha = \beta' \in R_+$. Then $U_\alpha, U_{\beta'} \subset H$ and $\beta' - \alpha \notin R$. Thus it remains to prove:

Lemma 6.2.— *Keep the notation of Exp. XXIII, 6.4. Suppose $p = 1$, that is, $\beta - \alpha$ is not a root. If H is a subgroup sheaf of B^u such that $U_\alpha, U_\beta \subset H$, then $U_{i\alpha+j\beta} \subset H$ whenever $i\alpha + j\beta \in R$, $i > 0$, $j > 0$.*

Distinguish four cases according as $q = 0, 1, 2, 3$. In what follows, x and y denote arbitrary sections of $\mathbb{G}_{a,S'}$, where $S' \to S$.

For $q = 0$, there is nothing to prove.

For $q = 1$, one has

$$p_{\alpha+\beta}(\pm x) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S'),$$

and hence $U_{\alpha+\beta} \subset H$.

For $q = 2$, one similarly has

$$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2 y) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S').$$

Thus, after changing some signs,

$$F(x,y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2 y) \in H(S').$$

Since $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ commute,

$$F(x,1)F(1,-x) = p_{2\alpha+\beta}(x^2 - x).$$

[28] If $a \in \mathbb{G}_a(S)$, the equation $X^2 - X = a$ defines a free surjective extension of S, namely $\operatorname{Spec} \mathcal{O}_S[X]/(X^2 - X - a)$. Consequently $p_{2\alpha+\beta}(a) \in H(S')$, so $U_{2\alpha+\beta} \subset H$, and then $U_{\alpha+\beta} \subset H$, since $F(1,a) \in H(S')$.

For $q = 3$, one similarly has

$$F(x,y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2 y)p_{3\alpha+\beta}(x^3 y)p_{3\alpha+2\beta}(x^3 y^2) \in H(S'),$$

and, since

$$p_{3\alpha+2\beta}(\pm x) = F(1,1)^{-1}p_\beta(-x)F(1,1)p_\beta(x) \in H(S'),$$

one obtains $U_{3\alpha+2\beta} \subset H$, and therefore

$$K(x,y) = p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2 y)p_{3\alpha+\beta}(x^3 y) \in H(S').$$

Computing

$$K(x+y,1)K(-x,1)K(1,y)^{-1}$$

modulo $U_{3\alpha+2\beta}$, one finds

$$p_{2\alpha+\beta}(2x^2 + y^2 + 2xy - y)p_{3\alpha+\beta}(y^3 + 3xy^2 + 3x^2 y - y) \in H(S').$$

If $a \in \mathbb{G}_a(S)$, the "equations"

$$x^2 = -xy - y + 1 - a, \qquad y^2 = 3y - 2 + 3a$$

define a free surjective extension of S. One then has $y^3 + 3xy^2 + 3x^2 y - y = 0$, and the preceding expression gives $p_{2\alpha+\beta}(a) \in H(S')$. [29] Thus H contains $U_{2\alpha+\beta}$ and $U_{3\alpha+2\beta}$, and

$$E(x,y) = p_{\alpha+\beta}(xy)p_{3\alpha+\beta}(x^3 y) \in H(S').$$

Since $p_{\alpha+\beta}(xy)$ and $p_{3\alpha+\beta}$ commute, the preceding computation applies:

$$E(1,x)E(1,-x) = p_{3\alpha+\beta}(x^3 - x).$$

Hence $U_{3\alpha+\beta} \subset H$, and then $U_{\alpha+\beta} \subset H$.

[28] *Editors' note.* We corrected $F(-1,x)$ to $F(1,-x)$.

[29] *Editors' note.* We corrected $p_{3\alpha+\beta}(a)$ to $p_{2\alpha+\beta}(a)$ and supplied the end of the argument.

Remark 6.2.1.— The preceding proof shows that the hypothesis "H contains U_α and U_β" could have been replaced by: "H contains U_α or U_β, and H is invariant under $\mathrm{int}(U_\alpha)$ and $\mathrm{int}(U_\beta)$."

Theorem 6.3.— *Let S be a scheme, let G and G' be two semisimple S-groups, and let B (respectively B') be a Borel subgroup of G (respectively G'). Every isomorphism $B \xrightarrow{\sim} B'$ is induced by a unique isomorphism $G \xrightarrow{\sim} G'$.*

The assertion is local for the étale topology, so we may suppose that G is split: $G = (G, T, M, R)$, with B defined by the system R_+ of positive roots of R. The given isomorphism $u : B \xrightarrow{\sim} B'$ induces an isomorphism from T onto a maximal torus T' of B', hence of G'. The given isomorphism $T \simeq D_S(M)$ gives an isomorphism $T' \simeq D_S(M)$ in which the elements of R_+ become the constant roots of B' relative to T', and hence the elements of R become the constant roots of G' relative to T'. Since G and G' are semisimple, the coroots are determined by duality (Exp. XXI, 1.2.5); this proves that (T', M, R) is a splitting of G' such that $\mathscr{R}(G) = \mathscr{R}(G')$.

Applying Exp. XXIII, 5.1 (the uniqueness theorem), one obtains a unique isomorphism $G \xrightarrow{\sim} G'$ agreeing with u on T and on the U_α, $\alpha \in \Delta$. By 5.1, its restriction to B is u.

Remark 6.3.1.— (i) Using Exp. XXII, 4.1.9 and arguing as in *loc. cit.*, 4.2.12, one may replace "isomorphism" in the theorem by "isogeny" (respectively, "central isogeny").

(ii) The theorem is false for reductive groups. For example, take $G = G' = \mathrm{SL}_2 \times \mathbb{G}_m$, identified with the group

$$\left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & h \end{pmatrix} \;\middle|\; ad - bc = 1,\ h \text{ invertible} \right\}.$$

take $B = B'$ to be the Borel subgroup defined by $c = 0$, and let u be the following automorphism of B:

$$u(a, b, d, h) = (a, b, d, ah).$$

Corollary 6.4.— *The functor*

$$(G, B) \longmapsto B$$

from the category of pairs (G, B), where G is an S-semisimple group and B is a Borel subgroup of G, to the category of S-group schemes, with isomorphisms as morphisms, is fully faithful.

Corollary 6.5.— *Let S be a scheme, G an S-semisimple group, B a Borel subgroup of G, and B' an S-group locally isomorphic to B for the* (fpqc) *topology. Then B' is locally isomorphic to B for the local finite étale topology,*[30] *and there exists an S-semisimple group G' of which B' is a Borel subgroup. Moreover, G' is unique up to a unique isomorphism inducing the identity on B'.*

Corollary 6.6.— *Let S be a scheme, G an S-semisimple group, and B a Borel subgroup of G. Then $\underline{\mathrm{Aut}}_{S\text{-gr.}}(B)$ is representable by an affine smooth S-scheme, $\underline{\mathrm{Autext}}(B)$ is representable by an étale finite S-scheme, and the evident morphisms induce an isomorphism of exact sequences.*

[30] *Editors' note.* This topology is denoted (étf) in Exposé IV, 6.3. In other words, for every $s \in S$ there exist an open neighborhood U of s and a finite surjective étale morphism $V \to U$ such that $B' \times_S V \simeq B \times_S V$.

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \underline{B}_{\mathrm{ad}}(G) & \longrightarrow & \underline{\mathrm{Aut}}_{S\text{-gr.}}(G,B) & \longrightarrow & \underline{\mathrm{Autext}}(G) & \longrightarrow & 1 \\ & & \big\downarrow{\wr} & & \big\downarrow{\wr} & & \big\downarrow{\wr} & & \\ 1 & \longrightarrow & \underline{B}_{\mathrm{ad}} & \longrightarrow & \underline{\mathrm{Aut}}_{S\text{-gr.}}(G) & \longrightarrow & \underline{\mathrm{Autext}}(B) & \longrightarrow & 1. \end{array}$$

This follows immediately from 2.1 and the preceding results. We leave to the reader the task of developing the analogues of the results of nos. 1, 2, 3, and 4 in the foregoing setting.

Remark 6.7.— If S is a scheme and B an S-group, then B can be a Borel subgroup of only one semisimple group, uniquely determined by B. It remains to characterize those S-groups B which are Borel subgroups of semisimple groups.[31]

7. Representability of the functors $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$, for G reductive

7.1. The split case

7.1.1.

Let S be a scheme, G a pinned S-group, T its maximal torus, and Δ the set of simple roots. For $\alpha \in \Delta$, let

$$u_\alpha \in U_\alpha^\times(S), \qquad w_\alpha \in \mathrm{Norm}_G(T)(S)$$

be the elements defined by the pinning.

Let H be an S-group scheme. We are interested in the functor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$, and more precisely in the morphism

$$q : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, H).$$

Let

$$f_T : T \longrightarrow H$$

be an S-group homomorphism and put $q^{-1}(f_T) = \mathscr{F}$. Thus

$$\mathscr{F}(S') = \Big\{ f \in \mathrm{Hom}_{S'\text{-gr.}}(G_{S'}, H_{S'}) \,\Big|\, f|_{T_{S'}} = (f_T)_{S'} \Big\}.$$

There is a morphism of S-functors

$$i : \mathscr{F} \longrightarrow H^{2n},$$

where $n = \mathrm{Card}(\Delta)$, defined on points by

$$i(f) = \big(f(u_\alpha), f(w_\alpha)\big)_{\alpha \in \Delta}.$$

By Exposé XXIII, 1.9, i is a monomorphism.

Proposition 7.1.2.— *If H is separated over S, then $\mathscr{F}$ is representable and i is a closed immersion.*

[31] *Editors' note.* One may ask whether every smooth affine S-group whose geometric fibers are Borel subgroups of semisimple groups is itself a Borel subgroup of a semisimple S-group. We have suppressed the counterexample in the original, with S the scheme of dual numbers over a field k, because it was erroneous, as M. Demazure pointed out.

The usual technique of relative representability[32] shows that it is enough to prove the following. Given sections

$$v_\alpha, h_\alpha \in H(S), \qquad \alpha \in \Delta,$$

the S-schemes S' for which there exists an S'-homomorphism

$$f : G_{S'} \longrightarrow H_{S'}$$

extending $(f_T)_{S'}$ and satisfying

$$f(u_\alpha) = v_\alpha, \qquad f(w_\alpha) = h_\alpha$$

are exactly those which factor through a certain closed subscheme of S. We may plainly suppose S affine.

For convenience, say that a morphism $Y \to X$[33] of affine schemes satisfies condition (L) if Y is the spectrum of an $\mathcal{O}(X)$-algebra which is a free $\mathcal{O}(X)$-module. Within the category of affine schemes, products and composites of morphisms satisfying (L) again satisfy (L), and (L) is stable under base change.

Lemma 7.1.3.— *Suppose that S is affine, and let $\alpha \in \Delta$. Consider the morphism*

$$a : T \times_S T \longrightarrow G_{a,S}, \qquad a(t,t') = \alpha(t) + \alpha(t').$$

Then:

(i) a is faithfully flat and of finite presentation;

(ii) if R is the fiber square of a, the structure morphism $R \to S$ satisfies (L).

The morphism a is surjective, since $\alpha : T \to G_{m,S}$ is surjective, and it is plainly of finite presentation. Since α satisfies (L), it is enough, for both assertions, to show that

$$u : G_{m,S}^2 \longrightarrow G_{a,S}, \qquad u(x,y) = x + y$$

satisfies (L). Equivalently, for every ring A, the ring

$$A[X, Y, X^{-1}, Y^{-1}]$$

must be free over its subring $A[X+Y]$.[34] Now $A[X,Y]$ is free with basis $\{1, X\}$ over

$$B = A[X+Y, XY],$$

because $X^2 - (X+Y)X + XY = 0$. Hence

$$A[X^{\pm 1}, Y^{\pm 1}] = A[X,Y]_{XY}$$

is free over

$$B_{XY} = A[X+Y, (XY)^{\pm 1}]$$

with basis $\{1, X\}$, and free over $A[X+Y]$ with basis

$$(XY)^p, \quad X(XY)^p, \qquad p \in \mathbb{Z}.$$

[32] *Editors' note.* See, for example, the proof of Exposé XXII, 5.8.1.
[33] *Editors' note.* We have corrected $X \to Y$ to $Y \to X$.
[34] *Editors' note.* We have simplified the proof in the original.

Lemma 7.1.4.— *Let $\alpha \in \Delta$, and let*

$$b : T \times_S T \longrightarrow H$$

be the morphism defined on points by

$$b(t, t') = \operatorname{int}(f_T(t))(v_\alpha)\ \operatorname{int}(f_T(t'))(v_\alpha).$$

Let $f_\alpha : U_\alpha \to H$ be an S-morphism. The following conditions are equivalent:[35]

(i) f_α *is a group homomorphism,* $f_\alpha(u_\alpha) = v_\alpha$, *and*

$$f_T(t) f_\alpha(x) f_T(t)^{-1} = f_\alpha(\alpha(t)x)$$

for all $t \in T(S')$, $x \in U_\alpha(S')$, and $S' \to S$;

(ii) $f_\alpha(u_\alpha) = v_\alpha$, *and*

$$f_\alpha(a(t, t')u_\alpha) = b(t, t')$$

for all $t, t' \in T(S')$ and $S' \to S$.

If f_α satisfies (i), then

$$f_\alpha(\alpha(t)u_\alpha) = \operatorname{int}(f_T(t))(v_\alpha),$$

which immediately gives (ii). Conversely, suppose (ii). Since a is an (fpqc) covering, to prove the last condition in (i) it suffices to show that, for $t, t', t'' \in T(S')$,

$$f_T(t)\, f_\alpha(a(t', t'')u_\alpha)\, f_T(t)^{-1} = f_\alpha(\alpha(t)a(t', t'')u_\alpha).$$

This is the same as

$$f_T(t) b(t', t'') f_T(t)^{-1} = b(tt', tt''),$$

which follows directly from the definition of b.

It remains to prove that f_α is a group homomorphism. The preceding property gives

$$\begin{aligned} f_\alpha(\alpha(t)u_\alpha) f_\alpha(\alpha(t')u_\alpha) &= \big(f_T(t) f_\alpha(u_\alpha) f_T(t)^{-1}\big)\big(f_T(t') f_\alpha(u_\alpha) f_T(t')^{-1}\big) \\ &= b(t, t') \\ &= f_\alpha\big(\alpha(t)u_\alpha + \alpha(t)\alpha(t')u_\alpha\big). \end{aligned}$$

Thus $f_\alpha(x + x') = f_\alpha(x) f_\alpha(x')$ whenever x, x' are sections of the schematically dense open subset $U_\alpha^\times \subset U_\alpha$. Exposé XVIII, 1.4 then gives the result.

7.1.5.

Fix provisionally an $\alpha \in \Delta$. Since a is faithfully flat and quasi-compact, $G_{a,S}$ is identified with the quotient of $T \times_S T$ by the equivalence relation R defined by a. Write $i_1, i_2 : R \to T \times_S T$ for its two projections.

$$R \underset{i_2}{\overset{i_1}{\rightrightarrows}} T \times_S T$$

[35] *Editors' note.* In what follows the group law of U_α is written additively. If $p_\alpha : G_{a,S} \xrightarrow{\sim} U_\alpha$ is the isomorphism with $p_\alpha(1) = u_\alpha$, and $x = p_\alpha(z)$, then $\alpha(t)x$ denotes $p_\alpha(\alpha(t)z)$, while $a(t, t')u_\alpha$ denotes $p_\alpha(a(t, t')) = \alpha(t)u_\alpha + \alpha(t')u_\alpha$.

For b to factor through a,[36] it is necessary and sufficient that

$$b \circ i_1 = b \circ i_2.$$

Let K be the equalizer of $b \circ i_1$ and $b \circ i_2$.

$$R \underset{b \circ i_2}{\overset{b \circ i_1}{\rightrightarrows}} H$$

Then the required condition is $K = R$. Since H is separated over S, K is a closed subscheme of R. On the other hand, R is essentially free over S, by 7.1.3.

7.1.6.

It follows that, for an S-scheme S', an f_α satisfying 7.1.4(i) exists over S', and is then unique, if and only if

$$K_{S'} = R_{S'}$$

and the resulting morphism satisfies

$$f_\alpha(u_\alpha) = v_\alpha.$$

The first condition is equivalent to requiring $S' \to S$ to factor through a certain closed subscheme of S (Exposé VIII, 6.4[37]). After replacing S by this closed subscheme, the second condition again defines a closed subscheme of S, since it is the equality of two sections of the separated S-scheme H.

After replacing S by the resulting closed subscheme, we may thus suppose that a morphism $f_\alpha : U_\alpha \to H$ satisfying 7.1.4(i) exists. Intersecting the closed subschemes obtained for all $\alpha \in \Delta$, we may suppose this condition holds for every $\alpha \in \Delta$.

7.1.7.

Similarly, for each $\alpha \in \Delta$, consider the two S-group homomorphisms

$$f_T \circ \mathrm{int}(w_\alpha), \qquad \mathrm{int}(h_\alpha) \circ f_T : T \longrightarrow H.$$

Since H is separated over S and T is essentially free over S, the same reasoning shows that, after replacing S by a closed subscheme, one may suppose that

$$f_T \circ \mathrm{int}(w_\alpha) = \mathrm{int}(h_\alpha) \circ f_T$$

for every $\alpha \in \Delta$.

7.1.8.

The generators-and-relations theorem (Exposé XXIII, 3.5) now shows that a group homomorphism $f : G \to H$ with the required properties exists if and only if a certain finite set of algebraic relations among the sections

$$h_\alpha, \qquad v_\alpha, \qquad f_T(\alpha^*(-1)), \qquad \alpha \in \Delta,$$

is satisfied:

$$R_i\Big((h_\alpha)_\alpha, (v_\alpha)_\alpha, \big(f_T(\alpha^*(-1))\big)_\alpha\Big) = e, \qquad i = 1, \ldots, n.$$

[36] *Editors' note.* That is, for condition 7.1.4(ii) to hold.

[37] *Editors' note.* See also the addition in Exposé $\mathrm{VI_B}$, 6.2.3.

Because H is separated over S, these relations again define a closed subscheme of S, proving Proposition 7.1.2.

Corollary 7.1.9.— *Let S be a scheme, G a split reductive S-group, T its maximal torus, and H a separated S-group scheme. Let $\mathcal{P}$ be a property of morphisms satisfying:*

(i) every closed immersion has $\mathcal{P}$;

(ii) a composite of morphisms having $\mathcal{P}$ has $\mathcal{P}$;

(iii) $\mathcal{P}$ is stable under base change;

(iv) the structure morphism $H \to S$ has $\mathcal{P}$.

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T,H)$$

is relatively representable by a separated morphism having $\mathcal{P}$.[38]

Indeed, we may suppose G pinned. The structure morphism $H^{2n} \to S$ has $\mathcal{P}$, and 7.1.2 gives the assertion.

Corollary 7.1.10.— *Let S be a scheme, G a split reductive S-group, and H a smooth S-group scheme with affine fibers.*[39] *Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H)$ is representable by an S-scheme locally of finite presentation and separated over S.*

Since H is smooth over S, its neutral component H^0 is defined; it has affine fibers and is smooth, separated, and of finite presentation over S (Exposé $\mathrm{VI_B}$, 3.10 and 5.5). Since G has connected fibers, H may be replaced by H^0. As G and H are then of finite presentation, we may suppose S noetherian and apply 7.1.9 with $\mathcal{P}$ equal to the property "of finite type". By Exposé XV, 8.9, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(T,H)$ is representable by a separated S-scheme locally of finite type.

Remark 7.1.11.— If H is affine over S, the reference to Exposé XV may be replaced by Exposé XI, 4.2.

7.2. General case

Proposition 7.2.1.— *Let S be a scheme, G a reductive S-group, and T a maximal torus of G. Let H be an S-group scheme whose structure morphism $H \to S$ satisfies:*

> (+) *Every point $s \in S$ has an open neighborhood U such that $H_U \to U$ is quasi-projective.*

Then the canonical morphism

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T,H)$$

is relatively representable by a morphism satisfying (+).

When G is split relative to T, apply 7.1.9 with $\mathcal{P} = (+)$. When G is locally isotrivial, for example when G is semisimple (4.2.2), note that the assertion is local for the local finite étale topology. Indeed, quasi-projectivity is local for the global finite étale topology and ensures

[38] *Editors' note.* More explicitly, for every representable $S' \to \underline{\mathrm{Hom}}_{S\text{-gr.}}(T,H)$, the fiber product $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H) \times S'$ is representable, and its morphism to S' is separated and has $\mathcal{P}$.

[39] *Editors' note.* We have suppressed the superfluous hypothesis that H be quasi-separated.

effectiveness of descent for that topology (SGA 1, VIII, 7.7). The general case follows from the next lemma.

Lemma 7.2.2.— *Let S be a scheme, G a reductive S-group, T a maximal torus of G, G' the derived group of G (Exposé XXII, 6.2), and $T' = T \cap G'$ the maximal torus of G' corresponding to T (Exposé XXII, 6.2.8). The natural diagram involving G, G', T, T' is a pushout in the category of S-group sheaves. Equivalently, for every S-group sheaf H, the corresponding square of Hom-functors is cartesian.*

$$\begin{tikzcd} G & G' \arrow[l] \\ T \arrow[u] & T' \arrow[l] \arrow[u] \end{tikzcd}$$

$$\begin{tikzcd} \underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H) \arrow[r] \arrow[d] & \underline{\mathrm{Hom}}_{S\text{-gr.}}(G',H) \arrow[d] \\ \underline{\mathrm{Hom}}_{S\text{-gr.}}(T,H) \arrow[r] & \underline{\mathrm{Hom}}_{S\text{-gr.}}(T',H) \end{tikzcd}$$

If $\mathrm{rad}(G)$ is the radical of G, then $\mathrm{rad}(G) \subset T$, so

$$\mathrm{rad}(G) \cap G' = \mathrm{rad}(G) \cap T' = K.$$

Multiplication in G induces isogenies (Exposé XXII, 6.2)

$$i : G' \times_S \mathrm{rad}(G) \longrightarrow G, \qquad j : T' \times_S \mathrm{rad}(G) \longrightarrow T.$$

Let

$$f_{G'} : G' \to H, \qquad f_T : T \to H, \qquad f_{T'} : T' \to H$$

be S-group homomorphisms such that

$$f_{G'}|_{T'} = f_T|_{T'} = f_{T'}.$$

We show that there is a unique S-group homomorphism $f : G \to H$ inducing $f_{G'}$ and f_T. Put

$$f_{\mathrm{rad}} = f_T|_{\mathrm{rad}(G)}$$

[40] and define

$$f_1 = f_{G'} \cdot f_{\mathrm{rad}} : G' \times_S \mathrm{rad}(G) \longrightarrow H.$$

For f to exist, and it is plainly then unique, it is necessary and sufficient that f_1 be trivial on the kernel of i, that is, that $f_{G'}$ and f_{rad} coincide on K. The existence of f_T similarly shows that $f_{T'}$ and f_{rad} coincide on K. Finally, $f_{G'}$ and $f_{T'}$ coincide on $K \subset T'$, proving the lemma.

Arguing as in 7.1.10, Proposition 7.2.1 gives:

Corollary 7.2.3.— *Let S be a scheme, G a reductive S-group, and H a smooth, quasi-projective S-group scheme with affine fibers. Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G,H)$ is representable by an S-scheme locally of finite presentation and separated over S.*

[40] *Editors' note.* We have replaced f_r by f_{rad}.

7.3. Phenomena specific to characteristic zero

If G and H are smooth S-groups, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, there is a canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie}}(\mathfrak{g}, \mathfrak{h}),$$

where the functor on the right has the evident meaning.

Proposition 7.3.1.— *Let S be a scheme of characteristic zero, that is, a $\mathbb{Q}$-scheme; let G be a reductive S-group, and H a smooth, quasi-projective S-group scheme with affine fibers.*

(i) $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ *is representable by a smooth separated S-scheme.*

(ii) If G is semisimple, this scheme is affine and of finite presentation over S.

(iii) If G is simply connected (Exposé XXII, 4.3.3), the canonical morphism

$$\mathrm{Lie} : \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-Lie}}(\mathfrak{g}, \mathfrak{h})$$

is an isomorphism.

Lemma 7.3.2.— *Let k be a field of characteristic zero, G a reductive k-group, V a finite-dimensional k-vector space, and $G \to \mathrm{GL}(V)$ a linear representation. Then*

$$H^0(G, V) = H^0(\mathfrak{g}, V), \qquad H^i(G, V) = 0 \quad (i > 0).$$

The first equality is valid in general for a smooth connected group.[41] For a reductive group it can also be proved as follows. We may suppose k algebraically closed, hence G split and generated by subgroups isomorphic to $G_{m,k}$,[42] and the assertion for this group is immediate.

The first equality shows that the functor $H^0(G, V)$ is exact in V when G is semisimple: $\mathfrak{g}$ is then a semisimple Lie algebra, and one applies [BLie], §I.6, Exercise 1(b). The assertion remains true when G is reductive. Indeed, introducing the radical R of G[43] and the semisimple quotient G/R, one has

$$H^0(G, V) = H^0(G/R, H^0(R, V)),$$

so $H^0(G, -)$ is a composite of two exact functors. Exposé I, 5.3.1 then gives $H^i(G, V) = 0$ for $i > 0$.

Remark 7.3.3.— Under the preceding hypotheses, if G is semisimple, then

$$H^1(\mathfrak{g}, V) = H^2(\mathfrak{g}, V) = 0;$$

see [BLie], *loc. cit.*, (b) and (d).

[41] *Editors' note.* The inclusion $V^G \subset V^{\mathfrak{g}}$ is clear. Conversely, $C = \mathrm{Centr}_G(V^{\mathfrak{g}})$ is a closed subgroup of G, smooth because $\mathrm{char}(k) = 0$. By Exposé II, 5.3.1, $\mathrm{Lie}(C) = \mathrm{Centr}_{\mathfrak{g}}(V^{\mathfrak{g}}) = \mathfrak{g}$; because C is smooth and G connected, $C = G$. Thus $V^{\mathfrak{g}} \subset V^G$. See also [DG70], §II.6, Proposition 2.1(c).

[42] *Editors' note.* The union of the maximal tori of G is dense in G; cf. *Bible*, §6.5, Theorem 5, or [Ch05], §6.6, Theorem 6.

[43] *Editors' note.* Observe that R is a torus.

7.3.4.

We prove the proposition. By 7.2.3, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is already representable by an S-scheme locally of finite presentation and separated over S. To show it smooth, it suffices to show it infinitesimally smooth (Exposé XI, 1.8),[44] and this follows from Exposé III, 2.8(i) and 7.3.2. This proves (i).

We next show that (ii) follows from (iii). First it is enough to prove that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is affine over S, since it is then of finite presentation, being smooth by (i). In any case,

$$\underline{\mathrm{Hom}}_{\mathscr{O}_S\text{-Lie}}(\mathfrak{g}, \mathfrak{h})$$

is represented by a closed subscheme of

$$\underline{\mathrm{Hom}}_{\mathscr{O}_S\text{-mod}}(\mathfrak{g}, \mathfrak{h}) \simeq \mathrm{W}(\mathfrak{g}^\vee \otimes_{\mathscr{O}_S} \mathfrak{h})\,,$$

which is an affine S-scheme. Thus the desired conclusion follows when G is simply connected.

In general, we may suppose G split, so that $G \simeq \mathrm{\acute{E}p}_S(\mathscr{R})$. Introduce the simply connected root datum $\mathrm{sc}(\mathscr{R})$ (Exposé XXI, 6.5.5). The existence theorem (Exposé XXV, 1.1) gives a simply connected S-group $\widetilde{G}$ and a central isogeny

$$\widetilde{G} \longrightarrow G.$$

Its kernel K is a finite diagonalizable S-group, hence a twisted constant S-group because S has characteristic zero. The functor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H)$ is plainly represented by a separated S-scheme. For example, if $K \simeq (\mathbb{Z}/n\mathbb{Z})_S$, then it is isomorphic to

$$\ker\Big(H \xrightarrow{n} H\Big)\,.$$

There is an exact sequence of pointed S-schemes

$$1 \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H) \xrightarrow{u} \underline{\mathrm{Hom}}_{S\text{-gr.}}(\widetilde{G}, H) \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(K, H).$$

Thus u is a closed immersion, and $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is affine over S.

7.3.5.

It remains to prove (iii). As in the proof of (i), and using 7.3.3, one shows that

$$\underline{\mathrm{Hom}}_{\mathscr{O}_S\text{-Lie}}(\mathfrak{g}, \mathfrak{h})$$

is smooth over S. We may therefore suppose $S = \mathrm{Spec}(k)$, where k is algebraically closed of characteristic zero. It is enough to show that Lie is bijective on k-points and induces a bijection on tangent spaces at corresponding points. First we show that

$$\mathrm{Hom}_{k\text{-gr.}}(G, H) \longrightarrow \mathrm{Hom}_{k\text{-Lie}}(\mathfrak{g}, \mathfrak{h})$$

is bijective.

If $u : G \to H$ is a k-group homomorphism, its graph is a connected subgroup of $G \times_k H$, determined by its Lie algebra, which is the graph of $\mathrm{Lie}(u)$. Hence the map is injective. Conversely, if $v : \mathfrak{g} \to \mathfrak{h}$ is a Lie algebra homomorphism, its graph $\mathfrak{k}$ is a Lie subalgebra of $\mathfrak{g} \times \mathfrak{h}$, isomorphic to $\mathfrak{g}$. Because $\mathfrak{g}$ is its own derived algebra, so is $\mathfrak{k}$. A theorem of Chevalley

[44] *Editors' note.* This means that for every local artinian S-scheme S', with closed point s', every homomorphism of $\kappa(s')$-groups $G_{s'} \to H_{s'}$ lifts to an S'-group homomorphism $G_{S'} \to H_{S'}$. By Exposé III, 2.8, this follows from the vanishing of $H^2(G_{s'}, V)$, where $V = \mathrm{Lie}(H_{s'}/\kappa(s'))$.

[Ch51], §14, Theorem 15, therefore shows that $\mathfrak{k}$ is the Lie algebra of a connected subgroup K of $G \times_k H$.[45]

Since $\mathfrak{k} \simeq \mathfrak{g}$ is semisimple, K is a semisimple k-group (the French prints G here; the argument and the following sentence require K). Moreover,

$$\dim K = \dim \mathfrak{k} = \dim \mathfrak{g} = \dim G,$$

and $K \cap H$ is finite because its Lie algebra is zero. Thus the canonical morphism

$$\mathrm{pr}_1 : K \longrightarrow G$$

is finite and dominant; because G is connected, it is surjective, and hence an isogeny. Since G is simply connected and K is semisimple, Exposé XXI, 6.2.7 shows that pr_1 is an isomorphism. Consequently K is the graph of a k-group homomorphism $u : G \to H$ with $\mathrm{Lie}(u) = v$.

Finally, let $u : G \to H$ be a k-group homomorphism. The tangent space at u to

$$\underline{\mathrm{Hom}}_{k\text{-gr.}}(G, H)$$

identifies with $Z^1(G, \mathfrak{h})$ (Exposé II, 4.2), where G acts on $\mathfrak{h}$ through $\mathrm{Ad}_H \circ u$. Similarly, the tangent space to

$$\underline{\mathrm{Hom}}_{k\text{-Lie}}(\mathfrak{g}, \mathfrak{h})$$

at $\mathrm{Lie}(u)$ identifies with $Z^1(\mathfrak{g}, \mathfrak{h})$. We must show that

$$Z^1(G, \mathfrak{h}) \longrightarrow Z^1(\mathfrak{g}, \mathfrak{h})$$

is bijective.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathfrak{h}^G & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(G,\mathfrak{h}) & \longrightarrow & H^1(G,\mathfrak{h}) \\
 & & \downarrow & & \| & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathfrak{h}^{\mathfrak{g}} & \longrightarrow & \mathfrak{h} & \longrightarrow & Z^1(\mathfrak{g},\mathfrak{h}) & \longrightarrow & H^1(\mathfrak{g},\mathfrak{h})
\end{array}$$

By 7.3.2 and 7.3.3,

$$\mathfrak{h}^G = \mathfrak{h}^{\mathfrak{g}}, \qquad H^1(G, \mathfrak{h}) = 0 = H^1(\mathfrak{g}, \mathfrak{h}),$$

which gives the desired conclusion.

Remark 7.3.6.— It is presumably true that, if S is locally noetherian, G a simply connected semisimple S-group, H a smooth S-group scheme, and $\widehat{G}$, $\widehat{H}$ their formal completions along the identity section, then every homomorphism $\widehat{G} \to \widehat{H}$ comes from a unique homomorphism $G \to H$. This would generalize 7.3.1(iii). When $S = \mathrm{Spec}(k)$ and H is affine and of finite type, this follows from a theorem of Dieudonné [Di57], §15, Theorem 4.[46]

[45] *Editors' note.* See also [Bo91], II 7.9. We have also added the following sentence.

[46] *Editors' note.* See Exposé $\mathrm{VII_B}$ for the definition of the formal groups $\widehat{G}$ and $\widehat{H}$. Suppose that $S = \mathrm{Spec}(k)$, where k is a field. By loc. cit., 2.2.1, giving a morphism of formal k-groups $v : \widehat{G} \to \widehat{H}$ is equivalent to giving a morphism of Hopf k-algebras $\phi : \mathrm{Dist}(G) \to \mathrm{Dist}(H)$, where $\mathrm{Dist}(G)$ denotes the algebra of distributions (at the origin) of G; see Exposé $\mathrm{VII_A}$, 2.1, [DG70], §II.4, 6.1, or [Ja87], I 7.7. It is called the hyperalgebra of G in [Di57] and [Ta75]. Theorem 4 of [Di57] (see also [Ta75], 0.3.4 (f) and (g)) generalizes in this context the theorem of Chevalley used in 7.3.5: one obtains a closed connected k-subgroup $K \subset G \times H$ such that $\widehat{K}$ equals the graph of v. Since $\mathrm{Dist}(K) \to \mathrm{Dist}(G)$ is an isomorphism, $K \to G$ is finite étale. It follows that K is semisimple and then, by Exposé XXI, 6.2.7, that $K \to G$ is an isomorphism (because G is simply connected; see also [Ta75], 1.8 and 2.2). More generally, loc. cit. studies the k-groups G having property (SC): every finite étale morphism of k-groups $G' \to G$, with G' connected, is an isomorphism. Finally, the preceding argument shows that every finite-dimensional $\mathrm{Dist}(G)$-module is, uniquely, a G-module; for an extension to a split reductive k-group G (or even to a Borel subgroup of G), see [Ja87], II 1.20 (and the references given there).

7.4. An example

As an example, we shall determine

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S}).$$

7.4.1.

Recall (Exposé XX, no. 5) that $\mathrm{SL}_{2,S}$ is the S-group scheme of matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

over S satisfying $ad - bc = 1$. A maximal torus T is defined by the monomorphism

$$\alpha^* : \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}, \qquad \alpha^*(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}.$$

A root of G with respect to T is defined by

$$\alpha(\alpha^*(z)) = z^2,$$

and a corresponding monomorphism is

$$p : \mathbb{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}, \qquad p(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Finally, putting $u = p(1)$, the corresponding representative of the Weyl group is

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Recall also (Exposé XX, 6.2) that $\mathrm{SL}_{2,S}$ is generated by T, $p(\mathbb{G}_{a,S})$, and w, subject to the relations

$$\alpha^*(z)p(x)\alpha^*(z^{-1}) = p(z^2x),$$
$$w\alpha^*(z)w^{-1} = \alpha^*(z^{-1}), \qquad w^2 = \alpha^*(-1), \qquad (wu)^3 = 1.$$

7.4.2.

Let f be an endomorphism of $\mathrm{SL}_{2,S}$. It first defines a homomorphism

$$f \circ \alpha^* : \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}.$$

The closed subgroup $\ker(f \circ \alpha^*) \subset \mathbb{G}_{m,S}$ is, locally on S, either $\mu_{n,S}$, with $n \geq 1$, or $\mathbb{G}_{m,S}$. After restricting S, we may therefore assume that there are an integer $n \in \mathbb{N}$ and a monomorphism

$$f' : \mathbb{G}_{m,S} \longrightarrow \mathrm{SL}_{2,S}$$

such that

$$f \circ \alpha^*(z) = f'(z^n).$$

By conjugacy of monomorphisms $\mathbb{G}_{m,S} \to \mathrm{SL}_{2,S}$, after an étale covering extension of the base there is a section g of G such that

$$f \circ \alpha^*(z) = \mathrm{int}(g) \circ \alpha^*(z^n).$$

Replacing f by an inner conjugate, we are thus reduced to the case

$$f \circ \alpha^*(z) = \alpha^*(z^n)$$

for some $n \in \mathbb{N}$.

7.4.3.

Consider now
$$f \circ p : \mathbb{G}_{a,S} \longrightarrow \mathrm{SL}_{2,S}.$$
It satisfies
$$\alpha^*(z^n)(f \circ p)(x)\alpha^*(z^n)^{-1} = (f \circ p)(z^2 x).$$
If $n = 0$, it follows at once that $f \circ p$ is invariant under the homotheties of $\mathbb{G}_{a,S}$, and hence is constant. If $n \neq 0$, Exposé XXII, 4.1.9 applies: the map $x \mapsto x^n$ is an endomorphism of $\mathbb{G}_{a,S}$, and there is a $\lambda \in \mathbb{G}_{m,S}$ such that
$$f \circ p(x) = p(\lambda x^n).$$
This relation is also valid for $n = 0$, on taking $\lambda = 1$. After extending S once more, choose $z \in \mathbb{G}_{m,S}$ with $z^2 = \lambda$. Replacing f by $\mathrm{int}(\alpha^*(z)) \circ f$, we are reduced to
$$f \circ \alpha^*(z) = \alpha^*(z^n), \qquad f \circ p(x) = p(x^n).$$

7.4.4.

Finally, one must have
$$f(w)\alpha^*(z^n)f(w)^{-1} = \alpha^*(z^n)^{-1}, \qquad (f(w)u)^3 = 1.$$
By Exposé XX, 3.8, this implies $f(w) = w^n$. Since G is generated by T, $p(\mathbb{G}_{a,S})$, and w, it follows that, for every $S' \to S$ and every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(S')$,
$$f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$

7.4.5.

Summarizing, locally on S for the étale topology, every endomorphism f of $\mathrm{SL}_{2,S}$ can be written
$$f = \phi \circ F_n,$$
where ϕ is an inner automorphism, $n \geq 0$, and
$$F_n\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^n & b^n \\ c^n & d^n \end{pmatrix}.$$
If $f = \phi \circ F_n$, the integer n is determined by any fiber f_s, for example by $\ker(f_s)$. Consequently n is a locally constant function on S.

7.4.6.

It follows that, for every endomorphism f of $\mathrm{SL}_{2,S}$, the scheme S has a canonical decomposition into open and closed subschemes
$$S_0, \qquad S_1, \qquad S_{p^n},$$
where p^n runs through the positive powers of prime numbers, such that:

(i) f_{S_0} is the zero morphism;

(ii) f_{S_1} is an isomorphism, necessarily an inner automorphism;

(iii) S_{p^n} has characteristic p, and $f_{S_{p^n}}$ has a unique decomposition

$$f_{S_{p^n}} = \phi \circ F_p^n,$$

where ϕ is an inner automorphism and F_p is the Frobenius endomorphism of $\mathrm{SL}_{2,\mathbb{F}_p}$.

7.4.7.

Equivalently,

$$\underline{\mathrm{Hom}}_{\mathbb{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbb{Z}}, \mathrm{SL}_{2,\mathbb{Z}})$$

is the disjoint-union scheme consisting of:

(i) one scheme isomorphic to $\mathrm{Spec}(\mathbb{Z})$;

(ii) one scheme isomorphic to

$$\underline{\mathrm{Aut}}_{\mathbb{Z}\text{-gr.}}(\mathrm{SL}_{2,\mathbb{Z}}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbb{Z}});$$

(iii) for each prime p and each integer $n > 0$, one scheme isomorphic to

$$\underline{\mathrm{Aut}}_{\mathbb{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbb{F}_p}) \simeq \mathrm{ad}(\mathrm{SL}_{2,\mathbb{F}_p}).$$

7.4.8.

In particular:

(i) $\underline{\mathrm{Hom}}_{\mathbb{F}_p\text{-gr.}}(\mathrm{SL}_{2,\mathbb{F}_p}, \mathrm{SL}_{2,\mathbb{F}_p})$ has infinitely many connected components and is therefore not quasi-compact.

(ii) If S is a scheme of unequal characteristics, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(\mathrm{SL}_{2,S}, \mathrm{SL}_{2,S})$ is not flat over S.

8. Appendix: Cohomology of a smooth group over a henselian ring. Cohomology and the functor $\prod$

Proposition 8.1.— *Let S be a henselian local scheme, s its closed point, and G a smooth S-group scheme such that every finite subset of G is contained in an affine open subset.**

(i) If P is a principal homogeneous bundle under G, there is a finite surjective étale morphism $S' \to S$ which trivializes P. Moreover,

$$\mathrm{Fib}(S, G) \simeq H^1_{\acute{e}t}(S, G).$$

(ii) For every finite surjective étale morphism $S' \to S$, the canonical map

$$H^1(S'/S, G) \longrightarrow H^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is bijective.

(iii) The canonical map

$$\mathrm{Fib}(S, G) \longrightarrow \mathrm{Fib}(\kappa(s), G_s)$$

is bijective.

*This last condition is in fact unnecessary; see A. Grothendieck, *Le groupe de Brauer III*, in *Dix Exposés sur la cohomologie des schémas*, North-Holland, 1968, Theorem 11.7 and Remarks 11.8.3.

8.1.2.

If K is a finite separable extension of $\kappa(s)$, there is a finite surjective étale morphism $S' \to S$ such that

$$K \simeq S' \otimes_S \kappa(s).$$

[47] If P is a principal homogeneous bundle under G, then P is smooth over S, hence P_s is smooth over $\kappa(s)$. Consequently there is a finite separable extension $K/\kappa(s)$ such that P_K has a section (EGA IV_4, 17.15.10). Representing K as above, Hensel's lemma (EGA IV_4, 18.5.17) shows that $P_{S'}$ has a section. This proves the first assertion of (i).

Conversely, if P is a principal homogeneous sheaf under G for the étale topology, then a finite surjective étale morphism $S' \to S$ trivializes P: every étale covering family of a henselian local scheme is dominated by a family consisting of one morphism of that form. By the descent hypothesis imposed on G, P is representable (SGA 1, VIII, 7.6). This completes the proof of (i), and shows that (ii) implies (iii). It remains to prove (ii).

8.1.3.

The map in (ii) is injective. Let P and Q be two principal homogeneous bundles under G, trivialized by S', and consider the S-group sheaf

$$\mathcal{H} = \underline{\mathrm{Isom}}_{G\text{-bundles}}(P, Q).$$

Since $\mathcal{H}_{S'}$ is isomorphic to $G_{S'}$, $\mathcal{H}$ is representable by the second hypothesis on G, as above. If $\mathcal{H}(\kappa(s)) \neq \varnothing$, Hensel's lemma gives $\mathcal{H}(S) \neq \varnothing$, and hence $P \simeq Q$.

8.1.4.

To prove surjectivity in (ii), it is equivalent to prove that

$$Z^1(S'/S, G) \longrightarrow Z^1(S' \otimes_S \kappa(s)/\kappa(s), G_s)$$

is surjective. Let Z^1 be the S-functor defined by

$$Z^1(T) = Z^1(S' \times_S T/T, G_T).$$

The preceding map is then

$$Z^1(S) \longrightarrow Z^1(\kappa(s)).$$

Another application of Hensel's lemma reduces the assertion to showing that Z^1 is representable by a smooth S-scheme.

8.1.5.

We first show that Z^1 is representable by an S-scheme locally of finite presentation. For $i = 0, 1, \ldots$, define the S-functor C_i by

$$C_i(T) = C_i(S' \times_S T/T, G_T).$$

Equivalently,

$$C_i(T) = G((S'/S)^{i+1} \times_S T), \qquad C_i = \underline{\mathrm{Hom}}_S((S'/S)^{i+1}, G).$$

Since Z^1 is obtained from C_1 and C_2 by fiber products, it is enough to prove that C_i, for $i = 1, 2$, is representable by an S-scheme locally of finite presentation.

[47] *Editors' note.* The scheme S' is still local and henselian. We have also retained the numbering of the original: there is no no. 8.1.1.

8.1.6.

Let $T \to S$ be a finite surjective étale morphism which splits S'. Then

$$C_i \times_S T = \underline{\mathrm{Hom}}_T((S' \times_S T/T)^{i+1}, G_T)$$

is represented by a product of n copies of G_T, where n is the degree of $(S'/S)^{i+1}$. The hypothesis on G, applied once again, shows that C_i is represented by an S-scheme locally of finite presentation (SGA 1, VIII, *loc. cit.*).

8.1.7.

It remains to prove that Z^1 is smooth. By definition, if T is an affine S-scheme and T_0 is the closed subscheme defined by a square-zero ideal J, we must show that

$$Z^1(T) \longrightarrow Z^1(T_0)$$

is surjective. Since G is smooth, the canonical map $G(T) \to G(T_0)$ is surjective, and it is enough to prove that

$$H^1(S' \times_S T/T, G_T) \longrightarrow H^1(S' \times_S T_0/T_0, G_{T_0})$$

is bijective. After a slight change of notation and a harmless generalization of the hypotheses, this reduces to the following lemma.

Lemma 8.1.8.— *Let S and S' be affine schemes, let $S' \to S$ be faithfully flat, let J be a square-zero ideal on S, and let S_0 be the closed subscheme which it defines. Let G be a smooth S-group. Put*

$$S'_0 = S' \times_S S_0, \qquad G_0 = G \times_S S_0.$$

Then the canonical map

$$H^1(S'/S, G) \longrightarrow H^1(S'_0/S_0, G_0)$$

is bijective.

Remark 8.1.9.— If G is commutative, the same assertion holds for every H^i, $i > 0$, with an analogous proof.

Proof. Let M_0 be the quasi-coherent $\mathcal{O}_{S_0}$-module

$$M_0 = \mathrm{Lie}(G_0/S_0) \otimes_{\mathcal{O}_{S_0}} J.$$

For every S_0-scheme T_0, put

$$M_0(T_0) = H^0\Big(T_0, M_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}\Big),$$

and define the S-group functors

$$M = \prod_{S_0/S} M_0, \qquad \overline{G} = \prod_{S_0/S} G_0$$

by

$$M(T) = M_0(T_0), \qquad \overline{G}(T) = G_0(T_0), \qquad T_0 = T \times_S S_0.$$

By Exposé III, 0.9 and (0.6.2), for every affine S-scheme T there is a functorial exact sequence

$$1 \longrightarrow M(T) \longrightarrow G(T) \longrightarrow \overline{G}(T) \longrightarrow 1.$$

We must study the map

$$H^1(S'/S, G) \longrightarrow H^1(S'_0/S_0, G_0) = H^1(S'/S, \overline{G}).$$

Suppose first that G is commutative. There is then an exact cohomology sequence

$$H^1(S'/S, M) \longrightarrow H^1(S'/S, G) \longrightarrow H^1(S'/S, \overline{G}) \longrightarrow H^2(S'/S, M).$$

Moreover,

$$H^i(S'/S, M) = H^i(S'_0/S_0, M_0) = H^i(S'_0/S_0, \mathcal{M}_0) = 0 \qquad (i \neq 0),$$

the last equality following from TDTE I, B, Lemma 1.1.

Now suppose that G is not commutative. We use the exact sequence of nonabelian cohomology. If $u \in Z^1(S'/S, \overline{G})$, the elements of $H^1(S'/S, G)$ whose image in $H^1(S'/S, \overline{G})$ equals the class of u lie in the image of the corresponding coboundary map

$$H^1(S'/S, M_u) \longrightarrow H^1(S'/S, G),$$

where M_u is M twisted by u. Similarly, for $v \in Z^1(S'/S, \overline{G})$, there is an obstruction

$$\Delta(v) \in H^2(S'/S, M_v)$$

which vanishes exactly when the class of v is in the image of $H^1(S'/S, G)$. It therefore suffices to prove

$$H^1(S'/S, M_u) = 0, \qquad H^2(S'/S, M_v) = 0.$$

Recall (Exposé III, 0.8) that the action of G on M defined by the exact sequence above is induced functorially by the adjoint representation

$$\mathrm{ad} : G_0 \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_{S_0}}(\mathrm{Lie}(G_0/S_0)).$$

The cocycle u, respectively v, therefore defines a descent datum on $\mathrm{Lie}(G_0/S_0)$; write L_u, respectively L_v, for the resulting quasi-coherent $\mathcal{O}_{S_0}$-module. For $T \to S$,

$$M_u(T) = H^0\Big(T_0, L_u \otimes_{\mathcal{O}_{S_0}} J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}\Big),$$

and similarly with v in place of u. Hence

$$H^1(S'/S, M_u) = H^1(S'_0/S_0, L_u \otimes J),$$

$$H^2(S'/S, M_v) = H^2(S'_0/S_0, L_v \otimes J),$$

and both groups vanish by the result already used. □

Proposition 8.2.— *Let $\mathcal{C}$ be a category with fiber products, endowed with a topology coarser than the canonical topology. Let $S' \to S$ be a morphism of $\mathcal{C}$, let G' be an S'-group sheaf, and let*

$$G = \prod_{S'/S} G'$$

be the corresponding S-group sheaf. Let

$$H^1_S(S', G') \subset H^1(S', G')$$

be the subset of classes of principal homogeneous sheaves under G' which are trivialized by a sieve on S' obtained by base change from a suitable covering sieve on S. The canonical map

$$H^1(S,G) \longrightarrow H^1(S',G'), \qquad P \longmapsto P \times_S S',$$

induces a bijection

$$H^1(S,G) \xrightarrow{\sim} H^1_S(S',G').$$

Its inverse is induced by

$$P' \longmapsto \prod_{S'/S} P'.$$

For every object X of $\mathcal{C}/S$, the definition gives an isomorphism, functorial in X,

$$\mathrm{Hom}_S(X,G) \xrightarrow{\sim} \mathrm{Hom}_{S'}(X \times_S S', G').$$

Thus, for every S-object T, there is a bijection functorial in T,

$$H^1(T/S,G) \xrightarrow{\sim} H^1(T'/S',G'), \qquad T' = T \times_S S'.$$

Replacing the single morphism $T \to S$ by an arbitrary covering family of S, and then passing to the inductive limit, gives the first part of the proposition. The second follows immediately.

Lemma 8.3.— *Under the hypotheses of 8.2, the equality*

$$H^1_S(S',G') = H^1(S',G')$$

is local on S. More precisely, suppose that there is a covering family $\{S_i \to S\}$ such that, for every i,

$$H^1_{S_i}(S' \times_S S_i, G') = H^1(S' \times_S S_i, G').$$

Then $H^1_S(S',G') = H^1(S',G')$.

Indeed, let P' be a principal homogeneous sheaf under G', and put

$$P'_i = P' \times_{S'} (S' \times_S S_i).$$

By hypothesis, there is a covering family $\{S_{ij} \to S_i\}$ such that, for each j,

$$P' \times_{S'} (S' \times_S S_{ij})$$

has a section. But $\{S_{ij} \to S\}$ is a covering family of S, and P' is therefore trivialized by the covering family of S' obtained from it by base change.

Proposition 8.4.— *Let $S' \to S$ be a finite étale morphism of schemes. Let G' be an S'-group sheaf and let*

$$G = \prod_{S'/S} G'$$

be the corresponding S-group sheaf. For the étale topology, respectively the local finite étale topology, respectively the (fpqc) *topology, the following functors induce mutually inverse bijections.*

$$\mathcal{P} \longmapsto \mathcal{P} \times_S S'$$

$$\prod_{S'/S} \mathcal{P}' \longleftarrow\!\shortmid \mathcal{P}'$$

In particular,

$$H^1(S, G) \simeq H^1(S', G').$$

By 8.2, it is enough to show

$$H^1_S(S', G') = H^1(S', G').$$

By 8.3, this may be checked locally for the local finite étale topology. We may therefore suppose that S' is a finite disjoint union of copies of S, say I_S, where I is a finite set. Then G' is given by a family $(G_i)_{i\in I}$ of sheaves on S, and

$$H^1(S', G') \simeq \prod_{i\in I} H^1(S, G_i).$$

On the other hand,

$$H^1(S, G) \simeq \prod_{i\in I} H^1(S, G_i).$$

It follows from 8.2 that $H^1_S(S', G') = H^1(S', G')$. □

Remarks 8.5.— Propositions 8.2 and 8.4 may be interpreted through the following exact sequence, where $f : S' \to S$ is the given morphism:

$$1 \longrightarrow H^1(S, f_*G') \longrightarrow H^1(S', G') \longrightarrow H^0(S, R^1f_*G').$$

In the commutative case, this exact sequence follows from the Leray spectral sequence. It remains valid in the noncommutative case.[48]

In this form one sees that the result remains valid when f is assumed only finite, or even merely integral, provided that the topology is the étale topology. Indeed, for every G',

$$R^1f_*(G') = \text{the final sheaf}$$

by SGA 4, VIII, 5.3.[49]

The result becomes false for a topology such as (fpqc) or (fppf), even if

$$S = \operatorname{Spec}(k), \qquad S' = \operatorname{Spec}(k[t]/t^2),$$

where k is an algebraically closed field of characteristic $p \neq 0$, and $G' = \mu_p$ or α_p.

Likewise, 8.2 becomes false, even for the étale topology, if the finiteness hypothesis on f is omitted. For example, take f to be an open immersion with

$$S = \operatorname{Spec}(V),$$

where V is a complete discrete valuation ring with algebraically closed residue field, take S' to be the open subscheme consisting of the generic point, and take G' to be the constant group $(\mathbb{Z}/n\mathbb{Z})_{S'}$, where n is prime to the residue characteristic of V. Then

$$H^1(S, G) = 0, \qquad H^1(S', G') \neq 0.$$

Replacing S' by $S \amalg S'$ gives an analogous example in which $S' \to S$ is étale and surjective, hence a cover for the topology under consideration.

[48] *Editors' note.* See §V.3.1.4 of [Gi71].
[49] *Editors' note.* This reference also points to §V.3.1.4 of [Gi71].

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[50] *Editors' note.* The editors have added, after these three references which appeared in the original, the references that follow.

Exposé XXV
The Existence Theorem

by M. Demazure

[51] For completeness, in this Exposé we give a proof of the existence theorem for split groups. Like Chevalley's original proof ("Sur certains schémas de groupes semi-simples", *Séminaire Bourbaki*, May 1961, no. 219), it rests on the existence of complex semisimple algebraic groups of every possible type. Cartier gave the principle of a more satisfactory proof, establishing directly the existence of a simply connected split semisimple $\mathbb{Z}$-group associated with a given Cartan matrix (unpublished).[52] Let us point out, however, that the difficulty is not to give an explicit construction of a group scheme, but to verify that the group so constructed really satisfies the required conditions—essentially, that its fibers are indeed smooth and reductive.

1. Statement of the theorem

Theorem 1.1.— *Let S be a nonempty scheme. The functor*

$$G \longmapsto R(G)$$

is an equivalence from the category of pinned reductive S-groups to the category of pinned reduced root data.

By the uniqueness theorem (Exposé XXIII, 4.1), the preceding statement is equivalent to the following.

Corollary 1.2 (Existence of split groups).— *For every reduced root datum R, there exists a split reductive $\mathbb{Z}$-group G such that $R(G) \simeq R$.*[53]

In particular:

Corollary 1.3.— *Let k be a field. For every splittable reductive k-group G_k, there exists a reductive $\mathbb{Z}$-group G such that*

$$G \otimes_{\mathbb{Z}} k \simeq G_k.$$

[51] Editors' note: Version of 13 October 2024.

[52] Editors' note: Such a principle was also sketched by B. Kostant *[Ko66]*; a complete proof, using quantum groups, was given more recently by G. Lusztig *[Lu09]*.

[53] Editors' note: The work *[BT84]* of F. Bruhat and J. Tits also contains a variant of Chevalley's construction (*loc. cit.*, 2.2.3–2.2.5 and §3.2), which in particular gives a smooth affine $\mathbb{Z}$-group G with connected fibers, possessing a split maximal torus, and whose generic fiber is a reductive $\mathbb{Q}$-group of type R. The reductivity of the geometric fibers of G follows from the study of the unipotent radical of a special fiber in *loc. cit.*, 4.6.12 and 4.6.15 (valid for more general G, associated with a valued root datum); but it is simpler to deduce it from the description of $\mathfrak{g} = \mathrm{Lie}(G)$, from Exposé XIX, 1.12(iii), and from the existence of the elements $w_\alpha(X)$ of Exposé XX, 3.1(iv); see *[BT84]*, 3.2.1.

Conversely, first observe that to prove 1.2 it is enough, by Exposé XXII, 4.3.1, and Exposé XXI, 6.5.10, to consider the case in which the root datum R is simply connected (and even irreducible, if desired, by Exposé XXI, 7.1.6).

On the other hand, under the hypotheses of 1.3 the group scheme G has constant type (since $\mathrm{Spec}(\mathbb{Z})$ is connected), hence type $R(G_k)$. By Exposé XXIII, 5.9, the validity of 1.3 for a given group G_k therefore entails the existence of a split $\mathbb{Z}$-group of type $R(G_k)$.[54]

Thus, to prove 1.2 and hence 1.1, it is enough to prove 1.3 when k has characteristic zero (for example, $k = \mathbb{C}$) and G_k is simply connected (and in particular semisimple), together with the following proposition.

Proposition 1.4.— *For every simply connected reduced root datum R, there exists a semisimple algebraic $\mathbb{C}$-group of type R.*

One may prove 1.4 as follows. First, there exists a complex semisimple Lie algebra $\mathfrak{g}$ of type R; see, for example, N. Jacobson, *Lie Algebras*, Chapter VII, Theorem 5.[55] Then $G = \mathrm{Aut}(\mathfrak{g})^0$ is a semisimple algebraic $\mathbb{C}$-group of type $\mathrm{ad}(R)$.[56] By *Bible*, §23.1, Proposition 1, it follows that there exists a semisimple $\mathbb{C}$-group of type R.

The remainder of this Exposé is devoted to the proof of 1.3 when k has characteristic zero and G_k is semisimple. The proof proceeds in two stages: the construction of a "piece of $\mathbb{Z}$-group scheme" (§2), and the study of the group obtained by applying "Weil's theorem" (§3).

To avoid confusion, in §2 we shall not use the abbreviated notation X_k for the k-scheme $X \otimes_{\mathbb{Z}} k$, where X is a $\mathbb{Z}$-scheme.

2. Existence theorem: construction of a piece of group

2.1.

Choose once and for all a splitting of G_k, denoted

$$(G_k, T_k, M, R)$$

(see Exposé XXII, 1.13), a system Δ of simple roots of R (defining the system R^+ of positive roots), and a Chevalley system $(X_{\alpha,k})_{\alpha\in R}$ of G_k (Exposé XXIII, 6.1 and 6.2) satisfying the additional condition (see Exposé XX, 2.6)

$$X_{\alpha,k}X_{-\alpha,k} = 1 \qquad (\alpha \in R).$$

[54] Editors' note: In fact, Exposé XXIII, 5.9 is unnecessary: the present Exposé constructs, for every semisimple $\mathbb{Q}$-group $G_{\mathbb{Q}}$, a semisimple $\mathbb{Z}$-group G of the same type as $G_{\mathbb{Q}}$, equipped with a split maximal torus T; hence G is split by Exposé XXII, 2.2.

[55] Editors' note: Let Δ be a base of R, and let $\widetilde{\mathfrak{g}}$ be the complex Lie algebra generated by $(e_\alpha, f_\alpha, h_\alpha)_{\alpha\in\Delta}$, subject to the relations $[h_\alpha, h_\beta] = 0$, $[e_\alpha, f_\beta] = h_\alpha$ if $\beta = \alpha$ and $= 0$ otherwise, $[h_\alpha, e_\beta] = (\alpha^*, \beta)e_\beta$, and $[h_\alpha, f_\beta] = -(\alpha^*, \beta)f_\beta$. In *loc. cit.*, $\mathfrak{g}$ is defined as the quotient of $\widetilde{\mathfrak{g}}$ by the intersection of the kernels of its finite-dimensional irreducible representations. In *[Se66]*, §VI.5, Theorem 9 (see also *[BLie]*, VIII, §4.3, Theorem 1), it is shown that $\mathfrak{g}$ is the quotient of $\widetilde{\mathfrak{g}}$ by the relations

$$\mathrm{ad}(e_\alpha)^{1-(\alpha^*,\beta)}(e_\beta) = 0, \qquad \mathrm{ad}(f_\alpha)^{1-(\alpha^*,\beta)}(f_\beta) = 0.$$

For an explicit description of the structure constants, and in particular the choice of signs, see Exposé XXIII, 6.5 and 6.7, as well as *[Ti66]*, §4, Theorem 1, and, for types A, D, E, *[Sp98]*, 10.2.5.

[56] Editors' note: One may suppose R irreducible, hence $\mathfrak{g}$ simple. By a general argument, $\mathrm{Lie}(G)$ is the complex Lie algebra of derivations of $\mathfrak{g}$; see *[DG70]*, §II.4, Proposition 2.3. These derivations are all inner; see *[BLie]*, I, §6.1, Corollary 3 to Proposition 1. Thus $\mathrm{Lie}(G) = \mathfrak{g}$, so G has no invariant subgroup of positive dimension and is semisimple. Its root system is consequently the same as that of $\mathfrak{g}$. Moreover, the center of G acts both trivially and faithfully on $\mathfrak{g}$, and is therefore trivial; hence G is adjoint.

Finally, choose on the subgroup of M generated by R a total order compatible with the group structure and such that the roots > 0 are the elements of R^+. We then write the roots as

$$-\alpha_n < -\alpha_{n-1} < \cdots < -\alpha_1 < \alpha_1 < \alpha_2 < \cdots < \alpha_n.$$

For $\alpha \in R$, let $U_{\alpha,k}$ denote the vector group corresponding to the root α, and let

$$p_{\alpha,k} \colon \mathbb{G}_{a,k} \xrightarrow{\sim} U_{\alpha,k}$$

be the isomorphism of vector groups defined by $X_{\alpha,k}$.

2.2.

The splitting of G_k includes, in particular, an isomorphism of k-groups

$$T_k \simeq D_k(M).$$

Put $T = D(M)$. This is a $\mathbb{Z}$-torus, and the preceding isomorphism may be regarded as an isomorphism

$$T_k \simeq T \otimes_{\mathbb{Z}} k.$$

We have

$$\mathrm{Hom}_{\mathbb{Z}\text{-gr.}}(T, \mathbb{G}_m) = M,$$

and shall regard the elements of $R \subset M$ as characters of T. Likewise, the elements of R^* will be regarded as morphisms of $\mathbb{Z}$-groups $\mathbb{G}_m \to T$.

2.3.

For each $\alpha \in R^+$, let $\mathbb{G}_a(\alpha)$ be a copy of $\mathbb{G}_a$, and consider the $\mathbb{Z}$-scheme

$$U = \mathbb{G}_a(\alpha_1) \times \cdots \times \mathbb{G}_a(\alpha_n).$$

If U_k denotes the unipotent part of the Borel subgroup B_k of G_k defined by R^+, let

$$a \colon U \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k$$

be the isomorphism of k-schemes defined by

$$a(x_1, \ldots, x_n) = p_{\alpha_1,k}(x_1) \cdots p_{\alpha_n,k}(x_n).$$

2.4.

The group law of U_k is expressed by relations of the form

$$a(x_1, \ldots, x_n) a(y_1, \ldots, y_n) = a(z_1, \ldots, z_n),$$

where, for $h = 1, \ldots, n$, each z_h is a polynomial

$$z_h = x_h + y_h + Q_h(x_1, \ldots, x_{h-1}, y_1, \ldots, y_{h-1}),$$

whose coefficients are integers (Exposé XXII, 5.5.8, and Exposé XXIII, 6.4). Moreover,

$$Q_h(x_1, \ldots, x_{h-1}, 0, \ldots, 0) = 0.$$

Equip U with the composition law defined by these formulas, which are indeed "defined over $\mathbb{Z}$". Since this law induces the group law on U_k, it is associative, and (0) is a unit element. Indeed, these two assertions are expressed by relations among the polynomials Q_h, and $\mathbb{Z} \to k$ is injective. This is a group law: if (x_i) is a section of U over an arbitrary scheme S, its inverse (y_i) is computed recursively by

$$y_i = -x_i - Q_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}),$$

formulas which are again "defined over $\mathbb{Z}$".

In summary, we have constructed a group law on U for which the isomorphism a above is an isomorphism of groups.

For each $\alpha \in R^+$, consider the morphism

$$p_\alpha\colon \mathbb{G}_a \longrightarrow U$$

defined by $p_\alpha(x) = (x_i)$, where

$$x_i = \begin{cases} x, & \alpha_i = \alpha, \\ 0, & \alpha_i \neq \alpha. \end{cases}$$

It is a closed immersion and a group homomorphism; denote its image by U_α. Since

$$(x_i) = p_{\alpha_1}(x_1) \cdots p_{\alpha_n}(x_n),$$

U is identified with the product

$$U = U_{\alpha_1} U_{\alpha_2} \cdots U_{\alpha_n}.$$

2.5.

Let $T = D(M)$ act on each U_α through the character α. One verifies immediately that this defines an action of T on the group U, and one may form the semidirect product $B = T \ltimes U$. There is a canonical isomorphism of k-groups

$$B \otimes_{\mathbb{Z}} k \xrightarrow{\sim} B_k.$$

If an arbitrary order is now placed on R^+, the morphism

$$\prod_{\alpha \in R^+} U_\alpha \longrightarrow U$$

defined by multiplication in U is still an isomorphism. Indeed, both sides are flat $\mathbb{Z}$-schemes of finite presentation, so it is enough to verify the assertion on geometric fibers. We are then reduced to Lazard's theory (*Bible*, §13.1, Proposition 1): regard U as a group with operators T, and use the fact that the U_α are pairwise nonisomorphic as groups with operators, since the characters $\alpha \in R^+$ of T are pairwise distinct on every fiber.

2.6.

Replacing R^+ by $R^- = -R^+$, one similarly constructs the groups U^-, B^-, and U_α for $\alpha \in R^-$, together with isomorphisms

$$p_\alpha\colon \mathbb{G}_a \xrightarrow{\sim} U_\alpha, \qquad \alpha \in R^-.$$

Finally introduce the product scheme

$$\Omega = U^- \times T \times U.$$

There is a canonical isomorphism of k-schemes

$$\Omega \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k^- \times_k T_k \times_k U_k \simeq \Omega_k,$$

where Ω_k is the "big cell" of G_k (Exposé XXII, 4.1.2).

From now on we identify $\Omega \otimes_{\mathbb{Z}} k$ with Ω_k by this isomorphism, and regard U^-, T, and U as subschemes of Ω through their unit sections. Write

$$e = ((0), e, (0))$$

for the "unit section" of Ω. Our aim is now to put a piece-of-group law on Ω.

Lemma 2.7.— *Let $\alpha \in \Delta$, and let $w_{\alpha,k}$ be the element of* $\mathrm{Norm}_{G_k}(T_k)(k)$ *defined by $X_{\alpha,k}$; recall that, by definition,*

$$w_{\alpha,k} = p_{-\alpha,k}(-1)p_{\alpha,k}(1)p_{-\alpha,k}(-1).$$

There exist an open subset V_α of Ω, containing the section e, and a morphism

$$h_\alpha \colon V_\alpha \longrightarrow \Omega$$

such that:

(i) $h_\alpha(e) = e$;

(ii) $(h_\alpha) \otimes_{\mathbb{Z}} k$ *coincides with the restriction of* $\mathrm{int}(w_{\alpha,k})$ *to* $V_\alpha \otimes_{\mathbb{Z}} k \subset G_k$;

(iii) $T \subset V_\alpha$, *and h_α sends T into T. For every $\beta \in R$, one has $U_\beta \subset V_\alpha$, and h_α sends U_β into $U_{s_\alpha(\beta)}$.*

By the definition of a Chevalley system (Exposé XXIII, 6.1), for every $\beta \in R$ there exists an integer $e_\beta = \pm 1$ such that

$$\mathrm{int}(w_{\alpha,k})p_{\beta,k}(x) = p_{s_\alpha(\beta),k}(e_\beta x)$$

for every $x \in \mathbb{G}_a(S)$ and every $S \to \mathrm{Spec}(k)$.

Let S be any scheme. Write an arbitrary element of $\Omega(S)$ in the form

$$\left(\left(\prod_{\substack{\beta\in R^-\\ \beta\neq-\alpha}} p_\beta(x_\beta)\right) p_{-\alpha}(x_{-\alpha}),\ t,\ p_\alpha(x_\alpha)\left(\prod_{\substack{\beta\in R^+\\ \beta\neq\alpha}} p_\beta(x_\beta)\right)\right),$$

where arbitrary orders have been chosen on $R^- \setminus \{-\alpha\}$ and $R^+ \setminus \{\alpha\}$, as permitted by 2.5. Define a morphism $d \colon \Omega \to \mathrm{Spec}(\mathbb{Z})$ by

$$d(u) = \alpha(t) + x_\alpha x_{-\alpha}.$$

Let $V_\alpha = \Omega_d$, the open subset of Ω defined by the invertibility of $d(u)$. It contains e, T, and every U_β, $\beta \in R$. Define

$$h_\alpha \colon V_\alpha \longrightarrow \Omega, \qquad h_\alpha(u) = (a(u), b(u), c(u)),$$

where

$$
\begin{aligned}
a(u) &= \left(\prod_{\substack{\beta\in R^-\\ \beta\neq-\alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta)\right) p_{-\alpha}\Big(-x_\alpha d(u)^{-1}\Big),\\
b(u) &= t\cdot\alpha^*(d(u)),\\
c(u) &= p_\alpha\Big(-x_{-\alpha}d(u)^{-1}\Big)\left(\prod_{\substack{\beta\in R^+\\ \beta\neq\alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta)\right).
\end{aligned}
$$

Since s_α permutes the positive roots distinct from α (respectively the negative roots distinct from $-\alpha$), $c(u)$ (respectively $a(u)$) is a section of U (respectively U^-), so the preceding morphism is well defined. It plainly satisfies (i) and (iii). Assertion (ii) follows immediately from the definition of the e_β, $\beta\in R$, and Exposé XX, 3.12.

Lemma 2.8.— *There exist open subsets V, V' of Ω and morphisms*

$$h\colon V\longrightarrow\Omega,\qquad h'\colon V'\longrightarrow\Omega$$

with the following properties:

(i) *V and V' contain e, and $h(e)=h'(e)=e$;*

(ii) *the morphism induced by*

$$h'\circ h\colon h^{-1}(V')\longrightarrow\Omega$$

is the restriction of the identity morphism;

(iii) *the k-morphisms $h\otimes_{\mathbb{Z}}k$ and $h'\otimes_{\mathbb{Z}}k$ are the restrictions, to $V\otimes_{\mathbb{Z}}k$ and $V'\otimes_{\mathbb{Z}}k$, of automorphisms of the group G_k;*

(iv) *V and V' contain U, T, U^-; the morphisms h, h' send U into U^-, U^- into U, and T into T.*

Let w^0 be the element of the Weyl group of G_k that sends R^+ to R^-. Write

$$w^0 = s_{\alpha_n}\cdots s_{\alpha_1},\qquad \alpha_i\in\Delta$$

(with no relation to the numbering in 2.1), and put

$$w_0 = w_{\alpha_n,k}\cdots w_{\alpha_1,k}\in \mathrm{Norm}_{G_k}(T_k)(k).$$

Define recursively, for $i\leq n$, an open subset V_i of Ω and a morphism $h_i\colon V_i\to\Omega$ by

$$V_0=\Omega,\qquad h_0=\mathrm{id},$$

and, for $i=0,\dots,n-1$,

$$V_{i+1}=h_i^{-1}(V_{\alpha_{i+1}}),\qquad h_{i+1}=h_{\alpha_{i+1}}\circ h_i,$$

where V_{α_j} and h_{α_j} are those of 2.7.

Take $V=V_n$ and $h=h_n$. The conditions (i), (iii), and (iv) concerning V and h hold: for (i) and (iii) this follows immediately from 2.8, and for (iv) from the fact that $h\otimes_{\mathbb{Z}}k$ is the restriction of $\mathrm{int}(w_0)$ to $V\otimes_{\mathbb{Z}}k$.

Since $(w^0)^2 = 1$, we also have

$$w^0 = s_{\alpha_1} \cdots s_{\alpha_n} = (s_{\alpha_1}^3) \cdots (s_{\alpha_n}^3).$$

Put

$$w_0' = (w_{\alpha_1,k})^3 \cdots (w_{\alpha_n,k})^3.$$

Repeating the construction above gives an open subset V' and a morphism h' that also satisfy (i), (iii), and (iv). Moreover, $h' \otimes_{\mathbb{Z}} k$ is the restriction of $\operatorname{int}(w_0')$ to $V' \otimes_{\mathbb{Z}} k$. For every simple root $\alpha \in \Delta$, however, $(w_{\alpha,k})^4 = e$ (Exposé XX, 3.1), hence $w_0' w_0 = e$. It follows that $h' \circ h$ induces the identity morphism on a nonempty open subset of $\Omega \otimes_{\mathbb{Z}} k$. Since Ω is smooth and of finite presentation over $\mathbb{Z}$, $\Omega \otimes_{\mathbb{Z}} k$ is schematically dense in Ω; this proves (ii).

Proposition 2.9.— *There exist an open subset V_1 of $\Omega \times \Omega$, an open subset V_2 of Ω, and morphisms*

$$\pi \colon V_1 \longrightarrow \Omega, \qquad \sigma \colon V_2 \longrightarrow \Omega$$

having the following properties:

(i) *if $x \in \Omega(S)$, then (e, x) and (x, e) are sections of V_1, and*

$$\pi(e, x) = \pi(x, e) = x;$$

(ii) *V_2 contains e, and $\sigma(e) = e$;*

(iii) *π_k and σ_k are the restrictions of the morphisms $G_k \times_k G_k \to G_k$ and $G_k \to G_k$ defined respectively by multiplication and inversion.*

Proof. Let (v, t, u) and (v', t', u') be two sections of Ω. By 2.8(iv), $h(u)$ is a section of U^- and $h(v')$ a section of U; thus $(h(u), e, h(v'))$ may be regarded as a section of Ω. Define V_1 as the open subset of $\Omega \times \Omega$ specified by

$$((v, t, u), (v', t', u')) \in V_1(S) \iff (h(u), e, h(v')) \in V'(S),$$

using the notation of 2.8. If the pair on the left is a section of V_1, then $h'(h(u), e, h(v'))$ is defined and is a section of Ω; write

$$h'(h(u), e, h(v')) = (v'', t'', u'').$$

Set

$$\pi\big((v, t, u), (v', t', u')\big) = (v \cdot tv''t^{-1},\ tt''t',\ t'^{-1}u''t' \cdot u').$$

Assertion (i) follows immediately from 2.8(ii). To verify the assertion about π in (iii), observe that

$$h'(h(u), e, h(v')) = uv' = v''t''u''$$

whenever $u \in U(S)$, $v' \in U^-(S)$, and $S \to k$, by 2.8(iii) and (ii).

The morphism σ is constructed similarly. If (v, t, u) is a section of Ω, then $h(u^{-1})$ is a section of U^-, $h(v^{-1})$ a section of U, and $h(t^{-1})$ a section of T. Thus

$$(h(u^{-1}), h(t^{-1}), h(v^{-1}))$$

is a section of Ω. Define an open subset V_2 of Ω by

$$(v, t, u) \in V_2(S) \iff (h(u^{-1}), h(t^{-1}), h(v^{-1})) \in V'(S),$$

and define

$$\sigma(v,t,u) = h'(h(u^{-1}), h(t^{-1}), h(v^{-1})).$$

The required properties of σ are verified in the same way.

Corollary 2.10.— *π is "generically associative" and σ is a "generic inverse": if $x, y, z \in \Omega(S)$, and if the expressions below are defined (as they always are over an open subset of Ω containing the unit section), then*

$$\pi(x, \pi(y,z)) = \pi(\pi(x,y), z), \qquad \pi(x, \sigma(x)) = e = \pi(\sigma(x), x).$$

Indeed, the two sides of each formula define morphisms between smooth $\mathbb{Z}$-schemes of finite presentation, and the morphisms agree on the generic fibers by 2.9(iii).

Corollary 2.11.— *Let $\alpha \in R$, and let S be a scheme. Suppose that $x, y \in \mathbb{G}_a(S)$ satisfy*

$$(p_\alpha(x), p_{-\alpha}(y)) \in V_1(S), \qquad 1 + xy \in \mathbb{G}_m(S).$$

These conditions define an open subset of $\mathbb{G}^2_{a,S}$ containing the section $(0,0)$. Then

$$\pi(p_\alpha(x), p_{-\alpha}(y)) = \left(p_{-\alpha}\left(\frac{y}{1+xy}\right), \alpha^*(1+xy), p_\alpha\left(\frac{x}{1+xy}\right)\right).$$

The proof is the same as above, by Exposé XX, 2.1.

3. Existence theorem: conclusion of the proof

For convenience, make the following definition.

Definition 3.1.— *Let S be a scheme and G an S-group scheme. The group G is called* admissible *if there is an open immersion of S-schemes*

$$i \colon \Omega_S = \Omega \times S \longrightarrow G$$

such that:

(i) *the diagram*

$$\begin{tikzcd} (V_1)_S \arrow[r] \arrow[d, "\pi"] & \Omega_S \times_S \Omega_S \arrow[r, "i \times_S i"] & G \times_S G \arrow[d, "\pi_G"] \\ \Omega_S \arrow[rr, "i_S"] & & G \end{tikzcd}$$

is commutative, where π_G denotes the multiplication morphism of G;

(ii) *there is a finite set of sections $a_j \in \Omega(S)$ such that the open subsets $i(a_j)\, i(\Omega_S)$ cover G.*

By "Weil's theorem" (Exposé XVIII, 3.13(iii) and (iv)), we have:

Lemma 3.2.— *If every admissible S-group is affine whenever S is étale and of finite type over $\mathbb{Z}$, then there exists an admissible affine $\mathbb{Z}$-group.*

We now have:

Lemma 3.3.— *Let S be a scheme and G an admissible S-group. Then G is smooth and of finite presentation over S, with affine, connected, semisimple fibers.*

Since Ω_S is smooth and of finite presentation over S, with connected fibers, the same is true of G by condition (ii). To prove 3.3, we may therefore suppose that S is the spectrum of a field K. Identify Ω_K with its image in G. Plainly Ω_K is the product

$$\left(\prod_{\alpha\in R^-} U_{\alpha,K}\right) T_K \left(\prod_{\alpha\in R^+} U_{\alpha,K}\right)$$

of the subgroups T_K and $U_{\alpha,K}$, $\alpha \in R$, of G. Thus the Lie algebra of G is identified with the direct sum

$$\mathrm{Lie}(T_K) \oplus \bigoplus_{\alpha\in R} \mathrm{Lie}(U_{\alpha,K}).$$

The inner automorphism defined by a section of T_K acts on $U_{\alpha,K}$, and hence on $\mathrm{Lie}(U_{\alpha,K})$, through the character

$$\alpha \in R \subset M \simeq \mathrm{Hom}_{K\text{-gr.}}(T_K, \mathbb{G}_{m,K}).$$

The preceding decomposition of $\mathrm{Lie}(G)$ is therefore exactly its decomposition under the adjoint action of T. Thus the roots of G_K with respect to T_K are the elements $\alpha \in R$.

Apply Exposé XIX, 1.13. Let T_α be the maximal torus of $\ker(\alpha) \subset T_K$, and put

$$Z_\alpha = \mathrm{Centr}_G(T_\alpha).$$

It is enough to prove that each Z_α is reductive. But

$$Z_\alpha \cap \Omega_K = \left(\prod_{\substack{\beta\in R^- \\ \beta|_{T_\alpha}=e}} U_{\beta,K}\right) T_K \left(\prod_{\substack{\beta\in R^+ \\ \beta|_{T_\alpha}=e}} U_{\beta,K}\right).$$

The roots vanishing on T_α are the rational multiples of α, hence precisely α and $-\alpha$. Consequently,

$$Z_\alpha \cap \Omega_K = U_{-\alpha,K}\, T_K\, U_{\alpha,K}.$$

By Exposé XX, 3.4, to prove Z_α reductive it is enough to show that $U_{\alpha,K}$ and $U_{-\alpha,K}$ do not commute, and this follows immediately from 2.11.

It follows from 3.2 and 3.3 that the proof will be complete once the following lemma is proved.

Lemma 3.4.— *Let S be a locally noetherian scheme of dimension at most* 1*, and let G be a smooth S-group of finite type, with affine, connected, semisimple fibers. Then G is affine (and hence semisimple).*

Nota. In Exposé XVI it was shown that 3.4 holds without any hypothesis on S, but the proof is comparatively delicate. Since only the special case 3.4 is needed here, we give a direct proof.

Consider the Lie algebra $\mathfrak{g}$ of G, a locally free $\mathcal{O}_S$-module, and the adjoint representation

$$\mathrm{Ad}\colon G \longrightarrow \mathrm{GL}_{\mathcal{O}_S}(\mathfrak{g}).$$

To prove that G is affine over S, it is enough to prove that Ad is affine. Since G is smooth with connected fibers, it is separated over S (Exposé $\mathrm{VI_B}$, 5.5), so Ad is separated. By the result proved in the Appendix (see 4.1), it suffices to show that Ad is quasi-finite. We are thus reduced to the case in which S is the spectrum of a field. In that case G is affine and therefore semisimple, and the assertion is Exposé XXII, 5.7.14.

4. Appendix

We used the following proposition in the course of the proof.

Proposition 4.1.— *Let S be a locally noetherian scheme of dimension at most 1; let G and H be two S-group schemes of finite type; and let $f\colon G \to H$ be a quasi-finite separated group homomorphism. If G is flat over S,*[57] *then f is affine.*

We shall give the proof only when G is smooth over S, which is indeed the hypothesis satisfied in the application above.

4.2.

By EGA II, 1.6.4, we may suppose that S is reduced. By the usual techniques of passage to the limit,[58] we may suppose S local. If $\dim(S) = 0$, the assertion is trivial.[59] Suppose therefore that $\dim(S) = 1$. By faithfully flat descent, we may suppose that S is complete with algebraically closed residue field. After replacing S by its normalization $\widetilde{S}$, if necessary, we may suppose S normal (EGA II, 6.7.1, and EGA 0_{IV}, 23.1.5).[60] We are thus reduced to the case in which S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field.

4.3.

Let η (respectively s) be the generic (respectively closed) point of S. Consider the image $f_\eta(G_\eta)$ of G_η in H_η. It is a closed subgroup scheme of H_η. Let H' be the schematic closure of $f_\eta(G_\eta)$ in H. Since $H' \to H$ is affine (it is a closed immersion), we may replace H by H', and thus suppose that H is flat over S and that f_η is surjective. Since f_η is finite and G, H are flat over S, we have

$$\dim(G_s) = \dim(G_\eta) = \dim(H_\eta) = \dim(H_s).$$

4.4.

Let $H_s^0, \dots, H_s^n$ be the irreducible components of H_s, where H_s^0 denotes the identity component, and let $z_0, \dots, z_n$ be their generic points. Since each local ring $\mathcal{O}_{H,z_i}$ has dimension at most 1, the morphism

$$G \times_H \mathcal{O}_{H,z_i} \longrightarrow \mathcal{O}_{H,z_i}$$

is affine, being quasi-finite and separated (see Exposé XVI, Lemma 4.2). Thus G is affine over H in a neighborhood of z_i. Let V be the largest open subset of H over which $G|_V$ is affine. Then V contains every z_i,[61] and therefore contains at least one closed point $y_i \in H_s^i$. Moreover, $\kappa(y_i) = \kappa(s)$, since $\kappa(s)$ is algebraically closed.

On the other hand, V is plainly stable under translation by any element $g \in G(S)$. We have $\dim(G_s) = \dim(H_s)$, and f_s is finite, so

$$f(s)\colon G_s^0(s) \longrightarrow H_s^0(s)$$

[57] Editors' note: We have added the flatness hypothesis, which had been omitted.

[58] Editors' note: See EGA IV_3, 8.10.5(viii).

[59] Editors' note: Indeed, if $S = \mathrm{Spec}(k)$, with k a field, then f is the composite of the projection $p\colon G \to G/\ker(f)$ and a closed immersion i. Since $\ker(f)$ is finite over k, p is finite (Exposé $\mathrm{VI_B}$, 9.2).

[60] Editors' note: Indeed, the irreducible components $S_1, \dots, S_r$ of S are complete integral noetherian local schemes. By a theorem of Nagata (EGA 0_{IV}, 23.1.5), the normalization $\widetilde{S}_i$ is finite over S_i, and hence $\widetilde{S}$ is finite over S. A theorem of Chevalley (EGA II, 6.7.1) then shows that if $f \times_S \widetilde{S}$ is affine, so is f.

[61] Editors' note: The following sentence has been added.

is surjective. Since A is complete and G is smooth over S, the canonical map $G^0(S) \to G^0_s(s)$ is surjective. Since $H^0_s(s)$ acts transitively on each $H^i_s(s)$, it follows that $V \supset H_s(s)$, and therefore, since $\kappa(s)$ is algebraically closed, $V \supset H_s$.[62]

Clearly $V \supset H_\eta$, since f_η is finite. Consequently, $V = H$. □

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[62] Editors' note: When G is not assumed smooth over S, one may argue as follows. Let h_0 be a rational point of H^0_s, and let g_0 be a rational point of G^0_s such that $f(g_0) = h_0$. By Exposé $\mathrm{VI_B}$, 5.6.1, there is a commutative diagram

$$\begin{tikzcd} S'' \arrow[r, "g"] \arrow[d, "\pi"'] \arrow[dr, "\phi"] & G \arrow[d] \\ S' \arrow[r, "w"'] & S \end{tikzcd}$$

in which w is étale and surjective, π is finite and surjective, and $\phi^{-1}(s)$ consists of a single point s'' such that $g(s'') = g_0$. Then $G_{S''} \to H_{S''}$ is affine above a neighborhood of $h_0 y_i$, and the same is true of $G_{S'} \to H_{S'}$ (EGA II, 6.7.1), and hence of $G \to H$ by faithfully flat descent (EGA IV_2, 2.7.1(xiii)). Thus $h_0 y_i \in V$, from which it follows that V contains $H_s(s)$, and therefore H_s.

[63] Editors' note: Additional references cited in this Exposé.

Exposé XXVI Parabolic Subgroups of Reductive Groups

M. Demazure

This Exposé studies the parabolic subgroups of an S-reductive group G. Its essential result is the conjugacy theorem (5.4). Its essential tool is the notion of the transversal position of two parabolic subgroups, a notion studied systematically in ğ4. Another fact plays an important role: the decomposition of the unipotent radical $\mathrm{rad}^u(P)$ of a parabolic subgroup P into successive extensions of vector groups (2.1).[64]

Various schemes associated with G are studied in ğ3; ğ6 deals with the split subtori[65] of G and their relations with parabolic subgroups.

Finally, in ğ7, we explain briefly how, over a semi-local base, one formulates the "relative theory" of reductive groups as presented over fields in the article of A. Borel and J. Tits, *Groupes réductifs*, Publications Mathématiques de l'IHÉS, no. 27. In that article, cited as [BT65] below, the reader will also find, for a base field, other results not touched upon here. [66]

1. Recollections. Levi subgroups

Definition 1.1.— *Let S be a scheme, G an S-reductive group, and P a sub-S-group-scheme of G. One says that P is a* parabolic subgroup *of G if:*

(i) *P is smooth over S;*

(ii) *for every $s \in S$, the quotient $\bar{s}$-scheme $G_{\bar{s}}/P_{\bar{s}}$ is proper (that is, by the Bible, ğ6.4, Theorem 4 = [Ch05], ğ6.5, Theorem 5, $P_{\bar{s}}$ contains a Borel subgroup of $G_{\bar{s}}$).*

Proposition 1.2 (Exp. XXII, 5.8.5).— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G. Then P is closed in G, has connected fibers, and*

$$P = \underline{\mathrm{Norm}}_G(P).$$

Moreover, the quotient sheaf G/P is representable by an S-scheme that is smooth and projective over S.

[64] *Editor's note.* Since the Levi subgroups of P form a torsor under $\mathrm{rad}^u(P)$ (1.9), this entails, when S is semi-local, that P has a Levi subgroup and hence a maximal torus (2.4).

[65] *Editor's note.* The terminology "trivial torus" has been replaced by "split torus".

[66] *Editor's note.* Version of 13 October 2024.

Proposition 1.3 (Exp. XXII, 5.3.9 and 5.3.11).— *Let S be a scheme, G an S-reductive group, and P, P' two parabolic subgroups of G. The following conditions are equivalent:*

(i) *P and P' are conjugate in G, locally for the étale* (respectively, fpqc) *topology.*

(ii) *For every $s \in S$, $P_{\bar{s}}$ and $P'_{\bar{s}}$ are conjugate by an element of $G(\bar{s})$.*

(iii) *The strict transporter $\underline{\mathrm{Transt}}_G(P, P')$ of P into P', defined for every $S' \to S$ by*

$$\underline{\mathrm{Transt}}_G(P, P')(S') = \{g \in G(S') \mid \mathrm{int}(g)P_{S'} = P'_{S'}\},$$

is a closed subscheme of G, smooth and of finite presentation over S, and is a principal homogeneous bundle on the right under P and on the left under P'.

Proposition 1.4.— *Let S be a nonempty scheme, let (G, T, M, R) be a split S-reductive group, and let R' be a subset of R. The following conditions on R' are equivalent:*

(i)
$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R'} \mathfrak{g}^\alpha$$

is the Lie algebra of a parabolic subgroup of G containing T (necessarily unique, Exp. XXII, 5.3.5).

(ii) *R' is of type* (R) (Exp. XXII, 5.4.2) *and contains a system of positive roots.*

(iii) *R' is a closed subset of R and satisfies: if $\alpha \in R - R'$, then $-\alpha \in R'$; equivalently, $R = R' \cup (-R')$.*

(iv) *There exist a system Δ of simple roots and a subset $A \subset \Delta$ such that R' is the union of the positive roots and the negative roots that are linear combinations of elements of A.*

(v) *R' contains a system of simple roots of R; moreover, if $\Delta \subset R'$ is a system of simple roots of R and*

$$A = (-R') \cap \Delta,$$

then R' is the union of the positive roots and the negative roots that are linear combinations of elements of A.

One has (i)⇔(ii) by Exp. XXII, 5.4.5(ii) and 5.5.1, and (iii)⇒(ii) by Exp. XXI, 3.3.6 and Exp. XXII, 5.4.7. Clearly (v)⇒(iv)⇒(iii), while (iii)⇒(v) follows from Exp. XXI, 3.3.6 and 3.3.10.

It remains to prove that (i) implies that R' is closed in R. This last assertion may be checked on any geometric fiber, so we may suppose that S is the spectrum of an algebraically closed field.

Let P be the parabolic subgroup of G containing T whose Lie algebra is $\mathfrak{g}_{R'}$. Since the Borel subgroups of P are the Borel subgroups of G contained in P, the Bible, ğ12.3, Theorem 1, and Exp. XXII, 5.4.5(i), show that, if U denotes the unipotent radical of P, then $T \cdot U$ is the subgroup of G containing T whose Lie algebra is

$$\mathfrak{g}_{R''} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R''} \mathfrak{g}^\alpha,$$

where R'' is the intersection of the systems of positive roots of R contained in R'. In particular, R'' is closed and $R'' \cap (-R'') = \varnothing$.

On the other hand, $H = P/U$ is reductive, the canonical image $\bar{T}$ of T is a maximal torus of H (and $T \to \bar{T}$ is an isomorphism), and there is an isomorphism of $\bar{T}$-modules, or equivalently of M-graded vector spaces,

$$\mathrm{Lie}(H) \simeq \mathfrak{t} \oplus \bigoplus_{\alpha \in R_s} \mathfrak{g}^\alpha,$$

where R_s is the complement of R'' in R'. Thus R_s identifies naturally with the roots of H relative to $\bar{T}$, and in particular $R_s = -R_s$. Consequently,

$$R'' = \{\alpha \in R' \mid -\alpha \notin R'\}, \qquad R_s = \{\alpha \in R' \mid -\alpha \in R'\}.$$

We now prove that R' is closed. Let $\alpha, \beta \in R'$ with $\alpha + \beta \in R$. If $\alpha, \beta \in R''$, then $\alpha + \beta \in R''$ because R'' is closed. If $\alpha \in R_s$, $\beta \in R''$, and $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$, whence

$$-\alpha = -(\alpha + \beta) + \beta \in R'',$$

because R'' is closed. This gives $-\alpha \in R'' \cap R_s$, a contradiction. It remains to consider $\alpha, \beta \in R_s$. If $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$. Since $\alpha + \beta \neq 0$, some system of positive roots of the root system R_s contains both α and β, and hence some Borel subgroup of $H = P/U$ contains the canonical images of U_α and U_β. Its inverse image in P is a Borel subgroup containing U_α, U_β, U, and therefore $U_{-(\alpha+\beta)}$, which is impossible.

Corollary 1.5.— *A parabolic subgroup of a reductive group is of type* (RC) (Exp. XXII, 5.11.1).

Proposition 1.6 (Exp. XXII, 5.11.4).— *Let S be a scheme, G an S-reductive group, and P a subgroup of G of type* (RC).[67]

(i) *P has a largest invariant sub-group-scheme that is smooth and of finite presentation over S, with connected unipotent geometric fibers. It is a characteristic subgroup of P, called the* unipotent radical *of P and denoted $\mathrm{rad}^u(P)$. The quotient sheaf $P/\mathrm{rad}^u(P)$ is representable by an S-reductive group.*

(ii) *If T is a maximal torus of P, then P has a reductive subgroup L containing T such that:*

  (a) *every reductive subgroup of P containing T is contained in L;*

  (b) *P is the semidirect product $L \cdot \mathrm{rad}^u(P)$, that is, the canonical morphism $L \to P/\mathrm{rad}^u(P)$ is an isomorphism.*

  Moreover, L is the unique subgroup (respectively, reductive subgroup) *of P containing T and satisfying (b)* (respectively, (a)). *Finally,*

$$\underline{\mathrm{Norm}}_P(L) = L, \qquad \underline{\mathrm{Norm}}_P(T) = \underline{\mathrm{Norm}}_L(T).$$

[67] *Editor's note.* "Parabolic subgroup" has been replaced by "subgroup of type (RC)", as in Exp. XXII, 5.11.4, because the statement will later be applied (4.5.1, 6.17) to $P \cap P'$ when P, P' are parabolic and $P \cap P'$ is of type (RC).

1.7. A subgroup L of P satisfying condition 1.6(ii)(b) is called a *Levi subgroup* of P. It is a maximal reductive subgroup of P. Indeed, it is reductive because it is isomorphic to $P/\operatorname{rad}^u(P)$. If L' is a reductive subgroup of P containing L, then, locally for the fpqc topology, L has a maximal torus T, and 1.6(ii) gives $L' = L$.

If L and L' are two Levi subgroups of P, they are conjugate in P, locally for the fpqc topology. Indeed, locally one may choose maximal tori $T \subset L$ and $T' \subset L'$; after a further fpqc localization they are conjugate in P, so one may assume $T = T'$, and then 1.6(ii) gives $L = L'$. Since $P = L \cdot \operatorname{rad}^u(P)$ and $\underline{\mathrm{Norm}}_P(L) = L$, one obtains:

Corollary 1.8.— *Let P be a subgroup of type* (RC) *of the S-reductive group G. If L, L' are two Levi subgroups of P, there is a unique $u \in \operatorname{rad}^u(P)(S)$ such that* $\operatorname{int}(u)L = L'$.

Let $\underline{\mathrm{Lev}}(P)$ denote the functor of Levi subgroups of P: for $S' \to S$, $\underline{\mathrm{Lev}}(P)(S')$ is the set of Levi subgroups of $P_{S'}$. Corollary 1.8 gives:

Corollary 1.9.— *Let P be a subgroup of type* (RC) *of the S-reductive group G. Then $\underline{\mathrm{Lev}}(P)$ is a principal homogeneous bundle under the S-group $\operatorname{rad}^u(P)$; in particular, it is representable by an S-scheme that is smooth and affine over S, with integral geometric fibers.*

Likewise, 1.6 immediately yields:

Corollary 1.10.— *Let P be a subgroup of type* (RC) *of the S-reductive group G. The functor $\underline{\mathrm{Tor}}(P)$ of maximal tori of P is representable by a smooth affine S-scheme. The relation $L \supset T$ defines a morphism*

$$\underline{\mathrm{Tor}}(P) \longrightarrow \underline{\mathrm{Lev}}(P),$$

whose fiber above $L \in \underline{\mathrm{Lev}}(P)(S)$ identifies with $\underline{\mathrm{Tor}}(L)$ (Exp. XXII, 5.8.3).

The first assertion of 1.10 follows from the other two and Exp. XXII, 5.8.3.

Definition 1.11.— *Let S be a nonempty scheme, G an S-reductive group, P a parabolic subgroup of G, and*

$$\mathcal{E} = (T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$$

a pinning of G. One says that $\mathcal{E}$ is adapted *to P, or is a* pinning of the pair (G, P), *if $P \supset T$ and the Lie algebra of P is of the form*

$$\mathfrak{t} \oplus \bigoplus_{\alpha\in R'} \mathfrak{g}^\alpha,$$

where $R' \subset R$ contains R^+.

In particular, if $T \subset B$ is the Killing pair defined by the pinning, then $T \subset B \subset P$. Under these conditions set $\Delta(P) = \Delta \cap (-R')$. Exp. XXII, 5.4.3 gives

$$\alpha \in \Delta(P) \iff \alpha \in \Delta \text{ and } U_{-\alpha} \subset P \iff \alpha \in \Delta \text{ and } U_{-\alpha} \cap P \neq e.$$

Proposition 1.4(v) and Exp. XXII, 5.11.3 and 5.10.6 now immediately give:

Proposition 1.12.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and $(T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$ a pinning of G adapted to P. Let $\Delta(P) \subset \Delta$ be as above.*

(i) *The unipotent radical of P is*

$$\operatorname{rad}^u(P) = U_{R''} = \prod_{\alpha\in R''} U_\alpha,$$

where R'' is the set of positive roots whose expansion in Δ contains at least one element of $\Delta - \Delta(P)$[68] *with a nonzero coefficient.*

[68] *Editor's note.* $\Delta(P)$ has been corrected to $\Delta - \Delta(P)$.

(ii) *The unique Levi subgroup L of P containing T is*

$$Z_{\Delta(P)} = \underline{\mathrm{Centr}}_G(T_{\Delta(P)}),$$

where $T_{\Delta(P)}$ is the maximal torus of $\bigcap_{\alpha\in\Delta(P)} \ker(\alpha)$. Moreover, $T_{\Delta(P)} = \mathrm{rad}(L)$.

Corollary 1.13.— *Every Levi subgroup L of a parabolic subgroup P of a reductive group G is a* critical subgroup *of G; that is* (Exp. XXII, 5.10.4),

$$L = \underline{\mathrm{Centr}}_G(\mathrm{rad}(L)).$$

This follows immediately from 1.12 and the following lemma, which is contained in 1.4 and Exp. XXII, 5.4.1:

Lemma 1.14.— *Locally for the étale topology, every pair (G, P), where P is a parabolic subgroup of the reductive group G, admits a pinning in the sense of 1.11.*

We also record:

Proposition 1.15.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and $\mathcal{E}, \mathcal{E}'$ two pinnings of G adapted to P. The unique inner automorphism of G over S that transforms $\mathcal{E}$ into $\mathcal{E}'$* (Exp. XXIV, 1.5) *comes from P, via*

$$P \longrightarrow P/\underline{\mathrm{Centr}}(P) = P/\underline{\mathrm{Centr}}(G) \longrightarrow G/\underline{\mathrm{Centr}}(G).$$

Indeed, it suffices to argue as in Exp. XXIV, 1.5, using:

Lemma 1.16 (Exp. XXII, 5.3.14 and 5.2.6).— *The maximal tori* (respectively, Borel subgroups, respectively, Killing pairs) *of a parabolic subgroup P of an S-reductive group G are conjugate in P, locally for the étale topology.*

Proposition 1.17.— *Let S be a scheme, G an S-reductive group, P, P' two parabolic subgroups of G, and B a Borel subgroup contained in both P and P'. If P and P' are conjugate in G, locally for the étale topology, then $P = P'$.*

Indeed, we may assume that $g \in G(S)$ satisfies $\mathrm{int}(g)P = P'$. The groups B and $\mathrm{int}(g)^{-1}B$ are then two Borel subgroups of P. After extending S, Lemma 1.16 lets us choose $p \in P(S)$ with $\mathrm{int}(p)\,\mathrm{int}(g^{-1})B = B$. Hence

$$pg^{-1} \in \underline{\mathrm{Norm}}_G(B)(S) = B(S),$$

so $g \in B(S) \cdot p \subset P(S)$, and therefore $P' = \mathrm{int}(g)P = P$.

Remark 1.18.— *If P, P' are two parabolic subgroups of G containing a common Borel subgroup, then $P \cap P'$ is again a parabolic subgroup of G. Indeed, it is smooth along the unit section* (Exp. XXII, 5.4.5) *and contains a Borel subgroup.*

Proposition 1.19.— *Let S be a scheme, G an S-reductive group, and G' its derived group* (Exp. XXII, 6.2.1).

(i) *The assignments*

$$P \longmapsto P' = P \cap G', \qquad P' \longmapsto P' \cdot \mathrm{rad}(G) = \underline{\mathrm{Norm}}_G(P')$$

are mutually inverse bijections between the parabolic subgroups of G and those of G'. Moreover, $\mathrm{rad}^u(P) = \mathrm{rad}^u(P')$.

(ii) *Let P be a parabolic subgroup of G and $P' = P \cap G'$. The assignments*

$$L \longmapsto L' = L \cap G' = L \cap P', \qquad L' \longmapsto L' \cdot \operatorname{rad}(G) = \underline{\operatorname{Centr}}_G(\operatorname{rad}(L'))$$

are mutually inverse bijections between the Levi subgroups of P and those of P'. Moreover,

$$\operatorname{rad}(L') = (\operatorname{rad}(L) \cap G')^0.$$

The proof, for example by reduction to the split case, is straightforward and is left to the reader, together with the immediate proof of:

Proposition 1.20.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and L a Levi subgroup of P. The assignments*

$$Q \longmapsto Q \cap L = Q', \qquad Q' \longmapsto Q' \cdot \operatorname{rad}^u(P)$$

are mutually inverse bijections between the parabolic subgroups of G contained in P and the parabolic subgroups of L. Moreover, the Levi subgroups of Q' are precisely the Levi subgroups of Q contained in L.

Proposition 1.6 may be completed as follows.

Proposition 1.21.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G.*

(i) *P has a largest invariant subgroup that is smooth and of finite presentation over S, with connected solvable geometric fibers. It is a characteristic subgroup of P, called the* radical *of P and denoted* $\operatorname{rad}(P)$. *The quotient sheaf $P/\operatorname{rad}(P)$ is representable by an S-semisimple group.*

(ii) *If L is a Levi subgroup of P, then* $\operatorname{rad}(P)$ *is the semidirect product of* $\operatorname{rad}^u(P)$ *and* $\operatorname{rad}(L)$*; one has*

$$\operatorname{rad}(L) = L \cap \operatorname{rad}(P), \qquad L = \underline{\operatorname{Centr}}_G(L \cap \operatorname{rad}(P)), \qquad P/\operatorname{rad}(P) \simeq L/\operatorname{rad}(L).$$

Indeed, assertion (i) being local, one may suppose that P has a Levi subgroup L. It is then immediate that

$$R = \operatorname{rad}^u(P) \cdot \operatorname{rad}(L)$$

has the properties asserted in (i). For (ii), it only remains to prove that $\operatorname{rad}(L) = L \cap \operatorname{rad}(P)$; this follows at once because $L \cap \operatorname{rad}(P)$ is smooth with connected fibers, L being the centralizer of a torus.[69]

2. Structure of the unipotent radical of a parabolic subgroup

Proposition 2.1.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and* $\operatorname{rad}^u(P)$ *its unipotent radical. There is a sequence of sub-group-schemes of* $\operatorname{rad}^u(P)$

$$\operatorname{rad}^u(P) = U_0 \supset U_1 \supset U_2 \supset \cdots \supset U_n \supset \cdots$$

having the following properties:

[69] *Editor's note.* See Exposé XIX, 1.4.

(i) *Every U_i is smooth, with connected fibers, characteristic and closed in P. The commutator of a section of U_i and a section of U_j is a section of U_{i+j+1}, over a varying $S' \to S$.*

(ii) *For every $i \geq 0$, there is a locally free $\mathcal{O}_S$-module $\mathcal{E}_i$ and an isomorphism of sheaves of S-groups*

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i).$$

Moreover, the automorphisms of P, over a varying $S' \to S$, act linearly on $W(\mathcal{E}_i)$.

(iii) *For every $s \in S$, one has $U_{n,s} = e$ whenever*

$$n > \dim(\mathrm{rad}^u(P)_s).$$

2.1.1. Suppose first that the pair (G, P) admits a pinning. Let $(T, M, R, \Delta, \ldots)$ be a pinning of G adapted to P, and let $\Delta(P) \subset \Delta$ be determined by P. Write $\alpha_1, \ldots, \alpha_p$ for the elements of $\Delta(P)$, and $\beta_1, \ldots, \beta_q$ for the elements of $\Delta - \Delta(P)$. Every root $\gamma \in R$ has a unique expression

$$\gamma = a_1\alpha_1 + \cdots + a_p\alpha_p + b_1\beta_1 + \cdots + b_q\beta_q.$$

Set

$$a(\gamma) = b_1 + \cdots + b_q. \tag{7}$$

[70] The definitions immediately give (cf. 1.12):

$$\begin{aligned} U_\gamma \subset P &\iff a(\gamma) \geq 0, \\ U_\gamma \subset \mathrm{rad}^u(P) &\iff a(\gamma) > 0, \\ a(n\gamma + m\gamma') &= n\,a(\gamma) + m\,a(\gamma') \qquad (n, m \in \mathbb{Z}). \end{aligned}$$

For $i \geq 0$, let R_i be the set of roots $\gamma \in R$ such that $a(\gamma) > i$. Each R_i is a closed set of roots satisfying $R_i \cap (-R_i) = \varnothing$. Consider (Exp. XXII, 5.6.5) the S-group

$$U_i = U_{R_i} = \prod_{\gamma \in R_i} U_\gamma.$$

It is a closed sub-group-scheme of G, smooth over S, with connected fibers.

Let $\alpha, \beta \in R$. Consider the commutation relation of Exp. XXII, 5.5.2:

$$p_\alpha(x)p_\beta(y)p_\alpha(-x) = p_\beta(y) \prod_{n,m \in \mathbb{N}^*} p_{n\alpha+m\beta}(C_{n,m,\alpha,\beta}x^n y^m),$$

where every

$$p_\gamma : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\gamma$$

is an isomorphism of vector groups. If $a(\alpha) > i$ and $a(\beta) > j$, then, for $n, m > 0$,

$$\begin{aligned} a(n\alpha + m\beta) &= n\,a(\alpha) + m\,a(\beta) \\ &\geq n(i+1) + m(j+1) > i + j + 1. \end{aligned}$$

Hence the commutator of a section of U_α and a section of U_β belongs to U_{i+j+1}; therefore

$$(U_i, U_j) \subset U_{i+j+1}.$$

[70] *Editor's note.* The original $a_1 + \cdots + a_p$ has been corrected to $b_1 + \cdots + b_q$.

In particular U_i/U_{i+1} is commutative, and identifies naturally with

$$U_i/U_{i+1} \simeq \prod_{a(\gamma)=i+1} U_\gamma \simeq W(\mathcal{E}_i),$$

where $\mathcal{E}_i$ is the direct sum of the $\mathfrak{g}^\gamma$ for which $a(\gamma) = i+1$.

Return to the commutation formula and suppose $a(\alpha) \geq 0$, $a(\beta) > i$. If $n, m \in \mathbb{N}^*$, then either

$$a(n\alpha + m\beta) > i + 1,$$

or $a(n\alpha + m\beta) = i+1$, in which case necessarily $m = 1$. This first proves that $\mathrm{int}(p_\alpha(x))$ preserves U_i, and hence also U_{i+1}. It then shows that, modulo U_{i+1}, the expression for $\mathrm{int}(p_\alpha(x))p_\beta(y)$ contains only terms

$$p_{n\alpha+\beta}(C_{n,1,\alpha,\beta}x^n y),$$

which are linear in y. Thus the inner automorphisms defined by sections of U_α act linearly on $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$. The same is clear for inner automorphisms defined by sections of T. Since P is generated by T and by the U_α with $a(\alpha) \geq 0$, it follows that:

(i) every U_i is invariant in P;

(ii) the inner automorphisms defined by sections of P act linearly on $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$.

2.1.2. Let $(T', M', R', \Delta', \ldots)$ now be another pinning of G adapted to P. By 1.15 there is an inner automorphism of G, coming from P, that carries the first pinning to the second. After extending S, it may be written $\mathrm{int}(p)$ for some $p \in P(S)$. Repeating the construction with the second pinning produces groups U_i' and vector-group isomorphisms

$$U_i'/U_{i+1}' \simeq W(\mathcal{E}_i')$$

obtained from U_i and $U_i/U_{i+1} \simeq W(\mathcal{E}_i)$ by transport of structure under $\mathrm{int}(p)$. The preceding invariance and linearity statements therefore give $U_i' = U_i$, and the two vector-space structures on

$$U_i/U_{i+1} = U_i'/U_{i+1}'$$

coincide.

Thus the U_i and the vector structures on their successive quotients are independent of the chosen pinning and, in particular, invariant under every automorphism of P. This proves the proposition when (G, P) admits a pinning. Assertion (iii) is immediate: by Exp. XXI, 3.1.2, the set $\{a(\gamma) \mid \gamma \in R\}$ is an interval in $\mathbb{Z}$, so $\dim U_{i,s} = \dim U_{i+1,s}$ cannot occur unless $U_{i,s} = e$.

2.1.3. In general there is an fpqc covering family $\{S_j \to S\}$ such that every pair (G_{S_j}, P_{S_j}) admits a pinning (1.14). What precedes gives descent data on the $\mathrm{rad}^u(P_{S_j})_i$, compatible with the vector structures on the quotients. The result follows by descent for closed subschemes and for locally free modules.[71]

Corollary 2.2.— *Let S be affine, G an S-reductive group, and P a parabolic subgroup of G. Then*

$$H^1(S, \mathrm{rad}^u(P)) = 0;$$

[71] *Editor's note.* See SGA 1, Exposé VIII, 1.3, 1.9, and 1.10.

that is, every principal homogeneous bundle under $\mathrm{rad}^u(P)$ *is trivial.*

Indeed, S is the disjoint union of subschemes on each of which $\mathrm{rad}^u(P)$ has constant relative dimension. By 2.1(iii) one may thus assume that $U_n = e$ for some n. Since

$$H^1(S, U_i/U_{i+1}) = H^1(S, W(\mathcal{E}_i)) = 0$$

by TDTE I, B, 1.1 (or SGA 1, Exposé XI, 5.1), the conclusion follows.

Corollary 2.3.— *Under the preceding hypotheses, P has a Levi subgroup L. If L is any Levi subgroup of P, the canonical map*

$$H^1(S, L) \longrightarrow H^1(S, P)$$

is bijective (see the introduction to Exp. XXIV for the definition of $H^1(S, -)$).

The first assertion follows from 2.2 and 1.9. The map $H^1(S, L) \to H^1(S, P)$ is surjective because $P = L \cdot \mathrm{rad}^u(P)$. For injectivity it is enough to show that

$$H^1(S, \mathrm{rad}^u(P)_Q) = 0$$

for every principal homogeneous L-bundle Q, where the subscript denotes twisting by Q. One may either repeat the proof of 2.2, using the L-invariance of the vector structures on U_i/U_{i+1}, or observe that $\mathrm{rad}^u(P)_Q$ is the unipotent radical of the parabolic subgroup P_Q of G_Q and apply 2.2 to P_Q.

Corollary 2.4.— *Let S be a semi-local scheme, G an S-reductive group, and P a parabolic subgroup of G. There is a maximal torus T of G contained in P.*

Indeed, by 2.3 the group P has a Levi subgroup L, and it suffices to know that L has a maximal torus, which follows from Exp. XIV, 3.20.

Corollary 2.5.— *Let S be affine, G an S-reductive group, and P a parabolic subgroup of G. There is a locally free $\mathcal{O}_S$-module $\mathcal{E}$ such that* $\mathrm{rad}^u(P)$*, as an S-scheme, is isomorphic to $W(\mathcal{E})$.*

We prove by induction on i that there is an isomorphism of S-schemes

$$\mathrm{rad}^u(P)/U_i \simeq W(\mathcal{E}_0 \oplus \mathcal{E}_1 \oplus \cdots \oplus \mathcal{E}_{i-1}).$$

This is clear for $i = 0$. If $i > 0$, then $\mathrm{rad}^u(P)/U_i$ is a principal homogeneous bundle over $\mathrm{rad}^u(P)/U_{i-1}$ under

$$(U_{i-1}/U_i)_{\mathrm{rad}^u(P)/U_{i-1}} \simeq W\Big(\mathcal{E}_{i-1} \otimes \mathcal{O}_{\mathrm{rad}^u(P)/U_{i-1}}\Big).$$

The base is affine, for example by induction, so the bundle is trivial (TDTE I or SGA 1, Exposé XI, *loc. cit.*). Hence

$$\mathrm{rad}^u(P)/U_i \simeq \big(\mathrm{rad}^u(P)/U_{i-1}\big) \times_S (U_{i-1}/U_i),$$

which completes the induction.

Corollary 2.6.— *Let S be semi-local, let $\{s_i\}$ be its closed points, let G be an S-reductive group, and let P be a parabolic subgroup of G. The canonical map*

$$\mathrm{rad}^u(P)(S) \longrightarrow \prod_i \mathrm{rad}^u(P)(\mathrm{Spec}\,\kappa(s_i))$$

is surjective.

Indeed, write $S = \operatorname{Spec} A$ and $\kappa(s_i) = A/\mathfrak{p}_i$. If $\mathcal{E}$ corresponds to the projective, hence flat, A-module E, it remains to prove that

$$E \longrightarrow \prod_i E \otimes_A A/\mathfrak{p}_i$$

is surjective. It suffices to do so for $E = A$, when it is well known (Bourbaki, *Algèbre commutative*, Chapter II, ğ1, no. 2, Proposition 5).

Corollary 2.7.— *Let k be an infinite field, G a k-reductive group, and P a parabolic subgroup of G. Then $\operatorname{rad}^u(P)(k)$ is dense in $\operatorname{rad}^u(P)$.*

Corollary 2.8.— *Let S be semi-local, let $\{s_i\}$ be its closed points, let G be an S-reductive group, let P be a parabolic subgroup of G, and, for every i, let L_i be a Levi subgroup of P_{s_i}. There is a Levi subgroup L of P whose fiber at every s_i is L_i.*

Indeed, let L_0 be a Levi subgroup of P (2.3). For every i, choose

$$u_i \in \operatorname{rad}^u(P)(\operatorname{Spec} \kappa(s_i)) \quad \text{with} \quad \operatorname{int}(u_i)L_{0,s_i} = L_i$$

by 1.8. If $u \in \operatorname{rad}^u(P)(S)$ induces the u_i, as 2.6 permits, then $L = \operatorname{int}(u)L_0$ has the required fibers.

Corollary 2.9.— *In the setting of 2.1, let H also be a sub-group-scheme of G, smooth and of finite presentation over S, with connected fibers, such that $P \cap H$ contains, locally for the fpqc topology, a maximal torus of G. For every $i \geq 0$ there is a locally direct-factor submodule $\mathcal{F}_i \subset \mathcal{E}_i$ for which the isomorphism*

$$U_i/U_{i+1} \xrightarrow{\sim} W(\mathcal{E}_i)$$

induces an isomorphism

$$(U_i \cap H)/(U_{i+1} \cap H) \xrightarrow{\sim} W(\mathcal{F}_i).$$

Indeed, H is a subgroup of G of type (R) (Exp. XXII, 5.2.1). The assertion is fpqc-local, so we may suppose that G is split relative to a maximal torus of $P \cap H$, and even reduce to 2.1.1 with H defined by a subset $R' \subset R$. In the notation used there, Exp. XXII, 5.6.7(ii) gives

$$U_i \cap H = \prod_{\alpha \in R_i \cap R'} U_\alpha.$$

Thus $(U_i \cap H)/(U_{i+1} \cap H)$ identifies with

$$\prod_{\substack{\alpha \in R' \\ a(\alpha) = i+1}} U_\alpha,$$

which proves the assertion.

Corollary 2.10.— *In the setting of 2.9, the conclusions of 2.2, 2.5, 2.6, and 2.7 remain valid with $\operatorname{rad}^u(P)$ replaced by $\operatorname{rad}^u(P) \cap H$.*

Corollary 2.11.— *Let*[72] *G be an S-reductive group, P a parabolic subgroup, and H a subgroup of P of type* (RC) *such that*

$$\operatorname{rad}^u(H) = \operatorname{rad}^u(P) \cap H.$$

Then statements 2.2 through 2.8 remain valid with P replaced by H.

[72] *Editor's note.* Corollary 2.11 has been added; it is used in 4.5.1 and 6.17.

3. Scheme of parabolic subgroups of a reductive group

3.1. Let E be a finite twisted constant S-scheme (Exp. X, 5.1). Consider the S-functor $\mathrm{Of}(E)$, where $\mathrm{Of}(E)(S')$ is the set of open-and-closed subschemes of $E_{S'}$, or equivalently the set of open-and-closed subsets of $E_{S'}$. Then $\mathrm{Of}(E)$ is representable by a finite twisted constant S-scheme. Indeed, if $E = A_S$, for a finite set A, then

$$\mathrm{Of}(E) \simeq \mathcal{P}(A)_S,$$

where $\mathcal{P}(A)$ denotes the power set of A; the general case follows by descent of open-and-closed subschemes. Plainly,

$$\mathrm{Of}(E_{S'}) = \mathrm{Of}(E)_{S'}, \qquad \mathrm{Of}(E \times_S E') = \mathrm{Of}(E) \times_S \mathrm{Of}(E').$$

3.2. Let S be a scheme and G an S-reductive group. Define the functor $\underline{\mathrm{Par}}(G)$ of parabolic subgroups of G by

$$\underline{\mathrm{Par}}(G)(S') = \{\text{parabolic subgroups of } G_{S'}\}.$$

In particular $G \in \underline{\mathrm{Par}}(G)(S)$, and $\underline{\mathrm{Bor}}(G) \subset \underline{\mathrm{Par}}(G)$. We shall define a morphism

$$t : \underline{\mathrm{Par}}(G) \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G))$$

with the following properties:

(i) t is functorial in G, with respect to isomorphisms, and commutes with base change.

(ii) If $(T, M, R, \Delta, \ldots)$ is a pinning of G adapted to a parabolic subgroup P, then the canonical isomorphism

$$\underline{\mathrm{Dyn}}(G) \simeq \Delta_S \qquad \text{(Exp. XXIV, 3.4(iii))}$$

carries $t(P)$ onto $\Delta(P)_S$, in the notation of 1.11 and 1.12.

First suppose that P is a parabolic subgroup of G and that $(T, M, R, \Delta, \ldots)$ is a pinning of G adapted to P. Define $t(P)$ by (ii). The resulting subscheme of $\underline{\mathrm{Dyn}}(G)$ is independent of the pinning. Indeed, if $(T', M', R', \Delta', \ldots)$ is another pinning adapted to P, then the unique inner automorphism of G carrying the first pinning to the second comes from P, by 1.15. The canonical isomorphism $\Delta \xrightarrow{\sim} \Delta'$ therefore carries $\Delta(P)$ onto $\Delta'(P)$.

Without assuming that (G, P) admits a pinning, 1.14 and the definition of $\underline{\mathrm{Dyn}}(G)$ (Exp. XXIV, 3.3) allow one to descend a unique open-and-closed subscheme $t(P)$ such that

$$t(P)_{S'} = t(P_{S'})$$

whenever $(G, P)_{S'}$ admits a pinning.

Theorem 3.3.— *Let S be a scheme, G an S-reductive group, and*

$$t : \underline{\mathrm{Par}}(G) \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G))$$

the morphism just defined.

(i) *Two parabolic subgroups P, P' of G are conjugate locally for the fpqc topology* (cf. 1.3) *if and only if $t(P) = t(P')$.*

(ii) $\underline{\mathrm{Par}}(G)$ *is representable, and* t *is smooth and projective, with integral geometric fibers.*

By 3.2(i), together with the fact that inner automorphisms of G act trivially on $\underline{\mathrm{Dyn}}(G)$ (Exp. XXIV, 3.4(iv)), conjugate parabolics have the same type. Conversely, suppose $t(P) = t(P')$. Locally one may first assume that both (G, P) and (G, P') admit pinnings (1.14). By the conjugacy of pinnings in G (Exp. XXIV, 1.5), one may then choose a single pinning (T, M, R, Δ) adapted to both P and P'. Equality of types gives $\Delta(P) = \Delta(P')$, hence $P = P'$ by 1.4(v). This proves (i).

For (ii), use the notation of Exp. XXII, 5.11.5.[73] There is a canonical morphism $\underline{\mathrm{Par}}(G) \to \mathscr{H}_c$, which fits into the cartesian square

$$\begin{tikzcd}
\underline{\mathrm{Par}}(G) \arrow[r, "t"] \arrow[d, hook] & \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \arrow[d, hook] \\
\mathscr{H}_c \arrow[r, "c\ell"] & \mathcal{C}\ell_c.
\end{tikzcd}$$

The vertical arrows are monomorphisms. By *loc. cit.*, $\mathscr{H}_c$ is representable and $c\ell$ is smooth, quasi-projective, and of finite presentation, with integral geometric fibers; hence the same is true of t.

It remains to prove that t is proper. This is fpqc-local, so one may reduce to a pinned group $G = (G, T, M, R, \Delta, \ldots)$. Then $\underline{\mathrm{Dyn}}(G) \simeq \Delta_S$, and it is enough to show that

$$t^{-1}((\Delta_1)_S)$$

is proper over S for every $\Delta_1 \subset \Delta$. Let P_1 be the parabolic containing T with $\Delta(P_1) = \Delta_1$. Part (i) shows that $g \mapsto \mathrm{int}(g)P_1$ induces an isomorphism

$$G/\underline{\mathrm{Norm}}_G(P_1) \xrightarrow{\sim} t^{-1}((\Delta_1)_S).$$

By 1.2, $G/\underline{\mathrm{Norm}}_G(P_1) = G/P_1$, which is projective over S.

Definition 3.4.— $\mathrm{Of}(\underline{\mathrm{Dyn}}(G))$ *is called the* scheme of types of parabolics *of* G*, and* $t(P)$ *is called the* type *of* P*.*

Corollary 3.5.— *The* S*-functor* $\underline{\mathrm{Par}}(G)$ *is representable by an* S*-scheme that is smooth and projective over* S*. The factorization*

$$\underline{\mathrm{Par}}(G) \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \longrightarrow S$$

is the Stein factorization (EGA III, 4.3.3) *of the structural morphism* $\underline{\mathrm{Par}}(G) \to S$*.*

Corollary 3.6.— *For every* $t \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S)$*, the* S*-scheme*

$$\underline{\mathrm{Par}}_t(G) = t^{-1}(t)$$

of parabolic subgroups of type t *is smooth and projective over* S *and homogeneous under* G*. If* P *is a parabolic subgroup of* G*, there is a canonical isomorphism*

$$G/P \xrightarrow{\sim} \underline{\mathrm{Par}}_{t(P)}(G).$$

[73] *Editor's note.* Recall that $\mathscr{H}_c = \mathscr{H}_c(G)$ denotes the functor of subgroups of G of type (RC), that $\mathcal{C}\ell_c = \mathcal{C}\ell_c(G)$ denotes the functor of their "conjugacy classes", and that $c\ell : \mathscr{H}_c \to \mathcal{C}\ell_c$ is the canonical projection.

Moreover,

$$\underline{\mathrm{Par}}_{\varnothing}(G) = \underline{\mathrm{Bor}}(G), \qquad \underline{\mathrm{Par}}_{\underline{\mathrm{Dyn}}(G)}(G) \xrightarrow{\sim} S.$$

Remark 3.7.— *The S-scheme* $\mathrm{Of}(\underline{\mathrm{Dyn}}(G))$ *has a natural order structure, the domination relation being set-theoretic inclusion of subschemes. It is a lattice; in particular, the infimum of two open-and-closed subschemes of* $\underline{\mathrm{Dyn}}(G)_{S'}$ *is their intersection.*

If B is a Borel subgroup of G, let X be the functor of parabolic subgroups containing B. The morphism

$$X \longrightarrow \mathrm{Of}(\underline{\mathrm{Dyn}}(G))$$

induced by t is an isomorphism of ordered schemes. This follows from $P \subset Q \Rightarrow t(P) \subset t(Q)$ and the next lemma.

Lemma 3.8.— *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, and*

$$t' \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S) \quad \textit{with} \quad t(P) \subset t'.$$

There is a unique parabolic subgroup P' of G containing P and satisfying $t(P') = t'$.

After extending the base, one may suppose that P contains a Borel subgroup B of G. Uniqueness then follows from 1.17. For existence one may work in the split case, where the assertion is immediate from ğ1.

Remarks 3.8.1.—

(i) *The analogue of 3.8 obtained by reversing the inclusions is false. It would imply, for example, that every group of type A_1 has a Borel subgroup, which is not so* (Exp. XX, ğ5).

(ii) *The preceding discussion shows that $t(P) \subset t(Q)$ means that locally for the fpqc, or étale, topology, P is conjugate to a subgroup of Q. It is enough to check this on geometric fibers. Section 5 will show that the étale topology may be replaced by the Zariski topology.*

3.9. The preceding discussion can be repeated for critical subgroups. Recall (Exp. XXII, 5.10.4 and 5.10.5) that a reductive subgroup H of a reductive group G is *critical* if

$$H = \underline{\mathrm{Centr}}_G(\mathrm{rad}(H)),$$

that a subtorus $Q \subset G$ is a *C-critical torus* if

$$Q = \mathrm{rad}(\underline{\mathrm{Centr}}_G(Q)),$$

and that critical subgroups and C-critical tori are in bijection via

$$H \longmapsto \mathrm{rad}(H), \qquad Q \longmapsto \underline{\mathrm{Centr}}_G(Q).$$

[74]

If (G, T, M, R) is split, the subgroup of G containing T and corresponding to $R' \subset R$ (Exp. XXII, 5.4.2) is critical if and only if R' is *vectorial*, meaning that it is the intersection of R with a vector subspace of $M \otimes \mathbb{Q}$ (Exp. XXII, 5.10.6).

[74] *Editor's note.* "Critical torus" has here been replaced by "C-critical torus". Below, "critical torus" is again used for "C-critical torus".

For arbitrary G, Exp. XXII, 5.11.5 defines a finite étale S-scheme $\mathcal{C}\ell_{\mathrm{crit}}$; in the split case it is the constant scheme associated with the vectorial subsets of R, modulo the action of the Weyl group. If $\underline{\mathrm{Crit}}(G)$ denotes the functor of critical subgroups, there is a canonical morphism

$$c\ell : \underline{\mathrm{Crit}}(G) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

in the cartesian square

$$\begin{tikzcd} \underline{\mathrm{Crit}}(G) \arrow[r, "c\ell"] \arrow[d, hook] & \mathcal{C}\ell_{\mathrm{crit}} \arrow[d, hook] \\ \mathscr{H}_c \arrow[r, "c\ell"] & \mathcal{C}\ell_c. \end{tikzcd}$$

[75] **Proposition 3.10.**— *Let S be a scheme, G an S-reductive group, $\underline{\mathrm{Crit}}(G)$ the functor of its critical subgroups, and let $\mathcal{C}\ell_{\mathrm{crit}}$ and*

$$c\ell : \underline{\mathrm{Crit}}(G) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

be as above.

(i) *Critical subgroups H, H' of G are conjugate, locally for the fpqc topology, if and only if $c\ell(H) = c\ell(H')$.*

(ii) $\underline{\mathrm{Crit}}(G)$ *is representable, and $c\ell$ is smooth and affine, with integral geometric fibers.*

The proof is the same as for 3.3 except for affineness of $c\ell$. It is enough to prove that $\underline{\mathrm{Crit}}(G)$ is affine over S. The functor $\underline{\mathrm{Crit}}(G)$ identifies naturally with the functor of critical tori of G, giving a canonical monomorphism

$$\underline{\mathrm{Crit}}(G) \longrightarrow \mathcal{M},$$

where $\mathcal{M}$ is the scheme of subgroups of multiplicative type of G (Exp. XI, 4.1). By Exp. XII, 5.3, it is enough to show that this is an open-and-closed immersion. After the base change $\mathcal{M} \to S$, this reduces to the following: if Q is a subgroup of multiplicative type in G, the $S' \to S$ for which $Q_{S'}$ is a critical torus of $G_{S'}$ are precisely those factoring through an open-and-closed subscheme of S. The condition is the conjunction of:

(1) Q is a torus;

(2) once Q is a torus, the torus $\mathrm{rad}(\underline{\mathrm{Centr}}_G(Q))$ has the same relative dimension as Q.

Both conditions are of the required kind.

Corollary 3.11.— *The S-functor $\underline{\mathrm{Crit}}(G)$ is representable by an S-scheme that is smooth and affine over S.*

Corollary 3.12.— *Let H be a critical subgroup of the S-reductive group G. Then $G/\underline{\mathrm{Norm}}_G(H)$ and G/H are representable by S-schemes that are affine and smooth over S.*

The first assertion follows from 3.11, and the second from the first and Exp. XXII, 5.10.2.

Corollary 3.13.— *Let Q be a subtorus of the S-reductive group G. Then $G/\underline{\mathrm{Centr}}_G(Q)$ is representable by an S-scheme that is smooth and affine over S. The same holds for $G/\underline{\mathrm{Norm}}_G(Q)$ when Q is a critical subtorus of G.*

Indeed, $H = \underline{\mathrm{Centr}}_G(Q)$ is critical (Exp. XXII, 5.10.5), and when Q is critical one has $\underline{\mathrm{Norm}}_G(H) = \underline{\mathrm{Norm}}_G(Q)$ (*loc. cit.*, 5.10.8).

[75] *Editor's note.* See 3.3, editor's note, for the notation $\mathscr{H}_c, \mathcal{C}\ell_c$.

3.14. The conjugacy of Levi subgroups of parabolic subgroups of G gives a unique morphism

$$u : \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

such that, for every parabolic subgroup P and every Levi subgroup $L \subset P$,

$$c\ell(L) = u(t(P)),$$

compatibly with every base change.

3.15. Let S be a scheme and G an S-reductive group. Consider the S-functors[76]

$$\begin{aligned} PL(S') &= \{(P,L) \mid P \supset L,\ P \text{ parabolic in } G_{S'},\ L \text{ a Levi subgroup of } P\}, \\ PT(S') &= \{(P,T) \mid P \supset T,\ P \text{ parabolic in } G_{S'},\ T \text{ a maximal torus of } P\}, \\ CT(S') &= \{(C,T) \mid C \supset T,\ C \text{ critical in } G_{S'},\ T \text{ a maximal torus of } C\}, \\ PLT(S') &= \{(P,L,T) \mid P \supset L \supset T,\ (P,T) \in PT(S'),\ L \text{ a Levi subgroup of } P\}. \end{aligned}$$

There are evident morphisms between these functors and $\underline{\mathrm{Par}}(G), \underline{\mathrm{Crit}}(G)$, and $\underline{\mathrm{Tor}}(G)$. They form the following commutative truncated cube.

$$\begin{tikzcd}
 & & \underline{CT} \arrow[dll, "a\ (\text{aff.})"'] \arrow[dd, "k\ (\text{ét.})"] & & \underline{PLT} \arrow[ll, "g\ (\text{ét.})"'] \arrow[dl, "b\ (\text{aff.})"] \arrow[dd, "e\ (\text{isom.})"] \\
\underline{\mathrm{Crit}}(G) \arrow[d, "c\ell\ (\text{aff.})"'] & & & \underline{PL} \arrow[lll, "f\ (\text{ét.})"'] \arrow[ddd, "d\ (\text{aff.})"] & \\
\mathcal{C}\ell_{\mathrm{crit}} & & \underline{\mathrm{Tor}}(G) \arrow[dl, "r\ (\text{aff.})"] & & \underline{PT} \arrow[ll, "h\ (\text{ét.})"'] \arrow[ddl, "c\ (\text{aff.})"] \\
 & S \arrow[ul, "q\ (\text{ét.})"'] & & & \\
\mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \arrow[uu, "u\ (\text{ét.})"] \arrow[ur, "p\ (\text{ét.})"'] & & & \underline{\mathrm{Par}}(G) \arrow[lll, "t\ (\text{proj.})"] &
\end{tikzcd}$$

Figure 56: The truncated cube of ğ3.15.

Theorem 3.16 (cf. Figure 3.15.1).—

(i) *Every morphism in the diagram is smooth, surjective, and of finite presentation.*

(ii) *Every morphism except t is affine; t is projective.*

(iii) *Every morphism is either finite étale or has integral geometric fibers: f, g, h, k, p, q, u are finite étale; $a, b, c, d, r, c\ell$ have integral geometric fibers; and e is an isomorphism.*

(iv) *The square (a, b, f, g) is cartesian.*

[76] *Editor's note.* The original LT has been changed to CT.

Proof. It is clear from 1.6(ii) that e is an isomorphism, and (iv) is immediate. The morphism a is smooth and affine with integral geometric fibers. Indeed, after the base change $\underline{\mathrm{Crit}}(G) \to S$, this reduces to the same statement for

$$\underline{\mathrm{Tor}}(L_0) \longrightarrow S,$$

where L_0 is the universal critical subgroup; L_0 is reductive by definition, so Exp. XXII, 5.8.3 applies. By (iv), b has the same properties.

The morphism d is likewise smooth and affine with integral geometric fibers, by 1.9; the same is therefore true of

$$c = dbe^{-1}.$$

The morphism r has these properties by Exp. XXII, 5.8.3, and $c\ell$ by 3.10.

We already know that p and q are finite étale and surjective (3.1 and 3.9). If f and k are finite étale and surjective, then so is g, by (iv), and so is $h = kge^{-1}$. Since the assertions about t were proved in 3.3(ii), it remains only to prove the claim for f and k. We do so for k, the proof for f being the same.

If T is a maximal torus of G, let $\mathcal{C}$ be the functor of critical subgroups containing T. We must show that $\mathcal{C}$ is represented by a finite étale S-scheme with nonempty fibers. We may suppose that G is split relative to T. If E is the set of vectorial subsets of the root system R, then $\mathcal{C}$ is represented by E_S, by 3.9. □

Corollary 3.17.— *All the functors in the preceding diagram are representable by S-schemes that are smooth over S, and all are affine over S except $\underline{\mathrm{Par}}(G)$.*

Remark 3.18.—

(i) *The fact that $f : PL \to \underline{\mathrm{Crit}}(G)$ is étale and surjective says that a subgroup of G is critical if and only if, locally for the étale topology, it is a Levi subgroup of a parabolic subgroup.*

In general the map $PL(S) \to \underline{\mathrm{Crit}}(G)(S)$ need not be surjective: a critical subgroup C may fail to arise over S from a parabolic subgroup. For example, a maximal torus need not be contained in a Borel subgroup, as for a nonsplit form of SL_2 (Exp. XX, ğ5).

(ii) *Likewise,*

$$u : \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \longrightarrow \mathcal{C}\ell_{\mathrm{crit}}$$

need not be an isomorphism. Two parabolic subgroups of different types may have Levi subgroups of the same type. In a group of type A_2, for instance, the two types of parabolics whose Levi subgroups have semisimple rank 1 *correspond to the two vertices of the diagram, whereas there is only one type of critical subgroup of rank* 1.

The analogous example for type A_3 shows that, even over an algebraically closed field, nonisomorphic parabolic subgroups may have Levi subgroups of the same type.[77]

We end this section with an application to principal bundles.

[77] *Editor's note.* Concretely, in GL_4, the parabolics

$$P = \left\{ \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \qquad P' = \left\{ \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

Lemma 3.20.— *Let*[78] *S be a scheme, G an S-reductive group, F a principal homogeneous bundle under G, and G_F the corresponding twisted form. Identify* $\underline{\mathrm{Dyn}}(G)$ *with* $\underline{\mathrm{Dyn}}(G_F)$ (Exp. XXIV, 3.5). *For every parabolic subgroup P of G, there is a canonical isomorphism*

$$F/P \xrightarrow{\sim} \underline{\mathrm{Par}}_{t(P)}(G_F).$$

In particular, the structural group of F reduces to P if and only if G_F has a parabolic subgroup of type $t(P)$.

The proof is exactly the same as that of Exp. XXIV, 4.2.1.

3.21. If S is a scheme, G an S-reductive group, and

$$t \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S),$$

write $H^1_t(S,G)$ for the subset of $H^1(S,G)$ consisting of the classes of principal G-bundles F for which the associated group G_F has a parabolic subgroup of type t. If G itself has a parabolic subgroup P of type t, then $H^1_t(S,G)$ is precisely the image of $H^1(S,P)$ in $H^1(S,G)$, and hence does not depend on the chosen P.

4. Relative position of two parabolic subgroups

4.1. A preliminary result

Lemma 4.1.1.— *Let k be a field, G a reductive k-group, and P, P' two parabolic subgroups of G.*

(i) *The intersection $P \cap P'$ is smooth, has the same reductive rank as G, and contains a maximal torus T of G.*

(ii) *The set R' of roots of $P \cap P'$ relative to T is a closed subset of the set R of roots of G relative to T.*

[79]

Proof. Suppose first that k is algebraically closed. Let B (respectively B') be a Borel subgroup of P (respectively P'). There is an element $g \in G(k)$ such that $\mathrm{int}(g)B = B'$. On the other hand, if T_0 is a maximal torus of B and $N = \underline{\mathrm{Norm}}_G(T_0)$, Bruhat's theorem (*Bible*, ğ13.4, Corollary 1 to Theorem 3) gives

$$G(k) = B(k)N(k)B(k).$$

Thus there are $b, b_1 \in B(k)$ and $n \in N(k)$ such that $g = bnb_1$, and hence $\mathrm{int}(b)\,\mathrm{int}(n)B = B'$. Consequently

$$P \cap P' \supset B \cap B' = \mathrm{int}(b)\big(B \cap \mathrm{int}(n)B\big) \supset \mathrm{int}(b)T_0.$$

are not isomorphic (their unipotent radicals are not isomorphic), while their Levi subgroups

$$L = \left\{ \begin{pmatrix} * & * & 0 & 0 \\ * & * & 0 & 0 \\ 0 & 0 & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}, \quad L' = \left\{ \begin{pmatrix} * & 0 & 0 & 0 \\ 0 & * & * & 0 \\ 0 & * & * & 0 \\ 0 & 0 & 0 & * \end{pmatrix} \right\}$$

are conjugate by $\begin{pmatrix} 0&0&1&0 \\ 0&1&0&0 \\ 1&0&0&0 \\ 0&0&0&1 \end{pmatrix}$.

[78] *Editor's note.* The numbering of the original is retained: there is no ğ3.19.

[79] *Editor's note.* Part (ii), which will be useful in 4.5.1, was added by the editors.

Now let k be arbitrary. The preceding result shows that $P_{\bar{k}} \cap P'_{\bar{k}}$ contains a maximal torus of $G_{\bar{k}}$. By Exp. XXII, 5.4.5, $(P \cap P')_{\bar{k}}$ is smooth along the identity section; since the base is a field, it is therefore smooth (Exp. $\mathrm{VI_A}$, 1.3.1), and so $P \cap P'$ is smooth. By Exp. $\mathrm{VI_A}$, 2.3.1, the identity component $(P \cap P')^0$ is an open subgroup of $P \cap P'$, smooth over k. Applying Exp. XIV, 1.1 to it proves (i).

Finally, the set R_P (respectively $R_{P'}$) of roots of P (respectively P') relative to T is closed, and

$$R' = R_P \cap R_{P'}.$$

This proves (ii). □

Remark 4.1.2.— *One can prove* ([BT65], 4.5) *that $P \cap P'$ is connected; this fact will be used in 4.5.1.*

80

Remark 4.1.3.— *The preceding lemma is false over an arbitrary scheme. For example, let G be a reductive group over an algebraically closed field k, and let B be a Borel subgroup of G. Take $S = G$ as base and consider the Borel subgroups B_1, B_2 of G_G, where*

$$B_1 = B_G, \qquad B_2 = \operatorname{int}(g_0)B_1,$$

and g_0 is the canonical (diagonal) section of G_G. For every $g \in G(k)$, the fiber of $B_1 \cap B_2$ at g is $B \cap \operatorname{int}(g)B$. If $B \neq G$, the dimension of this fiber varies with g, so $B_1 \cap B_2$ cannot be smooth over G.

4.2. Transversal position

Theorem 4.2.1.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G. The following conditions on the pair (P, Q) are equivalent:*

(i) $\mathrm{Lie}(P/S) + \mathrm{Lie}(Q/S) = \mathrm{Lie}(G/S)$.

(ii) *The canonical morphism $P \times_S Q \to G$ is smooth.*

(ii′) *The canonical morphism $P \to G/Q$ is smooth.*

(iii) *The canonical morphism $P \times_S Q \to G$ is open.*

(iii′) *The canonical morphism $P \to G/Q$ is open.*

(iv) *The canonical morphism $P \times_S Q \to G$ is fiberwise dominant.*

(iv′) *The canonical morphism $P \to G/Q$ is fiberwise dominant.*

(v) *For every $s \in S$, the group $P_s \cap Q_s$ has minimum dimension, that is,*

$$\dim(P_s \cap Q_s) = \dim P_s + \dim Q_s - \dim G_s.$$

(vi) *There is an étale covering family $\{S_i \to S\}$ and, for every i, a Borel subgroup B_i of P_{S_i} and a Borel subgroup B'_i of Q_{S_i}, such that $B_i \cap B'_i$ is a maximal torus of G_{S_i}.*

[80] *Editor's note.* In view of the details added in 4.5.1, the original wording "we shall not use this fact" was changed.

(vii) *There is an étale covering family $\{S_i \to S\}$ and, for every i, a splitting $(T_i, \dots, R_i)$ of G_{S_i} and a system R_i^+ of positive roots of R_i, such that P_{S_i} (respectively Q_{S_i}) is the subgroup of type* (R) *of G_{S_i} containing T_i and defined by a subset $R_i^{(1)}$ (respectively $R_i^{(2)}$) of R_i containing R_i^+ (respectively $-R_i^+$); see Exp. XXII, 5.4.2 and 5.2.1.*

Proof. We prove the theorem according to the following logical diagram:

$$\begin{array}{ccccccc} \text{(i)} & \Longleftrightarrow & \text{(vii)} & \Longleftrightarrow & & \text{(vi)} & \\ \Updownarrow & & & & & \nwarrow & \\ \text{(ii}'\text{)} & \Longrightarrow & \text{(iii}'\text{)} & \Longrightarrow & \text{(iv}'\text{)} & \Longrightarrow & \text{(v)} \\ \Updownarrow & & & & & \nearrow & \\ \text{(ii)} & \Longrightarrow & \text{(iii)} & \Longrightarrow & & \text{(iv).} & \end{array} \tag{20}$$

The implications (ii) $\Rightarrow$ (iii) and (ii′) $\Rightarrow$ (iii′) are immediate. If (iii) holds, the set-theoretic image of $P \times_S Q \to G$ is an open subset of G containing the identity section. Since the fibers of G are connected, this image is dense in every fiber; thus (iv) holds. The implication (iii′) $\Rightarrow$ (iv′) is identical.

Condition (ii′) implies (ii), by the cartesian diagram

$$\begin{tikzcd} P \times_S Q \arrow[r] \arrow[d, "\mathrm{pr}_1"'] & G \arrow[d] \\ P \arrow[r] & G/Q. \end{tikzcd} \tag{21}$$

On the other hand, (iv) or (iv′) implies (v) by dimension theory (EGA IV_2, 5.6.6). Indeed, one may assume $S = \mathrm{Spec}(k)$, with k algebraically closed; every nonempty fiber of the morphism in (iv) at a point of $G(k)$, or of the morphism in (iv′) at a point of $(G/Q)(k)$, is isomorphic to $P \cap Q$.

The implication (vi) $\Rightarrow$ (vii) follows from Exp. XXII, 5.5.1 (iv) and 5.9.2. Condition (vii) implies (i), because the latter may be checked fpqc-locally. Under (vii) we may therefore suppose that G is split, $P \supset B_{R^+}$, and $Q \supset B_{-R^+}$; then already

$$\mathrm{Lie}(B_{R^+}) + \mathrm{Lie}(B_{-R^+}) = \mathrm{Lie}(G).$$

Let us prove that (i) implies (ii′). Let $u : P \to G/Q$ be the canonical morphism. Smoothness can be checked on geometric fibers, since both P and G/Q are smooth over S; hence we may suppose that S is the spectrum of an algebraically closed field. The morphism u is compatible with the evident action of P: $u(pp') = p\,u(p')$. A translation argument (compare the proof of Exp. $\mathrm{VI_B}$, 1.3) therefore reduces smoothness to the point $e \in P(k)$. By SGA 1, II, 4.7, it is enough that the tangent map at e be surjective. This map is naturally identified with

$$\mathrm{Lie}(P) \longrightarrow \mathrm{Lie}(G)/\mathrm{Lie}(Q),$$

which is surjective under (i).

It remains to prove (v) $\Rightarrow$ (vi). Suppose first that S is the spectrum of an algebraically closed field. By 4.1.1 there is a maximal torus T contained in P and Q. Let R, R_1, and R_2 be the respective sets of roots of G, P, and Q relative to T. By Exp. XXII, 5.4.4 and 5.4.5,

$$\begin{aligned} \dim G &= \dim T + \mathrm{Card}(R), & \dim P &= \dim T + \mathrm{Card}(R_1), \\ \dim Q &= \dim T + \mathrm{Card}(R_2), & \dim(P \cap Q) &= \dim T + \mathrm{Card}(R_1 \cap R_2). \end{aligned}$$

Condition (v) is therefore equivalent to

$$\mathrm{Card}(R_1 \cap R_2) = \mathrm{Card}(R_1) + \mathrm{Card}(R_2) - \mathrm{Card}(R),$$

that is, to $R_1 \cup R_2 = R$. By Exp. XXII, 5.9.2 and 5.4.5, it is enough to prove that $R_1 \cap (-R_2)$ contains a system of positive roots of R. We are thus reduced to the following lemma.

Lemma 4.2.2.— *Let R be a "root system"—for example, the set of roots of a root datum in the sense of Exposé XXI. Let R_1, R_2 be two closed subsets of R, each containing a system of positive roots. If $R_1 \cup R_2 = R$, then $R_1 \cap (-R_2)$ contains a system of positive roots.*

Indeed, $R_3 = R_1 \cap (-R_2)$ is closed, so by Exp. XXI, 3.3.6 it is enough to show that $R_3 \cup (-R_3) = R$. Now

$$R_1 \cup (-R_1) = R = R_2 \cup (-R_2),$$

and the conclusion follows from the following elementary fact. If A, A', B, B' are four subsets of a set E, and

$$A \cup A' = B \cup B' = A \cup B = A' \cup B' = E,$$

then

$$(A \cap B') \cup (A' \cap B) = E.$$

This proves (v) ⇒ (vi) over the spectrum of an algebraically closed field. Return now to the general case and suppose (v) holds. Let $s \in S$. By the preceding argument, there is a Borel subgroup $\bar{B}$ of $P_{\bar{s}}$ and a Borel subgroup $\bar{B}'$ of $Q_{\bar{s}}$ such that $\bar{B} \cap \bar{B}'$ is a maximal torus of $G_{\bar{s}}$.

The S-scheme

$$\underline{\mathrm{Bor}}(P) \simeq \underline{\mathrm{Bor}}(P/\operatorname{rad}^u(P))$$

of Borel subgroups of P is smooth. Applying "Hensel's lemma" (Exp. XI, 1.10) and working étale-locally around s, we may suppose that there is a Borel subgroup B of P specializing to $\bar{B}$. Similarly, there is a Borel subgroup B' of Q specializing to $\bar{B}'$. Since $\bar{B} \cap \bar{B}'$ is a maximal torus of $G_{\bar{s}}$, some open neighborhood U of s has the property that $B_U \cap B'_U$ is a maximal torus of G_U (Exp. XXII, 5.9.4). This proves (vi). □

Definition 4.2.3.— *A pair (P, Q) satisfying the equivalent conditions* (i)–(vii) *of Theorem 4.2.1 is said to be in* transversal position. *One also says that P is in transversal position relative to Q, or, by an abuse of language, that P and Q are in (mutual) transversal position.*

In view of (vi), this definition agrees, for Borel subgroups, with that of Exp. XXII, 5.9.1.

Corollary 4.2.4.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G.*

(i) *The pair (P, Q) is in transversal position if and only if, for every $s \in S$, the pair $(P_{\bar{s}}, Q_{\bar{s}})$ is in transversal position. If $S' \to S$ is surjective and $(P_{S'}, Q_{S'})$ is in transversal position, then so is (P, Q).*

(ii) *There is an open subscheme $U \subset S$ with the following property: a morphism $S' \to S$ factors through U if and only if $(P_{S'}, Q_{S'})$ is in transversal position.*

(iii) *Consider the subfunctors*

$$\underline{\mathrm{Gen}}(G) \subset \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G), \qquad \underline{\mathrm{Gen}}(/Q) \subset \underline{\mathrm{Par}}(G), \qquad \underline{\mathrm{Gen}}(P/Q) \subset G,$$

defined as follows. For $S' \to S$, the set $\underline{\mathrm{Gen}}(G)(S')$ consists of pairs of parabolic subgroups of $G_{S'}$ in transversal position; $\underline{\mathrm{Gen}}(/Q)(S')$ consists of parabolic subgroups of

$G_{S'}$ in transversal position relative to $Q_{S'}$; and $\underline{\mathrm{Gen}}(P/Q)(S')$ consists of the $g \in G(S')$ such that $\mathrm{int}(g)P_{S'}$ is in transversal position relative to $Q_{S'}$.

Each of these functors is represented by an open subscheme, universally schematically dense over S, of the corresponding S-scheme

$$\underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G), \qquad \underline{\mathrm{Par}}(G), \qquad G,$$

respectively.

[81]

Assertion (i) follows immediately from condition 4.2.1(i). For (ii), take

$$U = S - \mathrm{Supp}(\mathrm{Coker}\, u),$$

where

$$u : \mathrm{Lie}(P) \oplus \mathrm{Lie}(Q) \longrightarrow \mathrm{Lie}(G)$$

is the canonical morphism.[82]

There are cartesian diagrams

$$\begin{tikzcd}
G \arrow[r, "f"] & \underline{\mathrm{Par}}(G) & \underline{\mathrm{Par}}(G) \arrow[r, "f'"] & \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G) \\
\underline{\mathrm{Gen}}(P/Q) \arrow[u, hook] \arrow[r] & \underline{\mathrm{Gen}}(/Q) \arrow[u, hook] & \underline{\mathrm{Gen}}(/Q) \arrow[u, hook] \arrow[r] & \underline{\mathrm{Gen}}(G), \arrow[u, hook]
\end{tikzcd} \tag{22}$$

where $f(g) = \mathrm{int}(g)P$ and $f'(R) = (R, Q)$. It therefore suffices to prove (iii) for $\underline{\mathrm{Gen}}(G)$.

Let P_0 be the canonical parabolic subgroup of $G_{\underline{\mathrm{Par}}(G)}$, and set

$$X = \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G), \qquad P = \mathrm{pr}_1^*(P_0), \qquad Q = \mathrm{pr}_2^*(P_0).$$

Applying (ii) to the parabolic subgroups P, Q of G_X constructs an open subscheme $U \subset X$, which is immediately identified with $\underline{\mathrm{Gen}}(G)$. The density assertion may be checked on geometric fibers,[83] so assume $S = \mathrm{Spec}(k)$, where k is algebraically closed. Since $\underline{\mathrm{Par}}(G)$ is smooth, it is enough to show that $\underline{\mathrm{Gen}}(G)$ meets every irreducible component of $\underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G)$. By 3.3, this amounts to the following: given $t, t' \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S)$, find a pair (P, P') in transversal position such that $t(P) = t$ and $t(P') = t'$.

Choose Borel subgroups B, B' of G such that $B \cap B'$ is a maximal torus (split G and apply Exp. XXII, 5.9.2). By 3.8 there are parabolic subgroups $P \supset B$ and $P' \supset B'$ with $t(P) = t$ and $t(P') = t'$. They have the required types and are in transversal position by 4.2.1(vi). □

Corollary 4.2.5.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G, with (P, Q) in transversal position.*

(i) *Let P', Q' be parabolic subgroups of G of the same types as P, Q, respectively. The pair (P', Q') is in transversal position if and only if, étale-locally, it is conjugate to (P, Q).* (In ğ5 we shall see that the étale topology can be replaced by the Zariski topology.)

[81] *Editor's note.* The original expression "relatively dense" was replaced by "universally schematically dense over S"; compare EGA IV_3, Definition 11.10.8.

[82] *Editor's note.* The second summand, printed as $\mathrm{Lie}(G)$, has been corrected to $\mathrm{Lie}(Q)$; this is the map used in Theorem 4.2.1(i).

[83] *Editor's note.* Compare EGA IV_3, 11.10.10.

(ii) *The canonical morphism $P \times_S Q \to G$ induces a smooth surjective morphism*

$$P \times_S Q \longrightarrow \underline{\mathrm{Gen}}(Q/P)$$

and an isomorphism

$$(P \times_S Q)/(P \cap Q) \xrightarrow{\sim} \underline{\mathrm{Gen}}(Q/P),$$

where $R = P \cap Q$ acts on $P \times_S Q$ by

$$(p, q)r = (pr, r^{-1}q).$$

(iii) *The canonical morphism*

$$P \longrightarrow \underline{\mathrm{Par}}_{t(Q)}(G), \qquad p \longmapsto \mathrm{int}(p)Q,$$

induces a smooth surjective morphism

$$P \longrightarrow \underline{\mathrm{Gen}}(/P) \cap \underline{\mathrm{Par}}_{t(Q)}(G)$$

and an isomorphism

$$P/(P \cap Q) \xrightarrow{\sim} \underline{\mathrm{Gen}}(/P) \cap \underline{\mathrm{Par}}_{t(Q)}(G).$$

(iv) *The canonical morphism*

$$G \longrightarrow \underline{\mathrm{Par}}_{t(P)}(G) \times_S \underline{\mathrm{Par}}_{t(Q)}(G), \qquad g \longmapsto \big(\mathrm{int}(g)P, \mathrm{int}(g)Q\big),$$

induces a smooth surjective morphism

$$G \longrightarrow \underline{\mathrm{Gen}}(G) \cap \big(\underline{\mathrm{Par}}_{t(P)}(G) \times_S \underline{\mathrm{Par}}_{t(Q)}(G)\big)$$

and an isomorphism

$$G/(P \cap Q) \xrightarrow{\sim} \underline{\mathrm{Gen}}(G) \cap \big(\underline{\mathrm{Par}}_{t(P)}(G) \times_S \underline{\mathrm{Par}}_{t(Q)}(G)\big).$$

Proof. Let us prove (i). The stated condition is plainly sufficient. Conversely, suppose (P', Q') is in transversal position. Since P and P' are conjugate étale-locally, we may assume $P = P'$. It is then enough to show that two parabolic subgroups Q, Q', of the same type and both in transversal position relative to P, are étale-locally conjugate by a section of P.

By 4.2.1(vi), after another étale localization there are Borel subgroups $B, B' \subset P$, $B_1 \subset Q$, and $B_1' \subset Q'$ such that

$$B \cap B_1 = T, \qquad B' \cap B_1' = T'$$

are maximal tori of G. The Killing pairs (B, T) and (B', T') of P are étale-locally conjugate in P (1.16). We may therefore assume $B = B'$ and $T = T'$. Then $B_1 = B_1'$ by Exp. XXII, 5.9.2, and hence $Q = Q'$ by 3.8.

Parts (ii), (iii), and (iv) are proved in parallel. We prove (ii). Let $g \in \underline{\mathrm{Gen}}(Q/P)(S)$; thus $g \in G(S)$ and $\mathrm{int}(g)Q$ is in transversal position relative to P. The preceding argument shows that Q and $\mathrm{int}(g)Q$ are, étale-locally, conjugate by a section of P. Working locally for this topology, choose $p \in P(S)$ such that $\mathrm{int}(g)Q = \mathrm{int}(p)Q$. Then

$$p^{-1}g \in \underline{\mathrm{Norm}}_G(Q)(S) = Q(S),$$

so $g = pq$ for some $q \in Q(S)$. Thus the morphism in (ii) is an étale covering. Comparing with 4.2.1(ii), it is smooth and surjective.

Finally, the equivalence relation on $P \times_S Q$ defined by $P \times_S Q \to G$ is exactly the relation associated with the action of $R = P \cap Q$,

$$(p, q)r = (pr, r^{-1}q).$$

This proves the quotient assertion, since a smooth surjective morphism is, in particular, an effective epimorphism. □

Remark 4.2.6.— *If P, Q are in transversal position, the open subset $\underline{\mathrm{Gen}}(Q/P)$ of G will often be denoted $P \cdot Q$; this notation is justified by 4.2.5*(ii)*.*

Proposition 4.2.7.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G, with (P, Q) in transversal position.*

(i) *The group $P \cap Q$ is smooth over S (and in fact has connected fibers by 4.1.2). Its identity component* $(P \cap Q)^0$ (Exp. $\mathrm{VI_B}$, 3.10) *is a subgroup of type* (RC) *of G* (Exp. XXII, 5.11.1)*, and its unipotent radical* (loc. cit., 5.11.4) *decomposes as the direct product*

$$\mathrm{rad}^u\big((P \cap Q)^0\big) = \big(\mathrm{rad}^u(P) \cap Q\big) \times_S \big(P \cap \mathrm{rad}^u(Q)\big).$$

(ii) *If S is affine, then*

$$H^1\big(S, \mathrm{rad}^u((P \cap Q)^0)\big) = 0.$$

If S is semilocal, $P \cap Q$ contains a maximal torus of G.

Proof. The group $P \cap Q$ is smooth by 4.2.1(ii) and the cartesian diagram

$$\begin{array}{ccc} P \times_S Q & \longrightarrow & G \\ \uparrow & & \uparrow \\ P \cap Q & \longrightarrow & S \end{array} \quad . \qquad (23)$$

The remaining assertions about $(P \cap Q)^0$ may be checked étale-locally. By 4.2.1(vii), choose a splitting (G, T, M, R) of G such that P, Q contain T and are defined by subsets R_1, R_2 of R, where R_1 contains a system R^+ of positive roots and R_2 contains the opposite system R^-. Let Δ be the set of simple roots of R^+, and put

$$A_1 = \Delta \cap (-R_1), \qquad A_2 = \Delta \cap R_2, \qquad A = A_1 \cap A_2.$$

By 1.4(v) and 1.12,

$$R_1 = R^+ \cup \big(R^- \cap (-\mathbb{N}A_1)\big), \qquad R_2 = \big(R^+ \cap \mathbb{N}A_2\big) \cup R^-,$$
$$\mathrm{rad}^u(P) = \prod_{\substack{\alpha \in R_1 \\ \alpha \notin -R_1}} U_\alpha, \qquad \mathrm{rad}^u(Q) = \prod_{\substack{\alpha \in R_2 \\ \alpha \notin -R_2}} U_\alpha.$$

Exp. XXII, 5.6.7 therefore gives

$$\mathrm{rad}^u(P) \cap Q = \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_1}} U_\alpha = \prod_{\alpha \in K_2} U_\alpha,$$
$$\mathrm{rad}^u(Q) \cap P = \prod_{\substack{\alpha \in R_1 \cap R_2 \\ \alpha \notin -R_2}} U_\alpha = \prod_{\alpha \in K_1} U_\alpha,$$

where K_2 consists of the positive roots that are linear combinations of elements of A_2, but not of elements of A, and K_1 consists of the negative roots that are linear combinations of elements of A_1, but not of elements of A. If $\alpha \in K_2$ and $\beta \in K_1$, then $\alpha + \beta$ is neither a root nor zero. Hence the two groups displayed above commute.

By Exp. XXII, 5.4.5, the group

$$H = (P \cap Q)^0$$

is defined by the set of roots

$$R_1 \cap R_2 = \big(R^+ \cap \mathbb{N}A_2\big) \cup \big(R^- \cap (-\mathbb{N}A_1)\big).$$

This set is closed, so H is of type (RC) by definition (Exp. XXII, 5.11.1). By loc. cit., 5.11.3 and 5.11.4,

$$\mathrm{rad}^u(H) = \prod_{\alpha \in K} U_\alpha,$$

where K consists of the $\alpha \in R_1 \cap R_2$ such that $\alpha \notin -(R_1 \cap R_2)$. The symmetric part of $R_1 \cap R_2$ is $R \cap \mathbb{Z}A$, and therefore $K = K_1 \cup K_2$. This proves (i).

The first assertion of (ii) follows from (i) and 2.10. For the second, the reductive group

$$(P \cap Q)^0 / \mathrm{rad}^u((P \cap Q)^0)$$

has a maximal torus $\bar{T}$ when the base is semilocal. The inverse image N of $\bar{T}$ in $(P \cap Q)^0$ is a subgroup of type (R) of G, with solvable fibers, and

$$N^u = \mathrm{rad}^u((P \cap Q)^0)$$

(Exp. XXII, 5.6.9). The scheme of maximal tori of N is a principal homogeneous bundle under N^u (loc. cit., 5.6.13), and therefore has a section because $H^1(S, N^u) = 0$. □

4.3. Opposite parabolic subgroups

4.3.1. Let G be an S-reductive group. Exp. XXIV, 3.16.6 defines a canonical "outer automorphism" s_G of G, of order at most 2.[84] It induces a canonical automorphism, again of order at most 2, of $\underline{\mathrm{Dyn}}(G)$, and hence of $\mathrm{Of}(\underline{\mathrm{Dyn}}(G))$. We denote all of these automorphisms by s_G, or simply by s. Two types of parabolic subgroups

$$t, t' \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S)$$

are said to be *opposite* if $t = s_G(t')$.

Theorem 4.3.2.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G.*

(a) *If L is a Levi subgroup of P, there is a unique parabolic subgroup P' of G such that $P \cap P' = L$.*

(b) *For every parabolic subgroup Q of G, the following conditions are equivalent:*

(i) *For every $s \in S$, the identity component $\big((P \cap Q)_{\bar{s}}\big)^0$, which is smooth by 4.1.1, is reductive.*

(ii) *The group $P \cap Q$ is a Levi subgroup of both P and Q.*

[84] *Editor's note.* "Of order 2" was replaced by "of order at most 2", since s_G may be trivial, for example when G has type $A_1, B_n, C_n, \ldots$.

(iii) *The groups P, Q have opposite types and the pair (P, Q) is in transversal position.*

(iv) *The groups P, Q have opposite types and* $\mathrm{rad}^u(P) \cap Q = e$.

(v)
$$\mathrm{rad}^u(P) \cap Q = \mathrm{rad}^u(Q) \cap P = e.$$

(vi) *The canonical morphism* $\mathrm{rad}^u(P) \times_S Q \to G$ *is an open immersion.*

(vi′) *The canonical morphism* $\mathrm{rad}^u(P) \to G/Q$ *is an open immersion.*

(vii) *There is an étale covering family $\{S_i \to S\}$ and, for every i, a splitting (T_i, M_i, R_i) of G_{S_i} and a subset $R_i^{(1)} \subset R_i$, such that P_{S_i} is the subgroup of type* (R) *containing T_i and defined by $R_i^{(1)}$, while Q_{S_i} is the subgroup defined by*
$$R_i^{(2)} = -R_i^{(1)}.$$

Proof. We first prove part (b). Conditions (iii) and (vii) are equivalent by 4.2.1(vii) and the definition of s_G in the split case (Exp. XXIV, 3.16.2(iv)). Plainly (ii) implies (i), and (vi′) implies (vi) by base change along $G \to G/Q$.

Suppose now that (vii) holds. Since every assertion to be proved is étale-local, assume that $G = (G, T, M, R)$ is split, that P is defined by $R' \subset R$, and that Q is defined by $-R'$. Let L be the subgroup of type (R) containing T and defined by $R' \cap (-R')$. By Exp. XXII, 5.11.3, L is a Levi subgroup of both P and Q. Moreover,
$$P = L \cdot \mathrm{rad}^u(P), \qquad Q = L \cdot \mathrm{rad}^u(Q),$$
and Exp. XXII, 5.6.7 gives
$$\mathrm{rad}^u(P) \cap Q = Q \cap \mathrm{rad}^u(P) = e.$$
Thus $P \cap Q = L$, proving (ii) and (v).

Since P, Q are in transversal position, the canonical morphism $P \to G/Q$ induces an open immersion
$$P/(P \cap Q) \longrightarrow G/Q$$
by 4.2.1. The canonical morphism
$$\mathrm{rad}^u(P) \longrightarrow P/(P \cap Q) = P/L$$
is an isomorphism, so (vi′) follows. Hence all the conditions follow from (vii).

It remains to show that each of (i), (iv), (v), and (vi) implies (vii). Since the equivalence of (ii) and (iii) is already known, this may be checked on geometric fibers. We may suppose that S is the spectrum of an algebraically closed field. By 4.1.1, a maximal torus T of G is contained in $P \cap Q$. Let R, R_1, R_2 be the sets of roots of G, P, Q, respectively, relative to T. Write
$$R_1^a = \{\alpha \in R_1 \mid -\alpha \notin R_1\},$$
and define R_2^a similarly. We must prove $R_1 = -R_2$.

- Condition (i) says that $R_1 \cap R_2$ is symmetric. If $\alpha \in R_1$ and $\alpha \notin R_2$, then $\alpha \in -R_2$; if $\alpha \in R_2$, then
$$\alpha \in R_1 \cap R_2 = -(R_1 \cap R_2) \subset -R_2.$$
Thus $R_1 \subset -R_2$, and symmetry gives $R_1 = -R_2$.

- Condition (iv) gives $\mathrm{Card}(R_1) = \mathrm{Card}(R_2)$ and $R_1^a \cap R_2 = \varnothing$. The latter condition is equivalent to $R_2 \subset -R_1$, and equality of cardinalities then gives $R_2 = -R_1$.

- Condition (v) gives
$$R_1^a \cap R_2 = R_2^a \cap R_1 = \varnothing.$$
Hence $R_2 \subset -R_1$ and $R_1 \subset -R_2$, so again $R_2 = -R_1$.

- Condition (vi) gives
$$\mathrm{Lie}(\mathrm{rad}^u(P)) \oplus \mathrm{Lie}(Q) = \mathrm{Lie}(G).$$
Thus R is the disjoint union of R_2 and R_1^a, which implies $R_2 = -R_1$.

This proves part (b). For part (a), the implication (vii) $\Rightarrow$ (ii) already gives the desired parabolic subgroup P' étale-locally. It remains only to prove uniqueness, and this too is étale-local. We may suppose that G is split relative to a maximal torus T of L, with P and P' defined by subsets R_1 and R_1' of R. By hypothesis $R_1 \cap R_1'$ is symmetric. The preceding argument gives $R_1' = -R_1$; hence P' is determined by P and L. □

Definition 4.3.3.— *Two parabolic subgroups of G satisfying the equivalent conditions* (i)–(vii) *of Theorem 4.3.2 are said to be* opposite. *If P is a parabolic subgroup of G and L is a Levi subgroup of P, the* parabolic subgroup opposite to P relative to L *is the unique parabolic subgroup Q of G such that $P \cap Q = L$.*

If T is a maximal torus of P, the parabolic subgroup opposite to P relative to T *is the unique Q such that $P \cap Q$ is the unique Levi subgroup of P containing T* (1.6)*; equivalently, it is the unique Q such that $P \cap Q$ contains T and P, Q are opposite.*

By 4.3.2(iii), Corollaries 4.2.4 and 4.2.5 immediately yield parallel results. We record one sample.

Corollary 4.3.4.— *Let S be a scheme, G an S-reductive group, and P, Q two parabolic subgroups of G.*

(i) *The groups P, Q are opposite if and only if $P_{\bar{s}}, Q_{\bar{s}}$ are opposite for every $s \in S$. If $S' \to S$ is surjective and $P_{S'}, Q_{S'}$ are opposite, then P, Q are opposite.*

(ii) *Let $\underline{\mathrm{Opp}}(G)(S')$ be the set of pairs of opposite parabolic subgroups of $G_{S'}$, for $S' \to S$. This functor is represented by an open subscheme of $\underline{\mathrm{Par}}(G)^2$. Let $\underline{\mathrm{Opp}}(/P)(S')$ be the set of parabolic subgroups of $G_{S'}$ opposite to $P_{S'}$. This functor is represented by an open subscheme, universally schematically dense over S, of*
$$\underline{\mathrm{Par}}_{s(t(P))}(G).$$

(iii) *Suppose P, Q are opposite, and let P', Q' be parabolic subgroups of G, with P' of the same type as P. The groups P', Q' are opposite if and only if, étale-locally, the pair (P', Q') is conjugate to (P, Q).* (In ğ5 the étale topology will be replaced by the Zariski topology.)

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Corollary 4.3.5.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G.*

[85] *Editor's note.* Here again, "relatively dense" was replaced by "universally schematically dense over S"; compare EGA IV_3, Definition 11.10.8.

(i) *The morphism*

$$\underline{\mathrm{Opp}}(/P) \longrightarrow \underline{\mathrm{Lev}}(P), \qquad Q \longmapsto P \cap Q,$$

is an isomorphism. The scheme $\underline{\mathrm{Opp}}(/P)$ *is a principal homogeneous bundle under* $\mathrm{rad}^u(P)$*, acting by inner automorphisms. If* S *is affine,* G *has a parabolic subgroup opposite to* P*.*

(ii) *Suppose* S *is semilocal, with closed points* $\{s_i\}$*. For each* i*, let* Q_i *be a parabolic subgroup of* G_{s_i} *opposite to* P_{s_i}*. There is a parabolic subgroup* Q *of* G*, opposite to* P*, such that* $Q_{s_i} = Q_i$ *for every* i*.*

(iii) *The morphism*

$$\underline{\mathrm{Opp}}(G) \longrightarrow \mathrm{PL}, \qquad (P, Q) \longmapsto (P, P \cap Q),$$

is an isomorphism.

These assertions follow from Theorem 4.3.2(a) and from 1.9, 2.3, and 2.8.

Remark 4.3.6.— *Let* P, Q *be opposite parabolic subgroups of* G*, and let* $P \cdot Q$ *be the open subscheme of* G*, introduced in 4.2.6, which is the sheaf-theoretic image of* $P \times_S Q$*. The product morphism* $G \times_S G \to G$ *induces isomorphisms*

$$\mathrm{rad}^u(P) \times_S Q \xrightarrow{\sim} P \cdot Q \xleftarrow{\sim} P \times_S \mathrm{rad}^u(Q).$$

Indeed, this follows from 4.3.2 (or 4.2.5(ii)) and from the fact that $P \cap Q$ is a Levi subgroup of both P and Q:

$$P = \mathrm{rad}^u(P) \cdot (P \cap Q), \qquad Q = \mathrm{rad}^u(Q) \cdot (P \cap Q).$$

Likewise, there is a commutative diagram

$$\begin{tikzcd}
\mathrm{rad}^u(P) \arrow[r, hook] \arrow[d, "\sim"'] & G/Q \arrow[d, "\sim"] \\
\underline{\mathrm{Opp}}(/P) \arrow[r, hook] & \underline{\mathrm{Par}}_{t(Q)}(G),
\end{tikzcd} \tag{24}$$

in which the vertical arrows are induced by $g \mapsto \mathrm{int}(g)Q$.

4.4. Osculating position

Proposition 4.4.1.— *Let* S *be a scheme,* G *an* S*-reductive group, and* P, Q *two parabolic subgroups of* G*. The following conditions are equivalent:*

(i) $P \cap Q$ *is a parabolic subgroup of* G*.*

(ii) *Étale-locally,* $P \cap Q$ *contains a Borel subgroup of* G*.*

(iii) *Étale-locally,* $P \cap Q$ *contains a maximal torus of* G*; moreover, for every* $S' \to S$ *and every maximal torus* T *of* $G_{S'}$ *contained in both* $P_{S'}$ *and* $Q_{S'}$*, the parabolic subgroup opposite to* $P_{S'}$ *relative to* T *is in transverse position relative to* $Q_{S'}$*.*

(iv) *There is an étale covering family* $\{S_i \to S\}$ *and, for each* i*, a maximal torus* T_i *of* G_{S_i}*, contained in both* P_{S_i} *and* Q_{S_i}*, such that the parabolic subgroup opposite to* P_{S_i} *relative to* T_i *is in transverse position relative to* Q_{S_i}*.*

(v) *There is an étale covering family* $\{S_i \to S\}$ *and, for each* i*, a splitting* (T_i, M_i, R_i) *of* G_{S_i} *such that* P_{S_i}*, respectively* Q_{S_i}*, is the subgroup of type* (R) *of* G_{S_i} *containing* T_i *and defined by a set of roots* $R_i^{(1)}$*, respectively* $R_i^{(2)}$*, with* $R_i^{(1)} \cap R_i^{(2)}$ *containing a system of positive roots of* R_i*.*

Moreover, when these conditions hold,

$$t(P \cap Q) = t(P) \cap t(Q)$$

in the notation of 3.2.

The implications (v)⇒(ii) and (iii)⇒(iv) are immediate. Moreover, (ii)⇒(i) follows from 1.18. To prove (iv)⇒(v), one may suppose that G is split and that P, respectively Q, is defined by a set of roots R_1, respectively R_2. The parabolic subgroup opposite to P is then defined by $-R_1$, and the assertion reduces to Lemma 4.2.2. The implication (i)⇒(iii) is proved in the same way after splitting. Finally, the last assertion may be proved étale-locally: one may suppose that $P \cap Q$ contains a Borel subgroup B of G, and the assertion then reduces to 3.7.

Definition 4.4.2.— *Two parabolic subgroups of* G *satisfying the equivalent conditions* (i)–(v) *of 4.4.1 are said to be in* osculating position.

Corollary 4.4.3.— *Let* P, Q *be parabolic subgroups in osculating position, and let* P', Q' *be parabolic subgroups of* G *of the same types as* P, Q*, respectively. The subgroups* P', Q' *are in osculating position if and only if, étale-locally, the pair* (P', Q') *is conjugate to the pair* (P, Q)*.*

It is enough to show that, if both P and P' are in osculating position relative to Q, then they are conjugate, étale-locally, by a section of Q. Now $P \cap Q$ and $P' \cap Q$ are parabolic subgroups of the same type contained in Q. By part (ii) of the lemma below, they are therefore conjugate étale-locally by a section of Q. We may consequently suppose $P \cap Q = P' \cap Q$; part (i) of that lemma then gives $P = P'$.

Lemma 4.4.4.— *Let* P, P', Q *be three parabolic subgroups of the* S*-reductive group* G*.*

(i) *The equality* $P = P'$ *holds if and only if* P *and* P' *are in osculating position and have the same type.*

(ii) *Suppose* $P \subset Q$ *and* $P' \subset Q$*. If* $g \in G(S)$ *is such that* $\mathrm{int}(g)P$ *and* P' *are in osculating position, then* $g \in Q(S)$*.*

Part (i) follows immediately from the last assertion of 4.4.1. To prove (ii), observe that Q and $\mathrm{int}(g)Q$ both contain $P' \cap \mathrm{int}(g)P$, and hence are in osculating position. They coincide by (i), so that

$$g \in \underline{\mathrm{Norm}}_G(Q)(S) = Q(S).$$

Assertions (iii) and (iv) of Proposition 4.4.1 give at once:

Corollary 4.4.5.— *Let* P, P', Q *be three parabolic subgroups of the* S*-reductive group* G*, containing the same maximal torus* T *of* G*. Suppose that* P *and* P' *are opposite relative to* T*. Then* Q *is in osculating position relative to* P *if and only if it is in transverse position relative to* P'*. Under these conditions,* $P \cap Q$ *is also in transverse position relative to* P'*.*

Corollary 4.4.6.— *Let P, Q be two parabolic subgroups of G containing the same maximal torus T. The subgroups P, Q are in transverse position if and only if there are parabolic subgroups $P' \subset P$ and $Q' \subset Q$ of G, opposite relative to T. One may even choose*

$$t(P') = t(P) \cap s(t(Q)).$$

[86]

The condition is plainly sufficient, by 4.2.1(i) and 4.3.2(iii). Conversely, let P^-, respectively Q^-, be the parabolic subgroup opposite to P, respectively Q, relative to T. By 4.4.5, $P^- \cap Q$ is in transverse position relative to both P and Q^-, and therefore also relative to $P \cap Q^-$, by another application of 4.4.5. Moreover,

$$\begin{aligned} t(P^- \cap Q) &= t(P^-) \cap t(Q) \\ &= s(t(P)) \cap s(t(Q^-)) \\ &= s\big(t(P) \cap t(Q^-)\big) \\ &= s\big(t(P \cap Q^-)\big). \end{aligned}$$

It follows that

$$P^- \cap Q = P', \qquad P \cap Q^- = Q'$$

are opposite, by 4.3.2(iii). Since $P' \cap Q' \supset T$, they are indeed opposite relative to T.

4.5. Standard position

In this subsection we briefly indicate how some of the preceding results generalize.

Proposition 4.5.1.— *Let P_1, P_2 be two parabolic subgroups of the S-reductive group G. The following conditions are equivalent:*

(i) *$P_1 \cap P_2$ is smooth.*

(ii) *$P_1 \cap P_2$ is a subgroup of type* (R)*, or of type* (RC)*, of G.*

(iii) *Fpqc-locally, $P_1 \cap P_2$ contains a maximal torus of G.*

(iv) *Zariski-locally, $P_1 \cap P_2$ contains a maximal torus of G.*

When S is semilocal, these conditions are also equivalent to:

(v) *$P_1 \cap P_2$ contains a maximal torus of G.*

Proof. Clearly, (iv) implies (iii), and (v) implies (iv) when S is semilocal; condition (ii) implies (i) by Exp. XXII, Definition 5.2.1. We shall prove

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \text{(iii)},$$

and then show that (iii) implies (i) and (iv), as well as (v) when S is semilocal. Put $K = P_1 \cap P_2$. By 4.1.1 and [BT65], 4.5, every geometric fiber $K_{\bar{s}}$ contains a maximal torus $T_{\bar{s}}$ of $G_{\bar{s}}$, is connected, and has, relative to $T_{\bar{s}}$, a closed subset of the roots of $G_{\bar{s}}$. Consequently, if K is smooth over S, then it is a subgroup of type (RC), hence a fortiori of type (R); see Exp. XXII, 5.2.1 and 5.11.1. Thus (i) and (ii) are equivalent.

If K is of type (R), it contains a maximal torus of G étale-locally, by Exp. XXII, 2.2 and Exp. XIX, 6.1. Hence (ii) implies (iii).

[86] *Editor's note.* The involution s was defined in 4.3.1.

Assume (iii), and let us prove that K is smooth. By fpqc descent we may suppose that

$$G = (G, T, M, R)$$

is split, where T is a maximal torus contained in $P_1 \cap P_2$, and that there are closed subsets R_1, R_2 of R such that

$$P_1 = H_{R_1}, \qquad P_2 = H_{R_2}.$$

Since K has connected fibers, Exp. XXII, 5.4.5 gives

$$K = H_{R_1 \cap R_2};$$

therefore K is smooth over S and of type (RC).

Let R_1^s be the symmetric part of R_1, and put $R_1^a = R_1 - R_1^s$; then

$$\mathrm{rad}^u(P_1) = H_{R_1^a}.$$

As observed in the proof of [BT65], 4.4,

$$R' = (R_1^s \cap R_2) \cup R_1^a$$

is a closed subset of R such that $R' \cup (-R') = R$. Hence, by Exp. XXI, 3.3.6, R' contains a system of positive roots of R. Moreover, the symmetric part of R' is $R_1^s \cap R_2^s$, which is contained in $R_1 \cap R_2$.

It follows that

$$P' = K \cdot \mathrm{rad}^u(P_1)$$

is a parabolic subgroup of G, and that K is a subgroup of type (RC) of P' satisfying

$$\mathrm{rad}^u(K) = K \cap \mathrm{rad}^u(P').$$

By 2.11, K therefore has a Levi subgroup L. Exp. XIV, 3.20 and 3.21 then show that K satisfies (iv), and also (v) when S is semilocal. This proves Proposition 4.5.1. □

[87]

Definition 4.5.1.1.— *When the preceding conditions hold, P_1 and P_2 are said to be in* mutual standard position*. This is the case, for example, when they are in transverse position, when they are in osculating position, or when the base is the spectrum of a field. The notion is stable under base extension and is fpqc-local.*

[88]

4.5.2. Let (P, Q) and (P', Q') be two pairs of parabolic subgroups of G in standard position. Define the subfunctor H of G by letting $H(S')$ be the set of $g \in G(S')$ such that

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)Q = Q'.$$

It is a closed subscheme of G, smooth over S and formally a principal homogeneous space under $P \cap Q$. It follows that the following conditions are equivalent:

(i) (P, Q) and (P', Q') are conjugate fpqc-locally;

(ii) (P, Q) and (P', Q') are conjugate étale-locally;

(iii) (P, Q) and (P', Q') are conjugate on every geometric fiber.

The pairs (P, Q) and (P', Q') are then said to have the same *type of mutual position*. This notion is stable under base change and is fpqc-local.

[87] *Editor's note.* The proof of the equivalence, together with condition (v), which was used implicitly in 6.17 of the original edition, was added in the Polo–Gille edition. The original 4.5.1 was consequently divided into Proposition 4.5.1 and Definition 4.5.1.1.

[88] *Editor's note.* See the preceding editorial note. For an example of parabolic subgroups that are not in standard position, see 7.11 below.

4.5.3. Let $\underline{\mathrm{Stand}}(G)$ be the subfunctor of $\underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G)$ formed by the pairs in mutual standard position. The functor $\underline{\mathrm{Stand}}(G)$ is representable. There is a finite étale S-scheme $\underline{\mathrm{TypeStand}}$, the *scheme of types of mutual standard position*, and a smooth morphism of finite presentation with irreducible geometric fibers, hence in particular faithfully flat,

$$t_2 : \underline{\mathrm{Stand}}(G) \longrightarrow \underline{\mathrm{TypeStand}}.$$

This morphism is a quotient of $\underline{\mathrm{Stand}}(G)$ by the action of G: two sections of $\underline{\mathrm{Stand}}(G)$, over any $S' \to S$, have the same type of mutual position if and only if they have the same image under t_2. There is a commutative diagram

$$\begin{tikzcd} \underline{\mathrm{Stand}}(G) \arrow[r, "t_2"] \arrow[d] & \underline{\mathrm{TypeStand}} \arrow[d, "q"] \\ \underline{\mathrm{Par}}(G) \times_S \underline{\mathrm{Par}}(G) \arrow[r, "t \times t"'] & \mathrm{Of}(\underline{\mathrm{Dyn}}(G)) \times_S \mathrm{Of}(\underline{\mathrm{Dyn}}(G)). \end{tikzcd} \tag{25}$$

The morphism q may be described by descent as follows. Let (P, Q) be a pair of parabolic subgroups of G in mutual standard position, and let T be a maximal torus of $P \cap Q$. The morphism

$$\underline{\mathrm{Norm}}_G(T) \longrightarrow \underline{\mathrm{Stand}}(G), \qquad n \longmapsto (P, \mathrm{int}(n)Q),$$

defined set-theoretically as displayed, induces an isomorphism

$$W_P(T) \backslash W_G(T) / W_Q(T) \xrightarrow{\sim} q^{-1}(t(P), t(Q)).$$

The left-hand side denotes the sheaf of double cosets. These assertions are immediate; note in particular that

$$t_2^{-1}(t_2(P, Q)) \simeq G/(P \cap Q).$$

4.5.4. Fix a parabolic subgroup P of G, and let $\underline{\mathrm{Par}}(G; P)$ be the functor of parabolic subgroups of G in standard position relative to P. For every

$$t \in \mathrm{Of}(\underline{\mathrm{Dyn}}(G))(S),$$

put

$$\underline{\mathrm{Par}}_t(G; P) = \underline{\mathrm{Par}}(G; P) \cap \underline{\mathrm{Par}}_t(G).$$

Both functors are obtained from $\underline{\mathrm{Stand}}(G)$ by fiber products, and are therefore represented by S-schemes, smooth and of finite presentation over S, with nonempty fibers. There is a canonical morphism induced by t_2,

$$t_P : \underline{\mathrm{Par}}_t(G; P) \longrightarrow q^{-1}(t(P), t), \qquad t_P(Q) = t_2(P, Q),$$

which is smooth and of finite presentation, with irreducible geometric fibers. The canonical morphism

$$\underline{\mathrm{Par}}_t(G; P) \longrightarrow \underline{\mathrm{Par}}_t(G)$$

is a surjective monomorphism, and may therefore be viewed as a *cell decomposition* of $\underline{\mathrm{Par}}_t(G)$, indexed by the set of connected components of $q^{-1}(t(P), t)$.

4.5.5. Suppose that the type t is of the form $t(Q)$, where Q is a parabolic subgroup of G in standard position relative to P, and suppose that $P \cap Q$ contains a maximal torus T. Then

$$\underline{\mathrm{Par}}_t(G) \simeq G/Q, \qquad q^{-1}(t(P),t) \simeq W_P(T)\backslash W_G(T)/W_Q(T),$$

and there is a diagram

$$\begin{array}{ccc} \underline{\mathrm{Par}}_{t(Q)}(G;P) & \xrightarrow{\ f\ } & W_P(T)\backslash W_G(T)/W_Q(T) \\ {\scriptstyle i}\big\downarrow & & \\ \underline{\mathrm{Par}}_{t(Q)}(G) & \xrightarrow[\sim]{\qquad} & G/Q. \end{array} \tag{26}$$

Here i is a surjective monomorphism, and f is smooth and of finite presentation, with irreducible geometric fibers. Moreover, if Q_1, Q_2 are two sections of $\underline{\mathrm{Par}}_{t(Q)}(G;P)$, over some $S' \to S$, that is, two parabolic subgroups of $G_{S'}$ which are fpqc-locally conjugate to Q and are in standard position relative to $P_{S'}$, then Q_1, Q_2 are fpqc-locally conjugate by a section of P if and only if

$$f(Q_1) = f(Q_2).$$

If S is the spectrum of an algebraically closed field k, this gives

$$P(k)\backslash G(k)/Q(k) \simeq W_P(T)(k)\backslash W_G(T)(k)/W_Q(T)(k).$$

More generally, suppose that the scheme $W_P(T)\backslash W_G(T)/W_Q(T)$ is constant, of the form E_S, as happens for example when G is split relative to T and S is connected. The subschemes $f^{-1}(e)$, for $e \in E$, then decompose $\underline{\mathrm{Par}}_{t(Q)}(G;P)$ into open and closed subschemes. They are homogeneous spaces under P, smooth and of finite presentation over S, with irreducible geometric fibers.

4.5.6. Return to the general situation of 4.5.4. The scheme $q^{-1}(t(P),t)$ always has two distinguished sections, corresponding respectively to the types "transverse position" and "osculating position". The inverse image of the first section is a relatively dense open subscheme of $\underline{\mathrm{Par}}_t(G;P)$, as seen above; it is the cell of greatest relative dimension in the decomposition.

The inverse image of the second section is presumably a closed subscheme of $\underline{\mathrm{Par}}_t(G;P)$; it is the cell of least relative dimension in the decomposition. [89]

5. Conjugacy theorem

Theorem 5.1.— *Let S be a semilocal scheme, G an S-reductive group, and P, P' two opposite parabolic subgroups of G* (4.3.3)*. Then*

$$\mathrm{rad}^u(P)(S) \cdot P' \cdot P = G.$$

In other words, as u runs through $\mathrm{rad}^u(P)(S)$*, the open subsets $uP' \cdot P$ of G* (4.3.6) *cover all of G.*

The proof is divided into several steps.

[89] *Editor's note.* In this paragraph the Polo–Gille edition corrects $t^{-1}(t(P),t)$ to $q^{-1}(t(P),t)$, and twice corrects $\underline{\mathrm{Par}}_t(G)$ to $\underline{\mathrm{Par}}_t(G;P)$.

5.1.1. It is enough to prove the theorem when S is the spectrum of a field k; this follows at once from 2.6.

5.1.2. Put $L = P \cap P'$. Suppose that L has a Borel subgroup B_L, and let T be a maximal torus of B_L, as in Exp. XXII, 5.9.7. One checks immediately that

$$B = B_L \cdot \mathrm{rad}^u(P)$$

is a Borel subgroup of P. Let B' be the Borel subgroup of G opposite to B relative to T, so that $B \cap B' = T$. After splitting G relative to T, one immediately sees that $B' \subset P'$. We claim that

$$B^u(S) \cdot B' \cdot B \subset \mathrm{rad}^u(P)(S) \cdot P' \cdot P. \tag{x}$$

Indeed,

$$B^u(S) \subset P(S) = \mathrm{rad}^u(P)(S) \cdot L(S) \subset \mathrm{rad}^u(P)(S) \cdot P'(S),$$

so it is enough to observe that $B' \cdot B \subset P' \cdot P$. Thus (x) reduces Theorem 5.1 to the pair (B, B').

5.1.3. The theorem holds when k is algebraically closed: the hypothesis of 5.1.2 is then satisfied, and Exp. XXII, 5.7.10 applies.

5.1.4. The theorem also holds when k is infinite. Indeed, by 2.7,

$$\mathrm{rad}^u(P)(k) \text{ is dense in } \mathrm{rad}^u(P)(\bar{k}),$$

and the theorem holds over $\bar{k}$.

5.1.5. We are therefore reduced to the case of a finite field k. The scheme $\underline{\mathrm{Bor}}(L)$ is a smooth homogeneous space under L; Lang's theorem (Amer. J. Math. **78** (1956)) therefore shows that L has a Borel subgroup B_L. [90] By 5.1.2, we may consequently suppose that $P = B$ and $P' = B'$ are Borel subgroups. We write $T = B \cap B'$.

5.1.6. Let K be an algebraic closure of k. Choose a pinning of (G_K, B_K, T_K), and let R^+, respectively Δ, be the set of positive, respectively simple, roots. By Exp. XXII, 5.7.2, it is enough to prove, for every $\alpha \in \Delta$, that

$$u_\alpha {B'}^u(K) \subset B^u(k) {B'}^u(K) B(K). \tag{1}$$

Let α_i be the distinct conjugates of α over k. They belong to Δ, since B is defined over k. Let R' be the set of roots which are linear combinations of the α_i, and put $R'_- = R_- \cap R'$. Since R' is defined over k, there is a subtorus Q of T such that Q_K is the maximal torus of the common kernel of the α_i.

Put

$$Z = \underline{\mathrm{Centr}}_G(Q), \qquad B_Z = B \cap Z, \qquad B'_Z = B' \cap Z$$

(Exp. XXII, 5.10.2). It is enough to verify the desired assertion in Z, namely

$$u_\alpha {B'}^u_Z(K) \subset B^u_Z(k) {B'}^u_Z(K) B_Z(K). \tag{2}$$

[90] *Editor's note.* See also [DG70], §III.5, 7.4.

Indeed,

$$(B'^u_Z)_K = \prod_{\beta \in R'_-} U_\beta.$$

Let $R'' = R - R'$, and set

$$V = \prod_{\beta \in R'' \cap R_-} U_\beta.$$

Then

$$(B'^u)_K = (B'^u_Z)_K \cdot V,$$

and $(B_Z)_K$ normalizes V, by Exp. XXII, 5.6.7. From (2) we therefore obtain successively

$$\begin{aligned} u_\alpha B'^u(K) &= u_\alpha B'^u_Z(K)V(K) \\ &\subset B^u_Z(k)B'^u_Z(K)B_Z(K)V(K) \\ &\subset B^u_Z(k)B'^u_Z(K)V(K)B_Z(K), \end{aligned}$$

which immediately gives (1). We are thus reduced to the case $G = Z$, that is, to the case in which $\mathrm{Gal}(K/k)$ acts transitively on the simple roots.

5.1.7. The assertion to be proved says equivalently that G/B is the union of the translates, under $B^u(k)$, of the open image of B'^u. This assertion is unchanged when G is replaced by its adjoint group, or indeed by any group having the same adjoint group. We may therefore suppose that G is adjoint.

5.1.8. Consider the Dynkin diagram of G_K. The Galois group acts transitively on it. But this Galois group has only cyclic quotients, whereas the Dynkin diagram has no cycles. The diagram is consequently of type nA_1, with $n \geq 0$, or mA_2, with $m \geq 0$. Using the canonical decomposition of Exp. XXIV, 5.9, we may write

$$G = \prod_{D/K} G_0,$$

where D is a finite K-scheme and G_0 is either a torus or is of type A_1 or A_2.

By Exp. XXIV, 5.12, the group B comes from a Borel subgroup B_0 of G_0, and T comes from a maximal torus T_0 of G_0; the group B' comes from the Borel subgroup B'_0 of G_0 opposite to B_0 relative to T_0. We have

$$B'^u(k) = B^u_0(D), \qquad B'^u \cdot T \cdot B^u = \prod_{D/k} B'^u_0 \cdot T_0 \cdot B^u_0.$$

It is therefore enough to prove the assertion for the triple (G_0, B_0, T_0).

5.1.9. We may consequently suppose that G has type $\varnothing$, A_1, or A_2. Since G has a Borel subgroup B, it is quasi-split relative to B, by Exp. XXIV, 3.9.1, and hence split if it has type $\varnothing$ or A_1. The theorem is already known in the split case, by Exp. XXII, 5.7.10, so only type A_2 remains.

By Exp. XXIV, 3.11, there is a morphism $E \to \mathrm{Spec}(k)$, a principal Galois bundle under the group $\mathbb{Z}/2\mathbb{Z}$ of automorphisms of the Dynkin diagram of type A_2, such that

$$G = \text{Q-Ép}_{E/\operatorname{Spec}(k)}(A_2).$$

If E has a section, G is split and the theorem follows. Otherwise $E = \mathrm{Spec}(k')$, where k'/k is a quadratic extension. Finally, as observed in 5.1.7, we may suppose that G is simply connected, that is, that G is a form of $\mathrm{SL}_{3,k}$.

5.1.10. We are now in the following situation: k is a finite field and k'/k is a quadratic extension. Pin $\mathrm{SL}_{3,k'}$, the group of 3×3 matrices of determinant 1, as follows. Its maximal torus is the group of diagonal matrices, its Borel subgroup is the group of upper triangular matrices, and its root vectors are

$$u_\alpha = \begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}, \qquad u_\beta = \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 1\\ 0 & 0 & 1\end{pmatrix}.$$

The big cell Ω is given by

$$\begin{pmatrix}a & b & c\\ d & e & f\\ g & h & i\end{pmatrix} \in \Omega(S) \quad\Longleftrightarrow\quad a \text{ and } ae-bd \text{ are invertible},$$

and

$$B^u(k) = \left\{ \begin{pmatrix}1 & \bar{x} & z\\ 0 & 1 & x\\ 0 & 0 & 1\end{pmatrix} \;\middle|\; x,z\in k',\ z+\bar{z} = x\bar{x} \right\}.$$

We must prove inclusion (1) of 5.1.6. Thus, for every $a,b,c\in K$, we must find $x,z\in k'$ such that $z+\bar{z}=x\bar{x}$ and

$$\begin{pmatrix}1 & \bar{x} & z\\ 0 & 1 & x\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ a & 1 & 0\\ b & c & 1\end{pmatrix} \in \Omega(K). \tag{3}$$

If $a\neq -1$, take $x=z=0$. If $a=-1$, the required conditions are

$$\begin{cases} z+\bar{z} = x\bar{x},\\ bz-\bar{x}\neq 0,\\ (b+c)\bar{z}+bx-1\neq 0. \end{cases} \tag{4}$$

Let $q\geq 2$ be the number of elements of k. For every $m\in k$, the equation $z+\bar{z}=m$, with $z\in k'$, has q solutions.

If $b=0$, take $x=1$. It remains to solve

$$z+\bar{z}=1, \qquad c\bar{z}\neq 1,$$

which is always possible by the preceding observation.

If $b\neq 0$, take $x=0$. It remains to solve

$$z+\bar{z}=0, \qquad z\neq 0, \qquad (b+c)z\neq 1.$$

This is always possible when $q\geq 3$. When $k=\mathbb{F}_2$, it is possible if $b+c\neq 1$, by taking $z=1$.

It remains to consider $k=\mathbb{F}_2$, $b+c=1$, and $b\neq 0$. The system becomes

$$z+\bar{z}=x\bar{x}, \qquad bz\neq\bar{x}, \qquad \bar{z}+bx\neq 1.$$

If $b=1$, or respectively $b\notin k'$, take $x=1$; the last two conditions become

$$z\neq b^{-1}, \qquad z\neq 1-b,$$

and follow from any solution of $z+\bar{z}=1$. Finally, if $b\in k'-k$, take $x=b$ and $z=\bar{b}$. This completes the proof. $\square$

Corollary 5.2.— *Let S be a semilocal scheme, G an S-reductive group, and P, P' two opposite parabolic subgroups of G. The canonical map*

$$\mathrm{rad}^u(P)(S) \cdot \mathrm{rad}^u(P')(S) \longrightarrow (G/P)(S)$$

is surjective. In particular,

$$(G/P)(S) = G(S)/P(S).$$

Every parabolic subgroup Q of G having the same type as P is of the form

$$Q = \mathrm{int}(uu')P$$

with $u \in \mathrm{rad}^u(P)(S)$ and $u' \in \mathrm{rad}^u(P')(S)$.

The second assertion is plainly equivalent to the first, so let us prove the latter. Let s_i be the closed points of S, let V be the open subscheme of G/P which is the image of P' (and is isomorphic to $\mathrm{rad}^u(P')$, by 4.3.6), and take $x \in (G/P)(S)$. By 5.1, for every i there is

$$u_i \in \mathrm{rad}^u(P)\big(\kappa(s_i)\big)$$

such that $u_i x_{s_i}$ is a section of V_{s_i}. Choose $u \in \mathrm{rad}^u(P)(S)$ lifting all the u_i, as permitted by 2.6. Then ux is a section of V, since this may be checked on the closed fibers. Finally,

$$\mathrm{rad}^u(P')(S) \xrightarrow{\sim} V(S)$$

is bijective, and the result follows.

Corollary 5.3.— *Let S be a semilocal scheme, G an S-reductive group, and P, P', Q three parabolic subgroups of G. There is a $g \in G(S)$ such that $\mathrm{int}(g)Q$ is in transverse position relative to both P and P'.*

In the notation of 4.2.4(ii), we must show that the universally schematically dense open subscheme

$$\underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P')$$

of G has a section over S. [91] Choose a parabolic subgroup Q_1 opposite to Q, by 4.3.5(i), and put

$$U = \mathrm{rad}^u(Q), \qquad U' = \mathrm{rad}^u(Q_1).$$

We shall find the desired g in $U(S)U'(S)$. In this form, 4.2.4(i) and 2.6 show that it suffices to work on the fibers over the closed points of S; we may therefore suppose $S = \mathrm{Spec}(k)$.

If k is algebraically closed, there is a $g \in G(k)$ with the required property. By 5.2 it may be written

$$g = uu'q, \qquad u \in U(k), \quad u' \in U'(k), \quad q \in Q(k),$$

and $\mathrm{int}(uu')Q = \mathrm{int}(g)Q$.

Suppose k is infinite. Let V be the open subset of $U \times_k U'$ defined by the cartesian diagram

$$\begin{tikzcd} U \times_k U' \arrow[rr] & & G \\ V \arrow[u, hook] \arrow[r] & \underline{\mathrm{Gen}}(Q/P) \cap \underline{\mathrm{Gen}}(Q/P'). \arrow[ur, hook] \end{tikzcd} \tag{27}$$

[91] *Editor's note.* The Polo–Gille edition replaces "relatively dense" by "universally schematically dense over S"; compare EGA IV$_3$, Definition 11.10.8.

Since $V(\bar{k}) \neq \varnothing$, as just shown, V is dense in $U \times_k U'$; it therefore has a k-point by 2.7.

Suppose finally that k is finite. Lang's theorem, as used in 5.1.5, shows that P, respectively P', has a Borel subgroup B, respectively B', since

$$\underline{\mathrm{Bor}}(P) \simeq \underline{\mathrm{Bor}}(P/\operatorname{rad}^u(P)), \qquad \underline{\mathrm{Bor}}(P') \simeq \underline{\mathrm{Bor}}(P'/\operatorname{rad}^u(P'))$$

are smooth. [92] Let B_1 be a Borel subgroup opposite to B. By 5.2 there are

$$a \in B^u(k), \qquad a_1 \in B_1^u(k)$$

such that $\operatorname{int}(aa_1)B = B'$. Then

$$B_0 = \operatorname{int}(aa_1)B_1 = \operatorname{int}(a)B_1$$

is opposite to both B' and B. Let Q_0 be the unique parabolic subgroup of G containing B_0 and having the same type as Q, as in 3.8. It is in transverse position relative to P and P', by 4.2.1(vi). On the other hand, 5.2 gives

$$Q_0 = \operatorname{int}(uu')Q, \qquad u \in U(k), \quad u' \in U'(k),$$

which proves the assertion.

Corollary 5.4.— *Let S be a semilocal scheme, G an S-reductive group, and P, Q two parabolic subgroups of G. There is a $g \in G(S)$ such that* $\operatorname{int}(g)P$ *is in osculating position relative to Q; equivalently,* $\operatorname{int}(g)P \cap Q$ *is a parabolic subgroup of G.*

Choose a parabolic subgroup P' opposite to P, by 4.3.5(i). By 5.3, there is a parabolic subgroup P_1' of the same type as P', in transverse position relative to both P and Q. Let T be a maximal torus of $P_1' \cap Q$, as in 4.2.7(ii), and let P_1 be the parabolic subgroup opposite to P_1' relative to T. Then P_1 and Q are in osculating position, by 4.4.5. Since P and P_1 are both opposite to P_1', there is a

$$g \in \operatorname{rad}^u(P_1')(S)$$

such that $\operatorname{int}(g)P = P_1$, by 4.3.5(i). □

For the same reason there is a $u \in \operatorname{rad}^u(P)(S)$ such that $\operatorname{int}(u)P' = P_1'$. Hence $g = \operatorname{int}(u)u'$ for some $u' \in \operatorname{rad}^u(P')(S)$, and

$$P_1 = \operatorname{int}(uu'u^{-1})P = \operatorname{int}(uu')P.$$

This also gives another proof of 5.2.

Corollaries 5.3 and 5.4 are the essential results of this section. We first record some consequences of 5.4.

Corollary 5.5.— *Let S be a semilocal scheme and G an S-reductive group.*

(i) *If P, Q are parabolic subgroups of G and $t(P) \subset t(Q)$, then there is a $g \in G(S)$ such that* $\operatorname{int}(g)P \subset Q$.

(ii) *Let*

$$P_1 \supset P_2 \supset \cdots \supset P_n, \qquad P_1' \supset P_2' \supset \cdots \supset P_n'$$

be two chains of parabolic subgroups of G, with $t(P_i) = t(P_i')$ for every i. There is a $g \in G(S)$ such that $\operatorname{int}(g)P_i = P_i'$ *for every i.*

[92] *Editor's note.* The displayed French source prints $\underline{\mathrm{Bor}}(Q)$ in the second isomorphism; the parallel argument requires $\underline{\mathrm{Bor}}(P')$, which is used here.

(iii) *Let P, Q, P', Q' be four parabolic subgroups of G, with $t(P) = t(P')$ and $t(Q) = t(Q')$. If the pairs (P, P') and (Q, Q') are both in transverse position, respectively both in osculating position, then there is a $g \in G(S)$ such that*

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)Q = Q'.$$

(iv) *Let P, P' be parabolic subgroups of the same type, and let L, respectively L', be a Levi subgroup of P, respectively P'. There is a $g \in G(S)$ such that*

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)L = L'.$$

Proof. Assertion (i) follows immediately from 5.4. Assertion (ii) is proved by induction on n, the case $n = 0$ being trivial. We may suppose that $P_i = P'_i$ for $i = 1, \ldots, n-1$. By 5.2 there is a $g \in G(S)$ such that $\mathrm{int}(g)P_n = P'_n$. But P_n and $\mathrm{int}(g)P_n$ are both contained in $P_{n-1} = P'_{n-1}$, and hence

$$g \in P_{n-1}(S)$$

by 4.4.4(ii). It follows that $\mathrm{int}(g)P_i = P'_i$ for every i.

Assertion (iv) follows at once from 5.2 and 1.8. Let us prove (iii) first in the transverse case. When the types of P and Q are opposite, the assertion follows from (iv) and 4.3.3(iii). In general, by 4.2.7(iii) and 4.4.6 one can find parabolic subgroups

$$P_1 \subset P, \quad P'_1 \subset P', \quad Q_1 \subset Q, \quad Q'_1 \subset Q'$$

such that P_1, P'_1 are opposite, Q_1, Q'_1 are opposite, and $t(P_1) = t(P'_1)$. There is consequently a $g \in G(S)$ such that

$$\mathrm{int}(g)P_1 = P'_1, \qquad \mathrm{int}(g)Q_1 = Q'_1.$$

After conjugating, we may suppose $P_1 = P'_1$ and $Q_1 = Q'_1$. Then P, P' are in osculating position and have the same type, so $P = P'$ by 4.4.4(i). Likewise $Q = Q'$.

It remains to prove (iii) in the osculating case. By 5.2 we may suppose $P = P'$. Again by 5.2 there is a $g \in G(S)$ such that $\mathrm{int}(g)Q = Q'$. But then $g \in P(S)$, by 4.4.4(ii), and therefore

$$\mathrm{int}(g)P = P = P'.$$

□

Definition 5.6.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G. The subgroup P is said to be* minimal *if every parabolic subgroup Q of G contained in P is equal to P.*

Notice that this notion is not, in general, stable under passage to the fibers.

Corollary 5.7.— *Let S be a semilocal scheme and G an S-reductive group.*

(i) *Let*

$$t, t' \in \mathrm{Of}(\mathrm{Dyn}(G))(S).$$

If G has a parabolic subgroup of type t and one of type t', then it has a parabolic subgroup of type $t \cap t'$. In particular, the set of $t(P)$, as P ranges over the parabolic subgroups of G, has a least element $t_{\min}$.

(ii) *Every parabolic subgroup of G contains a minimal parabolic subgroup. A parabolic subgroup of G is minimal if and only if it has type $t_{\min}$. Any two minimal parabolic subgroups of G are conjugate by an element of $G(S)$.*

This follows immediately from 5.4 and 5.5(i).

Remark 5.8.— *A parabolic subgroup opposite to a minimal parabolic subgroup is also minimal. Consequently,*

$$s(t_{\min}) = t_{\min}.$$

Corollary 5.9.— *Let S be a scheme, G an S-reductive group, and P a parabolic subgroup of G. The canonical morphism*

$$G \longrightarrow G/P = X$$

makes G a Zariski-locally trivial P_X-bundle over X. If L is a Levi subgroup of P, the canonical morphism (cf. 3.12)

$$G \longrightarrow G/L = Y$$

makes G a Zariski-locally trivial L_Y-bundle over Y.

It is enough to prove the following lifting assertion. Given a morphism $S' \to S$, with S' local, every morphism $S' \to X$ (respectively $S' \to Y$) lifts to a morphism $S' \to G$. In other words, we may assume that S is local, and must show that

$$G(S) \longrightarrow X(S), \qquad \text{respectively} \qquad G(S) \longrightarrow Y(S),$$

is surjective.

The first assertion was proved in 5.2. For the second, let $y \in Y(S)$. Its canonical image in $X(S)$ is the image of some $g \in G(S)$. The image y' of g in $Y(S)$ therefore has the same image in $X(S)$ as y. There is consequently a unique

$$u \in \operatorname{rad}^u(P)(S)$$

such that $y'u = y$, and the image of gu in $Y(S)$ is indeed y.

Corollary 5.10.— *Let S be a semilocal scheme and G an S-reductive group.*

(i) *Let P be a parabolic subgroup of G, and L a Levi subgroup of P. The canonical maps* (cf. 3.21) *induce bijections*

$$H^1(S, L) \xrightarrow{\sim} H^1(S, P) \xrightarrow{\sim} H^1_{t(P)}(S, G).$$

(ii) *If*

$$t, t' \in \operatorname{Of}(\operatorname{Dyn}(G))(S),$$

then (cf. 3.21)

$$H^1_t(S, G) \cap H^1_{t'}(S, G) = H^1_{t \cap t'}(S, G).$$

(iii) *If P and Q are two parabolic subgroups of G in osculating position, the following canonical diagram is cartesian and all its arrows are injective:*

$$\begin{array}{ccc} H^1(S, P) & \longrightarrow & H^1(S, G) \\ \uparrow & & \uparrow \\ H^1(S, P \cap Q) & \longrightarrow & H^1(S, Q) \end{array}$$

.

Let us prove (i). The map

$$H^1(S, L) \longrightarrow H^1(S, P)$$

is bijective by 2.3, while

$$H^1(S, P) \longrightarrow H^1_{t(P)}(S, G)$$

is surjective by 3.21. We must prove that it is injective, or equivalently that the canonical map

$$H^1(S, P) \longrightarrow H^1(S, G)$$

is injective. Let Q be a principal homogeneous bundle under P, let Q_1 be the associated principal homogeneous bundle under G, and let P' and G' be the corresponding twisted forms of P and G. Clearly G' is an S-reductive group and P' is a parabolic subgroup of G'.

The elements of $H^1(S, P)$ having the same image in $H^1(S, G)$ as the class of Q are naturally identified with the kernel of

$$H^1(S, P') \longrightarrow H^1(S, G').$$

By the exact sequence in cohomology, this kernel is in turn identified with the set of $G'(S)$-orbits in $(G'/P')(S)$.[93] But $G'(S)$ acts transitively on $(G'/P')(S)$, by 5.2.

For (ii), let Q be a principal homogeneous bundle under G, and let G_Q be the corresponding twisted form of G. By definition (3.21), we must prove that G_Q has a parabolic subgroup of type $t \cap t'$ if and only if it has parabolic subgroups of types t and t'. This is precisely the combination of 3.8 and 5.7(i). Finally, (iii) follows immediately from (i) and (ii).

We now state a consequence of 5.3.

Corollary 5.11.— *Let S be a semilocal scheme, G an S-reductive group, and P a parabolic subgroup of G. If $P \neq G$, there exist at least three distinct parabolic subgroups of G having the same type as P. In other words,*

$$P \neq G \qquad \Longrightarrow \qquad (G(S) : P(S)) \geq 3.$$

Indeed, let P' be a parabolic subgroup opposite to P, as in 4.3.5(i). Since $P \neq G$, one has $\mathrm{rad}^u(P') \neq e$, for example by 4.3.2. By 2.1, $\mathrm{rad}^u(P')(S) \neq e$; choose

$$u \in \mathrm{rad}^u(P')(S), \qquad u \neq e.$$

Then $\mathrm{int}(u)P \neq P$. By 5.3 there is a parabolic subgroup P_1 of the same type as P, opposite to both P and $\mathrm{int}(u)P$. Thus P_1, P, and $\mathrm{int}(u)P$ are three distinct parabolic subgroups of G, all of the same type. □

6. Parabolic subgroups and split tori

Proposition 6.1.— *Let S be a scheme, G an S-reductive group, $\mathfrak{g} = \mathrm{Lie}(G/S)$, and Q a split subtorus of G. Write $Q = D_S(M)$, and let*

$$\mathfrak{g} = \bigoplus_{\alpha \in M} \mathfrak{g}^\alpha$$

be the decomposition of $\mathfrak{g}$ under the action of Q. Let $M_1 \subset M$ satisfy

$$0 \in M_1, \qquad \alpha, \beta \in M_1 \Longrightarrow \alpha + \beta \in M_1.$$

[93] For these arguments in nonabelian cohomology, see Giraud's thesis, [Gi71], § III.3.

(i) *There is a unique smooth subgroup H_{M_1} of G, with connected fibers, which contains $\underline{\mathrm{Centr}}_G(Q)$ and whose Lie algebra is*

$$\bigoplus_{\alpha \in M_1} \mathfrak{g}^\alpha.$$

(ii) *The following implications hold:*

$$M_1 = \{0\} \Longrightarrow H_{M_1} = \underline{\mathrm{Centr}}_G(Q),$$
$$M_1 = -M_1 \Longrightarrow H_{M_1} \text{ is reductive},$$
$$M_1 \cup (-M_1) = M \Longrightarrow \begin{array}{c} H_{M_1} \text{ and } H_{-M_1} \text{ are opposite parabolic} \\ \text{subgroups of } G, \text{ with common Levi subgroup } H_{M_1 \cap (-M_1)}. \end{array}$$

Assertions (i) and (ii) are local for the fpqc topology. We may therefore suppose that Q is contained in a maximal torus T of G, and may moreover split G relative to T. Assertion (i) then follows immediately from Exposé XXII, 5.3.5, 5.4.5, and 5.4.7. The assertions in (ii) follow from Exposé XXII, 5.3.5, 5.10.1, and 5.11.3, together with 1.4 and 4.3.2 of the present Exposé.

Corollary 6.2.— *Let S be a scheme, G an S-reductive group, and Q a split subtorus of G. There is a parabolic subgroup of G having $\underline{\mathrm{Centr}}_G(Q)$ as a Levi subgroup.*

Indeed, write $Q = D_S(M)$, choose a total order on the group M, and let M_1 be the set of its positive elements. The subgroup H_{M_1} has the required property.

Corollary 6.3.— *If the S-reductive group G has a noncentral split subtorus, then it has a proper parabolic subgroup* (that is, one different from G).

Combining 6.2 with 5.9 and 5.10 gives the following result.

Corollary 6.4.— *Let S be a scheme, G an S-reductive group, and Q a split subtorus of G. The canonical morphism*

$$G \longrightarrow G/\underline{\mathrm{Centr}}_G(Q)$$

is a locally trivial fibration.

If S is semilocal, the map

$$G(S) \longrightarrow (G/\underline{\mathrm{Centr}}_G(Q))(S)$$

is surjective, and the map

$$H^1(S, \underline{\mathrm{Centr}}_G(Q)) \longrightarrow H^1(S, G)$$

is injective.

6.5.

Suppose that S is connected. If T is an S-torus and T', T'' are two split subtori of T, their product

$$T' \cdot T''$$

is again a split subtorus of T.[94] Indeed, it is identified with the quotient of $T' \times_S T''$ by $T' \cap T''$, and this quotient is split by Exposé IX, 2.11. It follows that T has a largest split subtorus, which we denote by T_{spl}.

[94] *Editor's note.* The Polo–Gille edition adopts multiplicative notation here, replacing the source's "sum $T' + T''$" by "product $T' \cdot T''$".

Lemma 6.6.— *Let S be a connected scheme, T an isotrivial S-torus, and T_{spl} its largest split subtorus. The following conditions are equivalent:*

(i) *There is a homomorphism*

$$T \longrightarrow \mathbf{G}_{m,S}$$

different from the trivial homomorphism.

(ii) $T_{\mathrm{spl}} \neq e$.

Since T is isotrivial, there are a finite group Γ, a connected principal Galois covering $S' \to S$ with group Γ, and an isomorphism

$$T_{S'} \simeq D_{S'}(M).$$

The group M is then a Γ-module, and there is a natural isomorphism

$$\mathrm{Hom}_S(T, \mathbf{G}_{m,S}) = H^0(\Gamma, M).$$

Let V be the vector subspace of $M \otimes \mathbf{Q}$ generated by the elements

$$\gamma(m) - m, \qquad \gamma \in \Gamma, \quad m \in M.$$

One checks immediately that

$$(T_{\mathrm{spl}})_{S'} \simeq D_{S'}\big(M/(M \cap V)\big).$$

Thus (i) is equivalent to

$$H^0(\Gamma, M) \neq 0,$$

or equivalently to

$$H^0(\Gamma, M \otimes \mathbf{Q}) \neq 0,$$

whereas (ii) is equivalent to $M \neq M \cap V$, or to $M \otimes \mathbf{Q} \neq V$. But

$$M \otimes \mathbf{Q} = H^0(\Gamma, M \otimes \mathbf{Q}) \oplus V.$$

This follows at once by considering the projector from $M \otimes \mathbf{Q}$ onto its invariants which sends x to the average of its transforms under Γ.

Lemma 6.7.— *Let S be a scheme, G an S-reductive group, $P \neq G$ a parabolic subgroup of G, L a Levi subgroup of P, and $Q = \mathrm{rad}(L)$. There is a homomorphism*

$$Q \longrightarrow \mathbf{G}_{m,S}$$

different from the trivial homomorphism.

Let $U = \mathrm{rad}^u(P)$. It is invariant under $\mathrm{int}(P)$, hence under $\mathrm{int}(Q)$. Consider the invertible $\mathcal{O}_S$-module

$$\det\big(\mathrm{Lie}(U)\big),$$

the highest exterior power of the locally free $\mathcal{O}_S$-module $\mathrm{Lie}(U)$. The adjoint representation defines a homomorphism

$$f : Q \longrightarrow \mathrm{Aut}\big(\det\big(\mathrm{Lie}(U)\big)\big) = \mathbf{G}_{m,S}.$$

Since $P \neq G$, one has $U \neq e$. Choose $s \in S$ such that $U_s \neq e$. After splitting G_s relative to a maximal torus containing Q_s, it is immediate that $f_s \neq e$.

Proposition 6.8.— *Let S be a connected semilocal scheme, G an S-reductive group, P a parabolic subgroup of G, L a Levi subgroup of P, and $Q = \operatorname{rad}(L)$. Let Q_{spl} be the largest split subtorus of Q, equivalently the largest central split subtorus of L. Then*

$$L = \underline{\operatorname{Centr}}_G(Q_{\text{spl}}).$$

Put $L' = \underline{\operatorname{Centr}}_G(Q_{\text{spl}})$. This is a reductive subgroup of G containing L, and

$$P' = P \cap L'$$

is a parabolic subgroup of L' with Levi subgroup L, by 1.20. If $L' \neq L$, then $L' \not\subset P$, since L is a maximal reductive subgroup of P by 1.7. Hence $P' \neq L'$.

Let G_1 be the derived group of G, and put $P_1 = P' \cap G_1$. By 1.19, P_1 is a parabolic subgroup of the semisimple group G_1, the group $L_1 = L \cap G_1$ is a Levi subgroup of P_1, and

$$Q_1 = \operatorname{rad}(L_1) = \big(\operatorname{rad}(L) \cap G_1\big)^0 = \big(Q \cap G_1\big)^0.$$

The group L_1 has a maximal torus T_1, by Exposé XIV, 3.20, and T_1 is isotrivial, by Exposé XXIV, 4.1.5. Therefore its subtorus Q_1 is also isotrivial, by Exposé IX, 2.11. Since $P_1 \neq G_1$, Lemmas 6.7 and 6.6 give

$$(Q_1)_{\text{spl}} \neq e.$$

It follows that

$$\big(Q_{\text{spl}} \cap G_1\big)^0 \neq e,$$

and hence that $Q_{\text{spl}} \not\subset \operatorname{rad}(L')$, because $\operatorname{rad}(L') \cap G_1$ is finite. This contradicts the definition of L'. □

Corollary 6.9.— *Let S be a connected semilocal scheme, G an S-reductive group, and Q a critical subtorus of G, that is, a subtorus such that*

$$\operatorname{rad}\big(\underline{\operatorname{Centr}}_G(Q)\big) = Q.$$

The group $\underline{\operatorname{Centr}}_G(Q)$ is a Levi subgroup of a parabolic subgroup of G if and only if

$$\underline{\operatorname{Centr}}_G(Q) = \underline{\operatorname{Centr}}_G(Q_{\text{spl}}),$$

or equivalently,

$$\operatorname{Lie}(G)^Q = \operatorname{Lie}(G)^{Q_{\text{spl}}}.$$

This follows from 6.2 and 6.8.

Corollary 6.10.— *Let S be a connected semilocal scheme, G an S-reductive group, and L a subgroup of G. The following conditions are equivalent:*

(i) *There is a parabolic subgroup of G having L as a Levi subgroup.*

(ii) *There is a split subtorus of G whose centralizer is L.*

(iii) *There is a homomorphism*

$$\mathbf{G}_{m,S} \longrightarrow G$$

whose centralizer is L.

The implication (i)⇒(ii) follows from 6.8, and (iii)⇒(i) from 6.1. It remains to prove (ii)⇒(iii). Suppose therefore that

$$L = \underline{\mathrm{Centr}}_G(Q), \qquad Q = D_S(M).$$

Write $\mathfrak{g} = \mathrm{Lie}(G/S)$ and

$$\mathfrak{g} = \bigoplus_{\alpha \in M} \mathfrak{g}^{\alpha},$$

and let R be the set of $\alpha \in M \setminus \{0\}$ for which $\mathfrak{g}^{\alpha} \neq 0$.

Since R is finite and does not contain 0, there is a homomorphism

$$u : M \longrightarrow \mathbf{Z}$$

such that $u(\alpha) \neq 0$ for every $\alpha \in R$. By duality, u yields a homomorphism $\mathbf{G}_{m,S} \to Q$, and hence a homomorphism

$$f : \mathbf{G}_{m,S} \longrightarrow G.$$

One has

$$\underline{\mathrm{Centr}}_G(f) \supset \underline{\mathrm{Centr}}_G(Q).$$

These are two smooth subgroups of G, with connected fibers, and their Lie algebras coincide, since both are equal to $\mathfrak{g}^0$. They are therefore equal, by the usual argument.

Corollary 6.11.— *Let S be a connected semilocal scheme and G an S-reductive group. The maps*

$$L \longmapsto \mathrm{rad}(L)_{\mathrm{spl}}, \qquad Q \longmapsto \underline{\mathrm{Centr}}_G(Q)$$

are mutually inverse, order-reversing bijections between:

- *the set of subgroups L of G which are Levi subgroups of parabolic subgroups of G; and*
- *the set of split subtori Q of G such that*

$$\mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big)_{\mathrm{spl}} = Q.$$

Corollary 6.12.— *Let S be a connected semilocal scheme and G an S-reductive group. Consider the following assertions:*

(i) *G has a parabolic subgroup different from G.*

(ii) *G has a noncentral split subtorus.*

(ii bis) *G has a noncentral split subtorus of relative dimension* 1.

(iii) *There is a group homomorphism*

$$\mathbf{G}_{a,S} \longrightarrow G$$

which is a closed immersion.

Then

$$\text{(i)} \iff \text{(ii)} \iff \text{(ii bis)} \implies \text{(iii)}.$$

The only new assertion is (i)⇒(iii). Let $P \neq G$ be a parabolic subgroup of G. Then $U = \mathrm{rad}^u(P) \neq e$. Consider the last nontrivial subgroup U_n in the composition series of U, as in 2.1. There is an isomorphism

$$U_n \simeq W(E_n),$$

where E_n is a locally free $\mathcal{O}_S$-module, and hence is free because S is semilocal and connected. Since $E_n \neq 0$, there is a monomorphism

$$\mathcal{O}_S \longrightarrow E_n$$

whose image is locally a direct summand. It induces a closed immersion

$$\mathbf{G}_{a,S} = W(\mathcal{O}_S) \hookrightarrow W(E_n) \simeq U_n,$$

which proves (iii).

Remark 6.12.1.— *When S is the spectrum of a field of characteristic* 0*, the Jacobson–Morozov theorem gives* (iii)⇒(ii bis)*. The four preceding conditions are then equivalent* ("Godement's criterion"; cf. [BT65], 8.5).[95]

Definition 6.13.— *Let S be a connected semilocal scheme and G an S-reductive group. The group G is said to be* anisotropic *if it contains no nontrivial split subtorus.*

Corollary 6.14.— *Let S be a connected semilocal scheme. An S-reductive group G is anisotropic if and only if it has no parabolic subgroup $P \neq G$ and its radical is anisotropic.*

Combining 6.6 with Exposé XXIV, 4.1.5, and Exposé XXII, 6.2 gives:

Corollary 6.15.— *Let S be a connected semilocal scheme and G an isotrivial S-reductive group* (for example, G semisimple, or S normal; cf. Exposé X, 5.16)*. The group G is anisotropic if and only if it has no parabolic subgroup $P \neq G$ and*

$$\mathrm{Hom}_{S\text{-gr}}(G, \mathbf{G}_{m,S}) = e.$$

Proposition 6.16.— *Let S be a connected semilocal scheme and G an S-reductive group. The maximal split subtori of G are precisely the largest central split subtori of the Levi subgroups of the minimal parabolic subgroups of G. Any two such tori are conjugate by an element of $G(S)$.*

Let Q be a maximal split subtorus of G. By 6.2,

$$L = \underline{\mathrm{Centr}}_G(Q)$$

is a Levi subgroup of a parabolic subgroup P of G. Since

$$Q \subset \mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big)_{\mathrm{spl}},$$

the maximality of Q gives

$$Q = \mathrm{rad}\big(\underline{\mathrm{Centr}}_G(Q)\big)_{\mathrm{spl}}.$$

By 6.11, L is minimal among the Levi subgroups of parabolic subgroups of G. Hence P is a minimal parabolic subgroup of G, by 1.20. It follows from 5.7 and 5.5(iv) that any two

[95] More generally, this remains true when S is the spectrum of a perfect field (Tits); see [BT71], Corollary 3.7.

such subtori Q are conjugate by an element of $G(S)$. The conjugacy of the tori Q and of the pairs (P, L) also proves the first assertion.

Corollary 6.17.— *Let S be a connected semilocal scheme, and let P, P' be two minimal parabolic subgroups in standard position* (4.5.1.1). *Then $P \cap P'$ contains a common Levi subgroup of P and P'.*

Indeed, $P \cap P'$ contains a maximal torus T of G, by 4.5.1(v). Let L be the unique Levi subgroup of P which contains T. By 1.21,

$$\mathrm{rad}(P) \cap T = \mathrm{rad}(P) \cap L = \mathrm{rad}(L).$$

Thus $\mathrm{rad}(P) \cap T$ contains $\mathrm{rad}(L)_{\mathrm{spl}}$, which is a maximal split subtorus of G, and is therefore necessarily equal to T_{spl}. Hence

$$L = \underline{\mathrm{Centr}}_G(T_{\mathrm{spl}}),$$

and, by symmetry, L is also a Levi subgroup of P'.

Remark 6.18.— *It follows from 1.21 that a parabolic subgroup P of G is minimal if and only if* $\mathrm{rad}(P)$ *contains a maximal split subtorus of G. In that case, if T is a maximal torus of P, the torus T_{spl} is a maximal split subtorus both of G and of* $\mathrm{rad}(P)$, *and* $\underline{\mathrm{Centr}}_G(T_{\mathrm{spl}})$ *is a Levi subgroup of P. Moreover, every Levi subgroup of P is obtained in this way.*

7. Relative root datum

Throughout this section, S denotes a nonempty connected semilocal scheme, G an S-reductive group, Q a maximal split subtorus of G, and L the centralizer of Q in G:

$$L = \underline{\mathrm{Centr}}_G(Q).$$

7.1.

Since Q is the largest central split subtorus of L, every element of $G(S)$ which normalizes L also normalizes Q. Thus (cf. 7.1.1)

$$\underline{\mathrm{Norm}}_G(L)(S) = \underline{\mathrm{Norm}}_G(Q)(S).$$

On the other hand, 6.4 shows that

$$G(S) \longrightarrow (G/L)(S)$$

is surjective. Consequently there is a canonical identification

$$W_G(Q)(S) = (\underline{\mathrm{Norm}}_G(Q)/\underline{\mathrm{Centr}}_G(Q))(S) \simeq \underline{\mathrm{Norm}}_G(L)(S)/L(S).$$

Let

$$M = \mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S}),$$

so that $Q \simeq D_S(M)$ canonically. Denote by W the group of automorphisms of M defined by $W_G(Q)(S)$. Thus

$$W \simeq W_G(Q)(S) \simeq \underline{\mathrm{Norm}}_G(L)(S)/L(S).$$

7.1.1.

In general one does not have $\underline{\mathrm{Norm}}_G(L) = \underline{\mathrm{Norm}}_G(Q)$. For example, let S be the spectrum of a field k having a quadratic extension k', and take for G the unitary group

$$^2A_{2,k'/k}$$

(cf. Exposé XXIV, 3.11.2). The minimal parabolic subgroups of G are its Borel subgroups, and their Levi subgroups are maximal tori; hence

$$(\underline{\mathrm{Norm}}_G(L)/L)(k) = \mathfrak{S}_3.$$

Since G is not split, its maximal split tori have dimension at most 1, and are therefore isomorphic to $\mathbf{G}_{m,k}$. The group $\underline{\mathrm{Norm}}_G(Q)/L$ acts faithfully on Q, and

$$(\underline{\mathrm{Norm}}_G(Q)/L)(k) = \mathbf{Z}/2\mathbf{Z}.$$

7.2.

Let P be a parabolic subgroup of G with Levi subgroup L; such a subgroup exists by 6.2. It is necessarily minimal, by 6.18. Using the conjugacy of the minimal parabolic subgroups of G (5.7), the conjugacy of the Levi subgroups of a parabolic subgroup (1.8), and the equalities

$$P = \underline{\mathrm{Norm}}_G(P), \qquad \underline{\mathrm{Norm}}_G(L) \cap P = L$$

from 1.6, we obtain the following conclusion: the set of (minimal) parabolic subgroups of G having Levi subgroup L is a principal homogeneous space under W.

7.3.

The Lie algebra of G decomposes under the action of Q as

$$\mathrm{Lie}(G) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R} \mathrm{Lie}(G)^\alpha,$$

where R is the set of nonzero characters α of Q such that $\mathrm{Lie}(G)^\alpha \neq 0$. These are the roots of G relative to Q.

Let

$$M^* = \mathrm{Hom}_{S\text{-gr}}(\mathbf{G}_{m,S}, Q).$$

It is in duality with M, and W acts on it naturally by transport of structure.

Theorem 7.4.— *With the notation of 7.3, there is a unique map*

$$R \longrightarrow M^*, \qquad \alpha \longmapsto \alpha^*,$$

which makes (M, M^*) into a root datum (Exposé XXI, 1.1) *with Weyl group W.*

Moreover, parabolic subgroups P of G having Levi subgroup L correspond bijectively to systems of positive roots R^+ in R by the relation

$$\mathrm{Lie}(P) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R^+} \mathrm{Lie}(G)^\alpha.$$

7.4.1.

Suppose first that the existence of the desired map $\alpha \mapsto \alpha^*$ has been proved. By Exposé XXI, 3.4.10, s_α is the unique element of W such that

$$s_\alpha(m) - m$$

is a rational multiple of α for every $m \in M$. Consequently s_α is determined by α. Since

$$\alpha^*(m)\alpha = m - s_\alpha(m),$$

[96] the coroot α^* is also determined by α, proving the uniqueness of the map.

7.4.2.

Let $\alpha \in R$. Denote by L_α, H_α, and $H_{-\alpha}$, respectively, the unique smooth subgroups of G with connected fibers which contain L and satisfy (cf. 6.1(i))

$$\begin{aligned}
\mathrm{Lie}(L_\alpha) &= \mathrm{Lie}(L) \oplus \bigoplus_{\gamma \in \mathbf{Z}\alpha \cap R} \mathrm{Lie}(G)^\gamma,\\
\mathrm{Lie}(H_\alpha) &= \mathrm{Lie}(L) \oplus \bigoplus_{\gamma \in \mathbf{N}\alpha \cap R} \mathrm{Lie}(G)^\gamma,\\
\mathrm{Lie}(H_{-\alpha}) &= \mathrm{Lie}(L) \oplus \bigoplus_{\gamma \in -\mathbf{N}\alpha \cap R} \mathrm{Lie}(G)^\gamma.
\end{aligned}$$

The group L_α is a reductive subgroup of G, and $H_\alpha, H_{-\alpha}$ are opposite parabolic subgroups of L_α, with common Levi subgroup L, by 6.1(ii).

By 7.2 there is therefore an element

$$s_\alpha \in \underline{\mathrm{Norm}}_{L_\alpha}(Q)(S)/L(S) \subset W$$

such that $s_\alpha(H_\alpha) = H_{-\alpha}$. One has

$$s_\alpha(\alpha) = -\alpha,$$

because α, respectively $-\alpha$, is the common divisor of the elements of R occurring in H_α, respectively $H_{-\alpha}$. Moreover $s_\alpha^2 = \mathrm{id}$, because $s_\alpha^2(H_\alpha)$ and H_α are both opposite to $H_{-\alpha}$ relative to L. We have thus constructed an element $s_\alpha \in W$ with the properties

$$s_\alpha(\alpha) = -\alpha, \qquad s_\alpha^2 = \mathrm{id}, \tag{x}$$

and

$$s_\alpha \text{ can be represented by an element of } L_\alpha(S). \tag{xx}$$

The construction is canonical in α; in particular,

$$w s_\alpha w^{-1} = s_{w(\alpha)} \qquad \text{for every } w \in W. \tag{xxx}$$

[96] *Editor's note.* Here and again in 7.4.4, the Polo–Gille edition restores the factor α omitted after $\alpha^*(m)$ in the original text.

7.4.3.

We now prove

$$s_\alpha(m) - m \in \mathbf{Z}\alpha \qquad \text{for every } m \in M. \tag{xxxx}$$

Since S is connected, this assertion is local for the fpqc topology. We may therefore suppose that

$$L_\alpha = G_1$$

is split relative to a maximal torus T_1 of L. Let (G_1, T_1, M_1, R_1) be such a splitting. The monomorphism $Q \to T_1$ identifies M with a quotient of M_1; denote the canonical map by

$$p : M_1 \longrightarrow M.$$

The image of R_1 under p, possibly including zero, consists of the roots of G_1 relative to Q, hence of the elements of R which are integral multiples of α. Thus

$$p(R_1) \subset \mathbf{Z}\alpha.$$

By (xx), an element of $\underline{\mathrm{Norm}}_{G_1}(L)(S)$ induces s_α on Q. Exposé XXII, 5.10.10, then gives a section

$$w \in (W_1)_S(S)$$

which induces s_α on Q, where W_1 is the Weyl group of the root datum $(M_1, R_1, \ldots)$. After restricting S if necessary, we may suppose that $w \in W_1$ and that

$$p(w(m_1)) = s_\alpha(p(m_1)) \qquad (m_1 \in M_1).$$

By definition, w is a product of reflections associated with elements of R_1. Hence $w(m_1) - m_1$ is an integral linear combination of elements of R_1. It follows that

$$s_\alpha(p(m_1)) - p(m_1)$$

is an integral linear combination of elements of $p(R_1) \subset \mathbf{Z}\alpha$, and is therefore an integral multiple of α, as required.

7.4.4.

We may now define $\alpha^* \in M^*$ by

$$\alpha^*(m)\alpha = m - s_\alpha(m).$$

By (x),

$$\langle \alpha^*, \alpha \rangle = 2.$$

Moreover, (xxx) implies that for every $(\alpha, \alpha') \in R \times R$,

$$s_\alpha(\alpha') \in R, \qquad s_\alpha(\alpha'^*) = s_\alpha(\alpha')^*.$$

By Exposé XXI, 1.1, the map $\alpha \mapsto \alpha^*$ therefore defines a root datum on (M, M^*).

7.4.5.

Let W' be the Weyl group of this root datum, that is, the group of transformations of M generated by the s_α. Then $W' \subset W$.

Choose a total order on the free abelian group M, and put

$$R^+ = \{\alpha \in R \mid \alpha \geq 0\}.$$

This is a system of positive roots in R. Let $w \in W$, and represent it by an element

$$n \in \underline{\mathrm{Norm}}_G(L)(S) = \underline{\mathrm{Norm}}_G(Q)(S).$$

Put $P = H_{R^+}$, in the notation of 6.1. Then P is a parabolic subgroup of G with Levi subgroup L, and

$$\mathrm{int}(n)P = H_{w(R^+)}.$$

It follows from 7.3 that $w(R^+) = R^+$ implies $w = e$. The group W' acts transitively on the systems of positive roots in R, by Exposé XXI, 3.3.7, while the stabilizer of R^+ in W is trivial. Therefore $W = W'$.

It follows as well that $W = W'$ acts simply transitively both on the systems of positive roots in R and on the parabolic subgroups of G having Levi subgroup L. This proves the last assertion of 7.4. □

7.5.

Let P, P_1 be two minimal parabolic subgroups of G, with Levi subgroups L, L_1, and let Q, Q_1 be the largest central split subtori of L, L_1, respectively. Then the pairs (P, Q) and (P_1, Q_1) are conjugate: there is a $g \in G(S)$ such that

$$\mathrm{int}(g)P = P_1, \qquad \mathrm{int}(g)Q = Q_1.$$

Indeed, P and P_1 are conjugate by 5.7, so we may suppose $P = P_1$. Then L and L_1 are conjugate by an element of $P(S)$, by 1.8. Moreover, if g, g' are two elements of $G(S)$ which conjugate (P, Q) to (P_1, Q_1), then $g'^{-1}g$ normalizes P and Q, hence also P and L. But

$$\underline{\mathrm{Norm}}_G(P) \cap \underline{\mathrm{Norm}}_G(L) = P \cap \underline{\mathrm{Norm}}_G(L) = L = \underline{\mathrm{Centr}}_G(Q).$$

Consequently the isomorphism

$$Q \xrightarrow{\sim} Q_1$$

induced by $\mathrm{int}(g)$ is independent of g.

Let $\mathcal{R}, \mathcal{R}_1$ be the root data defined by 7.4 in

$$\mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S}) \quad \text{and} \quad \mathrm{Hom}_{S\text{-gr}}(Q_1, \mathbf{G}_{m,S}),$$

and let R^+, R_1^+ be the systems of positive roots corresponding to P, P_1. The canonical isomorphism $Q \simeq Q_1$ just constructed carries $(\mathcal{R}, R^+)$ to $(\mathcal{R}_1, R_1^+)$. We may therefore define the *based relative root datum* of G over S, by identifying the various pairs $(\mathcal{R}, R^+)$ through the transitive system of isomorphisms above.[97]

From now on, denote this based root datum by

$$\mathcal{R}(G/S) = (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}^+).$$

For every pair $P \supset Q$ as above there is therefore a canonical isomorphism

$$\underline{M} \xrightarrow{\sim} \mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S})$$

which carries $\underline{R}$, respectively $\underline{R}^+$, to the roots of G, respectively of P, relative to Q, and carries $W(\mathcal{R})$ to $W_G(Q)(S)$.

[97] A based root datum is a root datum together with the choice of a system of positive roots, equivalently of simple roots; cf. Exposé XXIII, 1.5.

7.6.

Let again Q be a maximal split torus of G, let P be a minimal parabolic subgroup of G with Levi subgroup $\underline{\mathrm{Centr}}_G(Q)$, let

$$(M, M^*, R, R^*, R^+)$$

be the corresponding based root datum of 7.4, and let Δ be the set of simple roots in R^+. For every $A \subset \Delta$, let

$$R_A = R^+ \cup (\mathbf{Z}A \cap R^-).$$

Thus R_A consists of the positive roots together with the negative roots which are integral linear combinations of elements of A. It is a closed set of roots, by Exposé XXI, 3.1.4, and every closed set containing R^+ is uniquely of this form, by Exposé XXI, 3.3.10.

By 6.1 there is a unique smooth subgroup P_A of G, with connected fibers, which contains $\underline{\mathrm{Centr}}_G(Q)$ and satisfies

$$\mathrm{Lie}(P_A) = \mathrm{Lie}(G)^0 \oplus \bigoplus_{\alpha \in R_A} \mathrm{Lie}(G)^\alpha.$$

The following result now follows immediately from 6.1, the conjugacy of minimal parabolic subgroups, and the fact that the set of roots of a parabolic subgroup of G containing Q is closed (which follows from 1.4 after splitting).

Proposition 7.7.—

(i) *The map*

$$A \longmapsto P_A$$

is an inclusion-preserving bijection from the set of subsets of Δ onto the set of parabolic subgroups of G containing P.

(ii) *Every parabolic subgroup of G is conjugate by an element of $G(S)$ to a unique P_A.*

7.8.

Let $P \supset Q$ be as above. Consider the relative root datum of G over S defined in 7.5 and the canonical isomorphism

$$f : (M, M^*, R, R^*, R^+) \xrightarrow{\sim} (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}^+).$$

The set Δ of simple roots in R^+ is carried to the set $\underline{\Delta}$ of simple roots in $\underline{R}^+$. Thus every subset $A \subset \Delta$ is carried to a subset

$$f(A) \subset \underline{\Delta}.$$

Definition 7.9.0.— *Let H be any parabolic subgroup of G. By 7.7*(ii)*, it is conjugate to a unique P_A. Put*

$$t_r(H) = f(A) \subset \underline{\Delta}.$$

The conjugacy theorems show immediately that $t_r(H)$ is independent of the choice of the pair $P \supset Q$. It is called the relative type *of H.*

Proposition 7.9.—

(i) *The map $H \mapsto t_r(H)$ induces a bijection from the set of $G(S)$-conjugacy classes of parabolic subgroups of G onto the set of subsets of $\underline{\Delta}$.*

(ii) *Let H be a parabolic subgroup of G, P a minimal parabolic subgroup contained in H, Q the largest central split torus of a Levi subgroup of P, Δ the set of simple roots of P relative to Q, and*

$$f : \Delta \xrightarrow{\sim} \underline{\Delta}$$

the canonical isomorphism. For every $\alpha \in \Delta$,

$$f(\alpha) \in t_r(H) \iff \mathrm{Lie}(H)^{-\alpha} \neq 0,$$

and

$$H = P_A, \qquad A = f^{-1}(t_r(H)).$$

(iii) *If H, H' are two parabolic subgroups of G containing P, then* (see 3.8.1(ii) and 5.5(i) for other equivalent conditions)

$$t_r(H) \subset t_r(H') \iff t(H) \subset t(H').$$

7.10.

The relative positions of two minimal parabolic subgroups can be described as follows. We use the notation $t_2(P, P_1)$ of 4.5.2.

(1) Let P, P_1, P', P_1' be four minimal parabolic subgroups of G. Then

$$t_2(P, P_1) = t_2(P', P_1'),$$

that is, the pairs (P, P_1) and (P', P_1') are fpqc-locally conjugate, if and only if there is a $g \in G(S)$ such that

$$\mathrm{int}(g)P = P', \qquad \mathrm{int}(g)P_1 = P_1'.$$

(2) Fix a minimal parabolic subgroup P of G with Levi subgroup L, and let T be a maximal torus of L. Consider the scheme

$$\underline{\mathrm{Par}}_{t_{\min}}(G; P)$$

of minimal parabolic subgroups of G in standard position relative to P. There is a morphism (cf. 4.5.5)

$$f : \underline{\mathrm{Par}}_{t_{\min}}(G; P) \longrightarrow W_P(T)\backslash W_G(T)/W_P(T)$$

whose fibers are the orbits of P in $\underline{\mathrm{Par}}_{t_{\min}}(G; P)$. By (1), f therefore induces a monomorphism

$$P(S)\backslash \underline{\mathrm{Par}}_{t_{\min}}(G; P)(S) \hookrightarrow (W_P(T)\backslash W_G(T)/W_P(T))(S).$$

Its image is identified with W. This is the Bruhat theorem: each orbit of $P(S)$ in $\underline{\mathrm{Par}}_{t_{\min}}(G; P)(S)$ contains exactly one parabolic subgroup of G with Levi subgroup L, that is, one of the form

$$\mathrm{int}(n)P, \qquad n \in \underline{\mathrm{Norm}}_G(L)(S).$$

(3) Equivalently, let $E(S)$ be the set of elements $g \in G(S)$ such that $\mathrm{int}(g)P$ and P are in mutual standard position. Then $E(S)$ is partitioned into double cosets modulo $P(S)$, indexed by W:
$$E(S) = P(S)\, W\, P(S).$$
Putting $U = \mathrm{rad}^u(P)$, one may also write
$$E(S) = U(S) \cdot \underline{\mathrm{Norm}}_G(L)(S) \cdot U(S).$$
This exhibits a partition of $E(S)$ into double cosets modulo $U(S)$, indexed by $\underline{\mathrm{Norm}}_G(L)(S)$.

(4) If S is the spectrum of a field, then $E(S) = G(S)$, and one recovers [BT65], 5.15.

Counterexamples 7.11.— *Let*
$$S = \mathrm{Spec}(\mathbf{Z}/4\mathbf{Z}), \qquad G = \mathrm{SL}_{2,S}\,.$$
Let B be the usual Borel subgroup consisting of the matrices
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \textit{with } c = 0,$$
and put
$$g = \begin{pmatrix} -1 & 1 \\ 2 & 1 \end{pmatrix} \in G(S), \qquad B' = \mathrm{int}(g)B.$$
Then
$$B(S) = B'(S),$$
whereas $B \cap B'$ contains no maximal torus. [98]

This shows both that two distinct minimal parabolic subgroups may have the same group of sections and that, in general, one cannot decide from the groups $P(S)$ and $P'(S)$ alone whether two minimal parabolic subgroups P, P' are in standard position. In particular, the subset $E(S)$ of $G(S)$ does not appear to be definable solely from the configuration
$$\{G(S), P(S), \underline{\mathrm{Norm}}_G(L)(S)\}.$$
In the example above, this subset is defined by $c \neq 2$. [99]

7.12.

We now study how $\mathcal{R}(G/S)$ varies with S. Let S' be an S-scheme which is likewise nonempty, connected, and semilocal. If Q is a maximal split torus of G, then $Q_{S'}$ is a split torus of $G_{S'}$. Choose a maximal split torus Q' of $G_{S'}$ containing $Q_{S'}$, and put
$$\begin{aligned} M &= \mathrm{Hom}_{S\text{-gr}}(Q, \mathbf{G}_{m,S}) \simeq \mathrm{Hom}_{S'\text{-gr}}(Q_{S'}, \mathbf{G}_{m,S'}), \\ M' &= \mathrm{Hom}_{S'\text{-gr}}(Q', \mathbf{G}_{m,S'}). \end{aligned}$$

[98] Indeed, B' is the subgroup of G defined by $c = 2(a+b)$. Thus $B \cap B'$ is not flat over S, and hence contains no maximal torus, by 4.5.1.

[99] More generally, for every S-scheme S', the set $E(S')$ is defined by the condition that c is either zero or invertible.

The monomorphism $Q_{S'} \to Q'$ induces an epimorphism

$$u : M' \longrightarrow M.$$

Write

$$L = \underline{\mathrm{Centr}}_G(Q), \qquad L' = \underline{\mathrm{Centr}}_{G_{S'}}(Q').$$

Then $L' \subset L_{S'}$.

If H is a subgroup of G containing L, then $H_{S'}$ contains L', and

$$\mathrm{Lie}(H) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R_H} \mathrm{Lie}(H)^{\alpha},$$
$$\mathrm{Lie}(H_{S'}) = \mathrm{Lie}(L') \oplus \bigoplus_{\alpha' \in R'_{H_{S'}}} \mathrm{Lie}(H_{S'})^{\alpha'}.$$

Here R_H, respectively $R'_{H_{S'}}$, denotes the set of roots of H, respectively $H_{S'}$, relative to Q, respectively Q'. It follows immediately that

$$R_H \subset u(R'_{H_{S'}}) \subset R_H \cup \{0\}.$$

Taking $H = G$, we first obtain

$$R \subset u(R') \subset R \cup \{0\}.$$

Next take for H a minimal parabolic subgroup P with Levi subgroup L. Then $R'_{H_{S'}}$ contains a system of positive roots in R'. By 7.4, there is therefore a minimal parabolic subgroup P' of $G_{S'}$, with Levi subgroup L', contained in $P_{S'}$. We have constructed the diagram of inclusions

$$\begin{tikzcd}
Q_{S'} \arrow[r, hook] \arrow[d, hook] & L_{S'} \arrow[r, hook] & P_{S'} \\
Q' \arrow[r, hook] & L' \arrow[r, hook] \arrow[u, hook] & P' \arrow[u, hook]
\end{tikzcd}$$

.

If R^+, respectively Δ, is the system of positive, respectively simple, roots in R defined by P, and if R'^+, Δ' are defined similarly, then

$$R^+ \subset u(R'^+) \subset R^+ \cup \{0\}, \qquad \Delta \subset u(\Delta') \subset \Delta \cup \{0\}.$$

Now let

$$w \in W \simeq \underline{\mathrm{Norm}}_G(Q)(S)/\underline{\mathrm{Centr}}_G(Q)(S),$$

and choose a representative $n \in \underline{\mathrm{Norm}}_G(Q)(S)$. Since $\mathrm{int}(n)Q = Q$, one has $\mathrm{int}(n)L = L$, and hence $\mathrm{int}(n)L_{S'} = L_{S'}$. The tori Q' and $\mathrm{int}(n)Q'$ are therefore two maximal split tori of $L_{S'}$, so they are conjugate by an element $x \in L(S')$. We may choose x so that

$$\mathrm{int}(nx)Q' = Q'.$$

Thus $n' = nx$ belongs to $\underline{\mathrm{Norm}}_{G_{S'}}(Q')(S')$. Let w' be the image of n' in

$$W' \simeq \underline{\mathrm{Norm}}_{G_{S'}}(Q')(S')/\underline{\mathrm{Centr}}_{G_{S'}}(Q')(S').$$

The action of w' on M' is compatible with $u : M' \to M$, and the induced action on M is the action of w.

Using the definition of relative root data together with the conjugacy theorems gives the following result.

Theorem 7.13.— *Let S, S' be nonempty connected semilocal schemes, let $S' \to S$ be a morphism of schemes, let G be an S-reductive group, and write*

$$\mathcal{R}(G/S) = (\underline{M}, \underline{M}^*, \underline{R}, \underline{R}^*, \underline{R}^+),$$
$$\mathcal{R}(G_{S'}/S') = (\underline{M}', \underline{M}'^*, \underline{R}', \underline{R}'^*, \underline{R}'^+)$$

for the based relative root data. There is a canonical homomorphism

$$u : \underline{M}' \longrightarrow \underline{M}$$

with the following properties:

(i) *The homomorphism u is surjective.*

(ii) *For every $w \in W$, there is an element $w' \in W'$ compatible with u and inducing w on $\underline{M}$.*

(iii) *For every subset $X \subset \underline{M}$, put*

$$X^\wedge = X \cap (\underline{M} \setminus \{0\}).$$

Then

$$u(\underline{R}'^+)^\wedge = \underline{R}^+, \qquad u(\underline{\Delta}')^\wedge = \underline{\Delta}.$$

(iv) *For every parabolic subgroup H, consider*

$$t_r(H) \subset \underline{\Delta}, \qquad t_r(H_{S'}) \subset \underline{\Delta}'.$$

Then

$$\begin{aligned} t_r(H_{S'}) &= u^{-1}(t_r(H) \cup \{0\}) \cap \underline{\Delta}' \\ &= \{\alpha' \in \underline{\Delta}' \mid u(\alpha') \in t_r(H) \text{ or } u(\alpha') = 0\}. \end{aligned}$$

Remark 7.14.— *If G is splittable over S, its maximal split tori are maximal tori, and the relative notions introduced here coincide with the previously defined absolute notions. The preceding theorem therefore describes the relative root datum $\mathcal{R}(G/S)$ and the relative type t_r in terms of the absolute root datum and absolute type of $G_{S'}$, where S' is chosen so that $G_{S'}$ is splittable* (cf. Exposé XXIV, 4.4.1). *See* [BT65], *6.12 and following, for this description.*

7.15.

Let S be a Henselian local scheme, s_0 its closed point, and S_0 the spectrum of the residue field at s_0, identified with a closed subscheme of S. For every object X over S, write X_0 for its base change to S_0. Let G be an S-reductive group.

For every parabolic subgroup $\overline{P}$ of G, the group $\overline{P}_0$ is a parabolic subgroup of G_0. Conversely, for every parabolic subgroup P of G_0, there is a parabolic subgroup $\overline{P}$ of G such that

$$\overline{P}_0 = P.$$

This follows from Hensel's lemma and from the fact that $\underline{\mathrm{Par}}(G)$ is a smooth S-scheme. In particular, by 5.7, $\overline{P}$ is minimal if and only if $\overline{P}_0$ is minimal.

Once such a subgroup $\overline{P}$ has been chosen, the same argument shows that every maximal split subtorus of $\overline{P}_0$ is of the form T_0, where T is a maximal split subtorus of $\overline{P}$. It follows readily that the relative root data of G over S and of G_0 over S_0 are canonically isomorphic. Thus the theory of parabolic subgroups of G reduces to the corresponding theory for G_0.

Conversely, every S_0-reductive group is of the form G_0, by Exposé XXIV, Proposition 1.21. This makes it possible to reduce the study of the parabolic subgroups of an S_0-reductive group to the corresponding study over S.

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